

CLI Reference Guide

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Table of Contents

List of Tables.....	98
1 Getting started.....	104
2 Summary of changes.....	105
3 CLI usage.....	124
3.1 CLI structure.....	124
3.2 CLI contexts.....	124
4 Card and system management command reference.....	126
4.1 Card and system management command hierarchies.....	126
4.1.1 Card and system management configuration command hierarchies.....	126
4.1.1.1 Card configuration command hierarchy.....	126
4.1.1.2 Port configuration command hierarchy.....	126
4.1.1.3 Mobile-gateway system configuration command hierarchy.....	127
4.1.1.4 System security command hierarchy.....	128
4.1.1.5 Admin redundancy command hierarchy.....	129
4.1.1.6 Redundancy configuration command hierarchy.....	129
4.1.1.7 Accounting policy command hierarchy.....	130
4.1.2 Card and system management show command hierarchies.....	130
4.1.2.1 Show card command hierarchy.....	131
4.1.2.2 Show certificate command hierarchy.....	131
4.1.2.3 Show mobile gateway command hierarchy.....	131
4.1.2.4 Show redundancy command hierarchy.....	132
4.2 Card and system management command descriptions.....	132
4.2.1 Card and system management configuration command descriptions.....	132
4.2.1.1 Card configuration command descriptions.....	132
4.2.1.1.1 card.....	132
4.2.1.1.2 card-type.....	133
4.2.1.1.3 mda-type.....	134
4.2.1.2 Port configuration command descriptions.....	136
4.2.1.2.1 eth-pdu-port.....	136
4.2.1.3 Generic system configuration command descriptions.....	136
4.2.1.3.1 shutdown.....	136
4.2.1.4 Mobile-gateway system configuration command descriptions.....	137
4.2.1.4.1 system.....	137
4.2.1.4.2 call-insight.....	137
4.2.1.4.3 location.....	138
4.2.1.4.4 disable.....	138
4.2.1.4.5 limit.....	139

4.2.1.4.6	primary.....	139
4.2.1.4.7	max-files-number.....	140
4.2.1.4.8	disable-delayed-acct.....	140
4.2.1.4.9	error-indication-tx.....	141
4.2.1.4.10	group.....	141
4.2.1.4.11	card.....	143
4.2.1.4.12	pcmd-profile.....	144
4.2.1.4.13	override-profile.....	145
4.2.1.4.14	protect-active-delay.....	146
4.2.1.4.15	ipv6-path-mtu.....	146
4.2.1.4.16	ipv6-udp-checksum.....	147
4.2.1.4.17	non-terminating-rules.....	147
4.2.1.4.18	policer-distribution.....	148
4.2.1.4.19	resource-pool.....	148
4.2.1.4.20	thresholds.....	149
4.2.1.4.21	cam-utilization-threshold.....	150
4.2.1.4.22	system-mg-ism.....	151
4.2.1.4.23	counter.....	151
4.2.1.4.24	interval.....	152
4.2.1.4.25	ttc-accuracy.....	153
4.2.1.4.26	ue-fragmented-packets-handling.....	153
4.2.1.4.27	use-fixed-udp-src-port.....	154
4.2.1.4.28	security.....	155
4.2.1.4.29	dist-cpu-protection.....	155
4.2.1.4.30	policy.....	155
4.2.1.4.31	protocol.....	156
4.2.1.4.32	ssh.....	157
4.2.1.4.33	client-host-key-list.....	157
4.2.1.4.34	host-key.....	157
4.2.1.4.35	client-kex-list.....	159
4.2.1.4.36	kex.....	159
4.2.1.4.37	server-host-key-list.....	160
4.2.1.4.38	server-host-key-rsa-length.....	160
4.2.1.4.39	server-kex-list.....	161
4.2.1.4.40	tls.....	162
4.2.1.4.41	client-cipher-list.....	162
4.2.1.4.42	cipher.....	162
4.2.1.4.43	server-cipher-list.....	163
4.2.1.4.44	cipher.....	163
4.2.1.5	Admin redundancy command descriptions.....	164
4.2.1.5.1	redundancy.....	164
4.2.1.5.2	cpm-switchover.....	165
4.2.1.5.3	mc-master-lock.....	165
4.2.1.5.4	mc-mobile-switchover.....	166
4.2.1.5.5	mc-slave-lock.....	167
4.2.1.5.6	mg-switchover.....	168
4.2.1.5.7	group.....	169

4.2.1.6 Redundancy configuration command descriptions.....	170
4.2.1.6.1 multi-chassis.....	170
4.2.1.6.2 peer.....	170
4.2.1.6.3 mc-mobile.....	171
4.2.1.6.4 bfd-enable.....	171
4.2.1.6.5 diameter-connection-lost-detection.....	172
4.2.1.6.6 gx-failover.....	172
4.2.1.6.7 hold-on-neighbor-failure.....	173
4.2.1.6.8 keep-alive-interval.....	174
4.2.1.6.9 master-traffic-detection.....	174
4.2.1.6.10 mc-complete-ue-sync.....	175
4.2.1.6.11 mc-redirect.....	175
4.2.1.6.12 mobile-gateway.....	176
4.2.1.6.13 sync-mc-ipsec.....	177
4.2.1.6.14 tunnel-group.....	177
4.2.1.6.15 peer-connect.....	178
4.2.1.6.16 slave-traffic-delay-timer.....	178
4.2.1.6.17 traffic-detection.....	179
4.2.1.6.18 traffic-detection-poll-timer.....	180
4.2.1.7 Accounting policy command descriptions.....	181
4.2.1.7.1 record.....	181
4.2.2 Card and system management show command descriptions.....	187
4.2.2.1 Show card command descriptions.....	187
4.2.2.1.1 card.....	187
4.2.2.2 Show mobile gateway command descriptions.....	189
4.2.2.2.1 call-insight.....	189
4.2.2.2.2 all-ues.....	190
4.2.2.2.3 files.....	192
4.2.2.2.4 ue.....	193
4.2.2.2.5 ue-mask.....	195
4.2.2.2.6 connections.....	196
4.2.2.2.7 ism-mg.....	199
4.2.2.2.8 abs.....	200
4.2.2.2.9 cpu.....	201
4.2.2.2.10 memory-pools.....	204
4.2.2.2.11 sdf.....	207
4.2.2.2.12 mg-vm.....	209
4.2.2.2.13 cpu.....	209
4.2.2.2.14 memory-pools.....	211
4.2.2.2.15 system.....	214
4.2.2.3 Show redundancy command descriptions.....	219
4.2.2.3.1 mc-mobile.....	219
5 PDN command reference.....	233
5.1 PDN command hierarchies.....	233
5.1.1 PDN configuration command hierarchies.....	233
5.1.1.1 PDN command hierarchy.....	234

5.1.1.2 APN PDN command hierarchy.....	240
5.1.1.3 Bearer traffic PDN command hierarchy.....	254
5.1.1.4 Cloud database PDN command hierarchy.....	254
5.1.1.5 Charging PDN command hierarchy.....	255
5.1.1.6 Ga PDN command hierarchy.....	256
5.1.1.7 Gn PDN command hierarchy.....	256
5.1.1.8 Gp PDN command hierarchy.....	257
5.1.1.9 GTP load and overload control PDN command hierarchy.....	257
5.1.1.10 Gx PDN command hierarchy.....	259
5.1.1.11 Gy PDN command hierarchy.....	260
5.1.1.12 KPI-KCI PDN command hierarchy.....	261
5.1.1.13 N2 PDN command hierarchy.....	262
5.1.1.14 N3 PDN command hierarchy.....	262
5.1.1.15 N9 PDN command hierarchy.....	262
5.1.1.16 Network trace command hierarchy.....	263
5.1.1.17 NF profile attributes command hierarchy.....	263
5.1.1.18 NWu PDN command hierarchy.....	264
5.1.1.19 PFCP load and overload PDN command hierarchy.....	264
5.1.1.20 PGW selection PDN command hierarchy.....	265
5.1.1.21 Pi PDN command hierarchy.....	265
5.1.1.22 RADIUS PDN command hierarchy.....	266
5.1.1.23 Rf PDN command hierarchy.....	266
5.1.1.24 S1-u PDN command hierarchy.....	267
5.1.1.25 S11 PDN command hierarchy.....	268
5.1.1.26 S12 PDN command hierarchy.....	269
5.1.1.27 S2a PDN command hierarchy.....	269
5.1.1.28 S2b PDN command hierarchy.....	270
5.1.1.29 S4 PDN command hierarchy.....	270
5.1.1.30 S5 PDN command hierarchy.....	271
5.1.1.31 S6b PDN command hierarchy.....	272
5.1.1.32 S8 PDN command hierarchy.....	273
5.1.1.33 SBA client services PDN command hierarchy.....	274
5.1.1.34 SBA server services PDN command hierarchy.....	281
5.1.1.35 SBA service realm PDN command hierarchy.....	283
5.1.1.36 Sd PDN command hierarchy.....	283
5.1.1.37 Signaling PDN command hierarchy.....	284
5.1.1.38 STa PDN command hierarchy.....	285
5.1.1.39 SWm PDN command hierarchy.....	285
5.1.1.40 SWu PDN command hierarchy.....	286
5.1.1.41 SWw PDN command hierarchy.....	287
5.1.1.42 Sx-N4 PDN command hierarchy.....	287
5.1.1.43 T6c command hierarchy.....	288
5.1.1.44 Threshold PDN command hierarchy.....	288
5.1.1.45 Virtual APN selection PDN command hierarchy.....	289
5.1.2 PDN show command hierarchy.....	290
5.1.3 PDN clear command hierarchy.....	295
5.2 PDN command descriptions.....	297

5.2.1 PDN configuration command descriptions.....	298
5.2.1.1 Generic PDN command descriptions.....	299
5.2.1.1.1 description.....	299
5.2.1.1.2 shutdown.....	300
5.2.1.2 PDN command descriptions.....	301
5.2.1.2.1 pdn.....	301
5.2.1.2.2 5g-ims-fallback-timeout.....	301
5.2.1.2.3 aa-group.....	302
5.2.1.2.4 allow-duplicate-imei.....	303
5.2.1.2.5 allow-duplicate-msisdn.....	303
5.2.1.2.6 allow-multi-ue-session.....	304
5.2.1.2.7 allow-multiple-apn-pdn.....	305
5.2.1.2.8 apn.....	305
5.2.1.2.9 auto-set-handover-indication.....	307
5.2.1.2.10 call-procedure-retry.....	307
5.2.1.2.11 cam-optimization.....	308
5.2.1.2.12 dl-buffering.....	309
5.2.1.2.13 drop-oldest-pkt.....	310
5.2.1.2.14 max-pkt.....	310
5.2.1.2.15 max-pkt-size.....	311
5.2.1.2.16 total-buffer-capacity.....	312
5.2.1.2.17 total-buffer-pkts.....	312
5.2.1.2.18 create-procedure-retry.....	313
5.2.1.2.19 default-traffic-hash.....	314
5.2.1.2.20 del-bearer-reactivation-req-cause-pgw-failure.....	315
5.2.1.2.21 delay-nw-procedures.....	315
5.2.1.2.22 disable-hss-init-update-failure-delete.....	316
5.2.1.2.23 dns-client.....	317
5.2.1.2.24 cache-clear-no-such-name.....	317
5.2.1.2.25 dns-interface.....	317
5.2.1.2.26 dns-profile.....	318
5.2.1.2.27 enable-pcscf-low-ttl.....	319
5.2.1.2.28 refresh-interval.....	320
5.2.1.2.29 topon-match-first.....	321
5.2.1.2.30 ttl-zero.....	321
5.2.1.2.31 dnai-map.....	322
5.2.1.2.32 dns-error-mapping-profile.....	323
5.2.1.2.33 dual-connectivity.....	324
5.2.1.2.34 dual-connectivity-5g.....	324
5.2.1.2.35 enable-sgw-c-partitioning.....	325
5.2.1.2.36 epc-node.....	325
5.2.1.2.37 epdg-5gs-compatibility.....	326
5.2.1.2.38 exclude-error-indication-extension-header.....	327
5.2.1.2.39 fwd-gtpu-echo-req-to-mg.....	327
5.2.1.2.40 gtpc-init-msg-src-ip-validation-bypass.....	328
5.2.1.2.41 gtpu-drop-echo-threshold.....	328
5.2.1.2.42 gtpu-path-failure.....	329

5.2.1.2.43	gx-before-aaa.....	330
5.2.1.2.44	ho-apn-lookup.....	331
5.2.1.2.45	home-plmn-list.....	332
5.2.1.2.46	i-upf-selection.....	333
5.2.1.2.47	ims-emps-priority.....	333
5.2.1.2.48	imsi-plmn-list.....	334
5.2.1.2.49	instance-type.....	335
5.2.1.2.50	ipsmg-group.....	336
5.2.1.2.51	mg-group-call-failure-rate.....	337
5.2.1.2.52	non3gpp-pdn-connection-rejection-with-ip.....	338
5.2.1.2.53	handover-no-mip6-agent-info.....	338
5.2.1.2.54	pgw-handover-response-cc64.....	339
5.2.1.2.55	nsmf-congestion-control.....	339
5.2.1.2.56	enable-nsmf-congestion-control.....	340
5.2.1.2.57	max-activations.....	340
5.2.1.2.58	retry-after-timer.....	341
5.2.1.2.59	nsmf-overload-control.....	341
5.2.1.2.60	enable-nsmf-overload-control.....	342
5.2.1.2.61	overload-threshold.....	342
5.2.1.2.62	retry-after-timer.....	343
5.2.1.2.63	nswo-capable.....	344
5.2.1.2.64	oi-override-mapping.....	344
5.2.1.2.65	overload-control.....	345
5.2.1.2.66	cpu-sampling-interval.....	345
5.2.1.2.67	thresholds.....	346
5.2.1.2.68	critical.....	347
5.2.1.2.69	major.....	347
5.2.1.2.70	minor.....	348
5.2.1.2.71	override-profile.....	349
5.2.1.2.72	override.....	349
5.2.1.2.73	pcf-pcrf-based-pcscf-allocation.....	351
5.2.1.2.74	entry.....	352
5.2.1.2.75	pcmd.....	353
5.2.1.2.76	pcscf-recovery.....	353
5.2.1.2.77	pco-option.....	354
5.2.1.2.78	max-allocated-pcscf-addresses.....	355
5.2.1.2.79	pfcf-node-id-type.....	355
5.2.1.2.80	pfcf-cp-router-adv.....	356
5.2.1.2.81	reattach-timeout.....	357
5.2.1.2.82	relax-msb-check-on-path-management-messages.....	358
5.2.1.2.83	reserve-default-gateway-address.....	359
5.2.1.2.84	seamless-wlan-handover.....	359
5.2.1.2.85	send-mbr-for-bbai.....	360
5.2.1.2.86	send-mbr-for-bbai-emergency.....	360
5.2.1.2.87	serving-node-for-combo-sessions.....	361
5.2.1.2.88	session-deletion-rate.....	362
5.2.1.2.89	session-restoration-rate.....	362

5.2.1.2.90	slices.....	363
5.2.1.2.91	slice-instance-list.....	363
5.2.1.2.92	slice-list.....	364
5.2.1.2.93	supported-vendor.....	365
5.2.1.2.94	suspend.....	365
5.2.1.2.95	transit-sub.....	366
5.2.1.2.96	attach-response-wait-timer.....	366
5.2.1.2.97	attach-retry-hold-timer.....	367
5.2.1.2.98	data-triggered.....	368
5.2.1.2.99	twag-samog-radius-accounting.....	369
5.2.1.2.100	ue-cleanup-on-delete.....	369
5.2.1.2.101	umts-qos-policy.....	370
5.2.1.2.102	up-function-feature-options.....	371
5.2.1.2.103	exclude-adpdp.....	371
5.2.1.2.104	exclude-dlbd.....	372
5.2.1.2.105	exclude-dpdra.....	372
5.2.1.2.106	exclude-frrt.....	373
5.2.1.2.107	exclude-heeu.....	374
5.2.1.2.108	exclude-pdiu.....	374
5.2.1.2.109	exclude-ueip.....	375
5.2.1.2.110	up-peer-list.....	375
5.2.1.2.111	up-selection.....	376
5.2.1.2.112	pgw-ggsn-u-client-profile.....	376
5.2.1.2.113	sgw-u-client-profile.....	377
5.2.1.2.114	update-bearer-request-for-mvno.....	378
5.2.1.2.115	upf-acct-profile.....	378
5.2.1.2.116	url-encryption.....	379
5.2.1.2.117	user-plane-sync-for-delete-bearer.....	380
5.2.1.2.118	vsmf-profile.....	381
5.2.1.2.119	wlan-emergency.....	381
5.2.1.2.120	wps-profile.....	382
5.2.1.3	APN PDN command descriptions.....	383
5.2.1.3.1	aa-divert-per-bearer-qci-list.....	383
5.2.1.3.2	access-type.....	383
5.2.1.3.3	default-qos-profile.....	384
5.2.1.3.4	accounting.....	385
5.2.1.3.5	duplicate-server-group.....	386
5.2.1.3.6	selective-radius-accounting.....	387
5.2.1.3.7	accounting-priority.....	387
5.2.1.3.8	acct-fixed.....	388
5.2.1.3.9	aggr-rate-dl.....	389
5.2.1.3.10	aggr-rate-ul.....	389
5.2.1.3.11	allow-always-on.....	390
5.2.1.3.12	allow-dangling-rule-ded-bearer.....	390
5.2.1.3.13	allow-emergency.....	391
5.2.1.3.14	allow-wlan-ho-no-context.....	391
5.2.1.3.15	ambr-policing.....	392

5.2.1.3.16	apn-load-control.....	392
5.2.1.3.17	apn-network-id.....	393
5.2.1.3.18	apn-overload-control.....	395
5.2.1.3.19	overload-reduction.....	396
5.2.1.3.20	overload-threshold.....	396
5.2.1.3.21	validity-time.....	397
5.2.1.3.22	apn-rate-control.....	398
5.2.1.3.23	apn-type.....	398
5.2.1.3.24	application-id-mvi-policy.....	399
5.2.1.3.25	auth-fixed.....	400
5.2.1.3.26	authentication.....	401
5.2.1.3.27	bitrate-roundup.....	403
5.2.1.3.28	block-list.....	404
5.2.1.3.29	entry.....	404
5.2.1.3.30	call-sequence.....	405
5.2.1.3.31	cm-idle-timeout.....	406
5.2.1.3.32	converged-interface-support.....	406
5.2.1.3.33	converged-n10.....	407
5.2.1.3.34	converged-n40.....	408
5.2.1.3.35	converged-n40-gngp.....	408
5.2.1.3.36	converged-n7.....	409
5.2.1.3.37	converged-n7-gngp.....	410
5.2.1.3.38	dedicated-bearer-create-retry-count.....	411
5.2.1.3.39	dedicated-bearer-create-retry-wait-timer.....	411
5.2.1.3.40	dedicated-bearer-idle-timeout-profile.....	412
5.2.1.3.41	default-cc-group.....	413
5.2.1.3.42	default-qos-5g-profile.....	413
5.2.1.3.43	default-qos-application-modify.....	414
5.2.1.3.44	apn-ambr-attr.....	414
5.2.1.3.45	aggr-rate.....	415
5.2.1.3.46	arp.....	415
5.2.1.3.47	qci.....	416
5.2.1.3.48	qos-holding-time.....	417
5.2.1.3.49	retry-wait-timer.....	417
5.2.1.3.50	del-bearer-no-reactivation-req-cause.....	418
5.2.1.3.51	del-bearer-reactivation-req-cause.....	419
5.2.1.3.52	delay-tolerant-connection-suspension.....	420
5.2.1.3.53	delete-adc-dedicated-bearer-on-idle-timeout-only.....	420
5.2.1.3.54	destination-realm-gx.....	421
5.2.1.3.55	dhcp-profile.....	421
5.2.1.3.56	dhcpv4-server.....	422
5.2.1.3.57	dhcpv4-server-group.....	423
5.2.1.3.58	dhcpv6-server-group.....	423
5.2.1.3.59	dhcpv6-service.....	424
5.2.1.3.60	disable-bearer-bind-verification.....	425
5.2.1.3.61	disable-frame-route.....	425
5.2.1.3.62	disable-gbr-policing-qci.....	425

5.2.1.3.63	dnn-overload-control.....	426
5.2.1.3.64	n1-back-off-timer.....	427
5.2.1.3.65	retry-after-timer.....	427
5.2.1.3.66	dns-redirect-server.....	428
5.2.1.3.67	dns-snooping.....	428
5.2.1.3.68	dns-snooping-forward-to-aa.....	429
5.2.1.3.69	dynamic-rule-policing.....	430
5.2.1.3.70	dynamic-rule-to-internal-rule-map.....	431
5.2.1.3.71	enable-global-pcc-precedence.....	432
5.2.1.3.72	enable-gx-session-recovery.....	432
5.2.1.3.73	ethernet-action-dl.....	433
5.2.1.3.74	ethernet-action-up.....	434
5.2.1.3.75	ethernet-service-type.....	435
5.2.1.3.76	firewall-rules.....	436
5.2.1.3.77	abort.....	436
5.2.1.3.78	begin.....	436
5.2.1.3.79	commit.....	437
5.2.1.3.80	diff.....	437
5.2.1.3.81	policy-rule.....	438
5.2.1.3.82	policy-rule-base.....	438
5.2.1.3.83	flow-inactivity-timeout.....	439
5.2.1.3.84	fqdn-dns-redirect-server.....	439
5.2.1.3.85	fqdn-group.....	440
5.2.1.3.86	gbr-policing-direction.....	440
5.2.1.3.87	gtp-pdn-threshold.....	441
5.2.1.3.88	gtpu-source-port-dist.....	442
5.2.1.3.89	guard-timer.....	443
5.2.1.3.90	gx.....	444
5.2.1.3.91	gtp-precedence-mapping.....	444
5.2.1.3.92	include-rat-when-received.....	445
5.2.1.3.93	ho-failure-release-pdusession.....	445
5.2.1.3.94	ho-wrong-pdn-ip-stale-session-deletion.....	446
5.2.1.3.95	hybrid-access.....	446
5.2.1.3.96	bonding-support.....	447
5.2.1.3.97	link-switch.....	447
5.2.1.3.98	mode.....	447
5.2.1.3.99	uplink-traffic.....	448
5.2.1.3.100	hysteresis.....	448
5.2.1.3.101	monitoring-key.....	449
5.2.1.3.102	idle-timeout.....	449
5.2.1.3.103	idle-timer-direction-downlink.....	450
5.2.1.3.104	idle-timer-direction-uplink.....	451
5.2.1.3.105	ignore-up-ip-allocation.....	452
5.2.1.3.106	ip-alloc-allow-local.....	452
5.2.1.3.107	ip-alloc-extended-hss-allocation.....	453
5.2.1.3.108	ip-alloc-extended-radius-allocation.....	453
5.2.1.3.109	ip-alloc-method.....	454

5.2.1.3.110	ip-alloc-priority-order.....	455
5.2.1.3.111	ip-allocation-trace.....	456
5.2.1.3.112	ip-pool-exhaust-back-off-timer.....	457
5.2.1.3.113	kpi-kci.....	458
5.2.1.3.114	arp-traffic-stats.....	458
5.2.1.3.115	enhanced-traffic-stats.....	459
5.2.1.3.116	qci-apn-stats.....	460
5.2.1.3.117	sgw-apn-stats.....	460
5.2.1.3.118	ladn.....	461
5.2.1.3.119	enable-ladn.....	462
5.2.1.3.120	on-ue-move-outside-ladn.....	462
5.2.1.3.121	max-ambr-policer.....	463
5.2.1.3.122	max-pdn-connection.....	463
5.2.1.3.123	max-session-attach.....	464
5.2.1.3.124	min-pdn-connection.....	465
5.2.1.3.125	multi-ue-bonding.....	466
5.2.1.3.126	namf-profile.....	466
5.2.1.3.127	nemo-ha.....	467
5.2.1.3.128	non-ip-pdn.....	467
5.2.1.3.129	udp-tunnel-group.....	468
5.2.1.3.130	nsmf-dnn-threshold.....	468
5.2.1.3.131	nsmf-profile.....	469
5.2.1.3.132	optimize-gbr.....	470
5.2.1.3.133	origin-realm-per-application.....	470
5.2.1.3.134	realm-gx.....	471
5.2.1.3.135	realm-gy.....	471
5.2.1.3.136	realm-rf.....	472
5.2.1.3.137	realm-s6b.....	472
5.2.1.3.138	realm-sd.....	473
5.2.1.3.139	realm-swm.....	474
5.2.1.3.140	overload-back-off.....	474
5.2.1.3.141	exclude-arp-list.....	475
5.2.1.3.142	overload-back-off-timer.....	476
5.2.1.3.143	overload-control-profile.....	476
5.2.1.3.144	paging-bearer-binding.....	477
5.2.1.3.145	pcf-discovery.....	477
5.2.1.3.146	serving-scope.....	478
5.2.1.3.147	pdn-alloc-type.....	478
5.2.1.3.148	pdn-type.....	479
5.2.1.3.149	pdn-type-conversion.....	481
5.2.1.3.150	pdn-type-preferred-ipv6.....	481
5.2.1.3.151	pfcp-translate-tx.....	482
5.2.1.3.152	pfcp-user-plane-inactivity.....	482
5.2.1.3.153	policy-on-the-fly.....	483
5.2.1.3.154	ppi-mapping-profile.....	484
5.2.1.3.155	ppp-regeneration.....	485
5.2.1.3.156	exclude-dns-server-option.....	485

5.2.1.3.157	exclude-nbns-server-option.....	486
5.2.1.3.158	icrq-calling-number-type.....	487
5.2.1.3.159	include-hss-static-ip.....	488
5.2.1.3.160	l2tp-group.....	488
5.2.1.3.161	lcp-confreq-attributes.....	489
5.2.1.3.162	mobile-router.....	490
5.2.1.3.163	router.....	491
5.2.1.3.164	rx-connect-speed.....	492
5.2.1.3.165	tx-connect-speed.....	492
5.2.1.3.166	user-name.....	493
5.2.1.3.167	supported-vendor.....	495
5.2.1.3.168	pre-auth.....	496
5.2.1.3.169	pre-auth-fixed.....	498
5.2.1.3.170	preserve-volte.....	499
5.2.1.3.171	qci-policy.....	499
5.2.1.3.172	qfi-mapping-profile.....	500
5.2.1.3.173	radius-accounting-on-off.....	500
5.2.1.3.174	include-duplicate-servers.....	501
5.2.1.3.175	messages-sent.....	502
5.2.1.3.176	wait-for-ack.....	502
5.2.1.3.177	radius-auth-cache.....	503
5.2.1.3.178	radius-case-sensitive-called-station-id.....	503
5.2.1.3.179	radius-nas-identifier.....	504
5.2.1.3.180	radius-nas-ip-address.....	504
5.2.1.3.181	reactivation-requested.....	505
5.2.1.3.182	redirect.....	506
5.2.1.3.183	reject-foreign-subscribers.....	507
5.2.1.3.184	report-serving-node-supplied-cc-radius.....	507
5.2.1.3.185	report-serving-node-supplied-cc-s6b.....	508
5.2.1.3.186	restriction.....	508
5.2.1.3.187	roaming-handover.....	509
5.2.1.3.188	lte-qos-ambr.....	509
5.2.1.3.189	lte-qos-black-list.....	510
5.2.1.3.190	lte-qos-delay.....	511
5.2.1.3.191	router.....	511
5.2.1.3.192	ip-local-pool.....	512
5.2.1.3.193	tai-lai-list.....	513
5.2.1.3.194	router-advertisements.....	513
5.2.1.3.195	max-advertisement-interval.....	513
5.2.1.3.196	min-advertisement-interval.....	514
5.2.1.3.197	rulebase-selection.....	515
5.2.1.3.198	allow-rulebase.....	515
5.2.1.3.199	selected-dnn.....	516
5.2.1.3.200	selection-mode.....	517
5.2.1.3.201	session-timeout.....	518
5.2.1.3.202	shutdown-back-off-timer.....	518
5.2.1.3.203	slices.....	519

5.2.1.3.204	default-slice.....	520
5.2.1.3.205	slice-list.....	520
5.2.1.3.206	udm-based-slice-selection.....	521
5.2.1.3.207	ssc.....	522
5.2.1.3.208	allowed-ssc-modes.....	522
5.2.1.3.209	pdu-session-address-lifetime.....	523
5.2.1.3.210	pdu-session-reactivation-trigger.....	523
5.2.1.3.211	pcfcp-user-plane-inactivity-iupf-case.....	524
5.2.1.3.212	reject-not-allowed-ssc-mode.....	525
5.2.1.3.213	upf-reselection.....	525
5.2.1.3.214	uuid-mask-length.....	528
5.2.1.3.215	static-local-pcc.....	528
5.2.1.3.216	abort.....	529
5.2.1.3.217	begin.....	529
5.2.1.3.218	commit.....	530
5.2.1.3.219	dedicated-bearers.....	530
5.2.1.3.220	default-bearer-qci.....	530
5.2.1.3.221	policy-rule.....	531
5.2.1.3.222	policy-rule-base.....	532
5.2.1.3.223	diff.....	532
5.2.1.3.224	policy-rule.....	533
5.2.1.3.225	policy-rule-base.....	533
5.2.1.3.226	static-predefined-pcc.....	534
5.2.1.3.227	default-policy-rule-base.....	534
5.2.1.3.228	default-qci.....	535
5.2.1.3.229	steering.....	535
5.2.1.3.230	default-gw-addr-ipv4.....	536
5.2.1.3.231	default-gw-addr-ipv6.....	536
5.2.1.3.232	gw-signaling-addr.....	537
5.2.1.3.233	gw-addr-ipv4.....	538
5.2.1.3.234	gw-addr-ipv6.....	538
5.2.1.3.235	subscriber-mapping-profile.....	539
5.2.1.3.236	svc-steering.....	539
5.2.1.3.237	default-steering-profile.....	540
5.2.1.3.238	transport-byte-collection.....	540
5.2.1.3.239	imsi.....	541
5.2.1.3.240	msisdn.....	542
5.2.1.3.241	twag-radius-acct-feed.....	543
5.2.1.3.242	udm-ctrl-radius-auth.....	543
5.2.1.3.243	udm-sdm-disabled.....	544
5.2.1.3.244	udm-sdm-profile.....	544
5.2.1.3.245	udm-uecm-disabled.....	545
5.2.1.3.246	udm-uecm-profile.....	545
5.2.1.3.247	umts-qos-policy.....	546
5.2.1.3.248	up-security-policy.....	546
5.2.1.3.249	upf-acct-profile.....	547
5.2.1.3.250	upf-sel-before-auth.....	548

5.2.1.3.251 upf-selection.....	548
5.2.1.3.252 dnn.....	549
5.2.1.3.253 virtual-apn.....	549
5.2.1.3.254 virtual-apn-based-on-pco.....	550
5.2.1.3.255 virtual-apn-list.....	551
5.2.1.3.256 wait-accounting.....	551
5.2.1.4 APN PCF PDN command descriptions.....	552
5.2.1.4.1 pcf-selection.....	552
5.2.1.4.2 dnn.....	552
5.2.1.4.3 exclude-gpsi.....	553
5.2.1.4.4 exclude-snsais.....	553
5.2.1.4.5 exclude-supi.....	554
5.2.1.4.6 pd-alloc-method.....	554
5.2.1.4.7 nf-id-list.....	555
5.2.1.4.8 npcfc-profile.....	556
5.2.1.4.9 pcf-id.....	556
5.2.1.5 selection-type.....	557
5.2.1.6 user-range.....	559
5.2.1.7 APN PCO PDN command descriptions.....	559
5.2.1.7.1 pco-option.....	560
5.2.1.7.2 closest-v4-epdg-group.....	560
5.2.1.7.3 closest-v6-epdg-group.....	560
5.2.1.7.4 dns-server-ipv4.....	561
5.2.1.7.5 primary.....	561
5.2.1.7.6 secondary.....	562
5.2.1.7.7 dns-server-ipv6.....	562
5.2.1.7.8 primary.....	563
5.2.1.7.9 secondary.....	563
5.2.1.7.10 link-mtu.....	564
5.2.1.7.11 nbns-srvr-ipv4.....	564
5.2.1.7.12 primary.....	565
5.2.1.7.13 secondary.....	565
5.2.1.7.14 nbns-srvr-ipv6.....	566
5.2.1.7.15 primary.....	566
5.2.1.7.16 secondary.....	566
5.2.1.7.17 operator-pco.....	567
5.2.1.7.18 pcscf-srvr-ipv4.....	568
5.2.1.7.19 pcscf-srvr-ipv6.....	568
5.2.1.7.20 pcscf-srvr-v4-group.....	569
5.2.1.7.21 pcscf-srvr-v6-group.....	569
5.2.1.7.22 preferred-dns-address-type.....	570
5.2.1.7.23 selection-type.....	571
5.2.1.7.24 ue-location.....	572
5.2.1.7.25 tai-lai-list.....	572
5.2.1.8 user-range.....	573
5.2.1.9 APN PCRF PDN command descriptions.....	574
5.2.1.9.1 pcrf-selection.....	574

5.2.1.9.2	diameter-peer-group-list.....	575
5.2.1.9.3	dynamic-pcc.....	575
5.2.1.9.4	pcc-disable-per-rat.....	576
5.2.1.9.5	pcc-disable-per-rat-timer.....	577
5.2.1.9.6	primary-diameter-peer.....	578
5.2.1.9.7	secondary-diameter-peer.....	579
5.2.1.9.8	selection-type.....	579
5.2.1.9.9	suppress-ccr-with-no-qos-negotiation.....	580
5.2.1.10	user-range.....	580
5.2.1.11	Bearer traffic PDN command descriptions.....	582
5.2.1.11.1	bearer-traffic.....	582
5.2.1.11.2	gtp-u.....	582
5.2.1.11.3	sequence-number.....	583
5.2.1.11.4	udp-checksum.....	583
5.2.1.11.5	mss-adjust.....	584
5.2.1.11.6	non-3gpp-mss-adjust.....	585
5.2.1.12	Cloud database PDN command descriptions.....	585
5.2.1.12.1	cdbx.....	586
5.2.1.12.2	cloud-db-profile.....	586
5.2.1.12.3	interface.....	586
5.2.1.12.4	mss.....	587
5.2.1.13	Charging PDN command descriptions.....	588
5.2.1.13.1	charging.....	588
5.2.1.13.2	bill-shock-prevention.....	588
5.2.1.13.3	cc-ignore.....	589
5.2.1.13.4	cc-reject.....	590
5.2.1.13.5	cc-reject.....	590
5.2.1.13.6	cdr-avp-options-profile.....	591
5.2.1.13.7	cdr-file-separation.....	592
5.2.1.13.8	cdr-transfer.....	592
5.2.1.13.9	cdr-type.....	593
5.2.1.13.10	charging-prof.....	594
5.2.1.13.11	dcca-profile.....	594
5.2.1.13.12	diameter-peer-group-list.....	595
5.2.1.13.13	enable-nchf-charging.....	595
5.2.1.13.14	gtp-prime-group.....	596
5.2.1.13.15	incomplete-signaling.....	596
5.2.1.13.16	charging-method.....	597
5.2.1.13.17	rating-group.....	597
5.2.1.13.18	service-id.....	598
5.2.1.13.19	urr-id.....	598
5.2.1.13.20	nchf.....	599
5.2.1.13.21	chf-selection.....	599
5.2.1.13.22	chf-profile.....	600
5.2.1.13.23	nf-id-list.....	600
5.2.1.13.24	nchf-charging.....	601
5.2.1.13.25	charging-prof-selection-range.....	601

5.2.1.13.26	selection-type.....	602
5.2.1.13.27	user-range.....	603
5.2.1.13.28	multiple-pra.....	603
5.2.1.13.29	node-id.....	604
5.2.1.13.30	pgw-node-id.....	605
5.2.1.13.31	plmn-chrg-profile-map.....	605
5.2.1.13.32	pri-cdf-dia-peer.....	606
5.2.1.13.33	pri-ocs-dia-peer.....	607
5.2.1.13.34	sec-cdf-dia-peer.....	607
5.2.1.13.35	sec-ocs-dia-peer.....	608
5.2.1.13.36	remove-3gpp-headers.....	608
5.2.1.13.37	sgw-node-id.....	609
5.2.1.13.38	supported-vendor.....	609
5.2.1.13.39	tariff-time-change-gw-local-tz.....	610
5.2.1.13.40	time-limit-enh.....	610
5.2.1.13.41	use-last-known-uli.....	611
5.2.1.13.42	volume-limit-enh.....	612
5.2.1.14	Ga/Gn/Gp/Gx/Gy interface PDN command descriptions.....	612
5.2.1.14.1	ga.....	612
5.2.1.14.2	data-record-format-version.....	612
5.2.1.14.3	gtp-prime-group.....	613
5.2.1.14.4	interface-gtp-prime.....	614
5.2.1.14.5	gn.....	615
5.2.1.14.6	gp.....	615
5.2.1.14.7	interface.....	616
5.2.1.14.8	gtp-c.....	617
5.2.1.14.9	gtp-u.....	618
5.2.1.14.10	peer-list.....	619
5.2.1.14.11	signaling.....	620
5.2.1.14.12	gtp-c.....	620
5.2.1.14.13	profile.....	621
5.2.1.14.14	gtp-u.....	621
5.2.1.14.15	gtpu-keep-alive.....	622
5.2.1.14.16	profile.....	623
5.2.1.14.17	qci-policy.....	623
5.2.1.14.18	qfi-mapping-profile.....	624
5.2.1.14.19	gx.....	625
5.2.1.14.20	active-usage.....	625
5.2.1.14.21	alt-peer-rar.....	626
5.2.1.14.22	connection-failure-timer.....	627
5.2.1.14.23	doic.....	627
5.2.1.14.24	diameter-overload-control.....	628
5.2.1.14.25	dra-diameter-too-busy.....	629
5.2.1.14.26	gx-avp-options-profile.....	629
5.2.1.14.27	gx-profile.....	630
5.2.1.14.28	interface.....	631
5.2.1.14.29	diameter.....	631

5.2.1.14.30	multi-homing.....	632
5.2.1.14.31	diameter.....	633
5.2.1.14.32	origin-host.....	634
5.2.1.14.33	pcrf-selection.....	635
5.2.1.14.34	diameter-peer-group-list.....	635
5.2.1.14.35	primary-diameter-peer.....	636
5.2.1.14.36	secondary-diameter-peer.....	637
5.2.1.14.37	selection-priority.....	637
5.2.1.14.38	selection-type.....	638
5.2.1.14.39	user-range.....	639
5.2.1.14.40	retry-count.....	640
5.2.1.14.41	send-called-station-id.....	641
5.2.1.14.42	single-diameter-host.....	642
5.2.1.14.43	supported-vendor.....	642
5.2.1.14.44	suppress-ccr-t-unknown-session.....	643
5.2.1.14.45	tdf-info.....	644
5.2.1.14.46	tdf-ipv4.....	644
5.2.1.14.47	tdf-ipv6.....	645
5.2.1.14.48	tdf-realm.....	645
5.2.1.14.49	transaction-timer.....	646
5.2.1.14.50	usu-options-over-quota-traffic-handling.....	646
5.2.1.14.51	usu-options-reporting-threshold.....	647
5.2.1.14.52	usu-options-suppress-excess-quota.....	648
5.2.1.14.53	gy.....	648
5.2.1.14.54	alt-peer-rar-asr.....	649
5.2.1.14.55	dcca-profile.....	649
5.2.1.14.56	dia-routing-table.....	650
5.2.1.14.57	diameter-session.....	651
5.2.1.14.58	exclude-dest-host-pcrf-selected-server.....	652
5.2.1.14.59	interface.....	652
5.2.1.14.60	diameter.....	652
5.2.1.14.61	limit-session-id-length.....	653
5.2.1.14.62	ocs-selection.....	654
5.2.1.14.63	selection-type.....	654
5.2.1.14.64	user-range.....	655
5.2.1.14.65	outage.....	656
5.2.1.14.66	file-params.....	657
5.2.1.14.67	closure-rec-number.....	657
5.2.1.14.68	closure-size.....	658
5.2.1.14.69	closure-time.....	658
5.2.1.14.70	obsolete-time.....	659
5.2.1.14.71	location.....	659
5.2.1.14.72	limit.....	660
5.2.1.14.73	primary.....	661
5.2.1.14.74	primary-diameter-peer.....	661
5.2.1.14.75	secondary-diameter-peer.....	662
5.2.1.14.76	session-id-format.....	662

5.2.1.14.77 sig-rate-limit-profile.....	663
5.2.1.15 GTP load and overload control PDN command descriptions.....	664
5.2.1.15.1 gtp-load-control.....	664
5.2.1.15.2 enable-gtp-load-control.....	664
5.2.1.15.3 load-change-trigger.....	665
5.2.1.15.4 nth-message.....	665
5.2.1.15.5 gtp-overload-control.....	666
5.2.1.15.6 enable-gtp-overload-control.....	666
5.2.1.15.7 gtp-120-oci.....	667
5.2.1.15.8 gtp-73-120-reduction-metric.....	667
5.2.1.15.9 gtp-73-oci.....	668
5.2.1.15.10 local-throttling.....	668
5.2.1.15.11 critical.....	669
5.2.1.15.12 drop-only.....	669
5.2.1.15.13 exclude-arp-list.....	670
5.2.1.15.14 throttling-percent.....	671
5.2.1.15.15 local-overload-throttling.....	672
5.2.1.15.16 major.....	673
5.2.1.15.17 throttling-percent.....	673
5.2.1.15.18 minor.....	674
5.2.1.15.19 throttling-percent.....	674
5.2.1.15.20 throttling-alarms.....	675
5.2.1.15.21 overload-control-only-on-new-sessions.....	676
5.2.1.15.22 overload-reduction.....	676
5.2.1.15.23 critical.....	677
5.2.1.15.24 major.....	677
5.2.1.15.25 minor.....	678
5.2.1.15.26 s8.....	679
5.2.1.15.27 validity-time.....	679
5.2.1.15.28 validity-time-extra-time.....	680
5.2.1.15.29 pdn-threshold.....	680
5.2.1.15.30 max-s11-attach.....	680
5.2.1.16 KPI-KCI PDN command descriptions.....	681
5.2.1.16.1 kpi-kci.....	681
5.2.1.16.2 mass-deactivation.....	682
5.2.1.16.3 qci.....	682
5.2.1.16.4 qci-drop-stat.....	683
5.2.1.16.5 accounting-policy.....	684
5.2.1.16.6 arp-traffic-stats.....	684
5.2.1.16.7 collect-stats.....	685
5.2.1.16.8 enhanced-traffic-stats.....	685
5.2.1.16.9 ipv4-ipv6-stats.....	686
5.2.1.16.10 peer-traffic-stats.....	687
5.2.1.16.11 plmn-id-stats.....	687
5.2.1.16.12 sgw-only.....	688
5.2.1.17 N2 interface PDN command descriptions.....	688
5.2.1.17.1 n2.....	688

5.2.1.17.2 interface.....	689
5.2.1.17.3 n2-peer-profile.....	690
5.2.1.17.4 n3iwf-unique-id.....	690
5.2.1.17.5 supported-ta-item.....	691
5.2.1.17.6 tac.....	692
5.2.1.18 N3 interface PDN command descriptions.....	692
5.2.1.18.1 n3.....	692
5.2.1.18.2 dual-stack-uplane-pref.....	693
5.2.1.18.3 interface.....	694
5.2.1.18.4 gtp-u.....	694
5.2.1.18.5 signaling.....	696
5.2.1.18.6 gtp-u.....	696
5.2.1.18.7 profile.....	696
5.2.1.19 N9 interface PDN command descriptions.....	697
5.2.1.19.1 n9.....	697
5.2.1.19.2 dual-stack-uplane-pref.....	698
5.2.1.19.3 interface.....	699
5.2.1.19.4 gtp-u.....	699
5.2.1.19.5 signaling.....	700
5.2.1.19.6 gtp-u.....	701
5.2.1.19.7 profile.....	701
5.2.1.20 Network trace command descriptions.....	702
5.2.1.20.1 network-trace.....	702
5.2.1.20.2 instance.....	702
5.2.1.20.3 interface.....	703
5.2.1.20.4 network-trace-profile.....	704
5.2.1.20.5 shutdown.....	704
5.2.1.20.6 tapping-interface.....	705
5.2.1.21 NF profile attributes PDN command descriptions.....	705
5.2.1.21.1 nf-profile-attributes.....	705
5.2.1.21.2 capacity.....	706
5.2.1.21.3 exclude-ip-endpoints.....	707
5.2.1.21.4 fqdn.....	707
5.2.1.21.5 inter-plmn-fqdn.....	708
5.2.1.21.6 locality.....	708
5.2.1.21.7 nf-instance-id.....	709
5.2.1.21.8 priority.....	710
5.2.1.21.9 scp-domain.....	710
5.2.1.21.10 smf-serving-area-list.....	711
5.2.1.21.11 tai-list.....	712
5.2.1.21.12 vsmf-indication.....	712
5.2.1.22 NWu interface PDN command descriptions.....	713
5.2.1.22.1 nwu.....	713
5.2.1.22.2 interface.....	713
5.2.1.22.3 ipsec-profile.....	714
5.2.1.23 PFCP load and overload PDN command descriptions.....	715
5.2.1.23.1 pfcpl-load-control.....	715

5.2.1.23.2	enable-pfcp-load-control.....	715
5.2.1.23.3	load-change-trigger.....	716
5.2.1.23.4	nth-message-report.....	716
5.2.1.23.5	pfcp-overload-control.....	717
5.2.1.23.6	arp-to-message-priority-list.....	717
5.2.1.23.7	enable-pfcp-overload-control.....	718
5.2.1.23.8	critical.....	719
5.2.1.23.9	major.....	719
5.2.1.23.10	minor.....	720
5.2.1.23.11	validity-time.....	720
5.2.1.24	PGW selection PDN command descriptions.....	721
5.2.1.24.1	pgw-selection.....	721
5.2.1.24.2	dns-options.....	721
5.2.1.24.3	a-aaaa-pgw-discovery.....	722
5.2.1.24.4	apn-based-initial-attach.....	722
5.2.1.24.5	epdg-fqdn.....	723
5.2.1.24.6	epdg-fqdn-secondary.....	724
5.2.1.24.7	epdg-fqdn-tertiary.....	725
5.2.1.24.8	pgw-reselection-isolation.....	726
5.2.1.24.9	twag-fqdn.....	726
5.2.1.24.10	apn.....	727
5.2.1.24.11	pgw-selection-candidate-list.....	728
5.2.1.24.12	dns-results-preferred.....	728
5.2.1.24.13	topology-match-disable.....	729
5.2.1.24.14	profile.....	729
5.2.1.25	PI interface PDN command descriptions.....	730
5.2.1.25.1	pi.....	730
5.2.1.25.2	interface.....	730
5.2.1.25.3	mipv4.....	731
5.2.1.25.4	ip-in-ip-tunnel.....	732
5.2.1.25.5	peer-list.....	733
5.2.1.25.6	signaling.....	733
5.2.1.25.7	fa-ha-peer-list.....	734
5.2.1.25.8	profile.....	734
5.2.1.26	RADIUS PDN command descriptions.....	735
5.2.1.26.1	radius.....	735
5.2.1.26.2	accounting-on-off-level.....	735
5.2.1.26.3	delete-session-acct-interim-exh.....	736
5.2.1.26.4	interface.....	737
5.2.1.26.5	radius-accounting-on-off.....	738
5.2.1.26.6	messages-sent.....	738
5.2.1.26.7	radius-disconnect.....	739
5.2.1.26.8	rel-16-rat-type.....	739
5.2.1.27	RF interface PDN command descriptions.....	740
5.2.1.27.1	rf.....	740
5.2.1.27.2	acct-interim-interval.....	740
5.2.1.27.3	acct-level.....	740

5.2.1.27.4	charging-group-id.....	741
5.2.1.27.5	interface.....	742
5.2.1.27.6	diameter.....	742
5.2.1.27.7	operator-string.....	743
5.2.1.27.8	outage.....	744
5.2.1.27.9	acr-file-and-record-hdr-rel-id-ext.....	744
5.2.1.27.10	file-params.....	745
5.2.1.27.11	closure-rec-num.....	745
5.2.1.27.12	closure-size.....	746
5.2.1.27.13	closure-time.....	747
5.2.1.27.14	extension.....	747
5.2.1.27.15	obsolete-time.....	748
5.2.1.27.16	private-info.....	749
5.2.1.27.17	location.....	750
5.2.1.27.18	limit.....	750
5.2.1.27.19	primary.....	751
5.2.1.27.20	primary-diameter-peer.....	751
5.2.1.27.21	retry-count.....	752
5.2.1.27.22	secondary-diameter-peer.....	753
5.2.1.27.23	supported-vendor.....	754
5.2.1.27.24	transaction-timer.....	754
5.2.1.28	S1-U interface PDN command descriptions.....	755
5.2.1.28.1	s1-u.....	755
5.2.1.28.2	dual-stack-uplane-pref.....	756
5.2.1.28.3	interface.....	757
5.2.1.28.4	gtp-u.....	757
5.2.1.28.5	signaling.....	759
5.2.1.28.6	gtp-u.....	759
5.2.1.28.7	profile.....	759
5.2.1.29	S11 interface PDN command descriptions.....	760
5.2.1.29.1	s11.....	760
5.2.1.29.2	interface.....	761
5.2.1.29.3	gtp-c.....	761
5.2.1.29.4	gtp-u.....	763
5.2.1.29.5	mme-group.....	764
5.2.1.29.6	signaling.....	765
5.2.1.29.7	gtp-c.....	766
5.2.1.29.8	ddn-damp-timer.....	766
5.2.1.29.9	ddn-extended-buffering-time.....	767
5.2.1.29.10	ddn-throttling-arp-threshold.....	767
5.2.1.29.11	iot-restoration-time.....	768
5.2.1.29.12	legacy-create-idft-response-encoding.....	769
5.2.1.29.13	profile.....	770
5.2.1.29.14	restoration-time.....	770
5.2.1.29.15	ue-inactivity-timer.....	771
5.2.1.30	S12 interface PDN command descriptions.....	772
5.2.1.30.1	s12.....	772

5.2.1.30.2	dual-stack-uplane-pref.....	773
5.2.1.30.3	interface.....	774
5.2.1.30.4	gtp-u.....	774
5.2.1.30.5	signaling.....	775
5.2.1.30.6	gtp-u.....	776
5.2.1.30.7	profile.....	776
5.2.1.31	S2a and S2b interface PDN command descriptions.....	777
5.2.1.31.1	s2a.....	777
5.2.1.31.2	s2b.....	778
5.2.1.31.3	dual-stack-cplane-pref.....	779
5.2.1.31.4	dual-stack-uplane-pref.....	779
5.2.1.31.5	interface.....	780
5.2.1.31.6	gtp-c.....	781
5.2.1.31.7	gtp-u.....	782
5.2.1.31.8	pmipv6.....	783
5.2.1.31.9	match.....	784
5.2.1.31.10	apn-list.....	785
5.2.1.31.11	plmn-id-list.....	786
5.2.1.31.12	pgw-set-support.....	786
5.2.1.31.13	signaling.....	787
5.2.1.31.14	gtp-c.....	787
5.2.1.31.15	gtp-u.....	788
5.2.1.31.16	pmipv6.....	788
5.2.1.31.17	profile.....	789
5.2.1.31.18	s2b-error-mapping-profile.....	790
5.2.1.32	S4 interface PDN command descriptions.....	790
5.2.1.32.1	s4.....	790
5.2.1.32.2	interface.....	791
5.2.1.32.3	gtp-c.....	791
5.2.1.32.4	gtp-u.....	793
5.2.1.32.5	peer-list.....	794
5.2.1.32.6	signaling.....	795
5.2.1.32.7	gtp-c.....	795
5.2.1.32.8	ddn-damp-timer.....	795
5.2.1.32.9	ddn-extended-buffering-time.....	796
5.2.1.32.10	ddn-throttling-arp-threshold.....	797
5.2.1.32.11	gtp-u.....	798
5.2.1.32.12	delete-on-error-indication.....	798
5.2.1.32.13	profile.....	798
5.2.1.32.14	sgsn-group.....	799
5.2.1.33	S5 and S8 interface PDN command descriptions.....	800
5.2.1.33.1	s5.....	800
5.2.1.33.2	s8.....	801
5.2.1.33.3	dual-stack-cplane-pref.....	801
5.2.1.33.4	dual-stack-uplane-pref.....	802
5.2.1.33.5	interface.....	803
5.2.1.33.6	gtp-c.....	803

5.2.1.33.7	gtp-u.....	805
5.2.1.33.8	match.....	806
5.2.1.33.9	apn-list.....	807
5.2.1.33.10	plmn-id-list.....	807
5.2.1.33.11	peer-list.....	808
5.2.1.33.12	send-end-marker.....	809
5.2.1.33.13	signaling.....	809
5.2.1.33.14	gtp-c.....	810
5.2.1.33.15	enhanced-sgw-restoration.....	810
5.2.1.33.16	pause-charging.....	811
5.2.1.33.17	data-threshold.....	811
5.2.1.33.18	data-threshold-hold-time.....	812
5.2.1.33.19	pgw-pause-charging.....	813
5.2.1.33.20	radio-loss.....	814
5.2.1.33.21	profile.....	814
5.2.1.33.22	gtp-u.....	815
5.2.1.34	S6b interface PDN command descriptions.....	816
5.2.1.34.1	s6b.....	816
5.2.1.34.2	alt-peer-rar-asr.....	816
5.2.1.34.3	application-id.....	817
5.2.1.34.4	auth-mode.....	817
5.2.1.34.5	interface.....	818
5.2.1.34.6	diameter.....	818
5.2.1.34.7	node-identifier-list.....	819
5.2.1.34.8	peer-group.....	820
5.2.1.34.9	relax-aar-dest-realm.....	820
5.2.1.34.10	retry-count.....	821
5.2.1.34.11	s6b-profile.....	821
5.2.1.34.12	subscription-id-override.....	822
5.2.1.34.13	supported-vendor.....	822
5.2.1.34.14	transaction-timer.....	823
5.2.1.34.15	username-leading-digit.....	824
5.2.1.35	SBA client services PDN command descriptions.....	825
5.2.1.35.1	sba-client-services.....	825
5.2.1.35.2	addressing-priority.....	825
5.2.1.35.3	amf-client.....	826
5.2.1.35.4	namf-comm.....	826
5.2.1.35.5	callback-type-preference.....	826
5.2.1.35.6	fqdn.....	827
5.2.1.35.7	http2-profile.....	828
5.2.1.35.8	interface.....	829
5.2.1.35.9	mode.....	830
5.2.1.35.10	n1-profile.....	831
5.2.1.35.11	nf-id-list.....	832
5.2.1.35.12	oauth2-profile.....	832
5.2.1.35.13	paging-damp-timer.....	833
5.2.1.35.14	plmn-list.....	834

5.2.1.35.15	service-interface.....	835
5.2.1.35.16	immediate-report.....	836
5.2.1.35.17	udm-sdm-profile.....	836
5.2.1.35.18	udm-uecm-profile.....	837
5.2.1.35.19	namf-evts.....	838
5.2.1.35.20	http2-profile.....	838
5.2.1.35.21	chf-client.....	839
5.2.1.35.22	nchf-convergedcharging.....	839
5.2.1.35.23	alternative-api-version.....	840
5.2.1.35.24	api-version.....	840
5.2.1.35.25	http2-profile.....	841
5.2.1.35.26	converged-interface-support.....	842
5.2.1.35.27	converged-n10.....	842
5.2.1.35.28	converged-n40.....	843
5.2.1.35.29	converged-n40-gngp.....	843
5.2.1.35.30	converged-n7.....	844
5.2.1.35.31	converged-n7-gngp.....	845
5.2.1.35.32	pdu-session-id-base.....	845
5.2.1.35.33	http2-profile.....	846
5.2.1.35.34	nrf-client.....	847
5.2.1.35.35	nnrf-disc.....	847
5.2.1.35.36	blocklist-duration.....	848
5.2.1.35.37	chf-selection-options.....	848
5.2.1.35.38	exclude-gpsi.....	849
5.2.1.35.39	exclude-supi.....	849
5.2.1.35.40	include-preferred-api-version.....	850
5.2.1.35.41	exclude-snsais.....	850
5.2.1.35.42	http2-profile.....	851
5.2.1.35.43	mode.....	851
5.2.1.35.44	nrf-cache-interval.....	852
5.2.1.35.45	suppress-discovery.....	853
5.2.1.35.46	udm-selection-options.....	854
5.2.1.35.47	exclude-group-id.....	854
5.2.1.35.48	exclude-supi.....	854
5.2.1.35.49	use-locality.....	855
5.2.1.35.50	nnrf-nfm.....	856
5.2.1.35.51	blocklist-duration.....	856
5.2.1.35.52	heartbeat-nrf-blocklist.....	857
5.2.1.35.53	http2-profile.....	858
5.2.1.35.54	nfm-nrf-blocklist.....	858
5.2.1.35.55	nnrf-oauth2.....	859
5.2.1.35.56	http2-profile.....	860
5.2.1.35.57	pcf-client.....	860
5.2.1.35.58	npcf-smpolicycontrol.....	860
5.2.1.35.59	http2-profile.....	861
5.2.1.35.60	npcf-profile.....	861
5.2.1.35.61	scp-client.....	862

5.2.1.35.62	scp-service.....	862
5.2.1.35.63	block-duration.....	863
5.2.1.35.64	default-failure-action.....	863
5.2.1.35.65	fh-no-response.....	864
5.2.1.35.66	http-status-code.....	865
5.2.1.35.67	http2-profile.....	867
5.2.1.35.68	mode-fallback.....	867
5.2.1.35.69	scp-domain.....	868
5.2.1.35.70	sepp-client.....	869
5.2.1.35.71	sepp.....	869
5.2.1.35.72	http2-profile.....	870
5.2.1.35.73	smf-client.....	870
5.2.1.35.74	nsmf-pdusession.....	870
5.2.1.35.75	api-prefix.....	871
5.2.1.35.76	http2-profile.....	871
5.2.1.35.77	supported-features.....	872
5.2.1.35.78	dtssa.....	872
5.2.1.35.79	transaction-timer.....	873
5.2.1.35.80	version.....	873
5.2.1.35.81	udm-client.....	874
5.2.1.35.82	nudm-sdm.....	875
5.2.1.35.83	api-version.....	875
5.2.1.35.84	explicit-unsubscribe.....	876
5.2.1.35.85	http2-profile.....	877
5.2.1.35.86	nudm-uecm.....	877
5.2.1.35.87	http2-profile.....	878
5.2.1.35.88	user-agent-header.....	878
5.2.1.35.89	sbi-correlation-info-header.....	879
5.2.1.35.90	sbi-load-control.....	879
5.2.1.35.91	enable.....	880
5.2.1.35.92	load-change-trigger.....	880
5.2.1.35.93	nth-message.....	881
5.2.1.35.94	sbi-nf-peer-info-header.....	881
5.2.1.35.95	sbi-overload-control.....	882
5.2.1.35.96	enable.....	883
5.2.1.35.97	overload-reduction.....	883
5.2.1.35.98	critical.....	884
5.2.1.35.99	major.....	884
5.2.1.35.100	minor.....	885
5.2.1.35.101	validity-time.....	886
5.2.1.36	SBA server services PDN command descriptions.....	886
5.2.1.36.1	sba-server-services.....	886
5.2.1.36.2	fqdn.....	887
5.2.1.36.3	http2-profile.....	887
5.2.1.36.4	nbsf-management.....	888
5.2.1.36.5	allowed-nf-domains-list.....	888
5.2.1.36.6	allowed-nf-types.....	889

5.2.1.36.7	allowed-plmns.....	890
5.2.1.36.8	fqdn.....	890
5.2.1.36.9	http2-profile.....	891
5.2.1.36.10	interface.....	891
5.2.1.36.11	oauth2-profile.....	892
5.2.1.36.12	nsmf-event-exposure.....	893
5.2.1.36.13	allowed-nf-types.....	893
5.2.1.36.14	allowed-plmns.....	894
5.2.1.36.15	allowed-slices.....	895
5.2.1.36.16	default-subscription-timeout.....	896
5.2.1.36.17	fqdn.....	896
5.2.1.36.18	http2-profile.....	897
5.2.1.36.19	nsmf-profile.....	897
5.2.1.36.20	supported-features.....	898
5.2.1.36.21	communication-failure.....	898
5.2.1.36.22	ene-na.....	899
5.2.1.36.23	pdu-session-status.....	899
5.2.1.36.24	qfi-allocation.....	899
5.2.1.36.25	nsmf-pdusession.....	900
5.2.1.36.26	allowed-nf-types.....	900
5.2.1.36.27	allowed-plmns.....	901
5.2.1.36.28	fqdn.....	902
5.2.1.36.29	http2-profile.....	902
5.2.1.36.30	indirect-forwarding-timeout.....	903
5.2.1.36.31	n3-interface-realm.....	903
5.2.1.36.32	n9-interface-realm.....	904
5.2.1.36.33	n9-roaming-interface-realm.....	905
5.2.1.36.34	s5-ref-point.....	906
5.2.1.36.35	s8-ref-point.....	906
5.2.1.36.36	aasn.....	907
5.2.1.37	SBA service realm PDN command descriptions.....	907
5.2.1.37.1	sba-service-realm.....	907
5.2.1.37.2	client-service.....	908
5.2.1.37.3	server-service.....	908
5.2.1.38	Sd PDN command descriptions.....	909
5.2.1.38.1	sd.....	909
5.2.1.38.2	diameter-redirect.....	909
5.2.1.38.3	interface.....	910
5.2.1.38.4	diameter.....	910
5.2.1.38.5	multi-homing.....	911
5.2.1.38.6	diameter.....	912
5.2.1.38.7	peer-group.....	913
5.2.1.38.8	retry-count.....	913
5.2.1.38.9	sd-profile.....	914
5.2.1.38.10	transaction-timer.....	914
5.2.1.39	Signaling PDN command descriptions.....	915
5.2.1.39.1	signaling.....	915

5.2.1.39.2	diameter.....	915
5.2.1.39.3	mip6-agent-info.....	916
5.2.1.39.4	origin-host.....	917
5.2.1.39.5	origin-realm.....	917
5.2.1.39.6	profile.....	918
5.2.1.39.7	gtp-c.....	919
5.2.1.39.8	profile.....	919
5.2.1.39.9	gtp-u.....	920
5.2.1.39.10	profile.....	920
5.2.1.39.11	interface.....	921
5.2.1.39.12	mipv4.....	922
5.2.1.39.13	profile.....	923
5.2.1.39.14	multi-homing.....	923
5.2.1.39.15	interface.....	924
5.2.1.40	STa PDN command descriptions.....	925
5.2.1.40.1	sta.....	925
5.2.1.40.2	diameter-peer-group-list.....	925
5.2.1.40.3	interface.....	926
5.2.1.40.4	diameter.....	926
5.2.1.40.5	multi-homing.....	927
5.2.1.40.6	diameter.....	928
5.2.1.40.7	radius-server-group.....	929
5.2.1.40.8	retry-count.....	929
5.2.1.40.9	sta-profile.....	930
5.2.1.40.10	transaction-timer.....	930
5.2.1.41	SWm PDN command descriptions.....	931
5.2.1.41.1	swm.....	931
5.2.1.41.2	aar-call-flow.....	931
5.2.1.41.3	activate-aaa-failure-indication.....	932
5.2.1.41.4	alt-peer-rar-asr.....	933
5.2.1.41.5	dia-routing-table.....	933
5.2.1.41.6	swm-error-mapping-profile.....	934
5.2.1.41.7	interface.....	934
5.2.1.41.8	diameter.....	935
5.2.1.41.9	multi-homing.....	936
5.2.1.41.10	diameter.....	937
5.2.1.41.11	peer-group.....	937
5.2.1.41.12	retry-count.....	939
5.2.1.41.13	swm-profile.....	939
5.2.1.41.14	transaction timer.....	940
5.2.1.41.15	swm-error-mapping-profile.....	940
5.2.1.42	SWu PDN command descriptions.....	941
5.2.1.42.1	swu.....	941
5.2.1.42.2	interface.....	942
5.2.1.42.3	ipsec-profile.....	942
5.2.1.42.4	swu-ipsec-lockout.....	943
5.2.1.42.5	backoff-timer.....	943

5.2.1.42.6 configuration.....	945
5.2.1.42.7 function.....	946
5.2.1.42.8 swu-throttling.....	947
5.2.1.43 Sww PDN command descriptions.....	948
5.2.1.43.1 sww.....	948
5.2.1.43.2 interface.....	948
5.2.1.43.3 sww-acct.....	949
5.2.1.43.4 sww-auth.....	950
5.2.1.43.5 sww-coa.....	951
5.2.1.43.6 sww-rad-disconnect.....	952
5.2.1.43.7 sww-tunnel.....	953
5.2.1.43.8 signaling.....	955
5.2.1.43.9 sww-radius-profile.....	955
5.2.1.43.10 sww-tunnel-profile.....	956
5.2.1.44 Sx-N4 PDN command descriptions.....	956
5.2.1.44.1 sx-n4.....	956
5.2.1.44.2 enforced-pfcp-association-list.....	957
5.2.1.44.3 interface.....	958
5.2.1.44.4 gtp-u.....	958
5.2.1.44.5 pfcp.....	959
5.2.1.44.6 pfcp-association-list.....	960
5.2.1.44.7 signaling.....	960
5.2.1.44.8 pfcp.....	961
5.2.1.44.9 profile.....	961
5.2.1.45 T6c PDN command descriptions.....	962
5.2.1.45.1 t6c.....	962
5.2.1.45.2 http-port.....	963
5.2.1.45.3 interface.....	963
5.2.1.45.4 t6c-profile.....	964
5.2.1.45.5 uri.....	965
5.2.1.46 Threshold PDN command descriptions.....	965
5.2.1.46.1 thresholds.....	965
5.2.1.46.2 bearer-cf-failure.....	966
5.2.1.46.3 bearer-cf-success.....	966
5.2.1.46.4 bearer-mgmt-limit.....	966
5.2.1.46.5 bearer-traffic.....	967
5.2.1.46.6 dl-buff-utilization-threshold.....	967
5.2.1.46.7 path-mgmt.....	968
5.2.1.46.8 pdn-connections.....	968
5.2.1.46.9 counter.....	969
5.2.1.46.10 counter.....	969
5.2.1.46.11 counter.....	970
5.2.1.46.12 counter.....	971
5.2.1.46.13 counter.....	972
5.2.1.46.14 counter.....	973
5.2.1.46.15 interval.....	973
5.2.1.46.16 shutdown.....	974

5.2.1.47 Virtual APN selection PDN command descriptions.....	975
5.2.1.47.1 virtual-apn-selection.....	975
5.2.1.47.2 no-match-action.....	975
5.2.1.47.3 pco-fallback.....	976
5.2.1.47.4 pco-fallback-acct.....	977
5.2.1.47.5 prefer-pdn.....	978
5.2.1.47.6 selection-prio.....	978
5.2.1.47.7 selection-type.....	979
5.2.1.47.8 user-range.....	980
5.2.2 PDN show command descriptions.....	984
5.2.2.1 pdn.....	984
5.2.2.2 apn.....	985
5.2.2.3 accounting.....	1041
5.2.2.4 authentication.....	1043
5.2.2.5 charging.....	1044
5.2.2.6 ip-alloc.....	1046
5.2.2.7 l2tp.....	1048
5.2.2.8 pfcf.....	1049
5.2.2.9 policy.....	1050
5.2.2.10 virtual-apn.....	1051
5.2.2.11 apn-overload.....	1054
5.2.2.12 application-server-stats.....	1057
5.2.2.13 bearer-context.....	1059
5.2.2.14 cp-seid.....	1109
5.2.2.15 def-pfcf-u-sessions.....	1111
5.2.2.16 supi.....	1118
5.2.2.17 ue-ip.....	1120
5.2.2.18 up-seid.....	1122
5.2.2.19 call-flow-stats.....	1124
5.2.2.20 canary-test.....	1177
5.2.2.21 clear-status.....	1179
5.2.2.22 dnn-overload-control.....	1182
5.2.2.23 error-indication.....	1182
5.2.2.24 fixed-subscriber.....	1185
5.2.2.25 gtp-load-overload-control.....	1188
5.2.2.26 gtp-load-overload-control.....	1191
5.2.2.27 gtp-local-load-control.....	1196
5.2.2.28 gtp-local-overload-control.....	1197
5.2.2.29 local-storage-stats.....	1199
5.2.2.30 n2.....	1200
5.2.2.31 nf-peers.....	1201
5.2.2.32 nsmf-event-exposure-subscriptions.....	1206
5.2.2.33 nsmf-overload-control.....	1207
5.2.2.34 nwu.....	1208
5.2.2.35 path-mgmt-stats.....	1209
5.2.2.36 pcf-pcrf-based-pcscf-allocation.....	1228
5.2.2.37 pcmd-stats.....	1229

5.2.2.38	pdn-context.....	1233
5.2.2.39	cp-seid.....	1305
5.2.2.40	def-pfcp-u-sessions.....	1307
5.2.2.41	field-name.....	1313
5.2.2.42	supi.....	1317
5.2.2.43	ue-ip.....	1338
5.2.2.44	up-seid.....	1340
5.2.2.45	wrap.....	1345
5.2.2.46	pdu-session.....	1346
5.2.2.47	plmn-chrg-profile-map.....	1359
5.2.2.48	pfcp-load-overload-control.....	1360
5.2.2.49	pfcp-local-load-control.....	1363
5.2.2.50	pfcp-local-overload-control.....	1365
5.2.2.51	pgw-reselection-isolation.....	1366
5.2.2.52	qos-flow.....	1367
5.2.2.53	redirect-policy.....	1378
5.2.2.54	ref-point-peers.....	1379
5.2.2.55	ref-point-stats.....	1412
5.2.2.56	sbi-load-overload-control.....	1482
5.2.2.57	service-stats.....	1483
5.2.2.58	static-predefined-pcc.....	1512
5.2.2.59	statistics.....	1514
5.2.2.60	swu.....	1548
5.2.2.61	ue-context.....	1550
5.2.2.62	def-pfcp-u-sessions.....	1566
5.2.2.63	up-peer-list.....	1573
5.2.3	PDN clear command descriptions.....	1577
5.2.3.1	mobile-gateway.....	1577
5.2.3.2	pdn.....	1578
5.2.3.3	apn.....	1578
5.2.3.4	application-server-stats.....	1580
5.2.3.5	bearer-context.....	1580
5.2.3.6	call-flow-stats.....	1587
5.2.3.7	charging.....	1588
5.2.3.8	diameter-sessions.....	1588
5.2.3.9	dns-client-cache.....	1589
5.2.3.10	fixed-subscriber.....	1590
5.2.3.11	gtp-load-overload-control.....	1591
5.2.3.12	gtp-local-overload-control.....	1591
5.2.3.13	local-storage-stats.....	1592
5.2.3.14	ipsec-lockout.....	1592
5.2.3.15	nrf-client-cache.....	1593
5.2.3.16	nsmf-overload-control.....	1593
5.2.3.17	path-mgmt-fault.....	1594
5.2.3.18	path-mgmt-stats.....	1595
5.2.3.19	pdn-context.....	1596
5.2.3.20	pfcp-load-overload-control.....	1597

5.2.3.21	pfcp-local-load-control.....	1598
5.2.3.22	pfcp-local-overload-control.....	1598
5.2.3.23	pgw-reselection-isolation.....	1599
5.2.3.24	ref-point-stats.....	1599
5.2.3.25	service-stats.....	1601
5.2.3.26	statistics.....	1603
5.2.3.27	ue-context.....	1604
6	Profile command reference.....	1605
6.1	Profile command hierarchies.....	1605
6.1.1	Profile configuration command hierarchies.....	1605
6.1.1.1	Accounting priority profile command hierarchy.....	1607
6.1.1.2	Address list profile command hierarchy.....	1608
6.1.1.3	Call trace profile command hierarchy.....	1608
6.1.1.4	Canary test profile command hierarchy.....	1609
6.1.1.5	Charging profile command hierarchy.....	1609
6.1.1.6	CCFH charging profile command hierarchy.....	1611
6.1.1.7	CDR charging profile command hierarchy.....	1612
6.1.1.8	CHF charging profile command hierarchy.....	1616
6.1.1.9	ePDG charging profile command hierarchy.....	1617
6.1.1.10	FBC event triggers charging profile command hierarchy.....	1618
6.1.1.11	QBC event triggers charging profile command hierarchy.....	1620
6.1.1.12	Gy AVP options charging profile.....	1621
6.1.1.13	PGW charging profile command hierarchy.....	1622
6.1.1.14	SGW charging profile command hierarchy.....	1625
6.1.1.15	SMF charging profile command hierarchy.....	1626
6.1.1.16	TDF charging profile command hierarchy.....	1628
6.1.1.17	Cloud database profile command hierarchy.....	1629
6.1.1.18	DCCA profile command hierarchy.....	1629
6.1.1.19	Dedicated bearer idle timeout profile command hierarchy.....	1633
6.1.1.20	DHCP server group profile command hierarchy.....	1633
6.1.1.21	DHCPv4 server profile command hierarchy.....	1634
6.1.1.22	Diameter profile command hierarchy.....	1634
6.1.1.23	Diameter error handling profile command hierarchy.....	1635
6.1.1.24	Diameter peer profile command hierarchy.....	1635
6.1.1.25	DNS profile command hierarchy.....	1636
6.1.1.26	DNS redirect profile command hierarchy.....	1637
6.1.1.27	Emergency profile command hierarchy.....	1637
6.1.1.28	ePDG group profile command hierarchy.....	1638
6.1.1.29	Fa Ha peer list profile command hierarchy.....	1638
6.1.1.30	FQDN group profile command hierarchy.....	1638
6.1.1.31	GTP profile command hierarchy.....	1639
6.1.1.32	GTP prime group profile command hierarchy.....	1640
6.1.1.33	Gx profile command hierarchy.....	1642
6.1.1.34	HTTP/2 profile command hierarchy.....	1644
6.1.1.35	IPSec profile command hierarchy.....	1645
6.1.1.36	List profile command hierarchy.....	1647

6.1.1.37	MIPv4 profile command hierarchy.....	1649
6.1.1.38	MME group profile command hierarchy.....	1650
6.1.1.39	N1 profile command hierarchy.....	1650
6.1.1.40	N2 peer profile command hierarchy.....	1651
6.1.1.41	N7 attributes profile command hierarchy.....	1651
6.1.1.42	Namf profile command hierarchy.....	1651
6.1.1.43	Network trace profile command hierarchy.....	1652
6.1.1.44	Node selection profile command hierarchy.....	1652
6.1.1.45	NPCF profile command hierarchy.....	1654
6.1.1.46	NSMF profile command hierarchy.....	1655
6.1.1.47	OAuth2.0 authorization profile command hierarchy.....	1655
6.1.1.48	Overload control profile command hierarchy.....	1656
6.1.1.49	PCMD profile command hierarchy.....	1656
6.1.1.50	PCSCF group profile command hierarchy.....	1657
6.1.1.51	Peer list profile command hierarchy.....	1657
6.1.1.52	PFCP profile command hierarchy.....	1657
6.1.1.53	PMIPv6 profile command hierarchy.....	1660
6.1.1.54	Policy options profile command hierarchy.....	1661
6.1.1.55	PPI mapping profile command hierarchy.....	1668
6.1.1.56	QFI mapping profile command hierarchy.....	1668
6.1.1.57	QoS profile command hierarchy.....	1669
6.1.1.58	RADIUS profile command hierarchy.....	1669
6.1.1.59	S2b error mapping profile command hierarchy.....	1674
6.1.1.60	S6b profile command hierarchy.....	1674
6.1.1.61	SBI error mapping profile command hierarchy.....	1676
6.1.1.62	SCTP profile command hierarchy.....	1676
6.1.1.63	Sd profile command hierarchy.....	1677
6.1.1.64	SGSN profile command hierarchy.....	1677
6.1.1.65	Sig rate limit profile command hierarchy.....	1678
6.1.1.66	SLAAC host profile command hierarchy.....	1678
6.1.1.67	STa profile command hierarchy.....	1679
6.1.1.68	Subscriber mapping profile command hierarchy.....	1679
6.1.1.69	SWm error mapping profile command hierarchy.....	1679
6.1.1.70	SWm profile command hierarchy.....	1680
6.1.1.71	SWw profile command hierarchy.....	1680
6.1.1.72	T6c profile command hierarchy.....	1681
6.1.1.73	TAI and LAI list profile command hierarchy.....	1681
6.1.1.74	UDM-SDM profile command hierarchy.....	1682
6.1.1.75	UDM-UECM profile command hierarchy.....	1683
6.1.1.76	UDP tunnel group profile command hierarchy.....	1684
6.1.1.77	UMTS QoS policy profile command hierarchy.....	1684
6.1.1.78	UP security policy profile command hierarchy.....	1684
6.1.1.79	V-SMF profile command hierarchy.....	1685
6.1.1.80	WLAN emergency profile command hierarchy.....	1685
6.1.1.81	WPS profile command hierarchy.....	1686
6.1.2	Profile show command hierarchy.....	1686
6.2	Profile command descriptions.....	1689

6.2.1 Profile configuration command descriptions.....	1689
6.2.1.1 Generic profile command descriptions.....	1692
6.2.1.1.1 description.....	1692
6.2.1.1.2 shutdown.....	1694
6.2.1.1.3 profile.....	1695
6.2.1.2 Accounting priority profile command descriptions.....	1695
6.2.1.2.1 accounting-priority-profile.....	1695
6.2.1.2.2 priority.....	1696
6.2.1.2.3 single-wt-session.....	1696
6.2.1.2.4 match-id.....	1697
6.2.1.3 Address list profile command descriptions.....	1698
6.2.1.3.1 addr-list.....	1698
6.2.1.3.2 ip-address.....	1699
6.2.1.3.3 ip-address-group.....	1699
6.2.1.4 Call Trace profile command descriptions.....	1700
6.2.1.4.1 call-insight.....	1700
6.2.1.4.2 all-ues.....	1701
6.2.1.4.3 ue.....	1701
6.2.1.4.4 events.....	1702
6.2.1.4.5 format.....	1702
6.2.1.4.6 live-output.....	1703
6.2.1.4.7 path-management.....	1704
6.2.1.4.8 ref-point.....	1705
6.2.1.4.9 sba-service.....	1705
6.2.1.4.10 size-limit.....	1706
6.2.1.4.11 time-limit.....	1707
6.2.1.4.12 user-traffic.....	1707
6.2.1.5 Canary test profile command descriptions.....	1708
6.2.1.5.1 canary-test.....	1709
6.2.1.5.2 user-range.....	1709
6.2.1.6 ue-entry.....	1709
6.2.1.7 Charging profile ePDG, PGW, SGW, TDF and UP function command descriptions.....	1710
6.2.1.7.1 charging.....	1710
6.2.1.7.2 epdg-charging.....	1711
6.2.1.7.3 pgw-charging.....	1711
6.2.1.7.4 accounting-priority-profile.....	1712
6.2.1.7.5 sgw-charging.....	1712
6.2.1.7.6 data-volume-count-time.....	1713
6.2.1.7.7 tdf-charging.....	1713
6.2.1.7.8 upf-acct-profile.....	1714
6.2.1.7.9 faster-poll-remaining-quota-trigger.....	1714
6.2.1.7.10 force-sgw-cdr-in-combo-gw.....	1716
6.2.1.7.11 nchf-charging.....	1716
6.2.1.7.12 off-line.....	1717
6.2.1.7.13 charging-info-trigger.....	1717
6.2.1.7.14 apn-rate-control-change.....	1717
6.2.1.7.15 cgi-sai-change.....	1718

6.2.1.7.16	change-csg-cell.....	1719
6.2.1.7.17	change-csg-subscribed-hybrid-cell.....	1719
6.2.1.7.18	change-in-user-plane.....	1720
6.2.1.7.19	change-ue-presence-pra.....	1720
6.2.1.7.20	ecgi-change.....	1721
6.2.1.7.21	fh-close-cdr-containers.....	1722
6.2.1.7.22	fh-close-cdr-exclude-dcca-result-code-4010.....	1723
6.2.1.7.23	fh-close-cdr-exclude-dcca-result-code-4012.....	1723
6.2.1.7.24	fh-close-cdr-exclude-dcca-result-code-5005.....	1724
6.2.1.7.25	fh-close-cdr-exclude-dcca-result-code-5030.....	1724
6.2.1.7.26	gy-final.....	1725
6.2.1.7.27	gy-forced-reauthorization.....	1726
6.2.1.7.28	gy-qht.....	1726
6.2.1.7.29	gy-quota-exhausted.....	1727
6.2.1.7.30	gy-rating-condition-change.....	1728
6.2.1.7.31	gy-tariff-time-change.....	1728
6.2.1.7.32	gy-threshold-vqt.....	1729
6.2.1.7.33	gy-validity-time.....	1730
6.2.1.7.34	plmn-rate-control-change.....	1730
6.2.1.7.35	qos-change.....	1731
6.2.1.7.36	rai-change.....	1731
6.2.1.7.37	serving-node-change.....	1732
6.2.1.7.38	tai-change.....	1733
6.2.1.7.39	tariff-change-list.....	1733
6.2.1.7.40	tariff-time.....	1734
6.2.1.7.41	tariff-time-change.....	1735
6.2.1.7.42	time-limit-per-rating-grp.....	1736
6.2.1.7.43	user-loc-change.....	1736
6.2.1.7.44	uwan-loc-change.....	1737
6.2.1.7.45	volume-limit-per-rating-grp.....	1737
6.2.1.7.46	enable-offline.....	1738
6.2.1.7.47	enable-offline.....	1739
6.2.1.7.48	exclude-zero-vol-cdr.....	1739
6.2.1.7.49	exclude-zero-vol-complete-cdr.....	1740
6.2.1.7.50	include-bearer-charging.....	1741
6.2.1.7.51	partial-record-closure-trigger.....	1741
6.2.1.7.52	max-change-condition.....	1741
6.2.1.7.53	min-data-threshold.....	1742
6.2.1.7.54	mo-exception-data-receipt.....	1743
6.2.1.7.55	ms-time-zone-change.....	1743
6.2.1.7.56	plmn-change.....	1744
6.2.1.7.57	rat-change.....	1745
6.2.1.7.58	serving-node-change-limit.....	1745
6.2.1.7.59	time-limit.....	1746
6.2.1.7.60	time-limit-eutran.....	1747
6.2.1.7.61	time-limit-utran.....	1748
6.2.1.7.62	time-limit-eutran-nb-iot.....	1748

6.2.1.7.63	time-limit-lte-m.....	1749
6.2.1.7.64	volume-limit.....	1749
6.2.1.7.65	volume-limit-eutran.....	1750
6.2.1.7.66	volume-limit-eutran-nb-iot.....	1751
6.2.1.7.67	volume-limit-lte-m.....	1752
6.2.1.7.68	volume-limit-utran.....	1752
6.2.1.7.69	pri-cdf-dia-peer.....	1753
6.2.1.7.70	sd-container.....	1754
6.2.1.7.71	include-losd-with-sec-rat-usage-report.....	1754
6.2.1.7.72	include-zero-vol-sdc.....	1755
6.2.1.7.73	qos-info-first-sdc-of-cdr.....	1755
6.2.1.7.74	sec-cdf-dia-peer.....	1755
6.2.1.7.75	user-location-at-cdr-generation-time.....	1756
6.2.1.7.76	sgw-change-cause-record-closure.....	1757
6.2.1.7.77	on-line.....	1757
6.2.1.7.78	dcca-profile.....	1758
6.2.1.7.79	diameter-peer-group-list.....	1758
6.2.1.7.80	enable-online.....	1759
6.2.1.7.81	enable-online.....	1759
6.2.1.7.82	pri-ocs-dia-peer.....	1760
6.2.1.7.83	sec-ocs-dia-peer.....	1761
6.2.1.7.84	selection-type.....	1762
6.2.1.7.85	selection-type.....	1762
6.2.1.7.86	user-range.....	1763
6.2.1.7.87	user-range.....	1764
6.2.1.7.88	pcrf-selection.....	1765
6.2.1.7.89	dynamic-pcc.....	1766
6.2.1.7.90	pcc-disable-per-rat.....	1767
6.2.1.7.91	pcc-disable-per-rat-timer.....	1768
6.2.1.7.92	radius-accounting.....	1768
6.2.1.7.93	cgi-sai-change.....	1769
6.2.1.7.94	ecgi-change.....	1769
6.2.1.7.95	peer-change-trigger.....	1770
6.2.1.7.96	qos-change-trigger.....	1770
6.2.1.7.97	rai-change.....	1771
6.2.1.7.98	rat-change-trigger.....	1771
6.2.1.7.99	tai-change.....	1772
6.2.1.7.100	user-loc-change.....	1772
6.2.1.7.101	volume-limit-trigger.....	1773
6.2.1.7.102	report-serving-node-supplied-cc.....	1773
6.2.1.7.103	smf-legacy-nr-ratatype.....	1774
6.2.1.8	Charging profile CCFH command descriptions.....	1774
6.2.1.8.1	ccfh-profile.....	1775
6.2.1.8.2	assume-positive.....	1775
6.2.1.8.3	access-type.....	1775
6.2.1.8.4	message.....	1776
6.2.1.8.5	rating-group.....	1777

6.2.1.8.6	credit-pool-id.....	1777
6.2.1.8.7	enable.....	1778
6.2.1.8.8	orig-session-id-ccri.....	1778
6.2.1.8.9	cc-failure-handling.....	1779
6.2.1.8.10	cc-failure-handling-enh.....	1780
6.2.1.8.11	ccfh-fh-action.....	1780
6.2.1.8.12	disable-cc-session-failover.....	1781
6.2.1.8.13	fail-open.....	1781
6.2.1.8.14	fh-continue-action.....	1782
6.2.1.8.15	fh-session-continue-timer.....	1782
6.2.1.8.16	fh-volume-limit.....	1783
6.2.1.8.17	gy-to-gtp-error-mapping-profile.....	1784
6.2.1.8.18	message.....	1784
6.2.1.8.19	result-code.....	1785
6.2.1.8.20	mscc-result-code.....	1787
6.2.1.8.21	fui-terminate-session.....	1787
6.2.1.8.22	service-delay-timer.....	1788
6.2.1.8.23	nchf-cause-code.....	1789
6.2.1.8.24	nchf-http-code.....	1790
6.2.1.8.25	nchf-mui-result-code.....	1791
6.2.1.8.26	fui-terminate-session.....	1792
6.2.1.8.27	service-delay-timer.....	1793
6.2.1.8.28	result-code.....	1794
6.2.1.8.29	result-code-3xxx.....	1796
6.2.1.8.30	result-code-4xxx.....	1797
6.2.1.8.31	result-code-5xxx.....	1798
6.2.1.8.32	sbi-error-mapping-profile.....	1800
6.2.1.8.33	service-restore-max-offset.....	1800
6.2.1.8.34	session-restore-retry-timer.....	1801
6.2.1.8.35	store-and-replay.....	1802
6.2.1.8.36	buffer-playback-timer.....	1802
6.2.1.8.37	enable.....	1803
6.2.1.9	Charging profile CDR AVP command descriptions.....	1803
6.2.1.9.1	cdr-avp-options.....	1803
6.2.1.9.2	3gpp2-bsid.....	1804
6.2.1.9.3	apn-ambr.....	1804
6.2.1.9.4	apn-network-id.....	1805
6.2.1.9.5	apn-rate-control.....	1806
6.2.1.9.6	apn-selection-mode.....	1807
6.2.1.9.7	camel-information.....	1807
6.2.1.9.8	charging-char-select-mode.....	1808
6.2.1.9.9	cn-operator-selection.....	1809
6.2.1.9.10	cp-ciot-eps-optimisation-indicator.....	1809
6.2.1.9.11	diagnostics.....	1810
6.2.1.9.12	dualstack-epdg-address-ipv6.....	1811
6.2.1.9.13	dualstack-pgw-address-ipv6.....	1811
6.2.1.9.14	dualstack-serving-node-ipv6-addr.....	1812

6.2.1.9.15	dualstack-sgw-address-ipv6.....	1813
6.2.1.9.16	dualstack-twag-address-ipv6.....	1813
6.2.1.9.17	dynamic-addr-flag.....	1814
6.2.1.9.18	dynamic-addr-flag-extension.....	1815
6.2.1.9.19	enhanced-diagnostics.....	1815
6.2.1.9.20	extended-bw-nr.....	1816
6.2.1.9.21	imsi-unauthenticated-flag.....	1817
6.2.1.9.22	ims-sig-context.....	1817
6.2.1.9.23	imsi-unauthenticated-flag.....	1818
6.2.1.9.24	list-of-service-data.....	1819
6.2.1.9.25	list-traffic-data-volume.....	1820
6.2.1.9.26	local-record-seq-no.....	1820
6.2.1.9.27	low-priority-indicator.....	1821
6.2.1.9.28	max-requested-bandwidth.....	1822
6.2.1.9.29	mo-exception-data-counter.....	1822
6.2.1.9.30	ms-time-zone.....	1823
6.2.1.9.31	node-id.....	1824
6.2.1.9.32	nr-user-location-info.....	1824
6.2.1.9.33	nw-init-pdp-context.....	1825
6.2.1.9.34	pdn-connection-charging-id.....	1826
6.2.1.9.35	pdn-pdp-address-prefix-length.....	1826
6.2.1.9.36	pdp-pdn-type.....	1827
6.2.1.9.37	pdp-pdn-type-extension.....	1828
6.2.1.9.38	pgw-address.....	1828
6.2.1.9.39	pgw-plmn-id.....	1829
6.2.1.9.40	pra-info.....	1830
6.2.1.9.41	ps-furnish-charging-information.....	1830
6.2.1.9.42	originating-bsid.....	1831
6.2.1.9.43	originating-user-location-info.....	1832
6.2.1.9.44	ran-secondary-rat-usage-reports.....	1832
6.2.1.9.45	rat-type.....	1833
6.2.1.9.46	record-extension-visited-url.....	1834
6.2.1.9.47	sdh-charging-id.....	1835
6.2.1.9.48	sdh-pra-status.....	1835
6.2.1.9.49	sdh-qci.....	1836
6.2.1.9.50	sequence-number-for-complete-cdr.....	1837
6.2.1.9.51	served-imeisv.....	1837
6.2.1.9.52	served-mn-nai.....	1838
6.2.1.9.53	served-msisdh.....	1839
6.2.1.9.54	served-pdp-pdn-addr.....	1839
6.2.1.9.55	served-pdp-pdn-addr-extension.....	1840
6.2.1.9.56	serving-node-plmn-id.....	1841
6.2.1.9.57	serving-plmn-rate-control.....	1841
6.2.1.9.58	sgi-ptp-tunnelling-method.....	1842
6.2.1.9.59	service-context-id.....	1843
6.2.1.9.60	sgw-address.....	1843
6.2.1.9.61	sgw-change.....	1844

6.2.1.9.62	start-time.....	1845
6.2.1.9.63	stop-time.....	1845
6.2.1.9.64	tdv-charging-id.....	1846
6.2.1.9.65	tdv-pra-status.....	1847
6.2.1.9.66	twan-user-location-info.....	1848
6.2.1.9.67	twan-user-location-information-r16.....	1848
6.2.1.9.68	uni-pdu-cp-only-flag.....	1849
6.2.1.9.69	user-csg-info.....	1850
6.2.1.9.70	user-location-info.....	1850
6.2.1.9.71	user-location-info-time.....	1851
6.2.1.9.72	uwan-user-location-information.....	1852
6.2.1.9.73	uwan-user-location-information-r16.....	1852
6.2.1.10	Charging profile CHF command descriptions.....	1853
6.2.1.10.1	chf-profile.....	1853
6.2.1.10.2	activity-threshold.....	1853
6.2.1.10.3	ccfh-profile.....	1854
6.2.1.10.4	custom-online-offline-reporting.....	1854
6.2.1.10.5	default-rg-list.....	1856
6.2.1.10.6	implicit-vol-threshold.....	1856
6.2.1.10.7	ran-secondary-rat-usage-reports.....	1857
6.2.1.10.8	rating-group.....	1857
6.2.1.10.9	activity-threshold.....	1858
6.2.1.10.10	uuc-options-no-quota-initial.....	1858
6.2.1.10.11	uuc-options-no-quota-mid-session.....	1859
6.2.1.10.12	uuc-options-with-quota.....	1860
6.2.1.10.13	reauth-blocked-rg.....	1861
6.2.1.10.14	request-quota-before-traffic.....	1862
6.2.1.10.15	requested-unit-for-all-triggers.....	1862
6.2.1.10.16	requested-unit-total-volume.....	1863
6.2.1.10.17	sbi-srvc-operation.....	1864
6.2.1.10.18	retry-count.....	1864
6.2.1.10.19	transaction-timer.....	1865
6.2.1.10.20	smf-attributes-profile.....	1866
6.2.1.10.21	smf-generate-charging-data-ref.....	1866
6.2.1.10.22	supported-features.....	1866
6.2.1.10.23	support-2g3g.....	1867
6.2.1.10.24	supported-vendor.....	1867
6.2.1.10.25	unit-count-inactivity-timer.....	1868
6.2.1.10.26	uuc-reporting-enh.....	1868
6.2.1.11	Charging profile credit pool command descriptions.....	1869
6.2.1.11.1	credit-pool-id.....	1869
6.2.1.11.2	max-interim-quotas.....	1869
6.2.1.11.3	validity-time.....	1870
6.2.1.11.4	volume-quota.....	1871
6.2.1.12	Charging profile event triggers FBC command descriptions.....	1871
6.2.1.12.1	event-triggers-fbc.....	1871
6.2.1.12.2	3gpp-ps-data-off-status-change.....	1872

6.2.1.12.3	addition-of-upf.....	1872
6.2.1.12.4	default-qht.....	1873
6.2.1.12.5	handover-cancel.....	1873
6.2.1.12.6	handover-complete.....	1874
6.2.1.12.7	handover-start.....	1875
6.2.1.12.8	insertion-of-ismf.....	1875
6.2.1.12.9	max-change-condition.....	1876
6.2.1.12.10	plmn-change.....	1877
6.2.1.12.11	qos-change.....	1877
6.2.1.12.12	rat-type-change.....	1878
6.2.1.12.13	reallocation-of-ismf.....	1878
6.2.1.12.14	removal-of-ismf.....	1879
6.2.1.12.15	removal-of-upf.....	1879
6.2.1.12.16	serving-node-change.....	1880
6.2.1.12.17	serving-node-change-limit.....	1881
6.2.1.12.18	serving-node-type.....	1881
6.2.1.12.19	session-ambr-change.....	1882
6.2.1.12.20	tariff-change-list.....	1883
6.2.1.12.21	time-limit.....	1883
6.2.1.12.22	time-limit-eutran.....	1884
6.2.1.12.23	time-limit-nr.....	1885
6.2.1.12.24	time-limit-per-rating-grp.....	1886
6.2.1.12.25	trigger-override-option.....	1886
6.2.1.12.26	ue-in-pra-change.....	1887
6.2.1.12.27	ue-time-zone-change.....	1888
6.2.1.12.28	user-loc-change.....	1888
6.2.1.12.29	volume-limit.....	1889
6.2.1.12.30	volume-limit-eutran.....	1889
6.2.1.12.31	volume-limit-nr.....	1890
6.2.1.12.32	volume-limit-per-rating-grp.....	1891
6.2.1.13	Charging profile event triggers QBC command descriptions.....	1892
6.2.1.13.1	event-triggers-qbc.....	1892
6.2.1.13.2	addition-of-upf.....	1892
6.2.1.13.3	handover-cancel.....	1893
6.2.1.13.4	handover-complete.....	1893
6.2.1.13.5	handover-start.....	1894
6.2.1.13.6	max-change-condition.....	1894
6.2.1.13.7	plmn-change.....	1895
6.2.1.13.8	qos-change.....	1896
6.2.1.13.9	removal-of-upf.....	1896
6.2.1.13.10	serving-node-change.....	1897
6.2.1.13.11	serving-node-change-limit.....	1897
6.2.1.13.12	session-ambr-change.....	1898
6.2.1.13.13	tariff-change-list.....	1899
6.2.1.13.14	time-limit.....	1899
6.2.1.13.15	time-limit-per-qos-flow.....	1900
6.2.1.13.16	trigger-override-option.....	1901

6.2.1.13.17	ue-time-zone-change.....	1902
6.2.1.13.18	user-loc-change.....	1902
6.2.1.13.19	volume-limit.....	1903
6.2.1.13.20	volume-limit-per-qos-flow.....	1903
6.2.1.14	Charging profile Gy AVP options command descriptions.....	1904
6.2.1.14.1	gy-avp-options.....	1904
6.2.1.14.2	exclude-3gpp-charging-characteristics.....	1905
6.2.1.14.3	exclude-3gpp-charging-id.....	1905
6.2.1.14.4	exclude-3gpp-ggsn-mcc-mnc.....	1906
6.2.1.14.5	exclude-3gpp-imsi-mcc-mnc.....	1906
6.2.1.14.6	exclude-3gpp-ms-timezone.....	1906
6.2.1.14.7	exclude-3gpp-nsapi.....	1907
6.2.1.14.8	exclude-3gpp-pdp-type.....	1907
6.2.1.14.9	exclude-3gpp-rat-type.....	1908
6.2.1.14.10	exclude-3gpp-selection-mode.....	1908
6.2.1.14.11	exclude-3gpp-session-stop-indicator.....	1909
6.2.1.14.12	exclude-3gpp-sgsn-mcc-mnc.....	1909
6.2.1.14.13	exclude-3gpp-user-location-info.....	1909
6.2.1.14.14	exclude-3gpp2-bsid.....	1910
6.2.1.14.15	exclude-cc-correlation-id.....	1910
6.2.1.14.16	exclude-charging-characteristics-selection-mode.....	1911
6.2.1.14.17	exclude-dynamic-addr-flag.....	1911
6.2.1.14.18	exclude-dynamic-address-flag-extension.....	1912
6.2.1.14.19	exclude-event-timestamp.....	1912
6.2.1.14.20	exclude-ggsn-address.....	1912
6.2.1.14.21	exclude-ims-information.....	1913
6.2.1.14.22	exclude-imsi-unauthenticated-flag.....	1913
6.2.1.14.23	exclude-node-id.....	1914
6.2.1.14.24	exclude-origin-state-id.....	1914
6.2.1.14.25	exclude-pdn-connection-charging-id.....	1915
6.2.1.14.26	exclude-pdp-address.....	1915
6.2.1.14.27	exclude-pdp-address-extension.....	1915
6.2.1.14.28	exclude-pdp-address-prefix-length.....	1916
6.2.1.14.29	exclude-pdp-context-type.....	1916
6.2.1.14.30	exclude-service-id.....	1917
6.2.1.14.31	exclude-serving-node-type.....	1917
6.2.1.14.32	exclude-sgsn-address.....	1918
6.2.1.14.33	exclude-subscription-id-imsi.....	1918
6.2.1.14.34	exclude-subscription-id-msisdh.....	1918
6.2.1.14.35	exclude-subscription-id-nai.....	1919
6.2.1.14.36	exclude-user-equipment-info.....	1919
6.2.1.14.37	exclude-user-name.....	1920
6.2.1.14.38	include-3gpp-gprs-qos-negotiated-profile.....	1920
6.2.1.14.39	include-3gpp-user-location-info-5g.....	1921
6.2.1.14.40	include-cg-address.....	1921
6.2.1.14.41	include-charging-rule-base-name.....	1922
6.2.1.14.42	include-dualstack-ggsn-address-ipv6.....	1922

6.2.1.14.43	include-dualstack-sgsn-address-ipv6.....	1922
6.2.1.14.44	include-enhanced-diagnostics.....	1923
6.2.1.14.45	include-extended-bw-nr.....	1923
6.2.1.14.46	include-mscc-qos-information.....	1924
6.2.1.14.47	include-pra-info.....	1924
6.2.1.14.48	include-qos-information.....	1925
6.2.1.14.49	include-rg-in-raa.....	1925
6.2.1.14.50	include-twan-user-location-info.....	1926
6.2.1.14.51	include-twan-user-location-info-r16.....	1926
6.2.1.14.52	include-user-csg-info.....	1927
6.2.1.14.53	include-user-equipment-info-bcd-encode.....	1927
6.2.1.14.54	include-user-location-info-time.....	1928
6.2.1.14.55	include-user-name-msisdn-at-apn.....	1928
6.2.1.14.56	include-user-name-raa.....	1929
6.2.1.14.57	include-uan-user-location-information.....	1929
6.2.1.14.58	include-uan-user-location-information-r16.....	1929
6.2.1.15	Charging profile SMF command descriptions.....	1930
6.2.1.15.1	smf-attributes-profile.....	1930
6.2.1.15.2	ran-secondary-rat-usage-reports.....	1931
6.2.1.15.3	retransmission-indicator.....	1931
6.2.1.15.4	service-specific-information.....	1932
6.2.1.15.5	service-specific-information-value.....	1932
6.2.1.15.6	serving-network-include-source-ip.....	1933
6.2.1.15.7	serving-node-type.....	1934
6.2.1.15.8	smf-charging.....	1935
6.2.1.15.9	chf-selection.....	1935
6.2.1.15.10	chf-profile.....	1936
6.2.1.15.11	fail-open.....	1936
6.2.1.15.12	chf-profile.....	1936
6.2.1.15.13	nf-id-list.....	1937
6.2.1.15.14	nf-id-list.....	1938
6.2.1.15.15	default-charging-method.....	1938
6.2.1.15.16	event-triggers-fbc.....	1939
6.2.1.15.17	event-triggers-qbc.....	1939
6.2.1.15.18	qbc-charging.....	1940
6.2.1.15.19	quota-management.....	1940
6.2.1.15.20	final-unit-indication.....	1941
6.2.1.15.21	redirect-append-url.....	1941
6.2.1.15.22	redirect-filter-id.....	1941
6.2.1.15.23	redirection-start-send-rr-final.....	1942
6.2.1.15.24	redirection-start-send-rr-final-enh.....	1943
6.2.1.15.25	traffic-handling-options.....	1943
6.2.1.15.26	forced-re-authorization.....	1944
6.2.1.15.27	quota-exhausted-no-threshold.....	1944
6.2.1.15.28	quota-exhausted-threshold-active.....	1945
6.2.1.15.29	quota-unavailable.....	1945
6.2.1.15.30	rating-condition-change.....	1946

6.2.1.15.31	validity-time-expired.....	1947
6.2.1.15.32	radius-accounting.....	1947
6.2.1.15.33	ecgi-change.....	1948
6.2.1.15.34	peer-change.....	1948
6.2.1.15.35	qos-change.....	1949
6.2.1.15.36	rat-change.....	1949
6.2.1.15.37	tai-change.....	1950
6.2.1.15.38	user-loc-change.....	1950
6.2.1.15.39	volume-limit.....	1951
6.2.1.16	Charging profile TWAG command descriptions.....	1951
6.2.1.16.1	twag-charging.....	1951
6.2.1.16.2	radius-accounting.....	1952
6.2.1.16.3	wlan-ap-location-change.....	1952
6.2.1.17	Cloud database profile command descriptions.....	1953
6.2.1.17.1	cloud-db.....	1953
6.2.1.17.2	server.....	1953
6.2.1.18	DCCA profile command descriptions.....	1954
6.2.1.18.1	dcca.....	1955
6.2.1.18.2	activity-threshold.....	1955
6.2.1.18.3	append-url.....	1956
6.2.1.18.4	application-tx-timer.....	1956
6.2.1.18.5	cc-failover-handling.....	1957
6.2.1.18.6	cca-init-no-mscc-answer-ignore.....	1957
6.2.1.18.7	ccfh-fh-action.....	1958
6.2.1.18.8	ccfh-profile.....	1959
6.2.1.18.9	ccr-initial-on-first-online-rg.....	1959
6.2.1.18.10	ccr-initial-on-first-pkt.....	1960
6.2.1.18.11	ccr-mscc-limit.....	1961
6.2.1.18.12	ccrt-no-mscc.....	1961
6.2.1.18.13	command-level-fui.....	1962
6.2.1.18.14	default-qht.....	1962
6.2.1.18.15	default-rating-group.....	1963
6.2.1.18.16	default-rg-list.....	1964
6.2.1.18.17	rating-group.....	1964
6.2.1.18.18	destination-realm.....	1964
6.2.1.18.19	diam-sess-end-serialize-ccr.....	1965
6.2.1.18.20	disable-cc-session-failover.....	1966
6.2.1.18.21	disable-first-pkt-behavior.....	1966
6.2.1.18.22	disable-reauth-ccru-no-mscc.....	1967
6.2.1.18.23	event-triggers.....	1967
6.2.1.18.24	change-ue-presence-pra.....	1967
6.2.1.18.25	qos-apn-ambr-change.....	1968
6.2.1.18.26	qos-change.....	1968
6.2.1.18.27	qos-max-bitrate-dl-change.....	1969
6.2.1.18.28	qos-max-bitrate-ul-change.....	1969
6.2.1.18.29	rat-change.....	1970
6.2.1.18.30	serving-node-change.....	1970

6.2.1.18.31	trigger-override-option.....	1971
6.2.1.18.32	user-loc-change.....	1972
6.2.1.18.33	user-loc-change-cellid.....	1972
6.2.1.18.34	user-loc-change-ecgi.....	1973
6.2.1.18.35	user-loc-change-lac.....	1973
6.2.1.18.36	user-loc-change-rac.....	1974
6.2.1.18.37	user-loc-change-tac.....	1974
6.2.1.18.38	uwan-loc-change.....	1975
6.2.1.18.39	failure-handling.....	1975
6.2.1.18.40	mscc-result-code.....	1976
6.2.1.18.41	result-code.....	1977
6.2.1.18.42	service-delay-timer.....	1983
6.2.1.18.43	result-code-3xxx.....	1984
6.2.1.18.44	result-code-4xxx.....	1985
6.2.1.18.45	result-code-5xxx.....	1985
6.2.1.18.46	fh-session-continue-timer.....	1986
6.2.1.18.47	fh-volume-limit.....	1987
6.2.1.18.48	fui-validity-timer.....	1987
6.2.1.18.49	gy-avp-options-profile.....	1988
6.2.1.18.50	implicit-vol-threshold.....	1989
6.2.1.18.51	include-avp.....	1990
6.2.1.18.52	3gpp-gprs-qos-neg-prof.....	1990
6.2.1.18.53	qos-information.....	1990
6.2.1.18.54	include-rpt-reason-on-forced-reauth.....	1991
6.2.1.18.55	exclude-service-id.....	1991
6.2.1.18.56	offline-interwork.....	1992
6.2.1.18.57	one-time-redirect.....	1992
6.2.1.18.58	ps-fci-fh.....	1993
6.2.1.18.59	ps-fci-fh-container.....	1993
6.2.1.18.60	pre-emptive-quota.....	1994
6.2.1.18.61	queue-rar-pending-ccau.....	1995
6.2.1.18.62	rating-group.....	1995
6.2.1.18.63	rating-group-in-ch-rule-report.....	1996
6.2.1.18.64	reauth-blocked-rg.....	1996
6.2.1.18.65	redirect-filter-id.....	1997
6.2.1.18.66	redirection-start-send-rr-final.....	1997
6.2.1.18.67	retry-count.....	1998
6.2.1.18.68	rsu-total-octets.....	1999
6.2.1.18.69	serialize-ccr.....	1999
6.2.1.18.70	service-context-id.....	1999
6.2.1.18.71	service-id-level-credit-control.....	2000
6.2.1.18.72	service-type.....	2001
6.2.1.18.73	ccfh-profile.....	2001
6.2.1.18.74	session-restore-retry-timer.....	2002
6.2.1.18.75	supported-version.....	2003
6.2.1.18.76	suppress-multiple-ooc-notifications-per-rg.....	2004
6.2.1.18.77	traffic-handling-options.....	2004

6.2.1.18.78	forced-re-authorization.....	2005
6.2.1.18.79	quota-exhausted-no-threshold.....	2005
6.2.1.18.80	quota-exhausted-threshold-active.....	2006
6.2.1.18.81	quota-unavailable.....	2007
6.2.1.18.82	rating-condition-change.....	2007
6.2.1.18.83	validity-time-expired.....	2008
6.2.1.18.84	url-encryption.....	2009
6.2.1.18.85	usu-options-no-quota-initial.....	2010
6.2.1.18.86	usu-options-no-quota-mid-session.....	2011
6.2.1.18.87	usu-options-with-quota.....	2011
6.2.1.18.88	usu-reporting-enh.....	2013
6.2.1.18.89	validation-checks.....	2013
6.2.1.18.90	ignore-invalid-thresholds.....	2014
6.2.1.18.91	ignore-unsupported-triggers.....	2014
6.2.1.18.92	validity-timer-enh.....	2014
6.2.1.18.93	vol-threshold-reauth-enh.....	2015
6.2.1.18.94	wait-for-first-packet-on-fui-t-success.....	2016
6.2.1.18.95	wait-for-first-packet-on-no-gsu.....	2016
6.2.1.19	Dedicated bearer idle timeout profile command descriptions.....	2017
6.2.1.19.1	dedicated-bearer-idle-timeout-profile.....	2017
6.2.1.19.2	qci.....	2017
6.2.1.20	DHCP server group profile command descriptions.....	2018
6.2.1.20.1	dhcp-server-group.....	2018
6.2.1.20.2	client-identifier-msisdn-encoded.....	2019
6.2.1.20.3	deadtime.....	2020
6.2.1.20.4	interface.....	2021
6.2.1.20.5	peer.....	2022
6.2.1.20.6	priority.....	2023
6.2.1.20.7	vso-server-pool-name.....	2023
6.2.1.20.8	rebind-time-percent.....	2024
6.2.1.20.9	recommend-lease-time.....	2024
6.2.1.20.10	renew-time-percentage.....	2025
6.2.1.20.11	retry-count.....	2025
6.2.1.20.12	retry-timeout.....	2026
6.2.1.20.13	via-up-function.....	2026
6.2.1.21	DHCPv4 server profile command descriptions.....	2028
6.2.1.21.1	dhcpv4-server.....	2028
6.2.1.21.2	hold-time.....	2029
6.2.1.21.3	ignore-rapid-commit.....	2029
6.2.1.21.4	lease-time.....	2030
6.2.1.22	Diameter profile command descriptions.....	2030
6.2.1.22.1	diameter.....	2031
6.2.1.22.2	connection-timer.....	2031
6.2.1.22.3	dns-refresh-interval.....	2032
6.2.1.22.4	dpr-timeout.....	2032
6.2.1.22.5	fail-num-retries.....	2033
6.2.1.22.6	fail-retry-interval.....	2034

6.2.1.22.7 ip-dscp.....	2035
6.2.1.22.8 ip-ttl.....	2035
6.2.1.22.9 perm-fail-retry-timeout.....	2036
6.2.1.22.10 transaction-timer.....	2036
6.2.1.22.11 watchdog-timer.....	2037
6.2.1.23 Diameter error mapping profile command descriptions.....	2038
6.2.1.23.1 diameter-error-mapping-profile.....	2038
6.2.1.23.2 fh-tx-timeout.....	2038
6.2.1.23.3 result-code.....	2039
6.2.1.23.4 result-code-3xxx.....	2039
6.2.1.23.5 result-code-4xxx.....	2040
6.2.1.23.6 result-code-5xxx.....	2041
6.2.1.24 Diameter peer profile command descriptions.....	2041
6.2.1.24.1 diameter-peer.....	2042
6.2.1.24.2 application-type.....	2042
6.2.1.24.3 custom-vendor-id.....	2043
6.2.1.24.4 dest-host-in-initial-msg.....	2044
6.2.1.24.5 dest-host-table-threshold.....	2044
6.2.1.24.6 destination-realm.....	2045
6.2.1.24.7 diameter-profile.....	2045
6.2.1.24.8 disable-cea-auth-app-id-check.....	2046
6.2.1.24.9 disable-dwr-dwa-origin-host-length-check.....	2046
6.2.1.24.10 disable-learned-dest-realm.....	2047
6.2.1.24.11 exclude-dest-host-after-switchover.....	2048
6.2.1.24.12 interface.....	2048
6.2.1.24.13 last-known-dest-realm-after-switchover.....	2049
6.2.1.24.14 load-balance.....	2050
6.2.1.24.15 multi-homing.....	2050
6.2.1.24.16 interface.....	2051
6.2.1.24.17 origin-realm.....	2052
6.2.1.24.18 peer.....	2053
6.2.1.24.19 enable-dra.....	2053
6.2.1.24.20 sctp-alternate-address.....	2054
6.2.1.24.21 supported-host.....	2055
6.2.1.24.22 suspend.....	2056
6.2.1.24.23 reconnect-with-new-tcp-port.....	2057
6.2.1.24.24 sctp-profile.....	2057
6.2.1.24.25 sig-rate-limit-profile.....	2058
6.2.1.24.26 switchover-source-port-consistent.....	2058
6.2.1.24.27 transport.....	2059
6.2.1.24.28 validate-supported-host.....	2060
6.2.1.24.29 diameter-peer-group-list.....	2060
6.2.1.24.30 peer-group.....	2060
6.2.1.25 DNS profile command descriptions.....	2062
6.2.1.25.1 dns-error-mapping-profile.....	2062
6.2.1.25.2 backoff-timer.....	2062
6.2.1.25.3 swu-notify.....	2064

6.2.1.25.4	swu-send-info.....	2064
6.2.1.25.5	dns-profile.....	2065
6.2.1.25.6	ip-dscp.....	2065
6.2.1.25.7	ip-ttl.....	2066
6.2.1.25.8	message-retransmit.....	2067
6.2.1.25.9	primary-dns.....	2068
6.2.1.25.10	secondary-dns.....	2068
6.2.1.25.11	tertiary-dns.....	2069
6.2.1.26	DNS redirection profile command descriptions.....	2070
6.2.1.26.1	dns-redirect-server.....	2070
6.2.1.26.2	primary-dns.....	2071
6.2.1.27	Emergency profile command descriptions.....	2072
6.2.1.27.1	emergency-profile.....	2072
6.2.1.27.2	allow-udm-sdm.....	2072
6.2.1.27.3	block-udm-uecm.....	2073
6.2.1.27.4	default-5qi.....	2073
6.2.1.27.5	default-ambr.....	2074
6.2.1.27.6	default-arp.....	2074
6.2.1.27.7	sbi-message-priority.....	2075
6.2.1.28	ePDG group profile command descriptions.....	2076
6.2.1.28.1	epdg-group.....	2076
6.2.1.28.2	ip-address.....	2076
6.2.1.29	Fa Ha peer list profile command descriptions.....	2077
6.2.1.29.1	fa-ha-peer-list.....	2077
6.2.1.29.2	peer.....	2078
6.2.1.29.3	fa-ha-spi.....	2078
6.2.1.29.4	hash-algorithm.....	2079
6.2.1.29.5	secret.....	2079
6.2.1.30	FQDN group profile command descriptions.....	2080
6.2.1.30.1	fqdn-group.....	2080
6.2.1.30.2	dns-refresh-interval.....	2080
6.2.1.30.3	fqdn.....	2081
6.2.1.30.4	ip-expiry-refresh-count.....	2082
6.2.1.31	Gn/Gp profile command descriptions.....	2083
6.2.1.31.1	peer-list.....	2083
6.2.1.31.2	peer.....	2083
6.2.1.31.3	keep-alive.....	2084
6.2.1.31.4	plmn-id.....	2085
6.2.1.31.5	rat-type.....	2086
6.2.1.31.6	unlisted-peer.....	2086
6.2.1.32	GTP profile command descriptions.....	2087
6.2.1.32.1	gtp.....	2087
6.2.1.32.2	gtpv1-options.....	2087
6.2.1.32.3	exclude-apn-restrictions.....	2088
6.2.1.32.4	exclude-bcm.....	2088
6.2.1.32.5	exclude-chrg-gw-addr.....	2088
6.2.1.32.6	exclude-pco.....	2089

6.2.1.32.7	exclude-recovery.....	2090
6.2.1.32.8	include-common-flags.....	2090
6.2.1.32.9	include-nsapi.....	2090
6.2.1.32.10	gtpv1-session-rejection-cause.....	2091
6.2.1.32.11	apn-shutdown.....	2091
6.2.1.32.12	gw-suspend.....	2092
6.2.1.32.13	sx-connection-failure.....	2093
6.2.1.32.14	username-missing.....	2093
6.2.1.32.15	gtpv2-node-features.....	2094
6.2.1.32.16	include-ciot.....	2094
6.2.1.32.17	gtpv2-options.....	2095
6.2.1.32.18	exclude-aaa-node-identifier.....	2095
6.2.1.32.19	exclude-apco.....	2096
6.2.1.32.20	exclude-change-reporting-action.....	2096
6.2.1.32.21	exclude-cno-selection-entity.....	2097
6.2.1.32.22	exclude-csg-reporting-action.....	2097
6.2.1.32.23	exclude-ddn-arp.....	2097
6.2.1.32.24	exclude-ddn-fteid.....	2098
6.2.1.32.25	exclude-ddn-imsi.....	2098
6.2.1.32.26	exclude-lci.....	2099
6.2.1.32.27	exclude-maximum-wait-time.....	2099
6.2.1.32.28	exclude-node-features.....	2099
6.2.1.32.29	exclude-oci.....	2100
6.2.1.32.30	exclude-origination-timestamp.....	2100
6.2.1.32.31	exclude-paging-priority.....	2101
6.2.1.32.32	exclude-pco.....	2101
6.2.1.32.33	exclude-ran-nas-cause.....	2102
6.2.1.32.34	exclude-signaling-priority.....	2102
6.2.1.32.35	exclude-twan-identifier.....	2102
6.2.1.32.36	exclude-twan-identifier-timestamp.....	2103
6.2.1.32.37	exclude-uli-timestamp.....	2103
6.2.1.32.38	exclude-up-selection-indication.....	2104
6.2.1.32.39	include-ap-mac-address.....	2104
6.2.1.32.40	include-mo-exception-data-counter.....	2105
6.2.1.32.41	include-serving-plmn-rate-control.....	2105
6.2.1.32.42	non-3gpp-rat-type-pref.....	2105
6.2.1.32.43	gtpv2-session-rejection-cause.....	2106
6.2.1.32.44	apn-shutdown.....	2106
6.2.1.32.45	gw-suspend.....	2107
6.2.1.32.46	sx-connection-failure.....	2108
6.2.1.32.47	username-missing.....	2109
6.2.1.32.48	ip-dscp.....	2109
6.2.1.32.49	ip-ttl.....	2110
6.2.1.32.50	keep-alive.....	2111
6.2.1.32.51	message-retransmit.....	2112
6.2.1.33	GTP prime group profile command descriptions.....	2112
6.2.1.33.1	gtp-prime-group.....	2113

6.2.1.33.2	deadtime.....	2113
6.2.1.33.3	duplicate-prevention.....	2114
6.2.1.33.4	duplicate-prevention-storage.....	2114
6.2.1.33.5	fragment-drt.....	2115
6.2.1.33.6	gtp-prime-server.....	2115
6.2.1.33.7	echo-interval.....	2116
6.2.1.33.8	enable-gtpv2.....	2116
6.2.1.33.9	max-outstanding-requests.....	2117
6.2.1.33.10	node-alive.....	2118
6.2.1.33.11	path-protocol.....	2118
6.2.1.33.12	priority.....	2118
6.2.1.33.13	retries.....	2119
6.2.1.33.14	timeout.....	2119
6.2.1.33.15	inactive-time.....	2120
6.2.1.33.16	interface.....	2121
6.2.1.33.17	local-cdr-storage.....	2121
6.2.1.33.18	file-params.....	2122
6.2.1.33.19	cdr-file-compression.....	2122
6.2.1.33.20	closure-rec-num.....	2123
6.2.1.33.21	closure-size.....	2123
6.2.1.33.22	closure-time.....	2124
6.2.1.33.23	extension.....	2124
6.2.1.33.24	max-files-per-subdirectory.....	2125
6.2.1.33.25	max-number-subdirectories.....	2125
6.2.1.33.26	obsolete-time.....	2126
6.2.1.33.27	private-info.....	2126
6.2.1.33.28	location.....	2127
6.2.1.33.29	limit.....	2127
6.2.1.33.30	primary.....	2128
6.2.1.33.31	max-cdrs-per-pdu.....	2128
6.2.1.33.32	mtu.....	2129
6.2.1.33.33	queue-size.....	2130
6.2.1.33.34	redirection.....	2130
6.2.1.33.35	round-robin-load-balance.....	2130
6.2.1.34	Gx profile command descriptions.....	2131
6.2.1.34.1	gx.....	2132
6.2.1.34.2	disable-ccr-i-on-csreq.....	2132
6.2.1.34.3	event-triggers.....	2133
6.2.1.34.4	tethering-start.....	2133
6.2.1.34.5	tethering-stop.....	2134
6.2.1.34.6	allow-pcf-provided-chf-address.....	2134
6.2.1.34.7	dynamic-pcc-online-offline-flags-override.....	2135
6.2.1.34.8	fh-session-no-response.....	2136
6.2.1.34.9	fh-time-limit.....	2137
6.2.1.34.10	fh-volume-limit.....	2138
6.2.1.34.11	forced-triggers.....	2139
6.2.1.34.12	disable-ambr-modification-failure.....	2139

6.2.1.34.13	disable-default-eps-bearer-qos-change.....	2140
6.2.1.34.14	disable-default-eps-bearer-qos-modification-failure.....	2140
6.2.1.34.15	disable-qos-change.....	2141
6.2.1.34.16	disable-ue-local-ip-addr-change.....	2142
6.2.1.34.17	include-last-known-uli.....	2142
6.2.1.34.18	message.....	2143
6.2.1.34.19	gx-avp-options.....	2143
6.2.1.34.20	result-code.....	2144
6.2.1.34.21	result-code-3xxx.....	2148
6.2.1.34.22	result-code-4xxx.....	2149
6.2.1.34.23	result-code-5xxx.....	2151
6.2.1.34.24	retry-count.....	2152
6.2.1.34.25	transaction-timer.....	2152
6.2.1.34.26	pending-transaction-retry-count.....	2153
6.2.1.34.27	pending-transaction-timer.....	2154
6.2.1.34.28	qos-info-always-present.....	2155
6.2.1.34.29	report-serving-node-supplied-cc.....	2156
6.2.1.34.30	serialize-ccr.....	2156
6.2.1.34.31	session-restore-retry-timer.....	2157
6.2.1.34.32	supported-features.....	2157
6.2.1.34.33	adc.....	2158
6.2.1.34.34	app-based-charging.....	2158
6.2.1.34.35	clear-m-bit.....	2159
6.2.1.34.36	cno-uli.....	2160
6.2.1.34.37	cond-policy-info-default-qos.....	2160
6.2.1.34.38	conditional-policy-info.....	2161
6.2.1.34.39	disable-rel-10.....	2161
6.2.1.34.40	disable-rel-9.....	2162
6.2.1.34.41	exclude-usage.....	2163
6.2.1.34.42	extended-bw-nr.....	2163
6.2.1.34.43	include-rat-when-received.....	2164
6.2.1.34.44	mission-critical-qcis.....	2164
6.2.1.34.45	multiple-pra.....	2165
6.2.1.34.46	netloc.....	2166
6.2.1.34.47	netloc-untrusted-wlan.....	2166
6.2.1.34.48	pcscf-restoration.....	2167
6.2.1.34.49	pending-transaction.....	2167
6.2.1.34.50	ran-nas-cause.....	2168
6.2.1.34.51	rule-bound-to-default-bearer.....	2168
6.2.1.34.52	sgw-rest.....	2169
6.2.1.34.53	sponsored-connectivity.....	2170
6.2.1.34.54	time-based-usage-monitoring.....	2170
6.2.1.34.55	trusted-wlan.....	2171
6.2.1.34.56	user-equipment-info-extension.....	2171
6.2.1.34.57	usage-monitoring.....	2172
6.2.1.34.58	monitoring-key.....	2172
6.2.1.34.59	activity-threshold.....	2173

6.2.1.34.60	override-usage-report-value.....	2174
6.2.1.34.61	validation-checks.....	2174
6.2.1.34.62	raa-immediate.....	2175
6.2.1.35	HTTP/2 profile command descriptions.....	2175
6.2.1.35.1	http2.....	2175
6.2.1.35.2	clt-tls-profile.....	2176
6.2.1.35.3	connection-timer.....	2176
6.2.1.35.4	fail-num-retries.....	2177
6.2.1.35.5	fail-retry-interval.....	2178
6.2.1.35.6	header-table-size.....	2178
6.2.1.35.7	init-window-size.....	2179
6.2.1.35.8	ip-dscp.....	2179
6.2.1.35.9	max-frame-size.....	2180
6.2.1.35.10	max-header-list-size.....	2180
6.2.1.35.11	ping-period.....	2181
6.2.1.35.12	request-timeout.....	2182
6.2.1.35.13	srv-disable-sni-chk.....	2182
6.2.1.35.14	srv-max-streams-allowed.....	2183
6.2.1.35.15	srv-stream-lifetime.....	2183
6.2.1.35.16	srv-tls-profile.....	2184
6.2.1.35.17	strict-validation.....	2184
6.2.1.36	IPSec profile command descriptions.....	2185
6.2.1.36.1	ipsec-profile.....	2185
6.2.1.36.2	esp-sa.....	2186
6.2.1.36.3	lifetime.....	2186
6.2.1.36.4	nat-traversal-dynamic-updates.....	2187
6.2.1.36.5	proposal-entry.....	2188
6.2.1.36.6	dh-group.....	2188
6.2.1.36.7	encryption.....	2190
6.2.1.36.8	extended-seq-num.....	2191
6.2.1.36.9	integrity.....	2192
6.2.1.36.10	rekey-margin.....	2193
6.2.1.36.11	rekeying.....	2193
6.2.1.36.12	replay-window.....	2194
6.2.1.36.13	ignore-ue-rekey-requests.....	2195
6.2.1.36.14	ike-sa.....	2196
6.2.1.36.15	apn-black-white-list.....	2196
6.2.1.36.16	cert-entry.....	2197
6.2.1.36.17	epdg-cert.....	2198
6.2.1.36.18	priv-key.....	2199
6.2.1.36.19	send-chain.....	2199
6.2.1.36.20	url-base-address.....	2200
6.2.1.36.21	certreq-eap-only-prefer-certreq.....	2201
6.2.1.36.22	device-identity.....	2201
6.2.1.36.23	function.....	2201
6.2.1.36.24	reject-if-validation-fails.....	2202
6.2.1.36.25	type.....	2203

6.2.1.36.26	dpd-interval.....	2203
6.2.1.36.27	fragmentation.....	2204
6.2.1.36.28	global-half-open-sa-limit.....	2205
6.2.1.36.29	imsi-privacy-protection.....	2206
6.2.1.36.30	local-identity.....	2206
6.2.1.36.31	message-retransmit.....	2207
6.2.1.36.32	mobike.....	2208
6.2.1.36.33	natt-mode.....	2209
6.2.1.36.34	peer-half-open-sa-limit.....	2210
6.2.1.36.35	plmn-black-white-list.....	2210
6.2.1.36.36	backoff-timer.....	2211
6.2.1.36.37	enable-plmn-black-white-list.....	2212
6.2.1.36.38	error-code.....	2213
6.2.1.36.39	idi-realm-list.....	2213
6.2.1.36.40	plmn-list.....	2214
6.2.1.36.41	type.....	2215
6.2.1.36.42	prf.....	2215
6.2.1.36.43	redirect.....	2216
6.2.1.36.44	always-on.....	2217
6.2.1.36.45	idi-realm.....	2218
6.2.1.36.46	overload-redirection.....	2220
6.2.1.36.47	source-ip.....	2220
6.2.1.37	List profile command descriptions.....	2222
6.2.1.37.1	list.....	2222
6.2.1.37.2	aaa-host-realm.....	2222
6.2.1.37.3	id.....	2223
6.2.1.37.4	allowed-nf-domains-list.....	2223
6.2.1.37.5	allowed-nf-domain.....	2224
6.2.1.37.6	apn-list.....	2224
6.2.1.37.7	apn.....	2225
6.2.1.37.8	arp-list.....	2225
6.2.1.37.9	arp.....	2226
6.2.1.37.10	broadcast-plmn-list.....	2227
6.2.1.37.11	broadcast-plmn-item.....	2227
6.2.1.37.12	cell-id-list.....	2228
6.2.1.37.13	ncgi.....	2229
6.2.1.37.14	fqdn-list.....	2230
6.2.1.37.15	fqdn.....	2230
6.2.1.37.16	idi-realm-list.....	2231
6.2.1.37.17	idi-realm.....	2231
6.2.1.37.18	ike-redirect-server-list.....	2232
6.2.1.37.19	server.....	2233
6.2.1.37.20	ike-redirect-source-ip-list.....	2234
6.2.1.37.21	ip.....	2235
6.2.1.37.22	ip-address-list.....	2237
6.2.1.37.23	address.....	2237
6.2.1.37.24	nf-id-list.....	2238

6.2.1.37.25	nf-prof-id.....	2239
6.2.1.37.26	api-prefix.....	2239
6.2.1.37.27	api-version.....	2240
6.2.1.37.28	enable.....	2240
6.2.1.37.29	fqdn.....	2241
6.2.1.37.30	nf-priority.....	2241
6.2.1.37.31	nf-set.....	2242
6.2.1.37.32	peer-selection.....	2242
6.2.1.37.33	prioritized-address-list.....	2243
6.2.1.37.34	uuid.....	2243
6.2.1.37.35	oi-override-mapping.....	2244
6.2.1.37.36	subscriber-mcc.....	2244
6.2.1.37.37	pgw-selection-candidate-list.....	2245
6.2.1.37.38	candidate.....	2246
6.2.1.37.39	plmn.....	2247
6.2.1.37.40	mcc.....	2247
6.2.1.37.41	port-list.....	2248
6.2.1.37.42	port.....	2248
6.2.1.37.43	prioritized-ip-address-list.....	2249
6.2.1.37.44	address.....	2250
6.2.1.37.45	qci-list.....	2251
6.2.1.37.46	qci.....	2251
6.2.1.37.47	rat-type-list.....	2252
6.2.1.37.48	rat-type.....	2253
6.2.1.37.49	slice-instance-list.....	2253
6.2.1.37.50	slice-instance.....	2254
6.2.1.37.51	slice-list.....	2254
6.2.1.37.52	slice.....	2255
6.2.1.37.53	slice-instances.....	2256
6.2.1.37.54	suspend.....	2257
6.2.1.37.55	tai-list.....	2257
6.2.1.37.56	smf-serving-area-list.....	2258
6.2.1.37.57	smf-serving-area.....	2259
6.2.1.37.58	tai-list.....	2259
6.2.1.37.59	tariff-change-list.....	2260
6.2.1.37.60	time.....	2260
6.2.1.37.61	ue-usage-type-list.....	2261
6.2.1.37.62	ue-usage-type.....	2261
6.2.1.38	MIPv4 profile command descriptions.....	2262
6.2.1.38.1	mipv4.....	2262
6.2.1.38.2	dmu.....	2263
6.2.1.38.3	ip-dscp.....	2263
6.2.1.38.4	mn-aaa-auth.....	2264
6.2.1.38.5	mn-ha-auth.....	2264
6.2.1.38.6	mn-ha-spi.....	2265
6.2.1.38.7	secret.....	2266
6.2.1.38.8	hash-algorithm.....	2266

6.2.1.38.9 timestamp-tolerance.....	2267
6.2.1.38.10 registration-lifetime.....	2267
6.2.1.38.11 revocation.....	2268
6.2.1.38.12 tunnel-mode.....	2269
6.2.1.39 MME group profile command descriptions.....	2269
6.2.1.39.1 mme-group.....	2269
6.2.1.39.2 peer.....	2270
6.2.1.40 N1 profile command descriptions.....	2271
6.2.1.40.1 n1-profile.....	2271
6.2.1.40.2 cause-code.....	2271
6.2.1.40.3 apn-shutdown.....	2272
6.2.1.40.4 5gsm-cause-code.....	2272
6.2.1.40.5 application-error-code.....	2272
6.2.1.40.6 http-status-code.....	2273
6.2.1.40.7 dnn-subscription-removed.....	2274
6.2.1.40.8 5gsm-cause-code.....	2274
6.2.1.40.9 gw-suspend.....	2275
6.2.1.40.10 5gsm-cause-code.....	2275
6.2.1.40.11 application-error-code.....	2275
6.2.1.40.12 http-status-code.....	2276
6.2.1.40.13 n1-t3591.....	2276
6.2.1.40.14 n1-t3592.....	2277
6.2.1.41 N2 peer profile command descriptions.....	2278
6.2.1.41.1 n2-peer.....	2278
6.2.1.41.2 peer.....	2279
6.2.1.41.3 plmn-list.....	2279
6.2.1.41.4 sctp-profile.....	2280
6.2.1.42 N7 attributes profile command descriptions.....	2280
6.2.1.42.1 n7-attributes-profile.....	2280
6.2.1.43 ip-domain.....	2281
6.2.1.44 nokia-smf-indicator-as-vsa.....	2282
6.2.1.45 Namf profile command descriptions.....	2282
6.2.1.45.1 namf-profile.....	2282
6.2.1.45.2 extended-buffering-wait-time.....	2283
6.2.1.45.3 http-status-code.....	2284
6.2.1.45.4 max-peer-failover.....	2285
6.2.1.45.5 retry-count.....	2285
6.2.1.45.6 transaction-timer.....	2286
6.2.1.46 Network trace profile command descriptions.....	2287
6.2.1.46.1 network-trace-profile.....	2287
6.2.1.46.2 http2-profile.....	2288
6.2.1.46.3 packet-broker-server.....	2288
6.2.1.46.4 server-path.....	2289
6.2.1.47 Node selection profile command descriptions.....	2290
6.2.1.47.1 node-selection.....	2290
6.2.1.47.2 abort.....	2290
6.2.1.47.3 begin.....	2291

6.2.1.47.4	client-profile.....	2291
6.2.1.47.5	default-action.....	2291
6.2.1.47.6	set-target-profile.....	2292
6.2.1.47.7	entry.....	2292
6.2.1.47.8	action.....	2293
6.2.1.47.9	set-target-profile.....	2293
6.2.1.47.10	match.....	2294
6.2.1.47.11	apn.....	2294
6.2.1.47.12	cell-id-list.....	2295
6.2.1.47.13	iwki5gs.....	2295
6.2.1.47.14	nri5gc.....	2296
6.2.1.47.15	rat-type.....	2297
6.2.1.47.16	serving-node.....	2297
6.2.1.47.17	slice.....	2298
6.2.1.47.18	tai-lai.....	2298
6.2.1.47.19	ue-usage-type.....	2299
6.2.1.47.20	commit.....	2299
6.2.1.47.21	group-label.....	2300
6.2.1.47.22	target-profile.....	2300
6.2.1.47.23	default-action.....	2301
6.2.1.47.24	reject.....	2301
6.2.1.47.25	entry.....	2301
6.2.1.47.26	action.....	2302
6.2.1.47.27	reject.....	2302
6.2.1.47.28	match.....	2303
6.2.1.47.29	apn.....	2303
6.2.1.47.30	dns-based.....	2304
6.2.1.47.31	group-label.....	2304
6.2.1.47.32	nrf-based.....	2305
6.2.1.47.33	slice.....	2305
6.2.1.48	NPCF profile command descriptions.....	2306
6.2.1.48.1	npcf.....	2306
6.2.1.48.2	fh-time-limit.....	2307
6.2.1.48.3	fh-volume-limit.....	2307
6.2.1.48.4	ip-domain-type.....	2308
6.2.1.48.5	n7-attributes-profile.....	2309
6.2.1.48.6	npcf-error-mapping-profile.....	2310
6.2.1.48.7	sbi-srvc-operation.....	2310
6.2.1.48.8	http-status-code.....	2311
6.2.1.48.9	http-status-code-3xx.....	2314
6.2.1.48.10	http-status-code-4xx.....	2316
6.2.1.48.11	http-status-code-5xx.....	2317
6.2.1.48.12	retry-count.....	2318
6.2.1.48.13	transaction-timer.....	2319
6.2.1.48.14	supported-features.....	2320
6.2.1.48.15	adc.....	2320
6.2.1.48.16	af-charging-id.....	2320

6.2.1.48.17	eps-fallback-report.....	2321
6.2.1.48.18	netloc.....	2321
6.2.1.48.19	pcscf-restoration-enhancement.....	2322
6.2.1.48.20	pdu-session-rel-cause.....	2322
6.2.1.48.21	policy-decision-error-handling.....	2323
6.2.1.48.22	policy-update-when-ue-suspends.....	2323
6.2.1.48.23	pra.....	2324
6.2.1.48.24	prov-af-signal-flow.....	2324
6.2.1.48.25	resp-based-session-rel.....	2324
6.2.1.48.26	session-rule-error-handling.....	2325
6.2.1.48.27	sponsored-connectivity.....	2325
6.2.1.48.28	support-2g3g.....	2326
6.2.1.48.29	tsc.....	2326
6.2.1.48.30	umc.....	2327
6.2.1.48.31	supported-vendor.....	2327
6.2.1.48.32	ul-rq-timer.....	2328
6.2.1.49	use-last-auth-ambr.....	2329
6.2.1.50	NSMF profile command descriptions.....	2329
6.2.1.50.1	nsmf-profile.....	2329
6.2.1.50.2	http-status-code.....	2330
6.2.1.50.3	max-peer-failover.....	2331
6.2.1.50.4	retry-count.....	2331
6.2.1.50.5	transaction-timer.....	2332
6.2.1.51	OAuth2.0 authorization profile command descriptions.....	2333
6.2.1.51.1	oauth2-profile.....	2333
6.2.1.51.2	oauth2-nrf-certificate.....	2333
6.2.1.51.3	validate-consumer-uuid.....	2334
6.2.1.52	Overload control profile command descriptions.....	2335
6.2.1.52.1	overload-control-profile.....	2335
6.2.1.52.2	exclude-arp-priority-level.....	2335
6.2.1.52.3	session-rejection-rate.....	2336
6.2.1.53	PCFP profile command descriptions.....	2336
6.2.1.53.1	pfcf.....	2336
6.2.1.53.2	pfcf-association-peer-list.....	2337
6.2.1.53.3	node-id.....	2337
6.2.1.53.4	bfd-enable.....	2338
6.2.1.53.5	enable-partial-failure.....	2338
6.2.1.53.6	path-hold-time.....	2339
6.2.1.53.7	peer-entity.....	2340
6.2.1.53.8	ue-ip-allocation.....	2341
6.2.1.53.9	pfcf-peer.....	2342
6.2.1.53.10	bfd-enable.....	2342
6.2.1.53.11	partial-failure.....	2343
6.2.1.53.12	up-ip-allocation.....	2343
6.2.1.53.13	pfcf-error-mapping-profile.....	2344
6.2.1.53.14	pfcf-to-5gsm.....	2344
6.2.1.53.15	pfcf-to-gtp.....	2345

6.2.1.53.16	pfcf-to-http.....	2346
6.2.1.53.17	pfcf-options.....	2346
6.2.1.53.18	exclude-lci.....	2347
6.2.1.53.19	exclude-oci.....	2347
6.2.1.53.20	pfcf-profile.....	2348
6.2.1.53.21	association-retry-timer.....	2348
6.2.1.53.22	disable-pdi-optimization.....	2349
6.2.1.53.23	ext-urr-id.....	2349
6.2.1.53.24	heart-beat.....	2350
6.2.1.53.25	ip-dscp.....	2351
6.2.1.53.26	ip-ttl.....	2351
6.2.1.53.27	message-retransmit.....	2352
6.2.1.53.28	pfcf-error-mapping-profile.....	2353
6.2.1.53.29	pfcf-u-tunnel.....	2354
6.2.1.53.30	re-attempt-timer.....	2354
6.2.1.53.31	smf-set-support.....	2355
6.2.1.53.32	pfcf-translate-profile.....	2355
6.2.1.53.33	field.....	2356
6.2.1.53.34	up-peer-list.....	2357
6.2.1.53.35	peer.....	2357
6.2.1.53.36	apn.....	2358
6.2.1.53.37	block.....	2359
6.2.1.53.38	ip-pool-list.....	2359
6.2.1.53.39	ip-pool.....	2359
6.2.1.53.40	nas-ip.....	2360
6.2.1.53.41	network-realm.....	2361
6.2.1.53.42	pco-option.....	2361
6.2.1.53.43	dns-server-ipv4.....	2362
6.2.1.53.44	primary.....	2362
6.2.1.53.45	secondary.....	2362
6.2.1.53.46	dns-server-ipv6.....	2363
6.2.1.53.47	primary.....	2363
6.2.1.53.48	secondary.....	2364
6.2.1.53.49	tai-lai-list.....	2364
6.2.1.53.50	ip-pool.....	2365
6.2.1.53.51	disable-eps-iwk.....	2365
6.2.1.53.52	group-label.....	2366
6.2.1.53.53	network-realm.....	2366
6.2.1.53.54	fragment-distribution-size.....	2367
6.2.1.53.55	ip-pool.....	2368
6.2.1.53.56	group-name.....	2368
6.2.1.53.57	hold-timer-sec.....	2369
6.2.1.53.58	ipv4-prefix.....	2369
6.2.1.53.59	ipv6-prefix.....	2370
6.2.1.53.60	network-determined.....	2372
6.2.1.53.61	pool-usage-threshold.....	2372
6.2.1.53.62	priority.....	2373

6.2.1.53.63	remote.....	2374
6.2.1.53.64	reserved-ip-list.....	2374
6.2.1.53.65	round-robin.....	2375
6.2.1.53.66	suppress-dotzero-broadcast-address.....	2376
6.2.1.53.67	suppress-network-broadcast-address.....	2376
6.2.1.53.68	use-prefixes-immediately.....	2377
6.2.1.53.69	ip-pool-distribution.....	2378
6.2.1.53.70	override.....	2378
6.2.1.53.71	slice.....	2379
6.2.1.53.72	upf-id.....	2380
6.2.1.53.73	upf-selection.....	2381
6.2.1.54	PMIPv6 profile command descriptions.....	2381
6.2.1.54.1	pmipv6.....	2381
6.2.1.54.2	ip-dscp.....	2382
6.2.1.54.3	ip-ttl.....	2382
6.2.1.54.4	keep-alive.....	2383
6.2.1.54.5	message-retransmit.....	2384
6.2.1.54.6	supported-vendor.....	2385
6.2.1.55	PCMD profile command descriptions.....	2385
6.2.1.55.1	pcmd.....	2385
6.2.1.55.2	data-report.....	2386
6.2.1.55.3	constant-flow-period.....	2386
6.2.1.55.4	destination.....	2387
6.2.1.55.5	generation.....	2388
6.2.1.55.6	heartbeat.....	2389
6.2.1.55.7	session-report.....	2390
6.2.1.55.8	verbosity.....	2390
6.2.1.56	P-CSCF server group profile command descriptions.....	2391
6.2.1.56.1	pcscf-group.....	2391
6.2.1.56.2	pcscf-server.....	2392
6.2.1.56.3	refresh-interval.....	2393
6.2.1.57	Policy options profile command descriptions.....	2393
6.2.1.57.1	policy-options.....	2393
6.2.1.57.2	abort.....	2394
6.2.1.57.3	action-rule-unit.....	2394
6.2.1.57.4	aa-notify-action.....	2394
6.2.1.57.5	app-profile-name.....	2395
6.2.1.57.6	characteristic.....	2396
6.2.1.57.7	collect-display-stats.....	2396
6.2.1.57.8	custom-pco.....	2397
6.2.1.57.9	data-driven-dedicated-bearer.....	2398
6.2.1.57.10	disable-ubr-signaling.....	2399
6.2.1.57.11	dnai.....	2399
6.2.1.57.12	dnai-psa-select.....	2400
6.2.1.57.13	dns-redirect.....	2400
6.2.1.57.14	exclude-src-dest-ports-in-ue-tft.....	2401
6.2.1.57.15	gy-filter-id-action.....	2401

6.2.1.57.16	modify-qos-default-bearer.....	2404
6.2.1.57.17	mss-adjust.....	2405
6.2.1.57.18	record-extension-visited-url.....	2405
6.2.1.57.19	rule-level-redirect.....	2406
6.2.1.57.20	sdf-level-policing.....	2406
6.2.1.57.21	transport-byte-collection.....	2407
6.2.1.57.22	ap-action-profile.....	2407
6.2.1.57.23	policy-action.....	2408
6.2.1.57.24	apn-rate-control.....	2409
6.2.1.57.25	additional-ul-rate-limit-for-exception-data.....	2409
6.2.1.57.26	dl-rate-limit.....	2410
6.2.1.57.27	exclude-ul-exception-reports.....	2410
6.2.1.57.28	time-unit.....	2411
6.2.1.57.29	ul-rate-control-in-pgw.....	2412
6.2.1.57.30	ul-rate-limit.....	2412
6.2.1.57.31	begin.....	2413
6.2.1.57.32	charging-rule-unit.....	2413
6.2.1.57.33	app-srvc-provider-id.....	2413
6.2.1.57.34	charging-method.....	2414
6.2.1.57.35	custom-online-sru.....	2415
6.2.1.57.36	default-rg.....	2415
6.2.1.57.37	metering-method.....	2416
6.2.1.57.38	monitoring-key.....	2417
6.2.1.57.39	rating-group.....	2417
6.2.1.57.40	rating-group-offline.....	2418
6.2.1.57.41	reporting-level.....	2419
6.2.1.57.42	selective-radius-accounting.....	2419
6.2.1.57.43	service-identifier.....	2420
6.2.1.57.44	service-identifier-offline.....	2420
6.2.1.57.45	sponsor-id.....	2421
6.2.1.57.46	wait-for-ocs-response.....	2422
6.2.1.57.47	commit.....	2422
6.2.1.57.48	diff.....	2423
6.2.1.57.49	dynamic-rule-to-internal-rule-map.....	2423
6.2.1.57.50	entry.....	2424
6.2.1.57.51	action.....	2425
6.2.1.57.52	direction.....	2425
6.2.1.57.53	local-port-range.....	2426
6.2.1.57.54	precedence.....	2426
6.2.1.57.55	protocol.....	2427
6.2.1.57.56	remote-ip.....	2428
6.2.1.57.57	remote-port-range.....	2429
6.2.1.57.58	signal-internal-rule-tft.....	2429
6.2.1.57.59	match.....	2430
6.2.1.57.60	arp.....	2430
6.2.1.57.61	direction.....	2431
6.2.1.57.62	local-port-range.....	2432

6.2.1.57.63	protocol.....	2432
6.2.1.57.64	qci.....	2433
6.2.1.57.65	remote-ip.....	2434
6.2.1.57.66	remote-port-range.....	2435
6.2.1.57.67	gx-avp-options.....	2435
6.2.1.57.68	exclude-3gpp-charging-characteristics.....	2436
6.2.1.57.69	exclude-an-trusted.....	2436
6.2.1.57.70	exclude-bearer-usage.....	2437
6.2.1.57.71	exclude-pdn-connection-id.....	2437
6.2.1.57.72	exclude-subscription-id-imsi.....	2438
6.2.1.57.73	exclude-subscription-id-msisdn.....	2438
6.2.1.57.74	exclude-subscription-id-nai.....	2439
6.2.1.57.75	exclude-twan-identifier.....	2439
6.2.1.57.76	exclude-ue-local-ip-address.....	2439
6.2.1.57.77	exclude-ue-udp-port.....	2440
6.2.1.57.78	exclude-user-equipment-info.....	2440
6.2.1.57.79	include-framed-ip-address-ccrt.....	2441
6.2.1.57.80	include-framed-ipv6-prefix-ccrt.....	2442
6.2.1.57.81	include-ipdomain-upf-id.....	2442
6.2.1.57.82	include-ue-tcp-source-port.....	2442
6.2.1.57.83	policy-rule.....	2443
6.2.1.57.84	policy-rule-base.....	2444
6.2.1.57.85	aa-adv-functions.....	2445
6.2.1.57.86	app-profile-name.....	2445
6.2.1.57.87	characteristic.....	2446
6.2.1.57.88	apn-ambr-attr.....	2446
6.2.1.57.89	aggr-rate.....	2447
6.2.1.57.90	charging-profile-id.....	2448
6.2.1.57.91	ded-bearer-on-app-start.....	2448
6.2.1.57.92	disable-bearer-bind-verification.....	2449
6.2.1.57.93	policy-rule.....	2449
6.2.1.57.94	session-level-svc-steering.....	2450
6.2.1.57.95	policy-rule-unit.....	2450
6.2.1.57.96	aa-advanced-functions.....	2451
6.2.1.57.97	flow-description.....	2451
6.2.1.57.98	direction.....	2452
6.2.1.57.99	match.....	2453
6.2.1.57.100	aa-charging-group.....	2453
6.2.1.57.101	aa-inactive-charging-group.....	2454
6.2.1.57.102	application-id.....	2454
6.2.1.57.103	dns-fqdn-list.....	2455
6.2.1.57.104	dns-snooping.....	2455
6.2.1.57.105	fqdn-group.....	2456
6.2.1.57.106	local-ip.....	2457
6.2.1.57.107	local-port.....	2458
6.2.1.57.108	local-port-list.....	2459
6.2.1.57.109	protocol.....	2460

6.2.1.57.110	remote-ip.....	2460
6.2.1.57.111	remote-ip-address-group.....	2461
6.2.1.57.112	remote-ip-list.....	2462
6.2.1.57.113	remote-port.....	2463
6.2.1.57.114	remote-port-list.....	2463
6.2.1.57.115	tos-tc.....	2464
6.2.1.57.116	flow-gate-status.....	2465
6.2.1.57.117	mute-notification.....	2468
6.2.1.57.118	pdr-id.....	2468
6.2.1.57.119	qos.....	2469
6.2.1.57.120	down-link.....	2469
6.2.1.57.121	up-link.....	2470
6.2.1.57.122	service-class-indicator.....	2470
6.2.1.57.123	shallow-inspection-only.....	2471
6.2.1.57.124	svc-steering.....	2471
6.2.1.57.125	qci-policy.....	2472
6.2.1.57.126	qci.....	2473
6.2.1.57.127	dscp.....	2473
6.2.1.57.128	dscp-preserve.....	2474
6.2.1.57.129	fc.....	2475
6.2.1.57.130	profile.....	2476
6.2.1.57.131	qos-rule-unit.....	2477
6.2.1.57.132	average-window.....	2477
6.2.1.57.133	down-link.....	2478
6.2.1.57.134	max-data-burst-vol.....	2479
6.2.1.57.135	priority-level.....	2479
6.2.1.57.136	qer-id.....	2480
6.2.1.57.137	qfi.....	2481
6.2.1.57.138	qnc.....	2481
6.2.1.57.139	rqi.....	2482
6.2.1.57.140	sdf-policing.....	2482
6.2.1.57.141	service-class-indicator.....	2483
6.2.1.57.142	tos-tc.....	2483
6.2.1.57.143	up-link.....	2484
6.2.1.57.144	service-class-indicator.....	2485
6.2.1.57.145	service-class-value.....	2485
6.2.1.57.146	sru-list.....	2486
6.2.1.57.147	stat-rule-unit.....	2486
6.2.1.57.148	stat-rule-unit.....	2487
6.2.1.57.149	urr-id.....	2487
6.2.1.57.150	steering-profile.....	2488
6.2.1.57.151	traffic-redirect.....	2488
6.2.1.57.152	traffic-hash-profile.....	2489
6.2.1.57.153	traffic-distribution.....	2489
6.2.1.57.154	access-lte-percentage.....	2490
6.2.1.57.155	access-xdsl-percentage.....	2490
6.2.1.57.156	traffic-redirect-profile.....	2491

6.2.1.57.157	bfd-enable.....	2491
6.2.1.57.158	destination-nat.....	2492
6.2.1.57.159	destination-ipv4.....	2492
6.2.1.57.160	mg-group.....	2493
6.2.1.57.161	router.....	2493
6.2.1.57.162	traffic-redirect-target.....	2494
6.2.1.57.163	downlink-next-hop-ipv4.....	2494
6.2.1.57.164	downlink-next-hop-ipv6.....	2495
6.2.1.57.165	uplink-next-hop-ipv4.....	2496
6.2.1.57.166	uplink-next-hop-ipv6.....	2496
6.2.1.57.167	router.....	2497
6.2.1.57.168	threshold-target.....	2498
6.2.1.57.169	trigger-rule-unit.....	2498
6.2.1.57.170	access-type.....	2499
6.2.1.57.171	tai-lai-list.....	2500
6.2.1.57.172	urr-profile.....	2501
6.2.1.57.173	discard-on-fpt.....	2501
6.2.1.57.174	usage-monitoring-actions.....	2502
6.2.1.57.175	monitoring-action.....	2502
6.2.1.57.176	after-fill-delta-percent.....	2503
6.2.1.57.177	usage-monitoring-info.....	2503
6.2.1.57.178	monitoring-key.....	2504
6.2.1.57.179	fixed-access-fill-out.....	2504
6.2.1.57.180	fixed-access-sustained-rate.....	2505
6.2.1.57.181	quota-type.....	2506
6.2.1.57.182	traffic-hash.....	2506
6.2.1.57.183	validate-aa-charging-grp.....	2507
6.2.1.58	PPI mapping profile command descriptions.....	2507
6.2.1.58.1	ppi-mapping-profile.....	2507
6.2.1.58.2	dscp-to-ppi.....	2508
6.2.1.59	QFI mapping profile command descriptions.....	2509
6.2.1.59.1	qfi-mapping-profile.....	2509
6.2.1.59.2	5qi-as-qfi.....	2509
6.2.1.59.3	dynamic-mapping.....	2510
6.2.1.59.4	average-window.....	2510
6.2.1.59.5	max-data-burst-vol.....	2511
6.2.1.59.6	priority-level.....	2511
6.2.1.59.7	qnc.....	2512
6.2.1.60	QoS profile command descriptions.....	2512
6.2.1.60.1	qos-5g-profile.....	2512
6.2.1.60.2	5qi.....	2513
6.2.1.60.3	arp.....	2513
6.2.1.60.4	dl-ambr.....	2514
6.2.1.60.5	ul-ambr.....	2514
6.2.1.60.6	qos-profile.....	2515
6.2.1.60.7	arp.....	2515
6.2.1.60.8	dl-ambr.....	2516

6.2.1.60.9	qci.....	2516
6.2.1.60.10	ul-ambr.....	2517
6.2.1.61	RADIUS profile command descriptions.....	2517
6.2.1.61.1	radius.....	2517
6.2.1.61.2	acct-retry-count.....	2518
6.2.1.61.3	acct-retry-timeout.....	2519
6.2.1.61.4	apn-acct-on-off-initial-interval.....	2519
6.2.1.61.5	apn-acct-on-retry-interval.....	2520
6.2.1.61.6	auth-probe-interval.....	2521
6.2.1.61.7	auth-retry-count.....	2522
6.2.1.61.8	auth-retry-timeout.....	2523
6.2.1.61.9	deadtime.....	2523
6.2.1.61.10	delay-tolerance.....	2524
6.2.1.61.11	max-message-rate.....	2525
6.2.1.61.12	max-peer-reselections.....	2526
6.2.1.61.13	max-queue-size.....	2526
6.2.1.61.14	message-type-drop-order.....	2527
6.2.1.61.15	nas-port-type.....	2528
6.2.1.61.16	throttle-emergency.....	2528
6.2.1.61.17	radius-group.....	2529
6.2.1.61.18	acct-server-port.....	2529
6.2.1.61.19	auth-server-port.....	2530
6.2.1.61.20	failover-threshold.....	2531
6.2.1.61.21	failure-detection-time.....	2531
6.2.1.61.22	interface.....	2532
6.2.1.61.23	interim-update-interval.....	2533
6.2.1.61.24	limit-user-name-length.....	2533
6.2.1.61.25	peer.....	2534
6.2.1.61.26	priority.....	2535
6.2.1.61.27	radius-profile.....	2535
6.2.1.61.28	secret.....	2536
6.2.1.61.29	single-server-per-ue.....	2537
6.2.1.61.30	radius-avp-options.....	2538
6.2.1.61.31	acct-interim-avp-options.....	2538
6.2.1.61.32	exclude-3gpp-charging-characteristics.....	2538
6.2.1.61.33	exclude-3gpp-charging-ld.....	2539
6.2.1.61.34	exclude-3gpp-charging-ld.....	2539
6.2.1.61.35	exclude-3gpp-ggsn-address.....	2540
6.2.1.61.36	exclude-3gpp-ggsn-ipv6-address.....	2540
6.2.1.61.37	exclude-3gpp-ggsn-mcc-mnc.....	2541
6.2.1.61.38	exclude-3gpp-gprs-qos-negotiated-profile.....	2541
6.2.1.61.39	exclude-3gpp-imeisv.....	2542
6.2.1.61.40	exclude-3gpp-imsi.....	2542
6.2.1.61.41	exclude-3gpp-imsi.....	2543
6.2.1.61.42	exclude-3gpp-imsi-mcc-mnc.....	2544
6.2.1.61.43	exclude-3gpp-ms-timezone.....	2544
6.2.1.61.44	exclude-3gpp-nsapi.....	2545

6.2.1.61.45	exclude-3gpp-pdp-type.....	2545
6.2.1.61.46	exclude-3gpp-rat-type.....	2546
6.2.1.61.47	exclude-3gpp-selection-mode.....	2546
6.2.1.61.48	exclude-3gpp-session-stop-indicator.....	2547
6.2.1.61.49	exclude-3gpp-sgsn-address.....	2547
6.2.1.61.50	exclude-3gpp-sgsn-address.....	2548
6.2.1.61.51	exclude-3gpp-sgsn-ipv6-address.....	2548
6.2.1.61.52	exclude-3gpp-sgsn-mcc-mnc.....	2549
6.2.1.61.53	exclude-3gpp-user-location-info.....	2549
6.2.1.61.54	exclude-acct-authentic.....	2550
6.2.1.61.55	exclude-acct-delay-time.....	2550
6.2.1.61.56	exclude-acct-input-gigawords.....	2551
6.2.1.61.57	exclude-acct-input-octets.....	2551
6.2.1.61.58	exclude-acct-input-packets.....	2552
6.2.1.61.59	exclude-acct-output-gigawords.....	2552
6.2.1.61.60	exclude-acct-output-octets.....	2553
6.2.1.61.61	exclude-acct-output-packets.....	2553
6.2.1.61.62	exclude-acct-session-id.....	2554
6.2.1.61.63	exclude-acct-session-time.....	2554
6.2.1.61.64	exclude-called-station-id.....	2555
6.2.1.61.65	exclude-calling-station-id.....	2555
6.2.1.61.66	exclude-class.....	2556
6.2.1.61.67	exclude-framed-interface-id.....	2556
6.2.1.61.68	exclude-framed-ip-address.....	2557
6.2.1.61.69	exclude-framed-ip-netmask.....	2557
6.2.1.61.70	exclude-framed-ipv6-prefix.....	2558
6.2.1.61.71	exclude-framed-protocol.....	2558
6.2.1.61.72	exclude-nas-ip-address.....	2558
6.2.1.61.73	exclude-nas-ipv6-address.....	2559
6.2.1.61.74	exclude-nas-port-type.....	2559
6.2.1.61.75	exclude-service-type.....	2560
6.2.1.61.76	exclude-user-name.....	2560
6.2.1.61.77	include-3gpp-ip-addr-pool-info.....	2561
6.2.1.61.78	include-3gpp-supported-features.....	2561
6.2.1.61.79	include-event-timestamp.....	2562
6.2.1.61.80	include-nas-identifier.....	2562
6.2.1.61.81	include-nas-port-id.....	2563
6.2.1.61.82	disconnect-ack-avp-options.....	2563
6.2.1.61.83	exclude-acct-terminate-cause.....	2564
6.2.1.61.84	exclude-event-timestamp.....	2564
6.2.1.61.85	exclude-proxy-state.....	2565
6.2.1.61.86	disconnect-nak-avp-options.....	2565
6.2.1.61.87	exclude-error-cause.....	2565
6.2.1.61.88	exclude-proxy-state.....	2566
6.2.1.61.89	exclude-3gpp-allocate-ip-type.....	2566
6.2.1.61.90	include-acct-session-id-in-on-off-messages.....	2567
6.2.1.61.91	include-alc-sla-prof-str.....	2567

6.2.1.61.92	include-alc-subsc-id-str.....	2568
6.2.1.61.93	include-alc-subsc-prof-str.....	2568
6.2.1.61.94	include-called-station-id-apn-in-on-off-messages.....	2569
6.2.1.61.95	use-acct-interim-avp-options.....	2569
6.2.1.61.96	use-disconnect-ack-avp-options.....	2570
6.2.1.61.97	use-disconnect-nak-avp-options.....	2570
6.2.1.61.98	fh-session-no-response.....	2571
6.2.1.61.99	fh-time-limit.....	2571
6.2.1.61.100	fh-volume-limit.....	2572
6.2.1.61.101	ignore-acct-response.....	2572
6.2.1.61.102	server-type.....	2573
6.2.1.61.103	supported-features.....	2574
6.2.1.61.104	e-sess-ambr.....	2575
6.2.1.61.105	transaction-based-load-balancing.....	2575
6.2.1.61.106	via-up-function.....	2576
6.2.1.62	S2b error mapping profile command descriptions.....	2577
6.2.1.62.1	s2b-error-mapping-profile.....	2578
6.2.1.62.2	s2b-cause-code.....	2578
6.2.1.62.3	backoff-timer.....	2578
6.2.1.62.4	gtpv2-result-code-trigger-ipsec-lockout.....	2580
6.2.1.62.5	swu-notify.....	2581
6.2.1.62.6	swu-send-info.....	2581
6.2.1.63	S6b interface profile command descriptions.....	2582
6.2.1.63.1	s6b-profile.....	2582
6.2.1.63.2	drmp.....	2582
6.2.1.63.3	ehrpd-initial-attach.....	2583
6.2.1.63.4	ehrpd-initiated-detach.....	2583
6.2.1.63.5	ehrpd-lte-handover.....	2584
6.2.1.63.6	ehrpd-wifi-handover.....	2584
6.2.1.63.7	enable.....	2585
6.2.1.63.8	eutran-initial-attach.....	2585
6.2.1.63.9	eutran-initiated-detach.....	2586
6.2.1.63.10	lte-wifi-handover.....	2587
6.2.1.63.11	pgw-initiated-disconnect.....	2587
6.2.1.63.12	reauth-initiated.....	2588
6.2.1.63.13	wifi-initial-attach.....	2588
6.2.1.63.14	wifi-initiated-detach.....	2589
6.2.1.63.15	fh-session-no-response.....	2589
6.2.1.63.16	fh-time-limit.....	2590
6.2.1.63.17	message.....	2591
6.2.1.63.18	result-code.....	2591
6.2.1.63.19	result-code-3xxx.....	2592
6.2.1.63.20	result-code-4xxx.....	2593
6.2.1.63.21	result-code-5xxx.....	2594
6.2.1.63.22	s6b-avp-options.....	2594
6.2.1.63.23	exclude-emergency-service.....	2595
6.2.1.63.24	exclude-idle-timeout.....	2595

6.2.1.63.25	exclude-MIP6-agent-info.....	2595
6.2.1.63.26	exclude-MIP6-feature-vector.....	2596
6.2.1.63.27	exclude-rat-type.....	2596
6.2.1.63.28	exclude-supported-feature.....	2597
6.2.1.63.29	exclude-visited-network-identifier.....	2597
6.2.1.63.30	include-3gpp-allocate-ip-type.....	2597
6.2.1.63.31	include-3gpp-charging-characteristics.....	2598
6.2.1.63.32	include-3gpp-charging-id.....	2598
6.2.1.63.33	include-3gpp-ggsn-address.....	2599
6.2.1.63.34	include-3gpp-ggsn-ipv6-address.....	2599
6.2.1.63.35	include-3gpp-ggsn-mcc-mnc.....	2600
6.2.1.63.36	include-3gpp-gprs-negotiated-qos-profile.....	2600
6.2.1.63.37	include-3gpp-imeisv.....	2601
6.2.1.63.38	include-3gpp-imsi.....	2601
6.2.1.63.39	include-3gpp-ms-timezone.....	2602
6.2.1.63.40	include-3gpp-nsapi.....	2602
6.2.1.63.41	include-3gpp-pdp-type.....	2602
6.2.1.63.42	include-3gpp-rat-type.....	2603
6.2.1.63.43	include-3gpp-selection-mode.....	2603
6.2.1.63.44	include-3gpp-sgsn-address.....	2604
6.2.1.63.45	include-3gpp-sgsn-ipv6-address.....	2604
6.2.1.63.46	include-3gpp-sgsn-mcc-mnc.....	2605
6.2.1.63.47	include-3gpp-user-location-info.....	2605
6.2.1.63.48	include-auth-session-state.....	2606
6.2.1.63.49	include-framed-ip-address.....	2606
6.2.1.63.50	include-framed-ip-netmask.....	2606
6.2.1.63.51	include-framed-ipv6-prefix.....	2607
6.2.1.63.52	supported-features.....	2607
6.2.1.63.53	clear-m-bit.....	2608
6.2.1.63.54	wlan-pcscf-restoration.....	2608
6.2.1.64	SBI error mapping profile command descriptions.....	2609
6.2.1.64.1	sbi-error-mapping-profile.....	2609
6.2.1.64.2	http-to-5gsm.....	2609
6.2.1.64.3	http-to-gtp.....	2610
6.2.1.64.4	http-to-http.....	2611
6.2.1.65	SCTP profile command descriptions.....	2612
6.2.1.65.1	sctp.....	2612
6.2.1.65.2	adv-receiver-window-credit.....	2612
6.2.1.65.3	assoc-max-retrans.....	2613
6.2.1.65.4	cookie-life-timer.....	2614
6.2.1.65.5	heartbeat-interval.....	2614
6.2.1.65.6	initial-max-retrans.....	2615
6.2.1.65.7	mtu-size.....	2616
6.2.1.65.8	path-max-retrans.....	2616
6.2.1.65.9	rto-init-max-timer.....	2617
6.2.1.65.10	rto-init-timer.....	2618
6.2.1.65.11	rto-max-timer.....	2618

6.2.1.65.12	rto-min-timer.....	2619
6.2.1.65.13	sack-frequency.....	2620
6.2.1.65.14	sack-period.....	2620
6.2.1.66	Sd profile command descriptions.....	2621
6.2.1.66.1	sd.....	2621
6.2.1.66.2	pending-transaction-retry-count.....	2621
6.2.1.66.3	pending-transaction-timer.....	2622
6.2.1.66.4	supported-features.....	2623
6.2.1.66.5	cno-uli.....	2623
6.2.1.66.6	pending-transaction.....	2624
6.2.1.67	SGSN profile command descriptions.....	2624
6.2.1.67.1	sgsn-group.....	2624
6.2.1.67.2	peer.....	2625
6.2.1.68	Signaling rate limit profile command descriptions.....	2626
6.2.1.68.1	sig-rate-limit-profile.....	2626
6.2.1.68.2	delay-tolerance.....	2626
6.2.1.68.3	max-burst-size.....	2627
6.2.1.68.4	max-message-rate.....	2627
6.2.1.68.5	queue-threshold.....	2628
6.2.1.68.6	throttle-emergency.....	2629
6.2.1.69	SLAAC host profile command descriptions.....	2629
6.2.1.69.1	slaac-host.....	2629
6.2.1.69.2	router-advertisements.....	2630
6.2.1.69.3	force-unicast-ip.....	2630
6.2.1.69.4	include-rdnss.....	2631
6.2.1.69.5	max-rtr-adv-interval.....	2631
6.2.1.69.6	min-rtr-adv-interval.....	2632
6.2.1.69.7	mtu.....	2632
6.2.1.69.8	preferred-lifetime.....	2633
6.2.1.69.9	rdnss-lifetime.....	2633
6.2.1.69.10	reachable-time.....	2634
6.2.1.69.11	retransmit-time.....	2634
6.2.1.69.12	router-lifetime.....	2635
6.2.1.69.13	valid-lifetime.....	2635
6.2.1.70	STa profile command descriptions.....	2636
6.2.1.70.1	sta-profile.....	2636
6.2.1.70.2	diameter-optional-avps.....	2637
6.2.1.70.3	exclude-der-flags-avp.....	2637
6.2.1.71	Subscriber mapping profile command descriptions.....	2637
6.2.1.71.1	subscriber-mapping.....	2637
6.2.1.71.2	address.....	2638
6.2.1.72	SWm error mapping profile command descriptions.....	2639
6.2.1.72.1	swm-error-mapping-profile.....	2639
6.2.1.72.2	swm-error-code.....	2639
6.2.1.72.3	diameter-result-code-trigger-lockout.....	2640
6.2.1.72.4	swu-notify.....	2640
6.2.1.72.5	swu-send-info.....	2641

6.2.1.73 SWm profile command descriptions.....	2641
6.2.1.73.1 swm-profile.....	2642
6.2.1.73.2 swm-avp-options.....	2642
6.2.1.73.3 exclude-ue-local-ip-address.....	2642
6.2.1.73.4 include-aar-flags.....	2643
6.2.1.73.5 include-terminal-information.....	2643
6.2.1.74 SWw profile command descriptions.....	2644
6.2.1.74.1 sww-radius-profile.....	2644
6.2.1.74.2 access-accept-optional-attributes.....	2644
6.2.1.74.3 include-3gpp-charging-characteristics.....	2645
6.2.1.74.4 include-3gpp-ggsn-address.....	2645
6.2.1.74.5 include-3gpp-gprs-negotiated-qos-profile.....	2645
6.2.1.74.6 include-3gpp-imsi.....	2646
6.2.1.74.7 include-3gpp-msisdn.....	2646
6.2.1.74.8 include-3gpp-pdp-type.....	2647
6.2.1.74.9 include-alc-mgw-interface-type.....	2647
6.2.1.74.10 include-alc-msisdn.....	2648
6.2.1.74.11 include-alc-subsc-id-str.....	2648
6.2.1.74.12 include-alc-wlan-apn-name.....	2649
6.2.1.74.13 acct-triggered-mobility.....	2649
6.2.1.74.14 disconnect-optional-attributes.....	2649
6.2.1.74.15 exclude-framed-ip-address.....	2650
6.2.1.74.16 exclude-user-name.....	2650
6.2.1.74.17 disconnect-port.....	2651
6.2.1.74.18 fast-inter-ap-handover.....	2651
6.2.1.74.19 retry-count.....	2652
6.2.1.74.20 retry-timeout.....	2652
6.2.1.74.21 sww-tunnel-profile.....	2653
6.2.1.74.22 ap-mac-discovery.....	2653
6.2.1.74.23 default-access-type.....	2654
6.2.1.74.24 default-apn.....	2654
6.2.1.74.25 dhcpv4-server.....	2655
6.2.1.74.26 qtag.....	2655
6.2.1.74.27 slaac-host.....	2657
6.2.1.74.28 ssid.....	2657
6.2.1.74.29 ue-cache-time.....	2658
6.2.1.75 T6c profile command descriptions.....	2659
6.2.1.75.1 t6c-profile.....	2659
6.2.1.75.2 include-apn.....	2659
6.2.1.75.3 include-ip.....	2659
6.2.1.75.4 include-rat-type.....	2660
6.2.1.76 Tracking and location area identity profile commands descriptions.....	2660
6.2.1.76.1 tai-lai-list.....	2660
6.2.1.76.2 lai.....	2661
6.2.1.76.3 tai.....	2662
6.2.1.77 SDM-UDM profile command descriptions.....	2663
6.2.1.77.1 udm-sdm-profile.....	2663

6.2.1.77.2 fh-session-no-response.....	2663
6.2.1.77.3 block-timer.....	2664
6.2.1.77.4 continue-timer.....	2665
6.2.1.77.5 create-subscription.....	2666
6.2.1.77.6 http-status-code.....	2666
6.2.1.77.7 retry-count.....	2667
6.2.1.77.8 transaction-timer.....	2668
6.2.1.77.9 get-sm-subscription-data.....	2668
6.2.1.77.10 http-status-code.....	2669
6.2.1.77.11 exclude-plmn-id.....	2670
6.2.1.77.12 retry-count.....	2671
6.2.1.77.13 send-sd-with-reserved-value.....	2671
6.2.1.77.14 transaction-timer.....	2672
6.2.1.77.15 http-status-code.....	2672
6.2.1.77.16 max-peer-failover.....	2674
6.2.1.77.17 retry-count.....	2674
6.2.1.77.18 sbi-error-mapping-profile.....	2675
6.2.1.77.19 transaction-timer.....	2675
6.2.1.78 UDM-UECM profile command descriptions.....	2676
6.2.1.78.1 udm-uecm-profile.....	2676
6.2.1.78.2 register-smf.....	2677
6.2.1.78.3 retry-count.....	2677
6.2.1.78.4 transaction-timer.....	2677
6.2.1.79 sbi-error-mapping-profile.....	2678
6.2.1.80 UDP tunnel profile command descriptions.....	2679
6.2.1.80.1 udp-tunnel-group.....	2679
6.2.1.80.2 peer.....	2679
6.2.1.80.3 destination-port.....	2680
6.2.1.81 UMTS QoS policy profile commands descriptions.....	2681
6.2.1.81.1 umts-qos-policy.....	2681
6.2.1.81.2 background.....	2681
6.2.1.81.3 conversational.....	2682
6.2.1.81.4 interactive.....	2683
6.2.1.81.5 operator-qci.....	2683
6.2.1.81.6 streaming.....	2685
6.2.1.82 UP security policy profile commands descriptions.....	2686
6.2.1.82.1 up-security-policy.....	2686
6.2.1.82.2 override-udm-policy.....	2686
6.2.1.82.3 up-confidentiality.....	2687
6.2.1.82.4 up-integrity.....	2688
6.2.1.83 V-SMF profile command descriptions.....	2688
6.2.1.83.1 vsmf-profile.....	2688
6.2.1.83.2 allow-always-on.....	2689
6.2.1.83.3 up-confidentiality.....	2689
6.2.1.83.4 up-integrity.....	2690
6.2.1.83.5 vplmn-qos.....	2691
6.2.1.84 WLAN emergency profile command descriptions.....	2691

6.2.1.84.1	wlan-emergency.....	2691
6.2.1.84.2	ecd-table.....	2692
6.2.1.84.3	3gpp-ho-emergency-pgw-id.....	2693
6.2.1.84.4	apn.....	2693
6.2.1.84.5	apn-ambr-dl.....	2694
6.2.1.84.6	apn-ambr-ul.....	2694
6.2.1.84.7	pgw-id.....	2695
6.2.1.84.8	qos.....	2695
6.2.1.84.9	arp.....	2696
6.2.1.84.10	qci.....	2696
6.2.1.84.11	emergency-call-numbers-ikev2.....	2697
6.2.1.84.12	country-code.....	2697
6.2.1.84.13	digits.....	2698
6.2.1.84.14	ip-range-country-file-format.....	2699
6.2.1.84.15	idr-emergency-apn-name-list.....	2700
6.2.1.84.16	unauthenticated-uiccless-support.....	2701
6.2.1.85	WPS profile command descriptions.....	2702
6.2.1.85.1	wps-profile.....	2702
6.2.1.85.2	arp.....	2702
6.2.1.85.3	message-priority.....	2703
6.2.1.85.4	all.....	2703
6.2.1.85.5	gtpc.....	2704
6.2.1.85.6	pfcpc.....	2705
6.2.1.85.7	sbi.....	2705
6.2.1.85.8	up-dscp-marking.....	2706
6.2.1.85.9	down-link.....	2706
6.2.1.85.10	up-link.....	2707
6.2.2	Profile show command descriptions.....	2708
6.2.2.1	profile.....	2708
6.2.2.2	apn-rate-control.....	2708
6.2.2.3	call-insight.....	2710
6.2.2.4	all-ues.....	2710
6.2.2.5	ue.....	2712
6.2.2.6	charging.....	2714
6.2.2.7	ccfh-profile.....	2714
6.2.2.8	cdr-avp-options.....	2717
6.2.2.9	chf-profile.....	2723
6.2.2.10	epdg-charging.....	2726
6.2.2.11	event-triggers-fbc.....	2731
6.2.2.12	gy-avp-options.....	2736
6.2.2.13	pgw-charging.....	2741
6.2.2.14	sgw-charging.....	2747
6.2.2.15	smf-charging.....	2750
6.2.2.16	smf-profile.....	2752
6.2.2.17	tdf-charging.....	2752
6.2.2.18	twag-charging.....	2755
6.2.2.19	upf-acct-profile.....	2756

6.2.2.20	dcca.....	2757
6.2.2.21	dhcp-server-group.....	2767
6.2.2.22	dhcpv4-server.....	2767
6.2.2.23	diameter.....	2769
6.2.2.24	diameter-peer.....	2771
6.2.2.25	diameter-error-mapping-profile.....	2774
6.2.2.26	diameter-peer-group-list.....	2775
6.2.2.27	dns-error-mapping-profile.....	2777
6.2.2.28	dns-profile.....	2778
6.2.2.29	dns-redirect-server.....	2780
6.2.2.30	emergency-profile.....	2780
6.2.2.31	fa-ha-peer-list.....	2782
6.2.2.32	fqdn-group.....	2782
6.2.2.33	gtp.....	2784
6.2.2.34	gtp-prime-group.....	2789
6.2.2.35	gx.....	2793
6.2.2.36	http2.....	2801
6.2.2.37	ipsec-profile.....	2803
6.2.2.38	list.....	2822
6.2.2.39	aaa-host-realm.....	2822
6.2.2.40	apn-list.....	2823
6.2.2.41	arp-list.....	2824
6.2.2.42	broadcast-plmn-list.....	2826
6.2.2.43	cell-id-list.....	2829
6.2.2.44	fqdn-list.....	2830
6.2.2.45	nf-id-list.....	2832
6.2.2.46	oi-override-mapping.....	2833
6.2.2.47	pgw-selection-candidate-list.....	2835
6.2.2.48	plmn-list.....	2835
6.2.2.49	port-list.....	2836
6.2.2.50	prioritized-ip-address-list.....	2837
6.2.2.51	qci-list.....	2838
6.2.2.52	slice-instance-list.....	2839
6.2.2.53	slice-list.....	2841
6.2.2.54	ue-usage-type-list.....	2842
6.2.2.55	mipv4.....	2843
6.2.2.56	mme-group.....	2844
6.2.2.57	n1-profile.....	2846
6.2.2.58	n2-peer.....	2848
6.2.2.59	network-trace-profile.....	2850
6.2.2.60	node-selection.....	2851
6.2.2.61	client-profile.....	2851
6.2.2.62	group-label.....	2854
6.2.2.63	target-profile.....	2855
6.2.2.64	npcf.....	2857
6.2.2.65	overload-control-profile.....	2860
6.2.2.66	pcmd.....	2861

6.2.2.67	pcscf-group.....	2863
6.2.2.68	peer-list.....	2864
6.2.2.69	pfcf.....	2865
6.2.2.70	pfcf-association-peer-list.....	2865
6.2.2.71	pfcf-error-mapping-profile.....	2866
6.2.2.72	pfcf-profile.....	2867
6.2.2.73	pfcf-translate-profile.....	2870
6.2.2.74	up-peer-list.....	2871
6.2.2.75	pmipv6.....	2876
6.2.2.76	policy-options.....	2876
6.2.2.77	action-rule-unit.....	2878
6.2.2.78	ap-action-profile.....	2879
6.2.2.79	charging-rule-unit.....	2880
6.2.2.80	gx-avp-options.....	2881
6.2.2.81	policy-rule.....	2883
6.2.2.82	policy-rule-base.....	2884
6.2.2.83	policy-rule-unit.....	2885
6.2.2.84	qci-policy.....	2886
6.2.2.85	qos-rule-unit.....	2888
6.2.2.86	stat-rule-unit.....	2889
6.2.2.87	steering-profile.....	2890
6.2.2.88	traffic-redirect-profile.....	2891
6.2.2.89	trigger-rule-unit.....	2892
6.2.2.90	ppi-mapping-profile.....	2893
6.2.2.91	qfi-mapping-profile.....	2894
6.2.2.92	qos-profile.....	2894
6.2.2.93	radius.....	2895
6.2.2.94	radius-group.....	2896
6.2.2.95	s2b-error-mapping-profile.....	2898
6.2.2.96	s6b-profile.....	2900
6.2.2.97	sctp-profile.....	2903
6.2.2.98	sd.....	2904
6.2.2.99	sgsn-group.....	2905
6.2.2.100	sig-rate-limit-profile.....	2906
6.2.2.101	slaac-host.....	2907
6.2.2.102	swm-error-mapping-profile.....	2908
6.2.2.103	sww-radius-profile.....	2910
6.2.2.104	sww-tunnel-profile.....	2911
6.2.2.105	tai-lai-list.....	2912
6.2.2.106	udm-sdm-profile.....	2914
6.2.2.107	udm-uecm-profile.....	2914
6.2.2.108	udp-tunnel-group.....	2915
6.2.2.109	umts-qos-policy.....	2917
6.2.2.110	up-security-policy.....	2919
6.2.2.111	wlan-emergency.....	2920
6.2.2.112	wps-profile.....	2921

7 IP tunnel command reference.....	2923
7.1 IP tunnel command hierarchies.....	2923
7.1.1 IP tunnels configuration command hierarchies.....	2923
7.1.1.1 Hardware command hierarchy.....	2923
7.1.1.2 ISA command hierarchy.....	2923
7.1.1.3 IPSec command hierarchies.....	2924
7.1.1.3.1 Certificate profile command hierarchy.....	2924
7.1.1.3.2 Internet Key Exchange (IKE) policy command hierarchy.....	2925
7.1.1.3.3 Internet Key Exchange (IKE) transform command hierarchy.....	2925
7.1.1.3.4 IPSec transform command hierarchy.....	2926
7.1.1.3.5 RADIUS accounting policy command hierarchy.....	2926
7.1.1.3.6 IPSec static security association command hierarchy.....	2927
7.1.1.3.7 Trust anchor profile/TS list command hierarchy.....	2927
7.1.1.3.8 Tunnel template command hierarchy.....	2928
7.1.2 Services configuration command hierarchies.....	2928
7.1.2.1 IES service interface command hierarchy.....	2928
7.1.2.2 VPRN service command hierarchy.....	2930
7.1.2.2.1 VPRN service IPSec command hierarchy.....	2930
7.1.2.2.2 VPRN service interface command hierarchy.....	2931
7.1.2.2.3 SAP IPSec gateway command hierarchy.....	2932
7.1.2.2.4 SAP IPSec tunnel command hierarchy.....	2933
7.1.2.3 CMPv2 command hierarchies.....	2934
7.1.3 IP tunnels show command hierarchy.....	2935
7.1.4 IP tunnels debug command hierarchy.....	2936
7.2 IP tunnel command descriptions.....	2936
7.2.1 IP tunnels configuration command descriptions.....	2936
7.2.1.1 Generic command descriptions.....	2937
7.2.1.1.1 description.....	2937
7.2.1.1.2 shutdown.....	2937
7.2.1.2 Hardware command descriptions.....	2938
7.2.1.2.1 mda-type.....	2938
7.2.1.3 ISA command descriptions.....	2939
7.2.1.3.1 isa.....	2939
7.2.1.3.2 ipsmg-group.....	2939
7.2.1.3.3 mda.....	2940
7.2.1.3.4 tunnel-group.....	2940
7.2.1.3.5 active-mda-number.....	2941
7.2.1.3.6 backup.....	2942
7.2.1.3.7 mda.....	2943
7.2.1.3.8 multi-active.....	2943
7.2.1.3.9 primary.....	2944
7.2.1.3.10 reassembly.....	2945
7.2.1.4 IPSec command descriptions.....	2946
7.2.1.4.1 Certificate profile command descriptions.....	2946
7.2.1.4.2 Internet Key Exchange (IKE) command descriptions.....	2949
7.2.1.4.3 Internet Key Exchange (IKE) transform command descriptions.....	2962
7.2.1.4.4 IPSec transform command descriptions.....	2967

7.2.1.4.5	RADIUS accounting policy command descriptions.....	2970
7.2.1.4.6	IPSec static security association command descriptions.....	2976
7.2.1.4.7	Trust anchor profile/TS list command descriptions.....	2980
7.2.1.4.8	Tunnel template command descriptions.....	2983
7.2.1.5	IES and VPRN service command descriptions.....	2988
7.2.1.5.1	interface.....	2988
7.2.1.5.2	dynamic-tunnel-redundant-next-hop.....	2989
7.2.1.5.3	address.....	2990
7.2.1.5.4	link-local-address.....	2991
7.2.1.5.5	static-tunnel-redundant-next-hop.....	2992
7.2.1.5.6	sap.....	2993
7.2.1.5.7	ip-tunnel.....	2994
7.2.1.5.8	backup-remote-ip.....	2995
7.2.1.5.9	delivery-service.....	2995
7.2.1.5.10	dscp.....	2996
7.2.1.5.11	gre-header.....	2997
7.2.1.5.12	ip-mtu.....	2998
7.2.1.5.13	reassemble.....	2999
7.2.1.5.14	remote-ip.....	2999
7.2.1.5.15	source.....	3000
7.2.1.5.16	VPRN IPSec Commands.....	3000
7.2.1.5.17	SAP IPSec gateway commands.....	3006
7.2.1.5.18	SAP IPSec tunnel commands.....	3019
7.2.1.6	CMPv2 commands.....	3026
7.2.1.6.1	pki.....	3026
7.2.1.6.2	ca-profile.....	3026
7.2.1.6.3	certificate.....	3027
7.2.1.6.4	certificate-display-format.....	3028
7.2.1.6.5	cmpv2.....	3028
7.2.1.6.6	accept-unprotected-errormsg.....	3028
7.2.1.6.7	accept-unprotected-pkiconf.....	3029
7.2.1.6.8	always-set-sender-for-ir.....	3029
7.2.1.6.9	key-list.....	3030
7.2.1.6.10	key.....	3030
7.2.1.6.11	url.....	3031
7.2.1.6.12	revocation-check.....	3032
7.2.1.6.13	http-response-timeout.....	3033
7.2.1.6.14	http-version.....	3034
7.2.1.6.15	response-signing-cert.....	3034
7.2.1.6.16	same-recipnonce-for-pollreq.....	3035
7.2.1.6.17	cert-request.....	3035
7.2.1.6.18	clear-request.....	3036
7.2.1.6.19	initial-registration.....	3037
7.2.1.6.20	key-update.....	3038
7.2.1.6.21	poll.....	3039
7.2.1.6.22	show-request.....	3040
7.2.2	IP tunnel show command descriptions.....	3040

7.2.2.1	cert-profile.....	3040
7.2.2.2	certificate.....	3044
7.2.2.3	gateway.....	3047
7.2.2.4	ike-policy.....	3063
7.2.2.5	radius-accounting-policy.....	3064
7.2.2.6	radius-authentication-policy.....	3065
7.2.2.7	security-policy.....	3065
7.2.2.8	tunnel-template.....	3067
7.2.2.9	transform.....	3068
7.2.2.10	trust-anchor-profile.....	3068
7.2.2.11	ts-list.....	3070
7.2.2.12	tunnel.....	3071
7.2.3	IP tunnel debug command descriptions.....	3079
7.2.3.1	gateway.....	3079
7.2.3.2	tunnel.....	3080
7.2.3.3	certificate.....	3080
7.2.3.4	cmpv2.....	3081
7.2.3.5	ca-profile.....	3081
7.2.3.6	ocsp.....	3082
8	NAT command reference.....	3083
8.1	NAT command hierarchies.....	3083
8.1.1	NAT configuration command hierarchies.....	3083
8.1.1.1	ISA command hierarchy.....	3083
8.1.1.2	NAT service command hierarchy.....	3084
8.1.1.3	NAT syslog command hierarchy.....	3085
8.1.1.4	IPFlow information export protocol command hierarchy.....	3086
8.1.1.5	NAT VPRN command hierarchy.....	3086
8.1.1.6	NAT router command hierarchy.....	3088
8.1.1.7	NAT DNAT command hierarchy.....	3090
8.1.1.8	NAT admin command hierarchy.....	3091
8.1.2	NAT show command hierarchies.....	3091
8.1.3	NAT clear command hierarchy.....	3092
8.1.4	NAT tools command hierarchies.....	3092
8.1.5	NAT filter command hierarchy.....	3093
8.2	NAT command descriptions.....	3093
8.2.1	NAT configuration command descriptions.....	3094
8.2.1.1	Generic NAT command descriptions.....	3094
8.2.1.1.1	description.....	3094
8.2.1.1.2	shutdown.....	3095
8.2.1.2	ISA NAT command descriptions.....	3096
8.2.1.2.1	nat-group.....	3096
8.2.1.2.2	acct-policy-id.....	3096
8.2.1.2.3	active-mda-limit.....	3097
8.2.1.2.4	dynamic-mda.....	3097
8.2.1.2.5	local-nat.....	3098
8.2.1.2.6	mda.....	3099

8.2.1.2.7 radius-accounting-policy.....	3099
8.2.1.2.8 session-limits.....	3100
8.2.1.2.9 reserved.....	3100
8.2.1.2.10 watermarks.....	3101
8.2.1.3 Service NAT command descriptions.....	3102
8.2.1.3.1 deterministic-script.....	3102
8.2.1.3.2 location.....	3102
8.2.1.3.3 nat-policy.....	3103
8.2.1.3.4 block-limit.....	3104
8.2.1.3.5 filtering.....	3104
8.2.1.3.6 ipfix-export-policy.....	3105
8.2.1.3.7 pool.....	3105
8.2.1.3.8 port-limits.....	3106
8.2.1.3.9 forwarding.....	3106
8.2.1.3.10 reserved.....	3107
8.2.1.3.11 watermarks.....	3107
8.2.1.3.12 priority-sessions.....	3108
8.2.1.3.13 fc.....	3108
8.2.1.3.14 max.....	3109
8.2.1.3.15 tcp-mss-adjust.....	3109
8.2.1.3.16 timeouts.....	3110
8.2.1.3.17 icmp-query.....	3110
8.2.1.3.18 sip.....	3111
8.2.1.3.19 subscriber-retention.....	3111
8.2.1.3.20 tcp-established.....	3112
8.2.1.3.21 tcp-syn.....	3113
8.2.1.3.22 tcp-time-wait.....	3113
8.2.1.3.23 tcp-transitory.....	3114
8.2.1.3.24 udp.....	3115
8.2.1.3.25 udp-dns.....	3115
8.2.1.3.26 udp-initial.....	3116
8.2.1.3.27 udp-inbound-refresh.....	3117
8.2.1.4 Syslog NAT command descriptions.....	3117
8.2.1.4.1 syslog.....	3117
8.2.1.4.2 syslog-export-policy.....	3118
8.2.1.4.3 collector.....	3118
8.2.1.4.4 destination-port.....	3119
8.2.1.4.5 ipv4-source-address.....	3120
8.2.1.4.6 facility.....	3120
8.2.1.4.7 include.....	3121
8.2.1.4.8 destination-ip.....	3121
8.2.1.4.9 foreign-ip.....	3122
8.2.1.4.10 foreign-port.....	3122
8.2.1.4.11 nat-policy-name.....	3123
8.2.1.4.12 sub-id.....	3123
8.2.1.4.13 log-prefix.....	3123
8.2.1.4.14 max-tx-delay.....	3124

8.2.1.4.15	mtu.....	3125
8.2.1.4.16	rate-limit.....	3125
8.2.1.4.17	severity-level.....	3126
8.2.1.5	IPFlow information export protocol NAT command descriptions.....	3127
8.2.1.5.1	ipfix.....	3127
8.2.1.5.2	ipfix-export-policy.....	3127
8.2.1.5.3	collector.....	3128
8.2.1.5.4	mtu.....	3129
8.2.1.5.5	source-address.....	3129
8.2.1.5.6	template-refresh-timeout.....	3130
8.2.1.6	VPRN and router NAT command descriptions.....	3131
8.2.1.6.1	nat.....	3131
8.2.1.6.2	inside.....	3131
8.2.1.6.3	outside.....	3131
8.2.1.6.4	mtu.....	3132
8.2.1.6.5	nat-policy.....	3132
8.2.1.6.6	destination-prefix.....	3133
8.2.1.6.7	deterministic.....	3134
8.2.1.6.8	classic-lsn-max-subscriber-limit.....	3134
8.2.1.6.9	prefix.....	3135
8.2.1.6.10	map.....	3136
8.2.1.6.11	ip-fragmentation.....	3139
8.2.1.6.12	nat64.....	3140
8.2.1.6.13	destination-prefix.....	3141
8.2.1.6.14	drop-zero-ipv4-checksum.....	3142
8.2.1.6.15	ignore-tos.....	3142
8.2.1.6.16	insert-ipv6-fragment-header.....	3143
8.2.1.6.17	nat64.....	3143
8.2.1.6.18	drop-zero-ipv4-checksum.....	3144
8.2.1.6.19	insert-ipv6-fragment-header.....	3144
8.2.1.6.20	ipv6-mtu.....	3144
8.2.1.6.21	prefix.....	3145
8.2.1.6.22	set-tos.....	3146
8.2.1.6.23	subscriber-prefix-length.....	3147
8.2.1.6.24	redundancy.....	3147
8.2.1.6.25	peer.....	3148
8.2.1.6.26	peer6.....	3148
8.2.1.6.27	steering-route.....	3149
8.2.1.6.28	subscriber-identification.....	3150
8.2.1.6.29	attribute.....	3150
8.2.1.6.30	drop-unidentified-traffic.....	3152
8.2.1.6.31	radius-proxy-server.....	3152
8.2.1.6.32	mtu.....	3153
8.2.1.6.33	pool.....	3153
8.2.1.6.34	address-range.....	3154
8.2.1.6.35	drain.....	3155
8.2.1.6.36	mode.....	3155

8.2.1.6.37	port-forwarding-range.....	3156
8.2.1.6.38	deterministic.....	3157
8.2.1.6.39	port-reservation.....	3157
8.2.1.6.40	port-reservation.....	3158
8.2.1.6.41	mode.....	3159
8.2.1.6.42	port-forwarding-dyn-block-reservation.....	3159
8.2.1.6.43	redundancy.....	3160
8.2.1.6.44	export.....	3160
8.2.1.6.45	follow.....	3161
8.2.1.6.46	monitor.....	3162
8.2.1.6.47	subscriber-limit.....	3163
8.2.1.6.48	watermarks.....	3163
8.2.1.6.49	upstream-ip-filter.....	3164
8.2.1.7	DNAT NAT command descriptions.....	3164
8.2.1.7.1	dnat.....	3165
8.2.1.7.2	dnat-only.....	3165
8.2.1.7.3	dnat-only.....	3165
8.2.1.7.4	source-prefix-list.....	3166
8.2.1.7.5	route-limit.....	3167
8.2.1.7.6	classic-lsn-max-subscriber-limit.....	3167
8.2.1.7.7	nat-classifier.....	3168
8.2.1.7.8	nat-classifier.....	3169
8.2.1.7.9	nat-prefix-list.....	3170
8.2.1.7.10	nat-prefix-list.....	3170
8.2.1.7.11	prefix.....	3171
8.2.1.7.12	default-dnat-ip-address.....	3172
8.2.1.7.13	entry.....	3172
8.2.1.7.14	match.....	3173
8.2.1.7.15	dst-port-range.....	3174
8.2.1.7.16	action.....	3174
8.2.1.7.17	default-action.....	3175
8.2.1.7.18	default-dnat-ip-address.....	3176
8.2.1.8	Admin NAT command descriptions.....	3177
8.2.1.8.1	save-deterministic-script.....	3177
8.2.2	NAT show command descriptions.....	3177
8.2.2.1	nat-group.....	3177
8.2.2.2	nat-policy.....	3183
8.2.2.3	port-forwarding-entries.....	3186
8.2.2.4	lsn-blocks.....	3186
8.2.2.5	pool.....	3188
8.2.2.6	summary.....	3190
8.2.3	NAT clear command descriptions.....	3190
8.2.3.1	nat-group.....	3190
8.2.4	NAT tools command descriptions.....	3191
8.2.4.1	nat.....	3191
8.2.4.2	isa.....	3191
8.2.4.3	resources.....	3192

8.2.4.4 sessions.....	3193
8.2.4.5 histogram.....	3194
8.2.4.6 port-forwarding-action.....	3195
8.2.4.7 lsn.....	3195
9 Lawful Interception command reference.....	3197
9.1 Lawful Interception command hierarchies.....	3197
9.1.1 Lawful Interception configuration command hierarchies.....	3197
9.1.2 Lawful Interception show command hierarchies.....	3203
9.2 Lawful Interception command descriptions.....	3204
9.2.1 Lawful Interception configuration command descriptions.....	3204
9.2.1.1 Generic Lawful Interception Commands.....	3204
9.2.1.1.1 description.....	3204
9.2.1.1.2 shutdown.....	3205
9.2.1.2 Lawful Interception command descriptions.....	3207
9.2.1.2.1 li.....	3207
9.2.1.2.2 li-filter.....	3207
9.2.1.2.3 li-ip-filter.....	3207
9.2.1.2.4 entry.....	3208
9.2.1.2.5 match.....	3209
9.2.1.2.6 dst-ip.....	3211
9.2.1.2.7 dst-port.....	3212
9.2.1.2.8 src-ip.....	3213
9.2.1.2.9 src-port.....	3214
9.2.1.2.10 li-ipv6-filter.....	3215
9.2.1.2.11 match.....	3215
9.2.1.2.12 li-mac-filter.....	3216
9.2.1.2.13 match.....	3217
9.2.1.2.14 dst-mac.....	3218
9.2.1.2.15 src-mac.....	3219
9.2.1.2.16 li-filter-associations.....	3220
9.2.1.2.17 li-ip-filter.....	3220
9.2.1.2.18 ip-filter.....	3221
9.2.1.2.19 li-ipv6-filter.....	3221
9.2.1.2.20 ipv6-filter.....	3222
9.2.1.2.21 li-mac-filter.....	3222
9.2.1.2.22 mac-filter.....	3223
9.2.1.2.23 li-filter-block-reservation.....	3223
9.2.1.2.24 li-reserved-block.....	3224
9.2.1.2.25 ip-filter.....	3224
9.2.1.2.26 ipv6-filter.....	3225
9.2.1.2.27 mac-filter.....	3225
9.2.1.2.28 start-entry.....	3226
9.2.1.2.29 li-filter-lock-state.....	3226
9.2.1.2.30 li-source.....	3227
9.2.1.2.31 ip-filter.....	3228
9.2.1.2.32 ipv6-filter.....	3230

9.2.1.2.33	li-ip-filter.....	3232
9.2.1.2.34	li-ipv6-filter.....	3233
9.2.1.2.35	li-mac-filter.....	3235
9.2.1.2.36	mac-filter.....	3236
9.2.1.2.37	nat.....	3238
9.2.1.2.38	classic-lsn-sub.....	3238
9.2.1.2.39	intercept-id.....	3238
9.2.1.2.40	session-id.....	3240
9.2.1.2.41	dslite-lsn-sub.....	3241
9.2.1.2.42	ethernet-header.....	3242
9.2.1.2.43	l2-aware-sub.....	3242
9.2.1.2.44	sap.....	3243
9.2.1.2.45	subscriber.....	3245
9.2.1.2.46	log.....	3247
9.2.1.2.47	log-id.....	3247
9.2.1.2.48	filter.....	3248
9.2.1.2.49	from.....	3248
9.2.1.2.50	time-format.....	3249
9.2.1.2.51	to.....	3249
9.2.1.2.52	3gpp-5g-release.....	3250
9.2.1.2.53	3gpp-release.....	3251
9.2.1.2.54	allow-duplicate-msisdn-or-imei.....	3252
9.2.1.2.55	client-tls-profile.....	3252
9.2.1.2.56	custom-correlation-id-format.....	3252
9.2.1.2.57	df-peer.....	3253
9.2.1.2.58	df3-tcp-connections.....	3255
9.2.1.2.59	fan-out.....	3256
9.2.1.2.60	include-rai.....	3257
9.2.1.2.61	iri.....	3257
9.2.1.2.62	gw-function-iri-mode.....	3258
9.2.1.2.63	iri-per-mb-request.....	3258
9.2.1.2.64	neid.....	3259
9.2.1.2.65	te-id.....	3259
9.2.1.2.66	x2-iri-ice-type.....	3260
9.2.1.2.67	li-t3.....	3260
9.2.1.2.68	alpn.....	3260
9.2.1.2.69	li-t3-local-interface.....	3261
9.2.1.2.70	li-t3-mgmt-local-interface.....	3262
9.2.1.2.71	li-t3-up-peer.....	3263
9.2.1.2.72	sni.....	3265
9.2.1.2.73	li-x1.....	3265
9.2.1.2.74	admf-peer.....	3266
9.2.1.2.75	li-x1-keep-alive-time-p2.....	3267
9.2.1.2.76	li-x1-local-interface.....	3267
9.2.1.2.77	li-x1-mgmt.....	3269
9.2.1.2.78	li-x1-mgmt-admf-peer.....	3269
9.2.1.2.79	li-x1-mgmt-local-interface.....	3270

9.2.1.2.80	li-x2.....	3271
9.2.1.2.81	li-x2-keep-alive-time-p1.....	3271
9.2.1.2.82	li-x2-keep-alive-time-p2.....	3272
9.2.1.2.83	li-x3.....	3273
9.2.1.2.84	li-x3-keep-alive-time-p1.....	3273
9.2.1.2.85	li-x3-keep-alive-time-p2.....	3274
9.2.1.2.86	li-x3-sequence.....	3275
9.2.1.2.87	li-x3-tls.....	3275
9.2.1.2.88	local-interface.....	3276
9.2.1.2.89	nf-id-value.....	3277
9.2.1.2.90	num-conn-retries.....	3278
9.2.1.2.91	operator-id.....	3279
9.2.1.2.92	pcrf-shared-key.....	3279
9.2.1.2.93	per-bearer-seq-num.....	3280
9.2.1.2.94	pcfp-li-shared-key.....	3281
9.2.1.2.95	server-tls-profile.....	3281
9.2.1.2.96	target.....	3282
9.2.1.2.97	tcp-keepalive.....	3284
9.2.1.2.98	idle-time.....	3285
9.2.1.2.99	interval.....	3285
9.2.1.2.100	retries.....	3286
9.2.1.2.101	shutdown.....	3286
9.2.1.2.102	tls.....	3287
9.2.1.2.103	x2-iri-cache-size.....	3287
9.2.1.2.104	x2-iri-qos.....	3288
9.2.1.2.105	x2-keep-alive-time.....	3289
9.2.1.2.106	x3-cc-cache-size.....	3289
9.2.1.2.107	x3-cc-qos.....	3290
9.2.1.2.108	x3-keep-alive-time.....	3291
9.2.1.2.109	x3-transport.....	3291
9.2.1.2.110	save.....	3292
9.2.1.2.111	tech-support.....	3292
9.2.1.2.112	Other LI configuration commands.....	3293
9.2.2	Lawful Interception show command descriptions.....	3297
9.2.2.1	Lawful Interception show command descriptions.....	3297
9.2.2.1.1	debug.....	3297
9.2.2.1.2	active-subscribers.....	3299
9.2.2.1.3	service-using.....	3303
9.2.2.1.4	li.....	3305
9.2.2.1.5	li-source.....	3305
9.2.2.1.6	filter.....	3306
9.2.2.1.7	log.....	3307
9.2.2.1.8	mobile-gateway.....	3307
9.2.2.1.9	log-id.....	3318
9.2.2.1.10	status.....	3320

10 Application Assurance command reference..... 3321

10.1	Application Assurance command hierarchies.....	3321
10.1.1	Application Assurance configuration command hierarchies.....	3321
10.1.1.1	Hardware command hierarchy.....	3321
10.1.1.2	Admin command hierarchy.....	3322
10.1.1.3	ISA command hierarchy.....	3322
10.1.1.4	Application Assurance command hierarchy.....	3323
10.1.1.5	Application Assurance interface command hierarchy.....	3324
10.1.1.6	Application Assurance group command hierarchy.....	3325
10.1.1.7	Application Assurance group cflowd command hierarchy.....	3330
10.1.1.8	Application Assurance group policer command hierarchy.....	3332
10.1.1.9	Application Assurance group policy command hierarchy.....	3333
10.1.1.10	Application Assurance group session filter command hierarchy.....	3338
10.1.1.11	Application Assurance group statistics command hierarchy.....	3339
10.1.1.12	Application Assurance group URL filter and URL list command hierarchy.....	3340
10.1.2	Application Assurance show command hierarchy.....	3342
10.1.3	Application Assurance tools command hierarchy.....	3344
10.1.4	Application Assurance clear command hierarchy.....	3346
10.1.5	Application Assurance debug command hierarchy.....	3346
10.2	Application Assurance command descriptions.....	3347
10.2.1	Application Assurance configuration command descriptions.....	3347
10.2.1.1	Generic Application Assurance command descriptions.....	3348
10.2.1.1.1	description.....	3348
10.2.1.1.2	description.....	3349
10.2.1.1.3	shutdown.....	3350
10.2.1.2	Hardware Application Assurance command descriptions.....	3352
10.2.1.2.1	isa2-aa (mda-type).....	3352
10.2.1.3	Admin Application Assurance command descriptions.....	3352
10.2.1.3.1	application-assurance.....	3352
10.2.1.3.2	group.....	3353
10.2.1.3.3	url-list.....	3353
10.2.1.3.4	upgrade.....	3354
10.2.1.4	ISA Application Assurance command descriptions.....	3354
10.2.1.4.1	application-assurance-group.....	3354
10.2.1.4.2	dynamic-rating-expiry.....	3355
10.2.1.4.3	fail-to-open.....	3356
10.2.1.4.4	flow-attribute.....	3356
10.2.1.4.5	http-enrich-max-pkt.....	3358
10.2.1.4.6	isa-capacity-cost-high-threshold.....	3358
10.2.1.4.7	isa-capacity-cost-low-threshold.....	3359
10.2.1.4.8	isa-overload-cut-through.....	3359
10.2.1.4.9	minimum-isa-generation.....	3360
10.2.1.4.10	partitions.....	3361
10.2.1.4.11	primary.....	3361
10.2.1.4.12	qos.....	3362
10.2.1.4.13	egress.....	3362
10.2.1.4.14	from-subscriber.....	3363
10.2.1.4.15	pool.....	3363

10.2.1.4.16	resv-cbs.....	3364
10.2.1.4.17	slope-policy.....	3365
10.2.1.4.18	queue-policy.....	3365
10.2.1.4.19	wa-shared-high-wmark.....	3366
10.2.1.4.20	wa-shared-low-wmark.....	3366
10.2.1.4.21	port-scheduler-policy.....	3367
10.2.1.4.22	to-subscriber.....	3368
10.2.1.4.23	shared-resources.....	3368
10.2.1.4.24	bandwidth-shaper-buffer.....	3368
10.2.1.4.25	multi-device-detection.....	3369
10.2.1.4.26	tcp-adv-func.....	3369
10.2.1.4.27	web-service-url-filter.....	3370
10.2.1.4.28	transit-prefix-ipv4-entries.....	3370
10.2.1.4.29	transit-prefix-ipv4-remote-entries.....	3371
10.2.1.4.30	transit-prefix-ipv6-entries.....	3372
10.2.1.4.31	transit-prefix-ipv6-remote-entries.....	3373
10.2.1.4.32	vm-traffic-distribution-by-ip.....	3373
10.2.1.4.33	ingress.....	3374
10.2.1.5	Application Assurance command descriptions.....	3374
10.2.1.5.1	application-assurance.....	3374
10.2.1.5.2	bit-rate-high-wmark.....	3374
10.2.1.5.3	bit-rate-low-wmark.....	3375
10.2.1.5.4	datapath-cpu-high-wmark.....	3376
10.2.1.5.5	datapath-cpu-low-wmark.....	3376
10.2.1.5.6	flow-setup-high-wmark.....	3377
10.2.1.5.7	flow-setup-low-wmark.....	3377
10.2.1.5.8	flow-table-high-wmark.....	3378
10.2.1.5.9	flow-table-low-wmark.....	3379
10.2.1.5.10	packet-rate-high-wmark.....	3379
10.2.1.5.11	packet-rate-low-wmark.....	3380
10.2.1.5.12	protocol.....	3380
10.2.1.6	Application Assurance interface command descriptions.....	3381
10.2.1.6.1	aa-interface.....	3381
10.2.1.6.2	address.....	3381
10.2.1.6.3	ip-mtu.....	3382
10.2.1.6.4	sap.....	3383
10.2.1.6.5	egress.....	3383
10.2.1.6.6	ingress.....	3384
10.2.1.6.7	filter.....	3384
10.2.1.6.8	qos.....	3385
10.2.1.7	Application Assurance group command descriptions.....	3385
10.2.1.7.1	group.....	3385
10.2.1.7.2	aa-sub-congestion-detection.....	3386
10.2.1.7.3	rtt-threshold.....	3386
10.2.1.7.4	rtt-threshold-rat.....	3387
10.2.1.7.5	rtt-threshold-tolerance.....	3388
10.2.1.7.6	access-network-location.....	3389

10.2.1.7.7	source.....	3389
10.2.1.7.8	alternate-cdn-host-matching.....	3390
10.2.1.7.9	certificate-profile.....	3391
10.2.1.7.10	certificate.....	3391
10.2.1.7.11	key-file.....	3392
10.2.1.7.12	disable-deferred-billing.....	3392
10.2.1.7.13	dns-enrich.....	3393
10.2.1.7.14	gateway.....	3393
10.2.1.7.15	option.....	3394
10.2.1.7.16	packet-size-limit.....	3395
10.2.1.7.17	dns-ip-cache.....	3395
10.2.1.7.18	dns-match.....	3396
10.2.1.7.19	domain.....	3396
10.2.1.7.20	server-address.....	3397
10.2.1.7.21	ip-cache.....	3397
10.2.1.7.22	high-watermark.....	3398
10.2.1.7.23	low-watermark.....	3398
10.2.1.7.24	size.....	3399
10.2.1.7.25	static-address.....	3399
10.2.1.7.26	event-log.....	3400
10.2.1.7.27	buffer-type.....	3401
10.2.1.7.28	max-entries.....	3402
10.2.1.7.29	syslog.....	3402
10.2.1.7.30	address.....	3402
10.2.1.7.31	facility.....	3403
10.2.1.7.32	port.....	3404
10.2.1.7.33	severity.....	3404
10.2.1.7.34	vlan-id.....	3405
10.2.1.7.35	http-anti-fraud-cr-eol.....	3406
10.2.1.7.36	http-anti-fraud-policy.....	3406
10.2.1.7.37	http-enrich.....	3407
10.2.1.7.38	field.....	3407
10.2.1.7.39	aes-initialization-vector.....	3411
10.2.1.7.40	anti-spoof.....	3412
10.2.1.7.41	calling-line-id.....	3412
10.2.1.7.42	encode.....	3413
10.2.1.7.43	md5-salt.....	3414
10.2.1.7.44	name.....	3415
10.2.1.7.45	static-string.....	3415
10.2.1.7.46	rat-enrichment.....	3415
10.2.1.7.47	rat-type.....	3416
10.2.1.7.48	tls-extension.....	3416
10.2.1.7.49	extension-id.....	3417
10.2.1.7.50	subtype.....	3417
10.2.1.7.51	encode.....	3419
10.2.1.7.52	extension-id.....	3420
10.2.1.7.53	static-string.....	3420

10.2.1.7.54	http-error-redirect.....	3421
10.2.1.7.55	error-code.....	3421
10.2.1.7.56	http-host.....	3422
10.2.1.7.57	participant-id.....	3423
10.2.1.7.58	template.....	3423
10.2.1.7.59	http-match-all-requests.....	3424
10.2.1.7.60	http-notification.....	3425
10.2.1.7.61	interval.....	3425
10.2.1.7.62	template.....	3426
10.2.1.7.63	http-proxy.....	3426
10.2.1.7.64	description.....	3427
10.2.1.7.65	dns-server.....	3427
10.2.1.7.66	https.....	3428
10.2.1.7.67	intermediate-certificate-1.....	3428
10.2.1.7.68	intermediate-certificate-2.....	3429
10.2.1.7.69	root-certificate.....	3429
10.2.1.7.70	tls-profile.....	3430
10.2.1.7.71	url-mapping.....	3430
10.2.1.7.72	entry.....	3431
10.2.1.7.73	action.....	3431
10.2.1.7.74	http-enrich.....	3432
10.2.1.7.75	target-server.....	3432
10.2.1.7.76	uri-transform.....	3433
10.2.1.7.77	match.....	3434
10.2.1.7.78	expression.....	3434
10.2.1.7.79	shutdown.....	3435
10.2.1.7.80	vlan-id.....	3436
10.2.1.7.81	script-url.....	3436
10.2.1.7.82	template.....	3437
10.2.1.7.83	http-redirect.....	3437
10.2.1.7.84	redirect-https.....	3438
10.2.1.7.85	redirect-url.....	3438
10.2.1.7.86	tcp-client-reset.....	3439
10.2.1.7.87	template.....	3440
10.2.1.7.88	http-x-online-host.....	3441
10.2.1.7.89	ip-identification-assist.....	3441
10.2.1.7.90	passive-dns.....	3442
10.2.1.7.91	monitor.....	3442
10.2.1.7.92	trusted-server-address.....	3443
10.2.1.7.93	ip-identification-contribute.....	3443
10.2.1.7.94	ip-prefix-list.....	3443
10.2.1.7.95	prefix.....	3444
10.2.1.7.96	mp-tcp-proxy.....	3445
10.2.1.7.97	bonding-assist.....	3446
10.2.1.7.98	congestion-control.....	3446
10.2.1.7.99	default-scheduler-policy.....	3447
10.2.1.7.100	force-dss-checksums.....	3448

10.2.1.7.101	ip-address-profile.....	3449
10.2.1.7.102	path.....	3449
10.2.1.7.103	fixed.....	3450
10.2.1.7.104	mobile.....	3450
10.2.1.7.105	subflow-failure-indication.....	3451
10.2.1.7.106	multi-device-detection.....	3452
10.2.1.7.107	max-devices.....	3452
10.2.1.7.108	port-list.....	3452
10.2.1.7.109	port.....	3453
10.2.1.7.110	shallow-inspection.....	3454
10.2.1.7.111	tcp-optimizer.....	3454
10.2.1.7.112	dack-timeout.....	3455
10.2.1.7.113	high-cpu-backoff.....	3456
10.2.1.7.114	init-cwnd-size.....	3456
10.2.1.7.115	init-ss-threshold.....	3457
10.2.1.7.116	network-rtt-threshold.....	3457
10.2.1.7.117	tcp-stack.....	3458
10.2.1.7.118	tcp-validate.....	3459
10.2.1.7.119	event-log.....	3459
10.2.1.7.120	strict.....	3460
10.2.1.7.121	tethering-detection.....	3461
10.2.1.7.122	single-device.....	3461
10.2.1.7.123	expected-ttl.....	3462
10.2.1.7.124	invert-match.....	3462
10.2.1.7.125	tll-monitoring.....	3463
10.2.1.7.126	tcp-protocols.....	3463
10.2.1.7.127	udp-protocols.....	3463
10.2.1.7.128	wap1x.....	3464
10.2.1.8	Application Assurance group cflowd command descriptions.....	3464
10.2.1.8.1	cflowd.....	3465
10.2.1.8.2	collector.....	3465
10.2.1.8.3	comprehensive.....	3466
10.2.1.8.4	app-group.....	3466
10.2.1.8.5	application.....	3467
10.2.1.8.6	flow-rate.....	3467
10.2.1.8.7	flow-rate2.....	3468
10.2.1.8.8	template.....	3469
10.2.1.8.9	dynamic-fields.....	3469
10.2.1.8.10	field.....	3470
10.2.1.8.11	field-selection.....	3471
10.2.1.8.12	direct-export.....	3472
10.2.1.8.13	collector.....	3472
10.2.1.8.14	address.....	3473
10.2.1.8.15	vlan-id.....	3474
10.2.1.8.16	export-override.....	3474
10.2.1.8.17	interim-report-mode.....	3475
10.2.1.8.18	obfuscation.....	3476

10.2.1.8.19	aes-128-encryption.....	3477
10.2.1.8.20	aes-256-encryption.....	3478
10.2.1.8.21	pgw-edr.....	3478
10.2.1.8.22	flow-rate.....	3479
10.2.1.8.23	template-retransmit.....	3479
10.2.1.8.24	system-name.....	3480
10.2.1.8.25	tcp-performance.....	3481
10.2.1.8.26	app-group.....	3481
10.2.1.8.27	application.....	3482
10.2.1.8.28	volume.....	3482
10.2.1.8.29	rate.....	3483
10.2.1.9	Application Assurance group policer command descriptions.....	3483
10.2.1.9.1	policer.....	3483
10.2.1.9.2	action.....	3485
10.2.1.9.3	adaptation-rule.....	3486
10.2.1.9.4	congestion-override.....	3487
10.2.1.9.5	congestion-override-stage2.....	3487
10.2.1.9.6	cbs.....	3488
10.2.1.9.7	cir.....	3488
10.2.1.9.8	mbs.....	3489
10.2.1.9.9	pir.....	3490
10.2.1.9.10	flow-count.....	3490
10.2.1.9.11	rate.....	3491
10.2.1.9.12	rate-percentage.....	3492
10.2.1.9.13	tod-override.....	3492
10.2.1.9.14	time-range.....	3493
10.2.1.10	Application Assurance group policy command descriptions.....	3494
10.2.1.10.1	policy.....	3494
10.2.1.10.2	abort.....	3495
10.2.1.10.3	begin.....	3495
10.2.1.10.4	charging-filter.....	3495
10.2.1.10.5	entry.....	3496
10.2.1.10.6	charging-group.....	3496
10.2.1.10.7	match.....	3497
10.2.1.10.8	app-group.....	3497
10.2.1.10.9	application.....	3498
10.2.1.10.10	flow-attribute.....	3499
10.2.1.10.11	tethered-flow.....	3499
10.2.1.10.12	commit.....	3500
10.2.1.10.13	admin.....	3500
10.2.1.10.14	app-filter.....	3500
10.2.1.10.15	aa-sub-ip.....	3501
10.2.1.10.16	network-ip.....	3502
10.2.1.10.17	app-group.....	3503
10.2.1.10.18	app-profile.....	3504
10.2.1.10.19	app-qos-policy.....	3504
10.2.1.10.20	app-service-options.....	3505

10.2.1.10.21	charging-group.....	3505
10.2.1.10.22	characteristic.....	3506
10.2.1.10.23	value.....	3506
10.2.1.10.24	default-value.....	3507
10.2.1.10.25	application.....	3507
10.2.1.10.26	app-group.....	3508
10.2.1.10.27	charging-group.....	3508
10.2.1.10.28	charging-group.....	3509
10.2.1.10.29	notify-start-stop.....	3509
10.2.1.10.30	default-charging-group.....	3511
10.2.1.10.31	default-tethered-charging-group.....	3511
10.2.1.10.32	entry.....	3512
10.2.1.10.33	application.....	3513
10.2.1.10.34	dynamic-rating-action.....	3513
10.2.1.10.35	expression.....	3514
10.2.1.10.36	flow-setup-direction.....	3517
10.2.1.10.37	http-match-all-requests.....	3517
10.2.1.10.38	http-port.....	3518
10.2.1.10.39	ip-protocol-num.....	3519
10.2.1.10.40	server-address.....	3520
10.2.1.10.41	server-port.....	3521
10.2.1.10.42	protocol.....	3523
10.2.1.10.43	characteristic.....	3524
10.2.1.10.44	capacity-cost.....	3524
10.2.1.10.45	divert.....	3525
10.2.1.10.46	entry.....	3525
10.2.1.10.47	action.....	3526
10.2.1.10.48	abandon-tcp-optimization.....	3526
10.2.1.10.49	bandwidth-policer.....	3527
10.2.1.10.50	dns-enrich.....	3527
10.2.1.10.51	dns-ip-cache.....	3528
10.2.1.10.52	drop.....	3528
10.2.1.10.53	error-drop.....	3529
10.2.1.10.54	flow-count-limit.....	3529
10.2.1.10.55	flow-rate-limit.....	3530
10.2.1.10.56	fragment-drop.....	3531
10.2.1.10.57	http-enrich.....	3532
10.2.1.10.58	http-error-redirect.....	3532
10.2.1.10.59	http-notification.....	3533
10.2.1.10.60	http-redirect.....	3533
10.2.1.10.61	mirror-source.....	3535
10.2.1.10.62	mp-tcp-proxy.....	3535
10.2.1.10.63	mp-tcp-scheduler-policy-override.....	3536
10.2.1.10.64	offload.....	3538
10.2.1.10.65	overload-drop.....	3538
10.2.1.10.66	remark.....	3539
10.2.1.10.67	dscp.....	3539

10.2.1.10.68	fc.....	3540
10.2.1.10.69	priority.....	3540
10.2.1.10.70	session-filter.....	3541
10.2.1.10.71	tcp-mss-adjust.....	3542
10.2.1.10.72	tcp-validate.....	3542
10.2.1.10.73	tls-enrich.....	3543
10.2.1.10.74	url-filter.....	3544
10.2.1.10.75	match.....	3545
10.2.1.10.76	aa-sub.....	3545
10.2.1.10.77	aa-sub-tethering-state.....	3546
10.2.1.10.78	app-group.....	3547
10.2.1.10.79	application.....	3548
10.2.1.10.80	characteristic.....	3548
10.2.1.10.81	charging-group.....	3549
10.2.1.10.82	dscp.....	3550
10.2.1.10.83	dst-ip.....	3551
10.2.1.10.84	dst-port.....	3551
10.2.1.10.85	flow-attribute.....	3552
10.2.1.10.86	confidence.....	3553
10.2.1.10.87	ip-protocol-num.....	3554
10.2.1.10.88	multi-device-detection.....	3554
10.2.1.10.89	presence-reporting-area.....	3555
10.2.1.10.90	rat-type.....	3556
10.2.1.10.91	roaming-status.....	3557
10.2.1.10.92	src-ip.....	3557
10.2.1.10.93	src-port.....	3558
10.2.1.10.94	traffic-direction.....	3559
10.2.1.10.95	wireless-device-os.....	3560
10.2.1.10.96	wireless-device-type.....	3561
10.2.1.10.97	custom-protocol.....	3562
10.2.1.10.98	expression.....	3563
10.2.1.11	Application Assurance group session filter command descriptions.....	3564
10.2.1.11.1	session-filter.....	3564
10.2.1.11.2	default-action.....	3565
10.2.1.11.3	entry.....	3566
10.2.1.11.4	action.....	3566
10.2.1.11.5	action.....	3567
10.2.1.11.6	action.....	3568
10.2.1.11.7	match.....	3568
10.2.1.11.8	dst-ip.....	3568
10.2.1.11.9	dst-port.....	3569
10.2.1.11.10	ip-protocol-num.....	3570
10.2.1.11.11	src-ip.....	3571
10.2.1.11.12	src-port.....	3572
10.2.1.12	Application Assurance group statistics command descriptions.....	3573
10.2.1.12.1	statistics.....	3573
10.2.1.12.2	performance.....	3573

10.2.1.12.3	aa-admit-deny.....	3573
10.2.1.12.4	accounting-policy.....	3574
10.2.1.12.5	collect-stats.....	3574
10.2.1.12.6	policer-stats.....	3575
10.2.1.12.7	policer-stats-resources.....	3575
10.2.1.12.8	session-filter-stats.....	3576
10.2.1.12.9	tcp-validate-stats.....	3576
10.2.1.12.10	aa-partition.....	3577
10.2.1.12.11	aa-sub.....	3577
10.2.1.12.12	exclude-tcp-retrans.....	3577
10.2.1.12.13	app-group.....	3578
10.2.1.12.14	application.....	3578
10.2.1.12.15	protocol.....	3579
10.2.1.12.16	traffic-type.....	3579
10.2.1.12.17	threshold-crossing-alert.....	3580
10.2.1.12.18	error-drop.....	3580
10.2.1.12.19	high-wmark.....	3581
10.2.1.12.20	fragment-drop-all.....	3582
10.2.1.12.21	fragment-drop-out-of-order.....	3582
10.2.1.12.22	overload-drop.....	3583
10.2.1.12.23	policer.....	3583
10.2.1.12.24	session-filter.....	3584
10.2.1.12.25	default-action.....	3584
10.2.1.12.26	entry.....	3585
10.2.1.12.27	tcp-validate.....	3586
10.2.1.13	Application Assurance group URL filter and URL list command descriptions.....	3586
10.2.1.13.1	url-filter.....	3586
10.2.1.13.2	apply-function-specific-behavior.....	3587
10.2.1.13.3	default-action.....	3587
10.2.1.13.4	http-redirect.....	3589
10.2.1.13.5	http-request-filtering.....	3589
10.2.1.13.6	icap.....	3590
10.2.1.13.7	custom-x-header.....	3590
10.2.1.13.8	server.....	3590
10.2.1.13.9	vlan-id.....	3591
10.2.1.13.10	local-filtering.....	3591
10.2.1.13.11	allow-list.....	3592
10.2.1.13.12	deny-list.....	3592
10.2.1.13.13	web-service.....	3593
10.2.1.13.14	classification-overrides.....	3593
10.2.1.13.15	entry.....	3594
10.2.1.13.16	classifier.....	3594
10.2.1.13.17	default-profile.....	3595
10.2.1.13.18	dns-server.....	3595
10.2.1.13.19	fqdn.....	3596
10.2.1.13.20	profile.....	3597
10.2.1.13.21	category.....	3597

10.2.1.13.22	server.....	3598
10.2.1.13.23	vlan-id.....	3599
10.2.1.13.24	url-list.....	3599
10.2.1.13.25	decrypt-key.....	3600
10.2.1.13.26	expression-match.....	3600
10.2.1.13.27	file.....	3601
10.2.1.13.28	size.....	3601
10.2.2	Application Assurance show command descriptions.....	3602
10.2.2.1	application-assurance-group.....	3602
10.2.2.2	ip-reas-stats.....	3607
10.2.2.3	group.....	3609
10.2.2.4	aa-interface.....	3610
10.2.2.5	aa-sub.....	3611
10.2.2.6	aa-sub-list.....	3613
10.2.2.7	app-group.....	3615
10.2.2.8	application.....	3618
10.2.2.9	certificate-profile.....	3621
10.2.2.10	cflowd.....	3622
10.2.2.11	collector.....	3623
10.2.2.12	direct-export.....	3625
10.2.2.13	collector.....	3625
10.2.2.14	isa-collector.....	3627
10.2.2.15	status.....	3627
10.2.2.16	dns-enrich.....	3631
10.2.2.17	dns-ip-cache.....	3634
10.2.2.18	event-log.....	3637
10.2.2.19	http-enrich.....	3638
10.2.2.20	detail.....	3640
10.2.2.21	field.....	3640
10.2.2.22	fields.....	3641
10.2.2.23	http-notification.....	3641
10.2.2.24	http-proxy.....	3643
10.2.2.25	count.....	3645
10.2.2.26	admin.....	3649
10.2.2.27	app-filter.....	3650
10.2.2.28	app-group.....	3651
10.2.2.29	app-profile.....	3651
10.2.2.30	app-qos-policy.....	3651
10.2.2.31	app-service-option.....	3652
10.2.2.32	application.....	3652
10.2.2.33	charging-filter.....	3653
10.2.2.34	charging-group.....	3654
10.2.2.35	policers.....	3657
10.2.2.36	custom-protocol.....	3659
10.2.2.37	summary.....	3659
10.2.2.38	protocol.....	3659
10.2.2.39	summary.....	3661

10.2.2.40	ip-identification-assist.....	3663
10.2.2.41	mp-tcp-proxy.....	3664
10.2.2.42	partition.....	3666
10.2.2.43	policer.....	3666
10.2.2.44	policy.....	3669
10.2.2.45	http-redirect.....	3670
10.2.2.46	status.....	3672
10.2.2.47	tcp-optimizer.....	3679
10.2.2.48	tcp-validate.....	3681
10.2.2.49	traffic-type.....	3682
10.2.2.50	url-filter.....	3684
10.2.2.51	url-list.....	3689
10.2.2.52	http-enrich.....	3691
10.2.2.53	http-error-redirect.....	3692
10.2.2.54	error-codes.....	3693
10.2.2.55	template.....	3695
10.2.2.56	protocol.....	3696
10.2.2.57	session-filter.....	3701
10.2.2.58	threshold-crossing-alert.....	3703
10.2.2.59	version.....	3705
10.2.2.60	web-service.....	3705
10.2.2.61	category-set-id.....	3706
10.2.2.62	aa-sub-using.....	3706
10.2.2.63	app-profile.....	3707
10.2.3	Application Assurance tools command descriptions.....	3708
10.2.3.1	resources.....	3708
10.2.3.2	group.....	3708
10.2.3.3	aa-anl-list.....	3709
10.2.3.4	aa-sub-list.....	3711
10.2.3.5	aa-sub-search.....	3712
10.2.3.6	admit-deny-stats.....	3714
10.2.3.7	cflowd.....	3715
10.2.3.8	template.....	3715
10.2.3.9	dns-ip-cache.....	3716
10.2.3.10	dynamic-rating.....	3720
10.2.3.11	statistics.....	3720
10.2.3.12	event-log.....	3723
10.2.3.13	flow-record-search.....	3725
10.2.3.14	http-host-recorder detail.....	3730
10.2.3.15	http-host-recorder status.....	3732
10.2.3.16	http-host-recorder top.....	3733
10.2.3.17	http-proxy.....	3733
10.2.3.18	ip-identification-assist.....	3736
10.2.3.19	applications.....	3737
10.2.3.20	cache-contents.....	3739
10.2.3.21	passive-dns.....	3741
10.2.3.22	monitor.....	3741

10.2.3.23 mp-tcp-proxy.....	3743
10.2.3.24 port-recorder detail.....	3744
10.2.3.25 port-recorder status.....	3746
10.2.3.26 port-recorder top.....	3747
10.2.3.27 policer.....	3747
10.2.3.28 resources.....	3748
10.2.3.29 tethering.....	3749
10.2.3.30 traffic-capture.....	3749
10.2.4 Application Assurance clear command descriptions.....	3751
10.2.4.1 group.....	3751
10.2.5 Application Assurance debug command descriptions.....	3751
10.2.5.1 group.....	3751
10.2.5.2 http-host-recorder.....	3752
10.2.5.3 filter.....	3752
10.2.5.4 default-filter-action.....	3753
10.2.5.5 expression.....	3753
10.2.5.6 record.....	3754
10.2.5.7 rate.....	3755
10.2.5.8 shutdown.....	3756
10.2.5.9 port-recorder.....	3756
10.2.5.10 application.....	3756
10.2.5.11 traffic-capture.....	3757
10.2.5.12 match.....	3757
10.2.5.13 application.....	3758
10.2.5.14 client-ip.....	3758
10.2.5.15 client-port.....	3759
10.2.5.16 dst-ip.....	3759
10.2.5.17 dst-port.....	3759
10.2.5.18 ip-addr1.....	3760
10.2.5.19 ip-addr2.....	3760
10.2.5.20 ip-protocol-num.....	3760
10.2.5.21 port1.....	3761
10.2.5.22 port2.....	3761
10.2.5.23 server-ip.....	3761
10.2.5.24 server-port.....	3762
10.2.5.25 src-ip.....	3762
10.2.5.26 src-port.....	3762
10.2.5.27 record.....	3763
10.2.5.28 cut-through-packets.....	3763
10.2.5.29 limit.....	3764
10.2.5.30 start.....	3764
10.2.5.31 shutdown.....	3765
11 Tools, OAM, and ping command reference.....	3766
11.1 Tools, OAM, and ping command hierarchies.....	3766
11.1.1 Tools command hierarchy.....	3766
11.1.2 OAM command hierarchy.....	3768

11.1.3 Ping command hierarchy.....	3768
11.2 Tools, OAM and ping command descriptions.....	3768
11.2.1 Tools command descriptions.....	3769
11.2.1.1 tools.....	3769
11.2.1.2 dump.....	3769
11.2.1.3 dhcp.....	3769
11.2.1.4 test.....	3770
11.2.1.5 micronets.....	3771
11.2.1.6 mobile-gateway.....	3772
11.2.1.7 dedicated-bearer-monitoring.....	3773
11.2.1.8 diameter-peer-realm.....	3776
11.2.1.9 dns-client-cache.....	3777
11.2.1.10 dns-client-cache-candidates.....	3780
11.2.1.11 dns-client-cache-remove.....	3781
11.2.1.12 nf-profile.....	3781
11.2.1.13 pdn-session-db.....	3783
11.2.1.14 pfcf-audit.....	3784
11.2.1.15 pfcf-stale-session-audit-status.....	3785
11.2.1.16 policy-on-the-fly-update-stats.....	3786
11.2.1.17 radius.....	3787
11.2.1.18 test.....	3788
11.2.1.19 accounting.....	3788
11.2.1.20 authentication.....	3789
11.2.1.21 perform.....	3790
11.2.1.22 dhcp.....	3791
11.2.1.23 test.....	3791
11.2.1.24 mobile-gateway.....	3793
11.2.1.25 pdn.....	3793
11.2.1.26 apn.....	3794
11.2.1.27 dedicated-bearer-monitoring.....	3794
11.2.1.28 canary-test.....	3795
11.2.1.29 user-range.....	3795
11.2.1.30 dedicated-bearer-monitoring.....	3796
11.2.1.31 imsi.....	3796
11.2.1.32 shutdown.....	3797
11.2.1.33 force-clear.....	3797
11.2.1.34 bearer-context.....	3797
11.2.1.35 diameter-peer-destination-host.....	3798
11.2.1.36 path.....	3800
11.2.1.37 force-gtpu-echo-req-action.....	3802
11.2.1.38 nf-register.....	3802
11.2.1.39 pfcf-stale-session-audit-cancel.....	3803
11.2.1.40 pfcf-stale-session-audit-init.....	3803
11.2.1.41 profile.....	3804
11.2.1.42 diameter-peer.....	3804
11.2.1.43 refresh-dns.....	3805
11.2.1.44 fqdn-group.....	3805

11.2.1.45	clear-list.....	3806
11.2.1.46	refresh-list.....	3806
11.2.1.47	pcscf-group.....	3807
11.2.1.48	refresh-list.....	3807
11.2.1.49	radius.....	3807
11.2.1.50	test.....	3808
11.2.1.51	accounting.....	3808
11.2.1.52	authentication.....	3810
11.2.1.53	router.....	3811
11.2.1.54	ip-local-pool.....	3811
11.2.1.55	reserve-ipv4.....	3812
11.2.1.56	reserve-ipv6.....	3813
11.2.1.57	security.....	3813
11.2.1.58	authentication-server-check.....	3814
11.2.2	OAM command descriptions.....	3816
11.2.2.1	oam.....	3816
11.2.2.2	dns.....	3817
11.2.2.3	gtp-ping.....	3820
11.2.2.4	radius-ping.....	3822
11.2.2.5	ue-ping.....	3824
11.2.3	Ping command descriptions.....	3826
11.2.3.1	ping.....	3826
12	Debug command reference.....	3832
12.1	Debug command hierarchies.....	3832
12.1.1	Call trace debug command hierarchy.....	3832
12.1.2	Profile debug command hierarchy.....	3832
12.2	Debug command descriptions.....	3833
12.2.1	Debug commands.....	3833
12.2.2	Call trace debug command descriptions.....	3833
12.2.2.1	call-insight.....	3833
12.2.2.2	all-ues.....	3833
12.2.2.3	ue.....	3834
12.2.2.4	ue-mask.....	3836
12.2.3	Profile debug command descriptions.....	3837
12.2.3.1	profile.....	3837
12.2.3.2	emergency-db-errors.....	3837
13	VPRN and router command reference.....	3839
13.1	VPRN and router command hierarchies.....	3839
13.1.1	VPRN command hierarchy.....	3839
13.1.2	Router command hierarchy.....	3841
13.2	VPRN and router command descriptions.....	3843
13.2.1	VPRN and router configuration command descriptions.....	3843
13.2.1.1	router.....	3843
13.2.1.2	vpn.....	3844

13.2.1.3	bgp.....	3845
13.2.1.4	rapid-update.....	3845
13.2.1.5	enable-l2-access.....	3846
13.2.1.6	interface.....	3847
13.2.1.7	mc-redirect-mode.....	3847
13.2.1.8	dynamic-sap.....	3848
13.2.1.9	egress.....	3849
13.2.1.10	filter.....	3849
13.2.1.11	qos.....	3850
13.2.1.12	ingress.....	3851
13.2.1.13	filter.....	3851
13.2.1.14	qos.....	3852
13.2.1.15	steering.....	3853
13.2.1.16	ip-local-pool.....	3854
13.2.1.17	exclusive.....	3854
13.2.1.18	group-name.....	3855
13.2.1.19	hold-timer-sec.....	3855
13.2.1.20	ipv4-prefix.....	3856
13.2.1.21	ipv6-prefix.....	3857
13.2.1.22	next-hop-gw-ipv4.....	3858
13.2.1.23	next-hop-gw-ipv6.....	3859
13.2.1.24	remote.....	3860
13.2.1.25	reserved-ip-list.....	3860
13.2.1.26	reserved-ipv4.....	3861
13.2.1.27	reserved-ipv6.....	3862
13.2.1.28	round-robin.....	3863
13.2.1.29	suppress-dotzero-broadcast-address.....	3863
13.2.1.30	suppress-network-broadcast-address.....	3864
13.2.1.31	transit-sub.....	3864
13.2.1.32	ip-pool-distribution.....	3865
13.2.1.33	l2tp.....	3866
13.2.1.34	local-address.....	3866
13.2.1.35	group.....	3867
13.2.1.36	tunnel.....	3867
13.2.1.37	peer.....	3868
13.2.1.38	mg-group.....	3869
13.2.1.39	parent-router.....	3870
13.2.1.40	twamp-light.....	3871
13.2.1.41	reflector.....	3871
13.2.1.42	prefix.....	3872
13.2.1.42.1	description.....	3873
13.2.1.43	bgp.....	3873
13.2.1.44	rapid-update.....	3874
13.2.1.45	interface.....	3875
13.2.1.46	policy-options.....	3876
13.2.1.47	policy-statement.....	3876
13.2.1.48	entry.....	3877

13.2.1.49 from.....	3877
13.2.1.50 active-mg.....	3878
13.2.1.51 state.....	3878
13.2.1.52 selective-reassembly.....	3880
13.2.2 VPRN and router show command descriptions.....	3880
13.2.2.1 dynamic-sap-using.....	3880
13.2.2.2 mg-group-using.....	3883
13.2.2.3 ip-local-pool-stats.....	3884

List of Tables

Table 1:	Default collection interval for accounting records.....	183
Table 2:	Show card output fields.....	189
Table 3:	Show mobile profile call-insight output fields.....	191
Table 4:	Show connections fields.....	199
Table 5:	Show ISM-MG ABS fields.....	201
Table 6:	Show ISM-MG CPU utilization fields.....	204
Table 7:	Show ISM-MG memory utilization fields.....	206
Table 8:	Show mobile gateway ISM-MG and MG-VM SDF output fields.....	208
Table 9:	Show MG-VM CPU utilization fields.....	211
Table 10:	Show MG-VM memory utilization fields.....	213
Table 11:	Show mobile gateway system information.....	216
Table 12:	Show redundancy MC mobile output fields.....	223
Table 13:	Status of the admin MC-complete-ue-sync and oper MC-complete-ue-sync on the master and slave nodes	229
Table 14:	Values of the Last Oper Role Chg Rsn field.....	229
Table 15:	Show redundancy MC mobile red state TCP connection output fields.....	232
Table 16:	Show APN configuration output fields.....	989
Table 17:	Show APN detailed configuration output fields.....	1005
Table 18:	Show PDN APN statistics output fields.....	1016
Table 19:	Show DHCPv6 service output fields.....	1027
Table 20:	Show PDN APN eMPS advance priority call output fields.....	1030
Table 21:	Show PDN APN DNS redirect statistics output fields.....	1031
Table 22:	Show PDN APN FQDN group statistics output fields.....	1032
Table 23:	Show PDN APN override statistics output fields.....	1034
Table 24:	Show PDN APN P-CSCF statistics output fields.....	1035
Table 25:	Show PDN APN Gx assume positive statistics output fields.....	1037
Table 26:	Show PDN APN SNMP for PGW and combination gateway call flow statistics output fields...	1038
Table 27:	Show PDN APN transport statistics output fields.....	1041
Table 28:	Show APN IP address allocation configuration output fields.....	1047
Table 29:	Show PDN APN overload output fields.....	1056
Table 30:	Show PDN application server statistics output fields.....	1058
Table 31:	Show PDN bearer context PFCP-u shared tunnel session output fields.....	1116
Table 32:	Show PDN bearer context SUPI detail output fields.....	1119
Table 33:	Show PDN canary test output fields.....	1178
Table 34:	Show PDN clear status output fields.....	1181
Table 35:	Show PDN error indication output fields.....	1183
Table 36:	Show PDN fixed subscriber output fields.....	1186

Table 37:	Show PDN fixed subscriber detail output fields.....	1187
Table 38:	Show PDN GTP load and overload control information output fields.....	1189
Table 39:	Show PDN load and overload control information statistics output fields.....	1193
Table 40:	Show PDN local overload control information statistics output fields.....	1198
Table 41:	Show NF peer statistics.....	1205
Table 42:	Show PDN NWu output fields.....	1209
Table 43:	Show PDN path management statistics common output fields.....	1212
Table 44:	Show PDN path management statistics for N3 and N9.....	1215
Table 45:	Show PDN path management statistics for PFCP (Sx-N4).....	1217
Table 46:	Show PDN path management Rf, Gx, and Gy statistics output fields.....	1218
Table 47:	Show PDN path management S2a and S2b statistics output fields.....	1222
Table 48:	Show PDN path management S5, S5u, S8, Gn, and Gp statistics output fields.....	1224
Table 49:	Show PDN path management S6b statistics output fields.....	1226
Table 50:	Show PDN PCMD statistics output fields.....	1232
Table 51:	Show PDN context PFCP-u shared tunnel sessions output fields.....	1311
Table 52:	Show PDN context SUPI detail output fields.....	1320
Table 53:	Show PDU session Nchf account detail statistics.....	1349
Table 54:	Show PDU session ran-ue-ngap-id detail statistics.....	1354
Table 55:	Show PDU session ran-ue-ngap-id acct statistics.....	1358
Table 56:	Show PDN PLMN list charging profile output fields.....	1360
Table 57:	PFCP load and overload configuration output fields.....	1361
Table 58:	PFCP load and overload statistics output fields.....	1363
Table 59:	Local load control statistics for PFCP UP function output fields.....	1364
Table 60:	Local overload control statistics for PFCP output fields.....	1366
Table 61:	PGW reselection isolation output fields.....	1367
Table 62:	Show PDN QoS flow context SUPI detail output fields.....	1370
Table 63:	Show PDN QoS flow ran-ue-ngap-id detail statistics output fields.....	1378
Table 64:	Show PDN reference point peer common output fields.....	1383
Table 65:	Path Mgmt state values.....	1383
Table 66:	Show PDN Ga ref point peers output fields.....	1386
Table 67:	Show PDN Gx and Rf ref point peers output fields.....	1387
Table 68:	Show PDN CMG Gx ref point peers output fields.....	1389
Table 69:	Show PDN Gy Ref point peers output fields.....	1390
Table 70:	Gy Diameter routing table reference point peers output fields.....	1392
Table 71:	Show N2 reference point peer output fields.....	1394
Table 72:	Show PDN Pi ref point peers output fields.....	1396
Table 73:	Show PDN RADIUS ref point peers output fields.....	1397
Table 74:	Show PDN S1U, S2a-GTP, S2b, Gp, and Gn ref point peers output fields.....	1399
Table 75:	Show PDN S2a ref point peers output fields.....	1400

Table 76:	Show PDN S4 ref point peers output fields.....	1401
Table 77:	Show PDN S5 and S8 ref point peers output fields.....	1402
Table 78:	Show PDN S5u ref point peers output fields.....	1403
Table 79:	Show PDN S6b ref point peers output fields.....	1403
Table 80:	Show PDN S11u ref point peers output fields.....	1405
Table 81:	Show PDN STa ref point peers output fields.....	1406
Table 82:	Show PDN SWm and SWm Diameter routing point peers output fields.....	1407
Table 83:	Show PDN SWm Diameter routing point peers output fields.....	1409
Table 84:	Show PDN SWw tunnel ref point peers output fields.....	1410
Table 85:	Show PDN SWw RADIUS ref point peers output fields.....	1411
Table 86:	Show CHF service statistics output fields.....	1491
Table 87:	Show NRF service statistics output fields.....	1497
Table 88:	Show NRF OAuth2.0 service statistics output fields.....	1501
Table 89:	Show NRF OAuth2.0 failure codes service statistics output fields.....	1503
Table 90:	Show PCF service statistics output fields.....	1505
Table 91:	Show PDN static predefined PCC output fields.....	1514
Table 92:	Show PDN statistics output fields.....	1518
Table 93:	Show PDN attach failure ePDG statistics output fields.....	1543
Table 94:	Show PDN UPF statistics output fields.....	1545
Table 95:	Show PDN SWu output fields.....	1550
Table 96:	Show PDN UE context SUPI detail output fields.....	1557
Table 97:	Show PDN UE context PFCP-u shared tunnel sessions output fields.....	1571
Table 98:	3GPP result codes.....	1978
Table 99:	MSISDN hexadecimal encoding.....	2019
Table 100:	Gx profile CCFH action per call flow type.....	2131
Table 101:	3GPP result codes.....	2145
Table 102:	Possible combinations of the application error and the HTTP status code.....	2312
Table 103:	Show mobile profile APN rate control output fields.....	2709
Table 104:	Show mobile profile call-insight all UEs output fields.....	2711
Table 105:	Show mobile profile call-insight UE output fields.....	2713
Table 106:	Show mobile profile charging CCFH profile output fields.....	2716
Table 107:	Show mobile profile CDR AVP options output fields.....	2719
Table 108:	Show CHF charging profile output fields.....	2724
Table 109:	Show ePDG charging profile statistics output fields.....	2727
Table 110:	Show mobile profile output fields (event triggers FBC).....	2732
Table 111:	Show mobile profile charging output fields (Gy AVP).....	2737
Table 112:	PGW charging profile output fields.....	2743
Table 113:	SGW charging profile statistics output fields.....	2749
Table 114:	TDF charging profile statistics output fields.....	2753

Table 115:	TWAG charging profile statistics output fields.....	2756
Table 116:	UPF profile statistics output fields.....	2757
Table 117:	DCCA profile statistics output fields.....	2760
Table 118:	DHCPv4 server profile statistics output fields.....	2768
Table 119:	Diameter profile statistics output fields.....	2770
Table 120:	Show Diameter peer output fields.....	2773
Table 121:	Show Diameter error mapping profile statistics output fields.....	2775
Table 122:	Show Diameter peer group list output fields	2776
Table 123:	Show DNS error mapping profile output fields	2778
Table 124:	Show DNS profile output Fields	2779
Table 125:	Show emergency profile output fields	2781
Table 126:	Show mobile profile output fields.....	2783
Table 127:	Show GTP profile output fields	2785
Table 128:	Show GTP prime group profile output fields	2791
Table 129:	Show Gx profile output fields	2795
Table 130:	Show HTTP/2 profile output fields	2802
Table 131:	Show IPSec profile statistics output fields.....	2807
Table 132:	Show ARP list output fields with all the configured ARP Lists.....	2825
Table 133:	Show ARP list output fields with the excluded ARP names.....	2826
Table 134:	Show broadcast PLMN list information output fields.....	2827
Table 135:	Show ARP list output fields with the excluded ARP names.....	2828
Table 136:	Show cell-id list information output fields.....	2830
Table 137:	Show FQDN list output fields	2831
Table 138:	Show NF ID list output fields	2833
Table 139:	Show OI override mapping list output fields	2834
Table 140:	Show port list output fields	2837
Table 141:	Prioritized IP address list output fields	2838
Table 142:	Show slice instance list output fields	2840
Table 143:	Show slice list output fields	2842
Table 144:	Show mobile profile MME group output fields.....	2846
Table 145:	Show mobile N1 profile output fields.....	2847
Table 146:	Show N2 peer profile output fields.....	2849
Table 147:	Show network trace profile output fields.....	2851
Table 148:	Show mobile profile client profile output fields.....	2853
Table 149:	Show mobile profile group label output fields.....	2855
Table 150:	Show mobile profile target profile output fields.....	2856
Table 151:	Show mobile profile NPCF profile output fields.....	2858
Table 152:	Show mobile profile overload control profile output fields.....	2861
Table 153:	Show mobile profile PCMD output fields.....	2863

Table 154:	Show mobile profile PFCP output fields.....	2869
Table 155:	Show mobile profile PFCP translation output fields.....	2871
Table 156:	Show mobile profile PFCP UP peer list output fields.....	2873
Table 157:	Show mobile profile policy rule unit output fields.....	2877
Table 158:	Policy options profile Gx AVP options output fields.....	2882
Table 159:	Show mobile profile error mapping output fields.....	2900
Table 160:	Show mobile profile STCP profile output fields.....	2904
Table 161:	Show mobile profile SWm error mapping profile output fields.....	2909
Table 162:	Show mobile profile TAI LAI list output fields.....	2913
Table 163:	Show mobile profile UDP tunnel group output fields.....	2916
Table 164:	Show mobile profile UMTS QoS policy output fields.....	2918
Table 165:	Show IPSec certificate profile output fields.....	3042
Table 166:	Show IPSec certificate profile association output fields.....	3043
Table 167:	Show IPSec certificate output fields.....	3045
Table 168:	Show certificate output fields.....	3047
Table 169:	Show IPSec gateway name statistics output fields.....	3049
Table 170:	Show IPSec gateway service statistics output fields.....	3052
Table 171:	Show IPSec gateway tunnel detail statistics output fields.....	3055
Table 172:	Show IPSec gateway name and tunnel statistics output fields.....	3062
Table 173:	Show IPSec gateway tunnel count statistics output fields.....	3063
Table 174:	Show IPSec security policy output fields.....	3067
Table 175:	Show IPsec tunnel statistics output fields.....	3072
Table 176:	Show IPsec tunnel detail statistics output fields.....	3074
Table 177:	Show IPSec gateway tunnel count statistics output fields.....	3078
Table 178:	Match configuration protocols.....	3210
Table 179:	Destination MAC address mask formats.....	3219
Table 180:	Source MAC address mask formats.....	3220
Table 181:	Service using output fields.....	3304
Table 182:	LI target Gx and Gx IMSI ID output fields.....	3314
Table 183:	LI target IMSI ID detail output fields.....	3315
Table 184:	Show AA ISA group statistics output fields.....	3604
Table 185:	Show ISA AA group load-balancing output fields.....	3605
Table 186:	Show ISA AA group load-balancing unassigned output fields.....	3607
Table 187:	Show AA group application subscriber list output fields.....	3614
Table 188:	Show AA group application group output fields.....	3617
Table 189:	Show AA group application output fields.....	3620
Table 190:	Show AA group certificate profile output fields.....	3622
Table 191:	Show AA group Cflowd collector output fields.....	3624
Table 192:	Show AA group Cflowd collector output fields.....	3626

Table 193:	Show AA group Cflowd status output fields.....	3629
Table 194:	Show AA group partition Cflowd status output fields.....	3630
Table 195:	Show DNS Enrichment output fields.....	3632
Table 196:	Show DNS IP cache output fields.....	3635
Table 197:	Show AA group event log output fields.....	3638
Table 198:	Show AA group HTTP notification output fields.....	3643
Table 199:	Show AA group AA-sub count detail output fields.....	3648
Table 200:	Show AA group policy charging filter output fields.....	3654
Table 201:	Show AA group policy charging group output fields.....	3656
Table 202:	Show AA group AA-sub policers output fields.....	3658
Table 203:	Show policer output fields.....	3668
Table 204:	Show AA group HTTP redirect output fields.....	3671
Table 205:	Show AA traffic type TLS family output fields.....	3684
Table 206:	Show web-service URL filter output fields.....	3689
Table 207:	Show AA URL list output fields.....	3690
Table 208:	Show AA HTTP redirect error codes output fields.....	3695
Table 209:	Show AA group session filter output fields.....	3702
Table 210:	Show DNS IP cache debug output fields.....	3719
Table 211:	Show output fields of the dynamic rating statistics.....	3722
Table 212:	Dedicated dealer monitoring output fields.....	3775
Table 213:	Show service dynamic-sap-using output fields.....	3882
Table 214:	Show service mg-group-using output fields.....	3884

1 Getting started

Get general information about this guide.

About this guide

This guide describes features and functionality provided by the 7750 Service Router (SR) Mobile Gateway (7750 SR MG) and Cloud Mobile Gateway (CMG). It describes concepts and descriptions of the Command Line Interface (CLI) syntax and command reference.

The 7750 SR MG and the CMG products do not differ in network elements and configuration aspects, and the information described in this guide is generally applicable to both products, with the exception of the hardware information which is applicable to the 7750 SR MG only. For CMG-specific information, such as operation on physical hardware and virtual machine installation, refer to the *CMG Installation Guide*.



NOTE:

- Configuration outputs shown in this guide are examples only; actual displays may differ depending on supported functionality and user configuration.
- The CMG-specific administration, license, tools, and show commands are described in the *CMG Installation Guide*.
- This guide covers content for the release specified on the title page of the guide, and may also contain content that will be released in later maintenance loads. Refer to the applicable *7750 MG and CMG Release Notes* for information about features supported in each load of the release software.
- The information in this guide is intended to be used in conjunction with the other guides in the 7750 SR MG and CMG, and the 7750 SR configuration guides. The 7750 SR guides describe the SR OS service features that are supported by the 7750 SR MG and CMG and are mostly used without modification. These features apply to all mobile gateway types, unless explicitly noted.
- For more information about the acronyms included in this guide, refer to the *Acronyms and Terms*.

Audience

This guide is intended for network administrators who are responsible for configuring routers. It is assumed that the network administrators have an understanding of networking principles and configurations.

2 Summary of changes

Changes between guide issues.

For more information about the CMG features and enhancements, refer to the *CMG Feature Overview*.

Changes between issue 9-2 (25.3.R3) and issue 10-0 (25.7.R1)

Feature or enhancement	Description	See
3GPP baseline updating to Rel 17 for SMF/UPF Lawful Intercept (MG-4100)	Added support for configuring the 3GPP release 17 baseline for the 5G LI interfaces	<i>config>li>mobile>3gpp-5g-release</i> (added the rel-17 option)
4G MPS support on ePDG (MG-4655)	Added 4G MPS support on ePDG	• <i>config>mobile>pdn>wps-profile</i> • <i>config>mobile>profile>wps-profile</i> – <i>arp</i> • <i>message-priority</i> – <i>all</i> – <i>gtpc</i> – <i>up-dscp-marking</i> • <i>down-link</i> • <i>up-link</i> • <i>show>mobile>profile>wps-profile</i> • <i>show>mobile>pdn>ref-point-stats s2b</i> (enhanced the output) • <i>show>mobile>pdn>pdn-context imsi detail</i> (enhanced the output) • <i>show>mobile>pdn>statistics</i> (enhanced the output)
5GSA virtual-DNN support (CC-based) (MG-4470)	Added support for CC-based DNN override mechanism	<i>config>mobile>pdn>apn>upf-selection>dnn</i>

Feature or enhancement	Description	See
CM-IDLE timer enhancement on SMF (MG-4287)	Added support for CM-IDLE timer on the SMF	<code>config>mobile>pdn>apn>cm-idle-timeout</code>
Disable UDM services per APN (MG-4532)	Added support for disabling the Nudm_SubscriberData Management and Nudm_UEContextManagement services	• <code>config>mobile>pdn>apn>udm-sdm-disabled</code> • <code>config>mobile>pdn>apn>udm-uecm-disabled</code>
ePDG Diffie-Hellman group 19 (MG-4317)	Added support for Diffie-Hellman group 19, which is of type Elliptic Curve Prime, for both IKEv2 and ESP protocols; additionally, increased the number of IKEv2 and ESP security proposal entries to 20, for each protocol type	• <code>config>mobile>profile>ipsec-profile>esp-sa>proposal-entry</code> (updated the value range) – <code>dh-group</code> (updated the value range) • <code>config>mobile>profile>ipsec-profile>ike-sa>proposal-entry</code> (updated the value range) – <code>dh-group</code> (updated the value range) • <code>show>mobile>pdn>ref-point-stats nwu</code> • <code>show>mobile>pdn>ref-point-stats swu</code> • <code>show>mobile>pdn>ref-point-stats swu-nwu</code> • <code>clear>mobile>pdn>ref-point-stats swu [card slot-number]</code>
I-UPF without the I-SMF (MG-4420)	Added support to the SMF for controlling both the I-UPF and the anchor UPF without the I-SMF	<code>config>mobile>pdn>i-upf-selection</code>
IP wildcard support for MG addr-list ip-address and remote-ip entries (MG-3315)	Added support for using wildcards within IPv6 addresses when defining flow filters	• <code>config>mobile>profile>addr-list>ip-address-group</code> • <code>config>mobile>profile>policy-options>policy-rule-unit>flow-</code>

Feature or enhancement	Description	See
		<i>description>match>remote-ip-address-group</i>
Inserting the active PCC rulebases in the HTTP header (MG-2063)	Added support for enriching the HTTP header with the active PCC rulebase name	<i>config>mobile>profile>pfcf>pfcf-translate-profile>field</i> (added the <i>hdr-enrich-pcc-rulebase</i> option) <i>config>app-assure>group>http-enrich>field</i> (added the <i>pcc-rulebase</i> value to the <i>field-name</i> parameter)
N11 paging dampening timer (MG-3127)	Added support for paging dampening	<i>config>mobile>pdn>sba-client-services>amf-client>namf-comm>paging-damp-timer</i>
NCGI level UPF/PSA selection (MG-4475)	Added support for UPF/PSA selection at the cell-id level	<i>config>mobile>profile>list>cell-id-list</i> <i>config>mobile>profile>list>cell-id-list>ncgi</i> <i>config>mobile>profile>node-selection>client-profile>entry>match>cell-id-list</i> <i>show>mobile>profile>list>cell-id-list</i> <i>show>mobile>profile>node-selection>client-profile</i> (enhanced the output)
Nchf converged and failopen enhancements (MG-4594)	Added support for requesting a specific amount of volume quota when sending RequestedUnits to the CHF and for overriding the PCF-provided online and offline flags when the PCF installs a dynamic rule with both online and offline flags set to false	<i>config>mobile>profile>charging>chf-profile>requested-unit-total-volume</i> <i>config>mobile>profile>charging>chf-profile>rating-group>requested-unit-total-volume</i> <i>config>mobile>profile>npcf>dynamic-pcc-online-offline-flags-override</i>

Feature or enhancement	Description	See
		<code>config>mobile>profile>policy-options>policy-rule-base>requested-unit-total-volume</code>
Preferred pcfld in model A	Added support to the SMF not to use the received PCF ID from the AMF in any selection logic, when the provisioning of "usePcflid" is set to "ignore"	<code>config>mobile>pdn>apn>pcf-selection>pcf-id</code>
SMF configuration of query parameters for UDM sm-data (MG-4509)	Added support for excluding the PLMN ID when the session management subscription data is retrieved from the UDM	<code>config>mobile>profile>udm-sdm-profile>get-sm-subscription-data>exclude-plmn-id</code>
SMF to support PCF provided CHF addresses (MG-4194)	Added support for specifying if the PCF is allowed to send the CHF addresses within the ChargingInfo IE	<code>config>mobile>profile>npcf>allow-pcf-provided-chf-address</code>
SMF use of UDM immediate report (MG-4582)	Added support for additional recovery option for UDM failure handling where the failed service operation is reattempted	<code>config>mobile>pdn>sba-client-services>udm-client>nudm-sdm>supported-features</code> – <code>immediate-report</code> <code>config>mobile>pdn>sba-client-services>udm-client>nudm-sdm>supported-features>immediate-report</code> <code>config>mobile>profile>udm-sdm-profile>continue-timer</code> (added the retry value to the action command) <code>config>mobile>profile>udm-uecm-profile>continue-timer</code> (added the retry value to the action command)

Feature or enhancement	Description	See
Support of small IP pools per enterprise DNN with multiple MG groups on SMF (MG-4119)	Added support for setting an APN/ DNN type as small to assign the UE sessions of small enterprises to a single CP MG group	<code>config>mobile>pdn>apn>apn-type</code> <code>show>mobile>pdn>apn</code> (enhanced the output)
TLS Header Enrichment - Extension Types (MG-4241)	Added support to the CMG for HTTPS (TLS) header enrichment	<code>config>app-assure>group>http-enrich>tls-extension>subtype</code> (added the <i>plmn-id</i> and <i>static-string</i> values to the <i>tls extension subtype</i> variable). <code>config>app-assure>group>http-enrich>tls-extension>subtype></code> <code>aes-initialization-vector</code> <code>encode</code> <code>extension-id</code> <code>static-string</code>
UPF IP address allocation enhancements (MG-4248)	Added support for customized and pre-defined call-flow alternatives	<code>config>mobile>pdn>apn>call-sequence</code>
—	Added a limitation about the maximum number of slice lists and slices that can be configured	<code>config>mobile>profile>list>slice-list</code> <code>config>mobile>profile>list>slice-list>slice</code>
—	Added a note about the retransmit message	<code>config>mobile>profile>namf-profile>transaction-timer</code>
—	Added a note about the use of the command in live deployments	<code>debug>mobile>call-insight>all-ues</code>
—	Added the <code>callback-type-preference</code> to the following contexts: <code>config>mobile>pdn>sba-client-services>amf-client>namf-comm</code>	—

Feature or enhancement	Description	See
	• <code>config>mobile>pdn>sba-client-services>amf-client>namf-evts</code> • <code>config>mobile>pdn>sba-client-services>pcf-client>npcf-smpolicycontrol</code> • <code>config>mobile>pdn>sba-client-services>udm-client>nudm-uecm</code>	
—	Removed the PDN shutdown requirement from selected SBI configuration commands	• <i>SBA client services PDN command descriptions</i> • <i>SBA server services PDN command descriptions</i>
—	Updated the description of the <i>config>mobile>pdn>n2>supported-ta-item>tac</i> command	—
—	Updated <code>udp-port</code> to include the value range for 7750 SR MG	• <i>config>service>vprn>twamp-light>reflector</i> • <i>config>router>twamp-light>reflector</i>

Changes between issue 9-1 (25.3.R2) and issue 9-2 (25.3.R3)

Feature or enhancement	Description	See
CMG CNF ISSD for 1:1 (CUPS / 5GC SMF, UPF) (MG-4404)	Added ISSD support for 1:1 intra-CNF redundancy	<i>admin>redundancy>mg-switchover>group</i> (added the downgrade) option

Changes between issue 9-0 (25.3.R1) and issue 9-1 (25.3.R2)

Feature or enhancement	Description	See
Cflowd RTT field in microseconds and removal of 0 values (MG-4555)	Enhanced the AA comprehensive Cflowd reporting of client side and network side round-trip time during TCP session establishment	<code>config>app-assure>group>cflowd>comprehensive>template>dynamic-fields>field</code>
Support of additional 5G and other Cflowd fields (MG-4319)	Added support of additional configurable fields (5G and others) to be included in the AA comprehensive Cflowd records	<code>config>mobile>profile>pfcg>pfcg-translate-profile>field</code> <code>config>app-assure>group>cflowd>comprehensive>template>dynamic-fields>field</code> <code>config>app-assure>group>cflowd>system-name</code>
DNS enrichment (MG-4320)	Added support for enriching DNS Request messages with an additional extension to be used by the DNS servers for content filtering	<code>config>app-assure>group>dns-enrich</code> • <code>gateway</code> • <code>description</code> • <code>option</code> • <code>packet-size-limit</code> • <code>shutdown</code> <code>config>app-assure>group>policy>aqp>entry>action>dns-enrich</code> <code>config>app-assure>group>cflowd>comprehensive>template>dynamic-fields>field</code> (added the ednsSubPolicy Id value) <code>show>app-assure>group>dns-enrich</code>
Support for HTTP header encryption method encryption aes-256-gcm-sha384 salt (MG-4720)	Added support to AA for encrypting HTTP headers with the aes256 gcm-sha384 salt algorithm	<code>config>app-assure>group>http-enrich>field>encode</code>

Changes between issue 8-0 (24.11.R1) and issue 9-0 (25.3.R1)

Feature or enhancement	Description	See
4G WPS support on ePDG based on preconfigured ARP values (MG-4517)	Added 4G WPS support on ePDG	• <i>config>mobile>pdn>wps-profile</i> • <i>config>mobile>profile>wps-profile</i> – <i>arp</i> • <i>message-priority</i> – <i>all</i> – <i>gtpc</i> – <i>up-dscp-marking</i> • <i>down-link</i> • <i>up-link</i> • <i>show>mobile>profile>wps-profile</i> • <i>show>mobile>pdn>ref-point-stats s2b</i> (enhanced the output) • <i>show>mobile>pdn>pdn-context imsi detail</i> (enhanced the output) • <i>show>mobile>pdn>statistics</i> (enhanced the output)
base64 encoding of AES encrypted Header Enrichment fields (MG-4503)	Added support for encoding in base64 AES encrypted Header Enrichment fields	<i>config>app-assure>group>http-enrich>field>encode</i>
CMG adapts LI solutions towards 3rd party LIMS (MG-4615)	Added support for configuring how the Network ID (NEID) is populated	<i>config>li>mobile>iri>neid</i>
Control UPF Selection via N7 policy rules (MG-4549)	Added support for UPF selection of PDU session through the N7 policy rules	<i>config>mobile>profile>policy-options>action-rule-unit>dnai-psa-select</i>

Feature or enhancement	Description	See
SCP statistics (MG-3479)	Added support for new SCP statistics	<i>show>mobile>pdn>service-stats</i>
SMF service recovery after UDM failure (MG-4058)	Adds recovery actions for sessions established while the UDM was unavailable and defines UDM blocking to avoid signaling towards unavailable UDM	<i>config>mobile>profile>udm-usm-profile</i> • <i>block-timer</i> • <i>continue-timer</i> <i>config>mobile>profile>udm-uecm-profile</i> • <i>block-timer</i> • <i>continue-timer</i> Added the <i>block</i> parameter: • <i>fh-session-no-response</i> • <i>http-status-code</i> Added the <i>udm</i> parameter: • <i>show>mobile>pdn>nf-peers</i>
Start time parameter to be included in all CDRs (MG-4519)	Added the <i>always-include</i> option in the command	<i>config>mobile>profile>charging>cdr-avp-options>start-time</i>
Support of CDR files without headers (MG-4528)	Added support for excluding the CDR file headers and the CDR headers as defined in 3GPP TS 32.297	<i>config>mobile>pdn>charging>remove-3gpp-headers</i>
—	Added a note about disabling the <i>client-identifier-msisdn-encoded</i>	<i>tools>perform>dhcp>test</i>
—	Added a note about the aging of locally configured peers	<i>tools>perform>mobile>pdn>force-clear>diameter-peer-destination-host</i>

Feature or enhancement	Description	See
—	Updated the value range of the http-to-5gsm and the 5gsm-cause parameters	<i>config>mobile>profile>sbi-error-mapping-profile>http-to-5gsm</i>
—	Updated the value range of the http-to-gtp and the gtp-error parameters	<i>config>mobile>profile>sbi-error-mapping-profile>http-to-gtp</i>

Deprecated commands	
Command/parameter	Context
disable-skip-indicator	config>mobile>pdn>apn
qfi	config>mobile>profile>qfi-mapping-profile>static-mapping
5qi arp average-window max-data-burst-vol priority-level qnc	config>mobile>profile>qfi-mapping-profile>static-mapping>qfi
ue-type	config>mobile>profile>canary-test>user-range

Changes between issue 7-1 (24.7.R1) and issue 8-0 (24.11.R1)

Feature or enhancement	Description	See
4G and 5G WPS support on SGW/PGW/SMF based on preconfigured ARP and QCI_5QI values (MG-3045)	Added support for 4G and 5G WPS on SGW/PGW/SMF based on preconfigured ARP and QCI_5QI values	• <i>config>mobile>pdn>wps-profile</i> • <i>config>mobile>profile>wps-profile</i> – <i>arp</i> • <i>message-priority</i> – <i>all</i> – <i>gtpc</i> – <i>pfc</i> – <i>sbi</i> – <i>up-dscp-marking</i> • <i>down-link</i> • <i>up-link</i> • <i>show>mobile>profile>wps-profile</i> • <i>show>mobile>pdn>apn statistics</i> (enhanced the output) • <i>show>mobile>pdn>bearer-context</i> (enhanced the output) • <i>show>mobile>pdn>call-flow-stats</i> (enhanced the output) • <i>show>mobile>pdn>pdn-context imsi detail</i> (enhanced the output)
AA Cflowd: throughput and latency fields (MG-4301)	Enhanced Application Assurance Cflowd comprehensive template with new selectable fields	• <i>config>app-assure>group>policer</i> – <i>action</i> • <i>config>app-assure>group>cflowd>comprehensive>template>dynamic-fields>field</i>

Feature or enhancement	Description	See
CHF/N40 reporting behavior modification (MG-4439)	Added support for reporting charging data to the CHF (proprietary functionality)	<code>config>mobile>profile>charging>chf-profile>custom-online-offline-reporting</code> <code>config>mobile>profile>policy-options>charging-rule-unit>custome-online-sru</code>
Enhanced LI X2/X3 link down detection (MG-4456)	Added support for detecting X2 and X3 interface connection status	<code>config>li>mobile>tcp-keepalive</code> <code>idle-time</code> <code>interval</code> <code>retries</code> <code>shutdown</code>
Enabling N7 PCF CMG specific behavior in mixed vendor GW network (MG-4410)	Added support for configuring the Nokia proprietary IE "vendor Specific-006527": {"smfNokia": 2}	<code>config>mobile>profile>n7-attributes-profile>nokia-smf-indicator-as-vsa</code>
Enhance CMG capacity to handle GTP-U echo requests (MG-4206)	Added support for distributing the GTP-U Echo Requests across the MG groups	<code>config>mobile>pdn>fwd-gtpu-echo-req-to-mg</code>
ICR SWO based on Gx failover (MG-4223)	Added support for configuring the ICR Gx connection failure detection	<code>config>redundancy>multi-chassis>peer>mc-mobile>diameter-connection-lost-detection</code> – <code>gx-failover</code> <code>show>redundancy>multi-chassis>mc-mobile</code> (enhanced the output)
N40 failopen to use a separate offline CHF (MG-4488)	Added support for configuring the Fail-Open failure handling parameters on a separate CCFH profile	<code>config>mobile>profile>charging>ccfh-profile>fail-open</code> <code>config>mobile>profile>charging>smf-charging>chf-selection>fail-open</code> – <code>chf-profile</code>

Feature or enhancement	Description	See
		– <i>nf-id-list</i>
PGW usage of UDM (MG-3400)	Added support for enabling the N10 functionality in the PGW for sessions with disabled 5G interworking	• <i>config>mobile>pdn>apn>converged-interface-support>converged-n10</i> • <i>config>mobile>pdn>sba-client-services>converged-interface-support>converged-n10</i> • <i>config>mobile>pdn>apn>slices>udm-based-slice-selection</i> • <i>config>mobile>profile>list>slice-list>slice>tai-list</i> • <i>config>mobile>profile>udm-sdm-profile>get-sm-subscription-data>send-sd-with-reserved-value</i> • <i>show>mobile>pdn>apn</i> • <i>show>mobile>profile>list>slice-list</i> • <i>show>mobile>profile>udm-sdm-profile</i>
PLMN-based SEPP selection (MG-4116)	Adds support for SMF SEPP selection based on the target PLMN	<i>config>mobile>pdn>sba-client-services>sepp-client>sepp>plmn-list</i>
SMF support of TAI in UF discovery (MG-4168)	Adds support for including the TAI in the query parameter when the SMF uses the NRF-based UPF discovery	<i>config>mobile>profile>node-selection>target-profile>entry>match>nrf-based</i>
SMF supports optimal PSA UPF reselection (MG-3069)	Added support for an optimal PSA UPF reselection in 5GC when the UE moves out of the UPF serving area	• <i>config>mobile>pdn>apn>ssc>pdu-session-reactivation-trigger</i> • <i>config>mobile>pdn>apn>ssc>pfcf-user-plane-inactivity-iupf-case</i> • <i>config>mobile>pdn>apn>ssc>upf-reselection</i> (added the i -

Feature or enhancement	Description	See
		smf option to the <i>ssc-1-option</i> , <i>ssc-2-option</i> , and <i>ssc-3-option</i> parameters) • <i>config>mobile>pdn>apn>ssc>uuid-mask-length</i> • <i>show>mobile>pdn>apn</i> (added an output example)
Support of Protect Dedicated Bearer per UPF peer and optional APN (MG-4307)	Adds support for releasing sessions without dedicated bearer in a UPF peer and optionally for a specified APN	<i>clear>mobile>pdn>bearer-context[protect-dedicated-bearers] up-peer ipv4-address ipv6-address [apn name]</i>
UPF support of sdf-level QoS enforcement using dynamic PCC rules (MG-4232)	Added support for policing per service data flow for dynamic policy rules	<i>config>mobile>pdn>apn>dynamic-rule-policing</i>
UPF supports 5G LI_X3 Sequence Number (MG-4255)	Added support of the X3 Sequence Number IE in the UPF	<i>config>li>mobile>li-x3>li-x3-sequence</i>
—	Added the <i>relax-aar-dest-realm</i> command	<i>config>mobile>pdn>s6b>relax-aar-dest-realm</i>
—	Added a note about modifying on-the-fly an IP address list that is already referenced in a configuration	<i>config>mobile>profile>addr-list</i>
—	Added a note that this command is not supported in the current release	<i>config>mobile>profile>charging>cdr-avp-options>ims-sig-context</i>
—	Added the <i>ue-entry</i> command	<i>config>mobile>profile>canary-test>user-range>ue-entry</i>
—	Added the <i>canary-test</i> command	<i>show>mobile>pdn>canary-test</i>

Feature or enhancement	Description	See
—	Updated the note to indicate that a PDN shutdown is required when CMG operates in LB-less mode	<i>config>mobile>pdn>s11>signaling>gtp-c>ddn-extended-buffering-time</i>
—	7750 SR MG-specific content is removed in phases.	—

Changes between issue 7-0 (24.7.R1) and issue 7-1 (24.7.R1)

Feature or enhancement	Description	See
—	Added a note about PDN shutdown requirement	• <i>config>mobile>pdn>s1-u>signaling</i> • <i>config>mobile>pdn>s1-u>signaling>gtp-u</i> • <i>config>mobile>pdn>s1-u>signaling>gtp-u>profile</i>

Changes between issue 6-1 (24.3.R1) and issue 7-0 (24.7.R1)

Feature or enhancement	Description	See
5GS virtual DNN support (MG-4345)	Added virtual DNN support in 5GS	<i>config>mobile>pdn>apn>pcf-selection>dnn</i> (added the virtual and public options to the <i>type</i> parameter)
CMG support for network trace for non-SBI signaling messages and interfaces (MG-3499)	Added support for non-SBI interfaces in CMG and enhancements in vTAP solution	• <i>config>mobile>pdn>network-trace>instance</i> • <i>config>mobile>pdn>network-trace>instance>tapping-interface</i> • <i>config>mobile-gateway>profile>network-trace-profile>http2-profile</i>

Feature or enhancement	Description	See
		• <i>config>mobile>pdn>network-trace>instance>interface</i> (updated the context) • <i>config>mobile>pdn>network-trace>instance>network-trace-profile</i> (updated the context) • <i>config>mobile>pdn>network-trace>instance>shutdown</i> (updated the context) • <i>show>mobile>profile>network-trace-profile</i> (enhanced the output)
Configurable failure action if no response from UDM (MG-3486)	Added support for a configurable failure action if the UDM does not respond	• <i>clear>mobile>pdn>bearer-context>udm-no-get-response</i> • <i>config>mobile>profile>udm-sdm-profile>create-subscription>fh-session-no-response</i> • <i>config>mobile>profile>udm-sdm-profile>get-sm-subscription-data>fh-session-no-response</i> • <i>config>mobile>profile>udm-uecm-profile>register-smf>fh-session-no-response</i> • <i>show>mobile>profile>udm-sdm-profile</i> (enhanced the output) • <i>show>mobile>profile>udm-uecm-profile</i> (enhanced the output)
DHCP troubleshooting via CLI (MG-4093)	Added support for managing and monitoring the DHCP server status and performance using test CLI commands	• <i>tools>dump>dhcp>test</i> • <i>tools>perform>dhcp>test</i>
Domestic roaming in emergency (MG-4243)	Added support for domestic roaming support for LTE Data,	<i>config>mobile>profile>gx>supported-features>user-equipment-info-extension</i>

Feature or enhancement	Description	See
	Voice over LTE, and Emergency Call services	
ePDG as CNF: Support for geo-redundancy (UE-by-UE sync) (MG-3889)	Added support for geo-redundancy in ePDG as CNF	<i>config>redundancy>multi-chassis>peer>mc-mobile>mc-complete-ue-sync</i>
ePDG as VNF: ICR improvements – sync MG-ISM complete UE records for ePDG ICR (UE-by-UE) (MG-2655)	Added support for ICR UE-by-UE reconcile synchronization in ePDG as VNF	<i>config>redundancy>multi-chassis>peer>mc-mobile>mc-complete-ue-sync</i>
ePDG support for SMF set interworking (MG-4328)	Added support for ePDG-initiated PDN session restoration	• <i>config>mobile>pdn>session-restoration-rate</i> • <i>show>mobile>pdn>call-flow-stats</i> (enhanced the output)
Non-terminating policy rules (MG-2233)	Adds support for marking predefined policy rules as Non-Terminating	• <i>config>mobile>profile>policy-options>policy-rule>non-terminating</i> • <i>config>mobile>system>non-terminating-rules</i> • <i>show>mobile>pdn>statistics</i> (enhanced the output)
Preemptive RG indication on PCC rule or rulebase (MG-4125)	Added support for requesting preemptively quota for the rating group (or rating group and service identifier)	<i>config>mobile>profile>policy-options>charging-rule-unit>default-rg</i>
Serving node change limit (MG-3585)	Added support for disabling the serving node change limit	• <i>config>mobile>profile>charging>pgw-charging>off-line>part-rec-closure-trig>serving-node-change-limit</i> • <i>config>mobile>profile>charging>sgw-charging>off-line>part-rec-closure-trig>serving-node-change-limit</i>

Feature or enhancement	Description	See
		<code>config>mobile>profile>charging>event-triggers-fbc>serving-node-change-limit</code> <code>config>mobile>profile>charging>event-triggers-qbc>serving-node-change-limit</code>
SMF support including IpDomain in Npcf message (MG-3160)	Added support for including the ipDomain IE in the N7 and Gx interfaces	<code>config>mobile>profile>n7-attributes-profile</code> <code>config>mobile>profile>n7-attributes-profile>ip-domain</code> <code>config>mobile>profile>npcf>ip-domain-type</code> <code>config>mobile>profile>npcf>n7-attributes-profile</code> <code>config>mobile>profile>pfcf>up-peer-list>peer>upf-id</code> (added the string option) <code>config>mobile>profile>policy-options>gx-avp-options>include-ipdomain-upf-id</code>
—	Added the <code>cleanup-on-no-resp</code> option	<code>tools>perform>mobile>pdn>pfcf-stale-session-audit-init</code>
—	Added the IKE transform commands	<code>config>ipsec>ike-transform</code>
—	Added the <code>state</code> command	<code>config>router>policy-options>policy-statement>entry>from>state</code>
—	Updated the note about the S2a-PGW/S2b-PGW peers (TWAG/e PDG) exception	<code>tools>perform>mobile>pdn>force-clear>path</code>

Deprecated commands	
Command/parameter	Context
trace-tls-profile	config>app-assure>group>cflowdconfigure>mobile>profile>network-trace-profile>packet-broker-server

3 CLI usage

This section describes information about the CLI structure and CLI usage.

3.1 CLI structure

7750 SR MG and CMG CLI is a command-driven interface accessible through the console, Telnet, and SSH. The CLI can be used for the configuration and management of 7750 SR MG virtualized platform.

The CLI commands are organized in a hierarchical inverted tree structure. At the highest level is the ROOT level. Below this level are other tree levels with the major command groups; for example, configuration commands and show commands are levels below ROOT.

The CLI is organized so related commands with the same scope are at the same level or in the same context. Sublevels or subcontexts have related commands with a more refined scope. Each section in this guide describes a particular branch of the tree in hierarchical format, with full descriptions of the commands at all the sublevels and subcontexts.

3.2 CLI contexts

Use the CLI to access, configure, and manage the Nokia 7750 SR MG and CMG.

CLI commands are entered at the command line prompt. Access to specific CLI commands is controlled by the permissions set by your system administrator. Entering a CLI command makes navigation possible from one command context (or level) to another.

When you initially enter a CLI session, you are in the ROOT context. Navigate to another level by entering the name of successively lower contexts. For example, enter either the `configure` or `show` commands at the ROOT context to navigate to the `config` or `show` context, respectively. For example, at the command prompt (`#`), enter `config`. The active context displays the following in the command prompt:

```
# config
# config#
```

In a specific CLI context, enter commands at that context level by entering the text. It is also possible to include a command in a lower context as long as the command is formatted in the proper command and parameter syntax.

The following examples show two methods to navigate to a service SDP ingress level.

Example: Method 1

```
# configure mobile-gateway profile call-insight
config>mobile>profile>call-insight#
```


Example: Method 2

```
config# mobile
config>mobile# profile
config>mobile>profile# call-insight
config>mobile>profile>call-insight#
```

When the syntax is incorrect, the CLI returns the following error message:

```
config# mobile
Error: Bad command.
```

For more details and features of CLI, refer to *7450 ESS, 7750 SR, 7950 XRS, and VSR Basic System Configuration Guide*.

4 Card and system management command reference

The card, system, administration, configuration redundancy, security and certificate commands that are specifically supported for the 7750 SR MG.

The 7750 SR MG also supports most of the 7750 SR card and system commands. For descriptions of all supported 7750 SR card and system commands, refer to the *7450 ESS, 7750 SR, 7950 XRS, and VSR Interface Configuration Guide*, *7450 ESS, 7750 SR, 7950 XRS, and VSR Basic System Configuration Guide*, and *7450 ESS, 7750 SR, 7950 XRS, and VSR System Management Guide*.

4.1 Card and system management command hierarchies

- *Card and system management configuration command hierarchies*
- *Card and system management show command hierarchies*

4.1.1 Card and system management configuration command hierarchies

- *Card configuration command hierarchy*
- *Mobile-gateway system configuration command hierarchy*
- *System security command hierarchy*
- *Admin redundancy command hierarchy*
- *Redundancy configuration command hierarchy*
- *Accounting policy command hierarchy*
- *Port configuration command hierarchy*

4.1.1.1 Card configuration command hierarchy

```
configure
- [no] card slot-number
- card-type card-type
- no card-type
- mda-type mda-type
- no mda-type
```

4.1.1.2 Port configuration command hierarchy

```
configure
```

- port
 - ethernet
 - [no] *eth-pdu-port*

4.1.1.3 Mobile-gateway system configuration command hierarchy

```

configure
  – mobile-gateway
    – system
      – call-insight
        – location location
        – [no] location
          – [no] disable
          – limit limit
          – no limit
          – [no] primary
        – max-files-number max-files-number
        – no max-files-number
      – [no] disable-delayed-acct
      – [no] error-indication-tx
      – group group-number [redundancy type gateway gw-id] | [resource-pool resource-
pool-id]
        – no group
          – card slot-number [working | protect]
          – no card slot-number
          – pcmd-profile profile-name {inherit | none}
          – no pcmd-profile
          – override-profile override-profile-id
          – no override-profile
          – protect-active-delay protect-active-delay
          – no protect-active-delay
          – [no] shutdown
        – [no] ipv6-path-mtu
        – [no] ipv6-udp-checksum
          – non-terminating-rules {enable | disable}
          – no non-terminating-rules
        – policer-distribution dl-only dl-percentage
        – no policer-distribution
        – resource-pool resource-pool-id redundancy type gateway gw-id
        – no resource-pool resource-pool-id
          – card slot-number [working | protect]
          – no card slot-number
          – protect-active-delay protect-active-delay

```

```

    – no protect-active-delay
    – thresholds
    – cam-utilization-threshold high-threshold high-threshold [low-threshold low-threshold]
    – no cam-utilization-threshold
    – [no] system-mg-ism group-number
    – counter {cp-isa-cpu | cp-isa-memory | cp-isa-uplink | cp-isa-downlink}
    high-threshold threshold [low-threshold threshold]
    – no counter {cp-isa-cpu | cp-isa-memory | cp-isa-uplink | cp-isa-downlink}
    – interval minutes
    – no interval
    – [no] shutdown
    – ttc-accuracy {true | false}
    – no ttc-accuracy
    – [no] ue-fragmented-packets-handling
    – [no] use-fixed-udp-src-port

```

4.1.1.4 System security command hierarchy

This section describes MG-specific system security commands. For information about additional supported system security commands, refer to the *7450 ESS, 7750 SR, 7950 XRS, and VSR System Management Guide*.

```

configure
– system
– security
– dist-cpu-protection
– policy policy-name [create]
– no policy policy-name
– protocol protocol-name [create]
– no protocol protocol-name
– ssh
– client-host-key-list
– host-key index name key-name
– no host-key index
– client-kex-list
– kex index name kex-name
– no kex index
– server-host-key-list
– host-key index name key-name
– no host-key index
– server-host-key-rsa-length server-host-key-rsa-length
– no server-host-key-rsa-length
– server-kex-list

```

- *kex index* name *kex-name*
- no *kex index*
- *tls*
 - *client-cipher-list*
 - *cipher index* name *cipher-suite-code*
 - no *cipher index*
 - *server-cipher-list*
 - *cipher index* name *cipher-suite-code*
 - no *cipher index*

4.1.1.5 Admin redundancy command hierarchy

This section describes MG-specific administration redundancy commands. For information about additional supported admin redundancy commands, refer to the *7450 ESS, 7750 SR, 7950 XRS, and VSR Basic System Configuration Guide*.

admin

- *redundancy*
 - *cpm-switchover* [now]
 - [no] *mc-master-lock*
 - *mc-mobile-switchover* mobile-gateway *gw-id* peer *ip-address* [now] [abort] [reboot]
 - [no] *mc-slave-lock*
 - *mg-switchover*
 - *group* *mg-group-id* [now] [max-preparation-time *max-preparation-time*] [upgrade | downgrade]

4.1.1.6 Redundancy configuration command hierarchy

This section describes MG-specific configuration redundancy commands. For information about additional supported configuration redundancy commands, refer to the *7450 ESS, 7750 SR, 7950 XRS, and VSR Basic System Configuration Guide*.

configure

- *redundancy*
 - *multi-chassis*
 - [no] *peer ip-address*
 - [no] *mc-mobile*
 - *bfd-enable* [service *service-id*] interface *interface-name*
 - no *bfd-enable*
 - *diameter-connection-lost-detection*
 - *gx-failover* *guard-timer*
 - no *gx-failover*
 - *hold-on-neighbor-failure* *multiplier*

- no *hold-on-neighbor-failure*
- *keep-alive-interval interval*
- no *keep-alive-interval*
- *master-traffic-detection* {enable | disable}
- no *master-traffic-detection*
- *mc-complete-ue-sync* [partial-reconcile]
- no *mc-complete-ue-sync*
- [no] *mc-redirect*
- *mobile-gateway gw-id role role*
- no *mobile-gateway*
 - *sync-mc-ipsec*
 - [no] *tunnel-group tunnel-group-id*
- *peer-connect* {enable | disable}
- no *peer-connect*
- *slave-traffic-delay-timer delay-timer*
- no *slave-traffic-delay-timer*
- *traffic-detection* {strict | relaxed}
- no *traffic-detection*
- *traffic-detection-poll-timer seconds*
- no *traffic-detection-poll-timer*

4.1.1.7 Accounting policy command hierarchy

This section describes SR accounting policy commands, and includes the record command, which supports an option that is specific to the mobile gateway. For more information about accounting policy commands, refer to the *7450 ESS, 7750 SR, 7950 XRS, and VSR System Management Guide*.

configure

- log
 - accounting-policy *acct-policy-id*
 - no accounting-policy *acct-policy-id*
 - collection-interval *minutes*
 - no collection-interval
 - [no] default
 - description *description-string*
 - no description
 - [no] include-system-info
 - *record record-name*
 - no *record*
 - [no] shutdown
 - to file *log-file-id*

4.1.2 Card and system management show command hierarchies

- *Show card command hierarchy*
- *Show certificate command hierarchy*
- *Show mobile gateway command hierarchy*
- *Show redundancy command hierarchy*

4.1.2.1 Show card command hierarchy

```
show
  - card
  - card state
  - card [slot-number]
  - card [slot-number] detail
```

4.1.2.2 Show certificate command hierarchy

This section describes SR-specific certificate show commands applicable to ePDG. For more information about ePDG certificate area enhancements, refer to the *CMG ePDG Guide*.

```
show
  - certificate
  - ca-profile
```

4.1.2.3 Show mobile gateway command hierarchy

```
show
  - mobile-gateway
  - call-insight
    - all-ues
    - files [[long] | [finished [long]] | [running [long]]]
    - ue [[detail] | [imei imei [detail]] | [imsi imsi [detail]] | [mask-name mask-name [detail]] | [msisdn msisdn [detail]] | [mac-address mac-address [detail]]]
    - ue-mask [name name]
  - connections [detail]
  - ism-mg
    - abs [statistics]
    - cpu [slot number] [mda number] [repeat repeat-number]
    - cpu [repeat repeat-number] [vm vm-id] [vm-function vm-function-number]
    - memory-pools [slot number] [mda number] [repeat repeat-number]
  [process process-name]
```

```
– memory-pools [repeat repeat-number] [vm vm-id] [vm-function vm-function-number]  
– sdf [group number]  
– mg-vm  
– cpu [vm vm-id] [vm-function vm-function-number] [repeat repeat-number]  
– memory-pools [vm vm-id] [vm-function vm-function-number] [repeat repeat-number]  
– sdf [group-number]  
– system [gateway gw-id | group group-number | [card slot-number | vm vm-id]]  
[summary] [call-insight]
```

4.1.2.4 Show redundancy command hierarchy

```
show  
– redundancy  
– multi-chassis  
– mc-mobile peer {ip-address | ipv6-address}  
– mc-mobile peer {ip-address | ipv6-address} mc-red-tcp-stats
```

4.2 Card and system management command descriptions

- *Card and system management configuration command descriptions*
- *Card and system management show command descriptions*

4.2.1 Card and system management configuration command descriptions

- *Card configuration command descriptions*
- *Generic system configuration command descriptions*
- *Mobile-gateway system configuration command descriptions*
- *Admin redundancy command descriptions*
- *Redundancy configuration command descriptions*
- *Accounting policy command descriptions*

4.2.1.1 Card configuration command descriptions

4.2.1.1.1 card

Syntax

[no] card *slot-number*

Context

config

Description

This mandatory command enables access to the chassis card Input/Output Module (IOM), slot, and MDA CLI context.

The no form of this command removes the card from the configuration. All associated ports, services, and MDAs must be shutdown.

Default

No cards are configured.

Parameters

slot-number

the slot number of the card in the chassis

Values	1 to 10 depending on chassis model
–	1 to 20 for CMG

4.2.1.1.2 card-type

Syntax

card-type *card-type*

no card-type

Context

config>card

Description

This mandatory command adds an IOM to the device configuration for the slot. The card type can be pre-provisioned, meaning that the card does not need to be installed in the chassis.



NOTE:

This command describes configuration of the ISM-MG card. For CMG configuration examples, refer to the *CMG Installation Guide*.

A card must be provisioned before an MDA or port can be configured.

A card can only be provisioned in a slot that is vacant, meaning no other card can be provisioned (configured) for that particular slot. To reconfigure a slot position, use the no form of this command to remove the current information.

A card can only be provisioned in a slot if the card type is allowed in the slot. An error message is generated if an attempt is made to provision a card type that is not allowed.

If a card is inserted that does not match the configured card type for the slot, a medium severity alarm is raised. The alarm is cleared when the correct card type is installed or the configuration is modified.

A high severity alarm is raised if an administratively enabled card is removed from the chassis. The alarm is cleared when the correct card type is installed or the configuration is modified. A low severity trap is issued when a card is removed that is administratively disabled.

Because the IOM3-XP integrated card does not have the capability to install separate MDAs, the configuration of the MDA is automatic. This configuration only includes the default parameters such as default buffer policies. Commands to manage the MDA such as shut down, named buffer pool, and so on will remain in the MDA configuration context.

An appropriate alarm is raised if a partial or complete card failure is detected. The alarm is cleared when the error condition ceases.

No cards are preconfigured for any slots. The IOM card is equipped and pre-provisioned for slot 1.

The no form of this command removes the card from the configuration. It cannot be used for the IOM card which is fixed.

Parameters

card-type

the type of card to be configured and installed in that slot

Values: iom-20g | miom2-20g | iom-20g-b|iom3-xp | imm48-1gb-sfp | imm48-1gb-tx |
imm8-10gb-xfp|imm4-10gb-xfp | imm5-10gb-xfp | imm1-oc768-tun | imm1-40gb-tun |
imm1-100gb-cfp | imm12-10gb-sf+ | iom3-xp-b | imm48-1gb-sfp-b | imm3-40gb-qsfp |
iom3-xp-c | imm-1pac-fp3 | imm-2pac-fp3 | imm48-1gb-sfp-c | ism-mg-2pac-fp3 | iom-v |
iom-v-mg | iom-v-mgd

4.2.1.1.3 mda-type

Syntax

```
mda-type mda-type  
no mda-type
```

Context

```
config>card>mda
```

Description

This mandatory command provisions a specific MDA type to the device configuration for the slot. The MDA can be pre-provisioned but an MDA must be provisioned before ports can be configured. Ports can be configured once the MDA is properly provisioned.

A maximum of two MDAs can be provisioned on an IOMXMA. Only one MDA can be provisioned per IOM MDA slot. To modify an MDA slot, shut down all port associations.

A maximum of six MDAs or eight CMAs (or a combination) can be provisioned on a 7750 SR-c12. A maximum of two MDAs or four CMAs (or a combination) can be provisioned on a 7710 SR-c4. Only one MDA/CMA can be provisioned per MDA slot. To modify an MDA slot, shut down all port associations.

CMAs do not rely on MCM configuration and are provisioned without MCMs.

An MDA can only be provisioned in a slot if the MDA type is allowed in the MDA slot. An error message is generated when an MDA is provisioned in a slot where it is not allowed.

A medium severity alarm is generated if an MDA is inserted that does not match the MDA type configured for the slot. This alarm is cleared when the correct MDA is inserted or the configuration is modified.

A high severity alarm is raised when an administratively enabled MDA is removed from the chassis. This alarm is cleared if either the correct MDA type is inserted or the configuration is modified. A low severity trap is issued if an MDA is removed that is administratively disabled.

An alarm is raised if partial or complete MDA/CMA failure is detected. The alarm is cleared when the error condition ceases.

All parameters in the MDA context remain and if non-default values are required then their configuration remains as it is on all existing MDAs.

The no form of this command deletes the MDA/CMA from the configuration. The MDA/CMA must be administratively shut down before it can be deleted from the configuration.

**NOTE:**

- This command describes configuration of the MDA type for the ISM-MG card. For information about the CMG, refer to the *CMG Installation Guide*.

- CMAs/XMAs are provisioned using MDA commands. A medium severity alarm is generated if an MDA/CMA is inserted that does not match the MDA/CMA type configured for the slot. This alarm is cleared when the correct MDA/CMA is inserted or the configuration is modified. A high severity alarm is raised when an administratively enabled MDA/CMA is removed from the chassis. This alarm is cleared if either the correct MDA/CMA type is inserted or the configuration is modified. A low severity trap is issued if an MDA/CMA is removed that is administratively disabled.

Default

No MDA types are configured for any slots by default.

Parameters

mda-type

The type of MDA selected for the slot position. The supported MDA types depend on the card type. The available MDA types can be displayed via CLI for the configured card type.

4.2.1.2 Port configuration command descriptions

4.2.1.2.1 eth-pdu-port

Syntax

```
[no] eth-pdu-port
```

Context

```
config>port>ethernet
```

Description

This command enables the detection of DL ethernet frames and session matching based on C-VLAN and S-VLAN tagging on a specific port.

The no form of this command reverts to the default (disabled).

4.2.1.3 Generic system configuration command descriptions

4.2.1.3.1 shutdown

Syntax

[no] shutdown

Context

config>mobile>system>threshold>system-mg-ism

config>mobile>system>group

Description

This command administratively disables the entity. When disabled, an entity does not change, reset, or remove any configuration settings or statistics. Many entities must be explicitly enabled using the no shutdown command.

The shutdown command administratively disables an entity. The operational state of the entity is disabled as well as the operational state of any entities contained within. Many objects must be shut down before they may be deleted.

4.2.1.4 Mobile-gateway system configuration command descriptions

4.2.1.4.1 system

Syntax

system *number*

Context

config>mobile

Description

This command enables the context to configure mobile gateway profiles.

Parameters

number

specifies a profile number identifier

Values: 0 to 255

4.2.1.4.2 call-insight

Syntax

```
call-insight
```

Context

```
config>mobile>system
```

Description

This command enables the context to configure global call trace settings.

4.2.1.4.3 location

Syntax

```
location location
```

Context

```
config>mobile>system>call-insight
```

Description

This command configures the location of the Compact Flash (CF) call trace storage.

Parameters

location

specifies the location of the call trace storage

Values: cf1 | cf2

Default: cf1

4.2.1.4.4 disable

Syntax

```
[no] disable
```

Context

config>mobile>system>call-insight>location

Description

This command enables or disables the specified storage location for call trace files.

Default

no disable

4.2.1.4.5 limit

Syntax

limit *limit*

no limit

Context

config>mobile>system>call-insight>location

Description

This command limits the total captured call trace files size per CF.

Parameters

limit

specifies the total captured call trace files size in megabytes

Values: 0 to 65536 (0 means unlimited)

Default: 1000

4.2.1.4.6 primary

Syntax

[no] primary

Context

config>mobile>system>call-insight>location

Description

This command sets the primary CF. The other CF is automatically set to be secondary.

4.2.1.4.7 max-files-number

Syntax

`max-files-number` *max-files-number*

`no` `max-files-number`

Context

`config>mobile>system>call-insight`

Description

This command configures the maximum number of files for call trace output.

Parameters

max-files-number

specifies the maximum call trace files

Values: 0 to 1024

Default: 200

4.2.1.4.8 disable-delayed-acct

Syntax

`[no]` `disable-delayed-acct`

Context

`config>mobile>system`

Description

This command disables delayed accounting when AA Deep Packet Inspection (DPI) is deployed.

The no form of this command reverts to the default (delayed accounting is enabled).



NOTE:

When configuring the `disable-delayed-acct` command, some sessions may experience errors in charging data counts. Any delayed accounting reported after configuring the `disable-delayed-acct` command will be ignored and some data flows with application identification still in progress may be undercharged. To avoid this, perform the configuration change when the gateway is in the shutdown state.

Default

`no disable-delayed-acct`

4.2.1.4.9 error-indication-tx

Syntax

`[no] error-indication-tx`

Context

`config>mobile>system`

Description

This command enables GTP Error Indication messages toward the originating GTP-U peer for all GTP-U interfaces.

The `no` form of command disables generation of Error Indication messages.

Default

`no error-indication-tx`

4.2.1.4.10 group

Syntax

`group group-number [[redundancy type gateway gw-id] | [resource-pool resource-pool-id]]`

`no group group-number`

Context

`config>mobile>system`

Description

This command configures the mobile-gateway system group with one or more Mobility Service Module (MSM) cards on a 7750 SR MG platform, or MG-VMs (cards) on a CMG platform. The redundancy type and gateway ID are used for 1:1 redundant systems, and the resource pool is used for N:K redundant systems.

The no form of this command removes the specified group row entry, when the group is in shutdown.



NOTE:

- For 1:1 (one-to-one) redundant systems, configure the redundancy type and gateway ID options; the resource-pool option is not required.
- For N:K (many-to-many) redundant systems, configure the resource-pool option instead of the redundancy type and gateway ID. The resource pool must be configured prior to configuring the mobile-gateway system group; see the `resource-pool` command.
- An N:K redundant system is defined as: number of MG groups = N and (number of cards minus number of MG groups) = K.
- The card configuration for the mobile-gateway system group or resource pool depends on the redundancy method used; see the `card` command.
- A mobile-gateway system group that is used in an N:K redundant system requires a dynamic SAP on its VPRNs and a dynamic MDA in its NAT groups. For more information, see the `config>service>vprn>if>dynamic-sap`, `config>isa>nat-group`, and `config>isa>nat-group>dynamic-mds` commands.
- A no shutdown must be explicitly executed for new N:K redundant groups.

Parameters

group-number

specifies a mobile-gateway system group

Values: 1 to 15

type

specifies the redundancy type; none means the cards in this group are unprotected, and one-to-one specifies that each card has one redundant backup card.

Values: one-to-one

gw-id

specifies the mobile gateway to which this system group belongs

Values: 1 to 8



NOTE:

For N:K redundant systems, do not specify the `redundancy` or `gw-id` parameters. Use the `resource-pool` command to configure these parameters. The following example shows the syntax to use to configure a mobile-gateway system group for N:K redundancy:

```
configure mobile-gateway system group 1 resource-pool 2
```

resource-pool-id

specifies the ID of the resource pool

Values: 1 to 4



NOTE:

For N:K redundant systems, configure the resource pool before the mobile-gateway system group; see the `resource-pool` command.

4.2.1.4.11 card

Syntax

```
card slot-number [working | protect]
```

```
no card slot-number
```

Context

```
config>mobile>system>group
```

```
config>mobile>system>resource-pool
```

Description

This command configures the `slot-number` of the respective MSM card in the 7750 chassis, or VM for the CMG platform, for the mobile-gateway system group or resource pool contexts. The card/VM can be assigned as an active (`working`) or standby (`protect`) card/VM only when configured. Depending on the failure scenario, either card can be in an active or standby state. The required configuration depends on the redundancy method used for the group or resource pool:

- A non-redundant system uses an active (`working`) card/VM.
- A 1:1 redundant system uses an active (`working`) and a standby (`protect`) card/VM.
- An N:K redundant system does not use an active (`working`) or a standby (`protect`) VM (card) and these parameters should not be configured.

**NOTE:**

An N:K redundant system is defined as: number of MG groups = N and (number of cards minus number of MG groups) = K.

Parameters

slot-number

specifies the card/VM slot number

Values: 1 to 20 depending on chassis model

working

specifies that the card is the active card/VM; use for one-to-one or non-redundant systems only

protect

specifies that the card is the standby card/VM; use for one-to-one redundant systems only

4.2.1.4.12 *pcmd-profile*

Syntax

```
pcmd-profile profile-name {inherit | none}
```

```
no pcmd-profile
```

Context

```
config>mobile>system>group
```

Description

This command configures a PCMD profile for the group (specified by the *group-number*) which configures the session and data records destinations for this group.

Only the destinations of the PCMD profile configured here are used. Other PCMD parameters are defined in the PDN level *pcmd profile*.

If the profile is set to none, no PCMD is generated from this group.

The no form of this command reverts to the default.

Default

```
no pcmd-profile
```

Parameters

profile-name

specifies the name of the PCMD profile configured in the context
config>mobile>profile>pcmd that contains the destinations to be applied for this group, up to 32 characters

inherit

specifies if the PCMD profile applied to the group is the one configured in the config>mobile>pdn>pcmd context
If no profile is configured in the PDN, the PCMD records are not generated from this group.

none

specifies that the PCMD records are not generated from this group
Default: inherit

4.2.1.4.13 override-profile

Syntax

```
override-profile override-profile-id  
no override-profile
```

Context

```
config>mobile>system>group
```

Description

This command configures the override profile for each MG group for GTP-U redirection. To remove an override profile from a group, you must first shut it down using the shutdown command in the config>mobile>system>group context.

The no form of this command removes the selected override profile.



NOTE:

This command applies only to CMG.

Default

```
no override-profile
```

Parameters

override-profile-id

specifies the profile ID to configure an MG group-specific override profile

Values: 1 to 32

4.2.1.4.14 protect-active-delay

Syntax

protect-active-delay *protect-active-delay*

Context

config>mobile>system>group

config>mobile>system>resource-pool

Description

This command specifies the time to wait for the working card to come online before a standby card becomes active.

Parameters

protect-active-delay

specifies the wait time in seconds

Values: 1 to 30

Default: 10

4.2.1.4.15 ipv6-path-mtu

Syntax

[no] ipv6-path-mtu

Context

config>mobile>system

Description

This command enables IPv6 path MTU discovery to dynamically lower the path MTU if the ICMPv6 Packet Too Big is received.

Default

no ipv6-path-mtu



NOTE:

The IPv6 path MTU discovery is only supported over the S1u reference point interface.

4.2.1.4.16 ipv6-udp-checksum

Syntax

[no] ipv6-udp-checksum

Context

config>mobile>system

Description

This command enables UDP checksum calculation while sending IPv6 packets for GTP tunnel transport. The IPv6 UDP checksum remains enabled by default.

The no form of this command disables the IPv6 UDP checksum calculation.

Default

ipv6-udp-checksum

4.2.1.4.17 non-terminating-rules

Syntax

non-terminating-rules {enable | disable}

no non-terminating-rules

Context

config>mobile>system

Description

This command enables the Non-Terminating Policy Rules functionality at CMG level. When enabled, policy rules marked as non-terminating are taken into account when matching traffic to policy rules.

When disabled, all policy rules are terminating (a packet can only match to one policy rule according to precedence).

The no form of this command reverts to the default.

Default

disable

4.2.1.4.18 policer-distribution

Syntax

```
policer-distribution dl-only dl-percent  
no policer-distribution
```

Context

```
config>mobile>system
```

Description

This command provides the ability to distribute the total available policers between downlink only (dl-only) and uplink and downlink on the PGW/GGSN. This applies to APN AMBR policers for non-GBR bearers and to MBR policers for GBR bearers.

The percent for uplink and downlink is determined based on the percent configured for downlink only (dl-only *dl-percent*). For example, if the *dl-percent* is set to 80, the uplink and downlink percent is 20. The total of the *dl-percent* plus the uplink and downlink percent is always 100.

Parameters

dl-percent

specifies the downlink only (dl-only) policer percentage

Values: 10 to 100

Default: 100

4.2.1.4.19 resource-pool

Syntax

```
resource-pool resource-pool-id redundancy type gateway gw-id  
no resource-pool resource-pool-id
```


Context

config>mobile>system

Description

This command configures a resource pool consisting of cards (VMs) on the CMG platform and the redundancy method to use when the resource pool is associated with a mobile-gateway system group. Currently this command supports configuration of N:K (many-to-many) redundant systems. The resource pool must also be associated with a mobile-gateway system group when configuring N:K redundancy. For more details, see the group command.



NOTE:

- This command is supported on the CMG platform only.
- For N:K redundant systems, this configuration includes all VMs (cards) to be used across multiple mobile-gateway system groups. The cards in this context must not be configured as active (working) or standby (protect); see the card command.
- An N:K redundant system is defined as: number of MG groups = N and (number of cards minus number of MG groups) = K.
- This configuration is not currently supported for 1:1 redundant systems. 1:1 redundant systems are configured using the *group* command.

Parameters

resource-pool-id

specifies the ID of the resource pool

Values: 1 to 4

type

specifies the redundancy method

Values: many-to-many

gw-id

specifies the ID of the mobile gateway to which the resource pool belongs

Values: 1 to 8

4.2.1.4.20 thresholds

Syntax

thresholds

Context

```
config>mobile>system
```

Description

This command enables the context to configure monitoring thresholds.

4.2.1.4.21 cam-utilization-threshold

Syntax

```
cam-utilization-threshold high-threshold high-threshold [low-threshold low-threshold]
```

```
no cam-utilization-threshold
```

Context

```
config>mobile>system>thresholds
```

Description

This command configures the raising threshold for the major and minor CAM utilization alarms. The major alarm is raised when the CAM utilization exceeds the high threshold configured value. The minor alarm is raised when the CAM utilization exceeds the low threshold configured value. The no form of this command reverts to the default values of the high and low thresholds.

Default

```
no cam-utilization-threshold
```

Parameters

high-threshold

specifies the percentage of the CAM utilization that defines when the major alarm is raised

Values: 0 (disabled) to 100

Default: 90

low-threshold

specifies the percentage of the CAM utilization that defines when the minor alarm is raised

Values: 0 (disabled) to 100

Default: 70

**NOTE:**

The low threshold must be configured with a value of at least 10% lower than the high threshold.

4.2.1.4.22 system-mg-ism

Syntax

```
system-mg-ism group-number  
no system-mg-ism
```

Context

```
config>mobile>system>thresholds
```

Description

This command configures monitoring thresholds for ISM-MG.

The no form of this command removes the monitoring thresholds from the configuration of the profile.

Parameters

group-number

specifies group number

Values: 1 to 8

4.2.1.4.23 counter

Syntax

```
counter {cp-isa-cpu | cp-isa-memory | cp-isa-uplink | cp-isa-downlink} high-  
threshold threshold [low-threshold threshold]  
no counter {cp-isa-cpu | cp-isa-memory | cp-isa-uplink | cp-isa-downlink}
```

Context

```
config>mobile>system>thresholds>mg-ism
```

Description

This command configures the high and low thresholds for each object.

The no form of this command removes the high and low thresholds from the configuration of the profile.

Parameters

cp-isa-cpu

keyword to specify thresholds for the CPU, in percentage

cp-isa-memory

keyword to specify thresholds for the memory, in percentage

cp-isa-uplink

keyword to specify thresholds for the uplink, in abs Mbytes

cp-isa-downlink

keyword to specify thresholds for the downlink, in abs Mbytes

threshold

specifies the high threshold value for the object

Values: 1 to 100

Values: 1 to 20000000

low-threshold *threshold*

specifies the low threshold value for the object

Values: 1 to 100

Values: 1 to 20000000

4.2.1.4.24 interval

Syntax

interval *minutes*

no interval

Context

config>mobile>system>thresholds>mg-ism

Description

This command configures the interval values for ISM-MG.

The no form of this command removes the intervals from the configuration of the profile.

Parameters

minutes

specifies group number

Values: 5 to 60

Default: 5

4.2.1.4.25 ttc-accuracy

Syntax

ttc-accuracy {true | false}

no ttc-accuracy

Context

config>mobile>system

Description

This command enables optimizing tariff time switch volume reporting from the user plane. Reporting the user plane volume both before the tariff time change, as well as between the tariff switch and the actual reporting time, facilitates accurate reporting to the charging server. It is recommended that consecutive tariff switch times be at least 10 minutes apart.

The no form of this command reverts to the default (false).

Default

no ttc-accuracy

Parameters

true

enables the TTC accuracy optimization

false

disables the TTC accuracy optimization

Default: false

4.2.1.4.26 ue-fragmented-packets-handling

Syntax

[no] ue-fragmented-packets-handling

Context

config>mobile>system

Description

This command enables IP packet reassembly of the user-level traffic (inner IP) for packet classification, to the correct bearer through rule matching. The packets leave the system to the exact same size as originally received.

The no form of this command reverts to the default.

Default

no ue-fragmented-packets-handling



NOTE:

Apart from the CLI knob that enables the feature, specific configuration is required per MG card where fragmentation is expected to be realized, for example, "mda-type isa-ip-reas-v" should be set under every required card-type "iom-v-mg".

4.2.1.4.27 use-fixed-udp-src-port

Syntax

[no] use-fixed-udp-src-port

Context

config>mobile>system

Description

This command enables the user to select a fixed UDP source port (2152) for all egress GTP-U packets. If disabled, a dynamic UDP source port from range 61440 to 65535 is used.

The no form of this command reverts to the default (disabled).

Default

no use-fixed-udp-src-port



NOTE:

If enabled for the APN, the `gtpu-source-port-dist` command takes precedence over the `use-fixed-udp-src-port` command at the APN level.

4.2.1.4.28 security

Syntax

```
security
```

Context

```
config>system
```

Description

This command enables the context to configure security settings.

4.2.1.4.29 dist-cpu-protection

Syntax

```
dist-cpu-protection
```

Context

```
config>system>security
```

Description

This command enables the context to configure the Distributed CPU Protection (DCP) feature.

4.2.1.4.30 policy

Syntax

```
policy policy-name [create]
```

```
no policy policy-name
```

Context

```
config>system>security>dist-cpu-protection
```

Description

This command configures one of the maximum 16 Distributed CPU Protection policies. These policies can be applied to objects such as SAPs and network interfaces.

Parameters

policy-name

specifies the name of the policy to be configured

create

creates the policy

4.2.1.4.31 protocol

Syntax

```
protocol name [create]
```

```
no protocol name
```

Context

```
config>system>security>dist-cpu-protection>policy
```

Description

This command creates the protocol for control in the policy. For a complete description of the command, refer to the *7450 ESS, 7750 SR, 7950 XRS, and VSR System Management Guide*.

Additional protocol name:

- gtp-err-ind. This protocol name can be used to configure the rate limiting of incoming GTP Error Indication messages, before these reach the processing unit. It can be used to protect the CPU and the processing of other important signaling in case of a GTP Error Indication signaling storm. The gtp-err-ind protocol name is supported only with a static policer in the DCP policy. For more information about the static policer configuration, refer to the *7450 ESS, 7750 SR, 7950 XRS, and VSR System Management Guide* in the following context:

```
config>sys>security>dist-cpu-protection>policy>static-policer
```

Default

none

Parameters

name

specifies the protocol name

Values: arp | dhcp | http-redirect | icmp| icmp-ping-check | igmp | mld | ndis | pppoe-pppoa | all-unspecified | mpls-ttl | bfd-cpm | bgp | eth-cfm | isis | ldp | ospf | pim | rsvp

For more information, refer to the *7450 ESS, 7750 SR, 7950 XRS, and VSR System Management Guide*.

create

creates the protocol

4.2.1.4.32 ssh

Syntax

```
ssh
```

Context

```
config>system>security
```

Description

This command enables the context to configure the list of the SSH host keys.

4.2.1.4.33 client-host-key-list

Syntax

```
client-host-key-list
```

Context

```
config>system>security>ssh
```

Description

This command enables the context to configure the list of the acceptable SSH server host keys for the SSH client.

4.2.1.4.34 host-key

Syntax

```
host -key index name key-name  
no host -key index
```

Context

```
config>system>security>ssh>client-host-key-list  
config>system>security>ssh>server-host-key-list
```

Description

This command configures the SSH host key algorithm list for the SSH server and client.

If an empty list is configured, the following default list is used:

- ssh-rsa
- ssh-dss



NOTE:

To apply any host key list configuration changes on the server, the SSH server must be restarted (shutdown/no shutdown) via a telnet session. If a telnet session is not available, create the following script:

Configure the script to run automatically once under a scheduled script execution in the following contexts:

```
config>system>script-control
```

```
config>system>cron
```

The no form of this command removes the specified host-key index.

Parameters

index

specifies the index of the algorithm in the list

Values: 1 to 255

key-name

specifies the algorithm for host key generation

Values: ssh-dss | ssh-rsa

4.2.1.4.35 client-kex-list

Syntax

```
client-kex-list
```

Context

```
config>system>security>ssh
```

Description

This command enables the context to configure SSH KEX algorithms for 7750 SR MG or CMG as a client.

The default list that the SSH KEX advertises is an empty list which contains the following algorithms:

- diffie-hellman-group16-sha512
- diffie-hellman-group14-sha256
- diffie-hellman-group14-sha1
- diffie-hellman-group14-sha1
- diffie-hellman-group1-sha1

4.2.1.4.36 kex

Syntax

```
kex index name kex-name
```

```
no kex index
```

Context

```
config>system>security>ssh>client-kex-list
```

```
config>system>security>ssh>server-kex-list
```

Description

This command configures phase 1 SSHv2 KEX algorithms for 7750 SR MG or CMG as an SSH server or client. By default, the client and server lists are empty. If you configure the list, SSH uses the hard-coded list with the first-listed algorithm having the highest priority and so on. An empty server or client list is the default list and contains the following algorithms:

- diffie-hellman-group16-sha512

- diffie-hellman-group14-sha256
- diffie-hellman-group14-sha1
- diffie-hellman-group14-sha1
- diffie-hellman-group1-sha1

The no form of this command removes the specified kex index. If all KEX indexes are removed, the default list is used.

Parameters

index

specifies the index of the kex algorithm in the client list; the lowest index in the list is negotiated first on the SSH negotiation list, while the highest index is at the bottom of the SSH negotiation list.

Values: 1 to 255

kex-name

specifies the kex name of the algorithm in the client list for computing the shared secret key

Values: diffie-hellman-group16-sha512 | diffie-hellman-group14-sha256 | diffie-hellman-group14-sha1 | diffie-hellman-group-exchange-sha1 | diffie-hellman-group1-sha1

4.2.1.4.37 server-host-key-list

Syntax

```
server-host-key-list
```

Context

```
config>system>security>ssh
```

Description

This command enables the context to configure the list of the SSH server host keys.

4.2.1.4.38 server-host-key-rsa-length

Syntax

```
server-host-key-rsa-length server-host-key-rsa-length
```

```
no server-host-key-rsa-length
```

Context

```
config>system>security>ssh
```

Description

This command configures the SSH server host key length with any of the allowed values.



NOTE:

To apply any host key list configuration changes on the server, the SSH server must be restarted (shutdown/no shutdown) via a telnet session. If a telnet session is not available, create the following script:

Configure the script to run automatically once under a scheduled script execution in the following contexts:

```
config>system>script-control
```

```
config>system>cron
```

The no form of this command reverts to the default value of the RSA length.

Parameters

server-host-key-rsa-length

specifies the length of the RSA host key

Values: 1024 | 2048 | 4096

Default: 1024

4.2.1.4.39 server-kex-list

Syntax

```
server-kex-list
```

Context

```
config>system>security>ssh
```

Description

The default list that the SSH KEX advertises is an empty list which contains the following algorithms:

- diffie-hellman-group16-sha512

- diffie-hellman-group14-sha256
- diffie-hellman-group14-sha1
- diffie-hellman-group14-sha1
- diffie-hellman-group1-sha1

4.2.1.4.40 **tls**

Syntax

`tls`

Context

`config>system>security`

Description

This command enables the context to configure the TLS options.

4.2.1.4.41 **client-cipher-list**

Syntax

`client-cipher-list`

Context

`config>system>security>tls`

Description

This command enables the context to configure the client desired TLS ciphers.

4.2.1.4.42 **cipher**

Syntax

`cipher index name cipher-suite-code`

`no cipher index`

Context

config>system>security>tls>client-cipher-list

Description

This command configures the cipher suite to be negotiated by the client.

The no form of this command removes the specified cipher index.

Parameters

index

specifies the index number that provides the location of the cipher in the negotiation list; a low index is high in the negotiation list, while a high index is at the bottom of the negotiation list

Values: 1 to 255

cipher-suite-code

specifies the cipher suite code

Values: tls-rsa-with3des-edc-cbc-sha | tls-rsa-with-aes128-cbc-sha | tls-rsa-with-aes256-cbc-sha | tls-rsa-with-aes128-cbc-sha256 | tls-rsa-with-aes256-cbc-sha256 | tls-dhe-rsa-with-aes128-gcm-sha256 | tls-dhe-rsa-with-aes256-gcm-sha384 | tls-ecdhc-ecdsa-with-aes128-gcm-sha256 | tls-ecdhc-ecdsa-with-aes256-gcm-sha384 | tls-ecdhc-rsa-with-aes128-gcm-sha256 | tls-ecdhc-rsa-with-aes256-gcm-sha384 | tls-dhe-rsa-with-aes128-ccm | tls-dhe-rsa-with-aes256-gcm | tls13-aes256-gcm-sha384 | tls13-aes128-gcm-sha256 | tls13-aes128-ccm8-sha256 | tls13-aes128-ccm-sha256

4.2.1.4.43 server-cipher-list

Syntax

server-cipher-list

Context

config>system>security>tls

Description

This command enables the context to configure the server desired TLS ciphers.

4.2.1.4.44 cipher

Syntax

```
cipher index name cipher-suite-code  
no cipher index
```

Context

```
config>system>security>tls>server-cipher-list
```

Description

This command configures the cipher suite to be negotiated by the server.

The no form of this command removes the specified cipher index.

Parameters

index

specifies the index number that provides the location of the cipher in the negotiation list; a low index is high in the negotiation list, while a high index is at the bottom of the negotiation list

Values: 1 to 255

cipher-suite-code

specifies the cipher suite code

Values: tls-rsa-with3des-edc-cbc-sha | tls-rsa-with-aes128-cbc-sha | tls-rsa-with-aes256-cbc-sha | tls-rsa-with-aes128-cbc-sha256 | tls-rsa-with-aes256-cbc-sha256 | tls-dhe-rsa-with-aes128-gcm-sha256 | tls-dhe-rsa-with-aes256-gcm-sha384 | tls-ecdhc-ecdsa-with-aes128-gcm-sha256 | tls-ecdhc-ecdsa-with-aes256-gcm-sha384 | tls-ecdhc-rsa-with-aes128-gcm-sha256 | tls-ecdhc-rsa-with-aes256-gcm-sha384 | tls-dhe-rsa-with-aes128-ccm | tls-dhe-rsa-with-aes256-gcm | tls13-aes256-gcm-sha384 | tls13-aes128-gcm-sha256 | tls13-aes128-ccm8-sha256 | tls13-aes128-ccm-sha256

4.2.1.5 Admin redundancy command descriptions

This section describes admin redundancy commands specific to the mobile gateway. For descriptions of all supported admin redundancy commands, see the *7450 ESS, 7750 SR, 7950 XRS, and VSR Basic System Configuration Guide*

4.2.1.5.1 redundancy

Syntax

redundancy

Context

admin

config

Description

This command enables the context to allow the user to perform redundancy operations.

4.2.1.5.2 cpm-switchover

Syntax

cpm-switchover [now]

Context

admin>redundancy

Description

This command triggers a controlled CPM switchover.

Parameters

now

specifies to perform the switchover without requesting confirmation

4.2.1.5.3 mc-master-lock

Syntax

[no] mc-master - lock

Context

admin>redundancy

Description

This command locks the gateway in the master operational role when ICR/geo-redundancy is deployed. The gateway remains in the master operational role regardless of any ICR master/slave election criteria that may occur.

The no form of this command unlocks and restores the use of normal ICR master/slave election criteria.



NOTE:

- When enabled, the manual switchover command (`admin>redundancy>mc-mobile-switchover>mobile-gateway`) is also locked from being used and an error is returned if attempted while the `mc-master-lock` command is enabled.
- When enabled together with the Diameter connection lost detection command (`config>redundancy>multi-chassis>peer>mc-mobile>diameter-connection-lost-detection>gx-failover`) and all the Gx connections are lost, it does not block the Gx guard timer from running; it blocks the FSWO.



NOTICE:

In a Master-Master case, the command must only be used on one node. If executed on both nodes, the MCS sync fails and an alarm is issued.

Default

no mc-master-lock

4.2.1.5.4 mc-mobile-switchover

Syntax

```
mc-mobile-switchover mobile-gateway gw-id peer ip-address [{reboot | now}] [abort]
no mc-mobile-switchover
```

Context

admin>redundancy

Description

This command issues a manual switchover so the master gateway transitions to a slave role and its peer gateway transitions to the master. The manual switchover is not hitless, but synchronizes the complete charging information and state for each bearer, enabling the slave to transition with minimum disruption and data loss.

This command supports an option to force the switchover, if the reconciliation fail. If the `now` option is specified and the ICR is in an Up state, the manual switchover can be performed even when the Geo-Redundancy state is not hot.

**NOTICE:**

If the `now` option is specified, the switchover will be executed, but the new master node might have incomplete UE records. In some cases this may be service impacting.

The `abort` option can be executed to abort the switchover only for an ongoing switchover that has been in progress for at least 5 minutes. The user must respond to a confirmation request to proceed. If `yes` is specified, the system issues a disconnect to the slave, making it master, switches to the slave state, and then activates the MC redirect tunnel.

Parameters

mobile-gateway

mandatory keyword

gw-id

specifies the gateway ID

Values: 1 to 8

peer

mandatory keyword to specify the peer IP address

ip-address

specifies the peer IP address for the manual switchover

Values: ip-address | ipv6-address

reboot

optional keyword that reboots this node after mc-mobile switchover

now

optional keyword that forces the switchover without full reconciliation

If this option is specified:

- the new master node might have incomplete UE records. In some cases this may be service impacting
- if the ICR is in the up state, the manual switchover can be performed even when the Geo-Redundancy state is not hot

abort

optional keyword that aborts mc-mobile switchover in progress; can only be executed for an ongoing switchover that has been in progress for at least 5 minutes

4.2.1.5.5 mc-slave-lock

Syntax

```
[no] mc-slave-lock
```

Context

```
admin>redundancy
```

Description

This command locks the gateway in the slave operational role when ICR/geo-redundancy is deployed. The gateway remains in the slave operational role regardless of any ICR master/slave election criteria that may occur.

The no form of this command unlocks and restores the use of normal ICR master/slave election criteria.



NOTE:

- When enabled, the manual switchover command (admin>redundancy>mc-mobile-switchover>mobile-gateway) is also locked from being used and an error is returned if attempted while the mc-slave-lock command is enabled.
- When enabled together with the Diameter connection lost detection command (config>redundancy>multi-chassis>peer>mc-mobile>diameter-connection-lost-detection>gx-failover) and all the Gx connections are lost, it does not block the Gx guard timer from running; it blocks the FSWO.

Default

```
no mc-slave-lock
```

4.2.1.5.6 mg-switchover

Syntax

```
mg-switchover
```

Context

```
admin>redundancy
```

Description

This command enables the context to perform a controlled MG-VM switchover.

**NOTE:**

The command is only supported for 1:1 redundant systems.

4.2.1.5.7 group

Syntax

```
group mg-group-id [now] [max-preparation-time max-preparation-time] [upgrade |  
downgrade]
```

Context

```
admin>redundancy>mg-switchover
```

Description

This command triggers an MG-VM controlled switchover.

Parameters

mg-group-id

specifies the MG group ID for which to perform the controlled switchover

Values: 1 to 15

now

specifies to perform the controlled switchover without requesting confirmation

max-preparation-time

specifies the waiting time, in seconds, that the MG-VM uses to process all charging data before the standby MG-VM becomes active; when set to 0, the switchover proceeds without a maximum preparation time, until all charging data has been processed

**NOTE:**

Incoming control messages are not served during the preparation phase defined by the *max-preparation-time* parameter.

Values: 0 (disabled) to 60

Default: 5

upgrade

initiates the switchover from the active card (running the older software version) to the standby card (running the newer software version). When executed, the standby card

becomes the active card for the MG group with ID *mg-group-id* (and the initial active card becomes the new standby card for the group) and the group is upgraded. For more information, refer to the *In-service software upgrade (ISSU) with intra-CNF redundancy (N:K)* and *In-service software upgrade (ISSU) with intra-CNF redundancy (1:1)* sections in the *CMG Software Upgrade Guide*.

downgrade

initiates the switchover from the active card (running the newer software version) to the standby card (running the older software version). When executed, the standby card becomes the active card for the MG group with ID *mg-group-id* (and the initial active card becomes the new standby card for the group) and the group is downgraded. For more information, refer to the *In-service software downgrade (ISSD) with intra-CNF redundancy (1:1)* section in the *CMG Software Upgrade Guide*.



NOTE:

The options upgrade and downgrade are supported during the 1:1 and N:K ISSU/ISSD and only when the active and standby card of the MG group are in different software versions.

4.2.1.6 Redundancy configuration command descriptions

This section describes redundancy configuration commands specific to the mobile gateway. For descriptions of all supported redundancy configuration commands, see the *7450 ESS, 7750 SR, 7950 XRS, and VSR Basic System Configuration Guide*.

4.2.1.6.1 multi-chassis

Syntax

```
multi-chassis
```

Context

```
config>redundancy
```

Description

This command enables the context to configure multi-chassis redundancy information.

4.2.1.6.2 peer

Syntax

[no] peer *ip-address*

Context

config>redundancy>multi-chassis

Description

This command configures a multi-chassis redundancy peer.

Parameters

ip-address

specifies a peer IP address; multicast addresses are not allowed

4.2.1.6.3 mc-mobile

Syntax

[no] mc-mobile

Context

config>redundancy>multi-chassis>peer

Description

This command enables the context to configure MC mobile parameters.

Default

no mc-mobile

4.2.1.6.4 bfd-enable

Syntax

bfd-enable [service *service-id*] interface *interface-name*

no bfd-enable

Context

```
config>redundancy>multi-chassis>peer>mc-mobile
```

Description

This command enables the use of Bi-directional Forwarding Detection (BFD) to be associated with the peer. The mc-mobile redundancy protocol will use the BFD state to determine liveness of its peer. The parameters for the BFD session are set via the BFD command under the IP interface configuration.

Default

no bfd-enable

Parameters

service-id

specifies the service identifier string, up to 64 characters

Values: 1 to 2147483648

interface-name

specifies the interface name, up to 32 characters

4.2.1.6.5 diameter-connection-lost-detection

Syntax

```
diameter-connection-lost-detection
```

Context

```
config>redundancy>multi-chassis>peer>mc-mobile
```

Description

This command enables the context to configure the Diameter connection failure detection.

4.2.1.6.6 gx-failover

Syntax

```
gx-failover guard-timer
```

```
no gx-failover
```


Context

```
config>redundancy>multi-chassis>peer>mc-mobile>diameter-connection-lost-detection
```

Description

This command enables the ICR Gx connection failure detection. If all the Gx connections on the master CMG node are detected as lost while the MCS is up, this command triggers an ICR FSWO.



NOTE:

- The ICR Gx failover handling only applies when the ICR MCS (mc-mobile) is up. When the MCS (mc-mobile) is detected as down, the ICR traffic detection is used.
- You must first enable the ICR Gx failover handling on the ICR node that has the slave operational role and then enable it on the ICR node that has the master operational role.

Parameters

guard-timer

specifies the time, in minutes, to delay the ICR FSWO when all the Gx connections are detected as lost; if the Gx connections are detected again up before the timer expires, the CMG cancels the ICR FSWO and resumes monitoring the Gx connections

Values: 1 to 15

Default: 2

4.2.1.6.7 hold-on-neighbor-failure

Syntax

```
hold-on-neighbor-failure multiplier
```

```
no hold-on-neighbor-failure
```

Context

```
config>redundancy>multi-chassis>peer>mc-mobile
```

Description

This command specifies the number of keep-alive intervals that may expire before the local node decides that the peer has failed. A peer failure will be declared if no keep-alive responses are received after a `hold-on-neighbor-failure x keep-alive-interval`.

Parameters

multiplier

specifies the multiplier

Values: 2 to 25

Default: 3

4.2.1.6.8 keep-alive-interval

Syntax

```
keep-alive-interval interval
```

```
no keep-alive-interval
```

Context

```
config>redundancy>multi-chassis>peer>mc-mobile
```

Description

This command sets the interval at which keep-alive messages are sent to the peer when the BFD is not enabled or is down.

Parameters

interval

specifies the interval, in tenths of a second

Values: 5 to 500

Default: 10

4.2.1.6.9 master-traffic-detection

Syntax

```
master-traffic-detection {enable | disable}
```

```
no master-traffic-detection
```

Context

```
config>redundancy>multi-chassis>peer>mc-mobile
```

Description

This command enables or disables the traffic detection and only applies to the master operational role.

Parameters

enable

specifies to enable traffic detection

disable

specifies to disable traffic detection

Default: enable

4.2.1.6.10 mc-complete-ue-sync

Syntax

```
mc-complete-ue-sync [partial-reconcile]
```

```
no mc-complete-ue-sync
```

Context

```
config>redundancy>multi-chassis>peer>mc-mobile
```

Description

This command enables data synchronization between the master and the slave nodes in MC redundancy at the call flow level (synchronization in Hot state) and the UE level (synchronization in reconcile state).

When enabled, the MC mobile synchronization goes immediately into the ICR Hot state and the progress of the synchronization from the master to the slave node is completed UE by UE for all UE records. This allows a switchover to take place at any time if an issue arises, or if a failover occurs the number of UEs synced thus far will all survive.

Default

```
no mc-complete-ue-sync
```



NOTE:

The `mc-complete-ue-sync` command must be enabled for all ICR-configured mobile gateway systems except for the HRPD Home Agent.

4.2.1.6.11 mc-redirect

Syntax

[no] mc-redirect

Context

config>redundancy>multi-chassis>peer>mc-mobile

Description

This command enables the use of the MC redirect.

Default

no mc-redirect

4.2.1.6.12 mobile-gateway

Syntax

mobile-gateway *gw-id* role *role*

no mobile-gateway

Context

config>redundancy>multi-chassis>peer>mc-mobile

Description

This command sets whether the mobile gateway acts in a primary or secondary role. If set to primary, the peer must be configured as a secondary mobile gateway and vice-versa.



NOTE:

Multiple gateway IDs are supported per multi-chassis peer. However, only one *gw-id* per peer is supported in this release. Furthermore, only one MC mobile peer is supported in this release.

Parameters

gw-id

specifies the gateway ID

Values: 1 to 8

role

specifies the role of the gateway

Values: primary | secondary

4.2.1.6.13 sync-mc-ipsec

Syntax

sync-mc-ipsec

Context

config>redundancy>multi-chassis>peer>mc-mobile>mobile-gateway

Description

This command configures the association to the MC IPSec with the IPSec tunnel group that the MC mobile will use.

4.2.1.6.14 tunnel-group

Syntax

[no] tunnel-group *tunnel-group-id*

Context

config>redundancy>multi-chassis>peer>mc-mobile>mobile-gateway>sync-mc-ipsec

Description

This command configures the IPSec tunnel group number associated to the MC mobile and the mobile gateway.

**NOTE:**

The tunnel group ID must be configured under the config>redundancy>multi-chassis>peer>mc-ipsec>tunnel-group context.

Parameters***tunnel-group-id***

specifies the tunnel group ID used by for MC mobile and the mobile gateway

Values: 1 to 16

4.2.1.6.15 peer-connect

Syntax

```
peer-connect {enable | disable}
no peer-connect
```

Context

```
config>redundancy>multi-chassis>peer>mc-mobile
```

Description

This command is configured when the ICR routing and the MC mobile are left configured on the primary node, and the secondary node is removed. For example, when the ICR is used only for ISSU and, after the ISSU process, the secondary node is removed.



NOTE:

- This command must be set in enabled state during the normal ICR operation, that is, when both the primary and the secondary ICR nodes are deployed.
- This command is only supported on CMG and LB-less mode.

Parameters

enable

enables the peer connect state

disable

disables the peer connect state

Default: enable

4.2.1.6.16 slave-traffic-delay-timer

Syntax

```
slave-traffic-delay-timer delay-timer
no slave-traffic-delay-timer
```

Context

```
config>redundancy>multi-chassis>peer>mc-mobile
```

Description

This command configures a timer that is enabled when the MCS is detected as lost. The timer delays the start of traffic detection and thus, prevents switchovers when the MCS link is restored before the timer expires. If MC redirection is enabled (using the `mc-redirect` command), the slave node continues to try and route any traffic over the MC redirect tunnel.

The `no` form of this command disables the timer (the timer is set to 0 seconds).



NOTICE:

Before executing the command, make sure that a switchover is not in progress.

Default

```
no slave-traffic-delay-timer
```

Parameters

delay-timer

specifies the timer value, in seconds

Values: 0 to 60

Default: 0 (disabled)

4.2.1.6.17 traffic-detection

Syntax

```
traffic-detection {strict | relaxed}
```

```
no traffic-detection
```

Context

```
config>redundancy>multi-chassis>peer>mc-mobile
```

Description

This command configures the ICR traffic detection criteria.

**NOTE:**

- This command can apply to special 7750 SR MG and CMG ICR deployments, with a very small number of attached UEs, or very low, or no active user traffic. The `relaxed` option must not be used for any regular 7750 SR MG and CMG deployments.
- The mobile gateway ICR relaxed mode is only supported on 2G (GGSN) without any charging and policy or RADIUS deployed.

Parameters

strict

specifies that, when ICR traffic detection is enabled, three traffic types are required for the node to transition into or stay in the ICR Operational Role Master

relaxed

specifies that, when ICR traffic detection is enabled, one traffic type is sufficient for the node to transition into or stay in the ICR Operational Role Master

Default: `strict`

4.2.1.6.18 traffic-detection-poll-timer

Syntax

```
traffic-detection-poll-timer seconds
```

```
no traffic-detection-poll-timer
```

Context

```
config>redundancy>multi-chassis>peer>mc-mobile
```

Description

This command configures the traffic detection poll timer.

The `no` form of this command reverts to the default value.

**NOTE:**

Nokia recommends configuring the same value for the traffic detection poll timer in the primary and secondary gateways.

Default

no traffic-detection-poll-timer

Parameters

seconds

specifies the timer value, in seconds

Values: 5 to 1200

Default: 5

4.2.1.7 Accounting policy command descriptions

For descriptions of all additional accounting policy log commands, refer to the *7450 ESS*, *7750 SR*, *7950 XRS*, and *VSR System Management Guide*.

4.2.1.7.1 record

Syntax

[no] record *record-name*

Context

config>log>accounting-policy

Description

This command adds the accounting record type to the accounting policy to be forwarded to the configured accounting file. A record name can only be used in one accounting policy. To obtain a list of all record types that can be configured, use the `show log accounting-records` command.



NOTE:

aa, video and subscriber records are not applicable to the 7950 XRS.

```
# show log accounting-records
=====
Accounting Policy Records
=====
Record # Record Name                               Def. Interval
-----
1      service-ingress-octets                       5
2      service-egress-octets                        5
3      service-ingress-packets                      5
```

4	service-egress-packets	5
5	network-ingress-octets	15
6	network-egress-octets	15
7	network-ingress-packets	15
8	network-egress-packets	15
9	compact-service-ingress-octets	5
10	combined-service-ingress	5
11	combined-network-ing-egr-octets	15
12	combined-service-ing-egr-octets	5
13	complete-service-ingress-egress	5
14	combined-sdp-ingress-egress	5
15	complete-sdp-ingress-egress	5
16	complete-subscriber-ingress-egress	5
17	aa-protocol	15
18	aa-application	15
19	aa-app-group	15
20	aa-subscriber-protocol	15
21	aa-subscriber-application	15
23	custom-record-subscriber	5
24	custom-record-service	5
25	custom-record-aa-sub	15
26	queue-group-octets	15
27	queue-group-packets	15
28	combined-queue-group	15
29	combined-mpls-lsp-ingress	5
30	combined-mpls-lsp-egress	5
31	combined-ldp-lsp-egress	5
32	saa	5
33	video	10
34	kpi-system	5
35	kpi-bearer-mgmt	5
36	kpi-bearer-traffic	5
37	kpi-ref-point	5
38	kpi-path-mgmt	5
39	kci-iom-3	5
40	kci-system	5
41	kci-bearer-mgmt	5
42	kci-path-mgmt	5
43	complete-kpi	5
44	complete-kci	5
45	kpi-bearer-group	5
46	kpi-ref-path-group	5
47	kpi-kci-bearer-mgmt	5
48	kpi-kci-path-mgmt	5
49	kpi-kci-system	5
50	complete-kpi-kci	5
51	aa-performance	15
52	complete-ethernet-port	15
53	extended-service-ingress-egress	5
54	complete-network-ing-egr	15
55	aa-partition	15
56	complete-pm	5
58	kpi-bearer-traffic-gtp-endpoint	5
59	kpi-ip-reas	5
60	kpi-radius-group	5
61	kpi-ref-pt-failure-cause-code	5
62	kpi-dhcp-group	5

```
63      aa-admit-deny      15
64      nat-binding-port-block  5
=====
```

To configure an accounting policy for access ports, select a service record (for example, service-ingress-octets). To change the record name to another service record then the record command with the new record name can be entered and it will replace the old record name.

When configuring an accounting policy for network ports, a network record should be selected. When changing the record name to another network record, the record command with the new record name can be entered and it will replace the old record name.

If the change required modifies the record from network to service or from service to network, the old record name must be removed using the no form of this command.

Only one record may be configured in a single accounting policy. For example, if an accounting-policy is configured with a access-egress-octets record, to change it to service-ingress-octets, use the no record command under the accounting-policy to remove the old record and then enter the service-ingress-octets record.

**NOTE:**

Collecting excessive statistics can adversely affect the CPU utilization and take up large amounts of storage space.

The no form of this command removes the record type from the policy.

Default

no record

Parameters

record-name

Specifies the accounting record name. *Default collection interval for accounting records* lists the accounting record names available and the default collection interval.

Table 1: Default collection interval for accounting records

Record type	Accounting record name	Default interval
1	service-ingress-octets	5
2	service-egress-octets	5

Table 1: Default collection interval for accounting records (continued)

Record type	Accounting record name	Default interval
3	service-ingress-packets	5
4	service-egress-packets	5
5	network-ingress-octets	15
6	network-egress-octets	15
7	network-ingress-packets	15
8	network-egress-packets	15
9	compact-service-ingress-octets	5
10	combined-service-ingress	5
11	combined-network-ing-egr-octets	15
12	combined-service-ing-egr-octets	5
13	complete-service-ingress-egress	5
14	combined-sdp-ingress-egress	5
15	complete-sdp-ingress-egress	5
16	complete-subscriber-ingress-egress	5
17	aa-protocol	15
18	aa-application	15
19	aa-app-group	15

Table 1: Default collection interval for accounting records (continued)

Record type	Accounting record name	Default interval
20	aa-subscriber-protocol	15
21	aa-subscriber-application	15
23	custom-record-subscriber	5
24	custom-record-service	5
25	custom-record-aa-sub	15
26	queue-group-octets	15
27	queue-group-packets	15
28	combined-queue-group	15
29	combined-mpls-lsp-ingress	5
30	combined-mpls-lsp-egress	5
31	combined-ldp-lsp-egress	5
32	saa	5
33	video	10
34	kpi-system	5
35	kpi-bearer-mgmt	5
36	kpi-bearer-traffic	5
37	kpi-ref-point	5

Table 1: Default collection interval for accounting records (continued)

Record type	Accounting record name	Default interval
38	kpi-path-mgmt	5
39	kpi-iom-3	5
40	kci-system	5
41	kci-bearer-mgmt	5
42	kci-path-mgmt	5
43	complete-kpi	5
44	complete-kci	5
45	kpi-bearer-group	5
46	kpi-ref-path-group	5
47	kpi-kci-bearer-mgmt	5
48	kpi-kci-path-mgmt	5
49	kpi-kci-system	5
50	complete-kpi-kci	5
51	aa-performance	15
52	complete-ethernet-port	15
53	extended-service-ingress-egress	5
54	complete-network-ing-egr	15

Table 1: Default collection interval for accounting records (continued)

Record type	Accounting record name	Default interval
55	aa-partition	15
56	complete-pm	5
58	kpi-bearer-traffic-gtp-endpoint	5
59	kpi-ip-reas	5
60	kpi-radius-group	5
61	kpi-ref-pt-failure-cause-code	5
62	kpi-dhcp-group	5
63	aa-admit-deny	5
64	nat-binding-port-block	5

4.2.2 Card and system management show command descriptions

- *Show card command descriptions*
- *Show mobile gateway command descriptions*
- *Show redundancy command descriptions*

4.2.2.1 Show card command descriptions

4.2.2.1.1 card

Syntax

card state

`card [slot-number]`

`card [slot-number] detail`

Context

show

Description

This command displays physical card related information.

For further information about cards, MDAs, and ports, refer to the appropriate Interface Guide.

Parameters

slot-number

specifies the slot number

detail

displays detailed information about the specified card

Output

The following output example shows card information, and [Show card output fields](#) describes the fields.

Example output card statistics

```
# config>card# show card
=====
Card Summary
=====
Slot   Provisioned   Equipped   Admin   Operational   Application
Card-type Card-type   State     State        Type
-----
1      ism-mg-2pac-fp3 ism-mg-2pac-fp3 up      up/active     sgw
2      ism-mg-2pac-fp3 ism-mg-2pac-fp3 up      up/standby    sgw
5      iom2-20g                up      provisioned
6      iom2-20g      iom2-20g      up      up
9      iom3-xp       iom3-xp       up      up
A      sfm-400g      sfm-400g      up      up/active
B      sfm-400g      sfm-400g      up      up/standby
=====

# config>card#
=====
MDA Summary
=====
```


Slot	Provisioned Card-type	Equipped Card-type	Admin State	Operational State	Application Type
1	1isa-mg	isa-ms	up	up	
=====					
# config>card#					

Table 2: Show card output fields

Label	Description
Card Summary	
Slot	Indicates which slot number holds which card
Provisioned Card-type	Indicates the provisioned card type
Equipped Card-type	Indicates the equipped card type
Admin State	Indicates the preferred state of the card
Operational State	Indicates the present state of the card
Application Type	Specifies the type of application that can run with the card

4.2.2.2 Show mobile gateway command descriptions

4.2.2.2.1 call-insight

Syntax

call-insight

Context

show>mobile>system

Description

This command displays Call Trace information.

Output

The following output examples show Call Trace information.

Example output call trace system statistics

```
# show mobile-gateway system call-insight
=====
Mobile gateway system call-insight
=====
UE profile
-----
Primary CF : CF1                      Max files number : 200
CF1 state  : enabled                  CF1 volume limit : 1000 MB
CF2 state  : enabled                  CF2 volume limit : 1000 MB
-----
All-UEs profile
-----
Max. CPU   : 60
=====
```

4.2.2.2.2 all-ues

Syntax

all-ues

Context

show>mobile>call-insight

Description

This command displays Call Trace information for non-selective trace.

Output

The following output examples show call insight information for non-selective trace, and *Show mobile profile call-insight output fields* describes the fields.

Example output call trace all UEs statistics

```
# show mobile-gateway call-insight all-ues
=====
Call-insight All-UEs
```

```

=====
Status           : running           Capture format: pcap
Nr. of captured msgs : 47           Time limit    : 603800s
|--control-plane msgs: 47
|--events        : 0
Size of captured msgs: 12304B       Data limit    : unlimited
Started          : SEP 11 2017, 10:10:16 UTC
Ref-points       : all
Events           : none
Path management  : disabled
Live output addr/port: 77.77.2.2/7799
Live output router : Base
=====

```

Table 3: Show mobile profile call-insight output fields

Label	Description
Call-insight All-UEs	
Status	Indicates level of completion of the command
Capture format	Specifies format in which the capture is done
Nr. of captured msgs	Specifies total number of messages captured
Time limit	Provides limit on time permitted for the process
Control plane msgs	Specifies the number of control plane messages captured
Events	Specifies the number of events captured
Size of traced msgs	Indicates size of all messages captured
Data limit	Provides limit on data to be used for the process
Started	Provides date when the process began
Ref-points	Indicates what ref points are being used
Events	Indicates total number of events in the process

Table 3: Show mobile profile call-insight output fields (continued)

Label	Description
Path management	Indicates if path management is enabled
Live output addr/port	Specifies IP address of the port
Live output router	—

4.2.2.2.3 files

Syntax

```
files [long]
files finished [long]
files running [long]
```

Context

```
show>mobile>call-insight
```

Description

This command displays Call Trace information for UEs, all UEs and masked UEs.

Parameters

long

keyword to display long list format

finished

keyword to display completed Call Trace files

running

keyword to display active Call Trace files

Output

The following output examples show Call Trace file information.

Example output call trace files statistics

```
# show mobile-gateway call-insight files
=====
Call trace files of running jobs
=====
File path                                                    Size
-----
cf1-A:/calltrace/running/imsi_123456789012300_m_150327_1313.pcap      4194
cf1-A:/calltrace/running/imsi_223456789012300_m_150327_1313.pcap      4194
cf1-A:/calltrace/running/imsi_323456789012300_m_150327_1313.pcap      4194
=====
Call trace files of finished jobs
=====
File path                                                    Size
-----
=====
Call trace files storage limits for active CPM
=====
cf1-A available space :    304 MB          cf1-A volume limit :  1000 MB
Available file count  :    197          Maximal file count :   200
=====
```

4.2.2.2.4 ue

Syntax

```
ue [[detail] | [imei imei [detail]] | [imsi imsi [detail]] | [mask-name mask-name [detail]] |
[msisdn msisdn [detail]] | [mac-address mac-address [detail]]]
```

Context

```
show>mobile>call-insight
```

Description

This command displays Call Trace information for the specified UE type.

Parameters

detail

keyword to display detailed Call Trace information

imei

specifies an IMEI identification string, up to 16 characters

imsi

specifies an IMSI identification string, up to 15 characters

mask-name

specifies a mask name, up to 16 characters

msisdn

specifies an MSISDN identification string, up to 15 characters

mac-address

specifies the UE MAC-address that the packet capture is based on

Values: xx:xx:xx:xx:xx:xx or xx-xx-xx-xx-xx-xx [x - hexadecimal digit]

Output

The following output examples show Call Trace UE information.

- [Example output call trace UE statistics](#)
- [Example output call trace UE detail statistics](#)
- [Example output call trace UE IMSI detail statistics](#)

Example output call trace UE statistics

```
# show mobile-gateway call-insight ue
=====
Call-insight UEs
=====
Match      Value           Mask-name      Status      Msgs
-----
imsi       22222222222223  N/A           running    0
msisdn     17962480143     N/A           running    0
imei       505024101215080 N/A           running    0
imsi       22222222222221  N/A           running    0
-----
Number of call-insight debug jobs: 4
=====
```

Example output call trace UE detail statistics

```
# show mobile-gateway call-insight ue detail
=====
Call-insight UE detail
=====
IMSI       : 13456789012345      Status      : running
IMEI       : --             Match type   : imsi
MSISDN     : --             Capture format: simulated-p*
MAC address : --             Time limit   : 86400s
Nr. of captured msgs : 128      Data limit   : 10MB
|--control-plane msgs: 48
|--user-plane msgs   : 80
|--events            : 0
Size of captured msgs: 31770B
Started        : FEB 15 2018, 18:53:41 UTC
Ref-points     : all
User-traffic    : all
```

```

Events                : none
Mask-name             : N/A
Live output           : N/A
Output file base      : imsi_13456789012345_180215_1853
Detailed user-plane debug statistics
|--VMId:1 LB-VM ingress Uplink      packets: 40          size: 9440          B
|--VMId:1 LB-VM ingress Downlink    packets: 40          size: 9440          B
|--VMId:1 LB-VM egress  Uplink      packets: 40          size: 9440          B
|--VMId:1 LB-VM egress  Downlink    packets: 40          size: 9440          B
|--VMId:3 MG-VM   -    Uplink      packets: 30          size: 6000          B
|--VMId:3 MG-VM   -    Downlink    packets: 30          size: 6000          B
|--VMId:4 MG-VM   -    Uplink      packets: 10          size: 2000          B
|--VMId:4 MG-VM   -    Downlink    packets: 10          size: 2000          B
-----
Number of call-insight debug jobs: 1
Nr. of dropped user-plane packets: 0
-----
Note:Reference points field above is applicable only to control-plane messages.
=====

```

Example output call trace UE IMSI detail statistics

```

# show mobile-gateway call-insight ue imsi 310012001000001 detail
=====
Call-insight UE detail
=====
IMSI                : 310012001000001          Status        : running
IMEI                : --                      Match type     : imsi
MSISDN              : --                      Capture format: simulated-p*
MAC address         : --                      Time limit    : 86400s
Nr. of captured msgs : 21                     Data limit     : 10MB
|--control-plane msgs: 21
|--user-plane msgs   : 0
|--events            : 0
Size of captured msgs: 10407B
Started             : SEP 08 2017, 11:05:37 UTC
Ref-points          : all
User-traffic        : enabled
Events              : none
Mask-name           : N/A
Live output addr/port: 10.10.7.10/50001
Live output router   : Base
Output file base     : N/A
-----
Number of call-insight debug jobs: 1
Nr. of dropped user-plane packets: 0
-----
Note:Reference points field above is applicable only to control-plane messages.
=====
* indicates that the corresponding row element may have been truncated.

```

4.2.2.2.5 ue-mask

Syntax

ue-mask [name *name*]

Context

show>mobile>call-insight

Description

This command displays Call Trace information for masked UEs.

Parameters

name

specifies a string, up to 16 characters

Output

The following output example shows UE mask Call Trace information.

Example output call trace UE mask statistics

```
# show mobile-gateway call-insight ue-mask
=====
Call-insight UE-mask
=====
Mask-name           Mask (Type/Value)      UEs-active  Max-UEs    UEs-finished
-----
default             imsi/123*              0           50         0
default             imsi/223.5.7*          0           50         0
myMask              imei/323.5.7*          0           5          0
-----
Number of call-insight masks: 3
=====
```

4.2.2.2.6 connections

Syntax

connections [detail]

Context

show>mobile>connections

Description

This command displays UDP, TCP, and SCTP connections currently in progress on the ISM-MG.

Parameters

detail

keyword to display detailed connection information

Output

The following output example shows connection information and *Show connections fields* describes the output fields.

Example output connection detail statistics

```
# show mobile-gateway connections detail
=====
Connections (Detail)
=====
Prot RecvQ  TxmtQ   Local Address          State
      MSS  Remote Address          vRtrID
-----
TCP      0      0 10.20.1.3.64563        ESTABLISH
      1024 10.15.8.10.3868         1
TCP      0      0 10.20.1.3.64563        CLOSED
      1024 0.0.0.0.0              1
TCP      0      0 0.0.0.0.21059          LISTEN
      1024 0.0.0.0.0              4095
TCP      0      0 0.0.0.0.23             LISTEN
      1024 0.0.0.0.0              4095
SCTP     1      1 10.10.8.1.64512        COOKIE_WAIT
      1500 10.10.8.10.3868         0
-----
No. of SCTP Connections: 1
-----
No. of TCP Connections: 4
-----
TCP Statistics
-----
packets sent                : 136
data packets                 : 48 (26369 bytes)
data packet retransmitted    : 0 (0 bytes)
ack-only packets             : 56 (0 delayed)
URG only packet              : 0
window probe packet          : 0
window update packet         : 0
control packets              : 32
packets received             : 140
acks                         : 69 for (26407 bytes)
duplicate acks               : 17
ack for unsent data           : 0
packets received in-sequence : 54 (17760 bytes)
completely duplicate packet   : 0 (0 bytes)
```

```

packet with some dup. data      : 0 (0 bytes)
out-of-order packets           : 0 (0 bytes)
packet of data after window    : 0 (0 bytes)
window probe                   : 0
window update packet           : 0
packets received after close    : 0
discarded for bad checksum      : 0
discarded for bad header offset field : 0
discarded because packet too short : 0
connection request             : 6
connection accept              : 17
connections established (including accepts) : 22
connections closed             : 26 (including 4 drops)
embryonic connections dropped  : 1
segments updated rtt           : 74 (of 60 attempts)
retransmit timeouts            : 1
connections dropped by rexmit timeout : 0
persist timeouts              : 0
keepalive timeouts            : 0
keepalive probes sent          : 0
connections dropped by keepalive : 0
pcb cache lookups failed       : 0
connections dropped by bad md5 digest : 0
connections dropped by enhanced auth : 0

```

SCTP Statistics

```

normal chunks                  : 2
unordered chunks               : 0
fragmented chunks              : 0
out of order chunks            : 0
init sent                      : 131
init received                  : 0
init ack sent                  : 0
init ack received              : 2
cookie echo sent               : 3
cookie echo received           : 0
cookie ack sent                : 0
cookie ack received            : 2
heartbeat sent                 : 0
heartbeat received             : 0
heartbeat ack sent             : 0
heartbeat ack received         : 0
abort sent                     : 0
abort received                 : 1
shutdown sent                  : 1
shutdown received              : 1
shutdown ack sent              : 1
shutdown ack received          : 2
shutdown complete sent         : 2
shutdown complete received     : 1
OP ERROR sent                  : 0
OP ERROR received              : 0
OOTB sent                      : 0
OOTB received                  : 0
bad VTAGs                     : 0
bad CSUMs                     : 0

```

```
snd queue overflow      : 0
recv queue overflow    : 0
=====
```

Table 4: Show connections fields

Label	Description
CLOSED	Closed. The socket is not being used
LISTEN	Listening for incoming associations
ESTABLISHED	Association has been established
COOKIE-WAIT	INIT has been sent to the peer, awaiting acknowledgment
COOKIE-ECHOED	State cookie from the INIT-ACK has been sent to the peer, awaiting acknowledgment
SHUTDOWN-PENDING	SHUTDOWN has been received from the upper layer, awaiting acknowledgment of all outstanding DATA from the peer
SHUTDOWN-SENT	All outstanding data has been acknowledged in the SHUTDOWN-SENT state. SHUTDOWN has been sent to the peer, awaiting acknowledgment
SHUTDOWN-RECEIVED	SHUTDOWN has been received from the peer, awaiting acknowledgment of all outstanding DATA

4.2.2.2.7 ism-mg

Syntax

```
ism-mg
```

Context

```
show>mobile
```

Description

This command displays information about the ISM-MG.

4.2.2.2.8 abs

Syntax

```
abs [statistics]
```

Context

```
show>mobile>ism-mg
```

Description

The mobile gateway overload protection mechanism (ABS – Anti-Breakdown System) has a set of internal parameters to monitor signaling/processing latency and memory utilization on each of the ISM-MGs on the system independently. This command displays statistics and a table of ABS status per ISM-MG card (only MDA 1 is supported).

Parameters

statistics

shows ABS in more detailed statistics

Output

The following output examples show ABS information for the ISM-MG card and *Show ISM-MG ABS fields* describes the output fields.

- *Example output ISM-MG ABS*
- *Example output ISM-MG ABS statistics*

Example output ISM-MG ABS

```
# show mobile-gateway ism-mg abs
=====
ISM-MG ABS summary
=====
S /M /G  GwId Signal-Overload      Memory-Overload
-----
03/01/01 1    No                      No
04/01/01 1    No                      No
=====
```

Example output ISM-MG ABS statistics

```
# show mobile-gateway ism-mg abs statistics
=====
ISM-MG ABS statistics
=====
S /M /G  GwId First Occurrence (UTC)  Last Occurrence (UTC)  Tot
-----
03/01/01 1  -                -                0
04/01/01 1  -                -                0
=====
```

Table 5: Show ISM-MG ABS fields

Label	Description
S	Physical slot number for the ISM-MG
M	MDA number for the physical slot; only 1 is supported for the ISM-MG card
G	Group number
GW-ID	Gateway ID (1 to 8)
Signal-Overload	Indicates signaling overload
Memory-Overload	Indicates memory overload
First Occurrence	Time stamp of the first time that overload protection was triggered per ISM-MG
Last Occurrence	Time stamp of the last time that overload protection was triggered per ISM-MG
Tot	Indicates the total number that overload protection was triggered

4.2.2.2.9 cpu

Syntax

cpu [slot *number*] [mda *number*] [repeat *repeat-number*]

cpu [repeat *repeat-number*] [vm *vm-id*] [vm-function *vm-function-number*]

Context

show>mobile>ism-mg

Description

This command displays the CPU utilization for the ISM-MG card (only MDA 1 is supported).

Parameters

number

specifies the slot number of the ISM-MG card

mda number

specifies the MDA number within the ISM-MG slot

Values: 1 | 2 (only MDA 1 is supported for the ISM-MG)

repeat-number

specifies the sample period, the number of times to repeat the sampling

Values: 1 to 5

vm-id

specifies the VM ID

Values: 1 to 20

vm-function-number

specifies the VM function

Values: 1 | 2

Output

The following 7750 SR MG output examples show CPU utilization information for the ISM-MG card and *Show ISM-MG CPU utilization fields* describes the output fields.



NOTE:

The outputs of a CMG system are identical to the 7750 SR MG outputs except they have some CMG-specific fields displayed instead of hardware components (for example, VM is displayed instead of Card).

- *Example output ISM-MG CPU utilization summary statistics*
- *Example output ISM-MG CPU utilization statistics for a specified slot*

Example output ISM-MG CPU utilization summary statistics

```
# show mobile-gateway ism-mg cpu
=====
CPU utilization starting at 09/14/2018 21:26:09
=====
ISM-MG Summary
=====
Slot  Mda   Group gw-id   cpu utilization
-----
3     1     1     1           2.21%
4     1     1     1           1.99%
=====
```

Example output ISM-MG CPU utilization statistics for a specified slot

```
# show mobile-gateway ism-mg cpu slot 3
=====
CPU utilization starting at 09/14/2018 21:34:15
=====
ISM-MG Slot 3 Mda 1
=====
Name           CPU Usage
-----
AMS             0.01%
DHCP            ~ 0.00%
GA              ~ 0.00%
GX              0.02%
GY              0.02%
HA              0.10%
PFCP            0.03%
RADIUS          0.02%
RF              0.02%
S6B             0.03%
SESSMGR         0.04%
SIG             0.04%
SYSTEM          1.96%
TRACE           0.02%
RPY             0.01%
SWM             ~ 0.00%
LDNSC           ~ 0.00%
SD              0.01%
SDLB            ~ 0.00%
STA             ~ 0.00%
CDBMGR          ~ 0.00%
<NULL>         ~ 0.00%
HTTP            ~ 0.00%
```

```
-----
Total In Use:      2.32%
Idle:              97.68%
=====
```

Table 6: Show ISM-MG CPU utilization fields

Label	Description
Slot	Physical slot number for the ISM-MG
MDA	MDA number for the physical slot; only 1 is supported for the ISM-MG card
Group	Group number
GW-ID	Gateway ID (1 to 8)
CPU utilization	Percentage of CPU utilization per slot and MDA
Name	Protocol or interface name
CPU usage	Percentage of CPU usage per interface

4.2.2.2.10 memory-pools

Syntax

```
memory-pools [slot number] [mda number] [repeat repeat-number]
```

```
memory-pools [repeat repeat-number] [vm vm-id] [vm-function vm-function-number]
```

Context

```
show>mobile>ism-mg
```

Description

This command displays the memory utilization for the ISM-MG card (only MDA 1 is supported).

Parameters

number

specifies the slot number of the ISM-MG

mda number

specifies the MDA number within the ISM-MG slot

Values: 1 | 2 (only MDA 1 is supported for the ISM-MG)

repeat-number

specifies the sample period, the number of times to repeat the sampling

Values: 1 to 5

vm-id

specifies the VM ID

Values: 1 to 20

vm-function-number

specifies the VM function

Values: 1 | 2

Output

The following 7750 SR MGoutput examples show memory utilization information for the ISM-MG card and *Show ISM-MG memory utilization fields* describes the output fields.



NOTE:

The outputs of a CMG system are identical to the 7750 SR MG outputs except they have some CMG-specific fields displayed instead of hardware components (for example, VM is displayed instead of Card).

- *Example output ISM-MG memory statistics*
- *Example output ISM-MG memory statistics for a specified slot*

Example output ISM-MG memory statistics

```
# show mobile-gateway ism-mg memory-pools
=====
MEM utilization starting at 09/20/2018 00:03:19
=====
ISM-MG Summary
=====
Slot |Mda  |Group|gw-id|  Current Size|      In Use|Avail Virt Mem|Avail Phys Mem
-----+-----+-----+-----+-----+-----+-----+-----
3    |1     |1     |1     | 2,603,614,208| 2,306,386,904| 5,133,828,096| 5,133,828,096
4    |1     |1     |1     | 2,606,759,936| 2,310,874,000| 5,133,828,096| 5,133,828,096
=====
```

Example output ISM-MG memory statistics for a specified slot

```
# show mobile-gateway ism-mg memory-pools slot 3
=====
MEM utilization starting at 09/20/2018 00:03:30
=====
ISM-MG Slot 3 Mda 1
=====
Name                | Current Size| In Use
-----+-----+-----
AMS                 | 1,048,576| 158,264
DHCP                | 9,437,184| 5,566,256
GA                  | 2,097,152| 755,928
GX                  | 8,388,608| 6,203,808
GY                  | 6,291,456| 2,972,640
HA                  | 9,437,184| 1,816,176
PFCP                | 9,437,184| 2,748,104
RADIUS              | 13,631,488| 3,336,288
RF                  | 77,594,624| 68,117,896
S6B                 | 39,845,888| 34,266,368
SESSMGR             | 136,314,880| 82,402,904
SIG                 | 68,157,440| 46,028,992
SYSTEM              | 2,164,260,864| 2,017,004,704
TRACE               | 28,311,552| 21,164,912
RPY                 | 7,340,032| 3,420,904
SWM                 | 7,340,032| 3,415,472
LDNSC               | 2,097,152| 47,560
SD                  | 7,340,032| 4,600,416
SDLB                | 0| 0
STA                 | 5,242,880| 2,354,608
CDBMGR              | 0| 0
<NULL>              | 0| 0
HTTP                | 0| 0
-----+-----+-----
Total:                2,603,614,208 2,306,382,200
=====
available virtual memory: 5,133,828,096
available physical memory: 5,133,828,096
=====
```

Table 7: Show ISM-MG memory utilization fields

Label	Description
Slot	Physical slot number for the ISM-MG
MDA	MDA number for the physical slot; only 1 is supported for the ISM-MG Card

Table 7: Show ISM-MG memory utilization fields (continued)

Label	Description
Group	Group number
GW-ID	Gateway ID (1 to 8)
Current size	Total available memory size (per slot/MDA or interface)
In use	Total memory that is allocated (per slot/MDA or interface)
Avail Virt Mem	Total available virtual memory (per slot/MDA)
Avail Phys Mem	Total available physical memory (per slot/MDA)
Name	Protocol or interface name

4.2.2.2.11 sdf

Syntax

```
sdf [group-number]
```

Context

```
show>mobile>ism-mg
```

```
show>mobile>mg-vm
```

Description

This command displays the SDF tables utilization for the ISM-MG SDF group (7750 SR MG) or the MG-VM SDF group (CMG). The SDF is an aggregate set of packet flows carried through the PCEF that matches a service data flow template. The utilization of SDFs is tracked against the relative limits.

Parameters

group-number

specifies the number of the ISM-MG or MG-VM group

Values: 1 to 15

Output

The following two examples shows SDF utilization statistics for the ISM-MG and the MG-VM SDF groups. *Show mobile gateway ISM-MG and MG-VM SDF output fields* describes the output fields.

Example output SDF statistics

```
# show mobile-gateway ism-mg sdf
=====
ISM-MG SDF utilization summary
=====
Group                In use      FP Total    FP Utilization
                   Assigned    LCP Total    LCP Utilization
-----
1                    123422      8000000     1.54%
                   5436643     64000000     8.49%
=====

# show mobile-gateway mg-vm sdf
=====
MG-VM SDF utilization summary
=====
Group                In use      FP Total    FP Utilization
                   Assigned    LCP Total    LCP Utilization
-----
1                      0           8388608     0.00%
                      0           33554432     0.00%
=====
```

Table 8: Show mobile gateway ISM-MG and MG-VM SDF output fields

Label	Description
Group	The number of the ISM-MG SDF or the MG-VM SDF group
In use	The actual number of SDFs that are programmed in the Fast Path (FP)
Assigned	The number of SDFs that are provisioned on the LCP
FP Total	The total number of FP (hardware) boot time parameters. This is chosen depending on the card type. FP has a fixed eight million entries. SDFs from the LCP pool move to FP when traffic is seen and move back to the LCP pool when they are idle.

Table 8: Show mobile gateway ISM-MG and MG-VM SDF output fields (continued)

Label	Description
FP Utilization	The percentage of the hardware pool that is utilized, based on user traffic on sessions
LCP Total	The total number of LCP (IOM/MSM software) boot time parameters. This is chosen based on the card type. LCP is oversubscribed entries (64 million). SDFs from the LCP pool move to FP when traffic is seen and move back to the LCP pool when they are idle.
LCP Utilization	The percentage of the assigned LCP software pool that is utilized, based on the number of calls and polices

4.2.2.2.12 mg-vm

Syntax

```
mg - vm
```

Context

```
show>mobile
```

Description

This command displays information about the MG-VM for CMG systems.

4.2.2.2.13 cpu

Syntax

```
cpu [vm vm-id] [vm-function vm-function-number] [repeat repeat-number]
```

Context

```
show>mobile>mg-vm
```

Description

This command displays the CPU utilization of MG-VMs.

Parameters

vm-id

specifies the VM number

vm-function-number

specifies the MG-VM function number

Values: 1 | 2

repeat-number

specifies the sample period, the number of times to repeat the sampling

Values: 1 to 5

Output

The following output examples show CPU utilization information for MG-VMs and *Show MG-VM CPU utilization fields* describes the output fields.

- *Example output MG-VM CPU utilization summary statistics*
- *Example output MG-VM CPU utilization statistics for a specified MG-VM*

Example output MG-VM CPU utilization summary statistics

```
# show mobile-gateway mg-vm cpu
=====
CPU utilization starting at 09/14/2018 21:53:33
=====
ISM-MG Summary
=====
VM      VMF      Group gw-id      cpu utilization
-----
3       1       1       1       5.72%
4       1       1       1       1.85%
=====
```

Example output MG-VM CPU utilization statistics for a specified MG-VM

```
# show mobile-gateway mg-vm cpu vm 3
=====
CPU utilization starting at 09/14/2018 21:54:08
=====
ISM-MG VM 3 VMF 1
=====
Name          VM Usage
-----
AMS           0.01%
DHCP          ~ 0.00%
GA            ~ 0.00%
```

```

GX          0.01%
GY          0.02%
HA          0.07%
PFCP       0.02%
RADIUS      0.02%
RF          0.02%
S6B         0.03%
SESSMGR     0.03%
SIG         0.05%
SYSTEM      3.88%
TRACE       0.02%
RPY         0.01%
SWM         ~ 0.00%
LDNSC       ~ 0.00%
SD          0.01%
SDLB        ~ 0.00%
STA         ~ 0.00%
CDBMGR      0.09%
<NULL>      ~ 0.00%
HTTP        ~ 0.00%

```

```

-----
Total In Use:    4.27%
Idle:           95.73%
=====

```

Table 9: Show MG-VM CPU utilization fields

Label	Description
VM	MG-VM number
VMF	MG-VM function number
Group	Group number
GW-ID	Gateway ID (1 to 8)
CPU utilization	Percentage of CPU utilization per MG-VM number
Name	Protocol or interface name
CPU usage	Percentage of CPU usage per interface

4.2.2.2.14 memory-pools

Syntax

```
memory-pools [vm vm-id] [vm-function vm-function-number] [repeat repeat-number]
```

Context

```
show>mobile>mg-vm
```

Description

This command displays the memory utilization of MG-VMs.

Parameters

vm-id

specifies the VM number

vm-function-number

specifies the MG-VM function number

Values: 1 | 2

repeat-number

sample period, the number of times to repeat the sampling

Values: 1 to 5

Output

The following output examples show memory utilization information for MG-VMs and *Show MG-VM memory utilization fields* describes the output fields

- *Example output MG-VM memory statistics*
- *Example output MG-VM memory statistics for a specified MG-VM*

Example output MG-VM memory statistics

```
# show mobile-gateway mg-vm memory-pools
=====
MEM utilization starting at 09/20/2018 20:45:32
=====
ISM-MG Summary
=====
VM  |VME| |Group|gw-id|  Current Size|          In Use|Avail Virt Mem|Avail Phys Mem
-----+-----+-----+-----+-----+-----+-----+-----+-----
3   |1   |1   |1   | 9,289,334,784| 8,900,651,816| 6,677,331,968| 6,677,331,968
4   |1   |1   |1   | 9,242,148,864| 8,862,652,080| 6,811,549,696| 6,811,549,696
=====
```


Example output MG-VM memory statistics for a specified MG-VM

```
# show mobile-gateway mg-vm memory-pools vm 3
=====
MEM utilization starting at 09/20/2018 00:03:30
=====
ISM-MG VM 3 VMF 1
=====
Name | Current Size | In Use
-----+-----+-----
AMS | 1,048,576 | 158,264
DHCP | 9,437,184 | 5,566,256
GA | 2,097,152 | 755,928
GX | 8,388,608 | 6,203,808
GY | 6,291,456 | 2,972,640
HA | 9,437,184 | 1,816,176
PFCP | 9,437,184 | 2,748,104
RADIUS | 13,631,488 | 3,336,288
RF | 77,594,624 | 68,117,896
S6B | 39,845,888 | 34,266,368
SESSMGR | 136,314,880 | 82,402,904
SIG | 68,157,440 | 46,028,992
SYSTEM | 2,164,260,864 | 2,017,004,704
TRACE | 28,311,552 | 21,164,912
RPY | 7,340,032 | 3,420,904
SWM | 7,340,032 | 3,415,472
LDNSC | 2,097,152 | 47,560
SD | 7,340,032 | 4,600,416
SDLB | 0 | 0
STA | 5,242,880 | 2,354,608
CDBMGR | 0 | 0
<NULL> | 0 | 0
HTTP | 0 | 0
-----+-----+-----
Total: | 2,603,614,208 | 2,306,382,200
=====
available virtual memory: 5,133,828,096
available physical memory: 5,133,828,096
=====
```

Table 10: Show MG-VM memory utilization fields

Label	Description
VM	MG-VM number
VMF	MG-VM function number
Group	Group number

Table 10: Show MG-VM memory utilization fields (continued)

Label	Description
GW-ID	Gateway ID (1 to 8)
Current size	Total available memory size (per VM/VM function or interface)
In use	Total memory that is allocated (per VM/VM function or interface)
Avail Virt Mem	Total available virtual memory (per VM/VM function)
Avail Phys Mem	Total available physical memory (per VM/VM function)
Name	Protocol or interface name

4.2.2.2.15 system

Syntax

system [gateway *gw-id* | group *group-number* | [card *slot-number* | vm *vm-id*]] [summary]

system [call-insight]

Context

show>mobile

Description

This command displays mobile gateway system information.

Parameters

gw-id

specifies the gateway ID

Values: 1

group-number

specifies the group number

Values: 1 to 15

slot-number

specifies the card slot number (applicable only to 7750 SR MG)

Values: 1 to 20 | A | B

vm-id

specifies the VM ID (applicable only to CMG)

Values: 1 to 20 | A | B

summary

shows aggregated statistics per-peer level

Output

The following output examples show system information and *Show mobile gateway system information* describes the output fields.

- *Example output system information (one-to-one redundancy)*
- *Example output system information (many-to-many redundancy for CMG)*



NOTE:

CMG system outputs may be identical to the 7750 SR MG outputs, with some CMG-specific fields displayed instead of hardware components (for example, VM is displayed instead of Card).

Example output system information (one-to-one redundancy)

```
# show mobile-gateway system
=====
Mobile gateway system
=====
Use-fixed-udp-src-port : Disabled
Ue-fragmented-packets-handling : Disabled
ttc-accuracy : false
GTP-U Source Port Dist : Disabled
-----
Gateway : 1 Instance-type : None
Admin State : Down Oper state : Down
Suspended : No Suspended Since : N/A
Restart counter : N/A EPC Node :
Aa-grp/Part-id : 0/0 Def app profile :
parent-ctl-cc-na*:
Admin Uptime : N/A Gateway Uptime : N/A
-----
Number of gateways : 1
=====
* indicates that the corresponding row element may have been truncated.
```

Example output system information (many-to-many redundancy for CMG)

This output example is supported for the CMG platform only.

```
# show mobile-gateway system

=====
Mobile gateway system
=====
=====
Use-fixed-udp-src-port : Disabled
Ue-fragmented-packets-handling : Disabled
ttc-accuracy : false
GTP-U Source Port Dist : Disabled
-----
Gateway : 1 Instance-type : control
Admin State : Up Oper state : Up
Suspended : No Suspended Since : N/A
Restart counter : 1 EPC Node : 310.150.PGW.1.1.1
Aa-grp/Part-id : 0/0 Def app profile :
parent-ctl-cc-na*:
Admin Uptime : 03/14/2022 11:09:41 Gateway Uptime : 03/14/2022 11:09:41
PFCP Recover Time: 03/14/2022 11:09:41 UTC 3856244981#
Group : 1
Admin State : Up Oper state : Up
Override profile : N/A
Group Uptime : 03/14/2022 11:09:41
Redundancy : one-to-one Red State : Hot
Switch over count: 0 Switch over time : N/A
Switch over state: N/A
PCMD Profile : inherit PCMD oper. state : down
VM : 1/3 Role/Red State : working/active
VM : 1/4 Role/Red State : protect/standby
-----
Number of gateways : 1
=====
* indicates that the corresponding row element may have been truncated.
# indicates the number of seconds elapsed from epoch January 1 1900.
```

Table 11: Show mobile gateway system information

Label	Description
Use-fixed-udp-src-port	Indicates if a fixed UDP source port is used Enabled or disabled (default)
Ue-fragmented-packets-handling	Indicates if the IP packet reassembly of the user-level traffic for packet classification is used Enabled or disabled (default)

Table 11: Show mobile gateway system information (continued)

Label	Description
ttc-accuracy	TTC accuracy optimization
GTP-U Source Port Dist	Selects a UDP source port for GTP-U packets
Gateway	Gateway ID (1 to 8)
Instance type	User-plane node (UPF/SGW-U/PGW-U, combined SGW-U/PGW-U) Control-plane node (SMF/SGW-C/PGW-C, combined SGW-C/PGW-C)
Admin State	The administratively applied state of the gateway (Up or Down)
Oper State	The operational state of the gateway (Up or Down)
Suspend	Indicates if the gateway is in a suspended state, in which case it does not accept new session, but continues to support call-flow on existing sessions
Suspend since	The time at which the gateway was suspended
Restart counter	Restart counter value
EPC Node	EPC node name; can be 31 characters maximum composed of the following: <MCC>.<MNC>.<SGW PGW ePDG TWAG>.<Region String>.<Group Id>.<Node Id>
AA-grp/Part-id	The AA group/partition ID assigned to the mobile-gateway system
Def app profile	The default AA application profile assigned to the mobile-gateway system
parent-ctl-cc-na	The optional parental control ASO charging characteristic name, when the AA group and default application profile are configured
Admin Uptime	Time at which the gateway was administratively up

Table 11: Show mobile gateway system information (continued)

Label	Description
Gateway Uptime	Time the gateway was operationally up
PFCP Recover Time	Time of the PFCP recovery Note: It is displayed only in case CUPS is set up.
Group	Mobile-gateway system group number (1 to 8)
Admin State	The administratively applied state of the group (up or down)
Oper state	Operational state of the group (up, down, in transition, and scale-in in progress)
Override profile	Indicates if an override profile is configured, or not, for each group for GTP-U redirection; applicable to CMG only
Group Uptime	Time at which the group was operationally up
Red State	Redundancy state (hot or cold)
Switch over count	Indicates for the group, the number of switchovers for each {active to standby} transition of the VM/Card
Switch over time	Time of the last switchover
Switch over state	The state of the switchover
Resource pool	The resource pool assigned to the group defines the VM resources (cards) and the redundancy method used for the group; applicable to CMG only
Redundancy	Many-to-many ("M" active VMs are configured to have "N" standby/ spares to provide HOT redundancy) One-to-one (each group is configured to provide HOT redundancy per VM/Card)

Table 11: Show mobile gateway system information (continued)

Label	Description
PCMD Profile	Indicates if the group inherits the PCMD profile: inherit — the PCMD profile configured in the config>mobile>profile>pcmd context is applied to the group none — PCMD records are not generated for this group
PCMD Oper state	Operational state of the PCMD (up or down)
Card/VM	Card/VM slot number
Role/Red state	Indicates the working/active or protect/standby state of the VM/ Card

4.2.2.3 Show redundancy command descriptions

4.2.2.3.1 mc-mobile

Syntax

```
mc-mobile peer {ip-address | ipv6-address}  
mc-mobile peer {ip-address | ipv6-address} mc-red-tcp-stats
```

Context

```
show>redundancy>multi-chassis>mc-mobile
```

Description

This command displays LAG (Link Aggregation Group) information.

Parameters

ip-address

shows information for the specified IPv4 peer

ipv6-address

shows information for the specified IPv6 peer

mc-red-tcp-stats

displays MC redundancy red state TCP connection statistics

Output

The following output examples show Multi-Chassis (MC) redundancy information.

- [Example output MC redundancy statistics \(UE-by-UE reconcile synchronization disabled\)](#)
- [Example output MC redundancy statistics \(UE-by-UE reconcile synchronization enabled\)](#)
- [Example output MC redundancy statistics \(traffic detection enabled\)](#)
- [Example output MC redundancy red state TCP connection statistics](#)

Example output MC redundancy statistics (UE-by-UE reconcile synchronization disabled)

The following output example shows MC redundancy information when the UE-by-UE reconcile synchronization is disabled and [Show redundancy MC mobile output fields](#) explains the output fields.

```
# show redundancy multi-chassis mc-mobile peer 10.20.1.1
=====
Multi-chassis Peer Mc-Mobile Table
=====
Peer                : 10.20.1.1
Last State Change   : 05/21/2019 06:05:04
Admin State         : Up                Oper State         : Up
Peer Version        : TiMOS-MG-C-11.0.S65
Keep Alive          : 10 deci-sec       Hold On Nbr Fail    : 3
BFD Svc ID          : 0                BFD Interface Name  : system
MC-Redirect         : Disabled
Admin MC-complete-ue-sync : Disabled
Oper MC-complete-ue-sync  : Disabled
Slave Traffic Detection Delay Timer : Disabled
Traffic detection Poll timer           : 5 secs
Master Traffic detection                : Enabled
Traffic detection                      : Strict
-----
Gateway Id           : 1
-----
Admin Role           : Primary          Oper Role           : Master
Peer Admin Role      : Secondary        Peer Oper Role      : Slave
Admin State          : Up              Oper State          : Up
Advertised metric    : Master
Last Time Peer Connected : 05/21/2019 06:05:04
Last Oper Role Change   : 05/20/2019 11:38:39
Last Oper Role Chg Rsn  : Conf Primary
Geo-Redundancy State  : Hot
FSW0 synchronization in progress : no
Mc-redirect trap set   : no
Mc-master-lock         : disabled
Mc-slave-lock          : disabled
Remaining Slave Traffic Detection Delay Timer : Disabled
-----
```


FSW0 progress

```
-----
MG/OAM group Id      State      MCS sync use  FSW0 tasks synced
-----
OAM      : 0      Geo Redundancy : Hot      0%      0/0
MG Group : 1      Geo Redundancy : Hot      0%      0/0
-----
=====
```

Example output MC redundancy statistics (UE-by-UE reconcile synchronization enabled)

The following output example shows MC redundancy information when the UE-by-UE reconcile synchronization is enabled and *Show redundancy MC mobile output fields* explains the output fields.

```
# show redundancy multi-chassis mc-mobile peer 10.20.1.1
=====
Multi-chassis Peer Mc-Mobile Table
=====
Peer      : 10.20.1.1
Last State Change : 05/21/2019 06:05:04
Admin State : Up      Oper State : Up
Peer Version : TiMOS-MG-C-11.0.S65
Keep Alive : 10 deci-sec  Hold On Nbr Fail : 3
BFD Svc ID : 0      BFD Interface Name : system
MC-Redirect : Disabled
Admin MC-complete-ue-sync : Enabled
Oper MC-complete-ue-sync : Enabled
Slave Traffic Detection Delay Timer : Disabled
Traffic detection Poll timer : 5 secs
Master Traffic detection : Enabled
Traffic detection : Strict
-----
Gateway Id : 1
-----
Admin Role : Primary      Oper Role : Master
Peer Admin Role : Secondary Peer Oper Role : Slave
Admin State : Up      Oper State : Up
Advertised metric : Master
Last Time Peer Connected : 05/21/2019 06:05:04
Last Oper Role Change : 05/20/2019 11:38:39
Last Oper Role Chg Rsn : Conf Primary
Geo-Redundancy State : Hot
FSW0 synchronization in progress : no
Mc-redirect trap set : no
MC-complete-ue-sync status : 100%
Mc-master-lock : disabled
Mc-slave-lock : disabled
Remaining Slave Traffic Detection Delay Timer : Disabled
-----
FSW0 progress complete UE sync
-----
MG group Id      State      MCS sync use  Synced UEs
-----
CPM      : 0      Geo Redundancy : Hot      0%      100%
MG Group : 1      Geo Redundancy : Hot      0%      100%
-----
```

Example output MC redundancy statistics (traffic detection enabled)

The following output example shows MC redundancy information when traffic detection is enabled and *Show redundancy MC mobile output fields* explains the output fields.

```
# show redundancy multi-chassis mc-mobile peer 10.20.1.1
=====
Multi-chassis Peer Mc-Mobile Table
=====
Peer                : 10.20.1.1
Last State Change   : 02/25/2019 10:45:49
Admin State         : Up                Oper State           : Down
Peer Version        :
Keep Alive          : 10 deci-sec       Hold On Nbr Fail      : 3
BFD Svc ID          : 0                 BFD Interface Name    : system
MC-Redirect          : Disabled
Admin MC-complete-ue-sync : Disabled
Oper MC-complete-ue-sync  : Disabled
Slave Traffic Detection Delay Timer : Disabled
Traffic detection Poll timer           : 5 secs
Master Traffic detection                : Enabled
Traffic detection                      : Strict
-----
Gateway Id                : 1
-----
Admin Role                : Primary       Oper Role                : Master
Peer Admin Role           : Unknown       Peer Oper Role           : Unknown
Admin State               : Up            Oper State               : Down
Advertised metric         : Slave         Traffic detection        : No-traffic
Last Time Peer Connected  : 02/25/2019 10:19:59

Last Oper Role Change     : 02/25/2019 10:19:59
Last Oper Role Chg Rsn    : Conf Primary
Geo-Redundancy State      : Cold
FSW0 synchronization in progress : no
Mc-redirect trap set      : no
MC-complete-ue-sync status : 100%
Mc-master-lock            : disabled
Mc-slave-lock             : disabled
Remaining Slave Traffic Detection Delay Timer : Disabled
Remaining Traffic detection Poll timer         : 3 secs
Total remaining traffic detection Poll timer   : 1198 secs

-----
FSW0 progress
-----
MG/OAM group Id      State      MCS sync use  FSW0 tasks synced
-----
OAM      : 0      Geo Redundancy : Cold      0%      0/0
MG Group : 1      Geo Redundancy : Cold      0%      0/0
```

Table 12: Show redundancy MC mobile output fields

Label	Description
Peer	IP address of the ICR multi-chassis mobile peer
Last State Change	Date and time of the most recent state change of the peer
Admin State	Current administrative state
Oper State	Current operational state
Peer Version	Software version of the peer
Keep Alive	Configured keep alive time
Hold on NBR Fail	The number of keep-alive-intervals that may expire before the local node decides that the neighboring (NBR) peer has failed
BFD Svc ID	ID of the BFD service
BFD Interface Name	Name of the BFD interface
MC-Redirect	Indicates the status of MC redirection
Admin MC-complete-ue-sync	Indicates the status of the MC UE-by-UE ICR synchronization that is locally configured
Oper MC-complete-ue-sync	Indicates the operational synchronization that is currently running The master node dictates the ICR sync type that must be used
Slave Traffic Detection Delay Timer	Indicates the status when disabled and the configured value when enabled

Table 12: Show redundancy MC mobile output fields (continued)

Label	Description
Gateway ID	ID of the gateway
Admin Role	Administrative role of the gateway: Primary the gateway that is intended to be the master Secondary the gateway that is intended to be the slave
Oper Role	Current operational role of the gateway: Master the gateway that is currently forwarding mobile traffic and participating in signaling with the SGW, SGSN, and so on Slave the gateway that is ready to take over in case the master fails
Peer Admin Role	Administrative role of the peer gateway: Primary the gateway that is intended to be the master Secondary the gateway that is intended to be the slave
Peer Oper Role	Current operational role of the peer gateway: Master the gateway that is currently forwarding mobile traffic and participating in signaling with the SGW, SGSN, and so on

Table 12: Show redundancy MC mobile output fields (continued)

Label	Description
	Slave the gateway that is ready to take over in case the master fails
Advertised metric	Indicates if the advertised metrics are for the master, slave, pre-slave, or none This field is always present regardless of the MCS connection state
Traffic Detection	Indicates if all, partial, or no traffic is detected This field is displayed only when the MCS connection is lost
Diameter Connection Lost Detection	The status of the Diameter Connection Lost Detection
Gx Peer Guard Timer	The time, in minutes, and status of the Gx guard timer to delay the ICR FSWO when all the Gx connections are detected as lost
Last Time Peer Connected	Date and time of the last peer connection
Last Oper Role Change ¹	Date and time of the last operational role change

¹ The values of these fields change only when the election is done and there is a change in the operational role.

Scenario example:

- peer C is the master, peer A is the slave and ICR is HOT
- the MCS link between the geo-redundant nodes (C and A) goes down; because of this, the ICR goes COLD
- while the MCS link is down:
 - the old master (C) continues to receive all three types of traffic (UL control, UL data, and DL data) and stays as the master
 - the old slave (A) starts receiving all three types of traffic (UL control, UL data, and DL data) and takes up the role of the master
- when the MCS link is up, both geo-redundant nodes exchange their states and other parameters, for example:

Table 12: Show redundancy MC mobile output fields (continued)

Label	Description
Last Oper Role Chg Rsn ¹	Reason for the last operational role change
Geo-Redundancy State	Geo-redundant state: Hot MCS protocol is up and successfully syncing Warm MCS protocol is up, but not syncing (for example when the peer system is in the process of booting up) Cold MCS protocol is down
FSWO synchronization in progress	Indicates if a forced switchover is in progress
MC-master-lock	Indicates whether the gateway is locked in the master operational role
MC-slave-lock	Indicates whether the gateway is locked in the slave operational role
MC redirect trap set	Indicates whether an MC redirect trap is set
Remaining Slave Traffic Detection Delay Timer	Indicates the remaining time before the slave detection delay timer expires (if enabled)

-
- the master (C) sends its current PDN session count of 800K
 - the old slave that turned to master (A) sends its current PDN session count of 300K
 - as part of the election process, since C is the master with higher PDN session count, it takes up the role of master and A assumes the role of the slave
 - in this case, the geo-redundancy operational roles before and after the MCS link down events are the same (that is, C master and A slave)

In this example, since there is no change in the operational roles, there is no change in the Last Oper Role Chg Rsn and Last Oper Role Change fields in the output of the `show>redundancy>multi-chassis>mc-mobile>peer` command.

Table 12: Show redundancy MC mobile output fields (continued)

Label	Description
Traffic detection Poll timer	Indicates the configured value for the traffic detection poll timer
Master Traffic Detection	Indicates whether the ICR Traffic Detection is Enabled or Disabled in the master operational role The default value of this configuration is Enabled and is only applies to the gateway in the master operational role
Traffic Detection	Indicates the configured ICR traffic detection criteria Can apply to special 7750 SR MG and CMG ICR deployments, with a very small number of attached UEs, or very low, or no active user traffic. The relaxed option must not be used for any regular 7750 SR MG and CMG deployments. It is only supported on 2G (GGSN) without any charging and policy or RADIUS deployed.
Remaining Traffic Detection Poll Timer	Indicates the remaining time in the current traffic detection polling interval (the value configured using the traffic-detection-poll-timer command) This label is displayed in the CLI output only when the ICR traffic detection is running
Total Remaining Traffic Detection Poll Timer	Indicates the total remaining time in the traffic detection timer (1200 seconds); when the timer expires, the ICR traffic detection is disabled if no traffic is detected This label is displayed in the CLI output only when the ICR traffic detection is running
MC-Redirect Trap Details	
VPRN ID	VPRN ID for the MC redirect trap
Interface Name	Name of the VPRN interface for the MC redirect trap
Trap set count	Number of traps that have been set

Table 12: Show redundancy MC mobile output fields (continued)

Label	Description
Last trap timestamp	Date and time of the most recent trap
FSWO Progress	
MG/CPM group Id	ID of the gateway or CPM group for which the failed switchover is in progress
State	Geo-redundant state: Hot MCS protocol is up and successfully syncing Warm MCS protocol is up, but not syncing (for example when the peer system is in the process of booting up) Cold MCS protocol is down
MCS sync use	The percentage of MCS synchronization that is in use
FSWO tasks synced	The number of failed switchover tasks that have synchronized
Synced UEs	The number of UEs that have synchronized Visible only when the Oper MC-complete-ue-sync is enabled

Status of the admin MC-complete-ue-sync and oper MC-complete-ue-sync on the master and slave nodes lists the indicated status of the Admin MC-complete-ue-sync and Oper MC-complete-ue-sync fields on the master and slave nodes based on the status of the MC UE-by-UE reconcile synchronization feature.

Table 13: Status of the admin MC-complete-ue-sync and oper MC-complete-ue-sync on the master and slave nodes

MC-complete-ue-sync Status	Master node		Slave node	
	Admin MC-complete-ue-sync	Oper MC-complete-ue-sync	Admin MC-complete-ue-sync	Oper MC-complete-ue-sync
Disabled on master and slave	Disabled	Disabled	Disabled	Disabled
Enabled on master and disabled on slave	Enabled	Enabled	Disabled	Enabled
Disabled on the fly on master and disabled on slave	Disabled	Disabled	Disabled	Disabled
Disabled on master and enabled on slave	Disabled	Disabled	Enabled	Disabled

Values of the Last Oper Role Chg Rsn field describes the values of the Last Oper Role Chg Rsn field and their definitions on the master and slave nodes.

Table 14: Values of the Last Oper Role Chg Rsn field

Code	Description (Master)	Description (Slave)
No peer mcm	N/A	The node has transitioned to slave because: - the admin>multi-chassis>peer>shutdown command was issued, or - the MCS link is being lost
Cfg primary	The node is configured as primary and the peer is not master	N/A

Table 14: Values of the Last Oper Role Chg Rsn field (continued)

Code	Description (Master)	Description (Slave)
Cfg secondary	N/A	The node is configured as secondary and the peer is the master
Peer restart	The peer has restarted	—
Traffic event	The node has received all three types of traffic (UL control, UL data, and DL data) while traffic detection is enabled	N/A
Peer disc	The node has transitioned to master because: - the admin>multi-chassis>peer>shutdown command was issued on the peer node - the admin>redundancy>mc-mobile-switchover>mobile-gateway command with the now option was issued on the peer node	N/A
Manual switchover	The node has transitioned to master after the admin>redundancy>mc-mobile-switchover>mobile-gateway command was issued on the peer node	—
Traffic absent	N/A	The node did not receive any traffic while traffic monitoring was enabled and thus, was forced to transition to slave
Manual Switchover with MC-Redirect	The admin>redundancy>mc-mobile-switchover>mobile-gateway was issued on the peer and MC-redirect (shunt) is in use	N/A

Table 14: Values of the Last Oper Role Chg Rsn field (continued)

Code	Description (Master)	Description (Slave)
Master lock	The node is locked to be the master	The peer node is locked to be the master
Slave lock	The peer node is locked to be the slave	The node is locked to be the slave
Previous role	The node is the only one suitable to be the master due to its previous role	The node is not suitable to be the master due to its previous role
PDN shutdown	The node has transitioned to master because a PDN shutdown (using the config>mobile>pdn>shutdown command) is in progress on the peer node	The node has transitioned to slave because a PDN shutdown (using the config>mobile>pdn>shutdown command) is in progress on this node but the peer is operational
Session count	The node has transitioned to master because it has a higher number of sessions than its peer	The node has transitioned to slave because it has a lower number of sessions than its peer

Example output MC redundancy red state TCP connection statistics

The following output example shows MC redundancy red state TCP connection statistics and *Show redundancy MC mobile red state TCP connection output fields* explains the output fields.

```
# show redundancy multi-chassis mc-mobile peer 10.90.1.2 mc-red-tcp-stats
=====
Multi-chassis Redundancy TCP connection stats
=====
Local-address: 10.90.1.1
Peer address: 10.90.1.2
MG/CPM group Id Conn-state      Num-Retrans  Num-DupAck
-----
CPM      : 0    ESTABLISHED    0           0
MG Group : 1    ESTABLISHED    0           0
-----
Number of TCP connections: 2
=====
```

Table 15: Show redundancy MC mobile red state TCP connection output fields

Label	Description
Num-Retrans	The number of retransmissions
Num-DupAck	The number of duplicate acknowledgments

5 PDN command reference

This section describes the PDN-related commands.

5.1 PDN command hierarchies

- *PDN configuration command hierarchies*
- *PDN show command hierarchy*
- *PDN clear command hierarchy*

5.1.1 PDN configuration command hierarchies

- *PDN command hierarchy*
- *APN PDN command hierarchy*
- *Bearer traffic PDN command hierarchy*
- *Cloud database PDN command hierarchy*
- *Charging PDN command hierarchy*
- *Ga PDN command hierarchy*
- *Gn PDN command hierarchy*
- *Gp PDN command hierarchy*
- *GTP load and overload control PDN command hierarchy*
- *Gx PDN command hierarchy*
- *Gy PDN command hierarchy*
- *KPI-KCI PDN command hierarchy*
- *N2 PDN command hierarchy*
- *N3 PDN command hierarchy*
- *N9 PDN command hierarchy*
- *Network trace command hierarchy*
- *NF profile attributes command hierarchy*
- *Nwu PDN command hierarchy*
- *PFCP load and overload PDN command hierarchy*
- *PGW selection PDN command hierarchy*

- *Pi PDN command hierarchy*
- *RADIUS PDN command hierarchy*
- *Rf PDN command hierarchy*
- *S1-u PDN command hierarchy*
- *S11 PDN command hierarchy*
- *S12 PDN command hierarchy*
- *S2a PDN command hierarchy*
- *S2b PDN command hierarchy*
- *S4 PDN command hierarchy*
- *S5 PDN command hierarchy*
- *S6b PDN command hierarchy*
- *S8 PDN command hierarchy*
- *SBA client services PDN command hierarchy*
- *SBA server services PDN command hierarchy*
- *SBA service realm PDN command hierarchy*
- *Sd PDN command hierarchy*
- *Signaling PDN command hierarchy*
- *STa PDN command hierarchy*
- *SWm PDN command hierarchy*
- *SWu PDN command hierarchy*
- *SWw PDN command hierarchy*
- *Sx-N4 PDN command hierarchy*
- *T6c command hierarchy*
- *Threshold PDN command hierarchy*
- *Virtual APN selection PDN command hierarchy*

5.1.1.1 PDN command hierarchy

configure

- mobile-gateway
- [no] *pdn gw-id*
 - *5g-ims-fallback-timeout timeout*
 - no *5g-ims-fallback-timeout*

```

    - aa-group [aa-group-id[:partition-id]] [def-app-profile profile-name] [parent-ctl-cc-name value]
    - [no] allow-duplicate-imei
    - [no] allow-duplicate-msisdn
    - [no] allow-multi-ue-session
    - [no] allow-multiple-apn-pdn
    - [no] apn {apn-name | unknown}
    - [no] auto-set-handover-indication
    - bearer-traffic
    - call-procedure-retry retry-time seconds retry-attempts value [retry-delay retry-delay-msec]
    - no call-procedure-retry
    - cam-optimization {enable | disable}
    - no cam-optimization
    - cdbx
    - charging
    - dl-buffering
    - [no] drop-oldest-pkt
    - max-pkt limit
    - no max-pkt
    - max-pkt-size [1..2048]
    - no max-pkt-size
    - total-buffer-capacity value
    - no total-buffer-capacity
    - total-buffer-pkts value
    - no total-buffer-pkts
    - create-procedure-retry retry-time seconds retry-attempts attempts [retry-delay retry-attempt-msec]
    - no create-procedure-retry
    - default-traffic-hash profile-name
    - no default-traffic-hash
    - [no] del-bearer-reactivation-req-cause-pgw-failure
    - delay-nw-procedures timer-value-msec
    - no delay-nw-procedures
    - [no] disable-hss-init-update-failure-delete
    - dnai-map entry-id {[plmn-list plmn-list-name] [mnvo] [incoming-apn-list apn-list-name]} mapped-apn mapped-apn
    - no dnai-map entry-id
    - dns-client
    - [no] cache-clear-no-such-name
    - dns-interface interface-name [router router-instance]
    - no dns-interface
    - dns-profile profile-name
    - no dns-profile
    - [no] enable-pcscf-low-ttl

```

```

    - refresh-interval {pcscf | pgw | apn | all} refresh-interval-sec interval-in-seconds use-dns-ttl]
    - no refresh-interval {pcscf | pgw | apn | all}
    - [no] topon-match-first
    - [no] ttl-zero
    - dns-error-mapping-profile profile-name
    - no dns-error-mapping-profile
    - [no] dual-connectivity
    - [no] dual-connectivity-5g
    - [no] enable-sgw-c-partitioning
    - epc-node name
    - no epc-node
    - [no] epdg-5gs-compatibility [rel-17 {enable | disable}]
    - [no] exclude-error-indication-extension-header
    - [no] fwd-gtpu-echo-req-to-mg
    - ga
    - [no] gn [group-specific]
    - [no] gp [sig-ref-point-name] [group-specific]
    - gtp-load-control
    - gtp-overload-control
    - [no] gtpc-init-msg-src-ip-validation-bypass
    - gtpu-drop-echo-threshold queue-depth-threshold outstanding-echo-requests queue-threshold-sustain-time sustain-interval backoff-time backoff-time
    - no gtpu-drop-echo-threshold
    - gtpu-path-failure ref-point ref-point action action
    - no gtpu-path-failure ref-point ref-point
    - gx
    - [no] gx-before-aaa
    - gy
    - ho-apn-lookup {imsi-based-oi | ni-only}
    - no ho-apn-lookup
    - home-plmn-list list-name
    - no home-plmn-list
    - [no] i-upf-selection
    - ims-emps-priority cp-dscp-marking [value] up-uplink qci-policy-name up-downlink qci-policy-name
    - ims-emps-priority
    - no ims-emps-priority
    - imsi-plmn-list list-name
    - no imsi-plmn-list
    - instance-type instance-type-value
    - no instance-type
    - ipsmg-group ipsmg-group-id
    - no ipsmg-group
    - kpi-kci

```



```

– mass-deactivation
– qci qci-value [arp-pl arp-pl-value] session-delete-priority priority-value
– no qci qci-value [arp-pl arp-pl-value]
– mg-group-call-failure-rate call-failure-rate rate-percentage [monitor-time
timer]
– no mg-group-call-failure-rate
– n3 [sig-ref-point-name] [group-specific]
– n9 [sig-ref-point-name] [group-specific]
– nf-profile-attributes [bsf]
– network-trace
– non3gpp-pdn-connection-rejection-with-ip
– [no] handover-no-mip6-agent-info
– [no] pgw-handover-response-cc64
– [no] nsmf-congestion-control
– [no] enable-nsmf-congestion-control
– max-activations max-activations
– [no] max-activations
– retry-after-timer seconds
– [no] retry-after-timer
– [no] nsmf-overload-control
– [no] enable-nsmf-overload-control
– overload-threshold threshold-value
– [no] overload-threshold
– retry-after-timer seconds
– [no] retry-after-timer
– [no] nswo-capable
– oi-override-mapping list-name
– no oi-override-mapping
– overload-control
– cpu-sampling-interval [overload overload-interval] [overload-state-
change ovrld-state-change-intrvl] [no-overload no-overload-interval]
– no cpu-sampling-interval
– thresholds
– critical threshold-value
– no critical
– major threshold-value
– no major
– minor threshold-value
– no minor
– [no] override-profile override-profile-id
– description description-string
– no description
– [no] override ref-point-type ref-point-type ref-point-name to override
with ref-point-name to override with

```

```

    - override ref-point-type ref-point-type router router-instance to
      override interface-name to override with router-instance to override with interface-
name to override with
    - no override ref-point-type ref-point-type router router-instance to
      override interface-name to override
    - [no] pcf-pcrf-based-pcscf-allocation
    - entry entry-id apn apn-name {diameter-peer profile-name | nf-id uuid}
    {{pcscf-srvr-v4-group pcscf-group [secondary-pcscf-srvr-v4-group pcscf-group]
    [tertiary-pcscf-srvr-v4-group pcscf-group]} | {pcscf-srvr-v6-group pcscf-group
    [secondary-pcscf-srvr-v6-group pcscf-group] [tertiary-pcscf-srvr-v6-group pcscf-
group]}}
    - no entry entry-id
    - pcmd profile profile-name
    - no pcmd
    - pcscf-recovery [dns-cache-interval] [pco-reselection]
    - no pcscf-recovery
    - pco-option
    - max-allocated-pcscf-addresses number
    - no max-allocated-pcscf-addresses
    - pfcf-load-control
    - pfcf-node-id-type {fqdn | ip} [ip-type {ipv4 | ipv6}]
    - no pfcf-node-id-type
    - pfcf-overload-control
    - pfcf-cp-router-adv [application-id application-id-name]
    - no pfcf-cp-router-adv
    - pgw-selection
    - [no] pi sig-ref-point-name
    - qci-policy {qci-policy-name | uplink qci-policy-name [downlink qci-policy-name]
    | downlink qci-policy-name [uplink qci-policy-name]}
    - no qci-policy
    - qfi-mapping-profile profile-name
    - no qfi-mapping-profile
    - radius
    - reattach-timeout timeout
    - no reattach-timeout
    - [no] relax-msb-check-on-path-management-messages
    - [no] reserve-default-gateway-address
    - rf
    - s1-u [sig-ref-point-name] [group-specific]
    - no s1-u [sig-ref-point-name]
    - s11 [sig-ref-point-name] [group-specific]
    - no s11 [sig-ref-point-name]
    - s12 [sig-ref-point-name] [group-specific]
    - no s12 [sig-ref-point-name]
    - s2a [sig-ref-point-name] [group-specific]

```

- no *s2a* [*sig-ref-point-name*]
- *s2b* [*sig-ref-point-name*] [*group-specific*]
- no *s2b* [*sig-ref-point-name*]
- *s2b-error-mapping-profile* *profile-name*
- no *s2b-error-mapping-profile*
- [no] *s4* [*sig-ref-point-name*] [*group-specific*]
- *s5* [*sig-ref-point-name*] [*group-specific*]
- *s6b*
- *s8* [*sig-ref-point-name*] [*group-specific*]
- *sba-client-services*
- *sba-server-services*
- *sba-service-realm*
- *sd*
- [no] *seamless-wlan-handover*
- [no] *send-mbr-for-bbai*
- [no] *send-mbr-for-bbai-emergency*
- *serving-node-for-combo-sessions* {*mme* | *sgw*}
- no *serving-node-for-combo-sessions*
- *session-deletion-rate* *session-deletion-rate*
- no *session-deletion-rate*
- *session-restoration-rate* *session-restoration-rate*
- no *session-restoration-rate*
- [no] *shutdown*
- *signaling*
- *slices*
 - *slice-instance-list* *list-name*
 - no *slice-instance-list*
 - *slice-list* *list-name*
 - no *slice-list*
- *sta*
- *supported-vendor* {*custom-1* | *custom-2* | *custom-3*}
- no *supported-vendor*
- [no] *suspend*
- [no] *swm*
- *swm-error-mapping-profile* *profile-name*
- no *swm-error-mapping-profile*
- [no] *swu* [*sig-ref-point-name*]
- [no] *sww* [*ref-point-name*]
- [no] *sx-n4* [*ref-point-name*]
- *thresholds*
- *transit-sub*
 - *attach-response-wait-timer* *seconds*
 - no *attach-response-wait-timer*
 - *attach-retry-hold-timer* *seconds*
 - no *attach-retry-hold-timer*

- [no] *data-triggered*
- *twag-samog-radius-accounting radius-group*
- no *twag-samog-radius-accounting*
- *ue-cleanup-on-delete* {enable | disable}
- no *ue-cleanup-on-delete*
- *umts-qos-policy umts-qos-policy*
- no *umts-qos-policy*
- *up-function-feature-options*
 - [no] *exclude-adpdp*
 - [no] *exclude-dlbd*
 - [no] *exclude-dpdra*
 - [no] *exclude-frrt*
 - [no] *exclude-heeu*
 - [no] *exclude-pdiu*
 - [no] *exclude-ueip*
- *up-peer-list list-name*
- no *up-peer-list*
- *up-security-policy profile-name*
- no *up-security-policy*
- *up-selection*
 - *pgw-ggsn-u-client-profile client-profile-name*
 - no *pgw-ggsn-u-client-profile*
 - *sgw-u-client-profile client-profile-name*
 - no *sgw-u-client-profile*
- *update-bearer-request-for-mvno* {enable | disable}
- no *update-bearer-request-for-mvno*
- *upf-acct-profile profile-id*
- no *upf-acct-profile*
- *url-encryption* {aes-cbc-128} *passphrase string*
- no *url-encryption*
- [no] *user-plane-sync-for-delete-bearer*
- [no] *virtual-apn-selection*
- *vsmf-profile profile-name*
- no *vsmf-profile*
- *wlan-emergency profile-name*
- no *wlan-emergency*
- *wps-profile profile-name*
- no *wps-profile*

5.1.1.2 APN PDN command hierarchy

configure

- mobile-gateway
- [no] *pdn gw-id*

```

- [no] apn {apn-name | unknown}
- aa-divert-per-bearer-qci-list qci-list
- no aa-divert-per-bearer-qci-list
- aa-group [aa-group-id[:partition-id]] [def-app-profile profile-name] [parent-ctl-cc-name value]
- no aa-group
- [no] access-type access-type
- default-qos-profile profile-name
- no default-qos-profile
- accounting type radius username {imsi | msisdn | pco | pco-no-apn | imsi-apn | msisdn-apn} server-group server-group [suppress-imsi]
- no accounting
- [no] duplicate-server-group group-name
- [no] selective-radius-accounting
- accounting-priority profile-name
- no accounting-priority
- acct-fixed user-name long-description-string
- no acct-fixed
- aggr-rate-dl aggregate-rate-downlink
- no aggr-rate-dl
- aggr-rate-ul aggregate-rate-uplink
- no aggr-rate-ul
- [no] allow-always-on
- [no] allow-dangling-rule-ded-bearer
- allow-emergency [emergency-profile profile-name]
- no allow-emergency
- [no] allow-wlan-ho-no-context
- ambr-policing {dl-only | ul-and-dl | none}
- no ambr-policing
- [no] apn-load-control
- apn-network-id type {real [include-radius-auth | include-radius-pre-auth] [apn-ni-only [application application-name [application-name...(up to 5 max)]]] | public apn-name apn-name}
- no apn-network-id
- [no] apn-overload-control
- overload-reduction overload-reduction
- no overload-reduction
- overload-threshold overload-reduction
- no overload-threshold
- validity-time validity-time
- no validity-time
- apn-rate-control profile-name
- no apn-rate-control
- apn-type small
- no apn-type

```

```

    - [no] application-id-mvi-policy
    - auth-fixed user-name long-description-string password long-description-
string [hash | hash2]
    - no auth-fixed
    - authentication type {radius server-group group-name | diameter pri-dia-peer-
profile name sec-dia-peer-profile name} [user-name {imsi | msisdn | pco | fixed | apn |
pco-no-apn | imsi-apn | msisdn-apn}] [suppress-imsi] [pco-password] [password password
password-type {fixed | default} [hash | hash2]]
    - no authentication
    - [no] bitrate-roundup
    - block-list
    - entry entry-id group group pdn-type pdn-type
    - no entry entry-id
    - call-sequence {pfcf-first | local-upf-sel-before-auth}
    - no call-sequence
    - charging
    - cc-ignore {home | visiting | roaming | any} [inherit | enable | disable]
    - no cc-ignore [home | visiting | roaming | any]
    - cc-reject [inherit | enable | disable]
    - no cc-reject
    - cdr-avp-options-profile cdr-avr-options-profile-id
    - no cdr-avp-options-profile
    - charging-prof {home | visiting | roaming} charging-profile-id
    - no charging-prof [home | visiting | roaming]
    - dcca-profile profile-name
    - no dcca-profile
    - diameter-peer-group-list list-name
    - no diameter-peer-group-list
    - gtp-prime-group gtp-prime-group
    - no gtp-prime-group
    - incomplete-signaling
    - charging-method {online | offline | both | none}
    - no charging-method
    - rating-group rating-group
    - no rating-group
    - service-id service-id
    - no service-id
    - urr-id urr-id
    - no urr-id
    - nchf
    - chf-selection
    - chf-profile profile-name
    - no chf-profile
    - nf-id-list list-name
    - no nf-id-list

```

```

    - nchf-charging {enable | disable | inherit}
    - no nchf-charging
    - plmn-chrg-profile-map plmn-list-name list-name incoming-cc-value cc-value
charging-profile-id charging-profile-id
    - no plmn-chrg-profile-map plmn-list-name list-name [incoming-cc-value cc-
value]
    - pri-cdf-dia-peer profile-name
    - no pri-cdf-dia-peer
    - sec-cdf-dia-peer profile-name
    - no sec-cdf-dia-peer
    - sec-ocs-dia-peer profile-name
    - no sec-ocs-dia-peer
    - cm-idle-timeout cm-idle-timeout
    - no cm-idle-timeout
    - converged-interface-support
    - converged-n10 {enable | disable | inherit | 5gcnri}
    - no converged-n10
    - converged-n40 {enable | disable | inherit}
    - no converged-n40
    - converged-n40-gngp {enable | disable | inherit}
    - no converged-n40-gngp
    - converged-n7 {enable | disable | inherit}
    - no converged-n7
    - converged-n7-gngp {enable | disable | inherit}
    - no converged-n7-gngp
    - dedicated-bearer-create-retry-count count
    - no dedicated-bearer-create-retry-count
    - dedicated-bearer-create-retry-wait-timer wait-timer
    - no dedicated-bearer-create-retry-wait-timer
    - dedicated-bearer-idle-timeout-profile profile-name
    - no dedicated-bearer-idle-timeout-profile
    - default-qos-5g-profile profile-name
    - no default-qos-5g-profile
    - default-cc-group {rtr | uc | pp}
    - no default-cc-group
    - default-qos-application-modify
    - apn-ambr-attr
    - aggr-rate downlink-rate downlink-rate uplink-rate uplink-rate
    - no aggr-rate
    - arp arp-value
    - no arp
    - qci qci-value
    - no qci
    - qos-holding-time value
    - no qos-holding-time

```

```

    - retry-wait-timer value
    - no retry-wait-timer
    - del-bearer-no-reactivation-req-cause [idle-timeout]
    - no del-bearer-no-reactivation-req-cause
    - del-bearer-reactivation-req-cause {[fh-session-continue] [session-timeout]
[pdn-connection-clear] [pcscf-restoration] [all]}
    - no del-bearer-reactivation-req-cause
    - delay-tolerant-connection-suspension delay-tolerant-connection-suspension
    - no delay-tolerant-connection-suspension
    - [no] delete-adc-dedicated-bearer-on-idle-timeout-only
    - description description-string
    - no description
    - destination-realm-gx name
    - no destination-realm-gx
    - dhcp-profile name
    - dhcp-profile
    - dhcpv4-server profile-name
    - no dhcpv4-server
    - dhcpv4-server-group profile-name
    - no dhcpv4-server-group
    - dhcpv6-server-group profile-name
    - no dhcpv6-server-group
    - dhcpv6-service [enable | disable]
    - no dhcpv6-service
    - [no] disable-bearer-bind-verification
    - [no] disable-frame-route
    - [no] disable-gbr-policing-qci qci qci [arp arp-value]
    - [no] dnn-overload-control
        - n1-back-off-timer minutes
        - [no] n1-back-off-timer
        - retry-after-timer seconds
        - [no] retry-after-timer
    - dns-redirect-server profile-name
    - no dns-redirect-server
    - dns-snooping [dns-server-list addr-list] [snoop-all]
    - no dns-snooping
    - [no] dns-snooping-forward-to-aa
    - dynamic-rule-policing qci qci [arp arp-value] [direction direction]
[group group]
    - no dynamic-rule-policing qci qci [arp arp-value]
    - dynamic-rule-to-internal-rule-map profile-name
    - no dynamic-rule-to-internal-rule-map
    - [no] enable-global-pcc-precedence
    - [no] enable-gx-session-recovery
    - ethernet-action-dl action action-type [n3-c-vlan v-lan]

```


- no *ethernet-action-dl*
- *ethernet-action-up* action *action-type* tag-value *tag-value*
- no *ethernet-action-up*
- *ethernet-service-type* *service-type*
- no *ethernet-service-type*
- *firewall-rules*
 - *abort*
 - *begin*
 - *commit*
 - *diff*
 - [no] *policy-rule* *policy-rule-name*
 - [no] *policy-rule-base* *base-policy-name*
- *flow-inactivity-timeout* *inactivity-timeout*
- no *flow-inactivity-timeout*
- *fqdn-dns-redirect-server* *dns-server-name*
- no *fqdn-dns-redirect-server*
- [no] *fqdn-group* *fqdn-group-name*
- *gbr-policing-direction* *qci* *qci* [*arp arp-value*] {none | dl | bidir}
- no *gbr-policing-direction* *qci* *qci* [*arp arp-value*]
- *gtp-pdn-threshold* max-connection *max-connection* [alarm]
- no *gtp-pdn-threshold*
- [no] *gtpu-source-port-dist*
- *guard-timer* *guard* *timer*
- no *guard-timer*
- *gx*
 - *gtp-precedence-mapping* {no-override | standard-algorithm | VoLTE-algorithm}
 - no *gtp-precedence-mapping*
 - [no] *include-rat-when-received*
- *ho-failure-release-pdusession* [enable | disable]
- no *ho-failure-release-pdusession*
- *ho-wrong-pdn-ip-stale-session-deletion* [enable | disable]
- no *ho-wrong-pdn-ip-stale-session-deletion*
- *hybrid-access*
 - [no] *bonding-support*
 - *link-switch*
 - *mode*
 - *uplink-traffic*
 - *hysteresis* *time*
 - no *hysteresis*
- *monitoring-key* *monitoring-key-profile-name*
- no *monitoring-key*
- *idle-timeout* *idle-timeout*
- no *idle-timeout*
- [no] *idle-timer-direction-downlink*
- [no] *idle-timer-direction-uplink*

```

- [no] ignore-up-ip-allocation
- [no] ip-alloc-allow-local
- [no] ip-alloc-extended-hss-allocation
- [no] ip-alloc-extended-radius-allocation
- ip-alloc-method {local-pool | hss-static | aaa | dhcp-client | dhcp-relay}
[{{local-pool | hss-static | aaa | dhcp-client | dhcp-relay}...(up to 4 max)}}
- no ip-alloc-method
- ip-alloc-priority-order hss-static
- no ip-alloc-priority-order
- [no] ip-allocation-trace
- ip-pool-exhaust-back-off-timer timer-value
- [no] ip-pool-exhaust-back-off-timer
- kpi-kci
  - arp-traffic-stats [inherit| enable | disable]
  - no arp-traffic-stats
  - enhanced-traffic-stats [inherit| enable | disable]
  - no enhanced-traffic-stats
  - qci-apn-stats list-name
  - no qci-apn-stats
  - [no] sgw-apn-stats
- ladn
  - [no] enable-ladn
  - on-ue-move-outside-ladn {release-pdu | release-up}
  - no on-ue-move-outside-ladn
- [no] max-ambr-policer
- max-pdn-connection max-pdn-connections {alarm | enforce}
- no max-pdn-connection
- max-session-attach max-attach-limit [alarm] [include-emergency]
- no max-session-attach
- min-pdn-connection min-pdn-connections
- no min-pdn-connection
- [no] multi-ue-bonding
- namf-profile profile-name
- no namf-profile
- nemo-ha {enable | disable}
- no nemo-ha
- non-ip-pdn
  - nsmf-dnn-threshold tunnel-group-name
  - no nsmf-dnn-threshold
  - nsmf-dnn-threshold max-pdu-activations max-pdu-activations [alarm]
  - nsmf-dnn-threshold max-pdu-connections max-pdu-connections [alarm]
  - no nsmf-dnn-threshold
  - nsmf-profile profile-name
  - no nsmf-profile
- [no] optimize-gbr

```

- *origin-realm-per-application*
 - *realm-gx realm-gx*
 - *no realm-gx*
 - *realm-gy realm-gy*
 - *no realm-gy*
 - *realm-rf realm-rf*
 - *no realm-rf*
 - *realm-s6b realm-s6b*
 - *no realm-s6b*
 - *realm-sd realm-sd*
 - *no realm-sd*
 - *realm-swm realm-swm*
 - *no realm-swm*
- *overload-back-off*
 - *exclude-arp-list arp-list*
 - *no exclude-arp-list*
 - *overload-back-off-timer overload-back-off-timer*
 - *no overload-back-off-timer*
- *overload-control-profile profile-name*
- *no overload-control-profile*
- *[no] paging-bearer-binding*
- *pcf-discovery*
 - *serving-scope value*
 - *no serving-scope*
- *pcf-selection*
 - *dnn type type [apn-ni-only]*
 - *no dnn*
 - *[no] exclude-gpsi*
 - *exclude-snssais {enable | disable | inherit}*
 - *no exclude-snssais*
 - *[no] exclude-supi*
 - *nf-id-list list-name*
 - *no nf-id-list*
 - *npcf-profile list-name*
 - *no npcf-profile*
 - *pcf-id {use-as-target | ignore}*
 - *no pcf-id*
 - *selection-type {imsi|msisdn|imei} {prefix|full-id|suffix}*
 - *no selection-type*
 - *user-range user-range-name {start-id [end-id] | suffix} nf-id-list list-*
- name*
 - *no user-range user-range-name*
- *pco-fallback user-name user-name*
- *no pco-fallback*
- *pco-fallback-acct user-name user-name*

```

– no pco-fallback-acct
– pco-option
  – [no] closest-v4-epdg-group profile-name
  – [no] closest-v6-epdg-group profile-name
  – dns-server-ipv4
    – primary ip-address
    – no primary
    – secondary ip-address
    – no secondary
  – dns-server-ipv6
    – primary ipv6-address
    – no primary
    – secondary ipv6-address
    – no secondary
  – link-mtu octets
  – no link-mtu
  – nbns-srvr-ipv4
    – primary ip-address
    – no primary
    – secondary ip-address
    – no secondary
  – nbns-srvr-ipv6
    – primary ipv6-address
    – no primary
    – secondary ipv6-address
    – no secondary
  – operator-pco custom-1
  – no operator-pco
  – pcscf-srvr-ipv4 pcscf-server-ipv4
  – no pcscf-srvr-ipv4
  – pcscf-srvr-ipv6 pcscf-server-ipv6
  – no pcscf-srvr-ipv6
  – pcscf-srvr-v4-group pcscf-group [secondary pcscf-group] [tertiary pcscf-
group]
  – no pcscf-srvr-v4-group
  – pcscf-srvr-v6-group pcscf-group [secondary pcscf-group] [tertiary pcscf-
group]
  – no pcscf-srvr-v6-group
  – preferred-dns-address-type type type
  – no preferred-dns-address-type
  – selection-type {imsi | msisdn | imei} {prefix | full-id | suffix}
  – no selection-type
  – ue-location
    – tai-lai-list list-id tai-lai-list-name
    – no tai-lai-list list-id

```

```

    - pcscf-srvr-v4-group pcscf-group [secondary pcscf-group]
[tertiary pcscf-group]
    - no pcscf-srvr-v4-group
    - pcscf-srvr-v6-group pcscf-group [secondary pcscf-group]
[tertiary pcscf-group]
    - no pcscf-srvr-v6-group
    - user-range user-range-name {start-id [end-id] | suffix} pcscf-srvr-v4-
group pcscf-group [secondary-pcscf-srvr-v4-group pcscf-group] [tertiary-pcscf-srvr-
v4-group pcscf-group] pcscf-srvr-v6-group pcscf-group [secondary-pcscf-srvr-v6-
group pcscf-group] [tertiary-pcscf-srvr-v6-group pcscf-group]
    - no user-range user-range-name
    - pcrf-selection
    - diameter-peer-group-list list-name
    - no diameter-peer-group-list
    - dynamic-pcc [fh-session {enable | disable}] [fh-3xxx {enable | disable}]
[fh-4xxx {enable | disable}] [gx-profile gx-profile-name] [sd-profile sd-profile-name]
    - no dynamic-pcc
    - pcc-disable-per-rat [UTRAN] [GERAN] [WLAN] [GAN] [HSPA-Evolution] [EUTRAN]
[Virtual] [EUTRAN-NB-IoT]
    - no pcc-disable-per-rat
    - pcc-disable-per-rat-timer pcc-disable-per-rat-timer
    - no pcc-disable-per-rat-timer
    - primary-diameter-peer profile-name
    - no primary-diameter-peer
    - secondary-diameter-peer profile-name
    - no secondary-diameter-peer
    - selection-type {prefix | full-id | suffix}
    - no selection-type
    - [no] suppress-ccr-with-no-qos-negotiation
    - user-range user-range-name type {imsi | msisdn | imei} {start-id [end-id]
| suffix} diameter-peer-group-list list-name
    - no user-range user-range-name
    - pd-alloc-method {none | aaa}
    - no pd-alloc-method
    - pdn-alloc-type allocation type
    - pdn-type {ipv4 | ipv6 | ipv4v6 | non-ip | ethernet} [{ipv4 | ipv6 | ipv4v6 |
non-ip | ethernet}...(up to 4 max)]
    - no pdn-type
    - [no] pdn-type-conversion
    - [no] pdn-type-preferred-ipv6
    - pfcf-translate-tx profile-name
    - no pfcf-translate-tx
    - pfcf-user-plane-inactivity up-inactivity-timer
    - no pfcf-user-plane-inactivity
    - policy-on-the-fly {new-sessions | existing-sessions [reactivate-if-fail]}

```

```

- no policy-on-the-fly
- ppi-mapping-profile profile-name
- no ppi-mapping-profile
- [no] ppp-regeneration
  - [no] exclude-dns-server-option
  - [no] exclude-nbns-server-option
  - icrq-calling-number-type icrq-avp-identity
  - no icrq-calling-number-type
  - [no] include-hss-static-ip
  - l2tp-group tunnel-group-name
  - no l2tp-group
  - lcp-confreq-attributes {include | exclude} [mru-size] [auth-type auth-type]
  - no lcp-confreq-attributes
  - [no] mobile-router
  - router router-instance
  - no router
  - rx-connect-speed rx-connect-speed
  - no rx-connect-speed
  - tx-connect-speed tx-connect-speed
  - no tx-connect-speed
  - user-name {[msisdn | misidn-no-apn | imsi-no-apn | imsi]
[password password password-type {fixed | default}] | pco [allow-pco-missing]}
  - no user-name
  - supported-vendor custom-1
  - no supported-vendor
  - pre-auth type {radius [server-group server-group] [user-name user-name]
[suppress-imsi] [pco-password] [password password password-type {fixed | default}
[hash | hash2]] | diameter}
  - no pre-auth
  - pre-auth-fixed user-name long-description-string password long-description-
string [hash | hash2]
  - no pre-auth-fixed
  - [no] preserve-volte
  - qci-policy {qci-policy-name | uplink qci-policy-name [downlink qci-policy-
name] | downlink qci-policy-name [uplink qci-policy-name]}
  - no qci-policy
  - qfi-mapping-profile profile-name
  - no qfi-mapping-profile
  - radius-accounting-on-off
  - [no] include-duplicate-servers
  - messages-sent {none | on | on-off}
  - no messages-sent
  - [no] wait-for-ack
- [no] radius-auth-cache
- [no] radius-case-sensitive-called-station-id

```

```

    - radius-nas-identifier identifier-name
    - no radius-nas-identifier
    - radius-nas-ip-address ipv4-address
    - no radius-nas-ip-address
    - [no] reactivation-requested
    - redirect mobile-all [external-nexthopv4 ipv4-address] | [external-
nextthopv6 ipv6-address]
    - no redirect
    - [no] reject-foreign-subscribers
    - report-serving-node-supplied-cc-radius {enable | disable}
    - no report-serving-node-supplied-cc-radius
    - report-serving-node-supplied-cc-s6b {enable | disable}
    - no report-serving-node-supplied-cc-s6b
    - restriction {any | public1 | public2 | private1 | private2}
    - no restriction
    - roaming-handover
      - lte-qos-ambr downlink-rate downlink-rate uplink-rate uplink-rate
      - no lte-qos-ambr
      - lte-qos-black-list plmn-list-name
      - no lte-qos-black-list
      - lte-qos-delay delay
      - no lte-qos-delay
    - [no] router router-instance
      - [no] ip-local-pool pool-name
      - [no] tai-lai-list tai-lai-list-name
        - [no] ip-local-pool pool-name
    - router-advertisements
      - max-advertisement-interval seconds
      - no max-advertisement-interval
      - min-advertisement-interval seconds
      - no min-advertisement-interval
    - rulebase-selection
      - allow-rulebase {gy-gx-aaa-local | gx-gy-aaa-local}
      - no allow-rulebase
    - [no] seamless-wlan-handover
    - selected-dnn {real | virtual | amf-selected}
    - no selected-dnn
    - selection-mode {subscribed-apn | ms-provided-apn | network-provided-apn}
[{subscribed-apn | ms-provided-apn | network-provided-apn}...(up to 3 max)]
    - no selection-mode
    - session-timeout session-timeout
    - no session-timeout
    - [no] shutdown
    - shutdown-back-off-timer timer-value
    - no shutdown-back-off-timer

```

```

- slices
  - default-slice slice-name [slice-list slice-list-name]
  - no default-slice
  - slice-list list-name
  - no slice-list
  - udm-based-slice-selection {enable | disable}
  - no udm-based-slice-selection
- ssc
  - allowed-ssc-modes {ssc-1 | ssc-2 | ssc-3} [{ssc-1 | ssc-2 | ssc-3}...(up to
3 max)]
  - no allowed-ssc-modes
  - pdu-session-address-lifetime lifetime
  - no pdu-session-address-lifetime
  - pdu-session-reactivation-trigger trigger
  - no pdu-session-reactivation-trigger
  - pfcf-user-plane-inactivity-iupf-case up-inactivity-timer
  - no pfcf-user-plane-inactivity-iupf-case
  - [no] reject-not-allowed-ssc-mode
  - upf-reselection [ssc-1-option] [ssc-2 ssc-2-option] [ssc-3 ssc-3-option]
  - no upf-reselection
  - uuid-mask-length length
  - no uuid-mask-length
- static-local-pcc
  - abort
  - begin
  - commit
  - dedicated-bearers
    - [no] default-bearer-qci qci qci arp arp-value
      - [no] policy-rule policy-rule-name
      - [no] policy-rule-base policy-rule-base-name
  - diff
  - [no] policy-rule policy-rule-name
  - [no] policy-rule-base policy-rule-base-name
- static-predefined-pcc
  - [no] default-policy-rule-base base-policy-name
  - default-qci qci [arp arp-value]
  - no default-qci
- steering
  - default-gw-addr-ipv4 ip-address
  - no default-gw-addr-ipv4
  - default-gw-addr-ipv6 ip-address
  - no default-gw-addr-ipv6
  - [no] gw-signaling-addr addr-list-name
    - gw-addr-ipv4 ip-address
    - no gw-addr-ipv4

```



```

    - gw-addr-ipv6 ip-address
    - no gw-addr-ipv6
- subscriber-mapping-profile profile-name
- no subscriber-mapping-profile
- svc-steering
  - default-steering-profile profile-name
- transit-sub
  - attach-response-wait-timer seconds
  - no attach-response-wait-timer
  - attach-retry-hold-timer seconds
  - no attach-retry-hold-timer
  - [no] data-triggered
- transport-byte-collection
  - imsi imsi [static-rule static-rule] [dynamic-rule dynamic-rule]
  - no imsi imsi
  - msisdn msisdn [static-rule static-rule] [dynamic-rule dynamic-rule]
  - no msisdn msisdn
- [no] twag-radius-acct-feed
- [no] udm-ctrl-radius-auth
- udm-sdm-disabled {lte | non-3gpp | eps}
- no udm-sdm-disabled
- udm-sdm-profile profile-name
- no udm-sdm-profile
- [no] udm-uecm-disabled
- udm-uecm-profile profile-name
- no udm-uecm-profile
- umts-qos-policy umts-qos-policy
- no umts-qos-policy
- up-security-policy profile-name
- no up-security-policy
- upf-acct-profile profile-id
- upf-acct-profile inherit
- no upf-acct-profile
- [no] upf-sel-before-auth
- upf-selection
  - dnn type type [apn-ni-only]
  - no dnn
- virtual-apn apn-name
- no virtual-apn
- virtual-apn-based-on-pco [default-apn apn-name]
- no virtual-apn-based-on-pco
- [no] virtual-apn-list apn-list
- [no] virtual-apn-selection
  - no-match-action action [virtual-apn virtual-apn-name]
  - no no-match-action

```

```

    - [no] prefer-pdn
    - selection-prio UE type [UE type...up to 10 max]
    - no selection-prio
    - selection-type {prefix | full-id}
    - no selection-type
    - user-range user-range-name type [{imsi | msisdn | imei-tac} start-ue-id
    [end-ue-id] | [roaming-status roaming-status-type] | [cc start-cc start-cc-id [end-cc
    end-cc-id] | [imeisv-tacsvn start-ue-id [end-ue-id] svn svn-value] | [serving-
    node-plmn list-name] | [pdn-type ipv4 | ipv6 | ipv4v6] | [rat-type rat-type-list-name]
    | [ue-location ue-location-list-name]} {virtual-apn virtual-apn-name | {[append-
    string suffix] [pre-auth-type {radius server-group server-group-name [user-name {imsi
    | msisdn | pco | fixed}] [pco-password] [password password password-type {fixed |
    default} [hash | hash2}] | diameter} [suppress-imsi]}}
    - no user-range user-range-name
    - vsmf-profile profile-name
    - no vsmf-profile
    - [no] wait-accounting

```

5.1.1.3 Bearer traffic PDN command hierarchy

```

configure
- mobile-gateway
- [no] pdn gw-id
  - bearer-traffic
    - gtp-u
      - [no] sequence-number
      - [no] udp-checksum
    - mss-adjust segment-size
    - no mss-adjust
    - non-3gpp-mss-adjust mss-value
    - no non-3gpp-mss-adjust

```

5.1.1.4 Cloud database PDN command hierarchy

```

configure
- mobile-gateway
- [no] pdn gw-id
  - cdbx
    - cloud-db-profile profile-name
    - no cloud-db-profile
    - interface [router router-instance] interface-name
    - no interface

```

- *mss mss-value*
- no *mss*

5.1.1.5 Charging PDN command hierarchy

configure

- mobile-gateway
- [no] *pdn gw-id*
- *charging*
 - [no] *bill-shock-prevention*
 - *cc-ignore* {home | visiting | roaming | any} [inherit | enable | disable]
 - no *cc-ignore* [home | visiting | roaming | any]
 - [no] *cc-reject*
 - *cdr-avp-options-profile cdr-avr-options-profile-id*
 - no *cdr-avp-options-profile*
 - [no] *cdr-file-separation*
 - *cdr-transfer* {ga | bp}
 - no *cdr-transfer*
 - *cdr-type* {pgwCdr | gCdr | egCdr}
 - no *cdr-type*
 - *charging-prof* {home | visiting | roaming} *charging-profile-id*
 - no *charging-prof* [home | visiting | roaming]
 - *charging-prof-selection-range*
 - *selection-type imsi* {prefix | full-id}
 - no *selection-type*
 - *user-range user-range-name* {start-ue-id [end-ue-id]} gw-type {sgw}

charging-prof-id value

- no *user-range user-range-name*
- [no] *enable-nchf-charging*
- [no] *multiple-pra*
- *node-id node-id*
- no *node-id*
- *pgw-node-id pgw-node-id*
- no *pgw-node-id*
- *plmn-chrg-profile-map* plmn-list-name *list-name* incoming-cc-value *cc-value*

charging-profile-id profile-id

- no *plmn-chrg-profile-map* plmn-list-name *list-name* [incoming-cc-value *cc-*

value]

- [no] *remove-3gpp-headers*
- *sgw-node-id sgw-node-id*
- no *sgw-node-id*
- *supported-vendor custom-2*
- no *supported-vendor*
- *tariff-time-change-gw-local-tz* {enable | disable}

- no *tariff-time-change-gw-local-tz*
- [no] *time-limit-enh*
- [no] *use-last-known-uli*
- [no] *volume-limit-enh*

5.1.1.6 Ga PDN command hierarchy

```
configure
– mobile-gateway
– [no] pdn gw-id
– ga
– data-record-format-version string
– no data-record-format-version
– gtp-prime-group gtp-prime-group
– no gtp-prime-group
– interface-gtp-prime [router router-instance] interface-name
– no interface-gtp-prime
```

5.1.1.7 Gn PDN command hierarchy

```
configure
– mobile-gateway
– [no] pdn gw-id
– gn [sig-ref-point-name] [group-specific]
– no gn [sig-ref-point-name]
– interface
– gtp-c [router router-instance] interface-name [interface-realm realm-name]
– no gtp-c
– gtp-u [router router-instance] interface-name [interface-realm interface-
realm]
– no gtp-u
– peer-list list-name
– no peer-list
– signaling
– gtp-c
– profile profile-name
– no profile
– gtp-u
– [no] gtpu-keep-alive
– profile profile-name
– no profile
– shutdown list-name
```

- no *shutdown*

5.1.1.8 Gp PDN command hierarchy

```

configure
  – mobile-gateway
  – [no] pdn gw-id
    – gp [sig-ref-point-name] [group-specific]
    – no gp [sig-ref-point-name]
      – interface
        – gtp-c [router router-instance] interface-name [interface-realm interface-realm]
        – no gtp-c
        – gtp-u [router router-instance] interface-name [interface-realm realm-name]
        – no gtp-u
      – peer-list list-name
      – no peer-list
      – signaling
        – gtp-c
          – profile profile-name
          – no profile
        – gtp-u
          – [no] gtpu-keep-alive
          – profile profile-name
          – no profile
      – shutdown list-name
      – no shutdown

```

5.1.1.9 GTP load and overload control PDN command hierarchy

```

configure
  – mobile-gateway
  – [no] pdn gw-id
    – gtp-load-control
      – [no] enable-gtp-load-control
      – load-change-trigger percent
      – no load-change-trigger
      – nth-message number
      – no nth-message
    – gtp-overload-control
      – [no] enable-gtp-overload-control
      – [no] gtp-120-oci

```

- *gtp-73-120-reduction-metric* *metric*
- *no gtp-73-120-reduction-metric*
- *[no] gtp-73-oci*
- *local-throttling*
 - *critical*
 - *drop-only drop-type*
 - *no drop-only*
 - *exclude-arp-list list-name*
 - *no exclude-arp-list*
 - *throttling-percent percent-value*
 - *no throttling-percent*
 - *local-overload-throttling {enable | disable}*
 - *no local-overload-throttling*
- *major*
 - *drop-only drop-type*
 - *no drop-only*
 - *exclude-arp-list list-name*
 - *no exclude-arp-list*
 - *throttling-percent percent-value*
 - *no throttling-percent*
- *minor*
 - *drop-only drop-type*
 - *no drop-only*
 - *exclude-arp-list list-name*
 - *no exclude-arp-list*
 - *throttling-percent percent-value*
 - *no throttling-percent*
 - *throttling-alarms {enable | disable}*
 - *no throttling-alarms*
- *[no] overload-control-only-on-new-sessions*
- *overload-reduction*
 - *critical reduction-value*
 - *no critical*
 - *major reduction-value*
 - *no major*
 - *minor reduction-value*
 - *no minor*
- *[no] s8*
- *validity-time value*
- *no validity-time*
- *validity-time-extra-time time*
- *no validity-time-extra-time*
- *pdn-threshold*
 - *max-s11-attach max-attach-limit [alarm] [include-emergency]*
 - *no max-s11-attach*

5.1.1.10 Gx PDN command hierarchy

```

configure
- mobile-gateway
- [no] pdn gw-id
- gx
  - active-usage {disable | enable} active-threshold threshold-value
  - no active-usage
  - [no] alt-peer-rar
  - connection-failure-timer value
  - no connection-failure-timer
  - doic
    - [no] diameter-overload-control
  - dra-diameter-too-busy [re-router-timer minute-value]
  - no dra-diameter-too-busy
  - gx-avp-options-profile profile-number
  - no gx-avp-options-profile
  - gx-profile gx-profile
  - no gx-profile
  - interface
    - diameter [router router-instance] interface-name
    - no diameter
    - [no] multi-homing
      - diameter [router router-instance] secondary-interface-name
      - no diameter
  - origin-host name
  - no origin-host
  - pcrf-selection
    - diameter-peer-group-list list-name
    - no diameter-peer-group-list
    - primary-diameter-peer profile name
    - no primary-diameter-peer
    - secondary-diameter-peer profile name
    - no secondary-diameter-peer
    - selection-priority {imsi | msisdn | imei} [{imsi | msisdn | imei}...(up to
3 max)]
    - no selection-priority
    - selection-type {prefix | full-id | suffix}
    - no selection-type
    - user-range user-range-name type {imsi | msisdn | imei} {start-id [end-id]
| suffix} {primary-pcrf profile-name [secondary-pcrf profile-name] | diameter-peer-
group-list list-name}
    - no user-range user-range-name
  - retry-count value

```

- no *retry-count*
- *send-called-station-id* *format*
- no *send-called-station-id*
- [no] *single-diameter-host*
- *supported-vendor* {*custom-1* | *custom-2* | *custom-3*}
- no *supported-vendor*
- *tdf-info*
 - *tdf-ipv4* *ip-address*
 - no *tdf-ipv4*
 - *tdf-ipv6* *ip-address*
 - no *tdf-ipv6*
 - *tdf-realm* *realm-name* *tdf-host* *host-name*
 - no *tdf-realm*
- *transaction-timer* {[*value*] | [*timeout-msec* *value*]}
- no *transaction-timer*
- *usu-options-over-quota-traffic-handling* {*allow* | *discard*}
- no *usu-options-over-quota-traffic-handling*
- *usu-options-reporting-threshold* *percentage*
- no *usu-options-reporting-threshold*
- [no] *usu-options-suppress-excess-quota*

5.1.1.1 Gy PDN command hierarchy

configure

- mobile-gateway
 - [no] *pdn* *gw-id*
 - *gy*
 - [no] *alt-peer-rar-asr*
 - *dcca-profile* *profile-name*
 - no *dcca-profile*
 - *dia-routing-table* *host-index* *supported-host* *fqdn* *port* *port* *transport* {*tcp* | *sctp*}
 - no *dia-routing-table* [*host-index*]
 - *diameter-peer-group-list* *list-name*
 - no *diameter-peer-group-list*
 - *diameter-session* {*pdn-level* | *bearer-level*}
 - no *diameter-session*
 - [no] *exclude-dest-host-pcrf-selected-server*
 - *interface*
 - *diameter* [*router* *router-instance*] *interface-name*
 - no *diameter*
 - [no] *multi-homing*
 - *diameter* [*router* *router-instance*] *secondary-interface-name*
 - no *diameter*


```

- ocs-selection
- selection-type {imsi | msisdn} {prefix | full-id | suffix}
- no selection-type
- user-range user-range-name {start-id [end-id] | suffix} {primary-
ocs profile-name [secondary-ocs profile-name] | diameter-peer-group-list list-name}
[dcca-profile profile-name]
- no user-range user-range-name
- origin-host name
- no origin-host
- outage
- file-params
- closure-rec-number closure-rec-number
- [no] closure-rec-number
- closure-size closure-size
- [no] closure-size
- closure-time closure-time
- [no] closure-time
- obsolete-time obsolete-time
- [no] obsolete-time
- location {cf1 | cf2}
- limit limit
- no limit
- [no] primary
- primary-diameter-peer profile-name
- no primary-diameter-peer
- secondary-diameter-peer profile-name
- no secondary-diameter-peer
- session-id-format {type1 | type2}
- no session-id-format
- sig-rate-limit-profile profile-name
- no sig-rate-limit-profile
- [no] single-diameter-host
- [no] suppress-ccr-t-unknown-session

```

5.1.1.12 KPI-KCI PDN command hierarchy

```

configure
- mobile-gateway
- [no] pdn gw-id
- kpi-kci
- [no] accounting-policy acct-policy-id
- [no] arp-traffic-stats
- [no] collect-stats
- [no] enhanced-traffic-stats

```

- [no] *ipv4-ipv6-stats*
- *peer-traffic-stats* peer-ip-list *list-name*
- no *peer-traffic-stats*
- *plmn-id-stats* *plmn-list-name*
- no *plmn-id-stats*
- [no] *sgw-only*

5.1.1.13 N2 PDN command hierarchy

configure

- mobile-gateway
 - [no] *pdn* *gw-id*
 - [no] *n2* [*sig-ref-point-name*]
 - *interface* [router *router-instance*] *interface-name*
 - no *interface*
 - [no] *n2-peer-profile* *profile-name*
 - *n3iwf-unique-id* *unique-id*
 - [no] *n3iwf-unique-id*
 - *supported-ta-item*
 - *tac* *tac-value* broadcast-plmn-list *broadcast-plmn-list-name*
 - no *tac* *tac-value*

5.1.1.14 N3 PDN command hierarchy

configure

- mobile-gateway
 - [no] *pdn* *gw-id*
 - *n3* [*sig-ref-point-name*] [group-specific]
 - no *n3* [*sig-ref-point-name*]
 - *dual-stack-uplane-pref* {ipv4 | ipv6}
 - no *dual-stack-uplane-pref*
 - *interface*
 - *gtp-u* [router *router-instance*] *interface-name* [interface-realm *n3-realm-name*]
 - no *gtp-u*
 - *signaling*
 - *gtp-u*
 - *profile* *profile-name*
 - no *profile*

5.1.1.15 N9 PDN command hierarchy

```
configure
- mobile-gateway
- [no] pdn gw-id
  - n9 [sig-ref-point-name] [group-specific]
  - no n9 [sig-ref-point-name]
    - dual-stack-uplane-pref {ipv4 | ipv6}
    - no dual-stack-uplane-pref
    - interface
      - gtp-u [router router-instance] interface-name [interface-realm n9-realm-name]
      - no gtp-u
    - signaling
      - gtp-u
        - profile profile-name
        - no profile
```

5.1.1.16 Network trace command hierarchy

```
configure
- mobile-gateway
- [no] pdn gw-id
  - network-trace
    - instance instance-name
      - interface interface-name [router router-name]
      - no interface
      - network-trace-profile profile-name
      - no network-trace-profile
    -[no] shutdown
    -[no] tapping-interface [all | sba | non-sba]
```

5.1.1.17 NF profile attributes command hierarchy

```
configure
- mobile-gateway
- [no] pdn gw-id
  - nf-profile-attributes [bsf]
    - capacity {factor | total} capacity-value
    - no capacity
    - [no] exclude-ip-endpoints
    - fqdn fqdn
```

- no *fqdn*
- *inter-plmn-fqdn fqdn*
- no *inter-plmn-fqdn*
- *locality locality*
- no *locality*
- *nf-instance-id uuid*
- no *nf-instance-id*
- *priority priority-value*
- no *priority*
- *scp-domain domain-name*
- no *scp-domain*
- *smf-serving-area-list list-name*
- no *smf-serving-area-list*
- *tai-list list-name*
- no *tai-list*
- *vsmf-indication* {true | false | disabled}
- no *vsmf-indication*

5.1.1.18 NWu PDN command hierarchy

configure

- mobile-gateway
 - [no] *pdn gw-id*
 - [no] *nwu [sig-ref-point-name]*
 - *interface [router router-instance] interface-name*
 - no *interface*
 - *ipsec-profile profile-name*
 - no *ipsec-profile*

5.1.1.19 PFCP load and overload PDN command hierarchy

configure

- mobile-gateway
 - [no] *pdn gw-id*
 - *pfcpl-load-control*
 - [no] *enable-pfcpl-load-control*
 - *load-change-trigger percentage-value*
 - no *load-change-trigger*
 - *nth-message-report nth-message-value*
 - no *nth-message-report*
 - *pfcpl-overload-control*
 - *arp-to-message-priority-list list-name*

- no *arp-to-message-priority-list*
- *critical* *percentage-value*
- no *critical*
- *major* *percentage-value*
- no *major*
- *minor* *percentage-value*
- no *minor*
- *validity-time* *value*
- no *validity-time*

5.1.1.20 PGW selection PDN command hierarchy

configure

- mobile-gateway
 - [no] *pdn* *gw-id*
 - *pgw-selection*
 - *dns-options*
 - [no] *a-aaaa-pgw-discovery*
 - [no] *apn-based-initial-attach*
 - *epdg-fqdn* *epdg-fqdn-string*
 - no *epdg-fqdn*
 - *epdg-fqdn-secondary* *epdg-fqdn-string*
 - no *epdg-fqdn-secondary*
 - *epdg-fqdn-tertiary* *epdg-fqdn-string*
 - no *epdg-fqdn-tertiary*
 - *pgw-reselection-isolation* *cause-code* *cc* [*isolation-time* *time*]
 - no *pgw-reselection-isolation* *cause-code* *cc*
 - *twag-fqdn* *string*
 - no *twag-fqdn*
 - [no] *apn* *apn-name*
 - *pgw-selection-candidate-list* *list-name*
 - no *pgw-selection-candidate-list*
 - [no] *topology-match-disable*
 - *dns-results-preferred*
 - no *dns-results-preferred*

5.1.1.21 Pi PDN command hierarchy

configure

- mobile-gateway
 - [no] *pdn* *gw-id*
 - [no] *pi* *sig-ref-point-name*

- *interface*
 - *mipv4* *interface-name* [*router router-instance*]
 - *no mipv4* [*router router-instance*] *interface-name*
 - *ip-in-ip-tunnel* *router-instance*
 - *no ip-in-ip-tunnel*
- *peer-list* *list-name*
- *no peer-list*
- *signaling*
 - *mipv4*
 - *fa-ha-peer-list* *list-name*
 - *no fa-ha-peer-list*
 - [*no*] *profile* *mip4-profile-name*

5.1.1.22 RADIUS PDN command hierarchy

configure

- *mobile-gateway*
 - [*no*] *pdn gw-id*
 - *radius*
 - *accounting-on-off-level* {*system* | *apn*}
 - *no accounting-on-off-level*
 - [*no*] *delete-session-acct-interim-exh*
 - *interface* [*router router-instance*] *interface-name*
 - *no interface*
 - *radius-accounting-on-off*
 - *messages-sent* {*none* | *on* | *on-off*}
 - *no messages-sent*
 - [*no*] *radius-disconnect*
 - [*no*] *rel-16-rat-type*

5.1.1.23 Rf PDN command hierarchy

configure

- *mobile-gateway*
 - [*no*] *pdn gw-id*
 - *rf*
 - *acct-interim-interval* *acct-interim-interval*
 - *no acct-interim-interval*
 - *acct-level* {*pdn-level* | *qci-level*}
 - *no acct-level*
 - [*no*] *charging-group-id*
 - *interface*

```

– diameter [router router-instance] interface-name
– no diameter
– [no] multi-homing
  – diameter [router router-instance] secondary-interface-name
  – no diameter
– operator-string operator-string
– no operator-string
– origin-host name
– no origin-host
– outage
  – [no] acr-file-and-record-hdr-rel-id-ext
  – file-params
    – closure-rec-num closure-rec-num
    – no closure-rec-num
    – closure-size closure-size
    – no closure-size
    – closure-time closure-time
    – no closure-time
    – extension extension
    – no extension
    – obsolete-time obsolete-time
    – no obsolete-time
    – private-info private-info
    – no private-info
  – location location
    – limit limit
    – no limit
    – [no] primary
  – primary-diameter-peer profile-name
  – no primary-diameter-peer
  – retry-count retry-count
  – no retry-count
  – secondary-diameter-peer profile-name
  – no secondary-diameter-peer
  – supported-vendor custom-1
  – no supported-vendor
  – transaction-timer {[value] | [timeout-msec value]}
  – no transaction-timer

```

5.1.1.24 S1-u PDN command hierarchy

configure

- mobile-gateway
- [no] *pdn* gw-id

```

– s1-u [sig-ref-point-name] [group-specific]
– no s1-u [sig-ref-point-name]
– dual-stack-uplane-pref {ipv6 | ipv4}
– interface
– gtp-u [router router-instance] interface-name [interface-realm s1-u-realm-
name]
– no gtp-u
– signaling
– gtp-u
– profile profile-name
– no profile

```

5.1.1.25 S11 PDN command hierarchy

```

configure
– mobile-gateway
– [no] pdn gw-id
– s11 [sig-ref-point-name] [group-specific]
– no s11 [sig-ref-point-name]
– interface
– gtp-c [router router-instance] interface-name [s1-u binding-ref-pt-name]
[interface-realm s11-u-realm-name] [s1u-interface-realm binding-realm-name]
– no gtp-c
– gtp-u [router router-instance] interface-name [interface-realm s11-realm-
name]
– no gtp-u
– [no] mme-group MME-group-id
– signaling
– gtp-c
– ddn-damp-timer ddn-damp-timer
– no ddn-damp-timer
– ddn-extended-buffering-time ddn-extended-buffering-time
– no ddn-extended-buffering-time
– ddn-throttling-arp-threshold ddn-throttling-arp-threshold
– no ddn-throttling-arp-threshold
– iot-restoration-time time
– no iot-restoration-time
– [no] legacy-create-idft-response-encoding
– profile profile-name
– no profile
– restoration-time restoration-time [qci value]
– no restoration-time
– ue-inactivity-timer ue-activity-timer

```


– no *ue-inactivity-timer*

5.1.1.26 S12 PDN command hierarchy

configure

- mobile-gateway
 - [no] *pdn gw-id*
 - *s12* [*sig-ref-point-name*] [*group-specific*]
 - no *s12* [*sig-ref-point-name*]
 - *dual-stack-uplane-pref* {*ipv4* | *ipv6*}
 - *interface*
 - *gtp-u* [*router router-instance*] *interface-name* [*interface-realm realm-name*]
 - no *gtp-u*
 - *signaling*
 - *gtp-u*
 - *profile profile-name*
 - no *profile*

5.1.1.27 S2a PDN command hierarchy

configure

- mobile-gateway
 - [no] *pdn gw-id*
 - *s2a* [*ref-point-name*] [*group-specific*]
 - no *s2a* [*ref-point-name*]
 - *dual-stack-cplane-pref* {*ipv4* | *ipv6*}
 - no *dual-stack-cplane-pref*
 - *dual-stack-uplane-pref* {*ipv4* | *ipv6* | *use-cplane*}
 - no *dual-stack-uplane-pref*
 - *interface*
 - *gtp-c* [*router router-instance*] *interface-name* [*interface-realm realm-name*]
 - no *gtp-c*
 - *gtp-u* [*router router-instance*] *interface-name*
 - *gtp-u* [*router router-instance*] *interface-name* [*interface-realm realm-name*]
 - no *gtp-u*
 - *pmipv6 interface-name* [*router router-instance*]
 - no *pmipv6*
 - *match*
 - *apn-list list-name*
 - no *apn-list*
 - *plmn-id-list list-name*
 - no *plmn-id-list*

- *signaling*
 - *gtp-c*
 - *profile profile-name*
 - *no profile*
 - *gtp-u*
 - *profile profile-name*
 - *no profile*
- *pmipv6*
 - *profile profile-name*
 - *no profile*

5.1.1.28 S2b PDN command hierarchy

configure

- mobile-gateway
 - [no] *pdn gw-id*
 - *s2b [ref-point-name] [group-specific]*
 - *no s2b [ref-point-name]*
 - *dual-stack-cplane-pref {ipv4 | ipv6}*
 - *no dual-stack-cplane-pref*
 - *dual-stack-uplane-pref {ipv4 | ipv6 | use-cplane}*
 - *no dual-stack-uplane-pref*
 - *interface*
 - *gtp-c [router router-instance] interface-name [interface-realm realm-name]*
 - *no gtp-c*
 - *gtp-u [router router-instance] interface-name [interface-realm realm-name]*
 - *no gtp-u*
 - *match*
 - *apn-list list-name*
 - *no apn-list*
 - *plmn-id-list list-name*
 - *no plmn-id-list*
 - [no] *pgw-set-support*
 - *signaling*
 - *gtp-c*
 - *profile profile-name*
 - *no profile*
 - *gtp-u*
 - *profile profile-name*
 - *no profile*
 - *s2b-error-mapping-profile*

5.1.1.29 S4 PDN command hierarchy

```

configure
- mobile-gateway
- [no] pdn gw-id
  - s4 [sig-ref-point-name] [group-specific]
  - no s4 [sig-ref-point-name]
    - interface
      - gtp-c [router router-instance] interface-name [s12 binding-ref-pt-name]
[interface-realm s4-realm-name]
  - no gtp-c
  - gtp-u [router router-instance] interface-name [interface-realm realm-name]
  - no gtp-u
- peer-list list-name
- no peer-list
- signaling
  - gtp-c
    - ddn-damp-timer ddn-damp-timer
    - no ddn-damp-timer
    - ddn-extended-buffering-time ddn-extended-buffering-time
    - no ddn-extended-buffering-time
    - ddn-throttling-arp-threshold ddn-throttling-arp-threshold
    - no ddn-throttling-arp-threshold
    - [no] legacy-create-idft-response-encoding
    - profile profile-name
    - no profile
    - restoration-time restoration-time [qci value]
    - no restoration-time
  - gtp-u
    - [no] delete-on-error-indication
    - [no] gtpu-keep-alive
    - profile
    - no profile
- [no] sgsn-group SGSN-group-id

```

5.1.1.30 S5 PDN command hierarchy

```

configure
- mobile-gateway
- [no] pdn gw-id
  - s5 [sig-ref-point-name] [group-specific]
  - no s5 [sig-ref-point-name]
    - dual-stack-cplane-pref {ipv4 | ipv6}
    - no dual-stack-cplane-pref

```

```

– dual-stack-uplane-pref {ipv4 | ipv6}
– no dual-stack-uplane-pref
– interface
  – gtp-c [router router-instance] interface-name [interface-realm s5-u-realm-
name]
  – no gtp-c
  – gtp-u [router router-instance] interface-name
  – gtp-u [router router-instance] interface-name [interface-realm s5-u-realm-
name]
  – no gtp-u
– match
  – apn-list list-name
  – no apn-list
– peer-list list-name
– no peer-list
– send-end-marker
– no send-end-marker
– signaling
  – gtp-c
    – enhanced-sgw-restoration restoration-time
    – no enhanced-sgw-restoration
    – pause-charging
      – data-threshold packets
      – no data-threshold
      – data-threshold-hold-time seconds
      – no data-threshold-hold-time
      – pgw-pause-charging {enable | disable}
      – radio-loss {enable | disable}
    – profile profile-name
    – no profile
  – gtp-u
    – [no] gtpu-keep-alive
    – profile profile-name
    – no profile

```

5.1.1.31 S6b PDN command hierarchy

configure

- mobile-gateway
 - [no] *pdn* *gw-id*
 - [no] *s6b*
 - [no] *alt-peer-rar-asr*
 - *application-id* *value*
 - no *application-id*

```

– auth-mode [enable-non3gpp-only | enable-all]
– no auth-mode
– doic
  – [no] diameter-overload-control
– interface
  – diameter [router router-instance] interface-name
  – no diameter
  – [no] multi-homing
    – diameter [router router-instance] secondary-interface-name
    – no diameter
– node-identifier-list list-name
– no node-identifier-list
– origin-host name
– no origin-host
– peer-group group-id pri-dia-peer-profile profile-name sec-dia-peer-
profile profile-name
  – no peer-group
  – [no] relax-aar-dest-realm
  – retry-count value
  – no retry-count
  – s6b-profile name
  – no s6b-profile
  – [no] single-diameter-host
  – [no] subscription-id-override
  – supported-vendor custom-1
  – no supported-vendor
  – transaction-timer {[value] | [timeout-msec value]}
  – no transaction-timer
  – username-leading-digit [digit]
  – no username-leading-digit

```

5.1.1.32 S8 PDN command hierarchy

```

configure
– mobile-gateway
– [no] pdn gw-id
  – s8 [sig-ref-point-name] [group-specific]
  – no s8 [sig-ref-point-name]
  – dual-stack-cplane-pref {ipv4 | ipv6}
  – no dual-stack-cplane-pref
  – dual-stack-uplane-pref {ipv4 | ipv6}
  – no dual-stack-uplane-pref
  – interface
    – gtp-c [router router-instance] interface-name [interface-realm realm-name]

```

```

    - no gtp-c
    - gtp-u [router router-instance] interface-name [interface-realm realm-name]
    - no gtp-u
  - match
    - plmn-id-list plmn-list
    - no plmn-id-list
  - peer-list list-name
  - no peer-list
  - send-end-marker
  - no send-end-marker
  - signaling
    - gtp-c
      - enhanced-sgw-restoration restoration-time
      - no enhanced-sgw-restoration
      - pause-charging
        - data-threshold packets
        - no data-threshold
        - data-threshold-hold-time seconds
        - no data-threshold-hold-time
        - pgw-pause-charging {enable | disable}
        - no pgw-pause-charging
        - radio-loss {enable | disable}
        - no radio-loss
      - profile profile-name
      - no profile
    - gtp-u
      - [no] gtpu-keep-alive
      - profile profile-name
      - no profile

```

5.1.1.33 SBA client services PDN command hierarchy

configure

```

- mobile-gateway
- [no] pdn gw-id
  - sba-client-services
    - addressing-priority [fqdn | ip-endpoint | ip-endpoint-fallback]
    - no addressing-priority
    - amf-client
      - [no] namf-comm service-instance
        - callback-type-preference type
        - no callback-type-preference
        - fqdn fqdn-sni-string
        - no fqdn

```

```
– http2-profile http2-profile-name
– no http2-profile
– interface interface-name [router router-instance]
– no interface
– mode mode
– no mode
– n1-profile profile-name
– no n1-profile
– namf-profile profile-name
– no namf-profile
– nf-id-list list-name
– no nf-id-list
– oauth2-profile profile-name
– no oauth2-profile
– paging-damp-timer paging-damp-timer
– no paging-damp-timer
– service-interface ip-address | fqdn-name [port port] [scheme scheme]
– no service-interface
– [no] namf-evts service-instance
– callback-type-preference type
– no callback-type-preference
– fqdn fqdn-sni-string
– no fqdn
– http2-profile http2-profile-name
– no http2-profile
– interface interface-name [router router-instance]
– no interface
– mode mode
– no mode
– namf-profile profile-name
– no namf-profile
– nf-id-list list-name
– no nf-id-list
– oauth2-profile profile-name
– no oauth2-profile
– service-interface ip-address | fqdn-name [port port] [scheme scheme]
– no service-interface
– chf-client
– [no] nchf-convergedcharging instance-name
– alternative-api-version version
– no alternative-api-version
– api-version version [minor minor] [patch patch]
– no api-version
– fqdn fqdn-sni-string
– no fqdn
```

```

    - http2-profile http2-profile-name
    - no http2-profile
    - interface interface-name [router router-instance]
    - no interface
    - mode mode
    - no mode
    - nf-id-list list-name
    - no nf-id-list
    - oauth2-profile profile-name
    - no oauth2-profile
    - service-interface ip-address | fqdn-name [port port] [scheme scheme]
    - no service-interface
- converged-interface-support
  - converged-n10 {enable | disable | 5gcnri}
  - no converged-n10
  - converged-n40 {enable | disable}
  - no converged-n40
  - converged-n40-gngp {enable | disable}
  - no converged-n40-gngp
  - converged-n7 {enable | disable}
  - no converged-n7
  - converged-n7-gngp {enable | disable}
  - no converged-n7-gngp
  - pdu-session-id-base value
  - no pdu-session-id-base
- fqdn fqdn-sni-string
- no fqdn
- http2-profile http2-profile-name
- no http2-profile
- nrf-client
  - [no] nnrf-disc service-instance
    - blocklist-duration seconds
    - no blocklist-duration
    - chf-selection-options
      - [no] exclude-gpsi [vsmf-only]
      - [no] exclude-supi [vsmf-only]
      - [no] include-preferred-api-version
    - [no] exclude-snssais
    - fqdn fqdn-sni-string
    - no fqdn
    - http2-profile http2-profile-name
    - no http2-profile
    - interface interface-name [router router-instance]
    - no interface
    - mode mode

```


- no *mode*
- *nf-id-list* *list-name*
- no *nf-id-list*
- *nrf-cache-interval* *value*
- *nrf-cache-interval* *use-nrf-validity-period*
- no *nrf-cache-interval*
- *service-interface* *ip-address* | *fqdn-name* [*port port*] [*scheme scheme*]
- no *service-interface*
- [no] *suppress-discovery* {*amf* | *chf* | *pcf* | *udm*}
- *udm-selection-options*
 - [no] *exclude-group-id* [*vsmf-only*]
 - [no] *exclude-supi* [*vsmf-only*]
 - [no] *include-preferred-api-version*
- [no] *use-locality*
- [no] *nnrf-nfm* *service-instance*
 - *blocklist-duration* *seconds*
 - no *blocklist-duration*
 - *callback-type-preference* *type*
 - no *callback-type-preference*
 - *fqdn* *fqdn-sni-string*
 - no *fqdn*
 - *heartbeat-nrf-blocklist* *retry* *retry-count* *wait-timer* *seconds*
 - no *heartbeat-nrf-blocklist*
 - *http2-profile* *http2-profile-name*
 - no *http2-profile*
 - *interface* *interface-name* [*router router-instance*]
 - no *interface*
 - *mode* *mode*
 - no *mode*
 - *nf-id-list* *list-name*
 - no *nf-id-list*
 - *nfm-nrf-blocklist* *retry* *retry-count* *wait-timer* *seconds*
 - no *nfm-nrf-blocklist*
 - *service-interface* *ip-address* | *fqdn-name* [*port port*] [*scheme scheme*]
 - no *service-interface*
- [no] *nnrf-oauth2* *service-instance*
 - *fqdn* *fqdn-sni-string*
 - no *fqdn*
 - *http2-profile* *http2-profile-name*
 - no *http2-profile*
 - *interface* *interface-name* [*router router-instance*]
 - no *interface*
 - *nf-id-list* *list-name*
 - no *nf-id-list*
 - *service-interface* *ip-address* | *fqdn-name* [*port port*] [*scheme scheme*]

```

    - no service-interface
- pcf-client
  - [no] npcf-smpolicycontrol instance-name
    - callback-type-preference type
    - no callback-type-preference
    - fqdn fqdn-sni-string
    - no fqdn
    - http2-profile http2-profile-name
    - no http2-profile
    - interface interface-name [router router-instance]
    - no interface
    - mode mode
    - no mode
    - nf-id-list list-name
    - no nf-id-list
    - npcf-profile profile-name
    - no npcf-profile
    - oauth2-profile profile-name
    - no oauth2-profile
    - service-interface ip-address | fqdn-name [port port] [scheme scheme]
    - no service-interface
- [no] sbi-correlation-info-header
- sbi-nf-peer-info-header [scp] [sepp] [srvinst]
  - no sbi-nf-peer-info-header
- scp-client
  - [no] scp-service service-instance
    - block-duration value
    - no block-duration
    - default-failure-action failure-action
    - no default-failure-action
    - fh-no-response failure-action
    - no fh-no-response
    - fqdn fqdn-sni-string
    - no fqdn
    - http-status-code [status-code status-code-value [application-error app-error-code]] {failure-action}
    - no http-status-code [status-code status-code-value [application-error app-error-code]]
    - http2-profile http2-profile-name
    - no http2-profile
    - interface interface-name [router router-instance]
    - no interface
    - mode-fallback new-model
    - no mode-fallback
    - nf-id-list list-name

```

```
– no nf-id-list
– scp-domain domain-name
– no scp-domain
– service-interface ip-address | fqdn-name [port port] [scheme scheme]
– no service-interface
– sepp-client
– [no] sepp service-instance
– fqdn fqdn-sni-string
– no fqdn
– http2-profile http2-profile-name
– no http2-profile
– interface interface-name [router router-instance]
– no interface
– nf-id-list list-name
– no nf-id-list
– plmn-list list-name
– no plmn-list
– service-interface ip-address | fqdn-name [port port] [scheme scheme]
– no service-interface
– smf-client
– [no] nsmf-pdusession service-instance
– api-prefix string
– no api-prefix
– callback-type-preference type
– no callback-type-preference
– http2-profile http2-profile-name
– no http2-profile
– interface interface-name [router router-instance]
– no interface
– mode mode
– no mode
– nf-id-list list-name
– no nf-id-list
– nsmf-profile profile-name
– no nsmf-profile
– oauth2-profile profile-name
– no oauth2-profile
– service-interface ip-address | fqdn-name [port port] [scheme scheme]
– no service-interface
– supported-features
– [no] dtssa
– transaction-timer value
– no transaction-timer
– version major [minor minor] [patch patch]
– no version
```

```
– udm-client
– [no] nudm-sdm service-instance
  – api-version version
  – no api-version
  – callback-type-preference type
  – no callback-type-preference
  – explicit-unsubscribe [expiry-time expiry-time] [renew-on-expiry]
  – no explicit-unsubscribe
  – fqdn fqdn-sni-string
  – no fqdn
  – http2-profile http2-profile-name
  – no http2-profile
  – interface interface-name [router router-instance]
  – no interface
  – mode mode
  – no mode
  – nf-id-list list-name
  – no nf-id-list
  – oauth2-profile profile-name
  – no oauth2-profile
  – service-interface ip-address | fqdn-name [port port] [scheme scheme]
  – no service-interface
  – supported-features
    – immediate-report [skip-get]
    – no immediate-report
  – udm-sdm-profile profile-name
  – no udm-sdm-profile
– [no] nudm-uecm instance-name
  – callback-type-preference type
  – no callback-type-preference
  – fqdn fqdn-sni-string
  – no fqdn
  – http2-profile http2-profile-name
  – no http2-profile
  – interface interface-name [router router-instance]
  – no interface
  – mode mode
  – no mode
  – nf-id-list list-name
  – no nf-id-list
  – oauth2-profile profile-name
  – no oauth2-profile
  – service-interface ip-address | fqdn-name [port port] [scheme scheme]
  – no service-interface
  – udm-uecm-profile profile-name
```

- no *udm-uecm-profile*
- *user-agent-header* [*user-agent user-agent*]
- no *user-agent-header*

5.1.1.34 SBA server services PDN command hierarchy

configure

- mobile-gateway
 - [no] *pdn gw-id*
 - *sba-server-services*
 - *fqdn fqdn-sni-string*
 - no *fqdn*
 - *http2-profile http2-profile-name*
 - no *http2-profile*
 - [no] *nbsf-management service-instance*
 - *allowed-nf-domains-list list-name*
 - no *allowed-nf-domains-list*
 - *allowed-nf-types {pcf | nef | af | dra}*
 - no *allowed-nf-types [pcf | nef | af | dra]*
 - *allowed-plmns list-name*
 - no *allowed-plmns*
 - *fqdn fqdn-sni-string*
 - no *fqdn*
 - *http2-profile http2-profile-name*
 - no *http2-profile*
 - *interface interface-name [router router-instance] [port port-number]*
 - no *interface*
 - *oauth2-profile profile-name*
 - no *oauth2-profile*
 - *service-interface ip-address | fqdn-name [port port] [scheme scheme]*
 - no *service-interface*
 - [no] *nsmf-event-exposure service-instance*
 - *allowed-nf-domains-list list-name*
 - no *allowed-nf-domains-list*
 - [no] *allowed-nf-types {pcf | nef | af | amf | udm | dccf | nwdaf}*
 - *allowed-plmns list-name*
 - no *allowed-plmns*
 - *allowed-slices list-name*
 - no *allowed-slices*
 - *default-subscription-timeout subscription-timeout*
 - no *default-subscription-timeout*
 - *fqdn fqdn-sni-string*
 - no *fqdn*
 - *http2-profile http2-profile-name*

```
– no http2-profile
– interface interface-name [router router-instance] [port port-number]
– no interface
– mode mode
– no mode
– nsmf-profile profile-name
– no nsmf-profile
– oauth2-profile profile-name
– no oauth2-profile
– service-interface ip-address | fqdn-name [port port] [scheme scheme]
– no service-interface
– supported-features
  – [no] communication-failure
  – [no] ene-na
  – [no] pdu-session-status
  – [no] qfi-allocation
– [no] nsmf-pdusession service-instance
  – allowed-nf-domains-list list-name
  – no allowed-nf-domains-list
  – [no] allowed-nf-types {amf | smf}
  – allowed-plmns list-name
  – no allowed-plmns
  – api-prefix string
  – no api-prefix
  – fqdn fqdn-sni-string
  – no fqdn
  – http2-profile http2-profile-name
  – no http2-profile
  – indirect-forwarding-timeout indirect-forwarding-timeout
  – no indirect-forwarding-timeout
  – interface interface-name [router router-instance] [port port-number]
  – no interface
  – n3-interface-realm realm-name
  – no n3-interface-realm
  – n9-interface-realm realm-name
  – no n9-interface-realm
  – n9-roaming-interface-realm realm-name
  – no n9-roaming-interface-realm
  – oauth2-profile profile-name
  – no oauth2-profile
  – s5-ref-point sig-ref-point-name
  – no s5-ref-point
  – s8-ref-point sig-ref-point-name
  – no s8-ref-point
  – service-interface ip-address | fqdn-name [port port] [scheme scheme]
```

- no *service-interface*
- *supported-features*
 - [no] *aasn*
 - [no] *dtssa*
- *version major* [minor *minor*] [patch *patch*]
- no *version*
- *sbi-load-control*
 - [no] *enable*
 - *load-change-trigger percent*
 - no *load-change-trigger*
 - *nth-message number*
 - no *nth-message*
- *sbi-nf-peer-info-header* [*scp*] [*sepp*] [*srvinst*]
- no *sbi-nf-peer-info-header*
- *sbi-overload-control*
 - [no] *enable*
 - *overload-reduction*
 - *critical percent*
 - no *critical*
 - *major percent*
 - no *major*
 - *minor percent*
 - no *minor*
 - *validity-time validity-time*
 - no *validity-time*
- *user-agent-header* [user-agent *user-agent*]
- no *user-agent-header*

5.1.1.35 SBA service realm PDN command hierarchy

configure

- mobile-gateway
 - [no] *pdn gw-id*
 - [no] *sba-service-realm service-realm*
 - *client-service service-name service-instance service-instance*
- [priority value]
 - no *client-service service-name service-instance service-instance*
 - [no] *server-service service-name service-instance service-instance*

5.1.1.36 Sd PDN command hierarchy

configure

```

- mobile-gateway
- [no] pdn gw-id
  - sd
    - [no] diameter-redirect
    - doic
    - [no] diameter-overload-control
    - interface
      - diameter [router router-instance] interface-name
      - no diameter
      - [no] multi-homing
        - diameter [router router-instance] secondary-interface-name
        - no diameter
    - origin-host name
    - no origin-host
    - peer-group peer-group-id primary-pcrf profile-name [secondary-pcrf profile-
name]
    - no peer-group peer-group-id
    - retry-count value
    - no retry-count
    - [no] sd-profile profile-name
    - transaction-timer value
    - no transaction-timer

```

5.1.1.37 Signaling PDN command hierarchy

```

configure
- mobile-gateway
- [no] pdn gw-id
  - signaling
    - diameter
      - mip6-agent-info {use-fqdn fqdn use-realm realm | use-sig-ip-addr}
      - no mip6-agent-info
      - origin-host name
      - no origin-host
      - origin-realm name
      - no origin-realm
      - profile profile-name
      - no profile
    - gtp-c
      - profile profile-name
      - no profile
    - gtp-u
      - profile profile-name
      - no profile

```


- *interface* [router router-instance] interface-name
- no *interface*
- *mipv4*
 - *profile*
- *multi-homing*
 - *interface* [router router-instance] interface-name
 - no *interface*

5.1.1.38 STa PDN command hierarchy

configure

- mobile-gateway
 - [no] *pdn gw-id*
 - *sta*
 - *diameter-peer-group-list* group-list-name
 - no *diameter-peer-group-list*
 - *doic*
 - [no] *diameter-overload-control*
 - *interface*
 - *diameter* [router router-instance] interface-name
 - no *diameter*
 - [no] *multi-homing*
 - *diameter* [router router-instance] secondary-interface-name
 - no *diameter*
 - *origin-host* name
 - no *origin-host*
 - *radius-server-group* radius-group
 - no *radius-server-group*
 - *retry-count* value
 - no *retry-count*
 - *sta-profile* profile-name
 - no *sta-profile*
 - *transaction-timer* value
 - no *transaction-timer*

5.1.1.39 SWm PDN command hierarchy

configure

- mobile-gateway
 - [no] *pdn gw-id*
 - [no] *swm*
 - [no] *aar-call-flow*

```

    - activate-aaa-failure-indication {enable | disable}
    - no activate-aaa-failure-indication
    - [no] alt-peer-rar-asr
    - dia-routing-table idi-realm-id idi-realm-string diameter-peer-group-list
list-name
    - no dia-routing-table idi-realm-id
    - doic
      - [no] diameter-overload-control
    - interface
      - diameter [router router-instance] interface-name
      - no diameter
      - [no] multi-homing
        - diameter [router router-instance] secondary-interface-name
        - no diameter
    - peer-group group-id primary-diameter-peer profile-name [secondary-diameter-
peer profile-name] [priority priority-value]
      - origin-host name
      - no origin-host
      - no peer-group group-id
      - retry-count value
      - no retry-count
      - swm-profile swm-profile-name
      - no swm-profile
      - transaction timer value
      - no transaction timer
      - swm-error-mapping-profile profile-name
      - no swm-error-mapping-profile

```

5.1.1.40 SWu PDN command hierarchy

```

configure
- mobile-gateway
- [no] pdn gw-id
  - [no] swu [ref-point-name]
    - interface [router router-instance] interface-name
    - no interface
    - [no] ipsec-profile profile-name
  - swu-ipsec-lockout
    - backoff-timer [timer-value timer-value] [timer-unit timer-unit]
    - no backoff-timer
    - configuration count count time time lockout lockout-time
    - no configuration
    - function {enable | disable}
    - no function

```

- *swu-throttling* [level {best-effort | minimal | medium | aggressive}]
- no *swu-throttling*

5.1.1.41 SWw PDN command hierarchy

```

configure
  – mobile-gateway
  – [no] pdn gw-id
  – [no] sww [ref-point-name]
    – interface
      – sww-acct [router router-instance] interface-name port port secret secret
[hash | hash2]
      – no sww-acct
      – sww-auth [router router-instance] interface-name port port secret secret
[hash | hash2]
      – no sww-auth
      – sww-coa [router router-instance] interface-name port port
secret secret [hash | hash2]
      – no sww-coa
      – sww-rad-disconnect secret secret [hash | hash2]
      – no sww-rad-disconnect
      – sww-tunnel [router router-instance] interface-name [connect-method connect-
method]
      – no sww-tunnel
  – signaling
    – sww-radius-profile profile-name
    – no sww-radius-profile
    – sww-tunnel-profile profile-name
    – no sww-tunnel-profile

```

5.1.1.42 Sx-N4 PDN command hierarchy

```

configure
  – mobile-gateway
  – [no] pdn gw-id
  – [no] sx-n4 [ref-point-name]
    – [no] enforced-pfcp-association-list
    – interface
      – gtp-u [router router-instance] interface-name [interface-realm realm-name]
      – no gtp-u
      – pfcp [router router-instance] interface-name
      – no pfcp

```

- *pfcp-association-list list-name*
- *no pfcp-association-list*
- *signaling*
 - *pfcp*
 - *profile profile-name*
 - *no profile*

5.1.1.43 T6c command hierarchy

configure

- mobile-gateway
 - [no] *pdn gw-id*
 - *t6c [ref-point-name]*
 - *no t6c*
 - *http-port port-number*
 - *no http-port*
 - *interface interface-name [router router-instance]*
 - *no interface*
 - *t6c-profile profile-name*
 - *no t6c-profile*
 - *uri uri-name*
 - *no uri*

5.1.1.44 Threshold PDN command hierarchy

configure

- mobile-gateway
 - [no] *pdn gw-id*
 - *thresholds*
 - [no] *bearer-cf-failure*
 - *counter counter-name high-threshold threshold [low-threshold threshold]*
 - *no counter counter-name*
 - *interval minutes*
 - [no] *shutdown*
 - [no] *bearer-cf-success*
 - *counter counter-name high-threshold threshold [low-threshold threshold]*
 - *no counter counter-name*
 - *interval minutes*
 - [no] *shutdown*
 - [no] *bearer-mgmt-limit*
 - *counter counter-name high-threshold threshold [low-threshold threshold]*
 - *no counter counter-name*

- *interval minutes*
- *no interval*
- *[no] shutdown*
- *[no] bearer-traffic*
 - *counter counter-name high-threshold threshold [low-threshold threshold]*
 - *no counter counter-name*
 - *interval minutes*
 - *[no] shutdown*
- *dl-buff-utilization-threshold threshold*
- *no dl-buff-utilization-threshold*
- *[no] path-mgmt*
 - *counter counter-name high-threshold threshold [low-threshold threshold]*
 - *no counter counter-name*
 - *interval minutes*
 - *[no] shutdown*
- *[no] pdn-connections*
 - *counter counter-name high-threshold threshold [low-threshold threshold]*
 - *no counter counter-name*
 - *interval minutes*
 - *[no] shutdown*

5.1.1.45 Virtual APN selection PDN command hierarchy

configure

- *mobile-gateway*
 - *[no] pdn gw-id*
 - *[no] virtual-apn-selection*
 - *no-match-action action [virtual-apn apn]*
 - *no no-match-action*
 - *pco-fallback user-name user-name*
 - *[no] pco-fallback*
 - *selection-prio UE type [UE type...up to 10 max]*
 - *no selection-prio*
 - *selection-type {prefix | full-id}*
 - *no selection-type*
 - *user-range user-range-name type [{imsi | msisdn | imei-tac} start-ue-id [end-ue-id]] | [roaming-status roaming-status-type] | [cc start-cc start-cc-id [end-cc end-cc-id]] | [imeisv-tacsvn start-ue-id [end-ue-id] svn svn-value] | [serving-node-plmn list-name] | [pdn-type ipv4 | ipv6 | ipv4v6] | [rat-type rat-type-list-name] | [ue-location ue-location-list-name] | {virtual-apn virtual-apn-name | {[append-string suffix] [pre-auth-type {radius server-group server-group-name [user-name {imsi | msisdn | pco | fixed}] [password password password-type {fixed | default} [hash | hash2]] | diameter} [suppress-imsi]]}]}*

- no *user-range* *user-range-name*

5.1.2 PDN show command hierarchy

show

- mobile-gateway
- pdn
 - *apn* *apn-name* statistics [group *group-number*]
 - *apn* *apn-name* delete-session-statistics [gateway *gw-id* | group *group-number*]
- [summary]
 - *apn* *apn-name* attach-failure-statistics [gateway *gw-id* | group *group-number*]
- [summary]
 - *apn* *apn-name* dns-redirect-statistics [gateway *gw-id* | group *group-number*]
- [summary]
 - *apn* *apn-name* fqdn-group *name*
 - *apn* *apn-name* gy-ap-statistics
 - *apn* *apn-name* [detail]
 - *apn* *apn-name* apn-override-statistics [gateway *gw-id* | group *group-number*]
- [summary]
 - *apn* *apn-name* pcscf-statistics
 - *apn* *apn-name* gx-ap-statistics
 - *apn* *apn-name* call-flow-stats [gateway *gw-id* | group *group-number*] [summary]
 - *apn* *apn-name* access-type *access-type*
 - *accounting*
 - *authentication*
 - *charging*
 - *ip-alloc*
 - *l2tp*
 - *pfc*
 - *policy*
 - *virtual-apn* [detail]
 - *apn-overload* [apn-name]
 - *application-server-stats* [gateway *gw-id*] [udp-tunnel-group *udp-tunnel-group*]
- [peer *ip-address*]]
 - *bearer-context* [imsi | imei | msisdn] *string* [bearer *bearer-id*] [apn *name*]
- [gateway *gw-id* | group *group-number* | [card *slot-number* | vm *vm-id*]]
 - *bearer-context* imsi *string* [apn *name*] [bearer *bearer-id*] [gateway *gw-id* | group *group-number* | [card *slot-number* | vm *vm-id*]] [qci *qci-value* arp *arp-value*] {control | filter | bearer-traffic | detail}
 - *bearer-context* imsi *string* bearer *bearer-id* acct {ga | gy | rf} [detail]
 - *bearer-context* mac *ieee-address* [apn *name*] [bearer *bearer-id*] [gateway *gw-id* | group *group-number* | [card *slot-number* | vm *vm-id*]] [qci *qci-value* arp *arp-value*] [detail]

```

    - bearer-context default-session [def-session-pgw-ip pgw-ip-address] [apn name]
    [bearer bearer-id] [gateway gw-id | group group-number | [card slot-number | vm vm-id]]

    - bearer-context default-session def-session-pgw-ip pgw-ip-address apn name
    detail [bearer bearer-id] [gateway gw-id | group group-number | [card slot-number |
    vm vm-id]] [qci qci-value arp arp-value]

    - bearer-context default-session def-session-pgw-ip pgw-ip-address acct {ga | gy
    | rf} apn name [bearer bearer-id] [gateway gw-id] [detail]

    - bearer-context imsi | subscriber-id string [gateway gw-id] bearer bearer-id
    acct {stats}

        - cp-seid seid [cp-ipv4 ip-address] [cp-ipv6 ipv6-address]
        - cp-seid seid [cp-ipv4 ip-address] [cp-ipv6 ipv6-address] detail
        - def-pfcp-u-sessions [network-realm network-realm] [pfcp-u-peer-ip ip-address]
    [gateway gw-id] [apn apn-name] [card slot-number] [vm vm-id] [group group-number]
        - supi imsi-msisdn {control | filter | bearer-traffic | detail}
        - ue-ip ip-address network-instance apn [traffic-endpt-id traffic-endpt-id]
    detail [access-type {utran | geran | wlan | gan | hspa | eutran | ehrpdp | hrpd | 1xrtt
    | umb | virtual | eutran-nb-iot}]
        - up-seid seid
        - up-seid seid [traffic-endpt-id traffic-endpt-id] detail
    - bearer-context imsi | subscriber-id string [gateway gw-id] bearer bearer-id
    acct {stats}
        - cp-seid seid [cp-ipv4 ip-address] [cp-ipv6 ipv6-address]
        - call-flow-stats [gateway gw-id | group group-number | [card slot-number |
    vm vm-id]] [gw-type sgw | pgw | upf | smf] [summary]
        - canary-test gateway gw-id statistics
        - call-flow-stats oi-override-mapping [gateway gw-id] [plmn-mcc mcc-value plmn-
    mnc mnc-value]
        - charging
            - plmn-chrg-profile-map plmn-list-name list-name [gateway-id gw-id]
            - clear-status [gateway gw-id]
            - dnn-overload-control [dnn-name]
            - error-indication [gateway gw-id] [router vr-id] [peer src-address] [summary]
            - fixed-subscriber [subscriber-id] [apn apn-name] [gateway gw-id] [group group-
    number] [card slot-number] [vm vm-id]
            - fixed-subscriber subscriber-id detail [apn apn-name] [gateway gw-id]
    [group group-number] [card slot-number] [vm vm-id]
            - gtp-load-overload-control
            - gtp-load-overload-control stats [node-id]
            - gtp-local-load-control stats
            - gtp-local-overload-control stats
            - local-storage-stats [gtp-prime-group gtp-prime-group] ga
            - n2 [sig-ref-point-name] [gateway gw-id]
            - nf-peers nf-type [nf-id uuid] [gateway gw-id | group group-id] [instance |
    aggregate]

```

```

    - nsmf-event-exposure-subscriptions [supi supi] [gpsi gpsi] [subscription-
id subscription-id] [pdu-session-id pdu-session-id] [group-id group-id]
    - nsmf-overload-control stats
    - nwu [sig-ref-point-name] [gateway gw-id]
    - path-mgmt-stats {s11 | s1u | s12 | s4 | s5 | s8 | gx | rf | gn | ga | gp | gy
| s2a | s2b | s6b | swm | sww | sta | sx-n4 | n3 | n9 | s4u | s5u | s8u | gnu | gpu}
[peer ip-address [port port-number]] [gateway gw-id | group group-number | [card slot-
number | vm vm-id]] [instance | aggregate]
    - path-mgmt-stats {gn | gnu | gp | gpu} [ref-point ref-point-name] [peer ip-
address [port port-number]] [gateway gw-id | group group-number | [card slot-number |
vm vm-id]] [instance | aggregate] [active]
    - path-mgmt-stats {s5 | s5u | s8 | s8u | n9} [ref-point ref-point-name] [peer ip-
address [port port-number]] [gateway gw-id | group group-number | [card slot-number |
vm vm-id]] [instance | aggregate]
    - pcf-pcrf-based-pcscf-allocation [entry-id entry-id]
    - pcmd-stats [gateway gw-id | group group-number | [card slot-number | vm vm-id]]
[gw-type sgw | pgw] [summary]
    - pdn-context [imsi | imei | msisdn string detail] [epdg] [apn name] [gateway gw-
id | group group-number | card slot-number] [nemo]
    - pdn-context imsi | imei | msisdn string [apn name] [gateway gw-id] acct {rf |
gy} [detail]
    - pdn-context imsi | imei | msisdn string gx-usage-monitoring [gateway gw-id]
[apn name] [monitoring-key key-name]
    - pdn-context imsi | imei | msisdn string gx-time-of-day-info [gateway gw-id]
[apn name]
    - pdn-context imsi string detail {epdg | epdg-decrypt}
    - pdn-context mac ieee-address [apn name] [gateway gw-id | group group-number |
card slot-number] [detail]
    - pdn-context default-session [def-session-pgw-ip pgw-ip-address [apn name]
detail] [gateway gw-id | group group-number | card slot-number] [nemo]
    - pdn-context default-session acct {rf | gy} [def-session-pgw-ip pgw-ip-address]
[apn name] [gateway gw-id] [detail]
    - pdn-context ipsec-lockout [summary]
    - pdn-context ipsec-lockout idi IDi source-ip source-ip-address
    - pdn-context subscriber-id subscriber-id [apn name] [gateway gw-id |
group group-number | card slot-number] [access-type access-type] [nemo]
    - pdn-context subscriber-id subscriber-id detail acct {rf | gy | radius}
[apn name] [gateway gw-id]
    - pdn-context ue-ip ue-ip network-instance network-inst detail
    - pdn-context pdn-context up-seid string detail
    - pdn-context cp-seid string [cp-ipv4 ipv4-address cp-ipv6 ipv6-address] detail
    - pdn-context def-pfcp-u-sessions [network-realm realm-name] [pfcp-u-peer-ip ip-
address] [gateway gw-id | group group-number | vm vm-id] [apn name]
    - pdn-context field-name [gateway gw-id | group group-number | vm vm-id]
[apn name]

```



```

- pdn-context wrap [gateway gw-id | group group-number | vm vm-id] [apn name]
- cp-seid cp-seid seid [cp-ipv4 ip-address] [cp-ipv6 ipv6-address]
- cp-seid cp-seid seid [cp-ipv4 ip-address] [cp-ipv6 ipv6-address] detail
- def-pfcp-u-sessions [network-realm network-realm] [pfcp-u-peer-ip ip-address]
[gateway gw-id] [apn apn-name] [card slot-number] [vm vm-id] [group group-number]
- field-name {pcrf-ip | pcrf-dest-host | ue-ip | roaming-status | s6b-peer-ip |
s6b-peer-dest-host | p-cscf-ip | dns-ipv4 | dns-ipv6 | s5-pgw-gtpc-ipv4 | s5-pgw-gtpc-ipv6}
[gateway gw-id] [apn apn-name] [card slot-number] [vm vm-id] [group group-number]
[connected-time-gt connected-time-gt] [connected-time-leq connected-time-leq]
- supi imsi-msisdn {control | filter | bearer-traffic | detail}
- ue-ip ip-address network-instance network-instance [access-type {utran |
geran | wlan | gan | hspa | eutran | ehrpdp | hrpd | 1xrtt | umb | virtual | eutran-nb-iot}]}
- ue-ip ip-address network-instance network-instance detail acct stats urr-id
urr-id [access-type {utran | geran | wlan | gan | hspa | eutran | ehrpdp | hrpd |
1xrtt | umb | virtual | eutran-nb-iot}]}
- ue-ip ip-address network-instance network-instance detail [access-type {utran
| geran | wlan | gan | hspa | eutran | ehrpdp | hrpd | 1xrtt | umb | virtual | eutran-nb-iot}]}
- up-seid up-seid seid
- up-seid up-seid seid detail acct stats [urr-id urr-id]
- up-seid up-seid seid detail
- wrap [gateway gw-id] [apn apn-name] [card slot-number] [vm vm-id]
[group group-number]
- pdu-session
- pdu-session supi imsi-msisdn [detail]
- pdu-session supi imsi-msisdn acct {stats | nchf} [detail]
- pdu-session ran-ue-ngap-id ran-ue-ngap-id [pdu-session-id pdu-session-id]
- pdu-session ran-ue-ngap-id ran-ue-ngap-id detail [pdu-session-id pdu-session-id]
- pdu-session ran-ue-ngap-id ran-ue-ngap-id acct stats [pdu-session-id pdu-session-id] [qfi qfi]]
- pfcp-load-overload-control
- pfcp-load-overload-control stats
- pfcp-local-load-control stats
- pfcp-local-overload-control stats
- pgw-reselection-isolation
- qos-flow [qfi qfi] acct acct ran-ue-ngap-id ran-ue-ngap-id pdu-session-id pdu-session-id
- qos-flow [dnn dnn]
- qos-flow supi imsi-msisdn
- qos-flow supi imsi-msisdn detail
- redirect-policy [gateway gw-id] [apn apn]
- ref-point-peers ref-point-type [ref-point ref-point-name] [peer ip-address
[port port-number]] [gateway gw-id] [peer-type {sgw | pgw | twag | epdg}]

```

```

    - ref-point-peers {gn | gnu | gp | gpu} [ref-point ref-point-name] [peer ip-address [port port-number]] [gateway gw-id] [active] ref-point-peers {radius} [radius-group radius-group [peer ip-address]] [gateway gw-id]
    - ref-point-peers {radius} [radius-group radius-group [peer ip-address]]
    - ref-point-peers {s5 | s5u | s8 | s8u | n9} [ref-point ref-point-name] [peer ip-address [port port-number]] [gateway gw-id]
    - ref-point-peers {dhcp} [dhcp-group dhcp-group [peer ip-address]] [gateway gw-id]
    - ref-point-peers {pi} [pi-peer ipv4 [port port-number]] [gateway gw-id]
    - ref-point-peers {swm} swm-type [peer ip-address] [gateway gw-id] [ref-point ref-point-name]
    - ref-point-peers {cdbx} [peer ip-address]
    - ref-point-peers {swm | gy} [dia-routing-table] [gateway gw-id]
    - ref-point-peers {s6b | gx | swm | sta | sd} [gateway-id gw-id] [peer ip-address [port port-number]] doic
    - ref-point-peers {sx-n4 | n2} [peer ip-address] [gateway gw-id] [ref-point ref-point-name]
    - ref-point-peers {t6c} [ref-point ref-point-name] [peer ip-address [port port-number]] [gateway gw-id] [active]
    - ref-point-stats {s11 | s11u | s12 | s1u | s4 | s4u | s5 | s5u | s8 | s8u | gx | rf | gn | gnu | gp | gpu | ga | radius | gy | s2a | s2a-gtp | s2b | s6b | swm | swm | sta | sx-n4 | sd | dhcp | n3 | n9} [peer ip-address [port port-number]] [gateway gw-id | group group-number | card slot-number] [instance | aggregate] [peer-type {sgw | pgw | epdg}]
    - ref-point-stats stats {s11 | s4 | s5 | s5u | s8 | s8u | gx | gn | gnu | gp | gpu | gy | s2a | s2a-gtp | s2b | s6b | swm | sta | sx-n4} [peer ip-address [port port-number]] [gateway gw-id | group group-number | card slot-number] [instance | aggregate] {failure-codes}
    - ref-point-stats {gn | gnu | gp | gpu} [ref-point ref-point-name] [peer ip-address [port port-number]] [gateway gw-id | group group-number | card slot-number] [instance | aggregate] [active]
    - ref-point-stats {s5 | s5u | s8 | s8u | gn | gnu | gp | gpu | n9} [ref-point ref-point-name] [peer ip-address [port port-number]] [gateway gw-id | group group-number | card slot-number] [instance | aggregate]
    - ref-point-stats {dhcp} [dhcp-group dhcp-group [peer ip-address]] [gateway gw-id | group group-number | card slot-number] [instance | aggregate]
    - ref-point-stats {pi} [pi-peer ipv4 [port port-number]] [[gateway gw-id | group group-number | card slot-number]] [instance | aggregate]
    - ref-point-stats {radius} [radius-group radius-group [peer ip-address]] [gateway gw-id | group group-number | card slot-number] [instance | aggregate]
    - ref-point-stats {swu} [card slot-number] [aggregate]
    - ref-point-stats {swm} swm-type [peer ip-address] [gateway gw-id | group group-number | card slot-number] [instance | aggregate] [ref-point ref-point-name]
    - ref-point-stats {sx-n4} [ref-point ref-point-name] [peer ip-address] [gateway gw-id | group group-number | card slot-number] [instance | aggregate]

```

```

    - ref-point-stats {t6c} [ref-point ref-point-name] [peer ip-address [port port-number]] [peer-type scef] [gateway gw-id | group group-number | card slot-number] [instance | aggregate] [failure-codes]
    - sbi-load-overload-control
    - service-stats nf-type nf-type [service-realm service-realm-name] [service-name service-name] [service-instance service-instance] [remote-nf-id uuid | remote-nf-ip ip-address | fqdn] [gateway gw-id | group group-id] [failure-codes] [instance | aggregate]
    - static-predefined-pcc [apn name] [default-policy-rule-base base-policy-name] [gateway-id gw-id]
    - statistics [gw-type sgw | pgw | upf] [gateway gw-id | group group-number | [card slot-number | vm vm-id]] [summary]
    - statistics attach-failure-statistics [gateway gw-id | group group-number] [summary]
    - swu sig-ref-point-name
    - ue-context [imsi | imei | msisdn | supi | gpsi | pei | string [detail]] [gateway gw-id | group group-number | card slot-number | vm vm-id] [nemo]
    - ue-context mac ieee-address [gateway gw-id | group group-number | card slot-number | vm vm-id] [detail]
    - ue-context ran-ue-ngap-id ran-ue-ngap-id [detail]
    - def-pfcp-u-sessions [network-realm network-realm] [pfcp-u-peer-ip ip-address] [gateway gw-id] [card slot-number] [vm vm-id] [group group-number]
    - up-peer-list ip-pool-stats peer-ip peer-ip

```

5.1.3 PDN clear command hierarchy

```

clear
  - mobile-gateway
    - pdn gw-id
      - apn apn-name attach-failure-statistics [group group-number]
      - apn apn-name [group group-number] call-flow-stats
      - apn apn-name delete-session-statistics [group group-number]
      - apn apn-name [group group-number] apn-override-statistics
      - apn apn-name [group group-number] pcscf-statistics
      - application-server-stats [udp-tunnel-group udp-tunnel-group-name [peer ip-address]]
      - bearer-context {msisdn msisdn | imei imei | imsi imsi} [local-only] [apn name] [bearer bearer-id]
      - bearer-context {msisdn msisdn | imei imei | imsi imsi} [apn name] ] [cause-reactivation-req]
      - bearer-context mac ieee-address [local-only] [apn name]
      - bearer-context apn name [local-only]
      - bearer-context [session-deletion-rate session-deletion-rate] [group group] [protect-dedicated-bearers] [percentage percentage [ue-clear]] apn name

```

```

    - bearer-context [session-deletion-rate session-deletion-rate] [group group]
    [protect-dedicated-bearers] [percentage percentage]
    - bearer-context group group [session-deletion-rate session-deletion-rate]
    apn name pdn-type pdn-type
    - bearer-context group group apn name pdn-type pdn-type {cancel | data-packet-
    trigger}
    - bearer-context {idle-time-leq idle-time-leq | idle-time-gt idle-time-gt | idle-
    all} [session-deletion-rate session-deletion-rate] [group group] [apn name]
    - bearer-context {connected-time-leq connected-time-leq | connected-time-
    gt connected-time-gt} [session-deletion-rate session-deletion-rate] [group group]
    [apn name]
    - bearer-context [session-deletion-rate session-deletion-rate] [group group] ue-
    location ue-location ue-location-id ue-location-id [apn name]
    - bearer-context [session-deletion-rate session-deletion-rate] [group group]
    [protect-dedicated-bearers] [percentage percentage] {pcrf-ip ip-address [pcrf-
    destination-host fqdn] | p-cscf-ip ip-address} [apn name] [cause-reactivation-req]
    - bearer-context [session-deletion-rate session-deletion-rate] [group group]
    [protect-dedicated-bearers] [percentage percentage] pcrf-ip ip-address [pcrf-
    destination-host fqdn] [apn name] [cause-reactivation-req] {{idle-time-leq idle-time-
    leq | idle-time-gt idle-time-gt | idle-all} | {connected-time-leq connected-time-leq |
    connected-time-gt connected-time-gt}}
    - bearer-context [session-deletion-rate session-deletion-rate] [group group]
    [protect-dedicated-bearers] [percentage percentage] {slice slice-name | slice-
    instance slice-instance-name} [apn name]
    - bearer-context up-peer ipv4-address | ipv6-address apn apn-name [ipv4-prefix
    ipv4-address/mask | ipv6-prefix ipv6-address/prefix-length] [cancel]
    - bearer-context {rat-type {utran | geran | wlan | gan | hspa-e | eutran |
    virtual | eutran-nb | lte-m | nr | nr-redcap | wifi} | {user-range type {imsi | msisdn
    | imei-tac} start-ue-id end-ue-id}} [apn name] [local-only] [bearer bearer-id]
    - bearer-context {rat-type {utran | geran | wlan | gan | hspa-e | eutran |
    virtual | eutran-nb | lte-m | nr | nr-redcap | wifi} | {user-range type {imsi | msisdn
    | imei-tac} start-ue-id end-ue-id}} [apn name] [cause-reactivation-req]
    - bearer-context udm-no-get-response nf-instance-id [protect-dedicated-bearers]
    [cause-reactivation-req] [session-deletion-rate session-deletion-rate]
    - bearer-context [protect-dedicated-bearers] up-peer ipv4-address | ipv6-address
    [apn name]
    - call-flow-stats [group group-number | card slot-number]
    - charging imei string imsi string msisdn string [open-partial]
    - diameter-sessions peer {ip-addr | fqdn} [destination-host destination-host]
    [graceful] [port port] application-type {gx | gy}
    - dns-client-cache
    - fixed-subscriber subscriber-id [apn apn-name] [host ip-address] [local-only]
    - gtp-load-overload-control stats [node-id]
    - gtp-local-overload-control stats
    - local-storage-stats [gtp-prime-group gtp-prime-group] ga

```

```

- ipsec-lockout idi user-id source-ip source-ip-address
- nrf-client-cache [nf-type {upf | pcf | udm | amf}]
- nsmf-overload-control stats
- path-mgmt-fault {s5 | s8} [ref-point ref-point-name] [peer ip-address | fqdn]
[port port]
- path-mgmt-stats {s11 | s1u | s4 | s4u | s12 | s5 | s5u | s8 | s8u | gx | rf |
gn | gnu | gp | gpu | ga | gy | s2a | s2b | s6b | sx-4} [peer ip-address [port port]]
- path-mgmt-stats {s5 | s5u | s8 | s8u | gn | gnu | gp | gpu | n9} [ref-
point ref-point-name] [peer ip-address [port port]]
- pdn-context cp-seid seid [cp-ipv4 ip-address] [cp-ipv6 ipv6-address]
- pdn-context ue-ip ue-ip network-instance apn [access-type access-type]
- pdn-context up-seid seid
- pfcf-load-overload-control stats
- pfcf-local-load-control stats
- pfcf-local-overload-control stats
- pgw-reselection-isolation [peer ip-address]
- ref-point-stats {pi} [ref-point ref-point-name] [pi-peer ipv4 [port port]]
[group group-number | card slot-number]
- ref-point-stats {s5 | s8 | gx | rf | gn | gp | ga | gy | s2a | s2b | s6b | s1u
| s11 | s11u | s4 | s12 | sww | sww | sx-n4 | n3 | n9} [peer ip-address [port port]]
[group group-number | card slot-number]
- ref-point-stats swu [card slot-number]
- ref-point-stats {sww} sww-type [ref-point ref-point-name] [peer ip-address]
[group group-number | card slot-number]
- ref-point-stats {radius} [radius-group radius-group [peer ip-address]]
[group group-number | card slot-number]
- ref-point-stats {s5 | s8 | gn | gp | n9} [ref-point ref-point-name] [peer ip-
address [port port]] [group group-number | card slot-number]
- ref-point-stats {t6c} [ref-point ref-point-name] [peer ip-address [port port]]
[card slot-number] [peer-type scef]
- service-stats nf-type nf-type [service-realm service-realm-name] [service-
name service-name] [service-instance service-instance] [remote-nf-id uuid | remote-nf-
ip ip-address | fqdn] [gateway gw-id | group group-id]
- statistics attach-failure-statistics group group-number
- statistics [group group-number] ga-buffered-cdrs
- ue-context ran-ue-ngap-id ran-ue-ngap-id [local-only]

```

5.2 PDN command descriptions

- *PDN configuration command descriptions*
- *PDN show command descriptions*
- *PDN clear command descriptions*

5.2.1 PDN configuration command descriptions

- *Generic PDN command descriptions*
- *PDN command descriptions*
- *APN PDN command descriptions*
- *APN PCF PDN command descriptions*
- *APN PCO PDN command descriptions*
- *APN PCRF PDN command descriptions*
- *Bearer traffic PDN command descriptions*
- *Cloud database PDN command descriptions*
- *Charging PDN command descriptions*
- *Ga/Gn/Gp/Gx/Gy interface PDN command descriptions*
- *GTP load and overload Control PDN command descriptions*
- *KPI-KCI PDN command descriptions*
- *N2 interface PDN command descriptions*
- *N3 interface PDN command descriptions*
- *N9 interface PDN command descriptions*
- *Network trace command descriptions*
- *NWu interface PDN command descriptions*
- *NF profile attributes PDN command descriptions*
- *PFCP load and overload PDN command descriptions*
- *PGW selection PDN command descriptions*
- *PI interface PDN command descriptions*
- *RADIUS PDN command descriptions*
- *RF interface PDN command descriptions*
- *S1-U interface PDN command descriptions*
- *S11 interface PDN command descriptions*
- *S12 interface PDN command descriptions*
- *S2a and S2b interface PDN command descriptions*
- *S4 interface PDN command descriptions*
- *S5 and S8 interface PDN command descriptions*

- [S6b interface PDN command descriptions](#)
- [SBA client services PDN command descriptions](#)
- [SBA server services PDN command descriptions](#)
- [SBA service realm PDN command descriptions](#)
- [Sd PDN command descriptions](#)
- [Signaling PDN command descriptions](#)
- [STa PDN command descriptions](#)
- [SWm PDN command descriptions](#)
- [SWu PDN command descriptions](#)
- [SWw PDN command descriptions](#)
- [Sx-N4 PDN command descriptions](#)
- [T6c PDN command descriptions](#)
- [Threshold PDN command descriptions](#)
- [Virtual APN selection PDN command descriptions](#)

5.2.1.1 Generic PDN command descriptions

5.2.1.1.1 description

Syntax

`description description-string`

`no description`

Context

`config>mobile>pdn>apn`

Description

This command enables a text description stored in the configuration file for a configuration context.

The `description` command associates a text string with a configuration context to help identify the content in the configuration file.

The `no` form of this command removes the string from the configuration.

Default

no description

Parameters

description-string

the description character string; allowed values are any string up to 80 characters long composed of printable, 7-bit ASCII characters; if the string contains special characters (#, \$, spaces, and so on), the entire string must be enclosed within double quotes

5.2.1.1.2 shutdown

Syntax

[no] shutdown

Context

config>mobile>pdn

config>mobile>pdn>apn

config>mobile>pdn>gn

config>mobile>pdn>gp

config>mobile>pdn>threshold

config>aa>group[:partition]>wap1x

Description

This command administratively disables an entity (admin state down). When disabled, an entity does not change, reset, or remove any configuration settings or statistics. The operational state of the entity is disabled as well as the operational state of any entities contained within. Many entities must be shut down before they may be deleted.

Many entities must be explicitly enabled using the `no shutdown` command.



NOTICE:

The use of the `config>mobile>pdn>shutdown` command on the mobile gateway and in the Master Operation Role in ICR/geographical redundancy deployments will affect the service. When the command is issued a warning informs the administrator that the execution of the command will affect the service.

**NOTE:**

The shutdown command at the APN level only affects new sessions, meaning that operational state of APN related to existing sessions remains up while the administrative state is down. The gateway still forwards user traffic for the remaining sessions.

5.2.1.2 PDN command descriptions

5.2.1.2.1 pdn

Syntax

[no] pdn *gw-id*

Context

config>mobile

Description

This command configures a generic Packet Data Network (PDN) gateway instance. PDN accepts any type of session including SGW, PGW, GGSN and so on.

The no form of this command removes the gateway instance from the configuration.

Parameters

gw-id

specifies the gateway ID

Values: 1 to 8

5.2.1.2.2 5g-ims-fallback-timeout

Syntax

5g-ims-fallback-timeout *timeout*

no 5g-ims-fallback-timeout

Context

config>mobile>pdn

Description

This command configures the maximum time for the 5G IMS EPS fallback or RAT fallback for an IMS voice call, before the SMF/PGW aborts the setup of the IMS voice call.

Parameters

timeout

specifies the timeout, in seconds

Values: 1 to 20

Default: 5

5.2.1.2.3 aa-group

Syntax

```
aa-group [aa-group-id[:partition-id]] [def-app-profile profile-name] [parent-ctl-cc-name value]
```

```
no aa-group
```

Context

```
mobile>pdn
```

```
mobile>pdn>apn
```

Description

This command associates an aa-group with a PDN gw-id or with an individual APN.

When used at the PDN level, the aa-group names the aa-group that is associated with the *gw-id* and also all APNs that are supported for that gw-id. The partition-id is optional. If included the partition-id should be the default used for all APNs. It does not restrict the APNs from using their own partition-ids.

When used at the APN level, the aa-group-id identifies the aa-group and partition that should be used for this APN. The APN aa-group-id must match the aa-group-id specified at the PDN level. (Different mapping between APN and aa-group-ids may be possible in future releases.) The partition-id defines the partition to be used for this APN. The partition-id does not have to match the partition-id defined at the PDN level.



NOTE:

This command requires a PDN or an APN shutdown respectively.

Parameters

aa-group-id

specifies an integer that identifies an AA group

Values: 1 to 255

partition

specifies an integer that identifies a particular partition on this AA group

Values: 1 to 65535

profile-name

specifies an application profile name, up to 32 characters

value

specifies the optional parental control ASO charging characteristic name, when the AA group and the default application profile are configured, up to 32 characters

5.2.1.2.4 allow-duplicate-imei

Syntax

```
[no] allow-duplicate-imei
```

Context

```
config>mobile>pdn
```

Description

This command controls whether the gateway allows more than 1 UE to use a particular value of IMEI (SV). By default, the gateway will only allow 1 UE to use an IMEI. If this command is present the gateway will allow multiple UEs to attach with the same IMEI.



NOTE:

- If the `config>mobile>pdn>allow-duplicate-imei` command is enabled, the `show>mobile>pdn>pdn-context>imei` command does not show the existing sessions of the specified IMEI.
- This command requires a PDN shutdown.

Default

```
no allow-duplicate-imei
```

5.2.1.2.5 allow-duplicate-msisdn

Syntax

```
[no] allow-duplicate-msisdn
```

Context

```
config>mobile>pdn
```

Description

This command controls whether the gateway allows more than 1 UE to use a particular value of MSISDN (SV). By default, the gateway will only allow one UE to use an MSISDN. If this command is present the gateway will allow multiple UEs to attach with the same MSISDN.



NOTE:

- If the `config>mobile>pdn>allow-duplicate-imei` command is enabled, the `show>mobile>pdn>pdn-context>imei` command does not show the existing sessions of the specified MSISDN.
- This command requires a PDN shutdown.

Default

```
no allow-duplicate-msisdn
```

5.2.1.2.6 allow-multi-ue-session

Syntax

```
[no] allow-multi-ue-session
```

Context

```
config>mobile>pdn
```

Description

This command enables multiple UE session bonding support on the PDN gateway. To enable multiple UE session support, the `config>mobile>pdn>allow-duplicate-msisdn` command must be enabled and APN-level configuration is required to enable multiple UE bonding for a specific APN.

The no form of this command disables multiple UE session bonding.

**NOTE:**

- The `allow-multi-ue-session` command requires RADIUS support with IP subnets and UE IP address allocation received by the Framed-Route/Frame-IPv6-Route and the Framed-IP-Address/Framed-IPv6-Prefix attributes.
- This command requires a PDN shutdown.

Default

no allow-multi-ue-session

5.2.1.2.7 allow-multiple-apn-pdn

Syntax

[no] allow-multiple-apn-pdn

Context

config>mobile>pdn

Description

This command enables the SGW/PGW/GGSN to create multiple PDN sessions for the same (IMSI, APN). The handling for the PGW/GGSN and SGW is described in section *Enable Multiple PDNs per APN* in the *CMG Configuration Guide*; for more information, refer to tables *PGW/GGSN (S5/S8/Gn/Gp) Behavior* and *SGW (S11) Behavior*.

Default

no allow-multiple-apn-pdn

5.2.1.2.8 apn

Syntax

[no] apn {*apn-name* | unknown}

Context

config>mobile>pdn

Description

This command configures an Access Point Name (APN). Each APN represents a service provided by an upstream service provider. APNs can be classified as consumer APNs such as Internet, IMS, and Walled Garden) and corporate APNs (using L2TP, IPsec, GRE-IP VPNs).

The APN string may consist of NI + OI (upstream network provider identification + operator MCC and MNC identification) or just the NI part. The configured APN must match exactly to what is received as the real APN in the incoming session creation messages.

APN configuration in the CMG may use both APN-NI and full APN to as the identifier. When the PDP context is activated, CMG will use following steps to find the matching APN from the configuration:

1. If there is configured APN identified by APN-NI and it matches with the APN-NI in the Create PDP Context request, the matching configured APN is used to activate the PDP context.
2. If there is only one configured APN identified by the full APN where the APN-NI part matches with the APN-NI in the Create PDP Context request, the matching configured APN is used to activate the PDP context.
3. Two full APNs are constructed using the initial five and six digits of the authenticated IMSI and the APN-NI in the Create PDP Context request. If both of these constructed APN values matches with configured APNs the PDP context activation fails. If only one of the constructed APN values matches with configured APNs, the matching configured APN is used to activate the PDP context.
4. If no matching configured APN has not been found in the previous steps, but there is configuration for the unknown APN, the PDP context activation is done with this configuration. If there is no configuration for unknown APN, the PDP context activation fails.

To facilitate configuration and service provisioning, the real APNs can be further mapped to virtual APNs using various methods.

The no form of this command removes the APN from the configuration.

If the gateway is configured as user plane function using the `mobile>config>pdn>instance-type` command, this command specifies the network realm name associated for the APN/virtual router instance on the user plane function where the UE IP pools are located. This realm name is communicated in the Network-Instance IE part of the Sx/N4 interface session establishment/modification process together with the allocated UE IP address from the control plane function.



NOTE:

New APNs can be added without PDN shutdown. To delete an APN from the configuration, follow this procedure:

- shutdown the APN
- clear all sessions from the APN with `clear-mobile-gateway` command
- delete the APN

Parameters

apn-name

specifies an APN name, up to 80 characters



NOTE:

For UP function, the *apn-name* may be defined as APN-NI and APN-OI. The APN-NI must not exceed 63 characters.

unknown

specifies a reserved APN name that serves unknown APNs

5.2.1.2.9 auto-set-handover-indication

Syntax

[no] auto-set-handover-indication

Context

config>mobile>pdn

Description

This command enables the ePDG to automatically set the hand over indication bit, to better accommodate hand overs from LTE to WiFi. This is especially useful if a UE is performing the hand over without including its LTE IP address in the IKE request. Normally the ePDG checks the LTE IP address in the IKE request to detect hand overs. If the automatic hand over indication is enabled, the ePDG sets the hand over indication in Create Session request messages on the S2b. If the PGW rejects a request with a "Context Not Found" failure code, the ePDG will re-send it with the hand over indication set to bit_cleared_.



NOTE:

This command requires a PDN shutdown.

Default

no auto-set-handover-indication

5.2.1.2.10 call-procedure-retry

Syntax

```
call-procedure-retry retry-time seconds retry-attempts value [retry-delay retry-delay-msec]  
  
no call-procedure-retry
```

Context

```
config>mobile>pdn
```

Description

This command allows enabling and setting the retry time and the retry attempts for a local retry, which will be used instead of a trigger from a peering node (for example, the PCRF) on a call flow failure. The call procedure retry is supported for PGW/SMF-initiated procedures while the handover or TAU is in progress. The retries for delete bearer procedures for dedicated bearers are also supported.

The retry delay specifies for how long the PGW/SMF must wait after the TAU/HO, if the PGW/SMF retries the procedure immediately after the TAU/HO.

Default

```
no call-procedure-retry
```

Parameters

seconds

specifies the retry time, in seconds

Values: 1 to 20

Default: 1

value

specifies the number of retry attempts

Values: 0 to 5

Default: 0 (disabled)

retry-delay-msec

specifies the retry delay, in milliseconds

Values: 0 to 1000

Default: 0

5.2.1.2.11 cam-optimization

Syntax

```
cam-optimization {enable | disable}  
no cam-optimization
```

Context

```
config>mobile>pdn
```

Description

This command enables or disables CAM optimization. When enabled, a rule is split in two parts. The first part consists of the IP address and protocol and the second part consists of the source and destination ports that are programmed to a port list. Dynamic PCC rules with destination port but no protocol ID in flow description are rejected if the cam-optimization feature is disabled in SMF. Nokia recommends enabling the feature on both SMF and UPF.

The no form of this command reverts to the default (disable).

Default

```
no cam-optimization
```

Parameters

enable

enables the CAM optimization

disable

disables the CAM optimization

Default: disable

5.2.1.2.12 dl-buffering

Syntax

```
dl-buffering
```

Context

```
config>mobile>pdn
```

Description

This command enables the context to configure DL buffering per UE.

5.2.1.2.13 drop-oldest-pkt

Syntax

[no] drop-oldest-pkt

Context

config>mobile>pdn>dl-buffering

Description

This command enables dropping the oldest packet when the maximum number of DL buffering packets is reached.

The no form of this command reverts to the default behavior; that is, the incoming packet is dropped when the maximum number of DL buffering packets is reached.

Default

no drop-oldest-pkt

5.2.1.2.14 max-pkt

Syntax

max-pkt *limit*

no max-pkt

Context

config>mobile>pdn>dl-buffering

Description

This command configures the maximum number of packets for DL buffering per UE and is applicable to all paging scenarios. Until the DDN Acknowledge message with the DL Buffering Duration IE is received, the DL packets are buffered for a default duration of 30 seconds and up to the configured value of the max-pkt command.

The no form of this command reverts to the default.

**NOTE:**

This command applies only to CMG.

Parameters

limit

specifies the maximum number of packets for DL buffering

Values: 12 | 32 | 64 | 128

Default: 12

5.2.1.2.15 max-pkt-size

Syntax

max-pkt [*bytes*]

no max-pkt

Context

config>mobile>pdn>dl-buffering

Description

This command configures the maximum allowed size of the downlink buffered packet per UE. In addition to the UE payload, the outer headers and metadata of up to 78 bytes are taken into account while calculating the max-pkt-size.

The no form of this command reverts to the default value (2048 bytes).

**NOTE:**

- This command applies only to CMG.
- For a UE in idle state, a DL packet larger than the max-pkt-size and additional per-packet metadata will be dropped by CMG, but will still trigger the paging call flow.

Parameters

bytes

specifies the maximum allowed size, in bytes

Default: 2048

5.2.1.2.16 total-buffer-capacity

Syntax

```
total-buffer-capacity value  
no total-buffer-capacity
```

Context

```
config>mobile>pdn>dl-buffering
```

Description

This command configures the maximum allowed downlink buffering capacity per MG-VM.

The no form of this command reverts to the default value (128 Mbytes).



NOTE:

- This command applies only to CMG.
- This command configures system-level memory limits and on-the-fly changes are not allowed. Nokia recommends configuring this command during the initial CMG configuration. Any subsequent changes require a PDN shutdown.
- When the total-buffer-capacity limit is reached, any additional DL packets to be buffered will get dropped and this will not trigger the paging call flow.

Parameters

value

specifies the maximum allowed downlink buffering capacity, in Mbytes

Values: 4 to 65536 (in increments of 4)

Default: 128

5.2.1.2.17 total-buffer-pkts

Syntax

```
total-buffer-pkts value  
no total-buffer-pkts
```

Context

config>mobile>pdn>dl-buffering

Description

This command configures the maximum allowed downlink buffered packets per MG-VM.

The no form of this command reverts to the default value (48000 packets).



NOTE:

- This command applies only to CMG.
- When the total-buffer-pkts limit is reached, any additional DL packets to be buffered will get dropped, but will still trigger the paging call flow.

Parameters

value

specifies the maximum number of allowed downlink buffered packets

Values: 4000 to 2500000000

Default: 48000

5.2.1.2.18 create-procedure-retry

Syntax

create-procedure-retry retry-time *seconds* retry-attempts *attempts* [retry-delay
retry-attempt-msec]

no create-procedure-retry

Context

config>mobile>pdn

Description

This command configures the retry time and the retry attempts for the Create Bearer Request.

The no form of this command reverts to the default, that is apply the call-procedure-retry command to set the retry time and retry attempts.

Default

no create-procedure-retry

Parameters

seconds

specifies the retry time, in seconds

Values: 1 to 20

Default: 1

attempts

specifies the number of retry attempts

Values: 0 to 5

Default: 0 (disabled)

retry-attempt-msec

specifies the retry delay, in milliseconds

Values: 0 to 1000

Default: 0

5.2.1.2.19 default-traffic-hash

Syntax

default-traffic-hash *profile-name*

no default-traffic-hash

Context

config>mobile>pdn

Description

This command is used to specify a default traffic hash profile. The specified profile is applied to an SDF for a bonded session, when it is not explicitly instantiated with a traffic hash profile within a policy rule unit. This applies to any predefined (static) rules including a “any/any” (match-all) rule. The default traffic hash profile needs to be specified prior to bonded-support being enabled for an APN.

The no form of this command reverts to the default.



NOTE:

This command requires a PDN shutdown.

Default

no default-traffic-hash

Parameters

profile-name

specifies a profile name, up to 32 characters

5.2.1.2.20 del-bearer-reactivation-req-cause-pgw-failure

Syntax

[no] del-bearer-reactivation-req-cause-pgw-failure

Context

config>mobile>pdn

Description

This command enables the SGW to include a re-activation request in the Delete Bearer Request message if the PGW fails. The no form of this command reverts to the default where no re-activation request is included in the Delete Bearer Request message.

Default

no del-bearer-reactivation-req-cause-pgw-failure

5.2.1.2.21 delay-nw-procedures

Syntax

delay-nw-procedures *timer-value-msec*

no delay-nw-procedures

Context

config>mobile>pdn

Description

This command specifies the delay-time before the pending network-initiated modification procedure starts after the TAU or handover to 5G. The delay reduces the risk of the request from the SMF/PGW-C being rejected from the AMF because the TAU or handover has not yet completed in the serving node.

The no form of this command reverts to the default.



NOTE:

In some cases, the network-initiated modification procedure must be started right after the TAU or handover between 4G and 5G, or the 5G TAU or handover. This may occur, for example, due to the 5G QoS restoration, or when the temporarily canceled modification is pending for retry.

Default

no delay-nw-procedures

Parameters

timer-value-msec

specifies the delay-time in milliseconds to apply before the network-initiated modification procedure starts after the handover or TAU

Values: 0 to 1000

Default: 0

5.2.1.2.22 disable-hss-init-update-failure-delete

Syntax

```
[no] disable-hss-init-update-failure-delete
```

Context

```
config>mobile>pdn
```

Description

This command disables the 3GPP specified failure handling for the HSS-initiated QoS update. The PDN connection remains active if the MME rejects the Update Bearer Request message. The no form of this command reverts to the default.

Default

no disable-hss-init-update-failure-delete

5.2.1.2.23 dns-client

Syntax

```
dns-client
```

Context

```
config>mobile>pdn
```

Description

This command enables the context to configure the DNS client at the PDN level.

5.2.1.2.24 cache-clear-no-such-name

Syntax

```
[no] cache-clear-no-such-name
```

Context

```
config>mobile>pdn>dns-client
```

Description

This command forces the ePDG to clear a DNS cached entry, if the DNS server replies with the Reply Code: 3 "No Such Name" ON NAPTR, SRV or A/AAAA RESOLUTION. In this case, the ePDG removes the entry (when it pre-exists) and does not store the entry (if the query is for a non pre-existent entry).

The no form of this command reverts to the default DNS response handling.

Default

```
no cache-clear-no-such-name
```

5.2.1.2.25 dns-interface

Syntax

```
dns-interface interface-name [router router-instance]
```

```
no dns-interface
```

Context

config>mobile>pdn>dns-client

Description

This command configures the source interface to use with the DNS profile for DNS lookup on the mobile gateway. If the interface is not configured under the DNS client, the DNS configuration in the Boot Option File (BOF) is used.

The no form of this command reverts to the default DNS configuration in the BOF.



NOTE:

This command requires a PDN shutdown.

Default

no dns-interface

Parameters

interface-name

specifies an interface name, up to 32 characters and must start with a letter

router-instance

specifies the virtual router name or service ID

Values	router-name vprn-svc-id	
–	router-name	Base
–	vprn-svc-id	1 to 2147483647

Default: Base

5.2.1.2.26 dns-profile

Syntax

dns-profile *profile-name*

```
no dns-profile
```

Context

```
config>mobile>pdn>dns-client
```

Description

This command configures a DNS profile used for all DNS lookups on the control plane for the mobile gateway-specific reference point interfaces.

DNS lookups for reference point interfaces that are not specific to gateway functions continue to be done using the default DNS configuration in the Boot Option File (BOF).

The no form of this command reverts to the default DNS lookup configuration in the BOF.



NOTE:

This command requires a PDN shutdown.

Default

```
no dns-profile
```

Parameters

profile-name

specifies the name of the DNS profile, up to 32 characters

5.2.1.2.27 enable-pcscf-low-ttl

Syntax

```
[no] enable-pcscf-low-ttl
```

Context

```
config>mobile>pdn>dns-client
```

Description

This command specifies the refresh interval time (minimum two seconds) when the received DNS TTL is used.

Default

disabled

5.2.1.2.28 refresh-interval

Syntax

```
refresh-interval {pcscf | pgw | apn | all} refresh-interval-sec interval-in-seconds [use-dns-ttl]
```

```
no refresh-interval {pcscf | pgw | apn | all}
```

Context

```
config>mobile>pdn>dns-client
```

Description

This command specifies the DNS refresh interval at the PDN level. Filters support the ability to set a different refresh interval for DNS-based selections (P-CSCF, PGW, and APN), or to set the same refresh interval for all of them together.

An option can also be specified to refresh the DNS repository using the TTL provided in the DNS response for a given P-CSCF/PGW/APN server FQDN. The following restrictions apply:

- TTL based DNS refresh is supported for P-CSCF/PGW/APN FQDNs
- The shortest refresh time the PGW will support for P-CSCF FQDNs, is 30 seconds. The shortest refresh time that the ePDG supports for PGW/APN selection, is 300 seconds. For the PGW/APN if the received TTL<300 seconds, the system defaults to 300 seconds.
- A maximum of 100 P-CSCF FQDNs will be refreshed at 30 seconds. If the number of P-CSCF FQDNs is above 100, the PGW will use the configured refresh time (the minimum is 60 minutes).

The no form of this command reverts to the default.

Default

no refresh-interval

Parameters

pcscf

specifies the refresh interval for the dynamically learned P-CSCF server FQDN from AAA server over s6b interface

pgw

specifies the refresh level for the PGW

apn

specifies the refresh level for the APN

all

specifies the refresh level for all types of DNS-based selection. The minimum allowed configured value is 300 seconds.

interval-in-seconds

specifies the refresh interval time in seconds

Values: 300 to 86400

Default: 21600

use-dns-ttl

specifies the received DNS TTL

For the P-CSCF, if the received TTL<30 seconds, the system defaults to 30 seconds. For the PGW/APN if the received TTL<300 seconds, the system defaults to 300 seconds

5.2.1.2.29 topon-match-first

Syntax

[no] topon-match-first

Context

config>mobile>pdn>dns-client

Description

This command forces the ePDG to first run the topological matching algorithm and then, perform the A/AAAA resolution.

The no form of this command reverts to the default ePDG topological match logic.

Default

no topon-match-first

5.2.1.2.30 ttl-zero

Syntax

[no] ttl-zero

Context

```
config>mobile>pdn>dns-client
```

Description

This command allows the gateway not to cache the DNS provided answers, if the DNS provided with value TTL=0 (zero).

The no form of this command reverts to the default handling of the value of the DNS provided TTL=0, that rounds up to 300 seconds.



NOTICE:

Nokia recommends using the `ttl-zero` command only for lab testing activities, as the activation in live deployments leads to DNS signaling overload.

Default

```
no ttl-zero
```

5.2.1.2.31 dnai-map

Syntax

```
dnai-map entry-id {[plmn-list plmn-list-name] [mnvo] [incoming-apn-list apn-list-name] }  
mapped-apn mapped-apn
```

```
no dnai-map entry-id
```

Context

```
config>mobile>pdn
```

Description

This command configures an internal APN that is selected based on any one or more of the PLMN list, the incoming APN list, and mvno (for MVNO subscribers). The PLMN list contains one or more IMSI PLMN IDs (MCC+MNC) applicable to the operator.

The no form of this command deletes the DNAI mapping.



NOTE:

- The DNAI map that is configured with the lowest *entry-id* has precedence over multiple matching entries. For example, dnai-map 1 takes precedence over dnai-map 2. This allows the operator to choose the matching order by defining them in the preferred order of application.
- The PLMN list and APN list are defined in the `config>mobile>profile>list` context prior to applying this configuration.
- You can modify the PLMN list without shutting down the gateway.
- The idle timeout and session timeout are defined in the APN context.

Parameters

entry-id

specifies the entry ID for the DNAI map

Values: 1 to 100

plmn-list-name

specifies the PLMN list name for an operator, up to 32 characters, and is comprised of one or more MCC+MNC values applicable to the operator

mvno

keyword that specifies to map the incoming session to the mapped APN for MVNO subscribers

apn-list-name

specifies an APN list name, up to 32 characters, for matching the requested/received APN

mapped-apn

specifies the mapped-APN name, up to 80 characters, for LBO or MVNO sessions

5.2.1.2.32 dns-error-mapping-profile

Syntax

```
dns-error-mapping-profile profile-name
```

```
no dns-error-mapping-profile
```

Context

```
config>mobile>pdn
```

Description

This command allows the configuration of a specific DNS error mapping profile (already configured using the `config>mobile>profile>dns-error-mapping-profile` command) on the PDN.

The `no` form of this command removes the profile from the PDN and deactivates the functionality.

Default

no dns-error-mapping-profile

Parameters

profile-name

specifies the name of the error mapping profile, up to 32 characters

5.2.1.2.33 dual-connectivity

Syntax

[no] dual-connectivity

Context

config>mobile>pdn

Description

This command enables dual connectivity. When dual connectivity is enabled, you can use two different eNodeB peers in the Modify Bearer Request or Modify Access Bearers Request message.

Default

no dual-connectivity

5.2.1.2.34 dual-connectivity-5g

Syntax

[no] dual-connectivity-5g

Context

config>mobile>pdn

Description

This command enables dual connectivity in 5G. When dual connectivity is enabled, you can use two different gNB peers in the same PDU session.

Default

no dual-connectivity-5g

5.2.1.2.35 enable-sgw-c-partitioning

Syntax

[no] enable-sgw-c-partitioning

Context

config>mobile>pdn

Description

This command enables SGW-C partitioning at the PDN level to support UE relocation to an SGW-U that is associated with the same combined SGW-C/PGW-C. This applies for CMG CUPS deployments.

The CMG supports SGW-C partitioning by assigning one of multiple S11-C addresses per SGW partition that is treated as a legacy SGW by the MME. This supports the scenario, for example, where the combined SGW-C/PGW-C is centrally deployed to cover multiple service areas, while each SGW-U is locally deployed to cover a specific service area with an eNodeB connected to the SGW-U.

The SGW-C partition (S11 interface) covers only one service area, so the partition changes whenever an SGW-U relocation takes place. When a UE moves across the service area, a TAU or HO with the SGW relocation procedure is triggered, and the MME uses TAC-based DNS query to select a new SGW-C partition (S11-C address). After receiving a Create Session Request message, the SGW-C selects the new SGW-U based on the TAC list.

The no form of this command reverts to the default (no SGW-C partitioning).

Default

no enable-sgw-c-partitioning

5.2.1.2.36 epc-node

Syntax

epc-node *name*

no epc-node

Context

config>mobile>pdn

Description

This command configures the EPC node name.

The no form of this command removes the name from the configuration.

**NOTE:**

This command requires a PDN shutdown.

Default

no epc-node

Parameters

name

specifies the EPC node name, up to 31 characters

Syntax: <MCC>.<MNC>.<SGW | PGW | ePDG | TWAG>.<Region string>.<Group Id>.<Node Id>

5.2.1.2.37 epdg-5gs-compatibility

Syntax

[no] epdg-5gs-compatibility [rel-17 {enable | disable}]

Context

config>mobile>pdn

Description

This command configures the ePDG compatibility interworking with 5GS. When the epdg-5gs-compatibility command is specified, the ePDG functions according to the 3GPP pre-Rel-17 specification (processes and sends only the 5GSIWKI GTPv2 flag). When the rel-17 enable is configured, the ePDG functions according to the 3GPP Rel-17 specification (processes and sends the 5GSIWKI, 5GCNRI, and 5GCNRS GTPv2 flags).

**NOTE:**

Only pre-Rel-17 or Rel-17 configuration can be active at any time.

The no form of this command disables the ePDG compatibility interworking with 5GS.

Default

no epdg-5gs-compatibility

Parameters

rel-17 enable

specifies to enable the Rel-17 3GPP support for 5GS interworking

rel-17 disable

specifies to disable the Rel-17 3GPP support for 5GS interworking

5.2.1.2.38 exclude-error-indication-extension-header

Syntax

[no] exclude-error-indication-extension-header

Context

config>mobile>pdn

Description

This command includes or excludes the GTP extension headers from the GTPv1 Error Indication message.

Default

no exclude-error-indication-extension-header

5.2.1.2.39 fwd-gtpu-echo-req-to-mg

Syntax

[no] fwd-gtpu-echo-req-to-mg

Context

config>mobile>pdn

Description

This command forwards the GTP-U Echo Requests for distribution across the MG groups to improve the GTP-U Echo Response performance in LB-less CMG UP function deployments. The GTP-U Echo requests are forwarded and handled by the OAM or CPM.

The no form of this command does not distribute the GTP-U Echo Requests across the MG groups in LB-less CMG UP function deployments.

Default

no fwd-gtpu-echo-req-to-mg

5.2.1.2.40 gtpc-init-msg-src-ip-validation-bypass

Syntax

[no] gtpc-init-msg-src-ip-validation-bypass

Context

config>mobile>pdn

Description

This command enables the CMG to decouple GTP path validation process from the source IP address used in the incoming message. This allows the CMG receiving peer to successfully process an incoming message, whose source IP address is not used or known before, but holds a valid FTEID ID. The no form of this command reverts to the default.



NOTE:

The described functionality only applies to S2b interface and is about the PGW or PGW-C/SMF to ePDG GTPv2 messages.

Default

no gtpc-init-msg-src-ip-validation-bypass

5.2.1.2.41 gtpu-drop-echo-threshold

Syntax

gtpu-drop-echo-threshold queue-depth-threshold *outstanding-echo-requests* queue-threshold-sustain-time *sustain-interval* backoff-time *backoff-time*

no gtpu-drop-echo-threshold

Context

config>mobile>pdn

Description

This command configures CMG to drop the GTP-U Echo Request messages based on the thresholds of the system queue and sustain time.

The system goes into drop state when the system queue reaches the configured depth threshold, and the requests have persisted for the configured sustain time interval. An alarm and log event are generated. The system comes out of the drop state when the configured backoff time interval expires. An alarm and log event are generated.

The no form of this command does not drop the GTP-U Echo Requests.

Default

no gtpu-drop-echo-threshold

Parameters

outstanding-echo-requests

specifies the depth of the queue based on the number of the outstanding Echo Requests; when this limit is reached along with the sustain time interval, all the following GTP-U Echo Requests are dropped

Values: 2000 to 6000

Default: 0

sustain-interval

specifies the sustain time interval limit, in seconds, during which the number of outstanding GTP-U Echo Requests remain more than the depth threshold (configured with the queue-depth-threshold command) for the system to go into drop state, and all following GTP-U Echo Requests are dropped

Values: 0 to 120

Default: 0

backoff-time

specifies the backoff timer value, in seconds; this is the time interval after which the system comes out of the drop state and continues to process the GTP-U Echo Requests

Values: 0 to 600

Default: 0

5.2.1.2.42 gtpu-path-failure

Syntax

```
gtpu-path-failure ref-point ref-point action
no gtpu-path-failure ref-point ref-point
```

Context

```
config>mobile>pdn
```

Description

This command enables path management actions per GTP-U reference point.

The no form of this command disables the path management action for the specific reference point.

Default

```
no gtpu-path-failure
```

Parameters

- ref-point***
specifies the reference point type
Values: s1u | s4u | s5u | s8u | s12 | gnu | gpu
- action***
specifies the action taken upon path failure

Values	release-on-path-failure	specifies to release the S1u or S12 interface tunnels, that is, the UE goes into an IDLE state; after the release of the tunnels, the next downlink packet triggers paging
–	delete-on-path-failure	specifies to delete sessions or PDP contexts; available for the S4, S5, S8, Gn, and Gp interfaces

5.2.1.2.43 gx-before-aaa

Syntax

```
[no] gx-before-aaa
```

Context

```
config>mobile>pdn
```

Description

This command provides an alternative call flow where the initial Gx transaction happens before the AAA/RADIUS. This also means that the initial transaction does not include the IP address allocated for the UE, if the external dynamic allocation methods (RADIUS, LNS, and DHCP) are used. In these cases, the PCRF is notified about the allocated IP address via a separate transaction.

The no form of this command reverts to the default.

**NOTE:**

This command requires a PDN shutdown.

Default

```
no gx-before-aaa
```

5.2.1.2.44 ho-apn-lookup

Syntax

```
ho-apn-lookup {imsi-based-oi | ni-only}
```

```
no ho-apn-lookup
```

Context

```
config>mobile>pdn
```

Description

This command modifies the default design of the APN look up for HI=1 calls. The command allows the PGW/PGW-C to use the MCC+MNC information, extracted from the network-provided IMSI, in order to retrieve the OI and construct the full APN-NI+OI or make use of only the APN NI.

The no form of this command disables the construction of APN-NI+OI based on IMSI PLMN or the APN NI only approach.

Default

no ho-apn-lookup

Parameters

imsi-based-oi

specifies that the PGW/PGW-C replaces the incoming APN OI with the PLMN from IMSI to lookup for the already established call

ni-only

specifies the PGW/PGW-C uses only the incoming APN NI (ignore the OI) to lookup for the already established call

5.2.1.2.45 home-plmn-list

Syntax

home-plmn-list *list-name*

no home-plmn-list

Context

config>mobile>pdn

Description

This command configures the home Public Land Mobile Network (PLMN) list. This is used in comparing the newly acquired public land mobile network information with the roaming public mobile network table where roaming time is reduced by selectively registering a PLMN location only in a base station with a roaming function.



NOTE:

- This command requires a PDN shutdown.
- When you configure the nnrf-nfm client under the config>mobile>pdn>sba-service-realm context, this command accepts a PLMN list with a maximum of 32 entries.

Parameters

list-name

specifies the PLMN list name, up to 32 characters

5.2.1.2.46 i-upf-selection

Syntax

```
[no] i-upf-selection
```

Context

```
config>mobile>pdn
```

Description

This command enables the I-UPF selection as follows:

- the I-UPF is inserted when the anchor UPF cannot serve the TAI
- the I-UPF is removed when it cannot serve the current TAI and the anchor UPF can serve the TAI
- the I-UPF is changed when the anchor UPF and the current I-UPF cannot serve the TAI

The no command disables the I-UPF selection.

5.2.1.2.47 ims-emps-priority

Syntax

```
ims-emps-priority cp-dscp-marking [value] up-uplink qci-policy-name up-downlink qci-policy-name
```

```
ims-emps-priority
```

```
no ims-emps-priority
```

Context

```
config>mobile>pdn
```

Description

This command enables or disables the IMS eMPS priority. When enabled, the PCRF upgrades the PDN session and its bearers with eMPS priority or approves the PDN session with eMPS priority during the initial attach. This gives the PDN sessions/bearers a higher precedence against being throttled when the gateway is in an overload condition.

Optionally, this command also configures a specific DSCP transport marking value for eMPS sessions. When configured, any eMPS session packets on the transport are marked with the DSCP values specified for the CP DSCP marking on the S5-C and S11 reference points, and the DSCP value specified for the UP uplink and UP downlink on the S1-U, S5-U, and SGi reference points.

**NOTE:**

This command requires a PDN shutdown.

Default

ims-emps-priority

Parameters

value

specifies the IP DSCP/TOS value used for eMPS sessions

Values: 0 to 63

Default: 44

qci-policy-name

specifies the QCI policy name, configured in the config>mobile>profile>policy-options>qci-policy context, used for the uplink direction; up to 32 characters

up-downlink qci-policy-name

specifies the QCI policy name, configured in the config>mobile>profile>policy-options>qci-policy context, used for the downlink direction; up to 32 characters

5.2.1.2.48 imsi-plmn-list

Syntax

```
imsi-plmn-list list-name
```

```
no imsi-plmn-list
```

Context

```
config>mobile>pdn
```

Description

This command configures an IMSI PLMN list. The IMSI PLMN list defines the MNC length for certain operators not addressed by the ITU specifications.

**NOTE:**

- If a PLMN list is configured under the `home-plmn-list` command, it must not be configured under the `imsi-plmn-list` command (and the other way around) or any other application type, because changes on the fly are not supported for the `home-plmn-list` command. Therefore, if the same PLMN list is used for different application types then a change in the PLMN list will affect all.
- IMSI PLMN lists must be consistent across PDN instances; if not, list merging is applied.
- PLMN lists must be defined before the gateway goes live and should not be modified often. Modifying a PLMN list is allowed without shutting down the gateway. The affected sessions are terminated in the next Diameter transaction.

The no form of this command removes the list name from the configuration.

Parameters

list-name

specifies the PLMN list name, up to 32 characters

5.2.1.2.49 instance-type

Syntax

`instance-type instance-type-value`

`no instance-type`

Context

`config>mobile>pdn`

Description

This command configures the PDN instance as a 3GPP Control and User Plane Separation (CUPS) function.

The no form of this command will remove the instance from being configured as a CUPS instance.

**NOTICE:**

This command must not be executed on a live system. To change the instance type, use the shutdown command in the `config>mobile>pdn` context and then the `no shutdown` command under the same context to apply the change.



NOTE:

- This command must be configured prior to configuring any reference point interfaces.
- This command requires a PDN shutdown.

Default

no instance-type

Parameters

instance-type-value

specifies the function (control or user plane) of the gateway

Values	control	specifies the PDN instance as control plane function
–	user	specifies the PDN instance as user plane function

5.2.1.2.50 ipsmg-group

Syntax

ipsmg-group *ipsmg-group-id*
no ipsmg-group

Context

config>mobile>pdn

Description

This command associates an IPsec MG group with the ePDG.



NOTE:

This command requires a PDN shutdown.

Parameters

ipsmg-group-id

specifies the ID of the IPsec MG group to associate with the ePDG

Values: 1 to 8

5.2.1.2.51 mg-group-call-failure-rate

Syntax

```
mg-group-call-failure-rate call-failure-rate rate-percentage [monitor-time timer]
```

```
no mg-group-call-failure-rate
```

Context

```
config>mobile>pdn
```

Description

This command activates the monitoring of the call-failure-rate per MG group. If the call-failure-rate exceeds the configured value, and for time equal to monitor-time-before-action, the system proceeds with a switchover of the MG group. After the switchover, if the new active pod/node continues to report the same call-failure-rate figures, the MG group is set on a suspend status, and is not selected for new call requests.

After the MG group is instructed to switch over or suspend because of the call failure rate, a new alarm is generated that is cleared when the switchover or suspend operation is completed.

When the MG group is set on suspend state, a new critical alarm is generated. If all MG groups face call-failure-rate, all MG groups can be set under the suspend state. The new functionality does not support interworking with scale-in. The functionality is applicable to SMF and PGW-C type of gateways.

The no form of this command sets the call-failure-rate to value 0 (disabled).



NOTE:

- On a 1:1 MG group deployment scheme, if the standby card (node/pod) is not in hot standby status, the system performs a reboot on the failing card. After the reboot, if the issue persists, the system performs a card suspend according to the configured parameters of the rate percentage and the monitoring time.
- When an MG group is on suspend state, then this group can still be selected for new sessions of existing UEs that already have their sessions on the suspended group. This can lead to abnormal functionality for the new sessions. Therefore, Nokia recommends to use the `clear` commands to remove all sessions from the suspended group so that any new session is established over a non-suspended MG group.

Parameters

rate-percentage

specifies the call failure rate percentage

Values: 10 to 100

Default: 0

timer

specifies the time, in minutes, during which the system, when the failure rate is triggered, monitors continuously the call failure rate before generating the alarm

Values: 1 to 15

Default: 5

5.2.1.2.52 non3gpp-pdn-connection-rejection-with-ip

Syntax

```
non3gpp-pdn-connection-rejection-with-ip
```

Context

```
config>mobile>pdn
```

Description

This command enables the context to configure the ePDG to reject specific handover error case scenarios with SWu error code `pdn_connection_rejection` with an IP address included in the notification data field.

Default

disabled

5.2.1.2.53 handover-no-mip6-agent-info

Syntax

```
[no] handover-no-mip6-agent-info
```

Context

```
config>mobile>pdn>non3gpp-pdn-connection-rejection-with-ip
```

Description

This command rejects the handover when the ePDG does not receive the IP address with the MIP6 agent information or is missing the MIP6 agent information.

Default

no handover-no-mip6-agent-info

5.2.1.2.54 pgw-handover-response-cc64

Syntax

[no] pgw-handover - response-cc64

Context

config>mobile>pdn>non3gpp-pdn-connection-rejection-with-ip

Description

This command rejects the handover when the ePDG receives a cause code 64 on its S2b interface.



NOTE:

This command requires a PDN shutdown.

Default

no pgw-handover-response-cc64

5.2.1.2.55 nsmf-congestion-control

Syntax

[no] nsmf-congestion-control

Context

config>mobile>pdn

Description

This command enables the context to configure the Nsmf congestion control.

Default

no nsmf-congestion-control

5.2.1.2.56 enable-nsmf-congestion-control

Syntax

[no] enable-nsmf-congestion-control

Context

config>mobile>pdn>nsmf-congestion-control

Description

This command configures (when the max-activations per second threshold is reached) the SMF for any (non-emergency) subsequent Nsmf_PDUSession_SmContextCreateData or Nsmf_PDUSession_SmContextUpdateData to reject the request with HTTP 429 response including the error attribute/ProblemDetails set to NF_CONGESTION_RISK. If the HTTP header field Retry-After (in seconds) is also configured, this field will also be included in the HTTP 429 response.

The no form of this command disables the Nsmf NF congestion control toward the AMF.

Default

no enable-nsmf-congestion-control

5.2.1.2.57 max-activations

Syntax

max-activations *max-activations*

no max-activations

Context

config>mobile>pdn>nsmf-congestion-control

Description

This command configures the threshold of the maximum number of PDU connections or the maximum number of PDU activations per second for this DNN. The threshold is used for the Nsmf DNN NF congestion overload control. An alarm can be enabled or disabled to be sent when the maximum number of connections or the maximum number of activations is reached.

The no form of this command reverts to the default (disabled).

Default

no max-activations

Parameters

max-activations

specifies the maximum number of PDU activations per second

Values: 0 to 5000

Default: 0 (disabled)

5.2.1.2.58 retry-after-timer

Syntax

retry-after-timer *seconds*

no retry-after-timer

Context

config>mobile>pdn>nsmf-congestion-control

Description

This command configures the Retry-After field in the HTTP header of the 429 error respond when the max-activations threshold has been reached.

The no form of this command reverts to the default (disabled).

Default

no retry-after-timer

Parameters

seconds

specifies the retry timer in seconds

Values: 2 to 36000

Default: 0 (disabled)

5.2.1.2.59 nsmf-overload-control

Syntax

[no] nsmf-overload-control

Context

config>mobile>pdn

Description

This command enables the context to configure the Nsmf overload control.

Default

no nsmf-overload-control

5.2.1.2.60 enable-nsmf-overload-control

Syntax

[no] enable-nsmf-overload-control

Context

config>mobile>pdn>nsmf-overload-control

Description

This command configures the handling of node-level overload control toward the AMF.

This command configures (when the node-level overload threshold has been reached) an SNMP alarm to be triggered, and any (non-emergency) subsequent Nsmf_PDUSession_CreateSMContext with a PDU Session Establishment Create Request is rejected with an HTTP 503 response including the error attribute/ProblemDetails set to NF_CONGESTION. If the HTTP header field Retry-After (in seconds) is also configured, this field will also be included in the HTTP 503 response.

The no form of this command disables the handling of the Nsmf NF overload control toward the AMF.

Default

no enable-nsmf-overload-control

5.2.1.2.61 overload-threshold

Syntax

```
overload-threshold threshold-value  
no overload-threshold
```

Context

```
config>mobile>pdn>nsmf-overload-control
```

Description

This command configures the overload threshold value.

The no form of this command reverts to the default.

Default

no overload-threshold

Parameters

threshold-value

specifies the overload threshold value as a percent and when reached it triggers the Nsmf overload

Values: 50 to 97

Default: 92

5.2.1.2.62 retry-after-timer

Syntax

```
retry-after-timer seconds  
no retry-after-timer
```

Context

```
config>mobile>pdn>nsmf-overload-control
```

Description

This command configures the Retry-After field in the HTTP header of the 503 error respond when the node-level overload threshold has been reached.

The no form of this command reverts to the default (disabled).

Default

no retry-after-timer

Parameters

seconds

specifies the retry timer in seconds

Values: 2 to 36000

Default: 0 (disabled)

5.2.1.2.63 nswo-capable

Syntax

[no] nswo-capable

Context

config>mobile>pdn

Description

This command configures the system to run in Non-Seamless WLAN Offload (NSWO) mode.



NOTE:

This command requires a PDN shutdown.

Default

no nswo-capable

5.2.1.2.64 oi-override-mapping

Syntax

oi-override-mapping *list-name*

no oi-override-mapping

Context

```
config>mobile>pdn
```

Description

This command enables the ePDG to use the configured list of mapping pairs, between subscriber's PLMN ID (MCC/MNC) and PLMN ID replacement (MCC/MNC), to derive the Operator Identifier (OI) that will be used for constructing both the PGW DNS FQDN and the APN IE for the CSR toward the PGW. The list with the mapping pairs is configured in the `config>mobile>profile>list>oi-override-mapping` command. The `no` form of this command removes the override profile.

Parameters

list-name

specifies the list name, up to 32 characters

5.2.1.2.65 overload-control

Syntax

```
overload-control
```

Context

```
config>mobile>pdn
```

Description

This command enables the context to configure overload parameters used in GTP-C overload conditions.

5.2.1.2.66 cpu-sampling-interval

Syntax

```
cpu-sampling-interval [overload overload-interval] [overload-state-change ovrld-state-change-intvl] [no-overload no-overload-interval]  
no cpu-sampling-interval
```

Context

```
config>mobile>pdn>overload-control
```

Description

This command configures the CPU sampling interval window for the gateway node when the node state changes from:

- no overload to overload
- overload to different overload conditions (minor, major, and critical state)
- overload to no overload

The no form of this command returns the CPU sampling interval values to the default.

Parameters

overload-interval

specifies the CPU sampling period, in seconds, when transitioning from no overload to overload state

Values: 2 to 20

Default: 10

ovrld-state-change-intrvl

specifies the CPU sampling period, in seconds, when transitioning from overload to another overload state

Values: 2 to 20

Default: 2

no-overload-interval

specifies the CPU sampling period, in seconds, when transitioning from overload to no overload state

Values: 2 to 20

Default: 10

5.2.1.2.67 thresholds

Syntax

thresholds

Context

config>mobile>pdn>overload-control

Description

This command enables the context to configure the minor, major, and critical GTP-C overload thresholds.

**NOTE:**

The thresholds configured under this command are applicable for both the GTP-C and PFCP overload behaviors.

5.2.1.2.68 critical

Syntax

`critical threshold-value`

`no critical`

Context

`config>mobile>pdn>overload-control>thresholds`

Description

This command configures the threshold for the critical GTP-C overload level. For example, when the critical *threshold-value* is configured with value 97, the gateway reaches the critical GTP-C overload level when the gateway performance is between the configured major *threshold-value* and 97.

The no form of this command reverts to the default.

Default

no critical

Parameters

threshold-value

specifies the threshold value for the critical GTP-C overload level

Values: 50 to 97

Default: 97

5.2.1.2.69 major

Syntax

`major threshold-value`

`no major`

Context

config>mobile>pdn>overload-control>thresholds

Description

This command configures the threshold for the major GTP-C overload level. For example, when the major *threshold-value* is configured with value 92, the gateway reaches the major GTP-C overload level when the gateway performance is between the configured minor *threshold-value* and 92.

The no form of this command reverts to the default.

Default

no major

Parameters

threshold-value

specifies the threshold value for the major GTP-C overload level

Values: 50 to 97

Default: 92

5.2.1.2.70 minor

Syntax

minor *threshold-value*

no minor

Context

config>mobile>pdn>overload-control>thresholds

Description

This command configures the threshold for the minor GTP-C overload level. For example, when the minor *threshold-value* is configured with value 89, the gateway reaches the minor GTP-C overload level when the gateway performance is between 89 and the configured major *threshold-value*.

The no form of this command reverts to the default.

Default

no minor

Parameters

threshold-value

specifies the threshold for the minor GTP-C overload level

Values: 50 to 97

Default: 89

5.2.1.2.71 override-profile

Syntax

[no] override-profile *override-profile-id*

Context

config>mobile>pdn

Description

This command configures a non-default, group-specific reference point interface override profile for GTP-U redirection.

The no form of this command removes the override profile.



NOTE:

This command applies only to CMG.

Default

no override-profile

Parameters

override-profile-id

specifies the profile ID to configure MG group-specific reference point interfaces

Values: 1 to 32

5.2.1.2.72 override

Syntax

[no] override ref-point-type *ref-point-type* *ref-point-name* to override with *ref-point-name* to override with

override ref-point-type *ref-point-type* router *router-instance* to override interface-name to override with *router-instance* to override with interface-name to override with

no override ref-point-type *ref-point-type* router *router-instance* to override interface-name to override

Context

config>mobile>pdn>override-profile

Description

This command configures a non-default, group-specific reference point interface for each reference point interface type in an override profile.

The no form of this command removes the reference point specific interface from the override profile.



NOTE:

This command applies only to CMG.

Default

no override ref-point-type *ref-point-type* *ref-point-name* to override with *ref-point-name* to override with

no override ref-point-type *ref-point-type* router *router-instance* to override interface-name to override

Parameters

ref-point-type

specifies the reference point interface type for the override profile

Values: s11 | s12 | s1u | s4 | s5 | s8 | gn | gp | s2a-gtp | s2b | n3 | n9 | radius

ref-point-name to override

specifies the reference point that is overridden, up to 32 characters

ref-point-name-to override with

specifies the reference point used to override with, up to 32 characters

router-instance to override

specifies the router instance that is overridden

Values	router-name service-id	
–	router-name	Base
–	service-id	1 to 2147483647

Default: Base

router-instance to override with
specifies the router instance used to override with

Values	router-name service-id	
–	router-name	Base
–	service-id	1 to 2147483647

Default: Base

interface-name to override
specifies the interface name to override, up to 32 characters (must start with a letter)

interface-name to override with
specifies the interface name used to override with, up to 32 characters (must start with a letter)

5.2.1.2.73 pcf-pcrf-based-pcscf-allocation

Syntax

[no] pcf-pcrf-based-pcscf-allocation

Context

config>mobile>pdn

Description

This command enables the context to configure specific P-CSCF addresses to be sent to the UE, based on the PCRF/PCF peer that the PGW-C/SMF has communicated with.

The no form of this command deactivates the functionality and removes all entries.

5.2.1.2.74 entry

Syntax

```
entry entry-id apn apn-name {diameter-peer profile-name | nf-id uuid} {{pcscf-srvr-v4-group pcscf-group [secondary-pcscf-srvr-v4-group pcscf-group] [tertiary-pcscf-srvr-v4-group pcscf-group]} | {pcscf-srvr-v6-group pcscf-group [secondary-pcscf-srvr-v6-group pcscf-group] [tertiary-pcscf-srvr-v6-group pcscf-group]}}
```

```
no entry entry-id
```

Context

```
config>mobile>pdn>pcf-pcrf-based-pcscf-allocation
```

Description

This command configures specific P-CSCF addresses to be sent to the UE, based on the PCRF/PCF peer that the PGW-C/SMF has communicated with. Also, the command allows the creation of entries for these P-CSCF addresses. Each entry has one APN and per APN, there can be one Diameter Peer or NF ID entry.

One pcscf-srvr-v4-group and/or one pcscf-srvr-v6-group can be assigned in every Diameter Peer or NF ID, along with their optional secondary and tertiary groups. The P-CSCF server groups must be defined under the profile context, prior to associating it with a Diameter Peer or NF ID.

The no form of this command removes a specific entry ID configuration item.

Parameters

entry-id

specifies the entry ID for the PCF/PCRF based P-CSCF allocation

Values: 1 to 128

apn-name

specifies the APN name for matching the requested/received APN, up to 80 characters

profile-name

specifies the Diameter peer profile name, up to 32 characters

uuid

specifies the NF instance ID in UUIDv4 canonical textual representation

Values: xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxxxx - [0..F]H

pcscf-group

specifies the name of the v4 or v6 P-CSCF server groups (primary, secondary, tertiary), up to 32 characters

5.2.1.2.75 pcmd

Syntax

[no] pcmd profile *profile-name*

Context

config>mobile>pdn

Description

This command configures a PCMD profile for the PDN (specified by the *gw-id*) which enables sending generated PCMD records.

The configured profile in this command is by default applied in all MG groups for this gw-id unless, one, more or all groups that have a different profile configured in the context config>mobile>system>group>pcmd.

The no form of this command clears the PCMD configuration and disables the ability to generate extended PCMD records (if enabled for the profile report type parameter) and streaming of PCMD records. Supported gateway types include SGW, PGW/GGSN, combined SGW/PGW/GGSN, and ePDG.

Default

no pcmd profile

Parameters

profile-name

specifies the PCMD profile for the PDN, up to 32 characters. The profile is configured in the config>mobile>profile>pcmd context.

5.2.1.2.76 pcscf-recovery

Syntax

[no] pcscf-recovery [*dns-cache-interval*] [*pco-reselection*]

Context

```
config>mobile>pdn
```

Description

This command enables the `pcscf-recovery` feature whereby the PGW will allocate a new set of P-CSCF addresses to the UE upon receiving the P-CSCF Restoration Required indication from the MME or SGW.

The PGW will choose the new set of P-CSCF addresses from the P-CSCF group. If the P-CSCF group uses an FQDN to configure the P-CSCF, the PGW will consult the DNS cache to determine the list.

By default, the DNS cache will respect the TTL given by the DNS server. However, if the DNS cache interval is specified, the PGW will use this time instead of that returned by DNS.

Default

no PCSCF recovery

Parameters

dns-cache-interval

specifies the time of the DNS cache interval, in seconds, instead of the time returned by DNS

Values: 10 to 3600

Default: 0

pco-reselection

specifies whether the P-CSCF Re-selection value must be used from Protocol Configuration Options (PCO)

If the keyword is used, the PGW uses the `pco-reselection` support from the PCO container. If the keyword is not used, the PGW assumes the `pco-reselection` support for UE

5.2.1.2.77 *pco-option*

Syntax

```
pco-option
```

Context

```
config>mobile>pdn
```

```
config>mobile>pdn>apn
```

Description

This command enables the context to configure UE PCO options.

5.2.1.2.78 max-allocated-pcscf-addresses

Syntax

```
max-allocated-pcscf-addresses number  
no max-allocated-pcscf-addresses
```

Context

```
config>mobile>pdn>pco-option
```

Description

This command sets the maximum number of P-CSCF addresses that will be included in the Protocol Configuration Options. The actual number of P-CSCF addresses included will depend on the number of addresses listed explicitly in the pcscf-group or resolved from DNS using the FQDN. In other words, if the list contains less than the number specified by this command the PCO will allocate only the number of addresses in the list. If the list contains more than max-allocated-addresses, the system will load-balance the allocation of addresses from the list.

Parameters

number

specifies the maximum number of addresses sent to the PCO

Values: 1 to 3

Default: 1

5.2.1.2.79 pfcf-node-id-type

Syntax

```
pfcf-node-id-type {fqdn | ip} [ip-type {ipv4 | ipv6}]  
no pfcf-node-id-type
```

Context

```
config>mobile>pdn
```

Description

This command configures the PFCF node ID as an FQDN or IPv4/IPv6 address type.

The no form of this command specifies to use the default node ID type for the Sx/N4 interface.

**NOTE:**

This command requires a PDN shutdown.

Parameters

fqdn

specifies the PFCP node ID as an FQDN type and the FQDN to be sent is the one configured under the `config>mobile>pdn>epc-node` command

ip

specifies the PFCP node ID as an IP address type and the IP address to be sent is the one configured for the Sx/N4 interface

Default: fqdn

ipv4

specifies to use the IPv4 address for the PFCP node ID when dual stack is configured on the Sx-N4 reference point interface

ipv6

specifies to use the IPv6 address for the PFCP node ID when dual stack is configured on the Sx-N4 reference point interface

Default: ipv4

**NOTE:**

If dual stack is configured on the Sx-N4 reference point interface and the `pfc-node-id-type` is set with IP address, then the IPv4 or IPv6 address must also be configured.

**NOTICE:**

- The `pfc-node-id-type` command must not be executed on a live system.
- Use the shutdown command in the `config>mobile>pdn` context before changing the node ID type for the Sx/N4 interface. Use the `no shutdown` command under the same context to apply the change.

5.2.1.2.80 pfc-cp-router-adv

Syntax

[no] `pfc-cp-router-adv` [`application-id` *application-id-name*]

Context

```
config>mobile>pdn
```

Description

This command enables handling of ICMPv6 RS/RA and NS/NA messages on the CP function.

When the application ID option is configured, this command configures the CP function to instruct the UP function to use a predefined policy-rule-unit-name with SDF filters to detect ICMPv6 RS/RA and NS/NA messages, and DHCPv6 messages for APNs configured with DHCP service enabled. The application ID sent in the Create PDR > PDI > Application IE defines the policy-rule-unit-name.

When the application ID option is used, this command configures the application ID on the UP function. The UP function associates the application ID with the policy-rule-unit-name and the predefined SDFs/flow-descriptions that define how to detect ICMPv6 RS/NS (and DHCPv6) messages. See also the flow-description command in the config>mobile>profile>policy-options>policy-rule-unit context.

The no form of this command reverts to not handling any ICMPv6 RS/RA, NS/NA and DHCP messages directly on the CP function via the UP function.



NOTE:

- This functionality also uses the interface realm configured in the config>mobile>pdn>sx-n4>interface>gtp-u context, a PFCP-u tunnel reattempt timer configured in the config>mobile>profile>pfcpc>pfcpc-profile>pfcpc-u-tunnel context, and the maximum and minimum advertisement intervals configured in the config>mobile>pdn>apn>router-advertisements context.
- For systems using ICR, this functionality requires data synchronization between the master and the slave nodes in the MC-mobile redundancy. This is configured using the mc-complete-ue-sync command in the config>redundancy>multi-chassis>peer>mc-mobile context.
- When the application ID option is not used, this command should not be enabled on the UP function.

Default

```
no pfcpc-cp-router-adv
```

5.2.1.2.81 reattach-timeout

Syntax

```
reattach-timeout timeout
```

```
no reattach-timeout
```

Context

config>mobile>pdn

Description

This command defines the timeout for the reattach call flow. If the call flow is aborted, Cause code 94 is used in the Create Session Response message.

The no form of this command reverts the timeout to the default.

Default

no reattach-timeout

Parameters

timeout

specifies the value for the reattach call flow, in seconds

Values: 4 to 20

Default: 20

5.2.1.2.82 relax-msb-check-on-path-management-messages

Syntax

[no] relax-msb-check-on-path-management-messages

Context

config>mobile>pdn

Description

This command relaxes the check of the Most Significant Bit (MSB) of the Sequence Number (SN) field in the GTP-C header for path management messages. Even if the mobile gateway receives a GTP Echo Request with the MSB set to 1, the request is not discarded and the connection is kept alive.



NOTE:

This command is only applicable to GTPv2-C and not GTPv1.

Default

no relax-msb-check-on-path-management-messages

5.2.1.2.83 reserve-default-gateway-address

Syntax

[no] reserve-default-gateway-address

Context

config>mobile>pdn

Description

This command enables the reserved default gateway IP address option. When enabled for the TWAG, the address pool manager will reserve a single IP address for each prefix in the IP local pools from which it allocates IP addresses. The reserved IP address for a prefix is sent as the default gateway address for all sessions with IP addresses allocated from that prefix. The default gateway address for each prefix is automatically set to the last IP address in the prefix (except for the x.y.z.255 IP address). For example, for local pool IP address 10.1.0.0/16, the default gateway IP address will be 10.1.255.254.

**NOTE:**

This command requires a PDN shutdown.

Default

no reserve-default-gateway-address

5.2.1.2.84 seamless-wlan-handover

Syntax

[no] seamless-wlan-handover

Context

config>mobile>pdn

config>mobile>pdn>apn

Description

This command enables seamless handover of a UE that is connected to the PGW/GGSN via a WLAN gateway (for example, WiFi access) and moving over to 2G/3G access. Handover between WiFi and 2G/3G access will

be seamless only when the PDN connection over both access interfaces is established to the same PGW/GGSN node.

If this command is configured to the default at the APN level, the value configured for the gateway instance level is inherited.

If this command is disabled, the UE is re-attached on the PGW/GGSN during the handover from WiFi to 2G/3G access.

**NOTE:**

This command requires a PDN shutdown.

Default

seamless-wlan-handover

5.2.1.2.85 send-mbr-for-bbai

Syntax

```
[no] send-mbr-for-bbai
```

Context

```
config>mobile>pdn
```

Description

This command configures the ePDG to send a modify bearer request if the UE local IP address or UE UDP port changes. This is related to the Broadband Access Interworking (BBAI) function described by 3GPP.

The no form of this command specifies that the ePDG will not send a modify bearer request if the UE local IP address or UE UDP port changes.

Default

no send-mbr-for-bbai

5.2.1.2.86 send-mbr-for-bbai-emergency

Syntax

```
[no] send-mbr-for-bbai-emergency
```

Context

```
config>mobile>pdn
```

Description

This command configures the ePDG to send a modify bearer request for an emergency call if the UE local IP address, the UE UDP port, or the WLAN location information changes.

The no form of this command specifies that the ePDG does not send a modify bearer request for emergency calls, if the UE local IP address, the UE UDP port, or the WLAN location information changes.



NOTE:

This command requires a PDN shutdown.

Default

```
send-mbr-for-bbai-emergency
```

5.2.1.2.87 serving-node-for-combo-sessions

Syntax

```
serving-node-for-combo-sessions {mme | sgw}
```

```
no serving-node-for-combo-sessions
```

Context

```
config>mobile>pdn
```

Description

This command selects the IP address that the PGW sends as the serving node IP address to various interfaces for combined SGW/PGW sessions.

The no form of this command reverts to the default (that is, using the existing behavior of sending the MME IP address as the serving node).



NOTE:

This command can be re-configured at any time. When the IP address is changed, the next message sent out to the peer will reflect the changed address.

Default

no serving-node-for-combo-sessions

Parameters

mme

the PGW sends the MME IP address

sgw

the PGW sends the SGW IP address

Default: mme

5.2.1.2.88 session-deletion-rate

Syntax

session-deletion-rate *session-deletion-rate*

no session-deletion-rate

Context

config>mobile>pdn

Description

This command sets the maximum rate at which sessions are deleted from each MSCP group when a massive session deletion is triggered from CLI or automatically from the gateway. The configured session deletion rate is treated by the system as the maximum rate to be applied, not as the enforced rate for the session deletion.

The no form of this command reverts to the default.

Parameters

session-deletion-rate

specifies the number of sessions that are deleted per second from each MSCP group, and is treated as the maximum rate to be applied, not as the enforced rate for the session deletion

Values: 2 to 2048

Default: 512

5.2.1.2.89 session-restoration-rate

Syntax

```
session-restoration-rate session-restoration-rate  
no session-restoration-rate
```

Context

```
config>mobile>pdn
```

Description

This command sets the restoration rate of sessions per MSCP group per second when the system initiates session restoration to another PGW-C/SMF. The configured session restoration rate is the maximum rate that the mobile gateway applies and not the enforced rate for the session restoration.

The no form of this command reverts to the default.

Parameters

session-restoration-rate

specifies the maximum number of sessions that are restored per second from each MSCP group

Values: 2 to 2048

Default: 512

5.2.1.2.90 slices

Syntax

```
slices
```

Context

```
config>mobile>pdn
```

Description

This command enables the context to configure a slice list or a slice instance list that the SMF or the UPF supports.

5.2.1.2.91 slice-instance-list

Syntax

```
slice-instance-list list-name  
no slice-instance-list
```

Context

```
config>mobile>pdn>slices
```

Description

This command configures the slice instances that the SMF or the UPF supports.

The no form of this command reverts to the default.

Default

```
no slice-instance-list
```

Parameters

list-name

specifies the name of the slice instance list, up to 32 characters; the slice instance list is configured using the config>mobile>profile>list>slice-instance-list command

5.2.1.2.92 slice-list

Syntax

```
slice-list list-name  
no slice-list
```

Context

```
config>mobile>pdn>slices
```

Description

This command configures the slices that the SMF or the UPF supports.

The no form of this command reverts to the default.

Default

no slice-list

Parameters

list-name

specifies the name of the slice list, up to 32 characters; the slice list is configured using the `config>mobile>profile>list>slice-list` command

5.2.1.2.93 supported-vendor

Syntax

supported-vendor {custom-1 | custom-2 | custom-3}

no supported-vendor

Context

config>mobile>pdn

Description

This command configures the vendor number for which vendor-specific customizations are supported. Each number represents a specific customer.

The no form of this command reverts to the default.



NOTE:

This command requires a PDN shutdown.

Parameters

custom-1

specifies the customer-1 customer

custom-2

specifies the customer-2 customer

custom-3

specifies the customer-3 customer

Default: no customer

5.2.1.2.94 suspend

Syntax

[no] suspend

Context

config>mobile>pdn

Description

This command sets the gateway to a suspended state. While in a suspended state, the gateway rejects new sessions from being established, while continuing to serve existing sessions. In general, CSReq with zero-TEID are rejected.

All session management and mobility management procedures are served normally for existing UEs on the PGW. CSReq with zero-TEID and HI are accepted on the PGW.

No new sessions are accepted on the ePDG. For more information, refer to section *Gateway Suspend State*, in the *7750 SR MG and CMG Configuration Guide*.

All session management procedures on the SGW are served. CSReq with zero-TEID and HI are rejected on the SGW.

The no form of this command removes the gateway instance from the suspended state.

Default

no suspend

5.2.1.2.95 transit-sub

Syntax

transit-sub

Context

config>mobile>pdn

config>mobile>pdn>apn

Description

This command enables the context to create transit subscribers.

5.2.1.2.96 attach-response-wait-timer

Syntax

```
attach-response-wait-timer seconds  
no attach-response-wait-timer
```

Context

```
config>mobile>pdn>transit-sub  
config>mobile>pdn>apn>transit-sub
```

Description

This command configures the response wait time from the RADIUS server to create the subscriber context. If no response is received from the RADIUS server before the wait time expires, the subscriber context creation is put on hold.



NOTE:

This command requires a PDN shutdown, when used in the config>mobile>pdn>transit-sub context.

Default

```
no attach-response-wait-timer
```

Parameters

seconds

specifies the response wait time, in seconds

Values: 3 to 60

Default: 10

5.2.1.2.97 attach-retry-hold-timer

Syntax

```
attach-retry-hold-timer seconds  
no attach-retry-hold-timer
```

Context

```
config>mobile>pdn>transit-sub
```

```
config>mobile>pdn>apn>transit-sub
```

Description

This command configures the hold time before a subscriber context for a source IP address is initiated again after an initial failure.



NOTE:

This command requires a PDN shutdown, when used in the `config>mobile>pdn>transit-sub` context.

Default

no attach-retry-hold-timer

Parameters

seconds

specifies the hold time, in seconds

Values: 1 to 300

Default: 30 (disabled)

5.2.1.2.98 data-triggered

Syntax

```
[no] data-triggered
```

Context

```
config>mobile>pdn>transit-sub
```

```
config>mobile>pdn>apn>transit-sub
```

Description

This command enables the creation of transit subscribers using data trigger.



NOTE:

This command requires a PDN shutdown, when used in the `config>mobile>pdn>transit-sub` context.

Default

no data-triggered

5.2.1.2.99 twag-samog-radius-accounting

Syntax

`twag-samog-radius-accounting radius-group`

`no twag-samog-radius-accounting`

Context

`config>mobile>pdn`

Description

This command enables RADIUS accounting, using a specific RADIUS group, in TWAG SaMOG mode.



NOTE:

This command requires a PDN shutdown.

Default

no twag-samog-radius-accounting

Parameters

radius-group

specifies the name of a RADIUS group

5.2.1.2.100 ue-cleanup-on-delete

Syntax

`ue-cleanup-on-delete {enable | disable}`

`no ue-cleanup-on-delete`

Context

config>mobile>pdn

Description

This command enables or disables the cleanup of all the UE sessions when a delete request is received for any session for that UE. The cleanup of the UE sessions must be enabled while performing in-service migration to CUPS architecture so that overload during CUPS migration can be avoided.

The no form of this command reverts to the default (disable).

Default

no ue-cleanup-on-delete

Parameters

enable

enables the cleanup of UE sessions

disable

disables the cleanup of UE sessions

Default: disable

5.2.1.2.101 umts-qos-policy

Syntax

umts-qos-policy *umts-qos-policy*

no umts-qos-policy

Context

config>mobile>pdn

Description

This command specifies the name of Universal Mobile Telecommunications System (UMTS) Quality of Service (QoS) Policy on this Access Point Name (APN).



NOTE:

This command requires a PDN shutdown.

Parameters

umts-qos-policy

specifies the UMTS QoS policy on this APN, up to 32 characters

5.2.1.2.102 up-function-feature-options

Syntax

up-function-feature-options

Context

config>mobile>pdn

Description

This command enables the context to configure the bits in the UP Function Features IE. The UP function sends the UP Function Features IE in the PFCP association setup and update procedures.

5.2.1.2.103 exclude-adpdp

Syntax

[no] exclude-adpdp

Context

config>mobile>pdn>up-function-feature-options

Description

This command clears the ADPDP bit (7/2) in the UP Function Features IE (not set). Use this command in interoperability case scenarios where the CP function does not expect that the ADPDP bit in the UP Function Features IE is set.

The no form of this command reverts the ADPDP bit to the default (set).



NOTE:

- Regardless of the value of this bit, the Nokia UP function supports ADPDP.
- In a non-interoperability case (that is, when connected to the Nokia CP function), the ADPDP bit is set (no exclude-adpdp).

Default

no exclude-adpdp

5.2.1.2.104 exclude-dlbd

Syntax

[no] exclude-dlbd

Context

config>mobile>pdn>up-function-feature-options

Description

This command clears the DLBD bit (5/3) in the UP Function Features IE (not set). Use this command in interoperability case scenarios where the CP function does not expect that the DLBD bit in the UP Function Features IE is set.

The no form of this command reverts the DLBD bit to the default (set).



NOTE:

- Regardless of the value of this bit, the Nokia UP function supports DLBD.
- In a non-interoperability case (that is, when connected to the Nokia CP function), the DLBD bit is set (no exclude-dlbd).

Default

no exclude-dlbd

5.2.1.2.105 exclude-dpdra

Syntax

[no] exclude-dpdra

Context

config>mobile>pdn>up-function-feature-options

Description

This command clears the DPDRA bit (7/1) in the UP Function Features IE (not set). Use this command in interoperability case scenarios where the CP function does not expect that the DPDRA bit in the UP Function Features IE is set.

The no form of this command reverts the DPDRA bit to the default (set).



NOTE:

- Regardless of the value of this bit, the Nokia UP function supports DPDRA.
- In a non-interoperability case (that is, when connected to the Nokia CP function), the DPDRA bit is set (no exclude-dpdra).

Default

no exclude-dpdra

5.2.1.2.106 exclude-frrt

Syntax

[no] exclude-frrt

Context

config>mobile>pdn>up-function-feature-options

Description

This command clears the FRRT bit (6/6) in the UP Function Features IE (not set). Use this command in interoperability case scenarios where the CP function does not expect that the FRRT bit in the UP Function Features IE is set.

The no form of this command reverts the FRRT bit to the default (set).



NOTE:

- Regardless of the value of this bit, the Nokia UP function supports FRRT.
- In a non-interoperability case (that is, when connected to the Nokia CP function) with Framed Routing enabled in the CP function, the FRRT bit is set (no exclude-frrt). If the FRRT bit is not set, the Nokia CP function with Framed Routing enabled does not use Framed Routing toward the UP function.

Default

no exclude-frrt

5.2.1.2.107 exclude-heeu

Syntax

[no] exclude-heeu

Context

config>mobile>pdn>up-function-feature-options

Description

This command clears the HEEU bit (5/7) in the UP Function Features IE (not set). Use this command in interoperability case scenarios where the CP function does not expect that the HEEU bit in the UP Function Features IE is set.

The no form of this command reverts the HEEU bit to the default (set).



NOTE:

- Regardless of the value of this bit, the Nokia UP function supports HEEU.
- In a non-interoperability case (that is, when connected to the Nokia CP function), the HEEU bit is set (no exclude-heeu).

Default

no exclude-heeu

5.2.1.2.108 exclude-pdiu

Syntax

[no] exclude-pdiu

Context

config>mobile>pdn>up-function-feature-options

Description

This command clears the PDIU bit (6/2) in the UP Function Features IE (not set). Use this command in interoperability case scenarios where the CP function does not expect that the PDIU bit in the UP Function Features IE is set.

The no form of this command reverts the PDIU bit to the default (set).



NOTE:

- Regardless of the value of this bit, the Nokia UP function supports PDIU.
- In a non-interoperability case (that is, when connected to the Nokia CP function) with PDI optimized signaling enabled in the CP function, the PDIU bit is set (no exclude-pdiu). If the PDIU bit is not set, the Nokia CP function with PDI optimized signaling enabled does not use PDI optimized signaling toward the UP function.

Default

no exclude-pdiu

5.2.1.2.109 exclude-ueip

Syntax

[no] exclude-ueip

Context

config>mobile>pdn>up-function-feature-options

Description

This command clears the UEIP bit (7/3) in the UP Function Features IE (not set). Use this command in interoperability case scenarios where the CP function does not expect that the UEIP bit in the UP Function Features IE is set.

The no form of this command reverts the UEIP bit to the default (set).

Default

no exclude-ueip

5.2.1.2.110 up-peer-list

Syntax

```
up-peer-list list-name  
no up-peer-list
```

Context

```
config>mobile>pdn
```

Description

This command associates a UP peer selection list to the specific CP.

The no form of this command removes the association.

**NOTE:**

- This parameter is only used when the gateway is configured as a CP function using the `mobile>config>pdn>instance-type` command.
- This command requires a PDN shutdown.

Parameters

list-name

specifies the list name, up to 32 characters

5.2.1.2.111 up-selection

Syntax

```
up-selection
```

Context

```
config>mobile>pdn
```

Description

This command enables the context to configure a UP selection profile on the CP network instance.

5.2.1.2.112 pgw-ggsn-u-client-profile

Syntax

```
pgw-ggsn-u-client-profile client-profile-name  
no pgw-ggsn-u-client-profile
```

Context

```
config>mobile>pdn>up-selection
```

Description

This command configures the PGW-U and GGSN-U selection policy. The client profile is applied for PGW-U and GGSN-U node selection on the CP.

The no form of this command removes the node selection policy for PGW-U and GGSN-U.

Parameters

client-profile-name

specifies the client profile name, up to 32 characters

5.2.1.2.113 sgw-u-client-profile

Syntax

```
sgw-u-client-profile client-profile-name  
no sgw-u-client-profile
```

Context

```
config>mobile>pdn>up-selection
```

Description

This command configures the SGW-U selection policy. The client profile is applied for SGW-U node selection on the CP.

The no form of this command removes the node selection policy for SGW-U.

Parameters

client-profile-name

specifies the client profile name, up to 32 characters

5.2.1.2.114 update-bearer-request-for-mvno

Syntax

```
update-bearer-request-for-mvno {enable | disable}  
no update-bearer-request-for-mvno
```

Context

```
config>mobile>pdn
```

Description

This command enables or disables handling the bearer modification with the bearer QoS update functionality that is initiated from an MVNO PGW. This command requires supported-vendor customizations, that is, the command applies only when the *custom-2* parameter is enabled in the config>mobile>pdn>supported-vendor context.

The no form of this command disables the functionality.

Parameters

enable

specifies that an Update Bearer Request message from an MVNO PGW is transparently forwarded to the MME

disable

specifies that an Update Bearer Request message from an MVNO PGW is rejected
Default: disable

5.2.1.2.115 upf-acct-profile

Syntax

```
upf-acct-profile profile-id  
no upf-acct-profile
```

Context

```
config>mobile>pdn
```

Description

This command configures a CUPS UP function PDN gateway with the specified UP function profile. If a profile is also configured in the APN context, the profile under the APN takes precedence.

The no form of this command removes the profile from the PDN configuration.

Parameters

profile-id

specifies the profile ID of the UP function account, configured in the config>mobile>profile>charging context

Values: 0 to 255

Default: 0

5.2.1.2.116 url-encryption

Syntax

```
url-encryption {aes-cbc-128} passphrase string [hash | hash2]
```

```
no url-encryption
```

Context

```
config>mobile>pdn
```

Description

This command enables URL encryption for the UP function PDN gateway in 4G CUPS deployments. It also specifies the AES CBC 128 encryption type and optional passphrase to encrypt fields in the URL sent back to the UE for 302 redirect. This is used only when the URL is specified by the OCS.



NOTE:

The URL encryption configuration depends on the HTTP traffic redirection scenarios:

- for Gy charging response FUI Action Redirect:
 - in CUPS scenarios, URL encryption is based on the url-encryption command (configured for the UP PDN) in the config>mobile>pdn context
 - in non-CUPS scenarios, URL encryption is based on the url-encryption command in the config>mobile>profile>dcca context
- for Gy charging response with FUI Action Restrict and a filter ID PRB, URL encryption is based on the http-redirect and encryption options in the config>mobile>profile>policy-

options>action-rule-unit>gy-filter-id-action context, in CUPS and non-CUPS scenarios

- for Gx policy-rule based redirection, URL encryption is based on the http-redirect and encryption options in the config>mobile>profile>policy-options>policy-rule-unit>flow-gate-status context, in CUPS and non-CUPS scenarios

The no form of this command reverts to the default (disabled).

Default

no url-encryption

Parameters

aes-cbc-128

specifies AES CBC 128-bit encryption

string

encryption passphrase string, up to 64 characters

hash

specifies that the key is entered in an encrypted form

hash2

specifies that the key is entered in a more complex encrypted form

5.2.1.2.117 user-plane-sync-for-delete-bearer

Syntax

[no] user-plane-sync-for-delete-bearer

Context

config>mobile>pdn

Description

This command configures the control plane to wait for the user plane acknowledgment (de-programming the cross connect) prior to sending a delete bearer response for a UE-initiated bearer termination. For network-initiated bearer termination, the delete bearer signaling message and user plane cross connect deprogramming occur in parallel.

The no form of this command reverts to the default (disabled).

Default

no user-plane-sync-for-delete-bearer

5.2.1.2.118 vsmf-profile

Syntax

vsmf-profile *profile-name*

no vsmf-profile

Context

config>mobile>pdn

config>mobile>pdn>apn

Description

This command associates a V-SMF profile to the APN for LBO. The V-SMF profile under the PDN context is used in home-routed roaming.

The no form of this command removes the V-SMF profile association.

Parameters

profile-name

specifies a V-SMF profile name, up to 32 characters

5.2.1.2.119 wlan-emergency

Syntax

wlan-emergency *profile-name*

no wlan-emergency

Context

config>mobile>pdn

Description

This command assigns a unique wlan-emergency profile under a specific PDN and activates the emergency call function support on the ePDG.

The no form of this command removes the wlan-emergency profile from a specific PDN container and deactivates the emergency call support function from ePDG.

**NOTE:**

The no form of this command requires a PDN shutdown.

Parameters

profile-name

specifies a profile name, up to 32 characters

5.2.1.2.120 wps-profile

Syntax

wps-profile *profile-name*

no wps-profile

Context

config>mobile>pdn

Description

This command enables the Multimedia Priority Service (MPS) or Wireless Priority Service (WPS) configured using the wps-profile command under the config>mobile>profile context.

The no form of this command disables the WPS profile.

**NOTICE:**

You must not configure both the config>mobile>pdn>wps-profile and the config>mobile>pdn>ims-emps-priority commands. The CMG eMPS can be configured either as WPS (4G and 5G) or as eMPS (4G only).

The ims-emps-priority is enabled by default. Therefore, to enable the wps-profile, you must first disable the ims-emps-priority.

**NOTE:**

The WPS statistics and KPIs are displayed as eMPS sessions.

Parameters

profile-name

specifies the WPS profile name, up to 32 characters

5.2.1.3 APN PDN command descriptions

5.2.1.3.1 aa-divert-per-bearer-qci-list

Syntax

aa-divert-per-bearer-qci-list *qci-list*

no aa-divert-per-bearer-qci-list

Context

config>mobile>pdn>apn

Description

This command enables bearer traffic diversion to AA according to the bearer's QCI. When the QCI value of the bearer matches to a QCI value in the list, all bearer traffic is diverted to AA. When more than one bearer per session is diverted to AA, only one bearer can have AA rules installed.

The QCI list is configured in the config>mobile>profile>list>qci-list context. If there are no QCI values configured in the list, the default behavior applies, where only one bearer can be diverted to AA.

The no form of this command allows only one bearer to be diverted to AA.

Default

no aa-divert-per-bearer-qci-list

Parameters

qci-list

specifies the QCI list name, up to 32 characters

5.2.1.3.2 access-type

Syntax

[no] access-type *access-type*

Context

config>mobile>pdn>apn

Description

This command is used to specify a Radio Access Type (RAT), under an APN context, to associate a QoS profile for the specified RAT.

The no form of this command removes the specified access type.

Parameters

access-type

specifies the access type

Values: eutran | wlan | eutran-nb-iot | ehrrpd | utran | geran | gan | hspa-e

5.2.1.3.3 default-qos-profile

Syntax

default-qos-profile *profile-name*

no default-qos-profile

Context

config>mobile>pdn>apn>access-type

Description

This command is used to associate a QoS profile for the RAT specified for the access - type command. The specified QoS profile is applied to a session that goes into Assume Positive mode.



NOTE:

The specified QoS profile can also be applied when the PCC is disabled (for example, when the PCRF is not deployed).

The no form of this command removes the associated QoS profile for this access type.

Parameters

profile-name

specifies the name of the default QoS profile, configured under the

config>mobile>profile>qos-profile context, up to 32 characters

5.2.1.3.4 accounting

Syntax

```
accounting type radius username {imsi | msisdn | pco | pco-no-apn | imsi-apn | msisdn-  
apn} server-group server-group [suppress-imsi]  
  
no accounting
```

Context

```
config>mobile>pdn>apn
```

Description

This command configures the accounting type username and server group.

Parameters

type

specifies the accounting type used for accounting from an AAA server

Values	radius	specifies that the RADIUS accounting type is used for accounting from an AAA server
--------	--------	---

user-name

specifies the username used in accounting requests sent to an AAA server

Values	imsi	sets the username attribute to the IMSI
–	msisdn	sets the username attribute to the MSISDN
–	pco	sets the username attribute to the value received in the PCO
–	pco-no-apn	sets the username attribute to the value received in the PCO, but with special logic to include only the username part from the username@APN format;

this parameter is used together with the virtual APN selection based on the PCO feature, although there is no direct dependency. The parameter works only when the APN that exists in the PCO username (user@APN) is the current APN under which RADIUS accounting is performed

- | | | |
|---|------------|---|
| – | imsi-apn | sets the username attribute to the IMSI@APN |
| – | msisdn-apn | sets the username attribute to the MSISDN@APN |

Default: imsi

server-group

specifies the name of a RADIUS group or Diameter group

suppress-imsi

specifies that the 3GPP-IMSI attribute is not included



NOTE:

RADIUS accounting can be enabled or disabled on the fly (without service impact). The enabling and disabling can be combined with new or different values for server-group/user-name and so on, on the fly without shutdown. The new or changed values will only take effect for the new sessions. Old sessions should still use accounting as per the old parameters. Accounting enabled with new parameters does not disable the accounting for the old parameters since the it is still enabled under the APN (only the values have changed.)

When the accounting is disabled under the APN, the mobile gateway stops to send accounting messages (interims/stops). Re-enabling the accounting will only affect new sessions. The system still continues not to send accounting messages for old sessions. The only exception is in cases where there are no interim messages during disabling and enabling the accounting.

5.2.1.3.5 duplicate-server-group

Syntax

[no] duplicate-server-group *group-name*

Context

config>mobile>pdn>apn>accounting

Description

This command names a server group that is used for duplicate accounting. The PGW/GGSN sends the Accounting-Requests (Start, Stop, and Interim) separately to each duplicate server group, one message per group.



NOTE:

- This command requires an APN shutdown.
- Multiple duplicate server groups may be configured (up to 7).

Parameters

group-name

specifies the unique name of this duplicate RADIUS accounting server group, up to 32 characters

5.2.1.3.6 selective-radius-accounting

Syntax

[no] selective-radius-accounting

Context

config>mobile>pdn>apn>accounting

Description

This command enables selective RADIUS accounting. When enabled RADIUS accounting will only start for a UE if traffic from the UE is detected destined for any of the configured destination IP addresses.



NOTE:

When selective radius accounting is enabled the local charging-rule-unit and policy-rule-unit commands must also be configured under the config>mobile>profile>policy-options context.

5.2.1.3.7 accounting-priority

Syntax

```
accounting-priority profile-name  
no accounting-priority
```

Context

```
config>mobile>pdn>apn
```

Description

This command specifies the accounting priority profile used for this APN.

The no form of this command removes this configuration.

Default

```
no accounting-priority
```

Parameters

profile-name

specifies an existing accounting priority profile name, up to 32 characters

5.2.1.3.8 acct-fixed

Syntax

```
acct-fixed user-name long-description-string  
no acct-fixed
```

Context

```
config>mobile>pdn>apn
```

Description

This command configures the username when accounting is fixed.

The no form of this command removes this configuration.

Parameters

long-description-string

specifies the username, up to 32 characters

5.2.1.3.9 aggr-rate-dl

Syntax

```
aggr-rate-dl aggregate-rate-downlink  
no aggr-rate-dl
```

Context

```
config>mobile>pdn>apn
```

Description

This command configures the APN aggregate downlink rate.

The no form of this command removes the rate from the configuration.

Parameters

aggregate-rate-downlink

specifies the APN aggregate downlink rate in kb/s

Values: 0 to 25000000

Default: 0

5.2.1.3.10 aggr-rate-ul

Syntax

```
aggr-rate-ul aggregate-rate-uplink  
no aggr-rate-ul
```

Context

```
config>mobile>pdn>apn
```

Description

This command configures the APN aggregate uplink rate.

The no form of this command removes the rate from the configuration.

Parameters

aggregate-rate-uplink

specifies the APN aggregate uplink rate in kb/s

Values: 0 to 250000000

Default: 0

5.2.1.3.11 allow-always-on

Syntax

[no] allow-always-on

Context

config>mobile>pdn>apn

Description

This command enables always-on PDU sessions in the APN/DNN. When enabled, the SMF accepts the request for an always-on PDU session in the PDU session establishment request message.

The no form of this command reverts to the default (disabled).

Default

no allow-always-on

5.2.1.3.12 allow-dangling-rule-ded-bearer

Syntax

[no] allow-dangling-rule-ded-bearer

Context

config>mobile>pdn>apn

Description

This command enables dedicated bearers to be created for dynamic rules that are left dangling with a previous QCI/ARP, as a result of default bearer QCI/ARP modification. QCI/ARP changes of a default bearer with wildcard QCI/ARP PRB are supported. QCI/ARP modifications can be a result of handovers including 3G/4G, Wi-Fi/LTE, and so on, initiated by the PCRF, triggered as result of a handover with an MB command,

or initiated by the HSS. If dynamic rules end up with a previous QCI/ARP as a result of the QCI/ARP change, the PCRF is notified that these rules have been removed.

The no form of this command reverts to the default.

Default

no allow-dangling-rule-ded-bearer

5.2.1.3.13 allow-emergency

Syntax

```
allow-emergency [emergency-profile profile-name]  
no allow-emergency
```

Context

```
config>mobile>pdn>apn
```

Description

This command configures the APN as an emergency APN. This allows emergency PDN sessions to be established in this APN. The optional emergency profile is used to define additional parameters for the emergency calls.

The no form of this command reverts to the default.

Default

no allow-emergency

Parameters

profile-name

specifies an existing emergency profile name, up to 32 characters

5.2.1.3.14 allow-wlan-ho-no-context

Syntax

```
[no] allow-wlan-ho-no-context
```

Context

```
config>mobile>pdn>apn
```

Description

This command enables the PGW functionality to handle an S2a/S2b Handover Attach Request (HI=1), that does not exist at all, as a new attach instead of rejecting the request.

Default

```
no allow-wlan-ho-no-context
```

5.2.1.3.15 ambr-policing

Syntax

```
ambr-policing {dl-only | ul-and-dl | none}  
no ambr-policing
```

Context

```
config>mobile>pdn>apn
```

Description

This command specifies the type of AMBR policing to be enforced for this APN. By default, only downlink APN-AMBR policing is performed on the PGW/GGSN. For APNs that require both uplink and downlink policing, the `ul-and-dl` option should be specified. If there is no need for APN-AMBR policing (for example, policing is performed downstream at a TDF), the `none` option should be specified.

The no form of this command reverts to the default.

Parameters

dl-only

specifies that only downlink APN-AMBR policing is performed

ul-and-dl

specifies that both uplink and downlink APN-AMBR policing is performed

none

specifies that no APN-AMBR policing is performed

Default: dl-only

5.2.1.3.16 apn-load-control

Syntax

```
[no] apn-load-control
```

Context

```
config>mobile>pdn>apn
```

Description

This command enables the APN Load Control Information (LCI) IE to be sent to the consumer nodes when the `gtp-pdn-threshold` command is also configured. When enabled, the current resource utilization of `gtp-pdn-threshold max-connection` or `gtp-pdn-threshold max-activations` command is reported to the LCI IE piggy-packed in the messages sent to the MME.

**NOTE:**

- The GTP-C APN load control is only reported over GTPv2-C based on TS29.274.
- This command requires the node-level load control to be enabled in the context `config>mobile>pdn>gtp-load-control`.

5.2.1.3.17 apn-network-id

Syntax

```
apn-network-id type {real [include-radius-auth | include-radius-pre-auth] [apn-ni-only [application application-name [application-name ... (up to 8 max)]]] | public apn-name apn-name}
```

```
no apn-network-id
```

Context

```
config>mobile>pdn>apn
```

Description

This command specifies whether the real, public APN name, or virtual APN (if applicable) is sent as the value for the following:

- Called-Station-ID AVP of Gy and Gx Diameter messages and RADIUS accounting/authentication/pre-authentication messages
- Service-Selection AVP of S6b Diameter messages

- Called-Number AVP of L2TP ICRQ control message used for L2TP session establishment or the APN Network Identifier (APN NI) for Ga interfaces

When this command is not configured, the default value sent is the APN to which the user is attached.

For 5GC and 4G/5G interworking, this command allows to specify DNN NI-only included over N7/Npcf, N40/Nchf and N10/Nudm, using the command filter type `real`, `apn-ni-only` and application-name `n7`, `n40` and `n10`.

The no form of this command reverts to the default.

Default

no apn-network-id

Parameters

type

specifies the APN type

Values	real	uses the real APN from the GTP request or uses the DNN from the Nsmf_PDUSession service request
–	public	uses the APN name configured for apn-name



NOTE:

The public value does not apply to S6b Diameter messages.

include-radius-auth

specifies that the configuration is also applied to RADIUS authentication (applied by default to all other applications except RADIUS pre-authentication)

include-radius-pre-auth

specifies that the configuration is also applied to RADIUS pre-authentication

apn-ni-only

specifies that only the APN NI will be included in the Called-Station-ID AVP and for 5G that only DNN NI is included in the DDN data type over N7/Npcf or N40/Nchf or N10/Nudm

application-name

specifies up to five application names

Values	gy	specifies that the APN NI is included in the Called-Station-ID AVP for Gy applications
–	gx	specifies that the APN NI is included in the Called-Station-ID AVP for Gx applications
–	radius-acct	specifies that the APN NI is included in the Called-Station-ID AVP for RADIUS accounting
–	radius-auth	specifies that the APN NI is included in the Called-Station-ID AVP for RADIUS authentication
–	radius-pre-auth	specifies that the APN NI is included in the Called-Station-ID AVP for RADIUS pre-authentication
–	n7	specifies that only DNN NI is included in the Npcf_SMPolicyControl service requests sent to the PCF
–	n40	specifies that only DDN NI is included in the Nchf_ConvergedCharging service requests sent to the CHF
–	n10	specifies that only DNN NI is included in the Nudm_SubscriberDataManagement service request and in Nudm_UEContextManagement service requests sent to the UDM

apn-name

specifies the APN name is used when `public` is selected for `apn-network-id` type, up to 80 characters

5.2.1.3.18 apn-overload-control**Syntax**

`[no] apn-overload-control`

Context

`config>mobile>pdn>apn`

Description

This command enables the APN Overload Control Information (OCI) IE to be sent to the consumer nodes when the `gtp-pdn-threshold` command is also configured. When enabled and the value set by `gtp-pdn-threshold max-connection` or `gtp-pdn-threshold max-activations` command is reached, in a subsequent Create Session Request, the PGW rejects the request, responds with a Create Session Response with Cause Code APN Congestion (113), and includes the OCI IE.



NOTE:

- The GTP-C APN Overload control is only reported over GTPv2-C based on TS29.274.
- This command requires the node-level overload control to be enabled in the context config>mobile>pdn>gtp-overload-control.

Default

no apn-overload-control

5.2.1.3.19 overload-reduction

Syntax

```
overload-reduction overload-reduction  
no overload-reduction
```

Context

```
config>mobile>pdn>apn>apn-overload-control
```

Description

This command sets the percentage of the traffic reduction.

The no form of this command reverts to the default.

Parameters

overload-reduction

specifies the percentage of the traffic reduction

Values: 5 to 100

Default: 30

5.2.1.3.20 overload-threshold

Syntax

```
overload-threshold overload-threshold  
no overload-threshold
```

Context

```
config>mobile>pdn>apn>apn-overload-control
```

Description

This command sets the threshold percentage of the configured gtp-pdn-threshold for which the GTP-C OCI overload reduction will be triggered according to the configured overload-reduction value.

The no form of this command reverts to the default.



NOTE:

The overload threshold will rescale the GTP-C APN LCI IE load-metric when the overload-threshold is set to a value less than 100.

For example, if the value is set to 100000 and the value is set to 90%, from 0 to 90000 PDN connections represents the APN GTP-C LCI IE load-metric of 0 to 100%. When there are more than 90000 PDN connections, the APN GTP-C OCI IE with its critical configured traffic reduction value is sent.

Parameters

overload-threshold

specifies the percentage of the overload threshold

Values: 80 to 100

Default: 90

5.2.1.3.21 validity-time

Syntax

```
validity-time validity-time  
no validity-time
```

Context

```
config>mobile>pdn>apn>apn-overload-control
```

Description

This command specifies the length of time that the reported overload should be in effect.

The no form of this command reverts to the default.

Parameters

validity-time

specifies the time, in seconds

Values: 10 to 86400

Default: 600

5.2.1.3.22 apn-rate-control

Syntax

```
apn-rate-control profile-name
```

```
no apn-rate-control
```

Context

```
config>mobile>pdn>apn
```

Description

This command enables or disables the APN rate control.

The no form of this command disables the APN rate control.

Default

```
no apn-rate-control
```

Parameters

profile-name

specifies the name of the profile, up to 32 characters

5.2.1.3.23 apn-type

Syntax

```
apn-type small
no apn-type
```

Context

```
config>mobile>pdn>apn
```

Description

This command specifies how the APN/DNN UE sessions are load-balanced across available CP MG groups. It provides the option to assign all APN/DNN UE sessions to only one CP MG group by defining an APN/DNN as type small, that serves only few UE sessions instead of spreading APN/DNN UE sessions across all available CP MG groups. Session bouncing is avoided and performance is improved. Existing sessions must be manually deleted for an APN/DNN that is set to type small from regular.

The no form of this command reverts to the default (disabled).

Default

```
no apn-type
```

Parameters

```
small
    specifies the type of the APN as small
```

5.2.1.3.24 application-id-mvi-policy

Syntax

```
[no] application-id-mvi-policy
```

Context

```
config>mobile>pdn>apn
```

Description

This command configures the UP function to look for the Create PDR > PDI > Application-ID IE instead of the Create PDR > Activate Predefined Rules IE/Deactivate Predefined Rules IE to find the policy-option/policy-rule configured under the APN.

The no form of this command reverts back to the Create PDR > Activate Predefined Rules IE/Deactivate Predefined Rules IE.

**NOTE:**

- This command is not applicable to non-CUPS deployments.
- This configuration must only be done on the UP function.

Default

no application-id-mvi-policy

5.2.1.3.25 auth-fixed**Syntax**

```
auth-fixed user-name long-description-string password password [hash | hash2]  
no auth-fixed
```

Context

```
config>mobile>pdn>apn
```

Description

This command configures the username and password when authentication is fixed.

Default

no auth-fixed

Parameters***long-description-string***

specifies the username used in access requests sent to a AAA server when the authorized username is set to fixed, up to 32 characters

password

specifies the password used to authenticate requests sent to a AAA server when the username is set to fixed, up to 32 characters

hash

specifies the key is entered in an encrypted form

If the hash parameter is not used, the key is assumed to be in a non-encrypted, clear text form. For security, all keys are stored in encrypted form in the configuration file with the hash parameter specified.

hash2

specifies the key is entered in a more complex encrypted form

If the hash2 parameter is not used, the less encrypted hash form is assumed.

5.2.1.3.26 authentication

Syntax

```
authentication type {radius server-group group-name | diameter pri-dia-peer-
profile name sec-dia-peer-profile name} [user-name {imsi | msisdn | pco | fixed | apn |
pco-no-apn | imsi-apn | msisdn-apn}] [suppress-imsi] [pco-password] [password password
password-type {fixed | default} [hash | hash2]]

no authentication
```

Context

config>mobile>pdn>apn

Description

This command configures the RADIUS or Diameter authentication for the APN. RADIUS servers are identified by the server group. Diameter servers are identified by primary and secondary peers.

Parameters

type

specifies the authentication type used for authentication from an AAA server

Values	radius	specifies that the RADIUS authentication type is used for authentication from an AAA server
–	diameter	specifies that the Diameter authentication type is used for authentication from an AAA server

group-name

specifies the name of a radius-group

name

specifies the name of the primary Diameter peer profile that contains the peer IP address

sec-dia-peer-profile name

specifies the name of the secondary Diameter peer profile that contains the peer IP address

user-name

specifies the username used in access requests sent to a AAA server

Values	imsi	sets the User-Name attribute to the IMSI
–	msisdn	sets the User-Name attribute to the MSISDN
–	pco	sets the User-Name attribute to the value received in the PCO
–	fixed	use the name specified by the auth-fixed command
–	apn	sets the User-Name attribute to the APN
–	pco-no-apn	sets the User-Name attribute to the value received in the PCO, but with special logic to include only the username part from the username@APN format; this parameter is used together with the virtual APN selection based on the PCO feature, although there is no direct dependency. The parameter works only when the APN that exists in the PCO username (user@APN) is the current APN under which RADIUS accounting is performed
–	imsi-apn	sets the User-Name attribute to the IMSI@APN
–	msisdn-apn	sets the User-Name attribute to the MSISDN@APN

Default: imsi

suppress-imsi

specifies that the 3GPP-IMSI attribute is not included

Default: disabled

pco-password

specifies to set the password to the value received in the GTP PCO

password

specifies the PAP password used in access requests sent to a RADIUS server, up to 32 characters; up to 54 characters are supported for a hash password and up to 96 characters for a hash2 password

password-type

specifies the password type

Values	fixed	fixed password for overriding the PCO content or providing password with the IMSI, MSISDN or APN username (with PAP)
–	default	default password to be used together with the IMSI, MSISDN or APN username, and when missing from the PCO (with PAP)



NOTE:

If the hash and hash2 parameters are not used for the password, the key is assumed to be in a non-encrypted clear text form. For security purposes, all keys are automatically stored in an encrypted hash form in the configuration. The information saves and uses the configuration defined in the system>security>hash-control context.

hash

specifies that the key is entered in an encrypted form

hash2

specifies that the key is entered in a more complex encrypted form

5.2.1.3.27 bitrate-roundup

Syntax

[no] bitrate-roundup

Context

config>mobile>pdn>apn

Description

This command enables the gateway to send the PCRF provided QoS values rounded up according to NAS specification (24.301) to the LTE access (SGW/MME). The gateway internally keeps using the PCRF provided values.

The no form of this command reverts to the default (disabled).

Default

no bitrate-roundup

5.2.1.3.28 block-list

Syntax

block-list

Context

config>mobile>pdn>apn

Description

This command enables the context to create block-list entries.

5.2.1.3.29 entry

Syntax

entry *entry-id* group *group* pdn-type *pdn-type*

no entry *entry-id*

Context

config>mobile>pdn>apn>block-list

Description

This command creates a new block list entry for a specific group and for a specific PDN type on the configured APN.

The no form of this command deletes a created block list entry.

Parameters

entry-id

specifies a unique entry ID

Values: 1 to 64

group

specifies the mobile gateway group ID

Values	1 to 15	–
–	all	when this value is used, the block list entry blocks all new sessions in all groups for the specified PDN type

pdn-type

specifies the PDN type on the configured APN

This value is ignored for rejecting new sessions when the gateway is configured as a GW-U instance. This means that all sessions are rejected irrespectively of the configured value. However, the value is significant for dropping of user plane traffic and avoiding charging on blocked resources. It must be configured accordingly.

Values: ipv4 | ipv6

5.2.1.3.30 call-sequence

Syntax

```
call-sequence {pfcf-first | local-upf-sel-before-auth}
no call-sequence
```

Context

```
config>mobile>pdn>apn
```

Description

This command specifies the required call-flow sequence for the session.

The no form of this command disables the alternative call-flow sequence and reverts to the legacy order: aaa-auth, up-slc, gx, radius-acct, chr, pfcf.

Parameters

pfcp-first

specifies the order: aaa-auth, up-slc, pfcp, gx, radius-acct, chr

local-upf-sel-before-auth

specifies the order: up-slc, aaa-auth, pfcp, gx, radius-acct, chr

5.2.1.3.31 cm-idle-timeout

Syntax

`cm-idle-timeout cm-idle-timeout`

`no cm-idle-timeout`

Context

`config>mobile>pdn>apn`

Description

This command specifies the time a PDU session is allowed to remain idle before being deleted by the SMF.

Parameters

cm-idle-timeout

specifies the time in seconds a PDU session is allowed to remain idle before being deleted by the SMF

Values	0 5 to 2592000	emergency APN
–	0 60 to 2592000	APN

Default: 0

5.2.1.3.32 converged-interface-support

Syntax

`converged-interface-support`

Context

```
config>mobile>pdn>apn
```

Description

This command enables the context to configure the 5G SBI interface support for UEs without 5GS NAS support.

5.2.1.3.33 converged-n10

Syntax

```
converged-n10 {enable | disable | inherit | 5gcnri}
```

```
no converged-n10
```

Context

```
config>mobile>pdn>apn>converged-interface-support
```

Description

This command enables or disables the use of the 5G service-based N10 interface by the PGW-C/SMF for the UEs without 5GS NAS support. When enabled, the PGW-C/SMF uses the 5G service-based N10 interface for the UEs without 5GS NAS support.

When set to `enable` or `disable`, the system overrides the PDN-level configuration. When set to `inherit`, the system inherits the configuration from the PDN level. When set to `5gcnri` (5GC Not Restricted Indication), the N10 interface is enabled for the UEs without 5GS NAS support but the access to the 5GC is not restricted by the user subscription.

The `no` form of this command reverts to the default (`inherit`).

Parameters

enable

enables the use of the 5G service-based N10 interface by the PGW-C/SMF for the UEs without 5GS NAS support

disable

disables the use of the 5G service-based N10 interface by the PGW-C/SMF for the UEs without 5GS NAS support

inherit

specifies that the behavior for this APN is inherited from the PDN-level configuration

Default: `inherit`

5gcnri

enables the use of the 5G service-based N10 interface by the PGW-C/SMF for the UEs without 5GS NAS support but the access to the 5GC is not restricted by the user subscription

5.2.1.3.34 converged-n40

Syntax

```
converged-n40 {enable | disable | inherit}
no converged-n40
```

Context

```
config>mobile>pdn>apn>converged-interface-support
```

Description

This command enables or disables the use of the 5G service-based N40 interface by the PGW-C/SMF for the UEs without 5GS NAS support. When enabled, the PGW-C/SMF uses the 5G service-based N40 interface for the UEs without 5GS NAS support.

When set to `enable` or `disable`, the system overrides the PDN-level configuration. When set to `inherit`, the system inherits the configuration from the PDN level.

The `no` form of this command reverts to the default (`inherit`).

Parameters

enable

enables the use of the 5G service-based N40 interface by the PGW-C/SMF for the UEs without 5GS NAS support

disable

disables the use of the 5G service-based N40 interface by the PGW-C/SMF for the UEs without 5GS NAS support

inherit

specifies that the behavior for this APN is inherited from the PDN-level configuration
Default: `inherit`

5.2.1.3.35 converged-n40-gngp

Syntax

```
converged-n40-gngp {enable | disable | inherit}
```

```
no converged-n40-gngp
```

Context

```
config>mobile>pdn>apn>converged-interface-support
```

Description

This command enables or disables the use of the 5G service-based N40 interface by the PGW-C/SMF for the UEs in case of 2G/3G access. When enabled, the PGW-C/SMF uses the 5G service-based N40 interface for the UEs in case of 2G/3G access.

When set to `enable` or `disable`, the system overrides the PDN-level configuration. When set to `inherit`, the system inherits the configuration from the PDN level.

The `no` form of this command reverts to the default (`inherit`).

Parameters

`enable`

enables the use of the 5G service-based N40 interface by the PGW-C/SMF for the UEs in case of 2G/3G access

`disable`

disables the use of the 5G service-based N40 interface by the PGW-C/SMF for the UEs in case of 2G/3G access

`inherit`

specifies that the behavior for this APN is inherited from the PDN-level configuration
Default: `inherit`

5.2.1.3.36 converged-n7

Syntax

```
converged-n7 {enable | disable | inherit}  
no converged-n7
```

Context

```
config>mobile>pdn>apn>converged-interface-support
```

Description

This command enables or disables the use of the 5G service-based N7 interface by the PGW-C/SMF for the UEs without 5GS NAS support. When enabled, the PGW-C/SMF uses the 5G service-based N7 interface for the UEs without 5GS NAS support.

When set to `enable` or `disable`, the system overrides the PDN-level configuration. When set to `inherit`, the system inherits the configuration from the PDN level.

The no form of this command reverts to the default (`inherit`).

Parameters

`enable`

enables the use of the 5G service-based N7 interface by the PGW-C/SMF for the UEs without 5GS NAS support

`disable`

disables the use of the 5G service-based N7 interface by the PGW-C/SMF for the UEs without 5GS NAS support

`inherit`

specifies that the behavior for this APN is inherited from the PDN-level configuration

Default: `inherit`

5.2.1.3.37 converged-n7-gngp

Syntax

```
converged-n7-gngp {enable | disable | inherit}
```

```
no converged-n7-gngp
```

Context

```
config>mobile>pdn>apn>converged-interface-support
```

Description

This command enables or disables the use of the 5G service-based N7 interface by the PGW-C/SMF for the UEs in case of 2G/3G access. When enabled, the PGW-C/SMF uses the 5G service-based N7 interface for the UEs in case of 2G/3G access.

When set to `enable` or `disable`, the system overrides the PDN-level configuration. When set to `inherit`, the system inherits the configuration from the PDN level.

The no form of this command reverts to the default (`inherit`).

Parameters

enable

enables the use of the 5G service-based N7 interface by the PGW-C/SMF for the UEs in case of 2G/3G access

disable

disables the use of the 5G service-based N7 interface by the PGW-C/SMF for the UEs in case of 2G/3G access

inherit

specifies that the behavior for this APN is inherited from the PDN-level configuration

Default: inherit

5.2.1.3.38 dedicated-bearer-create-retry-count

Syntax

`dedicated-bearer-create-retry-count count`

`no dedicated-bearer-create-retry-count`

Context

`config>mobile>pdn>apn`

Description

This command is used to specify a dedicated bearer creation retry count. After the specified number of retries, the dedicated bearer creation is abandoned until the next active application period (subsequent to an application stop). It applies to AA rules installed with QCI/ARP that are different from the default bearer, in which case the mute-notification is enabled. When the application is detected, a dedicated bearer is created without further interaction with PCRF.

The no form of this command reverts to the default.

Parameters

count

specifies the number of times the dedicated bearer creation is retried upon failure

Values: 0 to 10

Default: 0

5.2.1.3.39 dedicated-bearer-create-retry-wait-timer

Syntax

```
dedicated-bearer-create-retry-wait-timer wait-timer  
no dedicated-bearer-create-retry-wait-timer
```

Context

```
config>mobile>pdn>apn
```

Description

This command specifies a dedicated bearer creation retry timer. Upon dedicated bearer creation failure for this application, the gateway waits the specified amount of time before retrying. It applies to AA rules installed with QCI/ARP that are different from the default bearer and then the mute-notification is enabled. When the application is detected, a dedicated bearer is created without further interaction with PCRF.

The no form of this command reverts to the default.

Parameters

wait-timer

specifies the time the gateway waits before retrying in minutes

Values: 0 to 60

Default: 0

5.2.1.3.40 dedicated-bearer-idle-timeout-profile

Syntax

```
dedicated-bearer-idle-timeout-profile profile-name  
no dedicated-bearer-idle-timeout
```

Context

```
config>mobile>pdn>apn
```

Description

This command configures the dedicated bearer idle-timeout profile for the APN.

Default

no dedicated-bearer-idle-timeout-profile

Parameters

profile-name

specifies the name of the dedicated bearer idle timeout profile, up to 32 characters

5.2.1.3.41 default-cc-group

Syntax

default-cc-group {rtr | uc | pp}

no default-cc-group

Context

config>mobile>pdn>apn

Description

This command specifies the default CC group to be used when a PCRF does not provide the CC group subscriber type for a subscriber PDN session. This command can also be used when Gx is not deployed or used. The no form of this command removes the configuration.

Parameters

rtr

specifies the service type as real-time report

uc

specifies the service type as usage control

pp

specifies the service type as prepaid

5.2.1.3.42 default-qos-5g-profile

Syntax

default-qos-5g-profile *profile-name*

no default-qos-5g-profile

Context

```
config>mobile>pdn>apn
```

Description

This command specifies the default 5G QoS profile. The profile must be configured if continued action is possible when the UDM rejects the retrieval of DNN subscription data. If a default 5G QoS profile is not defined, the PDU session activation fails without the DNN subscription data.

If the default 5G QoS profile is changed, the new profile is used only for new PDU sessions.

The no form of this command removes the default 5G QoS profile from the DNN.

Parameters

profile-name

specifies the name of the default 5G QoS profile, up to 32 characters

5.2.1.3.43 default-qos-application-modify

Syntax

```
default-qos-application-modify
```

Context

```
config>mobile>pdn>apn
```

Description

This command enables the context to configure QoS parameters required in case the application triggers the default bearer QoS update. The functionality is activated with the `modify-qos-default-bearer` command.

5.2.1.3.44 apn-ambr-attr

Syntax

```
apn-ambr-attr
```

Context

```
config>mobile>pdn>apn>default-qos-application-modify
```

Description

This command enables the context to configure APN Aggregate Maximum Bit-Rate (AMBR) attributes. It is used in case the application traffic triggers the default bearer QoS update.

5.2.1.3.45 **aggr-rate**

Syntax

```
aggr-rate downlink-rate downlink-rate uplink-rate uplink-rate  
no aggr-rate
```

Context

```
config>mobile>pdn>apn>default-qos-application-modify>apn-ambr-attr
```

Description

This command specifies the downlink and uplink rate for APN-AMBR. It is used if the application traffic triggers the default bearer QoS update.

The no form of this command removes the configured uplink and downlink rates.

If no apn-ambr-attr values are defined, the apn-ambr-attr values are not modified from the default bearer's previous values.

Default

no aggr-rate

Parameters

downlink-rate

specifies the downlink APN-AMBR in kb/s

Values: 0 to 25000000

uplink-rate

specifies the uplink APN-AMBR in kb/s

Values: 0 to 25000000

5.2.1.3.46 **arp**

Syntax

```
arp arp-value
```

```
no arp
```

Context

```
config>mobile>pdn>apn>default-qos-application-modify
```

Description

This command sets the ARP value required if the application traffic triggers the default bearer QoS update.

The no form of this command removes the configured ARP value.

If no ARP value is defined, the ARP is not modified from the default bearer's previous value.

Default

```
no arp
```

Parameters

arp-value

specifies the value of the ARP

Values: 1 to 15

5.2.1.3.47 qci

Syntax

```
qci qci-value
```

```
no qci
```

Context

```
config>mobile>pdn>apn>default-qos-application-modify
```

Description

This command sets the QCI value required if the application traffic triggers the default bearer QoS update.

The no form of this command reverts to the default.

If no QCI value is defined, the QCI is not modified from the default bearer's previous value.

Default

```
no qci
```

Parameters

qci-value

specifies the value of the QCI

Values: 1 to 9 | 65 | 66 | 69 | 70 | 128 to 254

5.2.1.3.48 qos-holding-time

Syntax

qos-holding-time *value*

no qos-holding-time

Context

config>mobile>pdn>apn>default-qos-application-modify

Description

This command sets the timeout for application traffic to trigger the QoS update back to the default QoS, if no usage is detected during this configured time.

The no form of this command reverts to the default.

If no value is configured, qos-holding-time uses a default value of 10 minutes.

Default

no qos-holding-time

Parameters

value

specifies the value, in minutes

Values: 1 to 60

Default: 10

5.2.1.3.49 retry-wait-timer

Syntax

retry-wait-timer *value*

no retry-wait-timer

Context

```
config>mobile>pdn>apn>default-qos-application-modify
```

Description

This command sets the time the gateway waits before retrying to update the default bearer QoS.

The no form of this command removes the configured value of the retry wait timer.

If no value is configured, the `retry-wait-timer` uses a default value of 1 minute.

Default

no `retry-wait-timer`

Parameters

value

specifies the value of `retry-wait-timer` in minutes

Values: 1 to 60

Default: 1

5.2.1.3.50 del-bearer-no-reactivation-req-cause

Syntax

```
del-bearer-no-reactivation-req-cause [idle-timeout]
```

```
no del-bearer-no-reactivation-req-cause
```

Context

```
config>mobile>pdn>apn
```

Description

This command configures the PGW to include the indication that the PDN reconnection to this APN is not allowed in the Delete Bearer Request message. The command overrides the `del-bearer-reactivation-req-cause` command.

The no form of this command reverts to the default behavior that uses the cause value: PDN connection inactivity timer expires (11).

Default

no `del-bearer-no-reactivation-req-cause`

Parameters

idle-timeout

specifies that the PDN reconnection to this APN disallowed cause value, is used if the PDN connection is released due to the `idle-timeout` command; this command is configured under the `config>mobile>pdn>apn` context

5.2.1.3.51 del-bearer-activation-req-cause

Syntax

```
del-bearer-activation-req-cause {[fh-session-continue] [session-timeout] [pdn-connection-clear] [pcscf-restoration] [all]}
```

```
no del-bearer-activation-req-cause
```

Context

```
config>mobile>pdn>apn
```

Description

This command enables the PGW to include a reactivation request in the delete bearer request that is sent to the UE for the specified scenarios.

The `no` form of this command reverts to the default behavior where no cause IE is included in the delete bearer request.

Default

```
no del-bearer-activation-req-cause
```

Parameters

fh-session-continue

specifies that a re-activation request is included in the Delete Bearer Request that is sent to the UE, when a Gy CCFH session continue timer expires

session-timeout

specifies that a re-activation request is included in the Delete Bearer Request/PDU Session Release Command message that are sent to the UE, when a session timeout expires (configured under the `config>mobile>pdn>apn` context)

pdn-connection-clear

specifies that a re-activation request is included in the Delete Bearer Request message that is sent to the UE, when the PDN connection is cleared, or if the session was cleared due to internal audit mechanism

pcscf-restoration

specifies that a re-activation request is used when the Delete Bearer Request message is triggered by the P-CSCF restoration

all

specifies the reactivation-requested cause code to be included, whenever the PGW deletes the PDN connection whether triggered externally (Gx, S6b, Gy, and so on) or internally via timeout or management intervention

5.2.1.3.52 delay-tolerant-connection-suspension

Syntax

`delay-tolerant-connection-suspension delay-tolerant-connection-suspension`
`no delay-tolerant-connection-suspension`

Context

`config>mobile>pdn>apn`

Description

This command is used to control the delay tolerant connection functionality in the PGW. If enabled, the PGW suspends the create/update procedures the MME/SGSN rejects, due to power saving. The command defines the maximum duration of the procedure suspension, but the MME/SGSN may resume the network initiated procedures before the configured time.

Parameters***delay-tolerant-connection-suspension***

specifies the maximum suspension time for delay tolerant procedure, in seconds

Values: 0 to 1000

Default: 0

5.2.1.3.53 delete-adc-dedicated-bearer-on-idle-timeout-only

Syntax

`[no] delete-adc-dedicated-bearer-on-idle-timeout-only`

Context

`config>mobile>pdn>apn`

Description

This command enables the deletion of the ADC dedicated bearer on idle timeout. When this command is enabled, a dedicated bearer created as a result of an AA application detection is terminated only when the dedicated bearer idle timeout expiration is reached (configured using the `config>mobile>pdn>apn>dedicated-bearer-idle-timeout-profile` command). The no form of this command reverts to the default (disabled; the gateway initiated dedicated bearer is terminated when an application STOP is detected).

Default

no delete-adc-dedicated-bearer-on-idle-timeout-only

5.2.1.3.54 destination-realm-gx

Syntax

`destination-realm-gx name`

`no destination-realm-gx`

Context

`config>mobile>pdn>apn`

Description

This command configures an APN-specific destination realm to use in the Gx interface. If configured, it overrides the destination realm configured in the `config>mobile>profile>diameter-peer` context.

If no value is configured, the destination-realm configured in the `diameter-peer` is used.

The no form of this command reverts to the default.

Default

no destination-realm-gx

Parameters

name

specifies the destination realm name, up to 80 characters

5.2.1.3.55 dhcp-profile

Syntax

```
dhcp-profile name  
no dhcp-profile
```

Context

```
config>mobile>pdn>apn
```

Description

This command configures the DHCP profile for the session.

The no form of this command removes the configuration.

Default

```
no dhcp-profile
```

Parameters

name

specifies the DHCP profile name, up to 32 characters

5.2.1.3.56 dhcpv4-server

Syntax

```
dhcpv4-server profile-name  
no dhcpv4-server
```

Context

```
config>mobile>pdn>apn
```

Description

This command configures the Dynamic Host Configuration Protocol version 4 (DHCPv4) server profile for this Access Point Name (APN).

The no form of this command removes the profile name from the configuration.

Parameters

profile-name

specifies the profile name, up to 32 characters

5.2.1.3.57 dhcpv4-server-group

Syntax

dhcpv4-server-group *profile-name*

no dhcpv4-server-group

Context

config>mobile>pdn>apn

Description

This command configures the DHCPv4 server group from which an IPv4 address would be allocated.

The no form of this command removes the profile name from the configuration.

Parameters

profile-name

specifies the profile name, up to 32 characters

5.2.1.3.58 dhcpv6-server-group

Syntax

dhcpv6-server-group *profile-name*

no dhcpv6-server-group

Context

config>mobile>pdn>apn

Description

This command configures the DHCPv6 server group from which an IPv6 address would be allocated.

The no form of this command removes the profile name from the configuration.

Default

no dhcpv6-server-group

Parameters

profile-name

specifies the profile name, up to 32 characters

5.2.1.3.59 dhcpv6-service

Syntax

dhcpv6-service [enable | disable]

no dhcpv6-service

Context

config>mobile>pdn>apn

Description

This command enables or disables the DHCPv6 service that provides DNSv6 addresses to the UE. When enabled, the PGW advertises this capability by setting the 'O' flag in the IPv6 router advertisements messages. PGW advertises this capability by setting the 'O' flag in the IPv6 router advertisements messages. This allows the UE to use the DHCPv6 service to obtain the DNSv6 addresses; the PGW processes the DHCPv6 Request message, and builds the DHCPv6 Reply message with the DNSv6 addresses toward the UE. When used in CUPS deployment, the command is only configured on the CP function, and the DHCPv6 service is handled on the CP function. Therefore, the CP function installs additional PFCP rules between the CP and UP function, and the PFCP sessions. These rules handle how the UP function detects, forwards, and tunnels the ICMPv6 RS/RA and DHCPv6 messages between the UE and the CP function for processing on the CP function.

The no form of this command removes the service.

Default

no dhcpv6-service

Parameters

enable

specifies that the DHCPv6 service is enabled

disable

specifies that the DHCPv6 service is disabled

5.2.1.3.60 disable-bearer-bind-verification

Syntax

```
[no] disable-bearer-bind-verification
```

Context

```
config>mobile>pdn>apn
```

Description

This command disables bearer binding verification for uplink traffic for a PDN session. By default, bearer binding verification is enabled as standard behavior.

This APN level command disables uplink bearer binding verification for the entire session. The command `disable-bearer-bind-verification` at the `policy-rule-base` level disables uplink bearer binding verification for the bearer on which the `policy-rule-base` is installed.

The no form of this command reverts to the default.

Default

no disable-bearer-bind-verification (uplink bearer binding verification is enabled)

5.2.1.3.61 disable-frame-route

Syntax

```
[no] disable-frame-route
```

Context

```
config>mobile>pdn>apn
```

Description

This command disables the support of Framed Routing on the specified APN/DNN.

The no form of this command reverts to the default (enabled).

5.2.1.3.62 disable-gbr-policing-qci

Syntax

```
[no] disable-gbr-policing-qci qci qci [arp arp-value]
```

Context

```
config>mobile>pdn>apn
```

Description

This command disables the GBR policing QCI.

Parameters

qci

specifies the QCI value

Values: 1 to 9 | 65 | 66 | 69 | 70 | 128 to 254 | *-wildcard

arp-value

specifies the ARP value

Values: 1 to 15 | *-wildcard

5.2.1.3.63 dnn-overload-control

Syntax

```
[no] dnn-overload-control
```

Context

```
config>mobile>pdn>apn
```

Description

This command enables the DNN overload control when the value set by the `nsmf-dnn-threshold max-pdu-connections` or the `nsmf-pdn-threshold max-pdu-activations` command is reached. The SMF in subsequent `Nsmf_PDUSession_CreateSMContext` with a PDU Session Establishment Create Request rejects the request and responds with HTTP 503 including the error attribute/ProblemDetails set to DNN_CONGESTION, the `n1SmMsg` set to PDU SESSION ESTABLISHMENT REJECT, and the N1 back-off timer set as configured. If the HTTP header field Retry-After (in seconds) is also configured, this field will also be included in the HTTP 503 response.

The `no` form of the command disables the DNN overload control.

Default

no dnn-overload-control

5.2.1.3.64 n1-back-off-timer

Syntax

n1-back-off-timer *minutes*

no n1-back-off-timer

Context

config>mobile>pdn>apn>dnn-overload-control

Description

This command configures the N1 back-off timer (GPRS timer 3) in the n1SmMsg of the PDU Session Establishment Reject message.

The no form of this command reverts to the default.

Default

no n1-back-off-timer

Parameters

minutes

specifies the n1SmMsg back-off timer (GPRS timer 3) in minutes

Values: 0 to 120

Default: 1

5.2.1.3.65 retry-after-timer

Syntax

retry-after-timer *seconds*

no retry-after-timer

Context

config>mobile>pdn>apn>dnn-overload-control

Description

This command configures the Retry-After field in the HTTP header of the 503 error respond when the DNN-level overload threshold has been reached.

The no form of this command reverts to the default (disabled).

Default

no retry-after-timer

Parameters

seconds

specifies the retry timer in seconds

Values: 2 to 36000

Default: 0 (disabled)

5.2.1.3.66 dns-redirect-server

Syntax

dns-redirect-server *profile-name*

no dns-redirect-server

Context

config>mobile>pdn>apn

Description

This command specifies the dns-redirect server profile associated for this APN.

Default

no dns-redirect-server

Parameters

profile-name

specifies the DNS redirect profile name, up to 32 characters

5.2.1.3.67 dns-snooping

Syntax

```
dns-snooping [dns-server-list addr-list] [snoop-all]  
no dns-snooping
```

Context

```
config>mobile>pdn>apn
```

Description

This command enables DNS snooping for the APN. When enabled, the UE DNS traffic is analyzed (DNS responses from validated DNS servers are snooped) and the IP addresses sent by the DNS server are stored to the mobile gateway. Any future traffic to these IP addresses, matches the corresponding policy rule filters with the domain names according to the lookup from the local storage.

DNS snooping can be configured to snoop all responses, using the `snoop-all` keyword, or responses from a DNS server IP address list (including the DNS servers defined in APN PCO), using the `dns-server-list` keyword. If no parameter is specified, DNS responses are snooped from the DNS servers defined in APN PCO.

The `no` form of this command disables DNS snooping.

Default

```
no dns-snooping
```

Parameters

addr-list

specifies the IP address list name of the DNS server, up to 32 characters; enables APN PCO and DNS server list DNS snooping (the *addr-list* is configured using the `config>mobile>profile>addr-list` command)

snoop-all

specifies to snoop all DNS queries

5.2.1.3.68 dns-snooping-forward-to-aa

Syntax

```
[no] dns-snooping-forward-to-aa
```

Context

config>mobile>pdn>apn

Description

This command enables forwarding of DNS traffic to AA when the dns-snooping command is enabled in the config>mobile>pdn>apn context.

The no form of this command disables forwarding traffic to AA when DNS snooping is enabled.

Default

no dns-snooping-forward-to-aa

5.2.1.3.69 dynamic-rule-policing

Syntax

dynamic-rule-policing qci *qci* [arp *arp-value*] [direction *direction*] [group *group*]

no dynamic-rule-policing qci *qci* [arp *arp-value*]

Context

config>mobile>pdn>apn

Description

This command enables policing per SDF for dynamic policy rules for GBR and non-GBR bearers and applies to LTE and 5G. If not configured, the MBR values sent in the non-GBR bearers dynamic policy rules are ignored.

The no form of this command disables the policing for new bearers or modification of the existing bearers.



NOTE:

- The CMG maximum number of policers is 8 million. The number of policers cannot be changed.
- In UPF, the ARP value must always be a wildcard.

Default

no dynamic-rule-policing

Parameters

qci

specifies the QCI value
Values: 1 to 255 | *-wildcard

arp-value

specifies the ARP value
Values: 1 to 15 | *-wildcard

direction

specifies the direction of the flow

Values	dl	specifies the downlink direction
–	bidir	specifies both uplink and downlink direction

Default: dl

group

specifies the level at which the rule is policed
Values: sdf
Default: sdf

5.2.1.3.70 dynamic-rule-to-internal-rule-map

Syntax

dynamic-rule-to-internal-rule-map *profile-name*
no dynamic-rule-to-internal-rule-map

Context

config>mobile>pdn>apn

Description

This command configures the APN to derive an internal static rule to use for flow classification. This rule facilitates using the CAM for PCC flows that are not specified as a 5-tuple when sent from the PCRF. Some applications such as Rich Chat Service (RCS) and Push To Talk (PTT) use dedicated application servers, in which case the remote IP/port information pertains to the dedicated servers.

Based on this information, an internal static rule is derived that applies across different chat sessions for all UEs. The flow description of the specified mapping profile is used to derive the internal static

rule. The mapping profile must be configured in the context `config>mobile>profile>policy-options>dynamic-rule-to-internal-rule-map` prior to associating it with the APN.

The no form of this command reverts to the default.

Default

no dynamic-rule-to-internal-rule-map

Parameters

profile-name

specifies the mapping profile of the flow description, up to 32 characters

5.2.1.3.71 enable-global-pcc-precedence

Syntax

[no] enable-global-pcc-precedence

Context

config>mobile>pdn>apn

Description

This command allows the user to use the precedence value of the policy rule to match the traffic against all installed rules, regardless of whether the rules are diverted to AA or not.

If enable-global-pcc-precedence is not configured, all policy rules handled by AA are given higher precedence (matched first) over non-AA rules.

If enable-global-pcc-precedence is configured, policy rule precedence is applied regardless of whether the policy rule is diverted to AA or not. If more than one rule with the same flow description is installed, traffic will match the policy rule according to the precedence.

The no form of this command reverts to the default (not configured).

Default

no enable-global-pcc-precedence

5.2.1.3.72 enable-gx-session-recovery

Syntax

[no] enable-gx-session-recovery

Context

```
config>mobile>pdn>apn
```

Description

This command enables Gx session discovery and recovery for the APN.

The no form of this command reverts to the default (disabled).

Default

no enable-gx-session-recovery

5.2.1.3.73 ethernet-action-dl

Syntax

```
ethernet-action-dl action action-type [n3-c-vlan vlan]  
no ethernet-action-dl
```

Context

```
config>mobile>pdn>apn
```

Description

This command specifies the action that must be applied on the DL frames.

The no form of this command reverts to the default.

Parameters

action-type

Values	pop	specifies that single un-tagging is applied on the DL frames
–	pop-pop	specifies that double un-tagging is applied on the DL frames
–	swap	specifies that single C-VLAN swapping is applied on the DL frames; the <i>n3-c-vlan</i> value must be provided

	–	swap-pop	specifies that C-VLAN swapping and single S-VLAN un-tagging is applied on the DL frames
	Default: pop		
vlan			
	Values: 0 to 4094		

5.2.1.3.74 ethernet-action-up

Syntax

```
ethernet-action-up action action-type tag-value tag-value  
no ethernet-action-up
```

Context

```
config>mobile>pdn>apn
```

Description

This command specifies the action that must be applied on the UL frames. The required VLAN tagging per action must be specified.

The no form of this command reverts to the default.

Parameters

<i>action-type</i>		
	Values	
	push	specifies that single tagging is applied on the UL frames
	–	
	push-push	specifies that double tagging is applied on the UL frames
	–	
	swap	specifies that single C-VLAN tagging is applied on the UL frames

	–	swap-push	specifies that single C-VLAN swapping and single S-VLAN tagging is applied on the UL frames
Default: push			
tag-value			
	Values	n7-c-vlan	specifies that C-VLAN single tagging is applied on the UL frames
	–	n7-s-vlan	specifies that S-VLAN single tagging is applied on the UL frames
	–	n7-c-vlan-s-vlan	specifies that C-VLAN and S-VLAN double tagging is applied on the UL frames
Default: none			

5.2.1.3.75 ethernet-service-type

Syntax

```
ethernet-service-type service-type
no ethernet-service-type
```

Context

```
config>mobile>pdn>apn
```

Description

This command specifies the ethernet service type of an ethernet PDU session. You can configure either eline or elan as a service type.

The no form of this command reverts to the default.

Parameters

service-type

Values	eline	specifies that eline (P2P service with no MAC learning) ethernet service is supported by this APN
–	elan	specifies that elan (P2M service with MAC learning) ethernet service is supported by this APN

Default: eline

5.2.1.3.76 firewall-rules

Syntax

```
firewall-rules
```

Context

```
config>mobile>pdn>apn
```

Description

This command enables the context to configure the firewall policy rules and firewall rules.

5.2.1.3.77 abort

Syntax

```
abort
```

Context

```
config>mobile>pdn>apn>firewall-rules
```

Description

This command ends the current policy editing session that begins when using the `config>mobile>pdn>apn>firewall-rules>begin` command, and discards any policy changes made during the editing session.

5.2.1.3.78 begin

Syntax

begin

Context

config>mobile>pdn>apn>firewall-rules

Description

This command begins a policy editing session. Changes made during the editing session apply when the config>mobile>pdn>apn>firewall-rules>commit command is used. The editing session continues until either the config>mobile>pdn>apn>firewall-rules>abort or the config>mobile>pdn>apn>firewall-rules>commit command is used, or until a control complex reset takes place. Editing sessions are not interrupted by high-availability activity switches or CLI session terminations, such as ending a Telnet session.

5.2.1.3.79 commit

Syntax

commit

Context

config>mobile>pdn>apn>firewall-rules

Description

This command saves the changes made during the current policy editing session (using the config>mobile>pdn>apn>firewall-rules>begin command). Any policy changes made during the editing session apply only when they are committed. If changes are successfully committed without errors detected during the commit or cross-reference verification against the existing policy configuration, the editing session will also be closed.

5.2.1.3.80 diff

Syntax

diff

Context

config>mobile>pdn>apn>firewall-rules

Description

This command displays the changes that were not committed during a policy editing session (changes made with the config>mobile>pdn>apn>firewall-rules>begin and config>mobile>pdn>apn>firewall-rules>commit commands).

5.2.1.3.81 policy-rule

Syntax

[no] policy-rule *policy-rule-name*

Context

config>mobile>pdn>apn>firewall-rules

Description

This command configures under the APN the firewall policy rules that are applied to PDN sessions locally by the gateway. The policy rules must be configured under the config>mobile>profile>policy-options>policy-rule context before they are applied under this CLI context.

Parameters

policy-rule-name

specifies the policy rule name, up to 64 characters

5.2.1.3.82 policy-rule-base

Syntax

[no] policy-rule-base *base-policy-name*

Context

config>mobile>pdn>apn>firewall-rules

Description

This command configures under the APN the firewall policy-rule bases that are applied locally to the PDN sessions by the gateway. The policy-rule bases must be configured under the

config>mobile>profile>policy-options>policy-rule-base context before they are applied under this CLI context.

Parameters

base-policy-name

specifies the name of the policy-rule base, up to 64 characters

5.2.1.3.83 flow-inactivity-timeout

Syntax

flow-inactivity-timeout *inactivity-timeout*

no flow-inactivity-timeout

Context

config>mobile>pdn>apn

Description

This command configures the flow inactivity timeout timer for AA offloaded flows. If no activity is detected during the configured time, the AA offloaded flow is deleted by the mobile gateway.

The no form of this command reverts to the default.

Parameters

inactivity-timeout

specifies the value, in seconds, of the inactivity timeout timer

Values: 60 to 600

Default: 60

5.2.1.3.84 fqdn-dns-redirect-server

Syntax

fqdn-dns-redirect-server *dns-server-name*

no fqdn-dns-redirect-server

Context

config>mobile>pdn>apn

Description

This command associates the specified DNS redirect server profile to this APN for FQDN DNS redirect.

Default

no fqdn-dns-redirect-server

Parameters

dns-server-name

specifies the name of the DNS server, up to 32 characters

5.2.1.3.85 fqdn-group

Syntax

fqdn-group *fqdn-group-name*

no fqdn-group

Context

config>mobile>pdn>apn

Description

This command associates the specified FQDN group to the APN. The FQDN group is configured in the profile context with a list of URLs. The URL list is resolved to a set of IP addresses (periodically refreshed) using the DNS server configured under the PCO for the APN. Multiple FQDN groups can be configured under an APN.

The no form of this command reverts to the default (removes the specified FQDN group from the APN configuration).

Parameters

fqdn-group-name

specifies the name of the FQDN group, up to 32 characters

5.2.1.3.86 gbr-policing-direction

Syntax

gbr-policing-direction qci *qci*[arp *arp-value*] {none | dl | bidir}

```
no gbr-policing-direction qci qci [arp arp-value]
```

Context

```
config>mobile>pdn>apn
```

Description

This command configures the policing that is performed in the APN for GBR dedicated bearers based on direction. The bandwidth of the GBR dedicated bearers is performed based on the aggregate of the MBR/GBR values defined in all dynamic rules bound to the bearer.

The no form of this command removes the configured policing of the GBR dedicated bearers in the APN.

Default

```
no gbr-policing-direction qci qci [arp arp-value]
```

Parameters

qci

specifies the QCI value

Values: 1 to 255 | *-wildcard

arp-value

specifies the optional ARP value

Values: 1 to 15 | *-wildcard

none

specifies no direction

dl

specifies downlink direction

bidir

specifies bidirectional direction

Default: dl

5.2.1.3.87 gtp-pdn-threshold

Syntax

```
gtp-pdn-threshold max-activations max-activations [alarm]
```

```
gtp-pdn-threshold max-connections max-connections [alarm]
```

```
no gtp-pdn-threshold
```

Context

```
config>mobile>pdn>apn
```

Description

This command configures the threshold of the maximum number of GTP-based PDN connections or the maximum number of GTP-based PDN activations per second for this APN. It is used for PGW APN-level congestion control (PGW Back-off timer or GTP-C Load and Overload).

An alarm can be enabled or disabled to be sent when the maximum number of connections or the maximum number of activations is reached.



NOTE:

This command must not be enabled when the `max-pdn-connection` command is enabled in the `conf>mobile>pdn>apn` context.

Default

```
no gtp-pdn-threshold
```

Parameters

max-activations

specifies the maximum number of PDN activations per second

Values: 0 to 5000 activations per second

Default: 0 (disabled)

max-connections

specifies the maximum number of PDN connections

Values: 1 to 20000000 connections

Default: 0

alarm

specifies if an alarm will be sent when the maximum number of connections or the maximum number of activations is reached

5.2.1.3.88 gtpu-source-port-dist

Syntax

```
[no] gtpu-source-port-dist
```

Context

```
config>mobile>pdn>apn
```

Description

This command enables a dynamic UDP source port in the range 57344 to 57599 that is used for all egress GTP-U packets on a flow basis.

The no form of this command reverts to the default (disabled).

Default

```
no gtpu-source-port-dist
```



NOTE:

When enabled, the `gtpu-source-port-dist` command takes precedence at the APN level over the `use-fixed-udp-src-port` command configured in the `config>mobile>system` context.

5.2.1.3.89 guard-timer

Syntax

```
guard-timer guard-timer
```

```
no guard-timer
```

Context

```
config>mobile>pdn>apn
```

Description

This command enables the configuration of a session setup timer on the PGW. If the session setup is delayed due to non response from the peering nodes for an extended duration (AAA/RADIUS, PCRF, OCS/OFCs), the session state is cleaned up and the response is sent with a failure cause code. This command applies to real APN.

Default

```
no guard-timer
```

Parameters

guard-timer

specifies the session setup time, in seconds

Values: 4 to 120

Default: 0 (disabled)

5.2.1.3.90 **gx**

Syntax

gx

Context

config>mobile>pdn>apn

Description

This command configures the Gx attributes for the APN.

5.2.1.3.91 **gtp-precedence-mapping**

Syntax

gtp-precedence-mapping {no-override | standard-algorithm | VoLTE-algorithm}

no gtp-precedence-mapping

Context

config>mobile>pdn>apn>gx

Description

This command configures the precedence mapping algorithm that maps the Gx PCC rule precedence to the (GTP) TFT precedence. The standard algorithm overrides the algorithm applied when the custom-1 option is selected for applicable CLI commands used for a specific operator. The VoLTE-algorithm specifies VoLTE precedence mapping.

The no form of this command reverts to the default (no-override).

Parameters

no-override | standard-algorithm | VoLTE-algorithm

specifies the GTP precedence mapping type

Default: no-override

5.2.1.3.92 include-rat-when-received

Syntax

```
[no] include-rat-when-received
```

Context

```
config>mobile>pdn>apn>gx
```

Description

This command enables the including of a RAT type AVP in the CCR-U to the PCRF if the message is due to any GTP signaling that includes the RAT type IE.

By default, a RAT type AVP is reported to the PCRF only in the CCR-I, and only when a RAT change event is detected.

The no form of this command reverts to the default (disabled).

Default

```
no include-rat-when-received
```

5.2.1.3.93 ho-failure-release-pdusession

Syntax

```
ho-failure-release-pdusession [enable | disable]  
no ho-failure-release-pdusession
```

Context

```
config>mobile>pdn>apn
```

Description

This command defines the policy when a handover fails in the RAN. When enabled, the SMF releases the PDU session after handover failure. If disabled, the SMF releases the user plane of the PDU session without releasing the whole PDU session.

Parameters

enable

specifies that the SMF releases the PDU session after a handover failure

disable

specifies that the SMF releases the user plane after a handover failure

Default: enabled

5.2.1.3.94 ho-wrong-pdn-ip-stale-session-deletion

Syntax

ho-wrong-pdn-ip-stale-session-deletion [enable | disable]

no ho-wrong-pdn-ip-stale-session-deletion

Context

config>mobile>pdn>apn

Description

This configuration defines the policy when a handover fails in the RAN. When the feature is enabled, the SMF will release the PDU session after handover failure. If the feature is disabled, the SMF will release the user plane of the PDU session without releasing the whole PDU session.

Parameters**enable**

specifies that the PGW deletes the stale session

disable

specifies that the PGW does not delete the stale session

Default: enable

5.2.1.3.95 hybrid-access

Syntax

hybrid-access

Context

config>mobile>pdn>apn

Description

This command enables the context to configure hybrid access parameters.

5.2.1.3.96 bonding-support

Syntax

[no] bonding-support

Context

config>mobile>pdn>apn>hybrid-access

Description

This command enables bonding support for the APN.

The no form of this command reverts to the default.

Default

no bonding-support

5.2.1.3.97 link-switch

Syntax

link-switch

Context

config>mobile>pdn>apn>hybrid-access

Description

This command enables the context to configure link switch parameters.

5.2.1.3.98 mode

Syntax

mode

Context

config>mobile>pdn>apn>hb-acc>lnk-sw

Description

This command enables the context to configure link switch mode parameters.

5.2.1.3.99 uplink-traffic

Syntax

```
uplink-traffic
```

Context

```
config>mobile>pdn>apn>hb-acc>lnk-sw>mode
```

Description

This command enables the context to configure uplink traffic parameters.

5.2.1.3.100 hysteresis

Syntax

```
hysteresis time
```

```
no hysteresis
```

Context

```
config>mobile>pdn>apn>hb-acc>lnk-sw>mode>uplink-traffic
```

Description

This command specifies the time that the uplink traffic detection waits before selecting another UE in the multiple UE session, after a new UE has just been detected and selected.

The no form of this command reverts to the default.

Default

no hysteresis

Parameters

time

specifies the hysteresis time, in milliseconds

Values: 200 to 2000

Default: 500

5.2.1.3.101 monitoring-key

Syntax

`monitoring-key monitoring-key-profile-name`

`no monitoring-key`

Context

`config>mobile>pdn>apn>hybrid-access`

Description

This command specifies a monitoring key profile for the hybrid access. The monitoring key profile is configured in the policy options usage monitoring information context; see the `usage-monitoring-info` command.

The `no` form of this command removes the specified monitoring key profile.

Parameters

monitoring-key-profile-name

specifies the profile name, up to a 32 characters

5.2.1.3.102 idle-timeout

Syntax

`idle-timeout idle-timeout`

`no idle-timeout`

Context

`config>mobile>pdn>apn`

Description

This command configures the length of time a PDP context is allowed to be idle before it is removed by the PGW. The `idle-timeout` value itself belongs to the PDN connection. The `idle-timeout` is applied per PDN context. If there is data traffic on any of the bearers (default or dedicated), the `idle-timeout` is reset.

If there is no traffic on any of the bearers and the timer expires, all bearers are removed for that PDN context. High idle timeout values are relevant for those scenarios where the APN is reserved for machine-type communication where the UE may remain idle for an extended period of time.

The idle timer is reset if the PGW receives any PDU corresponding to the PDP context (uplink or downlink). This includes GTP-U PDUs. The idle timeout is changed without a PDN shutdown and the change affects new sessions only.

The no form of this command reverts to the default.

Default

no idle-timeout

Parameters

idle-timeout

specifies the amount of time, in seconds, a bearer must be idle before it is released

Values	0 5 to 2592000 (emergency APN)
--------	----------------------------------

–	0 60 to 2592000 (normal APN)
---	--------------------------------

Default: 0 (disabled)

5.2.1.3.103 idle-timer-direction-downlink

Syntax

[no] idle-timer-direction-downlink

Context

config>mobile>pdn>apn

Description

This command enables the idle timer to monitor the downlink traffic.

The `idle-timer-direction-uplink` and the `idle-timer-direction-downlink` commands allow the following combinations:

- when the `idle-timer-direction-downlink` and the `idle-timer-direction-uplink` commands are enabled, both downlink and uplink traffic are monitored

- when the `idle-timer-direction-downlink` command is disabled, and the `idle-timer-direction-uplink` command is enabled, only the uplink traffic is monitored
- when the `idle-timer-direction-uplink` command is disabled, and the `idle-timer-direction-downlink` command is enabled, only the downlink traffic is monitored
- when the `idle-timer-direction-downlink` command and the `idle-timer-direction-uplink` commands are disabled, both downlink and uplink traffic are monitored

The no form of this command reverts to the default action (disabled).

**NOTE:**

In CUPS deployments, Nokia recommends to avoid using the `idle-timer-direction-downlink` or the `idle-timer-direction-uplink` command on high-scale APNs because separate uplink and downlink SDF filters are used in the Create PDRs over PFCP/Sx-N4 instead of bidirectional SDF filters.

Default

`no idle-timer-direction-downlink`

5.2.1.3.104 `idle-timer-direction-uplink`

Syntax

`[no] idle-timer-direction-uplink`

Context

`config>mobile>pdn>apn`

Description

This command enables the idle timer to monitor the uplink traffic.

The `idle-timer-direction-uplink` and the `idle-timer-direction-downlink` commands allow the following combinations:

- when the `idle-timer-direction-downlink` and the `idle-timer-direction-uplink` commands are enabled, both downlink and uplink traffic are monitored
- when the `idle-timer-direction-downlink` command is disabled, and the `idle-timer-direction-uplink` command is enabled, only the uplink traffic is monitored
- when the `idle-timer-direction-uplink` command is disabled, and the `idle-timer-direction-downlink` command is enabled, only the downlink traffic is monitored
- when the `idle-timer-direction-downlink` and the `idle-timer-direction-uplink` commands are disabled, both downlink and uplink traffic are monitored

The no form of this command reverts to the default action (disabled).

**NOTE:**

In CUPS deployments, Nokia recommends to avoid using the `idle-timer-direction-downlink` or the `idle-timer-direction-uplink` command on high-scale APNs because separate uplink and downlink SDF filters are used in the Create PDRs over PFCP/Sx-N4 instead of bidirectional SDF filters.

Default

no idle-timer-direction-uplink

5.2.1.3.105 ignore-up-ip-allocation

Syntax

[no] ignore-up-ip-allocation

Context

config>mobile>pdn>apn

Description

This command ignores the UP-based IP address allocation configuration on the UP peer and instructs the system to perform CP-based IP address allocation.

The no form of this command reverts to the default (disabled).

Default

no ignore-up-ip-allocation

5.2.1.3.106 ip-alloc-allow-local

Syntax

[no] ip-alloc-allow-local

Context

config>mobile>pdn>apn

Description

This command enables the allocation of a PDN IP address from a local pool when an AAA server provides only one PDN IP address in response to a request for a PDN IPv4v6 address. For an HSS-assigned static address, if only one PDN IP address is provided via GTP signaling, the missing PDN IP address is assigned from the local pool.

The no form of this command reverts to the default (disabled).



NOTE:

The mobile gateway must accept the IP address received from the AAA server before the secondary IP address is allocated.

Default

no ip-alloc-allow-local

5.2.1.3.107 ip-alloc-extended-hss-allocation

Syntax

[no] ip-alloc-extended-hss-allocation

Context

config>mobile>pdn>apn

Description

This command enables validation of the external IP addresses provided via GTP against the remote IP pools configured in the APN. The session is rejected if the static IPv4 address/IPv6 prefix provided does not match the remote IP pools configured in the APN. The cause of rejection is "User Authentication Failure (92)".

The no form of this command reverts to the default (disabled).

Default

no ip-alloc-extended-hss-allocation

5.2.1.3.108 ip-alloc-extended-radius-allocation

Syntax

[no] ip-alloc-extended-radius-allocation

Context

config>mobile>pdn>apn

Description

This command specifies the following extended logic for handling UE IP address allocation-related parameters in the Accept-Accept response message from the RADIUS server.

- If the Access-Accept message contains 255.255.255.254 or 255.255.255.255 in the Framed-IP-Address, or the Framed-IP-Address is missing from the Accept-Accept message, the system allocates an IPv4 address for the UE from any of the IPv4 pools associated with the APN. This choice is available when an IPv4 address is allocated for the IPv4 or dual stack session. By default, the session activation fails for this case.
- If the Access-Accept message contains the directed broadcast IP address of any of the IPv4 subnets associated with the APN, the system allocates the IPv4 address for the UE from this subnet. The directed broadcast IP address of a subnet is the last IP address of the subnet. The directed broadcast IP address of the subnet is never allocated to the UE. For example, if the subnet of an IP address pool is 192.168.1.128/25, the directed broadcast IP address is 192.168.1.255. This choice is available when an IPv4 address is allocated for the IPv4 or dual stack session. By default, the system allocates the directed broadcast IP address to the UE for this case.
- In a few special cases, the session is rejected and the GTP cause is set to User Authentication Failure (92), including:
 - the IPv4/IPv6 address is outside the range of the configured remote IP pools; by default, a separate /32 address or /64 prefix would be allocated in this case
 - the IPv4 address is a subnet network address
 - the IPv4/IPv6 address value contains all zeros
 - the same IPv4/v6 address would be assigned for different sessions (rejected by default, only cause code changes)

The no form of this command reverts to the default.

Default

no ip-alloc-extended-radius-allocation

5.2.1.3.109 ip-alloc-method

Syntax

```
ip-alloc-method {local-pool | hss-static | aaa | dhcp-client | dhcp-relay} [{local-pool | hss-static | aaa | dhcp-client | dhcp-relay}...(up to 4 max)]  
no ip-alloc-method
```

Context

```
config>mobile>pdn>apn
```

Description

This command specifies the IP address allocation scheme used by APN for assigning IP addresses to UE.



NOTE:

- In the current release, dhcp-relay is not supported.
- ip-alloc-method can be changed without PDN or APN shutdown

The no form of this command reverts to the default.

Parameters

local-pool

specifies that the UE is assigned by a IP address defined in the local-pool under VPRN

hss-static

specifies the static IP address assigned by HSS to MME and conveyed to PGW during Create Session Request

aaa

specifies that the UE IP address is allocated using RADIUS/DIAMETER server using the framed-ip and framed-pool-name

dhcp-client

specifies that the UE IP address is allocated using DHCP

dhcp-relay

reserved value, not supported in the current release

Default: local-pool

5.2.1.3.110 ip-alloc-priority-order

Syntax

```
ip-alloc-priority-order hss-static  
no ip-alloc-priority-order
```

Context

```
config>mobile>pdn>apn
```

Description

This command specifies the priority order in which the UE IP address allocation methods are evaluated.

The no form of this command removes the configuration, meaning that the UE IP address allocation is done using the default order.

Default

```
no ip-alloc-priority-order
```

Parameters

hss-static

specifies that the HLR/HSS-provided static IP address is used if included in the GTP Create PDP Context Request/Create Session Request message, and this APN will also allow dynamic IP address allocation methods which would be otherwise preferred over the static IP

5.2.1.3.111 ip-allocation-trace

Syntax

```
[no] ip-allocation-trace
```

Context

```
config>mobile>pdn>apn
```

Description

This command enables syslog reporting when an IP address is allocated or released for the PDN connection.

The no form of this command reverts to the default behavior (reporting disabled).

Default

no ip-allocation-trace

5.2.1.3.112 ip-pool-exhaust-back-off-timer

Syntax

`ip-pool-exhaust-back-off-timer timer-value`

`no ip-pool-exhaust-back-off-timer`

Context

config>mobile>pdn>apn

Description

This command enables the PGW/PGW-C+SMF/SMF to provide a Back-Off timer IE to the MME (PGW/PGW-C) or a GPRS timer 3, NAS level message, to the 5G UE (SMF) whenever the PGW/PGW-C+SMF/SMF rejects a PDN/PDU session cause of IP pool exhaustion.

The command is available only to the real APN/DNN configuration context, not the virtual APN/DNN. The logic applies to both real and any associated virtual APN/DNN.

If a real APN/DNN, on which the `ip-pool-exhaust-back-off-timer` is enabled, has more than one virtual APNs/DNNs configured and the IP address pool exhaustion takes place on any of the associated virtual APNs/DNNs, the CMG includes the PGW Back-Off Time IE or GPRS timer 3 in the response to the MME/AMF.



NOTE:

- The PGW Back-Off Time IE is only reported over the GTPv2-C based on 3GPP TS 29.274. The GPRS timer 3 is included on the N11 interface message response from the SMF to AMF, and is delivered directly to the UE.
- When the `ip-pool-exhaust-back-off-timer` is active, for any session setup rejection due to IP address pool exhaustion, the PGW rejects the session with GTPv2 cause code 113 “APN Congestion”. For N11 interface message response from SMF to AMF, the “5GSM cause” value sent to the UE is the “Insufficient resources”.

Default

no ip-pool-exhaust-back-off-timer

Parameters

timer-value

specifies the time during which the MME or 5G UE should refrain from sending subsequent PDN/PDU connection establishment requests to the PGW/PGW-C or SMF for the specific IP pool exhausted APN/DNN use case, for services other than service users and emergency services, in minutes

Values: 1 to 120

Default: 1

5.2.1.3.113 kpi-kci

Syntax

```
kpi-kci
```

Context

```
config>mobile>pdn>apn
```

Description

This command enables a context to configure KPI-KCI parameters per APN. A configuration change using this command requires the APN to be in a shutdown state.

5.2.1.3.114 arp-traffic-stats

Syntax

```
arp-traffic-stats [inherit | enable | disable]
```

```
no arp-traffic-stats
```

Context

```
config>mobile>pdn>apn>kpi-kci
```

Description

This command enables the generation of per ARP (Allocation and Retention Policy) and per QCI Traffic KPI-KCI statistics at the APN. The per ARP and per QCI statistics are collected by KPIBearerTrafficQciArpCP-ISA, KPIBearerManagementQciArpCP-ISA, and KCIBearerManagementQciArpCP-ISA modules. The respective counter module must also be configured in an accounting policy linked to this PDN under the config>mobile>pdn>kpi-kci>accounting-policy context.

If the APN level configuration is explicitly set, it overrides the PDN-level configuration. If not, it inherits the configuration from the PDN level.

The no form of this command reverts to the default.

Parameters

inherit

specifies that the behavior for this APN is inherited from the PDN-level configuration

enable

enables the generation of per ARP traffic statistics

disable

disables the generation of per ARP traffic statistics

Default: inherit

5.2.1.3.115 enhanced-traffic-stats

Syntax

```
enhanced-traffic-stats [inherit | enable | disable]
```

```
no enhanced-traffic-stats
```

Context

```
config>mobile>pdn>apn>kpi-kci
```

Description

This command enables the generation of an enhanced set of statistics at the APN level. When enabled, all traffic statistics are generated for all sessions on this APN.

The respective counter modules containing these counters must also be configured in the accounting policy linked to this PDN, for these counters to be generated.

For more information about which KPI Bearer Traffic modules this configuration applies to, refer to the section of user plane statistics in the *CMG KPI-KCI Counters Guide*.

The no form of this command reverts to the default.

Parameters

inherit

specifies that the behavior for this APN is inherited from the PDN-level configuration

enable

enables the enhanced generation of the traffic statistics

disable

disables the enhanced generation of the traffic statistics

Default: inherit

5.2.1.3.116 qci-apn-stats

Syntax

`qci-apn-stats list-name`

`no qci-apn-stats`

Context

`config>mobile>pdn>apn>kpi-kci`

Description

This command specifies the QCI statistics profile name under the APN. The QCI values in this list are accounted for this APN in the per APN, per QCI statistics.

**NOTE:**

When you configure a QCI list under the `qci-apn-stats` context, the wildcard value is not allowed to be configured in the QCI list. Doing so, an error indicating the same is displayed.

The no form of this command reverts to the default.

Default

`no qci-apn-stats`

Parameters***list-name***

specifies the name of the QCI list profile (configured in the `config>mobile>profile>list>qci-list` context), up to 32 characters

5.2.1.3.117 sgw-apn-stats

Syntax

`[no] sgw-apn-stats`

Context

```
config>mobile>pdn>apn>kpi-kci
```

Description

This command defines that this APN will be accounted in the per APN stats on the SGW. The respective counter modules (KCIBearerManagementSgwApn, KPIBearerManagementSgwApn, KPIBearerTrafficSgwApn) must also be configured in the accounting policy linked to this PDN for the per APN statistics to get generated on the gateway.

The no form of this command disables this APN from being measured in the per APN statistics on the gateway.

Deletion of the APN configuration from the gateway also disables the APN from being measured in the per APN statistics.



NOTE:

The available ePDG KPI/KCI per APN statistics are the following:

- VS.sessionActivationsSuccess
- VS.sessionActvFCC92UsrAuthFailed
- VS.sessionActivationsFailure
- VS.sessionActvFCC68ServNotSuprt
- VS.sessionActvFCC72SystemFailure
- VS.sessionActvFCC73NoRsources
- VS.sessionActvFCC78MissOrUkwnApn
- VS.sessionActvFCC83PrfPdnTypNoSprt
- VS.nwInitDedBearerActvSuccess
- VS.nwInitDedBearerActvFailure

Default

```
no sgw-apn-stats
```

5.2.1.3.118 ldn

Syntax

```
ldn
```

Context

```
config>mobile>pdn>apn
```

Description

This command enables the context to configure the LADN configuration. The context is not required when the LADN is not enabled in the APN.

5.2.1.3.119 enable-ladn

Syntax

```
[no] enable-ladn
```

Context

```
config>mobile>pdn>apn>ladn
```

Description

This command enables the LADN in the APN or the DNN.

The no form of this command disables the LADN in the APN or the DNN.

5.2.1.3.120 on-ue-move-outside-ladn

Syntax

```
on-ue-move-outside-ladn {release-pdu | release-up}
```

```
no on-ue-move-outside-ladn
```

Context

```
config>mobile>pdn>apn>ladn
```

Description

This command configures the PDU session action when the UE moves outside the LADN.

The no form of this command reverts to the default action.

Parameters

release-pdu

specifies that the PDU session is released

release-up

specifies that the UP of the PDU session is suspended

Default: release-pdu

5.2.1.3.121 max-ambr-policer

Syntax

[no] max-ambr-policer

Context

config>mobile>pdn>apn

Description

This command enables the AMBR policer rate to be internally set to a maximum value for this APN. This value overrides the negotiated APN-AMBR for the subscriber's IP-CAN session. The no form of this command reverts to the default (disabled).

When enabled, this command overrides all other APN-AMBR values, either locally configured or obtained from the PCRF.

Default

no max-ambr-policer

5.2.1.3.122 max-pdn-connection

Syntax

max-pdn-connection *max-pdn-connections* {alarm | enforce]

no max-pdn-connection

Context

config>mobile>pdn>apn

Description

This command configures the maximum number of PDN connections for this APN on the gateway, and enables or disables alarms to be sent when the maximum PDN connections are reached or configures send alarms and to not accept any new PDN connections when the *max-pdn-connections* value has been

reached. When the configured maximum number is reached, an alarm event is raised at the end of the pulling interval.

The no form of this command disables and removes the maximum connection limit and disables sending alarm even when the *max-pdn-connections* value is reached.

**NOTE:**

In CUPS deployment, the command is applicable only for the SMF.

Default

Alarm event will be sent when max-pdn-connections limit has been reached.

Parameters

max-pdn-connections

specifies the maximum number of PDN connections

Values: 0 to 20000000

Default: 0

alarm

specifies sending an alarm even when max-pdn-connections has been reached

enforce

specifies sending an alarm and not accept any new PDN connections when max-pdn-connections has been reached

5.2.1.3.123 max-session-attach

Syntax

```
max-session-attach max-attach-limit [alarm] [include-emergency]
```

```
no max-session-attach
```

Context

```
config>mobile>pdn>apn
```

Description

This command configures the threshold of the maximum number of GTP-based attach requests per second. The limit is applied to the new PDN (GTPv2) and PDP (GTPv1) session attach requests received. By default, the emergency attach requests are excluded from the threshold. When this command is enabled, the *alarm* keyword specifies to raise an alarm when the maximum number of attaches is reached.

The no form of this command reverts to the default (no limit is applied to the number of attach requests).

**NOTE:**

This command cannot be enabled when APN overload is enabled for the APN.

Default

no max-session-attach

Parameters***max-attach-limit***

specifies the maximum number of attach requests per second

Values: 0 to 100000

Default: 0

alarm

specifies to raise an alarm when the maximum number of attach requests per second is reached

Default: disabled

include-emergency

specifies to include the emergency attach requests to the threshold

Default: disabled

5.2.1.3.124 min-pdn-connection

Syntax

min-pdn-connection *min-pdn-connections*

no min-pdn-connection

Context

config>mobile>pdn>apn

Description

This command configures the minimum number of PDN connections that can be allocated on the gateway for this APN.

The no form of this command removes PDN connections allocated to this APN.

Parameters

min-pdn-connections

specifies the minimum number of PDN connections

Values: 0 to 20000000

Default: 0

5.2.1.3.125 multi-ue-bonding

Syntax

[no] multi-ue-bonding

Context

config>mobile>pdn>apn

Description

This command enables multiple UE session bonding support. This requires the Framed-Route attribute assigned by the RADIUS server to be set. When enabled, the PGW allows up to three individual UEs to be programmed with the same Framed-Route attribute information, making them routable on the SGi interface.

The PGW forwards uplink traffic from any of the UEs on the SGi interface. The downlink traffic received on the SGi interface, for any of the Framed-routes, the PGW forwards it on the UE that was last detected to send uplink traffic to.

The no form of this command disables multiple UE session bonding support.

Default

no multi-ue-bonding

5.2.1.3.126 namf-profile

Syntax

namf-profile *profile-name*

no namf-profile

Context

config>mobile>pdn>apn

```
config>mobile>pdn>sba-client-services>amf-client>namf-comm  
config>mobile>pdn>sba-client-services>amf-client>namf-evt
```

Description

This command associates an AMF profile (configured using the `config>mobile>profile>namf-profile` command) for defining the failure handling policy when the AMF services are used.

The `no` form of this command removes the AMF profile association.

Parameters

profile-name

specifies a profile name, up to 32 characters

5.2.1.3.127 nemo-ha

Syntax

```
nemo-ha {enable | disable}  
no nemo-ha
```

Context

```
config>mobile>pdn>apn
```

Description

This command enables Network Mobility (NEMO) support for this APN.

The `no` form of this command disables nemo.

Default

```
no nemo-ha
```

Parameters

enable

enables the Network Mobility (NEMO) support for the APN

disable

disables the Network Mobility (NEMO) support for the APN

5.2.1.3.128 non-ip-pdn

Syntax

```
non-ip-pdn
```

Context

```
config>mobile>pdn>apn
```

Description

This command enables the context for non-IP PDN tunneling over SGi.

5.2.1.3.129 udp-tunnel-group

Syntax

```
udp-tunnel-group tunnel-group-name
```

```
no udp-tunnel-group
```

Context

```
config>mobile>pdn>apn>udp-tunnel-group>non-ip-pdn
```

Description

This command associates the non-IP PDN type with a UDP tunnel group.

The no form of this command removes the association.

Parameters

tunnel-group-name

the name of a profile that contains UDP tunnel group configuration, up to 32 characters

5.2.1.3.130 nsmf-dnn-threshold

Syntax

```
nsmf-dnn-threshold max-pdu-activations max-pdu-activations [alarm]
```

```
nsmf-dnn-threshold max-pdu-connections max-pdu-connections [alarm]
```

```
no nsmf-dnn-threshold
```


Context

config>mobile>pdn>apn

Description

This command configures the threshold of the maximum number of PDU connections or the maximum number of PDU activations per second for this DNN. The threshold is used for the Nsmf DNN NF congestion overload control. An alarm can be enabled to be sent when the maximum number of connections or the maximum number of activations is reached.

The no form of this command removes the configuration.

Parameters

max-pdu-activations

specifies the maximum number of PDN activations per second

Values: 0 to 5000 (activations per second)

Default: 0 (disabled)

max-pdu-connections

specifies the maximum number of PDN connections

Values: 1 to 20000000

Default: 0

alarm

specifies whether an alarm will be sent when the maximum number of activations or the maximum number of connections is reached

5.2.1.3.131 nsmf-profile

Syntax

nsmf-profile *profile-name*

no nsmf-profile

Context

config>mobile>pdn>apn

config>mobile>pdn>sba-client-services>smf-client>nsmf-pdusession

Description

This command associates an SMF profile. The SMF profile defines the failure handling policy when the SMF services are used.

The no form of this command removes the SMF profile association.

Parameters

profile-name

specifies the profile name, up to 32 characters

5.2.1.3.132 optimize-gbr

Syntax

[no] optimize-gbr

Context

config>mobile>pdn>apn

Description

This command enables the GBR optimization capability. Normally, the GBR and MBR are calculated by the PGW based on the active and installed rules for that bearer, without taking into account scenarios such as Call Hold. In this scenario, the potential exists for a subscriber to put the call on hold and later add a second call into conference. The subscriber requires only the predefined amount of bandwidth for this. However, certain active rules with flow-status set to disabled may still be included in the GBR/MBR calculations, resulting in excess bandwidth on the RAN. If optimize-gbr is enabled, the PGW calculates the GBR and MBR while taking into account previously allocated RAN resources and disabled flows. At any given time, allocated RAN resources do not drop below previously allocated resources.

The no form of this command disables the GBR optimization capability.

Default

no optimize-gbr

5.2.1.3.133 origin-realm-per-application

Syntax

origin-realm-per-application

Context

```
config>mobile>pdn>apn
```

Description

This command enables the context to configure the origin realm per Diameter application.

5.2.1.3.134 realm-gx

Syntax

```
realm-gx realm-gx  
no realm-gx
```

Context

```
config>mobile>pdn>apn>origin-realm-per-application
```

Description

This command configures the originating realm or domain name of a Diameter node running the Gx application.

If the origin realm is configured at the Diameter peer level using the `config>mobile>profile>diameter-peer` command, the configuration of this command is ignored.

The no form of this command removes the name from the configuration. By default (not configured), the origin realm is selected according to the origin realm configuration at the PDN, or at the Diameter level.

Parameters

realm-gx

specifies the originating host name, up to 80 characters

5.2.1.3.135 realm-gy

Syntax

```
realm-gy realm-gy  
no realm-gy
```

Context

```
config>mobile>pdn>apn>origin-realm-per-application
```

Description

This command configures the originating realm or domain name of a Diameter node running the Gy application.

If the origin realm is configured at the Diameter peer level using the `config>mobile>profile>diameter-peer` command, the configuration of this command is ignored.

The no form of this command removes the name from the configuration. By default (not configured), the origin realm is selected according to the origin realm configuration at the PDN level, or at the Diameter level.

Parameters

realm-gy

specifies the originating host name, up to 80 characters

5.2.1.3.136 realm-rf

Syntax

`realm-rf realm-rf`

`no realm-rf`

Context

`config>mobile>pdn>apn>origin-realm-per-application`

Description

This command configures the originating realm or domain name of a Diameter node running the Rf application.

If the origin realm is configured at the Diameter peer level using the `config>mobile>profile>diameter-peer` command, the configuration of this command is ignored.

The no form of this command removes the name from the configuration. By default (not configured), the origin realm is selected according to the origin realm configuration at the PDN level, or at the Diameter level.

Parameters

realm-rf

specifies the originating host name, up to 80 characters

5.2.1.3.137 realm-s6b

Syntax

```
realm-s6b realm-s6b  
no realm-s6b
```

Context

```
config>mobile>pdn>apn>origin-realm-per-application
```

Description

This command configures the originating realm or domain name of a Diameter node running the S6b application.

If the origin realm is configured at the Diameter peer level using the `config>mobile>profile>diameter-peer` command, the configuration of this command is ignored.

The `no` form of this command removes the name from the configuration. By default (not configured), the origin realm is selected according to the origin realm configuration at the PDN level, or at the Diameter level.

Parameters

realm-s6b

specifies the originating host name, up to 80 characters

5.2.1.3.138 realm-sd

Syntax

```
realm-sd realm-sd  
no realm-sd
```

Context

```
config>mobile>pdn>apn>origin-realm-per-application
```

Description

This command configures the originating realm or domain name of a Diameter node running the Sd application.

If the origin realm is configured at the Diameter peer level using the `config>mobile>profile>diameter-peer` command, the configuration of this command is ignored.

The no form of this command removes the name from the configuration. By default (not configured), the origin realm is selected according to the origin realm configuration at the PDN level, or at the Diameter level.

Parameters

realm-sd

specifies the originating host name, up to 80 characters

5.2.1.3.139 realm-swm

Syntax

```
realm-swm realm-swm
```

```
no realm-swm
```

Context

```
config>mobile>pdn>apn>origin-realm-per-application
```

Description

This command configures the originating realm or domain name of a Diameter node running the Swm application.

If the origin realm is configured at the Diameter peer level using the config>mobile>profile>diameter-peer command, the configuration of this command is ignored.

The no form of this command removes the name from the configuration. By default (not configured), the origin realm is selected according to the origin realm configuration at the PDN level, or at the Diameter level.

Parameters

realm-swm

specifies the originating host name, up to 80 characters

5.2.1.3.140 overload-back-off

Syntax

```
overload-back-off-timer
```

Context

```
config>mobile>pdn>apn
```

Description

This command enables the context to configure the overload back-off timer settings.

5.2.1.3.141 exclude-arp-list

Syntax

```
exclude-arp-list arp-list
```

```
no exclude-arp-list
```

Context

```
config>mobile>pdn>apn>overload-back-off
```

Description

This command associates an ARP list (configured using the `config>mobile>profile>list>arp-list` command) with ARP values for which the PGW will still accept Create Session Requests (csreq), even when the thresholds configured using the `gtp-pdn-threshold`, `max-connection`, or `gtp-pdn-threshold max-activations` commands are reached.

For example, an APN is configured with thresholds using the `gtp-pdn-threshold` and `max-connection`, or the `max-activations` commands and the `exclude-arp-list` command is configured with ARP values of 6 and 7.

When the configured thresholds are reached:

- in any subsequent Create Session Request with an ARP value of 6 or 7, the PGW accepts the request
- in any subsequent Create Session Request with any ARP value other than 6 or 7, the PGW rejects the request, responds with a Create Session Response with Cause Code APN Congestion (113), and includes the PGW Back-off timer IE

The no form of this command reverts to the default value (no ARP values are excluded).

Default

```
no exclude-arp-list
```

Parameters

arp-list

specifies the name of the ARP list; up to 32 characters

5.2.1.3.142 overload-back-off-timer

Syntax

```
overload-back-off-timer overload-back-off-timer  
no overload-back-off-timer
```

Context

```
config>mobile>pdn>apn>overload-back-off
```

Description

This command enables the PGW Back-Off timer IE to be sent when a `gtp-pdn-threshold` value is also configured. When enabled and the configured values of `gtp-pdn-threshold`, `max-connection`, or `gtp-pdn-threshold max-activations` are reached, in a subsequent Create Session Request, the PGW rejects the request, responds with a Create Session Response with Cause Code APN Congestion (113), and includes the PGW Back-off timer IE.

**NOTE:**

The PGW Back-Off timer is only reported over GTPv2-C based on 3GPP TS 29.274.

Default

```
no overload-back-off-timer
```

Parameters

overload-back-off-timer

specifies the time in minutes during which the MME should refrain from sending subsequent PDN connection establishment requests to the PGW for the congested APN, for services other than service users and emergency services

Values: 1 to 120

Default: 0 (disabled)

5.2.1.3.143 overload-control-profile

Syntax

```
overload-control-profile profile-name  
no overload-control-profile
```


Context

config>mobile>pdn>apn

Description

This command specifies the overload control profile that is used in the APN.

The no form of this command reverts to the default (no profile).

Default

no overload-control-profile

Parameters

profile-name

specifies the name of the profile, up to 32 characters

5.2.1.3.144 paging-bearer-binding

Syntax

[no] paging-bearer-binding

Context

config>mobile>pdn>apn

Description

This command enables bearer binding during paging for a combined SGW and PGW session. If bearer binding during paging is disabled and bearer binding fails, the default bearer context is included in DDN.



NOTE:

Dynamic change is supported and shutdown of the gateway is not required.

Default

no paging-bearer-binding

5.2.1.3.145 pcf-discovery

Syntax

pcf-discovery

Context

config>mobile>pdn>apn

Description

This command enables the context to configure APN-specific options for PCF discovery.

5.2.1.3.146 serving-scope

Syntax

serving-scope *value*

no serving-scope

Context

config>mobile>pdn>apn>pcf-discovery

Description

This command enables and configures the use of serving scope in the query criteria when the SMF discovers the PCF using the NRF.

The no form of this command reverts to the default (serving scope is not used in the query criteria).

Default

no serving-scope

Parameters

value

specifies the serving scope value to use in the query criteria, up to 32 characters

5.2.1.3.147 pdn-alloc-type

Syntax

```
pdn-alloc-type allocation-type
no pdn-alloc-type
```

Context

```
config>mobile>pdn>apn
```

Description

This command specifies whether the User Equipment (UE) gets the IP address by RAN signaling or from its DHCP client functionality after the default bearer is established.

The no form of this command removes the allocation type from the configuration.

Default

```
no pdn-alloc-type
```

Parameters

allocation-type
specifies the allocation type

Values	ietf	specifies that the UE gets the IP address by the DHCP client functionality after the default bearer is established
–	ran-signaling	specifies that the UE gets the IP address by RAN signaling

Default: ran-signaling

5.2.1.3.148 pdn-type

Syntax

```
pdn-type {ipv4 | ipv6 | ipv4v6 | non-ip | ethernet} [{ipv4 | ipv6 | ipv4v6 | non-ip |  
ethernet}... (up to 4 max)]  
  
no pdn-type
```

Context

```
config>mobile>pdn>apn
```

Description

This command specifies the Packet Data Network (PDN) type. The PGW does not support a combination of static and dynamic addressing for PDN type IPv4v6. If the PGW receives a Create Session Request for PDP Type IPv4v6 and:

- there is a single IPv4 address in the PDN Address Allocation (PAA) IE, the PGW accepts and sets the PDN type to IPv4

or

- there is a single IPv6 address in the PAA IE, the PGW accepts and sets the PDN type to IPv6

or

- there are both an IPv4 and an IPv6 address in the PAA IE, the PGW accepts and leaves the PDN type as IPv4v6

The PGW rejects requests for one static and one dynamic address when using PDN type IPv4v6.

The no form of this command reverts to the default.

Parameters

ipv4

specifies that IPv4 PDN sessions are supported by this APN

ipv6

specifies that IPv6 PDN sessions are supported by this APN

ipv4v6

specifies that IPv4v6 PDN sessions are supported by this APN

non-ip

specifies that non-IP PDN sessions are supported by this APN

ethernet

specifies that ethernet PDU sessions are supported by this APN. This pdn-type cannot be configured along with the other pdn-types

Default: ipv4

5.2.1.3.149 pdn-type-conversion

Syntax

[no] pdn-type-conversion

Context

config>mobile>pdn>apn

Description

This command allows the PGW to create a session if the requested PDN type is dual stack, and either IPv4 or IPv6 is blocked on a specific APN and MG group. To perform the session creation, the IP allocation method must be able to allocate a single stack.

The no form of this command signifies that the PDN type fallback is not allowed.

Default

pdn-type-conversion

5.2.1.3.150 pdn-type-preferred-ipv6

Syntax

[no] pdn-type-preferred-ipv6

Context

config>mobile>pdn>apn

Description

This commands forces the use of an IPv6 address for the PDN connection when there is a UE-requested dual stack IPv4v6 address PDN type and the APN supports only IPv4 and IPv6 addresses.

The no form of this command reverts to the default.

Default

no pdn-type-preferred-ipv6

5.2.1.3.151 pfcp-translate-tx

Syntax

pfcp-translate-tx *profile-name*

no pfcp-translate-tx

Context

config>mobile>pdn>apn

Description

This commands associates a PFCP translation profile (configured in the config>mobile>profile>pfcp>pfcp-translate-profile context) for sending opaque values from the CP to the UP per APN.

The no form of this command removes the profile association.

Parameters

profile-name

specifies the name of the profile, up to 32 characters

5.2.1.3.152 pfcp-user-plane-inactivity

Syntax

pfcp-user-plane-inactivity *up-inactivity-timer*

no pfcp-user-plane-inactivity

Context

config>mobile>pdn>apn

Description

This command configures user plane inactivity detection and reporting for N4 (5GC) sessions, including the time period to monitor for user inactivity. When enabled, the CP function provisions the User Plane Inactivity Timer IE, which instructs the UP function to monitor the user plane activity for the established

PFCP session, and reports when no user plane packets are received for the time period exceeding the duration specified by this command.

If the CP function receives a PFCP Session Report with a User Plane Inactivity Report, it initiates a deactivation of the UP connection (for example, a data radio bearer and N3 tunnel) for the established PDU session. If the PDU session goes into idle, or a PDU session handover to a 4G PDN session occurs, the CP function stops the user plane inactivity timer on the UP function. The timer restarts only if the PDU session is reactivated, or another handover to a PDU session occurs.

The no form of this command disables the timer and stops including the User Plane Inactivity Timer IE in the PFCP Session Establishment and PFCP Session Modification Requests.

For more information about the supported IEs and messages, refer to the *7750 SR MG and CMG PFCP (Sx and N4) Interface Description*.

**NOTE:**

The PFCP user plane inactivity timer is only used for N4 (5GC) sessions.

Default

no pfcp-user-plane-inactivity

Parameters

up-inactivity-timer

specifies the user plane inactivity time to be monitored by the UP function, in seconds

Values: 5 to 360

Default: 0 (user plane inactivity detection and reporting is stopped)

5.2.1.3.153 policy-on-the-fly

Syntax

policy-on-the-fly {new-sessions | existing-sessions [reactivate-if-fail]}

no policy-on-the-fly

Context

config>mobile>pdn>apn

Description

This command specifies whether modifications to predefined policy rules on the gateway will be applied to new PDN sessions only or to both new and existing sessions, without terminating the existing session. If a

conflict occurs when existing session is specified, the policy is rolled back, unless the option `reactivate-if-fail` is also specified. When `reactivate-if-fail` is specified, the session will terminate with a “reactivation required” cause code and the updated policy will be applied upon reattachment.

This functionality supports the ability to modify a policy that is referenced under an APN or installed on a UE without removing it from the APN or UE session. It applies to policies configured in the following contexts:

- `config>mobile>profile>policy-options`
 - `charging-rule-unit`
 - `policy-rule`
 - `policy-rule-base`
 - `policy-rule-unit`
- `config>mobile>pdn>apn>static-local-pcc`
 - `policy-rule`
 - `policy-rule-base`

If a modification fails toward the UE, the PDN session will terminate with a “reactivation-required” cause code. Also, references to profiles within a policy rule unit can change, but the contents of those profiles cannot change.

The no form of this command reverts to the default.

Parameters

new-sessions

specifies that policy updates apply to new sessions only

existing-sessions

specifies that policy updates will be applied to both new and existing sessions

reactivate-if-fail

optionally specified when `existing-sessions` is also specified, to indicate that in the case of a conflict, the existing session will be terminated with a “reactivation required” cause code and policy updates will be applied upon reattachment

Default: existing sessions (new and existing sessions with policy rollback in case of conflict)

5.2.1.3.154 ppi-mapping-profile

Syntax

`ppi-mapping-profile` *profile-name*

`no ppi-mapping-profile`

Context

config>mobile>pdn>apn

Description

This command references a PPI mapping profile, configured under config>mobile>profile>ppi-mapping-profile, to the APN. It is used to define the PPI when the SMF initiates paging in the AMF.

The no form of this command reverts to the default (if no PPI mapping profile is defined, the PPI IE is not passed to the AMF).

Default

no ppi-mapping-profile

Parameters

profile-name

specifies the PPI profile name, up to 32 characters

5.2.1.3.155 ppp-regeneration

Syntax

[no] ppp-regeneration

Context

config>mobile>pdn>apn

Description

This command enables PPP regeneration on behalf of the sessions that terminate on this APN. This command must be present if LT2P (config>service>vprn>l2tp" or (config>router>l2tp) is also configured for the APN or RADIUS is configured to send L2TP tunnel attributes in Access-Accept for this APN. With ppp-regeneration enabled only PPP regeneration-based traffic is allowed on this APN.

The no form of this command reverts to the default.

Default

no ppp-regeneration

5.2.1.3.156 exclude-dns-server-option

Syntax

[no] exclude-dns-server-option

Context

config>mobile>pdn>apn>ppp-regeneration

Description

This command includes or excludes requesting the primary and secondary DNS IP address configuration option from the LNS in the PPP IPCP negotiation.

For CUPS/5GC deployments, this command on the CP function includes or excludes requesting in the PFCP Session Establishment Request to the UP function the primary and secondary DNS IP address configuration option from the LNS in the PPP IPCP negotiation.



NOTE:

This command is not applicable on the UP function.

Default

no exclude-dns-server-option

5.2.1.3.157 exclude-nbns-server-option

Syntax

[no] exclude-nbns-server-option

Context

config>mobile>pdn>apn>ppp-regeneration

Description

This command includes or excludes requesting the primary and secondary NBNS IP address configuration option from the LNS in the PPP IPCP negotiation.

For CUPS/5GC deployments, this command on the CP function includes or excludes requesting in the PFCP Session Establishment Request to the UP function the primary and secondary NBNS IP address configuration option from the LNS in the PPP IPCP negotiation.

**NOTE:**

This command is not applicable on the UP function.

Default

no exclude-nbns-server-option

5.2.1.3.158 icrq-calling-number-type

Syntax

```
icrq-calling-number-type icrq-avp-identity  
no icrq-calling-number-type
```

Context

```
config>mobile>pdn>apn>ppp-regeneration
```

Description

This command specifies the value that will be sent in the ICRQ Calling-Number AVP as the subscriber identity for the LNS. The Calling-Number can be configured to include either the IMSI or the MSISDN, as received in the corresponding GTP-C Create Session Request/PDP Context Request from the UE.

For CUPS/5GC deployments, the ICRQ Calling-Number AVP included is either the IMSI or the MSISDN in the User ID IE that the UP function receives in the PFCP Session Establishment Request.

**NOTE:**

- If the Create Session Request/PDP Context request does not include the IMSI/MSISDN, the anonymous is populated in the Calling-Number AVP and sent to the LNS.
- For CUPS/5GC if the PFCP Session Establishment Req does not include the IMSI/MSISDN, the anonymous is populated in the Calling-Number AVP and sent to the LNS.
- This command is not applicable on the CP function of CUPS/5GC deployments.

The no form of this command reverts to the default.

Parameters***icrq-avp-identity***

specifies the value to include in the ICRQ Calling-Number AVP

Values	msisdn	sets the Calling-Number to include the MSISDN
–	imsi	sets the Calling-Number to include the IMSI

Default: msisdn

5.2.1.3.159 include-hss-static-ip

Syntax

```
[no] include-hss-static-ip
```

Context

```
config>mobile>pdn>apn>ppp-regeneration
```

Description

This command includes or excludes the static UE IP address, if assigned by the HSS, to the LNS. When enabled, the GGSN/PGW will send the IPCP Configuration Request message with the HSS-assigned static UE IP address to the LNS. When disabled (default), the GGSN/PGW will always send 0.0.0.0 in the IPCP configuration request to the LNS.

For CUPS/5GC deployments, this command on the CP function includes or excludes the static UE IP address, if assigned by the HSS, to the PFCP Session Establishment Request to the UP function. When enabled, the CP function requests the HSS-assigned static UE IP address to the LNS in the PPP IPCP negotiation.



NOTE:

This command is not applicable on the UP function.

Default

```
no include-hss-static-ip
```

5.2.1.3.160 l2tp-group

Syntax

```
l2tp-group tunnel-group-name
```

```
no l2tp-group
```

Context

```
config>mobile>pdn>apn>ppp-regeneration
```

Description

This command specifies the L2TP group name that is to be associated with the APN. Therefore, the *tunnel-group-name* configured here must match one of the *l2tp-group-name* that has been configured in the `config>service>vprn>l2tp>group` context.

In case of dynamic L2TP tunnel and session setup using RADIUS only single *l2tp-group* is required and is used by the mobile gateway LAC function as a template to pick up any needed information missing from the received tunnel attributes from RADIUS.

In case of preconfigured/static configured L2TP tunnels, there can be 1 or more *l2tp-group-name* (one per LNS peer) configured in case of several LNSs connected under the same APN.



NOTE:

This command is not applicable on the CP function of CUPS/5GC deployments.

The no form of this command reverts to the default.

Parameters

tunnel-group-name

specifies the name of the L2TP tunnel group

5.2.1.3.161 lcp-confreq-attributes

Syntax

```
lcp-confreq-attributes {include | exclude} [mru-size] [auth-type auth-type]
```

```
no lcp-confreq-attributes
```

Context

```
config>mobile>pdn>apn>ppp-regeneration
```

Description

This command includes or excludes the Last Sent LCP Confreq and Last Received LCP Confreq attributes in the L2TP ICCN message to the LNS in case of CHAP authentication and optionally, in case of PAP authentication.

**NOTE:**

This command is not applicable on the CP function of CUPS/5GC deployments.

Parameters**include**

configures to include the LCP ConFreq attributes in the L2TP ICCN message sent by the GGSN/PGW

exclude

configures to exclude the LCP ConFreq attributes in the L2TP ICCN message sent by the GGSN/PGW

Default: include

mru-size

configures the MRU size in bytes sent in the Last Sent LCP Confreq and the Last Received LCP Confreq attributes

Values: 512 to 4096

Default: 1500

auth-type

configures to send the LCP Confreq attributes only in cases of CHAP authentication or for both PAP and CHAP authentication

Values: chap-only | both

5.2.1.3.162 mobile-router

Syntax

[no] mobile-router

Context

config>mobile>pdn>apn>ppp-regeneration

Description

This command specifies if mobile routers are allowed or not.

When the command `mobile-router` is enabled, the anti-spoofing for L2TP/PPP-regeneration sessions is disabled and the PGW/GGSN, PGW-U/GGSN-U/UPF does not verify the source IPv4 address of uplink packets or the destination IPv4 address of downlink packets arriving over the L2TP session. This allows the UEs to act as mobile routers and forward any IP host traffic over the L2TP session to the LNS. This also

allows the LNS to forward packets to the IP host (other than UE) over the L2TP session/GTP tunnel toward the UE.

When the command `mobile-router` is disabled, the anti-spoofing for L2TP/PPP-regeneration sessions is enabled, the PGW/GGSN, PGW-U/GGSN-U/UPF will only forward uplink IP packets with the source IPv4 address assigned to the UE and downlink IP packets with destination IPv4 address assigned to the UE. Any other uplink or downlink packets are dropped and not sent over the L2TP session to the LNS. Similarly, any other downlink packets are dropped and not sent over the GTP tunnel toward the UE.

The no form of this command reverts to the default.

**NOTE:**

- Enabling or disabling this command will take effect on any new attach after the command has been executed.
- This command is not applicable on the CP function of CUPS/5GC deployments.

Default

no mobile-router

5.2.1.3.163 router

Syntax

`router router-instance`

`no router`

Context

`config>mobile>pdn>apn>ppp-regeneration`

Description

This command specifies the router instance that is to be associated with the APN. The router instance configured here must match one of the VPRN IDs that has been configured under `config>service>vprn`. One router instance can be configured per APN.

The no form of this command reverts to the default.

**NOTE:**

This command is not applicable on the CP function of CUPS/5GC deployments.

Parameters

router-instance

specifies the router instance VPRN ID

Values: 1 to 2147483647

5.2.1.3.164 rx-connect-speed

Syntax

rx-connect-speed *rx-connect-speed*

no rx-connect-speed

Context

config>mobile>pdn>apn>ppp-regeneration

Description

This command configures the L2TP receive connection speed value that will be included in the optional ICCN Rx Connect Speed AVP (38).

The no form of this command specifies not to send the ICCN Rx Connect Speed AVP.



NOTE:

- The L2TP receive connection speed is configured independently of the APN-AMBR value.
- This command is not applicable on the CP function of CUPS/5GC deployments.

Default

no rx-connect-speed

Parameters

rx-connect-speed

specifies the L2TP receive connection speed, in b/s

Values: 0 to 4200000000

5.2.1.3.165 tx-connect-speed

Syntax

```
tx-connect-speed tx-connect-speed  
no tx-connect-speed
```

Context

```
config>mobile>pdn>apn>ppp-regeneration
```

Description

This command configures the L2TP transmit connection speed value that will be included in the mandatory ICCN Rx Connect Speed AVP (24).

The no form of this command reverts to the default.

**NOTE:**

- The L2TP receive connection speed is configured independently of the APN-AMBR value.
- This command is not applicable on the CP function of CUPS/5GC deployments.

Parameters

tx-connect-speed

specifies the L2TP transmit connection speed, in b/s

Values: 0 to 4200000000

Default: 0

5.2.1.3.166 user-name

Syntax

```
user-name {[msisdn | misisdn-no-apn | imsi-no-apn | imsi] [password password password-  
type {fixed | default}] | pco [allow-pco-missing]]  
no user-name
```

Context

```
config>mobile>pdn>apn>ppp-regeneration
```

Description

This command specifies the username.

When the `user - name` command is set to `pco`, both PAP and CHAP authentication methods are supported. The authentication method, username, and password/challenge are all based on the received PCO IE parameters in the received Create-Session-Request/PDP-Context-Request or in the PDU Session Request. For CUPS/5GC deployments, the CP function includes this information in the L2TP User Info IE in the PFCP Session Establishment Request to the UP function.

When the `allow-pco-missing` keyword is set, any Create Session Request or PDP Context Request received without the PCO IE parameters, or with empty username and password/challenge, are issued with an L2TP ICCN set with Proxy Auth Type 'No Authentication' and Last Sent LCP CONFREQ without Authentication Protocol. Therefore, PPP LCP Config with PPP PAP Auth is supported even when the username and password credentials are missing.

Furthermore, when the `allow-pco-missing` keyword is set and a Create-Session-Request/PDP-Context-Request is received with PCO Auth having an empty username and a non-empty password or a non-empty username and an empty password, an L2TP ICCN is issued with Proxy Auth Type "PAP" (3) and Last Sent LCP CONFREQ with PAP.

When the `user - name` command is set to `msisdn` or `imsi`, both PAP and CHAP authentication methods are supported. The authentication method and password/challenge are based on the received PCO IE parameters, and the username is MSISDN@APN, MSISDN or IMSI@APN or IMSI in the received Create-Session-Request/PDP-Context-Request or PDU Session Request. Any username in the received PCO IE is ignored in this case. In CUPS/5GC deployments, the CP function includes this information in the L2TP User Info IE in the PFCP Session Establishment Request to the UP function.

When the `user - name` command is set to `msisdn` or `imsi` together with a local password and the `password - type` is set to `fixed`, the PAP authentication method is supported. The username in this case is MSISDN@APN, MSISDN, IMSI@APN or IMSI in the received Create-Session-Request/PDP-Context-Request and the password is local configured as `fixed`. Any PCO Auth IE containers in the received Create-Session-Request/PDP-Context-Request is ignored. In CUPS/5GC deployments, the CP function includes this information in the L2TP User info IE in the PFCP Session Establishment Request to the UP function.

When the `user - name` command is set to `msisdn` or `imsi` together with a local password with `password - type` configured as `default`, the username is MSISDN@APN, MSISDN, IMSI or IMSI@APN in the received Create-Session-Request/PDP-Context-Request. The default password will only be used in case the PCO Auth IE containers is missing or the password/challenge is missing from the PCO Auth IE from the received Create-Session-Request/PDP-Context-Request. In this case, the L2TP/PPP will be sent as PAP authentications, based on the username and the default password. When the PCO Auth IE containers is included in the received Create-Session-Request/PDP-Context-Request, the L2TP/PPP regeneration will be sent as PAP or CHAP authentication, based on the configured username and the PCO Auth IE containers and default password are not used. In CUPS/5GC deployments, the CP function includes this information in the L2TP User info IE in the PFCP Session Establishment Request to the UP function.

The no form of this command reverts to the default.



NOTE:

This command is not applicable on the UP function.

Parameters

msisdn

sets the username attribute to the MSISDN, combining the MSISDN and APN received in the PDP Context Request (default value)

msisdn-no-apn

sets the username attribute to the MSISDN received in the Create Session Request/PDP Context Request

imsi-no-apn

sets the username attribute to the IMSI as received in the Create Session Request/PDP Context Request

imsi

sets the username attribute to the IMSI@APN, combining the IMSI and the APN as received in the Create Session Request/PDP Context Request

password

configures a local password to be used in the PAP Authentication, up to 32 characters

password-type

specifies the type of password

Values	fixed	configures the local password as fixed type
–	default	configures the local password as default type

pco

specifies that the username is sent as received in the PDP Context Request PCO IE

allow-pco-missing

specifies to allow handling when the PCO Auth is missing, or the username and password credentials are missing or are empty

5.2.1.3.167 supported-vendor

Syntax

supported-vendor *custom-1*

no supported-vendor

Context

```
config>mobile>pdn>apn>ppp-regeneration
```

Description

This command configures the use of Nokia vendor IEs or custom-1 vendor IEs for L2TP/PPP regeneration in CUPS/5GC deployments over the PFCP/Sx-N4 interface.

The no form of this command reverts to the use of Nokia L2TP/PPP vendor IEs.



NOTE:

This command is not applicable on the UP function. Thus, the UP function can dynamically handle both cases of L2TP supported vendor IEs.

Default

no supported-vendor

Parameters

custom-1

specifies to use the custom-1 L2TP/PPP vendor IEs

5.2.1.3.168 pre-auth

Syntax

```
pre-auth type {radius [server-group server-group] [user-name user-name] [suppress-  
imsi] [pco-password] [password password password-type {fixed | default} [hash | hash2]] |  
diameter}
```

```
no pre-auth
```

Context

```
config>mobile>pdn>apn
```

Description

This command configures the pre-authentication type used to support Virtual APNs. Either RADIUS (AAA) or Diameter (Gx) can be used to acquire the new virtual APN.

For Diameter, the peer selection follows the regular Gx configuration for the current APN.

The no form of this command removes the pre-authentication from the configuration.

Parameters

type

specifies the authentication type used for pre-authentication

Values	radius	specifies to use RADIUS
–	diameter	specifies to use Diameter

Default: radius

server-group

specifies a server group name, up to 32 characters

user-name

specifies the user name used in access requests sent to an AAA server

Values: imsi | msisdn | pco | fixed

Default: imsi

suppress-imsi

specifies that the 3GPP-IMSI attribute is not included, applicable for RADIUS

pco-password

specifies to set the password to the value received in the GTP PCO

password

specifies the PAP password used in access requests sent to a RADIUS server, up to 32 characters; up to 54 characters are supported for a hash password and up to 96 characters for a hash2 password

password-type

specifies the password type

Values	fixed	fixed password for overriding the PCO content or providing password with the IMSI, MSISDN or APN username (with PAP)
–	default	default password to be used together with the IMSI, MSISDN or APN username, and when missing from the PCO (with PAP)

**NOTE:**

If the hash and hash2 parameters are not used for the password, the key is assumed to be in a non-encrypted clear text form. For security purposes, all keys are automatically stored in an encrypted hash form in the configuration. The information saves and uses the configuration defined in the `system>security>hash-control` context.

hash

specifies that the key is entered in an encrypted form

hash2

specifies that the key is entered in a more complex encrypted form

5.2.1.3.169 pre-auth-fixed

Syntax

`pre-auth-fixed user-name long-description-string password long-description-string [hash | hash2]`

`no pre-auth`

Context

`config>mobile>pdn>apn`

Description

This command configures the username and password for pre-authorization.

Parameters***long-description-string***

specifies the user name used in access requests sent to a AAA server when the authorized username is set to fixed, up to 32 characters

password long-description-string

specifies the password used to authenticate requests sent to a AAA server when the user name is set to fixed, up to 32 characters

hash

specifies the key is entered in an encrypted form. If the hash parameter is not used, the key is assumed to be in a non-encrypted, clear text form. For security, all keys are stored in encrypted form in the configuration file with the hash parameter specified.

hash2

specifies the key is entered in a more complex encrypted form

If the hash2 parameter is not used, the less encrypted hash form is assumed

5.2.1.3.170 **preserve-volte**

Syntax

```
[no] preserve-volte
```

Context

```
config>mobile>pdn>apn
```

Description

This command preserves VoLTE traffic either when the PCRF is restarted and its origin ID is increased or when PCRF connectivity is lost and is not restored before the connection failure timer expires. VoLTE sessions are preserved until a DL packet is detected on the IMS signaling bearer (default bearer).

The no form of this command disables the behavior.

5.2.1.3.171 **qci-policy**

Syntax

```
qci-policy {qci-policy-name | uplink qci-policy-name [downlink qci-policy-name] | downlink  
qci-policy-name [uplink qci-policy-name]}
```

```
no qci-policy
```

Context

```
config>mobile>pdn>apn
```

Description

This command configures the QoS Class Identifier (QCI) policy profile name for uplink or downlink direction.

The no form of this command removes the policy name from the configuration.



NOTE:

This command requires an APN shutdown.

Default

no qci-policy

Parameters

qci-policy-name

specifies the QCI policy name

uplink *qci-policy-name*

specifies the QCI policy name for the uplink direction, up to 32 characters

downlink *qci-policy-name*

specifies the QCI policy name for the downlink direction, up to 32 characters

5.2.1.3.172 qfi-mapping-profile

Syntax

qfi-mapping-profile *profile-name*

no qfi-mapping-profile

Context

config>mobile>pdn>apn

Description

This command associates a QoS flow binding profile with the DDN for 5G systems. The profile needs to be created first before associating it for the specified DDN. QoS parameters specified within the profile are used for QoS flow binding. Profiles specified at the DDN level takes precedence over the one that is specified at the PDN level. When this optional profile is not configured, QFI is generated dynamically (in the range 1 to 64) using 5QI/ARP as default.

The no form of this command removes the profile name from the configuration.

Parameters

profile-name

specifies the QFI profile name, up to 32 characters

5.2.1.3.173 radius-accounting-on-off

Syntax

```
radius-accounting-on-off
```

Context

```
config>mobile>pdn>apn
```

Description

This command enables the context for controlling RADIUS accounting ON and OFF messaging.



NOTE:

- This command requires that RADIUS accounting is enabled.
- The command is possible to be changed on the fly but it will take effect only when APN is shutdown/no shutdown.

The commands in this context are effective only if the `config>mobile>pdn>radius>accounting-on-off - level` command is set to value `apn`.

5.2.1.3.174 include-duplicate-servers

Syntax

```
[no] include-duplicate-servers
```

Context

```
config>mobile>pdn>apn>radius-accounting-on-off
```

Description

This command configures the PGW/GGSN to send the RADIUS Accounting ON/OFF messages to the duplicate accounting servers too. The RADIUS Accounting ON/OFF messages are sent to the duplicate accounting servers only after a successful Acct-ON response has been received by the primary/secondary server. By default, these messages are not sent to the duplicate accounting servers.

The no form of this command reverts to the default.

Default

```
no include-duplicate-servers
```

5.2.1.3.175 messages-sent

Syntax

messages-sent {none | on | on-off}

no messages-sent

Context

config>mobile>pdn>apn>radius-accounting-on-off

Description

This command configures the PGW/GGSN sending of ON/OFF RADIUS Accounting messages per APN. The ON/OFF messages are optional and ON/OFF messaging must be explicitly enabled.

This command is effective only if the config>mobile>pdn>radius>accounting-on-off-level command is set to value apn.

The no form of this command reverts to the default.

Parameters

none

Accounting-ON or Accounting-OFF messages are not sent

on

only Accounting-ON messages are sent when the access point becomes active

on-off

both Accounting-ON and Accounting-OFF messages are sent

Default: none

5.2.1.3.176 wait-for-ack

Syntax

[no] wait-for-ack

Context

config>mobile>pdn>apn>radius-accounting-on-off

Description

This command configures the PGW/GGSN to wait for a response to Accounting-ON message from any server it has been sent to before allowing attaches to this APN. When this parameter is enabled, the system keeps on retrying infinitely by continuously sending Accounting-ON messages.

This command is effective only if the `config>mobile>pdn>radius>accounting-on-off-level` command is set to value `apn`.

The no form of this command reverts to the default.

Default

no wait-for-ack

5.2.1.3.177 radius-auth-cache

Syntax

[no] radius-auth-cache

Context

config>mobile>pdn>apn

Description

This command enables targeted bouncing to the MG group that serves the AAA-allocated IP pool(s). For dual-stack sessions, the session bouncing is performed to the respective MG group that has fragments/micronets/IP addresses for the provided IPv4 and IPv6 pools. After session bouncing, no separate RADIUS Authentication procedure is triggered by the new selected MG group, but the new MG group continues only with RADIUS Accounting (if configured on the used APN). The Acct-Session-ID AVP can be different across the authentication and accounting functions; this must be accepted by the customer AAA server.

The no form of this command reverts to the default.

Default

no radius-auth-cache

5.2.1.3.178 radius-case-sensitive-called-station-id

Syntax

[no] radius-case-sensitive-called-station-id

Context

config>mobile>pdn>apn

Description

This command specifies whether the Access Point Name (APN) is treated in a case sensitive manner in RADIUS.

The no form of this command reverts to the default.

Default

no radius-case-sensitive-called-station-id

5.2.1.3.179 radius-nas-identifier

Syntax

radius-nas-identifier *identifier-name*

no radius-nas-identifier

Context

config>mobile>pdn>apn

Description

This command specifies the NAS-Identifier to be used toward the RADIUS server. If not configured, the current v4 address of the interface defined in the radius-group is used.

The no form of this command reverts to the default.

Default

no radius-nas-identifier

Parameters

identifier-name

specifies the RADIUS NAS-Identifier, up to 32 characters

5.2.1.3.180 radius-nas-ip-address

Syntax

```
radius-nas-ip-address ipv4-address  
no radius-nas-ip-address
```

Context

```
config>mobile>pdn>apn
```

Description

This command configures the NAS-IP-Address that is used toward the RADIUS server. If an NAS-IP-Address is not configured, the current IPv4 address of the interface defined in the config>mobile>profile>radius-group context is used, or the MG group-specific IP address defined for the CMG LB-less architecture.

The no form of this command reverts to the default (RADIUS NAS-IP-Address disabled).

Default

```
no radius-nas-ip-address
```

Parameters

ipv4-address

specifies the RADIUS NAS-IP-Address

The special 0.0.0.0 address is not allowed because it is not a normal address.

Values: ipv4-address — a.b.c.d

5.2.1.3.181 reactivation-requested

Syntax

```
[no] reactivation-requested
```

Context

```
config>mobile>pdn>apn
```

Description

If this command is present, the PGW will send GTP-C cause #8 "reactivation requested" in Delete Bearer Request messages to disconnect the PDN. This cause code should be included for any case that the PGW deletes the PDN connection whether triggered externally (Gx, S6b, Gy, and so on) or internally via timeout or management intervention.

The no form of this command means the PGW sends the default cause code for the procedure.

Default

no reactivation-requested

5.2.1.3.182 redirect

Syntax

```
redirect mobile-all [external-nexthopv4 ipv4-address] | [external-nexthopv6 ipv6-address]  
no redirect
```

Context

```
config>mobile>pdn>apn
```

Description

This command configures the system to redirect all upstream mobile traffic toward the configured next-hops within this APN.

The no form of this command disables redirection.

Default

no redirect

Parameters

mobile-all

enables redirection

ipv4-address

specifies an IPv4 address to redirect traffic

ipv6-address

specifies an IPv6 address to redirect traffic

5.2.1.3.183 reject-foreign-subscribers

Syntax

```
[no] reject-foreign-subscribers
```

Context

```
config>mobile>pdn>apn
```

Description

When this command is enabled, the GGSN rejects Create PDP Context Requests for all foreign subscribers. A foreign subscriber is one whose PLMN ID (IMSI) does not match one of the GGSN PLMN IDs. When enabled, subscribers that are not in the home PLMN are rejected.

Default

```
no reject-foreign-subscribers
```

5.2.1.3.184 report-serving-node-supplied-cc-radius

Syntax

```
report-serving-node-supplied-cc-radius {enable | disable}  
no report-serving-node-supplied-cc-radius
```

Context

```
config>mobile>pdn>apn
```

Description

This command enables or disables the reporting of the serving node supplied charging characteristics value per APN. When enabled, the charging characteristics value received from the serving node is reported within the 3GPP-Charging-Characteristics AVP of RADIUS messages. When disabled, the internally used charging characteristics value is reported.

The no form of this command defines that the charging characteristics value from the charging profile is reported.

Default

```
report-serving-node-supplied-cc-radius enable
```

Parameters

enable

sets the reporting of the charging characteristics value received from the serving node

disable

sets the reporting of the charging characteristics value defined in the charging profile

5.2.1.3.185 report-serving-node-supplied-cc-s6b

Syntax

```
report-serving-node-supplied-cc-s6b {enable | disable}
```

```
no report-serving-node-supplied-cc-s6b
```

Context

```
config>mobile>pdn>apn
```

Description

This command enables or disables the reporting of the serving node supplied charging characteristics value reported per APN. When enabled, the charging characteristics value received from the serving node is reported within the 3GPP-Charging-Characteristics AVP of S6b messages. When disabled, the internally used charging characteristics value is reported.

The no form of this command defines that the charging characteristics value from the charging profile is reported.

Default

```
report-serving-node-supplied-cc-s6b enable
```

Parameters

enable

sets the reporting of the charging characteristics value received from the serving node

disable

sets the reporting of the charging characteristics value defined in the charging profile

5.2.1.3.186 restriction

Syntax

```
restriction {any | public1 | public2 | private1 | private2}  
no restriction
```

Context

```
config>mobile>pdn>apn
```

Description

This command specifies the level of restriction imposed on Evolved Packet System (EPS) bearer contexts created to the associated Access Point Name (APN). The value of this object is used to determine, on a per User Equipment (UE) basis, whether it is allowed to establish EPS bearers to other APNs.

The no form of this command reverts to the default.

Default

no restriction

Parameters

any | public1 | public2 | private1 | private2

specifies the level of restriction imposed on EPS bearer contexts created to the associated APN

5.2.1.3.187 roaming-handover

Syntax

```
roaming-handover
```

Context

```
config>mobile>pdn>apn
```

Description

This command enables the context to configure the supported features for the roaming handover.

5.2.1.3.188 **lte-qos-ambr**

Syntax

```
lte-qos-ambr downlink-rate downlink-rate uplink-rate uplink-rate  
no lte-qos-ambr
```

Context

```
config>mobile>pdn>apn>roaming-handover
```

Description

This command specifies the downlink and uplink APN-AMBR for roaming handovers to the LTE. If the APN-AMBR is specified, the PGW will initiate the APN-AMBR update after a RAT change from the UTRAN/GERAN to E-UTRAN and the UE is roaming (the S8 interface is used). If the MME sends the APN-AMBR in the Modify Bearer Command before the PGW has sent the Update Bearer Request, the PGW will send the Update Bearer Request as a reply to the Modify Bearer Command, using the APN-ABMR of the Update Bearer Request specified by this configuration. The configuration is allowed only if no PCRF is specified for the APN.

The no form of this command disables the APN-AMBR update.

Default

```
no lte-qos-ambr
```

Parameters

downlink-rate

specifies the downlink APN-AMBR in kb/s

Values: 1 to 1000000

uplink-rate

specifies the uplink APN-AMBR in kb/s

Values: 1 to 1000000

5.2.1.3.189 **lte-qos-black-list**

Syntax

```
lte-qos-black-list plmn-list-name  
no lte-qos-black-list
```

Context

config>mobile>pdn>apn>roaming-handover

Description

This command specifies the VPLMNs where the APN-ABMR update is disabled. There is no need to include HPLMN in the list, because the APN-AMBR update is never executed in the HPLMN.

Parameters

plmn-list-name

specifies the PLMN list name, up to 32 characters

5.2.1.3.190 lte-qos-delay

Syntax

lte-qos-delay *delay*

no lte-qos-delay

Context

config>mobile>pdn>apn>roaming-handover

Description

This command sets a delay before the PGW sends an Update Bearer Request based on the lte-qos-ambr.

The no form of this command reverts to the default.

Parameters

delay

specifies the delay, in seconds

Values: 0 to 5

Default: 0

5.2.1.3.191 router

Syntax

[no] router *router-instance*

Context

config>mobile>pdn>apn

Description

This command specifies the router to which this Access Point Name (APN) belongs. All subscribers belonging to this APN are mapped to this virtual router instance and therefore they all access the common services. Multiple router instances can be configured per APN for CMG LB-less architecture only.

The no form of this command reverts to the default.

Parameters

router-instance
specifies the router instance

Values	router-name service-id	
–	router-name	Base
–	service-id	1 to 2147483647

Default: Base

5.2.1.3.192 ip-local-pool

Syntax

[no] ip-local-pool *pool-name*

Context

config>mobile>pdn>apn>router
config>mobile>pdn>apn>router>tai-lai-list

Description

In the APN router context, this command configures an IP address pool in the Access Point Name (APN). In the TAI LAI list context, this command associates the IP address pool with the specified TAI LAI profile list.

A maximum of 8 IP pools can be configured for each APN.

The no form of this command removes the pool name from the configuration.

Parameters

pool-name

specifies a pool name, up to 32 characters

5.2.1.3.193 tai-lai-list

Syntax

[no] tai-lai-list *tai-lai-list-name*

Context

config>mobile>pdn>apn>router

Description

This command associates a Tracking Area Identity (TAI) or Location Area Identity (LAI) profile list in the APN.

The no form of this command removes the list from the configuration.

Parameters

tai-lai-list-name

specifies the name of a TAI or LAI profile list, up to 32 characters

5.2.1.3.194 router-advertisements

Syntax

router-advertisements

Context

config>mobile>pdn>apn

Description

This command enables the context to configure the IPv6 Stateless Address Auto Configuration (SLAAC) router advertisement parameters for PGW/GGSN.

5.2.1.3.195 max-advertisement-interval

Syntax

```
max-advertisement-interval seconds
no max-advertisement-interval
```

Context

```
config>mobile>pdn>apn>router-advertisements
```

Description

This command specifies the maximum time allowed between sending unsolicited multicast Router Advertisements as defined in 3GPP TS 29.061 and RFC 4861.

The no form of this command removes the configuration.

Parameters

seconds

specifies the maximum time, in seconds

Values	0 4 to 21600	–
–	The 0 value means that periodic sending is disabled.	–

Default: 21600

5.2.1.3.196 min-advertisement-interval

Syntax

```
min-advertisement-interval seconds
no min-advertisement-interval
```

Context

```
config>mobile>pdn>apn>router-advertisements
```

Description

This command specifies the minimum time allowed between sending unsolicited multicast Router Advertisements as defined in 3GPP TS 29.061 and RFC 4861.

The no form of this command removes the configuration.



NOTE:

The min-advertisement-interval value must not be greater than 0.75 times the max-advertisement-interval value.

Parameters

seconds

specifies the minimum time, in seconds

Values	0 3 to 16200	–
–	The 0 value means that periodic sending is disabled.	–

Default: 16200

5.2.1.3.197 rulebase-selection

Syntax

rulebase-selection

Context

config>mobile>pdn>apn

Description

This command enables the context for configuring priorities for the rulebase selection.

5.2.1.3.198 allow-rulebase

Syntax

```
allow-rulebase {gy-gx-aaa-local | gx-gy-aaa-local}  
no allow-rulebase
```

Context

```
config>mobile>pdn>apn
```

Description

This command configures the priorities for the rulebase selection. If not enabled, the default priorities are used.

Parameters

gy-gx-aaa-local

specifies that Gy has higher priority in setting the rulebase than Gx, Gx has higher priority than RADIUS, and RADIUS has higher priority than local configuration

gx-gy-aaa-local

specifies that Gx has higher priority in setting the rulebase than Gy, Gy has higher priority than RADIUS, and RADIUS has higher priority than local configuration

5.2.1.3.199 selected-dnn

Syntax

```
selected-dnn {real | virtual | amf-selected}  
no selected-dnn
```

Context

```
config>mobile>pdn>apn
```

Description

This command configures the DNN type in the PDU SESSION ESTABLISHMENT ACCEPT message.

The no form of this command reverts to the default.

Parameters

real

specifies to pass the real DNN in the DNN

virtual

specifies to pass the virtual DNN in the DNN; if the virtual DNN is not selected, the real DNN is passed in the DNN



NOTE:

To send the virtual DNN, execute the command in the virtual APN configuration.

amf-selected

specifies to pass the DNN selected by the AMF in the DNN; if the AMF does not select the DNN, the real DNN is passed in the DNN

Default: amf-selected

5.2.1.3.200 selection-mode

Syntax

```
selection-mode {subscribed-apn | ms-provided-apn | network-provided-apn}
[({subscribed-apn | ms-provided-apn | network-provided-apn}... (up to 3 max))]

no selection-mode
```

Context

```
config>mobile>pdn>apn
```

Description

This command ensures that the Mobile Station's (MS) subscription can use the APN. The setting specified for selection-mode is only examined during the processing of the APN contained in the Create Request and must match the corresponding setting on the SGSN/MME. If the two settings are not identical, the PGW/GGSN rejects the session with the cause code Cause: Service not supported (68).

The no form of this command returns the default.

Parameters

subscribed-apn

specifies that subscribed verified selection is allowed

ms-provided-apn

specifies that Mobile Station (MS) provided APN selection is allowed

network-provided-apn

specifies that network provided APN selection is allowed

Default: subscribed-apn

5.2.1.3.201 session-timeout

Syntax

`session-timeout session-timeout`

`no session-timeout`

Context

`config>mobile>pdn>apn`

Description

This command configures the absolute length of time that a PDN connection may exist before it is removed by the mobile gateway. The session timeout applies to all PDP/bearer contexts for the PDN connection simultaneously. Therefore, all PDP/ bearer contexts are removed upon reaching the session timeout. Also, it is supported in 5G. When the session-timeout expires, PDU session release procedure is initiated by the SMF. The session timeout is changed without a PDN shutdown and the change affects new sessions only.

When a session is created, it is added to a bucket that is determined by the window size and the configured session timeout value. The window size of the bucket is 300 seconds, this means, the session can be removed at any time within this time interval. The time interval starts at the time that is a multiple of the 300-seconds window size that is closer to the configured session timeout value. For example, if the configured session timeout is 1900 seconds, the session can be removed after 1800 seconds.

Default

`no session-timeout`

Parameters

session-timeout

specifies the timeout value, in seconds, of a session

Values: 0 (disabled), 1800 to 4294967295

Default: 86400

5.2.1.3.202 shutdown-back-off-timer

Syntax

```
shutdown-back-off-timer timer-value  
no shutdown-back-off-timer
```

Context

```
config>mobile>pdn>apn
```

Description

This command configures the back-off timer IE. When configured and the APN is in shutdown state, the PGW responds to new create session requests with cause code APN Congestion (113 for GTPv2 and 229 for GTPv1 messages) and includes the configured value of the back-off timer IE.

This configuration has the precedence over cause code configuration (configured using the `config>mobile>profile>gtp>gtpv2-session-rejection-cause` command) for a profile linked to a reference point.

The no form of this command disables this behavior.

Default

```
no shutdown-back-off-timer
```

Parameters

timer-value

specifies the timer value, in minutes

Values: 1 to 360

5.2.1.3.203 slices

Syntax

```
slices
```

Context

```
config>mobile>pdn>apn
```

Description

This command enters the context to configure slices for the APN.

5.2.1.3.204 default-slice

Syntax

```
default-slice slice-name [slice-list slice-list-name]  
no default-slice
```

Context

```
config>mobile>pdn>apn>slices
```

Description

This command configures the default slice of this APN/DNN. When the combined PGW-C/SMF cannot retrieve the UE subscribed S-NSSAIs from the UDM, the SMF fallback occurs to use the locally configured default S-NSSAI for this APN/DNN.

The no form of this command removes the default slice configuration from the APN.

Default

no default-slice

Parameters

slice-name

specifies the name of the slice, up to 32 characters

The slice name must be configured first to the slice-list for this APN/DNN

slice-list-name

specifies the name of the slice list that the slice name is configured, up to 32 characters

Nokia recommends configuring the *slice-list-name* to ensure unambiguous default slice configuration. If omitted, the default slice SST and SD are read from the first slice list that contains a slice with the given slice name.

5.2.1.3.205 slice-list

Syntax

```
slice-list list-name
```

```
no slice-list
```

Context

```
config>mobile>pdn>apn>slices
```

Description

This command configures the list of slices or a single slice for this APN/DNN. This configuration marks this APN as specific to this slice list. Registration to NRF includes the APN under the configured slices.

When set to `no slice-list`, the `slice-list` configured at the PDN level, applies to this APN.

The `no` form of this command reverts to the default.

Default

```
no slice-list
```

Parameters

list-name

Specifies the name of the slice list, up to 32 characters. The slice list is configured using the `config>mobile>profile>list>slice-list` command.

5.2.1.3.206 udm-based-slice-selection

Syntax

```
udm-based-slice-selection {enable | disable}
```

```
no udm-based-slice-selection
```

Context

```
config>mobile>pdn>apn>slices
```

Description

When a PDN connection is activated in the PGW-C, the PGW-C may need to select an S-NSSAI for the PDN connection when 5G interworking is enabled, or when the converged interfaces to other 5G NFs require that an S-NSSAI is defined. By default, the PGW-C uses the local slice configuration under the APN to select the S-NSSAI.

This command enables the UDM based S-NSSAI selection for the PGW-C. The PGW-C retrieves the subscription data of the UE from the UDM when the N10 interface is enabled. The returned subscription data from the UDM indicates which S-NSSAIs are subscribed in the APN, and the PGW selects the first S-NSSAI for the PDN connection.

The no form of this command reverts to the default.

Default

disable

Parameters

enable

enables the use of UDM subscription data in the S-NSSAI selection

disable

disables the use of UDM subscription data in the S-NSSAI selection

5.2.1.3.207 ssc

Syntax

ssc

Context

config>mobile>pdn>apn

Description

This command enables the context to configure Session and Service Continuity (SSC).

5.2.1.3.208 allowed-ssc-modes

Syntax

allowed-ssc-modes {ssc-1 | ssc-2 | ssc-3} [{ssc-1 | ssc-2 | ssc-3}... (up to 3 max)]

no allowed-ssc-modes

Context

config>mobile>pdn>apn>ssc

Description

This command controls the Data Network Name (DNN) supported SSC modes. It further restricts the supported SSC modes from the UDM subscription data but does not override them.

The no form of this command specifies that no SSC mode is allowed.

Default

no allowed-ssc-modes

Parameters

ssc-1

specifies that SSC mode 1 is allowed

ssc-2

specifies that SSC mode 2 is allowed

ssc-3

specifies that SSC mode 3 is allowed

5.2.1.3.209 pdu-session-address-lifetime

Syntax

pdu-session-address-lifetime *lifetime*

no pdu-session-address-lifetime

Context

config>mobile>pdn>apn>ssc

Description

This command configures the time duration that the network maintains the PDU session. The configured value is also communicated to the UE in the corresponding PCO container.

Default

no pdu-session-address-lifetime

Parameters

lifetime

specifies the value, in seconds, that the network maintains the PDU session

Values: 0 to 600

Default: 300

5.2.1.3.210 pdu-session-reactivation-trigger

Syntax

```
pdu-session-reactivation-trigger trigger
no pdu-session-reactivation-trigger
```

Context

```
config>mobile>pdn>apn>ssc
```

Description

This command defines the PDU session reactivation triggers.
The no form of this command reverts to the default.

Parameters

trigger
specifies the session reactivation trigger

Values	no-voice-call	specifies that no IMS voice calls QoS-flow (5QI=1) are active in the PDU session
–	idle	specifies the condition where the UPF indicated user inactivity
–	immediately	specifies PDU session reactivation procedure starts immediately after the handover procedure is completed

Default: immediately

5.2.1.3.211 pfcf-user-plane-inactivity-iupf-case

Syntax

```
pfcf-user-plane-inactivity-iupf-case up-inactivity-timer
no pfcf-user-plane-inactivity-iupf-case
```


Context

```
config>mobile>pdn>apn>ssc
```

Description

This command configures the User Plane inactivity detection and reporting for N4 (5GC) sessions (A-SMF to A-UPF), including the time period to monitor for user inactivity in the UL-CL/I-UPF use case.

The no form of this command disables the timer and stops including the user plane inactivity timer IE in the PFCP session establishment and PFCP session modification requests.

Default

```
no pfcp-user-plane-inactivity-iupf-case
```

Parameters

up-inactivity-timer

specifies the User Plane inactivity time monitored by the UP function, in seconds

Values: 5 to 360

Default: 0 (no user plane inactivity detection and reporting)

5.2.1.3.212 reject-not-allowed-ssc-mode

Syntax

```
[no] reject-not-allowed-ssc-mode
```

Context

```
config>mobile>pdn>apn>ssc
```

Description

This command configures the SMF to reject a new PDU session request if the SSC mode is incompatible with the allowed SSC modes.

The no form of this command configures the SMF to set the SSC mode to the default, if the requested SSC mode is not allowed.

Default

```
no reject-not-allowed-ssc-mode
```

5.2.1.3.213 upf-reselection

Syntax

```
upf-reselection [ssc-1-option] [ssc-2 ssc-2-option] [ssc-3 ssc-3-option]
no upf-reselection
```

Context

```
config>mobile>pdn>apn>ssc
```

Description

This command enables the UPF reselection.

The no form of this command disables the UPF reselection for active PDU sessions. The UPF is selected only during the PDU session activation time when there is no selective traffic offloading, that is using an UL-CL or BP.

Default

```
no upf-reselection
```

Parameters

ssc-1-option
specifies the SSC mode 1 configuration option

Values	disabled	specifies that the UPF reselection is disabled
–	location	specifies that the UPF reselection is completed when there is a location change and the new location cannot be served by the serving UPF
–	load	specifies that the UPF reselection is done when the load in the serving UPF reaches a predefined threshold, that is, 90%
–	i-smf	specifies that the UPF reselection is done when there is an I-SMF inserted that is located at a province different from the A-SMF

Default: disabled

ssc-2-option

specifies the SSC mode 2 configuration option

Values	disabled	specifies that the UPF reselection is disabled
–	location	specifies that the UPF reselection is completed when there is a location change and the new location cannot be served by the serving UPF
–	load	specifies that the UPF reselection is done when the load in the serving UPF reaches a predefined threshold, that is, 90%
–	i-smf	specifies that the UPF reselection is done when there is an I-SMF inserted that is located at a province different from the A-SMF

Default: disabled

ssc-3-option

specifies the SSC mode 3 configuration option

Values	disabled	specifies that the UPF reselection is disabled
–	location	specifies that the UPF reselection is completed when there is a location change and the new location cannot be served by the serving UPF
–	load	specifies that the UPF reselection is done when the load in the serving UPF reaches a predefined threshold, that is, 90%
–	i-smf	specifies that the UPF reselection is done when there is an I-SMF inserted that is located at a province different from the A-SMF

Default: disabled

5.2.1.3.214 uuid-mask-length

Syntax

uuid-mask-length *length*

no uuid-mask-length

Context

config>mobile>pdn>apn>ssc

Description

This command configures the mask length of the node part (six octets) of the UUID that is under comparison.

The no form of this command removes the configuration.

Default

no uuid-mask-length

Parameters

length

specifies the length, in bits

Values: 8 to 48

Default: 0

5.2.1.3.215 static-local-pcc

Syntax

static-local-pcc

Context

config>mobile>pdn>apn

Description

This command enables the context to configure static local policy rules and rule bases for the PDN. Whereas predefined rules locally configured on the gateway are activated and deactivated for a PDN session by the PCRF referencing the rule name and/or rule base name, the static policy rules locally configured on the gateway are activated and deactivated by the PCEF (gateway) for a PDN session. The PCRF is unaware of these static local policies.

The following assumptions apply.

- A static policy applied for a PDN session that is under PCRF control is applicable to the default bearer only.
- Static policy rules locally instantiated by the PCEF (gateway) are never communicated to the PCRF (for example, in a charging rule report).
- The Override Control feature applies to predefined and static rules.

5.2.1.3.216 abort

Syntax

```
abort
```

Context

```
config>mobile>pdn>apn>static-local-pcc
```

Description

This command ends the current policy editing session entered via the `begin` command and discards changes that have been made to policies during the editing session.

5.2.1.3.217 begin

Syntax

```
begin
```

Context

```
config>mobile>pdn>apn>static-local-pcc
```

Description

This command begins a policy editing session. Modifications done during the editing session will take effect when the `commit` command is issued. The editing session continues until either an `abort` or `commit`

command is issued, or a control complex reset takes place. Editing sessions are not interrupted by high-availability activity switches or CLI session terminations, such as closing a Telnet session.

5.2.1.3.218 **commit**

Syntax

```
commit
```

Context

```
config>mobile>pdn>apn>static-local-pcc
```

Description

This command saves changes made during the current policy editing session (see the `begin` command). Policy changes made during the editing session take effect only when they are committed. If the changes are successfully committed, with no errors detected during the commit or cross-reference verification against the existing policy configuration, the editing session will also be closed.

5.2.1.3.219 **dedicated-bearers**

Syntax

```
dedicated-bearers
```

Context

```
config>mobile>pdn>apn>static-local-pcc
```

Description

This command configures static rules and rule bases for the creation of dedicated bearers.



NOTE:

There can be a maximum of 10 static policy items per session, so the number of dedicated bearers triggered based on feature ADC Extended to Static PCC and VoIP Optimization is limited. One Layer 3/ Layer 4 data-driven rule is considered as a rule with two static policy items when data-driven dedicated bearer is active due to that rule. Local immediate dedicated bearer triggering rule is considered as a rule with one static policy item.

5.2.1.3.220 default-bearer-qci

Syntax

```
[no] default-bearer-qci qci qci arp arp-value
```

Context

```
config>mobile>pdn>apn>static-local-pcc>dedicated-bearers
```

Description

This command enables the configuration of static rules and rule bases that will create dedicated bearers for the configured default bearer QCI-ARP.

The no form of this command removes the configuration.

Parameters

qci

specifies the value of the QCI

Values: 1 to 9 | 65 | 66 | 69 | 70 | 128 to 254 | * (wildcard)

arp-value

specifies the value of the ARP

Values: 1 to 15 | * (wildcard)

5.2.1.3.221 policy-rule

Syntax

```
[no] policy-rule policy-rule-name
```

Context

```
config>mobile>pdn>apn>static-local-pcc>dedicated-bearers>default-bearer-qci
```

Description

This command configures the static local policy rules under the APN that are applied to PDN sessions locally by the gateway. These policy rules can create dedicated bearers when installed in a default bearer of the configured QCI-ARP.

The PCRF is not aware of the static policy that is applied for a PDN session.

By default, the system supports modifications to policies that are referenced under an APN or installed on a UE (via a PCRF or APN configuration), without having to remove them from the APN or the UE session. Changes are applied to existing sessions based on the APN policy-on-the-fly configuration.

The no form of this command disables the policy rule.

Parameters

policy-rule-name

specifies the name of the policy rule, up to 64 characters

5.2.1.3.222 policy-rule-base

Syntax

[no] policy-rule-base *policy-rule-base-name*

Context

config>mobile>pdn>apn>static-local-pcc>dedicated-bearers>default-bearer-qci

Description

This command configures the static local policy rule bases under the APN that are applied to PDN sessions locally by the gateway. These policy rule bases can create dedicated bearers when installed in a default bearer of the configured QCI-ARP.

The PCRF is not aware of the static policy that is applied for a PDN session.

By default, the system supports modifications to policies that are referenced under an APN or installed on a UE (via a PCRF or APN configuration), without having to remove them from the APN or the UE session. Changes will be applied to existing sessions based on the APN policy-on-the-fly configuration.

The no form of this command disables the policy rule.

Parameters

policy-rule-base-name

specifies the name of the policy rule base, up to 64 characters

5.2.1.3.223 diff

Syntax

diff

Context

```
config>mobile>pdn>apn>static-local-pcc
```

Description

This command shows the uncommitted changes made during a policy editing session (see the `begin` and `commit` commands).

5.2.1.3.224 policy-rule

Syntax

```
[no] policy-rule policy-rule-name
```

Context

```
config>mobile>pdn>apn>static-local-pcc
```

Description

This command configures static local policy rules under the APN that are applied to PDN sessions locally by the gateway. The PCRF is not aware of the static policy that is applied for a PDN session.

By default, the system supports modifications to policies that are referenced under an APN or installed on a UE (via PCRF or APN configuration), without having to remove them from the APN or the UE session. Changes will be applied to existing sessions based on the `policy-on-the-fly` configuration.

The `no` form of this command disables the policy rule.

Parameters

policy-rule-name

specifies the name of the policy rule, up to 64 characters

5.2.1.3.225 policy-rule-base

Syntax

```
[no] policy-rule-base policy-rule-base-name
```

Context

```
config>mobile>pdn>apn>static-local-pcc
```

Description

This command configures static local policy rule bases under an APN that are applied to PDN sessions locally by the gateway. The PCRF is not aware of the static policy that is applied for a PDN session.

By default, the system supports modifications to policies that are referenced under an APN or installed on a UE (via PCRF or APN configuration), without having to remove them from the APN or the UE session. Changes will be applied to existing sessions based on the APN *policy-on-the-fly* configuration.

The no form of this command disables the policy rule base.

Parameters

policy-rule-base-name

specifies the name of the policy rule base, up to 64 characters

5.2.1.3.226 static-predefined-pcc

Syntax

```
static-predefined-pcc
```

Context

```
config>mobile>pdn>apn
```

Description

This command enables the context to configure static policy parameters for PDN.

5.2.1.3.227 default-policy-rule-base

Syntax

```
[no] default-policy-rule-base base-policy-name
```

Context

```
config>mobile>pdn>apn>static-predefined-pcc
```

Description

This command configures policy-rule-bases to the APN in case PCRF is not in use.

Parameters

base-policy-name

specifies the name of the policy rule base, up to 64 characters

5.2.1.3.228 default-qci

Syntax

```
default-qci qci [arp arp-value]
```

```
no default-qci
```

Context

```
config>mobile>pdn>apn>static-predefined-pcc
```

Description

This command configures QCI and ARP override for default bearer/primary PDP context in case PCRF is not in use.

Parameters

qci

specifies the QCI value

Values: 1 to 64 | 67 to 255



NOTE:

Nokia recommends setting only non-GBR QCI values.

arp-value

specifies the ARP value

Values: 1 to 15

Default: matches all ARP values

5.2.1.3.229 steering

Syntax

```
steering
```

Context

```
config>mobile>pdn>apn
```

Description

This command enables the context to configure steering parameters.

5.2.1.3.230 default-gw-addr-ipv4

Syntax

```
default-gw-addr-ipv4 ip-address
```

```
no default-gw-addr-ipv4
```

Context

```
config>mobile>pdn>apn>steering>gw-signaling-addr
```

Description

This command configures the default IPv4 address of the default PGW/GGSN on a standalone SSG, in case no matching signaling interface address is configured. The PGW/GGSN address is used for forwarding downlink traffic based on the gateway signaling address received on the Sd interface.

The no form of this command reverts to the default.

Default

```
no default-gw-addr-ipv4
```

Parameters

ip-address

specifies the IPv4 IP address for the default PGW/GGSN

5.2.1.3.231 default-gw-addr-ipv6

Syntax

```
default-gw-addr-ipv6 ip-address
```

```
no default-gw-addr-ipv6
```

Context

```
config>mobile>pdn>apn>steering>gw-signaling-addr
```

Description

This command configures the default IPv6 address of the default PGW/GGSN on a standalone SSG, in case no matching signaling interface address is configured. The PGW/GGSN address is used for forwarding downlink traffic based on the gateway signaling address received on the Sd interface.

The no form of this command reverts to the default.

Default

```
no default-gw-addr-ipv6
```

Parameters

ip-address

specifies the IPv6 IP address for the default PGW/GGSN

5.2.1.3.232 gw-signaling-addr

Syntax

```
[no] gw-signaling-addr addr-list-name
```

Context

```
config>mobile>pdn>apn>steering
```

Description

This command configures a list of IP addresses on a standalone SSG to match the signaling interface addresses on the PGW/GGSN, for example, S5, S8, Gn, Gp, S2a, S2b, and so on. The PGW/GGSN signaling interface address is received on the standalone SSG via the Sd interface from the PCRF. The signaling interface address is used on the standalone SSG for selection of the PGW/GGSN for downlink traffic received for a UE that is assigned a static IP address.

The no form of this command reverts to the default.

Parameters

addr-list-name

specifies a list of IPv4 or IPv6 addresses to define a IP match list that can be used across features that require IP address list

5.2.1.3.233 gw-addr-ipv4

Syntax

```
gw-addr-ipv4 ip-address  
no gw-addr-ipv4
```

Context

```
config>mobile>pdn>apn>steering>tw-signaling-addr
```

Description

This command configures the IPv4 address of a PGW/GGSN on a standalone SSG. The PGW/GGSN address is used for forwarding downlink traffic based on the gateway signaling address received on the Sd interface.

The no form of this command reverts to the default.

Default

```
no gw-addr-ipv4
```

Parameters

ip-address

specifies the IPv4 address of a PGW/GGSN on standalone SSG

5.2.1.3.234 gw-addr-ipv6

Syntax

```
gw-addr-ipv6 ip-address  
no gw-addr-ipv6
```

Context

```
config>mobile>pdn>apn>steering>tw-signaling-addr
```

Description

This command configures the IPv6 address of a PGW/GGSN on a standalone SSG. The PGW/GGSN address is used for forwarding downlink traffic based on the gateway signaling address received on the Sd interface.

The no form of this command reverts to the default.

Default

no gw-addr-ipv6

Parameters

ip-address

specifies the IPv4 address of a PGW/GGSN on standalone SSG

5.2.1.3.235 subscriber-mapping-profile

Syntax

subscriber-mapping-profile *profile-name*

no subscriber-mapping-profile

Context

config>mobile>pdn>apn

Description

This command indicates the subscriber mapping profile used in this access point when SSG is deployed as a Multi-Path TCP (MP-TCP) Hybrid Access Gateway (HAG). If not configured with this command, the subscriber mapping profile name configured in the context config>mobile>profile>sub-id-mapping will be used.

Whether the subscriber context is created via RADIUS or PCRF needs to be specified with the transit-sub command in the config>service>vprn>ip-local-pool context.

The no form of this command reverts to the default (not used).

Parameters

profile-name

specifies the name of the subscriber mapping profile, up to 32 characters

5.2.1.3.236 svc-steering

Syntax

svc-steering

Context

```
config>mobile>pdn>apn
```

Description

This command configures service steering.

5.2.1.3.237 default-steering-profile

Syntax

```
default-steering-profile profile-name  
no default-steering-profile
```

Context

```
config>mobile>pdn>apn>svc-steering
```

Description

This command specifies a steering profile at the APN level. If all SDF's within an APN are steered to the same service chain, the steering profile at the APN level suffices. Alternately, different SDFs can be steered to different service chains by associating a steering profile at the SDF level (which overrides the APN level).

The no form of this command removes the associated steering profile for this APN.

Parameters

profile-name

specifies the name of the steering profile, up to 32 characters

5.2.1.3.238 transport-byte-collection

Syntax

```
transport-byte-collection
```

Context

```
config>mobile>pdn>apn
```


Description

This command configures the collection of transported bytes and, or packets for a specific IMSI and, or MSISDN.



NOTE:

- Up to 256 sessions (IMSI/MSISDN) per APN can be configured.
- The show commands work only when the feature is enabled and the session is up (before detach).
- In CUPS, CLI is configured in the PGW-C/SMF and statistics are collected from the PGW-U/UPF by a proprietary PFCP IE. Therefore, both PGW-C/SMF and PGW-U/UPF must be Nokia components.
- The feature does not work if the session has no charging and there is no acctmgr event configured, or in the case of light AcctMgr PDN case. However, the feature works with one rule with no charging, where the acctmgr event is configured.
- There is no persistence of data (DL/UL packets/bytes) after a IOM failover. The persistent data is in the internal table with the keys. If the feature is enabled and configured for an IMSI/MSISDN, a card/shelf failover occurs. In this case, the show command displays zero data. If the call is still attached, packet/bytes for any new traffic will be displayed after the failover.
- You must not track the same device with different identifiers. Non-deterministic behavior occurs when you invoke the `configure mobile-gateway pdn apn IMSI` and `configure mobile-gateway pdn apn MSISDN` commands back-to-back for the same IMSI/MSISDN pair.
- The feature is not supported when a duplicate MSISDN feature is enabled while invoking this feature. This can result in non-deterministic behavior.
- This feature must not be used together with the `collect-display-stats` command.

5.2.1.3.239 imsi

Syntax

```
imsi imsi [static-rule static-rule] [dynamic-rule dynamic-rule]
```

```
no imsi imsi
```

Context

```
config>mobile>pdn>apn>transport-byte-collection
```

Description

This command enables the collection of transport bytes/packets for a specific IMSI.

Default

no imsi

Parameters

imsi

specifies the IMSI, up to 15 digits

static-rule

specifies if the collection is enabled for static policy rules

Values: configured, all, none

Default: all

dynamic-rule

specifies if the collection is enabled for dynamic policy rules

Values: all, none

Default: all

5.2.1.3.240 msisdn

Syntax

`msisdn msisdn [static-rule static-rule] [dynamic-rule dynamic-rule]`

`no msisdn msisdn`

Context

config>mobile>pdn>apn>transport-byte-collection

Description

This command enables the collection of transport bytes/packets for a specific MSISDN.

Default

no msisdn

Parameters

msisdn

specifies the MSISDN, up to 15 digits

static-rule

specifies if the collection is enabled for static policy rules

Values: configured | all | none

Default: all

dynamic-rule

specifies if the collection is enabled for dynamic policy rules

Values: all | none

Default: all

5.2.1.3.241 twag-radius-acct-feed

Syntax

[no] twag-radius-acct-feed

Context

config>mobile>pdn>apn

Description

This command specifies for TWAG open SSID if the radius accounting information feed is to be used to inform the radius server when a new UE accesses (issuing DHCP request). The no form means HTTP header enrichment is used instead in the HTTP redirect to the captive portal.

Default

no twag-radius-acct-feed

5.2.1.3.242 udm-ctrl-radius-auth

Syntax

[no] udm-ctrl-radius-auth

Context

config>mobile>pdn>apn

Description

This command enables or disables the UDM-controlled RADIUS authentication. When disabled, the existing implementation applies, that is, the secondaryAuth attribute is ignored from the UDM, and local policies at the SMF are followed. When enabled, the command checks the secondaryAuth attribute from the UDM and proceeds accordingly, that is, if set as true, RADIUS authentication is required, but if set as false, RADIUS

authentication is not required. If absent, a secondary authentication is not required by the subscription, but it can be required by local policies at the SMF.

The no form of this command reverts to the default.

Default

no udm-ctrl-radius-auth

5.2.1.3.243 udm-sdm-disabled

Syntax

```
udm-sdm-disabled {lte | non-3gpp | eps}
no udm-sdm-disabled
```

Context

```
config>mobile>pdn>apn
```

Description

This command disables the Nudm_SubscriberDataManagement service in the APN/DNN. If there is access type change, the service remains disabled even if the service is allowed when the session is created in the new access type. If the service is disabled, the local configuration is used to define the session management subscription.

The no form of this command removes this configuration.

Parameters

lte

the Nudm_SubscriberDataManagement service is disabled if the session is created from MME

non-3gpp

the Nudm_SubscriberDataManagement service is disabled if the session is created from ePDG

eps

the Nudm_SubscriberDataManagement service is disabled if the session is created from MME or ePDG

5.2.1.3.244 udm-sdm-profile

Syntax

```
udm-sdm-profile profile-name  
no udm-sdm-profile
```

Context

```
config>mobile>pdn>apn
```

Description

This command specifies the UDM SDM profile for the Nudm_SubscriberDataManagement service.

The no form of this command removes the profile. If no profile is defined in the APN, the profile under the config>mobile>pdn>sba-client-services>udm-client>nudm-sdm context is used.

Parameters

profile-name

specifies the UDM SDM profile, up to 32 characters

5.2.1.3.245 udm-uecm-disabled

Syntax

```
[no] udm-uecm-disabled
```

Context

```
config>mobile>pdn>apn
```

Description

This command disables the Nudm_UEContextManagement service in the APN/DNN.

The no form of this command enables the Nudm_UEContextManagement service in the APN/DNN.

5.2.1.3.246 udm-uecm-profile

Syntax

```
udm-uecm-profile profile-name
```

```
no udm-uecm-profile
```

Context

```
config>mobile>pdn>apn
```

Description

This command specifies the UDM UECM profile for the Nudm_UEContextManagement service.

The no form of this command removes the profile. If no profile is defined in the APN, the profile under the config>mobile>pdn>sba-client-services>udm-client>nudm-uecm context is used.

Parameters

profile-name

specifies the UDM UECM profile, up to 32 characters

5.2.1.3.247 umts-qos-policy

Syntax

```
umts-qos-policy umts-qos-policy
```

```
no umts-qos-policy
```

Context

```
config>mobile>pdn>apn
```

Description

This command configures the name of Universal Mobile Telecommunications System (UMTS) Quality of Service (QoS) policy.

The no form of this command removes the policy name from the configuration.

Parameters

umts-qos-policy

specifies a UMTS QoS policy, up to 32 characters

5.2.1.3.248 up-security-policy

Syntax

```
up-security-policy profile-name  
no up-security-policy
```

Context

```
config>mobile>pdn  
config>mobile>pdn>apn
```

Description

This command configures the UP confidentiality protection policy. The configuration defines the policy when it is not configured by the UDM or when overriding the UDM policy has been enabled. If no policy is configured under the APN, the policy configured under the PDN is used.



NOTE:

The configuration is not applied in the V-SMF. The V-SMF uses the V-SMF profile to configure the policies.

The no form of this command disables the UP confidentiality protection policy.

Parameters

profile-name

specifies the name of the UP security policy profile, up to 32 characters

5.2.1.3.249 upf-acct-profile

Syntax

```
upf-acct-profile profile-id  
upf-acct-profile inherit  
no upf-acct-profile
```

Context

```
config>mobile>pdn>apn
```

Description

This command configures the specified UP function profile at the APN level. The profile configured for the APN takes precedence over the profile configured at the PDN level.

The no form of this command removes the profile from the APN configuration.

Parameters

profile-id

specifies a profile ID for the UP function account, configured in the config>mobile>profile>charging context

Values: 0 to 255

Default: 0

inherit

specifies the profile is inherited from the PDN level

5.2.1.3.250 upf-sel-before-auth

Syntax

[no] upf-sel-before-auth

Context

config>mobile>pdn>apn

Description

This command enables or disables the UPF selection before the RADIUS authentication.

Default

no upf-sel-before-auth

5.2.1.3.251 upf-selection

Syntax

upf-selection

Context

config>mobile>pdn>apn

Description

This command enables the context to configure UPF selection options.

5.2.1.3.252 dnn

Syntax

```
dnn type type [apn-ni-only]
no dnn
```

Context

```
config>mobile>pdn>apn>upf-selection
```

Description

This command specifies how the DNN is used in the UPF discovery query and provides the option to use one of the three DNN types (*real*, *virtual*, or *public*) in the DNN query parameter. The *apn-ni-only* option is configurable only when DNN type *real* is used.

When the type *real* is configured, the APN-NI and OI are used in the DNN query parameter; for example, ...\$dnn=internet.mncYYY.mccZZZ.gprs&....

When the type *real apn-ni-only* is configured, only the NI is used in the DNN query parameter; for example, ...\$dnn=internet&...

The *no* form of this command reverts to the default.

Parameters

type

specifies to use *real*, *virtual*, or *public* DNN in the UPF discovery query

Values: *real* | *virtual* | *public*

Default: *real*

apn-ni-only

specifies to use only the APN network identifier (APN-NI) in the UPF selection query

5.2.1.3.253 virtual-apn

Syntax

```
virtual-apn apn-name
no virtual-apn
```

Context

config>mobile>pdn>apn

Description

This command configures a static mapping from the APN received in the GTP create request message to the configured virtual APN.

Default

no virtual-apn

Parameters

apn-name

specifies an APN name, up to 80 characters

5.2.1.3.254 virtual-apn-based-on-pco

Syntax

virtual-apn-based-on-pco [default-apn *apn-name*]

no virtual-apn-based-on-pco

Context

config>mobile>pdn>apn

Description

This command enables a PCO-based virtual APN selection mechanism. When enabled, the system searches the APN and username, to be used in RADIUS messaging, information from the PCO in the GTP Create Session Request or Create PDP Context Request. The format of the PCO must be username@APN.

When the username is not available or is an empty string, a fallback APN is not used even if configured. The fallback APN is only used when a username either from CHAP or PAP is available, but does not contain a valid format username (that is, character @ is missing or exists but is followed by an empty string).

The no form of this command removes the configuration.

Default

no virtual-apn-based-on-pco

Parameters

apn-name

specifies an APN name to be used when this feature is enabled and the PCO does not contain a valid APN, up to 80 characters

5.2.1.3.255 virtual-apn-list

Syntax

```
[no] virtual-apn-list apn-list
```

Context

```
config>mobile>pdn>apn
```

Description

This command is used to associate a list of virtual APNs with a real APN.

The no form of this command removes the list of virtual APNs associated with a real APN.

Parameters

apn-list

specifies an APN list name, up to 32 characters

5.2.1.3.256 wait-accounting

Syntax

```
[no] wait-accounting
```

Context

```
config>mobile>pdn>apn
```

Description

This command configures the PGW/GGSN to wait for an accounting response before sending a GTP response. The PGW/GGSN is waiting for a RADIUS accounting-response and does not apply if RADIUS accounting is not enabled.

Default

no wait-accounting



NOTE:

If the wait-accounting command is disabled after the acct-start is sent out but before the response is received from the AAA mobile gateway, it conforms to its existing behavior and rejects the session.

5.2.1.4 APN PCF PDN command descriptions

5.2.1.4.1 pcf-selection

Syntax

pcf-selection

Context

config>mobile>pdn>apn

Description

This command enables the context to configure PCF selection options.

5.2.1.4.2 dnn

Syntax

dnn type *type* [apn-ni-only]

no dnn

Context

config>mobile>pdn>apn>pcf-selection

Description

This command specifies how the DNN is used in the PCF discovery query and provides the option to use one of the three DNN types (real, virtual, or public) in the DNN query parameter. The apn-ni-only option is configurable only when DNN type real is used.

When the type `real` is configured, the APN-NI and OI are used in the DNN query parameter; for example, `...&dnn=internet.mncYYY.mccZZZ.gprs&...`

When the type `real apn-ni-only` is configured, only the NI is used in the DNN query parameter; for example, `...&dnn=internet&...`

The no form of this command reverts to the default.

Parameters

type

specifies to use real, virtual, or public DNN in the PCF discovery query

Values: `real | virtual | public`

Default: `real`

apn-ni-only

specifies to use only the APN network identifier (APN-NI) in the PCF selection query

5.2.1.4.3 `exclude-gpsi`

Syntax

```
[no] exclude-gpsi
```

Context

```
config>mobile>pdn>apn>pcf-selection
```

Description

This command specifies whether GPSI is used or not in the query criteria when the SMF discovers the PCF using the NRF.

The no form of this command configures that GPSI is used in the query criteria.

Default

```
no exclude-gpsi
```

5.2.1.4.4 `exclude-snssais`

Syntax

```
exclude-snssais {enable | disable | inherit}
```

```
no exclude-snssais
```

Context

config>mobile>pdn>apn>pcf-selection

Description

This command specifies whether S-NSSAIs are used or not in the query criteria when the SMF discovers the PCF using the NRF.

The no form of this command reverts to the default.

Parameters

enable

specifies to use S-NSSAIs in the query criteria

disable

specifies not to use S-NSSAIs in the query criteria

inherit

specifies to inherit the PDN-level configuration

Default: inherit

5.2.1.4.5 exclude-supi

Syntax

[no] exclude-supi

Context

config>mobile>pdn>apn>pcf-selection

Description

This command specifies whether SUPI is used or not in the query criteria when the SMF discovers the PCF using the NRF.

The no form of this command configures that SUPI is used in the query criteria.

Default

no exclude-supi

5.2.1.4.6 pd-alloc-method

Syntax

```
pd-alloc-method {none | aaa}  
no pd-alloc-method
```

Context

```
config>mobile>pdn>apn
```

Description

This command enables and specifies the allocation scheme that the APN uses to assign the IPv6 prefix delegation to the UE/CPE after establishing the PDN connection.

The no form of this command reverts to the default.

Default

```
no pd-alloc-method
```

Parameters

aaa

specifies that the IPv6 delegated prefix is assigned using the RADIUS authentication server

none

disables the IPv6 prefix delegation

Default: none

5.2.1.4.7 nf-id-list

Syntax

```
nf-id-list list-name  
no nf-id-list
```

Context

```
config>mobile>pdn>apn>pcf-selection
```

Description

This command specifies an NF ID list for PCF selection.

The no form of this command removes the NF ID list from the configuration.

Default

no nf-id-list

Parameters

list-name

specifies the NF ID list name, up to 32 characters

5.2.1.4.8 npcfc-profile

Syntax

npcfc-profile *profile-name*

no npcfc-profile

Context

config>mobile>pdn>apn>pcfc-selection

Description

This command specifies an NPCFC profile for the PCFC selection at the APN.

The no form of this command removes the NPCFC profile from the configuration.

Default

no npcfc-profile

Parameters

profile-name

specifies the NPCFC profile name, up to 32 characters

5.2.1.4.9 pcfc-id

Syntax

pcfc-id {use-as-target | ignore}

no pcfc-id

Context

config>mobile>pdn>apn>pcf-selection

Description

This command configures the use of the PCF NF instance ID that the SMF receives from the AMF during the SM Context Create request for the PCF discovery. The PCF UUID can be either configured to be included in the query criteria when discovering the PCF, or it can be ignored.

The no form of this command reverts to default (the PCF NF instance ID is not selected in the configuration).

Default

no pcf-id

Parameters

use-as-target

specifies that the PCF UUID is used as the target NF instance ID

The AMF-received PCF ID is included in the discovery (that is, the received matching list only has this PCF ID or none); in case of fall back to static PCF selection, the SMF looks for that exact PCF ID in the local configuration and if it exists, uses it, if not, it fails

ignore

specifies to exclude the PCF UUID from the query

The SMF does not use the received PCF ID from the AMF in any selection logic (that is, not included in the discovery query and not used when looking for static PCF); the SMF chooses any appropriate PCF ID in those sets

Default: use-as-target

5.2.1.5 selection-type

Syntax

selection-type {imsi | msisdn | imei} {prefix | full-id | suffix}

no selection-type

Context

config>mobile>pdn>apn>pcf-selection

Description

This command specifies the PCF selection criteria which can be based on an IMSI, MSISDN, or IMEI range.

When you select the `full-id` or `prefix` option, enter a *start-id* for each user - range entry. The *end-id* is optional and is copied from the *start-id* entry, if not provided.

When you select the `suffix` option, you must enter a suffix digit.

If the IMSI/MSISDN/IMEI for a specific UE does not match any of the configured entries, the selection of the PCF follows the existing selection criteria.

The `no` form of this command removes the selection type for the user - range from the configuration.

Default

no selection-type

Parameters

imsi

specifies an IMSI range for the PCF selection

msisdn

specifies an MSISDN range for the PCF selection

imei

specifies an IMEI range for the PCF selection

prefix

specifies that the mapping is based on a prefix for the start and end identifiers for the user - range command

- for an IMSI range, the identifiers must be 5 to 15 characters long
- for an MSISDN range, the identifiers must be 4 to 15 characters long
- for an IMEI range, the identifiers must be 4 to 8 characters long

full-id

specifies that the mapping is based on a full identifier for the start and end identifiers for the user - range command

- for an IMSI range, the identifiers must be 10 to 15 characters long
- for an MSISDN range, the identifiers must be 9 to 15 characters long
- for an IMEI range, the identifiers must be 15 to 16 characters long

suffix

specifies an integer that selects the end digit for the user - range.

Default: full-id

5.2.1.6 user-range

Syntax

```
user-range user-range-name {start-id [end-id] | suffix} nf-id-list list-name
```

Context

```
config>mobile>pdn>apn>pcf-selection
```

Description

This command configures the peer user ranges. You can configure up to 32 user range entries.

The no form of this command removes the peer user ranges from the configuration.

Default

no user-range

Parameters

user-range-name

selects the name for the user range, up to 32 characters

start-id

selects the start identifier for the user range, from 4 to 16 characters; the number of characters depends on the selection type (prefix or full-id) and on the PCF selection (IMSI, MSISDN, or IMEI)

end-id

selects the end identifier for the user range, from 4 to 16 characters; the number of characters depends on the selection type (prefix or full-id) and on the PCF selection (IMSI, MSISDN, or IMEI)

suffix

specifies an integer that selects the end digit for the user - range; when you select the suffix option, the *start-id* and *end-id* are not used. In case of IMEI/IMEISV, the suffix matching is based on the last digit of the SNR.

Values: 0 to 9

list-name

specifies the PCF NF ID list name, up to 32 characters

5.2.1.7 APN PCO PDN command descriptions

5.2.1.7.1 pco-option

Syntax

pco-option

Context

config>mobile>pdn

config>mobile>pdn>apn

Description

This command enables the context to configure UE PCO options.

5.2.1.7.2 closest-v4-epdg-group

Syntax

[no] c losest -v4-epdg-group *profile-name*

Context

config>mobile>pdn>apn>pco

Description

This command configures the closest IPv4 ePDG group. For the closest IPv4 ePDG group, the command should list the list names configured under config>mobile>profile>epdg-group. If an administrator tries to assign a list name that contains IPv6 IP addresses, an error will be returned. Assigning an IP list activates the functionality. Round-Robin selection of IP addresses from the list should exist when provided to UE.

Parameters

profile-name

specifies the profile name, up to 32 characters

5.2.1.7.3 closest-v6-epdg-group

Syntax

```
[no] c losest -v6 -epdg -group profile-name
```

Context

```
config>mobile>pdn>apn>pco
```

Description

This command configures the closest IPv6 ePDG group. For the closest IPv6 ePDG group, the command should list the list names configured under config>mobile>profile>epdg-group. If an administrator tries to assign a list name that contains IPv4 IP addresses, an error will be returned. Assigning an IP address list activates the functionality. Round-Robin selection of IP addresses from the list must exist when provided to UE.

Parameters

profile-name

specifies the profile name, up to 32 characters

5.2.1.7.4 dns-server-ipv4

Syntax

```
dns-server-ipv4
```

Context

```
config>mobile>pdn>apn>pco
```

Description

This command configures the IPv4 address of the DNS server used by a User Equipment (UE) to resolve domain names to IP addresses.

5.2.1.7.5 primary

Syntax

```
primary ip-address
```

```
no primary
```

Context

```
config>mobile>pdn>apn>pco>dnsv4
```

Description

This command specifies the IPv4 address of the primary DNS server used by a User Equipment (UE) to resolve domain names to IP addresses.

Parameters

ip-address

specifies the IPv4 address

5.2.1.7.6 secondary

Syntax

```
secondary ip-address
```

```
no secondary
```

Context

```
config>mobile>pdn>apn>pco>dnsv4
```

Description

This command specifies the IPv4 address of the secondary DNS server used by a UE to resolve domain names to IP addresses.

Parameters

ip-address

specifies the IPv4 address

5.2.1.7.7 dns-server-ipv6

Syntax

```
dns-server-ipv6
```

Context

```
config>mobile>pdn>apn>pco
```

Description

This command specifies the IPv6 address of the DNS server used by a User Equipment (UE) to resolve domain names to IPv6 addresses.

5.2.1.7.8 primary

Syntax

`primary ipv6-address`

`no primary`

Context

`config>mobile>pdn>apn>pco>dnsv6`

Description

This command specifies the IPv6 address of the primary DNS server used by a User Equipment (UE) to resolve domain names to IP addresses.

The no form of this command removes the IP address from the configuration.

Parameters

ipv6-address

specifies the IPv6 address

5.2.1.7.9 secondary

Syntax

`secondary ipv6-address`

`no secondary`

Context

`config>mobile>pdn>apn>pco>dnsv6`

Description

This command specifies the IPv6 address of the secondary DNS server used by a User Equipment (UE) to resolve domain names to IP addresses.

The no form of this command removes the IP address from the configuration.

Parameters

ipv6-address

specifies the IPv6 address

5.2.1.7.10 link-mtu

Syntax

link-mtu *octets*

no link-mtu

Context

config>mobile>pdn>apn>pco

Description

This command causes the PGW/GGSN to send the IPv4 link MTU in the Protocol Configuration Options when a new PDN connection is created. This command also sets the MTU to be sent in the IPv6 router advertisement. If the link MTU is set to 0 (default) the MTU is not sent in the PCO or the router advertisement.

This functionality is also applicable to non-IP PDN connections with a minimum value of 128 octets for the link MTU. For IP PDN type connections the minimum link MTU value is 64 octets.

The no form of this command reverts to the default.

Parameters

octets

specifies the maximum number of addresses sent to the PCO or RA

Values: 0 | 64 to 65535

Default: 0 (MTU not sent)

5.2.1.7.11 nbns-srvr-ipv4

Syntax

nbns-srvr-ipv4

Context

config>mobile>pdn>apn>pco

Description

This command enables the context to configure IPv4 NBNS server parameters.

5.2.1.7.12 primary

Syntax

primary ip-address

no primary

Context

config>mobile>pdn>apn>pco>nbnsrv4

Description

This command specifies the IPv4 address of the primary NetBIOS Name Service (NBNS) server used by Microsoft NetBIOS clients to reach NetBIOS servers.

The no form of this command removes the IP address from the configuration.

Parameters

ip-address

specifies the IPv4 address of the NBNS server

5.2.1.7.13 secondary

Syntax

secondary ip-address

no secondary

Context

config>mobile>pdn>apn>pco>nbnsrv4

Description

This command specifies the IPv4 address of the secondary NetBIOS Name Service (NBNS) server used by Microsoft NetBIOS clients to reach netBIOS servers.

The no form of this command removes the IP address from the configuration.

Parameters

ip-address

specifies the IPv4 address of the NBNS server

5.2.1.7.14 nbns-srvr-ipv6

Syntax

```
nbns-srvr-ipv6
```

Context

```
config>mobile>pdn>apn>pco
```

Description

This command enables the context to configure IPv6 NBNS server parameters.

5.2.1.7.15 primary

Syntax

```
primary ipv6-address
```

```
no primary
```

Context

```
config>mobile>pdn>apn>pco>nbns6
```

Description

This command configures the primary IPv4 NBNS address.

The no form of this command removes the IP address from the configuration.

Parameters

ipv6-address

specifies the IPv6 address of the NBNS server

5.2.1.7.16 secondary

Syntax

```
secondary ipv6-address  
no secondary
```

Context

```
config>mobile>pdn>apn>pco>nbns6
```

Description

This command configures the secondary IPv6 NBNS address.

The no form of this command removes the IP address from the configuration.

Parameters

ipv6-address
specifies the IPv6 address of the NBNS server

5.2.1.7.17 operator-pco

Syntax

```
operator-pco custom-1  
no operator-pco
```

Context

```
config>mobile>pdn>apn>pco
```

Description

This command is used to enable the operator specific PCO container to be sent back to the UE.

The no form of this command reverts to the default.

Default

```
no operator-pco
```

Parameters

custom-1

specifies a PCO container for an operator

5.2.1.7.18 **pcscf-srvr-ipv4**

Syntax

```
pcscf-srvr-ipv4 pcscf-server-ipv4  
no pcscf-srvr-ipv4
```

Context

```
config>mobile>pdn>apn>pco
```

Description

This command configures the Proxy Call Session Control Function (P-CSCF) server IPv4 address.

The no form of this command removes the IP address from the configuration.

Parameters

pcscf-server-ipv4

specifies the IPv4 address of the P-CSCF server

5.2.1.7.19 **pcscf-srvr-ipv6**

Syntax

```
pcscf-srvr-ipv6 pcscf-server-ipv6  
no pcscf-srvr-ipv6
```

Context

```
config>mobile>pdn>apn>pco
```

Description

This command configures the Proxy Call Session Control Function (P-CSCF) server IPv6 address.

The no form of this command removes the IP address from the configuration.

Parameters

pcscf-server-ipv6

specifies the IPv6 address of the P-CSCF server

5.2.1.7.20 **pcscf-srvr-v4-group**

Syntax

```
pcscf-srvr-v4-group pcscf-group [secondary pcscf-group] [tertiary pcscf-group]  
no pcscf-srvr-v4-group
```

Context

```
config>mobile>pdn>apn>pco  
config>mobile>pdn>apn>pco>ue-location>tai-lai-list
```

Description

This command associates a v4 P-CSCF server group to an APN. The P-CSCF server groups must be defined under the profile context, prior to associating it with an APN. Command `pcscf-srvr-v4-group` is mutually exclusive with the `pcscf-srvr-ipv4` command under the PCO context.

The `no` form of this command removes the v4 P-CSCF server groups association with this APN.

Parameters

pcscf-group

specifies the name of the primary P-CSCF server group, up to 31 characters

secondary *pcscf-group*

specifies the secondary P-CSCF server group

tertiary *pcscf-group*

specifies the tertiary P-CSCF server group

5.2.1.7.21 **pcscf-srvr-v6-group**

Syntax

```
pcscf-srvr-v6-group pcscf-group [secondary pcscf-group] [tertiary pcscf-group]  
no pcscf-srvr-v6-group
```

Context

```
config>mobile>pdn>apn>pco  
config>mobile>pdn>apn>pco>ue-location>tai-lai-list
```

Description

This command associates a v6 P-CSCF server group to an APN. The P-CSCF server groups must be defined under the profile context, prior to associating it with an APN. Command `pcscf-srvr-v6-group` is mutually exclusive with the `pcscf-srvr-ipv6` command under the `pco` context.

The `no` form of this command removes the v6 P-CSCF server groups association with this APN.

Parameters

pcscf-group

specifies the name of the primary P-CSCF server group, up to 31 characters

secondary pcscf-group

specifies the secondary P-CSCF server group

tertiary pcscf-group

specifies the tertiary P-CSCF server group

5.2.1.722 preferred-dns-address-type

Syntax

`preferred-dns-address-type type type`

`no preferred-dns-address-type`

Context

`config>mobile>pdn>apn>pco`

Description

This command controls the DNS addresses sent to the UE in the PCO/APCO. In the default behavior, the UE requests are followed, but if IPv4 is preferred, the PGW/GGSN does not return an IPv6 DNS server address to the UE in PCO/APCO if the UE already has an IPv4 session to the same APN using IPv4 DNS server or if the UE is creating an IPv4v6 session. Instead, an IPv4 DNS server address is returned to UE.

The `no` form of this command disables the functionality.

The DNS addresses must be available, either locally configured or acquired from external source, for this feature to work.

Parameters

type

specifies the DNS addresses sent to the UE

Values	none	the system follows the UE requests
–	IPv4	the system prefers IPv4 DNS addresses

5.2.1.7.23 selection-type

Syntax

```
selection-type {imsi | msisdn | imei} {prefix | full-id | suffix}  
no selection-type
```

Context

```
config>mobile>pdn>apn>pco
```

Description

This command specifies the P-CSCF selection criteria which can be based on an IMSI, MSISDN, or IMEI range.

When you select the `full-id` option, enter a *start-id* and *end-id* for each user - range entry (when you select the `prefix` option, the *end-id* is optional).

When you select the `suffix` option, you must enter a suffix digit.

If the IMSI/MSISDN/IMEI for a specific UE does not match any of the configured entries, the selection of the P-CSCF follows the existing selection criteria at PDN level.

The `no` form of this command removes the selection type for the user - range from the configuration.

Default

```
no selection-type
```

Parameters

imsi

specifies an IMSI range for the P-CSCF selection

msisdn

specifies an MSISDN range for the P-CSCF selection

imei

specifies an IMEI range for the P-CSCF selection

prefix

specifies that the mapping is based on an IMSI, MSISDN, or IMEI prefix; up to 11 digits

full-id

specifies that the mapping is based on a full identifier for the start and end identifiers for the user - range command; the *start-id* and *end-id* must be full identifiers. For IMSI, the identifier must be 5 to 15 digits long. For MSISDN, the identifier must be 4 to 15 digits long.

suffix

specifies an integer that selects the end digit for the user - range.

Default: full-id

5.2.1.7.24 ue-location

Syntax

```
ue-location
```

Context

```
config>mobile>pdn>apn>pco
```

Description

This command enables the context to configure the allocation of P-CSCF addresses based on the UE location.

5.2.1.7.25 tai-lai-list

Syntax

```
tai-lai-list list-id tai-lai-list-name
```

```
no tai-lai-list list-id
```

Context

```
config>mobile>pdn>apn>pco>ue-location
```

Description

This command associates a Tracking Area Identity (TAI) or Location Area Identity (LAI) profile list to the UE's location for this APN.

The no form of this command removes the specified `tai-lai-list` from this configuration.

Parameters

list-id

specifies the static list ID; used only as a configuration reference

Values: 1 to 255

tai-lai-list-name

specifies the name of a TAI/LAI list profile, up to 32 characters

5.2.1.8 user-range

Syntax

```
user-range user-range-name {start-id [end-id] | suffix} pcscf-srvr-v4-group pcscf-group
[secondary-pcscf-srvr-v4-group pcscf-group] [tertiary-pcscf-srvr-v4-group pcscf-group]
pcscf-srvr-v6-group pcscf-group [secondary-pcscf-srvr-v6-group pcscf-group]
[tertiary-pcscf-srvr-v6-group pcscf-group]
```

```
no user-range user-range-name
```

Context

```
config>mobile>pdn>apn>pco
```

Description

This command configures the P-CSCF IP addresses sent to the UE, based on the IMSI, MSISDN, or IMEI mapping to the associated P-CSCF IP addresses.

You can assign one P-CSCF IPv4 server group and one P-CSCF IPv6 server group to every *user-range*, along with their optional secondary and tertiary groups. You must define the P-CSCF server groups under the profile context, before associating them with a *user-range*.

When you select the *prefix* option, you must enter an IMSI, MSISDN, or IMEI prefix up to 11 digits. For IMSI, the prefix length is from MCC+MNC to 11 digits. For MSISDN, the prefix length is from CC + NDC to 11 digits. For IMEI, the prefix length is from 4 to 11 digits.

When you select the *full-id* option, the *start-id* and *end-id* must be full identifiers.

A maximum of 32 entries per PLMN are supported and the ranges must not overlap.

When you select the *suffix* option, you must enter a 1-digit suffix.

The *no* form of this command removes the *user-range* configuration.

Default

```
no user-range
```

Parameters

user-range-name

selects the user range range, up to 32 characters

start-id

specifies the start identifier for the user range, 4 to 16 characters long

end-id

specifies the end identifier for the user range, 4 to 16 characters long

suffix

specifies an integer that selects the end digit for the user range. When you select the `suffix` option, the `start-id` and `end-id` are not used.

Values: 0 to 9

pcscf-group

specifies the name of the primary IPv4 P-CSCF server group, up to 32 characters

secondary-pcscf-srvr-v4-group pcscf-group

specifies the name of the secondary IPv4 P-CSCF server group, up to 32 characters

tertiary-pcscf-srvr-v4-group pcscf-group

specifies the name of the tertiary IPv4 P-CSCF server group, up to 32 characters

pcscf-group

specifies the name of the primary IPv6 P-CSCF server group, up to 32 characters

secondary-pcscf-srvr-v6-group pcscf-group

specifies the name of the secondary IPv6 P-CSCF server group, up to 32 characters

tertiary-pcscf-srvr-v6-group pcscf-group

specifies the name of the tertiary IPv6 P-CSCF server group, up to 32 characters

5.2.1.9 APN PCRF PDN command descriptions

5.2.1.9.1 pcrf-selection

Syntax

`pcrf-selection`

Context

`config>mobile>pdn>apn`

Description

This command configures Policy and Charging Rules Function (PCRF).

5.2.1.9.2 diameter-peer-group-list

Syntax

```
diameter-peer-group-list list-name  
no diameter-peer-group-list
```

Context

```
config>mobile>pdn>apn>pcrf
```

Description

This command associates a Diameter peer group list. It contains a list of Diameter peer groups. Each group is a set of a primary / secondary pair. The `diameter-peer-group-list` command can be used for PCRF/OCS selection based on the priority of the peer group.

Parameters

list-name

name of the Diameter peer group list, up to 32 characters

5.2.1.9.3 dynamic-pcc

Syntax

```
dynamic-pcc [fh-session {fh-session enable | disable}] [fh-3xxx {fh-3xxx enable |  
disable}] [fh-4xxx {fh-4xxx enable | disable}] [gx-profile gx-profile-name] [sd-profile  
sd-profile-name]  
no dynamic-pcc
```

Context

```
config>mobile>pdn>apn>pcrf
```

Description

This command enables dynamic Policy Control (PCC). When specified, the Gx or Sd profile is applied for this APN. All PDN sessions set up to this APN receive the treatment and behavior as per the specified profile.

The `no` form of this command disables the dynamic PCC interface (Gx or Sd) and removes the association of the specified Gx or Sd profile for this APN.

The Gx or Sd profile is specified as an option within the `dynamic-pcc` command and overrides the other `fh-session`, `fh-3xxx` & `fh-4xxx` control options, if specified. The operator should refrain from specifying both.

To prevent conflict between the Gx or Sd profile that can be specified at the PDN level (see `gx-profile`), and the `fh-session`, `fh-3xxx` and `fh-4xxx` control options within `dynamic-pcc`, it is recommended that the profile be specified at the APN level. This avoids possible conflict due to inheritance and overriding at the APN.

Parameters

fh-session

configures the failure handling session (on response from PCRF)

fh-3xxx

configures failure handling for 3xxx result code from PCRF

fh-4xxx

configures failure handling for 4xxx result code from PCRF

enable

allows new calls to go through upon PCRF failure

disable

blocks new calls from going through upon PCRF failure

gx-profile-name

configures the Gx profile name, up to 32 characters

The profile is configured in the mobile gateway Gx profile context; for more information, see [Gx profile command descriptions](#).

sd-profile-name

configures the Sd profile name, up to 32 characters

The profile is configured in the mobile gateway Sd profile context; for more information, see [Sd profile command descriptions](#).

5.2.1.9.4 pcc-disable-per-rat

Syntax

```
pcc-disable-per-rat [UTRAN] [GERAN] [WLAN] [GAN] [HSPA-Evolution] [EUTRAN] [Virtual]
[EUTRAN-NB-IoT]
```

```
no pcc-disable-per-rat
```

Context

config>mobile>pdn>apn>pcrf

Description

This command configures a set of RAT types which are not allowed to be under the control of the PCRF. The values are inherited from the `pcc-disable-per-rat` command under the `config>mobile>profile>charging>pgw-charging>pcrf-selection` context.

For example, if a charging profile disables the WLAN access, and the APN disables 2G and 3G access types, the UE using the configured charging profile will not be under the PCRF's control for WLAN, 2G, and 3G access. However, if a UE uses the same access point but a different charging profile, 2G and 3G access will not be allowed under the PCRF control.

If this command is not configured (no RAT types configured), all the RAT types are allowed under the PCRF's control when dynamic PCC is enabled for the session.

The no form of this command reverts to the default (not configured).

Parameters

UTRAN

keyword that specifies that UTRAN is not allowed to be under the control of the PCRF

GERAN

keyword that specifies that GERAN is not allowed to be under the control of the PCRF

WLAN

keyword that specifies that WLAN is not allowed to be under the control of the PCRF

GAN

keyword that specifies that GAN is not allowed to be under the control of the PCRF

HSPA-Evolution

keyword that specifies that HSPA-Evolution is not allowed to be under the control of the PCRF

EUTRAN

keyword that specifies that EUTRAN is not allowed to be under the control of the PCRF

Virtual

keyword that specifies that Virtual is not allowed to be under the control of the PCRF

EUTRAN-NB-IoT

keyword that specifies that EUTRAN-NB-IoT is not allowed to be under the control of the PCRF

5.2.1.9.5 pcc-disable-per-rat-timer

Syntax

```
pcc-disable-per-rat-timer pcc-disable-per-rat-timer  
no pcc-disable-per-rat-timer
```

Context

```
config>mobile>pdn>apn>pcrf
```

Description

This command configures the timer which is started when the UE moves into a RAT type which is not allowed under PCRF control. For RAT types configured at the charging profile level, the equivalent timer of the charging profile is used instead.

The no form of this command reverts to the default (not configured).

Parameters

pcc-disable-per-rat-timer

sets the timer, in seconds

Values: 0 to 7200

Default: 0 (tears down the PCRF control as soon as the RAT type change occurs)

5.2.1.9.6 primary-diameter-peer

Syntax

```
primary-diameter-peer profile-name  
no primary-diameter-peer
```

Context

```
config>mobile>pdn>apn>pcrf-selection
```

Description

This command configures the primary Diameter peer profile for the APN.

Default

For online charging over APN, a default primary OCS needs to be specified.

Parameters

profile-name

specifies the name of the Diameter profile associated with the primary OCS, up to 32 characters

5.2.1.9.7 secondary-diameter-peer

Syntax

secondary-diameter-peer *profile-name*

no secondary-diameter-peer

Context

config>mobile>pdn>apn>pcrf

Description

This command configures the secondary Diameter peer profile for the APN.

Default

For online charging over APN, a default secondary OCS needs to be specified.

Parameters

profile-name

specifies the name of the Diameter profile associated with the secondary OCS, up to 32 characters

5.2.1.9.8 selection-type

Syntax

selection-type {prefix | full-id | suffix}

no selection-type

Context

config>mobile>pdn>apn>pcrf

Description

This command configures the selection type for the peer user ranges.

The no form of this command removes the selection type for the user - range configuration.

Default

no selection-type

Parameters

prefix

specifies the prefix for the user range

full-id

specifies the full identifier for the user range

suffix

specifies an integer that selects the end digit for the user range.

Default: full-id

5.2.1.9.9 suppress-ccr-with-no-qos-negotiation

Syntax

[no] suppress-ccr-with-no-qos-negotiation

Context

config>mobile>pdn>apn>pcrf

Description

This command defines whether the CCR-U is sent to inform the PCRF about the QoS change received in the update of the PDP context request which also contained a No Qos negotiation indication.

Default

no suppress-ccr-with-no-qos-negotiation

5.2.1.10 user-range

Syntax

```
user-range user-range-name type {imsi | msisdn | imei} {start-id [end-id] | suffix} diameter-  
peer-group-list list-name
```

```
no user-range user-range-name
```

Context

```
config>mobile>pdn>apn>pcrf
```

Description

This command configures the peer user ranges. You can configure up to 32 user range entries.

The no form of this command removes the peer user ranges from the configuration.

Default

```
no user-range
```

Parameters

user-range-name

selects the user range name, up to 32 characters

imsi | msisdn | imei

selects the type of the user range

start-id

selects the start identifier for the user range, from 4 to 16 characters; when you select the prefix option:

- for an IMSI range, the start identifier must be 5 to 15 characters long
- for an MSISDN range, the start identifier must be 4 to 15 characters long
- for an IMEI range, the start identifier must be 4 to 8 characters long

when you select the full-id option:

- for an IMSI range, the start identifier must be 10 to 15 characters long
- for an MSISDN range, the start identifier must be 9 to 15 characters long
- for an IMEI range, the start identifier must be 15 to 16 characters long

end-id

selects the end identifier for the user range, from 4 to 16 characters; when you select the prefix option:

- for an IMSI range, the end identifier must be 5 to 15 characters long
- for an MSISDN range, the end identifier must be 4 to 15 characters long
- for an IMEI range, the end identifier must be 4 to 8 characters long

when you select the `full-id` option:

- for an IMSI range, the end identifier must be 10 to 15 characters long
- for an MSISDN range, the end identifier must be 9 to 15 characters long
- for an IMEI range, the end identifier must be 15 to 16 characters long

suffix

specifies an integer that selects the end digit for the user - range; when you select the `suffix` option, the *start-id* and *end-id* are not used. In case of IMEI/IMEISV, the suffix matching is based on the last digit of the SNR.

Values: 0 to 9

list-name

selects the Diameter peer group list name, up to 32 characters

5.2.1.11 Bearer traffic PDN command descriptions

5.2.1.11.1 bearer-traffic

Syntax

```
bearer-traffic
```

Context

```
config>mobile>pdn
```

Description

This command enables the context to configure bearer traffic parameters.

5.2.1.11.2 gtp-u

Syntax

```
gtp-u
```

Context

config>mobile>pdn>bearer

Description

This command configures gtp-u bearer-traffic.



NOTE:

This command requires a PDN shutdown.

5.2.1.11.3 sequence-number

Syntax

[no] sequence-number

Context

config>mobile>pdn>bearer>gtp-u

Description

This command specifies whether to ignore the sequence number in the GPRS Tunneling Protocol-User plane (GTP-U) header.

When enabled, GTP-U packets are reassembled based on the sequence number and complete packets are forwarded onto S5 interface. If this command is disabled, the sequence number is ignored and complete packets are forwarded onto S5 interface.



NOTE:

This command requires a PDN shutdown.

5.2.1.11.4 udp-checksum

Syntax

[no] udp-checksum

Context

config>mobile>pdn>bearer>gtp-u

Description

This command specifies whether or not to validate the UDP checksum of GPRS Tunneling Protocol-User plane (GTP-U) packets.

If the value of this object is set to enabled, the UDP checksum of the received GTP-U packets is verified and the packets which fail checksum verification are rejected.

If the value of this object is set to disabled, the UDP checksum of the received GTP-U packets is not verified.

The no form of this command disables the verification of the UDP checksum of the received GTP-U packets.



NOTE:

This command requires a PDN shutdown.

5.2.1.11.5 mss-adjust

Syntax

mss-adjust *segment-size*

no mss-adjust

Context

config>mobile>pdn>bearer

Description

This command configures the TCP maximum segment size (MSS) value to set in TCP SYN message for new user traffic sessions that have an MSS value larger than the configured value. The TCP MSS value is adjusted in both directions of flow.

The no form of this command sets MSS adjust to the default, that is no change in the MSS value.

For this command to work without diverting traffic to the AA, the Policy Rule Unit (PRU) must be configured to match protocol 6 (TCP), using the config>mobile>profile>policy-options>policy-rule-unit>flow-description>match>protocol 6 command.

Default

no mss-adjust

Parameters

segment-size

specifies the value to compare and set the TCP MSS for user traffic

Values: 0 | 160 to 10240

Default: 0 (no change)

5.2.1.11.6 non-3gpp-mss-adjust

Syntax

non-3gpp-mss-adjust *mss-value*

no non-3gpp-mss-adjust

Context

config>mobile>pdn>bearer

Description

This command configures the TCP maximum segment size (MSS) value to set in the TCP SYN message for new user traffic sessions that have an MSS value larger than the configured value. The TCP MSS value is adjusted in both directions of flow.

The no form of this command sets MSS adjust to the default, that is no change in the MSS value.



NOTE:

This command is applicable to non-3GPP type of gateway deployment.

Default

no non-3gpp-mss-adjust

Parameters

mss-value

specifies the value to compare and set the TCP MSS for user traffic

Values: 0 | 160 to 10240

Default: 0

5.2.1.12 Cloud database PDN command descriptions

5.2.1.12.1 cdbx

Syntax

`cdbx`

Context

`config>mobile>pdn`

Description

This command enables the context to configure cloud database profile parameters.

5.2.1.12.2 cloud-db-profile

Syntax

`cloud-db-profile profile-name`

`no cloud-db-profile`

Context

`config>mobile>pdn>cdbx`

Description

This command associates a cloud database profile to use as an interface to an external database.

The no form of this command removes the association.

Parameters

profile-name

specifies the profile name, up to 32 characters

5.2.1.12.3 interface

Syntax

`interface [router router-instance] interface-name`

no interface

Context

config>mobile>pdn>cdbx

Description

This command configures a local interface to use for communication with an external database.
The no form of this command removes the local interface association.

Parameters

router-instance

specifies the virtual router name or service ID

Values	router-name service-id	
–	router-name	Base
–	service-id	1 to 2147483647

Default: Base

interface-name

specifies an interface name, up to 32 characters and must start with a letter

5.2.1.12.4 mss

Syntax

mss *mss-value*
no mss

Context

config>mobile>pdn>cdbx

Description

This command configures the MSS size of the packets that the MG-VM forwards to the DB-VM. Nokia recommends increasing the MSS size based on the capabilities of the MTU interface.



NOTE:

- If no MSS value is specified, the default MSS size (1460) is used (derived from the default MTU-40).
- For an MTU value of 9000 bytes, the maximum CDBX MSS value is 8704 bytes for VNF deployments and 8448 bytes for CNF deployments.

Parameters

mss-value

specifies the MSS size

Values: 1460 to 8960

Default: 1460 (disabled)

5.2.1.13 Charging PDN command descriptions

5.2.1.13.1 charging

Syntax

charging

Context

config>mobile>pdn

config>mobile>pdn>apn

Description

This command enables the context to configure charging related profile parameters for the Packet Data Network (PDN) Gateway (PGW) or the Serving Gateway (SGW).

5.2.1.13.2 bill-shock-prevention

Syntax

```
[no] bill-shock-prevention
```

Context

```
config>mobile>pdn>charging
```

Description

This command enables or disables the Bill Shock Prevention (BSP) mechanism. When this command is enabled, the criteria precedence for selecting a charging profile is altered. The roaming status is used as a first priority and the HSS/MME supplied Charging Characteristics (CC) follows. The UE mobility causing a roaming status change will trigger the PDN connection deactivation procedure.

The no form of this command reverts to the default (the HSS/MME supplied CC is used as a first priority).

Default

```
no bill-shock-prevention
```

5.2.1.13.3 cc-ignore

Syntax

```
cc-ignore {home | visiting | roaming | any} [inherit | enable | disable]
```

```
no cc-ignore [home | visiting | roaming | any]
```

Context

```
config>mobile>pdn>apn>charging
```

```
config>mobile>pdn>charging
```

Description

This command configures the charging characteristics for Packet Data Network (PDN) Gateway (PGW) and Serving Gateway (SGW). The selected option ignores the HSS/MME-supplied charging characteristics for home, visiting, roaming scenarios or unconditionally (if “any” is selected), for this APN. Corresponding default profile is selected when the received charging characteristics are ignored.

The no form of this command removes the option from the configuration.

Parameters

home | visiting | roaming | any

indicates whether the charging characteristics received from the MME/HSS are to be ignored for home subscribers, visiting subscribers (local break-out), roamers (home-routed) or unconditionally (any)

inherit

specifies that the charging profile information is inherited

enable

specifies that the charging profile information is enabled

disable

specifies that the charging profile information is disabled

5.2.1.13.4 cc-reject

Syntax

[no] cc-reject

Context

config>mobile>pdn>charging

Description

This command enables or disables the rejection of the charging characteristics supplied by the HSS or the MME for a subscriber.

The no form of this command reverts to the corresponding default charging profile.

Default

no cc-reject

5.2.1.13.5 cc-reject

Syntax

cc-reject [inherit | enable | disable]

no cc-reject

Context

config>mobile>pdn>apn>charging

Description

This command specifies to reject HSS or MME supplied charging characteristics for a subscriber. The corresponding default charging profile is selected when the value of this object is disabled.

Default

By default, values are inherited from PDN level.

Parameters

inherit

specifies that the charging profile information is inherited

enable

specifies that the charging profile information is enabled

disable

specifies that the charging profile information is disabled

5.2.1.13.6 cdr-avp-options-profile

Syntax

cdr-avp-options-profile *cdr-avr-options-profile-id*
no cdr-avp-options-profile

Context

config>mobile>pdn>apn>charging

config>mobile>pdn>charging

Description

This command specifies the profile applicable to Ga CDR or Rf ACR content.

Parameters

cdr-avr-options-profile-id

indicates the profile number

Values: 1 to 255

5.2.1.13.7 cdr-file-separation

Syntax

```
[no] cdr-file-separation
```

Context

```
config>mobile>pdn>charging
```

Description

This command allows the generation of two separate CDR files, one with the SGW CDRs and the other with the PGW CDRs.

When enabled, the node ID used in the filename of the SGW CDRs is taken from the `sgw-node-id` specified in the same context. The node ID used in the filename of the PGW CDRs is taken from the `node-id` specified in the same context.

When disabled, by default CMG generates a single CDR file which contains both the SGW and PGW CDRs. The node ID used in the filename is taken from the `node-id` specified in the same context.

Default

```
no cdr-file-separation
```

5.2.1.13.8 cdr-transfer

Syntax

```
cdr-transfer {ga | bp}
```

```
no cdr-transfer
```

Context

```
config>mobile>pdn>charging
```

Description

This command allows the operator to choose the CDR delivery method using either the Ga interface (GTP prime) or the Bp interface (FTP/SFTP).

The no form of this command reverts to the default.

**NOTE:**

This command requires a PDN shutdown.

Parameters

ga

sets the delivery method using Ga interface (GTP')

bp

sets the delivery method using Bp interface (FTP/SFTP)

Default: ga

5.2.1.13.9 cdr-type

Syntax

```
cdr-type {pgwCdr | gCdr | egCdr}
```

```
no cdr-type
```

Context

```
config>mobile>pdn>charging
```

Description

This command determines what type of CDRs are generated by the system. Some cdr-avp-options are not compatible with all CDR types. For example, the ListofServiceData IE must only appear in eG-CDRs or PGW-CDRs since it is used for flow-based charging

**NOTE:**

This command requires a PDN shutdown.

Parameters

pgwCdr

generates PGW-CDRs

gCdr

generates G-CDRs

egCdr

generates eG-CDRs

Default: pgwCdr

5.2.1.13.10 charging-prof

Syntax

```
charging-prof {home | visiting | roaming | any} charging-prof-id  
charging-prof {home | visiting | roaming | any} [inherit]  
no charging-prof [home | visiting | roaming | any]
```

Context

```
config>mobile>pdn>apn>charging  
config>mobile>pdn>charging
```

Description

This command configures a charging profile to be used with a specific type of user. A charging profile can specify a type of user (home, visiting {local-breakout}, roaming {home-routed}), for this APN. If configured, this profile takes precedence over (overrides) the default profile specified at the gateway level.

The no form of this command reverts to the default.

Default

By default, values are inherited from PDN level.

Parameters

home | visiting | roaming | any

Indicates whether the charging profile is to be associated with home subscribers, visiting subscribers (local break-out) or roamers (home-routed).

charging-prof-id

specifies the charging profile ID number

Values: 0 to 255

inherit

specifies that the charging profile information is inherited

5.2.1.13.11 dcca-profile

Syntax

`dcca-profile profile-name`

Context

`config>mobile>pdn>apn>charging`

Description

This command configures the DCCA profile to be used for the primary and secondary OCS at the charging profile level.

Parameters

profile-name

specifies a DCCA profile name associated with the primary and secondary OCS, up to 32 characters

5.2.1.13.12 diameter-peer-group-list

Syntax

`diameter-peer-group-list list-name`

`no diameter-peer-group-list`

Context

`config>mobile>pdn>apn>charging`

Description

This command associates a Diameter peer group list. It contains a list of Diameter peer groups. Each group is a set of a primary and, or secondary pair. The `diameter-peer-group-list` command can be used for PCRF/OCS selection based on the priority of the peer group.

Parameters

list-name

specifies the name of the Diameter peer group list, up to 32 characters

5.2.1.13.13 enable-nchf-charging

Syntax

[no] enable-nchf-charging

Context

config>mobile>pdn>charging

Description

This command enables Nchf charging for the SMF for 5G-capable UEs only. The UE is assigned an SMF charging profile and an Nchf session is created toward the CHF. If this option is not enabled, the UE is assigned a PGW charging profile, and legacy charging interfaces are used for the session (Ga, Bp, Gy).

The no form of this command reverts to the default (disabled).

Default

no enable-nchf-charging

5.2.1.13.14 gtp-prime-group

Syntax

gtp-prime-group [*gtp-prime-group*]

Context

config>mobile>pdn>apn>charging

Description

This command displays GTP prime group information for the primary and secondary OCS at the charging profile level.

Parameters

gtp-prime-group

specifies the GTP prime group profile name, up to 32 characters

5.2.1.13.15 incomplete-signaling

Syntax

incomplete-signaling

Context

config>mobile>pdn>apn>charging

Description

This command enables the context to configure the incomplete-signaling functionality. It includes the configuration parameters for charging of the incomplete-signaling traffic. This functionality is available for PGW, SMF and UPF.

5.2.1.13.16 charging-method

Syntax

charging-method {online | offline | both / none}

Context

config>mobile>pdn>apn>charging>incomplete-signaling

Description

This command configures the charging method to be used for the incomplete-signaling functionality.

Parameters

online

incomplete-signaling is reported to online charging

offline

incomplete-signaling is reported to offline charging

both

incomplete-signaling is reported to both offline and online charging systems

none

incomplete-signaling is not reported to charging interfaces

Default: offline

5.2.1.13.17 rating-group

Syntax

rating-group *rating-group*

Context

config>mobile>pdn>apn>charging>incomplete-signaling

Description

This command configures the rating group to be used for the incomplete-signaling functionality.

Parameters

rating-group

specifies the rating group which is used for the incomplete-signaling functionality

Values: 0 to 4294967295

5.2.1.13.18 service-id

Syntax

service-id *service-id*

Context

config>mobile>pdn>apn>charging>incomplete-signaling

Description

This command configures the service identifier to be used for the incomplete-signaling functionality.

Parameters

service-id

specifies the service identifier which is used for the incomplete-signaling functionality

Values: 0 to 4294967295

5.2.1.13.19 **urr-id**

Syntax

`urr-id urr-id`

Context

`config>mobile>pdn>apn>charging>incomplete-signaling`

Description

This command configures the Usage-Reporting-Rule-Identifier (urr-id) to be used for the incomplete-signaling functionality.

Parameters

urr-id

specifies the urr-id which is used for the incomplete-signaling functionality

Values: 1 to 30000

5.2.1.13.20 **nchf**

Syntax

`nchf`

Context

`config>mobile>pdn>apn>charging`

Description

This command enables the context to configure the charging function selection.

5.2.1.13.21 **chf-selection**

Syntax

`chf-selection`

Context

`config>mobile>pdn>apn>charging>nchf`

Description

This command enables the context to configure the charging function selection.

5.2.13.22 chf-profile

Syntax

```
chf-profile profile-name  
no chf-profile
```

Context

```
config>mobile>pdn>apn>charging>nchf>chf-selection
```

Description

This command associates a charging function profile to be used by the charging function. The charging function profile is configured using the config>mobile>profile>charging>smf-charging>chf-selection>chf-profile command.

Parameters

profile-name

specifies the profile name, up to 32 characters

5.2.13.23 nf-id-list

Syntax

```
nf-id-list list-name  
no nf-id-list
```

Context

```
config>mobile>pdn>apn>charging>nchf>chf-selection
```

Description

This command associates a CHF ID list with the charging function. The CHF ID list is configured using the config>mobile>profile>list>nf-id-list command.

Parameters

list-name

specifies the list name, up to 32 characters

5.2.1.13.24 nchf-charging

Syntax

nchf-charging {enable | disable | inherit}

no nchf-charging

Context

config>mobile>pdn>apn>charging>nchf

Description

This command configures the LTE legacy charging at the APN level. When the nchf-charging is disabled (the LTE legacy charging is enabled), the 5G UEs and LTE UEs with 5GS subscription that attach to the SMF and PGW-C use the LTE legacy charging interfaces (Gy, Ga, Bp). By default (nchf-charging enabled), such sessions use the Nchf interface for charging.

Parameters

enable

specifies to enable the Nchf charging

disable

specifies to disable the Nchf charging and applies the SMF legacy charging

inherit

specifies that the Nchf charging inherits its configuration from the PDN configuration

Default: inherit

5.2.1.13.25 charging-prof-selection-range

Syntax

charging-prof-selection-range

Context

config>mobile>pdn>charging

Description

This command enables the context to configure the charging profile selection range.

5.2.13.26 selection-type

Syntax

```
selection-type imsi{prefix | full-id}
```

```
no selection-type
```

Context

```
config>mobile>pdn>charging>charging-prof-selection-range
```

Description

This command configures the charging profile selection criteria based on IMSI range.

If the IMSI for a UE does not match any of the configured entries, the selection of the charging profile falls back to other charging profile selection methods (for example charging characteristics, roaming status, PDN level configuration and so on).

The no form of this command removes the selection type for the user - range command.

Parameters

imsi

specifies the IMSI range for the charging profile selection, 5 to 15 digits

prefix

specifies that the mapping is based on the IMSI prefix, up to 11 digits



NOTE:

When you select the **prefix** option, the end-id is optional.

full-id

specifies that the mapping is based on a full identifier for the start and end identifiers of the user - range command, 10 to 15 characters.

For IMSI, the identifier length must be 10 to 15 digits.



NOTE:

When you select the `full-id` option, you must configure the `start-ue-id` and `end-ue-id` parameters for each entry of the `user-range` command (the `start-ue-id` and `end-ue-id` values must be full identifiers).

5.2.13.27 `user-range`

Syntax

```
user-range user-range-name {start-ue-id [end-ue-id]} gw-type {sgw} charging-prof-id value  
no user-range
```

Context

```
config>mobile>pdn>charging>charging-prof-selection-range
```

Description

This command selects an SGW charging profile based on IMSI range. Up to 32 user ranges are supported.

The no form of this command removes the selected *user-range-name*.

Parameters

user-range-name

specifies the user range name, up to 32 characters

start-id

specifies the start identifier for the user range, 5 to 15 characters

end-id

specifies the end identifier for the user range, 5 to 15 characters

sgw

specifies that the selection method is valid for the SGW charging profile

charging-prof-id

specifies the charging profile number

Values: 0 to 255

5.2.13.28 `multiple-pa`

Syntax

```
[no] multiple-pa
```

Context

config>mobile>pdn>charging

Description

This command enables support for the multiple PRA feature for both online and offline charging.

The no form of this command reverts to the default (disabled).

Default

no multiple-pra

5.2.1.13.29 node-id

Syntax

node-id *node-id*

no node-id

Context

config>mobile>pdn>charging

Description

This command specifies an operator configurable identifier for the node which is used over the Rf reference point or the Ga reference point.

The specified node ID is used as a prefix to the names of the files that are created when storing CCR, CDR or ACR charging records. The CDR file name convention has following structure: *<NodeID>_<RC>.<date>_<time>*. The node ID is used as a prefix as shown. The MSCP group ID is also added as suffix to the node ID, if configured, to specify the instance that produces the CDR file. RC is the running count, that increases by one for each new file. For Rf ACR charging records, the file name starts with gw *<gw-id>* as the node ID, if the node ID is not configured. For Gy CCR charging records, the file name always starts with AP-ASO and prefixed with the node ID, if configured. For example, if the node ID is configured as NOKIA-SAEGW01, the names of CDR files are in the form NOKIA-SAEGW01-AP-ASO-RC.*date-time*.

The node ID can be changed without a PDN shutdown, and this affects existing sessions. After the node ID changes, the above files are closed and new files are created using the new node ID as prefix. Also, in Gy interface, the new node ID is added to the corresponding AVP in CCR-U and CCR-T messages.

When a node identifier is used in Charging Data Records (CDRs) sent through a Ga reference point, only the first 18 characters are used.

The no form of this command removes the node identifier from the configuration.

Parameters

node-id

specifies the node identifier, up to 20 characters

5.2.1.13.30 pgw-node-id

Syntax

`pgw-node-id` *pgw-node-id*

`no` `pgw-node-id`

Context

`config>mobile>pdn>charging`

Description

This command specifies an operator configurable identifier for the PGW function, which is used over the Rf, Ga, or Gy reference point for charging records from the PGW. When a node identifier is used in Charging Data Records (CDR) sent through a Ga reference point, only the first 18 characters are used.

When configured, the PGW node ID overrides the node ID configured under `config>mobile>pdn>charging>node-id` for charging records from the PGW.

The `no` form of this command removes the PGW node identifier from the configuration.

Parameters

pgw-node-id

specifies the node identifier for the PGW functionality, up to 20 characters

5.2.1.13.31 plmn-chrg-profile-map

Syntax

`plmn-chrg-profile-map` `plmn-list-name` *list-name* `incoming-cc-value` *cc-value*
`charging-profile-id` *charging-profile-id*

`no` `plmn-chrg-profile-map` `plmn-list-name` *list-name* [`incoming-cc-value` *cc-value*]

Context

`config>mobile>pdn>charging`

`config>mobile>pdn>apn>charging`

Description

This command specifies the charging profiles on a per operator basis. The operator is identified using the PLMN identifier (MCC+MNC). The charging characteristics IE selects a locally configured profile on the gateway. Each of the bits B0 to B15 of a charging characteristics IE corresponds to a specific behavior for that operator and therefore, a corresponding charging profile. Each PLMN has a maximum of up to 16 charging profiles. A maximum of 254 PLMNs is supported.

The no form of this command deletes the mapping for the specified PLMN, when specifying the `plmn-list-name` parameter. If only the `plmn-list-name` parameter is specified, all entries are deleted. If the CC value is also specified, that specific entry is deleted.



NOTE:

For charging profile selection, if a matching PLMN corresponding to the received mobile's MCC+MNC is not found, the appropriate Home/Visiting/Roaming default profile is applied.

Parameters

list-name

specifies the PLMN list name for an operator; it comprises of one or more MCC+MNC values applicable to operator

cc-value

specifies the charging profile behavior identifier; it corresponds to the value of the Charging Characteristics IE received from HSS

charging-profile-id

specifies the charging profile number (configured using the `config>mobile>profile>charging` command)

5.2.1.13.32 pri-cdf-dia-peer

Syntax

```
pri-cdf-dia-peer profile-name
no pri-cdf-dia-peer
```

Context

```
config>mobile>pdn>apn>charging
```

Description

This command provides the primary CDF peer profile information for this APN. If no CDF is specified in the charging profile, the one specified at the APN level is applicable.

The no form of this command removes the profile name from the configuration.

Parameters

profile-name

specifies the profile name, up to 32 characters

5.2.1.13.33 pri-ocs-dia-peer

Syntax

```
pri-ocs-dia-peer profile-name
```

Context

```
config>mobile>pdn>apn>charging
```

Description

This command provides the primary OCS peer profile information.

Parameters

profile-name

specifies the profile name, up to 32 characters

5.2.1.13.34 sec-cdf-dia-peer

Syntax

```
sec-cdf-dia-peer profile-name
```

```
no sec-cdf-dia-peer
```

Context

```
config>mobile>pdn>apn>charging
```

Description

This command provides secondary CDF peer profile information for this APN. If no secondary CDF is specified in the charging profile, the one specified at the APN level is applicable.

Parameters

profile-name

specifies the profile name, up to 32 characters

5.2.1.13.35 sec-ocs-dia-peer

Syntax

```
sec-ocs-dia-peer profile-name
```

```
no sec-ocs-dia-peer
```

Context

```
config>mobile>pdn>apn>charging
```

Description

This command provides secondary OCS peer profile information for this APN. If no secondary OCS is specified in the charging profile, the one specified at the APN level is applicable.

Parameters

profile-name

specifies the profile name, up to 32 characters

5.2.1.13.36 remove-3gpp-headers

Syntax

```
[no] remove-3gpp-headers
```

Context

```
config>mobile>pdn>charging
```

Description

This command, when enabled, excludes the CDR file headers and the CDR headers (as defined in 3GPP TS 32.297) from the CDR files. In this case, the CDR files include only a list of ASN.1 encoded CDRs.

When disabled (default), the CDR files are generated according to 3GPP TS 32.297, with CDR file headers and CDR headers, followed by ASN.1 encoded CDRs.

Default

no remove-3gpp-headers

5.2.13.37 **sgw-node-id**

Syntax

sgw-node-id *sgw-node-id*

no sgw-node-id

Context

config>mobile>pdn>charging

Description

This command specifies an operator configurable identifier for the SGW function, which is used over the Rf reference point or the Ga reference point for charging records from the SGW. When a node identifier is used in Charging Data Records (CDR) sent through a Ga reference point, only the first 18 characters are used.

When configured, the SGW node ID overrides the node ID configured under config>mobile>pdn>charging>node-id for charging records from the SGW.

The no form of this command removes the SGW node identifier from the configuration.

Parameters

sgw-node-id

specifies the node identifier for the SGW functionality, up to 20 characters

5.2.13.38 **supported-vendor**

Syntax

supported-vendor *custom-2*

no supported-vendor

Context

config>mobile>pdn>charging

Description

This command configures the vendor number for which vendor-specific customizations are supported. Each option represents a specific customer.

The no form of this command reverts to the default.

Default

no supported-vendor

Parameters

custom-2

specifies a customer number

5.2.13.39 tariff-time-change-gw-local-tz

Syntax

```
tariff-time-change-gw-local-tz {enable | disable}
```

```
no tariff-time-change-gw-local-tz
```

Context

```
config>mobile>pdn>charging
```

Description

This command facilitates using the gateway local time zone (instead of the UE local time zone) for tariff time change event trigger for offline charging.

The no form of the command reverts to default (disable).

Parameters

enable

specifies to use the gateway local time zone

disable

specifies to use the UE local time zone

Default: disable

5.2.13.40 time-limit-enh

Syntax

[no] time-limit-enh

Context

config>mobile>pdn>charging

Description

This command forces time limit CDRs to have the duration of the configured time threshold configured using the time-limit command under the config>mobile>profile>charging>pgw-charging>off-line>partial-record-closure-trigger context. When enabled, time-based CDR generation takes place at the configured time after any previous CDR or after the PDN creation.

When disabled, time-based CDRs are generated at the configured minutes after the previous time-based CDR or after the PDN creation; therefore, a time-based CDR may have a smaller duration than the configured time threshold when there was one or more non-time-based CDRs generated before it.

Default

no time-limit-enh

5.2.13.41 use-last-known-uli

Syntax

[no] use-last-known-uli

Context

config>mobile>pdn>charging

Description

This command configures the PGW to use the last known ULI in updates toward online charging, offline charging, and RADIUS accounting.

The no form of this command disables the use of the ULI in updates toward online charging, offline charging, and RADIUS accounting.

Default

no use-last-known-uli

5.2.1.13.42 volume-limit-enh

Syntax

[no] volume-limit-enh

Context

config>mobile>pdn>charging

Description

This command facilitates custom volume-limit partial record enhancement for a customer. When enabled, a partial record is closed at the specified volume-limit.

Partial CDRs closed as a result of other triggers do not exceed the specified volume limit. Excess usage counts are carried over to the next container / reporting interval.

The no form of this command reverts to the default (disabled). This command applies to Ga, Bp, and Rf.

Default

no volume-limit-enh

5.2.1.14 Ga/Gn/Gp/Gx/Gy interface PDN command descriptions

5.2.1.14.1 ga

Syntax

ga

Context

config>mobile>pdn

Description

This command enables the context to configure Ga parameters.

5.2.14.2 data-record-format-version

Syntax

```
data-record-format-version string  
no data-record-format-version
```

Context

```
config>mobile>pdn>ga
```

Description

This command configures the Release and Version Identifier parts of the Data Record Format IE. Together they are the version of TS 32.298 that is used to create the CDR.

The Data Record Format Version defaults to the baseline release of TS 32.298 that is supported by the 7750 SR MG and CMG, but the operator can override the version with this command.

The contents of the CDRs are not affected by this command. Instead the individual IEs included in CDRs may be managed using the `cdr -avp -options` and related commands. The operator should ensure that the Data Record Format Version and the CDR AVP Options profile are configured to be consistent with each other.

Parameters

string

specifies the data record format version. The string must in the "xx.yy" format where xx is the release identifier and yy is the version identifier as shown on the cover sheet of TS 32.298. "xx" must be a number from 4 to 16 because the encoding offers only 4 bits. "yy" must be a number from 0 to 30. The system applies the encoding described in TS 32.295. (In other words, the system adds 1 to the version identifier.)

Values: xx.yy

xx [4 to 16]

yy [0 to 30]

5.2.14.3 gtp-prime-group

Syntax

```
gtp-prime-group gtp-prime-group  
no gtp-prime-group
```

Context

config>mobile>pdn>ga

Description

This command specifies the GTP prime group name.



NOTE:

This command requires a PDN shutdown.

Parameters

gtp-prime-group

specifies the GTP prime group name, up to 32 characters

5.2.1.14.4 interface-gtp-prime

Syntax

interface-gtp-prime [router *router-instance*] *interface-name*

no interface-gtp-prime

Context

config>mobile>pdn>ga

Description

This command specifies the local interface used for the GA reference point.



NOTE:

This command requires a PDN shutdown.

Parameters

router-instance

specifies the router instances

Values	router-name service-id	
–	router-name	Base
–	service-id	1 to 2147483647

Default: Base

interface-name

specifies the name of the local interface, up to 32 characters and must start with a letter

5.2.1.14.5 **gn**

Syntax

gn [*sig-ref-point-name*] [group-specific]

no gn [*sig-ref-point-name*]

Context

config>mobile>pdn

Description

This command configures the Gn parameters on all Gn interfaces, or for a specific Gn interface.



NOTE:

This command requires a PDN shutdown, unless the group-specific parameter is specified.

Parameters

sig-ref-point-name

specifies the Gn interface name, up to 32 characters

group-specific

specifies an MG group-specific reference point interface for GTP-U redirection; it is used when associating the reference point interface with an MG system group via an override profile and applies only to CMG

5.2.14.6 gp

Syntax

```
gp [sig-ref-point-name] [group-specific]  
no gp [sig-ref-point-name]
```

Context

```
config>mobile>pdn
```

Description

This command configures Gp parameters on all Gp interfaces, or for a specific Gp interface.



NOTE:

This command requires a PDN shutdown, unless the `group-specific` parameter is specified.

Parameters

sig-ref-point-name

specifies the Gp interface name, up to 32 characters

group-specific

specifies an MG group-specific reference point interface for GTP-U redirection; it is used when associating the reference point interface with an MG system group via an override profile and applies only to CMG

5.2.14.7 interface

Syntax

```
interface
```

Context

```
config>mobile>pdn>gn  
config>mobile>pdn>gp
```

Description

This command configures GTP-C and GTP-U parameters.



NOTE:

This command requires a PDN shutdown.

5.2.1.14.8 gtp-c

Syntax

```
gtp-c [router router-instance] interface-name [interface-realm interface-realm]  
no gtp-c
```

Context

```
config>mobile>pdn>gn>interface  
config>mobile>pdn>gp>interface
```

Description

This command configures the GTP-C profile parameters for the Gn or Gp interface.



NOTE:

This command requires a PDN shutdown.

Parameters

router-instance

specifies the virtual router name or service ID

Values	router-name service-id	
–	router-name	Base
–	service-id	1 to 2147483647

Default: Base

interface-name

specifies an interface name up to 32 characters, which must start with a letter

interface-realm

specifies a realm name, up to 32 characters

**NOTE:**

Only the following characters are permitted:

- alphabetic characters (A-Z and a-z)
- digits (0-9)
- hyphen (-)
- dot (.)

5.2.14.9 gtp-u**Syntax**

```
gtp-u [router router-instance] interface-name [interface-realm interface-realm]  
no gtp-u
```

Context

```
config>mobile>pdn>gn>interface
```

```
config>mobile>pdn>gp>interface
```

Description

This command configures the GTP-U profile parameters for the Gn or Gp interface.

**NOTE:**

This command requires a PDN shutdown.

Parameters***router-instance***

specifies the virtual router name or service ID

Values	router-name service-id
--------	--------------------------

–	router-name	Base
–	service-id	1 to 2147483647

Default: Base

interface-name

specifies an interface name up to 32 characters, which must start with a letter

interface-realm

specifies a realm name, up to 32 characters



NOTE:

- alphabetic characters (A-Z and a-z)
- digits (0-9)
- hyphen (-)
- dot (.)

5.2.14.10 peer-list

Syntax

```
peer-list list-name
```

```
no peer-list
```

Context

```
config>mobile>pdn>gn
```

```
config>mobile>pdn>gp
```

Description

This command configures the peer list name. The list consists of remote GTP peers where each GTP peer is identified by its IP address or subnet. If a matching GTP peer is found in the list, the additional GTP peer configuration can define attributes of the GTP peer (for example, a PLMN ID in case the remote GTP peer does not provide this information over GTP), and can either disable or enable the GTP path management for a specific GTP peer. The peer list can also define similar default configuration for those GTP peers which are not explicitly listed.

The no form of this command removes the list name from the configuration.

**NOTE:**

This command requires a PDN shutdown.

Default

no trusted-peer-list

Parameters***list-name***

specifies the list name, up to 32 characters

5.2.1.14.11 signaling

Syntax

signaling

Context

config>mobile>pdn>gn

config>mobile>pdn>gp

Description

This command enables the context to configure signaling parameters.

**NOTE:**

This command requires a PDN shutdown.

5.2.1.14.12 gtp-c

Syntax

gtp-c

Context

config>mobile>pdn>gn>signaling


```
config>mobile>pdn>gp
```

Description

This command specifies the GPRS Tunneling Protocol-Control plane (GTP-C) profile name. The GTP-C profile provides GTP-C signaling related configuration.



NOTE:

This command requires a PDN shutdown.

5.2.1.14.13 profile

Syntax

```
profile profile-name
```

```
no profile
```

Context

```
config>mobile>pdn>gn>signaling>gtp-c
```

```
config>mobile>pdn>gp>signaling>gtp-c
```

Description

This command specifies the GTP-C profile name.

The no form of this command removes the profile name from the configuration.



NOTE:

This command requires a PDN shutdown.

Default

```
no profile
```

Parameters

profile-name

specifies the GPRS Tunneling Protocol-Control plane (GTP-C) profile name

5.2.1.14.14 gtp-u

Syntax

```
gtp-u
```

Context

```
config>mobile>pdn>gn>signaling
```

```
config>mobile>pdn>gp>signaling
```

Description

This command configures the context to configure GTP-U-related parameters.

GTP-U is a IP-based tunneling protocol that permits many tunnels between each set of end points. When used in the UMTS, each subscriber can have one or more tunnel, one for each PDP context they have active. Separate tunnels are possible for specific connections with different QoS requirements.

The separate tunnels are identified by a Tunnel Endpoint Identifier (TEID) in the GTP-U messages, which is generally a dynamically allocated random number. If the number is encrypted quality then it provides some security against certain attacks. Even so, the requirement of the 3GPP standard is that all GTP traffic, including user data should be sent within secure private networks, not directly connected to the Internet.

**NOTE:**

This command requires a PDN shutdown.

5.2.1.14.15 gtpu-keep-alive

Syntax

```
[no] gtpu-keep-alive
```

Context

```
config>mobile>pdn>gn>signaling>gtp-u
```

```
config>mobile>pdn>gp>signaling>gtp-u
```

```
config>mobile>pdn>s4>signaling>gtp-u
```

```
config>mobile>pdn>s5>signaling>gtp-u
```

```
config>mobile>pdn>s8>signaling>gtp-u
```

Description

This command enables the GTP-U echo on the reference point. To send GTP-U echoes, a non-zero keep-alive value must also be configured in the `gtp-profile` command configured in the same context.

The no form of this command reverts to the default (GTP-U path management disabled).

Default

no gtpu-keep-alive

5.2.1.14.16 profile

Syntax

`profile profile-name`

`no profile`

Context

`config>mobile>pdn>gn>signaling>gtp-u`

`config>mobile>pdn>gp>signaling>gtp-u`

Description

This command specifies the GPRS Tunneling Protocol-User plane (GTP-U) profile name. The GTP-U profile provides GTP-U signaling related configuration.

The no form of this command removes the profile name from the configuration.



NOTE:

This command requires a PDN shutdown.

Parameters

profile-name

specifies the GPRS Tunneling Protocol-User plane (GTP-U) profile name

5.2.1.14.17 qci-policy

Syntax

```
qci-policy {qci-policy-name | uplink qci-policy-name [downlink qci-policy-name] | downlink  
qci-policy-name [uplink qci-policy-name]}  
  
no qci-policy
```

Context

```
config>mobile>pdn
```

Description

This command specifies the QoS Class Identifier (QCI) policy profile name for uplink and downlink directions.

QCI policies configured can be applied to both uplink and downlink bearer traffic or separate QCI policies can be configured in both uplink and downlink directions.

The no form of this command removes the policy name from the configuration.

Default

```
no qci-policy
```

Parameters

qci-policy-name

specifies a QCI policy name, up to 32 characters

uplink *qci-policy-name*

specifies the QCI policy profile name for uplink direction

downlink *qci-policy-name*

specifies the QCI policy profile name for downlink direction

5.2.1.14.18 qfi-mapping-profile

Syntax

```
qfi-mapping-profile profile-name  
  
no qfi-mapping-profile
```

Context

config>mobile>pdn

Description

This command associates an optional QoS flow binding profile for the PDN. The profile must be created first before associating it for the PDN. The QoS parameters specified within the profile are used for QoS flow binding. When this optional profile is not configured, QFI is generated dynamically (in the range 1 to 64) using 5QI/ARP as default.

The no form of this command removes the profile name from the configuration.

Parameters

profile-name

specifies the QFI profile name, up to 32 characters

5.2.1.14.19 gx

Syntax

gx

Context

config>mobile>pdn

Description

This command enables the context to configure GX signaling. Gx provides transfer of (QoS) policy and charging rules from PCRF to Policy and Charging Enforcement Function (PCEF) in the PGW. The Gx reference point enables a PCRF to have dynamic control over the PCC behavior at a PCEF. The Gx reference point enables the signaling of PCC decision, which governs the PCC behavior, and it supports the following functions:

- Request for PCC decision from PCEF to PCRF
- Provision of PCC decision from PCRF to PCEF
- Negotiation of IP-CAN bearer establishment mode (UE-only or UE/NW)
- Termination of Gx session (corresponding to an IP-CAN session) by PCEF or PCRF

5.2.14.20 active-usage

Syntax

```
active-usage {disable | enable} active-threshold threshold-value  
no active-usage
```

Context

```
config>mobile>pdn>gx
```

Description

This command provides the option to enable active-usage for time-consumption, for time-based usage monitoring. When enabled, the applicable activity-threshold is to be specified.

activity-threshold parameter defines an average data rate under which the subscriber-host is considered silent. As long as the effective rate of the application-usage (for a monitoring-key instance) will not exceed the rate defined by *activity-threshold*, given subscriber host will be considered silent and its corresponding quota will not be used. As long as the application-usage (for a monitoring-key instance) will exceed the rate, the corresponding time-quota will be consumed (in terms of time).

The no form of this command reverts to the default (disabled).

Parameters

disable

disables active usage for time consumption

enable

enables active usage for time consumption

threshold-value

specifies the threshold value

Values: 0 to 100000000

Default: 0

5.2.14.21 alt-peer-rar

Syntax

```
[no] alt-peer-rar
```

Context

```
config>mobile>pdn>gx
```

Description

This command enables the gateway to accept RAR messages from a PCRF other than the primary or secondary PCRF. When this command is enabled, a RAA message will be sent back to the peer that initiated the RAR. The IP-CAN session will continue to use the primary or secondary PCRF for the Gx session (that is, any CCR-U or CCR-T for the session will be sent to the destination peer that the last CCA was received from. The no form of this command reverts to the default (disabled).

Default

no alt-peer-rar

5.2.1.14.22 connection-failure-timer

Syntax

```
connection-failure-timer value  
no connection-failure-timer
```

Context

```
config>mobile>pdn>gx
```

Description

This command configures the time in seconds that the gateway waits after the connectivity failure is detected. When the timer expires, the preserve-volte command is applied, if enabled on the directly connected Diameter peer.

Parameters

value

specifies the time, in seconds
Values: 1 to 300
Default: 5

5.2.1.14.23 doic

Syntax

```
doic
```

Context

```
config>mobile>pdn>gx  
config>mobile>pdn>s6b  
config>mobile>pdn>sd  
config>mobile>pdn>sta  
config>mobile>pdn>swm
```

Description

This command enables the context to configure the Diameter Overload Indication Conveyance (DOIC) interface parameters.

5.2.1.14.24 diameter-overload-control

Syntax

```
[no] diameter-overload-control
```

Context

```
config>mobile>pdn>gx>doic  
config>mobile>pdn>s6b>doic  
config>mobile>pdn>sd>doic  
config>mobile>pdn>sta>doic  
config>mobile>pdn>swm>doic
```

Description

This command enables the Diameter Overload Indication Conveyance (DOIC) on the Diameter interface. When enabled, the PGW acts as the reactive node with capability announcement and overload action handling based on RFC 7683.

With Diameter overload enabled, the OC-Support-Features AVP grouped AVP is always sent with the OC-Feature-Vector AVP set to 0x000000000000000010 (OLR_DEFAULT_ALGO).

The no form of this command reverts to the default.



NOTE:

Initial Diameter requests are always realm-based routed. This means that in case of a host-level Diameter overload with multiple hosts associated with the reference point peer, the Diameter client will

not throttle any initial Diameter request until OC-OLR has been received from all the associated hosts to the reference point peer.

Default

no diameter-overload-control

5.2.1.14.25 dra-diameter-too-busy

Syntax

```
dra-diameter-too-busy [re-router-timer minute-value]  
no dra-diameter-too-busy
```

Context

```
config>mobile>pdn>gx
```

Description

This command configures the DRA overload handling based on receiving the result code Diameter_Too_Busy (3004), when the system is configured with Primary and Secondary DRA.

When enabled, and the PGW receives result code Diameter_Too_Busy from the DRA peer, the PGW reroutes all Diameter requests to the secondary DRA for the duration of the reroute timer. Any Diameter respond messages with result code Diameter_too_Busy from the remote PCRF server are not rerouted over the other DRA peer and follow failure handling with either continue (bypass), retry/terminate, or terminate as configured in the Gx profile.

The no form of this command reverts to the default (disabled).

Default

no dra-diameter-too-busy

Parameters

minute-value

specifies the time period, in minutes, that all subsequent Diameter messages are routed to the alternative DRA peer

Values: 1 to 30

Default: 5

5.2.14.26 **gx-avp-options-profile**

Syntax

```
gx-avp-options-profile profile-number  
no gx-avp-options-profile
```

Context

```
config>mobile>pdn>gx
```

Description

This command specifies the applicable Gx AVP options profile.

Parameters

profile-number

specifies the profile number

Values: 1 to 255

5.2.14.27 **gx-profile**

Syntax

```
gx-profile profile-name  
no gx-profile
```

Context

```
config>mobile>pdn>gx
```

Description

This command specifies a (default) Gx profile at the PDN level. If specified, the APN level Gx profile overrides the PDN level profile; see the `dynamic-pcc gx-profile` option. If no profile is specified at the APN level, the profile specified at the PDN level is inherited.

The no form of this command removes the association of the specified profile at the PDN level.

Parameters

profile-name

specifies the name of the Gx profile, up to 32 characters. The Gx profile is configured in the mobile gateway Gx profile context; see [Gx Profile command descriptions](#) for information.

5.2.14.28 interface

Syntax

```
interface
```

Context

```
config>mobile>pdn>gx
```

Description

This command enables the context to configure Diameter interface parameters

**NOTE:**

This command requires a PDN shutdown.

5.2.14.29 diameter

Syntax

```
diameter [router router-instance] interface-name
```

```
no diameter
```

Context

```
config>mobile>pdn>gx>if
```

Description

This command configures Diameter interface parameters. This command configures the virtual router in which the Diameter connections will be established by this Diameter policy.

The no form of this command removes the parameter from the configuration.

**NOTE:**

This command requires a PDN shutdown.

Parameters

router-instance

specifies the virtual router name or service ID

Values	router-name service-id	
–	router-name	Base
–	service-id	1 to 2147483647

Default: Base

interface-name

specifies an interface name, up to 32 characters

5.2.1.14.30 multi-homing

Syntax

multi-homing
no multi-homing

Context

config>mobile>pdn>gx>if
config>mobile>pdn>gy>if
config>mobile>pdn>rf>if
config>mobile>pdn>s6b>if

Description

This command enables multi-homing from the local peer. When multi-homing is enabled, a secondary Diameter interface can also be specified.

The no form of this command reverts to the default (disabled).



NOTE:

- Current multi-homing support requires primary and secondary interfaces to be in the same VPRN.

- This command requires a PDN shutdown.

Default

no multi-homing

5.2.1.14.31 diameter

Syntax

```
diameter [router router-instance] secondary-interface-name  
no diameter
```

Context

```
config>mobile>pdn>gx>if>multi-homing  
config>mobile>pdn>gy>if>multi-homing  
config>mobile>pdn>rf>if>multi-homing  
config>mobile>pdn>s6b>if>multi-homing
```

Description

This command configures Diameter secondary interface parameters and the virtual router in which the Diameter connection will be established by this Diameter policy.

The no form of this command removes the parameter from the configuration.



NOTE:

- Current multi-homing support requires primary and secondary interfaces to be in the same VPRN.
- This command requires a PDN shutdown.

Parameters

router-instance

specifies the virtual router name or service ID

Values	router-name service-id
--------	--------------------------

–	router-name	Base
–	service-id	1 to 2147483647

Default: Base

secondary-interface-name

specifies an interface name up to 32 characters

5.2.1.14.32 origin-host

Syntax

```
origin-host name  
no origin-host
```

Context

```
config>mobile>pdn>gx  
config>mobile>pdn>gy  
config>mobile>pdn>rf  
config>mobile>pdn>s6b  
config>mobile>pdn>sd  
config>mobile>pdn>sta  
config>mobile>pdn>swm
```

Description

This command specifies the originating host name used in the interface. The configured value is prepended with an MSCPxx prefix (where xx is the group ID of the redundant pair) and thus, the resulting origin host is unique for each redundant pair of ISM-MG cards or per MG-VM in CMG.

For example, for a single ISM-MG redundant pair in the chassis, if the configured origin host is `gx-7750mg.ipd-nokia.com`, the origin host sent over the connection is `mscp01.gx-7750mg.nokia.com`.

If no value is defined, the origin host parameter configured in the context `config>mobile>pdn>signaling>diameter>origin-host` is used instead.

The no form of this command reverts to the default (no value defined).

**NOTE:**

- The configured Gy origin host name cannot be removed if the `limit-session-id-length` command is enabled.
- This command requires a PDN shutdown.

Parameters

name

specifies the origin host name, up to 80 characters. If the `limit-session-id-length` command is enabled, the origin host name value must be up to 53 characters.

5.2.1.14.33 pcrf-selection

Syntax

```
pcrf-selection
```

Context

```
config>mobile>pdn>gx
```

Description

This command enables the context to configure Policy and Charging Rules Function (PCRF) selection parameters.

**NOTE:**

This command requires a PDN shutdown.

5.2.1.14.34 diameter-peer-group-list

Syntax

```
diameter-peer-group-list list-name
```

```
no diameter-peer-group-list
```

Context

```
config>mobile>pdn>gx>pcrf
```

```
config>mobile>pdn>gy
```

Description

This command associates a Diameter peer group list. It contains a list of Diameter peer groups. Each group is a set of a primary / secondary pair. The `diameter-peer-group-list` command can be used for PCRF/OCS selection based on the priority of the peer group.

Before associating the list, it must be created first using command `diameter-peer-group-list` under context `config>mobile>profile`.



NOTE:

This command requires a PDN shutdown, when used in the `config>mobile>pdn>gy` context.

Parameters

list-name

specifies the name of the Diameter peer group list (up to 32 characters)

5.2.1.14.35 primary-diameter-peer

Syntax

```
primary-diameter-peer profile-name
```

```
no primary-diameter-peer
```

Context

```
config>mobile>pdn>gx>pcrf
```

Description

This command configures default primary Diameter peer profile parameters.

The `no` form of this command removes the profile name from the configuration.



NOTE:

This command requires a PDN shutdown.

Default

```
no primary-diameter-peer
```


Parameters

profile-name

specifies a Diameter profile name, up to 32 characters

5.2.1.14.36 secondary-diameter-peer

Syntax

secondary-diameter-peer *profile-name*

no secondary-diameter-peer

Context

config>mobile>pdn>gx>pcrf

Description

This command configures default secondary Diameter peer profile parameters.

The no form of this command removes the profile name from the configuration.



NOTE:

This command requires a PDN shutdown.

Default

no secondary-diameter-peer

Parameters

profile-name

specifies a Diameter profile name, up to 32 characters

5.2.1.14.37 selection-priority

Syntax

selection-priority {imsi | msisdn | imei} [{imsi | msisdn | imei}... (up to 3 max)]

no selection-priority

Context

```
config>mobile>pdn>gx>pcrf
```

Description

This command configures the selection priority for the user-range of the peers.

The no form of this command removes the selection priority for the user-range from the configuration.

Default

no selection-priority

Parameters

imsi | msisdn | imei

selects the first (or second or third) selection for the user-range

Default: imsi msisdn imei

5.2.1.14.38 selection-type

Syntax

```
selection-type {prefix | full-id | suffix}
```

```
no selection-type
```

Context

```
config>mobile>pdn>gx>pcrf
```

Description

This command configures the selection type for the peer user ranges.

The no form of this command removes the selection type for the user - range configuration.

Default

no selection-type

Parameters

prefix

selects the prefix for the user range

full-id

selects the full identifier for the user range

suffix

specifies an integer that selects the end digit for the user range.

Default: full-id

5.2.1.14.39 user-range

Syntax

```
user-range user-range-name type {imsi | msisdn | imei} {start-id [end-id] | suffix} {primary-  
pcrf profile-name [secondary-pcrf profile-name] | diameter-peer-group-list list-name}  
  
no user-range user-range-name
```

Context

```
config>mobile>pdn>gx>pcrf
```

Description

This command configures the peer user ranges. You can configure up to 32 user range entries.

The no form of this command removes the peer user ranges from the configuration.

Default

```
no user-range
```

Parameters

user-range-name

selects the user range name, up to 32 characters

imsi | msisdn | imei

selects the type of the user range

start-id

selects the start identifier for the user range, 4 to 16 characters long; when you select the prefix option:

- for an IMSI range, the start identifier must be 5 to 15 characters long
- for an MSISDN range, the start identifier must be 4 to 15 characters long
- for an IMEI range, the start identifier must be 4 to 8 characters long

when you select the full-id option:

- for an IMSI range, the start identifier must be 10 to 15 characters long

- for an MSISDN range, the start identifier must be 9 to 15 characters long
- for an IMEI range, the start identifier must be 15 to 16 characters long

end-id

selects the end identifier for the user range, 4 to 16 characters long; when you select the *prefix* option:

- for an IMSI range, the end identifier must be 5 to 15 characters long
- for an MSISDN range, the end identifier must be 4 to 15 characters long
- for an IMEI range, the end identifier must be 4 to 8 characters long

when you select the *full-id* option:

- for an IMSI range, the end identifier must be 10 to 15 characters long
- for an MSISDN range, the end identifier must be 9 to 15 characters long
- for an IMEI range, the end identifier must be 15 to 16 characters long

suffix

specifies an integer that selects the end digit for the user range; when you select the *suffix* option, the *start-id* and *end-id* are not used. In case of IMEI/IMEISV, the suffix matching is based on the last digit of the SNR.

Values: 0 to 9

profile-name

selects the name for the primary PCRF profile, up to 32 characters

secondary-pcrf profile-name

selects the name for the secondary PCRF profile, up to 32 characters

list-name

selects the name of the Diameter peer group list, up to 32 characters

5.2.1.14.40 **retry-count**

Syntax

retry-count *value*

no retry-count

Context

config>mobile>pdn>gx

Description

This command configures the number of times the same message is re-tried before declaring a failed attempt.

The no form of this command removes the value from the configuration.

Default

no retry-count

Parameters

value

specifies the number of times the same message is re-tried before declaring a failed attempt

Values: 0 to 8

Default: 3

5.2.1.14.41 send-called-station-id

Syntax

send-called-station-id *format*

send-called-station-id

Context

config>mobile>pdn>gx

Description

This command configures the Gx Called-Station-Id AVP in the CCR-I toward the PCRF in a case-sensitive manner. This affects real APN, virtual APN, NI-only, public APN, and so on.

The no form of this command reverts to the default value (none); no change in the case format of the APN.

Default

no send-called-station-id

Parameters

format

specifies the case of the APN in the Called-Station-Id AVP

Values	none	specifies that the APN case remains unchanged
–	lower-case	specifies to implement APN in lower-case
–	upper-case	specifies to implement APN in upper-case
–	original-case	specifies to implement APN in original-case

Default: none

5.2.1.14.42 single-diameter-host

Syntax

```
[no] single-diameter-host
```

Context

```
config>mobile>pdn>gx
```

```
config>mobile>pdn>gy
```

```
config>mobile>pdn>s6b
```

Description

This command specifies whether the Diameter interface of the mobile gateway is seen by the Diameter server as a single Diameter client or as multiple Diameter clients. When enabled, the respective interface is seen as a single Diameter host by any Diameter server. When disabled, the respective interface is seen as one Diameter client per MG-ISM node.

The no form of this command reverts to the default (disabled).



NOTE:

This command requires a PDN shutdown, when used in the config>mobile>pdn>gy context.

Default

```
no single-diameter-host
```

5.2.1.14.43 supported-vendor

Syntax

```
supported-vendor {custom-1 | custom-2 | custom-3}
no supported-vendor
```

Context

```
config>mobile>pdn>gx
```

Description

This command configures the vendor number for which vendor-specific customizations are supported. Each number represents a specific customer.

The no form of this command reverts to the default.

Default

no supported-vendor (for example, enhancements related to custom-1 do not apply)

Parameters

custom-1

specifies the custom-1 customer

custom-2

specifies the custom-2 customer

custom-3

specifies the custom-3 customer

Default: no customer

5.2.1.14.44 suppress-ccr-t-unknown-session

Syntax

```
[no] suppress-ccr-t-unknown-session
```

Context

```
config>mobile>pdn>gy
```

Description

This command is used to suppress CCR-T messages sent from the gateway for unknown sessions. When the gateway receives a CCA (I/U) message and the session context is not found, the gateway normally sends a CCR-T message to the OCS. This command suppresses the CCR-T messages from being sent.

The no form of this command reverts to the default (disabled).

Default

no suppress-ccr-t-unknown-session

5.2.1.14.45 tdf-info

Syntax

tdf-info

Context

config>mobile>pdn>gx

Description

This command configures the TDF information.

5.2.1.14.46 tdf-ipv4

Syntax

tdf-ipv4 *ip-address*

no tdf-ipv4

Context

config>mobile>pdn>gx>tdf-info

Description

This command is used to configure a TDF IPv4 address. This information is sent by the PGW to PCRF over the Gx interface. Either the IP address or the realm and host should be configured.

Parameters

ip-address

specifies the IPv4 address of a TDF

Values: a.b.c.d

5.2.1.14.47 tdf-ipv6

Syntax

```
tdf-ipv6 ip-address
```

```
no tdf-ipv6
```

Context

```
config>mobile>pdn>gx>tdf-info
```

Description

This command is used to configure a TDF IPv6 address. This information is sent by the PGW to PCRF over the Gx interface. Either the IP address or the realm and host must be configured.

Parameters

ip-address

specifies the IPv6 address of a TDF, up to 64 characters

5.2.1.14.48 tdf-realm

Syntax

```
tdf-realm realm-name tdf-host host-name
```

```
no tdf-realm
```

Context

```
config>mobile>pdn>gx>tdf-info
```

Description

This command is used to configure a TDF realm and host. This information is sent by the PGW to PCRF over Gx interface. Either the IP address or the realm and host should be configured.

Parameters

realm-name

specifies the TDF realm, up to 80 characters

host-name

specifies the TDF host name, up to 80 characters

5.2.1.14.49 transaction-timer

Syntax

transaction-timer {[*value*] | [timeout-msec *value*]}

no transaction-timer

Context

config>mobile>pdn>gx

Description

This command sets maximum amount of time the node waits for a Diameter peer to respond before trying another. The value can be expressed in seconds or milliseconds. Setting the timer in milliseconds provides better timeout granularity.

The transaction timer configured by transaction-timer in the config>mobile>profile>gx>message context overrides the time specified for transaction-timer.

The no form of this command removes the configuration.

Parameters

value

specifies the maximum amount of time, in seconds, the node waits for a Diameter peer to respond before trying another peer

Values: 1 to 180 (increments in 1 s)

Default: 5

timeout-msec *value*

specifies the maximum amount of time, in milliseconds, the node waits for a Diameter peer to respond before trying another peer

Values: 1000 to 180000 (increments of 100 ms)

Default: 5000

5.2.14.50 usu-options-over-quota-traffic-handling

Syntax

```
usu-options-over-quota-traffic-handling {allow | discard}  
no usu-options-over-quota-traffic-handling
```

Context

```
config>mobile>pdn>gx
```

Description

This command specifies the option to allow or discard the traffic for the service, while waiting for the PCRF to grant more quota. It is only effective if the Gx usage monitoring is currently active.

The no form of this command reverts to the default setting.

Parameters

allow

specifies that traffic can flow through while waiting for new granted quota from the PCRF

discard

specifies that traffic is discarded while waiting for new granted quota from the PCRF

Default: allow

5.2.14.51 usu-options-reporting-threshold

Syntax

```
usu-options-reporting-threshold percentage  
no usu-options-reporting-threshold
```

Context

```
config>mobile>pdn>gx
```

Description

This command specifies when the usage report based on volume threshold is sent to the PCRF for Gx usage monitoring. When the accumulated usage reaches the percentage of the granted service units (GSU), the report is triggered. By default, there is no threshold set, meaning that the report is triggered when the accumulated usage reaches the GSU.

The no form of this command removes this configuration.

Parameters

percentage

specifies the percentage of the GSU that is used as the local threshold for triggering the reporting

Values: 70 to 95

5.2.1.14.52 **usu-options-suppress-excess-quota**

Syntax

[no] usu-options-suppress-excess-quota

Context

config>mobile>pdn>gx

Description

This command suppresses reporting of the Used Service Units (USU) which exceed the previously Granted Service Units (GSU) for the Gx usage monitoring.

This applies when:

- usage monitoring is explicitly disabled by the PCRF
- an IP-CAN session is terminated
- all the PCC rules associated with a monitoring key are removed or deactivated

The no form of this command removes this configuration.

Default

usu-options-suppress-excess-quota

5.2.1.14.53 **gy**

Syntax

gy

Context

config>mobile>pdn

Description

This command enables the context to configure Online Charging (GY) parameters.

5.2.1.14.54 alt-peer-rar-asr

Syntax

[no] alt-peer-rar-asr

Context

config>mobile>pdn>gy

Description

This command enables the Gateway to accept RAR and/or ASR messages from an alternate peer (other than the Primary and Secondary).

When enabled, a RAR message or ASA message as the case may be will be sent back to the peer that initiated the RAR or ASR respectively. The IP-CAN session will continue to use the Primary and Secondary for the Gy session (i.e., any CCR-U or CCR-T for the session will be sent to the destination peer the last CCA was received from).

The no form of this command reverts to the default (disabled).

5.2.1.14.55 dcca-profile

Syntax

dcca-profile *profile-name*

no dcca-profile

Context

config>mobile>pdn>apn>charging

config>mobile>pdn>gy

Description

This command configures the Diameter Credit Control Application (DCCA) profile to be used for the primary and secondary OCS at the charging profile level.

**NOTE:**

This command requires a PDN shutdown, when used in the `config>mobile>pdn>gy` context.

Default

For online charging over Gy, a default DCCA profile needs to be specified.

Parameters

profile-name

specifies a DCCA profile name associated with the primary and secondary OCS, up to 32 characters

5.2.1.14.56 dia-routing-table

Syntax

```
dia-routing-table host-index supported-host fqdn port port transport {tcp | sctp} realm  
dest-realm diameter-peer-group-list peer-list  
  
no dia-routing-table [host-index]
```

Context

`config>mobile>pdn>gy`

Description

This command configures a table with entries that associate the host with a destination realm, to be used toward the selected Diameter peer; for example, DRA. The PCRF-specified Primary-Event-Charging-Function-Name FQDN value (mapped to the supported-host, port, and transport parameters) can use the associated destination realm toward the Diameter peer. The no form of this command deletes a Diameter table entry (if *host-index* is also specified) or the Diameter table.

Default

`no dia-routing-table`

Parameters

host-index

specifies the host index number

Values: 1 to 512

fqdn

specifies the FQDN for the supported host, up to 255 characters

port

specifies the port

Values: 1 to 65535

tcp | sctp

specifies the transport used for the Diameter signaling

dest-realm

specifies the destination realm, up to 80 characters

peer-list

specifies a Diameter peer group list that has been configured in the context

configure>mobile>profile>list>diameter-peer-group-list, up to 32 characters

5.2.1.14.57 diameter-session

Syntax

```
diameter-session {pdn-level | bearer-level}
```

```
no diameter-session
```

Context

```
config>mobile>pdn>gy
```

Description

This command configures the Diameter session.



NOTE:

This command requires a PDN shutdown.

Default

For online charging over Gy, a default DCCA profile needs to be specified.

Parameters

pdn-level

specifies a Diameter session on the PDN level

bearer-level

specifies a Diameter session on the bearer level

5.2.14.58 **exclude-dest-host-pcrf-selected-server**

Syntax

```
[no] exclude-dest-host-pcrf-selected-server
```

Context

```
config>mobile>pdn>gy
```

Description

This command excludes the destination host from the CCR-I for an OCS server that is selected by the PCRF. In this scenario, the DRA selects the appropriate OCS server host based on the destination realm. The no form of this command includes the destination host in the CCR-I (default).

Default

```
no exclude-dest-host-pcrf-selected-server
```

5.2.14.59 **interface**

Syntax

```
interface
```

Context

```
config>mobile>pdn>gy
```

Description

This command enables the context configure a Gy interface.

5.2.14.60 diameter

Syntax

```
diameter [router router-instance] interface-name  
no diameter
```

Context

```
config>mobile>pdn>gy>interface
```

Description

This command specifies Diameter interface parameters.

**NOTE:**

This command requires a PDN shutdown.

Parameters

router-instance

specifies the virtual router instance

Values	router-name service-id	
–	router-name	Base
–	service-id	1 to 2147483647

interface-name

Specifies the interface on which to receive broadcast packets, up to 32 characters; if the string contains special characters (#, \$, spaces), the entire string must be enclosed within double quotes

5.2.14.61 limit-session-id-length

Syntax

```
[no] limit-session-id-length
```

Context

```
config>mobile>pdn>gy
```

Description

This command limits the length of the Session-Id AVP to 80 characters in the Gy messages sent by the PGW/GGSN. This is done by limiting the Gy application origin host name to 53 characters.

This command can only be used when a Gy application origin host name is configured. If the configured Gy application origin host name is more than 53 characters, it must be reconfigured before using this command.

Default

no limit-session-id-length

5.2.1.14.62 ocs-selection

Syntax

```
ocs-selection
```

Context

```
config>mobile>pdn>gy
```

Description

This command configures the Online Charging System (OCS) selection parameters.

5.2.1.14.63 selection-type

Syntax

```
selection-type {imsi | msisdn} {prefix | full-id | suffix}
```

```
no selection-type
```

Context

```
config>mobile>pdn>gy>ocs-selection
```

Description

This command specifies the OCS selection criteria which can be based on an IMSI or MSISDN range.

When you select the `full-id` option, enter a *start-id* and *end-id* for each user - range entry (when you select the `prefix` option, the *end-id* is optional).

When you select the `suffix` option, you must enter a suffix digit.

If the IMSI/MSISDN for a specific UE does not match any of the configured entries, the selection of OCS and DCCA profile follows the existing selection criteria (that is, APN, charging profile, PGW, and so on).

The no form of this command removes the selection type for the user - range.

Default

no selection-type

Parameters

imsi

specifies an IMSI range for the OCS selection

msisdn

specifies an MSISDN range for the OCS selection

prefix

specifies that the mapping is based on an IMSI or MSISDN prefix; up to 11 digits

full-id

specifies that the mapping is based on a full identifier for the start and end identifiers, for the user - range command. When you select the `full-id` option, the *start-id* and *end-id* must be full identifiers. For IMSI, the full identifier is 10 to 15 digits long and for MSISDN, 4 to 15 digits long.

suffix

specifies an integer that selects the end digit for the user range.

5.2.1.14.64 user-range

Syntax

```
user-range user-range-name {start-id [end-id] | suffix} {primary-ocs profile-name [secondary-ocs profile-name] | diameter-peer-group-list list-name} [dcca-profile profile-name]
```

Context

```
config>mobile>pdn>gy>ocs-selection
```

Description

This command specifies an IMSI or MSISDN mapping to the associated primary and secondary OCS peers. The specified DCCA profile applies to the configured primary and secondary OCS peers.

When you select the `prefix` option, enter an IMSI or MSISDN prefix up to 11 digits. For IMSI, the prefix length is from MCC+MNC to 11 digits and you can enter from 5 to 11 digits. For MSISDN, the prefix length is from CC + NDC to 11 digits and you can enter from 4 to 11 digits.

When you select the `full-id` option, the `start-id` and `end-id` must be full identifiers.

A maximum of up to 32 entries per PLMN are supported and the ranges must not overlap.

When you select the `suffix` option, you must enter a 1-digit suffix.

When the IMSI/MSISDN does not match any of the configured ranges, the primary and secondary OCS at the PDN gateway level, when configured, are used by default.

The no form of this command removes the user - range configuration.

**NOTE:**

The no form of this command requires a PDN shutdown.

Parameters

user-range-name

specifies the user range range, up to 32 characters

start-id

specifies the start identifier (IMSI or MSISDN) of the range; for IMSI, the subscriber's IMSI or IMSI prefix; for MSISDN, the subscriber's MSISDN or MSISDN prefix

end-id

specifies the end identifier (IMSI or MSISDN) of the range; for IMSI, the subscriber's IMSI or IMSI prefix; for MSISDN, the subscriber's MSISDN or MSISDN prefix

suffix

specifies an integer that selects the end digit for the user range; when you select the `suffix` option, the `start-id` and `end-id` are not used.

Values: 0 to 9

profile-name

specifies the OCS peer profile name, up to 32 characters

secondary-ocs profile-name

specifies the DCCA profile name associated with the OCS peers, up to 32 characters

list-name

specifies the Diameter peer group list name, up to 32 characters

5.2.14.65 outage

Syntax

outage

Context

config>mobile>pdn>gy

Description

This command enables the context to configure parameters for an outage scenario.

5.2.14.66 file-params

Syntax

file-params

Context

config>mobile>pdn>gy>outage

Description

This command provides the required context for detailed configuration of outage file parameters.

5.2.14.67 closure-rec-number

Syntax

closure-rec-number *closure-rec-number*

no closure-rec-number

Context

config>mobile>pdn>gy>outage>file-params

Description

This command specifies a limit for the number of CCRs that are stored in a file. The file is closed when it reaches the specified limit.

The no form of this command reverts to the default.

Parameters

closure-rec-number

specifies the maximum number of CCR messages to fit in a file

Values: 100 to 75000

Default: 50000

5.2.1.14.68 closure-size

Syntax

`closure-size closure-size`

`no closure-size`

Context

`config>mobile>pdn>gy>outage>file-params`

Description

This command specifies a maximum size for the file. The file is closed when it reaches the specified limit.

The no form of this command returns the setting to the default.

Parameters

closure-size

specifies the file closure size in megabytes

Values: 1 to 100

Default: 50

5.2.1.14.69 closure-time

Syntax

`closure-time closure-time`

`no closure-time`

Context

config>mobile>pdn>gy>outage>file-params

Description

This command specifies a file closure time in hours. The file is closed after the specified time limit is reached.

The no form of this command returns the setting to the default.

Parameters

closure-time

specifies the file closure time, in hours

Values: 1 to 24

Default: 1

5.2.1.14.70 *obsolete-time*

Syntax

obsolete-time obsolete-time

no *obsolete-time*

Context

config>mobile>pdn>gy>outage>file-params

Description

This command specifies a file duration, in days, after which the file is deleted from the primary or secondary directory.

The no form of this command reverts to the default.

Parameters

obsolete-time

specifies the duration, in days, after which the file is deemed obsolete and deleted

Values: 1 to 31

Default: 7

5.2.14.71 location

Syntax

```
location {cf1 | cf2}
```

Context

```
config>mobile>pdn>gy>outage
```

Description

This command specifies the primary compact flash that will be used for CCR storage. The options are cf1 or cf2; cf3 is reserved. If cf1 is chosen as the primary location, cf2 becomes the alternate compact flash that CCR messages will be written to when the primary is full, in the event of an outage.

Parameters

cf1 | cf2

specifies the primary compact flash to used for CCR storage

5.2.14.72 limit

Syntax

```
limit limit
```

```
no limit
```

Context

```
config>mobile>pdn>gy>outage>location
```

Description

This command specifies a limit, in megabytes, to the amount of memory that the compact flash can use for CCR storage.

The no form of this command reverts to the default.

Default

no limit

Parameters

limit

specifies the limit for the compact flash in megabytes

5.2.14.73 primary

Syntax

[no] primary

Context

config>mobile>pdn>gy>outage>location

Description

This command configures the compact flash specified in the config>mobile>pdn>gy>gy>outage>location command as the primary compact flash. The compact flash that is not specified in the location command becomes the secondary compact flash.

The no form of this command reverts to the default.

Default

cf1 is the primary compact flash; cf2 is the secondary compact flash

5.2.14.74 primary-diameter-peer

Syntax

primary-diameter-peer *profile-name*

no primary-diameter-peer

Context

config>mobile>pdn>gy

Description

This command configures the Diameter peer profile to be used for the primary Online Charging System (OCS) at the gateway level.



NOTE:

This command requires a PDN shutdown.

Default

For online charging over Gy, a default primary OCS needs to be specified.

Parameters

profile-name

specifies the name of the Diameter profile associated with the primary OCS, up to 32 characters

5.2.1.14.75 secondary-diameter-peer

Syntax

`secondary-diameter-peer profile-name`

`no secondary-diameter-peer`

Context

`config>mobile>pdn>gy`

Description

This command configures the Diameter peer profile to be used for the secondary Online Charging System (OCS) at the gateway level.



NOTE:

This command requires a PDN shutdown.

Default

For online charging over Gy, a default secondary OCS needs to be specified.

Parameters

profile-name

specifies the name of the Diameter profile associated with the secondary OCS, up to 32 characters

5.2.14.76 session-id-format

Syntax

```
session-id-format {type1 | type2}  
no session-id-format
```

Context

```
config>mobile>pdn>gy
```

Description

This command configures specific sections of the Session-Id AVP in the Gy messages that are sent from the PGW/GGSN to be recognized by the customer billing system (OCS). To apply changes to these sections, when the *type1* option is selected, the session creation time (Unix Time) is added in the second field (high 32 bits) in decimal format, the internal Diameter key in the third field while the optional fourth field is completely removed. If the `limit-session-id-length` functionality is enabled, this command cannot be used. A warning is displayed for the user to disable the `limit-session-id-length` before configuring the `session-id-format` command.

When the *type2* option is selected, the session creation time (in hex format) is added in the end of the session ID.



NOTE:

The *type2* parameter requires that the following command is enabled: `configure mobile-gateway profile dcca supported-vendor custom-1`.

Parameters

type1

specifies the format of the Session-Id AVP

type2

specifies a custom format of the Session-Id AVP

5.2.14.77 sig-rate-limit-profile

Syntax

```
sig-rate-limit-profile profile-name  
no sig-rate-limit-profile
```

Context

```
config>mobile>pdn>gy
```

Description

This command associates a rate limiting profile to be applied on the Gy interface. The same limit applies to all Diameter peers for this application. The profile must be configured first (using the command `sig-rate-limit-profile` in context `config>mobile>profile`) before being referenced by this command.

The `no` form of this command removes the association of the specified profile for the Gy interface.

Parameters

profile-name

specifies the name of the rate limiting profile, up to 32 characters

5.2.1.15 GTP load and overload control PDN command descriptions

5.2.1.15.1 gtp-load-control

Syntax

```
gtp-load-control
```

Context

```
config>mobile>pdn
```

Description

This command enables the context to configure GTP load control parameters.

5.2.1.15.2 enable-gtp-load-control

Syntax

```
[no] enable-gtp-load-control
```

Context

```
config>mobile>pdn>gtp-load-control
```

Description

This command enables GTP load control.



NOTE:

- When enabling the APN-level GTP-C load control, this command must also be configured.
- This command requires a PDN shutdown.

5.2.1.15.3 load-change-trigger

Syntax

load-change-trigger *percent*

no load-change-trigger

Context

config>mobile>pdn>gtp-load-control

Description

This command configures the load change value which will trigger a new load control update.



NOTE:

This command requires a PDN shutdown.

Parameters

percent

specifies the load change value percentage, from 5 to 20, in increments of 5

Values: 5 | 10 | 15 | 20

Default: 5

5.2.1.15.4 nth-message

Syntax

nth-message *number*

no nth-message

Context

config>mobile>pdn>gtp-load-control

Description

This command configures the message interval for sending load control information.

Parameters

number

specifies the interval to be used to include load control information

Values: 1 to 1000

Default: 50

5.2.1.15.5 gtp-overload-control

Syntax

gtp-overload-control

Context

config>mobile>pdn

Description

This command configures GTP overload control parameters.

5.2.1.15.6 enable-gtp-overload-control

Syntax

[no] enable-gtp-overload-control

Context

config>mobile>pdn>gtp-overload-control

Description

This command enables GTP overload control.

**NOTE:**

When enabling the APN-level GTP-C overload control, this command must also be configured.

5.2.1.15.7 gtp-120-oci

Syntax

```
[no] gtp-120-oci
```

Context

```
config>mobile>pdn>gtp-overload-control
```

Description

This command allows the ePDG to treat a PGW that provides the GTPv2 error code 120, as if the PGW is in an overload condition.

The no form of this command disables the PGW treatment.

Default

```
no gtp-120-oci
```

5.2.1.15.8 gtp-73-120-reduction-metric

Syntax

```
gtp-73-120-reduction-metric metric
```

```
no gtp-73-120-reduction-metric
```

Context

```
config>mobile>pdn>gtp-overload-control
```

Description

This command allows to set the overload reduction metric value that the ePDG applies for the PGW, providing the GTP error code 73 or 120. This command is valid only if the config>mobile>pdn>gtp-overload-control-gtp73-oci or gtp-120-oci is enabled.

The no form of this command reverts to the default.

**NOTE:**

This command is by default active, with a default value of 50% reduction metric. This ensures that the reduction metric is applied when the `gtp-73-oci` or `gtp-120-oci` is enabled.

Default

`no gtp-73-120-reduction-metric`

Parameters***metric***

specifies the percentage of the overload reduction metric that is applied

Values: 10 to 100 (in steps of 10)

Default: 50

5.2.1.15.9 gtp-73-oci**Syntax**

`[no] gtp-73-oci`

Context

`config>mobile>pdn>gtp-overload-control`

Description

This command allows the ePDG to treat a PGW that provides GTPv2 error code 73, as if the PGW is in an overload condition.

The `no` form of this command disables the PGW treatment.

Default

`no gtp-73-oci`

5.2.1.15.10 local-throttling**Syntax**

`local-throttling`

Context

```
config>mobile>pdn>gtp-overload-control
```

Description

This command configures the throttling (rejection) parameters of the call flows that add further load on the gateway when the gateway is in GTP-C overload.

5.2.1.15.11 critical

Syntax

```
critical
```

Context

```
config>mobile>pdn>gtp-overload-control>local-throttling
```

Description

This command enables the context to configure the parameters in the GTP-C critical overload state.

5.2.1.15.12 drop-only

Syntax

```
drop-only drop-type
```

```
no drop-only
```

Context

```
config>mobile>pdn>gtp-overload-control>local-throttling>critical
```

```
config>mobile>pdn>gtp-overload-control>local-throttling>major
```

```
config>mobile>pdn>gtp-overload-control>local-throttling>minor
```

Description

This command configures the type of GTP-C messages that should be throttled (rejected) if call flows add additional load when the gateway is already in different types of GTP-C overload levels.

Only the CSREQ with the *drop-type* option is supported. When the drop-only command is configured with the *drop-type* option csreq enabled, the Create Session Responses (CSResp) include the GTP-C cause-code: 73: No Resource Available

For example, if the gateway is in major overload, the major throttling percent is set to 60%, and the drop-only command is configured with the *drop-type* option csreq enabled, the gateway rejects six out of ten incoming Create Session Requests (CSREQs).

When the *drop-type* option is not configured, or the no drop-only command is configured, the gateway reverts to the default behavior, which is throttling the internal pre-mix of message types. For more details, refer to table *GTP-C Overload Condition and PGW Throttling Actions*, in the *CMG Configuration Guide*

**NOTE:**

The command is enabled only when local-overload-throttling is enabled and the throttling percentage value is set to non-zero.

Default

no drop-only

Parameters

drop-type

specifies the type of GTP-C message that is throttled during a GTP-C overload

Values: csreq (supported message type)

Default: no drop-only

5.2.1.15.13 exclude-arp-list

Syntax

```
exclude-arp-list list-name
```

```
no exclude-arp-list
```

Context

```
config>mobile>pdn>gtp-overload-control>local-throttling>critical
```

```
config>mobile>pdn>gtp-overload-control>local-throttling>major
```

```
config>mobile>pdn>gtp-overload-control>local-throttling>minor
```

Description

This command configures the ARP list to be used for excluding throttling of certain Create Session Requests (csreq) when the gateway is in different types of GTP-C overload levels.

For example, if the gateway is in major overload, the major throttling percent is set to 60%, the drop-only command is set with the option *csreq* enabled, and the list name includes ARP=6, 7, the gateway rejects six out of ten incoming CSREQs except those ARPs with values 6 and 7.

The no form of this command reverts to the default.

**NOTE:**

This command requires that the drop-only command with the *csreq* enabled is configured.

Default

no exclude-arp-list

Parameters

list-name

specifies the name of the ARP list; up to 32 characters

5.2.1.15.14 throttling-percent

Syntax

throttling-percent *percent-value*

no throttling-percent

Context

config>mobile>pdn>gtp-overload-control>local-throttling>critical

Description

This command configures the percentage of traffic that should be throttled (rejected) due to the call flows that add further load on the gateway when the gateway is in critical GTP-C overload level. For example, if the gateway is in critical overload and the critical throttling-value is set to 90%, the gateway rejects nine out of ten call flows.

The no form of this command disables the throttling of call flows when the gateway is in GTP-C critical overload.

**NOTE:**

This command takes effect when the local-overload-throttling command is enabled.

Default

no throttling-percent

Parameters

percent-value

specifies the percentage of the traffic that should be throttled when the gateway is in the critical GTP-C overload level

Values: 0 to 100

Default: 90

5.2.1.15.15 local-overload-throttling

Syntax

```
local-overload-throttling {enable | disable}
```

```
no local-overload-throttling
```

Context

```
config>mobile>pdn>gtp-overload-control>local-throttling
```

Description

This command configures the throttling (reject) of call flows that add extra load on the gateway when the gateway is in GTP-C overload state. When enabled, the gateway throttles (reject) more signaling loads based on the configured percentage of the minor, major and critical reduction values.

To set the throttling percentage for the minor, major and critical GTP overload levels, other commands must be configured in the context:

CLI Syntax: configure mobile-gateway pdn *gtp-overload-control local-throttling*

The no form of this command reverts to the default value, which enables throttling.



NOTE:

To enable the `local-overload-throttling` command, the `gtp-overload-control` command must also be enabled using the `enable-gtp-overload-control` command.

Parameters

enable | disable

enables or disables the parameters for local overload throttling

Default: enable

5.2.1.15.16 major

Syntax

major

Context

config>mobile>pdn>gtp-overload-control>local-throttling

Description

This command enables the context to configure the parameters for the major GTP-C overload level.

5.2.1.15.17 throttling-percent

Syntax

throttling-percent *percent-value*

no throttling-percent

Context

config>mobile>pdn>gtp-overload-control>local-throttling>major

Description

This command configures the percentage of the traffic that should be throttled (rejected) due to the call flows that add further load on the gateway when the gateway is in major GTP-C overload level. For example, if the gateway is in major overload and the major throttling-value is set to 60%, the gateway rejects six out of ten call flows.

The no form of this command reverts to the default.



NOTE:

This command requires that the local-overload-throttling command is enabled.

Default

no throttling-percent

Parameters

percent-value

specifies the percentage of the traffic that should be throttled when the gateway is in the major GTP-C overload level

Values: 0 to 100

Default: 60

5.2.1.15.18 minor

Syntax

minor

Context

config>mobile>pdn>gtp-overload-control>local-throttling

Description

This command enables the context to configure the parameters for the minor GTP-C overload level.

5.2.1.15.19 throttling-percent

Syntax

throttling-percent *percent-value*

no throttling-percent

Context

config>mobile>pdn>gtp-overload-control>local-throttling>minor

Description

This command configures the percentage of the traffic that should be throttled (rejected) due to the call flows that add further load on the gateway when the gateway is in minor GTP-C overload level. For example, if the gateway is in minor overload and the minor throttling-value is set to 30%, the gateway rejects three out of ten call flows.

The no form of this command disables to the default.



NOTE:

This command requires that the `local-overload-throttling` command is enabled.

Default

no throttling-percent

Parameters

percent-value

specifies the percentage of the traffic that should be throttled when the gateway is in the minor GTP-C overload level

Values: 0 to 100

Default: 0

5.2.1.15.20 throttling-alarms

Syntax

`throttling-alarms {enable | disable}`

`no throttling alarms`

Context

`config>mobile>pdn>gtp-overload-control>local-throttling`

Description

This command configures the throttling alarms. The alarms are raised when the first message is throttled (rejected) because of the call flows that add extra load on the gateway when the gateway is in GTP-C overload state. The alarms are cleared 30 seconds after the throttling (rejection) of the last message.

The no form of this command reverts to the default.



NOTE:

To enable throttling alarms, the commands must be configured as follows:

- the `local-overload-throttling` command must be enabled
- the `throttling-percent` command must be set in non-zero value
- the `drop-only` command with the `CSREQ` option must be configured under the contexts:
 - `config>mobile>pdn>gtp-overload-control>local-throttling>critical`
 - `config>mobile>pdn>gtp-overload-control>local-throttling>major`

```
– config>mobile>pdn>gtp-overload-control>local-throttling>minor
```

Once raised, the alarm will be cleared despite the state (enable/disable) of the throttling alarms CLI.

Default

no throttling-alarms

Parameters

enable | disable

enables or disables the parameters for throttling alarms

Default: disable

5.2.1.15.21 overload-control-only-on-new-sessions

Syntax

```
[no] overload-control-only-on-new-sessions
```

Context

```
config>mobile>pdn>gtp-overload-control
```

Description

This command allows to apply the overload control actions only for new sessions, leaving the existing sessions unaffected.

The no form of this command reverts to the default.

Default

no overload-control-only-on-new-sessions

5.2.1.15.22 overload-reduction

Syntax

```
overload-reduction
```


Context

```
config>mobile>pdn>gtp-overload-control
```

Description

This command configures the percentage of the traffic reduction requested for different GTP overload levels.

5.2.1.15.23 critical

Syntax

```
critical reduction-value
```

```
no critical
```

Context

```
config>mobile>pdn>gtp-overload-control>overload-reduction
```

Description

This command enables the context to configure the parameters for the critical GTP-C overload level.

The no form of this command reverts to the default.

Default

```
no critical
```

Parameters

reduction-value

specifies the traffic reduction percentage for critical overload

Values: 20 to 100

Default: 90

5.2.1.15.24 major

Syntax

```
major reduction-value
```

```
no major
```

Context

```
config>mobile>pdn>gtp-overload-control>overload-reduction
```

Description

This command configures the percentage of traffic reduction requested for major overload.

The no form of this command reverts to the default.

Default

no major

Parameters

reduction-value

specifies the traffic reduction percentage for major overload

Values: 20 to 100

Default: 60

5.2.1.15.25 minor

Syntax

```
minor reduction-value
```

```
no minor
```

Context

```
config>mobile>pdn>gtp-overload-control>overload-reduction
```

Description

This command configures the percentage of traffic reduction requested for minor overload.

The no form of this command reverts to the default.

Default

no minor

Parameters

reduction-value

specifies the traffic reduction percentage for minor overload

Values: 20 to 100

Default: 30

5.2.1.15.26 s8

Syntax

[no] s8

Context

config>mobile>pdn>gtp-overload-control

Description

This command configures whether to report overload information on S8.

Default

no s8

5.2.1.15.27 validity-time

Syntax

validity-time *value*

no validity-time

Context

config>mobile>pdn>gtp-overload-control

Description

This command configures the length of the time for the reported overload to be in effect.

Parameters

value

specifies the length of time, in seconds, for the overload to be in effect

Values: 10 to 86400

Default: 600

5.2.15.28 validity-time-extra-time

Syntax

```
validity-time-extra-time time
```

```
no validity-time-extra-time
```

Context

```
config>mobile>pdn>gtp-overload-control
```

Description

This command allows the ePDG to add an extra safeguard time on top of every PGW Overload Period of Validity Timer value provided.

The no form of this command disables this extra time.

Parameters

time

specifies the additional time in seconds that is added on top of PGW provided Overload Period of Validity Timer

Values: 30 to 3600

Default: 30

5.2.15.29 pdn-threshold

Syntax

```
pdn-threshold
```

Context

```
config>mobile>pdn
```

Description

This command enables the context to configure GTP threshold parameters.

5.2.1.15.30 max-s11-attach

Syntax

```
max-s11-attach max-attach-limit [alarm] [include-emergency]  
no max-s11-attach
```

Context

```
config>mobile>pdn>pdn-threshold
```

Description

This command configures the threshold of the maximum number of GTP-based attach requests per second. The limit is applied only on the attach requests received over S11. If enabled or disabled, an alarm is raised when the maximum number of attaches is reached. Emergency attach requests are excluded by default from the threshold, but it is also possible to apply the attach limit to emergency attach request.

The no form of this command reverts to the default where no limit is applied to the number of attach requests.

Parameters

max-attach-limit

specifies the number of attach requests per second over S11

Values: 0 to 5000

Default: 0 (disabled)

alarm

specifies if an alarm is sent when the maximum number of attach requests is reached

include-emergency

specifies the emergency attach requests to the threshold

5.2.1.16 KPI-KCI PDN command descriptions

5.2.1.16.1 kpi-kci

Syntax

```
kpi-kci
```

Context

config>mobile>pdn

config>mobile>pdn>apn

Description

This command enables the context to configure KPI-KCI parameters.

5.2.1.16.2 mass-deactivation

Syntax

mass-deactivation

Context

config>mobile>pdn

Description

This command enables the context to configure the mass deactivation parameters.

5.2.1.16.3 qci

Syntax

qci *qci-value* [arp-pl *arp-pl-value*] session-delete-priority *priority-value*

no qci *qci-value* [arp-pl *arp-pl-value*]

Context

config>mobile>pdn>mass-deactivation

Description

This command assigns a priority value for the QCI or QCI/ARP. If mass deactivation of the bearers is configured, the highest priority bearers are deactivated first.

If no priority value is configured for a specific QCI, normal priority is set for the QCI. If no priority value is configured for a specific QCI/ARP, the configuration for the QCI value is also set to the QCI/ARP. In this case, if there is no configuration, normal priority is set, as well.

The no form of this command removes the entry from the configuration, and sets the priority to the default value (normal).

Parameters

qci-value

specifies the QCI value

Values: 1 to 255

arp-pl-value

specifies the value for the ARP priority level

Values: 1 to 15

priority-value

specifies the priority for mass deactivation

Values	normal	lowest priority
–	medium	third highest priority
–	high	second highest priority
–	critical	highest priority

Default: normal

5.2.1.16.4 qci-drop-stat

Syntax

qci-drop-stat *list-name*
no qci-drop-stat

Context

config>mobile>pdn>apn>kpi-kci

Description

This command specifies the QoS Class Identifier (QCI) drop stat profile name.

The no form of this command reverts to the default.

Default

no qci-drop-stat

Parameters

list-name

specifies the name of the QCI list profile

5.2.1.16.5 accounting-policy

Syntax

accounting-policy *acct-policy-id*

no accounting-policy

Context

config>mobile>pdn>kpi-kci

Description

This command creates the accounting policy context.

The no form of this command removes the accounting policy association from the configuration.

Parameters

acct-policy-id

Enter the accounting *policy-id* as configured in the config>log>accounting-policy context.

Values: 1 to 99

5.2.1.16.6 arp-traffic-stats

Syntax

[no] arp-traffic-stats

Context

config>mobile>pdn>kpi-kci>accounting-policy

Description

This command enables the generation of per ARP (Allocation and Retention Policy) and per QCI Traffic KPI-KCI statistics. The per ARP and per QCI statistics are collected by KPIBearerTrafficQciArpCP-ISA, KPIBearerManagementQciArpCP-ISA, and KCIBearerManagementQciArpCP-ISA modules.

The respective counter module must also be configured in an accounting policy linked to this PDN under the `config>mobile>pdn>kpi-kci>accounting-policy` context.

If the `arp-traffic-stats` command is not configured, the KPI-KCI modules will not be collected even if the record is configured in the `accounting-policy` command.

The `no` form of this command disables the generation of per ARP traffic statistics.

Default

`no arp-traffic-stats`

5.2.16.7 collect-stats

Syntax

`[no] collect-stats`

Context

`config>mobile>pdn>kpi-kci>accounting-policy`

Description

This command enables accounting and statistical data collection.

The `no` form of this command disables accounting and statistical data collection.

Default

`collect-stats`

5.2.16.8 enhanced-traffic-stats

Syntax

`[no] enhanced-traffic-stats`

Context

`config>mobile>pdn>kpi-kci>accounting-policy`

Description

This command enables the generation of an enhanced set of statistics at the APN level. When enabled, all traffic statistics are generated for all sessions on the gateway, independently of the charging function. When this command is not enabled, the traffic statistics are accounted only for sessions that are being charged.

The respective counter modules containing these counters must also be configured in the accounting policy linked to this PDN, for these counters to be generated.

For more information about which KPI Bearer Traffic modules this configuration applies to, refer to the section on user plane statistics in the *CMG KPI-KCI Counters Guide*.

The no form of this command disables the generation of enhanced traffic statistics.

Default

no enhanced-traffic-stats

5.2.1.16.9 ipv4-ipv6-stats

Syntax

[no] ipv4-ipv6-stats

Context

config>mobile>pdn>kpi-kci>accounting-policy

Description

This command enables the separation of traffic volume statistics into IPv4 and IPv6. The counters having IPv6 in the name are populated. If this configuration is disabled, the IPv6 counters are not populated. This affects KPIBearerTrafficApnCP-ISA and KPIBearerTrafficMSMPGW counters.

Enabling the separation of traffic statistics into IPv4 and IPv6 means that traffic volumes are handled separately for IPv4 and IPv6 for every bearer of dual stack sessions in the system. The SDF usage is doubled for these sessions.

This is a PDN-level global configuration and changing it requires the PDN to be in a shutdown state. In 4G-CUPS, this configuration must be done in the User Plane gateway, which measures the separated IPv4/IPv6 traffic volumes.

The no form of this command reverts to the default.

Default

no ipv4-ipv6-stats

5.2.1.16.10 peer-traffic-stats

Syntax

```
peer-traffic-stats peer-ip-list list-name  
no peer-traffic-stats
```

Context

```
config>mobile>pdn>kpi-kci>accounting-policy
```

Description

This command enables the generation of per peer traffic statistics and configures the peer list for the peers to be accounted. The peer list is configured as a list of Control Plane IP addresses.

The no form of this command reverts to the default.

Default

```
no peer-traffic-stats
```

Parameters

list-name

specifies the name of the IP address list (configured in the context config>mobile>profile>list>ip-address-list) that contains the peer addresses, up to 32 characters

5.2.1.16.11 plmn-id-stats

Syntax

```
plmn-id-stats plmn-list-name  
no plmn-id-stats
```

Context

```
config>mobile>pdn>kpi-kci>accounting-policy
```

Description

This command enables the generation of per PLMN-ID statistics for the PLMN IDs that are included in the PLMN-ID list created in the config>mobile>profile>list>plmn context. The respective counter

modules must also be configured in the accounting policy linked to this PDN for the per PLMN-ID statistics generation.

The no form of this command disables the generation of per PLMN-ID statistics.

Default

no plmn-id-stats

Parameters

plmn-list-name

specifies the name of the PLMN-ID list

5.2.1.16.12 sgw-only

Syntax

[no] sgw-only

Context

config>mobile>pdn>kpi-kci>accounting-policy

Description

This command restricts the KPI/KCI collection to SGW statistics only. If only SGW statistics are enabled, the KPI/KCI is identified as an SGW instance.

The no form of this command removes the restriction for the KPI/KCI to collect only SGW statistics.

Default

no sgw-only

5.2.1.17 N2 interface PDN command descriptions

5.2.1.17.1 n2

Syntax

[no] n2 [*sig-ref-point-name*]

Context

config>mobile>pdn

Description

This command configures the N2 reference point name.

Parameters

sig-ref-point-name

specifies an N2 reference point name, up to 32 characters

5.2.1.17.2 interface

Syntax

interface [router *router-instance*] *interface-name*

no interface

Context

config>mobile>pdn>n2

Description

This command configures the N2 interface and the virtual router in which the N2 connections are established by this policy.



NOTE:

This command requires a PDN shutdown.

Parameters

interface-name

specifies an interface name, up to 32 characters (must start with a letter)

router-instance

specifies the virtual router name or service ID

Values

router-name | service-id

–	router-name	Base
–	service-id	1 to 2147483647

Default: Base

5.2.1.17.3 n2-peer-profile

Syntax

[no] n2-peer-profile *profile-name*

Context

config>mobile>pdn>n2

Description

This command specifies the N2 peer profile for the N2 reference point of the N3IWF interface. You can configure up to ten N2 peer profiles (the N2 peer profile is configured using the config>mobile>profile>n2-peer context). The AMF selection logic (N2-peer selection) is based on matching the configured PLMN list (of the N2-peer under the config>mobile>profile>n2-peer>plmn-list command) with the UE-provided selected PLMN list value.

To provide load sharing amongst multiple servers, you can configure multiple peer IDs. If multiple peer IDs are specified, and the system has more than one matching peer IDs, based on the configured PLMN list, the system uses round-robin selection to choose a peer group for each new N2 session.



NOTE:

This command requires a PDN shutdown.

Parameters

profile-name

specifies the name of an existing N2 peer profile, up to 32 characters

5.2.1.17.4 n3iwf-unique-id

Syntax

```
n3iwf-unique-id unique-id  
no n3iwf-unique-id
```

Context

```
config>mobile>pdn>n2
```

Description

This command configures the unique N3IWF identity value.

The no form of this command reverts to default value.

**NOTE:**

This value is used for the "N3IWF Id" generation which is part of the "Global N3IWF Id". When there are more than one N3IWF deployed in the network, each deployed N3IWF must have a different n3iwf-unique-id value.

Parameters

unique-id

specifies the unique ID of the N3IWF

Values: 1 to 64

Default: 1

5.2.1.17.5 supported-ta-item

Syntax

```
supported-ta-item
```

Context

```
config>mobile>pdn>n2
```

Description

This command enables the context to configure the supported TA for the N2 interface.

5.2.1.17.6 tac

Syntax

```
tac tac-value broadcast-plmn-list broadcast-plmn-list-name  
no tac tac-value
```

Context

config>mobile>pdn>n2>supported-ta-item

Description

This command configures the supported TAC (Tracking Area Code) and its associated broadcast PLMN list for the N2 interface.

The no form of this command removes the specified TAC along with its associated broadcast PLMN list.



NOTE:

- According to 3GPP, the N3IWF is allowed to have a single TAI (TAI=TAC+PLMN) configured.
- The no tac command is used for removing the configured values and adding new ones. If a TAC and its associated broadcast PLMN list profile is removed without adding new ones, this can potentially lead to system instability.

Parameters

tac-value

specifies a TAC value, 6-digit hexadecimal number

broadcast-plmn-list-name

specifies the name of an existing broadcast PLMN list profile, up to 32 characters

5.2.1.18 N3 interface PDN command descriptions

5.2.1.18.1 n3

Syntax

```
n3 [sig-ref-point-name] [group-specific]  
no n3 [sig-ref-point-name]
```


Context

config>mobile>pdn

Description

This command configures the N3 reference point name. The N3 interface provides UP tunneling between two UPFs and is based on the GTP protocol.



NOTE:

This command requires a PDN shutdown, unless the `group-specific` parameter is specified.

Parameters

sig-ref-point-name

specifies an N3 reference point name, up to 32 characters

group-specific

specifies an MG group-specific reference point interface for GTP-U redirection, which is used when the reference point interface is associated with an MG system group through an override profile, and applies only to CMG

5.2.1.18.2 dual-stack-uplane-pref

Syntax

dual-stack-uplane-pref {ipv4 | ipv6}

no dual-stack-uplane-pref

Context

config>mobile>pdn>n3

Description

This command specifies the preference in a dual stack when both IPv4 and IPv6 addresses are configured for the reference point interface.



NOTE:

This command requires a PDN shutdown.

Parameters

ipv4

specifies the IPv4 address family

ipv6

specifies the IPv6 address family

Default: ipv6

5.2.1.18.3 interface

Syntax

interface

Context

config>mobile>pdn>n3

Description

The command enables the context to configure the N3 interface.



NOTE:

This command requires a PDN shutdown.

5.2.1.18.4 gtp-u

Syntax

gtp-u *interface-name* [router *router-name*] [interface-realm *n3-realm-name*]

no gtp-u

Context

config>mobile>pdn>n3>if

Description

This command configures the GTP-U profile parameters for the N3 interface.

The no form of this command removes the parameter from the configuration.



NOTE:

This command requires a PDN shutdown.

Parameters

interface-name

specifies an interface name, up to 32 characters, that must start with a letter

router-name

specifies the virtual router name or service ID

Values	router-name service-id	
–	router-name	Base
–	service-id	1 to 2147483647

Default: Base

n3-realm-name

specifies the associated realm name for the virtual router instance and the N3 interface, up to 32 characters



NOTE:

Only the following characters are permitted:

- alphabetic characters (A-Z and a-z)
- digits (0-9)
- hyphen (-)
- dot (.)

The UP function uses the interface realm name to identify the correct router instance and N3 interface to use, based on the Network-Instance IE in the PFCP interface rules during the PFCP session establishment/modification process. The realm name must be unique for a given virtual routing instance.



NOTE:

This parameter is only used when the gateway is configured as a UP function using the `mobile>config>pdn>instance-type` command.

5.2.18.5 signaling

Syntax

```
signaling
```

Context

```
config>mobile>pdn>n3
```

Description

This command enables the context to configure signaling for the N3 interface.



NOTE:

This command requires a PDN shutdown.

5.2.18.6 gtp-u

Syntax

```
gtp-u
```

Context

```
config>mobile>pdn>n3>signaling
```

Description

This command enables the context to configure GTP-U for the N3 signaling interface.



NOTE:

This command requires a PDN shutdown.

5.2.1.18.7 profile

Syntax

```
profile profile-name  
no profile
```

Context

```
config>mobile>pdn>n3>signaling>gtp-u
```

Description

This command configures the GTP-U profile name for the N3 signaling interface.

The no form of this command removes the profile name from the configuration.

**NOTE:**

This command requires a PDN shutdown.

Default

no profile

Parameters

profile-name

specifies the GTP-U profile name, up to 32 characters

5.2.1.19 N9 interface PDN command descriptions

5.2.1.19.1 n9

Syntax

```
n9 [sig-ref-point-name] [group-specific]  
no n9 [sig-ref-point-name]
```

Context

```
config>mobile>pdn
```

Description

This command configures the N9 reference point name. The N9 interface provides UP tunneling between two UPFs and is based on the GTP protocol.

**NOTE:**

This command requires a PDN shutdown, unless the `group-specific` parameter is specified.

Parameters

sig-ref-point-name

specifies an N9 reference point name, up to 32 characters

group-specific

specifies an MG group-specific reference point interface for GTP-U redirection; it is used when associating the reference point interface with an MG system group via an override profile and applies only to CMG

5.2.1.19.2 dual-stack-uplane-pref

Syntax

```
dual-stack-uplane-pref {ipv4 | ipv6}
```

```
no dual-stack-uplane-pref
```

Context

```
config>mobile>pdn>n9
```

Description

This command specifies the preference in a dual stack when both IPv4 and IPv6 addresses are configured for the reference point interface.

**NOTE:**

This command requires a PDN shutdown.

Parameters

ipv4

specifies to use the IPv4 address family

ipv6

specifies to use the IPv6 address family

Default: ipv6

5.2.1.19.3 interface

Syntax

interface

Context

config>mobile>pdn>n9

Description

The command enables the context to configure the N9 interface.



NOTE:

This command requires a PDN shutdown.

5.2.1.19.4 gtp-u

Syntax

gtp-u *interface-name* [router *router-instance*] [interface-realm *n9-realm-name*]

no gtp-u

Context

config>mobile>pdn>n9>if

Description

This command configures the GTP-U profile parameters for the N9 interface.

The no form of this command removes the parameter from the configuration.



NOTE:

This command requires a PDN shutdown.

Parameters

- interface-name**
specifies an interface name, up to 32 characters
- router-instance**
specifies the virtual router name or service ID

Values	router-name service-id	
–	router-name	Base
–	service-id	1 to 2147483647

Default: Base

- n9-realm-name**
specifies the associated realm name for the virtual router instance and the N9 interface, up to 32 characters



NOTE:

Only the following characters are permitted:

- alphabetic characters (A-Z and a-z)
- digits (0-9)
- hyphen (-)
- dot (.)

The UP function uses the interface realm name to identify the correct router instance and N9 interface to use, based on the Network-Instance IE in the PFCP interface rules during the PFCP session establishment/modification process. The realm name must be unique for a given virtual routing instance.



NOTE:

This parameter is only used when the gateway is configured as a UP function using the `mobile>config>pdn>instance-type` command.

5.2.1.19.5 signaling

Syntax

signaling

Context

config>mobile>pdn>n9

Description

This command enables the context to configure signaling for the N9 interface.

**NOTE:**

This command requires a PDN shutdown.

5.2.1.19.6 gtp-u

Syntax

gtp-u

Context

config>mobile>pdn>n9>signaling

Description

This command enables the context to configure GTP-U for the N9 signaling interface.

**NOTE:**

This command requires a PDN shutdown.

5.2.1.19.7 profile

Syntax

profile *profile-name*

no profile

Context

```
config>mobile>pdn>n9>signaling>gtp-u
```

Description

This command configures the GTP-U profile name for the N9 signaling interface.

The no form of this command removes the profile name from the configuration.



NOTE:

This command requires a PDN shutdown.

Default

no profile

Parameters

profile-name

specifies the GTP-U profile name, up to 32 characters

5.2.1.20 Network trace command descriptions

5.2.1.20.1 network-trace

Syntax

```
network-trace
```

Context

```
config>mobile>pdn
```

Description

This command enables the context to configure the network trace client service.

5.2.1.20.2 instance

Syntax

```
[no] instance instance-name
```

Context

config>mobile>pdn>network-trace

Description

This command configures a network trace client service instance. Only one instance is supported.

The no form of this command removes the instance from the configuration.

Parameters

instance-name

specifies the instance name, up to 32 characters

5.2.1.20.3 interface

Syntax

interface *interface-name* [router *router-name*]

no interface

Context

config>mobile>pdn>network-trace>instance

Description

This command configures the interface in which the connection to the Packet Broker router is established for the specific instance.

The no form of this command reverts to the default.

Parameters

interface-name

specifies an existing interface name, up to 32 characters (must start with a letter)

router-name

specifies the router instance

Values router-name | service-id

–	router-name	Base
–	service-id	1 to 2147483647

Default: Base

5.2.1.20.4 network-trace-profile

Syntax

network-trace-profile *profile-name*

no network-trace-profile

Context

config>mobile>pdn>network-trace>instance

Description

This command associates a network trace profile, configured using the config>mobile>profile>network-trace-profile context, with the network trace client service.

The no form of this command removes the network trace profile from the configuration.

Parameters

profile-name

specifies the name of the network trace profile, up to 32 characters

5.2.1.20.5 shutdown

Syntax

[no] shutdown

Context

config>mobile>pdn>network-trace>instance

Description

This command administratively disables the instance of the network trace client service.

The no form of this command administratively enables the network trace client service instance.

5.2.1.20.6 tapping-interface

Syntax

```
[no] tapping-interface [all | sba | non-sba]
```

Context

```
config>mobile>pdn>network-trace>instance
```

Description

This command configures the interfaces that are tapped by the network trace client service

The no form of this command reverts to the default value.

Default

```
tapping-interface all
```

5.2.1.21 NF profile attributes PDN command descriptions

5.2.1.21.1 nf-profile-attributes

Syntax

```
nf-profile-attributes [bsf]
```

Context

```
config>mobile>pdn
```

Description

This command enables the context to configure NF profile attributes for SMF and BSF, or for UPF, according to the CMG deployment.

When bsf is specified, the command enables the context to configure attributes for the Binding Support Function (BSF) that is collocated with the SMF under the same gateway instance. If bsf is not specified, the

NF profile used for the BSF registration uses the same values as those defined for SMF, with the following exceptions:

NFInstanceID	A different NFInstanceID is generated
NFServices array	Populated with the configured BSF related services

**NOTE:**

The context to configure BSF attributes is enabled only when the gateway is configured as an SMF.

Parameters

bsf

specifies to enable the context to configure BSF attributes

5.2.1.21.2 capacity

Syntax

`capacity {factor | total} capacity-value`

`no capacity`

Context

`config>mobile>pdn>nf-profile-attributes`

Description

This command configures the capacity value that is used in the NF profile during NF registration and in the NF update to the NRF. If not configured, the capacity attribute is not populated.

The no form of this command reverts to the default.

Default

`no capacity`

Parameters

capacity-value

specifies the capacity value

Values: 0 to 65535

factor

specifies that the capacity value is the capacity of a single CMG component, such as a VM, and is calculated by multiplying the given value with the number of active CMG components

total

specifies that the capacity value is the total capacity of the NF instance

5.2.1.21.3 exclude-ip-endpoints

Syntax

[no] exclude-ip-endpoints

Context

config>mobile>pdn>nf-profile-attributes

Description

This command configures that the IP endpoints are not registered to the NRF in the NFProfile. To exclude the IP endpoints from the registration, an FQDN must be configured in the same context or in the sba-server-services context per server service.

The no form of this command specifies that IP endpoints are registered.

Default

no exclude-ip-endpoints

5.2.1.21.4 fqdn

Syntax

fqdn *fqdn*

no fqdn

Context

config>mobile>pdn>nf-profile-attributes

Description

This command configures the FQDN of the SMF that is used in the NF profile during NF registration and in the NF update to the NRF. If not configured, the FQDN attribute is not populated.

The no form of this command reverts to the default.

Default

no fqdn

Parameters

fqdn

specifies the SMF FQDN string, up to 255 characters

5.2.1.21.5 inter-plmn-fqdn

Syntax

inter-plmn-fqdn *fqdn*

no inter-plmn-fqdn

Context

config>mobile>pdn>nf-profile-attributes

Description

This command configures the inter-PLMN FQDN of the SMF that is used in the NF profile during NF registration and in the NF update to the NRF. If not configured, the inter-PLMN FQDN attribute is not populated.

The no form of this command disables the usage of inter-PLMN FQDN in the SMF registration to the NRF.

Default

no fqdn

Parameters

fqdn

specifies the SMF inter-PLMN FQDN string, up to 255 characters

5.2.1.21.6 locality

Syntax

locality *locality*


```
no locality
```

Context

```
config>mobile>pdn>nf-profile-attributes
```

Description

This command configures the locality that the SMF registers to the NRF. The locality can be a geographical description, data-center location, and so on. The no form of this command removes the configured locality.

Parameters

locality

a string that specifies the arbitrarily-defined location, such as the data center location or geographic location, up to 64 characters

5.2.1.21.7 nf-instance-id

Syntax

```
nf-instance-id uuid  
no nf-instance-id
```

Context

```
config>mobile>pdn>nf-profile-attributes
```

Description

This command configures the UUID of the SMF, UPF, or BSF instance. The UUID is registered to the NRF in the NFProfile as the NFInstanceId.



NOTE:

The newly configured UUID is used in the next NFRegister procedure only after an NFDeregister operation.

Default

An auto-generated value is used if this command is not configured.

Parameters

uuid

specifies the NF instance ID in UUIDv4 canonical textual representation

Values: xxxxxxxx-xxxx-4xxx-xxxx-xxxxxxxxxxxx

x: [0..F]H

5.2.1.21.8 priority

Syntax

`priority priority-value`

`no priority`

Context

`config>mobile>pdn>nf-profile-attributes`

Description

This command configures the priority value that is used in the NF profile during NF registration and in the NF update to the NRF. If not configured, the priority attribute is not populated.

The no form of this command reverts to the default.

Default

`no priority`

Parameters

priority-value

an integer that specifies the priority

Values: 0 to 65535

5.2.1.21.9 scp-domain

Syntax

`scp-domain domain-name`

`no scp-domain`

Context

config>mobile>pdn>nf-profile-attributes

Description

This command configures an SCP domain that is used in the NF profile during NF registration and in the NF update to the NRF. If not configured, the SCP domain attribute is not populated.

The no form of this command reverts to the default.

Default

no scp-domain

Parameters

domain-name

specifies the SCP domain name, up to 32 characters

5.2.1.21.10 smf-serving-area-list

Syntax

smf-serving-area-list *list-name*

no smf-serving-area-list

Context

config>mobile>pdn>nf-profile-attributes

Description

For the SMF, this command configures the SMF serving area associations to TAI lists. The associations are used to map TAI values to an SMF serving area for UPF discovery.

For the UPF, this command configures the SMF serving areas supported by the UPF. This list is registered to the NRF in the UPF NFProfile.

The no form of this command reverts to the default.

Default

no smf-serving-area-list

Parameters

list-name

specifies the name of the list

5.2.1.21.11 tai-list

Syntax

```
tai-list list-name
```

```
no tai-list
```

Context

```
config>mobile>pdn>nf-profile-attributes
```

Description

This command configures the TAI list or TAI range list that the SMF supports and registers to the NRF. If no configuration is added, the SMF supports all TAIs.

The no form of this command removes the configured list.

Parameters

list-name

specifies the name of the referenced TAI list, up to 32 characters; the TAI list names are configured under config>mobile>profile>tai-lai-list

5.2.1.21.12 vsmf-indication

Syntax

```
vsmf-indication {true | false | disabled}
```

```
no vsmf-indication
```

Context

```
config>mobile>pdn>nf-profile-attributes
```

Description

This command controls usage of vsmfSupportInd in the SMF registration to the NRF. The no form of this command reverts to the default.

Default

disabled

Parameters

true

vsmfSupportInd is set to 'true'

false

vsmfSupportInd is set to 'false'

disabled

vsmfSupportInd is not included in the SMF registration data

5.2.1.22 NWu interface PDN command descriptions

5.2.1.22.1 nwu

Syntax

[no] nwu [*sig-ref-point-name*]

Context

config>mobile>pdn

Description

This command configures the NWu reference point name. The system supports up to 16 NWu interfaces.



NOTE:

- This command requires a PDN shutdown.
- This command is not supported in the current release.

Parameters

sig-ref-point-name

specifies an NWu reference point name, up to 32 characters

5.2.1.22.2 interface

Syntax

```
interface [router router-instance] interface-name  
no interface
```

Context

```
config>mobile>pdn>nwu
```

Description

The command configures the interface of the NWu reference point.



NOTE:

This command is not supported in the current release.

Parameters

interface-name

specifies an interface name, up to 32 characters (must start with a letter)

router-instance

specifies the virtual router name or service ID

Values	router-name service-id	
–	router-name	Base
–	service-id	1 to 2147483647

Default: Base

5.2.1.22.3 ipsec-profile

Syntax

```
ipsec-profile profile-name
```

```
no ipsec-profile
```

Context

```
config>mobile>pdn>nwu
```

Description

This command configures the IPSec profile for this reference point.

The no form of this command removes the profile name from the configuration.



NOTE:

- This command requires a PDN shutdown.
- This command is not supported in the current release.

Parameters

profile-name

specifies the profile name, up to 32 characters

5.2.1.23 PFCP load and overload PDN command descriptions

5.2.1.23.1 pfcp-load-control

Syntax

```
pfcp-load-control
```

Context

```
config>mobile>pdn
```

Description

This command enables the context to configure PFCP load control.

5.2.1.23.2 enable-pfcp-load-control

Syntax

```
[no] enable-pfcp-load-control
```

Context

```
config>mobile>pdn>pfcf-load-control
```

Description

This command enables PFCF load control information handling. PFCF load control enables the UP function to send its load information to the CP function, which the CP function can then use to adaptively balance the PFCF session load across the UP functions, according to their effective load. Load control does not trigger overload mitigation actions, even if the UP function reports a high load.

The no form of this command reverts to the default (disabled).

Default

```
no enable-pfcf-load-control
```

5.2.1.23.3 load-change-trigger

Syntax

```
load-change-trigger percentage-value
```

```
no load-change-trigger
```

Context

```
config>mobile>pdn>pfcf-load-control
```

Description

This command configures the load change value that triggers a new load control update.

The no form of this command reverts to the default.

Parameters

percentage-value

specifies the load change value to trigger a new load control update, in increments of 5

Values: 5 | 10 | 15 | 20

Default: 5

5.2.1.23.4 nth-message-report

Syntax

```
nth-message-report nth-message-value  
no nth-message-report
```

Context

```
config>mobile>pdn>pfcf-load-control
```

Description

This command configures the number of messages between which a PFCF load control report message is sent, regardless of a load change or not.

The no form of this command reverts to the default.

Parameters

nth-message-value

specifies the number of messages between sending a PFCF load control report message

Values: 1 to 1000

Default: 50

5.2.1.23.5 pfcf-overload-control

Syntax

```
pfcf-overload-control
```

Context

```
config>mobile>pdn
```

Description

This command enables the context to configure PFCF overload control.

5.2.1.23.6 arp-to-message-priority-list

Syntax

```
arp-to-message-priority-list list-name  
no arp-to-message-priority-list
```

Context

```
config>mobile>pdn>pfcf-overload-control
```

Description

This command enables an ARP list that will be used on the CP function to set an MP flag and the message-priority in the PFCP message header of the PFCP Session Establishment request, the PFCP Session Modification request, and the PFCP Session Deletion request. The MP flag and the message-priority are based on the ARP value that is received in the corresponding GTP-C PDN session request, or the latest ARP value received from the PCRF message exchange.

The no form of this command disables the CP function from setting an MP Flag and the message-priority based on an ARP list.

Default

```
no arp-to-message-priority-list
```

Parameters

list-name

specifies the name of the ARP list; up to 32 characters

5.2.1.23.7 enable-pfcf-overload-control

Syntax

```
[no] enable-pfcf-overload-control
```

Context

```
config>mobile>pdn>pfcf-overload-control
```

Description

This command enables PFCP overload control information handling. PFCP overload control enables a UP function that is becoming or is already overloaded to gracefully reduce its incoming signaling load by

instructing the CP function to reduce sending traffic according to the provided overload information. A UP function is in overload when it operates over its capacity, which results in diminished performance (including impacts to handling of incoming and outgoing traffic).

Default

no enable-pfcp-load-control

5.2.1.23.8 critical

Syntax

```
critical-reduction percentage-value  
no critical-reduction
```

Context

```
config>mobile>pdn>pfcp-overload-control
```

Description

This command configures the percentage of traffic reduction for critical overload requests.

The no form of this command reverts to the default.

Parameters

percentage-value

specifies the percentage of traffic reduction for critical overload requests

Values: 20 to 100

Default: 90

5.2.1.23.9 major

Syntax

```
major percentage-value  
no major
```

Context

```
config>mobile>pdn>pfcp-overload-control
```

Description

This command configures the percentage of traffic reduction for major overload requests.

The no form of this command reverts to the default.

Parameters

percentage-value

specifies the percentage of traffic reduction for major overload requests

Values: 20 to 100

Default: 60

5.2.1.23.10 minor

Syntax

minor *percentage-value*

no minor

Context

config>mobile>pdn>pfcf-overload-control

Description

This command configures the percentage of traffic reduction for minor overload requests.

The no form of this command reverts to the default.

Parameters

percentage-value

specifies the percentage of traffic reduction for minor overload requests

Values: 20 to 100

Default: 30

5.2.1.23.11 validity-time

Syntax

validity-time *value*

no validity-time

Context

```
config>mobile>pdn>pfcf-overload-control
```

Description

This command configures the duration of time for which the reported overload will be in effect.

The no form of this command reverts to the default.

Parameters

value

specifies the duration of time for which the reported overload will be in effect, in seconds

Values: 10 to 86400

Default: 600

5.2.1.24 PGW selection PDN command descriptions

5.2.1.24.1 pgw-selection

Syntax

```
pgw-selection
```

Context

```
config>mobile>pdn
```

Description

This command enables the context to configure PGW selection PDN commands.



NOTE:

This command requires a PDN shutdown.

5.2.1.24.2 dns-options

Syntax

```
dns-options
```

Context

```
config>mobile>pdn>pgw-selection
```

Description

This command enables the context to configure DNS option commands.



NOTE:

This command requires a PDN shutdown.

5.2.1.24.3 a-aaaa-pgw-discovery

Syntax

```
[no] a-aaaa-pgw-discovery
```

Context

```
config>mobile>pdn>pgw-selection>dns-options
```

Description

This command specifies that when ePDG is using the MIP6-Info-agent, the ePDG discovers the PGW by using A or AAAA DNS type of query.



NOTE:

This command requires a PDN shutdown.

Default

```
no a-aaaa-pgw-discovery
```

5.2.1.24.4 apn-based-initial-attach

Syntax

```
[no] apn-based-initial-attach
```

Context

```
config>mobile>pdn>pgw-selection>dns-options
```

Description

This command specifies that on initial attach requests, the ePDG will always discover the PGW by performing a DNS query based on the APN provided by the UE (in this case, the ePDG ignores the AAA-provided PGW data through the MIP6-Agent-Info AVP). If the ePDG already has the APN name resolved toward a specific PGW in its DNS cache memory, the ePDG does not perform a new APN-based DNS query. NAPTR is used for this query.



NOTE:

- This command requires a PDN shutdown.
- When the `dns-results-preferred` command is not enabled, the ePDG attempts, as priority, to resolve the UE-provided APN based on the local APN configuration and its associated PGW selection candidate list. If this fails, the ePDG falls back to the DNS resolution method.

Default

no apn-based-initial-attach

5.2.1.24.5 epdg-fqdn

Syntax

`epdg-fqdn epdg-fqdn-string`

`no epdg-fqdn`

Context

`config>mobile>pdn>pgw-selection`

Description

This command sets the FQDNs assigned to the ePDG. Up to three ePDG FQDNs can be configured. If this command is present, the ePDG matches the configured FQDNs against the PGW candidate FQDNs to find the topologically closest PGW.

The ePDG tries to find the best match between primary, secondary, and tertiary configured ePDG FQDNs on a per longest match logic. If the primary ePDG FQDN has the same or better match (longer match) than the secondary and tertiary, the ePDG selects the corresponding PGW, using the primary configured FQDN. If multiple primary matches are equivalent, a round-robin logic is used across them to load-balance between the equivalent gateways.

The no form of this command removes the ePDG FQDN. The system chooses the PGW without considering the FQDN matching.

**NOTE:**

The ePDG always keeps the second longest matching DNS results, to be used as a fallback mechanism, in case the first longest matching results have failed to provide a successful session setup (PGW no response), when the second or third `epdg-fqdn` is configured by the operator.

Default

no `epdg-fqdn`

Parameters***epdg-fqdn-string***

specifies the ePDG FQDN string, up to 256 characters

5.2.1.24.6 `epdg-fqdn-secondary`**Syntax**

`epdg-fqdn-secondary epdg-fqdn-string`

no `epdg-fqdn`

Context

config>mobile>pdn>pgw-selection

Description

This command sets the secondary FQDN assigned to the ePDG. If this command is present, the ePDG matches this FQDN against the PGW candidate FQDNs to find the topologically closest PGW.

The primary ePDG FQDNs must already be configured before the secondary ePDG FQDN. If the secondary ePDG FQDN has a better match than the primary, and/or tertiary ePDG FQDN, the ePDG should select the corresponding PGW using secondary configured FQDN. If multiple secondary matches are equivalent, a round-robin is used across them to load-balance between the equivalent gateways.

The no form of this command removes the secondary ePDG FQDN. The system chooses the PGW without considering the FQDN matching.

**NOTE:**

The ePDG always keeps the second longest matching DNS results, to be used as a fallback mechanism, in case the first longest matching results have failed to provide a successful session setup (PGW no response), when the second or third `epdg-fqdn` is configured by the operator.

Default

no epdg-fqdn-secondary

Parameters

epdg-fqdn-string

specifies the ePDG secondary FQDN string, up to 256 characters

5.2.1.24.7 epdg-fqdn-tertiary

Syntax

epdg-fqdn-tertiary *epdg-fqdn-string*

no epdg-fqdn

Context

config>mobile>pdn>pgw-selection

Description

This command sets the tertiary FQDN assigned to the ePDG. If this command is present, the ePDG matches this FQDN against the PGW candidate FQDNs to find the topologically closest PGW.

The primary and secondary ePDG FQDNs must already be configured before the tertiary ePDG FQDN. If the tertiary ePDG FQDN has a better match than the primary, and the secondary ePDG FQDN, the ePDG should select the corresponding PGW using tertiary configured FQDN. If multiple tertiary matches are equivalent, a round-robin is used across them to load-balance between the equivalent gateways.

The no form of this command removes the tertiary ePDG FQDN. The system chooses the PGW without considering the FQDN matching.



NOTE:

The ePDG always keeps the second longest matching DNS results, to be used as a fallback mechanism, in case the first longest matching results have failed to provide a successful session setup (PGW no response), when the second or third epdg-fqdn is configured by the operator.

Default

no epdg-fqdn-tertiary

Parameters

epdg-fqdn-string

specifies the ePDG tertiary FQDN string, up to 256 characters

5.2.1.24.8 pgw-reselection-isolation

Syntax

```
pgw-reselection-isolation cause-code cc[isolation-time time]  
no pgw-reselection-isolation cause-code cc
```

Context

```
config>mobile>pdn>pgw-selection
```

Description

This command configures a specific "Rejection in a Response/triggered message" (3GPP TS 29.274) GTPv2 cause code that triggers the PGW/SMF reselection. Optionally, this command isolates (not used during the PGW selection process) the specific gateway responding with the specific cause code within a time ranging from 1 to 120 minutes.

Up to 16 entries can be configured in the system for a PDN gateway.

The no form of this command either totally disables the re-selection and isolation functionality, or removes a specific cause code from the configuration.

Default

```
no pgw-reselection-isolation
```

Parameters

cc

specifies a GTPv2 Rejection in a Response/triggered message cause code

Values: 64 to 239

time

specifies the isolation time, in minutes

Values: 1 to 120

Default: 5

5.2.1.24.9 twag-fqdn

Syntax

```
twag-fqdn string  
no twag-fqdn
```

Context

```
config>mobile>pdn>pgw-selection
```

Description

This command sets the FQDN assigned to the TWAG. If this command is present, the TWAG will match this FQDN against PGW candidate FQDNs to find the topologically closest PGW.

The no form of this command removes the TWAG FQDN. The system will choose the PGW without considering FQDN matching.

Default

```
no twag-fqdn
```

Parameters

string

specifies the TWAG FQDN, up to 256 characters

5.2.1.24.10 apn

Syntax

```
[no] apn apn-name
```

Context

```
config>mobile>pdn>pgw-selection
```

Description

This command creates a container for PGW selection information pertaining to a particular APN. The APN should be comprised of the APN-NI + APN-OI. The configured APN will be matched against the APN after the optional APN-OI replacement has been performed.

The no form of this command removes the APN configuration.

Parameters

apn-name

the name of the APN

5.2.1.24.11 pgw-selection-candidate-list

Syntax

```
pgw-selection-candidate-list list-name  
no pgw-selection-candidate-list
```

Context

```
config>mobile>pdn>pgw-selection>apn
```

Description

This command names the profile of a set of PGW candidates for this APN.

The no form of this command removes the list.

Parameters

list-name

the name of the pgw-selection-candidate-list profile

5.2.1.24.12 dns-results-preferred

Syntax

```
[no] dns-results-preferred
```

Context

```
config>mobile>pdn>pgw-selection>apn
```

Description

This command causes the system to prefer DNS resolved PGW candidates over static PGW candidates configured for the APN. PGW candidates configured with the PGW FQDN shall be considered as static candidates.

The no form of this command means that the system will prefer static PGW candidates over candidates resolved for the APN by DNS.

Default

no dns-results-preferred

5.2.1.24.13 topology-match-disable

Syntax

[no] topology-match-disable

Context

config>mobile>pdn>pgw-selection>apn

Description

This command configures the system to not perform FQDN-based topological matching for this particular APN.

The no form of this command means the system will prefer the best topological match with the epdg-fqdn and twag-fqdn. The system will match against only candidates specified by FQDN.

Default

no topology-match-disable

5.2.1.24.14 profile

Syntax

profile *profile-name*

no profile

Context

config>mobile>pdn>signaling>mipv4

Description

This command specifies the MIPv4 profile name. The profile provides signaling related configuration.

The no form of this command removes the profile name from the configuration.

Parameters

profile-name

specifies the MIPv4 profile name, up to 32 characters

5.2.1.25 PI interface PDN command descriptions

5.2.1.25.1 pi

Syntax

```
[no] pi [sig-ref-point-name]
```

Context

```
config>mobile>pdn
```

Description

This command configures parameters on all Pi interfaces, or for a specific Pi interface.



NOTE:

This command requires a PDN shutdown.

Parameters

sig-ref-point-name

specifies the Pi reference point name, up to 32 characters

5.2.1.25.2 interface

Syntax

```
interface
```

Context

```
config>mobile>pdn>pi
```

Description

This command enables the context to configure a Pi interface.



NOTE:

This command requires a PDN shutdown.

5.2.1.25.3 mipv4

Syntax

```
mipv4 [router router-instance] interface-name
no mipv4
```

Context

```
config>mobile>pdn>pi>interface
config>mobile>pdn>pi>signaling
```

Description

This command configures the mobile IPv4 (MIPv4) interface or signaling interface for Pi.
The no form of this command reverts to the default.



NOTE:

This command requires a PDN shutdown.

Default

```
no mipv4
```

Parameters

router-instance

specifies the router name or service ID

Values	router-name service-id	
–	router-name	Base
–	service-id	1 to 2147483647

Default: Base

interface-name

specifies the interface on which to receive broadcast packets; if the string contains special characters (#, \$, spaces, and so on), the entire string must be enclosed within double quotes, up to 32 characters

5.2.1.25.4 ip-in-ip-tunnel

Syntax

```
ip-in-ip-tunnel router-instance  
no ip-in-ip-tunnel
```

Context

```
config>mobile>pdn>pi>interface>mipv4
```

Description

This command is used to specify whether the IP-in-IP tunneling is used and the router ID is to be bound.



NOTE:

This command requires a PDN shutdown.

Default

```
no ip-in-ip-tunnel
```

Parameters

router-instance

specifies the virtual router name or service ID

Values	router-name service-id	
–	router-name	Base
–	service-id	1 to 2147483647

Default: Base

5.2.1.25.5 peer-list

Syntax

```
peer-list list-name  
no peer-list
```

Context

```
config>mobile>pdn>pi
```

Description

This command specifies the peer-list with the APN tied to this pi reference-point-name.

The no form of this command removes the peer-list from the configuration.



NOTE:

This command requires a PDN shutdown.

Default

```
no peer-list
```

Parameters

list-name

specifies a peer list name, up to 32 characters

5.2.1.25.6 signaling

Syntax

```
signaling
```

Context

```
config>mobile>pdn>pi
```

Description

This command enables the context to configure signaling parameters.

**NOTE:**

This command requires a PDN shutdown.

5.2.1.25.7 fa-ha-peer-list

Syntax

`fa-ha-peer-list peer-list-name`

`no fa-ha-peer-list`

Context

`config>mobile>pdn>pi>signaling>mipv4`

Description

This command specifies the fa-ha peer list name associated to this pi reference-point.

The no form of this command removes the fa-ha peer name from the configuration of this PI reference-point-name.

**NOTE:**

This command requires a PDN shutdown.

Default

`no fa-ha-peer-list`

Parameters

peer-list-name

specifies a fa-ha-peer-list name, up to 32 characters

5.2.1.25.8 profile

Syntax

```
profile profile-name  
no profile
```

Context

```
config>mobile>pdn>signaling>mipv4
```

Description

This command specifies the MIPv4 profile name. The profile provides signaling related configuration. The no form of this command removes the profile name from the configuration.

**NOTE:**

This command requires a PDN shutdown.

Parameters

profile-name

specifies the MIPv4 profile name, up to 32 characters

5.2.1.26 RADIUS PDN command descriptions

5.2.1.26.1 radius

Syntax

```
radius
```

Context

```
config>mobile>pdn
```

Description

This command enables the context to configure RADIUS parameters.

5.2.1.26.2 accounting-on-off-level

Syntax

```
accounting-on-off-level {system | apn}  
no accounting-on-off-level
```

Context

```
config>mobile>pdn>radius
```

Description

This command configures the RADIUS Accounting ON and OFF messaging to work on a per system or per APN level. With the per APN level alternative, the support must be separately configured per APN in the config>mobile>pdn>apn configuration context.

The no form of this command reverts to the default.

Parameters

system

single ON and OFF message is sent per system

apn

sending of ON and OFF messages is configurable per APN

Default: system

5.2.1.26.3 delete-session-acct-interim-exh

Syntax

```
[no] delete-session-acct-interim-exh
```

Context

```
config>mobile>pdn>radius
```

Description

This command enables the support of session deletion when the configurable amount of accounting interim retries are exhausted.

The no form of this command disables the support of session deletion upon exhaustion of accounting interim messages.

Default

no delete-session-acct-interim-exh

5.2.1.26.4 interface

Syntax

```
interface [router router-instance] interface-name
no interface
```

Context

config>mobile>pdn>radius

Description

This command configures a RADIUS interface.



NOTE:

This command requires a PDN shutdown.

Parameters

router-instance

specifies the router instance or service ID

Values	router-name service-id	
–	router-name	Base
–	service-id	1 to 2147483647

Default: Base

interface-name

specifies the interface which is used for this PDN gateway in case the RADIUS group interface is not configured; if the string contains special characters (#, \$, spaces, and so on), the entire string must be enclosed within double quotes, up to 32 character, must start with a letter

5.2.1.26.5 radius-accounting-on-off

Syntax

```
radius-accounting-on-off
```

Context

```
config>mobile>pdn>radius
```

Description

This command enables the context for controlling RADIUS accounting ON and OFF messaging on system level (pdn shut or cold restart).

5.2.1.26.6 messages-sent

Syntax

```
messages-sent {none | on | on-off}
```

```
no messages-sent
```

Context

```
config>mobile>pdn>radius>rad-accounting-on-off
```

Description

This command configures the PGW/GGSN sending ON/OFF RADIUS accounting messages on system level. The ON/OFF messages are sent by default.

The no form of this command reverts to the default.

Parameters

none

accounting ON or accounting OFF messages are not sent

on

only accounting ON messages are sent

on-off

both accounting ON and accounting OFF messages are sent

Default: on-off

5.2.1.26.7 radius-disconnect

Syntax

[no] radius-disconnect

Context

config>mobile>pdn>radius

Description

This command enables the support of RADIUS-initiated disconnect. When not present, the PGW/GGSN counts but otherwise ignore disconnect messages.

The no form of this command disables the RADIUS-initiated disconnect.



NOTE:

This command requires a PDN shutdown.

Default

no radius-disconnect

5.2.1.26.8 rel-16-rat-type

Syntax

[no] rel-16-rat-type

Context

config>mobile>pdn>radius

Description

This command enables or disables the 3GPP Release 16 RAT-type value in the RADIUS RAT-type attribute.

The no form of this command reverts to the default.

Default

no rel-16-rat-type

5.2.1.27 RF interface PDN command descriptions

5.2.1.27.1 rf

Syntax

```
rf
```

Context

```
config>mobile>pdn
```

Description

This command enables the context to configure RF.

5.2.1.27.2 acct-interim-interval

Syntax

```
acct-interim-interval acct-interim-interval  
no acct-interim-interval
```

Context

```
config>mobile>pdn>rf
```

Description

This command configures the accounting interim interval.

Parameters

acct-interim-interval

specifies the accounting interim interval, in minutes, to send interim accounting information to the CDF

Values: 1 to 1440

Default: 30

5.2.1.27.3 acct-level

Syntax

```
acct-level {pdn-level | qci-level}
no acct-level
```

Context

```
config>mobile>pdn>rf
```

Description

This command specifies whether PDN-level accounting or QCI-level accounting is applicable. When PDN-level accounting is specified, Diameter charging sessions to the CDF are setup on a per PDN connection basis. When QCI-level accounting is specified, Diameter charging sessions to the CDF are setup on a per bearer basis.

The no form of this command reverts to the default.



NOTE:

This command requires a PDN shutdown.

Parameters

pdn-level

specifies that Diameter charging sessions to the Charging Data Function (CDF) are set up on a per PDN connection basis

qci-level

specifies that Diameter charging sessions to the Charging Data Function (CDF) are set up on a per bearer basis

Default: qci-level

5.2.1.27.4 charging-group-id

Syntax

```
[no] charging-group-id
```

Context

```
config>mobile>pdn>rf
```

Description

This command enables the inclusion of Charging-Group-ID AVP in Accounting Requests (ACR) messages.

The no form of this command disables the command.



NOTE:

This command requires a PDN shutdown.

5.2.1.27.5 interface

Syntax

```
interface
```

Context

```
config>mobile>pdn>rf
```

Description

This command enables the context to configure interface Diameter parameters.



NOTE:

This command requires a PDN shutdown.

5.2.1.27.6 diameter

Syntax

```
diameter [router router-instance] interface-name
```

```
no diameter
```

Context

```
config>mobile>pdn>rf>if
```

Description

This command configures Diameter information parameters.

The no form of this command removes the parameters from the configuration.



NOTE:

This command requires a PDN shutdown.

Default

no diameter

Parameters

router-instance

specifies the virtual router name or service ID

Values	router-name service-id	
–	router-name	Base
–	service-id	1 to 2147483647

Default: Base

interface-name

specifies an interface name, up to 32 characters

5.2.1.27.7 operator-string

Syntax

operator-string *operator-string*
no operator-string

Context

config>mobile>pdn>rf

Description

This command configures the string to be included in operator-string AVP (non-standard) of an Accounting Requests (ACR) message.

The no form of this command removes the operator string from the configuration.

**NOTE:**

This command requires a PDN shutdown.

Default

no operator-string

Parameters***operator-string***

specifies the operator-string AVP of an ACR message

5.2.1.27.8 outage

Syntax

outage

Context

config>mobile>pdn>rf

Description

This command enables the context to configure parameters for an outage scenario.

**NOTE:**

This command requires a PDN shutdown.

5.2.1.27.9 acr-file-and-record-hdr-rel-id-ext

Syntax

[no] acr-file-and-record-hdr-rel-id-ext

Context

config>mobile>pdn>rf>outage

Description

This command is used to enable including the Release Identifier Extension field in the Accounting Requests (ACR) file header and record header (similar to how it is included in the CDR file header and header format as specified by 3GPP TS 32.297 - CDR File Format and Transfer). This command is applicable to the Diameter Rf interface.

The no form of this command reverts to the default.



NOTE:

This command requires a PDN shutdown.

Default

no acr-file-and-record-hdr-rel-id-ext

5.2.1.27.10 file-params

Syntax

file-params

Context

config>mobile>pdn>rf>outage

Description

This command provides the required context for detailed configuration of the outage file parameters.



NOTE:

This command requires a PDN shutdown.

5.2.1.27.11 closure-rec-num

Syntax

closure-rec-num *closure-rec-num*

no closure-rec-num

Context

```
config>mobile>pdn>rf>outage>file
```

```
config>mobile>profile>gtp-prime-group>cdr-storage>file
```

Description

This command specifies an operator configurable limit for the number of ACRs that are stored in a file. The file is closed after the specified limit is reached.



NOTE:

This command requires a PDN shutdown, when used in the `config>mobile>pdn>rf>outage>file-params` context.

Parameters

closure-rec-num

specifies the maximum number of ACR messages to fit in a file

Values: 100 to 75000

Default: 50000

5.2.1.27.12 closure-size

Syntax

```
closure-size closure-size
```

```
no closure-size
```

Context

```
config>mobile>pdn>rf>outage>file
```

```
config>mobile>profile>gtp-prime-group>cdr-storage>file
```

Description

This command specifies an operator configurable limit for the number of ACRs that are stored in a file. The file is closed after the specified limit is reached.



NOTE:

This command requires a PDN shutdown, when used in the `config>mobile>pdn>rf>outage>file-params` context.

Parameters

closure-size

specifies the file closure size limit in megabytes

Values: 1 to 100

Default: 50

5.2.1.27.13 closure-time

Syntax

`closure-time closure-time`

`no closure-time`

Context

`config>mobile>pdn>rf>outage>file`

`config>mobile>profile>gtp-prime-group>cdr-storage>file`

Description

This command specifies an operator configurable file closure size limit.



NOTE:

This command requires a PDN shutdown, when used in the `config>mobile>pdn>rf>outage>file-params` context.

Parameters

closure-time

specifies the file closure-time field, in hours

Values: 1 to 24

Default: 1

5.2.1.27.14 extension

Syntax

```
extension extension  
no extension
```

Context

```
config>mobile>pdn>rf>outage>file  
config>mobile>profile>gtp-prime-group>cdr-storage>file
```

Description

This command specifies an operator configurable file extension field that is used in the filename.

**NOTE:**

This command requires a PDN shutdown, when used in the config>mobile>pdn>rf>outage>file-params context.

Default

no extension

Parameters

extension

specifies the file extension field, up to 8 characters

5.2.1.27.15 obsolete-time

Syntax

```
obsolete-time obsolete-time  
no obsolete-time
```

Context

```
config>mobile>pdn>rf>outage>file  
config>mobile>profile>gtp-prime-group>cdr-storage>file
```


Description

This command specifies an operator configurable file duration, in days, after which the file is deleted.



NOTE:

This command requires a PDN shutdown, when used in the `config>mobile>pdn>rf>outage>file` context.

Parameters

obsolete-time

specifies the duration after which the file is deemed obsolete and deleted, in days

Values: 1 to 31

Default: 7

5.2.1.27.16 private-info

Syntax

```
private-info private-info
```

```
no private-info
```

Context

```
config>mobile>pdn>rf>outage>file
```

```
config>mobile>profile>gtp-prime-group>cdr-storage>file
```

Description

This command specifies an operator-configurable file private information field that is used in the filename.



NOTE:

This command requires a PDN shutdown, when used in the `config>mobile>pdn>rf>outage>file` context.

Default

no private-info

Parameters

private-info

specifies a private information field, up to 32 characters

5.2.1.27.17 location

Syntax

```
location location
```

Context

```
config>mobile>pdn>rf>outage
```

```
config>mobile>profile>gtp-prime-group>cdr-storage
```

Description

This command specifies the compact flash to which the ACRs are written.



NOTE:

This command requires a PDN shutdown, when used in the config>mobile>pdn>rf>outage context.

Parameters

location

Specifies the compact flash destination for ACRs

You can choose to write to cf1 or cf2 (cf3 is reserved). If cf1 is chosen as primary, cf2 becomes the alternate compact flash that ACR messages will be written to when the primary is full, in the event of an outage.

Values: cf1|cf2

5.2.1.27.18 limit

Syntax

```
limit limit
```

```
no limit
```

Context

```
config>mobile>pdn>rf>outage>location
```

```
config>mobile>profile>gtp-prime-group>cdr-storage>location
```

Description

This command specifies a limit for the amount of memory that can be used for ACR storage.



NOTE:

This command requires a PDN shutdown, when used in the `config>mobile>pdn>rf>outage>location` context.

Parameters

limit

specifies the limit for the compact flash

5.2.1.27.19 primary

Syntax

```
[no] primary
```

Context

```
config>mobile>pdn>rf>outage>location
```

```
config>mobile>profile>gtp-prime-group>cdr-storage>location
```

Description

This command indicates if the compact flash specified in the `location` command is the primary compact flash.



NOTE:

This command requires a PDN shutdown, when used in the `config>mobile>pdn>rf>outage>location` context.

5.2.1.27.20 primary-diameter-peer

Syntax

```
primary-diameter-peer profile-name  
no primary-diameter-peer
```

Context

```
config>mobile>pdn>rf
```

Description

This command configures the Diameter peer profile to be used for the primary Charging Data Function (CDF) at the Gateway level.

The no form of this command removes the profile name from the configuration.



NOTE:

This command requires a PDN shutdown.

Default

```
no primary-diameter-peer
```

Parameters

profile-name

specifies the Diameter peer profile to be used for the primary CDF at the gateway level, up to 32 characters

5.2.1.27.21 retry-count

Syntax

```
retry-count retry-count  
no retry-count
```

Context

```
config>mobile>pdn>rf
```

Description

This command configures the retry count for ACR messages. An ACR message will be retransmitted for the number of times specified, when an ACA is not received from the CDF server.

The no form of this command reverts to the default.

Parameters

retry-count

specifies the retry count for Accounting Requests (ACR) messages

An Accounting Requests (ACR) message is retransmitted for the number of times specified, when an Accounting Answers (ACA) is not received from the Charging Data Function (CDF) server.

Values: 0 to 8

Default: 3

5.2.1.27.22 secondary-diameter-peer

Syntax

`secondary-diameter-peer profile-name`

`no secondary-diameter-peer`

Context

`config>mobile>pdn>rf`

Description

This command configures the Diameter peer profile to be used for the secondary Charging Data Function (CDF) at the Gateway level.

The no form of this command removes the profile name from the configuration.



NOTE:

This command requires a PDN shutdown.

Default

`no secondary-diameter-peer`

Parameters

profile-name

specifies the Diameter peer profile to be used for the primary CDF at the gateway level, up to 32 characters

5.2.1.27.23 supported-vendor

Syntax

```
supported-vendor custom-1  
no supported-vendor
```

Context

```
config>mobile>pdn>rf
```

Description

This command enables the specified supported vendor.



NOTE:

This command requires a PDN shutdown.

Default

```
no supported-vendor
```

Parameters

custom-1

specifies the customer number

5.2.1.27.24 transaction-timer

Syntax

```
transaction-timer {[value] | [timeout-msec value]}  
no transaction-timer
```

Context

config>mobile>pdn>rf

Description

This command configures the application transaction timer for Accounting Requests (ACR) and Accounting Answers (ACA). The value can be expressed in seconds or milliseconds. Setting the timer in milliseconds provides better timeout granularity.

The no form of this command removes the configuration.

Parameters

value

specifies the maximum amount of time, in seconds, the node waits for a peer to respond before trying another peer

Values: 1 to 180 (increments of one second)

Default: 5

timeout-msec value

specifies the maximum amount of time, in milliseconds, the node waits for a peer to respond before trying another peer

Values: 1000 to 180000 (increments of 100 ms)

Default: 5000

5.2.1.28 S1-U interface PDN command descriptions

5.2.1.28.1 s1-u

Syntax

s1-u [*sig-ref-point-name*] [group-specific]

no s1-u [*sig-ref-point-name*]

Context

config>mobile>pdn

Description

This command configures the s1-u reference point name.

**NOTE:**

This command requires a PDN shutdown, unless the `group-specific` parameter is specified.

Parameters

sig-ref-point-name

specifies a signaling reference point name up to 32 characters

At least one S1-u interface must be configured on the gateway to allow peering with eNB nodes. A single S1-u interface configured on the gateway can form peering with multiple eNB nodes. Optionally, multiple S1-u interfaces can be configured on the gateway. If a signaling reference point name is not specified, the default instance of the S1-u interface is created.

Each S1-u interface must be associated with an S11 interface, configured using the `config>mobile>pdn>s11>if>gtp-c` command. If no S1-u interface is provided during S11 configuration, the S11 interface is associated with the default S1-u interface.

The S1-u interface is bound to an S11 interface to form a signaling group. For example, the default instance of the S1-u interface will be selected and assigned a TEID on receiving a Create Session Request over the default instance of an S11 interface.

group-specific

specifies an MG group-specific reference point interface for GTP-U redirection; it is used when associating the reference point interface with an MG system group via an override profile and applies only to CMG

5.2.1.28.2 dual-stack-uplane-pref

Syntax

```
dual-stack-uplane-pref {ipv4 | ipv6}
```

```
no dual-stack-uplane-pref
```

Context

```
config>mobile>pdn>s1-u
```

Description

The command specifies the preference in a dual stack when both IPv4 and IPv6 addresses are configured for the reference point interface.

**NOTE:**

This command requires a PDN shutdown.

Default

Use the same IP version as the control plane

5.2.1.28.3 interface

Syntax

```
interface
```

Context

```
config>mobile>pdn>s1-u
```

Description

The command enables the context to configure the s1-u interface.



NOTE:

This command requires a PDN shutdown.

5.2.1.28.4 gtp-u

Syntax

```
gtp-u [router router-instance] interface-name [interface-realm s1-u-realm-name]
```

```
no gtp-u
```

Context

```
config>mobile>pdn>s1-u>if
```

Description

This command configures the GTP-U profile parameters for the S1-U interface.

The no form of this command removes the parameter from the configuration.



NOTE:

This command requires a PDN shutdown.

Parameters

router-instance

specifies the virtual router name or service ID

Values	router-name service-id	
–	router-name	Base
–	service-id	1 to 2147483647

Default: Base

interface-name

specifies an interface name, up to 32 characters

s1-u-realm-name

specifies the associated realm name for the virtual router instance and the S1-U interface, up to 32 characters



NOTE:

Only the following characters are permitted:

- alphabetic characters (A-Z and a-z)
- digits (0-9)
- hyphen (-)
- dot (.)

The user plane function uses the interface realm name to identify the correct router instance and S1-U interface to use, based on the Network-Instance IE in the PFCP interface rules during the PFCP session establishment/modification process. The realm name must be unique for a given virtual routing instance.



NOTE:

This parameter is only used when the gateway is configured as a UP function using the `mobile>config>pdn>instance - type` command.

5.2.1.28.5 signaling

Syntax

signaling

Context

config>mobile>pdn>s1-u

Description

This command enables the context to configure signaling for the S1-U interface.



NOTE:

This command requires a PDN shutdown.

5.2.1.28.6 gtp-u

Syntax

gtp-u

Context

config>mobile>pdn>s1-u>signaling

Description

This command enables the context to configure GTP-U for the S1-U signaling interface.



NOTE:

This command requires a PDN shutdown.

5.2.1.28.7 profile

Syntax

profile *profile-name*

```
no profile
```

Context

```
config>mobile>pdn>s1-u>signaling>gtp-u
```

Description

This command configures the GTP-U profile name for the S1-u signaling interface.

The no form of this command removes the profile name from the configuration.



NOTE:

This command requires a PDN shutdown.

Default

```
no profile
```

Parameters

profile-name

specifies the GTP-U profile name, up to 32 characters

5.2.1.29 S11 interface PDN command descriptions

5.2.1.29.1 s11

Syntax

```
s11 [sig-ref-point-name] [group-specific]
```

```
no s11 [sig-ref-point-name]
```

Context

```
config>mobile>pdn
```

Description

This command configures the s11 reference point name.



NOTE:

This command requires a PDN shutdown, unless the `group-specific` parameter is specified.

Parameters

sig-ref-point-name

specifies a signaling reference point name up to 32 characters

At least one S11 interface must be configured on the gateway to allow peering with MME. A single S11 interface configured on the gateway can form peering with multiple MME nodes. Optionally, multiple S11 interfaces can be configured on the SGW. If a signaling reference point name is not specified, the default instance of S11 interface is created.

Each S11 interface must be associated with an S1-u interface, configured using the `config>mobile>pdn>s11>if>gtp-c` command. If no S1-u interface is provided during the S11 interface configuration, the S11 interface is associated with the default S1-u interface.

The S1-u interface is bound to an A11 interface to form a signaling group. For example, the default instance of the S1-u interface will be selected and assigned TEID on receiving a Create Session Request over the default instance of the S11 interface.

group-specific

specifies an MG group-specific reference point interface for GTP-U redirection; it is used when associating the reference point interface with an MG system group via an override profile and applies only to CMG

5.2.1.29.2 interface

Syntax

`interface`

Context

`config>mobile>pdn>s11`

Description

This command enables the context to configure the s11 interface.



NOTE:

This command requires a PDN shutdown.

5.2.1.29.3 gtp-c

Syntax

```
gtp-c [router router-instance] interface-name [s1-u binding-ref-pt-name] [interface-realm
s11-u-realm-name] [s1u-interface-realm binding-realm-name]

no gtp-c
```

Context

```
config>mobile>pdn>s11>if
```

Description

This command configures the GTP-C profile parameters for the S11 interface.



NOTE:

This command requires a PDN shutdown.

Parameters

router-instance

specifies the virtual router name or service ID

Values	router-name service-id	
–	router-name	Base
–	service-id	1 to 2147483647

Default: Base

interface-name

specifies an interface name up to 32 characters, which must start with a letter

binding-ref-point-name

specifies an S1-u interface signaling reference point name to bind, up to 32 characters

s11-u-realm-name

specifies the associated realm name for the virtual routing instance that has the S11/S1 GTP-U interface on the user plane function, up to 32 characters

**NOTE:**

Only the following characters are permitted:

- alphabetic characters (A-Z and a-z)
- digits (0-9)
- hyphen (-)
- dot (.)

The realm name must be unique for a given virtual routing instance on the user plane and is communicated in the Network-Instance IE part of the PFCP interface session establishment/modification process.

**NOTE:**

This parameter is only used when the gateway is configured as a CP function using the `mobile>config>pdn>instance-type` command.

binding-realm-name

specifies the associated realm name for the S1-U interface, up to 32 characters; relevant if the S1-U and the S11-U use different network realms

5.2.1.29.4 gtp-u

Syntax

```
gtp-u [router router-instance] interface-name [interface-realm s11-realm-name]  
no gtp-u
```

Context

```
config>mobile>pdn>s11>if
```

Description

This command configures the GTP-U profile parameters for the S11 interface.

**NOTE:**

This command requires a PDN shutdown.

Parameters

router-instance

specifies the virtual router name or service ID

Values	router-name service-id	
–	router-name	Base
–	service-id	1 to 2147483647

Default: Base

interface-name

specifies an interface name up to 32 characters, which must start with a letter

s11-realm-name

specifies the associated realm name for the virtual routing instance and the S11-U interface, up to 32 characters



NOTE:

Only the following characters are permitted:

- alphabetic characters (A-Z and a-z)
- digits (0-9)
- hyphen (-)
- dot (.)

The user plane function uses the interface realm name to identify the correct router instance and S11-U interface to use, based on the Network-Instance IE in the PFCP interface rules during the PFCP interface session establishment/modification process. The realm name must be unique for a given virtual routing instance.



NOTE:

This parameter is only used when the gateway is configured as a UP function using the `mobile>config>pdn>instance-type` command.

5.2.1.29.5 mme-group

Syntax

[no] mme-group *MME-group-id*

Context

config>mobile>pdn>s11

Description

This command disables or enables MME groups for the S11 reference point. There cannot be multiple MME groups which have any similarities in a peer list under the same S11 reference point.



NOTE:

This command requires a PDN shutdown.

Default

no mme-group

Parameters

MME-group-id

specifies the group name identifier

Values: 1 to 65535

5.2.1.29.6 signaling

Syntax

signaling

Context

config>mobile>pdn>s11

Description

This command enables the context to configure signaling for the S11 interface.

**NOTE:**

This command requires a PDN shutdown.

5.2.1.29.7 gtp-c

Syntax

```
gtp-c
```

Context

```
config>mobile>pdn>s11>signaling
```

Description

This command enables the context to configure GTP-C signaling for the S11 signaling interface.

**NOTE:**

This command requires a PDN shutdown.

5.2.1.29.8 ddn-damp-timer

Syntax

```
ddn-damp-timer ddn-damp-timer
```

```
no ddn-damp-timer
```

Context

```
config>mobile>pdn>s11>signaling>gtp-c
```

Description

This command configures the DDN dampening timer interval.

The no form of this command removes the dampening timer from the configuration.

**NOTE:**

This command requires a PDN shutdown.

Parameters

ddn-damp-timer

specifies a dampening timer interval, in seconds

Values: 0 | 10 to 3600

Default: 0

5.2.1.29.9 ddn-extended-buffering-time

Syntax

`ddn-extended-buffering-time ddn-extended-buffering-time`

`no ddn-extended-buffering-time`

Context

`config>mobile>pdn>s11>signaling>gtp-c`

Description

This command defines the maximum time for the extended buffering. If the value received from the MME is less than the configured value, the value received from the MME is used, otherwise the configured value defines how long the extended buffering is applied. If the maximum time is 0, the extended buffering is not supported. Buffering will last always at least 30 seconds.

The no form of this command removes the extended buffering timer from the configuration.



NOTE:

This command requires a PDN shutdown for a CMG operating in LB-less mode. In an LB-mode CMG, PDN shutdown is not required. You can change the value of the `ddn-extended-buffering-time` command before the DDN Ack is received and the new configured value is used.

Parameters

ddn-extended-buffering-time

specifies the maximum time for extended buffering, in seconds

Values: 0 | 30 to 86400

Default: 0

5.2.1.29.10 **ddn-throttling-arp-threshold**

Syntax

```
ddn-throttling-arp-threshold ddn-throttling-arp-threshold  
no ddn-throttling-arp-threshold
```

Context

```
config>mobile>pdn>s11>signaling>gtp-c
```

Description

This command configures the DDN ARP threshold.

The no form of this command removes the ARP threshold from the configuration.



NOTE:

This command requires a PDN shutdown.

Default

```
no ddn-throttling-arp
```

Parameters

ddn-throttling-arp-threshold

specifies a DDN ARP threshold

Values: 0 to 15

5.2.1.29.11 **iot-restoration-time**

Syntax

```
iot-restoration-time time  
no iot-restoration-time
```

Context

```
config>mobile>pdn>s11>signaling>gtp-c
```

Description

This command configures the restoration time limit that the SGW waits before it deletes the UE context for cellular IoT UEs only. When the time limit is set to a non-zero value, the command enables the MME restoration procedures for IoT UEs.

The timer starts when a failure in the S11 interface is detected. When the timer expires, the data and control plane resources are cleaned up for IoT UEs without an active S11 interface.



NOTE:

- The S11 interface path management is configured using the `config>mobile>pdn>s11>signaling>gtp-c>restoration-time` command.
- The GTP-C restoration time must be set to a non-zero value for the `iot-restoration-time` command to take effect.
- This command requires a PDN shutdown.

Parameters

time

specifies the value, in seconds, for the IoT restoration-time limit

Values: 3601 to 1116000

Default: 0

5.2.1.29.12 legacy-create-idft-response-encoding

Syntax

```
[no] legacy-create-idft-response-encoding
```

Context

```
config>mobile>pdn>s11>signaling>gtp-c
```

```
config>mobile>pdn>s4>signaling>gtp-c
```

Description

This command enables the legacy encoding of the Create Indirect Data Forwarding Tunnel Response where the F-TEID instances 0, 1, and 2 will use interface type values 1, 3, and 16 respectively.

The no form of this command selects the default encoding where the interface type value 23 is used for all instances of the F-TEID.

**NOTE:**

This command requires a PDN shutdown.

Default

no legacy-create-idft-response-encoding

5.2.1.29.13 profile

Syntax

profile *profile-name*

no profile

Context

config>mobile>pdn>s11>signaling>gtp-c

Description

This command specifies the GTP-C signaling profile name for the s11 signaling interface. The GTP-C profile provides GTP-C signaling related configuration

The no form of this command removes the profile name from the configuration.

**NOTE:**

This command requires a PDN shutdown.

Default

no profile

Parameters***profile-name***

specifies a profile name, up to 32 characters

5.2.1.29.14 restoration-time

Syntax

```
restoration-time time [qci value]  
no restoration-time
```

Context

```
config>mobile>pdn>s11>signaling>gtp-c  
config>mobile>pdn>s4>signaling>gtp-c
```

Description

This command specifies the restoration time limit that the SGW waits before it deletes the UE context. A non-zero value of this option enables MME/SGSN restoration procedures. The timer will trigger cleanup of resources for all active UEs that have no active S11/S4 interface. The timer will initiate after the failure of the S11/S4 interface is detected. When the timer expires, the data and control plane resources will be cleaned up for UEs without an active S11/S4 interface, and the SGW will cleanup remaining UEs that are in an idle state and do not have an active S11/S4 interface.

The optional QCI value must be applied only when the default bearers that have a matching QCI value, are eligible for restoration. If the list is not defined, all bearers are eligible for the restoration.



NOTE:

This command requires a PDN shutdown.

Parameters

time

specifies the restoration time limit value

Values: 0 to 3600

Default: 0

value

specifies the QCI value for the default bearers when restoration time is enabled

Values: 1 to 255

5.2.1.29.15 ue-inactivity-timer

Syntax

```
ue-inactivity-timer minutes  
no ue-inactivity-timer
```

Context

```
config>mobile>pdn>s11>signaling>gtp-c
```

Description

This command configures a timer to detect active status of UE after path management session on S11 reference point to a peer restarts without the peering node restarting. If the UE is idle for configured time after the path management session has restarted then the SGW will delete the PDN connection. This is done to remove stale sessions that may not be valid as they have moved to alternate SGW during SGW restoration procedure executed on MME.

The no form of this command disables UE inactivity timer configuration.

**NOTE:**

This command requires a PDN shutdown.

Parameters

minutes

specifies time interval, in minutes

Values: 0 | 30 to 2880

Default: 0

5.2.1.30 S12 interface PDN command descriptions

5.2.1.30.1 s12

Syntax

```
s12 [sig-ref-point-name] [group-specific]  
no s12 [sig-ref-point-name]
```


Context

```
config>mobile>pdn
```

Description

This command configures the reference point name for the s12 interface.



NOTE:

This command requires a PDN shutdown, unless the `group-specific` parameter is specified.

Parameters

sig-ref-point-name

specifies a signaling reference point name up to 32 characters

At least one S12 interface must be configured on the gateway to allow peering with RNC nodes. A single S12 interface configured on the gateway can form peering with multiple RNC nodes. Optionally, multiple S12 interfaces can be configured on the gateway. If a signaling reference point name is not provided, the default instance of the S12 interface is created.

Each S12 interface must be associated with an S4 interface, configured using the `config>mobile>pdn>s4>if>gtp-c` command. If no S12 interface is provided during the S4 interface configuration, the S4 interface is associated with the default S12 interface. The S12 interface is bound to an S4 interface to form a signaling group. For example, the default instance of the S12 interface will be selected and assigned TEID on receiving a Create Session Request over the default instance of the S4 interface.

group-specific

specifies an MG group-specific reference point interface for GTP-U redirection; it is used when associating the reference point interface with an MG system group via an override profile and applies only to CMG

5.2.1.30.2 dual-stack-uplane-pref

Syntax

```
dual-stack-uplane-pref {ipv4 | ipv6}
```

```
no dual-stack-uplane-pref
```

Context

```
config>mobile>pdn>s12
```

Description

The command specifies the preference in a dual stack when both IPv4 and IPv6 addresses are configured for the reference point interface.

**NOTE:**

This command requires a PDN shutdown.

Default

The same IP version as the control plane.

5.2.1.30.3 interface

Syntax

interface

Context

config>mobile>pdn>s12

Description

The command enables the context to configure the s12 interface.

**NOTE:**

This command requires a PDN shutdown.

5.2.1.30.4 gtp-u

Syntax

gtp-u [router *router-instance*] *interface-name* [interface-*realm* *realm-name*]

no gtp-u

Context

config>mobile>pdn>s12>if

Description

This command configures the GTP-U profile parameters for the S12 interface.

The no form of this command removes the parameter from the configuration.



NOTE:

This command requires a PDN shutdown.

Parameters

router-instance

specifies the router instance, up to 32 characters

interface-name

specifies an interface name, up to 32 characters

realm-name

specifies the associated realm name for the virtual routing instance and the S12-U interface, up to 32 characters



NOTE:

Only the following characters are permitted:

- alphabetic characters (A-Z and a-z)
- digits (0-9)
- hyphen (-)
- dot (.)

The user plane function uses the interface realm name to identify the correct router instance and S12-U interface to use, based on the Network-Instance IE in PFCP Rules during the PFCP interface session establishment/modification process. The realm name must be unique for a given virtual routing instance.



NOTE:

This parameter is only used when the gateway is configured as a UP function using the `mobile>config>pdn>instance-type` command.

5.2.1.30.5 signaling

Syntax

signaling

Context

config>mobile>pdn>s12

Description

This command enables the context to configure signaling for the s12 interface.

**NOTE:**

This command requires a PDN shutdown.

5.2.1.30.6 gtp-u

Syntax

gtp-u

Context

config>mobile>pdn>s12>signaling

Description

This command enables the context to configure GTP-U signaling for the s12 signaling interface.

**NOTE:**

This command requires a PDN shutdown.

5.2.1.30.7 profile

Syntax

profile *profile-name*

no profile

Context

```
config>mobile>pdn>s12>signaling>gtp-u
```

Description

This command specifies the GTP-U profile name for the s12 signaling interface. The GTP-U profile provides GTP-U signaling related configuration.

The no form of this command removes the profile name from the configuration.



NOTE:

This command requires a PDN shutdown.

Default

no profile

Parameters

profile-name

specifies a profile name, up to 32 characters

5.2.1.31 S2a and S2b interface PDN command descriptions

5.2.1.31.1 s2a

Syntax

```
s2a [ref-point-name] [group-specific]
```

```
no s2a [ref-point-name]
```

Context

```
config>mobile>pdn
```

Description

This command configures S2a interface parameters. The S2a GTP parameter provides tunneling and tunnel management between the TWAG and the PGW. The S2a PMIPv6 parameter provides tunneling between the eHRPD and the PGW.

**NOTE:**

- This keyword applies only to CMG and is supported for S2a GTP and not S2a PMIP.
- This command requires a PDN shutdown, unless the `group-specific` parameter is specified.

Parameters

ref-point-name

specifies a signaling reference point name, up to 32 characters

group-specific

specifies an MG group-specific reference point interface for GTP-U redirection; it is used when associating the reference point interface with an MG system group via an override profile

5.2.1.31.2 s2b

Syntax

s2b [*ref-point-name*] [*group-specific*]

no s2b [*ref-point-name*]

Context

config>mobile>pdn

Description

This command configures S2b interface parameters. The S2b parameters provide user plane tunneling and tunnel management between the EPDG and the PGW.

**NOTE:**

This command requires a PDN shutdown, unless the `group-specific` parameter is specified.

Parameters

ref-point-name

specifies a signaling reference point name, up to 32 characters

group-specific

specifies an MG group-specific reference point interface for GTP-U redirection. This is used when associating the reference-point interface with an MG system group via an override profile. This keyword applies to CMG systems only.

5.2.1.31.3 dual-stack-cplane-pref

Syntax

```
dual-stack-cplane-pref {ipv4 | ipv6}
no dual-stack-cplane-pref
```

Context

```
config>mobile>pdn>s2a
config>mobile>pdn>s2b
```

Description

This command sets the dual stack Cplane preference.



NOTE:

This command requires a PDN shutdown.

Parameters

ipv4

specifies the IPv4 address family

ipv6

specifies the IPv6 address family

Default: ipv6

5.2.1.31.4 dual-stack-uplane-pref

Syntax

```
dual-stack-uplane-pref {ipv4 | ipv6 | use-cplane}
no dual-stack-uplane-pref
```

Context

```
config>mobile>pdn>s2a
```

```
config>mobile>pdn>s2b
```

Description

This command specifies the preference in a dual stack when both IPv4 and IPv6 addresses are configured for the reference point interface.



NOTE:

This command requires a PDN shutdown.

Parameters

ipv4

specifies to use the IPv4 address family

ipv6

specifies to use the IPv6 address family

use-cplane

specifies the same IP version as the control plane

Default: use-cplane

5.2.1.31.5 interface

Syntax

```
interface
```

Context

```
config>mobile>pdn>s2a
```

```
config>mobile>pdn>s2b
```

Description

This command enables the context to configure S2a or S2b signaling.



NOTE:

This command requires a PDN shutdown.

5.2.1.31.6 gtp-c

Syntax

```
gtp-c [router router-instance] interface-name [interface-realm realm-name]  
no gtp-c
```

Context

```
config>mobile>pdn>s2a>if  
config>mobile>pdn>s2b>if
```

Description

This command configures the GTP-C profile parameters for the S2a or S2b interfaces.



NOTE:

This command requires a PDN shutdown.

Parameters

router-instance

specifies the router name or service ID

Values	router-name service-id	
–	router-name	Base
–	service-id	1 to 2147483647

Default: Base

interface-name

specifies the interface name, up to 32 characters

realm-name

specifies the associated realm name for the virtual routing instance that has the S2a or S2b GTP-U interface on the user plane function, up to 32 characters

**NOTE:**

Only the following characters are permitted:

- alphabetic characters (A-Z and a-z)
- digits (0-9)
- hyphen (-)
- dot (.)

The realm name must be unique for a given virtual routing instance on the user plane and is communicated in the Network-Instance IE part of the PFCP interface session establishment/modification process.

**NOTE:**

This parameter is only used when the gateway is configured as a CP function using the `mobile>config>pdn>instance-type` command.

5.2.1.31.7 gtp-u

Syntax

```
gtp-u [router router-instance] interface-name  
gtp-u [router router-instance] interface-name [interface-realm realm-name]  
no gtp-u
```

Context

```
config>mobile>pdn>s2a>if  
config>mobile>pdn>s2b>if
```

Description

This command configures the GTP-U profile parameters for the S2a or S2b interface.

**NOTE:**

This command requires a PDN shutdown.

Parameters

router-instance

specifies the router name or service ID

Values	router-name service-id	
–	router-name	Base
–	service-id	1 to 2147483647

Default: Base

interface-name

specifies the interface name, up to 32 characters

realm-name

specifies the associated realm name for the virtual routing instance and the S2a-U or S2b-U interface, up to 32 characters



NOTE:

Only the following characters are permitted:

- alphabetic characters (A-Z and a-z)
- digits (0-9)
- hyphen (-)
- dot (.)

The user plane function uses the interface realm name to identify the correct router instance and S2a-U or S2b-U interface to use, based on the Network-Instance IE in the PFCP interface rules during the PFCP session establishment/modification process. The realm name must be unique for a given virtual routing instance.



NOTE:

This parameter is only used when the gateway is configured as a UP function using the `mobile>config>pdn>instance-type` command.

5.2.1.31.8 pmipv6

Syntax

```
pmipv6 interface-name [router router-instance]
no pmipv6
```

Context

```
config>mobile>pdn>s2a>if
```

Description

This command configures the interface that is used to send and receive PMIPv6 messages. If this interface belongs to a VPRN other than the master instance, the command must include the VPRN name (interface names have VPRN scope).



NOTE:

This command requires a PDN shutdown.

Parameters

router-instance

specifies the router name or service ID

Values	router-name service-id	
–	router-name	Base
–	service-id	1 to 2147483647

Default: Base

interface-name

specifies the interface name, up to 32 characters

5.2.1.31.9 match

Syntax

```
match
```

Context

```
config>mobile>pdn>s2a
```

```
config>mobile>pdn>s2b
```

Description

This command enables the context to configure a match list for the reference point.

**NOTE:**

This command requires a PDN shutdown.

5.2.1.31.10 apn-list

Syntax

```
apn-list list-name
```

```
no apn-list
```

Context

```
config>mobile>pdn>s2a>match
```

```
config>mobile>pdn>s2b>match
```

Description

This command specifies an APN list instance to be used for the match criteria. The APN list is configured in the config>mobile>profile>list context.

The no form of this command disables the APN list matching.

**NOTE:**

This command requires a PDN shutdown.

Parameters

list-name

specifies the name of this APN list instance

5.2.1.31.11 plmn-id-list

Syntax

```
plmn-id-list list-name
```

```
no plmn-id-list
```

Context

```
config>mobile>pdn>s2a>match
```

```
config>mobile>pdn>s2b>match
```

Description

This command configures a Public Land Mobile Network (PLMN) list for S2a-GTP and S2b.



NOTE:

- For S2a GTP, the `plmn-id-list` command is only configurable under a non-default S2a interface (that is, s2a "<some name>").
- This command requires a PDN shutdown.

Parameters

list-name

specifies the name of the PLMN list

5.2.1.31.12 pgw-set-support

Syntax

```
[no] pgw-set-support
```

Context

```
config>mobile>pdn>s2b
```

Description

This command configures the support of "PGW Set Support" call flows as specified in 3GPP TS 23.007.

The no form of this command disables the ePDG support for PGW-C/SMF set.



NOTE:

In the current release, when the `pgw-set-support` command is enabled, the ePDG initiates the PDN connection session restoration procedure

Default

`no pgw-set-support`

5.2.1.31.13 signaling

Syntax

`signaling`

Context

`config>mobile>pdn>s2a`

`config>mobile>pdn>s2b`

Description

This command enables the context to configure signaling for an S2a or S2b interface.



NOTE:

This command requires a PDN shutdown.

5.2.1.31.14 gtp-c

Syntax

`gtp-c`

Context

`config>mobile>pdn>s2a>signaling`

```
config>mobile>pdn>s2b>signaling
```

Description

This command enables the context to configure GTP-C signaling for an S2a or S2b interface.



NOTE:

This command requires a PDN shutdown.

5.2.1.31.15 gtp-u

Syntax

```
gtp-u
```

Context

```
config>mobile>pdn>s2a>signaling
```

```
config>mobile>pdn>s2b>signaling
```

Description

This command enables the context to configure GTP-U signaling for an S2a or S2b interface.



NOTE:

This command requires a PDN shutdown.

5.2.1.31.16 pmipv6

Syntax

```
pmipv6
```

Context

```
config>mobile>pdn>s2a>signaling
```

```
config>mobile>pdn>s2b>signaling
```


Description

This command configures PMIPv6 signaling for an S2a or S2b interface.



NOTE:

This command requires a PDN shutdown.

5.2.1.31.17 profile

Syntax

`profile profile-name`

`no profile`

Context

`config>mobile>pdn>s2a>signaling>gtp-c`

`config>mobile>pdn>s2a>signaling>gtp-u`

`config>mobile>pdn>s2b>signaling>gtp-c`

`config>mobile>pdn>s2b>signaling>gtp-u`

`config>mobile>pdn>s2a>signaling>pmipv6`

Description

This command specifies the profile name to associate with the gateway. For PMIPv6, this command applies a PMIPv6 profile for the interface.

The no form of this command removes the profile name from the configuration.



NOTE:

This command requires a PDN shutdown.

Default

no profile

Parameters

profile-name

specifies a profile name, up to 32 characters

5.2.1.31.18 s2b-error-mapping-profile

Syntax

```
s2b-error-mapping-profile profile-name  
no s2b-error-mapping-profile
```

Context

```
config>mobile>pdn
```

Description

This command creates an S2b error mapping profile.

Parameters

profile-name

specifies a profile name, up to 32 characters

5.2.1.32 S4 interface PDN command descriptions

5.2.1.32.1 s4

Syntax

```
s4 [sig-ref-point-name] [group-specific]  
no s4 [sig-ref-point-name]
```

Context

```
config>mobile>pdn
```

Description

This command configures S4 signaling.



NOTE:

This command requires a PDN shutdown, unless the `group-specific` parameter is specified.

Parameters

sig-ref-point-name

Specifies a signaling reference point name up to 32 characters. At least one S4 interface must be configured on the gateway to allow peering with SGSN nodes. A single S4 interface configured on the gateway can form peering with multiple SGSN nodes. Optionally, multiple S4 interfaces can be configured on the gateway. If a signaling reference point name is not provided, the default instance of S4 interface is created.

The user plane related to an S4 interface can be handled by the S4-u or S12 interface. An S12 interface can be bound to an S4 interface to form a signaling group. If no S12 is provided during the S4 interface configuration, the S4 interface is associated with the default S12 interface for the scenarios where the user plane is handled by the S12 interface. For example, the default instance of the S12 interface will be selected and assigned a TEID on receiving a Create Session Request over the default instance of the S4 interface and direct tunnel is requested in Create Session Request.

group-specific

specifies an MG group-specific reference point interface for GTP-U redirection; it is used when associating the reference point interface with an MG system group via an override profile and applies only to CMG

5.2.1.32.2 interface

Syntax

```
interface
```

Context

```
config>mobile>pdn>s4
```

Description

This command enables the context to configure the S4 interface.



NOTE:

This command requires a PDN shutdown.

5.2.1.32.3 gtp-c

Syntax

```
gtp-c [router router-instance] interface-name [s12 binding-ref-pt-name] [interface-realm s4-  
realm-name]
```

```
no gtp-c
```

Context

```
config>mobile>pdn>s4>if
```

Description

This command configures the GTP-C profile parameters for an S4 interface.

**NOTE:**

This command requires a PDN shutdown.

Parameters

router-instance

specifies the router instance, up to 32 characters

interface-name

specifies an interface name, up to 32 characters

binding-ref-point-name

specifies the name of the s12 binding reference point, up to 32 characters

s4-realm-name

specifies the associated realm name for the virtual routing instance that has the S12 GTP-U interface on the user plane function, up to 32 characters

**NOTE:**

Only the following characters are permitted:

- alphabetic characters (A-Z and a-z)
- digits (0-9)
- hyphen (-)
- dot (.)

The realm name must be unique for a given virtual routing instance on the user plane and is communicated in the Network-Instance IE part of the PFCP interface session establishment/modification process.

**NOTE:**

This parameter is only used when the gateway is configured as a CP function using the `mobile>config>pdn>instance-type` command.

5.2.1.32.4 gtp-u

Syntax

```
gtp-u [router router-instance] interface-name [interface-realm realm-name]
```

```
no gtp-u
```

Context

```
config>mobile>pdn>s4>if
```

Description

This command configures the GTP-U profile parameters for an S4 interface.

**NOTE:**

This command requires a PDN shutdown.

Parameters

router-instance

specifies the router instance, up to 32 characters

interface-name

specifies an interface name, up to 32 characters

realm-name

specifies the associated realm name for the virtual routing instance and the S4-U interface, up to 32 characters

**NOTE:**

Only the following characters are permitted:

- alphabetic characters (A-Z and a-z)
- digits (0-9)
- hyphen (-)
- dot (.)

The UP function uses the interface realm name to identify the correct router instance and S4-U interface to use, based on the Network-Instance IE in the PFCP interface rules during the PFCP session establishment/modification process. The realm name must be unique for a given virtual routing instance.

**NOTE:**

This parameter is only used when the gateway is configured as a UP function using the `mobile>config>pdn>instance - type` command.

5.2.1.32.5 peer-list

Syntax

```
peer-list list-name  
no peer-list
```

Context

```
cconfig>mobile>pdn>s4
```

Description

This command configures the trusted peer list name. The list includes peer IPv4 or IPv6 address that are trusted. Separate trusted peer lists for eNodeBs, MMEs, and PGWs can be created for trusted peering. If no trusted-peer list has been defined on a given protocol, the protocol will accept messages from any peer. Conversely, when a trusted-peer list is defined, only messages from peers in the given list will be accepted. The operator can add new peers to a trusted-peer list at any time.

The no form of this command removes the list name from the configuration.

**NOTE:**

This command requires a PDN shutdown.

Default

```
no peer-list
```

Parameters

list-name

specifies the list name, up to 32 characters

5.2.1.32.6 signaling

Syntax

signaling

Context

config>mobile>pdn>s4

Description

This command enables the context to configure signaling for an S4 interface.



NOTE:

This command requires a PDN shutdown.

5.2.1.32.7 gtp-c

Syntax

gtp-c

Context

config>mobile>pdn>s4>signaling

Description

This command enables the context to configure GTP-C signaling for the s4 interface.



NOTE:

This command requires a PDN shutdown.

5.2.1.32.8 **ddn-damp-timer**

Syntax

```
ddn-damp-timer ddn-damp-timer  
no ddn-damp-timer
```

Context

```
config>mobile>pdn>s4>signaling>gtp-c
```

Description

This command configures the DDN dampening timer interval.

The no form of this command removes the dampening timer from the configuration.



NOTE:

This command requires a PDN shutdown.

Parameters

ddn-damp-timer

specifies a dampening timer interval, in seconds

Values: 0 | 10 to 3600

Default: 0

5.2.1.32.9 **ddn-extended-buffering-time**

Syntax

```
ddn-extended-buffering-time ddn-extended-buffering-time  
no ddn-extended-buffering-time
```

Context

```
config>mobile>pdn>s4>signaling>gtp-c
```

Description

This command defines the maximum time for the extended buffering. If the value received from the SGSN is less than the configured value, the value received from the SGSN is used, otherwise the configured value

defines how long the extended buffering is applied. If the maximum time is 0, the extended buffering is not supported. Buffering will last always at least 30 seconds.

**NOTE:**

This command requires a PDN shutdown.

Parameters

ddn-extended-buffering-time

specifies the maximum time for extended buffering, in seconds

Values: 0 | 30 to 86400

Default: 0

5.2.1.32.10 ddn-throttling-arp-threshold

Syntax

`ddn-throttling-arp-threshold ddn-throttling-arp-threshold`

`no ddn-throttling-arp-threshold`

Context

`config>mobile>pdn>s4>signaling>gtp-c`

Description

This command configures the DDN ARP threshold.

The no form of this command removes the ARP threshold from the configuration.

**NOTE:**

This command requires a PDN shutdown.

Default

`no ddn-throttling-arp`

Parameters

ddn-throttling-arp-threshold

specifies a DDN ARP threshold

Values: 0 to 15

5.2.1.32.11 gtp-u

Syntax

```
gtp-u
```

Context

```
config>mobile>pdn>s4>signaling
```

Description

This command enables the context to configure GTP-U signaling for the S4 interface.



NOTE:

This command requires a PDN shutdown.

5.2.1.32.12 delete-on-error-indication

Syntax

```
[no] delete-on-error-indication
```

Context

```
config>mobile>pdn>s4>signaling>gtp-u
```

Description

This configuration enables the deletion of the bearers when the error indication is received.

The no form of this command sets the default action, which is to release the user plane and send downlink data notification.

Default

```
no delete-on-error-indication
```

5.2.1.32.13 profile

Syntax

```
profile profile-name  
no profile
```

Context

```
config>mobile>pdn>s4>signaling>gtp-c  
config>mobile>pdn>s4>signaling>gtp-u
```

Description

This command specifies either the GTP-C or GTP-U profile name. For example, for GTP-C, this command applies a GTP-C profile for the interface. The profiles provide signaling related configuration.

The no form of this command removes the profile name from the configuration.

**NOTE:**

This command requires a PDN shutdown.

Default

no profile

Parameters

profile-name

specifies a profile name, up to 32 characters

5.2.1.32.14 sgsn-group

Syntax

```
[no] sgsn-group SGSN-group-id
```

Context

```
config>mobile>pdn>s4
```

Description

This command enables or disables the SGSN groups for the S4 interface. Multiple SGSN groups must not have any similarities in a peer list under the same S4 interface.

Parameters

SGSN-group-id

specifies the group name ID

Values: 1 to 65535

5.2.1.33 S5 and S8 interface PDN command descriptions

5.2.1.33.1 s5

Syntax

s5 [*sig-ref-point-name*] [group-specific]

no s5 [*sig-ref-point-name*]

Context

config>mobile>pdn

Description

This command enables the context to configure the S5 interface. S5 provides user plane tunneling and tunnel management between the SGW and the PGW. It is used for SGW relocation due to UE mobility and if the SGW needs to connect to a non-collocated PGW for the required PDN connectivity. The S5 is based on GTP protocol. The PMIP variant of the S5 interface is described in TS 23.402. Redundancy support on S5 reference points must be taken into account.



NOTE:

This command requires a PDN shutdown, unless the *group-specific* parameter is specified.

Parameters

sig-ref-point-name

specifies the S5 interface name, up to 32 characters

group-specific

specifies an MG group-specific reference point interface for GTP-U redirection; it is used when associating the reference-point interface with an MG system group via an override profile and applies only to CMG

5.2.1.33.2 s8

Syntax

```
s8 [sig-ref-point-name] [group-specific]
```

```
no s8 [sig-ref-point-name]
```

Context

```
config>mobile>pdn
```

Description

This command enables the context to configure the S8 interface. The S8 interface provides user plane tunneling and tunnel management between the SGW and the PGW.



NOTE:

This command requires a PDN shutdown, unless the `group-specific` parameter is specified.

Parameters

sig-ref-point-name

specifies the S8 interface name, up to 32 characters

group-specific

specifies an MG group-specific reference point interface for GTP-U redirection; it is used when associating the reference point interface with an MG system group via an override profile and applies only to CMG

5.2.1.33.3 dual-stack-cplane-pref

Syntax

```
dual-stack-cplane-pref {ipv4 | ipv6}
```

```
no dual-stack-cplane-pref
```

Context

config>mobile>pdn>s5

config>mobile>pdn>s8

Description

This command specifies the dual stack cplane preference.



NOTE:

This command requires a PDN shutdown.

Parameters

ipv4

specifies the IPv4 address family

ipv6

specifies the IPv4 address family

Default: ipv6

5.2.1.33.4 dual-stack-uplane-pref

Syntax

dual-stack-uplane-pref {ipv4 | ipv6}

no dual-stack-uplane-pref

Context

config>mobile>pdn>s5

config>mobile>pdn>s8

Description

The command specifies the preference in a dual stack when both IPv4 and IPv6 addresses are configured for the reference point interface.



NOTE:

This command requires a PDN shutdown.

Default

The same IP version as the control plane.

Parameters

ipv4

specifies the IPv4 address family

ipv6

specifies the IPv4 address family

5.2.1.33.5 interface

Syntax

```
interface
```

Context

```
config>mobile>pdn>s5
```

```
config>mobile>pdn>s8
```

Description

The command enables the context to configure interface parameters.



NOTE:

This command requires a PDN shutdown.

5.2.1.33.6 gtp-c

Syntax

```
gtp-c [router router-instance] interface-name [interface-realm realm-name]
```

```
no gtp-c
```

Context

```
config>mobile>pdn>s5>if
```

```
config>mobile>pdn>s8>if
```

Description

This command configures the GTP-C profile parameters for an S5 or S8 interface.

The no form of this command removes the parameter from the configuration.



NOTE:

This command requires a PDN shutdown.

Parameters

router-instance

specifies the virtual router name or service ID

Values	router-name service-id	
–	router-name	Base
–	service-id	1 to 2147483647

Default: Base

interface-name

specifies an interface name, up to 32 characters

realm-name

specifies the associated realm name for the virtual routing instance that has the S5/S8 GTP-U interface on the user plane function, up to 32 characters



NOTE:

Only the following characters are permitted:

- alphabetic characters (A-Z and a-z)
- digits (0-9)
- hyphen (-)
- dot (.)

The realm name must be unique for a given virtual routing instance on the user plane and is communicated in the Network-Instance IE part of the PFCP interface session establishment/modification process.

**NOTE:**

This parameter is only used when the gateway is configured as a CP function using the `mobile>config>pdn>instance-type` command.

5.2.1.33.7 gtp-u

Syntax

```
gtp-u [router router-instance] interface-name  
gtp-u [router router-instance] interface-name [interface-realm realm-name]  
no gtp-u
```

Context

```
config>mobile>pdn>s5>if  
config>mobile>pdn>s8>if
```

Description

This command configures GTP-U interface parameters.

The no form of this command removes the parameter from the configuration.

**NOTE:**

This command requires a PDN shutdown.

Parameters

router-instance

specifies the virtual router name or service ID

Values	router-name service-id	
–	router-name	Base
–	service-id	1 to 2147483647

Default: Base

interface-name

specifies an interface name, up to 32 characters

realm-name

specifies the associated realm name for the virtual routing instance and the S5-U/S8-U interface, up to 32 characters

**NOTE:**

Only the following characters are permitted:

- alphabetic characters (A-Z and a-z)
- digits (0-9)
- hyphen (-)
- dot (.)

The user plane function uses the interface realm name to identify the correct router instance and the S5-U/S8-U interface to use, based on the Network-Instance IE in the PFCP interface rules during the PFCP interface session establishment/modification process. The realm name must be unique for a given virtual routing instance.

**NOTE:**

This parameter is only used when the gateway is configured as a UP function using the `mobile>config>pdn>instance-type` command.

5.2.1.33.8 match

Syntax

match

Context

config>mobile>pdn>s5

config>mobile>pdn>s8

Description

This command enables the context to configure a match list for the reference point.

**NOTE:**

This command requires a PDN shutdown.

5.2.1.33.9 apn-list

Syntax

`apn-list list-name`

`no apn-list`

Context

`config>mobile>pdn>s5`

Description

This command specifies an APN list to be used for the match criteria. The APN list is configured in the mobile-gateway profile list context.

The no form of this command disables the APN list matching.

**NOTE:**

This command requires a PDN shutdown.

Parameters

list-name

specifies a name for this APN list instance, up to 32 characters

5.2.1.33.10 plmn-id-list

Syntax

`plmn-id-list list-name`

`no plmn-id-list`

Context

`config>mobile>pdn>s8`

Description

This command configures a Public Land Mobile Network (PLMN) list for the reference point.

**NOTE:**

This command requires a PDN shutdown.

Parameters

list-name

specifies the name of the PLMN list, up to 32 characters

5.2.1.33.11 peer-list

Syntax

```
peer-list list-name
```

```
no peer-list
```

Context

```
config>mobile>pdn>s5
```

```
config>mobile>pdn>s8
```

Description

This command configures the trusted peer list name. The list includes peer IPv4 or IPv6 address that are trusted. Separate trusted peer-list of eNBs, MMEs, and PGWs can be created for trusted peering. If no trusted-peer list has been defined on a given protocol, the protocol will accept messages from any peer. Conversely, when a trusted-peer list is defined, only messages from peers in the given list will be accepted. The operator should be able to add new peers to a trusted-peer list at any time

The no form of this command removes the list name from the configuration.

**NOTE:**

This command requires a PDN shutdown.

Default

no peer-list

Parameters

list-name

specifies the list name, up to 32 characters

5.2.1.33.12 send-end-marker

Syntax

[no] send-end-marker

Context

config>mobile>pdn>s5

config>mobile>pdn>s8

Description

This command enables or disables sending of an end marker from the PGW in the S5/S8 interface if there is an SGW change. This command should not be enabled if all SGWs in the S5/S8 interface are not supporting the end marker from the PGW.



NOTE:

This command requires a PDN shutdown.

Default

no send-end-marker

5.2.1.33.13 signaling

Syntax

signaling

Context

config>mobile>pdn>s5

config>mobile>pdn>s8

Description

This command enables the context to configure signaling for an S5 or an S8 interface.



NOTE:

This command requires a PDN shutdown.

5.2.1.33.14 gtp-c

Syntax

gtp-c

Context

config>mobile>pdn>s5>signaling

config>mobile>pdn>s8>signaling

Description

This command enables the context to configure GTP-C signaling for an S5 or an S8 signaling interface.



NOTE:

This command requires a PDN shutdown.

5.2.1.33.15 enhanced-sgw-restoration

Syntax

enhanced-sgw-restoration *restoration-time*

no enhanced-sgw-restoration

Context

config>mobile>pdn>s5>signaling>gtp-c

config>mobile>pdn>s8>signaling>gtp-c

Description

This command enables enhanced SGW restoration based on the optional option in TS23.007, called restoration of PDN connections after an SGW failure. This procedure makes it possible to restore PDN connections affected by an SGW failure with or without a restart. Setting the *restoration-time* parameter to 0 removes the configuration.

**NOTE:**

This command requires a PDN shutdown, when used in the `config>mobile>pdn>s5>signaling>gtp-c` context.

Parameters

restoration-time

specifies the length of time in seconds that the PGW will maintain the PDN connections and be ready to restore them by having the MME select either a new SGW or the same SGW (restart)

Values: 0 | 300 to 3600

Default: 0

5.2.1.33.16 pause-charging

Syntax

`pause-charging`

Context

`config>mobile>pdn>s5>signaling>gtp-c`

`config>mobile>pdn>s8>signaling>gtp-c`

Description

This command enables the context to configure pause of charging on the gateway.

**NOTE:**

This command requires a PDN shutdown.

5.2.1.33.17 data-threshold

Syntax

```
data-threshold packets  
no data-threshold
```

Context

```
config>mobile>pdn>s5>signaling>gtp-c>pause-charging  
config>mobile>pdn>s8>signaling>gtp-c>pause-charging
```

Description

This command configures the data threshold (number of packets) per UE that the SGW uses to trigger pause of charging on the PGW. The no form of this command disables the threshold-based pause charging.



NOTE:

This command requires a PDN shutdown.

Parameters

packets

specifies the data threshold in packets, where the total buffer size on MSCP per UE is 12 packets and the trigger is on 9th

Values: 0 (disabled) 9 packets

Default: 0

5.2.1.33.18 data-threshold-hold-time

Syntax

```
data-threshold-hold-time seconds  
no data-threshold-hold-time
```

Context

```
config>mobile>pdn>s5>signaling>gtp-c>pause-charging  
config>mobile>pdn>s8>signaling>gtp-c>pause-charging
```


Description

This command configures the number of seconds to wait before sending a pause of charging message to the PGW after packet drop is observed. Pause charging can lead to excessive signaling if drops are observed across many UEs. The hold timer start and stop records can be viewed for the gateway using the `show mobile-gateway PDN statistics` command. The `no` form of command disables the hold timer for initiating pause charging.

**NOTE:**

This command requires a PDN shutdown.

Parameters

seconds

specifies the wait time, in seconds

Values: 10 to 60 (0 disabled)

Default: 10

5.2.1.33.19 pgw-pause-charging

Syntax

```
pgw-pause-charging {enable | disable}
```

Context

```
config>mobile>pdn>s5>signaling>gtp-c>pause-charging
```

```
config>mobile>pdn>s8>signaling>gtp-c>pause-charging
```

Description

This command enables or disables pause of charging on the PGW when excessive downlink traffic is dropped because the UE is unreachable.

If a pause of charging indication is received from the SGW, the PGW stops accounting traffic to avoid charging packets that are dropped on SGW. If pause of charging is disabled on the PGW, the PDN Pause Support Indication flag in the Indication IE is set to 0 to indicate to the SGW that the PGW does not support pause of charging for PDN. The SGW will not generate further pause of charging messages toward PGW.

**NOTE:**

This command requires a PDN shutdown.

Parameters

enable

keyword to enable pause of charging on the PGW

disable

keyword to disable pause of charging on the PGW

Default: enable

5.2.1.33.20 radio-loss

Syntax

```
radio-loss {enable | disable}
```

Context

```
config>mobile>pdn>s5>signaling>gtp-c>pause-charging
```

```
config>mobile>pdn>s8>signaling>gtp-c>pause-charging
```

Description

This command enables support for pause of charging when there is an abnormal loss of radio resource.



NOTE:

This command requires a PDN shutdown.

Parameters

enable

keyword to enable support for pause of charging due to abnormal loss of radio resource

disable

keyword to disable support for pause of charging due to abnormal loss of radio resource

Default: enable

5.2.1.33.21 profile

Syntax

```
profile profile-name
```

```
no profile
```

Context

```
config>mobile>pdn>s5>signaling>gtp-c
config>mobile>pdn>s5>signaling>gtp-u
config>mobile>pdn>s8>signaling>gtp-c
config>mobile>pdn>s8>signaling>gtp-u
```

Description

This command specifies the profile name. For example, in the GTP-C context, this command applies a GTP-C profile for the interface. The profiles provide signaling related configuration.

The no form of this command removes the profile name from the configuration.



NOTE:

This command requires a PDN shutdown.

Default

no profile

Parameters

profile-name

specifies a profile name, up to 32 characters

5.2.1.33.22 gtp-u

Syntax

```
gtp-u
```

Context

```
config>mobile>pdn>s5
config>mobile>pdn>s8
```

Description

This command enables the context to configure interface parameters for GTP-U signaling for an s5 or s8 interface.

**NOTE:**

This command requires a PDN shutdown.

5.2.1.34 S6b interface PDN command descriptions

5.2.1.34.1 s6b

Syntax

s6b

Context

config>mobile>pdn

Description

This command enables the context to configure an s6b interface.

5.2.1.34.2 alt-peer-rar-asr

Syntax

[no] alt-peer-rar-asr

Context

config>mobile>pdn>s6b

Description

This command enables the gateway to accept RAR or ASR messages from an AAA other than the primary or secondary AAA. When this command is enabled, an RAA message is sent back to the peer that initiated the RAR. The IP-CAN session will continue to use the primary or secondary AAA for the S6b session (that is, any AAR-U or STR for the session will be sent to the destination peer that the last AAA was received from).

The no form of this command reverts to the default.

Default

no alt-peer-rar-asr

5.2.1.34.3 application-id

Syntax

```
application-id value  
no application-id
```

Context

```
config>mobile>pdn>s6b
```

Description

This command configures the Auth-Application-ID that will be sent in each message over S6b interface.

**NOTE:**

This command requires a PDN shutdown.

Parameters

value

specifies the application ID number
Default: 16777272
Values: 0 to 4294967295

5.2.1.34.4 auth-mode

Syntax

```
auth-mode [enable-non3gpp-only | enable-all]  
no auth-mode
```

Context

```
config>mobile>pdn>s6b
```

Description

This command configures the authorization mode of S6b for non-3GPP only or for all the sessions.

**NOTE:**

This command requires a PDN shutdown.

Parameters**enable-non3gpp-only**

S6b with authorization mode per TS29.273

enable-all

S6b with authorization mode per TS29.273 and TS29.061

Default: enable-non3gpp-only

5.2.1.34.5 interface

Syntax

interface

Context

config>mobile>pdn>s6b

Description

This command enables the context to configure the Diameter parameters for the interface.

**NOTE:**

This command requires a PDN shutdown.

5.2.1.34.6 diameter

Syntax

diameter [router *router-instance*] *interface-name*

no diameter

Context

config>mobile>pdn>s6b>interface

Description

This command configures the Diameter parameters for the interface.



NOTE:

This command requires a PDN shutdown.

Parameters

router-instance

specifies the router instance

Values	router-name service-id	
–	router-name	Base
–	service-id	1 to 2147483647

Default: Base

interface-name

specifies an existing interface name, up to 32 characters

5.2.1.34.7 node-identifier-list

Syntax

```
node-identifier-list list-name  
no node-identifier-list
```

Context

```
config>mobile>pdn>s6b
```

Description

This command configures the PGW to use the AAA node identifier received from the ePDG in CSRequest to connect to the same AAA that the ePDG has used for the UE/subscriber authentication and authorization.The no form of this command removes the association to the configured list and disables the functionality.

Parameters

list-name

specifies the list, configured using the `configure>mobile-gateway>profile>list>aaa-host-realm` command, that includes all the hosts and realms correlated to the `diameter-peer-group-list` command, up to 32 characters.

5.2.1.34.8 peer-group

Syntax

`peer-group group-id pri-dia-peer-profile profile-name sec-dia-peer-profile profile-name`

`no peer-group`

Context

`config>mobile>pdn>s6b`

Description

This command configures the list peer groups of primary Diameter and secondary peers for the S6b interface configured on the gateway level.



NOTE:

This command requires a PDN shutdown.

Parameters

profile-name

specifies the name of the primary Diameter peer profile, up to 32 characters

`sec-dia-peer-profile profile-name`

specifies the name of the primary Diameter peer profile, up to 32 characters

5.2.1.34.9 relax-aar-dest-realm

Syntax

`[no] relax-aar-dest-realm`

Context

config>mobile>pdn>s6b

Description

This command enables the S6b AA-Answer message to contain a different origin realm value than the one that PGW used as Destination Realm in the AA-Request message.



NOTE:

This command is only for lab testing activities.

The no form of this command disables this functionality.

Default

no relax-aar-dest-realm

5.2.1.34.10 retry-count

Syntax

retry-count *value*

no retry-count

Context

config>mobile>pdn>s6b

Description

This command specifies the number of times the system will retry to send an s6b message before declaring a failed attempt.

Parameters

value

specifies the number of times the system will retry

Values: 0 to 8

Default: 3

5.2.1.34.11 s6b-profile

Syntax

```
s6b-profile name  
no s6b-profile
```

Context

```
config>mobile>pdn>s6b
```

Description

This command specifies the s6b profile to associate with this mobile gateway.

Parameters

name
specifies the s6b profile name, up to 32 characters

5.2.1.34.12 subscription-id-override

Syntax

```
[no] subscription-id-override
```

Context

```
config>mobile>pdn>s6b
```

Description

When the S6b supported-vendor is set to custom-1, the PGW will use the Subscription-Id as the MSISDN for eHRPD attach/handover cases. This command causes the PGW to use Subscription-Id as the MSISDN for all attach types including LTE and WLAN. This command must only be used with supported-vendor custom-1.

The no form of this command returns to the default, which uses the GTP provided MSISDN.

Default

GTP provided MSISDN

5.2.1.34.13 supported-vendor

Syntax

```
supported-vendor custom-1
no supported-vendor
```

Context

```
config>mobile>pdn>s6b
```

Description

This command enables the specified supported vendor.



NOTE:

This command requires a PDN shutdown.

Default

```
no supported-vendor
```

Parameters

```
custom-1
    specifies the custom-1 customer
```

5.2.1.34.14 transaction-timer

Syntax

```
transaction-timer {[value] | [timeout-msec value]}
no transaction-timer
```

Context

```
config>mobile>pdn>s6b
```

Description

This command sets the maximum amount of time the node waits for a Diameter peer to respond before trying another peer. The value can be expressed in seconds or milliseconds. Setting the timer in milliseconds provides better timeout granularity.

The no form of this command removes the configuration.

Parameters

value

specifies the maximum amount of time, in seconds, the node waits for a Diameter peer to respond before trying another peer

Values: 1 to 180 (increments of one second)

Default: 5

timeout-msec value

specifies the maximum amount of time, in milliseconds, the node waits for a Diameter peer to respond before trying another peer

Values: 1000 to 180000 (increments of 100 ms)

Default: 5000

5.2.1.34.15 username-leading-digit

Syntax

```
username-leading-digit [digit]
```

```
no username-leading-digit
```

Context

```
config>mobile>pdn>s6b
```

Description

This command configures a leading digit in the User-Name AVP in the AA Request messages, when IMSI is used as username in the root NAI.

The no form of this command removes the leading digit from the User-Name AVP in the AA Request messages.

Default

```
no username-leading-digit
```

Parameters

digit

specifies the leading digit

Values: 0 to 9

5.2.1.35 SBA client services PDN command descriptions

5.2.1.35.1 sba-client-services

Syntax

```
sba-client-services
```

Context

```
config>mobile>pdn
```

Description

This command enables the context to configure the client-side options of CMG when the SMF uses SBA services as an NF consumer.

5.2.1.35.2 addressing-priority

Syntax

```
addressing-priority [fqdn | ip-endpoint | ip-endpoint-fallback]  
no addressing-priority
```

Context

```
config>mobile>pdn>sba-client-services
```

Description

This command configures the addressing method used for accessing the discovered services of peer NF producers when the discovered NFProfiles contain both FQDN and IP endpoints.

The no form of this command reverts to the default (IP endpoint).

Parameters

fqdn

specifies to use the FQDN, if present; if not, the IP endpoint is used

ip-endpoint

specifies to always use the IP endpoint

ip-endpoint-fallback

specifies to use the FQDN, if present; if not, the IP endpoint is used

If the FQDN is used and the resolved IP address is not accessible, the IP endpoint is used as a fallback

Default: ip-endpoint

5.2.1.35.3 amf-client

Syntax

```
amf-client
```

Context

```
config>mobile>pdn>sba-client-services
```

Description

This command enables the context to configure the SMF client-side options when the SMF uses AMF services.

5.2.1.35.4 namf-comm

Syntax

```
[no] namf-comm service-instance
```

Context

```
config>mobile>pdn>sba-client-services>amf-client
```

Description

This command configures an instance of the Namf_Communication service. At least one service instance must be configured to allow peering with the AMF.

The no form of this command removes the service instance from the configuration.

Parameters

service-instance

specifies the service instance name, up to 32 characters

5.2.1.35.5 callback-type-preference

Syntax

```
callback-type-preference type  
no callback-type-preference
```

Context

```
config>mobile>pdn>sba-client-services>amf-client>namf-comm  
config>mobile>pdn>sba-client-services>amf-client>namf-evts  
config>mobile>pdn>sba-client-services>nrf-client>nnrf-nfm  
config>mobile>pdn>sba-client-services>pcf-client>npcf-smpolicycontrol  
config>mobile>pdn>sba-client-services>smf-client>nsmf-pdusession  
config>mobile>pdn>sba-client-services>udm-client>nudm-sdm  
config>mobile>pdn>sba-client-services>udm-client>nudm-uecm
```

Description

This command defines the URI host type used in the callback URI.

The no form of this command reverts to the default.

Parameters

type

specifies the type of the URI host that contains an IPv4 or an IPv6 address

Values: ipv4 | ipv6

Default: ipv4

5.2.1.35.6 fqdn

Syntax

```
fqdn fqdn-sni-string  
no fqdn
```

Context

```
config>mobile>pdn>sba-client-services
```

```
config>mobile>pdn>sba-client-services>amf-client>namf-comm
config>mobile>pdn>sba-client-services>amf-client>namf-evts
config>mobile>pdn>sba-client-services>chf-client>nchf-convergedcharging
config>mobile>pdn>sba-client-services>nrf-client>nnrf-disc
config>mobile>pdn>sba-client-services>nrf-client>nnrf-nfm
config>mobile>pdn>sba-client-services>nrf-client>nnrf-oauth2
config>mobile>pdn>sba-client-services>nrf-client>nnrf-oauth2
config>mobile>pdn>sba-client-services>pcf-client>npcf-smpolicycontrol
config>mobile>pdn>sba-client-services>scp-client>scp-service
config>mobile>pdn>sba-client-services>sepp-client>sepp
config>mobile>pdn>sba-client-services>smf-client>nsmf-pdusession
config>mobile>pdn>sba-client-services>udm-client>nudm-sdm
config>mobile>pdn>sba-client-services>udm-client>nudm-uecm
```

Description

This command configures the FQDN which is used by CMG/SMF, when acting as a client in the TLS connections during notification requests from the NF producers. It is used by the producers as the SNI FQDN in the TLS connection, when these send notifications. A single FQDN can be configured for all client services in the `config>mobile>pdn>sba-client-services` context or per client service instance.

The `no` form of this command removes the FQDN from the configuration.

Parameters

fqdn-sni-string

specifies the FQDN name string, up to 255 characters

5.2.1.35.7 http2-profile

Syntax

```
http2-profile http2-profile-name
no http2-profile
```

Context

```
config>mobile>pdn>sba-client-services>amf-client>namf-comm
```


Description

This command associates an HTTP/2 profile, configured using the `config>mobile>profile>http2` command, to the Namf_Communication services.

The `no` form of this command removes the HTTP/2 profile from the configuration.

Parameters

http2-profile-name

specifies the name of the HTTP/2 profile, up to 32 characters

5.2.1.35.8 interface

Syntax

```
interface interface-name [router router-instance]
no interface
```

Context

```
config>mobile>pdn>sba-client-services>amf-client>namf-comm
config>mobile>pdn>sba-client-services>amf-client>namf-evts
config>mobile>pdn>sba-client-services>chf-client>nchf-convergedcharging
config>mobile>pdn>sba-client-services>nrf-client>nnrf-disc
config>mobile>pdn>sba-client-services>nrf-client>nnrf-nfm
config>mobile>pdn>sba-client-services>nrf-client>nnrf-oauth2
config>mobile>pdn>sba-client-services>pcf-client>npcf-smpolicycontrol
config>mobile>pdn>sba-client-services>scp-client>scp-service
config>mobile>pdn>sba-client-services>sepp-client>sepp
config>mobile>pdn>sba-client-services>smf-client>nsmf-pdusession
config>mobile>pdn>sba-client-services>udm-client>nudm-sdm
config>mobile>pdn>sba-client-services>udm-client>nudm-uecm
```

Description

This command configures the interface in which the HTTP/2 connections are established.

The `no` form of this command removes the interface from the configuration.

Parameters

- interface-name**
specifies an existing interface name and must start with a letter, up to 32 characters
- router-instance**
specifies the router instance

Values	router-name service-id	
–	router-name	Base
–	service-id	1 to 2147483647

Default: Base

5.2.1.35.9 mode

Syntax

- mode *mode*
- no mode

Context

- config>mobile>pdn>sba-client-services>amf-client>namf-comm
- config>mobile>pdn>sba-client-services>amf-client>namf-evts
- config>mobile>pdn>sba-client-services>chf-client>nchf-convergedcharging
- config>mobile>pdn>sba-client-services>pcf-client>npcf-smpolicycontrol
- config>mobile>pdn>sba-client-services>smf-client>nsmf-pdusession
- config>mobile>pdn>sba-client-services>udm-client>nudm-sdm
- config>mobile>pdn>sba-client-services>udm-client>nudm-uecm

Description

This command configures the used communication mode.



NOTE:

If the configuration is changed, the new requests use the new communication mode. The ongoing requests are processed according to the communication mode that was used when the request was transmitted.

Parameters

mode

specifies the communication mode

Values	direct	specifies that direct communication is used
–	indirect	specifies that indirect communication without delegated discovery is used
–	delegate	specifies that indirect communication with delegated discovery is enabled

Default: direct

5.2.1.35.10 n1-profile

Syntax

n1-profile *profile-name*

no n1-profile

Context

config>mobile>pdn>sba-client-services>amf-client>namf-comm

Description

This command associates an N1 profile with the service instance. The N1 profile is configured using the config>mobile>profile>n1-profile command.

Parameters

profile-name

specifies the name of the N1 profile, up to 32 characters

5.2.1.35.11 nf-id-list

Syntax

```
nf-id-list list-name  
no nf-id-list
```

Context

```
config>mobile>pdn>sba-client-services>amf-client>namf-comm  
config>mobile>pdn>sba-client-services>amf-client>namf-evts  
config>mobile>pdn>sba-client-services>chf-client>nchf-convergedcharging  
config>mobile>pdn>sba-client-services>nrf-client>nnrf-disc  
config>mobile>pdn>sba-client-services>nrf-client>nnrf-nfm  
config>mobile>pdn>sba-client-services>nrf-client>nnrf-oauth2  
config>mobile>pdn>sba-client-services>sepp-client>sepp  
config>mobile>pdn>sba-client-services>pcf-client>npcf-smpolicycontrol  
config>mobile>pdn>sba-client-services>scp-client>scp-service  
config>mobile>pdn>sba-client-services>smf-client>nsmf-pdusession  
config>mobile>pdn>sba-client-services>udm-client>nudm-sdm  
config>mobile>pdn>sba-client-services>udm-client>nudm-uecm
```

Description

This command defines the remote NF ID list to use, if the NRF is not used for the service discovery. The NF ID list is configured using the `config>mobile>profile>list>nf-id-list` command.

Parameters

list-name

specifies the list name, up to 32 characters

5.2.1.35.12 oauth2-profile

Syntax

```
oauth2-profile profile-name  
no oauth2-profile
```

Context

```
config>mobile>pdn>sba-client-services>amf-client>namf-comm  
config>mobile>pdn>sba-client-services>amf-client>namf-evts  
config>mobile>pdn>sba-client-services>chf-client>nchf-convergedcharging  
config>mobile>pdn>sba-client-services>pcf-client>npcf-smpolicycontrol  
config>mobile>pdn>sba-client-services>smf-client>nsmf-pdusession  
config>mobile>pdn>sba-client-services>udm-client>nudm-sdm  
config>mobile>pdn>sba-client-services>udm-client>nudm-uecm
```

Description

This command configures OAuth2.0 authorization for the requests sent to the specified service instance and defines the profile with OAuth2.0 behavioral aspects for the client side. A default profile can be configured.

When OAuth2.0 is configured, the SMF or BSF uses access tokens in the requests sent. If the SMF/BSF does not possess an access token, it initiates an access token GET request toward the NRF to get a new access token before making a service request.

The no form of this command disables the OAuth2.0 authorization in the specified client context.

Default

```
no oauth2-profile
```

Parameters

profile-name
specifies the client profile name, up to 32 characters

5.2.1.35.13 paging-damp-timer

Syntax

```
paging-damp-timer paging-damp-timer
```

```
no paging-damp-timer
```

Context

```
config>mobile>pdn>sba-client-services>amf-client>namf-comm
```

Description

This command defines the delay before the SMF initiates paging if the latest paging attempt failed.

The no form of this command reverts to the default value.

Parameters

paging-damp-timer

specifies the dampening interval, in seconds

Values: 0 to 3600

Default: 0 (disabled)

5.2.1.35.14 plmn-list

Syntax

```
plmn-list list-name
```

```
no plmn-list
```

Context

```
config>mobile>pdn>sba-client-services>sepp-client>sepp
```

Description

This command configures a PLMN list for SEPP selection.

The no form of the command removes the PLMN list from the configuration.

Default

```
no plmn-list
```

Parameters

list-name

specifies the PLMN list name, up to 32 characters

5.2.1.35.15 service-interface

Syntax

```
service-interface ip-address | fqdn-name [port port] [scheme scheme]  
no service-interface
```

Context

```
config>mobile>pdn>sba-client-services>amf-client>namf-comm  
config>mobile>pdn>sba-client-services>amf-client>namf-evts  
config>mobile>pdn>sba-client-services>chf-client>nchf-convergedcharging  
config>mobile>pdn>sba-client-services>nrf-client>nnrf-disc  
config>mobile>pdn>sba-client-services>nrf-client>nnrf-nfm  
config>mobile>pdn>sba-client-services>nrf-client>nnrf-oauth2  
config>mobile>pdn>sba-client-services>pcf-client>npcf-smpolicycontrol  
config>mobile>pdn>sba-client-services>sepp-client>sepp  
config>mobile>pdn>sba-client-services>scp-client>scp-service  
config>mobile>pdn>sba-client-services>smf-client>nsmf-pdusession  
config>mobile>pdn>sba-client-services>udm-client>nudm-sdm  
config>mobile>pdn>sba-client-services>udm-client>nudm-uecm  
config>mobile>pdn>sba-server-services>nbsf-management  
config>mobile>pdn>sba-server-services>nsmf-event-exposure  
config>mobile>pdn>sba-server-services>nsmf-pdusession
```

Description

This command configures the service routing interface, in the default Linux routing stack, using K8s CITM and, or NGINX HTTP/2 connections and Istio gateway load balancing.

**NOTE:**

When you configure this command in a context, the corresponding interface command under this context is not available.

Parameters

ip-address | *fqdn-name*

specifies the IP address and the FQDN name, up to 128 characters

port

specifies the port number

Values: 1 to 65535

Default: 80

scheme

specifies the use of HTTP or HTTPS (TLS)

Values: http | https

Default: http

5.2.1.35.16 immediate-report

Syntax

```
immediate-report [skip-get]
```

```
no immediate-report
```

Context

```
config>mobile>pdn>sba-client-services>udm-client>nudm-sdm>supported-features
```

Description

This command enables the optional feature of the nudm-sdm service to request an immediate report when a subscription is created in the UDM. You can use the optional `skip-get` parameter to skip the GET request when the PDU session is established.

The no form of this command reverts to default (disabled).

Default

```
no immediate-report
```

Parameters

skip-get

keyword that specifies that no GET request is sent when the PDU session is established

5.2.1.35.17 udm-sdm-profile

Syntax

```
udm-sdm-profile profile-name  
no udm-sdm-profile
```

Context

```
config>mobile>pdn>sba-client-services>udm-client>nudm-sdm
```

Description

This command associates a UDM SDM profile (configured in the config>mobile>profile>udm-sdm-profile context) for the Nudm_SubscriberDataManagement service.

The no form of this command removes the profile from the Nudm_SubscriberDataManagement service.

Default

```
no no udm-sdm-profile
```

Parameters

profile-name

specifies the profile name, up to 32 characters

5.2.1.35.18 udm-uecm-profile

Syntax

```
udm-uecm-profile profile-name  
no udm-uecm-profile
```

Context

```
config>mobile>pdn>sba-client-services>udm-client>nudm-uecm
```

Description

This command associates a UDM SDM profile (configured in the config>mobile>profile>udm-uecm-profile context) for the Nudm_UEContextManagement service.

The no form of this command removes the profile from the Nudm_UEContextManagement service.

Default

no no udm-uecm-profile

Parameters

profile-name

specifies the profile name, up to 32 characters

5.2.1.35.19 namf-evts

Syntax

[no] namf-evts *service-instance*

Context

config>mobile>pdn>sba-client-services>amf-client

Description

This command configures an instance of the Namf_EventExposure service. At least one service instance must be configured to enable usage of the AMF event exposure service.

The no form of this command removes the service instance from the configuration.

Parameters

service-instance

specifies the service instance name, up to 32 characters

5.2.1.35.20 http2-profile

Syntax

http2-profile *http2-profile-name*

no http2-profile

Context

config>mobile>pdn>sba-client-services>amf-client>namf-evts

Description

This command associates an HTTP/2 profile, configured using the `config>mobile>profile>http2` command, to the `Namf_EventExposure` service.

The `no` form of this command removes the HTTP/2 profile from the configuration.

Parameters

http2-profile-name

specifies the name of the HTTP/2 profile, up to 32 characters

5.2.1.35.21 chf-client

Syntax

```
chf-client
```

Context

```
config>mobile>pdn>sba-client-services
```

Description

This command enables the context to configure the SMF client-side options when the SMF uses CHF services.

5.2.1.35.22 nchf-convergedcharging

Syntax

```
[no] nchf-convergedcharging instance-name
```

Context

```
config>mobile>pdn>sba-client-services>chf-client
```

Description

This command configures an instance of the `Nchf_ConvergedCharging` service. At least one service instance must be configured to enable peering with the CHF.

Parameters

instance-name

specifies the service instance name, up to 32 characters

5.2.1.35.23 **alternative-api-version**

Syntax

`api-version version`

`no api-version`

Context

`config>mobile>pdn>sba-client-services>chf-client>nchf-convergedcharging`

Description

This command specifies the used major API version of the nchf-convergedcharging service if the CHF does not support the preferred major API version (defined by the `api-version` command) but supports the other possible major API versions.

The no form of this command reverts to the default where no alternative API version is defined.

Default

no alternative-api-version

Parameters

version

specifies the major API version

Values: 0 to 255

Default: 0 (no alternative API version is defined)

5.2.1.35.24 **api-version**

Syntax

`api-version version [minor minor] [patch patch]`

`no api-version`

Context

`config>mobile>pdn>sba-client-services>chf-client>nchf-convergedcharging`

Description

This command specifies the API version of the remote CHF peer. It is relevant in cases where the related API supports multiple major versions, or the minor and patch versions have impact to the API usage. By default, version 3 (release 16) is used.

The no form of this command reverts to the default value.

Default

no api-version

Parameters

version

specifies the major API version

Values: 0 to 255

Default: 0 (highest supported API version)

minor

specifies the minor API version

Values: 0 to 255

Default: 0

patch

specifies the patch API version

Values: 0 to 255

Default: 0

5.2.1.35.25 http2-profile

Syntax

http2-profile *http2-profile-name*

no http2-profile

Context

config>mobile>pdn>sba-client-services>chf-client>nchf-convergedcharging

Description

This command associates an HTTP/2 profile, configured using the config>mobile>profile>http2 command, to the Nchf_ConvergedCharging service.

The no form of this command removes the HTTP/2 profile from the configuration.

Parameters

http2-profile-name

specifies the name of the HTTP/2 profile, up to 32 characters

5.2.1.35.26 converged-interface-support

Syntax

```
converged-interface-support
```

Context

```
config>mobile>pdn>sba-client-services
```

Description

This command enables the context to configure the 5G SBI interface support for UEs without 5GS NAS support.

5.2.1.35.27 converged-n10

Syntax

```
converged-n10 {enable | disable | 5gcnri}
```

```
no converged-n10
```

Context

```
config>mobile>pdn>sba-client-services>converged-interface-support
```

Description

This command configures the use of the 5G service-based N10 interface by the PGW-C/SMF for the UEs without 5GS NAS support. When set to *5gcnri* (5GC Not Restricted Indication), the N10 interface is enabled for the UEs without 5GS NAS support but the access to the 5GC is not restricted by the user subscription.

The no form of this command reverts to the default (*disable*).

Parameters

enable

enables the use of the 5G service-based N10 interface by the PGW-C/SMF for the UEs without 5GS NAS support

disable

disables the use of the 5G service-based N10 interface by the PGW-C/SMF for the UEs without 5GS NAS support

Default: disable

5gcnri

enables the use of the 5G service-based N10 interface by the PGW-C/SMF for the UEs without 5GS NAS support but the access to the 5GC is not restricted by the user subscription

5.2.1.35.28 converged-n40

Syntax

converged-n40 {enable | disable}

no converged-n40

Context

config>mobile>pdn>sba-client-services>converged-interface-support

Description

This command configures the use of the 5G service-based N40 interface by the PGW-C/SMF for the UEs without 5GS NAS support.

The no form of this command reverts to the default (disable).

Parameters

enable

enables the use of the 5G service-based N40 interface by the PGW-C/SMF for the UEs without 5GS NAS support

disable

disables the use of the 5G service-based N40 interface by the PGW-C/SMF for the UEs without 5GS NAS support

Default: disable

5.2.1.35.29 converged-n40-gngp

Syntax

```
converged-n40-gngp {enable | disable}  
no converged-n40-gngp
```

Context

```
config>mobile>pdn>sba-client-services>converged-interface-support
```

Description

This command configures the use of the 5G service-based N40 interface by the PGW-C/SMF for the UEs in case of 2G/3G access. When enabled, the PGW-C/SMF uses the 5G service-based N40 interface for the UEs in case of 2G/3G access.

The no form of this command reverts to the default (disable).

Parameters

enable

enables the use of the 5G service-based N40 interface by the PGW-C/SMF for the UEs in case of 2G/3G access

disable

disables the use of the 5G service-based N40 interface by the PGW-C/SMF for the UEs in case of 2G/3G access

Default: disable

5.2.1.35.30 converged-n7

Syntax

```
converged-n7 {enable | disable}  
no converged-n7
```

Context

```
config>mobile>pdn>sba-client-services>converged-interface-support
```

Description

This command configures the use of the 5G service-based N7 interface by the PGW-C/SMF for the UEs without 5GS NAS support.

The no form of this command reverts to the default (disable).

Parameters

enable

enables the use of the 5G service-based N7 interface by the PGW-C/SMF for the UEs without 5GS NAS support

disable

disables the use of the 5G service-based N7 interface by the PGW-C/SMF for the UEs without 5GS NAS support

Default: disable

5.2.1.35.31 converged-n7-gngp

Syntax

```
converged-n7-gngp {enable | disable}
```

```
no converged-n7-gngp
```

Context

```
config>mobile>pdn>sba-client-services>converged-interface-support
```

Description

This command configures the use of the 5G service-based N7 interface by the PGW-C/SMF for the UEs in case of 2G/3G access. When enabled, the PGW-C/SMF uses the 5G service-based N7 interface for the UEs in case of 2G/3G access.

The no form of this command reverts to the default (disable).

Parameters

enable

enables the use of the 5G service-based N7 interface by the PGW-C/SMF for the UEs in case of 2G/3G access

disable

disables the use of the 5G service-based N7 interface by the PGW-C/SMF for the UEs in case of 2G/3G access

Default: disable

5.2.1.35.32 pdu-session-id-base

Syntax

```
pdu-session-id-base value  
no pdu-session-id-base
```

Context

```
config>mobile>pdn>sba-client-services>converged-interface-support
```

Description

This command specifies the PDU session ID base value to be used for the PDU session ID generated by the PGW-C/SMF when it uses the 5G service-based interfaces for the UEs without 5GS NAS support.

When the value is set to 0, the system defines the PDU session ID base value.

The no form of this command reverts to the default.



NOTE:

This command must be used for troubleshooting purposes only and it is recommended to let the system define the PDU session ID base value.

Parameters

value

specifies the value of the PDU session ID base

Values: 0 to 95

Default: 0

5.2.1.35.33 http2-profile

Syntax

```
http2-profile http2-profile-name  
no http2-profile
```

Context

```
config>mobile>pdn>sba-client-services
```

Description

This command associates an HTTP/2 profile, configured using the `config>mobile>profile>http2` command, to all SBA client services.

The no form of this command removes the HTTP/2 profile from the configuration.

Parameters

http2-profile-name

specifies the name of the HTTP/2 profile, up to 32 characters

5.2.1.35.34 nrf-client

Syntax

```
nrf-client
```

Context

```
config>mobile>pdn>sba-client-services
```

Description

This command enables the context to configure the SMF or UPF client-side options when the SMF or UPF uses NRF services.

5.2.1.35.35 nnrf-disc

Syntax

```
[no] nnrf-disc service-instance
```

Context

```
config>mobile>pdn>sba-client-services>nrf-client
```

Description

This command configures an instance of the Nnrf_NFDiscovery service.

Parameters

service-instance

specifies the service instance name, up to 32 characters

5.2.1.35.36 blocklist-duration

Syntax

```
blocklist-duration seconds  
no blocklist-duration
```

Context

```
config>mobile>pdn>sba-client-services>nrf-client>nnrf-disc
```

Description

This command configures the blocklist duration of NRF peers for the Nnrf_NFDDiscovery service operation. When an NRF peer is blocklisted, CMG attempts to reconnect to this NRF peer after the configured time expires.



NOTICE:

When the duration of the block list timer is configured with high values and all NRF peers are blocklisted, for example as a result of a short connectivity outage, there will not be any peers available until the blocklist timer expires. Nokia recommends to configure low values for the blocklist timer.

The no form of this command reverts to the default value.

Parameters

seconds

specifies the time, in seconds, between successive retries

Values: 0 to 3600

Default: 5

5.2.1.35.37 chf-selection-options

Syntax

```
chf-selection-options
```

Context

```
config>mobile>pdn>sba-client-services>nrf-client>nnrf-disc
```

Description

This command enables the context to configure the CHF selection options via the NRF.

5.2.1.35.38 **exclude-gpsi**

Syntax

```
[no] exclude-gpsi [vsmf-only]
```

Context

```
config>mobile>pdn>sba-client-services>nrf-client>nnrf-disc>chf-selection-options
```

Description

This command excludes GPSI from the NRF query for the CHF discovery.

The `vsmf-only` optional parameter excludes the GPSI only if the V-SMF is performing the discovery.

The `no` form of this command includes the GPSI in the CHF selection query.

Default

`no exclude-gpsi`

Parameters

vsmf-only

the V-SMF excludes the GPSI from the discovery

5.2.1.35.39 **exclude-supi**

Syntax

```
[no] exclude-supi [vsmf-only]
```

Context

```
config>mobile>pdn>sba-client-services>nrf-client>nnrf-disc>chf-selection-options
```

Description

This command excludes SUPI from the NRF query for the CHF discovery.

The `vsmf-only` optional parameter excludes the SUPI only if the V-SMF is performing the discovery.

The no form of this command includes the SUPI in the CHF selection query.

Default

no exclude-supi

Parameters

vsmf-only

the V-SMF excludes the SUPI from the discovery

5.2.1.35.40 include-preferred-api-version

Syntax

[no] include-preferred-api-version

Context

config>mobile>pdn>sba-client-services>nrf-client>nnrf-disc>chf-selection-options

config>mobile>pdn>sba-client-services>nrf-client>nnrf-disc>udm-selection-options

Description

This command enables the preferred-api-version query parameter in the NF discovery.

The no form of this command reverts to the default where the preferred-api-version query parameter is not used in the NF discovery.

Default

no include-preferred-api-version

5.2.1.35.41 exclude-snssais

Syntax

[no] exclude-snssais

Context

config>mobile>pdn>sba-client-services>nrf-client>nnrf-disc

Description

This command excludes the SNSSAI list from the query parameter lists in the NF discovery request. The no form of this command reverts to the default (the SNSSAI list is included in the NF discovery request).

Default

no exclude-snssais

5.2.1.35.42 http2-profile

Syntax

http2-profile *http2-profile-name*

no http2-profile

Context

config>mobile>pdn>sba-client-services>nrf-client>nnrf-disc

Description

This command associates an HTTP/2 profile, configured using the config>mobile>profile>http2 command, to the Nnrf_NFDiscovery service.

The no form of this command removes the HTTP/2 profile from the configuration.

Parameters

http2-profile-name

specifies the name of the HTTP/2 profile, up to 32 characters

5.2.1.35.43 mode

Syntax

mode *mode*

no mode

Context

config>mobile>pdn>sba-client-services>nrf-client>nnrf-disc

config>mobile>pdn>sba-client-services>nrf-client>nnrf-nfm

config>mobile>pdn>sba-server-services>nsmf-event-exposure

Description

This command configures the used communication mode.



NOTE:

If the configuration is changed, the new requests use the new communication mode. The ongoing requests are processed according to the communication mode used when the request was transmitted.

Parameters

mode

specifies the communication mode

Values	direct	direct communication is used
–	indirect	indirect communication without delegated discovery is used

Default: direct

5.2.1.35.44 nrf-cache-interval

Syntax

nrf-cache-interval *value*
nrf-cache-interval use-nrf-validity-period
no nrf-cache-interval

Context

config>mobile>pdn>sba-client-services>nrf-client>nnrf-disc

Description

This command configures the NRF discovery results validity period.

The no form of this command reverts to the default.

Default

nrf-cache-interval use-nrf-validity-period

Parameters

value

specifies the validity period, in hours

Values: 0 to 24

use-nrf-validity-period

specifies to use the validity period provided in the NRF discovery response

Default: use-nrf-validity-period

5.2.1.35.45 suppress-discovery

Syntax

[no] suppress-discovery {amf | chf | pcf | udm}

Context

config>mobile>pdn>sba-client-services>nrf-client>nnrf-disc

Description

This command configures an NF type to use the locally-configured NF ID lists for selecting a peer instead of using the NRF.

This configuration applies only to client services for which the mode command is configured with the `direct` option. For client services with the mode command set to `indirect` or `delegate`, the suppress discovery configuration is ignored.

The `no` form of this command configures that discovery via the NRF is used.

Parameters

amf

specifies that the AMF uses the NF ID lists for selecting a peer

chf

specifies that the CHF uses the NF ID lists for selecting a peer

pcf

specifies that the PCF uses the NF ID lists for selecting a peer

udm

specifies that the UDM uses the NF ID lists for selecting a peer

5.2.1.35.46 udm-selection-options

Syntax

```
udm-selection-options
```

Context

```
config>mobile>pdn>sba-client-services>nrf-client>nnrf-disc
```

Description

This command enables the context to configure UDM-specific discovery and selection options.

5.2.1.35.47 exclude-group-id

Syntax

```
[no] exclude-group-id [vsmf-only]
```

Context

```
config>mobile>pdn>sba-client-services>nrf-client>nnrf-disc>udm-selection-options
```

Description

This command excludes the group ID from the query parameter lists in the UDM discovery request. The no form of this command reverts to the default (group ID is included in the UDM discovery request).

The vsmf-only optional parameter excludes the group ID only if the V-SMF is performing the discovery.

Default

```
no exclude-group-id
```

Parameters

vsmf-only

the V-SMF excludes the group ID from the discovery

5.2.1.35.48 **exclude-supi**

Syntax

[no] exclude-supi [vsmf-only]

Context

config>mobile>pdn>sba-client-services>nrf-client>nnrf-disc>udm-selection-options

Description

This command excludes the SUPI from the query parameter lists in the UDM discovery request. The no form of this command reverts to the default (SUPI is included in the UDM discovery request).

The vsmf-only optional parameter excludes the SUPI only if the V-SMF is performing the discovery.

Default

no exclude-supi

Parameters

vsmf-only

the V-SMF excludes the SUPI from the discovery

5.2.1.35.49 **use-locality**

Syntax

[no] use-locality

Context

config>mobile>pdn>sba-client-services>nrf-client>nnrf-disc

Description

This command configures the discovery requests for PCF, UDM, and CHF peers to use the preferred locality. The no form of this command reverts to the default (preferred locality is not used in the discovery).



NOTE:

The NRF may adjust the priorities in the NF profiles when the preferred locality is used. The local cache contains the adjusted priorities that have been received by the NRF. If this configuration changes, Nokia recommends flushing the cache.

Default

no use-locality

5.2.1.35.50 nnrf-nfm

Syntax

[no] nnrf-disc *service-instance*

Context

config>mobile>pdn>sba-client-services>nrf-client

Description

This command configures an instance of the Nnrf_NFManagement service.

Parameters

service-instance

specifies the service instance name, up to 32 characters

5.2.1.35.51 blocklist-duration

Syntax

blocklist-duration *seconds*

no blocklist-duration

Context

config>mobile>pdn>sba-client-services>nrf-client>nnrf-nfm

Description

This command configures the blocklist duration of NRF peers for the Nnrf_NFManagement service operation. When an NRF peer is block-listed, CMG attempts to reconnect to this NRF peer after the configured time expires.

**NOTICE:**

When the duration of the blocklist timer is configured with high values and all NRF peers are blocklisted, for example as a result of a short connectivity outage, there will not be any peers available until the blocklist timer expires. Nokia recommends to configure low values for the blocklist timer.

The no form of this command reverts to the default value.

Parameters***seconds***

specifies the time, in seconds, between successive retries

Values: 0 to 3600

Default: 60

5.2.1.35.52 heartbeat-nrf-blocklist**Syntax**

```
heartbeat-nrf-blocklist retry retry-count wait-timer seconds
```

```
no heartbeat-nrf-blocklist
```

Context

```
config>mobile>pdn>sba-client-services>nrf-client>nnrf-nfm
```

Description

This command configures the heartbeat operation blocklist algorithm of NRF peers for the Nnrf_nfm service operations.

The retry keyword specifies the number of times a heartbeat is resent, when there is no response from the peer, after the timer configured using the wait-timer keyword expires.

The retry counter is also applied when specific HTTP error codes are received in the heartbeat response from the NRF. In this case, the new heartbeat is sent immediately after the response is received without waiting for the configured wait timer to expire.

For more information about the use of this command, refer to the section *NRF Failover and Multiple NRF Usage* of the *7750 SR MG and CMG 5G System Architecture and Configuration Guide*.

The no form of this command reverts to the default values.

Parameters***retry-count***

specifies the number of heartbeat retries

Values: 0 to 10

Default: 3

seconds

specifies the time, in seconds, between successive heartbeat retries

Values: 1 to 10

Default: 2

5.2.1.35.53 http2-profile

Syntax

`http2-profile http2-profile-name`

`no http2-profile`

Context

`config>mobile>pdn>sba-client-services>nrf-client>nnrf-nfm`

Description

This command associates an HTTP/2 profile, configured using the `config>mobile>profile>http2` command, to the Nnrf_NFManagement service.

The `no` form of this command removes the HTTP/2 profile from the configuration.

Parameters

http2-profile-name

specifies the name of the HTTP/2 profile, up to 32 characters

5.2.1.35.54 nfm-nrf-blocklist

Syntax

`nfm-nrf-blocklist retry retry-count wait-timer seconds`

`no nfm-nrf-blocklist`

Context

`config>mobile>pdn>sba-client-services>nrf-client>nnrf-nfm`

Description

This command configures the blocklist algorithm of NRF peers for the Nnrf_nfm service operations except the heartbeat.

The `retry` keyword specifies the number of times a request is resent, when there is no response from the peer, after the timer configured using the `wait - timer` keyword expires.

The retry counter is also applied when specific HTTP error codes are received in the heartbeat response from the NRF. In this case, a new request is sent immediately after the response is received without waiting for the configured wait timer to expire.

For more information about the use of this command, refer to the section *NRF Failover and Multiple NRF Usage* of the *CMG 5G System Architecture and Configuration Guide*.

The no form of this command reverts to the default values.

Parameters

retry-count

specifies the number of heartbeat retries

Values: 0 to 10

Default: 3

seconds

specifies the time, in seconds, between successive heartbeat retries

Values: 1 to 10

Default: 2

5.2.1.35.55 nnrf-oauth2

Syntax

```
[no] nnrf-oauth2 service-instance
```

Context

```
config>mobile>pdn>sba-client-services>nrf-client
```

Description

This command configures an instance of the Nnrf_AccessToken service. One service instance must be configured if OAuth2.0 is configured to be used when using other NF peer services.

The no form of this command removes the specified service instance from the configuration.

Parameters

service-instance

specifies the service instance name, up to 32 characters

5.2.1.35.56 http2-profile

Syntax

```
http2-profile http2-profile-name
```

```
no http2-profile
```

Context

```
config>mobile>pdn>sba-client-services>nrf-client>nnrf-oauth2
```

Description

This command associates an HTTP/2 profile, configured using the `config>mobile>profile>http2` command, to the `Nnrf_AccessToken` service.

The `no` form of this command removes the HTTP/2 profile from the configuration.

Parameters

http2-profile-name

specifies the name of the HTTP/2 profile, up to 32 characters

5.2.1.35.57 pcf-client

Syntax

```
pcf-client
```

Context

```
config>mobile>pdn>sba-client-services
```

Description

This command enables the context to configure the SMF client-side options when the SMF uses PCF services.

5.2.1.35.58 npcf-smpolicycontrol

Syntax

[no] npcf-smpolicycontrol *service-instance*

Context

config>mobile>pdn>sba-client-services>pcf-client

Description

This command configures an instance of the Npcf_SMPolicyControl service. At least one service instance must be configured to enable peering with the PCF.

Parameters

service-instance

specifies the service instance name, up to 32 characters

5.2.1.35.59 http2-profile

Syntax

http2-profile *http2-profile-name*

no http2-profile

Context

config>mobile>pdn>sba-client-services>pcf-client>npcf-smpolicycontrol

Description

This command associates an HTTP/2 profile, configured using the config>mobile>profile>http2 command, to the Npcf_SMPolicyControl service.

The no form of this command removes the HTTP/2 profile from the configuration.

Parameters

http2-profile-name

specifies the name of the HTTP/2 profile, up to 32 characters

5.2.1.35.60 npcf-profile

Syntax

```
npcf-profile profile-name  
no npcf-profile
```

Context

```
config>mobile>pdn>sba-client-services>pcf-client>npcf-smpolicycontrol
```

Description

This command configures a PCF profile for the PCF.

Parameters

profile-name
specifies the profile name, up to 32 characters

5.2.1.35.61 scp-client

Syntax

```
scp-client
```

Context

```
config>mobile>pdn>sba-client-services
```

Description

This command enables the context to configure the SCP options for indirect communication with or without delegated discovery.

5.2.1.35.62 scp-service

Syntax

```
[no] scp-service service-instance
```

Context

```
config>mobile>pdn>sba-client-services>scp-client
```

Description

This command configures an instance of the SCP client service. At least one service instance must be configured to allow peering with the SCP.

Parameters

service-instance

specifies the service instance name, up to 32 characters

5.2.1.35.63 block-duration

Syntax

block-duration *value*

no block-duration

Context

config>mobile>pdn>sba-client-services>scp-client>scp-service

Description

This command specifies the period that SCP remains blocked. If SCP is blocked, the SMF does not send requests to the SCP. If all SCPs are blocked, the SMF starts using direct communication.

Parameters

value

specifies the block duration, in seconds

Values: 0 to 3600

Default: 5

5.2.1.35.64 default-failure-action

Syntax

default-failure-action *failure-action*

no default-failure-action

Context

config>mobile>pdn>sba-client-services>scp-client>scp-service

Description

This command configures the default failure handling if the peer SCP rejects the request. The action is applied if the response from the peer SCP contains a Server header which indicates that SCP rejected the request.

Retransmissions and fail actions are controlled by the client service settings. For example, if the nudm-sdm request is sent via SCP, the nudm-sdm settings define the retransmissions and action, if the configured SCP failure action is `fail`.

The no form of this command reverts to the default action.

Parameters

failure-action

specifies the failure action

Values:

block

SCP is blocked and no requests are sent via the SCP. An alarm is raised if the SCP is blocked. The `block-duration` command specifies the duration that the SCP is blocked. Retransmission is allowed.

retry

Retransmission is allowed, and it may be performed via the same SCP.

failover

Retransmission is allowed, and it is performed via a different SCP.

fail

The related service operation fails, and the respective client service failure action defines failure action.

mode-fallback-a

There is fallback to a direct communication mode where the NF discovery is done from the local configuration. Any possible retransmissions use the new model. The `block-duration` command defines the duration that the new communication mode is used.

mode-fallback-b

There is fallback to a direct communication mode where the NF discovery is done with the NRF. Any possible retransmissions use the new model. The `block-duration` command defines the duration that the new communication mode is used.

Default: mode-fallback-a

5.2.1.35.65 fh-no-response

Syntax

```
fh-no-response failure-action
no fh-no-response
```

Context

```
config>mobile>pdn>sba-client-services>scp-client>scp-service
```

Description

This command configures the failure action if no response is received from the SCP.

Parameters

failure-action

specifies the failure action

Values:

block

SCP is blocked and no requests are sent via the SCP. An alarm is raised if the SCP is blocked. The `block-duration` command specifies how long the SCP is blocked. Retransmission is allowed.

retry

Retransmission is allowed, and it is performed via the same SCP.

failover

Retransmission is allowed, and it is performed via a different SCP.

The retransmissions are controlled by client service settings, for example, if a `nudm-request` is sent via the SCP, the `nudm-sdm` settings define the retransmissions.

Default: `retry`

5.2.1.35.66 http-status-code

Syntax

```
http-status-code [status-code status-code-value [application-error app-error-code] ]
failure-action

no http-status-code [status-code status-code-value [application-error app-error-code] ]
```

Context

config>mobile>pdn>sba-client-services>scp-client>scp-service

Description

This command specifies the failure action, if the SCP rejects the request. If an HTTP status code or application error is defined, the failure action is used only if the rejection uses a matching of HTTP status code and application error. If the command does not include an HTTP status code and application error, the defined failure action applies to any rejection.

The no form of this command removes the failure action and reverts to the default action (mode-fallback-a).

Parameters

status-code-value

specifies the three-digit status code value

Values: 300 to 599

app-error-code

specifies the application error code, up to 64 characters

failure-action

specifies the failure action

Values:

block

SCP is blocked and no requests are sent via the SCP. An alarm is raised if the SCP is blocked. The `block-duration` command specifies the period that the SCP remains blocked. Retransmission is allowed.

retry

retransmission is allowed, and it is performed via the same SCP

failover

retransmission is allowed, and it is performed via a different SCP

fail

the related service operation fails, and the respective client service failure action specifies the failure action

mode-fallback-a

a fallback to a direct communication mode is done where the NF discovery is performed from the local configuration. Possible retransmissions use the new model. The `block-duration` command specifies how long the new communication mode is used.

mode-fallback-b

a fallback to a direct communication mode is done where the NF discovery is performed through the NRF. Possible retransmissions use the new model. The `block-duration` command specifies how long the new communication mode is used.

Default: `mode-fallback-a`

5.2.1.35.67 `http2-profile`

Syntax

```
http2-profile http2-profile-name  
no http2-profile
```

Context

```
config>mobile>pdn>sba-client-services>scp-client>scp-service
```

Description

This command associates an HTTP/2 profile, configured using the `config>mobile>profile>http2` command, to the SCP client service.

The `no` form of this command removes the HTTP/2 profile from the configuration.

Parameters

http2-profile-name

specifies the name of the HTTP/2 profile, up to 32 characters

5.2.1.35.68 `mode-fallback`

Syntax

```
mode-fallback new-model  
no mode-fallback
```

Context

```
config>mobile>pdn>sba-client-services>scp-client>scp-service
```

Description

This command specifies the communication model to be used if there are no available SCPs or all the available SCPs are blocked.

The no form of this command reverts to the default.

Parameters

new-model

specifies the new communication model

Values:

a

fallback to direct communication mode with NF producer and NF discovery performed without NRF, using local configuration

b

fallback to direct communication mode with NF producer and NF discovery performed with NRF

Default: a

5.2.1.35.69 scp-domain

Syntax

scp-domain *domain-name*

no scp-domain

Context

config>mobile>pdn>sba-client-services>scp-client

Description

This command configures an SCP domain for the SCP client service.

The no form of this command reverts to the default.

Default

no scp-domain

Parameters

domain-name

specifies the SCP domain name, up to 32 characters

5.2.1.35.70 sepp-client

Syntax

`sepp-client`

Context

`config>mobile>pdn>sba-client-services`

Description

This command enables the context to configure the SEPP options for communication with NFs in another PLMN.

5.2.1.35.71 sepp

Syntax

`[no] sepp service-instance`

Context

`config>mobile>pdn>sba-client-services>sepp-client`

Description

This command configures a service instance to be used with SEPP. At least one service instance must be configured to enable peering with the SEPP.

If multiple SEPP instances are configured, only one of them can be defined without a PLMN list. If a PLMN list is defined, the SEPP instance can be used only if the remote PLMN is found in the specified PLMN list. The SEPP instance without a PLMN list is used in cases where a remote PLMN is not found from the PLMN lists of other SEPP instances.



NOTE:

This command requires a PDN shutdown only when an HTTP/2 profile is configured with TLS enabled.

Parameters

service-instance

specifies the service instance name, up to 32 characters

5.2.1.35.72 http2-profile

Syntax

```
http2-profile http2-profile-name  
no http2-profile
```

Context

```
config>mobile>pdn>sba-client-services>sepp-client>sepp
```

Description

This command associates an HTTP/2 profile, configured using the `config>mobile>profile>http2` command, to the SEPP client service.

The `no` form of this command removes the HTTP/2 profile from the configuration.

Parameters

http2-profile-name

specifies the name of the HTTP/2 profile, up to 32 characters

5.2.1.35.73 smf-client

Syntax

```
smf-client
```

Context

```
config>mobile>pdn>sba-client-services
```

Description

This command enables the context to configure the SMF client-side options when the SMF uses services of another SMF.

5.2.1.35.74 nsmf-pdusession

Syntax

```
[no] nudm-sdm service-instance
```

Context

```
config>mobile>pdn>sba-client-services>smf-client
```

Description

This command configures an instance of the Nsmf-pdusession service. At least one service instance must be configured to allow peering with the SMF.

Parameters

service-instance

specifies the service instance name, up to 32 characters

5.2.1.35.75 api-prefix

Syntax

```
api-prefix string
```

```
no api-prefix
```

Context

```
config>mobile>pdn>sba-client-services>smf-client>nsmf-pdusession
```

```
config>mobile>pdn>sba-server-services>smf-client>nsmf-pdusession
```

Description

This command configures the API prefix for the service that defines the optional path segment in the API root.

Parameters

string

specifies the API prefix, up to 32 characters

5.2.1.35.76 http2-profile

Syntax

```
http2-profile http2-profile-name
```

```
no http2-profile
```

Context

```
config>mobile>pdn>smf-client-services>nsmf-pdusession
```

Description

This command associates an HTTP/2 profile, configured using the `config>mobile>profile>http2` command, to the `Nsmf_pdusession` client service.

The `no` form of this command removes the HTTP/2 profile from the configuration.

Parameters

http2-profile-name

specifies the name of the HTTP/2 profile, up to 32 characters

5.2.1.35.77 supported-features

Syntax

```
supported-features
```

Context

```
config>mobile>pdn>sba-client-services>smf-client>nsmf-pdusession
```

```
config>mobile>pdn>sba-server-services>smf-client>nsmf-pdusession
```

```
config>mobile>pdn>sba-client-services>udm-client>nudm-sdm
```

Description

This command enables the context to configure the supported features of the service.

5.2.1.35.78 dtssa

Syntax

```
[no] dtssa
```

Context

```
config>mobile>pdn>sba-client-services>smf-client>nsmf-pdusession>supported-features
```

```
config>mobile>pdn>sba-server-services>smf-client>nsmf-pdusession>supported-features
```

Description

This command enables the optional DTSSA feature in the SMF PDU session service. The DTSSA feature must be enabled when deploying an I-SMF.

The no form of this command disables the DTSSA feature.

5.2.1.35.79 transaction-timer

Syntax

```
transaction-timer value  
no transaction-timer
```

Context

```
config>mobile>pdn>sba-client-services>smf-client>nsmf-pdusession  
config>mobile>pdn>sba-server-services>smf-client>nsmf-pdusession
```

Description

This command configures the time that the SMF waits for a response from the remote SMF.

The no form of this commands reverts to the default.

Parameters

value

specifies the time, in milliseconds
Values: 1000 to 20000 (in increments of 100)
Default: 5000

5.2.1.35.80 version

Syntax

```
version major [minor minor] [patch patch]  
no version
```

Context

```
config>mobile>pdn>sba-client-services>smf-client>nsmf-pdusession  
config>mobile>pdn>sba-server-services>smf-client>nsmf-pdusession
```

Description

This command configures the API version of the service. If the version is not defined, the used version is the one based on the latest fully supported API version.

The API version consists of three numbers:

1. major
2. minor
3. patch

Default

If no version is defined, version 1.0.2 is used.

Parameters

major

specifies the major version

Values: 1 to 255

Default: 1

minor

specifies the minor version

Values: 0 to 255

Default: 0

patch

specifies the patch version

Values: 0 to 255

Default: 2

5.2.1.35.81 udm-client

Syntax

```
udm-client
```

Context

```
config>mobile>pdn>sba-client-services
```

Description

This command enables the context to configure the SMF client-side options when the SMF uses UDM services.

5.2.1.35.82 nudm-sdm

Syntax

[no] nudm-sdm *service-instance*

Context

config>mobile>pdn>sba-client-services>udm-client

Description

This command configures an instance of the Nudm_SubscriberDataManagement service. At least one service instance must be configured to allow the SMF to get session management subscription data from the UDM.

Parameters

service-instance

specifies the service instance name, up to 32 characters

5.2.1.35.83 api-version

Syntax

api-version *version*

no api-version

Context

config>mobile>pdn>sba-client-services>udm-client>nudm-sdm

Description

This command specifies the major API version of the nudm-sdm service. The SMF prefers those nudm-sdm service instances with matching major version number.

The no form of this command reverts to the default where the SMF does not use the major API version to select the nudm-sdm service instances.

Default

no api-version

Parameters

version

specifies the major API version

Values: 0 to 255

Default: 0 (the SMF can use any major API version)

5.2.1.35.84 explicit-unsubscribe

Syntax

explicit-unsubscribe [expiry-time *expiry-time*] [renew-on-expiry]

no explicit-unsubscribe

Context

config>mobile>pdn>sba-client-services>udm-client>nudm-sdm

Description

This command sets the implicit unsubscribe attribute to the value false when a subscription is created in the UDM. The SMF sets the implicit unsubscribe attribute to the value true by default.

The optional *expiry-time* parameter defines the proposed expiry time of the subscription. The UDM may override the proposed expiry time when it creates the subscription.

When the optional *renew-on-expiry* parameter is configured, the SMF renews the subscription after it has expired and the PDU is still active. When the parameter is not configured, the SMF releases the PDU session after the subscription has expired.

The no form of this command reverts to the default.

Default

no explicit-unsubscribe

Parameters

expiry-time

specifies the suggested expiry time in seconds

Values: 1800 to 4294967295

Default: 4294967295

renew-on-expiry

specifies that the SMF must renew the subscription after it has expired and the PDU is still active

5.2.1.35.85 http2-profile

Syntax

`http2-profile http2-profile-name`

`no http2-profile`

Context

`config>mobile>pdn>sba-client-services>udm-client>nudm-sdm`

Description

This command associates an HTTP/2 profile, configured using the `config>mobile>profile>http2` command, to the Nudm_SubscriberDataManagement service.

The `no` form of this command removes the HTTP/2 profile from the configuration.

Parameters

http2-profile-name

specifies the name of the HTTP/2 profile, up to 32 characters

5.2.1.35.86 nudm-uecm

Syntax

`[no] nudm-uecm service-instance`

Context

`config>mobile>pdn>sba-client-services>udm-client`

Description

This command configures an instance of the Nudm_UEContextManagement service. At least one service instance must be configured to allow the SMF to register PDU sessions in the UDM.

Parameters

service-instance

specifies the service instance name, up to 32 characters

5.2.1.35.87 http2-profile

Syntax

http2-profile *http2-profile-name*

no http2-profile

Context

config>mobile>pdn>sba-client-services>udm-client>nudm-uecm

Description

This command associates an HTTP/2 profile, configured using the config>mobile>profile>http2 command, to the Nudm_UEContextManagement service.

The no form of this command removes the HTTP/2 profile from the configuration.

Parameters

http2-profile-name

specifies the name of the HTTP/2 profile, up to 32 characters

5.2.1.35.88 user-agent-header

Syntax

user-agent-header [user-agent *user-agent*]

no user-agent-header

Context

config>mobile>pdn>sba-client-services

config>mobile>pdn>sba-server-services

Description

This command enables the User-Agent header in outgoing SBI requests. The value of the header is defined with the *user-agent* parameter.

The no form of this command reverts to the default value.

Default

no user-agent-header (the User-Agent is not used in the outgoing SBI requests)

Parameters

user-agent

specifies the value of the header, up to 64 characters: the User-Agent value must always start with "SMF".

Default: SMF-*<NF instance ID>*

5.2.1.35.89 sbi-correlation-info-header

Syntax

[no] sbi-correlation-info-header

Context

config>mobile>pdn>sba-client-services

config>mobile>pdn>sba-server-services

Description

This command enables the 3gpp-Sbi-Correlation-Info header in the outgoing SBI messages.

The no form of this command reverts to the default value.

Default

no sbi-correlation-info-header (the 3gpp-Sbi-Correlation-Info-header header is not used in the outgoing SBI messages)

5.2.1.35.90 sbi-load-control

Syntax

sbi-load-control

Context

```
config>mobile>pdn>sba-server-services
```

Description

This command enables the context to configure the node-level SBI load control over the N11 and N16 interfaces.

5.2.1.35.91 enable

Syntax

```
[no] enable
```

Context

```
config>mobile>pdn>sba-server-services>sbi-load-control
```

Description

This command enables the load control support based on the LCI header.

The no form of this command removes the LCI header load control support.

Default

```
no enable
```

5.2.1.35.92 load-change-trigger

Syntax

```
load-change-trigger percent
```

```
no load-change-trigger
```

Context

```
config>mobile>pdn>sba-server-services>sbi-load-control
```

Description

This command configures the load change value which triggers a new load control update transmitted via the 3gpp-Sbi-Lci header.

The no form of this command configures the load change value to the default value (5).

Parameters

percent

specifies the percentage load change value

Values: 5 | 10 | 15 | 20

Default: 5

5.2.1.35.93 nth-message

Syntax

`nth-message number`

`no nth-message`

Context

`config>mobile>pdn>sba-server-services>sbi-load-control`

Description

This command configures the message interval for sending load control information via the 3gpp-Sbi-Lci header.

The no form of this command configures the message interval to the default value (50).

Parameters

number

specifies the interval used for including load control information

Values: 1 to 1000

Default: 50

5.2.1.35.94 sbi-nf-peer-info-header

Syntax

`sbi-nf-peer-info-header [scp] [sepp] [srvinst]`

`no sbi-nf-peer-info-header`

Context

`config>mobile>pdn>sba-client-services`

```
config>mobile>pdn>sba-server-services
```

Description

This command enables the 3gpp-Sbi-NF-Peer-Info header in the outgoing SBI messages. The header contains the NF instance ID of the SMF and the NF instance ID of the peer NF, if available.

The no form of this command reverts to the default value.

Default

no sbi-nf-peer-info-header (the 3gpp-Sbi-NF-Peer-Info header is not used in the outgoing SBI messages)

Parameters

scp

Enables the inclusion of the dstscp parameter in the 3gpp-Sbi-NF-Peer-Info of outgoing SBI messages to the SCP if the following conditions are true:

- the request is sent to the SCP and the FQDN of the SCP, if known
- the request associated with the sent response includes the scrscp parameter in the 3gpp-Sbi-Nf-Peer-Info header

sepp

Enables the inclusion of the dstsepp parameter in the 3gpp-Sbi-NF-Peer-Info header of outgoing SBI messages to the SEPP if the following conditions are true:

- the request is sent to the SEPP and the FQDN of the SEPP, if known
- the request associated with the sent response includes the scrsepp parameter in the 3gpp-Sbi-Nf-Peer-Info header

srvinst

Enables the inclusion of the srcservinst and dstservinst parameters in the 3gpp-Sbi-NF-Peer-Info header of outgoing SBI messages. The header contains the service instance value, if known.

5.2.1.35.95 sbi-overload-control

Syntax

```
sbi-overload-control
```

Context

```
config>mobile>pdn>sba-server-services
```

Description

This command enables the context to configure the node-level SBI overload control over the N11 and N16 interfaces.

5.2.1.35.96 enable

Syntax

[no] enable

Context

config>mobile>pdn>sba-server-services>sbi-overload-control

Description

This command enables the overload control support based on the OCI header.

The no form of this command removes the OCI header overload control support.



NOTE:

When you configure both the `sbi-overload-control` and `nsmf-overload-control` commands, the `minor`, `major`, and `critical` commands in the `config>mobile>pdn>nsmf-overload-control>overload-threshold` context must have a configured value higher than the `minor`, `major`, and `critical` commands in the `config>mobile>pdn>overload-control>thresholds` context used for SBI overload control.

Default

no enable

5.2.1.35.97 overload-reduction

Syntax

overload-reduction

Context

config>mobile>pdn>sba-server-services>sbi-overload-control

Description

This command enables the context to configure the percentage of the traffic reduction requested and transmitted over the N11 or N16 interface.

5.2.1.35.98 critical

Syntax

`critical percent`

`no critical`

Context

`config>mobile>pdn>sba-server-services>sbi-overload-control>overload-reduction`

Description

This command configures the traffic reduction percentage for the critical overload level of the N11 or N16 interface.

The no form of this command reverts to the default value (90).

Default

`no critical`

Parameters

percent

specifies the percentage of the traffic reduction

Values: 20 to 100

Default: 90

5.2.1.35.99 major

Syntax

`major percent`

`no major`

Context

config>mobile>pdn>sba-server-services>sbi-overload-control>overload-reduction

Description

This command configures the traffic reduction percentage for the major overload level of the N11 or N16 interface.

The no form of this command reverts to the default value (60).

Default

no major

Parameters

percent

specifies the percentage of the traffic reduction

Values: 20 to 100

Default: 60

5.2.1.35.100 minor

Syntax

minor *percent*

no minor

Context

config>mobile>pdn>sba-server-services>sbi-overload-control>overload-reduction

Description

This command configures the traffic reduction percentage for the minor overload level of the N11 or N16 interface.

The no form of this command reverts to the default value (30).

Default

no minor

Parameters

percent

specifies the percentage of the traffic reduction

Values: 20 to 100

Default: 30

5.2.1.35.101 *validity-time*

Syntax

`validity-time validity-time`

`no validity-time`

Context

`config>mobile>pdn>sba-server-services>sbi-overload-control`

Description

This command configures the length of the time for the reported overload to be in effect.

Parameters

validity-time

specifies the length of time, in seconds, for the overload to be in effect

Values: 10 to 86400

Default: 600

5.2.1.36 SBA server services PDN command descriptions

5.2.1.36.1 *sba-server-services*

Syntax

`sba-server-services`

Context

`config>mobile>pdn`

Description

This command enables the context to configure the server-side options provided by CMG as the NF producer.

5.2.1.36.2 fqdn

Syntax

`fqdn fqdn-sni-string`

`no fqdn`

Context

`config>mobile>pdn>sba-server-services`

Description

This command configures an FQDN which is used by all SBA server services.

The no form of this command removes the FQDN from the configuration.

v

5.2.1.36.3 http2-profile

Syntax

`http2-profile http2-profile-name`

`no http2-profile`

Context

`config>mobile>pdn>sba-server-services`

Description

This command associates an HTTP/2 profile, configured using the `config>mobile>profile>http2` command, to all SBA server services.

The no form of this command removes the HTTP/2 profile from the configuration.

Parameters

http2-profile-name

specifies the name of the HTTP/2 profile, up to 32 characters

5.2.1.36.4 nbsf-management

Syntax

```
[no] nbsf-management service-instance
```

Context

```
config>mobile>pdn>sba-server-services
```

Description

This command configures a service instance of the Nbsf_Management service. At least one service instance must be configured to allow other NFs to access the BSF services.

The no form of this command removes the service instance from the configuration.

Parameters

service-instance

specifies the service instance name, up to 32 characters

5.2.1.36.5 allowed-nf-domains-list

Syntax

```
allowed-nf-domains-list list-name  
no allowed-nf-domains-list
```

Context

```
config>mobile>pdn>sba-server-services>nbsf-management  
config>mobile>pdn>sba-server-services>nsmf-event-exposure  
config>mobile>pdn>sba-server-services>nsmf-pdusession
```

Description

This command configures the allowed NF domains per NF service instance. The allowed NF domain list is configured using the config>mobile>profile>list>allowed-nf-domains-list command.

These domains are used in the NFProfile during registration to the NRF. The configured values of the list are also used for static authorization that allows, or not, serving requests from the indicated NF domains.

The no form of this command reverts to the default value.

Default

no allowed-nf-domains-list

Parameters

list-name

specifies the allowed NF domain list name, up to 32 characters

5.2.1.36.6 allowed-nf-types

Syntax

allowed-nf-types {pcf | nef | af | dra}

no allowed-nf-types [pcf | nef | af | dra]

Context

config>mobile>pdn>sba-server-services>nbsf-management

Description

This command configures the allowed NF types per NF service instance. The NF types are used in the NFProfile in the registration to the NRF. The configured values are also used for static authorization that allows, or not, serving requests from the indicated NF types.

The no form of this command removes the specified values from the configuration.

Default

no allowed-nf-types

Parameters

pcf

specifies the PCF NF

nef

specifies the NEF NF

af

specifies the AF NF

dra

specifies the DRA NF

5.2.1.36.7 allowed-plmns

Syntax

```
allowed-plmns list-name  
no allowed-plmns
```

Context

```
config>mobile>pdn>sba-server-services>nbsf-management
```

Description

This command associates a PLMN list (configured under the `config>mobile>profile>list>plmn` context) to configure the discovery of the Nbsf_Management session service based on the PLMN of the NF performing the discovery. If the PLMN list is defined, the Nbsf_Management session is discoverable only from the PLMNs included in the list. If the PLMN list is not defined, the Nbsf_Management session service is discoverable from any PLMN.

The no form of this command removes the association of the PLMN list.



NOTE:

When you configure the nnrf-nfm client under the `config>mobile>pdn>sba-service-realm` context, this command accepts a PLMN list with a maximum of 32 entries.

Parameters

list-name

specifies the PLMN list name, up to 32 characters

5.2.1.36.8 fqdn

Syntax

```
fqdn fqdn-sni-string  
no fqdn
```

Context

```
config>mobile>pdn>sba-server-services>nbsf-management
```

Description

This command configures an FQDN which is used by the Nbsf_management server service.

The no form of this command removes the FQDN from the configuration.

Parameters

fqdn-sni-string

specifies the FQDN name string, up to 255 characters

5.2.1.36.9 http2-profile

Syntax

`http2-profile http2-profile-name`

`no http2-profile`

Context

`config>mobile>pdn>sba-server-services>nbsf-management`

Description

This command associates an HTTP/2 profile, configured using the `config>mobile>profile>http2` command, to the Nbsf_management server service.

The no form of this command removes the HTTP/2 profile from the configuration.

Parameters

http2-profile-name

specifies the name of the HTTP/2 profile, up to 32 characters

5.2.1.36.10 interface

Syntax

`interface interface-name [router router-instance] [port port-number]`

`no interface`

Context

`config>mobile>pdn>sba-server-services>nbsf-management`

```
config>mobile>pdn>sba-server-services>nsmf-event-exposure
config>mobile>pdn>sba-server-services>nsmf-pdusession
```

Description

This command configures the interface in which the HTTP/2 connections are established.
The no form of this command removes the interface from the configuration.

Parameters

interface-name

specifies an existing interface name and must start with a letter, up to 32 characters

router-instance

specifies the router instance

Values	router-name service-id	
–	router-name	Base
–	service-id	1 to 2147483647

Default: Base

port-number

specifies the port number
Values: 1 to 65535
Default: 80

5.2.1.36.11 oauth2-profile

Syntax

```
oauth2-profile profile-name
no oauth2-profile
```

Context

```
config>mobile>pdn>sba-server-services>nbsf-management
```



```
config>mobile>pdn>sba-server-services>nsmf-event-exposure  
config>mobile>pdn>sba-server-services>nsmf-pdusession
```

Description

This command configures OAuth2.0 authorization for the requests received in the specified service instance and defines the profile with OAuth2.0 behavioral aspects for the server side. A default profile can be configured.

When OAuth2.0 is configured, the SMF or BSF looks for an access token in the received requests and validates the token before serving the request.

The no form of this command disables the OAuth2.0 authorization in the specified server context.

Default

no oauth2-profile

Parameters

profile-name

specifies the server profile name, up to 32 characters

5.2.1.36.12 nsmf-event-exposure

Syntax

```
[no] nsmf-event-exposure service-instance
```

Context

```
config>mobile>pdn>sba-server-services
```

Description

This command configures a service instance of the Nsmf_EventExposure service. At least one service instance must be configured to allow peering with AMF.

The no form of this command removes the service instance from the configuration.

Parameters

service-name

specifies the service instance name, up to 32 characters

5.2.1.36.13 allowed-nf-types

Syntax

```
[no] allowed-nf-types {pcf | nef | af | amf | udm | dccf | nwdaf}
```

Context

```
config>mobile>pdn>sba-server-services>nsmf-event-exposure
```

Description

This command configures the allowed NF types per NF service instance. The NF types are used in the NFProfile in the registration to the NRF. The configured values are also used for static authorization that allows, or not, serving requests from the indicated NF types.

The no form of this command removes the specified values from the configuration.

Default

no allowed-nf-types

Parameters

pcf

specifies the PCF NF

nef

specifies the NEF NF

af

specifies the AF NF

amf

specifies the AMF NF

udm

specifies the UDM NF

dccf

specifies the DCCF NF

nwdaf

specifies the NWDAF NF

5.2.1.36.14 allowed-plmns

Syntax

```
allowed-plmns list-name  
no allowed-plmns
```

Context

```
config>mobile>pdn>sba-server-services>nsmf-event-exposure
```

Description

This command configures the discovery of the SMF event exposure service based on the PLMN of the NF performing the discovery. If the PLMN list is defined, the SMF event exposure is discoverable only from the PLMNs included in the list. If the PLMN list is not defined, the SMF event exposure is discoverable from any PLMN.

The no form of this command removes the allowed PLMN list from the configuration.



NOTE:

When you configure the nnrf-nfm client under the config>mobile>pdn>sba-service-realm context, this command accepts a PLMN list with a maximum of 32 entries.

Parameters

list-name

specifies the PLMN list name, up to 32 characters

5.2.1.36.15 allowed-slices

Syntax

```
allowed-slices list-name  
no allowed-slices
```

Context

```
config>mobile>pdn>sba-server-services>nsmf-event-exposure
```

Description

This command configures the discovery of the SMF event exposure service based on the S-NSSAI of the NF performing the discovery. If the slice list is defined, the SMF event exposure is discoverable only from the S-NSSAIs included in the list. If slice list is not defined, the SMF event exposure is discoverable from any S-NSSAI.

The no form of this command removes the allowed slice list from the configuration.

Parameters

list-name

specifies the slice list name, up to 32 characters

5.2.1.36.16 default-subscription-timeout

Syntax

```
default-subscription-timeout subscription-timeout
```

```
no default-subscription-timeout
```

Context

```
config>mobile>pdn>sba-server-services>nsmf-event-exposure
```

Description

This command configures the maximum duration of the SMF event exposure subscription.

The no form of this command reverts to the default subscription timeout value.

Parameters

subscription-timeout

specifies the maximum duration of the SMF event exposure subscription, in seconds

Values: 60 to 4294967295

Default: 84600

5.2.1.36.17 fqdn

Syntax

```
fqdn fqdn-sni-string
```

```
no fqdn
```

Context

```
config>mobile>pdn>sba-server-services>nsmf-event-exposure
```

Description

This command configures an FQDN that is used by the Nsmf_EventExposure server service.

The no form of this command removes the FQDN from the configuration.

Parameters

fqdn-sni-string

specifies the FQDN name string, up to 255 characters

5.2.1.36.18 http2-profile

Syntax

```
http2-profile http2-profile-name
```

```
no http2-profile
```

Context

```
config>mobile>pdn>sba-server-services>nsmf-event-exposure
```

Description

This command associates an HTTP/2 profile to the Nsmf_EventExposure server service. The HTTP/2 profile is configured using the config>mobile>profile>http2 command.

The no form of this command removes the HTTP/2 profile from the configuration.

Parameters

http2-profile-name

specifies the name of the HTTP/2 profile, up to 32 characters

5.2.1.36.19 nsmf-profile

Syntax

```
nsmf-profile profile-name
```

```
no nsmf-profile
```

Context

config>mobile>sba-server-services>nsmf-event-exposure

Description

This command adds a link to an SMF profile. Also, the command defines the failure handling policy when the SMF services are used.

The profile configured under an APN defines the failure handling in the nsmf-pdusession service for those PDU sessions which use the corresponding APN/DNN. If no profile is defined under an APN, the failure handling is defined by the SMF profile under the nsmf-pdusession service.

The no form of this command removes the link to the SMF profile.

Parameters

profile-name

specifies a profile name, up to 32 characters

5.2.1.36.20 supported-features

Syntax

supported-features

Context

config>mobile>pdn>sba-server-services>nsmf-event-exposure

Description

This command enables the context to configure the supported features of the service.

5.2.1.36.21 communication-failure

Syntax

[no] communication-failure

Context

config>mobile>pdn>sba-server-services>nsmf-event-exposure>supported-features

Description

This command enables the optional CommunicationFailure feature. The feature must be enabled if the NF consumers of the SMF subscribe to receive notifications about communication failures.

The no form of this command disables the feature.



NOTE:

This command is not supported in the current release.

5.2.1.36.22 ene-na

Syntax

[no] ene-na

Context

config>mobile>pdn>sba-server-services>nsmf-event-exposure>supported-features

Description

This command enables the optional EneNA feature. The feature must be enabled if the NF consumers of the SMF subscribe to receive notifications about RAT changes.

The no form of this command disables the feature.

5.2.1.36.23 pdu-session-status

Syntax

[no] pdu-session-status

Context

config>mobile>pdn>sba-server-services>nsmf-event-exposure>supported-features

Description

This command enables the optional PduSessionStatus feature. The feature must be enabled if the NF consumers of the SMF subscribe to receive notifications about PDU session establishment.

The no form of this command disables the feature.

5.2.1.36.24 qfi-allocation

Syntax

[no] qfi-allocation

Context

config>mobile>pdn>sba-server-services>nsmf-event-exposure>supported-features

Description

This command enables the optional QfiAllocation feature. The feature must be enabled if the NF consumers of the SMF subscribe to receive notifications about QFI allocation.

The no form of this command disables the feature.



NOTE:

This command is not supported in the current release.

5.2.1.36.25 nsmf-pdusession

Syntax

[no] nsmf-pdusession *service-name*

Context

config>mobile>pdn>sba-server-services

Description

This command configures a service instance of the Nsmf_PDUSession service. At least one service instance must be configured to allow peering with AMF.

The no form of this command removes the service instance from the configuration.

Parameters

service-name

specifies the service instance name, up to 32 characters

5.2.1.36.26 allowed-nf-types

Syntax

```
[no] allowed-nf-types {amf | smf}
```

Context

```
config>mobile>pdn>sba-server-services>nsmf-pdusession
```

Description

This command configures the allowed NF types per NF service instance. The NF types are used in the NFProfile in the registration to NRF. The configured values are also used for static authorization that allows or not serving requests from the indicated NF types.

The no form of this command removes the specified values from the configuration.

Default

```
no allowed-nf-domains-list
```

Parameters

amf

specifies the AMF NF

smf

specifies the SMF NF

5.2.1.36.27 allowed-plmns

Syntax

```
allowed-plmns list-name
```

```
no allowed-plmns
```

Context

```
config>mobile>pdn>sba-server-services>nsmf-pdusession
```

Description

This command associates a PLMN list (configured under the config>mobile>profile>list>plmn context) to configure the discovery of the SMF PDU session service based on the PLMN of the NF performing the discovery. If the PLMN list is defined, the SMF PDU session is discoverable only from the

PLMNs included in the list. If the PLMN list is not defined, the SMF PDU session service is discoverable from any PLMN.

The no form of this command removes the association of the PLMN list.

**NOTE:**

When you configure the nnrf-nfm client under the `config>mobile>pdn>sba-service-realm` context, this command accepts a PLMN list with a maximum of 32 entries.

Parameters

list-name

specifies the PLMN list name, up to 32 characters

5.2.1.36.28 fqdn

Syntax

`fqdn fqdn-sni-string`

`no fqdn`

Context

`config>mobile>pdn>sba-server-services>nsmf-pdusession`

Description

This command configures an FQDN which is used by the Nsmf_pdusession server service.

The no form of this command removes the FQDN from the configuration.

Parameters

fqdn-sni-string

specifies the FQDN name string, up to 255 characters

5.2.1.36.29 http2-profile

Syntax

`http2-profile http2-profile-name`

`no http2-profile`

Context

```
config>mobile>pdn>sba-server-services>nsmf-pdusession
```

Description

This command associates an HTTP/2 profile, configured using the `config>mobile>profile>http2` command, to the `Nsmf_pdusession` server service.

The `no` form of this command removes the HTTP/2 profile from the configuration.

Parameters

http2-profile-name

specifies the name of the HTTP/2 profile, up to 32 characters

5.2.1.36.30 indirect-forwarding-timeout

Syntax

```
indirect-forwarding-timeout indirect-forwarding-timeout  
no indirect-forwarding-timeout
```

Context

```
config>mobile>pdn>sba-server-services>nsmf-pdusession
```

Description

This command defines the maximum duration for the indirect data forwarding tunnel during the handovers. The SMF releases the indirect data forwarding tunnel after the timer expires.

The `no` form of this command reverts to the default (disabled).

Default

```
no indirect-forwarding-timeout
```

Parameters

indirect-forwarding-timeout

specifies the maximum duration for the indirect data forwarding tunnel, in milliseconds

Values: 100 to 5000

Default: 2000

5.2.1.36.31 n3-interface-realm

Syntax

```
n3-interface-realm realm-name  
no n3-interface-realm
```

Context

```
config>mobile>pdn>sba-server-services>nsmf-pdusession
```

Description

This command configures the network instance of the N3 reference point.

Parameters

realm-name

specifies the realm name, up to 32 characters



NOTE:

Only the following characters are permitted:

- alphabetic characters (A-Z and a-z)
- digits (0-9)
- hyphen (-)
- dot (.)

5.2.1.36.32 n9-interface-realm

Syntax

```
n9-interface-realm realm-name  
no n9-interface-realm
```

Context

```
config>mobile>pdn>sba-server-services>nsmf-pdusession
```

Description

This command defines the network instance of the N9 reference point.

Parameters

realm-name

specifies the realm name, up to 32 characters



NOTE:

Only the following characters are permitted:

- alphabetic characters (A-Z and a-z)
- digits (0-9)
- hyphen (-)
- dot (.)

5.2.1.36.33 n9-roaming-interface-realm

Syntax

n9-roaming-interface-realm *realm-name*

no n9-roaming-interface-realm

Context

config>mobile>pdn>sba-server-services>nsmf-pdusession

Description

This command defines the network instance of the inter-PLMN N9 reference point.

The no form of this command reverts to the default.

Default

n9-roaming-interface-realm

Parameters

realm-name

specifies the realm name, up to 32 characters



NOTE:

The following characters are permitted:

- alphabetic characters (A-Z and a-z)
- digits (0-9)
- hyphen (-)
- dot (.)

5.2.1.36.34 s5-ref-point

Syntax

```
s5-ref-point sig-ref-point-name  
no s5-ref-point
```

Context

```
config>mobile>pdn>sba-server-services>nsmf-pdusession
```

Description

This command configures the S5 reference point associated with this Nsmf_PDUSession service instance. The configuration is required for EPS interworking to define the S5 reference point used when there is handover from 5GS to EPS.

Parameters

sig-ref-point-name

specifies the reference point name, up to 32 characters

5.2.1.36.35 s8-ref-point

Syntax

```
s8-ref-point sig-ref-point-name  
no s8-ref-point
```

Context

```
config>mobile>pdn>sba-server-services>nsmf-pdusession
```

Description

This command configures the S8 reference point associated with this Nsmf_PDUSession service instance. The configuration is required for EPS interworking to define the S8 reference point used when there is handover from 5GS to EPS with home-routed roaming.

Parameters

sig-ref-point-name

specifies the reference point name, up to 32 characters

5.2.1.36.36 aasn

Syntax

[no] aasn

Context

config>mobile>pdn>sba-server-services>nsmf-pdusession>supported-features

Description

This command enables the optional AASN feature in the SMF PDU session service. The no form of this command disables the AASN feature.

5.2.1.37 SBA service realm PDN command descriptions

5.2.1.37.1 sba-service-realm

Syntax

[no] sba-service-realm *service-realm*

Context

config>mobile>pdn

Description

This command configures an SBA service realm on the SMF and the UPF. The service realm binds the server and client service instances.

The no form of this command removes the specified service realm from the configuration.

Parameters

service-realm

specifies the service realm name, up to 32 characters

5.2.1.37.2 client-service

Syntax

client-service *service-name* service-instance *service-instance* [priority *value*]

no client-service *service-name* service-instance *service-instance*

Context

config>mobile>pdn>sba-service-realm

Description

This command configures a client-side service instance on the SMF or the UPF under the service realm.

The no form of this command removes the specified client-side service instance from the configuration.

Parameters

service-name

specifies the client-side service name

Values: namf-comm | namf-evts | npcf-smpolicycontrol | nudm-sdm | nudm-uecm | nnrf-nfm
| nnrf-disc | nchf-convergedcharging | nsmf-pdusession | nnrf-oauth2 | scp | sepp

service-instance

specifies the service instance, up to 32 characters

value

specifies the priority value for the client service selection

Values: 1 to 10

Default: 1

5.2.1.37.3 server-service

Syntax

[no] server-service *service-name* service-instance *service-instance*

Context

config>mobile>pdn>sba-service-realm

Description

This command configures a server-side service instance on the SMF under the service realm.

The no form of this command removes the specified server-side service instance from the configuration.

Parameters

service-name

specifies the server side service name

Values: nsmf-pdusession | nbsf-management | nsmf-event-exposure

service-instance

specifies the service instance, up to 32 characters

5.2.1.38 Sd PDN command descriptions

5.2.1.38.1 sd

Syntax

sd

Context

config>mobile>pdn

Description

This command enables the context to configure the Sd interface.

5.2.1.38.2 diameter-redirect

Syntax

[no] diameter-redirect

Context

config>mobile>pdn>sd

Description

This command is used to enable base Diameter redirection for this interface. The no form of this command reverts to the default.

Default

no diameter-redirect

5.2.1.38.3 interface

Syntax

interface

Context

config>mobile>pdn>sd

Description

This command enables the context to configure Diameter interface parameters.



NOTE:

This command requires a PDN shutdown.

5.2.1.38.4 diameter

Syntax

diameter [router *router-instance*] *interface-name*

no diameter

Context

config>mobile>pdn>sd>interface

Description

This command configures Diameter interface parameters. It is used to configure the virtual router in which the Diameter connections will be established by this Diameter policy.

The no form of this command removes the parameter from the configuration.



NOTE:

This command requires a PDN shutdown.

Parameters

router-instance

specifies the virtual router name or service id

Values	router-name service-id	
–	router-name	Base
–	service-id	1 to 2147483647

Default: Base

interface-name

specifies the interface name, up to 32 characters and must start with a letter

5.2.1.38.5 multi-homing

Syntax

multi-homing
no multi-homing

Context

config>mobile>pdn>sd>interface

Description

This command enables multi-homing from a local peer. When multi-homing is enabled, a secondary Diameter interface can also be specified.

The no form of this command reverts to the default (disabled).



NOTE:

- Current multi-homing support requires primary and secondary interfaces to be in the same VPRN.
- This command requires a PDN shutdown.

Default

no multi-homing

5.2.1.38.6 diameter

Syntax

diameter [*router router-instance*] *secondary-interface-name*
no diameter

Context

config>mobile>pdn>sd>interface>multi-homing

Description

This command configures the Diameter secondary interface parameters and the virtual router in which the Diameter connections will be established by this Diameter policy.

The no form of this command removes the parameter from the configuration.



NOTE:

- Current multi-homing support requires primary and secondary interfaces to be in the same VPRN.
- This command requires a PDN shutdown.

Parameters

router-instance

specifies the virtual router name or service id

Values	router-name service-id	
–	router-name	Base
–	service-id	1 to 2147483647

Default: Base

secondary-interface-name

specifies the interface name, up to 32 characters and must start with a letter

5.2.1.38.7 peer-group

Syntax

peer-group *peer-group-id* primary-pcrf *profile-name* [secondary-pcrf *profile-name*]

no peer-group *peer-group-id*

Context

config>mobile>pdn>sd

Description

This command configures a list of peer groups of primary and secondary Diameter peers for the Sd interface.



NOTE:

This command requires a PDN shutdown.

Parameters

peer-group-id

specifies the peer group ID

Values: 1 to 10

profile-name

specifies the primary and secondary Diameter peer profile names, up to 32 characters

5.2.1.38.8 retry-count

Syntax

retry-count *value*

no retry-count

Context

config>mobile>pdn>sd

Description

This command configures the number of times the same message is re-tried before declaring a failed attempt.

The no form of this command reverts to the default. A value of 0 also disables retry.

Parameters

value

specifies the number of times the same message is re-tried before declaring a failed attempt

Values: 0 to 8

Default: 3

5.2.1.38.9 sd-profile

Syntax

[no] sd-profile *profile-name*

Context

config>mobile>pdn>sd

Description

This commands configures the Sd profile.

Parameters

profile-name

specifies the profile name for the Sd profile

5.2.1.38.10 transaction-timer

Syntax

transaction-timer *value*

no transaction-timer

Context

config>mobile>pdn>sd

Description

This command sets the maximum amount of time the node waits for a Diameter peer to respond before trying another.

The no form of this command reverts to the default.

Parameters

value

specifies the maximum amount of time, in seconds, the node waits for a Diameter peer to respond before trying another peer

Values: 1 to 180

Default: 5

5.2.1.39 Signaling PDN command descriptions

5.2.1.39.1 signaling

Syntax

signaling

Context

config>mobile>pdn

Description

This command enables the context to configure signaling.



NOTE:

- Reference Point PDN configuration for a specific interface has a higher priority than the signaling PDN configuration; for example, S1-u, S11, S12, S2a, S2b, S4, S5, and so on.
- This command requires a PDN shutdown.

5.2.1.39.2 diameter

Syntax

```
diameter
```

Context

```
config>mobile>pdn>signaling
```

Description

This command enables the context to configure Diameter signaling.

**NOTE:**

This command requires a PDN shutdown.

5.2.1.39.3 mip6-agent-info

Syntax

```
mip6-agent-info {use-fqdn fqdn use-realm realm | use-sig-ip-addr}  
no mip6-agent-info
```

Context

```
config>mobile>pdn>signaling>diameter
```

Description

This command specifies whether the S6b interface uses the FQDN or IP address of the signaling endpoint.

This command is also used to define the pgwFQDN attribute of the NF profile used when SMF registers to the NRF. The SMF pgwFQDN is configured using the `use-fqdn` option and specifying the FQDN value.

The `no` form of this command removes the configuration.

**NOTE:**

This command requires a PDN shutdown.

Parameters

fqdn

specifies the FQDN value

realm

specifies the realm value

use-sig-ip-addr

specifies use of the signaling IP address

5.2.1.39.4 origin-host

Syntax

origin-host *name*

no origin-host

Context

config>mobile>pdn>signaling>diameter

Description

This command specifies the originating host name of a Diameter node. The originating host information is sent in requests to a Diameter peer. The Origin-Host is unique for each redundant pair of ISM-MG cards. The configured Origin-Host is pre-pended with a mscpxx when the Diameter connection is established from the MSCP. (xx = the group-id of the redundant pair).

For example, for a single ISM-MG redundant pair in the chassis, if the configured origin-host is "7750mg.ipd-nok.com", the origin-host sent over the Diameter connection is "mscp01.7750mg.ipd-nok.com"

The no form of this command removes the name from the configuration.



NOTE:

This command requires a PDN shutdown.

Parameters

name

specifies the originating host name, up to 80 characters

5.2.1.39.5 origin-realm

Syntax

```
origin-realm name  
no origin-realm
```

Context

```
config>mobile>pdn>signaling>diameter
```

Description

This command specifies the originating realm or domain name of a Diameter node. The originating realm is included in messages exchanged with a Diameter peer.

The no form of this command removes the name from the configuration.



NOTE:

This command requires a PDN shutdown.

Parameters

name

specifies the originating host name, up to 80 characters

5.2.1.39.6 profile

Syntax

```
profile profile-name  
no profile
```

Context

```
config>mobile>pdn>signaling>diameter
```

Description

This command specifies the Diameter profile name. The Diameter profile provides Diameter connection related information.

The no form of this command reverts to the default.

**NOTE:**

This command requires a PDN shutdown.

Default

no profile

5.2.1.39.7 gtp-c**Syntax**

gtp-c

Context

config>mobile>pdn>signaling

Description

This command specifies the GPRS Tunneling Protocol-Control plane (GTP-C) profile name. The GTP-C profile provides GTP-C signaling related configuration.

**NOTE:**

This command requires a PDN shutdown.

5.2.1.39.8 profile**Syntax**

profile *profile-name*

no profile

Context

config>mobile>pdn>signaling>gtp-c

Description

This command specifies the GTP-C profile name.

The no form of this command removes the profile name from the configuration.

**NOTE:**

This command requires a PDN shutdown.

Default

no profile

Parameters***profile-name***

specifies the GTP-C profile name

5.2.1.39.9 gtp-u**Syntax**

gtp-u

Context

config>mobile>pdn>signaling

Description

This command configures the context to configure GTP-U-related parameters.

GTP-U is a IP-based tunneling protocol that permits many tunnels between each set of end points. When used in the UMTS, each subscriber can have one or more tunnel, one for each PDP context they have active. Separate tunnels are possible for specific connections with different QoS requirements.

The separate tunnels are identified by a Tunnel Endpoint Identifier (TEID) in the GTP-U messages, which is generally a dynamically allocated random number. If the number is encrypted quality then it provides some security against certain attacks. Even so, the requirement of the 3GPP standard is that all GTP traffic, including user data should be sent within secure private networks, not directly connected to the Internet.

**NOTE:**

This command requires a PDN shutdown.

5.2.1.39.10 profile

Syntax

```
profile profile-name  
no profile
```

Context

```
config>mobile>pdn>signaling>gtp-u
```

Description

This command specifies the GTP-U profile name. The GTP-U profile provides GTP-U signaling related configuration.

The no form of this command removes the profile name from the configuration.



NOTE:

This command requires a PDN shutdown.

Parameters

profile-name
specifies the GTP-U profile name

5.2.1.39.11 interface

Syntax

```
interface [router router-instance] interface-name  
no interface
```

Context

```
config>mobile>pdn>signaling
```

Description

This command specifies the virtual router to which the default signaling interface belongs.

The no form of this command removes the router instance or interface name from the configuration.



NOTE:

This command requires a PDN shutdown.

Parameters

router-instance

specifies the router instance

Values	router-name service-id	
–	router-name	Base
–	service-id	1 to 2147483647

Default: Base

interface-name

specifies an existing interface name, up to 32 characters

5.2.1.39.12 mipv4

Syntax

mipv4
no mipv4

Context

config>mobile>pdn>signaling

Description

This command configures to enable/disable SCTP multi-homing for Diameter signaling interfaces.
The no form of this command sets the value to the default.



NOTE:

This command requires a PDN shutdown.

Default

no mipv4

5.2.1.39.13 profile

Syntax

`profile profile-name`

`no profile`

Context

`config>mobile>pdn>signaling>mipv4`

Description

This command specifies the MIPv4 profile name. The profile provides signaling related configuration.

The no form of this command removes the profile name from the configuration.



NOTE:

This command requires a PDN shutdown.

Parameters

profile-name

specifies the MIPv4 profile name

5.2.1.39.14 multi-homing

Syntax

`[no] multi-homing`

Context

`config>mobile>pdn>signaling`

Description

This command configures to enable/disable SCTP multi-homing for Diameter signaling interfaces.

The no form of this command sets the value to default.



NOTE:

- Current multi-homing support requires primary and secondary interfaces to be in the same VPRN.
- This command requires a PDN shutdown.

Default

no multi-homing

5.2.1.39.15 interface

Syntax

interface [router *router-instance*] *interface-name*
no interface

Context

config>mobile>pdn>signaling>multi-homing

Description

This command configures Diameter secondary interface parameters and the virtual router in which the Diameter connections will be established.

The no form of this command removes the parameter from the configuration.



NOTE:

This command requires a PDN shutdown.

Parameters

router-instance

specifies the router instance

Values	router-name service-id		
–	router-name	Base	

– service-id 1 to 2147483647

Default: Base

interface-name

specifies an existing interface name, up to 32 characters and must start with a letter

5.2.1.40 STa PDN command descriptions

5.2.1.40.1 sta

Syntax

sta

Context

config>mobile>pdn

Description

This command enables the context to configure STa.

5.2.1.40.2 diameter-peer-group-list

Syntax

diameter-peer-group-list *group-list-name*

no diameter-peer-group-list

Context

config>mobile>pdn>sta

Description

This command configures the Diameter group to be used for Diameter based EAP Authentication.



NOTE:

- RADIUS or Diameter can be configured for the STa reference point interface, but they cannot be configured at the same time.
- This command requires a PDN shutdown.

Parameters

group-list-name

specifies the name of a Diameter group up to 32 characters maximum

5.2.1.40.3 interface

Syntax

interface

Context

config>mobile>pdn>sta

Description

This command enables the context to configure interface parameters.



NOTE:

This command requires a PDN shutdown.

5.2.1.40.4 diameter

Syntax

diameter [router *router-instance*] *interface-name*

no diameter

Context

config>mobile>pdn>sta>interface

Description

This command configures the interface to be used for STa.

The no form of this command removes the parameter from the configuration.

**NOTE:**

This command requires a PDN shutdown.

Parameters

router-instance

specifies the name of a router instance

Values	router-name service-id	
–	router-name	Base
–	service-id	1 to 2147483647

Default: Base

interface-name

specifies an existing interface name, up to 32 characters

5.2.1.40.5 multi-homing

Syntax

[no] multi-homing

Context

config>mobile>pdn>sta>interface

Description

This command enables and disables multi-homing for the interface. When multi-homing is enabled, a secondary Diameter interface can also be specified.

The no form of this command reverts to the default.

**NOTE:**

- Current multi-homing support requires primary and secondary interfaces to be in the same VPRN.

- This command requires a PDN shutdown.

Default

no multi-homing

5.2.1.40.6 diameter

Syntax

```
diameter [router router-instance] secondary-interface-name  
no diameter
```

Context

```
config>mobile>pdn>sta>interface>multi-homing
```

Description

This command configures Diameter secondary interface parameters and the virtual router in which the Diameter connection will be established by this Diameter policy.

The no form of this command removes the parameter from the configuration.



NOTE:

- Current multi-homing support requires primary and secondary interfaces to be in the same VPRN.
- This command requires a PDN shutdown.

Parameters

router-instance

specifies the virtual router name or service ID

Values	router-name service-id	
–	router-name	Base
–	service-id	1 to 2147483647

Default: Base

secondary-interface-name

specifies an interface name, up to 32 characters maximum

5.2.1.40.7 radius-server-group

Syntax

radius-server-group *radius-group*

no radius-server-group

Context

config>mobile>pdn>sta>

Description

This command configures the RADIUS group to be used for RADIUS based EAP Authentication.



NOTE:

- RADIUS or Diameter can be configured for the STa reference point interface, but they cannot be configured at the same time.
- This command requires a PDN shutdown.

Parameters

radius-group

specifies the name of a RADIUS group, up to 32 characters maximum

5.2.1.40.8 retry-count

Syntax

retry-count *value*

no retry-count

Context

config>mobile>pdn>sta>

Description

This is the number of times the system will retry to send an STa message before declaring a failed attempt.

The no form of this command reverts to the default.

Parameters

value

specifies the number of retries

Values: 0 to 8

Default: 3

5.2.1.40.9 sta-profile

Syntax

```
sta-profile profile-name
```

```
no sta-profile
```

Context

```
config>mobile>pdn>sta
```

Description

This command specifies an STa profile name to associate with the gateway.

The no form of this command removes the profile name from the configuration.

Parameters

profile-name

specifies an STa profile name, up to 32 characters maximum

5.2.1.40.10 transaction-timer

Syntax

```
transaction-timer value
```

```
no transaction-timer
```

Context

config>mobile>pdn>sta

Description

This command sets the maximum amount of time in seconds that the node waits for the STa peer to respond to an STa message.

The no form of this command reverts to the default.

Parameters

value

time in seconds

Values: 1 to 180

Default: 5

5.2.1.41 SWm PDN command descriptions

5.2.1.41.1 swm

Syntax

swm

Context

config>mobile>pdn

Description

This command enables the context to configure SWm parameters.

5.2.1.41.2 aar-call-flow

Syntax

[no] aar-call-flow

Context

config>mobile>pdn>swm

Description

This command enables support of the ePDG-initiated AAR Diameter message toward AAA/DRA.

The no form of this command disables support of the ePDG-initiated AAR Diameter message toward AAA/DRA.

The ePDG can initiate an AAR message if the ePDG detects a change of the outer IP address in the UE (MOBIKE), to update the new UE IP address toward the AAA and also to retrieve the latest WLAN location information (when applicable).

Default

no aar-call-flow

5.2.1.41.3 activate-aaa-failure-indication

Syntax

```
activate-aaa-failure-indication {enable | disable}
```

```
no activate-aaa-failure-indication
```

Context

```
config>mobile>pdn>swm
```

Description

This command enables or disables the AAA failure indication feature.

When this command is enabled, the AAA-Failure-Indication AVP specified by the SWm profile command `swm-avp-options` is automatically set to the Include value.

When this command is disabled, the AAA-Failure-Indication AVP specified by the SWm profile command `swm-avp-options` is automatically set to the Exclude value.

The no form of this command disables the functionality and sets the AAA-Failure-Indication AVP specified by the SWm profile command `swm-avp-options` to the Exclude value.

Default

no activate-aaa-failure-indication

Parameters

enable

the AAA-Failure-Indication AVP specified by the SWm profile command `swm-avp-options` is automatically set to the Include value

disable

the AAA-Failure-Indication AVP specified by the SWm profile command `swm-avp-options` is automatically set to the Exclude value

5.2.1.41.4 alt-peer-rar-asr

Syntax

[no] `alt-peer-rar-asr`

Context

`config>mobile>pdn>swm`

Description

This command enables the gateway to accept RAR and/or ASR messages from an alternate peer (other than the primary and the secondary). When enabled, an RAA message or an ASA message, will be sent back to the peer that initiated the RAR or the ASR respectively. The IP-CAN session will continue to use the primary and the secondary for the SWm session (for example, any STR or AAR for the session will be sent to the destination peer from which the last DEA was received).

The no form of this command reverts to the default.

Default

`no alt-peer-rar-asr`

5.2.1.41.5 dia-routing-table

Syntax

`dia-routing-table` *idi-realm-id* *idi-realm-string* `diameter-peer-group-list` *list-name*
`no dia-routing-table` *idi-realm-id*

Context

`config>mobile>pdn>swm`

Description

This command configures the correlation of specific IDi realms with destination Diameter servers.

**NOTE:**

The AAA servers assigned under `diameter-peer-group-list` follow the same rules for Destination-Realm AVP creation as those described for the AAA Destination-Realm AVP CLI.

Default

`no dia-routing-table`

Parameters***idi-realm-id***

specifies the index number

Values: 1 to 512

idi-realm-string

specifies the IDi realm, up to 255 characters

list-name

specifies an existing Diameter peer group list, up to 32 characters

5.2.1.41.6 swm-error-mapping-profile

Syntax

`swm-error-mapping-profile profile-name`

`no swm-error-mapping-profile`

Context

`config>mobile>pdn`

Description

This command creates an SWm error mapping profile.

Parameters***profile-name***

specifies a profile name, up to 32 characters

5.2.1.41.7 interface

Syntax

```
swm
```

Context

```
config>mobile>pdn>swm
```

Description

This command enables the context to configure SWm interface parameters.

**NOTE:**

This command requires a PDN shutdown.

5.2.1.41.8 diameter

Syntax

```
diameter [router router-instance] interface-name
```

```
no diameter
```

Context

```
config>mobile>pdn>swm>interface
```

```
config>mobile>pdn>swm>interface>multi-homing
```

Description

This command configures the Diameter interface for the SWm interface or the secondary Diameter interface for multi-homing and the virtual router in which the Diameter connections will be established by this Diameter policy.

**NOTE:**

- Current multi-homing support requires primary and secondary interfaces to be in the same VPRN.
- This command requires a PDN shutdown.

Parameters

router-instance
specifies the router instance

Values	router-name service-id	
–	router-name	Base
–	service-id	1 to 2147483647

Default: Base

interface-name
specifies an existing interface name, up to 32 characters and must start with a letter

5.2.1.41.9 multi-homing

Syntax

[no] multi-homing

Context

config>mobile>pdn>swm>interface

Description

This command enables multi-homing for the interface. When multi-homing is enabled, a secondary Diameter interface can also be specified.

The no form of this command reverts to the default (disabled).



NOTE:

- Current multi-homing support requires primary and secondary interfaces to be in the same VPRN.
- This command requires a PDN shutdown.

Default

no multi-homing

5.2.1.41.10 diameter

Syntax

```
diameter [router router-instance] secondary-interface-name  
no diameter
```

Context

```
config>mobile>pdn>swm>interface>multi-homing
```

Description

This command configures the Diameter secondary interface parameters and the virtual router in which the Diameter connections will be established by this Diameter policy.

The no form of this command removes the parameter from the configuration.



NOTE:

- Current multi-homing support requires primary and secondary interfaces to be in the same VPRN.
- This command requires a PDN shutdown.

Parameters

router-instance

specifies the virtual router name or service id

Values	router-name service-id	
–	router-name	Base
–	service-id	1 to 2147483647

Default: Base

secondary-interface-name

specifies the interface name, up to 32 characters and must start with a letter

5.2.1.41.11 peer-group

Syntax

```
peer-group group-id primary-diameter-peer profile-name [secondary-diameter-peer  
profile-name] [priority priority-value]
```

```
no peer-group group-id
```

Context

```
config>mobile>pdn>swm
```

Description

This command specifies the list of Diameter peer groups to create a set of primary and secondary peers for the SWm reference point for the ePDG. A peer-group may be a pair of AAA servers which are specified in primary and secondary order, so the primary server is normally used and the secondary server is used in case the primary fails. The peer group specified for the SWm reference point is limited to a primary Diameter peer profile and secondary Diameter peer profile, which in turn each refer to a single AAA server. For information about Diameter peer profiles, see [Sig Rate Limit Profile Command Hierarchy](#).

To provide load sharing amongst multiple servers, the operator can configure multiple peer groups. If multiple peer groups are specified, the system will use round-robin selection to choose a peer group for each new SWm session.

Each peer group can also be associated with a priority, to enable the ePDG to choose servers from a higher-priority peer group first and failover to peer groups with a lower priority.



NOTE:

This command requires a PDN shutdown.

Parameters

group-id

specifies the ID of the peer group

Values: 1 to 10

profile-name

specifies the name of an existing primary Diameter peer profile, up to 32 characters

For information about Diameter peer profiles, see [Sig Rate Limit Profile Command Hierarchy](#).

secondary-diameter-peer *profile-name*

specifies the name of an exiting secondary Diameter peer profile, up to 32 characters

priority-value

specifies the priority for the peer group

The ePDG will choose servers from a higher-priority peer group first and will failover to peer groups with a lower priority.

Values: 1 to 10 (highest is 1, lowest is 10)

Default: 10

5.2.1.41.12 **retry-count**

Syntax

`retry-count count`

`no retry-count`

Context

`config>mobile>pdn>swm`

Description

This command specifies the number of times the system will retry to send an SWm message before declaring a failed attempt.

Parameters

count

specifies the number of attempt, as an integer

Values: 0 to 8

Default: 3

5.2.1.41.13 **swm-profile**

Syntax

`swm-profile swm-profile-name`

`no swm-profile`

Context

`config>mobile>pdn>swm`

Description

This command specifies an SWm profile name to associate with the gateway.

The no form of this command removes the profile name from the configuration.

Default

no swm-profile

Parameters

swm-profile-name

specifies the profile name, up to 32 characters

5.2.1.41.14 transaction timer

Syntax

transaction-timer *value*

no transaction-timer

Context

config>mobile>pdn>swm

Description

This command sets maximum amount of time the node waits for the SWm peer to respond to an SWm message before reattempting the transaction. The value is expressed in seconds.

The no form of this command removes the configuration.

Parameters

value

specifies the maximum amount of time, in seconds, the node waits for a peer to respond before trying another peer

Values: 1 to 180 (increments in 1 s)

Default: 5

5.2.1.41.15 swm-error-mapping-profile

Syntax

```
swm-error-mapping-profile profile-name  
no swm-error-mapping-profile
```

Context

```
config>mobile>pdn
```

Description

This command creates an SWm error mapping profile.

Parameters

profile-name

specifies a profile name, up to 32 characters

5.2.1.42 SWu PDN command descriptions

5.2.1.42.1 swu

Syntax

```
swu [sig-ref-point-name]
```

Context

```
config>mobile>pdn
```

Description

This command enables the context to configure SWu interface parameters.

The system supports up to 16 SWu interfaces.



NOTE:

This command requires a PDN shutdown.

Parameters

sig-ref-point-name
specifies the name to be associated with this reference point, up to 32 characters

5.2.1.42.2 interface

Syntax

```
interface [router router-instance] interface-name  
no interface
```

Context

```
config>mobile>pdn>swu
```

Description

This command configures the interface to be used with the SWu reference point.



NOTE:

This command requires a PDN shutdown.

Parameters

router-instance
specifies the router instance

Values	router-name service-id	
–	router-name	Base
–	service-id	1 to 2147483647

Default: Base

interface-name
specifies an existing interface name, up to 32 characters

5.2.1.42.3 ipsec-profile

Syntax

```
[no] ipsec-profile profile-name
```

Context

```
config>mobile>pdn>swu
```

Description

This command sets the ipsec-profile to be used for this reference point.



NOTE:

This command requires a PDN shutdown.

Parameters

profile-name

specifies the name of the profile, up to 32 characters

5.2.1.42.4 swu-ipsec-lockout

Syntax

```
swu-ipsec-lockout
```

Context

```
config>mobile>pdn
```

Description

This command enables the context to configure IPsec authentication lockout functionality for the ePDG.

5.2.1.42.5 backoff-timer

Syntax

```
backoff-timer [timer-value timer-value] [timer-unit timer-unit]
```

```
no backoff-timer
```

Context

config>mobile>pdn>swu-ipsec-lockout

Description

This command specifies the value for the backoff timer that the ePDG can send to the UE along with the IKE error code.

The no form of this command deactivates the backoff timer from the IPsec client lockout functionality.

Default

no backoff-timer

Parameter

timer-value

specifies the value of the backoff timer

Values: 0 to 31

Default: 5

timer-unit

specifies the unit of the timer value unit

Values	deact	specifies that the timer is deactivated
–	2-s	specifies that the timer is incremented in multiples of 2 seconds
–	30-s	specifies that the timer is incremented in multiples of 30 seconds
–	1-min	specifies that the timer is incremented in multiples of 1 minute
–	10-min	specifies that the timer is incremented in multiples of 10 minutes
–	1-h	specifies that the timer is incremented in multiples of 1 hour

–	10-h	specifies that the timer is incremented in multiples of 10 hours
–	320-h	specifies that the timer is incremented in multiples of 320 hours

Default: 1-min



NOTE:

- A value of 0 indicates that the UE may retry the authentication procedure to an ePDG from the same PLMN.
- A value of deactivated indicates that the UE should not retry the authentication procedure to the same ePDG until it is switched off or the UICC containing the USIM is removed.

5.2.1.42.6 configuration

Syntax

```
configuration count count time time lockout lockout-time
no configuration
```

Context

```
config>mobile>pdn>swu-ipsec-lockout
```

Description

This command specifies the number of permitted authentication attempts, the authentication time period, and the lockout time period to be applied when the IPsec lockout functionality is enabled for the ePDG.

The no form of this command reverts to the default.

Default

no configuration

Parameters

count

mandatory parameter that specifies the maximum number of failed authentication attempts that are permitted for the specified authentication time period

Values: 1 to 64

Default: 3

time

mandatory parameter that specifies the period of time in minutes, during which the number of failed authentication attempts (specified by count) can be made before the user is locked-out

Values: 1 to 60

Default: 5

lockout-time

mandatory parameter that specifies the period of time in minutes, that the user will be locked-out from further authentication attempts

Values: 1 to 1440

Default: 10

5.2.1.42.7 function

Syntax

```
function {enable /disable}
```

```
no function
```

Context

```
config>mobile>pdn>swu-ipsec-lockout
```

Description

This command enables and disables the configured IPsec lockout functionality for the ePDG.

The no form of this command reverts to the default.

Default

no function

Parameters

enable

enables the IPsec lockout functionality for the ePDG

disable

disables the IPsec lockout functionality for the ePDG

Default: disable

5.2.1.42.8 swu-throttling

Syntax

```
swu-throttling [level{best-effort | minimal | medium | aggressive}]  
no swu-throttling
```

Context

```
config>mobile>pdn
```

Description

This command allows for the activation/deactivation of `swu-throttling` and the configuration of the throttling level functionality on the ePDG.

Parameters

level

allows for the configuration of the throttling level once `swu-throttling` is enabled

Values	best-effort	allows the IKE to drop IKE messages from the queue when there is 100% IKE Load and there is a delay of > 1500ms Best-effort only drops IKE messages from the queue
—		—
—	minimal	allows IKE tasks to stay under a heavy load for a longer period before entering a higher congestion level and thus allows higher latency for unprocessed messages
—	medium	allows IKE tasks to compromise between the IKE load and the latency monitored in the system

–	aggressive	allows the IKE to progress to a higher congestion level even before reaching 0% IKE idle time and tries to minimize unprocessed message latency to the lowest amount possible
---	------------	---

Default: best effort

5.2.1.43 SWw PDN command descriptions

5.2.1.43.1 sww

Syntax

[no] sww [*ref-point-name*]

Context

config>mobile>pdn

Description

This command configures SWw interface parameters. The sig-ref-point-name is used to configure multiple SWw interfaces for the TWAG.

The system will support up to 16 SWw interfaces.



NOTE:

This command requires a PDN shutdown.

Parameters

ref-point-name

specifies a name to associate with this reference point

5.2.1.43.2 interface

Syntax

interface

Context

```
config>mobile>pdn>sww
```

Description

This command enables the context to configure SWw interface parameters.



NOTE:

This command requires a PDN shutdown.

5.2.1.43.3 sww-acct

Syntax

```
sww-acct [router router-instance] interface-name port port secret secret [hash | hash2]  
no sww-acct
```

Context

```
config>mobile>pdn>sww>interface
```

Description

This command configures the interface to be used for SWw RADIUS accounting signaling toward the WLAN APs.



NOTE:

This command requires a PDN shutdown.

Parameters

router-instance

specifies the name of a router instance

Values	router-name service-id	
–	router-name	Base

– service-id 1 to 2147483647

Default: base

interface-name

specifies an existing interface name that must start with a letter, up to 32 characters

port

specifies the UDP port

Values: 1 to 65535

secret

specifies the shared secret key to be used with the WLAN AP

hash | hash2

specifies if the key is entered in an encrypted form

If the hash parameter is not used, the key is assumed to be in a non-encrypted, clear text form. For security, all keys are stored in an encrypted format.

5.2.1.43.4 sww-auth

Syntax

```
sww-auth [router router-instance] interface-name port port secret secret [hash | hash2]
```

```
no sww-auth
```

Context

```
config>mobile>pdn>sww>interface
```

Description

This command configures the interface to be used for SWw RADIUS authentication signaling toward the WLAN APs.



NOTE:

This command requires a PDN shutdown.

Parameters

router-instance

specifies the name of a router instance

Values	router-name service-id	
–	router-name	Base
–	service-id	1 to 2147483647

Default: base

interface-name

specifies an existing interface name that must start with a letter, up to 32 characters

port

specifies the UDP port

Values: 1 to 65535

secret

specifies the shared secret key to be used with the WLAN AP

hash | hash2

specifies if the key is entered in an encrypted form

If the hash parameter is not used, the key is assumed to be in a non-encrypted, clear text form. For security, all keys are stored in an encrypted format.

5.2.1.43.5 sww-coa

Syntax

```
sww-coa [router router-instance] interface-name port port secret secret [hash | hash2]  
no sww-coa
```

Context

```
config>mobile>pdn>sww>interface
```

Description

This command configures RADIUS CoA to be accepted when portal-based authentication (open SSID) is associated with the configured SWw reference point.



NOTE:

This command requires a PDN shutdown.

Parameters

router-instance

specifies the name of a router-instance

Values	router-name service-id	
–	router-name	Base
–	service-id	1 to 2147483647

Default: base

interface-name

specifies an existing interface name that must start with a letter, up to 32 characters

port

specifies the UDP port

Values: 1 to 65535

secret

specifies the shared secret key to be used with the RADIUS CoA server

hash | hash2

specifies if the key is entered in an encrypted form. If the hash parameter is not used, the key is assumed to be in a non-encrypted, clear text form. For security, all keys are stored in an encrypted format.

5.2.1.43.6 sww-rad-disconnect

Syntax

```
sww-rad-disconnect secret secret [hash | hash2]
no sww-rad-disconnect
```

Context

```
config>mobile>pdn>sww>interface
```

Description

This command specifies if the RADIUS disconnect will be sent to WiFi WLAN APs / WLCs, when a UE is disconnected on TWAG.



NOTE:

This command requires a PDN shutdown.

Default

```
no sww-rad-disconnect
```

Parameters

secret

specifies the shared secret key to be used with the WLAN AP

hash | hash2

specifies if the key is entered in an encrypted form

If the hash parameter is not used, the key is assumed to be in a non-encrypted, clear text form. For security, all keys are stored in an encrypted format.

5.2.1.43.7 sww-tunnel

Syntax

```
sww-tunnel [router router-instance] interface-name [connect-method connect-method]
```

```
no sww-tunnel
```

Context

```
config>mobile>pdn>sww>interface
```

Description

This command configures the SWw tunnel interface.



NOTE:

This command requires a PDN shutdown.

Parameters

router-instance

specifies the name of a router-instance

Values	router-name service-id	
–	router-name	Base
–	service-id	1 to 2147483647

Default: base

interface-name

specifies an existing interface name that must start with a letter, up to 32 characters

connect-method

specifies the connection method used on the SWw interface

Values	gre-tunnel	specifies that each AP/WLC is connected over a L2overGRE tunnel (soft GRE)
–	l2-access	specifies that each AP/WLC is directly connected on an L2 aggregated network

Default: gre-tunnel



NOTE:

- The L2-access requires one-to-one mapping between VPRN and the SWw reference point. For the L2-access method only one SWw reference point is allowed to be configured to the VPRN that holds the physical interface configuration under context config>service>vprn>interface toward the WLAN APs/L2 switch.
- For L2-access, the VLAN ID must be unique across all the physical ports/ interfaces configured under context config>service>vprn>interface

toward the WLAN APs/WLCs. For the L2-access method, each configured physical interface configured under context config>service>vprn must use different VLAN IDs.

- This command requires a PDN shutdown.

5.2.1.43.8 signaling

Syntax

signaling

Context

config>mobile>pdn>sww

Description

This command enables the context to configure signaling interface parameters.



NOTE:

This command requires a PDN shutdown.

5.2.1.43.9 sww-radius-profile

Syntax

sww-radius-profile *profile-name*

no sww-radius-profile

Context

config>mobile>pdn>sww>signaling

Description

This command specifies the SWw RADIUS profile tied to this SWw reference point name. The no form of this command removes the profile name from the configuration.



NOTE:

This command requires a PDN shutdown.

Default

no sww-radius-profile

Parameters

profile-name

specifies a profile name, up to 32 characters

5.2.1.43.10 sww-tunnel-profile

Syntax

sww-tunnel-profile *profile-name*

no sww-tunnel-profile

Context

config>mobile>pdn>sww>signaling

Description

This command specifies the SWW tunnel profile tied to this SWw reference point name. The no form of this command removes the profile name from the configuration.



NOTE:

This command requires a PDN shutdown.

Default

no sww-tunnel-profile

Parameters

profile-name

specifies a profile name, up to 32 characters

5.2.1.44 Sx-N4 PDN command descriptions

5.2.1.44.1 sx-n4

Syntax

[no] sx-n4 [*ref-point-name*]

Context

config>mobile>pdn

Description

This command enables the context to configure the Sx/N4 interface parameters.



NOTE:

This command requires a PDN shutdown.

Parameters

ref-point-name

specifies a name to associate with this reference point, up to 32 characters

5.2.1.44.2 enforced-pfcp-association-list

Syntax

[no] enforced-pfcp-association-list

Context

config>mobile>pdn>sx-n4

Description

This command enables the PFCP to accept only PFCP association setup from a peer that is configured in the PFCP association list (using the *pfcp-association-list* command).

The no form of this command reverts to the default, which accepts and dynamically adds PFCP association requests from a peer that is not configured in the PFCP association list.

Default

no enforced-pfcp-association-list

5.2.1.44.3 interface

Syntax

```
interface
```

Context

```
config>mobile>pdn>sx-n4
```

Description

This command enables the context to configure GTP-U and Packet Forwarding Control Protocol (PFCP) parameters for the Sx/N4 interface.

**NOTE:**

This command requires a PDN shutdown.

5.2.1.44.4 gtp-u

Syntax

```
gtp-u [router router-instance] interface-name [interface-realm realm-name]
```

```
no gtp-u
```

Context

```
config>mobile>pdn>sx-n4>if
```

Description

This command configures the GTP-U profile parameters for the PFCP (Sx/N4) interface.

The no form of this command removes the parameters from the configuration.

**NOTE:**

This command requires a PDN shutdown.

Parameters

router-instance

specifies the router instance, up to 32 characters

interface-name

specifies an interface name, up to 32 characters

realm-name

specifies the associated realm name of the network instance that has the PFCP GTP-U interface on the UP function, and is communicated in the Network-Instance IE part of the PFCP Session Establishment/Modification process



NOTE:

Only the following characters are permitted:

- alphabetic characters (A-Z and a-z)
- digits (0-9)
- hyphen (-)
- dot (.)

5.2.1.44.5 pfcf

Syntax

`pfcf [router router-instance] interface-name`

`no pfcf`

Context

`config>mobile>pdn>sx-n4>if`

Description

This command configures the Packet Forwarding Control Plane (PFCP) profile parameters for the Sx/N4 interface.

The no form of this command removes the parameter from the configuration.



NOTE:

This command requires a PDN shutdown.

Parameters

router-instance

specifies the router instance, up to 32 characters

interface-name

specifies an interface name, up to 32 characters

5.2.1.44.6 pfcf-association-list

Syntax

pfcf-association-list *list-name*

no pfcf-association-list

Context

config>mobile>pdn>sx-n4

Description

This command links the PFCF association peer list to the PDN (configured as CP or UP function using the mobile>config>pdn>instance-type command).

If the gateway is configured as a CP function, the command links the PFCF association peer list of the UP function peers that are to be associated with this Sx/N4 interface.

If the gateway is configured as a UP function, the command links the PFCF association peer list of the CP function peers that are to be associated with this Sx/N4 interface.



NOTE:

- If sessions exist on any peers in the specified PFCF association peer list, you must shutdown the PDN before removing or replacing the list.
- You can modify an existing PFCF association peer list on the fly (without a shutdown), to add or remove peers that do not have any sessions.

Parameters

list-name

specifies the name of the PFCF association peer list defined in the config>mobile>profile>pfcf context, up to 32 characters

5.2.1.44.7 signaling

Syntax

signaling

Context

config>mobile>pdn>sx-n4

Description

This command enables the context to configure the signaling parameters of the Sx/N4 interface.

**NOTE:**

This command requires a PDN shutdown.

5.2.1.44.8 pfcp

Syntax

pfcp

Context

config>mobile>pdn>sx-n4>signaling

Description

This command enables the context to configure PFCP signaling parameters.

**NOTE:**

This command requires a PDN shutdown.

5.2.1.44.9 profile

Syntax

profile *profile-name*

no profile

Context

```
config>mobile>pdn>sx-n4>signaling>pfcf
```

Description

This command configures the PFCF profile for use in the Sx/N4 interface.

The no form of this command removes the association.



NOTE:

This command requires a PDN shutdown.

Default

no profile

Parameters

profile-name

specifies a policy base profile name, up to 32 characters

5.2.1.45 T6c PDN command descriptions

5.2.1.45.1 t6c

Syntax

```
[no] t6c [ref-point-name]
```

Context

```
config>mobile>pdn
```

Description

This command enables the context to configure the T6c interface, which is the reference point between the PGW (PCEF) and the SCEF.

The no form of this command removes the reference point.



NOTE:

This command requires a PDN shutdown.

Parameters

ref-point-name

specifies a reference point name, up to 32 characters

5.2.1.45.2 http-port

Syntax

`http-port port-number`

`no http-port`

Context

`config>mobile>pdn>t6c`

Description

This command configures the HTTP port number for incoming requests.



NOTE:

This command requires a PDN shutdown.

Parameters

port-number

specifies the port number

Values: 80 | 8080

Default: 80

5.2.1.45.3 interface

Syntax

`interface interface-name [router router-instance]`

`no interface`

Context

`config>mobile>pdn>t6c`

Description

This command configures the virtual router in which the HTTP connections are accepted by this T6c reference point.



NOTE:

This command requires a PDN shutdown.

Parameters

interface-name

specifies an interface name that must start with a letter, up to 32 characters

router-instance

specifies the virtual router name or the service ID

Values	router-name service-id	
–	router-name	Base
–	service-id	1 to 2147483647

Default: Base

5.2.1.45.4 t6c-profile

Syntax

t6c-profile *profile-name*
no t6c-profile

Context

config>mobile>pdn>t6c

Description

This command associates the T6c profile (created in the config>mobile>profile>t6c context) with the T6c interface.

**NOTE:**

This command requires a PDN shutdown.

Parameters

profile-name

specifies the profile name, up to 32 characters

5.2.1.45.5 uri

Syntax

`uri uri-name`

`no uri`

Context

`config>mobile>pdn>t6c`

Description

This command configures the URI for the T6c reference point.

**NOTE:**

This command requires a PDN shutdown.

Parameters

uri-name

specifies the http://hostname/path (URI), up to 180 characters; the http://hostname is optional

Default: `http://localhost/epcpcc/QoS/v1/queryPCRF`

5.2.1.46 Threshold PDN command descriptions

5.2.1.46.1 thresholds

Syntax

thresholds

Context

config>mobile>pdn

Description

This command enables the context to configure monitoring thresholds.

5.2.1.46.2 bearer-cf-failure

Syntax

[no] bearer-cf-failure

Context

config>mobile>pdn>thresholds

Description

This command configures thresholds for bearer call flow failure.

5.2.1.46.3 bearer-cf-success

Syntax

[no] bearer-cf-success

Context

config>mobile>pdn>thresholds

Description

This command configures thresholds for bearer call flow successes.

5.2.1.46.4 bearer-mgmt-limit

Syntax

```
[no] bearer-mgmt-limit
```

Context

```
config>mobile>pdn>thresholds
```

Description

The command is used to configure the objects under the bearer-mgmt-limit for the complete gateway instance.

The high and low threshold values are absolute and configured in the group are checked during the end of each polling-interval. If the measured object thresholds are crossed then trap/alarm is raised at the end of the polling interval.

5.2.1.46.5 bearer-traffic

Syntax

```
[no] bearer-traffic
```

Context

```
config>mobile>pdn>thresholds
```

Description

This command configures thresholds for the bearer traffic.

The no form of this command disables the bearer traffic.

5.2.1.46.6 dl-buff-utilization-threshold

Syntax

```
dl-buff-utilization-threshold threshold  
no dl-buff-utilization-threshold
```

Context

```
config>mobile>pdn>thresholds
```

Description

This command configures the threshold for the downlink buffer utilization alarm. The `tmnxMobGwDlBufUtilAlarm` alarm is raised when the downlink buffer utilization (packets or memory) grows beyond the configured threshold value. The alarm is cleared when the downlink buffer utilization (packets or memory) falls below the configured threshold value.

The threshold can be changed on the fly, which may result in a notification generated for clearing or raising of the alarm.

The no form of this command reverts to the default. It disables downlink buffer utilization monitoring.

Default

no dl-buff-utilization-threshold

Parameters

threshold

specifies the threshold value, as a percentage

Values: 1 to 100

5.2.1.46.7 path-mgmt

Syntax

[no] path-mgmt

Context

config>mobile>pdn>thresholds

Description

This command configures thresholds for path management

The no form of this command disables path management.

5.2.1.46.8 pdn-connections

Syntax

[no] pdn-connections

Context

config>mobile>pdn>thresholds

Description

This command configures the threshold for the sum of all PDN connections from all the APNs with max-pdn-connection and/or min-pdn-connections admission control configured on the system.

The no form of this command disables connection limits.

5.2.1.46.9 counter

Syntax

counter *counter-name* high-threshold *high-threshold* [low-threshold *low-threshold*]

no counter *counter-name*

Context

config>mobile>pdn>threshold>bearer-cf-failure

Description

This command configures the high and low thresholds for each object.

The no form of this command removes the objects from the thresholds configuration.

Parameters

counter-name

specifies keywords for bearer call flow failure objects

Values: sgw-attach | attach | sgw-ded-br | ded-br | inter-sgw-rel | idle-user-paging | intra-inter-sgw-rel | srv-req

high-threshold

specifies the high threshold value

Values: 1 to 100%. Depending on the counter, the threshold value may be an absolute value or a percentage rate.

low-threshold

specifies the low threshold value

Values: 1 to 100%. Depending on the counter, the threshold value may be an absolute value or a percentage rate.

5.2.1.46.10 counter

Syntax

```
counter counter-name high-threshold high-threshold [low-threshold low-threshold]  
no counter counter-name
```

Context

```
config>mobile>pdn>threshold>bearer-cf-success
```

Description

This command configures the high and low thresholds for each object.

The no form of this command removes the objects from the thresholds configuration.

Parameters

counter-name

specifies keywords for bearer call flow success objects

Values: sgw-attach | attach | sgw-ded-br | ded-br | inter-sgw-rel | srv-req | intra-sgw-rel |
sgw-inter-sgw-rel | inter-sgw-idle-rel

high-threshold

specifies the high threshold value

Values: Depending on the counter, the threshold value may be an absolute value or a percentage rate.

low-threshold

specifies the low threshold value

Values: Depending on the counter, the threshold value may be an absolute value or a percentage rate.

5.2.1.46.11 counter

Syntax

```
counter counter-name high-threshold high-threshold [low-threshold low-threshold]  
no counter counter-name
```

Context

```
config>mobile>pdn>threshold>bearer-mgmt-limit
```

Description

This command configures the high and low thresholds for each object.

The no form of this command removes the objects from the thresholds configuration.

Parameters

counter-name

specifies keywords for bearer management limit objects

Values: sgw-br | br | sgw-def-br | def-br | sgw-ded-br | ded-br | ue | act-br | ue-paged

high-threshold

specifies the high threshold value

Values: Depending on the counter, the threshold value may be an absolute value or a percentage rate.

low-threshold

specifies the low threshold value

Values: Depending on the counter, the threshold value may be an absolute value or a percentage rate.

5.2.1.46.12 counter

Syntax

```
counter counter-name high-threshold high-threshold [low-threshold low-threshold]
```

```
no counter counter-name
```

Context

```
config>mobile>pdn>threshold>bearer-traffic
```

Description

This command configures the high and low thresholds for each object.

The no form of this command removes the objects from the thresholds configuration.

Parameters

counter-name

specifies keywords for bearer traffic objects

Values: uplink | downlink | s5-to-sgi | sgi-to-s5 | s8-to-sgi | sgi-to-s8 | gn-to-sgi | sgi-to-gn | gp-to-sgi | sgi-to-gp | s2b-to-sgi | sgi-to-s2b | s2a-to-sgi | sgi-to-s2a | s1u-to-sgi | sgi-to-s1u | s1u-to-s5 | s5-to-s1u | s1u-to-s8 | s8-to-s1u

high-threshold

specifies the high threshold value, delta Mbytes

Values: 1 to 20000000

Depending on the counter, the threshold value may be an absolute value or a percentage rate.

low-threshold

specifies the low threshold value, delta Mbytes

Values: 1 to 20000000

Depending on the counter, the threshold value may be an absolute value or a percentage rate.

5.2.1.46.13 counter

Syntax

```
counter counter-name high-threshold high-threshold [low-threshold low-threshold]
no counter counter-name
```

Context

```
config>mobile>pdn>threshold>path-mgmt
```

Description

This command configures the high and low thresholds for each object.

The no form of this command removes the objects from the thresholds configuration.

Parameters

counter-name

specifies keywords for bearer path management objects

Values: 5-path-fails5-restart | s5-noresp | s5-peers | s8-peers | gx-path-fail | rf-path-fail | s11-path-fail | s11-restart | s11-noresp | s11-peers | s1-u-peers

high-threshold

specifies the high threshold value

Values: Depending on the counter, the threshold value may be an absolute value or a percentage rate.

low-threshold

specifies the low threshold value

Values: Depending on the counter, the threshold value may be an absolute value or a percentage rate.

5.2.1.46.14 counter**Syntax**

```
counter counter-name high-threshold threshold [low-threshold threshold]
```

```
no counter counter-name
```

Context

```
config>mobile>pdn>threshold>pdn-connections
```

Description

This command configures the high and low thresholds of the max-pdn-connections and min-pdn-connections commands.

The no form of this command removes the objects from the thresholds configuration.

Parameters***counter-name***

specifies keywords for PDN connection limits objects

Values: max-pdn-connections | min-pdn-connections | serving-maxpdn-connections | serving-min-pdn-connections

high-threshold

specifies the high threshold value, in percentage rate

Values: 1 to 20000000

The threshold value is a percentage rate.

low-threshold

specifies the low threshold value, in percentage rate

Values: 1 to 20000000

The threshold value is a percentage rate.

5.2.1.46.15 interval

Syntax

`interval minutes`

Context

`config>mobile>pdn>threshold>bearer-cf-failure`
`config>mobile>pdn>threshold>bearer-cf-success`
`config>mobile>pdn>threshold>bearer-mgmt-limit`
`config>mobile>pdn>threshold>bearer-traffic`
`config>mobile>pdn>threshold>path-mgmt`
`config>mobile>pdn>threshold>pdn-connections`

Description

This command configures a common polling-interval for all objects under the group.

Parameters

minutes

specifies the interval

Values: 5 to 60

Default: 5

5.2.1.46.16 shutdown

Syntax

`[no] shutdown`

Context

`config>mobile>pdn>threshold>bearer-cf-failure`
`config>mobile>pdn>threshold>bearer-cf-success`
`config>mobile>pdn>threshold>bearer-mgmt-limit`
`config>mobile>pdn>threshold>bearer-traffic`
`config>mobile>pdn>threshold>path-mgmt`

```
config>mobile>pdn>threshold>pdn-connections
```

Description

This command administratively disables an entity (admin state down). When disabled, an entity does not change, reset, or remove any configuration settings or statistics. The operational state of the entity is disabled as well as the operational state of any entities contained within. Many entities must be shut down before they may be deleted.

Many entities must be explicitly enabled (admin state up) using the `no shutdown` command.



NOTE:

The shutdown command at the APN level only affects new sessions, meaning that operational state of APN related to existing sessions will remain enabled while administrative state will be disabled. The gateway still forwards user traffic for the remaining sessions.

The `no` form of this command administratively enables an entity.

5.2.1.47 Virtual APN selection PDN command descriptions

5.2.1.47.1 virtual-apn-selection

Syntax

```
[no] virtual-apn-selection
```

Context

```
config>mobile>pdn
```

```
config>mobile>pdn>apn
```

Description

This command configures the virtual APN selection criteria.

5.2.1.47.2 no-match-action

Syntax

```
no-match-action action [virtual-apn virtual-apn-name]
```

```
no no-match-action
```

Context

```
config>mobile>pdn>virtual-apn-selection
config>mobile>pdn>apn>virtual-apn-selection
```

Description

This command configures the action in cases where any defined user-range did not match.
The no form of this command reverts to the default.

Default

```
no no-match-action
```

Parameters

action

specifies the action

Values	continue	the session creation can be continued based on the real APN
–	reject	the session creation is rejected

Default: continue

virtual-apn-name

specifies the name for the virtual APN, up to 80 characters; only applicable when the *action* parameter is set to continue

5.2.1.47.3 pco-fallback

Syntax

```
pco-fallback user-name user-name
no pco-fallback
```

Context

```
config>mobile>pdn>virtual-apn-selection
```

```
config>mobile>pdn>apn
```

Description

This command configures the information used as the username in the pre-authentication and authentication requests when this information is expected to be received but it is missing from the PCO/APCO IE.

The no form of this command reverts to the default.

Default

no pco-fallback

Parameters

user-name

specifies the username

Values: msisdn | imsi | apn | fixed



NOTE:

The *fixed* option defines the use of the user-name that is specified by the *auth-fixed* command, as the fallback username used in the access requests when the username is expected to be received but it is missing from the PCO and, or APCO IE.

5.2.1.47.4 pco-fallback-acct

Syntax

```
pco-fallback-acct user-name user-name
```

```
no pco-fallback-acct
```

Context

```
config>mobile>pdn>apn
```

Description

This command configures the information used as the username in the accounting requests when this information is expected to be received but it is missing from the PCO and, or APCO IE.

The no form of this command reverts to the default.

**NOTE:**

This command requires an APN shutdown/no shutdown for change to take effect.

Default

no pco-fallback-acct

Parameters***user-name***

specifies the username

Values: msisdn | imsi | apn | fixed

5.2.1.47.5 prefer-pdn**Syntax**

[no] prefer-pdn

Context

config>mobile>pdn>apn>virtual-apn-selection

Description

This command specifies how the virtual APN selection configuration is evaluated. It only affects on the real APN and it is ignored for an APN used as a virtual APN for the new session.

By default, if `virtual-apn-selection` criteria are configured in the APN context (real APN in the Create Session Request), only those are evaluated. In case not configured at all, the PDN level `virtual-apn-selection` criteria are evaluated, meaning that the APN level configuration has priority.

When this command is enabled and `virtual-apn-selection` criteria are configured in the PDN context, they are evaluated first. In addition, if the `virtual-apn-selection` is configured in the final (real or virtual) APN, the APN level configuration may also be evaluated if the PDN level virtual APN selection criteria evaluation did not result in a match and the `no-match-action` is set to `continue`.

The no form of this command reverts to the default.

Default

no prefer-pdn

5.2.1.47.6 selection-prio

Syntax

```
selection-prio UE type [UE type ... up to 10 max]  
no selection-prio
```

Context

```
config>mobile>pdn>virtual-apn-selection  
config>mobile>pdn>apn>virtual-apn-selection
```

Description

This command configures the priority for selecting the criteria as configured under command `user-range`. Therefore, if more than one user range is defined that would match the same UE, the `selection-prio` command sets the priority for each and how the selection is processed. The priority between all the criteria used in the `user-range` has to be explicitly set.

The `no` form of this command removes the selection priority for the `user-range` command from the configuration.

Default

no selection-prio

Parameters

UE type

selects the first, second, third, fourth, fifth, sixth, seventh, eighth, ninth, and tenth selection for the `user-range` command

Values: `imsi` | `msisdn` | `imei` | `roaming-status` | `cc` | `imeisv-tacsv` | `serving-node-plmn` | `pdn-type` | `rat-type` | `ue-location`

Default: `imsi` (1st) | `msisdn` (2nd) | `imei` (3rd) | `roaming-status` (4th) | `cc` (5th) | `imeisv-tacsv` (6th) | `serving-node-plmn` (7th) | `pdn-type` (8th) | `rat-type` (9th) | `ue-location` (10th)

5.2.1.47.7 selection-type

Syntax

```
selection-type {prefix | full-id}  
no selection-type
```

Context

```
config>mobile>pdn>virtual-apn-selection
config>mobile>pdn>apn>virtual-apn-selection
```

Description

This command specifies that the mapping is based on prefix or full identifier being specified for the start and end identifier, for the user - range command.

The no form of this command removes the selection type for user-range from the configuration.

Parameters

prefix

specifies that the mapping is based on IMSI or MSISDN prefix (an IMSI or MSISDN prefix of up to 11 digits can be specified)

full-id

specifies that the mapping is based on a full identifier
Default: full-id

5.2.1.47.8 user-range

Syntax

```
user-range user-range-name type {[imsi | msisdn | imei-tac] start-ue-id [end-ue-id] |
[roaming-status roaming-status-type] | [cc start-cc start-cc-id [end-cc end-cc-id]] |
[imeisv-tacsvn start-ue-id [end-ue-id] svn svn-value] | [serving-node-plmn list-name] |
[pdn-type ipv4 | ipv6 | ipv4v6] | [rat-type rat-type-list-name] | [ue-location ue-
location-list-name]} {virtual-apn virtual-apn-name | {[append-string suffix] [pre-auth-type
{radius server-group server-group-name [user-name {imsi | msisdn | pco | fixed}] [pco-
password] [password password password-type {fixed | default} [hash | hash2]] | diameter}
[suppress-imsi] ]}}
```

```
no user-range user-range-name
```

Context

```
config>mobile>pdn>virtual-apn-selection
config>mobile>pdn>apn>virtual-apn-selection
```


Description

This command specifies an IMSI, MSISDN, IMEI-Type Allocation Code (TAC), IMEISV-TAC + Software Version Number (SVN), roaming status, Charging Characteristics (CC), serving node PLMN, PDN type, RAT type or UE location mapping to the associated virtual APN.

A combination of two matching criteria is supported. The criteria can be any type of {imsi | msisdn | imei-tac | imeisv-tacsvn}, roaming-status, cc, serving-node-plmn, pdn-type, rat-type, or ue-location. Only one type of {imsi | imei-tac | msisdn | imeisv-tacsvn} can be combined with the other types. The serving-node-plmn criterion can only be combined with pdn-type, rat-type, or ue-location. Both rat-type or ue-location criteria cannot be combined with imei-tac or imeisv-tacsvn.

When the `prefix` option is specified for the `selection-type` command, an IMSI or MSISDN prefix can be specified, up to 15 digits. For IMSI, a minimum of MCC+MNC and up to 15 digits can be specified. Therefore, you can specify from 5 to 15 digits. For MSISDN, a minimum of CC + NDC and up to 15 digits can be specified. Therefore, you can specify 4 to 15 digits.

For IMEI-TAC, the TAC part of the IMEI or IMEISV can be configured with up to eight digits. When the `prefix` option is specified for the `selection-type` command, you can specify from four to eight digits. The `prefix` option has no effect on the `roaming-status`, `cc`, `serving-node-plmn`, `pdn-type`, `rat-type`, or the `ue-location`.

For the `imeisv-tacsvn` keyword, the TAC and SVN parts of the IMEISV are configured using 10 digits, TAC (8 digits) + SVN (2 digits).

When the `full-id` option is specified for the `selection-type` command, the `start-ue-id` and `end-ue-id` must be full identifiers.

The two octet CC values are given in decimal format; for example:

- CC 0x0800 is given as 2048
- CC 0x0400 is given as 1024
- CC 0x0200 is given as 512
- CC 0x0100 is given as 256



NOTE:

The `user-range` command must have at least one of the following options defined: `append-string`, `virtual-apn`, or `pre-auth-type`.

The `no` form of this command removes the configured user range specified by the user range name.

Default

`no user-range`

Parameters

user-range-name

selects the name for the user range, up to 32 characters

type

selects the type of the user range

Values: imsi | msisdn | imei-tac | roaming-status | cc | imeisv-tacsv | serving-node-plmn |
pdn-type | rat-type | ue-location

start-ue-id

selects the start identifier for the user range, 4 to 15 characters

end-ue-id

selects the end identifier for the user range, 4 to 15 characters

roaming-status-type

specifies the roaming status type

Values: roaming | visiting, home

start-cc-id

specifies that the start value is charging characteristics

Values: 1 to 65535

end-cc-id

specifies that the end value is charging characteristics

Values: 1 to 65535

list-name

specifies the serving node PLMN ID which is used for virtual APN selection, up to 32 characters

pdn-type

specifies the PDN type

Values: ipv4 | ipv6 | ipv4v6

rat-type-list-name

specifies the RAT-Type list name that is used for the virtual APN selection, up to 32 characters

ue-location-list-name

specifies the UE location (TAC/RAC) list that is used for the virtual APN selection, up to 32 characters

virtual-apn-name

specifies the name for the virtual APN, up to 80 characters

It cannot be specified with either `append-string` or `pre-auth-type`.

suffix

used in the `config>mobile>pdn` context to add a suffix to the APN

This suffix is taken into account in the virtual APN selection and can co-exist with `pre-auth-type`. If the `pre-auth-type` is not defined, this is only supported in the `config>mobile>pdn` context. If the combined real APN and suffix exceeds the total supported character length for the APN string, the session creation fails.

pre-auth-type

triggers a pre-authentication used to support virtual APNs

Values	radius	specifies that the RADIUS authentication type is used for authentication from an AAA server
–	diameter	specifies that the Diameter authentication type is used for authentication from an AAA server

server-group-name

specifies the name of a RADIUS group, up to 32 characters

user-name

specifies the username used in access requests sent to a AAA server

Values	imsi	sets the User-Name attribute to the IMSI
–	msisdn	sets the User-Name attribute to the MSISDN
–	pco	sets the User-Name attribute to the value received in the PCO
–	fixed	use the name specified by the <code>pre-auth-fixed</code> command

suppress-imsi

specifies that the 3GPP-IMSI attribute is not included

pco-password

specifies to set the password to the value received in the GTP PCO

password

specifies the PAP password used in access requests sent to a RADIUS server, up to 32 characters; up to 54 characters are supported for a hash password and up to 96 characters for a hash2 password

password-type

specifies the password type

Values	fixed	fixed password for overriding the PCO content or providing password with the IMSI, MSISDN or APN username (with PAP)
–	default	default password to be used together with the IMSI, MSISDN or APN username, and when missing from the PCO (with PAP)



NOTE:

If the hash and hash2 parameters are not used for the password, the key is assumed to be in a non-encrypted clear text form. For security purposes, all keys are automatically stored in an encrypted hash form in the configuration. The information saves and uses the configuration defined in the `system>security>hash-control` context.

hash

specifies that key is entered in an encrypted form

hash2

specifies that the key is entered in a more complex encrypted form

svn-value

specifies the SVN

Values: 00 to 99

5.2.2 PDN show command descriptions

5.2.2.1 pdn

Syntax

pdn

Context

show>mobile

Description

This command enables the context to display configured PDN gateway profiles.

5.2.2.2 apn

Syntax

```

apn apn-name statistics [group group-number]
apn apn-name delete-session-statistics [gateway gw-id | group group-number] [summary]
apn apn-name attach-failure-statistics [gateway gw-id | group group-number] [summary]
apn apn-name dns-redirect-statistics [gateway gw-id | group group-number] [summary]
apn apn-name fqdn-group name
apn apn-name gy-ap-statistics
apn apn-name [detail]
apn apn-name apn-override-statistics [gateway gw-id | group group-number] [summary]
apn apn-name pcscf-statistics
apn apn-name gx-ap-statistics
apn apn-name call-flow-stats [gateway gw-id | group group-number] [summary]
apn apn-name l2tp
apn apn-name transport-statistics [imsi imsi | msisdn msisdn]

```

Context

show>mobile>pdn

Description

This command displays APN information, including statistics on the types of PDN sessions by access protocol and PDN type. APN statistics are tracked for real and virtual PDN connections. The virtual APN capability allows the PGW to change the APN from the one received via GTP-C (the "real" APN) to one that is used locally (the "virtual" APN). However a PDN session only counts statistics for the virtual APN to which it has been mapped. Furthermore an APN may be real for some PDN sessions and virtual for other PDNs. To track the usage of real APNs even though they are being mapped to a smaller set of virtual APNs, the PGW keeps statistics for the number of PDN connections that use the APN as a real APN. All other counters in the output are incremented only for those PDNs that use the APN as a virtual APN.

Parameters

apn-name

specifies the APN name

statistics

specifies aggregated statistics

summary

specifies summarized aggregated statistics

group-number

specifies the group number

Values: 1 to 8

delete-session-statistics

specifies delete session statistics

gw-id

specifies the gateway identifier number

Values: 1 to 8

attach-failure-statistics

specifies attach failure statistics

dns-redirect-statistics

specifies DNS redirect statistics

fqdn-group

specifies resolved IP addresses and expiry information

gy-ap-statistics

specifies Gy AP statistics

gx-ap-statistics

specifies Gx Assume Positive (AP) statistics

call-flow-stats

specifies the call flow statistics per APN

detail

specifies detailed APN configuration

pcscf-statistics

specifies P-CSCF statistics

override-statistics

specifies APN override statistics

l2tp

specifies L2TP statistics

transport-statistics

specifies transport-byte-collection statistics per APN

imsi

specifies the IMSI, up to 15 digits

msisdn

specifies the MSISDN, up to 15 digits

Output

The following output examples show APN Statistics.

- [Example output APN configuration](#)
- [Example output APN configuration](#)
- [Example output APN configuration](#)
- [Example output APN detail configuration](#)
- [Example output APN statistics](#)
- [Example output APN statistics for 5G sessions](#)
- [Example output PDN APN group statistics](#)
- [Example output APN statistics for DHCPv6 service](#)
- [Example output APN statistics for eMPS advance priority calls](#)
- [Example output APN DNS redirect statistics](#)
- [Example output APN FQDN group](#)
- [Example output APN override statistics](#)
- [Example output P-CSCF statistics](#)
- [Example output Gx assume positive statistics](#)
- [Example output PDN APN call flow statistics SNMP for PGW and combination gateway](#)
- [Example output PDN APN transport statistics](#)

Example output APN configuration

The following output example shows APN configuration information, and [Show APN configuration output fields](#) describes the output fields.

```
# show mobile-gateway pdn apn "ipd.nokia.1.1.com"
=====
APN Configuration
=====
APN Name           : "ipd.nokia.1.1.com"
APN Admin State    : up
Gateway            : 1
```

Misc

Ethernet

```

=====
DN Type           : ethernet
Service-type      : eline
Uplink Action     : push-push      Uplink Tag           : n7-c-vlan-s-vlan
Downlink Action   : pop-pop

```

```

Reactivation Requested : Disabled
Selection Mode         : subscribed-apn
Del Br React Rq Cause  : session-timeout
Del Br No React Rq Cause: idle-timeout
Idle Timeout (sec)     : Disabled      Session Timeout (sec) : 86400
Idle Timer Dir Uplink  : Enabled       Idle Timer Dir Downlink : Enabled
Cm idle timeout       : Disabled
Call sequence         : local-upf-sel-before-auth
Max PDN Connections   : No limit      Max PDN Conn Action    : alarm
Min PDN Connections   : No limit
Reject Foreign Subs    : Disabled      Guard Timer (sec)      : Disabled
Restriction           : any           Allow Emergency         : Enabled
Emergency Profile     : EmProfile1
NEMO HA               : Disabled      HO Wr PDN IP Stl Ses Del: Enabled
Allow WLAN HO No Ctx  : Disabled      Seamless WLAN HO       : Enabled
Allow Dangl Rule Ded Br : Disabled    Paging Br Binding      : Disabled
Multi UE Bonding       : Disabled
APN Load Control       : Disabled      Ovrld Back-Off Tmr (min): Disabled
Exclude-ARP-list      : arp-list1     IP Pool Exh Back-off Tmr: 5 (min)
Shut Back-Off Tmr (min) : Disabled
DHCPv6-service        : Enabled
PPI Mapping Profile    : none

```

```

APN Network ID       : real
Include Radius Auth   : Disabled
Include Rad Pre Auth  : Disabled
APN NI only          : gx gy n10 n40 n7
APN Overload Control  : Disabled
PCO Fallback          : imsi
PCO Fallback acct.    : none
Dedicated Bearer
  Crt Rtr Count       : 0              Crt Rtr Wait Tmr (min): 0
GTP PDN Threshold
  Max Connections     : Disabled      Max Actv (act/sec)    : Disabled
  Alarm               : Disabled
APN Network ID       : virtual
Hybrid Access
  Bonding Support      : Disabled
  Link Switch Mode
    Uplink
      Histeresis Tm (ms): 500
PCO Options
  DNS Server IPv4
    Primary           : 10.30.1.1
    Secondary          : 10.30.1.2

```



```

Pref DNS Addr Type : none          Link MTU (octets) : 0
Operator PCO       : none
KPI KCI
  ARP Traffic Stats : Inherit      SGW APN Stats      : Disabled
  Enh Traffic Stats : Inherit      QCI APN Stats      : "apn-qci-list"
PFCP
PFCP Translate Tx Prof : none
PFCP UP Inactivity    : Enabled      UP Inactivity Timer(sec): 30
NSSAI
  name: myslice-1     sst : 12      sd : 0x10203
  name: myslice-1     sst : 10      sd : 0x10204
Default S-NSSAI
  Slice Name          : my-default-slice-name
  Slice Service Type  : 12
  Slice Differentiator : 0x10203
-----
Number of entries      : 1
=====

```

Table 16: Show APN configuration output fields

Label	Description
APN Name	The name of the APN being viewed
APN Admin State	The APN administrative state
Gateway	The gateway ID
APN Type	The option that sets the APN/DNN type as small
CP MG-Grp-Id	The MG group that serves UE sessions assigned to a small type APN/DNN
Reactivation Requested	The option to send GTP-C cause #8 "reactivation requested" in Delete Bearer Request messages to disconnect the PDN
Selection Mode	The specified selection mode
Del Br React Rq Cause	The option to enable the PGW to include a reactivation request in the delete bearer request that is sent to the UE for the specified scenarios

Table 16: Show APN configuration output fields (continued)

Label	Description
Del Br No React Rq Cause	The option to enable the PGW to include an indication in the delete bearer request that the PDN reconnection to this APN is not allowed
Idle Timeout (sec)	The length of time, in seconds, a PDP context is allowed to be idle before it is removed by the PGW
Session Timeout (sec)	The length of time, in seconds, that a PDN connection may exist before it is removed by the mobile gateway
Idle Timer Dir Uplink	The idle timer which monitors the uplink traffic
Idle Timer Dir Downlink	The idle timer which monitors the downlink traffic
Cm idle timeout	The length of time, in seconds, a session is allowed to be idle before it is deleted by the SMF
Call sequence	The call-flow sequence for the session
Max PDN Connections	The maximum number of PDN connections for this APN on the gateway
Max PDN Conn Action	The option to send alarms when the maximum PDN connections are reached or to send alarms, but to not accept any new PDN connections when the maximum PDN connections are reached
Min PDN Connections	The minimum number of PDN connections for this APN on the gateway
Reject Foreign Subs	The option to reject Create PDP Context Requests for all foreign subscribers
Guard Timer (sec)	The length of time, in seconds, of the session setup timer on the PGW

Table 16: Show APN configuration output fields (continued)

Label	Description
Restriction	The level of restriction imposed on EPS bearer contexts created to the associated APN
Emergency Profile	The emergency profile name
Allow Emergency	The option to specify the APN as an emergency APN
NEMO HA	The option to enable NEMO support for this APN
HO Wr PDN IP Stl Ses Del	The stale session deletion behavior when the PGW rejects an HO (HI=1) CSReq that is received on the S2a/S2b interfaces with an incorrect PDN IP address
Allow WLAN HO No Ctx	The option to enable the PGW functionality to handle an S2a/S2b Handover Attach Request (HI=1) as a new attach
Seamless WLAN HO	The option to enable seamless handover of a UE that is connected to the PGW/GGSN via a WLAN gateway and moving over to 2G/3G access
Allow Dangling Rule Ded Br	The option to enable dedicated bearers to be created for dynamic rules that are left dangling with a previous QCI/ARP
Paging Br Binding	The option to enable bearer binding during paging for a combined SGW and PGW session
APN Load Control	The option to enable the APN Load Control Information (LCI) IE to be sent to the consumer nodes
Ovrld Back-Off Tmr (min)	The length of time, in minutes, of the PGW back-off timer IE
Exclude-ARP-list	The ARP list excluded from being rejected by the P-GW back-off timer IE

Table 16: Show APN configuration output fields (continued)

Label	Description
Shut Back-Off Tmr (min)	The configured back-off timer IE value sent in the Create Session Response together with the cause code APN Congestion, when the APN is in shutdown state
APN Overload Control	The option to enable the APN Overload Control Information (OCI) IE to be sent to the consumer nodes
PCO Fallback	The option to enable the use of the locally configured username information when this information is missing from the PCO/APCO IE
PCO Fallback acct.	The option to enable the use of the locally configured username in the accounting requests when this information is missing from the PCO/APCO IE
Dedicated Bearer	The Dedicated Bearer context
Crt Rtr Count	The number of retries before the dedicated bearer creation is abandoned until the next active application period
Crt Rtr Wait Tmr (min)	The length of time, in minutes, the gateway will wait before retrying dedicated bearer creation
GTP PDN Threshold	The GTP PDN Threshold context
Max Connections	The maximum number of PDN connections
Max Actv (act/sec)	The maximum number of PDN activations per second
Alarm	The option to send an alarm when the maximum number of connections is reached, or when the maximum number of activations is reached
APN Network ID	The specified APN network ID

Table 16: Show APN configuration output fields (continued)

Label	Description
Hybrid Access	The Hybrid Access context
Bonding Support	The option to enable bonding support for the APN
KPI KCI	The KPI KCI context
ARP Traffic Stats	The option to enable the generation of per ARP traffic statistics
SGW APN Stats	The option to enable this APN to be measured in the per APN statistics on the SGW
Enh Traffic Stats	The option to enable the generation of enhanced traffic statistics for this APN
QCI APN Stats	The option to specify the QoS Class Identifier drop statistics profile name
Number of entries	The number of entries retrieved
PCO Fallback	The fallback APN when the PCO does not contain a valid APN string
Include Radius Auth	The option to apply the APN network ID configuration to RADIUS authentication
Include Rad Pre Auth	—
Primary	The IPv4 address of the primary DNS server
Secondary	The IPv4 address of the secondary DNS server
Multi UE Session	The option to enable Multi UE Session support
PAA Overwrite	—

Table 16: Show APN configuration output fields (continued)

Label	Description
PCO Options	The PCO Options context
Pref DNS Addr Type	The specified DNS addresses sent to the UE
Link MTU (octets)	The maximum number of addresses sent to the PCO or RA
Operator PCO	The option to enable the operator specific PCO container to be sent back to the UE
PFCP Translate Tx Prof	Status of the PFCP translation profile (enabled/disabled) for sending opaque values from the CP to the UP
PFCP UP Inactivity	Status of the UP inactivity timer (enabled/disabled), used for 5GC N4 sessions only
UP Inactivity Timer	Time configured for the PFCP UP inactivity timer
DHCPv6-service	Status of DHCPv6 service configuration (enabled or disabled)
PPI Mapping Profile	The name of the PPI mapping profile

Example output APN configuration

The following output example shows APN configuration information.

```
#show mobile-gateway pdn apn "apn1"
```

```
=====
APN Configuration
=====
```

```
APN Name           : "apn1"
APN Admin State    : down
Gateway            : 1
```

```
Misc
=====
```

```

Reactivation Requested : Disabled
Selection Mode         : subscribed-apn
Del Br React Rq Cause  : none
Del Br No React Rq Cause: none
Idle Timeout (sec)     : Disabled      Session Timeout (sec) : 86400
Idle Timer Dir Uplink  : Disabled      Idle Timer Dir Downlink: Disabled
Max PDN Connections   : No limit       Max PDN Conn Action    : alarm
Min PDN Connections   : No limit
PDN Type Conversion    : Enabled
Reject Foreign Subs    : Disabled      Guard Timer (sec)      : Disabled
Restriction            : any           Allow Emergency         : Disabled
NEMO HA                : Disabled      HO Wr PDN IP Stl Ses Del: Enabled
Allow WLAN HO No Ctx   : Disabled      Seamless WLAN HO       : Enabled
Allow Dangl Rule Ded Br: Disabled      Paging Br Binding      : Disabled
Multi UE Bonding       : Disabled      RAD Case Sens Call St ID: Disabled
APN Load Control       : Disabled      Ovrld Back-Off Tmr (min): Disabled
Exclude-ARP-list       :               IP Pool Exh Back-off Tmr: Disabled
Shut Back-Off Tmr (min): Disabled
Allow always-on PDU Ses: Disabled
Selected DNN           : amf-selected
DNS Snooping           : Disabled
DNS Snooping AA Forward: Disabled
Dhcpv6 Service         : Disabled      Preserve VoLTE          : Disabled
Namf Profile           : (Not Specified)
Vsmf Profile           : (Not Specified)
Nsmf Profile           : (Not Specified)
Ignore Up IP Allocation: False

SSC
  Allowed Ssc Modes     : SSC-1 SSC-2
                        : SSC-3
  Upf Reselection SSC-1 : location      Rej No-Allow Ssc Mode : Disabled
  Upf Reselection SSC-2 : load          Upf Reselection SSC-3 : i-smf
  PDU Sess React Triggr : idle          PFCP UP Inact Timer   : 360 seconds
  UUID Mask Length      : 48 bits

APN Overload Control    : Disabled
PCO Fallback            : none
PCO Fallback Acct       : none

Dedicated Bearer
  Crt Rtr Count         : 0              Crt Rtr Wait Tmr (min): 0

GTP PDN Threshold
  Max Connections       : Disabled      Max Actv (act/sec)    : Disabled

```

```

Alarm                               : Disabled

APN Network ID                      : virtual

Hybrid Access
  Bonding Support                    : Disabled
  Link Switch Mode
    Uplink
      Hysteresis Tm (ms): 500

PCO Options
  Pref DNS Addr Type                : none          Link MTU (octets)      : 0
  Operator PCO                      : none

KPI KCI
  ARP Traffic Stats                 : Inherit        SGW APN Stats              : Disabled
  Enh Traffic Stats                 : Inherit

LADN                                : Disabled

UDM Based Slice Sel                 : Disabled

PCF Selection
  GPSI                             : Included        SUPI                      : Included
  PCF ID                           : use-as-target
  S-NSSAIs                         : Inherit
  DNN                              : type real

upf-acct-profile                    : inherit
upf-select before auth              : Disabled
udm-ctrl-radius-auth               : Disabled

Converged-interface-support
  Converged-n7                      : Inherit        Converged-n10              : Inherit
  Converged-n40                     : Inherit        Converged N7 Gn/Gp        : Inherit
  Converged N40 Gn/Gp               : Inherit

Udm Sdm Profile                    :                Udm Uecm Profile          :
Default 5G QoS Profile             :

Rulebase Selection
  Allow Rulebase                    : none
  AAA                              : Enabled        GX                      : Enabled
  GY                              : Enabled

```



```
-----
Number of entries      : 1
=====
```

Example output APN configuration

The following output example shows APN configuration information.

```
#show mobile-gateway pdn apn "apn.0.1.com"

=====
APN Configuration
=====
APN Name           : "apn.0.1.com"
APN Admin State    : up
Gateway            : 1
APN Type           : small          CP MG-Grp-Id      : 1

Misc
=====
Reactivation Requested : Disabled
Selection Mode         : subscribed-apn
Del Br React Rq Cause  : none
Del Br No React Rq Cause: none
Idle Timeout (sec)     : Disabled      Session Timeout (sec) : 86400
Idle Timer Dir Uplink  : Disabled      Idle Timer Dir Downlink : Disabled
Max PDN Connections   : No limit      Max PDN Conn Action    : alarm
Min PDN Connections   : No limit
PDN Type Conversion    : Enabled
Reject Foreign Subs    : Disabled      Guard Timer (sec)      : Disabled
Restriction           : any            Allow Emergency         : Disabled
NEMO HA               : Disabled      HO Wr PDN IP Stl Ses Del: Enabled
Allow WLAN HO No Ctx  : Disabled      Seamless WLAN HO       : Enabled
Allow Dangling Rule Ded Br : Disabled  Paging Br Binding      : Disabled
Multi UE Bonding      : Disabled      RAD Case Sens Call St ID: Disabled
APN Load Control      : Disabled      Ovrld Back-Off Tmr (min): Disabled
Exclude-ARP-list      :               IP Pool Exh Back-off Tmr: Disabled
Shut Back-Off Tmr (min) : Disabled
Allow always-on PDU Ses : Disabled
Selected DNN          : amf-selected
DNS Snooping          : Disabled
DNS Snooping AA Forward : Disabled
Dhcpv6 Service        : Disabled      Preserve VoLTE         : Disabled
Namf Profile          : (Not Specified)
Vsmf Profile          : (Not Specified)
Nsmf Profile          : (Not Specified)
Ignore Up IP Allocation : False
UDM-SDM-Disabled      : none          UDM-UECM-Disabled      : No
CM Idle Timeout (sec) : Disabled

SSC
  Allowed Ssc Modes    : SSC-1 SSC-2
                      : SSC-3
```

```

Upf Reselection SSC-1 : disabled      Rej No-Allow Ssc Mode : Disabled
Upf Reselection SSC-2 : disabled      Upf Reselection SSC-3 : disabled
PDU Sess React Trigr : immediately   PFCP UP Inact Timer   : 0 seconds
UUID Mask Length      : 0 bits

APN Overload Control   : Disabled
PCO Fallback           : none
PCO Fallback Acct      : none

Dedicated Bearer
  Crt Rtr Count        : 0              Crt Rtr Wait Tmr (min): 0

GTP PDN Threshold
  Max Connections      : Disabled       Max Actv (act/sec)    : Disabled
  Alarm                : Disabled

APN Network ID         : virtual

Hybrid Access
  Bonding Support      : Disabled
  Link Switch Mode
    Uplink
      Hysteresis Tm (ms): 500

PCO Options
  Pref DNS Addr Type   : none           Link MTU (octets)     : 0
  Operator PCO         : none

KPI KCI
  ARP Traffic Stats    : Inherit        SGW APN Stats         : Disabled
  Enh Traffic Stats    : Inherit

LADN                   : Disabled

UDM Based Slice Sel    : Disabled

PCF Selection
  GPSI                 : Included        SUPI                  : Included
  PCF ID               : use-as-target
  S-NSSAIs             : Inherit
  DNN                  : type real

upf-acct-profile        : inherit
upf-select before auth  : Disabled
udm-ctrl-radius-auth    : Disabled

Converged-interface-support
  Converged-n7          : Inherit        Converged-n10         : Inherit
  Converged-n40         : Inherit        Converged N7 Gn/Gp    : Inherit
  Converged N40 Gn/Gp   : Inherit

Udm Sdm Profile        :                Udm Uecm Profile      :
Default 5G QoS Profile :

Rulebase Selection
  Allow Rulebase        : none
  AAA                   : Enabled        GX                     : Enabled

```

```

GY                               : Enabled

Cm idle timeout                  : Disabled

Call sequence                    : none

-----
Number of entries                : 1
=====

```

Example output APN detail configuration

The following output examples show detailed APN configuration information, and *Show APN detailed configuration output fields* describes the output fields.

```

# show mobile-gateway pdn apn "internet" detail
=====
APN Configuration
=====
APN Name           : "internet"
APN Admin State    : down
Gateway            : 1

Misc
=====
Reactivation Requested : Disabled
Selection Mode         : subscribed-apn
Del Br React Rq Cause  : none
Idle Timeout (sec)     : Disabled      Session Timeout (sec) : 86400
Idle Timer Dir Uplink  : Enabled
Max PDN Connections   : No limit       Max PDN Conn Action    : alarm
Min PDN Connections   : No limit
PDN Type Conversion    : Enabled
Reject Foreign Subs   : Disabled        Guard Timer (sec)      : Disabled
Restriction           : any              Allow Emergency         : Disabled
NEMO HA               : Disabled        HO Wr PDN IP Stl Ses Del: Enabled
Allow WLAN HO No Ctx  : Disabled        Seamless WLAN HO       : Enabled
Allow Dangl Rule Ded Br : Disabled      Paging Br Binding       : Disabled
APN Load Control      : Disabled        Ovrld Back-Off Tmr (min): Disabled

APN Overload Control  : Disabled

Dedicated Bearer
  Crt Rtr Count       : 0                Crt Rtr Wait Tmr (min): 0
GTP PDN Threshold
  Max Connections     : Disabled          Max Actv (act/sec)     : Disabled
  Alarm               : Disabled

APN Network ID       : virtual

Hybrid Access
  Bonding Support      : Disabled
  Link Switch Mode
    Uplink
      Histeresis Tm (ms): 2000

```

PCO Options

Pref DNS Addr Type : none Link MTU (octets) : 0
 Operator PCO : none

Block List

Entry : 1
 Group : 1 Pdn Type : ipv4
 Entry : 2
 Group : 2 Pdn Type : ipv6

KPI KCI

ARP Traffic Stats : Inherit SGW APN Stats : Disabled
 Enh Traffic Stats : Inherit QCI APN Stats : "apn-qci-list"

Virtual APN

=====

Pre Auth : radius
 Username : imsi Password Type : none
 Supress Imsi : Disabled

Pre Auth Fixed

Username : "default"

Virtual APN based on PCO: Disabled

Virtual APN selection

Selection Type : full-id Prefer PDN : Disabled
 Selection Priority : imsi msisdn imei-tac roaming-status cc imeisv-tacsvn pdn-type
 No Match Action : continue
 PDN Type : IPv4v6

IP Allocation

=====

PDN Type : ipv4
 PDN Type Prefer IPv6 : Disabled PDN Alloc Type : ran-signaling
 IP Alloc Method : local-pool
 Allow Local : Disabled Extended Rad Alloc : Disabled
 Priority Order : none IP Allocation Trace : Disabled
 Extended HSS Alloc : Disabled Frame Route : Disabled

Router : "Base"
 MG-Group : none

Router Advertisements

Min Adv Interval (sec): 16200 Max Adv Interval (sec): 21600

Authentication

=====

Auth : radius
 Username : imsi Password Type : none
 Supress Imsi : Disabled

Auth Fixed

Username : "default"

L2TP

=====

```

PPP Regeneration      : Disabled
Router                : none
Username              : msisdn      Password Type          : none
ICRQ Call Num Type    : msisdn      Mobile Router         : Disabled
DNS Server Opt        : Included     NBNS Server Opt       : Included
Rx Connect Speed (bps): Disabled     Tx Connect Speed (bps): 0
LCP Conf Req Attr     : Included
Auth Type             : chap-only    MRU Size              : 1500
HSS Static IP         : excluded
Custom-1              : Disabled

```

Accounting

```

=====
Accounting Type       : radius
Username              : imsi        Suppress Imsi         : Disabled
Selective Rad Acct    : Disabled

```

Radius Accounting On-Off

```

Messages Sent        : none        Wait for ACK          : Disabled
Incl Dupl Servers    : Disabled

```

```

Wait Accounting       : Disabled
TWAG Radius Acct Feed : Disabled

```

Charging

```

=====
Charging Profile
Home                  : Inherit     Visiting              : Inherit
Roaming              : Inherit
CC Ignore
Home                  : Inherit     Visiting              : Inherit
Roaming              : Inherit     Any                   : Inherit

CC Reject             : Inherit

```

PFCP

```

=====
PFCP Translate Tx Prof : none
PFCP UP Inactivity     : Enabled    UP Inactivity Timer(sec): 30

```

Policy

```

=====
AMBR Policing         : dl-only     Max AMBR policer      : Disabled
Gx Session Recovery    : Disabled    Bearer Bind Verification: Enabled
Firewall Rules
Policy Rule           : "ABC"
Policy Rule Base      : "mDNS_policy-rule-base"
Optimize GBR          : Disabled     Bitrate Roundup       : Disabled
TWAG Unauth Sub Gx    : Disabled     Global PCC Precedence  : Disabled
Default CC Group      : none         Delay Tol Con Susp (sec): Disabled

PCRF Selection
PCC Dis RAT Tmr (sec) : 30          Sup CCR No QOS Neg     : Disabled
PCC Disable Per RAT   : none
Dynamic PCC           : Disabled

```

```

FH Session      : Disabled      FH 3xxx          : Disabled
FH 4xxx         : Disabled

Redirect        : none

Policy On The Fly : existing-sessions
Reactivate If Failed : Disabled

Transit Sub
  Att Rsp Wait Tmr (sec): 10      Att Rtr Hold Tmr (sec): 30
  Data Triggered       : Disabled

Default QoS Application Modify
  APN Ambr Attr
    Aggr DL Rate (kbps) : Disabled    Aggr UL Rate (kbps) : Disabled
    ARP                 : *           QCI                 : *
    QoS Hold Time (min) : 10          Retry Wait Timer (min): 1

Roaming Handover
  LTE QoS AMBR
    DownLink Rate (kbps): Disabled    Uplink Rate (kbps)  : Disabled
    LTE QoS Delay (sec) : 0

Aggregate Rate
  Downlink (kbps)      : Disabled    Uplink (kbps)       : Disabled

Static Predefined PCC
  Default QCI          : *           ARP                   : *

-----
Number of entries      : 1
=====

```

```

# show mobile-gateway pdn apn "cassandra7.mnc008.mcc244.gprs" detail
=====
APN Configuration
=====
APN Name           : "cassandra7.mnc008.mcc244.gprs"
APN Admin State    : up
Gateway            : 1
Description        : "APNConfiguration"
Misc
=====
Reactivation Requested : Disabled
Selection Mode        : subscribed-apn ms-provided-apn network-provided-apn
Del Br React Rq Cause : none
Idle Timeout (sec)    : Disabled      Session Timeout (sec) : 86400
Idle Timer Dir Uplink : Disabled
Max PDN Connections   : No limit      Max PDN Conn Action    : alarm
Min PDN Connections   : No limit
Reject Foreign Subs    : Disabled      Guard Timer (sec)      : Disabled
Restriction           : any            Allow Emergency         : Disabled
NEMO HA               : Disabled      HO Wr PDN IP Stl Ses Del: Enabled
Allow WLAN HO No Ctx  : Disabled      Seamless WLAN HO       : Enabled
Allow Dangl Rule Ded Br : Disabled    Paging Br Binding       : Disabled
Multi UE Bonding       : Disabled      RAD Case Sens Call St ID: Enabled

```

```

APN Load Control      : Disabled      Ovrld Back-Off Tmr (min): Disabled
Allow always-on PDU Ses : Disabled
RADIUS NAS-Identifier  : "abc def"
RADIUS NAS-IP-Address  : 1.2.3.4
APN Overload Control   : Disabled
PCO Fallback           : none
Dedicated Bearer
  Crt Rtr Count        : 0              Crt Rtr Wait Tmr (min): 0
GTP PDN Threshold
  Max Connections      : Disabled      Max Actv (act/sec)    : Disabled
  Alarm                : Disabled
APN Network ID         : virtual
Hybrid Access
  Bonding Support       : Disabled
  Link Switch Mode
    Uplink
      Hysteresis Tm (ms): 500
PCO Options
  Pref DNS Addr Type   : none          Link MTU (octets)     : 0
  Operator PCO         : none
KPI KCI
  ARP Traffic Stats    : Inherit       SGW APN Stats        : Disabled
  Enh Traffic Stats    : Inherit
Virtual APN
=====
Pre Auth               : radius
  Username             : imsi          Password Type        : none
  Supress Imsi         : Disabled
Pre Auth Fixed
  Username             : "default"
Virtual APN based on PCO: Disabled
Virtual APN selection
  Selection Type        : full-id       Prefer PDN           : Disabled
  Selection Priority    : imsi msisdn imei-tac roaming-status cc imeisv-tacsvn
  No Match Action      : continue
IP Allocation
=====
PDN Type               : ipv4
PDN Type Prefer IPv6   : Disabled      PDN Alloc Type        : ran-signaling
IP Alloc Method         : local-pool
Allow Local            : Disabled      Extended Rad Alloc     : Disabled
Priority Order          : none          IP Allocation Trace    : Disabled
Extended HSS Alloc     : Disabled      Frame Route           : Disabled

Router                 : "vprn121"
  MG-Group              : none
  IP Local Pools        : "ipv4-cassandra-pool-7"
Router Advertisements
  Min Adv Interval (sec): 16200         Max Adv Interval (sec): 21600

Authentication
=====
Auth                   : radius

```

```

Username       : imsi-apn      Password Type      : fixed
Password (hash) : "tiVTSjn3SXWZ8z6AtqUGVFsWms5ZQ=="
Server Group    : "Iliad-AAA-1"
Supress Imsi    : Disabled
Auth Fixed
  Username      : "default"
L2TP
=====
PPP Regeneration : Disabled
Router           : none
Username         : msisdn      Password Type      : none
ICRQ Call Num Type : msisdn      Mobile Router      : Disabled
DNS Server Opt   : Included     NBNS Server Opt    : Included
Rx Connect Speed (bps): Disabled  Tx Connect Speed (bps): 0
LCP Conf Req Attr : Included
  Auth Type      : chap-only    MRU Size           : 1500
HSS Static IP    : Excluded
Accounting
=====
Accounting Type   : radius
Username          : imsi-apn    Suppress Imsi      : Disabled
Selective Rad Acct : Disabled
Server Group      : "Iliad-AAA-1"
Radius Accounting On-Off
  Messages Sent   : none        Wait for ACK        : Disabled
  Incl Dupl Servers : Disabled
Wait Accounting   : Disabled
TWAG Radius Acct Feed : Disabled
Charging
=====
Charging Profile
  Home            : 8           Visiting            : 8
  Roaming         : 8
CC Ignore
  Home            : Inherit     Visiting            : Inherit
  Roaming         : Inherit     Any                 : Inherit
CC Reject        : Inherit
Policy
=====
AMBR Policing    : dl-only      Max AMBR policer    : Disabled
Gx Session Recovery : Disabled   Bearer Bind Verification: Enabled
Optimize GBR     : Disabled     Bitrate Roundup     : Disabled
TWAG Unauth Sub Gx : Disabled   Global PCC Precedence : Disabled
Default CC Group  : none        Delay Tol Con Susp (sec): Disabled
PCRF Selection
  PCC Dis RAT Tmr (sec) : 30      Sup CCR No QOS Neg   : Disabled
  PCC Disable Per RAT   : none
Dynamic PCC        : Disabled
  FH Session          : Disabled   FH 3xxx              : Disabled
  FH 4xxx             : Disabled
Redirect           : none
Policy On The Fly   : existing-sessions
  Reactivate If Failed : Disabled
Transit Sub
  Att Rsp Wait Tmr (sec): 10      Att Rtr Hold Tmr (sec): 30
  Data Triggered       : Disabled
Default QoS Application Modify

```



```

APN Ambr Attr
  Aggr DL Rate (kbps) : Disabled      Aggr UL Rate (kbps) : Disabled
  ARP                  : *            QCI                  : *
  QoS Hold Time (min) : 10            Retry Wait Timer (min): 1
Roaming Handover
  LTE QoS AMBR
    DownLink Rate (kbps): Disabled      Uplink Rate (kbps) : Disabled
  LTE QoS Delay (sec)  : 0
Aggregate Rate
  Downlink (kbps)      : Disabled      Uplink (kbps)      : Disabled
Firewall Rules
  Policy Rule           : "FirewallRule1"
                        "FirewallRule2"
  Policy Rule Base      : "FirewallRuleBase1"
Static Predefined PCC
  Default QCI           : *            ARP                  : *
PFCP
=====
PFCP Translate Tx Prof : none
UP Inactivity Timer(sec): Disabled
-----
Number of entries       : 1
=====

```

Table 17: Show APN detailed configuration output fields

Label	Description
Virtual APN	The Virtual APN context
Pre Auth	The pre-authentication type
Username	The user name used in access requests
Password Type	The password type
Supress Imsi	The option to include the 3GPP-IMSI attribute
Pre Auth Fixed	The Pre Auth Fixed context
Username	The user name for pre-authorization
Virtual APN based on PCO	The option to enable the PCO-based virtual APN selection mechanism

Table 17: Show APN detailed configuration output fields (continued)

Label	Description
Virtual APN selection	The Virtual APN selection context
Selection Type	The specified selection type
Prefer PDN	—
Selection Priority	The specified priority for selecting criteria under the user - range command
No Match Action	The action in cases where any defined user-range does not match
IP Allocation	The IP Allocation context
PDN Type	The PDN type
PDN Type Prefer IPv6	—
PDN Alloc Type	The PDN allocation type
IP Alloc Method	The specified IP address allocation scheme
Allow Local	The option to enable the allocation of a PDN IP address from a local pool
Extended Rad Alloc	The option to enable the handling of UE IP address allocation-related parameters in the Accept-Accept response message from the RADIUS server
Priority Order	The specified priority order in which the UE IP address allocation methods are evaluated
Frame Route	The option for Framed Routing support
Router	The router name

Table 17: Show APN detailed configuration output fields (continued)

Label	Description
MG-Group	The MG group associated with this APN
Router Advertisements	The Router Advertisements context
Min Adv Interval (sec)	The minimum length of time, in seconds, allowed between sending unsolicited multicast Router Advertisements
Max Adv Interval (sec)	The maximum length of time, in seconds, allowed between sending unsolicited multicast Router Advertisements
Authentication	The Authentication context
Auth	The authentication type being used
Username	The user name used in access requests sent to an AAA server
Password Type	The password type
Supress Imsi	The option to include the 3GPP-IMSI attribute
Auth Fixed	The Auth Fixed context
Username	The user name used in access requests sent to an AAA server when the authorized username is set to fixed
L2TP	The L2TP context
PPP Regeneration	The option to enable PPP regeneration on behalf of the sessions that terminate on this APN
Router	The router instance associated with the APN
Username	The username

Table 17: Show APN detailed configuration output fields (continued)

Label	Description
Password Type	The password type
ICRQ Call Num Type	The value that will be sent in the ICRQ Calling Number AVP as the subscriber identity for the LNS
Mobile Router	The option to enable mobile routers
DNS Server Opt	The option to include requesting the Primary and Secondary DNS address configuration option from the LNS in the PPP IPCP negotiation
NBNS Server Opt	The option to include requesting the Primary and Secondary NBNS address configuration option from the LNS in the PPP IPCP negotiation
Rx Connect Speed (bps)	The L2TP receive connection speed in b/s
Tx Connect Speed (bps)	The L2TP transmit connection speed in b/s
LCP Conf Req Attr	The option to include the Last Sent LCP Confreq and Last Received LCP Confreq attributes in the L2TP ICCN message to the LNS
Auth Type	The authentication type used when sending the LCP Confreq attributes
MRU Size	The MRU size, in bytes, sent in the LCP Confreq attributes
HSS Static IP	The option to include the HLR/HSS-provided static IP address in the GTP Create PDP Context Request/Create Session Request message
Accounting	The Accounting context
Accounting Type	The accounting type being used

Table 17: Show APN detailed configuration output fields (continued)

Label	Description
Username	The username
Suppress Imsi	The option to include the 3GPP-IMSI attribute
Selective Rad Acct	The option to enable selective RADIUS accounting
Radius Accounting On-Off	The Radius Accounting On-Off context
Messages Sent	The option to send ON/OFF RADIUS Accounting messages per APN
Wait for ACK	The option to enable the PGW/GGSN to wait for a response to Accounting-ON messages before allowing attaches to this APN
Incl Dupl Servers	The option to enable sending the RADIUS Accounting ON/OFF messages to the duplicate accounting servers
Wait Accounting	The option to enable the PGW/GGSN to wait for an accounting response before sending a GTP response
TWAG Radius Acct Feed	The option to enable using the RADIUS accounting feed for TWAG open SSID
Charging	The Charging context
Charging Profile	The Charging Profile context
Home	The option to inherit charging profile information from home subscribers
Visiting	The option to inherit charging profile information from visiting subscribers (local break-out)
Roaming	The option to inherit charging profile information from roamers (home-routed)

Table 17: Show APN detailed configuration output fields (continued)

Label	Description
CC Ignore	The CC Ignore context
Home	The option to enable the capability to ignore charging characteristics received from the MME/HSS for home subscribers
Visiting	The option to enable the capability to ignore charging characteristics received from the MME/HSS for visiting subscribers (local break-out)
Roaming	The option to enable the capability to ignore charging characteristics received from the MME/HSS for roamers (home-routed)
Any	The option to enable the capability to ignore charging characteristics received from the MME/HSS unconditionally
CC Reject	The option to enable the rejection of MME/HSS supplied charging characteristics
PFCP	PFCP Context
PFCP Translate Tx Prof	Status of the PFCP translation profile (enabled/disabled) for sending opaque values from the CP to the UP
PFCP UP Inactivity	Status of the UP inactivity timer (enabled/disabled), used for 5GC N4 sessions only
UP Inactivity Timer	Time configured for the PFCP UP inactivity timer
Policy	The Policy context
AMBR Policing	The type of AMBR policing being enforced for this APN
Max AMBR policer	The option to enable the AMBR policer rate to be set to a maximum value for this APN

Table 17: Show APN detailed configuration output fields (continued)

Label	Description
Gx Session Recovery	The option to enable Gx session discover and recovery for the APN
Bearer Bind Verification	The option to enable bearer binding verification for uplink traffic for a PDN session
Firewall Rules	The option to enable firewall rules
Policy Rule	Policy rule for this APN
Policy Rule Base	Policy rule base for this APN
Optimize GBR	The option to enable the GBR optimization capability
Bitrate Roundup	The option to enable the gateway to send the PCRF provided QoS values rounded up according to NAS specification (24. 301) to the LTE access (SGW/MME)
TWAG Unauth Sub Gx	The option to enable Gx interface support for unauthenticated subscribers in TWAG
Global PCC Precedence	The option to enable global PCC precedence for policy rules
Default CC Group	The default CC group used when a PCRF does not provide the CC group subscriber type for a subscriber PDN session
Delay Tol Con Susp (sec)	The maximum suspension time for the delay tolerant procedure, in seconds
PCRF Selection	The PCRF Selection context
PCC Dis RAT Tmr (sec)	The length of time, in seconds, before PCRF control is removed

Table 17: Show APN detailed configuration output fields (continued)

Label	Description
Sup CCR No QoS Neg	The option to enable sending the CCR-U to inform the PCRF about the QoS change received in the update of the PDP context request which also contained a No QoS negotiation indication
PCC Disable Per RAT	The RAT type that is not allowed to be under the control of the PCRF
Dynamic PCC	The option to enable dynamic PCC
FH Session	The option to enable the failure handling session
FH 3xxx	The option to enable failure handling for the 3xxx result code from the PCRF
FH 4xxx	The option to enable failure handling for the 4xxx result code from the PCRF
Redirect	The option to redirect all upstream mobile traffic toward the configured next-hop within this APN
Policy On The Fly	The specified PDN sessions to which modifications will be applied
Reactivate If Failed	The option to enable the termination of the existing session with a “reactivation required” cause code in the case of a conflict
Transit Sub	The Transit Sub context
Att Rsp Wait Tmr (sec)	The response wait time in seconds
Att Rtr Hold Tmr (sec)	The retry hold time in seconds
Data Triggered	The option to enable the creation of transit subscribers using the data trigger
Default QoS Application Modify	The Default QoS Application Modify context

Table 17: Show APN detailed configuration output fields (continued)

Label	Description
APN Ambr Attr	The APN Ambr Attr context
Aggr DL Rate (kbps)	The APN-AMBR downlink rate in kb/s
Aggr UL Rate (kbps)	The APN-AMBR uplink rate in kb/s
ARP	The ARP value
QCI	The QCI value
QoS Hold Time (min)	The length of time, in minutes, for application traffic to trigger the QoS updates back to the default QoS
Retry Wait Timer (min)	The length of time, in minutes, the gateway waits before retrying to update the default bearer QoS
Roaming Handover	The Roaming Handover context
LTE QoS AMBR	The LTE QoS AMBR context
DownLink Rate (kbps)	The APN-AMBR downlink rate in kb/s
UpLink Rate (kbps)	The APN-AMBR uplink rate in kb/s
LTE QoS Delay (sec)	The length of time, in seconds, before the PGW sends an Update Bearer Request
Aggregate Rate	The Aggregate Rate context
Downlink (kbps)	The APN aggregate downlink rate in kb/s
Uplink (kbps)	The APN aggregate uplink rate in kb/s

Table 17: Show APN detailed configuration output fields (continued)

Label	Description
Static Predefined PCC	The Static Predefined PCC context
Default QCI	The QCI value
ARP	The ARP value
Number of Entries	The number of entries being viewed

Example output APN statistics

The following output example shows APN statistics and *Show PDN APN statistics output fields* describes the output fields.

```
# show mobile-gateway pdn apn "ipd.nokia.com" statistics
=====
APN statistics summary
=====

APN Admin State      : up          APN Oper State      : up
APN                  : ipd.nokia.com
Homers               : 0            Visitors           : 0
Sess ePDG svrs req  : 0            Sess ePDG svrs all*: 0
Roamers              : 0            Session Timeouts   : 0
Idle Timeouts        : 0
PDNs w susp proc     : 0
Dedicated Bearers    : 0
Ipv4 PDN Sessions    : 0            Ipv6 PDN Sessions  : 0
Ipv4v6 PDN Sessions: 0            Non-Ip PDN Sessions: 0
Ipv4 PDP Contexts    : 0            Ipv6 PDP Contexts  : 0
Ipv4v6 PDP Contexts: 0            Secondary PDP Cont*: 0
Ipv4 PMIP Sessions   : 0            Ipv6 PMIP Sessions : 0
Ipv4v6 PMIP Sessio*: 0
Ipv4 S2a Sessions    : 0            Ipv6 S2a Sessions  : 0
Ipv4v6 S2a Sessions: 0            Ipv6 S2b Sessions  : 0
Ipv4 S2b Sessions    : 0
Ipv4v6 S2b Sessions: 0
L2TP Sessions        : 0
Ipv4 IP Local Pool   : 0            Ipv6 IP Local Pool : 0
Ipv4v6 IP Local Po*: 0
Ipv4 AAA Sessions    : 0            Ipv6 AAA Sessions  : 0
Ipv4v6 AAA Sessions: 0            Ipv6 HSS Sessions  : 0
Ipv4 HSS Sessions    : 0
Ipv4v6 HSS Sessions: 0
Ipv4 DHCP Sessions   : 0            Ipv6 DHCP Sessions : 0
Ipv4v6 DHCP Sessio*: 0            Ipv4 DHCP-Relay Se*: 0
```

```

Bonded IPv4 Sessio*: 0
Bonded IPv4v6 Sess*: 0
Pcrf-apn-ambr upda*: 0
Redirected Sessions: 0
Selective Radius a*: disabled
HRPD sessions : 0
LTE NEMO Sessions : 0
HA NEMO Sessions : 0
Ipv4 Steering Sess*: 0
Ipv4v6 Steering Se*: 0
TWAG Open SSID Ses*: 0
eMPS IMS Sessions : 0
eMPS IMS Ad Pri Ses: 0
PDN sessions having a dedicated bearer : 0
Bonded IPv6 Sessio*: 0
Tethered Sessions : 0
eHRPD NEMO Sessions: 0
NEMO IPv4 Subnets : 0
Ipv6 Steering Sess*: 0
eMPS ded bearers : 0

```

Statistics for APN used as real

```

Real APN PDN Sessi*: 0
Ipv4 PDN Sessions : 0
Ipv4v6 PDN Sessions: 0
Dedicated Bearers : 0
Ipv4 PDP Contexts : 0
Ipv4v6 PDP Contexts: 0
Ipv6 PDN Sessions : 0
Non-IP PDN Sessions: 0
Ipv6 PDP Contexts : 0
Secondary PDP Cont*: 0

```

```

Ipv4 S2a Sessions : 0
Ipv4v6 S2a Sessions: 0
Ipv4 S2b Sessions : 0
Ipv4v6 S2b Sessions: 0
Ipv6 S2a Sessions : 0
Ipv6 S2b Sessions : 0

```

```

Roaming LTE APN-AMBR updates
Successful : 0
Failed : 0
APN override triggers : 0
APN override triggers failed : 0
APN override establishments failed : 0

```

```

Restoration Priority Level : 1
Num of PDN Connections : 0

```

```

Restoration Priority Level : 2
Num of PDN Connections : 0

```

```

Restoration Priority Level : 3
Num of PDN Connections : 0

```

```

Restoration Priority Level : 4
Num of PDN Connections : 0

```

```

Restoration Priority Level : 5
Num of PDN Connections : 0

```

```

Restoration Priority Level : 6
Num of PDN Connections : 0

```

```

Restoration Priority Level : 7

```

```

Num of PDN Connections      : 0

Restoration Priority Level : 8
Num of PDN Connections      : 0

Inactivity Timer Sessions Distribution
0-30m      : 0          30m-1hr      : 0
1hr-3hr    : 0          3hr-6hr     : 0
6hr-12hr   : 0          12hr-24hr   : 0
1d-2d      : 0          2d-5d       : 0
5d-10d     : 0          10d+        : 0
TWAG SaMOG Sessions: 0      TWAG NSWO Sessions : 0

Router Solicitation Packets Rx      : 0
Router Solicitation Packets Dropped : 0
Router Advertisement Packets Tx     : 0
Neighbor Solicitation Packets Rx    : 0
Neighbor Solicitation Packets Droppd: 0
Neighbor Advertisement Packets Tx   : 0

UL Throughput (Mbps)      : 0
DL Throughput (Mbps)      : 0
-----
Number of cards : 1
=====
* indicates that the corresponding row element may have been truncated.

```

Table 18: Show PDN APN statistics output fields

Label	Description
APN Admin State	The APN administrative state
APN Oper State	The APN operational state
APN	The APN name
Homers	The number of home subscribers
Visitors	The number of visiting subscribers
Sess ePDG svrs req	The number of PDN sessions that are requested for the ePDG server address

Table 18: Show PDN APN statistics output fields (continued)

Label	Description
Sess ePDG svrs all*	The number of PDN sessions that the PGW provided for the ePDG server address
Roamers	The number of roaming subscribers
Idle Timeouts	The number of idle timeouts
Session Timeouts	The number of session timeouts
Dedicated Bearers	The number of dedicated bearers
Ipv4 PDN Sessions	The number of IPv4 sessions
Ipv6 PDN Sessions	The number of IPv6 sessions
Ipv4v6 PDN Sessions	The number of IPv4v6 sessions
Non-Ip PDN Sessions	The number of non-IP sessions
Ipv4 PDP Contexts	The number of IPv4 PDP contexts
Ipv6 PDP Contexts	The number of IPv6 PDP contexts
Ipv4v6 PDP Contexts	The number of IPv4v6 PDP contexts
Secondary PDP Cont*	The number of secondary PDP contexts
Ipv4 PMIP Sessions	The number of IPv4 PMIP sessions
Ipv6 PMIP Sessions	The number of IPv6 PMIP sessions
Ipv4v6 PMIP Sessio*	The number of IPv4v6 PMIP sessions

Table 18: Show PDN APN statistics output fields (continued)

Label	Description
Ipv4 S2a Sessions	The number of IPv4 S2a sessions
Ipv6 S2a Sessions	The number of IPv6 S2a sessions
Ipv4v6 S2a Sessions	The number of IPv4v6 S2a sessions
Ipv4 S2b Sessions	The number of IPv4 S2b sessions
Ipv6 S2b Sessions	The number of IPv6 S2b sessions
Ipv4v6 S2b Sessions	The number of IPv4v6 S2b sessions
L2TP Sessions	The number of L2TP sessions
Ipv4 IP Local Pool	The number of IPv4 local pools
Ipv6 IP Local Pool	The number of IPv6 local pools
Ipv4v6 IP Local Po*	The number of IPv4v6 local pools
Ipv4 AAA Sessions	The number of IPv4 AAA sessions
Ipv6 AAA Sessions	The number of IPv6 AAA sessions
Ipv4v6 AAA Sessions	The number of IPv4v6 AAA sessions
Ipv4 HSS Sessions	The number of IPv4 HSS sessions
Ipv6 HSS Sessions	The number of IPv6 HSS sessions
Ipv4v6 HSS Sessions	The number of IPv4v6 HSS sessions
Ipv4 DHCP Sessions	The number of IPv4 DHCP sessions

Table 18: Show PDN APN statistics output fields (continued)

Label	Description
Ipv6 DHCP Sessions	The number of IPv6 DHCP sessions
Ipv4v6 DHCP Sessio*	The number of IPv4v6 DHCP sessions
Ipv4 DHCP-Relay Se*	The number of IPv4 DHCP relay sessions
Bonded IPv4 Sessio*	The number of bonded IPv4 sessions
Bonded IPv6 Sessio*	The number of bonded IPv6 sessions
Bonded IPv4v6 Sess*	The number of bonded IPv4v6 sessions
Pcrf-apn-ambr upda*	—
Redirected Sessions	The number of redirected sessions
Tethered Sessions	The number of tethered sessions
Selective Radius a*	The status of selective RADIUS accounting
HRPD sessions	The number of HRPD sessions
LTE NEMO Sessions	The number of LTE NEMO sessions
eHRPD NEMO Sessions	The number of eHRPD NEMO sessions
HA NEMO Sessions	The number of HA NEMO sessions
NEMO IPv4 Subnets	The number of NEMO IPv4 sessions
Ipv4 Steering Sess*	The number of IPv4 steering sessions
Ipv6 Steering Sess*	The number of IPv6 steering sessions

Table 18: Show PDN APN statistics output fields (continued)

Label	Description
Ipv4v6 Steering Se*	The number of IPv4v6 steering sessions
TWAG Open SSID Ses*	The number of open SSID TWAG sessions
eMPS IMS Sessions	The number of eMPS IMS sessions
eMPS IMS Ad Pri Ses	—
eMPS ded bearers	The number of eMPS dedicated bearers
PDN sessions having a dedicated bearer	The number of PDN sessions having a dedicated bearer
Real APN PDN Sessi*	The number of real APN PDN sessions
Ipv4 PDN Sessions	The number of IPv4 sessions
Ipv6 PDN Sessions	The number of IPv6 sessions
Ipv4v6 PDN Sessions	The number of IPv4v6 sessions
Non-Ip PDN Sessions	The number of non-IP sessions
Dedicated Bearers	The number of dedicated bearers
Ipv4 PDP Contexts	The number of IPv4 PDP contexts
Ipv4v6 PDP Contexts	The number of IPv4v6 PDP contexts
Secondary PDP Cont*	The number of secondary PDP contexts
Ipv4 S2a Sessions	The number of IPv4 S2a sessions

Table 18: Show PDN APN statistics output fields (continued)

Label	Description
Ipv6 S2a Sessions	The number of IPv6 S2a sessions
Ipv4v6 S2a Sessions	The number of IPv4v6 S2a sessions
Ipv4 S2b Sessions	The number of IPv4 S2b sessions
Ipv6 S2b Sessions	The number of IPv6 S2b sessions
Ipv4v6 S2b Sessions	The number of IPv4v6 S2b sessions
APN override triggers	The number of APN override triggers
APN override triggers failed	The number of failed APN override triggers
APN override establishments failed	The number of failed APN override establishments
Restoration Priority Level	—
Num of PDN Connections	The number of PDN connections
Inactivity Timer Sessions Distribution	The distribution of the inactivity timer sessions
TWAG SaMOG Sessions	The number of TWAG SaMOG sessions
TWAG NSWOG Sessions	The number of TWAG NSWOG sessions
Router Solicitation Packets Rx	The number of router solicitation packets received
Router Solicitation Packets Dropped	The number of router solicitation packets dropped

Table 18: Show PDN APN statistics output fields (continued)

Label	Description
Router Advertisement Packets Tx	The number of router solicitation packets transmitted
Neighbor Solicitation Packets Rx	The number of neighbor solicitation packets received
Neighbor Solicitation Packets Droppd	The number of neighbor solicitation packets dropped
Neighbor Advertisement Packets Tx	The number of neighbor solicitation packets transmitted
UL Throughput (Mbps)	The uplink throughput in Mbps; this counter prints the instant throughput as it has been calculated in the latest 10 seconds interval, by sampling the forwarded data volume at the beginning and at the end of the interval
DL Throughput (Mbps)	The downlink throughput in Mbps; this counter prints the instant throughput as it has been calculated in the latest 10 seconds interval, by sampling the forwarded data volume at the beginning and at the end of the interval
Number of cards	The number of cards retrieved

Example output APN statistics for 5G sessions

The following output example shows APN statistics for 5G sessions.

```
# show mobile-gateway pdn apn "ipd.nokia.1.1.com" statistics
=====
APN statistics summary
=====

APN Admin State      : up          APN Oper State      : up
APN                  : ipd.nokia.1.1.com
Homers               : 0            Visitors            : 0
Sess ePDG svrs req  : 0            Sess ePDG svrs all*: 0
Roamers              : 0
Idle Timeouts        : 0            Session Timeouts    : 0
PDNs w susp proc     : 0
Dedicated Bearers    : 0
Ipv4 PDN Sessions    : 0            Ipv6 PDN Sessions   : 0
```

```

Ipv4v6 PDN Sessions: 0
Ipv4 PDP Contexts : 0
Ipv4v6 PDP Contexts: 0
Ipv4 PMIP Sessions : 0
Ipv4v6 PMIP Sessio*: 0
Ipv4 S2a Sessions : 0
Ipv4v6 S2a Sessions: 0
Ipv4 S2b Sessions : 0
Ipv4v6 S2b Sessions: 0
L2TP Sessions : 0
Ipv4 IP Local Pool : 0
Ipv4v6 IP Local Po*: 0
Ipv4 AAA Sessions : 0
Ipv4v6 AAA Sessions: 0
Ipv4 HSS Sessions : 0
Ipv4v6 HSS Sessions: 0
Ipv4 DHCP Sessions : 0
Ipv4v6 DHCP Sessio*: 0
Bonded IPv4 Sessio*: 0
Bonded IPv4v6 Sess*: 0
Pcrf-apn-ambr upda*: 0
Redirected Sessions: 0
Selective Radius a*: disabled
HRPD sessions : 0
LTE NEMO Sessions : 0
HA NEMO Sessions : 0
Ipv4 Steering Sess*: 0
Ipv4v6 Steering Se*: 0
TWAG Open SSID Ses*: 0
eMPS IMS Sessions : 0
eMPS IMS Ad Pri Ses: 0
Combo Ses by 5G UEs: 0
PDN sessions having a dedicated bearer : 0

Non-Ip PDN Sessions: 0
Ipv6 PDP Contexts : 0
Secondary PDP Cont*: 0
Ipv6 PMIP Sessions : 0
Ipv6 S2a Sessions : 0
Ipv6 S2b Sessions : 0
Ipv6 IP Local Pool : 0
Ipv6 AAA Sessions : 0
Ipv6 HSS Sessions : 0
Ipv6 DHCP Sessions : 0
Ipv4 DHCP-Relay Se*: 0
Bonded IPv6 Sessio*: 0
Tethered Sessions : 0
eHRPD NEMO Sessions: 0
NEMO IPv4 Subnets : 0
Ipv6 Steering Sess*: 0
eMPS ded bearers : 0
PGW Ses by 5G UEs : 0

Statistics for APN used as real
Real APN PDN Sessi*: 0
Ipv4 PDN Sessions : 0
Ipv4v6 PDN Sessions: 0
Dedicated Bearers : 0
Ipv4 PDP Contexts : 0
Ipv4v6 PDP Contexts: 0

Ipv6 PDN Sessions : 0
Non-Ip PDN Sessions: 0
Ipv6 PDP Contexts : 0
Secondary PDP Cont*: 0

Ipv6 S2a Sessions : 0
Ipv6 S2b Sessions : 0

Roaming LTE APN-AMBR updates
Successful : 0
Failed : 0
APN override triggers : 0
APN override triggers failed : 0
APN override establishments failed : 0

Restoration Priority Level : 1
Num of PDN Connections : 0

Restoration Priority Level : 2

```

```

Num of PDN Connections      : 0

Restoration Priority Level : 3
Num of PDN Connections      : 0

Restoration Priority Level : 4
Num of PDN Connections      : 0

Restoration Priority Level : 5
Num of PDN Connections      : 0

Restoration Priority Level : 6
Num of PDN Connections      : 0

Restoration Priority Level : 7
Num of PDN Connections      : 0

Restoration Priority Level : 8
Num of PDN Connections      : 0

Inactivity Timer Sessions Distribution
0-30m      : 0      30m-1hr      : 0
1hr-3hr    : 0      3hr-6hr     : 0
6hr-12hr   : 0      12hr-24hr  : 0
1d-2d      : 0      2d-5d      : 0
5d-10d     : 0      10d+       : 0
TWAG SaMOG Sessions: 0      TWAG NSWO Sessions : 0

Router Solicitation Packets Rx      : 0
Router Solicitation Packets Dropped : 0
Router Advertisement Packets Tx     : 0

UL Throughput (Mbps)      : 0
DL Throughput (Mbps)      : 0

Service Based Architecture (SBA) Statistics
IPv4 PDU Sessions : 0      IPv6 PDU Sessions : 0
IPv4v6 PDU Sessions: 0      IPv4 UDM PDU Sessi*: 0
IPv6 UDM PDU Sessi*: 0      IPv4v6 UDM PDU Ses*: 0
QoS Flows          : 0
PDU Sessions with more than one QoS Flow : 0
-----
Number of VMs : 1
=====
* indicates that the corresponding row element may have been truncated.

```

Example output PDN APN group statistics

```

# show mobile-gateway pdn apn "ipd.nokia.1.1.com" statistics group 1
=====
APN statistics summary
=====
Virtual APN      :
APN Admin State  : up      APN Oper State   : up
Router           : vprn1    PDN Type         : ipv4 ipv6 ipv4v6
Router           :          Ip Alloc Method          : local hss

```

```
Max-PDN-Connections: -
PPP Regeneration      : disabled
PPP User Name         : msisdn
PPP DNS-Server Opt    : included
L2TP Mobile Router    : enabled
L2TP ConFreq-attr     : included CHAP-only
PDN-Conn-enforced     : disabled
Restriction           : any
ICRQ Call Num Type    : imsi
GTP Prime Group       :
UDP tunnel Group      :
QCI Policy
Uplink                :
Static QoS Policy     :
Bonding Support       : disabled
Lte Session Guard *: 0
APN                   : ipd.nokia.1.1.com
Real APN PDN Sessi*: 0
Homers                : 0
Sess ePDG svrs req   : 0
Roamers               : 0
Idle Timeouts        : 0
Dedicated Bearers    : 0
Ipv4 PDN Sessions    : 0
Ipv4v6 PDN Sessions: 0
Ipv4 PDP Contexts    : 0
Ipv4v6 PDP Contexts: 0
Ipv4 PMIP Sessions   : 0
Ipv4v6 PMIP Sessio*: 0
Ipv4 S2a Sessions    : 0
Ipv4v6 S2a Sessions: 0
Ipv4 S2b Sessions    : 0
Ipv4v6 S2b Sessions: 0
L2TP Sessions        : 0
Ipv4 IP Local Pool    : 0
Ipv4v6 IP Local Po*: 0
Ipv4 AAA Sessions    : 0
Ipv4v6 AAA Sessions: 0
Ipv4 HSS Sessions    : 0
Ipv4v6 HSS Sessions: 0
Ipv4 DHCP Sessions   : 0
Ipv4v6 DHCP Sessio*: 0
Bonded IPv4 Sessio*: 0
Bonded IPv4v6 Sess*: 0
Pcrf-apn-ambr upda*: 0
Redirected Sessions   : 0
Selective RADIUS a*: disabled
HRPD sessions         : 0
LTE NEMO Sessions     : 0
HA NEMO Sessions      : 0
Ipv4 Steering Sess*: 0
Ipv4v6 Steering Se*: 0
TWAG Open SSID Ses*: 0
eMPS IMS Sessions     : 0

APN override triggers : 0
APN override triggers failed : 0

Min-PDN-Connections: -
L2TP Group Name      :
L2TP Router          :
PPP NBNS-Server Opt: included
L2TP ConFreq MRU     : 1500
Allow Wlan HO        : disabled
Selection Mode        : subscribed-apn

Downlink              :

Visitors              : 0
Sess ePDG svrs all*: 0

Session Timeouts      : 0

Ipv6 PDN Sessions     : 0
Non-IP PDN Sessions: 0
Ipv6 PDP Contexts     : 0

Ipv6 PMIP Sessions    : 0

Ipv6 S2a Sessions     : 0

Ipv6 S2b Sessions     : 0

Ipv6 IP Local Pool    : 0

Ipv6 AAA Sessions     : 0

Ipv6 HSS Sessions     : 0

Ipv6 DHCP Sessions    : 0
Ipv4 DHCP-Relay Se*: 0
Bonded IPv6 Sessio*: 0

Tethered Sessions     : 0

eHRPD NEMO Sessions: 0
NEMO IPv4 Subnets    : 0
Ipv6 Steering Sess*: 0
```

```

APN override establishments failed : 0

Restoration Priority Level : 1
Num of PDN Connections      : 0

Restoration Priority Level : 2
Num of PDN Connections      : 0

Restoration Priority Level : 3
Num of PDN Connections      : 0

Restoration Priority Level : 4
Num of PDN Connections      : 0

Restoration Priority Level : 5
Num of PDN Connections      : 0

Restoration Priority Level : 6
Num of PDN Connections      : 0

Restoration Priority Level : 7
Num of PDN Connections      : 0

Restoration Priority Level : 8
Num of PDN Connections      : 0

Inactivity Timer Sessions Distribution
0-30m           : 0           30m-1hr           : 0
1hr-3hr         : 0           3hr-6hr           : 0
6hr-12hr        : 0           12hr-24hr          : 0
1d-2d           : 0           2d-5d             : 0
5d-10d          : 0           10d+              : 0
TWAG SaMOG Sessions: 0       TWAG NSWO Sessions : 0

Router Solicitation Packets Rx      : 0
Router Solicitation Packets Dropped : 0
Router Advertisement Packets Tx     : 0
Neighbor Solicitation Packets Rx    : 0
Neighbor Solicitation Packets Droppd: 0
Neighbor Advertisement Packets Tx   : 0

UL Throughput (Mbps)      : 0
DL Throughput (Mbps)      : 0
-----
Number of cards : 1
=====
* indicates that the corresponding row element may have been truncated.

```

Example output APN statistics for DHCPv6 service

The following command output shows the DHCPv6 service enabled and *Show DHCPv6 service output fields* describes the output fields.

```
# show mobile-gateway pdn apn "ipd.nokia.1.1.com" statistics
=====
APN statistics summary
=====
Virtual APN :
APN Admin State : up APN Oper State : up
...
Router Solicitation Packets Rx : 0
Router Solicitation Packets Dropped : 0
Router Advertisement Packets Tx : 12
DHCPv6 Advertise messages:
DHCPv6 Information-Request Messages Received :
DHCPv6 Reply Messages Sent with DNS Address :
DHCPv6 Reply Messages Sent without DNS Address :
DHCPv6 Messages Dropped :
-----
Number of cards : 1
=====
```

Table 19: Show DHCPv6 service output fields

Label	Description
Router Solicitation Packets	The number of router solicitation packets received
Router Solicitation Packets Dropped	The number of router solicitation packets dropped
Router Advertisement Packets Tx	The number of router solicitation packets transmitted
DHCPv6 Advertise messages	The number of DHCPv6 messages advertised
DHCPv6 Information-Request Messages Received	The number of DHCPv6 Information-Request messages received
DHCPv6 Reply Messages Sent with DNS Address	The number of DHCPv6 Reply messages sent with a DNS address
DHCPv6 Reply Messages Sent without DNS Address	The number of DHCPv6 Reply messages sent without a DNS address

Table 19: Show DHCPv6 service output fields (continued)

Label	Description
DHCPv6 Messages Dropped	The number of DHCPv6 messages dropped
Number of cards	The number of cards retrieved

Example output APN Statistics for eMPS Advance Priority Calls

The following output example shows APN statistics for eMPS Advance Priority Calls and *Show PDN APN eMPS advance priority call output fields* describes the relevant output fields.

```
# show mobile-gateway pdn apn "ipd.nokia.1.1.com" statistics
=====
APN statistics summary
=====
Virtual APN      :
APN Admin State  : up                APN Oper State   : up
Router          : vprn1
PDN Type        : ipv4 ipv6 ipv4v6   Ip Alloc Method  : local hss
Ip Alloc Prio order: none
Max-PDN-Connections: -              Min-PDN-Connections: -
PPP Regeneration : disabled          PPP User Name    : msisdn
PPP DNS-Server Opt : included        PPP NBNS-Server Opt: included
L2TP Mobile Router : disabled
L2TP Group Name   :                  L2TP Router      :
L2TP ConFreq-attr : included CHAP-only L2TP ConFreq MRU : 1500
L2TP Tx Connect-Sp*: 0              L2TP Rx Connect-Sp*:
PDN-Conn-enforced : disabled         Allow Wlan HO    : disabled
HO Del Stale Sessi*: enabled
Restriction      : any               Selection Mode    : subscribed-apn
ICRQ Call Num Type : msisdn
GTP Prime Group   :
UDP tunnel Group  :
Destination Realm Gx:
QCI Policy
Uplink           :                  Downlink         :
Static QoS Policy :
Bonding Support   : disabled
Lte Session Guard *: 0
APN              :ipd.nokia.1.1.com
Real APN PDN Sessi*: 1
Homers           : 0                 Visitors          : 1
Sess ePDG svrs req : 0              Sess ePDG svrs all*: 0
Roamers          : 0
Idle Timeouts    : 0                 Session Timeouts  : 0
PDNs w susp proc : 0
Dedicated Bearers : 1
Ipv4 PDN Sessions : 0                Ipv6 PDN Sessions : 1
Ipv4v6 PDN Sessions: 0              Non-IP PDN Sessions: 0
```



```

Ipv4 PDP Contexts : 0
Ipv4v6 PDP Contexts: 0
Ipv4 PMIP Sessions : 0
Ipv4v6 PMIP Sessio*: 0
Ipv4 S2a Sessions : 0
Ipv4v6 S2a Sessions: 0
Ipv4 S2b Sessions : 0
Ipv4v6 S2b Sessions: 0
L2TP Sessions : 0
Ipv4 IP Local Pool : 0
Ipv4v6 IP Local Po*: 0
Ipv4 AAA Sessions : 0
Ipv4v6 AAA Sessions: 0
Ipv4 HSS Sessions : 0
Ipv4v6 HSS Sessions: 0
Ipv4 DHCP Sessions : 0
Ipv4v6 DHCP Sessio*: 0
Bonded IPv4 Sessio*: 0
Bonded IPv4v6 Sess*: 0
Pcrf-apn-ambr upda*: 0
Redirected Sessions: 0
Selective Radius a*: disabled
HRPD sessions : 0
LTE NEMO Sessions : 0
HA NEMO Sessions : 0
Ipv4 Steering Sess*: 0
Ipv4v6 Steering Se*: 0
TWAG Open SSID Ses*: 0
eMPS IMS Sessions : 0
eMPS IMS Ad Pri Ses: 1
PDN sessions having a dedicated bearer : 1

Roaming LTE APN-AMBR updates
Successful : 0
APN override triggers : 0
APN override triggers failed : 0
APN override establishments failed : 0

Failed : 0

Restoration Priority Level : 1
Num of PDN Connections : 0

Restoration Priority Level : 2
Num of PDN Connections : 0

Restoration Priority Level : 3
Num of PDN Connections : 0

Restoration Priority Level : 4
Num of PDN Connections : 0

Restoration Priority Level : 5
Num of PDN Connections : 0

Restoration Priority Level : 6
Num of PDN Connections : 0

Restoration Priority Level : 7

```

```

Num of PDN Connections      : 0

Restoration Priority Level : 8
Num of PDN Connections      : 0

Inactivity Timer Sessions Distribution
0-30m      : 1          30m-1hr      : 0
1hr-3hr    : 0          3hr-6hr     : 0
6hr-12hr   : 0          12hr-24hr   : 0
1d-2d      : 0          2d-5d       : 0
5d-10d     : 0          10d+        : 0
TWAG SaMOG Sessions: 0      TWAG NSWO Sessions : 0

Router Solicitation Packets Rx      : 0
Router Solicitation Packets Dropped : 0
Router Advertisement Packets Tx     : 4
-----
Number of cards : 1
=====
* indicates that the corresponding row element may have been truncated.
=====

```

Table 20: Show PDN APN eMPS advance priority call output fields

Label	Description
eMPS IMS Sessions	counts eMPS sessions that happen after regular PDN attach (where no GTP MP-flag is set and ARP value is greater than 2) via Gx: RAR with DRMP and ARP value set to 1, 2, or 3.
eMPS IMS Advance Priority Sessions	counts eMPS Advance priority sessions that happen at PDN attach/detach associated with default bearer where GTP MP-flag is set and ARP value is set to 1 or 2.
eMPS dedicated bearers	counts eMPS dedicated bearers that happen via Gx:RAR with DRMP and ARP value set to 1, 2 or 3.

Example output APN DNS redirect statistics

The following output example shows DNS redirect statistics for the APN and *Show PDN APN DNS redirect statistics output fields* describes the output fields.

```

# show mobile-gateway pdn apn "ipd.nokia.com" dns-redirect status
=====
DNS Redirect Server
=====
Redirect DNS Profile      : <dns-profile-name>
Redirect DNS Address     : 10.10.10.10

```

```

Active Time           : 03/12/2014 18:50:24
DNS Failover          : <No of times switching happened between DNS>

Primary DNS   : 10.10.10.10      State   : active
Secondary DNS : 20.20.20.20      State   : standby

Tx DNS messages : <no of packets transmitted>
Rx DNS messages : <no of Packets Received>

```

```

-----
Active redirect DNS at APN: 1
=====

```

Table 21: Show PDN APN DNS redirect statistics output fields

Label	Description
Redirect DNS Profile	The redirect DNS profile associated with this APN
Redirect DNS Address	The redirect DNS IP address
Active Time	Date and time when the DNS server was first active
DNS Failover	The number of DNS switches
Primary DNS	The IP address of the primary DNS server
Secondary DNS	The IP address of the secondary DNS server
State	The operational state of the DNS server
Tx DNS messages	The number of transmitted DNS packages
Rx DNS messages	The number of received DNS packages
Active redirect DNS at APN	The number of active redirect DNS at APN retrieved

Example output APN FQDN group

The following output example shows FQDN group statistics for the APN and *Show PDN APN FQDN group statistics output fields* describes the output fields.

```
# show mobile-gateway pdn apn "NOK_internal" fqdn-group "NOK_enrichment"
=====
FQDN Group                : NOK_enrichment
=====
DNS Refresh Interval      : 00:20:00 (hr:min:sec)
Next Refresh              : 00:10:11 (hr:min:sec)
IP Expiry Refresh Count  : 3

FQDN 1                    : fqdn1.NOK_enrichment.test.com
Resolved IP Addresses    :                Expiry Countdown   :
23.0.1.1                  3
23.0.1.2                  3
2300:1::1:0:0:0:1        2
2300:1::1:0:0:0:2        2

FQDN 2                    : fqdn2.NOK_enrichment.test.com
Resolved IP Addresses    :                Expiry Countdown   :
23.0.2.1                  3
23.0.2.2                  3
2300:1::2:0:0:0:1        1
2300:1::2:0:0:0:2        1

Total FQDN Count          : 2
Total IPv4 Addresses      : 4
Total IPv6 Addresses      : 4
=====
```

Table 22: Show PDN APN FQDN group statistics output fields

Label	Description
DNS Refresh Interval	The time in minutes that the node waits before sending a DNS query to refresh the URL list for the FQDN group
Next Refresh	The time in minutes when the next DNS query will be sent
IP Expiry Refresh Count	The DNS refresh count for resolved IP address expiration
FQDN #	The FQDN group number

Table 22: Show PDN APN FQDN group statistics output fields (continued)

Label	Description
Resolved IP Addresses	The list of the resolved IP addresses
Expiry Countdown	—
Total FQDN Count	The number of the FQDN
Total IPv4 Addresses	The number of IPv4 addresses retrieved
Total IPv6 Addresses	The number of IPv6 addresses retrieved

Example output APN override statistics

The following output example shows APN override statistics and *Show PDN APN override statistics output fields* describes the output fields.

```
# show mobile-gateway pdn apn "ipd.nokia.1.1.com" apn-override-
statistics [gateway gw-id | group group-number] [summary]
=====
APN override statistics
=====
Chassis/Card      : 1/4          Gateway      : 1
Static override triggers      : 0
Static override triggers failed : 0
Static override establishments failed : 0

RADIUS override triggers      : 0
RADIUS override triggers failed : 0
RADIUS override establishments failed : 0

Diameter/Gx override triggers      : 0
Diameter/Gx override triggers failed : 0
Diameter/Gx override establishments failed : 0

Diameter/S6b override triggers      : 0
Diameter/S6b override triggers failed : 0
Diameter/S6b override establishments failed : 0
-----
Number of cards: 1
=====
```

Table 23: Show PDN APN override statistics output fields

Label	Description
Chassis/Card	The chassis and card number
Gateway	The gateway ID
Static override triggers	The number of static override triggers
Static override triggers failed	The number of failed static override triggers
Static override establishments failed	The number of failed static override establishments
RADIUS override triggers	The number of RADIUS override triggers
RADIUS override triggers failed	The number of failed RADIUS override triggers
RADIUS override establishments failed	The number of failed RADIUS override establishments
Diameter/Gx override triggers	The number of Diameter/Gx override triggers
Diameter/Gx override triggers failed	The number of failed Diameter/Gx override triggers
Diameter/Gx override establishments failed	The number of failed Diameter/Gx override establishments
Diameter/S6b override triggers	The number of Diameter/S6b override triggers
Diameter/S6b override triggers failed	The number of failed Diameter/S6b override triggers
Diameter/S6b override establishments failed	The number of failed Diameter/S6b override establishments

Table 23: Show PDN APN override statistics output fields (continued)

Label	Description
Number of cards	The number of cards retrieved

Example output P-CSCF statistics

The following output example shows P-CSCF statistics for the APN and *Show PDN APN P-CSCF statistics output fields* describes the output fields.

```
# show mobile-gateway pdn apn "ipd.nokia.com" pcscf-statistics
=====
P-CSCF statistics summary
=====
UE-location based IPv4 P-CSCF server selection
Success                : 0
Failed (UE location not known) : 0
Failed (No matching location) : 0
IPv4 server selection
P-CSCF server selection successful : 0
P-CSCF server selection failed    : 0
UE-location based IPv6 P-CSCF server selection
Success                : 0
Failed (UE location not known) : 0
Failed (No matching location) : 0
IPv6 server selection
P-CSCF server selection successful : 0
P-CSCF server selection failed    : 0
=====
```

Table 24: Show PDN APN P-CSCF statistics output fields

Label	Description
UE Location-based IPv4/IPv6 P-CSCF Server Selection	
Success	The number of successful P-CSCF server selections
Failed (UE location not known)	The number of failed server selections because the UE location was unknown
Failed (No matching location)	The number of failed server selections because there was no matching location

Table 24: Show PDN APN P-CSCF statistics output fields (continued)

Label	Description
IPv4/IPv6 Server Selection	
P-CSCF server selection successful	The number of successful P-CSCF server selections (does not include possible UE location-based selection); includes the default selection if a match for UE location was not found
P-CSCF server selection failed	The number of failed P-CSCF server selections

Example output Gx assume positive statistics

The following output example shows Gx Assume Positive statistics and *Show PDN APN Gx assume positive statistics output fields* describes the output fields.

```
# show mobile-gateway pdn apn "ipd.nokia.1.1.com" gx-ap-statistics
=====
Gx Assume Positive Statistics
=====
Number of sessions in Gx Assume Positive: 0
Error conditions that triggered Gx Assume Positive
CCR-I Message Fail*: 0          CCR-U Message Fail*: 0
CCR-T Message Fail*: 0          Tx Time Out Failur*: 0
Other Failures      : 0          3002 Result Code F*: 0
3004 Result Code F*: 0          3xxx Result Code F*: 0
4xxx Result Code F*: 0          5012 Result Code F*: 0
5xxx Result Code F*: 0

Cumulative Statistics

Number of sessions in Gx Assume Positive: 0
Error conditions that triggered Gx Assume Positive
CCR-I Message Fail*: 0          CCR-U Message Fail*: 0
CCR-T Message Fail*: 0          Tx Time Out Failur*: 0
Other Failures      : 0          3002 Result Code F*: 0
3004 Result Code F*: 0          3xxx Result Code F*: 0
4xxx Result Code F*: 0          5012 Result Code F*: 0
5xxx Result Code F*: 0
Number of cards : 1
=====
```


Table 25: Show PDN APN Gx assume positive statistics output fields

Label	Description
Number of sessions in Gx Assume Positive	The number of total/cumulative sessions in Gx Assume Positive
CCR-I Message Fail*	The number of total/cumulative CCR-I message failures
CCR-U Message Fail*	The number of total/cumulative CCR-U message failures
CCR-T Message Fail*	The number of total/cumulative CCR-T message failures
Tx Time Out Failur*	The number of total/cumulative transmit time-out failures
Other Failures	The number of total/cumulative other failures
3002 Result Code F*	The number of total/cumulative 3002 (DIAMETER-UNABLE-TO-DELIVER) result code failures
3004 Result Code F*	The number of total/cumulative 3004 (DIAMETER-TOO-BUSY) result code failures
3xxx Result Code F*	The number of total/cumulative 3xxx result code failures
4xxx Result Code F*	The number of total/cumulative 4xxx result code failures
5012 Result Code F*	The number of total/cumulative 5012 (DIAMETER-UNABLE-TO-COMPLY) result code failures
5xxx Result Code F*	The number of total/cumulative 5xxx result code failures
Number of cards	The number of cards retrieved

Example output PDN APN call flow statistics SNMP for PGW and combination gateway

The following output example shows SNMP for PGW and Combination gateway statistics and *Show PDN APN SNMP for PGW and combination gateway call flow statistics output fields* describes the output fields.

```
# show mobile-gateway pdn 1 apn nokia.apn.com call-flow-stats
=====
APN call flow statistics
=====
VNF/VM           : 1/3      Gateway           : 1
Attach           : 0        Attach Fail        : 0
Detach           : 1        Detach Fail        : 0
NW Ded Bearer Actv : 0      NW Ded Brr Act Fail : 0
NW Ded Brr Deactv : 0      NW Ded Brr Deact Fl : 0
NW Ded Brr Modify : 0      NW Ded Brr Mdf Fail : 0
UE Ded Bearer Actv : 0      UE Ded Brr Act Fail : 0
UE Ded Brr Deactv : 0      UE Ded Brr Deact Fl : 0
UE Ded Brr Modify : 0      UE Ded Brr Mdf Fail : 0
MME Ded Brr Deact : 0      MME Ded Brr Deact F : 0
MS Bearer Modify : 0        MS Bearer Mdfy Fail : 0
HSS QoS Modify    : 0        HSS QoS Modify Fail : 0
PCRF QoS Modify   : 0        PCRF QoS Mdfy Fail  : 0
Reattach Success  : 0        Reattach Fail       : 0
NW Init Bearer Mdfy : 0      NW Init Brr Mdfy Fl : 0
ho-apn-lookup-imsi-based-oi : 0 ho-apn-lookup-ni-only : 0
-----
VNF/VM           : 1/A      Gateway           : 1
...
-----
Number of cards : 2
=====
```

Table 26: Show PDN APN SNMP for PGW and combination gateway call flow statistics output fields

Label	Description
VNF/VM	The VNF and VM numbers (applicable only to CMG)
Gateway	The gateway ID
Attach	The number of attaches
Attach Fail	The number of failed attaches
Detach	The number of detaches
Detach Fail	The number of failed detaches

Table 26: Show PDN APN SNMP for PGW and combination gateway call flow statistics output fields (continued)

Label	Description
NW Ded Bearer Actv	The number of network initiated dedicated bearer activations
NW Ded Brr Act Fail	The number of network initiated dedicated bearer activation failures
NW Ded Brr Deactv	The number of network dedicated bearer deactivations
NW Ded Brr Deact FI	The number of network dedicated bearer deactivation failures
NW Ded Brr Modify	The number of network dedicated bearer modifications
NW Ded Brr Mdf Fail	The number of network dedicated bearer modification failures
UE Ded Bearer Actv	The number of UE dedicated bearer activations
UE Ded Brr Act Fail	The number of UE dedicated bearer activation failures
UE Ded Brr Deactv	The number of UE dedicated bearer deactivations
UE Ded Brr Deact FI	The number of UE dedicated bearer deactivation failures
UE Ded Brr Modify	The number of UE dedicated bearer modifications
UE Ded Brr Mdf Fail	The number of UE dedicated bearer modification failures
MME Ded Brr Deact	The number of MME dedicated bearer deactivations
MME Ded Brr Deact F	The number of MME dedicated bearer deactivation failures
MS Bearer Modify	The number of MS bearer modifications
MS Bearer Mdfy Fail	The number of MS bearer modification failures
HSS QoS Modify	The number of HSS QoS modifications

Table 26: Show PDN APN SNMP for PGW and combination gateway call flow statistics output fields (continued)

Label	Description
HSS QoS Modify Fail	The number of HSS QoS modification failures
PCRF QoS Modify	The number of PCRF QoS modifications
PCRF QoS Mdfy Fail	The number of PCRF QoS modification failures
Reattach Success	The number of successful reattaches
Reattach Fail	The number of failed reattaches
NW Init Bearer Mdfy	The number of the network initiated bearer modifications
NW Init Brr Mdfy FI	The number of the network initiated bearer modification failures
ho-apn-lookup-imsi-based-oi	The number of HI=1 handover attaches where the IMSI PLMN was used for the APN OI during the attach process
ho-apn-lookup-ni-only	The number of HI=1 handover attaches where the APN NI was only used during the attach process
Number of cards	The number of cards retrieved

Example output PDN APN transport statistics

The following output example shows transport-byte-collection statistics and *Show PDN APN transport statistics output fields* describes the output fields.

```

=====
APN transport statistics
=====
Gateway          : 1                VNF/Grp ID/VM      : 1/1/3
IMSI             : 220016789012301
IP Address       : 3ffe:0:1:ff00::5:aa77
Policy Rule Name : onLine1100-100
URR ID           : 2684750337
Pkts DL          : 10                Pkts Up           : 10
Bytes DL         : 1000              Bytes Up          : 1000
=====

```

```

Policy Rule Name : onLine1Any5
URR ID          : 2684815873
  Pkts DL       : 5                Pkts Up       : 5
  Bytes DL      : 500             Bytes Up     : 500
-----
Number of IMSIs/MSISDNs: 1
=====

```

Table 27: Show PDN APN transport statistics output fields

Label	Description
Gateway	The gateway ID
VNF/Grp ID/VM	The VNF, Group ID and VM numbers (applicable only to CMG)
IMSI/MSISDN	The IMSI or MSISDN of the subscriber
IP Address	The IPv4 or IPv6 address of the subscriber
Policy Rule Name	The name of the policy rule
URR ID	The URR ID (applicable only in CUPS deployments)
Pkts DL	The number of downlink packets
Pkts Up	The number of uplink packets
Bytes DL	The number of downlink bytes
Bytes Up	The number of uplink bytes
Number of IMSIs/MSISDNs	The number of IMSIs and MSISDNs for which transport statistics have been retrieved

5.2.2.3 accounting

Syntax

accounting

Context

show>mobile>pdn>apn

Description

This command displays the APN accounting configuration.

Output

The following output example shows APN accounting configuration.

Example output APN accounting configuration

```
# show mobile-gateway pdn apn "internet" accounting
=====
APN Configuration
=====
APN Name           : "internet"
APN Admin State    : down
Gateway            : 1

Accounting
=====
Accounting Type     : radius
  Username          : imsi           Suppress Imsi       : Disabled
  Selective Rad Acct : Disabled

Radius Accounting On-Off
  Messages Sent     : none           Wait for ACK        : Disabled
  Incl Dupl Servers : Disabled

Wait Accounting     : Disabled
TWAG Radius Acct Feed : Disabled

-----
Number of entries   : 1
=====
```

```
#show mobile-gateway pdn apn "ipd.nokia.1.1.com.mnc012.mcc310.gprs" accounting
=====
APN Configuration
=====
APN Name           : "ipd.nokia.1.1.com.mnc012.mcc310.gprs"
APN Admin State    : up
Gateway            : 1
```

```

Accounting
=====
Accounting Type      : radius
Username            : msisdn-apn      Suppress Imsi      : Disabled
Selective Rad Acct   : Disabled
Server Group        : "Radius Group"

Radius Accounting On-Off
Messages Sent        : none           Wait for ACK         : Disabled
Incl Dupl Servers    : Disabled

Wait Accounting       : Disabled
TWAG Radius Acct Feed : Disabled

-----
Number of entries     : 1
=====

```

5.2.2.4 authentication

Syntax

```
authentication
```

Context

```
show>mobile>pdn>apn
```

Description

This command displays the APN authentication configuration.

Output

The following output example shows APN authentication configuration.

Example output APN authentication configuration

```

# show mobile-gateway pdn apn "internet" authentication
=====
APN Configuration
=====
APN Name           : "internet"
APN Admin State     : down
Gateway            : 1

Authentication
=====
Auth
Username           : imsi           Password Type       : none
Suppress Imsi      : Disabled       Radius Auth Cache   : Disabled

```

```
Auth Fixed
  Username          : "default"
```

```
-----
Number of entries   : 1
=====
```

```
#show mobile-
gateway pdn apn "ipd.nokia.1.1.com.mnc012.mcc310.gprs" authentication
```

```
=====
APN Configuration
```

```
=====
APN Name            : "ipd.nokia.1.1.com.mnc012.mcc310.gprs"
APN Admin State     : up
Gateway             : 1
```

```
Authentication
=====
```

```
Auth               : radius
  Username          : imsi-apn      Password Type       : none
  Server Group      : "Radius Group"
  Supress Imsi      : Disabled
```

```
Auth Fixed
  Username          : "default"
```

```
-----
Number of entries   : 1
=====
```

5.2.2.5 charging

Syntax

```
charging
```

Context

```
show>mobile>pdn>apn
```

Description

This command displays the APN charging configuration.

Output

The following output example shows APN charging configuration.

Example output APN charging configuration

```
# show mobile-gateway pdn apn "internet" charging
=====
APN Configuration
=====
APN Name           : "internet"
APN Admin State    : down
Gateway            : 1

Charging
=====
Charging Profile
  Home              : Inherit      Visiting          : Inherit
  Roaming           : Inherit

CC Ignore
  Home              : Inherit      Visiting          : Inherit
  Roaming           : Inherit      Any                : Inherit

CC Reject          : Inherit

-----
Number of entries   : 1
=====
```

Example output APN charging profile PLMN list configuration

```
# show mobile-gateway pdn apn "internet" charging
=====
APN Configuration
=====
APN                 : internet
Gateway            : 1

Charging
=====
Charging Profile
  Home              : Inherit      Visiting          : 3
  Roaming           : Inherit

CC Ignore
  Home              : Inherit      Visiting          : Inherit
  Roaming           : Inherit      Any                : Inherit

CC Reject          : Inherit
CDR AVP Options Profile : 200
DCCA Profile       : default
GTP-Prime Group    : chrg-gtp-pr-gr
CDF Diameter Peer
  Primary           : chrg-pri-cdf
  Secondary         : chrg-sec-cdf
OCS Diameter Peer
  Primary           : chrg-pri-ocs
  Secondary         : chrg-sec-ocs
PLMN charging profile map
  PLMN list name    : chrg-plmn-lst
  Incoming CC Value : 0x3          Charging Profile Id : 3
```

```

Incoming CC Value : 0x23          Charging Profile Id : 3
Incoming CC Value : 0x7530        Charging Profile Id : 3
PLMN list name    : chrg-plmn-lst-3
Incoming CC Value : 0xc           Charging Profile Id : 3
Incoming CC Value : 0x22          Charging Profile Id : 3
Incoming CC Value : 0x37          Charging Profile Id : 3
PLMN list name    : chrg-plmn-lst-4
Incoming CC Value : 0x4d          Charging Profile Id : 3

-----
Number of entries : 1
=====

```

5.2.2.6 ip-alloc

Syntax

```
ip-alloc
```

Context

```
show>mobile>pdn>apn
```

Description

This command displays the APN IP address allocation configuration.

Output

The following output example shows APN IP address allocation configuration and *Show APN IP address allocation configuration output fields* describes the output fields.

Example output APN IP address allocation configuration

```

# show mobile-gateway pdn apn "internet" ip-alloc
=====
APN Configuration
=====
APN Name          : "internet"
APN Admin State   : down
Gateway           : 1

IP Allocation
=====
PDN Type           : ipv4
PDN Type Prefer IPv6 : Disabled
IP Alloc Method    : local-pool
Allow Local        : Disabled
Priority Order      : none
Extended Hss Alloc : Disabled

PDN Alloc Type     : ran-signaling
PD Alloc Method    : none
Extended Rad Alloc : Disabled
IP Allocation Trace : Enabled

```

```

Router          : "Base"
MG-Group        : none

Router Advertisements
  Min Adv Interval (sec): 16200      Max Adv Interval (sec): 21600
SSC
  PDU Ses Addr Lftm(sec): 300

-----
Number of entries      : 1
=====

```

Table 28: Show APN IP address allocation configuration output fields

Label	Description
APN Name	The APN name
APN Admin State	The APN administrative state
Gateway	The gateway ID
IP Allocation	
PDN Type	The PDN IP type
PDN Type Prefer IPv6	The option to force the use of an IPv6 address for the PDN connection when there is a UE-requested dual stack IPv4v6 address PDN type and the APN supports only IPv4 and IPv6 addresses
PDN Alloc Type	The PDN allocation type
PD Alloc Method	The allocation scheme the APN uses to assign the IPv6 prefix delegation to the UE/CPE after the PDN connection
IP Alloc Method	The specified IP address allocation scheme
Allow Local	The option to enable the allocation of a PDN IP address from a local pool

Table 28: Show APN IP address allocation configuration output fields (continued)

Label	Description
Extended Rad Alloc	The option to enable the handling of UE IP address allocation-related parameters in the Accept-Accept response message from the RADIUS server
Priority Order	The specified priority order in which the UE IP address allocation methods are evaluated
IP Allocation Trace	The option to enable syslog reporting when an IP address is allocated or released for the PDN connection
Extended Hss Alloc	The option to validate the external IP addresses provided via GTP against the remote IP pools configured in the APN
Router	The router name
MG-Group	The MG group associated with this APN
Router Advertisements	The Router Advertisements context
Min Adv Interval (sec)	The minimum length of time, in seconds, allowed between sending unsolicited multicast Router Advertisements
Max Adv Interval (sec)	The maximum length of time, in seconds, allowed between sending unsolicited multicast Router Advertisements
SSC	Session and service continuity
PDU Ses Addr Lftm(sec)	Time in seconds for the PDU session address
Number of entries	The number of entries retrieved

5.2.2.7 l2tp

Syntax

l2tp

Context

show>mobile>pdn>apn

Description

This command displays the APN L2TP configuration.

Output

The following output example shows APN L2TP configuration.

Example output APN IP L2TP configuration

```
# show mobile-gateway pdn apn "internet" l2tp
=====
APN Configuration
=====
APN Name           : "internet"
APN Admin State    : down
Gateway            : 1

L2TP
=====
PPP Regeneration   : Disabled
Router             : none
Username           : msisdn           Password Type      : none
ICRQ Call Num Type : msisdn           Mobile Router       : Disabled
DNS Server Opt     : Included          NBNS Server Opt     : Included
Rx Connect Speed (bps): Disabled       Tx Connect Speed (bps): 0
LCP Conf Req Attr  : Included
Auth Type          : chap-only         MRU Size            : 1500
HSS Static IP      : excluded

-----
Number of entries   : 1
=====
```

5.2.2.8 pfcp

Syntax

pfcp

Context

```
show>mobile>pdn>apn
```

Description

This command displays the APN PFCP configuration.

Output

The following output example shows APN PFCP configuration.

Example output APN PFCP configuration

```
# show mobile-gateway pdn apn "apn1" pfc
=====
APN Configuration
=====
APN Name           : "apn1"
APN Admin State    : down
Gateway            : 1

PFCP
=====
PFCP Translate Tx Prof : pfc1
PFCP UP Inactivity    : Disabled/Enabled      UP Inactivity Timer(sec):

-----
Number of entries      : 1
=====
```

5.2.2.9 policy

Syntax

```
policy
```

Context

```
show>mobile>pdn>apn
```

Description

This command displays the APN policy configuration.

Output

The following output example shows APN policy configuration.

Example output APN policy configuration

```
# show mobile-gateway pdn apn "internet" policy
=====
APN Configuration
=====
APN Name           : "internet"
APN Admin State    : down
Gateway            : 1

Policy
=====
AMBR Policing      : dl-only           Max AMBR policer      : Disabled
Gx Session Recovery : Disabled         Bearer Bind Verification: Enabled
Optimize GBR       : Disabled         Bitrate Roundup       : Disabled
TWAG Unauth Sub Gx : Disabled         Global PCC Precedence  : Disabled
Default CC Group   : none             Delay Tol Con Susp (sec): Disabled

PCRF Selection
  PCC Dis RAT Tmr (sec) : 30           Sup CCR No QoS Neg    : Disabled
  PCC Disable Per RAT   : none
  Dynamic PCC           : Disabled
    FH Session          : Disabled      FH 3xxx              : Disabled
    FH 4xxx             : Disabled

Redirect            : none

Policy On The Fly   : existing-sessions
  Reactivate If Failed : Disabled

Transit Sub
  Att Rsp Wait Tmr (sec): 10           Att Rtr Hold Tmr (sec): 30
  Data Triggered        : Disabled

Default QoS Application Modify
  APN Ambr Attr
    Aggr DL Rate (kbps) : Disabled      Aggr UL Rate (kbps) : Disabled
  ARP                   : *             QCI                  : *
  QoS Hold Time (min)   : 10            Retry Wait Timer (min): 1

Roaming Handover
  LTE QoS AMBR
    DownLink Rate (kbps): Disabled      Uplink Rate (kbps)  : Disabled
  LTE QoS Delay (sec)   : 0

Aggregate Rate
  Downlink (kbps)       : Disabled      Uplink (kbps)       : Disabled

Static Predefined PCC
  Default QCI           : *             ARP                  : *

-----
Number of entries       : 1
=====
```

5.2.2.10 virtual-apn

Syntax

```
virtual-apn [detail]
```

Context

```
show>mobile>pdn>apn
```

Description

This command displays the APN policy configuration.

Parameters

detail

specifies to display detailed virtual APN configuration

Output

The following output examples show virtual APN configuration.

- *Example output virtual APN configuration*
- *Example output virtual APN with user range configuration*
- *Example output virtual APN with user range detail configuration*

Example output virtual APN configuration

```
# show mobile-gateway pdn apn "internet" virtual-apn
=====
APN Configuration
=====
APN Name           : "internet"
APN Admin State    : down
Gateway           : 1

Virtual APN
=====
Pre Auth           : radius
  Username          : imsi           Password Type       : none
  Supress Imsi      : Disabled

Pre Auth Fixed
  Username          : "default"

Virtual APN based on PC0: Disabled

Virtual APN selection
  Selection Type     : full-id        Prefer PDN         : Disabled
```



```

Selection Priority      : imsi msisdn imei-tac roaming-status cc imeisv-tacsvn
                        : serving-node-plmn pdn-type rat-type ue-location
No Match Action        : continue
User Range
    rat-type           : 1
-----
Number of entries      : 1
=====

```

Example output virtual APN with user range configuration

```

# show mobile-gateway pdn apn "apn1" virtual-apn
=====
APN Configuration
=====
APN Name                : "apn1"
APN Admin State         : up
Gateway                 : 1

Virtual APN
=====
Pre Auth                : radius
  Username               : imsi                Password Type       : default
  Password (hash2)       : "4sY.x6LqCl4nwXQSlhwmYU"
  Server Group           : "Rgrp1"
  Supress Imsi           : Disabled

Pre Auth Fixed
  Username               : "default"
Virtual APN based on PC0: Disabled

Virtual APN selection
  Selection Type          : full-id            Prefer PDN          : Disabled
  Selection Priority      : cc roaming-status imei-tac imeisv-tacsvn msisdn imsi
                        : serving-node-plmn
  No Match Action        : continue
  User Range
    roaming-status       : 1
    cc                   : 2
    imeisv-tacsvn        : 1
    cc msisdn            : 1
    serving-node-plmn    : 1
-----
Number of entries        : 1

```

Example output virtual APN with user range detail configuration

```

# show mobile-gateway pdn apn "apn1" virtual-apn detail
Type                    : default
  Password (hash2)       : "4sY.x6LqCl4nwXQSlhwmYU"
  Server Group           : "Rgrp1"
  Supress Imsi           : Disabled

Pre Auth Fixed
  Username               : "default"
Virtual APN based on PC0: Disabled

```

```

Virtual APN selection
Selection Type      : full-id      Prefer PDN      : Disabled
Selection Priority  : cc roaming-status imei-tac imeisv-tacsvn msisdn imsi
                    : serving-node-plmn
No Match Action     : continue
User Range Name     : "r1"
Types               : cc
Start CC ID         : 1             End CC ID         : 50000
Pre Auth            : radius
Username            : pco            Password Type     : default
Password (hash2)    : "4sY.x6LqCl4nwXQSlhwmYU"
Supp Imsi           : Disabled
Srv Grp             : "Rgrp1"
User Range Name     : "r2"
Types               : cc
Start CC ID         : 50001          End CC ID         : 65534
Pre Auth            : radius
Username            : pco            Password Type     : fixed
Password (hash2)    : "4sY.x6LqCl4nwXQSlhwmYU"
Supp Imsi           : Disabled
Srv Grp             : "Rgrp1"
User Range Name     : "snUR1"
Types               : serving-node-plmn
PLMN List Name      : snPlmn1
Virtual APN         : "apnVapn1"
User Range Name     : "imeiTacSvnRg1"
Types               : imeisv-tacsvn
Start UE ID         : 12345678
SVN                  : 10
Virtual APN         : "apnVapn2"
User Range Name     : "rs1"
Types               : roaming-status
Roam Stat Type      : roaming
Virtual APN         : "apnLocalSelectionVapn1"
User Range Name     : "msisdnCc"
Types               : cc msisdn
Start UE ID         : 123456789012345
End UE ID           : 123456789012346
Start CC ID         : 11
Virtual APN         : "apnStaticVapn"
-----
Number of entries    : 1
=====

```

5.2.2.11 apn-overload

Syntax

apn-over load [apn-name]

Context

```
show>mobile>pdn
```

Description

This command displays the APN overload statistics.

Parameters

apn

specifies the APN name, up to 80 characters

Output

The following output example shows APN overload statistics and [Show PDN APN overload output fields](#) describes the output fields.

Example output APN overload statistics

```
# show mobile-gateway pdn apn-overload
=====
APN Load/Overload and PDN-Back-Off timer info summary
=====
APN          : apn.0.1.com
gtp-thrsh-max-con  : 0          gtp-thrsh-max-act   : 0
pdn-gw-back-off   : disabled    pdn-gw-back-off-ti* : -
pdn-gw-back-off csreq throttling:
-----
Drop-only          ARP          Count
-----
Csreq              1            10
Csreq              2             5
Csreq              3            17
.....
Csreq              15           32
-----
load-control       : disabled
overload-control   : disabled    validity-time       : -
overload-reduction : -          overload-threshold  : -
current-apn-load   : 0%         current-apn-overlo* : no-overload
Cumulative CSReq D*: 0
-----
Number of APNs : 1
=====
* indicates that the corresponding row element may have been truncated.
```

Table 29: Show PDN APN overload output fields

Label	Description
APN	The APN name
gtp-thrsh-max-con	The maximum number of PDN connections
gtp-thrsh-max-act	The maximum number of PDN activations per second
pdn-gw-back-off	The status of the PGW Back-Off timer
pdn-gw-back-off-ti*	The time in minutes during which the MME refrains from sending subsequent PDN connection establishment requests to the PGW for the congested APN
pdn-gw-back-off csreq throttling	Shows the count of the csreq drops per ARP value Applies only when the <code>exclude-arp-list</code> command is configured
load-control	The status of the APN Load Control Information (LCI)
overload-control	The status of the APN Overload Control (OCI)
validity-time	The time in seconds that the reported overload should be in effect
overload-reduction	The percentage of the traffic reduction
overload-threshold	The percentage of the overload threshold
current-apn-load	The percentage of the current APN load
current-apn-overlo*	The current APN overload status
Cumulative CSReq D*	The cumulative number of Create Session Requests dropped
Number of APNs	The number of APNs retrieved

5.2.2.12 application-server-stats

Syntax

`application-server-stats [gateway gw-id] [udp-tunnel-group udp-tunnel-group [peer ip-address]]`

Context

`show>mobile>pdn`

Description

This command displays the statistics of the tunnel groups toward the Application Servers (AS) for non-IP PDN types.

Parameters

udp-tunnel-group

specifies a tunnel group

ip-address

specifies a peer within a tunnel group

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x: [0..FFFF]H
–	–	d: [0..255]D

gw-id

specifies the gateway ID

Values: 1 to 8

Output

The following output example shows tunnel group statistics and *Show PDN application server statistics output fields* describes the output fields.

Example output tunnel group statistics

```
# show mobile-gateway pdn application-server-stats
=====
Tunnel group statistics
=====
Tunnel Group           : Tunnel1
-----
Peer address           : 15.1.2.1

Number of tunnel establishments: 3
-----
Peer address           : 15.1.3.1

Number of tunnel establishments: 3
-----
Peer address           : 15.1.4.1

Number of tunnel establishments: 3
-----
Peer address           : 15.1.5.1

Number of tunnel establishments: 3
-----
Number of peer instances      : 4

-----
Number of tunnel groups      : 1
=====
```

Table 30: Show PDN application server statistics output fields

Label	Description
Tunnel Group	The tunnel group name
Peer address	The IP address of the peer within the tunnel group
Number of tunnel establishments	The number of tunnel establishments
Number of peer instances	The number of peer instances

Table 30: Show PDN application server statistics output fields (continued)

Label	Description
Number of tunnel groups	The number of tunnel groups

5.2.2.13 bearer-context

Syntax

```

bearer-context [imsi | imei | msisdn string [bearer bearer-id] [apn name] [gateway gw-id |
group group-number | [card slot-number | vm vm-id]]

bearer-context imsi string [apn name] [bearer bearer-id] [gateway gw-id | group group-
number | [card slot-number | vm vm-id]] [qci qci-value arp arp-value] {control | filter |
bearer-traffic | detail}

bearer-context imsi string bearer bearer-id acct {ga | gy | rf} [detail]

bearer-context mac ieee-address [apn name] [bearer bearer-id] [gateway gw-id | group group-
number | [card slot-number | vm vm-id]] [qci qci-value arp arp-value] [detail]

bearer-context default-session [def-session-pgw-ip pgw-ip-address] [apn name]
[bearer bearer-id] [gateway gw-id | group group-number | [card slot-number | vm vm-id]]

bearer-context default-session def-session-pgw-ip pgw-ip-address apn name detail
[bearer bearer-id] [gateway gw-id | group group-number | [card slot-number | vm vm-id]] [qci
qci-value arp arp-value]

bearer-context default-session def-session-pgw-ip pgw-ip-address acct {ga | gy | rf}
apn name [bearer bearer-id] [gateway gw-id] [detail]

bearer-context imsi | subscriber-id string [gateway gw-id] bearer bearer-id acct {stats}

bearer-context msisdn string [bearer bearer-id] acct gy [detail]

bearer-context def-pfcp-u-sessions [network-realm network-realm] [pfcp-u-peer-ip]
[gateway gw-id / group group-number | card slot-number | vm vm-id] [apn name]

```

Context

```
show>mobile>pdn
```

Description

This command displays bearer context information.

Parameters

arp-value

specifies the ARP value

Values: 0 to 15

bearer-id

specifies the bearer ID

Values: 5 to 15

control | filter | bearer-traffic

filters detailed show output information

default-session

displays default session statistics

detail

displays detailed information

ga | gy | rf

displays interface type output

group-number

specifies the group number

Values: 1 to 15

gw-id

specifies the gateway ID

Values: 1 to 8

ieee-address

specifies the UE MAC

name

specifies the APN name, up to 100 characters

network-realm

specifies the APN name, up to 100 characters

pgw-ip-address

displays information for the specified PGW IP address

Values	ipv4-address	a.b.c.d
–	ipv6-address	x::x::x::x::x::x (eight 16-bit pieces)

–	–	x:x:x:x:x:d.d.d
–	–	x: [0..FFFF]H
–	–	d: [0..255]D

qci-value

specifies the QCI value

Values: 1 to 9 | 65 | 66 | 69 | 70 | 128 to 254

slot-number

specifies the card slot number (applicable only to 7750 SR MG)

Values: 1 to 10 | A | B

stats

displays statistics information

string

specifies a string, up to 15 characters

subscriber-id string

displays information for the specified subscriber id, up to 256 characters

vm-id

specifies the VM ID (applicable only to CMG)

Values: 1 to 20 | A | B

Output

The following 7750 SR MG output examples show bearer context information.

**NOTE:**

- For the bearer-context accounting statistics, the aggregate counters represent the actual data traffic of the specified bearer on the current active card.
- Depending on the configured node function (UP or CP function), the description and value for some of the output fields differ. These fields, if present, are identified and described at the end of each example output.
- Depending on the configured node function (UP or CP function), the description and value for some of the output fields differ. These fields, if present, are identified and described at the end of each example output.

- *Example output bearer context statistics*
- *Example output bearer context statistics (CP function)*
- *Example output bearer context statistics (UP function)*
- *Example output bearer context IMEI statistics (UP function)*
- *Example output bearer context IMEI detail statistics (UP function)*
- *Example output bearer context IMSI statistics (UP function)*
- *Example output bearer context IMSI bearer statistics*
- *Example output bearer context IMSI detail statistics (UP function)*
- *Example output bearer context IMSI detail statistics (CP function)*
- *Example output bearer context IMSI detail statistics PGW*
- *Example output bearer context IMSI detail statistics SGW*
- *Example output bearer context IMSI detail statistics combined SGW/PGW*
- *Example output bearer context IMSI detail on session management function*
- *Example output bearer context IMSI detail statistics TWAG*
- *Example output bearer context IMSI detail statistics S2a*
- *Example output bearer context IMSI detail statistics S2b*
- *Example output bearer context IMSI detail statistics S11u*
- *Example output bearer context IMSI detail statistics Gn*
- *Example output bearer context IMSI detail statistics Gn/Gp IPv6 address support*
- *Example output bearer context IMSI detail statistics S2a PMIP*
- *Example output bearer context IMSI detail statistics S4/S12*
- *Example output bearer context IMSI detail statistics for eMPS advance priority calls*
- *Example output bearer context IMSI detail statistics for Non IP sessions*
- *Example output bearer context IMSI bearer traffic statistics*
- *Example output bearer context IMSI filter statistics*
- *Example output bearer context MAC detail statistics*
- *Example output bearer context IMSI default bearer account Ga statistics*
- *Example output bearer context IMSI dedicated bearer account Ga statistics*
- *Example output bearer context IMSI bearer account Ga detail statistics*
- *Example output bearer context IMSI bearer account Ga detail statistics for CUPS*
- *Example output bearer context IMSI bearer account Gy statistics combined SGW/PGW*

- *Example output bearer context IMSI bearer account Gy detail statistics*
- *Example output bearer context IMSI bearer account Gy detail statistics for CUPS*
- *Example output bearer context IMSI bearer account Rf statistics*
- *Example output bearer context default session APN account Ga detail statistics*
- *Example output bearer context default session APN account Gy detail statistics*
- *Example output bearer context MSISDN statistics (UP function)*
- *Example output bearer context MSISDN detail statistics (UP function)*
- *Example output UP function bearer context SEID statistics*
- *Example output UP function bearer context SEID detail statistics*

Example output bearer context statistics

```
# show mobile-gateway pdn bearer-context
=====
IMSI/IMEI(#)      APN              Bear* Type QCI/ARP  MBR/GBR  SDFs
/MSISDN(^)/MAC(`)
=====
123456789012926  nokia.corporate.* 5    Def  9/15    -        0
-----
Number of bearer contexts : 1
-----
# indicates that the corresponding row is IMEI entry.
^ indicates that the corresponding row is MSISDN entry.
` indicates that the corresponding row is MAC entry.
-----
* indicates that the corresponding row element may have been truncated.
```

Example output bearer context statistics (CP function)

```
# show mobile-gateway pdn bearer-context
=====
IMSI/IMEI(#)      APN              Bear* Type QCI/ARP  MBR/GBR  SDFs
/MSISDN(^)/MAC(`)
/SubId/SEID(~)
=====
777777777777700  ipd.nokiane.1.1.* 5    Def  9/15    -        0
-----
Number of bearer contexts : 1
-----
# indicates that the corresponding row is IMEI entry.
^ indicates that the corresponding row is MSISDN entry.
` indicates that the corresponding row is MAC entry.
~ indicates that the corresponding row is SEID entry.
-----
* indicates that the corresponding row element may have been truncated.
```

Example output bearer context statistics (UP function)

```
# show mobile-gateway pdn bearer-context
```

```

=====
IMSI/IMEI(#) APN Bear* Type QCI/ARP MBR/GBR SDFs
/MSISDN(^)/MAC(`)
/SubId/SEID(~)
=====
0x10100~ ipd.nokia.com 5 Def 9/NA - 0
460100001003204
-----
Number of bearer contexts : 1
-----
# indicates that the corresponding row is IMEI entry.
^ indicates that the corresponding row is MSISDN entry.
` indicates that the corresponding row is MAC entry.
~ indicates that the corresponding row is SEID entry.
-----
* indicates that the corresponding row element may have been truncated.

```

**NOTE:**

The bearer ID field in a node configured as the UP function, is the S5 interface traffic endpoint (S5 DL Trf End-Pt) which may be different from the bearer ID (Bearer ID).

Example output bearer context PFCP-u tunnel sessions

```

# show mobile-gateway pdn bearer-context imsi 777777777777777
=====
Bearer context
=====
IMSI/IMEI(#)      APN          Bearer  Type QCI/ARP  MBR/GBR  SDFs
-----
777777777777777  ipd.nokia* 5      Def  9/15     -         0
-----
Number of bearer contexts : 1
-----
# indicates that the corresponding row is IMEI entry.
=====
* indicates that the corresponding row element may have been truncated.

```

Example output bearer context IMEI statistics (UP function)

```

# show mobile-gateway pdn bearer-context imei 777777777777777
=====
IMSI/IMEI(#)      APN          Bear* Type QCI/ARP  MBR/GBR  SDFs
/MSISDN(^)/MAC(`)
/SubId/SEID(~)
=====
0x41110~          ipd.nokia.com 5      Def  6/NA     -         2
123456789012345
-----
Number of bearer contexts : 1
-----
# indicates that the corresponding row is IMEI entry.
^ indicates that the corresponding row is MSISDN entry.
` indicates that the corresponding row is MAC entry.

```

~ indicates that the corresponding row is SEID entry.

* indicates that the corresponding row element may have been truncated.

Example output bearer context IMEI detail statistics (UP function)

```
# show mobile-gateway pdn bearer-context imei 7777777777777777 detail
=====
Bearer context detail
=====
-----
IMSI           : 123456789012345
Gi Network Inst : ipd.nokia.com
Gn DL Trf End-Pt : 5                QCI           : 6

Up Time        : 0d 00:00:08

Gn PGW Data TEID : 0x40041115        Gn SGW Data TEID : 0x10000001
Gn PGW Data addr : 17.16.16.16
Gn PGW Data addr : 3ffe::1110:1010
Gn SGW Data addr : 10.23.15.2
Gn DL packets    : 0                  Gn DL bytes       : 0
SGI UL packets   : 0                  SGI UL bytes      : 0
Redirect SDF Cou*: 0

-----
SDF Rules Information
SDF Rule Count   : 1                  DSCP              : 0
-----
SDF rule name    : 2560
SDF rule ID      : 4096                SDF Precedence    : 65535

Charging Information
Reporting Level  : Rating Grp Level   Metering Method    : -
Billing Method   : -
Rating Group     : 1                  Service Id         : -
Sponsor Identity : -
Appl Svc Prov Id : -

Traffic Control Information
Flow Status      : Up/Downlink Enabled
Divert to AA     : No

PFCP Information
PDR ID UL        : 2560                PDR ID DL         : 35328
FAR ID UL        : 327680              FAR ID DL         : 537198592
QER ID           : 1                   QFI               : 0

Flow Information
SDF pkt filters  : 1

Filter id        : 0                    Filter direction   : BiDir
Filter protocol  : any                  Filter tos-tc      : -
Local address    : any
Remote addr      : any
-----
```

```
Number of bearer contexts : 1
```

```
* indicates that the corresponding row element may have been truncated.
```

Example output bearer context IMSI statistics (UP function)

```
# show mobile-gateway pdn bearer-context imsi 460100001003204
```

```
=====
IMSI/IMEI(#) APN Bear* Type QCI/ARP MBR/GBR SDFs
/MSISDN(^)/MAC(`)
/SubId/SEID(~)
=====
```

```
0x10100~ ipd.nokia.com 5 Def 9/NA - 0
460100001003204
```

```
-----
Number of bearer contexts : 1
```

```
-----
# indicates that the corresponding row is IMEI entry.
^ indicates that the corresponding row is MSISDN entry.
` indicates that the corresponding row is MAC entry.
~ indicates that the corresponding row is SEID entry.
-----
* indicates that the corresponding row element may have been truncated.
```

Example output bearer context IMSI bearer statistics

```
# show mobile-gateway pdn bearer-context imsi 7777777777777777 bearer 5
```

```
=====
Bearer context
```

```
=====
IMSI/IMEI(#)      APN          Bearer  Type QCI/ARP  MBR/GBR      SDFs
-----
7777777777777777 ipd.nokia* 5      Def  9/15      -             0
-----
```

```
Number of bearer contexts : 1
```

```
-----
# indicates that the corresponding row is IMEI entry.
=====
* indicates that the corresponding row element may have been truncated.
```

Example output bearer context IMSI detail statistics (UP function)

```
# show mobile-gateway pdn bearer-context imsi 7777777777777777 detail
```

```
=====
Bearer context detail
```

```
=====
IMSI              : 123456789012345
Gi Network Inst   : ipd.nokia.com
Gn DL Trf End-Pt : 5                QCI              : 6

Up Time           : 0d 00:00:07

Gn PGW Data TEID : 0x40041115      Gn SGW Data TEID : 0x10000001
Gn PGW Data addr : 17.16.16.16
Gn PGW Data addr : 3ffe::1110:1010
```

```

Gn SGW Data addr : 10.23.15.2
Gn DL packets   : 0           Gn DL bytes       : 0
SGI UL packets  : 0           SGI UL bytes      : 0
Redirect SDF Cou*: 0

-----
SDF Rules Information
SDF Rule Count   : 1           DSCP               : 0
-----
SDF rule name    : 2560
SDF rule ID      : 4096        SDF Precedence  : 65535

Charging Information
Reporting Level  : Rating Grp Level   Metering Method : -
Billing Method   : -
Rating Group     : 1                 Service Id      : -
Sponsor Identity : -
Appl Svc Prov Id : -

Traffic Control Information
Flow Status      : Up/Downlink Enabled
Divert to AA     : No

PFCP Information
PDR ID UL        : 2560           PDR ID DL       : 35328
FAR ID UL        : 327680        FAR ID DL       : 537198592
QER ID           : 1             QFI             : 0

Flow Information
SDF pkt filters  : 1

Filter id        : 0             Filter direction : BiDir
Filter protocol  : any           Filter tos-tc    : -
Local address    : any
Remote addr     : any

-----
Number of bearer contexts : 1
=====
* indicates that the corresponding row element may have been truncated.

```

Example output bearer context IMSI detail statistics (CP function)

```

# show mobile-gateway pdn bearer-context imsi 77777777777700 detail
=====
Bearer context detail
=====
IMSI           : 77777777777700
APN            : ipd.nokia.1.1.com
Bearer Id      : 5             Bearer type     : Default
QCI/ARP        : 9/11
Interworking    : Not Subscribed
Charg. Char. Rcvd: 0x0000
S1u DL Trf End-Pt: 69
Up Time        : 0d 00:01:19
eMPS IMS priority bearer : No
S1u downlink TEID: 0x1000002a   S1u SGW TEID    : 0x401181c5

```

```
S1u downlink addr: 6.6.6.1
S1u SGW address  : 3ffe::1146:1010
S1u SGW address  : 17.70.16.16
Def PFCP-u Sess  : False
CP SxN4 Data TEID: 0x70120          UP SxN4 Data TEID: 0x0
CP SxN4 Data addr: 33.33.1.1        UP SxN4 Data addr: -
CP SxN4 Data addr: 3ffe::2121:101
UP SxN4 Data addr: -
S11 QoS Modify   : 0
Charging server address: 10.23.13.2
Charging server *: Up                Charging Profile : 1
Charging triggers: 0                 Interim records  : 0
Rule Base Name   : mg2976Rb21To28AA
Rule Base Id     : 196816
Redirect SDF Cou*: 0

-----
SDF Rules Information
SDF Rule Count   : 2                  DSCP                : 0
-----
SDF rule name    : mg2976Pr21To28AA
SDF rule ID      : 39113              SDF Precedence      : 35228

Charging Information
Reporting Level  : Rating Grp Level   Metering Method     : Volume
Billing Method   : Online & Offline
Rating Group     : 13028               Service Id          : -
Sponsor Identity : -
Appl Svc Prov Id : -

Traffic Control Information
Flow Status      : Up/Downlink Enabled
Mute Notify      : Yes

PFCP Information
PDR ID UL        : 2562                PDR ID DL           : 35330
PDR ID S1u UL    : 18946
FAR ID UL        : 327680              FAR ID DL            : 537198592
QER ID           : 1                   QFI                  : 0

Flow Information
Num of AA chargi*: 8
AA Charging Group: mg2976CG_21
AA Charging Group: mg2976CG_22
AA Charging Group: mg2976CG_23
AA Charging Group: mg2976CG_24
AA Charging Group: mg2976CG_25
AA Charging Group: mg2976CG_26
AA Charging Group: mg2976CG_27
AA Charging Group: mg2976CG_28
-----
SDF rule name    : mg2976Pr31To38AA
SDF rule ID      : 39147              SDF Precedence      : 35338

Charging Information
Reporting Level  : Rating Grp Level   Metering Method     : Volume
Billing Method   : Online & Offline
```



```

Rating Group      : 16038          Service Id      : -
Sponsor Identity  : -
Appl Svc Prov Id : -

Traffic Control Information
Flow Status       : Up/Downlink Enabled
Mute Notify       : Yes

PFCP Information
PDR ID UL        : 2561           PDR ID DL        : 35329
PDR ID S1u UL    : 18945
FAR ID UL        : 327680        FAR ID DL        : 537198592
QER ID           : 1             QFI              : 0

Flow Information
Num of AA chargi*: 8
AA Charging Group: mg2976CG_31
AA Charging Group: mg2976CG_32
AA Charging Group: mg2976CG_33
AA Charging Group: mg2976CG_34
AA Charging Group: mg2976CG_35
AA Charging Group: mg2976CG_36
AA Charging Group: mg2976CG_37
AA Charging Group: mg2976CG_38
-----
Number of bearer contexts : 1
=====
* indicates that the corresponding row element may have been truncated.

```

Example output bearer context IMSI detail statistics PGW

```

# show mobile-gateway pdn bearer-context imsi 123457789012604 detail
=====
Bearer context detail
=====
IMSI              : 123457789012455
APN               : ipd.nokia.1.1.com
Bearer Id         : 5              Bearer type      : Default
QCI/ARP          : 9/11           SDFs             : 1
Charg. Char. Rcvd: 0x0000
Up Time          : 0d 00:06:06

Ref-pt/Sig-proto : S5/GTPv2

eMPS IMS priority bearer : No

S5 PGW Data TEID : 0x40015105      S5 SGW Data TEID : 0x11105
S5 PGW Data addr : 3ffe::1010:1010
S5 PGW Data addr : 16.16.16.16
S5 SGW Data addr : 3ffe::a0a:501

S5 DL packets    : 0              S5 DL bytes      : 0
SGI UL packets   : 0              SGI UL bytes     : 0
Charging server address: 10.23.13.2
Charging server *: Up              Charging Profile  : 1
Charging triggers: 0              Interim records   : 0

```

```

Rule Base Name   : otfPrb1
Rule Base Id     : 131081

SDF rule name    : multiQfiDefBrPriLevPr1
SDF rule ID      : 39003
Reporting Level  : Rating Grp Level   Metering Method : Volume
Billing Method   : Online & Offline   Flow Status      : Up/Downlink Enabled
Service Id       : -                  Rating Group     : 3
Sponsor Identity : -
Appl Svc Prov Id : -
SDF precedence   : 7002                SDF pkt filters  : 1
Divert to AA     : No
PDR ID UL        : 2561
PDR ID DL        : 35329
PDR ID S1u UL    : 18945
FAR ID UL        : 327680              FAR ID DL        : 537198592
QER ID           : 1
Filter id         : 0                  Filter direction : BiDir
Filter protocol   : udp (17)           Filter tos-tc    : -
Local address    : 0.0.0.0/0
Remote addr      : 15.1.1.1/32

```

```

-----
Number of bearer contexts : 1
=====

```

```

# show mobile-gateway pdn bearer-context imsi 123456789012353 detail

```

```

=====
Bearer context detail
=====

```

```

IMSI              : 123456789012353
APN               : ipd.nokia.com
Bearer Id         : 5                  Bearer type      : Default
Up Time          : 0d 00:05:15

QCI/ARP           : 8/15

S5 PGW Data TEID  : 0x40010525        S5 SGW Data TEID : 0x10115
S5 PGW Data addr  : 16.16.16.16
S5 SGW Data addr  : 10.10.5.1
S11u downlink TE*: 0x10000005        S11u SGW TEID   : 0x40010115
S11u downlink addr: 6.6.6.1
S11u SGW address  : 10.10.11.2

S5 UL packets     : 0                  S5 UL bytes      : 0
S11u DL packets   : 15                 S11u DL bytes    : 1320
S11u QoS Modify   : 0

```

```

-----
Number of bearer contexts : 1
=====

```

Example output bearer context IMSI detail statistics SGW

```

# show mobile-gateway pdn bearer-context imsi 1126000000000000 detail

```

```

=====
Bearer context detail

```

```

=====
IMSI           : 112600000000000
APN            : cg1.ipd.nokia.com
Bearer Id      : 5                      Bearer type      : Default

eMPS IMS priority bearer : No

Up Time        : 0d 00:07:13

QCI/ARP        : 9/15

S5 PGW Data TEID : 0x21000000          S5 SGW Data TEID : 0x30175
S5 PGW Data addr : 192.168.16.10
S5 SGW Data addr : 192.168.16.10
S5 SGW Data addr : 3ffe::105a:100a
S1u downlink TEID: 0x10000028          S1u SGW TEID      : 0x40030175
S1u downlink addr: 10.6.6.1
S1u SGW address  : 192.168.16.16
S1u SGW address  : 3ffe::1046:1010

S5 UL packets   : 0                    S5 UL bytes       : 0
S1u DL packets  : 0                    S1u DL bytes      : 0
S11 QoS Modify  : 0
Charging server address: 10.23.13.2
Charging server state : Up
Charging profile : 1
Charging triggers: 0                    Interim records   : 0
-----
Number of bearer contexts : 1
=====

```

Example output bearer context IMSI detail statistics combined SGW/PGW

```

# show mobile-gateway pdn bearer-context imsi 123456789012809 detail
=====
Bearer context detail
=====
IMSI           : 123456789012809
APN            : ipd.nokia.1.1.com
Bearer Id      : 5                      Bearer type      : Default
QCI/ARP        : 9/15                  SDFs             : 0
Charg. Char. Rcvd: 0x0000
Up Time        : 0d 00:04:59

S1u downlink TEID: 0x10000009          S1u SGW TEID      : 0x400155a5
S1u downlink addr: 6.6.6.1
S1u SGW address  : 16.70.16.16
S1u DL packets   : 0                    S1u DL bytes      : 0
SGI UL packets   : 0                    SGI UL bytes      : 0
S11 QoS Modify    : 0

-----
Number of bearer contexts : 1
=====

```

Example output bearer context IMSI detail on session management function

```
# show mobile-gateway pdn bearer-context imsi 11111111000000 detail
=====
Bearer context detail
=====
IMSI           : 11111111000000
APN            : ipd.nokia.1.2.com
Bearer Id      : 5                      Bearer type    : Default
QCI/ARP        : 9/15                  SDFs           : 0
Charg. Char. Rcvd: 0x0000
Up Time        : 0d 00:11:57
eMPS IMS priority bearer : No
S1u downlink TEID: 0x10000002          S1u SGW TEID   : 0x40010105
S1u downlink addr: 6.6.6.1
S1u SGW address : 3ffe::1146:1010
S1u SGW address : 17.70.16.16
S11 QoS Modify  : 0
Charging server address: 10.23.13.2
Charging server *: Up                  Charging Profile : 1
Charging triggers: 0                   Interim records  : 0
-----
IMSI           : 11111111000000
APN            : ipd.nokia.1.2.com
Bearer Id      : 6                      Bearer type    : Dedicated
QCI/ARP        : 3/2                  SDFs           : 1
QoS UL MBR     : 10 Kbps              QoS DL MBR     : 10 Kbps
QoS UL GBR     : 10 Kbps              QoS DL GBR     : 10 Kbps
Charg. Char. Rcvd: 0x0000
Up Time        : 0d 00:11:57
eMPS IMS priority bearer : No
S1u downlink TEID: 0x10000003          S1u SGW TEID   : 0x40010106
S1u downlink addr: 6.6.6.1
S1u SGW address : 3ffe::1146:1010
S1u SGW address : 17.70.16.16
S11 QoS Modify  : 0
Charging server address: 10.23.13.2
Charging server *: Up                  Charging Profile : 1
Charging triggers: 0                   Interim records  : 0
SDF rule name   : Rule_1
Reporting Level : Rating Grp Level    Metering Method  : -
Billing Method  : Online & Offline    Flow Status     : Up/Downlink Enabled
Service Id      : -                   Rating Group     : 2
Sponsor Identity : -
Appl Svc Prov Id : -
SDF precedence  : 100                  SDF pkt filters  : 2
SDF QoS UL MBR  : 10 Kbps              SDF QoS DL MBR  : 10 Kbps
SDF QoS UL GBR  : 10 Kbps              SDF QoS DL GBR  : 10 Kbps
Divert to AA    : No
PDR ID UL       : 6145                  PDR ID DL       : 38913
PDR ID S1u UL   : 22529
FAR ID UL       : 393216                FAR ID DL       : 537264128
QER ID          : 4102
Filter id       : 0                     Filter direction : DL
Filter protocol : udp (17)              Filter tos-tc    : 0
Local address   : 2.127.255.0/32
```

```

Local  ports      : 1234
Remote addr      : 15.1.1.2/32
Remote ports     : 4321
Filter id        : 1                Filter direction : UL
Filter protocol  : udp (17)         Filter tos-tc    : 0
Local  address   : 2.127.255.0/32
Local  ports     : 1234
Remote addr     : 15.1.1.2/32
Remote ports    : 4321
-----
Number of bearer contexts : 2
=====
* indicates that the corresponding row element may have been truncated.

```

Example output bearer context IMSI detail statistics TWAG

```

# show mobile-gateway pdn bearer-context imsi 3100120000000000 detail
=====
Bearer context detail
=====
IMSI/MAC : 3100120000000000/00:02:00:00:00:00
APN       : ipd.nokia.1.1.com.mnc012.mcc310.gprs
Bearer Id : 5                Bearer type   : Default
Up Time   : 0d 00:00:02

QCI/ARP   : 9/1
QoS UL MBR : 0 Kbps          QoS DL MBR   : 0 Kbps
QoS UL GBR : 0 Kbps          QoS DL GBR   : 0 Kbps

Sww TWAG tunnel addr : 16.81.16.17
Sww AP tunnel addr   : 180.1.3.2
Sww TWAG MAC address : 74:7f:ff:00:03:5b
Sww AP MAC address   : 99:02:00:00:00:00
S2a TWAG Data Teid   : 0x50d95
S2a PGW Data Teid    : 0x40041945
S2a TWAG V4/V6 Data Addr : 16.16.16.16
S2a PGW V4/V6 Data Addr : 10.10.5.1

S2a UL packets : 0                S2a UL bytes   : 0
Sww DL packets : 0                Sww DL bytes   : 0
Sww QoS Modify  : 0

SDF rule name : DHCP_DELAYED_IP_RULE
Reporting Level : Rating Grp Level  Metering Method : Volume
Billing Method  : -                  Flow Status      : Up/Downlink Enabled
Service Id     : -                  Rating Group     : 0
Sponsor Identity : -
Appl Svc Prov Id : -
SDF precedence  : 0                SDF pkt filters  : 1
Divert to AA    : No

Filter id       : 0                Filter direction : UL
Filter protocol : udp (17)         Filter tos-tc    : -
Local  address  : any
Remote addr    : any
Remote ports   : 67

```

```
-----
Number of bearer contexts : 1
=====
```

Example output bearer context IMSI detail statistics S2a

```
# show mobile-gateway pdn bearer-context imsi 22222222222200 detail
=====
Bearer context detail
=====
IMSI           : 22222222222200
APN            : ims.corporate.1.3.com
Bearer Id      : 5                Bearer type      : Default
QCI/ARP        : 9/1              SDFs            : 0
Charg. Char. Rcvd: 0x0000
Up Time        : 0d 00:01:11

Ref-pt/Sig-proto : S2a/GTPv2

S2a PGW Data TEID: 0x40060d75      S2a TWAG Data TE*: 0x30000019
S2a PGW Data addr : 16.16.16.16
S2a TWAG Data addr : 10.70.24.101

S2a DL packets   : 0                S2a DL bytes    : 0
SGI UL packets   : 0                SGI UL bytes    : 0
Charging server address: 10.23.13.2
Charging server *: Up                Charging Profile : 1
Charging triggers: 0                Interim records  : 0

-----
IMSI           : 22222222222200
APN            : ims.corporate.1.3.com
Bearer Id      : 6                Bearer type      : Dedicated
QCI/ARP        : 7/12             SDFs            : 1
Charg. Char. Rcvd: 0x0000
Up Time        : 0d 00:01:10

Ref-pt/Sig-proto : S2a/GTPv2

S2a PGW Data TEID: 0x40060d76      S2a TWAG Data TE*: 0x3000001a
S2a PGW Data addr : 16.16.16.16
S2a TWAG Data addr : 10.70.24.101

S2a DL packets   : 0                S2a DL bytes    : 0
SGI UL packets   : 0                SGI UL bytes    : 0
Charging server address: 10.23.13.2
Charging server *: Up                Charging Profile : 1
Charging triggers: 0                Interim records  : 0

SDF rule name    : rule for qci 7 arp 12 precedence 2000
Reporting Level  : Rating Grp Level  Metering Method  : -
Billing Method   : Online & Offline   Flow Status      : Up/Downlink Enabled
Service Id       : -                  Rating Group     : -
Sponsor Identity : -
Appl Svc Prov Id : -
SDF precedence   : 2000                SDF pkt filters  : 2
```

```

Divert to AA      : No

Filter id         : 0                      Filter direction : DL
Filter protocol   : icmp (1)              Filter tos-tc      : -
Local address     : 3ffe:0:3:ff00::2/64
Remote addr       : 3ffe::f01:102/128

Filter id         : 1                      Filter direction : UL
Filter protocol   : any                   Filter tos-tc      : -
Local address     : 3ffe:0:3:ff00::2/64
Remote addr       : 3ffe::f01:102/128
-----

IMSI              : 22222222222200
APN               : ims.corporate.1.3.com
Bearer Id         : 7                     Bearer type        : Dedicated
QCI/ARP           : 8/13                  SDFs               : 1
Charg. Char. Rcvd: 0x0000
Up Time           : 0d 00:01:10

Ref-pt/Sig-proto  : S2a/GTPv2

S2a PGW Data TEID: 0x40060d77             S2a TWAG Data TE*: 0x3000001b
S2a PGW Data addr : 16.16.16.16
S2a TWAG Data addr : 10.70.24.101

S2a DL packets    : 0                     S2a DL bytes       : 0
SGI UL packets    : 0                     SGI UL bytes       : 0
Charging server address: 10.23.13.2
Charging server *: Up                     Charging Profile    : 1
Charging triggers: 0                     Interim records     : 0

SDF rule name     : rule for qci 8 arp 13 precedence 3000
Reporting Level   : Rating Grp Level      Metering Method     : -
Billing Method    : Online & Offline      Flow Status         : Up/Downlink Enabled
Service Id        : -                     Rating Group        : -
Sponsor Identity  : -
Appl Svc Prov Id  : -
SDF precedence    : 3000                  SDF pkt filters     : 2
Divert to AA      : No

Filter id         : 0                      Filter direction : DL
Filter protocol   : icmp (1)              Filter tos-tc      : -
Local address     : 3ffe:0:3:ff00::2/64
Remote addr       : 3ffe::f01:103/128

Filter id         : 1                      Filter direction : UL
Filter protocol   : any                   Filter tos-tc      : -
Local address     : 3ffe:0:3:ff00::2/64
Remote addr       : 3ffe::f01:103/128
-----

IMSI              : 22222222222200
APN               : ims.corporate.1.4.com
Bearer Id         : 11                    Bearer type        : Default
QCI/ARP           : 5/2                  SDFs               : 0
Charg. Char. Rcvd: 0x0000
Up Time           : 0d 00:01:08

```

Ref-pt/Sig-proto : S2a/GTPv2

S2a PGW Data TEID: 0x40070d75 S2a TWAG Data TE*: 0x3000001c
S2a PGW Data addr : 16.16.16.16
S2a TWAG Data addr : 10.70.24.101

S2a DL packets : 0 S2a DL bytes : 0
SGI UL packets : 0 SGI UL bytes : 0
Charging server address: 10.23.13.2
Charging server *: Up Charging Profile : 1
Charging triggers: 0 Interim records : 0

IMSI : 22222222222200
APN : ims.corporate.1.4.com
Bearer Id : 8 Bearer type : Dedicated
QCI/ARP : 7/12 SDFs : 1
Charg. Char. Rcvd: 0x0000
Up Time : 0d 00:01:07

Ref-pt/Sig-proto : S2a/GTPv2

S2a PGW Data TEID: 0x40070d76 S2a TWAG Data TE*: 0x3000001d
S2a PGW Data addr : 16.16.16.16
S2a TWAG Data addr : 10.70.24.101

S2a DL packets : 0 S2a DL bytes : 0
SGI UL packets : 0 SGI UL bytes : 0
Charging server address: 10.23.13.2
Charging server *: Up Charging Profile : 1
Charging triggers: 0 Interim records : 0

SDF rule name : rule for qci 7 arp 12
Reporting Level : Rating Grp Level Metering Method : -
Billing Method : Online & Offline Flow Status : Up/Downlink Enabled
Service Id : - Rating Group : -
Sponsor Identity : -
Appl Svc Prov Id : -
SDF precedence : 2000 SDF pkt filters : 2
Divert to AA : No

Filter id : 0 Filter direction : DL
Filter protocol : any Filter tos-tc : -
Local address : 4.127.255.0/32
Remote addr : 15.1.1.2/32

Filter id : 1 Filter direction : UL
Filter protocol : any Filter tos-tc : -
Local address : 4.127.255.0/32
Remote addr : 15.1.1.2/32

IMSI : 22222222222200
APN : ims.corporate.1.4.com
Bearer Id : 9 Bearer type : Dedicated
QCI/ARP : 8/13 SDFs : 1
Charg. Char. Rcvd: 0x0000
Up Time : 0d 00:01:07


```

Ref-pt/Sig-proto : S2a/GTPv2

S2a PGW Data TEID: 0x40070d77          S2a TWAG Data TE*: 0x3000001e
S2a PGW Data addr : 16.16.16.16
S2a TWAG Data addr : 10.70.24.101

S2a DL packets   : 0                   S2a DL bytes      : 0
SGI UL packets   : 0                   SGI UL bytes      : 0
Charging server address: 10.23.13.2
Charging server *: Up                   Charging Profile : 1
Charging triggers: 0                   Interim records  : 0

SDF rule name    : rule for qci 8 arp 13
Reporting Level  : Rating Grp Level    Metering Method   : -
Billing Method   : Online & Offline    Flow Status       : Up/Downlink Enabled
Service Id       : -                   Rating Group      : -
Sponsor Identity : -
Appl Svc Prov Id : -
SDF precedence   : 3000                SDF pkt filters   : 2
Divert to AA     : No

Filter id        : 0                   Filter direction  : DL
Filter protocol  : any                 Filter tos-tc     : -
Local address    : 4.127.255.0/32
Remote addr      : 15.1.1.3/32

Filter id        : 1                   Filter direction  : UL
Filter protocol  : any                 Filter tos-tc     : -
Local address    : 4.127.255.0/32
Remote addr      : 15.1.1.3/32
-----
Number of bearer contexts : 6
=====
* indicates that the corresponding row element may have been truncated.

```

Example output bearer context IMSI detail statistics S2b

```

# show mobile-gateway pdn bearer-context imsi 22222222220000 detail
=====
Bearer context detail
=====
IMSI           : 22222222220000
APN            : ims.corporate.1.3.com
Bearer Id      : 5                   Bearer type      : Default
QCI/ARP        : 9/1                 SDFs             : 0
Charg. Char. Rcvd: 0x0000
Up Time        : 0d 00:01:05

Ref-pt/Sig-proto : S2b/GTPv2

S2b PGW Data TEID: 0x40070db5          S2b ePDG Data TE*: 0x10000019
S2b PGW Data addr : 16.70.16.19
S2b ePDG Data addr : 10.10.7.74

S2b DL packets   : 0                   S2b DL bytes      : 0

```

```

SGI UL packets   : 0          SGI UL bytes    : 0
Charging server address: 10.23.13.2
Charging server *: Up          Charging Profile : 1
Charging triggers: 0          Interim records  : 0

-----

IMSI             : 22222222220000
APN              : ims.corporate.1.3.com
Bearer Id        : 6          Bearer type     : Dedicated
QCI/ARP          : 7/12       SDFs          : 1
Charg. Char. Rcvd: 0x0000
Up Time          : 0d 00:01:04

Ref-pt/Sig-proto : S2b/GTPv2

S2b PGW Data TEID: 0x40070db6    S2b ePDG Data TE*: 0x1000001a
S2b PGW Data addr : 16.70.16.19
S2b ePDG Data addr : 10.10.7.74

S2b DL packets   : 0          S2b DL bytes    : 0
SGI UL packets   : 0          SGI UL bytes    : 0
Charging server address: 10.23.13.2
Charging server *: Up          Charging Profile : 1
Charging triggers: 0          Interim records  : 0

SDF rule name    : rule for qci 7 arp 12 precedence 2000
Reporting Level  : Rating Grp Level   Metering Method  : -
Billing Method   : Online & Offline   Flow Status      : Up/Downlink Enabled
Service Id       : -                  Rating Group     : -
Sponsor Identity : -
Appl Svc Prov Id : -
SDF precedence   : 2000              SDF pkt filters  : 2
Divert to AA     : No

Filter id         : 0          Filter direction : DL
Filter protocol   : icmp (1)   Filter tos-tc    : -
Local address     : 3ffe:0:3:ff01::2/64
Remote addr       : 3ffe::f01:102/128

Filter id         : 1          Filter direction : UL
Filter protocol   : any        Filter tos-tc    : -
Local address     : 3ffe:0:3:ff01::2/64
Remote addr       : 3ffe::f01:102/128

-----

IMSI             : 22222222220000
APN              : ims.corporate.1.3.com
Bearer Id        : 7          Bearer type     : Dedicated
QCI/ARP          : 8/13       SDFs          : 1
Charg. Char. Rcvd: 0x0000
Up Time          : 0d 00:01:04

Ref-pt/Sig-proto : S2b/GTPv2

S2b PGW Data TEID: 0x40070db7    S2b ePDG Data TE*: 0x1000001b
S2b PGW Data addr : 16.70.16.19
S2b ePDG Data addr : 10.10.7.74

```

```

S2b DL packets   : 0           S2b DL bytes    : 0
SGI UL packets   : 0           SGI UL bytes    : 0
Charging server address: 10.23.13.2
Charging server *: Up           Charging Profile : 1
Charging triggers: 0           Interim records : 0

SDF rule name    : rule for qci 8 arp 13 precedence 3000
Reporting Level  : Rating Grp Level   Metering Method : -
Billing Method   : Online & Offline   Flow Status     : Up/Downlink Enabled
Service Id       : -                 Rating Group    : -
Sponsor Identity : -
Appl Svc Prov Id : -
SDF precedence   : 3000              SDF pkt filters : 2
Divert to AA     : No

Filter id        : 0           Filter direction : DL
Filter protocol  : icmp (1)    Filter tos-tc   : -
Local address    : 3ffe:0:3:ff01::2/64
Remote addr      : 3ffe::f01:103/128

Filter id        : 1           Filter direction : UL
Filter protocol  : any         Filter tos-tc   : -
Local address    : 3ffe:0:3:ff01::2/64
Remote addr      : 3ffe::f01:103/128
-----
IMSI             : 22222222220000
APN              : ims.corporate.1.4.com
Bearer Id        : 11          Bearer type     : Default
QCI/ARP          : 5/2         SDFs            : 0
Charg. Char. Rcvd: 0x0000
Up Time          : 0d 00:01:02

Ref-pt/Sig-proto : S2b/GTPv2

S2b PGW Data TEID: 0x40080db5   S2b ePDG Data TE*: 0x1000001c
S2b PGW Data addr : 16.70.16.19
S2b ePDG Data addr : 10.10.7.74

S2b DL packets   : 0           S2b DL bytes    : 0
SGI UL packets   : 0           SGI UL bytes    : 0
Charging server address: 10.23.13.2
Charging server *: Up           Charging Profile : 1
Charging triggers: 0           Interim records : 0
-----
IMSI             : 22222222220000
APN              : ims.corporate.1.4.com
Bearer Id        : 8           Bearer type     : Dedicated
QCI/ARP          : 7/12        SDFs            : 1
Charg. Char. Rcvd: 0x0000
Up Time          : 0d 00:01:00

Ref-pt/Sig-proto : S2b/GTPv2

S2b PGW Data TEID: 0x40080db6   S2b ePDG Data TE*: 0x1000001d
S2b PGW Data addr : 16.70.16.19
S2b ePDG Data addr : 10.10.7.74

```

```

S2b DL packets   : 0                S2b DL bytes      : 0
SGI UL packets   : 0                SGI UL bytes      : 0
Charging server address: 10.23.13.2
Charging server *: Up                Charging Profile : 1
Charging triggers: 0                Interim records  : 0

SDF rule name    : rule for qci 7 arp 12
Reporting Level  : Rating Grp Level  Metering Method   : -
Billing Method   : Online & Offline  Flow Status       : Up/Downlink Enabled
Service Id       : -                 Rating Group      : -
Sponsor Identity : -
Appl Svc Prov Id : -
SDF precedence   : 2000              SDF pkt filters   : 2
Divert to AA     : No

Filter id        : 0                Filter direction  : DL
Filter protocol  : any              Filter tos-tc     : -
Local address    : 4.127.255.1/32
Remote addr      : 15.1.1.2/32

Filter id        : 1                Filter direction  : UL
Filter protocol  : any              Filter tos-tc     : -
Local address    : 4.127.255.1/32
Remote addr      : 15.1.1.2/32
-----
IMSI             : 22222222220000
APN              : ims.corporate.1.4.com
Bearer Id        : 9                Bearer type       : Dedicated
QCI/ARP          : 8/13             SDFs              : 1
Charg. Char. Rcvd: 0x0000
Up Time          : 0d 00:01:00

Ref-pt/Sig-proto : S2b/GTPv2

S2b PGW Data TEID: 0x40080db7      S2b ePDG Data TE*: 0x1000001e
S2b PGW Data addr : 16.70.16.19
S2b ePDG Data addr : 10.10.7.74

S2b DL packets   : 0                S2b DL bytes      : 0
SGI UL packets   : 0                SGI UL bytes      : 0
Charging server address: 10.23.13.2
Charging server *: Up                Charging Profile : 1
Charging triggers: 0                Interim records  : 0

SDF rule name    : rule for qci 8 arp 13
Reporting Level  : Rating Grp Level  Metering Method   : -
Billing Method   : Online & Offline  Flow Status       : Up/Downlink Enabled
Service Id       : -                 Rating Group      : -
Sponsor Identity : -
Appl Svc Prov Id : -
SDF precedence   : 3000              SDF pkt filters   : 2
Divert to AA     : No

Filter id        : 0                Filter direction  : DL
Filter protocol  : any              Filter tos-tc     : -
Local address    : 4.127.255.1/32

```

```

Remote addr      : 15.1.1.3/32

Filter id        : 1                      Filter direction : UL
Filter protocol  : any                    Filter tos-tc     : -
Local address    : 4.127.255.1/32
Remote addr      : 15.1.1.3/32
-----
Number of bearer contexts : 6
=====
* indicates that the corresponding row element may have been truncated.

```

Example output bearer context IMSI detail statistics S11u

```

# show mobile-gateway pdn bearer-context imsi 123456789012571 detail
=====
Bearer context detail
=====
IMSI              : 123456789012571
APN               : ipd.nokia.1.1.com
Bearer Id         : 5                      Bearer type      : Default
QCI/ARP           : 8/15                   SDFs             : 0
Charg. Char. Rcvd: 0x0000
Up Time           : 0d 00:03:59

S11u downlink TE*: 0x10000006             S11u SGW TEID    : 0x40010535
S11u downlink addr: 6.6.6.1
S11u SGW address  : 16.70.16.17

S11u DL packets   : 0                      S11u DL bytes    : 0
SGI UL packets    : 0                      SGI UL bytes     : 0
S11u QoS Modify   : 0
Charging server address: 10.23.13.2
Charging server *: Up                      Charging Profile  : 1
Charging triggers: 0                      Interim records   : 0

-----
Number of bearer contexts : 1
=====

```

Example output bearer context IMSI detail statistics Gn

```

# show mobile-gateway pdn bearer-context imsi 333333333330000 detail
=====
Bearer context detail
=====
IMSI              : 333333333330000
APN               : ims.corporate.0.3.com
Bearer Id         : 5                      Bearer type      : Default
QCI/ARP           : 9/1                   SDFs             : 0
Charg. Char. Rcvd: 0x0000
Qos Byte Dump     : (length=16)
                   : 87 00 0d 01 23 91 1f
                   : 92 99 96 c5 74 fb ff
                   : ff 00
Up Time           : 0d 00:00:59

```

Ref-pt/Sig-proto : Gn/GTPv1

Gn PGW Data TEID : 0x40060d85 Gn SGSN Data TEID: 0x10000019
 Gn PGW Data addr : 16.16.16.16
 Gn SGSN Data addr : 10.23.15.2

Gn DL packets : 0 Gn DL bytes : 0
 SGI UL packets : 0 SGI UL bytes : 0
 Charging server address: 10.23.13.2
 Charging server *: Up Charging Profile : 1
 Charging triggers: 0 Interim records : 0

 IMSI : 333333333330000
 APN : ims.corporate.0.3.com
 Bearer Id : 6 Bearer type : Dedicated
 QCI/ARP : 5/1 SDFs : 1
 Charg. Char. Rcvd: 0x0000
 Qos Byte Dump : (length=16)
 : 87 00 0d 01 0b 91 1f
 : 72 99 96 c5 74 f9 ff
 : ff 10
 Up Time : 0d 00:00:49

Ref-pt/Sig-proto : Gn/GTPv1

Gn PGW Data TEID : 0x40060d86 Gn SGSN Data TEID: 0x1000001a
 Gn PGW Data addr : 16.16.16.16
 Gn SGSN Data addr : 10.23.15.2

Gn DL packets : 0 Gn DL bytes : 0
 SGI UL packets : 0 SGI UL bytes : 0
 Charging server address: 10.23.13.2
 Charging server *: Up Charging Profile : 1
 Charging triggers: 0 Interim records : 0

SDF rule name : rule-1-qci:5-arp:1
 Reporting Level : Rating Grp Level Metering Method : -
 Billing Method : Online & Offline Flow Status : Up/Downlink Enabled
 Service Id : - Rating Group : -
 Sponsor Identity : -
 Appl Svc Prov Id : -
 SDF precedence : 3 SDF pkt filters : 4
 Divert to AA : No

Filter id : 0 Filter direction : DL
 Filter protocol : udp (17) Filter tos-tc : -
 Local address : 3.0.255.0/32
 Remote addr : 15.1.1.2/32

Filter id : 2 Filter direction : DL
 Filter protocol : udp (17) Filter tos-tc : -
 Local address : 3.0.255.0/32
 Remote addr : 15.1.1.2/32

Filter id : 1 Filter direction : UL
 Filter protocol : udp (17) Filter tos-tc : -

```

Local address : 3.0.255.0/32
Remote addr   : 15.1.1.2/32

Filter id      : 3                      Filter direction : UL
Filter protocol : udp (17)              Filter tos-tc     : -
Local address  : 3.0.255.0/32
Remote addr    : 15.1.1.2/32
-----
IMSI           : 333333333330000
APN            : ims.corporate.0.3.com
Bearer Id      : 7                      Bearer type       : Dedicated
QCI/ARP        : 6/1                    SDFs              : 1
Charg. Char. Rcvd: 0x0000
Qos Byte Dump  : (length=16)
                  : 87 00 0d 01 0b 91 1f
                  : 72 99 96 c5 74 f9 ff
                  : ff 00
Up Time        : 0d 00:00:39

Ref-pt/Sig-proto : Gn/GTPv1

Gn PGW Data TEID : 0x40060d87          Gn SGSN Data TEID: 0x1000001b
Gn PGW Data addr : 16.16.16.16
Gn SGSN Data addr : 10.23.15.2

Gn DL packets   : 0                    Gn DL bytes       : 0
SGI UL packets  : 0                    SGI UL bytes      : 0
Charging server address: 10.23.13.2
Charging server *: Up                  Charging Profile   : 1
Charging triggers: 0                   Interim records    : 0

SDF rule name    : rule-1-qci:6-arp:1
Reporting Level  : Rating Grp Level    Metering Method    : -
Billing Method   : Online & Offline    Flow Status        : Up/Downlink Enabled
Service Id       : -                   Rating Group        : -
Sponsor Identity : -
Appl Svc Prov Id : -
SDF precedence   : 3                   SDF pkt filters    : 4
Divert to AA     : No

Filter id        : 0                      Filter direction : DL
Filter protocol  : udp (17)              Filter tos-tc     : -
Local address    : 3.0.255.0/32
Remote addr      : 15.1.1.3/32

Filter id        : 2                      Filter direction : DL
Filter protocol  : udp (17)              Filter tos-tc     : -
Local address    : 3.0.255.0/32
Remote addr      : 15.1.1.3/32

Filter id        : 1                      Filter direction : UL
Filter protocol  : udp (17)              Filter tos-tc     : -
Local address    : 3.0.255.0/32
Remote addr      : 15.1.1.3/32

Filter id        : 3                      Filter direction : UL
Filter protocol  : udp (17)              Filter tos-tc     : -

```

```

Local address : 3.0.255.0/32
Remote addr   : 15.1.1.3/32
-----
IMSI          : 33333333330000
APN           : ims.corporate.0.4.com
Bearer Id     : 11                      Bearer type    : Default
QCI/ARP       : 9/1                      SDFs           : 0
Charg. Char. Rcvd: 0x0000
Qos Byte Dump : (length=16)
               : 87 00 0d 01 23 91 1f
               : 92 99 96 c5 74 fb ff
               : ff 00
Up Time       : 0d 00:00:33

Ref-pt/Sig-proto : Gn/GTPv1

Gn PGW Data TEID : 0x40070d85          Gn SGSN Data TEID: 0x1000001c
Gn PGW Data addr : 16.16.16.16
Gn SGSN Data addr : 10.23.15.2

Gn DL packets   : 0                      Gn DL bytes     : 0
SGI UL packets  : 0                      SGI UL bytes    : 0
Charging server address: 10.23.13.2
Charging server *: Up                      Charging Profile : 1
Charging triggers: 0                      Interim records  : 0
-----
IMSI          : 33333333330000
APN           : ims.corporate.0.4.com
Bearer Id     : 8                      Bearer type    : Dedicated
QCI/ARP       : 5/1                      SDFs           : 1
Charg. Char. Rcvd: 0x0000
Qos Byte Dump : (length=16)
               : 87 00 0d 01 0b 91 1f
               : 72 99 96 c5 74 f9 ff
               : ff 10
Up Time       : 0d 00:00:22

Ref-pt/Sig-proto : Gn/GTPv1

Gn PGW Data TEID : 0x40070d86          Gn SGSN Data TEID: 0x1000001d
Gn PGW Data addr : 16.16.16.16
Gn SGSN Data addr : 10.23.15.2

Gn DL packets   : 0                      Gn DL bytes     : 0
SGI UL packets  : 0                      SGI UL bytes    : 0
Charging server address: 10.23.13.2
Charging server *: Up                      Charging Profile : 1
Charging triggers: 0                      Interim records  : 0

SDF rule name   : rule-1-qci:5-arp:1
Reporting Level : Rating Grp Level      Metering Method  : -
Billing Method  : Online & Offline      Flow Status     : Up/Downlink Enabled
Service Id     : -                      Rating Group     : -
Sponsor Identity : -
Appl Svc Prov Id : -
SDF precedence  : 3                      SDF pkt filters : 4

```



```

Divert to AA      : No

Filter id         : 0                Filter direction : DL
Filter protocol   : udp (17)         Filter tos-tc      : -
Local address     : 4.0.255.0/32
Remote addr       : 15.1.1.2/32

Filter id         : 2                Filter direction : DL
Filter protocol   : udp (17)         Filter tos-tc      : -
Local address     : 4.0.255.0/32
Remote addr       : 15.1.1.2/32

Filter id         : 1                Filter direction : UL
Filter protocol   : udp (17)         Filter tos-tc      : -
Local address     : 4.0.255.0/32
Remote addr       : 15.1.1.2/32

Filter id         : 3                Filter direction : UL
Filter protocol   : udp (17)         Filter tos-tc      : -
Local address     : 4.0.255.0/32
Remote addr       : 15.1.1.2/32
-----
IMSI              : 333333333330000
APN               : ims.corporate.0.4.com
Bearer Id         : 9                Bearer type        : Dedicated
QCI/ARP           : 6/1              SDFs               : 1
Charg. Char. Rcvd: 0x0000
Qos Byte Dump     : (length=16)
                  : 87 00 0d 01 0b 91 1f
                  : 72 99 96 c5 74 f9 ff
                  : ff 00
Up Time           : 0d 00:00:12

Ref-pt/Sig-proto : Gn/GTPv1

Gn PGW Data TEID : 0x40070d87        Gn SGSN Data TEID: 0x1000001e
Gn PGW Data addr : 16.16.16.16
Gn SGSN Data addr : 10.23.15.2

Gn DL packets     : 0                Gn DL bytes        : 0
SGI UL packets    : 0                SGI UL bytes       : 0
Charging server address: 10.23.13.2
Charging server *: Up                Charging Profile    : 1
Charging triggers: 0                Interim records     : 0

SDF rule name     : rule-1-qci:6-arp:1
Reporting Level    : Rating Grp Level Metering Method     : -
Billing Method     : Online & Offline  Flow Status         : Up/Downlink Enabled
Service Id         : -                 Rating Group         : -
Sponsor Identity   : -
Appl Svc Prov Id   : -
SDF precedence     : 3                SDF pkt filters     : 4
Divert to AA       : No

Filter id         : 0                Filter direction : DL
Filter protocol   : udp (17)         Filter tos-tc      : -
Local address     : 4.0.255.0/32

```

```

Remote addr      : 15.1.1.3/32

Filter id        : 2                Filter direction : DL
Filter protocol  : udp (17)         Filter tos-tc    : -
Local address    : 4.0.255.0/32
Remote addr      : 15.1.1.3/32

Filter id        : 1                Filter direction : UL
Filter protocol  : udp (17)         Filter tos-tc    : -
Local address    : 4.0.255.0/32
Remote addr      : 15.1.1.3/32

Filter id        : 3                Filter direction : UL
Filter protocol  : udp (17)         Filter tos-tc    : -
Local address    : 4.0.255.0/32
Remote addr      : 15.1.1.3/32
-----
Number of bearer contexts : 6
=====
* indicates that the corresponding row element may have been truncated.

```

Example output bearer context IMSI detail statistics Gn/Gp IPv6 address support

```

# show mobile-gateway pdn bearer-context imsi 310150123456789 detail
=====
Bearer context detail
=====
IMSI              : 310150123456789
APN               : ipd.nokia.com
Bearer Id         : 5                Bearer type      : Default
QCI/ARP           : 6/1             SDFs             : 0
Charg. Char. Rcvd: 0x0000
Qos Byte Dump     : (length=16)
                   : 87 00 0d 01 0b 61 1f
                   : 6b 96 fe fe 74 f9 ff
                   : ff 00
Up Time           : 0d 00:01:23

Ref-pt/Sig-proto : Gn/GTPv1

eMPS IMS priority bearer : No

Gn PGW Data TEID : 0x40020505        Gn SGSN Data TEID: 0x10000006
Gn PGW Data addr : 3ffe::1010:1010
Gn PGW Data addr : 16.16.16.16
Gn SGSN Data addr : 3ffe::a17:f02

Gn DL packets     : 0                Gn DL bytes      : 0
SGI UL packets    : 0                SGI UL bytes     : 0
Charging server address: 10.23.13.2
Charging server *: Up           Charging Profile : 1
Charging triggers: 0            Interim records  : 0
-----
Number of bearer contexts : 1
=====

```

Example output bearer context IMSI detail statistics S2a PMIP

```
# show mobile-gateway pdn bearer-context imsi 44444444440000 detail
=====
Bearer context detail
=====
IMSI           : 44444444440000
APN            : ims.corporate.1.1.com
Bearer Id      : 5                      Bearer type    : Default
QCI/ARP        : 9/1                    SDFs           : 0
Charg. Char. Rcvd: 0x0000
Up Time        : 0d 00:00:05

Ref-pt/Sig-proto : S2a/PMIPv6

PGW GRE Key    : 0x40070da0             MAG GRE Key    : 0x8000000a
PGW Data addr  : 3ffe::b14:103
MAG Data addr  : 3ffe::a0a:2b17

S2a DL Packets : 0                      S2a DL Bytes   : 0
SGI UL packets : 0                      SGI UL bytes   : 0
Charging server address: 10.23.13.2
Charging server *: Up                   Charging Profile : 1
Charging triggers: 0                    Interim records  : 0

-----
IMSI           : 44444444440000
APN            : ims.corporate.1.1.com
Bearer Id      : 6                      Bearer type    : Dedicated
QCI/ARP        : 7/2                    SDFs           : 1
Charg. Char. Rcvd: 0x0000
Up Time        : 0d 00:00:05

Ref-pt/Sig-proto : S2a/PMIPv6

PGW GRE Key    : 0x40070da0             MAG GRE Key    : 0x8000000a
PGW Data addr  : 3ffe::b14:103
MAG Data addr  : 3ffe::a0a:2b17

S2a DL Packets : 0                      S2a DL Bytes   : 0
SGI UL packets : 0                      SGI UL bytes   : 0
Charging server address: 10.23.13.2
Charging server *: Up                   Charging Profile : 1
Charging triggers: 0                    Interim records  : 0

SDF rule name   : Rule_1
Reporting Level  : Rating Grp Level      Metering Method : -
Billing Method   : Online & Offline      Flow Status     : Up/Downlink Enabled
Service Id       : -                     Rating Group    : -
Sponsor Identity : -
Appl Svc Prov Id : -
SDF precedence   : 140                   SDF pkt filters : 2
Divert to AA     : No

Filter id        : 0                      Filter direction : DL
Filter protocol  : udp (17)               Filter tos-tc    : -
```

```

Local address : 1.127.255.3/32
Remote addr   : 15.1.1.2/32

Filter id      : 1                      Filter direction : UL
Filter protocol : udp (17)              Filter tos-tc    : -
Local address  : 1.127.255.3/32
Remote addr    : 15.1.1.2/32
-----
IMSI          : 444444444440000
APN           : ims.corporate.1.1.com
Bearer Id     : 7                      Bearer type      : Dedicated
QCI/ARP       : 8/3                    SDFs             : 1
Charg. Char. Rcvd: 0x0000
Up Time       : 0d 00:00:05

Ref-pt/Sig-proto : S2a/PMIPv6

PGW GRE Key   : 0x40070da0             MAG GRE Key      : 0x8000000a
PGW Data addr : 3ffe::b14:103
MAG Data addr  : 3ffe::a0a:2b17

S2a DL Packets : 0                     S2a DL Bytes     : 0
SGI UL packets : 0                     SGI UL bytes     : 0
Charging server address: 10.23.13.2
Charging server *: Up                  Charging Profile : 1
Charging triggers: 0                   Interim records  : 0

SDF rule name  : Rule_2
Reporting Level : Rating Grp Level      Metering Method   : -
Billing Method  : Online & Offline      Flow Status       : Up/Downlink Enabled
Service Id      : -                     Rating Group      : -
Sponsor Identity : -
Appl Svc Prov Id : -
SDF precedence  : 130                   SDF pkt filters   : 2
Divert to AA    : No

Filter id      : 0                      Filter direction : DL
Filter protocol : udp (17)              Filter tos-tc    : -
Local address  : 1.127.255.3/32
Remote addr    : 15.1.1.3/32

Filter id      : 1                      Filter direction : UL
Filter protocol : udp (17)              Filter tos-tc    : -
Local address  : 1.127.255.3/32
Remote addr    : 15.1.1.3/32
-----
IMSI          : 444444444440000
APN           : ims.corporate.1.2.com
Bearer Id     : 5                      Bearer type      : Default
QCI/ARP       : 9/1                    SDFs             : 0
Charg. Char. Rcvd: 0x0000
Up Time       : 0d 00:00:02

Ref-pt/Sig-proto : S2a/PMIPv6

PGW GRE Key   : 0x40080da0             MAG GRE Key      : 0x8000000b
PGW Data addr : 3ffe::b14:103

```

```

MAG Data addr      : 3ffe::a0a:2b17

S2a DL Packets    : 0                S2a DL Bytes      : 0
SGI UL packets    : 0                SGI UL bytes     : 0
Charging server address: 10.23.13.2
Charging server *: Up                Charging Profile : 1
Charging triggers: 0                Interim records  : 0

-----

IMSI              : 44444444440000
APN               : ims.corporate.1.2.com
Bearer Id         : 6                Bearer type      : Dedicated
QCI/ARP           : 7/2              SDFs             : 1
Charg. Char. Rcvd: 0x0000
Up Time          : 0d 00:00:02

Ref-pt/Sig-proto  : S2a/PMIPv6

PGW GRE Key       : 0x40080da0       MAG GRE Key      : 0x8000000b
PGW Data addr     : 3ffe::b14:103
MAG Data addr     : 3ffe::a0a:2b17

S2a DL Packets    : 0                S2a DL Bytes      : 0
SGI UL packets    : 0                SGI UL bytes     : 0
Charging server address: 10.23.13.2
Charging server *: Up                Charging Profile : 1
Charging triggers: 0                Interim records  : 0

SDF rule name     : Rule_1
Reporting Level   : Rating Grp Level Metering Method   : -
Billing Method    : Online & Offline  Flow Status      : Up/Downlink Enabled
Service Id       : -                  Rating Group     : -
Sponsor Identity  : -
Appl Svc Prov Id : -
SDF precedence    : 140                SDF pkt filters  : 2
Divert to AA     : No

Filter id         : 0                Filter direction : DL
Filter protocol   : udp (17)         Filter tos-tc    : -
Local address     : 2.127.255.0/32
Remote addr      : 15.1.1.2/32

Filter id         : 1                Filter direction : UL
Filter protocol   : udp (17)         Filter tos-tc    : -
Local address     : 2.127.255.0/32
Remote addr      : 15.1.1.2/32

-----

IMSI              : 44444444440000
APN               : ims.corporate.1.2.com
Bearer Id         : 7                Bearer type      : Dedicated
QCI/ARP           : 8/3              SDFs             : 1
Charg. Char. Rcvd: 0x0000
Up Time          : 0d 00:00:02

Ref-pt/Sig-proto  : S2a/PMIPv6

PGW GRE Key       : 0x40080da0       MAG GRE Key      : 0x8000000b

```

```

PGW Data addr      : 3ffe::b14:103
MAG Data addr      : 3ffe::a0a:2b17

S2a DL Packets     : 0                S2a DL Bytes       : 0
SGI UL packets     : 0                SGI UL bytes       : 0
Charging server address: 10.23.13.2
Charging server *: Up                Charging Profile   : 1
Charging triggers: 0                Interim records    : 0

SDF rule name      : Rule_2
Reporting Level    : Rating Grp Level  Metering Method    : -
Billing Method     : Online & Offline   Flow Status        : Up/Downlink Enabled
Service Id         : -                  Rating Group       : -
Sponsor Identity   : -
Appl Svc Prov Id   : -
SDF precedence     : 130                SDF pkt filters    : 2
Divert to AA       : No

Filter id           : 0                Filter direction   : DL
Filter protocol     : udp (17)         Filter tos-tc      : -
Local address       : 2.127.255.0/32
Remote addr         : 15.1.1.3/32

Filter id           : 1                Filter direction   : UL
Filter protocol     : udp (17)         Filter tos-tc      : -
Local address       : 2.127.255.0/32
Remote addr         : 15.1.1.3/32
-----
Number of bearer contexts : 6
=====
* indicates that the corresponding row element may have been truncated.

```

Example output bearer context IMSI detail statistics S4/S12

```

# show mobile-gateway pdn bearer-context imsi 22222222222200 detail
=====
Bearer context detail
=====
IMSI                : 111111111110000
APN                 : ims.corporate.0.1.com
Bearer Id           : 5                Bearer type        : Default
QCI/ARP             : 9/1              SDFs               : 0
Charg. Char. Rcvd   : 0x0000
Up Time             : 0d 00:01:17

S4 downlink TEID    : 0x1000007d       S4 SGW TEID        : 0x40060d55
S4 downlink addr    : 6.6.6.1
S4 SGW address      : 16.70.16.16

S4 DL packets       : 0                S4 DL bytes        : 0
SGI UL packets      : 0                SGI UL bytes       : 0
S4 QoS Modify       : 0
Charging server address: 10.23.13.2
Charging server *: Up                Charging Profile    : 1
Charging triggers: 0                Interim records     : 0

```

```

-----
IMSI           : 111111111110000
APN            : ims.corporate.0.1.com
Bearer Id      : 6                      Bearer type    : Dedicated
QCI/ARP        : 7/12                  SDFs           : 1
Charg. Char. Rcvd: 0x0000
Up Time        : 0d 00:01:16

S4 downlink TEID : 0x1000007e          S4 SGW TEID    : 0x40060d56
S4 downlink addr : 6.6.6.1
S4 SGW address   : 16.70.16.16

S4 DL packets    : 0                   S4 DL bytes     : 0
SGI UL packets   : 0                   SGI UL bytes    : 0
S4 QoS Modify    : 0
Charging server address: 10.23.13.2
Charging server *: Up                  Charging Profile : 1
Charging triggers: 0                   Interim records  : 0

SDF rule name    : rule for qci 7 arp 12
Reporting Level  : Rating Grp Level    Metering Method  : -
Billing Method   : Online & Offline    Flow Status      : Up/Downlink Enabled
Service Id       : -                   Rating Group     : -
Sponsor Identity : -
Appl Svc Prov Id : -
SDF precedence   : 2000                SDF pkt filters  : 2
Divert to AA     : No

Filter id        : 0                   Filter direction : DL
Filter protocol  : any                 Filter tos-tc    : -
Local address    : 1.127.255.0/32
Remote addr      : 15.1.1.2/32

Filter id        : 1                   Filter direction : UL
Filter protocol  : any                 Filter tos-tc    : -
Local address    : 1.127.255.0/32
Remote addr      : 15.1.1.2/32
-----
IMSI           : 111111111110000
APN            : ims.corporate.0.1.com
Bearer Id      : 7                      Bearer type    : Dedicated
QCI/ARP        : 8/13                  SDFs           : 1
Charg. Char. Rcvd: 0x0000
Up Time        : 0d 00:01:16

S4 downlink TEID : 0x1000007f          S4 SGW TEID    : 0x40060d57
S4 downlink addr : 6.6.6.1
S4 SGW address   : 16.70.16.16

S4 DL packets    : 0                   S4 DL bytes     : 0
SGI UL packets   : 0                   SGI UL bytes    : 0
S4 QoS Modify    : 0
Charging server address: 10.23.13.2
Charging server *: Up                  Charging Profile : 1
Charging triggers: 0                   Interim records  : 0

SDF rule name    : rule for qci 8 arp 13

```

```

Reporting Level : Rating Grp Level
Billing Method  : Online & Offline
Service Id     : -
Sponsor Identity : -
Appl Svc Prov Id : -
SDF precedence : 3000
Divert to AA   : No
Metering Method : -
Flow Status    : Up/Downlink Enabled
Rating Group   : -
SDF pkt filters : 2

```

```

Filter id       : 0
Filter protocol : any
Local address   : 1.127.255.0/32
Remote addr     : 15.1.1.3/32
Filter direction : DL
Filter tos-tc   : -

```

```

Filter id       : 1
Filter protocol : any
Local address   : 1.127.255.0/32
Remote addr     : 15.1.1.3/32
Filter direction : UL
Filter tos-tc   : -

```

```

-----
IMSI           : 111111111110000
APN            : ims.corporate.0.2.com
Bearer Id      : 11
QCI/ARP        : 5/2
Charg. Char. Rcvd: 0x0000
Up Time        : 0d 00:01:15
Bearer type     : Default
SDFs           : 0

S4 downlink TEID : 0x10000080
S4 downlink addr : 6.6.6.1
S4 SGW address   : 16.70.16.16
S4 SGW TEID      : 0x40070d55

S4 DL packets    : 0
SGI UL packets   : 0
S4 QoS Modify    : 0
Charging server address: 10.23.13.2
Charging server *: Up
Charging triggers: 0
Charging Profile : 1
Interim records  : 0

```

```

-----
IMSI           : 111111111110000
APN            : ims.corporate.0.2.com
Bearer Id      : 8
QCI/ARP        : 7/12
Charg. Char. Rcvd: 0x0000
Up Time        : 0d 00:01:13
Bearer type     : Dedicated
SDFs           : 1

S4 downlink TEID : 0x10000081
S4 downlink addr : 6.6.6.1
S4 SGW address   : 16.70.16.16
S4 SGW TEID      : 0x40070d56

S4 DL packets    : 0
SGI UL packets   : 0
S4 QoS Modify    : 0
Charging server address: 10.23.13.2
Charging server *: Up
Charging triggers: 0
Charging Profile : 1
Interim records  : 0

```



```

SDF rule name      : rule for qci 7 arp 12 precedence 2000
Reporting Level    : Rating Grp Level      Metering Method    : -
Billing Method     : Online & Offline      Flow Status        : Up/Downlink Enabled
Service Id        : -                     Rating Group       : -
Sponsor Identity   : -
Appl Svc Prov Id   : -
SDF precedence     : 2000                  SDF pkt filters    : 2
Divert to AA       : No

Filter id          : 0                     Filter direction   : DL
Filter protocol    : icmp (1)              Filter tos-tc      : -
Local address      : 3ffe:0:2:ff00::2/64
Remote addr        : 3ffe::f01:102/128

Filter id          : 1                     Filter direction   : UL
Filter protocol    : any                   Filter tos-tc      : -
Local address      : 3ffe:0:2:ff00::2/64
Remote addr        : 3ffe::f01:102/128
-----
IMSI               : 111111111110000
APN                : ims.corporate.0.2.com
Bearer Id          : 9                     Bearer type        : Dedicated
QCI/ARP            : 8/13                  SDFs                : 1
Charg. Char. Rcvd : 0x0000
Up Time           : 0d 00:01:13

S4 downlink TEID   : 0x10000082            S4 SGW TEID        : 0x40070d57
S4 downlink addr   : 6.6.6.1
S4 SGW address     : 16.70.16.16

S4 DL packets      : 0                     S4 DL bytes        : 0
SGI UL packets     : 0                     SGI UL bytes       : 0
S4 QoS Modify      : 0
Charging server address: 10.23.13.2
Charging server *: Up           Charging Profile    : 1
Charging triggers: 0            Interim records    : 0

SDF rule name      : rule for qci 8 arp 13 precedence 3000
Reporting Level    : Rating Grp Level      Metering Method    : -
Billing Method     : Online & Offline      Flow Status        : Up/Downlink Enabled
Service Id        : -                     Rating Group       : -
Sponsor Identity   : -
Appl Svc Prov Id   : -
SDF precedence     : 3000                  SDF pkt filters    : 2
Divert to AA       : No

Filter id          : 0                     Filter direction   : DL
Filter protocol    : icmp (1)              Filter tos-tc      : -
Local address      : 3ffe:0:2:ff00::2/64
Remote addr        : 3ffe::f01:103/128

Filter id          : 1                     Filter direction   : UL
Filter protocol    : any                   Filter tos-tc      : -
Local address      : 3ffe:0:2:ff00::2/64
Remote addr        : 3ffe::f01:103/128
-----
Number of bearer contexts : 6

```

=====

* indicates that the corresponding row element may have been truncated.

Example output bearer context IMSI detail statistics for eMPS advance priority calls

```
# show mobile-gateway pdn bearer-context imsi 2222222222222222 detail
=====
Bearer context detail
=====
IMSI           : 2222222222222222
APN            : ipd.nokia.1.1.com
Bearer Id      : 5                      Bearer type      : Default
QCI/ARP       : 5/2                      SDFs             : 0
Charg. Char. Rcvd: 0x0000
Up Time       : 0d 00:01:35
eMPS IMS priority bearer : Yes
S1u downlink TEID: 0x10000001          S1u SGW TEID    : 0x40010105
S1u downlink addr: 6.6.6.1
S1u SGW address : 16.70.16.16
S1u SGW address : 3ffe::1046:1010
S1u DL packets  : 0                      S1u DL bytes    : 0
SGI UL packets  : 0                      SGI UL bytes    : 0
S11 QoS Modify  : 0
-----
IMSI           : 2222222222222222
APN            : ipd.nokia.1.1.com
Bearer Id      : 6                      Bearer type      : Dedicated
QCI/ARP       : 1/1                      SDFs             : 1
QoS UL MBR    : 9 Kbps                   QoS DL MBR      : 9 Kbps
QoS UL GBR    : 9 Kbps                   QoS DL GBR      : 9 Kbps
Charg. Char. Rcvd: 0x0000
Up Time       : 0d 00:01:30
eMPS IMS priority bearer : Yes

S1u downlink TEID: 0x10000002          S1u SGW TEID    : 0x40010106
S1u downlink addr: 6.6.6.1
S1u SGW address : 16.70.16.16
S1u SGW address : 3ffe::1046:1010

S1u DL packets  : 0                      S1u DL bytes    : 0
SGI UL packets  : 0                      SGI UL bytes    : 0
S11 QoS Modify  : 0

SDF rule name   : ruleQci1
Reporting Level : Rating Grp Level      Metering Method  : -
Billing Method  : Online & Offline      Flow Status     : Up/Downlink Enabled
Service Id     : -                      Rating Group     : -
Sponsor Identity : -
Appl Svc Prov Id : -
SDF precedence  : 1                      SDF pkt filters : 2
SDF QoS UL MBR  : 9 Kbps                 SDF QoS DL MBR  : 9 Kbps
SDF QoS UL GBR  : 9 Kbps                 SDF QoS DL GBR  : 9 Kbps
Divert to AA    : No

Filter id       : 0                      Filter direction : DL
```

```

Filter protocol : udp (17)           Filter tos-tc : 0
Local address  : 3ffe:0:1:ff00::ffff:e3cc/64
Local ports    : 2100
Remote addr    : 3ffe::f01:1/128
Remote ports   : 4100

Filter id      : 1                   Filter direction : UL
Filter protocol : udp (17)           Filter tos-tc : 0
Local address  : 3ffe:0:1:ff00::ffff:e3cc/64
Local ports    : 2100
Remote addr    : 3ffe::f01:1/128
Remote ports   : 4100

```

```

-----
Number of bearer contexts : 2
=====

```

Example output bearer context IMSI detail statistics for non IP sessions

```

# show mobile-gateway pdn bearer-context imsi 1020000000000001 detail
=====
Bearer context detail
=====
IMSI           : 1020000000000001
APN            : ipd.nokia.1.1.com
Bearer Id      : 5                   Bearer type      : Default
QCI/ARP        : 9/11               SDFs             : 0
Charg. Char. Rcvd: 0x0000
Up Time        : 0d 00:18:36

UDP Tunnel Peer : 15.1.3.1
S1u downlink TEID: 0x10000003       S1u SGW TEID    : 0x40015525
S1u downlink addr: 6.6.6.1
S1u SGW address  : 16.70.16.16
S1u DL packets   : 1                 S1u DL bytes    : 64
SGI UL packets   : 1                 SGI UL bytes    : 64
S11 QoS Modify   : 0

```

Example output bearer context IMSI bearer traffic statistics

```

# show mobile-gateway pdn bearer-context imsi 7777777777777777 bearer-traffic
=====
Bearer context detail
=====
IMSI           : 7777777777777777
APN            : ipd.nokia.com
Bearer Id      : 5                   Bearer type      : Default
QCI            : 9                   ARP              : 15

S5 DL packets   : 0                 S5 DL bytes      : 0
SGI UL packets   : 0                 SGI UL bytes      : 0
-----
Number of bearer contexts : 1
=====

```

```
# show mobile-gateway pdn bearer-context imsi 123456789012301 bearer-traffic
```

```
=====
IMSI           : 123456789012301
APN            : ipd.nokia.com
Bearer Id      : 6                Bearer type      : Dedicated
QCI           : 8                ARP              : 12

Gn DL packets  : 1                Gn DL bytes    : 64
SGI UL packets : 1                SGI UL bytes   : 64
-----
Number of bearer contexts : 1
=====
```

Example output bearer context IMSI filter statistics

```
# show mobile-gateway pdn bearer-context imsi 123456789012301 filter
```

```
=====
Bearer context detail
=====
```

```
IMSI           : 123456789012301
APN            : ipd.nokia.com
Bearer Id      : 6                Bearer type      : Dedicated
QCI           : 8                ARP              : 12

SDF rule name  : rule for qci 8 arp 12
Reporting Level : -                Metering Method  : -
Billing Method  : -                Flow Status     : -
Service Id     : 0                Rating Group    : 0
AF Charging Id  : -
SDF precedence  : 3                SDF pkt filters : 2
SDF QoS UL MBR : 0 Kbps           SDF QoS DL MBR  : 0 Kbps
SDF QoS UL GBR : 0 Kbps           SDF QoS DL GBR  : 0 Kbps
Filter id      : 0                Filter direction : DL
Filter protocol : any
Source address  : 15.1.1.1/32
Destination addr : 5.64.1.0/32
Filter id      : 1                Filter direction : UL
Filter protocol : any
Source address  : 5.64.1.0/32
Destination addr : 15.1.1.1/32
-----
```

```
Number of bearer contexts : 1
=====
```

Example output bearer context MSISDN detail statistics

```
# show mobile-gateway pdn bearer-context msisdn 1234567890 detail
```

```
=====
Bearer context detail
=====
```

```
MSISDN         : 1234567890
APN            : ipd.nokia.1.1.com
NAI            : 1234567890@nnnn.com
Bearer Id      : 5                Bearer type      : Default
QCI/ARP       : 9/15             SDFs             : 1
```

Charg. Char. Rcvd: 0x0000

Up Time : 0d 00:13:59

Ref-pt/Sig-proto : Pi/MIPv4

PGW GRE Key : 0x40013700 FA GRE Key : 0x80000005
PGW Data addr : 16.16.16.16
FA Data addr : 10.23.15.2

Pi DL Packets : 0 Pi DL Bytes : 0
SGI UL packets : 0 SGI UL bytes : 0

Rule Base Name : dnsRb

SDF rule name : dnsR
DNS Redirect : enabled
Reporting Level : Rating Grp Level Metering Method : Volume
Billing Method : Online & Offline Flow Status : Up/Downlink Enabled
Service Id : - Rating Group : 0
AF Charging Id : -
SDF precedence : 0 SDF pkt filters : 1
SDF QoS UL MBR : 0 Kbps SDF QoS DL MBR : 0 Kbps
SDF QoS UL GBR : 0 Kbps SDF QoS DL GBR : 0 Kbps

Filter id : 0 Filter direction : BiDir
Filter protocol : any
Local address : 0.0.0.0/0
Remote addr : 15.1.1.1/32

Number of bearer contexts : 1
=====

Example output bearer context MAC detail statistics

show mobile-gateway pdn bearer-context mac 00:02:00:00:00:00 detail

=====

Bearer context detail

=====

MAC : 00:02:00:00:00:00
APN : ipd.nokia.2.1.com
Bearer Id : 5 Bearer type : Default
QCI/ARP : 9/10
Up Time : 0d 00:06:09

SWw TWAG V4 tunnel addr : 201.3.1.1
SWw AP V4 tunnel addr : 201.3.1.10
SWw TWAG MAC address : b2:40:ff:00:03:5b
SWw AP MAC address : 00:ff:04:00:00:01

SWw DL packets : 0 SWw DL bytes : 0
SGI UL packets : 0 SGI UL bytes : 0
SWw QoS Modify : 0

```

Rule Base Name : DHCP_DELAYED_IP_RULE
                : redirectRuleBase

SDF rule name : DHCP_DELAYED_IP_RULE
Reporting Level : Rating Grp Level    Metering Method : Volume
Billing Method : -                    Flow Status      : Uplink Enabled
Service Id     : -                    Rating Group    : 0
Sponsor Identity : -
Appl Svc Prov Id : -
SDF precedence : 0                    SDF pkt filters : 1
Divert to AA   : No

Filter id      : 0                    Filter direction : UL
Filter protocol : udp (17)
Local address  : any
Remote addr    : any
Remote ports   : 67
SDF rule name  : allow_portal
Reporting Level : Rating Grp Level    Metering Method : Volume
Billing Method : -                    Flow Status      : Up/Downlink Enabled
Service Id     : -                    Rating Group    : 1
Sponsor Identity : -
Appl Svc Prov Id : -
SDF precedence : 82                    SDF pkt filters : 1
Divert to AA   : No

SDF rule name : redirect_all_http
Reporting Level : Rating Grp Level    Metering Method : Volume
Billing Method : -                    Flow Status      : Redirect
Service Id     : -                    Rating Group    : 50
Sponsor Identity : -
Appl Svc Prov Id : -
SDF precedence : 100                    SDF pkt filters : 1
Divert to AA   : No

SDF rule name : allow_dns
Reporting Level : Rating Grp Level    Metering Method : Volume
Billing Method : -                    Flow Status      : Up/Downlink Enabled
Service Id     : -                    Rating Group    : 9999
Sponsor Identity : -
Appl Svc Prov Id : -
SDF precedence : 101                    SDF pkt filters : 1
Divert to AA   : No

-----
Number of bearer contexts : 1
=====

# show mobile-gateway pdn bearer-context mac 00:02:00:00:00:00 detail
=====
Bearer context detail
=====
IMSI/MAC : 3100120000000000/00:02:00:00:00:00
APN       : ipd.nokia.1.1.com.mnc012.mcc310.gprs
Bearer Id  : 5                      Bearer type   : Default
Up Time    : 0d 00:00:51

```

```

QCI/ARP          : 8/9
QoS UL MBR       : 0 Kbps          QoS DL MBR       : 0 Kbps
QoS UL GBR       : 0 Kbps          QoS DL GBR       : 0 Kbps

Sww TWAG tunnel addr : 201.3.2.1
Sww AP tunnel addr  : 201.3.2.10
Sww TWAG MAC address : 4a:4a:ff:00:03:5b
Sww AP MAC address  : 99:02:00:00:00:00
S2a TWAG Data Teid  : 0x14505
S2a PGW Data Teid   : 0x40012905
S2a TWAG V4/V6 Data Addr : 16.16.16.16
S2a PGW V4/V6 Data Addr : 10.10.5.1

S2a UL packets    : 0          S2a UL bytes     : 0
Sww DL packets    : 0          Sww DL bytes     : 0
Sww QoS Modify    : 0

SDF rule name     : DHCP_DELAYED_IP_RULE
Reporting Level   : Rating Grp Level   Metering Method   : Volume
Billing Method    : -                 Flow Status       : Up/Downlink Enabled
Service Id        : -                 Rating Group      : 0
Sponsor Identity  : -
Appl Svc Prov Id  : -
SDF precedence    : 0                 SDF pkt filters   : 1
Divert to AA      : No

Filter id         : 0          Filter direction : UL
Filter protocol   : udp (17)   Filter tos-tc    : -
Local address     : any
Remote addr       : any
Remote ports      : 67

-----
Number of bearer contexts : 1
=====

```

Example output bearer context IMSI default bearer account Ga statistics

```
# show mobile-gateway pdn bearer-context imsi 7777777777777777 acct ga bearer-id 5
```

```

=====
Bearer context Accounting Ga Brief
=====
IMSI              : 7777777777777777
APN               : ipd.nokia.com
Bearer Id         : 5          Bearer type       : Default
Charging ID       : 1543503872 Sustained Rate (5 min) : 0
Aggr. UL bytes    : 0          Aggr. DL bytes    : 0
Aggr. UL packets  : 0          Aggr. DL packets  : 0

Number of Partial CDRs : 0
Number of Containers   : 0
Number of Max Changes  : 0

```

```

Number of Rating Groups          : 1
Number of Rating Groups, Service IDs : 0
-----
Number of bearer contexts : 1
=====

```

Example output bearer context IMSI dedicated bearer account Ga statistics

```

# show mobile-gateway pdn bearer-context imsi 123456789012301 bearer 5
acct ga
=====
Bearer context Accounting Ga Brief
=====
IMSI                : 123456789012301
APN                 : N/A
Bearer Id           : 0                Bearer type           : Dedicated
Charging ID         : 0                Sustained Rate (5 min) : 0
Aggr. UL bytes      : 0                Aggr. DL bytes         : 0
Aggr. UL packets    : 0                Aggr. DL packets       : 0

Number of Partial CDRs          : 0
Number of Containers            : 0
Number of Max Changes           : 0
Number of Rating Groups         : 0
Number of Rating Groups, Service IDs : 0
-----
Number of bearer contexts : 1
=====

```

Example output bearer context IMSI bearer account Ga detail statistics

```

# show mobile-gateway pdn bearer-context imsi 7777777777777777
acct ga bearer 5 detail
=====
Bearer context Accounting Ga Detail
=====
IMSI                : 7777777777777777
APN                 : ipd.nokia.com
Bearer Id           : 5                Bearer type           : Default
Charging ID         : 1543503872       Sustained Rate (5 min) : 0
Aggr. UL bytes      : 0                Aggr. DL bytes         : 0
Aggr. UL packets    : 0                Aggr. DL packets       : 0

Number of Partial CDRs          : 0
Number of Containers            : 0
Number of Max Changes           : 0
Number of Rating Groups         : 1
Number of Rating Groups, Service IDs : 0
-----
Rating Group         : 0                Service ID            : 0
Online Charging      : On              Offline Charging       : On
UL bytes             : 0                DL bytes              : 0
UL packets           : 0                DL packets            : 0

```



```
-----
Number of bearer contexts : 1
=====
```

```
# show mobile-gateway pdn bearer-context imsi 123456789012301 bearer 5
acct ga detail
=====
```

Bearer context Accounting Ga Detail

```
=====
IMSI                : 123456789012301
APN                 : N/A
Bearer Id           : 0                Bearer type           : Dedicated
Charging ID         : 0                Sustained Rate (5 min) : 0
Aggr. UL bytes      : 0                Aggr. DL bytes         : 0
Aggr. UL packets    : 0                Aggr. DL packets       : 0
```

```
Number of Partial CDRs           : 0
Number of Containers              : 0
Number of Max Changes             : 0
Number of Rating Groups           : 0
Number of Rating Groups, Service IDs : 0
-----
```

```
Number of bearer contexts : 1
=====
```

Example output bearer context IMSI bearer account Ga detail statistics for CUPS

```
# show mobile-gateway pdn bearer-
context acct ga detail imsi 111111111000000 bearer 5
=====
```

Bearer context Accounting Ga Detail

```
=====
IMSI                : 111111111000000
APN                 : ipd.nokia.1.2.com
Bearer Id           : 5                Bearer type           : Default
URR ID              : 1073741829
Charging ID         : 17825792
Aggr. UL bytes      : 0                Aggr. DL bytes         : 0
Aggr. UL packets    : 0                Aggr. DL packets       : 0
Number of Partial CDRs           : 0
Number of Containers              : 0
Number of Max Changes             : 0
Number of Rating Groups           : 1
Number of Rating Groups, Service IDs : 0
Number of Rating Groups, Sponsor/ASP IDs: 0
-----
```

```
Rating Group        : 0
URR ID              : 1610612737
Online Charging     : On                Offline Charging        : On
UL bytes            : 0                DL bytes                : 0
UL packets          : 0                DL packets              : 0
-----
```

```
Number of bearer contexts : 1
```

Example output bearer context IMSI bearer account Gy statistics combined SGW/PGW

```
# show mobile-gateway pdn bearer-
context imsi 7777777777777777 detail acct gy bearer 5
=====
Bearer context Accounting Gy Detail
=====
IMSI                : 7777777777777777
APN                 : nokia.corporate.com
Bearer Id           : 5                Bearer type           : Default
Charging ID         : 19922944
Aggr. UL bytes      : 0                Aggr. DL bytes        : 0
Aggr. UL packets    : 0                Aggr. DL packets     : 0

DCCA Profile        : DCCA-Prof1
Primary Diam-peer Profile : Gy-Peer-Prof1
Secondary Diam-peer Profile :
Peer-profile In-use : Gy-Peer-Prof1
Failover Count      : 0
OCS Peer Origin     : Local Configuration
OCS Hostname        : (Not Specified)
OCS Address         : 10.23.18.10
OCS Port            : 3868
OCS Destination Realm : nokia.com
OCS Destination Host : pgw1.gy.nokia.com
Local Address       : 16.16.16.24

Failure Handling     : Terminate (L)
Session Failover    : Enabled (L)
Number of FUI Redirections : 0
Number Of Rating Groups : 0
Number of Rating Groups, Service IDs : 0
CCFH Time Remaining : 0 (Secs)
CCFH Volume Remaining : 0 (Bytes)
Restore Time Remaining : 0 (Secs)
-----
Number of bearer contexts : 1
=====
```

Example output bearer context IMSI bearer account Gy detail statistics

```
# show mobile-gateway pdn bearer-context imsi 220016789012301 bearer 6
detail acct gy
=====
Bearer context Accounting Gy Detail
=====
IMSI                : 220016789012301
APN                 : ipd.nokia.1.1.com
Bearer Id           : 6                Bearer type           : Dedicated
Charging ID         : 84934659
Aggr. UL bytes      : 1000            Aggr. DL bytes        : 1000
Aggr. UL packets    : 10             Aggr. DL packets     : 10
DCCA Profile        : DCCA-Prof1
Primary Diam-peer Profile : Gy-Peer-Prof1
```

```

Secondary Diam-peer Profile :
Peer-profile In-use       : Gy-Peer-Prof1
Failover Count           : 0
OCS Peer Origin           : Local Configuration
OCS Hostname              : (Not Specified)
OCS Address               : 10.23.18.10
OCS Port                  : 3868
OCS Destination Host      : pgw1.gy.nokia.com
Local Address             : 16.16.16.24
Failure Handling           : Terminate (L)
Session Failover          : Enabled (L)
Number of FUI Redirections : 0
Number of times Credits replenished : 1
Time of last credit replenishment : Feb 18 22:38:04 2016
Number Of Rating Groups : 2
Number of Rating Groups, Service IDs : 0
CCFH Time Remaining       : 0 (Secs)
CCFH Volume Remaining      : 0 (Bytes)
Restore Time Remaining     : 0 (Secs)
-----
Rating Group               : 3000
Used Input Octets          : 0           Granted Input Octets : -
Used Output Octets         : 0           Granted Output Octets : -
Used Total Octets          : 0           Granted Total Octets : 0
Time Used (secs)           : 0           Time Granted (secs)   : -
Billing Method              : Online & Off* Reporting Level      : Rating Grp L*
Online disabled             : No           FUI Enabled           : Restrict
Diameter Code               : 2001         Credit Pool Id        : -
Rating Group's State        : Allow restricted rules
Filter-Id count             : 1
Filter-Id (1)              : filterId1DedBr
Activity Threshold Timer (secs) : 0
OCS Triggers                : -
-----

```

```

Number of bearer contexts : 1
=====

```

```
#show mobile-gateway pdn bearer context imsi 123456789012301 bearer 5 acct gy detail
```

```
=====
Bearer context Accounting Gy Detail
=====
```

```

IMSI                      : 123456789012301
APN                       : N/A
Bearer Id                  : 0           Bearer type             : Dedicated
Charging ID                : 0           Sustained Rate (5 min)  : 0
Aggr. UL bytes             : 0           Aggr. DL bytes          : 0
Aggr. UL packets           : 0           Aggr. DL packets        : 0

```

```

Number of Redirections      : 0
Number of times Credits replenished : 0
Time of Last Redirection time : 0
Number of Rating Groups     : 0
Number of Rating Groups, Service IDs : 0
-----

```

```

Number of bearer contexts : 1

```

Example output bearer context IMSI bearer account Gy detail statistics for CUPS

```
# show mobile-gateway pdn bearer-
context imsi 111111111000000 bearer 6 acct gy detail
=====
Bearer context Accounting Gy Detail
=====
IMSI                : 111111111000000
APN                 : ipd.nokia.1.2.com
Bearer Id           : 6                Bearer type           : Dedicated
URR ID              : 1073741830
Charging ID         : 17825793
Aggr. UL bytes      : 0                Aggr. DL bytes        : 0
Aggr. UL packets    : 0                Aggr. DL packets      : 0
DCCA Profile        : DCCA-Prof1
Primary Diam-peer Profile : Gy-Peer-Prof1
Secondary Diam-peer Profile :
Peer-profile In-use : Gy-Peer-Prof1
Failover Count      : 0
OCS Peer Origin     : Local Configuration
OCS Hostname        : (Not Specified)
OCS Address         : 10.23.18.10
OCS Port            : 3868
OCS Destination Realm : nokia.com
OCS Destination Host : pgw1.gy.nokia.com
Local Address       : 16.16.16.24
Failure Handling     : Terminate (L)
Session Failover    : Enabled (L)
Number of FUI Redirections : 0
Number of times Credits replenished : 0
Number Of Rating Groups : 0
Number of Rating Groups, Service IDs : 0
CCFH Time Remaining : 0 (Secs)
CCFH Volume Remaining : 0 (Bytes)
Restore Time Remaining : 0 (Secs)
-----
Number of bearer contexts : 1
=====
```

Example output bearer context IMSI bearer account Rf statistics

```
# show mobile-gateway pdn bearer-context imsi 777777777777777 acct rf
bearer 5
=====
Bearer context Accounting Rf Brief
=====
No Matching Entries
=====
#show mobile-gateway pdn bearer-context imsi 777777777777777 acct rf bearer 5 detail
MINOR: CLI Bearer-Context rf acct detail is not supported.
=====
```

```
# show mobile-gateway pdn bearer-context imsi 123456789012301 bearer 5
acct rf
=====
Bearer context Accounting Rf Brief
=====
IMSI                : 123456789012301
APN                 : N/A
Bearer Id           : 0                Bearer type           : Dedicated
Charging ID         : 0                Sustained Rate (5 min) : 0
Aggr. UL bytes      : 0                Aggr. DL bytes         : 0
Aggr. UL packets    : 0                Aggr. DL packets       : 0

Number of Interims Sent           : 0
Number of Rating Groups           : 0
Number of Rating Groups, Service IDs : 0
-----
Number of bearer contexts : 1
=====
```

Example output bearer context default session APN account Ga detail statistics

```
# show mobile-gateway pdn bearer-context default-session def-session-pgw-
ip 31.1.1.2 apn "ipd.nokia.1.1.com" acct ga detail
=====
Bearer context Accounting Ga Detail
=====
IMSI                : 000010006182011
APN                 : ipd.nokia.1.1.com
Bearer Id           : 5                Bearer type           : Default
Charging ID         : 427819028
Aggr. UL bytes      : 0                Aggr. DL bytes         : 0
Aggr. UL packets    : 0                Aggr. DL packets       : 0
Aggr. DL steered bytes : 0            Aggr. DL steered packets: 0

Number of Partial CDRs           : 0
Number of Containers              : 0
Number of Max Changes             : 0
Number of Rating Groups           : 0
Number of Rating Groups, Service IDs : 0
Number of Rating Groups, Sponsor/ASP IDs: 0
-----
Number of bearer contexts : 1
=====
```

Example output bearer context default session APN account Gy detail statistics

```
# show mobile-gateway pdn bearer-context default-session def-session-pgw-
ip 31.1.1.2 apn "ipd.nokia.1.1.com" acct gy detail
=====
Bearer context Accounting Gy Detail
=====
IMSI                : 000010006182011
APN                 : ipd.nokia.1.1.com
```

```

Bearer Id           : 5           Bearer type           : Default
Charging ID        : 427819028
Aggr. UL bytes     : 0           Aggr. DL bytes      : 0
Aggr. UL packets   : 0           Aggr. DL packets   : 0
Aggr. DL steered bytes : 0       Aggr. DL steered packets: 0

DCCA Profile       :
Primary Diam-peer Profile :
Secondary Diam-peer Profile :
Peer-profile In-use :
Failover Count     :
OCS Peer Origin    :
OCS Hostname       : (Not Specified)
OCS Address        :
OCS Port          :
OCS Destination Realm : (Not Specified)
OCS Destination Host : (Not Specified)
Local Address      :
CCFH-Profile       : 0
Failure Handling    :
Session Failover   :
Number of FUJ Redirections : 0
Number of times Credits replenished : 0
Number Of Rating Groups : 0
Number of Rating Groups, Service IDs : 0
CCFH Time Remaining : 0 (Secs)
CCFH Volume Remaining : 0 (Bytes)
Restore Time Remaining : 0 (Secs)
-----
Number of bearer contexts : 1
=====

```

Example output bearer context MSISDN statistics (UP function)

```

# show mobile-gateway pdn bearer-context msisdn 7777777777777777
=====
IMSI/IMEI(#)      APN              Bear* Type QCI/ARP  MBR/GBR      SDFs
/MSISDN(^)/MAC(`)
/SubId/SEID(~)
=====
0x41110~         ipd.nokia.com      5      Def  6/NA      -           2
123456789012345
-----
Number of bearer contexts : 1
-----
# indicates that the corresponding row is IMEI entry.
^ indicates that the corresponding row is MSISDN entry.
` indicates that the corresponding row is MAC entry.
~ indicates that the corresponding row is SEID entry.
-----
* indicates that the corresponding row element may have been truncated.

```

Example output bearer context MSISDN detail statistics (UP function)

```

# show mobile-gateway pdn bearer-context msisdn 7777777777777777 detail
=====

```

Bearer context detail

```

=====
-----
IMSI           : 123456789012345
Gi Network Inst : ipd.nokia.com
Gn DL Trf End-Pt : 5                QCI           : 6

Up Time        : 0d 00:00:09

Gn PGW Data TEID : 0x40041115        Gn SGW Data TEID : 0x10000001
Gn PGW Data addr : 17.16.16.16
Gn PGW Data addr : 3ffe::1110:1010
Gn SGW Data addr : 10.23.15.2
Gn DL packets    : 0                  Gn DL bytes      : 0
SGI UL packets   : 0                  SGI UL bytes     : 0
Redirect SDF Cou*: 0

```

SDF Rules Information

```

-----
SDF Rule Count   : 1                DSCP           : 0
-----

```

```

SDF rule name    : 2560
SDF rule ID      : 4096            SDF Precedence  : 65535

```

Charging Information

```

Reporting Level  : Rating Grp Level  Metering Method : -
Billing Method   : -
Rating Group     : 1                 Service Id      : -
Sponsor Identity : -
Appl Svc Prov Id : -

```

Traffic Control Information

```

Flow Status      : Up/Downlink Enabled
Divert to AA     : No

```

PFCP Information

```

PDR ID UL       : 2560                PDR ID DL       : 35328
FAR ID UL       : 327680             FAR ID DL       : 537198592
QER ID          : 1                  QFI             : 0

```

Flow Information

```

SDF pkt filters  : 1

```

```

Filter id        : 0                  Filter direction : BiDir
Filter protocol  : any                Filter tos-tc    : -
Local address    : any
Remote addr      : any

```

```

-----
Number of bearer contexts : 1
=====

```

* indicates that the corresponding row element may have been truncated.

Example output UP function bearer context SEID statistics

```

# show mobile-gateway pdn bearer-context up-seid 0x41110
=====

```

IMSI/IMEI(#) /MSISDN(^)/MAC(`) /SubId/SEID(~)	APN	Bear*	Type	QCI/ARP	MBR/GBR	SDFs
---	-----	-------	------	---------	---------	------

0x41110~ 123456789012345	ipd.nokia.com	5	Def	6/NA	-	2
-----------------------------	---------------	---	-----	------	---	---

Number of bearer contexts : 1

indicates that the corresponding row is IMEI entry.

^ indicates that the corresponding row is MSISDN entry.

` indicates that the corresponding row is MAC entry.

~ indicates that the corresponding row is SEID entry.

* indicates that the corresponding row element may have been truncated.

Example output UP function bearer context SEID detail statistics

```
show mobile-gateway pdn bearer-context up-seid 0x41110 detail
```

Bearer context detail

IMSI : 123456789012345

Gi Network Inst : ipd.nokia.com

Gn DL Trf End-Pt : 5 QCI : 6

Up Time : 0d 00:00:07

Gn PGW Data TEID : 0x40041115

Gn SGW Data TEID : 0x10000001

Gn PGW Data addr : 17.16.16.16

Gn PGW Data addr : 3ffe::1110:1010

Gn SGW Data addr : 10.23.15.2

Gn DL packets : 0

Gn DL bytes : 0

SGI UL packets : 0

SGI UL bytes : 0

Redirect SDF Cou*: 0

SDF Rules Information

SDF Rule Count : 1 DSCP : 0

SDF rule name : 2560

SDF rule ID : 4096 SDF Precedence : 65535

Charging Information

Reporting Level : Rating Grp Level Metering Method : -

Billing Method : -

Rating Group : 1 Service Id : -

Sponsor Identity : -

Appl Svc Prov Id : -

Traffic Control Information

Flow Status : Up/Downlink Enabled

Divert to AA : No

PFPCP Information


```

PDR ID UL      : 2560          PDR ID DL      : 35328
FAR ID UL      : 327680       FAR ID DL      : 537198592
QER ID         : 1            QFI              : 0

```

Flow Information

```
SDF pkt filters : 1
```

```

Filter id       : 0            Filter direction : BiDir
Filter protocol : any          Filter tos-tc    : -
Local address   : any
Remote addr     : any

```

```
-----
Number of bearer contexts : 1
=====
```

```
* indicates that the corresponding row element may have been truncated.
```

5.2.2.14 cp-seid

Syntax

```

cp-seid seid [cp-ipv4 ip-address] [cp-ipv6 ipv6-address]
cp-seid seid [cp-ipv4 ip-address] [cp-ipv6 ipv6-address] detail

```

Context

```
show>mobile>pdn>bearer-context
```

Description

This command displays CP session context information related to a specific SEID.

Parameters

seid

specifies the SEID for the CP or UP session context, up to 40 characters

ip-address

specifies the CP function's Sx/N4 reference point IPv4 address

Values: a.b.c.d

ipv6-address

specifies the CP function's Sx/N4 reference point IPv6 address, up to 64 characters

detail

displays detailed information

Output

The following output examples show CP session bearer context information for a SEID.

- [Example output CP session bearer context SEID statistics](#)
- [Example output CP session bearer context SEID detail statistics](#)

Example output CP session bearer context SEID statistics

```
# show mobile-gateway pdn bearer-context cp-seid 0x10100 cp-ipv4 33.33.1.1 cp-
ipv6 3ffe::2121:101
=====
IMSI/IMEI(#)      APN              Bear* Type QCI/ARP  MBR/GBR      SDFs
/MSISDN(^)/MAC(`)
/SubId/SEID(~)
=====
0x10100~          ipd.nokiane.1.2.* 5      Def  9/NA      N/A          3
0x10100~          ipd.nokiane.1.2.* 6      Ded  3/NA      N/A          3
-----
Number of bearer contexts : 2
-----
# indicates that the corresponding row is IMEI entry.
^ indicates that the corresponding row is MSISDN entry.
` indicates that the corresponding row is MAC entry.
~ indicates that the corresponding row is SEID entry.
-----
* indicates that the corresponding row element may have been truncated.
```

Example output CP session bearer context SEID detail statistics

```
# show mobile-gateway pdn bearer-context cp-seid 0x10110 cp-ipv4 33.33.1.1 cp-
ipv6 3ffe::2121:101 detail
=====
Bearer context detail
=====
-----
Gi Network Inst   : ipd.nokia.1.2.com
Traffic End-pt ID: 5                DSCP                : 5
Up Time          : 0d 00:09:51
S1u downlink TEID: 0x10000002        S1u SGW TEID        : 0x40010105
S1u downlink addr: 6.6.6.1
S1u SGW address   : 17.70.16.16
S1u SGW address   : 3ffe::1146:1010
S1u DL packets    : 0                S1u DL bytes        : 0
SGI UL packets    : 0                SGI UL bytes        : 0
SDF rule name     : 5120
Reporting Level   : Rating Grp Level Metering Method      : -
Billing Method    : -                Flow Status          : Up/Downlink Enabled
Service Id        : -                Rating Group         : 1
Sponsor Identity  : -
Appl Svc Prov Id  : -
SDF precedence    : 65535            SDF pkt filters     : 1
Divert to AA      : No
PDR ID UL         : 5120            PDR ID DL           : 37888
PDR ID S1u UL     : 21504
```

```

FAR ID UL      : 327680          FAR ID DL      : 537198592
QER ID        : 1
Filter id      : 0              Filter direction : BiDir
Filter protocol : any          Filter tos-tc    : 0
Local address  : any
Remote addr    : any
-----
-----
Gi Network Inst : ipd.nokia.1.2.com
Traffic End-pt ID: 6          DSCP              : 6
Up Time        : 0d 00:09:51
S1u downlink TEID: 0x10000003  S1u SGW TEID  : 0x40010106
S1u downlink addr: 6.6.6.1
S1u SGW address : 17.70.16.16
S1u SGW address : 3ffe::1146:1010
S1u DL packets  : 0          S1u DL bytes   : 0
SGI UL packets  : 0          SGI UL bytes   : 0
SDF rule name   : 6145
Reporting Level : Rating Grp Level  Metering Method : -
Billing Method  : -          Flow Status      : Up/Downlink Enabled
Service Id     : -          Rating Group     : 2
Sponsor Identity : -
Appl Svc Prov Id : -
SDF precedence  : 100        SDF pkt filters : 1
SDF QoS UL MBR  : 10 Kbps    SDF QoS DL MBR  : 10 Kbps
SDF QoS UL GBR  : 10 Kbps    SDF QoS DL GBR  : 10 Kbps
Divert to AA    : No
PDR ID UL       : 6145       PDR ID DL       : 38913
PDR ID S1u UL   : 22529
FAR ID UL       : 393216     FAR ID DL       : 537264128
QER ID          : 4102
Filter id       : 0          Filter direction : BiDir
Filter protocol : udp (17)   Filter tos-tc    : 0
Local address   : 2.127.255.0/32
Local ports     : 1234
Remote addr     : 15.1.1.2/32
Remote ports    : 4321
-----
Number of bearer contexts : 2
=====

```

5.2.2.15 def-pfcp-u-sessions

Syntax

```
def-pfcp-u-sessions [network-realm realm-name] [pfcp-u-peer-ip ip-address] [gateway
gw-id] [apn apn-name] [card slot-number] [vm vm-id] [group group-number]
```

Context

```
show>mobile>pdn>bearer-context
```

Description

This command displays the default PFCP-u shared tunnel session information, when PFCP-u (GTP-U) tunnels are configured to handle ICMPv6 RS/RA and NS/NA message exchange between the CP function and the UP function. When only `def - p f c p - u` is specified, results for all default PFCP-u sessions are displayed. The `network - rea lm` option may also be specified to identify the realm name of the network instance that has the PFCP GTP-U interface on the UP function, and is communicated in the Network-Instance IE part of the PFCP Session Establishment/Modification process. The `p f c p - u - p e e r - i p` option can also be used to display information for a specific PFCP-u Peer IP address.

Parameters

realm-name

specifies the realm name of the network instance that has the PFCP GTP-U interface on the UP function, up to 80 characters

ip-address

specifies the PFCP-u peer IP address, up to 64 characters

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x: [0..FFFF]H
–	–	d: [0..255]D

gw-id

specifies the gateway ID

Values: 1 to 8

apn-name

specifies the APN name, up to 80 characters

slot-number

specifies the card slot number (applicable only to 7750 SR MG)

Values: 1 to 20 | A | B

vm-id

specifies the VM ID (applicable only to CMG)

Values: 1 to 20 | A | B

group-number

specifies the group number

Values: 1 to 15

Output

The following output examples show default PFCP-u shared tunnel sessions information for the PDN bearer context and *Show PDN bearer context PFCP-u shared tunnel session output fields* describes the output fields.

- *Example output PDN bearer context default statistics PFCP-u shared tunnel sessions*
- *Example output PDN bearer context network realm statistics PFCP-u shared tunnel sessions*
- *Example output PDN bearer context peer IP address statistics PFCP-u shared tunnel sessions*
- *Example output PDN bearer context network realm and peer IP address statistics PFCP-u shared tunnel sessions*

Example output PDN bearer context default statistics PFCP-u shared tunnel sessions

```
# show mobile-gateway pdn bearer-context def-pfcp-u sessions
=====
IMSI/IMEI( #)      APN      Bear* Type QCI/ARP  MBR/GBR  SDFs
/MSISDN(^)/MAC(`)
/SubId/SEID(~)
=====
999999888100211    5      Ded  0/0      -        0
999999888100211    5      Ded  0/0      -        0
999999888100211    5      Ded  0/0      -        0
999999888100211    5      Ded  0/0      -        0
999999888100211    5      Ded  0/0      -        0
999999888100211    5      Ded  0/0      -        0
999999888100211    5      Ded  0/0      -        0
999999888100211    5      Ded  0/0      -        0
999999888100211    5      Ded  0/0      -        0
999999888100210    5      Ded  0/0      -        0
999999888100210    5      Ded  0/0      -        0
999999888100210    5      Ded  0/0      -        0
999999888100210    5      Ded  0/0      -        0
999999888100210    5      Ded  0/0      -        0
999999888100210    5      Ded  0/0      -        0
999999888100210    5      Ded  0/0      -        0
999999888100210    5      Ded  0/0      -        0
999999888100210    5      Ded  0/0      -        0
999999888100209    5      Ded  0/0      -        0
999999888100212    5      Ded  0/0      -        0
999999888100212    5      Ded  0/0      -        0
999999888100212    5      Ded  0/0      -        0
999999888100212    5      Ded  0/0      -        0
999999888100212    5      Ded  0/0      -        0
999999888100212    5      Ded  0/0      -        0
999999888100212    5      Ded  0/0      -        0
999999888100212    5      Ded  0/0      -        0
999999888100212    5      Ded  0/0      -        0
```

99999888100213	5	Ded	0/0	-	0
99999888100213	5	Ded	0/0	-	0
99999888100213	5	Ded	0/0	-	0
99999888100213	5	Ded	0/0	-	0
99999888100213	5	Ded	0/0	-	0
99999888100213	5	Ded	0/0	-	0
99999888100213	5	Ded	0/0	-	0
99999888100213	5	Ded	0/0	-	0
99999888100213	5	Ded	0/0	-	0
99999888100211	5	Ded	0/0	-	0
99999888100211	5	Ded	0/0	-	0
99999888100211	5	Ded	0/0	-	0
99999888100211	5	Ded	0/0	-	0
99999888100211	5	Ded	0/0	-	0
99999888100211	5	Ded	0/0	-	0
99999888100211	5	Ded	0/0	-	0
99999888100211	5	Ded	0/0	-	0
99999888100210	5	Ded	0/0	-	0
99999888100210	5	Ded	0/0	-	0
99999888100210	5	Ded	0/0	-	0
99999888100210	5	Ded	0/0	-	0
99999888100210	5	Ded	0/0	-	0
99999888100210	5	Ded	0/0	-	0
99999888100210	5	Ded	0/0	-	0
99999888100210	5	Ded	0/0	-	0
99999888100210	5	Ded	0/0	-	0
99999888100210	5	Ded	0/0	-	0
99999888100209	5	Ded	0/0	-	0
99999888100209	5	Ded	0/0	-	0
99999888100212	5	Ded	0/0	-	0
99999888100212	5	Ded	0/0	-	0
99999888100212	5	Ded	0/0	-	0
99999888100212	5	Ded	0/0	-	0
99999888100212	5	Ded	0/0	-	0
99999888100212	5	Ded	0/0	-	0
99999888100212	5	Ded	0/0	-	0
99999888100212	5	Ded	0/0	-	0
99999888100213	5	Ded	0/0	-	0
99999888100213	5	Ded	0/0	-	0
99999888100213	5	Ded	0/0	-	0
99999888100213	5	Ded	0/0	-	0
99999888100213	5	Ded	0/0	-	0
99999888100213	5	Ded	0/0	-	0
99999888100213	5	Ded	0/0	-	0
99999888100211	5	Ded	0/0	-	0
99999888100211	5	Ded	0/0	-	0
99999888100211	5	Ded	0/0	-	0
99999888100211	5	Ded	0/0	-	0
99999888100211	5	Ded	0/0	-	0
99999888100211	5	Ded	0/0	-	0
99999888100211	5	Ded	0/0	-	0
99999888100210	5	Ded	0/0	-	0
99999888100210	5	Ded	0/0	-	0
99999888100210	5	Ded	0/0	-	0
99999888100210	5	Ded	0/0	-	0
99999888100210	5	Ded	0/0	-	0

99999888100210	5	Ded	0/0	-	0
99999888100210	5	Ded	0/0	-	0
99999888100210	5	Ded	0/0	-	0
99999888100209	5	Ded	0/0	-	0
99999888100209	5	Ded	0/0	-	0
99999888100212	5	Ded	0/0	-	0
99999888100212	5	Ded	0/0	-	0
99999888100212	5	Ded	0/0	-	0
99999888100212	5	Ded	0/0	-	0
99999888100212	5	Ded	0/0	-	0
99999888100212	5	Ded	0/0	-	0
99999888100212	5	Ded	0/0	-	0
99999888100212	5	Ded	0/0	-	0
99999888100212	5	Ded	0/0	-	0
99999888100213	5	Ded	0/0	-	0
99999888100213	5	Ded	0/0	-	0
99999888100213	5	Ded	0/0	-	0
99999888100213	5	Ded	0/0	-	0
99999888100213	5	Ded	0/0	-	0
99999888100213	5	Ded	0/0	-	0
99999888100213	5	Ded	0/0	-	0
99999888100213	5	Ded	0/0	-	0
99999888100213	5	Ded	0/0	-	0
99999888100213	5	Ded	0/0	-	0
99999888100211	5	Ded	0/0	-	0
99999888100211	5	Ded	0/0	-	0
99999888100211	5	Ded	0/0	-	0
99999888100211	5	Ded	0/0	-	0
99999888100211	5	Ded	0/0	-	0
99999888100211	5	Ded	0/0	-	0
99999888100211	5	Ded	0/0	-	0
99999888100210	5	Ded	0/0	-	0
99999888100210	5	Ded	0/0	-	0
99999888100210	5	Ded	0/0	-	0
99999888100210	5	Ded	0/0	-	0
99999888100210	5	Ded	0/0	-	0
99999888100210	5	Ded	0/0	-	0
99999888100210	5	Ded	0/0	-	0
99999888100210	5	Ded	0/0	-	0
99999888100209	5	Ded	0/0	-	0
99999888100209	5	Ded	0/0	-	0
99999888100209	5	Ded	0/0	-	0
99999888100212	5	Ded	0/0	-	0
99999888100212	5	Ded	0/0	-	0
99999888100212	5	Ded	0/0	-	0
99999888100212	5	Ded	0/0	-	0
99999888100212	5	Ded	0/0	-	0
99999888100212	5	Ded	0/0	-	0
99999888100212	5	Ded	0/0	-	0
99999888100212	5	Ded	0/0	-	0
99999888100212	5	Ded	0/0	-	0
99999888100213	5	Ded	0/0	-	0
99999888100213	5	Ded	0/0	-	0
99999888100213	5	Ded	0/0	-	0
99999888100213	5	Ded	0/0	-	0
99999888100213	5	Ded	0/0	-	0
99999888100213	5	Ded	0/0	-	0
99999888100213	5	Ded	0/0	-	0

```
-----
Number of bearer contexts : 140
-----
```

```
# indicates that the corresponding row is IMEI entry.
```

```
^ indicates that the corresponding row is MSISDN entry.
```

```
` indicates that the corresponding row is MAC entry.
```

```
~ indicates that the corresponding row is SEID entry.
```

```
-----
* indicates that the corresponding row element may have been truncated.
```

Table 31: Show PDN bearer context PFCP-u shared tunnel session output fields

Label	Description
IMSI/IMEI/MSISDN/MAC/SubId/SEID	The subscriber, equipment, or session identifier type
APN	The APN name
Bearer	The bearer number
Type	The bearer type
QCI/ARP	The QCI/ARP rules
MBR/GBR	The MBR/GBR rules
SDFs	The number of SDFs

Example output PDN bearer context network realm statistics PFCP-u shared tunnel sessions

```
# show mobile-gateway pdn bearer-context def-pfcp-u-sessions network-realm s5-
default
```

```
=====
IMSI/IMEI(#)      APN      Bear* Type QCI/ARP  MBR/GBR  SDFs
/MSISDN(^)/MAC(`)
/SubId/SEID(~)
=====
```

```
999999888100212      5      Ded  0/0      -      0
999999888100211      5      Ded  0/0      -      0
999999888100213      5      Ded  0/0      -      0
999999888100210      5      Ded  0/0      -      0
999999888100209      5      Ded  0/0      -      0
-----
```

```
Number of bearer contexts : 5
```



```
# indicates that the corresponding row is IMEI entry.
^ indicates that the corresponding row is MSISDN entry.
` indicates that the corresponding row is MAC entry.
~ indicates that the corresponding row is SEID entry.
-----
* indicates that the corresponding row element may have been truncated.
```

Example output PDN bearer context peer IP address statistics PFCP-u shared tunnel sessions

```
# show mobile-gateway pdn bearer-context def-pfcp-u-sessions pfcp-u-peer-  
ip 33.34.1.1
```

[illegible]

Number of bearer contexts : 33

```
# indicates that the corresponding row is IMEI entry.
^ indicates that the corresponding row is MSISDN entry.
\ indicates that the corresponding row is MAC entry.
```

~ indicates that the corresponding row is SEID entry.

Example output PDN bearer context network realm and peer IP address statistics PFCP-u shared tunnel sessions

```
# show mobile-gateway pdn bearer-context def-pfcp-u-sessions pfcp-u-peer-
ip 33.34.1.1 network-realm s5-default
=====
IMSI/IMEI(#)      APN      Bear* Type QCI/ARP  MBR/GBR  SDFs
/MSISDN(^)/MAC(`)
/SubId/SEID(~)
=====
999999888100210      5      Ded  0/0      -        0
-----
Number of bearer contexts : 1
-----
# indicates that the corresponding row is IMEI entry.
^ indicates that the corresponding row is MSISDN entry.
` indicates that the corresponding row is MAC entry.
~ indicates that the corresponding row is SEID entry.
-----
* indicates that the corresponding row element may have been truncated.
```

5.2.2.16 supi

Syntax

```
supi imsi-msisdn
```

```
supi imsi-msisdn {control | filter | bearer-traffic | detail}
```

Context

```
show>mobile>pdn>bearer-context
```

Description

This command displays bearer context information for the specified SUPI for 5G sessions.

Parameters

imsi-msisdn

specifies the SUPI identifier, up to 15 characters

control

specifies the control information filter

filter

specifies the filter information

bearer-traffic

specifies the bearer-traffic information filter

detail

specifies detail information

Output

The following output example shows SUPI information for a specified IMSI-MSISDN in the bearer context and *Show PDN bearer context SUPI detail output fields* describes the output fields.

**NOTE:**

- The `supi` option displays information related to EPS interworking.
- The output supports a limited number of fields for 5G sessions and displays elements applicable for 4G deployments; for example, APN and IMSI, rather than DNN and SUPI.

Example output PDN bearer context SUPI detail statistics (5G sessions)

```
# show mobile-gateway pdn bearer-context supi 460100001003204 detail
=====
Bearer context detail
=====
IMSI           : 460100001003204
APN            : ipd.nokia.com
Bearer Id      : 5                Bearer type      : Default
QCI/ARP        : 5/9              SDFs           : 0
Interworking    : Not Subscribed
Charg. Char. Rcvd: 0x123a
Up Time        : 0d 00:10:40
Ref-pt/Sig-proto : N11/http2
eMPS IMS priority bearer : No
PGW GRE Key     : 0x0              MAG GRE Key     : 0x0
PGW Data addr   : 3ffe::1146:1010
PGW Data addr   : 17.70.16.16
MAG Data addr   : ::
Redirect SDF Cou*: 0
-----
Number of bearer contexts : 1
=====
* indicates that the corresponding row element may have been truncated.
```

Table 32: Show PDN bearer context SUPI detail output fields

Label	Description
IMSI	The SUPI IMSI number

Table 32: Show PDN bearer context SUPI detail output fields (continued)

Label	Description
APN	The APN name
Bearer Id	The bearer ID
Bearer type	The bearer type
Interworking	The 3G/4G EPS interworking status
Charg. Char. Rcvd	The charging characters received
Uptime	The uptime of the bearer
emPS IMS priority bearer	The eMPS IMS priority bearer status
PGW GRE Key	The PGW GRE key
MAG GRE Key	The MAG GRE key
PGW Data addr	The PGW data IP address
MAG Data addr	The MAG data IP address
Redirect SDF Cou*	The redirect count

5.2.2.17 ue-ip

Syntax

```
ue-ip ip-address network-instance apn [traffic-endpt-id traffic-endpt-id] detail
[access-type {utran | geran | wlan | gan | hspa | eutran | ehrpd | hrpd | 1xrtt | umb | virtual
eutran-nb-iot}]
```

```
ue-ip ip-address network-instance apn [traffic-endpt-id traffic-endpt-id] detail  
[access-type detail {utran | geran | wlan | gan | hspa | eutran | ehrpd | hrpd | 1xrtt | umb |  
virtual | eutran-nb-iot}]
```

Context

show>mobile>pdn>bearer-context

Description

This command displays bearer context information related to a specific UE IP address.

Parameters

ip-address

specifies the UE IP address

Values		
	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x: [0..FFFF]H
–	–	d: [0..255]D

apn

specifies the APN name associated for the Virtual Router instance on the UP peer where the UE IP address is located and is communicated over Sx/N4 in the Network-Instance IE, up to 80 characters

traffic-endpt-id

specifies the traffic endpoint ID

Values: 0 to 4294967295

detail

displays detailed information

access-type

specifies access type

Output

The following output example shows bearer context information for a specific UE IP address.

Example output bearer context Information for UE IP address

```
# show mobile-gateway pdn bearer-context ue-
ip 2.127.255.0 network instance ipd.nokia.1.2.com

=====
IMSI/IMEI(#)      APN              Bear* Type QCI/ARP  MBR/GBR    SDFs
/MSISDN(^)/MAC(`)
/SubId/SEID(~)
=====
0x10100~          ipd.nokia.1.2.* 5      Def  9/NA     N/A        3
0x10100~          ipd.nokia.1.2.* 6      Ded  3/NA     N/A        3
-----
Number of bearer contexts : 2
-----
# indicates that the corresponding row is IMEI entry.
^ indicates that the corresponding row is MSISDN entry.
` indicates that the corresponding row is MAC entry.
~ indicates that the corresponding row is SEID entry.
-----
* indicates that the corresponding row element may have been truncated.
```



NOTE:

- On the UP function, the SDF count is equal to the number of PDRs for PGW and combined SGW/PGW sessions. For an SGW session, the SDF count is shown as 0
- In the UP function, the ARP value is indicated as N/A (not supported by the PFCP protocol).

5.2.2.18 up-seid

Syntax

```
up-seid seid
```

```
up-seid seid [traffic-endpt-id traffic-endpt-id] detail
```

Context

```
show>mobile>pdn>bearer-context
```

Description

This command displays bearer context information related to a specific Session Endpoint Identifier (SEID) for the UP session context.

Parameters

seid

specifies the Session Endpoint Identifier (SEID) for the CP or UP session context, up to 40 characters

traffic-endpt-id

specifies the traffic endpoint ID

Values: 0 to 4294967295

detail

displays detailed information

Output

The following output example shows detailed UP session bearer context information for a specified SEID.

Example output UP session bearer context detail information for specified SEID

```
# show mobile-gateway pdn bearer-context up-seid 0x10100 detail
=====
Bearer context detail
=====
-----
Gi Network Inst   : ipd.nokia.com
S5 DL Trf End-Pt : 5                QCI                : 9
Up Time          : 0d 00:00:43

S5 PGW Data TEID : 0x40010105        S5 SGW Data TEID : 0x10105
S5 PGW Data addr : 3ffe::1110:1010
S5 PGW Data addr : 17.16.16.16
S5 SGW Data addr : 3ffe::a0a:501
S5 DL packets    : 0                  S5 DL bytes        : 0
SGI UL packets   : 0                  SGI UL bytes       : 0
SDF rule name    : 5120
Reporting Level  : Rating Grp Level   Metering Method    : -
Billing Method   : -                  Flow Status        : Up/Downlink Enabled
Service Id       : -                  Rating Group       : 1
Sponsor Identity : -
Appl Svc Prov Id : -
SDF precedence   : 65535              SDF pkt filters    : 1
Divert to AA     : No
PDR ID UL        : 5120              PDR ID DL          : 37888
FAR ID UL        : 327680            FAR ID DL           : 537198592
QER ID           : 1

Filter id         : 0                  Filter direction   : BiDir
Filter protocol   : any                Filter tos-tc       : -
Local address     : any
Remote addr       : any
-----
Number of bearer contexts : 1
=====
```

**NOTE:**

- If a dynamic rule is installed for the session, the CP function only sends the SDF filters to the UP function (not the policy rule name) and the SDF rule name field indicates the PDR ID (UP function). If a static rule is installed for the session, the SDF rule name field indicates the policy rule name in both the CP and UP functions.
- For the CP function, the Rating Group field indicates an internally generated ID for each of the URRs. In the UP function, the Rating Group field is not available.

5.2.2.19 call-flow-stats

Syntax

```
call-flow-stats [gateway gw-id | group group-number | [card slot-number | vm vm-id]] [gw-type sgw | pgw | upf | smf] summary
```

```
call-flow-stats oi-override-mapping [gateway gw-id] [p lmn-mcc mcc-value p lmn-mnc mnc-value]
```

Context

```
show>mobile>pdn
```

Description

This command displays call flow statistics for each card.

When displaying Operator Identifier (OI) mapping statistics, it shows the number of sessions established (initial and handover) as a summary per gateway ID, OI override mapping list, or per subscriber PLMN entry that the ePDG used OI or serving network PLMN replacement from the OI override mapping table.

Parameters

gw-id

specifies the gateway ID

Values: 1 to 8

group-number

specifies the group number

Values: 1 to 8

slot-number

specifies the card slot number (applicable only to 7750 SR MG)

Values: 1 to 10 | A | B

vm-id

specifies the VM ID (applicable only to CMG)

Values: 1 to 20 | A | B

summary

displays summarized information.

mcc-value

specifies an MCC value

mnc value

specifies an MNC value

gw-type

specifies the gateway type

Values: sgw | pgw | upf | smf

Output

The following 7750 SR MG output examples show call flow statistics information.

**NOTE:**

The outputs of a CMG system are identical to the 7750 SR MG outputs except they have some CMG-specific fields displayed instead of hardware components (for example, VM is displayed instead of Card).

- *Example output call flow statistics PDN gateway (active CPM)*
- *Example output call flow statistics PDN gateway (active card of MG group)*
- *Example output call flow statistics PDN gateway (active card slot of MG group)*
- *Example output call flow statistics ePDG/PGW combination gateways*
- *Example output call flow statistics on session management function*
- *Example output call flow statistics on user plane function*
- *Example output call flow statistics on control plane function for 5G sessions*
- *Example output call flow gateway statistics PDN gateway*
- *Example output call flow group statistics PDN gateway*
- *Example output call flow card statistics PDN gateway*
- *Example output call flow summary statistics PDN gateway*
- *Example output PDN gateway call flow card summary statistics*
- *Example output call flow statistics serving gateway*

- [Example output call flow statistics OI override mapping](#)
- [Example output call flow UPF statistics](#)
- [Example output call flow NamfEventExposure service statistics](#)
- [Example output call flow UDM service statistics](#)
- [Example output call flow statistics for 5G sessions](#)

**NOTE:**

Any 4G-related content is not included in the 5G-related outputs.

Example output call flow statistics PDN gateway (active CPM)

This example has been truncated to show only statistics of the active CPM.

```
# show mobile-gateway pdn call-flow-stats
=====
PDN gateway call flow statistics
=====
Chassis/Card           : 1/A      Gateway           : 1
Attach                 : 0        Attach Fail       : 0
Detach                 : 0        Detach Fail       : 0
NW Ded Bearer Actv     : 0        NW Ded Bearer Actv Fail : 0
NW Ded Bearer Deactv   : 0        NW Ded Bearer Deactv Fail : 0
UE Ded Bearer Actv     : 0        UE Ded Bearer Actv Fail : 0
UE Ded Bearer Deactv   : 0        UE Ded Bearer Deactv Fail : 0
UE Ded Bearer Modify   : 0        UE Ded Bearer Modify Fail : 0
MME Ded Bearer Deact   : 0        MME Ded Bearer Deact Fail : 0
MS Bearer Modify       : 0        MS Bearer Modify Fail : 0
HSS QoS Modify         : 0        HSS QoS Modify Fail : 0
PCRF QoS Modify        : 0        PCRF QoS Modify Fail : 0
SGSN/SGW Reloc        : 0        SGSN/SGW Reloc Fail : 0
PGW/GGSN PDN Sess Del : 0        PGW/GGSN PDN Sess Del Fail : 0
Attach Piggy Backing   : 0
eHRPD attach success   : 0        eHRPD attach failure : 0
eHRPD detach success   : 0        eHRPD detach failure : 0
eHRPD/LTE handover success : 0    eHRPD/LTE handover failure : 0
LTE/eHRPD handover success : 0    LTE/eHRPD handover failure : 0
Inter HSGW handover success: 0    Inter HSGW handover failure: 0
PDN suspend notifications : 0    PDN resume notifications : 0
Implicit Resume Service Re*: 0    Emergency Attach Proc succ*: 0
Reattach Success       : 0        Reattach Fail      : 0
eHRPD Reattach Success : 0        eHRPD Reattach Fail : 0
HRPD attach success     : 0        HRPD attach failure : 0
HRPD detach success     : 0        HRPD detach failure : 0
Inter PDSN handover success: 0    Inter PDSN handover failure: 0
eMPS IMS session Actv   : 0        eMPS IMS session Deactv : 0
ho-apn-lookup-imsi-based-oi: 0    ho-apn-lookup-ni-only : 0
PGW/GGSN initiated 3GPP session deactivations
Due to session-timeout : 0        Due to idle-timeout : 0
PGW initiated trusted-non-3GPP session deactivations
Due to session-timeout : 0        Due to idle-timeout : 0
PGW initiated S2b non-3GPP session deactivations
```

```

Due to session-timeout      : 0      Due to idle-timeout      : 0
PGW/GGSN initiated sd interface session deactivations
Due to session-timeout      : 0      Due to idle-timeout      : 0
Inter RAT handover
3g to EUTRAN success        : 0      3g to EUTRAN failure     : 0
EUTRAN to 3g success        : 0      EUTRAN to 3g failure     : 0
S2b attach success          : 0      S2b attach failure       : 0
S2b detach success          : 0      S2b detach failure       : 0
S2b/LTE handover success    : 0      S2b/LTE handover failure : 0
LTE/S2b handover success    : 0      LTE/S2b handover failure : 0
S2b PGW PDN Sess Del        : 0      S2b PGW PDN Sess Del Fail : 0
S2b/UMTS handover success   : 0      S2b/UMTS handover failure : 0
UMTS/S2b handover success   : 0      UMTS/S2b handover failure : 0
S2b/S2b handover success    : 0      S2b/S2b handover failure : 0
Network Requested Bearer Deactivation
GTP S2b Success             : 0      GTP S2b Failure          : 0
S1 release procedures        : 0      S1 release failures       : 0
UE init service req          : 0      UE init svc req failures  : 0
Intra SGW idle mode TAU      : 0      Paging timeout expiration : 0
Inter SGW idle mode TAU      : 0      Inter SGW idle TAU fails  : 0
Inter SGW S1 HO with InTnl   : 0      S1 HO with Ind Tunnel fails : 0
Inter SGW S1 HO w/o InTnl    : 0      S1 HO w/o Ind Tunnel fails : 0
Inter SGW X2 handovers       : 0      Inter SGW X2 HO fails     : 0
Inter SGW relocation (Out)    : 0      Intra SGW idle TAU fails  : 0
eHRPD to LTE handovers      : 0      eHRPD to LTE HO fails     : 0
PDN Sess De-activation       : 0      PDN Sess De-act fails     : 0
Paging Attempts             : 0      Paging fails              : 0
Intra SGW X2+S1 HOs          : 0      Intra SGW X2+S1 HO fails  : 0
Intra SGW S1 handvr InTnl    : 0      S1 handvr Ind Tunnel fails : 0
Attach Proc Fail Piggy Back : 0
NW init bearer modify        : 0      NW init bearer modify fail : 0
Inter MME/S4-SGSN
Idle TAU                     : 0      Idle TAU fails            : 0
X2+S1 Reloc                  : 0      X2+S1 Reloc fails         : 0
S1 Reloc Ind Tnl             : 0      S1 Reloc fails            : 0
Only Relocations             : 0
ePDG attach success          : 0      ePDG attach failure       : 0
ePDG detach success          : 0      ePDG detach failure       : 0
ePDG reattach success        : 0      ePDG reattach failure     : 0
ePDG PDN Sess Del            : 0      ePDG PDN Sess Del Fail    : 0
ePDG NW Ded Bearer Actv      : 0      ePDG NW Ded Bearer Actv Fa*: 0
ePDG NW Ded Bearer Deactv    : 0      ePDG NW Ded Bearer Deactv *: 0
ePDG QoS Modify              : 0      ePDG QoS Modify Fail      : 0
ePDG PCRF QoS Modify         : 0      ePDG PCRF QoS Modify Fail : 0
ePDG to LTE HO success       : 0      ePDG to LTE HO failure    : 0
LTE to ePDG HO success       : 0      LTE to ePDG HO failure    : 0
S2b NW Ded Bearer Actv       : 0      S2b NW Ded Bearer Actv Fail: 0
S2b NW Ded Bearer Deactv     : 0      S2b NW Ded Bearer Deactv F*: 0
S2b QoS Modify               : 0      S2b QoS Modify Fail       : 0
S2b PCRF QoS Modify          : 0      S2b PCRF QoS Modify Fail  : 0
ePDG MOBIKE success          : 0      ePDG MOBIKE failure       : 0
Imsi-Privacy Unenc attaches : 0      Imsi-Privacy Unencrypted HO: 0
Imsi-Privacy Encr attaches  : 0      Imsi-Privacy Encrypted HO : 0
UE idle error ind            : 0      Session deleted error ind  : 0
Bearer deleted error ind      : 0
LTE/ePDG HO Fail no-mip6     : 0      LTE/ePDG HO Fail resp-cc64 : 0
TWAG portal access (Open SSID)

```

```

DORA attach : 0 DORA attach failure : 0
Rapid commit attach : 0 Rapid commit attach failure: 0
DHCP lease-time expire : 0 NW init detach : 0
DHCP release : 0 DHCP reattach : 0
CoA access connect : 0 CoA access disconnect : 0
Inter AP HO data trigger : 0 Inter AP HO DHCP : 0
DHCP hold timeout : 0
SLAAC attach : 0 SLAAC attach failure : 0
TWAG eap access (Closed SSID)
DORA attach : 0 DORA attach failure : 0
Rapid commit attach : 0 Rapid commit attach failure: 0
DHCP lease-time expire : 0 Network init detach : 0
DHCP release : 0 DHCP reattach : 0
EAP access attach success : 0 EAP access attach failure : 0
Inter-AP HO data trigger : 0 Inter-AP HO DHCP : 0
Re-auth from new AP : 0
ACCT from new AP : 0
Fast-Inter-AP HO Re-auth : 0
Fast-Inter-AP HO ACCT : 0
Handover Failure : 0
DHCP hold timeout : 0
SLAAC attach : 0 SLAAC attach failure : 0
TDF attach success : 0 TDF attach failure : 0
TDF detach success : 0 TDF detach failure : 0
TDF reattach success : 0 TDF reattach failure : 0
S2b/eHPRD handover success : 0 S2b/eHPRD handover failure : 0
eHPRD/S2b handover success : 0 eHPRD/S2b handover failure : 0
E-SGW Restoration
PDN connections restored : 0 PDN connections deleted : 0
S2a attach success : 0 S2a attach failure : 0
S2a detach success : 0 S2a detach failure : 0
S2a/LTE handover success : 0 S2a/LTE handover failure : 0
LTE/S2a handover success : 0 LTE/S2a handover failure : 0
S2a PGW PDN Sess Del : 0 S2a PGW PDN Sess Del Fail : 0
S2a/UMTS handover success : 0 S2a/UMTS handover failure : 0
UMTS/S2a handover success : 0 UMTS/S2a handover failure : 0
S2a/S2a handover success : 0 S2a/S2a handover failure : 0
S2a NW Ded Bearer Actv : 0 S2a NW Ded Bearer Actv Fail: 0
S2a NW Ded Bearer Deactv : 0 S2a NW Ded Bearer Deactv F*: 0
S2a QoS Modify : 0 S2a QoS Modify Fail : 0
S2a PCRF QoS Modify : 0 S2a PCRF QoS Modify Fail : 0
Session cleaned setup guar*: 0
S11-u tunnel activations : 0 S11-u tunnel deactivations : 0
S11-u to S1-u transitions : 0 S1-u to S11-u transitions : 0
Unsuccessful Emergency Cal*: 0 Successful Emergency Calls : 0
Unsuccessful Emergency Calls due to
User Auth rejection : 0 Issues on S2b interface : 0
Issues on DNS res Emer APN : 0 Missing Emergency Configur*: 0
Issues on DNS res non-Emer : 0

```

Number of cards : 1

=====

* indicates that the corresponding row element may have been truncated.

Example output call flow statistics PDN gateway (active card of MG group)

The call-flow-stats command will show statistics for all active cards or VMs of the MG group. This example has been truncated to show statistics of only two active cards.

```
# show mobile-gateway pdn call-flow-stats
=====
PDN gateway call flow statistics
=====
```

Chassis/Card	: 1/3	Gateway	: 1
Attach	: 0	Attach Fail	: 0
Detach	: 0	Detach Fail	: 0
NW Ded Bearer Actv	: 0	NW Ded Bearer Actv Fail	: 0
NW Ded Bearer Deactv	: 0	NW Ded Bearer Deactv Fail	: 0
NW Ded Bearer Modify	: 0	NW Ded Bearer Modify Fail	: 0
UE Ded Bearer Actv	: 0	UE Ded Bearer Actv Fail	: 0
UE Ded Bearer Deactv	: 0	UE Ded Bearer Deactv Fail	: 0
UE Ded Bearer Modify	: 0	UE Ded Bearer Modify Fail	: 0
HSS QoS Modify	: 0	HSS QoS Modify Fail	: 0
PCRF QoS Modify	: 0	PCRF QoS Modify Fail	: 0
SGSN/SGW Reloc	: 0	SGSN/SGW Reloc Fail	: 0
PGW/GGSN PDN Sess Del	: 0	PGW/GGSN PDN Sess Del Fail	: 0
Attach Piggy Backing	: 0		
eHRPD attach success	: 1	eHRPD attach failure	: 1
eHRPD detach success	: 0	eHRPD detach failure	: 0
eHRPD/LTE handover success	: 0	eHRPD/LTE handover failure	: 0
LTE/eHRPD handover success	: 0	LTE/eHRPD handover failure	: 0
Inter HSGW handover success	: 0	Inter HSGW handover failure	: 0
PDN suspend notifications	: 0	PDN resume notifications	: 0
MS Bearer Modify	: 0	MS Bearer Modify Fail	: 0
Inter RAT handover			
3g to EUTRAN success	: 0	3g to EUTRAN failure	: 0
EUTRAN to 3g success	: 0	EUTRAN to 3g failure	: 0
Implicit Resume Service Re*	: 0	Emergency Attach Proc succ*	: 0

```
-----
```

Chassis/Card	: 1/4	Gateway	: 1
Attach	: 0	Attach Fail	: 0
Detach	: 0	Detach Fail	: 0
NW Ded Bearer Actv	: 0	NW Ded Bearer Actv Fail	: 0
NW Ded Bearer Deactv	: 0	NW Ded Bearer Deactv Fail	: 0
NW Ded Bearer Modify	: 0	NW Ded Bearer Modify Fail	: 0
UE Ded Bearer Actv	: 0	UE Ded Bearer Actv Fail	: 0
UE Ded Bearer Deactv	: 0	UE Ded Bearer Deactv Fail	: 0
UE Ded Bearer Modify	: 0	UE Ded Bearer Modify Fail	: 0
HSS QoS Modify	: 0	HSS QoS Modify Fail	: 0
PCRF QoS Modify	: 0	PCRF QoS Modify Fail	: 0
SGSN/SGW Reloc	: 0	SGSN/SGW Reloc Fail	: 0
PGW/GGSN PDN Sess Del	: 0	PGW/GGSN PDN Sess Del Fail	: 0
Attach Piggy Backing	: 0		
eHRPD attach success	: 1	eHRPD attach failure	: 1
eHRPD detach success	: 1	eHRPD detach failure	: 0
eHRPD/LTE handover success	: 0	eHRPD/LTE handover failure	: 0
LTE/eHRPD handover success	: 0	LTE/eHRPD handover failure	: 0
Inter HSGW handover success	: 0	Inter HSGW handover failure	: 0
PDN suspend notifications	: 0	PDN resume notifications	: 0

```
Implicit Resume Service Re*: 0      Emergency Attach Proc succ*: 0
```

```
-----
Number of cards : 2
=====
```

```
* indicates that the corresponding row element may have been truncated.
```

Example output call flow statistics PDN gateway (active card slot of MG group)

The call-flow-stats command when used with the active card slot will give a truncated output for that card.

```
# show mobile-gateway pdn call-flow-stats card 3
=====
PDN gateway call flow statistics
=====
Chassis/Card          : 1/3      Gateway          : 2

Attach                : 0      Attach Fail      : 0
Detach                : 0      Detach Fail      : 0
NW Ded Bearer Actv    : 0      NW Ded Bearer Actv Fail : 0
NW Ded Bearer Deactv  : 0      NW Ded Bearer Deactv Fail : 0
UE Ded Bearer Actv    : 0      UE Ded Bearer Actv Fail : 0
UE Ded Bearer Deactv  : 0      UE Ded Bearer Deactv Fail : 0
UE Ded Bearer Modify  : 0      UE Ded Bearer Modify Fail : 0
MME Ded Bearer Deact  : 0      MME Ded Bearer Deact Fail : 0
MS Bearer Modify      : 0      MS Bearer Modify Fail : 0
HSS QoS Modify        : 0      HSS QoS Modify Fail : 0
PCRF QoS Modify       : 0      PCRF QoS Modify Fail : 0
SGSN/SGW Reloc       : 0      SGSN/SGW Reloc Fail : 0
PGW/GGSN PDN Sess Del : 0      PGW/GGSN PDN Sess Del Fail : 0
Attach Piggy Backing  : 0
eHRPD attach success  : 0      eHRPD attach failure : 0
eHRPD detach success  : 0      eHRPD detach failure : 0
eHRPD/LTE handover success : 0      eHRPD/LTE handover failure : 0
LTE/eHRPD handover success : 0      LTE/eHRPD handover failure : 0
Inter HSGW handover success : 0      Inter HSGW handover failure : 0
PDN suspend notifications : 0      PDN resume notifications : 0
Implicit Resume Service Re*: 0      Emergency Attach Proc succ*: 0
Reattach Success      : 0      Reattach Fail      : 0
eHRPD Reattach Success : 0      eHRPD Reattach Fail : 0
HRPD attach success   : 0      HRPD attach failure : 0
HRPD detach success   : 0      HRPD detach failure : 0
Inter PDSN handover success : 0      Inter PDSN handover failure : 0
eMPS IMS session Actv : 0      eMPS IMS session Deactv : 0
eMPS IMS Adv Pri Ses Actv : 0      eMPS IMS Adv Pri Ses Deactv : 0
eMPS Ded Bearer Actv   : 0      eMPS Ded Bearer Deactv : 0

PGW/GGSN initiated 3GPP session deactivations
Due to session-timeout : 0      Due to idle-timeout : 0

PGW initiated trusted-non-3GPP session deactivations
Due to session-timeout : 0      Due to idle-timeout : 0

PGW initiated S2b non-3GPP session deactivations
Due to session-timeout : 0      Due to idle-timeout : 0
```

```

PGW/GGSN initiated sd interface session deactivations
Due to session-timeout      : 0          Due to idle-timeout      : 0

Inter RAT handover
3g to EUTRAN success       : 0          3g to EUTRAN failure     : 0
EUTRAN to 3g success       : 0          EUTRAN to 3g failure     : 0
S2b attach success         : 0          S2b attach failure       : 0
S2b detach success         : 0          S2b detach failure       : 0
S2b/LTE handover success   : 0          S2b/LTE handover failure : 0
LTE/S2b handover success   : 0          LTE/S2b handover failure : 0
S2b PGW PDN Sess Del       : 0          S2b PGW PDN Sess Del Fail : 0
S2b/UMTS handover success  : 0          S2b/UMTS handover failure : 0
UMTS/S2b handover success  : 0          UMTS/S2b handover failure : 0
S2b/S2b handover success   : 0          S2b/S2b handover failure : 0
S2b/eHPRD handover success : 0          S2b/eHPRD handover failure : 0
eHPRD/S2b handover success : 0          eHPRD/S2b handover failure : 0

Network Requested Bearer Deactivation
GTP S2b Success            : 0          GTP S2b Failure          : 0
S1 release procedures       : 0          S1 release failures       : 0
UE init service req        : 0          UE init svc req failures  : 0
Intra SGW idle mode TAU    : 0          Paging timeout expiration : 0
Inter SGW idle mode TAU    : 0          Inter SGW idle TAU fails  : 0
Inter SGW S1 HO with InTnl : 0          S1 HO with Ind Tunnel fails: 0
Inter SGW S1 HO w/o InTnl  : 0          S1 HO w/o Ind Tunnel fails : 0
Inter SGW X2 handovers     : 0          Inter SGW X2 HO fails     : 0
Inter SGW relocation (Out) : 0          Intra SGW idle TAU fails  : 0
eHPRD to LTE handovers     : 0          eHPRD to LTE HO fails     : 0
PDN Sess De-activation     : 0          PDN Sess De-act fails     : 0
Paging Attempts            : 0          Paging fails              : 0
Intra SGW X2+S1 HOs        : 0          Intra SGW X2+S1 HO fails  : 0
Intra SGW S1 handvr InTnl  : 0          S1 handvr Ind Tunnel fails : 0
Attach Proc Fail Piggy Back: 0
NW init bearer modify      : 0          NW init bearer modify fail : 0

Inter MME/S4-SGSN
Idle TAU                    : 0          Idle TAU fails            : 0
X2+S1 Reloc                 : 0          X2+S1 Reloc fails         : 0
S1 Reloc Ind Tnl            : 0          S1 Reloc fails            : 0
Only Relocations            : 0

ePDG attach success        : 1          ePDG attach failure       : 2
ePDG detach success        : 0          ePDG detach failure       : 0
ePDG reattach success      : 0          ePDG reattach failure     : 0
ePDG PDN Sess Del          : 1          ePDG PDN Sess Del Fail    : 0
ePDG NW Ded Bearer Actv    : 0          ePDG NW Ded Bearer Actv Fa*: 0
ePDG NW Ded Bearer Deactv  : 0          ePDG NW Ded Bearer Deactv *: 0
ePDG QoS Modify            : 0          ePDG QoS Modify Fail      : 0
ePDG PCRF QoS Modify       : 0          ePDG PCRF QoS Modify Fail : 0
ePDG to LTE HO success     : 0          ePDG to LTE HO failure    : 0
LTE to ePDG HO success     : 0          LTE to ePDG HO failure    : 0
S2b NW Ded Bearer Actv     : 0          S2b NW Ded Bearer Actv Fail: 0
S2b NW Ded Bearer Deactv   : 0          S2b NW Ded Bearer Deactv F*: 0
S2b QoS Modify             : 0          S2b QoS Modify Fail       : 0
S2b PCRF QoS Modify        : 0          S2b PCRF QoS Modify Fail  : 0
ePDG MOBIKE success        : 0          ePDG MOBIKE failure       : 0
Imsi-Privacy Unenc attaches: 0          Imsi-Privacy Unencrypted HO: 0

```

```

Imsi-Privacy Encr attaches : 1      Imsi-Privacy Encrypted HO : 0
UE idle error ind          : 0      Session deleted error ind : 0
Bearer deleted error ind   : 0
LTE/ePDG HO Fail no-mip6   : 0      LTE/ePDG HO Fail resp-cc64 : 0
ASR alt peer session del   : 0

TWAG portal access (Open SSID)
DORA attach                : 0      DORA attach failure        : 0
Rapid commit attach        : 0      Rapid commit attach failure: 0
DHCP lease-time expire     : 0      NW init detach             : 0
DHCP release               : 0      Handover Failure           : 0
CoA access connect         : 0      CoA access disconnect      : 0
Inter AP HO data trigger   : 0      Inter AP HO DHCP           : 0
DHCP hold timeout          : 0
SLAAC attach               : 0      SLAAC attach failure       : 0
Portal to EAP transition   : 0
Inter-AP HO Fail Data Rogu*: 0      Inter-AP HO Fail DHCP Rogu*: 0
Inter-AP HO Fail Router So*: 0

TWAG eap access (Closed SSID)
DORA attach                : 0      DORA attach failure        : 0
Rapid commit attach        : 0      Rapid commit attach failure: 0
DHCP lease-time expire     : 0      Network init detach        : 0
DHCP release               : 0      Handover Failure           : 0
EAP access attach success  : 0      EAP access attach failure  : 0
Inter-AP HO data trigger   : 0      Inter-AP HO DHCP           : 0
Re-auth from new AP        : 0      ACCT from new AP           : 0
DHCP hold timeout          : 0
SLAAC attach               : 0      SLAAC attach failure       : 0
EAP to Portal transition   : 0
Fast-Inter-AP HO Re-auth   : 0      Fast-Inter-AP HO ACCT      : 0
PMK Cache Timeout          : 0

TDF attach success         : 0      TDF attach failure         : 0
TDF detach success         : 0      TDF detach failure         : 0
TDF reattach success       : 0      TDF reattach failure       : 0

E-SGW Restoration
PDN connections restored   : 0      PDN connections deleted    : 0

S2a attach success         : 0      S2a attach failure         : 0
S2a detach success         : 0      S2a detach failure         : 0
S2a/LTE handover success   : 0      S2a/LTE handover failure   : 0
LTE/S2a handover success   : 0      LTE/S2a handover failure   : 0
S2a PGW PDN Sess Del       : 0      S2a PGW PDN Sess Del Fail  : 0
S2a/UMTS handover success  : 0      S2a/UMTS handover failure  : 0
UMTS/S2a handover success  : 0      UMTS/S2a handover failure  : 0
S2a/S2a handover success   : 0      S2a/S2a handover failure   : 0

S2a NW Ded Bearer Actv     : 0      S2a NW Ded Bearer Actv Fail: 0
S2a NW Ded Bearer Deactv   : 0      S2a NW Ded Bearer Deactv F*: 0
S2a QoS Modify             : 0      S2a QoS Modify Fail        : 0
S2a PCRF QoS Modify        : 0      S2a PCRF QoS Modify Fail   : 0

Session cleaned setup guar*: 0

S11-u tunnel activations   : 0      S11-u tunnel deactivations : 0

```



```

S11-u to S1-u transitions : 0          S1-u to S11-u transitions : 0

Unsuccessful Emergency Cal*: 0          Successful Emergency Calls : 0
Unsuccessful Emergency Calls due to
User Auth rejection : 0                Issues on S2b interface : 0
Issues on DNS res Emer APN : 0          Missing Emergency Configur*: 0
Issues on DNS res non-Emer : 0
Emergency Calls detected due to
IDr value in Emergency APN*: 0

ePDG Emergency Call Requests
Unauthenticated Init Served: 0          Unauthenticated HO Served : 0
Unauthenticated Init Rej : 0            Unauthenticated HO Rejected: 0
Uicc-less Initial Success : 0           Uicc-less Handover Success : 0
Uicc-less Initial Rejected : 0          Uicc-less Handover Rejected: 0

```

```

-----
Number of cards : 1
=====

```

* indicates that the corresponding row element may have been truncated.

Example output call flow statistics ePDG/PGW combination gateways

```

# show mobile-gateway pdn call-flow-stats
=====
PDN gateway call flow statistics
=====
Chassis/Card          : 1/3          Gateway          : 1

Attach                : 0            Attach Fail      : 0
Detach                : 0            Detach Fail      : 0
NW Ded Bearer Actv    : 0            NW Ded Bearer Actv Fail : 0
NW Ded Bearer Deactv  : 0            NW Ded Bearer Deactv Fail : 0
UE Ded Bearer Actv    : 0            UE Ded Bearer Actv Fail : 0
UE Ded Bearer Deactv  : 0            UE Ded Bearer Deactv Fail : 0
UE Ded Bearer Modify  : 0            UE Ded Bearer Modify Fail : 0
MME Ded Bearer Deact  : 0            MME Ded Bearer Deact Fail : 0
MS Bearer Modify      : 0            MS Bearer Modify Fail : 0
HSS QoS Modify        : 0            HSS QoS Modify Fail : 0
PCRF QoS Modify       : 0            PCRF QoS Modify Fail : 0
SGSN/SGW Reloc       : 0            SGSN/SGW Reloc Fail : 0
PGW/GGSN PDN Sess Del : 0            PGW/GGSN PDN Sess Del Fail : 0
Attach Piggy Backing  : 0
eHRPD attach success  : 0            eHRPD attach failure : 0
eHRPD detach success  : 0            eHRPD detach failure : 0
eHRPD/LTE handover success : 0        eHRPD/LTE handover failure : 0
LTE/eHRPD handover success : 0        LTE/eHRPD handover failure : 0
Inter HSGW handover success : 0        Inter HSGW handover failure : 0
PDN suspend notifications : 0        PDN resume notifications : 0
Implicit Resume Service Re* : 0        Emergency Attach Proc succ* : 0
Reattach Success      : 0            Reattach Fail     : 0
eHRPD Reattach Success : 0            eHRPD Reattach Fail : 0
HRPD attach success    : 0            HRPD attach failure : 0
HRPD detach success    : 0            HRPD detach failure : 0
Inter PDSN handover success : 0        Inter PDSN handover failure : 0

PGW/GGSN initiated 3GPP session deactivations

```

```

Due to session-timeout      : 0          Due to idle-timeout      : 0

PGW initiated trusted-non-3GPP session deactivations
Due to session-timeout      : 0          Due to idle-timeout      : 0

PGW initiated S2b non-3GPP session deactivations
Due to session-timeout      : 0          Due to idle-timeout      : 0

PGW/GGSN initiated sd interface session deactivations
Due to session-timeout      : 0          Due to idle-timeout      : 0

Inter RAT handover
3g to EUTRAN success        : 0          3g to EUTRAN failure      : 0
EUTRAN to 3g success        : 0          EUTRAN to 3g failure      : 0
S2b attach success          : 1          S2b attach failure        : 0
S2b detach success          : 0          S2b detach failure        : 0
S2b/LTE handover success    : 0          S2b/LTE handover failure  : 0
LTE/S2b handover success    : 0          LTE/S2b handover failure  : 0
S2b PGW PDN Sess Del        : 0          S2b PGW PDN Sess Del Fail : 0
S2b/UMTS handover success   : 0          S2b/UMTS handover failure : 0
UMTS/S2b handover success   : 0          UMTS/S2b handover failure : 0
S2b/S2b handover success    : 0          S2b/S2b handover failure  : 0

Network Requested Bearer Deactivation
GTP S2b Success             : 0          GTP S2b Failure           : 0
S1 release procedures        : 0          S1 release failures        : 0
UE init service req          : 0          UE init svc req failures   : 0
Intra SGW idle mode TAU      : 0          Paging timeout expiration  : 0
Inter SGW idle mode TAU      : 0          Inter SGW idle TAU fails   : 0
Inter SGW S1 HO with InTnl   : 0          S1 HO with Ind Tunnel fails : 0
Inter SGW S1 HO w/o InTnl    : 0          S1 HO w/o Ind Tunnel fails : 0
Inter SGW X2 handovers        : 0          Inter SGW X2 HO fails      : 0
Inter SGW relocation (Out)    : 0          Intra SGW idle TAU fails   : 0
eHRPD to LTE handovers       : 0          eHRPD to LTE HO fails      : 0
PDN Sess De-activation       : 0          PDN Sess De-act fails      : 0
Paging Attempts              : 0          Paging fails                : 0
Intra SGW X2+S1 HOs          : 0          Intra SGW X2+S1 HO fails   : 0
Intra SGW S1 handvr InTnl    : 0          S1 handvr Ind Tunnel fails  : 0
Attach Proc Fail Piggy Back   : 0
NW init bearer modify         : 0          NW init bearer modify fail : 0

Inter MME/S4-SGSN
Idle TAU                     : 0          Idle TAU fails              : 0
X2+S1 Reloc                  : 0          X2+S1 Reloc fails           : 0
S1 Reloc Ind Tnl              : 0          S1 Reloc fails              : 0
Only Relocations              : 0

ePDG attach success          : 0          ePDG attach failure         : 0
ePDG detach success          : 0          ePDG detach failure         : 0
ePDG reattach success        : 0          ePDG reattach failure       : 0
ePDG PDN Sess Del            : 0          ePDG PDN Sess Del Fail      : 0
ePDG NW Ded Bearer Actv      : 0          ePDG NW Ded Bearer Actv Fa* : 0
ePDG NW Ded Bearer Deactv    : 0          ePDG NW Ded Bearer Deactv*  : 0
ePDG QoS Modify              : 0          ePDG QoS Modify Fail        : 0
ePDG PCRF QoS Modify         : 0          ePDG PCRF QoS Modify Fail   : 0
ePDG to LTE HO success       : 0          ePDG to LTE HO failure      : 0
LTE to ePDG HO success       : 0          LTE to ePDG HO failure      : 0

```

S2b NW Ded Bearer Actv	: 0	S2b NW Ded Bearer Actv Fail	: 0
S2b NW Ded Bearer Deactv	: 0	S2b NW Ded Bearer Deactv F*	: 0
S2b QoS Modify	: 0	S2b QoS Modify Fail	: 0
S2b PCRF QoS Modify	: 0	S2b PCRF QoS Modify Fail	: 0
ePDG MOBIKE success	: 0	ePDG MOBIKE failure	: 0
Imsi-Privacy Unenc attaches	: 0	Imsi-Privacy Unencrypted H0	: 0
Imsi-Privacy Encr attaches	: 0	Imsi-Privacy Encrypted H0	: 0
UE idle error ind	: 0	Session deleted error ind	: 0
Bearer deleted error ind	: 0		
LTE/ePDG H0 Fail no-mip6	: 0	LTE/ePDG H0 Fail resp-cc64	: 0
ePDG init success restoration:	20	ePDG init failed restoration:	6
ASR alt peer session del	: 0		
TWAG portal access (Open SSID)			
DORA attach	: 0	DORA attach failure	: 0
Rapid commit attach	: 0	Rapid commit attach failure	: 0
DHCP lease-time expire	: 0	NW init detach	: 0
DHCP release	: 0	DHCP reattach	: 0
CoA access connect	: 0	CoA access disconnect	: 0
Inter AP H0 data trigger	: 0	Inter AP H0 DHCP	: 0
Inter-AP H0 re-auth	: 0		
DHCP hold timeout	: 0		
SLAAC attach	: 0	SLAAC attach failure	: 0
TDF attach success	: 0	TDF attach failure	: 0
TDF detach success	: 0	TDF detach failure	: 0
TDF reattach success	: 0	TDF reattach failure	: 0
S2b/eHPRD handover success	: 0	S2b/eHPRD handover failure	: 0
eHPRD/S2b handover success	: 0	eHPRD/S2b handover failure	: 0
E-SGW Restoration			
PDN connections restored	: 0	PDN connections deleted	: 0
S2a attach success	: 0	S2a attach failure	: 0
S2a detach success	: 0	S2a detach failure	: 0
S2a/LTE handover success	: 0	S2a/LTE handover failure	: 0
LTE/S2a handover success	: 0	LTE/S2a handover failure	: 0
S2a PGW PDN Sess Del	: 0	S2a PGW PDN Sess Del Fail	: 0
S2a/UMTS handover success	: 0	S2a/UMTS handover failure	: 0
UMTS/S2a handover success	: 0	UMTS/S2a handover failure	: 0
S2a/S2a handover success	: 0	S2a/S2a handover failure	: 0
S2a NW Ded Bearer Actv	: 0	S2a NW Ded Bearer Actv Fail	: 0
S2a NW Ded Bearer Deactv	: 0	S2a NW Ded Bearer Deactv F*	: 0
S2a QoS Modify	: 0	S2a QoS Modify Fail	: 0
S2a PCRF QoS Modify	: 0	S2a PCRF QoS Modify Fail	: 0
Session cleaned setup guar*	: 0		
S11-u tunnel activations	: 0	S11-u tunnel deactivations	: 0
S11-u to S1-u transitions	: 0	S1-u to S11-u transitions	: 0
Unsuccessful Emergency Cal*	: 0	Successful Emergency Calls	: 0
Unsuccessful Emergency Calls due to			
User Auth rejection	: 0	Issues on S2b interface	: 0
Issues on DNS res Emer APN	: 0		
Issues on DNS res non-Emer	: 0		

Number of cards : 1

* indicates that the corresponding row element may have been truncated.

Example output call flow statistics on session management function

```
# show mobile-gateway pdn call-flow-stats
```

```
PDN gateway call flow statistics
```

VNF/VM	: 1/3	Gateway	: 1
Attach	: 1	Attach Fail	: 0
Detach	: 0	Detach Fail	: 0
NW Ded Bearer Actv	: 0	NW Ded Bearer Actv Fail	: 0
NW Ded Bearer Deactv	: 0	NW Ded Bearer Deactv Fail	: 0
UE Ded Bearer Actv	: 0	UE Ded Bearer Actv Fail	: 0
UE Ded Bearer Deactv	: 0	UE Ded Bearer Deactv Fail	: 0
UE Ded Bearer Modify	: 0	UE Ded Bearer Modify Fail	: 0
MME Ded Bearer Deact	: 0	MME Ded Bearer Deact Fail	: 0
MS Bearer Modify	: 0	MS Bearer Modify Fail	: 0
HSS QoS Modify	: 0	HSS QoS Modify Fail	: 0
PCRF QoS Modify	: 0	PCRF QoS Modify Fail	: 0
SGSN/SGW Reloc	: 0	SGSN/SGW Reloc Fail	: 0
PGW/GGSN PDN Sess Del	: 0	PGW/GGSN PDN Sess Del Fail	: 0
Attach Piggy Backing	: 0		
eHRPD attach success	: 0	eHRPD attach failure	: 0
eHRPD detach success	: 0	eHRPD detach failure	: 0
eHRPD/LTE handover success	: 0	eHRPD/LTE handover failure	: 0
LTE/eHRPD handover success	: 0	LTE/eHRPD handover failure	: 0
Inter HSGW handover success	: 0	Inter HSGW handover failure	: 0
PDN suspend notifications	: 0	PDN resume notifications	: 0
Implicit Resume Service Re*	: 0	Emergency Attach Proc succ*	: 0
Reattach Success	: 0	Reattach Fail	: 0
eHRPD Reattach Success	: 0	eHRPD Reattach Fail	: 0
HRPD attach success	: 0	HRPD attach failure	: 0
HRPD detach success	: 0	HRPD detach failure	: 0
Inter PDSN handover success	: 0	Inter PDSN handover failure	: 0
eMPS IMS session Actv	: 0	eMPS IMS session Deactv	: 0
eMPS IMS Adv Pri Ses Actv	: 0	eMPS IMS Adv Pri Ses Deactv	: 0
eMPS Ded Bearer Actv	: 0	eMPS Ded Bearer Deactv	: 0
5GC/EPC(PGW ses) HO Sc	: 0	5GC/EPC(PGW ses) HO Fl	: 0
5GC/EPC(Cmb ses) HO Sc	: 0	5GC/EPC(Cmb ses) HO Fl	: 0
5GC/EPC(PGW ses) Idl Mob Sc	: 0	5GC/EPC(PGW ses) Idl Mob Fl	: 0
5GC/EPC(Cmb ses) Idl Mob Sc	: 0	5GC/EPC(Cmb ses) Idl Mob Fl	: 0

```
PGW/GGSN initiated 3GPP session deactivations
```

```
Due to session-timeout : 0 Due to idle-timeout : 0
```

```
PGW initiated trusted-non-3GPP session deactivations
```

```
Due to session-timeout : 0 Due to idle-timeout : 0
```

```
PGW initiated S2b non-3GPP session deactivations
```

```
Due to session-timeout : 0 Due to idle-timeout : 0
```

```
PGW/GGSN initiated sd interface session deactivations
```

```
Due to session-timeout      : 0          Due to idle-timeout      : 0

Inter RAT handover
3g to EUTRAN success        : 0          3g to EUTRAN failure      : 0
EUTRAN to 3g success        : 0          EUTRAN to 3g failure      : 0
S2b attach success          : 0          S2b attach failure        : 0
S2b detach success          : 0          S2b detach failure        : 0
S2b/LTE handover success    : 0          S2b/LTE handover failure  : 0
LTE/S2b handover success    : 0          LTE/S2b handover failure  : 0
S2b PGW PDN Sess Del        : 0          S2b PGW PDN Sess Del Fail : 0
S2b/UMTS handover success   : 0          S2b/UMTS handover failure : 0
UMTS/S2b handover success   : 0          UMTS/S2b handover failure : 0
S2b/S2b handover success    : 0          S2b/S2b handover failure  : 0
S2b/eHRPD handover success  : 0          S2b/eHRPD handover failure : 0
eHRPD/S2b handover success  : 0          eHRPD/S2b handover failure : 0

Network Requested Bearer Deactivation
GTP S2b Success              : 0          GTP S2b Failure           : 0
S1 release procedures        : 0          S1 release failures        : 0
UE init service req          : 0          UE init svc req failures   : 0
Intra SGW idle mode TAU      : 0          Paging timeout expiration  : 0
Inter SGW idle mode TAU      : 0          Inter SGW idle TAU fails   : 0
Inter SGW S1 HO with InTnl   : 0          S1 HO with Ind Tunnel fails : 0
Inter SGW S1 HO w/o InTnl    : 0          S1 HO w/o Ind Tunnel fails : 0
Inter SGW X2 handovers       : 0          Inter SGW X2 HO fails      : 0
Inter SGW relocation (Out)    : 0          Intra SGW idle TAU fails   : 0
eHRPD to LTE handovers       : 0          eHRPD to LTE HO fails      : 0
PDN Sess De-activation       : 0          PDN Sess De-act fails      : 0
Paging Attempts              : 0          Paging fails                : 0
Intra SGW X2+S1 HOs          : 0          Intra SGW X2+S1 HO fails   : 0
Intra SGW S1 handvr InTnl    : 0          S1 handvr Ind Tunnel fails : 0
Attach Proc Fail Piggy Back : 0
NW init bearer modify        : 0          NW init bearer modify fail : 0

Inter MME/S4-SGSN
Idle TAU                     : 0          Idle TAU fails             : 0
X2+S1 Reloc                  : 0          X2+S1 Reloc fails          : 0
S1 Reloc Ind Tnl             : 0          S1 Reloc fails             : 0
Only Relocations             : 0

ePDG attach success          : 0          ePDG attach failure        : 0
ePDG detach success          : 0          ePDG detach failure        : 0
ePDG reattach success        : 0          ePDG reattach failure      : 0
ePDG PDN Sess Del            : 0          ePDG PDN Sess Del Fail     : 0
ePDG NW Ded Bearer Actv      : 0          ePDG NW Ded Bearer Actv Fa*: 0
ePDG NW Ded Bearer Deactv    : 0          ePDG NW Ded Bearer Deactv *: 0
ePDG QoS Modify              : 0          ePDG QoS Modify Fail       : 0
ePDG PCRF QoS Modify         : 0          ePDG PCRF QoS Modify Fail  : 0
ePDG to LTE HO success       : 0          ePDG to LTE HO failure     : 0
LTE to ePDG HO success       : 0          LTE to ePDG HO failure     : 0
S2b NW Ded Bearer Actv       : 0          S2b NW Ded Bearer Actv Fail: 0
S2b NW Ded Bearer Deactv     : 0          S2b NW Ded Bearer Deactv F*: 0
S2b QoS Modify               : 0          S2b QoS Modify Fail        : 0
S2b PCRF QoS Modify          : 0          S2b PCRF QoS Modify Fail   : 0
ePDG MOBIKE success          : 0          ePDG MOBIKE failure        : 0
ePDG attach success CERTREQ : 0
ePDG Attach Requests over PGW-C+SMF
```

```

Initial Attach Success      : 0      Initial Attach Failure    : 0
HO Attach Success          : 0      HO Attach Failure         : 0
Imsi-Privacy Unenc attaches: 0      Imsi-Privacy Unencrypted HO: 0
Imsi-Privacy Encr attaches : 0      Imsi-Privacy Encrypted HO  : 0
UE idle error ind         : 0      Session deleted error ind  : 0
Bearer deleted error ind   : 0
LTE/ePDG HO Fail no-mip6   : 0      LTE/ePDG HO Fail resp-cc64 : 0
ASR alt peer session del   : 0

TWAG portal access (Open SSID)
DORA attach                : 0      DORA attach failure       : 0
Rapid commit attach        : 0      Rapid commit attach failure: 0
DHCP lease-time expire     : 0      NW init detach            : 0
DHCP release               : 0      Handover Failure          : 0
CoA access connect         : 0      CoA access disconnect     : 0
Inter AP HO data trigger   : 0      Inter AP HO DHCP          : 0
DHCP hold timeout         : 0
SLAAC attach               : 0      SLAAC attach failure       : 0
Portal to EAP transition   : 0

TWAG eap access (Closed SSID)
DORA attach                : 0      DORA attach failure       : 0
Rapid commit attach        : 0      Rapid commit attach failure: 0
DHCP lease-time expire     : 0      Network init detach       : 0
DHCP release               : 0      Handover Failure          : 0
EAP access attach success  : 0      EAP access attach failure  : 0
Inter-AP HO data trigger   : 0      Inter-AP HO DHCP          : 0
Re-auth from new AP        : 0      ACCT from new AP          : 0
DHCP hold timeout         : 0
SLAAC attach               : 0      SLAAC attach failure       : 0
EAP to Portal transition   : 0
Fast-Inter-AP HO Re-auth   : 0      Fast-Inter-AP HO ACCT     : 0
PMK Cache Timeout         : 0

TDF attach success         : 0      TDF attach failure        : 0
TDF detach success         : 0      TDF detach failure        : 0
TDF reattach success       : 0      TDF reattach failure      : 0

E-SGW Restoration
PDN connections restored   : 0      PDN connections deleted   : 0

S2a attach success         : 0      S2a attach failure        : 0
S2a detach success         : 0      S2a detach failure        : 0
S2a/LTE handover success   : 0      S2a/LTE handover failure  : 0
LTE/S2a handover success   : 0      LTE/S2a handover failure  : 0
S2a PGW PDN Sess Del       : 0      S2a PGW PDN Sess Del Fail : 0
S2a/UMTS handover success  : 0      S2a/UMTS handover failure : 0
UMTS/S2a handover success  : 0      UMTS/S2a handover failure : 0
S2a/S2a handover success   : 0      S2a/S2a handover failure  : 0

S2a NW Ded Bearer Actv     : 0      S2a NW Ded Bearer Actv Fail: 0
S2a NW Ded Bearer Deactv   : 0      S2a NW Ded Bearer Deactv F*: 0
S2a QoS Modify             : 0      S2a QoS Modify Fail       : 0
S2a PCRF QoS Modify        : 0      S2a PCRF QoS Modify Fail   : 0

Session cleaned setup guar*: 0

```

S11-u tunnel activations : 0	S11-u tunnel deactivations : 0
S11-u to S1-u transitions : 0	S1-u to S11-u transitions : 0
Unsuccessful Emergency Cal*: 0	Successful Emergency Calls : 0
Unsuccessful Emergency Calls due to	
User Auth rejection : 0	Issues on S2b interface : 0
Issues on DNS res Emer APN : 0	Missing Emergency Configur*: 0
Issues on DNS res non-Emer : 0	
Emergency Calls detected due to	
IDr value in Emergency APN*: 0	
ePDG Emergency Call Requests	
Unauthenticated Init Served: 0	Unauthenticated HO Served : 0
Unauthenticated Init Rej : 0	Unauthenticated HO Rejected: 0
Uicc-less Initial Success : 0	Uicc-less Handover Success : 0
Uicc-less Initial Rejected : 0	Uicc-less Handover Rejected: 0
PFCP	
Association : 0	Association Fail : 0
Association Upd : 0	Association Upd Fail : 0
Association Release : 0	Association Release Fail : 0
Node Report : 0	Node Report Fail : 0

VNF/VM : 1/A	Gateway : 1
Attach : 0	Attach Fail : 0
Detach : 0	Detach Fail : 0
NW Ded Bearer Actv : 0	NW Ded Bearer Actv Fail : 0
NW Ded Bearer Deactv : 0	NW Ded Bearer Deactv Fail : 0
UE Ded Bearer Actv : 0	UE Ded Bearer Actv Fail : 0
UE Ded Bearer Deactv : 0	UE Ded Bearer Deactv Fail : 0
UE Ded Bearer Modify : 0	UE Ded Bearer Modify Fail : 0
MME Ded Bearer Deact : 0	MME Ded Bearer Deact Fail : 0
MS Bearer Modify : 0	MS Bearer Modify Fail : 0
HSS QoS Modify : 0	HSS QoS Modify Fail : 0
PCRF QoS Modify : 0	PCRF QoS Modify Fail : 0
SGSN/SGW Reloc : 0	SGSN/SGW Reloc Fail : 0
PGW/GGSN PDN Sess Del : 0	PGW/GGSN PDN Sess Del Fail : 0
Attach Piggy Backing : 0	
eHRPD attach success : 0	eHRPD attach failure : 0
eHRPD detach success : 0	eHRPD detach failure : 0
eHRPD/LTE handover success : 0	eHRPD/LTE handover failure : 0
LTE/eHRPD handover success : 0	LTE/eHRPD handover failure : 0
Inter HSGW handover success: 0	Inter HSGW handover failure: 0
PDN suspend notifications : 0	PDN resume notifications : 0
Implicit Resume Service Re*: 0	Emergency Attach Proc succ*: 0
Reattach Success : 0	Reattach Fail : 0
eHRPD Reattach Success : 0	eHRPD Reattach Fail : 0
HRPD attach success : 0	HRPD attach failure : 0
HRPD detach success : 0	HRPD detach failure : 0
Inter PDSN handover success: 0	Inter PDSN handover failure: 0
eMPS IMS session Actv : 0	eMPS IMS session Deactv : 0
eMPS IMS Adv Pri Ses Actv : 0	eMPS IMS Adv Pri Ses Deactv: 0
eMPS Ded Bearer Actv : 0	eMPS Ded Bearer Deactv : 0
5GC/EPC(PGW ses) HO Sc : 0	5GC/EPC(PGW ses) HO Fl : 0
5GC/EPC(Cmb ses) HO Sc : 0	5GC/EPC(Cmb ses) HO Fl : 0
5GC/EPC(PGW ses) Idl Mob Sc: 0	5GC/EPC(PGW ses) Idl Mob Fl: 0

```

5GC/EPC(Cmb ses) Idl Mob Sc: 0          5GC/EPC(Cmb ses) Idl Mob Fl: 0

PGW/GGSN initiated 3GPP session deactivations
Due to session-timeout      : 0          Due to idle-timeout      : 0

PGW initiated trusted-non-3GPP session deactivations
Due to session-timeout      : 0          Due to idle-timeout      : 0

PGW initiated S2b non-3GPP session deactivations
Due to session-timeout      : 0          Due to idle-timeout      : 0

PGW/GGSN initiated sd interface session deactivations
Due to session-timeout      : 0          Due to idle-timeout      : 0

Inter RAT handover
3g to EUTRAN success        : 0          3g to EUTRAN failure     : 0
EUTRAN to 3g success        : 0          EUTRAN to 3g failure     : 0
S2b attach success          : 0          S2b attach failure       : 0
S2b detach success          : 0          S2b detach failure       : 0
S2b/LTE handover success    : 0          S2b/LTE handover failure : 0
LTE/S2b handover success    : 0          LTE/S2b handover failure : 0
S2b PGW PDN Sess Del        : 0          S2b PGW PDN Sess Del Fail : 0
S2b/UMTS handover success   : 0          S2b/UMTS handover failure : 0
UMTS/S2b handover success   : 0          UMTS/S2b handover failure : 0
S2b/S2b handover success    : 0          S2b/S2b handover failure : 0
S2b/eHPRD handover success  : 0          S2b/eHPRD handover failure : 0
eHPRD/S2b handover success  : 0          eHPRD/S2b handover failure : 0

Network Requested Bearer Deactivation
GTP S2b Success             : 0          GTP S2b Failure          : 0
S1 release procedures        : 0          S1 release failures       : 0
UE init service req          : 0          UE init svc req failures  : 0
Intra SGW idle mode TAU      : 0          Paging timeout expiration : 0
Inter SGW idle mode TAU      : 0          Inter SGW idle TAU fails  : 0
Inter SGW S1 HO with InTnl   : 0          S1 HO with Ind Tunnel fails: 0
Inter SGW S1 HO w/o InTnl    : 0          S1 HO w/o Ind Tunnel fails : 0
Inter SGW X2 handovers       : 0          Inter SGW X2 HO fails     : 0
Inter SGW relocation (Out)   : 0          Intra SGW idle TAU fails  : 0
eHPRD to LTE handovers       : 0          eHPRD to LTE HO fails     : 0
PDN Sess De-activation       : 0          PDN Sess De-act fails     : 0
Paging Attempts              : 0          Paging fails              : 0
Intra SGW X2+S1 HOs          : 0          Intra SGW X2+S1 HO fails  : 0
Intra SGW S1 handvr InTnl    : 0          S1 handvr Ind Tunnel fails : 0
Attach Proc Fail Piggy Back: 0
NW init bearer modify        : 0          NW init bearer modify fail : 0

Inter MME/S4-SGSN
Idle TAU                     : 0          Idle TAU fails            : 0
X2+S1 Reloc                  : 0          X2+S1 Reloc fails         : 0
S1 Reloc Ind Tnl             : 0          S1 Reloc fails            : 0
Only Relocations              : 0

ePDG attach success          : 0          ePDG attach failure       : 0
ePDG detach success          : 0          ePDG detach failure       : 0
ePDG reattach success        : 0          ePDG reattach failure     : 0
ePDG PDN Sess Del            : 0          ePDG PDN Sess Del Fail    : 0
ePDG NW Ded Bearer Actv      : 0          ePDG NW Ded Bearer Actv Fa*: 0

```


ePDG NW Ded Bearer Deactiv	: 0	ePDG NW Ded Bearer Deactiv *	: 0
ePDG QoS Modify	: 0	ePDG QoS Modify Fail	: 0
ePDG PCRF QoS Modify	: 0	ePDG PCRF QoS Modify Fail	: 0
ePDG to LTE H0 success	: 0	ePDG to LTE H0 failure	: 0
LTE to ePDG H0 success	: 0	LTE to ePDG H0 failure	: 0
S2b NW Ded Bearer Activ	: 0	S2b NW Ded Bearer Activ Fail	: 0
S2b NW Ded Bearer Deactiv	: 0	S2b NW Ded Bearer Deactiv F*	: 0
S2b QoS Modify	: 0	S2b QoS Modify Fail	: 0
S2b PCRF QoS Modify	: 0	S2b PCRF QoS Modify Fail	: 0
ePDG MOBIKE success	: 0	ePDG MOBIKE failure	: 0
ePDG attach success CERTREQ	: 0		
ePDG Attach Requests over PGW-C+SMF			
Initial Attach Success	: 0	Initial Attach Failure	: 0
H0 Attach Success	: 0	H0 Attach Failure	: 0
Imsi-Privacy Unenc attaches	: 0	Imsi-Privacy Unencrypted H0	: 0
Imsi-Privacy Encr attaches	: 0	Imsi-Privacy Encrypted H0	: 0
UE idle error ind	: 0	Session deleted error ind	: 0
Bearer deleted error ind	: 0		
LTE/ePDG H0 Fail no-mip6	: 0	LTE/ePDG H0 Fail resp-cc64	: 0
ASR alt peer session del	: 0		
TWAG portal access (Open SSID)			
DORA attach	: 0	DORA attach failure	: 0
Rapid commit attach	: 0	Rapid commit attach failure	: 0
DHCP lease-time expire	: 0	NW init detach	: 0
DHCP release	: 0	Handover Failure	: 0
CoA access connect	: 0	CoA access disconnect	: 0
Inter AP H0 data trigger	: 0	Inter AP H0 DHCP	: 0
DHCP hold timeout	: 0		
SLAAC attach	: 0	SLAAC attach failure	: 0
Portal to EAP transition	: 0		
TWAG eap access (Closed SSID)			
DORA attach	: 0	DORA attach failure	: 0
Rapid commit attach	: 0	Rapid commit attach failure	: 0
DHCP lease-time expire	: 0	Network init detach	: 0
DHCP release	: 0	Handover Failure	: 0
EAP access attach success	: 0	EAP access attach failure	: 0
Inter-AP H0 data trigger	: 0	Inter-AP H0 DHCP	: 0
Re-auth from new AP	: 0	ACCT from new AP	: 0
DHCP hold timeout	: 0		
SLAAC attach	: 0	SLAAC attach failure	: 0
EAP to Portal transition	: 0		
Fast-Inter-AP H0 Re-auth	: 0	Fast-Inter-AP H0 ACCT	: 0
PMK Cache Timeout	: 0		
TDF attach success			
TDF detach success	: 0	TDF attach failure	: 0
TDF reattach success	: 0	TDF detach failure	: 0
		TDF reattach failure	: 0
E-SGW Restoration			
PDN connections restored	: 0	PDN connections deleted	: 0
S2a attach success			
S2a detach success	: 0	S2a attach failure	: 0
S2a/LTE handover success	: 0	S2a detach failure	: 0
LTE/S2a handover success	: 0	S2a/LTE handover failure	: 0
		LTE/S2a handover failure	: 0

```

S2a PGW PDN Sess Del      : 0      S2a PGW PDN Sess Del Fail : 0
S2a/UMTS handover success : 0      S2a/UMTS handover failure : 0
UMTS/S2a handover success : 0      UMTS/S2a handover failure : 0
S2a/S2a handover success  : 0      S2a/S2a handover failure  : 0

S2a NW Ded Bearer Actv    : 0      S2a NW Ded Bearer Actv Fail: 0
S2a NW Ded Bearer Deactv  : 0      S2a NW Ded Bearer Deactv F*: 0
S2a QoS Modify            : 0      S2a QoS Modify Fail       : 0
S2a PCRF QoS Modify       : 0      S2a PCRF QoS Modify Fail  : 0

Session cleaned setup guar*: 0

S11-u tunnel activations  : 0      S11-u tunnel deactivations : 0
S11-u to S11-u transitions : 0      S11-u to S11-u transitions  : 0

Unsuccessful Emergency Cal*: 0      Successful Emergency Calls : 0
Unsuccessful Emergency Calls due to
User Auth rejection       : 0      Issues on S2b interface    : 0
Issues on DNS res Emer APN : 0      Missing Emergency Configur*: 0
Issues on DNS res non-Emer : 0
Emergency Calls detected due to
IDr value in Emergency APN*: 0

ePDG Emergency Call Requests
Unauthenticated Init Served: 0      Unauthenticated HO Served  : 0
Unauthenticated Init Rej   : 0      Unauthenticated HO Rejected: 0
Uicc-less Initial Success  : 0      Uicc-less Handover Success : 0
Uicc-less Initial Rejected : 0      Uicc-less Handover Rejected: 0

PFCP
Association                : 1      Association Fail            : 1
Association Upd            : 0      Association Upd Fail        : 0
Association Release        : 0      Association Release Fail    : 0
Node Report                : 0      Node Report Fail            : 0
-----
Number of VMs : 2
=====
* indicates that the corresponding row element may have been truncated.
=====
SMF call flow statistics
=====
VNF/VM          : 1/3      Gateway          : 1

Establish PDU Session Succ : 0      Establish PDU Session Fail : 0
UE Init PDU Sess Rel Succ  : 0      UE Init PDU Sess Rel Fail  : 0
NW Init PDU Sess Rel Succ  : 0      NW Init PDU Sess Rel Fail  : 0
AMF Init PDU Sess Rel Succ : 0      AMF Init PDU Sess Rel Fail : 0
UE Triggered Serv Req Succ : 0      UE Triggered Serv Req Fail : 0
NW Triggered Serv Req Succ : 0      NW Triggered Serv Req Fail : 0
AN Release              : 0
Xn Handover Success     : 0      Xn Handover Fail          : 0
N2 Handover Success     : 0      N2 Handover Fail          : 0
RAN Init PDU Sess Mod Succ : 0      RAN Init PDU Sess Mod Fail : 0
UE Init PDU Sess Mod Succ : 0      UE Init PDU Sess Mod Fail  : 0
NW Init PDU Sess Mod Succ : 0      NW Init PDU Sess Mod Fail  : 0
EPC(PGW ses)/5GC HO Sc  : 0      EPC(PGW ses)/5GC HO Fl   : 0
EPC(Cmb ses)/5GC HO Sc  : 0      EPC(Cmb ses)/5GC HO Fl   : 0

```

```

EPC(PGW ses)/5GC Idl Mob Sc: 0      EPC(PGW ses)/5GC Idl Mob Fl: 0
EPC(Cmb ses)/5GC Idl Mob Sc: 0      EPC(Cmb ses)/5GC Idl Mob Fl: 0
AMF Create Event Subsc Succ: 0      AMF Create Event Subsc Fail: 0
AMF Update Event Subsc Succ: 0      AMF Update Event Subsc Fail: 0
AMF Delete Event Subsc Succ: 0      AMF Delete Event Subsc Fail: 0
AMF Event Notific Received : 0      AMF Event Subscr Expiration: 0
-----
VNF/VM                : 1/A      Gateway                : 1

Establish PDU Session Succ : 0      Establish PDU Session Fail : 0
UE Init PDU Sess Rel Succ  : 0      UE Init PDU Sess Rel Fail  : 0
NW Init PDU Sess Rel Succ  : 0      NW Init PDU Sess Rel Fail  : 0
AMF Init PDU Sess Rel Succ : 0      AMF Init PDU Sess Rel Fail : 0
UE Triggered Serv Req Succ : 0      UE Triggered Serv Req Fail : 0
NW Triggered Serv Req Succ : 0      NW Triggered Serv Req Fail : 0
AN Release                : 0
Xn Handover Success       : 0      Xn Handover Fail          : 0
N2 Handover Success       : 0      N2 Handover Fail          : 0
RAN Init PDU Sess Mod Succ : 0      RAN Init PDU Sess Mod Fail : 0
UE Init PDU Sess Mod Succ  : 0      UE Init PDU Sess Mod Fail  : 0
NW Init PDU Sess Mod Succ  : 0      NW Init PDU Sess Mod Fail  : 0
EPC(PGW ses)/5GC HO Sc    : 0      EPC(PGW ses)/5GC HO Fl    : 0
EPC(Cmb ses)/5GC HO Sc    : 0      EPC(Cmb ses)/5GC HO Fl    : 0
EPC(PGW ses)/5GC Idl Mob Sc: 0      EPC(PGW ses)/5GC Idl Mob Fl: 0
EPC(Cmb ses)/5GC Idl Mob Sc: 0      EPC(Cmb ses)/5GC Idl Mob Fl: 0
AMF Create Event Subsc Succ: 0      AMF Create Event Subsc Fail: 0
AMF Update Event Subsc Succ: 0      AMF Update Event Subsc Fail: 0
AMF Delete Event Subsc Succ: 0      AMF Delete Event Subsc Fail: 0
AMF Event Notific Received : 0      AMF Event Subscr Expiration: 0
PDU Sess Mod Fail Ho      : 0      PDU Sess Mod Fail Rjst    : 0
-----
Number of VMs : 2
=====

```

Example output call flow statistics on user plane function

```

# show mobile-gateway pdn call-flow-stats
=====
PDN gateway call flow statistics
=====
VNF/VM                : 1/3      Gateway                : 1

Attach                : 0      Attach Fail            : 0
Detach                : 0      Detach Fail            : 0
NW Ded Bearer Actv    : 0      NW Ded Bearer Actv Fail : 0
NW Ded Bearer Deactv  : 0      NW Ded Bearer Deactv Fail : 0
UE Ded Bearer Actv    : 0      UE Ded Bearer Actv Fail : 0
UE Ded Bearer Deactv  : 0      UE Ded Bearer Deactv Fail : 0
UE Ded Bearer Modify  : 0      UE Ded Bearer Modify Fail : 0
MME Ded Bearer Deact  : 0      MME Ded Bearer Deact Fail : 0
MS Bearer Modify      : 0      MS Bearer Modify Fail    : 0
HSS QoS Modify        : 0      HSS QoS Modify Fail      : 0
PCRF QoS Modify       : 0      PCRF QoS Modify Fail     : 0
SGSN/SGW Reloc       : 0      SGSN/SGW Reloc Fail      : 0
PGW/GGSN PDN Sess Del : 0      PGW/GGSN PDN Sess Del Fail : 0
Attach Piggy Backing  : 0

```

```

eHRPD attach success      : 0      eHRPD attach failure      : 0
eHRPD detach success      : 0      eHRPD detach failure      : 0
eHRPD/LTE handover success : 0      eHRPD/LTE handover failure : 0
LTE/eHRPD handover success : 0      LTE/eHRPD handover failure : 0
Inter HSGW handover success: 0      Inter HSGW handover failure: 0
PDN suspend notifications  : 0      PDN resume notifications  : 0
Implicit Resume Service Re*: 0      Emergency Attach Proc succ*: 0
Reattach Success          : 0      Reattach Fail              : 0
eHRPD Reattach Success    : 0      eHRPD Reattach Fail        : 0
HRPD attach success        : 0      HRPD attach failure        : 0
HRPD detach success        : 0      HRPD detach failure        : 0
Inter PDSN handover success: 0      Inter PDSN handover failure: 0
eMPS IMS session Actv     : 0      eMPS IMS session Deactv    : 0
eMPS IMS Adv Pri Ses Actv  : 0      eMPS IMS Adv Pri Ses Deactv: 0
eMPS Ded Bearer Actv       : 0      eMPS Ded Bearer Deactv     : 0
5GC/EPC(PGW ses) HO Sc    : 0      5GC/EPC(PGW ses) HO Fl    : 0
5GC/EPC(Cmb ses) HO Sc    : 0      5GC/EPC(Cmb ses) HO Fl    : 0
5GC/EPC(PGW ses) Idl Mob Sc: 0      5GC/EPC(PGW ses) Idl Mob Fl: 0
5GC/EPC(Cmb ses) Idl Mob Sc: 0      5GC/EPC(Cmb ses) Idl Mob Fl: 0

PGW/GGSN initiated 3GPP session deactivations
Due to session-timeout     : 0      Due to idle-timeout        : 0

PGW initiated trusted-non-3GPP session deactivations
Due to session-timeout     : 0      Due to idle-timeout        : 0

PGW initiated S2b non-3GPP session deactivations
Due to session-timeout     : 0      Due to idle-timeout        : 0

PGW/GGSN initiated sd interface session deactivations
Due to session-timeout     : 0      Due to idle-timeout        : 0

Inter RAT handover
3g to EUTRAN success       : 0      3g to EUTRAN failure       : 0
EUTRAN to 3g success       : 0      EUTRAN to 3g failure       : 0
S2b attach success         : 0      S2b attach failure         : 0
S2b detach success         : 0      S2b detach failure         : 0
S2b/LTE handover success   : 0      S2b/LTE handover failure   : 0
LTE/S2b handover success   : 0      LTE/S2b handover failure   : 0
S2b PGW PDN Sess Del       : 0      S2b PGW PDN Sess Del Fail  : 0
S2b/UMTS handover success  : 0      S2b/UMTS handover failure  : 0
UMTS/S2b handover success  : 0      UMTS/S2b handover failure  : 0
S2b/S2b handover success   : 0      S2b/S2b handover failure   : 0
S2b/eHRPD handover success : 0      S2b/eHRPD handover failure : 0
eHRPD/S2b handover success : 0      eHRPD/S2b handover failure : 0

Network Requested Bearer Deactivation
GTP S2b Success            : 0      GTP S2b Failure            : 0
S1 release procedures       : 0      S1 release failures        : 0
UE init service req         : 0      UE init svc req failures   : 0
Intra SGW idle mode TAU     : 0      Paging timeout expiration  : 0
Inter SGW idle mode TAU     : 0      Inter SGW idle TAU fails   : 0
Inter SGW S1 HO with InTnl  : 0      S1 HO with Ind Tunnel fails: 0
Inter SGW S1 HO w/o InTnl   : 0      S1 HO w/o Ind Tunnel fails : 0
Inter SGW X2 handovers      : 0      Inter SGW X2 HO fails      : 0
Inter SGW relocation (Out)  : 0      Intra SGW idle TAU fails   : 0
eHRPD to LTE handovers      : 0      eHRPD to LTE HO fails     : 0

```

PDN Sess De-activation	: 0	PDN Sess De-act fails	: 0
Paging Attempts	: 0	Paging fails	: 0
Intra SGW X2+S1 HOs	: 0	Intra SGW X2+S1 HO fails	: 0
Intra SGW S1 handvr InTnl	: 0	S1 handvr Ind Tunnel fails	: 0
Attach Proc Fail Piggy Back	: 0		
NW init bearer modify	: 0	NW init bearer modify fail	: 0
Inter MME/S4-SGSN			
Idle TAU	: 0	Idle TAU fails	: 0
X2+S1 Reloc	: 0	X2+S1 Reloc fails	: 0
S1 Reloc Ind Tnl	: 0	S1 Reloc fails	: 0
Only Relocations	: 0		
ePDG attach success			
ePDG detach success	: 0	ePDG attach failure	: 0
ePDG reattach success	: 0	ePDG detach failure	: 0
ePDG PDN Sess Del	: 0	ePDG reattach failure	: 0
ePDG NW Ded Bearer Actv	: 0	ePDG PDN Sess Del Fail	: 0
ePDG NW Ded Bearer Deactv	: 0	ePDG NW Ded Bearer Actv Fa*	: 0
ePDG QoS Modify	: 0	ePDG NW Ded Bearer Deactv *	: 0
ePDG PCRF QoS Modify	: 0	ePDG QoS Modify Fail	: 0
ePDG to LTE HO success	: 0	ePDG PCRF QoS Modify Fail	: 0
LTE to ePDG HO success	: 0	ePDG to LTE HO failure	: 0
S2b NW Ded Bearer Actv	: 0	LTE to ePDG HO failure	: 0
S2b NW Ded Bearer Deactv	: 0	S2b NW Ded Bearer Actv Fail	: 0
S2b QoS Modify	: 0	S2b NW Ded Bearer Deactv F*	: 0
S2b PCRF QoS Modify	: 0	S2b QoS Modify Fail	: 0
ePDG MOBIKE success	: 0	S2b PCRF QoS Modify Fail	: 0
ePDG attach success CERTREQ	: 0	ePDG MOBIKE failure	: 0
ePDG Attach Requests over PGW-C+SMF			
Initial Attach Success	: 0	Initial Attach Failure	: 0
HO Attach Success	: 0	HO Attach Failure	: 0
Imsi-Privacy Unenc attaches	: 0	Imsi-Privacy Unencrypted HO	: 0
Imsi-Privacy Encr attaches	: 0	Imsi-Privacy Encrypted HO	: 0
UE idle error ind	: 0	Session deleted error ind	: 0
Bearer deleted error ind	: 0		
LTE/ePDG HO Fail no-mip6	: 0	LTE/ePDG HO Fail resp-cc64	: 0
ASR alt peer session del	: 0		
TWAG portal access (Open SSID)			
DORA attach	: 0	DORA attach failure	: 0
Rapid commit attach	: 0	Rapid commit attach failure	: 0
DHCP lease-time expire	: 0	NW init detach	: 0
DHCP release	: 0	Handover Failure	: 0
CoA access connect	: 0	CoA access disconnect	: 0
Inter AP HO data trigger	: 0	Inter AP HO DHCP	: 0
DHCP hold timeout	: 0		
SLAAC attach	: 0	SLAAC attach failure	: 0
Portal to EAP transition	: 0		
TWAG eap access (Closed SSID)			
DORA attach	: 0	DORA attach failure	: 0
Rapid commit attach	: 0	Rapid commit attach failure	: 0
DHCP lease-time expire	: 0	Network init detach	: 0
DHCP release	: 0	Handover Failure	: 0
EAP access attach success	: 0	EAP access attach failure	: 0
Inter-AP HO data trigger	: 0	Inter-AP HO DHCP	: 0

Re-auth from new AP	: 0	ACCT from new AP	: 0
DHCP hold timeout	: 0		
SLAAC attach	: 0	SLAAC attach failure	: 0
EAP to Portal transition	: 0		
Fast-Inter-AP HO Re-auth	: 0	Fast-Inter-AP HO ACCT	: 0
PMK Cache Timeout	: 0		
TDF attach success	: 0	TDF attach failure	: 0
TDF detach success	: 0	TDF detach failure	: 0
TDF reattach success	: 0	TDF reattach failure	: 0
E-SGW Restoration			
PDN connections restored	: 0	PDN connections deleted	: 0
S2a attach success	: 0	S2a attach failure	: 0
S2a detach success	: 0	S2a detach failure	: 0
S2a/LTE handover success	: 0	S2a/LTE handover failure	: 0
LTE/S2a handover success	: 0	LTE/S2a handover failure	: 0
S2a PGW PDN Sess Del	: 0	S2a PGW PDN Sess Del Fail	: 0
S2a/UMTS handover success	: 0	S2a/UMTS handover failure	: 0
UMTS/S2a handover success	: 0	UMTS/S2a handover failure	: 0
S2a/S2a handover success	: 0	S2a/S2a handover failure	: 0
S2a NW Ded Bearer Actv	: 0	S2a NW Ded Bearer Actv Fail	: 0
S2a NW Ded Bearer Deactv	: 0	S2a NW Ded Bearer Deactv F*	: 0
S2a QoS Modify	: 0	S2a QoS Modify Fail	: 0
S2a PCRF QoS Modify	: 0	S2a PCRF QoS Modify Fail	: 0
Session cleaned setup guar*	: 0		
S11-u tunnel activations	: 0	S11-u tunnel deactivations	: 0
S11-u to S1-u transitions	: 0	S1-u to S11-u transitions	: 0
Unsuccessful Emergency Cal*	: 0	Successful Emergency Calls	: 0
Unsuccessful Emergency Calls due to			
User Auth rejection	: 0	Issues on S2b interface	: 0
Issues on DNS res Emer APN	: 0	Missing Emergency Configur*	: 0
Issues on DNS res non-Emer	: 0		
Emergency Calls detected due to			
IDr value in Emergency APN*	: 0		
ePDG Emergency Call Requests			
Unauthenticated Init Served	: 0	Unauthenticated HO Served	: 0
Unauthenticated Init Rej	: 0	Unauthenticated HO Rejected	: 0
Uicc-less Initial Success	: 0	Uicc-less Handover Success	: 0
Uicc-less Initial Rejected	: 0	Uicc-less Handover Rejected	: 0
PFCP			
Session Establishment	: 1	Session Establishment Fail	: 0
Session Modification	: 0	Session Modification Fail	: 0
Session Deletion	: 0	Session Deletion Fail	: 0
Session Report	: 0	Session Report Fail	: 0
Association	: 0	Association Fail	: 0
Association Upd	: 0	Association Upd Fail	: 0
Association Release	: 0	Association Release Fail	: 0
Node Report	: 0	Node Report Fail	: 0

VNF/VM	: 1/A	Gateway	: 1
Attach	: 0	Attach Fail	: 0
Detach	: 0	Detach Fail	: 0
NW Ded Bearer Actv	: 0	NW Ded Bearer Actv Fail	: 0
NW Ded Bearer Deactv	: 0	NW Ded Bearer Deactv Fail	: 0
UE Ded Bearer Actv	: 0	UE Ded Bearer Actv Fail	: 0
UE Ded Bearer Deactv	: 0	UE Ded Bearer Deactv Fail	: 0
UE Ded Bearer Modify	: 0	UE Ded Bearer Modify Fail	: 0
MME Ded Bearer Deact	: 0	MME Ded Bearer Deact Fail	: 0
MS Bearer Modify	: 0	MS Bearer Modify Fail	: 0
HSS QoS Modify	: 0	HSS QoS Modify Fail	: 0
PCRF QoS Modify	: 0	PCRF QoS Modify Fail	: 0
SGSN/SGW Reloc	: 0	SGSN/SGW Reloc Fail	: 0
PGW/GGSN PDN Sess Del	: 0	PGW/GGSN PDN Sess Del Fail	: 0
Attach Piggy Backing	: 0		
eHRPD attach success	: 0	eHRPD attach failure	: 0
eHRPD detach success	: 0	eHRPD detach failure	: 0
eHRPD/LTE handover success	: 0	eHRPD/LTE handover failure	: 0
LTE/eHRPD handover success	: 0	LTE/eHRPD handover failure	: 0
Inter HSGW handover success	: 0	Inter HSGW handover failure	: 0
PDN suspend notifications	: 0	PDN resume notifications	: 0
Implicit Resume Service Re*	: 0	Emergency Attach Proc succ*	: 0
Reattach Success	: 0	Reattach Fail	: 0
eHRPD Reattach Success	: 0	eHRPD Reattach Fail	: 0
HRPD attach success	: 0	HRPD attach failure	: 0
HRPD detach success	: 0	HRPD detach failure	: 0
Inter PDSN handover success	: 0	Inter PDSN handover failure	: 0
eMPS IMS session Actv	: 0	eMPS IMS session Deactv	: 0
eMPS IMS Adv Pri Ses Actv	: 0	eMPS IMS Adv Pri Ses Deactv	: 0
eMPS Ded Bearer Actv	: 0	eMPS Ded Bearer Deactv	: 0
5GC/EPC(PGW ses) HO Sc	: 0	5GC/EPC(PGW ses) HO Fl	: 0
5GC/EPC(Cmb ses) HO Sc	: 0	5GC/EPC(Cmb ses) HO Fl	: 0
5GC/EPC(PGW ses) Idl Mob Sc	: 0	5GC/EPC(PGW ses) Idl Mob Fl	: 0
5GC/EPC(Cmb ses) Idl Mob Sc	: 0	5GC/EPC(Cmb ses) Idl Mob Fl	: 0
PGW/GGSN initiated 3GPP session deactivations			
Due to session-timeout	: 0	Due to idle-timeout	: 0
PGW initiated trusted-non-3GPP session deactivations			
Due to session-timeout	: 0	Due to idle-timeout	: 0
PGW initiated S2b non-3GPP session deactivations			
Due to session-timeout	: 0	Due to idle-timeout	: 0
PGW/GGSN initiated sd interface session deactivations			
Due to session-timeout	: 0	Due to idle-timeout	: 0
Inter RAT handover			
3g to EUTRAN success	: 0	3g to EUTRAN failure	: 0
EUTRAN to 3g success	: 0	EUTRAN to 3g failure	: 0
S2b attach success	: 0	S2b attach failure	: 0
S2b detach success	: 0	S2b detach failure	: 0
S2b/LTE handover success	: 0	S2b/LTE handover failure	: 0
LTE/S2b handover success	: 0	LTE/S2b handover failure	: 0
S2b PGW PDN Sess Del	: 0	S2b PGW PDN Sess Del Fail	: 0
S2b/UMTS handover success	: 0	S2b/UMTS handover failure	: 0

```

UMTS/S2b handover success : 0      UMTS/S2b handover failure : 0
S2b/S2b handover success : 0      S2b/S2b handover failure : 0
S2b/eHPRD handover success : 0    S2b/eHPRD handover failure : 0
eHPRD/S2b handover success : 0    eHPRD/S2b handover failure : 0

Network Requested Bearer Deactivation
GTP S2b Success : 0      GTP S2b Failure : 0
S1 release procedures : 0  S1 release failures : 0
UE init service req : 0  UE init svc req failures : 0
Intra SGW idle mode TAU : 0  Paging timeout expiration : 0
Inter SGW idle mode TAU : 0  Inter SGW idle TAU fails : 0
Inter SGW S1 HO with InTnl : 0  S1 HO with Ind Tunnel fails: 0
Inter SGW S1 HO w/o InTnl : 0  S1 HO w/o Ind Tunnel fails : 0
Inter SGW X2 handovers : 0  Inter SGW X2 HO fails : 0
Inter SGW relocation (Out) : 0  Intra SGW idle TAU fails : 0
eHPRD to LTE handovers : 0  eHPRD to LTE HO fails : 0
PDN Sess De-activation : 0  PDN Sess De-act fails : 0
Paging Attempts : 0  Paging fails : 0
Intra SGW X2+S1 HOs : 0  Intra SGW X2+S1 HO fails : 0
Intra SGW S1 handvr InTnl : 0  S1 handvr Ind Tunnel fails : 0
Attach Proc Fail Piggy Back: 0
NW init bearer modify : 0      NW init bearer modify fail : 0

Inter MME/S4-SGSN
Idle TAU : 0      Idle TAU fails : 0
X2+S1 Reloc : 0  X2+S1 Reloc fails : 0
S1 Reloc Ind Tnl : 0  S1 Reloc fails : 0
Only Relocations : 0

ePDG attach success : 0      ePDG attach failure : 0
ePDG detach success : 0      ePDG detach failure : 0
ePDG reattach success : 0    ePDG reattach failure : 0
ePDG PDN Sess Del : 0      ePDG PDN Sess Del Fail : 0
ePDG NW Ded Bearer Actv : 0  ePDG NW Ded Bearer Actv Fa*: 0
ePDG NW Ded Bearer Deactv : 0  ePDG NW Ded Bearer Deactv *: 0
ePDG QoS Modify : 0      ePDG QoS Modify Fail : 0
ePDG PCRF QoS Modify : 0    ePDG PCRF QoS Modify Fail : 0
ePDG to LTE HO success : 0    ePDG to LTE HO failure : 0
LTE to ePDG HO success : 0    LTE to ePDG HO failure : 0
S2b NW Ded Bearer Actv : 0    S2b NW Ded Bearer Actv Fail: 0
S2b NW Ded Bearer Deactv : 0  S2b NW Ded Bearer Deactv F*: 0
S2b QoS Modify : 0      S2b QoS Modify Fail : 0
S2b PCRF QoS Modify : 0    S2b PCRF QoS Modify Fail : 0
ePDG MOBIKE success : 0      ePDG MOBIKE failure : 0
ePDG attach success CERTREQ: 0
ePDG Attach Requests over PGW-C+SMF
Initial Attach Success : 0      Initial Attach Failure : 0
HO Attach Success : 0      HO Attach Failure : 0
Imsi-Privacy Unenc attaches: 0  Imsi-Privacy Unencrypted HO: 0
Imsi-Privacy Encr attaches : 0  Imsi-Privacy Encrypted HO : 0
UE idle error ind : 0      Session deleted error ind : 0
Bearer deleted error ind : 0
LTE/ePDG HO Fail no-mip6 : 0    LTE/ePDG HO Fail resp-cc64 : 0
ASR alt peer session del : 0

TWAG portal access (Open SSID)
DORA attach : 0      DORA attach failure : 0

```


Rapid commit attach	: 0	Rapid commit attach failure:	0
DHCP lease-time expire	: 0	NW init detach	: 0
DHCP release	: 0	Handover Failure	: 0
CoA access connect	: 0	CoA access disconnect	: 0
Inter AP HO data trigger	: 0	Inter AP HO DHCP	: 0
DHCP hold timeout	: 0		
SLAAC attach	: 0	SLAAC attach failure	: 0
Portal to EAP transition	: 0		
TWAG eap access (Closed SSID)			
DORA attach	: 0	DORA attach failure	: 0
Rapid commit attach	: 0	Rapid commit attach failure:	0
DHCP lease-time expire	: 0	Network init detach	: 0
DHCP release	: 0	Handover Failure	: 0
EAP access attach success	: 0	EAP access attach failure	: 0
Inter-AP HO data trigger	: 0	Inter-AP HO DHCP	: 0
Re-auth from new AP	: 0	ACCT from new AP	: 0
DHCP hold timeout	: 0		
SLAAC attach	: 0	SLAAC attach failure	: 0
EAP to Portal transition	: 0		
Fast-Inter-AP HO Re-auth	: 0	Fast-Inter-AP HO ACCT	: 0
PMK Cache Timeout	: 0		
TDF attach success : 0			
TDF detach success	: 0	TDF attach failure	: 0
TDF reattach success	: 0	TDF detach failure	: 0
		TDF reattach failure	: 0
E-SGW Restoration			
PDN connections restored	: 0	PDN connections deleted	: 0
S2a attach success : 0			
S2a detach success	: 0	S2a attach failure	: 0
S2a/LTE handover success	: 0	S2a detach failure	: 0
LTE/S2a handover success	: 0	S2a/LTE handover failure	: 0
S2a PGW PDN Sess Del	: 0	LTE/S2a handover failure	: 0
S2a/UMTS handover success	: 0	S2a PGW PDN Sess Del Fail	: 0
UMTS/S2a handover success	: 0	S2a/UMTS handover failure	: 0
S2a/S2a handover success	: 0	UMTS/S2a handover failure	: 0
		S2a/S2a handover failure	: 0
S2a NW Ded Bearer Actv	: 0	S2a NW Ded Bearer Actv Fail:	0
S2a NW Ded Bearer Deactv	: 0	S2a NW Ded Bearer Deactv F*:	0
S2a QoS Modify	: 0	S2a QoS Modify Fail	: 0
S2a PCRF QoS Modify	: 0	S2a PCRF QoS Modify Fail	: 0
Session cleaned setup guar*: 0			
S11-u tunnel activations : 0			
S11-u to S1-u transitions	: 0	S11-u tunnel deactivations	: 0
		S1-u to S11-u transitions	: 0
Unsuccessful Emergency Cal*: 0			
Unsuccessful Emergency Calls due to		Successful Emergency Calls	: 0
User Auth rejection	: 0	Issues on S2b interface	: 0
Issues on DNS res Emer APN	: 0	Missing Emergency Configur*:	0
Issues on DNS res non-Emer	: 0		
Emergency Calls detected due to			
IDr value in Emergency APN*:	0		

```
ePDG Emergency Call Requests
Unauthenticated Init Served: 0          Unauthenticated HO Served : 0
Unauthenticated Init Rej : 0          Unauthenticated HO Rejected: 0
Uicc-less Initial Success : 0        Uicc-less Handover Success : 0
Uicc-less Initial Rejected : 0      Uicc-less Handover Rejected: 0
```

```
PFCP
Session Establishment : 0          Session Establishment Fail : 0
Session Modification : 0          Session Modification Fail : 0
Session Deletion : 0          Session Deletion Fail : 0
Session Report : 0          Session Report Fail : 0
Association : 1          Association Fail : 1
Association Upd : 0          Association Upd Fail : 0
Association Release : 0          Association Release Fail : 0
Node Report : 0          Node Report Fail : 0
```

```
-----
Number of VMs : 2
=====
```

Example output call flow statistics on control plane function for 5G sessions

```
# show mobile-gateway pdn call-flow-stats
```

```
=====
SMF call flow statistics
=====
```

```
VNF/VM : 1/A          Gateway : 1

Establish PDU Session Succ : 0          Establish PDU Session Fail : 0
Re-establish PDU Sess Succ : 0          Re-establish PDU Sess Fail : 0
UE Init PDU Sess Rel Succ : 0          UE Init PDU Sess Rel Fail : 0
NW Init PDU Sess Rel Succ : 0          NW Init PDU Sess Rel Fail : 0
UDM Init PDU Sess Rel Succ : 0          UDM Init PDU Sess Rel Fail : 0
AMF Init PDU Sess Rel Succ : 0          AMF Init PDU Sess Rel Fail : 0
UE Triggered Serv Req Succ : 0          UE Triggered Serv Req Fail : 0
NW Triggered Serv Req Succ : 0          NW Triggered Serv Req Fail : 0
AN Release : 0
Xn Handover Success : 0          Xn Handover Fail : 0
N2 Handover Success : 0          N2 Handover Fail : 0
RAN Init PDU Sess Mod Succ : 0          RAN Init PDU Sess Mod Fail : 0
UE Init PDU Sess Mod Succ : 0          UE Init PDU Sess Mod Fail : 0
SMF Init PDU Sess Mod Succ : 0          SMF Init PDU Sess Mod Fail : 0
PCF Init PDU Sess Mod Succ : 0          PCF Init PDU Sess Mod Fail : 0
UDM Init PDU Sess Mod Succ : 0          UDM Init PDU Sess Mod Fail : 0
EPC(PGW ses)/5GC HO Sc : 0          EPC(PGW ses)/5GC HO Fl : 0
EPC(Cmb ses)/5GC HO Sc : 0          EPC(Cmb ses)/5GC HO Fl : 0
EPC(PGW ses)/5GC Idl Mob Sc : 0          EPC(PGW ses)/5GC Idl Mob Fl : 0
EPC(Cmb ses)/5GC Idl Mob Sc : 0          EPC(Cmb ses)/5GC Idl Mob Fl : 0
AMF Create Event Subsc Succ : 0          AMF Create Event Subsc Fail : 0
AMF Update Event Subsc Succ : 0          AMF Update Event Subsc Fail : 0
AMF Delete Event Subsc Succ : 0          AMF Delete Event Subsc Fail : 0
AMF Event Notific Received : 0          AMF Event Subscr Expiration : 0
PDU Sess Mod Rej IMS Fllbck : 0
LADN PDU Sess Released : 0
LADN UP Suspended : 0          LADN UP Resumed : 0
```

```
Successful Emergency Attaches with
```

```
No SUPI          : 0          Unauthenticated SUPI      : 0
Authenticated SUPI : 0
```

```
Successful Emergency Handovers to 5GC with
No SUPI          : 0          Unauthenticated SUPI      : 0
Authenticated SUPI : 0
```

```
Unsuccessful Emergency Calls due to
Missing Policy In APN : 0          PCF Failure          : 0
UDM Failure          : 0          Other Failure         : 0
```

```
Unsuccessful Emergency Call Handovers to 5GC due to
Missing Policy In APN : 0          PCF Failure          : 0
UDM Failure          : 0          Other Failure         : 0
```

```
PDU Sess Mod Fail Ho      : 0          PDU Sess Mod Fail Reg : 0
```

```
Successful P-CSCF Restorat*: 0
Un-successful P-CSCF Restoration
5G Sessions              : 0
Un-successful P-CSCF Resto*: 0
```

```
-----
Number of VMS : 1
=====
```

* indicates that the corresponding row element may have been truncated.

```
=====
I-SMF call flow statistics
=====
```

```
VNF/VM          : 1/A          Gateway          : 1
```

```
Establish PDU Session Succ : 0          Establish PDU Session Fail : 0
AMF Init PDU Sess Rel Succ : 0          AMF Init PDU Sess Rel Fail : 0
UE Init PDU Sess Rel Succ  : 0          UE Init PDU Sess Rel Fail  : 0
I-SMF Init PDU Sess Rel Sc : 0          I-SMF Init PDU Sess Rel Fl : 0
SMF Init PDU Sess Rel Succ : 0          SMF Init PDU Sess Rel Fail : 0
UE Triggered Serv Req Succ : 0          UE Triggered Serv Req Fail : 0
NW Triggered Serv Req Succ : 0          NW Triggered Serv Req Fail : 0
AN Release              : 0
Xn Handover Success     : 0          Xn Handover Fail          : 0
N2 Handover Success     : 0          N2 Handover Fail          : 0
AMF Change Success      : 0          AMF Change Fail           : 0
EPC/5GC Handover Success : 0          EPC/5GC Handover Fail     : 0
EPC/5GC Mobility Success : 0          EPC/5GC Mobility Fail     : 0
UE Init PDU Sess Mod Succ : 0          UE Init PDU Sess Mod Fail  : 0
RAN Init PDU Sess Mod Succ : 0          RAN Init PDU Sess Mod Fail : 0
SMF Init PDU Sess Mod Succ : 0          SMF Init PDU Sess Mod Fail : 0
```

```
-----
Number of VMS : 1
=====
```

```
=====
SMF with I-SMF call flow statistics
=====
```

```
VNF/VM          : 1/A          Gateway          : 1
```

```
I-SMF Init PDU Sess Rel Sc : 0          I-SMF Init PDU Sess Rel Fl : 0
```

```

SMF Init PDU Sess Rel Succ : 0      SMF Init PDU Sess Rel Fail : 0
UE Init PDU Sess Mod Succ  : 0      UE Init PDU Sess Mod Fail  : 0
RAN Init PDU Sess Mod Succ : 0      RAN Init PDU Sess Mod Fail : 0
PCF Init PDU Sess Mod Succ : 0      PCF Init PDU Sess Mod Fail : 0
SMF Init PDU Sess Mod Succ : 0      SMF Init PDU Sess Mod Fail : 0
UDM Init PDU Sess Mod Succ : 0      UDM Init PDU Sess Mod Fail : 0

```

```

-----
Number of VMs : 1
=====

```

```

=====
SMF with I-SMF added call flow statistics
=====

```

```

VNF/VM                : 1/A      Gateway                : 1

I-SMF Added Success    : 0      I-SMF Added Fail      : 0
Establish PDU Session  : 0      Establish PDU Session : 0
UE Triggered Serv Req  : 0      UE Triggered Serv Req : 0
N2 Handover Success    : 0      N2 Handover Fail      : 0
AMF Change Success     : 0      AMF Change Fail        : 0
EPC/5GC(PGW sess) HO  : 0      EPC/5GC(PGW sess) HO : 0
EPC/5GC(Cmb sess) HO  : 0      EPC/5GC(Cmb sess) HO : 0
EPC/5GC(PGW sess) Mob : 0      EPC/5GC(PGW sess) Mob : 0
EPC/5GC(Cmb sess) Mob : 0      EPC/5GC(Cmb sess) Mob : 0

```

```

-----
Number of VMs : 1
=====

```

```

=====
SMF with I-SMF change call flow statistics
=====

```

```

VNF/VM                : 1/A      Gateway                : 1

I-SMF Change Success   : 0      I-SMF Change Fail     : 0
AMF Change Success     : 0      AMF Change Fail        : 0

```

```

-----
Number of VMs : 1
=====

```

```

=====
SMF with I-SMF deleted call flow statistics
=====

```

```

VNF/VM                : 1/A      Gateway                : 1

UE Triggered Serv Req  : 0      UE Triggered Serv Req : 0
Xn Handover Success    : 0      Xn Handover Fail      : 0
N2 Handover Success    : 0      N2 Handover Fail      : 0
AMF Change Success     : 0      AMF Change Fail        : 0
EPC/5GC(PGW sess) HO  : 0      EPC/5GC(PGW sess) HO : 0
EPC/5GC(Cmb sess) HO  : 0      EPC/5GC(Cmb sess) HO : 0
EPC/5GC(PGW sess) Mob : 0      EPC/5GC(PGW sess) Mob : 0
EPC/5GC(Cmb sess) Mob : 0      EPC/5GC(Cmb sess) Mob : 0

```

```

-----
Number of VMs : 1
=====
=====

```

H-SMF call flow statistics

```

=====
VNF/VM                : 1/A      Gateway                : 1

Establish PDU Session Succ : 0      Establish PDU Session Fail : 0
V-SMF Init PDU Sess Rel Sc : 0      V-SMF Init PDU Sess Rel Fl : 0
H-SMF Init PDU Sess Rel Sc : 0      H-SMF Init PDU Sess Rel Fl : 0
V-SMF Init PDU Sess Mod Sc : 0      V-SMF Init PDU Sess Mod Fl : 0
PCF Init PDU Sess Mod Succ : 0      PCF Init PDU Sess Mod Fail : 0
H-SMF Init PDU Sess Mod Sc : 0      H-SMF Init PDU Sess Mod Fl : 0
UDM Init PDU Sess Mod Succ : 0      UDM Init PDU Sess Mod Fail : 0
=====

```

```

-----
Number of VMs : 1
=====

```

V-SMF call flow statistics

```

=====
VNF/VM                : 1/A      Gateway                : 1

Establish PDU Session Succ : 0      Establish PDU Session Fail : 0
AMF Init PDU Sess Rel Succ : 0      AMF Init PDU Sess Rel Fail : 0
UE Init PDU Sess Rel Succ  : 0      UE Init PDU Sess Rel Fail  : 0
V-SMF Init PDU Sess Rel Sc : 0      V-SMF Init PDU Sess Rel Fl : 0
H-SMF Init PDU Sess Rel Sc : 0      H-SMF Init PDU Sess Rel Fl : 0
UE Triggered Serv Req Succ : 0      UE Triggered Serv Req Fail : 0
NW Triggered Serv Req Succ : 0      NW Triggered Serv Req Fail : 0
An Release                : 0
Xn Handover Success       : 0      Xn Handover Fail           : 0
N2 Handover Success       : 0      N2 Handover Fail           : 0
UE Init PDU Sess Mod Succ : 0      UE Init PDU Sess Mod Fail  : 0
RAN Init PDU Sess Mod Succ : 0      RAN Init PDU Sess Mod Fail : 0
H-SMF Init PDU Sess Mod Sc : 0      H-SMF Init PDU Sess Mod Fl : 0
=====

```

```

-----
Number of VMs : 1
=====

```

Example output call flow gateway statistics PDN gateway

```

# show mobile-gateway pdn call-flow-stats gateway 1

```

PDN gateway call flow statistics

```

=====
Chassis/Card          : 1/3      Gateway                : 1

Attach                : 1      Attach Fail             : 0
Detach                : 0      Detach Fail             : 0
NW Ded Bearer Actv    : 1      NW Ded Bearer Actv Fail : 0
NW Ded Bearer Deactv   : 0      NW Ded Bearer Deactv Fail : 0
NW Ded Bearer Modify   : 0      NW Ded Bearer Modify Fail : 0
UE Ded Bearer Actv     : 0      UE Ded Bearer Actv Fail  : 0
UE Ded Bearer Deactv   : 0      UE Ded Bearer Deactv Fail : 0
UE Ded Bearer Modify   : 0      UE Ded Bearer Modify Fail : 0
HSS QoS Modify        : 0      HSS QoS Modify Fail     : 0
PCRF QoS Modify       : 0      PCRF QoS Modify Fail     : 0
SGSN/SGW Reloc        : 0      SGSN/SGW Reloc Fail     : 0
=====

```

```

PGW/GGSN PDN Sess Del      : 0          PGW/GGSN PDN Sess Del Fail : 0
Attach Piggy Backing       : 0
eHRPD attach success       : 0          eHRPD attach failure      : 0
eHRPD detach success       : 0          eHRPD detach failure      : 0
eHRPD/LTE handover success : 0          eHRPD/LTE handover failure : 0
LTE/eHRPD handover success : 0          LTE/eHRPD handover failure : 0
Inter HSGW handover success: 0          Inter HSGW handover failure: 0
PDN suspend notifications  : 0          PDN resume notifications   : 0
Implicit Resume Service Re*: 0          Emergency Attach Proc succ*: 0

```

```

-----
Number of cards : 1
=====

```

* indicates that the corresponding row element may have been truncated.

Example output call flow group statistics PDN gateway

```

# show mobile-gateway pdn call-flow-stats group 1
=====
PDN gateway call flow statistics
=====
Chassis/Card      : 1/3      Gateway      : 1

Attach            : 1        Attach Fail    : 0
Detach            : 0        Detach Fail    : 0
NW Ded Bearer Actv : 1        NW Ded Bearer Actv Fail : 0
NW Ded Bearer Deactv : 0      NW Ded Bearer Deactv Fail : 0
NW Ded Bearer Modify : 0      NW Ded Bearer Modify Fail : 0
UE Ded Bearer Actv  : 0        UE Ded Bearer Actv Fail  : 0
UE Ded Bearer Deactv : 0      UE Ded Bearer Deactv Fail : 0
UE Ded Bearer Modify : 0      UE Ded Bearer Modify Fail : 0
HSS QoS Modify     : 0        HSS QoS Modify Fail     : 0
PCRF QoS Modify     : 0        PCRF QoS Modify Fail     : 0
SGSN/SGW Reloc     : 0        SGSN/SGW Reloc Fail     : 0
PGW/GGSN PDN Sess Del : 0      PGW/GGSN PDN Sess Del Fail : 0
Attach Piggy Backing : 0
eHRPD attach success : 0      eHRPD attach failure     : 0
eHRPD detach success : 0      eHRPD detach failure     : 0
eHRPD/LTE handover success : 0    eHRPD/LTE handover failure : 0
LTE/eHRPD handover success : 0    LTE/eHRPD handover failure : 0
Inter HSGW handover success: 0    Inter HSGW handover failure: 0
PDN suspend notifications : 0      PDN resume notifications   : 0
Implicit Resume Service Re*: 0      Emergency Attach Proc succ*: 0
-----
Number of cards : 1
=====

```

* indicates that the corresponding row element may have been truncated.

Example output call flow card statistics PDN gateway

```

# show mobile-gateway pdn call-flow-stats card 3
=====
PDN gateway call flow statistics
=====
Chassis/Card      : 1/3      Gateway      : 1

Attach            : 1        Attach Fail    : 0

```

Detach	: 0	Detach Fail	: 0
NW Ded Bearer Actv	: 1	NW Ded Bearer Actv Fail	: 0
NW Ded Bearer Deactv	: 0	NW Ded Bearer Deactv Fail	: 0
NW Ded Bearer Modify	: 0	NW Ded Bearer Modify Fail	: 0
UE Ded Bearer Actv	: 0	UE Ded Bearer Actv Fail	: 0
UE Ded Bearer Deactv	: 0	UE Ded Bearer Deactv Fail	: 0
UE Ded Bearer Modify	: 0	UE Ded Bearer Modify Fail	: 0
HSS QoS Modify	: 0	HSS QoS Modify Fail	: 0
PCRF QoS Modify	: 0	PCRF QoS Modify Fail	: 0
SGSN/SGW Reloc	: 0	SGSN/SGW Reloc Fail	: 0
PGW/GGSN PDN Sess Del	: 0	PGW/GGSN PDN Sess Del Fail	: 0
Attach Piggy Backing	: 0		
eHRPD attach success	: 0	eHRPD attach failure	: 0
eHRPD detach success	: 0	eHRPD detach failure	: 0
eHRPD/LTE handover success	: 0	eHRPD/LTE handover failure	: 0
LTE/eHRPD handover success	: 0	LTE/eHRPD handover failure	: 0
Inter HSGW handover success	: 0	Inter HSGW handover failure	: 0
PDN suspend notifications	: 0	PDN resume notifications	: 0
Implicit Resume Service Re*	: 0	Emergency Attach Proc succ*	: 0

Number of cards : 1

=====

* indicates that the corresponding row element may have been truncated.

Example output call flow summary statistics PDN gateway

```
# show mobile-gateway pdn call-flow-stats summary gw-type pgw
```

=====

PDN gateway call flow statistics summary

=====

Attach	: 1	Attach Fail	: 0
Detach	: 0	Detach Fail	: 0
NW Ded Bearer Actv	: 0	NW Ded Bearer Actv Fail	: 0
NW Ded Bearer Deactv	: 0	NW Ded Bearer Deactv Fail	: 0
UE Ded Bearer Actv	: 0	UE Ded Bearer Actv Fail	: 0
UE Ded Bearer Deactv	: 0	UE Ded Bearer Deactv Fail	: 0
UE Ded Bearer Modify	: 0	UE Ded Bearer Modify Fail	: 0
MME Ded Bearer Deact	: 0	MME Ded Bearer Deact Fail	: 0
MS Bearer Modify	: 0	MS Bearer Modify Fail	: 0
HSS QoS Modify	: 0	HSS QoS Modify Fail	: 0
PCRF QoS Modify	: 0	PCRF QoS Modify Fail	: 0
SGSN/SGW Reloc	: 0	SGSN/SGW Reloc Fail	: 0
PGW/GGSN PDN Sess Del	: 0	PGW/GGSN PDN Sess Del Fail	: 0
Attach Piggy Backing	: 0		
eHRPD attach success	: 0	eHRPD attach failure	: 0
eHRPD detach success	: 0	eHRPD detach failure	: 0
eHRPD/LTE handover success	: 0	eHRPD/LTE handover failure	: 0
LTE/eHRPD handover success	: 0	LTE/eHRPD handover failure	: 0
Inter HSGW handover success	: 0	Inter HSGW handover failure	: 0
PDN suspend notifications	: 0	PDN resume notifications	: 0
Implicit Resume Service Re*	: 0	Emergency Attach Proc succ*	: 0
Reattach Success	: 0	Reattach Fail	: 0
eHRPD Reattach Success	: 0	eHRPD Reattach Fail	: 0
HRPD attach success	: 0	HRPD attach failure	: 0
HRPD detach success	: 0	HRPD detach failure	: 0

```

Inter PDSN handover success: 0          Inter PDSN handover failure: 0
eMPS IMS session Actv      : 0          eMPS IMS session Deactv      : 0
eMPS IMS Adv Pri Ses Actv  : 0          eMPS IMS Adv Pri Ses Deactv : 0
eMPS Ded Bearer Actv      : 0          eMPS Ded Bearer Deactv     : 0
5GC/EPC(PGW ses) HO Sc    : 0          5GC/EPC(PGW ses) HO Fl    : 0
5GC/EPC(Cmb ses) HO Sc    : 0          5GC/EPC(Cmb ses) HO Fl    : 0
5GC/EPC(PGW ses) Idl Mob Sc: 0          5GC/EPC(PGW ses) Idl Mob Fl: 0
5GC/EPC(Cmb ses) Idl Mob Sc: 0          5GC/EPC(Cmb ses) Idl Mob Fl: 0

PGW/GGSN initiated 3GPP session deactivations
Due to session-timeout      : 0          Due to idle-timeout        : 0

PGW initiated trusted-non-3GPP session deactivations
Due to session-timeout      : 0          Due to idle-timeout        : 0

PGW initiated S2b non-3GPP session deactivations
Due to session-timeout      : 0          Due to idle-timeout        : 0

PGW/GGSN initiated sd interface session deactivations
Due to session-timeout      : 0          Due to idle-timeout        : 0

Inter RAT handover
3g to EUTRAN success       : 0          3g to EUTRAN failure      : 0
EUTRAN to 3g success       : 0          EUTRAN to 3g failure      : 0
S2b attach success        : 0          S2b attach failure        : 0
S2b detach success        : 0          S2b detach failure        : 0
S2b/LTE handover success   : 0          S2b/LTE handover failure   : 0
LTE/S2b handover success   : 0          LTE/S2b handover failure   : 0
S2b PGW PDN Sess Del       : 0          S2b PGW PDN Sess Del Fail : 0
S2b/UMTS handover success  : 0          S2b/UMTS handover failure  : 0
UMTS/S2b handover success  : 0          UMTS/S2b handover failure  : 0
S2b/S2b handover success   : 0          S2b/S2b handover failure   : 0
S2b/eHRPD handover success : 0          S2b/eHRPD handover failure : 0
eHRPD/S2b handover success : 0          eHRPD/S2b handover failure : 0

Network Requested Bearer Deactivation
GTP S2b Success           : 0          GTP S2b Failure           : 0
S1 release procedures      : 0          S1 release failures       : 0
UE init service req       : 0          UE init svc req failures  : 0
Intra SGW idle mode TAU    : 0          Paging timeout expiration : 0
Inter SGW idle mode TAU    : 0          Inter SGW idle TAU fails   : 0
Inter SGW S1 HO with InTnl : 0          S1 HO with Ind Tunnel fails: 0
Inter SGW S1 HO w/o InTnl  : 0          S1 HO w/o Ind Tunnel fails : 0
Inter SGW X2 handovers     : 0          Inter SGW X2 HO fails     : 0
Inter SGW relocation (Out) : 0          Intra SGW idle TAU fails   : 0
eHRPD to LTE handovers     : 0          eHRPD to LTE HO fails     : 0
PDN Sess De-activation     : 0          PDN Sess De-act fails     : 0
Paging Attempts           : 0          Paging fails               : 0
Intra SGW X2+S1 HOs        : 0          Intra SGW X2+S1 HO fails   : 0
Intra SGW S1 handvr InTnl  : 0          S1 handvr Ind Tunnel fails : 0
Attach Proc Fail Piggy Back: 0
NW init bearer modify      : 0          NW init bearer modify fail : 0

Inter MME/S4-SGSN
Idle TAU                   : 0          Idle TAU fails             : 0
X2+S1 Reloc                : 0          X2+S1 Reloc fails          : 0
S1 Reloc Ind Tnl           : 0          S1 Reloc fails             : 0

```



```

Only Relocations          : 0

ePDG attach success      : 0      ePDG attach failure      : 0
ePDG detach success      : 0      ePDG detach failure      : 0
ePDG reattach success    : 0      ePDG reattach failure    : 0
ePDG PDN Sess Del        : 0      ePDG PDN Sess Del Fail   : 0
ePDG NW Ded Bearer Actv   : 0      ePDG NW Ded Bearer Actv Fa*: 0
ePDG NW Ded Bearer Deactv : 0      ePDG NW Ded Bearer Deactv *: 0
ePDG QoS Modify          : 0      ePDG QoS Modify Fail     : 0
ePDG PCRF QoS Modify      : 0      ePDG PCRF QoS Modify Fail : 0
ePDG to LTE H0 success    : 0      ePDG to LTE H0 failure    : 0
LTE to ePDG H0 success    : 0      LTE to ePDG H0 failure    : 0
S2b NW Ded Bearer Actv    : 0      S2b NW Ded Bearer Actv Fail: 0
S2b NW Ded Bearer Deactv  : 0      S2b NW Ded Bearer Deactv F*: 0
S2b QoS Modify           : 0      S2b QoS Modify Fail      : 0
S2b PCRF QoS Modify       : 0      S2b PCRF QoS Modify Fail  : 0
ePDG MOBIKE success      : 0      ePDG MOBIKE failure      : 0
ePDG attach success CERTREQ: 0
ePDG Attach Requests over PGW-C+SMF

Initial Attach Success    : 0      Initial Attach Failure    : 0
H0 Attach Success         : 0      H0 Attach Failure         : 0
Imsi-Privacy Unenc attaches: 0      Imsi-Privacy Unencrypted H0: 0
Imsi-Privacy Encr attaches : 0      Imsi-Privacy Encrypted H0 : 0
UE idle error ind        : 0      Session deleted error ind : 0
Bearer deleted error ind  : 0
LTE/ePDG H0 Fail no-mip6  : 0      LTE/ePDG H0 Fail resp-cc64 : 0
ASR alt peer session del  : 0

TWAG portal access (Open SSID)

DORA attach              : 0      DORA attach failure      : 0
Rapid commit attach      : 0      Rapid commit attach failure: 0
DHCP lease-time expire   : 0      NW init detach           : 0
DHCP release             : 0      Handover Failure         : 0
CoA access connect       : 0      CoA access disconnect    : 0
Inter AP H0 data trigger : 0      Inter AP H0 DHCP         : 0
DHCP hold timeout        : 0
SLAAC attach             : 0      SLAAC attach failure     : 0
Portal to EAP transition  : 0

TWAG eap access (Closed SSID)

DORA attach              : 0      DORA attach failure      : 0
Rapid commit attach      : 0      Rapid commit attach failure: 0
DHCP lease-time expire   : 0      Network init detach      : 0
DHCP release             : 0      Handover Failure         : 0
EAP access attach success : 0      EAP access attach failure : 0
Inter-AP H0 data trigger : 0      Inter-AP H0 DHCP         : 0
Re-auth from new AP      : 0      ACCT from new AP         : 0
DHCP hold timeout        : 0
SLAAC attach             : 0      SLAAC attach failure     : 0
EAP to Portal transition  : 0
Fast-Inter-AP H0 Re-auth : 0      Fast-Inter-AP H0 ACCT    : 0
PMK Cache Timeout        : 0

TDF attach success       : 0      TDF attach failure       : 0
TDF detach success       : 0      TDF detach failure       : 0
TDF reattach success     : 0      TDF reattach failure     : 0

```

```

E-SGW Restoration
PDN connections restored      : 0          PDN connections deleted      : 0

S2a attach success           : 0          S2a attach failure             : 0
S2a detach success           : 0          S2a detach failure             : 0
S2a/LTE handover success     : 0          S2a/LTE handover failure       : 0
LTE/S2a handover success     : 0          LTE/S2a handover failure       : 0
S2a PGW PDN Sess Del         : 0          S2a PGW PDN Sess Del Fail     : 0
S2a/UMTS handover success    : 0          S2a/UMTS handover failure     : 0
UMTS/S2a handover success    : 0          UMTS/S2a handover failure     : 0
S2a/S2a handover success     : 0          S2a/S2a handover failure     : 0

S2a NW Ded Bearer Actv       : 0          S2a NW Ded Bearer Actv Fail: 0
S2a NW Ded Bearer Deactv     : 0          S2a NW Ded Bearer Deactv F*: 0
S2a QoS Modify               : 0          S2a QoS Modify Fail           : 0
S2a PCRF QoS Modify          : 0          S2a PCRF QoS Modify Fail     : 0

Session cleaned setup guar*: 0

S11-u tunnel activations     : 0          S11-u tunnel deactivations    : 0
S11-u to S1-u transitions    : 0          S1-u to S11-u transitions     : 0

Unsuccessful Emergency Cal*: 0          Successful Emergency Calls    : 0
Unsuccessful Emergency Calls due to
User Auth rejection          : 0          Issues on S2b interface       : 0
Issues on DNS res Emer APN   : 0          Missing Emergency Configur*: 0
Issues on DNS res non-Emer   : 0
Emergency Calls detected due to
IDr value in Emergency APN*: 0

ePDG Emergency Call Requests
Unauthenticated Init Served: 0          Unauthenticated HO Served    : 0
Unauthenticated Init Rej   : 0          Unauthenticated HO Rejected: 0
Uicc-less Initial Success  : 0          Uicc-less Handover Success   : 0
Uicc-less Initial Rejected : 0          Uicc-less Handover Rejected: 0

PFCP
Association                  : 1          Association Fail              : 1
Association Upd               : 0          Association Upd Fail          : 0
Association Release           : 0          Association Release Fail      : 0
Node Report                   : 0          Node Report Fail              : 0
-----
Number of VMs : 2
=====
* indicates that the corresponding row element may have been truncated.

```

Example output PDN gateway call flow card summary statistics

```

# show mobile-gateway pdn call-flow-stats card 3 summary
=====
PDN gateway call flow statistics summary
=====

Attach                : 1          Attach Fail            : 0
Detach                : 0          Detach Fail            : 0
NW Ded Bearer Actv    : 1          NW Ded Bearer Actv Fail : 0

```

```

NW Ded Bearer Deactiv      : 0      NW Ded Bearer Deactiv Fail : 0
NW Ded Bearer Modify       : 0      NW Ded Bearer Modify Fail  : 0
UE Ded Bearer Activ        : 0      UE Ded Bearer Activ Fail   : 0
UE Ded Bearer Deactiv      : 0      UE Ded Bearer Deactiv Fail : 0
UE Ded Bearer Modify       : 0      UE Ded Bearer Modify Fail  : 0
HSS QoS Modify             : 0      HSS QoS Modify Fail       : 0
PCRF QoS Modify            : 0      PCRF QoS Modify Fail      : 0
SGSN/SGW Reloc            : 0      SGSN/SGW Reloc Fail      : 0
PGW/GGSN PDN Sess Del     : 0      PGW/GGSN PDN Sess Del Fail : 0
Attach Piggy Backing       : 0
eHRPD attach success       : 0      eHRPD attach failure      : 0
eHRPD detach success       : 0      eHRPD detach failure      : 0
eHRPD/LTE handover success : 0      eHRPD/LTE handover failure : 0
LTE/eHRPD handover success : 0      LTE/eHRPD handover failure : 0
Inter HSGW handover success: 0      Inter HSGW handover failure: 0
PDN suspend notifications  : 0      PDN resume notifications  : 0
Implicit Resume Service Re*: 0      Emergency Attach Proc succ*: 0

```

Number of cards : 1

=====

* indicates that the corresponding row element may have been truncated.

Example output call flow statistics serving gateway

```

# show mobile-gateway pdn call-flow-stats sgw
=====
PDN gateway call flow statistics
=====
Chassis/Card      : 1/A      Gateway      : 1

Attach            : 0      Attach Fail   : 0
Detach            : 0      Detach Fail   : 0
NW Ded Bearer Activ      : 0      NW Ded Bearer Activ Fail   : 0
NW Ded Bearer Deactiv    : 0      NW Ded Bearer Deactiv Fail : 0
UE Ded Bearer Activ      : 0      UE Ded Bearer Activ Fail   : 0
UE Ded Bearer Deactiv    : 0      UE Ded Bearer Deactiv Fail : 0
UE Ded Bearer Modify     : 0      UE Ded Bearer Modify Fail  : 0
MME Ded Bearer Deact     : 0      MME Ded Bearer Deact Fail  : 0
MS Bearer Modify        : 0      MS Bearer Modify Fail      : 0
HSS QoS Modify          : 0      HSS QoS Modify Fail       : 0
PCRF QoS Modify          : 0      PCRF QoS Modify Fail      : 0
SGSN/SGW Reloc          : 0      SGSN/SGW Reloc Fail      : 0
PGW/GGSN PDN Sess Del   : 0      PGW/GGSN PDN Sess Del Fail : 0
Attach Piggy Backing     : 0
eHRPD attach success     : 0      eHRPD attach failure      : 0
eHRPD detach success     : 0      eHRPD detach failure      : 0
eHRPD/LTE handover success : 0      eHRPD/LTE handover failure : 0
LTE/eHRPD handover success : 0      LTE/eHRPD handover failure : 0
Inter HSGW handover success: 0      Inter HSGW handover failure: 0
PDN suspend notifications : 0      PDN resume notifications  : 0
Implicit Resume Service Re*: 0      Emergency Attach Proc succ*: 0
Reattach Success        : 0      Reattach Fail   : 0
eHRPD Reattach Success   : 0      eHRPD Reattach Fail : 0
HRPD attach success      : 0      HRPD attach failure : 0
HRPD detach success      : 0      HRPD detach failure : 0
Inter PDSN handover success: 0      Inter PDSN handover failure: 0

```

```

eMPS IMS session Actv      : 0          eMPS IMS session Deactv    : 0
eMPS IMS Adv Pri Ses Actv  : 0          eMPS IMS Adv Pri Ses Deactv: 0
eMPS Ded Bearer Actv      : 0          eMPS Ded Bearer Deactv    : 0

PGW/GGSN initiated 3GPP session deactivations
Due to session-timeout     : 0          Due to idle-timeout       : 0

PGW initiated trusted-non-3GPP session deactivations
Due to session-timeout     : 0          Due to idle-timeout       : 0

PGW initiated S2b non-3GPP session deactivations
Due to session-timeout     : 0          Due to idle-timeout       : 0

PGW/GGSN initiated sd interface session deactivations
Due to session-timeout     : 0          Due to idle-timeout       : 0

Inter RAT handover
3g to EUTRAN success       : 0          3g to EUTRAN failure     : 0
EUTRAN to 3g success       : 0          EUTRAN to 3g failure     : 0
S2b attach success         : 0          S2b attach failure       : 0
S2b detach success         : 0          S2b detach failure       : 0
S2b/LTE handover success   : 0          S2b/LTE handover failure : 0
LTE/S2b handover success   : 0          LTE/S2b handover failure : 0
S2b PGW PDN Sess Del       : 0          S2b PGW PDN Sess Del Fail : 0
S2b/UMTS handover success  : 0          S2b/UMTS handover failure : 0
UMTS/S2b handover success  : 0          UMTS/S2b handover failure : 0
S2b/S2b handover success   : 0          S2b/S2b handover failure : 0
S2b/eHRPD handover success : 0          S2b/eHRPD handover failure : 0
eHRPD/S2b handover success : 0          eHRPD/S2b handover failure : 0

Network Requested Bearer Deactivation
GTP S2b Success            : 0          GTP S2b Failure          : 0
S1 release procedures       : 0          S1 release failures       : 0
UE init service req         : 0          UE init svc req failures  : 0
Intra SGW idle mode TAU     : 0          Paging timeout expiration : 0
Inter SGW idle mode TAU     : 0          Inter SGW idle TAU fails  : 0
Inter SGW S1 HO with InTnl  : 0          S1 HO with Ind Tunnel fails: 0
Inter SGW S1 HO w/o InTnl   : 0          S1 HO w/o Ind Tunnel fails : 0
Inter SGW X2 handovers      : 0          Inter SGW X2 HO fails     : 0
Inter SGW relocation (Out)  : 0          Intra SGW idle TAU fails  : 0
eHRPD to LTE handovers     : 0          eHRPD to LTE HO fails     : 0
PDN Sess De-activation      : 0          PDN Sess De-act fails     : 0
Paging Attempts            : 0          Paging fails              : 0
Intra SGW X2+S1 HOs         : 0          Intra SGW X2+S1 HO fails  : 0
Intra SGW S1 handvr InTnl   : 0          S1 handvr Ind Tunnel fails : 0
Attach Proc Fail Piggy Back : 0
NW init bearer modify       : 0          NW init bearer modify fail : 0

Inter MME/S4-SGSN
Idle TAU                    : 0          Idle TAU fails            : 0
X2+S1 Reloc                 : 0          X2+S1 Reloc fails         : 0
S1 Reloc Ind Tnl            : 0          S1 Reloc fails            : 0
Only Relocations            : 0

ePDG attach success         : 0          ePDG attach failure       : 0
ePDG detach success         : 0          ePDG detach failure       : 0
ePDG reattach success       : 0          ePDG reattach failure     : 0

```

ePDG PDN Sess Del	: 0	ePDG PDN Sess Del Fail	: 0
ePDG NW Ded Bearer Actv	: 0	ePDG NW Ded Bearer Actv Fa*	: 0
ePDG NW Ded Bearer Deactv	: 0	ePDG NW Ded Bearer Deactv *	: 0
ePDG QoS Modify	: 0	ePDG QoS Modify Fail	: 0
ePDG PCRF QoS Modify	: 0	ePDG PCRF QoS Modify Fail	: 0
ePDG to LTE H0 success	: 0	ePDG to LTE H0 failure	: 0
LTE to ePDG H0 success	: 0	LTE to ePDG H0 failure	: 0
S2b NW Ded Bearer Actv	: 0	S2b NW Ded Bearer Actv Fail	: 0
S2b NW Ded Bearer Deactv	: 0	S2b NW Ded Bearer Deactv F*	: 0
S2b QoS Modify	: 0	S2b QoS Modify Fail	: 0
S2b PCRF QoS Modify	: 0	S2b PCRF QoS Modify Fail	: 0
ePDG MOBIKE success	: 0	ePDG MOBIKE failure	: 0
Imsi-Privacy Unenc attaches	: 0	Imsi-Privacy Unencrypted H0	: 0
Imsi-Privacy Encr attaches	: 0	Imsi-Privacy Encrypted H0	: 0
UE idle error ind	: 0	Session deleted error ind	: 0
Bearer deleted error ind	: 0		
LTE/ePDG H0 Fail no-mip6	: 0	LTE/ePDG H0 Fail resp-cc64	: 0
ASR alt peer session del	: 0		
TWAG portal access (Open SSID)			
DORA attach	: 0	DORA attach failure	: 0
Rapid commit attach	: 0	Rapid commit attach failure	: 0
DHCP lease-time expire	: 0	NW init detach	: 0
DHCP release	: 0	Handover Failure	: 0
CoA access connect	: 0	CoA access disconnect	: 0
Inter AP H0 data trigger	: 0	Inter AP H0 DHCP	: 0
DHCP hold timeout	: 0		
SLAAC attach	: 0	SLAAC attach failure	: 0
Portal to EAP transition	: 0		
TWAG eap access (Closed SSID)			
DORA attach	: 0	DORA attach failure	: 0
Rapid commit attach	: 0	Rapid commit attach failure	: 0
DHCP lease-time expire	: 0	Network init detach	: 0
DHCP release	: 0	Handover Failure	: 0
EAP access attach success	: 0	EAP access attach failure	: 0
Inter-AP H0 data trigger	: 0	Inter-AP H0 DHCP	: 0
Re-auth from new AP	: 0	ACCT from new AP	: 0
DHCP hold timeout	: 0		
SLAAC attach	: 0	SLAAC attach failure	: 0
EAP to Portal transition	: 0		
Fast-Inter-AP H0 Re-auth	: 0	Fast-Inter-AP H0 ACCT	: 0
PMK Cache Timeout	: 0		
TDF attach success			
TDF detach success	: 0	TDF attach failure	: 0
TDF reattach success	: 0	TDF detach failure	: 0
		TDF reattach failure	: 0
E-SGW Restoration			
PDN connections restored	: 0	PDN connections deleted	: 0
S2a attach success			
S2a detach success	: 0	S2a attach failure	: 0
S2a/LTE handover success	: 0	S2a detach failure	: 0
LTE/S2a handover success	: 0	S2a/LTE handover failure	: 0
S2a PGW PDN Sess Del	: 0	LTE/S2a handover failure	: 0
S2a/UMTS handover success	: 0	S2a PGW PDN Sess Del Fail	: 0
		S2a/UMTS handover failure	: 0

```

UMTS/S2a handover success : 0          UMTS/S2a handover failure : 0
S2a/S2a handover success  : 0          S2a/S2a handover failure  : 0

S2a NW Ded Bearer Actv    : 0          S2a NW Ded Bearer Actv Fail: 0
S2a NW Ded Bearer Deactv  : 0          S2a NW Ded Bearer Deactv F*: 0
S2a QoS Modify            : 0          S2a QoS Modify Fail       : 0
S2a PCRF QoS Modify       : 0          S2a PCRF QoS Modify Fail  : 0

Session cleaned setup guar*: 0

S11-u tunnel activations  : 0          S11-u tunnel deactivations : 0
S11-u to S11-u transitions : 0          S11-u to S11-u transitions : 0

Unsuccessful Emergency Cal*: 0          Successful Emergency Calls : 0
Unsuccessful Emergency Calls due to
User Auth rejection       : 0          Issues on S2b interface    : 0
Issues on DNS res Emer APN : 0          Missing Emergency Configur*: 0
Issues on DNS res non-Emer : 0
Emergency Calls detected due to
IDr value in Emergency APN*: 0

ePDG Emergency Call Requests
Unauthenticated Init Served: 0          Unauthenticated HO Served  : 0
Unauthenticated Init Rej   : 0          Unauthenticated HO Rejected: 0
Uicc-less Initial Success  : 0          Uicc-less Handover Success : 0
Uicc-less Initial Rejected : 0          Uicc-less Handover Rejected: 0
-----
Number of cards : 1
=====
* indicates that the corresponding row element may have been truncated.

```

Example output call flow statistics for eMPS advance priority calls

```

# show mobile-gateway pdn call-flow-stats
=====
PDN gateway call flow statistics
=====
VNF/VM          : 1/3          Gateway          : 1

Attach          : 0          Attach Fail      : 0
Detach          : 0          Detach Fail      : 0
NW Ded Bearer Actv : 0          NW Ded Bearer Actv Fail : 0
NW Ded Bearer Deactv : 0          NW Ded Bearer Deactv Fail : 0
UE Ded Bearer Actv : 0          UE Ded Bearer Actv Fail : 0
UE Ded Bearer Deactv : 0          UE Ded Bearer Deactv Fail : 0
UE Ded Bearer Modify : 0          UE Ded Bearer Modify Fail : 0
MME Ded Bearer Deact : 0          MME Ded Bearer Deact Fail : 0
MS Bearer Modify : 0          MS Bearer Modify Fail : 0
HSS QoS Modify   : 0          HSS QoS Modify Fail : 0
PCRF QoS Modify  : 0          PCRF QoS Modify Fail : 0
SGSN/SGW Reloc   : 0          SGSN/SGW Reloc Fail : 0
PGW/GGSN PDN Sess Del : 0          PGW/GGSN PDN Sess Del Fail : 0
Attach Piggy Backing : 0
eHRPD attach success : 0          eHRPD attach failure : 0
eHRPD detach success : 0          eHRPD detach failure : 0
eHRPD/LTE handover success : 0          eHRPD/LTE handover failure : 0

```

```

LTE/eHRPD handover success : 0          LTE/eHRPD handover failure : 0
Inter HSGW handover success: 0          Inter HSGW handover failure: 0
PDN suspend notifications  : 0          PDN resume notifications   : 0
Implicit Resume Service Re*: 0          Emergency Attach Proc succ*: 0
Reattach Success           : 0          Reattach Fail              : 0
eHRPD Reattach Success     : 0          eHRPD Reattach Fail        : 0
HRPD attach success        : 0          HRPD attach failure        : 0
HRPD detach success        : 0          HRPD detach failure        : 0
Inter PDSN handover success: 0          Inter PDSN handover failure: 0
eMPS IMS session Actv      : 0          eMPS IMS session Deactv    : 0
eMPS IMS Adv Pri Ses Actv  : 0          eMPS IMS Adv Pri Ses Deactv: 0
eMPS Ded Bearer Actv       : 0          eMPS Ded Bearer Deactv     : 0
5GC/EPC(PGW ses) HO Sc    : 0          5GC/EPC(PGW ses) HO Fl    : 0
5GC/EPC(Cmb ses) HO Sc    : 0          5GC/EPC(Cmb ses) HO Fl    : 0
5GC/EPC(PGW ses) Idl Mob Sc: 0          5GC/EPC(PGW ses) Idl Mob Fl: 0
5GC/EPC(Cmb ses) Idl Mob Sc: 0          5GC/EPC(Cmb ses) Idl Mob Fl: 0

PGW/GGSN initiated 3GPP session deactivations
Due to session-timeout      : 0          Due to idle-timeout        : 0

PGW initiated trusted-non-3GPP session deactivations
Due to session-timeout      : 0          Due to idle-timeout        : 0

PGW initiated S2b non-3GPP session deactivations
Due to session-timeout      : 0          Due to idle-timeout        : 0

PGW/GGSN initiated sd interface session deactivations
Due to session-timeout      : 0          Due to idle-timeout        : 0

Inter RAT handover
3g to EUTRAN success        : 0          3g to EUTRAN failure      : 0
EUTRAN to 3g success        : 0          EUTRAN to 3g failure      : 0
S2b attach success          : 0          S2b attach failure        : 0
S2b detach success          : 0          S2b detach failure        : 0
S2b/LTE handover success    : 0          S2b/LTE handover failure  : 0
LTE/S2b handover success    : 0          LTE/S2b handover failure  : 0
S2b PGW PDN Sess Del        : 0          S2b PGW PDN Sess Del Fail : 0
S2b/UMTS handover success   : 0          S2b/UMTS handover failure : 0
UMTS/S2b handover success   : 0          UMTS/S2b handover failure : 0
S2b/S2b handover success    : 0          S2b/S2b handover failure  : 0
S2b/eHRPD handover success  : 0          S2b/eHRPD handover failure: 0
eHRPD/S2b handover success  : 0          eHRPD/S2b handover failure: 0

Network Requested Bearer Deactivation
GTP S2b Success             : 0          GTP S2b Failure           : 0
S1 release procedures        : 0          S1 release failures        : 0
UE init service req          : 0          UE init svc req failures   : 0
Intra SGW idle mode TAU      : 0          Paging timeout expiration  : 0
Inter SGW idle mode TAU      : 0          Inter SGW idle TAU fails   : 0
Inter SGW S1 HO with InTnl   : 0          S1 HO with Ind Tunnel fails: 0
Inter SGW S1 HO w/o InTnl    : 0          S1 HO w/o Ind Tunnel fails : 0
Inter SGW X2 handovers       : 0          Inter SGW X2 HO fails      : 0
Inter SGW relocation (Out)    : 0          Intra SGW idle TAU fails   : 0
eHRPD to LTE handovers       : 0          eHRPD to LTE HO fails      : 0
PDN Sess De-activation       : 0          PDN Sess De-act fails      : 0
Paging Attempts              : 0          Paging fails                : 0
Intra SGW X2+S1 HOs          : 0          Intra SGW X2+S1 HO fails   : 0

```

Intra SGW S1 handvr InTnl : 0	S1 handvr Ind Tunnel fails : 0
Attach Proc Fail Piggy Back: 0	
NW init bearer modify : 0	NW init bearer modify fail : 0
Inter MME/S4-SGSN	
Idle TAU : 0	Idle TAU fails : 0
X2+S1 Reloc : 0	X2+S1 Reloc fails : 0
S1 Reloc Ind Tnl : 0	S1 Reloc fails : 0
Only Relocations : 0	
ePDG attach success : 0	
ePDG detach success : 0	ePDG detach failure : 0
ePDG reattach success : 0	ePDG reattach failure : 0
ePDG PDN Sess Del : 0	ePDG PDN Sess Del Fail : 0
ePDG NW Ded Bearer Actv : 0	ePDG NW Ded Bearer Actv Fa*: 0
ePDG NW Ded Bearer Deactv : 0	ePDG NW Ded Bearer Deactv *: 0
ePDG QoS Modify : 0	ePDG QoS Modify Fail : 0
ePDG PCRF QoS Modify : 0	ePDG PCRF QoS Modify Fail : 0
ePDG to LTE H0 success : 0	ePDG to LTE H0 failure : 0
LTE to ePDG H0 success : 0	LTE to ePDG H0 failure : 0
S2b NW Ded Bearer Actv : 0	S2b NW Ded Bearer Actv Fail: 0
S2b NW Ded Bearer Deactv : 0	S2b NW Ded Bearer Deactv F*: 0
S2b QoS Modify : 0	S2b QoS Modify Fail : 0
S2b PCRF QoS Modify : 0	S2b PCRF QoS Modify Fail : 0
ePDG MOBIKE success : 0	ePDG MOBIKE failure : 0
ePDG attach success CERTREQ: 0	
ePDG Attach Requests over PGW-C+SMF	
Initial Attach Success : 0	Initial Attach Failure : 0
H0 Attach Success : 0	H0 Attach Failure : 0
Imsi-Privacy Unenc attaches: 0	Imsi-Privacy Unencrypted H0: 0
Imsi-Privacy Encr attaches : 0	Imsi-Privacy Encrypted H0 : 0
UE idle error ind : 0	Session deleted error ind : 0
Bearer deleted error ind : 0	
LTE/ePDG H0 Fail no-mip6 : 0	LTE/ePDG H0 Fail resp-cc64 : 0
ASR alt peer session del : 0	
TWAG portal access (Open SSID)	
DORA attach : 0	DORA attach failure : 0
Rapid commit attach : 0	Rapid commit attach failure: 0
DHCP lease-time expire : 0	NW init detach : 0
DHCP release : 0	Handover Failure : 0
CoA access connect : 0	CoA access disconnect : 0
Inter AP H0 data trigger : 0	Inter AP H0 DHCP : 0
DHCP hold timeout : 0	
SLAAC attach : 0	SLAAC attach failure : 0
Portal to EAP transition : 0	
TWAG eap access (Closed SSID)	
DORA attach : 0	DORA attach failure : 0
Rapid commit attach : 0	Rapid commit attach failure: 0
DHCP lease-time expire : 0	Network init detach : 0
DHCP release : 0	Handover Failure : 0
EAP access attach success : 0	EAP access attach failure : 0
Inter-AP H0 data trigger : 0	Inter-AP H0 DHCP : 0
Re-auth from new AP : 0	ACCT from new AP : 0
DHCP hold timeout : 0	
SLAAC attach : 0	SLAAC attach failure : 0

EAP to Portal transition	: 0		
Fast-Inter-AP HO Re-auth	: 0	Fast-Inter-AP HO ACCT	: 0
PMK Cache Timeout	: 0		
TDF attach success	: 0	TDF attach failure	: 0
TDF detach success	: 0	TDF detach failure	: 0
TDF reattach success	: 0	TDF reattach failure	: 0
E-SGW Restoration			
PDN connections restored	: 0	PDN connections deleted	: 0
S2a attach success	: 0	S2a attach failure	: 0
S2a detach success	: 0	S2a detach failure	: 0
S2a/LTE handover success	: 0	S2a/LTE handover failure	: 0
LTE/S2a handover success	: 0	LTE/S2a handover failure	: 0
S2a PGW PDN Sess Del	: 0	S2a PGW PDN Sess Del Fail	: 0
S2a/UMTS handover success	: 0	S2a/UMTS handover failure	: 0
UMTS/S2a handover success	: 0	UMTS/S2a handover failure	: 0
S2a/S2a handover success	: 0	S2a/S2a handover failure	: 0
S2a NW Ded Bearer Actv	: 0	S2a NW Ded Bearer Actv Fail	: 0
S2a NW Ded Bearer Deactv	: 0	S2a NW Ded Bearer Deactv F*	: 0
S2a QoS Modify	: 0	S2a QoS Modify Fail	: 0
S2a PCRF QoS Modify	: 0	S2a PCRF QoS Modify Fail	: 0
Session cleaned setup guar*	: 0		
S11-u tunnel activations	: 0	S11-u tunnel deactivations	: 0
S11-u to S1-u transitions	: 0	S1-u to S11-u transitions	: 0
Unsuccessful Emergency Cal*	: 0	Successful Emergency Calls	: 0
Unsuccessful Emergency Calls due to			
User Auth rejection	: 0	Issues on S2b interface	: 0
Issues on DNS res Emer APN	: 0	Missing Emergency Configur*	: 0
Issues on DNS res non-Emer	: 0		
Emergency Calls detected due to			
IDr value in Emergency APN*	: 0		
ePDG Emergency Call Requests			
Unauthenticated Init Served	: 0	Unauthenticated HO Served	: 0
Unauthenticated Init Rej	: 0	Unauthenticated HO Rejected	: 0
Uicc-less Initial Success	: 0	Uicc-less Handover Success	: 0
Uicc-less Initial Rejected	: 0	Uicc-less Handover Rejected	: 0
PFCP			
Session Establishment	: 1	Session Establishment Fail	: 0
Session Modification	: 0	Session Modification Fail	: 0
Session Deletion	: 0	Session Deletion Fail	: 0
Session Report	: 0	Session Report Fail	: 0
Association	: 0	Association Fail	: 0
Association Upd	: 0	Association Upd Fail	: 0
Association Release	: 0	Association Release Fail	: 0
Node Report	: 0	Node Report Fail	: 0

VNF/VM	: 1/A	Gateway	: 1
Attach	: 0	Attach Fail	: 0

Detach	: 0	Detach Fail	: 0
NW Ded Bearer Actv	: 0	NW Ded Bearer Actv Fail	: 0
NW Ded Bearer Deactv	: 0	NW Ded Bearer Deactv Fail	: 0
UE Ded Bearer Actv	: 0	UE Ded Bearer Actv Fail	: 0
UE Ded Bearer Deactv	: 0	UE Ded Bearer Deactv Fail	: 0
UE Ded Bearer Modify	: 0	UE Ded Bearer Modify Fail	: 0
MME Ded Bearer Deact	: 0	MME Ded Bearer Deact Fail	: 0
MS Bearer Modify	: 0	MS Bearer Modify Fail	: 0
HSS QoS Modify	: 0	HSS QoS Modify Fail	: 0
PCRF QoS Modify	: 0	PCRF QoS Modify Fail	: 0
SGSN/SGW Reloc	: 0	SGSN/SGW Reloc Fail	: 0
PGW/GGSN PDN Sess Del	: 0	PGW/GGSN PDN Sess Del Fail	: 0
Attach Piggy Backing	: 0		
eHRPD attach success	: 0	eHRPD attach failure	: 0
eHRPD detach success	: 0	eHRPD detach failure	: 0
eHRPD/LTE handover success	: 0	eHRPD/LTE handover failure	: 0
LTE/eHRPD handover success	: 0	LTE/eHRPD handover failure	: 0
Inter HSGW handover success	: 0	Inter HSGW handover failure	: 0
PDN suspend notifications	: 0	PDN resume notifications	: 0
Implicit Resume Service Re*	: 0	Emergency Attach Proc succ*	: 0
Reattach Success	: 0	Reattach Fail	: 0
eHRPD Reattach Success	: 0	eHRPD Reattach Fail	: 0
HRPD attach success	: 0	HRPD attach failure	: 0
HRPD detach success	: 0	HRPD detach failure	: 0
Inter PDSN handover success	: 0	Inter PDSN handover failure	: 0
eMPS IMS session Actv	: 0	eMPS IMS session Deactv	: 0
eMPS IMS Adv Pri Ses Actv	: 1	eMPS IMS Adv Pri Ses Deactv	: 1
eMPS Ded Bearer Actv	: 0	eMPS Ded Bearer Deactv	: 0
5GC/EPC(PGW ses) HO Sc	: 0	5GC/EPC(PGW ses) HO Fl	: 0
5GC/EPC(Cmb ses) HO Sc	: 0	5GC/EPC(Cmb ses) HO Fl	: 0
5GC/EPC(PGW ses) Idl Mob Sc	: 0	5GC/EPC(PGW ses) Idl Mob Fl	: 0
5GC/EPC(Cmb ses) Idl Mob Sc	: 0	5GC/EPC(Cmb ses) Idl Mob Fl	: 0
PGW/GGSN initiated 3GPP session deactivations			
Due to session-timeout	: 0	Due to idle-timeout	: 0
PGW initiated trusted-non-3GPP session deactivations			
Due to session-timeout	: 0	Due to idle-timeout	: 0
PGW initiated S2b non-3GPP session deactivations			
Due to session-timeout	: 0	Due to idle-timeout	: 0
PGW/GGSN initiated sd interface session deactivations			
Due to session-timeout	: 0	Due to idle-timeout	: 0
Inter RAT handover			
3g to EUTRAN success	: 0	3g to EUTRAN failure	: 0
EUTRAN to 3g success	: 0	EUTRAN to 3g failure	: 0
S2b attach success	: 0	S2b attach failure	: 0
S2b detach success	: 0	S2b detach failure	: 0
S2b/LTE handover success	: 0	S2b/LTE handover failure	: 0
LTE/S2b handover success	: 0	LTE/S2b handover failure	: 0
S2b PGW PDN Sess Del	: 0	S2b PGW PDN Sess Del Fail	: 0
S2b/UMTS handover success	: 0	S2b/UMTS handover failure	: 0
UMTS/S2b handover success	: 0	UMTS/S2b handover failure	: 0
S2b/S2b handover success	: 0	S2b/S2b handover failure	: 0
S2b/eHRPD handover success	: 0	S2b/eHRPD handover failure	: 0

```

eHRPD/S2b handover success : 0          eHRPD/S2b handover failure : 0

Network Requested Bearer Deactivation
GTP S2b Success           : 0          GTP S2b Failure           : 0
S1 release procedures     : 0          S1 release failures       : 0
UE init service req       : 0          UE init svc req failures  : 0
Intra SGW idle mode TAU   : 0          Paging timeout expiration : 0
Inter SGW idle mode TAU   : 0          Inter SGW idle TAU fails  : 0
Inter SGW S1 HO with InTnl : 0        S1 HO with Ind Tunnel fails: 0
Inter SGW S1 HO w/o InTnl : 0          S1 HO w/o Ind Tunnel fails : 0
Inter SGW X2 handovers    : 0          Inter SGW X2 HO fails     : 0
Inter SGW relocation (Out) : 0         Intra SGW idle TAU fails  : 0
eHRPD to LTE handovers    : 0          eHRPD to LTE HO fails     : 0
PDN Sess De-activation    : 0          PDN Sess De-act fails     : 0
Paging Attempts          : 0          Paging fails              : 0
Intra SGW X2+S1 HOs       : 0          Intra SGW X2+S1 HO fails  : 0
Intra SGW S1 handvr InTnl : 0          S1 handvr Ind Tunnel fails : 0
Attach Proc Fail Piggy Back: 0
NW init bearer modify     : 0          NW init bearer modify fail : 0

Inter MME/S4-SGSN
Idle TAU                  : 0          Idle TAU fails            : 0
X2+S1 Reloc               : 0          X2+S1 Reloc fails        : 0
S1 Reloc Ind Tnl          : 0          S1 Reloc fails           : 0
Only Relocations          : 0

ePDG attach success       : 0          ePDG attach failure       : 0
ePDG detach success       : 0          ePDG detach failure       : 0
ePDG reattach success     : 0          ePDG reattach failure     : 0
ePDG PDN Sess Del         : 0          ePDG PDN Sess Del Fail    : 0
ePDG NW Ded Bearer Actv   : 0          ePDG NW Ded Bearer Actv Fa*: 0
ePDG NW Ded Bearer Deactv : 0         ePDG NW Ded Bearer Deactv *: 0
ePDG QoS Modify           : 0          ePDG QoS Modify Fail     : 0
ePDG PCRF QoS Modify      : 0          ePDG PCRF QoS Modify Fail : 0
ePDG to LTE HO success    : 0          ePDG to LTE HO failure    : 0
LTE to ePDG HO success    : 0          LTE to ePDG HO failure    : 0
S2b NW Ded Bearer Actv    : 0          S2b NW Ded Bearer Actv Fail: 0
S2b NW Ded Bearer Deactv  : 0         S2b NW Ded Bearer Deactv F*: 0
S2b QoS Modify            : 0          S2b QoS Modify Fail      : 0
S2b PCRF QoS Modify       : 0          S2b PCRF QoS Modify Fail  : 0
ePDG MOBIKE success       : 0          ePDG MOBIKE failure       : 0
ePDG attach success CERTREQ: 0
ePDG Attach Requests over PGW-C+SMF
Initial Attach Success     : 0          Initial Attach Failure     : 0
HO Attach Success          : 0          HO Attach Failure          : 0
Imsi-Privacy Unenc attaches: 0         Imsi-Privacy Unencrypted HO: 0
Imsi-Privacy Encr attaches : 0         Imsi-Privacy Encrypted HO : 0
UE idle error ind         : 0          Session deleted error ind  : 0
Bearer deleted error ind   : 0
LTE/ePDG HO Fail no-mip6   : 0         LTE/ePDG HO Fail resp-cc64 : 0
ASR alt peer session del   : 0

TWAG portal access (Open SSID)
DORA attach               : 0          DORA attach failure        : 0
Rapid commit attach       : 0          Rapid commit attach failure: 0
DHCP lease-time expire    : 0          NW init detach             : 0
DHCP release              : 0          Handover Failure           : 0

```

CoA access connect	: 0	CoA access disconnect	: 0
Inter AP HO data trigger	: 0	Inter AP HO DHCP	: 0
DHCP hold timeout	: 0		
SLAAC attach	: 0	SLAAC attach failure	: 0
Portal to EAP transition	: 0		
TWAG eap access (Closed SSID)			
DORA attach	: 0	DORA attach failure	: 0
Rapid commit attach	: 0	Rapid commit attach failure:	0
DHCP lease-time expire	: 0	Network init detach	: 0
DHCP release	: 0	Handover Failure	: 0
EAP access attach success	: 0	EAP access attach failure	: 0
Inter-AP HO data trigger	: 0	Inter-AP HO DHCP	: 0
Re-auth from new AP	: 0	ACCT from new AP	: 0
DHCP hold timeout	: 0		
SLAAC attach	: 0	SLAAC attach failure	: 0
EAP to Portal transition	: 0		
Fast-Inter-AP HO Re-auth	: 0	Fast-Inter-AP HO ACCT	: 0
PMK Cache Timeout	: 0		
TDF attach success	: 0	TDF attach failure	: 0
TDF detach success	: 0	TDF detach failure	: 0
TDF reattach success	: 0	TDF reattach failure	: 0
E-SGW Restoration			
PDN connections restored	: 0	PDN connections deleted	: 0
S2a attach success	: 0	S2a attach failure	: 0
S2a detach success	: 0	S2a detach failure	: 0
S2a/LTE handover success	: 0	S2a/LTE handover failure	: 0
LTE/S2a handover success	: 0	LTE/S2a handover failure	: 0
S2a PGW PDN Sess Del	: 0	S2a PGW PDN Sess Del Fail	: 0
S2a/UMTS handover success	: 0	S2a/UMTS handover failure	: 0
UMTS/S2a handover success	: 0	UMTS/S2a handover failure	: 0
S2a/S2a handover success	: 0	S2a/S2a handover failure	: 0
S2a NW Ded Bearer Actv	: 0	S2a NW Ded Bearer Actv Fail:	0
S2a NW Ded Bearer Deactv	: 0	S2a NW Ded Bearer Deactv F*:	0
S2a QoS Modify	: 0	S2a QoS Modify Fail	: 0
S2a PCRF QoS Modify	: 0	S2a PCRF QoS Modify Fail	: 0
Session cleaned setup guar*: 0			
S11-u tunnel activations	: 0	S11-u tunnel deactivations	: 0
S11-u to S1-u transitions	: 0	S1-u to S11-u transitions	: 0
Unsuccessful Emergency Cal*:	0	Successful Emergency Calls	: 0
Unsuccessful Emergency Calls due to			
User Auth rejection	: 0	Issues on S2b interface	: 0
Issues on DNS res Emer APN	: 0	Missing Emergency Configur*:	0
Issues on DNS res non-Emer	: 0		
Emergency Calls detected due to			
IDr value in Emergency APN*:	0		
ePDG Emergency Call Requests			
Unauthenticated Init Served:	0	Unauthenticated HO Served	: 0
Unauthenticated Init Rej	: 0	Unauthenticated HO Rejected:	0

```
Uicc-less Initial Success : 0      Uicc-less Handover Success : 0
Uicc-less Initial Rejected : 0    Uicc-less Handover Rejected: 0
```

PFCP

```
Session Establishment      : 0      Session Establishment Fail : 0
Session Modification       : 0      Session Modification Fail  : 0
Session Deletion          : 0      Session Deletion Fail     : 0
Session Report            : 0      Session Report Fail       : 0
Association                : 1      Association Fail           : 1
Association Upd            : 0      Association Upd Fail      : 0
Association Release        : 0      Association Release Fail  : 0
Node Report               : 0      Node Report Fail         : 0
```

```
-----
Number of VMs : 2
=====
```

Example output call flow statistics OI override mapping

```
# show mobile-gateway pdn call-flow-stats OI-override-mapping
=====
PDN gateway call flow statistics - OI-Override-Mapping
=====
-----
Gateway: 1          List Name: override1
-----
Subscriber-PLMN  Number of OI Replacements  Number of Serv Network Replacements
-----
MCC=100 MNC=200      10                5
MCC=110 MNC=210      20                8
-----
-----
Gateway: 1          List Name: test0
-----
Subscriber-PLMN  Number of OI Replacements  Number of Serv Network Replacements
-----
MCC=120 MNC=220      40                5
-----
-----
Gateway: 2          List Name: test0
-----
Subscriber-PLMN  Number of OI Replacements  Number of Serv Network Replacements
-----
MCC=200 MNC=300      30                8
-----
=====
```

Example output call flow UPF statistics

```
# show mobile-gateway pdn call-flow-stats gw-type upf
=====
UPF gateway call flow statistics
=====
VNF/VM          : 1/3      Gateway          : 1

PFCP
Session Establishment : 1      Session Establishment Fail : 0
```

```

Session Modification      : 0      Session Modification Fail : 0
Session Deletion         : 0      Session Deletion Fail    : 0
Session Report           : 0      Session Report Fail      : 0
Association               : 0      Association Fail          : 0
Association Upd           : 0      Association Upd Fail     : 0
Association Release       : 0      Association Release Fail  : 0
Node Report              : 0      Node Report Fail         : 0
-----

```

```

VNF/VM                   : 1/A      Gateway                      : 1

```

PFCP

```

Session Establishment    : 0      Session Establishment Fail : 0
Session Modification     : 0      Session Modification Fail  : 0
Session Deletion         : 0      Session Deletion Fail      : 0
Session Report           : 0      Session Report Fail        : 0
Association               : 1      Association Fail            : 1
Association Upd           : 0      Association Upd Fail       : 0
Association Release       : 0      Association Release Fail    : 0
Node Report              : 0      Node Report Fail           : 0
-----

```

```

Number of cards : 2
=====

```

Example output call flow NamfEventExposure service statistics

```

# show mobile-gateway pdn call-flow-stats
=====
SMF call flow statistics
=====
VNF/VM                   : 1/3      Gateway                      : 1
.....
AMF Create Event Subsc Succ: 0      AMF Create Event Subsc Fail: 0
AMF Update Event Subsc Succ: 0      AMF Update Event Subsc Fail: 0
AMF Delete Event Subsc Succ: 0      AMF Delete Event Subsc Fail: 0
AMF Event Notific Received : 0      AMF Event Subscr Expiration: 0

```

Example output call flow UDM service statistics

```

# show mobile-gateway pdn call-flow-stats
=====
SMF call flow statistics
=====
VNF/VM                   : 1/3      Gateway                      : 1
.....
UDM Init PDU Sess Rel Succ : 0      UDM Init PDU Sess Rel Fail : 0
.....
UDM Init PDU Sess Mod Succ : 0      UDM Init PDU Sess Mod Fail : 0
.....

Successful Emergency Handovers to 5GC with
No SUPI                  : 0      Unauthenticated SUPI       : 0
Authenticated SUPI       : 0

Unsuccessful Emergency Calls due to
.....
UDM Failure               : 0      Other Failure               : 0

```

```

Unsuccessful Emergency Call Handovers to 5GC due to
.....
UDM Failure           : 0           Other Failure           : 0

```

```

=====
SMF with I-SMF call flow statistics
=====
VNF/VM                : 1/3          Gateway                : 1
.....
UDM Init PDU Sess Mod Succ : 0          UDM Init PDU Sess Mod Fail : 0

```

```

=====
H-SMF call flow statistics
=====
VNF/VM                : 1/3          Gateway                : 1
.....
UDM Init PDU Sess Mod Succ : 0          UDM Init PDU Sess Mod Fail : 0

```

Example output call flow statistics for 5G sessions

```

# show mobile-gateway pdn call-flow-stats
=====
SMF call flow statistics
=====
VNF/VM                : 1/3          Gateway                : 1

Establish PDU Session Succ : 0          Establish PDU Session Fail : 0
Re-establish PDU Sess Succ : 0          Re-establish PDU Sess Fail : 0
UE Init PDU Sess Rel Succ  : 0          UE Init PDU Sess Rel Fail  : 0
NW Init PDU Sess Rel Succ  : 0          NW Init PDU Sess Rel Fail  : 0
UDM Init PDU Sess Rel Succ : 0          UDM Init PDU Sess Rel Fail : 0
AMF Init PDU Sess Rel Succ : 0          AMF Init PDU Sess Rel Fail : 0
UE Triggered Serv Req Succ : 0          UE Triggered Serv Req Fail : 0
NW Triggered Serv Req Succ : 0          NW Triggered Serv Req Fail : 0
AN Release                : 0
Xn Handover Success       : 0          Xn Handover Fail          : 0
N2 Handover Success       : 0          N2 Handover Fail          : 0
RAN Init PDU Sess Mod Succ : 0          RAN Init PDU Sess Mod Fail : 0
UE Init PDU Sess Mod Succ : 0          UE Init PDU Sess Mod Fail : 0
SMF Init PDU Sess Mod Succ : 0          SMF Init PDU Sess Mod Fail : 0
PCF Init PDU Sess Mod Succ : 0          PCF Init PDU Sess Mod Fail : 0
UDM Init PDU Sess Mod Succ : 0          UDM Init PDU Sess Mod Fail : 0
EPC(PGW ses)/5GC H0 Sc    : 0          EPC(PGW ses)/5GC H0 Fl    : 0
EPC(Cmb ses)/5GC H0 Sc    : 0          EPC(Cmb ses)/5GC H0 Fl    : 0
EPC(PGW ses)/5GC Idl Mob Sc : 0        EPC(PGW ses)/5GC Idl Mob Fl : 0
EPC(Cmb ses)/5GC Idl Mob Sc : 0        EPC(Cmb ses)/5GC Idl Mob Fl : 0
AMF Create Event Subsc Succ : 0        AMF Create Event Subsc Fail : 0
AMF Update Event Subsc Succ : 0        AMF Update Event Subsc Fail : 0
AMF Delete Event Subsc Succ : 0        AMF Delete Event Subsc Fail : 0
AMF Event Notific Received : 0          AMF Event Subscr Expiration : 0
PDU Sess Mod Rej IMS Flbck : 0
LADN PDU Sess Released    : 0
LADN UP Suspended         : 0          LADN UP Resumed           : 0

```

```

Successful Emergency Attaches with
No SUPI                : 0          Unauthenticated SUPI      : 0
Authenticated SUPI     : 0

Successful Emergency Handovers to 5GC with
No SUPI                : 0          Unauthenticated SUPI      : 0
Authenticated SUPI     : 0

Unsuccessful Emergency Calls due to
Missing Policy In APN   : 0          PCF Failure                : 0
UDM Failure             : 0          Other Failure              : 0

Unsuccessful Emergency Call Handovers to 5GC due to
Missing Policy In APN   : 0          PCF Failure                : 0
UDM Failure             : 0          Other Failure              : 0

PDU Sess Mod Fail Ho    : 0          PDU Sess Mod Fail Reg     : 0

Successful P-CSCF Restorat*: 0
Un-successful P-CSCF Restoration
5G Sessions             : 0
Un-successful P-CSCF Resto*: 0
-----
VNF/VM                  : 1/A        Gateway                  : 1

Establish PDU Session Succ : 0      Establish PDU Session Fail : 0
Re-establish PDU Sess Succ : 0      Re-establish PDU Sess Fail : 0
UE Init PDU Sess Rel Succ  : 0      UE Init PDU Sess Rel Fail  : 0
NW Init PDU Sess Rel Succ  : 0      NW Init PDU Sess Rel Fail  : 0
UDM Init PDU Sess Rel Succ : 0      UDM Init PDU Sess Rel Fail : 0
AMF Init PDU Sess Rel Succ : 0      AMF Init PDU Sess Rel Fail : 0
UE Triggered Serv Req Succ : 0      UE Triggered Serv Req Fail : 0
NW Triggered Serv Req Succ : 0      NW Triggered Serv Req Fail : 0
AN Release                : 0
Xn Handover Success       : 0      Xn Handover Fail          : 0
N2 Handover Success       : 0      N2 Handover Fail          : 0
RAN Init PDU Sess Mod Succ : 0      RAN Init PDU Sess Mod Fail : 0
UE Init PDU Sess Mod Succ : 0      UE Init PDU Sess Mod Fail  : 0
SMF Init PDU Sess Mod Succ : 0      SMF Init PDU Sess Mod Fail : 0
PCF Init PDU Sess Mod Succ : 0      PCF Init PDU Sess Mod Fail : 0
UDM Init PDU Sess Mod Succ : 0      UDM Init PDU Sess Mod Fail : 0
EPC(PGW ses)/5GC HO Sc   : 0      EPC(PGW ses)/5GC HO Fl    : 0
EPC(Cmb ses)/5GC HO Sc   : 0      EPC(Cmb ses)/5GC HO Fl    : 0
EPC(PGW ses)/5GC Idl Mob Sc: 0      EPC(PGW ses)/5GC Idl Mob Fl: 0
EPC(Cmb ses)/5GC Idl Mob Sc: 0      EPC(Cmb ses)/5GC Idl Mob Fl: 0
AMF Create Event Subsc Succ: 0      AMF Create Event Subsc Fail: 0
AMF Update Event Subsc Succ: 0      AMF Update Event Subsc Fail: 0
AMF Delete Event Subsc Succ: 0      AMF Delete Event Subsc Fail: 0
AMF Event Notific Received : 0      AMF Event Subscr Expiration: 0
PDU Sess Mod Rej IMS Fllbck: 0
LADN PDU Sess Released    : 0
LADN UP Suspended         : 0      LADN UP Resumed           : 0

Successful Emergency Attaches with
No SUPI                : 0          Unauthenticated SUPI      : 0
Authenticated SUPI     : 0

```



```

Successful Emergency Handovers to 5GC with
No SUPI                : 0          Unauthenticated SUPI      : 0
Authenticated SUPI     : 0

Unsuccessful Emergency Calls due to
Missing Policy In APN   : 0          PCF Failure              : 0
UDM Failure             : 0          Other Failure             : 0

Unsuccessful Emergency Call Handovers to 5GC due to
Missing Policy In APN   : 0          PCF Failure              : 0
UDM Failure             : 0          Other Failure             : 0

PDU Sess Mod Fail Ho    : 0          PDU Sess Mod Fail Reg    : 0

Successful P-CSCF Restorat*: 0
Un-successful P-CSCF Restoration
LTE Sessions            : 0          S2b Sessions              : 0
Un-successful P-CSCF Resto*: 0

```

```

-----
Number of VMs : 2
=====

```

```

* indicates that the corresponding row element may have been truncated.
=====

```

I-SMF call flow statistics

```

=====
VNF/VM                : 1/3          Gateway                  : 1

Establish PDU Session Succ : 0          Establish PDU Session Fail : 0
AMF Init PDU Sess Rel Succ : 0          AMF Init PDU Sess Rel Fail : 0
UE Init PDU Sess Rel Succ  : 0          UE Init PDU Sess Rel Fail  : 0
I-SMF Init PDU Sess Rel Sc : 0          I-SMF Init PDU Sess Rel Fl : 0
SMF Init PDU Sess Rel Succ : 0          SMF Init PDU Sess Rel Fail : 0
UE Triggered Serv Req Succ : 0          UE Triggered Serv Req Fail : 0
NW Triggered Serv Req Succ : 0          NW Triggered Serv Req Fail : 0
AN Release                : 0
Xn Handover Success       : 0          Xn Handover Fail           : 0
N2 Handover Success       : 0          N2 Handover Fail           : 0
AMF Change Success        : 0          AMF Change Fail            : 0
EPC/5GC Handover Success  : 0          EPC/5GC Handover Fail     : 0
EPC/5GC Mobility Success  : 0          EPC/5GC Mobility Fail     : 0
UE Init PDU Sess Mod Succ  : 0          UE Init PDU Sess Mod Fail  : 0
RAN Init PDU Sess Mod Succ : 0          RAN Init PDU Sess Mod Fail : 0
SMF Init PDU Sess Mod Succ : 0          SMF Init PDU Sess Mod Fail : 0

```

```

-----
VNF/VM                : 1/A          Gateway                  : 1

Establish PDU Session Succ : 0          Establish PDU Session Fail : 0
AMF Init PDU Sess Rel Succ : 0          AMF Init PDU Sess Rel Fail : 0
UE Init PDU Sess Rel Succ  : 0          UE Init PDU Sess Rel Fail  : 0
I-SMF Init PDU Sess Rel Sc : 0          I-SMF Init PDU Sess Rel Fl : 0
SMF Init PDU Sess Rel Succ : 0          SMF Init PDU Sess Rel Fail : 0
UE Triggered Serv Req Succ : 0          UE Triggered Serv Req Fail : 0
NW Triggered Serv Req Succ : 0          NW Triggered Serv Req Fail : 0
AN Release                : 0
Xn Handover Success       : 0          Xn Handover Fail           : 0
N2 Handover Success       : 0          N2 Handover Fail           : 0

```

```

AMF Change Success      : 0          AMF Change Fail        : 0
EPC/5GC Handover Success : 0          EPC/5GC Handover Fail    : 0
EPC/5GC Mobility Success : 0          EPC/5GC Mobility Fail    : 0
UE Init PDU Sess Mod Succ : 0          UE Init PDU Sess Mod Fail : 0
RAN Init PDU Sess Mod Succ : 0          RAN Init PDU Sess Mod Fail : 0
SMF Init PDU Sess Mod Succ : 0          SMF Init PDU Sess Mod Fail : 0

```

```

-----
Number of VMs : 2
=====

```

```

=====
SMF with I-SMF call flow statistics
=====

```

```

VNF/VM          : 1/3          Gateway          : 1

I-SMF Init PDU Sess Rel Sc : 0          I-SMF Init PDU Sess Rel Fl : 0
SMF Init PDU Sess Rel Succ : 0          SMF Init PDU Sess Rel Fail : 0
UE Init PDU Sess Mod Succ  : 0          UE Init PDU Sess Mod Fail  : 0
RAN Init PDU Sess Mod Succ : 0          RAN Init PDU Sess Mod Fail : 0
PCF Init PDU Sess Mod Succ : 0          PCF Init PDU Sess Mod Fail : 0
SMF Init PDU Sess Mod Succ : 0          SMF Init PDU Sess Mod Fail : 0
UDM Init PDU Sess Mod Succ : 0          UDM Init PDU Sess Mod Fail : 0

```

```

-----
VNF/VM          : 1/A          Gateway          : 1

I-SMF Init PDU Sess Rel Sc : 0          I-SMF Init PDU Sess Rel Fl : 0
SMF Init PDU Sess Rel Succ : 0          SMF Init PDU Sess Rel Fail : 0
UE Init PDU Sess Mod Succ  : 0          UE Init PDU Sess Mod Fail  : 0
RAN Init PDU Sess Mod Succ : 0          RAN Init PDU Sess Mod Fail : 0
PCF Init PDU Sess Mod Succ : 0          PCF Init PDU Sess Mod Fail : 0
SMF Init PDU Sess Mod Succ : 0          SMF Init PDU Sess Mod Fail : 0
UDM Init PDU Sess Mod Succ : 0          UDM Init PDU Sess Mod Fail : 0

```

```

-----
Number of VMs : 2
=====

```

```

=====
SMF with I-SMF added call flow statistics
=====

```

```

VNF/VM          : 1/3          Gateway          : 1

I-SMF Added Success      : 0          I-SMF Added Fail        : 0
Establish PDU Session Succ : 0          Establish PDU Session Fail : 0
UE Triggered Serv Req Succ : 0          UE Triggered Serv Req Fail : 0
N2 Handover Success      : 0          N2 Handover Fail        : 0
AMF Change Success      : 0          AMF Change Fail        : 0
EPC/5GC(PGW sess) HO Succ : 0          EPC/5GC(PGW sess) HO Fail : 0
EPC/5GC(Cmb sess) HO Succ : 0          EPC/5GC(Cmb sess) HO Fail : 0
EPC/5GC(PGW sess) Mob Succ : 0          EPC/5GC(PGW sess) Mob Fail : 0
EPC/5GC(Cmb sess) Mob Succ : 0          EPC/5GC(Cmb sess) Mob Fail : 0

```

```

-----
VNF/VM          : 1/A          Gateway          : 1

I-SMF Added Success      : 0          I-SMF Added Fail        : 0
Establish PDU Session Succ : 0          Establish PDU Session Fail : 0
UE Triggered Serv Req Succ : 0          UE Triggered Serv Req Fail : 0
N2 Handover Success      : 0          N2 Handover Fail        : 0

```

```

AMF Change Success      : 0          AMF Change Fail      : 0
EPC/5GC(PGW sess) HO Succ : 0          EPC/5GC(PGW sess) HO Fail : 0
EPC/5GC(Cmb sess) HO Succ : 0          EPC/5GC(Cmb sess) HO Fail : 0
EPC/5GC(PGW sess) Mob Succ : 0          EPC/5GC(PGW sess) Mob Fail : 0
EPC/5GC(Cmb sess) Mob Succ : 0          EPC/5GC(Cmb sess) Mob Fail : 0

```

```

-----
Number of VMs : 2
=====

```

```

=====
SMF with I-SMF change call flow statistics
=====

```

```

VNF/VM      : 1/3      Gateway      : 1

I-SMF Change Success      : 0          I-SMF Change Fail      : 0
AMF Change Success      : 0          AMF Change Fail      : 0

```

```

-----
VNF/VM      : 1/A      Gateway      : 1

I-SMF Change Success      : 0          I-SMF Change Fail      : 0
AMF Change Success      : 0          AMF Change Fail      : 0

```

```

-----
Number of VMs : 2
=====

```

```

=====
SMF with I-SMF deleted call flow statistics
=====

```

```

VNF/VM      : 1/3      Gateway      : 1

UE Triggered Serv Req Succ : 0          UE Triggered Serv Req Fail : 0
Xn Handover Success      : 0          Xn Handover Fail      : 0
N2 Handover Success      : 0          N2 Handover Fail      : 0
AMF Change Success      : 0          AMF Change Fail      : 0
EPC/5GC(PGW sess) HO Succ : 0          EPC/5GC(PGW sess) HO Fail : 0
EPC/5GC(Cmb sess) HO Succ : 0          EPC/5GC(Cmb sess) HO Fail : 0
EPC/5GC(PGW sess) Mob Succ : 0          EPC/5GC(PGW sess) Mob Fail : 0
EPC/5GC(Cmb sess) Mob Succ : 0          EPC/5GC(Cmb sess) Mob Fail : 0

```

```

-----
VNF/VM      : 1/A      Gateway      : 1

UE Triggered Serv Req Succ : 0          UE Triggered Serv Req Fail : 0
Xn Handover Success      : 0          Xn Handover Fail      : 0
N2 Handover Success      : 0          N2 Handover Fail      : 0
AMF Change Success      : 0          AMF Change Fail      : 0
EPC/5GC(PGW sess) HO Succ : 0          EPC/5GC(PGW sess) HO Fail : 0
EPC/5GC(Cmb sess) HO Succ : 0          EPC/5GC(Cmb sess) HO Fail : 0
EPC/5GC(PGW sess) Mob Succ : 0          EPC/5GC(PGW sess) Mob Fail : 0
EPC/5GC(Cmb sess) Mob Succ : 0          EPC/5GC(Cmb sess) Mob Fail : 0

```

```

-----
Number of VMs : 2
=====

```

```

=====
H-SMF call flow statistics
=====

```

```

VNF/VM      : 1/3      Gateway      : 1

```

```

Establish PDU Session Succ : 0          Establish PDU Session Fail : 0
V-SMF Init PDU Sess Rel Sc : 0          V-SMF Init PDU Sess Rel Fl : 0
H-SMF Init PDU Sess Rel Sc : 0          H-SMF Init PDU Sess Rel Fl : 0
V-SMF Init PDU Sess Mod Sc : 0          V-SMF Init PDU Sess Mod Fl : 0
PCF Init PDU Sess Mod Succ : 0          PCF Init PDU Sess Mod Fail : 0
H-SMF Init PDU Sess Mod Sc : 0          H-SMF Init PDU Sess Mod Fl : 0
UDM Init PDU Sess Mod Succ : 0          UDM Init PDU Sess Mod Fail : 0

```

```

-----
VNF/VM                : 1/A          Gateway                : 1

```

```

Establish PDU Session Succ : 0          Establish PDU Session Fail : 0
V-SMF Init PDU Sess Rel Sc : 0          V-SMF Init PDU Sess Rel Fl : 0
H-SMF Init PDU Sess Rel Sc : 0          H-SMF Init PDU Sess Rel Fl : 0
V-SMF Init PDU Sess Mod Sc : 0          V-SMF Init PDU Sess Mod Fl : 0
PCF Init PDU Sess Mod Succ : 0          PCF Init PDU Sess Mod Fail : 0
H-SMF Init PDU Sess Mod Sc : 0          H-SMF Init PDU Sess Mod Fl : 0
UDM Init PDU Sess Mod Succ : 0          UDM Init PDU Sess Mod Fail : 0

```

```

-----
Number of VMs : 2
=====

```

V-SMF call flow statistics

```

=====
VNF/VM                : 1/3          Gateway                : 1

```

```

Establish PDU Session Succ : 0          Establish PDU Session Fail : 0
AMF Init PDU Sess Rel Succ : 0          AMF Init PDU Sess Rel Fail : 0
UE Init PDU Sess Rel Succ  : 0          UE Init PDU Sess Rel Fail  : 0
V-SMF Init PDU Sess Rel Sc : 0          V-SMF Init PDU Sess Rel Fl : 0
H-SMF Init PDU Sess Rel Sc : 0          H-SMF Init PDU Sess Rel Fl : 0
UE Triggered Serv Req Succ : 0          UE Triggered Serv Req Fail : 0
NW Triggered Serv Req Succ : 0          NW Triggered Serv Req Fail : 0
An Release                 : 0
Xn Handover Success        : 0          Xn Handover Fail          : 0
N2 Handover Success        : 0          N2 Handover Fail          : 0
UE Init PDU Sess Mod Succ  : 0          UE Init PDU Sess Mod Fail  : 0
RAN Init PDU Sess Mod Succ : 0          RAN Init PDU Sess Mod Fail : 0
H-SMF Init PDU Sess Mod Sc : 0          H-SMF Init PDU Sess Mod Fl : 0

```

```

-----
VNF/VM                : 1/A          Gateway                : 1

```

```

Establish PDU Session Succ : 0          Establish PDU Session Fail : 0
AMF Init PDU Sess Rel Succ : 0          AMF Init PDU Sess Rel Fail : 0
UE Init PDU Sess Rel Succ  : 0          UE Init PDU Sess Rel Fail  : 0
V-SMF Init PDU Sess Rel Sc : 0          V-SMF Init PDU Sess Rel Fl : 0
H-SMF Init PDU Sess Rel Sc : 0          H-SMF Init PDU Sess Rel Fl : 0
UE Triggered Serv Req Succ : 0          UE Triggered Serv Req Fail : 0
NW Triggered Serv Req Succ : 0          NW Triggered Serv Req Fail : 0
An Release                 : 0
Xn Handover Success        : 0          Xn Handover Fail          : 0
N2 Handover Success        : 0          N2 Handover Fail          : 0
UE Init PDU Sess Mod Succ  : 0          UE Init PDU Sess Mod Fail  : 0
RAN Init PDU Sess Mod Succ : 0          RAN Init PDU Sess Mod Fail : 0
H-SMF Init PDU Sess Mod Sc : 0          H-SMF Init PDU Sess Mod Fl : 0

```

```
Number of VMs : 2
=====
```

5.2.2.20 canary-test

Syntax

```
canary-test gateway gw-id statistics
```

Context

```
show>mobile>pdn
```

Description

This command displays the statistics for a Canary test.

Parameters

gw-id

specifies the gateway ID

Values: 1 to 8

statistics

displays summarized information

Output

The following outputs example shows Canary test information and *Show PDN canary test output fields* describes the output fields.

Example output canary test statistics

```
# show mobile-gateway pdn canary-test gateway 1 statistics
=====
Canary Test Statistics
=====
Gateway           : 1
Group             : 2          Card           : 3

Status            : In-progress
Targets           : 25          CanaryAttach Attempts: 14
                                   Grp Specific Attempts: 0
                                   ACCUM. CanaryAttach  : 1515
                                   ACCUM. GroupSpecific  : 61

User Range        : ur1
=====
```

Type	Start-id	End-id		

imsi	460100003000000	460100012000000		
imsi	460100013000000	460100023000000		
imsi	500000000000000	580000000000000		
imsi	600000000000000	700000000000000		
imsi	800000000000000	850000000000000		
imsi	310012123450000	310012123450100		
msisdn	20000000000	30000000000		
msisdn	40000000000	50000000000		
msisdn	70000000000	75000000000		
msisdn	6970000000	6975000000		

History Logs				

TimeStamp	UserRange	Targets	CanaryAttach	GroupSpecfic

09/12/2024 03:36:20	ur1	19	19	0
09/12/2024 03:42:36	ur1	10	9	0
09/12/2024 03:45:00	ur2	15	8	0
09/12/2024 03:46:47	ur2	20	20	0
09/12/2024 03:53:10	ur1	10	9	0
09/12/2024 03:58:18	ur1	20	19	0
09/12/2024 04:04:47	ur1	25	25	0
09/12/2024 04:08:53	ur1	15	15	0
09/12/2024 04:19:28	ur1	15	14	0
09/12/2024 04:27:33	ur1	17	16	0
09/12/2024 04:38:34	ur1	25	21	0
09/12/2024 04:43:14	ur1	28	28	0
09/12/2024 05:26:28	ur1	215	86	0
09/12/2024 05:33:58	ur1	15	14	0
09/12/2024 05:45:36	ur1	17	16	0

=====				

Table 33: Show PDN canary test output fields

Label	Description
Gateway	The gateway ID
Group	The Canary group
Card	The Canary card

Table 33: Show PDN canary test output fields (continued)

Label	Description
Status	The Canary test status; it can be "Ready", "In-progress", "Done", or "None"
Targets	The configured number of subscribers for the Canary testing
CanaryAttach Attempts	The IMSI/MSISDN based canary attach attempts for the current test; reset to zero for a new Canary test
Grp Specific Attempts	The number of attaches (such as re-attach) routed to the Canary group; reset to zero for a new Canary test (applies only on the SMF)
ACCUM. CanaryAttach	The accumulated number of IMSI/MSISDN based Canary attach attempts; not displayed when zero
ACCUM. GroupSpecific	The accumulated number of attaches (such as re-attach) routed to the Canary group (applies only on the SMF); not displayed when zero
User Range	The user range name
Type	The UE type
Start-id	The start identifier for the user range
End-id	The end identifier for the user range
History Logs	The history log of the performed Canary tests; a maximum number of 15 rows in the output is supported
TimeStamp	The time at which the Canary test ends

5.2.2.21 clear-status

Syntax

```
clear-status [gateway gw-id]
```

Context

```
show>mobile>pdn
```

Description

This command shows the status of the last administrative `clear` command. The user can monitor an ongoing `clear` command using this output.

Parameters

gw-id

specifies the gateway ID

Output

The following outputs are examples of `clear` command status messages and *Show PDN clear status output fields* describes the output fields.

Example output mobile gateway clear status

```
# show mobile-gateway pdn clear-status
=====
Mobile gateway clear status
=====
Gateway                               : 1
Suspended                             : No
Number of Sessions                     : 3547
Num of Sessions having a dedicated bearer : 827

Last administrative sessions clearing stats :
Started                                : 08/15/2017 02:39:28
Finished                               : 08/15/2017 02:40:33
Total Sessions to delete                : 650
Sessions deleted from APN                : 650
Sessions deleted                         : 1454

Ongoing Session Deletions
1.clearing on APN: apn1, Group: 1, PDN-type: ipv4, slow-loop
2.clearing on APN: apn2, Group: 2, PDN-type: ipv4, data-packet-trigger
3.clearing on APN: apn2, Group: 2, PDN-type: ipv4, data-packet-trigger
```



```

-----
no UP-Peer                APN                IP-prefix
-----
1.clearing on UP-Peer: 1.2.3.4, APN: apn4, IPv4-prefix: 10.10.10.1/255
2.clearing on UP-Peer: 1.2.3.5, APN: apn5, IPv6-prefix: 10:10::/128
-----

Number of gateways : 1
-----

Note:"Sessions deleted from APN" is only applicable when both "percentage" and "apn"
are used in the clear command.
=====

```

Table 34: Show PDN clear status output fields

Label	Description
Gateway	The gateway ID
Suspended	Indicates if the clear status is suspended
Number of Sessions	The number of sessions on the gateway
Num of Sessions having a dedicated bearer	The number of sessions with a dedicated bearer on the gateway
Started	Date and time when the the last administrative sessions clearing started
Finished	Date and time when the last administrative sessions clearing was completed; if not, the field is blank
Total Sessions to delete	The total number of sessions targeted for deletion, used only with the percentage option
Sessions deleted from APN	The number of sessions deleted so far from the selected APN, used only with the percentage option
Sessions deleted	The number of sessions deleted so far in the administrative sessions cleaning
Number of gateways	The number of gateways retrieved

5.2.2.22 dnn-overload-control

Syntax

```
dnn-overload-control [dnn-name]
```

Context

```
show>mobile>pdn
```

Description

This command displays the DNN overload control information.

Parameters

dnn-name

specifies the DNN name, up to 80 characters

Output

The following output example shows detailed DNN overload control information for a specified DNN.

Example output DNN overload control information

```
# show mobile-gateway pdn dnn-overload-control dnn1
=====
Nsmf Services DNN Overload information summary
=====
DNN                               : dnn1
Overload control                  : disabled
Nsmf-thrsh-max-connections       : 30000
Nsmf-thrsh-max-activations       : 50000
Retry-after timer                : disable / 300 seconds
N1-Back-Off-timer               : 60 seconds
Current dnn-overload state       : no overload/overload
Cum PDU Sess Est Req dropped    : 500
-----
Number of APNs : 1
=====
```

5.2.2.23 error-indication

Syntax

```
error-indication [gateway gw-id] [router vr-id] [peer src-address] [summary]
```

Context

show>mobile>pdn

Description

This command displays information about error indication messages generated per peer IP address.

Output

The following output examples show error indication messages information and *Show PDN error indication output fields* describes the output fields.

- [Example error indication statistics](#)
- [Example error indication detailed statistics](#)
- [Example error indication summary statistics](#)

Example Error Indication Statistics

```
# show mobile-gateway pdn error-indication
=====
PDN GTP error indication statistics
=====
Gateway                : 1
Router                 : vprn20071
Peer address   : 16.80.16.10
local address : 16.90.16.10
Packets Tx      : 8           User Equipments Idle      : 0
Bearers Deleted : 0           Sessions Deleted       : 0
Invalid Bundle Tx : 0
-----
Number of entries : 1
=====
```

Table 35: Show PDN error indication output fields

Label	Description
Gateway	The gateway ID
Router	The virtual ID of the router
Peer address	The IP source address of the peer
local address	The local IP address

Table 35: Show PDN error indication output fields (continued)

Label	Description
Packets Tx	The number of error indication packets transmitted
Invalid Bundle Tx	The number of error indications due to Unknown TEID. Invalid Bundle Tx is a subset of Packets TX.
User Equipments Idle	—
Bearer Deleted	The number of deleted bearers
Sessions Deleted	The number of deleted sessions
Number of entries	The number of entries retrieved

Example error indication detailed statistics

```
# show mobile-gateway pdn error-indication gateway 1 router 20071 peer 6.6.13.1
=====
PDN GTP error indication statistics
=====
Gateway                : 1
Router                 : vprn20071
Peer address  : 6.6.13.1
Local address : 16.70.16.16
Packets Tx       : 7           User Equipments Idle       : 0
Bearer Deleted   : 0           Sessions Deleted       : 0
Invalid Bundle Tx : 0
-----
Number of entries : 1
=====
```

Example error indication summary statistics

```
# show mobile-gateway pdn error-indication gateway 1 router 20071 peer 16.80.16.10
summary
=====
PDN GTP error indication statistics summary
=====
Packets Tx       : 8           User Equipments Idle       : 0
Bearer Deleted   : 0           Sessions Deleted       : 0
-----
Number of entries : 1
```

5.2.2.24 fixed-subscriber

Syntax

```
fixed-subscriber [subscriber-id] [apn apn-name] [gateway gw-id] [group group-number] [card  
slot-number] [vm vm-id]
```

```
fixed-subscriber subscriber-id detail [apn apn-name] [gateway gw-id] [group group-number]  
[card slot-number] [vm vm-id]
```

Context

```
show>mobile>pdn
```

Description

This command displays subscriber context with single or multiple hosts connected to a fixed network over DSL, residential broadband, or business leased line.

Parameters

subscriber-id

specifies the string value (up to 32 characters) stored in the Subscriber Profile Repository (SPR) that identifies each subscriber in the network. The value is assigned and maintained by individual operators and varies in format.

apn-name

specifies the string value (up to 80 characters) of the APN configured to serve fixed subscribers

detail

specifies to show detailed information

gw-id

specifies the gateway ID

Values: 1 to 8

group-number

specifies the group number

Values: 1 to 8

slot-number

specifies the slot number

Values: 1 to 10 | A | B

vm-id

specifies the VM ID

Values: 1 to 22

Output

The following output examples show fixed subscriber information.

- [Example output fixed subscriber information](#)
- [Example output fixed subscriber detail information](#)

Example output fixed subscriber information

The following output shows fixed subscriber information and [Show PDN fixed subscriber output fields](#) describes the output fields.

```
# show mobile-gateway pdn fixed-subscriber
=====
Subscriber-Id      State      APNs      Hosts
=====
goldUsr101        ACTIVE    1         1
-----
Number of Subscribers: 1
-----
```

Table 36: Show PDN fixed subscriber output fields

Label	Description
Subscriber-id	The subscriber ID in the SPR
State	The state of the subscriber
APNs	The number of APNs
Hosts	The number of hosts
Number of Subscribers	The number of subscribers retrieved

Example output fixed subscriber detail information

The following outputs show fixed subscriber detail information and *Show PDN fixed subscriber detail output fields* describes the output fields.

```
# show mobile-gateway pdn fixed-subscriber 10.0.0.2@ipd.nokia.1.1.com detail
=====
Fixed subscriber context detail
=====
Subscriber      : 10.0.0.2@ipd.nokia.1.1.com
State           : ACTIVE                Session/Host      : 2
RAT             : Virtual
-----
Subscriber Hosts:
-----
APN             : ipd.nokia.1.1.com
Card/VM         : 1/1/3                Ref-pt           : AUTO CREATE
IP address      : 10.0.0.2
Policy Rule Name : N/A
APN             : ipd.nokia.1.1.com
Card/VM         : 1/1/3                Ref-pt           : MPTCP
IP address      : 20.0.0.2
Policy Rule Name : N/A
-----
=====
```

Table 37: Show PDN fixed subscriber detail output fields

Label	Description
Subscriber	The subscriber ID in the SPR
State	The state of the subscriber
Session/Host	The number of sessions/hosts of the subscriber
RAT	The RAT type of the subscriber
APN	The APN name
Card/VM	The card or VM number
IP address (v4/v6)	The IP address of the subscriber
Ref-pt	The reference point used

Table 37: Show PDN fixed subscriber detail output fields (continued)

Label	Description
Policy Rule Name	The name of the policy rule applied

5.2.2.25 gtp-load-overload-control

Syntax

```
gtp-load-overload-control
```

Context

```
show>mobile>pdn
```

Description

This command displays the current overload and load status of the gateway and the configuration for this feature.

Output

The following output example shows GTP-C load and overload information and *Show PDN GTP load and overload control information output fields* describes the output fields.

Example output GTP load and overload control information

```
# show mobile-gateway pdn gtp-load-overload-control
=====
GTP Load and over load control information
=====
Gateway                : 1

Load control           : Disabled
Load change trigger value : 5 %      Nth msg load report update : 50
Overload control       : Enabled
Minor threshold value   : 70 %      Major threshold value       : 85 %
Critical threshold value : 90 %
Minor reduction value    : 30 %      Major reduction value      : 60 %
Critical reduction value : 90 %

local overload throttling : Enabled  throttling alarms           : Disabled
Minor local overload throttling
value                     : 0 %      drop-only                     : -
exclude-arp-list          : -
Major local overload throttling
```



```

value          : 100 %    drop-only          : csreq
exclude-arp-list : -
Critical local overload throttling
value          : 100 %    drop-only          : csreq
exclude-arp-list : arpTestList

Overload Validity time      : 600 sec  Overload reporting over S8 : No
CPU sampling interval
Overload                   : 10 sec   Overload state change      : 2 sec
No-overload                : 10 sec

Current load                : 0 %      Current overload state    : critical
GTP-73-OCI                 : Disabled GTP-120-OCI                 : Disabled
Validity-Time-Extra-time   : 30 sec
gtp-73-120-reduction-metric : 50 %
ovld-control-only-on-new-sess: Disabled
-----
=====
* indicates that the corresponding row element may have been truncated.

```

Table 38: Show PDN GTP load and overload control information output fields

Label	Description
Gateway	The gateway ID
Load control	The status of the GTP load control
Load change trigger value	The load change value which will trigger a new load control update
Nth msg load report update	The message interval for sending Load Control Information) LCI
Overload control	The status of the GTP overload control
Minor threshold value	The threshold value of the GTP minor overload control
Major threshold value	The threshold value of the GTP major overload control
Critical threshold value	The threshold value of the GTP critical overload control
local overload throttling	The status of the local overload throttling

Table 38: Show PDN GTP load and overload control information output fields (continued)

Label	Description
Minor local overload throttling value	The throttling percentage when the gateway is in GTP minor overload
Major local overload throttling value	The throttling percentage when the gateway is in GTP major overload
Critical local overload throttling value	The throttling percentage when the gateway is in GTP critical overload
exclude-arp-list	ARP list used to exclude throttling of the CSREQs when the gateway is in different types of GTP-C overload levels
Overload Validity time	The length of time in seconds for the overload to be in effect
Overload reporting over S8	The status of the overload information report on the S8 interface
Minor reduction value	The traffic reduction percentage for minor overload
Major reduction value	The traffic reduction percentage for major overload
Critical reduction value	The traffic reduction percentage for critical overload
CPU sampling interval	The sampling interval
Overload	The CPU sampling period, in seconds, when transitioning from no overload to overload state
Overload state change	The CPU sampling period, in seconds, when transitioning from overload to another overload state
No-overload	The CPU sampling period, in seconds, when transitioning from overload to no overload state
Current load	The percentage of the current GTP load

Table 38: Show PDN GTP load and overload control information output fields (continued)

Label	Description
Current overload state	The status of the current overload state
GTP-73-OCI	The GTPv2 error code 73 is configured to be handled as OCI
GTP-120-OCI	The GTPv2 error code 120 is configured to be handled as OCI
Validity-Time-Extra-time	The additional time on top of the provided PGW Overload Period of Validity Timer
gtp-73-120-reduction-metric	The percentage of overload reduction metric applied for GTPv2 errors 73 and 120 when these are handled as OCI
overload-control-only-on-new*	The overload control actions only for new sessions; existing sessions remain unaffected

**NOTE:**

The `exclude-arp-list` and `drop-only` commands are displayed only when the operator explicitly configures for minor, major or critical local overload throttling the commands: `drop-only csreq` and `exclude-arp-list list-name`.

5.2.2.26 gtp-load-overload-control

Syntax

```
gtp-load-overload-control stats [node-id]
```

Context

```
show>mobile>pdn
```

Description

This command displays the GTP-C load and overload information received from peer nodes. The fields populated in the output vary according to the peer element type. For example, a PGW can receive only

Overload Control Information (OCI) IEs from MME, SGW, and ePDG node types. The load metric field will always be shown as zero in this case. The PGW itself can send out both Load Control Information (LCI) and OCI, so other node types can also show this information when peering with PGWs.

Parameters

stats

keyword used to show aggregated statistics

node-id

specifies the peer node type

Values: MME | SGW | PGW | EPDG

Output

The following output example shows GTP-C load and overload information and *Show PDN load and overload control information statistics output fields* describes the output fields.

Example output GTP overload control statistics

In this example, there are two MMEs and one SGW in overload.

```
# show mobile-gateway pdn gtp-load-overload-control stats
=====
PDN gateway gtp overload statistics per node
=====
Node address          : 10.10.65.10
Node Id               : PGW
Router                : vprn20033
Load metric           : 0
Last overload report time : 09/27/2019 07:19:33
Current overload       : Yes      Traffic reduction value : 60%
Remaining validity time : 7sec
Cumulative CB Req dropped : 0      Cumulative UB Req dropped : 0
Cumulative DB Req dropped : 0      Cumulative Attach Req rej : 0
Cum mobility Attach Req rej : 0     Cum AAA/HSS init Modify rej : 0
Cum MOBIKE MB Req rejected : 0
Last overload CB Req dropped: 0      Last overload UB Req dropped: 0
Last overload DB Req dropped: 0      Last overload Attach Req rej: 0
Lst ovl Mobility Att Req rej: 0      Last ovld AAA/HSS Modify rej: 0
Last ovld MOBIKE MB Req rej : 0
Cumulative Att Req rej 73 : 4      Last Ovld Att Req rej 73 : 4
Cumulative Att Req rej 120 : 0     Last Ovld Att Req rej 120 : 0
Cum mobility Att Req rej 73 : 0     Last Ovld mob Att Req rej 73: 0
Cum mobility Att Req rej 120: 0     Last Ovl mob Att Req rej 120: 0
Cum AAA/HSS init Mod rej 73 : 0     Last Ovld AAA/HSS Mod rej 73: 0
Cum AAA/HSS init Mod rej 120: 0    Last Ovl AAA/HSS Mod rej 120: 0
Cum MOBIKE MBReq rej 73 : 0        Last Ovl MOBIKE MBReq rej 73: 0
Cum MOBIKE MBReq rej 120 : 0       Lst Ovl MOBIKE MBReq rej 120: 0
# of 73 with BACKOFF_TIMER : 0      # of 73 w/o BACKOFF_TIMER : 1
# of 120 with BACKOFF_TIMER : 0     # of 120 w/o BACKOFF_TIMER : 0
-----
Number of nodes : 1
```

Table 39: Show PDN load and overload control information statistics output fields

Label	Description
Node address	The node IP address
Node Id	The node ID
Router	The router name
Load metric	The load metric
Last overload report time	Date and time of the last overload report
Current overload	The current overload state
Traffic reduction value	The percentage of traffic reduction for minor overload
Remaining validity time	The length of time in seconds remaining for the overload to be in effect
Cumulative CB Req dropped	The cumulative number of Create Bearer Request messages dropped
Cumulative UB Req dropped	The cumulative number of Update Bearer Request messages dropped
Cumulative DB Req dropped	The cumulative number of Delete Bearer Request messages dropped
Cumulative Attach Req rej	The cumulative number of Attach Request messages rejected
Cum mobility Attach Req rej	The cumulative number of mobility Attach Request messages rejected
Cum AAA/HSS init Modify rej	The cumulative number of AAA/HSS initial Modify messages rejected

Table 39: Show PDN load and overload control information statistics output fields (continued)

Label	Description
Last overload CB Req dropped	The number of Update Bearer Request messages dropped during the last overload period
Last overload UB Req dropped	The number of Update Bearer Request messages dropped during the last overload period
Last overload DB Req dropped	The number of Delete Bearer Request messages dropped during the last overload period
Last overload Attach Req rej	The number of Attach Request messages rejected during the last overload period
Lst ovl Mobility Att Req rej	The number of mobility Attach Request messages rejected during the last overload period
Last ovl AAA/HSS Modify rej	The number of AAA/HSS initial Modify messages rejected during the last overload period
Last ovld MOBIKE MB Req rej	The number of MOBIKE Modify Bearer Request messages rejected during the last overload period
Cumulative Att Req rej 73	The cumulative number of Attach Request messages rejected with GTP error code 73
Last Ovl Att Req rej 73	The number of Attach Request messages rejected during the last overload period with GTP error code 73
Cumulative Att Req rej 120	The cumulative number of Attach Request messages rejected with GTP error code 120
Last Ovl mob Att Req rej 120	The number of Attach Request messages rejected during the last overload period with GTP error code 120
Cum mobility Att Req rej 73	The cumulative number of mobility Attach Request messages rejected with GTP error code 73

Table 39: Show PDN load and overload control information statistics output fields (continued)

Label	Description
Last Ovld mob Att Req rej 73	The number of mobility Attach Request messages rejected during the last overload period with GTP error code 73
Cum mobility Att Req rej 120	The cumulative number of mobility Attach Request messages rejected with GTP error code 120
Last Ovl mob Att Req rej 120	The number of mobility Attach Request messages rejected during the last overload period with GTP error code 120
Cum AAA/HSS init Mod rej 73	The cumulative number of AAA/HSS initial Modify messages rejected with GTP error code 73
Last Ovld AAA/HSS Mod rej 73	The number of AAA/HSS initial Modify messages rejected during the last overload period with GTP error code 73
Cum AAA/HSS init Mod rej 120	The cumulative number of AAA/HSS initial Modify messages rejected with GTP error code 120
Last Ovl AAA/HSS Mod rej 120	The number of AAA/HSS initial Modify messages rejected during the last overload period with GTP error code 120
Cum MOBIKE MBReq rej 73	The cumulative number of MOBIKE Modify Bearer Request messages rejected with GTP error code 73
Last Ovl MOBIKE MBReq rej 73	The number of MOBIKE Modify Bearer Request messages rejected during the last overload period with GTP error code 73
Cum MOBIKE MBReq rej 120	The cumulative number of MOBIKE Modify Bearer Request messages rejected with GTP error code 120
Lst Ovl MOBIKE MBReq rej 120	The number of MOBIKE Modify Bearer Request messages rejected during the last overload period with GTP error code 120
# of 73 with BACKOFF_TIMER	The number of GTP messages with cause 73 and with GTP BACKOFF_TIMER IE

Table 39: Show PDN load and overload control information statistics output fields (continued)

Label	Description
# of 73 w/o BACKOFF_TIMER	The number of GTP messages with cause 120 and with GTP BACKOFF_TIMER IE.
# of 120 with BACKOFF_TIMER	The number of GTP messages with cause 73, without GTP BACKOFF_TIMER IE
# of 120 w/o BACKOFF_TIMER	The number of GTP messages with cause 120, without GTP BACKOFF_TIMER IE

5.2.2.27 gtp-local-load-control

Syntax

```
gtp-load-load-control stats
```

Context

```
show>mobile>pdn
```

Description

This command displays local load information for a PGW.

Parameters

stats

keyword to show aggregated statistics

Output

The following output example shows GTP-C local load information.

Example output GTP local load control statistics

```
# show mobile-gateway pdn gtp-local-load-control stats
=====
PDN gateway gtp local load statistics
=====
Last twenty 100% load report events:
```



```

-----
Time                LCI reported                Slot
-----
10/02/2018 23:19:17          100                1
10/02/2018 23:19:21          <100               1
10/02/2018 23:19:23          100                1
10/02/2018 23:19:28          <100               1
10/02/2018 23:19:30          100                1
10/02/2018 23:19:34          <100               1
10/02/2018 23:19:36          100                1
10/02/2018 23:19:41          <100               1
-----
Current GTP-C load (LCI) reported : 88
-----
Number of nodes : 1
=====

```

5.2.2.28 gtp-local-overload-control

Syntax

```
gtp-load-overload-control stats
```

Context

```
show>mobile>pdn
```

Description

This command displays local overload information for a PGW.

Parameters

stats

keyword to show aggregated statistics

Output

The following output example shows GTP-C local overload information and *Show PDN local overload control information statistics output fields* describes the output fields.

Example output GTP local overload control statistics

```

# show mobile-gateway pdn gtp-local-overload-control stats
=====
PDN gateway gtp local overload statistics
=====
Last overload report time   : 10/02/2018 23:19:39
Last twenty overload report events:
-----

```

```

Time                Overload level      OCI reported      Slot
-----
10/02/2018 23:19:17  major          40                1
10/02/2018 23:19:19  no-overload    0                 1
10/02/2018 23:19:23  major          40                1
10/02/2018 23:19:26  no-overload    0                 1
10/02/2018 23:19:30  minor          20                1
10/02/2018 23:19:32  no-overload    0                 1
10/02/2018 23:19:36  critical       60                1
10/02/2018 23:19:39  no-overload    0                 1
-----
Current GTP-C overload      : No      Traffic reduction value      : 0%
Local throttling value      : 100 %
Remaining validity time     : 0 sec

Local overload throttling:
-----
Drop-only      ARP      Count
-----
Csreq          1        #_
Csreq          2        #_
Csreq          3        #_
.
.
Csreq          15       #_
Cumulative CS Req dropped   : 0      Cumulative MB Com dropped   : 0
Cumulative BR Com dropped   : 0      Cumulative CB Req dropped   : 0
Cumulative UB Req dropped   : 0

-----
Number of nodes : 1
=====

```

**NOTE:**

The drop-only messages are displayed only when you explicitly configure the drop-only command for any one or all cases of minor, major, and critical local overload throttling.

Table 40: Show PDN local overload control information statistics output fields

Label	Description
Current GTP-C overload	The current status of the GTP-C overload
Traffic reduction value	The percentage of traffic reduction for minor overload

Table 40: Show PDN local overload control information statistics output fields (continued)

Label	Description
Drop-only	Messages to be throttled
ARP	The ARP used as match criterion
Count	The number of messages for either case of local overload throttling
Csreq	Supported type of messages to be throttled
Cumulative CS Req dropped	The cumulative number of Create Session Request messages dropped
Cumulative MB Com dropped	The cumulative number of Modify Bearer commands dropped
Cumulative BR Com dropped	The cumulative number of Bearer Resource commands dropped
Cumulative CB Req dropped	The cumulative number of Create Bearer Request messages dropped
Cumulative UB Req dropped	The cumulative number of Update Bearer Request messages dropped
Remaining validity time	The length of time in seconds remaining for the overload to be in effect
Number of nodes	The number of nodes retrieved

5.2.2.29 local-storage-stats

Syntax

```
local-storage-stats [gtp-prime-group gtp-prime-group] ga
```

Context

```
show>mobile>pdn
```

Description

This command displays statistics for locally stored CDR files in Ga interface.

Parameters

gtp-prime-group

specifies the GTP prime group profile name, up to 32 characters

ga

specifies the Ga interface

Output

The following output example shows CDR file storage statistics for Ga interface.

Example output CDR file storage Ga statistics

```
# show mobile-gateway pdn local-storage-stats ga
=====
PDN Ga local storage
=====
Number of CDR files generated      : 18918
Number of CDRs received           : 18912141
Number of Minor disk space alarms CF1 : 0
Number of Major disk space alarms CF1 : 0
Number of Minor disk space alarms CF2 : 0
Number of Major disk space alarms CF2 : 0
-----
Number of GTP prime groups: 1
=====
```

5.2.2.30 n2

Syntax

n2 [*sig-ref-point-name*] [gateway *gw-id*]

Context

show>mobile>pdn

Description

This command displays the PDN-level N2 peer information.

Parameters

sig-ref-point-name

specifies the signaling reference point name, up to 32 characters

gw-id

specifies the gateway ID

Values: 1 to 8

Output

Example output PDN N2 peer information

```
# show mobile-gateway pdn n2
```

```
=====
N2 Reference Point Configuration
=====
```

```
Reference Point : default
Gateway Id      : 1
Router          : Base
Interface Name  : system
```

```
N2 Peer Profile(s) :
  n2_peer_prof_1
  n2_peer_prof_2
```

```
Supported TA Item
```

```
-----
TAC          Broadcast PLMN List
-----
0x02ae       plmn_list_1
0x01ae       broadcast_plmn_2
=====
```

5.2.2.31 nf-peers

Syntax

```
nf-peers nf-type [nf-id uuid] [gateway gw-id] group group-id [instance | aggregate]
```

Context

```
show>mobile>pdn
```

Description

This command displays NF peer statistics.

Parameters

nf-type

specifies the NF type

Values: amf | scp | udm

gw-id

specifies the gateway ID

Values: 1 to 8

nf-id

specifies the UUID of the remote NF

Values: xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx (x: [0..F]H)

group-number

specifies the group number

Values: 1 to 15

instance

shows statistics per peer per VM

aggregate

shows aggregated statistics

Output

The following output examples show NF peer statistics and *Show NF peer statistics* describes the output fields.

- *Example output AMF peer statistics*
- *Example output SCP peer statistics*
- *Example output AMF peer instance statistics*
- *Example output SCP peer instance statistics*
- *Example output UDM peer statistics per peer instance*

Example output AMF peer statistics

```
# show mobile-gateway pdn nf-peers amf

=====
AMF peer statistics
=====
NF instance ID       : 12345678-9012-3456-7890-123456789011
Gateway Id          : 1

Connection failures  : 223           Failovers           : 3
-----
NF instance ID       : 12345678-9012-3456-7890-123456789012
```

```

Gateway Id           : 1

Connection failures   : 243           Failovers           : 7
-----
Number of peer instances : 4
=====

```

Example output SCP peer statistics

```

# show mobile-gateway pdn nf-peers scp

=====
SCP peer statistics
=====
NF instance ID       : 3296d21c-3f5a-4c35-bc79-344dc38e3f32
Gateway Id          : 1

Blocked              : false           Blocked Count        : 0
-----
NF instance ID       : 3296d21d-3f5a-4c35-bc79-344dc38e3f32
Gateway Id          : 1

Blocked              : false           Blocked Count        : 0
-----
Number of peer instances : 4
=====

```

Example output AMF peer instance statistics

```

# show mobile-gateway pdn nf-peers amf instance

=====
AMF peer statistics per peer instance
=====
NF instance ID       : 12345678-9012-3456-7890-123456789011
Gateway Id          : 1           VNF/Group Id/VM       : 1/1/3

Connection failures   : 111           Failovers             : 1
-----
NF instance ID       : 12345678-9012-3456-7890-123456789011
Gateway Id          : 1           VNF/Group Id/VM       : 1/2/5

Connection failures   : 112           Failovers             : 2
-----
NF instance ID       : 12345678-9012-3456-7890-123456789012
Gateway Id          : 1           VNF/Group Id/VM       : 1/1/3

Connection failures   : 121           Failovers             : 3
-----
NF instance ID       : 12345678-9012-3456-7890-123456789012
Gateway Id          : 1           VNF/Group Id/VM       : 1/2/5

Connection failures   : 122           Failovers             : 4
-----
Number of peer instances : 4

```

Example output SCP peer instance statistics

```
# show mobile-gateway pdn nf-peers scp instance

SCP peer statistics per peer instance
=====
NF instance ID      : 3296d21c-3f5a-4c35-bc79-344dc38e3f32
Gateway Id         : 1                VNF/Group Id/VM       : 1/0/21

Blocked            : false            Blocked Count        : 0
-----
NF instance ID      : 3296d21c-3f5a-4c35-bc79-344dc38e3f32
Gateway Id         : 1                VNF/Group Id/VM       : 1/1/3

Blocked            : true             Blocked Count        : 1
-----
NF instance ID      : 3296d21d-3f5a-4c35-bc79-344dc38e3f32
Gateway Id         : 1                VNF/Group Id/VM       : 1/0/21

Blocked            : false            Blocked Count        : 0
-----
NF instance ID      : 3296d21d-3f5a-4c35-bc79-344dc38e3f32
Gateway Id         : 1                VNF/Group Id/VM       : 1/1/3

Blocked            : false            Blocked Count        : 0
-----
Number of peer instances : 4
=====
Gateway Id         : 1                VNF/Group Id/VM       : 1/0/21

Blocked            : false            Blocked Count        : 0
-----
NF instance ID      : 3296d21d-3f5a-4c35-bc79-344dc38e3f32
Gateway Id         : 1                VNF/Group Id/VM       : 1/1/3

Blocked            : false            Blocked Count        : 0
-----
Number of peer instances : 4
=====
```

Example output UDM peer statistics per peer instance

```
# show mobile-gateway pdn nf-peers udm instance

=====
UDM peer statistics per peer instance
=====
NF instance ID : 12345678-9012-3456-7890-123456789011
Gateway Id : 1 VNF/Group Id/VM : 1/1/3
Blocked by signaling failure : <true | false>
Blocked by NRF : <true | false>
```


Block count : 0

Table 41: Show NF peer statistics

Label	Description
AMF peer statistics per peer instance	
NF instance ID	The NF instance ID
Gateway Id	The gateway ID
VNF/Group Id/VM	The VNF, group ID, and VM number
Connection failures	The number of connection failures
Failovers	The number of failovers
Number of peer instances	The number of peer instances
SCP peer statistics per peer instance	
NF instance ID	The NF instance ID
Gateway Id	The gateway ID
VNF/Group Id/VM	The VNF, group ID, and VM number
Blocked	Indicates if SCP is currently blocked
Blocked Count	The number of SCPs blocked
Number of peer instance	The number of peer instances
UDM peer statistics per peer instance	

Table 41: Show NF peer statistics (continued)

Label	Description
Blocked by signalling failure	Indicates if the UDM is currently blocked based on signalling failure
Blocked by NRF	Indicates if the UDM is currently blocked based on NF status notification from NRF
Block count	Indicates how many times the UDM has been blocked due to signalling failure

5.2.2.32 nsmf-event-exposure-subscriptions

Syntax

```
nsmf-event-exposure-subscriptions [supi supi] [gpsi gpsi] [subscription-id  
subscription-id] [pdu-session-id pdu-session-id] [group-id group-id]
```

Context

```
show>mobile>pdn
```

Description

This command displays configuration information or statistics about the Nsmf event exposure function.



NOTE:

This command is not supported in the current release.

Parameters

supi

specifies to show only the subscriptions with a matching SUPI, up to 15 digits

gpsi

specifies to show only the subscriptions with a matching GPSI, up to 15 digits

subscription-id

specifies to show only the subscriptions with a matching subscription ID, up to 31 characters

pdu-session-id

specifies to show only the subscriptions with a matching PDU session ID, up to 255 characters; this filter cannot be used without SUPI or GPSI

group-id

specifies to show only the subscriptions with a matching group ID, up to 20 characters

5.2.2.33 nsmf-overload-control

Syntax

```
nsmf-overload-control stats
```

Context

```
show>mobile>pdn
```

Description

This command displays configuration information or statistics for the Nsmf overload control function.

Parameters

stats

specifies to show aggregated statistics

Output

The following output examples show the Nsmf overload control configuration and statistics respectively.

Example output of the Nsmf overload control configuration

```
# show mobile-gateway pdn nsmf-overload-control
=====
Nsmf Overload control information
=====
Gateway           : 1
Overload control   : Enabled/Disabled
Retry-after-timer  : 300 seconds
Current overload state : none / no-overload / overload
-----
* indicates that the corresponding row element may have been truncated.
```

Example output of the Nsmf overload control statistics

```
# show mobile-gateway pdn nsmf-overload-control stats
=====
Nsmf Overload control statistics
```

```

=====
Last overload report time      : 01/23/2020 01:19:26
Last twenty overload report events:
-----
Time                          UTC                      Overload
-----
10/02/2019                    23:19:17         overload
12/19/2019                    02:34:54         no-overload
01/02/2020                    04:12:13         overload
01/23/2020                    01:19:26         no-overload
-----
Current Nsmf overload         : Yes/No   Traffic reduction value : 20%
Remaining validity time       : 60 sec
Cum PDU Sess Est Req dropped  : 800
-----
Number of nodes : 1
=====

```

5.2.2.34 nwu

Syntax

nwu [*sig-ref-point-name*] [gateway *gw-id*]

Context

show>mobile>pdn

Description

This command displays the NWu configuration information.



NOTE:

This command is not supported in the current release.

Parameters

sig-ref-point-name

specifies the NWu reference point name, up to 32 characters

gw-id

specifies the gateway ID

Values: 1 to 8

Output

Example output of the NWu configuration

The following output example shows the NWu context information and *Show PDN NWu output fields* describes the output fields.

```
# show mobile-gateway pdn nwu

=====
NWu Configuration
=====
Sig Ref Point Name : default
Gateway Id         : 1
Router Instance    : Base
Interface Name      : system
IPsec Profile Name : n3iwfProfile
=====
```

Table 42: Show PDN NWu output fields

Label	Description
Sig Ref Point Name	The name of signaling reference point
Gateway Id	The ID of the gateway
Router Instance	The name of the router instance
Interface Name	The name of the interface
IPsec Profile Name	The name of the IPSec profile

5.2.2.35 path-mgmt-stats

Syntax

```
path-mgmt-stats ref-point-type peer ip-address [port port-number] [gateway gw-id | group
group-number | [card slot-number | vm vm-id]] [instance | aggregate]
```

```
path-mgmt-stats {gn | gnu | gp | gpu} [ref-point ref-point-name] [peer ip-address [port port-number]] [gateway gw-id | group group-number | [card slot-number | vm vm-id]] [instance | aggregate] [active]

path-mgmt-stats {s5 | s5u | s8 | s8u | n9} [ref-point ref-point-name] [peer ip-address [port port-number]] [gateway gw-id | group group-number | card slot-number | vm vm-id]] [instance | aggregate]
```

Context

show>mobile>pdn

Description

This command displays path management statistics.

Parameters

ref-point-type

specifies the interface type

Values: s11 | s1u | s12 | s4 | s5 | s8 | gx | rf | gn | ga | gp | gy | s2a | s2b | s6b | swm | sww | sta | sx-n4 | n3 | n9 | s4u | s5u | s8u | gnu | gpu

ip-address

specifies an existing peer IP address

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x: [0..FFFF]H
–	–	d: [0..255]D
–	fqdn	FQDN

port-number

specifies a port ID

Values: 1 to 65535

slot-number

specifies the card slot number (applicable only to 7750 SR MG)

Values: 1 to 20 | A | B

gw-id

specifies the gateway ID

Values: 1 to 8

group-number

specifies the group number

Values: 1 to 15

vm-id

specifies the VM ID (applicable only to CMG)

Values: 1 to 20 | A | B

keyword

specifies the interface type

Values: s5 | s5u | s8 | s8u | n9 | gn | gnu | gp | gpu

ref-point-name

specifies the reference point name, up to 32 characters

instance

shows routing instance statistics

aggregate

shows aggregated global statistics

active

shows active peer statistics

Output

The following 7750 SR MG output examples show path management statistics and [Show PDN path management statistics common output fields](#) describes the common output fields. Descriptions of reference-point specific fields are also provided in the applicable sections.

**NOTE:**

CMG system outputs may display CMG-specific fields instead of the 7750 SR MG hardware components (for example, VM is displayed instead of Card).

- [Example output path management statistics Gn](#)
- [Example output path management statistics Gp](#)
- [Example output path management statistics Gx](#)

- *Example output path management statistics Gy*
- *Example output path management statistics N3 and N9*
- *Example output path management statistics PCFP (Sx-N4)*
- *Example output path management statistics Rf*
- *Example output path management statistics S11*
- *Example output path management statistics S12*
- *Example output path management statistics S1u*
- *Example output path management statistics S2a and S2b*
- *Example output path management statistics S4*
- *Example output path management statistics S5*
- *Example output path management statistics S5u*
- *Example output path management statistics S6b*
- *Example output path management statistics S8*

Table 43: Show PDN path management statistics common output fields

Label	Description
Peer address	The IP address of the peer
Router	The router name
Port	The port number
Reference point	The reference point name
Peer restarts	The number of peer restarts
Path management failures	The number of path management failures
Peer restart count	The peer restart count
Number of peer instances	The number of peer instances retrieved

Example output path management statistics Gn

The following examples shows path management statistics for Gn and *Show PDN path management S5, S5u, S8, Gn, and Gp statistics output fields* describes the output fields. *Show PDN path management statistics common output fields* describes additional common path management output fields.

```
# show mobile-gateway pdn path-mgmt-stats gn
=====
PDN gateway Gn path management statistics
=====
Peer address : 10.10.8.10
Router       : Base
Port        : 2123
Reference point : default

Peer restarts      : 0          Path management failures : 0
Peer restart count : 123
Echo requests Rx   : 0          Echo responses Tx      : 0
Echo requests Tx   : 17        Echo responses Rx      : 17
-----
Number of peer instances : 1
=====

# show mobile-gateway pdn path-mgmt-stats gn active
=====
PDN gateway Gn path management statistics
=====
Peer address : 10.10.8.10
Router       : Base
Port        : 2123
Reference point : default

Peer restarts      : 0          Path management failures : 0
Peer restart count : 123
Echo requests Rx   : 0          Echo responses Tx      : 0
Echo requests Tx   : 18        Echo responses Rx      : 18
-----
Number of peer instances : 1
=====
```

Example output path management statistics Gp

The following example shows path management statistics for Gp and *Show PDN path management S5, S5u, S8, Gn, and Gp statistics output fields* describes the output fields. *Show PDN path management statistics common output fields* describes additional common path management output fields.

```
# show mobile-gateway pdn path-mgmt-stats gp
=====
PDN gateway Gp path management statistics
=====
Peer address : 10.10.8.10
Reference point : default

Peer restarts      : 0          Path management failures : 0
```

```

Peer restart count      : 123
Echo requests Rx       : 0      Echo responses Tx       : 0
Echo requests Tx       : 2      Echo responses Rx       : 2

```

```

-----
Number of peer instances : 1
=====

```

Example output path management statistics Gx

The following example shows path management statistics for Gx and *Show PDN path management Rf, Gx, and Gy statistics output fields* describes the output. *Show PDN path management statistics common output fields* describes additional common path management output fields.

```

# show mobile-gateway pdn path-mgmt-stats gx
=====
PDN gateway Gx path management statistics
=====
Peer address : 10.10.8.10
Router       : Base
Port        : 3868

Messages Tx      : 6      Messages Rx      : 6
CER messages Tx  : 1      CEA messages Rx  : 1
DPR messages Tx  : 0      DPR messages Rx  : 0
DPA messages Tx  : 0      DPA messages Rx  : 0
DWR messages Tx  : 1      DWR messages Rx  : 61
DWA messages Tx  : 61     DWA messages Rx  : 1
Connection attempts : 1      Connection failures : 0
Transport disc msgs Rx : 0      Unexpected ver msgs Rx : 0
Msgs too big Rx   : 0      Msgs too small Rx  : 0
Invalid CEA msgs Rx : 0      Retransmit msgs Tx  : 0
-----
Number of peer instances : 1
=====

```

Example output path management statistics Gy

The following example shows path management statistics for Gy and *Show PDN path management Rf, Gx, and Gy statistics output fields* describes the output. *Show PDN path management statistics common output fields* describes additional common path management output fields.

```

# show mobile-gateway pdn path-mgmt-stats gy
=====
PDN gateway Gy path management statistics
=====
Peer address : 10.10.18.10
Router       : Base
Port        : 3868

Messages Tx      : 3      Messages Rx      : 3
CER messages Tx  : 1      CEA messages Rx  : 1
DPR messages Tx  : 0      DPR messages Rx  : 0
DPA messages Tx  : 0      DPA messages Rx  : 0
DWR messages Tx  : 1      DWR messages Rx  : 63
DWA messages Tx  : 63     DWA messages Rx  : 1
Connection attempts : 1      Connection failures : 0

```

```

Transport disc msgs Rx      : 0          Unexpected ver msgs Rx      : 0
Msgs too big Rx            : 0          Msgs too small Rx            : 0
Invalid CEA msgs Rx        : 0          Retransmit msgs Tx          : 0
-----

```

```

Number of peer instances : 1
=====

```

Example output path management statistics N3 and N9

The following output examples show N3 and N9 path management statistics and *Show PDN path management statistics for N3 and N9* describes the output fields.

```
# show mobile-gateway pdn path-mgmt-stats n3
```

```
=====
PDN gateway N3 path management statistics
=====
```

```
Peer address : 192.168.1.1
```

```
Reference point      : default
```

```
Path management failures : 0
```

```
Echo requests Rx      : 0          Echo responses Tx          : 0
```

```
Echo requests Tx      : 0          Echo responses Rx          : 0
-----

```

```
Number of peer instances : 1
=====
```

```
# show mobile-gateway pdn path-mgmt-stats n9
```

```
=====
PDN gateway N9 path management statistics
=====
```

```
Peer address : 2001:db8::211:ff:11
```

```
Reference point      : default
```

```
Path management failures : 0
```

```
Echo requests Rx      : 0          Echo responses Tx          : 0
```

```
Echo requests Tx      : 4          Echo responses Rx          : 4
-----

```

```
Number of peer instances : 1
=====
```

Table 44: Show PDN path management statistics for N3 and N9

Label	Description
Peer address	The IP address of the peer
Reference point	The reference point name
Path management failures	The number of path management failures

Table 44: Show PDN path management statistics for N3 and N9 (continued)

Label	Description
Echo requests Rx	The number of echo requests received
Echo responses Tx	The number of echo responses transmitted
Echo requests Tx	The number of echo requests transmitted
Echo responses Rx	The number of echo responses received
Number of peer instances	The number of peer instances retrieved

Example output path management statistics PCFP (Sx-N4)

The following output examples show Sx-N4 path management statistics for SMF/CP function and UPF/UP function respectively, and *Show PDN path management statistics for PCFP (Sx-N4)* describes the output fields.

```
# show mobile-gateway pdn path-mgmt-stats sx-n4
=====
PCFP path management statistics
=====
Peer address           : 192.168.34.1
Router                 : vprn20091
Peer restarts          : 0           Path management failures   : 0
Peer-Timestamp         : 04/17/2020 07:18:06 UTC 3796114686#
PCFP Heartbeat Requests Rx : 0           PCFP Heartbeat Responses Tx: 0
PCFP Heartbeat requests Tx : 16          PCFP Heartbeat responses Rx: 16
Path Down Assoc Rel Req : 0
-----
Number of peer instances : 1
=====
# indicates the number of seconds elapsed from epoch January 1 1900.
```

```
# show mobile-gateway pdn path-mgmt-stats sx-n4
=====
PCFP path management statistics
=====
Peer address           : 192.168.33.1
Router                 : vprn20091
Peer restarts          : 0           Path management failures   : 0
Peer-Timestamp         : 04/17/2020 07:18:06 UTC 3796114686#
PCFP Heartbeat Requests Rx : 16          PCFP Heartbeat Responses Tx: 16
PCFP Heartbeat requests Tx : 0           PCFP Heartbeat responses Rx: 0
Path Down Assoc Rel Req : 0
```

```
-----
Number of peer instances : 1
=====
```

```
# indicates the number of seconds elapsed from epoch January 1 1900.
```

Table 45: Show PDN path management statistics for PFCP (Sx-N4)

Label	Description
Peer address	The IP address of the peer
Router	The router name
Peer restarts	The number of peer restarts
Path management failures	The number of path management failures
Peer-Timestamp	The date and time the peer restarted or came operationally up, both in local time zone and UTC formats
PFCP Heartbeat Requests Rx	The number of heartbeat requests received
PFCP Heartbeat Responses Tx	The number of heartbeat responses transmitted
PFCP Heartbeat Requests Tx	The number of heartbeat requests transmitted
PFCP Heartbeat responses Rx	The number of heartbeat responses received
Path Down Assoc Rel Req	The number of times this path went down due to Sx Association Release Request messages
Number of peer instances	The number of peer instances retrieved

Example output path management statistics Rf

The following example shows Rf path management statistics and *Show PDN path management Rf, Gx, and Gy statistics output fields* describes the output. *Show PDN path management statistics common output fields* describes additional common path management output fields.

```
# show mobile-gateway pdn path-mgmt-stats rf
```

```

=====
PDN gateway Rf path management statistics
=====
Peer address : 10.11.7.10
Router       : Base
Port        : 3868

Messages Tx      : 0      Messages Rx      : 0
CER messages Tx  : 0      CEA messages Rx  : 0
DPR messages Tx  : 0      DPR messages Rx  : 0
DPA messages Tx  : 0      DPA messages Rx  : 0
DWR messages Tx  : 0      DWR messages Rx  : 66
DWA messages Tx  : 66     DWA messages Rx  : 0
Connection attempts : 0      Connection failures : 0
Transport disc msgs Rx : 0      Unexpected ver msgs Rx : 0
Msgs too big Rx   : 0      Msgs too small Rx   : 0
Invalid CEA msgs Rx : 0      Retransmit msgs Tx   : 0
-----
Number of peer instances : 1
=====

```

Table 46: Show PDN path management Rf, Gx, and Gy statistics output fields

Label	Description
Messages Tx	The number of messages transmitted
Messages Rx	The number of messages received
CER messages Tx	The number of CER messages transmitted
CEA messages Rx	The number of CER messages received
DPR messages Tx	The number of DPR messages transmitted
DPR messages Rx	The number of DPR messages received
DPA messages Tx	The number of DPA messages transmitted
DPA messages Rx	The number of DPR messages received

Table 46: Show PDN path management Rf, Gx, and Gy statistics output fields (continued)

Label	Description
DWR messages Tx	The number of DWR messages transmitted
DWR messages Rx	The number of DWR messages received
DWA messages Tx	The number of DWA messages transmitted
DWA messages Rx	The number of DWA messages received
Connection attempts	The number of connection attempts
Connection failures	The number of connection failures
Transport disc msgs Rx	The number of transport disconnect messages received
Unexpected ver msgs Rx	The number of unexpected version messages received
Msgs too big Rx	The number of Messages too big received
Msgs too small Rx	The number of Messages too big transmitted
Invalid CEA msgs Rx	The number of invalid CEA messages received
Retransmit msgs Tx	The number of retransmit messages transmitted

Example output path management statistics S11

The following output example shows S11 path management statistics and *Show PDN path management statistics common output fields* describes the common path management output fields.

```
# show mobile-gateway pdn path-mgmt-stats s11
=====
PDN gateway S11 path management statistics
=====
Peer address : 192.168.0.1
Peer type    : MME
Reference point      : default
```

```

Peer restarts           : 0           Path management failures : 0
Peer restart count      : 1
Echo requests Rx        : 0           Echo responses Tx       : 0
Echo requests Tx        : 0           Echo responses Rx       : 0
Pgw restart notify      : 0           Pgw restart notify ack  : 0
Malformed packets Rx    : 0

```

```

-----
Peer address : 2001:db8::211:ff:11
Peer type    : MME
Reference point      : default

```

```

Peer restarts           : 0           Path management failures : 0
Peer restart count      : 123
Echo requests Rx        : 0           Echo responses Tx       : 0
Echo requests Tx        : 0           Echo responses Rx       : 0
Pgw restart notify      : 0           Pgw restart notify ack  : 0
Malformed packets Rx    : 0

```

```

-----
Number of peer instances : 2
=====

```

Example output path management statistics S12

The following output example shows S12 path management statistics and *Show PDN path management statistics common output fields* describes the common path management output fields.

```

# show mobile-gateway pdn path-mgmt-stats s12
=====
PDN gateway S12 path management statistics
=====
Peer address : 192.168.0.1
Reference point      : default

Peer restarts           : 0           Path management failures : 0
Peer restart count      : 0
Echo requests Rx        : 0           Echo responses Tx       : 0
Echo requests Tx        : 0           Echo responses Rx       : 0
-----
Number of peer instances : 1
=====

```

Example output path management statistics S1u

The following output example shows S1u path management statistics and *Show PDN path management statistics common output fields* describes common path management output fields.

```

# show mobile-gateway pdn path-mgmt-stats s1u
=====
PDN gateway S1u path management statistics
=====
Peer address : 192.168.0.1
Reference point      : default

Peer restarts           : 0           Path management failures : 0
Peer restart count      : 0
Echo requests Rx        : 0           Echo responses Tx       : 0

```



```

Echo requests Tx           : 0           Echo responses Rx           : 0
-----
Peer address : 2001:db8::211:ff:11
Reference point : default

Peer restarts           : 0           Path management failures       : 0
Peer restart count      : 0
Echo requests Rx        : 0           Echo responses Tx              : 0
Echo requests Tx        : 16          Echo responses Rx              : 16
-----
Number of peer instances : 2
=====

```

Example output path management statistics S2a and S2b

The following output examples show S2a and S2b path management statistics and *Show PDN path management statistics S2a and S2b statistics output fields* describes the output fields. *Show PDN path management statistics common output fields* describes additional common path management output fields.

```

# show mobile-gateway pdn path-mgmt-stats s2a

=====
PDN gateway S2a path management statistics
=====
Peer address : 3FFE::A10:80A
Router       : Base
Port         : 0

Heartbeat Compatible : YES
Peer restarts        : 0           Path management failures       : 0
Peer restart count   : 1
Tx Heartbeat responses : 2           Rx Heartbeat requests         : 2
Rx Heartbeat responses : 15          Tx Heartbeat requests         : 15
-----
Number of peer instances : 1
=====

# show mobile-gateway pdn path-mgmt-stats s2a-gtp

=====
PDN gateway S2a path management statistics
=====
Peer address : 10.23.23.2
Peer type    : TWAG
Reference point : default

Peer restarts           : 0           Path management failures       : 0
Peer restart count      : 0
Echo requests Rx        : 0           Echo responses Tx              : 0
Echo requests Tx        : 0           Echo responses Rx              : 0
Pgw restart notify      : 0           Pgw restart notify ack        : 0
Malformed packets Rx    : 0
-----

```

```

Number of peer instances : 1
=====

# show mobile-gateway pdn path-mgmt-stats s2b
=====
PDN gateway S2b path management statistics
=====
Peer address : 10.70.24.101
Peer type    : ePDG
Reference point      : default

Peer restarts          : 0          Path management failures : 0
Peer restart count     : 1
Echo requests Rx       : 0          Echo responses Tx       : 0
Echo requests Tx       : 0          Echo responses Rx       : 0
Pgw restart notify     : 0          Pgw restart notify ack  : 0
Malformed packets Rx   : 0

-----
Number of peer instances : 1
=====

```

Table 47: Show PDN path management S2a and S2b statistics output fields

Label	Description
S2a	
Heartbeat Compatible	The number of messages received
Tx Heartbeat responses	The number of heartbeat responses transmitted
Rx Heartbeat requests	The number of heartbeat requests received
Rx Heartbeat responses	The number of heartbeat responses received
Tx Heartbeat requests	The number of heartbeat requests transmitted
S2a-GTP and S2b	
Peer type	The peer type
Echo requests Rx	The number of echo requests received

Table 47: Show PDN path management S2a and S2b statistics output fields (continued)

Label	Description
Echo responses Tx	The number of echo responses transmitted
Echo requests Tx	The number of echo requests transmitted
Echo responses Rx	The number of echo responses received
Pgw restart notify	The number of PGW restart notify messages
Pgw restart notify ack	The number of PGW restart notify acknowledgment messages
Malformed packets Rx	The number of malformed packets received

Example output path management statistics S4

The following output example shows S4 path management statistics and *Show PDN path management statistics common output fields* describes the common path management output fields.

```
# show mobile-gateway pdn path-mgmt-stats s4
=====
PDN gateway S4 path management statistics
=====
Peer address : 10.10.7.73
Peer type    : SGSN
Reference point      : default

Peer restarts      : 0          Path management failures : 0
Peer restart count : 1
Echo requests Rx   : 0          Echo responses Tx      : 0
Echo requests Tx   : 0          Echo responses Rx      : 0
Pgw restart notify : 0          Pgw restart notify ack : 0
Malformed packets Rx : 0

-----
Number of peer instances : 1
=====
```

Example output path management statistics S5

The following example shows path management statistics for S5 and *Show PDN path management S5, S5u, S8, Gn, and Gp statistics output fields* describes the output fields. *Show PDN path management statistics common output fields* describes additional common path management output fields.

```
# show mobile-gateway pdn path-mgmt-stats s5
```

```

=====
PDN gateway S5 path management statistics
=====
Peer address : 10.20.1.1
Router       : Base
Port        : 2123

Peer restarts      : 0          Path management failures : 0
Peer restart count : 2
Echo requests Rx   : 1          Echo responses Tx      : 1
Echo requests Tx   : 9          Echo responses Rx      : 9
-----
Number of peer instances : 1
=====

```

Table 48: Show PDN path management S5, S5u, S8, Gn, and Gp statistics output fields

Label	Description
Echo requests Rx	The number of echo requests received
Echo responses Tx	The number of echo responses transmitted
Echo requests Tx	The number of echo requests transmitted
Echo responses Rx	The number of echo responses received

Example output path management statistics S5u

The following example shows path management statistics for S5u and *Show PDN path management S5, S5u, S8, Gn, and Gp statistics output fields* describes the output fields. *Show PDN path management statistics common output fields* describes additional common path management output fields.

```

# show mobile-gateway pdn path-mgmt-stats s5u
=====
PDN gateway S5u path management statistics
=====
Peer address      : 10.10.5.1
Reference point   : default
Path management failures : 0
Echo requests Rx  : 0          Echo responses Tx      : 0
Echo requests Tx  : 0          Echo responses Rx      : 0
-----
Number of peer instances : 1
=====

```

Example output path management statistics S6b

The following output examples show S6b path management statistics and *Show PDN path management S6b statistics output fields* describes the output fields. *Show PDN path management statistics common output fields* describes additional common path management output fields.

```
# show mobile-gateway pdn path-mgmt-stats s6b
=====
PDN gateway S6b path management statistics
=====
Peer address : 10.10.19.10
Router       : Base
Port        : 3868

Messages Tx      : 3      Messages Rx      : 3
CER messages Tx  : 1      CEA messages Rx  : 1
DPR messages Tx  : 0      DPR messages Rx  : 0
DPA messages Tx  : 0      DPA messages Rx  : 0
DWR messages Tx  : 0      DWR messages Rx  : 66
DWA messages Tx  : 66     DWA messages Rx  : 0
Connection attempts : 1      Connection failures : 0
Transport disc msgs Rx : 0      Unexpected ver msgs Rx : 0
Msgs too big Rx   : 0      Msgs too small Rx   : 0
Invalid CEA msgs Rx : 0      Retransmit msgs Tx   : 0
-----
Number of peer instances : 1
=====

# show mobile-gateway pdn path-mgmt-stats s6b instance
=====
PDN gateway S6b path management statistics per-peer instance
=====
Peer address : 10.10.19.10
Router       : Base
Port        : 3868      Gateway/Chassis/Card : 1/1/3

Messages Tx      : 3      Messages Rx      : 3
CER messages Tx  : 1      CEA messages Rx  : 1
DPR messages Tx  : 0      DPR messages Rx  : 0
DPA messages Tx  : 0      DPA messages Rx  : 0
DWR messages Tx  : 0      DWR messages Rx  : 67
DWA messages Tx  : 67     DWA messages Rx  : 0
Connection attempts : 1      Connection failures : 0
Transport disc msgs Rx : 0      Unexpected ver msgs Rx : 0
Msgs too big Rx   : 0      Msgs too small Rx   : 0
Invalid CEA msgs Rx : 0      Retransmit msgs Tx   : 0
Rate Limit AAR Drops : 20
Rate Limit STR Drops : 22
Rate Limit ASA Drops : 15
Rate Limit RAA Drops : 08
-----
Number of peer instances : 1
=====

# show mobile-gateway pdn path-mgmt-stats s6b aggregate
=====
```

PDN gateway S6b path management statistics aggregate

```

=====
Messages Tx           : 3           Messages Rx           : 3
CER messages Tx       : 1           CEA messages Rx       : 1
DPR messages Tx       : 0           DPR messages Rx       : 0
DPA messages Tx       : 0           DPA messages Rx       : 0
DWR messages Tx       : 0           DWR messages Rx       : 67
DWA messages Tx       : 67          DWA messages Rx       : 0
Connection attempts   : 1           Connection failures    : 0
Transport disc msgs Rx : 0           Unexpected ver msgs Rx : 0
Msgs too big Rx       : 0           Msgs too small Rx     : 0
Invalid CEA msgs Rx   : 0           Retransmit msgs Tx    : 0
=====

```

```

-----
Number of peer instances : 1
=====

```

show mobile-gateway pdn path-mgmt-stats s6b

Pdn gateway S6b path management statistics

```

=====
Peer address : 19.20.21.23
Port : xxxx Chassis/Card : 1/3
CER messages Tx : 0 CEA messages Rx : 0
DPR messages Rx : 0 DPA messages Tx : 0
DWR messages Tx : 0 DWA messages Rx : 0
-----

```

```

Number of peer instances : 1
=====

```

Table 49: Show PDN path management S6b statistics output fields

Label	Description
S6b	
Messages Tx	The number of messages transmitted
Messages Rx	The number of messages received
CER messages Tx	The number of CER messages transmitted
CEA messages Rx	The number of CER messages received
DPR messages Tx	The number of DPR messages transmitted

Table 49: Show PDN path management S6b statistics output fields (continued)

Label	Description
DPR messages Rx	The number of DPR messages received
DPA messages Tx	The number of DPA messages transmitted
DPA messages Rx	The number of DPR messages received
DWR messages Tx	The number of DWR messages transmitted
DWR messages Rx	The number of DWR messages received
DWA messages Tx	The number of DWA messages transmitted
DWA messages Rx	The number of DWA messages received
Connection attempts	The number of connection attempts
Connection failures	The number of connection failures
Transport disc msgs Rx	The number of transport disconnect messages received
Unexpected ver msgs Rx	The number of unexpected version messages received
Msgs too big Rx	The number of Messages too big received
Msgs too small Rx	The number of Messages too big transmitted
Invalid CEA msgs Rx	The number of invalid CEA messages received
Retransmit msgs Tx	The number of retransmit messages transmitted
S6b Instance	
Rate Limit AAR Drops	The number of AAR rate limit drops

Table 49: Show PDN path management S6b statistics output fields (continued)

Label	Description
Rate Limit STR Drops	The number of STR rate limit drops
Rate Limit ASA Drops	The number of ASA rate limit drops
Rate Limit RAA Drops	The number of RAA rate limit drops

Example output path management statistics S8

The following example shows path management statistics for S8 and *Show PDN path management S5, S5u, S8, Gn, and Gp statistics output fields* describes the output fields. *Show PDN path management statistics common output fields* describes additional common path management output fields.

```
# show mobile-gateway pdn path-mgmt-stats s8
=====
PDN gateway S8 path management statistics
=====
Peer address : 10.20.1.1
Router       : Base
Port        : 2123

Peer restarts      : 0          Path management failures : 0
Peer restart count : 2
Echo requests Rx   : 1          Echo responses Tx      : 1
Echo requests Tx   : 9          Echo responses Rx      : 9
-----
Number of peer instances : 1
=====
```

5.2.2.36 pcf-pcrf-based-pcscf-allocation

Syntax

```
pcf-pcrf-based-pcscf-allocation [entry-id entry-id]
```

Context

```
show>mobile>pdn
```

Description

This command displays the PDN PCF PCRF based P-CSCF allocation configuration data.

Parameters

entry-id

specifies the entry ID

Values: 1 to 128

Output

The following output example shows PDN PCF PCRF based P-CSCF allocation configuration data.

Example output PCF PCRF based P-CSCF allocation statistics

```
# show mobile-gateway pdn pcf-pcrf-based-pcscf-allocation
=====
PCF-PCRF based P-CSCF Allocation
=====
Entry-id: 1
APN : apn.0.1.com
Diameter-Peer/nf-id :          pcscf-srvr-v4-group :    pcscf-srvr-v6-
group :
1 pcrf-1-1                  pcscf-group-1-1      pcscf-group-2-1
-----
Entry-id: 2
APN : apn.ims.3.com
Diameter-Peer/nf-id :          pcscf-srvr-v4-group :    pcscf-srvr-v6-
group :
1 pcrf-5-1                  pcscf-group-5-1      pcscf-group-5-1

# show mobile-gateway pdn pcf-pcrf-based-pcscf-allocation entry-id 2
=====
PCF-PCRF based P-CSCF Allocation
=====
Entry-id: 2
APN : apn.ims.3.com
Diameter-Peer/nf-id :          pcscf-srvr-v4-group :    pcscf-srvr-v6-
group :
1 pcrf-5-1                  pcscf-group-5-1      pcscf-group-5-1
```

5.2.2.37 pcmd-stats

Syntax

```
pcmd-stats [gateway gw-id | group group-number | card slot-number | vm vm-id]] [gw-type sgw
| pgw] [summary]
```

Context

```
show>mobile>pdn
```

Description

This command displays PCMD statistics for the specified gateway group or card and type.

Parameters

gw-id

specifies the gateway ID

Values: 1 to 8

group-number

specifies the group number

Values: 1 to 15

slot-number

specifies the card slot number (applicable only to 7750 SR MG)

Values	1 to 5	SR7
–	1 to 10	SR12

vm-id

specifies the VM ID (applicable only to CMG)

Values: 1 to 20

sgw

specifies the SGW or PGW gateway type

summary

specifies summarized aggregated statistics

- Success records

Number of session records generated with Procedure Result being success. Those are presented per PCMD profile and per gateway lifetime.

Success records for the gateway lifetime are being counted even when the records is not transmitted, that is no destination configured in the PCMD profile.
- Failure records

Number of session records generated with Procedure Result being failure. Those are presented per pcmd profile and per gateway lifetime.

Failure records for the gateway lifetime are being counted even when the records is not transmitted, that is no destination configured in the PCMD profile.

- **Success extended**
Number of session records generated with extended format.
- **Success limited**
Number of session records generated with standard format, while extended format has been configured, due to CPU threshold exceeded.
- **Total Records**
Total number of records generated. This is presented for both session and data records.
- **Total Bytes**
Total number of bytes transmitted. This is presented for both session and data records.

Output

The following 7750 SR MG output examples show PCMD statistics and *Show PDN PCMD statistics output fields* describes the output fields.



NOTE:

The outputs of a CMG system are identical to the 7750 SR MG outputs except they have some CMG-specific fields displayed instead of hardware components (for example, VM is displayed instead of Card).

- *Example output PCMD statistics*
- *Example output PCMD summary statistics*

Example output PCMD statistics

```
# show mobile-gateway pdn pcmd-stats
=====
PDN gateway PCMD statistics
=====
Chassis/Card      : 1/3                Gateway      : 2

PDN PCMD Profile name : test          Profile duration : 0d 00:08:50
PDN PCMD oper. state  : up
Group PCMD Profile name : test-1
Group PCMD oper. state : partial-up
Session records      Data Records
-----
Success records      : 300              Total records      : 100
Failure records      : 405              Total bytes        : 9700
Success extended     : 0                Total suppressed   : 0
Success limited      : 0
Total records        : 705
Total bytes          : 122540
-----
Gateway lifetime statistics
-----
```

```

Success records   : 300                      Failure records : 405
-----
Number of cards   : 1
-----
NOTE: Gateway lifetime statistics represent all generated session records count
      since gateway creation regardless profile is assigned to gateway or not.
=====

```

Example output PCMD summary statistics

```

# show mobile-gateway pdn pcmd-stats summary
=====
PDN gateway PCMD statistics summary
=====
Session records                               Data Records
-----
Success records   : 0                      Total records    : 0
Failure records   : 0                      Total bytes      : 0
Success extended  : 0                      Total suppressed : 0
Success limited   : 0
Total records     : 0
Total bytes       : 0
-----
Gateway lifetime statistics
-----
Success records   : 0                      Failure records   : 0
-----
Number of cards   : 1
-----
NOTE: Gateway lifetime statistics represent all generated session records count
      since gateway creation regardless profile is assigned to gateway or not.
=====

```

Table 50: Show PDN PCMD statistics output fields

Label	Description
Chassis/Card	The chassis and card numbers (applicable only to 7750 SR MG)
VNF/VM	The VNF and VM numbers (applicable only to CMG)
Gateway	The gateway ID
Profile name	The profile name
Profile duration	The duration of the PCMD profile

Table 50: Show PDN PCMD statistics output fields (continued)

Label	Description
PCMD oper. state	The PCMD operational state
Success records	The number of session records generated with Procedure Result being success
Total records	The number of total records generated (session and data records)
Failure records	The number of session records generated with Procedure Result being failure
Total bytes	The number of total bytes transmitted (session and data records)
Success extended	The number of session records generated with extended format
Total suppressed	The number of CDATA records not generated, due to CPU threshold exceeded
Success limited	The number of session records generated with standard format, while extended format has been configured, due to CPU threshold exceeded
Number of cards	The number of cards retrieved (applicable only to 7750 SR MG)
Number of VMs	The number of VMs retrieved (applicable only to CMG)

5.2.2.38 pdn-context

Syntax

```
pdn-context [imsi | imei | msisdn string detail] [apn name] [epdg] [gateway gw-id | group
group-number | card slot-number] [nemo]
```

```
pdn-context imsi | imei | msisdn string [apn name] [gateway gw-id] acct {rf | gy | radius}
[detail]
```

```

pdn-context imsi | imei | msisdn string gx-usage-monitoring [gateway gw-id] [apn name]
[monitoring-key key-name]

pdn-context imsi | imei | msisdn string gx-time-of-day-info [gateway gw-id] [apn name]

pdn-context imsi string detail {epdg | epdg-decrypt}

pdn-context mac ieee-address [apn name] [gateway gw-id] group group-number | card slot-
number] [detail]

pdn-context default-session [def-session-pgw-ip pgw-ip-address [apn name] detail]
[gateway gw-id] group group-number | card slot-number] [nemo]

pdn-context default-session acct {rf | gy} [def-session-pgw-ip pgw-ip-address] [apn
name] [gateway gw-id] [detail]

pdn-context ipsec-lockout [summary]

pdn-context ipsec-lockout idi IDi source-ip source-ip-address

pdn-context subscriber-id subscriber-id [apn name] [gateway gw-id] group group-number |
card slot-number] [access-type access-type] [nemo]

pdn-context subscriber-id subscriber-id detail acct {rf | gy | radius} [apn name]
[gateway gw-id]

pdn-context default-session [apn name] [gateway gw-id] group group-number | card slot-
number] [def-session-pgw-ip pgw-ip-address] [session-type {transit-pass-through |
transit-capture}] [detail]

pdn-context imsi | subscriber-id string [gateway gw-id] [apn name] acct {stats}

pdn-context ue-ip ue-ip network-instance network-inst detail

pdn-context up-seid string detail

pdn-context cp-seid string [cp-ipv4 ipv4-address cp-ipv6 ipv6-address] detail

pdn-context def-pfcp-u-sessions [network-realm realm-name] [pfcp-u-peer-ip ip-
address] [gateway gw-id] group group-number | vm vm-id] [apn name]

pdn-context wrap [gateway gw-id] group group-number | vm vm-id] [apn name]

pdn-context field-name [gateway gw-id] group group-number | vm vm-id] [apn name]

```

Context

show>mobile>pdn

Description

This command displays the current PDN contexts for each card.



NOTE:

- If the `config>mobile>pdn>allow-duplicate-msisdn` command is enabled, the `show>mobile>pdn>pdn-context>msisdn` command does not show the existing sessions of the specified MSISDN.
- If the `config>mobile>pdn>allow-duplicate-imei` command is enabled, the `show>mobile>pdn>pdn-context>imei` command does not show the existing sessions of the specified IMEI.
- The `eMPS IMS priority session` field is updated from both the MPS and the WPS detected calls; currently, for ePDG this field is updated only by the WPS calls.

Parameters

- gw-id*

specifies the gateway ID

Values: 1 to 8
- name*

specifies the APN name for the context, up to 80 characters
- slot-number*

specifies the card slot number
- group-number*

specifies the group number

Values: 1 to 8
- Idi*

specifies the initiator ID for the IPsec authentication lockout list
- ieee-address*

specifies the UE MAC
- source-ip-address*

specifies the source IP address for the IPsec authentication lockout list

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x: [0..FFFF]H

– – d: [0..255]D

imsi | imei | msisdn *string*

specifies the UE's IMSI or IMEI or MSISDN

subscriber-id

string value, up to 32 characters, stored in the Subscriber Profile Repository (SPR) that identifies each subscriber in the network; the value is assigned and maintained by individual operators and varies in format

acct rf

displays the RF context statistics

acct radius

displays the RADIUS accounting statistics

detail

displays the detailed summary information for the PDN context

epdg-decrypt

displays the decryption keys for both IKE and ESP SAs

gx-usage-monitoring

displays the Gx usage monitoring information for the specified IMSI, IMEI, or MSISDN

Displayed information for each IP-CAN session includes the usage monitoring instances and the list of associated usage monitoring-keys.

ipsec-lockout

displays IPsec lockout statistics for the PDN context

monitoring-key

displays detailed information for the monitoring key instance (scope, GSU, GSU-scope, usage-reported, number of threshold reports and on-demand reports)

When the usage monitoring key scope is at the PCC rule level, the associated PCC rule names are also displayed.

gx-time-of-day-info

displays the Gx Time of Day procedures related information for the specified IMSI for each IP-CAN session including the current Revalidation-Time applicable for the session, Rule-Activation-Time, Rule-Deactivation-Time and the corresponding applicable rules

up-seid *string*

specifies the Session Endpoint Identifier for the UE session context, up to 40 characters

cp-seid *string*

specifies the Session Endpoint Identifier for the CP session context, up to 40 characters

realm-name

specifies the realm name of the network instance that has the PFCP GTP-U interface on the UP function, up to 80 characters

ip-address

specifies the CP function's Sx/N4 reference point IPv4 address

Values: a.b.c.d

ipv6-address

specifies the CP function's Sx/N4 reference point IPv6 address, up to 64 characters

summary

displays summary information for IPsec lockout statistics

field-name

sets for printing the list of UEs together with the selected field

wrap

specifies that the UE IP address column is not truncated and the full IP address is visible by wrapping the output in the next line displays

Output

The following output examples show PDN context information.

**NOTE:**

Depending on the configured node function (UP or CP function), the description and value for some of the output fields differ. These fields, if present, are identified and described at the end of each example output.

- [Example output PDN context statistics](#)
- [Example output PDN context statistics \(CP function\)](#)
- [Example output PDN context statistics detail \(CP function\)](#)
- [Example output PDN context statistics \(UP function\)](#)
- [Example output PDN context statistics on session management function](#)
- [Example output PDN context statistics combined SGW/PGW](#)
- [Example output PDN context statistics detail combined SGW/PGW](#)
- [Example output PDN context statistics detail SGW](#)
- [Example output PDN context detail statistics TWAG](#)
- [Example output PDN context detail statistics S4/S12](#)
- [Example output PDN context detail statistics S2a](#)

- *Example output PDN context detail statistics S2b*
- *Example output PDN context detail statistics Gn*
- *Example output PDN context detail statistics Gn/Gp IPv6 address support*
- *Example output PDN context detail statistics S2a PMIP*
- *Example output PDN context detail SMF*
- *Example output PDN context detail I-SMF*
- *Example output PDN context statistics detail Ethernet PDU session SMF*
- *Example output PDN context statistics detail Ethernet PDU session UPF*
- *Example output PDN context statistics Gy detail combined SGW/PGW*
- *Example output PDN context statistics Rf detail*
- *Example output PDN context statistics RADIUS detail*
- *Example output PDN context IMEI statistics (UP function)*
- *Example output PDN context IMEI detail statistics (UP function)*
- *Example output PDN context IMEI detail account statistics (UP function)*
- *Example output PDN context IMSI statistics (UP function)*
- *Example output PDN context IMSI detail statistics (UP function)*
- *Example output PDN context IMSI detail account statistics (UP function)*
- *Example output PDN context MSISDN statistics (UP function)*
- *Example output PDN context MSISDN detail statistics (UP function)*
- *Example output PDN context MSISDN detail account statistics (UP function)*
- *Example output PDN context statistics detail NEMO session*
- *Example output PDN context statistics detail eMPS advance priority calls*
- *Example output PDN context statistics detail virtual APN selection*
- *Example output PDN context statistics dedicated bearer*
- *Example output PDN context statistics Gx*
- *Example output PDN context Gx time of day information statistics (time-conditioned policy information)*
- *Example output PDN context statistics MAC detail*
- *Example output PDN context statistic (MAPCON)*
- *Example output PDN context combination ePDG/PGW*
- *Example output PDN context IMSI detail combination ePDG/PGW*

- [Example output PDN context ePDG/ decrypt IMSI Detail](#)
- [Example output PDN context IMSI/IMEI detail ePDG for emergency calls over WLAN](#)
- [Example output PDN context IPsec lockout summary](#)
- [Example output PDN context IPsec lockout IDi source IP](#)
- [Example output PDN context APN](#)
- [Example output PDN context subscriber ID](#)
- [Example output PDN context subscriber ID detail](#)
- [Example output PDN context RADIUS accounting detail](#)
- [Example output PDN context MSISDN detail](#)
- [Example output PDN context field name](#)
- [Example output UP function PDN context SEID \(UP function\)](#)
- [Example output UP function PDN context detail SEID \(UP function\)](#)
- [Example output PDN context wrap](#)

Example output PDN context statistics

```
# show mobile-gateway pdn pdn-context imsi 3100120000000000
=====
IMSI/IMEI(#)      APN      Type    Bearers  UE Address (IPv4/IPv6)  Ref-pt/Si*
/MSISDN(^)
=====
3100120000000000  ipd.noki* IPv4    1        5.64.0.2/-              S2a/GTPv2
-----
Number of PDN instances : 1
-----
# indicates that the corresponding row is IMEI entry.
^ indicates that the corresponding row is MSISDN entry.
-----
* indicates that the corresponding row element may have been truncated.
```

Example output PDN context statistics (CP function)

```
# show mobile-gateway pdn pdn-context
=====
IMSI/IMEI(#)      APN      Type    Beare* UE Address (IPv4/IPv6)  Ref-pt/Si*
/MSISDN(^)/MAC(`)
/SubId/SEID(~)
=====
7777777777777700  nokia.ne* IPv4    1        5.0.255.0/-             S5/GTPv2
-----
Number of PDN instances : 1
-----
# indicates that the corresponding row is IMEI entry.
^ indicates that the corresponding row is MSISDN entry.
` indicates that the corresponding row is MAC entry.
~ indicates that the corresponding row is SEID entry.
```

 * indicates that the corresponding row element may have been truncated.

Example output PDN context statistics detail (CP function)

```
# show mobile-gateway pdn pdn-context imsi 123456789012048 detail
=====
PDN context detail
=====
IMSI           : 123456789012048
APN            : ipd.apn1.com
Public APN     : N/A
Virtual APN    : virtual-100
Virtual APN sel : rat-type
Virtual APN sel user-range type : rat-type
Virtual APN sel scope : apn
Auth Status    : IMSI authenticated
IMEI           : 987654321012399
MSISDN         : 12345098761
Data Session Id : 0x00010100
PDN type       : IPv6                      LBI           : 5
Ref-point/Sig-pr*: S5/GTPv2              APN restriction : 0
PRA ID         : 0x0                      PRA Status    : -
UL APN AMBR    : 5000 Kbps                DL APN AMBR    : 5000 Kbps
QFIs           : 0
Bearer contexts : 1                      SDFs           : 0
Piggyback bearers requested : 0
Piggyback bearers creation successful : 0

Primary DNS IPv4 address : 192.168.10.110
Secondary DNS IPv4 address : 0.0.0.0
Primary DNS IPv6 address : ::
Secondary DNS IPv6 address : ::

Pri NBNS IPv4 ad*: 0.0.0.0                Sec NBNS IPv4 ad*: 0.0.0.0
Pri NBNS IPv6 ad*: ::                     Sec NBNS IPv6 ad*: ::

Charging bearer *: Visiting

UE IPv6 address : 2001:db8::211:ff:11
UE IPv6 address source : Local Pool

S5 PGW Ctrl Teid : 0x10100                S5 SGW Ctrl Teid : 0x10100
S5 PGW V4 Ctl Adr: 192.168.10.1           S5 SGW V4 Ctl Adr: 192.168.10.2
S5 PGW V6 Ctl Adr: 2001:db8::1010:1010
S5 SGW V6 Ctl Adr: 2001:db8::a0a:501

PCRF triggers : N/A
Gx primary peer profile : Peer-Profile1
Gx secondary peer profile : Peer-Profile3
Gx peer profile in use : Peer-Profile1
Gx failover count : 0
Gx peer origin : Local configuration
PCRF hostname :
PCRF address : 192.168.43.1
PCRF port : 3868
```

```

PCRF destination host      : primary-pcrf.dest.com
Gx local address          : 192.168.16.19
PCRF enabled Gy Event Reporting :

Charging profile : 1                      Interim records *: 0
RF triggers      : 0

S6b primary peer profile :
S6b secondary peer profile :
S6b peer profile in use :
S6b failover count      : 0
S6b peer hostname       :
S6b peer address        : ::
S6b peer port           : 0
S6b peer dest host      :
S6b peer origin         :
S6b local address       : ::

Framed Routes      : 0
Redir Mob Ipv4 Addr : ::
Redir Mob Ipv6 Addr : ::
L2TP Connection *: 0

Restoration Priority Level : 0

P-CSCF Reselection      : Y
Registered P-CSCF Addr  : ::

eMPS IMS priority session : No

Alloc ePDG svr Ipv4 Addr : ::
Alloc ePDG svr Ipv6 Addr : ::

Override Rating Group      : N/A
Override Online Billing Method : Disabled
Override Offline Billing Method : Disabled
Serving PLMN rat*: disabled
CPOPCI      : disabled          EPCOSI      : disabled
PGW UL Rate Control      : disabled
MO Exception Cou*: 0          MO Exception Tim*: 0

CP Sx-N4 Seid      : 0x10100          UP Sx-N4 Seid      : 0x10100
CP Sx-N4 Ctl V4 *: 192.168.33.1      UP Sx-N4 Ctl V4 *: 192.168.34.1
CP Sx-N4 Ctl V6 *: 2001:db8::2121:101  UP Sx-N4 Ctl V6 *: 2001:db8::2122:101
PDRs      : 2                      FARS      : 2
QERs      : 1                      BAR ID    : 5
-----
Number of PDN contexts : 1
=====
* indicates that the corresponding row element may have been truncated.

```

Example output PDN context statistics (UP function)

```

# show mobile-gateway pdn pdn-context
=====
IMSI/IMEI(##)      APN      Type      Beare* UE Address (IPv4/IPv6)  Ref-pt/Si*

```

```

/MSISDN(^)/MAC(`)
/SubId/SEID(~)
=====
0x54110~          ipd.nokia* IPv4    1      1.127.255.0/-      N/A
460100001003204
-----
Number of PDN instances : 1
-----
# indicates that the corresponding row is IMEI entry.
^ indicates that the corresponding row is MSISDN entry.
` indicates that the corresponding row is MAC entry.
~ indicates that the corresponding row is SEID entry.
-----
* indicates that the corresponding row element may have been truncated.

```

Example output PDN context statistics on session management function

```

# show mobile-gateway pdn pdn-context
=====
IMSI/IMEI(#)      APN          Type    Beare* UE Address (IPv4/IPv6)  Ref-pt/Si*
/MSISDN(^)/MAC(`)
/SubId/SEID(~)
=====
111111114000000  ipd.noki* IPv4    2      1.127.255.0/-      S5/GTPv2
111111115000000  ipd.noki* IPv4    2      2.127.255.1/-      NA
111111112100000  ipd.noki* IPv4    2      1.127.255.2/-      S2a/GTPv2
111111111100000  ipd.noki* IPv4    2      2.127.255.0/-      NA
111111116000000  cg2.ipd.* IPv4    2      6.6.6.6/-          S5/GTPv2
111111112000000  cg1.ipd.* IPv4    2      5.5.5.5/-          S5/GTPv2
111111117000000  cg2.ipd.* IPv4    2      6.6.6.7/-          S5/GTPv2
111111113000000  cg1.ipd.* IPv4    2      5.5.5.6/-          S5/GTPv2
111111118000000  ipd.noki* IPv4    2      1.127.255.1/-      Gn/GTPv1
111111120000000  ipd.noki* IPv4    2      3.127.255.0/-      S2b/GTPv2
-----
Number of PDN instances : 10
-----
# indicates that the corresponding row is IMEI entry.
^ indicates that the corresponding row is MSISDN entry.
` indicates that the corresponding row is MAC entry.
~ indicates that the corresponding row is SEID entry.
-----
* indicates that the corresponding row element may have been truncated.

```

Example output PDN context statistics combined SGW/PGW

```

# show mobile-gateway pdn pdn-context
=====
IMSI/IMEI(#)      APN          Type    Beare* UE Address (IPv4/IPv6)  Ref-pt/Si*
/MSISDN(^)/MAC(`)
/SubId/SEID(~)
=====
123456789012926  nokia.co* IPv6    1      -/3ffe:0:1:ff02::2    NA
-----
Number of PDN instances : 1
-----
# indicates that the corresponding row is IMEI entry.
^ indicates that the corresponding row is MSISDN entry.

```

` indicates that the corresponding row is MAC entry.

* indicates that the corresponding row element may have been truncated.

Example output PDN context statistics detail combined SGW/PGW

```
# show mobile-gateway pdn pdn-context imsi 110100000000000 detail
=====
PDN context detail
=====
IMSI           : 110100000000000
APN            : apn.1.1.com
Auth Status    : IMSI authenticated
IMEI           : 110100000000000
MSISDN         : 110100000000
PDN type       : IPv4                      LBI           : 5
S5 sig protocol : GTP                      APN restriction : 0
UL APN AMBR    : 5000 Kbps                 DL APN AMBR    : 5000 Kbps
Bearer contexts : 3

eMPS IMS priority session : No

UE IPv4 address  : 10.10.255.0

S5 PGW Ctrl Teid : 0x10100                S5 SGW Ctrl Teid : 0x10100
S5 PGW V4 Ctl Adr: 192.168.5.1             S5 SGW V4 Ctl Adr: 192.168.16.16
S5 PGW V6 Ctl Adr: 3ffe::a0a:501
S5 SGW V6 Ctl Adr: 3ffe::1010:1010

Charging level  : Bearer                    Bearer type     : Home
-----
IMSI           : 110100000000000
APN            : apn.1.2.com
Public APN     : N/A
Virtual APN    : N/A
Virtual APN sel : -
Virtual APN sel user-range type : -
Virtual APN sel scope : N/A
Auth Status    : IMSI authenticated
IMEI           : 110100000000000
MSISDN         : 110100000000
Data Session Id : 0x00020100                AA Session Id   : 0x00040002
PDN type       : IPv6                      LBI           : 11
Ref-point/Sig-pr*: N/A                    APN restriction : 0
PCRF Subscribed PRAs
PRA ID         : 0x112233                  PRA Status     : Inactive
PRA ID         : 0x445566                  PRA Status     : In
PRA ID         : 0x778899                  PRA Status     : Out
OCS Subscribed PRAs
PRA ID         : 0x112233                  PRA Status     : Inactive
PRA ID         : 0x445566                  PRA Status     : In
PRA ID         : 0x778899                  PRA Status     : Out
UL APN AMBR    : 5000 Kbps                 DL APN AMBR    : 5000 Kbps
Bearer contexts : 3                        SDFs           : 2
Piggyback bearers requested                : 0
Piggyback bearers creation successful       : 0
```

```
Primary DNS IPv4 address   : ::
Secondary DNS IPv4 address : ::
Primary DNS IPv6 address   : ::
Secondary DNS IPv6 address : ::

Pri NBNS IPv4 ad*: ::          Sec NBNS IPv4 ad*: ::
Pri NBNS IPv6 ad*: ::          Sec NBNS IPv6 ad*: ::

Charging bearer *: Visiting

UE IPv6 address           : 3ffe:0:2:ff00::ffff:d8d7
UE IPv6 address source    : Local Pool

PCRF triggers      : SGSN QOS RAT BR_LOSS BR_RVRY RAI ULI IPCAN_CHANGE
                   DEF_BR_QOS AN_GW RSRG_ALLOC RSRG_MOD TZ OUT_CRDT
                   REALLOC_CRDT REVAL_TO PLMN
Gx primary peer profile : Peer-Profile1
Gx secondary peer profile : Peer-Profile3
Gx peer profile in use   : Peer-Profile1
Gx failover count        : 0
Gx peer origin           : Local configuration
PCRF hostname            :
PCRF address             : 10.10.43.2
PCRF port                : 3868
PCRF destination host    : primary-pcrf.nokia.com
Gx local address         : 192.168.16.19
PCRF enabled Gy Event Reporting :

Charging profile : 1              Interim records *: 0
RF triggers      : 0

S6b primary peer profile :
S6b secondary peer profile :
S6b peer profile in use   :
S6b failover count        : 0
S6b peer hostname         :
S6b peer address          : ::
S6b peer port             : 0
S6b peer dest host        :
S6b peer origin           :
S6b local address         : ::

Framed Routes      : 0
Redir Mob Ipv4 Addr : ::
Redir Mob Ipv6 Addr : ::
L2TP Connection *: 0

Restoration Priority Level : 0

P-CSCF Reselection      : Y
Registered P-CSCF Addr   : ::

eMPS IMS priority session : No

Alloc ePDG svr Ipv4 Addr : ::
```



```

Alloc ePDG svr Ipv6 Addr   : ::

Override Rating Group      : N/A
Override Online Billing Method : Disabled
Override Offline Billing Method : Disabled
Serving PLMN rat*: disabled
CPOPCI      : disabled          EPCOSI      : disabled
PGW UL Rate Control      : disabled
DL APN rate lmt Drp Pkt   : 0
UL APN rate lmt Drp Pkt   : 0
PLMN rate lmt Drp Pkt     : 0
MO Exception Cou*: 0          MO Exception Tim*: 0
-----
Number of PDN contexts : 2
=====
* indicates that the corresponding row element may have been truncated.

```

Example output PDN context statistics detail SGW

```

# show mobile-gateway pdn pdn-context imsi 112600000000000 detail
=====
PDN context detail
=====
IMSI           : 112600000000000
APN            : cg1.ipd.nokia.com
Auth Status    : IMSI authenticated
IMEI          : 112600000000000
MSISDN        : 112600000000
PDN type       : IPv4                LBI           : 5
S5 sig protocol : GTP                APN restriction : 0
UL APN AMBR    : 5000 Kbps           DL APN AMBR    : 5000 Kbps
Bearer contexts : 1

eMPS IMS priority session : No

UE IPv4 address : 10.15.5.5

S5 PGW Ctrl Teid : 0x31000000          S5 SGW Ctrl Teid : 0x30170
S5 PGW V4 Ctl Adr: 16.80.16.10         S5 SGW V4 Ctl Adr: 16.90.16.10
S5 SGW V6 Ctl Adr: 3ffe::105a:100a

Charging level  : Bearer                Bearer type    : Home
-----
Number of PDN contexts : 1
=====

```

Example output PDN context detail statistics TWAG

```

# show mobile-gateway pdn pdn-context imsi 310012000000000 detail
=====
PDN context detail
=====
IMSI/MAC : 310012000000000/00:02:00:00:00:00
APN       : ipd.nokia.1.1.com.mnc012.mcc310.gprs
Auth Status : IMSI authenticated
IMEI       : N/A

```

```

PDN type       : IPv4                LBI                : 5
Ref-point/Sig-pr*: SWw/Soft-GRE      APN restriction   : 255
UL APN AMBR    : 5000 Kbps           DL APN AMBR       : 5000 Kbps
UE IPv4 address : 5.0.255.2
TWAG
Access type    : eap
STA/RADIUS     : STA
SaMOG/NSW0/Combo : SaMOG
TWAG MAC address : 74:7f:ff:00:03:5b
AP MAC address  : 99:02:00:00:00:00
SWw TWAG tunnel addr : 16.81.16.17
SWw AP tunnel addr  : 180.1.3.2
SWw RADIUS TWAG Auth Addr : 16.81.16.16
SWw RADIUS AP Auth Addr  : 180.1.3.2
STa TWAG Addr      : 16.82.16.16
STa AAA Addr       : 180.1.2.200
RADIUS AAA Addr    : 0.0.0.0
RADIUS Acct addr   : 0.0.0.0
S2a TWAG Ctrl Teid : 0x50d90
S2a PGW Ctrl Teid  : 0x41940
S2a TWAG V4 Ctl Adr : 16.16.16.16      S2a PGW V4 Ctl Adr : 10.10.5.1
S2a TWAG V6 Ctl Adr: -
S2a PGW V6 Ctl Adr : -
DHCP remaining lease time (sec): 0
Remaining cache time (sec)      : 3755
-----
Number of PDN contexts : 1
=====
* indicates that the corresponding row element may have been truncated.

```

Example output PDN context detail statistics S4/S12

```

# show mobile-gateway pdn pdn-context imsi 1105000000000000 detail
=====
PDN context detail
=====
IMSI           : 1105000000000000
APN            : apn.1.2.com
Public APN     : N/A
Virtual APN    : N/A
Virtual APN sel : -
Virtual APN sel user-range type : -
Virtual APN sel scope : N/A
Auth Status    : IMSI authenticated
IMEI           : 1105000000000000
MSISDN        : 110500000000
Data Session Id : 0x00020170      AA Session Id : 0x0005c002
PDN type       : IPv6            LBI                : 11
Ref-point/Sig-pr*: N/A          APN restriction   : 0
PCRF Subscribed PRAs
PRA ID        : 0x112233          PRA Status      : Inactive
PRA ID        : 0x445566          PRA Status      : In
PRA ID        : 0x778899          PRA Status      : Out
OCS Subscribed PRAs
PRA ID        : 0x112233          PRA Status      : Inactive
PRA ID        : 0x445566          PRA Status      : In

```

```

PRA ID          : 0x778899          PRA Status      : Out
UL APN AMBR     : 5000 Kbps         DL APN AMBR     : 5000 Kbps
Bearer contexts  : 3                 SDFs            : 2
Piggyback bearers requested      : 0
Piggyback bearers creation successful : 0

Primary DNS IPv4 address  : ::
Secondary DNS IPv4 address : ::
Primary DNS IPv6 address  : ::
Secondary DNS IPv6 address : ::

Pri NBNS IPv4 ad*: ::          Sec NBNS IPv4 ad*: ::
Pri NBNS IPv6 ad*: ::          Sec NBNS IPv6 ad*: ::

Charging bearer *: Visiting

UE IPv6 address          : 3ffe:0:2:ff03::ffff:dfe4
UE IPv6 address source   : Local Pool

PCRF triggers           : SGSN QOS RAT BR_LOSS BR_RVRY RAI ULI IPCAN_CHANGE
                        DEF_BR_QOS AN_GW RSRC_ALLOC RSRC_MOD TZ OUT_CRDT
                        REALLOC_CRDT REVAL_TO PLMN
Gx primary peer profile :
Gx secondary peer profile :
Gx peer profile in use   :
Gx failover count        : 0
Gx peer origin           : (Not Specified)
PCRF hostname            : (Not Specified)
PCRF address             : ::
PCRF port                : 0
PCRF destination host    : (Not Specified)
Gx local address         : ::
PCRF enabled Gy Event Reporting :

Charging profile : 1          Interim records *: 0
RF triggers      : 0

S6b primary peer profile :
S6b secondary peer profile :
S6b peer profile in use   :
S6b failover count        : 0
S6b peer hostname         :
S6b peer address          : ::
S6b peer port             : 0
S6b peer dest host        :
S6b peer origin           :
S6b local address         : ::

Framed Routes           : 0
Redir Mob Ipv4 Addr     : ::
Redir Mob Ipv6 Addr     : ::
L2TP Connection *: 0

Restoration Priority Level : 0

P-CSCF Reselection      : Y

```

```

Registered P-CSCF Addr      : ::

eMPS IMS priority session   : No

Alloc ePDG svr Ipv4 Addr    : ::
Alloc ePDG svr Ipv6 Addr    : ::

Override Rating Group       : N/A
Override Online Billing Method : Disabled
Override Offline Billing Method : Disabled
Serving PLMN rat*: disabled
CPOPCI      : disabled          EPCOSI      : disabled
PGW UL Rate Control         : disabled
DL APN rate lmt Drp Pkt     : 0
UL APN rate lmt Drp Pkt     : 0
PLMN rate lmt Drp Pkt       : 0
MO Exception Cou*: 0         MO Exception Tim*: 0
-----
Number of PDN contexts : 1
=====
* indicates that the corresponding row element may have been truncated.

```

Example output PDN context detail statistics S2a

```

# show mobile-gateway pdn pdn-context imsi 1109000000000000 detail
=====
PDN context detail
=====
IMSI           : 1109000000000000
APN            : apn.1.2.com
Public APN     : N/A
Virtual APN    : N/A
Virtual APN sel           : -
Virtual APN sel user-range type : -
Virtual APN sel scope     : N/A
Auth Status      : IMSI authenticated
IMEI            : 1109000000000000
MSISDN         : 110900000000
Data Session Id  : 0x000201a0      AA Session Id   : 0x00068002
PDN type        : IPv6             LBI             : 11
Ref-point/Sig-pr*: S2a/GTPv2      APN restriction  : 0
PCRF Subscribed PRAs
OCS Subscribed PRAs
UL APN AMBR     : 5000 Kbps        DL APN AMBR     : 5000 Kbps
Bearer contexts : 3                SDFs           : 2
Piggyback bearers requested       : 0
Piggyback bearers creation successful : 0

Primary DNS IPv4 address : ::
Secondary DNS IPv4 address : ::
Primary DNS IPv6 address  : ::
Secondary DNS IPv6 address : ::

Pri NBNS IPv4 ad*: ::          Sec NBNS IPv4 ad*: ::
Pri NBNS IPv6 ad*: ::          Sec NBNS IPv6 ad*: ::

```

Charging bearer *: Visiting

UE IPv6 address : 3ffe:0:2:ff07::ffff:ed5a
 UE IPv6 address source : Local Pool

S2a PGW Ctrl Teid: 0x201a0 S2a TWAG Ctrl Te*: 0xd0000002
 S2a PGW V4 Ctl A*: 10.60.16.16 S2a TWAG V4 Ctl *: 10.70.24.101
 S2a PGW V6 Ctl Adr: 3ffe::1010:1010

PCRF triggers : SGSN QOS RAT BR_LOSS BR_RVRY RAI ULI IPCAN_CHANGE
 DEF_BR_QOS AN_GW RSRC_ALLOC RSRC_MOD TZ OUT_CRDT
 REALLOC_CRDT REVAL_TO PLMN

Gx primary peer profile :
 Gx secondary peer profile :
 Gx peer profile in use :
 Gx failover count : 0
 Gx peer origin : (Not Specified)
 PCRF hostname : (Not Specified)
 PCRF address : ::
 PCRF port : 0
 PCRF destination host : (Not Specified)
 Gx local address : ::
 PCRF enabled Gy Event Reporting :

Charging profile : 1 Interim records *: 0
 RF triggers : 0

S6b primary peer profile : s6b_Peer-Profile1
 S6b secondary peer profile : s6b_Peer-Profile2
 S6b peer profile in use : s6b_Peer-Profile1
 S6b failover count : 0
 S6b peer hostname :
 S6b peer address : ::
 S6b peer port : 0
 S6b peer dest host : aaa.nokia.com
 S6b peer origin : mscpl01.diam.origRealm.com
 S6b local address : ::

Framed Routes : 0
 Redir Mob Ipv4 Addr : ::
 Redir Mob Ipv6 Addr : ::
 L2TP Connection *: 0

Restoration Priority Level : 0

P-CSCF Reselection : Y
 Registered P-CSCF Addr : ::

eMPS IMS priority session : No

Alloc ePDG svr Ipv4 Addr : ::
 Alloc ePDG svr Ipv6 Addr : ::

Override Rating Group : N/A
 Override Online Billing Method : Disabled
 Override Offline Billing Method : Disabled
 Serving PLMN rat*: disabled

```

CPOPCI          : disabled          EPCOSI          : disabled
PGW UL Rate Control : disabled
DL APN rate lmt Drp Pkt : 0
UL APN rate lmt Drp Pkt : 0
PLMN rate lmt Drp Pkt : 0
MO Exception Cou*: 0                MO Exception Tim*: 0
-----
Number of PDN contexts : 1
=====
* indicates that the corresponding row element may have been truncated.

```

Example output PDN context detail statistics S2b

```

# show mobile-gateway pdn pdn-context imsi 111000000000000 detail
=====
PDN context detail
=====
IMSI           : 111000000000000
APN            : apn.1.2.com
Public APN     : N/A
Virtual APN    : N/A
Virtual APN sel : -
Virtual APN sel user-range type : -
Virtual APN sel scope : N/A
Auth Status    : IMSI authenticated
IMEI           : 111000000000000
MSISDN         : 111000000000
Data Session Id : 0x000201b0      AA Session Id : 0x0006c002
PDN type       : IPv6             LBI           : 11
Ref-point/Sig-pr*: S2b/GTPv2     APN restriction : 0
PCRF Subscribed PRAs
OCS Subscribed PRAs
UL APN AMBR    : 5000 Kbps        DL APN AMBR    : 5000 Kbps
Bearer contexts : 3                SDFs          : 2
Piggyback bearers requested : 0
Piggyback bearers creation successful : 0

Primary DNS IPv4 address : ::
Secondary DNS IPv4 address : ::
Primary DNS IPv6 address : ::
Secondary DNS IPv6 address : ::

Pri NBNS IPv4 ad*: ::             Sec NBNS IPv4 ad*: ::
Pri NBNS IPv6 ad*: ::             Sec NBNS IPv6 ad*: ::

Charging bearer *: Visiting

UE IPv6 address : 3ffe:0:2:ff08::ffff:ef4c
UE IPv6 address source : Local Pool

S2b PGW Ctrl Teid: 0x201b0        S2b ePDG Ctrl Te*: 0xc0000002
S2b PGW V4 Ctl A*: 10.70.16.19    S2b ePDG V4 Ctl *: 10.10.7.74
S2b PGW V6 Ctl Adr: 3ffe::1046:1013

PCRF triggers : SGSN QOS RAT BR_LOSS BR_RVRY RAI ULI IPCAN_CHANGE
               DEF_BR_QOS AN_GW RSRC_ALLOC RSRC_MOD TZ OUT_CRDT

```

```

REALLOC_CRDT REVAL_TO PLMN
Gx primary peer profile      :
Gx secondary peer profile    :
Gx peer profile in use       :
Gx failover count            : 0
Gx peer origin                : (Not Specified)
PCRF hostname                 : (Not Specified)
PCRF address                  : ::
PCRF port                     : 0
PCRF destination host        : (Not Specified)
Gx local address              : ::
PCRF enabled Gy Event Reporting :

Charging profile : 1                      Interim records *: 0
RF triggers      : 0

S6b primary peer profile : s6b_Peer-Profile1
S6b secondary peer profile : s6b_Peer-Profile2
S6b peer profile in use : s6b_Peer-Profile1
S6b failover count      : 0
S6b peer hostname       :
S6b peer address        : ::
S6b peer port           : 0
S6b peer dest host      : aaa.nokia.com
S6b peer origin         : mscpl01.diam.origRealm.com
S6b local address       : ::

Framed Routes : 0
Redir Mob Ipv4 Addr : ::
Redir Mob Ipv6 Addr : ::
L2TP Connection *: 0

Restoration Priority Level : 0

P-CSCF Reselection : Y
Registered P-CSCF Addr : ::

eMPS IMS priority session : No

Alloc ePDG svr Ipv4 Addr : ::
Alloc ePDG svr Ipv6 Addr : ::

Override Rating Group : N/A
Override Online Billing Method : Disabled
Override Offline Billing Method : Disabled
Serving PLMN rat*: disabled
CPOPCI : disabled          EPCOSI : disabled
PGW UL Rate Control : disabled
DL APN rate lmt Drp Pkt : 0
UL APN rate lmt Drp Pkt : 0
PLMN rate lmt Drp Pkt : 0
MO Exception Cou*: 0          MO Exception Tim*: 0

WLAN Location Information
SSID : (Not Specified)
BSSID :
Civic-address : (Not Specified)

```

Operator-Name : (Not Specified)

Number of PDN contexts : 1

* indicates that the corresponding row element may have been truncated.

Example output PDN context detail statistics Gn

```
# show mobile-gateway pdn pdn-context imsi 110600000000000 detail
=====
PDN context detail
=====
IMSI           : 110600000000000
APN            : apn.1.2.com
Public APN     : N/A
Virtual APN    : N/A
Virtual APN sel                : -
Virtual APN sel user-range type : -
Virtual APN sel scope          : N/A
Auth Status      : IMSI authenticated
IMEI             : 110600000000000
MSISDN          : 11060000000
Data Session Id  : 0x00020140      AA Session Id   : 0x00050002
PDN type         : IPv6             APN restriction  : 0
PCRF Subscribed PRAs
OCS Subscribed PRAs
UL APN AMBR      : 1984 Kbps        DL APN AMBR      : 4992 Kbps
Bearer contexts  : 3                SDFs             : 2
Piggyback bearers requested      : 0
Piggyback bearers creation successful : 0

Primary DNS IPv4 address : ::
Secondary DNS IPv4 address : ::
Primary DNS IPv6 address : ::
Secondary DNS IPv6 address : ::

Pri NBNS IPv4 ad*: ::          Sec NBNS IPv4 ad*: ::
Pri NBNS IPv6 ad*: ::          Sec NBNS IPv6 ad*: ::

Charging bearer *: Visiting

UE IPv6 address      : 3ffe:0:2:ff04::ffff:e49a
UE IPv6 address source : Local Pool

Gn PGW Ctrl Teid : 0x20140      Gn SGSN Ctrl Teid: 0xb0000002
Gn PGW V4 Ctl Adr: 10.16.16.16  Gn SGSN V4 Ctl A*: 10.23.15.2
Gn PGW V6 Ctl Adr: 3ffe::1010:1010

PCRF triggers       : N/A
Gx primary peer profile :
Gx secondary peer profile :
Gx peer profile in use :
Gx failover count     : 0
Gx peer origin         : (Not Specified)
PCRF hostname          : (Not Specified)
PCRF address           : ::
```



```

PCRF port                : 0
PCRF destination host    : (Not Specified)
Gx local address         : ::
PCRF enabled Gy Event Reporting :

Charging profile : 1                      Interim records *: 0
RF triggers      : 0

S6b primary peer profile :
S6b secondary peer profile :
S6b peer profile in use :
S6b failover count      : 0
S6b peer hostname       :
S6b peer address        : ::
S6b peer port           : 0
S6b peer dest host      :
S6b peer origin         :
S6b local address       : ::

Framed Routes : 0
Redir Mob Ipv4 Addr : ::
Redir Mob Ipv6 Addr : ::
L2TP Connection *: 0

Restoration Priority Level : 0

P-CSCF Reselection : Y
Registered P-CSCF Addr : ::

eMPS IMS priority session : No

Alloc ePDG svr Ipv4 Addr : ::
Alloc ePDG svr Ipv6 Addr : ::

Override Rating Group : N/A
Override Online Billing Method : Disabled
Override Offline Billing Method : Disabled
MO Exception Cou*: 0      MO Exception Tim*: 0
-----
Number of PDN contexts : 1
=====
* indicates that the corresponding row element may have been truncated.

```

Example output PDN context detail statistics Gn/Gp IPv6 address support

```

# show mobile-gateway pdn pdn-context imsi 310150123456789 detail
=====
PDN context detail
=====
IMSI           : 310150123456789
APN            : ipd.nokia.com
Public APN     : N/A
Virtual APN    : N/A
Virtual APN sel : -
Virtual APN sel user-range type : -
Virtual APN sel scope : N/A

```

```

Auth Status      : IMSI authenticated
IMEI             : 987654321012399
MSISDN           : 12345098761
Data Session Id  : 0x00020500      AA Session Id   : 0x00140002
PDN type         : IPv4              APN restriction  : 0
PRA ID           : 0x0              PRA Status      : -
UL APN AMBR      : 8640 Kbps        DL APN AMBR     : 8640 Kbps
QFIs             : 0
Bearer contexts  : 1                SDFs            : 0
Piggyback bearers requested      : 0
Piggyback bearers creation successful : 0

Primary DNS IPv4 address : 0.0.0.0
Secondary DNS IPv4 address : 0.0.0.0
Primary DNS IPv6 address  : ::
Secondary DNS IPv6 address : ::

Pri NBNS IPv4 ad*: 0.0.0.0      Sec NBNS IPv4 ad*: 0.0.0.0
Pri NBNS IPv6 ad*: ::          Sec NBNS IPv6 ad*: ::

Charging bearer *: Home

UE IPv4 address      : 1.127.255.5
UE IPv4 address source : Local Pool

Gn PGW Ctrl Teid : 0x20500      Gn SGSN Ctrl Teid: 0xb0000006
Gn PGW V4 Ctl Adr: 16.16.16.16  Gn SGSN V4 Ctl A*: -
Gn PGW V6 Ctl Adr: 3ffe::1010:1010
Gn SGSN V6 Ctl Adr: 3ffe::a17:f02

PCRF triggers      : SGSN QOS RAT BR_LOSS BR_RVRY RAI ULI IPCAN_CHANGE
                   DEF_BR_QOS AN_GW RSRG_ALLOC RSRG_MOD TZ OUT_CRDT
                   REALLOC_CRDT REVAL_TO PLMN
Gx primary peer profile : Peer-Profile1
Gx secondary peer profile : Peer-Profile3
Gx peer profile in use   : Peer-Profile1
Gx failover count        : 0
Gx peer origin           : Local configuration
PCRF hostname            :
PCRF address              : 10.10.43.2
PCRF port                 : 3868
PCRF destination host     : primary-pcrf.nokia.com
Gx local address          : 16.16.16.19
PCRF enabled Gy Event Reporting :

Charging profile : 1              Interim records *: 0
RF triggers      : 0

S6b primary peer profile :
S6b secondary peer profile :
S6b peer profile in use   :
S6b failover count        : 0
S6b peer hostname         :
S6b peer address          : ::
S6b peer port             : 0
S6b peer dest host        :
S6b peer origin           :

```

```

S6b local address      : ::

Framed Routes         : 0
Redir Mob Ipv4 Addr   : ::
Redir Mob Ipv6 Addr   : ::
L2TP Connection *    : 0

Restoration Priority Level : 0

P-CSCF Reselection    : Y
Registered P-CSCF Addr : ::

eMPS IMS priority session : No

Alloc ePDG svr Ipv4 Addr : ::
Alloc ePDG svr Ipv6 Addr : ::

Override Rating Group      : N/A
Override Online Billing Method : Disabled
Override Offline Billing Method : Disabled
MO Exception Cou*: 0          MO Exception Tim*: 0
-----
Number of PDN contexts : 1
=====

```

Example output PDN context detail statistics S2a PMIP

```

# show mobile-gateway pdn pdn-context imsi 444444444440000 detail
=====
PDN context detail
=====
IMSI           : 444444444440000
APN            : ims.corporate.1.1.com
Virtual APN     : N/A
Virtual APN sel : -
PDN type       : IPv4                APN restriction : 0
Access-Type    : NON 3GPP            Selection mode  : Subscribed
Ref-point/Sig-pr*: S2a/PMIPv6
UL APN AMBR    : 5000 Kbps           DL APN AMBR     : 5000 Kbps
Bearer contexts : 1                  SDFs            : 2

Pri DNS IPv4 addr: 0.0.0.0           Sec DNS IPv4 addr: 0.0.0.0
Pri DNS IPv6 addr: ::                Sec DNS IPv6 addr: ::

Roaming status  : Visiting

UE IPv4 address      : 1.127.255.3
UE IPv4 address source : AAA|HSS|Local Pool|DHCP|L2TP

PGW GRE UL Key      : 0x40070da0      MAG GRE DL Key    : 0x8000000a
PGW Ctrl Addr       : 3ffe::b14:103    MAG Ctrl Addr     : 3ffe::a0a:2b17
PGW data Addr       : 3ffe::b14:103    MAG data Addr     : 3ffe::a0a:2b17
Gx PCRF addr        : 10.10.43.2
Gx PGW addr         : 16.16.16.19

S2a DL packets      : 0                S2a DL bytes      : 0

```

```

SGI UL packets   : 0                SGI UL bytes      : 0

PCRF triggers    : RAT AN_GW
Gx primary peer profile : Peer-Profile1
Gx secondary peer profile : Peer-Profile3
Gx peer profile in use   : Peer-Profile1
Gx failover count      : 0
Gx peer origin          : Local configuration
PCRF hostname          :
PCRF address            : 10.10.43.2
PCRF port                : 3868
PCRF destination host   : primary-pcrf.nokia.com
Gx local address        : 16.16.16.19

S6b primary peer profile : s6b_Peer-Profile1
S6b secondary peer profile : s6b_Peer-Profile2
S6b peer profile in use   : s6b_Peer-Profile1
S6b failover count      : 0
S6b peer hostname        :
S6b peer address         : 10.23.19.2
S6b peer port            : 3868
S6b peer dest host       : aaa.nokia.com
S6b peer origin          : diam.origRealm.com
S6b local address        : 16.16.16.25

RF server addr   : ::
RF server state   : disabled           Charging level   : Bearer
Charging profile : 1                   Interim records *: 0
RF triggers       : 0

Redir Mob Ipv4 Addr : ::
Redir Mob Ipv6 Addr : ::
PMIP session lifetime : 9996
PMIP session remaining lifetime : 9987
Restoration Priority Level : 0

Alloc ePDG svr Ipv4 Addr : ::
Alloc ePDG svr Ipv6 Addr : ::
-----
Number of PDN contexts : 1
=====
* indicates that the corresponding row element may have been truncated.

```

Example output PDN context detail SMF

```

*A:Dut-E# show mobile-gateway pdn pdn-context imsi 1000000000000001 detail
=====
PDN context detail
=====
SUPI           : 1000000000000001
DNN             : ipd.nokia.com
PEI             : 987654321012345
GPSI            : 10000098761
RAT             : NR
Interworking     : Not Subscribed
Data Session Id  : 0x00010130

```

```

PDU type      : IPv6
SMF type: SMF
Ref-point/Sig-pr*: N/A/http2      APN restriction : 0
5G Capability
  RQI         : Disabled           Multihoming      : Disabled
PDU Session ID : 1
PCRF Subscribed PRAs
OCS Subscribed PRAs
UL APN AMBR    : 200.00 Mbps      DL APN AMBR    : 50.00 Mbps
Bearer contexts : 1               SDFs           : 0

Primary DNS IPv4 address : -
Secondary DNS IPv4 address : -
Primary DNS IPv6 address : -
Secondary DNS IPv6 address : -

Pri NBNS IPv4 ad*: -           Sec NBNS IPv4 ad*: -
Pri NBNS IPv6 ad*: -
Sec NBNS IPv6 ad*: -

Charging bearer *: Visiting

UE IPv6 address : 3ffe:0:101:1f00::5:921c
UE IPv6 address source : Local Pool

Charging profile : 1           Interim records *: 0
RF triggers      : 0

Framed Routes    : 0

P-CSCF Reselection : Y

Override Rating Group : N/A
Override Online Billing Method : Disabled
Override Offline Billing Method : Disabled
MO Exception Cou*: 0         MO Exception Tim*: 0

ULI Timestmp(sec): 0         SSC Mode          : 1
N3 State          : -
LADN state        : Inactive

NSSAI
SST               : 11         SD                : 0xd143a5

Service Based Interface (SBI) Information
-----
NF Service      : nsmf-pdusession
NF Role         : producer
NF ID           : 00010000-2031-417e-9003-00000203117e
URI             : http://16.70.16.17:6545/nsmf-pdusession/v1/sm-contexts/
                  10010130
NF Service      : nsmf-pdusession
NF Role         : producer
NF ID           : 00010000-2031-417e-9003-00000203117e
URI             : http://16.70.16.17:6545/nsmf-pdusession/v1/pdusessions/
                  10010130
NF Service      : nsmf-pdusession

```

```

NF Role       : consumer
NF ID        : 57010000-5f21-417e-9572-100005f2117e (I-SMF NF ID)
URI         : {ismfPduSession} URI received from I-SMF

NF Service   : namf-comm
NF ID        : 00000000-0000-0000-0000-000000000000

NF Service   : namf-evts
NF ID        : 00000000-0000-0000-0000-000000000000

NF Service   : npcfsmpolicycontrol
NF ID        : 34635855-c481-4ee2-b571-a151579dc866
URI         : http://10.10.7.75:7890/npcf-smpolicycontrol/v1/sm-policies/
              100000000000001_1
Service Instance : policy1

NF Service   : nudm-sdm
NF ID        : 44f08cba-9a89-77fe-ab7d-2a0066584f65
URI         : http://10.10.7.78:7788/nudm-sdm/v1/imsi-100000000000001/sdm-
              subscriptions/100000000000001_0
Service Instance : sdm1

NF Service   : nudm-uecm
NF ID        : 44f08cba-9a89-77fe-ab7d-2a0066584f85
Service Instance : uecm1

```

QoS Information

```

-----
QFI : 5      QFI Type : 5QI      QoS Flow Type : Default
-----

```

QoS Profile

```

5QI/ARP      : 5/9      Priority Level      : 0
Avg Window (ms) : 0      Max Bst Vol (bytes) : 0
QNC          : Disabled  RQI                  : Disabled
UL MBR (kbps) : 0      DL MBR (kbps)      : 0
UL GBR (kbps) : 0      DL GBR (kbps)      : 0
-----

```

User Plane (UP) Information

```

-----
UP Type      : Original PSA
UP Sx-N4 Ctl V4 : 33.34.1.3
UP Sx-N4 Ctl V6 : 3ffe::2122:103
CP Sx-N4 Ctl V4 : 33.33.1.2
CP Sx-N4 Ctl V6 : 3ffe::2121:102
CP Sx-N4 Seid  : 0x10130
UP Sx-N4 Seid  : 0x11130

```

```

QFI(s)      : (Not Specified)
-----

```

```

-----
Number of PDN contexts : 1
=====

```

* indicates that the corresponding row element may have been truncated.

Example output PDN context detail I-SMF

```
# show mobile-gateway pdn pdn-context imsi 1000000000000001 detail
=====
PDN context detail
=====
SUPI           : 1000000000000001
DNN            : ipd.nokia.com
PEI           : 987654321012345
GPSI          : 10000098761
RAT           : NR
Interworking   : Not Subscribed
Data Session Id : 0x00010130
PDU type      : IPv6
SMF type      : I-SMF
Ref-point/Sig-pr*: N11/http2          APN restriction : 0
5G Capability
  RQI         : Disabled                Multihoming   : Disabled
PDU Session ID : 1
PRA ID        : 0x0                   PRA Status    : -
UL APN AMBR   : 200.00 Mbps           DL APN AMBR   : 50.00 Mbps
Bearer contexts : 1                   SDFs          : 0

Primary DNS IPv4 address : -
Secondary DNS IPv4 address : -
Primary DNS IPv6 address : -
Secondary DNS IPv6 address : -

Pri NBNS IPv4 ad*: -                Sec NBNS IPv4 ad*: -
Pri NBNS IPv6 ad*: -
Sec NBNS IPv6 ad*: -

Charging bearer *: Visiting

UE IPv6 address : 3ffe:0:101:1f00::
UE IPv6 address source : unknown

Charging profile : 1                Interim records *: 0
RF triggers      : 0

Framed Routes    : 0

P-CSCF Reselection : Y

Override Rating Group : N/A
Override Online Billing Method : Disabled
Override Offline Billing Method : Disabled
MO Exception Cou*: 0                MO Exception Tim*: 0

ULI Timestmp(sec): 0                SSC Mode       : 1
N3 State          : Activ
LADN state        : Inactive

NSSAI
```

```

SST          : 11                      SD          : 0xd143a5

Service Based Interface (SBI) Information
-----
Service Realm : sbaRealm1

NF Service    : nsmf-pdusession
NF Role       : producer
NF ID         : 57010000-5f21-417e-9572-100005f2117e
URI           : http://16.70.16.16:65454/nsmf-pdusession/v1/sm-contexts/
               00010130
Service Instance : pdu1
NF Service    : nsmf-pdusession
NF Role       : producer
NF ID         : 57010000-5f21-417e-9572-100005f2117e
URI           : {ismfPduSession} URI sent to SMF
Service Instance : pdu1
NF Service    : nsmf-pdusession
NF Role       : consumer
NF ID         : 00010000-2031-417e-9003-00000203117e (SMF NF ID)
URI           : http://16.70.16.17:65455/nsmf-pdusession/v1/pdusessions/
               10010130
Service Instance : pdu1

NF Service    : namf-comm
NF ID         : 44f08cba-9a89-46fe-ab7d-2a0066584f55
Service Instance : comm1

NF Service    : namf-evts
NF ID         : 44f08cba-9a89-46fe-ab7d-2a0066584f55
Service Instance : evts1
-----

QoS Information
-----
QFI : 5      QFI Type : 5QI      QoS Flow Type : Default
-----

QoS Profile
5QI/ARP      : 5/9      Priority Level      : 0
Avg Window (ms) : 0      Max Bst Vol (bytes) : 0
QNC          : Disabled  RQI          : Disabled
UL MBR (kbps) : 0      DL MBR (kbps)      : 0
UL GBR (kbps) : 0      DL GBR (kbps)      : 0
-----

User Plane (UP) Information
-----

UP Type      : Original PSA
UP Sx-N4 Ctl V4 : 33.34.1.1
UP Sx-N4 Ctl V6 : 3ffe::2122:101
CP Sx-N4 Ctl V4 : 33.33.1.1
CP Sx-N4 Ctl V6 : 3ffe::2121:101
CP Sx-N4 Seid   : 0x10130
UP Sx-N4 Seid   : 0x11130

QFI(s)       : (Not Specified)

```



```
-----
-----
Number of PDN contexts : 1
=====
```

```
* indicates that the corresponding row element may have been truncated.
```

Example output PDN context statistics detail Ethernet PDU session SMF

```
# show mobile-gateway pdn pdn-context imsi 460100001003204 detail

=====
PDN context detail
=====
SUPI           : 460100001003204
DNN            : ethernetApn1
Public DNN     : N/A
Virtual DNN    : ipd.nokia.com
Virtual DNN sel : static
Virtual DNN sel user-range type      : -
Virtual DNN sel scope                 : N/A
Auth Status    : IMSI authenticated
PEI           : 987654321012345
GPSI          : 12345678901
RAT            : NR
Interworking   : Not Subscribed
DN Type       : ethernet
SMF type      : SMF
Ref-point/Sig-pr*: N11/http2          DNN restriction : 0
5G Capability
  RQI         : Disabled                Multihoming   : Disabled
PDU Session ID : 1
UL DNN AMBR    : 10.00 Mbps             DL DNN AMBR    : 10.00 Mbps
Bearer contexts : 1                     SDFs          : 1

Primary DNS IPv4 address : -
Secondary DNS IPv4 address : -
Primary DNS IPv6 address : -
Secondary DNS IPv6 address : -

Pri NBNS IPv4 ad*: -                Sec NBNS IPv4 ad*: -
Pri NBNS IPv6 ad*: -
Sec NBNS IPv6 ad*: -

Charging bearer *: Home

UE IPv4 address : -
UE IPv4 address source : unknown

UE IPv6 address : -
UE IPv6 address source : unknown

Charging profile : 1                Interim records *: 0
RF triggers      : 0
```

```

Framed Routes      : 0

P-CSCF Reselection      : Y

Override Rating Group    : N/A
Override Online Billing Method : Disabled
Override Offline Billing Method : Disabled
MO Exception Cou*: 0      MO Exception Tim*: 0

ULI Timestmp(sec): 0      SSC Mode      : 1
N3 State      : Active
LADN state    : Inactive

NSSAI
SST          : 11      SD          : 0xd143a5

Service Based Interface (SBI) Information
-----
Service Realm      : sbaRealm1

NF Service      : nsmf-pdusession
NF Role         : producer
NF ID           : 44f08cba-9a89-46fe-ab7d-2a0066584f55
URI             : http://16.70.16.16:65397/nsmf-pdusession/v1/sm-contexts/
                  00011110
Service Instance : pdu1

NF Service      : namf-comm
NF Role         : consumer
NF ID           : 44f08cba-9a89-46fe-ab7d-2a0066584f55
Service Instance : comm1

NF Service      : namf-evts
NF Role         : consumer
NF ID           : 44f08cba-9a89-46fe-ab7d-2a0066584f55
URI             : http://16.70.16.16:65397/smfPduSession/00011110
Service Instance : evts1

NF Service      : nchf-convergedcharging
NF Role         : consumer
NF ID           : 4f85e924-1509-47b4-907f-9778afdb2ffb
URI             : http://10.10.7.85:5566/nchf-convergedcharging/v3/
                  chargingdata/0
Service Instance : chfClient1

NF Service      : npcfsmpolicycontrol
NF Role         : consumer
NF ID           : 1720b601-fae2-4fc4-b535-8df9da3c64a7
URI             : http://10.10.7.76:7890/npcf-smpolicycontrol/v1/sm-policies/
                  460100001003204_1
Service Instance : policy1

NF Service      : nudm-sdm
NF Role         : consumer
NF ID           : 3296d21b-3f5a-4c35-bc79-344dc38e3f32
URI             : http://10.10.7.74:7788/nudm-sdm/v1/imsi-460100001003204/sdm-
                  subscriptions/460100001003204_0

```

```

Service Instance : sdm1

NF Service      : nudm-uecm
NF Role        : consumer
NF ID          : 3296d21b-3f5a-4c35-ec79-344dc38e3f32
Service Instance : uecm1
-----

QoS Information
-----
QFI : 1    QFI Type : Dynamic    QoS Flow Type : Default
-----

QoS Profile
5QI/ARP      : 5/9                Priority Level      : 0
Avg Window (ms) : 0                Max Bst Vol (bytes) : 0
QNC          : Disabled           RQI                  : Disabled
UL MBR (kbps) : 0                DL MBR (kbps)       : 0
UL GBR (kbps) : 0                DL GBR (kbps)       : 0
-----

User Plane (UP) Information
-----

UP Type      : Original PSA
UP Sx-N4 Ctl V4 : 33.34.1.1
UP Sx-N4 Ctl V6 : 3ffe::2122:101
CP Sx-N4 Ctl V4 : 33.33.1.1
CP Sx-N4 Ctl V6 : 3ffe::2121:101
CP Sx-N4 Seid   : 0x11110
UP Sx-N4 Seid   : 0x15110
gNb N3 Data TEID : 0x10000001
gNb N3 IPv4 Address : 6.7.9.1
UP N3 Data TEID  : 0x40015115
UP N3 IPv4 Address : 30.1.1.6
UP N3 IPv6 Address : 3ffe::1e01:106

QFI(s)      : 1
-----

Def PFCP-u Sess : False
Network/Itf Realm: N/A
-----

Number of PDN contexts : 1
=====
* indicates that the corresponding row element may have been truncated.

```

Example output PDN context statistics detail Ethernet PDU session UPF

```

# show mobile-gateway pdn pdn-context imsi 460100001003204 detail

=====
PDN context detail
=====
SUPI          : 460100001003204
N6 Network Inst : ipd.nokia.com

```

```

DN Type           : ethernet
UL DNN AMBR       : 10.00 Mbps          DL DNN AMBR       : 10.00 Mbps
GW/Group/VM       : 1/2/4
UE IPv4 Address   : -
UE IPv6 Address   : -
CP Sx-N4 Seid     : 0x11110             UP Sx-N4 Seid     : 0x15110
CP Sx-N4 Ctl V4   *: 33.33.1.1          UP Sx-N4 Ctl V4   *: 33.34.1.1
CP Sx-N4 Ctl V6   *: 3ffe::2122:101
UP Sx-N4 Ctl V6   *: 3ffe::2121:101
PDRs              : 2                   FARS              : 2
QERs              : 2
State             : CONNECTED           AA Session Id     : 0x01444001
Def PFCP-u Sess   : False
Network/Itf Realm: N/A
DL DNN rate lmt Drp Pkt : 0
UL DNN rate lmt Drp Pkt : 0
PLMN rate lmt Drp Pkt  : 0
L2TP Connection ID   : 0

```

```

NSSAI
SST              : 11                   SD                : 0xd143a5

```

```

-----
Number of PDN contexts : 1
=====

```

* indicates that the corresponding row element may have been truncated.

Example output PDN context statistics Gy detail combined SGW/PGW

```

# show mobile-gateway pdn pdn-context imsi 7777777777777777 acct gy detail
=====
PDN context Accounting Gy Detail
=====
IMSI              : 7777777777777777
APN               : nokia.corporate.com
Virtual APN       : N/A
PDN type          : IPv4                LBI                : 5
UL APN AMBR       : 4992 Kbps           DL APN AMBR         : 4992 Kbps
Bearer contexts   : 1                   Charging ID         : 83886081

UE IPv4 address   : 1.127.255.0

Aggregate UL bytes : 0                   Aggregate DL bytes   : 0
Aggregate UL packets : 0                 Aggregate DL packets : 0

DCCA Profile      : DCCA-Prof1
Primary Diam-peer Profile : Gy-Peer-Prof1
Secondary Diam-peer Profile :
Peer-profile In-use : Gy-Peer-Prof1
Failover Count    : 0
OCS Peer Origin   : Local Configuration
OCS Hostname      : (Not Specified)
OCS Address       : 10.23.18.10
OCS Port          : 3868
OCS Destination Realm : nokia.com
OCS Destination Host : pgw1.gy.nokia.com
Local Address     : 16.16.16.24

```

```

Failure Handling           : Terminate (L)
Session Failover          : Enabled (L)
Number of FUI Redirections : 0
Number of times Credits replenished : 0
Number Of Rating Groups : 0
Number of Rating Groups, Service IDs : 0
CCFH Time Remaining       : 0 (Secs)
CCFH Volume Remaining      : 0 (Bytes)
Restore Time Remaining     : 0 (Secs)

```

```

-----
Number of PDN contexts : 1
=====

```

Example output PDN context statistics Rf detail

```
# show mobile-gateway pdn pdn-context imsi 7777777777777777 acct rf detail
```

```
=====
PDN Context Accounting Rf Detail
=====
```

```

IMSI : 220011234567800
APN : offlineonline
Virtual APN : N/A
PDN type : IPv4                LBI : 5
UL APN AMBR : 4096 kbps        DL APN AMBR : 4096 kbps
Bearer contexts : 2            Charging ID : 298844160

```

```
UE IPv4 address : 1.0.0.1
```

```

Aggregate UL bytes : 6000      Aggregate DL bytes : 6000
Aggregate UL packets : 60      Aggregate DL packets : 60

```

```

Rf Peer State : Enabled        Charging Level : Pdn
Primary Diam-peer Profile : Peer-Profile2
Secondary Diam-peer Profile :
Peer-profile In-use : Peer-Profile2
Rf Failover Count : 0
Rf Peer Hostname :
Rf Peer Address : 10.23.13.2
Rf Peer Port : 3868
Rf Destination Host : nokia-ccfsim
Rf Local Address : 10.42.1.3
Rating Groups : 0              Rating Group,Service IDs: 2
RF triggers : 0               Rating Group,Sponsor IDs: 0

```

```

-----
Rating Group : 100
Online Charging : On           Offline Charging : On
UL bytes : 3000               DL bytes : 3000
UL packets : 30               DL packets : 30
-----

```

```

Rating Group : 200
Online Charging : On           Offline Charging : On
UL bytes : 3000                DL bytes : 3000
UL packets : 30                DL packets : 30

```

```

-----
Number of PDN contexts : 1
=====

```

* indicates that the corresponding row element may have been truncated.

Example output PDN context statistics RADIUS detail

```
# show mobile-gateway pdn pdn-context imsi 123456789012168 acct radius detail
```

```
=====
PDN Context Accounting Rf Detail
=====
```

```

IMSI           : 220011234567800
APN            : nokia.com
Virtual APN    : N/A
PDN type       : IPv6           LBI : 5
UL APN AMBR    : 4096 kbps      DL APN AMBR : 4096 kbps
Bearer contexts : 1             Charging ID : 288358400

```

```

UE IPv4 address : N/A
UE IPv6 address : 3ffe:0:1:ff01::5e3c

```

```

Aggregate UL bytes : 0           Aggregate DL bytes : 0
Aggregate UL packets : 0         Aggregate DL packets : 0

```

```

Radius Group      : Radius Group 1
Suppress IMSI     : no           Authenticated via Radius : no
Selective Accounting : no        Username Type : Imsi

```

```

Interim Accounting
Accounting Interval : 0 s
Event Triggers      : peer, uli, ecgi, tai, rai, cgi/sai
Messages Tx         : 0

```

```

-----
Number of PDN contexts : 1
=====

```

* indicates that the corresponding row element may have been truncated.

Example output PDN context IMEI statistics (UP function)

```
# show mobile-gateway pdn pdn-context imei 7777777777777777
```

```

IMSI/IMEI(#)      APN      Type      Beare* UE Address (IPv4/IPv6)  Ref-pt/Si*
/MSISDN(^)/MAC(`)
/SubId/SEID(~)
=====
0x41110~         ipd.nokia.com    IPv4      1      1.127.255.0/-              N/A
123456789012345
-----
Number of PDN instances : 1
-----
# indicates that the corresponding row is IMEI entry.
^ indicates that the corresponding row is MSISDN entry.
` indicates that the corresponding row is MAC entry.
~ indicates that the corresponding row is SEID entry.
-----
* indicates that the corresponding row element may have been truncated.

```

Example output PDN context IMEI detail statistics (UP function)

```

show mobile-gateway pdn pdn-context imei 7777777777777777 detail
=====
PDN context detail
=====
IMSI              : 123456789012345
Gi Network Inst   : ipd.nokia.com
PDN Type          : IPv4
UL APN AMBR       : 8640 Kbps           DL APN AMBR       : 8640 Kbps
Bearer contexts   : 1                   GW/Group/VM       : 1/1/3
UE IPv4 Address   : 1.127.255.0
CP Sx-N4 Seid     : 0x11120             UP Sx-N4 Seid     : 0x41110
CP Sx-N4 Ctl V4 * : 33.33.1.1           UP Sx-N4 Ctl V4 * : 33.34.1.1
CP Sx-N4 Ctl V6 * : 3ffe::2122:101
UP Sx-N4 Ctl V6 * : 3ffe::2121:101
PDRs              : 2                   FARs              : 2
QERS              : 1
State             : CONNECTED           AA Session Id     : 0x00444004
DL APN rate lmt Drp Pkt : 0
UL APN rate lmt Drp Pkt : 0
PLMN rate lmt Drp Pkt  : 0
-----
Number of PDN contexts : 1
=====
* indicates that the corresponding row element may have been truncated.

```

Example output PDN context IMEI detail account statistics (UP function)

```

# show mobile-gateway pdn pdn-context imei 187654321013001 acct stats detail
=====
PDN Context Accounting Statistics Detail
=====
URR ID            : 1073741825          Sequence No       : 1
Measurement Method : Volume
-----
Reporting Triggers                               Subscribed
-----
Linked Usage Reporting                             N
Dropped DL Traffic Threshold                       N

```

Stop of Traffic		N	
Start of Traffic		Y	
QHT		N	
Time Threshold		N	
Volume Threshold		N	
Periodic Reporting		N	
Time Quota		N	
Volume Quota		N	

Measure Period	: 0		
Volume Threshold			
UL	: 0	DL	: 0
Total	: 0		
Subsequent Volume Threshold			
UL	: 0	DL	: 0
Total	: 0		
Volume Quota			
UL	: 0	DL	: 0
Total	: 0		
Volume Usage			
UL	: 7200	DL	: 7200
Time Threshold	: 0 secs	Subsequent Time Thresho*	: 0 secs
Time Quota	: 0 secs	Time Usage	: 0 secs
Quota Holding Time	: 0 secs	Drp DL Traffic Threshold:	0
Monitoring Time	: 0 secs	Inactivity Detection Ti*	: 0 secs
Parent URR	: 0		
Measurement Information			
Before QOS	: No	Inactivate Measurement	: No

URR ID	: 1073741829	Sequence No	: 0
Measurement Method	: Volume		

Reporting Triggers		Subscribed	

Linked Usage Reporting		Y	
Dropped DL Traffic Threshold		N	
Stop of Traffic		N	
Start of Traffic		N	
QHT		N	
Time Threshold		N	
Volume Threshold		Y	
Periodic Reporting		N	
Time Quota		N	
Volume Quota		N	

Measure Period	: 0		
Volume Threshold			
UL	: 0	DL	: 0
Total	: 4191904		
Subsequent Volume Threshold			
UL	: 0	DL	: 0
Total	: 0		
Volume Quota			
UL	: 0	DL	: 0
Total	: 0		
Volume Usage			
UL	: 7200	DL	: 7200


```

Time Threshold      : 0 secs      Subsequent Time Thresho*: 0 secs
Time Quota          : 0 secs      Time Usage                : 0 secs
Quota Holding Time  : 0 secs      Drp DL Traffic Threshold: 0
Monitoring Time     : 0 secs      Inactivity Detection Ti*: 0 secs
Parent URR          : 1073741825

```

```

Measurement Information
Before QOS          : No          Inactivate Measurement : No

```

```

-----
URR ID              : 1610612737  Sequence No              : 0
Measurement Method   : Volume

```

```

-----
Reporting Triggers                               Subscribed
-----
Linked Usage Reporting                           Y
Dropped DL Traffic Threshold                     N
Stop of Traffic                                 N
Start of Traffic                                 N
QHT                                               N
Time Threshold                                   N
Volume Threshold                                 N
Periodic Reporting                              N
Time Quota                                       N
Volume Quota                                    N

```

```

-----
Measure Period      : 0
Volume Threshold
UL                  : 0          DL                  : 0
Total               : 0
Subsequent Volume Threshold
UL                  : 0          DL                  : 0
Total               : 0
Volume Quota
UL                  : 0          DL                  : 0
Total               : 0
Volume Usage
UL                  : 1800       DL                  : 1800
Time Threshold      : 0 secs     Subsequent Time Thresho*: 0 secs
Time Quota          : 0 secs     Time Usage                : 0 secs
Quota Holding Time  : 0 secs     Drp DL Traffic Threshold: 0
Monitoring Time     : 0 secs     Inactivity Detection Ti*: 0 secs
Parent URR          : 1073741829
Measurement Information
Before QOS          : No          Inactivate Measurement : No

```

```

-----
URR ID              : 1610612738  Sequence No              : 0
Measurement Method   : Duration Volume

```

```

-----
Reporting Triggers                               Subscribed
-----
Linked Usage Reporting                           Y
Dropped DL Traffic Threshold                     N
Stop of Traffic                                 N
Start of Traffic                                 N
QHT                                               N
Time Threshold                                   N
Volume Threshold                                 Y
Periodic Reporting                              N

```

```

Time Quota                                     N
Volume Quota                                  N
-----
Measure Period          : 0
Volume Threshold
UL                      : 0          DL          : 0
Total                   : 2097152
Subsequent Volume Threshold
UL                      : 0          DL          : 0
Total                   : 0
Volume Quota
UL                      : 0          DL          : 0
Total                   : 0
Volume Usage
UL                      : 1800        DL          : 1800
Time Threshold          : 0 secs      Subsequent Time Thresho*: 0 secs
Time Quota              : 0 secs      Time Usage              : 63 secs
Quota Holding Time      : 0 secs      Drp DL Traffic Threshold: 0
Monitoring Time         : 0 secs      Inactivity Detection Ti*: 60 secs
Parent URR              : 1073741829
Measurement Information
Before QOS              : No          Inactivate Measurement : No
-----
URR ID                  : 1610612739   Sequence No              : 2
Measurement Method      : Volume
-----
Reporting Triggers          Subscribed
-----
Linked Usage Reporting      Y
Dropped DL Traffic Threshold N
Stop of Traffic             N
Start of Traffic            N
QHT                         N
Time Threshold              N
Volume Threshold            Y
Periodic Reporting          N
Time Quota                  N
Volume Quota                 N
-----
Measure Period          : 0
Volume Threshold
UL                      : 0          DL          : 0
Total                   : 1600
Subsequent Volume Threshold
UL                      : 0          DL          : 0
Total                   : 0
Volume Quota
UL                      : 0          DL          : 0
Total                   : 0
Volume Usage
UL                      : 600        DL          : 600
Time Threshold          : 0 secs      Subsequent Time Thresho*: 0 secs
Time Quota              : 0 secs      Time Usage              : 0 secs
Quota Holding Time      : 0 secs      Drp DL Traffic Threshold: 0
Monitoring Time         : 0 secs      Inactivity Detection Ti*: 0 secs
Parent URR              : 1073741829
Measurement Information

```

```

Before QOS          : No          Inactivate Measurement : No
-----
URR ID              : 1610612740   Sequence No         : 2
Measurement Method   : Duration Volume
-----
Reporting Triggers                                     Subscribed
-----
Linked Usage Reporting                                Y
Dropped DL Traffic Threshold                          N
Stop of Traffic                                       N
Start of Traffic                                      N
QHT                                                    N
Time Threshold                                        N
Volume Threshold                                      Y
Periodic Reporting                                    N
Time Quota                                            N
Volume Quota                                          N
-----
Measure Period      : 0
Volume Threshold
UL                  : 0          DL                  : 0
Total              : 1600
Subsequent Volume Threshold
UL                  : 0          DL                  : 0
Total              : 0
Volume Quota
UL                  : 0          DL                  : 0
Total              : 0
Volume Usage
UL                  : 600        DL                  : 600
Time Threshold      : 0 secs    Subsequent Time Thresho*: 0 secs
Time Quota          : 0 secs    Time Usage              : 18 secs
Quota Holding Time  : 0 secs    Drp DL Traffic Threshold: 0
Monitoring Time     : 0 secs    Inactivity Detection Ti*: 60 secs
Parent URR          : 1073741829
Measurement Information
Before QOS          : No          Inactivate Measurement : No
-----
Number of PDN contexts : 6
=====
* indicates that the corresponding row element may have been truncated.

```

Example output PDN context IMSI statistics (UP function)

```

# show mobile-gateway pdn pdn-context imsi 123456789012345
=====
IMSI/IMEI(#)      APN      Type      Beare* UE Address (IPv4/IPv6)  Ref-pt/Si*
/MSISDN(^)/MAC(`)
/SubId/SEID(~)
=====
0x411110~        ipd.nokia.com    IPv4      1      1.127.255.0/-          N/A
123456789012345
-----
Number of PDN instances : 1
-----
# indicates that the corresponding row is IMEI entry.

```

^ indicates that the corresponding row is MSISDN entry.
 ` indicates that the corresponding row is MAC entry.
 ~ indicates that the corresponding row is SEID entry.

 * indicates that the corresponding row element may have been truncated.

Example output PDN context IMSI detail statistics (UP function)

```
# show mobile-gateway pdn pdn-context imsi 123456789012345 detail
=====
PDN context detail
=====
IMSI                : 123456789012345
Gi Network Inst     : ipd.nokia.com
PDN Type            : IPv4
UL APN AMBR         : 8640 Kbps           DL APN AMBR         : 8640 Kbps
Bearer contexts     : 1                   GW/Group/VM          : 1/1/3
UE IPv4 Address     : 1.127.255.0
CP Sx-N4 Seid       : 0x11120             UP Sx-N4 Seid        : 0x41110
CP Sx-N4 Ctl V4 *   : 33.33.1.1           UP Sx-N4 Ctl V4 *    : 33.34.1.1
CP Sx-N4 Ctl V6 *   : 3ffe::2122:101
UP Sx-N4 Ctl V6 *   : 3ffe::2121:101
PDRs                : 2                   FARs                 : 2
QERs                : 1
State               : CONNECTED            AA Session Id        : 0x00444004
DL APN rate lmt Drp Pkt : 0
UL APN rate lmt Drp Pkt : 0
PLMN rate lmt Drp Pkt  : 0
-----
Number of PDN contexts : 1
=====
* indicates that the corresponding row element may have been truncated.
```

Example output PDN context IMSI detail account statistics (UP function)

```
# show mobile-gateway pdn pdn-context imsi 822016789012301 acct stats detail
=====
PDN Context Accounting Statistics Detail
=====
URR ID              : 1073741825   Sequence No          : 1
Measurement Method   : Volume
-----
Reporting Triggers           Subscribed
-----
Linked Usage Reporting       N
Dropped DL Traffic Threshold N
Stop of Traffic              Y
Start of Traffic              Y
QHT                          N
Time Threshold               N
Volume Threshold             N
Periodic Reporting           N
Time Quota                   N
Volume Quota                 N
-----
Measure Period            : 0
```

```

Volume Threshold
UL                : 0          DL                : 0
Total            : 0
Subsequent Volume Threshold
UL                : 0          DL                : 0
Total            : 0
Volume Quota
UL                : 0          DL                : 0
Total            : 0
Volume Usage
UL                : 7200       DL                : 7200
Time Threshold   : 0 secs     Subsequent Time Thresho*: 0 secs
Time Quota       : 0 secs     Time Usage           : 0 secs
Quota Holding Time : 0 secs   Drp DL Traffic Threshold: 0
Monitoring Time  : 0 secs     Inactivity Detection Ti*: 0 secs
Parent URR       : 0
Measurement Information
Before QOS       : No        Inactivate Measurement : No

```

```

-----
URR ID           : 1073741829  Sequence No       : 0
Measurement Method : Volume

```

```

-----
Reporting Triggers                               Subscribed
-----
Linked Usage Reporting                           Y
Dropped DL Traffic Threshold                     N
Stop of Traffic                                  N
Start of Traffic                                  N
QHT                                                N
Time Threshold                                   N
Volume Threshold                                 Y
Periodic Reporting                              N
Time Quota                                       N
Volume Quota                                     N

```

```

-----
Measure Period      : 0
Volume Threshold
UL                : 0          DL                : 0
Total            : 4191904
Subsequent Volume Threshold
UL                : 0          DL                : 0
Total            : 0
Volume Quota
UL                : 0          DL                : 0
Total            : 0
Volume Usage
UL                : 7200       DL                : 7200
Time Threshold   : 0 secs     Subsequent Time Thresho*: 0 secs
Time Quota       : 0 secs     Time Usage           : 0 secs
Quota Holding Time : 0 secs   Drp DL Traffic Threshold: 0
Monitoring Time  : 0 secs     Inactivity Detection Ti*: 0 secs
Parent URR       : 1073741825
Measurement Information
Before QOS       : No        Inactivate Measurement : No

```

```

-----
URR ID           : 1610612737  Sequence No       : 0
Measurement Method : Volume

```

Reporting Triggers		Subscribed	
Linked Usage Reporting		Y	
Dropped DL Traffic Threshold		N	
Stop of Traffic		N	
Start of Traffic		N	
QHT		N	
Time Threshold		N	
Volume Threshold		N	
Periodic Reporting		N	
Time Quota		N	
Volume Quota		N	
Measure Period		: 0	
Volume Threshold			
UL	: 0	DL	: 0
Total	: 0		
Subsequent Volume Threshold			
UL	: 0	DL	: 0
Total	: 0		
Volume Quota			
UL	: 0	DL	: 0
Total	: 0		
Volume Usage			
UL	: 1800	DL	: 1800
Time Threshold	: 0 secs	Subsequent Time Thresho*	: 0 secs
Time Quota	: 0 secs	Time Usage	: 0 secs
Quota Holding Time	: 0 secs	Drp DL Traffic Threshold:	0
Monitoring Time	: 0 secs	Inactivity Detection Ti*	: 0 secs
Parent URR	: 1073741829		
Measurement Information			
Before QOS	: No	Inactivate Measurement	: No
URR ID	: 1610612738	Sequence No	: 0
Measurement Method	: Duration Volume		
Reporting Triggers		Subscribed	
Linked Usage Reporting		Y	
Dropped DL Traffic Threshold		N	
Stop of Traffic		N	
Start of Traffic		N	
QHT		N	
Time Threshold		N	
Volume Threshold		Y	
Periodic Reporting		N	
Time Quota		N	
Volume Quota		N	
Measure Period		: 0	
Volume Threshold			
UL	: 0	DL	: 0
Total	: 2097152		
Subsequent Volume Threshold			
UL	: 0	DL	: 0
Total	: 0		

```

Volume Quota
UL                : 0                DL                : 0
Total            : 0
Volume Usage
UL                : 1800             DL                : 1800
Time Threshold   : 0 secs           Subsequent Time Thresho*: 0 secs
Time Quota       : 0 secs           Time Usage       : 63 secs
Quota Holding Time : 0 secs         Drp DL Traffic Threshold: 0
Monitoring Time  : 0 secs           Inactivity Detection Ti*: 60 secs
Parent URR       : 1073741829
Measurement Information
Before QOS       : No               Inactivate Measurement : No
-----
URR ID           : 1610612739       Sequence No      : 2
Measurement Method : Volume
-----
Reporting Triggers                Subscribed
-----
Linked Usage Reporting            Y
Dropped DL Traffic Threshold      N
Stop of Traffic                   N
Start of Traffic                  N
QHT                               N
Time Threshold                    N
Volume Threshold                  Y
Periodic Reporting                N
Time Quota                        N
Volume Quota                      N
-----
Measure Period      : 0
Volume Threshold
UL                  : 0                DL                : 0
Total              : 1600
Subsequent Volume Threshold
UL                  : 0                DL                : 0
Total              : 0
Volume Quota
UL                  : 0                DL                : 0
Total              : 0
Volume Usage
UL                  : 600             DL                : 600
Time Threshold     : 0 secs           Subsequent Time Thresho*: 0 secs
Time Quota         : 0 secs           Time Usage         : 0 secs
Quota Holding Time : 0 secs           Drp DL Traffic Threshold: 0
Monitoring Time    : 0 secs           Inactivity Detection Ti*: 0 secs
Parent URR         : 1073741829
Measurement Information
Before QOS         : No               Inactivate Measurement : No
-----
URR ID             : 1610612740       Sequence No      : 2
Measurement Method  : Duration Volume
-----
Reporting Triggers                Subscribed
-----
Linked Usage Reporting            Y
Dropped DL Traffic Threshold      N
Stop of Traffic                   N

```

```

Start of Traffic                N
QHT                            N
Time Threshold                  N
Volume Threshold                Y
Periodic Reporting              N
Time Quota                     N
Volume Quota                   N
-----
Measure Period                  : 0
Volume Threshold
UL                             : 0          DL                      : 0
Total                          : 1600
Subsequent Volume Threshold
UL                             : 0          DL                      : 0
Total                          : 0
Volume Quota
UL                             : 0          DL                      : 0
Total                          : 0
Volume Usage
UL                             : 600        DL                      : 600
Time Threshold                  : 0 secs    Subsequent Time Thresho*: 0 secs
Time Quota                     : 0 secs    Time Usage                : 18 secs
Quota Holding Time              : 0 secs    Drp DL Traffic Threshold: 0
Monitoring Time                 : 0 secs    Inactivity Detection Ti*: 60 secs
Parent URR                     : 1073741829
Measurement Information
Before QOS                     : No         Inactivate Measurement   : No
-----
Number of PDN contexts : 6
=====
* indicates that the corresponding row element may have been truncated.

```

Example output PDN context MSISDN statistics (UP function)

```

# show mobile-gateway pdn pdn-context msisdn 7777777777777777
=====
IMSI/IMEI(#)      APN      Type      Beare* UE Address (IPv4/IPv6)  Ref-pt/Si*
/MSISDN(^)/MAC(`)
/SubId/SEID(~)
=====
0x41110~         ipd.nokia.com    IPv4      1      1.127.255.0/-          N/A
123456789012345
-----
Number of PDN instances : 1
-----
# indicates that the corresponding row is IMEI entry.
^ indicates that the corresponding row is MSISDN entry.
` indicates that the corresponding row is MAC entry.
~ indicates that the corresponding row is SEID entry.
-----
* indicates that the corresponding row element may have been truncated.

```

Example output PDN context MSISDN detail statistics (UP function)

```

# show mobile-gateway pdn pdn-context msisdn 7777777777777777 detail
=====

```



```

PDN context detail
=====
IMSI           : 123456789012345
Gi Network Inst : ipd.nokia.com
PDN Type       : IPv4
UL APN AMBR    : 8640 Kbps          DL APN AMBR    : 8640 Kbps
Bearer contexts : 1                 GW/Group/VM   : 1/1/3
UE IPv4 Address : 1.127.255.0
CP Sx-N4 Seid  : 0x11120           UP Sx-N4 Seid  : 0x41110
CP Sx-N4 Ctl V4 *: 33.33.1.1       UP Sx-N4 Ctl V4 *: 33.34.1.1
CP Sx-N4 Ctl V6 *: 3ffe::2122:101
UP Sx-N4 Ctl V6 *: 3ffe::2121:101
PDRs           : 2                 FARS          : 2
QERs           : 1
State          : CONNECTED         AA Session Id  : 0x00444004
DL APN rate lmt Drp Pkt : 0
UL APN rate lmt Drp Pkt : 0
PLMN rate lmt Drp Pkt   : 0
-----
Number of PDN contexts : 1
=====
* indicates that the corresponding row element may have been truncated.

```

Example output PDN context MSISDN detail account statistics (UP function)

```

# show mobile-gateway pdn pdn-context msisdn 12345098768 acct stats detail
=====
PDN Context Accounting Statistics Detail
=====
URR ID           : 1073741825      Sequence No      : 1
Measurement Method : Volume
-----
Reporting Triggers          Subscribed
-----
Linked Usage Reporting      N
Dropped DL Traffic Threshold N
Stop of Traffic             N
Start of Traffic            Y
QHT                          N
Time Threshold              N
Volume Threshold            N
Periodic Reporting          N
Time Quota                  N
Volume Quota                N
-----
Measure Period           : 0
Volume Threshold
UL                       : 0          DL                       : 0
Total                    : 0
Subsequent Volume Threshold
UL                       : 0          DL                       : 0
Total                    : 0
Volume Quota
UL                       : 0          DL                       : 0
Total                    : 0
Volume Usage

```

```

UL : 7200 DL : 7200
Time Threshold : 0 secs Subsequent Time Thresho*: 0 secs
Time Quota : 0 secs Time Usage : 0 secs
Quota Holding Time : 0 secs Drp DL Traffic Threshold: 0
Monitoring Time : 0 secs Inactivity Detection Ti*: 0 secs
Parent URR : 0
Measurement Information
Before QOS : No Inactivate Measurement : No

```

```

-----
URR ID : 1073741829 Sequence No : 0
Measurement Method : Volume

```

```

-----
Reporting Triggers Subscribed
-----
Linked Usage Reporting Y
Dropped DL Traffic Threshold N
Stop of Traffic N
Start of Traffic N
QHT N
Time Threshold N
Volume Threshold Y
Periodic Reporting N
Time Quota N
Volume Quota N

```

```

-----
Measure Period : 0
Volume Threshold
UL : 0 DL : 0
Total : 4191904
Subsequent Volume Threshold
UL : 0 DL : 0
Total : 0
Volume Quota
UL : 0 DL : 0
Total : 0
Volume Usage
UL : 7200 DL : 7200
Time Threshold : 0 secs Subsequent Time Thresho*: 0 secs
Time Quota : 0 secs Time Usage : 0 secs
Quota Holding Time : 0 secs Drp DL Traffic Threshold: 0
Monitoring Time : 0 secs Inactivity Detection Ti*: 0 secs
Parent URR : 1073741825
Measurement Information
Before QOS : No Inactivate Measurement : No

```

```

-----
URR ID : 1610612737 Sequence No : 0
Measurement Method : Volume

```

```

-----
Reporting Triggers Subscribed
-----
Linked Usage Reporting Y
Dropped DL Traffic Threshold N
Stop of Traffic N
Start of Traffic N
QHT N
Time Threshold N
Volume Threshold N

```

Periodic Reporting		N	
Time Quota		N	
Volume Quota		N	

Measure Period	: 0		
Volume Threshold			
UL	: 0	DL	: 0
Total	: 0		
Subsequent Volume Threshold			
UL	: 0	DL	: 0
Total	: 0		
Volume Quota			
UL	: 0	DL	: 0
Total	: 0		
Volume Usage			
UL	: 1800	DL	: 1800
Time Threshold	: 0 secs	Subsequent Time Thresho*	: 0 secs
Time Quota	: 0 secs	Time Usage	: 0 secs
Quota Holding Time	: 0 secs	Drp DL Traffic Threshold:	0
Monitoring Time	: 0 secs	Inactivity Detection Ti*	: 0 secs
Parent URR	: 1073741829		
Measurement Information			
Before QOS	: No	Inactivate Measurement	: No

URR ID	: 1610612738	Sequence No	: 0
Measurement Method	: Duration Volume		

Reporting Triggers		Subscribed	

Linked Usage Reporting		Y	
Dropped DL Traffic Threshold		N	
Stop of Traffic		N	
Start of Traffic		N	
QHT		N	
Time Threshold		N	
Volume Threshold		Y	
Periodic Reporting		N	
Time Quota		N	
Volume Quota		N	

Measure Period	: 0		
Volume Threshold			
UL	: 0	DL	: 0
Total	: 2097152		
Subsequent Volume Threshold			
UL	: 0	DL	: 0
Total	: 0		
Volume Quota			
UL	: 0	DL	: 0
Total	: 0		
Volume Usage			
UL	: 1800	DL	: 1800
Time Threshold	: 0 secs	Subsequent Time Thresho*	: 0 secs
Time Quota	: 0 secs	Time Usage	: 63 secs
Quota Holding Time	: 0 secs	Drp DL Traffic Threshold:	0
Monitoring Time	: 0 secs	Inactivity Detection Ti*	: 60 secs
Parent URR	: 1073741829		

```

Measurement Information
Before QOS                : No                Inactivate Measurement : No
-----
URR ID                    : 1610612739        Sequence No              : 2
Measurement Method        : Volume
-----
Reporting Triggers                Subscribed
-----
Linked Usage Reporting           Y
Dropped DL Traffic Threshold     N
Stop of Traffic                  N
Start of Traffic                 N
QHT                              N
Time Threshold                   N
Volume Threshold                 Y
Periodic Reporting               N
Time Quota                       N
Volume Quota                     N
-----
Measure Period                : 0
Volume Threshold
UL                            : 0                DL                : 0
Total                        : 1600
Subsequent Volume Threshold
UL                            : 0                DL                : 0
Total                        : 0
Volume Quota
UL                            : 0                DL                : 0
Total                        : 0
Volume Usage
UL                            : 600                DL                : 600
Time Threshold                : 0 secs            Subsequent Time Thresho* : 0 secs
Time Quota                    : 0 secs            Time Usage                : 0 secs
Quota Holding Time            : 0 secs            Drp DL Traffic Threshold: 0
Monitoring Time               : 0 secs            Inactivity Detection Ti* : 0 secs
Parent URR                    : 1073741829
Measurement Information
Before QOS                : No                Inactivate Measurement : No
-----
URR ID                    : 1610612740        Sequence No              : 2
Measurement Method        : Duration Volume
-----
Reporting Triggers                Subscribed
-----
Linked Usage Reporting           Y
Dropped DL Traffic Threshold     N
Stop of Traffic                  N
Start of Traffic                 N
QHT                              N
Time Threshold                   N
Volume Threshold                 Y
Periodic Reporting               N
Time Quota                       N
Volume Quota                     N
-----
Measure Period                : 0
Volume Threshold

```

```

UL          : 0          DL          : 0
Total       : 1600
Subsequent Volume Threshold
UL          : 0          DL          : 0
Total       : 0
Volume Quota
UL          : 0          DL          : 0
Total       : 0
Volume Usage
UL          : 600        DL          : 600
Time Threshold : 0 secs  Subsequent Time Thresho*: 0 secs
Time Quota     : 0 secs  Time Usage       : 18 secs
Quota Holding Time : 0 secs  Drp DL Traffic Threshold: 0
Monitoring Time  : 0 secs  Inactivity Detection Ti*: 60 secs
Parent URR      : 1073741829
Measurement Information
Before QOS      : No      Inactivate Measurement : No
-----
Number of PDN contexts : 6
=====
* indicates that the corresponding row element may have been truncated.

```

Example output PDN context statistics detail NEMO session

The following example illustrates IMSI detail statistics over S5, but it is also applicable over S2a (eHRPD) and Pi (HRPD).

```

# show mobile-gateway pdn pdn-context imsi 310012123450000 detail
=====
PDN context detail
=====
IMSI          : 310012123450000
APN           : ipd.1000.nokia.1.1.com
Virtual APN   : N/A
Virtual APN sel : -
Auth Status   : IMSI authenticated
IMEI          : 987654321012399
MSISDN        : 123456789000
PDN type      : IPv4          LBI          : 6
Ref-point/Sig-pr*: S5/GTPv2  APN restriction : 0
UL APN AMBR   : 2000 Kbps     DL APN AMBR   : 5000 Kbps
Bearer contexts : 1          SDFs          : 0
Piggyback bearers requested : 0
Piggyback bearers creation successful : 0

Pri DNS IPv4 addr: 10.30.1.1      Sec DNS IPv4 addr: 0.0.0.0
Pri DNS IPv6 addr: ::             Sec DNS IPv6 addr: ::

Pri NBNS IPv4 ad*: 0.0.0.0        Sec NBNS IPv4 ad*: 0.0.0.0
Pri NBNS IPv6 ad*: ::             Sec NBNS IPv6 ad*: ::

Charging bearer *: Home

UE IPv4 address      : 1.7.209.0
UE IPv4 address source : AAA|HSS|Local Pool|DHCP|L2TP

```

```
S5 PGW Ctrl Teid : 0x54500          S5 SGW Ctrl Teid : 0x11120
S5 PGW V4 Ctl Adr: 16.16.16.16      S5 SGW V4 Ctl Adr: 10.10.5.1

PCRF triggers      : SGW QOS RAT BR_LOSS BR_RVRY RAI ULI IPCAN_CHANGE
                    DEF_BR_QOS AN_GW RSRC_ALLOC RSRC_MOD TZ OUT_CRDT
                    REALLOC_CRDT REVAL_TO PLMN
Gx primary peer profile      : Peer-Profile1
Gx secondary peer profile    :
Gx peer profile in use       : Peer-Profile1
Gx failover count            : 0
Gx peer origin                : Local configuration
PCRF hostname                :
PCRF address                  : 10.10.43.2
PCRF port                     : 3868
PCRF destination host        : primary-pcrf.nokia.com
Gx local address              : 16.16.16.19
PCRF enabled Gy Event Reporting :

Charging profile : 1                Interim records *: 0
RF triggers      : 0

S6b primary peer profile      :
S6b secondary peer profile    :
S6b peer profile in use       :
S6b failover count            : 0
S6b peer hostname             :
S6b peer address               : ::
S6b peer port                  : 0
S6b peer dest host             :
S6b peer origin                :
S6b local address              : ::

Framed Routes      : 0
Redir Mob Ipv4 Addr : ::
Redir Mob Ipv6 Addr : ::
L2TP Connection *: 0

Restoration Priority Level : 0

P-CSCF Reselection      : Y
Registered P-CSCF Addr   : ::

eMPS IMS priority session : No

Alloc ePDG svr Ipv4 Addr  : ::
Alloc ePDG svr Ipv6 Addr  : ::

Override Rating Group      : N/A
Override Online Billing Method : Disabled
Override Offline Billing Method : Disabled
-----
IMSI      : 310012123450000
APN        : ipd.1000.nokia.1.1.com
Access-Type : NON 3GPP          Selection mode : NA
Ref-point/Sig-pr*: Pi/MIPv4
Nemo HA Ctrl Addr  : 31.31.31.31      FA Ctrl Addr   : 1.7.209.0
Nemo HA data Addr  : 31.31.31.31      FA data Addr   : 1.7.209.0
```

```

Pi DL packets      : 0                Pi DL bytes       : 0
SGI UL packets     : 0                SGI UL bytes      : 0

Nemo Type: CSC0 NSVE / RFC5177
Framed Routes      : 8 101.1.0.0/24      1(no metric provided)
  102.1.0.0/24      1(no metric provided)
  103.1.0.0/24      1(no metric provided)
  104.1.0.0/24      1(no metric provided)
  105.1.0.0/24      1(no metric provided)
  106.1.0.0/24      1(no metric provided)
  107.1.0.0/24      1(no metric provided)
  108.1.0.0/24      1(no metric provided)
-----
Number of PDN contexts : 2
=====

```

Example output PDN context statistics detail eMPS advance priority calls

```

# show mobile-gateway pdn pdn-context imsi 222222222222222 detail
=====
PDN context detail
=====
IMSI                : 222222222222222
APN                  : ipd.nokia.1.1.com
Virtual APN          : N/A
Public APN           : N/A
Virtual APN sel      : -
Auth Status          : IMSI authenticated
IMEI                 : 222222222222222
MSISDN               : 22222222222
Data Session Id      : 0x00010100      AA Session Id    : 0x00040001
PDN type              : IPv6             LBI               : 5
Ref-point/Sig-pr*    : N/A              APN restriction    : 0
UL APN AMBR          : 8640 Kbps         DL APN AMBR        : 8640 Kbps
Bearer contexts       : 2                SDFs               : 1
Piggyback bearers requested : 0
Piggyback bearers creation successful : 0

Primary DNS IPv4 address : 10.30.1.1
Secondary DNS IPv4 address : 0.0.0.0
Primary DNS IPv6 address  : ::
Secondary DNS IPv6 address : ::

Pri NBNS IPv4 ad*: 0.0.0.0      Sec NBNS IPv4 ad*: 0.0.0.0
Pri NBNS IPv6 ad*: ::          Sec NBNS IPv6 ad*: ::

Charging bearer *: Visiting

UE IPv6 address        : 3ffe:0:1:ff00::ffff:e3cc
UE IPv6 address source : Local Pool

PCRF triggers          : RSRG_ALLOC
Gx primary peer profile : Peer-Profile1
Gx secondary peer profile : Peer-Profile3

```

```

Gx peer profile in use      : Peer-Profile1
Gx failover count          : 0
Gx peer origin              : Local configuration
PCRF hostname               :
PCRF address                : 10.10.43.2
PCRF port                   : 3868
PCRF destination host      : primary-pcrf.nokia.com
Gx local address            : 16.16.16.19
PCRF enabled Gy Event Reporting :

Charging profile : 1                      Interim records *: 0
RF triggers      : 0
S6b primary peer profile : s6b_Peer-Profile1
S6b secondary peer profile : s6b_Peer-Profile2
S6b peer profile in use   : s6b_Peer-Profile1
S6b failover count       : 0
S6b peer hostname        :
S6b peer address         : 10.23.19.2
S6b peer port            : 3868
S6b peer dest host       : aaa.nokia.com
S6b peer origin          : mscpl01.diam.origRealm.com
S6b local address        : 16.16.16.25

Framed Routes      : 0
Redir Mob Ipv4 Addr : ::
Redir Mob Ipv6 Addr : ::
L2TP Connection *: 0

Restoration Priority Level : 0

P-CSCF Reselection      : Y
Registered P-CSCF Addr  : ::

eMPS IMS priority session : Advance

Alloc ePDG svr Ipv4 Addr : ::
Alloc ePDG svr Ipv6 Addr : ::

Override Rating Group    : N/A
Override Online Billing Method : Disabled
Override Offline Billing Method : Disabled
Serving PLMN rat*: disabled
CPOPCI                   : disabled          EPCOSI                   : disabled
APN rate lmt Drp*: 0      PLMN rate lmt Dr*: 0
MO Exception Cou*: 0      MO Exception Tim*: 0
-----
Number of PDN contexts : 1
=====

```

Example output PDN context statistics detail virtual APN selection

```

# show mobile-gateway pdn pdn-context imsi 123456789012301 detail
=====
PDN context detail
=====
IMSI : 123456789012301

```



```
APN : apn.1.1.com
Public APN : N/A
Intermediate Virtual APN : apn.1.1.com-premium01
Intermediate Virtual APN sel : append-string
Intermediate Virtual APN sel user-range type : imsi cc
Virtual APN : apn.1.2.com
Virtual APN sel : Radius Pre-Auth
Virtual APN sel user-range type : msisdn
Virtual APN sel scope : pdn apn
Auth Status : IMSI authenticated
IMEI : 187654321012301
MSISDN : 123456789112301
Data Session Id : 0x00031110 AA Session Id : 0x00444003
PDN type : IPv4 LBI : 5
Ref-point/Sig-pr*: S5/GTPv2 APN restriction : 0
UL APN AMBR : 5000 Kbps DL APN AMBR : 5000 Kbps
Bearer contexts : 1 SDFs : 0
Piggyback bearers requested : 0
Piggyback bearers creation successful : 0
Primary DNS IPv4 address : 30.40.1.1
Secondary DNS IPv4 address : 30.40.1.2
Primary DNS IPv6 address : ::
Secondary DNS IPv6 address : ::
Pri NBNS IPv4 ad*: 0.0.0.0 Sec NBNS IPv4 ad*: 0.0.0.0
Pri NBNS IPv6 ad*: :: Sec NBNS IPv6 ad*: ::
Charging bearer *: Visiting
UE IPv4 address : 2.127.255.0
UE IPv4 address source : Local Pool
S5 PGW Ctrl Teid : 0x31110 S5 SGW Ctrl Teid : 0x20120
S5 PGW V4 Ctl Adr: 16.16.16.16 S5 SGW V4 Ctl Adr: 10.10.5.1
PCRF triggers : N/A
Gx primary peer profile : Peer-Profile1
Gx secondary peer profile : Peer-Profile3
Gx peer profile in use : Peer-Profile1
Gx failover count : 0
Gx peer origin : Local configuration
PCRF hostname :
PCRF address : 10.10.43.2
PCRF port : 3868
PCRF destination host : primary-pcrf.nokia.com
Gx local address : 16.16.16.19
PCRF enabled Gy Event Reporting :
Charging profile : 1 Interim records *: 0
RF triggers : 0
S6b primary peer profile :
S6b secondary peer profile :
S6b peer profile in use :
S6b failover count : 0
S6b peer hostname :
S6b peer address : ::
S6b peer port : 0
S6b peer dest host :
S6b peer origin :
S6b local address : ::
Framed Routes : 0
Redir Mob Ipv4 Addr : ::
Redir Mob Ipv6 Addr : ::
```

```

L2TP Connection *: 0
Restoration Priority Level : 0
P-CSCF Reselection : Y
  Registered P-CSCF Addr : ::
eMPS IMS priority session : No
Alloc ePDG svr Ipv4 Addr : ::
  Alloc ePDG svr Ipv6 Addr : ::
Override Rating Group : N/A
  Override Online Billing Method : Disabled
  Override Offline Billing Method : Disabled
Serving PLMN rat*: disabled
CPOPCI : disabled EPCOSI : disabled
PGW UL Rate Control : disabled
DL APN rate lmt Drp Pkt : 0
UL APN rate lmt Drp Pkt : 0
PLMN rate lmt Drp Pkt : 0
MO Exception Cou*: 0 MO Exception Tim*: 0
-----
Number of PDN contexts : 1
=====
indicates that the corresponding row element may have been truncated.

```

Example output PDN context statistics dedicated bearer

Session level example:

```

# show mobile-gateway pdn pdn-context {imsi|imei|msisdn} string dedicated-bearer-
monitoring [gateway <gw-id>] [apn <name>]
=====
PDN Context Dedicated Bearer Monitoring
=====
IMSI           : 123456789012953
APN            : ipd.nokia.com
PDN Type       : IPv6             LBI           : 5
Bearer Id      : 4
Status         :
Timestamp      :                  # for activation, actv-attempt or deactivation
Setup duration :
Active Duration :                  # filled only if activated (duration up to now)
                                   # or deactivated (total duration)
P-CSCF IP Address :                  # report the one at the time of deact or failure
Deactivation Reason :
Deactivation CC-GTP : GTP sent / received CC
Deactivation CC-Dia*: Diameter sent received CC
Setup Fail Gx CC  :
Detailed Failure Gx :
Setup Fail GTP CC :
Detailed Failure GTP:
-----
Number of PDN contexts : 1
=====
* indicates that the corresponding row element may have been truncated
#

```

Example output PDN context statistics Gx

Session level example:

```
# show mobile-gateway pdn pdn-context imsi 123456789012301 gx-usage-
monitoring monitoring-key 6533221
=====
PDN Context Gx Usage Monitoring Detail
=====
IMSI                : 123456789012301
APN                 : ipd.nokia.com
PDN Type            : IPv4                      LBI                : 5
UL APN AMBR         : 8640 Kbps                 DL APN AMBR         : 8640 Kbps
Bearer Contexts     : 1
UE IPv4 Address     : 5.127.255.0
Num Usage Mon Inst* : 1                        Session State      : Normal
Over Quota Traffic  : Allow                     Suppress Excess Qu* : Disabled
-----
Monitoring Key      : 6533221                   Scope              : Session Level
GSU Scope           : Total
GSU Threshold       : 100000
Current USU         : 0
Threshold Reports   : 0                        On Demand Reports   : 0
Local Threshold     : 0
Local Threshold Re* : 0
-----
Number of PDN contexts : 1
=====
* indicates that the corresponding row element may have been truncated.
```

Rule level example:

```
# show mobile-gateway pdn pdn-context imsi 321011234567800 gx-usage-
monitoring monitoring-key 123456789
=====
PDN Context Gx Usage Monitoring Detail
=====
IMSI                : 321011234567800
APN                 : onlineoffline
PDN Type            : IPv4                      LBI                : 5
UL APN AMBR         : 2048 Kbps                 DL APN AMBR         : 2048 Kbps
Bearer Contexts     : 1
UE IPv4 Address     : 1.127.255.2
Num Usage Mon Inst* : 1                        Session State      : Normal
Over Quota Traffic  : Allow                     Suppress Excess Qu* : Disabled
-----
Monitoring Key      : 123456789                   Scope              : Rule Level
GSU Scope           : Total
GSU Threshold       : 100000
Current USU         : 0
Threshold Reports   : 0                        On Demand Reports   : 0
Local Threshold     : 0
Local Threshold Re* : 0
-----
PCC Rules                                                    Excluded
-----
Rule 2 : qci 9 arp 11 : port 63502 62502                    False
-----
```

```
Number of PDN contexts : 1
```

```
=====
* indicates that the corresponding row element may have been truncated.
```

Example output PDN context Gx Time of Day information statistics (time-conditioned policy information)



NOTE:

The `gx-time-of-day-info` command displays both APN-AMBR and QCI/ARP values, regardless of whether they are both installed. For example, if only QCI/ARP is enabled the APN-AMBR indicates 0 and vice versa.

```
# show mobile-gateway pdn pdn-context imsi 123457789012609 gx-time-of-day-info
```

```
=====
PDN Context Gx Time Of Day
```

```
=====
IMSI           : 123457789012609
APN            : ipd.nokia.1.1.com
PDN type       : IPv4                LBI           : 5
UL APN AMBR    : 90 Kbps              DL APN AMBR   : 50.00 Mbps
UE IPv4 address : 1.127.255.1
Revalidation Time : -
Revalidation Timeo*: 0
Time-conditioned Policy Info:
  Execution-Time : 10/26/2018 05:35:46
  UL APN AMBR    : 90 Kbps              DL APN AMBR   : 50.00 Mbps
  Default-QoS QCI : 7                  Default-QoS ARP : 10
  Execution-Time : 10/26/2018 05:35:56
  UL APN AMBR    : 9 Kbps               DL APN AMBR   : 1000.00 Mbps
  Default-QoS QCI : 8                  Default-QoS ARP : 10
-----
```

```
Number of PDN contexts : 1
=====
```

Example output PDN context statistics MAC detail

```
# show mobile-gateway pdn pdn-context mac 00:02:00:00:00:00 detail
```

```
=====
PDN context detail
```

```
=====
IMSI/MAC : 3100120000000000/00:02:00:00:00:00
APN       : ipd.nokia.1.1.com.mnc012.mcc310.gprs
Auth Status : IMSI authenticated
IMEI      : N/A
PDN type   : IPv4                LBI           : 5
Ref-point/Sig-pr*: SWw/Soft-GRE  APN restriction : 255
UL APN AMBR : 5000 Kbps          DL APN AMBR   : 5000 Kbps
UE IPv4 address : 5.0.255.0
TWAG
Access type : eap
STA/RADIUS  : STA
SaMOG/NSW0/Combo : SaMOG
TWAG MAC address : 4a:4a:ff:00:03:5b
AP MAC address : 99:02:00:00:00:00
```

```

Sww TWAG tunnel addr : 201.3.2.1
Sww AP tunnel addr   : 201.3.2.10
Sww RADIUS TWAG Auth Addr : 9.9.9.10
Sww RADIUS AP Auth Addr  : 201.3.2.10
STa TWAG Addr         : 180.1.2.1
STa AAA Addr          : 180.1.2.200
RADIUS AAA Addr        : 0.0.0.0
RADIUS Acct addr       : 0.0.0.0
S2a TWAG Ctrl Teid    : 0x14500
S2a PGW Ctrl Teid     : 0x12900
S2a TWAG V4 Ctl Adr   : 16.16.16.16      S2a PGW V4 Ctl Adr   : 10.10.5.1
S2a TWAG V6 Ctl Adr   : 3ffe::1010:1010
S2a PGW V6 Ctl Adr    : 3ffe::a0a:501
DHCP remaining lease time (sec): 7176
Remaining cache time (sec)   : 3636
-----
Number of PDN contexts : 1
=====

```

Example output PDN context statistic (MAPCON)

```

# show mobile-gateway pdn pdn-context
=====
IMSI/IMEI( # )      APN      Type   Beare* UE Address (IPv4/IPv6)  Ref-pt/Si*
/MSISDN(^)/MAC(`)
=====
123456789012024    cg1.ipd.* IPv4    1      5.5.5.85/-                  S5/GTPv2
123456789012024    ipd.noki* IPv4    1      1.127.255.0/-              S2b/GTPv2
-----
Number of PDN instances : 2
-----
# indicates that the corresponding row is IMEI entry.
^ indicates that the corresponding row is MSISDN entry.
` indicates that the corresponding row is MAC entry.

```

```

# show mobile-gateway pdn pdn-context imsi 123456789012024 detail
=====
PDN context detail
=====
IMSI      : 123456789012024
APN       : cg1.ipd.nokia.com
Virtual APN : N/A
Auth Status : IMSI authenticated
IMEI      : 987654321012399
MSISDN    : 12345098761
PDN type  : IPv4                LBI                : 5
S5 sig protocol : GTP                APN restriction   : 0
UL APN AMBR : 5000 Kbps            DL APN AMBR       : 5000 Kbps
Bearer contexts : 1

UE IPv4 address      : 5.5.5.85
UE IPv4 address source : AAA|HSS|Local Pool|DHCP|L2TP

S5 PGW Ctrl Teid : 0x31000050      S5 SGW Ctrl Teid : 0x12120

```

```
S5 PGW V4 Ctl Adr: 16.80.16.10      S5 SGW V4 Ctl Adr: 16.90.16.10
S5 SGW V6 Ctl Adr: 3ffe::105a:100a
```

```
Charging level   : Pdn                Bearer type       : Home
Charging server address: 10.23.13.2
Charging server state : Up
Charging prof    : 1
Charging triggers: 0                  Interim records   : 0
```

```
-----
IMSI             : 123456789012024
APN              : ipd.nokia.1.1.com
Virtual APN      : N/A
Auth Status      : IMSI authenticated
IMEI             : 987654321012399
MSISDN           : 12345098761
PDN type         : IPv4                LBI              : 5
Ref-point/Sig-pr*: S2b/GTPv2          APN restriction   : 0
UL APN AMBR      : 5000 Kbps          DL APN AMBR       : 5000 Kbps
Bearer contexts  : 1                  SDFs              : 0
Piggyback bearers requested           : 0
Piggyback bearers creation successful : 0
```

```
Charging bearer *: Visiting
```

```
UE IPv4 address      : 1.127.255.0
```

```
S2b PGW Ctrl Teid: 0x22120          S2b ePDG Ctrl Te*: 0xc0000056
S2b PGW V4 Ctl A*: 16.70.16.19      S2b ePDG V4 Ctl *: 10.70.24.25
S2b PGW V6 Ctl Adr: 3ffe::1046:1013
```

```
PCRF triggers      : QOS RAT IPCAN_CHANGE DEF_BR_QOS AN_GW
Gx primary peer profile : Peer-Profile1
Gx secondary peer profile : Peer-Profile3
Gx peer profile in use   : Peer-Profile1
Gx failover count       : 0
Gx peer origin          : Local configuration
PCRF hostname          :
PCRF address           : 10.10.43.2
PCRF port              : 3868
PCRF destination host   : primary-pcrf.nokia.com
Gx local address        : 16.16.16.19
PCRF enabled Gy Event Reporting :
```

```
Charging profile : 1                  Interim records *: 0
RF triggers       : 0
```

```
S6b primary peer profile : s6b_Peer-Profile1
S6b secondary peer profile : s6b_Peer-Profile2
S6b peer profile in use   : s6b_Peer-Profile1
S6b failover count       : 0
S6b peer hostname        :
S6b peer address         : 10.23.19.2
S6b peer port            : 3868
S6b peer dest host       : aaa.nokia.com
S6b peer origin          : diam.origRealm.com
S6b local address        : 16.16.16.25
```

```

Framed Routes      : 0
Redir Mob Ipv4 Addr : ::
Redir Mob Ipv6 Addr : ::
L2TP Connection *: 0

Restoration Priority Level : 0

P-CSCF Reselection      : Y
Registered P-CSCF Addr   : ::

Override Rating Group      : N/A
Override Online Billing Method : Disabled
Override Offline Billing Method : Disabled
-----
Number of PDN contexts : 2
=====
* indicates that the corresponding row element may have been truncated.

```

Example output PDN context combination ePDG/PGW

```

# show mobile-gateway pdn pdn-context
=====
IMSI/IMEI(#)      APN      Type      Beare* UE Address (IPv4/IPv6)  Ref-pt/Si*
/MSISDN(^)/MAC(`)
=====
3100120000000000  ipd.nok* IPv4      1      1.127.255.0/-      S2b/GTPv2
3100120000000000  ipd.nok* IPv4      1      1.127.255.0/-      S2b/GTPv2
-----
Number of PDN instances : 2
-----
# indicates that the corresponding row is IMEI entry.
^ indicates that the corresponding row is MSISDN entry.
` indicates that the corresponding row is MAC entry.
-----
* indicates that the corresponding row element may have been truncated.

# show mobile-gateway pdn pdn-context imsi 3100120000000000
=====
IMSI/IMEI(#)      APN      Type      Beare* UE Address (IPv4/IPv6)  Ref-pt/Si*
/MSISDN(^)/MAC(`)
=====
3100120000000000  ipd.nok* IPv4      1      1.127.255.0/-      S2b/GTPv2
3100120000000000  ipd.nok* IPv4      1      1.127.255.0/-      S2b/GTPv2
-----
Number of PDN instances : 2
-----
# indicates that the corresponding row is IMEI entry.
^ indicates that the corresponding row is MSISDN entry.
` indicates that the corresponding row is MAC entry.
-----
* indicates that the corresponding row element may have been truncated.

```

Example output PDN context IMSI detail combination ePDG/PGW

```
# show mobile-gateway pdn pdn-context imsi 123456789012809 detail
=====
PDN context detail
=====
IMSI           : 123456789012809
APN            : ipd.nokia.1.1.com
Virtual APN    : N/A
Virtual APN sel : -
Auth Status    : IMSI authenticated
IMEI           : 123560789012809
MSISDN        : 123104104098761
PDN type       : IPv6                      LBI           : 5
Ref-point/Sig-pr* :                      APN restriction : 0
UL APN AMBR    : 5000 Kbps                DL APN AMBR    : 5000 Kbps
Bearer contexts : 1                      SDFs           : 0
Piggyback bearers requested : 0
Piggyback bearers creation successful : 0

Pri DNS IPv4 addr: 10.30.1.1              Sec DNS IPv4 addr: 0.0.0.0
Pri DNS IPv6 addr: ::                     Sec DNS IPv6 addr: ::

Pri NBNS IPv4 ad*: 0.0.0.0                Sec NBNS IPv4 ad*: 0.0.0.0
Pri NBNS IPv6 ad*: ::                     Sec NBNS IPv6 ad*: ::

Charging bearer *: Visiting

UE IPv6 address      : 3ffe:0:1:ff02::2
UE IPv4 address source : AAA|HSS|Local Pool|DHCP|L2TP

PCRF triggers       : N/A
Gx primary peer profile : Peer-Profile1
Gx secondary peer profile : Peer-Profile3
Gx peer profile in use : Peer-Profile1
Gx failover count    : 0
Gx peer origin       : Local configuration
PCRF hostname        :
PCRF address         : 10.10.43.2
PCRF port            : 3868
PCRF destination host : primary-pcrf.nokia.com
Gx local address     : 16.16.16.19
PCRF enabled Gy Event Reporting :
Charging profile : 1                      Interim records *: 0
RF triggers       : 0

S6b primary peer profile : s6b_Peer-Profile1
S6b secondary peer profile : s6b_Peer-Profile2
S6b peer profile in use : s6b_Peer-Profile1
S6b failover count    : 0
S6b peer hostname     :
S6b peer address      : 10.23.19.2
S6b peer port         : 3868
S6b peer dest host    : aaa.nokia.com
S6b peer origin       : diam.origRealm.com
S6b local address     : 16.16.16.25
```



```

Framed Routes      : 0
Redir Mob Ipv4 Addr : ::
Redir Mob Ipv6 Addr : ::
L2TP Connection *: 0

Restoration Priority Level : 0

P-CSCF Reselection      : Y
Registered P-CSCF Addr   : ::

eMPS IMS priority session : No

Alloc ePDG svr Ipv4 Addr : ::
Alloc ePDG svr Ipv6 Addr : ::

Override Rating Group      : N/A
Override Online Billing Method : Disabled
Override Offline Billing Method : Disabled

```

```

-----
Number of PDN contexts : 1
=====

```

```
* indicates that the corresponding row element may have been truncated.
```

```
# show mobile-gateway pdn pdn-context imsi 310012123456789 detail
```

```
=====
PDN context detail
=====
```

```

IMSI           : 310012123456789
APN            : ipd.nokia.1.1.com
Virtual APN     : N/A
Auth Status    : IMSI authenticated
IMEI          : 987654321012399
MSISDN        : 12345098761
PDN type       : IPv4                      LBI           : 5
Ref-point/Sig-pr*: S2a/GTPv2              APN restriction : 0
UL APN AMBR    : 5000 Kbps                 DL APN AMBR     : 5000 Kbps
Bearer contexts : 1                       SDFs           : 0
Piggyback bearers requested                : 0
Piggyback bearers creation successful       : 0

```

```
Charging bearer *: Home
```

```

UE IPv4 address      : 1.0.255.0
UE IPv4 address source : AAA|HSS|Local Pool|DHCP|L2TP

```

```

S2a PGW Ctrl Teid: 0x12d00                S2a TWAG Ctrl Te*: 0xc0000001
S2a PGW V4 Ctl A*: 10.40.1.3              S2a TWAG V4 Ctl *: 10.23.23.2

```

```

PCRF triggers      : N/A
Gx primary peer profile : Peer-Profile1
Gx secondary peer profile :
Gx peer profile in use : Peer-Profile1
Gx failover count    : 0
Gx peer origin       : Local configuration
PCRF hostname        :

```

```

PCRF address          : 10.10.43.2
PCRF port             : 3868
PCRF destination host : primary-pcrf.nokia.com
Gx local address      : 16.16.16.19
PCRF enabled Gy Event Reporting :

Charging profile : 1                      Interim records *: 0
RF triggers      : 0

S6b primary peer profile : s6b_Peer-Profile1
S6b secondary peer profile : s6b_Peer-Profile2
S6b peer profile in use : s6b_Peer-Profile1
S6b failover count      : 0
S6b peer hostname       :
S6b peer address        : ::
S6b peer port           : 900
S6b peer dest host      : aaa.nokia.com
S6b peer origin         : diam.origRealm.com
S6b local address       : ::

Framed Routes          : 0
Redir Mob Ipv4 Addr    : ::
Redir Mob Ipv6 Addr    : ::
L2TP Connection *: 0

Restoration Priority Level : 0

P-CSCF Reselection      : Y
Registered P-CSCF Addr  : ::

Override Rating Group    : N/A
Override Online Billing Method : Disabled
Override Offline Billing Method : Disabled
-----
Number of PDN contexts : 1
=====
* indicates that the corresponding row element may have been truncated.

# show mobile-gateway pdn pdn-context imsi 3100120000000000 detail
=====
PDN context detail
=====
IMSI          : 3100120000000000
APN           : ipd.nokia.1.1.com.mnc012.mcc310.gprs
Virtual APN    : N/A
Auth Status   : IMSI authenticated
IMEI          : N/A
MSISDN        : N/A
PDN type      : IPV4                      LBI          : 5
Ref-point/Sig-pr*: S2b/GTPv2             APN restriction : 0
UL APN AMBR   : 5000 kbps                 DL APN AMBR   : 5000 kbps
Bearer contexts : 1                      SDFs          : 0
Piggyback bearers requested              : 0
Piggyback bearers creation successful    : 0

Charging bearer *: Home

```

```
UE IPv4 address          : 1.127.255.0
UE IPv4 address source   : AAA|HSS|Local Pool|DHCP|L2TP

S2b PGW Ctrl Teid: 0x13900          S2b ePDG Ctrl Te*: 0x13d00
S2b PGW V4 Ctl A*: 16.70.16.19      S2b ePDG V4 Ctl *: 16.70.16.24

PCRF triggers           : N/A
Gx primary peer profile : Peer-Profile1
Gx secondary peer profile : Peer-Profile3
Gx peer profile in use  : Peer-Profile1
Gx failover count       : 0
Gx peer origin          : Local configuration
PCRF hostname           :
PCRF address            : 10.10.43.2
PCRF port               : 3868
PCRF destination host   : primary-pcrf.nokia.com
Gx local address        : 16.16.16.19
PCRF enabled Gy Event Reporting :

Charging profile : 1                      Interim records *: 0
RF triggers      : 0

S6b primary peer profile : s6b_Peer-Profile1
S6b secondary peer profile : s6b_Peer-Profile2
S6b peer profile in use  : s6b_Peer-Profile1
S6b failover count       : 0
S6b peer hostname        :
S6b peer address         : 10.23.19.2
S6b peer port            : 3868
S6b peer dest host       : aaa.nokia.com
S6b peer origin          : diam.origRealm.com
S6b local address        : 16.16.16.25

Framed Routes           : 0
Redir Mob Ipv4 Addr : ::
Redir Mob Ipv6 Addr : ::
L2TP Connection *: 0

Restoration Priority Level : 0

Override Rating Group    : N/A
Override Online Billing Method : Disabled
Override Offline Billing Method : Disabled
-----
IMSI                     : 3100120000000000
APN                      : ipd.nokia.1.1.com.mnc012.mcc310.gprs
Virtual APN              : N/A
Auth Status              : IMSI authenticated
IMEI                     : N/A
MSISDN                   : N/A
PDN type                  : IPv4                      LBI                      : 5
S2b sig protocol         : GTP                      APN restriction         : 255
UL APN AMBR              : 5000 kbps                 DL APN AMBR             : 5000 kbps
Bearer contexts          : 1

UE IPv4 address          : 1.127.255.0
```

```

S2b PGW Ctrl Teid: 0x13900          S2b ePDG Ctrl Te*: 0x13d00
S2b PGW V4 Ctl A*: 16.70.16.19      S2b ePDG V4 Ctl *: 16.70.16.24

SWu UE Ctrl SPI : 68be5efaf62102f1  SWu ePDG Ctrl SPI: 576103d02f813d0e
SWu UE Ctrl Addr : 10.168.0.1         SWu ePDG Ctrl Ad*: 16.70.16.28

```

```

Charging level : Bearer          Bearer type : Home
-----

```

```

Number of PDN contexts : 2
=====

```

```

* indicates that the corresponding row element may have been truncated.

```

Example output PDN context ePDG/ decrypt IMSI detail

```

# show mobile-gateway pdn pdn-context imsi 3100120000000000 detail epdg
=====
PDN context detail
=====
IMSI           : 3100120000000000
APN            : ipd.nokia.1.1.com.mnc012.mcc310.gprs
Virtual APN    : N/A

ike-sa
Local address  : 16.70.16.28          Remote Address  : 10.168.0.1
Local UDP port : 4500                 Remote UDP port : 4500
Auth          : hmac-sha1
Encryption    : aes128
DH-group      : modp1024
Local SPI     : 68be5efaf62102f1      Remote SPI      : 576103d02f813d0e
Lifetime      : 23h20m00s
Established time : 04/14/2016 17:01:29
Expiry time    : 04/15/2016 16:21:29
Number of rekeys : 0
Rx packets     : 4                    Tx packets     : 4
Rx errors      : 0                    Tx errors      : 0
Rx DPD         : 0                    Tx DPD         : 0
Rx DPD ACK     : 0                    Tx DPD ACK     : 0
Rx retransmits : 0                    Tx retransmits : 0

esp-sa
Extended-seq-num : Disabled
TS-type         : IPv4
Auth           : hmac-sha1
Encryption     : aes256
DH-group       : none
Local/Inbound SPI: 3d000000           Remote/Outbound *: 10000100
Lifetime       : 23h20m00s
Established time : 04/14/2016 17:01:29
Expiry time     : 04/15/2016 16:21:29
Decoded packets : 0                   Encoded packets: : 0
Decoded bytes   : 0                   Encoded bytes:   : 0
Decoded errors  : 0                   Encoded errors:  : 0
Replay errors   : 0
-----
Number of PDN contexts : 1

```

=====

* indicates that the corresponding row element may have been truncated.

=====

```
# show mobile-gateway pdn pdn-context imsi 310012000000000 detail epdg-decrypt
```

=====

PDN context detail

=====

```
IMSI           : 310012000000000
APN            : ehrrpd.mnc012.mcc310.gprs
Emergency call : No
ike-sa
Local address  : 10.0.1.118           Remote Address  : 168.0.0.1
Local UDP port : 4500                 Remote UDP port : 4500
Integrity      : hmac-sha1
Pseudo-Random-Fu*: prf-sha1
Encryption     : aes128cbc
DH-group       : modp1024
Local SPI      : 3325a602fbb3a1e      Remote SPI      : 9037724d45706116
Lifetime       : 23h20m00s
Established time : 10/12/2020 05:26:41
Expiry time    : 10/13/2020 04:46:41
Number of rekeys : 0
Rx packets     : 4                   Tx packets     : 4
Rx errors      : 0                   Tx errors      : 0
Rx DPD         : 0                   Tx DPD         : 0
Rx DPD ACK     : 0                   Tx DPD ACK     : 0
Rx retransmits : 0                   Tx retransmits : 0
Authentication/Decryption Keys
SK_ai          : 0x1636fe074e203d74b6cb719a14d9b5c9b25886ad
SK_ar          : 0x67c158b5c4dfb372a9a376c690795f5c3c1760c4
SK_ei          : 0xc9ab21f47a38408e05a0920f5e8d02e5
SK_er          : 0x7ed9a3e49cffae0e1b0be8f2aa489fa9
esp-sa
Extended-seq-num : Disabled
TS-type         : IPV4
Integrity       : hmac-sha1
Encryption      : aes256cbc
DH-group        : none
Local/Inbound SPI: 61000001          Remote/Outbound *: 00000100
Lifetime        : 23h20m00s
Established time : 10/12/2020 05:26:41
Expiry time     : 10/13/2020 04:46:41
Decoded packets : 0                   Encoded packets: : 0
Decoded bytes   : 0                   Encoded bytes:   : 0
Decoded errors  : 0                   Encoded errors:  : 0
Replay errors   : 0
Authentication/Decryption Keys
In IK           : 0xd15ad6cb4ce5dde8af320309a2260a4bb7d6d057
In CK           : 0x290ccea8fb027955a8f682bde54b58877ae1da8a89b4c46a9c0e705fe2
                d6f515
Out IK          : 0xb4399b6e5ae943a2be2569100d432b8687c7a0b9
Out CK          : 0xb499ad0b64f55feb4890f6d271e5629086f95ab55b04204918085386d0
                3114e8
-----
```

Example output PDN context IMSI/IMEI detail ePDG for emergency calls over WLAN

```
# show mobile-gateway pdn pdn-context imsi 310012000000000 detail epdg
=====
PDN context detail
=====
IMSI           : 310012000000001
APN            : ehrrpd.mnc012.mcc310.gprs
Auth Status    : IMSI authenticated
IMEI           : 1234567000000100
MSISDN         : 421445221
PDN type       : IPv4                      LBI:5
S2b sig protocol : GTP                    APN restriction : 255
UL APN AMBR     : 100.00 Mbps              DL APN AMBR      : 100.00 Mbps
Bearer contexts : 1

UE IPv4 address : 1.0.255.15

S2b PGW Ctrl Teid: 0x201f0                S2b ePDG Ctrl Te*: 0x18100
S2b PGW V4 Ctl A*: 10.10.6.3              S2b ePDG V4 Ctl *: 10.10.53.1
S2b PGW V6 Ctl Adr: 3ffd::abc:1
S2b ePDG V6 Ctl Adr: 3ffa::c0:1

SWu UE Ctrl SPI : 7984bfd830f62203        SWu ePDG Ctrl SPI: a78c4d46c75a8109
SWu UE Ctrl Addr : 217.33.24.193          SWu ePDG Ctrl Ad*: 10.10.51.2

Swm primary peer profile : swm_Peer-Profile1
Swm secondary peer profile : swm_Peer-Profile2
Swm peer profile in use : swm_Peer-Profile1
Swm failover count      : 0
Swm peer hostname       : (Not Specified)
Swm peer address        : 10.10.68.2
Swm peer port           : 3868
Swm peer dest realm     : nokia.com
Swm peer dest host      : aaa.nokia.com
Swm peer origin         : msc01.diam.origRealm.com
Swm local address       : 10.10.51.3

Charging level: Pdn                      Bearer type      : Home
Charging server address:
Charging server state:
Charging prof: 1
Charging triggers: 0                      Interim records  : 0
```

WLAN Location Information

```

SSID           : apEpdg
BSSID          : 01:02:03:04:05:06
Civic-address   : 000023456478924342
Operator-Name  : nokia.com

```

```

-----
Number of PDN contexts : 1
=====

```

```

* indicates that the corresponding row element may have been truncated.

```

Example output PDN context IPsec lockout summary

```

# show mobile-gateway pdn pdn-context ipsec-lockout

```

```

=====
IDi

```

```

Source IP RT (mins) Subsequent lockout
=====

```

```

0611356789012345@wlan.mnc012.mcc310.3gppnetwork.org
7.50.4.51 37 minutes 1x
0673156789012345@wlan.mnc012.mcc310.3gppnetwork.org
7.50.4.51 24 minutes 1x
0154656789012345@wlan.mnc012.mcc310.3gppnetwork.org
7.50.4.51 17 minutes 1x
0157156789012345@wlan.mnc012.mcc310.3gppnetwork.org
7.50.4.51 50 minutes 1x
0389456789012345@wlan.mnc012.mcc310.3gppnetwork.org
7.50.4.51 11 minutes 1x
0466756789012345@wlan.mnc012.mcc310.3gppnetwork.org
7.50.4.51 27 minutes 1x
0347156789012345@wlan.mnc012.mcc310.3gppnetwork.org
7.50.4.51 29 minutes 1x

```

```

# show mobile-gateway pdn pdn-context ipsec-lockout summary

```

```

=====
Ipsec Lockout statistics
=====

```

```

Number of locked user : 150
=====

```

Example output PDN context IPsec lockout IDi source IP

```

# show mobile-gateway pdn pdn-context ipsec-lockout idi

```

```

0611356789012345@wlan.mnc012.mcc310.3gppnetwork.org source-ip 7.50.4.51
=====

```

```

IDi

```

```

Source IP RT (mins) Subsequent lockout
=====

```

```

0611356789012345@wlan.mnc012.mcc310.3gppnetwork.org
7.50.4.51 34 minutes 1x
-----

```

```

Number of locked users: 1
-----

```

Example output PDN context APN

```
# show mobile-gateway pdn pdn-context apn "ipd.nokia.1.1.com"
=====
IMSI/IMEI(#)      APN      Type      Beare* UE Address (IPv4/IPv6)  Ref-pt/Si*
/MSISDN(^)/MAC(`)
/SubId
=====
123456789012575   ipd.noki* IPv6      1      -/3ffe:0:1:ff00::2      S5/GTPv2
123456789012621   ipd.noki* v4v6      1      1.127.255.1/3ffe:0:1:f* S5/GTPv2
123456789012036   ipd.noki* IPv4      1      1.127.255.2/-          NA
123456789012809   ipd.noki* IPv6      1      -/3ffe:0:1:ff02::2      NA
123456789012891   ipd.noki* IPv4      1      1.127.255.0/-          S5/GTPv2
-----
Number of PDN instances : 5
-----
# indicates that the corresponding row is IMEI entry.
^ indicates that the corresponding row is MSISDN entry.
` indicates that the corresponding row is MAC entry.
-----
* indicates that the corresponding row element may have been truncated.
```

Example output PDN context subscriber ID

```
# show mobile-gateway pdn pdn-context subscriber-id goldUsr101
=====
IMSI/IMEI(#)      APN      Type      Beare* UE Address (IPv4/IPv6)  Ref-pt/Si*
/MSISDN(^)/MAC(`)
/SubId
=====
goldUsr101        ipd.nokia* IPv4      1      9.0.0.1/-              radius/-
-----
Number of PDN instances : 1
-----
# indicates that the corresponding row is IMEI entry.
^ indicates that the corresponding row is MSISDN entry.
` indicates that the corresponding row is MAC entry.
-----
* indicates that the corresponding row element may have been truncated.
```

Example output PDN context subscriber ID detail

```
# show mobile-gateway pdn pdn-context subscriber-id goldUsr101 detail
=====
PDN context detail
=====
Subscribe-Id : goldUsr101
APN           : ipd.nokia.1.2.com
Virtual APN    : N/A
Public APN     : N/A
Virtual APN sel : -
Auth Status    : N/A
IMEI           : N/A
MSISDN         : N/A
PDN type       : IPv4          APN restriction : 0
UL APN AMBR    : 6000 Kbps     DL APN AMBR     : 6000 Kbps
```



```

Bearer contexts : 1                      SDFs : 4
Primary DNS IPv4 address : ::
Secondary DNS IPv4 address : ::
Primary DNS IPv6 address : ::
Secondary DNS IPv6 address : ::
Pri NBNS IPv4 ad*: ::                  Sec NBNS IPv4 ad*: ::
Pri NBNS IPv6 ad*: ::                  Sec NBNS IPv6 ad*: ::
Charging bearer *: Visiting

UE IPv4 address : 9.0.0.1
EPDG V4 Ctl Adr : -                    TWAG V4 Ctl Adr : 31.1.2.2
EPDG V6 Ctl Adr : -
TWAG V6 Ctl Adr : -
PCRF triggers : N/A
Sd primary peer profile :
Sd secondary peer profile :
Sd peer profile in use :
Sd failover count : 0
Sd peer origin : (Not Specified)
PCRF hostname : (Not Specified)
PCRF address : ::
PCRF port : 0
PCRF destination host : (Not Specified)
Sd local address : ::
PCRF enabled Gy Event Reporting :

Charging profile : 0                    Interim records *: 0
RF triggers : 0

Framed Routes : 0
Redir Mob Ipv4 Addr : ::
Redir Mob Ipv6 Addr : ::
Restoration Priority Level : 0
P-CSCF Reselection : Y
Registered P-CSCF Addr : ::
eMPS IMS priority session : No
Alloc ePDG svr Ipv4 Addr : ::
Alloc ePDG svr Ipv6 Addr : ::
Override Rating Group : N/A
Override Online Billing Method : Disabled
Override Offline Billing Method : Disabled
-----
Number of PDN contexts : 1
=====
* indicates that the corresponding row element may have been truncated.

```

Example output PDN context RADIUS accounting detail

```

# show mobile-gateway pdn pdn-context imsi 310012123456789 acct radius detail
=====
PDN Context Accounting Radius Detail
=====
IMSI : 310012123456789
APN : apn.nonip.1.1.com
Virtual APN : N/A
PDN type : non-ip          LBI : 5

```

```

URR ID          : 1073741825
UL APN AMBR     : 5000 Kbps      DL APN AMBR       : 5000 Kbps
Bearer Contexts : 1              Charging ID        : 1627389959

UE IPv4 Address  : 10.27.255.0
UE IPv6 Address  : N/A

Aggr UL Bytes    : 0              Aggr DL Bytes     : 0
Aggr UL Packets  : 0              Aggr DL Packets   : 0

Radius Group     : Radius Group
Supress IMSI     : no             Authenticated via Radius: no
Selective Accounting : no         Username Type      : Imsi

Interim Accounting
Accounting Interval : 60 s        Volume Limit      : 0 kB
Event Triggers      : N/A
Messages Tx         : 0
-----
Number of PDN contexts : 1
=====

```

Example output PDN context MSISDN detail

```

# show mobile-gateway pdn pdn-context msisdn 12345123456 detail
=====
PDN context detail
=====
IMSI          : 123456789012024
APN           : ipd.nokia.1.1.com
Public APN    : N/A
Virtual APN   : N/A
Virtual APN sel           : -
Virtual APN sel user-range type : -
Virtual APN sel scope     : N/A
Auth Status    : IMSI authenticated
IMEI           : 987654321012399
MSISDN         : 12345123456
Data Session Id : 0x00012110      AA Session Id   : 0x00844001
PDN type       : v4v6              LBI             : 5
Ref-point/Sig-pr*: N/A            APN restriction  : 0
PRA ID        : 0x0                PRA Status       : -
UL APN AMBR    : 5000 Kbps          DL APN AMBR      : 5000 Kbps
Bearer contexts : 1                 SDFs             : 0
Piggyback bearers requested       : 0
Piggyback bearers creation successful : 0
Primary DNS IPv4 address  : 10.30.1.1
Secondary DNS IPv4 address : 0.0.0.0
Primary DNS IPv6 address  : ::
Secondary DNS IPv6 address : ::
Pri NBNS IPv4 ad*: 0.0.0.0      Sec NBNS IPv4 ad*: 0.0.0.0
Pri NBNS IPv6 ad*: ::           Sec NBNS IPv6 ad*: ::
Charging bearer *: Visiting

UE IPv4 address      : 11.0.0.9
UE IPv4 address source : AAA

```

```

UE IPv6 address      : 20fe:2::2
UE IPv6 address source : AAA

PCRF triggers       : N/A
Gx primary peer profile : Peer-Profile1
Gx secondary peer profile : Peer-Profile3
Gx peer profile in use : Peer-Profile1
Gx failover count     : 0
Gx peer origin         : Local configuration
PCRF hostname         :
PCRF address           : 10.10.43.2
PCRF port              : 3868
PCRF destination host : primary-pcrf.nokia.com
Gx local address       : 16.16.16.19
PCRF enabled Gy Event Reporting :

Charging profile : 1                      Interim records *: 0
RF triggers      : 0

S6b primary peer profile :
S6b secondary peer profile :
S6b peer profile in use :
S6b failover count      : 0
S6b peer hostname       :
S6b peer address        : ::
S6b peer port           : 0
S6b peer dest host      :
S6b peer origin         :
S6b local address       : ::

Framed Routes          : 6
  7.0.2.0/24           : 1
  7.0.3.0/24           : 1
  7.0.4.0/24           : 1
  20fe:2::/56          : 1
  30fe:2::/56          : 1
  40fe:2::/56          : 1
Redir Mob Ipv4 Addr    : ::
Redir Mob Ipv6 Addr    : ::
L2TP Connection *      : 0

Restoration Priority Level : 0

P-CSCF Reselection     : Y
Registered P-CSCF Addr  : ::
eMPS IMS priority session : No
Alloc ePDG svr Ipv4 Addr : ::
Alloc ePDG svr Ipv6 Addr : ::
Override Rating Group   : N/A
Override Online Billing Method : Disabled
Override Offline Billing Method : Disabled
Serving PLMN rat*       : disabled
CPOPCI                  : disabled          EPCOSI          : disabled
PGW UL Rate Control     : disabled
DL APN rate lmt Drp Pkt : 0
UL APN rate lmt Drp Pkt : 0
PLMN rate lmt Drp Pkt   : 0

```

MO Exception Cou*: 0 MO Exception Tim*: 0

Number of PDN contexts : 1
=====

* indicates that the corresponding row element may have been truncated.

Example output PDN context field name

show mobile-gateway pdn pdn-context field-name ueip

=====

IMSI/IMEI(#)	UE Address (IPv4/IPv6)
1111111111	10.10.25.1
2222222222	10.10.25.2
3333333333	10.10.25.3
...	

Example output UP function PDN context SEID (UP function)

show mobile-gateway pdn pdn-context up-seid 0x41110

=====

IMSI/IMEI(#)	APN	Type	Beare*	UE Address (IPv4/IPv6)	Ref-pt/Si*
/MSISDN(^)/MAC(`)					
/SubId/SEID(~)					
0x41110~	ipd.nokia.com	IPv4	1	1.127.255.0/-	N/A
123456789012345					

Number of PDN instances : 1

indicates that the corresponding row is IMEI entry.

^ indicates that the corresponding row is MSISDN entry.

` indicates that the corresponding row is MAC entry.

~ indicates that the corresponding row is SEID entry.

* indicates that the corresponding row element may have been truncated.

Example output UP function PDN context detail SEID (UP function)

show mobile-gateway pdn pdn-context up-seid 0x41110 detail

=====

PDN context detail

=====

IMSI	: 123456789012345		
Gi Network Inst	: ipd.nokia.com		
PDN Type	: IPv4		
UL APN AMBR	: 8640 Kbps	DL APN AMBR	: 8640 Kbps
Bearer contexts	: 1	GW/Group/VM	: 1/1/3
UE IPv4 Address	: 1.127.255.0		
CP Sx-N4 Seid	: 0x11120	UP Sx-N4 Seid	: 0x41110
CP Sx-N4 Ctl V4 *	: 33.33.1.1	UP Sx-N4 Ctl V4 *	: 33.34.1.1
CP Sx-N4 Ctl V6 *	: 3ffe::2122:101		
UP Sx-N4 Ctl V6 *	: 3ffe::2121:101		
PDRs	: 2	FARS	: 2
QERs	: 1		

=====

```

State           : CONNECTED           AA Session Id    : 0x00444004
DL APN rate lmt Drp Pkt : 0
UL APN rate lmt Drp Pkt : 0
PLMN rate lmt Drp Pkt   : 0
L2TP Connection ID      : 0
-----

```

```

Number of PDN contexts : 1
=====

```

```

* indicates that the corresponding row element may have been truncated.

```

Example output PDN context wrap

```

# show mobile-gateway pdn pdn-context wrap
=====
IMSI/IMEI(#)      APN          Type    Beare* UE Address (IPv4/IPv6)  Ref-pt/Si*
/MSISDN(^)/MAC(`)
/SubId/SEID(~)
=====
98765432101      ipd.nokia.com    IPv6    1  -/3ffe:0:5:ff00::15:53a S5/GTPv2
                el.1.1.c*        3
-----

```

5.2.2.39 cp-seid

Syntax

```
cp-seid seid [cp-ipv4 ip-address] [cp-ipv6 ipv6-address]
```

```
cp-seid seid [cp-ipv4 ip-address] [cp-ipv6 ipv6-address] detail
```

Context

```
show>mobile>pdn>pdn-context
```

Description

This command displays CP session context information related to a specific SEID.

Parameters

seid

specifies the SEID for the CP or UP session context, up to 40 characters

ip-address

specifies the CP function's Sx/N4 reference point IPv4 address

Values: a.b.c.d

ipv6-address

specifies the CP function's Sx/N4 reference point IPv6 address, up to 64 characters

detail

displays detailed information

Output

The following output example shows CP session context information for a specified SEID.

- *Example output CP session PDN context information for specified SEID*
- *Example output CP session PDN context detail information for SEID*

Example output CP session PDN context information for specified SEID

```
# show mobile-gateway pdn pdn-context cp-seid 0x10100 cp-ipv4 192.168.33.1
=====
IMSI/IMEI(#)      APN      Type    Beare* UE Address (IPv4/IPv6)  Ref-pt/Si*
/MSISDN(^)/MAC(`)
/SubId/SEID(~)
=====
0x10100~          ipd.apn* IPv6     1      -/2001:db8::211:ff:11    N/A
-----
Number of PDN instances : 1
-----
# indicates that the corresponding row is IMEI entry.
^ indicates that the corresponding row is MSISDN entry.
` indicates that the corresponding row is MAC entry.
~ indicates that the corresponding row is SEID entry.
-----
* indicates that the corresponding row element may have been truncated.
```

Example output CP session PDN context detail information for SEID

```
# show mobile-gateway pdn pdn-context up-seid 0x10100 detail
=====
PDN context detail
=====
Gi Network Inst   : ipd.gi-net.com
PDN Type          : IPv6
UL APN AMBR       : 5000 Kbps          DL APN AMBR       : 5000 Kbps
Bearer contexts   : 1                GW/Group/VM       : 1/1/3
UE IPv6 Address   : 2001:db8::211:22ff:fe33:1111
CP Sx-N4 Seid     : 0x10100          UP Sx-N4 Seid     : 0x10100
CP Sx-N4 Ctl V4 * : 192.168.33.1      UP Sx-N4 Ctl V4 * : 192.168.34.1
CP Sx-N4 Ctl V6 * : 2001:db8::2121:101  UP Sx-N4 Ctl V6 * : 2001:db8::2122:101
PDRs              : 2                FARS              : 2
QERS              : 1
State             : CONNECTED        AA Session Id     : 0x00040001
DL APN rate lmt Drp Pkt : 0
UL APN rate lmt Drp Pkt : 0
PLMN rate lmt Drp Pkt  : 0
-----
Number of PDN contexts : 1
=====
* indicates that the corresponding row element may have been truncated.
```

5.2.2.40 def-pfcp-u-sessions

Syntax

```
def-pfcp-u-sessions [network-realm realm-name] [pfcp-u-peer-ip ip-address] [gateway  
gw-id] [apn apn-name] [card slot-number] [vm vm-id] [group group-number]
```

Context

```
show>mobile>pdn>pdn-context
```

Description

This command displays the default PFCP-u shared tunnel session information, when PFCP-u (GTP-U) tunnels are configured to handle ICMPv6 RS/RA and NS/NA message exchange between the CP function and the UP function. When only `def-pfcp-u` is specified, results for all default PFCP-u sessions are displayed. The `network-realm` option may also be specified to identify the realm name of the network instance that has the PFCP GTP-U interface on the UP function, and is communicated in the Network-Instance IE part of the PFCP Session Establishment/Modification process. The `pfcp-u-peer-ip` option can also be used to display information for a specific PFCP-u Peer IP address.

Parameters

realm-name

specifies the realm name of the network instance that has the PFCP GTP-U interface on the UP function, up to 80 characters

ip-address

specifies the PFCP-u peer IP address, up to 64 characters

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x: [0..FFFF]H
–	–	d: [0..255]D

gw-id

specifies the gateway ID

Values: 1 to 8

apn-name

specifies the APN name, up to 80 characters

slot-number

specifies the card slot number (applicable only to 7750 SR MG)

Values: 1 to 20 | A | B

vm-id

specifies the VM ID (applicable only to CMG)

Values: 1 to 20 | A | B

group-number

specifies the group number

Values: 1 to 15

Output

The following output example shows default PFCP-u s0hared tunnel sessions information for the PDN bearer context and *Show PDN context PFCP-u shared tunnel sessions output fields* describes the output fields.

- *Example output PDN context default statistics PFCP-u shared tunnel sessions*
- *Example output PDN context network realm statistics PFCP-u shared tunnel sessions*
- *Example output PDN context peer IP address statistics PFCP-u shared tunnel sessions*
- *Example output PDN context network realm and peer IP address statistics PFCP-u shared tunnel sessions*

Example output PDN context default statistics PFCP-u shared tunnel sessions

```
# show mobile-gateway pdn pdn-context def-pfcp-u-sessions
=====
IMSI/IMEI( #)      APN      Type  Beare* UE Address (IPv4/IPv6)  Ref-pt/Si*
/MSISDN(^)/MAC(`)
/SubId/SEID(~)
=====
999999888100211    -      1      -      S5/GTPv2
999999888100211    -      1      -      S5/GTPv2
999999888100211    -      1      -      S5/GTPv2
999999888100211    -      1      -      S5/GTPv2
999999888100211    -      1      -      S5/GTPv2
999999888100211    -      1      -      S5/GTPv2
999999888100211    -      1      -      S5/GTPv2
999999888100211    -      1      -      S5/GTPv2
999999888100211    -      1      -      S5/GTPv2
999999888100210    -      1      -      S5/GTPv2
```


999999888100210	-	1	-	S5/GTPv2
999999888100210	-	1	-	S5/GTPv2
999999888100210	-	1	-	S5/GTPv2
999999888100210	-	1	-	S5/GTPv2
999999888100210	-	1	-	S5/GTPv2
999999888100210	-	1	-	S5/GTPv2
999999888100210	-	1	-	S5/GTPv2
999999888100209	-	1	-	S5/GTPv2
999999888100212	-	1	-	S5/GTPv2
999999888100212	-	1	-	S5/GTPv2
999999888100212	-	1	-	S5/GTPv2
999999888100212	-	1	-	S5/GTPv2
999999888100212	-	1	-	S5/GTPv2
999999888100212	-	1	-	S5/GTPv2
999999888100212	-	1	-	S5/GTPv2
999999888100212	-	1	-	S5/GTPv2
999999888100213	-	1	-	S5/GTPv2
999999888100213	-	1	-	S5/GTPv2
999999888100213	-	1	-	S5/GTPv2
999999888100213	-	1	-	S5/GTPv2
999999888100213	-	1	-	S5/GTPv2
999999888100213	-	1	-	S5/GTPv2
999999888100213	-	1	-	S5/GTPv2
999999888100213	-	1	-	S5/GTPv2
999999888100213	-	1	-	S5/GTPv2
999999888100211	-	1	-	S5/GTPv2
999999888100211	-	1	-	S5/GTPv2
999999888100211	-	1	-	S5/GTPv2
999999888100211	-	1	-	S5/GTPv2
999999888100211	-	1	-	S5/GTPv2
999999888100211	-	1	-	S5/GTPv2
999999888100211	-	1	-	S5/GTPv2
999999888100211	-	1	-	S5/GTPv2
999999888100210	-	1	-	S5/GTPv2
999999888100210	-	1	-	S5/GTPv2
999999888100210	-	1	-	S5/GTPv2
999999888100210	-	1	-	S5/GTPv2
999999888100210	-	1	-	S5/GTPv2
999999888100210	-	1	-	S5/GTPv2
999999888100210	-	1	-	S5/GTPv2
999999888100210	-	1	-	S5/GTPv2
999999888100209	-	1	-	S5/GTPv2
999999888100209	-	1	-	S5/GTPv2
999999888100212	-	1	-	S5/GTPv2
999999888100212	-	1	-	S5/GTPv2
999999888100212	-	1	-	S5/GTPv2
999999888100212	-	1	-	S5/GTPv2
999999888100212	-	1	-	S5/GTPv2
999999888100212	-	1	-	S5/GTPv2
999999888100212	-	1	-	S5/GTPv2
999999888100212	-	1	-	S5/GTPv2
999999888100213	-	1	-	S5/GTPv2
999999888100213	-	1	-	S5/GTPv2
999999888100213	-	1	-	S5/GTPv2
999999888100213	-	1	-	S5/GTPv2
999999888100213	-	1	-	S5/GTPv2

99999888100213	-	1	-	S5/GTPv2
99999888100213	-	1	-	S5/GTPv2
99999888100213	-	1	-	S5/GTPv2
99999888100211	-	1	-	S5/GTPv2
99999888100211	-	1	-	S5/GTPv2
99999888100211	-	1	-	S5/GTPv2
99999888100211	-	1	-	S5/GTPv2
99999888100211	-	1	-	S5/GTPv2
99999888100211	-	1	-	S5/GTPv2
99999888100211	-	1	-	S5/GTPv2
99999888100211	-	1	-	S5/GTPv2
99999888100210	-	1	-	S5/GTPv2
99999888100210	-	1	-	S5/GTPv2
99999888100210	-	1	-	S5/GTPv2
99999888100210	-	1	-	S5/GTPv2
99999888100210	-	1	-	S5/GTPv2
99999888100210	-	1	-	S5/GTPv2
99999888100210	-	1	-	S5/GTPv2
99999888100210	-	1	-	S5/GTPv2
99999888100210	-	1	-	S5/GTPv2
99999888100210	-	1	-	S5/GTPv2
99999888100209	-	1	-	S5/GTPv2
99999888100209	-	1	-	S5/GTPv2
99999888100212	-	1	-	S5/GTPv2
99999888100212	-	1	-	S5/GTPv2
99999888100212	-	1	-	S5/GTPv2
99999888100212	-	1	-	S5/GTPv2
99999888100212	-	1	-	S5/GTPv2
99999888100212	-	1	-	S5/GTPv2
99999888100212	-	1	-	S5/GTPv2
99999888100212	-	1	-	S5/GTPv2
99999888100213	-	1	-	S5/GTPv2
99999888100213	-	1	-	S5/GTPv2
99999888100213	-	1	-	S5/GTPv2
99999888100213	-	1	-	S5/GTPv2
99999888100213	-	1	-	S5/GTPv2
99999888100213	-	1	-	S5/GTPv2
99999888100213	-	1	-	S5/GTPv2
99999888100213	-	1	-	S5/GTPv2
99999888100211	-	1	-	S5/GTPv2
99999888100211	-	1	-	S5/GTPv2
99999888100211	-	1	-	S5/GTPv2
99999888100211	-	1	-	S5/GTPv2
99999888100211	-	1	-	S5/GTPv2
99999888100211	-	1	-	S5/GTPv2
99999888100210	-	1	-	S5/GTPv2
99999888100210	-	1	-	S5/GTPv2
99999888100210	-	1	-	S5/GTPv2
99999888100210	-	1	-	S5/GTPv2
99999888100210	-	1	-	S5/GTPv2
99999888100210	-	1	-	S5/GTPv2
99999888100210	-	1	-	S5/GTPv2
99999888100210	-	1	-	S5/GTPv2
99999888100210	-	1	-	S5/GTPv2
99999888100209	-	1	-	S5/GTPv2
99999888100209	-	1	-	S5/GTPv2
99999888100209	-	1	-	S5/GTPv2

```

999999888100212      -      1      -      S5/GTPv2
999999888100212      -      1      -      S5/GTPv2
999999888100212      -      1      -      S5/GTPv2
999999888100212      -      1      -      S5/GTPv2
999999888100212      -      1      -      S5/GTPv2
999999888100212      -      1      -      S5/GTPv2
999999888100212      -      1      -      S5/GTPv2
999999888100212      -      1      -      S5/GTPv2
999999888100212      -      1      -      S5/GTPv2
999999888100212      -      1      -      S5/GTPv2
999999888100213      -      1      -      S5/GTPv2
999999888100213      -      1      -      S5/GTPv2
999999888100213      -      1      -      S5/GTPv2
999999888100213      -      1      -      S5/GTPv2
999999888100213      -      1      -      S5/GTPv2
999999888100213      -      1      -      S5/GTPv2
999999888100213      -      1      -      S5/GTPv2
999999888100213      -      1      -      S5/GTPv2

```

Number of PDN instances : 140

indicates that the corresponding row is IMEI entry.
^ indicates that the corresponding row is MSISDN entry.
\ indicates that the corresponding row is MAC entry.
~ indicates that the corresponding row is SEID entry.

* indicates that the corresponding row element may have been truncated.

Table 51: Show PDN context PFCP-u shared tunnel sessions output fields

Label	Description
IMSI/IMEI/MSISDN/MAC/SubId/SEID	The subscriber, equipment, or session identifier type
APN	The APN name
Type	The bearer type
Bearer	The bearer number
UE Address	The IP address of the UE
IPv4/IPv6	The type of IP address, either IPv4 or IPv6
Ref-pt/Si	The reference point or interface

Example output PDN context network realm statistics PFCP-u shared tunnel sessions

```
# show mobile-gateway pdn pdn-context def-pfcp-u-sessions network-realm s5-default
=====
IMSI/IMEI(#)          APN          Type      Beare*  UE Address (IPv4/IPv6)  Ref-pt/Si*
/MSISDN(^)/MAC(`)
/SubId/SEID(~)
=====
999999888100212      -          1          -          S5/GTPv2
999999888100211      -          1          -          S5/GTPv2
999999888100213      -          1          -          S5/GTPv2
999999888100210      -          1          -          S5/GTPv2
999999888100209      -          1          -          S5/GTPv2
-----
Number of PDN instances : 5
-----
# indicates that the corresponding row is IMEI entry.
^ indicates that the corresponding row is MSISDN entry.
` indicates that the corresponding row is MAC entry.
~ indicates that the corresponding row is SEID entry.
-----
* indicates that the corresponding row element may have been truncated.
```

Example output PDN context peer IP address statistics PFCP-u shared tunnel sessions

[illegible]

```

999999888100210      -      1      -      S5/GTPv2
999999888100210      -      1      -      S5/GTPv2
999999888100210      -      1      -      S5/GTPv2
999999888100210      -      1      -      S5/GTPv2
999999888100210      -      1      -      S5/GTPv2
999999888100210      -      1      -      S5/GTPv2
999999888100210      -      1      -      S5/GTPv2
999999888100210      -      1      -      S5/GTPv2

```

```
-----
Number of PDN instances : 33
-----
```

```

# indicates that the corresponding row is IMEI entry.
^ indicates that the corresponding row is MSISDN entry.
` indicates that the corresponding row is MAC entry.
~ indicates that the corresponding row is SEID entry.
-----
* indicates that the corresponding row element may have been truncated.

```

Example output PDN context network realm and peer IP address statistics PFCP-u shared tunnel sessions

```

# show mobile-gateway pdn pdn-context def-pfcp-u-sessions pfcp-u-peer-
ip 33.34.1.1 network-realm s5-default
=====
IMSI/IMEI(#)      APN      Type      Beare* UE Address (IPv4/IPv6)  Ref-pt/Si*
/MSISDN(^)/MAC(`)
/SubId/SEID(~)
=====
999999888100210      -      1      -      S5/GTPv2
-----
Number of PDN instances : 1
-----
# indicates that the corresponding row is IMEI entry.
^ indicates that the corresponding row is MSISDN entry.
` indicates that the corresponding row is MAC entry.
~ indicates that the corresponding row is SEID entry.
-----
* indicates that the corresponding row element may have been truncated.

```

5.2.2.41 field-name

Syntax

field-name {pcrf-ip | pcrf-dest-host | ue-ip | roaming-status status | s6b-peer-ip | s6b-peer-dest-host | p-cscf-ip | dns-ipv4 | dns-ipv6 | s5-pgw-gtpc-ipv4 | s5-pgw-gtpc-ipv6} [gateway *gw-id*] [apn *apn-name*] [card *slot-number*] [vm *vm-id*] [group *group-number*] [connected-time-gt *connected-time-gt*] [connected-time-leq *connected-time-leq*]

Context

show>mobile>pdn>pdn-context

Description

This command displays the specified fields for printing.

Parameters

pcrf-ip

keyword specifies the PCRF IP address

pcrf-dest-host

keyword specifies the PCRF destination host

ue-ip

keyword specifies the UE IP address

roaming-status

keyword specifies the roaming status

s6b-peer-ip

specifies the peer IP address for the S6B interface

s6b-peer-dest-host

specifies the peer destination host for the S6B interface

p-cscf-ip

specifies the P-CSCF IP address

dns-ipv4

specifies the IPv4 DNS address

dns-ipv6

specifies the IPv6 DNS address

s5-pgw-gtpc-ipv4

specifies the IPv4 address for the S5 PGW GTPC

s5-pgw-gtpc-ipv6

specifies the IPv6 address for the S5 PGW GTPC

gw-id

specifies the gateway ID

Values: 1 to 8

apn-name

specifies the context's APN name

group-number

specifies the group number

Values: 1 to 15

slot-number

specifies the card slot number

Values: 1 to 20 | A | B

vm-id

specifies the VM ID (applicable only to CMG)

Values: 1 to 20 | A | B

connected-time-gt

specifies the connected time "greater than" in seconds, as a criterion for the bearer-contexts (UEs at the PDN-level command)

Values: 1 to 2592000 (30 days)

connected-time-leq

specifies the connected time "less than or equal" in seconds, as a criterion for the bearer-contexts (UEs at the PDN-level command)

Values: 1 to 2592000 (30 days)

Output

The following output examples show information for the PDN context field name.

- *Example output PDN context field name*
- *Example output PDN context field name S6B peer IP*
- *Example output PDN context field name P-CSCF IP*
- *Example output PDN context field name DNS IPv4*
- *Example output PDN context field name S5 PGW GTPC IPv4*
- *Example output PDN context field name S5 PGW GTPC IPv6*

Example output PDN context field name

```
show mobile-gateway pdn pdn-context field-name ue-ip
```

```
=====
IMSI/IMEI(##)      UE Address (IPv4/IPv6)
=====
11111111111        10.10.25.1
22222222222        10.10.25.2
33333333333        10.10.25.3
...
=====
```

Example output PDN context field name S6B peer IP

```
show mobile-gateway pdn pdn-context field-name s6b-peer-ip
```

```
=====
IMSI/IMEI(##)      S6b Peer IP Address
=====
98765432101        1.2.3.4
=====
```

```

Number of PDN instances : 1
-----
# indicates that the corresponding row is IMEI entry.
^ indicates that the corresponding row is MSISDN entry.
` indicates that the corresponding row is MAC entry.
~ indicates that the corresponding row is SEID entry.
-----
*A:CMG# show mobile-gateway pdn pdn-context field-name s6b-peer-dest-host
=====
IMSI/IMEI(#)          S6b Peer Destination Host
=====
98765432101          s6bpeer.nokia-dra.host
-----
Number of PDN instances : 1
-----
# indicates that the corresponding row is IMEI entry.
^ indicates that the corresponding row is MSISDN entry.
` indicates that the corresponding row is MAC entry.
~ indicates that the corresponding row is SEID entry.
-----

```

Example output PDN context field name P-CSCF IP

```

show mobile-gateway pdn pdn-context field-name p-cscf-ip
=====
IMSI/IMEI(#)          P-CSCF IP Address
=====
98765432101          1.2.3.4
98765432101          100.101.102.103
98765432101          4.3.2.1
-----
Number of PDN instances : 1
-----
# indicates that the corresponding row is IMEI entry.
^ indicates that the corresponding row is MSISDN entry.
` indicates that the corresponding row is MAC entry.
~ indicates that the corresponding row is SEID entry.
-----

```

Example output PDN context field name DNS IPv4

```

show mobile-gateway pdn pdn-context field-name dns-ipv4
=====
IMSI/IMEI(#)          DNS IPv4 Address
=====
98765432101          1.2.3.4
98765432101          4.3.2.1
-----
Number of PDN instances : 1
-----
# indicates that the corresponding row is IMEI entry.
^ indicates that the corresponding row is MSISDN entry.
` indicates that the corresponding row is MAC entry.
~ indicates that the corresponding row is SEID entry.
-----

```


Example output PDN context field name S5 PGW GTPC IPv4

```
show mobile-gateway pdn pdn-context field-name s5-pgw-gtpc-ipv4
=====
IMSI/IMEI(#)      S5 PGW IPv4 Control Address
=====
98765432101      10.10.5.1
-----
Number of PDN instances : 1
-----
# indicates that the corresponding row is IMEI entry.
^ indicates that the corresponding row is MSISDN entry.
\ indicates that the corresponding row is MAC entry.
~ indicates that the corresponding row is SEID entry.
-----
```

Example output PDN context field name S5 PGW GTPC IPv6

```
show mobile-gateway pdn pdn-context field-name s5-pgw-gtpc-ipv6
=====
IMSI/IMEI(#)      S5 PGW IPv6 Control Address
=====
98765432101      2001:db8:85a3:8d3:1319:8a2e:370:7348
-----
Number of PDN instances : 1
-----
# indicates that the corresponding row is IMEI entry.
^ indicates that the corresponding row is MSISDN entry.
\ indicates that the corresponding row is MAC entry.
~ indicates that the corresponding row is SEID entry.
-----
```

5.2.2.42 supi**Syntax**

supi imsi-msisdn

supi imsi-msisdn detail

Context

show>mobile>pdn>pdn-context

Description

This command displays PDN context information for the specified SUPI for 5G sessions.

Parameters

imsi-msisdn

specifies the SUPI identifier, up to 15 characters

detail

specifies detail information

Output

The following output example shows information for the specified SUPI IMSI-MSISDN in the PDN context, and *Show PDN context SUPI detail output fields* describes the output fields.

- *Example output PDN context SUPI detail statistics (5G sessions)*
- *Example output PDN context SUPI detail statistics I-UPF (5G sessions)*
- *Example output PDN context SUPI detail statistics combined ULCL (5G sessions)*
- *Example output PDN context SUPI detail statistics standalone ULCL (5G sessions)*
- *Example output PDN context SUPI detail statistics combined branching point (5G sessions)*
- *Example output PDN context SUPI detail statistics standalone branching point (5G sessions)*



NOTE:

When the `supi` option is used, information related to EPS interworking is displayed; for specific details, see the notes in *Show PDN context SUPI detail output fields*.

Example output PDN context SUPI detail statistics (5G Sessions)

```
# show mobile-gateway pdn pdn-context supi 460100001003204 detail
=====
PDN context detail
=====
SUPI           : 460100001003204
DNN            : ipd.nokia.com
PEI            : 987654321012345
GPSI           : 12345678901
RAT            : -
Interworking    : Not Subscribed
Data Session Id : 0x00010130
Ref-point/Sig-pr*: N11/http2          APN restriction : 0
5G Capability
  RQI          : Disabled              Multihoming      : Disabled
PDU Session ID : 1
PCRF Subscribed PRAs
OCS Subscribed PRAs
UL APN AMBR     : 4000 Kbps           DL APN AMBR      : 2000 Kbps
Bearer contexts : 1                   SDFs             : 0

Primary DNS IPv4 address : ::
Secondary DNS IPv4 address : ::
Primary DNS IPv6 address : ::
Secondary DNS IPv6 address : ::
```

```

Pri NBNS IPv4 ad*: ::          Sec NBNS IPv4 ad*: ::
Pri NBNS IPv6 ad*: ::          Sec NBNS IPv6 ad*: ::

Charging bearer *: Visiting

UE IPv4 address      : 1.127.255.3
UE IPv4 address source : Local Pool

Charging profile : 1          Interim records *: 0
RF triggers      : 0

Framed Routes      : 0

P-CSCF Reselection : Y

Override Rating Group : N/A
Override Online Billing Method : Disabled
Override Offline Billing Method : Disabled
MO Exception Cou*: 0          MO Exception Tim*: 0
ULI Timestmp(sec): 0          SSC Mode : 1
N3 State : Actv
NSSAI
SST : 11          SD : 0xd143a5
AMF PCF ID : 00000000-0000-0000-0000-000000000000

Service Based Interface (SBI) Information
NF Service : npcfsmpolicycontrol
NF Role : consumer
NF ID : 34635855-c481-4ee2-b571-a151579dc866
URI : http://10.10.7.75:7890/npcf-smpolicycontrol/v1/sm-policies/
    460100001003204_1
PCF enabled triggers:
PCF Subscribed PRAs

NF Service : nudm-sdm
NF Role : consumer
NF ID : 44f08cba-9a89-77fe-ab7d-2a0066584f65
URI : http://10.10.7.78:7788/nudm-sdm/v1/imsi-460100001003204/sdm-
    subscriptions/460100001003204_1
Service Instance : sdm1

NF Service : nudm-uecm
NF Role : consumer
NF ID : 44f08cba-9a89-77fe-ab7d-2a0066584f85
Service Instance : uecm1
QoS Information
-----
QFI : 5          QFI Type : Dynamic          QoS Flow Type : Default
-----
QoS Profile
5QI/ARP : 5/9          Priority Level : 0
Avg Window (ms) : 0          Max Bst Vol (bytes) : 0
QNC : Disabled          RQI : Disabled
UL MBR (kbps) : 0          DL MBR (kbps) : 0
UL GBR (kbps) : 0          DL GBR (kbps) : 0

```

User Plane (UP) Information

```

UP Type           : Original PSA
UP Sx-N4 Ctl V4   : 33.34.1.1
UP Sx-N4 Ctl V6   : 3ffe::2122:101
CP Sx-N4 Ctl V4   : 33.33.1.1
CP Sx-N4 Ctl V6   : 3ffe::2121:101
CP Sx-N4 Seid     : 0x10130
UP Sx-N4 Seid     : 0x10130
gNb N3 Data TEID  : 0x20000003
gNb N3 IPv4 Address : 6.7.10.1
UP N3 Data TEID   : 0x40010135
UP N3 IPv4 Address : 17.70.16.16
UP N3 IPv6 Address : 3ffe::1146:1010
QFI(s)           : 5

```

Number of PDN contexts : 1

* indicates that the corresponding row element may have been truncated.

Table 52: Show PDN context SUPI detail output fields

Label	Description
SUPI	SUPI identifier
DNN	DNN name
PEI	PEI identifier
GPSI	GPSI identifier
RAT ²	RAT type
Interworking	3G/4G EPS interworking status
Data Session ID	Charging characters received
Ref-point/sig-pr*	Uptime of the bearer
APN restriction	APN restriction

² The RAT type is not supported for EPS interworking information.

Table 52: Show PDN context SUPI detail output fields (continued)

Label	Description
5G Capability	
RQI	RQI status (enabled/disabled)
Multihoming ³	Multihoming status (enabled/disabled)
PDU Session ID	PDU session identifier
PCRF Subscribed PRA	PCRF subscribed PRAs
OCS Subscribed PRA	OCS subscribed PRAs
UL APN AMBR	Uplink APN AMBR
DL APN AMBR	Downlink APN AMBR
Bearer Contexts	Bearer contexts
SDF	SDFs
Primary DNS IPV4 address	Primary DNS IPv4 address
Secondary DNS IPV4 address	Secondary DNS IPv4 address
Primary DNS IPV6 address	Primary DNS IPv6 address
Secondary DNS IPV6 address	Secondary DNS IPv6 address
Pri NBNS IPv4 ad*	Primary NBNS IPv4 address
Sec NBNS IPv4 ad*	Secondary NBNS IPv4 address

³ Only one multihoming IP address is displayed for EPS interworking information, if multihoming is enabled.

Table 52: Show PDN context SUPI detail output fields (continued)

Label	Description
Pri NBNS IPv6 ad*	Primary NBNS IPv6 address
Sec NBNS IPv6 ad*	Secondary NBNS IPv6 address
Charging bearer *	Charging bearer type
UE IPv4 address	UE IPv4 address
UE IPv4 address source	UE IPv4 address source
Charging profile	Charging profile name or number
Interim records *	Number of interim charging records
RF triggers	Number of RF triggers
Framed routes	Number of framed routes
P-CSCF Reselection	P-CSCF reselection status (enabled/disabled)
Override Rating Group	Override rating group status
Override Online Billing Method	Override online billing method status (enabled/disabled)
Override Offline Billing Method	Override offline billing method status (enabled/disabled)
Mo Exception Cou*	Mo exception count
Mo Exception Tim*	Mo exception time
ULI_Timestamp (sec)	ULI timestamp in seconds
SSC Mode	SSC Mode

Table 52: Show PDN context SUPI detail output fields (continued)

Label	Description
N3 state	N3 active state
NSSAI	
SST	Number of SSTs
SD	SD identifier
AMF PCF ID	AMF PCF identifier
Service Based Interface (SBI) Information	
NF Service ⁴	NF service type
NF Role	NF consumer or producer role
NF ID	NF service ID
URI	NF service URI
Service Instance	NF service instance
QoS Information	
QFI	QFI value associated with a static QFI mapping profile
QFI Type	QFI type (dynamic or static)

⁴ The following fields are not supported under SBI services for EPS interworking information.

- NF Service: nsmf-pdusession
- NF Service: namf-comm
- NF Service: namf-evts

Table 52: Show PDN context SUPI detail output fields (continued)

Label	Description
QoS Flow Type	QoS flow type
QoS Profile	
5QI/ARP	QFI generated dynamically using 5QI/ARP as default (in the range 1 to 64)
Priority Level	Priority level in scheduling resources among QoS flows
Avg Window (ms)	Average window (duration over which the guaranteed and maximum bitrate is calculated) for the QFI value within a static QFI mapping profile
Max Bst Vol (bytes)	Maximum burst volume associated with the QFI value within a static QFI mapping profile
QNC	QNC status, for notifications from the NG-RAN when GFBR can no longer be guaranteed during the life of the QoS flow for the SDF (enabled/disabled)
RQI	RQI status (enabled/disabled)
UL MBR	Uplink MBR
DL MBR	Downlink MBR
UL GBR	Uplink GBR
DL GBR	Downlink GBR
User Plane (UP) Information	
UP Type	UP uplink type

Table 52: Show PDN context SUPI detail output fields (continued)

Label	Description
UP Sx-N4 Ctl V4	Sx-N4 (PFCP) UP CTL IPv4 address
CP Sx-N4 Ctl V4	Sx-N4 (PFCP) CP CTL IPv4 address
CP Sx-N4 Ctl V6	Sx-N4 (PFCP) CP CTL IPv6 address
CP Sx-N4 Seid	Sx-N4 (PFCP) CP SEID
UP Sx-N4 Seid	Sx-N4 (PFCP) UP SEID
gNb N3 Data TEID	gNb N3 data TEID
gNb N3 IPv4 Address	gNb N3 IPv4 address
UP N3 Data TEID	UP N3 data TEID
UP N3 IPv4 Address	UP N3 IPv4 IP address
UP N3 IPv6 Address	UP N3 IPv6 IP address
UP Lcl N9 Data TEID	UP LCI N9 data TEID
UP Lcl IPv4 N9 Addr	UP LCI N9 IPv4 address
UP Lcl IPv6 N9 Addr	UP LCI N9 IPv6 address
UP Rmt N9 Data TEID	UP RMT N9 data TEID
UP Rmt IPv4 N9 Addr	UP RMT N9 IPv4 address
UP Rmt IPv6 N9 Addr	UP RMT N9 IPv6 address
QFI(s)	QFI number or numbers

Example output PDN context SUPI detail statistics I-UPF (5G sessions)

```
# show mobile-gateway pdn pdn-context supi 460100001003204 detail
=====
PDN context detail
=====
SUPI           : 460100001003204
DNN            : ipd.nokia.com
PEI           : 987654321012345
GPSI          : 12345678901
RAT            : -
Interworking   : Not Subscribed
Data Session Id : 0x00010130
Ref-point/Sig-pr*: N11/http2          APN restriction : 0
5G Capability
  RQI          : Disabled                Multihoming   : Disabled
PDU Session ID : 1
PCRF Subscribed PRAs
OCS Subscribed PRAs
UL APN AMBR    : 4000 Kbps              DL APN AMBR    : 2000 Kbps
Bearer contexts : 1                    SDFs           : 0

Primary DNS IPv4 address : ::
Secondary DNS IPv4 address : ::
Primary DNS IPv6 address : ::
Secondary DNS IPv6 address : ::

Pri NBNS IPv4 ad*: ::                Sec NBNS IPv4 ad*: ::
Pri NBNS IPv6 ad*: ::                Sec NBNS IPv6 ad*: ::

Charging bearer *: Visiting

UE IPv4 address      : 1.127.255.3
UE IPv4 address source : Local Pool

Charging profile : 1                  Interim records *: 0
RF triggers      : 0

Framed Routes      : 0

P-CSCF Reselection : Y

Override Rating Group : N/A
Override Online Billing Method : Disabled
Override Offline Billing Method : Disabled
MO Exception Cou*: 0          MO Exception Tim*: 0

ULI Timestmp(sec): 0          SSC Mode          : 1
N3 State          : Actv

NSSAI
SST              : 11          SD                  : 0xd143a5
```

AMF PCF ID : 00000000-0000-0000-0000-000000000000

Service Based Interface (SBI) Information

NF Service : npcfsmpolicycontrol
 NF Role : consumer
 NF ID : 34635855-c481-4ee2-b571-a151579dc866
 URI : http://10.10.7.75:7890/npcf-smpolicycontrol/v1/sm-policies/
 460100001003204_1

PCF enabled triggers:

PCF Subscribed PRAs

NF Service : nudm-sdm
 NF Role : consumer
 NF ID : 44f08cba-9a89-77fe-ab7d-2a0066584f65
 URI : http://10.10.7.78:7788/nudm-sdm/v1/imsi-460100001003204/sdm-
 subscriptions/460100001003204_1

Service Instance : sdm1

NF Service : nudm-uecm
 NF Role : consumer
 NF ID : 44f08cba-9a89-77fe-ab7d-2a0066584f85
 Service Instance : uecm1

QoS Information

 QFI : 5 QFI Type : Dynamic QoS Flow Type : Default

QoS Profile

5QI/ARP	: 5/9	Priority Level	: 0
Avg Window (ms)	: 0	Max Bst Vol (bytes)	: 0
QNC	: Disabled	RQI	: Disabled
UL MBR (kbps)	: 0	DL MBR (kbps)	: 0
UL GBR (kbps)	: 0	DL GBR (kbps)	: 0

User Plane (UP) Information

 UP Type : Original PSA
 UP Sx-N4 Ctl V4 : 33.34.1.1
 UP Sx-N4 Ctl V6 : 3ffe::2122:101
 CP Sx-N4 Ctl V4 : 33.33.1.1
 CP Sx-N4 Ctl V6 : 3ffe::2121:101
 CP Sx-N4 Seid : 0x10130
 UP Sx-N4 Seid : 0x10130
 UP Lcl N9 Data TEID : 0x40010135
 UP Lcl IPv4 N9 Addr : 17.70.16.17
 UP Lcl IPv6 N9 Addr : 3ffe::1146:1011
 UP Rmt N9 Data TEID : 0x10115
 UP Rmt IPv4 N9 Addr : 3ffe::a46:501
 UP Rmt IPv6 N9 Addr : 10.70.5.1

QFI(s) : 5

```

-----
UP Type                : Intermediate
UP Sx-N4 Ctl V4        : 33.34.1.2
CP Sx-N4 Ctl V4        : 33.33.1.1
CP Sx-N4 Ctl V6        : 3ffe::2121:101
CP Sx-N4 Seid          : 0x20130
UP Sx-N4 Seid          : 0x10110
gNb N3 Data TEID       : 0x10000003
gNb N3 IPv4 Address    : 6.7.9.1
UP N3 Data TEID        : 0x40010115
UP N3 IPv4 Address     : 10.70.5.1
UP N3 IPv6 Address     : 3ffe::a46:501
UP Lcl N9 Data TEID    : 0x10115
UP Lcl IPv4 N9 Addr    : 10.70.5.1
UP Lcl IPv6 N9 Addr    : 3ffe::a46:501

```

```

UP Rmt N9 Data TEID    : 0x40010135
UP Rmt IPv4 N9 Addr    : 3ffe::1146:1011
UP Rmt IPv6 N9 Addr    : 17.70.16.17

```

```

QFI(s)                : 5

```

```

-----
Number of PDN contexts : 1
=====

```

* indicates that the corresponding row element may have been truncated.

Example output PDN context SUPI detail statistics combined ULCL (5G sessions)

```

show mobile-gateway pdn pdn-context supi 460100001003204 detail
=====
PDN context detail
=====
SUPI                : 460100001003204
DNN                 : ulcl.noNat.staticPolicy.1.com
PEI                 : 987654321012345
GPSI                : 12345678901
RAT                 : -
Interworking        : Not Subscribed
Data Session Id     : 0x00020100
Ref-point/Sig-pr*: N11/http2          APN restriction : 0
5G Capability
  RQI                : Disabled          Multihoming    : Disabled
PDU Session ID      : 1
PCRF Subscribed PRAs
OCS Subscribed PRAs
UL APN AMBR         : 4000 Kbps          DL APN AMBR      : 2000 Kbps
Bearer contexts      : 1                  SDFs             : 2

```

```

Primary DNS IPv4 address : ::
Secondary DNS IPv4 address : ::
Primary DNS IPv6 address : ::
Secondary DNS IPv6 address : ::

Pri NBNS IPv4 ad*: ::          Sec NBNS IPv4 ad*: ::
Pri NBNS IPv6 ad*: ::          Sec NBNS IPv6 ad*: ::

Charging bearer *: Visiting

UE IPv4 address           : 1.127.255.4
UE IPv4 address source    : Local Pool

Charging profile : 1           Interim records *: 0
RF triggers      : 0

Framed Routes      : 0

P-CSCF Reselection : Y

Override Rating Group : N/A
Override Online Billing Method : Disabled
Override Offline Billing Method : Disabled
MO Exception Cou*: 0      MO Exception Tim*: 0
ULI Timestmp(sec): 0      SSC Mode : 1
N3 State : Actv
NSSAI
SST : 11          SD : 0xd143a5
AMF PCF ID : 00000000-0000-0000-0000-000000000000

Service Based Interface (SBI) Information
NF Service : npcfsmpolicycontrol
NF Role : consumer
NF ID : 1720b601-fae2-4fc4-b535-8df9da3c64a7
URI : http://10.10.7.76:7890/npcf-smpolicycontrol/v1/sm-policies/
460100001003204_1
PCF enabled triggers:
PCF Subscribed PRAs

NF Service : nudm-sdm
NF Role : consumer
NF ID : 44f08cba-9a89-77fe-ab7d-2a0066584f55
URI : http://10.10.7.74:7788/nudm-sdm/v1/imsi-460100001003204/sdm-
subscriptions/460100001003204_1
Service Instance : sdm1

NF Service : nudm-uecm
NF Role : consumer
NF ID : 44f08cba-9a89-77fe-ab7d-2a0066584f75
Service Instance : uecm1
QoS Information
-----
QFI : 1      QFI Type : Dynamic      QoS Flow Type : Default
-----
QoS Profile
5QI/ARP : 5/9      Priority Level : 0
Avg Window (ms) : 0      Max Bst Vol (bytes) : 0

```

```

QNC                : Disabled          RQI                : Disabled
UL MBR (kbps)      : 0                 DL MBR (kbps)       : 0
UL GBR (kbps)      : 0                 DL GBR (kbps)       : 0

```

```

-----
QFI : 2           QFI Type : Dynamic      QoS Flow Type : Non-GBR
-----

```

QoS Profile

```

5QI/ARP           : 5/9                 Priority Level      : 3
Avg Window (ms)   : 1000                Max Bst Vol (bytes) : 640
QNC               : Disabled            RQI               : Disabled
UL MBR (kbps)     : 0                   DL MBR (kbps)     : 0
UL GBR (kbps)     : 0                   DL GBR (kbps)     : 0

```

User Plane (UP) Information

```

-----
UP Type           : Original PSA
UP Sx-N4 Ctl V4   : 33.34.1.1
UP Sx-N4 Ctl V6   : 3ffe::2122:101
CP Sx-N4 Ctl V4   : 33.33.1.1
CP Sx-N4 Ctl V6   : 3ffe::2121:101
CP Sx-N4 Seid     : 0x20100
UP Sx-N4 Seid     : 0x20100
UP Lcl N9 Data TEID : 0x40020105
UP Lcl IPv4 N9 Addr : 17.70.16.17
UP Lcl IPv6 N9 Addr : 3ffe::1146:1011
UP Rmt N9 Data TEID : 0x10125
UP Rmt IPv4 N9 Addr : 3ffe::a46:501
UP Rmt IPv6 N9 Addr : 10.70.5.1
QFI(s)           : 1

```

```

-----
UP Type           : ULCL+Original P*
UP Sx-N4 Ctl V4   : 33.34.1.2
CP Sx-N4 Ctl V4   : 33.33.1.1
CP Sx-N4 Ctl V6   : 3ffe::2121:101
CP Sx-N4 Seid     : 0x30100
UP Sx-N4 Seid     : 0x10120
gNb N3 Data TEID  : 0x10000004
gNb N3 IPv4 Address : 6.7.9.1
UP N3 Data TEID   : 0x40010125
UP N3 IPv4 Address : 10.70.5.1
UP N3 IPv6 Address : 3ffe::a46:501
UP Lcl N9 Data TEID : 0x10125
UP Lcl IPv4 N9 Addr : 10.70.5.1
UP Lcl IPv6 N9 Addr : 3ffe::a46:501
UP Rmt N9 Data TEID : 0x40020105
UP Rmt IPv4 N9 Addr : 3ffe::1146:1011
UP Rmt IPv6 N9 Addr : 17.70.16.17
QFI(s)           : 1, 2

```

* indicates that the corresponding row element may have been truncated.

```

-----
Number of PDN contexts : 1
=====

```

* indicates that the corresponding row element may have been truncated.

Example output PDN context SUPI detail statistics standalone ULCL (5G sessions)

```
# show mobile-gateway pdn pdn-context supi 460100001003204 detail
=====
PDN context detail
=====
SUPI           : 460100001003204
DNN            : ulcl.noNat.staticPolicy.1.com
PEI           : 987654321012345
GPSI          : 12345678901
RAT            : -
Interworking   : Not Subscribed
Data Session Id : 0x00020110
Ref-point/Sig-pr*: N11/http2          APN restriction : 0
5G Capability
  RQI          : Disabled              Multihoming    : Disabled
PDU Session ID : 1
PCRF Subscribed PRAs
OCS Subscribed PRAs
UL APN AMBR    : 4000 Kbps             DL APN AMBR    : 2000 Kbps
Bearer contexts : 1                    SDFs           : 2

Primary DNS IPv4 address : ::
Secondary DNS IPv4 address : ::
Primary DNS IPv6 address : ::
Secondary DNS IPv6 address : ::

Pri NBNS IPv4 ad*: ::              Sec NBNS IPv4 ad*: ::
Pri NBNS IPv6 ad*: ::              Sec NBNS IPv6 ad*: ::

Charging bearer *: Visiting

UE IPv4 address      : 1.127.255.5
UE IPv4 address source : Local Pool

Charging profile : 1                Interim records *: 0
RF triggers      : 0

Framed Routes      : 0

P-CSCF Reselection : Y

Override Rating Group : N/A
Override Online Billing Method : Disabled
Override Offline Billing Method : Disabled
MO Exception Cou*: 0             MO Exception Tim*: 0
ULI Timestmp(sec): 0             SSC Mode           : 1
N3 State           : Activ
NSSAI
SST                : 11            SD                : 0xd143a5
AMF PCF ID         : 00000000-0000-0000-0000-000000000000

Service Based Interface (SBI) Information
NF Service         : npcfsmpolicycontrol
NF Role            : consumer
NF ID              : e1d44f4f-a1c2-45a3-b9c4-126c2e7e1620
```

```

URI          : http://10.10.7.77:7890/npcf-smpolicycontrol/v1/sm-policies/
              460100001003204_1
PCF enabled triggers:
PCF Subscribed PRAs

NF Service    : nudm-sdm
NF Role       : consumer
NF ID         : 44f08cba-9a89-77fe-ab7d-2a0066584f65
URI          : http://10.10.7.78:7788/nudm-sdm/v1/imsi-460100001003204/sdm-
              subscriptions/460100001003204_2
Service Instance : sdm1

NF Service    : nudm-uecm
NF Role       : consumer
NF ID         : 44f08cba-9a89-77fe-ab7d-2a0066584f85
Service Instance : uecm1
QoS Information
-----
QFI : 1      QFI Type : Dynamic      QoS Flow Type : Default
-----
QoS Profile
5QI/ARP      : 5/9                    Priority Level      : 0
Avg Window (ms) : 0                    Max Bst Vol (bytes) : 0
QNC          : Disabled                RQI                : Disabled
UL MBR (kbps) : 0                      DL MBR (kbps)      : 0
UL GBR (kbps) : 0                      DL GBR (kbps)      : 0
-----
QFI : 2      QFI Type : Dynamic      QoS Flow Type : Non-GBR
-----
QoS Profile
5QI/ARP      : 5/9                    Priority Level      : 3
Avg Window (ms) : 1000                 Max Bst Vol (bytes) : 640
QNC          : Disabled                RQI                : Disabled
UL MBR (kbps) : 0                      DL MBR (kbps)      : 0
UL GBR (kbps) : 0                      DL GBR (kbps)      : 0
-----
User Plane (UP) Information
-----
UP Type      : Original PSA
UP Sx-N4 Ctl V4 : 33.34.1.1
UP Sx-N4 Ctl V6 : 3ffe::2122:101
CP Sx-N4 Ctl V4 : 33.33.1.1
CP Sx-N4 Ctl V6 : 3ffe::2121:101
CP Sx-N4 Seid   : 0x20110
UP Sx-N4 Seid   : 0x20110
UP Lcl N9 Data TEID : 0x40020115
UP Lcl IPv4 N9 Addr : 17.70.16.17
UP Lcl IPv6 N9 Addr : 3ffe::1146:1011
UP Rmt N9 Data TEID : 0x10106
UP Rmt IPv4 N9 Addr : 3ffe::1246:1011
UP Rmt IPv6 N9 Addr : 18.70.16.17
QFI(s)       : 1
-----
UP Type      : PSA 1
UP Sx-N4 Ctl V4 : 33.34.1.2
CP Sx-N4 Ctl V4 : 33.33.1.1

```



```

CP Sx-N4 Ctl V6      : 3ffe::2121:101
CP Sx-N4 Seid        : 0x30110
UP Sx-N4 Seid        : 0x10130
UP Lcl N9 Data TEID  : 0x40010135
UP Lcl IPv4 N9 Addr  : 10.70.5.1
UP Lcl IPv6 N9 Addr  : 3ffe::a46:501
UP Rmt N9 Data TEID  : 0x10106
UP Rmt IPv4 N9 Addr  : 3ffe::1246:1011
UP Rmt IPv6 N9 Addr  : 18.70.16.17
QFI(s)               : 2
-----
UP Type              : ULCL
UP Sx-N4 Ctl V4      : 33.34.1.3
CP Sx-N4 Ctl V4      : 33.33.1.1
CP Sx-N4 Ctl V6      : 3ffe::2121:101
CP Sx-N4 Seid        : 0x40110
UP Sx-N4 Seid        : 0x10100
gNb N3 Data TEID     : 0x20000005
gNb N3 IPv4 Address   : 6.7.10.1
UP N3 Data TEID      : 0x40010105
UP N3 IPv4 Address   : 18.70.16.16
UP N3 IPv6 Address   : 3ffe::1246:1010
UP Lcl N9 Data TEID  : 0x10106
UP Lcl IPv4 N9 Addr  : 18.70.16.17
UP Lcl IPv6 N9 Addr  : 3ffe::1246:1011
UP Rmt N9 Data TEID  : 0x40020115
UP Rmt IPv4 N9 Addr  : 3ffe::1146:1011
UP Rmt IPv6 N9 Addr  : 17.70.16.17
UP Rmt N9 Data TEID  : 0x40010135
UP Rmt IPv4 N9 Addr  : 3ffe::a46:501
UP Rmt IPv6 N9 Addr  : 10.70.5.1
DNAI                 : ulcl.noNat.1.1.com

```

```

QFI(s)               : 1, 2
-----

```

```

Number of PDN contexts : 1
=====

```

* indicates that the corresponding row element may have been truncated.

Example output PDN context SUPI detail statistics combined branching point (5G sessions)

```

# show mobile-gateway pdn pdn-context supi 460100001003204 detail
=====

```

```

PDN context detail
=====

```

```

SUPI          : 460100001003204
DNN           : dynamicQfiMapping.1.com
PEI           : 987654321012345
GPSI          : 12345678901
RAT           : -
Interworking   : Not Subscribed
Data Session Id : 0x00030130
Ref-point/Sig-pr*: N11/http2          APN restriction : 0
5G Capability
  RQI          : Disabled              Multihoming    : Enabled
PDU Session ID : 1

```

```

PCRF Subscribed PRAs
OCS Subscribed PRAs
UL APN AMBR      : 4000 Kbps          DL APN AMBR      : 2000 Kbps
Bearer contexts  : 1                  SDFs             : 0

Primary DNS IPv4 address  : ::
Secondary DNS IPv4 address : ::
Primary DNS IPv6 address  : ::
Secondary DNS IPv6 address : ::

Pri NBNS IPv4 ad*: ::          Sec NBNS IPv4 ad*: ::
Pri NBNS IPv6 ad*: ::          Sec NBNS IPv6 ad*: ::

Charging bearer *: Visiting

UE IPv6 address      : 3ffe:0:1:ff01::9:fb81
UE IPv6 address source : Local Pool

Charging profile : 1          Interim records *: 0
RF triggers      : 0

Framed Routes      : 0

P-CSCF Reselection : Y

Override Rating Group      : N/A
Override Online Billing Method : Disabled
Override Offline Billing Method : Disabled
MO Exception Cou*: 0        MO Exception Tim*: 0
ULI Timestmp(sec): 0        SSC Mode          : 3
N3 State                  : Actv
NSSAI
SST                        : 11                SD                        : 0xd143a5
AMF PCF ID                 : 00000000-0000-0000-0000-000000000000

Service Based Interface (SBI) Information
NF Service      : npcfsmpolicycontrol
NF Role         : consumer
NF ID           : 1720b601-fae2-4fc4-b535-8df9da3c64a7
URI             : http://10.10.7.76:7890/npcf-smpolicycontrol/v1/sm-policies/
                  460100001003204_1

PCF enabled triggers:
PCF Subscribed PRAs

NF Service      : nudm-sdm
NF Role         : consumer
NF ID           : 44f08cba-9a89-77fe-ab7d-2a0066584f65
URI             : http://10.10.7.78:7788/nudm-sdm/v1/imsi-460100001003204/sdm-
                  subscriptions/460100001003204_3
Service Instance : sdm1

NF Service      : nudm-uecm
NF Role         : consumer
NF ID           : 44f08cba-9a89-77fe-ab7d-2a0066584f85
Service Instance : uecm1
QoS Information
-----

```

```

QFI : 1      QFI Type : Dynamic      QoS Flow Type : Default
-----
QoS Profile
5QI/ARP      : 5/9      Priority Level      : 0
Avg Window (ms) : 0      Max Bst Vol (bytes) : 0
QNC          : Disabled  RQI              : Disabled
UL MBR (kbps) : 0      DL MBR (kbps)    : 0
UL GBR (kbps) : 0      DL GBR (kbps)    : 0
-----
User Plane (UP) Information
-----
UP Type      : Original PSA
UP Sx-N4 Ctl V4 : 33.34.1.1
UP Sx-N4 Ctl V6 : 3ffe::2122:101
CP Sx-N4 Ctl V4 : 33.33.1.1
CP Sx-N4 Ctl V6 : 3ffe::2121:101
CP Sx-N4 Seid   : 0x30130
UP Sx-N4 Seid   : 0x20130
UP Lcl N9 Data TEID : 0x40020135
UP Lcl IPv4 N9 Addr : 17.70.16.17
UP Lcl IPv6 N9 Addr : 3ffe::1146:1011
UP Rmt N9 Data TEID : 0x20116
UP Rmt IPv4 N9 Addr : 3ffe::a46:501
UP Rmt IPv6 N9 Addr : 10.70.5.1
QFI(s)       : 1
-----
UP Type      : BP+Original PSA
IPv6 Prefix   : 3ffe:0:5:ff01::a:13c/64
UP Sx-N4 Ctl V4 : 33.34.1.2
CP Sx-N4 Ctl V4 : 33.33.1.1
CP Sx-N4 Ctl V6 : 3ffe::2121:101
CP Sx-N4 Seid   : 0x40130
UP Sx-N4 Seid   : 0x20110
gNb N3 Data TEID : 0x20000007
gNb N3 IPv4 Address : 6.7.10.1
UP N3 Data TEID : 0x40020115
UP N3 IPv4 Address : 10.70.5.1
UP N3 IPv6 Address : 3ffe::a46:501
UP Lcl N9 Data TEID : 0x20116
UP Lcl IPv4 N9 Addr : 10.70.5.1
UP Lcl IPv6 N9 Addr : 3ffe::a46:501
UP Rmt N9 Data TEID : 0x40020135
UP Rmt IPv4 N9 Addr : 3ffe::1146:1011
UP Rmt IPv6 N9 Addr : 17.70.16.17
QFI(s)       : 1
-----
Number of PDN contexts : 1
=====
* indicates that the corresponding row element may have been truncated.

```

Example output PDN context SUPI detail statistics standalone branching point (5G sessions)

```

# show mobile-gateway pdn pdn-context supi 460100001003204 detail
=====
PDN context detail
=====

```

```

SUPI           : 4601000001003204
DNN            : dynamicQfiMapping.1.com
PEI           : 987654321012345
GPSI          : 12345678901
RAT           : -
Interworking   : Not Subscribed
Data Session Id : 0x00030120
Ref-point/Sig-pr*: N11/http2          APN restriction : 0
5G Capability
  RQI          : Disabled              Multihoming      : Enabled
PDU Session ID : 1
PCRF Subscribed PRAs
OCS Subscribed PRAs
UL APN AMBR    : 4000 Kbps            DL APN AMBR      : 2000 Kbps
Bearer contexts : 1                  SDFs            : 0

Primary DNS IPv4 address : ::
Secondary DNS IPv4 address : ::
Primary DNS IPv6 address : ::
Secondary DNS IPv6 address : ::

Pri NBNS IPv4 ad*: ::              Sec NBNS IPv4 ad*: ::
Pri NBNS IPv6 ad*: ::              Sec NBNS IPv6 ad*: ::

Charging bearer *: Visiting

UE IPv6 address      : 3ffe:0:1:ff00::9:ccf2
UE IPv6 address source : Local Pool

Charging profile : 1                Interim records *: 0
RF triggers      : 0

Framed Routes      : 0

P-CSCF Reselection : Y

Override Rating Group : N/A
Override Online Billing Method : Disabled
Override Offline Billing Method : Disabled
MO Exception Cou*: 0              MO Exception Tim*: 0
ULI Timestmp(sec): 0              SSC Mode          : 3
N3 State           : Actv
NSSAI
SST                : 11              SD                : 0xd143a5
AMF PCF ID         : 00000000-0000-0000-0000-000000000000

Service Based Interface (SBI) Information
NF Service         : npcfsmpolicycontrol
NF Role            : consumer
NF ID              : 34635855-c481-4ee2-b571-a151579dc866
URI                : http://10.10.7.75:7890/npcf-smpolicycontrol/v1/sm-policies/
                    4601000001003204_1
PCF enabled triggers:
PCF Subscribed PRAs

NF Service         : nudm-sdm
NF Role            : consumer

```

```

NF ID       : 44f08cba-9a89-77fe-ab7d-2a0066584f55
URI         : http://10.10.7.74:7788/nudm-sdm/v1/imsi-460100001003204/sdm-
             subscriptions/460100001003204_2
Service Instance : sdm1

NF Service   : nudm-uecm
NF Role      : consumer
NF ID       : 44f08cba-9a89-77fe-ab7d-2a0066584f75
Service Instance : uecm1
QoS Information
-----
QFI : 1      QFI Type : Dynamic      QoS Flow Type : Default
-----
QoS Profile
5QI/ARP      : 5/9                  Priority Level      : 0
Avg Window (ms) : 0                  Max Bst Vol (bytes) : 0
QNC          : Disabled             RQI                : Disabled
UL MBR (kbps) : 0                   DL MBR (kbps)      : 0
UL GBR (kbps) : 0                   DL GBR (kbps)      : 0
-----
User Plane (UP) Information
-----
UP Type      : Original PSA
UP Sx-N4 Ctl V4 : 33.34.1.1
UP Sx-N4 Ctl V6 : 3ffe::2122:101
CP Sx-N4 Ctl V4 : 33.33.1.1
CP Sx-N4 Ctl V6 : 3ffe::2121:101
CP Sx-N4 Seid   : 0x30120
UP Sx-N4 Seid   : 0x20120
UP Lcl N9 Data TEID : 0x40020125
UP Lcl IPv4 N9 Addr : 17.70.16.17
UP Lcl IPv6 N9 Addr : 3ffe::1146:1011
UP Rmt N9 Data TEID : 0x10116
UP Rmt IPv4 N9 Addr : 3ffe::1246:1011
UP Rmt IPv6 N9 Addr : 18.70.16.17
QFI(s)        : 1
-----
UP Type      : PSA 1
UP Sx-N4 Ctl V4 : 33.34.1.2
CP Sx-N4 Ctl V4 : 33.33.1.1
CP Sx-N4 Ctl V6 : 3ffe::2121:101
CP Sx-N4 Seid   : 0x40120
UP Sx-N4 Seid   : 0x20100
UP Lcl N9 Data TEID : 0x40020105
UP Lcl IPv4 N9 Addr : 10.70.5.1
UP Lcl IPv6 N9 Addr : 3ffe::a46:501
UP Rmt N9 Data TEID : 0x10116
UP Rmt IPv4 N9 Addr : 3ffe::1246:1011
UP Rmt IPv6 N9 Addr : 18.70.16.17
QFI(s)        : (Not Specified)
-----
UP Type      : BP
IPv6 Prefix   : 3ffe:0:5:ff00::9:d7ef/64
UP Sx-N4 Ctl V4 : 33.34.1.3
CP Sx-N4 Ctl V4 : 33.33.1.1
CP Sx-N4 Ctl V6 : 3ffe::2121:101
CP Sx-N4 Seid   : 0x50120

```

```

UP Sx-N4 Seid      : 0x10110
gNb N3 Data TEID   : 0x10000006
gNb N3 IPv4 Address : 6.7.9.1
UP N3 Data TEID    : 0x40010115
UP N3 IPv4 Address : 18.70.16.16
UP N3 IPv6 Address : 3ffe::1246:1010
UP Lcl N9 Data TEID : 0x10116
UP Lcl IPv4 N9 Addr : 18.70.16.17
UP Lcl IPv6 N9 Addr : 3ffe::1246:1011
UP Rmt N9 Data TEID : 0x40020125
UP Rmt IPv4 N9 Addr : 3ffe::1146:1011
UP Rmt IPv6 N9 Addr : 17.70.16.17
UP Rmt N9 Data TEID : 0x40020105
UP Rmt IPv4 N9 Addr : 3ffe::a46:501
UP Rmt IPv6 N9 Addr : 10.70.5.1
QFI(s)             : 1

```

```

-----
Number of PDN contexts : 1
=====

```

* indicates that the corresponding row element may have been truncated.

5.2.2.43 ue-ip

Syntax

ue-ip *ip-address* network-instance *network-instance* [access-type {utran | geran | wlan | gan | hspa | eutran | ehrpd | hrpd | 1xrtt | umb | virtual | eutran-nb-iot}]

ue-ip *ip-address* network-instance *network-instance* detail acct stats urr-id *urr-id* [access-type {utran | geran | wlan | gan | hspa | eutran | ehrpd | hrpd | 1xrtt | umb | virtual | eutran-nb-iot}]

ue-ip *ip-address* network-instance *network-instance* detail [access-type {utran | geran | wlan | gan | hspa | eutran | ehrpd | hrpd | 1xrtt | umb | virtual | eutran-nb-iot}]

Context

```
show>mobile>pdn>pdn-context
```

Description

This command displays context information related to a specific UE IP address.

Parameters

ip-address

specifies the UE IP address

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x: [0..FFFF]H
–	–	d: [0..255]D

network-instance

specifies the network instance, up to 80 characters

urr-id

specifies a unique identity for the usage rule report for this session

Values: 0 to 4294967295

detail

displays detailed information

access-type

specifies access type

Output

The following output example shows PDN context information for a specific UE IP address.

Example output PDN context information for specified UE IP address

```
# show mobile-gateway pdn pdn-context ue-ip 2.127.255.0 network-
instance ipd.nokia.1.2.com
=====
IMSI/IMEI(#)      APN      Type      Beare*  UE Address (IPv4/IPv6)  Ref-pt/Si*
 /MSISDN(^)/MAC(`)
 /SubId/SEID(~)
=====
0x10100~          ipd.noki* IPv4      2        2.127.255.0/-          N/A
-----
Number of PDN instances : 1
-----
# indicates that the corresponding row is IMEI entry.
^ indicates that the corresponding row is MSISDN entry.
` indicates that the corresponding row is MAC entry.
~ indicates that the corresponding row is SEID entry.
-----
* indicates that the corresponding row element may have been truncated.
```

Example output PDN context detail information for specified UE IP address

```
# show mobile-gateway pdn pdn-context ue-ip 2.127.255.0 network-
instance ipd.nokia.1.2.com detail
=====
PDN context detail
=====
IMSI                : 7777777777777711
Gi Network Inst     : ipd.nokia.1.1.com
PDN Type            : IPv4
UL APN AMBR         : 5000 Kbps           DL APN AMBR : 5000 Kbps
Bearer contexts     : 1                   GW/Group/VM : 1/1/3
UE IPv4 Address     : 1.127.255.0
CP Sx-N4 Seid       : 0x12110            UP Sx-N4 Seid : 0x71110
CP Sx-N4 Ctl V4 *   : 33.33.1.1          UP Sx-N4 Ctl V4 *: 33.34.1.1
CP Sx-N4 Ctl V6 *   : 3ffe::2122:101
UP Sx-N4 Ctl V6 *   : 3ffe::2121:101
PDRs : 3 FARs       : 2
QERS                : 1
BAR ID              : 5
State               : CONNECTED           AA Session Id : 0x00444007
DL APN rate limit Drp Pkt : 0
UL APN rate limit Drp Pkt : 0
PLMN rate limit Drp Pkt  : 0
L2TP Connection ID     : 0
-----
Number of PDN contexts : 1
=====
* indicates that the corresponding row element may have been truncated.
```

5.2.2.44 up-seid

Syntax

```
up-seid seid
up-seid seid detail acct stats [urr-id urr-id]
up-seid seid detail
```

Context

```
show>mobile>pdn>pdn-context
```

Description

This command displays context information related to a specific Session Endpoint Identifier for the UP session context.

Parameters

seid

specifies the Session Endpoint Identifier for the CP or UP session context, up to 40 characters

urr-id

specifies a unique identity for the usage rule report for this session

Values: 0 to 4294967295

detail

displays detailed information

Output

The following output examples show UP session context information for a specified Session Endpoint Identifier.

- [Example output session endpoint identifier for UP](#)
- [Example output PDN context session endpoint ID for UP detail](#)
- [Example output PDN context session endpoint ID UP detail acct statistics](#)

Example output session endpoint identifier for UP

```
# show mobile-gateway pdn pdn-context up-seid 0x10100
=====
IMSI/IMEI(#)      APN      Type    Beare* UE Address (IPv4/IPv6)  Ref-pt/Si*
/MSISDN(^)/MAC(`)
/SubId/SEID(~)
=====
0x10100~          ipd.noki* IPv4      2      2.127.255.0/-              N/A
-----
Number of PDN instances : 1
-----
# indicates that the corresponding row is IMEI entry.
^ indicates that the corresponding row is MSISDN entry.
` indicates that the corresponding row is MAC entry.
~ indicates that the corresponding row is SEID entry.
-----
* indicates that the corresponding row element may have been truncated.
```

Example output PDN context session endpoint ID for UP detail

```
# show mobile-gateway pdn pdn-context up-seid 0x1a562100 detail
=====
PDN context detail
=====
Gi Network Inst  : apn-1.nokia.com-visiting.mnc480.mcc311.gprs
PDN Type         : IPv4
UL APN AMBR      : 75.00 Mbps                DL APN AMBR      : 75.00 Mbps
Bearer contexts  : 1                    GW/Group/VM     : 1/1/3
UE IPv4 Address  : 82.127.75.136
```

```

CP Sx-N4 Seid      : 0x2e440b90      UP Sx-N4 Seid      : 0x1a562100
CP Sx-N4 Ctl V4 * : 192.168.1.1      UP Sx-N4 Ctl V4 * : 172.16.1.1
CP Sx-N4 Ctl V6 * : 3ffe::2121:101    UP Sx-N4 Ctl V6 * : 3ffe::2121:102
PDRs               : 15               FARS              : 2
QERs               : 1
BAR ID             : 5
State              : IDLE              AA Session Id      : 0x00841a56
DL APN rate lmt Drp Pkt : 0
UL APN rate lmt Drp Pkt : 0
PLMN rate lmt Drp Pkt  : 0
L2TP Connection ID   : 0
-----
Number of PDN contexts : 1
=====

```

Example output PDN context session endpoint ID UP detail acct statistics

```

# show mobile-gateway pdn pdn-context up-seid 0x10100 detail acct stats
=====
PDN Context Accounting Statistics Detail
=====
URR ID           : 1073741825      Sequence No      : 0
Measurement Method : Volume
-----
Reporting Triggers                Subscribed
-----
Linked Usage Reporting            N
Dropped DL Traffic Threshold      N
Stop of Traffic                   N
Start of Traffic                  Y
QHT                               N
Time Threshold                    N
Volume Threshold                  N
Periodic Reporting                N
Time Quota                       N
Volume Quota                      N
-----
Measure Period      : 0
Volume Threshold
UL                  : 0          DL                  : 0
Total               : 0
Subsequent Volume Threshold
UL                  : 0          DL                  : 0
Total               : 0
Volume Quota
UL                  : 0          DL                  : 0
Total               : 0
Volume Usage
UL                  : 0          DL                  : 0
Time Threshold      : 0 secs    Subsequent Time Thresho*: 0 secs
Time Quota          : 0 secs    Time Usage           : 0 secs
Quota Holding Time  : 0 secs    Drp DL Traffic Threshold: 0
Monitoring Time     : 0 secs    Inactivity Detection Ti*: 0 secs
Parent URR          : 0
Measurement Information
Before QOS          : No        Inactivate Measurement : Yes

```

```

-----
URR ID           : 1073741829   Sequence No       : 0
Measurement Method : Volume
-----
Reporting Triggers                               Subscribed
-----
Linked Usage Reporting                           Y
Dropped DL Traffic Threshold                     N
Stop of Traffic                                 N
Start of Traffic                                N
QHT                                              N
Time Threshold                                  N
Volume Threshold                                Y
Periodic Reporting                              N
Time Quota                                      N
Volume Quota                                    N
-----
Measure Period      : 0
Volume Threshold
UL                  : 0          DL                  : 0
Total               : 4194304
Subsequent Volume Threshold
UL                  : 0          DL                  : 0
Total               : 0
Volume Quota
UL                  : 0          DL                  : 0
Total               : 0
Volume Usage
UL                  : 0          DL                  : 0
Time Threshold      : 0 secs    Subsequent Time Thresho*: 0 secs
Time Quota          : 0 secs    Time Usage           : 0 secs
Quota Holding Time  : 0 secs    Drp DL Traffic Threshold: 0
Monitoring Time     : 0 secs    Inactivity Detection Ti*: 0 secs
Parent URR          : 1073741825
Measurement Information
Before QOS          : No        Inactivate Measurement : No
-----
URR ID           : 1073741830   Sequence No       : 0
Measurement Method : Volume
-----
Reporting Triggers                               Subscribed
-----
Linked Usage Reporting                           Y
Dropped DL Traffic Threshold                     N
Stop of Traffic                                 N
Start of Traffic                                N
QHT                                              N
Time Threshold                                  N
Volume Threshold                                Y
Periodic Reporting                              N
Time Quota                                      N
Volume Quota                                    N
-----
Measure Period      : 0
Volume Threshold
UL                  : 0          DL                  : 0
Total               : 4194304

```

```

Subsequent Volume Threshold
UL                : 0                DL                : 0
Total             : 0
Volume Quota
UL                : 0                DL                : 0
Total             : 0
Volume Usage
UL                : 0                DL                : 0
Time Threshold    : 0 secs           Subsequent Time Thresho*: 0 secs
Time Quota        : 0 secs           Time Usage         : 0 secs
Quota Holding Time : 0 secs           Drp DL Traffic Threshold: 0
Monitoring Time   : 0 secs           Inactivity Detection Ti*: 0 secs
Parent URR        : 1073741825
Measurement Information
Before QOS        : No                Inactivate Measurement : No
-----
URR ID            : 1610612737       Sequence No        : 0
Measurement Method : Duration Volume
-----

```

Reporting Triggers	Subscribed
Linked Usage Reporting	Y
Dropped DL Traffic Threshold	N
Stop of Traffic	N
Start of Traffic	Y
QHT	N
Time Threshold	N
Volume Threshold	Y
Periodic Reporting	N
Time Quota	N
Volume Quota	N

```

Measure Period    : 0
Volume Threshold
UL                : 0                DL                : 0
Total             : 2097152
Subsequent Volume Threshold
UL                : 0                DL                : 0
Total             : 0
Volume Quota
UL                : 0                DL                : 0
Total             : 0
Volume Usage
UL                : 0                DL                : 0
Time Threshold    : 0 secs           Subsequent Time Thresho*: 0 secs
Time Quota        : 0 secs           Time Usage         : 0 secs
Quota Holding Time : 0 secs           Drp DL Traffic Threshold: 0
Monitoring Time   : 0 secs           Inactivity Detection Ti*: 60 secs
Parent URR        : 1073741829
Measurement Information
Before QOS        : No                Inactivate Measurement : No
-----
URR ID            : 1610612738       Sequence No        : 0
Measurement Method : Duration Volume
-----

```

Reporting Triggers	Subscribed
--------------------	------------

```

Linked Usage Reporting                Y
Dropped DL Traffic Threshold          N
Stop of Traffic                      N
Start of Traffic                     Y
QHT                                  N
Time Threshold                       N
Volume Threshold                     Y
Periodic Reporting                   N
Time Quota                           N
Volume Quota                         N
-----
Measure Period      : 0
Volume Threshold
UL                  : 0          DL                  : 0
Total               : 2097152
Subsequent Volume Threshold
UL                  : 0          DL                  : 0
Total               : 0
Volume Quota
UL                  : 0          DL                  : 0
Total               : 0
Volume Usage
UL                  : 0          DL                  : 0
Time Threshold      : 0 secs    Subsequent Time Thresho*: 0 secs
Time Quota          : 0 secs    Time Usage           : 0 secs
Quota Holding Time  : 0 secs    Drp DL Traffic Threshold: 0
Monitoring Time     : 0 secs    Inactivity Detection Ti*: 60 secs
Parent URR          : 1073741830
Measurement Information
Before QOS          : No        Inactivate Measurement : No
-----
Number of URRs      : 5
=====

```

* indicates that the corresponding row element may have been truncated.

5.2.2.45 wrap

Syntax

```
wrap [gateway gw-id] [apn apn-name] [card slot-number] [vm vm-id] [group group-number]
```

Context

```
show>mobile>pdn>pdn-context
```

Description

This command specifies that the UE IP address column is not truncated and the full IP address is visible by wrapping the output in the next line displays.

Parameters

gw-id

specifies the gateway ID

Values: 1 to 8

apn

specifies the context's APN name

group-number

specifies the group number

Values: 1 to 15

slot-number

specifies the card slot number

Values: 1 to 20 | A | B

vm-id

specifies the VM ID (applicable only to CMG)

Values: 1 to 20 | A | B

Output

The following output example shows wrapped output for the UE IP address.

Example output PDN context wrap

```
# show mobile-gateway pdn pdn-context wrap
=====
IMSI/IMEI(#)      APN          Type   Beare*  UE Address (IPv4/IPv6)  Ref-pt/Si*
/MSISDN(^)/MAC(`)
/SubId/SEID(~)
=====
98765432101      ipd.nokia*  IPv6    1       -/3ffe:0:5:ff00::15:53a S5/GTPv2
                  el.1.1.c*              3
-----
```

5.2.2.46 pdu-session

Syntax

pdu-session

pdu-session supi *imsi-msisdn* [detail]

pdu-session supi *imsi-msisdn* acct {*stats* | *nchf*} [detail]

pdu-session ran-ue-ngap-id *ran-ue-ngap-id* [pdu-session-id *pdu-session-id*]

pdu-session ran-ue-ngap-id *ran-ue-ngap-id* detail [pdu-session-id *pdu-session-id*]

```
pdu-session ran-ue-ngap-id ran-ue-ngap-id acct stats [pdu-session-id pdu-session-id
[qfi qfi]]
```

Context

```
show>mobile>pdn
```

Description

This command displays context information related to the PDU session SUPI, RAN-UE-NGAP-ID, and Nchf service.

Parameters

imsi-msisdn

specifies the SUPI identifier, up to 15 characters

detail

specifies detail information

nchf

specifies Nchf information

ran-ue-ngap-id

specifies the identity of the RAN UE NGAP ID

Values: 0 to 2^{32-1} (HEX)

qfi

specifies the QFI identifier

Values: 0 to 63

Output

The following output examples show PDU session information for the specified SUPI ID, RAN-UE-NGAP-ID, and Nchf account. The *Show PDU session Nchf account detail statistics*, *Show PDU session ran-ue-ngap-id detail statistics*, and *Show PDU session ran-ue-ngap-id acct statistics* describe the output fields.

- *Example output PDU session Nchf account statistics*
- *Example output PDU session Nchf account detail statistics*
- *Example output PDU session ran-ue-ngap-id detail statistics*
- *Example output PDU session ran-ue-ngap-id acct statistics*

Example output PDU session Nchf account statistics

```
# show mobile-gateway pdn pdu-session supi 310012001003204 acct nchf
=====
PDN context Accounting Nchf
```

```

=====
PDN context Accounting Nchf Detail
=====
IMSI      : 310012001003204
MSISDN    : 12345678901
IMEI      : 987654321012345
APN       : ipd.nokia.com
URR Id    : 1073741825      Charging Id    : 553648130
Charging Profile : 1        Roaming Status   : Home
CHF Profile : 1            CCFH Profile    : 0
PDN type   : IPv4
UE IPv4 address : 1.127.255.0
Service Realm : sbaRealm1
Service Instance : chfClient1
CHF URI     : http://10.10.7.85:5566/nchf-convergedcharging/v1/
              chargingdata/0
CHF NF Instance Id : 4f85e924-1509-47b4-907f-9778afdb2ffb
CHF Peer Origin  : Local configuration
Failure Handling : Terminate (L)
Session Failover : Enabled (L)
Num URR        : 3          Num FAR          : 0
Num 5QI/ARP    : 1
Aggr. UL bytes : 2000       Aggr. DL bytes : 2000
-----
5QI/ARP
-----
5QI          : 5            ARP              : 9
URR Id       : 1073741829
Num RG       : 1            Num RG/Service-Id : 0
Aggr. UL bytes : 2000       Aggr. DL bytes : 2000
QFI          : 5
=====
Number of PDN contexts : 1

```

Example output PDU session Nchf account detail statistics

```
# show mobile-gateway pdn pdu-session supi 310012001003204 acct nchf detail
```

```
PDN context Accounting Nchf Detail
```

```

=====
IMSI      : 310012001003204
MSISDN    : 12345678901
IMEI      : 987654321012345
APN       : ipd.nokia.com
URR Id    : 1073741825      Charging Id    : 553648130
Charging Profile : 1        Roaming Status   : Home
CHF Profile : 1            CCFH Profile    : 0
PDN type   : IPv4
UE IPv4 address : 1.127.255.0
Service Realm : sbaRealm1
Service Instance : chfClient1
CHF URI     : http://10.10.7.85:5566/nchf-convergedcharging/v1/
              chargingdata/0
CHF NF Instance Id : 4f85e924-1509-47b4-907f-9778afdb2ffb
CHF Peer Origin  : Local configuration

```



```

Failure Handling : Terminate (L)
Session Failover : Enabled (L)
Num URR : 3 Num FAR : 0
Num 5QI/ARP : 1
Aggr. UL bytes : 2000 Aggr. DL bytes : 2000
Time Limit : 3600 Time Limit Category: Immediate
Volume Limit : 4194304 Vol Limit Category : Immediate

```

URRs

```

URR Id : 1073741825
Measurement Method : volume
Reporting Triggers : vol-thresh
Vol Threshold Total: 4194304

```

```

URR Id : 1073741829
Measurement Method : volume
Reporting Triggers : linked-usg
Linked URR Id : 1073741825

```

```

URR Id : 1610612737
Measurement Method : time volume
Reporting Triggers : vol-thresh linked-usg
Vol Threshold Total: 4000
Inact Detect Time : 60
Linked URR Id : 1073741829

```

5QI/ARP

```

5QI : 5 ARP : 9
URR Id : 1073741829
Num RG : 1 Num RG/Service-Id : 0
Aggr. UL bytes : 2000 Aggr. DL bytes : 2000
QFI : 5

```

Rating Groups

```

Rating Group : 0
URR Id : 1610612737
Granted Total Vol : 4000
Time Limit : 1800 Time Limit Category: Deferred
Volume Limit : 2097152 Vol Limit Category : Deferred

```

```

=====
Number of PDN contexts : 1

```

Table 53: Show PDU session Nchf account detail statistics

Label	Description
IMSI	IMSI identifier

Table 53: Show PDU session Nchf account detail statistics (continued)

Label	Description
MSISDN	MSISDN identifier
IMEI	IMEI identifier
APN	APN identifier
URR Id	URR ID
Charging Id	Charging ID
Charging Profile	Charging profile ID
Roaming status	Roaming status
CHF Profile	CHF profile ID
CCFH Profile	CCFH profile ID
PDN type	PDN IP type
UE IPv4 address	UE IPv4 address
Service Realm	Service realm name
Service Instance	Service instance name
CHF URI	CHF URI address
CHF NF Instance Id	CHF NF instance ID
CHF Peer Origin	Origin of the peer
Failure Handling	Failure Handling

Table 53: Show PDU session Nchf account detail statistics (continued)

Label	Description
Session Failover	Session failover status (enabled/disabled)
Num URR	Number of URR rules
Num FAR	Number of FAR rules
Num 5Q/ARP	Number of 5QI/ARP rules
Aggr. UL bytes	Aggregate uplink bytes
Aggr. DL bytes	Aggregate downlink bytes
Time Limit	Time limit
Time Limit Category	Time limit category (immediate)
Volume limit	Volume limit
Volume Limit Category	Volume limit category (for example, immediate)
URRs	
URR ID	URR ID
Measurement Method	Measurement method (for example, time, or volume)
Reporting Triggers	Reporting triggers (for example, volume threshold, or linked USQ)
Volume threshold total	Total volume threshold
Inact Detect Time	Inactivity detection time
Linked URR ID	Linked URR ID

Table 53: Show PDU session Nchf account detail statistics (continued)

Label	Description
5QI/ARP	
5QI	5QI ID
ARP	ARP ID
URR Id	URR ID
Num RG	Number of RGs
Num RG/Service-ID	The number of RG Service IDs
Aggr. UL bytes	Aggregate uplink bytes
Aggr. DL bytes	Aggregate downlink bytes
QFI	QFI ID
Rating Group	
URR Id	URR ID
Granted total volume	Total granted volume for the RG
Time Limit	Time limit
Time Limit Category	Time limit category (for example, immediate)
Volume Limit	Volume limit
Volume Limit Category	Volume limit category (for example, deferred)

Example output PDU session ran-ue-ngap-id detail statistics

```
# show mobile-gateway pdn pdu-session ran-ue-ngap-id 0x10100 detail
```

```
=====
PDU session detail
=====
```

```
PDU Session ID      : 1
```

```
RAT                  : VIRTUAL
```

```
Data Session ID     : 0x00020100
```

```
Ref Pnt/Sig Proto:  NWu/IKE
```

```
UL DNN AMBR         : 4000 Kbps
```

```
DL DNN AMBR         : 2000 Kbps
```

```
NSSAI
```

```
SST                  : 11
```

```
SD                   : 0xd143a5
```

```
QoS Information
-----
```

```
QFI : 5    QoS Flow Type : Non-GBR
-----
```

```
QoS Profile
```

```
5QI/ARP              : 5/9
```

```
Priority Level        : 0
```

```
Avg Window (ms)      : 0
```

```
Max Bst Vol (bytes)   : 0
```

```
QNC                   : Disabled
```

```
RQI                   : Disabled
```

```
UL MBR (kbps)         : 0
```

```
DL MBR (kbps)         : 0
```

```
UL GBR (kbps)         : 0
```

```
DL GBR (kbps)         : 0
-----
```

```
User Plane (UP) Information
-----
```

```
N3 Data TEID         : 0x20105
```

```
N3 IPv4 Addr          : 17.70.16.16
```

```
N3 IPv6 Addr          : 3ffe::1146:1010
-----
```

```
DATA ESP SA Information
-----
```

```
UP IPv4 address       : 14.0.0.0
```

```
UP IPv6 address       : -
```

```
QFI(s)                : 5
-----
```

```
Extended Seq Num      : Disabled
```

```

TS Type           : IPv4
Integrity         : hmac-sha1
Encryption        : aes192ctr
DH Group          : none
Local/Inbound SPI: 01000001          Remote/Outbnd SPI: 40000101
Lifetime          : 23h20m00s
Established Time  : 01/31/2023 12:34:46
Expiry Time       : 02/01/2023 11:54:46
Number of Rekeys  : 0
Decoded Packets   : 1                Encoded Packets   : 1
Decoded Bytes     : 188              Encoded Bytes     : 200
Decoded Errors    : 0                Encoded Errors    : 0
Replay Errors     : 0

```

Child DATA ESP SA Information

```

-----
Extended Seq Num : Disabled
TS Type           : IPv4
Integrity         : hmac-sha1
Encryption        : aes192ctr
DH Group          : none
Local/Inbound SPI: 01000001          Remote/Outbnd SPI: 40000101
Lifetime          : 23h20m00s
Established Time  : 01/31/2023 12:34:46
Expiry Time       : 02/01/2023 11:54:46
Number of Rekeys  : 0
Decoded Packets   : 1                Encoded Packets   : 1
Decoded Bytes     : 188              Encoded Bytes     : 200
Decoded Errors    : 0                Encoded Errors    : 0
Replay Errors     : 0

```

```

-----
Number of PDU sessions : 1
=====

```

Table 54: Show PDU session ran-ue-ngap-id detail statistics

Label	Description
PDU Session ID	The identity of the PDU session
RAT	The radio access type

Table 54: Show PDU session ran-ue-ngap-id detail statistics (continued)

Label	Description
Data Session ID	The identity of the data session
Ref Pnt/Sig Proto	There reference point name and the signaling protocol used
UL DNN AMBR	The uplink AMBR for the DNN
DL DNN AMBR	The downlink AMBR for the DNN
SST	Slice/service type
SD	Slice differentiator
QFI	QoS flow ID
QoS Flow Type	QoS flow type
5QI/ARP	5G QoS Identifier / Allocation Retention Priority
Priority Level	Priority Level
Avg Window (ms)	Average window (duration over which the guaranteed and maximum bitrate is calculated) for the QFI value within a static QFI mapping profile
Max Bst Vol (bytes)	Maximum burst volume associated with the QFI value within a static QFI mapping profile
QNC	QNC status, for notifications from the NG-RAN when GFBR can no longer be guaranteed during the life of the QoS flow for the SDF (enabled/disabled)
RQI	RQI status (enabled/disabled)
UL MBR	Uplink MBR

Table 54: Show PDU session ran-ue-ngap-id detail statistics (continued)

Label	Description
DL MBR	Downlink MBR
UL GBR	Uplink GBR
DL GBR	Downlink GBR
N3 Data TEID	N3 Data TEID
N3 IPv4 Addr	N3 IPv4 address
N3 IPv6 Addr	N3 IPv6 address
UP IPv4 address	User Plane IPv4 address
UP IPv6 address	User Plane IPv6 address
QFI(s)	QoS Flow Id(s)
Extended Seq Num	Extended Sequence Number
TS Type	Traffic Selector type
Integrity	Integrity algorithm
Encryption	Encryption algorithm
DH Group	Diffie-Hellman group
Local/Inbound SPI	Local/Inbound SPI
Remote/Outbnd SPI	Remote/Outbound SPI
Lifetime	Lifetime of the security association

Table 54: Show PDU session ran-ue-ngap-id detail statistics (continued)

Label	Description
Established Time	Established time of the security association
Expiry Time	Expiry time of the security association
Number of Rekeys	Number of Rekeys
Decoded Packets	Number of decoded packets
Encoded Packets	Number of encoded packets
Decoded Bytes	Number of decoded bytes
Encoded Bytes	Number of encoded bytes
Decoded Errors	Number of decoded errors
Encoded Errors	Number of encoded errors
Replay Errors	Number of replay errors
Number of PDU sessions	Number of PDU sessions

Example output PDU session ran-ue-ngap-id acct statistics

```
# show mobile-gateway pdn pdu-session ran-ue-ngap-id 0x10100 acct stats
```

```
=====
PDU session accounting statistics brief
=====
```

```
RAN UE NGAP ID   : 0x10100
PDU Session ID   : 1
QFI Contexts     : 1
Aggr UL Bytes    : 156                Aggr DL Bytes    : 128
Aggr UL Packets  : 1                  Aggr DL Packets  : 1
-----
```

```

Number of PDU sessions : 1
=====

*A:Dut-A# show mobile-gateway pdn pdu-session acct stats ran-ue-ngap-id
0x10100 pdu-session-id 1 qfi 1

=====

PDU session accounting statistics brief
=====

RAN UE NGAP ID      : 0x10100
PDU Session ID      : 1
QFI Contexts        : 1
UL Bytes            : 128                DL Bytes            : 128
UL Packets          : 1                DL Packets            : 1
-----

Number of PDU sessions : 1
=====

*A:Dut-A# show mobile-gateway pdn pdu-session acct stats ran-ue-ngap-id
0x10100 pdu-session-id 1 qfi 5

=====

PDU session accounting statistics brief
=====

RAN UE NGAP ID      : 0x10100
PDU Session ID      : 1
QFI Contexts        : 1
UL Bytes            : 256                DL Bytes            : 256
UL Packets          : 2                DL Packets            : 2
-----

Number of PDU sessions : 1
=====

```

Table 55: Show PDU session ran-ue-ngap-id acct statistics

Label	Description
RAN UE NGAP ID	RAN UE NGAP Identity
PDU Session ID	PDU Session Identity
QFI Contexts	Number of QFI contexts

Table 55: Show PDU session ran-ue-ngap-id acct statistics (continued)

Label	Description
Aggr UL Bytes	Aggregate uplink bytes
Aggr DL Bytes	Aggregate downlink bytes
Aggr UL Packets	Aggregate uplink packets
Aggr DL Packets	Aggregate downlink packets
Number of PDU sessions	Number of PDU sessions

5.2.2.47 plmn-chrg-profile-map

Syntax

```
plmn-chrg-profile-map plmn-list-name list-name [gateway-Id gw-id]
```

Context

```
show>mobile>pdn>charging
```

Description

Displays the configured charging profile map for a specified PLMN list name.

Parameters

list-name

specifies PLMN list name, up to 32 characters

gw-id

specifies the gateway ID

Output

The following output examples show PDN context information and *Show PDN PLMN list charging profile output fields* describes the output fields.

Example output PLMN charging profile map

```
# show mobile-gateway pdn charging plmn-chrg-profile-map plmn-list-name "basicPlmn"

=====
PLMN charging profile map
=====
PLMN list name : basicPlmn
ccValue       : 0x64           chargingProfileId: 1
ccValue       : 0xc8           chargingProfileId: 2
ccValue       : 0x100          chargingProfileId: 1
ccValue       : 0xc12c          chargingProfileId: 1
ccValue       : 0xc200          chargingProfileId: 2
=====
```

Table 56: Show PDN PLMN list charging profile output fields

Label	Description
PLMN list name	The PLMN list name
ccValue	The charging profile behavior identifier
chargingProfileId	The charging profile ID

5.2.2.48 pfcf-load-overload-control

Syntax

```
pfcf-load-overload-control
pfcf-load-overload-control stats
```

Context

```
show>mobile>pdn
```

Description

This command displays PFCF load and overload control configuration information or statistics per UP node for the CP function gateway.

Parameters

stats

keyword to display statistics information

Output

The following output examples show load and overload configuration and statistics per UP node for the CP function gateway and *PFCP load and overload configuration output fields* and *PFCP load and overload statistics output fields* describe the output fields.

- *Example output PFCP load and overload control configuration*
- *Example output PFCP load and overload control statistics*

Example output PFCP load and overload control configuration

```
# show mobile-gateway pdn pfcp-load-overload-control
=====
PFCP load and overload control information
=====
Gateway                : 1
Load control           : Disabled
Load change trigger value : 5 %      Nth msg load report update : 50
arp-to-message-priority-list : list-name
Overload control       : Disabled Overload Validity time       : 600 sec
Minor reduction value  : 30 %      Major reduction value     : 60 %
Critical reduction value : 90 %

CPU sampling interval
Overload                : 10 sec   Overload state change       : 2 sec
No-overload             : 10 sec

Current load            : 0 %      Current overload state      : no-over*
-----
```

Table 57: PFCP load and overload configuration output fields

Label	Description
Gateway	Gateway ID
Load control	Status of the PFCP load control
Load change trigger value	Percentage load change value that will trigger a new load control update

Table 57: PFCP load and overload configuration output fields (continued)

Label	Description
Nth msg load report update	The number of messages between which a PFCP load control report message will be sent, regardless of load change or not
arp-to-message-priority-list	ARP list to be used on the CP Function for setting an MP Flag and the message-priority in the PFCP message header
Overload control	Status of the PFCP load control
Overload validity time	The duration of time for which the reported overload will be in effect
Minor reduction value	Percentage of traffic reduction for minor overload requests
Major reduction value	Percentage of traffic reduction for major overload requests
Critical reduction value	Percentage of traffic reduction for critical overload requests
CPU sampling interval	
Overload	Overload time
Overload state change	Overload state change time
No-overload	No overload time
Current load	Current load state
Current overload state	Current overload state

Example output PFCP load and overload control statistics

```
# show mobile-gateway pdn pfcpl-load-overload-control stats
=====
CP function gateway PFCP load and overload statistics per UP node
=====
Gateway : 1
```

```

Router : vprn20091
Node address : 10.34.1.1
Load metric : 100
Last overload report time :04/10/2019 13:11:33
Current overload : Yes Traffic reduction value :30%
Remaining validity time : 589 sec
PFCP sessions
Session Establishment Req dropped : 0
-----
Number of nodes : 1
=====

```

Table 58: PFCP load and overload statistics output fields

Label	Description
Gateway	Gateway ID
Router	Router ID
Node address	Node IP address
Load metric	Current load level of the sending node as a percentage
Last overload report time	Date and time of most recently reported overload
Current overload	Current overload status
Traffic reduction value	Percentage of the requested traffic reduction value
Remaining validity time	Time remaining of the current configured validity time, during which the overload condition specified by the OCI IE is valid (unless overridden by subsequent new overload control information)
Session Establishment Req dropped	The number of dropped session establishment requests due to the overload control condition

5.2.2.49 pfcp-local-load-control

Syntax

```
pfcp-local-load-control stats
```

Context

```
show>mobile>pdn
```

Description

This command displays configuration information or statistics for local load control for the PFCP UP function.

Parameters

stats

keyword to display statistics information

Output

The following output example shows PFCP local load control statistics for the UP function and [Local load control statistics for PFCP UP function output fields](#) describes the output fields.

Example output local load control statistics for PFCP UP function

```
# show mobile-gateway pdn pfcp-local-load-control stats
=====
UP Function PFCP local load statistics
=====
Last twenty 100% load report events:
  Time                LCI reported          Slot
03/19/18 18:46:33      100                    3
03/19/18 18:46:37      <100                   3
03/19/18 18:47:15      100                    3
03/19/18 18:47:23      <100                   3
Current pfcp load (LCI) reported : 86
=====
```

Table 59: Local load control statistics for PFCP UP function output fields

Label	Description
Time	The time of the most recent 20 load reports

Table 59: Local load control statistics for PFCP UP function output fields (continued)

Label	Description
LCI reported	Load percentage reported via the LCI IE
Slot	UP function slot or VM
Current PFCP load (LCI) reported	Current total PFCP load reports via LCI IE

5.2.2.50 pfcpl-local-overload-control

Syntax

```
pfcpl-local-overload-control stats
```

Context

```
show>mobile>pdn
```

Description

This command displays configuration information or statistics for PFCP local overload control for the UP function.

Parameters

stats

keyword to display statistics information

Output

The following output example shows PFCP local overload control statistics for the UP function and *Local overload control statistics for PFCP output fields* describes the output fields.

Example output local overload control statistics for PFCP UP function

```
# show mobile-gateway pdn pfcpl-local-overload-control stats
=====
UP Function PFCP local overload statistics
=====
Last twenty overload report events:
Time           Overload level   OCI reported      Slot
03/19/18 18:47:15   Minor             30                 3
```

```

03/19/18 18:48:22    no overload           0           3
03/19/18 18:50:21    Minor             30          3
03/19/18 18:51:21    Major             60          3
03/19/18 18:57:21    Minor             30          3
03/19/18 18:59:22    no overload           0           3
Current PFCP overload : No           Traffic reduction value :
Remaining validity time : 386sec
=====

```

Table 60: Local overload control statistics for PFCP output fields

Label	Description
Time	Time of the most recent 20 load reports
Overload level	Indicates the overload level
OCI reported	Percentage of the reported overload
Slot	UP function slot or VM
Current PFCP overload	Current overload or not
Traffic reduction value	Percentage of traffic reduction requested by the sender of the OCI request; 0 indicates that the UP function is not in overload
Remaining validity time	Time remaining for this overload condition to be valid

5.2.2.51 pgw-reselection-isolation

Syntax

```
pgw-reselection-isolation
```

Context

```
show>mobile>pdn
```

Description

This command displays the configuration information for the PGW reselection isolation.

Output

The following output example shows the PGW reselection isolation configuration and the *PGW reselection isolation output fields* describes the output fields.

Example output PGW reselection isolation

```
# show mobile-gateway pdn pgw-reselection-isolation
=====
PGW Reselection & Isolation List
=====
CC          PGW/SMF peer          Conf. Isol. Time      Rem. Isol. Time
94          172.65.12.3                10                    4
129         172.65.12.3                120                   45
=====
```

Table 61: PGW reselection isolation output fields

Label	Description
CC	Cause code
PGW/SMF peer	IP address of the peer
Conf. Isol. Time	Configured isolation time
Rem. Isol. Time	Remaining isolation time

5.2.2.52 qos-flow

Syntax

```
qos-flow [qfi qfi] acct acct ran-ue-ngap-id ran-ue-ngap-id pdu-session-id pdu-session-id
qos-flow [dnn dnn]
qos-flow supi supi [dnn dnn]
qos-flow supi supi detail [qfi qfi] [dnn dnn]
```

Context

```
show>mobile>pdn
```

Description

This command displays QoS flow information for the specified SUPI for 5G sessions.

Parameters

acct

specifies the accounting information

Value: stats

ran-ue-ngap-id

specifies the RAN UE NGAP ID

Values: 0 to $2^{32}-1$ (HEX)

pdu-session-id

specifies the PDU session ID

Values: 0 to 255

dnn

specifies the DNN, up to 80 characters

supi

specifies the SUPI identifier, up to 15 characters

detail

specifies detail information

qfi

specifies the QFI identifier

Values: 0 to 63

Output

The following output examples show QoS flow information for a specified SUPI or RAN UE NGAP ID. The *Show PDN QoS flow context SUPI detail output fields* and *Show PDN QoS flow ran-ue-ngap-id detail statistics output fields* describe the output fields.

- *Example output QoS flow SUPI detail statistics (5G sessions)*
- *Example output QoS flow SUPI detail statistics multiple QFIs (5G sessions)*
- *Example output QoS flow ran-ue-ngap-id detail statistics (5G sessions)*



NOTE:

When the `supi` option is used, information related to EPS interworking is displayed; for specific details, see the notes in *Show PDN QoS flow context SUPI detail output fields*.

Example output QoS flow SUPI detail statistics (5G sessions)

```
# show mobile-gateway pdn qos-flow supi 460100001003204 detail
=====
Qos Flow detail
=====
SUPI           : 460100001003204
DNN            : ipd.nokia.com
QFI           : 5                QOS Flow type : Default/Gbr/Non-Gbr
EPS Bearer Id  : 5                Policy Ctrl Interface : Npcf
SDFs          : 2
Charg. Char. Rcvd : 0x123a
State         : CONNECTED
Up Time       : 1d 04:43:02
UP Sx-N4 Seid :
Qos Profile:
  5QI/ARP      : 5/11
  Preemption capability : NOT_PREEMPT  Preempt vulnerability : NOT_PREEMPTABLE
  Priority Level : 10                UL MBR           : 0 kbps
  Average Window : 0 ms              DL MBR           : 0 kbps
  Max Burst Volume : 0 bytes          UL GBR           : 0 kbps
  RQI            : 0                UL GBR           : 0 kbps
  QNC            : Disabled          QNC state        : Guaranteed

Redirect SDF Cou*: 0
-----
Policy Rules Information
PCC Rule Count      : 2                DSCP              : af
-----
PCC rule name       : multiQfiDefBrPriLevPr1
SDF ID              : 39003            PCC precedence     : 7002

N1 Information
QoS rule ID         : 1                QoS rule preced*   : 7002
Signaling opcode    : Create

Qos Information
qosId               : qosData5_9
MBR UL              : 0 kbps           MBR DL             : 0 kbps
GBR UL              : 0 kbps           GBR DL             : 0 kbps
QNC                 : False           RQI                : False

Charging Information
chgId               : rg10_on_off
Reporting Level     : Rating Grp Level  Metering Method    : -
Billing Method      : Online & Offline
Rating Group        : 10              Service Id         : -
Sponsor Identity    : -               Appl Svc Prov Id   : -
```

```

Traffic Control Information
tcId          : TC_ENABLED
Divert to AA   : No                Flow Status      : Up/Downlink Enable
DNAI          : N/A                Mute Notify     : False
PFCP Information
PDR ID UL     : 2561                PDR ID DL      : 35329
PDR ID S1u UL : 18945
FAR ID UL     : 327680              FAR ID DL      : 537198592
QER ID        : 1                  QFI            : 5
Flow Information
Pkt filters   : 1
Filter id     : 0                  Filter direction : DL
Filter protocol : udp (17)         Filter tos-tc    : -
Local address  : 1.127.255.0/32    Remote addr      : 16.1.1.1/32
Local ports    : 6666              Remote ports     : 5555
Signaling opcode : Create
-----
Number of qos flows : 1
=====
* indicates that the corresponding row element may have been truncated.

```

Table 62: Show PDN QoS flow context SUPI detail output fields

Label	Description
SUPI	SUPI IMSI number
DNN	5G DNN (APN) name
QFI	QFI
QOS Flow Type	QoS flow type
Policy Ctrl Interface	Policy control interface
EPS Bearer Id	EPS bearer ID
SDFs	Number of SDFs
Charg. Char. Rcvd	Charging characters received
State	State

Table 62: Show PDN QoS flow context SUPI detail output fields (continued)

Label	Description
Up Time	Uptime of the bearer
UP Sx-N4 Seid ⁵	UP Sx-N4 (PFCP) SEID
QoS Profile	Qos flow profile associated with the DDN
5QI/ARP ⁵	The QFI generated dynamically using 5QI/ARP as default (in the range 1 to 64)
Preemption capability	Preemption capability state
Preempt vulnerability	Preemption vulnerability state
UL MBR	Uplink MBR of the QoS flow profile
Priority Level	Priority level characteristic of the QoS flow
DL MBR	Downlink MBR of the QoS flow
Average Window	Average window characteristic of the QoS flow
UL GBR	Uplink GBR of the QoS flow
Max Burst Volume	Maximum burst volume
DL GBR	Downlink GBR of the QoS flow
QNC	QNC status, for notifications from the NG-RAN when GGBR can no longer be guaranteed during the life of the QoS flow for the SDF
QNC state	QNC state

⁵ In the case of EPS interworking information, if there are no SDFs per QFI, the 5QI/ARP and UP Sx-N4 SEID fields display "0" instead of the actual value.

Table 62: Show PDN QoS flow context SUPI detail output fields (continued)

Label	Description
RQI	RQI status (enabled/disabled)
Redirect SDF Count	SDF redirect count
PCC Rule Count	Number of PCC rules
DSCP	DSCP name
PCC rule name	PCC rule name
SDF ID	SDF ID
QoS rule ID	QoS rule ID
Signaling opcode	Signaling operational state
QoS rule precede*	QoS rule precedence
chglId	Charging ID
Reporting Level	Reporting level
Metering Method	Metering method
Billing method	Billing method
Flow Status	Flow status
Service Id	Service identifier
Rating Group	Rating group
Sponsor Identity	Sponsor identifier

Table 62: Show PDN QoS flow context SUPI detail output fields (continued)

Label	Description
Appl Svc Prov Id	APPL Service Provider identifier
RQI	RQI
QNC	QNC
DNAI	DNAI status
PCC precedence	PCC precedence
pkt filters	Number of packet filters
tcId	TC ID
Divert to AA	Divert to AA status
PDR ID UL	Uplink PDR ID
PDR ID DL	Downlink PDR ID
PDR ID S1u UL	Uplink S1u PDR ID
FAR ID UL	Uplink FAR ID
FAR ID DL	Downlink FAR ID
QER ID	QER ID
QFI	QFI value associated with a static QFI mapping profile
Filter id	Filter ID
Filter direction	Filter direction specification

Table 62: Show PDN QoS flow context SUPI detail output fields (continued)

Label	Description
Filter protocol	Filter protocol
Filter tos-tc	Filter TOS-TC
Local address	Local IP address
Remote addr	Remote IP address
Local ports	Local ports
Remote ports	Remote ports
Signaling opcode	Signaling operational state

Example output QoS flow SUPI detail statistics multiple QFIs (5G sessions)

```

show mobile-gateway pdn qos-flow supi 460100001003204 detail
=====
QoS flow detail
=====
SUPI           : 460100001003204
DNN            : ulcl.nonat.staticpolicy.1.com
QFI            : 1                QoS Flow Type      : Default
EPS Bearer Id  : 5                Policy Ctrl Interface : Npcf
SDFs           : 1
Charg. Char. Rcvd : 0x123a
State          : CONNECTED

Up Time        : 0d 00:02:22
UP Sx-N4 Seid  : 0x20100

Qos Profile:
  5QI/ARP       : 5/9              UL MBR           : 0 kbps
  Preemption capability : NOT_PREEMPT  Preempt vulnerability : NOT_PREEMPTABL
E
  Priority Level  : 0              DL MBR           : 0 kbps
  Average Window  : 0 ms           UL GBR           : 0 kbps
  Max Burst Volume : 0 bytes        DL GBR           : 0 kbps
  RQI            : 0              UL GBR           : 0 kbps
  QNC            : Disabled        QNC state        : Guaranteed

```

Redirect SDF Count : 0

Policy Rules Information

PCC Rule Count : 2 DSCP : af

PCC rule name : pr_ulcl_anyAny
SDF ID : 39003

N1 Information

QoS rule ID : 4096 QoS rule precede*: 65081
Signaling opcode : Create

Qos Information

qosId : qosData5_9
MBR UL : 0 kbps MBR DL : 0 kbps
GBR UL : 0 kbps GBR DL : 0 kbps
QNC : False RQI : False

Charging Information

chgId : rg10_on_off
Reporting Level : Rating Grp Level Metering Method : Volume
Billing Method : - Flow Status : Up/Downlink Enabled
Rating Group : 1 Service ID : -
Sponsor Identity : - Appl Svc Prov Id : -

Traffic Control Information

tcId : TC_ENABLED
Divert to AA : No
DNAI : N/A

PFCP Information

PDR ID UL : 2561 PDR ID DL : 35329
PDR ID S1u UL : 18945
FAR ID UL : 327680 FAR ID DL : 537198592
QER ID : 1 QFI : 1

Flow Information

Pkt filters : 1
Filter id : 0 Filter direction : BiDir
Filter protocol : any Filter tos-tc : -
Local address : any Remote addr : any
Local ports : 6666 Remote ports : 5555
Signaling opcode : Create

SUPI : 460100001003204
DNN : ulcl.nonat.staticpolicy.1.com
QFI : 2 QOS Flow Type : Non-GBR
EPS Bearer Id : 5 Policy Ctrl Interface : Npcf
SDFs : 1
Charg. Char. Rcvd : 0x123a
State : CONNECTED

Up Time : 0d 00:02:22
UP Sx-N4 Seid : 0x20100

Qos Profile:

5QI/ARP : 5/9 UL MBR : 0 kbps
Preemption capability : NOT_PREEMPT Preempt vulnerability : NOT_PREEMPTABLE
E
Priority Level : 3 DL MBR : 0 kbps
Average Window : 1000 ms UL GBR : 0 kbps
Max Burst Volume : 640 bytes DL GBR : 0 kbps
RQI : 0 UL GBR : 0 kbps
QNC : Disabled QNC state : Guaranteed

Redirect SDF Count : 0

Policy Rules Information

PCC Rule Count : 2 DSCP : af

PCC rule name : pr_ulcl_noNat_1
SDF ID : 39003

N1 Information

QoS rule ID : 38986 QoS rule precede*: 65021
Signaling opcode : Create

Qos Information

qosId : qosData5_9
MBR UL : 0 kbps MBR DL : 0 kbps
GBR UL : 0 kbps GBR DL : 0 kbps
QNC : False RQI : False

Charging Information

chgId : rg10_on_off
Reporting Level : Rating Grp Level Metering Method : Volume
Billing Method : - Flow Status : Up/Downlink Enabled
Rating Group : 1 Service ID : -
Sponsor Identity : - Appl Svc Prov Id : -

Traffic Control Information

```
tcId      : TC_ENABLED
Divert to AA : No
DNAI      : ulcl.noNat.1.1.com

PFCEP Information
PDR ID UL      : 2562          PDR ID DL      : 3533
PDR ID S1u UL   : 18946
FAR ID UL      : 327680       FAR ID DL      : 537198592
QER ID         : 1           QFI             : 2

Flow Information
Pkt filters    : 1
Filter id      : 0           Filter direction : BiDir
Filter protocol : any        Filter tos-tc    : -
Local address   : 0.0.0.0/0   Remote addr     : 15.1.1.7/0
Local ports    : 6667        Remote ports    : 5556
Signaling opcode : Create

-----
Number of QoS flows : 2
=====
* indicates that the corresponding row element may have been truncated.
```

Example output QoS flow ran-ue-ngap-id detail statistics (5G sessions)

```
# show mobile-gateway pdn qos-flow ran-ue-ngap-id 0x10100 acct stats pdu-
session-id 1
```

QoS flow accounting statistics brief

```
=====
RAN UE NGAP ID   : 0x10100
PDU Session ID   : 1
QFI              : 4           QFI type         : Non-GBR
UL Bytes         : 120        DL Bytes         : 100
UL Packets       : 2           DL Packets       : 2
-----
```

```
RAN UE NGAP ID   : 0x10100
PDU Session ID   : 1
QFI              : 5           QFI type         : Non-GBR
UL Bytes         : 156        DL Bytes         : 128
UL Packets       : 1           DL Packets       : 1
-----
```

```
Number of QoS flows : 2
=====
```

```
# show mobile-gateway pdn qos-flow ran-ue-ngap-id 0x10100 acct stats pdu-
session-id 1 qfi 5
```

```
=====
QoS flow accounting statistics brief
=====
```

```
RAN UE NGAP ID   : 0x10100
PDU Session ID   : 1
QFI              : 5                QFI type           : Non-GBR
UL Bytes         : 156             DL Bytes           : 128
UL Packets       : 1                DL Packets         : 1
```

```
-----
Number of QoS flows : 1
=====
```

Table 63: Show PDN QoS flow ran-ue-ngap-id detail statistics output fields

Label	Description
RAN UE NGAP ID	RAN UE NGAP Identity
PDU Session ID	The PDU session ID
QFI	The QFI value
QFI type	The QFI type
UL Bytes	Number of uplink bytes
DL Bytes	Number of downlink bytes
UL Packets	Number of uplink packets
DL Packets	Number of downlink packets
Number of QoS flows	The number of qos flows

5.2.2.53 redirect-policy

Syntax

```
redirect-policy [gateway gw-id] [apn apn]
```

Context

```
show>mobile>pdn
```

Description

This command displays redirection information.

Parameters

apn-name

specifies the APN name

gw-id

specifies the gateway ID

Output

The following output is an example of policy redirection information.

Example output redirect policy

```
# show mobile-gateway pdn redirect-policy gateway 1

=====
Redirect Policy
=====
-----
No Matching Entries
=====
```

5.2.2.54 ref-point-peers

Syntax

```
ref-point-peers ref-point-type [ref-point ref-point-name] [peer ip-address [port port-number]]  
[gateway gw-id] [peer-type sgw | pgw | twag | epdg]
```

```
ref-point-peers {gn | gnu | gp | gpu} [ref-point ref-point-name] [peer ip-address [port port-number]]  
[gateway gw-id] [active] ref-point-peers {radius} [radius-group radius-group]  
[peer ip-address] [gateway gw-id]
```

```

ref-point-peers {radius} [radius-group radius-group [peer ip-address]] [gateway gw-id]
ref-point-peers {s5 | s5u | s8 | s8u | n9} [ref-point ref-point-name] [peer ip-address [port port-number]] [gateway gw-id]
ref-point-peers {dhcp} [dhcp-group dhcp-group [peer ip-address]] [gateway gw-id]
ref-point-peers {pi} [pi-peer ipv4 [port port-number]] [gateway gw-id]
ref-point-peers {swm} swm-type [peer ip-address] [gateway gw-id] [ref-point ref-point-name]
ref-point-peers {cdbx} [peer ip-address]
ref-point-peers {swm | gy} [dia-routing-table] [gateway gw-id]
ref-point-peers {s6b | gx | swm | sta | sd} [gateway-id gw-id] [peer ip-address [port port-number]] doic
ref-point-peer {sx-n4 | n2} [ref-point ref-point-name] [peer ip-address] [gateway gw-id]
ref-point-peer {t6c} [ref-point ref-point-name] [peer ip-address [port port-number]] [gateway gw-id] [active]

```

Context

show>mobile>pdn

Description

This command displays the reference point peers for the router instance.

Parameters

s11 | s11u | s12 | s1u | s4 | s4u | s5 | s5u | s8 | s8u | gx | rf | gn | gnu | gp | gpu | ga | radius | gy | s2a | s2a-gtp | s2b | s6b | swm | swm | sta | sx-n4 | sd | dhcp | n3 | n9

specifies the reference point type

ip-address

specifies the IP address of the peer

Values: a.b.c.d or x:x:x:x:x:x:x

port-number

specifies the port number

Values: 1 to 65535

gw-id

specifies the gateway ID

Values: 1 to 8

peer-type

specifies the peer type

group-name

specifies the RADIUS group name

ref-point-name

specifies the reference point name

sww-type

specifies the SWw type

active

specifies the active peers

radius

specifies the RADIUS as the peer

dhcp

specifies the DHCP peers

group-name

specifies the DHCP group name

dia-routing-table

specifies the configuration of the Diameter routing table

cdbx

specifies the cloud database peers

doic

specifies the realms and hosts with valid Diameter Overload Indication Conveyance (DOIC) parameters

vm-id

specifies the VM ID

Values: 1 to 22

Output

The following output examples show reference point peer statistics and *Show PDN reference point peer common output fields* describes the common output fields. Descriptions of reference-point specific fields are also provided in the applicable sections.

- *Example output cloud database reference point peers*
- *Example output Ga reference point peers*
- *Example output Gn reference point peers*
- *Example output Gp reference point peers*
- *Example output Gx reference point peers*

- *Example output for CMG Gx reference point peers*
- *Example output Gy reference point peers*
- *Example output Gy Diameter routing table reference point peers*
- *Example output N2 reference point peer*
- *Example output N3 and N9 reference point peers*
- *Example output Pi reference point peers*
- *Example output RADIUS reference point peers*
- *Example output Rf reference point peers*
- *Example output S1u reference point peers*
- *Example output S2a-GTP reference point peers*
- *Example output S2a reference point peers*
- *Example output S2b reference point peers*
- *Example output S4 reference point peers*
- *Example output S5 reference point peers*
- *Example output S5u reference point peers*
- *Example output S6b reference point peers*
- *Example output S6b reference point peers Diameter overload indication conveyance*
- *Example output S11u reference point peers*
- *Example output S8 reference point peers*
- *Example output STa reference point peers*
- *Example output SWm reference point peers*
- *Example output SWm Diameter routing table reference point peers*
- *Example output SWw tunnel reference point peers*
- *Example output SWw RADIUS reference point peers*
- *Example output for Sx-N4 reference point peers on session management function*
- *Example output for Sx-N4 reference point peers on user plane function*
- *Example output for Sx-N4 reference point peers on user plane function*
- *Example output for Sx-N4 reference point peers on user plane function (MPAS feature)*

Table 64: Show PDN reference point peer common output fields

Label	Description
Peer address	The IP address of the peer
Peer type	The IP address of the peer
Router	The router name
Port	The port number
Reference point	The reference point name
Path Mgmt State	The path management state For GTP ref-point-peers (s11, s11u, s12, s1u, s4, s4u, s5, s5u, s8, s8u, gn, gnu, gp, gpu, s2a, s2a-gtp, s2b, s2bu, n3, n9) the values are described in Path Mgmt state values These values do not apply to Radius and Diameter reference point types
Create Time	Date and time of peer creation
Gateway Id	The gateway ID
Number of peers	The number of peers retrieved

Table 65: Path Mgmt state values

Peer State	Transition State	Description
disabled	Initial state	A peer remains in disabled state until the path management starts on this peer. The path management starts when a peer changes to up state. One exception is the S2b-PGW and S2a-PGW

Table 65: Path Mgmt state values (continued)

Peer State	Transition State	Description
		peers, where the path management starts immediately after a peer creation.
up	disabled -> up idle -> up	A peer changes to up state when the first session is established on this peer. For an SGW peer, this state change happens when a CS Req is received. For PGW peers, this state change happens when a CS Resp from the PGW peer is received. This is to make sure that a path up trap is generated only for real PGW peers in case of GTP-C redirection. A peer remains in up state as long as it has at least one session.
timeout/retry	up -> timeout/retry	Echo keep alive interval has timed out. The gateway is retrying Echo messages at a faster rate.
fault	timeout/retry -> fault	A peer changes to fault state when a path management failure is detected. It is a temporary state as the peer changes to idle state immediately.
idle	fault -> idle up -> idle	A peer changes to idle state when session cleanup is done due to a path management failure and also when the last call has drained off the peer due to a Detach/Admin delete. A peer does not have any sessions when it is in idle state.
restart	up -> restart idle -> restart	A peer changes to restart state when a peer restart is detected from the

Table 65: Path Mgmt state values (continued)

Peer State	Transition State	Description
		recovery value carried in the incoming messages on the CPM.

Example output cloud database reference point peers

The following example shows cloud database reference point peers statistics.

```
# show mobile-gateway pdn ref-point-peers cdbx
=====
PDN Cdbx reference point peers
=====
Peer address      : 10.100.11.24
Router            : Base                               Port            : 5678
Vm                : 3
Admin State       : Up                                Oper State       : Up
Create Time       : 11/08/2018 17:56:01                Gateway Id       : 1
cloud-db-prof     : prof1
Interface         : dbNetwork
DB Sync state     : HOT                                VM Sync state    : HOT
Last status chng: 11/08/2018 17:56:01
-----
Number of peers : 1
=====
```

Example output Ga reference point peers

The following example shows Ga reference point peer statistics and *Show PDN Ga ref point peers output fields* describes the output fields. *Show PDN reference point peer common output fields* describes additional common output fields.

```
# show mobile-gateway pdn ref-point-peers ga
=====
PDN Ga reference point peers
=====
Peer address      : 10.11.7.101
Router            : Base                               Port            : 3386
Create Time       : 06/03/2013 18:26:04                Gateway Id       : 1
GTP Prim Grp     : gaProfileComboGw_pdn                Card slot num    : 3
Up Time          : 06/03/2013 18:26:04                Oper state       : active
Interface         : system
-----
Peer address      : 10.11.7.102
Router            : Base                               Port            : 3386
Create Time       : 06/03/2013 18:26:04                Gateway Id       : 1
GTP Prim Grp     : gaProfileComboGw_pdn                Card slot num    : 3
Up Time          : 06/03/2013 18:26:04                Oper state       : up
Interface         : system
-----
```

```
Number of peers : 2
=====
```

Table 66: Show PDN Ga ref point peers output fields

Label	Description
GTP Prim Grp	The GTP prime group name
Card slot num	The card slot number
Up Time	The peer up time
Oper state	The operational state of the peer
Interface	The name of the interface

Example output Gn reference point peers

The following example shows Gn reference point peer statistics and *Show PDN S1U, S2a-GTP, S2b, Gp, and Gn ref point peers output fields* describes the output fields. *Show PDN reference point peer common output fields* describes additional common output fields.

```
# show mobile-gateway pdn ref-point-peers gn
=====
PDN gateway Gn reference point peers
=====
Peer address      : 10.11.8.10
Router            : Base                               Port          : 2123
Reference Point   : default
Path Mgmt State   : up
Create Time       : 06/03/2013 18:31:08                Gateway Id     : 1
GTP Version       : GTPv1
-----
Number of peers : 1
=====
```

Example output Gp reference point peers

The following example shows Gp reference point peer statistics and *Show PDN S1U, S2a-GTP, S2b, Gp, and Gn ref point peers output fields* describes the output fields. *Show PDN reference point peer common output fields* describes additional common output fields.

```
# show mobile-gateway pdn ref-point-peers gp
=====
PDN gateway Gp reference point peers
```

```

=====
Peer address      : 10.23.15.2
Peer type        : SGSN
Router           : vprn20003
Reference point  : default
Path Mgmt State  : up
Create Time      : 09/02/2016 17:32:23      Gateway Id    : 1
GTP Version      : GTPv2
Node features    : none
-----
Number of peers  : 1
=====

```

Example output Gx reference point peers

The following example shows Gx reference point peer statistics and *Show PDN Gx and Rf ref point peers output fields* describes the output fields. *Show PDN reference point peer common output fields* describes additional common output fields.

```

# show mobile-gateway pdn ref-point-peers gx
=====
PDN Gx reference point peers
=====
Peer address      : 10.10.8.10
Multi-homing peer address : 0.0.0.0
Multi-homing      : No
Router            : Base      Port          : 3868
Slot              : 3
Path Mgmt State   : Active    Detail State : Open
Create Time       : 07/08/2016 17:37:59      Gateway Id   : 1
Dia Peer Prof     : Peer-Profile1
Interface         : gx
Multi-homing secondary interface : -
Rate limit buffer utilization : 0%
Last rate limit low utilization time : 07/08/2016 18:05:01
Last rate limit high utilization time : 07/08/2016 18:05:06
Last rate limit full utilization time : 07/08/2016 18:05:07
-----
Number of peers  : 1
=====

```

Table 67: Show PDN Gx and Rf ref point peers output fields

Label	Description
Multi-homing peer address	The multi-homing peer IP address
Multi-homing	The local peer multi-homing status

Table 67: Show PDN Gx and Rf ref point peers output fields (continued)

Label	Description
Slot	The slot number
Detail State	The detail state
Dia Peer Prof	The Diameter peer profile name
Interface	The interface name
Multi-homing secondary interface	The secondary multi-homing interface name
Rate limit buffer utilization	The percentage of the rate limit buffer utilization
Last rate limit low utilization time	Date and time of the last rate low limit utilization
Last rate limit high utilization time	Date and time of the last rate high limit utilization
Last rate limit full utilization time	Date and time of the last rate full limit utilization

Example output for CMG Gx reference point peers

The following example shows CMG Gx reference point peers statistics and *Show PDN CMG Gx ref point peers output fields* describes the output fields. *Show PDN reference point peer common output fields* describes additional common output fields.

```
# show mobile-gateway pdn ref-point-peers gx
=====
PDN Gx reference point peers
=====
Peer address      : 10.10.8.10
Multi-homing peer address : 0.0.0.0
Multi-homing      : No
Router            : Base                      Port            : 3868
VM                : 4
Path Mgmt State   : Active                    Detail State      : Open
Create Time       : 03/01/2017 20:45:15       Gateway Id        : 1
Dia Peer Prof     : Peer-Profile1
Interface         : gx
```



```
Multi-homing secondary interface      : -
-----
Number of peers : 1
=====
```

Table 68: Show PDN CMG Gx ref point peers output fields

Label	Description
Multi-homing peer address	The multi-homing peer IP address
Multi-homing	The local peer multi-homing status
VM	The VM ID
Detail State	The detail state
Create Time	Date and time of the interface creation
Dia Peer Prof	The Diameter peer profile name
Interface	The interface name
Multi-homing secondary interface	The secondary multi-homing interface name

Example output Gy reference point peers

The following example shows Gy reference point peers statistics and *Show PDN Gy Ref point peers output fields* describes the output fields. *Show PDN reference point peer common output fields* describes additional common output fields.

```
# show mobile-gateway pdn ref-point-peers gy
=====
PDN Gy reference point peers
=====
Peer address      : 10.10.18.10
Multi-homing peer address : 0.0.0.0
Multi-homing      : No
Router            : Base                      Port            : 3868
Slot              : 3
Path Mgmt State   : Active                    Detail State      : Open
Create Time       : 07/08/2016 17:37:59       Gateway Id       : 1
```

```

Dia Peer Prof   : Gy-Peer-Prof1
Interface       : gy
Multi-homing secondary interface : -
Rate limit buffer utilization      : 0%
Last rate limit low utilization time : 07/08/2016 18:01:33
Last rate limit high utilization time : 07/08/2016 18:04:01
Last rate limit full utilization time : 07/08/2016 18:04:04
-----
Number of peers : 1
=====

```

Table 69: Show PDN Gy Ref point peers output fields

Label	Description
Multi-homing peer address	The multi-homing peer IP address
Multi-homing	The local peer multi-homing status
Slot	The slot number
Detail State	The detail state
Dia Peer Prof	The Diameter peer profile name
Interface	The interface name
Multi-homing secondary interface	The secondary multi-homing interface name
Rate limit buffer utilization	The percentage of the rate limit buffer utilization
Last rate limit low utilization time	Date and time of the last rate low limit utilization
Last rate limit high utilization time	Date and time of the last rate high limit utilization
Last rate limit full utilization time	Date and time of the last rate full limit utilization

Example output Gy Diameter routing table reference point peers

The following example shows Gy Diameter routing table reference point peers statistics and *Gy Diameter routing table reference point peers output fields* describes the output fields. *Show PDN reference point peer common output fields* describes additional common output fields.

```
# show mobile-gateway pdn ref-point-peers gy dia-routing-table
=====
PDN Gy diameter routing table
=====
host-index : 1
supported-host : ipd.nokia.ocs.fqdn.1
port : 3868
transport-type : tcp
realm : nokia.1.com
peer-group-list : gyPeerList10
host-index : 2
supported-host : ipd.nokia.ocs.fqdn.1
port : 3870
transport-type : tcp
realm : nokia.2.com
peer-group-list : gyPeerList10
host-index : 3
supported-host : ipd.nokia.ocs.fqdn.1
port : 3868
transport-type : sctp
realm : nokia.3.com
peer-group-list : gyPeerList10
host-index : 4
supported-host : ipd.nokia.ocs.fqdn.4
port : 3868
transport-type : tcp
realm : nokia.4.com
peer-group-list : gyPeerList10
host-index : 5
supported-host : ipd.nokia.ocs.fqdn.5
port : 3868
transport-type : tcp
realm : nokia.1.com
peer-group-list : gyPeerList10
host-index : 6
supported-host : ipd.nokia.ocs.fqdn.6
port : 3868
transport-type : tcp
realm : nokia.1.com
peer-group-list : gyPeerList2
host-index : 7
supported-host : ipd.nokia.ocs.fqdn.7
port : 3868
transport-type : tcp
realm : nokia.7.com
peer-group-list : gyPeerList11
host-index : 8
supported-host : ipd.nokia.ocs.fqdn.8
port : 3868
transport-type : tcp
realm : nokia.8.com
peer-group-list : gyPeerList11
```

```

host-index : 9
supported-host : ipd.nokia.ocs.fqdn.8
port : 3870
transport-type : tcp
realm : nokia.8.com
peer-group-list : gyPeerList2
Total Count : 9
=====

```

Table 70: Gy Diameter routing table reference point peers output fields

Label	Description
Host Index	Host index entry number
Supported-host	Supported host FQDN
Port	The configured port
Transport-type	The configured transport type for Diameter signaling; TCP or SCTP
Realm	The configured destination realm
Peer-group-list	The configured Diameter peer group list
Total Count	The total number of entries in the table

Example output N2 reference point peer

The following example shows N2 reference point peer statistics and *Show N2 reference point peer output fields* describes the common output fields.

```
# show mobile-gateway pdn ref-point-peers n2
```

```
=====
PDN Gateway N2 Reference Point Peers
=====
```

```

Peer IPv4 Address      : 10.10.7.101
Local IPv4 Address     : 17.70.16.18
Router                 : vprn20071
Reference Point        : default
Gateway ID             : 1
AMF Name               : N/A

```

Create Time : 02/22/2024 11:55:43

N3IWF Peer Associations

Global N3IWF ID : MCC 310 MNC 12 N3IWF ID 256
 SCTP Association State : established
 NGAP State : setup
 Local Port : 65416
 Group : 1
 Instance : 0
 UE Records : 0

Global N3IWF ID : MCC 310 MNC 12 N3IWF ID 257
 SCTP Association State : established
 NGAP State : setup
 Local Port : 65417
 Group : 1
 Instance : 1
 UE Records : 0

Global N3IWF ID : MCC 310 MNC 12 N3IWF ID 258
 SCTP Association State : established
 NGAP State : setup
 Local Port : 65418
 Group : 1
 Instance : 2
 UE Records : 0

Global N3IWF ID : MCC 310 MNC 12 N3IWF ID 259
 SCTP Association State : established
 NGAP State : setup
 Local Port : 65419
 Group : 1
 Instance : 3
 UE Records : 0

Table 71: Show N2 reference point peer output fields

Label	Description
Peer IPv4 address	The IPv4 address of the peer
Local IPv4 address	The local IPv4 address of the N2 interface of the N3IWF
Peer IPv6 address	The IPv6 address of the peer
Local IPv6 address	The local IPv6 address of the N2 interface of the N3IWF
Router	The local router instance used for the peer communication
Reference Point	The reference point used
Gateway ID	The gateway ID
AMF Name	The name of the AMF as this is received by the AMF
Global N3IWF ID	The global name of the N3IWF as this is provided by N3IWF to AMF
SCTP Association State	The status of the SCTP level association connection Possible values: • established • closed
NGAP State	The status of the SCTP level association connection Possible values: • setup • setup-in-progress • closed
Local Port	The local port used for the communication with AMF

Table 71: Show N2 reference point peer output fields (continued)

Label	Description
Group	The MSCP Group ID
Instance	Identity of N3IWF signaling instance
UE Records	Number of registered UEs per signaling instance
N3IWF ID	The N3IWF ID created based on the "Group Id", "Instance Id", and "n3iwf-unique-id"

Example output N3 and N9 reference point peers

The following example shows N3 and N9 reference point peers statistics and the *Show PDN reference point peer common output fields* describes the common output fields.

```
# show mobile-gateway pdn ref-point-peers n3
=====
PDN gateway n3 reference point peers
=====
Peer address      : 192.168.0.1
Peer type         : gNodeB
Router            : vprn20071
Reference point   : default
Path Mgmt State   : up
Create Time       : 09/26/2019 14:43:26      Gateway Id    : 1
GTP Version       : GTPv1
-----
Number of peers : 1
=====

# show mobile-gateway pdn ref-point-peers n9
=====
PDN gateway n9 reference point peers
=====
Peer address      : 2001:db8::211:ff:11
Peer type         : -
Router            : vprn20071
Reference point   : default
Path Mgmt State   : up
Create Time       : 09/26/2019 14:43:42      Gateway Id    : 1
GTP Version       : GTPv1
-----
Number of peers : 1
=====
```

Example output Pi reference point peers

The following example shows Pi reference point peers statistics and *Show PDN Pi ref point peers output fields* describes the output fields. *Show PDN reference point peer common output fields* describes additional common output fields.

```
# show mobile-gateway pdn ref-point-peer pi ref-point <ref-point-name>
=====
PDN gateway pi reference point peers
=====
Peer address :
Router : router-instance Port :

pi interface : sig-ref-point-name

Create Time : date / time                Gateway Id : number

mipv4 profile : profile-name
fa-ha-peer-list: list-name

-----
Number of peers : 1
=====
```

Table 72: Show PDN Pi ref point peers output fields

Label	Description
pi interface	The Pi interface name
mipv4 profile	The MIPv4 profile name
fa-ha-peer-list	The FA peer list name

Example output RADIUS reference point peers

The following example shows RADIUS reference point peers statistics and *Show PDN RADIUS ref point peers output fields* describes the output fields. *Show PDN reference point peer common output fields* describes additional common output fields.

```
# show mobile-gateway pdn ref-point-peers radius
=====
```


PDN Radius peers

```

=====
Group Name      : Radius Group 1
Peer address    : 10.11.12.10
Interface       : radius1                Router      : Base
Radius Profile  : bb                     Priority     : 1
Auth Serv Port  : 1812                   Acct Serv Port: 1813
Operation State : Up                     Admin State  : up
Op State Since  : 04/02/2020 11:55:30
-----
Group Name      : Radius Group 1
Peer address    : 10.11.13.10
Interface       : radius1                Router      : Base
Radius Profile  : cc                     Priority     : 2
Auth Serv Port  : 1812                   Acct Serv Port: 1813
Operation State : Up                     Admin State  : up
Op State Since  : 04/02/2020 11:55:30
-----
Group Name      : Radius Group 1
Peer address    : 10.11.14.10
Interface       : radius1                Router      : Base
Radius Profile  : dd                     Priority     : 3
Auth Serv Port  : 1812                   Acct Serv Port: 1813
Operation State : Down                   Admin State  : up
Op State Since  : 04/02/2020 12:06:19
-----
Number of peers : 3
=====

```

Table 73: Show PDN RADIUS ref point peers output fields

Label	Description
Group Name	The RADIUS group name
Interface	The interface name
Radius Profile	The RADIUS profile name
Priority	The RADIUS server priority
Auth Serv Port	The destination UDP port for the RADIUS authentication server
Acct Serv Port	The destination UDP port for the RADIUS accounting server
Operation State	The RADIUS interface operational state

Table 73: Show PDN RADIUS ref point peers output fields (continued)

Label	Description
Admin State	The RADIUS administrative state
Op State Since	Date and time of the current operational state

Example output Rf reference point peers

The following example shows Rf reference point peer statistics and *Show PDN Gx and Rf ref point peers output fields* describes the output fields. *Show PDN reference point peer common output fields* describes additional common output fields.

```
# show mobile-gateway pdn ref-point-peers rf
=====
PDN gateway RF reference point peers
=====
Peer address      : 10.11.7.10
Multi-homing peer addresses:
Router           : Base                      Port           : 3868
Slot            : 3
Path Mgmt State  : Active                    Detail State    : Open
Create Time     : 06/03/2013 18:26:06        Gateway Id     : 1
Dia Peer Prof   : peerProfileComboGwservergrf1
Interface       : system
Multi-homing secondary interface:
-----
Number of peers : 1
=====
```

Example output S1u reference point peers

The following example shows S1u reference point peer statistics and *Show PDN S1U, S2a-GTP, S2b, Gp, and Gn ref point peers output fields* describes the output fields. *Show PDN reference point peer common output fields* describes additional common output fields.

```
# show mobile-gateway pdn ref-point-peers s1u
=====
PDN gateway s1u reference point peers
=====
Peer address      : 6.6.6.1
Peer type         : eNodeB
Router           : vprn20071
Reference point   : default
Path Mgmt State  : up
Create Time     : 10/07/2019 08:48:22        Gateway Id     : 1
GTP Version      : GTPv1
PMTU             : 0
-----
```

```
Number of peers : 1
=====
```

Table 74: Show PDN S1U, S2a-GTP, S2b, Gp, and Gn ref point peers output fields

Label	Description
GTP Version	The peer GTP version
Node features	—
PMTU	—

Example output S2a-GTP reference point peers

The following example shows S2a-gtp reference point peer statistics and *Show PDN S1U, S2a-GTP, S2b, Gp, and Gn ref point peers output fields* describes the output fields. *Show PDN reference point peer common output fields* describes additional common output fields.

```
# show mobile-gateway pdn ref-point-peers s2a-gtp
=====
PDN gateway s2a-gtp reference point peers
=====
Peer address      : 10.23.23.2
Peer type         : TWAG
Router            : vprn20003
Reference point   : default
Path Mgmt State   : up
Create Time       : 09/02/2016 17:32:23      Gateway Id    : 1
GTP Version       : GTPv2
Node features     : none
-----
Number of peers : 1
=====
```

Example output S2a reference point peers

The following example shows S2a reference point peers statistics and *Show PDN S2a ref point peers output fields* describes the output fields. *Show PDN reference point peer common output fields* describes additional common output fields.

```
# show mobile-gateway pdn ref-point-peers s2a
=====
PDN gateway S2a reference point peers
=====
Peer address      : 3FFE::A10:80A
Router            : Base                      Port           : 0
Path Mgmt State   : up
```

```

Create Time      : 06/03/2013 18:33:59          Gateway Id   : 1
PMIP profile     : default
PMIP interface   : system
Heartbeat Compatible : YES
-----

```

```

Number of peers : 1
=====

```

Table 75: Show PDN S2a ref point peers output fields

Label	Description
PMIP profile	The PMIP profile name
PMIP interface	The PMIP interface name
Heartbeat Compatible	The heartbeat compatibility

Example output S2b reference point peers

The following example shows S2b reference point peer statistics and *Show PDN S2a ref point peers output fields* describes the output fields. *Show PDN reference point peer common output fields* describes additional common output fields.

```

# show mobile-gateway pdn ref-point-peers s2b
=====
PDN gateway S2b reference point peers
=====
Peer address      : 10.70.24.151
Peer type         : ePDG
Router           : vprn20071
Reference point   : default
Path Mgmt State   : up
Create Time       : 09/02/2016 17:32:23          Gateway Id   : 1
GTP Version       : GTPv2
Node features     : none
-----
Number of peers   : 1
=====

```

Example output S4 reference point peers

The following example shows S4 reference point peer statistics and *Show PDN S4 ref point peers output fields* describes the output fields. *Show PDN reference point peer common output fields* describes additional common output fields.

```
# show mobile-gateway pdn ref-point-peers s4
=====
PDN gateway s4 reference point peers
=====
Peer address      : 10.10.7.10
Peer type         : SGSN
Router            : vprn20071
Reference point   : default
Path Mgmt State   : idle
Create Time       : 02/04/2020 10:55:14          Gateway Id    : 1
GTP Version       : GTPv2
Node features     : NTSR
SGSN pool-id      : 0x000000041300
-----
Peer address      : 10.10.7.10
Peer type         : SGSN
Router            : vprn101
Reference point   : redir_s4_Grp1
Path Mgmt State   : up
Create Time       : 02/04/2020 10:55:14          Gateway Id    : 1
GTP Version       : GTPv2
Node features     : NTSR
SGSN pool-id      : 0x000000041300
-----
Number of peers : 2
=====
```

Table 76: Show PDN S4 ref point peers output fields

Label	Description
GTP Version	The peer GTP version
Node features	The features supported by MME/S4-SGSN
SGSN pool-id	The ID of the SGSN pool

Example output S5 reference point peers

The following example shows S5 reference point peer statistics and *Show PDN S5 and S8 ref point peers output fields* describes the output fields. *Show PDN reference point peer common output fields* describes additional common output fields.

```
# show mobile-gateway pdn ref-point-peers s5
```

```

=====
PDN gateway S5 reference point peers
=====
Peer address      : 10.70.1.1
Router           : vprn2                               Port           : 2123
Reference point   : default
Path Mgmt State   : up
Create Time       : 06/03/2013 18:27:44                 Gateway Id      : 1
Enhanced SGW restoration: Yes/No                       restoration-pdn-timer:
GTP Version       : GTPv2
Node features     : PRN MABR NTSR
-----
Number of peers : 1
=====

```

Table 77: Show PDN S5 and S8 ref point peers output fields

Label	Description
Enhanced SGW restoration	The state of the enhanced SGW restoration
restoration-pdn-timer	The time in seconds that the PGW will maintain the PDN connections and be ready to restore them by having the MME select either a new SGW or the same SGW
GTP Version	The peer GTP version
Node features	—

Example output S5u reference point peers

The following example shows S5u reference point peer statistics and *Show PDN S5u ref point peers output fields* describes the output fields. *Show PDN reference point peer common output fields* describes additional common output fields.

```

# show mobile-gateway pdn ref-point-peers s5u
=====
PDN gateway s5u reference point peers
=====
Peer address      : 10.10.5.1
Peer type         : PGW
Router            : vprn20003
Reference point    : default
Path Mgmt State   : up
Create Time       : 10/07/2019 08:58:13                 Gateway Id      : 1
GTP Version       : GTPv1
-----

```

```
Number of peers : 1
=====
```

Table 78: Show PDN S5u ref point peers output fields

Label	Description
GTP Version	The peer GTP version

Example output S6b reference point peers

The following examples show S6b reference point peers statistics and *Show PDN S6b ref point peers output fields* describes the output fields. *Show PDN reference point peer common output fields* describes additional common output fields.

```
# show mobile-gateway pdn ref-point-peers s6b
=====
PDN S6b reference point peers
=====
Peer address      : 10.10.19.10
Router           : Base                               Port           : 3868
Slot            : 3
Path Mgmt State  : Active                             Detail State    : Open
Create Time     : 01/18/2017 16:45:55                 Gateway Id     : 1
Dia Peer Prof   : peerProfileComboGwpdns6b1
Rate limit buffer utilization      : 34%
Last rate limit low utilization time : 01/18/2017 16:45:55
Last rate limit high utilization time : 01/18/2017 16:45:55
Last rate limit full utilization time : 01/18/2017 16:45:56
-----
Number of peers : 1
=====
```

Table 79: Show PDN S6b ref point peers output fields

Label	Description
Multi-homing peer address	The multi-homing peer IP address
Multi-homing	The local peer multi-homing status
Slot	The slot number

Table 79: Show PDN S6b ref point peers output fields (continued)

Label	Description
Detail State	The detail state
Dia Peer Prof	The Diameter peer profile name
Interface	The interface name
Multi-homing secondary interface	The secondary multi-homing interface name
Rate limit buffer utilization	The percentage of the rate limit buffer utilization
Last rate limit low utilization time	Date and time of the last rate low limit utilization
Last rate limit high utilization time	Date and time of the last rate high limit utilization
Last rate limit full utilization time	Date and time of the last rate full limit utilization

Example output S6b reference point peers Diameter overload indication conveyance

The following examples show S6b reference point peers statistics for Diameter Overload Indication Conveyance.

```
# show mobile-gateway pdn ref-point-peers s6b peer 10.23.19.14 doic
=====
PDN s6b reference point peers - Diameter Overload Indication Conveyance
=====
Peer address      : 10.23.19.14
Slot              : 3
Realm based report
Realm             : nokia.com
Traffic reduction value: 50          Remaining validity time: 463
Sequence-Number   : 1
Number of entries : 1
=====
```

Example output S11u reference point peers

The following examples show S11u reference point peers statistics and *Show PDN S11u ref point peers output fields* describes the output fields.

```
# show mobile-gateway pdn ref-point-peers s11u
=====
PDN gateway s11u reference point peers
=====
Peer address : 6.6.6.1
Peer type : MME
Router : vprn20032
Reference point : default
Path Mgmt State : disabled
Create Time : 06/27/2017 07:44:37 Gateway Id : 2
GTP Version : GTPv1
-----
Number of peers : 1
=====
```

Table 80: Show PDN S11u ref point peers output fields

Label	Description
Peer type	Specifies MME as peer type
Path Mgmt State	Always indicates that path management is disabled

Example output S8 reference point peers

The following example shows S8 reference point peer statistics and *Show PDN S5 and S8 ref point peers output fields* describes the output fields. *Show PDN reference point peer common output fields* describes additional common output fields.

```
# show mobile-gateway pdn ref-point-peers s8
=====
PDN gateway S8 reference point peers
=====
Peer address : 10.70.2.1
Router : Base Port : 2123
Reference point : default
Path Mgmt State : up
Create Time : 06/03/2013 18:27:44 Gateway Id : 1
Enhanced SGW restoration: Yes restoration-pdn-timer: 300
-----
Number of peers : 1
=====
```

Example output STa reference point peers

The following example shows STa reference point peers statistics and *Show PDN STa ref point peers output fields* describes the output fields. *Show PDN reference point peer common output fields* describes additional common output fields.

```
# show mobile-gateway pdn ref-point-peers sta
=====
PDN Sta reference point peers
=====
Peer address      : 180.1.2.200
Multi-homing peer address : 0.0.0.0
Multi-homing      : No
Router            : vprn20005          Port            : 3868
Slot              : 3
Path Mgmt State   : Active              Detail State    : Open
Create Time       : 01/18/2017 16:45:55 Gateway Id      : 1
Dia Peer Prof     : Sta_Peer-Profile1
Interface         : STa-interface
Rate limit buffer utilization : 34%
Last rate limit low utilization time : 01/18/2017 16:45:55
Last rate limit high utilization time : 01/18/2017 16:45:55
Last rate limit full utilization time : 01/18/2017 16:45:56
-----
Number of peers : 1
=====
```

Table 81: Show PDN STa ref point peers output fields

Label	Description
Multi-homing peer address	The multi-homing peer IP address
Multi-homing	The local peer multi-homing status
Slot	The slot number
Detail State	The detail state
Dia Peer Prof	The Diameter peer profile name
Interface	The interface name
Multi-homing secondary interface	The secondary multi-homing interface name

Table 81: Show PDN STa ref point peers output fields (continued)

Label	Description
Rate limit buffer utilization	The percentage of the rate limit buffer utilization
Last rate limit low utilization time	Date and time of the last rate low limit utilization
Last rate limit high utilization time	Date and time of the last rate high limit utilization
Last rate limit full utilization time	Date and time of the last rate full limit utilization

Example output SWm reference point peers

The following example shows SWm reference point peers statistics and *Show PDN SWm and SWm Diameter routing point peers output fields* describes the output fields. *Show PDN reference point peer common output fields* describes additional common output fields.

```
# show mobile-gateway pdn ref-point-peers swm
=====
PDN Swm reference point peers
=====
Peer address      : 10.10.80.10
Multi-homing peer address : 10.10.80.20
Multi-homing      : Yes
Router            : Base                      Port            : 3868
Slot              : 3
Path Mgmt State   : Active                    Detail State      : Open
Create Time       : 01/18/2017 16:45:56       Gateway Id       : 1
Dia Peer Prof     : swm_Peer-Profile1
Rate limit buffer utilization : 34%
Last rate limit low utilization time : 01/18/2017 16:45:55
Last rate limit high utilization time : 01/18/2017 16:45:55
Last rate limit full utilization time : 01/18/2017 16:45:56
-----
Number of peers : 1
=====
```

Table 82: Show PDN SWm and SWm Diameter routing point peers output fields

Label	Description
Multi-homing peer address	The multi-homing peer IP address

Table 82: Show PDN SWm and SWm Diameter routing point peers output fields (continued)

Label	Description
Multi-homing	The local peer multi-homing status
Slot	The slot number
Detail State	The detail state
Dia Peer Prof	The Diameter peer profile name
Interface	The interface name
Multi-homing secondary interface	The secondary multi-homing interface name
Rate limit buffer utilization	The percentage of the rate limit buffer utilization
Last rate limit low utilization time	Date and time of the last rate low limit utilization
Last rate limit high utilization time	Date and time of the last rate high limit utilization
Last rate limit full utilization time	Date and time of the last rate full limit utilization

Example output SWm Diameter routing table reference point peers

The following example shows SWm Diameter routing table reference point peers statistics and *Show PDN SWm Diameter routing point peers output fields* describes the output fields. *Show PDN reference point peer common output fields* describes additional common output fields.

```
# show mobile-gateway pdn ref-point-peers swm dia-routing-table
=====
PDN SWm Diameter routing table
=====
Entry 1
IDi Realm : nai.epc.mnc012.mcc310.3gppnetwork.org
Peer Group List : swmList1
Entry 2
IDi Realm : nai.epc.mnc013.mcc310.3gppnetwork.org
Peer Group List : swmList2
```

```

Entry 3
IDi Realm : nai.epc.mnc014.mcc310.3gppnetwork.org
Peer Group List : swmList3
Entry 4
IDi Realm : nai.epc.mnc015.mcc310.3gppnetwork.org
Peer Group List : swmList4
Entry 5
IDi Realm : nai.epc.mnc016.mcc310.3gppnetwork.org
Peer Group List : swmList5
Entry 6
IDi Realm : nai.epc.mnc017.mcc310.3gppnetwork.org
Peer Group List : swmList6
Entry 7
IDi Realm : nai.epc.mnc018.mcc310.3gppnetwork.org
Peer Group List : swmList7
Entry 8
IDi Realm : nai.epc.mnc019.mcc310.3gppnetwork.org
Peer Group List : swmList8
Total Count : 8
=====

```

Table 83: Show PDN SWm Diameter routing point peers output fields

Label	Description
Entry	The entry number
IDi Realm	The IDi realm
Peer Group List	The peer group list name
Total Count	The total number of entries

Example output SWw tunnel reference point peers

The following example shows SWw tunnel reference point peers statistics and *Show PDN SWw tunnel ref point peers output fields* describes the output fields. *Show PDN reference point peer common output fields* describes additional common output fields.

```

# show mobile-gateway pdn ref-point-peers sww tunnel
=====
PDN gateway SWw tunnel reference point peers
=====
Peer Address      : 201.3.1.10
Peer MAC Address  : 00:00:00:00:00:00
Peer Type         : 0
Router           : vprn3003
Reference Point   : swwrefpt1

```

```

Operational State: 0
Create Time       : 11/09/2015 16:48:19 Gateway Id       : 1
Tunnel Type      : Soft Gre Tunnel
-----

```

```

Number of peers : 1
=====

```

Table 84: Show PDN SWw tunnel ref point peers output fields

Label	Description
Peer MAC Address	The peer MAC address
Reference Point	The reference point name
Operation State	The interface operational state
Create Time	Date and time of the interface creation
Tunnel Type	The connection method used on the SWw interface

Example output SWw RADIUS reference point peers

The following example shows SWw RADIUS reference point peers statistics and *Show PDN SWw RADIUS ref point peers output fields* describes the output fields. *Show PDN reference point peer common output fields* describes additional common output fields.

```

# show mobile-gateway pdn ref-point-peers sww radius
=====
PDN gateway SWw radius reference point peers
=====
Peer Address      : 201.3.2.10
Interface         : listening_radius_se* Router           : vprn3003
sww-radius-profi* : Administrative S*: Up
Auth-server-port  : 1812 Gateway Id                     : 1
Acct-server-port  : 0 radius-disconnect: enabled
Operational State: Up Create Time                       : 09/14/2016 16:19:01
-----
Number of peers : 1
=====
* indicates that the corresponding row element may have been truncated.

```

Table 85: Show PDN SWw RADIUS ref point peers output fields

Label	Description
Interface	The interface name
sww-radius-profi	The SWw RADIUS profile name
Auth-server-port	The destination UDP port for the RADIUS authentication server
Acct-server-port	The destination UDP port for the RADIUS accounting server
radius-disconnect	The status of the RADIUS-initiated disconnect
Operation State	The SWw RADIUS operational state
Create Time	Date and time of the SWw RADIUS interface creation

Example output for Sx-N4 reference point peers on session management function

```
# show mobile-gateway pdn ref-point-peers sx-n4
=====
PFCP reference point peers
=====
Peer address      : 192.168.34.1
Router            : vprn20091
Path Mgmt State   : up
Create Time       : 05/21/2019 17:51:50 Gateway Id       : 1
UP Features       : FTUP HEEU EMPU PDIU DLBD PDIU QUOAC FRRT DPDRA ADPDP
UP Association     : up                               Last Change Time : 05/21/2019 17:52:20
Enforced PFCP association list: Yes / No
-----
Number of peers : 1
=====
```

Example output for Sx-N4 reference point peers on user plane function

```
show mobile-gateway pdn ref-point-peers sx-n4
=====
PFCP reference point peers
=====
Peer address      : 10.10.10.10
Router            : Base
Path Mgmt State   : path-hold-InProgress
Restoration
Timer Expiry      : 08/21/2023 16:51:14
```

```
Create Time      : 02/14/2023 15:16:40 Gateway Id      : 1
UP Features     : FTUP PDIU MNOP IP6PL
UP BBF Features : PPPOE IPOE LCPKO
UP Association   : up                               Last Change Time :
UP Selection    : False
Enforced PFCP association list : No
```

```
Peer address    : 10.10.20.10
Router          : Base
Path Mgmt State : path_down
Create Time     : 02/14/2023 15:16:40 Gateway Id      : 1
UP Features     : FTUP PDIU MNOP IP6PL
UP BBF Features : PPPOE IPOE LCPKO
UP Association   : False                           Last Change Time :
UP Selection    : False
Enforced PFCP association list : No
```

```
-----
Number of peers : 2
=====
```

Example output for Sx-N4 reference point peers on user plane function

```
# show mobile-gateway pdn ref-point-peers sx-n4
=====
PFCP reference point peers
=====
Peer address    : 192.168.33.1
Router          : vprn20091
Path Mgmt State : up
Create Time     : 05/21/2019 17:51:50 Gateway Id      : 1
CP Association   : up                               Last Change Time : 05/21/2019 17:52:20
-----
Number of peers : 1
=====
```

Example output for Sx-N4 reference point peers on user plane function (MPAS feature)

```
# show mobile-gateway pdn ref-point-peers sx-n4
=====
PFCP reference point peers
=====
Peer address    : 192.168.33.1
Router          : vprn20091
Path Mgmt State : up
Create Time     : 05/21/2023 17:51:50 Gateway Id      : 1
UP Features     : ... MPAS
UP Association   : up                               Last Change Time : 05/21/2023 17:52:20
...
SMF Set ID      : 10.10.43.2
-----
Number of peers : 1
=====
```


5.2.2.55 ref-point-stats

Syntax

```

ref-point-stats peer ip-address [port port-number] [gateway gw-id | group group-number |
card slot-number] [instance | aggregate] [peer-type {sgw | pgw | epdg}]

ref-point-stats {s11 | s11u | s12 | s1u | s4 | s4u | s5 | s5u | s8 | s8u | gx | rf | gn | gnu | ga | gp |
gpu | gy | s2a | s2a-gtp | s2b | s2bu | s6b | swm | sww | sta | sd | sx-n4 | n3 | n9} [peer ip-address
[port port-number]] [gateway gw-id | group group-number | card slot-number] [instance |
aggregate] {failure-codes}

ref-point-stats {gn | gnu | gp | gpu} [ref-point ref-point-name] [peer ip-address [port port-
number]] [gateway gw-id | group group-number | card slot-number] [instance | aggregate]
[active]

ref-point-stats {s5 | s5u | s8 | s8u | gn | gnu | gp | gpu | n9} [ref-point ref-point-name]
[peer ip-address [port port-number]] [gateway gw-id | group group-number | card slot-number]
[instance | aggregate]

ref-point-stats {dhcp} [dhcp-group dhcp-group [peer ip-address]] [gateway gw-id | group
group-number | card slot-number] [instance | aggregate]

ref-point-stats {pi} [pi-peer ipv4 [port port-number]] [gateway gw-id | group group-number
| card slot-number] [instance | aggregate]

ref-point-stats {radius} [radius-group radius-group [peer ip-address]] [gateway gw-id |
group group-number | card slot-number] [instance | aggregate]

ref-point-stats {swu} [throttling | plmn-black-white-list] [card slot-number]
[aggregate] [failure-codes]

ref-point-stats {sww} sww-type [peer ip-address] [gateway gw-id | group group-number | card
slot-number] [instance | aggregate] [ref-point ref-point-name]

ref-point-stats {cdbx} [peer ip-address] [vm vm-id]

ref-point-stats {sx-n4} [ref-point ref-point-name] [peer ip-address] [gateway gw-id | group
group-number | card slot-number] [instance | aggregate]

ref-point-stats {t6c} [ref-point ref-point-name] [peer ip-address [port port-number]]
[peer-type scef] [gateway gw-id | group group-number | card slot-number] [instance |
aggregate] [failure-codes]

```

Context

```
show>mobile>pdn
```

Description

This command displays reference point statistics. For more information about reference point type definitions, refer to section *3GPP Reference Points* in the *7750 SR MG and CMG Configuration Guide*.

Parameters

s11 | s11u | s12 | s1u | s4 | s4u | s5 | s5u | s8 | s8u | gx | rf | gn | gnu | ga lgp | gpu | gy | s2a | s2a-gtp | s2b | s2bu | s6b | swm | sww | sta | sd | sx-n4 | n3 | n9

specifies the reference point type



NOTE:

In the show outputs,

- the *Number of Bearers* of the gnu/gpu/s1u/s8u/s12 reference point types, is shown only, where the peer filter is used. For the s4u/s5u/s11u reference point types, the *Number of Bearers* is shown even without the peer filter
- the eMPS MP-Flag-priority req and eMPS MP-Flag-priority resp fields count both the MPS and the WPS detected calls; currently, for ePDG these fields count only the WPS calls

ip-address / fqdn

specifies an existing peer IP address/FQDN

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x: [0..FFFF]H
–	–	d: [0..255]D
–	fqdn	[255 characters maximum]

port

specifies a port ID

Values: 1 to 65535

slot-number

specifies the card slot number

Values: 1 to 10 | A | B

gw-id

specifies the gateway ID

Values: 1 to 8

group-number

specifies the group number

Values: 1 to 8

instance

keyword to show statistics per-peer per-card

aggregate

shows aggregated statistics globally

failure-codes

keyword to show failure-code statistics

active

specifies the active peers

radius

specifies the RADIUS reference point

dhcp

specifies the DHCP reference point

radius-group

specifies the RADIUS group

dhcp-group

specifies the DHCP group

peer-type

specifies the peer type

sww-type

specifies the SWw type

vm-id

specifies the VM ID

Values: 1 to 22

cdbx

specifies the cloud database reference point

pi

keyword to specify PI reference point

ipv4

specifies the PI peer IP address

Values: ipv4-address — a.b.c.d

swu

specifies the SWu reference point

throttling

shows the relevant statistics related to IKE overload control state

plmn-black-white-list

specifies the statistics of the PLMN denylist/allowlist

t6c

specifies the reference point between PGW (PCEF) and SCEF

scef

specifies the peer type, *scef* is the only applicable peer type for t6c

Output

The following output examples show reference point peer information.

- *Example output cloud database reference point statistics*
- *Example output Ga reference point statistics*
- *Example output Ga reference point instance statistics*
- *Example output Ga reference point aggregate statistics*
- *Example output Ga reference point card statistics*
- *Example output Gn reference point statistics*
- *Example output Gnu reference point peers*
- *Example output Gnu reference point statistics*
- *Example output Gpu reference point peers*
- *Example output Gpu reference point statistics*
- *Example output Gx reference point statistics*
- *Example output Gx reference point instance statistics*
- *Example output Gx reference point card statistics*
- *Example output Gx reference point failure code statistics*
- *Example output Gx reference point peer statistics*
- *Example output Gx reference point eMPS advance priority calls*
- *Example output Gy reference point statistics*

- *Example output Gy reference point instance statistics*
- *Example output Gy reference point failure code statistics*
- *Example output N3 and N9 reference point statistics*
- *Example output PI reference point statistics*
- *Example output RADIUS reference point statistics*
- *Example output RADIUS reference point aggregate statistics*
- *Example output RADIUS reference point instance statistics*
- *Example output RF reference point instance statistics*
- *Example output RF reference point aggregate statistics*
- *Example output RF reference point card statistics*
- *Example output S1u reference point peers*
- *Example output S1u reference point statistics*
- *Example output S2a reference point statistics*
- *Example output S2a reference point statistics*
- *Example output S2a reference point statistics*
- *Example output S2a reference point statistics*
- *Example output S2a reference point card statistics*
- *Example output S2a-GTP reference point statistics*
- *Example output S2a-GTP reference point failure code statistics*
- *Example output S2b reference point statistics ePDG/PGW combination gateways*
- *Example output S2b reference point failure code Statistics ePDG/PGW combination gateways*
- *Example output S2bu reference point peers*
- *Example output S2bu peer reference point statistics*
- *Example output S4 reference point statistics*
- *Example output S4u reference point seers*
- *Example output S4u reference point statistics*
- *Example output S4u Peer reference point statistics*
- *Example output S5 reference point card statistics*
- *Example output S5 reference point failure code statistics*
- *Example output S5 reference point peers*
- *Example output S5 reference point aggregate statistics*

- *Example output S5 reference point P-CSCF recovery statistics*
- *Example output S5 reference point eMPS advance priority calls*
- *Example output S5u reference point peers*
- *Example output S5u reference point statistics*
- *Example output S5u peer reference point statistics*
- *Example output S6b reference point statistics*
- *Example output S6b reference point summary statistics*
- *Example output S6b reference point card statistics*
- *Example output S6b reference point failure code statistics*
- *Example output S6b reference point statistics per-peer instance*
- *Example output S6b reference point statistics per-peer Diameter overload indication conveyance*
- *Example output S8 reference point statistics*
- *Example output S8 peer reference point statistics*
- *Example output S8u reference point peers*
- *Example output S8u reference point statistics*
- *Example output S11 reference point P-CSF recovery statistics*
- *Example output S11 reference point eMPS advance priority calls*
- *Example output S11u reference point statistics*
- *Example output S12 reference point peers*
- *Example output S12 reference point statistics*
- *Example output SWm reference point statistics*
- *Example output SWm failure codes statistics*
- *Example output SWu reference point statistics*
- *Example output SWu reference point failure codes statistics*
- *Example output SWu reference point Statistics ePDG/PGW combination gateways*
- *Example output SWu reference point failure codes statistics ePDG/PGW combination gateways*
- *Example output SWu reference point statistics for PLMN-Black-White-List*
- *Example output SWu reference point statistics throttling state*
- *Example output SWw and SWw tunnel/RADIUS reference point statistics*
- *Example output STa reference point statistics*
- *Example output STa reference point failure code statistics*

- *Example output Sx-N4 reference point statistics on CP function/session management function*
- *Example output Sx-N4 reference point statistics on user plane function*
- *Example output Sx-N4 reference point aggregate statistics*
- *Example output Sx-N4 reference point failure code statistics*
- *Example output T6c reference point statistics*
- *Example output T6c reference point statistics per-peer instance*
- *Example output T6c reference point statistics aggregate*
- *Example output T6c failure code statistics*
- *Example output T6c failure code statistics per-peer instance*
- *Example output T6c failure code statistics aggregate*
- *Example output T6c reference point peers*

Example output cloud database reference point statistics

```
# show mobile-gateway pdn ref-point-stats cdbx
=====
PDN gateway Cdbx reference point statistics
=====
Peer address : 192.168.130.24
Router       : Base
Port        : 5678

WRITE requests      : 99846
QUERY requests     : 0
QUERY responses     : 0
DELETE requests     : 0
NUM UEs in flush queue : 617
NUM UEs synched to DB : 99836
-----
Number of peer instances : 1
=====
```

Example output Ga reference point statistics

```
# show mobile-gateway pdn ref-point-stats ga
=====
PDN gateway Ga reference point statistics
=====
Peer address : 10.11.7.101
Router       : vprn20004
Port        : 3386

DRT request Tx      : 2          DRT request retried      : 1
DRT response Rx     : 1          DRT request unacked     : 0
Redirection request Rx : 0        Redirection response Tx  : 0
Invalid messages Rx  : 0          Version not supported Rx  : 0
DRT request (dup) Tx : 0          GTP-prime failures       : 1
CDRs Tx             : 1          Version not supported Tx  : 0
```

```

DRT request (empty) Tx      : 1      DRT request (cancel) Tx      : 0
DRT request (release) Tx    : 0      DRT response (dup) Rx       : 0
DRT response (empty) Rx     : 1      DRT response (cancel) Rx    : 0
DRT response (release) Rx   : 0
Triggers Tx
Termination                 : 1      Max change condition       : 0
MS Time Zone Change         : 0      RAT Change                 : 0
Time Limit                  : 0      Volume Limit               : 0
PLMN change                 : 0      Serving Node Change Limit  : 0
Management Intervention     : 0
Causes Rx
Request accepted            : 0      No resources available     : 0
Request not fulfilled       : 0      Request already fulfilled  : 0
Dup Req already fulfilled   : 1      Invalid message format     : 0
Version not supported       : 0      Service not supported      : 0
Mandatory IE incorrect     : 0      Mandatory IE missing       : 0
Optional IE incorrect      : 0      System failure             : 0
Fragmented DRT request Tx   : 0      DRT Fragments              : 0
Fragmented DRT responses Rx : 0
-----
Number of peer instances : 1
=====

```

Example output Ga reference point instance statistics

```

# show mobile-gateway pdn ref-point-stats ga instance
=====
PDN gateway Ga reference point statistics per-peer instance
=====
Peer address : 10.11.7.104
Router       : Base
Port        : 3386      Gateway/Chassis/Card      : 1/1/3

DRT request Tx      : 0      DRT request retried      : 0
DRT response Rx     : 0      DRT request unacked      : 0
Redirection request Rx : 0      Redirection response Tx  : 0
Invalid messages Rx  : 0      Version not supported Rx  : 0
DRT request (dup) Tx : 0      GTP-prime failures       : 0

CDRs Tx            : 0      Version not supported Tx  : 0

Triggers Tx
Termination        : 0      Max change condition     : 0
MS Time Zone Change : 0      RAT Change               : 0
Time Limit         : 0      Volume Limit             : 0
PLMN change        : 0      Serving Node Change Limit : 0
Management Intervention : 0

Causes Rx
Request accepted    : 0      No resources available    : 0
Request not fulfilled : 0      Request already fulfilled : 0
Dup Req already fulfilled : 0      Invalid message format    : 0
Version not supported : 0      Service not supported     : 0
Mandatory IE incorrect : 0      Mandatory IE missing      : 0
Optional IE incorrect : 0      System failure            : 0
-----

```



```
Number of peer instances : 1
=====
```

Example output Ga reference point aggregate statistics

```
# show mobile-gateway pdn ref-point-stats ga aggregate
=====
PDN gateway Ga reference point statistics aggregate
=====
DRT request Tx           : 0          DRT request retried       : 0
DRT response Rx          : 0          DRT request unacked       : 0
Redirection request Rx   : 0          Redirection response Tx   : 0
Invalid messages Rx      : 0          Version not supported Rx   : 0
DRT request (dup) Tx     : 0          GTP-prime failures        : 0
CDRs Tx                  : 0          Version not supported Tx   : 0

Triggers Tx
Termination              : 0          Max change condition       : 0
MS Time Zone Change      : 0          RAT Change                 : 0
Time Limit               : 0          Volume Limit               : 0
PLMN change              : 0          Serving Node Change Limit  : 0
Management Intervention  : 0

Causes Rx
Request accepted         : 0          No resources available     : 0
Request not fulfilled    : 0          Request already fulfilled   : 0
Dup Req already fulfilled : 0          Invalid message format     : 0

Version not supported    : 0          Service not supported      : 0
Mandatory IE incorrect  : 0          Mandatory IE missing       : 0
Optional IE incorrect   : 0          System failure             : 0
-----
Number of peer instances : 1
=====
```

Example output Ga reference point card statistics

```
# show mobile-gateway pdn ref-point-stats ga card 3
=====
PDN gateway Ga reference point statistics
=====
Peer address : 10.11.7.101
Router       : Base
Port         : 3386

DRT request Tx           : 0          DRT request retried       : 0
DRT response Rx          : 0          DRT request unacked       : 0
Redirection request Rx   : 0          Redirection response Tx   : 0
Invalid messages Rx      : 0          Version not supported Rx   : 0
DRT request (dup) Tx     : 0          GTP-prime failures        : 0
CDRs Tx                  : 0          Version not supported Tx   : 0

Triggers Tx
Termination              : 0          Max change condition       : 0
MS Time Zone Change      : 0          RAT Change                 : 0
Time Limit               : 0          Volume Limit               : 0
```

```

PLMN change          : 0          Serving node change limit : 0
Management Intervention : 0

Causes Rx
Request accepted      : 0          No resources available    : 0
Request not fulfilled : 0          Request already fulfilled : 0
Dup Req already fulfilled : 0      Invalid message format   : 0
Version not supported : 0          Service not supported     : 0
Mandatory IE incorrect : 0        Mandatory IE missing      : 0
Optional IE incorrect  : 0        System failure            : 0
-----
Number of peer instances : 1
=====

```

Example output Gn reference point statistics

```

# show mobile-gateway pdn ref-point-stats gn
=====
PDN gateway Gn reference point statistics
=====
Peer address : 10.23.15.2
Reference point : default
Create PDP Request : 2          Initiate PDP Request : 0
Create PDP Request Redirect: 0
Create PDP Resp Succ : 2          Initiate PDP Resp Succ : 0
Create PDP Resp Fail : 0          Initiate PDP Resp Fail : 0
Tx Del PDP Req : 0          Rx Del PDP Req : 0
Tx Del PDP Resp Succ : 0          Rx Del PDP Resp Succ : 0
Tx Del PDP Resp Fail : 0          Rx Del PDP Resp Fail : 0
Tx Upd PDP Req : 0          Rx Upd PDP Req : 0
Tx Upd PDP Resp Succ : 0          Rx Upd PDP Resp Succ : 0
Tx Upd PDP Resp Fail : 0          Rx Upd PDP Resp Fail : 0
Tx MS Info Chg Ntf Rsp Succ: 2      Rx MS Info Change Ntf Req : 2
Tx MS Info Chg Ntf Rsp Fail: 0      Tx Upd PDP Req with ErrInd : 0
Malformed packets Rx : 0          Unknown message pkts Rx : 0
Missing mand IE pkts Rx : 0
Number of sessions : 2
-----
Number of peer instances : 2
=====

```

Example output Gnu reference point peers

```

# show mobile-gateway pdn ref-point-peers gnu
=====
PDN gateway gnu reference point peers
=====
Peer address : 10.23.15.2
Peer type : SGSN
Router : vprn20003
Reference point : default
Path Mgmt State : disabled
Create Time : 03/20/2020 18:18:56          Gateway Id : 1
GTP Version : GTPv1
-----
Number of peers : 1

```

Example output Gnu reference point statistics

```
=====
# show mobile-gateway pdn ref-point-stats gnu peer 10.23.15.2
=====
PDN gateway Gnu reference point statistics
=====
Peer address : 10.23.15.2
Reference point      : default

Number of Bearers      : 6
-----
Number of peer instances : 2
=====
```

Example output Gpu reference point peers

```
=====
# show mobile-gateway pdn ref-point-peers gpu
=====
PDN gateway gpu reference point peers
=====
Peer address      : 10.23.15.2
Peer type         : SGSN
Router            : vprn20003
Reference point   : default
Path Mgmt State   : disabled
Create Time       : 03/20/2020 18:30:16      Gateway Id      : 1
GTP Version       : GTPv1
-----
Number of peers : 1
=====
```

Example output Gpu reference point statistics

```
=====
# show mobile-gateway pdn ref-point-stats gpu peer 10.23.15.2
=====
PDN gateway Gpu reference point statistics
=====
Peer address : 10.23.15.2
Reference point      : default

Number of Bearers      : 6
-----
Number of peer instances : 2
=====
```

Example output Gx reference point statistics

```
=====
show mobile-gateway pdn ref-point-stats gx
=====
PDN gateway Gx reference point statistics
=====
Peer address : 10.12.43.2
Router       : vprn20002
=====
```

```

Port                               : 3868

Active Sessions                    : 0
CCR Initial messages Tx           : 0      CCA Initial messages Rx       : 0
CCA Initial msgs Rx fail         : 0
CCR Update messages Tx           : 0      CCA Update messages Rx       : 0
CCA Update msgs Rx fail         : 0
CCR Terminate messages Tx        : 0      CCA Terminate messages Rx   : 0
CCA Terminate msgs Rx fail      : 0
CCR Initial msg failures         : 0      CCR Update msg failures    : 0
CCR Terminate msg failures      : 0      RAA Nack messages Tx       : 0
RAR messages Rx                  : 0      RAA messages Tx            : 0
RAR eMPS messages Rx            : 0
CCR Initial message Re-Tx        : 0      CCR Update message Re-Tx   : 0
CCR Terminate message Re-Tx     : 0
Malformed packets Rx             : 0      Event Triggers              : 0
CCA Init malformed pkts Rx      : 0      CCA Updt malformed pkts Rx : 0
CCA Term malformed pkts Rx      : 0      RAR malformed pkts Rx      : 0
Unknown packets Rx              : 0      CCA Init unknown packets Rx: 0
CCA Updt unknown packets Rx     : 0      CCA Term unknown packets Rx: 0
RAR Unknown packets Rx          : 0      Missing AVP packets Rx     : 0
CCA-I missing AVP pkts Rx       : 0      CCA-U missing AVP pkts Rx  : 0
CCA-T missing AVP pkts Rx       : 0      RAR missing AVP pkts Rx    : 0
CCA-I unknown session pkts      : 0      CCA-U unknown session pkts : 0
CCA-T unknown session pkts      : 0      RAR unknown session pkts   : 0
CCR redirection Tx               : 0      CCA redirection Rx         : 0
CCA Unknown redirect host *: 0      CCR Redirection Fail Tx    : 0
Pri OCS addr Rx                 : 0      Sec OCS addr Rx            : 0
Invalid Pri OCS addr Rx         : 0      Invalid Sec OCS addr Rx    : 0
SDR messages Rx                 : 0      SDR malformed pkts Rx      : 0
SDR missing AVP pkts Rx         : 0      SDR unknown session pkts Rx: 0
SDA Success Msgs Tx             : 0      SDA Failure Msgs Tx        : 0
CCR Update AN_GW Failed Tx      : 0      RAA AN_GW Failed Tx        : 0
RAA DIAMETER_AN_GW Failed *: 0
CCR-I Pending Trans Re-Tx       : 0      CCR-U Pending Trans Re-Tx  : 0
RAR with P-CSCF Recovery        : 0
CCR-I dropped due to OC-OLR: 0
CCR-I re-route-dra-too-busy: 0      CCR-U re-route-dra-too-busy: 0
CCR-T re-route-dra-too-busy: 0

```

Number of peer instances : 1

=====

* indicates that the corresponding row element may have been truncated.

Example output Gx reference point instance statistics

```

# show mobile-gateway pdn ref-point-stats gx instance
=====
PDN gateway Gx reference point statistics per-peer instance
=====
Peer address : 10.10.8.10
Router                               : Base
Port                                 : 3868      Gateway/Chassis/Card       : 1/1/3

Active Sessions                      : 1
CCR Initial messages Tx              : 471687   CCA Initial messages Rx    : 370889

```

```

CCA Initial msgs Rx fail      : 0
CCR Update messages Tx       : 47242
CCA Update msgs Rx fail      : 0
CCR Terminate messages Tx    : 255803
CCA Terminate msgs Rx fail   : 0
CCR Initial msg failures     : 4250
CCR Terminate msg failures   : 7847
RAR messages Rx              : 0
CCR Initial message Re-Tx    : 487912
CCR Terminate message Re-Tx  : 629720
Malformed packets Rx        : 0
CCA Init malformed pkts Rx   : 0
CCA Term malformed pkts Rx   : 0
Unknown packets Rx          : 0
CCA Updt unknown packets Rx : 33209
RAR Unknown packets Rx       : 0
CCA-I missing AVP pkts Rx   : 0
CCA-T missing AVP pkts Rx   : 0
CCA-I unknown session pkts  : 12065
CCA-T unknown session pkts  : 27979
CCR redirection Tx           : 0
CCA Unknown redirect host *: 0
Pri OCS addr Rx             : 0
Invalid Pri OCS addr Rx     : 0
SDR messages Rx             : 52175
SDR missing AVP pkts Rx     : 923
SDA Success Msgs Tx         : 50596
CCR Update AN_GW Failed Tx  : 0
RAA DIAMETER_AN_GW Failed *: 0
Rate limit CCR(I) Drops     : 447605
Rate limit CCR(T) Drops     : 675624
Rate limit SDA Drops        : 37254
CCR-I Pending Trans Re-Tx   : 0
RAR with P-CSCF Recovery    : 0

CCA Update messages Rx      : 36051
CCA Terminate messages Rx   : 94347
CCR Update msg failures     : 0
RAA Nack messages Tx       : 81596
RAA messages Tx            : 0
CCR Update message Re-Tx    : 136827
Event Triggers              : 47242
CCA Updt malformed pkts Rx : 0
RAR malformed pkts Rx       : 0
CCA Init unknown packets Rx: 37324
CCA Term unknown packets Rx: 2322
Missing AVP packets Rx      : 0
CCA-U missing AVP pkts Rx   : 0
RAR missing AVP pkts Rx     : 0
CCA-U unknown session pkts  : 0
RAR unknown session pkts    : 81596
CCA redirection Rx          : 0
CCR Redirection Fail Tx     : 0
Sec OCS addr Rx            : 0
Invalid Sec OCS addr Rx     : 0
SDR malformed pkts Rx       : 0
SDR unknown session pkts Rx: 0
SDA Failure Msgs Tx         : 1579
RAA AN_GW Failed Tx         : 0
Rate limit CCR(U) Drops     : 148018
Rate limit RAA Drops        : 61713
CCR-U Pending Trans Re-Tx   : 0

```

Example output Gx reference point card statistics

```

# show mobile-gateway pdn ref-point-stats gx card 3
=====
PDN gateway Gx reference point statistics
=====
Peer address : 10.10.8.10
Router       : Base
Port         : 3868

CCR Initial messages Tx      : 2
CCA Initial msgs Rx fail    : 0
CCR Update messages Tx      : 2
CCA Update msgs Rx fail     : 0
CCR Terminate messages Tx   : 2
CCA Terminate msgs Rx fail  : 0
CCR Initial msg failures    : 0
CCR Terminate msg failures  : 0
RAR messages Rx             : 0
CCR Initial message Re-Tx   : 0

CCA Initial messages Rx      : 2
CCA Update messages Rx      : 5
CCA Terminate messages Rx   : 2
CCR Update msg failures     : 1
RAA Nack messages Tx        : 0
RAA messages Tx             : 0
CCR Update message Re-Tx    : 3

```

```

CCR Terminate message Re-Tx: 0
Malformed packets Rx      : 0      Event Triggers           : 6
CCA Init malformed pkts Rx : 0      CCA Updt malformed pkts Rx : 0
CCA Term malformed pkts Rx : 0      RAR malformed pkts Rx      : 0
Unknown packets Rx        : 0      CCA Init unknown packets Rx: 0
CCA Updt unknown packets Rx: 4      CCA Term unknown packets Rx: 0
RAR Unknown packets Rx     : 0      Missing AVP packets Rx     : 0
CCA-I missing AVP pkts Rx  : 0      CCA-U missing AVP pkts Rx  : 0
CCA-T missing AVP pkts Rx  : 0      RAR missing AVP pkts Rx    : 0
CCA-I unknown session pkts : 0      CCA-U unknown session pkts : 0
CCA-T unknown session pkts : 0      RAR unknown session pkts   : 0
CCR redirection Tx         : 0      CCA redirection Rx         : 0
CCA Unknown redirect host *: 0      CCR Redirection Fail Tx    : 0
Pri OCS addr Rx           : 0      Sec OCS addr Rx            : 0
Invalid Pri OCS addr Rx   : 0      Invalid Sec OCS addr Rx    : 0
SDR messages Rx           : 0      SDR malformed pkts Rx      : 0
SDR missing AVP pkts Rx    : 0      SDR unknown session pkts Rx: 0
SDA Success Msgs Tx       : 0      SDA Failure Msgs Tx        : 0
CCR Update AN_GW Failed Tx : 0      RAA AN_GW Failed Tx        : 0
RAA DIAMETER_AN_GW Failed *: 0

```

```

-----
Number of peer instances : 1
=====

```

Example output Gx reference point failure code statistics

```

# show mobile-gateway pdn ref-point-stats gx failure-codes
=====
PDN gateway Gx Failure Code statistics
=====
Peer address : 10.10.43.2
Router       : vprn20002
Port        : 3868

Rx Error Codes (CCA)
3001 Command Unsupported : 0      3002 Unable to deliver : 0
3004 Too busy            : 0
4141 PCC Bearer Event    : 0      4144 Pending-Transaction : 0
4197 Session recovery req. : 0
5001 AVP unsupported      : 0      5002 Unknown-session-id : 0
5003 Authorization rejected: 0      5004 Invalid AVP value  : 0
5005 Missing AVP         : 1      5006 Resources exceeded : 1
5008 AVP not allowed     : 0      5140 Initial parameters : 0
5141 Trigger event       : 0      5142 PCC Rule event      : 0
5143 Bearer not authorized : 0      5144 Traffic mapping reject: 0
5147 Conflicting request  : 0      5198 Overload control    : 0
5199 Newer session detected: 0
Other-3xxx               : 4      Other-4xxx               : 2
Other-5xxx               : 0

Tx Error Codes (RAA)
5001 AVP unsupported      : 0      5002 Unknown-Session-Id : 0
5008 AVP not allowed     : 0      5012 Unable-To-Comply    : 0
5142 PCC Rule event       : 0      4143 DIAMETER AN GW FAILED : 0
4144 Pending-Transaction  : 0

Tx Termination Causes (CCR-T)
Bad answer                : 0      Administrative           : 0

```

```

Tx Error Codes (SDA)
4196 Session not found      : 0          5001 AVP unsupported      : 0
5004 Invalid AVP value      : 0          5005 Missing AVP          : 0
5008 AVP not allowed        : 0          5009 AVP Too Many Times  : 0
5012 Unable-To-Comply       : 0
-----

```

Example output Gx reference point peer statistics

```

# show mobile-gateway pdn ref-point-stats gx peer 10.10.43.10
=====
PDN gateway Gx reference point statistics
=====
Peer address : 10.10.43.10
Router       : vprn20002
Port        : 3868

Active Sessions      : 0
CCR Initial messages Tx : 1          CCA Initial messages Rx : 1
CCA Initial msgs Rx fail : 1
CCR Update messages Tx : 0          CCA Update messages Rx : 0
CCA Update msgs Rx fail : 0
CCR Terminate messages Tx : 0        CCA Terminate messages Rx : 0
CCA Terminate msgs Rx fail : 0
CCR Initial msg failures : 0          CCR Update msg failures : 0
CCR Terminate msg failures : 0        RAA Nack messages Tx : 0
RAR messages Rx       : 0          RAA messages Tx : 0
RAR eMPS messages Rx  : 0
CCR Initial message Re-Tx : 0        CCR Update message Re-Tx : 0
CCR Terminate message Re-Tx: 0
Malformed packets Rx   : 0          Event Triggers : 0
CCA Init malformed pkts Rx : 0        CCA Updt malformed pkts Rx : 0
CCA Term malformed pkts Rx : 0        RAR malformed pkts Rx : 0
Unknown packets Rx     : 0          CCA Init unknown packets Rx: 0
CCA Updt unknown packets Rx: 0        CCA Term unknown packets Rx: 0
RAR Unknown packets Rx  : 0          Missing AVP packets Rx : 0
CCA-I missing AVP pkts Rx : 0        CCA-U missing AVP pkts Rx : 0
CCA-T missing AVP pkts Rx : 0        RAR missing AVP pkts Rx : 0
CCA-I unknown session pkts : 0       CCA-U unknown session pkts : 0
CCA-T unknown session pkts : 0        RAR unknown session pkts : 0
CCR redirection Tx      : 0          CCA redirection Rx : 0
CCA Unknown redirect host *: 0        CCR Redirection Fail Tx : 0
Pri OCS addr Rx        : 0          Sec OCS addr Rx : 0
Invalid Pri OCS addr Rx : 0          Invalid Sec OCS addr Rx : 0
SDR messages Rx        : 0          SDR malformed pkts Rx : 0
SDR missing AVP pkts Rx : 0          SDR unknown session pkts Rx: 0
SDA Success Msgs Tx    : 0          SDA Failure Msgs Tx : 0
CCR Update AN_GW Failed Tx : 0        RAA AN_GW Failed Tx : 0
RAA DIAMETER_AN_GW Failed *: 0
CCR-I Pending Trans Re-Tx : 0         CCR-U Pending Trans Re-Tx : 0
RAR with P-CSCF Recovery : 0
CCR-I dropped due to OC-OLR: 0
CCR-I re-route-dra-too-busy: 0        CCR-U re-route-dra-too-busy: 0
CCR-T re-route-dra-too-busy: 0
-----
Number of peer instances : 1

```

=====

* indicates that the corresponding row element may have been truncated.

Example output Gx reference point eMPS advance priority calls

```
# show mobile-gateway pdn ref-point-stats gx peer 10.10.43.2
=====
PDN gateway Gx reference point statistics
=====
Peer address : 10.10.43.2
Router          : vprn20002
Port            : 3868

Active Sessions          : 1
CCR Initial messages Tx  : 1          CCA Initial messages Rx   : 1
CCA Initial msgs Rx fail : 0
CCR Update messages Tx   : 1          CCA Update messages Rx   : 1
CCA Update msgs Rx fail  : 0
CCR Terminate messages Tx : 0          CCA Terminate messages Rx : 0
CCA Terminate msgs Rx fail : 0
CCR Initial msg failures  : 0          CCR Update msg failures   : 0
CCR Terminate msg failures : 0          RAA Nack messages Tx      : 0
RAR messages Rx          : 1          RAA messages Tx          : 1
RAR eMPS messages Rx     : 1
CCR Initial message Re-Tx : 0          CCR Update message Re-Tx  : 0
CCR Terminate message Re-Tx : 0
Malformed packets Rx     : 0          Event Triggers            : 1
CCA Init malformed pkts Rx : 0          CCA Updt malformed pkts Rx : 0
CCA Term malformed pkts Rx : 0          RAR malformed pkts Rx     : 0
Unknown packets Rx       : 0          CCA Init unknown packets Rx : 0
CCA Updt unknown packets Rx : 0          CCA Term unknown packets Rx : 0
RAR Unknown packets Rx   : 0          Missing AVP packets Rx    : 0
CCA-I missing AVP pkts Rx : 0          CCA-U missing AVP pkts Rx : 0
CCA-T missing AVP pkts Rx : 0          RAR missing AVP pkts Rx   : 0
CCA-I unknown session pkts : 0          CCA-U unknown session pkts : 0
CCA-T unknown session pkts : 0          RAR unknown session pkts  : 0
CCR redirection Tx        : 0          CCA redirection Rx        : 0
CCA Unknown redirect host *: 0          CCR Redirection Fail Tx   : 0
Pri OCS addr Rx          : 0          Sec OCS addr Rx           : 0
Invalid Pri OCS addr Rx  : 0          Invalid Sec OCS addr Rx   : 0
SDR messages Rx          : 0          SDR malformed pkts Rx     : 0
SDR missing AVP pkts Rx  : 0          SDR unknown session pkts Rx : 0
SDA Success Msgs Tx      : 0          SDA Failure Msgs Tx       : 0
CCR Update AN_GW Failed Tx : 0          RAA AN_GW Failed Tx       : 0
RAA DIAMETER_AN_GW Failed *: 0
CCR-I Pending Trans Re-Tx : 0          CCR-U Pending Trans Re-Tx : 0
RAR with P-CSCF Recovery  : 0
CCR-I dropped due to OC-OLR : 0

-----
Number of peer instances : 1
=====
* indicates that the corresponding row element may have been truncated.
```

Example output Gy reference point statistics

```
# show mobile-gateway pdn ref-point-stats gy
```



```

=====
PDN gateway Gy reference point statistics
=====
Peer address : 10.23.18.10
Router       : vprn20005
Port        : 3868

Active Sessions      : 1
CCR Granted messages : 5      CCR Denied messages      : 0
CCR Initial messages Tx : 1    CCA Initial messages Rx : 1
CCR Update messages Tx : 4    CCA Update messages Rx : 4
CCR Terminate messages Tx : 0  CCA Terminate messages Rx : 0
ASA messages Tx      : 0      ASR messages Rx        : 0
RAR messages Rx      : 0      RAA messages Tx        : 0
ASA Nack messages Tx : 0      RAA Nack messages Tx   : 0
CCA MSCC level SUCCESS : 0    CCA MSCC level LIMIT SUCCE*: 0
CCR Initial msg failures : 0   CCR Update msg failures : 0
CCR Terminate msg failures : 0 ASR unknown session pkts : 0
CCA-I unknown session pkts : 0 CCA-U unknown session pkts : 0
CCA-T unknown session pkts : 0 RAR unknown session pkts : 0
Malformed packets Rx   : 0    ASR Malformed packets Rx : 0
CCA Init malformed pkts Rx : 0 CCA Updt malformed pkts Rx : 0
CCA Term malformed pkts Rx : 0 RAR malformed pkts Rx   : 0
Unknown packets Rx     : 0    CCA Init unknown packets Rx: 0
CCA Updt unknown packets Rx: 0 CCA Term unknown packets Rx: 0
RAR Unknown packets Rx : 0    ASR Unknown packets Rx   : 0
Missing AVP packets Rx : 0    ASR Missing AVP packets Rx : 0
CCA-I missing AVP pkts Rx : 0 CCA-U missing AVP pkts Rx : 0
CCA-T missing AVP pkts Rx : 0 RAR missing AVP pkts Rx : 0
CCR Initial message Re-Tx : 0  CCR Update message Re-Tx : 0
CCR Terminate message Re-Tx: 0 CCR-E messages Re-Tx    : 0
CCR-E messages Tx       : 0    CCA-E messages Rx       : 0
CCR-E msg failures      : 0    CCA-E Nack messages Rx  : 0
CCA-E malformed packets Rx : 0  CCA-E missing AVP packets *: 0
CCA-E unknown packets Rx : 0    CCA Initial Fail msgs Rx : 0
CCA Update Fail msgs Rx  : 0    CCA Terminate Fail msgs Rx : 0
RG FUI-Redirect-recv     : 0
RG FUI-Terminate-recv    : 0
RG FUI-Restrict-recv     : 0
RG FUI-Redirect-applied  : 0
RG FUI-Terminate-applied : 0
RG FUI-Restrict-applied  : 0
Cmd-level FUI-Redirect   : 0
Gy Sess Converted To Offline Due To Fh-action-offline : 0
Gy Sess Converted To Allow Due To Fh-action-allow      : 0
Gy Sess Converted To Block Due To Fh-action-block      : 0

```

```

-----
Number of peer instances : 1
=====

```

```

* indicates that the corresponding row element may have been truncated.

```

Example output Gy reference point instance statistics

```

# show mobile-gateway pdn ref-point-stats gy instance
=====
PDN gateway Gy reference point statistics per-peer instance

```

```

=====
Peer address : 10.10.18.10
Router       : Base
Port        : 3868      Gateway/Chassis/Card : 1/1/3

Active Sessions : 0
CCR Granted messages : 0      CCR Denied messages : 0
CCR Initial messages Tx : 0    CCA Initial messages Rx : 0
CCR Update messages Tx : 0    CCA Update messages Rx : 0
CCR Terminate messages Tx : 0  CCA Terminate messages Rx : 0
ASA messages Tx : 0          ASR messages Rx : 0
RAR messages Rx : 0          RAA messages Tx : 0
ASA Nack messages Tx : 0     RAA Nack messages Tx : 0
CCA MSCC level SUCCESS : 0    CCA MSCC level LIMIT SUCCE*: 0
CCR Initial msg failures : 0  CCR Update msg failures : 0
CCR Terminate msg failures : 0 ASR unknown session pkts : 0
CCA-I unknown session pkts : 0 CCA-U unknown session pkts : 0
CCA-T unknown session pkts : 0 RAR unknown session pkts : 0
Malformed packets Rx : 0     ASR Malformed packets Rx : 0
CCA Init malformed pkts Rx : 0 CCA Updt malformed pkts Rx : 0
CCA Term malformed pkts Rx : 0 RAR malformed pkts Rx : 0
Unknown packets Rx : 0       CCA Init unknown packets Rx: 0
CCA Updt unknown packets Rx: 0 CCA Term unknown packets Rx: 0
RAR Unknown packets Rx : 0    ASR Unknown packets Rx : 0
Missing AVP packets Rx : 0    ASR Missing AVP packets Rx : 0
CCA-I missing AVP pkts Rx : 0 CCA-U missing AVP pkts Rx : 0
CCA-T missing AVP pkts Rx : 0 RAR missing AVP pkts Rx : 0
CCR Initial message Re-Tx : 0  CCR Update message Re-Tx : 0
CCR Terminate message Re-Tx: 0 CCR-E messages Re-Tx : 0
CCR-E messages Tx : 0         CCA-E messages Rx : 0
CCR-E msg failures : 0        CCA-E Nack messages Rx : 0
CCA-E malformed packets Rx : 0 CCA-E missing AVP packets *: 0
CCA-E unknown packets Rx : 0
RG FUI-Redirect-recv : 0
RG FUI-Terminate-recv : 0
RG FUI-Restrict-recv : 0
RG FUI-Redirect-applied : 0
RG FUI-Terminate-applied : 0
RG FUI-Restrict-applied : 0
Cmd-level FUI-Redirect : 0
Gy Sess Converted To Offline Due To Fh-action-offline : 0
Gy Sess Converted To Allow Due To Fh-action-allow : 0
Gy Sess Converted To Block Due To Fh-action-block : 0

```

```

-----
Number of peer instances : 1
=====

```

* indicates that the corresponding row element may have been truncated.

Example output Gy reference point failure code statistics

```

# show mobile-gateway pdn ref-point-stats gy failure-codes

```

```

=====
PDN gateway Gy Failure Code Statistics
=====

```

```

Peer address : 10.23.18.10

```

```

Router       : vprn20005

```

```

Port : 3868
Rx Error Codes (CCA)
3002 Unable-To-Deliver : 0      3004 Too-Busy : 0
4002 Out-Of-Space : 0      4010 enduser-service-denied: 0
4011 CC-not-applicable : 0    4012 credit-limit-reached : 0
5001 Avp-Unsupported : 0      5002 Unknown-Session-Id : 0
5003 Auth-Rejected : 0      5004 Invalid-Avp-Value : 0
5005 Missing-Avp : 0      5012 Unable-To-Comply : 0
5030 User-Unknown : 0      5031 Rating-Failed : 0
Other-3xxx : 0      Other-4xxx : 0
Other-5xxx : 0
Rx Error Codes (MSCC)
4010 Enduser-service-denied: 0    4011 CC-not-applicable : 0
4012 credit-limit-reached : 0    5003 Authorization-Rejected: 0
5031 Rating-Failed : 0
Other-3xxx : 0      Other-4xxx : 0
Other-5xxx : 0
Tx Error Codes (ASA/RAA)
5002 Unknown-Session-Id : 0      5012 Unable-To-Comply : 0
Tx Termination Causes (CCR)
Bad-Answer : 0      Administrative : 0
-----
Number of peer instances : 1
=====

```

Example output N3 and N9 reference point statistics

The following example shows N3 and N9 reference point statistics.

```

# show mobile-gateway pdn ref-point-stats n3
=====
PDN gateway N3 reference point statistics
=====
Peer address : 192.168.0.1
Reference point : default
-----
Number of peer instances : 2
=====

# show mobile-gateway pdn ref-point-stats n9
=====
PDN gateway N9 reference point statistics
=====
Peer address : 2001:db8::211:ff:11
Reference point : default
-----
Number of peer instances : 2
=====

```

Example output PI reference point statistics

```

# show mobile-gateway pdn ref-point-stats pi ref-point default2
=====
PDN gateway Pi reference point statistics

```

```

=====
Peer Address      : 10.23.15.50
Router           : vprn20003
Port             : 434
Reference Point   : default2

Initial Registration Requests Stats:
Received          : 135      Accepted          : 135
Denied           : 0

Renew Registration Requests Stats:
Received          : 1        Accepted          : 1
Denied           : 0

De-registration Requests Stats:
Received          : 135      Accepted          : 135
Denied           : 0

Registration Reply Stats:
Total             : 271      Accepted Registration : 136
Accepted De-registration : 135

Handoff Requests Stats:
Received          : 0        Accepted          : 0
Denied           : 0        Unspecified Reason : 0
Unknown CVSE Rcvd : 0        UDP Encap Unavailable : 0

Registration Revocation Stats:
Sent Total        : 0        Retries Sent       : 0
Ack Rcvd          : 0        Not Acknowledged   : 0
Rcvd Total        : 0        Ack Sent           : 0

-----
Number of peer instances : 1
=====

```

Example output RADIUS reference point statistics

```

# show mobile-gateway pdn ref-point-stats radius
=====
PDN Gateway Radius Interface Statistics
=====
RADIUS Group      : Radius Group 1
Access request Tx  : 131040    Access Accept Rx    : 131040
Access Reject Rx   : 0        Access Challenge Rx  : 0

```

```

Acct Req Start Tx      : 0      Acct Req Interim Tx      : 0
Acct Req Stop Tx       : 0      Acct Response Rx         : 0
Acct Req On Tx         : 0      Acct Req Off Tx          : 0
Mand. Attr Missing     : 65520  Mand. Attr Errors        : 0
Option Attr Errors     : 0      Authenticator Errors     : 0
Unexpected Code        : 0      Unsupported Errors       : 0
Resp Time < 1 Sec      : 131040 Resp Time 1-4 Sec        : 0
Resp Time > 4 Sec      : 0      Retries                  : 0
Peer Final Timeout     : 0      Msg Final Timeout        : 0

```

```

Rate Limit AuthReq Drops - Init      : 0
Rate Limit AcctReqStart Drops - Init : 0
Rate Limit AcctReqStop Drops - Init  : 0
Rate Limit AcctReqInter Drops - Init : 0
Rate Limit AuthReq Drops - Retr      : 0
Rate Limit AcctReqStart Drops - Retr : 0
Rate Limit AcctReqStop Drops - Retr  : 0
Rate Limit AcctReqInter Drops - Retr : 0

```

Disconnect statistics

```

Disc Req Rx           : 0      Disc Unsupported Attr      : 0
Disc ACK Tx           : 0      Disc Session Not Found    : 0
Disc NAK Tx           : 0      Authenticator error       : 0
Mandatory attr missing : 0      Malformed message         : 0

```

CoA statistics

```

CoA Req Rx           : 0      CoA Unsupported Attr      : 0
CoA ACK Tx           : 0      CoA Session Not Found    : 0
CoA NAK Tx           : 0      Authenticator error       : 0
Mandatory attr missing : 0      Malformed message         : 0
Internal Errors(CoA&Disc) : 0

```

```

Number of RADIUS groups : 1
=====

```

Example output RADIUS reference point aggregate statistics

```
# show mobile-gateway pdn ref-point-stats radius aggregate
```

PDN Gateway Radius Interface Statistics

```

Access request Tx      : 2      Access Accept Rx         : 2
Access Reject Rx       : 0      Access Challenge Rx      : 0
Acct Req Start Tx      : 0      Acct Req Interim Tx      : 0
Acct Req Stop Tx       : 0      Acct Response Rx         : 0
Acct Req On Tx         : 0      Acct Req Off Tx          : 0
Mand. Attr Missing     : 0      Mand. Attr Errors        : 0
Option Attr Errors     : 0      Authenticator Errors     : 0
Unexpected Code        : 0      Unsupported Errors       : 0
Resp Time < 1 Sec      : 2      Resp Time 1-4 Sec        : 0
Resp Time > 4 Sec      : 0      Retries                  : 0
Peer Final Timeout     : 0      Msg Final Timeout        : 0

```

Disconnect statistics

```

Disc Req Rx           : 0      Disc Unsupported Attr      : 0
Disc ACK Tx           : 0      Disc Session Not Found    : 0
Disc NAK Tx           : 0      Authenticator error       : 0

```

```

Mandatory attr missing      : 0          Malformed message      : 0
CoA statistics
CoA Req Rx                  : 0          CoA Unsupported Attr    : 0
CoA ACK Tx                  : 0          CoA Session Not Found    : 0
CoA NAK Tx                  : 0          Authenticator error      : 0
Mandatory attr missing      : 0          Malformed message      : 0
Internal Errors(CoA&Disc)   : 0
=====

```

Example output RADIUS reference point instance statistics

```

# show mobile-gateway pdn ref-point-stats radius instance
=====
PDN Gateway Radius Interface Statistics
=====
RADIUS Group                : Radius Group
Peer address                 : 10.23.19.2
Gateway/Chassis/Card         : 1/1/3

Access request Tx           : 0          Access Accept Rx         : 0
Access Reject Rx           : 0          Access Challenge Rx      : 0
Acct Req Start Tx          : 1          Acct Req Interim Tx      : 0
Acct Req Stop Tx           : 0          Acct Response Rx        : 0

Mand. Attr Missing         : 0          Mand. Attr Errors        : 0
Option Attr Errors         : 0          Authenticator Errors     : 0
Unexpected Code            : 0          Unsupported Errors       : 0

Resp Time < 1 Sec          : 0          Resp Time 1-4 Sec       : 0
Resp Time > 4 Sec          : 0          Retries                  : 1
Peer Final Timeout         : 1

-----
Number of RADIUS peers : 1
=====

```

Example output RF reference point instance statistics

```

# show mobile-gateway pdn ref-point-stats rf instance
=====
PDN gateway Rf reference point statistics per-peer instance
=====
Peer address : 10.15.8.10
Router       : Base
Port         : 3868          Gateway/Chassis/Card      : 1/1/3

ACR Start messages Tx      : 1          ACR Interim messages Tx  : 1
ACR Stop messages Tx       : 0

-----
Number of peer instances : 1
=====

```

Example output RF reference point aggregate statistics

```

# show mobile-gateway pdn ref-point-stats rf aggregate
=====
PDN gateway Rf reference point statistics aggregate

```

```

=====
ACR Start messages Tx      : 1          ACR Interim messages Tx    : 1
ACR Stop messages Tx      : 0
-----
Number of peer instances : 1
=====

```

Example output RF reference point card statistics

```

# show mobile-gateway pdn ref-point-stats rf card 3
=====
PDN gateway Rf reference point statistics
=====
Peer address : 10.15.8.10
Router       : Base
Port        : 3868

ACR Start messages Tx      : 0          ACR Interim messages Tx    : 0
ACR Stop messages Tx      : 0
-----
Number of peer instances : 1
=====

```

Example output S1u reference point peers

```

# show mobile-gateway pdn ref-point-peers s1u
=====
PDN gateway s1u reference point peers
=====
Peer address   : 6.6.6.1
Peer type      : eNodeB
Router        : vprn20032
Reference point : default
Path Mgmt State : up
Create Time    : 03/20/2020 18:01:27      Gateway Id      : 2
GTP Version    : GTPv1
PMTU          : 0
-----
Number of peers : 1
=====

```

Example output S1u reference point statistics

```

# show mobile-gateway pdn ref-point-stats s1u peer 6.6.6.1
=====
PDN gateway s1u reference point statistics
=====
Peer address : 6.6.6.1
Reference point : default

Number of Bearers : 6
-----
Number of peer instances : 2

```

Example output S2a reference point statistics

```

=====
# show mobile-gateway pdn ref-point-stats s2a
=====
PDN gateway S2a reference point statistics
=====
Peer address          : 3FFE::A10:80A
Router                : Base
Port                  : 0

Rx Proxy Binding Updates : 5          Tx Proxy Binding Acks(Succ*: 0/0
Tx Binding Revocation Ind : 0          Rx Binding Revocation Ack(*: 0/0
Rx Malformed packets     : 0          Rx Unknown message pkts    : 0
Rx Missing mand IE pkts  : 0

-----
Number of peer instances : 1
=====
* indicates that the corresponding row element may have been truncated.

```

Example output S2a reference point statistics

```

=====
# show mobile-gateway pdn ref-point-stats s2a
=====
PDN gateway S2a reference point statistics
=====
Peer address : 3FFE::A10:80A
Router       : Base
Port         : 0

Rx PBU          : 4          Tx PBA (Succ/Fail)      : 2/0
Tx BRI          : 0          Rx BRA (Succ/Fail)      : 0/0
Rx Malformed packets : 0      Rx Unknown message pkts : 0
Rx Missing mand IE pkts : 0

-----
Number of peer instances : 1
=====

```

Example output S2a reference point statistics

```

=====
# show mobile-gateway pdn ref-point-stats s2a
=====
PDN gateway S2a reference point statistics
=====
Peer address : 3FFE::A10:80A
Router       : Base
Port         : 0

Rx PBU          : 2          Tx PBA (Succ/Fail)      : 1/0

```



```

Tx BRI                : 0          Rx BRA (Succ/Fail)      : 0/0
Rx Malformed packets  : 0          Rx Unknown message pkts : 0
Rx Missing mand IE pkts : 0

```

```

-----
Number of peer instances : 1
=====

```

Example output S2a reference point statistics

```

# show mobile-gateway pdn ref-point-stats s2a-gtp
=====
PDN gateway S2a reference point statistics
=====
Peer address : 10.23.23.2
Peer type    : TWAG
Reference point      : default

Create session requests      : 0
Create session resp succ     : 0          Create session resp fail : 0
CS req teid non zero        : 0
CS unique req teid non zero : 0
DSCP value                  : 0
Delete session requests     : 0
Delete session resp succ    : 0          Delete session resp fail  : 0
Create bearer requests      : 0
Create bearer resp succ     : 0          Create bearer resp fail   : 0
Delete bearer requests      : 0
Delete bearer resp succ     : 0          Delete bearer resp fail   : 0
Update bearer requests      : 0
Update bearer resp succ     : 0          Update bearer resp fail   : 0
Modify bearer requests      : 0
Modify bearer resp succ     : 0          Modify bearer resp fail   : 0
Missing mand IE pkts Rx     : 0
Malformed packets Rx        : 0          Unknown message pkts Rx   : 0
Modify bearer commands      : 0          Modify bearer fail ind    : 0
Delete bearer commands      : 0          Delete bearer fail ind    : 0
Bearer resource commands    : 0          Bearer resource fail ind  : 0
Suspend notification reqs   : 0
Suspend notification ack s* : 0          Suspend notification ack f* : 0
Resume notification reqs    : 0
Resume notification ack su* : 0          Resume notification ack fa* : 0
MBReq IE P-CSCF recovery    : 0
Number of sessions:         : 1
-----
Number of peer instances : 1
=====

```

Example output S2a reference point card statistics

```

# show mobile-gateway pdn ref-point-stats s2a card 3
=====
PDN gateway S2a reference point statistics
=====
Peer address : 3FFE::A10:80A
Router       : Base
Port         : 0

```

```

Rx PBU                : 0          Tx PBA (Succ/Fail)      : 0/0
Tx BRI                : 0          Rx BRA (Succ/Fail)      : 0/0
Rx Malformed packets  : 0          Rx Unknown message pkts : 0
Rx Missing mand IE pkts : 0

```

```

-----
Number of peer instances : 1
=====

```

Example output S2a-GTP reference point statistics

```

# show mobile-gateway pdn ref-point-stats s2a-gtp
=====
PDN gateway S2a reference point statistics
=====
Peer address : 10.10.5.1
Peer type    : PGW
Reference point      : default

Create session requests      : 1
Create session resp succ    : 1          Create session resp fail : 0
CS req teid non zero        : 0
CS unique req teid non zero: 0
DSCP value                  : 0
Delete session requests     : 0
Delete session resp succ    : 0          Delete session resp fail : 0
Create bearer requests      : 0
Create bearer resp succ     : 0          Create bearer resp fail  : 0
Delete bearer requests      : 0
Delete bearer resp succ     : 0          Delete bearer resp fail  : 0
Update bearer requests      : 0
Update bearer resp succ     : 0          Update bearer resp fail  : 0
Modify bearer requests      : 0
Modify bearer resp succ     : 0          Modify bearer resp fail  : 0
Missing mand IE pkts Rx     : 0
Malformed packets Rx        : 0          Unknown message pkts Rx  : 0
Modify bearer commands      : 0          Modify bearer fail ind   : 0
Delete bearer commands      : 0          Delete bearer fail ind   : 0
Bearer resource commands    : 0          Bearer resource fail ind : 0
Suspend notification reqs   : 0
Suspend notification ack s* : 0          Suspend notification ack f* : 0
Resume notification reqs    : 0
Resume notification ack su* : 0          Resume notification ack fa* : 0
MBReq IE P-CSCF recovery    : 0
Number of sessions:         : 1
-----
Number of peer instances : 2
=====
* indicates that the corresponding row element may have been truncated.

```

Example output S2a-GTP reference point failure code statistics

```

# show mobile-gateway pdn ref-point-stats s2a-gtp failure-codes

```

```
=====
PDN gateway S2a Failure Code statistics
=====
```

```
Peer address : 10.10.5.1
```

```
Peer type    : PGW
```

```
Reference point      : default
```

Failure code	Rx count	Tx count

Request Accepted	1	1
Request Rejected	0	0
Mandatory IE Incorrect	0	0
Mandatory IE Missing	0	0
Context Not Found	0	0
Invalid Length	0	0
Remote Peer Not Responding	0	0
No Resources Available	0	0
User Authentication Failed	0	0
Request accepted partially	0	0
New PDN type due to network preference	0	0
New PDN type due to single address bearer only	0	0
Missing or unknown APN	0	0
Preferred PDN type not supported	0	0
All dynamic addresses are occupied	0	0
APN access denied - no subscription	0	0
Service not supported	0	0
Unable to page UE due to Suspension	0	0
Conditional IE missing	0	0
Temporarily rejected due to H0 proc in progress	0	0
UE context without TFT already activated	0	0
Semantic error in the TFT operation	0	0
Syntactic error in the TFT operation	0	0
Semantic errors in packet filters	0	0
Syntactic errors in packet filters	0	0
Version not supported by next peer	0	0
Protocol type not supported	0	0
APN congestion	0	0
Denied in RAT	0	0
APN Restriction type incompatible	0	0
UE already re-attached	0	0
GRE key not found	0	0
MPDN connections for a given APN not allowed	0	0
Modifications not limited to S1-U bearers	0	0
Network Failure	0	0
QoS parameter mismatch	0	0
ISR Deactivation	0	0
RAT chgd 3GPP to non-3GPP	0	0
Reactivation requested	0	0
PDN reconnection to this APN disallowed	0	0
PDN connection inactivity timer expires	0	0
Access changed from Non-3GPP to 3GPP	0	0
Others	0	0
Invalid message format	0	0
System failure	0	0
UE not responding	0	0
UE refuses	0	0
Unable to page UE	0	0

```

No memory available          0          0
IMSI/IMEI not known         0          0
Semantic error in the TAD operation 0          0
Syntactic error in the TAD operation 0          0
Invalid length of response/piggybacked init msg 0          0
Invalid reply from remote peer 0          0
Invalid peer                 0          0
Bearer handling not supported 0          0
Idle recover                 0          0
-----
Number of peer instances : 2
=====

```

Example output S2b reference point statistics ePDG/PGW combination gateways

```

# show mobile-gateway pdn ref-point-stats s2b
=====
PDN gateway S2b reference point statistics
=====
Peer address : 16.70.16.24
Peer type    : ePDG
Reference point      : default

Create session requests      : 1
Create session resp succ    : 1          Create session resp fail    : 0
CS req teid non zero        : 0
CS unique req teid non zero: 0
DSCP value                  : 0
Delete session requests     : 0
Delete session unique req   : 0
Delete session resp succ    : 0          Delete session resp fail    : 0
Create bearer requests      : 0
Create bearer resp succ     : 0          Create bearer resp fail    : 0
Delete bearer requests      : 1
Delete bearer resp succ     : 1          Delete bearer resp fail    : 0
Update bearer requests      : 0
Update bearer resp succ     : 0          Update bearer resp fail    : 0
Modify bearer requests      : 0
Modify bearer resp succ     : 0          Modify bearer resp fail    : 0
Modify bearer req bbai      : 0          Modify bearer req bbai emer: 0
Missing mand IE pkts Rx     : 0
Malformed packets Rx        : 0          Unknown message pkts Rx    : 0
Modify bearer commands      : 0          Modify bearer fail ind     : 0
Delete bearer commands      : 0          Delete bearer fail ind     : 0
Bearer resource commands    : 0          Bearer resource fail ind   : 0
Suspend notification reqs   : 0
Suspend notification ack s*: 0          Suspend notification ack f*: 0
Resume notification reqs    : 0
Resume notification ack su*: 0          Resume notification ack fa*: 0
MBReq IE P-CSCF recovery    : 0          UBReq with P-CSCF servers  : 0
P-CSCF Recovery DBReq       : 1
Number of sessions:         : 0

```

Create Session Requests with HI bit

```

-----
Ctxt not found init attach : 0
Context not found          : 0
Ctxt found with diff IP add: 0
-----

Peer address : 16.70.16.19
Peer type    : PGW
Reference point      : default

Create session requests      : 1
Create session resp succ     : 1          Create session resp fail : 0
CS req teid non zero        : 0
CS unique req teid non zero: 0
DSCP value                   : 0
Delete session requests      : 0
Delete session unique req    : 0
Delete session resp succ     : 0          Delete session resp fail : 0
Create bearer requests       : 0
Create bearer resp succ      : 0          Create bearer resp fail  : 0
Delete bearer requests       : 1
Delete bearer resp succ      : 1          Delete bearer resp fail  : 0
Update bearer requests       : 0
Update bearer resp succ      : 0          Update bearer resp fail  : 0
Modify bearer requests       : 0
Modify bearer resp succ      : 0          Modify bearer resp fail  : 0
Missing mand IE pkts Rx     : 0
Malformed packets Rx        : 0          Unknown message pkts Rx  : 0
Modify bearer commands       : 0          Modify bearer fail ind   : 0
Delete bearer commands       : 0          Delete bearer fail ind   : 0
Bearer resource commands     : 0          Bearer resource fail ind : 0
Suspend notification reqs    : 0
Suspend notification ack s*: 0          Suspend notification ack f*: 0
Resume notification reqs     : 0
Resume notification ack su*: 0          Resume notification ack fa*: 0
MBReq IE P-CSCF recovery    : 0          UBReq with P-CSCF servers : 0
P-CSCF Recovery DBReq       : 1
Number of sessions:         : 0

```

Create Session Requests with HI bit

```

-----
Ctxt not found init attach : 0
Context not found          : 0
Ctxt found with diff IP add: 0
-----

Number of peer instances : 4
=====
* indicates that the corresponding row element may have been truncated.

```

Example output S2b reference point failure code statistics ePDG/PGW combination gateways

```

# show mobile-gateway pdn ref-point-stats s2b failure-codes
=====
PDN gateway S2b Failure Code statistics
=====

```

Peer address : 16.70.16.24

Peer type : ePDG

Reference point : default

Failure code	Rx count	Tx count
Request Accepted	1	1
Request Rejected	0	0
Mandatory IE Incorrect	0	0
Mandatory IE Missing	0	0
Context Not Found	0	0
Invalid Length	0	0
Remote Peer Not Responding	0	0
No Resources Available	0	0
User Authentication Failed	0	0
Request accepted partially	0	0
New PDN type due to network preference	0	0
New PDN type due to single address bearer only	0	0
Missing or unknown APN	0	0
Preferred PDN type not supported	0	0
All dynamic addresses are occupied	0	0
APN access denied - no subscription	0	0
Service not supported	0	0
Unable to page UE due to Suspension	0	0
Conditional IE missing	0	0
Temporarily rejected due to H0 proc in progress	0	0
UE context without TFT already activated	0	0
Semantic error in the TFT operation	0	0
Syntactic error in the TFT operation	0	0
Semantic errors in packet filters	0	0
Syntactic errors in packet filters	0	0
Version not supported by next peer	0	0
Protocol type not supported	0	0
APN congestion	0	0
Denied in RAT	0	0
APN Restriction type incompatible	0	0
UE already re-attached	0	0
GRE key not found	0	0
MPDN connections for a given APN not allowed	0	0
Modifications not limited to S1-U bearers	0	0
Network Failure	0	0
QoS parameter mismatch	0	0
ISR Deactivation	0	0
RAT chgd 3GPP to non-3GPP	0	0
Reactivation requested	0	0
PDN reconnection to this APN disallowed	0	0
PDN connection inactivity timer expires	0	1
Access changed from Non-3GPP to 3GPP	0	0
Invalid message format	0	0
System failure	0	0
UE not responding	0	0
UE refuses	0	0
Unable to page UE	0	0
No memory available	0	0
IMSI/IMEI not known	0	0
Semantic error in the TAD operation	0	0
Syntactic error in the TAD operation	0	0

Invalid length of response/piggybacked init msg	0	0
Invalid reply from remote peer	0	0
Invalid peer	0	0
Bearer handling not supported	0	0
Idle recover	0	0
Others	0	0

Peer address : 16.70.16.19		
Peer type : PGW		
Reference point : default		

Failure code	Rx count	Tx count

Request Accepted	1	1
Request Rejected	0	0
Mandatory IE Incorrect	0	0
Mandatory IE Missing	0	0
Context Not Found	0	0
Invalid Length	0	0
Remote Peer Not Responding	0	0
No Resources Available	0	0
User Authentication Failed	0	0
Request accepted partially	0	0
New PDN type due to network preference	0	0
New PDN type due to single address bearer only	0	0
Missing or unknown APN	0	0
Preferred PDN type not supported	0	0
All dynamic addresses are occupied	0	0
APN access denied - no subscription	0	0
Service not supported	0	0
Unable to page UE due to Suspension	0	0
Conditional IE missing	0	0
Temporarily rejected due to H0 proc in progress	0	0
UE context without TFT already activated	0	0
Semantic error in the TFT operation	0	0
Syntactic error in the TFT operation	0	0
Semantic errors in packet filters	0	0
Syntactic errors in packet filters	0	0
Version not supported by next peer	0	0
Protocol type not supported	0	0
APN congestion	0	0
Denied in RAT	0	0
APN Restriction type incompatible	0	0
UE already re-attached	0	0
GRE key not found	0	0
MPDN connections for a given APN not allowed	0	0
Modifications not limited to S1-U bearers	0	0
Network Failure	0	0
QoS parameter mismatch	0	0
ISR Deactivation	0	0
RAT chgd 3GPP to non-3GPP	0	0
Reactivation requested	0	0
PDN reconnection to this APN disallowed	0	0
PDN connection inactivity timer expires	1	0
Access changed from Non-3GPP to 3GPP	0	0
Invalid message format	0	0
System failure	0	0

UE not responding	0	0
UE refuses	0	0
Unable to page UE	0	0
No memory available	0	0
IMSI/IMEI not known	0	0
Semantic error in the TAD operation	0	0
Syntactic error in the TAD operation	0	0
Invalid length of response/piggybacked init msg	0	0
Invalid reply from remote peer	0	0
Invalid peer	0	0
Bearer handling not supported	0	0
Idle recover	0	0

Others	0	0
Number of peer instances : 2		
=====		

Example output S2bu reference point peers

```
# show mobile-gateway pdn ref-point-peers s2bu
=====
PDN gateway s2bu reference point peers
=====
Peer address      : 10.70.24.151
Peer type         : ePDG
Router            : vprn20003
Reference point   : errInd
Path Mgmt State   : disabled
Create Time       : 03/20/2020 18:14:00      Gateway Id    : 1
GTP Version       : GTPv1
-----
Number of peers : 1
=====
```

Example output S2bu peer reference point statistics

```
# show mobile-gateway pdn ref-point-stats s2bu peer 10.70.24.151
=====
PDN gateway S2bu reference point statistics
=====
Peer address : 10.70.24.151
Reference point : errInd

Number of Bearers : 6
-----
Number of peer instances : 2
=====
```

Example output S4 reference point statistics

```
# show mobile-gateway pdn ref-point-stats s4
=====
PDN gateway S4 reference point statistics
=====
Peer address : 10.10.37.10
Peer type    : SGSN
```


Reference point : default

Example output S4u reference point peers

```
# show mobile-gateway pdn ref-point-peers s4u
=====
PDN gateway s4u reference point peers
=====
Peer address      : 6.6.6.1
Peer type         : SGSN
Router            : vprn20071
Reference point   : default
Path Mgmt State   : disabled
Create Time       : 03/20/2020 18:05:31      Gateway Id    : 1
GTP Version       : GTPv1
-----
Number of peers : 1
=====
```

Example output S4u reference point statistics

```
# show mobile-gateway pdn ref-point-stats s4u
=====
PDN gateway S4u reference point statistics
=====
Peer address      : 6.6.6.1
Reference point   : default

Number of Bearers: 6
-----
Number of peer instances : 2
=====
```

Example output S4u peer reference point statistics

```
show mobile-gateway pdn ref-point-stats s4u peer 6.6.6.1
=====
PDN gateway S4u reference point statistics
=====
Peer address : 6.6.6.1
Reference point      : default

Number of Bearers      : 6
-----
Number of peer instances : 2
=====
```

Example output S5 reference point failure code statistics

```
# show mobile-gateway pdn ref-point-stats s5 failure-codes
=====
PDN gateway S5 Failure Code statistics
=====
```

Peer address : 10.20.1.1		
Peer type : SGW		
Reference point : default		
Failure code	Rx count	Tx count

Request Accepted	1	1
Request Rejected	0	0
Mandatory IE Incorrect	0	0
Mandatory IE Missing	0	0
Context Not Found	0	0
Invalid Length	0	0
Remote Peer Not Responding	0	0
No Resources Available	0	0
User Authentication Failed	0	0
Request accepted partially	0	0
New PDN type due to network preference	0	0
New PDN type due to single address bearer only	0	0
Missing or unknown APN	0	0
Preferred PDN type not supported	0	0
All dynamic addresses are occupied	0	0
APN access denied - no subscription	0	0
Service not supported	0	1
Unable to page UE due to Suspension	0	0
Conditional IE missing	0	0
Temporarily rejected due to H0 proc in progress	0	0
UE context without TFT already activated	0	0
Semantic error in the TFT operation	0	0
Syntactic error in the TFT operation	0	0
Semantic errors in packet filters	0	0
Syntactic errors in packet filters	0	0
Version not supported by next peer	0	0
Protocol type not supported	0	0
APN congestion	0	0
Denied in RAT	0	0
APN Restriction type incompatible	0	0
UE already re-attached	0	0
GRE key not found	0	0
MPDN connections for a given APN not allowed	0	0
Modifications not limited to S1-U bearers	0	0
Network Failure	0	0
QoS parameter mismatch	0	0
ISR Deactivation	0	0
RAT chgd 3GPP to non-3GPP	0	0
Reactivation requested	0	0
PDN reconnection to this APN disallowed	0	0
PDN connection inactivity timer expires	0	0
Invalid message format	0	0
System failure	0	0
UE not responding	0	0
UE refuses	0	0
Unable to page UE	0	0
No memory available	0	0
IMSI/IMEI not known	0	0
Semantic error in the TAD operation	0	0
Syntactic error in the TAD operation	0	0
Invalid length of response/piggybacked init msg	0	0
Invalid reply from remote peer	0	0

```

Invalid peer                                0          0
Bearer handling not supported                0          0
Idle recover                                0          0
-----
Others                                      0          0
Number of peer instances : 1
=====

```

Example output S5 reference point peers

```

# show mobile-gateway pdn ref-point-peers s5 instance
=====
PDN gateway S5 reference point statistics per-peer instance
=====
Peer address : 10.70.1.1
Router          : Base
Port            : 2123      Chassis/Card          : 1/A
Reference point : default

Create session requests      : 2
Create session resp succ    : 0      Create session resp fail : 0
Delete session requests     : 0
Delete session resp succ    : 0      Delete session resp fail : 0
Create bearer requests      : 0
Create bearer resp succ     : 0      Create bearer resp fail  : 0
Delete bearer requests      : 0
Delete bearer resp succ     : 0      Delete bearer resp fail  : 0
Update bearer requests      : 0
Update bearer resp succ     : 0      Update bearer resp fail  : 0
Modify bearer requests      : 0
Modify bearer resp succ     : 0      Modify bearer resp fail  : 0
Malformed packets Rx       : 0      Unknown message pkts Rx  : 0
Modify bearer commands      : 0      Modify bearer fail ind   : 0
Delete bearer commands      : 0      Delete bearer fail ind   : 0
Missing mand IE pkts Rx    : 0
Bearer resource commands    : 0      Bearer resource fail ind : 0
Suspend notification reqs   : 0      Suspend notification acks : 0
Resume notification reqs    : 0      Resume notification acks  : 0
Number of sessions:        : 0

-----
Number of peer instances : 1
=====

```

Example output S5 reference point card statistics

```

# show mobile-gateway pdn ref-point-stats s5 card 3
=====
PDN gateway S5 reference point statistics
=====
Peer address : 10.20.1.1
Router          : Base
Port            : 2123

Create session requests      : 0
Create session resp succ    : 1      Create session resp fail : 0
Delete session requests     : 0

```

```

Delete session resp succ : 0      Delete session resp fail : 0
Create bearer requests   : 1
Create bearer resp succ  : 1      Create bearer resp fail  : 0
Delete bearer requests   : 0
Delete bearer resp succ  : 0      Delete bearer resp fail  : 0
Update bearer requests   : 0
Update bearer resp succ  : 0      Update bearer resp fail  : 0
Modify bearer requests   : 0
Modify bearer resp succ  : 0      Modify bearer resp fail  : 0
Malformed packets Rx    : 0      Unknown message pkts Rx : 0
Modify bearer commands   : 0      Modify bearer fail ind   : 0
Delete bearer commands   : 0      Delete bearer fail ind   : 0
Missing mand IE pkts Rx : 0
Bearer resource commands : 0      Bearer resource fail ind : 0
Suspend notification reqs : 0     Suspend notification acks : 0
Resume notification reqs  : 0     Resume notification acks  : 0
-----
Number of peer instances : 1
=====

```

Example output S5 reference point statistics

```

# show mobile-gateway pdn ref-point-stats s5
=====
PDN gateway S5 reference point statistics
=====
Peer address : 10.70.24.23
Peer type    : SGW
Reference point : default
Create session requests : 5
Create session resp succ : 1      Create session resp fail : 1
CS req teid non zero    : 0
CS unique req teid non zero: 0
DSCP value              : 0
Delete session requests : 0
Delete session unique req : 0
Delete session resp succ : 0      Delete session resp fail : 0
Create bearer requests   : 0
Create bearer resp succ  : 0      Create bearer resp fail   : 0
Delete bearer requests   : 0
Delete bearer resp succ  : 0      Delete bearer resp fail   : 0
Update bearer requests   : 0
Update bearer resp succ  : 0      Update bearer resp fail   : 0
Modify bearer requests   : 0
Modify bearer unique req : 0
Modify bearer resp succ  : 0      Modify bearer resp fail   : 0
Missing mand IE pkts Rx : 0
Malformed packets Rx    : 0      Unknown message pkts Rx : 0
Modify bearer commands   : 0      Modify bearer fail ind   : 0
Delete bearer commands   : 0      Delete bearer fail ind   : 0
Bearer resource commands : 0      Bearer resource fail ind : 0
Suspend notification reqs : 0
Suspend notification ack s*: 0     Suspend notification ack f*: 0
Resume notification reqs  : 0
Resume notification ack su*: 0     Resume notification ack fa*: 0
MBReq with pause charging : 0      MBReq with unpause charging: 0

```

```

MBReq IE P-CSCF recovery : 0
MBReq with CSG           : 0
Number of sessions:      : 0
MBReq for SGW restoration : 0
-----

```

Example output S5 reference point aggregate statistics

```

# show mobile-gateway pdn ref-point-stats s5 aggregate
=====
PDN gateway S5 reference point statistics aggregate
=====
Create session requests      : 3
Create session resp succ    : 3      Create session resp fail : 0
Delete session requests     : 1
Delete session resp succ    : 1      Delete session resp fail : 0
Create bearer requests      : 0
Create bearer resp succ     : 0      Create bearer resp fail  : 0
Delete bearer requests      : 0
Delete bearer resp succ     : 0      Delete bearer resp fail  : 0
Update bearer requests      : 0
Update bearer resp succ     : 0      Update bearer resp fail  : 0
Modify bearer requests      : 0
Modify bearer resp succ     : 0      Modify bearer resp fail  : 0
Malformed packets Rx       : 0      Unknown message pkts Rx  : 0
Modify bearer commands      : 0      Modify bearer fail ind   : 0
Delete bearer commands      : 0      Delete bearer fail ind   : 0
Missing mand IE pkts Rx    : 0
Bearer resource commands    : 0      Bearer resource fail ind : 0
Suspend notification reqs   : 0      Suspend notification acks : 0
Resume notification reqs    : 0      Resume notification acks  : 0
Number of sessions:        : 2
-----
Number of peer instances : 1
=====

```

Example output S5 reference point P-CSCF recovery statistics

This example output shows MBRs sent with private IE indicating P-CSCF recovery.

```

# show mobile-gateway pdn ref-point-stats s5
=====
PDN gateway S5 reference point statistics
=====
Peer address : 16.16.16.16
Peer type    : PGW
Reference point : default

Create session requests      : 3      Create session unique requ*: 0
Create session resp succ    : 3      Create session resp fail : 0
Delete session requests     : 3      Delete session unique requ*: 0
Delete session resp succ    : 3      Delete session resp fail : 0
Create bearer requests      : 0
Create bearer resp succ     : 0      Create bearer resp fail  : 0

```

```

Delete bearer requests      : 0
Delete bearer resp succ    : 0      Delete bearer resp fail    : 0
Update bearer requests     : 0
Update bearer resp succ    : 0      Update bearer resp fail    : 0
Modify bearer requests     : 0      Modify bearer unique reque*: 0
Modify bearer resp succ    : 0      Modify bearer resp fail    : 0
Missing mand IE pkts Rx   : 0
Malformed packets Rx      : 0      Unknown message pkts Rx   : 0
Modify bearer commands     : 0      Modify bearer fail ind     : 0
Delete bearer commands     : 0      Delete bearer fail ind     : 0
Bearer resource commands   : 0      Bearer resource fail ind   : 0
Suspend notification reqs  : 0
Suspend notification ack s*: 0      Suspend notification ack f*: 0
Resume notification reqs   : 0
Resume notification ack su*: 0      Resume notification ack fa*: 0
MBReq with pause charging  : 0      MBReq with unpause charging: 0
MBReq IE P-CSCF recovery   : 0
MBReq with CSG             : 0
Number of sessions:        : 0

```

```
-----
Number of peer instances : 1
=====
```

* indicates that the corresponding row element may have been truncated.

Example output S5 reference point eMPS advance priority calls

```

# show mobile-gateway pdn ref-point-stats s5
=====
PDN gateway S5 reference point statistics
=====
Peer address : 10.10.5.1
Peer type    : SGW
Reference point      : default

Create session requests      : 1
Create session resp succ    : 1      Create session resp fail    : 0
CS req teid non zero        : 0
CS unique req teid non zero: 0
DSCP value                  : 0
Delete session requests     : 0
Delete session unique req   : 0
Delete session resp succ    : 0      Delete session resp fail    : 0
Create bearer requests      : 1
Create bearer resp succ     : 1      Create bearer resp fail     : 0
Delete bearer requests      : 0
Delete bearer resp succ     : 0      Delete bearer resp fail     : 0
Update bearer requests      : 0
Update bearer resp succ     : 0      Update bearer resp fail     : 0
Modify bearer requests      : 0
Modify bearer unique req    : 0
Modify bearer resp succ     : 0      Modify bearer resp fail     : 0
eMPS MP-Flag-priority req   : 2      eMPS MP-Flag-priority resp : 2
Missing mand IE pkts Rx    : 0

```

```

Malformed packets Rx      : 0          Unknown message pkts Rx      : 0
Modify bearer commands    : 0          Modify bearer fail ind    : 0
Delete bearer commands    : 0          Delete bearer fail ind    : 0
Bearer resource commands  : 0          Bearer resource fail ind   : 0
Suspend notification reqs : 0
Suspend notification ack s*: 0          Suspend notification ack f*: 0
Resume notification reqs  : 0
Resume notification ack su*: 0          Resume notification ack fa*: 0
MBReq with pause charging : 0          MBReq with unpause charging: 0
MBReq IE P-CSCF recovery  : 0          UBReq with P-CSCF servers  : 0
MBReq with CSG            : 0          P-CSCF Recovery DBReq     : 0
MBReq for SGW restoration : 0
Number of sessions:        : 1

```

```

-----
Number of peer instances : 1
=====

```

* indicates that the corresponding row element may have been truncated.

Example output S5u reference point peers

```

# show mobile-gateway pdn ref-point-peers s5u
=====
PDN gateway S5u reference point peers
=====
Peer address      : 3ffe::a0a:501
Peer type         : SGW
Router           : vprn20003
Reference point   : default
Path Mgmt State   : disabled
Create Time       : 03/20/2020 18:01:27          Gateway Id      : 1
GTP Version       : GTPv1
-----
Number of peers : 1
=====

```

Example output S5u reference point statistics

```

# show mobile-gateway pdn ref-point-stats s5u
=====
PDN gateway S5u reference point statistics
=====
Peer address      : 3ffe::a0a:501
Reference point    : default

Number of Bearers      : 6
-----
Number of peer instances : 2
=====

```

Example output S5u peer reference point statistics

```

# show mobile-gateway pdn ref-point-stats s5u peer 3ffe::a0a:501
=====
PDN gateway S5u reference point statistics
=====
Peer address : 3ffe::a0a:501

```

```
Reference point          : default
```

```
Number of Bearers       : 6
```

```
-----
Number of peer instances : 2
=====
```

Example output S6b reference point statistics

```
# show mobile-gateway pdn reference-point-stats s6b
=====
PDN gateway S6b reference point statistics
=====
Peer address : 10.40.46.213
Router       : vprn2651
Port        : 3868
AAR Initial Attach Tx      : 276      AAR Extension Tx          : 3
AAR Detach Tx              : 0         AAR Reauth Tx             : 0
AAA Initial Attach Rx      : 276      AAA Extension Rx          : 3
AAA Detach Rx              : 0         AAA Reauth Rx             : 0
AAA Success Rx             : 275      AAA Reject Rx            : 4
RA Request Rx              : 0         RA Answer Tx              : 0
ST Request Tx              : 271      ST Answer Rx              : 271
AS Request Rx              : 0         AS Answer Tx              : 0
AAA Missing AVP Pkts Rx    : 0         RAR Missing AVP Pkts Rx   : 0
STA Missing AVP Pkts Rx    : 0         ASR Missing AVP Pkts Rx   : 0
AAA Unknown Sess Pkts Rx   : 0         AAR Retries               : 0
STR Retries                : 0         AAR Final Timeout Tx      : 0
STR Unk Sess Tx            : 0         STR Final Timeout Tx      : 0
ASA Unk Sess Tx            : 0         RAA Unk Sess Tx           : 0
AAA Malformed Pkts Rx      : 0         AAA Bad AVP Val Rx        : 0
STA Malformed Pkts Rx      : 0         STA Bad AVP Val Rx        : 0
RAR Bad AVP Val Rx         : 0         RAR Duplicate Rx          : 0
ASR Bad AVP Val Rx         : 0         ASR Duplicate Rx          : 0
AAR Redirection Tx         : 0         AAA Redirection Rx        : 0
AAA Unk Redir Host Rx      : 0         AAR Redirection Fail Tx   : 0
RAR with P-CSCF Recovery   : 0
NodeIdentifier match       : 10         NodeIdentifier no-match    : 5
Node Identifier Match AAR Initial Attach Tx
Success                    : 0         Failure                    : 0
-----
Number of peer instances : 1
```

Example output S6b reference point summary statistics

```
# show mobile-gateway pdn reference-point-stats s6b summary
=====
PDN gateway S6b reference point statistics summary
=====
Diameter profile : prof1
Latest:
Router           : Base
Port             : 3868

AAR Initial Attach Tx      : 0         AAR Extension Tx          : 0
AAR Detach Tx              : 0         AAR Reauth Tx             : 0
```


AAA Initial Attach Rx	: 0	AAA Extension Rx	: 0
AAA Detach Rx	: 0	AAA Reauth Rx	: 0
AAA Success Rx	: 0	AAA Reject Rx	: 0
RA Request Rx	: 0	RA Answer Tx	: 0
ST Request Tx	: 0	ST Answer Rx	: 0
AS Request Rx	: 0	AS Answer Tx	: 0
AAA Missing AVP Pkts Rx	: 0	RAR Missing AVP Pkts Rx	: 0
STA Missing AVP Pkts Rx	: 0	ASR Missing AVP Pkts Rx	: 0
AAA Unknown Sess Pkts Rx	: 0	AAR Retries	: 0
STR Retries	: 0	AAR Final Timeout Tx	: 0
STR Unk Sess Tx	: 0	STR Final Timeout Tx	: 0
ASA Unk Sess Tx	: 0	RAA Unk Sess Tx	: 0
AAA Malformed Pkts Rx	: 0	AAA Bad AVP Val Rx	: 0
STA Malformed Pkts Rx	: 0	STA Bad AVP Val Rx	: 0
RAR Bad AVP Val Rx	: 0	RAR Duplicate Rx	: 0
ASR Bad AVP Val Rx	: 0	ASR Duplicate Rx	: 0
AAR Redirection Tx	: 0	AAA Redirection Rx	: 0
AAA Unk Redir Host Rx	: 0	AAR Redirection Fail Tx	: 0
Rate Limit AAR Drops : 20			
Rate Limit STR Drops : 22			
Rate Limit ASA Drops : 15			
Rate Limit RAA Drops : 08			

Example output S6b reference point card statistics

```
# show mobile-gateway pdn ref-point-stats s6b card 3
=====
PDN gateway S6b reference point statistics
=====
Peer address : 10.10.19.10
Router          : Base
Port            : 3868

AAR Initial Attach Tx      : 0      AAR Extension Tx          : 0
AAR Detach Tx              : 0      AAR Reauth Tx             : 0
AAA Initial Attach Rx      : 0      AAA Extension Rx          : 0
AAA Detach Rx              : 0      AAA Reauth Rx             : 0
AAA Success Rx             : 0      AAA Reject Rx             : 0
RA Request Rx              : 0      RA Answer Tx              : 0
ST Request Tx              : 0      ST Answer Rx              : 0
AS Request Rx              : 0      AS Answer Tx              : 0
AAA Missing AVP Pkts Tx    : 0      RAR Missing AVP Pkts Rx   : 0
STA Missing AVP Pkts Tx    : 0      ASR Missing AVP Pkts Rx   : 0
AAA Unknown Sess Pkts Rx   : 0      AAR Retries               : 0
STR Retries                : 0
RAR with P-CSCF Recovery   : 1

-----
Number of peer instances : 1
=====
```

Example output S6b reference point failure code statistics

```
# show mobile-gateway pdn ref-point-stats s6b failure-codes
=====
```

PDN gateway S6b Failure Code Statistics

```

=====
Peer address : 10.23.19.2
Router       : vprn20005
Port        : 3868

Rx Error Codes (AAA)
3002 Unable to deliver      : 0          3004 Too busy          : 0
3005 Loop detected         : 0          3007 Application unsupport*: 0
3009 Invalid AVP bits      : 0
4001 Authentication reject*: 0          5001 AVP unsupported       : 0
5002 Unknown-session-id    : 0          5003 Authorization rejected: 0
5004 Invalid AVP value     : 0          5005 Missing AVP          : 0
5006 Resources exceeded    : 0          5008 AVP not allowed      : 0
5012 Unable to comply      : 0
Other-3xxx                 : 0          Other-4xxx                 : 0
Other-5xxx                 : 0

Rx Experimental Error Codes (AAA)
5001 User unknown          : 0          5003 Identity not register*: 0
5004 Roaming Not Allowed   : 0          5005 Identity already regi*: 0
5198 Overload control      : 0          5199 Newer session detected: 0
5450 No Non 3GPP subscript*: 0          5451 No APN subscription   : 0
5452 RAT type not allowed  : 0

Tx Error Codes (ASA)
5002 Unknown-session-id    : 0          5012 Unable to comply      : 0
Tx Error Codes (RAA)
5002 Unknown-session-id    : 0          5012 Unable to comply      : 0
Rx Error Codes (STA)
3002 Unable to deliver      : 0          3004 Too busy          : 0
4002 Diameter Out Of Space : 0          5001 AVP unsupported       : 0
5002 Unknown-session-id    : 0          5004 Invalid AVP value     : 0
5005 Missing AVP           : 0          5006 Resources exceeded    : 0
5008 AVP not allowed       : 0          5012 Unable to Comply      : 0
Other-3xxx                 : 0          Other-4xxx                 : 0
Other-5xxx                 : 0

Rx Experimental Error Codes (STA)
5005 Identity already regi*: 0          5198 Overload control      : 0
Other-3xxx                 : 0          Other-4xxx                 : 0
Other-5xxx                 : 0

Tx Termination Causes (STR)
Logout                     : 0          Service Not Provided       : 0
Bad answer                 : 0          Administrative             : 0
Session timeout            : 0

```

```

-----
Number of peer instances : 1
=====

```

Example output S6b reference point statistics per-peer Instance

```

# show mobile-gateway pdn ref-point-stats s6b peer 10.23.19.14 instance

```

```

=====
PDN gateway S6b reference point statistics per-peer instance
=====

```

```

Peer address : 10.23.19.14
Router       : vprn20005
Port        : 3868
Gateway/Chassis/Card : 1/1/4

```

```

AAR Initial Attach Tx      : 0          AAR Extension Tx          : 0
AAR Detach Tx              : 0          AAR Reauth Tx              : 0
AAA Initial Attach Rx      : 0          AAA Extension Rx           : 0
AAA Detach Rx              : 0          AAA Reauth Rx              : 0
AAA Success Rx             : 0          AAA Reject Rx              : 0
RA Request Rx              : 0          RA Answer Tx               : 0
ST Request Tx              : 0          ST Answer Rx               : 0
AS Request Rx              : 0          AS Answer Tx               : 0
AAA Missing AVP Pkts Rx    : 0          RAR Missing AVP Pkts Rx    : 0
STA Missing AVP Pkts Rx    : 0          ASR Missing AVP Pkts Rx    : 0
AAA Unknown Sess Pkts Rx   : 0          AAR Retries                : 0
STR Retries                : 0          AAR Final Timeout Tx       : 0
STR Unk Sess Tx            : 0          STR Final Timeout Tx       : 0
ASA Unk Sess Tx            : 0          RAA Unk Sess Tx            : 0
AAA Malformed Pkts Rx      : 0          AAA Bad AVP Val Rx         : 0
STA Malformed Pkts Rx      : 0          STA Bad AVP Val Rx         : 0
RAR Bad AVP Val Rx         : 0          RAR Duplicate Rx           : 0
ASR Bad AVP Val Rx         : 0          ASR Duplicate Rx           : 0
AAR Redirection Tx         : 0          AAA Redirection Rx         : 0
AAA Unk Redir Host Rx      : 0          AAR Redirection Fail Tx    : 0
RAR with P-CSCF Recovery   : 0
Rate Limit AAR Drops       : 0
Rate Limit STR Drops       : 0
Rate Limit ASA Drops       : 0
Rate Limit RAA Drops       : 0

```

```

-----
Number of peer instances : 1
=====

```

Example output S6b reference point statistics per-peer Diameter overload indication conveyance

```

# show mobile-gateway pdn ref-point-stats s6b peer 10.23.19.14
=====
PDN gateway S6b reference point statistics
=====
Peer address : 10.23.19.14
Router                : vprn20005
Port                  : 3868

AAR Initial Attach Tx      : 494          AAR Extension Tx          : 0
AAR Detach Tx              : 0            AAR Reauth Tx              : 0
AAA Initial Attach Rx      : 494          AAA Extension Rx           : 0
AAA Detach Rx              : 0            AAA Reauth Rx              : 0
AAA Success Rx             : 494          AAA Reject Rx              : 0
RA Request Rx              : 0            RA Answer Tx               : 0
ST Request Tx              : 369          ST Answer Rx               : 368
AS Request Rx              : 0            AS Answer Tx               : 0
AAA Missing AVP Pkts Rx    : 0            RAR Missing AVP Pkts Rx    : 0
STA Missing AVP Pkts Rx    : 0            ASR Missing AVP Pkts Rx    : 0
AAA Unknown Sess Pkts Rx   : 0            AAR Retries                : 0
STR Retries                : 0            AAR Final Timeout Tx       : 0
STR Unk Sess Tx            : 0            STR Final Timeout Tx       : 0
ASA Unk Sess Tx            : 0            RAA Unk Sess Tx            : 0
AAA Malformed Pkts Rx      : 0            AAA Bad AVP Val Rx         : 0
STA Malformed Pkts Rx      : 0            STA Bad AVP Val Rx         : 0

```

```

RAR Bad AVP Val Rx      : 0      RAR Duplicate Rx      : 0
ASR Bad AVP Val Rx      : 0      ASR Duplicate Rx      : 0
AAR Redirection Tx      : 0      AAA Redirection Rx     : 0
AAA Unk Redir Host Rx   : 0      AAR Redirection Fail Tx : 0
RAR with P-CSCF Recovery : 0
AAR-I dropped due to OC-OLR: 506
Rate Limit AAR Drops    : 0
Rate Limit STR Drops    : 0
Rate Limit ASA Drops    : 0
Rate Limit RAA Drops    : 0

```

```

-----
Number of peer instances : 1
=====

```

Example output S8 reference point statistics

```

# show mobile-gateway pdn ref-point-stats s8
=====
PDN gateway S8 reference point statistics
=====
Peer address : 10.10.6.1
Router      : Base
Port        : 2123

Create session requests      : 1
Create session resp succ    : 1      Create session resp fail : 0
Delete session requests     : 1
Delete session resp succ    : 1      Delete session resp fail : 0
Create bearer requests      : 0
Create bearer resp succ     : 0      Create bearer resp fail  : 0
Delete bearer requests      : 0
Delete bearer resp succ     : 0      Delete bearer resp fail  : 0
Update bearer requests      : 0
Update bearer resp          : 0      Update bearer resp       : 0
Modify bearer requests      : 0
Modify bearer resp succ     : 0      Modify bearer resp fail  : 0
Malformed packets Rx        : 0      Unknown message pkts Rx : 0
Modify bearer commands      : 0      Modify bearer fail ind   : 0
Delete bearer commands      : 0      Delete bearer fail ind   : 0
Missing mand IE pkts Rx     : 0
Bearer resource commands    : 0      Bearer resource fail ind : 0
MBReq for SGW restoration   : 0
-----
Number of peer instances : 1
=====

```

Example output S8 Peer reference point statistics

```

# show mobile-gateway pdn ref-point-stats s8 peer 10.10.6.3 port 2123 router Base
=====
Serving gateway S8 reference point statistics
=====
Peer address : 10.10.6.3
Router      : Base

```

```

Port                               : 2123

Create session requests             : 1
Create session resp succ            : 1      Create session resp fail    : 0
Delete session requests             : 0
Delete session resp succ            : 0      Delete session resp fail    : 0
Create bearer requests              : 0
Create bearer resp succ             : 0      Create bearer resp fail     : 0
Delete bearer requests              : 0
Delete bearer resp succ             : 0      Delete bearer resp fail     : 0
Update bearer requests              : 0
Update bearer resp                  : 0      Update bearer resp          : 0
Modify bearer requests              : 0
Modify bearer resp succ             : 0      Modify bearer resp fail     : 0
Malformed packets Rx                : 0      Unknown message pkts Rx    : 0
Modify bearer commands              : 0      Modify bearer fail ind      : 0
Delete bearer commands              : 0      Delete bearer fail ind      : 0
Missing mand IE pkts Rx            : 0
Bearer resource commands            : 0      Bearer resource fail ind    : 0
-----
Number of peer instances : 1
=====

```

Example output S8u reference point peers

```

# show mobile-gateway pdn ref-point-peers s8u
=====
PDN gateway s8u reference point peers
=====
Peer address      : 16.80.16.10
Peer type         : PGW
Router            : vprn20071
Reference point   : simPgwS8
Path Mgmt State   : disabled
Create Time       : 03/20/2020 18:22:47      Gateway Id       : 1
GTP Version       : GTPv1
-----
Number of peers : 1
=====

```

Example output S8u reference point statistics

```

# show mobile-gateway pdn ref-point-stats s8u peer 16.80.16.10
=====
PDN gateway S8u reference point statistics
=====
Peer address      : 16.80.16.10
Reference point    : simPgwS8

Number of Bearers      : 4
-----
Number of peer instances : 2
=====

```

Example output S11 reference point P-CSF recovery statistics

This example output shows MBRs received with private IE indicating P-CSCF recovery.

```
# show mobile-gateway pdn ref-point-stats s11
=====
PDN gateway S11 reference point statistics
=====
Peer address : 10.10.7.10
Peer type    : MME
Reference point      : default

Create session requests      : 3          Create session req redirect: 0
Create session resp succ    : 3          Create session resp fail   : 0
Delete session requests     : 0
Delete session unique req   : 0
Delete session resp succ    : 0          Delete session resp fail   : 0
Create bearer requests      : 0
Create bearer resp succ     : 0          Create bearer resp fail    : 0
Delete bearer requests      : 0
Delete bearer resp succ     : 0          Delete bearer resp fail    : 0
Update bearer requests      : 0
Update bearer resp succ     : 0          Update bearer resp fail    : 0
Modify bearer requests      : 3
Modify bearer unique req    : 3
Modify bearer resp succ     : 3          Modify bearer resp fail    : 0
Release access bearers req  : 0
Rel access bearrs rsp succ  : 0          Rel access bearrs rsp fail : 0
Modify access bearers req   : 0
Mod acc bearers resp succ   : 0          Mod acc bearers resp fail  : 0
Create indirect tnl req     : 0
Create indirect tnl rsp suc : 0          Create indirect tnl rsp fl : 0
Delete indirect tnl req     : 0
Delete indirect tnl rsp suc : 0          Delete indirect tnl rsp fl : 0
Downlink notifications Tx   : 0
Downlink acks Rx succ      : 0          Downlink acks Rx fail     : 0
DL fail notifications Rx    : 0
DDN retries due to MME chg  : 0
Missing mand IE pkts Rx    : 0
Malformed packets Rx       : 0          Unknown message pkts Rx   : 0
Modify bearer commands      : 0          Modify bearer fail ind    : 0
Delete bearer commands      : 0          Delete bearer fail ind    : 0
Bearer resource commands    : 0          Bearer resource fail ind  : 0
Suspend notification reqs   : 0
Suspend notification ack s* : 0          Suspend notification ack f* : 0
Resume notification reqs    : 0
Resume notification ack su* : 0          Resume notification ack fa* : 0
Rel Acc Brr with radio loss : 0
MBReq IE P-CSCF recovery    : 1          UBReq with P-CSCF servers : 1
MBReq with CSG              : 0          P-CSCF Recovery DBReq     : 0
Number of UEs               : 1

-----
Number of peer instances : 1
=====
```

Example output S11 reference point eMPS advance priority calls

```
# show mobile-gateway pdn ref-point-stats s11
=====
PDN gateway S11 reference point statistics
=====
Peer address : 10.10.7.10
Peer type    : MME
Reference point      : default

Create session requests      : 1
Create session resp succ    : 1          Create session resp fail    : 0
CS req teid non zero        : 0
CS unique req teid non zero: 0
DSCP value                  : 0
Delete session requests     : 0
Delete session unique req   : 0
Delete session resp succ    : 0          Delete session resp fail    : 0
Create bearer requests      : 1
Create bearer resp succ     : 1          Create bearer resp fail    : 0
Delete bearer requests      : 0
Delete bearer resp succ     : 0          Delete bearer resp fail    : 0
Update bearer requests      : 0
Update bearer resp succ     : 0          Update bearer resp fail    : 0
Modify bearer requests      : 1
Modify bearer unique req    : 1
Modify bearer resp succ     : 1          Modify bearer resp fail    : 0
Release access bearers req  : 0
Rel access bearrs rsp succ  : 0          Rel access bearrs rsp fail : 0
Modify access bearers req   : 0
Mod acc bearers resp succ   : 0          Mod acc bearers resp fail  : 0
Create indirect tnl req     : 0
Create indirect tnl rsp suc : 0          Create indirect tnl rsp fl : 0
Delete indirect tnl req     : 0
Delete indirect tnl rsp suc : 0          Delete indirect tnl rsp fl : 0
Downlink notifications Tx   : 0
Downlink acks Rx succ      : 0          Downlink acks Rx fail     : 0
DDN Ack with extended buff  : 0
DL fail notifications Rx    : 0
DDN retries due to MME chg  : 0
eMPS MP-Flag-priority req   : 3          eMPS MP-Flag-priority resp : 3
Missing mand IE pkts Rx    : 0
Malformed packets Rx       : 0          Unknown message pkts Rx   : 0
Modify bearer commands      : 0          Modify bearer fail ind     : 0
Delete bearer commands      : 0          Delete bearer fail ind     : 0
Bearer resource commands    : 0          Bearer resource fail ind   : 0
Suspend notification reqs   : 0
Suspend notification ack s* : 0          Suspend notification ack f* : 0
Resume notification reqs    : 0
Resume notification ack su* : 0          Resume notification ack fa* : 0
Rel Acc Brr with radio loss : 0
MBReq IE P-CSCF recovery    : 0          UBReq with P-CSCF servers  : 0
MBReq with CSG              : 0          P-CSCF Recovery DBReq     : 0
Number of UEs               : 1
-----
Number of peer instances : 1
```

```
=====
* indicates that the corresponding row element may have been truncated.
```

Example output S11u reference point statistics

```
# show mobile-gateway pdn ref-point-stats s11u
=====
PDN gateway S11u reference point statistics
=====
Peer address : 6.6.6.1
Reference point: default
Number of bearers : 142
-----
Number of peer instances : 2
=====
```

Example output S12 reference point peers

```
# show mobile-gateway pdn ref-point-peers s12
=====
PDN gateway s12 reference point peers
=====
Peer address : 6.6.6.1
Peer type : RNC
Router : vprn20071
Reference point : default
Path Mgmt State : up
Create Time : 03/20/2020 18:09:23 Gateway Id : 1
GTP Version : GTPv1
-----
Number of peers : 1
=====
```

Example output S12 reference point statistics

```
#show mobile-gateway pdn ref-point-stats s12 peer 6.6.6.1
=====
PDN gateway S12 reference point statistics
=====
Peer address : 6.6.6.1
Reference point : default

Number of Bearers : 6
-----
Number of peer instances : 2
=====
```

Example output SWm reference point statistics

```
# show mobile-gateway pdn ref-point-stats swm
=====
PDN gateway Swm reference point statistics
=====
Peer address : 10.10.80.10
Router : Base
```



```

Port                               : 3868

DER Initial Attach Tx              : 1      DER Subsequent Attach Tx      : 1
DEA Multi-round-auth Rx            : 1      DEA Other-result-code Rx      : 0
AAR Extension Tx                   : 0      AAA Extension Rx              : 0
AAR Reauth Tx                      : 0      AAA Reauth Rx                 : 0
RA Request Rx                      : 0      RA Answer Tx                  : 0
ST Request Tx                      : 0      ST Answer Rx                  : 0
AS Request Rx                      : 0      AS Answer Tx                  : 0
DEA Success Rx                     : 1
AAA Success Rx                     : 0      AAA Reject Rx                 : 0
STA Success Rx                     : 0
DER Retries                        : 0      AAR Retries                   : 0
STR Retries                        : 0      DER Final Timeout Tx         : 0
AAR Final Timeout Tx               : 0      STR Final Timeout Tx         : 0
STR Unk Sess DEA Tx                : 0      STR Unk Sess AAA Tx          : 0
ASA Unk Sess Tx                    : 0      RAA Unk Sess Tx              : 0
DEA Invalid Pkts Rx                : 0      AAA Invalid Pkts Rx          : 0
STA Invalid Pkts Rx                : 0      RAR Invalid Pkts Rx          : 0
ASR Invalid Pkts Rx                : 0      DEA Redirection Rx           : 0
DEA Redirection Fail Rx            : 0      DER Failure Indication Tx    : 1
ASR Unassociated Session Rx: 0      RAR Unassociated Session Rx: 0
Rate Limit DER Drops               : 100
Rate Limit AAR Drops               : 54
Rate Limit STR Drops               : 67
Rate Limit ASA Drops               : 0
Rate Limit RAA Drops               : 5

```

Example output SWm failure codes statistics

```

# show mobile-gateway pdn ref-point-stats swm failure-codes
=====
PDN gateway Swm Failure Code Statistics
=====
Peer address : 10.10.80.10
Router                               : Base
Port                                 : 3868

Rx Error Codes (DEA)
3002 Unable to deliver                : 0      3004 Too busy                  : 0
3005 Loop detected                    : 0      3007 Application unsupport*: 0
3009 Invalid AVP bits                 : 0
4001 Authentication reject*: 0      5001 AVP unsupported           : 0
5002 Unknown-session-id               : 0      5003 Authorization rejected: 0
5004 Invalid AVP value                : 0      5005 Missing AVP              : 0
5006 Resources exceeded               : 0      5008 AVP not allowed          : 0
5012 Unable to comply                 : 0
Other 3xxx                            : 0      Other-4xxx                     : 0
Other 5xxx                            : 0

Rx Experimental Error Codes (DEA)
5001 User unknown                    : 0      5003 Identity not register*: 0
5004 Roaming Not Allowed              : 0      5005 Identity already regi*: 0
5450 No Non 3GPP subscript*: 0      5451 No APN subscription      : 0
5452 RAT type not allowed             : 0
Rx Error Codes (AAA)
3002 Unable to deliver                : 0      3004 Too busy                  : 0

```

```

3005 Loop detected           : 0          3006 Redirect indication    : 0
3007 Application unsupport*: 0          3009 Invalid AVP bits       : 0
4001 Authentication reject*: 0          5001 AVP unsupported       : 0
5002 Unknown-session-id     : 0          5003 Authorization rejected: 0
5004 Invalid AVP value      : 0          5005 Missing AVP           : 0
5006 Resources exceeded     : 0          5008 AVP not allowed       : 0
5012 Unable to comply       : 0
Other 3xxx                  : 0          Other-4xxx                 : 0
Other 5xxx                  : 0
Rx Experimental Error Codes (AAA)
5001 User unknown           : 0          5003 Identity not register*: 0
5004 Roaming Not Allowed    : 0          5005 Identity already regi*: 0
5450 No Non 3GPP subscript*: 0          5451 No APN subscription   : 0
5452 RAT type not allowed   : 0
Tx Error Codes (ASA)
5002 Unknown-session-id     : 0          5012 Unable to comply       : 0
Tx Error Codes (RAA)
5002 Unknown-session-id     : 0          5012 Unable to comply       : 0
Rx Error Codes (STA)
3002 Unable to deliver      : 0          3004 Too busy              : 0
4002 Diameter Out Of Space : 0          5001 AVP unsupported       : 0
5002 Unknown-session-id     : 0          5004 Invalid AVP value     : 0
5005 Missing AVP            : 0          5006 Resources exceeded    : 0
5008 AVP not allowed        : 0          5012 Unable to Comply      : 0
Other-3xxx                  : 0          Other-4xxx                 : 0
Other-5xxx                  : 0
Rx Experimental Error Codes (STA)
5005 Identity already regi*: 0          5198 Overload control      : 0
Other-3xxx                  : 0          Other-4xxx                 : 0
Other-5xxx                  : 0
Tx Termination Causes (STR)
Logout                      : 1          Service Not Provided       : 0
Bad answer                  : 0          Administrative             : 0
Session timeout             : 0
-----
Number of peer instances : 1
=====

```

Example output SWu reference point statistics

```

# show mobile-gateway pdn ref-point-stats swu
=====
PDN gateway Swu reference point statistics
=====
Chassis/Card                : 1/3          Gateway                    : 2
Number of IKE SA             : 0            Number of ESP SA          : 0
Established IKE SA           : 0            Established ESP SA        : 0
IKE SA re-keys               : 0            ESP SA re-keys           : 0
UE initiated IKE SA re-keys: 0            UE initiated ESP SA re-keys: 0
ePDG initiated IKE SA re-k*: 0            ePDG initiated ESP SA re-k*: 0
IKE SA re-keying collisions: 0            ESP SA re-keying collisions: 0
Half-opened IKE SA          : 0            Esta. IKE SA supp. REDIRECT: 10
Number of Redirected UEs    : 8
Received IKE messages
-----
IKE msgs                     : 0            Discarded msgs            : 0

```

```

IKE fragments                : 0          Fragments discarded (TO) : 0
Fragments discarded (OF)    : 0          Fragments discarded (BF) : 0
Fragments discarded (NFS)   : 0
NAT keep-alive msgs         : 0          IKE SA INIT req msgs    : 0
Create child SA req msgs    : 0          Create child SA resp msgs : 0
Informational req msgs      : 0          Informational resp msgs   : 0
IKE auth req msgs           : 0          Retransmitted msgs       : 0
Dead peer detect req msgs   : 0          Dead peer detect rsp msgs : 0
MOBIKE update SA req msgs   : 0          MOBIKE ret route chck resps: 0
IMEI/IMEISV rsp msgs        : 0
Transmitted IKE messages
-----
IKE msgs                     : 0          Unsuccessful msgs        : 0
IKE fragments                : 0
NAT keep-alive msgs         : 0          IKE SA INIT resp msgs    : 0
Create child SA req msgs    : 0          Create child SA resp msgs : 0
Informational req msgs      : 0          Informational resp msgs   : 0
IKE auth req msgs           : 0          Locally retransmit msgs   : 0
Dead peer detect req msgs   : 0          Dead peer detect rsp msgs : 0
MOBIKE update SA rsp msgs   : 0          MOBIKE ret route chck reqs : 0
IMEI/IMEISV req msgs        : 0
Further IKE statistics
-----
Unanswered request IKE messages : 0
Internal errors during processing of IKE messages : 0
IKE messages with missing non-ESP marker : 0
NAT keepalive msgs with invalid length : 0
IKE messages with invalid length : 0
IKE messages with invalid header format : 0
IKE messages with invalid payload format : 0
IKE_SA_INIT messages with non-zero message ID : 0
Discarded messages due to many half-opened IKE SA : 0
IKE request messages with invalid SPI : 0
IKE response messages with invalid SPI : 0
Received late retransmissions of IKE messages : 0
Received retransmissions of too old IKE messages : 0
Received invalid IKE response messages : 0
Received duplicated IKE response messages : 0
Not authenticated IKE messages : 0
Received messages with missing encryption payload : 0
Expired IKE_SA_INIT exchanges : 0
Received IKE dynamic address updates : 0
Received ESP dynamic address updates : 0
Rejected Attaches cause of White List : 0
Rejected Attaches cause of Black List : 0
IKE messages with missing IMEI/IMEISV : 0
IKE messages with invalid IMEI/IMEISV : 0
Emergency Call Req : 0
ePDG initiated IKE SA delete due to re-key disabled : 0
ePDG initiated ESP SA delete due to re-key disabled : 0
Rejected Attaches cause of user in locked out state : 0
Redirects cause of always-on-existing sessions : 0
Redirects cause of always-on act-to-idle-sessions : 0
Redirects cause of overload conditions-existing sessions: 0
Redirects cause of Source IP-existing sessions(MOBIKE) : 0
Redirects cause of realm part of IDi : 0
Emergency_Call_Numbers Notify Payload sent to Ue : 0

```

```

UE attach with CERTREQ and EAP_ONLY_AUTHENTICATION : 0
UE 5GS Capable Initial Attach Request : 0
UE 5GS Capable Handover Attach Request : 0
IPSMG statistics
-----
Tx ESP and GTP packets : 0
Tx ESP and GTP bytes : 0
Tx ESP packets : 0
Tx ESP bytes : 0
Tx inbound ESP packets : 0
Tx outbound GTP packets : 0
Rx ESP and GTP packets : 0
Rx ESP and GTP bytes : 0
Rx ESP packets : 0
Rx ESP bytes : 0
Rx fragmented ESP packets : 0
Rx non-IP packets : 0
Rx IP packets with error : 0
Rx IP packets with L4 error : 0
Rx IP packets with exception : 0
Rx IP packets with invalid length : 0
Rx IP packets on wrong internal port : 0
Rx IP packets with internal error : 0
Rx inbound IP packets with a crypto error : 0
Rx inbound IP packets with an authentication error : 0
Rx inbound IP packets with missing crypto function args : 0
Rx inbound IP packets with improper crypto packet len : 0
Rx inbound IP packets with improper crypto conf key len : 0
Rx inbound IP packets with improper crypto MAC secret : 0
Rx inbound IP packets with other decrypt/auth failure : 0
Rx inbound IP packets with an invalid pad length : 0
Rx inbound IP packets with an invalid pad byte : 0
Rx outbound fragmented IP packets : 0
Rx outbound IP packets with invalid IPv4 header length : 0
Rx outbound IP packets with invalid IPv6 header length : 0
Rx outbound IP packets with invalid UDP header length : 0
Rx outbound IP packets with invalid UDP port : 0
Rx outbound IP packets with invalid IP protocol : 0
Rx outbound IP packets with invalid SADB entry : 0
Rx outbound IP packets on an inactive SA : 0
Rx outbound IP packets with zero sequence number : 0
Rx outbound IP packets with a crypto error : 0
Rx outbound IP packets with missing crypto function args: 0
Rx outbound IP packets with improper crypto packet len : 0
Rx outbound IP packets with improper crypto conf key len: 0
Rx outbound IP packets with improper crypto MAC secret : 0
Rx outbound IP packets with other decrypt/auth failure : 0
Rx ESP packets with invalid IPv4 header length : 0
Rx ESP packets with invalid IPv6 header length : 0
Rx ESP packets with invalid UDP header length : 0
Rx ESP packets with invalid UDP port : 0
Rx ESP packets with invalid IPv4 protocol : 0
Rx ESP packets with invalid ESP length : 0
Rx ESP packets with invalid ESP SPI : 0
Rx ESP packets with invalid ESP sequence number : 0
Rx ESP packets with invalid SADB entry : 0
Rx ESP packets with invalid encapsulation : 0

```

```

Rx ESP packets with stale ESP sequence number      : 0
Rx ESP packets with duplicate ESP sequence number  : 0
Rx ESP packets with ESP SPI pointing to inactive SA : 0
Rx ESP packets with invalid tunnel endpoint address type: 0
Rx ESP packets with invalid remote tunnel endpoint addr : 0
Rx ESP packets with invalid remote tunnel endpoint port : 0
Rx ESP packets with invalid local tunnel endpoint addr : 0
Unsuccessfully transmitted packets from data plane card : 0

```

```
-----
Chassis/Card          : 1/A      Gateway          : 2

```

CPM IKE statistics

```

-----
Rx messages                      : 0
Rx discarded messages            : 0
Rx IKE_SA_INIT request messages : 0
Tx messages                      : 0
Tx error messages                : 0
Tx socket full messages         : 0
Tx IKE_SA_INIT response messages : 0
Remote request retried messages : 0
Messages forwarded to control plane ISA/VM : 0
Messages with audit clean       : 0
IP fragment messages            : 0
Bad UDP checksum messages       : 0
Invalid destination IP address messages : 0
Gateway down messages           : 0
Internal error messages         : 0
NATT missing non-ESP marker messages : 0
NATT invalid non-ESP marker messages : 0
NATT keepalive messages with invalid length : 0
Messages with invalid length    : 0
Messages with invalid header    : 0
Messages with invalid payloads  : 0
IKE_SA_INIT messages with non-zero ID : 0
IKE_SA_INIT request messages with non-null response SPI : 0
Too many larval IKE SA messages : 0
Too many larval IKE SA messages from peer : 0
Request messages with invalid SPI : 0
Messages with cookie mismatch   : 0
IKE_SA_INIT expired messages    : 0
Redirects cause of always-on-new-attach : 0
Redirects cause of overload conditions-new attach : 0
Redirects cause of Source IP-new attach : 0
REDIRECTED_FROM received       : 0

```

```
-----
Number of cards : 2

```

```
=====
* indicates that the corresponding row element may have been truncated.

```

Example output SWu reference point failure codes statistics

```
# show mobile-gateway pdn ref-point-stats swu failure-codes
```

PDN gateway SWu Failure Code statistics

```
-----
Chassis/Card          : 1/3      Gateway          : 2

```

```

Attach reject
-----
IKE SA proposal not chosen                : 0
ESP SA proposal not chosen                : 0
ESP SA proposal not chosen (emergency attach) : 0
IKE SA invalid KE payload                 : 0
Traffic Selector doesn't match            : 0
Traffic Selector doesn't match (emergency attach) : 0
PDN type not supported                    : 0
APN access denied - no AAA subscription   : 0
Unauthenticated_emergency_not_supported  : 0
IMEI_not_accepted                         : 0
Re-keying reject
-----
ePDG IKE SA proposal not chosen           : 0
ePDG ESP SA proposal not chosen           : 0
ePDG IKE SA invalid KE payload            : 0
ePDG ESP SA invalid KE payload            : 0
ePDG ignored IKE SA rekey request         : 0
ePDG ignored ESP SA rekey request         : 0
UE IKE SA proposal not chosen             : 0
UE ESP SA proposal not chosen             : 0
UE IKE SA invalid KE payload              : 0
UE ESP SA invalid KE payload              : 0
MOBIKE reject
-----
UPDATE_SA_ADDRESSES received while MOBIKE not negotiated : 0
UPDATE_SA_ADDRESSES received with invalid source address : 0
Return routability check failed due to timeout           : 0
Return routability check failed due to COOKIE2 mismatch : 0
-----
Number of cards : 1
=====

```

Example output SWu reference point statistics ePDG/PGW combination gateways

```

# show mobile-gateway pdn ref-point-stats swu
=====
PDN gateway Swu reference point statistics
=====
Chassis/Card          : 1/3      Gateway          : 1
Number of IKE SA      : 0        Number of ESP SA   : 0
Established IKE SA    : 0        Established ESP SA  : 0
IKE SA re-keys        : 0        ESP SA re-keys     : 0
IKE SA re-keying collisions: 0    ESP SA re-keying collisions: 0
Half-opened IKE SA    : 0
Received IKE messages
-----
IKE msgs              : 4        Discarded msgs     : 0
IKE fragments         : 4        Fragments discarded (T0) : 0
Fragments discarded (OF) : 0
NAT keep-alive msgs   : 0        IKE SA INIT req msgs : 1
Create child SA req msgs : 0      Create child SA resp msgs : 0
Informational req msgs : 0        Informational resp msgs : 0
IKE auth req msgs     : 0        Retransmitted msgs  : 0
Dead peer detect req msgs : 0    Dead peer detect rsp msgs : 0

```

```

MOBIKE update SA req msgs : 0          MOBIKE ret route chk resp: 0
Transmitted IKE messages
-----
IKE msgs : 4          Unsuccessful msgs : 0
IKE fragments : 4
NAT keep-alive msgs : 0          IKE SA INIT resp msgs : 1
Create child SA req msgs : 0      Create child SA resp msgs : 0
Informational req msgs : 0        Informational resp msgs : 0
IKE auth req msgs : 3            Locally retransmit msgs : 0
Dead peer detect req msgs : 0      Dead peer detect rsp msgs : 0
MOBIKE update SA rsp msgs : 0      MOBIKE ret route chk reqs : 0
Further IKE statistics
-----
Unanswered request IKE messages : 0
Internal errors during processing of IKE messages : 0
IKE messages with missing non-ESP marker : 0
NAT keepalive msgs with invalid length : 0
IKE messages with invalid length : 0
IKE messages with invalid header format : 0
IKE messages with invalid payload format : 0
IKE_SA_INIT messages with non-zero message ID : 0
Discarded messages due to many half-opened IKE SA : 0
IKE request messages with invalid SPI : 0
IKE response messages with invalid SPI : 0
Received late retransmissions of IKE messages : 0
Received retransmissions of too old IKE messages : 0
Received invalid IKE response messages : 0
Received duplicated IKE response messages : 0
Not authenticated IKE messages : 0
Received messages with missing encryption payload : 0
Expired IKE_SA_INIT exchanges : 0
Received IKE dynamic address updates : 0
Received ESP dynamic address updates : 0
IPSMG statistics
-----
Tx ESP and GTP packets : 0
Tx ESP and GTP bytes : 0
Tx ESP packets : 0
Tx ESP bytes : 0
Tx inbound ESP packets : 0
Tx outbound GTP packets : 0
Rx ESP and GTP packets : 0
Rx ESP and GTP bytes : 0
Rx ESP packets : 0
Rx ESP bytes : 0
Rx fragmented ESP packets : 0
Rx non-IP packets : 0
Rx IP packets with error : 0
Rx IP packets with L4 error : 0
Rx IP packets with exception : 0
Rx IP packets with invalid length : 0
Rx IP packets on wrong internal port : 0
Rx IP packets with internal error : 0
Rx inbound IP packets with a crypto error : 0
Rx inbound IP packets with an authentication error : 0
Rx inbound IP packets with missing crypto function args : 0
Rx inbound IP packets with improper crypto packet len : 0

```

```

Rx inbound IP packets with improper crypto conf key len : 0
Rx inbound IP packets with improper crypto MAC secret : 0
Rx inbound IP packets with other decrypt/auth failure : 0
Rx inbound IP packets with an invalid pad length : 0
Rx inbound IP packets with an invalid pad byte : 0
Rx outbound fragmented IP packets : 0
Rx outbound IP packets with invalid IPv4 header length : 0
Rx outbound IP packets with invalid IPv6 header length : 0
Rx outbound IP packets with invalid UDP header length : 0
Rx outbound IP packets with invalid UDP port : 0
Rx outbound IP packets with invalid IP protocol : 0
Rx outbound IP packets with invalid SADB entry : 0
Rx outbound IP packets on an inactive SA : 0
Rx outbound IP packets with zero sequence number : 0
Rx outbound IP packets with a crypto error : 0
Rx outbound IP packets with missing crypto function args: 0
Rx outbound IP packets with improper crypto packet len : 0
Rx outbound IP packets with improper crypto conf key len: 0
Rx outbound IP packets with improper crypto MAC secret : 0
Rx outbound IP packets with other decrypt/auth failure : 0
Rx ESP packets with invalid IPv4 header length : 0
Rx ESP packets with invalid IPv6 header length : 0
Rx ESP packets with invalid UDP header length : 0
Rx ESP packets with invalid UDP port : 0
Rx ESP packets with invalid IPv4 protocol : 0
Rx ESP packets with invalid ESP length : 0
Rx ESP packets with invalid ESP SPI : 0
Rx ESP packets with invalid ESP sequence number : 0
Rx ESP packets with invalid SADB entry : 0
Rx ESP packets with invalid encapsulation : 0
Rx ESP packets with stale ESP sequence number : 0
Rx ESP packets with duplicate ESP sequence number : 0
Rx ESP packets with ESP SPI pointing to inactive SA : 0
Rx ESP packets with invalid tunnel endpoint address type: 0
Rx ESP packets with invalid remote tunnel endpoint addr : 0
Rx ESP packets with invalid remote tunnel endpoint port : 0
Rx ESP packets with invalid local tunnel endpoint addr : 0
Unsuccessfully transmitted packets from data plane card : 0

```

```

-----
Chassis/Card : 1/4 Gateway : 2

Number of IKE SA : 0 Number of ESP SA : 0
Established IKE SA : 0 Established ESP SA : 0
IKE SA re-keys : 0 ESP SA re-keys : 0
IKE SA re-keying collisions: 0 ESP SA re-keying collisions: 0
Half-opened IKE SA : 0
Received IKE messages
-----
IKE msgs : 5 Discarded msgs : 0
NAT keep-alive msgs : 0 IKE SA INIT req msgs : 1
Create child SA req msgs : 0 Create child SA resp msgs : 0
Informational req msgs : 0 Informational resp msgs : 1
IKE auth req msgs : 3 Retransmitted msgs : 0
Dead peer detect req msgs : 0 Dead peer detect rsp msgs : 0
MOBIKE update SA req msgs : 0 MOBIKE ret route chck resps: 0
Transmitted IKE messages
-----

```



```

IKE msgs                : 5           Unsuccessful msgs        : 0
NAT keep-alive msgs     : 0           IKE SA INIT resp msgs   : 1
Create child SA req msgs : 0           Create child SA resp msgs : 0
Informational req msgs   : 1           Informational resp msgs   : 0
IKE auth req msgs        : 3           Locally retransmit msgs   : 0
Dead peer detect req msgs : 0          Dead peer detect rsp msgs : 0
MOBIKE update SA rsp msgs : 0          MOBIKE ret route chck reqs : 0
Further IKE statistics
-----
Unanswered request IKE messages                : 0
Internal errors during processing of IKE messages : 0
IKE messages with missing non-ESP marker         : 0
NAT keepalive msgs with invalid length          : 0
IKE messages with invalid length                : 0
IKE messages with invalid header format         : 0
IKE messages with invalid payload format        : 0
IKE_SA_INIT messages with non-zero message ID   : 0
Discarded messages due to many half-opened IKE SA : 0
IKE request messages with invalid SPI           : 0
IKE response messages with invalid SPI          : 0
Received late retransmissions of IKE messages   : 0
Received retransmissions of too old IKE messages : 0
Received invalid IKE response messages          : 0
Received duplicated IKE response messages       : 0
Not authenticated IKE messages                  : 0
Received messages with missing encryption payload : 0
Expired IKE_SA_INIT exchanges                   : 0
Received IKE dynamic address updates             : 0
Received ESP dynamic address updates             : 0
IPSMG statistics
-----
Tx ESP and GTP packets                : 0
Tx ESP and GTP bytes                   : 0
Tx ESP packets                        : 0
Tx ESP bytes                          : 0
Tx inbound ESP packets                : 0
Tx outbound GTP packets               : 0
Rx ESP and GTP packets                : 0
Rx ESP and GTP bytes                  : 0
Rx ESP packets                       : 0
Rx ESP bytes                         : 0
Rx fragmented ESP packets             : 0
Rx non-IP packets                    : 0
Rx IP packets with error              : 0
Rx IP packets with L4 error           : 0
Rx IP packets with exception          : 0
Rx IP packets with invalid length     : 0
Rx IP packets on wrong internal port   : 0
Rx IP packets with internal error     : 0
Rx inbound IP packets with a crypto error : 0
Rx inbound IP packets with an authentication error : 0
Rx inbound IP packets with missing crypto function args : 0
Rx inbound IP packets with improper crypto packet len : 0
Rx inbound IP packets with improper crypto conf key len : 0
Rx inbound IP packets with improper crypto MAC secret : 0
Rx inbound IP packets with other decrypt/auth failure : 0
Rx inbound IP packets with an invalid pad length : 0

```

```

Rx inbound IP packets with an invalid pad byte           : 0
Rx outbound fragmented IP packets                       : 0
Rx outbound IP packets with invalid IPv4 header length  : 0
Rx outbound IP packets with invalid IPv6 header length  : 0
Rx outbound IP packets with invalid UDP header length   : 0
Rx outbound IP packets with invalid UDP port            : 0
Rx outbound IP packets with invalid IP protocol         : 0
Rx outbound IP packets with invalid SADB entry          : 0
Rx outbound IP packets on an inactive SA                : 0
Rx outbound IP packets with zero sequence number        : 0
Rx outbound IP packets with a crypto error              : 0
Rx outbound IP packets with missing crypto function args: 0
Rx outbound IP packets with improper crypto packet len  : 0
Rx outbound IP packets with improper crypto conf key len: 0
Rx outbound IP packets with improper crypto MAC secret  : 0
Rx outbound IP packets with other decrypt/auth failure  : 0
Rx ESP packets with invalid IPv4 header length          : 0
Rx ESP packets with invalid IPv6 header length          : 0
Rx ESP packets with invalid UDP header length           : 0
Rx ESP packets with invalid UDP port                   : 0
Rx ESP packets with invalid IPv4 protocol               : 0
Rx ESP packets with invalid ESP length                  : 0
Rx ESP packets with invalid ESP SPI                     : 0
Rx ESP packets with invalid ESP sequence number         : 0
Rx ESP packets with invalid SADB entry                  : 0
Rx ESP packets with invalid encapsulation               : 0
Rx ESP packets with stale ESP sequence number           : 0
Rx ESP packets with duplicate ESP sequence number       : 0
Rx ESP packets with ESP SPI pointing to inactive SA     : 0
Rx ESP packets with invalid tunnel endpoint address type: 0
Rx ESP packets with invalid remote tunnel endpoint addr : 0
Rx ESP packets with invalid remote tunnel endpoint port : 0
Rx ESP packets with invalid local tunnel endpoint addr  : 0
Unsuccessfully transmitted packets from data plane card : 0
-----
Chassis/Card      : 1/A      Gateway      : 1
CPM IKE statistics
-----
Rx messages              : 1
Rx discarded messages    : 0
Rx IKE_SA_INIT request messages : 1
Tx messages              : 0
Tx error messages        : 0
Tx socket full messages  : 0
Tx IKE_SA_INIT response messages : 0
Remote request retried messages : 0
Messages forwarded to control plane ISA/VM : 1
Messages with audit clean : 0
IP fragment messages     : 0
Bad UDP checksum messages : 0
Invalid destination IP address messages : 0
Gateway down messages    : 0
Internal error messages  : 0
NATT missing non-ESP marker messages : 0
NATT invalid non-ESP marker messages : 0
NATT keepalive messages with invalid length : 0
Messages with invalid length : 0

```

```

Messages with invalid header           : 0
Messages with invalid payloads         : 0
IKE_SA_INIT messages with non-zero ID  : 0
IKE_SA_INIT request messages with non-null response SPI : 0
Too many larval IKE SA messages        : 0
Too many larval IKE SA messages from peer : 0
Request messages with invalid SPI      : 0
Messages with cookie mismatch          : 0
IKE_SA_INIT expired messages           : 0
-----
Number of cards : 3
=====

```

Example output SWu reference point failure codes statistics ePDG/PGW combination gateways

```

# show mobile-gateway pdn ref-point-stats swu failure-codes
=====
PDN gateway SWu Failure Code Statistics
=====
Chassis/Card      : 1/3      Gateway      : 1
Attach reject
-----
IKE SA proposal not chosen           : 0
ESP SA proposal not chosen           : 0
IKE SA invalid KE payload            : 0
Traffic Selector doesn't match       : 0
PDN type not supported               : 0
APN access denied - no AAA subscription : 0
Unauthenticated Emergency not supported : 0
IMEI not accepted                    : 0
Re-keying reject
-----
ePDG IKE SA proposal not chosen      : 0
ePDG ESP SA proposal not chosen      : 0
ePDG IKE SA invalid KE payload       : 0
ePDG ESP SA invalid KE payload       : 0
UE IKE SA proposal not chosen        : 0
UE ESP SA proposal not chosen        : 0
UE IKE SA invalid KE payload         : 0
UE ESP SA invalid KE payload         : 0
MOBIKE reject
-----
UPDATE_SA_ADDRESSES received while MOBIKE not negotiated : 0
UPDATE_SA_ADDRESSES received with invalid source address : 0
Return routability check failed due to timeout           : 0
Return routability check failed due to COOKIE2 mismatch : 0
-----
Chassis/Card      : 1/4      Gateway      : 2
Attach reject
-----
IKE SA proposal not chosen           : 0
ESP SA proposal not chosen           : 0
IKE SA invalid KE payload            : 0
Traffic Selector doesn't match       : 0
PDN type not supported               : 0
APN access denied - no AAA subscription : 0

```

```

Re-keying reject
-----
ePDG IKE SA proposal not chosen           : 0
ePDG ESP SA proposal not chosen           : 0
ePDG IKE SA invalid KE payload            : 0
ePDG ESP SA invalid KE payload            : 0
UE IKE SA proposal not chosen             : 0
UE ESP SA proposal not chosen             : 0
UE IKE SA invalid KE payload              : 0
UE ESP SA invalid KE payload              : 0
MOBIKE reject
-----
UPDATE_SA_ADDRESSES received while MOBIKE not negotiated : 0
UPDATE_SA_ADDRESSES received with invalid source address : 0
Return routability check failed due to timeout           : 0
Return routability check failed due to COOKIE2 mismatch  : 0
-----
Number of cards : 2
=====

```

Example output SWu reference point statistics for PLMN-Black-White-List

```

# show mobile-gateway pdn ref-point-stats swu plmn-black-white-list
=====
PDN gateway Swu reference point statistics for PLMN-black-white-list
=====
Chassis/Card      : 1/3  Gateway           : 1
Rejected Attaches cause of PLMN White List           : 0
Rejected Attaches cause of PLMN Black List            : 0
Rejected Attaches cause of IDI-REALM White List       : 0
Rejected Attaches cause of IDI-REALM Black List       : 0
-----
Number of cards : 1
=====

```

Example output SWu reference point statistics throttling state

```

# show mobile-gateway pdn ref-point-stats swu throttling
=====
PDN gateway Swu reference point throttling state and statistics
=====
Chassis/Card      : 1/3  Gateway           : 2
IKE OLC state & statistics
-----
IKE congestion level      : none
IKE CPU Idle              : 100%
IKE message delay         : 0ms
Number of suppressed ePDG initiated IKE rekeys       : 0
Number of redirected ePDG initiated IKE rekeys       : 0
Number of suppressed ePDG initiated ESP rekeys       : 0
Number of redirected ePDG initiated ESP rekeys       : 0
Number of suppressed UE initiated IKE rekeys (IKE+ESP) : 0
Number of redirected UE initiated IKE rekeys (IKE+ESP) : 0
Number of dropped IKE-SA INIT requests                : 0
Number of redirected IKE-SA INIT requests             : 0
Number of dropped IKE AUTH requests                   : 0

```

```

Number of dropped IKE messages with best-effort      : 0
Number of dropped INFORMATIONAL requests due to OLC  : 0
-----
Chassis/Card          : 1/A          Gateway          : 2
IKE CPM's OLC statistics
-----
Number of redirected IKE-SA INIT requests due to OLC : 0
Number of dropped IKE-SA INIT requests due to OLC    : 0
-----
Number of cards : 2
=====

```

Example output SWw and SWw tunnel/RADIUS reference point statistics

```

show mobile-gateway pdn ref-point-stats sww tunnel
=====
PDN gateway SWw-tunnel reference point statistics
=====
Tunnel Type          : Soft Gre *
Peer Address         : 3ffe::c903:10a
Peer Type            : AP
Reference Point       : swwrefpt1

Discover Packets Rx   : 1           Request Packets Rx       : 1
Inform Packets Rx     : 0
Release Packets Rx    : 0           Decline Packets Rx       : 0
Offer Packets Tx      : 1           Ack Packets Tx           : 1
Nak Packets Tx        : 0

Unknown Host Packets Rx : 0
Malformed Packets Rx    : 0           Unknown Message Pkts Rx : 0
Number of UE           : 0
Router Sol Packets Rx   : 0           Neighbor Sol Packets Rx  : 0
Router Adv Packets Tx   : 0           Neighbor Adv Packets Tx  : 0
UE ARP Requests Rx     : 0           UE ARP Responses Tx      : 0
AP MAC Discovery
ARP-Requests-over-GRE Tx : 0           ARP-Responses-over-GRE Rx : 0
Neighbor Sol over GRE Tx : 0           Neighbor Adv over GRE Rx  : 0
SHCV
ARP Requests Tx        : 0           ARP Responses Rx         : 0
Neighbor Sol Packets Tx : 0           Neighbor Adv Packets Rx   : 0
-----
Number of peer instances : 1
=====

```

```

# show mobile-gateway pdn ref-point-stats sww radius
=====
PDN gateway SWw radius reference point statistics
=====
Local IP Address      : 9.9.9.10
Remote IP Address     : 201.3.2.10
Gateway/Chassis/Card  : 0/0/0
Access Requests Rx    : 2           Access Challenge Tx      : 1
Access Reject Tx      : 0           Access Accept Tx         : 1
Mand Attr Missing     : 0           Mand Attr Error          : 0
Opt Attr Error        : 0           Auth Error               : 0

```

```

Unexp Code           : 0           Unsupported Error       : 0
Accounting Request Start Rx : 0       Accounting Request Stop Rx : 0
Accounting Request Interim Rx: 0
Accounting Request On Rx   : 0       Accounting Request Off Rx  : 0
Accounting Response Tx     : 0       Disconnect Request Tx      : 0
Disconnect Ack Rx         : 0       Disconnect Nak Rx         : 0
-----
Number of peer instances : 1
=====

```

Example output STa reference point statistics

```

# show mobile-gateway pdn ref-point-stats sta
=====
PDN gateway Sta reference point statistics
=====
Peer address : 180.1.2.200
Router       : vprn20005
Port        : 3868

DER Initial Attach Tx      : 1          DER Subsequent Attach Tx : 1
DEA Multi-round-auth Rx   : 1          DEA Other-result-code Rx : 0
DER Re-authentication Tx  : 0
AAR Extension Tx          : 0          AAA Extension Rx        : 0
AAR Reauth Tx             : 0          AAA Reauth Rx           : 0
RA Request Rx             : 0          RA Answer Tx            : 0
ST Request Tx             : 0          ST Answer Rx            : 0
AS Request Rx             : 0          AS Answer Tx            : 0
DEA Success Rx            : 1
AAA Success Rx            : 0          AAA Reject Rx           : 0
STA Success Rx            : 0
DER Retries               : 0          AAR Retries             : 0
STR Retries               : 0          DER Final Timeout Tx    : 0
AAR Final Timeout Tx      : 0          STR Final Timeout Tx    : 0
STR Unk Sess DEA Tx       : 0          STR Unk Sess AAA Tx     : 0
ASA Unk Sess Tx           : 0          RAA Unk Sess Tx         : 0
DEA Invalid Pkts Rx       : 0          AAA Invalid Pkts Rx     : 0
STA Invalid Pkts Rx       : 0          RAR Invalid Pkts Rx     : 0
ASR Invalid Pkts Rx       : 0          DEA Redirection Rx      : 0
DEA Redirection Fail Rx   : 0
Rate Limit DER Drops      : 100
Rate Limit AAR Drops      : 54
Rate Limit STR Drops      : 67
Rate Limit ASA Drops      : 0
Rate Limit RAA Drops      : 5
-----
Number of peer instances : 1
=====

```

Example output STa reference point failure code statistics

```

# show mobile-gateway pdn ref-point-stats sta failure-codes
=====
PDN gateway STa Failure Code Statistics
=====
Peer address : 180.1.2.200

```

```

Router          : vprn20005
Port            : 3868

Rx Error Codes (DEA)
3002 Unable to deliver      : 0      3004 Too busy          : 0
3005 Loop detected          : 0      3007 Application unsupport*: 0
3009 Invalid AVP bits       : 0
4001 Authentication reject.: 0      5001 AVP unsupported        : 0
5002 Unknown session-id     : 0      5003 Authorization rejected: 0
5004 Invalid AVP value      : 0      5005 Missing AVP           : 0
5006 Resources exceeded     : 0      5008 AVP not allowed        : 0
5012 Unable to comply       : 0
Other 3xxx                 : 0      Other 4xxx                  : 0
Other 5xxx                 : 0

Rx Experimental Error Codes (DEA)
5001 User unknown           : 0      5003 Identity not register.: 0
5004 Roaming not allowed    : 0      5005 already Identity regi*: 0
5450 No non 3GPP subscript.: 0      5451 User no APN subscript.: 0
5452 RAT type not allowed   : 0

Rx Error Codes (AAA)
3002 Unable to deliver      : 0      3004 Too busy          : 0
3005 Loop detected          : 0      3006 Redirect indication   : 0
3007 Application unsupport*: 0      3009 Invalid AVP bits       : 0
4001 Authentication reject.: 0      5001 AVP unsupported        : 0
5002 Unknown session-id     : 0      5003 Authorization rejected: 0
5004 Invalid AVP value      : 0      5005 Missing AVP           : 0
5006 Resources exceeded     : 0      5008 AVP not allowed        : 0
5012 Unable to comply       : 0
Other 3xxx                 : 0      Other 4xxx                  : 0
Other 5xxx                 : 0

Rx Experimental Error Codes (AAA)
5001 User unknown           : 0      5003 Identity not register.: 0
5004 Roaming not allowed    : 0      5005 Identity already regi*: 0
5450 No non 3GPP subscript.: 0      5451 User no APN subscript.: 0
5452 RAT type not allowed   : 0

Tx Error Codes (ASA)
5002 Unknown session-id     : 0      5012 Unable to comply       : 0
Tx Error Codes (RAA)
5002 Unknown session-id     : 0      5012 Unable to comply       : 0
Rx Error Codes (STA)
3002 Unable to deliver      : 0      3004 Too busy          : 0
4002 Diameter Out Of Space : 0      5001 AVP unsupported        : 0
5002 Unknown-session-id     : 0      5004 Invalid AVP value      : 0
5005 Missing AVP            : 0      5006 Resources exceeded     : 0
5008 AVP not allowed        : 0      5012 Unable to Comply       : 0
Other-3xxx                  : 0      Other-4xxx                  : 0
Other-5xxx                  : 0

Rx Experimental Error Codes (STA)
5005 Identity already regi*: 0      5198 Overload control       : 0
Other-3xxx                  : 0      Other-4xxx                  : 0
Other-5xxx                  : 0

Tx Termination Causes (STR)
Logout                      : 0      Service not provided         : 0
Bad answer                  : 0      Administrative               : 0
Session timeout             : 0

-----
Number of peer instances : 1

```

=====

* indicates that the corresponding row element may have been truncated.

Example output Sx-N4 reference point statistics on CP function/session management function

```
# show mobile-gateway pdn ref-point-stats sx-n4
=====
PCF reference point statistics
=====

Peer address           : 192.168.34.1
Peer type              : User
Reference point        : default

PCF Session Related Messages

Session Establish Req   : 1           Session Establish Resp   : 1
Session Establish Resp fail: 0
Session Modify Req      : 0           Session Modify Resp      : 0
Session Modify Resp fail: 0
Session Deletion Req    : 0           Session Deletion Resp    : 0
Session Deletion Resp fail: 0
Session Report Req      : 0           Session Report Resp      : 0
Session Report Resp fail: 0
Session Report DDR Req  : 0           Session Report DDR Resp  : 0
Session Report DDR Rsp fail: 0
Session Report UR Req   : 0           Session Report UR Resp   : 0
Session Report UR Rsp fail: 0
Session Report EIR Req  : 0           Session Report EIR Resp  : 0
Session Report EIR Rsp fail: 0
Session Report UPIR Req : 0           Session Report UPIR Rsp  : 0
Session Report UPIR Rs fail: 0

PCF Session Audit Related Messages**

Session Modify Req      : 0           Session Modify Resp      : 0
Session Modify Resp fail: 0
Session Report Req      : 0           Session Report Resp      : 0
Session Report Rsp fail : 0

PCF-u Shared Tunnel Session Related Messages
Session Establish Req    : 33           Session Establish Resp    : 33
Session Establish Resp fail: 0

PCF Node Related Messages

Node Report Req         : 0           Node Report Resp         : 0
Node Report Resp fail   : 0
Assoc Setup Req Rx      : 0           Assoc Setup Resp Rx      : 1
Assoc Setup Resp fail Rx: 0
Assoc Setup Req Tx      : 5           Assoc Setup Resp Tx      : 0
Assoc Setup Resp fail Tx: 0
Assoc Upd Req Rx        : 0           Assoc Upd Resp Rx        : 0
Assoc Upd Resp fail Rx  : 0
```



```

Assoc Upd Req Tx      : 0          Assoc Upd Resp Tx      : 0
Assoc Upd Resp fail Tx : 0
Assoc Upd Req with ARR : 0          Assoc Upd Req with GRP : 0
Assoc Upd Req ARR & GRP : 0
Assoc Release Req     : 0          Assoc Release Resp     : 0
Assoc Release Resp fail : 0
Missing mand IE pkts Rx : 0        Malformed packets Rx   : 0
Unknown message pkts Rx : 0
Number of Sessions    : 1

```

```
-----
Number of peer instances : 2
-----
```

```

** These are all CUSTOM designed messages implemented for specific purpose:
to AUDIT the PFCP sessions on the control-plane and user-plane nodes at the
request of ADMIN initiated "tools" commands.
=====

```

Example output Sx-N4 reference point statistics on user plane function

```

# show mobile-gateway pdn ref-point-stats sx-n4
=====
PFCP reference point statistics
=====

Peer address      : 192.168.33.1
Peer type         : Control
Reference point   : default

PFCP Session Related Messages

Session Establish Req      : 1          Session Establish Resp      : 1
Session Establish Resp fail: 0
Session Modify Req        : 0          Session Modify Resp         : 0
Session Modify Resp fail  : 0
Session Deletion Req      : 0          Session Deletion Resp       : 0
Session Deletion Resp fail: 0
Session Report Req        : 0          Session Report Resp         : 0
Session Report Resp fail  : 0
Session Report DDR Req    : 0          Session Report DDR Rsp      : 0
Session Report DDR Rsp fail: 0
Session Report UR Req     : 0          Session Report UR Rsp       : 0
Session Report UR Rsp fail: 0
Session Report EIR Req    : 0          Session Report EIR Rsp      : 0
Session Report EIR Rsp fail: 0
Session Report UPIR Req   : 0          Session Report UPIR Rsp     : 0
Session Report UPIR Rs fail: 0

PFCP Session Audit Related Messages**

Session Modify Req        : 0          Session Modify Resp         : 0
Session Modify Resp fail  : 0
Session Report Req        : 0          Session Report Resp         : 0
Session Report Rsp fail   : 0

PFCP-u Shared Tunnel Session Related Messages
Session Establish Req      : 33          Session Establish Resp      : 33

```

```
Session Establish Resp fail: 0
```

PCFP Node Related Messages

Node Report Req	: 0	Node Report Resp	: 0
Node Report Resp fail	: 0		
Assoc Setup Req Rx	: 1	Assoc Setup Resp Rx	: 0
Assoc Setup Resp fail Rx	: 0		
Assoc Setup Req Tx	: 4	Assoc Setup Resp Tx	: 1
Assoc Setup Resp fail Tx	: 0		
Assoc Upd Req Rx	: 0	Assoc Upd Resp Rx	: 0
Assoc Upd Resp fail Rx	: 0		
Assoc Upd Req Tx	: 0	Assoc Upd Resp Tx	: 0
Assoc Upd Resp fail Tx	: 0		
Assoc Upd Req with ARR	: 0	Assoc Upd Req with GRP	: 0
Assoc Upd Req ARR & GRP	: 0		
Assoc Release Req	: 0	Assoc Release Resp	: 0
Assoc Release Resp fail	: 0		
Missing mand IE pkts Rx	: 0	Malformed packets Rx	: 0
Unknown message pkts Rx	: 0		
Number of Sessions	: 1		

```
-----
Number of peer instances : 2
-----
```

** These are all CUSTOM designed messages implemented for specific purpose:
to AUDIT the PCFP sessions on the control-plane and user-plane nodes at the
request of ADMIN initiated "tools" commands.

```
=====
```

Example output Sx-N4 reference point aggregate statistics

```
# show mobile-gateway pdn ref-point-stats sx-n4 aggregate
```

PCFP reference point statistics aggregate

PCFP Session Related Messages

Session Establish Req	: 0	Session Establish Resp	: 0
Session Establish Resp fail:	0		
Session Modify Req	: 0	Session Modify Resp	: 0
Session Modify Resp fail	: 0		
Session Deletion Req	: 0	Session Deletion Resp	: 0
Session Deletion Resp fail	: 0		
Session Report Req	: 0	Session Report Resp	: 0
Session Report Resp fail	: 0		
Session Report DDR Req	: 0	Session Report DDR Rsp	: 0
Session Report DDR Rsp fail:	0		
Session Report UR Req	: 0	Session Report UR Rsp	: 0
Session Report UR Rsp fail	: 0		
Session Report EIR Req	: 0	Session Report EIR Rsp	: 0
Session Report EIR Rsp fail:	0		

PCFP Session Audit Related Messages**

Session Modify Req	: 0	Session Modify Resp	: 0
Session Modify Resp fail	: 0		

```

Session Report Req      : 0          Session Report Rsp      : 0
Session Report Rsp fail : 0

PFCP-u Shared Tunnel Session Related Messages
Session Establish Req   : 33          Session Establish Rsp   : 33
Session Establish Rsp fail: 0

PFCP Node Related Messages
Node Report Req        : 0          Node Report Rsp        : 0
Node Report Rsp fail   : 0
Assoc Setup Req Rx     : 1          Assoc Setup Rsp Rx     : 1
Assoc Setup Rsp fail Rx : 0
Assoc Setup Req Tx     : 213        Assoc Setup Rsp Tx     : 1
Assoc Setup Rsp fail Tx : 0
Assoc Upd Req Rx       : 0          Assoc Upd Rsp Rx       : 0
Assoc Upd Rsp fail Rx  : 0
Assoc Upd Req Tx       : 0          Assoc Upd Rsp Tx       : 1
Assoc Upd Rsp fail Tx  : 0
Assoc Upd Req with ARR : 0          Assoc Upd Req with GRP : 0
Assoc Upd Req ARR & GRP : 1
Assoc Release Req      : 1          Assoc Release Rsp     : 1
Assoc Release Rsp fail : 0
Missing mand IE pkts Rx : 0          Malformed packets Rx  : 0
Unknown message pkts Rx : 0
Number of Sessions     : 0

-----
Number of peer instances : 4
-----

** These are all CUSTOM designed messages implemented for specific purpose:
to AUDIT the PFCP sessions on the control-plane and user-plane nodes at the
request of ADMIN initiated "tools" commands.
=====

```

Example output Sx-N4 reference point failure code statistics

The following output examples show Sx-N4 reference point failure code statistics for SMF/CP function and UPF/UP function respectively.

```

# show mobile-gateway pdn ref-point-stats sx-n4 failure-codes
=====
PFCP Failure Code Statistics
=====
Peer address      : 192.168.34.1
Peer type        : User
Reference point   : default

Failure code                      Rx count    Tx count
-----
Request Rejected (reason not specified)    0           0
Session context not found                  0           0
Mandatory IE Missing                       0           0
Conditional IE Missing                     0           0
Invalid length                             0           0
Mandatory IE incorrect                     0           0

```

```

Invalid Forwarding Policy          0          0
Invalid F-TEID allocation option    0          0
No established PFCP Association      0          0
Rule creation/modification Failure  0          0
No resources available              0          0
Service not supported               0          0
System failure                     0          0
Others                             0          0
-----
Number of peer associations : 2
=====

```

```

# show mobile-gateway pdn ref-point-stats sx-n4 failure-codes
=====
PFCP Failure Code Statistics
=====
Peer address      : 192.168.33.1
Peer type         : Control
Reference point   : default

Failure code              Rx count    Tx count
-----
Request Rejected (reason not specified)  0          0
Session context not found                0          0
Mandatory IE Missing                    0          0
Conditional IE Missing                   0          0
Invalid length                          0          0
Mandatory IE incorrect                   0          0
Invalid Forwarding Policy                 0          0
Invalid F-TEID allocation option          0          0
No established PFCP Association            0          0
Rule creation/modification Failure        0          0
No resources available                    0          0
Service not supported                     0          0
System failure                           0          0
Others                                   0          0
-----
Number of peer associations : 2
=====

```

Example output T6c reference point statistics

```

# show mobile pdn ref-point-stats t6c
=====
T6c reference point statistics
=====

Peer Address      : 1.1.1.1
Peer port         : 10
Peer type         : scef
Reference point   : reference_point_1
Router           : Base
Port             : 80
HTTP requests     : 1          HTTP responses success : 2

```

```
-----
Number of peer instances : 1
=====
```

Example output T6c reference point statistics per-peer instance

```
# show mobile-gateway pdn ref-point-stats t6c instance
=====
T6c reference point statistics per-peer instance
=====

Peer Address      : 1.1.1.1
Peer port         : 10
Peer type        : scef
Chassis/Card      : 1/3      Gateway          : 1
Reference point   : reference_point_1
Router           : Base
Port             : 80
HTTP requests     : 1        HTTP responses success : 2
-----
Number of peer instances : 1
=====
```

Example output T6c reference point statistics aggregate

```
# show mobile-gateway pdn ref-point-stats t6c aggregate
=====
T6c reference point statistics aggregate
=====
HTTP requests      : 34      HTTP responses success : 38
-----
Number of peer instances : 4
=====
```

Example output T6c failure code statistics

```
# show mobile-gateway pdn ref-point-stats t6c failure-codes
=====
T6c Failure Code Statistics
=====
Peer Address      : 4.4.4.4
Peer port         : 14
Peer type        : scef
Reference point   : reference_point_1
Router           : Base
Port             : 80
Error result codes
400 Wrong request : 10      404 Not found          : 11
500 Internal error : 12
-----
Number of peer instances : 1
=====
```

Example output T6c failure code statistics per-peer instance

```
# show mobile-gateway pdn ref-point-stats t6c failure-codes instance
=====
T6c Failure Code Statistics per-peer instance
=====
Peer Address      : 4.4.4.4
Peer port         : 14
Peer type         : scef
Chassis/Card      : 1/3          Gateway          : 1
Reference point   : reference_point_1
Router           : Base
Port             : 80
Error result codes
400 Wrong request : 10          404 Not found          : 11
500 Internal error : 12
-----
Number of peer instances : 1
=====
```

Example output T6c failure code statistics aggregate

```
# show mobile-gateway pdn ref-point-stats t6c failure-codes aggregate
=====
T6c Failure Code Statistics aggregate
=====
Error result codes
400 Wrong request : 58          404 Not found          : 62
500 Internal error : 66
-----
Number of peer instances : 4
=====
```

Example output T6c reference point peers

```
# show mobile-gateway pdn ref-point-stats t6c
=====
T6c reference point peers
=====
Peer Address      : 5.5.5.5
Peer port         : 16
Peer type         : scef
Router           : Base
Reference point   : reference_point_1
Path Mgmt State   : up
Create Time       : 06/15/2017 22:15:09 Gateway Id       : 1
Last Request Rcvd: 06/15/2017 22:15:09
-----
Number of peers : 1
=====
```

5.2.2.56 sbi-load-overload-control

Syntax

```
sbi-load-overload-control
```

Context

```
show>mobile>pdn
```

Description

This command displays SBI load and overload control information.

Output

The following output example shows SBI load and overload control information.

Example output SBI load and overload control information

```
# show mobile-gateway pdn sbi-load-overload-control

=====
SMF Node level Load/Overload control over LCI/OCI Headers
=====
Gateway                : 1
Load control           : Enabled
Load change trigger value : 20 %      Nth msg load report update : 1000

Overload control       : Enabled      Overload Validity time       : 86400 sec
Minor reduction value  : 70 %        Major reduction value       : 30 %
Critical reduction value : 40 %

CPU sampling interval
Overload               : 10          Overload state change       : 2

Current load           : 10 %
Current overload state  : critical
=====
```

5.2.2.57 service-stats

Syntax

```
service-stats nf-type nf-type [service-realm service-realm-name] [service-name service-name] [service-instance service-instance] [remote-nf-id uuid] remote-nf-ip ip-address | fqdn] [gateway gw-id] group group-id] [failure-codes] [instance | aggregate]
```

Context

show>mobile>pdn

Description

This command displays SBA statistics.

Parameters

nf-type

specifies the NF type
Values: amf | bsf | nrf | pcf | udm | chf | sepp | scp

service-realm-name

specifies the service realm name, up to 32 characters

service-name

specifies the service name
Values: nsmf-pdusession | namf-comm | namf-evts | nbsf-management | npcfsmpolicycontrol | nudm-sdm | nudm-uecm | nchf-convergedcharging | nnrf-oauth2 | nsmf-event-exposure | sepp | scp

service-instance

specifies the service instance name, up to 32 characters

uuid

specifies the UUID of the remote NF
Values: xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx (x: [0..F]H)

ip-address / fqdn

specifies the IP address/FQDN of the remote NF

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

– fqdn FQDN, up to 255 characters

gw-id

specifies the gateway ID

Values: 1 to 8

group-id

specifies the group ID

Values: 0 (CPM statistics) | 1 to 15

failure-codes

specifies to display failure code statistics

instance

specifies to display statistics per remote NF per VM

aggregate

specifies to display aggregated statistics per service realm

Output

The following output examples show SBA statistics and *Show NRF service statistics output fields* describes the output fields.

- *Example output AMF service statistics*
- *Example output AMF failure codes service statistics*
- *Example output BSF Nbsf_Management service statistics*
- *Example output BSF failure codes service statistics*
- *Example output CHF service statistics*
- *Example output CHF session service statistics*
- *Example output CHF failure codes service statistics*
- *Example output NRF service statistics*
- *Example output NRF failure codes service statistics*
- *Example output NRF OAuth2.0 service statistics*
- *Example output NRF OAuth2.0 failure codes service statistics*
- *Example output PCF service statistics*
- *Example output PCF failure codes service statistics*
- *Example output SCP service statistics*

- *Example output SEPP service statistics*
- *Example output SMF event exposure service statistics*
- *Example output UDM service statistics*
- *Example output UDM failure codes service statistics*

Example output AMF service statistics

```
# show mobile-gateway pdn service-stats nf-type amf
=====
AMF peer service statistics
=====
Service Realm      : sbaRealm1                      Gateway Id   : 1
Service Name       : namf-comm
-----
Service Instance   : comm1
NF instance ID     : 44f08cba-9a89-46fe-ab7d-2a0066584f55
N1N2 Message Transfer : 1
  N1 PDU Session Establishment Reject      : 0
  N1 PDU Session Modification Command      : 0
  N1 PDU Session Release Command          : 0
  N2 PDU Session Resource Setup Request    : 1
  N1 PDU Session Establishment Accept      : 1
  N2 PDU Session Resource Modify Request   : 0
  N2 PDU Session Resource Release Command  : 0
N1N2 Msg Transf Succ : 1          N1N2 Msg Transf Fail : 0
N1N2 Transf Fail Notif : 0
EBI Assignment Req   : 0          EBI Assignment Succ   : 0
EBI Assignment Fail  : 0
Number Of Sessions  : 0
-----
Number of instances : 1
-----
Service Realm      : sbaRealm1                      Gateway Id   : 1
Service Name       : namf-evts
-----
Service Instance   : evts1
NF instance ID     : 44f08cba-9a89-46fe-ab7d-2a0066584f55
Subscription Cre Req : 0          Subscription Cre Succ : 0
Subscription Cre Fail : 0
Subscription Upd Req : 0          Subscription Upd Succ : 0
Subscription Upd Fail : 0
Subscription Del Req : 0          Subscription Del Succ : 0
Subscription Del Fail : 0
Event Notifications : 0          Event Subscript Expir : 0
-----
Number of instances : 1
-----
Service Realm      : sbaRealm1                      Gateway Id   : 1
Service Name       : nsmf-pdusession
-----
Service Instance   : pdu1
NF instance ID     : 44f08cba-9a89-46fe-ab7d-2a0066584f55
Create SM Ctx Req  : 1
  N1 PDU Session Establishment Request      : 1
```

```

N2 Handover Required : 0
N2 Path Switch Request : 0
Create SM Ctx Succ : 1
N2 PDU Session Resource Setup Request : 0
N2 Path Switch Request Acknowledge : 0
N2 Path Switch Request Unsuccessful : 0
Create SM Ctx Fail : 0
N1 PDU Session Establishment Reject : 0
N1 5GSM Status : 0
Update SM Ctx Req : 1
N1 PDU Session Modification Request : 0
N1 PDU Session Modification Complete : 0
N1 PDU Session Release Request : 0
N1 PDU Session Release Complete : 0
N1 5GSM Status : 0
N2 PDU Session Resource Setup Response : 1
N2 PDU Session Resource Setup Unsuccessful : 0
N2 PDU Session Resource Notify : 0
N2 PDU Session Resource Modify Indication : 0
N2 PDU Session Resource Modify Response : 0
N2 PDU Session Resource Modify Unsuccessful : 0
N2 PDU Session Resource Release Response : 0
N2 PDU Session Resource Notify Released : 0
N2 Path Switch Request : 0
N2 Path Switch Request Setup Failed : 0
N2 Handover Required : 0
N2 Handover Request Acknowledge : 0
N2 Handover Resource Allocation Unsuccessful : 0
With HoState Completed : 0
With HoState Cancelled : 0
With upCnxState Activating : 0
With upCnxState Deactivated : 0
With amfId change : 0
With dataForwarding True : 0
With dataForwarding False : 0
With release due to reactivation : 0
With release due to duplicate session Id : 0
With release due to slice not available : 0
Update SM Ctx Succ : 1
N1 PDU Session Modification Command : 0
N1 PDU Session Release Command : 0
N2 PDU Session Resource Setup Request : 0
N2 PDU Session Resource Modify Confirm : 0
N2 PDU Session Resource Modify Request : 0
N2 PDU Session Resource Release Command : 0
N2 Path Switch Request Acknowledge : 0
N2 Handover Command : 0
With upCnxState Deactivated : 0
Update SM Ctx Fail : 0
N1 PDU Session Modification Reject : 0
N1 PDU Session Release Reject : 0
N1 PDU Session Establishment Reject : 0
N1 5GSM Status : 0
N2 PDU Session Resource Modify Indication Unsuccessful : 0
N2 Path Switch Request Unsuccessful : 0
N2 Handover Preparation Unsuccessful : 0
Release SM Ctx Req : 0

```

```

With PDU session status mismatch          : 0
Release SM Ctx Succ      : 0              Release SM Ctx Fail    : 0
SM Ctx Stat Notif       : 0              SM Ctx Stat Notif Succ : 0
SM Ctx Stat Notif Fail  : 0
Retrieve SM Ctx Req      : 0              Retrieve SM Ctx Succ   : 0
Retrieve SM Ctx Fail     : 0
-----
Number of instances      : 1
=====

```

Example output AMF failure codes service statistics

```

show mobile-gateway pdn service-stats nf-type amf failure-codes
=====
AMF peer service statistics
=====
Service Realm      : sbaRealm1              Gateway Id   : 1
Service Name       : namf-comm
-----
Service Instance   : comm1
NF instance ID     : 44f08cba-9a89-46fe-ab7d-2a0066584f55
N1N2 Msg Transf Fail : 0
  400 Bad Request   : 0                    401 Unauthorized    : 0
  403 Forbidden     : 0                    404 Not Found       : 0
  409 Conflict      : 0                    411 Length Required : 0
  429 Too Many Requests: 0                500 Server Error    : 0
  503 Service Unavail : 0
EBI Assignment Fail : 0
  400 Bad Request   : 0                    401 Unauthorized    : 0
  403 Forbidden     : 0                    404 Not Found       : 0
  409 Conflict      : 0                    411 Length Required : 0
  429 Too Many Requests: 0                500 Server Error    : 0
  503 Service Unavail : 0
-----
Number of instances : 1
-----
Service Realm      : sbaRealm1              Gateway Id   : 1
Service Name       : namf-evts
-----
Service Instance   : evts1
NF instance ID     : 44f08cba-9a89-46fe-ab7d-2a0066584f55
Subscription Cre Fail : 0
  400 Bad Request   : 0                    401 Unauthorized    : 0
  404 Not Found     : 0                    500 Server Error    : 0
Subscription Upd Fail : 0
  400 Bad Request   : 0                    401 Unauthorized    : 0
  404 Not Found     : 0                    500 Server Error    : 0
Subscription Del Fail : 0
  400 Bad Request   : 0                    401 Unauthorized    : 0
  404 Not Found     : 0                    500 Server Error    : 0
-----
Number of instances : 1
-----
Service Realm      : sbaRealm1              Gateway Id   : 1
Service Name       : nsmf-pdusession
-----

```

```

Service Instance      : pdu1
NF instance ID       : 44f08cba-9a89-46fe-ab7d-2a0066584f55
Create SM Ctx Fail   : 0
  400 Bad Request     : 0
  403 Forbidden       : 0
  500 Server Error    : 0
  401 Unauthorized    : 0
  404 Not Found       : 0
  503 Service Unavail : 0
Update SM Ctx Fail   : 0
  400 Bad Request     : 0
  403 Forbidden       : 0
  500 Server Error    : 0
  401 Unauthorized    : 0
  404 Not Found       : 0
  503 Service Unavail : 0
Release SM Ctx Fail  : 0
  400 Bad Request     : 0
  404 Not Found       : 0
  500 Server Error    : 0
  401 Unauthorized    : 0
  503 Service Unavail : 0
SM Ctx Stat Notif Fail : 0
  400 Bad Request     : 0
  404 Not Found       : 0
  500 Server Error    : 0
  401 Unauthorized    : 0
  503 Service Unavail : 0
Retrieve SM Ctx Fail  : 0
  400 Bad Request     : 0
  403 Forbidden       : 0
  500 Server Error    : 0
  401 Unauthorized    : 0
  404 Not Found       : 0
  503 Service Unavail : 0

-----
Number of instances   : 1
=====

```

Example output BSF Nbsf_Management service statistics

```

show mobile-gateway pdn service-stats nf-type bsf service-name nbsf-management
=====
BSF peer service statistics
=====
Service Realm      : sbaRealm1                      Gateway Id   : 1
Service Name       : nbsf-management
-----
Service Instance   : nbsfService1
NF Instance IP/FQDN : 10.10.43.2
NF Instance Port    : 60613
PCF Binding Requests : 1                      PCF Binding Req Succ : 0
PCF Binding Req Fail : 1
-----
Number of instances : 1
=====

```

Example output BSF failure codes service statistics

```

# show mobile-gateway pdn service-stats nf-type bsf failure-codes
=====
BSF peer service statistics
=====
Service Realm      : sbaRealm1                      Gateway Id   : 1
Service Name       : nbsf-management
-----
Service Instance   : nbsfService1
NF Instance IP/FQDN : 10.10.43.2
NF Instance Port    : 35901
PCF Binding Req Fail : 1
  400 Bad Request     : 0
  401 Unauthorized    : 0

```

```

404 Not Found      : 1          500 Server Error      : 0
-----
Number of instances : 1
=====

```

Example output CHF service statistics

The following example shows CHF service statistics output and *Show CHF service statistics output fields* describes the output fields.

```

# show mobile-gateway pdn service-stats nf-type type chf
=====
CHF peer service statistics
=====
Service Realm      : sbaRealm1          Gateway Id   : 1
Service Name       : nchf-Convergedcharging
-----
Service Instance   : chfClient1
NF instance ID     : 4f85e924-1509-47b4-907f-9778afdb2ffb
Active Sessions    : 0
ConvergedCharging Create      : 11
ConvergedCharging Create Succ : 11
ConvergedCharging Create Fail : 0
ConvergedCharging Update     : 3
ConvergedCharging Update Succ : 3
ConvergedCharging Update Fail : 0
ConvergedCharging Release    : 10
ConvergedCharging Release Succ : 0
ConvergedCharging Release Fail : 10
ConvergedCharging Notify     : 0
ConvergedCharging Notify Succ : 0
ConvergedCharging Notify Fail : 0
-----
Service Instance   : chfClient1
NF instance ID     : e899b215-384c-472b-b8c6-fb9dee15653e
Active Sessions    : 0
ConvergedCharging Create      : 0
ConvergedCharging Create Succ : 0
ConvergedCharging Create Fail : 0
ConvergedCharging Update     : 0
ConvergedCharging Update Succ : 0
ConvergedCharging Update Fail : 0
ConvergedCharging Release    : 0
ConvergedCharging Release Succ : 0
ConvergedCharging Release Fail : 0
ConvergedCharging Notify     : 0
ConvergedCharging Notify Succ : 0
ConvergedCharging Notify Fail : 0
-----
Number of instances : 2

```

Table 86: Show CHF service statistics output fields

Label	Description
Service Realm	The service realm of the NRF
Service Name	The service name of the NRF
Gateway Id	The gateway ID of the NRF
Service Instance	The service instance of the CHF
NF Instance ID	The NF instance IP address
Active Sessions	The number of active sessions
ConvergedCharging Create	The number of Converged Charging Create instances
ConvergedCharging Create Succ	The number of Converged Charging Create Success instances
ConvergedCharging Update	The number of Converged Charging Update instances
ConvergedCharging Update Succ	The number of Converged Charging Update Success instances
ConvergedCharging Update Fail	The number of Converged Charging Update Fail instances
ConvergedCharging Release	The number of Converged Charging Release instances
ConvergedCharging Release Succ	The number of Converged Charging Release Success instances
ConvergedCharging Release Fail	The number of Converged Charging Release Fail instances
ConvergedCharging Notify	The number of Converged Charging Notify instances
ConvergedCharging Notify Succ	The number of Converged Charging Notify Success instances
ConvergedCharging Notify Fail	The number of Converged Charging Notify Fail instances

Example output CHF session service statistics

```
# show mobile-gateway pdn service-stats nf-type chf
=====
CHF peer service statistics
=====
Service Realm          : sbaRealm1                      Gateway Id   : 1
Service Name           : nchf-Convergedcharging
-----
Service Instance       : chfClient1
NF instance ID         : 4f85e924-1509-47b4-907f-9778afdb2fffb
ConvergedCharging Create
  Request Tx           : 0
  Response Success Rx  : 0
  Response Fail Rx     : 0
  Request Failures     : 0
  Request ReTx         : 0
  Malformed Response Rx : 0
  Unknown Session Response Rx : 0
ConvergedCharging Update
  Request Tx           : 0
  Response Success Rx  : 0
  Response Fail Rx     : 0
  Request Failures     : 0
  Request ReTx         : 0
  Malformed Response Rx : 0
  Unknown Session Response Rx : 0
ConvergedCharging Release
  Request Tx           : 0
  Response Success Rx  : 0
  Response Fail Rx     : 0
  Request Failures     : 0
  Request ReTx         : 0
  Malformed Response Rx : 0
  Unknown Session Response Rx : 0
ConvergedCharging Notify
  Request Rx           : 0
  Response Success Tx   : 0
  Response Fail Tx     : 0
  Notify Unknown Session Request Rx : 0
RG FUI-Redirect-recv   : 0
RG FUI-Terminate-recv : 0
RG FUI-Restrict-recv   : 0
RG FUI-Redirect-applied : 0
RG FUI-Terminate-applied : 0
RG FUI-Restrict-applied : 0
-----
Number of instances    : 1
=====
```

Example output CHF failure codes service statistics

```
# show mobile-gateway pdn service-stats nf-type chf failure-codes
=====
CHF peer service statistics
=====
```



```

Service Realm      : sbaRealm1                      Gateway Id   : 1
Service Name       : nchf-Convergedcharging
-----
Service Instance   : chfClient1
NF instance ID     : 4f85e924-1509-47b4-907f-9778afdb2ffb
ConvergedCharging Create
Response Fail Rx   : 0
  400 Bad Request   : 0
  401 Unauthorized   : 0
  403 Forbidden      : 0
  404 Not Found      : 0
  408 Request Timeout : 0
  429 Too Many Requests : 0
  500 Internal Server Error : 0
  503 Service Unavailable : 0
  Other 3xx          : 0
  Other 4xx          : 0
  Other 5xx          : 0
ConvergedCharging Update
Response Fail Rx   : 0
  400 Bad Request   : 0
  401 Unauthorized   : 0
  403 Forbidden      : 0
  404 Not Found      : 0
  408 Request Timeout : 0
  429 Too Many Requests : 0
  500 Internal Server Error : 0
  503 Service Unavailable : 0
  Other 3xx          : 0
  Other 4xx          : 0
  Other 5xx          : 0
ConvergedCharging Release
Response Fail Rx   : 0
  400 Bad Request   : 0
  401 Unauthorized   : 0
  403 Forbidden      : 0
  404 Not Found      : 0
  408 Request Timeout : 0
  429 Too Many Requests : 0
  500 Internal Server Error : 0
  503 Service Unavailable : 0
  Other 3xx          : 0
  Other 4xx          : 0
  Other 5xx          : 0
ConvergedCharging Notify
Response Fail Tx    : 0
  400 Bad Request   : 0
  404 Not Found      : 0
  500 Internal Server Error : 0
-----
Number of instances : 1
=====

```

Example output NRF service statistics

```
# show mobile-gateway pdn service-stats nf-type nrf
```

```
=====
NRF peer service statistics
=====
```

```
Service Realm      : realm1                      Gateway Id   : 1
Service Name       : nnrf-nfm
```

```
-----
Service Instance   : nnrf-nfm1
NF instance IP     : 1.1.1.1
Port               : 8080
```

```
Registered         : Yes, 05/30/2019 05:59:51
Heartbeat Interval : 1
Cache Validity receive : 1           Cache Validity applied : 1
```

```
NF Register        : 2
NF Register Failures : 2
```

```
NF Update Partial   : 0
NF Update Partial Fail : 0
```

```
NF Heartbeat        : 0
NF Heartbeat Failures : 0
```

```
NF Update Full      : 0
NF Update Full Failures: 0
```

```
NF Deregister       : 0
NF Deregister Failures : 0
```

```
NF Status Subscribe : 0
NF Status Subscrib Fail : 0
AMF Status Subscribe : 0
UDM Status Subscribe : 0
```

```
PCF Status Subscribe : 0
```

```
NF Status UnSubscribe : 0
NF Status UnSubsc Fail : 0
AMF Status UnSubscribe : 0
UDM Status UnSubscribe : 0
```

```
PCF Status UnSubscribe : 0
```

```
NF Status Notify     : 0
NF Status Notify Fail : 0
AMF Deregister Notify : 0
AMF Restart Notify    : 0
PCF Deregister Notify : 0
PCF Restart Notify    : 0
UDM Deregister Notify : 0
UDM Restart Notify    : 0
```

```
AMF Failure Notify      : 0
AMF Service Fail Notify: 0
PCF Failure Notify      : 0
PCF Service Fail Notify: 0
UDM Failure Notify      : 0
UDM Service Fail Notify: 0
```

```
-----
Service Instance   : nnrf-nfm1
NF instance IP     : 1.1.1.1
Port               : 80
```

```
Registered         : Yes, 05/30/2019 05:59:51
Heartbeat Interval : 1
Cache Validity receive : 1           Cache Validity applied : 1
```

```
NF Register        : 2
```

```

NF Register Failures      : 2

NF Update Partial        : 0
NF Update Partial Fail   : 0

NF Heartbeat             : 0
NF Heartbeat Failures    : 0

NF Update Full           : 0
NF Update Full Failures  : 0

NF Deregister            : 0
NF Deregister Failures   : 0

NF Status Subscribe      : 0
NF Status Subscrib Fail  : 0
AMF Status Subscribe     : 0          PCF Status Subscribe    : 0
UDM Status Subscribe     : 0

NF Status UnSubscribe    : 0
NF Status UnSubsc Fail   : 0
AMF Status UnSubscribe   : 0          PCF Status UnSubscribe  : 0
UDM Status UnSubscribe   : 0

NF Status Notify         : 0
NF Status Notify Fail    : 0
AMF Deregister Notify    : 0          AMF Failure Notify      : 0
AMF Restart Notify       : 0          AMF Service Fail Notify: 0
PCF Deregister Notify    : 0          PCF Failure Notify      : 0
PCF Restart Notify       : 0          PCF Service Fail Notify: 0
UDM Deregister Notify    : 0          UDM Failure Notify      : 0
UDM Restart Notify       : 0          UDM Service Fail Notify: 0
-----
Number of NF instances : 2
-----
Service Realm           : realm1          Gateway Id    : 1
Service Name            : nnrf-disc
-----
Service Instance        : nnrf-disc1
NF instance IP          : 1.1.1.1
Port                    : 80

NF Discovery            : 2
NF Discovery Failures   : 2
AMF Discovery           : 2
AMF Discovery Failures  : 2
PCF Discovery           : 0
PCF Discovery Failures  : 0
UDM Discovery           : 0
UDM Discovery Failures  : 0
UPF Discovery           : 0
UPF Discovery Failures  : 0
-----
Number of NF instances : 1
=====

```

Example output NRF failure codes service statistics

```
# show mobile-gateway pdn service-stats nf-type nrf failure-codes
```

```
=====
NRF peer service statistics
=====
```

```
Service Realm      : sbaRealm1                      Gateway Id   : 1
Service Name       : nnrf-nfm
-----
Service Instance   : nfm1
NF Instance IP/FQDN : 11.127.5.5
NF Instance Port    : 8080
Registered         : Yes, 04/21/2021 10:35:05
Heartbeat Interval  : 0
NF Register Failures : 0
    Other 3xx       : 0                               400 Bad Request    : 0
    404 Not Found    : 0                               Other 4xx         : 0
    500 Server Error : 0                               Other 5xx         : 0
    HTTP2 timeouts   : 0
NF Update Partial Fail : 0
    Other 3xx       : 0                               400 Bad Request    : 0
    404 Not Found    : 0                               Other 4xx         : 0
    500 Server Error : 0                               Other 5xx         : 0
    HTTP2 timeouts   : 0
NF Heartbeat Failures : 0
    Other 3xx       : 0                               400 Bad Request    : 0
    404 Not Found    : 0                               Other 4xx         : 0
    500 Server Error : 0                               Other 5xx         : 0
    HTTP2 timeouts   : 0
NF Update Full Fail   : 0
    Other 3xx       : 0                               400 Bad Request    : 0
    404 Not Found    : 0                               Other 4xx         : 0
    500 Server Error : 0                               Other 5xx         : 0
    HTTP2 timeouts   : 0
NF Deregister Failures : 0
    Other 3xx       : 0                               400 Bad Request    : 0
    404 Not Found    : 0                               Other 4xx         : 0
    500 Server Error : 0                               Other 5xx         : 0
    HTTP2 timeouts   : 0
NF Status Subscrib Fail : 0
    Other 3xx       : 0                               400 Bad Request    : 0
    404 Not Found    : 0                               Other 4xx         : 0
    500 Server Error : 0                               Other 5xx         : 0
    HTTP2 timeouts   : 0
NF Status UnSubsc Fail : 0
    Other 3xx       : 0                               400 Bad Request    : 0
    404 Not Found    : 0                               Other 4xx         : 0
    500 Server Error : 0                               Other 5xx         : 0
    HTTP2 timeouts   : 0
NF Status Notify Fail  : 0
    Other 3xx       : 0                               400 Bad Request    : 0
    404 Not Found    : 0                               Other 4xx         : 0
    500 Server Error : 0                               Other 5xx         : 0
    HTTP2 timeouts   : 0
-----
```

```

Number of instances      : 2
-----
Service Realm           : sbaRealm1                      Gateway Id    : 1
Service Name            : nnrf-disc
-----
Service Instance        : disc1
NF Instance IP/FQDN     : 11.127.5.5
NF Instance Port        : 8080
Cache Validity receive  : 0                               Cache Validity applied : 0
NF Discovery Failures   : 0
  Other 3xx              : 0                               400 Bad Request       : 0
  403 Forbidden          : 0                               404 Not Found        : 0
  Other 4xx              : 0                               500 Server Error     : 0
  Other 5xx              : 0                               HTTP2 timeouts       : 0
  Rsp dropped oversize   : 0
AMF Discovery Failures  : 0
  Other 3xx              : 0                               400 Bad Request       : 0
  403 Forbidden          : 0                               404 Not Found        : 0
  Other 4xx              : 0                               500 Server Error     : 0
  Other 5xx              : 0                               HTTP2 timeouts       : 0
  Rsp dropped oversize   : 0
PCF Discovery Failures  : 0
  Other 3xx              : 0                               400 Bad Request       : 0
  403 Forbidden          : 0                               404 Not Found        : 0
  Other 4xx              : 0                               500 Server Error     : 0
  Other 5xx              : 0                               HTTP2 timeouts       : 0
  Rsp dropped oversize   : 0
UDM Discovery Failures  : 0
  Other 3xx              : 0                               400 Bad Request       : 0
  403 Forbidden          : 0                               404 Not Found        : 0
  Other 4xx              : 0                               500 Server Error     : 0
  Other 5xx              : 0                               HTTP2 timeouts       : 0
  Rsp dropped oversize   : 0
UPF Discovery Failures  : 0
  Other 3xx              : 0                               400 Bad Request       : 0
  403 Forbidden          : 0                               404 Not Found        : 0
  Other 4xx              : 0                               500 Server Error     : 0
  Other 5xx              : 0                               HTTP2 timeouts       : 0
  Rsp dropped oversize   : 0
-----
Number of instances      : 1
=====

```

Table 87: Show NRF service statistics output fields

Label	Description
Service Realm	The service realm of the NRF
Service Name	The service name of the NRF

Table 87: Show NRF service statistics output fields (continued)

Label	Description
Gateway Id	The gateway ID of the NRF
Service Instance	The service instance of the NRF
NF Instance ID	The NF instance IP address
Port	The port number
Registered	The registration status including date and time
Heartbeat Interval	The heartbeat interval
Cache Validity receive	The cache validity received
Cache Validity applied	The cache validity applied
NF Register	The number of NF Register Request messages that the CMG has sent
NF Register Failures	The number of NF Register operation failures
400 Bad Request	The number of responses received with this error code
404 Not found	The number of responses received with this error code
Rsp dropped oversize	The number of responses dropped because of oversized payload (larger than 2 MB)
500 Internal	The number of responses received with this error code
NF Update Partial	The number of NF Update Partial Request messages the CMG has sent
NF Update Partial Fail	The number of NF Update Partial operation failures

Table 87: Show NRF service statistics output fields (continued)

Label	Description
NF Heartbeat	The number of Heartbeat Request messages the CMG has sent
NF Heartbeat Failures	The number of Heartbeat failures
NF Update Full	The number of NF Update Full Request messages the CMG has sent
NF Update Full Failures	The number of NF Update Full operation failures
NF Deregister	The number of NF Deregister Request messages the CMG has sent
NF Deregister Failures	The number of NF Deregister operation failures
NF Status Subscribe	The number of NF Status Subscribe Request messages the CMG has sent
NF Status Subscb Fail	The number of NF Status Subscribe operation failures
AMF/PCF/UDM Status Subscribe	The number of NF Status Subscribe Request messages the CMG has sent for AMF/PCF/UDM peer
NF Status UnSubscribe	The number of NF Status UnSubscribe Request messages the CMG has sent
NF Status UnSubscribe Fail	The number of NF Status UnSubscribe operation failures
AMF/PCF/UDM Status UnSubscribe	The number of NF Status UnSubscribe Request messages the CMG has sent for AMF/PCF/UDM peer
NF Status Notify	The number of NF Status Notify Request messages the CMG has received
NF Status Notify Fail	The number of NF Status Notify operation failures

Table 87: Show NRF service statistics output fields (continued)

Label	Description
AMF/PCF/UDM Deregister Notify	The number of NF Status Notify Request messages the CMG has received for an AMF/PCF/UDM de-registration event
AMF/PCF/UDM Failure Notify	The number of NF Status Notify Request messages the CMG has received for an AMF/PCF/UDM NF Failure event
AMF/PCF/UDM Restart Notify	The number of NF Status Notify Request messages the CMG has received for an AMF/PCF/UDM NF Restart event
AMF/PCF/UDM Service Fail Notify	The number of NF Status Notify Request messages the CMG has received for AMF/PCF/UDM Service Failure event
NF Discovery	The number of NF Discovery Request messages the CMG has sent
NF Discovery Failures	The number of NF Discovery operation failures
AMF/PCF/UDM/UPF Discovery	The number of NF Discovery Request messages the CMG has sent for AMF/PCF/UDM/UPF discovery
AMF/PCF/UDM/UPF Discovery Failures	The number of NF Discovery operation failures for AMD/PCF/UDM/UPF peer discovery

Example output NRF OAuth2.0 service statistics

The following example shows the NRF OAuth2.0 service statistics output and [Show NRF OAuth2.0 service statistics output fields](#) describes the output fields.

```
# show mobile-gateway pdn service-stats nf-type nrf service-name nnrf-oauth2
=====
NRF peer service statistics
=====
Service Realm      : realm1                      Gateway Id   : 1
Service Name       : nnrf-oauth2
-----
Service Instance   : nnrf-oauth1
NF Instance IP/FQDN : 1.2.3.4
NF Instance Port    : 1
OAuth Token Get Req : 1
OAuth Token Get Fls : 1
```



```

AMF OAuth Token Get Req: 1
AMF OAuth Token Get Fls: 1
CHF OAuth Token Get Req: 1
CHF OAuth Token Get Fls: 1
SMF OAuth Token GET Req: 1
SMF OAuth Token GET Fls: 1
PCF OAuth Token Get Req: 1
PCF OAuth Token Get Fls: 1
UDM OAuth Token Get Req: 1
UDM OAuth Token Get Fls: 1

```

```

-----
Number of instances      : 1
=====

```

Table 88: Show NRF OAuth2.0 service statistics output fields

Label	Description
Service Realm	The service realm of the NRF
Service Name	The service name of the NRF
Gateway Id	The gateway ID of the NRF
Service Instance	The service instance of the NRF
NF Instance IP/FQDN	The NF instance IP address
NF Instance Port	The NF instance port number
OAuth Token Get Req	The total number of oauth access token get requests sent
OAuth Token Get Fls	The total number of failures in responses of access token get requests
AMF/CHF/PCF/SMF/ UDM OAuth Token Get Req	The number of oauth access token get requests sent with an AMF/CHF/PCF/SMF/UDM as token audience, that is, the target NF producer

Table 88: Show NRF OAuth2.0 service statistics output fields (continued)

Label	Description
AMF/CHF/PCF/SMF/ UDM OAuth Token Get Fls	The number of failures in responses of oauth access token get requests sent with an AMF/CHF/PCF/SMF/UDM as token audience, that is, the target NF producer

Example output NRF OAuth2.0 failure codes service statistics

The following example shows the NRF OAuth2.0 failure codes service statistics output and *Show NRF OAuth2.0 failure codes service statistics output fields* describes the output fields.

```
# show mobile-gateway pdn service-stats nf-type nrf service-name nnrf-
oauth2 failure-codes
=====
NRF peer service statistics
=====
Service Realm      : realm1                      Gateway Id   : 1
Service Name       : nnrf-oauth2
-----
Service Instance   : nnrf-oauth1
NF Instance IP/FQDN : 1.2.3.4
NF Instance Port    : 1
OAuth Token Get Fls : 1
  400 Bad Request   : 1
    Invalid Request : 1
    Invalid Client  : 1
    Unauthorized Client: 1
    Invalid Scope   : 1
AMF OAuth Token Get Fls: 1
  400 Bad Request   : 1
    Invalid Request : 1
    Invalid Client  : 1
    Unauthorized Client: 1
    Invalid Scope   : 1
CHF OAuth Token Get Fls: 1
  400 Bad Request   : 1
    Invalid Request : 1
    Invalid Client  : 1
    Unauthorized Client: 1
    Invalid Scope   : 1
SMF OAuth Token GET Fls: 1
  400 Bad Request   : 1
    Invalid Request : 1
    Invalid Client  : 1
    Unauthorized Client: 1
    Invalid Scope   : 1
PCF OAuth Token Get Fls: 1
  400 Bad Request   : 1
    Invalid Request : 1
```

```

Invalid Client      : 1
Unauthorized Client: 1
Invalid Scope      : 1
UDM OAuth Token Get Fls: 1
  400 Bad Request   : 1
  Invalid Request   : 1
  Invalid Client    : 1
  Unauthorized Client: 1
  Invalid Scope     : 1
-----
Number of instances : 1
=====

```

Table 89: Show NRF OAuth2.0 failure codes service statistics output fields

Label	Description
Service Realm	The service realm of the NRF
Service Name	The service name of the NRF
Gateway Id	The gateway ID of the NRF
Service Instance	The service instance of the NRF
NF Instance IP/FQDN	The NF instance IP address
NF Instance Port	The NF instance port number
OAuth Token Get Fls	The total number of failures in responses of access token get requests
AMF/CHF/PCF/SMF/ UDM OAuth Token Get Fls	The number of failures in responses of oauth access token get requests sent with an AMF/CHF/PCF/SMF/UDM as token audience, that is, the target NF producer
400 Bad Request	The number of HTTP 400 status code received in oauth access token get responses
Invalid Request	The number of oauth access token get responses received with this error code

Table 89: Show NRF OAuth2.0 failure codes service statistics output fields (continued)

Label	Description
Invalid Client	The number of oauth access token get responses received with this error code
Unauthorized Client	The number of oauth access token get responses received with this error code
Invalid Scope	The number of oauth access token get responses received with this error code

Example output PCF service statistics

The following example shows PCF service statistics output and [Show PCF service statistics output fields](#) describes the output fields.

```
show mobile-gateway pdn service-stats nf-type pcf service-name npcfsmpolicycontrol
=====
PCF peer service statistics
=====
Service Realm      : sbaRealm1                      Gateway Id   : 1
Service Name       : npcfsmpolicycontrol
-----
Service Instance   : policy1
NF instance ID     : 1720b601-fae2-4fc4-b535-8df9da3c64a7
SMPolicyControl Create      : 0
SMPolicyControl Create Succ : 0
SMPolicyControl Create Fail : 0
SMPolicyControl Update     : 0
SMPolicyControl Update Succ : 0
SMPolicyControl Update Fail : 0
SMPolicyControl Delete     : 0
SMPolicyControl Delete Succ : 0
SMPolicyControl Delete Fail : 0
SMPolicyControl Update Notify : 0
SMPolicyControl Update Notify Succ : 0
SMPolicyControl Update Notify Fail : 0
SMPolicyControl Terminate Notify : 0
SMPolicyControl Terminate Notify Succ : 0
SMPolicyControl Terminate Notify Fail : 0
-----
Service Instance   : policy1
NF instance ID     : 34635855-c481-4ee2-b571-a151579dc866
SMPolicyControl Create      : 0
SMPolicyControl Create Succ : 0
SMPolicyControl Create Fail : 0
SMPolicyControl Update     : 0
```

```

SMPolicyControl Update Succ           : 0
SMPolicyControl Update Fail           : 0
SMPolicyControl Delete                : 0
SMPolicyControl Delete Succ           : 0
SMPolicyControl Delete Fail           : 0
SMPolicyControl Update Notify         : 0
SMPolicyControl Update Notify Succ    : 0
SMPolicyControl Update Notify Fail    : 0
SMPolicyControl Terminate Notify      : 0
SMPolicyControl Terminate Notify Succ : 0
SMPolicyControl Terminate Notify Fail : 0
-----
Service Instance                      : policy1
NF instance ID                       : e1d44f4f-a1c2-45a3-b9c4-126c2e7e1620
SMPolicyControl Create               : 0
SMPolicyControl Create Succ          : 0
SMPolicyControl Create Fail          : 0
SMPolicyControl Update               : 0
SMPolicyControl Update Succ          : 0
SMPolicyControl Update Fail          : 0
SMPolicyControl Delete               : 0
SMPolicyControl Delete Succ          : 0
SMPolicyControl Delete Fail          : 0
SMPolicyControl Update Notify        : 0
SMPolicyControl Update Notify Succ   : 0
SMPolicyControl Update Notify Fail   : 0
SMPolicyControl Terminate Notify     : 0
SMPolicyControl Terminate Notify Succ : 0
SMPolicyControl Terminate Notify Fail : 0
-----
Number of instances      : 3
=====

```

Table 90: Show PCF service statistics output fields

Label	Description
Service Realm	The SBA service realm to which the PCF client service is bound
Service Name	The name of the (PCF) client service (npcf-smpolicycontrol)
Gateway Id	The gateway ID associated with the PCF
Service Instance	The name of the PCF service instance
NF Instance ID	The NF (PCF) instance IP address

Table 90: Show PCF service statistics output fields (continued)

Label	Description
SMPolicyControl Create	The number of SMPolicyControl Create requests
SMPolicyControl Create Succ	The number of SMPolicyControl Create successes
SMPolicyControl Create Fail	The number of SMPolicyControlCreate failures
SMPolicyControl Update	The number of SMPolicyControl Update requests
SMPolicyControl Update Succ	The number of SMPolicyControl Update successes
SMPolicyControl Update Fail	The number of SMPolicyControl Update failures
SMPolicyControl Delete	The number of SMPolicyControl Delete requests
SMPolicyControl Delete Succ	The number of SMPolicyControl Delete successes
SMPolicyControl Delete Fail	The number of SMPolicyControl Delete failures
SMPolicyControl Terminate Notify	The number of SMPolicyControl Terminate Notify requests
ConvergedSMPolicyControl Terminate Notify Succ	The number of SMPolicyControl Terminate Notify successes
SMPolicyControl Terminate Notify Fail	The number of SMPolicyControl Terminate Notify failures

Example output PCF failure codes service statistics

```
# show mobile-gateway pdn service-stats nf-type pcf failure-codes
=====
PCF peer service statistics
=====
Service Realm      : sbaRealm1                      Gateway Id   : 1
Service Name       : npcfc-SMPolicyControl
-----
```

```

Service Instance           : policy1
NF instance ID             : 1720b601-fae2-4fc4-b535-8df9da3c64a7

SMPolicyControl Create Fail : 0
  400 Bad Request           : 0
  401 Unauthorized          : 0
  403 Forbidden              : 0
  404 Not Found              : 0
  500 Internal Server Error  : 0
SMPolicyControl Update Fail : 0
  400 Bad Request           : 0
  401 Unauthorized          : 0
  403 Forbidden              : 0
  404 Not Found              : 0
  500 Internal Server Error  : 0
SMPolicyControl Delete Fail : 0
  400 Bad Request           : 0
  401 Unauthorized          : 0
  404 Not Found              : 0
  500 Internal Server Error  : 0
SMPolicyControl Update Notify Fail : 0
  400 Bad Request           : 0
  404 Not Found              : 0
  500 Internal Server Error  : 0
SMPolicyControl Terminate Notify Fail : 0
  400 Bad Request           : 0
  404 Not Found              : 0
  500 Internal Server Error  : 0

-----
Number of instances      : 1
=====

```

Example output SCP service statistics

```

# show mobile-gateway pdn service-stats nf-type scp peer-nf all
=====
SCP peer service statistics
=====
Service Realm : sbaRealm1 Gateway Id : 1
Service Name : scp
-----
Service Instance : scp
NF Instance IP/FQDN : 10.10.43.80
NF Instance Port : 7890
SCP Request UDM Create Tx : 0 SCP Request UDM Create Rx : 0
SCP Request UDM Update Tx : 0 SCP Request UDM Update Rx : 0
SCP Request UDM Delete Tx : 0 SCP Request UDM Delete Rx : 0

SCP Request PCF Create Tx : 0 SCP Request PCF Create Rx : 0
SCP Request PCF Update Tx : 0 SCP Request PCF Update Rx : 0
SCP Request PCF Delete Tx : 0 SCP Request PCF Delete Rx : 0

```

```

SCP Request CHF Create Tx : 0 SCP Request CHF Create Rx : 0
SCP Request CHF Update Tx : 0 SCP Request CHF Update Rx : 0
SCP Request CHF Delete Tx : 0 SCP Request CHF Delete Rx : 0

SCP Failures UDM : 0
SCP Failures PCF : 0
SCP Failures CHF : 0
SCP Failures AMF : 0
SCP TPS UDM Min : 0 Max : 0
SCP TPS PCF Min : 0 Max : 0
SCP TPS CHF Min : 0 Max : 0
SCP TPS AMF Min : 0 Max : 0

Total TPS Min : 0 Max : 0 Last Max TPS : <Time Stamp>

Malformed packets Rx : 0
Missing mand IE pkts Rx : 0

Resp Time < 1 Sec : 0 Resp Time 1-4 Sec : 0
Resp Time > 4 Sec : 0 Retries : 0

```

Example output SEPP service statistics

```

# show mobile-gateway pdn service-stats nf-type sepp
=====
SEPP peer service statistics
=====
Service Realm      : myRealm                      Gateway Id   : 1
Service Name       : sepp
-----
Service instance   : sepp1
NF Instance IP     : 1.1.1.1
NF Instance Port   : 80
SBI Request Tx     : 0                          SBI Request Rx : 0
SBI Response Tx    : 0                          SBI Response Rx : 0
SEPP failures      : 0

```

Example output SCP service statistics

```

=====
SCP peer service statistics
=====
Service Realm      : sbaRealm1 Gateway Id : 1
Service Name       : scp
-----
Service Instance   : scp

```



```

NF Instance IP/FQDN      : 10.50.5.1
NF Instance Port         : 8888
SCP Request Tx           : 2 SCP Request Rx : 0
SCP Response Tx          : 0 SCP Response Rx : 2
SCP Failures             : 0
SCP Re transmissions     : 0 SCP fallbacks : 0

```

Example output SMF event exposure service statistics

```
# show mobile-gateway pdn service-stats service-name nsmf-event-exposure
```

```
=====
SMF event exposure service statistics
=====
```

```
Service Realm      : sbaRealm1                      Gateway Id   : 1
Service Name       : nsmf-event-exposure
-----
```

```
Service Instance   : insta-1
Subscr Req         : 0                               Subscr Req Succ      : 0
Subscr Req Fail    : 0
Subscr Mod         : 0                               Subscr Mod Req Succ  : 0
Subscr Mod Fail    : 0
Unsubscr Req       : 0                               Unsubscr Req Succ    : 0
Unsubscr Req Fail  : 0                               Unsubscr Implicit    : 0
Notifications      : 0                               Notification Req Succ : 0
Notification Req Fail : 0
-----
```

```
Number of NF instances : 1
-----
```

Example output UDM service statistics

```
# show mobile-gateway pdn service-stats nf-type udm
```

```
=====
UDM peer service statistics
=====
```

```
Service Realm      : urllc                      Gateway Id   : 1
Service Name       : nudm-sdm
-----
```

```
Service Instance   : insta-1
NF instance ID      : 12345678-9012-3456-7890-123456789015
Get Session Management Subscription Data Req : 14
Get Session Management Subscription Data Succ : 8
Get Session Management Subscription Data Fail : 6
Subscribe SDM Notification Req : 8
Subscribe SDM Notification Succ : 6
Subscribe SDM Notification Fail : 2
Unsubscribe SDM Notification Req : 2
Unsubscribe SDM Notification Succ : 1
Unsubscribe SDM Notification Fail : 1

```

```

SDM Notify: 10
Number of SDM Subscriptions: 4
-----
Number of NF instances : 1
-----
Service Realm      : urllc                               Gateway Id   : 1
Service Name       : nudm-uecm
-----
Service Instance   : insta-1
NF instance ID      : 12345678-9012-3456-7890-123456789015
SMF Registration Req : 14
SMF Registration Succ : 8
SMF Registration Fail : 6
SMF Deregistration Req : 8
SMF Deregistration Succ : 7
SMF Deregistration Fail : 1
-----
Number of NF instances : 1
-----

```

Example output UDM aggregate service statistics

```

# show mobile-gateway pdn service-stats nf-type udm aggregate
=====
UDM peer service statistics aggregate
=====
Service Realm      : sbaRealm1                               Gateway Id   : 1
Service Name       : nudm-sdm
-----
Get SM Subscription Data
Request Tx          : 1
Response Success Rx : 1
Response Fail Rx    : 0
Request Failure Tx  : 0
Request Timeout Tx  : 0
Request Retransmission Tx : 0
Malformed Response Rx : 0
Unknown Session Response Rx : 0
Peer Failover       : 0
Subscribe SDM Notification
Request Tx          : 1
Response Success Rx : 1
Response Fail Rx    : 0
Request Failure Tx  : 0
Request Timeout Tx  : 0
Request Retransmission Tx : 0
Malformed Response Rx : 0
Unknown Session Response Rx : 0
Peer Failover       : 0
Unsubscribe SDM Notification
Request Tx          : 0
Response Success Rx : 0
Response Fail Rx    : 0
Request Failure Tx  : 0
Request Timeout Tx  : 0
Request Retransmission Tx : 0

```

```

Malformed Response Rx          : 0
Unknown Session Response Rx    : 0
Peer Failover                  : 0

SDM Notification
Request Rx                     : 0
Response Success Tx            : 0
Response Fail Tx               : 0
Malformed Request Rx           : 0
Unknown Session Request Rx     : 0
Number of SDM Subscriptions    : 0
-----
Number of instances      : 3
-----
Service Realm           : sbaRealm1          Gateway Id   : 1
Service Name            : nudm-uecm
-----
SMF Registration
Request Tx              : 1
Response Success Rx     : 1
Response Fail Rx        : 0
Request Failure Tx      : 0
Request Timeout Tx      : 0
Request Retransmission Tx : 0
Malformed Response Rx   : 0
Unknown Session Response Rx : 0
Peer Failover           : 0
SMF Deregistration
Request Tx              : 1
Response Success Rx     : 1
Response Fail Rx        : 0
Request Failure Tx      : 0
Request Timeout Tx      : 0
Request Retransmission Tx : 0
Malformed Response Rx   : 0
Unknown Session Response Rx : 0
Peer Failover           : 0
-----
Number of instances      : 3
=====

```

Example output UDM failure codes service statistics

```

# show mobile-gateway pdn service-stats nf-type udm failure-codes
=====
UDM peer service statistics
=====
Service Realm           : sbaRealm1          Gateway Id   : 1
Service Name            : nudm-sdm
-----
Service Instance        : sdm1
NF instance ID          : 44f08cba-9a89-77fe-ab7d-
                        2a0066584f55
Session Management Subscription Data Fail : 0
401 Unauthorized        : 0                  404 User Not Found : 0

```

```

404 Data Not Found      : 0
Subscribe SDM Notification Fail      : 0
404 User Not Found      : 0          404 Context Not Found      : 0
501 Not Implemented      : 0
-----
Service Instance          : sdm1
NF instance ID            : 44f08cba-9a89-77fe-ab7d-
                           2a0066584f65

Session Management Subscription Data Fail : 0
401 Unauthorized          : 0          404 User Not Found      : 0
404 Data Not Found        : 0
Subscribe SDM Notification Fail      : 0
404 User Not Found        : 0          404 Context Not Found      : 0
501 Not Implemented        : 0
-----
Number of instances      : 2
-----
Service Realm             : sbaRealm1          Gateway Id      : 1
Service Name              : nudm-uecm
-----
Service Instance          : uecm1
NF instance ID            : 44f08cba-9a89-77fe-ab7d-
                           2a0066584f75

SMF Registration Fail      : 0
401 Unauthorized          : 0          403 Forbidden        : 0
404 Not Found              : 0
-----
Service Instance          : uecm1
NF instance ID            : 44f08cba-9a89-77fe-ab7d-
                           2a0066584f85

SMF Registration Fail      : 0
401 Unauthorized          : 0          403 Forbidden        : 0
404 Not Found              : 0
-----
Number of instances      : 2
=====

```

5.2.2.58 static-predefined-pcc

Syntax

```
static-predefined-pcc [apn apn-name] [default-policy-rule-base base-policy-name]
[gateway-id gw-id]
```

Context

```
show>mobile>pdn
```

Description

This command displays PCC information.

Parameters

apn-name

specifies the APN name

base-policy-name

specifies the policy name

gw-id

specifies the gateway ID

Output

The following output example shows static PCC information and *Show PDN static predefined PCC output fields* describes the output fields.

Example output static PCC

```
# show mobile-gateway pdn static-predefined-pcc
=====
Static PCC
=====
Gateway ID      : 1
APN             : apn2.nokia.com

APN            : ipd.nokia.0.1.com
APN            : ipd.nokia.0.2.com
APN            : ipd.nokia.0.3.com
APN            : ipd.nokia.0.4.com
APN            : ipd.nokia.0.5.com
APN            : ipd.nokia.0.6.com
APN            : ipd.nokia.0.7.com
APN            : ipd.nokia.0.8.com
APN            : ipd.nokia.0.9.com
APN            : ipd.nokia.0.10.com
APN            : ipd.nokia.0.11.com
APN            : ipd.nokia.com
APN            : ipd.nokia.com.mnc012.mcc345.gprs
```

```
-----  
Number of policies : 0  
=====
```

Table 91: Show PDN static predefined PCC output fields

Label	Description
Gateway ID	The gateway ID
APN	The APN name
Number of policies	The number of policies retrieved

5.2.2.59 statistics

Syntax

```
statistics [gw-type sgw | pgw | upf] [gateway gw-id | group group-number | [card slot-number |  
vm vm-id]] [summary] [suppress-zero]
```

```
statistics attach-failure-statistics [gateway gw-id | group group-number] [summary]
```

```
statistics [summary]
```

Context

```
show>mobile>pdn
```

Description

This command displays statistics for each card.

Parameters

gw-type

specifies the gateway type

Values: *sgw* | *pgw* | *upf*

gw-id

specifies the gateway ID

Values: 1 to 8

group-number

specifies the group number

Values: 1 to 8

slot-number

specifies the card slot number (applicable only to 7750 SR MG)

Values: 1 to 10

vm-id

specifies the VM ID (applicable only to CMG)

Values: 1 to 20

summary

displays summarized information

attach-failure-statistics

displays the attach failure statistics

Output

The following 7750 SR MG output examples show PDN statistics information and *Show PDN statistics output fields* describes the output fields.

**NOTE:**

The outputs of a CMG system are identical to the 7750 SR MG outputs except they have some CMG-specific fields displayed instead of hardware components (for example, VM is displayed instead of Card).

- *Example output PDN statistics*
- *Example output PDN SMF statistics*
- *Example output PDN gateway statistics*
- *Example output PDN group SMF statistics*
- *Example output PDN group UPF statistics*
- *Example output PDN card statistics*
- *Example output PDN card summary statistics*
- *Example output PDN PGW statistics*
- *Example output PDN summary statistics*
- *Example output PDN attach failure ePDG statistics*
- *Example output PDN UPF statistics*

Example output PDN statistics

```
# show mobile-gateway pdn statistics
=====
PDN gateway statistics
=====
VNF/VM                : 1/3      Gateway                : 1

Real APNs              : 43      Roamers                 : 0
Homers                 : 0        Visitors                : 0
VPRNs                  : 20      IP Local Pools          : 23
PDN Sessions           : 0        Ipv4 PDN Sessions       : 0
Ipv6 PDN Sessions      : 0        Ipv4v6 PDN Sessions     : 0
Non-IP PDN sessions    : 0
Bearers                : 0        Default Bearers         : 0
Dedicated Bearers      : 0        Ipv4 Bearers            : 0
Ipv6 Bearers           : 0        Ipv4v6 Bearers          : 0
SDFs                   : 0
IPv4 CAM entries in use : 0      IPv6 CAM entries in use : 0
HSS Static IPv4 PDN Sessio*: 0    HSS Static IPv6 PDN Sessio*: 0
HSS Static IPv4v6 PDN Sess*: 0    eHRPD PDN sessions      : 0
LTE PDN sessions       : 0        2G/3G PDN sessions      : 0
PDN sessions in suspend st*: 0    Emergency PDN sessions  : 0
Rf Peers               : 1        Rf Acct start buffered  : 0
Rf Acct Int buffered   : 0        Rf Acct stop buffered   : 0
Ga DRT Req in queue    : 0        Ga DRT Req max queued   : 0
Ga CDRs buffered       : 0        S2b Sessions            : 0
Redirected Sessions    : 0        Tethered Sessions       : 0
Overload - Dropped requests: 0
Paging in progress     : 0        Allocated Paging Buffers : 0
Available Paging Buffers : 8191    Paging Buffers n/a Errors : 0
Paging Buffer Curr packets : 3800000    Paging buffer Max Packets : 64000000
Paging buffer Curr memory : 10GB     Paging buffer Max memory  : 64GB
Charging paused sessions : 0
Number of ENBs         : 0        Number of MMEs          : 0
Total Number of UEs     : 0        Total Number of Idle UEs : 0
UEs in suspended state  : 0        Paging drops             : 0
Buff Pkts drop tmr expired : 0      Buff Bytes drop tmr expired: 0 B
Buff Pkts drop max pkt lmt : 0      Buff Bytes drop max pkt lmt: 0 B
Buff Pkts drop max cap lmt : 0      Buff Bytes drop max cap lmt: 0 B
Number of RNCs         : 0        Number of SGSNs          : 0
Combo Bearers          : 0        Combo Default bearers    : 0
Combo Active Bearers   : 0        Combo Idle bearers       : 0
Combo Dedicated bearers : 0        Combo Ipv4 bearers       : 0
Combo Ipv6 bearers     : 0        Combo Ipv4v6 bearers     : 0
Combo sessions         : 0        Combo Ipv4 sessions      : 0
Combo Ipv6 sessions    : 0        Combo Ipv4v6 sessions    : 0
Combo Non-IP sessions  : 0
IPv4 Home Agent Sessions : 0      HRPD PDN sessions       : 0
LTE NEMO Sessions      : 0        eHRPD NEMO Sessions     : 0
HA NEMO Sessions       : 0        NEMO IPv4 Subnets       : 0
Ga Big DTRs buffered   : 0        ePDG sessions           : 0
Steering sessions      : 0
S2a Sessions           : 0
Non Terminating Sessions : 0
TWAG Open SSID portal sess : 0
```



```

TWAG SaMOG Sessions      : 0          TWAG NSWO sessions      : 0
ePDG 5GS total sessions  : 0
ePDG 5GS Init Att Sessions : 0
ePDG 5GS HI Att Sessions  : 0
Steering default sessions : 0          Gy CCRs buffered      : 0
Low Priority Combo Sessions: 0          Low Priority PGW Sessions : 0
Active ePDG Emergency Calls: 0         Unsol. SSG Sess In Att Ret*: 0
PDN sessions w dedic bear : 0
Number of PFCP Peers     : 0          PDN sessions in Idle state : 0
eMPS IMS Sessions        : 0          eMPS IMS Advance Prio Sess : 0
eMPS dedicated bearers    : 0
Combo Sessions by 5G UEs  : 0          PGW Sessions by 5G UEs     : 0
PFCP-u Shared Tnl Sess   : 0          PFCP-u Shared Tnl Sess Ipv4: 0
PFCP-u Shared Tnl Sess Ipv6: 0

UL Throughput (Mbps)      : 0
DL Throughput (Mbps)      : 0

-----
Number of VMs : 1
=====
* indicates that the corresponding row element may have been truncated.
=====
Serving gateway statistics
=====
VNF/VM      : 1/3      Gateway      : 1

APNs        : 43      Bearers      : 0
Default bearers : 0      Dedicated bearers : 0
Ipv4 bearers   : 0      Ipv6 bearers   : 0
Ipv4v6 bearers : 0      Roamers        : 0
Active bearers : 0      Idle bearers    : 0

Number of MMEs : 0      Number of PGWs   : 0
Rf Peers       : 1      Number of ENBs   : 0
Number of RNCs : 0

Paging in progress : 0      Allocated Paging Buffers : 0
Available Paging Buffers : 8191      Paging Buffers n/a Errors : 0
Pause chrg hold timer start: 0      Pause chrg hold timer stop : 0
Number of Homers   : 0      Number of Visitors   : 0
Total Number of UEs : 0      Total Number of Idle UEs : 0
Active Serving UEs : 0
Rf Acct start buffered : 0      Rf Acct Int buffered : 0
Rf Acct stop buffered : 0      Emergency PDN sessions : 0
UEs in suspended state : 0      Paging drops : 0
Ga DRT Req in queue : 0      Ga DRT Req max queued : 0
Ga CDRs buffered    : 0      Number of SGSNs : 0
Overload - Dropped requests: 0
Ga Big DRTs Buffered : 0      Number of MMEs using S11-U : 0
Low Priority SGW Sessions : 0
PDN sessions w dedic bear : 0
eMPS IMS Sessions      : 0      eMPS dedicated bearers : 0
SGW Sessions by 5G UEs : 0
Local Breakout Sessions : 0      Local Breakout bearers : 0

UL Throughput (Mbps) : 0
DL Throughput (Mbps) : 0

```

```
-----
Number of VMs : 1
=====
```

Table 92: Show PDN statistics output fields

Label	Description
Chassis/Card	The chassis and card number
Gateway	The gateway ID
Real APNs	The number of real APNs
Roamers	The number of roaming subscribers
Homers	The number of home subscribers
Visitors	The number of visiting subscribers
VPRNs	The number of VPRNs
IP Local Pools	The number of IP local pools
PDN Sessions	The number of PGW only sessions
Ipv4 PDN Sessions	The number of IPv4 PGW only sessions
Ipv6 PDN Sessions	The number of IPv6 PGW only sessions
Ipv4v6 PDN Sessions	The number of IPv4v6 PGW only sessions
Non-Ip PDN sessions	The number of non-IP PGW only sessions
Bearers	The sum of PGW only bearers
Default Bearers	The number of default PGW only bearers

Table 92: Show PDN statistics output fields (continued)

Label	Description
Dedicated Bearers	The number of dedicated PGW only bearers
Ipv4 Bearers	The number of IPv4 PGW only bearers
Ipv6 Bearers	The number of IPv6 PGW only bearers
Ipv4v6 Bearers	The number of IPv4v6 PGW only Bearers
SDFs	The number of service data flows
IPv4 CAM entries in use	The number of IPv4 CAM entries in use
IPv6 CAM entries in use	The number of IPv6 CAM entries in use
HSS Static IPv4 PDN Sessio*	The option to include the HLR/HSS-provided static IPv4 PGW only sessions in the GTP Create PDP Context Request/Create Session Request message
HSS Static IPv6 PDN Sessio*	The option to include the HLR/HSS-provided static IPv6 PGW only sessions in the GTP Create PDP Context Request/Create Session Request message
HSS Static IPv4v6 PDN Sess*	The option to include the HLR/HSS-provided static IPv4v6 PGW only sessions in the GTP Create PDP Context Request/Create Session Request message
eHRPD PDN sessions	The number of eHRPD PDN PGW only sessions
LTE PDN sessions	The sum of LTE PGW only and combined SGW & PGW sessions
2G/3G PDN sessions	The number of 2G/3G PDN sessions
PDN sessions in suspend st*	The number of PDN sessions that are in suspend state

Table 92: Show PDN statistics output fields (continued)

Label	Description
Emergency PDN sessions	The number of emergency PDN sessions
Rf Peers	The number of Rf peers
Rf Acct start buffered	The number of Rf accounting start buffered messages
Rf Acct Int buffered	The number of Rf accounting interim buffered messages
Rf Acct stop buffered	The number of Rf accounting stop buffered messages
Ga DRT Req in queue	The number of Ga DRT request messages in queue
Ga DRT Req max queued	The maximum number of Ga DRT request messages in queue
Ga CDRs buffered	The number of Ga CDRs buffered
S2b Sessions	The number of S2b Sessions
Redirected Sessions	The number of redirected sessions
Tethered Sessions	The number of tethered sessions
Overload - Dropped requests	The number of the dropped requests due to overload
Paging in progress	The number of paging procedure in progress
Allocated Paging Buffers	The number of allocated paging buffers, for 7750 MG HW / IOM buffers
Available Paging Buffers	The number of available paging buffers, for 7750 MG HW / IOM buffers
Paging Buffers n/a Errors	The number of errors non available paging buffers

Table 92: Show PDN statistics output fields (continued)

Label	Description
Paging Buffer Curr packets	The number of DL packets that are currently on the paging buffer, for CMG
Paging buffer Max Packets	The maximum number of packets that the paging buffer can hold, for CMG
Paging buffer Curr memory	The current amount of memory used for buffering DL packets, for CMG
Paging buffer Max memory	The maximum amount of memory used for buffering DL packets, for CMG
Charging paused sessions	The number of the charging paused sessions
Number of ENBs	The number of eNodeB peers
Number of MMEs	The number of MME peers
Total Number of UEs	The total number of UEs
Total Number of Idle UEs ⁶	The total number of idle UEs
UEs in suspended state	The number of UEs in suspended state
Buff Pkts drop tmr expired	The number of dropped packets due to paging time expiration
Buff Bytes drop tmr expired	The number of bytes of the dropped packets due to paging time expiration
Buff Pkts drop max pkt lmt	The number of dropped packets due to maximum buffered packets limit reached

⁶ For the SMF, this statistic is invalid and displays incorrect results.

Table 92: Show PDN statistics output fields (continued)

Label	Description
Buff Bytes drop max pkt lmt	The number of bytes of the dropped packets due to maximum buffered packet limit reached
Buff Pkts drop max cap lmt	The number of dropped packets due to maximum buffering capacity limit reached
Buff Bytes drop max cap lmt	The number of bytes of the dropped packets due to maximum buffering capacity limit reached
Paging drops	The number of paging drops
Number of RNCs	The number of RNCs
Number of SGSNs	The number of SGSNs
Combo bearers	The number of combined SGW and PGW bearers
Combo Default bearers	The number of the default combined SGW and PGW bearers
Combo Active bearers	The number of the active combined SGW and PGW bearers
Combo Idle bearers	The number of the idle combined SGW and PGW bearers
Combo Dedicated bearers	The number of the dedicated combined SGW and PGW bearers
Combo Ipv4 bearers	The number of combined SGW and PGW IPv4 bearers
Combo Ipv6 bearers	The number of combined SGW and PGW IPv6 bearers
Combo Ipv4v6 bearers	The number of combined SGW and PGW IPv4v6 bearers
Combo sessions	The number of combined SGW and PGW sessions

Table 92: Show PDN statistics output fields (continued)

Label	Description
Combo Ipv4 sessions	The number of combined SGW and PGW IPv4 sessions
Combo Ipv6 sessions	The number of combined SGW and PGW IPv6 sessions
Combo Ipv4v6 sessions	The number of combined SGW and PGW IPv4v6 sessions
Combo Non-Ip sessions	The number of non IP combined SGW and PGW sessions
IPv4 Home Agent Sessions	The number of IPv4 home agent sessions
HRPD PDN sessions	The number of HRPD PDN sessions
LTE NEMO Sessions	The number of LTE NEMO sessions
eHRPD NEMO Sessions	The number of eHRPD NEMO sessions
HA NEMO Sessions	The number of HA NEMO sessions
NEMO IPv4 Subnets	The number of NEMO IPv4 subnets
Ga Big DTRs buffered	The number of Ga Big DTRs buffered sessions
ePDG sessions	The number of ePDG sessions
Steering sessions	The number of steering sessions
S2a Sessions	The number of S2a sessions
Non Terminating Sessions	The number of non-terminating sessions
TWAG Open SSID portal sess	The number of open SSID TWAG portal sessions
TWAG SaMOG Sessions	The number of TWAG SaMOG sessions

Table 92: Show PDN statistics output fields (continued)

Label	Description
TWAG NSWO sessions	The number of TWAG NSWO sessions
ePDG 5GS total sessions	The total number of 5GS sessions active in ePDG
ePDG 5GS Init Att Sessions	The total number of 5GS sessions active in ePDG due to initial type of attach
ePDG 5GS HI Att Sessions	The total number of 5GS sessions active in ePDG due to Handover type of attach
Steering default sessions	The number of the default steering sessions
Gy CCRs buffered	The number of the buffered Gy CCR sessions
Low Priority Combo Sessions	The number of low priority combined SGW and PGW sessions
Low Priority PGW Sessions	The number of low priority PGW sessions
Active ePDG Emergency Calls	The number of active ePDG emergency calls
Unsol. SSG Sess In Att Ret	—
PDN sessions w dedic bear	The number of PDN sessions with a dedicated bearer
Number of PFCP Peers	The number of PFCP peers
PDN sessions in Idle state	The number of PDN sessions in idle state
eMPS IMS Sessions	The number of eMPS IMS sessions It is updated from both the MPS and the WPS detected calls; currently, for ePDG this field is updated only by the WPS calls.
eMPS IMS Advance Prio Sess	The number of eMPS IMS advanced priority session

Table 92: Show PDN statistics output fields (continued)

Label	Description
eMPS dedicated bearers	The number of eMPS dedicated bearers
Combo Sessions by 5G UEs	The number of combined SGW and PGW sessions by 5G UEs
PGW Sessions by 5G UEs	The number of PGW sessions by 5G UEs
PFCP-u Shared Tnl Sess	The number of PFCP-u (GTP-U) shared tunnel sessions, when ICMPv6 RS/RA and NS/NA message handling is enabled for the CP function (required in 5G)
PFCP-u Shared Tnl Sess Ipv4	The number of IPv4 PFCP-u (GTP-U) shared tunnel sessions, when ICMPv6 RS/RA and NS/NA message handling is enabled for the CP function (required in 5G)
PFCP-u Shared Tnl Sess Ipv6	The number of IPv6 PFCP-u (GTP-U) shared tunnel sessions, when ICMPv6 RS/RA and NS/NA message handling is enabled for the CP function (required in 5G)
Local Breakout Sessions	The number of LBO sessions
Local Breakout bearers	The number of LBO bearers
UL Throughput (Mbps)	The uplink throughput in Mbps; this counter prints the instant throughput as it has been calculated in the latest 10 seconds interval, by sampling the forwarded data volume at the beginning and at the end of the interval
DL Throughput (Mbps)	The downlink throughput in Mbps; this counter prints the instant throughput as it has been calculated in the latest 10 seconds interval, by sampling the forwarded data volume at the beginning and at the end of the interval
Number of cards	The number of cards retrieved

Example output PDN SMF statistics

```
# show mobile-gateway pdn statistics
```

PDN gateway statistics

```
=====
VNF/VM                : 1/3      Gateway                : 1
Real APNs              : 41      Roamers                : 0
Homers                 : 0        Visitors               : 1
VPRNs                  : 28      IP Local Pools         : 23
PDN Sessions           : 0        Ipv4 PDN Sessions     : 0
Ipv6 PDN Sessions      : 0        Ipv4v6 PDN Sessions   : 0
Non-IP PDN sessions    : 0        Default Bearers       : 0
Bearers                : 0        Ipv4 Bearers          : 0
Dedicated Bearers      : 0        Ipv4v6 Bearers        : 0
Ipv6 Bearers           : 0
SDFs                   : 0
IPv4 CAM entries in use : 36      IPv6 CAM entries in use : 36
HSS Static IPv4 PDN Sessio*: 0    HSS Static IPv6 PDN Sessio*: 0
HSS Static IPv4v6 PDN Sess*: 0    eHRPD PDN sessions     : 0
LTE PDN sessions       : 0        2G/3G PDN sessions    : 0
PDN sessions in suspend st*: 0    Emergency PDN sessions : 0
Rf Peers               : 1        Rf Acct start buffered : 0
Rf Acct Int buffered   : 0        Rf Acct stop buffered  : 0
Ga DRT Req in queue    : 0        Ga DRT Req max queued  : 0
Ga CDRs buffered       : 0        S2b Sessions           : 0
Redirected Sessions    : 0        Tethered Sessions      : 0
Overload - Dropped requests: 0
Paging in progress     : 0        Allocated Paging Buffers : 0
Available Paging Buffers : 8191    Paging Buffers n/a Errors : 0
Paging Buffer Curr packets : 0      Paging buffer Max Packets : 48000
Paging Buffer Curr memory : 4.000 MB  Paging buffer Max memory  : 128.000 MB
Charging paused sessions : 0
Number of ENBs         : 0        Number of MMEs         : 0
Total Number of UEs     : 1        Total Number of Idle UEs : 0
UEs in suspended state  : 0        Paging drops            : 0
Buff Pkts drop tmr expired : 0      Buff Bytes drop tmr expired: 0 B
Buff Pkts drop max pkt lmt : 0      Buff Bytes drop max pkt lmt: 0 B
Buff Pkts drop max cap lmt : 0      Buff Bytes drop max cap lmt: 0 B
Number of RNCs         : 0
Combo Bearers          : 0        Number of SGSNs        : 0
Combo Active Bearers    : 0        Combo Default bearers   : 0
Combo Dedicated bearers : 0        Combo Idle bearers      : 0
Combo Ipv6 bearers      : 0        Combo Ipv4 bearers      : 0
Combo sessions         : 0        Combo Ipv4v6 bearers    : 0
Combo Ipv6 sessions     : 0        Combo Ipv4 sessions     : 0
Combo Non-IP sessions   : 0        Combo Ipv4v6 sessions   : 0
IPv4 Home Agent Sessions : 0
LTE NEMO Sessions      : 0        HRPD PDN sessions      : 0
HA NEMO Sessions       : 0        eHRPD NEMO Sessions    : 0
Ga Big DTRs buffered   : 0        NEMO IPv4 Subnets     : 0
Steering sessions      : 0        ePDG sessions          : 0
S2a Sessions           : 0
TWAG Open SSID portal sess : 0
TWAG SaM0G Sessions    : 0        TWAG NSW0 sessions     : 0
ePDG 5GS Total Sessions : 0        ePDG 5GS Init At Sessions : 0
=====
```

```

ePDG 5GS HO At Sessions      : 0
Steering default sessions    : 0      Gy CCRs buffered           : 0
Low Priority Combo Sessions: 0      Low Priority PGW Sessions  : 0
Active ePDG Emergency Calls: 0      Unsol. SSG Sess In Att Ret*: 0
PDN sessions w dedic bear   : 0
Number of PFCP Peers        : 2      PDN sessions in Idle state : 0
eMPS IMS Sessions           : 0      eMPS IMS Advance Prio Sess : 0
eMPS dedicated bearers      : 0
Combo Sessions by 5G UEs    : 0      PGW Sessions by 5G UEs    : 0
PFCP-u Shared Tnl Sess     : 33      PFCP-u Shared Tnl Sess Ipv4: 33
PFCP-u Shared Tnl Sess Ipv6: 33
UL Throughput (Mbps)        : 0
DL Throughput (Mbps)        : 0

```

```
-----
Number of VMs : 1
=====
```

```
* indicates that the corresponding row element may have been truncated.
=====
```

Serving gateway statistics

```

=====
VNF/VM          : 1/3      Gateway                : 1
APNs            : 41      Bearers                : 0
Default bearers : 0      Dedicated bearers     : 0
Ipv4 bearers    : 0      Ipv6 bearers          : 0
Ipv4v6 bearers  : 0      Roamers               : 0
Active bearers  : 0      Idle bearers          : 0
Number of MMEs  : 0      Number of PGWs        : 0
Rf Peers        : 1      Number of ENBs        : 0
Number of RNCs  : 0
Paging in progress : 0    Allocated Paging Buffers : 0
Available Paging Buffers : 8191  Paging Buffers n/a Errors : 0
Pause chrg hold timer start: 0    Pause chrg hold timer stop : 0
Number of Homers : 0      Number of Visitors     : 0
Total Number of UEs : 1    Total Number of Idle UEs : 0
Active Serving UEs : 1
Rf Acct start buffered : 0      Rf Acct Int buffered    : 0
Rf Acct stop buffered : 0      Emergency PDN sessions  : 0
UEs in suspended state : 0      Paging drops            : 0
Ga DRT Req in queue  : 0      Ga DRT Req max queued   : 0
Ga CDRs buffered     : 0      Number of SGSNs         : 0
Overload - Dropped requests: 0
Ga Big DRTs Buffered : 0      Number of MMEs using S11-U : 0
Low Priority SGW Sessions : 0
PDN sessions w dedic bear : 0
eMPS IMS Sessions    : 0      eMPS dedicated bearers   : 0
SGW Sessions by 5G UEs : 0
Local breakout sessions : 0      Local breakout bearers   : 0
UL Throughput (Mbps)  : 0
DL Throughput (Mbps)  : 0

```

```
-----
Number of VMs : 1
=====
```

Session Management Function statistics

```

=====
VNF/VM          : 1/3      Gateway                : 1
Real DNNs       : 0      Roamers                : 0

```

```

Homers                : 0          Visitors                : 0
VPRNs                 : 0          IP Local Pools          : 0
PDU Sessions          : 1          Ipv4 PDU Sessions     : 1
Ipv6 PDU Sessions     : 0          Ipv4v6 PDU Sessions  : 0
UDM Static IPv4 PDU Sessio*: 0      UDM Static IPv6 PDU Sessio*: 0
UDM Static IPv4v6 PDU Sess*: 0
Active PDU Sessions   : 0          Idle PDU Sessions     : 0
QoS Flows              : 1          Dedicated QoS Flows  : 0
Default QoS Flows     : 0          Allocated Paging Buffers : 0
Paging in progress    : 0          Paging Buffers n/a Errors : 0
Available Paging Buffers : 0        Number of AMFs        : 0
Number of GNBs        : 0          Total Number of Idle UEs : 0 #This statistic is invalid and
Total Number of UEs    : 1          displays incorrect results.#
Num of UEs SGW Serv Node : 0        Num of UEs ePDG Serv Node : 0
Num of UEs AMF Serv Node : 0        Num of 5G Capble UEs 4G Ses: 0
Paging drops          : 0          Emergency PDU Sessions : 0
-----
Number of VMs : 1
=====
* indicates that the corresponding row element may have been truncated.

```

Example output PDN gateway statistics

```

# show mobile-gateway pdn statistics gateway 1
=====
PDN gateway statistics
=====
Chassis/Card          : 1/3          Gateway                : 1

Real APNs              : 43          Roamers                 : 0
Homers                 : 0          Visitors                : 0
VPRNs                  : 20          IP Local Pools          : 23
PDU Sessions           : 0          Ipv4 PDU Sessions       : 0
Ipv6 PDU Sessions      : 0          Ipv4v6 PDU Sessions     : 0
Non-Ip PDU sessions    : 0          Default Bearers         : 0
Bearers                : 0          Ipv4 Bearers            : 0
Dedicated Bearers      : 0          Ipv4v6 Bearers          : 0
Ipv6 Bearers           : 0
SDFs                   : 0
HSS Static IPv4 PDU Sessio*: 0      HSS Static IPv6 PDU Sessio*: 0
HSS Static IPv4v6 PDU Sess*: 0      eHRPD PDU sessions      : 0
LTE PDU sessions       : 0          2G/3G PDU sessions     : 0
PDU sessions in suspend st*: 0      Emergency PDU sessions  : 0
Rf Peers               : 1          Rf Acct start buffered  : 0
Rf Acct Int buffered   : 0          Rf Acct stop buffered   : 0
Ga DRT Req in queue    : 0          Ga DRT Req max queued   : 0
Ga CDRs buffered       : 0          S2b Sessions            : 0
Redirected Sessions    : 0          Tethered Sessions       : 0
Overload - Dropped requests: 0
Paging in progress     : 0          Allocated Paging Buffers : 0
Available Paging Buffers : 8191      Paging Buffers n/a Errors : 0
Charging paused sessions : 0
Number of ENBs         : 0          Number of MMEs          : 0
Total Number of UEs    : 0          Total Number of Idle UEs : 0
UEs in suspended state : 0          Paging drops             : 0

```

```

Buff Pkts drop tmr expired : 0      Buff Bytes drop tmr expired: 400 B
Buff Pkts drop max pkt lmt : 0      Buff Bytes drop max pkt lmt: 38.147 MB
Buff Pkts drop max cap lmt : 0      Buff Bytes drop max cap lmt: 3.725 GB
Number of RNCs                : 0      Number of SGSNs                : 0
Combo Bearers                  : 0      Combo Default bearers          : 0
Combo Active Bearers           : 0      Combo Idle bearers             : 0
Combo Dedicated bearers        : 0      Combo Ipv4 bearers             : 0
Combo Ipv6 bearers             : 0      Combo Ipv4v6 bearers          : 0
Combo sessions                 : 0      Combo Ipv4 sessions           : 0
Combo Ipv6 sessions            : 0      Combo Ipv4v6 sessions          : 0
Combo Non-IP sessions          : 0
IPv4 Home Agent Sessions       : 0      HRPD PDN sessions              : 0
LTE NEMO Sessions              : 0      eHRPD NEMO Sessions           : 0
HA NEMO Sessions               : 0      NEMO IPv4 Subnets            : 0
Ga Big DTRs buffered           : 0      ePDG sessions                  : 0
Steering sessions              : 0
S2a Sessions                   : 0
TWAG Open SSID portal sess : 0
TWAG SaMOG Sessions           : 0      TWAG NSWO sessions            : 0
Steering default sessions      : 0      Gy CCRs buffered              : 0
Low Priority Combo Sessions: 0      Low Priority PGW Sessions      : 0
Active ePDG Emergency Calls: 0      Unsol. SSG Sess In Att Ret*: 0
PDN sessions w dedic bear     : 0
Number of PFCP Peers           : 0      PDN sessions in Idle state : 0
eMPS IMS Sessions              : 0      eMPS IMS Advance Prio Sess : 0
eMPS dedicated bearers         : 0
Combo Sessions by 5G UEs       : 0      PGW Sessions by 5G UEs       : 0
PFCP-u Shared Tnl Sess        : 0      PFCP-u Shared Tnl Sess Ipv4: 0
PFCP-u Shared Tnl Sess Ipv6: 0

UL Throughput (Mbps)           : 0
DL Throughput (Mbps)           : 0

```

Number of cards : 1

=====

* indicates that the corresponding row element may have been truncated.

=====

Serving gateway statistics

=====

Chassis/Card	: 1/3	Gateway	: 1
APNs	: 43	Bearers	: 0
Default bearers	: 0	Dedicated bearers	: 0
Ipv4 bearers	: 0	Ipv6 bearers	: 0
Ipv4v6 bearers	: 0	Roamers	: 0
Active bearers	: 0	Idle bearers	: 0
Number of MMEs	: 0	Number of PGWs	: 0
Rf Peers	: 1	Number of ENBs	: 0
Number of RNCs	: 0		
Paging in progress	: 0	Allocated Paging Buffers	: 0
Available Paging Buffers	: 8191	Paging Buffers n/a Errors	: 0
Pause chrg hold timer start:	: 0	Pause chrg hold timer stop	: 0
Number of Homers	: 0	Number of Visitors	: 0
Total Number of UEs	: 0	Total Number of Idle UEs	: 0

```

Active Serving UEs          : 0
Rf Acct start buffered     : 0      Rf Acct Int buffered      : 0
Rf Acct stop buffered      : 0      Emergency PDN sessions   : 0
UEs in suspended state    : 0      Paging drops             : 0
Ga DRT Req in queue       : 0      Ga DRT Req max queued    : 0
Ga CDRs buffered          : 0      Number of SGSNs          : 0
Overload - Dropped requests: 0
Ga Big DRTs Buffered       : 0      Number of MMEs using S11-U : 0
Low Priority SGW Sessions  : 0
PDN sessions w dedic bear  : 0
eMPS IMS Sessions         : 0      eMPS dedicated bearers    : 0
SGW Sessions by 5G UEs    : 0

UL Throughput (Mbps)       : 0
DL Throughput (Mbps)       : 0

```

```

-----
Number of cards : 1
=====

```

Example output PDN group SMF statistics

A:Dut-C# show mobile-gateway pdn statistics group 1

PDN gateway statistics

```

=====
VNF/VM          : 1/3      Gateway          : 1

Real APNs       : 46      Roamers          : 0
Homers          : 0      Visitors         : 0
VPRNs          : 43      IP Local Pools   : 38
PDN Sessions    : 0      Ipv4 PDN Sessions : 0
Ipv6 PDN Sessions : 0      Ipv4v6 PDN Sessions : 0
Non-Ip PDN sessions : 0
Bearers         : 0      Default Bearers   : 0
Dedicated Bearers : 0      Ipv4 Bearers      : 0
Ipv6 Bearers    : 0      Ipv4v6 Bearers    : 0
SDFs            : 0
IPv4 CAM entries in use : 36      IPv6 CAM entries in use : 36
HSS Static IPv4 PDN Sessio*: 0      HSS Static IPv6 PDN Sessio*: 0
HSS Static IPv4v6 PDN Sess*: 0      eHRPD PDN sessions : 0
LTE PDN sessions : 0      2G/3G PDN sessions : 0
PDN sessions in suspend st*: 0      Emergency PDN sessions : 0
Rf Peers        : 1      Rf Acct start buffered : 0
Rf Acct Int buffered : 0      Rf Acct stop buffered : 0
Ga DRT Req in queue : 0      Ga DRT Req max queued : 0
Ga CDRs buffered   : 0      S2b Sessions        : 0
Redirected Sessions : 0      Tethered Sessions    : 0
Overload - Dropped requests: 0
Paging in progress : 0      Allocated Paging Buffers : 0
Available Paging Buffers : 8191      Paging Buffers n/a Errors : 0
Paging Buffer Curr packets : 0      Paging buffer Max Packets : 48000
Paging Buffer Curr memory : 0 B      Paging buffer Max memory : 128.000 MB
Charging paused sessions : 0
Number of ENBs     : 0      Number of MMEs      : 0
Total Number of UEs : 0      Total Number of Idle UEs : 0

```

```

UEs in suspended state      : 0      Paging drops                  : 0
Buff Pkts drop tmr expired  : 0      Buff Bytes drop tmr expired: 0 B
Buff Pkts drop max pkt lmt : 0      Buff Bytes drop max pkt lmt: 0 B
Buff Pkts drop max cap lmt : 0      Buff Bytes drop max cap lmt: 0 B
Number of RNCs              : 0      Number of SGSNs              : 0
Combo Bearers               : 0      Combo Default bearers        : 0
Combo Active Bearers        : 0      Combo Idle bearers           : 0
Combo Dedicated bearers     : 0      Combo Ipv4 bearers           : 0
Combo Ipv6 bearers          : 0      Combo Ipv4v6 bearers         : 0
Combo sessions              : 0      Combo Ipv4 sessions          : 0
Combo Ipv6 sessions         : 0      Combo Ipv4v6 sessions        : 0
Combo Non-IP sessions       : 0
IPv4 Home Agent Sessions    : 0      HRPD PDN sessions            : 0
LTE NEMO Sessions           : 0      eHRPD NEMO Sessions          : 0
HA NEMO Sessions            : 0      NEMO IPv4 Subnets           : 0
Ga Big DTRs buffered        : 0      ePDG sessions                : 0
Steering sessions           : 0
S2a Sessions                : 0
TWAG Open SSID portal sess : 0
TWAG SaMOG Sessions         : 0      TWAG NSWO sessions           : 0
ePDG 5GS Total Sessions     : 0      ePDG 5GS Init At Sessions    : 0
ePDG 5GS HO At Sessions     : 0
Steering default sessions   : 0      Gy CCRs buffered             : 0
Low Priority Combo Sessions  : 0      Low Priority PGW Sessions     : 0
Active ePDG Emergency Calls: 0      Unsol. SSG Sess In Att Ret*: 0
PDN sessions w dedic bear   : 0
Number of PFCP Peers        : 3      PDN sessions in Idle state   : 0
eMPS IMS Sessions           : 0      eMPS IMS Advance Prio Sess    : 0
eMPS dedicated bearers      : 0
Combo Sessions by 5G UEs    : 0      PGW Sessions by 5G UEs       : 0
PFCP-u Shared Tnl Sess      : 37     PFCP-u Shared Tnl Sess Ipv4: 37
PFCP-u Shared Tnl Sess Ipv6: 37

```

```

UL Throughput (Mbps)        : 0
DL Throughput (Mbps)        : 0

```

```

-----
Number of VMs : 1
=====

```

* indicates that the corresponding row element may have been truncated.

Serving gateway statistics

```

=====
VNF/VM          : 1/3      Gateway                  : 1
APNs             : 46      Bearers                  : 0
Default bearers  : 0      Dedicated bearers        : 0
Ipv4 bearers     : 0      Ipv6 bearers             : 0
Ipv4v6 bearers   : 0      Roamers                  : 0
Active bearers   : 0      Idle bearers             : 0

Number of MMEs   : 0      Number of PGWs           : 0
Rf Peers         : 1      Number of ENBs           : 0
Number of RNCs   : 0

Paging in progress : 0      Allocated Paging Buffers : 0
Available Paging Buffers : 8191  Paging Buffers n/a Errors : 0

```

```

Pause chrg hold timer start: 0          Pause chrg hold timer stop : 0
Number of Homers                : 0      Number of Visitors         : 0
Total Number of UEs             : 0      Total Number of Idle UEs   : 0
Active Serving UEs              : 0
Rf Acct start buffered          : 0      Rf Acct Int buffered       : 0
Rf Acct stop buffered           : 0      Emergency PDN sessions     : 0
UEs in suspended state          : 0      Paging drops               : 0
Ga DRT Req in queue             : 0      Ga DRT Req max queued      : 0
Ga CDRs buffered                : 0      Number of SGSNs            : 0
Overload - Dropped requests: 0
Ga Big DRTs Buffered            : 0      Number of MMEs using S11-U : 0
Low Priority SGW Sessions        : 0
PDN sessions w dedic bear       : 0
eMPS IMS Sessions               : 0      eMPS dedicated bearers     : 0
SGW Sessions by 5G UEs          : 0
Local breakout sessions         : 0      Local breakout bearers     : 0

UL Throughput (Mbps)            : 0
DL Throughput (Mbps)            : 0

-----
Number of VMs : 1
=====

=====
Session Management Function statistics
=====
VNF/VM                : 1/3      Gateway                    : 1
Real DNNs              : 0
H-SMF Roamers          : 0      V-SMF Roamers              : 0
Homers                 : 0      Visitors                    : 0
VPRNs                  : 0      IP Local Pools              : 0
PDU Sessions           : 0      Ipv4 PDU Sessions           : 0
Ipv6 PDU Sessions      : 0      Ipv4v6 PDU Sessions         : 0
UDM Static IPv4 PDU Sessio*: 0    UDM Static IPv6 PDU Sessio*: 0
UDM Static IPv4v6 PDU Sess*: 0
Active PDU Sessions     : 0      Idle PDU Sessions           : 0
Ethernet PDU Sessions   : 0
QoS Flows               : 0
Default QoS Flows       : 0      Dedicated QoS Flows         : 0
Paging in progress      : 0      Allocated Paging Buffers    : 0
Available Paging Buffers : 0      Paging Buffers n/a Errors   : 0
Number of GNBs          : 0      Number of AMFs              : 0
Total Number of UEs     : 0      Total Number of Idle UEs    : 0 #This statistic is invalid and
displays incorrect results.#
Num of UEs SGW Serv Node : 0      Num of UEs ePDG Serv Node  : 0
Num of UEs AMF Serv Node : 0      Num of 5G Capble UEs 4G Ses: 0
Paging drops            : 0      Emergency PDU Sessions      : 0

-----
Number of VMs : 1
=====
* indicates that the corresponding row element may have been truncated.

```

Example output PDN group UPF statistics

```
A:Dut-A# show mobile-gateway pdn statistics group 1
```


=====

```

PDN gateway statistics
=====
VNF/VM          : 1/3      Gateway          : 1

Real APNs       : 46      Roamers         : 0
Homers          : 0      Visitors        : 0
VPRNs           : 21      IP Local Pools  : 23
PDN Sessions    : 0      Ipv4 PDN Sessions : 0
Ipv6 PDN Sessions : 0      Ipv4v6 PDN Sessions : 0
Non-IP PDN sessions : 0
Bearers         : 0      Default Bearers : 0
Dedicated Bearers : 0      Ipv4 Bearers    : 0
Ipv6 Bearers    : 0      Ipv4v6 Bearers  : 0
SDFs            : 0
IPv4 CAM entries in use : 36      IPv6 CAM entries in use : 36
HSS Static IPv4 PDN Sessio*: 0      HSS Static IPv6 PDN Sessio*: 0
HSS Static IPv4v6 PDN Sess*: 0      eHRPD PDN sessions : 0
LTE PDN sessions : 0      2G/3G PDN sessions : 0
PDN sessions in suspend st*: 0      Emergency PDN sessions : 0
Rf Peers        : 1      Rf Acct start buffered : 0
Rf Acct Int buffered : 0      Rf Acct stop buffered : 0
Ga DRT Req in queue : 0      Ga DRT Req max queued : 0
Ga CDRs buffered : 0      S2b Sessions        : 0
Redirected Sessions : 0      Tethered Sessions    : 0
Overload - Dropped requests: 0
Paging in progress : 0      Allocated Paging Buffers : 0
Available Paging Buffers : 8191      Paging Buffers n/a Errors : 0
Paging Buffer Curr packets : 0      Paging buffer Max Packets : 48000
Paging Buffer Curr memory : 0 B      Paging buffer Max memory : 128.000 MB
Charging paused sessions : 0
Number of ENBs    : 0      Number of MMES      : 0
Total Number of UEs : 0      Total Number of Idle UEs : 0
UEs in suspended state : 0      Paging drops         : 0
Buff Pkts drop tmr expired : 0      Buff Bytes drop tmr expired: 0 B
Buff Pkts drop max pkt lmt : 0      Buff Bytes drop max pkt lmt: 0 B
Buff Pkts drop max cap lmt : 0      Buff Bytes drop max cap lmt: 0 B
Number of RNCs    : 0      Number of SGSNs      : 0
Combo Bearers     : 0      Combo Default bearers : 0
Combo Active Bearers : 0      Combo Idle bearers    : 0
Combo Dedicated bearers : 0      Combo Ipv4 bearers    : 0
Combo Ipv6 bearers : 0      Combo Ipv4v6 bearers  : 0
Combo sessions    : 0      Combo Ipv4 sessions   : 0
Combo Ipv6 sessions : 0      Combo Ipv4v6 sessions : 0
Combo Non-IP sessions : 0
IPv4 Home Agent Sessions : 0      HRPD PDN sessions    : 0
LTE NEMO Sessions : 0      eHRPD NEMO Sessions  : 0
HA NEMO Sessions  : 0      NEMO IPv4 Subnets    : 0
Ga Big DTRs buffered : 0      ePDG sessions         : 0
Steering sessions : 0
S2a Sessions      : 0
TWAG Open SSID portal sess : 0
TWAG SaMOG Sessions : 0      TWAG NSW0 sessions    : 0
ePDG 5GS Total Sessions : 0      ePDG 5GS Init At Sessions : 0
ePDG 5GS HO At Sessions : 0
Steering default sessions : 0      Gy CCRs buffered      : 0
Low Priority Combo Sessions: 0      Low Priority PGW Sessions : 0

```

```

Active ePDG Emergency Calls: 0          Unsol. SSG Sess In Att Ret*: 0
PDN sessions w dedic bear  : 0
Number of PFCP Peers      : 1          PDN sessions in Idle state : 0
eMPS IMS Sessions        : 0          eMPS IMS Advance Prio Sess : 0
eMPS dedicated bearers    : 0
Combo Sessions by 5G UEs  : 0          PGW Sessions by 5G UEs    : 0
PFCP-u Shared Tnl Sess   : 37         PFCP-u Shared Tnl Sess Ipv4: 37
PFCP-u Shared Tnl Sess Ipv6: 37

```

```

UL Throughput (Mbps)      : 0
DL Throughput (Mbps)      : 0

```

```

-----
Number of VMs : 1
=====

```

* indicates that the corresponding row element may have been truncated.

```

=====
Serving gateway statistics
=====

```

```

VNF/VM          : 1/3          Gateway          : 1

APNs             : 46          Bearers         : 0
Default bearers  : 0          Dedicated bearers : 0
Ipv4 bearers     : 0          Ipv6 bearers     : 0
Ipv4v6 bearers   : 0          Roamers          : 0
Active bearers   : 0          Idle bearers     : 0

Number of MMEs   : 0          Number of PGWs   : 0
Rf Peers         : 1          Number of ENBs   : 0
Number of RNCs   : 0

Paging in progress : 0          Allocated Paging Buffers : 0
Available Paging Buffers : 8191  Paging Buffers n/a Errors : 0
Pause chrg hold timer start: 0    Pause chrg hold timer stop : 0
Number of Homers   : 0          Number of Visitors : 0
Total Number of UEs : 0          Total Number of Idle UEs : 0
Active Serving UEs : 0

Rf Acct start buffered : 0        Rf Acct Int buffered : 0
Rf Acct stop buffered  : 0        Emergency PDN sessions : 0
UEs in suspended state : 0        Paging drops          : 0
Ga DRT Req in queue    : 0        Ga DRT Req max queued : 0
Ga CDRs buffered       : 0        Number of SGSNs       : 0
Overload - Dropped requests: 0

Ga Big DRTs Buffered   : 0        Number of MMEs using S11-U : 0
Low Priority SGW Sessions : 0
PDN sessions w dedic bear : 0
eMPS IMS Sessions      : 0          eMPS dedicated bearers : 0
SGW Sessions by 5G UEs : 0
Local breakout sessions : 0          Local breakout bearers : 0

UL Throughput (Mbps) : 0
DL Throughput (Mbps) : 0

```

```

-----
Number of VMs : 1
=====
=====

```

UPF gateway statistics

```

=====
VNF/VM                : 1/3      Gateway                : 1

Real APNs              : 46
UPF PDN Sessions       : 0        IPv4 UPF PDN Sessions   : 0
IPv6 UPF PDN Sessions  : 0        IPv4v6 UPF PDN Sessions : 0
Static IPv4 UPF PDN Sessio*: 0    Static IPv6 UPF PDN Sessio*: 0
Static IPv4v6 UPF PDN Sess*: 0    Non-IP UPF PDN Sessions : 0
Ethernet Sessions     : 0
UPF Bearers            : 0        Default UPF Bearers    : 0
Dedicated UPF Bearers  : 0        Ipv4 UPF Bearers       : 0
Ipv6 UPF Bearers       : 0        Ipv4v6 UPF Bearers     : 0
PDN sessions in suspend st*: 0    PDN sessions in Idle state : 0
LTE UPF PDN sessions   : 0        3G UPF PDN sessions    : 0
S2b UPF PDN sessions   : 0        S2a UPF PDN sessions    : 0
5G PDU Sessions        : 0
Active PDU Sessions    : 0        Idle PDU Sessions      : 0
QoS Flows              : 0
Default QoS Flows      : 0        Dedicated QoS Flows    : 0
Paging in progress     : 0        Allocated Paging Buffers : 0
Available Paging Buffers : 8191    Paging Buffers n/a Errors : 0
Number of ENBs         : 0        Paging drops           : 0
Number of RNCs         : 0        Number of SGSNs        : 0
Total Number of UEs     : 0        UEs with 5G Session     : 0
Num of Sessions W/O UE Id : 0
Combo Non-IP UPF sessions : 0    Combo UPF Bearers      : 0
Combo Default UPF bearers : 0    Combo Active UPF Bearers : 0
Combo Idle UPF bearers   : 0    Combo Dedicated UPF bearers: 0
Combo Ipv4 UPF bearers   : 0    Combo Ipv6 UPF bearers   : 0
Combo Ipv4v6 UPF bearers : 0    Combo UPF PDN sessions  : 0
Combo Ipv4 UPF PDN sessions: 0    Combo Ipv6 UPF PDN sessions: 0
Combo Ipv4v6 UPF PDN sessi*: 0    PDN sessions w dedic bear : 0

UL Throughput (Mbps)    : 0
DL Throughput (Mbps)    : 0
=====

```

Number of VMs : 1

* indicates that the corresponding row element may have been truncated.

Example output PDN card statistics

```
# show mobile-gateway pdn statistics card 3
```

PDN gateway statistics

```

=====
Chassis/Card          : 1/3      Gateway                : 1

Real APNs              : 43        Roamers                 : 0
Homers                 : 0         Visitors                : 0
VPRNs                  : 20        IP Local Pools          : 23
PDN Sessions           : 0         Ipv4 PDN Sessions       : 0
Ipv6 PDN Sessions      : 0         Ipv4v6 PDN Sessions     : 0
Non-IP PDN sessions    : 0
Bearers                : 0         Default Bearers         : 0
=====

```

```

Dedicated Bearers          : 0          Ipv4 Bearers          : 0
Ipv6 Bearers               : 0          Ipv4v6 Bearers        : 0
SDFs                      : 0
HSS Static IPv4 PDN Sessio*: 0        HSS Static IPv6 PDN Sessio*: 0
HSS Static IPv4v6 PDN Sess*: 0        eHRPD PDN sessions    : 0
LTE PDN sessions           : 0        2G/3G PDN sessions    : 0
PDN sessions in suspend st*: 0        Emergency PDN sessions : 0
Rf Peers                   : 1          Rf Acct start buffered : 0
Rf Acct Int buffered       : 0          Rf Acct stop buffered  : 0
Ga DRT Req in queue        : 0          Ga DRT Req max queued  : 0
Ga CDRs buffered           : 0          S2b Sessions           : 0
Redirected Sessions        : 0          Tethered Sessions      : 0
Overload - Dropped requests: 0
Paging in progress         : 0          Allocated Paging Buffers : 0
Available Paging Buffers   : 8191       Paging Buffers n/a Errors : 0
Charging paused sessions   : 0
Number of ENBs             : 0          Number of MMEs         : 0
Total Number of UEs        : 0          Total Number of Idle UEs : 0
UEs in suspended state     : 0          Paging drops            : 0
Buff Pkts drop tmr expired : 0          Buff Bytes drop tmr expired: 400 B
Buff Pkts drop max pkt lmt : 0          Buff Bytes drop max pkt lmt: 38.147 MB
Buff Pkts drop max cap lmt : 0          Buff Bytes drop max cap lmt: 3.725 GB
Number of RNCs             : 0          Number of SGSNs        : 0
Combo Bearers              : 0          Combo Default bearers   : 0
Combo Active Bearers       : 0          Combo Idle bearers      : 0
Combo Dedicated bearers    : 0          Combo Ipv4 bearers      : 0
Combo Ipv6 bearers         : 0          Combo Ipv4v6 bearers    : 0
Combo sessions             : 0          Combo Ipv4 sessions     : 0
Combo Ipv6 sessions        : 0          Combo Ipv4v6 sessions   : 0
Combo Non-IP sessions      : 0
IPv4 Home Agent Sessions   : 0          HRPD PDN sessions      : 0
LTE NEMO Sessions          : 0          eHRPD NEMO Sessions    : 0
HA NEMO Sessions           : 0          NEMO IPv4 Subnets     : 0
Ga Big DTRs buffered       : 0          ePDG sessions          : 0
Steering sessions          : 0
S2a Sessions               : 0
TWAG Open SSID portal sess : 0
TWAG SaMOG Sessions        : 0          TWAG NSWO sessions     : 0
Steering default sessions  : 0          Gy CCRs buffered       : 0
Low Priority Combo Sessions: 0          Low Priority PGW Sessions : 0
Active ePDG Emergency Calls: 0          Unsol. SSG Sess In Att Ret*: 0
PDN sessions w dedic bear  : 0          PDN sessions in Idle state : 0
Number of PFCP Peers       : 0          eMPS IMS Advance Prio Sess : 0
eMPS IMS Sessions          : 0          PGW Sessions by 5G UEs   : 0
eMPS dedicated bearers     : 0          PFCP-u Shared Tnl Sess Ipv4: 0
Combo Sessions by 5G UEs   : 0
PFCP-u Shared Tnl Sess     : 0
PFCP-u Shared Tnl Sess Ipv6: 0

UL Throughput (Mbps)       : 0
DL Throughput (Mbps)       : 0

```

Number of cards : 1

=====

* indicates that the corresponding row element may have been truncated.

=====

Serving gateway statistics

```

=====
Chassis/Card           : 1/3      Gateway           : 1

APNs                   : 43        Bearers            : 0
Default bearers        : 0        Dedicated bearers  : 0
Ipv4 bearers           : 0        Ipv6 bearers       : 0
Ipv4v6 bearers         : 0        Roamers            : 0
Active bearers         : 0        Idle bearers       : 0

Number of MMEs         : 0        Number of PGWs     : 0
Rf Peers               : 1        Number of ENBs     : 0
Number of RNCs         : 0

Paging in progress     : 0        Allocated Paging Buffers : 0
Available Paging Buffers : 8191    Paging Buffers n/a Errors : 0
Pause chrg hold timer start: 0      Pause chrg hold timer stop : 0
Number of Homers       : 0        Number of Visitors   : 0
Total Number of UEs    : 0        Total Number of Idle UEs : 0
Active Serving UEs     : 0

Rf Acct start buffered : 0        Rf Acct Int buffered   : 0
Rf Acct stop buffered  : 0        Emergency PDN sessions : 0
UEs in suspended state : 0        Paging drops           : 0
Buff Pkts drop tmr expired : 0      Buff Bytes drop tmr expired: 400 B
Buff Pkts drop max pkt lmt : 0      Buff Bytes drop max pkt lmt: 38.147 MB
Buff Pkts drop max cap lmt : 0      Buff Bytes drop max cap lmt: 3.725 GB
Ga DRT Req in queue    : 0        Ga DRT Req max queued   : 0
Ga CDRs buffered       : 0        Number of SGSNs        : 0
Overload - Dropped requests: 0
Ga Big DRTs Buffered   : 0        Number of MMEs using S11-U : 0
Low Priority SGW Sessions : 0
PDN sessions w dedic bear : 0
eMPS IMS Sessions      : 0        eMPS dedicated bearers   : 0
SGW Sessions by 5G UEs : 0

UL Throughput (Mbps)   : 0
DL Throughput (Mbps)   : 0
-----
Number of cards : 1
=====

```

Example output PDN card summary statistics

```

# show mobile-gateway pdn statistics card 3 summary
=====
PDN gateway statistics summary
=====

Real APNs              : 43        Roamers            : 0
Homers                 : 0        Visitors           : 0
VPRNs                 : 20        IP Local Pools     : 23
PDN Sessions           : 0        Ipv4 PDN Sessions  : 0
Ipv6 PDN Sessions      : 0        Ipv4v6 PDN Sessions : 0
Non-Ip PDN sessions    : 0
Bearers                : 0        Default Bearers    : 0
Dedicated Bearers      : 0        Ipv4 Bearers       : 0

```

```

Ipv6 Bearers           : 0          Ipv4v6 Bearers           : 0
SDFs                   : 0
HSS Static IPv4 PDN Sessio*: 0      HSS Static IPv6 PDN Sessio*: 0
HSS Static IPv4v6 PDN Sess*: 0      eHRPD PDN sessions       : 0
LTE PDN sessions       : 0          2G/3G PDN sessions       : 0
PDN sessions in suspend st*: 0      Emergency PDN sessions   : 0
Rf Peers               : 2          Rf Acct start buffered   : 0
Rf Acct Int buffered   : 0          Rf Acct stop buffered    : 0
Ga DRT Req in queue    : 0          Ga DRT Req max queued    : 0
Ga CDRs buffered       : 0          S2b Sessions             : 0
Redirected Sessions     : 0          Tethered Sessions        : 0
Overload - Dropped requests: 0
Paging in progress     : 0          Allocated Paging Buffers : 0
Available Paging Buffers : 8191      Paging Buffers n/a Errors : 0
Charging paused sessions : 0
Number of ENBs         : 0          Number of MMEs           : 0
Total Number of UEs     : 0          Total Number of Idle UEs : 0
UEs in suspended state : 0          Paging drops              : 0
Buff Pkts drop tmr expired : 0      Buff Bytes drop tmr expired: 400 B
Buff Pkts drop max pkt lmt : 0      Buff Bytes drop max pkt lmt: 38.147 MB
Buff Pkts drop max cap lmt : 0      Buff Bytes drop max cap lmt: 3.725 GB
Number of RNCs         : 0          Number of SGSNs          : 0
Combo Bearers          : 0          Combo Default bearers    : 0
Combo Active Bearers   : 0          Combo Idle bearers       : 0
Combo Dedicated bearers : 0          Combo Ipv4 bearers       : 0
Combo Ipv6 bearers     : 0          Combo Ipv4v6 bearers     : 0
Combo sessions         : 0          Combo Ipv4 sessions      : 0
Combo Ipv6 sessions    : 0          Combo Ipv4v6 sessions    : 0
Combo Non-IP sessions  : 0
IPv4 Home Agent Sessions : 0        HRPD PDN sessions        : 0
LTE NEMO Sessions      : 0          eHRPD NEMO Sessions      : 0
HA NEMO Sessions       : 0          NEMO IPv4 Subnets       : 0
Ga Big DTRs buffered   : 0          ePDG sessions            : 0
Steering sessions      : 0
S2a Sessions           : 0
TWAG Open SSID portal sess : 0
TWAG SaMOG Sessions    : 0          TWAG NSW0 sessions       : 0
Steering default sessions : 0        Gy CCRs buffered        : 0
Low Priority Combo Sessions: 0        Low Priority PGW Sessions : 0
Active ePDG Emergency Calls: 0        Unsol. SSG Sess In Att Ret*: 0
PDN sessions w dedic bear : 0
Number of PFCP Peers   : 0          PDN sessions in Idle state : 0
eMPS IMS Sessions      : 0          eMPS IMS Advance Prio Sess : 0
eMPS dedicated bearers : 0
Combo Sessions by 5G UEs : 0          PGW Sessions by 5G UEs   : 0

UL Throughput (Mbps)    : 0
DL Throughput (Mbps)    : 0

```

Number of cards : 1

=====

* indicates that the corresponding row element may have been truncated.

=====

Serving gateway statistics summary

=====

```

APNs                : 43      Bearers                : 0
Default bearers     : 0      Dedicated bearers    : 0
Ipv4 bearers        : 0      Ipv6 bearers        : 0
Ipv4v6 bearers      : 0      Roamers              : 0
Active bearers      : 0      Idle bearers        : 0

Number of MMEs      : 0      Number of PGWs       : 0
Rf Peers            : 1      Number of ENBs       : 0
Number of RNCs      : 0

Paging in progress  : 0      Allocated Paging Buffers : 0
Available Paging Buffers : 8191  Paging Buffers n/a Errors : 0
Pause chrg hold timer start: 0    Pause chrg hold timer stop : 0
Number of Homers    : 0      Number of Visitors     : 0
Total Number of UEs : 0      Total Number of Idle UEs : 0
Active Serving UEs  : 0

Rf Acct start buffered : 0      Rf Acct Int buffered    : 0
Rf Acct stop buffered  : 0      Emergency PDN sessions  : 0
UEs in suspended state : 0      Paging drops            : 0
Buff Pkts drop tmr expired : 0    Buff Bytes drop tmr expired: 400 B
Buff Pkts drop max pkt lmt : 0    Buff Bytes drop max pkt lmt: 38.147 MB
Buff Pkts drop max cap lmt : 0    Buff Bytes drop max cap lmt: 3.725 GB
Ga DRT Req in queue   : 0      Ga DRT Req max queued   : 0
Ga CDRs buffered      : 0      Number of SGSNs         : 0
Overload - Dropped requests: 0

Ga Big DRTs Buffered  : 0      Number of MMEs using S11-U : 0
Low Priority SGW Sessions : 0
PDN sessions w dedic bear : 0
eMPS IMS Sessions     : 0      eMPS dedicated bearers    : 0
SGW Sessions by 5G UEs : 0

UL Throughput (Mbps)  : 0
DL Throughput (Mbps)  : 0

-----
Number of cards : 1
=====

```

Example output PDN PGW statistics

```

# show mobile-gateway pdn statistics gw-type pgw summary
=====
PDN gateway statistics summary
=====

Real APNs           : 43      Roamers              : 0
Homers              : 0      Visitors             : 0
VPRNs               : 20     IP Local Pools       : 23
PDN Sessions        : 0      Ipv4 PDN Sessions    : 0
Ipv6 PDN Sessions   : 0      Ipv4v6 PDN Sessions  : 0
Non-Ip PDN sessions : 0

Bearers             : 0      Default Bearers       : 0
Dedicated Bearers   : 0      Ipv4 Bearers          : 0
Ipv6 Bearers        : 0      Ipv4v6 Bearers        : 0
SDFs                : 0

HSS Static IPv4 PDN Sessio*: 0    HSS Static IPv6 PDN Sessio*: 0
HSS Static IPv4v6 PDN Sess*: 0    eHRPD PDN sessions    : 0

```

```

LTE PDN sessions           : 0          2G/3G PDN sessions       : 0
PDN sessions in suspend st*: 0          Emergency PDN sessions   : 0
Rf Peers                   : 1          Rf Acct start buffered    : 0
Rf Acct Int buffered       : 0          Rf Acct stop buffered     : 0
Ga DRT Req in queue        : 0          Ga DRT Req max queued     : 0
Ga CDRs buffered           : 0          S2b Sessions              : 0
Redirected Sessions        : 0          Tethered Sessions         : 0
Overload - Dropped requests: 0
Paging in progress         : 0          Allocated Paging Buffers  : 0
Available Paging Buffers    : 8191      Paging Buffers n/a Errors  : 0
Charging paused sessions    : 0
Number of ENBs             : 0          Number of MMEs            : 0
Total Number of UEs         : 0          Total Number of Idle UEs  : 0
UEs in suspended state     : 0          Paging drops               : 0
Buff Pkts drop tmr expired : 0          Buff Bytes drop tmr expired: 400 B
Buff Pkts drop max pkt lmt : 0          Buff Bytes drop max pkt lmt: 38.147 MB
Buff Pkts drop max cap lmt : 0          Buff Bytes drop max cap lmt: 3.725 GB
Number of RNCs             : 0          Number of SGSNs           : 0
Combo Bearers              : 0          Combo Default bearers     : 0
Combo Active Bearers       : 0          Combo Idle bearers        : 0
Combo Dedicated bearers    : 0          Combo Ipv4 bearers        : 0
Combo Ipv6 bearers         : 0          Combo Ipv4v6 bearers      : 0
Combo sessions             : 0          Combo Ipv4 sessions       : 0
Combo Ipv6 sessions        : 0          Combo Ipv4v6 sessions     : 0
Combo Non-IP sessions      : 0
IPv4 Home Agent Sessions   : 0          HRPD PDN sessions         : 0
LTE NEMO Sessions          : 0          eHRPD NEMO Sessions       : 0
HA NEMO Sessions           : 0          NEMO IPv4 Subnets        : 0
Ga Big DTRs buffered       : 0          ePDG sessions             : 0
Steering sessions         : 0
S2a Sessions               : 0
TWAG Open SSID portal sess : 0
TWAG SaMOG Sessions        : 0          TWAG NSW0 sessions        : 0
Steering default sessions  : 0          Gy CCRs buffered          : 0
Low Priority Combo Sessions : 0          Low Priority PGW Sessions  : 0
Active ePDG Emergency Calls: 0          Unsol. SSG Sess In Att Ret*: 0
PDN sessions w dedic bear  : 0          PDN sessions in Idle state : 0
Number of PFCP Peers       : 0          eMPS IMS Advance Prio Sess : 0
eMPS IMS Sessions          : 0
eMPS dedicated bearers     : 0          PGW Sessions by 5G UEs    : 0
Combo Sessions by 5G UEs   : 0

UL Throughput (Mbps)       : 0
DL Throughput (Mbps)       : 0

```

Number of cards : 1

=====

* indicates that the corresponding row element may have been truncated.

Example output PDN summary statistics

```
# show mobile-gateway pdn statistics summary
```

```
=====
```

PDN gateway statistics summary

```
=====
```



```
Real APNs                : 43      Roamers                : 0
Homers                   : 0        Visitors                : 0
VPRNs                   : 20       IP Local Pools          : 23
PDN Sessions             : 0        Ipv4 PDN Sessions       : 0
Ipv6 PDN Sessions        : 0        Ipv4v6 PDN Sessions     : 0
Non-IP PDN sessions      : 0
Bearers                  : 0        Default Bearers         : 0
Dedicated Bearers        : 0        Ipv4 Bearers            : 0
Ipv6 Bearers             : 0        Ipv4v6 Bearers          : 0
SDFs                     : 0
HSS Static IPv4 PDN Sessio*: 0      HSS Static IPv6 PDN Sessio*: 0
HSS Static IPv4v6 PDN Sess*: 0      eHRPD PDN sessions      : 0
LTE PDN sessions         : 0        2G/3G PDN sessions      : 0
PDN sessions in suspend st*: 0      Emergency PDN sessions  : 0
Rf Peers                 : 1        Rf Acct start buffered  : 0
Rf Acct Int buffered     : 0        Rf Acct stop buffered   : 0
Ga DRT Req in queue      : 0        Ga DRT Req max queued   : 0
Ga CDRs buffered         : 0        S2b Sessions            : 0
Redirected Sessions       : 0        Tethered Sessions       : 0
Overload - Dropped requests: 0
Paging in progress       : 0        Allocated Paging Buffers : 0
Available Paging Buffers : 8191     Paging Buffers n/a Errors : 0
Charging paused sessions : 0
Number of ENBs           : 0        Number of MMES          : 0
Total Number of UEs      : 0        Total Number of Idle UEs : 0
UEs in suspended state   : 0        Paging drops             : 0
Buff Pkts drop tmr expired : 0      Buff Bytes drop tmr expired: 400 B
Buff Pkts drop max pkt lmt : 0      Buff Bytes drop max pkt lmt: 38.147 MB
Buff Pkts drop max cap lmt : 0      Buff Bytes drop max cap lmt: 3.725 GB
Number of RNCs           : 0        Number of SGSNs         : 0
Combo Bearers            : 0        Combo Default bearers    : 0
Combo Active Bearers     : 0        Combo Idle bearers       : 0
Combo Dedicated bearers  : 0        Combo Ipv4 bearers       : 0
Combo Ipv6 bearers       : 0        Combo Ipv4v6 bearers     : 0
Combo sessions           : 0        Combo Ipv4 sessions      : 0
Combo Ipv6 sessions      : 0        Combo Ipv4v6 sessions    : 0
Combo Non-IP sessions    : 0
IPv4 Home Agent Sessions : 0        HRPD PDN sessions       : 0
LTE NEMO Sessions        : 0        eHRPD NEMO Sessions     : 0
HA NEMO Sessions         : 0        NEMO IPv4 Subnets       : 0
Ga Big DTRs buffered     : 0        ePDG sessions           : 0
Steering sessions        : 0
S2a Sessions             : 0
TWAG Open SSID portal sess : 0
TWAG SaMOG Sessions      : 0        TWAG NSW0 sessions      : 0
Steering default sessions : 0        Gy CCRs buffered        : 0
Low Priority Combo Sessions: 0        Low Priority PGW Sessions : 0
Active ePDG Emergency Calls: 0      Unsol. SSG Sess In Att Ret*: 0
PDN sessions w dedic bear : 0
Number of PFCP Peers     : 0        PDN sessions in Idle state : 0
eMPS IMS Sessions        : 0        eMPS IMS Advance Prio Sess : 0
eMPS dedicated bearers   : 0
Combo Sessions by 5G UEs : 0        PGW Sessions by 5G UEs   : 0

UL Throughput (Mbps)      : 0
DL Throughput (Mbps)      : 0
-----
```

```

Number of cards : 1
=====
* indicates that the corresponding row element may have been truncated.

=====
Serving gateway statistics summary
=====

APNs                : 43          Bearers              : 0
Default bearers     : 0          Dedicated bearers    : 0
Ipv4 bearers        : 0          Ipv6 bearers        : 0
Ipv4v6 bearers      : 0          Roamers              : 0
Active bearers      : 0          Idle bearers         : 0

Number of MMEs      : 0          Number of PGWs       : 0
Rf Peers            : 1          Number of ENBs       : 0
Number of RNCs      : 0

Paging in progress  : 0          Allocated Paging Buffers : 0
Available Paging Buffers : 8191    Paging Buffers n/a Errors : 0
Pause chrg hold timer start: 0      Pause chrg hold timer stop : 0
Number of Homers    : 0          Number of Visitors     : 0
Total Number of UEs : 0          Total Number of Idle UEs : 0
Active Serving UEs  : 0

Rf Acct start buffered : 0        Rf Acct Int buffered    : 0
Rf Acct stop buffered  : 0        Emergency PDN sessions  : 0
UEs in suspended state : 0        Paging drops             : 0
Ga DRT Req in queue    : 0        Ga DRT Req max queued   : 0
Ga CDRs buffered       : 0        Number of SGSNs         : 0
Overload - Dropped requests: 0
Ga Big DRTs Buffered   : 0        Number of MMEs using S11-U : 0
Low Priority SGW Sessions : 0
PDN sessions w dedic bear : 0
eMPS IMS Sessions     : 0        eMPS dedicated bearers   : 0
SGW Sessions by 5G UEs : 0

UL Throughput (Mbps)   : 0
DL Throughput (Mbps)   : 0
-----
Number of cards : 1
=====

```

Example output PDN attach failure ePDG statistics

The following output example shows PDN Attach Failure ePDG Statistics and *Show PDN attach failure ePDG statistics output fields* describes the output fields.

```

# show mobile-gateway pdn statistics attach-failure-statistics
=====
PDN gateway session attach failure statistics
=====
Chassis/Card        : 1/3          Gateway              : 1

Create Packet Issue  : 0          GTP No Resource      : 0
Peer Down            : 0          PMIP No Resource     : 0
Peer Admin Down      : 0          Peer Recovery Count Changed: 0

```

```

Missing Invalid IE      : 0      Packet Drop           : 0
Internal Error          : 0      Invalid Peer           : 0
Ip Addr Not Available   : 0      Peer PGW overload      : 0
User Authentication Failed : 0    S11 Attach rate lmt-Rej req: 0
-----

```

Table 93: Show PDN attach failure ePDG statistics output fields

Label	Description
Chassis/Card	The chassis and card number
Gateway	The gateway ID
Create Packet Issue	Invalid GTP/PMIP request
GTP No Resource	Temporary unavailability of the resources to process the GTP request
Peer Down	The peer is down
PMIP No Resource	Temporary unavailability of the resources to process the PMIP request
Peer Admin Down	The peer administrator is down
Peer Recovery Count Changed	The peer recovery count has changed
Missing Invalid IE	The missing invalid IE
Packet Drop	The packet drop
Internal Error	The internal error
Invalid Peer	The invalid peer
Ip Addr Not Available	The IP address is not available
Peer PGW overload	The peer PGW is overload

Table 93: Show PDN attach failure ePDG statistics output fields (continued)

Label	Description
User Authentication Failed	The user authentication has failed
S11 Attach rate lmt-Rej req	The cumulative number of attach requests rejected due to attach rate limit

Example output PDN UPF statistics

The following output example shows PDN UPF Statistics and *Show PDN UPF statistics output fields* describes the output fields.

```
# show mobile-gateway pdn statistics gw-type upf
=====
UPF gateway statistics
=====
Chassis/Card           : 1/3      Gateway           : 1

Real APNs              : 43
UPF PDN Sessions       : 0        IPv4 UPF PDN Sessions : 0
IPv6 UPF PDN Sessions  : 0        IPv4v6 UPF PDN Sessions : 0
Static IPv4 UPF PDN Sessio*: 0      Static IPv6 UPF PDN Sessio*: 0
Static IPv4v6 UPF PDN Sess*: 0      Non-IP UPF PDN Sessions : 0
UPF Bearers            : 0        Default UPF Bearers    : 0
Dedicated UPF Bearers  : 0        Ipv4 UPF Bearers       : 0
Ipv6 UPF Bearers       : 0        Ipv4v6 UPF Bearers     : 0
PDN sessions in suspend st*: 0      PDN sessions in Idle state : 0
LTE UPF PDN sessions   : 0        3G UPF PDN sessions    : 0
S2b UPF PDN sessions   : 0        S2a UPF PDN sessions   : 0
5G PDU Sessions        : 0
Active PDU Sessions    : 0        Idle PDU Sessions      : 0
QoS Flows              : 0
Default QoS Flows      : 0        Dedicated QoS Flows    : 0
Paging in progress     : 0        Allocated Paging Buffers : 0
Available Paging Buffers : 8191    Paging Buffers n/a Errors : 0
Number of ENBs         : 0        Paging drops           : 0
Number of RNCs         : 0        Number of SGSNs        : 0
Total Number of UEs    : 0        UEs with 5G session    : 0
Num of Sessions W/O UE Id : 0
Combo Non-IP UPF sessions : 0      Combo UPF Bearers      : 0
Combo Default UPF bearers : 0      Combo Active UPF Bearers : 0
Combo Idle UPF bearers   : 0      Combo Dedicated UPF bearers: 0
Combo Ipv4 UPF bearers   : 0      Combo Ipv6 UPF bearers   : 0
Combo Ipv4v6 UPF bearers : 0      Combo UPF PDN sessions  : 0
Combo Ipv4 UPF PDN sessions: 0      Combo Ipv6 UPF PDN sessions: 0
Combo Ipv4v6 UPF PDN sessi*: 0      PDN sessions w dedic bear : 0

UL Throughput (Mbps)    : 0
DL Throughput (Mbps)    : 0
```

 Number of cards : 1
 =====

* indicates that the corresponding row element may have been truncated.

Table 94: Show PDN UPF statistics output fields

Label	Description
Chassis/Card	The chassis and card number
Gateway	The gateway ID
Real APNs	The number of real APNs
UPF PDN Sessions	The number of UPF sessions
IPv4 UPF PDN Sessions	The number of IPv4 UPF sessions
IPv6 UPF PDN Sessions	The number of IPv6 UPF sessions
IPv4v6 UPF PDN Sessions	The number of IPv4v6 UPF sessions
Static IPv4 UPF PDN Sessio*	The number of static IPv4 UPF sessions
Static IPv6 UPF PDN Sessio*	The number of static IPv6 UPF sessions
Static IPv4v6 UPF PDN Sess*	The number of static IPv4v6 UPF sessions
Non-IP UPF PDN Sessions	The number of non-IP UPF sessions
UPF Bearers	The number UPF bearers
Default UPF Bearers	The number of the default UPF bearers
Dedicated UPF Bearers	The number of the dedicated UPF bearers
Ipv4 UPF Bearers	The number of IPv4 UPF bearers

Table 94: Show PDN UPF statistics output fields (continued)

Label	Description
Ipv6 UPF Bearers	The number of IPv6 UPF bearers
Ipv4v6 UPF Bearers	The number of IPv4v6 UPF bearers
PDN sessions in suspend st*	The number of PDN sessions in suspend state
PDN sessions in Idle state	The number of PDN sessions in idle state
LTE UPF PDN sessions	The number of LTE UPF PDN sessions
3G UPF PDN sessions	The number of 3G UPF PDN sessions
S2b UPF PDN sessions	The number of S2b UPF PDN sessions
S2a UPF PDN sessions	The number of S2a UPF PDN sessions
5G PDU Sessions	The number of 5G PDU sessions
Active PDU Sessions	The number of the active PDU sessions
Idle PDU Sessions	The number of the idle PDU sessions
QoS Flows	The number of the QoS flows
Default QoS Flows	The number of the QoS flows with default QoS characteristics
Dedicated QoS Flows	The number of QoS flows with service specific QoS characteristics
Paging in progress	The number of paging procedure in progress
Allocated Paging Buffers	The number of allocated paging buffers
Available Paging Buffers	The number of available paging buffers

Table 94: Show PDN UPF statistics output fields (continued)

Label	Description
Paging Buffers n/a Errors	The number of errors non available paging buffers
Number of ENBs	The number of eNodeB peers
Paging drops	The number of paging drops
Number of RNCs	The number of RNCs
Total Number of UEs	The total number of UEs served on the UPF, only UEs for which the UE ID is known are counted
UEs with 5G session	The number of UEs served on the UPF, which have a session with N3 or N9 interface type
Num of Sessions W/O UE Id	The number of sessions on the UPF, for which the UE ID is unknown to the UPF
Number of SGSNs	The number of SGSNs
Combo Non-Ip UPF sessions	The number of non IP combined UPF sessions
Combo UPF Bearers	The number of combined UPF bearers
Combo Default UPF bearers	The number of combined default UPF bearers
Combo Active UPF Bearers	The number of combined active UPF bearers
Combo Idle UPF bearers	The number of combined idle UPF bearers
Combo Dedicated UPF bearers	The number of combined dedicated UPF bearers
Combo Ipv4 UPF bearers	The number of combined IPv4 UPF bearers

Table 94: Show PDN UPF statistics output fields (continued)

Label	Description
Combo Ipv6 UPF bearers	The number of combined IPv6 UPF bearers
Combo Ipv4v6 UPF bearers	The number of combined IPv4v6 UPF bearers
Combo UPF PDN sessions	The number of combined UPF PDN sessions
Combo Ipv4 UPF PDN sessions	The number of combined IPv4 UPF PDN sessions
Combo Ipv6 UPF PDN sessions	The number of combined IPv6 UPF PDN sessions
Combo Ipv4v6 UPF PDN sessi*	The number of combined IPv4v6 UPF PDN sessions
PDN sessions w dedic bear	The number of PDN sessions with a dedicated bearer
UL Throughput (Mbps)	The uplink throughput in Mbps; this counter prints the instant throughput as it has been calculated in the latest 10 seconds interval, by sampling the forwarded data volume at the beginning and at the end of the interval
DL Throughput (Mbps)	The downlink throughput in Mbps; this counter prints the instant throughput as it has been calculated in the latest 10 seconds interval, by sampling the forwarded data volume at the beginning and at the end of the interval
Number of cards	The number of cards retrieved

5.2.2.60 swu

Syntax

swu sig-ref-point-name

Context

```
show>mobile>pdn
```

Description

This command displays SWu configuration information.

Parameters

sig-ref-point-name

specifies the SWu reference point name, up to 32 characters

Output

The following output examples show SWu context information and *Show PDN SWu output fields* describes the output fields.

Example output SWu configuration

```
# show mobile-gateway pdn swu
=====
SWU configuration
=====
Global IPsec-lockout parameters
Status          : disabled
Count           : 3
Time            : 5 mins
Lockout-time    : 10 mins
Backoff-Timer Value: 1
Backoff-Timer Unit : 10-h
Global SWu-throttling parameters
Status          : disabled
Throttling level : best-effort
=====
# show mobile-gateway pdn swu "epdg"
=====
SWU configuration
=====
sig-ref-point-name : epdg
router-instance    : vprn20031
interface-name     : 5
ipsec-profile-name : epdgProfile
Global IPsec-lockout parameters
Status            : disabled
Count             : 3
Time              : 5 mins
Lockout-time      : 10 mins
Backoff-Timer Value: 1
Backoff-Timer Unit : 10-h
Global SWu-throttling parameters
Status            : Enabled
Throttling-Level  : Major
```

Table 95: Show PDN SWu output fields

Label	Description
Status	The status of the SWu reference point
Count	The maximum number of failed authentication attempts
Time	The time in minutes during which the number of failed authentication attempts can be made before the user is locked-out
Lockout-time	The time in minutes that the user will be locked-out from further authentication attempts
Backoff-Timer Value	The value of the backoff timer
Backoff-Timer Unit	The unit to be used for the timer value
Throttling level	The throttling level of the SWu interface on the ePDG
sig-ref-point-name	The reference point name of the SWu interface
router-instance	The router instance
interface-name	The interface name of the SWu reference point

5.2.2.61 ue-context

Syntax

```
ue-context [imsi | imei | msisdn | supi | gpsi | pei | string [detail]] [gateway gw-id | group
group-number | [card slot-number | vm vm-id] [nemo]
```

```
ue-context mac mac [gateway gw-id | group group-number | [card slot-number | vm vm-id]
[detail]
```

```
ue-context ran-ue-ngap-id ran-ue-ngap-id [detail]
```

Context

show>mobile>pdn

Description

This command displays the individual UE contexts for each session.

Parameters

string

specifies the UE’s IMSI, IMEI, MSISDN, SUPI, GPSI, PEI, or RAN-UE-NGAP-ID

Values	IMSI MSISDN SUPI GPSI	1 to 15
–	IMEI PEI	1 to 16
–	RAN-UE-NGAP-ID	0 to 2 ³²⁻¹ (HEX)

detail

displays the detailed summary information for the UE context

gw-id

specifies the gateway ID

Values: 1 to 8

group-number

specifies the group number

Values: 1 to 8

slot-number

specifies the card slot number (applicable only to 7750 SR MG)

Values: 1 to 10 | A | B

vm-id

specifies the VM ID (applicable only to CMG)

Values: 1 to 20 | A | B

Output

The following output examples show UE context information.

**NOTE:**

- The outputs of a CMG system are identical to the 7750 SR MG outputs except they have some CMG-specific fields displayed instead of hardware components (for example, VM is displayed instead of Card).
- Depending on the configured node function (UP or CP function), the description and value for some of the output fields differ. These fields, if present, are identified and described at the end of each example output.

- *Example output PDN UE context statistics*
- *Example output PDN UE context statistics (CP Function)*
- *Example output PDN UE context statistics (UP Function)*
- *Example output PDN UE context statistics PGW*
- *Example output PDN UE context statistics SGW*
- *Example output PDN UE context combined statistics SGW/PGW*
- *Example output PDN UE context IMEI statistics (UP function)*
- *Example output PDN UE context IMSI statistics (UP function)*
- *Example output PDN UE context IMSI detail statistics (CP function)*
- *Example output PDN UE context MSISDN statistics (UP function)*
- *Example output PDN UE context detail SUPI statistics (5G sessions)*
- *Example output PDN UE context detail statistics PGW*
- *Example output PDN UE context detail statistics SGW*
- *Example output PDN UE context detail statistics combined SGW/PGW*
- *Example output PDN UE context detail statistics TWAG*
- *Example output PDN UE context detail statistics S4/S12*
- *Example output PDN UE context detail statistics S2a*
- *Example output PDN UE context detail statistics S2b*
- *Example output PDN UE context detail statistics Gn*
- *Example output PDN UE context PMIP detail statistics S2a*
- *Example output PDN UE context MAC detail statistics*
- *Example output UE context statistics combination ePDG/PGW*
- *Example output PDN UE context IMSI detail statistics combination ePDG/PGW*
- *Example output PDN UE context statistics N3IWF*

- *Example output PDN UE context detail ran-ue-ngap-id statistics N3IWF*
- *Example output PDN UE context default statistics PFCP-u shared tunnel sessions*
- *Example output PDN UE context network realm statistics PFCP-u shared tunnel sessions*
- *Example output PDN UE context peer IP address statistics PFCP-u shared tunnel sessions*
- *Example output PDN UE context network realm and peer IP address statistics PFCP-u shared tunnel sessions*
- *Example output IP address pool statistics for peer*

Example output PDN UE context statistics

```
# show mobile-gateway pdn ue-context
=====
IMSI/IMEI(#)      MSISDN           State  APNs  Bearers
/MSISDN(^)/MAC(`)
=====
777777777777700  123100100098761  ACTIVE  1     1
-----
Number of UEs : 1
-----
# indicates that the corresponding row is IMEI entry.
^ indicates that the corresponding row is MSISDN entry.
` indicates that the corresponding row is MAC entry.
~ indicates that the corresponding row is SEID entry.
& indicates that the corresponding row is IMSI entry for
UPF session received via Sx-N4 ref-point.
-----
```

Example output PDN UE context statistics (CP function)

```
# show mobile-gateway pdn ue-context
=====
IMSI/IMEI(#)      MSISDN           State  APNs  Bearers
/MSISDN(^)/MAC(`)
=====
777777777777700  123100100098761  ACTIVE  1     1
-----
Number of UEs : 1
-----
# indicates that the corresponding row is IMEI entry.
^ indicates that the corresponding row is MSISDN entry.
` indicates that the corresponding row is MAC entry.
~ indicates that the corresponding row is SEID entry.
& indicates that the corresponding row is IMSI entry for
UPF session received via Sx-N4 ref-point.
-----
```

Example output PDN UE context statistics (UP function)

```
# show mobile-gateway pdn ue-context
=====
IMSI/IMEI(#)      MSISDN           State  APNs  Bearers
/MSISDN(^)/MAC(`)
=====
```

```

/MSISDN(^)/MAC(`)
=====
77777777777700&      -          ACTIVE  1      1
-----
Number of UEs : 1
-----
# indicates that the corresponding row is IMEI entry.
^ indicates that the corresponding row is MSISDN entry.
` indicates that the corresponding row is MAC entry.
~ indicates that the corresponding row is SEID entry.
& indicates that the corresponding row is IMSI entry for
UPF session received via Sx-N4 ref-point.
-----

```

Example output PDN UE context statistics PGW

```

# show mobile-gateway pdn ue-context
=====
IMSI/IMEI(#)      MSISDN      State  APNs  Bearers
/MSISDN(^)/MAC(`)
=====
123456789012803   12345098761   ACTIVE  1      1
-----
Number of UEs : 1
-----
# indicates that the corresponding row is IMEI entry.
^ indicates that the corresponding row is MSISDN entry.
` indicates that the corresponding row is MAC entry.
-----

```

Example output PDN UE context statistics SGW

```

# show mobile-gateway pdn ue-context
=====
IMSI/IMEI(#)      MSISDN      State  APNs  Bearers
/MSISDN(^)/MAC(`)
=====
123456789012803   12345098761   ACTIVE  1      1
-----
Number of UEs : 1
-----
# indicates that the corresponding row is IMEI entry.
^ indicates that the corresponding row is MSISDN entry.
` indicates that the corresponding row is MAC entry.
-----

```

Example output PDN UE context combined statistics SGW/PGW

```

# show mobile-gateway pdn ue-context
=====
IMSI/IMEI(#)      MSISDN      State  APNs  Bearers
/MSISDN(^)/MAC(`)
=====
123456789012926   12345098761   ACTIVE  1      1
-----
Number of UEs : 1

```

```

-----
# indicates that the corresponding row is IMEI entry.
^ indicates that the corresponding row is MSISDN entry.
` indicates that the corresponding row is MAC entry.
-----

```

Example output PDN UE context IMEI statistics (UP function)

```

# show mobile-gateway pdn ue-context imei 123450987612345
=====
IMSI/IMEI(#)      MSISDN      State  APNs  Bearers
/MSISDN(^)/MAC(`)
=====
123456789012345&  123450987612345  ACTIVE  1     1
-----
Number of UEs : 1
-----
# indicates that the corresponding row is IMEI entry.
^ indicates that the corresponding row is MSISDN entry.
` indicates that the corresponding row is MAC entry.
~ indicates that the corresponding row is SEID entry.
& indicates that the corresponding row is IMSI entry for
UPF session received via Sx ref-point.
-----

```

Example output PDN UE context IMSI Statistics (UP function)

```

# show mobile-gateway pdn ue-context imsi 777777777777777
=====
IMSI/IMEI(#)      MSISDN      State  APNs  Bearers
/MSISDN(^)/MAC(`)
=====
123456789012345&  12345098761      ACTIVE  1     1
-----
Number of UEs : 1
-----
# indicates that the corresponding row is IMEI entry.
^ indicates that the corresponding row is MSISDN entry.
` indicates that the corresponding row is MAC entry.
~ indicates that the corresponding row is SEID entry.
& indicates that the corresponding row is IMSI entry for
UPF session received via Sx ref-point.
-----

```

Example output PDN UE context IMSI detail statistics (CP function)

```

# show mobile-gateway pdn ue-context imsi 777777777777700 detail
=====
UE context detail
=====
IMSI      : 777777777777700      State      : ACTIVE
MEI       : 777877777777700      MSISDN     : 123100100098761
Network MCC : 310                Network MNC : 012
Tracking area id : 0xffff        Cell Id     : 0x1234567
NAI       : 077777777777700@nai.epc.mnc077.mcc777.3gppnetwork.org

```

```

GW/Group/VM      : 1/1/3          RAT          : EUTRAN
PDN contexts     : 1              Bearers       : 1

S11 MME ctrl TEID: 0xb0000003      S11 SGW ctrl TEID: 0x40010100
S11 MME ctrl addr: 10.10.7.10
S11-U tunnel     : false
S11 SGW ctrl addr : 10.70.16.16

Inter ENB X2 HO  : 0              Inter ENB S1 HO  : 0
S1 release procs : 0              Paging Attempts : 0
UE init svc req  : 0              DDN buff duration: 0/0
-----
Number of UEs : 1
=====

```

Example output PDN UE context MSISDN statistics (UP function)

```

# show mobile-gateway pdn ue-context msisdn 123456789012345
=====
IMSI/IMEI(#)      MSISDN          State  APNs  Bearers
  /MSISDN(^)/MAC(`)
=====
123456789012345& 12345098761      ACTIVE  1     1
-----
Number of UEs : 1
-----
# indicates that the corresponding row is IMEI entry.
^ indicates that the corresponding row is MSISDN entry.
` indicates that the corresponding row is MAC entry.
~ indicates that the corresponding row is SEID entry.
& indicates that the corresponding row is IMSI entry for
UPF session received via Sx ref-point.
-----

```

Example output PDN UE context detail SUPI statistics (5G sessions)

The following example output shows detailed SUPI statistics for 5G sessions in UE context and *Show PDN UE context SUPI detail output fields* describes the output fields.

```

*A:Dut-A# show mobile-gateway pdn ue-context imsi "460100001003204" detail
=====
UE context detail
=====
SUPI           : 460100001003204      State          : ACTIVE
PEI            : 987654321012345      GPSI           : 12345678901
Network MCC    : 460                  Network MNC     : 10
Tracking area id : 0x123410           NR Cell Id     : 0x225bd6007
Csg Id         : 0x0
Access Mode    : CLOSED                CMI            : N/A
NAI            : 0460100001003204@nai.epc.mnc010.mcc460.3gppnetwork.org

GW/Group/VM    : 1/1/3
PDU sessions   : 1                    5G MAPCON      : NOT ACTIVE

Def PFCP-u Sess : False

```



```
N11 AMF Network Instance Id: 44f08cba-9a89-46fe-ab7d-2a0066584f55
N11 AMF GUAMI : 310 012 0x112233
```

```
Paging Attempts : 0                      DL buff duration : 0/0
```

```
-----
Number of UEs : 1
=====
```

```
*A: Dut -A#
```

Table 96: Show PDN UE context SUPI detail output fields

Label	Description
SUPI	SUPI IMSI number
State	State of the UE
PEI	PEI of the UE
GPSI	GPSI of the UE
Network MCC	Network MCC number
Network MNC	Network MNC number
Tracking area id	Tracking area identifier
NR Cell Id	NR cell identifier
Csg Id	CSG identifier
Access Mode	Access mode
CMI	CMI
NAI	NAI
Gw/Group/VM	The associated gateway, group and VM (or card)

Table 96: Show PDN UE context SUPI detail output fields (continued)

Label	Description
PDU sessions	Number of PDU sessions
5G MAPCON	5G MAPCON status
N11 AMF Network Instance Id	N11 AMF network instance ID
N11 AMF GUAMI	N11 AMF GUAMI
Paging Attempts	Number of paging attempts
DL buff duration	Downlink buffering duration

Example output PDN UE context detail statistics PGW

```
# show mobile-gateway pdn ue-context imsi 123456789012803 detail
=====
UE context detail
=====
IMSI           : 123456789012803      State           : ACTIVE
MEI            : 987654321012399      MSISDN          : 12345098761
Network MCC    : 310                  Network MNC      : 012
Tracking area id : 0xffff              Cell Id         : 0x1234567
Csg Id         : 0x0
Access Mode    : CLOSED               CMI             : NA
NAI            : 0123456789012803@nai.epc.mnc045.mcc123.3gppnetwork.org

PDN contexts   : 1                    RAT             : EUTRAN

-----
Number of UEs : 1

=====
```

Example output PDN UE context detail statistics SGW

```
# show mobile-gateway pdn ue-context imsi 123456789012803 detail
=====
UE context detail
=====
IMSI           : 123456789012803      State           : ACTIVE
MEI            : 987654321012399      MSISDN          : 12345098761
```

```

Network MCC      : 310           Network MNC      : 012
Tracking area id : 0xffff        Cell Id        : 0x1234567
NAI              : 0123456789012803@nai.epc.mnc045.mcc123.3gppnetwork.org

GW/Group/Card    : 2/1/2         RAT              : EUTRAN
PDN contexts     : 1             Bearers       : 1

S11 MME ctrl TEID: 0xb0000002    S11 SGW ctrl TEID: 0x40010100
S11 MME ctrl addr: 10.10.7.10     S11-U tunnel : true
S11 SGW ctrl addr : 10.10.11.2

Inter ENB X2 HO  : 0             Inter ENB S1 HO  : 0
S1 release procs : 0             Paging Attempts  : 0
UE init svc req  : 0

-----
Number of UEs : 1
=====

```

Example output PDN UE context detail statistics combined SGW/PGW

```

# show mobile-gateway pdn ue-context imsi 123456789012926 detail
=====
UE context detail
=====
IMSI           : 123456789012926   State          : ACTIVE
MEI            : 987654321012399   MSISDN         : 12345098761
Network MCC    : 310              Network MNC     : 012
Tracking area id : 0xffff          Cell Id        : 0x1234567
Csg Id         : 0x0
Access Mode    : CLOSED           CMI            : NA
NAI            : 0123456789012926@nai.epc.mnc045.mcc123.3gppnetwork.org

GW/Group/Card  : 1/1/3            Bearers        : 1
PDN contexts   : 1               RAT           : EUTRAN

S11 MME ctrl TEID: 0xb0000003    S11 SGW ctrl TEID: 0x40010120
S11 MME ctrl addr: 10.10.7.10     S11-U tunnel : true
S11 SGW ctrl addr: 16.70.16.16

Inter ENB X2 HO : 0              Inter ENB S1 HO : 0
S1 release procs : 0             Paging Attempts  : 0
UE init svc req  : 0

-----
Number of UEs : 1
=====

```

Example output PDN UE context detail statistics TWAG

```

# show mobile-gateway pdn ue-context imsi 3100120000000000 detail
=====
UE context detail
=====
IMSI/MAC : 3100120000000000/00:02:00:00:00:00
State     : ACTIVE
MEI       : N/A
Network MCC : 310           Network MNC : 150

```

```

SSIDNOKwifi
NAI                : 0310012000000000@nai.epc.mnc012.mcc310.3gppnetwork.org

GW/Group/Card      : 1/1/3                RAT                : WLAN
PDN contexts       : 1                    Bearers            : 1

Sww AP IP addr: 180.1.3.2
-----
Number of UEs : 1
=====

```

Example output PDN UE context detail statistics S4/S12

```

# show mobile-gateway pdn ue-context imsi 11111111110000 detail
=====
UE context detail
=====
IMSI                : 11111111110000      State                : ACTIVE
MEI                 : 11111111110000      MSISDN                : 11111110000
Network MCC         : 310                 Network MNC            : 012
Tracking area id    : 0x2222              Cell Id                : 0x5432
Csg Id              : 0x0
Access Mode         : CLOSED              CMI                    : NA
NAI                 : 011111111110000@nai.epc.mnc011.mcc111.3gppnetwork.org

GW/Group/Card       : 1/1/3                Bearers                : 6
PDN contexts        : 2                    RAT                    : UTRAN

S4 SGSN ctrl TEID: 0xb000003a             S4 SGW ctrl TEID : 0x40040d50
S4 SGSN ctrl addr: 10.10.7.10
S11-U tunnel        : false
S4 SGW ctrl addr : 16.70.16.16

Inter ENB X2 HO : 0                      Inter ENB S1 HO : 0
S1 release procs : 0                     Paging Attempts : 0
UE init svc req  : 0

-----
Number of UEs : 1
=====

```

Example output PDN UE context detail statistics S2a

```

# show mobile-gateway pdn ue-context imsi 22222222222200 detail
=====
UE context detail
=====
IMSI                : 22222222222200      State                : ACTIVE
MEI                 : 22222222222200      MSISDN                : 22222222200
Network MCC         : 310                 Network MNC            : 012
Tracking area id    : 0x0                 Cell Id                : 0x0
Csg Id              : 0x0
Access Mode         : CLOSED              CMI                    : NA
NAI                 : 02222222222200@nai.epc.mnc022.mcc222.3gppnetwork.org

PDN contexts        : 2                    RAT                    : WLAN

```

```
-----
Number of UEs : 1
=====
```

Example output PDN UE context detail statistics S2b

```
# show mobile-gateway pdn ue-context imsi 22222222220000 detail
=====
UE context detail
=====
IMSI           : 22222222220000      State           : ACTIVE
MEI            : 22222222220000      MSISDN          : 2222220000
Network MCC    : 310                 Network MNC      : 012
Tracking area id : 0x0                 Cell Id          : 0x0
Csg Id         : 0x0
Access Mode    : CLOSED                CMI             : NA
NAI            : 02222222220000@nai.epc.mnc022.mcc222.3gppnetwork.org

PDN contexts   : 2                     RAT              : VIRTUAL

-----
Number of UEs : 1
=====
```

Example output PDN UE context detail statistics Gn

```
# show mobile-gateway pdn ue-context imsi 33333333330000 detail
=====
UE context detail
=====
IMSI           : 33333333330000      State           : ACTIVE
MEI            : 33333333330000      MSISDN          : 3333330000
Network MCC    : 000                 Network MNC      : 000
Routing area id : 0x0                 Cell Id          : 0xff00
Csg Id         : 0x0
Access Mode    : CLOSED                CMI             : NA
NAI            : 03333333330000@nai.epc.mnc033.mcc333.3gppnetwork.org

PDN contexts   : 2                     RAT              : UTRAN

-----
Number of UEs : 1
=====
```

Example output PDN UE context PMIP detail statistics S2a

```
# show mobile-gateway pdn ue-context imsi 44444444440000 detail
=====
UE context detail
=====
IMSI           : 44444444440000      State           : ACTIVE
MEI            : N/A
Network MCC    : 000                 Network MNC      : 000
Tracking area id : 0x0                 Cell Id          : 0x0
Csg Id         : 0x0
Access Mode    : CLOSED                CMI             : NA
```

NAI : 0444444444440000@nai.epc.mnc044.mcc444.3gppnetwork.org

PDN contexts : 2 RAT : EHRPD

Number of UEs : 1
=====

Example output PDN UE context MAC detail statistics

```
# show mobile-gateway pdn ue-context mac 00:02:00:00:00:00 detail
```

=====

UE context detail

=====

IMSI/MAC : 3100120000000000/00:02:00:00:00:00

State : ACTIVE

MEI : N/A

Network MCC : 310 Network MNC : 150

SSIDNOKwifi

NAI : 0310012000000000@nai.epc.mnc012.mcc310.3gppnetwork.org

GW/Group/Card : 1/1/3 RAT : WLAN

PDN contexts : 1 Bearers : 1

SWw AP IP addr: 201.3.2.10

Number of UEs : 1
=====

Example output UE context statistics combination ePDG/PGW

```
# show mobile-gateway pdn ue-context
```

=====

IMSI/IMEI(#)	MSISDN	State	APNs	Bearers
/MSISDN(^)/MAC(`)				

=====

3100120000000000	-	ACTIVE	1	1
------------------	---	--------	---	---

3100120000000000	-	ACTIVE	1	1
------------------	---	--------	---	---

3100120000000001	-	ACTIVE	1	1
------------------	---	--------	---	---

3100120000000001	-	ACTIVE	1	1
------------------	---	--------	---	---

Number of UEs : 2

indicates that the corresponding row is IMEI entry.

^ indicates that the corresponding row is MSISDN entry.

` indicates that the corresponding row is MAC entry.

~ indicates that the corresponding row is SEID entry.

show mobile-gateway pdn ue-context imsi 3100120000000000

=====

IMSI/IMEI(#)	MSISDN	State	APNs	Bearers
/MSISDN(^)/MAC(`)				

=====

3100120000000000	-	ACTIVE	1	1
------------------	---	--------	---	---

3100120000000000	-	ACTIVE	1	1
------------------	---	--------	---	---

```
-----
Number of UEs : 2
-----
```

```
# indicates that the corresponding row is IMEI entry.
^ indicates that the corresponding row is MSISDN entry.
` indicates that the corresponding row is MAC entry.
-----
```

Example output PDN UE context IMSI detail statistics combination ePDG/PGW

```
# show mobile-gateway pdn ue-context imsi 3100120000000000 detail
=====
UE context detail
=====
IMSI           : 3100120000000000      State           : ACTIVE
MEI            : N/A
Network MCC    : 310                   Network MNC       : 012
Tracking area id : 0x0                 Cell Id          : 0x0
Csg Id         : 0x0
Access Mode    : CLOSED                CMI              : NA
NAI            : 03100120000000000@nai.epc.mnc012.mcc310.3gppnetwork.org

PDN contexts   : 1                     RAT              : WLAN

-----
IMSI           : 3100120000000000      State           : ACTIVE
MEI            : N/A
Network MCC    : 310                   Network MNC       : 012
Tracking area id : 0x0                 Cell Id          : 0x0
NAI            : 03100120000000000@nai.epc.mnc012.mcc310.3gppnetwork.org

GW/Group/Card  : 2/2/4                 RAT              : WLAN
PDN contexts   : 1                     Bearers          : 1

Swu remote UE addr: 10.168.0.1
Swu remote UE port: 4500
-----
Number of UEs : 2
=====
```

Example output PDN UE context statistics N3IWF

```
# show mobile-gateway pdn ue-context
=====
IMSI/IMEI(#)    MSISDN                State   APNs   Bearers
  /MSISDN(^)/MAC(`)
  /RAN-UE-NGAP-ID(@)
=====
@0x10100       -                ACTIVE  1      1
-----
Number of UEs : 1
```

```

-----
# indicates that the corresponding row is IMEI entry.
^ indicates that the corresponding row is MSISDN entry.
` indicates that the corresponding row is MAC entry.
~ indicates that the corresponding row is SEID entry.
& indicates that the corresponding row is IMSI entry for
  UPF session received via Sx ref-point.
@ indicates that the corresponding row is RAN-UE-NGAP-ID
  entry.
-----

```

Example output PDN UE context detail ran-ue-ngap-id statistics N3IWF

```
# show mobile-gateway pdn ue-context ran-ue-ngap-id 0x10100 detail
```

```
=====
UE context detail
=====
```

```
GW/Group/VM      : 1/1/3
```

```
PDU Sessions     : 1
```

```
UE Lcl IPv4 Addr : 10.168.0.1
```

```
UE Lcl IPv6 Addr : -
```

```
UE Int IPv4 Addr : 15.0.0.0
```

```
UE Int IPv6 Addr : -
```

```
NAS IPv4 Addr    : 15.1.1.1
```

```
NAS IPv6 Addr    : -
```

```
NAS TCP Port     : 32768
```

```
Req NSSAI
```

```
SST              : 11                      SD              : 0xd143a5
```

```
IKE SA Information
```

```
-----
Local IPv4 Addr   : 10.11.51.2
```

```
Remote IPv4 Addr  : 10.168.0.1
```

```
Local UDP Port    : 4500
```

```
Remote UDP Port   : 4500
```

```
Integrity         : hmac-sha1
```

```
Pseudo Rand Func  : prf-sha1
```

```
Encryption        : aes192ctr
```

```
DH Group          : modp1024
```

```
Local SPI         : 013ecaa2d5610100      Remote SPI        : d69b18017a037917
```

```
Lifetime          : 23h20m00s
```



```

Established Time : 01/31/2023 12:34:34
Expiry Time      : 02/01/2023 11:54:34
Number of Rekeys : 0
Rx Packets       : 7                      Tx Packets       : 7
Rx Errors        : 0                      Tx Errors        : 0
Rx DPD           : 0                      Tx DPD           : 0
Rx DPD ACK       : 0                      Tx DPD ACK       : 0
Rx Retransmits   : 0                      Tx Retransmits   : 0

```

Child IKE SA Information

```

-----
Local IPv4 Addr  : 10.11.51.2
Remote IPv4 Addr : 10.168.0.1
Local UDP Port   : 4500                  Remote UDP Port   : 4500
Integrity        : hmac-sha1
Pseudo Rand Func : prf-sha1
Encryption       : aes192ctr
DH Group         : modp1024
Local SPI        : 013ecaa2d5610100      Remote SPI        : d69b18017a037917
Lifetime         : 23h20m00s
Established Time : 01/31/2023 12:34:34
Expiry Time      : 02/01/2023 11:54:34
Number of Rekeys : 0
Rx Packets       : 7                      Tx Packets       : 7
Rx Errors        : 0                      Tx Errors        : 0
Rx DPD           : 0                      Tx DPD           : 0
Rx DPD ACK       : 0                      Tx DPD ACK       : 0
Rx Retransmits   : 0                      Tx Retransmits   : 0

```

Control ESP SA Information

```

-----
Extended Seq Num : Disabled
TS Type          : IPv4
Integrity        : hmac-sha1
Encryption       : aes192ctr
DH Group         : none
Local/Inbound SPI: 01000000              Remote/Outbnd SPI: c0000100
Lifetime         : 23h20m00s
Established Time : 01/31/2023 12:34:34
Expiry Time      : 02/01/2023 11:54:34
Number of Rekeys : 0
Decoded Packets  : 5                      Encoded Packets   : 5
Decoded Bytes    : 444                    Encoded Bytes     : 608
Decoded Errors   : 0                      Encoded Errors    : 0
Replay Errors    : 0

```

Child Control ESP SA Information

```

-----
Extended Seq Num : Disabled
TS Type          : IPv4
Integrity        : hmac-sha1
Encryption       : aes192ctr
DH Group         : none
Local/Inbound SPI: 01000000          Remote/Outbnd SPI: c0000100
Lifetime         : 23h20m00s
Established Time : 01/31/2023 12:34:34
Expiry Time      : 02/01/2023 11:54:34
Number of Rekeys : 0
Decoded Packets  : 5                  Encoded Packets   : 5
Decoded Bytes    : 444                Encoded Bytes     : 608
Decoded Errors   : 0                  Encoded Errors    : 0
Replay Errors    : 0

```

NGAP Information

```

-----
RAN UE NGAP ID   : 0x10100
RAN AMF NGAP ID  : 0x003e8
Global N3IWF ID  : MCC 310 MNC 012      N3IWF ID          : 1
Peer IPv4 Addr   : 10.10.7.102
Local IPv4 Addr  : 17.70.16.18
Peer IPv6 Addr   : 0000:0000:0000:0000:0000:0000:0000:0001
Local IPv6 Addr  : 0000:0000:0000:0000:0000:0000:0000:0002

AMF Name         : -

```

```

-----
Number of UEs : 1
=====

```

5.2.2.62 def-pfcp-u-sessions

Syntax

```
def-pfcp-u-sessions [network-realm realm-name] [pfcp-u-peer-ip ip-address] [gateway
gw-id] [card slot-number] [vm vm-id] [group group-number]
```

Context

```
show>mobile>pdn>ue-context
```

Description

This command displays the default PFCP-u shared tunnel session information, when PFCP-u (GTP-U) tunnels are configured to handle ICMPv6 RS/RA and NS/NA message exchange between the CP function and the UP function. When only `def - p f c p - u` is specified, results for all default PFCP-u sessions are displayed. The `network - rea lm` option may also be specified to identify the realm name of the network instance that has the PFCP GTP-U interface on the UP function, and is communicated in the Network-Instance IE part of the PFCP Session Establishment/Modification process. The `p f c p - u - p e e r - i p` option can also be used to display information for a specific PFCP-u Peer IP address.

Parameters

realm-name

specifies the realm name of the network instance that has the PFCP GTP-U interface on the UP function, up to 80 characters

ip-address

specifies the PFCP-u peer IP address, up to 64 characters

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x: [0..FFFF]H
–	–	d: [0..255]D

gw-id

specifies the gateway ID

Values: 1 to 8

slot-number

specifies the card slot number (applicable only to 7750 SR MG)

Values: 1 to 20 | A | B

vm-id

specifies the VM ID (applicable only to CMG)

Values: 1 to 20 | A | B

group-number

specifies the group number

Values: 1 to 15

Output

The following output examples show default PFCP-u shared tunnel sessions information for the PDN bearer context and *Show PDN UE context PFCP-u shared tunnel sessions output fields* describes the output fields.

- *Example output PDN UE context default statistics PFCP-u shared tunnel sessions*
- *Example output PDN UE context network realm statistics PFCP-u shared tunnel sessions*
- *Example output PDN UE context peer IP address statistics PFCP-u shared tunnel sessions*
- *Example output PDN UE context network realm and peer IP address statistics PFCP-u shared tunnel sessions*

Example output PDN UE context default statistics PFCP-u shared tunnel sessions

```
# show mobile-gateway pdn ue-context def-pfcp-u-sessions
=====
IMSI/IMEI( #)      MSISDN      State  APNs  Bearers
/MSISDN(^)/MAC(`)
=====
999999888100211    -          ACTIVE  1     1
999999888100211    -          ACTIVE  1     1
999999888100211    -          ACTIVE  1     1
999999888100211    -          ACTIVE  1     1
999999888100211    -          ACTIVE  1     1
999999888100211    -          ACTIVE  1     1
999999888100211    -          ACTIVE  1     1
999999888100211    -          ACTIVE  1     1
999999888100211    -          ACTIVE  1     1
999999888100210    -          ACTIVE  1     1
999999888100210    -          ACTIVE  1     1
999999888100210    -          ACTIVE  1     1
999999888100210    -          ACTIVE  1     1
999999888100210    -          ACTIVE  1     1
999999888100210    -          ACTIVE  1     1
999999888100210    -          ACTIVE  1     1
999999888100210    -          ACTIVE  1     1
999999888100210    -          ACTIVE  1     1
999999888100209    -          ACTIVE  1     1
999999888100212    -          ACTIVE  1     1
999999888100212    -          ACTIVE  1     1
999999888100212    -          ACTIVE  1     1
999999888100212    -          ACTIVE  1     1
999999888100212    -          ACTIVE  1     1
999999888100212    -          ACTIVE  1     1
999999888100212    -          ACTIVE  1     1
999999888100212    -          ACTIVE  1     1
999999888100212    -          ACTIVE  1     1
999999888100212    -          ACTIVE  1     1
999999888100213    -          ACTIVE  1     1
999999888100213    -          ACTIVE  1     1
999999888100213    -          ACTIVE  1     1
999999888100213    -          ACTIVE  1     1
```

999999888100213	-	ACTIVE	1	1
999999888100213	-	ACTIVE	1	1
999999888100213	-	ACTIVE	1	1
999999888100213	-	ACTIVE	1	1
999999888100213	-	ACTIVE	1	1
999999888100211	-	ACTIVE	1	1
999999888100211	-	ACTIVE	1	1
999999888100211	-	ACTIVE	1	1
999999888100211	-	ACTIVE	1	1
999999888100211	-	ACTIVE	1	1
999999888100211	-	ACTIVE	1	1
999999888100211	-	ACTIVE	1	1
999999888100211	-	ACTIVE	1	1
999999888100210	-	ACTIVE	1	1
999999888100210	-	ACTIVE	1	1
999999888100210	-	ACTIVE	1	1
999999888100210	-	ACTIVE	1	1
999999888100210	-	ACTIVE	1	1
999999888100210	-	ACTIVE	1	1
999999888100210	-	ACTIVE	1	1
999999888100210	-	ACTIVE	1	1
999999888100210	-	ACTIVE	1	1
999999888100210	-	ACTIVE	1	1
999999888100210	-	ACTIVE	1	1
999999888100209	-	ACTIVE	1	1
999999888100209	-	ACTIVE	1	1
999999888100212	-	ACTIVE	1	1
999999888100212	-	ACTIVE	1	1
999999888100212	-	ACTIVE	1	1
999999888100212	-	ACTIVE	1	1
999999888100212	-	ACTIVE	1	1
999999888100212	-	ACTIVE	1	1
999999888100212	-	ACTIVE	1	1
999999888100212	-	ACTIVE	1	1
999999888100212	-	ACTIVE	1	1
999999888100213	-	ACTIVE	1	1
999999888100213	-	ACTIVE	1	1
999999888100213	-	ACTIVE	1	1
999999888100213	-	ACTIVE	1	1
999999888100213	-	ACTIVE	1	1
999999888100213	-	ACTIVE	1	1
999999888100213	-	ACTIVE	1	1
999999888100211	-	ACTIVE	1	1
999999888100211	-	ACTIVE	1	1
999999888100211	-	ACTIVE	1	1
999999888100211	-	ACTIVE	1	1
999999888100211	-	ACTIVE	1	1
999999888100211	-	ACTIVE	1	1
999999888100211	-	ACTIVE	1	1
999999888100210	-	ACTIVE	1	1
999999888100210	-	ACTIVE	1	1
999999888100210	-	ACTIVE	1	1
999999888100210	-	ACTIVE	1	1
999999888100210	-	ACTIVE	1	1
999999888100210	-	ACTIVE	1	1
999999888100210	-	ACTIVE	1	1
999999888100210	-	ACTIVE	1	1
999999888100210	-	ACTIVE	1	1
999999888100209	-	ACTIVE	1	1

999999888100209	-	ACTIVE	1	1
999999888100212	-	ACTIVE	1	1
999999888100212	-	ACTIVE	1	1
999999888100212	-	ACTIVE	1	1
999999888100212	-	ACTIVE	1	1
999999888100212	-	ACTIVE	1	1
999999888100212	-	ACTIVE	1	1
999999888100212	-	ACTIVE	1	1
999999888100212	-	ACTIVE	1	1
999999888100212	-	ACTIVE	1	1
999999888100213	-	ACTIVE	1	1
999999888100213	-	ACTIVE	1	1
999999888100213	-	ACTIVE	1	1
999999888100213	-	ACTIVE	1	1
999999888100213	-	ACTIVE	1	1
999999888100213	-	ACTIVE	1	1
999999888100213	-	ACTIVE	1	1
999999888100213	-	ACTIVE	1	1
999999888100213	-	ACTIVE	1	1
999999888100213	-	ACTIVE	1	1
999999888100211	-	ACTIVE	1	1
999999888100211	-	ACTIVE	1	1
999999888100211	-	ACTIVE	1	1
999999888100211	-	ACTIVE	1	1
999999888100211	-	ACTIVE	1	1
999999888100211	-	ACTIVE	1	1
999999888100211	-	ACTIVE	1	1
999999888100211	-	ACTIVE	1	1
999999888100210	-	ACTIVE	1	1
999999888100210	-	ACTIVE	1	1
999999888100210	-	ACTIVE	1	1
999999888100210	-	ACTIVE	1	1
999999888100210	-	ACTIVE	1	1
999999888100210	-	ACTIVE	1	1
999999888100210	-	ACTIVE	1	1
999999888100210	-	ACTIVE	1	1
999999888100209	-	ACTIVE	1	1
999999888100209	-	ACTIVE	1	1
999999888100209	-	ACTIVE	1	1
999999888100212	-	ACTIVE	1	1
999999888100212	-	ACTIVE	1	1
999999888100212	-	ACTIVE	1	1
999999888100212	-	ACTIVE	1	1
999999888100212	-	ACTIVE	1	1
999999888100212	-	ACTIVE	1	1
999999888100212	-	ACTIVE	1	1
999999888100212	-	ACTIVE	1	1
999999888100213	-	ACTIVE	1	1
999999888100213	-	ACTIVE	1	1
999999888100213	-	ACTIVE	1	1
999999888100213	-	ACTIVE	1	1
999999888100213	-	ACTIVE	1	1
999999888100213	-	ACTIVE	1	1

Number of UEs : 140

indicates that the corresponding row is IMEI entry.

^ indicates that the corresponding row is MSISDN entry.
 ` indicates that the corresponding row is MAC entry.
 ~ indicates that the corresponding row is SEID entry.
 & indicates that the corresponding row is IMSI entry for
 UPF session received via Sx ref-point.

Table 97: Show PDN UE context PFCP-u shared tunnel sessions output fields

Label	Description
IMSI/IMEI/MAC/SubId/SEID	The subscriber, equipment, or session identifier type
MSISDN	The MSISDN number, if applicable
State	The state of the session
APNs	The number of APNs
Type	The bearer type
Bearer	The bearer number
UE Address	The IP address of the UE
IPv4/IPv6	The type of IP address, either IPv4 or IPv6
Ref-pt/Si	The reference point or interface

Example output PDN UE context network realm statistics PFCP-u shared tunnel sessions

```

# show mobile-gateway pdn ue-context def-pfcp-u-sessions network-realm s5-default
=====
IMSI/IMEI( #)      MSISDN      State  APNs  Bearers
  /MSISDN(^)/MAC(`)
=====
999999888100212    -          ACTIVE  1     1
999999888100211    -          ACTIVE  1     1
999999888100213    -          ACTIVE  1     1
999999888100210    -          ACTIVE  1     1
999999888100209    -          ACTIVE  1     1
-----
Number of UEs : 5
  
```

```
# indicates that the corresponding row is IMEI entry.
^ indicates that the corresponding row is MSISDN entry.
` indicates that the corresponding row is MAC entry.
~ indicates that the corresponding row is SEID entry.
& indicates that the corresponding row is IMSI entry for
UPF session received via Sx ref-point.
```

Example output PDN UE context peer IP address statistics PFCP-u shared tunnel sessions

```
# show mobile-gateway pdn ue-context def-pfcp-u-sessions pfcp-u-peer-ip 33.34.1.1
```

[illegible]

Number of UEs : 33

```
# indicates that the corresponding row is IMEI entry.
^ indicates that the corresponding row is MSISDN entry.
` indicates that the corresponding row is MAC entry.
~ indicates that the corresponding row is SEID entry.
```


& indicates that the corresponding row is IMSI entry for UPF session received via Sx ref-point.

Example output PDN UE context network realm and peer IP address statistics PFCP-u shared tunnel sessions

```
*# show mobile-gateway pdn ue-context def-pfcp-u-sessions pfcp-u-peer-
ip 33.34.1.1 network-realm s5-default
```

```
=====
IMSI/IMEI(#)      MSISDN      State  APNs  Bearers
/MSISDN(^)/MAC(`)
=====
999999888100210  -          ACTIVE  1     1
-----
```

Number of UEs : 1

```
# indicates that the corresponding row is IMEI entry.
^ indicates that the corresponding row is MSISDN entry.
` indicates that the corresponding row is MAC entry.
~ indicates that the corresponding row is SEID entry.
& indicates that the corresponding row is IMSI entry for
UPF session received via Sx ref-point.
-----
```

5.2.2.63 up-peer-list

Syntax

```
up-peer-list ip-pool-stats peer-ip peer-ip
```

Context

```
show>mobile>pdn
```

Description

This command displays the IP address pool statistics of a peer.

Parameters

ip-pool-stats

keyword to display IP address pool statistics

peer-ip

specifies the IP address of a peer

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x: [0..FFFF]H
–	–	d: [0..255]D

Output

The following output example shows IP address pool statistics for a peer.

Example output IP address pool statistics for peer

```
# show mobile-gateway pdn up-peer-list ip-pool-stats peer-ip 33.34.1.1
=====
UP Peer list IP Local Pool Stats
=====
Peer address      : 33.34.1.1
-----
Network-realm    : realm-for-pool1.1.0
-----
pool-name        : pool1.1.0
Apn-Exclusive    : disabled
Network-Determ   : enabled
Remote-Pool      : disabled
-----
Prefix
      Block  Force  Used  Free  Held  Util%
-----
1.0.0.0/9
      No     No     3    8388603  0    0.00
3ffe:0:1::/48
      No     No     0    65536  0    0.00
-----
Total
      Used  Free  Held  Util%
-----
Ipv4      3    8388603  0    0.00
Ipv6      0    65536  0    0.00
-----
=====
Network-realm    : realm-for-pool1.2.0
-----
pool-name        : pool1.2.0
Apn-Exclusive    : disabled
Network-Determ   : enabled
Remote-Pool      : disabled
-----
Prefix
```

```

                Block    Force    Used    Free    Held    Util%
-----
2.0.0.0/9
                No      No      2      8388604  0      0.00
3ffe:0:2::/48
                No      No      0      65536   0      0.00
-----
Total
                Used    Free    Held    Util%
-----
Ipv4
                2      8388604  0      0.00
Ipv6
                0      65536   0      0.00
-----
=====
Network-realm   : realm-for-pool1.3.0
-----
pool-name       : pool1.3.0
Apn-Exclusive   : disabled
Remote-Pool     : disabled
Network-Determ  : enabled
-----
Prefix
                Block    Force    Used    Free    Held    Util%
-----
3.0.0.0/9
                No      No      1      8388605  0      0.00
3ffe:0:3::/48
                No      No      0      65536   0      0.00
-----
Total
                Used    Free    Held    Util%
-----
Ipv4
                1      8388605  0      0.00
Ipv6
                0      65536   0      0.00
-----
=====
Network-realm   : realm-for-pool1.4.0
-----
pool-name       : pool1.4.0
Apn-Exclusive   : disabled
Remote-Pool     : disabled
Network-Determ  : enabled
-----
Prefix
                Block    Force    Used    Free    Held    Util%
-----
4.0.0.0/9
                No      No      0      8388606  0      0.00
3ffe:0:4::/48
                No      No      0      65536   0      0.00
-----
Total
                Used    Free    Held    Util%
-----
Ipv4
                0      8388606  0      0.00
Ipv6
                0      65536   0      0.00
-----
=====
Network-realm   : realm-for-pool0.1.0
-----
pool-name       : pool0.1.0
Apn-Exclusive   : disabled
Remote-Pool     : disabled

```

Network-Determ : enabled

Prefix	Block	Force	Used	Free	Held	Util%
18.0.0.0/9	No	No	0	8388606	0	0.00
3ffe:0:12::/48	No	No	0	65536	0	0.00
Total			Used	Free	Held	Util%
Ipv4			0	8388606	0	0.00
Ipv6			0	65536	0	0.00

Network-realm : realm-for-pool0.2.0

pool-name : pool0.2.0
 Apn-Exclusive : disabled Remote-Pool : disabled
 Network-Determ : enabled

Prefix	Block	Force	Used	Free	Held	Util%
19.0.0.0/9	No	No	0	8388606	0	0.00
3ffe:0:13::/48	No	No	0	65536	0	0.00
Total			Used	Free	Held	Util%
Ipv4			0	8388606	0	0.00
Ipv6			0	65536	0	0.00

Network-realm : realm-for-pool0.3.0

pool-name : pool0.3.0
 Apn-Exclusive : disabled Remote-Pool : disabled
 Network-Determ : enabled

Prefix	Block	Force	Used	Free	Held	Util%
20.0.0.0/9	No	No	0	8388606	0	0.00
3ffe:0:14::/48	No	No	0	65536	0	0.00
Total			Used	Free	Held	Util%
Ipv4			0	8388606	0	0.00
Ipv6			0	65536	0	0.00

Network-realm : realm-for-pool0.4.0

```
-----
pool-name       : pool0.4.0
Apn-Exclusive   : disabled
Network-Determ  : enabled
Remote-Pool     : disabled
-----
```

Prefix	Block	Force	Used	Free	Held	Util%
21.0.0.0/9	No	No	0	8388606	0	0.00
3ffe:0:15::/48	No	No	0	65536	0	0.00
Total			Used	Free	Held	Util%
Ipv4			0	8388606	0	0.00
Ipv6			0	65536	0	0.00

```
=====
Reserved IP List : ipList1
-----
```

```
IPv4 Address: 1.2.3.4
IPv4 Address: 6.7.8.9
IPv4 Address: 4.5.6.7
```

```
IPv6 Address: 1:2:3:4:5:6:7:8
IPv6 Address: 3:4:5:6:7:8:9:0
IPv6 Address: 2:3:4:5:6:7:8:7
```

```
=====
Number of ip-pools : 8
=====
```

```
* indicates that the corresponding row element may have been truncated.
```

5.2.3 PDN clear command descriptions

5.2.3.1 mobile-gateway

Syntax

```
mobile-gateway
```

Context

```
clear
```

Description

This command enables the context to clear mobile gateway information.

5.2.3.2 pdn

Syntax

```
pdn gw-id
```

Context

```
clear>mobile
```

Description

This command clears PDN gateway related information.

Parameters

gw-id

specifies the gateway ID

Values: 1 to 8

5.2.3.3 apn

Syntax

```
apn apn-name attach-failure-statistics [group group-number]
```

```
apn apn-name [group group-number] call-flow-stats
```

```
apn apn-name delete-session-statistics [group group-number]
```

```
apn apn-name [group group-number] apn-override-statistics
```

```
apn apn-name [group group-number] pcscf-statistics
```

```
apn apn-name [group group-number] pcscf-statistics
```

```
apn apn-name transport-statistics [imsi imsi] msisdn msisdn]
```

Context

```
clear>mobile>pdn
```

Description

This command clears APN information.

Parameters

apn-name

specifies the APN name

group-number

specifies the group number

Values: 1 to 8

attach-failure-statistics

specifies the attach failure statistics

group-number

specifies the group number

Values: 1 to 8

call-flow-stats

specifies the deletion of call-flow statistics

group-number

specifies the group number

Values: 1 to 8

delete-session-statistics

specifies the deletion session statistics

group-number

specifies the group number

Values: 1 to 8

apn-override-statistics

specifies the APN override statistics

group-number

specifies the group number

Values: 1 to 8

pcscf-statistics

specifies the P-CSCF statistics

group-number

specifies the group number

Values: 1 to 8

transport-statistics

specifies transport-byte-collection statistics per APN

imsi

specifies the IMSI, up to 15 digits

msisdn

specifies the MSISDN, up to 15 digits

5.2.3.4 application-server-stats

Syntax

application-server-stats [udp-tunnel-group *udp-tunnel-group-name* [peer *ip-address*]]

Context

clear>mobile>pdn

Description

This command clears the UDP tunnel group statistics.

Parameters

udp-tunnel-group-name

specifies the tunnel group name

ip-address

specifies a peer within a tunnel group

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x: [0..FFFF]H
–	–	d: [0..255]D

5.2.3.5 bearer-context

Syntax

```

bearer-context {msisdn msisdn | imei imei | imsi imsi} [local-only] [apn name] [bearer
bearer-id]

bearer-context {msisdn msisdn | imei imei | imsi imsi} [apn name] [cause-reactivation-
req]

bearer-context mac ieee-address [local-only] [apn name]

bearer-context apn name [local-only]

bearer-context [session-deletion-rate session-deletion-rate] [group group] [protect-
dedicated-bearers] [percentage percentage [ue-clear]] apn name

bearer-context [session-deletion-rate session-deletion-rate] [group group] [protect-
dedicated-bearers] [percentage percentage]

bearer-context group group [session-deletion-rate session-deletion-rate] apn name pdn-
type pdn-type

bearer-context group group apn name pdn-type pdn-type {cancel | data-packet-trigger}

bearer-context {idle-time-leq idle-time-leq | idle-time-gt idle-time-gt | idle-all}
[session-deletion-rate session-deletion-rate] [group group] [apn name]

bearer-context {connected-time-leq connected-time-leq | connected-time-gt connected-
time-gt} [session-deletion-rate session-deletion-rate] [group group] [apn name]

bearer-context [session-deletion-rate session-deletion-rate] [group group] ue-location
ue-location ue-location-id ue-location-id [apn name]

bearer-context [session-deletion-rate session-deletion-rate] [group group] [protect-
dedicated-bearers] [percentage percentage] {pcrf-ip ip-address [pcrf-destination-
host fqdn] | p-cscf-ip ip-address} [apn name] [cause-reactivation-req]

bearer-context [session-deletion-rate session-deletion-rate] [group group] [protect-
dedicated-bearers] [percentage percentage] {slice slice-name | slice-instance slice-
instance-name} [apn name]

bearer-context [session-deletion-rate session-deletion-rate] [group group] [protect-
dedicated-bearers] [percentage percentage] pcrf-ip ip-address pcrf-destination-host
fqdn [apn name] [cause-reactivation-req] {{idle-time-leq idle-time-leq | idle-time-gt
idle-time-gt | idle-all} | {connected-time-leq connected-time-leq | connected-time-gt
connected-time-gt}}

bearer-context up-peer ipv4-address | ipv6-address apn apn-name [ipv4-prefix ipv4-
address/mask | ipv6-prefix ipv6-address/prefix-length] [cancel]

```

```
bearer-context {rat-type {utran | geran | wlan | gan | hspa-e | eutran | virtual | eutran-nb | lte-m | nr | nr-redcap | wifi} | {user-range type {imsi | msisdn | imei-tac} start-ue-id end-ue-id}} [apn name] [local-only] [bearer bearer-id]
```

```
bearer-context {rat-type {utran | geran | wlan | gan | hspa-e | eutran | virtual | eutran-nb | lte-m | nr | nr-redcap | wifi} | {user-range type {imsi | msisdn | imei-tac} start-ue-id end-ue-id}} [apn name] [cause-reactivation-req]
```

```
bearer-context udm-no-get-response nf-instance-id [protect-dedicated-bearers] [protect-dedicated-bearers] [cause-reactivation-req] [session-deletion-rate session-deletion-rate]
```

```
bearer-context [protect-dedicated-bearers] up-peer ipv4-address | ipv6-address [apn name]
```

Context

clear>mobile>pdn

Description

This command clears bearer-context information.



NOTE:

The optional and mandatory parameters must be given at the exact location indicated in the syntax.

When the command is given at the PDN level:

- bearer and session deletion is initiated for all APNs configured under the specified gateway instance
- the parameters used with the command to filter the sessions are matched against the whole UE; if they match, all UE sessions are deleted
- the session deletion rate is treated by the system as the maximum rate to be applied, not as an enforced rate for the session deletion

The rate configured under the config>mobile>pdn>session-deletion-rate command is used if a value is not specified in the command.

A subsequent clear command is not accepted while another one is still ongoing, except in the following cases:

- the intent is to increase/decrease the deletion rate
- the intent is to clear all the sessions for the same entity (gateway or APN) given at the ongoing deletion
- the intent is to change the percentage value, by giving the exact same command with different percentage value

The following optional parameters select entities and can be combined arbitrarily:

- apn
- slice
- slice-instance (cannot be used together with slice)
- group

The following optional parameters are filters. Combining more than one filter in the same command is not supported; except for the pcrf-ip (PCRF or DRA address), protect-dedicated-bearers and percentage filters which can be combined in the commands they exist.

- protect-dedicated-bearers
- percentage
- idle-time-leq
- idle-time-gt
- idle-all
- connected-time-leq
- connected-time-gt
- ue-location
- ue-location-id
- pdn-type
- pcrf-destination-host

**NOTE:**

Before using the `clear` command, refer to *Commands for administrative session deletion*, in the *CMG Configuration Guide*.

Parameters

msisdn

specifies the MSISDN, up to 15 characters

imei

specifies an IMEI, up to 16 characters

imsi

specifies the IMSI, up to 15 characters; supported in PGW, SGW, and ePDG

local-only

specifies an internal cleanup

name

specifies the APN name

bearer-id

Specifies the EPS Bearer Identifier of the specific UE. If more than one bearers exist with the same EBI (for example, for S2b and S5), both bearers are deleted. If the given EBI is the default bearer, all bearers are deleted.

Values: 5 to 15

cause-reactivation-req

keyword to set the Request Cause of the release signaling to 8 (Reactivation Requested) in GTPv2 and to 6 (Reactivation Requested) in GTPv1. In 5G deployment, this keyword sets that SMF will send the 5GSM cause value #39 “reactivation requested” in the PDU SESSION RELEASE COMMAND message.

ieee-address

specifies the UE MAC

session-deletion-rate

sets the deletion rate of sessions per MSCP group per second only for this command execution. This is treated by the system as the maximum rate to be applied, not as an enforced rate for the session deletion. This parameter cannot be used together with the `local-only` parameter. When this parameter is not set, the transmission rate configured in the context `config>mobile>pdn>session-deletion-rate` is applied.

Values: 2 to 2048

group

specifies the group number as a criterion for the bearer-contexts to be cleared

Values: 1 to 8

protect-dedicated-bearers

specifies that when this option is set, the sessions (or the UEs at the PDN level) with at least one dedicated bearer are not deleted. They will be deleted automatically (without a separate clear command), when the dedicated bearers are torn down (normally by the network); for example, when the VoLTE call ends. This parameter cannot be used together with the `local-only` option.

When the parameter is used with the percentage option, the command execution finishes when the requested percentage of sessions has been deleted.

percentage

specifies the percentage of sessions to get deleted

Values: 1 to 100

ue-clear

specifies that all sessions (in all APNs) of a UE that has a session on the selected APN, will be deleted. This option can be used only with the percentage option.

pdn-type

specifies the PDN type of the sessions to be deleted

Values	ipv4	clears the sessions with an IPv4 address
–	ipv6	clears the sessions with an IPv6 address
–	ipv4ipv6	clears the sessions with either IPv4 or IPv6, or both types of addresses

cancel

This option cancels an ongoing command. It is supported and can be used only in the commands shown in the syntax. The ongoing deletion instance to be canceled is identified by using the same set of parameters and their values that were used initially when the `clear` command was initiated.

data-packet-trigger

specifies that the sessions, included to the selected entity (apn, group) and to the specific pdn - type are deleted when an uplink data packet is being received for these sessions

idle-time-leq / idle-time-gt

specifies the `idle-time` in seconds, as a criterion for the bearer-contexts (UEs at the PDN level command) to be cleared

Values: 60 to 2592000 (30 days)

idle-all

specifies that all bearer-contexts (UEs at the PDN level command) which are not having traffic are selected for deletion

**NOTE:**

The `idle-time-leq`, `idle-time-gt`, and `idle-all` options are supported only when charging is enabled on the gateway.

connected-time-gt

specifies the connected time "greater than" in seconds, as a criterion for the bearer-contexts (UEs at the PDN level command) to be cleared

Values: 1 to 2592000 (30 days)

connected-time-leq

specifies the connected time "less than or equal" in seconds, as a criterion for the bearer-contexts (UEs at the PDN level command) to be cleared

Values: 1 to 2592000 (30 days)

ue-location

specifies the UE location type for the user location filter

Values: LTE access | tracking-area-code

slice-name

specifies the slice name

slice-instance-name

specifies the slice instance name

ue-location-id

specifies the location ID is selected, the TAC is set here.

Values: 4-digit hexadecimal number for TAC

ip-address

specifies the IPv4 or IPv6 address of the peer server (a PCRF or a DRA)

Values	ipv4 address	a.b.c.d
–	ipv6 address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

fqdn

specifies the FQDN (destination host) when the peer is a DRA, up to 255 characters

ipv4-address

specifies the IPv4 address, up to 64 characters

Values: a.b.c.d

mask

specifies the IPv4 address mask

Values: 9 to 30

prefix-length

specifies the IPv6 prefix length

Values: 41 to 56

rat-type

specifies the RAT type

Values: *utran* | *geran* | *wlan* | *gan* | *hspa-e* | *eutran* | *virtual* | *eutran-nb* | *lte-m* | *nr* | *nr-redcap* | *wifi*

user-range type

specifies the type of the user range

Values: *imsi* | *msisdn* | *imei-tac*

start-ue-id

specifies the user range start identifier, 4 to 15 characters

end-ue-id

specifies the user range end identifier, 4 to 15 characters

udm-no-get-response

specifies to release the sessions which were activated when the UDM did not return the session management subscription data

nf-instance-id

specifies the UUID of the remote NF

Values: xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx (x: [0..F]H)

5.2.3.6 call-flow-stats

Syntax

```
call-flow-stats [group group-number | card slot-number]
```

Context

```
clear>mobile>pdn
```

Description

This command clears call flow statistics.

Parameters

group-number

specifies the group number

Values: 1 to 8

slot-number

specifies the card slot number

Values: 1 to 10 | A | B

local-only

specifies an internal cleanup

5.2.3.7 charging

Syntax

```
charging imei string imsi string msisdn string [open-partial]
```

Context

```
clear>mobile>pdn
```

Description

This command clears bearer statistics.

Parameters

string

specifies the IMEI string, 15 digits maximum

imsi *string*

specifies the IMSI string, 15 digits maximum

msisdn *string*

specifies the MSISDN string, 15 digits maximum

open-partial

specifies the open-partial keyword

5.2.3.8 diameter-sessions

Syntax

```
diameter-sessions peer {ip-addr | fqdn} [destination-host destination-host] [graceful]  
[port port] application-type {gx | gy}
```

Context

```
clear>mobile>pdn
```

Description

This command clears all sessions from the Diameter peer. If the Diameter peer is directly connected, the configured peer must be set to suspend before executing this command. If DRAs are deployed between the mobile gateway and the diameter peer, the destination-host command is used.



NOTE:

The `destination-host` parameter is used when DRA proxy is deployed between the mobile gateway and the diameter peer. In this case, the configured `destination-host` is the string included in the origin-host AVP that is used by the diameter peer initiated requests (RAR) and answers (CCA).

Parameters

ip-addr

specifies the IPv4 or IPv6 address of the diameter peer

Values	ipv4 address	a.b.c.d
–	ipv6 address	x:x:x:x:x:x:x (eight 16-bit pieces)
–		x:x:x:x:x:d.d.d.d
–		x - [0..FFFF]H
–		d - [0..255]D

fqdn

specifies the FQDN, up to 255 characters

destination-host

the host FQDN, up to 255 characters

graceful

configures graceful session shutdown by sending CCR-T messages from the gateway to the Diameter server

port

Values: 1 to 65535

gx

specifies the gx application type running on the Diameter peer

gy

specifies the gy application type running on the Diameter peer

5.2.3.9 dns-client-cache

Syntax

`dns-client-cache`

Context

`clear>mobile>pdn`

Description

This command clears DNS client cache statistics.

5.2.3.10 fixed-subscriber

Syntax

`fixed-subscriber subscriber-id [apn apn-name] [host ip-address] [local-only]`

Context

`clear>mobile>pdn`

Description

This command clears fixed network transit sessions associated with the subscriber ID.

If no other option is specified then all sessions with the specified subscriber ID will be cleared. If an APN name is specified, the transit sessions associated with the subscriber ID created under the APN will be cleared. If an IP address is specified, the transit session associated with the subscriber ID for the host with the specified IP address is cleared.

Parameters

subscriber-id

specifies the subscriber ID, up to 32 characters

apn-name

specifies the APN name, up to 80 characters

ip-address

specifies the host IP address

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x: [0..FFFF]H
–	–	d: [0..255]D

local-only
keyword for internal cleanup

5.2.3.11 gtp-load-overload-control

Syntax
gtp-load-overload-control stats [node-id]

Context
clear>mobile>pdn

Description
This command clears GTP-C load and overload control statistics.

Parameters

stats
specifies to clear aggregated stats

node-id
specifies the node ID to clear statistics
Values: MME | SGW | PGW | EPDG

5.2.3.12 gtp-local-overload-control

Syntax

```
gtp-local-overload-control stats
```

Context

```
clear>mobile>pdn
```

Description

This command clears GTP-C local overload control statistics.

Parameters

stats

specifies to clear aggregated stats

5.2.3.13 local-storage-stats

Syntax

```
local-storage-stats [gtp-prime-group gtp-prime-group] ga
```

Context

```
clear>mobile>pdn
```

Description

This command clears statistics for locally stored CDR files in Ga interface.

Parameters

gtp-prime-group

specifies the GTP prime group profile name, up to 32 characters

ga

specifies the Ga interface

5.2.3.14 ipsec-lockout

Syntax

```
ipsec-lockout idi user-id source-ip source-ip-address
```

Context

```
clear>mobile>pdn
```

Description

This command clears a user from the IPsec lockout list by providing the user ID and source IP address.

Parameters

user-id

specifies the ID of the locked out user that is shown on the IPsec lockout list

source-ip-address

specifies the source IP address of the locked out user as shown on the IPsec lockout list

5.2.3.15 nrf-client-cache

Syntax

```
nrf-client-cache [nf-type {upf | pcf | udm | amf}]
```

Context

```
clear>mobile>pdn
```

Description

This command clears the NRF cache. The optional nf - type filter defines the discovery target NF type for which the cache entries are cleared.

Parameters

nf - type

Values: upf | pcf | udm | amf

5.2.3.16 nsmf-overload-control

Syntax

```
nsmf-overload-control stats
```

Context

```
clear>mobile>pdn
```

Description

This command clears the Nsmf overload control statistics.

Parameters

stats

specifies to clear statistics

5.2.3.17 path-mgmt-fault

Syntax

```
path-mgmt-fault {s5 | s8} [ref-point ref-point-name] [peer ip-address / fqdn] [port port]
```

Context

```
clear>mobile>pdn
```

Description

This command clears path management fault statistics.

Parameters

s5 | s8

specifies the reference point type

ref-point-name

specifies the reference point name

ip-address | fqdn

specifies an existing peer IP address

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x: [0..FFFF]H
–	–	d: [0..255]D
–	fqdn	FQDN, up to 255 characters

port
specifies a port ID
Values: 1 to 65535

5.2.3.18 path-mgmt-stats

Syntax

```
path-mgmt-stats {s11 | s1u | s4 | s4u | s12 | s5 | s5u | s8 | s8u | gx | rf | gn | gnu | gp | gpu | ga | gy | s2a | s2b | s6b | sx-4} [peer ip-address [port port]]  
  
path-mgmt-stats {s5 | s5u | s8 | s8u | gn | gnu | gp | gpu | n9} [ref-point ref-point-name] [peer ip-address [port port]]
```

Context

```
clear>mobile>pdn
```

Description

This command clears path management statistics.

Parameters

s11 | s1u | s4 | s4u | s12 | s5 | s5u | s8 | s8u | gx | rf | gn | gnu | gp | gpu | ga | gy | s2a | s2b | s6b | sx-n4 | n9

specifies the reference point type

ref-point-name

specifies the reference point name

ip-address | *fqdn*

specifies an existing peer IP address

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x: [0..FFFF]H
–	–	d: [0..255]D
–	fqdn	FQDN, up to 255 characters

port

specifies a port ID

Values: 1 to 65535

5.2.3.19 pdn-context

Syntax

```
pdn-context cp-seid seid [cp-ipv4 ip-address] [cp-ipv6 ipv6-address]  
pdn-context ue-ip ue-ip network-instance apn [access-type access-type]  
pdn-context up-seid seid
```

Context

```
clear>mobile>pdn
```

Description

This command clears the UP session context locally on the UP function.

Parameters

ue-ip

specifies the UE IP address

Values		
	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x: [0..FFFF]H
–	–	d: [0..255]D

apn

specifies the APN name associated for the Virtual Router instance on the UP peer where the UE IP address is located and is communicated over Sx/N4 in the Network-Instance IE, up to 80 characters

access-type

specifies the access type

Values: utran | geran | wlan | gan | hspa | eutran | ehrpd | hrpd | 1xrtt | umb | virtual | eutran-nb-iot

seid

specifies the Session Endpoint Identifier for the CP or the UP session context, up to 40 characters

ip-address

specifies the CP function's Sx/N4 reference point IPv4 address

Values: a.b.c.d

ipv6-address

specifies the CP function's Sx/N4 reference point IPv6 address, up to 64 characters

5.2.3.20 pfcpl-load-overload-control

Syntax

pfcpl-load-overload-control stats

Context

clear>mobile>pdn

Description

This command clears PFCP load and overload control statistics for the CP function.

Parameters

stats

keyword to clear statistics

5.2.3.21 pfcpl-local-load-control

Syntax

pfcpl-local-load-control stats

Context

clear>mobile>pdn

Description

This command clears PFCP local load control statistics for the UP function.

Parameters

stats

keyword to clear statistics

5.2.3.22 pfcpl-local-overload-control

Syntax

pfcpl-local-overload-control stats

Context

clear>mobile>pdn

Description

This command clears PFCP local overload control statistics for the UP function.

Parameters

stats

keyword to clear local statistics

5.2.3.23 pgw-reselection-isolation

Syntax

pgw-reselection-isolation [peer ip-address]

Context

clear>mobile>pdn

Description

This command removes a specific peer that is in isolation status (under the show>mobile>pdn>pgw-selection>pgw-reselection-isolation command) before the expiration of the isolation timer. The peer becomes available for selection.

Parameters

ip-address

specifies the IPv4 or IPv6 address for an isolated peer that is listed under the show>mobile>pdn>pgw-selection>pgw-reselection-isolation command

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x: [0..FFFF]H
–	–	d: [0..255]D

5.2.3.24 ref-point-stats

Syntax

```
ref-point-stats {s5 | s8 | gx | rf | gn | gp | ga | gy | s2a | s2b | s6b | pi | s1u | s11 | s11u | s4 |
s12 | swm | sww | sta | sd | s2a-gtp | sx-n4} [peer ip-address [port port]] [group group-number |
card slot-number]

ref-point-stats {radius} [radius-group radius-group [peer ip-address]] [group group-
number | card slot-number]

ref-point-stats {s5 | s8 | gn | gp | n9} [ref-point ref-point-name] [peer ip-address [port
port]] [group group-number | card slot-number]

ref-point-stats {pi} [ref-point ref-point-name [pi-peer ipv4 [port port]]] [group group-
number | card slot-number]

ref-point-stats swu [card slot-number]

ref-point-stats {sww} sww-type [ref-point ref-point-name] [peer ip-address] [group group-
number | card slot-number]

ref-point-stats {t6c} [ref-point ref-point-name] [peer ip-address [port port]] [card slot-
number] [peer-type scef]
```

Context

```
clear>mobile>pdn
```

Description

This command clears reference point statistics.

Parameters

- s5 | s8 | gx | rf | gn | gp | ga | gy | s2a | s2b | s6b | pi | s1u | s11 | s11u | s4 | s12 | swm | sww | sta | sd | s2a-gtp | sx-n4 | n9
- specifies the reference point type
- ip-address | fqdn
- specifies an existing peer IP address/FQDN

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)

–	–	x:x:x:x:x:d.d.d.d
–	–	x: [0..FFFF]H
–	–	d: [0..255]D
–	fqdn	FQDN

port

specifies a port ID

Values: 1 to 65535

group-number

specifies the group number

Values: 1 to 8

slot-number

specifies the card slot number

1 to 10 | A | B

ref-point-name

specifies the reference point name

radius

specifies the RADIUS reference point

radius-group

specifies the RADIUS group

pi

specifies the PI reference point

swu

specifies the SWu reference point

sww

specifies the SWw type

t6c

specifies the reference point between PGW (PCEF) and SCEF

scef

specifies the peer type, *scef* is the only applicable peer type for t6c

5.2.3.25 service-stats

Syntax

```
service-stats nf-type nf-type [service-realm service-realm-name] [service-name service-name] [service-instance service-instance] [remote-nf-id uuid | remote-nf-ip ip-address | fqdn] [gateway gw-id | group group-id]
```

Context

clear>mobile>pdn

Description

This command clears SBA statistics.

Parameters

nf-type

specifies the NF type
Values: amf | bsf | nrf | pcf | udm | chf | smf | sepp

service-realm-name

specifies the service realm name, up to 32 characters

service-name

specifies the service name
Values: nsmf-pdusession | namf-comm | namf-evts | nbsf-management | nnrf-nfm | nnrf-disc | npcfsmpolicycontrol | nudm-sdm | nudm-uecm | nchf-convergedcharging | nnrf-oauth2 | nsmf-event-exposure | sepp

service-instance

specifies the service instance name, up to 32 characters

uuid

specifies the UUID of the remote NF
Values: xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx (x: [0..F]H)

ip-address / fqdn

specifies the IP address/FQDN of the remote NF

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)

–	–	x:x:x:x:x:d.d.d.d
–	–	x: [0..FFFF]H
–	–	d: [0..255]D
–	fqdn	FQDN, up to 255 characters

gw-id

specifies the gateway ID

group-id

specifies the group ID

Values: 0 (CPM statistics) | 1 to 15

5.2.3.26 statistics**Syntax**

```
statistics attach-failure-statistics group group-number
statistics [group group-number] ga-buffered-cdrs
```

Context

```
clear>mobile>pdn
```

Description

This command clears statistics for each card.

Parameters***group-number***

specifies the group number

Values: 1 to 15

attach-failure-statistics

clears the attach failure statistics

ga-buffered-cdrs

clears the counter for Ga buffered CDR files

5.2.3.27 ue-context

Syntax

```
ue-context ran-ue-ngap-id ran-ue-ngap-id [local-only]
```

Context

```
clear>mobile>pdn
```

Description

This command clears the UE context sessions for a specific RAN UE NGAP ID.

Parameters

ran-ue-ngap-id

specifies the RAN UE NGAP ID

Values: 0 to $2^{32}-1$ (HEX)

local-only

specifies to release locally all resources in N3IWF without notifying any external node (AMF/UPF/UE)

6 Profile command reference

This section describes the profile-related commands.

6.1 Profile command hierarchies

- *Profile configuration command hierarchies*
- *Profile show command hierarchy*

6.1.1 Profile configuration command hierarchies

- *Accounting priority profile command hierarchy*
- *Address list profile command hierarchy*
- *Call trace profile command hierarchy*
- *Canary test profile command hierarchy*
- *Charging profile command hierarchy*
- *CCFH charging profile command hierarchy*
- *CDR charging profile command hierarchy*
- *CHF charging profile command hierarchy*
- *ePDG charging profile command hierarchy*
- *FBC event triggers charging profile command hierarchy*
- *Gy AVP options charging profile*
- *PGW charging profile command hierarchy*
- *SGW charging profile command hierarchy*
- *SMF charging profile command hierarchy*
- *TDF charging profile command hierarchy*
- *Cloud database profile command hierarchy*
- *DCCA profile command hierarchy*
- *Dedicated bearer idle timeout profile command hierarchy*
- *DHCP server group profile command hierarchy*
- *DHCPv4 server profile command hierarchy*
- *Diameter error handling profile command hierarchy*

- *Diameter profile command hierarchy*
- *Diameter peer profile command hierarchy*
- *DNS profile command hierarchy*
- *DNS redirect profile command hierarchy*
- *Emergency profile command hierarchy*
- *ePDG group profile command hierarchy*
- *Fa Ha peer list profile command hierarchy*
- *FQDN group profile command hierarchy*
- *GTP profile command hierarchy*
- *GTP prime group profile command hierarchy*
- *Gx profile command hierarchy*
- *HTTP/2 profile command hierarchy*
- *IPSec profile command hierarchy*
- *List profile command hierarchy*
- *MIPv4 profile command hierarchy*
- *MME group profile command hierarchy*
- *N1 profile command hierarchy*
- *N2 peer profile command hierarchy*
- *N7 attributes profile command hierarchy*
- *Namf profile command hierarchy*
- *Network trace profile command hierarchy*
- *Node selection profile command hierarchy*
- *NPCF profile command hierarchy*
- *NSMF profile command hierarchy*
- *OAuth2.0 authorization profile command hierarchy*
- *Overload control profile command hierarchy*
- *PCMD profile command hierarchy*
- *PCSCF group profile command hierarchy*
- *Peer list profile command hierarchy*
- *PFCP profile command hierarchy*
- *PMIPv6 profile command hierarchy*

- [Policy options profile command hierarchy](#)
- [PPI mapping profile command hierarchy](#)
- [QFI mapping profile command hierarchy](#)
- [QoS profile command hierarchy](#)
- [RADIUS profile command hierarchy](#)
- [S2b error mapping profile command hierarchy](#)
- [S6b profile command hierarchy](#)
- [SBI error mapping profile command hierarchy](#)
- [SCTP profile command hierarchy](#)
- [Sd profile command hierarchy](#)
- [SGSN profile command hierarchy](#)
- [Sig rate limit profile command hierarchy](#)
- [SLAAC host profile command hierarchy](#)
- [STa profile command hierarchy](#)
- [Subscriber mapping profile command hierarchy](#)
- [SWm error mapping profile command hierarchy](#)
- [SWm profile command hierarchy](#)
- [SWw profile command hierarchy](#)
- [T6c profile command hierarchy](#)
- [Tracking area identity \(TAI\) and location area identity \(LAI\) list profile command hierarchy](#)
- [UDM-SDM profile command hierarchy](#)
- [UDM-UECM profile command hierarchy](#)
- [UDP tunnel group profile command hierarchy](#)
- [UMTS QoS policy profile command hierarchy](#)
- [UP security policy profile command hierarchy](#)
- [V-SMF profile command hierarchy](#)
- [WLAN emergency profile command hierarchy](#)
- [WPS profile command hierarchy](#)

6.1.1.1 Accounting priority profile command hierarchy

configure

```

– mobile-gateway
  – profile
    – [no] accounting-priority-profile profile-name
      – description description-string
      – no description
    – [no] priority priority
      – match-id match-id [roaming-status status] [start-rating-group group-id
[end-rating-group [0..4294967295]] [monitoring-key key] [start-qci qci-value] [end-qci
[1..254]]
      – no match-id match-id
      – single-wt-session {enable | disable}
      – no single-wt-session

```

6.1.1.2 Address list profile command hierarchy

```

configure
– mobile-gateway
  – profile
    – [no] addr-list addr-list-name
    – [no] ip-address ip-address/prefix-length
    – [no] ip-address-group ipv6-prefix/prefix-length

```

6.1.1.3 Call trace profile command hierarchy

```

configure
– mobile-gateway
  – profile
    – call-insight
      – [no] all-ues profile-name
        – [no] events {all | acctmgr | apm | bmc | dcm | dhcp | dth | slaac | ga |
gal | gtp | gtpv2 | gxc | gxl | ike | lrdm | mip | msngr | pmip | rfl | rol | s6b |
sctp | sd | sessmgr | sigmgr | swm | sww | sta | rpy | dpy | pfcf | nserv | n3iwf-
ngap}
        – [no] format [pcap | simulated-pcap]
        – [no] live-output ip-address | fqdn [port port] [router router-instance]
        – [no] path-management [ref-point {all | ga | gn | gp | gx | gxc | rf | ro |
s1 | s11 | s12 | s2a | s2b | s4 | s5 | s6b | s8 | sd | sta | swm | sx-n4 | n3 | n9 |
n2} | sba-service {all | nnrf-disc | nnrf-nfm | nnrf-oauth2}]
        – [no] ref-point {all | dhcp | ga | gn | gp | gx | gxc | l2tp | pi | radius
| rf | ro | s1 | s11 | s12 | s2a | s2b | s4 | s5 | s6b | s8 | sd | swm | swu | swu-
cleartext | sww | sta | dns | swu-esp-drop | sx-n4 | n2}

```

```

    - [no] sba-service {all | namf-comm | namf-evts | nchf-convergedcharging |
npcf-smpolicycontrol | nsmf-event-exposure | nsmf-pdusession | nudm-sdm | nudm-uecm}
    - size-limit size-limit
    - no size-limit
    - time-limit time-limit
    - no size-limit
  - [no] ue profile-name
    - [no] events {all | acctmgr | apm | bmc | dcm | dhcp | dth | slaac | ga |
gal | gtp | gtpv2 | gxc | gxl | ike | lrdm | mip | msngr | pmip | rfl | rol | s6b |
sctp | sd | sessmgr | sigmgr | swm | sww | sta | rpy | dpy | pfcf | nserv}
    - [no] format {pcap | simulated-pcap}
    - [no] live-output ip-address | fqdn [port port] [router router-instance]
    - [no] ref-point {all | dhcp | ga | gn | gp | gx | gxc | l2tp | pi | radius
| rf | ro | s1 | s11 | s12 | s2a | s2b | s4 | s5 | s6b | s8 | sd | swm | swu | swu-
cleartext | sww | sta | dns | swu-esp-drop | sx-n4 | n7 | n10 | n11}
    - [no] sba-service {all | namf-comm | namf-evts | nchf-convergedcharging |
npcf-smpolicycontrol | nsmf-event-exposure | nsmf-pdusession | nudm-sdm | nudm-uecm}
    - [no] size-limit size-limit
    - [no] time-limit time-limit
    - [no] user-traffic [{internal | statistics | ingress | egress | metadata}]

```

6.1.1.4 Canary test profile command hierarchy

```

configure
  - mobile-gateway
    - profile
      - canary-test
        - [no] user-range user-range-name
        - ue-entry ue-entry-id ue-type ue-type start-id start-id [end-
id end-id]
        - no ue-entry ue-entry-id [ue-type ue-type] [start-id start-id]
        [end-id end-id]

```

6.1.1.5 Charging profile command hierarchy

This section lists the main branches of the charging profile command hierarchy. See the following for the commands in each branch:

- *CCFH charging profile command hierarchy*
- *CDR charging profile command hierarchy*
- *CHF charging profile command hierarchy*
- *ePDG charging profile command hierarchy*

- *FBC event triggers charging profile command hierarchy*
- *QBC event triggers charging profile command hierarchy*
- *Gy AVP options charging profile*
- *PGW charging profile command hierarchy*
- *SGW charging profile command hierarchy*
- *SMF charging profile command hierarchy*
- *TDF charging profile command hierarchy*

configure

- mobile-gateway
 - *profile*
 - *charging*
 - [no] *ccfh-profile charging-profile-id*
 - [no] *cdr-avp-options charging-profile-id*
 - [no] *chf-profile charging-profile-id*
 - [no] *credit-pool-id pool-id*
 - *max-interim-quotas number*
 - no *max-interim-quotas*
 - *validity-time seconds*
 - no *validity-time*
 - *volume-quota kilobytes*
 - no *volume-quota*
 - [no] *epdg-charging charging-profile-id*
 - [no] *event-triggers-fbc profile-id*
 - [no] *event-triggers-qbc profile-id*
 - [no] *gy-avp-options gy-avp-options-profile-id*
 - [no] *pgw-charging charging-profile-id*
 - [no] *sgw-charging charging-profile-id*
 - [no] *smf-attributes-profile profile-id*
 - *description description-string*
 - no *description*
 - *ran-secondary-rat-usage-reports {include | exclude}*
 - no *ran-secondary-rat-usage-reports*
 - *retransmission-indicator {include | exclude}*
 - no *retransmission-indicator*
 - *service-specific-information {include | exclude}*
 - no *service-specific-information*
 - *service-specific-information-value information*
 - no *service-specific-information-value*
 - *serving-network-include-source-ip {enable | disable}*
 - no *serving-network-include-source-ip*
 - *serving-node-type {r15 | r16 | r17}*

- no *serving-node-type*
- [no] *smf-charging profile-id*
- [no] *tdf-charging charging-profile-id*
- [no] *twag-charging charging-profile-id*
 - *description description-string*
- no *description*
- *radius-accounting*
 - [no] *wlan-ap-location-change*
- [no] *upf-acct-profile profile-id*
 - *faster-poll-remaining-quota-trigger value*
 - no *faster-poll-remaining-quota-trigger*

6.1.1.6 CCFH charging profile command hierarchy

configure

- mobile-gateway
 - *profile*
 - *charging*
 - [no] *ccfh-profile profile-id*
 - *assume-positive*
 - [no] *access-type {lte | wifi | ehrpd | utran | all}*
 - *message [ccr-i] [ccr-u] [ccr-t]*
 - no *message*
 - [no] *rating-group rating-group | **
 - *credit-pool-id pool-id*
 - no *credit-pool-id*
 - [no] *enable*
 - [no] *orig-session-id-ccri*
 - *cc-failure-handling {terminate | continue | retry-and-terminate}*
 - no *cc-failure-handling*
 - [no] *cc-failure-handling-enh*
 - *ccfh-fh-action {terminate | re-start}*
 - no *ccfh-fh-action*
 - [no] *disable-cc-session-failover*
 - [no] *fail-open*
 - *fh-continue-action {allow | block | offline}*
 - no *fh-continue-action*
 - *fh-session-continue-timer fh-session-continue-timer*
 - no *fh-session-continue-timer*
 - *fh-volume-limit volume-limit*
 - no *fh-volume-limit*
 - *gy-to-gtp-error-mapping-profile profile-name*
 - no *gy-to-gtp-error-mapping-profile*
 - [no] *message {ccr-i | ccr-u}*

```

    - result-code result-code-value result-error-code [restore]
    - no result-code result-code-value
  - [no] mscc-result-code result-code-value
    - [no] fui-terminate-session
    - service-delay-timer minutes
    - no service-delay-timer
  - nchf-cause-code application-error-code action {terminate | continue |
retry-and-terminate | retry-and-continue} [continue-action {allow | block}]
    - no nchf-cause-code application-error-code
  - nchf-http-code value action [allow|block]
  - no nchf-http-code value
  - [no] nchf-mui-result-code string-result-code
    - [no] fui-terminate-session
    - service-delay-timer minutes
    - no service-delay-timer
  - result-code result-code-value result-error-code [restore]
  - no result-code result-code-value
  - result-code-3xxx result-error-code [restore]
  - no result-code-3xxx
  - result-code-4xxx result-error-code [restore]
  - no result-code-4xxx
  - result-code-5xxx result-error-code [restore]
  - no result-code-5xxx
  - sbi-error-mapping-profile profile-name
  - no sbi-error-mapping-profile
  - service-restore-max-offset max-offset
  - no service-restore-max-offset
  - session-restore-retry-timer retry-timer
  - no session-restore-retry-timer
  - store-and-replay
    - buffer-playback-timer time
    - no buffer-playback-timer
    - [no] enable

```

6.1.1.7 CDR charging profile command hierarchy

configure

```

- mobile-gateway
- profile
- charging
  - [no] cdr-avp-options profile-id
  - 3gpp2-bsid {include | exclude}
  - no 3gpp2-bsid
  - apn-ambr {include | exclude}

```


- *no apn-ambr*
- *description description-string*
- *no description*
- *apn-network-id* {include | exclude}
- *no apn-network-id*
- *apn-rate-control* {include | exclude}
- *no apn-rate-control*
- *apn-selection-mode* {include | exclude}
- *no apn-selection-mode*
- *camel-information* {include | exclude}
- *no camel-information*
- *charging-char-select-mode* {include | exclude}
- *no charging-char-select-mode*
- *cn-operator-selection* {include | exclude}
- *no cn-operator-selection*
- *cp-ciot-eps-optimisation-indicator* {include | exclude}
- *no cp-ciot-eps-optimisation-indicator*
- *diagnostics* {include | exclude}
- *no diagnostics*
- *dualstack-epdg-address-ipv6* {include | exclude}
- *no dualstack-epdg-address-ipv6*
- *dualstack-pgw-address-ipv6* {include | exclude}
- *no dualstack-pgw-address-ipv6*
- *dualstack-serving-node-ipv6-addr* {include | exclude}
- *no dualstack-serving-node-ipv6-addr*
- *dualstack-sgw-address-ipv6* {include | exclude}
- *no dualstack-sgw-address-ipv6*
- *dualstack-twag-address-ipv6* {include | exclude}
- *no dualstack-twag-address-ipv6*
- *dynamic-addr-flag* {include | exclude}
- *no dynamic-addr-flag*
- *dynamic-addr-flag-extension* {include | exclude}
- *no dynamic-addr-flag-extension*
- *enhanced-diagnostics* {include | exclude}
- *no enhanced-diagnostics*
- *extended-bw-nr* {include | exclude} [pre-extended]
- *no extended-bw-nr*
- *ims-sig-context* {include | exclude}
- *no ims-sig-context*
- *imsi-unauthenticated-flag* {include | exclude}
- *no imsi-unauthenticated-flag*
- *list-of-service-data* {include | exclude}
- *no list-of-service-data*
- *list-traffic-data-volume* {include | exclude}
- *no list-traffic-data-volume*

```
- local-record-seq-no {include | exclude}
- no local-record-seq-no
- low-priority-indicator {include | exclude}
- no low-priority-indicator
- max-requested-bandwidth {include | exclude}
- no max-requested-bandwidth
- mo-exception-data-counter {include | exclude}
- no mo-exception-data-counter
- ms-sig-context {include | exclude}
- no ms-sig-context
- ms-time-zone {include | exclude}
- no ms-time-zone
- node-id {include | exclude}
- no node-id
- nr-user-location-info {include | exclude}
- no nr-user-location-info
- nw-init-pdp-context {include | exclude}
- no nw-init-pdp-context
- originating-bsid {include | exclude}
- no originating-bsid
- originating-user-location-info {include | exclude}
- no originating-user-location-info
- pdn-connection-charging-id {include | exclude}
- no pdn-connection-charging-id
- pdn-pdp-address-prefix-length {include | exclude}
- no pdn-pdp-address-prefix-length
- pdp-pdn-type {include | exclude}
- no pdp-pdn-type
- pdp-pdn-type-extension {include | exclude}
- no pdp-pdn-type-extension
- pgw-address {include | exclude}
- no pgw-address
- pgw-plmn-id {include | exclude}
- no pgw-plmn-id
- pra-info {include | exclude}
- no pra-info
- ps-furnish-charging-information {include | exclude}
- no ps-furnish-charging-information
- ran-secondary-rat-usage-reports {include | exclude}
- no ran-secondary-rat-usage-reports
- rat-type {include | exclude}
- no rat-type
- record-extension-visited-url {include | exclude}
- no record-extension-visited-url
- sdc-charging-id {include | exclude}
```

```
- no sdg-charging-id
- sdg-pra-status {include | exclude}
- no sdg-pra-status
- sdg-qci {include | exclude}
- no sdg-qci
- sequence-number-for-complete-cdr {include | exclude}
- no sequence-number-for-complete-cdr
- served-imeisv include [acr-interim] [acr-stop]
- served-imeisv exclude
- no served-imeisv
- served-mn-nai {include | exclude}
- no served-mn-nai
- served-msisdn {include | exclude}
- no served-msisdn
- served-pdp-pdn-addr {include | exclude}
- no served-pdp-pdn-addr
- served-pdp-pdn-addr-extension {include | exclude}
- no served-pdp-pdn-addr-extension
- service-context-id {include | exclude}
- no service-context-id
- serving-node-plmn-id {include | exclude}
- no serving-node-plmn-id
- serving-plmn-rate-control {include | exclude}
- no serving-plmn-rate-control
- sgi-ptp-tunnelling-method {include | exclude}
- no sgi-ptp-tunnelling-method
- sgw-address {include | exclude}
- no sgw-address
- sgw-change {include | exclude}
- no sgw-change
- start-time {include | exclude | always-include}
- no start-time
- stop-time {include | exclude}
- no stop-time
- tdv-charging-id {include | exclude}
- no tdv-charging-id
- tdv-pra-status {include | exclude}
- no tdv-pra-status
- twan-user-location-info {include | exclude}
- no twan-user-location-info
- twan-user-location-information-r16 {include | exclude}
- no twan-user-location-information-r16
- uni-pdu-cp-only-flag {include | exclude}
- no uni-pdu-cp-only-flag
- user-csg-info {include | exclude}
```

- no *user-csg-info*
- *user-location-info* {include | exclude}
- no *user-location-info*
- *user-location-info-time* {include | exclude}
- no *user-location-info-time*
- *uwan-user-location-information* {include | exclude}
- no *uwan-user-location-information*
- *uwan-user-location-information-r16* {include | exclude}
- no *uwan-user-location-information-r16*

6.1.1.8 CHF charging profile command hierarchy

configure

- mobile-gateway
 - *profile*
 - *charging*
 - [no] *chf-profile* *charging-profile-id*
 - *activity-threshold* *activity-threshold*
 - no *activity-threshold*
 - *ccfh-profile* *profile-id*
 - no *ccfh-profile*
 - *custom-online-offline-reporting* [enable | disable]
 - no *custom-online-offline-reporting*
 - *default-rg-list*
 - [no] *rating-group* *rating-group*
 - *description* *description-string*
 - no *description*
 - *implicit-vol-threshold* *implicit-vol-threshold* *reason* *reason*
 - no *implicit-vol-threshold*
 - *ran-secondary-rat-usage-reports* {include | exclude}
 - no *ran-secondary-rat-usage-reports*
 - [no] *rating-group* *rating-group*
 - *activity-threshold* [activity-threshold] | [inherit]
 - no *activity-threshold*
 - *requested-unit-total-volume* *requested-unit-total-volume*
 - [no] *requested-unit-total-volume*
 - *uuc-options-no-quota-initial* [total] [uplink] [downlink] [time]
- [suppress] [inherit]
 - no *uuc-options-no-quota-initial*
 - *uuc-options-no-quota-mid-session* [suppress] [do-not-suppress] [inherit]
 - no *uuc-options-no-quota-mid-session*
 - *uuc-options-with-quota* [none] [total] [uplink] [downlink] [time]
- [inherit]
 - no *uuc-options-with-quota*

- [no] *reauth-blocked-rg*
- [no] *request-quota-before-traffic*
- [no] *requested-unit-for-all-triggers*
- *requested-unit-total-volume* *requested-unit-total-volume*
- [no] *requested-unit-total-volume*
- *sbi-srvc-operation*
 - *retry-count* *value*
 - no *retry-count*
 - *transaction-timer* *value*
 - no *transaction-timer*
- *smf-attributes-profile* *profile-id*
- no *smf-attributes-profile*
- [no] *smf-generate-charging-data-ref*
- *supported-features*
 - [no] *support-2g3g*
- *supported-vendor* *custom-3*
- no *supported-vendor*
- *unit-count-inactivity-timer* *timer-value*
- no *unit-count-inactivity-timer*
- *uuc-options-no-quota-initial* [total] [uplink] [downlink] [time] [suppress]
- no *uuc-options-no-quota-initial*
- *uuc-options-no-quota-mid-session* [suppress] [do-not-suppress]
- no *uuc-options-no-quota-mid-session*
- *uuc-options-with-quota* [none] [total] [uplink] [downlink] [time]
- no *uuc-options-with-quota*
- [no] *uuc-reporting-enh*

6.1.1.9 ePDG charging profile command hierarchy

configure

- mobile-gateway
 - *profile*
 - *charging*
 - [no] *epdg-charging* *charging-profile-id*
 - *description* *description-string*
 - *off-line*
 - *charging-info-trigger*
 - [no] *qos-change*
 - *tariff-change-list* *prof-name*
 - no *tariff-change-list*
 - *tariff-time* *start time* *end time*
 - [no] *tariff-time-change*
 - [no] *uwan-loc-change*
 - [no] *enable-offline*

- [no] *exclude-zero-vol-cdr*
- [no] *exclude-zero-vol-complete-cdr*
- [no] *partial-record-closure-trigger*
 - *max-change-condition* *max-change-condition*
 - no *max-change-condition*
 - *min-data-threshold* *min-data-threshold*
 - no *min-data-threshold*
 - *time-limit* *time-limit*
 - no *time-limit*
 - *volume-limit* *volume-limit*
 - no *volume-limit*
- *pri-cdf-dia-peer* *profile-name*
- no *pri-cdf-dia-peer*
- *sec-cdf-dia-peer* *profile-name*
- no *sec-cdf-dia-peer*

6.1.1.10 FBC event triggers charging profile command hierarchy

configure

- mobile-gateway
 - *profile*
 - *charging*
 - [no] *event-triggers-fbc* *profile-id*
 - *3gpp-ps-data-off-status-change* [category {immediate | deferred}]
 - no *3gpp-ps-data-off-status-change*
 - *addition-of-upf* [category {immediate | deferred}]
 - no *addition-of-upf*
 - *default-qht* *default-qht*
 - no *default-qht*
 - *description* *description-string*
 - no *description*
 - *handover-cancel* [category {immediate | deferred}]
 - no *handover-cancel*
 - *handover-complete* [category {immediate | deferred}]
 - no *handover-complete*
 - *handover-start* [category {immediate | deferred}]
 - no *handover-start*
 - *insertion-of-ismf* [category {immediate | deferred}]
 - no *insertion-of-ismf*
 - *max-change-condition* *max-change-condition*
 - no *max-change-condition*
 - *plmn-change* [category {immediate | deferred}]
 - no *plmn-change*
 - *qos-change* [category {immediate | deferred}]

```

- no qos-change
- rat-type-change [category {immediate | deferred}]
- no rat-type-change
- reallocation-of-ismf [category {immediate | deferred}]
- no reallocation-of-ismf
- removal-of-ismf [category {immediate | deferred}]
- no removal-of-ismf
- removal-of-upf [category {immediate | deferred}]
- no removal-of-upf
- serving-node-change [category {immediate | deferred}]
- no serving-node-change
- serving-node-change-limit serving-node-change-limit
- no serving-node-change-limit
- serving-node-type {r15 | r16}
- no serving-node-type
- session-ambr-change [category {immediate | deferred}]
- no session-ambr-change
- tariff-change-list list-name
- no tariff-change-list
- time-limit time-limit
- no time-limit
- time-limit-eutran time-limit-eutran
- no time-limit-eutran
- time-limit-nr time-limit-nr
- no time-limit-nr
- time-limit-per-rating-grp time-limit-per-rating-grp [category {immediate |
deferred}]
- no time-limit-per-rating-grp
- trigger-override-option {local | chf | local-and-chf}
- no trigger-override-option
- ue-in-pra-change [category {immediate | deferred}]
- no ue-in-pra-change
- ue-time-zone-change [category {immediate | deferred}]
- no ue-time-zone-change
- user-loc-change [category {immediate | deferred}]
- no user-loc-change
- volume-limit volume-limit
- no volume-limit
- volume-limit-eutran volume-limit-eutran
- no volume-limit-eutran
- volume-limit-nr volume-limit-nr
- no volume-limit-nr
- volume-limit-per-rating-grp vol-limit-per-rating-grp [category {immediate |
deferred}]
- no volume-limit-per-rating-grp

```

6.1.1.11 QBC event triggers charging profile command hierarchy

```

configure
- mobile-gateway
- profile
- charging
- [no] event-triggers-qbc profile-id
- addition-of-upf [category {immediate | deferred}]
- no addition-of-upf
- description description-string
- no description
- handover-cancel [category {immediate | deferred}]
- no handover-cancel
- handover-complete [category {immediate | deferred}]
- no handover-complete
- handover-start [category {immediate | deferred}]
- no handover-start
- max-change-condition max-change-condition
- no max-change-condition
- plmn-change [category {immediate | deferred}]
- no plmn-change
- qos-change [category {immediate | deferred}]
- no qos-change
- removal-of-upf [category {immediate | deferred}]
- no removal-of-upf
- serving-node-change [category {immediate | deferred}]
- no serving-node-change
- serving-node-change-limit serving-node-change-limit
- no serving-node-change-limit
- session-ambr-change [category {immediate | deferred}]
- no session-ambr-change
- tariff-change-list list-name
- no tariff-change-list
- time-limit time-limit [category {immediate | deferred}]
- no time-limit
- time-limit-per-qos-flow time-limit-per-qos-flow [category {immediate |
deferred}]
- no time-limit-per-qos-flow
- trigger-override-option {local | chf | local-and-chf}
- no trigger-override-option
- ue-time-zone-change [category {immediate | deferred}]
- no ue-time-zone-change
- user-loc-change [category {immediate | deferred}]
- no user-loc-change

```


- *volume-limit* *volume-limit* [category {immediate | deferred}]
- no *volume-limit*
- *volume-limit-per-qos-flow* *volume-limit-per-qos-flow* [category {immediate | deferred}]
- no *volume-limit-per-qos-flow*

6.1.1.12 Gy AVP options charging profile

configure

- mobile-gateway
 - *profile*
 - *charging*
 - [no] *gy-avp-options* *gy-avp-options-profile-id*
 - *description* *description-string*
 - no *description*
 - [no] *exclude-3gpp-charging-characteristics*
 - [no] *exclude-3gpp-charging-id*
 - [no] *exclude-3gpp-ggsn-mcc-mnc*
 - [no] *exclude-3gpp-imsi-mcc-mnc*
 - [no] *exclude-3gpp-ms-timezone*
 - [no] *exclude-3gpp-nsapi*
 - [no] *exclude-3gpp-pdp-type*
 - [no] *exclude-3gpp-rat-type*
 - [no] *exclude-3gpp-selection-mode*
 - [no] *exclude-3gpp-session-stop-indicator*
 - [no] *exclude-3gpp-sgsn-mcc-mnc*
 - [no] *exclude-3gpp-user-location-info*
 - [no] *exclude-3gpp2-bsid*
 - [no] *exclude-cc-correlation-id*
 - [no] *exclude-charging-characteristics-selection-mode*
 - [no] *exclude-dynamic-addr-flag*
 - [no] *exclude-dynamic-address-flag-extension*
 - [no] *exclude-event-timestamp*
 - [no] *exclude-ggsn-address*
 - [no] *exclude-imsi-unauthenticated-flag*
 - [no] *exclude-node-id*
 - [no] *exclude-origin-state-id*
 - [no] *exclude-pdn-connection-charging-id*
 - [no] *exclude-pdp-address*
 - [no] *exclude-pdp-address-extension*
 - [no] *exclude-pdp-address-prefix-length*
 - [no] *exclude-pdp-context-type*
 - [no] *exclude-service-id*
 - [no] *exclude-serving-node-type*

- [no] *exclude-sgsn-address*
- [no] *exclude-subscription-id-imsi*
- [no] *exclude-subscription-id-msisdn*
- [no] *exclude-subscription-id-nai*
- [no] *exclude-user-equipment-info*
- [no] *exclude-user-name*
- [no] *include-3gpp-gprs-qos-negotiated-profile*
- [no] *include-3gpp-user-location-info-5g*
- [no] *include-cg-address*
- [no] *include-charging-rule-base-name*
- [no] *include-dualstack-ggsn-address-ipv6*
- [no] *include-dualstack-sgsn-address-ipv6*
- [no] *include-enhanced-diagnostics*
- [no] *include-extended-bw-nr*
- [no] *include-mscc-qos-information*
- [no] *include-pra-info*
- [no] *include-qos-information*
- [no] *include-rg-in-raa*
- [no] *include-twan-user-location-info*
- [no] *include-twan-user-location-info-r16*
- [no] *include-user-csg-info*
- [no] *include-user-equipment-info-bcd-encode*
- [no] *include-user-location-info-time*
- [no] *include-user-name-msisdn-at-apn*
- [no] *include-user-name-raa*
- [no] *include-uan-user-location-information*
- [no] *include-uan-user-location-information-r16*

6.1.1.13 PGW charging profile command hierarchy

configure

- mobile-gateway
 - *profile*
 - *charging*
 - [no] *pgw-charging charging-profile-id*
 - [no] *accounting-priority-profile profile-name*
 - *description description-string*
 - no *description*
 - *faster-poll-remaining-quota-trigger value*
 - no *faster-poll-remaining-quota-trigger*
 - [no] *force-sgw-cdr-in-combo-gw*
 - *nchf-charging {enable | disable}*
 - no *nchf-charging*
 - *off-line*

- *charging-info-trigger*
 - [no] *apn-rate-control-change*
 - [no] *cgi-sai-change*
 - [no] *change-csg-cell*
 - [no] *change-csg-subscribed-hybrid-cell*
 - [no] *change-ue-presence-pra*
 - [no] *ecgi-change*
 - [no] *fh-close-cdr-containers*
 - [no] *fh-close-cdr-exclude-dcca-result-code-4010*
 - [no] *fh-close-cdr-exclude-dcca-result-code-4012*
 - [no] *fh-close-cdr-exclude-dcca-result-code-5005*
 - [no] *fh-close-cdr-exclude-dcca-result-code-5030*
 - [no] *gy-final*
 - [no] *gy-forced-reauthorization*
 - [no] *gy-qht*
 - [no] *gy-quota-exhausted*
 - [no] *gy-rating-condition-change*
 - [no] *gy-tariff-time-change*
 - [no] *gy-threshold-vqt*
 - [no] *gy-validity-time*
 - [no] *plmn-rate-control-change*
 - [no] *qos-change*
 - [no] *rai-change*
 - [no] *serving-node-change-limit*
 - [no] *tai-change*
 - *tariff-change-list* *prof-name*
 - no *tariff-change-list*
 - *tariff-time* *start time end time*
 - [no] *tariff-time-change*
 - *time-limit-per-rating-grp* *time-limit-per-rating-grp*
 - no *time-limit-per-rating-grp*
 - *user-loc-change* [*cc-user-loc-change*]
 - no *user-loc-change*
 - [no] *uwan-loc-change*
 - *volume-limit-per-rating-grp* *volume-limit-per-rating-grp*
 - no *volume-limit-per-rating-grp*
- [no] *enable-offline*
- [no] *exclude-zero-vol-cdr*
- [no] *exclude-zero-vol-complete-cdr*
- [no] *include-bearer-charging*
- *partial-record-closure-trigger*
 - *max-change-condition* *max-change-condition*
 - no *max-change-condition*
 - *min-data-threshold* *min-data-threshold*
 - no *min-data-threshold*

- [no] *mo-exception-data-receipt*
- [no] *ms-time-zone-change*
- [no] *plmn-change*
- [no] *rat-change*
- *serving-node-change-limit* *serving-node-change-limit*
- *time-limit* *time-limit*
- no *time-limit*
- *time-limit-eutran* *time-limit-eutran*
- no *time-limit-eutran*
- *time-limit-utran* *time-limit-utran*
- no *time-limit-utran*
- *time-limit-eutran-nb-iot* *time-limit-eutran-nb-iot*
- no *time-limit-eutran-nb-iot*
- *time-limit-lte-m* *time-limit*
- no *time-limit-lte-m*
- *volume-limit* *volume-limit*
- no *volume-limit*
- *volume-limit-eutran* *volume-limit-eutran*
- no *volume-limit-eutran*
- *volume-limit-utran* *volume-limit-utran*
- no *volume-limit-utran*
- *volume-limit-eutran-nb-iot* *volume-limit-eutran-nb-iot*
- no *volume-limit-eutran-nb-iot*
- *volume-limit-lte-m* *volume-limit*
- no *volume-limit-lte-m*
- *pri-cdf-dia-peer* *profile-name*
- no *pri-cdf-dia-peer*
- *sd-container*
 - [no] *include-losd-with-sec-rat-usage-report*
 - [no] *include-zero-vol-sdc*
 - [no] *qos-info-first-sdc-of-cdr*
- *sec-cdf-dia-peer* *profile-name*
- no *sec-cdf-dia-peer*
- *user-location-at-cdr-generation-time* {enable | disable}
- no *user-location-at-cdr-generation-time*
- [no] *on-line*
 - [no] *dcca-profile* *profile-name*
 - *diameter-peer-group-list* *list-name*
 - no *diameter-peer-group-list*
 - [no] *enable-online*
 - [no] *pri-ocs-dia-peer* *profile*
 - [no] *sec-ocs-dia-peer* *profile*
 - *selection-type* {imsi | msisdn} {prefix | full-id | suffix}
 - no *selection-type*

```

    - user-range user-range-name {start-id [end-id] | suffix} {diameter-peer-
group-list list-name}
    - no user-range user-range-name
    - pcrf-selection
    - dynamic-pcc {enable | disable | inherit}
    - no dynamic-pcc
    - pcc-disable-per-rat [UTRAN] [GERAN] [WLAN] [GAN] [HSPA-Evolution]
[EUTRAN] [Virtual] [EUTRAN-NB-IoT]
    - no pcc-disable-per-rat
    - pcc-disable-per-rat-timer pcc-disable-per-rat-timer
    - no pcc-disable-per-rat-timer
    - radius-accounting
    - [no] cgi-sai-change
    - [no] ecgi-change
    - [no] peer-change-trigger
    - [no] qos-change-trigger
    - [no] rai-change
    - [no] rat-change-trigger
    - [no] tai-change
    - [no] user-loc-change
    - volume-limit-trigger value
    - no volume-limit-trigger
    - [no] report-serving-node-supplied-cc
    - smf-legacy-nr-rattype {reserved | utran | geran | wlan | gan | hspa |
eutran | ehrpd | hrpd | one-xrtt | umb | virtual | eutran-nb-iot | nr}
    - no smf-legacy-nr-rattype

```

6.1.1.14 SGW charging profile command hierarchy

```

configure
- mobile-gateway
- profile
- charging
- [no] sgw-charging charging-profile-id
- data-volume-count-time start-time time end-time time
- description description-string
- no description
- nchf-charging {enable | disable}
- no nchf-charging
- off-line
- charging-info-trigger
- [no] cgi-sai-change
- [no] change-csg-cell
- [no] change-csg-subscribed-hybrid-cell

```

```

    - [no] change-in-user-plane
    - [no] change-ue-presence-pra
    - [no] ecgi-change
    - [no] plmn-rate-control-change
    - [no] qos-change
    - [no] rai-change
    - [no] tai-change
    - tariff-change-list prof-name
    - no tariff-change-list
    - tariff-time start time end time
    - [no] tariff-time-change
    - user-loc-change [cc-user-loc-change]
    - no user-loc-change
  - [no] enable-offline
  - [no] exclude-zero-vol-cdr
  - [no] exclude-zero-vol-complete-cdr
  - partial-record-closure-trigger
    - max-change-condition max-change-condition
    - no max-change-condition
    - min-data-threshold min-data-threshold
    - no min-data-threshold
    - [no] mo-exception-data-receipt
    - [no] ms-time-zone-change
    - [no] plmn-change
    - [no] rat-change
    - serving-node-change-limit serving-node-change-limit
    - no serving-node-change-limit
    - volume-limit volume-limit
    - no volume-limit
  - pri-cdf-dia-peer volume-limit
  - no pri-cdf-dia-peer
  - sec-cdf-dia-peer profile-name
  - no sec-cdf-dia-peer
  - [no] sgw-change-cause-record-closure
  - user-location-at-cdr-generation-time {enable | disable}
  - no user-location-at-cdr-generation-time
- [no] report-serving-node-supplied-cc
- smf-legacy-nr-ratype {reserved | utran | geran | wlan | gan | hspa |
eutran | ehrpd | hrpd | one-xrtt | umb | virtual | eutran-nb-iot | nr}
- no smf-legacy-nr-ratype

```

6.1.1.15 SMF charging profile command hierarchy

configure

- mobile-gateway
 - *profile*
 - *charging*
 - [no] *smf-charging charging-profile-id*
 - *chf-selection*
 - *chf-profile charging-profile-id*
 - no *chf-profile*
 - *fail-open*
 - *chf-profile charging-profile-id*
 - no *chf-profile*
 - *nf-id-list list-name*
 - no *nf-id-list*
 - [no] *nf-id-list list-name*
 - no *nf-id-list*
 - *default-charging-method*
 - *description description-string*
 - no *description*
 - *event-triggers-fbc profile-id*
 - no *event-triggers-fbc*
 - *event-triggers-qbc profile-id*
 - no *event-triggers-qbc*
 - *qbc-charging {enable | disable}*
 - no *qbc-charging*
 - *quota-management*
 - *final-unit-indication*
 - [no] *redirect-append-url*
 - *redirect-filter-id base-policy-name*
 - no *redirect-filter-id*
 - [no] *redirection-start-send-rr-final*
 - [no] *redirection-start-send-rr-final-enh*
 - *traffic-handling-options*
 - *forced-re-authorization {allow | discard}*
 - no *forced-re-authorization*
 - *quota-exhausted-no-threshold {allow | discard}*
 - no *quota-exhausted-no-threshold*
 - *quota-exhausted-threshold-active {allow | discard}*
 - no *quota-exhausted-threshold-active*
 - *quota-unavailable {allow | discard}*
 - no *quota-unavailable*
 - *rating-condition-change {allow | discard}*
 - no *rating-condition-change*
 - *validity-time-expired {allow | discard}*
 - no *validity-time-expired*
 - *radius-accounting*
 - [no] *ecgi-change*

- [no] *peer-change*
- [no] *qos-change*
- [no] *rat-change*
- [no] *tai-change*
- [no] *user-loc-change*
- *volume-limit* *value*
- no *volume-limit*
- [no] *report-serving-node-supplied-cc*

6.1.1.16 TDF charging profile command hierarchy

configure

- mobile-gateway
 - *profile*
 - *charging*
 - [no] *tdf-charging charging-profile-id*
 - *description* *description-string*
 - no *description*
 - *off-line*
 - *charging-info-trigger*
 - [no] *cgi-sai-change*
 - [no] *change-ue-presence-pra*
 - [no] *ecgi-change*
 - [no] *rai-change*
 - [no] *serving-node-change*
 - [no] *tai-change*
 - *tariff-change-list* *list-name*
 - no *tariff-change-list*
 - *tariff-time* *start time* *end time*
 - [no] *tariff-time-change*
 - *user-loc-change* [*cc-user-loc-change*]
 - no *user-loc-change*
 - *enable-offline* [*session-level* | *fbc* | *both*]
 - [no] *exclude-zero-vol-cdr*
 - [no] *exclude-zero-vol-complete-cdr*
 - *partial-record-closure-trigger*
 - *max-change-condition* *max-change-condition*
 - no *max-change-condition*
 - *min-data-threshold* *min-data-threshold*
 - no *min-data-threshold*
 - [no] *ms-time-zone-change*
 - [no] *plmn-change*
 - [no] *rat-change*
 - *time-limit* *time-limit*


```

    - no time-limit
    - volume-limit volume-limit
    - no volume-limit
    - pri-cdf-dia-peer profile-name
    - no pri-cdf-dia-peer
    - sec-cdf-dia-peer profile-name
    - no sec-cdf-dia-peer
  - on-line
    - dcca-profile profile-name
    - no dcca-profile
    - diameter-peer-group-list list-name
    - no diameter-peer-group-list
    - enable-online [session-level | rg-level]
    - no enable-online
    - pri-ocs-dia-peer profile-name
    - no pri-ocs-dia-peer
    - sec-ocs-dia-peer profile-name
    - no sec-ocs-dia-peer
    - selection-type {imsi | msisdn} {prefix | full-id}
    - no selection-type
    - user-range user-range-name start-id [end-id] {diameter-peer-group-
list list-name}
    - no user-range user-range-name

```

6.1.1.17 Cloud database profile command hierarchy

```

configure
- mobile-gateway
- profile
  - [no] cloud-db profile-name
  - description description-string
  - no description
  - [no] server ip-addr | fqdn [port port]
  - [no] shutdown

```

6.1.1.18 DCCA profile command hierarchy

```

configure
- mobile-gateway
- profile
  - [no] dcca profile-name
  - activity-threshold activity-threshold

```

- no *activity-threshold*
- [no] *append-url*
- *application-tx-timer* {value | timeout-msec value}
- no *application-tx-timer*
- *cc-failover-handling* {terminate | continue | retry-and-terminate}
- no *cc-failover-handling*
- [no] *cca-init-no-mscc-answer-ignore*
- *ccfh-fh-action* {terminate | re-start}
- no *ccfh-fh-action*
- *ccfh-profile* profile-id
- no *ccfh-profile*
- [no] *ccr-initial-on-first-online-rg*
- [no] *ccr-initial-on-first-pkt*
- *ccr-mscc-limit* num-mscc
- no *ccr-mscc-limit*
- [no] *ccrt-no-mscc*
- [no] *command-level-fui*
- *default-qht* minutes
- no *default-qht*
- *default-rg-list*
 - [no] *rating-group* rating-group
- *description* description-string
- no *description*
- *destination-realm* name
- no *destination-realm*
- [no] *diam-sess-end-serialize-ccr*
- [no] *disable-cc-session-failover*
- [no] *disable-first-pkt-behavior*
- [no] *disable-reauth-ccru-no-mscc*
- *event-triggers*
 - [no] *change-ue-presence-pra*
 - [no] *qos-apn-ambr-change*
 - [no] *qos-change*
 - [no] *qos-max-bitrate-dl-change*
 - [no] *qos-max-bitrate-ul-change*
 - [no] *rat-change*
 - [no] *serving-node-change*
 - *trigger-override-option* {local | ocs | local-and-ocs}
 - no *trigger-override-option*
 - [no] *user-loc-change*
 - [no] *user-loc-change-cellid*
 - [no] *user-loc-change-ecgi*
 - [no] *user-loc-change-lac*
 - [no] *user-loc-change-rac*
 - [no] *user-loc-change-tac*

```

    - [no] uwan-loc-change
  - failure-handling
    - [no] mscc-result-code mscc-result-code-value
      - service-delay-timer minutes
      - no service-delay-timer
    - result-code result-code-value {terminate | continue | retry-and-terminate}
    - no result-code result-code-value
    - result-code-3xxx {terminate | continue | retry-and-terminate}
    - no result-code-3xxx
    - result-code-4xxx {terminate | continue | retry-and-terminate}
    - no result-code-4xxx
    - result-code-5xxx {terminate | continue | retry-and-terminate}
    - no result-code-5xxx
  - fh-session-continue-timer minutes
  - no fh-session-continue-timer
  - fh-volume-limit volume-limit
  - no fh-volume-limit
  - fui-validity-timer minutes [precedence]
  - no fui-validity-timer
  - gy-avp-options-profile gy-avp-options-profile-id
  - no gy-avp-options-profile
  - implicit-vol-threshold percent reason [threshold | exhaustion]
  - no implicit-vol-threshold
  - include-avp
    - [no] 3gpp-gprs-qos-neg-prof
    - [no] exclude-service-id
    - [no] qos-information
  - [no] include-rpt-reason-on-forced-reauth
  - offline-interwork
    - [no] ps-fci-fh
    - [no] ps-fci-fh-container
  - one-time-redirect welcome-page-rg rg-value alert-page-rg rg-value
    - no one-time-redirect
  - [no] pre-emptive-quota
  - [no] queue-rar-pending-ccau
  - rating-group rating-group-id
  - no rating-group
    - activity-threshold [activity-threshold | inherit]
    - no activity-threshold
    - usu-options-no-quota-initial [cc-total] [cc-input] [cc-output] [cc-time]
    - [suppress] [inherit]
    - no usu-options-no-quota-initial
    - usu-options-no-quota-mid-session [suppress | do-not-suppress] [inherit]
    - no usu-options-no-quota-mid-session

```

```

    - usu-options-with-quota [none] [cc-total] [cc-input] [cc-output] [cc-time]
[inherit] [override-ocs]
    - no usu-options-with-quota
    - [no] rating-group-in-ch-rule-report
    - [no] reauth-blocked-rg
    - redirect-filter-id base-policy-name
    - no redirect-filter-id
    - [no] redirection-start-send-rr-final
    - retry-count retry-count
    - no retry-count
    - rsu-total-octets value
    - no rsu-total-octets
    - [no] serialize-ccr
    - service-context-id string
    - no service-context-id
    - [no] service-id-level-credit-control
    - [no] service-type {rtr | uc | pp | all}
        - [no] ccfh-profile profile-id
    - session-restore-retry-timer retry-timer
    - no session-restore-retry-timer
    - supported-version {3gpp | version2}
    - no supported-version
    - [no] suppress-multiple-ooc-notifications-per-rg
    - traffic-handling-options
        - forced-re-authorization {allow | discard}
        - no forced-re-authorization
        - quota-exhausted-no-threshold {allow | discard}
        - no quota-exhausted-no-threshold
        - quota-exhausted-threshold-active {allow | discard}
        - no quota-exhausted-threshold-active
        - quota-unavailable {allow | discard | discard-on-ftp}
        - no quota-unavailable
        - rating-condition-change {allow | discard}
        - no rating-condition-change
        - validity-time-expired {allow | discard}
        - no validity-time-expired
    - url-encryption {aes-cbc-128} passphrase string
    - no url-encryption
    - usu-options-no-quota-initial [cc-total] [cc-input] [cc-output] [cc-time]
[suppress]
    - no usu-options-no-quota-initial
    - usu-options-no-quota-mid-session [suppress | do-not-suppress]
    - no usu-options-no-quota-mid-session
    - usu-options-with-quota [none] [cc-total] [cc-input] [cc-output] [cc-time]
[override-ocs]

```

- no *usu-options-with-quota*
- *usu-reporting-enh* [discard-overage]
- no *usu-reporting-enh*
- [no] *validation-checks*
 - [no] *ignore-invalid-thresholds*
 - [no] *ignore-unsupported-triggers*
- [no] *validity-timer-enh*
- [no] *vol-threshold-reauth-enh*
- [no] *wait-for-first-packet-on-fui-t-success*
- [no] *wait-for-first-packet-on-no-gsu*

6.1.1.19 Dedicated bearer idle timeout profile command hierarchy

configure

- mobile-gateway
 - *profile*
 - [no] *dedicated-bearer-idle-timeout-profile profile-name*
 - *qci qci* timeout seconds
 - no *qci qci*

6.1.1.20 DHCP server group profile command hierarchy

configure

- mobile-gateway
 - *profile*
 - [no] *dhcp-server-group server-group-name*
 - *client-identifier-msisdn-encoded* [with-apn]
 - no *client-identifier-msisdn-encoded*
 - *deadtime* seconds
 - no *deadtime*
 - *description* description-string
 - no *description*
 - *interface* [router-instance router-instance] interface-name
 - no *interface*
 - *peer* ipv4 | ipv6
 - no *peer*
 - *priority* value
 - no *priority*
 - *vso-server-pool-name* server-pool-name
 - no *vso-server-pool-name*
 - *rebind-time-percent* value
 - no *rebind-time-percent*

- *recommend-lease-time* *seconds*
- *no recommend-lease-time*
- *renew-time-percentage* *value*
- *no renew-time-percentage*
- *retry-count* *number*
- *no retry-count*
- *retry-timeout* *seconds*
- *no retry-timeout*
- *via-up-function* *apn apn-name* [*up-peer-ip-address ip-address* | *ipv6-address*]
- *no via-up-function*

6.1.1.21 DHCPv4 server profile command hierarchy

configure

- *mobile-gateway*
 - *profile*
 - [no] *dhcpv4-server profile-name*
 - *description* *description-string*
 - *no description*
 - *hold-time* *mins minutes secs seconds*
 - *no hold-time*
 - [no] *ignore-rapid-commit*
 - *lease-time* *hrs hours mins minutes secs seconds*
 - *no lease-time*

6.1.1.22 Diameter profile command hierarchy

configure

- *mobile-gateway*
 - *profile*
 - [no] *diameter profile-name*
 - *connection-timer* *value*
 - *no connection-timer*
 - *description* *description-string*
 - *no description*
 - *dns-refresh-interval* *value*
 - *no dns-refresh-interval*
 - *dpr-timeout* *value*
 - *no dpr-timeout*
 - *fail-num-retries* *value*
 - *no fail-num-retries*
 - *fail-retry-interval* *value*

- no *fail-retry-interval*
- *ip-dscp* value
- no *ip-dscp*
- *ip-ttl* value
- no *ip-ttl*
- *perm-fail-retry-timeout* value
- no *perm-fail-retry-timeout*
- *transaction-timer* value
- no *transaction-timer*
- *watchdog-timer* value
- no *watchdog-timer*

6.1.1.23 Diameter error handling profile command hierarchy

configure

- mobile-gateway
 - *profile*
 - *diameter-error-mapping-profile* profile-name
 - no *diameter-error-mapping-profile*
 - *fh-tx-timeout* gtp-error-code gtp-value
 - no *fh-tx-timeout*
 - *result-code* result-code-value gtp-error-code gtp-value
 - no *result-code* result-code-value
 - *result-code-3xxx* gtp-error-code gtp-value
 - no *result-code-3xxx*
 - *result-code-4xxx* gtp-error-code gtp-value
 - no *result-code-4xxx*
 - *result-code-5xxx* gtp-error-code gtp-value
 - *result-code-5xxx*

6.1.1.24 Diameter peer profile command hierarchy

configure

- mobile-gateway
 - *profile*
 - [no] *diameter-peer* profile-name
 - *application-type* {gx | rf | gy | s6b | swm | sd | sta}
 - no *application-type*
 - *custom-vendor-id* vendor-id
 - no *custom-vendor-id*
 - *description* description-string
 - no *description*

```

– [no] dest-host-in-initial-msg
– dest-host-table-threshold threshold-percent
– no dest-host-table-threshold
– destination-realm name
– no destination-realm
– diameter-profile profile-name
– no diameter-profile
– [no] disable-cea-auth-app-id-check
– [no] disable-learnt-dest-realm
– [no] exclude-dest-host-after-switchover
– interface [router router-instance] interface-name
– interface
– [no] last-known-dest-realm-after-switchover
– [no] load-balance
– [no] multi-homing
– interface [router router-instance] interface-name
– no interface
– origin-realm realm
– no origin-realm
– [no] peer ip-addr | fqdn [port port]
– [no] enable-dra
– sctp-alternate-address ip-address
– [no] sctp-alternate-address
– [no] shutdown
– supported-host name host-index
– no supported-host host-index
– [no] suspend
– [no] reconnect-with-new-tcp-port
– sctp-profile profile-name
– no sctp-profile
– sig-rate-limit-profile profile-name
– no sig-rate-limit-profile
– [no] switchover-source-port-consistent
– transport {tcp | sctp}
– no transport
– [no] validate-supported-host
– [no] diameter-peer-group-list list-name
– peer-group peer-group-id primary-diameter-peer profile-name [secondary-
diameter-peer profile-name] [priority priority]
– no peer-group peer-group-id

```

6.1.1.25 DNS profile command hierarchy

configure

- mobile-gateway
 - *profile*
 - [no] *dns-error-mapping-profile profile-name*
 - *backoff-timer* [timer-value timer-value] [timer-unit timer-unit]
 - no *backoff-timer*
 - *description description-string*
 - no *description*
 - *swu-notify swu-notify-value* [description description-string]
 - no *swu-notify*
 - *swu-send-info* [notify]
 - no *swu-send-info*
 - [no] *dns-profile profile-name*
 - *ip-dscp value*
 - no *ip-dscp*
 - *ip-ttl value*
 - no *ip-ttl*
 - *message-retransmit* [timeout seconds] [retry-count value]
 - no *message-retransmit*
 - *primary-dns ip-address*
 - no *primary-dns*
 - *secondary-dns ip-address*
 - no *secondary-dns*
 - *tertiary-dns ip-address*
 - no *tertiary-dns*

6.1.1.26 DNS redirect profile command hierarchy

configure

- mobile-gateway
 - *profile*
 - [no] *dns-redirect-server profile-name*
 - *primary-dns primary-dns* [secondary-dns secondary-dns]

6.1.1.27 Emergency profile command hierarchy

configure

- mobile-gateway
 - *profile*
 - [no] *emergency-profile profile-name*
 - [no] *allow-udm-sdm*
 - [no] *block-udm-uecm*
 - *default-5qi value*

- no *default-5qi*
- *default-ambr* *ul-rate* *ul-rate* *dl-rate* *dl-rate*
- no *default-ambr*
- *default-arp* *value*
- no *default-arp*
- *sbi-message-priority* *value*
- no *sbi-message-priority*

6.1.1.28 ePDG group profile command hierarchy

configure

- mobile-gateway
 - *profile*
 - [no] *epdg-group* *profile-name*
 - [no] *ip-address* *ipv4* | *ipv6*

6.1.1.29 Fa Ha peer list profile command hierarchy

configure

- mobile-gateway
 - *profile*
 - [no] *fa-ha-peer-list* *profile-name*
 - *description* *description-string*
 - no *description*
 - [no] *peer* *ip-address/prefix-length*
 - [no] *fa-ha-spi* *value*
 - *description* *description-string*
 - no *description*
 - *hash-algorithm* {*md5* | *hmac-md5* | *rfc2002-md5*}
 - no *hash-algorithm*
 - *secret* *secret*
 - no *secret*

6.1.1.30 FQDN group profile command hierarchy

configure

- mobile-gateway
 - *profile*
 - [no] *fqdn-group* *fqdn-group-name*
 - *dns-refresh-interval* *dns-refresh-interval*
 - no *dns-refresh-interval*

- *fqdn fqdn-name-id fqdn-name-string*
- *no fqdn fqdn-name-id*
- *ip-expiry-refresh-count ip-expiry-refresh-count*
- *no ip-expiry-refresh-count*

6.1.1.31 GTP profile command hierarchy

configure

- mobile-gateway
 - *profile*
 - [no] *gtp profile-name*
 - *description description-string*
 - *no description*
 - [no] *gtpv1-options*
 - [no] *exclude-apn-restrictions*
 - [no] *exclude-bcm*
 - [no] *exclude-chrg-gw-addr*
 - [no] *exclude-pco*
 - [no] *exclude-recovery*
 - [no] *include-common-flags*
 - [no] *include-nsapi*
 - *gtpv1-session-rejection-cause*
 - *apn-shutdown cause-code-value*
 - *no apn-shutdown*
 - *gw-suspend cause-code-value*
 - *no gw-suspend*
 - *sx-connection-failure cause-code-value*
 - *no sx-connection-failure*
 - *username-missing cause-code-value*
 - *no username-missing*
 - *gtpv2-node-features*
 - [no] *include-ciot*
 - [no] *gtpv2-options*
 - [no] *exclude-aaa-node-identifier*
 - [no] *exclude-apco*
 - [no] *exclude-change-reporting-action*
 - [no] *exclude-cno-selection-entity*
 - [no] *exclude-csg-reporting-action*
 - [no] *exclude-ddn-arp*
 - [no] *exclude-ddn-ftaid*
 - [no] *exclude-ddn-imsi*
 - [no] *exclude-lci*
 - [no] *exclude-maximum-wait-time*
 - [no] *exclude-node-features*

```

    - [no] exclude-oci
    - [no] exclude-origination-timestamp
    - [no] exclude-paging-priority
    - [no] exclude-pco
    - [no] exclude-ran-nas-cause
    - [no] exclude-signaling-priority
    - [no] exclude-twan-identifier
    - [no] exclude-twan-identifier-timestamp
    - [no] exclude-uli-timestamp
    - [no] exclude-up-selection-indication
    - [no] include-ap-mac-address
    - [no] include-mo-exception-data-counter
    - [no] include-serving-plmn-rate-control
    - non-3gpp-rat-type-pref {wlan | virtual}
    - no non-3gpp-rat-type-pref
- gtpv2-session-rejection-cause
  - apn-shutdown cause-code-value
  - no apn-shutdown
  - gw-suspend cause-code-value
  - no gw-suspend
  - sx-connection-failure cause-code-value
  - no sx-connection-failure
  - username-missing cause-code-value
  - no username-missing
- ip-dscp value
- no ip-dscp
- ip-ttl value
- no ip-ttl
- keep-alive [interval interval] [retry-count retry-count] [timeout timeout]
- no keep-alive
- message-retransmit [timeout timeout] [retry-count value] [timeout-
msec timeout-msec]
  - no message-retransmit

```

6.1.1.32 GTP prime group profile command hierarchy

```

configure
- mobile-gateway
- profile
  - [no] gtp-prime-group gtp-prime-group
  - deadtime deadtime
  - no deadtime
  - description description-string
  - no description

```

- [no] *duplicate-prevention*
- [no] *duplicate-prevention-storage*
- [no] *fragment-drt*
- [no] *gtp-prime-server ip-address [port port]*
 - *echo-interval echo-interval*
 - no *echo-interval*
 - [no] *enable-gtpv2*
 - *max-outstanding-requests max-requests*
 - no *max-outstanding-requests*
 - [no] *node-alive*
 - *path-protocol path-protocol*
 - no *path-protocol*
 - *priority priority*
 - no *priority*
 - *retries retries*
 - no *retries*
 - [no] *shutdown*
 - *timeout timeout*
 - no *timeout*
- *inactive-time inactive-time*
- no *inactive-time*
- *interface [router router-instance] interface-name*
- no *interface*
- *local-cdr-storage*
 - *file-params*
 - *cdr-file-compression {enable | disable}*
 - no *cdr-file-compression*
 - *closure-rec-num closure-rec-num*
 - no *closure-rec-num*
 - *closure-rec-num closure-size*
 - no *closure-rec-num*
 - *closure-time closure-time*
 - no *closure-time*
 - *extension extension*
 - no *extension*
 - *max-files-per-subdirectory max-files-per-subdirectory*
 - no *max-files-per-subdirectory*
 - *max-number-subdirectories max-number-subdirectories*
 - no *max-number-subdirectories*
 - *obsolete-time obsolete-time*
 - no *obsolete-time*
 - *private-info private-info*
 - no *private-info*
- *location location*
- *limit limit*

- no *limit*
- [no] *primary*
- *max-cdrs-per-pdu* *max-cdrs-per-pdu*
- no *max-cdrs-per-pdu*
- *mtu* *mtu-size*
- no *mtu*
- *queue-size* *queue-size*
- no *queue-size*
- [no] *redirection*
- [no] *round-robin-load-balance*
- [no] *shutdown*

6.1.1.33 Gx profile command hierarchy

configure

- mobile-gateway
 - *profile*
 - [no] *gx* *profile-name*
 - [no] *disable-ccr-i-on-csreq*
 - *event-triggers*
 - *tethering-start* *value*
 - no *tethering-start*
 - *tethering-stop* *value*
 - no *tethering-stop*
 - *fh-session-no-response* {*terminate* | *continue* | *ap-continue* | *terminate-next-attempt*} [*ap-action-profile* *profile-name*]
 - no *fh-session-no-response*
 - *fh-time-limit* *time-limit*
 - no *fh-time-limit*
 - *fh-volume-limit* *volume-limit*
 - no *fh-volume-limit*
 - *forced-triggers*
 - [no] *disable-ambr-modification-failure*
 - [no] *disable-default-eps-bearer-qos-change*
 - [no] *disable-default-eps-bearer-qos-modification-failure*
 - [no] *disable-qos-change*
 - [no] *disable-ue-local-ip-addr-change*
 - [no] *include-last-known-uli*
 - [no] *message* {*ccr-i* | *ccr-u* | *ccr-t*}
 - *gx-avp-options* *profile-number*
 - no *gx-avp-options*
 - *result-code* *result-code-value* {*terminate* | *continue* | *retry-and-terminate*}
 - *result-code* *result-code-value* {*ap-continue* | *retry-and-ap-continue*} [*ap-action-profile* *profile-name*]

```

    - no result-code result-code-value
    - result-code-3xxx {terminate | continue | retry-and-terminate}
    - result-code-3xxx result-code-value {ap-continue | retry-and-ap-continue}
[ap-action-profile profile-name]
    - no result-code-3xxx
    - result-code-4xxx {terminate | continue | retry-and-terminate}
    - result-code-4xxx result-code-value {ap-continue | retry-and-ap-continue}
[ap-action-profile profile-name]
    - no result-code-4xxx
    - result-code-5xxx {terminate | continue | retry-and-terminate}
    - result-code-5xxx result-code-value {ap-continue | retry-and-ap-continue}
[ap-action-profile profile-name]
    - no result-code-5xxx
    - retry-count value
    - no retry-count
    - transaction-timer {[value] | [timeout-msec value]}
    - no transaction-timer
    - pending-transaction-retry-count value
    - no pending-transaction-retry-count
    - pending-transaction-timer msec-value
    - no pending-transaction-timer
    - [no] qos-info-always-present
    - [no] report-serving-node-supplied-cc
    - [no] serialize-ccr
    - session-restore-retry-timer retry-timer
    - no session-restore-retry-timer
    - supported-features
      - [no] adc
      - [no] app-based-charging
      - [no] clear-m-bit
      - [no] cno-uli
      - [no] cond-policy-info-default-qos
      - [no] conditional-policy-info
      - [no] disable-rel-10
      - [no] disable-rel-9
      - [no] exclude-usage
      - [no] extended-bw-nr
      - [no] include-rat-when-received
      - [no] mission-critical-qcis
      - [no] multiple-pra
      - [no] netloc
      - [no] netloc-untrusted-wlan
      - [no] pcscf-restoration
      - [no] pending-transaction
      - [no] ran-nas-cause

```

- [no] *rule-bound-to-default-bearer*
- [no] *sgw-rest*
- [no] *sponsored-connectivity*
- [no] *time-based-usage-monitoring*
- [no] *trusted-wlan*
- [no] *user-equipment-info-extension*
- *usage-monitoring*
 - [no] *monitoring-key monitoring-key*
 - *activity-threshold threshold*
 - no *activity-threshold*
 - [no] *override-usage-report-value*
- *validation-checks*
 - [no] *raa-immediate*

6.1.1.34 HTTP/2 profile command hierarchy

configure

- mobile-gateway
 - *profile*
 - [no] *http2 profile-name*
 - *clt-tls-profile profile-name*
 - no *clt-tls-profile*
 - *connection-timer value*
 - no *connection-timer*
 - *fail-num-retries value*
 - no *fail-num-retries*
 - *fail-retry-interval value*
 - no *fail-retry-interval*
 - *header-table-size number*
 - no *header-table-size*
 - *init-window-size number*
 - no *init-window-size*
 - *ip-dscp value*
 - no *ip-dscp*
 - *max-frame-size number*
 - no *max-frame-size*
 - *max-header-list-size number*
 - no *max-header-list-size*
 - *ping-period number*
 - no *ping-period*
 - *request-timeout time*
 - no *request-timeout*
 - [no] *srv-disable-sni-chk*
 - *srv-max-streams-allowed number*

- *no srv-max-streams-allowed*
- *srv-stream-lifetime* *number*
- *no srv-stream-lifetime*
- *srv-tls-profile* *profile-name*
- *no srv-tls-profile*
- *strict-validation* [*dns-id*] [*uuid*] [*keyUsage*] [*extKeyUsage*] [*flags*]
- *no strict-validation*

6.1.1.35 IPSec profile command hierarchy

configure

- *mobile-gateway*
 - *profile*
 - [*no*] *ipsec-profile* *profile-name*
 - [*no*] *description* *string*
 - *esp-sa*
 - *lifetime* *seconds*
 - *no lifetime*
 - [*no*] *nat-traversal-dynamic-updates*
 - [*no*] *proposal-entry* *entry-id*
 - *dh-group* *Diffie-Hellman group*
 - *no dh-group*
 - *encryption* *encryption algorithm*
 - *no encryption*
 - [*no*] *extended-seq-num*
 - *integrity* *authentication algorithm*
 - *no integrity*
 - *rekey-margin* *percentage*
 - *no rekey-margin*
 - *rekeying* {*enable* | *disable*}
 - *no rekeying*
 - *replay-window* *replay-window*
 - *no replay-window*
 - [*no*] *ignore-ue-rekey-requests*
 - *ike-sa*
 - *apn-black-white-list* *type type* *apn-list list-name* [*error-code error-code*]
 - *no apn-black-white-list*
 - [*no*] *cert-entry* *cert-entry-id*
 - *epdg-cert* *epdg-cert-name*
 - *no epdg-cert*
 - *priv-key* *epdg-priv-key*
 - *no priv-key*
 - *send-chain* *number-of-chains*
 - *no send-chain*

```
– shutdown
– url-base-address url
– no url-base-address
– certreq-eap-only-prefer-certreq url
– no certreq-eap-only-prefer-certreq
– [no] device-identity
– function [enable | disable]
– no function
– [no] reject-if-validation-fails
– type {imeisv | imei}
– no type
– dpd-interval seconds
– no dpd-interval
– fragmentation [ipv4-mtu ipv4-mtu] [ipv6-mtu ipv6-mtu]
– no fragmentation
– global-half-open-sa-limit global-half-open-sa-limit
– no global-half-open-sa-limit
– [no] imsi-privacy-protection
– lifetime seconds
– no lifetime
– local-identity ip-addr | fqdn
– no local-identity
– message-retransmit [timeout seconds] [retry-count num-retries]
– no message-retransmit
– mobike [check-return-routability]
– no mobike
– [no] nat-traversal-dynamic-updates
– natt-mode {auto | force}
– no natt-mode
– peer-half-open-sa-limit peer-half-open-sa-limit
– no peer-half-open-sa-limit
– plmn-black-white-list
– backoff-timer [timer-value timer-value] [timer-unit timer-unit]
– no backoff-timer
– [no] enable-plmn-black-white-list
– error-code error-code
– no error-code
– idi-realm-list list-name
– no idi-realm-list
– plmn-list list-name
– no plmn-list
– type type
– no type
– [no] proposal-entry entry-id
– dh-group Diffie-Hellman-group
```

```

    - no dh-group
    - encryption encryption algorithm
    - no encryption
    - integrity authentication algorithm
    - no integrity
    - prf prf algorithm
    - no prf
  - redirect
    - always-on server-list ike-redirect-server-list-name [established-
sessions-redirect-rate rate-value] [redirect-established-emergency] [redirect-when-
idle]
    - no always-on
    - idi-realm idi-realm-list-name type-of-list-screening {white white-list-
redir-epdg {ipv4 | ipv6 | fqdn} ipv4 | ipv6 | fqdn | black}
    - no idi-realm
    - overload-redirect server-list ike-redirect-server-list-name [redirect-
established-emergency]
    - no overload-redirect
    - source-ip source-ip-list-name type-of-list-screening {white white-list-
redir-epdg {ipv4 | ipv6 | fqdn} ipv4 | ipv6 | fqdn | black}
    - no source-ip
    - rekey-margin percentage
    - no rekey-margin
    - rekeying {enable | disable}
    - no rekeying

```

6.1.1.36 List profile command hierarchy

```

configure
  - mobile-gateway
    - profile
      - list
        - [no] aaa-host-realm list-name
        - id id realm realm-string host host-string diameter-peer-group-list group-
list-name
        - no id id
        - [no] allowed-nf-domains-list list-name
        - [no] allowed-nf-domain string
        - [no] apn-list list-name
        - [no] apn apn-name
        - [no] arp-list arp-list-name
        - [no] arp arp-value [message-priority msg-priority] [gtp-message-
priority gtp-msg-priority]
        - [no] broadcast-plmn-list list-name

```

```

    - broadcast-plmn-item bplmn-id tai-slice-support-list slice-list-name plmn-
mcc mcc value plmn-mnc mnc value
    - no broadcast-plmn-item bplmn-id
    - [no] cell-id-list cell-id-list-name
    - [no] ncgi mcc mcc-value mnc mnc-value nci nci-value
    - [no] ncgi mcc mcc-value mnc mnc-value nci-range-start nci-range-start-value
nci-range-end nci-range-end-value
    - [no] fqdn-list fqdn-list-name
    - fqdn fqdn-name-id fqdn-name-string
    - no fqdn fqdn-name-id
    - [no] idi-realm-list list-name
    - idi-realm idi-realm-id realm-name [redirected-epdg {ipv4 | ipv6 | fqdn}]
[ipv4 | ipv6 | fqdn]
    - no idi-realm idi-realm-id
    - [no] ike-redirect-server-list list-name
    - server [1 to 20] redirected-epdg {ipv4 | ipv6 | fqdn} ipv4 | ipv6 | fqdn
    - no server [1 to 20]
    - [no] ike-redirect-source-ip-list list-name
    - ip [1 to 512] address address redirected-epdg {ipv4 | ipv6 | fqdn} ipv4 |
ipv6 | fqdn
    - ip [1 to 512] start-address start-ip-address end-address ip-address
redirected-epdg {ipv4 | ipv6 | fqdn} ipv4 | ipv6 | fqdn
    - no ip [1 to 512]
    - [no] ip-address-list list-name
    - [no] address ip-address
    - [no] nf-id-list list-name
    - [no] nf-prof-id nf-id
    - api-prefix api-prefix-value
    - no api-prefix
    - api-version version [minor minor] [patch patch]
    - no api-version
    - [no] enable
    - fqdn fqdn-sni-string
    - no fqdn
    - nf-priority priority-value
    - no nf-priority
    - nf-set list-name
    - no nf-set
    - peer-selection {enabled | disabled}
    - no peer-selection
    - prioritized-address-list list-name
    - no prioritized-address-list
    - uuid uuid
    - no uuid
    - [no] oi-override-mapping list-name

```

```

    - subscriber-mcc mcc value subscriber-mnc mnc value replacement-mcc mcc value
replacement-mnc mnc value
    - no subscriber-mcc mcc value subscriber-mnc mnc value
    - [no] pgw-selection-candidate-list list-name
    - candidate ipv4 | ipv6 | fqdn [priority priority-value] [weight weight-
value] [ref-pt {s2a | s2b}]
    - no candidate ipv4 | ipv6 | fqdn
    - [no] plmn list-name
    - [no] mcc mcc value mnc mnc value
    - [no] port-list list-name
    - [no] port port-number
    - [no] port range start-port-number end-port-number
    - [no] prioritized-ip-address-list list-name
    - address {ip-address | fqdn} [priority value] [port port]
    - no address {ip-address | fqdn} [port port]
    - [no] qci-list list-name
    - [no] qci qci-value
    - rat-type-list rat-type-list-name
    - no rat-type-list rat-type-list-name
    - [no] rat-type
    - [no] slice-instance-list list-name
    - slice-instance name [suspend {enable | disable}]
    - no slice-instance name
    - [no] slice-list list-name
    - slice name sst sst-value [sd sd-value]
    - no slice name
    - slice-instances list-name
    - no slice-instances
    - [no] suspend
    - tai-list list-name
    - [no] tai-list
    - [no] smf-serving-area-list list-name
    - [no] smf-serving-area string
    - tai-list profile-name
    - no tai-list
    - [no] tariff-change-list list-name
    - time id hh:mm
    - no time id
    - [no] ue-usage-type-list list-name
    - [no] ue-usage-type usage-type

```

6.1.1.37 MIPv4 profile command hierarchy

configure

```

– mobile-gateway
  – profile
    – [no] mipv4 profile-name
    – [no] description string
    – [no] dmu hash-algorithm [md5 | hmac-md5 | rfc2002-md5] timestamp-
tolerance seconds
    – [no] ip-dscp value
    – [no] mn-aaa-auth [initial-reg-only | always | allow-noauth]
    – [no] mn-ha-auth [always | allow-noauth]
    – [no] mn-ha-spi value
    – description string
    – hash-algorithm [md5 | hmac-md5 | rfc2002-md5]
    – secret string
    – timestamp-tolerance seconds
    – [no] registration-lifetime seconds
    – [no] revocation i-bit-negotiation [enable | disable] max-
retransmission number retransmission-timeout seconds
    – tunnel-mode {ip-in-ip | gre}

```

6.1.1.38 MME group profile command hierarchy

```

configure
– mobile-gateway
  – profile
    – [no] mme-group MME-group-id
    – [no] peer ipv4 | ipv6

```

6.1.1.39 N1 profile command hierarchy

```

configure
– mobile-gateway
  – profile
    – [no] n1-profile profile-name
    – cause-code
    – apn-shutdown
    – 5gsm-cause-code 5gsm-cause-code
    – no 5gsm-cause-code
    – application-error-code application-error-code
    – no application-error-code
    – http-status-code http-status-code
    – no http-status-code
    – dnn-subscription-removed

```

- *5gsm-cause-code 5gsm-cause-code*
- *no 5gsm-cause-code*
- *gw-suspend*
- *5gsm-cause-code 5gsm-cause-code*
- *no 5gsm-cause-code*
- *application-error-code application-error-code*
- *no application-error-code*
- *http-status-code http-status-code*
- *no http-status-code*
- *n1-t3591 timeout [retry-count retry-count]*
- *no n1-t3591*
- *n1-t3592 timeout [retry-count retry-count]*
- *no n1-t3592*

6.1.1.40 N2 peer profile command hierarchy

```

configure
– mobile-gateway
– profile
  – [no] n2-peer profile-name
    – peer ip-address | ipv6-address
    – no peer
  – plmn-list list-name
  – no plmn-list
  – sctp-profile profile-name
  – no sctp-profile

```

6.1.1.41 N7 attributes profile command hierarchy

```

configure
– mobile-gateway
– profile
  – [no] n7-attributes-profile profile-name
    – ip-domain {included | excluded}
    – no ip-domain
  – nokia-smf-indicator-as-vsa {included | excluded}
  – no nokia-smf-indicator-as-vsa

```

6.1.1.42 Namf profile command hierarchy

```

configure

```

```

– mobile-gateway
– profile
  – [no] namf-profile profile-name
    – extended-buffering-wait-time value
    – no extended-buffering-wait-time
    – http-status-code status-code-value [application-error app-error-code] failure-action
    – no http-status-code status-code-value [application-error app-error-code]
    – max-peer-failover value
    – no max-peer-failover
    – retry-count [failure-action failure-action] value
    – no retry-count
    – transaction-timer value
    – no transaction-timer

```

6.1.1.43 Network trace profile command hierarchy

```

configure
– mobile-gateway
– profile
  – [no] network-trace-profile profile-name
  – http2-profile profile-name
  – no http2-profile
  – [no] packet-broker-server {ip-address} [port port]
  – [no] server-path server-path-string

```

6.1.1.44 Node selection profile command hierarchy

```

configure
– mobile-gateway
– profile
  – node-selection
    – abort
    – begin
  – [no] client-profile profile-name
  – default-action
    – set-target-profile profile-name
    – no set-target-profile
  – description description-string
  – no description
  – [no] entry id
  – action

```



```

    - set-target-profile profile-name
    - no set-target-profile
  - description description-string
  - no description
  - match
    - [no] apn apn-name
    - cell-id-list cell-id-list-name
    - no cell-id-list
    - [no] iwki5gs
    - [no] nri5gc
    - rat-type {utran | eutran}
    - no rat-type
    - rat-type {utran | eutran}
    - no rat-type
    - serving-node ip-address-list
    - no serving-node
    - [no] slice list-name
    - tai-lai list-name
    - no tai-lai
    - ue-usage-type list-name
    - no ue-usage-type
  - commit
  - [no] group-label profile-name
    - description description-string
    - no description
  - [no] target-profile profile-name
    - default-action
    - [no] reject
    - description description-string
  - [no] entry id
    - action
    - [no] reject
    - description description-string
    - no description
  - match
    - apn client-apn
    - no apn
    - dns-based [nr-capable]
    - no dns-based
    - [no] group-label group-name
    - nrf-based [smf-serving-area | tai]
    - no nrf-based
    - slice client-slice
    - no slice

```

6.1.1.45 NPCF profile command hierarchy

```

configure
- mobile-gateway
  - profile
    - [no] npcf profile-name
      - [no] allow-pcf-provided-chf-address
      - dynamic-pcc-online-offline-flags-override {offline | both | none}
      - [no] dynamic-pcc-online-offline-flags-override
      - fh-session-no-response {terminate | continue | terminate-next-attempt}
      - fh-session-no-response ap-continue [ap-action-profile profile-name]
      - no fh-session-no-response
      - fh-time-limit time-limit
      - no fh-time-limit
      - fh-volume-limit volume-limit
      - fh-volume-limit
      - ip-domain-type {[network-realm-name] [psa-upf-id] [upf-string] [ip-pool-
name]]}
      - ip-domain-type none
      - no ip-domain-type
      - n7-attributes-profile profile-name
      - no n7-attributes-profile
      - npcf-error-mapping-profile profile-name
      - no npcf-error-mapping-profile
      - [no] sbi-srvc-operation {create | update | delete}
        - http-status-code status-code-value {terminate | continue | retry-and-
terminate | ap-continue | retry-and-ap-continue} [ap-action-profile profile-name]
[application-error app-error-code]
        - no http-status-code status-code-value [application-error app-error-code]
        - http-status-code-3xx {terminate | continue | retry-and-terminate | ap-
continue | retry-and-ap-continue} [ap-action-profile profile-name] [application-
error app-error-code]
        - no http-status-code-3xx [application-error app-error-code]
        - http-status-code-4xx {terminate | continue | retry-and-terminate | ap-
continue | retry-and-ap-continue} [ap-action-profile profile-name] [application-
error app-error-code]
        - no http-status-code-4xx [application-error app-error-code]
        - http-status-code-5xx {terminate | continue | retry-and-terminate | ap-
continue | retry-and-ap-continue} [ap-action-profile profile-name] [application-
error app-error-code]
        - no http-status-code-5xx [application-error app-error-code]
      - no retry-count
      - transaction-timer value
      - no transaction-timer

```

- *supported-features*
 - [no] *adc*
 - [no] *af-charging-id*
 - [no] *eps-fallback-report*
 - [no] *netloc*
 - [no] *pcscf-restoration-enhancement*
 - [no] *pdu-session-rel-cause*
 - [no] *policy-decision-error-handling*
 - [no] *policy-update-when-ue-suspends*
 - [no] *pra*
 - [no] *prov-af-signal-flow*
 - [no] *resp-based-session-rel*
 - [no] *session-rule-error-handling*
 - [no] *sponsored-connectivity*
 - [no] *support-2g3g*
 - [no] *tsc*
 - [no] *umc*
- *supported-vendor* {custom-1 | custom-2 | custom-3}
- no *supported-vendor*
- *ul-rq-timer* value
- no *ul-rq-timer*
- [no] *use-last-auth-ambr*

6.1.1.46 NSMF profile command hierarchy

```
configure
  – mobile-gateway
    – profile
      – [no] nsmf-profile profile-name
        – http-status-code status-code-value [application-error app-error-
code] failure-action
          – no http-status-code status-code-value [application-error app-error-code]
          – max-peer-failover value
          – no max-peer-failover
          – retry-count [failure-action failure-action] value
          – no retry-count
          – transaction-timer value
          – no transaction-timer
```

6.1.1.47 OAuth2.0 authorization profile command hierarchy

```
configure
```

- mobile-gateway
 - *profile*
 - [no] *oauth2-profile profile-name*
 - *oauth2-nrf-certificate cert-filename*
 - [no] *oauth2-nrf-certificate*
 - [no] *validate-consumer-uuid*

6.1.1.48 Overload control profile command hierarchy

configure

- mobile-gateway
 - *profile*
 - [no] *overload-control-profile profile-name*
 - [no] *exclude-arp-priority-level arp-value*
 - *session-rejection-rate rejection-rate*
 - no *session-rejection-rate*

6.1.1.49 PCMD profile command hierarchy

configure

- mobile-gateway
 - *profile*
 - [no] *pcmd profile-name*
 - *description description-string*
 - no *description*
 - *data-report*
 - *constant-flow-period period*
 - no *constant-flow-period*
 - *destination {{ip-addr | fqdn} [port port] [router router-instance]}* |

inherit

- no *destination*
- *generation {disabled | event-based | constant-flow}*
- no *generation*
- *destination {{ip-addr | fqdn} [port port] [router router-instance]}* | inherit
- no *destination*
- *heartbeat [period period]*
- no *heartbeat*
- *session-report*
 - *destination {{ip-addr | fqdn} [port port] [router router-instance]}* |

inherit

- no *destination*
- *verbosity [failure-only | standard | extended]*

- no *verbosity*

6.1.1.50 PCSCF group profile command hierarchy

configure

- mobile-gateway
 - *profile*
 - [no] *pcscf-group pcscf-group*
 - [no] *pcscf-server* {*ip-address* {*weight*} | *fqdn* {*ipv4* | *ipv6*}}
 - *refresh-interval* *hours*
 - no *refresh-interval*

6.1.1.51 Peer list profile command hierarchy

configure

- mobile-gateway
 - *profile*
 - [no] *peer-list profile-name*
 - *description* *description-string*
 - no *description*
 - [no] *peer ip-address* | *prefix-length*
 - *description* *description-string*
 - no *description*
 - [no] *keep-alive*
 - *plmn-id* *foreign*
 - *plmn-id* *mcc mcc value mnc mnc value*
 - no *plmn-id*
 - *rat-type* {*geran* | *utran* | *eutran*}
 - no *rat-type*
 - [no] *shutdown*
 - *unlisted-peer*
 - [no] *keep-alive*
 - *plmn-id* *foreign*
 - *plmn-id* *mcc mcc value mnc mnc value*
 - no *plmn-id*
 - *rat-type* {*geran* | *utran* | *eutran*}
 - no *rat-type*
 - [no] *shutdown*

6.1.1.52 PFCP profile command hierarchy

```

configure
- mobile-gateway
- profile
- pfc
  - [no] pfc-association-peer-list profile-name
  - description description-string
  - no description
  - [no] node-id node-id
  - [no] bfd-enable
  - [no] enable-partial-failure
  - path-hold-time minutes
  - no path-hold-time
  - [no] peer-entity ip-address
  - [no] ue-ip-allocation
  - [no] pfc-peer ipv4 | ipv6
  - [no] bfd-enable
  - [no] partial-failure
  - path-hold-time minutes
  - no path-hold-time
  - [no] up-ip-allocation
  - [no] pfc-error-mapping-profile profile-name
  - pfc-to-5gsm pfc-cause-code 5gsm-cause 5gsm-cause-code
  - no pfc-to-5gsm pfc-cause-code
  - pfc-to-gtp pfc-cause-code gtp-error gtp-cause-code
  - no pfc-to-gtp pfc-cause-code
  - pfc-to-http pfc-cause-code http-error http-status-code-out application-
error-out app-error-code-out
  - no pfc-to-http pfc-cause-code
  - pfc-options
  - [no] exclude-lci
  - [no] exclude-oci
  - [no] pfc-profile profile-name
  - association-retry-timer retry-time
  - no association-retry-timer
  - description description-string
  - no description
  - [no] disable-pdi-optimization
  - ext-urr-id {enable | disable}
  - [no] ext-urr-id
  - heart-beat [interval seconds] [retry-count retry-count-value]
[timeout timeout-value]
  - no heart-beat
  - ip-dscp ip-dscp-value

```

```

    - no ip-dscp
    - ip-ttl ip-ttl-value
    - no ip-ttl
    - message-retransmit [timeout timeout] [timeout-msec timeout-msec] [retry-
count retry-count-value]
    - no message-retransmit
    - pfc-p-error-mapping-profile profile-name
    - no pfc-p-error-mapping-profile
    - pfc-p-u-tunnel
      - re-attempt-timer reattempt-timer
      - no re-attempt-timer
    - [no] smf-set-support
    - [no ] pfc-p-translate-profile profile-name
      - description description-string
      - no description
      - [no] field {hdr-enrich-imsi | hdr-enrich-msisdn | hdr-enrich-imei-sv |
hdr-enrich-rat-type | hdr-enrich-charging-id | hdr-enrich-user-location | hdr-enrich-
billing-type | hdr-enrich-roaming-status | hdr-enrich-slice-id | hdr-enrich-amf-ip |
hdr-enrich-pcc-rulebase}
    - [no] up-peer-list list-name
      - description description-string
      - no description
    - [no] peer ipv4 | ipv6
      - [no] apn apn-name
        - [no] block
        - ip-pool-list
          - [no] ip-pool pool-name
        - nas-ip nas-ip-address
        - no nas-ip
        - network-realm network-realm-name
        - no network-realm
      - pco-option
        - dns-server-ipv4
          - primary ip-address
          - no primary
          - secondary ip-address
          - no secondary
        - dns-server-ipv6
          - primary ipv6-address
          - no primary
          - secondary ipv6-address
          - no secondary
      - [no] tai-lai-list tai-lai-list-name
        - [no] ip-pool pool-name
    - [no] disable-eps-iwk

```

```

– [no] group-label profile-name
– [no] network-realm network-realm
– fragment-distribution-size fragment-distribution-size-value
– [no] fragment-distribution-size
– [no] ip-pool pool-name
– group-name group-name
– no group-name
– hold-timer-sec hold-timer-in-sec [hold-for-pdn]
– no hold-timer-sec
– ipv4-prefix ipv4-address/mask [block [force]] [prefix-fragment-
size prefix-fragment-size-value]
– no ipv4-prefix ipv4-address/mask
– ipv6-prefix ipv6-address/prefix-length [block [force]] [prefix-
fragment-size prefix-fragment-size-value]
– no ipv6-prefix ipv6-address/prefix-length
– pool-usage-threshold high-threshold high-threshold [low-
threshold low-threshold]
– [no] network-determined
– no pool-usage-threshold
– priority priority-value
– no priority
– [no] remote
– reserved-ip-list ip-address-list-name
– no reserved-ip-list
– [no] round-robin
– [no] suppress-dotzero-broadcast-address
– [no] suppress-network-broadcast-address
– [no] use-prefixes-immediately
– [no] ip-pool-distribution
– override radius-server-group radius-group interface [router router-
name] interface-name
– no override radius-server-group radius-group
– slice list-name
– no slice
– upf-id uuid uuid [string string]
– no upf-id
– upf-selection {true | false}
– no upf-selection

```

6.1.1.53 PMIPv6 profile command hierarchy

configure

- mobile-gateway
- *profile*

- [no] *pmipv6 profile-name*
- *description description-string*
- no *description*
- *ip-dscp value*
- no *ip-dscp*
- *ip-ttl value*
- no *ip-ttl*
- *keep-alive* [interval seconds] [retry-count value] [timeout timeout]
- no *keep-alive*
- *message-retransmit* [timeout seconds] [retry-count value]
- no *message-retransmit*
- *supported-vendor custom-1*
- no *supported-vendor*

6.1.1.54 Policy options profile command hierarchy

configure

- mobile-gateway
 - *profile*
 - *policy-options*
 - *abort*
 - *begin*
 - *commit*
 - *diff*
 - [no] *action-rule-unit aru-name*
 - *aa-notify-action action-value*
 - no *aa-notify-action*
 - *app-profile-name profile-name*
 - no *app-profile-name*
 - *characteristic name value aa-attr-value*
 - no *characteristic name*
 - [no] *collect-display-stats*
 - *custom-pco* {none | normal | terminate | restrict-access | redirect | rate-throttling | self-activation | pre-paid}
 - no *custom-pco*
 - [no] *data-driven-dedicated-bearer*
 - [no] *disable-ubr-signaling*
 - *dnai dnai*
 - no *dnai*
 - *dnai-psa-select* {ulcl | anhor-upf}
 - no *dnai-psa-select*
 - *dns-redirect* [access-type access-type]
 - no *dns-redirect*
 - [no] *exclude-src-dest-ports-in-ue-tft*

```

    - gy-filter-id-action http-redirect url-string [encryption {aes-cbc-128}]
    [passphrase passphrase] [hash | hash2]
    - gy-filter-id-action {none | drop}
    - gy-filter-id-action traffic-redirect profile-name
    - no gy-filter-id-action
    - [no] modify-qos-default-bearer
    - [no] record-extension-visited-url
    - [no] mss-adjust
    - [no] rule-level-redirect
    - sdf-level-policing [direction direction] [group group]
    - no sdf-level-policing
    - [no] transport-byte-collection
    - [no] ap-action-profile profile-name
    - policy-action {pre-defined-policy | pcrf-policy-and-pre-defined-policy |
pcrf-policy}
    - no policy-action
    - [no] apn-rate-control profile-name
    - additional-ul-rate-limit-for-exception-data additional-ul-rate-limit-for-
exception-data
    - no additional-ul-rate-limit-for-exception-data
    - description description-string
    - no description
    - dl-rate-limit dl-rate-limit
    - no dl-rate-limit
    - [no] exclude-ul-exception-reports
    - time-unit time-unit
    - no time-unit
    - [no] ul-rate-control-in-pgw
    - ul-rate-limit ul-rate-limit
    - no ul-rate-limit
    - [no] charging-rule-unit policy-name
    - app-srvc-provider-id app-srvc-provider-id
    - no app-srvc-provider-id
    - charging-method {online | offline | both | none}
    - no charging-method
    - custom-online-sru sru-name
    - no custom-online-sru
    - default-rg {enable | disable}
    - no default-rg
    - description description-string
    - no description
    - metering-method {time-based | volume-based | both}
    - no metering-method
    - monitoring-key key-identifier [sru sru-name]
    - no monitoring-key

```

- *rating-group* *group-id* [*sru sru-name*]
- *no rating-group*
- *rating-group-offline* *group-id* [*sru sru-name*]
- *no rating-group-offline*
- *reporting-level* {*rating-group* | *service-id* | *sponsor-connectivity*}
- *no reporting-level*
- [*no*] *selective-radius-accounting*
- *service-identifier* *service-id* [*sru sru-name*]
- *no service-identifier*
- *service-identifier-offline* *service-id* [*sru sru-name*]
- *no service-identifier-offline*
- *sponsor-id* *sponsor-id* [*sru sru-name*]
- *no sponsor-id*
- *wait-for-ocs-response* {*enable* | *disable*}
- *no wait-for-ocs-response*
- [*no*] *dynamic-rule-to-internal-rule-map* *profile-name*
- *entry id* [*priority priority*]
- *no entry id*
- *action*
 - *direction* {*down-link* | *up-link* | *both*}
 - *no direction*
 - *local-port-range* *start start-port end end-port*
 - *no local-port-range*
 - *precedence* *precedence*
 - *no precedence*
 - *protocol* *protocol-id*
 - *no protocol*
 - *remote-ip* *ip-prefix/ip-prefix-length*
 - *no remote-ip*
 - *remote-port-range* *start start-port end end-port*
 - *no remote-port-range*
 - [*no*] *signal-internal-rule-tft*
- *match*
 - *arp arp*
 - *no arp*
 - *direction* {*down-link* | *up-link* | *both*}
 - *no direction*
 - *local-port-range* *start start-port end end-port*
 - *no local-port-range*
 - *protocol* *protocol-id*
 - *no protocol*
 - *qci qci*
 - *no qci*
 - *remote-ip* *ip-prefix/ip-prefix-length*
 - *no remote-ip*

```

    - remote-port-range start start-port end end-port
    - no remote-port-range
- [no] gx-avp-options profile-number
  - description description-string
  - no description
  - [no] exclude-3gpp-charging-characteristics
  - [no] exclude-an-trusted
  - [no] exclude-bearer-usage
  - [no] exclude-pdn-connection-id
  - [no] exclude-subscription-id-imsi
  - [no] exclude-subscription-id-msisdn
  - [no] exclude-subscription-id-nai
  - [no] exclude-twan-identifier
  - [no] exclude-ue-local-ip-address
  - [no] exclude-ue-udp-port
  - [no] exclude-user-equipment-info
  - [no] include-framed-ip-address-ccrt
  - [no] include-framed-ipv6-prefix-ccrt
  - [no] include-ipdomain-upf-id
  - [no] include-ue-tcp-source-port
  - policy-rule policy-rule-name policy-rule-unit policy-rule-unit-name
    [charging-rule-unit charging-rule-unit-policy-name] [qci qci] [arp arp-value]
    precedence precedence [action-rule-unit aru-name] [trigger-rule-unit tru-name] [stat-
    rule-unit-list sru-list-name] [qos-rule-unit gru-name] [non-terminating]
    - no policy-rule policy-rule-name
  - [no] policy-rule-base base-policy-name
    - aa-adv-functions
      - app-profile-name profile-name
      - no app-profile-name
      - characteristic characteristic-name value value-name
      - no characteristic characteristic-name
    - apn-ambr-attr
      - aggr-rate downlink downlink-rate uplink uplink-rate
      - no aggr-rate
    - charging-profile-id charging-profile-id
    - no charging-profile-id
    - [no] ded-bearer-on-app-start
    - [no] collect-display-stats
    - [no] disable-bearer-bind-verification
    - [no] policy-rule policy-rule-name
    - [no] requested-unit-total-volume requested-unit-total-volume
    - no requested-unit-total-volume
    - [no] session-level-svc-steering
  - [no] policy-rule-unit policy-name
    - [no] aa-advanced-functions

```

```

- description description-string
- no description
- [no] flow-description entry-id
  - direction {up-link | down-link | both}
  - no direction
  - match
    - aa-charging-group aa-charging-group-name [sru sru-name]
    - no aa-charging-group
    - [no] aa-inactive-charging-group
    - application-id application-id-name
    - no application-id
    - dns-fqdn-list fqdn-list-name
    - no dns-fqdn-list
    - dns-snooping fqdn-list-name
    - no dns-snooping
    - fqdn-group fqdn-group-name
    - no fqdn-group
    - local-ip ip-prefix/ip-prefix-length
    - no local-ip
    - local-port {lt | gt | eq} lcl-port-number
    - local-port range start end
    - no local-port
    - local-port-list list-name
    - no local-port-list
    - protocol protocol-id
    - no protocol
    - remote-ip ip-prefix/ip-prefix-length
    - no remote-ip
    - remote-ip-address-group {ipv6-prefix/prefix-length}
    - no remote-ip-address-group
    - remote-ip-list list-name
    - no remote-ip-list
    - remote-port {lt | gt | eq} rmt-port-number
    - remote-port range start end
    - no remote-port
    - remote-port-list list-name
    - no remote-port-list
    - tos-tc dscp-value
    - no tos-tc
  - flow-gate-status flow-gate-status
  - flow-gate-status traffic-redirect traffic-redirect-profile-id
  - flow-gate-status http-redirect url-string [encryption {aes-cbc-128}]
[passphrase passphrase] [hash | hash2]
  - flow-gate-status traffic-hash traffic-hash-profile-name
  - no flow-gate-status

```

```

    - [no] mute-notification
    - pdr-id pdr-id
    - no pdr-id
    - qos
      - down-link gbr gbr mbr mbr
      - no down-link
      - up-link gbr gbr mbr mbr
      - no up-link
    - [no] service-class-indicator service-class-id
    - shallow-inspection-only
    - no shallow-inspection-only
    - svc-steering [steering-profile profile-name]
    - no svc-steering
  - [no] qci-policy policy-name
    - description description-string
    - no description
    - qci qci-value arp arp-value
      - dscp {dscp-name | in dscp-name [out dscp-name] | out dscp-name [in dscp-
name]]}
      - no dscp
      - [no] dscp-preserve [preserve-user-plane]
      - fc fc-name
      - no fc
      - profile {in | out | apply-cir}
      - no profile
  - [no] qos-rule-unit qru-name
    - average-window average-window-value
    - no average-window
    - down-link gbr gbr-value mbr mbr-value
    - no down-link
    - max-data-burst-vol max-data-burst-vol
    - no max-data-burst-vol
    - priority-level priority-level
    - no priority-level
    - qer-id qer-id-value
    - no qer-id
    - qfi qfi-value
    - no qfi
    - [no] qnc
    - [no] rqi
    - sdf-policing direction direction
    - no sdf-policing
    - service-class-indicator service-class-id
    - no service-class-indicator
    - tos-tc dscp-value

```

- no *tos-tc*
- *up-link* gbr *gbr-value* mbr *mbr-value*
- no *up-link*
- [no] *service-class-indicator* *service-class-id*
- [no] *service-class-value* *sci-value*
- [no] *sru-list* *list-name*
- [no] *stat-rule-unit* *sru-name*
- [no] *stat-rule-unit* *sru-name*
- *urr-id* *urr-id-value* *urr-profile* *profile-name*
- no *urr-id*
- [no] *steering-profile* *steering-profile-name*
- *traffic-redirect* *traffic-redirect-profile-id*
- no *traffic-redirect*
- [no] *traffic-hash-profile* *profile-name*
- *traffic-distribution*
 - *access-lte-percentage* *lte-percent-value*
 - no *access-lte-percentage*
 - *access-xdsl-percentage* *xdsl-percent-value*
 - no *access-fixed-percentage*
- [no] *traffic-redirect-profile* *traffic-redirection-profile-name*
- [no] *bfd-enable*
- *destination-nat*
 - *destination-ipv4* *ipv4-address*
 - no *destination-ipv4*
- [no] *mg-group* *group-id*
 - *router* [ul *router-name* [dl *router-name*]]
 - [no] *router*
 - [no] *traffic-redirect-target* *traffic-redirect-target-id*
 - *downlink-next-hop-ipv4* *ip-address*
 - [no] *downlink-next-hop-ipv4*
 - *downlink-next-hop-ipv6* *ipv6-address*
 - [no] *downlink-next-hop-ipv6*
 - *uplink-next-hop-ipv4* *ip-address*
 - [no] *uplink-next-hop-ipv4*
 - *uplink-next-hop-ipv6* *ipv6-address*
 - [no] *uplink-next-hop-ipv6*
- *router* *router-instance*
- no *router*
- *threshold-target* *hold-time* *seconds* *start* *start-targets* *stop* *stop-targets*
- no *threshold-target*
- [no] *traffic-redirect-target* *traffic-redirect-target-id*
 - *downlink-next-hop-ipv4* *ipv4-address*
 - no *downlink-next-hop-ipv4*
 - *downlink-next-hop-ipv6* *ipv6-address*
 - no *downlink-next-hop-ipv6*

```

    – uplink-next-hop-ipv4 ipv4-address
    – no uplink-next-hop-ipv4
    – uplink-next-hop-ipv6 ipv6-address
    – no uplink-next-hop-ipv6
  – [no] trigger-rule-unit tru-name
    – access-type {[eutran] [wlan] [eutran-nb-iot] [ehrpdp] [utran] [geran] [gan]
[hspa-e]}
    – no access-type
    – tai-lai-list tai-lai-list-name
    – no tai-lai-list
  – [no] urr-profile urr-profile-name
    – [no] discard-on-fpt
  – [no] usage-monitoring-actions
    – [no] monitoring-action monitoring-action-profile-name
      – after-fill-delta-percent delta-percent-value
      – no after-fill-delta-percent
    – [no] usage-monitoring-info
      – [no] monitoring-key monitoring-key-profile-name
        – fixed-access-fill-out low-threshold-percent low-threshold-value high-
threshold-percent high-threshold-value
        – no fixed-access-fill-out
        – fixed-access-sustained-rate above-low-threshold above-low-threshold-value
below-low-threshold below-low-threshold-value
        – no fixed-access-sustained-rate
      – monitoring-action monitoring-action-profile-name
      – no monitoring-action
      – quota-type rate
      – no quota-type
      – traffic-hash traffic-hash-profile-name
      – no traffic-hash
  – validate-aa-charging-grp {enable | disable}
  – no validate-aa-charging-grp

```

6.1.1.55 PPI mapping profile command hierarchy

```

configure
– mobile-gateway
  – profile
    – [no] ppi-mapping-profile profile-name
      – dscp-to-ppi dscp-value ppi ppi-value
      – no dscp-to-ppi dscp-value

```


6.1.1.56 QFI mapping profile command hierarchy

```
configure
- mobile-gateway
- profile
  - [no] qfi-mapping-profile profile-name
  - [no] 5qi-as-qfi
  - dynamic-mapping
    - [no] average-window
    - [no] max-data-burst-vol
    - [no] priority-level
    - [no] qnc
```

6.1.1.57 QoS profile command hierarchy

```
configure
- mobile-gateway
- profile
  - [no] qos-5g-profile profile-name
    - 5qi value
    - no 5qi
    - arp arp-value
    - no arp
    - dl-ambr ambr-value
    - no dl-ambr
    - ul-ambr ambr-value
    - no ul-ambr
  - [no] qos-profile profile-name
    - arp arp-value
    - no arp
    - dl-ambr ambr-value
    - no dl-ambr
    - qci qci-value
    - no qci
    - ul-ambr ambr-value
    - no ul-ambr
```

6.1.1.58 RADIUS profile command hierarchy

```
configure
- mobile-gateway
```

- *profile*
 - [no] *radius profile-name*
 - *acct-retry-count* value
 - no *acct-retry-count*
 - *acct-retry-timeout* seconds
 - no *acct-retry-timeout*
 - *apn-acct-on-off-initial-interval* value
 - no *apn-acct-on-off-initial-interval*
 - *apn-acct-on-retry-interval* value
 - no *apn-acct-on-retry-interval*
 - *auth-probe-interval* value
 - no *auth-probe-interval*
 - *auth-retry-count* value
 - no *auth-retry-count*
 - *auth-retry-timeout* seconds
 - no *auth-retry-timeout*
 - *deadtime* deadtime [drop-acct-messages] [send-auth-messages]
 - no *deadtime*
 - *delay-tolerance* dt-value
 - no *delay-tolerance*
 - *description* description-string
 - no *description*
 - *max-message-rate* max-rate
 - no *max-message-rate*
 - *max-peer-reselections* count [wrap]
 - no *max-peer-reselections*
 - *max-queue-size* queue-size
 - no *max-queue-size*
 - *message-type-drop-order* message-order
 - no *message-type-drop-order*
 - *nas-port-type* nas-port-type
 - no *nas-port-type*
 - [no] *throttle-emergency*
 - [no] *radius-group radius-group*
 - *acct-server-port* port
 - no *acct-server-port*
 - *auth-server-port* port
 - no *auth-server-port*
 - *description* description-string
 - no *description*
 - *fh-session-no-response* {terminate | continue}
 - no *fh-session-no-response*
 - *fh-time-limit* time-limit
 - no *fh-time-limit*
 - *fh-volume-limit* volume-limit

- *no fh-volume-limit*
- *ignore-acct-response*
- *no ignore-acct-response*
- *interface* [router router-instance] interface-name
- *no interface*
- *interim-update-interval* value
- *no interim-update-interval*
- [no] *limit-user-name-length*
- [no] *peer* {ipv4-address | ipv6-address}
 - *acct-server-port* port
 - *no acct-server-port*
 - *auth-server-port* port
 - *no auth-server-port*
 - *failover-threshold* count
 - *no failover-threshold*
 - *failure-detection-time* value
 - *no failure-detection-time*
 - *priority* value
 - *no priority*
 - *radius-profile* profile-name
 - *no radius-profile*
 - *secret* secret [hash | hash2]
 - *no secret*
- [no] *single-server-per-ue*
- *radius-avp-options*
 - [no] *acct-interim-avp-options*
 - [no] *exclude-3gpp-charging-characteristics*
 - [no] *exclude-3gpp-charging-Id*
 - [no] *exclude-3gpp-ggsn-address*
 - [no] *exclude-3gpp-ggsn-ipv6-address*
 - [no] *exclude-3gpp-ggsn-mcc-mnc*
 - [no] *exclude-3gpp-gprs-qos-negotiated-profile*
 - [no] *exclude-3gpp-imeisv*
 - [no] *exclude-3gpp-imsi*
 - [no] *exclude-3gpp-imsi-mcc-mnc*
 - [no] *exclude-3gpp-ms-timezone*
 - [no] *exclude-3gpp-nsapi*
 - [no] *exclude-3gpp-pdp-type*
 - [no] *exclude-3gpp-rat-type*
 - [no] *exclude-3gpp-selection-mode*
 - [no] *exclude-3gpp-sgsn-address*
 - [no] *exclude-3gpp-sgsn-ipv6-address*
 - [no] *exclude-3gpp-sgsn-mcc-mnc*
 - [no] *exclude-3gpp-user-location-info*
 - [no] *exclude-acct-authentic*

- [no] *exclude-acct-delay-time*
- [no] *exclude-acct-input-gigawords*
- [no] *exclude-acct-input-octets*
- [no] *exclude-acct-input-packets*
- [no] *exclude-acct-output-gigawords*
- [no] *exclude-acct-output-octets*
- [no] *exclude-acct-output-packets*
- [no] *exclude-acct-session-id*
- [no] *exclude-acct-session-time*
- [no] *exclude-called-station-id*
- [no] *exclude-calling-station-id*
- [no] *exclude-class*
- [no] *exclude-framed-interface-id*
- [no] *exclude-framed-ip-address*
- [no] *exclude-framed-ipv6-prefix*
- [no] *exclude-framed-protocol*
- [no] *exclude-nas-ip-address*
- [no] *exclude-nas-ipv6-address*
- [no] *exclude-nas-port-type*
- [no] *exclude-service-type*
- [no] *exclude-user-name*
- [no] *include-3gpp-ip-addr-pool-info*
- [no] *include-event-timestamp*
- [no] *include-nas-identifier*
- [no] *include-nas-port-id*
- [no] *disconnect-ack-avp-options*
 - [no] *exclude-acct-terminate-cause*
 - [no] *exclude-event-timestamp*
 - [no] *exclude-proxy-state*
- [no] *disconnect-nak-avp-options*
 - [no] *exclude-error-cause*
 - [no] *exclude-proxy-state*
- [no] *exclude-3gpp-allocate-ip-type*
- [no] *exclude-3gpp-charging-characteristics*
- *exclude-3gpp-charging-Id* [accounting | authentication | both]
- no *exclude-3gpp-charging-Id*
- [no] *exclude-3gpp-ggsn-address*
- [no] *exclude-3gpp-ggsn-ipv6-address*
- [no] *exclude-3gpp-ggsn-mcc-mnc*
- [no] *exclude-3gpp-gprs-qos-negotiated-profile*
- [no] *exclude-3gpp-imei*
- *exclude-3gpp-imsi* [accounting | authentication | both]
- no *exclude-3gpp-imsi*
- [no] *exclude-3gpp-imsi-mcc-mnc*
- [no] *exclude-3gpp-ms-timezone*

- [no] *exclude-3gpp-nsapi*
- [no] *exclude-3gpp-pdp-type*
- [no] *exclude-3gpp-rat-type*
- [no] *exclude-3gpp-selection-mode*
- [no] *exclude-3gpp-session-stop-indicator*
- *exclude-3gpp-sgsn-address* [accounting | authentication | both]
- no *exclude-3gpp-sgsn-address*
- [no] *exclude-3gpp-sgsn-ipv6-address*
- [no] *exclude-3gpp-sgsn-mcc-mnc*
- [no] *exclude-3gpp-user-location-info*
- [no] *exclude-acct-authentic*
- [no] *exclude-acct-delay-time*
- [no] *exclude-acct-input-gigawords*
- [no] *exclude-acct-input-octets*
- [no] *exclude-acct-input-packets*
- [no] *exclude-acct-output-gigawords*
- [no] *exclude-acct-output-octets*
- [no] *exclude-acct-output-packets*
- [no] *exclude-acct-session-id*
- [no] *exclude-acct-session-time*
- [no] *exclude-called-station-id*
- [no] *exclude-calling-station-id*
- [no] *exclude-class*
- [no] *exclude-framed-interface-id*
- [no] *exclude-framed-ip-address*
- [no] *exclude-framed-ip-netmask*
- [no] *exclude-framed-ipv6-prefix*
- [no] *exclude-framed-protocol*
- [no] *exclude-nas-port-type*
- [no] *exclude-service-type*
- [no] *exclude-user-name*
- [no] *include-3gpp-ip-addr-pool-info*
- [no] *include-3gpp-supported-features*
- [no] *include-acct-session-id-in-on-off-messages*
- [no] *include-alc-sla-prof-str*
- [no] *include-alc-subsc-id-str*
- [no] *include-alc-subsc-prof-str*
- [no] *include-called-station-id-apn-in-on-off-messages*
- [no] *include-event-timestamp*
- [no] *include-nas-identifier*
- [no] *include-nas-port-id*
- [no] *use-acct-interim-avp-options*
- [no] *use-disconnect-ack-avp-options*
- [no] *use-disconnect-nak-avp-options*
- *radius-profile profile-name*

- *no radius-profile*
- *secret* *secret* [*hash* | *hash2*]
- *server-type* *server-type*
- *no server-type*
- *supported-features*
 - [*no*] *e-sess-ambr*
- [*no*] *transaction-based-load-balancing*
- *via-up-function* *apn* *apn-name* *all*
- *via-up-function* *apn* *apn-name* [*up-peer-ip-address* *ip-address* | *ipv6-address*]
- *no via-up-function*

6.1.1.59 S2b error mapping profile command hierarchy

configure

- mobile-gateway
 - *profile*
 - [*no*] *s2b-error-mapping-profile* *profile-name*
 - [*no*] *s2b-cause-code* *s2b-cause-code-value*
 - *backoff-timer* [*timer-value* *timer-value*] [*timer-unit* *timer-unit*]
 - *no backoff-timer*
 - *description* *description-string*
 - *no description*
 - [*no*] *gtpv2-result-code-trigger-ipsec-lockout*
 - *swu-notify* *swu-notify-value* [*description* *description-string*]
 - *no swu-notify*
 - *swu-send-info* [*notify*]
 - *no swu-send-info*

6.1.1.60 S6b profile command hierarchy

configure

- mobile-gateway
 - *profile*
 - [*no*] *s6b-profile* *s6b-profile-name*
 - *description* *description-string*
 - *no description*
 - *drmp*
 - *ehrpd-initial-attach* *priority-value*
 - *no ehrpd-initial-attach*
 - *ehrpd-initiated-detach* *priority-value*
 - *no ehrpd-initiated-detach*
 - *ehrpd-lte-handover* *priority-value*

- no *ehrpd-lte-handover*
- *ehrpd-wifi-handover* *priority-value*
- no *ehrpd-wifi-handover*
- [no] *enable*
- *eutran-initial-attach* *priority-value*
- no *eutran-initial-attach*
- *eutran-initiated-detach* *priority-value*
- no *eutran-initiated-detach*
- *lte-wifi-handover* *priority-value*
- no *lte-wifi-handover*
- *pgw-initiated-disconnect* *priority-value*
- no *pgw-initiated-disconnect*
- *reauth-initiated* *priority-value*
- no *reauth-initiated*
- *wifi-initial-attach* *priority-value*
- no *wifi-initial-attach*
- *wifi-initiated-detach* *priority-value*
- no *wifi-initiated-detach*
- *fh-session-no-response* {*terminate* | *continue-emps*}
- no *fh-session-no-response*
- *fh-time-limit* *time-limit*
- no *fh-time-limit*
- [no] *message* {*aar-i* | *aar-u*}
- *result-code* *result-code-value* {*terminate* | *continue-emps*}
- no *result-code* *result-code-value*
- *result-code-3xxx* {*terminate* | *continue-emps*}
- no *result-code-3xxx*
- *result-code-4xxx* {*terminate* | *continue-emps*}
- no *result-code-4xxx*
- *result-code-5xxx* {*terminate* | *continue-emps*}
- no *result-code-5xxx*
- *s6b-avp-options*
 - [no] *exclude-emergency-service*
 - [no] *exclude-idle-timeout*
 - [no] *exclude-MIP6-agent-info*
 - [no] *exclude-MIP6-feature-vector*
 - [no] *exclude-rat-type*
 - [no] *exclude-supported-feature*
 - [no] *exclude-visited-network-identifier*
 - [no] *include-3gpp-allocate-ip-type*
 - [no] *include-3gpp-charging-characteristics*
 - [no] *include-3gpp-charging-id*
 - [no] *include-3gpp-ggsn-address*
 - [no] *include-3gpp-ggsn-ipv6-address*
 - [no] *include-3gpp-ggsn-mcc-mnc*

- [no] *include-3gpp-gprs-negotiated-qos-profile*
- [no] *include-3gpp-imeisv*
- [no] *include-3gpp-imsi*
- [no] *include-3gpp-ms-timezone*
- [no] *include-3gpp-nsapi*
- [no] *include-3gpp-pdp-type*
- [no] *include-3gpp-rat-type*
- [no] *include-3gpp-selection-mode*
- [no] *include-3gpp-sgsn-address*
- [no] *include-3gpp-sgsn-ipv6-address*
- [no] *include-3gpp-sgsn-mcc-mnc*
- [no] *include-3gpp-user-location-info*
- [no] *include-auth-session-state*
- [no] *include-framed-ip-address*
- [no] *include-framed-ip-netmask*
- [no] *include-framed-ipv6-prefix*
- [no] *supported-features*
- [no] *clear-m-bit*
- [no] *wlan-pcscf-restoration*

6.1.1.61 SBI error mapping profile command hierarchy

configure

- mobile-gateway
 - *profile*
 - [no] *sbi-error-mapping-profile profile-name*
 - *http-to-5gsm http-status-code [application-error app-error-code] 5gsm-cause 5gsm-cause-code*
 - no *http-to-5gsm http-status-code [application-error app-error-code]*
 - *http-to-gtp http-status-code [application-error app-error-code] gtp-error gtp-cause-code*
 - no *http-to-gtp http-status-code [application-error app-error-code]*
 - *http-to-http http-status-code-in [application-error-in app-error-code-in] http-error http-status-code-out application-error-out app-error-code-out*
 - no *http-to-http http-status-code-in [application-error-in app-error-code-in]*

6.1.1.62 SCTP profile command hierarchy

configure

- mobile-gateway
 - *profile*
 - [no] *sctp profile-name*

- *adv-receiver-window-credit* value
- no *adv-receiver-window-credit*
- *assoc-max-retrans* value
- no *assoc-max-retrans*
- *cookie-life-timer* seconds
- no *cookie-life-timer*
- *heartbeat-interval* seconds
- no *heartbeat-interval*
- *initial-max-retrans* value
- no *initial-max-retrans*
- *mtu-size* bytes
- no *mtu-size*
- *path-max-retrans* value
- no *path-max-retrans*
- *rto-init-max-timer* timer
- no *rto-init-max-timer*
- *rto-init-timer* timer
- no *rto-init-timer*
- *rto-max-timer* timer
- no *rto-max-timer*
- *rto-min-timer* timer
- no *rto-min-timer*
- *sack-frequency* value
- no *sack-frequency*
- *sack-period* timer
- no *sack-period*

6.1.1.63 Sd profile command hierarchy

configure

- mobile-gateway
 - *profile*
 - [no] *sd profile-name*
 - *description* description-string
 - no *description*
 - *pending-transaction-retry-count* value
 - no *pending-transaction-retry-count*
 - *pending-transaction-timer* msec-value
 - no *pending-transaction-timer*
 - *supported-features*
 - [no] *cno-uli*
 - [no] *pending-transaction*

6.1.1.64 SGSN profile command hierarchy

```
configure
- mobile-gateway
- profile
  - [no] sgsn-group SGSN-group-id
  - [no] peer ipv4 | ipv6
```

6.1.1.65 Sig rate limit profile command hierarchy

```
configure
- mobile-gateway
- profile
  - [no] sig-rate-limit-profile profile-name
    - delay-tolerance dt-value
    - no delay-tolerance
    - max-burst-size size
    - no max-burst-size
    - max-message-rate max-rate
    - no max-message-rate
    - queue-threshold low low-value high high-value
    - no queue-threshold
  - [no] throttle-emergency
```

6.1.1.66 SLAAC host profile command hierarchy

```
configure
- mobile-gateway
- profile
  - [no] slaac-host profile-name
    - description description-string
    - no description
    - router-advertisements
      - [no] force-unicast-ip
      - [no] include-rdss
      - max-rtr-adv-interval seconds
      - no max-rtr-adv-interval
      - min-rtr-adv-interval seconds
      - no min-rtr-adv-interval
      - mtu bytes
      - no mtu
    - preferred-lifetime seconds
```

- no *preferred-lifetime*
- *rdnss-lifetime* seconds
- no *rdnss-lifetime*
- *reachable-time* milliseconds
- no *reachable-time*
- *retransmit-time* milliseconds
- no *retransmit-time*
- *router-lifetime* seconds
- no *router-lifetime*
- *valid-lifetime* seconds
- no *valid-lifetime*

6.1.1.67 STa profile command hierarchy

configure

- mobile-gateway
 - *profile*
 - [no] *sta-profile* profile-name
 - *description* description-string
 - no *description*
 - *diameter-optional-avps*
 - [no] *exclude-der-flags-avp*

6.1.1.68 Subscriber mapping profile command hierarchy

configure

- mobile-gateway
 - *profile*
 - [no] *subscriber-mapping* profile-name
 - [no] *address* ip-address subscriber-id subscriber-string

6.1.1.69 SWm error mapping profile command hierarchy

configure

- mobile-gateway
 - *profile*
 - [no] *swm-error-mapping-profile* profile-name
 - [no] *swm-error-code* swm-error-code-value [experimental]
 - *backoff-timer* [timer-value timer-value] [timer-unit timer-unit]
 - no *backoff-timer*
 - *description* description-string

- no *description*
- [no] *diameter-result-code-trigger-lockout*
- *swu-notify* *swu-notify-value* [*description description-string*]
- no *swu-notify*
- *swu-send-info* [*notify*]
- no *swu-send-info*

6.1.1.70 SWm profile command hierarchy

configure

- mobile-gateway
 - *profile*
 - [no] *swm-profile profile-name*
 - *description description-string*
 - no *description*
 - *swm-avp-options*
 - [no] *exclude-ue-local-ip-address*
 - [no] *include-aar-flags*
 - [no] *include-terminal-information*

6.1.1.71 SWw profile command hierarchy

configure

- mobile-gateway
 - *profile*
 - [no] *sww-radius-profile profile-name*
 - *access-accept-optional-attributes*
 - [no] *include-3gpp-charging-characteristics*
 - [no] *include-3gpp-ggsn-address*
 - [no] *include-3gpp-gprs-negotiated-qos-profile*
 - [no] *include-3gpp-imsi*
 - [no] *include-3gpp-msisdn*
 - [no] *include-3gpp-pdp-type*
 - [no] *include-alc-mgw-interface-type*
 - [no] *include-alc-msisdn*
 - [no] *include-alc-subsc-id-str*
 - [no] *include-alc-wlan-apn-name*
 - [no] *acct-triggered-mobility*
 - *description description-string*
 - no *description*
 - *disconnect-optional-attributes*
 - [no] *exclude-framed-ip-address*

- [no] *exclude-user-name*
- *disconnect-port* *port*
- no *disconnect-port*
- [no] *fast-inter-ap-handover*
- *retry-count* *value*
- no *retry-count*
- *retry-timeout* *value*
- no *retry-timeout*
- [no] *swt-tunnel-profile* *profile-name*
- [no] *ap-mac-discovery*
- *default-access-type* [*portal* | *eap*]
- no *default-access-type*
- *default-apn* *apn-name* [*portal* | *eap*]
- no *default-apn* [*portal* | *eap*]
- *description* *description-string*
- no *description*
- *dhcpv4-server* *profile-name*
- no *dhcpv4-server*
- *qtag* *qtag-value* *apn* *apn-name* [*portal* | *eap*] [*ssid* *ssid-name*]
- no *qtag* *qtag-value*
- *slaac-host* *profile-name*
- no *slaac-host*
- *ssid* *ssid-value* *apn* *apn-name* [*portal* | *eap*]
- no *ssid* *ssid-value*
- *ue-cache-time* *seconds*
- no *ue-cache-time*

6.1.1.72 T6c profile command hierarchy

configure

- mobile-gateway
 - *profile*
 - [no] *t6c-profile* *profile-name*
 - [no] *include-apn*
 - [no] *include-ip*
 - [no] *include-rat-type*

6.1.1.73 TAI and LAI list profile command hierarchy

configure

- mobile-gateway
 - *profile*

```

    - [no] tai-lai-list tai-lai-list-name
    - [no] lai mcc mcc-value mnc mnc-value lac lac-value
    - [no] lai mcc mcc-value mnc mnc-value lac-range-start lac-range-start value
lac-range-end lac-range-end value
    - [no] tai mcc mcc-value mnc mnc-value tac tac-value
    - [no] tai mcc mcc-value mnc mnc-value tac-range-start tac-range-start value
tac-range-end tac-range-end value

```

6.1.1.74 UDM-SDM profile command hierarchy

```

configure
- mobile-gateway
- profile
  - [no] udm-sdm-profile profile-name
  - block-timer interval
  - no block-timer
  - continue-timer interval action action
  - no continue-timer
  - create-subscription
    - fh-session-no-response {terminate | continue | retry-and-terminate | retry-
and-continue} [block]
    - no fh-session-no-response
    - http-status-code status-code-value [application-error app-error-code]
action {terminate | continue | retry-and-terminate | retry-and-continue} [block]
  - no http-status-code status-code-value [application-error app-error-code]
  - retry-count retry-count
  - no retry-count
  - transaction-timer value
  - no transaction-timer
  - fh-session-no-response {terminate | continue | retry-and-terminate | retry-
and-continue} [block]
  - no fh-session-no-response
  - get-sm-subscription-data
    - [no] exclude-plmn-id
    - fh-session-no-response {terminate | continue | retry-and-terminate | retry-
and-continue} [block]
    - no fh-session-no-response
    - http-status-code status-code-value [application-error app-error-code]
[access-type {5g | eps | epdg}] action {terminate | continue | retry-and-terminate |
retry-and-continue} [block]
  - no http-status-code status-code-value [application-error app-error-code]
[access-type {5g | eps | epdg}]
  - retry-count retry-count
  - no retry-count

```

```

    – send-sd-with-reserved-value {enable | disable}
    – no send-sd-with-reserved-value
    – transaction-timer value
    – no transaction-timer
    – http-status-code status-code-value [application-error app-error-code] action
{terminate | continue | retry-and-terminate | retry-and-continue} [block]
    – no http-status-code status-code-value [application-error app-error-code]
    – max-peer-failover value
    – no max-peer-failover
    – retry-count retry-count
    – no retry-count
    – sbi-error-mapping-profile profile-name
    – no sbi-error-mapping-profile
    – transaction-timer value
    – no transaction-timer

```

6.1.1.75 UDM-UECM profile command hierarchy

```

configure
– mobile-gateway
– profile
    – [no] udm-uecm-profile profile-name
    – block-timer interval
    – no block-timer
    – continue-timer interval action action
    – no continue-timer
    – fh-session-no-response {terminate | continue | retry-and-terminate | retry-
and-continue} [block]
    – no fh-session-no-response
    – http-status-code status-code-value [application-error app-error-code] action
{terminate | continue | retry-and-terminate | retry-and-continue} [block]
    – no http-status-code status-code-value [application-error app-error-code]
    – max-peer-failover value
    – register-smf
    – fh-session-no-response {terminate | continue | retry-and-terminate | retry-
and-continue} [block]
    – no fh-session-no-response
    – http-status-code status-code-value [application-error app-error-code]
action {terminate | continue | retry-and-terminate | retry-and-continue} [block]
    – no http-status-code status-code-value [application-error app-error-code]
    – retry-count retry-count
    – no retry-count
    – transaction-timer value
    – no transaction-timer

```

- *retry-count* *retry-count*
- *no* *retry-count*
- *sbi-error-mapping-profile* *profile-name*
- *no* *sbi-error-mapping-profile*
- *transaction-timer* *value*
- *no* *transaction-timer*

6.1.1.76 UDP tunnel group profile command hierarchy

configure

- *mobile-gateway*
 - *profile*
 - *[no]* *udp-tunnel-group* *tunnel-group-name*
 - *description* *description-string*
 - *no* *description*
 - *[no]* *peer* *ipv4* | *ipv6*
 - *destination-port* *port*
 - *no* *destination-port*

6.1.1.77 UMTS QoS policy profile command hierarchy

configure

- *mobile-gateway*
 - *profile*
 - *[no]* *umts-qos-policy* *umts-qos-policy*
 - *background* *qci* *qci*
 - *no* *background*
 - *conversational* *ssd* *ssd* [*delay* *delay*] *qci* *qci*
 - *no* *conversational* *ssd* *ssd* [*delay* *delay*]
 - *interactive* *priority* *priority* [*signaling-indication*] *qci* *qci*
 - *no* *interactive* *priority* *priority* [*signaling-indication*]
 - *operator-qci* *operator-qci* *traffic-class* *traffic-class* [*traffic-priority* *traffic-priority*] [*signaling-indication* *signaling-indication*] [*ssd* *ssd*] [*delay* *delay*]
 - *no* *operator-qci* *operator-qci*
 - *streaming* *qci* *qci*
 - *no* *streaming*

6.1.1.78 UP security policy profile command hierarchy

configure

- mobile-gateway
 - *profile*
 - [no] *up-security-policy profile-name*
 - [no] *override-udm-policy*
 - *up-confidentiality* {*required* | *preferred* | *not-needed*}
 - no *up-confidentiality*
 - *up-integrity* {*required* | *preferred* | *not-needed*}
 - no *up-integrity*

6.1.1.79 V-SMF profile command hierarchy

configure

- mobile-gateway
 - *profile*
 - [no] *vsmf-profile profile-name*
 - [no] *allow-always-on*
 - *up-confidentiality* {*required* | *preferred* | *not-needed*}
 - no *up-confidentiality*
 - *up-integrity* {*required* | *preferred* | *not-needed*}
 - no *up-integrity*
 - *vplmn-qos profile-name*
 - no *vplmn-qos*

6.1.1.80 WLAN emergency profile command hierarchy

configure

- mobile-gateway
 - *profile*
 - [no] *wlan-emergency profile-name*
 - [no] *ecd-table*
 - *3gpp-ho-emergency-pgw-id list-name*
 - no *3gpp-ho-emergency-pgw-id*
 - *apn apn-name*
 - no *apn*
 - *apn-ambr-dl apn-ambr-dl*
 - no *apn-ambr-dl*
 - *apn-ambr-ul apn-ambr-ul*
 - no *apn-ambr-ul*
 - *pgw-id list-name*
 - no *pgw-id*
 - *qos*
 - *arp arp-value*

```

    - no arp
    - qci qci
    - no qci
  - emergency-call-numbers-ikev2
  - country-code country-code
  - no country-code {country-code | all}
  - digits digits [category category [category...(up to 5 max)]]
  - no digits digits
  - ip-range-country-file-format dbip url-dbip url-string
  - ip-range-country-file-format geolite url-geolite-v4 url-string url-geolite-
cc url-string [url-geolite-v6 url-string]
  - ip-range-country-file-format ip2loc url-ip2loc-v4 url-string [url-ip2loc-
v6 url-string]
  - no ip-range-country-file-format
  - idr-emergency-apn-name-list apn-list
  - no idr-emergency-apn-name-list
  - [no] unauthenticated-uiccless-support

```

6.1.1.81 WPS profile command hierarchy

```

configure
- mobile-gateway
- profile
- [no] wps-profile profile-name
- [no] arp value
- [no] message-priority
  - all value
  - no all
  - gtpc value
  - no gtpc
  - pfcp value
  - no pfcp
  - sbi value
  - no sbi
- up-dscp-marking
  - down-link qci-policy-name
  - no down-link
  - up-link qci-policy-name
  - no up-link

```

6.1.2 Profile show command hierarchy

show

- mobile-gateway
 - profile
 - call-insight
 - all-ues
 - ue
 - charging
 - ccfh-profile *profile-id*
 - cdr-avp-options [*cdr-avp-options-profile-id*]
 - chf-profile [*chf-charging-profile-id*]
 - epdg-charging [*charging-profile-id*]
 - event-triggers-fbc [*profile-id*]
 - gy-avp-options [*gy-avp-options-profile-id*]
 - pgw-charging [*charging-profile-id*]
 - sgw-charging [*charging-profile-id*]
 - smf-charging [*charging-profile-id*]
 - smf-profile [*charging-profile-id*]
 - tdf-charging [*charging-profile-id*]
 - twag-charging [*charging-profile-id*]
 - upf-acct-profile [*profile-id*]
 - dcca [*profile-name*]
 - dhcp-server-group [*profile-name*]
 - dhcpv4-server [*profile-name*]
 - diameter [*profile-name*]
 - diameter-error-mapping-profile [*profile-name*]
 - diameter-peer [*profile-name*]
 - diameter-peer-group-list [*list-name*]
 - dns-error-mapping-profile *profile-name*
 - dns-profile [*profile-name*]
 - dns-redirect-server [*profile-name*]
 - emergency-profile [*profile-name*]
 - fa-ha-peer-list [*profile-name*]
 - fqdn-group [*profile-name*]
 - gtp [*profile-name*]
 - gx [*profile-name*]
 - http2 [*profile-name*]
 - gtp-prime-group [*profile-name*]
 - ipsec-profile [*profile-name*]
 - list
 - aaa-host-realm [*list-name*]
 - apn-list [*list-name*]
 - arp-list [*arp-list-name*]
 - broadcast-plmn-list [*list-name*]

```
– cell-id-list [cell-id-list-name]
– fqdn-list [list-name]
– nf-id-list [list-name]
– oi-override-mapping [list-name]
– pgw-selection-candidate-list [list-name]
– plmn-list [list-name]
– port-list [list-name]
– prioritized-ip-address-list [list-name]
– qci-list [list-name]
– slice-instance-list [list-name]
– slice-list [list-name]
– ue-usage-type-list [list-name]
– mipv4 [profile-name]
– mme-group [MME-group-id]
– n1-profile profile-name
– n2-peer profile-name
– network-trace-profile profile-name
– node-selection
  – client-profile [profile-name]
  – group-label [profile-name]
  – target-profile [profile-name]
– npcfc [profile-name]
– pcmd [profile-name]
– pcscf-group [profile-name]
– peer-list [profile-name]
– pfcpc
  – pfcpc-association-peer-list [profile-name]
  – pfcpc-error-mapping-profile [profile-name]
  – pfcpc-profile [profile-name]
  – pfcpc-translate-profile [profile-name]
  – up-peer-list [profile-name]
– pmipv6 [profile-name]
– policy-options
  – action-rule-unit [action-rule-unit-name]
  – ap-action-profile [profile-name]
  – charging-rule-unit [policy-name]
  – gx-avp-options [policy-name]
  – policy-rule [policy-name]
  – policy-rule-base [policy-name]
  – policy-rule-unit [policy-name]
  – qci-policy [policy-name]
  – qos-rule-unit [profile-name]
  – stat-rule-unit [profile-name]
  – steering-profile [steering-profile-name]
  – traffic-redirect-profile [traffic-redirection-profile-id]
```

```

– trigger-rule-unit [tru-name]
– ppi-mapping-profile [profile-name]
– qfi-mapping-profile [profile-name]
– qos-profile [profile-name]
– radius [profile-name]
– radius-group [profile-name]
– s2b-error-mapping-profile [profile-name]
– s6b-profile [profile-name]
– sctp-profile [profile-name]
– sd [profile-name]
– sgsn-group [SGSN-group-id]
– sig-rate-limit-profile [profile-name]
– slaac-host [profile-name]
– swm-error-mapping-profile [profile-name]
– sww-radius-profile [profile-name]
– sww-tunnel-profile [profile-name]
– tai-lai-list [tai-lai-list-name]
– udm-sdm-profile [profile-name]
– udm-uecm-profile [profile-name]
– udp-tunnel-group [tunnel-group-name]
– umts-qos-policy [profile-name]
– up-security-policy [profile-name]
– wlan-emergency [profile-name]
– wlan-emergency profile-name country-code country-code
– wps-profile [profile-name]

```

6.2 Profile command descriptions

- *Profile configuration command descriptions*
- *Profile show command descriptions*

6.2.1 Profile configuration command descriptions

- *Generic profile command descriptions*
- *Accounting priority profile command descriptions*
- *Address list profile command descriptions*
- *Call trace profile command descriptions*
- *Charging profile CCFH command descriptions*
- *Charging profile CDR AVP command descriptions*

- [Charging profile CHF command descriptions](#)
- [Charging profile credit pool command descriptions](#)
- [Charging profile ePDG, PGW, SGW, TDF and UP function command descriptions](#)
- [Charging profile event triggers FBC command descriptions](#)
- [Charging profile event triggers QBC command descriptions](#)
- [Charging profile Gy AVP options command descriptions](#)
- [Charging profile SMF command descriptions](#)
- [Charging profile TWAG command descriptions](#)
- [Cloud database profile command descriptions](#)
- [DCCA profile command descriptions](#)
- [Dedicated bearer idle timeout profile command descriptions](#)
- [DHCP server group profile command descriptions](#)
- [DHCPv4 server profile command descriptions](#)
- [Diameter error mapping profile command descriptions](#)
- [Diameter profile command descriptions](#)
- [Diameter peer profile command descriptions](#)
- [DNS profile command descriptions](#)
- [DNS redirection profile command descriptions](#)
- [Emergency profile command descriptions](#)
- [ePDG group profile command descriptions](#)
- [Fa Ha peer list profile command descriptions](#)
- [FQDN group profile command descriptions](#)
- [Gn/Gp profile command descriptions](#)
- [GTP profile command descriptions](#)
- [GTP prime group profile command descriptions](#)
- [Gx profile command descriptions](#)
- [HTTP/2 profile command descriptions](#)
- [IPSec profile command descriptions](#)
- [List profile command descriptions](#)
- [MIPv4 profile command descriptions](#)
- [MME group profile command descriptions](#)

- *N1 profile command descriptions*
- *N2 peer profile command descriptions*
- *Namf profile command descriptions*
- *Network trace profile command descriptions*
- *Node selection profile command descriptions*
- *NPCF profile command descriptions*
- *NSMF profile command descriptions*
- *OAuth2.0 authorization profile command descriptions*
- *Overload control profile command descriptions*
- *PFCP profile command descriptions*
- *PMIPv6 profile command descriptions*
- *PCMD profile command descriptions*
- *PCSCF server group profile command descriptions*
- *Policy options profile command descriptions*
- *PPI mapping profile command descriptions*
- *QFI mapping profile command descriptions*
- *QoS profile command descriptions*
- *RADIUS profile command descriptions*
- *S2b error mapping profile command descriptions*
- *S6b interface profile command descriptions*
- *SCTP profile command descriptions*
- *Sd profile command descriptions*
- *SGSN profile command descriptions*
- *Signaling rate limit profile command descriptions*
- *SLAAC host profile command descriptions*
- *STa profile command descriptions*
- *Subscriber mapping profile command descriptions*
- *SWm error mapping profile command descriptions*
- *SWm profile command descriptions*
- *SWw profile command descriptions*
- *T6c profile command descriptions*

- [Tracking and location area identity profile commands descriptions](#)
- [UDP tunnel profile command descriptions](#)
- [UMTS QoS policy profile commands descriptions](#)
- [UP security policy profile commands descriptions](#)
- [V-SMF profile command descriptions](#)
- [WLAN emergency profile command descriptions](#)
- [WPS profile command descriptions](#)

6.2.1.1 Generic profile command descriptions

6.2.1.1.1 description

Syntax

```
description description-string  
no description
```

Context

```
config>mobile>profile  
config>mobile>profile>accounting-priority-profile  
config>mobile>profile>charging>cdr-avp-options-profile  
config>mobile>profile>charging>chf-profile  
config>mobile>profile>charging>epdg-charging  
config>mobile>profile>charging>event-triggers-fbc  
config>mobile>profile>charging>event-triggers-qbc  
config>mobile>profile>charging>gy-avp-options  
config>mobile>profile>charging>pgw-charging  
config>mobile>profile>charging>sgw-charging  
config>mobile>profile>charging>smf-charging  
config>mobile>profile>charging>smf-attributes-profile  
config>mobile>profile>charging>tdf-charging  
config>mobile>profile>charging>twag-charging  
config>mobile>profile>call-insight>all-ues
```



```
config>mobile>profile>call-insight>ue
config>mobile>profile>cloud-db
config>mobile>profile>dcca
config>mobile>profile>dhcp-server-group
config>mobile>profile>dhcpv4-server
config>mobile>profile>diameter
config>mobile>profile>diameter-peer
config>mobile>profile>dns-error-mapping-profile
config>mobile>profile>fa-ha-peer-list
config>mobile>profile>fa-ha-peer-list>peer>fa-ha-spi
config>mobile>profile>ipsec-profile
config>mobile>profile>mipv4>mn-ha-spi
config>mobile>profile>node-selection>client-profile
config>mobile>profile>node-selection>client-profile>entry
config>mobile>profile>node-selection>group-label
config>mobile>profile>pcmd
config>mobile>profile>peer-list
config>mobile>profile>pfcf>pfcf-association-peer-list
config>mobile>profile>pfcf>pfcf-profile
config>mobile>profile>pfcf>pfcf-translate-profile
config>mobile>profile>pfcf>up-peer-list
config>mobile>profile>pmipv6
config>mobile>profile>policy-options>apn-rate-control
config>mobile>profile>policy-options>gx-avp-options
config>mobile>profile>policy-options>policy-rule-unit
config>mobile>profile>policy-options>qci-policy
config>mobile>profile>radius
config>mobile>profile>radius-group
config>mobile>profile>s6b-profile
config>mobile>profile>sd
config>mobile>profile>slaac-host
```

```
config>mobile>profile>sta-profile  
config>mobile>profile>swm-radius-profile  
config>mobile>profile>swm-tunnel-profile  
config>mobile>profile>udp-tunnel-group
```

Description

This command enables a text description stored in the configuration file for a configuration context. The `description` command associates a text string with a configuration context to help identify the content in the configuration file.

The `no` form of this command removes the string from the configuration.

Default

no description

Parameters

description-string

specifies the description character string. Allowed values are any string up to 80 characters long composed of printable, 7-bit ASCII characters. If the string contains special characters (#, \$, spaces, and so on), the entire string must be enclosed within double quotes.

6.2.1.1.2 shutdown

Syntax

```
[no] shutdown
```

Context

```
config>mobile>profile  
config>mobile>profile>cloud-db>server  
config>mobile>profile>dhcp-server-group>peer  
config>mobile>profile>diameter-peer  
config>mobile>profile>ipsec-profile>ike-sa>cert-entry  
config>mobile>profile>peer-list>peer  
config>mobile>profile>peer-list>unlisted-peer
```

Description

This command administratively disables the entity. When disabled, an entity does not change, reset, or remove any configuration settings or statistics. Many entities must be explicitly enabled using the no shutdown command. The shutdown command administratively disables an entity. The operational state of the entity is disabled as well as the operational state of any entities contained within. Many objects must be shut down before they may be deleted.

6.2.1.1.3 profile

Syntax

```
profile
```

Context

```
config>mobile
```

Description

This command enables the context to configure mobile gateway profiles.

6.2.1.2 Accounting priority profile command descriptions

6.2.1.2.1 accounting-priority-profile

Syntax

```
[no] accounting-priority-profile profile-name
```

Context

```
config>mobile>profile
```

Description

This command configures an accounting priority profile. The no form of this command removes this configuration.

Parameters

profile-name

specifies the accounting priority profile name, up to 32 characters

6.2.1.2.2 priority

Syntax

```
[no] priority priority
```

Context

```
config>mobile>profile>accounting-priority-profile
```

Description

This command enables the context to configure matching criteria for the given accounting priority. Matching is evaluated starting from the highest priority context, falling through to the next priority context until a match is found or no further match criteria is available.

The no form of this command removes this configuration including all parameters in the context.

Parameters

priority

specifies the priority; a higher number indicates a higher priority

Values: 0 to 3

6.2.1.2.3 single-wt-session

Syntax

```
single-wt-session {enable | disable}
```

```
no single-wt-session
```

Context

```
config>mobile>profile>accounting-priority-profile
```

Description

This command facilitates the allocation of a single WT for UP packet processing for a PDN session. The single WT mode of operation applies from the start until the end of a session.

The no form of this command reverts to default (disable).

Parameters

enable

specifies to enable the allocation of a single WT for UP packet processing for a PDN session

disable

specifies to disable the allocation of a single WT for UP packet processing for a PDN session

6.2.1.2.4 match-id

Syntax

```
match-id match-id [roaming-status status] [start-rating-group group-id] [end-rating-group [0..4294967295]] [monitoring-key key] [start-qci qci-value] [end-qci [1..254]]
```

```
no match-id match-id
```

Context

```
config>mobile>profile>accounting-priority-profile>priority
```

Description

This command configures the mapping of the combination of roaming status, rating group, monitoring key, and/or QCI to the accounting priority. All the parameters are optional, but a single configuration line must have at least one parameter. If not specified, the parameters default to a wildcard matching all possible values. When several parameters are specified, all must match for the full line to match. Multiple *match-id* lines can be configured under the same priority. Any matching line will resolve into using the specific priority of the context.

The no form of this command removes this configuration.

Default

no match-id

Parameters

match-id

an index value identifying the match criteria

status

specifies the roaming status

Values: roaming

group-id

specifies the identification number of the rating group; configuring only the `start-rating-group` parameter allows for single values, optionally configuring also the `end-rating-group` parameter allows the use of a range between these values

Values: 0 to 4294967295

key

specifies the key string which is required for Gx usage monitoring to be supported for predefined rules, up to 32 characters

qci-value

specifies the value of the QCI; configuring only the `start-qci` parameter allows for single values; optionally configuring also the `end-qci` parameter allows the use of a range between these values

Values: 1 to 9 | 65 | 66 | 69 | 70 | 128 to 254

6.2.1.3 Address list profile command descriptions

6.2.1.3.1 addr-list

Syntax

```
[no] addr-list addr-list-name
```

Context

```
config>mobile>profile
```

Description

This command configures a list of IPv4 or IPv6 addresses to define an IP match list that can be used across features that require IP address lists.

**NOTE:**

An IP address list cannot be modified on-the-fly when it is already referenced in a configuration. You must first remove the address list from the existing configuration and reconfigure it to the new reference point.

The `no` form of this command disables the address list.

Parameters

addr-list-name

specifies the name of the profile

6.2.1.3.2 ip-address

Syntax

`ip-address ip-address/prefix-length`

Context

`config>mobile>profile>addr-list`

Description

This command configures the IPv4 or IPv6 address subnet for IP address list.

Parameters

ip-address/prefix-length
specifies the address

Values	ipv4 address	a.b.c.d
–	ipv4 prefix-length	[1..32]
–	mask	[1..32]
–	ipv6 address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D
–	ipv6 prefix-length	[1..128]

6.2.1.3.3 ip-address-group

Syntax

ip-address-group *ipv6-prefix/prefix-length*

Context

config>mobile>profile>addr-list

Description

This command enables the use of the asterisk (*) character as wildcard within the destination IPv6 address. Only full quartets can be wildcarded. At least one and up to eight consecutive wildcards are supported. Wildcards can be applied to any subnet. For example, 2608:7700:0:*:*:82d3:bc84/128, 2608:7700::*:*:*:1100:0/104.

The no form of this command unsets this configuration.

Parameters

ipv6-prefix/prefix-length
specifies the IPv6 address group

Values	IPv6 prefix	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x - [0..FFFF]H
–	Prefix length	1 to 128

6.2.1.4 Call Trace profile command descriptions

6.2.1.4.1 call-insight

Syntax

[no] call-insight

Context

config>mobile>profile

Description

This command enables the context to configure call trace profile parameters.

6.2.1.4.2 all-ues

Syntax

[no] all-ues *profile-name*

Context

config>mobile>profile>call-insight

Description

This command specifies a call trace profile for all UEs. The profile includes configuration options to support tracing of specified events, reference point interfaces to capture control messages related to the traced UE, data volume control, output file size, time limit, and output file format.

Parameters

profile-name

specifies a profile name, up to 32 characters

6.2.1.4.3 ue

Syntax

[no] ue *profile-name*

Context

config>mobile>profile>call-insight

Description

This command specifies a call trace profile for the UE. The profile includes configuration to support tracing of specified events, reference point interfaces to capture control messages related to the traced UE user traffic, data volume control, output file size, time limit, and output file format.

Default

no ue

Parameters

profile-name

specifies a profile name, up to 32 characters

6.2.1.4.4 events

Syntax

```
[no] events {all | acctmgr | apm | bmc | dcm | dhcp | dth | slaac | ga | gal | gtp | gtpv2 | gxc |  
gxl | ike | lrdm | mip | msngr | pmip | rfl | rol | s6b | sctp | sd | sessmgr | sigmgr | swm | sww |  
sta | rpy | dpy | pfcf | nserv | n3iwf-ngap}
```

Context

```
config>mobile>profile>call-insight>all-ues
```

```
config>mobile>profile>call-insight>ue
```

Description

This command configures events capturing for advanced debugging capabilities. Only use events configuration if needed and as instructed by Nokia Technical Support.

Parameters

**all | acctmgr | apm | bmc | dcm | dhcp | dth | slaac | ga | gal | gtp | gtpv2 | gxc | gxl | ike | lrdm | mip |
msngr | pmip | rfl | rol | s6b | sctp | sd | sessmgr | sigmgr | swm | sww | sta | rpy | dpy | pfcf | nserv |
n3iwf-ngap**

specifies the events



NOTE:

The n3iwf-ngap is valid only with the config>mobile>profile>call-insight>all-ues context.

6.2.1.4.5 format

Syntax

```
[no] format [pcap | simulated-pcap]
```

Context

```
config>mobile>profile>call-insight>all-ues
```

```
config>mobile>profile>call-insight>ue
```

Description

This command configures output file format.

Parameters

pcap

specifies a pcap requiring plugin

simulated-pcap

specifies a pcap without plugin

Default: simulated-pcap

6.2.1.4.6 live-output

Syntax

```
[no] live-output ip-address | fqdn [port port] [router router-instance]
```

Context

```
config>mobile>profile>call-insight>all-ues
```

```
config>mobile>profile>call-insight>ue
```

Description

This command configures where to resend each captured message/event.

Parameters

ip-address* | *fqdn

specifies the address

Values	ipv4-address	a.b.c.d (host bits must be 0)
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D
–	fqdn	255 chars max (per label max 63)

port

specifies the port
Values: 1 to 65535
Default: 29770

router-instance

specifies the router

Values	router-name	base
–	service-id	1 to 2147483647

Default: Base

6.2.1.4.7 path-management

Syntax

[no] path-management [ref-point {all | ga | gn | gp | gx | gxc | rf | ro | s1 | s11 | s12 | s2a | s2b | s4 | s5 | s6b | s8 | sd | sta | swm | sx-n4 | n3 | n9 | n2} | sba-service {all | nnrf-disc | nnrf-nfm | nnrf-oauth2}]

Context

```
config>mobile>profile>call-insight>all-ues
```

Description

This command enables path management messages capture.

6.2.1.4.8 ref-point

Syntax

```
[no] ref-point {all | dhcp | ga | gn | gp | gx | gxc | l2tp | pi | radius | rf | ro | s1 | s11 | s12 |  
s2a | s2b | s4 | s5 | s6b | s8 | sd | swm | swu | swu-cleartext | sww | sta | dns | swu-esp-drop |  
sx-n4 | n2}
```

Context

```
config>mobile>profile>call-insight>all-ues
```

```
config>mobile>profile>call-insight>ue
```

Description

This command configures interfaces to capture control messages related to the traced UE or for all-ues. The call trace debug commands can be used to enable tracing for the configured reference points; refer to the *Call trace and packet capture* section, in *CMG Configuration Guide*.

Parameters

**all | dhcp | ga | gn | gp | gx | gxc | l2tp | pi | radius | rf | ro | s1 | s11 | s12 | s2a | s2b | s4 | s5 | s6b | s8 |
sd | swm | swu | swu-cleartext | sww | sta | dns | swu-esp-drop | sx-n4 | n2**

specifies the reference point type



NOTE:

The l2tp capturing and swu-esp-drop parameters apply only to an all-ues profile. Capturing includes all L2TP control packets from the tunnel establish to the tunnel tear down. For ZLB, HELLO and all the messages after CDN and before ICRQ, the GW-Id and UE-Id attributes are not filled in the custom header.

Default: all-ues: none | ue: all

6.2.1.4.9 sba-service

Syntax

```
[no] sba-service {all | namf-comm | namf-evts | nchf-convergedcharging | npcf-smpolicycontrol | nsmf-event-exposure | nsmf-pdusession | nudm-sdm | nudm-uecm}
```

Context

```
config>mobile>profile>call-insight>all-ues
```

```
config>mobile>profile>call-insight>ue
```

Description

This command configures SBA services to capture control messages related to the traced UE or for all UEs. The call trace debug commands can enable tracing for the configured reference points; refer to the *Call Trace Debug Commands* section, in *CMG Configuration Guide*.

Parameters

all | namf-comm | namf-evts | nchf-convergedcharging | npcf-smpolicycontrol | nsmf-event-exposure | nsmf-pdusession | nudm-sdm | nudm-uecm

specifies the reference point type

Default: all-ues: none | ue: all

6.2.1.4.10 size-limit

Syntax

```
[no] size-limit size-limit
```

Context

```
config>mobile>profile>call-insight>all-ues
```

```
config>mobile>profile>call-insight>ue
```

Description

This command configures maximum data volume in output.

Parameters

size-limit

specifies the size limit in MB

Values	all-ues	[0..1000]; 0 means unlimited
–	ue	[1..1000]

Default: all-ues 1000
ue: 10

6.2.1.4.11 time-limit

Syntax

[no] time-limit *time-limit*

Context

config>mobile>profile>call-insight>all-ues
config>mobile>profile>call-insight>ue

Description

This command configures the maximum time limit for output.

Parameters

time-limit
specifies the size limit in seconds
Values: 0 to 604800; 0 means unlimited
Default: 86400

6.2.1.4.12 user-traffic

Syntax

[no] user-traffic [{internal | statistics | ingress | egress | metadata}]

Context

config>mobile>profile>call-insight>ue

Description

This command enables user plane call trace and statistics. User plane call trace and statistics can be used for debugging forwarding plane issues for end-to-end application by capturing user plane traffic in PCAP format and collecting statistics. This captured user-traffic information contains the following:

1. a set of counters at ingress and egress direction of the LB-VMs and internally in the MG-VMs
2. a PCAP file containing user-plane packets from all the reference points captured once and without the tunneling and transport headers internally in MG-VM
3. a PCAP file containing user-plane packets at ingress and egress direction of the LB VMs

Only one PCAP file is created with the `call-trace debug` command, so it is not recommended to use all the user-traffic attributes together, for example, ingress, egress and internal at the same time. The PCAP file contains the same packet several times.

The `no` form of this command disables the user plane capturing in the profile.

Default

`no user-traffic`

Parameters

`internal`

specifies that user-plane packets are captured once, without tunneling and transport headers, and user-plane packets statistics are also collected

`statistics`

specifies that only debug statistics are captured

`ingress`

Specifies the user plane packets that are captured at ingress point of LB-VM. Capturing is done with transport and tunneling headers. User plane packets statistics are also collected.

`egress`

Specifies the user plane packets that are captured at egress point of LB-VM. Capturing is done with transport and tunneling headers. User plane packets statistics are also collected.

`metadata`

specifies the enabling of policy and charging control metadata in user plane PCAP files



NOTE:

Parameters `internal`, `statistics`, `ingress`, `egress`, and `metadata` are not supported in hardware.

6.2.1.5 Canary test profile command descriptions

6.2.1.5.1 canary-test

Syntax

canary-test

Context

config>mobile>profile

Description

This command enables the context to configure the canary test profile parameters.

6.2.1.5.2 user-range

Syntax

[no] user-range *user-range-name*

Context

config>mobile>profile>canary-test

Description

This command configures a user range name for the canary testing.

Parameters

user-range-name

specifies the user range name for the canary testing, up to 32 characters

Default

None

6.2.1.6 ue-entry

Syntax

```
ue-entry ue-entry-id ue-type ue-type start-id start-id [end-id end-id]  
no ue-entry ue-entry-id [ue-type ue-type] [start-id start-id] [end-id end-id]
```

Context

```
config>mobile>profile>canary-test>user-range>ue-entry
```

Description

This command configures multiple UE ranges based on the IMSI/MSISDN types.

Parameters

ue-entry-id

specifies the UE entry id

Values: 1 to 256

ue-type

specifies the UE type

Values: imsi | msisdn

start-id

specifies the start identifier for the user range, up to 15 characters; IMSIs must only have 15 characters and MSISDNs must have a minimum length of nine characters

end-id

specifies the end identifier for the user range, up to 15 characters; IMSIs must only have 15 characters and MSISDNs must have a minimum length of nine characters

Default

None

6.2.1.7 Charging profile ePDG, PGW, SGW, TDF and UP function command descriptions

6.2.1.7.1 charging

Syntax

charging

Context

config>mobile>profile

Description

This command enables the context to configure the charging profile for the mobile gateway.

6.2.1.7.2 epdg-charging

Syntax

[no] epdg-charging *charging-profile-id*

Context

config>mobile>profile>charging

Description

This command configures an ePDG charging profile for configuring ePDG charging options.

The no form of this command reverts to the default.

Parameters

charging-profile-id

specifies a charging profile identifier

Values: 0 to 255

6.2.1.7.3 pgw-charging

Syntax

[no] pgw-charging *charging-profile-id*

Context

config>mobile>profile>charging

Description

This command configures a PGW charging profile for configuring PGW charging options. When a PGW charging profile is no longer used in shutdown state, it is deleted without a PDN shutdown.

Parameters

charging-profile-id

specifies a profile charging identifier

Values: 0 to 255

This is the Octet 1 as received in the Charging Characteristics IE.

6.2.1.7.4 accounting-priority-profile

Syntax

[no] accounting-priority-profile *profile-name*

Context

config>mobile>profile>charging>pgw-charging

Description

This command configures an accounting priority profile.

The no form of this command removes the specified profile from the configuration.

Parameters

profile-name

specifies the accounting priority profile name, up to 32 characters

6.2.1.7.5 sgw-charging

Syntax

[no] sgw-charging *charging-profile-id*

Context

config>mobile>profile>charging

Description

This command configures an SGW charging profile for configuring SGW charging options. When an SGW charging profile is no longer used in shutdown state, it is deleted without a PDN shutdown.

Parameters

charging-profile-id

specifies a profile charging identifier

Values: 0 to 255

This is the Octet 1 as received in the Charging Characteristics IE.

6.2.1.7.6 data-volume-count-time

Syntax

data-volume-count-time start-time *time* end-time *time*

Context

config>mobile>profile>charging>sgw-charging

Description

This command configures a time bracket during which the customized functionality applies and is only valid when *custom-2* is configured in the config>mobile>pdn>gx>supported-vendor command.

Parameters

start-time *time*

specifies the start time of data volume count

Values: [hh:mm]

Default: 00:00

end-time *time*

specifies the end time of data volume count

Values: [hh:mm]

Default: 00:00

6.2.1.7.7 tdf-charging

Syntax

[no] tdf-charging *charging-profile-id*

Context

config>mobile>profile>charging

Description

This command configures a TDF charging profile for configuring the TDF charging options.

Parameters

charging-profile-id

specifies a profile charging identifier

Values: 0 to 255

Default: 0

6.2.1.7.8 upf-acct-profile

Syntax

[no] upf-acct-profile *profile-id*

Context

config>mobile>profile>charging

Description

This command configures a UP function charging profile for configuring UP function charging options.

Parameters

profile-id

specifies a UP function charging profile identifier

Values: 0 to 255

6.2.1.7.9 faster-poll-remaining-quota-trigger

Syntax

```
faster-poll-remaining-quota-trigger value  
no faster-poll-remaining-quota-trigger
```

Context

```
config>mobile>profile>charging>pgw-charging  
config>mobile>profile>charging>upf-acct-profile
```

Description

This command configures an absolute remaining quota threshold used for CMG enhanced statistics polling instead of the default percentage (80%). When the remaining quota is less than or equal to the configured threshold value, faster polling is applied to the statistics object.

The no form of this command reverts to the default (disabled), that is, the default threshold (80%) for enhanced statistics polling.

Default

no faster-poll-remaining-quota-trigger



NOTE:

- This command is supported for non-CUPS and CUPS UP function gateways. For non-CUPS deployments, this command is configured under the config>mobile>profile>charging>pgw-charging context. For CUPS UP function, this command is configured under the config>mobile>profile>charging>upf-acct-profile context, and the profile is applied at the PDN or APN level.
- This feature is only applicable when the CMG has a dedicated core allocated for statistics polling. For more information, refer to the *Allocating a core for enhanced statistics polling* section in the *CMG Installation Guide*.

Parameters

value

specifies the absolute remaining quota threshold in megabytes, used for statistics polling instead of the default percentage (80%)

Values: 0 to 100

Default: 0

6.2.1.710 force-sgw-cdr-in-combo-gw

Syntax

```
[no] force-sgw-cdr-in-combo-gw
```

Context

```
config>mobile>profile>charging>pgw-charging
```

Description

This command enables the generation of SGW-CDRs and PGW-CDRs for the same bearer. When this command is enabled, sessions that use both the SGW and PGW interfaces of the combined SGW/PGW, generate SGW-CDRs and PGW-CDRs. When this command is disabled, the SGW/PGW will generate only PGW CDRs per session according to the applicable charging profile configuration.

6.2.1.711 nchf-charging

Syntax

```
nchf-charging {enable | disable}  
no nchf-charging
```

Context

```
config>mobile>profile>charging>pgw-charging  
config>mobile>profile>charging>sgw-charging
```

Description

This command configures the LTE legacy charging at the charging profile level. When the `nchf-charging` command is disabled (that is, the LTE legacy charging is enabled), the 5G UEs and LTE UEs with 5GS subscription that attach to the SMF and the PGW-C or SGW-C use the LTE legacy charging interfaces (Gy, Ga, and Bp in case of PGW, or Ga and Bp in case of SGW). By default, such sessions use the Nchf interface for charging (the `nchf-charging` command is enabled).

Parameters

enable

specifies to enable the Nchf charging

disable

specifies to disable the Nchf charging and applies the SMF legacy charging

Default: disable

6.2.1.7.12 off-line

Syntax

off-line

Context

config>mobile>profile>charging>epdg-charging

config>mobile>profile>charging>pgw-charging

config>mobile>profile>charging>sgw-charging

config>mobile>profile>charging>tdf-charging

Description

This command configures offline charging for this charging profile.

6.2.1.7.13 charging-info-trigger

Syntax

charging-info-trigger

Context

config>mobile>profile>charging>epdg-charging>off-line

config>mobile>profile>charging>pgw-charging>off-line

config>mobile>profile>charging>sgw-charging>off-line

config>mobile>profile>charging>tdf-charging>off-line

Description

This command enables the context to configure triggers.

6.2.1.7.14 apn-rate-control-change

Syntax

[no] apn-rate-control-change

Context

config>mobile>profile>charging>pgw-charging>off-line>cit

Description

This command enables the APN rate control change event trigger. The current traffic data container in the charging record is closed and a new container is opened when an APN rate control change occurs.



NOTE:

APN rate control changes are allowed when the gateway is up and affect new sessions only. The existing sessions will continue to operate under the previous settings and the charging APN rate control change trigger will not occur.

The no form of this command reverts to the default (disabled).

Default

no apn-rate-control-change

6.2.1.7.15 cgi-sai-change

Syntax

[no] cgi-sai-change

Context

config>mobile>profile>charging>pgw-charging>off-line>cit

config>mobile>profile>charging>sgw-charging>off-line>cit

config>mobile>profile>charging>tdf-charging>off-line>cit

Description

This command is used to enable CGI-SAI change location event trigger. CGI-SAI-change is a charging information addition trigger. User Location Information contains the location (for example CGI/SAI, ECGI/TAI or RAI) where the UE is located and used during the transfer of the data volume captured by the

container (applicable only to the SGW-CDR). This is included in the Traffic data container only if previous container's change condition is "user location change". User location information contains the user location (for example CGI/SAI, ECGI/TAI or RAI) where the UE was located during the service data container (applicable to PGW-CDR & TDF-CDR) recording interval. This is included in the service data container only if previous container's change condition is "user location change" or one of the "CGI/SAI, ECGI or TAI or RAI Change". When this (CGI-SAI-change) event trigger occurs, the max-number of change conditions partial record trigger is incremented. For Ga, a partial CDR is closed if the configured max number of change conditions is met. In the case of Rf, an ACR interim is sent out to the OFCS if the configured max number of change conditions is met.

The no form of this command reverts to default (disabled).

Default

no cgi-sai-change

6.2.1.7.16 change-csg-cell

Syntax

[no] change-csg-cell

Context

config>mobile>profile>charging>pgw-charging>off-line>cit

config>mobile>profile>charging>sgw-charging>off-line>cit

Description

This command is used to enable the gateway to report user CSG information change to the OFCS when the UE enters, leaves, or accesses via a CSG cell. This results in SDC (PGW) or TDV (SGW) containers being added to the CDR. For the PGW context, the PCRF specified CSG information reporting AVP takes precedence over locally configured event triggers for offline charging.

The no form of this command reverts to default (disabled).

Default

no change-csg-cell

6.2.1.7.17 change-csg-subscribed-hybrid-cell

Syntax

[no] change-csg-subscribed-hybrid-cell

Context

```
config>mobile>profile>charging>pgw-charging>off-line>cit
```

```
config>mobile>profile>charging>sgw-charging>off-line>cit
```

Description

This command is used to enable the gateway to report user CSG information change to the OFCS when the UE enters, leaves, or accesses via a hybrid cell in which the subscriber is a CSG member. This results in SDC containers (PGW) or TDV containers (SGW) being added to the CDR. For the PGW context, the PCRF specified CSG Information Reporting AVP takes precedence over locally configured event trigger for offline charging.

The no form of this command reverts to default (disabled).

Default

no change-csg-subscribed-hybrid-cell

6.2.1.7.18 change-in-user-plane

Syntax

```
[no] change-in-user-plane
```

Context

```
config>mobile>profile>charging>sgw-charging>off-line>cit
```

Description

This command specifies whether change in UP to UE trigger is enabled. This results in TDV containers (SGW) being added to the CDR if the value of the CP Clot EPS optimization indicator changes.

The no form of this command reverts to the default (disabled).

Default

no change-in-user-plane

6.2.1.7.19 change-ue-presence-pra

Syntax

```
[no] change-ue-presence-pra
```

Context

```
config>mobile>profile>charging>pgw-charging>off-line>cit
```

```
config>mobile>profile>charging>sgw-charging>off-line>cit
```

```
config>mobile>profile>charging>tdf-charging>off-line>cit
```

Description

This command is used to enable the change in UE presence in the Presence Reporting Area (PRA) charging information addition trigger for offline charging. During IP-CAN session establishment, a PRA for the PDN connection is determined. The PCRF, when deployed, may provide a PRA to the PGW. The PCRF-selected PRA is then delivered to the TDF and OCS. When the PCRF does not provide a PRA, the OCS may provide the PRA to the PGW. Only one PRA may be selected per PDN connection. For this offline charging trigger to be supported, either PCRF or OCS needs to provide the PRA Identifier to the gateway.

The no form of this command reverts to the default (disabled).

Default

no change-ue-presence-pra

6.2.1.7.20 ecgi-change

Syntax

```
[no] ecgi-change
```

Context

```
config>mobile>profile>charging>pgw-charging>off-line>cit
```

```
config>mobile>profile>charging>sgw-charging>off-line>cit
```

```
config>mobile>profile>charging>tdf-charging>off-line>cit
```

Description

This command is used to enable eCGI change location event trigger. eCGI-change is a charging information addition trigger. User Location Information contains the location (for example CGI/SAI, ECGI/TAI or RAI) where the UE is located and used during the transfer of the data volume captured by the container (applicable only to the SGW-CDR). This is included in the Traffic data container only if previous container's change condition is "user location change". User location information contains the user location (for example CGI/SAI, ECGI/TAI or RAI) where the UE was located during the service data container (applicable to PGW-CDR and TDF-CDR) recording interval. This is included in the service data container only if previous container's change condition is "user location change" or one of the "CGI/SAI, ECGI or TAI or RAI Change". When this (eCGI-change) event trigger occurs, the max-number of change conditions partial record trigger

is incremented. For Ga, a partial CDR is closed if the configured max number of change conditions is met. In the case of Rf, an ACR interim is sent out to the OFCS if the configured max number of change conditions is met.

The no form of this command reverts to default (disabled).

Default

no ecgi-change

6.2.1.7.21 fh-close-cdr-containers

Syntax

[no] fh-close-cdr-containers

Context

config>mobile>profile>charging>pgw-charging>off-line>cit

Description

When this command is enabled, if the PDN session is under online charging control and a failure handling begins, the PGW closes the CDR containers. The closed PGW-CDR containers include one of the following reasons in ServiceConditionChange:

- dCCAContinueOngoingSession (18)
- dCCATerminateOngoingSession (20)



NOTE:

FailureHandlingContinue CDR field is not present in those closed containers.

If DCCA failure handling is active, all the subsequent CDRs are marked with the FailureHandlingContinue field. This allows the operator to use offline charging for data that was potentially missed by online charging due to DCCA failure handling. When the failure handling mechanism stops (for example, OCS is reachable again), when this command is enabled, all the existing open CDR containers are closed by setting the configuration change bit configurationChange (8) in the serviceConditionChange field. The CDR containers closed by this reason also include the failureHandlingContinue field, but all the subsequent containers do not include, since what they report is also reported to OCS via the Gy.

The no form of this command reverts to the default state (PGW-CDR containers are not closed when failure handling starts).

Default

no fh-close-cdr-containers

6.2.1.7.22 fh-close-cdr-exclude-dcca-result-code-4010

Syntax

[no] fh-close-cdr-exclude-dcca-result-code-4010

Context

config>mobile>profile>charging>pgw-charging>off-line>cit

Description

This command configures the mobile gateway to mark the corresponding PGW-CDR with ServiceConditionChange dccaContinueOngoingSession(18), dCCARetryAndTerminateOngoingSession(19), and dCCATerminateOngoingSession(20), when the OCS sends the result code 4010.

This command is effective only when the fh-close-cdr-containers command is enabled.

By default, when the fh-close-cdr-containers command is enabled, the PGW-CDRs are marked with ServiceConditionChange when failure handling is triggered due to an OCS error.

The no form of this command reverts to the default state (disabled).

Default

no fh-close-cdr-exclude-dcca-result-code-4010

6.2.1.7.23 fh-close-cdr-exclude-dcca-result-code-4012

Syntax

[no] fh-close-cdr-exclude-dcca-result-code-4012

Context

config>mobile>profile>charging>pgw-charging>off-line>cit

Description

This command configures the mobile gateway to mark the corresponding PGW-CDR with ServiceConditionChange dccaContinueOngoingSession(18), dCCARetryAndTerminateOngoingSession(19), and dCCATerminateOngoingSession(20), when the OCS sends the result code 4012.

This command is effective only when the `fh-close-cdr-containers` command is enabled.

By default, when the `fh-close-cdr-containers` command is enabled, the PGW-CDRs are marked with `ServiceConditionChange` when failure handling is triggered due to an OCS error.

The no form of this command reverts to the default state (disabled).

Default

`no fh-close-cdr-exclude-dcca-result-code-4012`

6.2.1.7.24 fh-close-cdr-exclude-dcca-result-code-5005

Syntax

`[no] fh-close-cdr-exclude-dcca-result-code-5005`

Context

`config>mobile>profile>charging>pgw-charging>off-line>cit`

Description

This command configures the mobile gateway to mark the corresponding PGW-CDR with `ServiceConditionChange dccaContinueOngoingSession(18)`, `dCCARetryAndTerminateOngoingSession(19)`, and `dCCATerminateOngoingSession(20)`, when the OCS sends the result code 5005.

This command is effective only when the `fh-close-cdr-containers` command is enabled.

By default, when the `fh-close-cdr-containers` command is enabled, the PGW-CDRs are marked with `ServiceConditionChange` when failure handling is triggered due to an OCS error.

The no form of this command reverts to the default state (disabled).

Default

`no fh-close-cdr-exclude-dcca-result-code-5005`

6.2.1.7.25 fh-close-cdr-exclude-dcca-result-code-5030

Syntax

`[no] fh-close-cdr-exclude-dcca-result-code-5030`

Context

`config>mobile>profile>charging>pgw-charging>off-line>cit`

Description

This command configures the mobile gateway to mark the corresponding PGW-CDR with ServiceConditionChange dccaContinueOngoingSession(18), dCCARetryAndTerminateOngoingSession(19), and dCCATerminateOngoingSession(20), when the OCS sends the result code 5030.

This command is effective only when the `fh-close-cdr-containers` command is enabled.

By default, when the `fh-close-cdr-containers` command is enabled, the PGW-CDRs are marked with ServiceConditionChange when failure handling is triggered due to an OCS error.

The no form of this command reverts to the default state (disabled).

Default

no fh-close-cdr-exclude-dcca-result-code-5030

6.2.1.7.26 gy-final

Syntax

[no] gy-final

Context

config>mobile>profile>charging>pgw-charging>off-line>cit

Description

This command enables the online and offline tight integration trigger. When a CCR-U is sent to the OCS for a rating group with a “Reporting-Reason = Final”, the corresponding SDC containers for that rating group are cached. As a result, the maximum number of change conditions, set under the `partial-record-closure-trigger` command, is incremented. An ACR interim message is sent to the OFCS if the configured maximum number of change conditions is met.



NOTE:

- This trigger is supported for the Rf interface only and must not be used for Ga charging.
- Online and offline charging must be enabled for the subscriber session (`enable-offline/enable-online`).

The no form of this command disables the integration trigger.

Default

no gy-final

6.2.1.7.27 gy-forced-reauthorization

Syntax

[no] gy-forced-reauthorization

Context

config>mobile>profile>charging>pgw-charging>off-line>cit

Description

This command enables the online and offline tight integration trigger. When a CCR-U is sent to the OCS for a rating group with a “Reporting-Reason = Forced-Reauthorization”, the corresponding SDC containers for that rating group are cached. As a result, the maximum number of change conditions, set under the partial-record-closure-trigger command, is incremented. An ACR interim message is sent to the OFCS if the configured maximum number of change conditions is met.



NOTE:

- This trigger is supported for the Rf interface only and must not be used for Ga charging.
- Online and offline charging must be enabled for the subscriber session (enable-offline/enable-online).

The no form of this command disables the integration trigger.

Default

no gy-forced-reauthorization

6.2.1.7.28 gy-qht

Syntax

[no] gy-qht

Context

config>mobile>profile>charging>pgw-charging>off-line>cit

Description

This command enables the online and offline tight integration trigger. When a CCR-U is sent to the OCS for a rating group with a “Reporting-Reason = QHT” (Quota Holding Time), the corresponding SDC containers for that rating group are cached. As a result, the maximum number of change conditions, set under the `partial-record-closure-trigger` command, is incremented. An ACR interim message is sent to the OFCS if the configured maximum number of change conditions is met.

**NOTE:**

- This trigger is supported for the Rf interface only and must not be used for Ga charging.
- Online and offline charging must be enabled for the subscriber session (`enable-offline/enable-online`).

The `no` form of this command disables the integration trigger.

Default

`no gy-qht`

6.2.1.7.29 gy-quota-exhausted

Syntax

`[no] gy-quota-exhausted`

Context

`config>mobile>profile>charging>pgw-charging>off-line>cit`

Description

This command is used to enable an online and offline tight integration trigger. When a CCR-U is sent to the OCS for a rating group with a Reporting-Reason = QUOTA-EXHAUSTED, the corresponding SDC containers for that rating group are cached. As a result, the maximum number of change conditions, set under the `partial-record-closure-trigger` command, is incremented. An ACR interim is sent out to the OFCS if the configured maximum number of change conditions is met.

**NOTE:**

- This trigger is supported for the Rf interface only and must not be used for Ga charging.
- Online and offline charging must be enabled for the subscriber session (`enable-offline/enable-online`).

The no form of this command disables the integration trigger.

Default

no-gy-quota-exhausted

6.2.1.730 gy-rating-condition-change

Syntax

[no] gy-rating-condition-change

Context

config>mobile>profile>charging>pgw-charging>off-line>cit

Description

This command enables the online and offline tight integration trigger. When a CCR-U is sent to the OCS for a rating group with a “Reporting-Reason = Rating-Condition-Change”, the corresponding SDC containers for that rating group are cached. As a result, the maximum number of change conditions, set under the partial-record-closure-trigger command, is incremented. An ACR interim message is sent to the OFCS if the configured maximum number of change conditions is met.



NOTE:

- This trigger is supported for the Rf interface only and must not be used for Ga charging.
- Online and offline charging must be enabled for the subscriber session (enable-offline/enable-online).

The no form of this command disables the integration trigger.

Default

no gy-rating-condition-change

6.2.1.731 gy-tariff-time-change

Syntax

[no] gy-tariff-time-change

Context

```
config>mobile>profile>charging>pgw-charging>off-line>cit
```

Description

This command is used to enable the Tariff-Time Change (TTC) online/offline tight integration trigger. The OCS subscribes to the TTC in a CCA. When the TTC occurs, the corresponding SDC containers for that rating group are closed and cached. As a result, the maximum number of change conditions partial record trigger is incremented. An ACR interim is sent out to the OFCS if the configured maximum number of change conditions is met. Two instances of USU (one for usage before TTC and one for usage after TTC) are included in the next CCR to the OCS as per the existing implementation.



NOTE:

- This trigger is supported for the Rf interface only and must not be used for Ga charging.
- Online and offline charging must be enabled for the subscriber session (enable-le-offline/enable-online).

The no form of this command disables the integration trigger.

Default

```
no gy-tariff-time-change
```

6.2.1.7.32 gy-threshold-vqt

Syntax

```
[no] gy-threshold-vqt
```

Context

```
config>mobile>profile>charging>pgw-charging>off-line>cit
```

Description

This command is used to enable an online and offline tight integration trigger. When a CCR-U is sent to the OCS for a rating group with a Reporting-Reason = THRESHOLD, the corresponding SDC containers for that gating group are cached. As a result of this, the maximum number of change conditions partial record trigger is incremented. An ACR interim is sent out to the OFCS if the configured maximum number of change conditions is met.



NOTE:

- This trigger is supported for the Rf interface only and must not be used for Ga charging.
- Online and offline charging must be enabled for the subscriber session (enable-offline/enable-online).

The no form of this command disables the integration trigger.

Default

no gy-threshold-vqt

6.2.1.733 gy-validity-time

Syntax

[no] gy-validity-time

Context

config>mobile>profile>charging>pgw-charging>off-line>cit

Description

This command enables an online/offline tight integration trigger. When a CCR-U is sent to the OCS for a rating group with a “Reporting-Reason = Validity-Time”, the corresponding SDC containers for that rating group are cached. As a result, the maximum number of change conditions, set under the partial-record-closure-trigger command, is incremented. An ACR interim message is sent to the OFCS if the configured maximum number of change conditions is met.



NOTE:

- This trigger is supported for the Rf interface only and must not be used for Ga charging.
- Online and offline charging must be enabled for the subscriber session (enable-offline/enable-online).

The no form of this command disables the integration trigger.

Default

no gy-validity-time

6.2.1.734 plmn-rate-control-change

Syntax

[no] plmn-rate-control-change

Context

config>mobile>profile>charging>pgw-charging>off-line>cit

config>mobile>profile>charging>sgw-charging>off-line>cit

Description

This command enables the PLMN rate control change event trigger. The current traffic data container in the charging record is closed and a new container is opened when a PLMN rate control change occurs.

The no form of this command reverts to the default (disabled).

Default

no plmn-rate-control-change

6.2.1.735 qos-change

Syntax

[no] qos-change

Context

config>mobile>profile>charging>epdg-charging>off-line>cit

config>mobile>profile>charging>pgw-charging>off-line>cit

config>mobile>profile>charging>sgw-charging>off-line>cit

Description

This command specifies whether the QoS change trigger is enabled, which results in List of traffic data volumes container charging information added to the charging data record (CDR) when there is a change in QoS.

The no form of this command disables the QoS change trigger.

6.2.1.7.36 rai-change

Syntax

[no] rai-change

Context

config>mobile>profile>charging>pgw-charging>off-line>cit

config>mobile>profile>charging>sgw-charging>off-line>cit

config>mobile>profile>charging>tdf-charging>off-line>cit

Description

This command is used to enable RAI change location event trigger. RAI-change is a charging info addition trigger. User Location Information contains the location (for example CGI/SAI, ECGI/TAI or RAI) where the UE is located and used during the transfer of the data volume captured by the container (applicable only to the SGW-CDR). This is included in the Traffic data container only if previous container's change condition is "user location change". User location information contains the user location (for example CGI/SAI, ECGI/TAI or RAI) where the UE was located during the service data container (applicable to PGW-CDR & TDF-CDR) recording interval. This is included in the service data container only if previous container's change condition is "user location change" or one of the "CGI/SAI, ECGI or TAI or RAI Change". When this (RAI-change) event trigger occurs, the maximum number of change conditions partial record trigger is incremented. For Ga, a partial CDR is closed if the configured maximum number of change conditions is met. In the case of Rf, an ACR interim is sent out to the OFCS if the configured maximum number of change conditions is met.

The no form of this command reverts to default (disabled).

Default

no rai-change

6.2.1.7.37 serving-node-change

Syntax

[no] serving-node-change

Context

config>mobile>profile>charging>tdf-charging>off-line>cit

Description

This command specifies whether the serving-node change trigger is enabled, which results in a list of service data container charging information added to the charging data record (CDR) when there is a change in serving node (SGSN, SGW, or ePDG).

The no form of this command reverts to the default (disabled).

Default

no serving-node-change

6.2.1.7.38 tai-change

Syntax

[no] tai-change

Context

config>mobile>profile>charging>pgw-charging>off-line>cit

config>mobile>profile>charging>sgw-charging>off-line>cit

config>mobile>profile>charging>tdf-charging>off-line>cit

Description

This command is used to enable the TAI change location event trigger. TAI-change is a charging information addition trigger. User Location Information contains the location (for example CGI/SAI, ECGI/TAI, or RAI) where the UE is located and used during the transfer of the data volume captured by the container (applicable only to the SGW-CDR). This is included in the Traffic data container only if the previous container's change condition is "user location change". User location information contains the user location (for example CGI/SAI, ECGI/TAI, or RAI) where the UE was located during the service data container (applicable to PGW-CDR and TDF-CDR) recording interval. This is included in the service data container only if previous container's change condition is "user location change" or one of the "CGI/SAI, ECGI or TAI or RAI Change". When this (TAI-change) event trigger occurs, the maximum number of change conditions partial record trigger is incremented. For Ga, a partial CDR is closed if the configured maximum number of change conditions is met. In the case of Rf, an ACR interim is sent out to the OFCS if the configured maximum number of change conditions is met.

The no form of this command reverts to default (disabled).

Default

no tai change

6.2.1.739 tariff-change-list

Syntax

```
tariff-change-list list-name  
no tariff-change-list
```

Context

```
config>mobile>profile>charging>epdg-charging>off-line>cit  
config>mobile>profile>charging>event-triggers-fbc  
config>mobile>profile>charging>pgw-charging>off-line>cit  
config>mobile>profile>charging>sgw-charging>off-line>cit  
config>mobile>profile>charging>tdf-charging>off-line>cit
```

Description

This command is used to specify a tariff change list profile to be applied for the tariff change charging info trigger. Multiple tariff change times can be specified within the profile to facilitate differentiated charging at different times of day. Each tariff change entry must have a unique time of day. The (UE) time of day is then used to generate a tariff change event for charging records. Tariff change is a charging info addition trigger. The usage data containers (TDV and SDC containers) are added to the CDR at the specified time. As a result of this, the maximum number of change conditions partial record trigger is incremented. For Ga, a partial CDR is closed if the configured max number of change conditions is met. In the case of Rf, an ACR interim is sent out to the OFCS if the configured maximum number of change conditions is met.

The no form of this command removes the specified tariff change-list profile association for this context.

Parameters

prof-name

specifies the name of the profile, up to 32 characters

6.2.1.740 tariff-time

Syntax

```
tariff-time start time end time
```

Context

```
config>mobile>profile>charging>epdg-charging>off-line>cit
```

```
config>mobile>profile>charging>pgw-charging>off-line>cit
```

```
config>mobile>profile>charging>sgw-charging>off-line>cit
```

```
config>mobile>profile>charging>tdf-charging>off-line>cit
```

Description

This command configures the tariff time for charging profile.

The no form of this command reverts to the default.

Parameters

time

specifies the start of the tariff time for the CDR

Values: [hh:mm]

end time

specifies the end of the tariff time for the CDR

Values: [hh:mm]

Default: 0

6.2.1.7.41 tariff-time-change

Syntax

```
[no] tariff-time-change
```

Context

```
config>mobile>profile>charging>epdg-charging>off-line>cit
```

```
config>mobile>profile>charging>pgw-charging>off-line>cit
```

```
config>mobile>profile>charging>sgw-charging>off-line>cit
```

```
config>mobile>profile>charging>tdf-charging>off-line>cit
```

Description

This command enables the tariff time change trigger, which results in a List of Traffic Data Volumes container charging information added to the CDR when there is a change in tariff time.

The no form of this command disables the tariff time change trigger.

Default

no tariff-time-change

6.2.1.7.42 time-limit-per-rating-grp

Syntax

```
time-limit-per-rating-grp time-limit-per-rating-grp  
no time-limit-per-rating-grp
```

Context

```
config>mobile>profile>charging>pgw-charging>off-line>cit
```

Description

This command specifies the SDC time limit (in minutes). When the specific time limit is reached, the currently open Service Data Container is closed and added in the PGW CDR. Setting the parameter to 0 disables the closure of the SDC due to the time limit.

Parameters

time-limit-per-rating-grp

specifies the SDC time limit (in minutes)

Values: 0 | 5 to 1440

Default: 30

6.2.1.7.43 user-loc-change

Syntax

```
[no] user-loc-change [cc-user-loc-change]
```

Context

```
config>mobile>profile>charging>pgw-charging>off-line>cit
```

```
config>mobile>profile>charging>sgw-charging>off-line>cit
```

```
config>mobile>profile>charging>tdf-charging>off-line>cit
```

Description

This command specifies whether the user location change trigger is enabled, which results in a list of traffic data volumes container charging information added to the CDR when there is a change in QoL or user location.

The no form of this command reverts to the default (disabled).

Default

no user-loc-change

Parameters

cc-user-loc-change

specifies that the change condition populated for the generic user-loc-change event trigger is user-location-change

6.2.1.7.44 uwan-loc-change

Syntax

[no] uwan-loc-change

Context

config>mobile>profile>charging>epdg-charging>off-line>cit

config>mobile>profile>charging>pgw-charging>off-line>cit

Description

This command enables or disables the UWAN location change trigger. When enabled, the UWAN User Location Information is added to the CDR when there is a change in the UE location.

The no form of this command reverts to the default.

Default

no uwan-loc-change

6.2.1.7.45 volume-limit-per-rating-grp

Syntax

volume-limit-per-rating-grp *volume-limit-per-rating-grp*

```
no volume-limit-per-rating-grp
```

Context

```
config>mobile>profile>charging>pgw-charging>off-line>cit
```

Description

This command specifies the SDC volume limit (in kbytes). When the specified volume limit is reached, the currently open service data container is closed and added in the PGW CDR.

Parameters

volume-limit-per-rating-grp

specifies the SDC volume limit (in kbytes)

Values: 0 to 104857600

Default: 0

6.2.1.7.46 enable-offline

Syntax

```
[no] enable-offline
```

Context

```
config>mobile>profile>charging>epdg-charging>off-line
```

```
config>mobile>profile>charging>pgw-charging>off-line
```

```
config>mobile>profile>charging>sgw-charging>off-line
```

Description

This command specifies whether offline charging is enabled for this charging profile. Offline charging is enabled or disabled for this charging profile without a PDN shutdown in SGW charging and PGW charging. On the fly, enabling or disabling of offline charging affects new sessions only.

The no form of this command reverts to the default.



NOTE:

This command requires a PDN shutdown, when used in the `config>mobile>profile>charging>epdg-charging` context.

Default

no enable-offline

6.2.1.7.47 enable-offline

Syntax

enable-offline [session-level | fbc | both]

no enable-offline

Context

config>mobile>profile>charging>tdf-charging>off-line

Description

This command specifies whether offline charging is enabled for this charging profile. Flow-based or session-level charging or both can be enabled.

The no form of this command disables offline charging for this charging profile.



NOTE:

This command requires a PDN shutdown.

Default

no enable-offline

Parameters

session-level

specifies that session level offline charging is enabled

fbc

specifies that FBC offline charging is enabled

both

specifies that both FBC and session level offline charging are enabled

Default: fbc

6.2.1.748 exclude-zero-vol-cdr

Syntax

```
[no] exclude-zero-vol-cdr
```

Context

```
config>mobile>profile>charging>epdg-charging>off-line
```

```
config>mobile>profile>charging>pgw-charging>off-line
```

```
config>mobile>profile>charging>sgw-charging>off-line
```

```
config>mobile>profile>charging>tdf-charging>off-line
```

Description

This command specifies whether partial CDRs that have zero volume will be generated. If enabled, partial CDRs with zero volume will not be generated.

The no form of this command reverts to the default (CDRs with zero volume are generated).

Default

no exclude-zero-vol-cdr

6.2.1.749 exclude-zero-vol-complete-cdr

Syntax

```
[no] exclude-zero-vol-complete-cdr
```

Context

```
config>mobile>profile>charging>epdg-charging>off-line
```

```
config>mobile>profile>charging>pgw-charging>off-line
```

```
config>mobile>profile>charging>sgw-charging>off-line
```

```
config>mobile>profile>charging>tdf-charging>off-line
```

Description

This command specifies whether zero volume CDRs (with zero or non-zero duration) are generated when the last CDR is also the first CDR of the session. If enabled, those CDRs are not generated. The no form of this command reverts to the default (CDRs with zero volume which also are the only CDRs for the session are generated).

Default

no exclude-zero-vol-complete-cdr

6.2.1.7.50 include-bearer-charging

Syntax

[no] include-bearer-charging

Context

config>mobile>profile>charging>pgw-charging>off-line

Description

This command enables the reporting of IP CAN bearer charging in eG-CDRs and PGW-CDRs within a List-Of-Service-Data CDR container, using the reserved rating group 0. The other rating groups are not affected and keep reporting Flow Based Charging.

The no form of this command reverts to the default behavior.

Default

no include-bearer-charging

6.2.1.7.51 partial-record-closure-trigger

Syntax

partial-record-closure-trigger

Context

config>mobile>profile>charging>epdg-charging>off-line

config>mobile>profile>charging>pgw-charging>off-line

config>mobile>profile>charging>sgw-charging>off-line

config>mobile>profile>charging>tdf-charging>off-line

Description

This command enables the context to configure the Partial Record Closure Trigger parameters.

6.2.1.7.52 max-change-condition

Syntax

```
max-change-condition max-change-condition  
no max-change-condition
```

Context

```
config>mobile>profile>charging>epdg-charging>off-line>part-rec-closure-trig  
config>mobile>profile>charging>pgw-charging>off-line>part-rec-closure-trig  
config>mobile>profile>charging>sgw-charging>off-line>part-rec-closure-trig  
config>mobile>profile>charging>tdf-charging>off-line>part-rec-closure-trig
```

Description

This command configures the maximum number of change conditions that can occur, such as a tariff change, location change, or QoS change, before a partial record is created.

Parameters

max-change-condition

specifies the maximum number of change conditions that can occur
Values: 0 to 32
Default: 0

6.2.1.7.53 min-data-threshold

Syntax

```
min-data-threshold min-data-threshold  
no min-data-threshold
```

Context

```
config>mobile>profile>charging>epdg-charging>off-line>part-rec-closure-trig  
config>mobile>profile>charging>pgw-charging>off-line>part-rec-closure-trig  
config>mobile>profile>charging>sgw-charging>off-line>part-rec-closure-trig  
config>mobile>profile>charging>tdf-charging>off-line>part-rec-closure-trig
```

Description

This command specifies the minimum data threshold in kilobytes for triggering time-based ePDG, PGW, SGW, or TDF CDRs. If this command is configured, time-based CDRs will be generated only if the total volume of the CDR is equal to or greater than the configured value. If this command is not configured, time-based CDRs will be generated regardless of the volume.

The no form of this command reverts to the default; that is, no value is set (time-based CDRs are generated independently of the volume reported).

Default

no min-data-threshold

Parameters

min-data-threshold

specifies the minimum data threshold value in kilobytes

Values: 0 to 1000000

Default: 0

6.2.1.7.54 mo-exception-data-receipt

Syntax

[no] mo-exception-data-receipt

Context

config>mobile>profile>charging>pgw-charging>off-line>part-rec-closure-trig

config>mobile>profile>charging>sgw-charging>off-line>part-rec-closure-trig

Description

This command enables the MO exception data receipt event trigger. A partial record is created when an MO exception data receipt event is detected.

The no form of this command reverts to the default (disabled).

Default

no mo-exception-data-receipt

6.2.1.7.55 ms-time-zone-change

Syntax

[no] ms-time-zone-change

Context

config>mobile>profile>charging>pgw-charging>off-line>part-rec-closure-trig

config>mobile>profile>charging>sgw-charging>off-line>part-rec-closure-trig

config>mobile>profile>charging>tdf-charging>off-line>part-rec-closure-tdf

Description

This command specifies whether the MS time zone change partial record trigger is enabled. Upon encountering an MS time zone change, a partial record is created.

Default

no ms-time-zone-change

6.2.1.7.56 plmn-change

Syntax

[no] plmn-change

Context

config>mobile>profile>charging>pgw-charging>off-line>part-rec-closure-trig

config>mobile>profile>charging>sgw-charging>off-line>part-rec-closure-trig

config>mobile>profile>charging>tdf-charging>off-line>part-rec-closure-trig

Description

This command enables the PLMN change partial record trigger. Upon encountering a PLMN change, a partial record is created.

The no form of this command disables the PLMN change partial record trigger.

Default

no plmn-change

6.2.1.757 rat-change

Syntax

[no] rat-change

Context

config>mobile>profile>charging>pgw-charging>off-line>part-rec-closure-trig

config>mobile>profile>charging>sgw-charging>off-line>part-rec-closure-trig

config>mobile>profile>charging>tdf-charging>off-line>part-rec-closure-trig

Description

This command enables the RAT change partial record trigger. Upon encountering a RAT change, a partial record is created.

The no form of this command disables the RAT change partial record trigger.

Default

no rat-change

6.2.1.758 serving-node-change-limit

Syntax

serving-node-change-limit *serving-node-change-limit*

no serving-node-change-limit

Context

config>mobile>profile>charging>pgw-charging>off-line>part-rec-closure-trig

config>mobile>profile>charging>sgw-charging>off-line>part-rec-closure-trig

Description

This command specifies the maximum limit for the change in serving node. A partial record is created when the specified change limit is satisfied.



NOTE:

If you set the parameter to 0, a maximum of 16 ServingNodeAddress/ServingNodeType IEs are included in the SGW CDRs on message level. The last address is always the most recent one. If two subsequent addresses/types have the same value, only one entry is included in the list.

Parameters

serving-node-change-limit

specifies the change limit for the partial record closure trigger

Values: 0 to 15

Default: 15

6.2.1.759 time-limit

Syntax

```
time-limit time-limit
```

```
no time-limit
```

Context

```
config>mobile>profile>charging>epdg-charging>off-line>part-rec-closure-trig
```

```
config>mobile>profile>charging>pgw-charging>off-line>part-rec-closure-trig
```

```
config>mobile>profile>charging>sgw-charging>off-line>part-rec-closure-trig
```

```
config>mobile>profile>charging>tdf-charging>off-line>part-rec-closure-trig
```

Description

This command configures a time limit applicable per session or bearer depending on the accounting level selected, for example PDN-level accounting or QCI and bearer-level accounting. A partial record is created when the specified time limit is satisfied. When the `time-limit-eutran` command is enabled, it applies to LTE and the `time-limit` command applies to all other RATs. When the `time-limit-eutran` is enabled, it applies to UTRAN Radio Access Technology, and the `time-limit` command applies to all other RATs that do not have an explicit time trigger. When the `time-limit-eutran` command is not enabled, the value specified for the `time-limit` command applies to E-UTRAN as well. Similarly, when the `time-limit-utran` command is not enabled, the value specified for the `time-limit` applies to UTRAN as well.

The no form of this command reverts to the default.

Default

no time-limit

Parameters

time-limit
specifies a time limit, in minutes, applicable per session or bearer

Values	PGW and SGW	5 to 1440
–	ePDG and TDF	0 to 1440
Default	PGW, SGW, and TDF	60
–	ePDG	0 (disabled)

6.2.1.7.60 **time-limit-eutran**

Syntax

`time-limit-eutran time-limit-eutran`
`no time-limit-eutran`

Context

`config>mobile>profile>charging>pgw-charging>off-line>part-rec-closure-trig`

Description

This command specifies a time limit applicable for E-UTRAN radio access technology. A partial record is created when the specified time-limit is reached for this RAT. When the `time-limit-eutran` command is enabled, it applies to LTE and the `time-limit` command applies to all other RATs. When the `time-limit-eutran` command is not enabled, the value specified for the `time-limit` command applies to E-UTRAN as well.

The no form of this command reverts to the default.

Parameters

time-limit-eutran
specifies a time limit, in minutes, applicable per session or bearer

Values: 0 to 1440

Default: 0

6.2.1.7.61 time-limit-utran

Syntax

```
time-limit-utran time-limit-utran
```

```
no time-limit-utran
```

Context

```
config>mobile>profile>charging>pgw-charging>off-line>part-rec-closure-trig
```

Description

This command specifies a time limit applicable for UTRAN radio access technology. A partial record is created when the specified time-limit is reached for this RAT. When the `time-limit-utran` command is enabled, it applies to UTRAN Radio Access Technology and the `time-limit` command applies to all other RATs that do not have an explicit time trigger. When the `time-limit-utran` command is not enabled, the value specified for the `time-limit` command applies to UTRAN as well.

The no form of this command reverts to the default.

Parameters

time-limit-utran

specifies a time limit, in minutes, applicable per session or bearer

Values: 0 to 1440

Default: 0

6.2.1.7.62 time-limit-eutran-nb-iot

Syntax

```
time-limit-eutran-nb-iot time-limit-eutran-nb-iot
```

```
no time-limit-eutran-nb-iot
```

Context

```
config>mobile>profile>charging>pgw-charging>off-line>part-rec-closure-trig
```


Description

This command specifies a time limit applicable for E-UTRAN radio access technology. A partial record is created when the specified time-limit is reached for this RAT. When the `time-limit-eutran-nb-iot` command is enabled, it applies to LTE and the `time-limit` command applies to all other RATs. When the `time-limit-eutran-nb-iot` command is not enabled, the value specified for the `time-limit` command applies to E-UTRAN as well.

The no form of this command reverts to the default.

Parameters

time-limit-eutran-nb-iot

specifies a time limit, in minutes, applicable per session or bearer

Values: 0 to 1440

Default: 0

6.2.1.7.63 time-limit-lte-m

Syntax

```
time-limit-lte-m time-limit
```

```
no time-limit-lte-m
```

Context

```
config>mobile>profile>charging>pgw-charging>off-line>part-rec-closure-trig
```

Description

This command specifies a time limit that applies to the LTE-M Radio Access Technology. A partial record is created when the specified time limit is reached.

The no form of this command reverts to the default.

Parameters

time-limit

specifies a time limit, in minutes, applicable per session or bearer

Values: 0 to 1440

Default: 0 (use the generic time limit when specified)

6.2.1.7.64 volume-limit

Syntax

```
volume-limit volume-limit  
no volume-limit
```

Context

```
config>mobile>profile>charging>epdg-charging>off-line>part-rec-closure-trig  
config>mobile>profile>charging>pgw-charging>off-line>part-rec-closure-trig  
config>mobile>profile>charging>sgw-charging>off-line>part-rec-closure-trig  
config>mobile>profile>charging>tdf-charging>off-line>part-rec-closure-trig
```

Description

This command specifies a data volume limit applicable per session or bearer depending on the accounting level selected like PDN-level accounting or QCI/bearer-level accounting. A partial record is created when the specified volume limit is satisfied. When the `volume-limit-eutran` command is enabled, it applies to LTE and the `volume-limit` command applies to all other RATs. When the `volume-limit-utran` command is enabled, it applies to UTRAN Radio Access Technology and the `volume-limit` command applies to all other RATs that do not have an explicit volume limit trigger. When the `volume-limit-eutran` command is not enabled, the `volume-limit` command applies to E-UTRAN as well. Similarly, when the `volume-limit-utran` command is not enabled, the `volume-limit` applies to UTRAN as well.

The no form of this command reverts to the default.

Parameters

volume-limit

specifies a data volume limit, in kilobytes

Values: 0 to 104857600

Default	PGW, SGW, and FBC	4096
–	ePDG and TDF	0

6.2.1.765 volume-limit-eutran

Syntax

```
volume-limit-eutran volume-limit-eutran  
no volume-limit-eutran
```

Context

```
config>mobile>profile>charging>pgw-charging>off-line>part-rec-closure-trig
```

Description

This command specifies a data volume limit applicable for E-UTRAN radio access technology. A partial record is created when the specified volume limit is reached. When the `volume-limit-eutran` command is enabled, it applies to LTE and the `volume-limit` command applies to all other RATs. When the `volume-limit-eutran` command is not enabled, the value specified for the generic `volume-limit` command applies to E-UTRAN as well.

The no form of this command reverts to the default.

Parameters

volume-limit-eutran

specifies a data volume limit

Values: 0 to 104857600

Default: 0 (use the generic volume limit when specified)

6.2.1.766 volume-limit-eutran-nb-iot

Syntax

```
volume-limit-eutran-nb-iot volume-limit-eutran-nb-iot  
no volume-limit-eutran-nb-iot
```

Context

```
config>mobile>profile>charging>pgw-charging>off-line>part-rec-closure-trig
```

Description

This command specifies a data volume limit applicable for E-UTRAN radio access technology. A partial record is created when the specified volume limit is reached. When the `volume-limit-eutran-nb-iot` command is enabled, it applies to LTE and the `volume-limit` command applies to all other RATs.

When the `volume-limit-eutran-nb-iot` command is not enabled, the value specified for the generic `volume-limit` command applies to E-UTRAN as well.

The no form of this command reverts to the default.

Parameters

volume-limit-eutran-nb-iot

specifies a data volume limit

Values: 0 to 104857600

Default: 0 (use the generic volume limit when specified)

6.2.1.7.67 volume-limit-lte-m

Syntax

```
volume-limit-lte-m volume-limit
```

```
no volume-limit-lte-m
```

Context

```
config>mobile>profile>charging>pgw-charging>off-line>part-rec-closure-trig
```

Description

This command specifies a data volume limit that applies to the LTE-M Radio Access Technology. A partial record is created when the specified volume limit is reached.

The no form of this command reverts to the default.

Parameters

volume-limit

specifies a data volume limit

Values: 0 to 104857600

Default: 0 (use the generic volume limit when specified)

6.2.1.7.68 volume-limit-utran

Syntax

```
volume-limit-utran volume-limit-utran
```

```
no volume-limit-utran
```

Context

config>mobile>profile>charging>pgw-charging>off-line>part-rec-closure-trig

Description

This command specifies a data volume limit applicable for UTRAN radio access technology. A partial record is created when the specified volume limit is reached. When the `volume-limit-utran` command is enabled, it applies to UTRAN Radio Access Technology and the generic `volume-limit` partial record trigger applies to all other RATs that do not have an explicit volume limit trigger. When the `volume-limit-utran` command is not enabled, the value specified for the `volume-limit` command applies to UTRAN as well.

The no form of this command reverts to the default.

Parameters

volume-limit-utran

specifies a data volume limit

Values: 0 to 104857600

Default: 0 (use the generic volume limit when specified)

6.2.1.7.69 pri-cdf-dia-peer

Syntax

pri-cdf-dia-peer *profile-name*

no pri-cdf-dia-peer

Context

config>mobile>profile>charging>epdg-charging>off-line

config>mobile>profile>charging>pgw-charging>off-line

config>mobile>profile>charging>sgw-charging>off-line

config>mobile>profile>charging>tdf-charging>off-line

Description

This command specifies the primary CDF Diameter peer profile for this charging profile.

Parameters

profile-name

specifies a profile name, up to 32 characters

6.2.1.770 sd-container

Syntax

```
sd-container
```

Context

```
config>mobile>profile>charging>pgw-charging>off-line
```

Description

This command enables the context to configure container-level settings for the SDCs.

6.2.1.771 include-lostd-with-sec-rat-usage-report

Syntax

```
[no] include-lost-with-sec-rat-usage-report
```

Context

```
config>mobile>profile>charging>pgw-charging>off-line>sd-container
```

Description

This command enables the PGW-CDRs to include ListOfServiceData container in case of split CDR. If the PGW-CDR is split due to the maximum number of containers, each split PGW-CDR includes one ListOfServiceData container. If the command is not enabled, when CMG receives more than 10 SecondaryRATUsageReport containers per bearer, a split PGW-CDR can be generated without ListOfServiceData containers and only with SecondaryRATUsageReport containers.



NOTE:

- The command can be enabled when the `include-zero-vol-sdc` command is enabled, too.
- The operator must enable an offline (or online and offline) "anyany" rule. In case there is only the SecondaryRATUsageReport container, and no active group(s), CMG includes one ListOfServiceData container reporting the "anyany" rating group with zero volume.

The no form of this command reverts to the default.

Default

no include-losd-with-sec-rat-usage-report

6.2.1.772 include-zero-vol-sdc

Syntax

[no] include-zero-vol-sdc

Context

config>mobile>profile>charging>pgw-charging>off-line>sd-container

Description

This command enables the sending of zero volume (SDC) in the PGW-CDRs for the OFCS.

The no form of this command reverts to the default (zero volume SDCs are not included).

Default

no include-zero-vol-sdc

6.2.1.773 qos-info-first-sdc-of-cdr

Syntax

[no] qos-info-first-sdc-of-cdr

Context

config>mobile>profile>charging>pgw-charging>off-line>sd-container

Description

When this command is enabled, the QoS-Information AVP will always be included in the first SDC for an RG/SVC-ID in a CDR/ACR. (This applies across CDRs for the same session).

The no form of this command reverts to the default (disabled).

Default

no qos-info-first-sdc-of-cdr

6.2.1.7.74 sec-cdf-dia-peer

Syntax

```
sec-cdf-dia-peer profile-name  
no sec-cdf-dia-peer
```

Context

```
config>mobile>profile>charging>epdg-charging>off-line  
config>mobile>profile>charging>pgw-charging>off-line  
config>mobile>profile>charging>sgw-charging>off-line  
config>mobile>profile>charging>tdf-charging>off-line
```

Description

This command specifies the secondary CDF Diameter peer profile for this charging profile.

Parameters

profile-name
specifies a profile name, up to 32 characters

6.2.1.7.75 user-location-at-cdr-generation-time

Syntax

```
user-location-at-cdr-generation-time {enable | disable}  
no user-location-at-cdr-generation-time
```

Context

```
config>mobile>profile>charging>pgw-charging>off-line  
config>mobile>profile>charging>sgw-charging>off-line
```

Description

This command configures whether the user location visible in SGW-CDRs and PGW-CDRs is the location at the time when the CDR was opened or the last known location when the CDR was generated. If the CDR is generated due to a bearer update, the location in the CDR is the last value before the bearer update.

**NOTE:**

This command requires the use - last - known - u- command to be configured.

Parameters**enable**

specifies the option that the user location visible in SGW-CDRs and PGW-CDRs is the last known location when the CDR was generated

disable

specifies the option that the user location visible in SGW-CDRs and PGW-CDRs is the location at the time when the CDR was opened

Default: disable

6.2.1.776 sgw-change-cause-record-closure

Syntax

[no] sgw-change-cause-record-closure

Context

config>mobile>profile>charging>sgw-charging>off-line

Description

This command enables the population of SGW Change as the reason for record closure from the source SGW in the case of SGW relocation, for Ga interface. In the case of Rf charging, this command enables the population of 'SGW Change' as the change condition in a ACR STOP message from the source SGW in the case of SGW relocation.

The no form of this command reverts to the default (disabled).

Default

no sgw-change-cause-record-closure

6.2.1.777 on-line

Syntax

on-line

Context

```
config>mobile>profile>charging>pgw-charging
```

```
config>mobile>profile>charging>tdf-charging
```

Description

This parameter configures online charging. By default, online charging is disabled.

6.2.1.778 dcca-profile

Syntax

```
dcca-profile profile-name
```

```
no dcca-profile
```

Context

```
config>mobile>profile>charging>pgw-charging>on-line
```

```
config>mobile>profile>charging>tdf-charging>on-line
```

Description

This command configures the DCCA profile for the primary and secondary OCS at the charging profile level.

The no form of this command means there is no DCCA profile associated with the primary and secondary OCS at the charging profile level.

Default

```
no dcca-profile
```

Parameters

profile-name

specifies the name of the DCCA profile associated with the primary and secondary OCS, up to 32 characters

6.2.1.779 diameter-peer-group-list

Syntax

```
diameter-peer-group-list list-name
```

```
no diameter-peer-group-list
```

Context

```
config>mobile>profile>charging>pgw-charging>on-line
```

```
config>mobile>profile>charging>tdf-charging>on-line
```

Description

This command associates a Diameter peer group list. It contains a list of Diameter peer groups. Each group is a set of a primary/secondary pair. The `diameter-peer-group-list` command can be used for PCRF/OCS selection when an IMSI/MSISDN/IMEI based selection is used. The list is associated with a user range.

Parameters

list-name

specifies the name of the Diameter peer group list, up to 32 characters

6.2.1.780 enable-online

Syntax

```
enable-online
```

```
no enable-online
```

Context

```
config>mobile>profile>charging>pgw-charging>on-line
```

Description

This command activates online charging. Online charging is enabled or disabled for this charging profile without a PDN shutdown in SGW charging and PGW charging. On the fly, enabling or disabling of online charging affects new sessions only.

The `no` form of this command disables online charging.

Default

```
no enable-online
```

6.2.1.7.81 enable-online

Syntax

```
enable-online [session-level | rg-level]
no enable-online
```

Context

```
config>mobile>profile>charging>tdf-charging>on-line
```

Description

This command specifies whether online charging is enabled for this charging profile. Currently, only rating-group level online charging can be enabled.

The no form of this command disables online charging for this charging profile.



NOTE:

This command requires a PDN shutdown.

Default

no enable-online

Parameters

session-level

specifies that session-level online charging is enabled

rg-level

specifies that RG-level online charging is enabled

Default: rg-level

6.2.1.7.82 pri-ocs-dia-peer

Syntax

```
primary-ocs-dia-peer profile-name
no primary-ocs-dia-peer
```

Context

config>mobile>profile>charging>pgw-charging>on-line

config>mobile>profile>charging>tdf-charging>on-line

Description

This command specifies the primary OCS Diameter peer profile for this charging profile. By default, no peer profile is associated with the primary OCS.

Default

no primary-ocs-dia-peer

Parameters

profile-name

specifies the profile name, up to 32 characters

6.2.1.7.83 sec-ocs-dia-peer

Syntax

secondary-ocs-dia-peer *profile-name*

no secondary-ocs-dia-peer

Context

config>mobile>profile>charging>pgw-charging>on-line

config>mobile>profile>charging>tdf-charging>on-line

Description

This command specifies the secondary OCS Diameter peer profile for this charging profile. By default, no peer profile is associated with the secondary OCS.

Default

no secondary-ocs-dia-peer

Parameters

profile-name

specifies the profile name, up to 32 characters

6.2.1.784 selection-type

Syntax

```
selection-type {imsi | msisdn} {prefix | full-id | suffix}  
no selection-type
```

Context

```
config>mobile>profile>charging>pgw-charging>on-line
```

Description

This command specifies the OCS selection criteria which can be based on an IMSI or MSISDN range.

When you select the `full-id` option, enter the *start-id* and *end-id* for each user - range entry (when you select the `prefix` option, the *end-id* is optional).

When you select the `suffix` option, you must enter a suffix digit.

If the IMSI or MSISDN for a UE does not match any of the configured entries, the selection of the OCS and DCCA profile fallbacks or defaults to the APN-level and the PDN-level OCS peer configuration.

The `no` form of this command removes the selection type for the user - range configuration.

Parameters

imsi

specifies an IMSI range for the OCS selection

msisdn

specifies an MSISDN range for the OCS selection

prefix

specifies that the mapping is based on the IMSI or MSISDN prefix, up to 11 digits

full-id

specifies that the mapping is based on a full identifier for the start and end identifiers of the user - range command. When you select the `full-id` option, the *start-id* and *end-id* must be full identifiers. For IMSI, the identifier length must be from 9 to 15 digits and for MSISDN, from 4 to 15 digits.

suffix

specifies an integer that selects the end digit for the user - range.

6.2.1.785 selection-type

Syntax

```
selection-type {imsi | msisdn} {prefix | full-id}
no selection-type
```

Context

```
config>mobile>profile>charging>tdf-charging>on-line
```

Description

This command specifies whether the OCS selection criteria is based on an IMSI or MSISDN range. The no form of this command reverts to the default. When the *full-id* option type is selected, the *start-id* and *end-id* parameters of the user - range command both need to be entered for each user - range entry (unlike the *prefix* selection type option where the *end-id* is optional). If the IMSI or MSISDN for a UE does not match any of the configured entries, the selection of the OCS and DCCA profile fallbacks or defaults to the APN-level and the PDN-level OCS peer configuration.

Parameters

imsi

specifies the OCS selection based on the IMSI range

msisdn

specifies the OCS selection based on the MSISDN range

prefix

specifies that the mapping is based on the IMSI or MSISDN prefix, up to 11 digits

full-id

specifies that the mapping is based on a full identifier for the start and end identifiers of the user - range command. When the *full-id* selection option type is chosen, the *start-id* and *end-id* need to be full identifiers. For IMSI, this will be between 5 to 15 digits and for MSISDN this will be between 4 to 15 digits.

6.2.1.786 user-range

Syntax

```
user-range user-range-name {start-id [end-id] | suffix} {diameter-peer-group-list list-name}
no user-range user-range-name
```

Context

config>mobile>profile>charging>pgw-charging>on-line

Description

This command is used to select an OCS based on an IMSI or MSISDN user range. This command allows the selection of an OCS based on a combination of charging characteristics and IMSI/MSISDN; the selection criteria must be within a charging profile.



NOTE:

- The diameter-peer-group-list parameter must be created prior to associating it under the context: config>mobile>profile>charging>pgw-charging>on-line
- The diameter-peer-group-list parameter, when specified, is used as a fallback for the OCS selection based on the user range.
- When an OCS cannot be selected within the charging profile (for example, there is a possible misconfiguration), the selection will fallback or default to the APN-level and then the PDN-level OCS peer configuration.
- The no form of this command requires a PDN shutdown, when used in the config>mobile>profile>charging>pgw-charging>on-line context.

Parameters

user-range-name

specifies the user range name, up to 32 characters

start-id

specifies the start identifier for the user range, from 4 to 15 characters

end-id

specifies the end identifier for the user range, from 4 to 15 characters

suffix

specifies an integer that selects the end digit for the user range. When you select the suffix option, the *start-id* and *end-id* are not used.

Values: 0 to 9

list-name

specifies the Diameter peer group list name, up to 32 characters

6.2.1.7.87 user-range

Syntax

```
user-range user-range-name start-id [end-id] {diameter-peer-group-list list-name}  
no user-range user-range-name
```

Context

```
config>mobile>profile>charging>tdf-charging>on-line
```

Description

This command is used to select an OCS based on an IMSI or MSISDN user range. This command allows the selection of an OCS based on a combination of charging characteristics and IMSI/MSISDN; the selection criteria must be within a charging profile.

**NOTE:**

- The *diameter-peer-group-list* parameter must be created prior to associating it under the context: `config>mobile>profile>charging>tdf-charging>on-line`
- The *diameter-peer-group-list* parameter, when specified, is used as a fallback for the OCS selection based on the user range.
- When an OCS cannot be selected within the charging profile (for example, there is a possible misconfiguration), the selection will fallback or default to the APN-level and then the PDN-level OCS peer configuration.

Parameters

user-range-name

specifies the name for the user-range, up to 32 characters

start-id

specifies the start identifier for the user-range, from 4 to 15 characters

end-id

specifies the end identifier for the user-range, from 4 to 15 characters

suffix

specifies an integer that selects the end digit(s) for the user-range. If a suffix is selected, the *start-id* and *end-id* are not used.

list-name

specifies the name of the *diameter-peer-group-list*, up to 32 characters

6.2.1.788 pcrf-selection

Syntax

```
[no] pcrf-selection
```

Context

```
config>mobile>profile>charging>pgw-charging
```

Description

This command enables or disables PCRF selection for the PGW charging profile. The operator can configure the PGW/GGSN to select the charging profile based on the charging characteristics. The charging-profile can then be used to enable and disable the use of Gx.

6.2.1.789 dynamic-pcc

Syntax

```
dynamic-pcc {enable /disable /inherit}  
no dynamic-pcc
```

Context

```
config>mobile>profile>charging>pgw-charging>pcrf-selection
```

Description

This command is used to enable or disable policy and charging control (PCC) interface (Gx) for a subscriber's IP-CAN session. When enabled, a Gx session is set up with a PCRF. When disabled, there is no Gx session for the IP-CAN session. If the command is set to `inherit`, the system will use the PCRF selection command from the APN level. Any local static policy specified under APN configuration applies.

The no form of this command reverts to default.

Default

```
no dynamic-pcc
```

Parameters

enable

specifies the Gx session that is set up with a PCRF

disable

specifies that there is no Gx session for the IP-CAN session

inherit

specifies the system to use the PCRF selection command from the APN level instead

6.2.1.790 pcc-disable-per-rat

Syntax

```
pcc-disable-per-rat [UTRAN] [GERAN] [WLAN] [GAN] [HSPA-Evolution] [EUTRAN] [Virtual]  
[EUTRAN-NB-IoT]
```

```
no pcc-disable-per-rat
```

Context

```
config>mobile>profile>charging>pgw-charging>pcrf-selection
```

Description

This command configures a set of RAT types which are not allowed to be under the control of the PCRF. When an IP CAN session creation occurs and a UE is under a forbidden RAT type, PCRF control is not started. If the UE moves to an allowed RAT type, the PCRF is started at that instance. If the UE moves back to a RAT type that is not allowed, the PCRF session is terminated after the allotted value configured using the `pcc-disable-per-rat-timer` command. If this command is not configured (no RAT types configured), all the RAT types are allowed under the PCRF control when dynamic PCC is enabled for the session.

The no form of this command reverts the configuration to the default (not configured).

Parameters

UTRAN

specifies that UTRAN is not allowed to be under the control of the PCRF

GERAN

specifies that GERAN is not allowed to be under the control of the PCRF

WLAN

specifies that WLAN is not allowed to be under the control of the PCRF

GAN

specifies that GAN is not allowed to be under the control of the PCRF

HSPA-Evolution

specifies that HSPA-Evolution is not allowed to be under the control of the PCRF

EUTRAN

specifies that EUTRAN is not allowed to be under the control of the PCRF

Virtual

specifies that Virtual is not allowed to be under the control of the PCRF

EUTRAN-NB-IoT

specifies that EUTRAN-NB-IoT is not allowed to be under the control of the PCRF

6.2.1.7.91 pcc-disable-per-rat-timer

Syntax

`pcc-disable-per-rat-timer` *pcc-disable-per-rat-timer*

`no pcc-disable-per-rat-timer`

Context

`config>mobile>profile>charging>pgw-charging>pcrf-selection`

Description

This command configures the timer which is started when the UE moves into a RAT type which is not allowed under PCRF control. When the timer expires and the UE is still under a forbidden RAT type, the PCRF session is terminated.

The no form of this command reverts the configuration to the default (not configured).

Parameters***pcc-disable-per-rat-timer***

sets the timer, in seconds

Values: 0 to 7200

Default: 30

6.2.1.7.92 radius-accounting

Syntax

`radius-accounting`

Context

`config>mobile>profile>charging>pgw-charging`

Description

This command enables the context for configuring the RADIUS accounting.

6.2.1.7.93 cgi-sai-change

Syntax

```
[no] cgi-sai-change
```

Context

```
config>mobile>profile>charging>pgw-charging>radius-accounting
```

Description

This command enables the RADIUS interim accounting to be triggered, based on any CGI-SAI change location event. Triggering an interim accounting message due to any event, re-initializes the periodic accounting timer and/or volume-based threshold (if enabled).

The no form of this command reverts to the default.

Default

```
no cgi-sai-change
```

6.2.1.7.94 ecgi-change

Syntax

```
[no] ecgi-change
```

Context

```
config>mobile>profile>charging>pgw-charging>radius-accounting
```

Description

This command enables the RADIUS interim accounting to be triggered, based on an eCGI change location event. Triggering an interim accounting message due to any event, re-initializes the periodic accounting timer and/or volume-based threshold (if enabled).

The no form of this command reverts to the default.

Default

no ecgi-change

6.2.1.7.95 peer-change-trigger

Syntax

[no] peer-change-trigger

Context

config>mobile>profile>charging>pgw-charging>radius-accounting

Description

This command enables the RADIUS interim accounting to be triggered, based on a peer change event for the default bearer. For the GGSN this peer is the SGSN, and for the PGW this is the SGW. Triggering an interim accounting message due to any event, re-initializes the periodic accounting timer and/or volume-based threshold (if enabled).

The no form of this command reverts to the default.

Default

no peer-change-trigger

6.2.1.7.96 qos-change-trigger

Syntax

[no] qos-change-trigger

Context

config>mobile>profile>charging>pgw-charging>radius-accounting

Description

This command enables the RADIUS interim accounting to be triggered, based on a QoS change event for the default bearer. Triggering an interim accounting message due to any event, re-initializes the periodic accounting timer and/or volume-based threshold (if enabled).

The no form of this command reverts to the default.

Default

no qos-change-trigger

6.2.1.7.97 rai-change

Syntax

[no] rai-change

Context

config>mobile>profile>charging>pgw-charging>radius-accounting

Description

This command enables the RADIUS interim accounting to be triggered, based on a RAI change location event. Triggering an interim accounting message due to any event, re-initializes the periodic accounting timer and/or volume-based threshold (if enabled).

The no form of this command reverts to the default.

Default

no rai-change

6.2.1.7.98 rat-change-trigger

Syntax

[no] rat-change-trigger

Context

config>mobile>profile>charging>pgw-charging>radius-accounting

Description

This command enables the RADIUS interim accounting to be triggered, based on a RAT change event. Triggering an interim accounting message due to a RAT change, re-initializes the periodic accounting timer (if enabled). Triggering an interim accounting message due to any event, re-initializes the periodic accounting timer and/or volume-based threshold (if enabled).

The no form of this command reverts to the default.

Default

no rat-change-trigger

6.2.1.7.99 tai-change**Syntax**

[no] tai-change

Context

config>mobile>profile>charging>pgw-charging>radius-accounting

Description

This command enables the RADIUS interim accounting to be triggered, based on a TAI change location event. Triggering an interim accounting message due to any event, re-initializes the periodic accounting timer and/or volume-based threshold (if enabled).

The no form of this command reverts to the default.

Default

no tai-change

6.2.1.7.100 user-loc-change**Syntax**

[no] user-loc-change

Context

config>mobile>profile>charging>pgw-charging>radius-accounting

Description

This command enables the RADIUS interim accounting to be triggered, based on any change location event. Triggering an interim accounting message due to any event, re-initializes the periodic accounting timer and/or volume-based threshold (if enabled).

The no form of this command reverts to the default.

Default

no user-loc-change

6.2.1.7.101 volume-limit-trigger

Syntax

volume-limit-trigger *value*

no volume-limit-trigger

Context

config>mobile>profile>charging>pgw-charging>radius-accounting

Description

This command specifies the volume threshold for interim accounting. Triggering an interim accounting message due to any event, re-initializes the periodic accounting timer and/or volume-based threshold (if enabled). Setting a low value can cause a RADIUS server to be overloaded.

The no form of this command reverts to the default.

Default

no volume-limit-trigger

Parameters

value

specifies the volume threshold, in kbytes

Values: 0 to 104857600

Default: 0 (disabled)

6.2.1.7.102 report-serving-node-supplied-cc

Syntax

[no] report-serving-node-supplied-cc

Context

config>mobile>profile>charging>pgw-charging

```
config>mobile>profile>charging>sgw-charging  
config>mobile>profile>charging>smf-charging
```

Description

This command enables the report of the charging characteristic value received from the serving node to the charging interfaces (SGW-CDRs, PGW-CDRs, Gy, Rf for LTE, and Nchf for SMF). This command has no influence on the charging profile selection precedence used to determine which charging profile is selected.

The no form of this command reverts to the default (the internally used charging characteristic value is reported).

Default

no report-serving-node-supplied-cc

6.2.1.7.103 smf-legacy-nr-ratatype

Syntax

```
smf-legacy-nr-ratatype {reserved | utran | geran | wlan | gan | hspa | eutran | ehrpd | hrpd  
| one-xrtt | umb | virtual | eutran-nb-iot | nr}  
no smf-legacy-nr-ratatype
```

Context

```
config>mobile>profile>charging>pgw-charging  
config>mobile>profile>charging>sgw-charging
```

Description

This command configures the RAT type sent when a UE that uses the 5G NR RAT type is charged using the legacy-charging interfaces (Gy, Ga, and Bp).

The no form of this command reverts to the default value.

Parameters

**reserved | utran | geran | wlan | gan | hspa | eutran | ehrpd | hrpd | one-xrtt | umb | virtual
| eutran-nb-iot | nr**

specifies the RAT type

Default: utran

6.2.1.8 Charging profile CCFH command descriptions

6.2.1.8.1 ccfh-profile

Syntax

[no] ccfh-profile *profile-id*

Context

config>mobile>profile>charging

Description

This command creates a CCHF profile. The created profile can be associated with a service type using the config>mobile>profile>dcca>service-type command or with a charging function profile using the config>mobile>profile>charging>chf-profile command.

The no form of this command removes the specified CCFH profile.

Parameters

profile-id

specifies the profile ID for this CCFH profile

Values: 0 to 255

Default: 0

6.2.1.8.2 assume-positive

Syntax

assume-positive

Context

config>mobile>profile>charging>ccfh-profile

Description

This command enables the context to configure Assume Positive parameters for Gy CCFH profiles.

6.2.1.8.3 access-type

Syntax

```
[no] access-type {lte | wifi | ehrpd | utran | all}
```

Context

```
config>mobile>profile>charging>ccfh-profile>assume-positive
```

Description

This command is used to specify an access type for Assume Positive behavior. The no form of this command reverts to the default (all - applicable to all access types). When the access type is specified as a parameter for the no access-type command, the configuration associated with the access type is removed and it reverts to the default (none).

Parameters

lte

specifies the access type as eUTRAN

wifi

specifies the access type as WLAN

ehrpd

specifies the access type as eHRPD

utran

specifies the access type as UTRAN

all

defines the access type as a generic wildcard (applies to all access types)

6.2.1.8.4 message

Syntax

```
message [ccr-i] [ccr-u] [ccr-t]
```

```
no message
```

Context

```
config>mobile>profile>charging>ccfh-profile>assume-positive>access-type
```

Description

This command is used to specify a message type (initial, update, or terminate) for which the Assume Positive behavior is applied.

The no form of this command reverts to the default (all, applies to all message types).

Default

all message types

Parameters

ccr -i

specifies the message type as CCR initial

ccr -t

specifies the message type as CCR terminate

ccr -u

specifies the message type as CCR update

6.2.1.8.5 rating-group

Syntax

[no] rating-group *rating-group*

Context

config>mobile>profile>charging>ccfh-profile>assume-positive>access-type

Description

This command is used to specify a rating group for which the volume quota and/or time quota will apply while in Assume Positive behavior.

The no form of this command removes the configuration associated with the rating group.

Parameters

rating-group

specifies a rating group

Values: 0 to 4294967295 | * (wildcard that applies to any rating group)

6.2.1.8.6 credit-pool-id

Syntax

```
credit-pool-id pool-id  
no credit-pool-id
```

Context

```
config>mobile>profile>charging>ccfh-profile>assume-positive>access-type>rating-group
```

Description

This command is used to specify a credit pool ID for this rating group. The volume quota, time quota, and number of interim quotas configured for this pool ID are applied to this rating group.

The no form of this command reverts to the default (none).

Parameters

pool-id

specifies a credit pool ID (defined under config>mobile>profile>charging context)

Values: 1 to 16

6.2.1.8.7 enable

Syntax

```
[no] enable
```

Context

```
config>mobile>profile>charging>ccfh-profile>assume-positive
```

Description

This command enables the context for the Assume Positive behavior for online charging.

6.2.1.8.8 orig-session-id-ccri

Syntax

```
[no] orig-session-id-ccri
```

Context

```
config>mobile>profile>charging>ccfh-profile>assume-positive
```

Description

This command allows the original session ID in a CCR-I sent to the OCS while the PGW is in Assume Positive mode.

The no form of this command reverts to the default (disabled).

Default

```
no orig-session-id-ccri
```

6.2.1.8.9 cc-failure-handling

Syntax

```
cc-failure-handling {terminate | continue | retry-and-terminate | retry-and-continue}
```

```
no cc-failure-handling
```

Context

```
config>mobile>profile>charging>ccfh-profile
```

Description

This command enables CC failure handling.

Parameters

terminate

specifies to terminate the connection

continue

specifies to continue the connection

retry-and-terminate

specifies to retry connecting and if unsuccessful, terminate the connection

retry-and-continue

specifies that if failover to secondary is supported, the secondary OCS is tried before the session is continued under Assume Positive

Default: terminate

6.2.1.8.10 cc-failure-handling-enh

Syntax

```
[no] cc-failure-handling-enh
```

Context

```
config>mobile>profile>charging>ccfh-profile
```

Description

This command configures failure handling of the result codes on all command levels. When the command is not configured, failure handling applies to some command level result codes.

**NOTE:**

When the command is configured, failure handling applies to all command level result codes, except for 4011 result code.

The no form of this command reverts to the default (disabled).

Default

```
no cc-failure-handling-enh
```

6.2.1.8.11 ccfh-fh-action

Syntax

```
ccfh-fh-action {terminate | re-start}
```

```
no ccfh-fh-action
```

Context

```
config>mobile>profile>charging>ccfh-profile
```

Description

This command configures the action to take, upon the expiration of the fh-session-continue-timer or when the limit set by fh-volume-limit is reached (whichever occurs first).

**NOTE:**

The CCR-I does not include any requests for quota (including for any configured default rating group). The installed rating groups are put in first packet mode, wherein the request for quota (CCR-U) is upon receiving a first packet for that rating group.

The no form of this command reverts to the default value `terminate` (IP-CAN session terminated).

Parameters**terminate**

the IP-CAN session is terminated

re-start

a CCR-I is sent to the OCS upon the expiration of the fh session continue timer or when the limit set by the fh volume limit is reached (whichever occurs first); if the OCS is still unavailable, the session is then terminated

Default: `terminate`

6.2.1.8.12 `disable-cc-session-failover`

Syntax

```
[no] disable-cc-session-failover
```

Context

```
config>mobile>profile>charging>ccfh-profile
```

Description

This command allows the cc session failover to be disabled.

The no form of this command enables cc session failover.

6.2.1.8.13 `fail-open`

Syntax

```
[no] fail-open
```

Context

```
config>mobile>profile>charging>ccfh-profile
```

Description

This command enables the use of the CCFH profile for configuring the Fail-Open failure handling parameters.

The no form of this command reverts to the default (disabled).

6.2.1.8.14 fh-continue-action

Syntax

```
fh-continue-action {allow | block | offline}
no fh-continue-action
```

Context

```
config>mobile>profile>charging>ccfh-profile
```

Description

This command specifies the Failure Handling Continue action.

The no form of this command reverts to the default (allow).

Parameters

allow

the user can access all online-charged flows that have not been explicitly blocked or denied by the OCS, for the duration of the bearer

block

all online-charged flows are blocked and packets are dropped for the duration of the bearer

offline

the session is converted to a normal postpaid session and CDRs are created

Default: allow

6.2.1.8.15 fh-session-continue-timer

Syntax

```
fh-session-continue-timer fh-session-continue-timer
no fh-session-continue-timer
```

Context

config>mobile>profile>charging>ccfh-profile

Description

This command specifies the Failure Handling Session Continue timer. When the CCFH = Continue, the credit control session is considered closed; however, the service is granted for the configured period of time. When `fh-session-continue-timer` and `fh-volume-limit` commands are active (CCFH = Continue), the IP-CAN session is terminated (unless configured to reauthorize) upon either condition being met (whichever occurs first). If the CCFH profile is associated with the service type, the time limit takes precedence over the time limit under the DCCA profile.

Parameters

fh-session-continue-timer

specifies the Failure Handling Session Continue timer in minutes

Values: 0 to 1440

Default: 30

6.2.1.8.16 fh-volume-limit

Syntax

`fh-volume-limit volume-limit`

`no fh-volume-limit`

Context

config>mobile>profile>charging>ccfh-profile

Description

This command specifies the volume limit for CCFH (Credit Control Failure Handling). When `fh-session-continue-timer` and `fh-volume-limit` commands are active (CCFH = Continue), the IP-CAN session is terminated (unless configured to reauthorize) upon either condition being met (whichever occurs first). If the CCFH profile is associated with the service type, the volume limit takes precedence over the volume limit under the DCCA profile.

The no form of this command reverts to default (disabled).

Parameters

volume-limit

specifies the volume limit in kilobytes

Values: 0 to 100000000

Default: 0

6.2.1.8.17 gy-to-gtp-error-mapping-profile

Syntax

```
gy-to-gtp-error-mapping-profile profile-name
```

```
no gy-to-gtp-error-mapping-profile
```

Context

```
config>mobile>profile>charging>ccfh-profile
```

Description

This command specifies the Gy to GTP error mapping profile.

The no form of this command removes the reference/association of the specified profile for this Gy profile.

Parameters

profile-name

specifies the profile name, up to 32 characters.

6.2.1.8.18 message

Syntax

```
[no] message {ccr-i | ccr-u}
```

Context

```
config>mobile>profile>charging>ccfh-profile
```

Description

This command enables the context to configure options for the specified message type.

The no form of this command removes the configuration for the specified message type.

Parameters

ccr-i

specifies the CCR initial message

ccr-u

specifies the CCR update message

6.2.1.8.19 result-code

Syntax

`result-code` *result-code-value* *result-error-code* [restore]

`no result-code` *result-code-value*

Context

`config>mobile>profile>charging>ccfh-profile>message`

Description

This command configures the failure handling for the specified result code.

The no form of this command removes the configuration for the specified result code.



NOTE:

When same behavior needs to be applied to all message types, use the `result-code` command in the `config>mobile>profile>charging>ccfh-profile` context. The context configuration takes precedence over the context configuration. For example, the following sets of two commands given in the context result in the same configuration:

- `message ccr-i result-code 5996 continuemessage ccr-u result-code 5996 terminate`
- `message ccr-i result-code 5996 continueresult-code 5996 terminate`

Parameters

result-code-value

specifies the four-digit result code

Values: 3000 to 5999



NOTE:

The configuration of the following error codes is not allowed:

- 4011 — Diameter credit control not applicable

- 4012 — Diameter credit limit reached

result-error-code

specifies the failure handling error code for the specified result code

Values:

terminate

specifies the error code for failure handling "terminate"

continue

specifies the error code for failure handling "continue"

retry-and-terminate

specifies the error code for failure handling with the action "retry-and-terminate"

retry-and-continue

if failover to a secondary OCS is supported, the secondary OCS is tried before the session is continued under Assume Positive, for the specified result-code

continue-allow

specifies the error code for failure handling "continue" with the action "allow"

continue-block

specifies the error code for failure handling "continue" with the action "block"

continue-offline

specifies the error code for failure handling "continue" with the action "offline"

retry-and-cont-allow

specifies the error code for failure handling "retry-and-continue" with the action "allow"

retry-and-cont-block

specifies the error code for failure handling "retry-and-continue" with the action "block"

retry-and-cont-offline

specifies the error code for failure handling "retry-and-continue" with the action "offline"

restore

specifies that session restore applies as per the session-restore-retry-timer configuration in the config>gateway>profile>charging>ccfh-profile context and for the following *result-error-code* actions:

- continue
- continue-allow
- continue-block
- continue-offline

- `retry-and-continue`
- `retry-and-cont-allow`
- `retry-and-cont-block`
- `retry-and-cont-offline`

**NOTE:**

To maintain existing functionality, the following error codes trigger session restore regardless of the restore option:

- 3002 — Diameter Unable to Deliver
- 3004 — Diameter too Busy
- 3005 — Diameter Loop Detected

6.2.1.8.20 `mscc-result-code`

Syntax

`[no] msc-result-code result-code-value`

Context

`config>mobile>profile>charging>ccfh-profile`

Description

This command configures the MSCC result code.

Parameters

result-code-value

specifies the four-digit result code

Values: 3000 to 5999

6.2.1.8.21 `fui-terminate-session`

Syntax

`[no] fui-terminate-session`

Context

```
config>mobile>profile>charging>ccfh-profile>mscc-result-code
```

Description

This command, when enabled, immediately terminates the PDP-Context/Bearer upon receiving a FUA (Final-Unit-Action) = Terminate in CCA (Credit-Control-Answer) for the configured MSCC result code (for example, 4012). The PDP-Context/Bearer is terminated immediately when an FUA = Terminate request is received for the configured MSCC result code (for example, 4012), and no GSU or GSU with zero units and no Validity Time unit are included. The PDP-Context/Bearer is terminated regardless of whether other RGs have ongoing accounting tasks.

When FPT (First Packet Trigger) mode is disabled, the gateway requests a quota for installed RGs in CCR-I. Upon receiving a CCA-I with an FUA = Terminate in CCA for the configured MSCC result code in any MSCC, the PDP-Context/Bearer activation is rejected. A CCR-T is not sent to the OCS because the Gy session was not set up successfully.

When FPT (First Packet Trigger) mode is enabled, the gateway requests a quota for the RG that is associated with the first packet in a CCR-U. Upon receiving a CCA-U with an FUA = Terminate in CCA for the configured MSCC result code, the PDP-Context/Bearer is terminated. A CCR-T is sent to the OCS to terminate the Gy session. This behavior also applies for any subsequent CCA-Us. A CCR-U with an MSCC having a GSU>0, FUI = Terminate, and a RC=2001 results in a CCR-U (Final) when exhausting the last-slice.

A CCA-U response with a GSU=0 (or no GSU), FUI = Terminate, and RC = xxxx (except 2001/2002) will not result in the PDP-Context/Bearer Termination even when the `fui-terminate-session` command is enabled for a given MSCC; this is because the received MSCC is ignored after any CCR-U FINAL.

The no form of this command reverts to the default.

Default

```
no fui-terminate-session
```

6.2.1.8.22 service-delay-timer

Syntax

```
service-delay-timer minutes
```

```
no service-delay-timer
```

Context

```
config>mobile>profile>charging>ccfh-profile>mscc-result-code
```


Description

This command configures the service delay timer.

Parameters

minutes

specifies the minutes to set for the service delay timer

Values: 0 to 1440

Default: 0

6.2.1.8.23 nchf-cause-code

Syntax

```
nchf-cause-code application-error-code action {terminate | continue | retry-and-terminate | retry-and-continue} [continue-action {allow | block}]
```

```
no nchf-cause-code application-error-code
```

Context

```
config>mobile>profile>charging>ccfh-profile
```

Description

This command configures the 5G NCHF cause code value for the CCFH profile and maps it to an action for failure handling. If the Nchf cause code received from the CHF is not configured, the SMF uses the corresponding HTTP code configured in the CCFH profile (`nchf-http-code` command). If the corresponding HTTP code is not configured, the SMF uses the CC failure handling configured in the CCFH profile (`cc-failure-handling` command).

The no form of this command reverts to the default (disabled).

Default

```
no nchf-cause-code application-error-code
```

Parameters

application-error-code

specifies the application error code sent by the CHF in the Cause attribute within Problem Details, up to 32 characters

Values: any standard or custom application error code can be mapped to a failure handling action

**NOTE:**

- The CHARGING_NOT_APPLICABLE application error code for the *application-error-code* parameter is configured only with parameters *terminate* and *continue*. The default action when CHARGING_NOT_APPLICABLE error code is not configured is *continue*.
- If the CMG receives the CHARGING_NOT_APPLICABLE application error with the HTTP status code "403 Forbidden" from the CHF, and *nchf-cause-code* CHARGING_NOT_APPLICABLE is not configured, the default failure handling action is *continue*.

action

specifies the action for failure handling

terminate

specifies terminate for failure handling

continue

specifies continue for failure handling

retry-and-terminate

specifies retry and terminate for failure handling

retry-and-continue

If failover to a secondary OCS is supported, the secondary OCS is tried before the session is continued

allow

specifies failure handling continue with action "allow"

If not configured, the failure handling continue action is inherited from the *fh-continue-action* command.

block

specifies failure handling continue with action "block"

If not configured, the failure handling continue action is inherited from the *fh-continue-action* command.

6.2.1.8.24 nchf-http-code

Syntax

```
nchf-http-code value action [allow | block]
```

```
no nchf-http-code value
```

Context

config>mobile>profile>charging>ccfh-profile

Description

This command configures the failure handling action for the specified 5G Nchf HTTP code. If an HTTP code received from the CHF is not configured, the SMF uses the CC failure handling configured in the CCFH profile (cc-failure-handling command).

The no form of this command reverts to the default (disabled).

Default

no nchf-http-code *value*

Parameters

value

specifies a three-digit HTTP code

Values: 300 to 599

action

a keyword that specifies the failure handling for the HTTP code

Values: terminate | continue | retry-and-terminate | retry-and-continue |

allow

specifies failure handling continue with action “allow”

If not configured, the failure handling continue action is inherited from the fh-continue-action command.

block

specifies failure handling continue with action “block”

If not configured, the failure handling continue action is inherited from the fh-continue-action command.

6.2.1.8.25 nchf-mui-result-code

Syntax

[no] nchf-mui-result-code *string-result-code*

Context

config>mobile>profile>charging>ccfh-profile

Description

This command configures the behavior for the nchf multiple-unit-information result codes. If not configured, the default behavior, according to the error code, is as follows:

- END_USER_SERVICE_REJECTED and RATING_FAILED blocks the RG indefinitely (service-delay-timer is not started)
- QUOTA_MANAGEMENT_NOT_APPLICABLE is not considered an error code, it moves the RG to be offline (no quota management)
- Other result codes use a default service-delay-timer of 3 minutes

The no form of this command reverts to the default.

Default

no nchf-mui-result-code

Parameters

string-result-code

specifies the result code, up to 32 characters; lower case letters are not allowed

6.2.1.8.26 fui-terminate-session

Syntax

[no] fui-terminate-session

Context

config>mobile>profile>charging>ccfh-profile>nchf-mui-result-code

Description

This command, when enabled, immediately terminates the PDU session upon receiving a FUA (Final-Unit-Action) = Terminate in Nchf_CDResponse for the configured Result Code within Multiple-Unit-Information IE (for example, END_USER_DENIED); no GSU, or GSU with zero units and no Validity Time units, are included. The PDU session is terminated regardless of whether other RGs have ongoing accounting tasks.

When the FPT (First Packet Trigger) mode is disabled, the gateway requests a quota for the installed RGs in the CDRequest-Create message. Upon receiving a CDResponse-Create message with an FUA = Terminate in CDResp for the configured Multiple-Unit-Information (MUI) result code in any MUI, the PDU session activation is rejected. A CDReq-Terminate is not sent to the CHF because the Nchf session was not set up successfully.

When the FPT (First Packet Trigger) mode is enabled, the gateway requests a quota for the RG that is associated with the first packet in a CDReq-Update. Upon receiving a CDResp-Update with an FUA =

Terminate in MUI for the configured MUI result code, the PDU session is terminated. A CDReq-Terminate is sent to the CHF to terminate the Nchf session. This behavior also applies for any subsequent CDReq-Updates. A CDReq-Update with an MUI having a GSU>0, FUI = Terminate, and a RC=SUCCESS results in a CDReq-Update (Final) when exhausting the last quota slice.

A CDResp-Update response with a GSU=0 (or no GSU), FUI = Terminate, and RC = xxxx (except SUCCESS) does not result in the PDU session termination even when the `fui-terminate-session` command is enabled for a given MSCC. This is because the received MUI is ignored after any CDReq-Update FINAL.

The no form of this command reverts to the default.

Default

no fui-terminate-session

6.2.1.8.27 service-delay-timer

Syntax

`service-delay-timer minutes`

`no service-delay-timer`

Context

`config>mobile>profile>charging>ccfh-profile>nchf-mui-result-code`

Description

This command configures a service delay timer per MUI result code. Service is denied for as long as the configured `service-delay-timer` defines, when the CHF sends an error code within the Multiple-Unit-Information container.

`service-delay-timer` set to 0 means that the SMF will never request quota again for this MUI (permanently blocked).

When not configured, a default value of 3 minutes or a permanent block is used for the `service-delay-timer`, depending on the type of the error.

The no form of this command reverts to the default.

Default

no service-delay-timer

Parameters

minutes

specifies the value of the delay timer in minutes

Values: 0 to 1440

Default: 0

6.2.1.8.28 result-code

Syntax

`result-code result-code-value result-error-code [restore]`

`no result-code result-code-value`

Context

`config>mobile>profile>charging>ccfh-profile`

Description

This command sets the result code value for the CCFH profile.

Parameters

result-code-value

specifies the four-digit result code value to set

Values: 3000 to 5999



NOTE:

The configuration of the following error codes is not allowed:

- 4011 — Diameter credit control not applicable
- 4012 — Diameter credit limit reached

result-error-code

specifies the error code

Values:

terminate

specifies the error code for failure handling “terminate”

continue

specifies the error code for failure handling “continue”

continue-allow

specifies the error code for failure handling "continue" with the action "allow"

continue-block

specifies the error code for failure handling "continue" with the action "block"

continue-offline

specifies the error code for failure handling "continue" with the action "offline"

retry-and-terminate

specifies the error code for failure handling with the action "retry-and-terminate"

retry-and-continue

if failover to a secondary OCS is supported, the secondary OCS is tried before the session is continued under Assume Positive, for the specified result-code

retry-and-cont-allow

specifies the error code for failure handling "retry-and-continue" with the action "allow"

retry-and-cont-block

specifies the error code for failure handling "retry-and-continue" with the action "block"

retry-and-cont-offline

specifies the error code for failure handling "retry-and-continue" with the action "offline"

restore

specifies that session restore applies as per the session-restore-retry-timer configuration in the config>gateway>profile>charging>ccfh-profile context and for the following *result-error-code* actions:

- continue
- continue-allow
- continue-block
- continue-offline
- retry-and-continue
- retry-and-cont-allow
- retry-and-cont-block
- retry-and-cont-offline

**NOTE:**

To maintain existing functionality, the following error codes trigger session restore regardless of the restore option:

- 3002 — Diameter Unable to Deliver
- 3004 — Diameter too Busy

- 3005 — Diameter Loop Detected

6.2.1.8.29 result-code-3xxx

Syntax

```
result-code-3xxx result-error-code [restore]
no result-code-3xxx
```

Context

```
config>mobile>profile>charging>ccfh-profile
```

Description

This command sets the result code value for the CCFH profile.

Parameters

result-error-code

specifies the error code

Values:

terminate

specifies the error code for failure handling "terminate"

continue

specifies the error code for failure handling "continue"

continue-allow

specifies the error code for failure handling "continue" with the action "allow"

continue-block

specifies the error code for failure handling "continue" with the action "block"

continue-offline

specifies the error code for failure handling "continue" with the action "offline"

retry-and-terminate

specifies the error code for failure handling with the action "retry-and-terminate"

retry-and-continue

if failover to a secondary OCS is supported, the secondary OCS is tried before the session is continued under Assume Positive, for the specified result-code

retry-and-cont-allow

specifies the error code for failure handling "retry-and-continue" with the action "allow"

retry-and-cont-block

specifies the error code for failure handling "retry-and-continue" with the action "block"

retry-and-cont-offline

specifies the error code for failure handling "retry-and-continue" with the action "offline"

restore

specifies that session restore applies as per the *session-restore-retry-timer* configuration and for the following *result-error-code* actions:

- continue
- continue-allow
- continue-block
- continue-offline
- retry-and-continue
- retry-and-cont-allow
- retry-and-cont-block
- retry-and-cont-offline

6.2.1.8.30 result-code-4xxx

Syntax

result-code-4xxx *result-error-code* [restore]

no result-code-4xxx

Context

config>mobile>profile>charging>ccfh-profile

Description

This command sets the result code value 4xxx for the CCFH profile.

Parameters***result-error-code***

specifies the error code

Values:

terminate

specifies the error code for failure handling "terminate"

continue

specifies the error code for failure handling "continue"

continue-allow

specifies the error code for failure handling "continue" with the action "allow"

continue-block

specifies the error code for failure handling "continue" with the action "block"

continue-offline

specifies the error code for failure handling "continue" with the action "offline"

retry-and-terminate

specifies the error code for failure handling with the action "retry-and-terminate"

retry-and-continue

if failover to a secondary OCS is supported, the secondary OCS is tried before the session is continued under Assume Positive, for the specified result-code

retry-and-cont-allow

specifies the error code for failure handling "retry-and-continue" with the action "allow"

retry-and-cont-block

specifies the error code for failure handling "retry-and-continue" with the action "block"

retry-and-cont-offline

specifies the error code for failure handling "retry-and-continue" with the action "offline"

restore

specifies that session restore applies as per the [session-restore-retry-timer](#) configuration and for the following *result-error-code* actions:

- continue
- continue-allow
- continue-block
- continue-offline
- retry-and-continue
- retry-and-cont-allow
- retry-and-cont-block
- retry-and-cont-offline

6.2.1.8.31 result-code-5xxx

Syntax

```
result-code-5xxx result-error-code [restore]  
no result-code-5xxx
```

Context

```
config>mobile>profile>charging>ccfh-profile
```

Description

This command sets the result code value 5xxx for the CCFH profile.

Parameters

result-error-code

specifies the error code

Values:

terminate

specifies the error code for failure handling "terminate"

continue

specifies the error code for failure handling "continue"

continue-allow

specifies the error code for failure handling "continue" with the action "allow"

continue-block

specifies the error code for failure handling "continue" with the action "block"

continue-offline

specifies the error code for failure handling "continue" with the action "offline"

retry-and-terminate

specifies the error code for failure handling with the action "retry-and-terminate"

retry-and-continue

if failover to a secondary OCS is supported, the secondary OCS is tried before the session is continued under Assume Positive, for the specified result-code

retry-and-cont-allow

specifies the error code for failure handling "retry-and-continue" with the action "allow"

retry-and-cont-block

specifies the error code for failure handling "retry-and-continue" with the action "block"

retry-and-cont-offline

specifies the error code for failure handling "retry-and-continue" with the action "offline"

restore

specifies that session restore applies as per the [session-restore-retry-timer](#) configuration and for the following *result-error-code* actions:

- continue
- continue-allow
- continue-block
- continue-offline
- retry-and-continue
- retry-and-cont-allow
- retry-and-cont-block
- retry-and-cont-offline

6.2.1.8.32 sbi-error-mapping-profile

Syntax

`sbi-error-mapping-profile profile-name`

`no sbi-error-mapping-profile`

Context

`config>mobile>profile>charging>ccfh-profile`

Description

This command configures the SBI error mapping profile for rejections received from CHF.

The no form of this command removes the configuration.

Parameters

profile-name

specifies the SBI error mapping profile name, up to 32 characters

6.2.1.8.33 service-restore-max-offset

Syntax

```
service-restore-max-offset max-offset  
no service-restore-max-offset
```

Context

```
config>gateway>profile>charging>ccfh-profile
```

Description

This command configures a time range to distribute all CCR-Init due to Credit Control session restore messages. When the session restore retry timer expires, all CCR-Init due to credit control session restore messages are triggered. With the command configured, the CCR-Init messages are distributed within the specified time range, that is, session-restore-retry timer + [0 - max-offset].

The no form of this command reverts to the default (0); no offset is used.

Parameters

max-offset

specifies the time range in minutes

Values: 0 to 60 minutes

Default: 0 (disabled)

6.2.1.8.34 session-restore-retry-timer

Syntax

```
session-restore-retry-timer retry-timer  
no session-restore-retry-timer
```

Context

```
config>gateway>profile>charging>ccfh-profile
```

Description

This command specifies a session restore retry timer. When the fh-session-continue-timer and/or fh-volume-limit commands are active (CCFH = Continue), the gateway periodically sends a CCR-I to restore the Gy session with the OCS. If the OCS sends a successful CCA-I, the Gy session starts again with the installed rating groups in first packet mode. Any usage and unused quotas from the previous session

are discarded and not reported to the OCS. The `session-restore-timer` must be smaller than the `fh-session-continue-timer`, when `fh-session-continue-timer` is used (non-zero). The CCR-I does not include a request for quota (for any configured default rating group). The installed rating groups operate in first packet mode, whereby the request for quota (CCR-U) is issued when the first packet for that rating group is received.

The no form of this command reverts to the default.

**NOTE:**

If the CCFH Continue is initiated due to an error result code, the session restore retry timer is activated only for the 3002, 3004, and 3005 result codes.

Parameters

retry-timer

specifies the retry time in minutes

Values: 0 to 60 minutes

Default: 0 (disabled)

6.2.1.8.35 store-and-replay

Syntax

`store-and-replay`

Context

`config>gateway>profile>charging>ccfh-profile`

Description

This command enables the context to configure the Nchf Buffer and Playback functionality.

6.2.1.8.36 buffer-playback-timer

Syntax

`buffer-playback-timer time`

`no buffer-playback-timer`

Context

config>gateway>profile>charging>ccfh-profile>store-and-replay

Description

This command specifies a buffer playback timer. When the timer expires, CMG attempts to play back the buffered messages to the CHF. If the timer is not configured, and the Nchf Buffer and Playback functionality is enabled, the default value of five minutes is used. If the timer value is set to zero, the default value of five minutes is used.

The no form of this command reverts to the default.

Parameters

time

specifies the buffer playback time, in minutes

Values: 0 to 60

Default: 5

6.2.1.8.37 enable

Syntax

[no] enable

Context

config>gateway>profile>charging>ccfh-profile>store-and-replay

Description

This command enables the Nchf Buffer and Playback functionality. By default, the functionality is disabled.

6.2.1.9 Charging profile CDR AVP command descriptions

6.2.1.9.1 cdr-avp-options

Syntax

[no] cdr-avp-options *charging-profile-id*

Context

config>mobile>profile>charging

Description

This command enables the context to configure charging parameters for the CDR AVP options profile. This is used to configure and manage individual IEs included in CDRs.

Parameters

charging-profile-id

specifies a profile charging identifier

Values: 0 to 255

6.2.1.9.2 3gpp2-bsid

Syntax

3gpp2-bsid {include | exclude}

no 3gpp2-bsid

Context

config>mobile>profile>charging>cdr-avp-options

Description

This command specifies that the User Location Information for 3gpp2 access (for example, 3gpp2 BSID for eHRPD). Applicable to PGW-CDR only.

The no form of this command reverts to the default.

Default

no 3gpp2-bsid

Parameters

include

includes this IE in the CDR

exclude

excludes this IE from the CDR

6.2.1.9.3 apn-ambr

Syntax

```
apn-ambr {include | exclude}
no apn-ambr
```

Context

```
config>mobile>profile>charging>cdr-avp-options
```

Description

This command specifies the option to include or exclude the aPNAggregateMaxBitrateUL and aPNAggregateMaxBitrateDL fields of the EPCQoSInformation IE in the PGW-CDRs.

The no form of this command reverts to the default (include).

Default

```
no apn-ambr
```

Parameters

include

includes the aPNAggregateMaxBitrateUL and aPNAggregateMaxBitrateDL fields in the PGW-CDRs

exclude

excludes the aPNAggregateMaxBitrateUL and aPNAggregateMaxBitrateDL fields from the PGW-CDRs

6.2.1.9.4 apn-network-id

Syntax

```
apn-network-id {include | exclude}
no apn-network-id
```

Context

```
config>mobile>profile>charging>cdr-avp-options
```

Description

This command specifies that the Access Point Name Network Identifier be included or not included in CDRs. Indicates the logical name (network identifier part of the APN) of the connected access point. Always included in the CDR, if provisioned. Applicable to SGW, PGW, and GGSN CDRs.

The no form of this command reverts to the default.

Default

no apn-network-id

Parameters

include

includes this IE in the CDR

exclude

excludes this IE from the CDR

virtual

includes the virtual APN

6.2.1.9.5 apn-rate-control

Syntax

```
apn-rate-control {include | exclude}
```

```
no apn-rate-control
```

Context

```
config>mobile>profile>charging>cdr-avp-options
```

Description

This command specifies the option to include or exclude the APN rate control CDR field in the PGW-CDRs.

The no form of this command reverts to the default.

Default

no apn-rate-control

Parameters

include

includes this IE in the CDR

exclude

excludes this IE from the CDR

6.2.1.9.6 apn-selection-mode

Syntax

```
apn-selection-mode {include | exclude}
```

```
no apn-selection-mode
```

Context

```
config>mobile>profile>charging>cdr-avp-options
```

Description

This command specifies that the APN Selection Mode be included or not included in CDRs. An index indicating how the APN was selected. Always included in the CDR, if provisioned. Applicable to SGW, PGW, and GGSN CDRs.

The no form of this command reverts to the default.

Default

```
no apn-selection-mode
```

Parameters**include**

includes this IE in the CDR

exclude

excludes this IE from the CDR

6.2.1.9.7 camel-information

Syntax

```
camel-information {include | exclude}
```

```
no camel-information
```

Context

config>mobile>profile>charging>cdr-avp-options

Description

This command set of CAMEL information related to IP CAN bearer. Applies only to GGSN / PGW CDRs.

The no form of this command reverts to the default.

Default

no camel-information

Parameters

include

includes this IE in the CDR

exclude

excludes this IE from the CDR

6.2.1.9.8 charging-char-select-mode

Syntax

charging-char-select-mode {include | exclude}

no charging-char-select-mode

Context

config>mobile>profile>charging>cdr-avp-options

Description

This command specifies whether the Charging Characteristics Selection Mode is included or not in CDRs. Always included in the CDR, if provisioned. Applicable to SGW, PGW, and GGSN CDRs.

The no form of this command reverts to the default.

Default

no charging-char-select-mode

Parameters

include

includes this IE in the CDR

exclude

excludes this IE from the CDR

6.2.1.9.9 cn-operator-selection

Syntax

cn-operator-selection {include | exclude}

no cn-operator-selection

Context

config>mobile>profile>charging>cdr-avp-options

Description

This command specifies the option to include or exclude the CN Operator Selection CDR field in CDRs. Applicable to SGW and PGW CDRs.

The no form of this command reverts to the default.

Default

no cn-operator-selection

Parameters**include**

includes this IE in the CDR

exclude

excludes this IE from the CDR

6.2.1.9.10 cp-ciot-eps-optimisation-indicator

Syntax

cp-ciot-eps-optimisation-indicator {include | exclude}

no cp-ciot-eps-optimisation-indicator

Context

config>mobile>profile>charging>cdr-avp-options

Description

This command specifies that the Cp Ciot EPS Optimisation Indicator be included or not included in CDRs. It will indicate whether S11-U is used or not.

The no form of this command reverts to the default.

Default

no cp-ciot-eps-optimisation-indicator

Parameters

include

includes this IE in the CDR

exclude

excludes this IE from the CDR

6.2.1.9.11 diagnostics

Syntax

diagnostics {include | exclude}

no diagnostics

Context

config>mobile>profile>charging>cdr-avp-options

Description

This command specifies that the “Diagnostics” be included or not included in CDRs. Detailed reason for the release of the connection. Included in the CDR, by default. Applicable to SGW, PGW and GGSN CDRs.

The no form of this command reverts to the default.

Default

no diagnostics

Parameters

include

includes this IE in the CDR

exclude

excludes this IE from the CDR

6.2.1.9.12 dualstack-epdg-address-ipv6

Syntax

```
dualstack-epdg-address-ipv6 {include | exclude}
no dualstack-epdg-address-ipv6
```

Context

```
config>mobile>profile>charging>cdr-avp-options
```

Description

This command specifies the option to include or exclude the ePDG IPv6 address in case of IPv4v6 dual stack, in ePDG-CDRs. Applicable to Ga and Rf interfaces.

The no form of this command reverts to the default.

Default

```
no dualstack-epdg-address-ipv6
```

Parameters

include

includes this IE in the CDR for Ga and ACR for Rf

exclude

excludes this IE from the CDR for Ga and ACR for Rf

6.2.1.9.13 dualstack-pgw-address-ipv6

Syntax

```
dualstack-pgw-address-ipv6 {include | exclude}
no dualstack-pgw-address-ipv6
```

Context

```
config>mobile>profile>charging>cdr-avp-options
```

Description

This command specifies the option to include or exclude the PGW IPv6 address in case of IPv4v6 dual stack, in SGW/PGW CDRs. Applicable to Ga and Rf interfaces.

The no form of this command reverts to the default.

Default

no dualstack-pgw-address-ipv6

Parameters

include

includes this IE in the CDR for Ga and ACR for Rf

exclude

excludes this IE from the CDR for Ga and ACR for Rf

6.2.1.9.14 dualstack-serving-node-ipv6-addr

Syntax

dualstack-serving-node-ipv6-addr {include | exclude}

no dualstack-serving-node-ipv6-addr

Context

config>mobile>profile>charging>cdr-avp-options

Description

This command specifies the option to include or exclude the serving node IPv6 address in case of IPv4v6 dual stack, in SGW/PGW CDRs. Applicable to Ga and Rf interfaces.

The no form of this command reverts to the default.



NOTE:

From a SGW perspective, a serving node can be an MME or an S4-SGSN. From a PGW perspective, a serving node can be an SGW, an HSGW, an SGSN, or an ePDG.

Default

no dualstack-serving-node-ipv6-address

Parameters

include

includes this IE in the CDR for Ga and ACR for Rf

exclude

excludes this IE from the CDR for Ga and ACR for Rf

6.2.1.9.15 dualstack-sgw-address-ipv6

Syntax

```
dualstack-sgw-address-ipv6 {include | exclude}
```

```
no dualstack-sgw-address-ipv6
```

Context

```
config>mobile>profile>charging>cdr-avp-options
```

Description

This command specifies the option to include or exclude the SGW IPv6 address in case of IPv4v6 dual stack, in SGW-CDRs. Applicable to Ga and Rf interfaces.

The no form of this command reverts to the default.

Default

```
no dualstack-sgw-address-ipv6
```

Parameters

include

includes this IE in the CDR for Ga and ACR for Rf

exclude

excludes this IE from the CDR for Ga and ACR for Rf

6.2.1.9.16 dualstack-twag-address-ipv6

Syntax

```
dualstack-twag-address-ipv6 {include | exclude}
```

```
no dualstack-twag-address-ipv6
```

Context

```
config>mobile>profile>charging>cdr-avp-options
```

Description

This command specifies the option to include or exclude a TWAG control plane IPv6 address, in case of IPv4v6 dual stack, within a PGW-CDR.

The no form of this command reverts to the default.

Default

```
no dualstack-twag-address-ipv6
```

Parameters

include

includes the TWAG control plane IPv6 address

exclude

excludes the TWAG control plane IPv6 address

6.2.1.9.17 dynamic-addr-flag

Syntax

```
dynamic-addr-flag {include | exclude}
```

```
no dynamic-addr-flag
```

Context

```
config>mobile>profile>charging>cdr-avp-options
```

Description

This command indicates if the PDN address is dynamically allocated. Applicable to SGW, PGW, and GGSN CDRs.

The no form of this command reverts to the default.

Default

```
no dynamic-addr-flag
```

Parameters

include

includes this IE in the CDR

exclude

excludes this IE from the CDR

6.2.1.9.18 dynamic-addr-flag-extension

Syntax

```
dynamic-addr-flag-extension {include | exclude}
```

```
no dynamic-addr-flag-extension
```

Context

```
config>mobile>profile>charging>cdr-avp-options
```

Description

This command indicates if the PDN address extension is dynamically allocated. Applicable to SGW, PGW, and GGSN CDRs.

The no form of this command reverts to the default.

Default

```
no dynamic-addr-flag-extension
```

Parameters

include

includes this IE in the CDR

exclude

excludes this IE from the CDR

6.2.1.9.19 enhanced-diagnostics

Syntax

```
enhanced-diagnostics {include | exclude}
```

```
no enhanced-diagnostics
```

Context

config>mobile>profile>charging>cdr-avp-options

Description

This command specifies that the enhanced-diagnostics be included or not included in CDRs. It contains the RAN/NAS cause value received from MME, ePDG, or TWAG.

The no form of this command reverts to the default.

Default

no enhanced-diagnostics

Parameters

include

includes this IE in the CDR

exclude

excludes this IE from the CDR

6.2.1.9.20 extended-bw-nr

Syntax

extended-bw-nr {include | exclude} [pre-extended]

no extended-bw-nr

Context

config>mobile>profile>charging>cdr-avp-options

Description

This command configures whether the following extended IEs used for high data rates in option 3x NR, are sent to the OFCS as part of EPC QOS information:

- extendedMaxRequestedBWUL
- extendedMaxRequestedBWDL
- extendedGBRUL
- extendedGBRDL
- extendedAPNAMBRUL

- extendedAPNAMBRDL

The no form of this command reverts to the default (disabled).

Default

no extended-bw-nr

Parameters

include | exclude

specifies inclusion or exclusion of these IEs within the CDR for the Ga interface and within the ACR for the RF interface

Default: exclude

pre-extended

specifies the pre-extended qualifier. When the qualifier is enabled, and if an extended QoS field is included, the corresponding regular field is also included with the maximum value of $2^{32}-1$ bps.

6.2.1.9.21 imsi-unauthenticated-flag

Syntax

```
imsi-unauthenticated-flag {include | exclude}
```

```
no imsi-unauthenticated-flag
```

Context

```
config>mobile>profile>charging>cdr-avp-options
```

Description

This command specifies to include or exclude the IMSI unauthenticated flag.

The no form of this command reverts to the default (excluded).

Parameters

include | exclude

specifies inclusion or exclusion of this IE in the CDR

Default: exclude

6.2.1.9.22 ims-sig-context

Syntax

```
ims-sig-context {include | exclude}
no ims-sig-context
```

Context

```
config>mobile>profile>charging>cdr-avp-options
```

Description

This command specifies that the IMS Signaling context be included or not included in CDRs. Indicates if the IP-CAN bearer is used for IMS signaling. Applicable to SGW, PGW, and GGSN CDRs.



NOTE:

In the current release, this command is not supported.

The no form of this command reverts to the default.

Default

```
no ims-sig-context
```

Parameters

include

includes this IE in the CDR

exclude

excludes this IE from the CDR

6.2.1.9.23 imsi-unauthenticated-flag

Syntax

```
imsi-unauthenticated-flag {include | exclude}
no imsi-unauthenticated-flag
```

Context

```
config>mobile>profile>charging>cdr-avp-options
```

Description

This command indicates if the served IMSI is not authenticated (emergency bearer). Applicable to SGW-CDR / PGW-CDR only.

The no form of this command reverts to the default.

Default

no imsi-unauthenticated-flag

Parameters

include

includes this IE in the CDR

exclude

excludes this IE from the CDR

6.2.1.9.24 list-of-service-data

Syntax

```
list-of-service-data {include | exclude}
```

```
no list-of-service-data
```

Context

```
config>mobile>profile>charging>cdr-avp-options
```

Description

This command specifies that the List of Service Data containers be included or not included in CDRs. A list of changes in charging conditions for all SDFs within the IP-CAN bearer categorized per rating group or per combination of rating group and service ID. Always included in the CDR, if provisioned. Applicable to eG-CDR / PGW-CDR only.

The no form of this command reverts to the default.

Default

no list-of-service-data

Parameters

include

includes this IE in the CDR

exclude

excludes this IE from the CDR

6.2.1.9.25 list-traffic-data-volume

Syntax

```
list-traffic-data-volume {include | exclude}  
no list-traffic-data-volume
```

Context

config>mobile>profile>charging>cdr-avp-options

Description

This command specifies that the List of Traffic Data Volumes be included or not included in CDRs. Always included in the CDR, if provisioned. Applicable to SGW / GGSN CDRs.

The no form of this command reverts to the default.

Default

no list-traffic-data-volume

Parameters**include**

includes this IE in the CDR

exclude

excludes this IE from the CDR

6.2.1.9.26 local-record-seq-no

Syntax

```
local-record-seq-no {include | exclude}  
no local-record-seq-no
```

Context

config>mobile>profile>charging>cdr-avp-options

Description

This command specifies that the Local Record sequence Number be included or not included in CDRs. Indicates a unique record number created by this node. Always included in the CDR, if provisioned. Applicable to SGW, PGW, and GGSN CDRs.

The no form of this command reverts to the default.

Default

no local-record-seq-no

Parameters

include

includes this IE in the CDR

exclude

excludes this IE from the CDR

6.2.1.9.27 low-priority-indicator

Syntax

```
low-priority-indicator {include | exclude}
no low-priority-indicator
```

Context

```
config>mobile>profile>charging>cdr-avp-options
```

Description

This command indicates if this IP-CAN session has a low priority, for instance, for machine type communication. Applicable to SGW and PGW CDRs.

The no form of this command reverts to the default.

Default

no low-priority-indicator

Parameters

include

includes this IE in the CDR

exclude

excludes this IE from the CDR

6.2.1.9.28 max-requested-bandwidth

Syntax

max-requested-bandwidth {include | exclude}

no max-requested-bandwidth

Context

config>mobile>profile>charging>cdr-avp-options

Description

This command specifies the option to include or exclude the maxRequestedBandwidthUL and maxRequestedBandwidthDL fields of the EPCQoSInformation IE in the PGW-CDRs.

The no form of this command reverts to the default (include).

Default

no max-requested-bandwidth

Parameters**include**

includes the maxRequestedBandwidthUL and maxRequestedBandwidthDL fields in the PGW-CDRs

exclude

excludes the maxRequestedBandwidthUL and maxRequestedBandwidthDL fields from the PGW-CDRs

6.2.1.9.29 mo-exception-data-counter

Syntax

mo-exception-data-counter {include | exclude}

no mo-exception-data-counter

Context

config>mobile>profile>charging>cdr-avp-options

Description

This command specifies the option to include or exclude the MO exception data counter CDR field in the PGW and SGW CDRs.

The no form of this command reverts to the default.

Default

no mo-exception-data-counter

Parameters

include

includes this IE in the CDR

exclude

excludes this IE from the CDR

6.2.1.9.30 ms-time-zone

Syntax

ms-time-zone {include | exclude}

no ms-time-zone

Context

config>mobile>profile>charging>cdr-avp-options

Description

This command specifies the timezone the UE is located in. Included if available (provided by MME). Applicable to SGW, PGW, and GGSN CDRs.

The no form of this command reverts to the default.

Default

no ms-time-zone

Parameters

include

includes this IE in the CDR

exclude

excludes this IE from the CDR

6.2.1.9.31 node-id

Syntax

```
node-id {include | exclude}
```

```
no node-id
```

Context

```
config>mobile>profile>charging>cdr-avp-options
```

Description

This command specifies that the Node ID be included or not included in CDRs. Name of the recording entity. Always included in the CDR, if provisioned. Applicable to SGW, PGW, and GGSN CDRs.

The no form of this command reverts to the default.

Default

```
no node-id
```

Parameters

include

includes this IE in the CDR

exclude

excludes this IE from the CDR

6.2.1.9.32 nr-user-location-info

Syntax

```
nr-user-location-info {include | exclude}
```

```
no nr-user-location-info
```

Context

```
config>mobile>profile>charging>cdr-avp-options
```

Description

This command specifies whether the proprietary NR User Location Information is included or not in the PGW-CDRs.

The no form of this command reverts to the default.

Default

```
no nr-user-location-info
```

Parameters

include

includes the proprietary 5G Location CDR field in the PGW-CDR RecordExtensions when SMF legacy charging is enabled

exclude

excludes the 5G location from the PGW-CDRs when SMF Legacy charging is enabled

6.2.1.9.33 nw-init-pdp-context

Syntax

```
nw-init-pdp-context {include | exclude}  
no nw-init-pdp-context
```

Context

```
config>mobile>profile>charging>cdr-avp-options
```

Description

This command indicates if it is a network initiated pdp context. Applicable to G-CDR / eG-CDR only.

The no form of this command reverts to the default.

Default

```
no nw-init-pdp-context
```

Parameters

include

includes this IE in the CDR

exclude

excludes this IE from the CDR

6.2.1.9.34 pdn-connection-charging-id

Syntax

```
pdn-connection-charging-id {include | exclude}
```

```
no pdn-connection-charging-id
```

Context

```
config>mobile>profile>charging>cdr-avp-options
```

Description

This command specifies the PDN connection identifier, used to identify different records belonging to the same PDN connection. This field includes Charging Id of first IP-CAN bearer activated within the PDN connection. Applicable to SGW and PGW CDRs.

The no form of this command reverts to the default.

Default

```
no pdn-connection-charging-id
```

Parameters

include

includes this IE in the CDR

exclude

excludes this IE from the CDR

6.2.1.9.35 pdn-pdp-address-prefix-length

Syntax

```
pdn-pdp-address-prefix-length {include | exclude}
```

```
no pdn-pdp-address-prefix-length
```

Context

config>mobile>profile>charging>cdr-avp-options

Description

This command specifies the option to include or exclude the prefix length of an IPv6-typed Served PDP/PDN Address. The field is not available for prefix length of 64 bits, as in this case the 64 bit prefix length default will be assumed.

The no form of this command reverts to the default.

Default

no pdn-pdp-address-prefix-length

Parameters

include

includes this IE in the CDR

exclude

excludes this IE from the CDR

6.2.1.9.36 pdp-pdn-type

Syntax

pdp-pdn-type {include | exclude}

no pdp-pdn-type

Context

config>mobile>profile>charging>cdr-avp-options

Description

This command indicates the PDN type (IPv4, IPv6 or IPv4v6). Always included in the CDR, if provisioned. Applicable to SGW, PGW, and GGSN CDRs.

The no form of this command reverts to the default.

Default

no pdp-pdn-type

Parameters

include

includes this IE in the CDR

exclude

excludes this IE from the CDR

6.2.1.9.37 pdp-pdn-type-extension

Syntax

pdp-pdn-type-extension {include | exclude}

no pdp-pdn-type-extension

Context

config>mobile>profile>charging>cdr-avp-options

Description

This command specifies the option to include or exclude the PDP PDN type extension CDR field in the PGW and SGW CDRs.

The no form of this command reverts to the default.

Default

no pdp-pdn-type-extension

Parameters

include

includes this IE in the CDR

exclude

excludes this IE from the CDR

6.2.1.9.38 pgw-address

Syntax

pgw-address {include | exclude}

no pgw-address

Context

config>mobile>profile>charging>cdr-avp-options

Description

This command specifies the Control Plane IP address of the PGW used. Applicable to SGW-CDR.

The no form of this command reverts to the default.

Default

no pgw-address

Parameters

include

includes this IE in the CDR

exclude

excludes this IE from the CDR

6.2.1.9.39 pgw-plmn-id

Syntax

pgw-plmn-id {include | exclude}

no pgw-plmn-id

Context

config>mobile>profile>charging>cdr-avp-options

Description

This command specifies the PLMN Identifier (MCC & MNC) of the PGW used. Applicable to SGW and PGW CDRs.

The no form of this command reverts to the default.

Default

no pgw-plmn-id

Parameters

include

includes this IE in the CDR

excl ude

excludes this IE from the CDR

6.2.1.9.40 pra-info

Syntax

pra-info {include | exclude}

no pra-info

Context

config>mobile>profile>charging>cdr-avp-options

Description

This command specifies the option to include or exclude the PRA information toward an offline charging server. The change-ue-presence-pra event trigger must be enabled for offline charging.

The no form of this command reverts to the default.

Default

no pra-info

Parameters**inc lude**

includes this IE in the CDR

excl ude

excludes this IE from the CDR

6.2.1.9.41 ps-furnish-charging-information

Syntax

ps-furnish-charging-information {include | exclude}

no ps-furnish-charging-information

Context

config>mobile>profile>charging>cdr-avp-options

Description

This command specifies the online charging session information. Applicable to eG-CDR and PGW-CDR only.

The no form of this command reverts to the default.

Default

no ps-furnish-charging-information

Parameters

include

includes this IE in the CDR

exclude

excludes this IE from the CDR

6.2.1.9.42 originating-bsid

Syntax

originating-bsid {include | exclude}

no originating-bsid

Context

config>mobile>profile>charging>cdr-avp-options

Description

This command is used to enable sending the originating BSID to the OFCS. When enabled, the information is sent to the OFCS in the ACR (Start/Interim/Stop) messages, and is the value of the 3GPP-BSID AVP at the PS level when the session was initiated, or for WiFi, when the session attaches to the network.

The no form of this command reverts to the default (disabled).



NOTE:

This command is applicable to Diameter RF deployments only.

Default

no originating-bsid

Parameters

include

enables sending the parameter to the OFCS

exclude}

excludes the parameter from being sent to the OFCS

6.2.1.9.43 originating-user-location-info

Syntax

```
originating-user-location-info {include | exclude}
```

```
no originating-user-location-info
```

Context

```
config>mobile>profile>charging>cdr-avp-options
```

Description

This command is used to enable sending the originating user location information to the OFCS. When enabled, the information is sent to the OFCS in the ACR (Start/Interim/Stop) messages, and is the value of the 3GPP-User-Location-Inf AVP at the PS level when the session was initiated, or for WiFi, when the session attaches to the network.

The no form of this command reverts to the default (disabled).



NOTE:

This command is applicable to Diameter RF deployments only.

Default

```
no originating-user-location-inf
```

Parameters

include

enables sending the parameter to the OFCS

exclude}

excludes the parameter from being sent to the OFCS

6.2.1.9.44 ran-secondary-rat-usage-reports

Syntax

```
ran-secondary-rat-usage-reports {include | exclude}
no ran-secondary-rat-usage-reports
```

Context

```
config>mobile>profile>charging>cdr-avp-options
```

Description

This command enables sending secondary RAT usage reports for 5G NR to the offline charging server. For the GTP' GA interface, the List of RAN Secondary RAT Usage Report field is populated within the SGW-CDR and PGW-CDR as applicable. For the Diameter Rf interface, the SGW and PGW send the RAN Secondary RAT Usage Report IE (within PS-Information) in ACR (INTERIM) and ACR (STOP) messages as applicable, and the OFCS generates the CDRs.

The no form of this command reverts to the default (excluded).

Default

```
no ran-secondary-rat-usage-reports
```

Parameters

include

includes the IE within the CDR for GA and Diameter RF interfaces

exclude

excludes the IE within the CDR for GA and Diameter RF interfaces

6.2.1.9.45 rat-type

Syntax

```
rat-type {include | exclude}
no rat-type
```

Context

```
config>mobile>profile>charging>cdr-avp-options
```

Description

This command specifies that the RAT type be included or not included in CDRs. Indicates the Radio Access Technology type used by the UE. Included if available. Applicable to SGW, PGW, and GGSN CDRs.

The no form of this command resets the configuration to the default.

Default

no rat-type

Parameters

include

includes this IE in the CDR

exclude

excludes this IE from the CDR

6.2.1.9.46 record-extension-visited-url

Syntax

record-extension-visited-url {include | exclude}

no record-extension-visited-url

Context

config>mobile>profile>charging>cdr-avp-options

Description

This command is used to control the inclusion or exclusion of a proprietary PGW-CDR field within the recordExtensions IE used to report traffic per visited URL domain.

The no form of this command resets the configuration to the default.

Default

no record-extension-visited-url

Parameters

include

includes the proprietary PGW-CDR field within the recordExtensions IE

exclude

excludes the proprietary PGW-CDR field from the recordExtensions IE

6.2.1.9.47 sdc-charging-id

Syntax

```
sdc-charging-id {include | exclude}
no sdc-charging-id
```

Context

```
config>mobile>profile>charging>cdr-avp-options
```

Description

This command is used to control the inclusion or exclusion of 3gpp-charging-id within a service data container (SDC) AVP. By default, 3gpp-charging-id is excluded from an SDC.

The no form of this command resets the configuration to the default.

Parameters

include

includes 3gpp-charging-id within an SDC container

exclude

excludes 3gpp-charging-id from an SDC container

Default: exclude

6.2.1.9.48 sdc-pra-status

Syntax

```
sdc-pra-status {include | exclude}
no sdc-pra-status
```

Context

```
config>mobile>profile>charging>cdr-avp-options
```

Description

This command specifies the option to include or exclude the Presence Reporting Area (PRA) status within a Service Data Container (SDC), toward an offline charging server. PRA status contains the status of the UE

presence (that is, whether the UE is inside or outside) in the PRA identified by the PRA identifier contained in the PGW-CDR or TDF-CDR main level. This is included in the service data container when the initial status is reported after the PGW-CDR or TDF-CDR is opened, or if the previous container's change condition is Presence in Presence Reporting Area Change. The change-ue-presence-pra charging-info-addition trigger must also be enabled for offline charging.

The no form of this command reverts to the default (excluded).

Default

no sdc-pra-status

Parameters

include

includes this IE in the CDR

exclude

excludes this IE from the CDR

6.2.1.9.49 sdc-qci

Syntax

```
sdc-qci {include | exclude}
```

```
no sdc-qci
```

Context

```
config>mobile>profile>charging>cdr-avp-options
```

Description

This command is used to control the inclusion and exclusion of the QoS class identifier (QCI) within an SDC AVP. By default, QCI is included in an SDC container.

The no form of this command resets the configuration to the default.

Parameters

include

QCI is included within an SDC container

exclude

QCI is excluded from an SDC container

Default: include

6.2.1.9.50 sequence-number-for-complete-cdr

Syntax

```
sequence-number-for-complete-cdr {include | exclude}  
no sequence-number-for-complete-cdr
```

Context

```
config>mobile>profile>charging>cdr-avp-options
```

Description

This command configures the behavior of the RecordSequenceNumber CDR field.

By default (exclude), the RecordSequenceNumber is written to the SGW-CDRs and PGW-CDRs except for the case when the CDR generated due to a session release is also the first CDR for the session (session with a single CDR).

If configured, the RecordSequenceNumber is written to SGW-CDRs and PGW-CDRs also in the case of a session with a single CDR. In this case, the RecordSequenceNumber value is 1.

The no form of this command resets the configuration to the default (exclude).

Default

```
no sequence-number-for-complete-cdr
```

Parameters

include

includes the RecordSequenceNumber for sessions with a single CDR

exclude

excludes the RecordSequenceNumber for sessions with a single CDR

6.2.1.9.51 served-imeisv

Syntax

```
served-imeisv include [acr-interim] [acr-stop]  
served-imeisv exclude  
no served-imeisv
```

Context

```
config>mobile>profile>charging>cdr-avp-options
```

Description

This command specifies the IMEISV of the UE to be included, if available. Applicable to SGW, PGW, and GGSN CDRs.

The no form of this command reverts to the default.

Default

no served-index

Parameters

include

includes this IE in the CDR

exclude

excludes this IE from the CDR

acr-interim

includes IMEISV in the ACR-Interim message

acr-stop

includes IMEISV in the ACR-Stop message

6.2.1.9.52 served-mn-nai

Syntax

```
served-mn-nai {include | exclude}
```

```
no served-mn-nai
```

Context

```
config>mobile>profile>charging>cdr-avp-options
```

Description

This command specifies the mobile node identifier in NAI format (based on IMSI). Applicable to PGW-CDR only.

The no form of this command reverts to the default.

Default

no served-mn-nai

Parameters

include

includes this IE in the CDR

exclude

excludes this IE from the CDR

6.2.1.9.53 served-msisdn

Syntax

served-msisdn {include | exclude}

no served-msisdn

Context

config>mobile>profile>charging>cdr-avp-options

Description

This command specifies the MSISDN of the UE to be included, if available. Applicable to SGW, PGW, and GGSN CDRs.

The no form of this command reverts to the default.

Default

no served-msisdn

Parameters

include

includes this IE in the CDR

exclude

excludes this IE from the CDR

6.2.1.9.54 served-pdp-pdn-addr

Syntax

```
served-pdp-pdn-addr {include | exclude}
no served-pdp-pdn-addr
```

Context

```
config>mobile>profile>charging>cdr-avp-options
```

Description

This command specifies the IP address allocated to the PDN connection. IPv4 when PDN Type is IPv4 or IPv6 when PDN Type is IPv6 or IPv4v6. Applicable to SGW, PGW, and GGSN CDRs.

The no form of this command reverts to the default.

Default

```
no served-pdp-pdn-addr
```

Parameters

include

includes this IE in the CDR

exclude

excludes this IE from the CDR

6.2.1.9.55 served-pdp-pdn-addr-extension

Syntax

```
served-pdp-pdn-addr-extension {include | exclude}
no served-pdp-pdn-addr-extension
```

Context

```
config>mobile>profile>charging>cdr-avp-options
```

Description

This command specifies the IP address allocated to the PDN connection. This field holds IPv4 address of the served IMSI, if available, when PDN type is IPv4v6. Applicable to SGW / PGW CDRs.

The no form of this command reverts to the default.

Default

no served-pdp-pdn-addr-extension

Parameters

include

includes this IE in the CDR

exclude

excludes this IE from the CDR

6.2.1.9.56 serving-node-plmn-id

Syntax

serving-node-plmn-id {include | exclude}

no serving-node-plmn-id

Context

config>mobile>profile>charging>cdr-avp-options

Description

This command specifies the SGW MCC+MNC used. Included in the CDR, if provisioned. Applicable to SGW, PGW, and GGSN CDRs.

The no form of this command reverts to the default.

Default

no serving-node-plmn-id

Parameters

include

includes this IE in the CDR

exclude

excludes this IE from the CDR

6.2.1.9.57 serving-plmn-rate-control

Syntax

```
serving-plmn-rate-control {include | exclude}
no serving-plmn-rate-control
```

Context

```
config>mobile>profile>charging>cdr-avp-options
```

Description

This command specifies the option to include or exclude the serving PLMN rate control CDR field in the SGW and PGW CDRs.

The no form of this command reverts to the default.

Default

```
no serving plmn-rate-control
```

Parameters

include

includes this IE in the CDR

exclude

excludes this IE from the CDR

6.2.1.9.58 sgi-ptp-tunnelling-method

Syntax

```
sgi-ptp-tunnelling-method {include | exclude}
no sgi-ptp-tunnelling-method
```

Context

```
config>mobile>profile>charging>cdr-avp-options
```

Description

This command specifies the option to include or exclude the SGi PTP tunneling method CDR field in the PGW-CDRs.

The no form of this command reverts to the default.

Default

no sgi-ptp-tunnelling-method

Parameters

include

includes this IE in the CDR

exclude

excludes this IE from the CDR

6.2.1.9.59 service-context-id

Syntax

service-context-id {include | exclude}

no service-context-id

Context

config>mobile>profile>charging>cdr-avp-options

Description

This command is used to configure the service-context-id sent to the OCS. While 3GPP specifies a particular format for the service-context-id, some operators expect custom service-context-id to be sent to OCS.

The no form of this command reverts to the default.

Default

3gpp compliant service-context-id

Parameters

include

includes this IE in the CDR

exclude

excludes this IE from the CDR

6.2.1.9.60 sgw-address

Syntax

```
sgw-address {include | exclude}
no sgw-address
```

Context

```
config>mobile>profile>charging>cdr-avp-options
```

Description

This command specifies the IP address of the serving node used.

The no form of this command reverts to the default.

Parameters

include

includes this IE in the CDR

exclude

excludes this IE from the CDR

Default: exclude

6.2.1.9.61 sgw-change

Syntax

```
sgw-change {include | exclude}
no sgw-change
```

Context

```
config>mobile>profile>charging>cdr-avp-options
```

Description

This command specifies that this is the first record after a change from another serving node (for example, SGW, ePDG or HSGW). It is applicable to SGW and ePDG CDRs.

The no form of this command reverts to the default.

Parameters

include

includes this IE in the CDR

exclude

excludes this IE from the CDR

Default: **exclude**

6.2.1.9.62 start-time

Syntax

start-time {**include** | **exclude** | **always-include**}

no start-time

Context

config>mobile>profile>charging>cdr-avp-options

Description

This command specifies the start time of the user IP-CAN session. Applicable to SGW and PGW CDRs.

The no form of this command reverts to the default.

Default

no start-time

Parameters

include

includes this IE in the CDR

exclude

excludes this IE from the CDR

always-include

includes this IE in every partial CDR of the default bearer

6.2.1.9.63 stop-time

Syntax

```
stop-time {include | exclude}
no stop-time
```

Context

```
config>mobile>profile>charging>cdr-avp-options
```

Description

This command specifies the stop time of the user IP-CAN session. Applicable to SGW/PGW CDRs. The no form of this command reverts to the default.

Default

```
no stop-time
```

Parameters

include

includes this IE in the CDR

exclude

excludes this IE from the CDR

6.2.1.9.64 tdv-charging-id

Syntax

```
tdv-charging-id {include | exclude}
no tdv-charging-id
```

Context

```
config>mobile>profile>charging>cdr-avp-options
```

Description

This command specifies the option to include or exclude the 3gpp-charging-id within the traffic data volumes container. By default, 3gpp-charging-id is excluded from the TDV.

The no form of this command resets the configuration to the default.

Default

no tdv-charging-id

Parameters

include

includes the 3gpp-charging-id within a TDV container

exclude

excludes the 3gpp-charging-id within a TDV container

6.2.1.9.65 tdv-pra-status

Syntax

tdv-pra-status {include | exclude}

no tdv-pra-status

Context

config>mobile>profile>charging>cdr-avp-options

Description

This command specifies the option to include or exclude the PRA status within a TDV container, toward an offline charging server. PRA status contains the status of the UE presence (that is, whether the UE is inside or outside) in the PRA identified by the PRA identifier contained in the SGW-CDR main level. This is included in the traffic data container when the initial status is reported after the SGW-CDR is opened, or if the previous container's change condition is Presence in Presence Reporting Area Change. The change-ue-presence-pra charging-info-addition trigger must also be enabled for offline-charging.

The no form of this command reverts to the default (excluded).

Default

no tdv-pra-status

Parameters

include

includes this IE in the CDR

exclude

excludes this IE from the CDR

6.2.1.9.66 twan-user-location-info

Syntax

```
twan-user-location-info {include | exclude}
no twan-user-location-info
```

Context

```
config>mobile>profile>charging>cdr-avp-options
```

Description

This command specifies the option to include or exclude the TWAN user location information in the PGW-CDRs. Applicable to Ga and Rf interfaces.

The no form of this command reverts to the default.

Default

```
no twan-user-location-info
```

Parameters

include

includes this IE in the CDR for Ga and ACR for Rf

exclude

excludes this IE from the CDR Ga and ACR for Rf

6.2.1.9.67 twan-user-location-information-r16

Syntax

```
twan-user-location-information-r16 {include | exclude}
no twan-user-location-information-r16
```

Context

```
config>mobile>profile>charging>cdr-avp-options
```

Description

This command specifies if the TWAN-User-Location-Information AVP with 3GPP release 16 format is included or not in the CDRs.

The no form of this command reverts to the default (excluded).



NOTE:

This command is not supported in the current release.

Parameters

include

includes this IE in the CDR

exclude

excludes this IE in the CDR

Default: exclude

6.2.1.9.68 uni-pdu-cp-only-flag

Syntax

uni-pdu-cp-only-flag {include | exclude}

no uni-pdu-cp-only-flag

Context

config>mobile>profile>charging>cdr-avp-options

Description

This command specifies the option to include or exclude the UNI PDU CP-only flag CDR field in the SGW and PGW CDRs.

The no form of this command reverts to the default.

Default

no uni-pdu-cp-only-flag

Parameters

include

includes this IE in the CDR

exclude

excludes this IE from the CDR

6.2.1.9.69 user-csg-info

Syntax

```
user-csg-info {include | exclude}
```

```
no user-csg-info
```

Context

```
config>mobile>profile>charging>cdr-avp-options
```

Description

This command specifies the option to include or exclude the user CSG information in the SGW and PGW CDRs. Applicable to Ga and Rf interfaces.

The no form of this command reverts to the default.

Default

```
no user-csg-info
```

Parameters**include**

includes this IE in the CDR for Ga and ACR for Rf

exclude

excludes this IE from the CDR Ga and ACR for Rf

6.2.1.9.70 user-location-info

Syntax

```
user-location-info {include | exclude}
```

```
no user-location-info
```

Context

```
config>mobile>profile>charging>cdr-avp-options
```

Description

This command specifies the option to include or exclude the user location information. Applicable to PGW / GGSN CDRs.

The no form of this command reverts to the default.

Default

no user-location-info

Parameters

include

includes this IE in the CDR

exclude

excludes this IE from the CDR

6.2.1.9.71 user-location-info-time

Syntax

```
user-location-info-time {include | exclude}
```

```
no user-location-info-time
```

Context

```
config>mobile>profile>charging>cdr-avp-options
```

Description

This command specifies the option to include or exclude the user location information time in the SGW/ PGW CDRs. Applicable to Ga and Rf interfaces.

The no form of this command reverts to the default.

Default

no user-location-info-time

Parameters

include

includes this IE in the CDR for Ga and ACR for Rf

exclude

excludes this IE from the CDR for Ga and ACR for Rf

6.2.1.9.72 uwan-user-location-information

Syntax

```
uwan-user-location-information {include | exclude}
no uwan-user-location-information
```

Context

```
config>mobile>profile>charging>cdr-avp-options
```

Description

This command specifies if the UWAN-User-Location-Information AVP is included or not in the CDRs. It contains the UE local IP address and the UDP port.

The no form of this command reverts to the default.

Parameters

include

includes this IE in the CDR

exclude

excludes this IE from the CDR

Default: exclude

6.2.1.9.73 uwan-user-location-information-r16

Syntax

```
uwan-user-location-information-r16 {include | exclude}
no uwan-user-location-information-r16
```

Context

```
config>mobile>profile>charging>cdr-avp-options
```

Description

This command specifies if the UWAN-User-Location-Information AVP with 3GPP release 16 format is included or not in the CDRs.

The no form of this command reverts to the default (excluded).

Parameters

include | exclude

specifies inclusion or exclusion of this IE in the CDR

Default: exclude

include | exclude

specifies inclusion or exclusion of this IE in the CDR

Default: exclude

6.2.1.10 Charging profile CHF command descriptions

6.2.1.10.1 chf-profile

Syntax

[no] *chf-profile charging-profile-id*

Context

config>mobile>profile>charging

Description

This command configures a charging function profile ID.

The no form of this command removes the profile from the configuration.

Parameters

charging-profile-id

specifies a charging profile ID

Values: 0 to 255

6.2.1.10.2 activity-threshold

Syntax

activity-threshold activity-threshold

no *activity-threshold*

Context

config>mobile>profile>charging>chf-profile

Description

This command configures the activity threshold for time-based metering.

Parameters

activity-threshold

specifies the activity threshold value in kb/s

Values: 0 to 1000000000

Default: 0

6.2.1.10.3 ccfh-profile

Syntax

[no] ccfh-profile *profile-id*

Context

config>mobile>profile>charging>chf-profile

Description

This command associates a CCFH profile to be used in a charging function profile. The CCFH profile is configured using the config>mobile>profile>charging>ccfh-profile command.

The no form of this command removes the association.

Parameters

profile-id

specifies a CCFH profile

Values: 0 to 255

6.2.1.10.4 custom-online-offline-reporting

Syntax

custom-online-offline-reporting [enable | disable]

no custom-online-offline-reporting

Context

config>mobile>profile>charging>chf-profile

Description

This command configures a customized reporting. When this command is enabled and Nchf_ConvergedCharging_Update is issued, the following apply:

- all online charged SDFs are reported at the Rating Group level
- all offline charged SDFs are reported at the Service ID level
- when the trigger is an offline charging trigger (time, volume limit, or both), only the offline SDFs are reported
- when the trigger is an online charging trigger (for example, quota exhausted), only the online SDFs are reported
- when the trigger is an online and offline trigger (for example, RAT change and final), all the online and offline SDFs are reported; additionally, a USU container reports the accumulated online SDFs



NOTE:

In the current release, the following triggers can be used with this functionality:

- volume-limit
- time-limit
- volume-limit-per-rg
- time-limit-per-rg
- quota-exhausted
- validity-time
- quota-holding-time
- final

The no form of this command disables the customized reporting.

Parameters

enable

specifies to enable the customized reporting

disable

specifies to disable the customized reporting

Default: disable

6.2.1.10.5 default-rg-list

Syntax

```
default-rg-list
```

Context

```
config>mobile>profile>charging>chf-profile
```

Description

This command enables the context to configure the default rating group list.

6.2.1.10.6 implicit-vol-threshold

Syntax

```
implicit-vol-threshold implicit-vol-threshold reason reason  
no implicit-vol-threshold
```

Context

```
config>mobile>profile>charging>chf-profile
```

Description

This command configures an implicit volume threshold, which is used when the allocated quota does not include a quota threshold from the CHF. This threshold value is used to notify the CHF when the specified threshold is reached. The threshold is specified as a percentage of the received granted units (volume quota). For example, when the implicit-vol-threshold value is configured at 20%, the CHF is notified when less than or equal to 20% of the allocated quota is remaining.

The no form of this command reverts to the default (disabled).

Parameters

implicit-vol-threshold

specifies the threshold percent of the allocated quota

Values: 0 to 99

Default: 0

reason

specifies the reported trigger type

Values: threshold | exhaustion

Default: threshold

6.2.1.10.7 ran-secondary-rat-usage-reports

Syntax

ran-secondary-rat-usage-reports {include | exclude}

no ran-secondary-rat-usage-reports

Context

config>mobile>profile>charging>chf-profile

Description

This command enables sending secondary RAT usage reports for 5G NR from the SMF to the Nchf interface.

The no form of this command reverts to the default (excluded).

Default

no ran-secondary-rat-usage-reports

Parameters**include**

includes the IE in the Nchf interface

exclude

excludes the IE from the Nchf interface

6.2.1.10.8 rating-group

Syntax

[no] rating-group *rating-group*

Context

config>mobile>profile>charging>chf-profile>default-rg-list

```
config>mobile>profile>charging>chf-profile
```

Description

This command adds a rating group in the default rating-group list, or specifies a rating group for which activity-threshold and UUC options can be configured, depending on the context.

The no form of this command removes the specified rating group from the default rating-group list.

Parameters

rating-group

specifies a rating group

Values: 0 to 4294967295

6.2.1.10.9 activity-threshold

Syntax

```
activity-threshold [activity-threshold] | [inherit]
```

```
no activity-threshold
```

Context

```
config>mobile>profile>charging>chf-profile>rating-group
```

Description

This command configures the activity threshold for time-based metering.

Parameters

activity-threshold

specifies the activity threshold value, in kb/s

Values: 0 to 100000000

inherit

specifies to inherit the activity threshold value configured in the
configure>mobile>profile>charging>chf-profile>activity-threshold
context

6.2.1.10.10 uuc-options-no-quota-initial

Syntax

```
uuc-options-no-quota-initial [total] [uplink] [downlink] [time] [suppress] [inherit]  
no uuc-options-no-quota-initial
```

Context

```
config>mobile>profile>charging>chf-profile  
config>mobile>profile>charging>chf-profile>rating-group
```

Description

This command specifies the UUC options for categories for which the CHF has not allocated any initial quota. The UUC for these categories will include the configured units.

The no form of this command reverts to the default, suppress (UUC not included).

Default

```
no uuc-options-no-quota-initial
```

Parameters

total

specifies to include the total volume

uplink

specifies to include the uplink volume

downlink

specifies to include the downlink volume

time

specifies to include the time

suppress

specifies not to include UUC categories for which the CHF has not allocated any initial quota

inherit

specifies that the rating group will inherit the values from the CHF profile level; applicable only under the rating-group context

Default: inherit

6.2.1.10.11 uuc-options-no-quota-mid-session

Syntax

```
uuc-options-no-quota-mid-session [suppress | do-not-suppress | inherit]
no uuc-options-no-quota-mid-session
```

Context

```
config>mobile>profile>charging>chf-profile
config>mobile>profile>charging>chf-profile>rating-group
```

Description

This command configures whether to include the UUC options for categories for which the CHF has allocated quota, but has run out of credit mid-stream; in general, when no valid granted quota exists for a rating group.

The no form of this command reverts to the default, suppress (UUC not included).

Default

no uuc-options-no-quota-mid-session

Parameters

suppress

specifies not to include the UUC for categories for which the CHF has allocated quota but has run out of credit mid-stream

do-not-suppress

specifies to include the UUC for quota types initially allocated by the CHF

inherit

specifies that the rating group inherits the value from the CHF profile level; applicable only under the rating-group context

Default: inherit

6.2.1.10.12 uuc-options-with-quota

Syntax

```
uuc-options-with-quota [none] [total] [uplink] [downlink] [time] [inherit]
no uuc-options-with-quota
```


Context

```
config>mobile>profile>charging>chf-profile
```

```
config>mobile>profile>charging>chf-profile>rating-group
```

Description

This command configures the UUC options for categories for which the CHF allocates quota. The superset that is applied includes the locally configured options and everything that is received from the CHF. For example, if the uplink volume and downlink volume are locally configured, and the CHF sends only the total volume for the granted units, the UUC equals the sum of the uplink volume, the downlink volume, and the total volume.

The no form of this command reverts to the default (the quota sent by the CHF only).

Default

no uuc-options-with-quota

Parameters

none

specifies to include the default behavior of returning what the CHF requests

total

specifies to include the total volume

uplink

specifies to include the uplink volume

downlink

specifies to include the downlink volume

time

specifies to include the time

inherit

specifies that the rating group inherits the values from the CHF profile level; applicable only under the rating-group context

Default: `inherit`

6.2.1.10.13 reauth-blocked-rg

Syntax

```
[no] reauth-blocked-rg
```

Context

```
config>mobile>profile>charging>chf-profile
```

Description

This command sets a blocked rating-group as active when a Nchf Charging Notify Request (Re-authorization) message is received from the CHF. When enabled, the service delay timer is stopped and the blocked rating group is included in the Charging Data Request (Update) that follows the Re-authorization message.

The no form of this command reverts to the default behavior (disabled); the rating group is excluded from the Charging Data Request message when blocked.

6.2.1.10.14 request-quota-before-traffic

Syntax

```
[no] request-quota-before-traffic
```

Context

```
config>mobile>profile>charging>chf-profile
```

Description

This command enables the SMF to request quota for an RG upon activation of an SDF (policy rule activation) belonging to an RG for which no quota exists. When this command is disabled, the SMF requests quota for an RG when it detects the first packet for an SDF that belongs to an RG for which no quota exists.

The no form of this command reverts to the default (disabled).

Default

```
no request-quota-before-traffic
```

6.2.1.10.15 requested-unit-for-all-triggers

Syntax

```
[no] requested-unit-for-all-triggers
```

Context

```
config>mobile>profile>charging>chf-profile
```

Description

This command specifies when the Requested-Unit IE is present in Charging Data Request.

When enabled, the RequestedUnit IE is present in rating groups that require quota management for all NCH triggers.

When disabled, the RequestedUnit IE is present in rating groups that require quota management for all NCH triggers except NCH time limit and NCH volume limit.

The no form of this command reverts to the default (disabled).



NOTE:

When the IE is included, empty "RequestedUnit" {} is sent.

Default

no requested-unit-for-all-trigger

6.2.1.10.16 requested-unit-total-volume

Syntax

`requested-unit-total-volume requested-unit-total-volume`

`no requested-unit-total-volume`

Context

`config>mobile>profile>charging>chf-profile`

`config>mobile>profile>charging>chf-profile>rating-group`

`config>mobile>profile>policy-options>policy-rule-base`

Description

This command specifies whether to request a specific amount of volume quota when sending RequestedUnits to the CHF.

When disabled, CMG sends an empty RequestedUnits attribute to the CHF when the CHF requests quota.

When you configure this command at the policy rule base level using the `config>mobile>profile>policy-options>policy-rule-base` context, it applies to all the rating groups configured in the session.

When you configure this command at the rating group level using the `config>mobile>profile>charging>chf-profile>rating-group` context, it applies to the specified rating group.

When you configure this command at the CHF profile level, it applies to those rating groups of the CHF profile which do not have this command configured at rating-group level.

The value configured at the policy rule base takes precedence over the other configurations.

If you configure more than one policy rule base for a session, the smallest value configured across all installed rule bases will be used.

**NOTE:**

You can configure up to 200 rating groups.

The no form of this command reverts to the default (disabled).

Default

no requested-unit-total-volume

Parameters***requested-unit-total-volume***

specifies the volume in Kbytes when requesting quota from the CHF

Values: 1 to 104857600

6.2.1.10.17 sbi-srvc-operation

Syntax

sbi-srvc-operation

Context

config>mobile>profile>charging>chf-profile

Description

This command enables the context to configure the SBI service operation parameters that detect failures at the Nchf interface.

6.2.1.10.18 retry-count

Syntax

retry-count *value*

no retry-count

Context

config>mobile>profile>charging>chf-profile>sbi-srv-operation

Description

This command configures the retry count used to detect connection failures to the configured CHF. If a message is sent to CHF with no reply after the time specified with the `transaction-timer` command, the message is sent again. After the message has been resent without success for the configured retry count, the connection to the CHF is considered down and the failure handling is triggered.

Parameters

value

specifies the number of retry attempts

Values: 0 to 10

Default: 0

6.2.1.10.19 transaction-timer

Syntax

transaction-timer *value*

no transaction-timer

Context

config>mobile>profile>charging>chf-profile>sbi-srv-operation

Description

This command configures the transaction time used to detect connection failures to the configured CHF. If there is no reply when a message is sent to the CHF, the message is sent again after the time configured for this command. This is repeated for the number of times configured for the `retry-count` command, after which the connection to the CHF is considered down and failure handling is triggered.

Parameters

value

specifies the transaction timer value, in milliseconds

Values: 1000 to 180000

Default: 5000

6.2.1.10.20 smf-attributes-profile

Syntax

```
smf-attributes-profile profile-id  
no smf-attributes-profile
```

Context

```
config>mobile>profile>charging>chf-profile
```

Description

This command references an `smf-attributes-profile` configured in the `config>mobile>profile>charging` context to specify which attributes are included or excluded in the Nchf messages sent by the SMF.

The no form of this command removes the association with the SMF attribute profile.

Parameters

profile-id

specifies the profile identifier

Values: 1 to 255

6.2.1.10.21 smf-generate-charging-data-ref

Syntax

```
[no] smf-generate-charging-data-ref
```

Context

```
config>mobile>profile>charging>chf-profile
```

Description

This command enables the generation of charging data reference by SMF. It is used in Nchf Buffer and Playback functionality.

The no form of this command reverts to the default.

6.2.1.10.22 supported-features

Syntax

supported-features

Context

config>mobile>profile>charging>chf-profile

Description

This command enables the context to configure the supported features for the CHF profile.

6.2.1.10.23 support-2g3g

Syntax

[no] support-2g3g

Context

config>mobile>profile>charging>chf-profile>supported-features

Description

This command informs the CHF about the support of GERAN and UTRAN access over N40 interface by including TEI17_NIESGU (9) flag in the Supported-Features attribute.

The no form of this command reverts to the default.

Default

no support-2g3g

6.2.1.10.24 supported-vendor

Syntax

supported-vendor *custom-3*

no supported-vendor

Context

```
config>mobile>profile>charging>chf-profile
```

Description

This command configures vendor-specific customizations to be enabled for the Nchf interface. Each option represents a specific customer.

The no form of this command reverts to the default.

Default

no supported-vendor

Parameters

custom-3

specifies a customer number

6.2.1.10.25 unit-count-inactivity-timer

Syntax

```
unit-count-inactivity-timer timer-value
```

```
no unit-count-inactivity-timer
```

Context

```
config>mobile>profile>charging>chf-profile
```

Description

This command specifies the unit count inactivity timer.

Parameters

timer-value

specifies the inactivity timer value, in seconds

Values: 1 to 3600

Default: 0

6.2.1.10.26 uuc-reporting-enh

Syntax

[no] uuc-reporting-enh

Context

config>mobile>profile>charging>chf-profile

Description

This command enables the UUC reporting enhancement.

6.2.1.11 Charging profile credit pool command descriptions

6.2.1.11.1 credit-pool-id

Syntax

[no] credit-pool-id *pool-id*

Context

config>mobile>profile>charging

Description

This command configures a credit pool ID. Each pool ID includes the volume quota, time quota, and the maximum number of times the specified interim quota can be allocated. A configured pool ID is associated with a rating group within the DCCA profile for Assume Positive behavior.

The no form of this command removes the specified pool ID configuration.

Parameters

pool-id

specifies a credit pool identifier

Values: 0 to 16

6.2.1.11.2 max-interim-quotas

Syntax

```
max-interim-quotas number  
no max-interim-quotas
```

Context

```
config>mobile>profile>charging>credit-pool-id
```

Description

This command specifies the maximum number of times an interim quota is allocated while in Assume Positive mode.

The no form of this command returns the setting to disabled (0).

Parameters

number

specifies the number of times an interim quota is allocated

Values: 0 to 65535

Default: 0

6.2.1.11.3 validity-time

Syntax

```
validity-time seconds  
no validity-time
```

Context

```
config>mobile>profile>charging>credit-pool-id
```

Description

This command configures the time quota for a credit pool used for AP behavior.

The no form of this command returns the setting to disabled (0).

Parameters

seconds

specifies the time quota in seconds

Values: 0 to 21600

Default: 0

6.2.1.11.4 volume-quota

Syntax

volume-quota *kilobytes*

no volume-quota

Context

config>mobile>profile>charging>credit-pool-id

Description

This command specifies the volume quota for a credit pool used for Assume Positive mode.

The no form of this command returns the setting to disabled (0).

Parameters

kilobytes

specifies the volume quota in kilobytes

Values: 0 to 10000000

Default: 0

6.2.1.12 Charging profile event triggers FBC command descriptions

6.2.1.12.1 event-triggers-fbc

Syntax

[no] event-triggers-fbc *profile-id*

Context

config>mobile>profile>charging

Description

This command configures an event trigger profile for flow-based charging.

The no form of this command removes the specified profile from the configuration.

Parameters

profile-id

specifies a charging profile identifier

Values: 0 to 255

6.2.1.12.2 3gpp-ps-data-off-status-change

Syntax

```
3gpp-ps-data-off-status-change [category {immediate | deferred}]
```

```
no 3gpp-ps-data-off-status-change
```

Context

```
config>mobile>profile>charging>event-triggers-fbc
```

Description

This command enables the 3GPP-PS-Data-Off trigger.

The no form of this command disables the 3GPP-PS-Data-Off trigger.

Parameters

immediate

specifies an immediate CDreq is sent to CHF

deferred

specifies the UUC is stored and sent later

Default: deferred

6.2.1.12.3 addition-of-upf

Syntax

```
addition-of-upf [category {immediate | deferred}]
```

```
no addition-of-upf
```

Context

config>mobile>profile>charging>event-triggers-fbc

Description

This command enables the addition of UPF trigger.

The no form of this command disables the addition of UPF trigger.

Parameters

immediate

specifies an immediate CDreq is sent to CHF

deferred

specifies the UUC is stored and sent later

Default: immediate

6.2.1.12.4 default-qht

Syntax

default-qht *default-qht*

no default-qht

Context

config>mobile>profile>charging>event-triggers-fbc

Description

This command configures the default QHT.

The no form of this command disables the trigger.

Parameters

default-qht

specifies the QHT, in minutes

Values: 0 to 30

Default: 0

6.2.1.12.5 handover-cancel

Syntax

```
handover-cancel [category {immediate | deferred}]  
no handover-cancel
```

Context

```
config>mobile>profile>charging>event-triggers-fbc
```

Description

This command enables the handover-cancel event trigger and optionally defines the trigger category. If not enabled, the CHF overwrites the local configuration. If the category is not configured, the 3GPP-defined category for the trigger is applied. By default, this parameter is not enabled (unless CHF enables it).

The no form of this command disables the trigger.

Parameters

immediate

sets the category for the event trigger as immediate

deferred

sets the category for the event trigger as deferred

Default: immediate

6.2.1.12.6 handover-complete

Syntax

```
handover-complete [category {immediate | deferred}]  
no handover-complete
```

Context

```
config>mobile>profile>charging>event-triggers-fbc
```

Description

This command enables the handover-complete event trigger and optionally defines the trigger category. If not enabled, the CHF overwrites the local configuration. If the category is not configured, the 3GPP-defined category for the trigger is applied. By default, this parameter is not enabled (unless CHF enables it).

The no form of this command disables the trigger.

Parameters

immediate

sets the category for the event trigger as immediate

deferred

sets the category for the event trigger as deferred

Default: immediate

6.2.1.12.7 handover-start

Syntax

```
handover-start [category {immediate | deferred}]
```

```
no handover-start
```

Context

```
config>mobile>profile>charging>event-triggers-fbc
```

Description

This command enables the handover-start event trigger and optionally defines the trigger category. If not enabled, the CHF overwrites the local configuration. If the category is not configured, the 3GPP-defined category for the trigger is applied. By default, this parameter is not enabled (unless CHF enables it).

The no form of this command disables the handover start event trigger.

Parameters

immediate

sets the category for the event trigger as immediate

deferred

sets the category for the event trigger as deferred

Default: immediate

6.2.1.12.8 insertion-of-ismf

Syntax

```
insertion-of-ismf [category {immediate | deferred}]
```

```
no insertion-of-ismf
```

Context

```
config>mobile>profile>charging>event-triggers-fbc
```

Description

This command configures the insertion of ISMF trigger.

The no form of this command disables the trigger.

Parameters

immediate

specifies an immediate CDreq is sent to CHF

deferred

specifies the UUC is stored and sent later

Default: deferred

6.2.1.12.9 max-change-condition

Syntax

```
max-change-condition max-change-condition
```

```
no max-change-condition
```

Context

```
config>mobile>profile>charging>event-triggers-fbc
```

Description

This command configures the maximum number of change conditions, for example, a tariff change, location change, or QoS change, that can occur before a CDreq is created.

The no form of this command disables the trigger.

Parameters

max-change-condition

specifies the maximum number of change conditions that can occur

Values: 0 to 32

Default: 0

6.2.1.12.10 plmn-change

Syntax

```
plmn-change [category {immediate | deferred}]  
no plmn-change
```

Context

```
config>mobile>profile>charging>event-triggers-fbc
```

Description

This command enables the PLMN change trigger.

The no form of this command disables the PLMN change partial record trigger.

Default

```
no plmn-change
```

Parameters

immediate

specifies an immediate CDreq is sent to CHF

deferred

specifies the UUC is stored and sent later

Default: immediate

6.2.1.12.11 qos-change

Syntax

```
qos-change [category {immediate | deferred}]  
no qos-change
```

Context

```
config>mobile>profile>charging>event-triggers-fbc
```

Description

This command specifies whether the QoS change trigger is enabled.

The no form of this command disables the QoS change trigger.

Parameters

immediate

specifies an immediate CDreq is sent to CHF

deferred

specifies the UUC is stored and sent later

Default: deferred

6.2.1.12.12 rat-type-change

Syntax

```
rat-type-change [category {immediate | deferred}]
```

```
no rat-type-change
```

Context

```
config>mobile>profile>charging>event-triggers-fbc
```

Description

This command enables the RAT type change trigger.

The no form of this command disables the RAT type change trigger.

Parameters

immediate

specifies an immediate CDreq is sent to CHF

deferred

specifies the UUC is stored and sent later

Default: immediate

6.2.1.12.13 reallocation-of-ismf

Syntax

```
reallocation-of-ismf [category {immediate | deferred}]
```

```
no reallocation-of-ismf
```

Context

config>mobile>profile>charging>event-triggers-fbc

Description

This command configures the reallocation of ISMF trigger.

The no form of this command disables the trigger.

Parameters

immediate

specifies an immediate CDreq is sent to CHF

deferred

specifies the UUC is stored and sent later

Default: deferred

6.2.1.12.14 removal-of-ismf

Syntax

removal-of-ismf [category {immediate | deferred}]

no removal-of-ismf

Context

config>mobile>profile>charging>event-triggers-fbc

Description

This command configures the category as immediate or deferred for the removal of ISMF trigger.

The no form of this command disables the trigger.

Parameters

immediate

specifies an immediate CDreq is sent to CHF

deferred

specifies the UUC is stored and sent later

Default: deferred

6.2.1.12.15 removal-of-upf

Syntax

```
removal-of-upf [category {immediate | deferred}]  
no removal-of-upf
```

Context

```
config>mobile>profile>charging>event-triggers-fbc
```

Description

This command enables the removal of UPF trigger.

The no form of this command disables the removal of UPF trigger.

Parameters

immediate

specifies an immediate CDreq is sent to CHF

deferred

specifies the UUC is stored and sent later

Default: immediate

6.2.1.12.16 serving-node-change

Syntax

```
serving-node-change [category {immediate | deferred}]  
no serving-node-change
```

Context

```
config>mobile>profile>charging>event-triggers-fbc
```

Description

This command enables the serving node change trigger.

The no form of this command disables the serving node change trigger.

Parameters

immediate

specifies an immediate CDreq is sent to CHF

deferred

specifies the UUC is stored and sent later

Default: deferred

6.2.1.12.17 **serving-node-change-limit**

Syntax

serving-node-change-limit *serving-node-change-limit*

no serving-node-change-limit

Context

config>mobile>profile>charging>event-triggers-fbc

Description

This command specifies the maximum limit for the change in serving node. A charging data request is generated when the specified number of serving node changes is reached.



NOTE:

If you set the parameter to 0, a maximum of 16 ServingNodeAddress/ServingNodeType IEs are included in the Nchf message. The last address is always the most recent one. If two subsequent addresses/types have the same value, only one entry is included in the list.

Parameters

serving-node-change-limit

specifies the change limit for the partial record trigger

Values: 0 to 15

Default: 15

6.2.1.12.18 serving-node-type

Syntax

```
serving-node-type {r15 | r16}  
no serving-node-type
```

Context

```
config>mobile>profile>charging>event-triggers-fbc
```

Description

This command configures the serving node type that is sent to the CHF in the case of EPC Interworking. The serving node type options are always AMF (r15), or AMF or SGW (r16). SGW is the serving node of the SMF for EPC Interworking. However, 3GPP Release 15 defines only AMF for the serving node, and 3GPP Release 16 also defines SGW as an option for the serving node.

The no form of this command reverts to the default (r15).



NOTE:

The Serving Node Type is encoded in the field NodeFunctionality of the ServingNetworkFunctionID attribute.

Parameters

r15

specifies that the SMF sends AMF as the value of NodeFunctionality

r16

specifies that the SMF sends AMF or SGW as the value of NodeFunctionality

Default: r15

6.2.1.12.19 session-ambr-change

Syntax

```
session-ambr-change [category {immediate | deferred}]  
no session-ambr-change
```

Context

```
config>mobile>profile>charging>event-triggers-fbc
```

Description

This command enables the session AMBR change trigger.

The no form of this command disables the session AMBR change trigger.

Parameters

immediate

specifies an immediate CDreq is sent to CHF

deferred

specifies the UUC is stored and sent later

Default: immediate

6.2.1.12.20 tariff-change-list

Syntax

```
tariff-change-list list-name
```

```
no tariff-change-list
```

Context

```
config>mobile>profile>charging>event-triggers-fbc
```

Description

This command configures a tariff change list profile to be applied for the tariff change charging info trigger. Multiple tariff change times can be specified within the profile to facilitate differentiated charging at different times of day. Each tariff change entry must have a unique time of day. The (UE) time of day is used to generate a tariff change event for charging records. Tariff change is a deferred trigger whose category cannot be changed. The UUC containers are stored at the specified time and are included in the next CDreq that is sent.

The no form of this command removes the specified tariff-change list profile association for this context.

Parameters

list-name

specifies the name of the list, up to 32 characters

6.2.1.12.21 time-limit

Syntax

```
time-limit time-limit  
no time-limit
```

Context

```
config>mobile>profile>charging>event-triggers-fbc
```

Description

This command configures a time limit that applies per session or bearer depending on the accounting level selected, for example PDN-level accounting or QCI and bearer-level accounting. A partial record is created when the specified time limit is satisfied. The time-limit trigger applies to all RAT types for which no explicit time limit has been configured. If the `time-limit-nr` command or the `time-limit-eutran` command is also configured, the `time-limit` command applies to all other RAT types.

The no form of this command reverts to the default.

Default

```
no time-limit
```

Parameters

time-limit

specifies a time limit, in minutes, applicable per session or bearer

Values: 5 to 1440

Default: 0

6.2.1.12.22 time-limit-eutran

Syntax

```
time-limit-eutran time-limit-eutran  
no time-limit-eutran
```

Context

```
config>mobile>profile>charging>event-triggers-fbc
```


Description

This command configures a time limit that applies for the E-UTRAN RAT type. A partial record is created when the specified time limit is reached for this RAT. When configured, the `time-limit-eutran` command applies to the E-UTRAN RAT type and the `time-limit` command applies to all other RAT types for which no specific time limit is configured. If the `time-limit-eutran` command is not enabled, the value configured for the `time-limit` command applies to the E-UTRAN RAT type as well.

The no form of this command reverts to the default.

Parameters

time-limit-eutran

specifies a time limit, in minutes, applicable per session or bearer

Values: 0 to 1440

Default: 0

6.2.1.12.23 time-limit-nr

Syntax

```
time-limit-nr time-limit-nr  
no time-limit-nr
```

Context

```
config>mobile>profile>charging>event-triggers-fbc
```

Description

This command configures a time limit for the NR RAT type. A partial record is created when the specified time limit is reached for this RAT. When configured, the `time-limit-nr` command applies to the NR RAT type and the `time-limit` command applies to all other RAT types for which no specific time limit is configured. When the `time-limit-nr` command is not configured, the `time-limit` command applies to the NR RAT type as well.

The no form of this command disables the trigger.

Parameters

time-limit-nr

specifies the NR time limit, in minutes

Values: 0 to 1440

Default: 0

6.2.1.12.24 time-limit-per-rating-grp

Syntax

```
time-limit-per-rating-grp time-limit-per-rating-grp [category {immediate | deferred}]  
no time-limit-per-rating-grp
```

Context

```
config>mobile>profile>charging>event-triggers-fbc
```

Description

This command configures the time limit (in seconds) that applies per rating group or bearer depending on the accounting level selected, such as PDN-level accounting or QCI/bearer-level accounting. A partial record is created when the specified time limit is satisfied.

The no form of this command disables the trigger.

Parameters

time-limit-per-rating-grp

specifies the time limit (in minutes) applicable per rating group or bearer

Values: 5 to 1440

Default: 0

immediate

specifies an immediate CDreq is sent to CHF

deferred

specifies the UUC is stored and sent later

Default: deferred

6.2.1.12.25 trigger-override-option

Syntax

```
trigger-override-option {local | chf | local-and-chf}  
no trigger-override-option
```

Context

```
config>mobile>profile>charging>event-triggers-fbc
```

Description

This command configures the trigger override behavior at the SMF, for flow-based charging. Optionally, the CHF overrides the locally configured triggers, the SMF ignores the triggers the CHF sends, or the CHF-subscribed triggers are armed in addition to the locally configured triggers.

The no form of this command disables the trigger.

Parameters

local

specifies that only the locally enabled triggers are enforced at the SMF and CHF-subscribed triggers are ignored

chf

specifies that all locally configured triggers at the SMF are disabled and only the CHF subscribed triggers are enforced

local-and-chf

specifies that locally configured triggers are enabled and CHF-subscribed triggers are also enforced, if applicable

Default: chf

6.2.1.12.26 ue-in-pra-change

Syntax

```
ue-in-pra-change [category {immediate | deferred}]
```

```
no ue-in-pra-change
```

Context

```
config>mobile>profile>charging>event-triggers-fbc
```

Description

This command enables the session UE in PRA change trigger.

The no form of this command disables the UE in PRA change trigger.

Default

```
no ue-in-pra-change
```

Parameters

immediate

specifies an immediate CDreq is sent to CHF

deferred

specifies the UUC is stored and sent later

Default: deferred

6.2.1.12.27 ue-time-zone-change

Syntax

```
ue-time-zone-change [category {immediate | deferred}]
```

```
no ue-time-zone-change
```

Context

```
config>mobile>profile>charging>event-triggers-fbc
```

Description

This command enables the UE time zone change trigger.

The no form of this command disables the UE time zone change trigger.

Parameters**immediate**

specifies an immediate CDreq is sent to CHF

deferred

specifies the UUC is stored and sent later

Default: immediate

6.2.1.12.28 user-loc-change

Syntax

```
user-loc-change [category {immediate | deferred}]
```

```
no user-loc-change
```

Context

```
config>mobile>profile>charging>event-triggers-fbc
```

Description

This command enables or disables the user location change event trigger.

The no form of this command disables the user location change event trigger.

Parameters

immediate

specifies an immediate CDreq is sent to CHF

deferred

specifies the UUC is stored and sent later

Default: deferred

6.2.1.12.29 volume-limit

Syntax

`volume-limit volume-limit`

`no volume-limit`

Context

`config>mobile>profile>charging>event-triggers-fbc`

Description

This command configures a data volume limit that applies per session or bearer depending on the accounting level selected like PDN-level accounting or QCI/bearer-level accounting. A partial record is created when the specified volume limit is satisfied. The volume-limit trigger applies to all RAT types for which no explicit volume limit has been configured. If the `volume-limit-nr` command or the `volume-limit-eutran` command is also configured, the `volume-limit` command applies to all other RAT types.

The no form of this command reverts to the default.

Parameters

volume-limit

specifies a data volume limit, in kilobytes

Values: 0 to 104857600

Default: 0

6.2.1.12.30 volume-limit-eutran

Syntax

```
volume-limit-eutran volume-limit-eutran  
no volume-limit-eutran
```

Context

```
config>mobile>profile>charging>event-triggers-fbc
```

Description

This command configures a data volume limit that applies for the E-UTRAN RAT type. A partial record is created when the specified volume limit is reached. When configured, the `volume-limit-eutran` command applies to E-UTRAN RAT type and the `volume-limit` command applies to all other RAT types for which no specific volume limit is configured. If the `volume-limit-eutran` command is not enabled, the value configured for the `volume-limit` command applies to the E-UTRAN RAT type as well.

The no form of this command reverts to the default.

Parameters

volume-limit-eutran

specifies a data volume limit

Values: 0 to 104857600

Default: 0 (use the generic volume limit when specified)

6.2.1.12.31 volume-limit-nr

Syntax

```
volume-limit-nr volume-limit-nr  
no volume-limit-nr
```

Context

```
config>mobile>profile>charging>event-triggers-fbc
```

Description

This command configures a data volume limit that applies for the NR RAT type. A partial record is created when the specified volume limit is reached. When configured, the `volume-limit-nr` command applies to NR and the `volume-limit` command applies to all other RAT types for which no specific time limit is

configured. When the `volume-limit-nr` command is not configured, the `volume-limit` command applies to the NR RAT type as well.

The no form of this command reverts to the default.

Parameters

volume-limit-nr

specifies a data volume limit, in Kbytes

Values: 0 to 104857600

Default: 0 (use the generic volume limit when specified)

6.2.1.12.32 volume-limit-per-rating-grp

Syntax

```
volume-limit-per-rating-grp volume-limit-per-rating-grp [category {immediate |  
deferred}]
```

```
no volume-limit-per-rating-grp
```

Context

```
config>mobile>profile>charging>event-triggers-fbc
```

Description

This command configures the volume limit (in kbytes) that applies per rating group or bearer depending on the accounting level selected, such as PDN-level accounting or QCI/bearer-level accounting. A partial record is created when the specified volume limit is satisfied.

The no form of this command disables the trigger.

Parameters

vol-limit-per-rating-grp

specifies the volume limit (in kbytes) applicable per rating group or bearer

Values: 0 to 104857600

Default: 0

immediate

specifies an immediate CDreq is sent to CHF

deferred

specifies the UUC is stored and sent later

Default: deferred

6.2.1.13 Charging profile event triggers QBC command descriptions

6.2.1.13.1 event-triggers-qbc

Syntax

```
[no] event-triggers-qbc profile-id
```

Context

```
config>mobile>profile>charging
```

Description

This command enables the context to configure a QBC event trigger profile. The specified profile is assigned to a session by referencing the profile ID from the smf-charging profile.

The no form of this command removes the specified profile from the configuration.

Parameters

profile-id

specifies a charging profile identifier

Values: 0 to 255

6.2.1.13.2 addition-of-upf

Syntax

```
addition-of-upf [category {immediate | deferred}]
```

```
no addition-of-upf
```

Context

```
config>mobile>profile>charging>event-triggers-qbc
```

Description

This command enables the addition of a UPF QBC trigger.

The no form of this command disables the trigger.

Parameters

immediate

specifies that an immediate CDreq is sent to CHF

deferred

specifies that the UUC is stored and sent later

Default: immediate

6.2.1.13.3 handover-cancel

Syntax

```
handover -cancel [category {immediate | deferred}]
```

```
no handover -cancel
```

Context

```
config>mobile>profile>charging>event-triggers-qbc
```

Description

This command enables the handover cancel QBC trigger.

The no form of this command disables the trigger.

Parameters

immediate

specifies that an immediate CDreq is sent to CHF

deferred

specifies that the UUC is stored and sent later

Default: immediate

6.2.1.13.4 handover-complete

Syntax

```
handover -complete [category {immediate | deferred}]
```

```
no handover -complete
```

Context

config>mobile>profile>charging>event-triggers-qbc

Description

This command enables the handover complete QBC trigger.

The no form of this command disables the trigger.

Parameters

immediate

specifies that an immediate CDreq is sent to CHF

deferred

specifies that the UUC is stored and sent later

Default: immediate

6.2.1.13.5 handover-start

Syntax

handover-start [category {immediate | deferred}]

no handover-start

Context

config>mobile>profile>charging>event-triggers-qbc

Description

This command enables the handover start QBC trigger.

The no form of this command disables the trigger.

Parameters

immediate

specifies that an immediate CDreq is sent to CHF

deferred

specifies that the UUC is stored and sent later

Default: immediate

6.2.1.13.6 max-change-condition

Syntax

```
max-change-condition max-change-condition  
no max-change-condition
```

Context

```
config>mobile>profile>charging>event-triggers-qbc
```

Description

This command configures the maximum number of change conditions, for example, a tariff change, location change, or QoS change that can occur before a CDreq is created.

The no form of this command disables the trigger.

Parameters

max-change-condition

specifies the maximum number of change conditions that can occur

Values: 0 to 32

Default: 0

6.2.1.13.7 plmn-change

Syntax

```
plmn-change [category {immediate | deferred}]  
no plmn-change
```

Context

```
config>mobile>profile>charging>event-triggers-qbc
```

Description

This command enables the PLMN change trigger.

The no form of this command disables the PLMN change partial record trigger.

Default

```
no plmn-change
```

Parameters

immediate

specifies that an immediate CDreq is sent to CHF

deferred

specifies that the UUC is stored and sent later

Default: immediate

6.2.1.13.8 qos-change

Syntax

```
qos-change [category {immediate | deferred}]
```

```
no qos-change
```

Context

```
config>mobile>profile>charging>event-triggers-qbc
```

Description

This command enables the QoS change trigger.

The no form of this command disables the QoS change trigger.

Parameters

immediate

specifies that an immediate CDreq is sent to CHF

deferred

specifies that the UUC is stored and sent later

Default: deferred

6.2.1.13.9 removal-of-upf

Syntax

```
removal-of-upf [category {immediate | deferred}]
```

```
no removal-of-upf
```

Context

config>mobile>profile>charging>event-triggers-qbc

Description

This command enables the removal of UPF QBC trigger.

The no form of this command disables the trigger.

Parameters

immediate

specifies that an immediate CDreq is sent to CHF

deferred

specifies that the UUC is stored and sent later

Default: immediate

6.2.1.13.10 serving-node-change

Syntax

serving-node-change [category {immediate | deferred}]

no serving-node-change

Context

config>mobile>profile>charging>event-triggers-qbc

Description

This command enables the serving node change trigger.

The no form of this command disables the serving node change trigger.

Parameters

immediate

specifies that an immediate CDreq is sent to CHF

deferred

specifies that the UUC is stored and sent later

Default: deferred

6.2.1.13.11 *serving-node-change-limit*

Syntax

```
serving-node-change-limit serving-node-change-limit  
no serving-node-change-limit
```

Context

```
config>mobile>profile>charging>event-triggers-qbc
```

Description

This command specifies a maximum limit for the change in the serving node. A charging data request is generated when the specified number of the serving node changes is reached.

**NOTE:**

If you set the parameter to 0, a maximum of 16 ServingNodeAddress/ServingNodeType IEs are included in the Nchf message. The last address is always the most recent one. If two subsequent addresses/types have the same value, only one entry is included in the list.

Parameters

serving-node-change-limit

specifies the change limit for the serving node

Values: 0 to 15

Default: 15

6.2.1.13.12 *session-ambr-change*

Syntax

```
session-ambr-change [category {immediate | deferred}]  
no session-ambr-change
```

Context

```
config>mobile>profile>charging>event-triggers-qbc
```

Description

This command enables the session AMBR change trigger.

The no form of this command disables the session AMBR change trigger.

Parameters

immediate

specifies that an immediate CDreq is sent to CHF

deferred

specifies that the UUC is stored and sent later

Default: immediate

6.2.1.13.13 tariff-change-list

Syntax

tariff-change-list *list-name*

no tariff-change-list

Context

config>mobile>profile>charging>event-triggers-qbc

Description

This command configures a tariff change list profile to apply for the tariff change charging info trigger. Multiple tariff change times can be specified within the profile to facilitate differentiated charging at different times of day. Each tariff change must have a unique time of day. The (UE) time of day is used to generate a tariff change event for charging records. Tariff change is a deferred trigger whose category cannot be changed. The UUC containers are stored at the specified time and are included in the next CDreq that is sent.

The no form of this command removes the specified tariff-change list profile association for this context.

Parameters

list-name

specifies the list name, up to 32 characters

6.2.1.13.14 time-limit

Syntax

time-limit *time-limit* [category {immediate | deferred}]

no time-limit

Context

config>mobile>profile>charging>event-triggers-qbc

Description

This command configures a time limit that applies to the PDU session. The time limit trigger applies to all RAT types for which no explicit time limit has been configured. By default, the time limit is not configured.

The no form of this command reverts to the default.

Default

no time-limit

Parameters

time-limit

specifies a time limit, in minutes, applicable to the session

Values: 5 to 1440

Default: 0

immediate

specifies that an immediate CDreq is sent to CHF

deferred

specifies that the UUC is stored and sent later

Default: immediate

6.2.1.13.15 time-limit-per-qos-flow

Syntax

time-limit-per-qos-flow *time-limit-per-qos-flow* [category {immediate | deferred}]

no time-limit-per-qos-flow

Context

config>mobile>profile>charging>event-triggers-qbc

Description

This command configures the time limit that applies per QoS flow.

The no form of this command disables the trigger.

Parameters

time-limit-per-qos-flow

specifies the time limit, in minutes, applicable per QoS flow

Values: 5 to 1440

Default: 0

immediate

specifies that an immediate CDreq is sent to CHF

deferred

specifies that the UUC is stored and sent later

Default: deferred

6.2.1.13.16 trigger-override-option

Syntax

`trigger-override-option {local | chf | local-and-chf}`

`no trigger-override-option`

Context

`config>mobile>profile>charging>event-triggers-qbc`

Description

This command configures the trigger override behavior at the SMF, for QBC-based charging. Optionally, the CHF can override the locally configured triggers, the SMF can ignore the triggers the CHF sends, or the CHF-subscribed triggers can be configured in addition to the locally configured triggers.

The no form of this command disables the trigger.

Parameters

local

specifies that only the locally enabled triggers are applied at the SMF and the CHF-subscribed triggers are ignored

chf

specifies that all locally configured triggers at the SMF are disabled and only the CHF subscribed triggers are applied

local-and-chf

specifies that locally configured triggers are enabled and the CHF-subscribed triggers are also applied, if applicable

Default: chf

6.2.1.13.17 ue-time-zone-change

Syntax

```
ue-time-zone-change [category {immediate | deferred}]  
no ue-time-zone-change
```

Context

```
config>mobile>profile>charging>event-triggers-qbc
```

Description

This command enables the UE time zone change trigger.

The no form of this command disables the UE time zone change trigger.

Parameters

immediate

specifies that an immediate CDreq is sent to CHF

deferred

specifies that the UUC is stored and sent later

Default: immediate

6.2.1.13.18 user-loc-change

Syntax

```
user-loc-change [category {immediate | deferred}]  
no user-loc-change
```

Context

```
config>mobile>profile>charging>event-triggers-qbc
```

Description

This command enables the user location change event trigger.

The no form of this command disables the user location change event trigger.

Parameters

immediate

specifies that an immediate CDreq is sent to CHF

deferred

specifies that the UUC is stored and sent later

Default: deferred

6.2.1.13.19 volume-limit

Syntax

```
volume-limit volume-limit [category {immediate | deferred}]
```

```
no volume-limit
```

Context

```
config>mobile>profile>charging>event-triggers-qbc
```

Description

This command configures a data volume limit that applies to the PDU session. By default, the volume limit is not configured. The volume limit trigger applies to all RAT types for which no explicit volume limit has been configured.

The no form of this command reverts to the default.

Parameters

volume-limit

specifies a data volume limit, in kilobytes, applicable to the session

Values: 0 to 104857600

Default: 0

immediate

specifies that an immediate CDreq is sent to CHF

deferred

specifies that the UUC is stored and sent later

Default: immediate

6.2.1.13.20 volume-limit-per-qos-flow

Syntax

```
volume-limit-per-qos-flow volume-limit-per-qos-flow [category {immediate | deferred}]  
no volume-limit-per-qos-flow
```

Context

```
config>mobile>profile>charging>event-triggers-qbc
```

Description

This command configures a data volume limit that applies per QoS flow.

The no form of this command reverts to the disables the trigger.

Parameters

volume-limit-per-qos-flow

specifies a data volume limit, in kilobytes, applicable per QoS flow

Values: 0 to 104857600

Default: 0

immediate

specifies that an immediate CDreq is sent to CHF

deferred

specifies that the UUC is stored and sent later

Default: deferred

6.2.1.14 Charging profile Gy AVP options command descriptions

6.2.1.14.1 gy-avp-options

Syntax

```
[no] gy-avp-options gy-avp-options-profile-id
```

Context

```
config>mobile>profile>charging
```

Description

This command enables the context to configure what AVP IEs are included or excluded in the Gy messages sent by the PGW/GGSN.

Parameters

gy-avp-options-profile-id

specifies a profile charging identifier

Values: 0 to 255

6.2.1.14.2 exclude-3gpp-charging-characteristics

Syntax

```
[no] exclude-3gpp-charging-characteristics
```

Context

```
config>mobile>profile>charging>gy-avp-options
```

Description

This command includes or excludes the 3GPP charging characteristics IE in the Gy messages sent by the PGW/GGSN.

Default

```
no exclude-3gpp-charging-characteristics
```

6.2.1.14.3 exclude-3gpp-charging-id

Syntax

```
[no] exclude-3gpp-charging-id
```

Context

```
config>mobile>profile>charging>gy-avp-options
```

Description

This command includes or excludes the 3GPP charging identifier IE in the Gy messages sent by the PGW/GGSN.

Default

no exclude-3gpp-charging-id

6.2.1.14.4 exclude-3gpp-ggsn-mcc-mnc**Syntax**

[no] exclude-3gpp-ggsn-mcc-mnc

Context

config>mobile>profile>charging>gy-avp-options

Description

This command includes or excludes the 3GPP GGSN MCC and MNC IE in the Gy messages sent by the PGW/GGSN.

Default

no exclude-3gpp-ggsn-mcc-mnc

6.2.1.14.5 exclude-3gpp-imsi-mcc-mnc**Syntax**

[no] exclude-3gpp-imsi-mcc-mnc

Context

config>mobile>profile>charging>gy-avp-options

Description

This command includes or excludes the 3GPP IMSI MCC and MNC IE in the Gy messages sent by the PGW/GGSN.

Default

no exclude-3gpp-imsi-mcc-mnc

6.2.14.6 exclude-3gpp-ms-timezone

Syntax

[no] exclude-3gpp-ms-timezone

Context

config>mobile>profile>charging>gy-avp-options

Description

This command includes or excludes the 3GPP MS time zone IE in the Gy messages sent by the PGW/GGSN.

Default

no exclude-3gpp-ms-timezone

6.2.14.7 exclude-3gpp-nsapi

Syntax

[no] exclude-3gpp-nsapi

Context

config>mobile>profile>charging>gy-avp-options

Description

This command includes or excludes the 3GPP NS API IE in the Gy messages sent by the PGW/GGSN.

Default

no exclude-3gpp-nsapi

6.2.14.8 exclude-3gpp-pdp-type

Syntax

[no] exclude-3gpp-pdp-type

Context

config>mobile>profile>charging>gy-avp-options

Description

This command includes or excludes the 3GPP Packet Data Protocol type IE in the Gy messages sent by the PGW/GGSN.

Default

no exclude-3gpp-pdp-type

6.2.1.14.9 exclude-3gpp-rat-type

Syntax

[no] exclude-3gpp-rat-type

Context

config>mobile>profile>charging>gy-avp-options

Description

This command includes or excludes the 3GPP radio access technology IE in the Gy messages sent by the PGW/GGSN.

Default

no exclude-3gpp-rat-type

6.2.1.14.10 exclude-3gpp-selection-mode

Syntax

[no] exclude-3gpp-selection-mode

Context

config>mobile>profile>charging>gy-avp-options

Description

This command includes or excludes the 3GPP selection mode IE in the Gy messages sent by the PGW/GGSN.

Default

no exclude-3gpp-selection-mode

6.2.1.14.11 exclude-3gpp-session-stop-indicator

Syntax

[no] exclude-3gpp-session-stop-indicator

Context

config>mobile>profile>charging>gy-avp-options

Description

This command includes or excludes the 3GPP session stop IE in the Gy messages sent by the PGW/GGSN.

Default

no exclude-3gpp-session-stop-indicator

6.2.1.14.12 exclude-3gpp-sgsn-mcc-mnc

Syntax

[no] exclude-3gpp-sgsn-mcc-mnc

Context

config>mobile>profile>charging>gy-avp-options

Description

This command includes or excludes the 3GPP SGSN MCC and MNC IE in the Gy messages sent by the PGW/GGSN.

Default

no exclude-3gpp-sgsn-mcc-mnc

6.2.1.14.13 exclude-3gpp-user-location-info

Syntax

[no] exclude-3gpp-user-location-info

Context

config>mobile>profile>charging>gy-avp-options

Description

This command includes or excludes the 3GPP user location IE in the Gy messages sent by the PGW/GGSN.

Default

no exclude-3gpp-user-location-info

6.2.1.14.14 exclude-3gpp2-bsid

Syntax

[no] exclude-3gpp-bsid

Context

config>mobile>profile>charging>gy-avp-options

Description

This command excludes the 3GPP2 BSID IE in the Gy messages sent by the PGW/GGSN.

The no form of this command reverts to the default (include).

Default

no exclude-3gpp-bsid

6.2.1.14.15 exclude-cc-correlation-id

Syntax

[no] exclude-cc-correlation-id

Context

```
config>mobile>profile>charging>gy-avp-options
```

Description

This command includes or excludes the CC correlation identifier IE in the Gy messages sent by the PGW/GGSN.

Default

no exclude-cc-correlation-id

6.2.1.14.16 exclude-charging-characteristics-selection-mode

Syntax

```
[no] exclude-charging-characteristics-selection-mode
```

Context

```
config>mobile>profile>charging>gy-avp-options
```

Description

This command includes or excludes charging characteristics selection IE in the Gy messages sent by the PGW/GGSN.

Default

no exclude-charging-characteristics-selection-mode

6.2.1.14.17 exclude-dynamic-addr-flag

Syntax

```
[no] exclude-dynamic-addr-flag
```

Context

```
config>mobile>profile>charging>gy-avp-options
```

Description

This command includes or excludes dynamic address flag IE in the Gy messages sent by the PGW/GGSN.

Default

no exclude-dynamic-addr-flag

6.2.1.14.18 exclude-dynamic-address-flag-extension**Syntax**

[no] exclude-dynamic-address-flag-extension

Context

config>mobile>profile>charging>gy-avp-options

Description

This command includes or excludes the dynamic address flag extension IE in the Gy messages sent by the PGW/GGSN.

Default

no exclude-dynamic-address-flag-extension

6.2.1.14.19 exclude-event-timestamp**Syntax**

[no] exclude-event-timestamp

Context

config>mobile>profile>charging>gy-avp-options

Description

This command includes or excludes event timestamp IE in the Gy messages sent by the PGW/GGSN.

Default

no exclude-event-timestamp

6.2.1.14.20 **exclude-ggsn-address**

Syntax

[no] exclude-ggsn-address

Context

config>mobile>profile>charging>gy-avp-options

Description

This command includes or excludes GGSN address IE in the Gy messages sent by the PGW/GGSN.

Default

no exclude-ggsn-address

6.2.1.14.21 **exclude-ims-information**

Syntax

[no] exclude-ims-information

Context

config>mobile>profile>charging>gy-avp-options

Description

This command includes or excludes IMS IE in the Gy messages sent by the PGW/GGSN. When included, only the 'node functionality' sub-AVP information under the IMS is sent.

Default

no exclude-ims-information

6.2.1.14.22 **exclude-imsi-unauthenticated-flag**

Syntax

[no] exclude-imsi-unauthenticated-flag

Context

```
config>mobile>profile>charging>gy-avp-options
```

Description

This command includes or excludes IMSI unauthenticated flag IE in the Gy messages sent by the PGW/GGSN.

Default

no exclude-imsi-unauthenticated-flag

6.2.1.14.23 exclude-node-id

Syntax

```
[no] exclude-node-id
```

Context

```
config>mobile>profile>charging>gy-avp-options
```

Description

This command includes or excludes node identifier IE in the Gy messages sent by the PGW/GGSN.

Default

no exclude-node-id

6.2.1.14.24 exclude-origin-state-id

Syntax

```
[no] exclude-origin-state-id
```

Context

```
config>mobile>profile>charging>gy-avp-options
```

Description

This command includes or excludes origin state identifier IE in the Gy messages sent by the PGW/GGSN.

Default

no exclude-origin-state-id

6.2.14.25 exclude-pdn-connection-charging-id**Syntax**

[no] exclude-pdn-connection-charging-id

Context

config>mobile>profile>charging>gy-avp-options

Description

This command includes or excludes PDN connection charging identifier IE in the Gy messages sent by the PGW/GGSN.

Default

no exclude-pdn-connection-charging-id

6.2.14.26 exclude-pdp-address**Syntax**

[no] exclude-pdp-address

Context

config>mobile>profile>charging>gy-avp-options

Description

This command includes or excludes PDF address IE in the Gy messages sent by the PGW/GGSN.

Default

no exclude-pdp-address

6.2.14.27 exclude-pdp-address-extension

Syntax

[no] exclude-pdp-address-extension

Context

config>mobile>profile>charging>gy-avp-options

Description

This command includes or excludes PDP address extension IE in the Gy messages sent by the PGW/GGSN.

Default

no exclude-pdp-address-extension

6.2.14.28 exclude-pdp-address-prefix-length

Syntax

[no] exclude-pdp-address-prefix-length

Context

config>mobile>profile>charging>gy-avp-options

Description

This command includes or excludes PDP address prefix length IE in the Gy messages sent by the PGW/GGSN.

Default

no exclude-pdp-address-prefix-length

6.2.14.29 exclude-pdp-context-type

Syntax

[no] exclude-pdp-context-type

Context

```
config>mobile>profile>charging>gy-avp-options
```

Description

This command includes or excludes PDP context type IE in the Gy messages sent by the PGW/GGSN.

Default

```
no exclude-pdp-context-type
```

6.2.14.30 exclude-service-id

Syntax

```
[no] exclude-service-id
```

Context

```
config>mobile>profile>charging>gy-avp-options
```

Description

This command includes or excludes service identifier IE in the Gy messages sent by the PGW/GGSN.

Default

```
no exclude-service-id
```

6.2.14.31 exclude-serving-node-type

Syntax

```
[no] exclude-serving-node-type
```

Context

```
config>mobile>profile>charging>gy-avp-options
```

Description

This command includes or excludes server node type IE in the Gy messages sent by the PGW/GGSN.

Default

no exclude-serving-node-type

6.2.14.32 exclude-sgsn-address**Syntax**

[no] exclude-sgsn-address

Context

config>mobile>profile>charging>gy-avp-options

Description

This command includes or excludes SGSN address IE in the Gy messages sent by the PGW/GGSN.

Default

no exclude-sgsn-address

6.2.14.33 exclude-subscription-id-imsi**Syntax**

[no] exclude-subscription-id-imsi

Context

config>mobile>profile>charging>gy-avp-options

Description

This command includes or excludes IMSI subscription identifier IE in the Gy messages sent by the PGW/GGSN.

Default

no exclude-subscription-id-imsi

6.2.14.34 exclude-subscription-id-msisdn

Syntax

[no] exclude-subscription-id-msisdn

Context

config>mobile>profile>charging>gy-avp-options

Description

This command includes or excludes MSISDN subscription identifier IE in the Gy messages sent by the PGW/GGSN.

Default

no exclude-subscription-id-msisdn

6.2.14.35 exclude-subscription-id-nai

Syntax

[no] exclude-subscription-id-nai

Context

config>mobile>profile>charging>gy-avp-options

Description

This command includes or excludes NAI subscription identifier IE in the Gy messages sent by the PGW/GGSN.

Default

no exclude-subscription-id-nai

6.2.14.36 exclude-user-equipment-info

Syntax

[no] exclude-user-equipment-info

Context

```
config>mobile>profile>charging>gy-avp-options
```

Description

This command includes or excludes user equipment IE in the Gy messages sent by the PGW/GGSN. The command is mutually exclusive with `include-user-equipment-info-bcd-encode` command within the Gy AVP options charging profile.

Default

```
no exclude-user-equipment-info
```

6.2.1.14.37 exclude-user-name

Syntax

```
[no] exclude-user-name
```

Context

```
config>mobile>profile>charging>gy-avp-options
```

Description

This command includes or excludes user name IE in the Gy messages sent by the PGW/GGSN.

Default

```
no exclude-user-name
```

6.2.1.14.38 include-3gpp-gprs-qos-negotiated-profile

Syntax

```
[no] include-3gpp-gprs-qos-negotiated-profile
```

Context

```
config>mobile>profile>charging>gy-avp-options
```

Description

This command includes or excludes the 3GPP GPRS QoS-negotiated profile IE in the Gy messages sent by the PGW/GGSN.

Default

no include-3gpp-gprs-qos-negotiated-profile

6.2.1.14.39 include-3gpp-user-location-info-5g

Syntax

[no] include-3gpp-user-location-info-5g

Context

config>mobile>profile>charging>gy-avp-options

Description

This command includes or excludes the 5G attributes in the 3GPP user location IE in the Gy messages. This is valid only for the SMF legacy charging functionality, when the UE is in 5G radio and the SMF has an active Gy interface toward an OCS.

Default

no include-3gpp-user-location-info-5g

6.2.1.14.40 include-cg-address

Syntax

[no] include-cg-address

Context

config>mobile>profile>charging>gy-avp-options

Description

This command includes or excludes the CG address IE in the Gy messages sent by the PGW/GGSN.

Default

no include-cg-address

6.2.1.14.41 include-charging-rule-base-name**Syntax**

[no] include-charging-rule-base-name

Context

config>mobile>profile>charging>gy-avp-options

Description

This command includes or excludes the charging rule base name IE in the Gy messages sent by the PGW/GGSN.

Default

no include-charging-rule-base-name

6.2.1.14.42 include-dualstack-ggsn-address-ipv6**Syntax**

[no] include-dualstack-ggsn-address-ipv6

Context

config>mobile>profile>charging>gy-avp-options

Description

This command includes or excludes the dualstack GGSN IPv6 address IE in the Gy messages sent by the PGW/GGSN.

Default

no include-dualstack-ggsn-address-ipv6

6.2.1.14.43 include-dualstack-sgsn-address-ipv6

Syntax

[no] include-dualstack-sgsn-address-ipv6

Context

config>mobile>profile>charging>gy-avp-options

Description

This command includes or excludes the dualstack SGSN IPv6 address IE in the Gy messages sent by the PGW/GGSN.

Default

no include-dualstack-sgsn-address-ipv6

6.2.1.14.44 include-enhanced-diagnostics

Syntax

[no] include-enhanced-diagnostics

Context

config>mobile>profile>charging>gy-avp-options

Description

This command includes or excludes the 3GPP enhanced diagnostics AVP in the Gy messages sent by the PGW. It contains the RAN/NAS cause value received from MME, ePDG, or TWAG.

Default

no include-enhanced-diagnostics

6.2.1.14.45 include-extended-bw-nr

Syntax

[no] include-extended-bw-nr

Context

```
config>mobile>profile>charging>gy-avp-options
```

Description

This command supports sending to the OCS, the following extended information IEs used for high data rates for option-3x NR as part of EPC QOS information:

- Extended-Max-Requested-BW-UL
- Extended-Max-Requested-BW-DL
- Extended-GBR-UL
- Extended-GBR-DL
- Extended-APN-AMBR-UL
- Extended-APN-AMBR-DL

The no form of this command reverts to the default (exclude).

Default

```
no include-extended-bw-nr
```

6.2.1.14.46 include-mscc-qos-information

Syntax

```
[no] include-mscc-qos-information
```

Context

```
config>mobile>profile>charging>gy-avp-options
```

Description

This command includes the qos-information AVP within the MSCC.

The no form of this command reverts to the default (exclude).

Default

```
no include-mscc-qos-information
```


6.2.14.47 include-pra-info

Syntax

[no] include-pra-info

Context

config>mobile>profile>charging>gy-avp-options

Description

This command includes the PRA information sent to an online charging server. The change-ue-presence-pra command must be enabled as needed for online charging; for example, via OCS subscription or local configuration.

The no form of this command reverts to the default (exclude).

Default

no include-pra-info

6.2.14.48 include-qos-information

Syntax

[no] include-qos-information

Context

config>mobile>profile>charging>gy-avp-options

Description

This command includes or excludes QoS information IE in the Gy messages sent by the PGW/GGSN.

Default

no include-qos-information

6.2.14.49 include-rg-in-raa

Syntax

[no] include-rg-in-raa

Context

```
config>mobile>profile>charging>gy-avp-options
```

Description

This command enables the mobile gateway to include in the RAA reply, the rating group AVP with the same value as the one received from the OCS if the rating group AVP is already included.

The no form of this command reverts to the default, that is, the rating group AVP is not included.

Default

```
no include-rg-in-raa
```

6.2.1.14.50 include-twan-user-location-info

Syntax

```
[no] include-twan-user-location-info
```

Context

```
config>mobile>profile>charging>gy-avp-options
```

Description

This command includes or excludes the TWAN user location information IE in the Gy messages sent by the PGW/GGSN.

Default

```
no include-twan-user-location-info
```

6.2.1.14.51 include-twan-user-location-info-r16

Syntax

```
[no] include-twan-user-location-info-r16
```

Context

```
config>mobile>profile>charging>gy-avp-options
```

Description

This command includes or excludes the TWAN user location information IE with 3GPP release 16 format for online charging.

The no form of this command reverts to the default (exclude).



NOTE:

This command is not supported in the current release.

Default

no include-twan-user-location-info-r16

6.2.1.14.52 include-user-csg-info

Syntax

[no] include-user-csg-info

Context

config>mobile>profile>charging>gy-avp-options

Description

This command includes or excludes the user CSG information IE in the Gy messages sent by the PGW/GGSN.

Default

no include-user-csg-info

6.2.1.14.53 include-user-equipment-info-bcd-encode

Syntax

[no] include-user-equipment-info-bcd-encode

Context

config>mobile>profile>charging>gy-avp-options

Description

This command specifies the BCD encoding type. The command is mutually exclusive with the `exclude-user-equipment-info` command within the Gy AVP options charging profile.

The no form of this command reverts to the default (utf8 encoding).

Default

no include-user-equipment-info-bcd-encode

6.2.14.54 include-user-location-info-time

Syntax

[no] include-user-location-info-time

Context

config>mobile>profile>charging>gy-avp-options

Description

This command includes the user location information and time.

The no form of this command reverts to the default (exclude).

Default

no include-user-location-info-time

6.2.14.55 include-user-name-msisdn-at-apn

Syntax

[no] include-user-name-msisdn-at-apn

Context

config>mobile>profile>charging>gy-avp-options

Description

This command includes the User-Name AVP in the MSISDN@APN format. The APN is the same as the APN in the Called-Station-Id AVP sent to the OCS. This command and the `exclude-user-name` command in the `config>mobile>profile>charging>gy-avp-options` context are mutually exclusive.

The no form of this command reverts to the default (exclude).

Default

no include-user-name-msisdn-at-apn

6.2.14.56 include-user-name-raa

Syntax

```
[no] include-user-name-raa
```

Context

```
config>mobile>profile>charging>gy-avp-options
```

Description

The command includes or excludes the User-Name AVP in the RAA message that is sent to OCS.

Default

no include-user-name-raa

6.2.14.57 include-uan-user-location-information

Syntax

```
[no] include-uan-user-location-information
```

Context

```
config>mobile>profile>charging>gy-avp-options
```

Description

This command specifies to include or exclude the UWAN user location information for online charging.

The no form of this command reverts to the default (exclude).

Default

no include-uan-user-location-information

6.2.14.58 include-uan-user-location-information-r16

Syntax

```
[no] include-uan-user-location-information-r16
```

Context

```
config>mobile>profile>charging>gy-avp-options
```

Description

This command includes or excludes the UWAN user location information with 3GPP release 16 format for online charging.

The no form of this command reverts to the default (exclude).

Default

```
no include-uan-user-location-information-r16
```

6.2.15 Charging profile SMF command descriptions

6.2.15.1 smf-attributes-profile

Syntax

```
[no] smf-attributes-profile profile-id
```

Context

```
config>mobile>profile>charging
```

Description

This command enables the context to configure what attributes are included or excluded in the Nchf messages sent by the SMF. It is referenced from the config>mobile>profile>charging>chf-profile context.

The no form of this command removes the specified attributes profile from the configuration.

Parameters

profile-id

specifies a profile identifier

Values: 1 to 255

6.2.15.2 ran-secondary-rat-usage-reports

Syntax

```
ran-secondary-rat-usage-reports {include | exclude}
no ran-secondary-rat-usage-reports
```

Context

```
config>mobile>profile>charging>smf-attributes-profile
```

Description

This command enables sending secondary RAT usage reports for 5G NR to the OFCS. For the GTP' GA interface, the List of RAN Secondary RAT Usage Report field is populated within the SGW-CDR and PGW-CDR as applicable. For the Diameter Rf interface, the SGW and PGW send the RAN Secondary RAT Usage Report IE (within the PS-Information AVP) in ACR (INTERIM) and ACR (STOP) messages as applicable, and the OFCS generates the CDRs.

The no form of this command reverts to the default (excluded).

Parameters

include

includes the IE within the CDR for Ga and Diameter RF interfaces

exclude

excludes the IE within the CDR for Ga and Diameter RF interfaces

Default: exclude

6.2.15.3 retransmission-indicator

Syntax

```
retransmission-indicator {include | exclude}
no retransmission-indicator
```

Context

```
config>mobile>profile>charging>smf-attributes-profile
```

Description

This command specifies to include or exclude the Retransmission-Indicator attribute in the Nchf.

The no form of this command reverts to the default (excluded).

Parameters

include

includes the IE in the Nchf

exclude

excludes the IE in the Nchf

Default: exclude

6.2.1.15.4 service-specific-information

Syntax

```
service-specific-information {include | exclude}
```

```
no service-specific-information
```

Context

```
config>mobile>profile>charging>smf-attributes-profile
```

Description

This command specifies to include or exclude the Service-Specific-Information attribute in the Nchf. By default, it is excluded.

The no form of this command reverts to the default (excluded).

Default

```
no service-specific-information
```

Parameters

include

includes the IE in the Nchf

exclude

excludes the IE in the Nchf

6.2.1.15.5 service-specific-information-value

Syntax

```
service-specific-information-value information  
no service-specific-information-value
```

Context

```
config>mobile>profile>charging>smf-attributes-profile
```

Description

This command specifies the string value to be used in the Service-Specific-Information attribute in the Nchf. If not configured, the TS 32.255 Release 16 value is used.

The no form of this command reverts to the default.

Default

```
no service-specific-information
```

Parameters

information

specifies the string value to be used in the Service-Specific-Information attribute in the Nchf, up to 64 characters

6.2.1.15.6 serving-network-include-source-ip

Syntax

```
serving-network-include-source-ip {enable | disable}  
no serving-network-include-source-ip
```

Context

```
config>mobile>profile>charging>smf-attributes-profile
```

Description

This command, when enabled, always includes the available IP address (nFIPv4address or nFIPv6address) of the serving node in the Nchf ServingNetworkFunctionID IE. The IP address corresponds to the serving node IP address only when there is a direct communication between the serving node and the SMF.

When disabled, the Nchf ServingNetworkFunctionID IE includes the IP address, only if it is available in the SMF. The serving node IP address is optional in 5G networks.

Parameters

enable

specifies that the IP address of the serving node within the ServingNetworkFunction IE is always included

disable

specifies that the IP address of the serving node within the ServingNetworkFunction IE is only included when it is available in the SMF

Default

disable

6.2.1.15.7 serving-node-type

Syntax

```
serving-node-type {r15 | r16 | r17}
```

```
no serving-node-type
```

Context

```
config>mobile>profile>charging>smf-attributes-profile
```

Description

This command configures the serving node type that is sent to the CHF in the case of EPC interworking or 2G/3G access. The serving node types are always AMF for r15, AMF or SGW for r16, and AMF or SGW or SGSN for r17.

The SGW is the serving node of the SMF for EPC interworking. However, 3GPP Release 15 defines only AMF for the serving node; 3GPP Release 16 also defines SGW as an option for the serving node, and 3GPP Release 17 also defines SGSN as an option for the serving node in case of 2G/3G access.

The no form of this command reverts to the default (r15).



NOTE:

The Serving Node Type attribute is encoded in the field NodeFunctionality of the ServingNetworkFunctionID attribute.

Parameters

r15

specifies that the SMF sends AMF as the value of NodeFunctionality

r16

specifies that the SMF sends AMF or SGW as the value of NodeFunctionality

r17

specifies that the SMF sends AMF, SGW, or SGSN as the value of NodeFunctionality

Default: r15

6.2.1.15.8 smf-charging

Syntax

[no] smf-charging *charging-profile-id*

Context

config>mobile>profile>charging

Description

This command configures an SMF charging profile for configuring SMF charging options.

The no form of this command removes the specified charging profile from the configuration.

Parameters

charging-profile-id

specifies a profile charging identifier

Values: 0 to 255

This is the Octet 1 as received in the Charging Characteristics IE.

6.2.1.15.9 chf-selection

Syntax

chf-selection

Context

config>mobile>profile>charging>smf-charging

Description

This command enables the context to configure charging selection options for SMF charging.

6.2.15.10 chf-profile

Syntax

```
chf-profile charging-profile-id  
no chf-profile
```

Context

```
config>mobile>profile>charging>smf-charging>chf-selection
```

Description

This command configures the charging profile for SMF charging selection.

Parameters

charging-profile-id

specifies a profile ID

Values: 0 to 255

6.2.15.11 fail-open

Syntax

```
fail-open
```

Context

```
config>mobile>profile>charging>smf-charging>chf-selection
```

Description

This command enables the context to configure the Fail-Open failure handling parameters.

6.2.1.15.12 chf-profile

Syntax

```
chf-profile charging-profile-id  
no chf-profile
```

Context

```
config>mobile>profile>charging>smf-charging>chf-selection>fail-open
```

Description

This command configures the CHF charging profile to be used to control the fail-open CHF. The selected CHF profile refers to the CCFH profile configured under the `config>mobile>profile>charging>chf>ccfh-profile` context that enables the Fail-Open functionality.

Parameters

charging-profile-id

specifies the CHF profile ID

Values: 0 to 255

6.2.1.15.13 nf-id-list

Syntax

```
nf-id-list list-name  
no nf-id-list
```

Context

```
config>mobile>profile>charging>smf-charging>chf-selection>fail-open
```

Description

This command configures the prioritized fail-open NF ID list for the CHFs assigned to the session.

Parameters

list-name

specifies the name of the NF ID list, up to 32 characters

6.2.1.15.14 nf-id-list

Syntax

```
nf-id-list list-name
```

```
no nf-id-list
```

Context

```
config>mobile>profile>charging>smf-charging>chf-selection
```

Description

This command configures the NF ID list of the CHF function for SMF charging. The NF ID list is configured using the `config>mobile>profile>list>nf-id-list` command.

Parameters

list-name

specifies the name of the NF ID list, up to 32 characters

6.2.1.15.15 default-charging-method

Syntax

```
default-charging-method {online | offline | disabled}
```

```
no default-charging-method
```

Context

```
config>mobile>profile>charging>smf-charging
```

Description

This command configures the default charging method for SMF charging.

The no form of this command reverts to the default.

Parameters

online

specifies online charging as the default method

offline

specifies offline charging as the default method

disabled

specifies that default charging method is disabled

Default: disabled

6.2.1.15.16 event-triggers-fbc

Syntax

event-triggers-fbc *profile-id*

no event-triggers-fbc

Context

config>mobile>profile>charging>smf-charging

Description

This command configures the FBC events trigger profile to apply for the session. The specified profile must be configured under the config>mobile>profile>charging>event-triggers-qbc context.

The no form of this command removes the specified event-trigger profile ID from the smf-charging context.

Parameters***profile-id***

specifies a charging profile identifier

Values: 1 to 255

6.2.1.15.17 event-triggers-qbc

Syntax

event-triggers-qbc *profile-id*

no event-triggers-qbc

Context

config>mobile>profile>charging>smf-charging

Description

This command configures the QBC events trigger profile to apply for the session. The specified profile must be configured under the `config>mobile>profile>charging>event-triggers-qbc` context.

The `no` form of this command removes the specified event trigger profile ID from the `smf-charging` context.

Default

`no event-triggers-qbc`

Parameters

profile-id

specifies a charging profile ID

Values: 1 to 255

6.2.1.15.18 qbc-charging

Syntax

`qbc-charging {enable | disable}`

`no qbc-charging`

Context

`config>mobile>profile>charging>smf-charging`

Description

This command enables the reporting of the QBC charging information by the V-SMF for roaming sessions. The H-SMF reports the QBC charging information in addition to FBC charging information.

The `no` form of this command disables QBC charging.

Parameters

enable

enables QBC charging

disable

disables QBC charging

Default: `disable`

6.2.1.15.19 quota-management

Syntax

```
quota-management
```

Context

```
config>mobile>profile>charging>smf-charging
```

Description

This command enables the context to configure quota management for SMF charging.

6.2.1.15.20 final-unit-indication

Syntax

```
final-unit-indication
```

Context

```
config>mobile>profile>charging>smf-charging>quota-management
```

Description

This command enables the context to configure FUI for SMF charging quota management.

6.2.1.15.21 redirect-append-url

Syntax

```
[no] redirect-append-url
```

Context

```
config>mobile>profile>charging>smf-charging>quota-management>final-unit-indication
```

Description

This command enables redirect append URL for the quota management FUI for SMF charging.

The no form of this command disables redirect append URL.

6.2.15.22 redirect-filter-id

Syntax

```
redirect-filter-id base-policy-name  
no redirect-filter-id
```

Context

```
config>mobile>profile>charging>smf-charging>quota-management>final-unit-indication
```

Description

This command configures a redirect filter ID for quota management FUI for SMF charging.

Parameters

base-policy-name

specifies the base policy name for the redirect filter ID, up to 64 characters

6.2.15.23 redirection-start-send-rr-final

Syntax

```
[no] redirection-start-send-rr-final
```

Context

```
config>mobile>profile>charging>smf-charging>quota-management>final-unit-indication
```

Description

This command enables the option to send a CDReq message with the Reporting-Reason FINAL for active rating groups when redirection starts.



NOTE:

- CMG considers a Rating Group to be active if the UE has generated traffic for that rating group that has not been dropped (for example, due to out of quota)
- If you do not configure the `redirection-start-send-rr-final` command or the `redirection-start-send-rr-final-en` command, the default behavior applies. Starting redirection does not trigger a CDReq with Reporting-Reason FINAL.

The no form of this command disables sending Reporting-Reason FINAL when redirection starts.

6.2.1.15.24 redirection-start-send-rr-final-enh

Syntax

```
[no] redirection-start-send-rr-final-enh
```

Context

```
config>mobile>profile>charging>smf-charging>quota-management>final-unit-indication
```

Description

This command enables the option to send a CDReq message with the Reporting-Reason AVP set to FINAL for active and inactive rating groups when redirection starts.



NOTE:

- CMG considers a rating group to be active if the UE has generated traffic for that rating group that has not been dropped (for example, due to out of quota)
- If you do not configure the `redirection-start-send-rr-final` command or the `redirection-start-send-rr-final-enh` command, starting redirection does not trigger a CDRequest message with the Reporting-Reason AVP set to FINAL (default behavior).
- To enable the `redirection-start-send-rr-final-enh` command, you must first enable the `redirection-start-send-rr-final` command.

The no form of this command disables sending the Reporting-Reason AVP set to FINAL when redirection starts.

6.2.1.15.25 traffic-handling-options

Syntax

```
traffic-handling-options
```

Context

```
config>mobile>profile>charging>smf-charging>quota-management
```

Description

This command enables the context to configure traffic handling options for SMF charging quota management.

6.2.1.15.26 forced-re-authorization

Syntax

```
forced-re-authorization {allow | discard}  
no forced-re-authorization
```

Context

```
config>mobile>profile>charging>smf-charging>quota-management>traffic-handling-options
```

Description

This command configures forced reauthorization for SMF charging traffic handling. This command specifies the option to allow or discard the traffic for the service while waiting for a CCA message from the CHF, when a CCR message is sent to the CHF as a result of forced reauthorization by the CHF.

The no form of this command disabled forced reauthorization.

Parameters

allow

allows traffic to flow while waiting for a CCA from the CHF

discard

discards traffic while waiting for a CCA from the CHF

Default: allow

6.2.1.15.27 quota-exhausted-no-threshold

Syntax

```
quota-exhausted-no-threshold {allow | discard}  
no quota-exhausted-no-threshold
```

Context

```
config>mobile>profile>charging>smf-charging>quota-management>traffic-handling-options
```

Description

This command configures the option to allow or discard the traffic for the service while waiting for CCA response message from the CHF when no threshold is active and the quota gets exhausted.

The no form of this command disables exhausted quota threshold.

Parameters

allow

allows traffic to flow while waiting for a CCA message from the CHF

discard

discards traffic while waiting for a CCA message from the CHF

Default: allow

6.2.1.15.28 quota-exhausted-threshold-active

Syntax

```
quota-exhausted-threshold-active {allow | discard}
```

```
no quota-exhausted-threshold-active
```

Context

```
config>mobile>profile>charging>smf-charging>quota-management>traffic-handling-options
```

Description

This command configures the option to allow or discard the traffic for the service while waiting for a CCA response message from the CHF when there is an active threshold, but the quota gets exhausted while waiting for a CCA upon hitting the threshold.

The no form of this command disables active quota exhausted threshold active for SMF charging traffic handling.

Parameters

allow

allows traffic to flow while waiting for a CCA message from the CHF

discard

discards traffic while waiting for a CCA message from the CHF

Default: allow

6.2.15.29 quota-unavailable

Syntax

```
quota-unavailable {allow | discard}  
no quota-unavailable
```

Context

```
config>mobile>profile>charging>smf-charging>quota-management>traffic-handling-options
```

Description

This command specifies the option to allow or discard the traffic for the service while waiting for a CCA response message from the CHF when no quota is available for a rating group.

The no form of this command disables quota unavailable for SMF charging traffic handling.

Parameters

allow

allows traffic to flow while waiting for a CCA message from the CHF

discard

discards traffic while waiting for a CCA message from the CHF

Default: allow

6.2.15.30 rating-condition-change

Syntax

```
rating-condition-change {allow | discard}  
no rating-condition-change
```

Context

```
config>mobile>profile>charging>smf-charging>quota-management>traffic-handling-options
```

Description

This command configures the option to allow traffic to flow or be discarded for the service, while waiting for a Charging Data Response (CDresp) message after a Charging Data Request (CDreq) message is sent to the CHF due to a detected Rating-Condition change.

Parameters

allow

allows traffic to flow while waiting for a CDresp from the CHF

discard

discards traffic while waiting for a CDresp from the CHF

Default: allow

6.2.1.15.31 validity-time-expired

Syntax

```
validity-time-expired {allow | discard}
```

```
no validity-time-expired
```

Context

```
config>mobile>profile>charging>smf-charging>quota-management>traffic-handling-options
```

Description

This command specifies the option to allow or discard the traffic for the service while waiting for a CCA response message from the CHF when the validity timer expires.

The no form of this command disables validity time expired for SMF charging traffic handling.

Parameters

allow

allows traffic to flow while waiting for a CCA from the CHF

discard

discards traffic while waiting for a CCA from the CHF

Default: allow

6.2.1.15.32 radius-accounting

Syntax

```
radius-accounting
```

Context

```
config>mobile>profile>charging>smf-charging
```

Description

This command enables the context for configuring the RADIUS accounting.

6.2.1.15.33 ecgi-change

Syntax

```
[no] ecgi-change
```

Context

```
config>mobile>profile>charging>smf-charging>radius-accounting
```

Description

This command enables the RADIUS interim accounting to be triggered, based on an eCGI change location event. Triggering an interim accounting message due to any event, re-initializes the periodic accounting timer and/or volume-based threshold (if enabled).

The no form of this command reverts to the default.

Default

```
no ecgi-change
```

6.2.1.15.34 peer-change

Syntax

```
[no] peer-change
```

Context

```
config>mobile>profile>charging>smf-charging>radius-accounting
```

Description

This command enables the RADIUS interim accounting to be triggered based on a peer change event for the default bearer. Triggering an interim accounting message due to any event, re-initializes the periodic accounting timer and/or volume-based threshold (if enabled).

The no form of this command reverts to the default.

Default

no peer-change

6.2.1.15.35 qos-change

Syntax

[no] qos-change

Context

config>mobile>profile>charging>smf-charging>radius-accounting

Description

This command enables the RADIUS interim accounting to be triggered, based on a QoS change event for the default bearer. Triggering an interim accounting message due to any event, re-initializes the periodic accounting timer and/or volume-based threshold (if enabled).

The no form of this command reverts to the default.

Default

no qos-change

6.2.1.15.36 rat-change

Syntax

[no] rat-change

Context

config>mobile>profile>charging>smf-charging>radius-accounting

Description

This command enables the RADIUS interim accounting to be triggered based on a RAT change event. Triggering an interim accounting message due to a RAT change, re-initializes the periodic accounting timer (if enabled). Triggering an interim accounting message due to any event, re-initializes the periodic accounting timer and/or volume-based threshold (if enabled).

The no form of this command reverts to the default.

Default

no rat-change

6.2.15.37 tai-change

Syntax

[no] tai-change

Context

config>mobile>profile>charging>smf-charging>radius-accounting

Description

This command enables the RADIUS interim accounting to be triggered based on a TAI change location event. Triggering an interim accounting message due to any event, re-initializes the periodic accounting timer and/or volume-based threshold (if enabled).

The no form of this command reverts to the default.

Default

no tai-change

6.2.15.38 user-loc-change

Syntax

[no] user-loc-change

Context

config>mobile>profile>charging>smf-charging>radius-accounting

Description

This command enables the RADIUS interim accounting to be triggered based on any change location event. Triggering an interim accounting message due to any event, re-initializes the periodic accounting timer and/or volume-based threshold (if enabled).

The no form of this command reverts to the default.

Default

no user-loc-change

6.2.15.39 volume-limit

Syntax

volume-limit *value*

no volume-limit

Context

config>mobile>profile>charging>smf-charging>radius-accounting

Description

This command configures the volume threshold for interim accounting. Triggering an interim accounting message due to any event, re-initializes the periodic accounting timer and/or volume-based threshold (if enabled). Setting a low value can cause overload to the RADIUS server.

The no form of this command reverts to the default.

Default

no volume-limit

Parameters

value

specifies the volume threshold, in kbytes

Values: 0 to 104857600

Default: 0 (disabled)

6.2.16 Charging profile TWAG command descriptions

6.2.16.1 twag-charging

Syntax

[no] twag-charging *charging-profile-id*

Context

config>mobile>profile>charging

Description

This command configures a TWAG charging profile for configuring TWAG charging options.

Parameters

charging-profile-id

specifies a profile charging identifier

Values: 0 to 255

Default: 0

6.2.1.16.2 radius-accounting

Syntax

radius-accounting

Context

config>mobile>profile>charging>twag-charging

Description

This command enables the context to configure RADIUS accounting.

6.2.1.16.3 wlan-ap-location-change

Syntax

[no] wlan-ap- location-change

Context

config>mobile>profile>charging>twag-charging>radius-accounting

Description

This command enables or disables the triggering of RADIUS Accounting-Interim updates to the AAA server when the TWAG detects a change in the UE WLAN AP location (due to mobility). The RADIUS Accounting-Interim updates will include the Called-Station-Id with the MAC@ of the new WLAN AP as well as the NAS-

IP-Address WAN IP@ of the new WLAN AP (in case of Soft-GRE). If the WLAN APs are configured to send the RADIUS Accounting requests to inform the TWAG of the UE being attached to it, the `acct-triggered-mobility` command must also be enabled in the `config>mobile>profile>sww-profile` context.

The no form of this command reverts to the default (disabled).

**NOTE:**

If L2 access is used (no Soft-GRE tunneling), this command will only be supported for the following messages:

- L2 Closed SSID: SWw RADIUS Accounting and SWw RADIUS Authentication
- L2 Open SSID: DHCP (only if option 82 is provided)

Default

`no wlan-ap-location-change`

6.2.1.17 Cloud database profile command descriptions

6.2.1.17.1 cloud-db

Syntax

`[no] cloud-db profile-name`

Context

`config>mobile>profile`

Description

This command configures a cloud database profile. It creates an interface with an external database used for storage of subscriber/PDN context.

The no form of this command removes the profile from the configuration.

Parameters***profile-name***

specifies a profile name, up to 32 characters

6.2.1.17.2 server

Syntax

[no] server *ip-addr* | *fqdn* [port *port*]

Context

config>mobile>profile>cloud-db

Description

This command configures a cloud database server for the external database.
The no form of this command removes the cloud database server from the configuration.

Parameters

ip-addr
specifies the IPv4 or IPv6 address

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x
–	–	x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

fqdn
specifies the Fully Qualified Domain Name, up to 255 characters

port
specifies the port number
Values: 1 to 65535
Default: 5678

6.2.1.18 DCCA profile command descriptions

6.2.1.18.1 dcca

Syntax

[no] dcca *profile-name*

Context

config>mobile>profile

Description

This command configures a Diameter Credit Control Application (DCCA) profile used for the primary and secondary online Charging System (OCS) for this Online charging (Gy) profile.

The no form of this command removes the profile name from the configuration.

Parameters

profile-name

specifies a profile name, up to 32 characters

6.2.1.18.2 activity-threshold

Syntax

activity-threshold *activity-threshold*

no activity-threshold

Context

config>mobile>profile>dcca

config>mobile>profile>dcca>rating-group

Description

This command specifies the activity-threshold for time-based metering.

Parameters

activity-threshold

specifies the activity-threshold for time-based metering in kilobits per second.

Values: 1 to 100000000

6.2.18.3 append-url

Syntax

```
[no] append-url
```

Context

```
config>mobile>profile>dcca
```

Description

This command specifies whether the UE URL is appended to the Redirect-Server URL AVP in case of FUI redirection, when the Append-URL AVP is not received from the OCS. If enabled, the UE URL is appended to the Redirect-Server URL. When the URL-Append AVP, that is sent by the OCS, is received, the URL-Append AVP overwrites the configurable default value. If the command is not enabled, the Redirect-Server URL is used according to 3GPP standards.

The no form of this command reverts to the default (the Redirect-Server URL is used).

Default

```
no append-url
```

6.2.18.4 application-tx-timer

Syntax

```
application-tx-timer {value | timeout-msec value}
```

```
no application-tx-timer
```

Context

```
config>mobile>profile>dcca
```

Description

This command specifies the application transaction timer for CCR and CCA.

Parameters

value

specifies the application transaction timer, in seconds

Values: 1 to 180 (increments in 1 second)

Default: 5

timeout-msec *value*

specifies the application transaction timer, in milliseconds

Values: 1000 to 180000 (increments in 100 milliseconds)

Default: 5000

6.2.1.18.5 cc-failover-handling

Syntax

```
cc-failover-handling {terminate | continue | retry-and-terminate}
```

```
no cc-failover-handling
```

Context

```
config>mobile>profile>dcca
```

Description

This command configures the CCFH support option. It changes without a PDN shutdown.



NOTE:

The behavior configured for specific error codes in the context `config>mobile>profile>dcca>failure-handling (result-code, result-code-3xxx, result-code-4xxx, and result-code-5xxx)` overrides the CCFH command.

Parameters

terminate

configures the failover handling to result in termination

continue

configures the failover handling to result in continue

retry-and-terminate

configures the failover handling to result in retry and terminate

Default: terminate

6.2.1.18.6 cca-init-no-mscc-answer-ignore

Syntax

```
[no] cca-init-no-mscc-answer-ignore
```

Context

```
config>mobile>profile>dcca
```

Description

This command specifies the CMG behavior if no MSCC instance exists at the CCA-Init when a request for quota for that MSCC has been made. If enabled, when CMG request for quota is received, the CMG treats it as if 0 quota is allocated, if the MSCC instance is not present in the CCA-Init. Further quota is requested for that MSCC only when the first UE packet matches the MSCC. If the command is disabled, the CCA-Init is treated as Bad-Answer, when it does not include the MSCC instance for which quota was requested. By default the command is disabled.

The no form of this command reverts to the default (disabled).

Default

```
no cca-init-no-mscc-answer-ignore
```

6.2.1.18.7 ccfh-fh-action

Syntax

```
ccfh-fh-action {terminate | restart}  
no ccfh-fh-action
```

Context

```
config>mobile>profile>dcca
```

Description

This command configures the action to take, upon the expiration of the fh-session-continue-timer or when the limit set by fh-volume-limit is reached (whichever occurs first).



NOTE:

The CCR-I does not include any requests for quota (including for any configured default rating group). The installed rating groups are put in first packet mode, wherein the request for quota (CCR-U) is upon receiving a first packet for that rating group.

The no form of this command reverts to the default value terminate (IP-CAN session terminated).

Parameters

terminate

the IP-CAN session is terminated

re-start

a CCR-I is sent to the OCS upon the expiration of the fh-session-continue-timer or when the limit set by fh-volume-limit is reached (whichever occurs first); if the OCS is still unavailable, the session is then terminated

Default: terminate

6.2.1.18.8 ccfh-profile

Syntax

ccfh-profile *profile-id*

no ccfh-profile

Context

config>mobile>profile>dcca

Description

This command associates a DCCA session to a CCFH profile. If a valid CCFH profile ID is configured, the CCFH values from that CCFH profile are used (the CCFH values under the DCCA profile are ignored). If a CCFH profile ID is not configured, the CCFH parameters configured under the DCCA profile are used instead.

Parameters

profile-id

specifies the profile ID for this CCFH profile

Values: 0 to 255

6.2.1.18.9 ccr-initial-on-first-online-rg

Syntax

```
[no] ccr-initial-on-first-online-rg
```

Context

```
config>mobile>profile>dcca
```

Description

This command enables sending a CCR initial message on detection of a first online rating group. When enabled, a Gy online charging session is started toward the OCS when a PCC rule with the `online` command enabled is installed for the bearer. For example, a PDN session with a default bearer exists without online charging enabled. When a dedicated bearer with online charging enabled (PCC-rule) is established, a CCR-I is sent to the OCS. Online charging toward the OCS continues as long as the bearer exists. The Gy session is terminated when the corresponding bearer is terminated. This option cannot be enabled when the `ccr-initial-on-first-pkt` command is enabled.



NOTE:

If the DCCA profile is referenced in the configuration, this command requires a PDN shutdown.

Default

```
no ccr-initial-on-first-online-rg
```

6.2.1.18.10 ccr-initial-on-first-pkt

Syntax

```
[no] ccr-initial-on-first-pkt
```

Context

```
config>mobile>profile>dcca
```

Description

This command enables sending a CCR initial message on detection of a first packet. When enabled, the PDP context/bearer is set up without OCS authorization. A CCR-Initial message is sent when the first packet is detected for the PDP context/bearer. The CCR-Initial message includes an MSCC corresponding to the rating group that the first packet maps to. For subsequent SDFs that map to a different rating group, a CCR-U is sent requesting a quota when traffic is detected. First packet behavior must not be disabled if this

command is used. This option cannot be enabled when `ccr-initial-on-first-online-rg` command is enabled.

**NOTE:**

If the DCCA profile is referenced in the configuration, this command requires a PDN shutdown.

Default

no ccr-initial-on-first-pkt

6.2.1.18.11 ccr-mscc-limit

Syntax

`ccr-mscc-limit num-mscc`

`no ccr-mscc-limit`

Context

`config>mobile>profile>dcca`

Description

This command configures the maximum number of MSCC instances in a CCR message.

Parameters***num-mscc***

specifies the number of MSCC instances

Values: 1 to 16

Default: 16

6.2.1.18.12 ccrt-no-mscc

Syntax

`[no] ccrt-no-mscc`

Context

`config>mobile>profile>dcca`

Description

This command enables the report of an MSCC in the CCR-U message (used to report Final Units), upon session release. The CCR-T message that follows does not include any MSCC instances. If disabled, when the DCCA session is released, the CCR-T message includes all the active MSCC instances.

The no form of this command reverts to the default (disabled).

Default

no ccrt-no-mscc

6.2.1.18.13 command-level-fui

Syntax

[no] command-level-fui

Context

config>mobile>profile>dcca

Description

This command is used to enable and disable the support for the Final-Unit-Indication (FUI) AVP at the command level of a CCA message. Only Final-Unit-Action (FUA) = Terminate / Redirect is supported with FUI at the command level. When enabled, command level FUI is supported in CCA-I and CCA-U messages.

The no form of this command reverts to the default (disabled).



NOTE:

VT (command-level) expiration will result in CCR-U with MSCC being sent for the default RG (if configured) or the first online RG installed for online charging. The session gets out of redirect when there is a successful quota allocation in CCA-U for the RG. The session can also come out of redirect with a forced reauthorization (RAR) from the online-charging server, followed by successful quota allocation in CCA-U for the default RG or first online RG. Other RGs for the session will be put in first packet mode, after which traffic detected for an RG will result in a quota request for that RG.

Default

no command-level-fui

6.2.18.14 default-qht

Syntax

```
default-qht minutes  
no default-qht
```

Context

```
config>mobile>profile>dcca
```

Description

This command specifies the default Quota Holding Timer (QHT) that would be applicable, when OCS does not specify a QHT. When OCS specifies a QHT, it overrides the default-QHT configured on the gateway.

Parameters

minutes

specifies the Quota Holding Timer in minutes

Values: 0 to 30

Default: 0

6.2.18.15 default-rating-group

Syntax

```
default-rating-group rating-group  
no default-rating-group
```

Context

```
config>mobile>profile>dcca
```

Description

This command configures the default rating group.

Parameters

rating-group

specifies the rating group

Values: 0 to 4294967295

Default: 0

6.2.1.18.16 default-rg-list

Syntax

```
default-rg-list
```

Context

```
config>mobile>profile>dcca
```

Description

This command enables the context to configure the default rating-group list.

6.2.1.18.17 rating-group

Syntax

```
[no] rating-group rating-group
```

Context

```
config>mobile>profile>dcca>default-rg-list
```

Description

This command configures the default rating-group list. If enabled, the PGW requests quota for the rating groups defined in the rating group list at the CCR-Init. If not enabled, the CCR-Init does not request quota.

Default

no rating-group

Parameters

rating-group

specifies the rating group

Values: 0 to 4294967295

6.2.1.18.18 destination-realm

Syntax

```
destination-realm name  
no destination-realm
```

Context

```
config>mobile>profile>dcca
```

Description

This command configures a destination realm that will override the destination realm configured for the Diameter peer profile in the config>mobile>profile>diameter-peer context.

When the no form of this command is specified, the destination realm configured for the Diameter peer profile is used (default).

Default

```
no destination-realm
```

Parameters

name

specifies the name of the destination realm, up to 80 characters

6.2.1.18.19 diam-sess-end-serialize-ccr

Syntax

```
[no] diam-sess-end-serialize-ccr
```

Context

```
config>mobile>profile>dcca
```

Description

This command configures the serialization of CCR messages when a Diameter session is terminated. When the configuration is enabled, the gateway waits for outstanding CCA messages before sending a next CCR message during Diameter session termination.

The no form of this command reverts to the default (disabled).

Default

no diam-sess-end-serialize-ccr

6.2.18.20 disable-cc-session-failover

Syntax

[no] disable-cc-session-failover

Context

config>mobile>profile>dcca

Description

This command configures the disable-cc-session failover support option. It changes without a PDN shutdown.

6.2.18.21 disable-first-pkt-behavior

Syntax

[no] disable-first-pkt-behavior

Context

config>mobile>profile>dcca

Description

This command configures the disable first packet behavior. First packet behavior is enabled by default. When enabled, CCR-I is sent for OCS authorization without including any MSCC AVP. CCR-U with MSCC (including rating group/Service-Id and RSU) is sent upon detection of the first packet for an SDF belonging to a rating group for which no quota exists. When disabled, CCR-I / CCR-U includes a request for quota (MSCCs with rating group and RSU) in conjunction with the with the activation of the PCC rule; when no quota currently exists for that rating group.



NOTE:

- This command requires a PDN shutdown.
- To enable the one-time redirection option, the first packet behavior must not be disabled. For more information, see `one-time-redirect`.

6.2.1.18.22 disable-reauth-ccru-no-mscc

Syntax

```
[no] disable-reauth-ccru-no-mscc
```

Context

```
config>mobile>profile>dcca
```

Description

This command disables the sending of an empty CCR-U message (with no MSCC instances) after a session-level RAR message. The command is used for collision scenarios or scenarios where no rating groups are suitable to be reauthorized. These scenarios include the following cases:

- receiving a session-level RAR message from the OCS while a CCR-U for the only active rating group is in progress
- receiving a session-level RAR message from the OCS when no active rating groups for the session exist at the Gateway (the rating groups are blocked or inactive)

The no form of the command reverts to the default, that is, sending an empty CCR-U message for the collision scenario.

Default

```
no disable-reauth-ccru-no-mscc
```

6.2.1.18.23 event-triggers

Syntax

```
event-triggers
```

Context

```
config>mobile>profile>dcca
```

Description

This command enables the context to configure local event-triggers for the DCCA profile for online charging.

6.2.1.18.24 change-ue-presence-pra

Syntax

[no] change-ue-presence-pra

Context

config>mobile>profile>dcca>event-triggers

Description

This command locally enables the change in UE presence in a PRA event trigger for online charging.

The no form of this command reverts to the default (disabled).

Default

no change-ue-presence-pra

6.2.1.18.25 qos-apn-ambr-change

Syntax

[no] qos-apn-ambr-change

Context

config>mobile>profile>dcca>event-triggers

Description

This command enables the Qos APN AMBR change event trigger locally for online charging. A change to the configuration of this command applies to new sessions only. Event triggers provided by the OCS override locally configured event triggers.

The no form of this command reverts to the default (disabled).

Default

no qos-apn-ambr-change

6.2.1.18.26 qos-change

Syntax

[no] qos-change

Context

config>mobile>profile>dcca>event-triggers

Description

This command enables or disables the QoS change event trigger locally for online charging. A change to the configuration of this command applies to new sessions only. OCS provided event triggers override locally configured ones.

The no form of this command reverts to the default (disabled).

6.2.1.18.27 qos-max-bitrate-dl-change

Syntax

[no] qos-max-bitrate-dl-change

Context

config>mobile>profile>dcca>event-triggers

Description

This command enables the qos-max-bitrate-downlink-change event trigger locally for online charging. A change to the configuration of this command applies to new sessions only. The Event triggers provided by the OCS override locally configured event triggers.

The no form of this command reverts to the default (disabled).

Default

no qos-max-bitrate-dl-change

6.2.1.18.28 qos-max-bitrate-ul-change

Syntax

[no] qos-max-bitrate-ul-change

Context

```
config>mobile>profile>dcca>event-triggers
```

Description

This command enables the `qos-max-bitrate-uplink-change` event trigger locally for online charging. A change to the configuration of this command applies to new sessions only. The Event triggers provided by the OCS override locally configured event triggers.

The no form of this command reverts to the default (disabled).

Default

```
no qos-max-bitrate-ul-change
```

6.2.1.18.29 rat-change

Syntax

```
[no] rat-change
```

Context

```
config>mobile>profile>dcca>event-triggers
```

Description

This command enables or disables the RAT change event trigger locally for online charging. A change to the configuration of this command applies to new sessions only. OCS provided event triggers override locally configured ones.

The no form of this command reverts to the default (disabled).

Default

```
no rat-change
```

6.2.1.18.30 serving-node-change

Syntax

```
[no] serving-node-change
```

Context

```
config>mobile>profile>dcca>event-triggers
```

Description

This command enables or disables the serving node change event trigger locally for online charging. A change to the configuration of this command applies to new sessions only. OCS provided event triggers override locally configured ones.

The no form of this command reverts to the default (disabled).

Default

no serving-node-change

6.2.1.18.31 trigger-override-option

Syntax

```
trigger-override-option {local | ocs | local-and-ocs}
```

```
no trigger-override-option
```

Context

```
config>mobile>profile>dcca>event-triggers
```

Description

This command is used to select the event trigger override option. The trigger override options that can be enforced at the gateway include locally configured event triggers only (ignoring the OCS subscribed triggers), OCS-subscribed event triggers only (ignoring the locally configured triggers), or a super set of both locally configured and OCS subscribed triggers.

The no form of this command reverts to the default (OCS only).

Parameters

local

Only locally-enabled triggers are enforced at the gateway; OCS-subscribed triggers are ignored

ocs

Only OCS-subscribed triggers are enforced at the gateway; locally-enabled triggers are ignored

local-and-ocs

Both locally-enabled and OCS-subscribed triggers are enforced at the gateway

Default: ocs

6.2.1.18.32 user-loc-change

Syntax

[no] user - loc - change

Context

config>mobile>profile>dcca>event-triggers

Description

This command enables or disables the user location change event trigger locally for online charging. A change to the configuration of this command applies to new sessions only. OCS provided event triggers override locally configured ones.

The no form of this command reverts to the default (disabled).

Default

no user-loc-change

6.2.1.18.33 user-loc-change-cellid

Syntax

[no] user - loc - change - cellid

Context

config>mobile>profile>dcca>event-triggers

Description

This command enables the change in location CELLID (cell identity) event trigger locally on the gateway.



NOTE:

This command must not be enabled in conjunction with the generic user - loc - change event trigger.

The no form of this command reverts to the default (disabled).

Default

no user-loc-change-cellid

6.2.1.18.34 user-loc-change-ecgi

Syntax

[no] user - loc - change - ecgi

Context

config>mobile>profile>dcca>event-triggers

Description

This command enables the change in location ECGI event trigger locally on the gateway.



NOTE:

This command must not be enabled in conjunction with the generic user - loc - change event trigger.

The no form of this command reverts to the default (disabled).

Default

no user-loc-change-ecgi

6.2.1.18.35 user-loc-change-lac

Syntax

[no] user - loc - change - lac

Context

config>mobile>profile>dcca>event-triggers

Description

This command enables the change in location LAC event trigger locally on the gateway.



NOTE:

This command must not be enabled in conjunction with the generic user - loc - change event trigger.

The no form of this command reverts to the default (disabled).

Default

no user-loc-change-lac

6.2.1.18.36 user-loc-change-rac

Syntax

[no] user - loc - change - rac

Context

config>mobile>profile>dcca>event-triggers

Description

This command enables the change in location RAC event trigger locally on the gateway.



NOTE:

This command must not be enabled in conjunction with the generic user - loc - change event trigger.

The no form of this command reverts to the default (disabled).

Default

no user-loc-change-rac

6.2.1.18.37 user-loc-change-tac

Syntax

[no] user - loc - change - tac

Context

config>mobile>profile>dcca>event-triggers

Description

This command enables the change in location TAC event trigger locally on the gateway.



NOTE:

This command must not be enabled in conjunction with the generic user - loc - change event trigger.

The no form of this command reverts to the default (disabled).

Default

no user-loc-change-tac

6.2.1.18.38 uwan-loc-change

Syntax

[no] uwan - loc - change

Context

config>mobile>profile>dcca>event-triggers

Description

This command enables or disables the UWAN location change event trigger locally for online charging. The OCS provided event triggers override locally configured sessions.

The no form of this command reverts to the default.

Default

no uwan-loc-change

6.2.1.18.39 failure-handling

Syntax

failure-handling

Context

config>mobile>profile>dcca

Description

This command enables the context to configure failure-handling multi-service credit control and result codes.

6.2.1.18.40 mscc-result-code

Syntax

```
[no] mscc-result-code mscc-result-code-value
```

Context

```
config>mobile>profile>dcca>failure-handling
```

Description

This command specifies a result code for the configurable delay timer during which service is denied, on an error code basis. Subsequent to the timer expiry, traffic for an RG results in a CCR-U to the OCS requesting the quota.

The no form of this command removes the result code and associated timer entry and reverts to the current behavior.

Default

no mscc-result-code (current behavior applies)

Parameters

mscc-result-code-value

specifies the four digit result code value for the service-delay-timer command applicable at the MSCC level; result code values include 3GPP specified values and some operator custom result codes



NOTE:

As a means to future proof, other (arbitrary) values will be allowed as well.

Values	4010	DIAMETER_END_USER_SERVICE_DENIED
–	4012	DIAMETER_CREDIT_LIMIT_REACHED

–	4201	USER-NEVER-USED
–	4202	USER-EXPIRED
–	4203	USER-INVALID
–	4204	USER-NOT-EXIST

6.2.1.18.41 result-code

Syntax

```
result-code result-code-value {terminate | continue | retry-and-terminate}  
no result-code
```

Context

```
config>mobile>profile>dcca>failure-handling
```

Description

This command specifies the failure handling behavior on a per error code basis.

The no form of this command removes the error-specific failure handling for the specified result code and reverts to the behavior for the generic command.



NOTE:

Configuration behavior for specific error codes overrides the generic CCFH command (cc-failover-handling).

Default

no result-code (behavior of the generic command at the DCCA main level applies)

Parameters

result-code-value

specifies the result code values applicable at the command level. Result code values include 3GPP specified values and some operator custom result codes.

**NOTE:**

- For all other values, the generic CCFH command under the DCCA profile applies. Configuration behavior for specific error codes overrides the generic CCFH command.
- To ensure future functionality, other (arbitrary) values will be allowed.
- For result codes other than 3002, 3004, and 3005, a retry and terminate defaults to terminate (for example, no alternate peer is tried before terminating).

Values: *3GPP result codes* lists the supported 3GPP result codes.

**NOTE:**

The result codes 4010, 4011, 4012, and 5030 are NOT allowed as input to the *result-code-value* parameter and the current behavior applies.

terminate

specifies that the session terminates without trying an alternate peer

continue

specifies that the session continues

retry-and-terminate

specifies the retry and terminate option for codes 3002, 3004, and 3005 by default. All others are treated as terminate

Table 98: 3GPP result codes

Result code	Diameter command result code name	Default behavior	Result-code behavior
3001	DIAMETER-COMMAND-UNSUPPORTED	Follow CCFH	Overrides the generic CCFH command
3002	DIAMETER-UNABLE-TO-DELIVER	(Failover Enabled) Sent to the secondary OCS or to the CCFH if there is	Overrides the generic CCFH command

Table 98: 3GPP result codes (continued)

Result code	Diameter command result code name	Default behavior	Result-code behavior
		no secondary OCS	
3003	DIAMETER-REALM-NOT-SERVED	Follow CCFH	Overrides the generic CCFH command
3004	DIAMETER-TOO-BUSY	(Failover Enabled) Sent to the secondary OCS or to the CCFH if there is no secondary OCS	Overrides the generic CCFH command
3005	DIAMETER-LOOP-DETECTED	(Failover Enabled) Sent to the secondary OCS or to the CCFH if there is no secondary OCS	Overrides the generic CCFH command
3006	DIAMETER-REDIRECT-INDICATION	Follow CCFH	Overrides the generic CCFH command
3007	DIAMETER-APPLICATION-UNSUPPORTED	Follow CCFH	Overrides the generic CCFH command
3008	DIAMETER-INVALID-HDR-BITS	Follow CCFH	Overrides the generic CCFH command

Table 98: 3GPP result codes (continued)

Result code	Diameter command result code name	Default behavior	Result-code behavior
3009	DIAMETER-INVALID-AVP-BITS	Follow CCFH	Overrides the generic CCFH command
3010	DIAMETER-UNKNOWN-PEER	Follow CCFH	Overrides the generic CCFH command
4001	DIAMETER-AUTHENTICATION-REJECTED	Follow CCFH	Overrides the generic CCFH command
4002	DIAMETER-OUT-OF-SPACE	Follow CCFH	Overrides the generic CCFH command
4003	ELECTION-LOST	Follow CCFH	Overrides the generic CCFH command
4010	DIAMETER_END_USER_SERVICE_DENIED	Terminate the session; when received in CCA-U, CCR-T is sent	Same as before
4011	DIAMETER_CREDIT_CONTROL_NOT_APPLICABLE	Gy user session allowed	Same as before
4012	DIAMETER-CREDIT_LIMIT_REACHED	Terminate the session; when received in CCA-U, CCR-T is sent	Same as before

Table 98: 3GPP result codes (continued)

Result code	Diameter command result code name	Default behavior	Result-code behavior
4201	USER-NEVER-USED	Follow CCFH	Overrides the generic CCFH command
4202	USER-EXPIRED	Follow CCFH	Overrides the generic CCFH command
4203	USER-INVALID	Follow CCFH	Overrides the generic CCFH command
4204	USER-NOT-EXIST	Follow CCFH	Overrides the generic CCFH command
5001	DIAMETER-AVP-UNSUPPORTED	Follow CCFH	Overrides the generic CCFH command
5002	DIAMETER-UNKNOWN-SESSION-ID	Follow CCFH	Overrides the generic CCFH command
5003	DIAMETER-AUTHORIZATION-REJECTED	Terminate the session. When received in CCA-U, CCR-T is sent	Same as before
5004	DIAMETER-INVALID-AVP-VALUE	Follow CCFH	Overrides the generic CCFH command

Table 98: 3GPP result codes (continued)

Result code	Diameter command result code name	Default behavior	Result-code behavior
5005	DIAMETER-MISSING-AVP	Follow CCFH	Overrides the generic CCFH command
5006	DIAMETER-RESOURCES-EXCEEDED	Follow CCFH	Overrides the generic CCFH command
5007	DIAMETER-CONTRADICTING-AVP	Follow CCFH	Overrides the generic CCFH command
5008	DIAMETER-AVP-NOT-ALLOWED	Follow CCFH	Overrides the generic CCFH command
5009	DIAMETER-AVP-OCCURS-TOO-MANY-TIMES	Follow CCFH	Overrides the generic CCFH command
5010	DIAMETER-NO-COMMON-APPLICATION	Follow CCFH	Overrides the generic CCFH command
5011	DIAMETER-UNSUPPORTED-VERSION	Follow CCFH	Overrides the generic CCFH command
5012	DIAMETER-UNABLE-TO-COMPLY	Follow CCFH	Overrides the generic CCFH command
5013	DIAMETER-INVALID-BIT-IN-HEADER	Follow CCFH	Overrides the generic CCFH command

Table 98: 3GPP result codes (continued)

Result code	Diameter command result code name	Default behavior	Result-code behavior
5014	DIAMETER-INVALID-AVP-LENGTH	Follow CCFH	Overrides the generic CCFH command
5015	DIAMETER-INVALID-MESSAGE-LENGTH	Follow CCFH	Overrides the generic CCFH command
5016	DIAMETER-INVALID-AVP-BIT-COMMAND	Follow CCFH	Overrides the generic CCFH command
5017	DIAMETER-NO-COMMON-SECURITY	Follow CCFH	Overrides the generic CCFH command
5030	DIAMETER-USER-UNKNOWN	Terminate the session	Same as before
5031	DIAMETER-RATING-FAILED	Follow CCFH	Overrides the generic CCFH command

6.2.18.42 service-delay-timer

Syntax

service-delay-timer *minutes*

no service-delay-timer

Context

config>mobile>profile>dcca>failure-handling>mscc-result-code

Description

This command specifies a value for the service delay timer. Service is denied for the specified duration. Result codes 4010 and 4012 by default have a 3 minute delay timer (current behavior). For any other result code that is not configured, the existing behavior of blocking the RG applies, and no timer is started.

The no form of this command reverts to the default.

Parameters

minutes

specifies the value of the delay timer in minutes

Values: 1 to 1440

Default: 0

6.2.1.18.43 result-code-3xxx

Syntax

```
result-code-3xxx {terminate | continue | retry-and-terminate}
```

```
no result-code-3xxx
```

Context

```
config>mobile>profile>dcca>failure-handling
```

Description

This command specifies the CCFH behavior for 3xxx series of error codes in general. This configuration behavior for specific error-codes overrides the generic CCFH command (`cc-failover-handling`).

The no form of this command removes the error-specific failure handling for the 3xxx series of error code and reverts to the behavior for the generic command at the DCCA main level.

Default

disabled (the behavior of the generic DCCA command at the main level applies)

Parameters

terminate

specifies that the session will terminate without trying an alternate peer

continue

specifies that the session will continue

retry-and-terminate

specifies the retry and terminate option for codes 3002, 3004, and 3005 by default. All others are treated as terminate.

6.2.1.18.44 result-code-4xxx**Syntax**

```
result-code-4xxx {terminate | continue | retry-and-terminate}
```

```
no result-code-4xxx
```

Context

```
config>mobile>profile>dcca>failure-handling
```

Description

This command specifies the CCFH behavior for the 4xxx series of error codes in general. This configuration behavior for specific error-codes overrides the generic CCFH command (`cc-failover-handling`).

The no form of this command removes the error-specific failure handling for the 4xxx series of error code and reverts to the behavior for the generic command at the DCCA main level.

Default

no result-code-4xxx (the behavior of the generic DCCA command at the main level applies)

Parameters**terminate**

specifies that the session will terminate without trying an alternate peer

continue

specifies that the session will continue

retry-and-terminate

specifies the retry and terminate option for codes 3002, 3004, and 3005 by default. All others are treated as terminate.

6.2.1.18.45 result-code-5xxx**Syntax**

```
result-code-5xxx {terminate | continue | retry-and-terminate}
```

```
no result-code-5xxx
```

Context

```
config>mobile>profile>dcca>failure-handling
```

Description

This command specifies the failure-handling behavior for the 5xxx series of error codes in general. This configuration behavior for specific error-codes overrides the generic CCFH command (`cc-failover-handling`).

The `no` form of this command removes the error-specific failure handling for the 5xxx series of error code and reverts to the behavior for the generic DCCA command.

Default

`no result-code-5xxx` (the behavior of the generic DCCA command at the main level applies)

Parameters

`terminate`

specifies that the session will terminate without trying an alternate peer

`continue`

specifies that the session will continue

`retry-and-terminate`

specifies the retry and terminate option for codes 3002, 3004, and 3005 by default. All others are treated as terminate.

6.2.1.18.46 fh-session-continue-timer

Syntax

```
fh-session-continue-timer minutes
```

```
no fh-session-continue-timer
```

Context

```
config>mobile>profile>dcca
```

Description

This command specifies the Failure Handling Session Continue timer. When CCFH = Continue, the credit control session is considered closed; however, the service is granted for the configured period of time.

When `fh-session-continue-timer` and `fh-volume-limit` commands are active (CCFH = Continue), the IP-CAN session is terminated (unless configured to reauthorize) upon either condition being met (whichever occurs first).

Parameters

minutes

specifies the Failure Handling Session continue timer in minutes

Values: 1 to 1440

Default: 30

6.2.1.18.47 fh-volume-limit

Syntax

```
fh-volume-limit volume-limit
```

```
no fh-volume-limit
```

Context

```
config>mobile>profile>dcca
```

Description

This command specifies the volume limit for CCFH. When `fh-session-continue-timer` and `fh-volume-limit` are active (CCFH = Continue), the IP-CAN session is terminated (unless configured to reauthorize) upon either condition being met (whichever occurs first).

The no form of this command reverts to the default (disabled).

Parameters

volume-limit

specifies the volume limit in number of kilobytes

Values: 0 to 10000000

Default: 0

6.2.1.18.48 fui-validity-timer

Syntax

```
fui-validity-timer minutes [precedence]
```

```
no fui-validity-timer
```

Context

```
config>mobile>profile>dcca
```

Description

This command specifies the time that the Final Unit Action (Terminate, Restrict-Access, and Redirect) is active. When the timer expires, a CCR-U is triggered to the OCS to request quota. The UE traffic generated when the timer is active will be dropped (FUI = terminate), restricted (FUI = restrict-access) or redirected (FUI = redirect). The Validity-Time AVP, if sent by the OCS, will overwrite the local configuration. This parameter is used only in the case when the OCS does not support the Validity-Time AVP, otherwise the last AVP value is used.

When the precedence parameter is not configured (default), and if OCS sends an MSCC without the Validity-Time AVP, CMG uses the last known value of the Validity-Time AVP that is sent from the OCS in a previous MSCC. When the precedence parameter is configured, and if OCS sends an MSCC without the Validity-Time AVP, CMG uses the local configuration of fui-validity-timer; the previous values of the Validity-Time AVP are ignored.

The no form of this command reverts to the default (value not set).

Parameters

minutes

specifies the redirection validity time in minutes

Values: 1 to 1440

Default: 0

precedence

specifies that the local configuration of fui-validity-timer takes precedence over the last known value of the Validity-Time AVP when OCS sends an MSCC without the Validity-Time AVP

6.2.1.18.49 gy-avp-options-profile

Syntax

```
gy-avp-options-profile gy-avp-options-profile-id
```

```
no gy-avp-options-profile
```

Context

```
config>mobile>profile>dcca
```


Description

This command enables the context to configure what AVP IE included or excluded in the Gy messages sent by the PGW/GGSN. It changes without a PDN shutdown.

Parameters

gy-avp-options-profile-id

specifies a profile identifier

Values: 1 to 255

6.2.1.18.50 implicit-vol-threshold

Syntax

```
implicit-vol-threshold percent reason [threshold | exhaustion]
```

```
no implicit-vol-threshold
```

Context

```
config>gateway>profile>dcca
```

Description

This command configures an implicit-volume threshold which is used when the allocated quota does not include a quota threshold from the OCS. This threshold value is used to notify the OCS when the specified threshold is reached. The threshold is specified as a percentage of the received GSU (volume quota). For example, when the implicit-vol-threshold value is configured at 20%, the OCS is notified when less than or equal to 20% of the allocated quota is remaining.

The no form of this command reverts to the default (disabled).

Default

```
no implicit-vol-threshold
```

Parameters

percent

specifies the threshold percent of the allocated quota.

Values: 0 to 99

Default: 0

threshold

specifies reporting-reason is threshold

exhaustion

specifies reporting-reason is exhaustion

6.2.18.51 include-avp**Syntax**

```
include-avp
```

Context

```
config>gateway>profile>dcca
```

Description

This command enables the context to configure AVP options.

6.2.18.52 3gpp-gprs-qos-neg-prof**Syntax**

```
[no] 3gpp-gprs-qos-neg-prof
```

Context

```
config>gateway>profile>dcca>include-avp
```

Description

This command specifies whether the 3GPP-QoS-Negotiated-Profile AVP is included in the CCR. This command cannot be enabled if the `qos-information` command is enabled. It changes without a PDN shutdown.

The no form of this command removes the 3GPP-QoS-Negotiated-Profile AVP from the CCR.

Default

```
no 3gpp-gprs-qos-neg-prof
```

6.2.18.53 qos-information**Syntax**

```
[no] qos-information
```

Context

```
config>gateway>profile>dcca>include-avp
```

Description

This command specifies whether QoS-Information AVP is included in the CCR. Only either 3GPP-GPRS-QoS-Negotiated-Profile or QoS-Information must be enabled. It changes without a PDN shutdown.

The no form of this command disables QoS-Information AVP to be included in the CCR.

6.2.18.54 include-rpt-reason-on-forced-reauth

Syntax

```
[no] include-rpt-reason-on-forced-reauth
```

Context

```
config>gateway>profile>dcca
```

Description

This command includes the Reporting-Reason AVP with value Forced-Reauthorisation (7) when the mobile gateway sends a CCR-U with the RSU AVP for a rating group that is under redirection.

The no form of this command reverts to the default, that is, the RSU AVP is not included.

Default

```
no include-rpt-reason-on-forced-reauth
```

6.2.18.55 exclude-service-id

Syntax

```
[no] exclude-service-id
```

Context

```
config>mobile>profile>dcca>include-avp
```

Description

This command includes or excludes service identifier IE in the DCCA messages sent by the PGW/GGSN. It changes without a PDN shutdown.

Default

no exclude-service-id

6.2.18.56 offline-interwork

Syntax

offline-interwork

Context

config>mobile>profile>dcca

Description

This command configures offline interwork parameters.

6.2.18.57 one-time-redirect

Syntax

one-time-redirect *welcome-page-rg welcome-page-rg alert-page-rg alert-page-rg*
no one-time-redirect

Context

config>mobile>profile>dcca

Description

This command specifies the rating groups associated with 'welcome page' and 'alert page' traffic. The UE is redirected once to the welcome and alert page. Subsequent HTTP requests are not redirected.

The no form of this command reverts to the default (disabled).



NOTE:

- This command requires a PDN shutdown.
- To enable one-time redirect, the first-packet trigger mode must be enabled. First packet mode is enabled by default, so when a first packet is detected for a rating group, the gateway sends a CCR-U to request the quota. For more information, see `disable-first-pkt-behavior`.

Default

no one-time-redirect

Parameters

welcome-page-rg

specifies the rating group for the welcome page redirection. The rating group is configured via the `rating-group` command.

alert-page-rg

specifies the rating group for the alert page redirection. The rating group is configured via the `rating-group` command.

Values: 0 to 4294967295

Default: 0

6.2.1.18.58 ps-fci-fh

Syntax

[no] `ps-fci-fh`

Context

`config>mobile>profile>dcca>offline-interwork`

Description

This command is used to enable/disable custom failure handling applicable to a specific operator. When enabled, for scenarios that result in Tx timer expiration or CC session termination initiated by the OCS, the CDR is closed with a reason = Management Intervention (20), a new CDR is opened and the PS-Free-Format-Data within the PS-Furnish-Charging-Information at the main level of the CDR is populated with a value of 11.

The no form of this command reverts to the default (custom behavior disabled).



NOTE:

This command requires a PDN shutdown.

Default

no `ps-fci-fh`

6.2.1.18.59 ps-fci-fh-container

Syntax

[no] ps-fci-fh-container

Context

config>mobile>profile>dcca>offline-interwork

Description

This command enables the custom failure handling mechanism applicable to a specific operator. When enabled, and the ps-fci-fh command is also enabled, when FH starts, a partial CDR is closed and a new partial CDR is opened. In every PGW-CDR container, the FCI field value of the partial CDR is set to 16, if the CDR container includes charging data which has already been reported to the OCS. If the CDR container was not reported to the OCS, the FCI field value is set to 11.

The no form of this command reverts to the default (disabled).

Default

no ps-fci-fh-container

6.2.1.18.60 pre-emptive-quota

Syntax

[no] pre-emptive-quota

Context

config>mobile>profile>dcca

Description

This command is used to enable support for pre-emptive quota handling. When enabled, the gateway includes RSU (Requested Service Units) AVP at command level in a CCR, to indicate support for pre-emptive quota handling to the OCS. OCS can allocate quota and send MSCCs in a CCA prior to the gateway explicitly asking for quota for an RG in a CCR.

The no form of this command disables support for pre-emptive quota handling



NOTE:

This command requires a PDN shutdown.

6.2.1.18.61 queue-rar-pending-ccau

Syntax

[no] queue-rar-pending-ccau

Context

config>mobile>profile>dcca

Description

This command specifies CMG to queue the RAR message, and wait for the CCA-U before CMG sends a RAA message. This applies when a session-level RAR is received from the OCS and a CCR-U message is in progress. In a collision scenario, configuring this command facilitates CMG to include all active rating groups in a CCR-U that is sent to OCS following a RAR message.

The no form of this command reverts to the default.

Default

no queue-rar-pending-ccau

6.2.1.18.62 rating-group

Syntax

rating-group *rating-group-id*

no rating-group

Context

config>mobile>profile>dcca

Description

This command specifies a particular rating-group for which activity-threshold and USU options can be configured.

Parameters

rating-group-id

specifies the rating group

Values: 0 to 4294967295

6.2.18.63 rating-group-in-ch-rule-report

Syntax

```
[no] rating-group-in-ch-rule-report
```

Context

```
config>mobile>profile>dcca
```

Description

This command enables a customized Charging-Rule-Report AVP to the PCRF, which reports the Rating-Group AVP instead of the Charging-Rule-Name AVP, for the event triggers Out-of-Credit (15) and Reallocation-of-Credit (16). The Charging-Rule-Report AVP includes the Rating-Group AVP (code: 432) and the Final-Unit-Indication AVP (with only the Final-Unit-Action AVP), and omits the Charging-Rule-Name AVP and the PCC-Rule-Status AVP.

**NOTE:**

This option is only valid for the event triggers Out-Of-Credit (15) and Reallocation-of-Credit (16).

The no form of this command reverts to the default behavior; the Charging-Rule-Report AVP aligns to 3GPP specifications and includes the Charging-Rule-Name, PCC-Rule-Status, and Final-Unit-Indication AVPs.

Default

```
no rating-group-in-ch-rule-report
```

6.2.18.64 reauth-blocked-rg

Syntax

```
[no] reauth-blocked-rg
```

Context

```
config>mobile>profile>dcca
```

Description

This command sets a blocked rating-group as active when a RAR message is received from the OCS. When enabled, the service delay timer is stopped, and the blocked rating-group is included in the CCR-U that follows the RAA message.

The no form of this command reverts to the default behavior (disabled); the rating-group is excluded from the CCR-U.

Default

no reauth-blocked-rg

6.2.1.18.65 redirect-filter-id

Syntax

`redirect-filter-id base-policy-name`

`no redirect-filter-id`

Context

`config>mobile>profile>dcca`

Description

This command is used to specify a policy rule base that gets locally activated when a rating group is put into an FUI redirect state, provided that the OCS doesn't specify the filter ID. This rule base will be deactivated when the RG gets out of the redirect state. The rule base can, for example, include rules to allow DNS traffic to be allowed without charging.

The no form of this command removes the configured policy rule base for this DCCA profile.

Parameters

base-policy-name

specifies the name of the policy rule base, up to 64 characters

6.2.1.18.66 redirection-start-send-rr-final

Syntax

`[no] redirection-start-send-rr-final`

Context

`config>mobile>profile>dcca`

Description

This command enables the reporting of an MSCC, which is under redirection, when a DCCA session is terminated. If enabled, when the MSCC redirection starts, a CCR-U is sent to the OCS to indicate that the MSCC starts redirection. The MSCC includes the Used-Service-Units AVP (USU) but not the Requested-Service-Units AVP (RSU). The Reporting-Reason AVP at the MSCC level has the FINAL value (the flow is terminated from OCS point of view). When a DCCA session is terminated, the CCR-T sent to the OCS does not include the MSCC under redirection state. If the command is disabled (default behavior), when the MSCC redirection starts, a CCR-U is sent to the OCS to indicate that the MSCC starts redirection. The MSCC includes the Used-Service-Units AVP (USU) but not the Requested-Service-Units AVP (RSU). The Reporting-Reason AVP at MSCC level is QUOTA-EXHAUSTED (the OCS is informed that MSCC is active but under redirection state). When a DCCA session is terminated, the CCR-T which is sent to the OCS includes the MSCC under redirection state with Reporting-Reason FINAL. The MSCC does not include the User-Service-Units (USU).

The no form of this command reverts to the default behavior (disabled).

Default

no redirection-start-send-rr-final

6.2.1.18.67 retry-count

Syntax

retry-count *retry-count*

no retry-count

Context

config>mobile>profile>dcca

Description

This command configures the retry count for CCR messages. A CCR message will be retransmitted for the number of times specified, when a CCA is not received from the OCS server.

The no form of this command reverts to the default.

Parameters

retry-count

specifies the number of retransmit times

Values: 0 to 3

Default: 0

6.2.18.68 **rsu-total-octets**

Syntax

```
rsu-total-octets value  
no rsu-total-octets
```

Context

```
config>mobile>profile>dcca
```

Description

This command when enabled sets the value of the CC-Total-Octets AVP that CMG sends in a CCR message to OCS as part of the Requested-Service-Unit AVP.

The no form of this command reverts to the default.

Parameters

value

specifies the value of the CC-Total-Octets in bytes

Values: 0 to 4000000000

Default: 0 (disabled; an empty RSU is sent as per existing behavior)

6.2.18.69 **serialize-ccr**

Syntax

```
[no] serialize-ccr
```

Context

```
config>mobile>profile>dcca
```

Description

This command serializes the CCR messages sent to the OCS for a Gy session. When enabled, the gateway waits for a CCA response from the OCS for a CCR before sending the next CCR.

The no form of this command reverts to the default (disabled).

6.2.18.70 service-context-id

Syntax

```
service-context-id string  
no service-context-id
```

Context

```
config>mobile>profile>dcca
```

Description

This command is used to configure the service-context-id sent to the OCS. While 3GPP specifies a particular format for the service-context-id, some operators expect custom service-context-id values to be sent to the OCS.

The no form of this command reverts to the default.



NOTE:

This command requires a PDN shutdown.

Default

3gpp compliant service-context-id

Parameters

string

specifies the service context-id, up to 64 characters

6.2.18.71 service-id-level-credit-control

Syntax

```
[no] service-id-level-credit-control
```

Context

```
config>mobile>profile>dcca
```

Description

This command configures that quota allocation and usage reporting are based on the combination of rating group and service ID.

The no form of this command reverts to the default (quota allocation and usage reporting are based on the rating group).

Default

no service-id-level-credit-control

6.2.18.72 service-type

Syntax

```
[no] service-type {rtr | uc | pp | all}
```

Context

```
config>mobile>profile>dcca
```

Description

This command specifies a service type for Assume Positive behavior.

The no form of this command returns the setting to the default. When a service type parameter is specified for the no form of this command, that service type is removed and no service type is set.

Parameters

rtr

specifies the service type as real-time reporting

uc

specifies the service type as usage control

pp

specifies the service type as prepaid

all

specifies the service type as a generic wildcard (applies to all service types)

Default: all

6.2.1.18.73 ccfh-profile

Syntax

```
ccfh-profile profile-id  
no ccfh-profile profile-id
```

Context

```
config>mobile>profile>dcca>service-type
```

Description

This command is used to associate the specified CCFH profile to a service type. Service type 'all' applies when the same profile applies to all service types or as a generic wildcard when the service type is not applicable. The CCFH profile needs to be created under the config>mobile>profile>charging context before it can be associated with a service type.

The no form of this command removes the specified CCFH profile association with the service type. When no ccfh-profile is specified, the default profile applies.

Parameters

profile-id

specifies the profile ID for the CCFH profile

Values: 0 to 255

Default: 0

6.2.1.18.74 session-restore-retry-timer

Syntax

```
session-restore-retry-timer retry-timer  
no session-restore-retry-timer
```

Context

```
config>gateway>profile>dcca
```

Description

This command specifies a session restore retry timer. When fh-session-continue-timer and/or fh-volume-limit commands are active (CCFH = Continue), the gateway periodically sends a CCR-I to restore the Gy session with the OCS. If the OCS sends a successful CCA-I, the Gy session starts again with

the installed rating groups in first packet mode. Any usage and unused quotas from the previous session are discarded and not reported to the OCS. The `session-restore-timer` command must be smaller than the `fh-session-continue-timer` command, when the `fh-session-continue-timer` is used (non-zero). The CCR-I does not include a request for quota (for any configured default rating group). The installed rating groups operate in first packet mode, whereby the request for quota (CCR-U) is issued when the first packet for that rating group is received.

The no form of this command reverts to the default.

**NOTE:**

If the CCFH Continue is initiated due to an error result code, the session restore retry timer is activated only for the 3002, 3004, and 3005 result codes.

Parameters

retry-timer

specifies the retry time in minutes

Values: 0 to 60

Default: 0

6.2.1.18.75 supported-version

Syntax

`supported-version {3gpp | version2}`

`no supported-version`

Context

`config>gateway>profile>dcca`

Description

This command is used to specify the VF DCCA version.

The no form of this command reverts to the default.

**NOTE:**

This command requires a PDN shutdown.

Parameters

3gpp

specifies 3GPP as the version

version2

specifies version 2 as the version (VF-CLCI 2.0)

Default: 3gpp

6.2.1.18.76 suppress-multiple-ooc-notifications-per-rg

Syntax

```
[no] suppress-multiple-ooc-notifications-per-rg
```

Context

```
config>gateway>profile>dcca
```

Description

This command enables the mobile gateway to send Out Of Credit notifications to the PCRF only once per rating group.

The no form of this command reverts to the default. To enable this command, the rating-group-in-ch-rule-report command must also be enabled.



NOTE:

If t command is not enabled, the suppress-multiple-ooc-notifications-per-rg command is ignored.

Default

```
no suppress-multiple-ooc-notifications-per-rg
```

6.2.1.18.77 traffic-handling-options

Syntax

```
traffic-handling-options
```

Context

```
config>gateway>profile>dcca
```


Description

This command enables the context to configure traffic handling options for DCCA.

6.2.1.18.78 forced-re-authorization

Syntax

```
forced-re-authorization {allow | discard}
no forced-re-authorization
```

Context

```
config>gateway>profile>dcca>traffic-handling-options
```

Description

This command specifies the option to allow or discard the traffic for the service while waiting for a CCA response message from the OCS when a CCR message is sent to the OCS as a result of forced reauthorization by the OCS.



NOTE:

This command requires a PDN shutdown.

Parameters

allow

Traffic is allowed to flow through while waiting for a CCA from the OCS.

discard

Traffic is discarded while waiting for a CCA from the OCS.

Default: allow

6.2.1.18.79 quota-exhausted-no-threshold

Syntax

```
quota-exhausted-no-threshold {allow | discard}
no quota-exhausted-no-threshold
```

Context

```
config>gateway>profile>dcca>traffic-handling-options
```

Description

This command specifies the option to allow or discard the traffic for the service while waiting for a CCA response message from the OCS when no threshold is active and the quota gets exhausted.



NOTE:

This command requires a PDN shutdown.

Parameters

allow

Traffic is allowed to flow through while waiting for a CCA from the OCS.

discard

Traffic is discarded while waiting for a CCA from the OCS.

Default: allow

6.2.1.18.80 quota-exhausted-threshold-active

Syntax

```
quota-exhausted-threshold-active {allow | discard}
```

```
no quota-exhausted-threshold-active
```

Context

```
config>gateway>profile>dcca>traffic-handling-options
```

Description

This command specifies the option to allow or discard the traffic for the service while waiting for a CCA response message from the OCS when there is an active threshold, but the quota gets exhausted while waiting for a CCA upon hitting the threshold.



NOTE:

This command requires a PDN shutdown.

Parameters

allow

Traffic is allowed to flow through while waiting for a CCA from the OCS.

discard

Traffic is discarded while waiting for a CCA from the OCS.

Default: allow

6.2.1.18.81 quota-unavailable

Syntax

```
quota-unavailable {allow | discard | discard-on-fpt}
```

```
no quota-unavailable
```

Context

```
config>gateway>profile>dcca>traffic-handling-options
```

Description

This command specifies the option to allow or discard the traffic for the service while waiting for a CCA response message from the OCS when no quota is available for a rating group.



NOTE:

This command requires a PDN shutdown.

Parameters

allow

Traffic is allowed to flow through while waiting for a CCA from the OCS.

discard

Traffic is discarded while waiting for a CCA from the OCS.

discard-on-fpt

Traffic is discarded upon detection of a first packet trigger while waiting for a CCA from the OCS.

Default: allow

6.2.1.18.82 rating-condition-change

Syntax

```
rating-condition-change {allow | discard}  
no rating-condition-change
```

Context

```
config>gateway>profile>dcca>traffic-handling-options
```

Description

This command specifies the option to allow or discard the traffic for the service while waiting for a CCA response message from the OCS when a CCR message is sent to the OCS due to the Rating-Condition change being detected and waiting for a CCA response message from the OCS.



NOTE:

This command requires a PDN shutdown.

Parameters

allow

Traffic is allowed to flow through while waiting for a CCA from the OCS.

discard

Traffic is discarded while waiting for a CCA from the OCS.

Default: allow

6.2.1.18.83 validity-time-expired

Syntax

```
validity-time-expired {allow | discard}  
no validity-time-expired
```

Context

```
config>gateway>profile>dcca>traffic-handling-options
```

Description

This command specifies the option to allow or discard the traffic for the service while waiting for a CCA response message from the OCS when the validity timer expires.

**NOTE:**

This command requires a PDN shutdown.

Parameters

allow

Traffic is allowed to flow through while waiting for a CCA from the OCS.

discard

Traffic is discarded while waiting for a CCA from the OCS.

Default: allow

6.2.1.18.84 url-encryption

Syntax

```
url-encryption {aes-cbc-128} passphrase string
```

```
no url-encryption
```

Context

```
config>mobile>profile>dcca
```

Description

This command specifies the URL encryption type and passphrase to encrypt fields in the URL sent back to the UE for 302 redirect. This is used only when the URL is specified by the OCS.

The no form of this command reverts to the default.

**NOTE:**

The URL encryption configuration depends on the following HTTP traffic redirection scenarios:

- for Gy charging response FUI Action Redirect:
 - in non-CUPS scenarios, URL encryption is based on the url-encryption command in the config>mobile>profile>dcca context

– in CUPS scenarios, URL encryption is based on the `url-encryption` command (configured for the UP PDN) in the `config>mobile>pdn` context

- for Gy charging response with FUI Action Restrict and a filter ID PRB, URL encryption is based on the `http-redirect` and encryption options in the `config>mobile>profile>policy-options>action-rule-unit>gy-filter-id-action` context, in CUPS and non-CUPS scenarios
- for Gx policy-rule based redirection, URL encryption is based on the `http-redirect` and encryption options in the `config>mobile>profile>policy-options>policy-rule-unit>flow-gate-status` context, in CUPS and non-CUPS scenarios

Default

no url-encryption

Parameters

aes-cbc-128

specifies AES CBC 128-bit encryption

passphrase *string*

encryption passphrase string, up to 64 characters

6.2.18.85 usu-options-no-quota-initial

Syntax

```
usu-options-no-quota-initial [cc-total] [cc-input] [cc-output] [cc-time] [suppress]  
no usu-options-no-quota-initial
```

Context

`config>gateway>profile>dcca`

`config>gateway>profile>dcca>rating-group`

Description

This command specifies the USU options for categories that the OCS has not allocated any quota to start with. USU for such categories will include the configured units.

The no form of this command defaults to suppress (USU not included).

Parameters

cc-total

include cc-total-octets

cc-input

include cc-input-octets

cc-output

include cc-output-octets

cc-time

include cc-time

suppress

USU not included for categories that OCS has not allocated any quota to start

Default: suppress

6.2.1.18.86 usu-options-no-quota-mid-session

Syntax

usu-options-no-quota-mid-session [suppress | do-not-suppress]

no usu-options-no-quota-mid-session

Context

config>gateway>profile>dcca

config>gateway>profile>dcca>rating-group

Description

This command specifies whether USU options are included or not for categories that the OCS has allocated quota to but run out of credit mid-stream; in general, when no valid granted quota exists for an RG. USU for such categories will include the configured units.

The no form of this command defaults to suppress (USU not included).

Parameters

do-not-suppress

USU included (for quota type initially allocated by the OCS)

suppress

USU not included for categories that OCS has allocated quota to start with but runs out of credit mid-stream; in general, when no valid granted quota exists for an RG

Default: suppress

6.2.1.18.87 **usu-options-with-quota**

Syntax

```
usu-options-with-quota [none] [cc-total] [cc-input] [cc-output] [cc-time] [inherit]
[override-ocs]

no usu-options-with-quota
```

Context

```
config>gateway>profile>dcca
config>gateway>profile>dcca>rating-group
```

Description

This command specifies the USU options for categories that the OCS allocates quota. The superset based on what is locally configured and what is received from the OCS is applied. For example, if cc-input and cc-output are locally configured and we get only Granted Service Units (GSU) = cc-total from OCS, USU = cc-input + cc-output + cc-total. The `override-ocs` parameter can be used to override the OCS specified GSU units with the locally specified configuration. When this option is specified, the local configuration takes precedence over the OCS GSU units. For example, if cc-input and cc-output are specified locally along with the `override-ocs` option, and a GSU = cc-total-octets is received from the OCS, the USU is reported with cc-input-octets and cc-output-octets.

The no form of this command defaults to what is sent by the OCS – if GSU = cc-total, USU = cc-total octets.

Parameters

none

includes the default behavior of sending back what the OCS requests

cc-total

includes cc-total-octets

cc-input

includes cc-input-octets

cc-output

includes cc-output-octets

cc-time

includes cc-time

inherit

specifies inherit

override-ocs

specifies to override the OCS specified GSU units with the locally specified configuration.

Default: none

6.2.18.88 usu-reporting-enh

Syntax

```
usu-reporting-enh [discard-overage]
no usu-reporting-enh
```

Context

```
config>gateway>profile>dcca
```

Description

This command enables back-to-back reporting of USU with multiple CCR-U/CCA-U transactions, when the observed USU is greater than the Granted-Service-Units (GSU). When the discard-overage option is specified, any excess usage (USU greater than GSU) is not reported.

The no form of this command reverts to the default.

Default

```
no usu-reporting-enh
```

Parameters

discard-overage

specifies that the USU reported does not exceed the GSU; any overage (excess usage) is discarded

6.2.18.89 validation-checks

Syntax

```
validation-checks
```

Context

```
config>gateway>profile>dcca
```

Description

This command enables the context for commands that can be used to enable or disable validation checks.

6.2.1.18.90 ignore-invalid-thresholds

Syntax

```
[no] ignore-invalid-thresholds
```

Context

```
config>gateway>profile>dcca>validation-checks
```

Description

When this command is enabled, the gateway ignores an invalid threshold sent by the OCS (for example when the OCS sends a CCA with GSU less than the VQT). When this command is disabled, if the OCS sends an invalid threshold (deviating from the standard), the gateway terminates the session with a DIAMETER-BAD-ANSWER cause.

The no form of this command reverts to the default.

Default

no ignore-invalid-threshold

6.2.1.18.91 ignore-unsupported-triggers

Syntax

```
[no] ignore-unsupported-triggers
```

Context

```
config>gateway>profile>dcca>validation-checks
```

Description

When this command is enabled, the gateway ignores unsupported event triggers to which the OCS subscribes. When this command is disabled, if the OCS subscribes to an unsupported event trigger, the gateway terminates the session with a DIAMETER-BAD-ANSWER cause.

The no form of this command reverts to the default (disabled).

Default

no ignore-unsupported-triggers

6.2.1.18.92 validity-timer-enh

Syntax

```
[no] validity-timer-enh
```

Context

```
config>gateway>profile>dcca
```

Description

This command, when enabled, allows the operator to reduce the average amount of CCR-U messages. This is achieved by combining in the same CCR-U message all the MSCC instances which expire due to the validity time in less than two seconds. If not enabled, when an MSCC instance expires due to the validity time, it triggers a CCR-U message. When a second MSCC instance in the same session expires due to the validity time a few milliseconds later, it triggers a second CCR-U message.

The no form of this command reverts to the default (disabled).

Default

```
no validity-timer-enh
```

6.2.1.18.93 vol-threshold-reauth-enh

Syntax

```
[no] vol-threshold-reauth-enh
```

Context

```
config>gateway>profile>dcca
```

Description

This command when enabled facilitates the gateway sending a CCR-U to the OCS for volume threshold reauthorization by comparing the delta of the quota consumption between the last and recent polling. If the delta is equal to or higher than the remaining quota until the next threshold, a quota reauthorization is triggered.

The no form of this command reverts to the default (disabled).



NOTE:

This command applies to the implicit volume threshold (configured using the `implicit-volume-threshold` command) or to the Volume-Quota-Threshold attribute provided by the OCS.

6.2.18.94 wait-for-first-packet-on-fui-t-success

Syntax

```
[no] wait-for-first-packet-on-fui-t-success
```

Context

```
config>gateway>profile>dcca
```

Description

This command applies a certain behavior, when the OCS terminates an MSCC instance with FUI=Terminate and result code Success, for the traffic that is generated after the MSCC instance is terminated. If the command is enabled, the mobile gateway does not block traffic. Therefore, when the first packet which matches the MSCC instance, arrives, the mobile gateway triggers a CCR-U message to request for quota. If the command is disabled, the mobile gateway blocks traffic for the configured inactivity timer.

Default

```
no wait-for-first-packet-on-fui-t-success
```

6.2.18.95 wait-for-first-packet-on-no-gsu

Syntax

```
[no] wait-for-first-packet-on-no-gsu
```

Context

```
config>gateway>profile>dcca
```

Description

This command causes an MSCC instance to terminate if the OCS sends a CCA-U with an empty MSCC (result code success without a GSU). This ensures that the UE is not blocked, so when the first packet matching the same flow is detected, a new CCR-U is trigger to request quota for the MSCC. If the command is not enabled (default), the gateway behavior is based on 3GPP specifications that considers an empty MSCC (result code success without a GSU) as a bad answer.

**NOTE:**

If a final MSCC with USU needs to be sent when the MSCC is terminated, the `usu-options-no-quota-mid-session` command must be configured with the `do-not-suppress` keyword.

The `no` form of this command reverts to the default (disabled).

Default

`no wait-for-first-packet-on-no-gsu`

6.2.1.19 Dedicated bearer idle timeout profile command descriptions

6.2.1.19.1 dedicated-bearer-idle-timeout-profile

Syntax

`[no] dedicated-bearer-idle-timeout-profile profile-name`

Context

`config>mobile>profile`

Description

This command configures dedicated bearer idle-timeout profile.

Parameters

profile-name

specifies the name of the dedicated bearer idle timeout profile, up to 32 characters

6.2.1.19.2 qci

Syntax

`qci qci timeout seconds`

`no qci`

Context

`config>mobile>profile>dedicated-bearer-idle-timeout-profile`

Description

This command configures the length of time an established dedicated bearer is allowed to be idle before it is removed by the PGW. The idle-timeout is applied to the established dedicated bearer session. If there is data traffic on the dedicated bearer, `idle-dedicated-bearer-timeout` command timer will be reset. If there is no traffic activity and timer expires, the dedicated bearer will be removed.

The no form of this command removes the QCI value.

Parameters

qci

specifies the QCI value for which the idle-time is specified

If the * (wildcard) is specified, all dedicated bearers not identified with any specific QCI value will get this idle-timeout time.

Values: 1 to 255, *-wildcard

seconds

specifies the amount of time, in seconds, a dedicated bearer must be idle before it is released

Values: 1 to 86400

6.2.1.20 DHCP server group profile command descriptions

6.2.1.20.1 dhcp-server-group

Syntax

[no] `dhcp-server-group server-group-name`

Context

`config>gateway>profile`

Description

This command creates a new DHCP server group. The DHCP server group may contain a set of DHCPv4 or DHCPv6 servers, but the DHCPv4 and DHCPv6 servers cannot be mixed within a group.

Parameters

server-group-name

specifies the name of this DHCP server group, up to 32 characters

6.2.1.20.2 client-identifier-msisdn-encoded

Syntax

```
client-identifier-msisdn-encoded [with-apn]
```

```
no client-identifier-msisdn-encoded
```

Context

```
config>mobile>profile>dhcp-server-group
```

Description

This command enables the inclusion of the Client-Identifier (61) option in DHCPv4 and the OPTION_CLIENTID (1) in DHCPv6. For DHCPv4, the Client-Identifier (61) is included in the DHCPv4 discovery, requests, and release messages sent to the DHCPv4 server. For DHCPv6, the OPTION_CLIENTID (1) is included in the DHCPv6 solicit, renew, rebind, and release messages sent to the DHCPv6 server. The Client-Identifier content is configurable to include the MSISDN encoded in hexadecimal, the MSISDN encoded in hexadecimal:APN, or the IMSI:APN (default). When the MSISDN is included in the Client-Identifier (61) option, it is encoded in hexadecimal as follows:

1. the first byte added is always set to 00
2. every following byte has the MSISDN number in hexadecimal representation of ASCII characters for digits, according to *MSISDN hexadecimal encoding*

The *MSISDN hexadecimal encoding* lists the MSISDN hexadecimal encoding.

Table 99: MSISDN hexadecimal encoding

ASCII	Hexadecimal	Symbol/Digit
48	30	0
49	31	1
50	32	2
51	33	3
52	34	4

Table 99: MSISDN hexadecimal encoding (continued)

ASCII	Hexadecimal	Symbol/Digit
53	35	5
54	36	6
55	37	7
56	38	8
57	39	9

Example

A GTP-C CSReq with an MSISDN value of 44 12 345 678 (length is 10), is encoded in the DHCP client-identifier (61) IE with the payload as a hexadecimal representation of ASCII characters for digits to 00 34 34 31 32 33 34 35 36 37 38 (length = 11). The no form of this command reverts to the default (includes the client identifier (61) option with IMSI:APN).



NOTE:

This command is only applicable to DHCPv4.

Default

no client-identifier-msisdn-encoded

Parameters

with-apn

specifies to include the DHCP Discovery client-identifier (61) option with the APN encoded

6.2.1.20.3 **deadtime**

Syntax

deadtime *seconds*

no deadtime

Context

```
config>mobile>profile>dhcp-server-group
```

Description

When the system fails to reach a particular server after a configurable number of retries, the server is marked dead and another server is selected. This command sets the time that a server continues to be considered dead before the system attempts to use it for IP address allocation. Once the dead timer expires, the system may once again send DHCP messages. The server is subject to the retry-timeout and retry-count mechanism to determine if the server is still in the failed state. In case all DHCP servers are unreachable, the system will continuously attempt to send messages to the highest priority server, even if it is during the dead time period.

Parameters

seconds

specifies the time interval that a server is considered dead, in seconds

Values: 300 to 3600

Default: 600

6.2.1.20.4 interface

Syntax

```
interface [router -instance router-instance] interface-name
```

```
no interface
```

Context

```
config>mobile>profile>dhcp-server-group
```

Description

This command configures the interface that will be used to send and receive DHCP messages. Therefore, this command indirectly determines the source IP address. In case this interface belongs to a VPRN other than the master instance, the command must include the VPRN name. The IP address for this interface is used as the gi-addr for DHCPv4.

Parameters

router-instance

specifies the virtual router name or service ID

Values	router-name service-id	
–	router-name	Base
–	service-id	1 to 2147483647

Default: Base

interface-name

specifies an interface, up to 32 characters

6.2.1.20.5 peer

Syntax

[no] peer *ipv4* | *ipv6*

Context

config>mobile>profile>dhcp-server-group

Description

This command identifies a DHCP server and enters the server context to configure parameters specific to the identified server. This command may configure a single server by its IPv4 or IPv6 address. All servers (peers) belonging to a DHCP server group must use the same IP version and the IP version must match that of the source interface. Up to 15 peers may be configured per group.

Parameters

ipv4* | *ipv6

specifies the peer’s address, either an IPv4 address or an IPv6 address

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d

–	–	x - [0..FFFF]H
–	–	d - [0..255]D

6.2.1.20.6 priority

Syntax

priority *value*

no priority

Context

config>mobile>profile>dhcp-server-group>peer

Description

This command configures the priority value for each DHCP peer address. The value range is from 1 to 15 with 1 considered the highest priority. The system will attempt to use the highest priority server available at the time. The system will automatically use round-robin load-balancing among servers configured with the same priority. The priority of all DHCP servers must be either all the same or all different.

Parameters

value

specifies the priority value

Values: 1 to 15

Default: 1

6.2.1.20.7 vso-server-pool-name

Syntax

vso-server-pool-name *server-pool-name*

no vso-server-pool-name

Context

config>mobile>profile>dhcp-server-group>peer

Description

This command configures the DHCPv6 server pool name. The command is available in this context only when the DHCP server group peer is configured with an IPv6 address.

Default

no vso-server-pool-name

Parameters

server-pool-name

specifies the server-pool name for IPv6 peers, up to 32 characters

6.2.1.20.8 rebind-time-percent

Syntax

rebind-time-percent *value*

no rebind-time-percent

Context

config>mobile>profile>dhcp-server-group

Description

This command configures the rebind time (T2) as a percentage of the lease time.

Parameters

value

specifies the rebind time as a percentage of the lease time

Values: 67 to 90

Default: 88

6.2.1.20.9 recommend-lease-time

Syntax

recommend-lease-time *seconds*

no recommend-lease-time

Context

config>mobile>profile>dhcp-server-group

Description

This command configures the lease time that the client will recommend to the server.

Parameters

seconds

specifies the rebind time, in seconds

Values: 0 | 3600 to 86400

Default: 0

6.2.1.20.10 renew-time-percentage

Syntax

renew-time-percentage *value*

no renew-time-percentage

Context

config>mobile>profile>dhcp-server-group

Description

This command configures the renew time (T1) as a percentage of the lease time.

Parameters

value

specifies the renew time, as a percentage of the lease time

Values: 40 to 66

Default: 50

6.2.1.20.11 retry-count

Syntax

retry-count *number*

```
no retry-count
```

Context

```
config>mobile>profile>dhcp-server-group
```

Description

This command configures the number of times the system will attempt to send a DHCP message to the server.

Parameters

number

specifies the number of attempts as an integer

Values: 0 to 3

Default: 1

6.2.1.20.12 retry-timeout

Syntax

```
retry-timeout seconds  
no retry-timeout
```

Context

```
config>mobile>profile>dhcp-server-group
```

Description

This command configures the interval between retries.

Parameters

seconds

specifies the interval between retries, in seconds

Values: 2 to 15

Default: 2

6.2.1.20.13 via-up-function

Syntax

```
via-up-function apn apn-name [up-peer-ip-address ip-address | ipv6-address]  
no via-up-function
```

Context

```
config>mobile>profile>dhcp-server-group
```

Description

For CUPS deployments, this command specifies the server group DHCP traffic to be tunnelled through PFCP-u and routed via a UP function to the DHCP server. The specific UP peer function can be optionally configured with the `up-peer-ip-address` keyword. If the `up-peer-ip-address` keyword is not configured, the UP peer function is selected from the UP peer list (configured using the `config>mobile>profile>pfcu>up-peer-list` command) that this APN is configured with. If there is more than one UP peer functions configured with this APN, the first matched peer in the configured UP peer list with this APN is selected as the UP peer function to route all DHCP traffic for this server group.



NOTE:

- When the `up-peer-ip-address` is specified, this peer must also be configured using the `config>mobile>profile>pfcu>up-peer-list` command.
- The APN specified in this command must also be configured under a peer using the `config>mobile>profile>pfcu>up-peer-list` and the `config>mobile>pdn>apn` commands. If the APN is not configured, the default PFCP-u tunnel between the CP and UP functions is not established and the DHCP traffic for this server group profile fails.
- If the server group profile is configured to use a reserved IP address for the DHCP client VPRN/loopback address from one of the UP peers configured using the `config>mobile>profile>pfcu>up-peer-list>peer>apn` and the `config>mobile>profile>pfcu>up-peer-list>peer>network-realm>ip-pool>reserved-ip-list` commands, the `up-peer-ip-address` keyword must be specified with the same UP peer IP address or not be configured. In case the `up-peer-ip-address` keyword is not configured, the system finds the peer configured in the `up-peer-list` by matching both the APN and the reserved IP address configured using the `reserved-ip-list` command.

The `no` form of this command removes this configuration.

Default

```
no via-up-function
```

Parameters

apn-name

specifies the APN to associate with the PFCP-u tunnel on the UP function (it can be the same as the UE APN or a separate APN configured for this DHCP server group when shared with other UE APNs), up to 80 characters

ip-address | *ipv6-address*

specifies the IP address of the UP function

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

6.2.1.21 DHCPv4 server profile command descriptions

6.2.1.21.1 dhcpv4-server

Syntax

[no] dhcpv4-server *profile-name*

Context

config>mobile>profile

Description

This command creates a new DHCPv4 server profile which is used for Trusted WLAN access over the SWw interface.

Parameters

profile-name

specifies the name of this profile

6.2.1.21.2 hold-time

Syntax

hold-time mins *minutes* secs *seconds*
no hold-time

Context

config>mobile>profile>dhcpv4-server

Description

This command configures the hold time for which the client must respond to the DHCP offer. Thus, if the server does not get a response from the client within this time, the session is torn down and the offered IP address can be re-assigned.

Parameters

minutes

specifies the hold-time, in minutes
Values: 0 to 59
Default: 1

seconds

specifies the hold-time, in seconds
Values: 0 to 59
Default: 0

6.2.1.21.3 ignore-rapid-commit

Syntax

[no] ignore-rapid-commit

Context

config>mobile>profile>dhcpv4-server

Description

This command configures to enable or disable responding to the rapid commit option (per RFC 4039) if received by the client.

Default

no ignore-rapid-commit

6.2.1.21.4 lease-time

Syntax

lease-time hrs *hours* mins *minutes* secs *seconds*

no lease-time

Context

config>mobile>profile>dhcpv4-server

Description

This command configures the lease time which the local DHCPv4 server will use in the DHCPv4 process of trusted WLAN access over the SWw interface.

Parameters

hours

specifies the number of initial lease time, in hours

Values: 0 to 8759

Default: 24

minutes

specifies the number of initial lease time, in minutes

Values: 0 to 59

Default: 0

seconds

specifies the number of initial lease time, in seconds

Values: 0 to 59

Default: 0

6.2.1.22 Diameter profile command descriptions

6.2.1.22.1 diameter

Syntax

[no] diameter *profile-name*

Context

config>mobile>profile

Description

This command configures a Diameter profile to specify connection-related parameters such as timers, and DSCP. A Diameter profile is referenced through the Diameter peer profiles. Multiple Diameter peer profiles can reference the same Diameter profile. The Diameter local configuration is provisioned first before the Diameter profiles.

The no form of this command removes the profile name from the configuration. A Diameter profile cannot be deleted if it is referenced by Diameter applications.

Parameters

profile-name

configures a Diameter profile name, up to 32 characters

6.2.1.22.2 connection-timer

Syntax

connection-timer *value*

no connection-timer

Context

config>mobile>profile>diameter

Description

This command sets maximum amount of time the node attempts to reconnect to a Diameter peer after a connection to the peer has been brought down due to a transport failure. There are certain exceptions to this rule, such as peer which terminated the transport connection indicating that it does not wish to

communicate. The value of 0 means that the connection will not be retried. The configuration at peer level overrules the value configured at Diameter-base level for the given peer.

Parameters

value

specifies the connection timer (TC), in seconds, used by this Diameter policy. It defines the time the systems waits before attempting to reconnect to a peer after the connection was lost.

Values: 0 to 180

Default: 30

The default value at peer level is taken from the Diameter-base.

6.2.1.22.3 dns-refresh-interval

Syntax

```
dns-refresh-interval value
```

```
no dns-refresh-interval
```

Context

```
config>mobile>profile>diameter
```

Description

This command sets the amount of time the node waits before sending a DNS query to refresh the FQDN resolution to a list of IP addresses. The TTL received in a DNS response is not used at the gateway. The value of 0 means that the DNS refresh timer is disabled.

The no form of this command reverts to the default.

Parameters

value

the amount of time in hours that the node waits before sending a DNS query

Values: 0 to 24

Default: 6

6.2.1.22.4 dpr-timeout

Syntax

```
dpr-timeout value  
no dpr-timeout
```

Context

```
config>mobile>profile>diameter
```

Description

This command configures a timer used to re-start a Diameter connection setup when the remote side intentionally disconnects by sending a DPR. Reception of a DPR indicates that the receiver must not periodically attempt to setup a Diameter connection. This timer will provide a provision that will force the Diameter client to retry the connection setup. When a DPR is received, the first connection attempt (CER) is performed on the interval specified for the `dpr - timeout` command. If a TCP connection is established but no CEA is received, upon expiration of the transaction timer a new TCP/CER connection is attempted after the length of time equal to the configured DPR timeout.

Parameters

value

specifies the DPR timeout value, in seconds

Values: 1 to 3600

Default: 1800

6.2.1.22.5 fail-num-retries

Syntax

```
fail-num-retries value  
no fail-num-retries
```

Context

```
config>mobile>profile>diameter
```

Description

This command configures the number of re-attempts made to establish a transport connection with a Diameter server. If the command is not configured, re-establishing the Diameter connection is attempted

every Tc seconds (30s by default) as per the Diameter Base Protocol RFC. When the configured value is reached, the fail-retry mechanism restart after Tc seconds.

The no form of this command reverts to the default (disabled).

Default

no fail-num-retries

Parameters

value

specifies the number of attempts to re-establish the connection with the Diameter server

Values: 0 to 50

Default: 0

6.2.1.22.6 fail-retry-interval

Syntax

fail-retry-interval *value*

no fail-retry-interval

Context

config>mobile>profile>diameter

Description

This command configures the time after which the mobile gateway attempts to re-establish a connection with a Diameter server, when the fail-num-retries command is enabled. For more information about the fail-retry functionality, refer to the *Fail-Retry functionality for TCP/SCTP transport* section, in the *CMG Configuration Guide*.

The no form of this command reverts to the default (disabled).

Default

no fail-retry-interval

Parameters

value

specifies the time, in tenths of a second (0.1 seconds), after which the mobile gateway re-attempts to establish connection with a Diameter server

Values: 1 to 50

Default: 5

6.2.1.22.7 ip-dscp

Syntax

`ip-dscp value`

`no ip-dscp`

Context

`config>mobile>profile>diameter`

Description

This command configures the DSCP value in the IP header for Diameter signaling messages sent, so it is treated as network control packets ahead of Expedited Forwarding (EF) packets.

Default

`no ip-dscp`

Parameters

value

specifies the forwarding class for Diameter signaling packets sent

Values: 0 to 63

Default: 56

6.2.1.22.8 ip-ttl

Syntax

`ip-ttl value`

`no ip-ttl`

Context

`config>mobile>profile>diameter`

Description

This command configures the IP TTL value for Diameter signaling messages. Restricting the number of hops between signaling end-points is from a security point of view.

Parameters

value

specifies the IP TTL value for Diameter signaling packets

Values: 1 to 255

Default: 255

6.2.1.22.9 perm-fail-retry-timeout

Syntax

```
perm-fail-retry-timeout value
```

```
no perm-fail-retry-timeout
```

Context

```
config>mobile>profile>diameter
```

Description

This command specifies the amount of time the node waits before retrying Diameter connection setup to start the connection dynamically without the operator's intervention in case of permanent failures from the remote peer. The value of 0 means that no attempt will be made to re-establish the connection.

Parameters

value

the amount of time, in minutes, the node waits before retrying a Diameter connection setup

Values: 0 to 1440

Default: 30

6.2.1.22.10 transaction-timer

Syntax

```
transaction-timer value
```

```
no transaction-timer
```


Context

config>mobile>profile>diameter

Description

This command configures the maximum amount of time the node waits for a Diameter peer to respond before trying another peer. The configuration at peer level overrules the value configured at Diameter-base level for the given peer. This timer must only be for peer-level commands such as DWR/DWA; it does not affect application-level commands such as ACR/ACA for Rf and CCR/CCA for Gx. The configuration at the peer level does not overrule the value configured at the Diameter-base level since the timers are for different messages.

The no form of this command removes the configuration.

Parameters

value

specifies the maximum amount of time, in seconds, the node waits for a peer to respond before trying another peer

Values: 1 to 180 (increments in 1 second)

Default: 5

6.2.1.22.11 watchdog-timer

Syntax

watchdog-timer *value*

no watchdog-timer

Context

config>mobile>profile>diameter

Description

This command configures the maximum amount of time the node waits for a Diameter peer to respond to a watchdog packet. When the watchdog timer expires, a DWR is sent to the Diameter peer and the watchdog timer is reset. If a DWA is not received before the next expiration of the watchdog timer, transport failure to the Diameter peer has occurred. The configuration at the peer level overrides the value configured at the Diameter-base level for the given peer.

Parameters

value

specifies the maximum amount of time in seconds, that the node waits for a Diameter peer to respond to a watchdog packet

**NOTE:**

If the time configured is greater than 3 seconds, the configured value is randomly adjusted up or down by up to 2 seconds whenever the watchdog timer is used.

Values: 0 to 180

Default: 30

6.2.1.23 Diameter error mapping profile command descriptions

6.2.1.23.1 diameter-error-mapping-profile

Syntax

```
diameter-error-mapping-profile profile-name
```

```
no diameter-error-mapping-profile
```

Context

```
config>mobile>profile
```

Description

This command creates a Diameter error mapping profile.

The no form of this command removes the specified profile.

Parameters

profile-name

specifies the profile name, up to 32 characters

6.2.1.23.2 fh-tx-timeout

Syntax

```
fh-tx-timeout gtp-error-code gtp-value
```

```
no fh-tx-timeout
```

Context

config>mobile>profile>diameter-error-mapping-profile

Description

This command configures the GTP error code that is sent back to the access node when no response is received from the peer node.

The no form of this command removes the specified error code.

Parameters

gtp-value

specifies the cause value sent back to the access node

Values: 0 to 15 | 64 to 253 | 255 (non-success GTP-C causes)

6.2.1.23.3 result-code

Syntax

result-code *result-code-value* gtp-error-code *gtp-value*

no result-code *result-code-value*

Context

config>mobile>profile>diameter-error-mapping-profile

Description

This command creates a mapping between a Diameter error code (for example, an error code received from the OCS) and the GTP error code sent back to the access node.

The no form of this command removes the previously specified mapping for the result code.

Parameters

result-code-value

specifies the result code value from the Diameter peer, for example OCS

Values: 3000 to 5999

gtp-value

specifies the cause value sent back to the access node

Values: 0 to 15 | 64 to 253 | 255 (non-success GTP-C causes)

6.2.1.23.4 result-code-3xxx

Syntax

```
result-code-3xxx gtp-error-code gtp-value  
no result-code-3xxx
```

Context

```
config>mobile>profile>diameter-error-mapping-profile
```

Description

This command creates a mapping between a generic 3xxx Diameter error code (for example, an error code received from the OCS) and the GTP error code sent back to the access node.



NOTE:

Specific result code mapping takes precedence over generic 3xxx mapping.

The no form of this command removes the previously specified mapping for the generic 3xxx result code.

Parameters

gtp-value

specifies the cause value sent back to the access node

Values: 0 to 15 | 64 to 253 | 255 (non-success GTP-C causes)

6.2.1.23.5 result-code-4xxx

Syntax

```
result-code-4xxx gtp-error-code gtp-value  
no result-code-4xxx
```

Context

```
config>mobile>profile>diameter-error-mapping-profile
```

Description

This command creates a mapping between a generic 4xxx Diameter error code (for example, an error code received from the OCS) and the GTP error code sent back to the access node.

**NOTE:**

Specific result code mapping takes precedence over generic 4xxx mapping.

The no form of this command removes the previously specified mapping for the generic 4xxx result code.

Parameters

gtp-value

specifies the cause value sent back to the access node

Values: 0 to 15 | 64 to 253 | 255 (non-success GTP-C causes)

6.2.1.23.6 result-code-5xxx

Syntax

```
result-code-5xxx gtp-error-code gtp-value
```

```
no result-code-5xxx
```

Context

```
config>mobile>profile>diameter-error-mapping-profile
```

Description

This command creates a mapping between a generic 5xxx Diameter error code (for example, an error code received from the OCS) and the GTP error code sent back to the access node.

**NOTE:**

Specific result code mapping takes precedence over generic 5xxx mapping.

The no form of this command removes the previously specified mapping for the generic 5xxx result code.

Parameters

gtp-value

specifies the cause value sent back to the access node

Values: 0 to 15 | 64 to 253 | 255 (non-success GTP-C causes)

6.2.1.24 Diameter peer profile command descriptions

6.2.1.24.1 diameter-peer

Syntax

[no] diameter-peer *profile-name*

Context

config>mobile>profile

Description

This command specifies a Diameter peer profile for peer-related information for Diameter applications. The Diameter peer profile cannot be deleted when it is still referenced by Diameter applications.

The no form of this command removes the profile from the configuration.

Parameters

profile-name

specifies the name of the peer within a Diameter peer profile, up to 32 characters

6.2.1.24.2 application-type

Syntax

application-type {gx | rf | gy | s6b | swm | sd | sta}

no application-type

Context

config>mobile>profile>diameter-peer

Description

This command specifies the type of application where this profile is used.



NOTE:

This command requires that the diameter-peer is not referenced and that the peers that are configured for this diameter-peer are shutdown.

Parameters

gx	specifies the gx application
rf	specifies the rf application
gy	specifies the gy application
s6b	specifies the s6b application
swm	specifies the swm application
sd	specifies the sd application
sta	specifies the sta application

6.2.1.24.3 custom-vendor-id

Syntax

```
custom-vendor-id vendor-id  
no custom-vendor-id
```

Context

```
config>mobile>profile>diameter-peer
```

Description

This command configures a supported custom vendor ID that is included during capabilities exchange negotiation with a Diameter peer.

The no form of this command reverts to the default (custom vendor ID is not included).

Default

```
no custom-vendor-id
```

Parameters

vendor-id

specifies the Diameter-supported vendor ID

Values: 1 to 4294967295

6.2.1.24.4 dest-host-in-initial-msg

Syntax

```
[no] dest-host-in-initial-msg
```

Context

```
config>mobile>profile>diameter-peer
```

Description

This command enables sending the destination-host AVP in an initial message (for example, CCR-Initial) from the gateway. Normally the destination-host is not included in an initial message, because any host within a server could serve this session. The origin-host in the response message is used as the destination host to ensure that a subsequent message for the session terminates on the same host that serves the session. When this command is enabled, the destination-host in the initial message is populated with the value configured for the first supported host configured under the `config>mobile>profile>diameter-peer>peer>supported-host>` context.

The no form of this command reverts to the default.

Default

```
no dest-host-in-initial-msg
```

6.2.1.24.5 dest-host-table-threshold

Syntax

```
dest-host-table-threshold threshold-percent
```

```
no dest-host-table-threshold
```

Context

```
config>mobile>profile>diameter-peer
```

Description

This command configures the threshold above which this alarm is triggered to indicate that the destination host table exceeded a specific number of hosts.

The no form of this command reverts to the default.

Parameters

threshold-percent

specifies the threshold level for the alarm trigger, in percentage

Values: 1 to 100

Default: 80

6.2.1.24.6 destination-realm

Syntax

destination-realm *name*

no destination-realm

Context

config>mobile>profile>diameter-peer

Description

This command configures the destination domain (realm) to be included in messages exchanged with a Diameter peer. For the ePDG, when the value of the destination-realm (name parameter) is left empty (null), the ePDG will construct the Destination-Realm AVP value based on the realm value of the User-Name AVP. When the value of destination-realm is configured with a specific value, that value is used for the Destination-Realm AVP toward the AAA.

The no form of this command removes the destination realm configuration from the diameter-peer.



NOTE:

This command requires a peer shutdown.

Default

no destination-realm

Parameters

name

specifies the destination realm name, up to 80 characters

6.2.1.24.7 diameter-profile

Syntax

```
diameter-profile profile-name  
no diameter-profile
```

Context

```
config>mobile>profile>diameter-peer
```

Description

This command configures peer-related information to the Diameter protocol-specific timer values. A Diameter profile cannot be deleted when it is referenced by Diameter applications.

Parameters

profile-name

specifies a profile name, up to 32 characters

6.2.1.24.8 disable-cea-auth-app-id-check

Syntax

```
[no] disable-cea-auth-app-id-check
```

Context

```
config>mobile>profile>diameter-peer
```

Description

This command disables the validation check of the Auth-Application-Id received in a CEA from the peer. This applies to Gy online charging only.

The no form of this command reverts to the default (Auth-Application-Id validation is performed).

Default

```
no disable-cea-auth-app-id-check
```

6.2.1.24.9 disable-dwr-dwa-origin-host-length-check

Syntax

```
[no] disable-dwr-dwa-origin-host-length-check
```

Context

```
config>mobile>profile>diameter-peer
```

Description

This command disables the checking for an invalid AVP length of the origin host and the origin realm in Diameter DWR/DWA messages. The length of the origin host in the Diameter CEA message is assumed to be correct.

The no form of this command reverts to the default.

Default

```
no disable-dwr-dwa-origin-host-length-check
```

6.2.1.24.10 disable-learnt-dest-realm

Syntax

```
[no] disable-learnt-dest-realm
```

Context

```
config>mobile>profile>diameter-peer
```

Description

This command disables sending the learned realm (from the origin realm received in the latest CCA, RAR, or ASR message) of a Diameter peer within the destination-realm AVP in subsequent messages to a peer. The configured realm of the peer is used to populate the destination realm.



NOTE:

This command must not be enabled if the `last-known-dest-realm-after-switchover` command needs to be enabled. An on-the-fly change to this command results in the next transaction picking up the updated behavior. Enabling the `disable-learnt-dest-realm` command disables dynamic learning of destination realms from then on. For existing sessions that have already learned a destination realm, the realm that was last learned is used from then on.

The no form of this command reverts to the default (the learned destination realm of the peer is sent in subsequent messages).

Default

no disable-learned-dest-realm

6.2.1.24.11 exclude-dest-host-after-switchover

Syntax

[no] exclude-dest-host-after-switchover

Context

config>mobile>profile>diameter-peer

Description

This command configures the population of the destination host in the first message sent to an alternate server after a switchover to a secondary server after the primary failure. This applies to the Gx and Gy Diameter interfaces. If this command is not enabled, the destination host is populated by default.

The no form of this command reverts to the default.

Default

no exclude-dest-host-after-switchover

6.2.1.24.12 interface

Syntax

interface [router *router-instance*] *interface-name*

no interface

Context

config>mobile>profile>diameter-peer

Description

This command configures Diameter interface parameters and the virtual router in which the Diameter connections are established.

The no form of this command removes the parameter from the configuration.



NOTE:

This command requires a peer shutdown.

Parameters

router-instance

specifies the virtual router name or service ID

Values	router-name service-id	
–	router-name	Base
–	service-id	1 to 2147483647

Default: Base

interface-name

specifies an interface name, up to 32 characters; must start with a letter

6.2.1.24.13 last-known-dest-realm-after-switchover

Syntax

[no] last-known-dest-realm-after-switchover

Context

config>mobile>profile>diameter-peer

Description

This command enables sending the last known realm (from the origin realm received in the latest CCA, RAR, or ASR message) of a Diameter peer within the destination realm AVP after a switchover to an alternate peer.



NOTE:

A pre-condition for this command to be enabled is not to disable learning the destination realm; that is, do not enable the `disable-learned-destination-realm` command. The CLI cross check applies to the same peer profile.

An on-the-fly change to this command results in the next transaction picking up the updated behavior.

The no form of this command reverts to the default (the configured destination realm of the peer is sent in a CCR message after switchover).

Default

no last-known-destination-realm-after-switchover

6.2.1.24.14 load-balance

Syntax

[no] load-balance

Context

config>mobile>profile>diameter-peer

Description

This command enables load balancing. The Diameter peer profiles for Rf Diameter applications can have up to 20 peer addresses. If the peer addresses are FQDN, the DNS a-record resolution obtains the IP addresses and the first 20 IP addresses are used for peer connection. If the load balance option is enabled, all peers are used in round-robin fashion, otherwise only the first peer is used for Diameter sessions.



NOTE:

For Gx and Gy Diameter applications, if load balancing is enabled and multiple peers are configured under the Diameter peer profiles, up to two failovers can take place. In case of a failover, the active peers of the primary profile are first selected.

Default

no load-balance

6.2.1.24.15 multi-homing

Syntax

```
[no] multi-homing
```

Context

```
config>mobile>profile>diameter-peer
```

Description

This command allows multi-homing from a remote peer.

The no form of this command reverts to the default.

**NOTE:**

- Currently, multi-homing support requires primary and secondary interfaces to be in the same VPRN.
- This command requires a PDN shutdown.

Default

```
no multi-homing
```

6.2.1.24.16 interface

Syntax

```
interface [router router-instance] interface-name
```

```
no interface
```

Context

```
config>mobile>profile>diameter-peer>multi-homing
```

Description

This command configures Diameter secondary interface parameters and the virtual router in which the Diameter connections are established.

The no form of this command removes the parameter from the configuration.



NOTE:

This command requires a peer shutdown.

Parameters

router-instance

specifies the virtual router name or service ID

Values	router-name service-id	
–	router-name	Base
–	service-id	1 to 2147483647

Default: Base

interface-name

specifies an interface name, up to 32 characters; must start with a letter

6.2.1.24.17 origin-realm

Syntax

origin-realm *realm*
no origin-realm

Context

config>mobile>profile>diameter-peer

Description

This command specifies the originating realm or domain name of a Diameter node. The originating realm is included in the messages exchanged with a Diameter peer.

The no form of this command removes the name from the configuration.

By default (not configured), the origin-realm is selected according to the origin-realm configuration at the APN or PDN level. The origin-realm from the APN level is used for application messages. For Diameter base messages, if this value is not configured, the PDN level is used.

Parameters

realm

specifies the originating host name, up to 80 characters

6.2.1.24.18 peer

Syntax

```
[no] peer ip-addr | fqdn [port port]
```

Context

```
config>mobile>profile>diameter-peer
```

Description

This command configures a Diameter peer. The peer address can be an IP address or FQDN. If an FQDN is specified, the IP address for the peer is obtained by a DNS a-record query. The Diameter per profile for Gx can have only one peer address provisioned. If the peer address is an FQDN, the first IP address from DNS resolution is used. This command applies to all Diameter interfaces on the gateway (Rf, Gx, Gy, and S6b). However, FQDN resolution to multiple IP addresses and load balancing are supported for the Rf interface only. An Rf Diameter peer within a Diameter profile may be modified without a gateway shutdown. In the case of Rf, for existing sessions, a change in Diameter peer likely results in an incomplete record generated at the leCCF. While the modification is being done on the primary Diameter peer, new sessions are sent to the secondary server (if configured). If a secondary server is not configured, new records are written to flash as per the Rf outage configuration settings.



NOTE:

This command requires a peer shutdown.

Parameters

ip-address

specifies an IPv4 or IPv6 address

fqdn

specifies FQDN, up to 255 characters

port

specifies a port value

Values: 1 to 65535

6.2.1.24.19 enable-dra

Syntax

[no] enable-dra

Context

config>mobile>profile>diameter-peer>peer

Description

This command enhances the support for specific DRA-based network environments. It eliminates the need for configuring supported hosts, to support Diameter Redirect call flows in cases where the same host exists under multiple Diameter peers or when Redirect needs to take place on Redirect-Host AVP sent by the DRA must be in URI format hosts under the same Diameter Peer.

- Redirect-Host AVP sent by the DRA must be in URI format
- The received URI must be configured as supported host under a Diameter peer

The supported hosts that are already configured under the config>mobile>profile>diameter-peer>peer>supported-host context are ignored.



NOTE:

The configuration for this command applies after the peer restart.

Default

no enable-dra

6.2.1.24.20 sctp-alternate-address

Syntax

sctp-alternate-address *ip-address*

no sctp-alternate-address

Context

config>mobile>profile>diameter-peer>peer

Description

This command configures an alternate SCTP address for the Diameter peer in case of SCTP multi-homing. When the connection to the Diameter peer is unsuccessful to the main address, or the Diameter peer main address does not reply in Init-Ack with an alternate multi-homing address, then the locally configured SCTP alternate address is used. If there is a mismatch between the Diameter peer main address and the local configuration, the alternate address of the Init-Ack is used and the local configuration is ignored.



NOTE:

This command requires a peer shutdown.

Default

no sctp-alternate-address

Parameters

ip-address

specifies the alternate IPv4 or IPv6 address of the SCTP

Values	ipv4 address	a.b.c.d
–	ipv6 address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

6.2.1.24.21 supported-host

Syntax

supported-host *name host-index*

no supported-host *host-index*

Context

```
config>mobile>profile>diameter-peer>peer
```

Description

This command configures a supported host list for a Diameter peer. When there is no DRA deployed, it includes a single host. However, when a DRA is deployed in the network, the list can support a maximum of up to 16 hosts. With PCRF specified OCS or OFCS addresses, the addresses are validated against the supported hosts configured with this command.



NOTE:

This command requires a peer shutdown.

Parameters

name

specifies the supported hosts that are associated with the Diameter peer, up to 255 characters

host-index

specifies an integer
Values: 1 to 16

6.2.1.24.22 suspend

Syntax

```
[no] suspend
```

Context

```
config>mobile>profile>diameter-peer>peer
```

Description

This command configures that no new sessions are allowed to be established while the existing sessions continue. The purpose is to smoothly offload a diameter peer from its sessions, for example, maintenance reasons or planning. If DRAs (relays/proxies) are deployed between the mobile gateway and final diameter peer, the suspend operation on the mobile gateway cannot avoid or affect the final diameter peer selection for new session establishments. The deployed DRAs perform the diameter peer selection for new sessions and must be configured accordingly.

Default

no suspend

6.2.1.24.23 reconnect-with-new-tcp-port

Syntax

[no] reconnect-with-new-tcp-port

Context

config>mobile>profile>diameter-peer

Description

This command defines how the source port is selected when the TCP connection is reopened after the previous TCP was closed. If this command is enabled, the TCP connection uses a different source port than before.

The no form of this command reverts to the default and the TCP connection uses the same source port as before.

Default

no reconnect-with-new-tcp-port

6.2.1.24.24 sctp-profile

Syntax

sctp-profile *profile-name*

no sctp-profile

Context

config>mobile>profile>diameter-peer

Description

This command configures the SCTP profile for the Diameter peer.



NOTE:

This command requires a peer shutdown.

Parameters

profile-name

specifies a profile name, up to 32 characters

6.2.1.24.25 sig-rate-limit-profile

Syntax

```
sig-rate-limit-profile profile-name
```

```
no sig-rate-limit-profile
```

Context

```
config>mobile>profile>diameter-peer
```

Description

This command associates a rate limiting profile to be applied in this Diameter peer for the specified application. The specified throttling rate is per ISM-MG/peer. The profile must be configured first (using the `sig-rate-limit-profile` command in the `config>mobile>profile` context) before being referenced here.

The no form of this command removes the association of the specified profile for this interface/peer.



NOTE:

As an example, a specific Diameter peer could see “max-rate X times the number of active ISM-MG groups” from a gateway node.

Parameters

profile-name

name of the rate limiting profile, up to 32 characters

6.2.1.24.26 switchover-source-port-consistent

Syntax

```
[no] switchover-source-port-consistent
```

Context

```
config>mobile>profile>diameter-peer
```

Description

This command enables keeping the same port for a TCP session as previous on an intra-VNF or chassis-HA switchover.

The no form of this command enables change of the source port for the TCP session on an intra-VNF or chassis-HA switchover.

Default

no switchover-source-port-consistent

6.2.1.24.27 transport

Syntax

```
transport {tcp | sctp}
```

```
no transport
```

Context

```
config>mobile>profile>diameter-peer
```

Description

This command specifies whether the Diameter signaling is TCP or SCTP-based.



NOTE:

This command requires a peer shutdown.

Default

no transport

Parameters

tcp

specifies the TCP Diameter signaling

sctp

specifies the SCTP Diameter signaling

6.2.1.24.28 validate-supported-host

Syntax

```
[no] validate-supported-host
```

Context

```
config>mobile>profile>diameter-peer
```

Description

This command enables validation of a received origin-host from the server (for example, the PCRF) with the supported hosts that are locally configured within the Diameter peer profile.

The no form of this command reverts to the default (disabled).

Default

```
no validate-supported-host
```

6.2.1.24.29 diameter-peer-group-list

Syntax

```
[no] diameter-peer-group-list list-name
```

Context

```
config>mobile>profile
```

Description

This command creates a Diameter peer group list profile. It contains a list of Diameter peer groups. Each group is a set of a primary/secondary pair. The diameter-peer-group-list command can be used for PCRF/OCS selection based on the priority of the peer group. The Diameter peer group list profile can be applied to Gx and Gy interfaces using the diameter-peer-group-list command in the config>mobile>pdn>gx>pcrf-selection and config>mobile>pdn>gy contexts respectively.

Parameters

list-name

specifies the name of the Diameter peer group list, up to 32 characters

6.2.1.24.30 peer-group

Syntax

```
peer-group peer-group-id primary-diameter-peer profile-name [secondary-diameter-peer  
profile-name] [priority priority]
```

```
no peer-group peer-group-id
```

Context

```
config>mobile>profile>diameter-peer-group-list
```

Description

This command specifies the list of Diameter peer groups that can be used to create a set of primary and secondary peers. This list can be used for Gx and Gy interfaces. This command and existing commands that specify primary and secondary PRCF/OCS are mutually exclusive.



NOTE:

- Each Diameter peer group can support a primary and a secondary pair. For example, S1 acts as secondary for P1 and S2 acts as secondary for P2.
- To support a single cluster, the secondary peer is not specified.
- When priority is not specified or priority is the same for more than one group, load balancing (round-robin) is applied across the groups.
- Traffic goes over the connection with highest priority.
- In a single group/cluster, if a secondary Diameter peer group is not specified, there is no failover to a secondary peer group.

Parameters

peer-group-id

specifies the ID of the peer group

Values: 1 to 32

profile-name

specifies the name of an existing Diameter peer profile, up to 32 characters

priority

specifies the priority for the peer group

Values: 0 to 32

Default: 32

**NOTE:**

A peer-group-id with priority 0 within a diameter-peer-list is considered as disabled.

6.2.1.25 DNS profile command descriptions

6.2.1.25.1 dns-error-mapping-profile

Syntax

[no] dns-error-mapping-profile *profile-name*

Context

config>mobile>profile

Description

This command allows the creation of a DNS error mapping profile.

The no form of this command removes the specified profile from the configuration.

Default

no dns-error-mapping-profile

Parameters

profile-name

specifies the name of the error mapping profile, up to 32 characters

6.2.1.25.2 backoff-timer

Syntax

backoff-timer [timer-value *timer-value*] [timer-unit *timer-unit*]

no backoff-timer

Context

config>mobile>profile>dns-error-mapping-profile

Description

This command configures the backoff timer status type of the notify payload. The backoff timer is sent without any error code notify payload, if both the unit and value parameters are configured. The unit and value fields can be configured independently since they have default values.

The no form of this command removes the backoff timer from the existing mapping code.

Parameters

timer-value

specifies the value of the backoff timer

Values: 0 to 31

Default: 5



NOTE:

A value of zero means that the UE may retry the authentication procedure to an ePDG from the same PLMN.

timer-unit

specifies the unit for the timer-value parameter

Values	deact	the value indicates that the timer is deactivated; the authentication must not retry the procedure to the same ePDG until the UE is switched off or the UICC containing the USIM is removed
–	2-s	the value is incremented in multiples of 2 seconds
–	30-s	the value is incremented in multiples of 30 seconds
–	1-min	the value is incremented in multiples of 1 minute
–	10-min	the value is incremented in multiples of 10 minutes
–	1-h	the value is incremented in multiples of 1 hour
–	10-h	the value is incremented in multiples of 10 hours

– 320-h the value is incremented in multiples of 320 hours

Default: 1-min

6.2.1.25.3 swu-notify

Syntax

```
swu-notify swu-notify-value [description description-string]  
no swu-notify
```

Context

```
config>mobile>profile>dns-error-mapping-profile
```

Description

This command configures the SWu notify error message type.

The no form of this command sets the transmitted IKE error code to the default (36: INTERNAL_ADDRESS_FAILURE).

Default

```
no swu-notify
```

Parameters

swu-notify-value

specifies the SWu notify type

Values: 1 to 16383

description-string

specifies the description of the DNS error code, up to 80 characters

6.2.1.25.4 swu-send-info

Syntax

```
swu-send-info [notify]  
no swu-send-info
```

Context

config>mobile>profile>dns-error-mapping-profile

Description

This command enables the transmission of an IKE INFORMATIONAL request (with DELETE payload).

The no form of this command removes any SWu INFORMATIONAL action (that is, no IKE INFORMATIONAL request message is sent).

Default

no swu-send-info

Parameters

notify

specifies if the ePDG, along with the IKE INFORMATIONAL request message (with DELETE payload), includes a NOTIFY payload with the same IKE error code value, as the one defined with the swu-notify command. If the notify keyword is not configured with the swu-send-info command, only the INFORMATIONAL request is sent with DELETE payload.

6.2.1.25.5 dns-profile

Syntax

[no] dns-profile *profile-name*

Context

config>mobile>profile

Description

This command creates a DNS profile for the mobile gateway. The DNS profile can be used in a DNS lookup for the control plane mechanism on the gateway. If the DNS profile is not configured, the DNS lookup is done using the default DNS configuration via the Boot Option File (BOF) on the gateway.

The no form of this command deletes the DNS profile.

Parameters

profile-name

specifies the name of the DNS profile, up to 32 characters

6.2.1.25.6 ip-dscp

Syntax

```
ip-dscp value  
no ip-dscp
```

Context

```
config>mobile>profile>dns-profile
```

Description

This command configures the IP DSCP/TOS value in the IP header of a DNS request. By default, the DSCP/TOS value for DNS lookup is configured as the network control packet.

The no form of this command configures the default value of the IP DSCP/TOS.

Default

```
no ip-dscp
```

Parameters

value

specifies the IP DSCP/TOS value

Values: 0 to 63

Default: 56

6.2.1.25.7 ip-ttl

Syntax

```
ip-ttl value  
no ip-ttl
```

Context

```
config>mobile>profile>dns-profile
```

Description

This command configures the IP Time To Live (TTL) value for DNS lookup messages from the mobile gateway.

The no form of this command configures the default value of the IP TTL.

Default

no ip-ttl

Parameters

value

specifies the IP TTL value

Values: 0 to 255

Default: 255

6.2.1.25.8 message-retransmit

Syntax

message-retransmit [timeout *seconds*] [retry-count *value*]

no message-retransmit

Context

config>mobile>profile>dns-profile

Description

This command configures the DNS request timeout and retry values for DNS lookup messages from the mobile gateway.

The no form of this command configures the default values of the timeout and retry counts.

Default

no message-retransmit

Parameters

seconds

specifies the time, in seconds, before a DNS request message is retransmitted

Values: 1 to 30

Default: 5

value

specifies the number of retries before a DNS request message to the server fails

Values: 1 to 8

Default: 3

6.2.1.25.9 primary-dns

Syntax

primary-dns *ip-address*

no primary-dns

Context

config>mobile>profile>dns-profile

Description

This command configures the primary DNS IPv4 or IPv6 address for DNS lookup on the mobile gateway.

The no form of this command removes the primary DNS server address from the DNS profile.

Default

no primary-dns

Parameters

ip-address

specifies the IPv4 or IPv6 address of the primary DNS server

Values	ipv4 address	a.b.c.d
–	ipv6 address	x::x::x::x::x::x (eight 16-bit pieces)
–	–	x::x::x::x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

6.2.1.25.10 secondary-dns

Syntax

```
secondary-dns ip-address  
no secondary-dns
```

Context

```
config>mobile>profile>dns-profile
```

Description

This command configures the secondary DNS IPv4 or IPv6 address for DNS lookup on the mobile gateway. The no form of this command removes the secondary DNS server address from the DNS profile.

Default

```
no secondary-dns
```

Parameters

ip-address

specifies the IPv4 or IPv6 address of the secondary DNS server

Values		
	ipv4 address	a.b.c.d
–	ipv6 address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

6.2.1.25.11 tertiary-dns

Syntax

```
tertiary-dns ip-address
no tertiary-dns
```

Context

```
config>mobile>profile>dns-profile
```

Description

This command configures the tertiary DNS IPv4 or IPv6 address for DNS lookup on the mobile gateway. The no form of this command removes the tertiary DNS server address from the DNS profile.

Default

```
no tertiary-dns
```

Parameters

ip-address
specifies the IPv4 or IPv6 address of the tertiary DNS server

Values	ipv4 address	a.b.c.d
–	ipv6 address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

6.2.1.26 DNS redirection profile command descriptions

6.2.1.26.1 dns-redirect-server

Syntax

[no] dns-redirect-server *profile-name*

Context

config>mobile>profile

Description

This command specifies the dns-redirect option as part of the policy-rule-unit of a PCC rule. The profile created as part of the policy option is applied to the PCC rule. The dns-redirect option only uses the policy-rule-unit multi entry flow-description->match->remote-ip rule which defines the DNS server destination IP/prefix for which all DNS request are redirected.

Parameters

profile-name
specifies the name of the profile, up to 32 characters

6.2.1.26.2 primary-dns

Syntax

primary-dns *primary-dns* [secondary-dns *secondary-dns*]

Context

config>mobile>profile>dns-redirect-server

Description

This command configures the IP address of the primary and secondary DNS server.

Parameters

primary-dns
specifies the address of the primary DNS server

Values	ipv4 address	a.b.c.d
--------	--------------	---------

secondary-dns

specifies the address of the secondary DNS server

Values	ipv4 address	a.b.c.d
--------	--------------	---------

6.2.1.27 Emergency profile command descriptions

6.2.1.27.1 emergency-profile

Syntax

[no] emergency-profile *profile-name*

Context

config>mobile>profile

Description

This command configures an emergency profile.

The no form of this command removes the profile from the configuration.

Parameters

profile-name

specifies the name of the profile, up to 32 characters

6.2.1.27.2 allow-udm-sdm

Syntax

[no] allow-udm-sdm

Context

config>mobile>profile>emergency-profile

Description

This command enables the use of the UDM SDM service for emergency calls with an authenticated SUPI. If the SUPI is not available or it is not authenticated, the SMF does not try to get session management subscription data from the UDM.

The no form of this command reverts to the default, where the SMF does not try to get session management subscription data from the UDM for emergency calls.

6.2.1.27.3 block-udm-uecm

Syntax

```
[no] block-udm-uecm
```

Context

```
config>mobile>profile>emergency-profile
```

Description

This command disables the use of the UDM SDM service for the authenticated SUPI. Therefore, the SMF does not register the PDU session to the UDM.

The no form of this command reverts to the default, where the SMF registers the PDU session to the UDM if the SUPI is available and authenticated.

6.2.1.27.4 default-5qi

Syntax

```
default-5qi value
```

```
no default-5qi
```

Context

```
config>mobile>profile>emergency-profile
```

Description

This command configures the default 5QI value for the 5G emergency calls. The value may be overridden by the value received from the UDM.

The no form of this command reverts to the default.

Parameters

value

specifies the default 5QI value

Values: 1 to 255

Default: 5

6.2.1.27.5 default-ambr

Syntax

```
default-ambr ul-rate ul-rate dl-rate dl-rate
```

```
no default-ambr
```

Context

```
config>mobile>profile>emergency-profile
```

Description

This command configures the default session AMBR for the 5G emergency calls. The value may be overridden by the value received from the UDM.

The no form of this command reverts to the default.

Parameters

ul-rate

specifies the uplink bitrate for the AMBR session in kb/s

Values: 1 to 300000

Default: 4096

dl-rate

specifies the downlink bitrate for the session AMBR in kb/s

Values: 1 to 300000

Default: 4096

6.2.1.27.6 default-arp

Syntax

```
default-arp value
```

```
no default-arp
```

Context

```
config>mobile>profile>emergency-profile
```

Description

This command configures the default ARP priority level for the 5G emergency calls. The value may be overridden by the value received from the UDM.

The no form of this command reverts to the default.

Parameters

value

specifies the default ARP priority level

Values: 1 to 15

Default: 1

6.2.1.27.7 sbi-message-priority

Syntax

```
sbi-message-priority value
```

```
no sbi-message-priority
```

Context

```
config>mobile>profile>emergency-profile
```

Description

This command configures the message priority in SBI signaling associated with the emergency calls.

The no form of this command disables the setting of the message priority in SBI signaling (the priority is set to the default value).

Parameters

value

specifies the SBI message priority

Values: 0 to 31

Default: 24

6.2.1.28 ePDG group profile command descriptions

6.2.1.28.1 epdg-group

Syntax

[no] epdg-group *profile-name*

Context

config>mobile>profile

Description

This command configures an IP address for the ePDG list. A maximum number of 20 IP entries per profile can be configured. The type of the first IP entry dictates the type of the rest too.

The no form of this command removes the list name from the configuration.

Parameters

profile-name

specifies the name of the profile, up to 32 characters

6.2.1.28.2 ip-address

Syntax

[no] ip-address *ipv4* | *ipv6*

Context

config>mobile>profile>epdg-group

Description

This command configures the IP address for the ePDG list.

Parameters

ipv4

specifies the IPv4 address for the ePDG list

ipv6

specifies the IPv6 address for the ePDG list

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

6.2.1.29 Fa Ha peer list profile command descriptions

6.2.1.29.1 fa-ha-peer-list

Syntax

[no] fa-ha-peer-list *list-name*

Context

config>mobile>profile

Description

This command configures the FA peer list name. The list consists of peer IPv4 address that the Home Agent has a secure association with. If no trusted-peer list has been defined, the protocol accepts messages from any peer. Conversely, when a fa-ha-peer list is defined, only messages from peers in the given list are accepted. The operator can add new peers to a fa-ha-peer list at any time.

The no form of this command removes the list name from the configuration.

Default

no fa-ha-peer-list

Parameters

list-name

specifies the name of the FA peer list, up to 32 characters

6.2.1.29.2 peer

Syntax

[no] peer *ip-address/prefix-length*

Context

config>mobile>profile>fa-ha-peer-list

Description

This command configures the IP address of the FA/FAs including network prefix. This is a multiple entry command.

The no form of this command removes the specified IP address of the FA/FAs from the configuration.

Parameters

ip-address

specifies the peer's address, up to 64 characters

prefix-length

specifies the set of peers via a subnet, up to 64 characters

6.2.1.29.3 fa-ha-spi

Syntax

[no] fa-ha-spi *value*

Context

config>mobile>profile>fa-ha-peer-list>peer

Description

This command configures the key per FA peer and HA. This is a multiple entry command that allows configuring multiple keys indexed by security parameter indexes (SPIs) per FA peer.

The no form of this command removes the key per FA peer and HA.

Parameters

value

configures the SPI which indicates a security context between the FA and the HA in accordance with RFC 2002

Values: 256 to 4294967295

6.2.1.29.4 hash-algorithm

Syntax

```
hash-algorithm {md5 | rfc2002-md5 | hmac-md5}
```

Context

```
config>mobile>profile>fa-ha-peer-list>peer>fa-ha-spi
```

Description

This command configures the hash-algorithm.

Parameters

md5

configures the hash-algorithm to implement MD5 per RFC 1321

hmac-md5

configures the hash-algorithm to implement HMAC-MD5 per RFC 3220

rfc2002-md5

configures the hash-algorithm to implement keyed-MD5 per RFC 2002

Default: rfc2002-md5

6.2.1.29.5 secret

Syntax

```
secret secret
```

```
no secret
```

Context

```
config>mobile>profile>fa-ha-peer-list>peer>fa-ha-spi
```

Description

This command configures the shared-secret between the FA and the HA.

Parameters

secret

shared-secret between the FA and the HA

Values: 1 to 254 characters (alphanumeric case sensitive)

6.2.1.30 FQDN group profile command descriptions

6.2.1.30.1 fqdn-group

Syntax

[no] fqdn-group *fqdn-group-name*

Context

config>mobile>profile

Description

This command defines an FQDN group for the profile. The FQDN group can be configured with a list of FQDNs, which is used to derive IP filters dynamically using DNS resolution. The FQDN group is specified under a policy rule unit; the FQDN list is periodically refreshed for resolution to a set of IP addresses. Traffic matching the IP addresses is diverted to the AA card, if AA advanced functions is enabled under that policy rule unit; see [aa-advanced-functions](#).

The no form of this command reverts to the default (removes the specified FQDN group configuration).

Default

no fqdn-group

Parameters

fqdn-group-name

specifies the name of the FQDN group used to specify URLs to resolve the remote IP addresses

Values: 1 to 32 characters (alphanumeric case sensitive)

6.2.1.30.2 dns-refresh-interval

Syntax

```
dns-refresh-interval dns-refresh-interval  
no dns-refresh-interval
```

Context

```
config>mobile>profile>fqdn-group
```

Description

This command specifies the time that the node waits before sending a DNS query to refresh the FQDN group URL list to one or more IP addresses. The TTL received in a DNS response is not used at the gateway. A value of 0 disables the DNS refresh timer.

The no form of this command reverts to the default.

Parameters

dns-refresh-interval

specifies the time in minutes that the node waits before sending a DNS query to refresh the URL list for the FQDN group

Values: 0 | 10 to 1440

Default: 30

6.2.1.30.3 fqdn

Syntax

```
fqdn fqdn-name-id fqdn-name-string  
no fqdn fqdn-name-id
```

Context

```
config>mobile>profile>fqdn-group
```

Description

This command specifies a FQDN within an FQDN group. The FQDN is resolved to one or more IP addresses and is periodically refreshed. The list of FQDNs is used to derive the IP-filters dynamically using DNS resolution. Traffic matching the IP addresses is diverted to the AA card, if the AA advanced functions option is enabled under that policy rule unit; see [aa-advanced-functions](#).

The no form of this command removes the specified FDQN from the FQDN group. To remove the FQDN the corresponding index is specified.

Default

no fqdn-group

Parameters

fqdn-name-id

specifies index of the URL

Values: 1 to 2000

fqdn-name-string

specifies the name of the FQDN

Values: 1 to 254 (alphanumeric characters)

6.2.1.30.4 ip-expiry-refresh-count

Syntax

ip-expiry-refresh-count *ip-expiry-refresh-count*

no ip-expiry-refresh-count

Context

config>mobile>profile>fqdn-group

Description

This command specifies the refresh count for resolved IP address expiration (aging). The count is decreased by one, each time a resolved IP address is not found during a DNS refresh. The IP address is removed (aged out) from the resolved list if it is not found after the refresh count reaches zero. A value of 0 disables the aging mechanism.

The no form of this command reverts to the default.

Parameters

ip-expiry-refresh-count

specifies the DNS refresh count for resolved IP address expiration

Values: 0 to 127

Default: 5

6.2.1.31 Gn/Gp profile command descriptions

6.2.1.31.1 peer-list

Syntax

`peer-list list-name`

`no peer-list`

Context

`config>mobile>profile`

Description

When used at the profile level, this command creates a list of GTP peers. When the profile is used at the PDN Gn or Gp level it applies a peer list. There is a limit of 1000 peers in a peer list.

The no form of this command removes the peer list name from the configuration.

Parameters

list-name

specifies the name of the peer-list, up to 32 characters

6.2.1.31.2 peer

Syntax

`[no] peer ip-address/prefix-length`

Context

`config>mobile>profile>peer-list`

Description

This command configures a GTP peer. For the GGSN case, this would typically be an SGSN. There is a limit of 1000 peers in a peer list.

Parameters

ip-address

specifies the peer’s address

Values	ipv4-address	a.b.c.d
–	mask	1 to 32
–	ipv6-address	x::x::x::x::x::x (eight 16-bit pieces)
–	–	x::x::x::x::d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

prefix-length

used as a shortcut to describe a set of peers via a subnet

Values	ipv4-prefix-length	1 to 32
–	ipv6-prefix-length	1 to 128

6.2.1.31.3 keep-alive

Syntax

[no] keep-alive

Context

config>mobile>profile>peer-list>peer
config>mobile>profile>peer-list>unlisted-peer

Description

This command disables GTP echo for this peer. This overrides any GTP echo configuration that may have been applied to the signaling interface elsewhere.

Default

no keep-alive

6.2.1.31.4 plmn-id

Syntax

```
plmn-id foreign
plmn-id mcc mcc value mnc mnc value
no plmn-id
```

Context

```
config>mobile>profile>peer-list>peer
config>mobile>profile>peer-list>unlisted-peer
```

Description

This command sets the PLMN ID that must be associated with the peer if the PLMN ID cannot be determined from GTP. This value does not override the value of PLMN ID that is sent via GTP. The PLMN ID is used to determine whether the UE is roaming. The SGSN or SGW PLMN ID is matched against the PGW list of PLMN IDs to determine roaming status. Optionally this command allows the operator to identify the GTP peer as a foreign node if the PLMN ID cannot be determined via GTP.

The no form of this command reverts to the default.

Default

The GTP shall be considered a home node if the PLMN ID is not present in GTP.

Parameters

foreign

specifies that the GTP peer is a foreign node

mcc value

specifies the MCC portion of the PLMN ID

Values: 000 to 999

mnc value

specifies the MNC portion of the PLMN ID

Values: 00 to 999

6.2.1.31.5 rat-type

Syntax

```
rat-type {geran | utran | eutran}
```

```
no rat-type
```

Context

```
config>mobile>profile>peer-list>peer
```

```
config>mobile>profile>peer-list>unlisted-peer
```

Description

This command specifies the RAT type that is being served by the peer. This is the RAT type that must be used if the RAT type cannot be determined via the GTP message itself.

Default

utran

Parameters**geran**

specifies that the GERAN RAT type

utran

specifies the UTRAN RAT type

eutran

specifies the EUTRAN RAT type

6.2.1.31.6 unlisted-peer

Syntax

```
unlisted-peer
```

Context

config>mobile>profile>peer-list

Description

This command enables the context to configure peers that are not explicitly listed by the peer statements. For example, this command can be used to disable GTP echo for all peers not explicitly configured.

6.2.1.32 GTP profile command descriptions

6.2.1.32.1 gtp

Syntax

[no] gtp *profile-name*

Context

config>mobile>profile

Description

This command enables the context to configure GTP parameters.

Parameters

profile-name

specifies the name of the GTP profile, up to 32 characters

6.2.1.32.2 gtpv1-options

Syntax

[no] gtpv1-options

Context

config>mobile>profile>gtp

Description

This command enables the IE configuration for GTPv1 parameters.

6.2.1.32.3 exclude-apn-restrictions

Syntax

```
[no] exclude-apn-restrictions
```

Context

```
config>mobile>profile>gtp>gtpv1-options
```

Description

This command includes or excludes APN restriction IE in the PGW/GGSN create PDP context response, update PDP context request and update PDP context response messages. When this command is set to include this option, the PGW/GGSN may or may not include the option depending on the protocol operation.

Default

```
no exclude-apn-restrictions
```

6.2.1.32.4 exclude-bcm

Syntax

```
[no] exclude-bcm
```

Context

```
config>mobile>profile>gtp>gtpv1-options
```

Description

This command includes or excludes BCM IE in the PGW/GGSN messages. When this command is set to include this option, the PGW/GGSN may or may not include the option depending on the protocol operation.

Default

```
no exclude-bcm
```

6.2.1.32.5 exclude-chrg-gw-addr

Syntax

[no] exclude-chrg-gw-addr

Context

config>mobile>profile>gtp>gtpv1-options

Description

This command includes or excludes the charging gateway address in the PGW/GGSN create PDP context response and update PDP context response messages. When this command is set to include, the PGW/GGSN may still not include the charging gateway address, depending on the protocol operation. For example, the PGW/GGSN does not include the IE if no charging gateway address is selected.

Default

no exclude-chrg-gw-addr

6.2.1.32.6 exclude-pco

Syntax

[no] exclude-pco

Context

config>mobile>profile>gtp>gtpv1-options

Description

This command includes or excludes Protocol Configuration Options (PCO) in the following PGW/GGSN IE messages:

- initiate PDP context request.
- create PDP context response
- update PDP context response

When this command is set to include the PCO, the PGW/GGSN may or may not include the option, depending on the protocol operation. For example, the PGW/GGSN may not include the PCO IE in a Create PDP Context Response, if it did not receive a PCO in the PDP Context Request.

Default

no exclude-pco

6.2.1.32.7 exclude-recovery**Syntax**

[no] exclude-recovery

Context

config>mobile>profile>gtp>gtpv1-options

Description

This command includes or excludes the recovery IE from the PGW/GGSN create PDP context response, update PDP context request and update PDP context response messages.

Default

no exclude-recovery

6.2.1.32.8 include-common-flags**Syntax**

[no] include-common-flags

Context

config>mobile>profile>gtp>gtpv1-options

Description

This command includes or excludes the common flags IE in the PGW/GGSN create PDP context response and in the update PDP context response messages.

Default

no include-common-flags

6.2.1.32.9 include-nsapi

Syntax

```
[no] include-nsapi
```

Context

```
config>mobile>profile>gtp>gtpv1-options
```

Description

This command includes or excludes the NSAPI IE in the PGW/GGSN create PDP context response messages.

Default

```
include-nsapi
```

6.2.1.32.10 gtpv1-session-rejection-cause

Syntax

```
gtpv1-session-rejection-cause
```

Context

```
config>mobile>profile>gtp
```

Description

This command enables the context to configure the supported features for the GTPv1 session rejection cause.

6.2.1.32.11 apn-shutdown

Syntax

```
apn-shutdown cause-code-value
```

```
no apn-shutdown
```

Context

```
config>mobile>profile>gtp>gtpv1-session-rejection-cause
```

Description

This command defines the value for the Cause IE in the Create PDP Context Response message if the PDP Context Activation message is rejected because the APN is in shutdown state.

The no form of this command reverts to the default.



NOTE:

Setting the GTPv1 session rejection cause code to the default value is equivalent to setting [no] command. For example, setting the apn-shutdown command with the 222 value is equivalent to setting the no apn-shutdown command.

Parameters

cause-code-value

specifies the value for the Cause IE

Values: 192 to 240

Default: 222 (APN access denied — no subscription)

6.2.1.32.12 gw-suspend

Syntax

gw-suspend *cause-code-value*

no gw-suspend

Context

config>mobile>profile>gtp>gtpv1-session-rejection-cause

Description

This command defines the value for the Cause IE in the Create PDP Context Response message if the PDP Context Activation message is rejected because the gateway is in suspended state.

The no form of this command reverts to the default.



NOTE:

This command requires a PDN shutdown.

Parameters

cause-code-value

specifies the value for the Cause IE

Values: 192 to 240

Default: 204 (system failure)

6.2.1.32.13 **sx-connection-failure**

Syntax

`sx-connection-failure cause-code-value`

`no sx-connection-failure`

Context

`config>mobile>profile>gtp>gtpv1-session-rejection-cause`

Description

This command defines the value for the Cause IE in the Create PDP Context Response message if the PDP Context Activation message is rejected due to Sx connection failure.

The no form of this command reverts to the default.



NOTE:

This command requires a PDN shutdown.

Parameters

cause-code-value

specifies the value for the Cause IE

Values: 192 to 240

Default: 209 (user authentication failure)

6.2.1.32.14 **username-missing**

Syntax

`username-missing cause-code-value`

`no username-missing`

Context

```
config>mobile>profile>gtp>gtpv1-session-rejection-cause
```

Description

This command defines the value for the Cause IE in the Create PDP Context Response message if the PDP Context Activation message is rejected because a username or password is required for the session activation but neither of these is included in the PCO IE of the Create PDP Context Request.

The no form of this command reverts to the default.



NOTE:

This command requires a PDN shutdown.

Parameters

cause-code-value

specifies the value for the Cause IE

Values: 192 to 240

Default: 202 (mandatory IE missing)

6.2.1.32.15 gtpv2-node-features

Syntax

```
gtpv2-node-features
```

Context

```
config>mobile>profile>gtp
```

Description

This command enables the context to configure GTPv2 node features.

6.2.1.32.16 include-ciot

Syntax

```
[no] include-ciot
```

Context

```
config>mobile>profile>gtp>gtpv2-node-features
```

Description

This command includes or excludes the CIOT flag in the Node Features IE. If the Node Features IE is excluded from GTPv2 messages, the command is irrelevant. The CIOT flag is excluded by default.

Default

no include-ciot

6.2.1.32.17 gtpv2-options

Syntax

```
[no] gtpv2-options
```

Context

```
config>mobile>profile>gtp
```

Description

This command enables the IE configuration for GTPv2 parameters.

6.2.1.32.18 exclude-aaa-node-identifier

Syntax

```
[no] exclude-aaa-node-identifier
```

Context

```
config>mobile>profile>gtp>gtpv2-options
```

Description

This command excludes the node identifier IE, containing the AAA host and realm, toward the PGW.



NOTE:

Per GTP protocol design, the maximum length that can be carried over is 254 bytes for the Node Name (origin host) and 254 bytes for the Node Realm (origin realm). If the AAA provided origin host and origin realm exceed this limit, the ePDG does not send the node identifier IE.

Default

no exclude-aaa-node-identifier

6.2.1.32.19 exclude-apco

Syntax

[no] exclude-apco

Context

config>mobile>profile>gtp>gtpv2-options

Description

This command excludes the APCO IE in the GTPv2 messages. The APCO IE is included by default.

Default

no exclude-apco

6.2.1.32.20 exclude-change-reporting-action

Syntax

[no] exclude-change-reporting-action

Context

config>mobile>profile>gtp>gtpv2-options

Description

This command excludes the change reporting action IE in GTPv2 messages.

Default

no exclude-change-reporting-action

6.2.1.32.21 `exclude-cno-selection-entity`

Syntax

```
[no] exclude-cno-selection-entity
```

Context

```
config>mobile>profile>gtp>gtpv2-options
```

Description

This command excludes the CN Operator Selection Entity IE in GTPv2 messages.

Default

```
no exclude-cno-selection-entity
```

6.2.1.32.22 `exclude-csg-reporting-action`

Syntax

```
[no] exclude-csg-reporting-action
```

Context

```
config>mobile>profile>gtp>gtpv2-options
```

Description

This command excludes the CSG reporting action IE in GTPv2 messages.

Default

```
no exclude-csg-reporting-action
```

6.2.1.32.23 `exclude-ddn-arp`

Syntax

```
[no] exclude-ddn-arp
```

Context

```
config>mobile>profile>gtp>gtpv2-options
```

Description

This command excludes the ARP IE in the PGW/GGSN messages.

Default

The ARP IE is included by default.

6.2.1.32.24 exclude-ddn-fteid

Syntax

```
[no] exclude-ddn-fteid
```

Context

```
config>mobile>profile>gtp>gtpv2-options
```

Description

This command excludes the DDN FTEID IE in the PGW/GGSN messages.

Default

```
no exclude-ddn-fteid
```

6.2.1.32.25 exclude-ddn-imsi

Syntax

```
[no] exclude-ddn-imsi
```

Context

```
config>mobile>profile>gtp>gtpv2-options
```

Description

This command excludes the DDN IMSI IE in the PGW/GGSN messages.

Default

no exclude-ddn-imsi

6.2.1.32.26 exclude-lci**Syntax**

[no] exclude-lci

Context

config>mobile>profile>gtp>gtpv2-options

Description

This command excludes the Load Control Indicator (LCI) IE in the GTPv2 messages.

Default

no exclude-lci

6.2.1.32.27 exclude-maximum-wait-time**Syntax**

[no] exclude-maximum-wait-time

Context

config>mobile>profile>gtp>gtpv2-options

Description

This command excludes the Maximum Wait Time IE in the GTPv2 messages.

Default

no exclude-maximum-wait-time

6.2.1.32.28 exclude-node-features

Syntax

[no] exclude-node-features

Context

config>mobile>profile>gtp>gtpv2-options

Description

This command excludes the node features IE in the PGW/GGSN messages.

Default

no exclude-node-features

6.2.1.32.29 exclude-oci

Syntax

[no] exclude-oci

Context

config>mobile>profile>gtp>gtpv2-options

Description

This command excludes the Overload Control Indicator (OCI) IE in the GTPv2 messages

Default

no exclude-oci

6.2.1.32.30 exclude-origination-timestamp

Syntax

[no] exclude-origination-timestamp

Context

config>mobile>profile>gtp>gtpv2-options

Description

This command excludes the Origination Timestamp IE in the GTPv2 messages.

Default

no exclude-origination-timestamp

6.2.1.32.31 exclude-paging-priority

Syntax

[no] exclude-paging-priority

Context

config>mobile>profile>gtp>gtpv2-options

Description

This command excludes the paging and service information IE within a DDN message on the S11 interface. The paging policy indication value in octet 'm' is encoded as the DSCP marking received in a user packet during paging.

Default

no exclude-paging-priority

6.2.1.32.32 exclude-pco

Syntax

[no] exclude-pco

Context

config>mobile>profile>gtp>gtpv2-options

Description

This command excludes the PCO/ePCO IE in the GTPv2 messages. The PCO/ePCO IE is included by default.

Default

no exclude-pco

6.2.1.32.33 **exclude-ran-nas-cause**

Syntax

[no] exclude-ran-nas-cause

Context

config>mobile>profile>gtp>gtpv2-options

Description

This command excludes the RAN/NAS cause IE in the GTPv2 messages.

Default

no exclude-ran-nas-cause

6.2.1.32.34 **exclude-signaling-priority**

Syntax

[no] exclude-signaling-priority

Context

config>mobile>profile>gtp>gtpv2-options

Description

This command excludes the signaling priority indication IE on the S5 and S8 interfaces.

Default

no exclude-signaling-priority

6.2.1.32.35 **exclude-twan-identifier**

Syntax

[no] exclude-twan-identifier

Context

```
config>mobile>profile>gtp>gtpv2-options
```

Description

This command excludes the TWAN identifier AVP from GTPv2 messages sent to the PGW. The TWAN identifier indicates the UE location in a trusted or untrusted WLAN access network.

The no form of this command reverts to the default; that is, the TWAN identifier AVP is included in GTPv2 messages sent to the PGW.

Default

```
no exclude-twan-identifier
```

6.2.1.32.36 exclude-twan-identifier-timestamp

Syntax

```
[no] exclude-twan-identifier-timestamp
```

Context

```
config>mobile>profile>gtp>gtpv2-options
```

Description

This command excludes the TWAN identifier-timestamp IE from GTPv2 messages sent to the PGW. The TWAN identifier-timestamp indicates the WLAN location timestamp if available.

The no form of this command reverts to the default; that is, the TWAN identifier AVP is included in GTPv2 messages sent to the PGW.

Default

```
no exclude-twan-identifier-timestamp
```

6.2.1.32.37 exclude-uli-timestamp

Syntax

```
[no] exclude-uli-timestamp
```

Context

```
config>mobile>profile>gtp>gtpv2-options
```

Description

This command excludes the ULI Timestamp IE in GTPv2 messages.

Default

```
no exclude-uli-timestamp
```

6.2.1.32.38 exclude-up-selection-indication

Syntax

```
[no] exclude-up-selection-indication
```

Context

```
config>mobile>profile>gtp>gtpv2-options
```

Description

This command excludes the UP Function Selection Indication IE in GTPv2 messages.

The no form of this command reverts to the default (include).

Default

```
no exclude-up-selection-indication
```

6.2.1.32.39 include-ap-mac-address

Syntax

```
[no] include-ap-mac-address
```

Context

```
config>mobile>profile>gtp>gtpv2-options
```

Description

This command causes the ePDG to send the WiFi Access Point MAC address if learned from the IDi format.

The no form of this command causes the ePDG to exclude this IE.

Default

no include-ap-mac-address

6.2.1.32.40 include-mo-exception-data-counter

Syntax

[no] include-mo-exception-data-counter

Context

config>mobile>profile>gtp>gtpv2-options

Description

This command includes the MO Exception Data Counter IE in GTPv2 messages. The IE is excluded by default.

Default

no include-mo-exception-data-counter

6.2.1.32.41 include-serving-plmn-rate-control

Syntax

[no] include-serving-plmn-rate-control

Context

config>mobile>profile>gtp>gtpv2-options

Description

This command includes the serving PLMN rate control IE in GTPv2 messages. The IE is excluded by default.

Default

no include-serving-plmn-rate-control

6.2.1.32.42 non-3gpp-rat-type-pref

Syntax

```
non-3gpp-rat-type-pref {wlan | virtual}  
no non-3gpp-rat-type-pref
```

Context

```
config>mobile>profile>gtp>gtpv2-options
```

Description

This command configures the RAT type value that the ePDG sends to the PGW.
The no form of this command reverts to the default.

Default

```
no non-3gpp-rat-type-pref
```

Parameters

wlan

specifies the RAT type with the wlan value

virtual

specifies the RAT type with the virtual value

Default: wlan

6.2.1.32.43 gtpv2-session-rejection-cause

Syntax

```
gtpv2-session-rejection-cause
```

Context

```
config>mobile>profile>gtp
```

Description

This command enables the context to configure the supported features for the GTPv2 session rejection cause.

6.2.1.32.44 apn-shutdown

Syntax

```
apn-shutdown cause-code-value  
no apn-shutdown
```

Context

```
config>mobile>profile>gtp>gtpv2-session-rejection-cause
```

Description

This command defines the value for the Cause IE in the Create Session Response if the PDN connection activation is rejected because the APN is in shutdown state.

The no form of this command reverts to the default.



NOTE:

This command requires a PDN shutdown.

Parameters

cause-code-value

specifies the value for the Cause IE

Values: 64 to 239

Default: 93 (APN access denied - no subscription)

6.2.1.32.45 gw-suspend

Syntax

```
gw-suspend cause-code-value  
no gw-suspend
```

Context

```
config>mobile>profile>gtp>gtpv2-session-rejection-cause
```

Description

This command defines the value for the Cause IE, if the gateway is in suspend state. The default value applies for both GTPv2 and GTPv1. If the gateway is in suspend state, the session rejection on the Gn/Gp interfaces carries Cause Value 89. If a different value is required for the Gn/Gp interfaces, it must be configured.

The no form of this command reverts to the default.

**NOTE:**

This command requires a PDN shutdown.

Parameters

cause-code-value

specifies the value for the Cause IE

Values: 64 to 239

Default: 89 (service denied)

6.2.1.32.46 **sx-connection-failure**

Syntax

`sx-connection-failure cause-code-value`

`no sx-connection-failure`

Context

`config>mobile>profile>gtp>gtpv2-session-rejection-cause`

Description

This command defines the value for the Cause IE in the Create Session Response message if the PDN Connection Activation message is rejected due to Sx connection failure.

The no form of this command reverts to the default.

**NOTE:**

This command requires a PDN shutdown.

Parameters

cause-code-value

specifies the value for the Cause IE

Values: 64 to 239

Default: 94 (Request message rejected — reason not specified)

6.2.1.32.47 username-missing

Syntax

`username-missing cause-code-value`

`no username-missing`

Context

`config>mobile>profile>gtp>gtpv2-session-rejection-cause`

Description

This command defines the value for the Cause IE in the Create Session Response, if the PDN connection activation is rejected because a username or password is required for the session activation, but was not included in the PCO/APCO/ePCO of the Create Session Request.

The no form of this command reverts to the default.



NOTE:

This command requires a PDN shutdown.

Parameters

cause-code-value

specifies the value for the Cause IE

Values: 64 to 239

Default: 70 (mandatory IE missing)

6.2.1.32.48 ip-dscp

Syntax

`ip-dscp value`

```
no ip-dscp
```

Context

```
config>mobile>profile>gtp
```

Description

This command specifies the DSCP value in the IP header. This value can be configured to treat a packet as Network Control (NC) packet ahead of Expedited Forwarding (EF) packets.

Default

```
no ip-dscp
```

Parameters

value

specifies Differentiated Services Code Point (DSCP) value

Values: 0 to 63

Default: 56

6.2.1.32.49 ip-ttl

Syntax

```
ip-ttl value
```

```
no ip-ttl
```

Context

```
config>mobile>profile>gtp
```

Description

This command specifies the IP Time-To-Live (TTL) value for GTP signaling messages.

Parameters

value

specifies IP Time-To-Live (TTL) value

Values: 1 to 255

Default: 255

6.2.1.32.50 keep-alive

Syntax

```
keep-alive [interval interval] [retry-count retry-count] [timeout timeout]  
no keep-alive
```

Context

```
config>mobile>profile>gtp
```

Description

This command configures the keep alive timeout attributes for the GTP profile. Keep alive (GTPv2 Echo messages) are sent when a GTPv2 path comes up (one or more UEs attach on the reference point). The keep alive interval timer is based on the attributes configured for keep-alive. The gateway sends a message at the time specified for *interval*. If a GTP-C or GTP-U Echo Response is not received within the time specified for *timeout*, the gateway retries the timeout wait period, for the number of times specified for *retry-count*. If the gateway hears a GTP-C or GTP-U Echo Response from the remote end, it aborts the retry procedure and restarts the keep alive interval timer for this peer. If no response is received before the end of the retry-timeout sequence, the path is declared to be down. For GTP-U, to enable the keep-alive command in the GTP-U part of the Gn, Gp, S4, S5, and S8 reference points, the *gtpu-keep-alive* command must be configured in addition to a non-zero value of the keep alive interval. For example, for the S5 interface, use the `config>mobile>pdn>s5>signaling>gtp-u>gtpu-keep-alive` command. To enable GTP-U on the S1u, S12, N3, and N9 reference points, the no separate *gtpu-keep-alive* command is not required.

Parameters

interval

specifies the keep alive message interval, in seconds

Values: 0 | 60 to 600

Default: 60

retry-count

specifies the number of retries

Values: 1 to 15

Default: 4

timeout

specifies the timeout interval, in seconds

Values: 1 to 30

Default: 5

6.2.1.32.51 message-retransmit

Syntax

```
message-retransmit [timeout timeout] [retry-count value] [timeout-msec timeout-msec]  
no message-retransmit
```

Context

```
config>mobile>profile>gtp
```

Description

This command configures the GTP message retransmit timeout and retry count values. Retransmit timeout consists of T3 and N3 values. T3 is the time period a network element waits before re-attempting to send a message, if there is no response to an initial message. T3 can be configured either in seconds (*timeout*) or milliseconds (*timeout-msec*). N3 is the maximum number of times a network element re-attempts to send a message. N3 can be configured by changing the value of *value*.

Parameters

timeout

specifies the time period, in seconds, that a network element waits before resending a message if there is no response to an initial message. The *timeout* and *timeout-msec* parameters cannot be both set at the same time.

Values: 1 to 30

Default: 5

value

specifies the number of times the NE re-attempts to send a message

Values: 1 to 8

Default: 3

timeout-msec

specifies the time period in milliseconds that a network element waits before resending a message if there is no response to the initial message. The *timeout* and *timeout-msec* parameters cannot be both set at the same time.

Values: 1000 to 30000 (increments of 100 milliseconds)

Default: 5000

6.2.1.33 GTP prime group profile command descriptions

6.2.1.33.1 gtp-prime-group

Syntax

[no] gtp-prime-group *gtp-prime-group*

Context

config>mobile>profile

Description

This command specifies the unique name of this GPRS Tunneling Protocol (GTP) Prime Server Group. When a GTP prime server is no longer used in a shutdown state, it is deleted without a PDN shutdown.



NOTE:

The no form of this command requires a PDN shutdown.

Parameters

gtp-prime-group

specifies the name of the GTP prime server group

6.2.1.33.2 deadtime

Syntax

deadtime *deadtime*

no deadtime

Context

config>mobile>profile>gtp-prime-group

Description

This command specifies the time that a server must be considered dead before it may be used again.

Parameters

deadtime

specifies the time interval that a server is considered dead, in seconds

Values: 0 to 3600

Default: 0

6.2.1.33.3 duplicate-prevention

Syntax

[no] duplicate-prevention

Context

config>mobile>profile>gtp-prime-group

Description

This command specifies if the duplicate prevention algorithm based on GTP prime signaling (3GPP 32.295, sections 5.2.2.2, 5.2.2.3) is enabled. If disabled, the CDRs are probably marked, so duplicate prevention is performed by the Charging Gateway Function (CGF) instead.

The no form reverts to the default (disabled).

Default

no duplicate-prevention

6.2.1.33.4 duplicate-prevention-storage

Syntax

[no] duplicate-prevention-storage

Context

config>mobile>profile>gtp-prime-group

Description

This command specifies whether the CDRs are sent to the CGF and are waiting for acknowledgment (including possible duplicates) are stored into permanent storage in case of failure. If disabled, the possible duplicate CDRs are not stored in permanent storage in case of failure.

The no form reverts to the default (disabled).

Default

no duplicate-prevention-storage

6.2.1.33.5 fragment-drt

Syntax

[no] fragment -drt

Context

config>mobile>profile>gtp-prime-group

Description

This command specifies that large CDRs generated when the gateway is configured to use Ga, are sent and fragmentation is permitted for this GTP prime group profile. You can store the CDR in flash (see `mtu`) or send the CDR and permit fragmentation to occur on the egress interface (by enabling the `fragment -drt` command).

6.2.1.33.6 gtp-prime-server

Syntax

[no] gtp-prime-server *ip-address* [port *port*]

Context

config>mobile>profile>gtp-prime-group

Description

This command specifies the IP address and the destination TCP or UDP port number for GTP Prime Server.

Parameters

ip-address

Specifies the IP address of the peer for which this entry contains information. Either an IPv4 or an IPv6 address for the same CGF can be configured under the *ip-address* parameter. Only one IP address type is allowed.

Values	ipv4-address	a.b.c.d
–	ipv6-address	x::x::x::x::x::x (eight 16-bit pieces)
–	–	x::x::x::x::d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

port
specifies the destination TCP or UDP port number for GTP Prime Server
Values: 1 to 65535
Default: 3386

6.2.1.33.7 echo-interval

Syntax

```
echo-interval echo-interval  
no echo-interval
```

Context

```
config>mobile>profile>gtp-prime-group>gtp-prime-server
```

Description

This command specifies the interval at which the system must send echo-requests for the GTP Prime Server PDUs used to send the CDRs.

Parameters

echo-interval
specifies echo interval
Values: 1 to 3600
Default: 60

6.2.1.33.8 enable-gtpv2

Syntax

[no] enable-gtpv2

Context

config>mobile>profile>gtp-prime-group>gtp-prime-server

Description

This command configures the GTP' version to be used when sending Ga messages to a peer server. By default the gateway uses GTP'v1 to send Ga messages. However, if GTP'v2 is enabled, the gateway uses GTP'v2 to send initial messages (such as an Echo Request). If the charging server peer does not support GTP'v2, the peer responds with a Version Not Supported message. The gateway then automatically reverts to the version indicated by the message (GTP'v1 or GTP'v0).

Default

no enable-gtpv2

6.2.1.33.9 max-outstanding-requests

Syntax

max-outstanding-requests *max-requests*

no max-outstanding-requests

Context

config>mobile>profile>gtp-prime-group>gtp-prime-server

Description

The value specifies the maximum number of unacknowledged GTP PDUs before the system stops sending the CDRs.

Parameters

max-request

specifies the maximum number of requests

Values: 1 to 512

Default: 256

6.2.1.33.10 node-alive

Syntax

```
[no] node-alive
```

Context

```
config>mobile>profile>gtp-prime-group>gtp-prime-server
```

Description

This command enables the GTP' Node Alive mechanism. The system sends a Node Alive Request to the CGFs after a restart to indicate that it is operational again. This mechanism results in a quicker restart than the echo protocol. This command also enables the system to respond to Node Alive Requests from a CGF.

Default

```
no node-alive
```

6.2.1.33.11 path-protocol

Syntax

```
path-protocol udp  
no path-protocol
```

Context

```
config>mobile>profile>gtp-prime-group>gtp-prime-server
```

Description

This command specifies the path protocol used for the GTP primary server.

6.2.1.33.12 priority

Syntax

```
priority priority  
no priority
```

Context

config>mobile>profile>gtp-prime-group>gtp-prime-server

Description

This command specifies the priority of the GTP primary server.

Parameters

priority

specifies the priority number, where the higher priority corresponds to the higher value, for example, value 2 has greater priority than value 1

Values: 0 to 100

Default: 0

6.2.1.33.13 retries

Syntax

retries *retries*

no retries

Context

config>mobile>profile>gtp-prime-group>gtp-prime-server

Description

This command specifies the number of times the system tries to send a GPRS GTP Prime Server PDU to a CGF.

Parameters

retries

specifies number of retries

Values: 1 to 8

Default: 4

6.2.1.33.14 timeout

Syntax

```
timeout timeout  
no timeout
```

Context

```
config>mobile>profile>gtp-prime-group>gtp-prime-server
```

Description

This command configures the interval between GPRS GTP Prime Server PDU retries.

The no form of this command reverts to the default.

Parameters

timeout

specifies the interval between retries

Values: 1 to 180

Default: 20

6.2.1.33.15 inactive-time

Syntax

```
inactive-time inactive-time  
no inactive-time
```

Context

```
config>mobile>profile>gtp-prime-group
```

Description

This command configures the amount of time the peer must remain inactive to store all the cached GPRS GTP packets in the flash.

Parameters

inactive-time

specifies the amount of time the peer must remain inactive to store all the cached GPRS Tunneling Protocol (GTP) packets in the flash

Values: 1 to 60

Default: 10

6.2.1.33.16 interface

Syntax

```
interface [router router-instance] interface-name
```

Context

```
config>mobile>profile>gtp-prime-group
```

Description

This command configures the default interface.

Parameters

router-instance

specifies the virtual router name or service ID

Values	router-name service-id	
–	router-name	Base
–	service-id	1 to 2147483647

Default: Base

interface-name

specifies an interface name, up to 32 characters

6.2.1.33.17 local-cdr-storage

Syntax

```
local-cdr-storage
```

Context

```
config>mobile>profile>gtp-prime-group
```

Description

This command enables the context to configure local CDR storage parameters. The local CDR storage is used when the set of charging gateways is unreachable or when the `config>mobile>pdn>charging>cdr-transfer bp` command is specified.

6.2.1.33.18 file-params

Syntax

```
file-params
```

Context

```
config>mobile>profile>gtp-prime-group>local-cdr-storage
```

Description

This command provides the required context for detailed configuration of the outage file parameters.

6.2.1.33.19 cdr-file-compression

Syntax

```
cdr-file-compression {enable | disable}  
no cdr-file-compression
```

Context

```
config>mobile>profile>gtp-prime-group>local-cdr-storage>file-params
```

Description

This command enables the CDR file compression before storing the files on the disk.

Parameters

enable

specifies to enable the CDR file compression

disable

specifies to disable the CDR file compression

6.2.1.33.20 closure-rec-num

Syntax

closure-rec-num *closure-rec-num*

no closure-rec-num

Context

config>mobile>profile>gtp-prime-group>local-cdr-storage>file-params

Description

This command specifies an operator configurable limit for the number of ACRs that are stored in a file. The file is closed after the specified limit is reached.

Parameters

closure-rec-num

specifies the file closure number field

Values: 100 to 75000

Default: 50000

6.2.1.33.21 closure-size

Syntax

closure-size *closure-size*

no closure-size

Context

config>mobile>profile>gtp-prime-group>local-cdr-storage>file-params

Description

The command specifies an operator configurable file closure size limit.

Parameters

closure-size

specifies the file closure size limit, in Mbytes

Values: 1 to 100

Default: 50

6.2.1.33.22 closure-time

Syntax

`closure-time closure-time`

`no closure-time`

Context

`config>mobile>profile>gtp-prime-group>local-cdr-storage>file-params`

Description

The command configures an operator configurable file closure lifetime. The file is closed after the specified duration.

Parameters

closure-time

specifies the file closure time limit, in seconds

Values: 30 to 86400

Default: 3600

6.2.1.33.23 extension

Syntax

`extension extension`

`no extension`

Context

`config>mobile>profile>gtp-prime-group>local-cdr-storage>file-params`

Description

The command configures an operator configurable file extension field that is used in the filename.

Parameters

extension

specifies the file extension field

Values: 0 to 8

6.2.1.33.24 max-files-per-subdirectory

Syntax

`max-files-per-subdirectory` *max-files-per-subdirectory*

`no max-files-per-subdirectory`

Context

`config>mobile>profile>gtp-prime-group>local-cdr-storage>file-params`

Description

This command configures the maximum number of files per subdirectory used for storing CDR files. The *max-files-per-subdirectory* parameter is used together with the *max-number-subdirectories* parameter for the distribution of CDR files into several directories. When a subdirectory is full, the remaining CDR files are stored in the next subdirectory. This command is valid only in CMG, not in 7750 SR MG.

The no form of this command reverts to the default.

Parameters

max-files-per-subdirectory

specifies the maximum number of files per subdirectory

Values: 1000 to 50000

Default: 1000

6.2.1.33.25 max-number-subdirectories

Syntax

`max-number-subdirectories` *max-number-subdirectories*

`no max-number-subdirectories`

Context

`config>mobile>profile>gtp-prime-group>local-cdr-storage>file-params`

Description

This command configures the maximum number of subdirectories used for storing CDR files. The *max-number-subdirectories* parameter is used together with the *max-files-per-subdirectory* parameter for the distribution of CDR files into several directories. When a subdirectory is full, the remaining CDR files are stored in the next subdirectory. If *max-number-subdirectories* is configured to 1, all CDR files are stored in the same subdirectory regardless of the maximum number of files per subdirectory (*max-files-per-subdirectory* value is ignored). This command is valid only in CMG, not in 7750 SR MG.

The no form of this command reverts to the default.

Parameters

max-number-subdirectories

specifies the maximum number of subdirectories

Values: 1 to 200

Default: 150

6.2.1.33.26 **obsolete-time**

Syntax

`obsolete-time obsolete-time`

`no obsolete-time`

Context

`config>mobile>profile>gtp-prime-group>local-cdr-storage>file-params`

Description

This command configures an operator configurable file duration, after which the file is deleted.

Parameters

obsolete-time

specifies the file duration time limit, in days

Values: 1 to 31

Default: 7

6.2.1.33.27 private-info

Syntax

```
private-info private-info  
no private-info
```

Context

```
config>mobile>profile>gtp-prime-group>local-cdr-storage>file-params
```

Description

The command configures an operator configurable file private information field that is used in the filename.

Parameters

private-info

specifies the contents of the private information field, up to 32 characters

6.2.1.33.28 location

Syntax

```
location {cf1 | cf2}
```

Context

```
config>mobile>profile>gtp-prime-group>local-cdr-storage
```

Description

This command configures the compact flash slots for use as ACRs storage.

Parameters

cf1

keyword to configure compact flash slot #1

cf2

keyword to configure compact flash slot #2

6.2.1.33.29 limit

Syntax

```
limit limit  
no limit
```

Context

```
config>mobile>profile>gtp-prime-group>local-cdr-storage>location
```

Description

The command specifies the limit (in megabytes) for an amount of memory on the first compact flash that can be used for the ACRs storage. Once the threshold has been reached, the system can no longer support accurate charging.

Parameters

limit
specifies the ACR storage limit, in megabytes

6.2.1.33.30 primary

Syntax

```
[no] primary
```

Context

```
config>mobile>profile>gtp-prime-group>local-cdr-storage>location
```

Description

This command specifies which compact flash is used as a primary storage for the ACRs storage. If the user tries to set the cf1, and, if the cf1 is not available, the cf2 will be set as primary storage. If the user tries to set the cf2, and, if the cf2 is not available, the cf1 will be set as primary storage.

6.2.1.33.31 max-cdrs-per-pdu

Syntax

```
max-cdrs-per-pdu max-cdrs-per-pdu
```

no max-cdrs-per-pdu

Context

config>mobile>profile>gtp-prime-group

Description

This command configures the maximum number of the CDRs that can be placed in a single GTP Prime Server PDU. Also, the number of CDRs are bound by the size of the GTP Prime Server PDU.

Parameters

max-cdrs-per-pdu

specifies the maximum number of CDRs per PDU limit

Values: 0 to 100

Default: 0

6.2.1.33.32 mtu

Syntax

mtu *mtu*

no mtu

Context

config>mobile>profile>gtp-prime-group

Description

This command specifies the size of the MTU for this GTP prime group profile, for local storage of large CDRs that would otherwise be sent when the gateway is configured to use Ga. You can store the CDR in flash (by configuring *mtu*) or send the CDR and permit fragmentation to occur on the egress interface (see *fragment-drt*).

Parameters

mtu

specifies the MTU size in octets

Values: 512 to 4096

Default: 1500

6.2.1.33.33 queue-size

Syntax

`queue-size queue-size`

`no queue-size`

Context

`config>mobile>profile>gtp-prime-group`

Description

This command configures the maximum number of unsent GTP packets cached in the mobile gateway, waiting for at least one CGF to come up. Once this threshold is reached, all the cached GTP packets are stored in the flash.

Parameters

queue-size

specifies the queue size (number of GTP packets)

Values: 10000 to 500000

Default: 100000

6.2.1.33.34 redirection

Syntax

`[no] redirection`

Context

`config>mobile>profile>gtp-prime-group`

Description

This command specifies whether the GTP Prime Server redirection function is enabled. Enabling the redirection function allows the CGF to inform the system when it is about to go out of service and cause the system to select a different CGF.

6.2.1.33.35 round-robin-load-balance

Syntax

[no] round-robin-load-balance

Context

config>mobile>profile>gtp-prime-group

Description

This command enables the use of the round robin algorithm to load balance the traffic among servers with equal priority. When disabled, one CGF is selected to send all traffic to.

Default

no round-robin-load-balance

6.2.1.34 Gx profile command descriptions

The *Gx profile CCFH action per call flow type* describes the expected behavior for Gx profile based failure handling, when the system receives a CCA-U with a failure code from the PCRF.

Table 100: Gx profile CCFH action per call flow type

Call flow type	Gx profile CCFH setting	
	Continue	AP-Continue
Handover	Terminate Session	Activate Gx AP and Continue
ULI Change		
SGW Relocation		
Failure code 5002 (DIAMETER_UNKNOWN_SESSION_ID)	Terminate Session	Terminate Session

6.2.1.34.1 gx

Syntax

[no] gx *profile-name*

Context

config>mobile>profile

Description

This command defines a Gx profile that can be applied on a PDN or APN basis; see the `gx-profile` command. The profile includes configuration to support the number of retries and the transaction timer on a per message basis. In addition, failure handling behavior can be configured on a per error code basis as well as the GTP-C cause code sent per result-code to the peer node. It also supports enabling and disabling of optional features sent in the supported feature AVP. For new sessions that fall back to static policy without further PCRF control as a result of failure handling knobs being enabled, time and volume limits can be configured to indicate how long the session stays active before termination. In the case of a Diameter race condition at the PCRF, the configured transaction timer and retry count values on the gateway to be used.



NOTE:

See *Gx profile CCFH action per call flow type* for the expected behavior of Gx profile-based CCFH, when the system receives a CCA-U with a failure code from the PCRF.

Parameters

profile-name

specifies the Gx profile name, up to 32 characters

6.2.1.34.2 disable-ccr-i-on-csreq

Syntax

[no] disable-ccr-i-on-csreq

Context

config>mobile>profile>gx

Description

This command configures the custom Gxc interface to send a CCR-Init to a PCRF after a Modify Bearer Request message is received during an S1 handover with SGW relocation where CMG acts as an SGW. When the command is not configured, the CCR-Init is sent after a Create Session Request message is received. This command requires the custom Gxc interface to be configured by enabling the custom-2 parameter in the config>mobile>pdn>gx>supported-vendor context

Default

no disable-ccr-i-on-csreq

6.2.1.34.3 event-triggers

Syntax

event-triggers

Context

config>mobile>profile>gx

Description

This command enables the context to configure event triggers for the Gx profile.

6.2.1.34.4 tethering-start

Syntax

tethering-start *tethering-start-value*

no tethering-start

Context

config>mobile>profile>gx>event-triggers

Description

This command is used to assign a value for the tethering-start event trigger. Upon PCRF subscription, the configured event-trigger value is used to notify the PCRF of a tethering-start event.

The no form of this command reverts to the default (disabled).

Default

no tethering-start

Parameters

tethering-start-value

specifies the value of the event trigger used to notify the PCRF of a tethering-start event

Values: 127 to 4294967295

Default: 1501

6.2.1.34.5 tethering-stop

Syntax

tethering-stop *tethering-stop-value*

no tethering-stop

Context

config>mobile>profile>gx>event-triggers

Description

This command is used to assign a value for the tethering-stop event trigger. Upon PCRF subscription, the configured event-trigger value is used to notify the PCRF of a tethering-stop event.

The no form of this command reverts to the default (disabled).

Default

no tethering-stop

Parameters

tethering-stop-value

specifies the value of the event-trigger used to notify the PCRF of a tethering-stop event

Values: 127 to 4294967295

Default: 1502

6.2.1.34.6 allow-pcf-provided-chf-address

Syntax

```
[no] allow-pcf-provided-chf-address
```

Context

```
config>mobile>profile>npcf
```

Description

This command specifies if the PCF is allowed to send the CHF addresses within the ChargingInfo IE.

When you enable this feature, the PCF-provided CHF addresses override other CHF selection methods (NRF, UDM, and default SMF configuration). In this case, the following N7 IEs are supported:

- primaryChfAddress
- secondaryChfAddress

When this feature is disabled, the PCF-provided CHF addresses are ignored for the CHF selection.



NOTE:

In the current release, the following IEs are not supported:

- primaryChfSetId
- secondaryChfSetId
- primaryChfInstanceId
- secondaryChfInstanceId

The no form of this command reverts to the default (disabled).

6.2.1.34.7 dynamic-pcc-online-offline-flags-override

Syntax

```
dynamic-pcc-online-offline-flags-override {offline | both | none}
```

```
no dynamic-pcc-online-offline-flags-override
```

Context

```
config>mobile>profile>npcf
```

Description

This command overrides the PCF-provided online and offline flags when the PCF installs a dynamic rule with both flags set to false (online:false; offline:false). For any other combination this configuration is ignored (for example, for predefined and dynamic rules when the PCF sends online or offline flags as true).

When disabled, CMG applies the PCF-provided online and offline flags to the policy rule.

The no form of this command reverts to the default (disabled).

Parameters

offline

specifies that when the PCF sends offline:false, CMG overrides it to offline:true

both

specifies that when the PCF sends offline:false and online:false, CMG overrides it to offline:true and online: true

none

specifies that the PCF-provided online and offline flags apply to the policy rule

Default: none

6.2.1.34.8 fh-session-no-response

Syntax

```
fh-session-no-response {terminate | continue | terminate-next-attempt} [ap-action-profile profile-name]
```

```
fh-session-no-response [ap-action-profile profile-name]
```

```
no fh-session-no-response
```

Context

```
config>mobile>profile>gx
```

```
config>mobile>profile>npcf
```

Description

This command specifies failure handling behavior for new and ongoing sessions that fail to get a response from the PCRF. If a secondary PCRF is configured, an attempt to get a response from it is tried before applying the selected behavior.

The no form of this command reverts to the default.

**NOTE:**

See *Gx profile CCFH action per call flow type* for the expected behavior of Gx profile-based CCFH, when the system receives a CCA-U with a failure code from the PCRF.

Parameters

terminate

specifies that the session terminates after a failed response

continue

specifies that the session continues after a failed response

ap-continue

specifies that the session goes into AP mode and the specified AP action profile (config>mobile>profile>policy-options>ap-action-profile) is then applied to this session after a failed response

terminate-next-attempt

specifies to terminate the session at the next failed signaling attempt

Default: terminate

profile-name

specifies the name of the AP action profile

6.2.1.34.9 fh-time-limit

Syntax

```
fh-time-limit time-limit
```

```
no fh-time-limit
```

Context

```
config>mobile>profile>gx
```

```
config>mobile>profile>npcf
```

Description

This command specifies the time limit for new and ongoing sessions that get set up and continue without further PCRF control (fall back to static policy) as a result of failure handling.(for example, PCRF no response, and so on). The PDN session is terminated upon reaching the specified time limit. When enabled, the fh-session, fh-3xxx and fh-4xxx CLI commands under APN configuration allow new sessions to

fall back to static policy and continue without any further PCRF control. When the `fh-time-limit` and the `fh-volume-limit` are both enabled, the session is terminated upon whichever condition occurs first.

The no form of this command reverts to the default.

**NOTE:**

See *Gx profile CCFH action per call flow type* for the expected behavior of Gx profile-based CCFH, when the system receives a CCA-U with a failure code from the PCRF.

Default

no fh-time-limit

Parameters

time-limit

specifies that the session continues after a failed response, in minutes

Values: 0 to 2880

Default: 0 (disabled)

6.2.1.34.10 fh-volume-limit

Syntax

fh-volume-limit *volume-limit*

no fh-volume-limit

Context

config>mobile>profile>gx

config>mobile>profile>npcf

Description

This command specifies the volume limit for new and ongoing sessions that are set up and continue without further PCRF control (fall back to static policy) as a result of failure handling (for example, PCRF no response). The PDN session is terminated upon reaching the specified volume limit.

The no form of this command reverts to the default (disabled).

**NOTE:**

See *Gx profile CCFH action per call flow type* for the expected behavior of Gx profile-based CCFH, when the system receives a CCA-U with a failure code from the PCRF.

Parameters

volume-limit

specifies the volume limit in number of kilobytes

Values: 0 to 10000000

Default: 0 (disabled)

6.2.1.34.11 forced-triggers

Syntax

```
gx-forced-triggers
```

Context

```
config>mobile>profile>gx
```

Description

This command enables the context to disable the event triggers which are enabled by default.

6.2.1.34.12 disable-ambr-modification-failure

Syntax

```
[no] disable-ambr-modification-failure
```

Context

```
config>mobile>profile>gx>forced-triggers
```

Description

This command disables the reporting to PCRF in case of an AMBR-Modification-Failure event. According to 3GPP, an AMBR-Modification-Failure event is reported to the PCRF even if the PCRF does not subscribe to this event. This command disables this event trigger. If configured, the AMBR-Modification-Failure event is reported to the PCRF only if the PCRF subscribes to this event trigger. If the command is not configured, the event trigger is enabled by default without the need for the PCRF to subscribe to it (3GPP behavior).

The no form of this command reverts to the default.

**NOTE:**

See *Gx profile CCFH action per call flow type* for the expected behavior of Gx profile-based CCFH, when the system receives a CCA-U with a failure code from the PCRF.

Default

no disable-ambr-modification-failure

6.2.1.34.13 disable-default-eps-bearer-qos-change

Syntax

[no] disable-default-eps-bearer-qos-change

Context

config>mobile>profile>gx>forced-triggers

Description

This command disables the reporting to PCRF in case of a Default-EPS-Bearer-QoS-change event. According to 3GPP, a Default-EPS-Bearer-QoS-change event is reported to the PCRF even if the PCRF does not subscribe to this event. This command disables this event trigger. If configured, the Default-EPS-Bearer-QoS-change event is reported to the PCRF only if the PCRF subscribes to this event trigger. If the command is not configured, the event trigger is enabled by default, without the need for the PCRF to subscribe to it (3GPP behavior).

The no form of this command reverts to the default.

**NOTE:**

See *Gx profile CCFH action per call flow type* for the expected behavior of Gx profile-based CCFH, when the system receives a CCA-U with a failure code from the PCRF.

Default

no disable-default-eps-bearer-qos-change

6.2.1.34.14 disable-default-eps-bearer-qos-modification-failure

Syntax

```
[no] disable-default-eps-bearer-qos-modification-failure
```

Context

```
config>mobile>profile>gx>forced-triggers
```

Description

This command disables the reporting to the PCRF in case of a Default-EPS-Bearer-QoS-Modification-Failure. According to 3GPP, a Default-EPS-Bearer-QoS-Modification-Failure event is reported to the PCRF even if the PCRF does not subscribe to this event. This command disables this event trigger. If configured, the Default-EPS-Bearer-QoS-Modification-Failure is reported to the PCRF only if the PCRF subscribes to this event trigger. If the command is not configured, the event trigger is enabled without the need for PCRF to subscribe to it (3GPP behavior).

The no form of this command reverts to the default.



NOTE:

See *Gx profile CCFH action per call flow type* for the expected behavior of Gx profile-based CCFH, when the system receives a CCA-U with a failure code from the PCRF.

Default

```
no disable-default-eps-bearer-qos-modification-failure
```

6.2.1.34.15 disable-qos-change

Syntax

```
[no] disable-qos-change
```

Context

```
config>mobile>profile>gx>forced-triggers
```

Description

This command disables the reporting to the PCRF in case of a QoS change. According to 3GPP, a QoS change event is reported to the PCRF even if the PCRF does not subscribe to this event. This command disables this event trigger. If configured, the QoS change is reported to the PCRF only if the PCRF

subscribes to this event trigger. If the command is not configured, the event trigger is enabled without the need for PCRF to subscribe to it (3GPP behavior).

The no form of this command reverts to the default.

**NOTE:**

See *Gx profile CCFH action per call flow type* for the expected behavior of Gx profile-based CCFH, when the system receives a CCA-U with a failure code from the PCRF.

Default

no disable-qos-change

6.2.1.34.16 disable-ue-local-ip-addr-change

Syntax

```
[no] disable-ue-local-ip-addr-change
```

Context

```
config>mobile>profile>gx>forced-triggers
```

Description

This command disables the UE-LOCAL-IP-ADDRESS-CHANGE event trigger (that requires no subscription from the PCRF) on the Gx interface to the PCRF.

The no form of this command reverts to the default.

Default

no disable-ue-local-ip-addr-change

6.2.1.34.17 include-last-known-uli

Syntax

```
[no] include-last-known-uli
```

Context

```
config>mobile>profile>gx
```

Description

This command facilitates CMG to include the last known ULI in a CCR-U sent toward the PCRF. The CMG includes whatever is the last known ULI to the GW without any need for subscribing to the location change reporting toward the MME.

The no form of this command reverts to the default (disables sending the last known ULI in updates toward the PCRF).

Default

no include-last-known-uli

6.2.1.34.18 message

Syntax

[no] message {ccr -i | ccr -u | ccr -t}

Context

config>mobile>profile>gx

Description

This command specifies the message type and enables the context to configure options for the message type including the retry count, transaction-timer and failure handling.



NOTE:

See *Gx profile CCFH action per call flow type* for the expected behavior of Gx profile-based CCFH, when the system receives a CCA-U with a failure code from the PCRF.

Parameters

ccr -i

specifies the CCR initial message

ccr -u

specifies the CCR update message

ccr -t

specifies the CCR termination message

6.2.1.34.19 gx-avp-options

Syntax

```
gx-avp-options profile-number  
no gx-avp-options
```

Context

```
config>mobile>profile>gx>message
```

Description

This command enables the context to configure GX AVP profile options for the message type.

Parameters

profile-number

specifies a profile charging identifier

Values: 0 to 255

6.2.1.34.20 result-code

Syntax

```
result-code result-code-value {terminate | continue | retry-and-terminate  
result-code result-code-value { ap-continue | retry-and-ap-continue} [ap-action-  
profile profile-name]  
no result-code result-code-value
```

Context

```
config>mobile>profile>gx>message
```

Description

This command specifies failure handling behavior on a per error code basis.

The no form of this command removes the error-specific failure handling for the specified result code and reverts to the behavior for the generic command. This configuration behavior for specific error codes overrides the generic command.



NOTE:

See *Gx profile CCFH action per call flow type* for the expected behavior of Gx profile-based CCFH, when the system receives a CCA-U with a failure code from the PCRF.

Default

no result-code

Parameters

result-code-value

specifies the result code values applicable at the command level. Result code values include 3GPP specified values and some operator custom result codes

For all other values, the generic command applies. The configuration behavior for specific error codes overrides the generic command. As a means to future proof, other (arbitrary) values are to be allowed as well.

Values: *3GPP result codes* lists the supported 3GPP result codes.

terminate

specifies that the PDN session is terminated

continue

specifies that the PDN session continues for CCR-I/CCR-A and for CCA-U the session continues but the requested modification fails

retry-and-terminate

specifies the retry and terminate for codes 3001, 3002, 3003, 3004, and 5012. After the PDN session is retried with the secondary PCRF, it is terminated if the result is a failure. For all other result-codes, the treatment is terminate.

Table 101: 3GPP result codes

Result code	Diameter command result code name
3001	DIAMETER-COMMAND-UNSUPPORTED
3002	DIAMETER-UNABLE-TO-DELIVER
3003	DIAMETER-REALM-NOT-SERVED
3004	DIAMETER-TOO-BUSY
3005	DIAMETER-LOOP-DETECTED

Table 101: 3GPP result codes (continued)

Result code	Diameter command result code name
3006	DIAMETER-REDIRECT-INDICATION
3007	DIAMETER-APPLICATION-UNSUPPORTED
3008	DIAMETER-INVALID-HDR-BITS
3009	DIAMETER-INVALID-AVP-BITS
3010	DIAMETER-UNKNOWN-PEER
4001	DIAMETER-AUTHENTICATION-REJECTED
4002	DIAMETER-OUT-OF-SPACE
4003	ELECTION-LOST
4141	DIAMETER_PCC_BEARER_EVENT
4143	DIAMETER_AN_GW_FAILED
5001	DIAMETER-AVP-UNSUPPORTED
5002	DIAMETER-UNKNOWN-SESSION-ID
5003	DIAMETER-AUTHORIZATION-REJECTED
5004	DIAMETER-INVALID-AVP-VALUE
5005	DIAMETER-MISSING-AVP
5006	DIAMETER-RESOURCES-EXCEEDED
5007	DIAMETER-CONTRADICTING-AVP

Table 101: 3GPP result codes (continued)

Result code	Diameter command result code name
5008	DIAMETER-AVP-NOT-ALLOWED
5009	DIAMETER-AVP-OCCURS-TOO-MANY-TIMES
5010	DIAMETER-NO-COMMON-APPLICATION
5011	DIAMETER-UNSUPPORTED-VERSION
5012	DIAMETER-UNABLE-TO-COMPLY
5013	DIAMETER-INVALID-BIT-IN-HEADER
5014	DIAMETER-INVALID-AVP-LENGTH
5015	DIAMETER-INVALID-MESSAGE-LENGTH
5016	DIAMETER-INVALID-AVP-BIT-COMMAND
5017	DIAMETER-NO-COMMON-SECURITY
5030	DIAMETER-USER-UNKNOWN
5140	DIAMETER_ERROR_INITIAL_PARAMETERS
5141	DIAMETER_ERROR_TRIGGER_EVENT
5142	DIAMETER_PCC_RULE_EVENT
5143	DIAMETER_ERROR_BEARER_NOT_AUTHORIZED
5144	DIAMETER_ERROR_TRAFFIC_MAPPING_INFO_REJECTED

Table 101: 3GPP result codes (continued)

Result code	Diameter command result code name
5147	DIAMETER_ERROR_CONFLICTING_REQUEST

ap-continue

specifies that upon receiving this error code from the Diameter server peer, the session goes into Assume Positive (AP) mode whereupon the specified AP action profile (ap-action-profile) is applied for this session.

retry-and-ap-continue

specifies that upon receiving this error code from the primary Diameter server peer, the message is sent to the secondary peer and then goes into AP mode if it fails. The specified AP action profile (ap-action-profile) is then applied for this session while in AP mode.

profile-name

specifies the AP action profile name

6.2.1.34.21 result-code-3xxx

Syntax

```
result-code-3xxx {terminate | continue | retry-and-terminate}
```

```
result-code-3xxx result-code-value {ap-continue | retry-and-ap-continue} [ap-action-profile profile-name]
```

```
no result-code-3xxx
```

Context

```
config>mobile>profile>gx>message
```

Description

This command specifies failure handling behavior for the 3xxx series of error codes in general. The configuration behavior for specific error codes overrides the generic command.

The no form of this command reverts to the default.

**NOTE:**

See *Gx profile CCFH action per call flow type* for the expected behavior of Gx profile-based CCFH, when the system receives a CCA-U with a failure code from the PCRF.

Parameters

terminate

specifies that the PDN session is terminated

continue

specifies that the PDN session continues for CCR-I and CCR-A, and the PDN session continues but the requested modification fails for CCA-U

retry-and-terminate

specifies retry and terminate for codes 3001, 3002, 3003, and 3004. After the PDN session is retried with the secondary PCRF, it is terminated if the result is a failure. For all other result-codes, the treatment is terminate.

result-code-value

specifies the result code values applicable at the command level. Result code values include 3GPP specified values and some operator custom result codes

Values: *3GPP result codes* lists the supported 3GPP result codes.

ap-continue

specifies that upon receiving this error code from the Diameter server peer, the session goes into Assume Positive (AP) mode whereupon the specified AP action profile (ap-action-profile) is applied for this session

retry-and-ap-continue

specifies that upon receiving this error code from the primary Diameter server peer, the message is sent to the secondary peer and then goes into AP mode if it fails. The specified AP action profile (ap-action-profile) is then applied for this session while in AP mode.

profile-name

specifies the AP action profile name

Default: terminate

6.2.1.34.22 result-code-4xxx

Syntax

result-code-4xxx {terminate | continue | retry-and-terminate} [ap-action-profile *profile-name*]

result-code-4xxx {ap-continue | retry-and-ap-continue} [ap-action-profile *profile-name*]

no result-code-4xxx

Context

config>mobile>profile>gx>message

Description

This command specifies failure handling behavior for the 4xxx series of error codes in general. This configuration behavior for specific error codes overrides the generic command.

The no form of this command reverts to the default.



NOTE:

See *Gx profile CCFH action per call flow type* for the expected behavior of Gx profile-based CCFH, when the system receives a CCA-U with a failure code from the PCRF.

Parameters

terminate

specifies that the PDN session is terminated

continue

specifies that the PDN session continues for CCR-I and CCR-A, and the PDN session continues but the requested modification fails for CCA-U

retry-and-terminate

specifies retry and terminate for codes 3001, 3002, 3003, 3004, and 5012. After the PDN session is retried with the secondary PCRF, it is terminated if the result is a failure. 4xxx result codes are not affected by this parameter.

result-code-value

specifies the result code values applicable at the command level. Result code values include 3GPP specified values and some operator custom result codes

Values: *3GPP result codes* lists the supported 3GPP result codes.

ap-continue

specifies that upon receiving this error code from the Diameter server peer, the session goes into Assume Positive (AP) mode whereupon the specified AP action profile (ap-action-profile) is then applied for this session

retry-and-ap-continue

specifies that upon receiving this error code from the primary Diameter server peer, the message is sent to the secondary peer and then goes into AP mode if it fails. The specified AP action profile (ap-action-profile) is then applied for this session while in AP mode.

profile-name

specifies the AP action profile name

Default: terminate

6.2.1.34.23 result-code-5xxx

Syntax

```
result-code-5xxx {terminate | continue | retry-and-terminate}
result-code-5xxx {ap-continue | retry-and-ap-continue} [ap-action-profile profile-name]
no result-code-5xxx
```

Context

```
config>mobile>profile>gx>message
```

Description

This command specifies the failure-handling behavior for the 5xxx series of error codes in general. The configuration behavior for specific error-codes overrides the generic command.

The no form of this comm reverts to the default.



NOTE:

See *Gx profile CCFH action per call flow type* for the expected behavior of Gx profile-based CCFH, when the system receives a CCA-U with a failure code from the PCRF.

Parameters

terminate

specifies that the PDN session is terminated

continue

specifies that the PDN session continues for CCR-I and CCR-A, and the PDN session continues but the requested modification fails for CCA-U

retry-and-terminate

specifies retry and terminate for 5xxx result codes. After the PDN session is retried with the secondary PCRF, it is terminated if the result is a failure. For all other result-codes, the treatment is terminate.

result-code-value

specifies the result code values applicable at the command level. Result code values include 3GPP specified values and some operator custom result codes

Values: *3GPP result codes* lists the supported 3GPP result codes.

ap-continue

specifies that upon receiving this error code from the Diameter server peer, the session goes into Assume Positive (AP) mode whereupon the specified AP action profile (`ap-action-profile`) is then applied for this session

retry-and-ap-continue

specifies that upon receiving this error code from the primary Diameter server peer, the message is sent to the secondary peer and then goes into AP mode if it fails. The specified AP action profile (`ap-action-profile`) is then applied for this session while in AP mode.

profile-name

specifies the AP action profile name

Default: `terminate`

6.2.1.34.24 retry-count

Syntax

`retry-count` *retry-count*

`no` `retry-count`

Context

`config>mobile>profile>gx>message`

Description

This command configures the retry count for a message type.

The `no` form of this command reverts to the default.



NOTE:

See *Gx profile CCFH action per call flow type* for the expected behavior of Gx profile-based CCFH, when the system receives a CCA-U with a failure code from the PCRF.

Parameters

retry-count

specifies the number of retransmit times

Values: 0 to 8

Default: 3

6.2.1.34.25 transaction-timer

Syntax

```
transaction-timer {[value] | [timeout-msec value]}  
no transaction-timer
```

Context

```
config>mobile>profile>gx>message
```

Description

This command specifies the transaction timer for the Gx message profile. The value can be expressed in seconds or milliseconds. Setting the timer in milliseconds provides better timeout granularity. The transaction timer from this command overrides the time specified for `transaction-timer` under the `config>mobile>pdn>gx` context.

The no form of this command removes the configuration.



NOTE:

See *Gx profile CCFH action per call flow type* for the expected behavior of Gx profile-based CCFH, when the system receives a CCA-U with a failure code from the PCRF.

Parameters

transaction-timer value

specifies the maximum amount of time, in seconds, the node waits for a Diameter peer to respond before trying another peer

Values: 1 to 180 (increments in 1 seconds)

Default: 5

timeout-msec value

specifies the maximum amount of time, in milliseconds, the node waits for a Diameter peer to respond before trying another peer

Values: 1000 to 180000 (increments of 100 milliseconds)

Default: 5000

6.2.1.34.26 pending-transaction-retry-count

Syntax

```
pending-transaction-retry-count value  
no pending-transaction-retry-count
```

Context

```
config>mobile>profile>gx
```

Description

This command is used to specify the pending transaction retry count for the Diameter race condition handling feature. If the PCRF sends back a DIAMETER-PENDING-TRANSACTION response to a request message sent from the gateway, the gateway retries the request after the pending transaction timer expiration. The number of times the same request is re-attempted due to DIAMETER_PENDING_TRANSACTION is configured using this command. For the retries to apply for the Diameter race condition handling feature, command `pending-transaction` in the context `config>mobile>profile>gx>supported-features` must be enabled.

The `no` form of this command reverts to the default.



NOTE:

See *Gx profile CCFH action per call flow type* for the expected behavior of Gx profile-based CCFH, when the system receives a CCA-U with a failure code from the PCRF.

Default

```
no pending-transaction-retry-count
```

Parameters

value

specifies the number of retries

Values: 0 to 8

Default: 3

6.2.1.34.27 pending-transaction-timer

Syntax

```
pending-transaction-timer msec-value
```

```
no pending-transaction-timer
```

Context

```
config>mobile>profile>gx>
```

Description

This command is used to specify the pending transaction timer for the Diameter race condition handling feature. If the PCRF sends back a DIAMETER-PENDING-TRANSACTION response to a request message sent from the gateway, the gateway retries the request after the pending transaction timer expiration. A value of 0 disables the timer. For the timer to apply for the Diameter race condition handling feature, command `pending-transaction` in the context `config>mobile>profile>gx>supported-features` must be enabled.

The no form of this command reverts to the default.



NOTE:

See *Gx profile CCFH action per call flow type* for the expected behavior of Gx profile-based CCFH, when the system receives a CCA-U with a failure code from the PCRF.

Default

```
no pending-transaction-timer
```

Parameters

msec-value

timer value in milliseconds

Values: 0 to 180000

Default: 100

6.2.1.34.28 qos-info-always-present

Syntax

```
[no] qos-info-always-present
```

Context

```
config>mobile>profile>gx
```

Description

This command configures the QoS-Information AVP presence in every CCR-U message. If not configured, QoS information is included only with QoS change.

The no form of this command reverts to the default.



NOTE:

See *Gx profile CCFH action per call flow type* for the expected behavior of Gx profile-based CCFH, when the system receives a CCA-U with a failure code from the PCRF.

Default

no qos-info-always-present

6.2.1.34.29 report-serving-node-supplied-cc

Syntax

[no] report-serving-node-supplied-cc

Context

config>mobile>profile>gx

Description

This command enables the required reporting. When enabled, the charging characteristics value received from the serving node is reported within the 3GPP-Charging-Characteristics AVP of the Gx messages.

The no form of this command reverts to the default (internally used charging characteristics value).

Default

no report-serving-node-supplied-cc

6.2.1.34.30 serialize-ccr

Syntax

[no] serialize-ccr

Context

```
config>mobile>profile>gx
```

Description

This command serializes the CCR messages sent to the PCRF for a Gx session. When enabled, the gateway waits for a CCA response from the PCRF for an outstanding CCR before sending the next CCR for this Gx session.

The no form of this command reverts to the default (disabled).

Default

```
no serialize-ccr
```

6.2.1.34.31 session-restore-retry-timer

Syntax

```
session-restore-retry-timer retry-timer
```

```
no session-restore-retry-timer
```

Context

```
config>mobile>profile>gx
```

Description

This command configures a session restore retry timer. When the `fh-time-limit` and/or the `fh-volume-limit` commands are active and a session falls into assume positive mode (for example, failure handling = continue), the gateway sends a retry message (CCR-I or CCR-U depending on when the session went into assume positive) every time the configured `session-restore-retry-timer` expires. A successful response from the PCRF results in the Gx session getting restored. The `session-restore-retry-timer` must be smaller than the `fh-time-limit` when the `fh-time-limit` is used (non-zero).

The no form of this command reverts to the default.

Parameters

retry-timer

specifies the retry timer in minutes

Values: 0 to 60

Default: 0

6.2.1.34.32 supported-features

Syntax

supported-features

Context

config>mobile>profile>gx

Description

This command enables the context to configure supported features for the gx profile. For more information about the Gx interface Supported-Features AVP, refer to *Gx Interface Supported-Features AVP* section, in the *CMG Configuration Guide*.

6.2.1.34.33 adc

Syntax

[no] adc

Context

config>mobile>profile>gx>supported-features

Description

This command enables the ADC (application detection and control) optional feature within the supported features AVP.

The no form of this command reverts to the default.



NOTE:

See *Gx profile CCFH action per call flow type* for the expected behavior of Gx profile-based CCFH, when the system receives a CCA-U with a failure code from the PCRF.

Default

no adc

6.2.1.34.34 app-based-charging

Syntax

[no] app-based-charging

Context

config>mobile>profile>gx>supported-features

Description

This command enables the Application Based Charging (ABC) feature bit setting for supported features AVP sent to the PCRF.

The no form of this command disables the feature bit setting for supported features AVP sent to the PCRF.



NOTE:

See *Gx profile CCFH action per call flow type* for the expected behavior of Gx profile-based CCFH, when the system receives a CCA-U with a failure code from the PCRF.

6.2.1.34.35 clear-m-bit

Syntax

[no] clear-m-bit

Context

config>mobile>profile>gx>supported-features

Description

This command specifies the M-Bit setting for the supported features AVP sent to the PCRF. When this command is enabled, the M-bit is cleared (0) in the supported features AVP sent to the PCRF.

The no form of this command reverts to the default.



NOTE:

See *Gx profile CCFH action per call flow type* for the expected behavior of Gx profile-based CCFH, when the system receives a CCA-U with a failure code from the PCRF.

Default

no clear-m-bit

6.2.1.34.36 cno-uli

Syntax

[no] cno-uli

Context

config>mobile>profile>gx>supported-features

Description

This command facilitates the gateway in populating the CNO-ULI feature bit within the supported features AVP sent to the PCRF. The CNO-ULI feature indicates support for PRA information reporting.

The no form of this command reverts to the default (disabled).



NOTE:

See *Gx profile CCFH action per call flow type* for the expected behavior of Gx profile-based CCFH, when the system receives a CCA-U with a failure code from the PCRF.

Default

no cno-uli

6.2.1.34.37 cond-policy-info-default-qos

Syntax

[no] cond-policy-info-default-qos

Context

config>mobile>profile>gx>supported-features

Description

This command enables the Time-conditioned Default EPS Bearer QoS change handling feature within the supported-features AVP sent to PCRF. The PCRF has the ability to specify up to four instances of the Conditional-Policy-Information AVP, each of which includes APN-AMBR and/or Default-EPS-Bearer-QoS and

an Execution-Time. At the specified time, the gateway updates and enforces the corresponding Default EPS Bearer QoS change handling, without interacting with the PCRF. The updated default bearer QCI/ARP is signaled toward the UE. This command can only be enabled after the Rule Bound to Bearer feature is enabled using the `rule-bound-to-default-bearer` command.

The no form of this command reverts to the default (disabled).

**NOTE:**

See *Gx profile CCFH action per call flow type* for the expected behavior of Gx profile-based CCFH, when the system receives a CCA-U with a failure code from the PCRF.

Default

no cond-policy-info-default-qos

6.2.1.34.38 conditional-policy-info

Syntax

[no] conditional-policy-info

Context

config>mobile>profile>gx>supported-features

Description

This command enables the Conditional Policy Information optional feature within the supported features AVP sent to the PCRF.

The no form of this command reverts to the default.

**NOTE:**

See *Gx profile CCFH action per call flow type* for the expected behavior of Gx profile-based CCFH, when the system receives a CCA-U with a failure code from the PCRF.

Default

no conditional-policy-info

6.2.1.34.39 disable-rel-10

Syntax

[no] disable-rel-10

Context

config>mobile>profile>gx>supported-features

Description

This command disables population of Release 10 feature bits within the supported features AVP sent to the PCRF in the CCR-I for the Gx interface. Release 10 feature bits are populated by default, unless this option is enabled.

The no form of this command reverts to the default.



NOTE:

See *Gx profile CCFH action per call flow type* for the expected behavior of Gx profile-based CCFH, when the system receives a CCA-U with a failure code from the PCRF.

Default

no disable-rel-10

6.2.1.34.40 disable-rel-9

Syntax

[no] disable-rel-9

Context

config>mobile>profile>gx>supported-features

Description

This command disables population of Release 9 feature bits within the supported features AVP sent to the PCRF in the CCR-I for the Gx interface. Release 9 feature bits are populated by default, unless this option is enabled.

The no form of this command reverts to the default.

**NOTE:**

See *Gx profile CCFH action per call flow type* for the expected behavior of Gx profile-based CCFH, when the system receives a CCA-U with a failure code from the PCRF.

Default

no disable-rel-9

6.2.1.34.41 exclude-usage

Syntax

[no] exclude-usage

Context

config>mobile>profile>gx>supported-features

Description

This command enables excluding the corresponding service data flow for the volume and/or time measurement on an IP-CAN session within the supported features AVP sent to the PCRF.

The no form of this command reverts to the default.

**NOTE:**

See *Gx profile CCFH action per call flow type* for the expected behavior of Gx profile-based CCFH, when the system receives a CCA-U with a failure code from the PCRF.

Default

no exclude-usage

6.2.1.34.42 extended-bw-nr

Syntax

[no] extended-bw-nr

Context

config>mobile>profile>gx>supported-features

Description

This command enables support for high data rates for option-3x NR. When enabled, the PGW indicates support for this feature by populating the appropriate feature bit in the supported-features AVP sent to the PCRF.

The no form of this command reverts to the default.



NOTE:

See *Gx profile CCFH action per call flow type* for the expected behavior of Gx profile-based CCFH, when the system receives a CCA-U with a failure code from the PCRF.

Default

no extended-bw-nr

6.2.1.34.43 include-rat-when-received

Syntax

```
[no] include-rat-when-received
```

Context

```
config>mobile>profile>gx>supported-features
```

Description

This command enables the including of a RAT type AVP in the CCR-U to the PCRF if the message is due to any GTP signaling that includes the RAT type IE. By default, a RAT type AVP is reported to the PCRF only in the CCR-I, and only when a RAT change event is detected.

The no form of this command reverts to the default (disabled).

Default

no include-rat-when-received

6.2.1.34.44 mission-critical-qcis

Syntax

```
[no] mission-critical-qcis
```


Context

config>mobile>profile>gx>supported-features

Description

This command enables the mission critical QCI optional feature within the supported-features AVP sent to the PCRF.

The no form of this command reverts to the default (disabled).



NOTE:

See *Gx profile CCFH action per call flow type* for the expected behavior of Gx profile-based CCFH, when the system receives a CCA-U with a failure code from the PCRF.

Default

no mission-critical-qcis

6.2.1.34.45 multiple-pra

Syntax

[no] multiple-pra

Context

config>mobile>profile>gx>supported-features

Description

This command enables support for the multiple PRA feature. When enabled, the PGW indicates support for the multiple PRA feature by populating the appropriate feature bit in the supported-features AVP sent to the PCRF.

The no form of this command reverts to the default.



NOTE:

See *Gx profile CCFH action per call flow type* for the expected behavior of Gx profile-based CCFH, when the system receives a CCA-U with a failure code from the PCRF.

Default

no multiple-pra

6.2.1.34.46 netloc

Syntax

[no] net loc

Context

config>mobile>profile>gx>supported-features

Description

This command enables the NetLoc optional feature within the Supported-Features AVP sent to the PCRF. The NetLoc feature indicates the support of Access Network Information Reporting for GPRS and EPS.

The no form of this command reverts to the default.



NOTE:

See *Gx profile CCFH action per call flow type* for the expected behavior of Gx profile-based CCFH, when the system receives a CCA-U with a failure code from the PCRF.

Default

no netloc

6.2.1.34.47 netloc-untrusted-wlan

Syntax

[no] net loc-untrusted-wlan

Context

config>mobile>profile>gx>supported-features

Description

This command enables the NetLoc untrusted WLAN feature within the Supported-Features AVP sent to the PCRF.

The no form of this command reverts to the default (disabled).



NOTE:

See *Gx profile CCFH action per call flow type* for the expected behavior of Gx profile-based CCFH, when the system receives a CCA-U with a failure code from the PCRF.

Default

no netloc-untrusted-wlan

6.2.1.34.48 pcscf-restoration

Syntax

[no] pcscf-restoration

Context

config>mobile>profile>gx>supported-features

Description

This command enables the P-CSCF Restoration Enhancement feature bit setting within the supported features AVP sent to the PCRF.

The no form of this command disables sending the P-CSCF Restoration Enhancement.



NOTE:

See *Gx profile CCFH action per call flow type* for the expected behavior of Gx profile-based CCFH, when the system receives a CCA-U with a failure code from the PCRF.

Default

no pcscf-restoration

6.2.1.34.49 pending-transaction

Syntax

[no] pending-transaction

Context

config>mobile>profile>gx>supported-features

Description

This command, when enabled, facilitates the gateway populating the PendingTransaction feature bit within the supported features AVP sent to the PCRF.

The no form of this command reverts to the default.



NOTE:

See *Gx profile CCFH action per call flow type* for the expected behavior of Gx profile-based CCFH, when the system receives a CCA-U with a failure code from the PCRF.

Default

no pending-transaction

6.2.1.34.50 ran-nas-cause

Syntax

[no] ran-nas-cause

Context

config>mobile>profile>gx>supported-features

Description

This command enables the RAN-NAS cause optional feature within the supported features of AVP. If the PGW receives the RAN/NAS cause value for bearer deletion from the TWAG, ePDG, or MME, and this feature is enabled, the PGW passes the RAN/NAS cause value to the PCRF.

The no form of this command reverts to the default.



NOTE:

See *Gx profile CCFH action per call flow type* for the expected behavior of Gx profile-based CCFH, when the system receives a CCA-U with a failure code from the PCRF.

Default

no ran-nas-cause

6.2.1.34.51 rule-bound-to-default-bearer

Syntax

[no] rule-bound-to-default-bearer

Context

config>mobile>profile>gx>supported-features

Description

This command enables the Rule Bound to Default Bearer optional feature within the supported-features AVP sent to PCRF. When enabled, the PCRF indicates to the PCEF that a PCC rule is to be bound to, and remain on the default bearer. The PCEF binds the indicated PCC rule to the default bearer, until the PCC rule is removed or the PCRF modifies the PCC rule to set the Default-Bearer-Indication AVP to the value BIND_TO_APPLICABLE_BEARER. This option must be enabled prior to enabling the Time-conditioned Default EPS Bearer QoS change handling feature using the cond-policy-info-default-qos command.

The no form of this command reverts to the default.



NOTE:

See *Gx profile CCFH action per call flow type* for the expected behavior of Gx profile-based CCFH, when the system receives a CCA-U with a failure code from the PCRF.

Default

no rule-bound-to-default-bearer

6.2.1.34.52 sgw-rest

Syntax

[no] sgw-rest

Context

config>mobile>profile>gx>supported-features

Description

This command indicates support for SGW restoration procedures as defined in 3GPP TS 23.007 to the supported features AVP.

The no form of this command removes the configuration.

**NOTE:**

See *Gx profile CCFH action per call flow type* for the expected behavior of Gx profile-based CCFH, when the system receives a CCA-U with a failure code from the PCRF.

Default

no sgw-rest

6.2.1.34.53 sponsored-connectivity

Syntax

[no] sponsored-connectivity

Context

config>mobile>profile>gx>supported-features

Description

This command enables the sponsored connectivity optional feature within the supported features AVP.

The no form of this command reverts to the default.

**NOTE:**

See *Gx profile CCFH action per call flow type* for the expected behavior of Gx profile-based CCFH, when the system receives a CCA-U with a failure code from the PCRF.

Default

no sponsored-connectivity

6.2.1.34.54 time-based-usage-monitoring

Syntax

[no] time-based-usage-monitoring

Context

config>mobile>profile>gx>supported-features

Description

This command enables the time-based usage monitoring optional feature within the supported features AVP sent to the PCRF.



NOTE:

See *Gx profile CCFH action per call flow type* for the expected behavior of Gx profile-based CCFH, when the system receives a CCA-U with a failure code from the PCRF.

Default

no time-based-usage-monitoring

6.2.1.34.55 trusted-wlan

Syntax

[no] trusted-wlan

Context

config>mobile>profile>gx>supported-features

Description

This command enables the trusted WLAN optional feature within the supported features AVP.

The no form of this command reverts to the default.



NOTE:

See *Gx profile CCFH action per call flow type* for the expected behavior of Gx profile-based CCFH, when the system receives a CCA-U with a failure code from the PCRF.

Default

no trusted-wlan

6.2.1.34.56 user-equipment-info-extension

Syntax

[no] user-equipment-info-extension

Context

```
config>mobile>profile>gx>supported-features
```

Description

This command enables the User-Equipment-Info-Extension AVP within the Suuported-Features AVP sent to the PCRF.

The no form of this command reverts to the default (disabled).

Default

```
no user-equipment-info-extension
```

6.2.1.34.57 usage-monitoring

Syntax

```
usage-monitoring
```

Context

```
config>mobile>profile>gx
```

Description

This command enables the context to set Gx usage monitoring settings.

6.2.1.34.58 monitoring-key

Syntax

```
[no] monitoring-key monitoring-key
```

Context

```
config>mobile>profile>gx>usage-monitoring
```

Description

This command enables the context to specify the activity threshold on a per monitoring-key basis. The activity threshold is used to define active usage for time-based usage monitoring. The PCRF specified QCT is ignored and the specified activity threshold is used instead.

**NOTE:**

See *Gx profile CCFH action per call flow type* for the expected behavior of Gx profile-based CCFH, when the system receives a CCA-U with a failure code from the PCRF.

Parameters

monitoring-key

specifies the monitoring key for which to configure the activity threshold, up to 80 characters

6.2.1.34.59 activity-threshold

Syntax

activity-threshold *threshold-value*

no activity-threshold

Context

config>mobile>profile>gx>usage-monitoring>monitoring-key

Description

This command configures the activity threshold that defines an average data rate under which the subscriber host is considered silent. As long as the effective rate of the application usage (for a monitoring-key instance) does not exceed the rate defined by the activity threshold, the given subscriber host will be considered silent and its corresponding quota will not be used. If the application usage (for a monitoring-key instance) does exceed the rate defined by the activity threshold, the corresponding time quota will be consumed (in terms of time).

**NOTE:**

See *Gx profile CCFH action per call flow type* for the expected behavior of Gx profile-based CCFH, when the system receives a CCA-U with a failure code from the PCRF.

Parameters

threshold-value

specifies the activity threshold value that defines an average data rate under which the subscriber host is considered silent

Values: 0 to 100000000

6.2.1.34.60 override-usage-report-value

Syntax

```
[no] override-usage-report-value
```

Context

```
config>mobile>profile>gx>usage-monitoring
```

Description

This command overrides of the default usage report event trigger value. When enabled, the USAGE_REPORT event trigger value will be 26 instead of 33. In 3GPP Release 9, the value of the USAGE_REPORT event trigger changed from 26 to 33. Enabling the override option helps interoperation with PCRF vendors that continue to use the USAGE_REPORT event trigger value of 26 for usage monitoring. Enabling this command prevents gateway support for the TAI_CHANGE event trigger, which uses a value of 26. If both TAI_CHANGE and usage monitoring are deployed, the PCRF supports both the TAI_CHANGE (value 26) and USAGE_REPORT (value 33) event triggers accordingly.

The no form of this command disables the override.



NOTE:

See *Gx profile CCFH action per call flow type* for the expected behavior of Gx profile-based CCFH, when the system receives a CCA-U with a failure code from the PCRF.

Default

```
no override-usage-report-value
```

6.2.1.34.61 validation-checks

Syntax

```
validation-checks
```

Context

```
config>mobile>profile>gx
```

Description

This command enables the context to perform Gx validation checks.

6.2.1.34.62 raa-immediate

Syntax

```
[no] raa-immediate
```

Context

```
config>mobile>profile>gx>validation-checks
```

Description

When this command is enabled, upon receiving a RAR message from the PCRF, the PGW sends a RAA message immediately prior to validating the rules and/or rule bases sent in the RAR message. If a fault is found during validation, a CCR-U message is sent to the PCRF.

The no form of this command reverts to the default (disabled).

**NOTE:**

See *Gx profile CCFH action per call flow type* for the expected behavior of Gx profile-based CCFH, when the system receives a CCA-U with a failure code from the PCRF.

Default

```
no raa-immediate
```

6.2.1.35 HTTP/2 profile command descriptions

6.2.1.35.1 http2

Syntax

```
[no] http2 profile-name
```

Context

```
config>mobile>profile
```

Description

This command configures an HTTP/2 profile that is used when establishing an HTTP/2 connection.

The no form of this command removes the specified profile from the configuration.

Default

no http2

Parameters

profile-name

specifies the name of the HTTP/2 profile, up to 32 characters

6.2.1.35.2 clt-tls-profile

Syntax

clt-tls-profile *profile-name*

no clt-tls-profile

Context

config>mobile>profile>http2

Description

This command specifies the TLS client profile. When configured, the HTTP/2 connection uses TLS for client connections. The client TLS profile is configured in the config>system>security>tls>client-tls-profile context.

The no form of this command removes the specified client TLS profile from the HTTP/2 profile configuration.

Default

no clt-tls-profile

Parameters

profile-name

specifies the name of the client TLS profile, up to 32 characters

6.2.1.35.3 connection-timer

Syntax

connection-timer *value*

no connection-timer

Context

config>mobile>profile>http2

Description

This command specifies how long the SMF waits before it starts to re-establish an SBI transport connection, after the connection has failed and the attempts to restore it have reached the limit defined by the `fail-num-retries` command with no success. When the limit is reached, the number of retries is reset and the SMF attempts to re-establish the connection based on the `fail-retry-interval` and `fail-num-retries` commands.

Parameters

value

specifies the connection timer, in seconds

Values: 1 to 180

Default: 20

6.2.1.35.4 fail-num-retries

Syntax

fail-num-retries *value*

no fail-num-retries

Context

config>mobile>profile>http2

Description

This command specifies the number of attempts made to re-establish a transport connection with a remote NF.

The no form of this command reverts to the default (disabled).

Parameters

value

specifies the number of attempts to re-establish a connection

Values: 0 to 50

Default: 0

6.2.1.35.5 fail-retry-interval

Syntax

```
fail-retry-interval value  
no fail-retry-interval
```

Context

```
config>mobile>profile>http2
```

Description

This command specifies the time that the mobile gateway waits before it attempts to re-establish a connection with a remote NF, when the `fail-num-retries` command is enabled.

The no form of this command reverts to the default (disabled).

Parameters

value

specifies the time, after which the SMF attempts to re-establish a connection, in tenths of a second

Values: 1 to 50

Default: 5

6.2.1.35.6 header-table-size

Syntax

```
header-table-size number  
no header-table-size
```

Context

```
config>mobile>profile>http2
```

Description

This command configures the HTTP2 header table size in the HTTP connection setup. If it is set to zero, it indicates that no compression is done.

The no form of this command reverts to the default.

Parameters

number

specifies the header table size value, in octets. The 0 value means no compression

Values: 0 to 4096

Default: 4096

6.2.1.35.7 init-window-size

Syntax

init-window-size *number*

no init-window-size

Context

config>mobile>profile>http2

Description

This command configures the initial window size in the HTTP connection setup.

The no form of this command reverts to the default.

Parameters

number

specifies the initial window size value, in octets

Values: 1 to 1048576

Default: 524287

6.2.1.35.8 ip-dscp

Syntax

ip-dscp *value*

no ip-dscp

Context

config>mobile>profile>http2

Description

This command configures the IPv4 DSCP or the IPv6 Traffic Class DS in the IP header for the HTTP2 signaling messages sent, so it is treated with better priority.

The no form of this command reverts to the default.

Parameters

value

specifies the DSCP value or traffic class

Values: 0 to 63

Default: 56

6.2.1.35.9 max-frame-size

Syntax

max-frame-size *number*

no max-frame-size

Context

config>mobile>profile>http2

Description

This command configures the maximum frame size that can be received.

The no form of this command reverts to the default.

Parameters

number

specifies the maximum frame size value, in octets

Values: 4096 to 16777215

Default: 16384

6.2.1.35.10 max-header-list-size

Syntax

max-header-list-size *number*


```
no max-header-list-size
```

Context

```
config>mobile>profile>http2
```

Description

This command configures the maximum header list size in the HTTP connection setup.

The no form of this command reverts to the default.

Parameters

number

specifies the maximum header list size value, in octets

Values: 1024 to 4294967295

Default: 4294967295

6.2.1.35.11 ping-period

Syntax

```
ping-period number
```

```
no ping-period
```

Context

```
config>mobile>profile>http2
```

Description

This command defines the frequency of sending ping frames after a quiet period. The timer for sending ping frames is reset every time there is activity. This configuration also serves as a configuration for the idle time, after which the ping will start.

The no form of this command reverts to the default.

Parameters

number

specifies the ping period value, in seconds

Values: 1 to 60

Default: 60

6.2.1.35.12 request-timeout

Syntax

```
request-timeout time  
no request-timeout
```

Context

```
config>mobile>profile>http2
```

Description

This command configures the time that an HTTP/2 client waits for an HTTP/2 response from a server before timing out.

The no form of this commands reverts to the default value.

Parameters

time

specifies the time, in seconds

Values: 5 to 30

Default: 20

6.2.1.35.13 srv-disable-sni-chk

Syntax

```
[no] srv-disable-sni-chk
```

Context

```
config>mobile>profile>http2
```

Description

This command disables checking for the presence of the SNI in the Client Hello message. TLS tunnel establishment proceeds if the SNI is not present.

The no form of this commands enables checking for the presence of the SNI in the Client Hello message.

Default

no srv-disable-sni-chk

6.2.1.35.14 srv-max-streams-allowed

Syntax

srv-max-streams-allowed *number*

no srv-max-streams-allowed

Context

config>mobile>profile>http2

Description

This command configures the maximum number of concurrent streams that can be sent to an HTTP/2 client in the SETTING message.

The no form of this commands reverts to the default value.

Parameters

number

specifies the maximum number of concurrent streams

Values: 50 to 5120

Default: 256

6.2.1.35.15 srv-stream-lifetime

Syntax

srv-stream-lifetime *number*

no srv-stream-lifetime

Context

config>mobile>profile>http2

Description

This command configures the maximum default lifetime of an HTTP/2 server stream. The server resets the stream and sends an HTTP 500 as a response to the client, if no response is received from the application within this timer. This timer can be overwritten during call processing.

The no form of this command reverts to the default.

Parameters

number

specifies the server stream lifetime value, in seconds

Values: 5 to 30

Default: 15

6.2.1.35.16 **srv-tls-profile**

Syntax

```
srv-tls-profile profile-name
```

```
no srv-tls-profile
```

Context

```
config>mobile>profile>http2
```

Description

This command specifies the TLS server profile. When configured, the HTTP/2 connection uses TLS for server connections. The server TLS profile is configured in the config>system>security>tls>server-tls-profile context.

The no form of this command removes the specified server TLS profile from the HTTP/2 profile configuration.

Default

```
no srv-tls-profile
```

Parameters

profile-name

specifies the name of the server TLS profile, up to 32 characters

6.2.1.35.17 strict-validation

Syntax

```
strict-validation [dns-id] [uuid] [keyUsage] [extKeyUsage] [flags]  
no strict-validation
```

Context

```
config>mobile>profile>http2
```

Description

This command enables performing additional validation checks on the received certificate.

Default

no strict-validation

Parameters

dns-id

specifies to validate the DNS ID in the Subject Alternative Name certificate

uuid

specifies to check for the UUID presence and validates its format in the Subject Alternative Name certificate

keyUsage

specifies to validate the keyUsage extension

extKeyUsage

specifies to validate the extkeyUsage extension

flags

specifies to validate the mandatory and critical flags

6.2.1.36 IPSec profile command descriptions

6.2.1.36.1 ipsec-profile

Syntax

```
[no] ipsec-profile profile-name
```

Context

config>mobile>profile

Description

This command creates an IPSec profile that applies to IPSec tunnels that are used for the ePDG or N3IWF function. The system supports up to 16 IPSec profiles.

The no form of this command disables the profile.

Default

no ipsec-profile

Parameters

profile-name

specifies a unique name, up to 32 characters

6.2.1.36.2 esp-sa

Syntax

esp-sa

Context

config>mobile>profile>ipsec-profile

Description

This command enables the context to configure the Encapsulating Security Payload (ESP SA) parameters.

6.2.1.36.3 lifetime

Syntax

lifetime *seconds*

no lifetime

Context

config>mobile>profile>ipsec-profile>esp-sa

```
config>mobile>profile>ipsec-profile>ike-sa
```

Description

This command specifies the lifetime of the ESP SA or the IKE SA.

The no form of this command reverts to the default.



NOTE:

This command requires a PDN shutdown.

Parameters

seconds

specifies the value in seconds of the ESP SA or IKE SA lifetime

Values: 60 to 172800

Default: 1200 for esp-sa, 3600 for ike-sa

6.2.1.36.4 nat-traversal-dynamic-updates

Syntax

```
[no] nat-traversal-dynamic-updates
```

Context

```
config>mobile>profile>ipsec-profile>esp-sa
```

```
config>mobile>profile>ipsec-profile>ike-sa
```

Description

This command enables the ePDG to dynamically update the source IP address and port for the ESP SA or IKE SA. This is used if there is a NAT between the UE and ePDG (and the UE is behind the NAT) and the UE source IP address and/or UDP port change from the perspective of the ePDG. The ESP SA *replay-window* command must be enabled if this command is enabled.

The no form of this command disables dynamic updates.



NOTE:

This command requires a PDN shutdown.

Default

no nat-traversal-dynamic-updates

6.2.1.36.5 proposal-entry

Syntax

proposal-entry *entry-id*

no proposal-entry

Context

config>mobile>profile>ipsec-profile>esp-sa

config>mobile>profile>ipsec-profile>ike-sa

Description

This command configures an IKE or ESP proposal that the ePDG or N3IWF will accept from a peer. The proposal consists of a combination of authentication and encryption algorithms. You can configure up to ten algorithm suites (proposals), to allow different algorithm suites. The IKE or the ESP selects the first matching proposal, using a match order of '1 -> 2 -> 3 -> 4 -> 5 -> 6 -> 7 -> 8 -> 9 -> 10 -> ... 20'. The proposals must be configured in decreasing order of preference, with the most preferred suite as proposal 1, the second preferred suite as proposal 2, and the least preferred suite as proposals 3, 4, 5, 6, 7, 8, 9, 10 to 20. Proposal 1 can also be thought of as the "preferred" proposal, proposal 2 as the "acceptable" proposal, and proposals 3 - 20 as the "last resort" proposals. If none of the proposals offered by the peer (mobile device) match any of the proposals configured on the ePDG or N3IWF, the ePDG or N3IWF sends a NO_PROPOSAL_CHOSEN error notification back to the peer.

The no form of this command removes the ESP or IKE proposal.



NOTE:

This command requires a PDN shutdown.

Parameters

entry-id

an index to identify the proposal

Values: 1 to 20

6.2.1.36.6 dh-group

Syntax

```
dh-group Diffie-Hellman group  
no dh-group
```

Context

```
config>mobile>profile>ipsec-profile>esp-sa>proposal-entry  
config>mobile>profile>ipsec-profile>ike-sa>proposal-entry
```

Description

This command specifies the Diffie-Hellman group.

The no form of this command reverts to the default.



NOTE:

This command requires a PDN shutdown.

Default

```
no dh-group
```

Parameters

Diffie-Hellman group

specifies a value for the Diffie Hellman group, in bits

Values	1	1 to 768
–	2	2 to 1024
–	5	5 to 1536
–	14	14 to 2048
–	15	15 to 3072

–

19

19 to ECP 256

Default: none for ESP-SA, 2 for IKE-SA

6.2.1.36.7 encryption

Syntax

encryption *encryption algorithm*
no encryption

Context

config>mobile>profile>ipsec-profile>esp-sa>proposal-entry
config>mobile>profile>ipsec-profile>ike-sa>proposal-entry

Description

This command specifies the encryption algorithm to use for the IKE session.
The no form of this command removes the encryption algorithm from the configuration.

Parameters

encryption algorithm



NOTE:

- This command requires a PDN shutdown.
- This option is not applicable for IKE SA.

Values	none	Specifies that no encryption is used.
–	3des	Specifies the 3-DES algorithm for encryption. This is a modified application of the DES algorithm that uses multiple DES operations for more security.

- | | | |
|---|---------------------|--|
| – | aes128cbc (default) | Specifies the AES-CBC algorithm with 128 bits long key. This is the mandatory implementation size for AES. |
| – | aes192cbc | Specifies the AES-CBC algorithm with 192 bits long key. This is a stronger version of AES. |
| – | aes256cbc | Specifies the AES-CBC algorithm with 256 bits long key. This is the strongest available version of AES. |
| – | des | Specifies the DES algorithm for encryption. |
| – | aes128ctr | Specifies the AES-CTR algorithm with 128 bits long key. This is the mandatory implementation size for AES. |
| – | aes192ctr | Specifies the AES-CTR algorithm with 192 bits long key. This is a stronger version of AES. |
| – | aes256ctr | Specifies the AES-CTR algorithm with 256 bits long key. This is the strongest available version of AES. |

6.2.1.36.8 extended-seq-num

Syntax

[no] extended-seq-num

Context

config>mobile>profile>ipsec-profile>esp-sa>proposal-entry

Description

This command enables the use of extended sequence numbers.

The no form of this command disables extended sequence numbers.

Default

no extended-seq-num

6.2.1.36.9 integrity

Syntax

integrity authentication algorithm

no integrity

Context

config>mobile>profile>ipsec-profile>esp-sa>proposal-entry

config>mobile>profile>ipsec-profile>ike-sa>proposal-entry

Description

This command specifies the hashing algorithm to use for the authentication function.

The no form of this command removes the specified hashing algorithm from the configuration.



NOTE:

This command requires a PDN shutdown.

Parameters

authentication algorithm

Values		
md5		Configures the hash-algorithm to implement MD5 per RFC 1321
–	sha1	Specifies the hmac-sha1 algorithm for authentication
–	sha256	Specifies the sha256 algorithm for authentication
–	sha384	Specifies the sha384 algorithm for authentication
–	sha512	Specifies the sha512 algorithm for authentication

– aes-xcbc Specifies the aes-xcbc algorithm for authentication

Default: sha1

6.2.1.36.10 rekey-margin

Syntax

rekey-margin *percentage*

no rekey-margin

Context

config>mobile>profile>ipsec-profile>esp-sa

config>mobile>profile>ipsec-profile>ike-sa

Description

This system will attempt to re-key the SA once this percentage of the lifetime has expired.

The no form of this command reverts to the default.



NOTE:

This command requires a PDN shutdown.

Parameters

percentage

specifies the percentage of the lifetime to expire before the system attempts to re-key the SA

Values: 10 to 90

Default: 10

6.2.1.36.11 rekeying

Syntax

rekeying {enable | disable}

no rekeying

Context

config>mobile>profile>ipsec-profile>esp-sa

config>mobile>profile>ipsec-profile>ike-sa

Description

When this parameter is disabled, upon lifetime expiration the ePDG or N3IWF will delete the ESP SA. In this case, the UE has a period of 20 seconds to create a new ESP SA. If the UE does not do so in 20 seconds, the ePDG or N3IWF also deletes the IKE SA, and therefore its associated S2b and SWm sessions. Child SAs must also be deleted. When this command is disabled, the ePDG or N3IWF ignores the re-key margin logic. When this command is enabled, the ePDG or N3IWF makes use of the re-key percentage of lifetime logic and configured values (existing functionality). The rekey of emergency calls will use values from the ESP SA and IKE SA profiles for rekey margin and lifetime, independent of the setting specified for rekeying.



NOTE:

This command requires a PDN shutdown.

Parameters

enable

enables the re-key percentage of lifetime logic and configured values

disable

deletes the ESP SA/IKE SA sessions

Default: enable

6.2.1.36.12 replay-window

Syntax

replay-window *replay-window*

no replay-window

Context

config>mobile>profile>ipsec-profile>esp-sa

Description

This command enables the anti-replay service and establishes the window size. This command must be enabled if the ESP SA *nat-traversal-dynamic-updates* command is enabled.

The no form of this command disables the anti-replay service



NOTE:

This command requires a PDN shutdown.

Default

no replay-window

Parameters

replay-window

specifies the size of the anti-replay window. It is represented by values in geometric progression $32 * 2^n$ up to 1024 packets.

Values: 0 | 32 | 64 | 128 | 256 | 512 | 1024

Default: 0

6.2.1.36.13 ignore-ue-rekey-requests

Syntax

[no] ignore-ue-rekey-requests

Context

config>mobile>profile>ipsec-profile

Description

This command enables or disables the ePDG or N3IWF behavior when receiving UE initiated re-key requests for either IKE or ESP SA. When this command is enabled, the ePDG or N3IWF will not respond to the UE initiated received re-key requests, for either IKE or ESP SA. When this command is disabled, the ePDG or N3IWF will properly process and answer the UE initiated re-key requests for both IKE or ESP SA. To avoid breaking the existing Emergency active calls, the default behavior of the ePDG or N3IWF is to accept and answer UE initiated rekey requests for emergency calls. This is independent of the setting specified for the ignore-ue-rekey-requests.



NOTE:

This command requires a PDN shutdown.

Default

no ignore-ue-rekey-requests (disabled)

6.2.1.36.14 ike-sa

Syntax

```
ike-sa
```

Context

```
config>mobile>profile>ipsec-profile
```

Description

This command enables the context to configure the IKE SA parameters.

6.2.1.36.15 apn-black-white-list

Syntax

```
apn-black-white-list type type apn-list list-name [error-code error-code]
```

```
no apn-black-white-list
```

Context

```
config>mobile>profile>ipsec-profile>ike-sa
```

Description

This command configures the option to accept or reject a UE attached, based on the requested APN value. The command specifies the APN list, the type of APN screening to be done (black or white), and the error code to be sent on the UE when a call is rejected.



NOTE:

This command applies only to the ePDG type of untrusted non-3GPP access gateway and is not used when the IPSec profile is assigned using the `config>mobile>pdn>nwu>ipsec-profile` command.

The no form of this command disables the white-black-list functionality.

Default

no apn-black-white-list

Parameters

type

mandatory parameter specifies the type of the list (black or white) to be configured

Values: black | white

Default: white

apn-list

mandatory parameter specifies the name of the APN list to be configured for either the black or white list. The APN list or lists must be configured using the `apn-list` command in the `config>mobile>profile>list` context. The ePDG offers the available configured apn-list names. Up to 32 characters.

error-code

optional parameter specifies the value of the error code that the ePDG sends

Values: 1 to 16383

Default: 8192 (PDN_CONNECTION_REJECTION)

6.2.1.36.16 cert-entry

Syntax

[no] `cert-entry` *cert-entry-id*

Context

`config>mobile>profile>ipsec-profile>ike-sa`

Description

This command creates a certificate entry profile. The profile includes the ePDG or N3IWF certificate, the ePDG or N3IWF private key, and the number of certificates to be send when the chain is available and the URL base address used when the HASH_AND_URL method of the certificate exchange is used. Up to five certificate entry profiles can be configured.

The `no` form of this command removes a specific certificate entry profile from the configuration. When the `no shutdown` command is given, the system runs revocation and validation checks. The certificate entry profile operational status must be checked using the `show>mobile>profile>ipsec-profile` and viewing `ipsec-prof` and `operation state` fields. If the Operation State has a value different than Up (for example, Invalid Cert, Invalid Key, Key/Cert mismatch, or CRL/Chain invalid), the certificate entry profile creation has failed. Check the errors and debug the issue using the `show>log>log-id 101` command.

**NOTE:**

- Before configuring the certificate profile, it is recommended to check the certificate status using the `show>certificate>ca-profile` command. If the CA profile fields Oper State and Admin State is other than Up, the certificate import has failed. Check the `show>log>log-id 101` command for more information.
- The ePDG or N3IWF certificate functionality reuses a set of SR security and certificate commands. For more information, refer to the SR OS documentation.

Default

no cert-entry

Parameters

cert-entry-id

specifies the certificate profile entry ID

Values: 1 to 5

6.2.1.36.17 epdg-cert

Syntax

`epdg-cert epdg-cert-name`

`no epdg-cert`

Context

`config>mobile>profile>ipsec-profile>ike-sa>cert-entry`

Description

This command specifies the ePDG or N3IWF certificate name. The *epdg-cert-name* parameter value must be the ePDG or N3IWF certificate name defined via the output option of the `admin>certificate>import>type>cert` command which is used for the ePDG or N3IWF certificate import.

The no form of this command removes the ePDG or N3IWF certificate entry. If the certificate is removed from the certificate entry profile, the whole profile is not used by the ePDG or N3IWF.

Default

no epdg-cert

Parameters

epdg-cert-name

specifies the ePDG or N3IWF certificate name used during the import process, up to 95 characters

6.2.1.36.18 **priv-key**

Syntax

`priv-key epdg-priv-key`

`no priv-key`

Context

`config>mobile>profile>ipsec-profile>ike-sa>cert-entry`

Description

This command specifies the ePDG or N3IWF private key. The *epdg-priv-key* parameter value needs to be the ePDG or N3IWF private key name defined via the output option of the `admin>certificate>import>type>key` command, which is used for the ePDG or N3IWF private key import.

The no form of this command removes the private key entry. If the private key is removed from the certificate entry profile, the whole profile is not used by the ePDG or N3IWF.

Default

`no priv-key`

Parameters

epdg-priv-key

specifies the ePDG or N3IWF private key name used during the import, up to 95 characters

6.2.1.36.19 **send-chain**

Syntax

`send-chain number-of-chains`

`no send-chain`

Context

```
config>mobile>profile>ipsec-profile>ike-sa>cert-entry
```

Description

This command specifies the number of certificates to be sent when there is a match of the UE-requested certificate (via the CERTREQ IKE payload) and a configured certificate chain. When the `send-chain` command is configured with the value of 1, only the ePDG or N3IWF own certificate is transmitted to the UE.

The no form of this command reverts to the default.

Default

```
no send-chain
```

Parameters

number-of-chains

specifies the number of certificates to be sent

Values: 1 to 7

Default: 1

6.2.1.36.20 url-base-address

Syntax

```
url-base-address url
```

```
no url-base-address
```

Context

```
config>mobile>profile>ipsec-profile>ike-sa>cert-entry
```

Description

This command specifies the URL base address used by the ePDG or N3IWF when the certificates transmission is done based on the HASH_AND_URL method. The address is used for all possible certificates of a chain configuration.

The no form of this command reverts to the default.

Default

```
no url-base-address
```

Parameters

url

specifies the remote URL returned to the UE, up to 180 characters

6.2.1.36.21 certreq-eap-only-prefer-certreq

Syntax

```
[no] certreq-eap-only-prefer-certreq
```

Context

```
config>mobile>profile>ipsec-profile>ike-sa
```

Description

When this command is enabled under the specified IPsec profile, the ePDG or N3IWF considers only the CERTREQ IKE payload and ignores the EAP-ONLY payload, if the UE provides both payloads in the IKE_AUTH_REQUEST message.

The no form of this command returns to the default state, where the ePDG or N3IWF considers only the EAP-ONLY payload and ignores the CERTREQ IKE payload, if the UE provides both in the IKE_AUTH_REQUEST messages.

Default

```
no certreq-eap-only-prefer-certreq
```

6.2.1.36.22 device-identity

Syntax

```
[no] device-identity
```

Context

```
config>mobile>profile>ipsec-profile>ike-sa
```

Description

This command enables the context to configure the device identity for the IKE SA.

6.2.1.36.23 function

Syntax

```
function [enable | disable]
no function
```

Context

```
config>mobile>profile>ipsec-profile>ike-sa>device-identity
```

Description

This command enables/disables the device identity functionality. When the command is enabled, the ePDG or N3IWF enables the functionality of querying UE for its Identity IMEI or IMEISV.

The no form of this command disables the Device-Identity functionality.

**NOTE:**

This command requires a PDN shutdown.

Default

```
no function
```

Parameters

enable

enables the device identity functionality

disable

disables device identity functionality

6.2.1.36.24 reject-if-validation-fails

Syntax

```
[no] reject-if-validation-fails
```

Context

```
config>mobile>profile>ipsec-profile>ike-sa>device-identity
```

Description

This command allows the operator to decide if a failed validation of IMEI/IMEISV received from UE will result on call reject or call process. When the ePDG or N3IWF detects that a received IMEI or IMEISV validation has failed, and the command parameter is enabled, it rejects the call toward UE with code `invalid_syntax`. When this parameter is disabled, regardless of the validation outcome of the received IMEI or IMEISV, the call is processed. When the UE initiates an emergency call attach (initial or handover), the IMEI validation check is not performed.

The no form of this command allows processing of the call regardless of validation outcome of received IMEI or IMEISV.

Default

no reject-if-validation-fails

6.2.1.36.25 type

Syntax

```
type {imeisv | imei}
```

```
no type
```

Context

```
config>mobile>profile>ipsec-profile>ike-sa>device-identity
```

Description

This command configures the type of device identity asked from ePDG or N3IWF toward the UE.

The no form of this command sets the device identity type to the default value of `imeisv`.

Parameters

imeisv

sets the device identity type to IMEISV

imei

sets the device identity type to IMEI

Default: `imeisv`

6.2.1.36.26 dpd-interval

Syntax

```
dpd-interval seconds  
no dpd-interval
```

Context

```
config>mobile>profile>ipsec-profile>ike-sa
```

Description

This command enables dead peer detection for the ePDG or N3IWF using the specified interval. DPD is disabled if the value is 0.

The no form of this command reverts to the default.



NOTE:

- The DPD mechanism in the ePDG or N3IWF is active when there is no control plane (IKEv2) or user plane (IPsec) traffic detected from the UE. To avoid unnecessary internal as well as external signaling, the interval between DPD messages can be up to twice the configured value.
- This command requires a PDN shutdown.

Parameters

seconds

specifies the interval, in seconds, at which the ePDG or N3IWF will send DPD messages

Values: 0 | 60 to 3600

Default: 60

6.2.1.36.27 fragmentation

Syntax

```
fragmentation [ipv4-mtu bytes] [ipv6-mtu bytes]  
no fragmentation
```

Context

```
config>mobile>profile>ipsec-profile>ike-sa
```


Description

This command enables the use of IKE fragmentation as described in RFC 7383.

The no form of this command disables the IKE fragmentation.



NOTE:

This command requires a PDN shutdown.

Default

no fragmentation

Parameters

ipv4-mtu

specifies the IPv4 PDU size that governs the maximum size of an IKE fragment. It is the size after IP/UDP encapsulation. By default the IPv4 MTU is 576 bytes when IKE fragmentation is enabled.

Values: 176 to 4000

Default: 576

ipv6-mtu

specifies the IPv6 PDU size that governs the maximum size of an IKE fragment. It is the size after IP/UDP encapsulation. By default the IPv6 MTU is 1280 bytes when IKE fragmentation is enabled.

Values: 880 to 4000

Default: 1280

6.2.1.36.28 global-half-open-sa-limit

Syntax

global-half-open-sa-limit *global-half-open-sa-limit*

no global-half-open-sa-limit

Context

config>mobile>profile>ipsec-profile>ike-sa

Description

This command specifies the maximum number of global larval SA. When this threshold is exceeded, the ePDG or N3IWF will raise an alarm and send COOKIE challenge responses.

The no form of this command reverts to the default.



NOTE:

This command requires a PDN shutdown.

Parameters

global-half-open-sa-limit

specifies the threshold for the global larval SA

Values: 100 to 10000

Default: 2000

6.2.1.36.29 imsi-privacy-protection

Syntax

```
[no] imsi-privacy-protection
```

Context

```
config>mobile>profile>ipsec-profile>ike-sa
```

Description

This command enables support for Apple's IMSI privacy protection in the SWu interface.



NOTE:

This command applies only to the ePDG type of untrusted non-3GPP access gateway and is not used when the IPSec profile is assigned using the `config>mobile>pdn>nwu>ipsec-profile` command.

The no form of this command reverts to the default.

Default

```
no imsi-privacy-protection
```

6.2.1.36.30 local-identity

Syntax

```
local-identity ip-addr | fqdn
no local-identity
```

Context

```
config>mobile>profile>ipsec-profile>ike-sa
```

Description

This is the local identity that the ePDG or N3IWF will present to the UE.

The no form of this command removes the local identity configuration.



NOTE:

This command requires a PDN shutdown.

Parameters

ip-addr

specifies the IPv4 or IPv6 address for the local identity

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x
–	–	x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

fqdn

specifies the fully qualified domain name, up to 255 characters (up to 63 characters per label)

6.2.1.36.31 message-retransmit

Syntax

```
message-retransmit [timeout seconds] [retry-count num-retries]  
no message-retransmit
```

Context

```
config>mobile>profile>ipsec-profile>ike-sa
```

Description

This command sets retransmission and retries for IKEv2 control messages.

The no form of this command reverts to the default.



NOTE:

This command requires a PDN shutdown.

Parameters

seconds

specifies the number, in seconds the system waits before resending a message when there is no response

Values: 1 to 10

Default: 2

num-retries

specifies the number of times the system will resend a control message when there has been no response

Values: 0 to 10

Default: 3

6.2.1.36.32 mobike

Syntax

```
mobike [check-return-routability]  
no mobike
```

Context

```
config>mobile>profile>ipsec-profile>ike-sa
```

Description

This command enables MOBIKE support on the ePDG or N3IWF. This allows the UE to change the IP address for its end of the SA if it moves to a new network access point, without requiring the SA to be re-established.

The no form of this command disables the MOBIKE.



NOTE:

This command requires a PDN shutdown.

Default

no mobike

6.2.1.36.33 natt-mode

Syntax

```
natt-mode {auto | force}
```

```
no natt-mode
```

Context

```
config>mobile>profile>ipsec-profile>ike-sa
```

Description

This command enables the IKE NAT traversal procedures. When the ePDG or N3IWF detects the use of a NAT between itself and the UE it will start using UDP encapsulation, per RFC 3948. The system may also send periodic NAT keepalive requests, to maintain the binding in a NAT.

The no form of this command reverts to the default (disabled).



NOTE:

This command requires a PDN shutdown.

Parameters

auto

specifies that the ePDG or N3IWF only uses UDP encapsulation after it detects a NAT using the mechanism described by IKEv2

force

specifies that the ePDG or N3IWF always uses UDP encapsulation and sends NAT keepalive messages, without using NAT detection

Default: auto

6.2.1.36.34 peer-half-open-sa-limit

Syntax

peer-half-open-sa-limit *peer-half-open-sa-limit*

no peer-half-open-sa-limit

Context

config>mobile>profile>ipsec-profile>ike-sa

Description

This command specifies the maximum number of larval SA per peer. When this threshold is exceeded, the ePDG or N3IWF will raise an alarm and send COOKIE challenge responses.

The no form of this command reverts to the default.



NOTE:

This command requires a PDN shutdown.

Parameters

peer-half-open-sa-limit

specifies the threshold for the global larval SA

Values: 100 to 10000

Default: 200

6.2.1.36.35 plmn-black-white-list

Syntax

```
plmn-black-white-list
```

Context

```
config>mobile>profile>ipsec-profile>ike-sa
```

Description

This command enables the context to create and configure denylists and allowlists with UE-provided PLMNs and IDi realm values.

6.2.1.36.36 backoff-timer

Syntax

```
backoff-timer [timer-value timer-value] [timer-unit timer-unit]
```

```
no backoff-timer
```

Context

```
config>mobile>profile>ipsec-profile>ike-sa>plmn-black-white-list
```

Description

This command specifies the value of the backoff timer that the ePDG or N3IWF sends to the UE when rejecting a request.

The no form of this command reverts to the default.

Parameters

timer-value

configures the value of the backoff timer

Values: 0 to 31

Default: 5



NOTE:

A value of zero means that the UE may retry the authentication procedure to an ePDG or N3IWF from the same PLMN.

timer-unit

specifies the unit for the *timer-value* parameter

Values	deact	the value indicates that the timer is deactivated; the authentication must not retry the procedure to the same ePDG or N3IWF until the UE is switched off or the UICC containing the USIM is removed
–	2-s	the value is incremented in multiples of 2 seconds
–	30-s	the value is incremented in multiples of 30 seconds
–	1-min	the value is incremented in multiples of 1 minute
–	10-min	the value is incremented in multiples of 10 minutes
–	1-h	the value is incremented in multiples of 1 hour
–	10-h	the value is incremented in multiples of 10 hours
–	320-h	the value is incremented in multiples of 320 hours

Default: 1-min

6.2.1.36.37 enable-plmn-black-white-list

Syntax

[no] enable-plmn-black-white-list

Context

config>mobile>profile>ipsec-profile>ike-sa>plmn-black-white-list

Description

This command enables or disables the PLMN denylist/allowlist functionality. The `plmn-list` or the `idi-realm-list` command must be configured before enabling this command.

The `no` form of this command deactivates the functionality.

Default

no enable-plmn-black-white-list

6.2.1.36.38 error-code

Syntax

error-code *error-code*

no error-code

Context

config>mobile>profile>ipsec-profile>ike-sa>plmn-black-white-list

Description

This command specifies the value of the error code that the ePDG or N3IWF sends when rejecting a UE attach request.

The no form of this command reverts to the default.

Parameters

error-code

specifies the value of the error code that the ePDG or N3IWF sends

Values: 1 to 16383

Default: 11011 (PLMN_NOT_ALLOWED)

6.2.1.36.39 idi-realm-list

Syntax

idi-realm-list *list-name*

no idi-realm-list

Context

config>mobile>profile>ipsec-profile>ike-sa>plmn-black-white-list

Description

This command specifies the name of the IDi realm list that can be configured and checked by the ePDG, if the PLMN list is not configured or the ePDG received a PLMN from the IDi realm and did not match against the configured PLMN list. If a match is found, it is treated based on the screening type (black or white) configured in the `config>mobile>profile>ipsec-profile>ike-sa>plmn-black-white-list>type` command.

**NOTE:**

- The list name referenced in this command must be already configured under the `config>mobile>profile>list>idi-realm-list` command.
- This command applies only to the ePDG type of untrusted non-3GPP access gateway and is not used when the IPSec profile is assigned using the `config>mobile>pdn>nwu>ipsec-profile` command.

The no form of this command reverts to the default.

Parameters

list-name

specifies the name of the list, up to 32 characters

6.2.1.36.40 plmn-list

Syntax

```
plmn-list list-name
```

```
no plmn-list
```

Context

```
config>mobile>profile>ipsec-profile>ike-sa>plmn-black-white-list
```

Description

This command specifies the name of the PLMN list to be configured for either the denylist or the allowlist.

**NOTE:**

The list name referenced in this command must be already configured under the `config>mobile>profile>list>plmn` command.

The no form of this command reverts to the default.

Default

no plmn-list

Parameters

list-name

specifies the name of the list, up to 32 characters

6.2.1.36.41 type

Syntax

type *type*

no type

Context

config>mobile>profile>ipsec-profile>ike-sa>plmn-black-white-list

Description

This command specifies the type of the list (black or white). An allowlist (the type of the list is white) allows UE attach requests to go through if the incoming, PLMN or IDi-realm is matched to the configured list. If no match is found, the attach is rejected. A denylist (the type of the list is black) rejects UE attach requests when the incoming, PLMN or IDi-realm is matched against the configured list. If no match is found, the attach is allowed to go through.

The no form of this command reverts to the default.

Parameters

type

the type of the list

Values: black | white

Default: white

6.2.1.36.42 prf

Syntax

prf *prf algorithm*

no prf

Context

```
config>mobile>profile>ipsec-profile>ike-sa>proposal-entry
```

Description

This command specifies the Pseudo-Random-Function (PRF) algorithm for IKE-SA.

The no form of this command removes the specified hashing algorithm from the configuration.



NOTE:

This command requires a PDN shutdown.

Parameters

prf algorithm

Values	md5	Configures the hash-algorithm to implement MD5 per RFC 1321
–	sha1	Specifies the hmac-sha1 algorithm for authentication
–	sha256	Specifies the sha256 algorithm for authentication
–	sha384	Specifies the sha384 algorithm for authentication
–	sha512	Specifies the sha512 algorithm for authentication
–	aes-xcbc	Specifies the aes-xcbc algorithm for authentication

Default: sha1

6.2.1.36.43 redirect

Syntax

```
redirect
```

Context

```
config>mobile>profile>ipsec-profile>ike-sa
```

Description

This command enables the context for IKEv2 redirect configuration, including enabling and disabling the functionality and setting different types of redirection. Different IKEv2 redirection types can be configured and enabled at the same time.

6.2.1.36.44 always-on

Syntax

```
always-on server-list ike-redirect-server-list-name [established-sessions-redirect-  
rate] rate-value [redirect-established-emergency] [redirect-when-idle]  
  
no always-on
```

Context

```
config>mobile>profile>ipsec-profile>ike-sa>redirect
```

Description

This command enables of the IKEv2 redirect for all the sessions, new and already established. If the `always-on` type of redirection is enabled, it takes precedence over all the other types of redirect. The command also allows configuring the rate (UEs per second) at which the already established sessions will be redirected from the ePDG or N3IWF to indirectly control the number of attaches toward the new ePDGs or N3IWFs. The behavior of the ePDG or N3IWF, for the already established emergency call sessions, can be configured to not redirect (and wait for emergency calls completion before redirecting a session) or redirect emergency calls (dropping the call).

The `no` form of this command disables the `always-on` functionality.

Default

```
no always-on
```

Parameters

ike-redirect-server-list-name

specifies the list name of the IKE redirect server list that is already configured in the `config>mobile>profile>list>ike-redirect` context, up to 32 characters

**NOTE:**

An empty `ike-redirect-server-list` is not allowed to be configured in the `server-list` parameter.

rate-value

specifies the redirection rate value of the established sessions in UEs per second

Values: 100 to 2000

Default: 500

redirect-established-emergency

specifies the redirection of the established emergency calls

redirect-when-idle

specifies if a UE is redirected only when its sessions are idle (no active dedicated bearers)

**NOTE:**

The option `redirect-when-idle` cannot be activated if the `always-on` redirect type is already active. The `always-on` must be deactivated and then the `always-on` with the `redirect-when-idle` options can be activated. The same logic applies when the `always-on` with the `redirect-when-idle` options are active and the `redirect-when-idle` option must be deactivated. The existing `always-on` with `redirect-when-idle` options must be deactivated and then the `always-on` option can be activated.

6.2.1.36.45 **idi-realm**

Syntax

```
idi-realm idi-realm-list-name type-of-list-screening {white white-list-redir-epdg  
{ipv4 | ipv6 | fqdn} ipv4 | ipv6 | fqdn | black}  
no idi-realm
```

Context

```
config>mobile>profile>ipsec-profile>ike-sa>redirect
```

Description

This command allows the ePDG to redirect a UE request based on the realm part of the provided IDi. The list of the realms must already be configured under the `config>mobile>profile>list>idi-realm` context. The type of the IDi realm list screening is also configurable. If there is a matched redirection

action based on both the IDi realm and the source IP address, the IDi realm-based redirection will take precedence.



NOTE:

This command applies only to the ePDG type of untrusted non-3GPP access gateway and is not used when the IPSec profile is assigned using the `config>mobile>pdn>nwu>ipsec-profile` command.

The no form of this command disables the IDi realm redirect functionality.

Default

no idi-realm

Parameters

idi-realm-list-name

specifies the list name of an IDi realm list that is already configured, up to 32 characters

type-of-list-screening

specifies the type of the IDi realm list screening

Values	white	if there is a match found between the received IDi and the IDi realm list, the UE request is not redirected and the call is processed as normal; When the type-of-list-screening is set to white then the CLI performs a check on the applied list (idi-realm-list-name) and: • if all entries on the list have the same type and value, the CLI allows the execution of the command • if all entries do not have the same type and value, the CLI rejects the command with the error Cannot accept redirect list, list records does not contain same type and value
–	black	if there is a match found between the received IDi and the IDi realm list, the UE request is immediately redirected
Default: white		

6.2.1.36.46 overload-redirect

Syntax

```
overload-redirect server list ike-redirect-server-list-name [redirect-  
established-emergency]  
  
no overload-redirect
```

Context

```
config>mobile>profile>ipsec-profile>ike-sa>redirect
```

Description

This command allows the redirection based on the ePDG or N3IWF overload conditions. The behavior of the ePDG or N3IWF for the already established emergency call sessions, can be configured to not redirect (and wait for emergency calls completion before redirecting a session) or redirect emergency calls (dropping the call).

The no form of this command disables the overload-redirect functionality.

Default

no overload-redirect

Parameters

ike-redirect-server-list-name

specifies the IKE redirect server list name, up to 32 characters



NOTE:

An empty `ike-redirect-server-list` is not allowed to be configured in the `server-list` parameter.

redirect-established-emergency

specifies the redirection of emergency calls

6.2.1.36.47 source-ip

Syntax

```
source-ip source-ip-list-name type-of-list-screening {white white-list-redir-epdg  
{ipv4 | ipv6 | fqdn} ipv4 | ipv6 | fqdn | black}]
```


no source-ip

Context

config>mobile>profile>ipsec-profile>ike-sa>redirect

Description

This command allows the redirection based on the source IP address of the incoming UE-originated IKE packets. The list of the realms must already be configured under the config>mobile>profile>list>source-ip context. The type of the source IP address list screening is also configurable.

The no form of this command disables the source IP address redirect functionality.

Default

no source-ip

Parameters

source-ip-list-name

specifies the source IP address list name, up to 32 characters

type-of-list-screening

specifies the type of the IDi realm list screening

Values		
white		if there is a match found between the received source IP address and the entries of the list, the UE request is not redirected and the call is processed as normal; When the type-of-list-screening is set to white then the CLI performs a check on the applied list (idi-realm-list-name) and: • if all entries on the list have the same type and value, the CLI allows the execution of the command • if all entries do not have the same type and value, the CLI rejects the command with the error Cannot accept redirect list, list records does not contain same type and value
–	white-list-redir-epdg	it allows to define the redirect ePDG or N3IWF information that is provided to the UE when the type

of list screening is white and there is no match found.
type and value must be configured

– black
if there is a match found between the received IDi
and the IDi realm list, the UE request is immediately
redirected

Default: white

6.2.1.37 List profile command descriptions

6.2.1.37.1 list

Syntax

list

Context

config>mobile>profile

Description

This command enables the context to configure or delete mobile gateway lists.

6.2.1.37.2 aaa-host-realm

Syntax

[no] aaa-host-realm *list-name*

Context

config>mobile>profile>list

Description

This command configures an AAA host realm list that can be used in the node identifier functionality of the S6b interface.

Parameters

list-name

specifies the list name, up to 32 characters

6.2.1.37.3 id

Syntax

```
id id realm realm-string host host-string diameter-peer-group-list group-list-name  
no id id
```

Context

config>mobile>profile>list>aaa-host-realm

Description

This command configures combinations of realms, hosts, and Diameter peer group lists. The Diameter peer group list must be an existing list configured using the config>mobile-gateway>profile>diameter-peer-group-list command.

Parameters

id

specifies the ID of the host realm entry

Values: 1 to 512

realm-string

specifies the realms FQDN name of the AAA network element, up to 255 characters

host-string

specifies the host FQDN name of the AAA network element, up to 255 characters

group-list-name

specifies the name of the Diameter peer group list to be used, up to four entries of Diameter peer groups are allowed; up to 32 characters

6.2.1.37.4 allowed-nf-domains-list

Syntax

```
[no] allowed-nf-domains-list list-name
```

Context

config>mobile>profile>list

Description

This command configures an allowed NF domain list.

The no form of this command removes the specified list from the configuration.

Parameters

list-name

specifies the list name, up to 32 characters

6.2.1.37.5 allowed-nf-domain

Syntax

[no] allowed-nf-domain *string*

Context

config>mobile>profile>list>allowed-nf-domains-list

Description

This command configures an allowed NF domain in the allowed NF domain list.

The no form of this command removes the specified allowed NF domain from the list.

Parameters

string

specifies the value of the allowed NF domain or the domain pattern, up to 32 characters

6.2.1.37.6 apn-list

Syntax

[no] apn-list *list-name*

Context

config>mobile>profile>list

Description

This command configures an Access Point Name (APN) list. An APN list consists of multiple APN names. APN list can be referenced by modules that use match criteria as per APN name, for example, S5 selection based on APN.

The no form of this command removes the list name from configuration.

Parameters

list-name

specifies the list name of the APN, up to 32 characters

6.2.1.37.7 apn

Syntax

[no] apn *apn-name*

Context

config>mobile>profile>list>apn

Description

This command configures Access Point Name (APN) name in APN list.

The no form of this command removes the APN name from APN list configuration.

Parameters

apn-name

specifies the APN name, up to 80 characters

6.2.1.37.8 arp-list

Syntax

[no] arp-list *arp-list-name*

Context

config>mobile>profile>list

Description

This command configures an ARP list. An ARP list consists of maximum 15 ARP values. These values can later be referenced by modules that use match criteria as per ARP value. For example, match criteria can exclude local node-level overload throttling based on ARP values.

The no form of this command removes the list name from the configuration.

Default

no arp-list *arp-list-name*

Parameters

arp-list-name

specifies the ARP name, up to 32 characters

6.2.1.37.9 arp

Syntax

[no] arp *arp-value* [message-priority *msg-priority*] [gtp-message-priority *gtp-msg-priority*]

Context

config>mobile>profile>list>arp-list

Description

This command configures an ARP value in the ARP list. If the mobile gateway is configured as CP function, the message priority can be optionally set; this is sent in the PFCP message header of the PFCP Session Establishment request, the PFCP Session Modification request, and the PFCP Session Deletion request to translate the ARP value to the PFCP message priority value.

The no form of this command removes the ARP value (and message priority if configured/applicable) from the ARP list.

Default

no arp *arp-value*



NOTE:

- The optional gtp-message-priority is currently used as custom only feature, and the same gtp-message-priority value must be only configured once in the list.

- To configure the arp-to-message-priority-list, the same value for both the arp and gtp-message-priority must be linked only to one message-priority (PFCP) value.

Parameters

arp-value

specifies the ARP value

Values: 1 to 15

msg-priority

specifies the message priority associated with the ARP value

Values: 0 to 15

Default: 0

gtp-msg-priority

specifies the GTPv2 message priority associated with the ARP value (custom feature only)

Values: 0 to 15

6.2.1.37.10 broadcast-plmn-list

Syntax

```
[no] broadcast-plmn-list list-name
```

Context

```
config>mobile>profile>list
```

Description

This command configures the name for the broadcast PLMN list.

The no form of this command removes the list name from the configuration.

Parameters

list-name

specifies the broadcast PLMN list name, up to 32 characters

6.2.1.37.11 broadcast-plmn-item

Syntax

```
broadcast-plmn-item bplmn-id tai-slice-support-list slice-list-name plmn-mcc mcc value  
plmn-mnc mnc value  
  
no broadcast-plmn-item bplmn-id
```

Context

```
config>mobile>profile>list>broadcast-plmn-list
```

Description

This command configures the broadcast PLMN item. Each broadcast PLMN item consists of a PLMN ID (MCC +MNC values) and a TAI slice support list. The `slice-instances` parameter of the slice list name assigned to the TAI slice support list, is not considered for the broadcast PLMN list.

The no form of this command removes the broadcast PLMN item from the broadcast PLMN list.

Parameters

bplmn-id

specifies the ID for the broadcast PLMN item

Values: 1 to 12

slice-list-name

specifies the name of the slice list (the slice list must be already configured under the `config>mobile>profile>list>slice-list` context), up to 32 characters

mcc value

specifies the MCC value

Values: 000 to 999

mnc value

specifies the MNC value

Values: 000 to 999 (2 or 3 digit number)

6.2.1.37.12 cell-id-list

Syntax

```
[no] cell-id-list cell-id-list-name
```


Context

config>mobile>profile>list

Description

This command creates a new profile list of cell-id values. The list can include NCGI values and NCGI ranges. Other cell identities can be added in the future (for example, CGI, ECGI).

The no form of this command removes the specified list from the configuration.

Parameters

cell-id-list-name

specifies the cell-id profile list name, up to 32 characters

6.2.1.37.13 ncgi

Syntax

[no] ncgi mcc *mcc-value* mnc *mnc-value* nci *nci-value*

[no] ncgi mcc *mcc-value* mnc *mnc-value* nci-range-start *nci-range-start-value* nci-range-end *nci-range-end-value*

Context

config>mobile>profile>list>cell-id-list

Description

This command adds a new NCGI and NCGI range to the specified cell-id list profile.

The no form of this command removes the specified entry from the list.

Parameters

mcc-value

specifies the MCC value of the NCGI

Values: 000 to 999

mnc-value

specifies the MNC value of the NCGI

Values: 00 to 999

nci-value

specifies the NR cell-id value of the NCGI, 9-digit hexadecimal number

nci-range-start-value

specifies the start value of the NR cell-id range, 9-digit hexadecimal number

nci-range-end-value

specifies the end value of the NR cell-id range, 9-digit hexadecimal number

6.2.1.37.14 fqdn-list

Syntax

[no] fqdn *fqdn-list-name*

Context

config>mobile>profile>list

Description

This command configures an FQDN list. An FQDN list consists of multiple FQDN names and it is referenced by the dns-fqdn-list command in config>mobile>profile>policy-options>policy-rule-unit>flow-description>match context.

The no form of this command removes the list name from the configuration.

Parameters

fqdn-list-name

specifies the FQDN list name of the APN, up to 32 characters

6.2.1.37.15 fqdn

Syntax

fqdn *fqdn-name-id fqdn-name-string*

no fqdn *fqdn-name-id*

Context

config>mobile>profile>list>fqdn-list

Description

This command configures FQDNs name in an FQDN list.

The no form of this command removes the FQDN name from the FQDN list configuration.

Parameters

fqdn-name-id

specifies the FQDN name ID

Values: 1 to 2000

fqdn-name-string

specifies the FQDN name string, up to 255 characters

6.2.1.37.16 idi-realm-list

Syntax

```
[no] idi-realm-list list-name
```

Context

```
config>mobile>profile>list
```

Description

This command creates an IDi realm list. A maximum of 512 entries per list is allowed.

The no form of this command removes the list from the configuration.

Parameters

list-name

specifies the name of the list to be created, up to 32 characters

6.2.1.37.17 idi-realm

Syntax

```
idi-realm idi-realm-id realm-name [redirected-epdg {ipv4 | ipv6 | fqdn}] [ipv4 / ipv6 / fqdn]
```

```
no idi-realm idi-realm-id
```

Context

```
config>mobile>profile>list>idi-realm-list
```

Description

This command specifies the realm ID and the realm name. The IDi realms that are used for IKE-Redirect or plmn-black-white-list based screening. When list is configured for applying in black type of IKE-Redirect then it is mandatory that redirected-ePDG option is also configured (in the form of IPv4 or IPv6 or FQDN). When the list is configured for applying in plmn-black-white-list based screening or IKE-redirect allow list, the redirected ePDG option is not configured. Since a UE can indicate an emergency call by inserting the SOS label as part of its IDi realm, it is highly recommended to not configure a realm value with an SOS label. A specific list is only assigned to one functionality, either the ike-redirect or the plmn-black-white-list.

The no form of this command removes a specific entry of the list.

Parameters

idi-realm-id

specifies the IDi realm ID

Values: 1 to 512

realm-name

specifies the realm name, up to 256 characters

redirected-epdg

Specifies the type (IPv4 or IPv6 or FQDN) and the value of the ePDG server. This is an optional parameter and is used only when the defined list is configured for use in the black type of the IDi-Realm IKE redirect functionality.

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D
–	fqdn	FQDN, up to 255 characters

Default: IPv4

6.2.1.37.18 ike-redirect-server-list

Syntax

```
[no] ike-redirect-server-list list-name
```

Context

```
config>mobile>profile>list
```

Description

This command creates an IKE redirect server list. A maximum of 20 entries per list is allowed.

The no form of this command removes the list from the configuration.

Parameters

list-name

specifies the name of the IKE redirect server list, up to 32 characters

6.2.1.37.19 server

Syntax

```
server [1..20] redirected-epdg {ipv4 | ipv6 | fqdn} ipv4 / ipv6 / fqdn
```

```
no server [1..20]
```

Context

```
config>mobile>profile>list>ike-redirect-server-list
```

Description

This command creates a redirect server.

The no form of this command removes the list from the configuration.

Parameters

1..20

specifies the server ID

redirected-epdg

Specifies the type (IPv4 or IPv6 or FQDN) and the value of the ePDG server. This is an optional parameter and is used only when the defined list is configured for use in the black type of the IDi-Realm IKE redirect functionality.

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D
–	fqdn	FQDN, up to 255 characters

6.2.1.37.20 ike-redirect-source-ip-list

Syntax

[no] ike-redirect-source-ip-list *list-name*

Context

config>mobile>profile>list

Description

This command creates an IKE redirect source IP address list. A maximum of 512 entries per list must be allowed. Single IP address or IP address range configuration overlap is not allowed. The CLI performs a check when the operator tries to modify a list that is already assigned to a white type list. In this case, it is allowed to add an item (IP address) that can have the same type and redirect the ePDG value as the existing entries. If this is not the case, the CLI returns an error that the list is used as an allowlist and thus it must meet this restriction.

The no form of this command removes list from the configuration.

Parameters

list-name

specifies the name of the the IKE redirect source IP address list, up to 32 characters

6.2.1.37.21 ip

Syntax

```
ip [1..512] address address redirected-epdg {ipv4 | ipv6 | fqdn} ipv4 / ipv6 / fqdn  
ip [1..512] start-address start-ip address end-address ip-address redirected-epdg {ipv4 |  
ipv6 | fqdn} ipv4 / ipv6 / fqdn  
no ip [1..512]
```

Context

```
config>mobile>profile>list>ike-redirect-source-ip-list
```

Description

This command creates a source IP address. Allowed values are single IPv4 or IPv6 addresses or IP address range.
The no form of this command removes the specified entry from the list.

Parameters

- 1 to 512**
specifies the IP address ID
- address**
specifies the IP address

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

redirected-epdg

Specifies the type (IPv4 or IPv6 or FQDN) and the value of the ePDG server. This is an optional parameter and is used only when the defined list is configured for use in the black type of the IDi-Realm IKE redirect functionality.

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D
–	fqdn	FQDN, up to 255 characters

Default: IPv4

start-ip address

specifies the source start IP address

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

ip address

specifies the source end IP address

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

6.2.1.37.22 ip-address-list

Syntax

[no] ip-address-list *list-name*

Context

config>mobile>profile>list

Description

This command creates an IP address list.

The no form of this command removes the specified list from the configuration.

Parameters

list-name

specifies the name of the IP address list, up to 32 characters

6.2.1.37.23 address

Syntax

[no] address *ip-address*

Context

config>mobile>profile>list>ip-address-list

Description

This command configures an IP address for the specified IP address list.

The no form of this command removes the IP address from the list.

Parameters

ip-address
specifies the IP address

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

6.2.1.37.24 nf-id-list

Syntax

[no] nf-id-list *list-name*

Context

config>mobile>profile>list

Description

This command configures an NF ID list.

The no form of this command removes the specified list from the configuration.

Parameters

list-name
specifies the name of the NF ID list, up to 32 characters

6.2.1.37.25 **nf-prof-id**

Syntax

[no] **nf-prof-id** *nf-id*

Context

config>mobile>profile>list>nf-id-list

Description

This command adds a new NF instance to the list of NF peers.

The no form of this command removes the NF instance from the list.

Parameters

nf-id

specifies the internal ID of the NF peer

Values: 1 to 4294967295

6.2.1.37.26 **api-prefix**

Syntax

api-prefix *api-prefix-value*

no **api-prefix**

Context

config>mobile>profile>list>nf-id-list>nf-prof-id

Description

This command specifies the API prefix to be used after the authority part of the API URI endpoint.

The no form of this command removes the API prefix.

Parameters

api-prefix-value

specifies the API prefix, up to 64 characters

may contain more than one path component separated by a slash “/” character; for example, “nokia/3gpp/services”

6.2.1.37.27 api-version

Syntax

```
api-version version [minor minor] [patch patch]  
no api-version
```

Context

```
config>mobile>profile>list>nf-id-list>nf-prof-id
```

Description

This command specifies the API version of the remote NF peer. It is relevant in cases where the related API supports multiple major versions, or the minor and patch versions have impact to the API usage.

The no version of the command reverts to the default value.

Parameters

version

specifies the major API version

Values: 0 to 255

Default: 0 (highest supported API version)

minor

specifies the minor API version

Values: 0 to 255

Default: 0

patch

specifies the patch API version

Values: 0 to 255

Default: 0

6.2.1.37.28 enable

Syntax

```
[no] enable
```

Context

```
config>mobile>profile>list>nf-id-list>nf-prof-id
```

Description

This command enables the NF profile. To enable the NF profile, the UUID of the NF profile must be configured (using the `config>mobile>profile>list>nf-id-list>nf-prof-id>uuid` command).

The no version of the command disables the NF profile.

Default

no enable

6.2.1.37.29 fqdn

Syntax

```
fqdn fqdn-sni-string
```

```
no fqdn
```

Context

```
config>mobile>profile>list>nf-id-list>nf-prof-id
```

Description

This command configures an FQDN for the NF instance. The FQDN is used as the SNI for the client services.

The no form of this command removes the FQDN.

Parameters

fqdn-sni-string

specifies the FQDN name string, up to 255 characters

6.2.1.37.30 nf-priority

Syntax

```
nf-priority priority-value
```

```
no nf-priority
```

Context

config>mobile>profile>list>nf-id-list>nf-prof-id

Description

This command configures the priority associated with the specific NF instance.

The no form of this command removes the priority from the configuration.

Parameters

priority-value

specifies the priority

Values: 0 to 65535

Default: 65535

6.2.1.37.31 nf-set

Syntax

nf-set *list-name*

no nf-set

Context

config>mobile>profile>list>nf-id-list>nf-prof-id

Description

This command configures the name of the NF ID list containing the NFs that belong to the NF set.

Parameters

list-name

specifies the NF ID list name

6.2.1.37.32 peer-selection

Syntax

peer-selection {enabled | disabled}

no peer-selection

Context

config>mobile>profile>list>nf-id-list>nf-prof-id

Description

This command configures whether a configured peer is enabled for selection, or it is manually disabled. This applies only to NRF peers.

Parameters

enabled

specifies that the peer is available for selection

disabled

specifies that the peer is disabled

Default: enabled

6.2.1.37.33 prioritized-address-list

Syntax

prioritized-address-list *list-name*

no prioritized-address-list

Context

config>mobile>profile>list>nf-id-list>nf-prof-id

Description

This command associates a configured list to be the prioritized address list for this NF instance. The lists are configured using the config>mobile>profile>list>prioritized-ip-address-list command.

Parameters

list-name

specifies the IP address list name, up to 32 characters

6.2.1.37.34 uuid

Syntax

```
uuid uuid  
no uuid
```

Context

```
config>mobile>profile>list>nf-id-list>nf-prof-id
```

Description

This command adds the NF instance ID in UUIDv4 canonical textual representation.

Parameters

uuid

specifies the NF instance ID

Values: xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx, x: [0..f]H

6.2.1.37.35 oi-override-mapping

Syntax

```
[no] oi-override-mapping list-name
```

Context

```
config>mobile>profile>list
```

Description

This command creates an Operator Identifier (OI) override mapping list.

The no form of this command removes the specified list from the configuration.

Parameters

list-name

specifies the name of the PLMN ID mapping list, up to 32 characters

6.2.1.37.36 subscriber-mcc

Syntax

```
subscriber-mcc mcc value subscriber-mnc mnc value {[replacement-mcc mcc value  
replacement-mnc mnc value] [serving-network-mcc mcc value serving-network-mnc mnc  
value]}
```

```
no subscriber-mcc mcc value subscriber-mnc mnc value
```

Context

```
config>mobile>profile>list>oi-override-mapping
```

Description

The command configures the mapping pairs of subscriber PLMN ID (MCC/MNC) and PLMN ID replacement (MCC/MNC), and/or serving network ID replacement (MCC/MNC). The subscriber PLMNs must be already configured under config>mobile>pdn>home-plmn-list (up to 1000 entries per list). The no form of this command deletes an entry from the list.

Parameters

mcc value

specifies the subscriber MCC or replacement MCC value or serving network MCC value

Values: 000 to 999

mnc value

specifies the subscriber MNC or replacement MNC value or serving network MNC value

Values: 00 to 999 - 2 or 3 digit number

6.2.1.37.37 pgw-selection-candidate-list

Syntax

```
[no] pgw-selection-candidate-list list-name
```

Context

```
config>mobile>profile>list
```

Description

This command creates a container for a list of PGW candidates. This profile may be applied to the entity that must select a set of PGWs.

The no form of this command removes the profile.

Parameters

list-name

specifies the name of the pgw-selection-candidate-list profile

6.2.1.37.38 candidate

Syntax

```
candidate ipv4 | ipv6 | fqdn [priority priority-value] [weight weight-value] [ref-pt {s2a | s2b}]  
no candidate ipv4 | ipv6 | fqdn
```

Context

```
config>mobile>profile>list>pgw-sel-cand-list
```

Description

This command identifies a PGW and optionally associates a priority, weight and reference-point.

The no form of this command removes the PGW.

Parameters

ipv4

the PGW is identified by an IPv4 address

ipv6

the PGW is identified by an IPv6 address

fqdn

the PGW is identified by an FQDN

priority *priority-value*

the priority of the PGW in the candidate list; the peer with the lowest-numbered priority is selected first

Values: 0 to 65535

Default: 0

weight *weight-value*

the weight of the PGW in the candidate list; among peers with the same priority, the peer with a higher-numbered weight is more preferable

Values: 0 to 65535

Default: 0

ref-pt s2a

specifies that the PGW is a candidate for S2a selection

ref-pt s2b

specifies that the PGW is a candidate for S2b selection

Default: s2b

6.2.1.37.39 plmn

Syntax

[no] p lmn *list-name*

Context

config>mobile>profile>list

Description

This command configures a PLMN list. A PLMN list consists of a list of MCC and MNC IDs. The PLMN list is changed without PDN shutdown. If the PLMN configuration changes lead to the roaming status change of a UE, the DNS is configured properly so that the MME resolves the right PGW, matching up the roaming status. On the PGW, different IP addresses are configured for S5 and S8 interfaces and are published in the DNS for this purpose. The same stands for the Gn/Gp interfaces.

The no form of this command removes the list name from the configuration.

Parameters

list-name

specifies the name of this PLMN list, up to 32 characters

6.2.1.37.40 mcc

Syntax

[no] mcc *mcc-value* mnc *mnc-value*

Context

config>mobile>profile>list>plmn

Description

This command specifies the MCC and MNC for this PLMN.

The no form of this command removes the values from the configuration.

Parameters

mcc-value

specifies the MCC of this PLMN

Values: 000 to 999

mnc-value

specifies the MNC of this PLMN

Values: 00 to 999

6.2.1.37.41 port-list

Syntax

[no] port - list *list-name*

Context

config>mobile>profile>list

Description

This command configures a list of ports that can be used for a more advanced filtering in PCC rules.

The no form of command removes the list name from the configuration.

Default

no port-list

Parameters

list-name

specifies the name of the port list profile, up to 32 characters

6.2.1.37.42 port

Syntax

[no] port *port-number*

[no] port *`1range start-port-number end-port-number*

Context

config>mobile>profile>list>port-list

Description

This command specifies the TCP or UDP port number to use in the port list definition.

The no form of this command restores the default by removing the port number/range from the port list.

Default

no port

Parameters

port-number

specifies the port number

Values: 0 to 65535

start-port-number

specifies the start port number if a range is used

Values: 0 to 65534

end-port-number

specifies the end port number if a range is used

Values: 1 to 65535

6.2.1.37.43 prioritized-ip-address-list

Syntax

[no] prioritized-ip-address-list *list-name*

Context

config>mobile>profile>list

Description

This command creates a new prioritized IP address list.

The no form of this command removes the configured list.

Parameters

list-name

specifies the name of the prioritized IP address list, up to 32 characters

6.2.1.37.44 address

Syntax

address {ip-address / fqdn} [priority value] [port port]

no address {ip-address / fqdn} [port port]

Context

config>mobile>profile>list>prioritized-ip-address-list

Description

This command configures an IP address in the configured prioritized IP address list. The address can optionally be assigned with a priority.

The no form of this command removes the IP address from the list.

Parameters

ip-address

specifies the IP address

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H

– – d - [0..255]D

fqdn

specifies the FQDN, up to 255 characters

value

specifies the priority

Values: 0 to 8

Default: 8

port

specifies the port

Values: 0 to 65535

Default: 80

6.2.1.37.45 qci-list

Syntax

[no] qci-list *list-name*

Context

config>mobile>profile>list

Description

This command configures a list of QCI values that can be used across features that require QCI lists, for example, drop statistics per QCI.

Default

no qci-list

Parameters

list-name

specifies the name of the QCI list profile, up to 32 characters

6.2.1.37.46 qci

Syntax

[no] qci *qci-value*

Context

config>mobile>profile>list>qci-list

Description

This command configures QCI values for a QCI list.

Default

no qci

Parameters

qci-value

specifies the QCI values for the QCI list

Values: 1 to 254, *-wildcard



NOTE:

The wildcard value is not allowed when you configure a QCI list under the qci-apn-stats context. Doing so, an error indicating the same is displayed.

6.2.1.37.47 rat-type-list

Syntax

rat-type-list *rat-type-list-name*

no rat-type-list *rat-type-list-name*

Context

config>mobile>profile>list

Description

This command configures RAT type list that consists of multiple RAT type values. This list can be referenced by modules that use match criteria as per RAT type value, for example, virtual APN selection.

The no form of this command removes the list name from the configuration.

Default

no rat-type-list

Parameters

rat-type-list-name

specifies the RAT type list name

Values: rat-type

6.2.1.37.48 rat-type

Syntax

[no] rat -type

Context

config>mobile>profile>list>rat-type-list

Description

This command configures a RAT type value in a RAT type list.

The no form of this command removes the RAT type value in the RAT type list.

Parameters

keyword

specifies the rat-type value

Values: utran | geran | wlan | gan | hspa-e | eutran | virtual | eutran-nb | lte-m | nr | ehrpd |
hrpd | cdma-1x | umb | wifi | wbbf-a

Default: utran

6.2.1.37.49 slice-instance-list

Syntax

[no] slice-instance-list *list-name*

Context

```
config>mobile>profile>list
```

Description

This command configures a slice instance list. The slice instance list can be referenced from a slice list.

The no form of this command removes the configured slice instance list.

Parameters

list-name

specifies the name of the slice instance list, up to 32 characters

6.2.1.37.50 slice-instance

Syntax

```
slice-instance name [suspend {enable | disable}]
```

```
no slice-instance name
```

Context

```
config>mobile>profile>list>slice-instance-list
```

Description

This command configures a slice instance in the configured slice instance list. The suspend parameter is used to put the slice instance to suspended state. While a slice instance is in suspended state, the gateway rejects new PDU sessions on this slice instance from being established, while continuing to serve existing sessions on this slice.

The no form of this command removes the slice instance from the list.

Parameters

name

specifies the name of the slice instance, up to 128 characters

suspend

specifies a slice instance to suspended state

Values: enable | disable

Default: disable

6.2.1.37.51 slice-list

Syntax

```
[no] slice-list list-name
```

Context

```
config>mobile>profile>list
```

Description

This command configures a slice list. The slice list created can be referenced from the PDN-level configuration.

**NOTE:**

Up to 120 slice lists can be configured. Each list can have maximum 100 slices. The total slices for all slice lists cannot be more than 1024.

If the `slice-list` is configured under PDN/APN (reported to NRF), it cannot have more than 100 slices due to the size of the message.

The `no` form of this command removes the configured slice list.

Parameters

list-name

specifies the name of the slice list, up to 32 characters

6.2.1.37.52 slice

Syntax

```
slice name sst sst-value [sd sd-value]
```

```
no slice name
```

Context

```
config>mobile>profile>list>slice-list
```

Description

This command configures a slice in the configured slice list. The slice must be associated to a SST. Optionally, the slice is associated with a Service Differentiator (SD). SD is optional information that complements the SST to differentiate among multiple network slices of the same SST.

**NOTE:**

Up to 120 slice lists can be configured. Each list can have maximum 100 slices. The total slices for all slice lists cannot be more than 1024.

If the `slice-list` is configured under PDN/APN (reported to NRF), it cannot have more than 100 slices due to the size of the message.

The `no` form of this command removes the slice from the list.

Parameters

name

specifies the name of the slice, up to 32 characters

sst-value

specifies the SST identifier for the created slice

Values: 0 to 255 (integer)

standardized SSTs are found in the following table, values from 128 to 255 correspond to operator specific SSTs:

- 1 - eMBB
- 2 - URLLC
- 3 - MIoT

sd-value

specifies the SD identifier for the created slice, representing the SD, a 3 octet string in hexadecimal representation

Values: 6-byte string, allowed characters 0 to 9 or A to F

For example, 0xD143A5 is encoded to string as "D143A5".

6.2.1.37.53 slice-instances

Syntax

```
slice-instances list-name
```

```
no slice-instances
```

Context

config>mobile>profile>list>slice-list>slice

Description

This command configures a list of slice instances which are instantiated for the specific slice. A slice instance can belong to lists of more than one slice.

The no form of this command removes the slice instances list.

Parameters

list-name

specifies the name of the referenced slice instances list, up to 32 characters

Values: configured slice instance lists under the config>mobile>profile>list>slice-instance-list context

6.2.1.37.54 suspend

Syntax

[no] suspend

Context

config>mobile>profile>list>slice-list>slice

Description

This command sets the slice to suspended state. While a slice is in suspended state, the gateway rejects new PDU sessions on this slice from being established, while continuing to serve existing sessions on this slice.

The no form of this command removes the slice from suspended state.

Default

no suspend

6.2.1.37.55 tai-list

Syntax

tai-list *list-name*

[no] tai-list

Context

config>mobile>profile>list>slice-list>slice

Description

This command defines the TAI list of the S-NSSAI. When the UDM returns the subscribed S-NSSAI of the APN, the PGW-C selects the first S-NSSAI which is available in the current TAI. If the TAI list of the S-NSSAI is undefined, the S-NSSAI is available in any TAI. If the TAI list of the S-NSSAI is defined, it lists the TAI where the S-NSSAI is available.



NOTE:

This command applies when the UDM-based S-NSSAI selection is enabled in the PGW-C (using the `udm-based-slice-selection` command under the APN configuration).

Default

no suspend

Parameters

list-name

specifies the list name, up to 32 characters

6.2.1.37.56 smf-serving-area-list

Syntax

[no] smf-serving-area-list *list-name*

Context

config>mobile>profile>list

Description

This command configures an SMF serving area list.

The no form of this command removes the specified list from the configuration.

Parameters

list-name

specifies the list name, up to 32 characters

6.2.1.37.57 smf-serving-area

Syntax

[no] smf-serving-area *string*

Context

config>mobile>profile>list>smf-serving-area-list

Description

This command configures an SMF serving area in the SMF serving area list.

The no form of this command removes the specified SMF serving area from the list.

Parameters

string

specifies the SMF serving area name, up to 32 characters

6.2.1.37.58 tai-list

Syntax

tai-list *profile-name*

no tai-list

Context

config>mobile>profile>list>smf-serving-area-list>smf-serving-area

Description

This command associates a TAI list to the SMF serving area. The TAI list is used in the SMF to map a TAI value to an SMF serving area, when the UPF discovery based on the SMF serving area needs to be performed.

The no form of this command removes the TAI list from the SMF serving area.

**NOTE:**

The TAI list association to the SMF serving area is not used for the UPF.

Parameters***profile-name***

specifies the TAI list name, up to 32 characters

6.2.1.37.59 tariff-change-list**Syntax**

[no] tariff-change-list *list-name*

Context

config>mobile>profile>list

Description

This command creates or removes a tariff change list profile that can be applied to an offline charging information trigger (config>mobile>profile>charging>{epdg-charging | pgw-charging | sgw-charging | tdf-charging}>off-line>cit).

Parameters***list-name***

specifies the list name, up to 32 characters

6.2.1.37.60 time**Syntax**

time *id hh:mm*

no time *id*

Context

config>mobile>profile>list>tariff-change-list

Description

This command is used to specify a tariff-change time for a tariff-change charging-info trigger. Multiple tariff change times can be specified within the profile to facilitate differentiated charging at different times of day. Each tariff change entry must have a unique time of day. The (UE) time of day is then used to generate a tariff change event for charging records. Tariff change is a charging info addition trigger. The usage data containers (TDV / SDC containers) are added to the CDR at the specified time. As a result of this, the max-number of change conditions partial record trigger is incremented. For Ga, a partial CDR is closed if the configured max number of change conditions is met. In the case of Rf, an ACR interim is sent out to the OFCS if the configured max number of change conditions is met.

The no form of this command removes the specified tariff change time entry.

Parameters

<i>id</i>	specifies a tariff change index (Range: 1 - 24)
<i>hh</i>	time in hours
<i>mm</i>	time in minutes

6.2.1.37.61 ue-usage-type-list

Syntax

[no] ue-usage-type-list *list-name*

Context

config>mobile>profile>list

Description

This command configures a UE usage type list. A UE usage type list consists of multiple type values.

The no form of this command removes the UE usage type list from the configuration.

Parameters

<i>list-name</i>	specifies the list name of the UE usage type, up to 32 characters
-------------------------	---

6.2.1.37.62 ue-usage-type

Syntax

[no] ue-usage-type *usage-type*

Context

config>mobile>profile>list>ue-usage-type-list

Description

This command configures the UE usage type values in the UE usage type list.

The no form of this command removes the specified UE usage type value from the UE usage type list configuration.

Parameters

usage-type

specifies the UE usage type value, up to 80 characters

Values: 0 to 255

6.2.1.38 MIPv4 profile command descriptions

6.2.1.38.1 mipv4

Syntax

no mipv4 *profile-name*

Context

config>mobile>profile

Description

This command creates a named Mobile IPv4 profile.

Parameters

profile-name

specifies the unique name for the profile

6.2.1.38.2 dmdu

Syntax

```
dmdu hash-algorithm [md5 | hmac-md5 | rfc2002-md5 | none] timestamp-tolerance seconds
```

Context

```
config>mobile>profile>mipv4
```

Description

This command configures the Home Agent to use Dynamic MIP Key Updates (RFC4784). This means that the HA retrieve the MN-HA key from the AAA during the initial call setup or in case of a key refresh has occurred. With dmdu enabled any locally configured mn-ha-keys in the mip profile will be ignored. Therefore, only mn-ha key from AAA will be used.

Default

rfc2002-md5

Parameters

md5

configures the hash-algorithm to implement MD5 per RFC 1321

hmac-md5

configures the hash-algorithm to implement HMAC-MD5 per RFC 2002bis

rfc2002-md5

configures the hash-algorithm to implement keyed-MD5 per RFC 2002

timestamp-tolerance

configures the allowable difference (tolerance) in timestamps that is acceptable. If the difference is exceeded, the session is rejected. If set to 0, the timestamp tolerance check is disabled. Timestamp-tolerance is measured in seconds.

Values: 0 to 65500

Default: 0

6.2.1.38.3 ip-dscp

Syntax

```
ip-dscp value
```

Context

config>mobile>profile>mipv4

Description

This command sets the DSCP used when sending Mobile IPv4 control PDUs.

Parameters

value

specifies the DSCP to be set

Values: 0 to 63

Default: 56 (network control)

6.2.1.38.4 mn-aaa-auth

Syntax

mn-aaa-auth [initial-reg-only | always | allow-noauth]

Context

config>mobile>profile>mipv4

Description

This command configures whether or not the Home Agent looks for the NM-AAA authentication extension in the MIP RRQ.

Parameters

always

specifies that the HA service performs mn-aaa authentication with RADIUS always: that is, each time a mobile node registers / re-registers/ de-registers.

initial-reg-only

specifies that the HA service performs mn-aaa authentication with RADIUS for mobile initial registration only

allow-noauth

accepts a MIP RRQ that does not contain the mn-aaa authentication extension

Default: always

6.2.1.38.5 mn-ha-auth

Syntax

```
mn-ha-auth [always | allow-noauth]
```

Context

```
config>mobile>profile>mipv4
```

Description

This command configures whether or not the Home Agent looks for the NM-HA authentication extension in the MIP RRQ.

Parameters

always

MIP RRQ must always contain the mn-ha authentication extension to be accepted and authenticate all registrations (that is, initial-registration / re-registrations/ de-registration).

allow-noauth

accepts a MIP RRQ that does not contain the mn-ha authentication extension

Default: always

6.2.1.38.6 mn-ha-spi

Syntax

```
[no] mn-ha-spi value
```

Context

```
config>mobile>profile>mipv4
```

Description

This command configures the key between the HA and the mobile node (MN). This only used when dmU is disabled. This is a multi entry command to allow to configure multiple keys indexed by SPIs.

The no form of the command removes the key.

Parameters

value

configures the SPI which indicates a security context between MN and the HA in accordance with RFC 2002

Values: 256 to 4294967295

6.2.1.38.7 secret

Syntax

`secret string`

Context

`config>mobile>profile>mip4>mn-ha-spi`

Description

This command configures the shared-secret between MN and the HA.

Parameters

string

shared-secret between MN and the HA

Values: 1 to 254 (alphanumeric characters case sensitive)

6.2.1.38.8 hash-algorithm

Syntax

`hash-algorithm [md5 | hmac-md5 | rfc2002-md5]`

Context

`config>mobile>profile>mip4>mn-ha-spi`

Description

This command configures the hash-algorithm.

Parameters

md5

configures the hash-algorithm to implement MD5 per RFC 1321

hmac-md5

configures the hash-algorithm to implement HMAC-MD5 per RFC 2002bis

rfc2002-md5

configures the hash-algorithm to implement keyed-MD5 per RFC 2002

Default: rfc2002-md5

6.2.1.38.9 timestamp-tolerance

Syntax

`timestamp-tolerance seconds`

Context

`config>mobile>profile>mip4>mn-ha-spi`

Description

This command configures allowable difference (tolerance) in timestamps that is acceptable. If the difference is exceeded, the session is rejected. If set to 0, the timestamp tolerance check is disabled.

Parameters

seconds

specifies the timestamp tolerance, in seconds

Values: 0 to 65500

Default: 0

6.2.1.38.10 registration-lifetime

Syntax

`registration-lifetime seconds`

Context

`config>mobile>profile>mip4`

Description

This command configures the shortest registration lifetime that the HA service will allow in any Registration Request message from the mobile node. If the mobile node requests a shorter lifetime than what is specified, the HA will use this value.

Parameters

seconds

specifies the shortest allowed registration lifetime, in seconds

Values: 1 to 65500

Default: 7200

6.2.1.38.11 revocation

Syntax

```
revocation i-bit-negotiation [enable | disable] max-retransmission number  
retransmission-timeout seconds
```

```
no revocation
```

Context

```
config>mobile>profile>mipv4
```

Description

This command configures the MIP revocation support to be enabled or disabled on the Home Agent.

Default

no revocation

Parameters

i-bit-negotiation [enable | disable]

Enable or disable the i-bit negotiation in the MIP RRQ/RRP per RFC 3543.

number

configures the maximum number of retransmissions of a Revocation message before the revocation fails

Values: 0 to 10

Default: 3

seconds

configures the number of seconds to wait for a Revocation Acknowledgment from the FA before retransmitting the Revocation message

Values: 1 to 10

Default: 3

6.2.1.38.12 tunnel-mode

Syntax

```
mipv4-tunnel-mode {ip-in-ip | gre}
```

Context

```
config>mobile>profile>mipv4
```

Description

This command sets the MIPv4 tunnel mode.

Parameters

ip-in-ip

specifies MIPv4 tunnel to be ip-in-ip

gre

specifies MIPv4 tunnel to be ip-gre-ip

Default: gre

6.2.1.39 MME group profile command descriptions

6.2.1.39.1 mme-group

Syntax

```
[no] mme-group MME-group-id
```

Context

```
config>mobile>profile
```

Description

This command creates a new MME group for network-triggered service restoration.



NOTE:

Overlapping subnets are not allowed in MME group.

Parameters

MME-group-id

specifies the group name identifier

Values: 1 to 65535

6.2.1.39.2 peer

Syntax

[no] peer *ipv4* | *ipv6*

Context

config>mobile>profile>mme-group>peer

Description

This command identifies nodes that may be used for network-triggered service restoration. When the NTSR procedure is triggered by MME failure, the MME group that contains the failed MME node will be selected.



NOTE:

- When modifying an already referenced MME group, a peer IP that already exists to another MME group under the same S11 referenced point cannot be added.
- When modifying an already referenced MME group, the changes reflect to the S11 reference points that this MME group is enabled.

Parameters

ipv4 | *ipv6*

specifies the peer's IPv4 or IPv6 address

Values	ipv4	a.b.c.d[/mask]
–	–	mask - [27..32]
–	ipv6	x:x:x:x:x:x:x (eight 16-bit pieces)

–	–	x:x:x:x:x:d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

6.2.1.40 N1 profile command descriptions

6.2.1.40.1 n1-profile

Syntax

[no] n1-profile *profile-name*

Context

config>mobile>profile

Description

This command configures an N1 profile with the specified profile name.

The no form of this command removes the specified N1 profile.

Parameters

profile-name

specifies the name of the N1 profile, up to 32 characters

6.2.1.40.2 cause-code

Syntax

cause-code

Context

config>mobile>profile>n1-profile

Description

This command enables the context used to define cause codes.

6.2.1.40.3 apn-shutdown

Syntax

```
apn-shutdown
```

Context

```
config>mobile>profile>n1-profile>cause-code
```

Description

This command enables the context used to define cause codes for APN shutdown.

6.2.1.40.4 5gsm-cause-code

Syntax

```
5gsm-cause-code 5gsm-cause-code  
no 5gsm-cause-code
```

Context

```
config>mobile>profile>n1-profile>cause-code>apn-shutdown
```

Description

This command specifies the 5GSM Cause IE in the N1 PDU Session Establishment Reject procedure when a new PDU session activation is rejected because the APN is in shutdown state.

The no form of this command reverts to the default.

Parameters

5gsm-cause-code

specifies the N1 cause value

Values: 0 to 255

Default: 8 (operator-determined barring)

6.2.1.40.5 application-error-code

Syntax

```
application-error-code application-error-code  
no application-error-code
```

Context

```
config>mobile>profile>n1-profile>cause-code>apn-shutdown
```

Description

This command specifies the application error in the N11 Create SM Context Response or N16/N16a Create Response procedures when a new PDU session activation is rejected because the APN is in shutdown state.

The no form of this command reverts to the default.

Parameters

application-error-code

specifies the application error, up to 64 characters

Default: DNN_NOT_SUPPORTED

6.2.1.40.6 http-status-code

Syntax

```
http-status-code http-status-code  
no http-status-code
```

Context

```
config>mobile>profile>n1-profile>cause-code>apn-shutdown
```

Description

This command specifies the HTTP status code in the N11 Create SM Context Response or N16/N16a Create Response procedures when a new PDU session activation is rejected due to APN shutdown.

The no form of this command reverts to the default.

Parameters

http-status-code

specifies the HTTP status code

Values: 400 | 403 | 404 | 500 | 503 | 504

Default: 403 (Forbidden)

6.2.1.40.7 dnn-subscription-removed

Syntax

dnn-subscription-removed

Context

config>mobile>profile>n1-profile>cause-code

Description

This command specifies the cause values that are used when the DNN subscription is removed from the UDM.

6.2.1.40.8 5gsm-cause-code

Syntax

5gsm-cause-code *5gsm-cause-code*

no 5gsm-cause-code

Context

config>mobile>profile>n1-profile>cause-code>dnn-subscription-removed

Description

This command specifies the 5GSM Cause IE value in the PDU Session Release Command, if an existing PDU session is released because the UDM has sent a notification about the removal of the subscription to the DNN.

The no form of this command reverts to the default.

Parameters

5gsm-cause-code

Values: 0 to 255

Default: 29 (user authentication or authorization failed)

6.2.1.40.9 gw-suspend

Syntax

gw-suspend

Context

config>mobile>profile>n1-profile>cause-code

Description

This command specifies the context used to define cause codes when the gateway is suspended.

6.2.1.40.10 5gsm-cause-code

Syntax

5gsm-cause-code *5gsm-cause-code*

no 5gsm-cause-code

Context

config>mobile>profile>n1-profile>cause-code>gw-suspend

Description

This command specifies the 5GSM Cause IE in the N1 PDU Session Establishment Reject procedure when a new PDU session activation is rejected because the gateway is suspended.

The no form of this command reverts to the default.

Parameters

5gsm-cause-code

Values: 0 to 255

Default: 8 (operator-determined barring)

6.2.1.40.11 application-error-code

Syntax

application-error-code *application-error-code*

Context

```
config>mobile>profile>n1-profile>cause-code>gw-suspend
```

Description

This command specifies the application error in the N11 Create SM Context Response or N16/N16a Create Response procedures when a new PDU session activation is rejected because the gateway is suspended.

The no form of this command reverts to the default.

Parameters

application-error-code

specifies the application error, up to 64 characters

Default: NF_FAILOVER

6.2.1.40.12 http-status-code

Syntax

```
http-status-code http-status-code
```

```
no http-status-code
```

Context

```
config>mobile>profile>n1-profile>cause-code>gw-suspend
```

Description

This command specifies the HTTP status code in the N11 Create SM Context Response or N16/N16a Create Response procedures when a new PDU session activation is rejected because the gateway is suspended.

The no form of this command reverts to the default.

Parameters

http-status-code

specifies the HTTP status code

Values: 400 | 403 | 404 | 500 | 503 | 504

Default: 500 (Internal Server Error)

6.2.1.40.13 n1-t3591

Syntax

```
n1-t3591 timeout [retry-count retry-count]  
no n1-t3591
```

Context

```
config>mobile>profile>n1-profile
```

Description

This command configures the T3591 timer specified in 3GPP TS 24.501. The T3591 timer defines the retransmission timeout for the PDU SESSION MODIFICATION COMMAND message.

The no form of this command reverts to the default.

Parameters

timeout

specifies the time period in deci seconds that a network element waits before resending a PDU SESSION MODIFICATION COMMAND message over the N11 interface if there is no response to an initial message, in deciseconds

Values: 10 to 300

Default: 160

retry-count

specifies the number of retransmissions

Values: 0 to 10

Default: 4

6.2.1.40.14 n1-t3592

Syntax

```
n1-t3592 timeout [retry-count retry-count]  
no n1-t3592
```

Context

```
config>mobile>profile>n1-profile
```

Description

This command configures the T3592 timer specified in 3GPP TS 24.501. The T3592 timer defines the retransmission timeout for the PDU SESSION RELEASE COMMAND message.

The no form of this command reverts to the default.

Parameters

timeout

specifies the time period in deci seconds that a network element waits before resending a PDU SESSION RELEASE COMMAND message over the N11 interface if there is no response to an initial message, in deciseconds

Values: 10 to 300

Default: 160

retry-count

specifies the number of retransmissions

Values: 0 to 10

Default: 4

6.2.1.41 N2 peer profile command descriptions

6.2.1.41.1 n2-peer

Syntax

[no] n2-peer *profile-name*

Context

config>mobile>profile

Description

This command specifies an N2 peer (AMF peer) profile for peer-related information for the N2 NGAP applications. You cannot delete an N2 peer profile when it is referenced by an N2 NGAP application.

The no form of this command removes the specified profile from the configuration.

Parameters

profile-name

specifies the name of the peer within a N2 peer profile, up to 32 characters

6.2.1.41.2 peer

Syntax

```
peer ip-address | ipv6-address
no peer
```

Context

```
config>mobile>profile>n2-peer
```

Description

This command configures an N2 (AMF) peer. The peer address can be an IP address. The no form of this command removes the peer IP address from the configuration.

Parameters

ip-address
specifies the peer IPv4 or IPv6 address

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

6.2.1.41.3 plmn-list

Syntax

```
plmn-list list-name
no plmn-list
```

Context

config>mobile>profile>n2-peer

Description

This command configures the PLMN list for the N2 peer (the PLMN list must be already configured under the config>mobile>profile>list>plmn context). The PLMN list data is considered during the AMF selection under the config>mobile>pdn>n2>n2-peer context.

Parameters

ip-address

specifies the name of the PLMN list, up to 32 characters

6.2.1.41.4 sctp-profile

Syntax

sctp-profile *profile-name*

no sctp-profile

Context

config>mobile>profile>n2-peer

Description

This command configures the SCTP profile for the N2 peer.

The no form of this command removes the specified profile from the configuration.

Parameters

profile-name

specifies the SCTP profile name, up to 32 characters

6.2.1.42 N7 attributes profile command descriptions

6.2.1.42.1 n7-attributes-profile

Syntax

```
[no] n7-attributes-profile profile-name
```

Context

```
config>mobile>profile
```

Description

This command configures a profile with a set of attributes that are included or excluded from the N7 interface. This profile is referenced by the NPCF profile.

The no form of this command removes the N7 attributes profile from the configuration.

Parameters

profile-name

specifies the name of the N7 attributes profile, up to 64 characters

6.2.1.43 ip-domain

Syntax

```
ip-domain {included | excluded }
```

```
no ip-domain
```

Context

```
config>mobile>profile>n7-attributes-profile
```

Description

This command includes or excludes the ipDomain IE in the N7 interface.

The no form of this command reverts to the default.

Parameters

included

includes the ipDomain IE in the N7 interface

excluded

excludes the ipDomain IE from the N7 interface

Default: `excl uded`

6.2.1.44 `nokia-smf-indicator-as-vsa`

Syntax

```
nokia-smf-indicator-as-vsa {incl uded | excl uded }  
no nokia-smf-indicator-as-vsa
```

Context

```
config>mobile>profile>n7-attributes-profile
```

Description

This command specifies if the Nokia vendor-specific attribute `smfNokia` is included in the N7 Create Session message to the PCF. When enabled, the N7 Create Session Request message includes the proprietary IE: "vendorSpecific-006527": {"smfNokia": 2}.

The `no` form of this command excludes the proprietary IE from the N7 interface.

Parameters

`incl uded`

includes the Nokia IE "vendorSpecific-006527" in the N7 interface

`excl uded`

excludes the Nokia IE "vendorSpecific-006527" from the N7 interface

Default: `excl uded`

6.2.1.45 `Namf profile command descriptions`

6.2.1.45.1 `namf-profile`

Syntax

```
[no] namf-profile profile-name
```

Context

```
config>mobile>profile
```

Description

This command configures an AMF profile which defines the failure handling policy when the AMF services are used.

The no form of this command removes the AMF profile from the configuration.

Parameters

profile-name

specifies the name of the AMF profile, up to 32 characters

6.2.1.45.2 extended-buffering-wait-time

Syntax

extended-buffering-wait-time *value*

no buffering-wait-time

Context

config>mobile>profile>namf-profile

Description

This command configures the maximum wait time for the extended buffering in 5G. If the value received from the AMF is lower than the configured value, the value received from the AMF is used. If the AMF-received value is higher than the configured value, the configured value defines the duration of the extended buffering.

If the maximum wait time is 0, the extended buffering is disabled in 5G.



NOTE:

Buffering always lasts for at least 30 seconds.

The no form of this command disables the extended buffering in 5G.

Parameters

value

specifies the maximum wait time for the extended buffering, in seconds

Values: 0 | 30 to 86400

Default: 0

6.2.1.45.3 http-status-code

Syntax

```
http-status-code status-code-value [application-error app-error-code] failure-action  
no http-status-code status-code-value [application-error app-error-code]
```

Context

```
config>mobile>profile>namf-profile
```

Description

This command configures the AMF failure handling policy. The defined action is applied, if the response from the AMF matches the configured HTTP status code. If an optional application error is included, the response must have a matching application error.

The no form of this command removes the error handling for the specified HTTP status code and reverts to the default.

Parameters

status-code-value

specifies the three-digit status code value

Values: 300 to 599

app-error-code

specifies the application error code, up to 64 characters

failure-action

specifies the failure action

Values:

terminate

The PDU session is terminated

continue

The PDU session is not terminated

retry-continue

The service operation is reattempted with another AMF peer; when the maximum number of peer failovers (configured using the `max-peer-failover` command) has been reached, the PDU session is not terminated

retry-terminate

The service operation is reattempted with another AMF peer; when the maximum number of peer failovers (configured using the `max-peer-failover` command) has been reached, the PDU session is terminated

Default: continue

6.2.1.45.4 max-peer-failover

Syntax

`max-peer-failover value`

`no max-peer-failover`

Context

`config>mobile>profile>namf-profile`

Description

This command configures the maximum number of times failover to another AMF in the AMF set.

The no form of this command reverts to the default.

Parameters

value

specifies the maximum number of failovers

Values: 0 to 3

Default: 1

6.2.1.45.5 retry-count

Syntax

`retry-count value [failure-action failure-action]`

`no retry-count`

Context

`config>mobile>profile>namf-profile`

Description

This command configures the maximum number of retransmissions.

The no form of this command reverts to the default.

Parameters

value

specifies the maximum number of retransmissions

Values: 0 to 8

Default: 0

failure-action

specifies the action, if no response is received

Values	terminate	The PDU session is terminated
–	continue	The PDU session is not terminated
–	retry-continue	The service operation is reattempted with another AMF peer; when the maximum number of peer failovers (configured using the max-peer-failover command) has been reached, the PDU session is not terminated
–	retry-terminate	The service operation is reattempted with another AMF peer; when the maximum number of peer failovers (configured using the max-peer-failover command) has been reached, the PDU session is terminated

Default: continue

6.2.1.45.6 transaction-timer

Syntax

transaction-timer *milliseconds*
no transaction-timer

Context

config>mobile>profile>namf-profile

Description

This command configures the time that the SMF waits a response from the AMF.

The no form of this command reverts to the default.



NOTE:

When there is no response from the primary SCP between the SMF and the AMF and the failure action of the `config>mobile>pdn>sba-client-services>scp-client>scp-service>fh-no-response` command is set to `failover`, a delay of 500 msec is added before sending the retransmit message.

Parameters

milliseconds

specifies the transaction timer, in milliseconds (increments of 100 milliseconds)

Values: 1000 to 20000

Default: 5000

6.2.1.46 Network trace profile command descriptions

6.2.1.46.1 network-trace-profile

Syntax

[no] `network-trace-profile` *profile-name*

Context

`configure>mobile>profile`

Description

This command configures a network trace profile used when the network trace functionality is enabled.

The no form of this command (default) removes the specified profile from the configuration.

Default

`no network-trace-profile`

Parameters

profile-name

specifies the name of the network trace profile, up to 32 characters

6.2.1.46.2 http2-profile

Syntax

`http2-profile profile-name`

`no http2-profile`

Context

`configure>mobile>profile>network-trace-profile`

Description

This command associates an HTTP/2 profile, configured using the `config>mobile>profile>http2` context, with the network trace functionality.

The `no` form of this command removes the HTTP/2 profile from the configuration.

Default

`no http2-profile`

Parameters

profile-name

specifies the name of the HTTP/2 profile, up to 32 characters

6.2.1.46.3 packet-broker-server

Syntax

`[no] packet-broker-server {ip-address | fqdn } [port port]`

Context

`configure>mobile>profile>network-trace-profile`

Description

This command specifies the Packet Broker server used in the network trace profile. You can specify the Packet Broker server using an IP address (IPv6 or IPv4).

The no form of this command removes the specified Packet Broker server from the network trace profile..

Parameters

ip-address

specifies the CMG local IP address

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

port

specifies the port

Values: 0 to 65535

Default: 80

6.2.1.46.4 server-path

Syntax

[no] server-path *server-path-string*

Context

configure>mobile>profile>network-trace-profile>packet-broker-server

Description

This command configures the Packet Broker server path used for the communication with the network trace client service.

Default

no server-path

Parameters

server-path-string

specifies the Packet Broker server path string, up to 255 characters

6.2.1.47 Node selection profile command descriptions

6.2.1.47.1 node-selection

Syntax

node-selection

Context

config>mobile>profile

Description

This command enables the context to configure the policy for node selection.

6.2.1.47.2 abort

Syntax

abort

Context

config>mobile>profile>node-selection

Description

This command ends the current policy editing session entered via the begin command and discards any changes that have been made to policies during the editing session.

6.2.1.47.3 begin

Syntax

```
begin
```

Context

```
config>mobile>profile>node-selection
```

Description

This command begins a policy editing session. This is required for all configuration changes under the `config>mobile>profile>node-selection` context. The editing session continues until an `abort` or a `commit` command is issued, or a control complex reset takes place. Any modifications made during the editing session, will take effect when the `commit` command is issued. Editing sessions are not interrupted by high availability activity switches or CLI session terminations, such as closing a Telnet session.

6.2.1.47.4 client-profile

Syntax

```
[no] client-profile profile-name
```

Context

```
config>mobile>profile>node-selection
```

Description

This command configures a client selection profile for policy-based node selection. The profile is applied for node selection based on the node type (for example, SGW-U or PGW-U/GGSN-U).

The `no` form of this command removes the node selection client profile from the configuration.

Parameters

profile-name

specifies the client selection profile name, up to 32 characters

6.2.1.47.5 default-action

Syntax

```
default-action
```

Context

```
config>mobile>profile>node-selection>client-profile
```

Description

This command enables the context to configure the default action to apply if no match is found across all entries in the client selection profile.

6.2.1.47.6 set-target-profile

Syntax

```
set-target-profile profile-name
```

```
no set-target-profile
```

Context

```
config>mobile>profile>node-selection>client-profile>default-action
```

Description

This command configures the selection target profile to apply per selection client profile if no match is found across all the entries in the selection client profile. The selection target profile defines the rules for node selection. If there is no selection target profile configured under the `default-action` command, the selection profile is not considered; that is, the configuration falls back to the default policy (the default policy is to reject UP selection).

The `no` form of this command removes the selected target profile from the default action.

Parameters

profile-name

specifies the target profile name, up to 32 characters

6.2.1.47.7 entry

Syntax

[no] entry *id*

Context

config>mobile>profile>node-selection>client-profile

Description

This command configures the client selection entry for a client selection profile. A selection client profile can have multiple entries with each entry matching the client/source parameters (for example, PDN/subscriber parameters) to apply for the target selection. Each entry has a selection priority processing based on the numeric value of the ID associated with the entry. The parameters configured per entry are applied collectively. If no parameters are configured within an entry, it is considered as matching all clients (that is, PDN and UE).

The no form of this command removes the selected entry from the client profile.

Parameters

id

specifies the entry ID

Values: 0 to 4294967295

6.2.1.47.8 action

Syntax

action

Context

config>mobile>profile>node-selection>client-profile>entry

Description

This command enables the context to configure the action to apply on match of the client selection entry.

6.2.1.47.9 set-target-profile

Syntax

```
set-target-profile profile-name  
no set-target-profile
```

Context

```
config>mobile>profile>node-selection>client-profile>entry
```

Description

This command configures the selection target profile to apply per selection client entry on match. The selection target profile defines the rules for node selection. If no selection target profile is configured for an entry, that entry is not considered during the selection procedure.

The no form of this command removes the selected target profile from the selected entry.

Parameters

profile-name

specifies the target profile name, up to 32 characters

6.2.1.47.10 match

Syntax

```
match
```

Context

```
config>mobile>profile>node-selection>client-profile>entry
```

Description

This command enables the context to configure the match criteria of the client selection entry.

6.2.1.47.11 apn

Syntax

```
[no] apn apn-name
```

Context

config>mobile>profile>node-selection>client-profile>entry>match

Description

This command configures an APN per selection client profile entry. The client selection policy applies the selection rule for the node selection if the APN name of the PDN session matches at least one APN name configured per entry. A maximum of 5 APNs can be configured per entry. A partial APN name can be configured to match multiple APNs.

The no form of this command removes the APN from the entry.

Parameters

apn-name

specifies the APN name, up to 80 characters

6.2.1.47.12 cell-id-list

Syntax

cell-id-list *cell-id-list-name*

no cell-id-list

Context

config>mobile>profile>node-selection>client-profile>entry>match

Description

This command configures a list of cell-id values that are used as match criteria in the node selection rules.

The no form of this command removes the cell-id list from the entry.

Parameters

cell-id-list-name

specifies the cell-id list name, up to 32 characters

6.2.1.47.13 iwk5gs

Syntax

[no] iwk5gs

Context

```
config>mobile>profile>node-selection>client-profile>entry>match
```

Description

This command configures the 5GSIWKI flag (5GS Interworking Indication) to match and apply to the selection rule for the node selection.

The no form of this command removes the IWKI5GS (5GSIWKI flag) from the entry.



NOTE:

The matching criteria design works based on the AND condition. When both the IWKI5GS and the NRI5GC parameters are set under a matching criteria set, and if they are received with values 5GSIWKI=0 and 5GCNRI=1, the matching entry is not considered as TRUE, therefore, it is a MATCH.

Default

no iwiki5gs

6.2.1.47.14 nri5gc

Syntax

```
[no] nri5gc
```

Context

```
config>mobile>profile>node-selection>client-profile>entry>match
```

Description

This command configures the 5GCNRI flag (5GC Not Restricted Indication) to match and apply to the selection rule for the node selection.

The no form of this command removes the NRI5GC (5GCNRI indication flag) from the entry.



NOTE:

The matching criteria design works based on the AND condition. When both the IWKI5GS and the NRI5GC parameters are set under a matching criteria set, and if they are received with values 5GSIWKI=0 and 5GCNRI=1, the matching entry is not considered as TRUE, therefore, it is a MATCH.

Default

no nri5gc

6.2.1.47.15 rat-type

Syntax

```
rat-type {utran | eutran}  
no rat-type
```

Context

```
config>mobile>profile>node-selection>client-profile>entry>match
```

Description

This command configures the RAT type to match and apply to the selection rule for node selection.
The no form of this command removes the RAT type from the entry.

Parameters

utran

applicable to 3G (UTRAN) and 2G (GERAN)

eutran

applicable to 4G (LTE)

6.2.1.47.16 serving-node

Syntax

```
serving-node ip-address-list  
no serving-node
```

Context

```
config>mobile>profile>node-selection>client-profile>entry>match
```

Description

This command associates a list of serving nodes to match and apply the selection rule for node selection configured using the config>mobile>profile>list>ip-address-list command.

The no form of this command removes the serving node list from the entry.

Parameters

ip-address-list

specifies the IP address list name, up to 32 characters

6.2.1.47.17 slice

Syntax

[no] slice *list-name*

Context

config>mobile>profile>node-selection>client-profile>entry>match

Description

This multi-entry command configures network slice selection for a client profile entry. The client selection policy applies the selection rule for the node selection if the S-NSSAI of the PDU session matches at least one network slice configured per entry. A maximum of five network slices can be configured per entry.

The no form of this command removes the network slices from the entry.

Parameters

list-name

specifies the network slice list name, up to 32 characters

6.2.1.47.18 tai-lai

Syntax

tai-lai *list-name*

no tai-lai

Context

config>mobile>profile>node-selection>client-profile>entry>match

Description

This command configures a list of TAI and LAI values to match and apply the selection rule for node selection.

The no form of this command removes the TAI/LAI list from the entry.

Parameters

list-name

specifies the TAI/LAI list name, up to 32 characters

6.2.1.47.19 ue-usage-type

Syntax

```
ue-usage-type list-name  
no ue-usage-type
```

Context

```
config>mobile>profile>node-selection>client-profile>entry>match
```

Description

This command associates a list of UE usage type values to match and apply the selection rule for node selection configured using the config>mobile>profile>list>ue-usage-type-list command.

The no form of this command removes the UE usage type list from the entry.



NOTE:

When a UE usage type is used, the target selection must be DNS-based.

Parameters

list-name

specifies the UE usage type list name, up to 32 characters

6.2.1.47.20 commit

Syntax

```
commit
```

Context

config>mobile>profile>node-selection

Description

This command saves the changes made during the current node selection editing session (see the `begin` command). Any policy changes made during the editing session will only take effect when they are committed. When the changes are successfully committed and there are no errors detected during the commit or cross-reference verification against the existing policy configuration, the editing session is closed.

6.2.1.47.21 group-label

Syntax

[no] group-label *profile-name*

Context

config>mobile>profile>node-selection

Description

This command configures a group label for policy-based node selection. The group label is used to create a group of nodes (for example, UP) to identify a set of nodes based on a common criterion among the group. The no form of this command removes the selected group label from the configuration.

Parameters

profile-name

specifies the profile name, up to 32 characters

6.2.1.47.22 target-profile

Syntax

[no] target-profile *profile-name*

Context

config>mobile>profile>node-selection

Description

This command configures a node selection target profile. The selection target profile is applied individually to a selection client profile entry.

The no form of this command removes the selected node selection target profile from the configuration.

Parameters

profile-name

specifies the profile name, up to 32 characters

6.2.1.47.23 default-action

Syntax

default-action

Context

config>mobile>profile>node-selection>target-profile

Description

This command enables the context to configure the default action to apply if no match is found across all entries in the target selection profile.

6.2.1.47.24 reject

Syntax

[no] reject

Context

config>mobile>profile>node-selection>target-profile>default-action

Description

This command enables call rejection across all entries in the selection target profile, when no match is found during the target selection.

The no form of this command disables call rejection across all entries in the selection target profile, when no match is found during target selection.

6.2.1.47.25 entry

Syntax

[no] entry *id*

Context

config>mobile>profile>node-selection>target-profile

Description

This command configures the target selection entry for a target selection profile. The entry matches the target profiles (for example, node load) for node selection. Each entry has a selection priority processing based on the numeric value of the ID associated with the entry.

The no form of this command removes the selected entry from the target profile.

Parameters

id

specifies the entry ID

Values: 0 to 4294967295

6.2.1.47.26 action

Syntax

action

Context

config>mobile>profile>node-selection>target-profile>entry

Description

This command enables the context to configure the action to apply when there is no match of the target selection entry.

6.2.1.47.27 reject

Syntax

[no] reject

Context

```
config>mobile>profile>node-selection>target-profile>entry>action
```

Description

This command stops the target profile evaluation and all the following target profile entries are not processed. Selection of the UPF is based on round robin mechanism for all the active UP peers for the SGW and for the UP peers serving the APN/DNN for the PGW.

The no form of this command re-establishes the target profile evaluation.

6.2.1.47.28 match

Syntax

```
match
```

Context

```
config>mobile>profile>node-selection>target-profile>entry
```

Description

This command enables the context to configure the match criteria of the target selection entry.

6.2.1.47.29 apn

Syntax

```
apn client-apn
```

```
no apn
```

Context

```
config>mobile>profile>node-selection>target-profile>entry>match
```

Description

This command configures the target selection based on the APN match.

The no form of this command disables the target selection based on the APN.

Default

```
no apn
```

Parameters

client-apn

specifies to match the APN name of the client PDN

6.2.1.47.30 dns-based

Syntax

dns-based [nr-capable]

no dns-based

Context

config>mobile>profile>node-selection>target-profile>entry>match

Description

This command configures target selection using DNS lookup. DNS-based lookup is mutually exclusive with other commands under the match command; for example, either DNS-based UP selection is applied for an entry or profile-based selection.

The no form of this command disables target selection using DNS lookup.

Default

no dns-based

Parameters

nr-capable

specifies UP selection for NR-capable UP

6.2.1.47.31 group-label

Syntax

[no] group-label *group-name*

Context

config>mobile>profile>node-selection>target-profile>entry>match

Description

This command configures the target selection based on a set of nodes included in a group. A maximum of five group labels can be configured per entry.

The no form of this command removes the group label as a criterion for node selection.

Default

no group-label

Parameters

group-name

specifies the group name, up to 32 characters

6.2.1.47.32 nrf-based

Syntax

nrf-based [smf-serving-area | tai]

no nrf-based

Context

config>mobile>profile>node-selection>target-profile>entry>match

Description

This command configures the target selection based on NRF discovery.

The no form of this command reverts to the default (disabled).

Default

no nrf-based

Parameters

smf-serving-area

specifies to use the SMF serving area in the query criteria, when the SMF discovers the UPF using the NRF

tai

specifies to use the TAI in the query criteria, when the SMF discovers the UPF using the NRF

6.2.1.47.33 slice

Syntax

```
slice client-slice
no slice
```

Context

```
config>mobile>profile>node-selection>target-profile>entry>match
```

Description

This command configures the target selection based on the network slice match.

The no form of this command disables the target selection based on the network slice.

Default

no slice

Parameters

client-slice

specifies to match the network slice of the client PDN

6.2.1.48 NPCF profile command descriptions

6.2.1.48.1 npcf

Syntax

```
[no] npcf profile-name
```

Context

```
config>mobile>profile
```

Description

This command configures an NPCF profile that can be applied on a PDN or APN basis, for 5G configurations. The profile includes configuration to support failure handling behavior, SBI service operation handling, and supported-vendor features.

The no form of this command removes the profile from the configuration.

Parameters

profile-name

specifies the NPCF profile name, up to 32 characters

6.2.1.48.2 fh-time-limit

Syntax

```
fh-time-limit time-limit
```

```
no fh-time-limit
```

Context

```
config>mobile>profile>npcf
```

Description

This command specifies the time limit for both new and ongoing sessions that are set up and continue without further PCF control (fall-back to static policy) as a result of failure handling. For example, there is a PCF no response, and so on. The PDN session is terminated upon reaching the specified time limit. When *fh-time-limit* and *fh-volume-limit* are both enabled, the session is terminated upon whichever condition occurs first.

The no form of this command reverts to the default.

Default

```
no fh-time-limit
```

Parameters

time-limit

specifies the time limit, in minutes

Values: 0 to 2880

Default: 0 (disabled)

6.2.1.48.3 fh-volume-limit

Syntax

```
fh-volume-limit volume-limit
```

```
no fh-volume-limit
```

Context

config>mobile>profile>npcf

Description

This command specifies the volume limit for new and ongoing sessions that are set up and continue without further PCF control (fall back to static policy) as a result of failure handling (for example, PCF no response). The PDN session is terminated upon reaching the specified volume limit.

The no form of this command reverts to the default (disabled).

Parameters

volume-limit

specifies the volume limit, in kilobytes

Values: 0 to 10000000

Default: 0 (disabled)

6.2.1.48.4 ip-domain-type

Syntax

ip-domain-type {[network-realm-name] [psa-upf-id] [upf-string] [ip-pool-name]}

ip-domain-type none

no ip-domain-type

Context

config>mobile>profile>npcf

Description

This command specifies the content type of the N7 ipDomain IE. If more than one type is configured, the ipDomain IE includes the selected content types concatenated. For example, if the ip-domain-type upf-string network-realm-name is configured, the CMG populates the ipDomain attribute with the concatenated string value <upf-string> <network-realm-name>.

The no form of this command reverts to the default.

Default

no ip-domain-type

Parameters

network-realm-name

sets the ipDomain value with the VPRN name as configured in the UPF with the
config>mobile>profile>pfc>up-peer-list>peer>apn>network-realm
command

psa-upf-id

sets the ipDomain value with the UPF ID configured with the
config>mobile>profile>pfc>up-peer-list>peer>upf-id command



NOTE:

In case of multiple UPFs per session, the upf-id is the PSA anchor upf-id.

upf-string

sets the ipDomain value with the UPF ID as a string identifier configured with the
config>mobile>profile>pfc>up-peer-list>peer>upf-id uuid command

ip-pool-name

sets the ipDomain attribute with the IP pool name

6.2.1.48.5 n7-attributes-profile

Syntax

n7-attributes-profile *profile-name*

no n7-attributes-profile

Context

config>mobile>profile>npcf

Description

This command configures the N7 attributes profile for a session.

The no form of this command reverts to the default.

Default

no n7-attributes-profile

Parameters

profile-name

specifies the profile name, up to 64 characters

6.2.1.48.6 npcf-error-mapping-profile

Syntax

```
npcf-error-mapping-profile profile-name  
no npcf-error-mapping-profile
```

Context

```
config>mobile>profile>npcf
```

Description

This command configures the NPCF error mapping profile.

The no form of this command removes the configuration.

Parameters

profile-name

specifies the profile name, up to 32 characters

6.2.1.48.7 sbi-srvc-operation

Syntax

```
[no] sbi-srvc-operation {create | update | delete}
```

Context

```
config>mobile>profile>npcf
```

Description

This command configures the option for SBI service operation handling for the NPCF profile.

The no form of this command removes the configuration.

Parameters

create

keyword that specifies the create action

update

keyword that specifies the update action

delete

keyword that specifies the delete action

6.2.1.48.8 http-status-code

Syntax

```
http-status-code status-code-value {terminate | continue | retry-and-terminate | ap-continue | retry-and-ap-continue} [ap-action-profile profile-name] [application-error app-error-code]
```

```
no http-status-code status-code-value [application-error app-error-code]
```

Context

```
config>mobile>profile>npcf>sbi-srvc-operation
```

Description

This command specifies the failure handling behavior.

The no form of this command removes the error handling for the specified HTTP status code and reverts to the default action (terminate).

**NOTE:**

When multiple peers are configured, failover is supported for a total of three times (including the primary) for no response scenarios. For PCF failure handling, only one customer-specific application error is supported.

Default

```
no http-status-code status-code-value (terminate)
```

Parameters

status-code-value

specifies the three-digit status code value

Values: 300 to 599

terminate

specifies to terminate the session after a failed response

continue

specifies to continue the session after a failed response

retry-and-terminate

specifies to retry, and if no response is received from the peer, to terminate the session

ap-continue

specifies to apply the configured AP action profile (config>mobile>profile>policy-options>ap-action-profile) to the session after a failed response

retry-and-ap-continue

specifies to retry, and if no response is received from the peer, to apply the configured AP action profile (config>mobile>profile>policy-options>ap-action-profile) to the session

Default: terminate

profile-name

specifies the name of the AP action profile, up to 32 characters

app-error-code

specifies the application error that the PCF has sent, up to 36 characters; when the optional application error is specified, the resulting action (for example, terminate) is applied for the specified combination of the http-status-code and application-error parameters;

Possible combinations of the application error and the HTTP status code describes possible combinations of the application error and the HTTP status code

Values: invalid-api | invalid-msg-format | mandatory-query-param-incorrect | optional-query-param-incorrect | mandatory-query-param-missing | mandatory-ie-incorrect | optional-ie-incorrect | mandatory-ie-missing | unspecified-msg-failure | modification-not-allowed | subscription-not-found | resource-uri-structure-not-found | incorrect-length | nf-congestion-risk | insufficient-resources | unspecified-nf-failure | system-failure | user-unknown | error-initial-parameters | error-trigger-event | error-traffic-mapping-info-rejected | error-conflicting-request | policy-context-denied | any customer-specific application errors

Table 102: Possible combinations of the application error and the HTTP status code

Application error	HTTP status code	Description
USER_UNKNOWN	400 Bad Request	The HTTP request is rejected because the end user specified in the request is unknown to the PCF ¹

Table 102: Possible combinations of the application error and the HTTP status code (continued)

Application error	HTTP status code	Description
ERROR_INITIAL_PARAMETERS	400 Bad Request	The HTTP request is rejected because the set of session or subscriber information required by the PCF for rule selection is incomplete, erroneous, or not available for the decision to be made (For example, QoS, RAT type, subscriber information) 1 , 2
ERROR_TRIGGER_EVENT	400 Bad Request	The HTTP request is rejected because the set of session information in a message originated from a trigger that is met, is incoherent with the previous set of session information for the same session (For example, the trigger that is met, is RAT changed, and the RAT notified is the same as before) 2
ERROR_TRAFFIC_MAPPING_INFO_REJECTED	403 Forbidden	The HTTP request is rejected because the PCF does not accept one or more of the traffic mapping filters provided by the SMF in a PCC Request 2
ERROR_CONFLICTING_REQUEST	403 Forbidden	The HTTP request is rejected because the PCF cannot accept the UE-initiated resource request when a network-initiated resource allocation is already in progress and has packet filters that cover the packet

Table 102: Possible combinations of the application error and the HTTP status code
(continued)

Application error	HTTP status code	Description
		filters in the received UE-initiated resource request The SMF rejects the attempt for the UE-initiated resource request 2
POLICY_CONTEXT_DENIED	403 Forbidden	The HTTP request is rejected because the PCF does not accept the SMF request due to operator policies, local configuration, or both. 1

**NOTE:**

1. These application errors are used by the create service operation and included in the responses to the POST request.
2. These application errors are used by the update service operation and included in the responses to the POST request.

6.2.1.48.9 http-status-code-3xx

Syntax

```
http-status-code-3xx {terminate | continue | retry-and-terminate | ap-continue |  
retry-and-ap-continue} [ap-action-profile profile-name] [application-error app-  
error-code]
```

```
no http-status-code-3xx [application-error app-error-code]
```

Context

```
config>mobile>profile>npcf>sbi-srvc-operation
```

Description

This command specifies the failure handling behavior.

The no form of this command removes the error handling for the HTTP status code 3xx.

**NOTE:**

When multiple peers are configured, failover is supported for a total of three times (including the primary) for no response scenarios.

Default

no http-status-code-3xx (terminate)

Parameters

terminate

specifies to terminate the session after a failed response

continue

specifies to continue the session after a failed response

retry-and-terminate

specifies to retry, and if no response is received from the peer, to terminate the session

ap-continue

specifies to apply the configured AP action profile (config>mobile>profile>policy-options>ap-action-profile) to the session after a failed response

retry-and-ap-continue

specifies to retry, and if no response is received from the peer, to apply the configured AP action profile (config>mobile>profile>policy-options>ap-action-profile) to the session

Default: terminate

profile-name

specifies the name of the AP action profile, up to 32 characters

app-error-code

specifies the application error that the PCF has sent; when the optional application error is specified, the resulting action (for example, terminate) is

applied for the specified combination of http-status-code and application-error; *Possible combinations of the application error and the HTTP status code* describes possible combinations of the application error and the HTTP status code

Values: invalid-api | invalid-msg-format | mandatory-query-param-incorrect | optional-query-param-incorrect | mandatory-query-param-missing | mandatory-ie-incorrect | optional-ie-incorrect | mandatory-ie-missing | unspecified-msg-failure | modification-not-allowed | subscription-not-found | resource-uri-structure-not-found | incorrect-length | nf-congestion-risk | insufficient-resources | unspecified-nf-failure | system-failure | user-

unknown | error-initial-parameters | error-trigger-event | error-traffic-mapping-info-rejected
| error-conflicting-request | policy-context-denied

6.2.1.48.10 http-status-code-4xx

Syntax

```
http-status-code-4xx {terminate | continue | retry-and-terminate | ap-continue |  
retry-and-ap-continue} [ap-action-profile profile-name] [application-error app-  
error-code]
```

```
no http-status-code-4xx [application-error app-error-code]
```

Context

```
config>mobile>profile>npcf>sbi-srvc-operation
```

Description

This command specifies the failure handling behavior for HTTP status code 4xx.

The no form of this command removes the error handling for the HTTP status code 4xx.



NOTE:

When multiple peers are configured, failover is supported for a total of three times (including the primary) for no response scenarios.

Default

```
no http-status-code-4xx (terminate)
```

Parameters

terminate

specifies to terminate the session after a failed response

continue

specifies to continue the session after a failed response

retry-and-terminate

specifies to retry, and if no response is received from the peer, to terminate the session

ap-continue

specifies to apply the AP action profile (config>mobile>profile>policy-options>ap-action-profile) to the session after a failed response

retry-and-ap-continue

specifies to retry, and if no response is received from the peer, to apply the AP action profile (config>mobile>profile>policy-options>ap-action-profile) to the session

Default: terminate

profile-name

specifies the name of the AP action profile, up to 32 characters

app-error-code

specifies the application error that the PCF has sent; when the optional application error is specified, the resulting action (for example, terminate) is

applied for the specified combination of http-status-code and application-error; *Possible combinations of the application error and the HTTP status code* describes possible combinations of the application error and the HTTP status code

Values: invalid-api | invalid-msg-format | mandatory-query-param-incorrect | optional-query-param-incorrect | mandatory-query-param-missing | mandatory-ie-incorrect | optional-ie-incorrect | mandatory-ie-missing | unspecified-msg-failure | modification-not-allowed | subscription-not-found | resource-uri-structure-not-found | incorrect-length | nf-congestion-risk | insufficient-resources | unspecified-nf-failure | system-failure | user-unknown | error-initial-parameters | error-trigger-event | error-traffic-mapping-info-rejected | error-conflicting-request | policy-context-denied

6.2.1.48.11 http-status-code-5xx

Syntax

```
http-status-code-5xx {terminate | continue | retry-and-terminate | ap-continue |
retry-and-ap-continue} [ap-action-profile profile-name] [application-error app-
error-code]
```

```
no http-status-code-5xx [application-error app-error-code]
```

Context

```
config>mobile>profile>npcf>sbi-srv-operation
```

Description

This command specifies the failure handling behavior for HTTP status code 5xx.

The no form of this command removes the error handling for the HTTP status code 5xx.



NOTE:

When multiple peers are configured, failover is supported for a total of three times (including the primary) for no response scenarios.

Default

no http-status-code-5xx (terminate)

Parameters

terminate

specifies terminate the session after a failed response

continue

specifies continue the session after a failed response

retry-and-terminate

specifies to retry, and if no response is received from the peer, to terminate the session

ap-continue

specifies to apply the AP action profile (config>mobile>profile>policy-options>ap-action-profile) to the session after a failed response

retry-and-ap-continue

specifies to retry, and if no response is received from the peer, to apply the AP action profile (config>mobile>profile>policy-options>ap-action-profile) to the session

Default: terminate

profile-name

specifies the name of the AP action profile, up to 32 characters

app-error-code

specifies the application error that the PCF has sent; when the optional application error is specified, the resulting action (for example, terminate) is

applied for the specified combination of http-status-code and application-error; *Possible combinations of the application error and the HTTP status code* describes possible combinations of the application error and the HTTP status code

Values: invalid-api | invalid-msg-format | mandatory-query-param-incorrect | optional-query-param-incorrect | mandatory-query-param-missing | mandatory-ie-incorrect | optional-ie-incorrect | mandatory-ie-missing | unspecified-msg-failure | modification-not-allowed | subscription-not-found | resource-uri-structure-not-found | incorrect-length | nf-congestion-risk | insufficient-resources | unspecified-nf-failure | system-failure | user-unknown | error-initial-parameters | error-trigger-event | error-traffic-mapping-info-rejected | error-conflicting-request | policy-context-denied

6.2.1.48.12 retry-count

Syntax

retry-count *value*

```
no retry-count
```

Context

```
config>mobile>profile>npcf>sbi-srvc-operation
```

Description

This command configures the retry count used to detect connection failures to the PCF. If a message is sent to PCF with no reply after the time specified for the `transaction-timer` command, the message is sent again. When the message has been resent without success for the configured retry count, the connection to the PCF is considered down and failure handling is triggered.

Parameters

value

specifies the number of retry counts

Values: 0 to 8 (0 indicates no retry)

Default: 3

6.2.1.48.13 transaction-timer

Syntax

```
transaction-timer value
```

```
no transaction-timer
```

Context

```
config>mobile>profile>npcf>sbi-srvc-operation
```

Description

This command configures the transaction time used to detect connection failures to the configured PCF. If there is no reply when a message is sent to the PCF, the message is sent again after the time configured for this command. This is repeated for the number of times configured for the `retry-count` command, after which the connection to the PCF is considered down and failure handling is triggered.

Parameters

value

specifies the transaction timer value, in milliseconds

Values: 1000 to 180000

Default: 5000

6.2.1.48.14 supported-features

Syntax

supported-features

Context

config>mobile>profile>npcf

Description

This command enables the context to configure supported features for the NPCF profile.

6.2.1.48.15 adc

Syntax

[no] adc

Context

config>mobile>profile>npcf>supported-features

Description

This command enables the ADC option within the Supported-Features AVP.

The no form of this command reverts to the default.

Default

no adc

6.2.1.48.16 af-charging-id

Syntax

[no] af-charging-id

Context

config>mobile>profile>npcf>supported-features

Description

This command enables the string encoding of the application function charging ID. The support is indicated in the Supported-Features AVP.

The no form of this command reverts to the default.

Default

no af-charging-id

6.2.1.48.17 eps-fallback-report

Syntax

[no] eps-fallback-report

Context

config>mobile>profile>npcf>supported-features

Description

This command enables support for the EPS fallback report. The support is indicated in the Supported-Features AVP.

The no form of this command reverts to the default.

Default

no eps-fallback-report

6.2.1.48.18 netloc

Syntax

[no] net loc

Context

config>mobile>profile>npcf>supported-features

Description

This command enables the NetLoc option within the Supported-Features AVP. The NetLoc feature indicates the support of Access Network Information Reporting.

The no form of this command reverts to the default.

Default

no netloc

6.2.1.48.19 pcscf-restoration-enhancement

Syntax

[no] pcscf-restoration-enhancement

Context

config>mobile>profile>npcf>supported-features

Description

This command enables the P-CSCF Restoration Enhancement feature bit setting within the Supported-Features AVP.

The no form of this command disables sending the P-CSCF Restoration Enhancement.

Default

no pcscf-restoration

6.2.1.48.20 pdu-session-rel-cause

Syntax

[no] pdu-session-rel-cause

Context

config>mobile>profile>npcf>supported-features

Description

This command enables support for the PDU session release cause, which indicates that the PDU session is terminated due to a PS to CS handover. The support is indicated in the Supported-Features AVP.

The no form of this command reverts to the default.

Default

no pdu-session-rel-cause

6.2.1.48.21 policy-decision-error-handling

Syntax

[no] policy-decision-error-handling

Context

config>mobile>profile>npcf>supported-features

Description

This command enables support for the policy decision error handling. The support is indicated in the Supported-Features AVP.

The no form of this command reverts to the default.

Default

no policy-decision-error-handling

6.2.1.48.22 policy-update-when-ue-suspends

Syntax

[no] policy-update-when-ue-suspends

Context

config>mobile>profile>npcf>supported-features

Description

This command enables the option for policy updates when the UE is suspended, within the Supported-Features AVP.

The no form of this command reverts to the default.

Default

no policy-update-when-ue-suspends

6.2.1.48.23 pra

Syntax

[no] pra

Context

config>mobile>profile>npcf>supported-features

Description

This command enables the PRA option within the Supported-Features AVP.

The no form of this command reverts to the default.

Default

no pra

6.2.1.48.24 prov-af-signal-flow

Syntax

[no] prov-af-signal-flow

Context

config>mobile>profile>npcf>supported-features

Description

This command enables the provisional AF signal flow option within the Supported-Features AVP.

The no form of this command reverts to the default.

Default

no prov-af-signal-flow

6.2.1.48.25 resp-based-session-rel

Syntax

[no] resp-based-session-rel

Context

config>mobile>profile>npcf>supported-features

Description

This command enables support for the response based session release. The support is indicated in the Supported-Features AVP.

The no form of this command reverts to the default.

Default

no resp-based-session-rel

6.2.1.48.26 session-rule-error-handling

Syntax

[no] session-rule-error-handling

Context

config>mobile>profile>npcf>supported-features

Description

This command enables the session rule error handling option within the Supported-Features AVP.

The no form of this command reverts to the default.

Default

no session-rule-error-handling

6.2.1.48.27 sponsored-connectivity

Syntax

[no] sponsored-connectivity

Context

config>mobile>profile>npcf>supported-features

Description

This command enables the sponsored connectivity option within the Supported-Features AVP.

The no form of this command reverts to the default.

Default

no sponsored-connectivity

6.2.1.48.28 support-2g3g

Syntax

[no] support -2g3g

Context

config>mobile>profile>npcf>supported-features

Description

This command informs the PCF about the support of GERAN and UTRAN access over N7 interface by including the flag 2G3GIWK(50) in the Supported-Features attribute.

The no form of this command reverts to the default.

Default

no support-2g3g

6.2.1.48.29 tsc

Syntax

[no] tsc

Context

config>mobile>profile>npcf>supported-features

Description

This command enables the TSC option within the Supported-Features AVP.

The no form of this command reverts to the default.

Default

no tsc

6.2.1.48.30 umc

Syntax

[no] umc

Context

config>mobile>profile>npcf>supported-features

Description

This command enables the UMC option within the Supported-Features AVP.

The no form of this command reverts to the default.

Default

no umc

6.2.1.48.31 supported-vendor

Syntax

supported-vendor {custom-1 | custom-2 | custom-3}

no supported-vendor

Context

config>mobile>profile>npcf

Description

This command enables vendor-specific customizations for the Npcf interface. Each option represents a specific customer.

The no form of this command reverts to the default (disabled).

Default

no supported-vendor

Parameters

custom-1 | custom-2 | custom-3

specifies a customer number

6.2.1.48.32 ul-rq-timer

Syntax

`ul-rq-timer value`

`no ul-rq-timer`

Context

config>mobile>profile>npcf

Description

This command specifies a reflective QoS timer for the UL traffic. When the PCF disables the reflective QoS for an SDF, the UPF does not set the RQI bit in the encapsulation header on the N3 reference point for the corresponding DL traffic; the UL packets for this SDF are not dropped for the duration of the RQ time from the PCRF plus the specified time. The corresponding timer at the UE is one minute (default) that applies when no RQ timer is sent from the PCRF. For this case, the SMF timer expires at 62 seconds (60 seconds for the UE plus 2 seconds for the SMF by default). After the timer expires, the uplink packets for this SDF are not accepted by the UPF over the QoS flow on which the reflective QoS was applied for this SDF; the UL PDR for this SDF is removed from that QoS flow.



NOTE:

This command is not supported in the current release.

Parameters

value

specifies the timer value, in seconds

Values: 2 to 60

Default: 2

**NOTE:**

By default this timer is two seconds longer than the RQ timer. This is to ensure that no UL packet is dropped until the UE-derived QoS rule is deleted by the UE.

6.2.1.49 use-last-auth-ambr

Syntax

[no] use - last - auth - ambr

Context

config>mobile>profile>npcf

Description

This command configures the SMF behavior when the PCF does not provide the authorized AMBR.

If the use - last - auth - ambr command is configured, when the SMF triggers a PCF update to modify the authorized AMBR and the PCF replies without providing the authorized AMBR, the SMF assumes that the proposed AMBR is rejected (the AMBR is not modified).

If the use - last - auth - ambr command is not configured, when the SMF triggers a PCF update to modify the authorized AMBR and the PCF replies without providing the authorized AMBR, the SMF assumes that the proposed AMBR is accepted (the AMBR is modified).

Default

no use-last-auth-ambr

6.2.1.50 NSMF profile command descriptions

6.2.1.50.1 nsmf-profile

Syntax

[no] nsmf - profile *profile-name*

Context

config>mobile>profile

Description

This command configures an SMF profile to define the failure handling policy when the SMF services are used.

The no form of this command removes the SMF profile from the configuration.

Parameters

profile-name

specifies the SMF profile name, up to 32 characters

6.2.1.50.2 http-status-code

Syntax

```
http-status-code status-code-value application-error app-error-code failure-action  
no http-status-code status-code-value [application-error app-error-code]
```

Context

```
config>mobile>profile>nsmf-profile
```

Description

This command specifies the SMF failure handling policy. The defined action is applied if the response from the SMF matches with the configured HTTP status code. If the optional application error is included, the response must have a matching application error.

The no form of this command removes the error handling for the specified HTTP status code and reverts to the default action (continue).

Parameters

status-code-value

specifies the three-digit status code value

Values: 300 to 599

app-error-code

specifies the application error; a string value up to 64 characters

failure-action

specifies the failure action

Values:

terminate

the PDU session is terminated

continue

the PDU session is not terminated

retry-continue

the service operation is reattempted with another SMF peer; when the maximum number of peer failovers (configured using the `max-peer-failover` command) is reached, the PDU session is not terminated

retry-terminate

the service operation is reattempted with another SMF peer; when the maximum number of peer failovers (configured using the `max-peer-failover` command) is reached, the PDU session is terminated

Default: `continue`

6.2.1.50.3 max-peer-failover

Syntax

`max-peer-failover` *value*

`no max-peer-failover`

Context

`config>mobile>profile>nsmf-profile`

Description

This command configures the maximum number of failovers to another SMF in the SMF set.

The `no` form of this command reverts to the default.

**NOTE:**

This command is not supported in the current release.

Parameters

value

specifies the maximum number of failovers

Values: 0 to 3

Default: 1

6.2.1.50.4 retry-count

Syntax

```
retry-count [failure-action failure-action] value  
no retry-count
```

Context

config>mobile>profile>nsmf-profile

Description

This command configures the maximum number of retransmissions.
The no form of this command reverts to the default.

Parameters

failure-action

specifies the action, if no response is received

Values	terminate	The PDU session is terminated
–	continue	The PDU session is not terminated

Default: continue

value

specifies the maximum number of retransmissions
Values: 0 to 8
Default: 0

6.2.1.50.5 transaction-timer

Syntax

```
transaction-timer value  
no transaction-timer
```


Context

config>mobile>profile>nsmf-profile

Description

This command configures the time that the SMF waits for a response from the peer SMF.

The no form of this command reverts to the default.

Parameters

milliseconds

specifies the transaction timer, in milliseconds (increments of 100 milliseconds)

Values: 1000 to 20000

Default: 5000

6.2.1.51 OAuth2.0 authorization profile command descriptions

6.2.1.51.1 oauth2-profile

Syntax

[no] oauth2-profile *profile-name*

Context

config>mobile>profile

Description

This command configures an OAuth2.0 authorization profile that specifies behavioral aspects of OAuth both for client and producer sides.

The no form of this command removes the specified profile from the configuration.

Parameters

profile-name

specifies the name of the profile, up to 32 characters

6.2.1.51.2 oauth2-nrf-certificate

Syntax

```
oauth2-nrf-certificate cert-filename  
no oauth2-nrf-certificate
```

Context

```
config>mobile>profile>oauth2-profile
```

Description

This command configures the certificate of the NRF acting as an authorization server. When the NRF certificate is configured, the SMF, as a service producer, uses the public key of this certificate to validate the integrity of the access token it receives in incoming service requests. If the UUID of the NRF is present in the certificate, the SMF, as a service producer, uses this UUID when validating the issuer claim in the access token.

The no form of this command removes the configured certificate.

Parameters

cert-filename

specifies the filename of an imported certificate via the `admin>certificate>import` command, up to 256 characters

6.2.1.51.3 validate-consumer-uuid

Syntax

```
[no] validate-consumer-uuid
```

Context

```
config>mobile>profile>oauth2-profile
```

Description

This command specifies that token proof of possession is checked. When this validation is set, the SMF, as a service producer, validates that the UUID of the NF consumer, which is found in the sub-claim of the access token request of the incoming request, matches to the UUID that the NF consumer has been authenticated with. Authentication is provided in the TLS layer.

The no form of this command disables this validation.

Default

no validate-consumer-uuid

6.2.1.52 Overload control profile command descriptions

6.2.1.52.1 overload-control-profile

Syntax

[no] overload-control-profile *profile-name*

Context

config>mobile>profile

Description

This command specifies an overload control profile.

The no form of this command removes the profile.

Parameters

profile-name

specifies the name of the profile, up to 32 characters

6.2.1.52.2 exclude-arp-priority-level

Syntax

[no] exclude-arp-priority-level *arp-value*

Context

config>mobile>profile>overload-control-profile

Description

This command configures a new ARP priority level for an overload control profile. The sessions that match the ARP priority level are excluded from the manual restriction, and the APN level congestion control. The configuration is not applied to GTPv1.

The no form of this command removes the ARP priority level from the overload control profile.

Parameters

arp-value

specifies the ARP priority level

Values: 1 to 15

6.2.1.52.3 session-rejection-rate

Syntax

session-rejection-rate *rejection-rate*

no session-rejection-rate

Context

config>mobile>profile>overload-control-profile

Description

This command specifies the rejection rate as a percentage for the new sessions in the PGW. The PGW accepts all sessions by default.

The no form of this command reverts to the default (rejection rate set to 0).

Default

no session-rejection-rate

Parameters

rejection-rate

specifies the rejection rate

Values: 0 to 100

6.2.1.53 PFCP profile command descriptions

6.2.1.53.1 pfc

Syntax

pfc

Context

config>mobile>profile

Description

This command enables the context to configure a PFCP profile.

6.2.1.53.2 pfc-association-peer-list

Syntax

[no] pfc-association-peer-list *profile-name*

Context

config>mobile>profile>pfc

Description

This command creates a PFCP association peer list profile.

The no form of this command removes the profile from the configuration.

Parameters

profile-name

specifies the name for the profile, up to 32 characters

6.2.1.53.3 node-id

Syntax

[no] node-id *node-id*

Context

config>mobile>profile>pfc>pfc-association-peer-list

Description

This multi-entry command configures a node ID string of the remote PFCP peer.

The no form of this command removes the specified node ID of the remote PFCP peer from the configuration.

Parameters

node-id

specifies the node ID of the remote PFCP peer, up to 31 characters

6.2.1.53.4 bfd-enable

Syntax

[no] bfd-enable

Context

config>mobile>profile>pfcp>pfcp-association-peer-list>node-id

Description

This command enables the BFD association with the PFCP peer. The node (CP or UP) uses the BFD state to determine the route convergence toward itself, before triggering an Sx Association Setup Request toward the PFCP peers.

The no form of this command disables the BFD association with the PFCP peer.



NOTE:

- When this feature is used on the UP function, only a single IP address peer can be configured.
- For the BFD state to be Up per PFCP peer, you must configure it as follows:
 - on both the CP and UP peering nodes, use the pfcp-peer-association-peer-list command
 - on the Sx loopback interface, apply the configuration in the config>service>vprn>if context
- To monitor the BFD status for PFCP interfaces, use the show >router>bfd session command with the appropriate router ID.

Default

no bfd-enable

6.2.1.53.5 enable-partial-failure

Syntax

[no] enable-partial-failure

Context

config>mobile>profile>pfcf>pfcf-association-peer-list>node-id

Description

This command enables the partial failover of the remote PFCF peer.

The no form of this command reverts to the default (disabled).

Default

no enable-partial-failure

6.2.1.53.6 path-hold-time

Syntax

path-hold-time *minutes*

no path-hold-time

Context

config>mobile>profile>pfcf>pfcf-association-peer-list>node-id

config>mobile>profile>pfcf>pfcf-association-peer-list>pfcf-peer

Description

This command configures the path hold time for the peer and must be configured on both the CP and UP functions for which a path hold time is configured.

When a PFCF path hold time is configured:

- If the PFCF path management detects a peer down and then up with the same timestamp, and the path hold time is still running and has not expired, the PFCF path session deletion process is not started. When the peer is detected to be up, all the call flows/sessions and the new session setup resume.
- If the PFCF path management detects the peer still down, and the path hold time expires or comes back up with a new timestamp, the PFCF path session deletion process deletes all sessions.

When two peer entities (PFCP peers) per node are used, the `path-hold-time` must be configured under the `config>mobile>profile>pfcf>pfcf-association-peer-list>node-id` context. If a single peer entity (PFCP peer) is used per node, the `path-hold-time` can be configured under the `config>mobile>profile>pfcf>pfcf-association-peer-list>node-id` or `config>mobile>profile>pfcf>pfcf-association-peer-list>pfcf-peer` context.

The `no` form of this command disables the restoration time.

**NOTE:**

- While the path hold timer is running, any new PDN/PDU attach or update requests to any existing PDN/PDU sessions are rejected and deleted on the CP function.
- While the path hold timer is running, the UP function deletes locally the existing sessions for which it receives GTP-u error indications from the gNB/eNB.
- While the path hold timer is running, the charging reporting may be lost and the given quota may be exceeded.
- The `path-hold-time` must not be enabled on the UP functions with IMS APNs/DNNs or other APNs/DNNs that require accurate charging and LI handling.
- In case of a LOAM/OAM active/standby failover or in case of an ICR deployment with an active/standby gateway failover, the path hold timer is restarted so that it runs longer.

Parameters

minutes

specifies the number of minutes of the path hold timer

Values: 5 to 1440

Default

`no path-hold-time`

6.2.1.53.7 peer-entity

Syntax

`[no] peer-entity ip-address`

Context

`config>mobile>profile>pfcf>pfcf-association-peer-list>node-id`

Description

This multi-entry command configures the IPv4 or the IPv6 address of the PFCP entity for the remote PFCP peer.

The no form of this command removes the specified IPv4 or the IPv6 address of the PFCP entity for the remote PFCP peer from the configuration.

Parameters

ip-address

specifies the peer IPv4 or IPv6 address

Values		
	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

6.2.1.53.8 ue-ip-allocation

Syntax

[no] ue-ip-allocation

Context

config>mobile>profile>pfcp>pfcp-association-peer-list>pfcp-peer>node-id

Description

This command enables the UP-based IP address allocation between the CP and the UP. The command must be enabled on the CP.

The no form of this command disables the UP-based IP address allocation.

Default

no ue-ip-allocation

6.2.1.53.9 pfc-p-peer

Syntax

[no] pfc-p-peer ipv4 | ipv6

Context

config>mobile>profile>pfc-p>pfc-p-association-peer-list

Description

This multi-entry command configures the IPv4 or the IPv6 address of the remote PFCP peer.

The no form of this command removes the specified IPv4 or the IPv6 address of the remote PFCP peer from the configuration.



NOTE:

When this feature is used on the UP function, only a single peer’s IP address can be configured.

Parameters

ipv4 | ipv6

specifies the peer’s IP address

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

6.2.1.53.10 bfd-enable

Syntax

[no] bfd-enable

Context

config>mobile>profile>pfcf>pfcf-association-peer-list>pfcf-peer

Description

This command enables the BFD association with the PFCF peer. The node (CP or UP) uses the BFD state to determine route convergence toward itself, before triggering an Sx Association Setup Request toward the PFCF peers.

The no form of this command disables the BFD association with the PFCF peer.

Default

no bfd-enable

6.2.1.53.11 partial-failure

Syntax

[no] partial-failure

Context

config>mobile>profile>pfcf>pfcf-association-peer-list>pfcf-peer

Description

This command enables partial failure handling between the CP and the UP. The command must be enabled on the CP.

The no form of this command disables the partial failure handling.

Default

no partial-failure

6.2.1.53.12 up-ip-allocation

Syntax

[no] up-ip-allocation

Context

config>mobile>profile>pfcf>pfcf-association-peer-list>pfcf-peer

Description

This command enables the UP-based IP address allocation handling between the CP and UP. The command must be enabled on the CP.

The no form of this command disables the UP-based IP address allocation handling.

Default

no up-ip-allocation

6.2.1.53.13 pfcf-error-mapping-profile

Syntax

[no] pfcf-error-mapping-profile *profile-name*

Context

config>mobile>profile>pfcf>

Description

This command configures the PFCF error mapping profile.

The no form of this command removes the specific profile from the configuration.

Parameters

profile-name

specifies the PFCF error mapping profile name, up to 32 characters

6.2.1.53.14 pfcf-to-5gsm

Syntax

```
pfcf-to-5gsm pfcf-cause-code 5gsm-cause gsm-cause-code  
no pfcf-to-5gsm pfcf-cause-code
```

Context

```
config>mobile>profile>pfcf>pfcf-error-mapping-profile>
```

Description

This command configures the mapping between the PFCF cause code and the 5GSM cause code.
The no form of this command removes the mapping.

Parameters

pfcf-cause-code

specifies the PFCF cause code

Values: 64 to 255

5gsm-cause-code

specifies the 5GSM cause code

Values: 0 (disables N1 mapping) to 255

6.2.1.53.15 pfcf-to-gtp

Syntax

```
pfcf-to-gtp pfcf-cause-code gtp-error gtp-cause-code  
no pfcf-to-gtp pfcf-cause-code
```

Context

```
config>mobile>profile>pfcf>pfcf-error-mapping-profile>
```

Description

This command configures the mapping between the PFCF cause code and the GTP cause code.
The no form of this command removes the mapping.

Parameters

pfcp-cause-code

specifies the PFCP cause code

Values: 64 to 255

gtp-cause-code

specifies the GTP cause code

Values: 64 to 255

6.2.1.53.16 pfcp-to-http

Syntax

```
pfcp-to-http pfcp-cause-code http-error http-status-code-out application-error-out  
app-error-code-out
```

```
no pfcp-to-http pfcp-cause-code
```

Context

```
config>mobile>profile>pfcp>pfcp-error-mapping-profile>
```

Description

This command configures the mapping between the PFCP cause code, the HTTP cause code, and the application error. The HTTP cause code and application error are used in the N11 when the SMF rejects the request.

The no form of this command removes the mapping.

Parameters

pfcp-cause-code

specifies the PFCP cause code

Values: 64 to 255

http-status-code-out

specifies the three-digit HTTP status code value

Values: 400 | 403 | 404 | 413 | 415 | 429 | 500 | 503 | 504

app-error-code-out

specifies the application error, up to 36 characters

6.2.1.53.17 pfc-p-options

Syntax

pfc-p-options

Context

config>mobile>profile>pfc-p

Description

This command enables the context for the PFCP options in the PFCP messages.

6.2.1.53.18 exclude-lci

Syntax

[no] exclude-lci

Context

config>mobile>profile>pfc-p>pfc-p-profile>pfc-p-options

Description

This command includes or excludes the LCI IE in the PFCP messages.

Default

no exclude-lci

6.2.1.53.19 exclude-oci

Syntax

[no] exclude-oci

Context

config>mobile>profile>pfc-p>pfc-p-profile>pfc-p-options

Description

This command includes or excludes the OCI IE in the PFCP messages.

Default

no exclude-oci

6.2.1.53.20 pfcf-profile

Syntax

[no] pfcf-profile *profile-name*

Context

config>mobile>profile>pfcf

Description

This command creates a PFCF profile.

The no form of this command removes the profile from the configuration.

Parameters

profile-name

specifies the name for the profile, up to 32 characters

6.2.1.53.21 association-retry-timer

Syntax

association-retry-timer *retry-time*

no association-retry-timer

Context

config>mobile>profile>pfcf>pfcf-profile

Description

This command configures the retry time between sending an association request in case of an Sx/N4 interface path fault or if the initial association setup request fails.

Parameters

retry-time

specifies the time between retries for sending an association request, in seconds

Values: 1 to 36000

Default: 10

6.2.1.53.22 disable-pdi-optimization

Syntax

```
[no] disable-pdi-optimization
```

Context

```
config>mobile>profile>pfcf>pfcf-profile
```

Description

This command configures the PFCF PDI optimization support.

The no form of this command reverts to the default (PDI optimization support is enabled).



NOTE:

- Nokia recommends disabling the PDI optimization only in interoperability cases where the PFCF support level is limited.
- In the current release, disabling PDI optimization for SGW sessions is not supported. PDI optimization can only be disabled in interoperability cases for the following deployment scenarios:
 - PGW-C/GGSN-C and PGW-U/GGSN-U
 - SMF and UPF
 - PGW-C/SMF and UPF/PGW-U

Default

```
no disable-pdi-optimization
```

6.2.1.53.23 ext-urr-id

Syntax

```
ext-urr-id {enable | disable}
```

```
no ext-urr-id
```

Context

```
config>mobile>profile>pfcf>pfcf-profile
```

Description

This command enables or disables support for the extended URR ID range for pre-defined URR IDs. When disabled, the pre-defined URR ID value range is limited.

The no form of this command reverts to the default.

Default

ext-urr-id disable

Parameters

enable

specifies the extended URR ID value range, that is, from 0x00000001 to 0x7fffffff (1 to 2147483647)

disable

specifies the limited URR ID value range, that is, from 0x00000001 to 0x00007530 (1 to 30000)

6.2.1.53.24 heart-beat

Syntax

```
heart-beat [interval seconds] [retry-count retry-count-value] [timeout timeout-value]  
no heart-beat
```

Context

```
config>mobile>profile>pfcf>pfcf-profile
```

Description

This command configures the heartbeat attributes for the PFCP profile. When configured, heartbeats are sent when the PFCP node association has been successfully set up. The gateway will send a heartbeat message at the time specified in the `interval` parameter. If a PFCP Heartbeat Response is not received within the time specified in the `timeout` parameter, the gateway retries the timeout wait period, for the number of times specified in the `retry-count` parameter. If the gateway hears a PFCP Heartbeat Response from the remote end, it aborts the retry procedure and restarts the heartbeat interval timer for this peer. If no response is received before the end of the retry-timeout sequence, the PFCP peer is declared to be down.

Heartbeats are turned off by setting the interval value to 0. In this case, the gateway only responds to heartbeat requests received from the remote PFCP peer.

Parameters

seconds

specifies the heartbeat message interval, in seconds

Values: 0 | 5 to 180

Default: 60

retry-count-value

specifies the number of retries

Values: 1 to 15

Default: 4

timeout-value

specifies the timeout interval, in seconds

Values: 1 to 20

Default: 5

6.2.1.53.25 ip-dscp

Syntax

```
ip-dscp ip-dscp-value
```

```
no ip-dscp
```

Context

```
config>mobile>profile>pfcpc>pfcpc-profile
```

Description

This command configures the DSCP value in the IP Header used for PFCP signaling messages.

Parameters

ip-dscp-value

specifies the DSCP value

Values: 0 to 63

Default: 56

6.2.1.53.26 ip-ttl

Syntax

```
ip-ttl ip-ttl-value  
no ip-ttl
```

Context

```
config>mobile>profile>pfcf>pfcf-profile
```

Description

This command configures the IP TTL value used for PFCP signaling messages.

Parameters

ip-ttl-value

specifies the IP TTL value

Values: 0 to 255

Default: 255

6.2.1.53.27 message-retransmit

Syntax

```
message-retransmit [timeout timeout] [timeout-msec timeout-msec] [retry-count retry-count-value]  
no message-retransmit
```

Context

```
config>mobile>profile>pfcf>pfcf-profile
```

Description

This command configures the PFCP message retransmit timeout and retry count values. The retransmit timeout consists of T1 and N1 values. T1 is the time period that the PFCP waits before re-attempting the request message if there is no response to an initial request message. T1 can be configured either in seconds (*timeout*) or milliseconds (*timeout-msec*). N1 is the maximum number of times that the PFCP attempts to send a message. N1 can be configured by specifying the value of the *retry-count* keyword.

Parameters

timeout

specifies the time that the PFCP waits before resending a message if there is no response to an initial message, in seconds

Values: 1 to 30

Default: 5

timeout-msec

specifies the time that the PFCP waits before resending a message if there is no response to an initial message, in milliseconds

Values: 1000 to 3000 (increments of 100)

Default: 5000



NOTE:

You can specify either the `timeout` or the `timeout-msec` keywords.

retry-count-value

specifies the number of times the PFCP attempts to send a message

Values: 1 to 8

Default: 3

6.2.1.53.28 pfcpc-error-mapping-profile

Syntax

`pfcpc-error-mapping-profile` *profile-name*

`no pfcpc-error-mapping-profile`

Context

`config>mobile>profile>pfcpc>pfcpc-profile`

Description

This command configures the PFCP error mapping profile for rejections received from the UPF.

The `no` form of this command removes the configuration.

Parameters

profile-name

specifies the PFCP error mapping profile name, up to 32 characters

6.2.1.53.29 pfc-p-u-tunnel

Syntax

```
pfc-p-u-tunnel
```

Context

```
config>mobile>profile>pfc-p>pfc-p-profile
```

Description

This command enables the context to configure a PFCP session message time interval between attempts to re-establish a shared PFCP-u tunnel session between the CP function and the UP function that is detected to have failed.

6.2.1.53.30 re-attempt-timer

Syntax

```
re-attempt-timer re-attempt-timer
```

```
no re-attempt-timer
```

Context

```
config>mobile>profile>pfc-p>pfc-p-profile>pfc-p-u-tunnel
```

Description

This command configures the interval between attempts to re-establish a shared PFCP-u tunnel session between the CP function and the UP function. The timer is activated after the PFCP retransmission mechanism takes place, as defined by the `timeout` and `retry-count-value` parameters in the `config>mobile>profile>pfc-p>pfc-p-profile>message-retransmit` context. The CP function then continues to send retransmissions indefinitely, at the interval configured for the `re-attempt-timer`. The `no` form of this command reverts to the default value, which turns off the re-attempt timer. No further attempts are made to establish a shared PFCP-u tunnel after the initial attempts defined by the `message-retransmit` command.

Parameters

re-attempt-timer

specifies the interval that the PFCP waits before re-attempting to establish the PFCP-u tunnel again, in seconds

Values: 5 to 3600

Default: 0 (turned off)

6.2.1.53.31 smf-set-support

Syntax

[no] smf-set-support

Context

config>mobile>profile>pfcpc>pfcpc-profile

Description

This command enables SMF set support on the UP function. When enabled, the UP function also sets the MPAS flag in the UP Function Features IE to indicate the support.

The no form of this command reverts to the default (disabled).

Default

no smf-set-support

6.2.1.53.32 pfcpc-translate-profile

Syntax

[no] pfcpc-translate-profile *profile-name*

Context

config>mobile>profile>pfcpc

Description

This command creates a PFCP message translation function profile for sending selective opaque values in the IEs.

The no form of this command removes the specified profile from the configuration.

Parameters

profile-name

specifies the name of the profile, up to 32 characters

6.2.1.53.33 field

Syntax

```
[no] field {hdr-enrich-imsi | hdr-enrich-msisdn | hdr-enrich-imei-sv | hdr-enrich-rat-type | hdr-enrich-charging-id | hdr-enrich-user-location | hdr-enrich-billing-type | hdr-enrich-roaming-status | hdr-enrich-slice-id | hdr-enrich-amf-ip | hdr-enrich-pcc-rulebase}
```

Context

```
config>mobile>profile>pfcf>pfcf-translate-profile
```

Description

This command configures the field values to be forwarded to the UP.



NOTE:

The default profile forwards all fields to the UP.

The no form of this command removes the selected field from being forwarded to the UP.

Parameters

hdr-enrich-imsi

specifies to include the IMSI field in header enrichment

hdr-enrich-msisdn

specifies to include the MSISDN field in header enrichment

hdr-enrich-imei-sv

specifies to include the IMEI SV field in header enrichment

hdr-enrich-rat-type

specifies to include the RAT type field in header enrichment

hdr-enrich-charging-id

specifies to include the charging Id field in header enrichment

hdr-enrich-user-location

specifies to include the user location field in header enrichment

hdr-enrich-billing-type

specifies to include the billing type field in header enrichment

hdr-enrich-roaming-status

specifies to include the roaming status field in header enrichment

hdr-enrich-slice-id

specifies to include the slice Id field in header enrichment

hdr-enrich-amf-ip

specifies to include the AMF or MME IP address (IPv4 or IPv6) in header enrichment

hdr-enrich-pcc-rulebase

specifies to include the active PCC rulebase in header enrichment

6.2.1.53.34 up-peer-list

Syntax

[no] up-peer-list *list-name*

Context

config>mobile>profile>pfcf

Description

This command creates a list of UPs with the supported APNs and the UE IP address pools.

The no form of this command removes the list from the configuration.

**NOTE:**

This command is only used if the gateway is configured as CP function using the config>mobile>pdn>instance-type command.

Parameters

list-name

specifies the name for the list, up to 32 characters

6.2.1.53.35 peer

Syntax

[no] peer *ipv4* | *ipv6*

Context

config>mobile>profile>pfcf>up-peer-list

Description

This multi-entry command specifies the IPv4 or the IPv6 address of the user plane peer.

Parameters

ipv4 | *ipv6*
specifies the user plane peer’s IP address

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

6.2.1.53.36 apn

Syntax

[no] apn *apn-name*

Context

config>mobile>profile>pfcf>up-peer-list>peer

Description

This multi-entry command specifies the APN names associated with the user plane peer.

Parameters

apn-name

specifies an APN name, up to 32 characters

6.2.1.53.37 block

Syntax

[no] block

Context

config>mobile>profile>pfcf>up-peer-list>peer>apn

Description

This command blocks a specific peer from being selected for new sessions on this APN. Existing sessions on a blocked UP peer are served normally.

The no form of this command reverts to the default.

Default

no block

6.2.1.53.38 ip-pool-list

Syntax

ip-pool-list

Context

config>mobile>profile>pfcf>up-peer-list>peer>apn

Description

This command enables the context to configure IP address pools.

6.2.1.53.39 ip-pool

Syntax

[no] ip-pool *pool-name*

Context

```
config>mobile>profile>pfcf>up-peer-list>peer>apn>ip-pool-list
```

Description

This multi-entry command specifies the IP address pool name for the APN. A maximum of eight IP address pools can be configured for an APN. When more than eight IP address pools are required, this field can be left empty; only under the APN, the network realm can be configured from where the IP pools would be inherited.

Parameters

pool-name
specifies the IP address pool name, up to 32 characters

6.2.1.53.40 nas-ip

Syntax

```
nas-ip nas-ip-address
no nas-ip
```

Context

```
config>mobile>profile>pfcf>up-peer-list>peer>apn
```

Description

This command configures the NAS-IP-Address AVP for each configured APN that is used toward the RADIUS server.

Parameters

nas-ip-address
specifies the RADIUS NAS-IP-Address

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)

–	–	x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

6.2.1.53.41 network-realm

Syntax

```
network-realm network-realm  
no network-realm
```

Context

```
config>mobile>profile>pfcf>up-peer-list>peer>apn
```

Description

This multi-entry command specifies the network realm name associated to this APN.

Parameters

network-realm
specifies the network realm name, up to 80 characters

6.2.1.53.42 pco-option

Syntax

```
pco-option
```

Context

```
config>mobile>profile>pfcf>up-peer-list>peer>apn
```

Description

This command enables the context to configure UE PCO options per UP peer.

6.2.1.53.43 dns-server-ipv4

Syntax

dns-server-ipv4

Context

config>mobile>profile>pfcf>up-peer-list>peer>apn>pco-option

Description

This command configures the IPv4 address of the DNS server used by a UE to resolve domain names to IP addresses.

6.2.1.53.44 primary

Syntax

primary *ip-address*

no primary

Context

config>mobile>profile>pfcf>up-peer-list>peer>apn>pco-option>dns-server-ipv4

Description

This command specifies the IPv4 address of the primary DNS server used by a UE to resolve domain names to IP addresses for a specific UP peer.

Parameters

ip-address

specifies the IPv4 address

Values: a.b.c.d

6.2.1.53.45 secondary

Syntax

secondary *ip-address*

no secondary

Context

config>mobile>profile>pfcf>up-peer-list>peer>apn>pco-option>dns-server-ipv4

Description

This command specifies the IPv4 address of the secondary DNS server used by a UE to resolve domain names to IP addresses for a specific UP peer.

Parameters

ip-address

specifies the IPv4 address

Values: a.b.c.d

6.2.1.53.46 dns-server-ipv6

Syntax

dns-server-ipv4

Context

config>mobile>profile>pfcf>up-peer-list>peer>apn>pco-option

Description

This command configures the IPv6 address of the DNS server used by a UE to resolve domain names to IP addresses.

6.2.1.53.47 primary

Syntax

primary *ipv6-address*

no primary

Context

config>mobile>profile>pfcf>up-peer-list>peer>apn>pco-option>dns-server-ipv6

Description

This command specifies the IPv6 address of the primary DNS server used by a UE to resolve domain names to IP addresses for a specific UP peer.

Parameters

ipv6-address

specifies the IPv6 address, up to 64 characters

6.2.1.53.48 secondary

Syntax

secondary *ipv6-address*

no secondary

Context

config>mobile>profile>pfcf>up-peer-list>peer>apn>pco-option>dns-server-ipv6

Description

This command specifies the IPv6 address of the secondary DNS server used by a UE to resolve domain names to IP addresses for a specific UP peer.

Parameters

ipv6-address

specifies the IPv6 address, up to 64 characters

6.2.1.53.49 tai-lai-list

Syntax

[no] tai-lai-list *tai-lai-list-list-name*

Context

config>mobile>profile>pfcf>up-peer-list>peer>apn

Description

This command associates a TAI or a LAI profile list to the APN, when the initial attach is within a specific TAI/LAI area.

The no form of this command removes the specified list from the configuration.

Parameters

tai-lai-list-list-name

specifies the TAI or LAI list name, up to 32 characters

6.2.1.53.50 ip-pool

Syntax

[no] ip-pool *pool-name*

Context

config>mobile>profile>pfcp>up-peer-list>peer>apn>tai-lai-list

Description

This multi-entry command specifies the IP address pool names for the APN and the associated TAI or LAI. A maximum of eight IP address pools can be configured for each APN.

The no form of this command removes the specified IP address pool from the configuration.

Parameters

pool-name

specifies the IP address pool name, up to 32 characters

6.2.1.53.51 disable-eps-iwk

Syntax

[no] disable-eps-iwk

Context

config>mobile>profile>pfcp>up-peer-list>peer

Description

This command disables EPS interworking support in the user plane peer.



NOTE:

Disabling the EPS interworking impacts only the user plane selection of the *new* sessions. EPS interworking is not disabled for existing sessions that use the user plane peer.

The no form of this command reverts to the default, that is, EPS interworking is supported.

Default

no disable-ews-iwk

6.2.1.53.52 group-label

Syntax

[no] group-label *profile-name*

Context

config>mobile>profile>pfcf>up-peer-list>peer

Description

This multi-entry command associates a group for the UP instance. The group association creates a logical grouping of target nodes with selection criteria based on different parameters (APN, TAI, LAI, serving node, IP address, and so on). A maximum of 5 groups can be associated with a PFCF peer.

The no form of this command removes the group association from the PFCF peer.

Parameters

profile-name

specifies the group name, up to 32 characters

6.2.1.53.53 network-realm

Syntax

[no] network-realm *network-realm*

Context

```
config>mobile>profile>pfcf>up-peer-list>peer
```

Description

This multi-entry command specifies the network realm names associated with the virtual router instance on the user plane peer where the UE IP address pools are located. This is also communicated over the Session Establishment process together with the allocated UE IP address to the user plane peer.

The no form of this command removes the specified network realm name from the configuration.

Parameters

network-realm

specifies the network realm name as an APN-IN and APN-OI, up to 80 characters



NOTE:

The APN-IN must not exceed 63 characters.

6.2.1.53.54 fragment-distribution-size

Syntax

```
fragment-distribution-size fragment-distribution-size-value
```

```
[no] fragment-distribution-size
```

Context

```
config>mobile>profile>pfcf>up-peer-list>peer>network-realm
```

Description

This command specifies the number of UP MG groups in CP, and applies to all configured IP-pools under the network-realm context. Also, the command is used to start assigning or using immediately the prefixes required for the centralized IP-pool allocation in UPF LB-less mode.

The no form of this command reverts to the default.

Default

```
no fragment-distribution-size
```

Parameters

fragment-distribution-size-value

specifies the value of the fragment distribution size

Values: 0 to 15

Default: 0 (disabled)

6.2.1.53.55 ip-pool

Syntax

[no] ip-pool *pool-name*

Context

config>mobile>profile>pfcf>up-peer-list>peer>network-realm

Description

This multi-entry command specifies the IP address pool name that the control plane will use to allocate UE IPv4/IPv6 address and IPv6 prefixes from.

Parameters

pool-name

specifies IP address pool name, up to 32 characters

6.2.1.53.56 group-name

Syntax

[no] group-name *group-name*

Context

config>mobile>profile>pfcf>up-peer-list>peer>network-realm>ip-pool

Description

This command configures the group name required for selecting IP address pools used for load balancing. The IP address pools which are configured with the same group name belong to the same IP address group.

The no form of this command removes the group name from the configuration.

Parameters

group-name

specifies a group name, up to 32 characters

6.2.1.53.57 hold-timer-sec

Syntax

`hold-timer-sec hold-timer-in-sec [hold-for-pdn]`

`no hold-timer-sec`

Context

`config>mobile>profile>pfcf>up-peer-list>peer>network-realm>ip-pool`

Description

This command configures the IP address pool hold timer. This is the time, after the UE or the session is released, that the gateway waits before assigning the IP address to another UE or the session. When you specify the `hold-for-pdn` keyword, the gateway reassigns the reserved IP address to the UE or the session connection that released it if the UE or the session connection reattaches before the timer expires.

The `no` form of this command removes the configured hold timer value from the configuration.

Parameters

hold-timer-in-sec

specifies the hold timer, in seconds

Values: 0 to 864000

Default: 180

hold-for-pdn

keyword to reallocate a held IP address to the UE/session that released it



NOTICE:

Configuring a high value for the hold timer can lead to IP address exhaustion. If you configure a high value for the hold timer, make sure you have several IP addresses available or enable the `hold-for-pdn` keyword.

6.2.1.53.58 ipv4-prefix

Syntax

```
ipv4-prefix ipv4-address/mask [block [force]] [prefix-fragment-size prefix-fragment-size-value]
```

```
no ipv4-prefix ipv4-address/mask
```

Context

```
config>mobile>profile>pfcp>up-peer-list>peer>network-realm>ip-pool
```

Description

This command configures the IPv4 address prefix parameters. The prefix fragment size is defined for parent IP pools in LB-less mode deployments (non-CUPS). The IP pools are divided into fragments of this length/size. The system can internally lease fragments to individual MG groups at runtime to optimize IP address usage. A mobile host route is installed in each MG group specific child VPRN for each different fragment assigned to each MG group. Also, this is visible/advertised to site routers for attracting DL traffic.

Parameters

ipv4-address

specifies the IPv4 address

Values: a.b.c.d (host bits must be 0)

mask

specifies the IPv4 address prefix length

Values: 9 to 30

block

blocks any new IP address allocation from this prefix

force

clears all sessions that run in the blocked IP pool

prefix-fragment-size-value

specifies the value of the prefix fragment size

Values: 0 | 14 to 24

Default: 0 (disabled)

6.2.1.53.59 ipv6-prefix

Syntax

```
ipv6-prefix ipv6-address/prefix-length [block [force]] [prefix-fragment-size prefix-fragment-size-value]  
  
no ipv6-prefix ipv6-address/prefix-length
```

Context

```
config>mobile>profile>pfcp>up-peer-list>peer>network-realm>ip-pool
```

Description

This command configures the IPv6 prefix parameters. The prefix fragment size is defined for parent IP pools in LB-less mode deployments (non-CUPS). The IP pools are divided into fragments of this length/size. The system can internally lease fragments to individual MG groups at runtime to optimize IP address usage. A mobile host route is installed in each MG group specific child VPRN for each different fragment assigned to each MG group. Also, this is visible/advertised to site routers for attracting DL traffic.

Parameters

ipv6-address
specifies the IPv6 address

Values	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

prefix-length
specifies the IPv6 prefix length
Values: 41 to 56

block
blocks any new IP address allocation from this prefix

force

clears all sessions that run in the blocked IP pool

prefix-fragment-size-value

specifies the value of the prefix fragment size

Values: 0 | 46 to 56

Default: 0 (disabled)

6.2.1.53.60 network-determined

Syntax

[no] network-determined

Context

config>mobile>profile>pfc>up-peer-list>peer>network-realm>ip-pool

config>service>vprn>ip-local-pool

config>router>ip-local-pool

Description

This command specifies the IP address allocation method that the IP address pool supports. When enabled, the IP allocation method is determined based on the response of the peer RADIUS server during the initial attach. If the RADIUS server response provides an IPv4 address/IPv6 prefix, the IP address pool is (internally) considered as remote and can only be used for external IP address allocation from RADIUS. If not, the IP address pool can only be used for local IP address allocation.

The no form of this command reverts to the default value.



NOTE:

This command is only valid for IP address pools that are not configured for remote allocation (using the no remote command).

Default

no network-determined

6.2.1.53.61 pool-usage-threshold

Syntax

pool-usage-threshold high-threshold *high-threshold* [low-threshold *low-threshold*]


```
no pool-usage-threshold
```

Context

```
config>mobile>profile>pfc>up-peer-list>peer>network-realm>ip-pool
```

```
config>service>vprn>ip-local-pool
```

```
config>router>ip-local-pool
```

Description

This command configures the thresholds for IP address pool usage. When the thresholds are reached, alarms are generated.

The no form of this command reverts the threshold limits to their default values.

Default

```
no pool-usage-threshold
```

Parameters

high-threshold

specifies the high threshold value, in percentage rate

Values: 0 | 10 to 100

Default: 95

low-threshold

specifies the low threshold value, in percentage rate

Values: 0 | 10 to 100

Default: 85



NOTE:

The 0 value is allowed to indicate the disabling of the respective threshold. The value of the *high-threshold* parameter must be greater or equal to the value of the *low-threshold* parameter plus 10.

6.2.1.53.62 priority

Syntax

```
priority priority-value
```

```
no priority
```

Context

config>mobile>profile>pfc>up-peer-list>peer>network-realm>ip-pool

Description

This command configures the priority value of an IP address pool. The system attempts to use the highest priority IP address pool available and applies Round-robin load balancing (if configured) among the IP address pools with the same priority.

The no form of this command reverts to the default value.

Default

no priority

Parameters

priority-value

specifies the priority of the IP pool

Values: low | high

Default: none

6.2.1.53.63 remote

Syntax

[no] remote

Context

config>mobile>profile>pfc>up-peer-list>peer>network-realm>ip-pool

Description

This command configures an IP pool for remote allocation.

The no form of this command removes the value from the configuration.

Default

no remote

6.2.1.53.64 reserved-ip-list

Syntax

```
reserved-ip-list ip-address-list-name  
no reserved-ip-list
```

Context

```
config>mobile>profile>pfc>up-peer-list>peer>network-realm>ip-pool
```

Description

This command configures on the CP function a list of IP addresses as reserved addresses in the local IP address pool; it is not used for IP address allocation. Up to three IPv4 addresses and/or IPv6 addresses can be reserved from the configured IP address pool. The reserved IP addresses can then be used for RADIUS or DHCP loopback/client IP addresses. They can also be used in the ping command under the same apn>vprn-id/base. The config>profile>ip-profile list must be configured first. The list cannot be empty. The IP addresses configured in the *ip-address-list-name* parameter must belong to the configured prefixes in the local IP address pool and must not be in use.

The no form of this command removes the IP address list and releases the reserved IP addresses back to the IP address pool.



NOTE:

- When the reserved IP addresses are configured and used for RADIUS or DHCP loopback/client IP addresses, or they are used with the ping command, the no form of this command must not be used.
- In CUPS, the ping command runs from the UP function. Therefore, the reserved IP address must also be reserved on the UP function using the config>service>vprn>ip-local-pool>reserved-ipv4 | reserved-ipv6 command.
- This command is not allowed if any IP address in the reserved-ip-list is already allocated.

Parameters

ip-address-list-name

specifies the list name of the reserved IP addresses under the config>profile>ip-profile command, up to 32 characters

6.2.1.53.65 round-robin

Syntax

[no] round-robin

Context

config>mobile>profile>pfc>up-peer-list>peer>network-realm>ip-pool

Description

This command enables the round robin selection mechanism to allocate the IP address. The round robin algorithm is performed across all IP pools and prefixes configured on the APN/VPRN so that an IP address is not reused unless all other IPs under that APN/VPRN are already used.

The no form of this command reverts to the default.

Default

no round-robin

6.2.1.53.66 suppress-dotzero-broadcast-address

Syntax

[no] suppress-dotzero-broadcast-address

Context

config>mobile>profile>pfc>up-peer-list>peer>network-realm>ip-pool

Description

This command configures an IP address pool to suppress allocation of 0 and x.x.x.255 IP addresses from the local IP address pool to mobile customers.

The no form of this command removes the value from the configuration.

Default

no suppress-dotzero-broadcast-address

6.2.1.53.67 suppress-network-broadcast-address

Syntax

[no] suppress-network-broadcast-address

Context

config>mobile>profile>pfcp>up-peer-list>peer>network-realm>ip-pool

Description

This command configures an IP address pool to suppress allocation of network (1st IP address of the prefix) and broadcast (last IP address of the prefix) addresses from the local IP address pool to mobile subscribers for IPv4 prefixes which have 128 or fewer IP addresses.

The no form of this command reverts to the default.

Default

no suppress-network-broadcast-address

6.2.1.53.68 use-prefixes-immediately

Syntax

[no] use-prefixes-immediately

Context

config>mobile>profile>pfcp>up-peer-list>peer>network-realm>ip-pool

config>router>pool

config>service>vprn>pool

Description

This command specifies if the added prefixes to the IP address pool are forced to be used immediately. The next attach or PDN connection activation considers the added prefix as part of the available/candidate IP addresses for IP address allocation.

The no form of this command reverts to the default.

Default

no use-prefixes-immediately

6.2.1.53.69 ip-pool-distribution

Syntax

```
[no] ip-pool-distribution
```

Context

```
config>mobile>profile>pfcf>up-peer-list>peer>network-realm
```

Description

This command enables the UE IP address allocation to be distributed across the configured IP address pools and prefixes with the IP address pools. Thus, when configured, the CP function will distribute (round-robin) the allocation of the UE IP addresses across the configured IP address pools and the prefixes configured under these IP address pools for the network realm. For example, when two IP address pools are configured under a network realm and each IP address pool has two subnets/prefixes configured, the first PDN connection gets its IP address from the first IP address pool and the first subnet/prefix, the second PDN connection from the second IP address pool and the first subnet/prefix, the third PDN connection from the first IP address pool and the second subnet/prefix, the fourth PDN connection from the second IP address pool and the second subnet/prefix, and so on.

Default

```
no ip-pool-distribution
```

6.2.1.53.70 override

Syntax

```
override radius-server-group radius-group interface [router router-name] interface-name
```

```
no override radius-server-group radius-group
```

Context

```
config>mobile>profile>pfcf>up-peer-list>peer
```

Description

This command overrides the radius-server-group interface IP address with a UPF specific radius-server-group interface IP address.

Parameters

- radius-group**
specifies the name of a RADIUS group; up to 32 characters
- interface**
keyword to specify an interface name
- router**
keyword to specify a router name or service ID
- router-name**
specifies the router name or service ID

Values	router-name service-id	
–	router-name	Base
–	service-id	1 to 2147483647

Default: Base

- interface-name**
specifies an interface name, up to 32 characters and must start with a letter

6.2.1.53.71 slice

Syntax

slice *list-name*
[no] slice

Context

config>mobile>profile>pfcp>up-peer-list>peer

Description

This command specifies the network slice name that is associated with the user plane peer.
The no form of this command disassociates the network slice name from the user plane peer.

Parameters

list-name

specifies the network slice list name, up to 32 characters

6.2.1.53.72 upf-id

Syntax

```
upf-id uuid uuid [string string]
```

```
no upf-id
```

Context

```
config>mobile>profile>pfcf>up-peer-list>peer
```

Description

This command configures a UPF ID that the SMF includes in the charging records it sends over the Nchf SBI, to allocate quota and report usage per UPF peer. This allows the SMF to manage quota per UPF, when UPF selection is based on the local configuration. If the SMF uses the NRF for UPF discovery and selection, it uses the UPF ID from the NRF discovery instead of the locally configured UPF ID.



NOTE:

When the 5G LI functionality is enabled, the configuration of the `upf-id` is mandatory.

The no form of this command reverts to the default (disabled).

Default

```
no upf-id
```

Parameters

uuid

specifies the UPF ID in UUID canonical textual representation, that is, 16 octets represented as 32 hexadecimal (base-16) digits, displayed in 5 groups separated by hyphens, in the form 8-4-4-4-12 for a total of 36 characters (32 alphanumeric characters and 4 hyphens)

Values: xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx (x: [0..F]H)

string

specifies a string identifier in the ipDomain attribute for the selected UPF peer, used when the configured `ip-domain-type` is `upf-string`, up to 80 characters

6.2.1.53.73 upf-selection

Syntax

```
upf-selection {true | false}  
no upf-selection
```

Context

```
config>mobile>profile>pfcf>up-peer-list>peer
```

Description

This command configures a peer that is defined in the UP peer list to be selectable (`upf-selection true`) or not selectable (`upf-selection false`). This enables UPF selection to be suspended, to support on-the-fly configuration of the PFCP association list and UP peer list. When UPF selection is set to false, the CP function does not select the peer for any new PDN/PDU sessions, thus enabling the peer to gracefully drain its existing sessions before it is taken out of service. While in this mode, the CP function continues to handle existing sessions and to run PFCP path management with the peer.

The no form of this command reverts to the default state; UPF selection is true (selectable).



NOTE:

- The default state for the UPF selection of a new peer depends on the method used to add the peer to the UP peer list:
 1. If using CLI configuration through a console, UPF selection of a new peer is automatically false. This ensures that the peer is not selected for new PDN/PDU sessions until it has been fully added and is ready to be selected.
 2. If executing the admin save config file without UPF selection configured in it, UPF selection is automatically allowed (true).
- Changing the UPF selection is only allowed for existing peers. You must not add a new peer and set the UP selection in the same command execution (in a single line). Changing the UPF selection state is permitted after the peer has been added.

Default

```
no upf-selection
```

6.2.1.54 PMIPv6 profile command descriptions

6.2.1.54.1 pmipv6

Syntax

[no] pmipv6 *profile-name*

Context

config>mobile>profile

Description

This command creates a named PMIPv6 profile.

Parameters

profile-name

specifies the unique name for the profile

6.2.1.54.2 ip-dscp

Syntax

ip-dscp *value*

no ip-dscp

Context

config>mobile>profile>pmipv6

Description

This command sets the DSCP used when sending PMIPv6 control PDUs.

Parameters

value

specifies the DSCP to be set

Values: 0 to 63

Default: 56 (network control)

6.2.1.54.3 ip-ttl

Syntax

```
ip-ttl value  
no ip-ttl
```

Context

```
config>mobile>profile>pmipv6
```

Description

This command is used to set the TTL used when sending PMIPv6 control PDUs.

Parameters

value

specifies the TTL to be set

Values: 1 to 255

Default: 255 (network control)

6.2.1.54.4 keep-alive

Syntax

```
keep-alive [interval seconds] [retry-count value] [timeout timeout]  
no keep-alive
```

Context

```
config>mobile>profile>pmipv6
```

Description

This command is used to set parameters for the PMIPv6 heartbeat mechanism. If the interval is set to zero (0), the system will not send heartbeat messages; however, the system will always respond to heartbeat messages.

Parameters

seconds

specifies the interval between heartbeat messages, in seconds

Values: 0 | 60 to 180

Default: 60

value

specifies the number of times the system will resend a heartbeat message when there has been no response in seconds

Values: 1 to 15

Default: 4

timeout

specifies the amount of time the system awaits a response before sending the message again in seconds

Values: 1 to 20

Default: 5

6.2.1.54.5 message-retransmit

Syntax

message-retransmit [timeout *seconds*] [retry-count *value*]

no message-retransmit

Context

config>mobile>profile>pmipv6

Description

This command is used to set the timeout and retry for PMIPv6 control messages.

Parameters

seconds

specifies the number of seconds between control message resend attempts when there has been no response

Values: 1 to 30

Default: 5

value

specifies the amount of time the system awaits a response before sending the message again

Values: 1 to 8

Default: 3

6.2.1.54.6 supported-vendor

Syntax

supported-vendor *custom-1*
no supported-vendor

Context

config>mobile>profile>pmipv6

Description

This command enables the specified supported vendor function and attribute values.

The no form of this command enables per 3GPP standard function and attribute values.

Default

no supported-vendor

Parameters

custom-1

specifies the name of the supported vendor

6.2.1.55 PCMD profile command descriptions

6.2.1.55.1 pcmd

Syntax

[no] pcmd *profile-name*

Context

config>mobile>profile

Description

This command specifies a PCMD profile. Once created, you can define parameters related to PCMD. Profile names are case-sensitive and must be unique. Specifying a new PCMD profile name creates a new PCMD profile. Specifying an existing PCMD profile name enables the context for editing and configuration.

The no form of this command removes the PCMD profile and all associated configurations. To remove or modify a PCMD profile, you must first remove associations with any PCMD object.

Default

no pcmd

Parameters

profile-name

specifies the unique name for this pcmd profile, up to 32 characters maximum

6.2.1.55.2 data-report

Syntax

data-report t

Context

config>mobile>profile>pcmd

Description

This command enables the context to enter commands to configure the data report generation, such as the type of data reports to be generated and the period.

6.2.1.55.3 constant-flow-period

Syntax

constant-flow-period *period*

no constant-flow-period

Context

config>mobile>profile>pcmd>data-report

Description

This command configures the period for which the PCMD CDATA records is generated and reported. The command is only relevant when constant - flow command has been configured in the data-report>generation command in the PCMD profile. The no form of this command reverts to the default.

Parameters

period

specifies the value of the CDATA records transmission period in minutes

Values: 5 to 20

Default: 15

6.2.1.55.4 destination

Syntax

`destination {{ip-addr / fqdn} [port port] [router router-instance]} | inherit`

`no destination`

Context

`config>mobile>profile>pcmd`

`config>mobile>profile>pcmd>data-report`

`config>mobile>profile>pcmd>session-report`

Description

This command configures the destination where PCMD records are to be sent. The configured destination in the context `config>mobile>profile>pcmd` is by default inherited for both data and session PCMD reports, unless a different destination has been configured under the contexts `config>mobile>profile>pcmd>data-report` and/or `config>mobile>profile>pcmd>session-report`.

The `no` form of this command clears the configured destination for the context `config>mobile>profile>pcmd`.

The `no` form of this command sets the destination to the default value in the `config>mobile>profile>pcmd>data-report` and `config>mobile>profile>pcmd>session-report` contexts.

Default

`no destination`, in `pcmd` context

`inherit`, in `data-report` and `session-report` contexts

Parameters

ip-addr

specifies the IPv4 or IPv6 address for the local identity

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x
–	–	x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

fqdn
specifies the fully qualified domain name; DNS resolution is performed only once, either when the PCMD profile is assigned to a PDN or when the PDN is set to the no shutdown state.
Values: up to 255 characters

port
specifies the port number
Values: 1 to 65535
Default: 29780

router-instance
specifies the virtual router name or service ID

Values	router-name service-id	
–	router-name	Base
–	service-id	1 to 2147483647

Default: Base

inherit
specifies the destination configured under contexts:
config>mobile>profile>pcmd>data-report and
config>mobile>profile>pcmd>session-report are inherited from the parent context
config>mobile>profile>pcmd

6.2.1.55.5 generation

Syntax

```
generation {disabled | event-based | constant-flow}
no generation
```

Context

```
config>mobile>profile>pcmd>data-report
```

Description

This command configures the method that the data PCMD records get generated and the type of the data record.

The no form of this command reverts to the default.

Parameters

disabled

specifies the PCMD data record generation to turn off; data records are not reported

event-based

specifies the PCMD DATA records to be generated according to the charging events; this value requires the presence of charging functionality and its configuration on the gateway

constant-flow

specifies the PCMD CDATA records to be generated in a periodic manner; the PCMD CDATA records are reported periodically, independent of the charging functionality on the gateway

Default: event-based

6.2.1.55.6 heartbeat

Syntax

```
heartbeat [period period]
no heartbeat
```

Context

```
config>mobile>profile>pcmd
```

Description

This command enables the heartbeat transmission with the default period 10s. The period parameter enables the heartbeat transmission with the given period value.

The no form of this command disables the heartbeat transmission.

Default

no heartbeat

Parameters

period

specifies the period in seconds for the heartbeat transmission in seconds

Values: 1 to 120

Default: 10

6.2.1.55.7 session-report

Syntax

session-report

Context

config>mobile>profile>pcmd

Description

This command enables the context to configure the session report generation, such as the verbosity of the reports.

6.2.1.55.8 verbosity

Syntax

verbosity [failure-only | standard | extended]

no verbosity

Context

config>mobile>profile>pcmd>session-report

Description

This command configures the type of session reports being sent, such as failure only session reports, extended containers, and so on.

The no form of this command reverts to the default.

Parameters

failure-only

specifies reporting of only failed cases that use an extended structure for PCMD records; mandatory, conditional, optional and conditional-optional fields are reported

standard

specifies reporting of failed cases that use an extended structure of the PCMD records and successful cases that use the basic structure for the PCMD records; only mandatory and conditional fields are included

extend

specifies reporting of both failed and successful cases that use an extended structure of PCMD records

Default: standard

6.2.1.56 P-CSCF server group profile command descriptions

6.2.1.56.1 pcscf-group

Syntax

[no] pcscf-group *pcscf-group*

Context

config>mobile>profile

Description

This command configures the name of the P-CSCF server group.

The no form of this command removes the P-CSCF server group from the configuration.

Parameters

pcscf-group

specifies the name for the P-CSCF server group, up to 32 characters

6.2.1.56.2 pcscf-server

Syntax

[no] pcscf - server {ip-address {weight} | fqdn {ipv4 | ipv6}}

Context

config>mobile>profile>pcscf-group

Description

This command specifies a P-CSCF server as part of a P-CSCF server group. The P-CSCF server is specified by an IP address (IPv6 or IPv4) or an FQDN. If an IP address is used to identify a P-CSCF the weight parameter is used for load-balancing the allocation of P-CSCF to the set of UEs that use the P-CSCF group. The system allocates P-CSCFs in a ratio of the configured weight and the sum of all weights. If an FQDN is specified, the FQDN is resolved into a list of IP addresses by a DNS query. The ‘type’ specified when FQDN is configured indicates the DNS query to be launched (v4 or v6).The P-CSCF servers are load balanced by allocating a P-CSCF server (to a UE request) from the list using round robin algorithm. When FQDN is used, only one entry must be specified.

The no form of this command removes the specified P-CSCF server from the P-CSCF server group.

Parameters

ip-address

specifies the peer’s address

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

weight

specifies the weight

Values: 1 to 100

Default: 1

fqdn

specifies FQDN, up to 255 characters

ipv4

specifies the IPv4 address type for the FQDN

ipv6

specifies the IPv6 address type for the FQDN

6.2.1.56.3 refresh-interval

Syntax

`refresh-interval hours`

`no refresh-interval`

Context

`config>mobile>profile>pcscf-group`

Description

This command sets the amount of time the node waits before sending a DNS query to refresh the P-CSCF server list. A value of 0 means that the DNS refresh timer is disabled.

The no form of this command reverts to the default.

Parameters**hours**

the amount of time, in hours, that the node waits before sending a Domain Name Server (DNS) query

Values: 0 to 24

6.2.1.57 Policy options profile command descriptions

6.2.1.57.1 policy-options

Syntax

`policy-options`

Context

config>mobile>profile

Description

This command configures and creates mobile gateway policies.

6.2.1.57.2 abort

Syntax

abort

Context

config>mobile>profile>policy-options

Description

This command ends the current policy editing session entered via the *begin* command and discards changes that have been made to policies during the editing session.

6.2.1.57.3 action-rule-unit

Syntax

[no] action-rule-unit *aru-name*

Context

config>mobile>profile>policy-options

Description

This command configures an ARU to be associated with a predefined PCC rule.

The no form of this command removes the specified configuration.

Parameters

aru-name

specifies the ARU name, up to 32 characters

6.2.1.57.4 aa-notify-action

Syntax

```
aa-notify-action action-value  
no aa-notify-action
```

Context

```
config>mobile>profile>policy-options>action-rule-unit
```

Description

This command overrides the reporting granularity which is configured in AA.

The no form of this command reverts to the granularity configured in AA.

Parameters

flow-level

requests notifications on flow level

charging-group

requests notifications on charging-group level

Default: none

6.2.1.57.5 app-profile-name

Syntax

```
app-profile-name profile-name  
no app-profile-name
```

Context

```
config>mobile>profile>policy-options>action-rule-unit
```

Description

This command specifies an existing AA application profile name. The specified application profile (configured as part of a pre-defined rule) overrides any default application profile configured for the APN, when the rule is activated for the IP-CAN session. When no application profile is specified for the instantiated rule for the IP-CAN session, the default application profile configured for the APN applies.

The no form of this command reverts to the default.

Parameters

profile-name

specifies the name of the AA application profile, up to 32 characters

6.2.1.57.6 characteristic

Syntax

`characteristic name value aa-attr-value`

`no characteristic name`

Context

`config>mobile>profile>policy-options>action-rule-unit`

Description

This command specifies an existing ASO characteristic name and an associated value. The specified ASO characteristic and value override the corresponding characteristic's value in the applied application profile. No new characteristic can be added to an existing profile. Only the value of the specified attribute (characteristic) in the profile is overridden.

The no form of this command reverts to the default; that is, no specified ASO characteristic within the rule.

Parameters

name

specifies the name of an existing ASO characteristic, up to 32 characters

aa-attr-value

specifies an existing value of the specified ASO characteristic, up to 32 characters

6.2.1.57.7 collect-display-stats

Syntax

`[no] collect-display-stats`

Context

`config>mobile>profile>policy-options>action-rule-unit`

`config>mobile>profile>policy-options>policy-rule-base`

Description

This command enables the collection of PR, PRU, and CRU traffic statistics for the policy rule and/or policy rule base. When the PCRF installs a rule base (by rule base name), and the `collect-display-stats` command is enabled by this rule base configuration, traffic statistics are collected for all the policy rules that belong to the rule base of this action-rule-unit. The statistics are grouped per PRB, PR, PRU, and CRU. Also, when the PCRF installs a policy rule (by policy rule name), and this policy rule configuration uses an action-rule-unit that enables the `collect-display-stats` command, traffic statistics are collected for that policy rule. The statistics are grouped per PR, PRU, and CRU. Counters can be collected for up to 45 rule bases, 200 rules, 200 PRUs, and 200 CRUs.

The no form of this command reverts to the default; that is, no PR, PRU and CRU traffic statistics are collected.

Default

no collect-display-stats

6.2.1.57.8 custom-pco

Syntax

```
custom-pco {none | normal | terminate | restrict-access | redirect | rate-throttling |  
self-activation | pre-paid}  
  
no custom-pco
```

Context

```
config>mobile>profile>policy-options>action-rule-unit
```

Description

This command configures a custom PCO action value. This is a vendor-specific custom IE. The custom PCO value that is configured can be sent back to the UE with the PCO IE. When a device attaches to the network or connects to the PDN, the network responds with the PCO IE to the device. A vendor-specific value (custom to a specific operator) configured in the PCO IE can indicate termination, redirection, restricted access, or throttling.

The no form of this command reverts to the default.

Default

no custom-pco

Parameters

none

specifies the action none for the PCO action

normal

specifies no restriction on the account for the PCO action. Data can be used as normal. If the device does not receive the PCO, it is treated as normal.

terminate

specifies terminate for the PCO action. The data session is terminated due to no data usage available to the account. The device can reconnect to the PDN only for customer-initiated data.

restrict-access

specifies restrict access for the PCO action. The data session is restricted to customer-initiated foreground data traffic only. The device cannot initiate any background data traffic.

redirect

specifies redirect for the PCO action. The data session is redirected due to no data usage available to the account. The device only allows customer-initiated foreground data traffic and cannot initiate any background data traffic.

rate-throttling

specifies rate throttling for the PCO action

self-activation

specifies self-activation for the PCO action

pre-paid

specifies that the subscriber is a pre-paid customer (maps to a value of 11 in the custom PCO sent to the UE)

Default: none

6.2.1.57.9 data-driven-dedicated-bearer

Syntax

[no] data-driven-dedicated-bearer

Context

config>mobile>profile>policy-options>action-rule-unit

Description

This command enables a data driven dedicated bearer. That is, if the policy rule requires a dedicated bearer and this command is configured, the dedicated bearer will be created when traffic matching the rule is

detected. If this command is not configured, the dedicated bearer will be created when the policy rule is installed.

The no form of this command removes the configuration.

6.2.1.57.10 disable-ubr-signaling

Syntax

```
[no] disable-ubr-signaling
```

Context

```
config>mobile>profile>policy-options>action-rule-unit
```

Description

This command disables the sending of a UBR with an operator-specific PCO container.

The no form of this command reverts to the default (UBR is sent).



NOTE:

- This command applies only when the operator-pco command is configured under the APN PCO options (config>mobile>pdn>apn>pco).
- If this command is enabled for at least one of the rules installed for the session, a UBR is sent.

Default

disabled (UBR is sent by default)

6.2.1.57.11 dnai

Syntax

```
dnai dnai
```

```
no dnai
```

Context

```
config>mobile>profile>policy-options>action-rule-unit
```

Description

This command installs the rule for the specified DNAI when the location of the UE (ULI) matches any entry in the `tai-lai-list` within the `trigger-rule-unit`.

The `no` form of this command removes the DNAI from the configuration.

Parameters

dnai

specifies the DNAI, up to 80 characters

6.2.1.57.12 dnai-psa-select

Syntax

```
dnai-psa-select {ulcl | anchor-upf}
no dnai-psa-select
```

Context

```
config>mobile>profile>policy-options>action-rule-unit
```

Description

This command specifies whether a combined UL-CL + L-PSA or an A-PSA must be added to the PDU session for MEC access.

The `no` form of this command reverts to the default. A combined UL-CL + L-PSA is added to the PDU session for MEC access.

Parameters

ulcl

specifies to use a combined UL-CL + L-PSA for the PDU session

anchor-upf

specifies to use an A-PSA for the PDU session

Default: `ulcl`

6.2.1.57.13 dns-redirect

Syntax

```
dns-redirect [access-type access-type]
```

```
no dns-redirect
```

Context

```
config>mobile>profile>policy-options>action-rule-unit
```

Description

This command configures the DNS redirect action value based on the flow description match of the FQDN or the remote IP under the policy rule unit.

Default

```
no dns-redirect
```

Parameters

access-type

specifies if the ARU is for all accesses or for HRPD access only

Values: all | hrpd

Default: all

6.2.1.57.14 exclude-src-dest-ports-in-ue-tft

Syntax

```
[no] exclude-src-dest-ports-in-ue-tft
```

Context

```
config>mobile>profile>policy-options>action-rule-unit
```

Description

This command excludes the source and destination ports in the UE TFT IP address filters sent to the UE for the detected AA application.

The no form of this command reverts to the default (disabled; 5-tuple is sent to the UE).

Default

```
no exclude-src-dest-ports-in-ue-tft
```

6.2.1.57.15 gy-filter-id-action

Syntax

```
gy-filter-id-action http-redirect url-string [encryption {aes-cbc-128}] [passphrase passphrase] [hash | hash2]

gy-filter-id-action {none | drop}

gy-filter-id-action traffic-redirect profile-name

no gy-filter-id-action
```

Context

```
config>mobile>profile>policy-options>action-rule-unit
```

Description

This command specifies the action for a filter ID sent by the OCS. The action is applied to already installed rules associated with this rating group. Typically, the flow description (IP filters) within the policy rule selected by the filter ID allows traffic to specific sites while the FUI action (FUA = restrict-access/redirect) is in effect. The policy rule selected by a filter ID is activated and deactivated locally by the gateway. It is activated when the OCS refers to the filter ID and deactivated when the FUA condition is removed. The filter ID policy rule is activated prior to reporting an OOC to a PCRF. When the action is drop/redirect, the policy rule is identified as a post-processing rule. The optional encryption and passphrase parameters can be used to specify encryption as needed. New macros such as \$ENCRYPTED_IMEI, \$ENCRYPTED_IMSI, and so on, are supported for the URL string.



NOTE:

The URL encryption configuration depends on the following HTTP traffic redirection scenarios:

- for Gy charging response with FUI Action Restrict and a filter ID PRB, URL encryption is based on the http-redirect and encryption options in the config>mobile>profile>policy-options>action-rule-unit>gy-filter-id-action context, in CUPS and non-CUPS scenarios
- for Gy charging response FUI Action Redirect:
 - in CUPS scenarios, URL encryption is based on the url-encryption command (configured for the UP PDN) in the config>mobile>pdn context
 - in non-CUPS scenarios, URL encryption is based on the url-encryption command in the config>mobile>profile>dcca context
- for Gx policy-rule based redirection, URL encryption is based on the http-redirect and encryption options in the config>mobile>profile>policy-options>policy-rule-unit>flow-gate-status context, in CUPS and non-CUPS scenarios

The no form of this command reverts to the default.

Default

no gy-filter-id-action (rule behaves normally)

Parameters

url-string

specifies the web portal redirect server URL, up to 255 characters

Values	\$APN	Access Point Name
–	\$IMEI	International Mobile Station Equipment Identity
–	\$IMSI	International Mobile Subscriber Identity
–	\$IP	IP address
–	\$MSISDN	Mobile Station International Subscriber Directory Number
–	\$PLMN	Public Land Mobile Network
–	\$URL	The Request-URI in the HTTP GET Request received
–	\$ENCRYPTED_IMSI	Encrypted International Mobile Subscriber Identity
–	\$ENCRYPTED_IMEI	Encrypted International Mobile Station Equipment Identity
–	\$ENCRYPTED_MSISDN	Encrypted Mobile Station International Subscriber Directory Number
–	\$RID	Agent Remote-ID
–	\$MAC	User Equipment MAC address
–	\$CID	Agent Circuit-ID

–	\$SUB	User Equipment MAC address
–	\$SAP	Service Access Point Identifier

encryption {aes-cbc-128}

specifies encryption; if specified, a passphrase string must also be specified

passphrase

specifies the encryption passphrase string, up to 64 characters; mandatory if encryption {aes-cbc-128} is specified

hash | hash2

specifies the hashing scheme used in the hashed key

none | drop

specifies whether the policy rule will behave normally (none) or drop traffic for existing rules for the session (drop); when the action is to drop/redirect, the policy rule is identified as a post-processing rule

profile-name

specifies a traffic redirection profile, up to 32 characters

6.2.1.57.16 modify-qos-default-bearer**Syntax**

```
[no] modify-qos-default-bearer
```

Context

```
config>mobile>profile>policy-options>action-rule-unit
```

Description

This command specifies if the application associated with this rule will trigger the modification of the default bearer QoS.

The no form of this command reverts to the default (disabled).

Default

```
no modify-qos-default-bearer
```

```
http-to-htt
```


6.2.1.57.17 mss-adjust

Syntax

[no] mss-adjust

Context

config>mobile>profile>policy-options>action-rule-unit

Description

This command configures TCP MSS adjust based on flow description matches. MSS adjust is performed in TCP SYN messages for each flow. TCP SYN messages detected with a matching flow are forwarded to the AA module with a configured MS value (configured with the config>mobile>pdn>bearer-traffic>mss-adjust command).

The no form of this command reverts to the default.

Default

no mss-adjust

6.2.1.57.18 record-extension-visited-url

Syntax

[no] record-extension-visited-url

Context

config>mobile>profile>policy-options>action-rule-unit

Description

This command enables a proprietary PGW-CDR field within recordExtensions IE which reports traffic per visited URL domain. When the feature is enabled for an action-rule-unit, the URL domains for the policy rules using that action-rule-unit are recorded and included in PGW-CDRs within the recordExtensions field. Additionally, the traffic per URL, rating group, and service ID are included. The URL name visible in the CDR is the aa-charging-group name. The charging group name reflects the URL domain that it maps to.

The no form of this command resets the configuration to the default.

Default

no record-extension-visited-url

6.2.1.57.19 rule-level-redirect

Syntax

[no] rule-level-redirect

Context

config>mobile>profile>policy-options>action-rule-unit

Description

This command enables, for IP-filter-based classification rules, the redirection to a specified URL of any traffic that matches this rule (pre-defined/static). If the redirect rule is associated with an ARU with rule-level redirection enabled, the mobile gateway performs a rule-level redirection for that rule.

The no form of this command reverts to default behavior (disabled).

Default

no rule-level-redirect

6.2.1.57.20 sdf-level-policing

Syntax

sdf-level-policing [direction *direction*] [group *group*]

no sdf-level-policing

Context

config>mobile>profile>policy-options>action-rule-unit

Description

This command specifies the level and direction at which the rule is policed. The rule can be policed at the SDF-level, the RG-level, or at a combination of the RG and the service ID levels. The rule can be policed in the downlink direction, the uplink direction, or bi-directionally.

Parameters

direction

specifies the direction in which the rule is policed

Values: downlink | bi-direction | uplink

group

specifies the level at which the rule is policed

Values: sdf | rg | rgsvcid

6.2.1.57.21 transport-byte-collection

Syntax

```
[no] transport-byte-collection
```

Context

```
config>mobile>profile>policy-options>action-rule-unit
```

Description

This command enables the collection of transport bytes/packets for a specific IMSI/MSISDN.



NOTE:

- The feature is not supported when a duplicate MSISDN feature is enabled while invoking this feature. This can result in non-deterministic behavior.
- The feature is supported in LTE and 5G.

Default

```
no transport-byte-collection
```

6.2.1.57.22 ap-action-profile

Syntax

```
[no] ap-action-profile profile-name
```

Context

```
config>mobile>profile>policy-options
```

Description

This command is used to define an Assume Positive (AP) action profile. The AP action profile is then associated with CCFH failure handling.

The no form of this command removes the specified action profile.

Parameters

profile-name

specifies the AP action profile name, up to 32 characters

6.2.1.57.23 policy-action

Syntax

```
policy-action {pre-defined-policy | pcrf-policy-and-pre-defined-policy | pcrf-policy}
```

```
no policy-action
```

Context

```
config>mobile>profile>policy-options>ap-action-profile
```

Description

This command is used to specify the policy action to be applied while in AP mode.

The no form of this command reverts to the default.

The no `policy-action` defaults to applying the `pre-defined-policy` parameter (specified under APN) for a failure to set up a Gx session, that is, a CCR-I failure/no-response. For a CCR-U failure, it defaults to applying the `pcrf-policy` parameter, that is, the policy that the PCRF has installed for this session (including any pre-defined rules that the PCRF may have activated for this session).

Default

```
no policy-action
```

Parameters

pre-defined-policy

specifies that the session continues with a pre-defined policy (specified under APN) while in AP mode

pcrf-policy-and-pre-defined-policy

specifies that the session continues with a PCRF installed policy in addition to a pre-defined policy (specified under APN) while in AP mode

pcrf-policy

specifies that the session continues with a PCRF installed policy while in AP mode

6.2.1.57.24 apn-rate-control

Syntax

[no] apn-rate-control *profile-name*

Context

config>mobile>profile>policy-options

Description

This command configures the profile for the APN rate control. There is no profile configured by default. The no form of this command removes the specified profile name.

Parameters

profile-name

specifies the profile name, up to 32 characters

6.2.1.57.25 additional-ul-rate-limit-for-exception-data

Syntax

additional-ul-rate-limit-for-exception-data *additional-ul-rate-limit-for-exception-data*

no additional-ul-rate-limit-for-exception-data

Context

config>mobile>profile>policy-options>apn-rate-control

Description

This command specifies the additional uplink rate limit for exception data for the APN rate control.

The no form of this command reverts to the default (disables the additional uplink rate limit option for exception data).

Default

no additional-ul-rate-limit-for-exception-data

Parameters

additional-ul-rate-limit-for-exception-data

specifies the additional rate limit value

Values: 0 to 65535

6.2.1.57.26 dl-rate-limit

Syntax

dl-rate-limit *dl-rate-limit*

no dl-rate-limit

Context

config>mobile>profile>policy-options>apn-rate-control

Description

This command defines the downlink rate limit for the APN rate control. There is no limit by default.

Parameters

dl-rate-limit

specifies the downlink rate limit

Values: 0 to 16777215

Default: 0 (no limit)

6.2.1.57.27 exclude-ul-exception-reports

Syntax

[no] exclude-ul-exception-reports

Context

config>mobile>profile>policy-options>apn-rate-control

Description

This command defines whether exceptions reports are included or excluded from the APN rate control. The exception reports are included by default.

Default

no exclude-ul-exception-reports

6.2.1.57.28 time-unit

Syntax

time-unit *time-unit*
no time-unit

Context

config>mobile>profile>policy-options>apn-rate-control

Description

This command defines the time unit for the APN rate control. There is no limit by default.

Parameters

time-unit
specifies the time unit for the APN rate control

Values	0	unrestricted (no limit)
–	1	limits the time unit to a minute
–	2	limits the time unit to an hour
–	3	limits the time unit to a day
–	4	limits the time unit to a week

Default: 0

6.2.1.57.29 ul-rate-control-in-pgw

Syntax

```
ul-rate-control-in-pgw
no ul-rate-control-in-pgw
```

Context

```
config>mobile>profile>policy-options>apn-rate-control
```

Description

This command defines whether uplink APN rate control in PGW is used.

The no form of this command reverts to the default (disables the uplink APN rate control).

Default

```
no ul-rate-control-in-pgw
```

6.2.1.57.30 ul-rate-limit

Syntax

```
ul-rate-limit ul-rate-limit
no ul-rate-limit
```

Context

```
config>mobile>profile>policy-options>apn-rate-control
```

Description

This command defines the uplink rate limit for the APN rate control. There is no limit by default.

Parameters

ul-rate-limit

specifies the uplink rate limit

Values: 0 to 16777215

Default: 0

6.2.1.57.31 begin

Syntax

begin

Context

config>mobile>profile>policy-options

Description

This command begins a policy editing session. It is required for all configuration changes under the `policy-options` branch. The editing session continues until an *abort* or *commit* command is issued, or a control complex reset takes place. Modifications done during the editing session will take effect when the *commit* command is issued. Editing sessions are not interrupted by high-availability activity switches or CLI session terminations, such as closing a Telnet session.

6.2.1.57.32 charging-rule-unit

Syntax

[no] charging-rule-unit *policy-name*

Context

config>mobile>profile>policy-options

Description

This command configures the charging policy. By default, the system supports modifications to policies that are referenced under an APN or installed on a UE (via PCRF or APN configuration), without having to remove them from the APN or the UE session. Changes will be applied to existing sessions based on the APN policy-on-the-fly configuration.

Parameters

policy-name

specifies the charging policy unit name, up to 32 characters

6.2.1.57.33 app-srvc-provider-id

Syntax

```
app-srvc-provider-id app-srvc-provider-id  
no app-srvc-provider-id
```

Context

```
config>mobile>profile>policy-options>charging-rule-unit
```

Description

This command specifies the applicable application service provider identity for sponsored connectivity-level reporting. When the reporting level is set to sponsor-connectivity, the application service provider identity must be specified.

Parameters

app-srvc-provider-id

specifies the application service provider identity, up to 32 characters

6.2.1.57.34 charging-method

Syntax

```
charging-method {online | offline | both | both}  
no charging-method
```

Context

```
config>mobile>profile>policy-options>charging-rule-unit
```

Description

This command configures the charging method for the PCC rule.

Parameters

online

specifies online charging

offline

specifies offline charging

both

specifies both online and offline charging

none

specifies no charging

6.2.1.57.35 custom-online-sru

Syntax

custom-online-sru *sru-name*

no custom-online-sru

Context

config>mobile>profile>policy-options>charging-rule-unit

Description

This command configures the SRU that manages the online triggers for the rating group when the custom-online-offline-reporting is enabled under the config>mobile>profile>charging>chf-profile context. Otherwise, the SRU is ignored.

Parameters

sru-name

specifies a SRU name, up to 32 characters

6.2.1.57.36 default-rg

Syntax

default-rg {enable | disable}

no default-rg

Context

config>mobile>profile>policy-options>charging-rule-unit

Description

This command specifies whether quota for the rating group (or for the rating group and the service identifier) is requested preemptively. When disabled, the request of quota in advance is done according to the default rating group list.



NOTE:

- If you use the default rating group lists (configured using the `config>mobile>profile>charging>dcca>default-rg-list>rating-group` and `config>mobile>profile>charging>chf-profile>default-rg-list>rating-group` commands) for a session in addition to policy rules with rating groups that request for quota preemptively, all rating groups within the default rating group list request preemptive quota in addition to the preemptive rating groups associated with policy rules per charging rule unit.
- At CCR-Initial (in case of Gy) or Charging-Data-Request-Initial (in case of N40), the CMG requests for quota in advance for all the installed preemptive rating groups, regardless of whether these policy rules were installed by a local rulebase, a PCF/PCRF rule or a PCF/PCRF rulebase.

Parameters

enable

specifies requesting quota in advance for the configured rating group (or for the rating group and the service identifier)

disable

specifies requesting quota in advance according to the default rating group list

Default: disable

6.2.1.57.37 metering-method

Syntax

`metering-method {time-based | volume-based | both}`

`no metering-method`

Context

`config>mobile>profile>policy-options>charging-rule-unit`

Description

This command configures the metering method for the PCC rule.

Parameters

time-based

specifies time-based metering

volume-based

specifies volume-based metering

both

specifies both time-based and volume-based metering

Default: volume-based

6.2.1.57.38 monitoring-key

Syntax

monitoring-key *key-identifier* [*sru sru-name*]

no monitoring-key

Context

config>mobile>profile>policy-options>charging-rule-unit

Description

This command specifies the monitoring-key as part of the charging-rule-unit of a PCC rule. This is required for Gx usage monitoring to be supported for predefined rules. The associated usage-monitoring-information (thresholds and so on) is provided by the PCRF when the predefined rule is activated. A monitoring key identifies a usage monitoring control instance. This command is also used to associate a SRU to a monitoring key, used for Gx usage monitoring.

Parameters

key-identifier

specifies the key string, up to 80 characters

sru-name

specifies an SRU, up to 32 characters

6.2.1.57.39 rating-group

Syntax

rating-group *group-id* [*sru sru-name*]

no rating-group

Context

config>mobile>profile>policy-options>charging-rule-unit

Description

This command configures the rating group for the PCC rule and associates an SRU to a rating group.

Parameters

group-id

specifies the identification number of the group

Values: 0 to 4294967295

Default: 0

sru-name

specifies an SRU, up to 32 characters

6.2.1.57.40 rating-group-offline

Syntax

rating-group-offline *group-id* [sru *sru-name*]

no rating-group-offline

Context

config>mobile>profile>policy-options>charging-rule-unit

Description

This command configures the rating group for the offline PCC rule, and associates an SRU to a rating group.



NOTE:

If the rating-group-offline command is not configured, the rating-group configuration applies to both online and offline charging.

Parameters

group-id

specifies the identification number of the group

Values: 0 to 4294967295

Default: 0

sr-u-name

specifies an SRU, up to 32 characters

6.2.1.57.41 reporting-level

Syntax

```
reporting-level {rating-group | service-id | sponsor-connectivity}  
no reporting-level
```

Context

```
config>mobile>profile>policy-options>charging-rule-unit
```

Description

This command configures the reporting level for the PCC rule.

Parameters

rating-group

specifies rating group reporting level

service-id

specifies rating group plus service ID reporting level

sponsor-connectivity

specifies sponsored connectivity-level reporting. Separate service data container for a combination of rating-group, sponsor-identity and application-service-provider identity.

Default: rating-group

6.2.1.57.42 selective-radius-accounting

Syntax

```
[no] selective-radius-accounting
```

Context

```
config>mobile>profile>policy-options>policy-rule-unit>charging-rule-unit
```

Description

This command specifies the selective-radius-accounting enabled part of the charging-rule-unit. This also requires the policy-rule-unit of the policy-rule to be specified with flow-descriptor->match rule for what remote IP addresses the selective-radius-accounting must be triggered.

Default

no selective-radius-accounting

6.2.1.57.43 service-identifier

Syntax

```
service-identifier service-id [sru sru-name]  
no service-identifier
```

Context

config>mobile>profile>policy-options>charging-rule-unit

Description

This command configures the service ID for the PCC rule, and associates an SRU with the service ID.

Parameters

service-id

specifies the identification number of the service

Values: 0 to 4294967295

Default: 0

sru-name

specifies an SRU, up to 32 characters

6.2.1.57.44 service-identifier-offline

Syntax

```
service-identifier-offline service-id [sru sru-name]  
no service-identifier-offline
```


Context

config>mobile>profile>policy-options>charging-rule-unit

Description

This command configures the service ID for the PCC rule to use in offline charging, and associates an SRU with the service ID.



NOTE:

If the `service-identifier-offline` command is not configured, the `service-identifier` configuration applies to both online and offline charging.

Parameters

service-id

specifies the identification number of the service

Values: 0 to 4294967295

Default: 0

sru-name

specifies an SRU, up to 32 characters

6.2.1.57.45 sponsor-id

Syntax

`sponsor-id sponsor-id [sru sru-name]`

`no sponsor-id`

Context

config>mobile>profile>policy-options>charging-rule-unit

Description

This command specifies the applicable sponsor identity for sponsored connectivity level reporting and associates an SRU with the sponsor identifier. When command `reporting-level` is set to `sponsor-connectivity`, the sponsor identity must be specified.

Parameters

sponsor-id

specifies the identification number of the sponsor

Values: 0 to 32 characters

sru-name

specifies an SRU, up to 32 characters

6.2.1.57.46 wait-for-ocs-response

Syntax

```
wait-for-ocs-response {enable | disable}
```

```
no wait-for-ocs-response
```

Context

```
config>mobile>profile>policy-options>charging-rule-unit
```

Description

This command specifies whether to wait or not for an OCS response for setting up a session or a bearer when the PCC rule for this session or bearer is installed. During a Create Session Request, if all data flows governed by the PCC rules for that session are indicated with `online=enabled` with `OnlineResponseRequired = DONT WAIT`, the PGW returns the Create Session Response without waiting for a Gy (CCA-I) response. During a Create Bearer Request, if all data flows governed by the PCC rules for that bearer are indicated with `online=enabled` with `OnlineResponseRequired = DONT WAIT`, the PGW continues with the dedicated bearer setup without waiting for a Gy response. When there is at least one rule with `OnlineResponseRequired = Wait`, the PGW waits for the OCS response before setting up a session or a bearer creation.

The no form of this command reverts to default.

Parameters

enable

specifies to wait for the OCS response

disable

specifies not to wait for the OCS response

Default: enable

6.2.1.57.47 commit

Syntax

```
commit
```

Context

```
config>mobile>profile>policy-options
```

Description

This command saves changes made during the current policy editing session (see the `begin` command). Policy changes made during the editing session take effect only when they are committed. If the changes are successfully committed, with no errors detected during the commit or cross-reference verification against the existing policy configuration, the editing session will also be closed.

6.2.1.57.48 diff

Syntax

```
diff
```

Context

```
config>mobile>profile>policy-options
```

Description

This command shows the uncommitted changes made during a policy editing session (see the `begin` and `commit` commands).

6.2.1.57.49 dynamic-rule-to-internal-rule-map

Syntax

```
[no] dynamic-rule-to-internal-rule-map profile-name
```

Context

```
config>mobile>profile>policy-options
```

Description

This command configures a flow description mapping profile to derive an internal static rule for flow classification. This mitigates CAM usage for PCC flows that are not specified as a 5-tuple when sent from the PCRF. Some applications such as Rich Chat Service (RCS) and Push To Talk (PTT) use dedicated application servers and therefore the remote IP/port information is related to these dedicated servers. This information is used to derive the internal static rule that can be used across different chat sessions for all UEs.

The no form of this command removes the profile.

Parameters

profile-name

specifies the name of the flow description mapping profile, up to 32 characters

6.2.1.57.50 entry

Syntax

entry *id* [priority *priority*]

[no] entry *id*

Context

config>mobile>profile>policy-options>dynamic-rule-to-internal-rule-map

Description

This command configures an entry (with the associated priority) within the flow description mapping profile. Each entry has criteria that are used to match the received information within a PCC rule to the configured criteria, such as a QCI/ARP, a remote IP, or a port. Also, the match criteria specify the information fields from which the internal static rule is derived for flow classification. Each entry is evaluated in the order of the unique priority within the profile. A maximum number of 64 entries can be supported.

The no form of this command removes the specified entry from the profile.

Parameters

id

specifies the ID of an entry within the flow description mapping profile

Values: 1 to 64

priority

specifies the unique priority of an entry within the flow description mapping profile; entries within the profile are evaluated in the order of priority

Values: 1 to 255

**NOTE:**

The priority parameter is mandatory when creating an entry in the mapping profile, and optional after the entry ID is configured.

6.2.1.57.51 action

Syntax

```
action
```

Context

```
config>mobile>profile>policy-options>dynamic-rule-to-internal-rule-map>entry
```

Description

This command enables the context to configure the action values for the given entry.

6.2.1.57.52 direction

Syntax

```
direction {down-link | up-link | both}  
no direction
```

Context

```
config>mobile>profile>policy-options>dynamic-rule-to-internal-rule-map>entry>action
```

Description

This command configures the direction for an IP filter that is used to derive the internal static rule within the flow description mapping profile.

The no form of this command reverts to the default.

Parameters

down-link

specifies the direction from the server to the UE

up-link

specifies the direction from the UE to the sever

both

specifies both the up-link and down-link direction

Default: both

6.2.1.57.53 local-port-range

Syntax

local-port-range start *start-port* end *end-port*

no local-port-range start

Context

config>mobile>profile>policy-options>dynamic-rule-to-internal-rule-map>entry>action

Description

This command configures a local port range that is used to derive the static rule for flow classification. If specified, the command can override the range from the match criterion.

The no form of this command removes the local port from the action to take.

Parameters

start-port

specifies the starting point of an inclusive range for port numbers to be used as match criterion; expressed as a decimal integer

Values: 1 to 65535

Default: 0

end-port

specifies the ending point of an inclusive range for port numbers to be used as match criterion; expressed as a decimal integer

Values: 1 to 65535

Default: 0

6.2.1.57.54 precedence

Syntax

precedence *precedence*
no precedence

Context

config>mobile>profile>policy-options>dynamic-rule-to-internal-rule-map>entry>action

Description

This command configures the precedence that is used to derive the static rule for the flow classification. The no form of this command removes the precedence from the action to take.

Parameters

precedence

specifies the precedence to set as action criteria
Values: 0 to 65535

6.2.1.57.55 protocol

Syntax

protocol *protocol-id*
no protocol

Context

config>mobile>profile>policy-options>dynamic-rule-to-internal-rule-map>entry>action

Description

This command configures a protocol ID that is used to derive the static rule for flow classification. The no form of this command removes the protocol from the action to take.

Parameters

protocol-id

specifies the protocol ID to set as action criteria

Values: 0 to 255 (protocol numbers accepted only in decimals)

Keywords: none | crtp | crudp | egp | eigrp | encap | ether-ip | gre | icmp | idrp | igmp | igp | ip | ipv6 | ipv6-frag | ipv6-icmp | ipv6-no-nxt | ipv6-opts | ipv6-route | isis | iso-ip | l2tp | ospf-igp | pim | pnni | ptp | rdp | rsvp | stp | tcp | udp | vrrp | sctp

Default: any protocol

6.2.1.57.56 remote-ip

Syntax

```
remote-ip ip-prefix/ip-prefix-length
remote-ip
```

Context

```
config>mobile>profile>policy-options>dynamic-rule-to-internal-rule-map>entry>action
```

Description

This command configures the remote IP that is used to derive the internal static rule for flow classification. The optional IP address, if specified, overrides the IP selected from the match criterion.

The no form of this command removes the remote IP from the action that is taken to derive the internal static rule.

Parameters

ip-prefix
specifies the prefix for the IP match criterion in a dotted decimal notation

Values	ipv4-address	a.b.c.d. (host bits must be 0)
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

ip-prefix-length

specifies the prefix length for the IP address; expressed as a decimal integer

Values	ipv4-address	1 to 32
–	ipv6-address	1 to 128

6.2.1.57.57 remote-port-range**Syntax**

remote-port-range start *start-port* end *end-port*

no remote-port-range

Context

config>mobile>profile>policy-options>dynamic-rule-to-internal-rule-map>entry>action

Description

This command configures a remote port range that is used to derive the static rule for flow classification. If specified, it can override the range from the match criterion.

The no form of this command removes the remote port from the action to take.

Parameters***start***

specifies the starting point of an inclusive range for port numbers to be used as match criterion; expressed as a decimal integer

Values: 1 to 65535

Default: 0

end

specifies the ending point of an inclusive range for port numbers to be used as match criterion; expressed as a decimal integer

Values: 1 to 65535

Default: 0

6.2.1.57.58 signal-internal-rule-tft

Syntax

```
[no] signal-internal-rule-tft
```

Context

```
config>mobile>profile>policy-options>dynamic-rule-to-internal-rule-map>entry>action
```

Description

This command enables sending the IP filter, which is derived from the internal static rule, to the UE; instead of sending the flow description received from PCRF as part of the PCC rule. Therefore, a generic IP filter is signaled to the UE, and the number of flow descriptions to be sent is optimized.

The no form of this command reverts to the default (disabled).

Default

```
no signal-internal-rule-tft
```

6.2.1.57.59 match

Syntax

```
match
```

Context

```
config>mobile>profile>policy-options>dynamic-rule-to-internal-rule-map>entry
```

Description

This command enables the context to configure match criteria values for the given entry.

6.2.1.57.60 arp

Syntax

```
arp arp
```

```
no arp
```

Context

```
config>mobile>profile>policy-options>dynamic-rule-to-internal-rule-map>entry>match
```

Description

This command configures the ARP to be used as match criterion of an entry within the flow description matching profile. The ARP of an incoming PCC rule from the PCRF is matched to the configured value.

The no form of this command reverts to the default.

Parameters

arp

specifies the ARP to match

Values: 1 to 15

Default: 0

6.2.1.57.61 direction

Syntax

```
direction {down-link | up-link | both}
```

```
no direction
```

Context

```
config>mobile>profile>policy-options>dynamic-rule-to-internal-rule-map>entry>match
```

Description

This command configures the direction for an IP filter as match criterion within the flow description mapping profile.

The no form of this command reverts to the default.

Parameters

down-link

specifies the direction from the server to the UE

up-link

specifies the direction from the UE to the sever

both

specifies both the up-link and down-link direction

Default: both

6.2.1.57.62 local-port-range

Syntax

```
local-port-range start start-port end end-port  
no local-port-range
```

Context

```
config>mobile>profile>policy-options>dynamic-rule-to-internal-rule-map>entry>match
```

Description

This command configures a local port range for an IP filter as match criterion. The end user (UE) is considered the local port.

The no form of this command removes the local port match criterion.

Parameters

start-port

specifies the starting point of an inclusive range for port numbers to be used as match criterion; expressed as a decimal integer

Values: 1 to 65535

Default: 0

end-port

specifies the ending point of an inclusive range for port numbers to be used as match criterion; expressed as a decimal integer

Values: 1 to 65535

Default: 0

6.2.1.57.63 protocol

Syntax

```
protocol protocol-id  
no protocol
```

Context

config>mobile>profile>policy-options>dynamic-rule-to-internal-rule-map>entry>match

Description

This command configures the protocol ID that is used for flow description match criterion.

The no form of this command removes the protocol from the action to take.

Parameters

protocol-id

specifies the protocol ID to set a match criterion

Values: 0 to 255 (protocol numbers accepted only in decimals)

Keywords: none | crtp | crudp | egp | eigrp | encap | ether-ip | gre | icmp | idrp | igmp | igp | ip | ipv6 | ipv6-frag | ipv6-icmp | ipv6-no-nxt | ipv6-opts | ipv6-route | isis | iso-ip | l2tp | ospf-igp | pim | pnni | ptp | rdp | rsvp | stp | tcp | udp | vrrp | sctp

Default: any protocol

6.2.1.57.64 qci

Syntax

qci *qci*

no qci

Context

config>mobile>profile>policy-options>dynamic-rule-to-internal-rule-map>entry>match

Description

This command configures the QCI to be used in the match criteria of an entry within the flow description matching profile. The QCI of an incoming PCC rule from PCRF is matched to the configured value.

The no form of this command removes the QCI from the match criteria.

Parameters

qci

specifies the QCI to match

Values: 1 to 255

Default: 0

6.2.1.57.65 remote-ip

Syntax

```
remote-ip ip-prefix/ip-prefix-length
remote-ip
```

Context

```
config>mobile>profile>policy-options>dynamic-rule-to-internal-rule-map>entry>match
```

Description

This command configures the remote IP that is used as an IP filter match criterion. The optional IP address, if specified, overrides the IP selected from the match criterion.

The no form of this command removes the remote IP match criterion.

Parameters

ip-prefix
specifies the prefix for the IP match criterion in a dotted decimal notation

Values	ipv4-address	a.b.c.d. (host bits must be 0)
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

ip-prefix-length
specifies the prefix length for the IP address; expressed as a decimal integer

Values	ipv4-address	1 to 32
--------	--------------	---------

– ipv6-address 1 to 128

6.2.1.57.66 remote-port-range

Syntax

```
remote-port-range start start-port end end-port  
no remote-port-range
```

Context

```
config>mobile>profile>policy-options>dynamic-rule-to-internal-rule-map>entry>match
```

Description

This command configures a remote port range for an IP filter match criterion.

The no form of this command removes the remote port match criterion.

Parameters

start

specifies the starting point of an inclusive range for port numbers to be used as match criterion; expressed as a decimal integer

Values: 1 to 65535

Default: 0

end

specifies the ending point of an inclusive range for port numbers to be used as match criterion; expressed as a decimal integer

Values: 1 to 65535

Default: 0

6.2.1.57.67 gx-avp-options

Syntax

```
[no] gx-avp-options profile-number
```

Context

config>mobile>profile>policy-options

Description

This command configures the AVP IE for the Gx interface profile that can be applied at the PGW/GGSN.

Parameters

profile-number

specifies the Gx profile number

Values: 1 to 255

6.2.1.57.68 **exclude-3gpp-charging-characteristics**

Syntax

[no] exclude-3gpp-charging-characteristics

Context

config>mobile>profile>policy-options>gx-avp-options

Description

This command excludes sending the 3GPP-charging-characteristics AVP over Gx interface.

The no form of this command reverts to the default.

Default

no exclude-3gpp-charging-characteristics

6.2.1.57.69 **exclude-an-trusted**

Syntax

[no] exclude-an-trusted

Context

config>mobile>profile>policy-options>gx-avp-options

Description

This command excludes the “AN-Trusted” AVP in a CCR message sent to the PCRF. The “AN-Trusted” AVP indicates whether the Access Network is trusted or untrusted for a non-3GPP access network.

The no form of this command reverts to the default (included).

Default

no exclude-an-trusted

6.2.1.57.70 exclude-bearer-usage

Syntax

[no] exclude-bearer-usage

Context

config>mobile>profile>policy-options>gx-avp-options

Description

This command excludes the bearer usage AVP in a CCR message sent to the PCRF.

The no form of this command reverts to the default (included).

Default

no exclude-bearer-usage

6.2.1.57.71 exclude-pdn-connection-id

Syntax

[no] exclude-pdn-connection-id

Context

config>mobile>profile>policy-options>gx-avp-options

Description

This command excludes the PDN connection ID in a CCR message sent to the PCRF. The PDN connection ID is the default bearer charging ID allocated for a PDN session. The AVP is included by default.

The no form of this command reverts to the default (included).

**NOTE:**

The Gx profile allows for the specified AVP to be excluded from CCR-I/CCR-U or CCR-T as needed. For this AVP, the “M” bit is not set per 29.212; so per standard, if the PCRF receives the AVP and does not understand, it ignores the AVP.

Default

no exclude-pdn-connection-id

6.2.1.57.72 exclude-subscription-id-imsi**Syntax**

[no] exclude-subscription-id-imsi

Context

config>mobile>profile>policy-options>gx-avp-options

Description

This command excludes IMSI in the CCR message AVP.

Default

no exclude-subscription-id-imsi

6.2.1.57.73 exclude-subscription-id-msisdn**Syntax**

[no] exclude-subscription-id-msisdn

Context

config>mobile>profile>policy-options>gx-avp-options

Description

This command excludes MSISDN in the CCR message AVP.

Default

no exclude-subscription-id-msisdn

6.2.1.57.74 exclude-subscription-id-nai

Syntax

[no] exclude-subscription-id-nai

Context

config>mobile>profile>policy-options>gx-avp-options

Description

This command excludes NAI in the CCR message AVP.

Default

no exclude-subscription-id-nai

6.2.1.57.75 exclude-twan-identifier

Syntax

[no] exclude-twan-identifier

Context

config>mobile>profile>policy-options>gx-avp-options

Description

This command excludes the TWAN-Identifier AVP in a CCR message sent to the PCRF. The TWAN-Identifier indicates the UE location in a Trusted WLAN Access Network.

The no form of this command reverts to the default (included).

Default

no exclude-twan-identifier

6.2.1.57.76 exclude-ue-local-ip-address

Syntax

[no] exclude-ue-local-ip-address

Context

config>mobile>profile>policy-options>gx-avp-options

Description

This command causes the PGW to exclude the UE Local IP Address IE. This IE is used for Broadband access interworking.

The no form of this command causes the PGW to include this IE.

Default

no exclude-ue-local-ip-address

6.2.1.57.77 exclude-ue-udp-port

Syntax

[no] exclude-ue-udp-port

Context

config>mobile>profile>policy-options>gx-avp-options

Description

This command causes the PGW to exclude the UE UDP Port IE. This IE is used for Broadband access interworking.

The no form of this command causes the PGW to include this IE.

Default

no exclude-ue-udp-port

6.2.1.57.78 exclude-user-equipment-info

Syntax

[no] exclude-user-equipment-info

Context

config>mobile>profile>policy-options>gx-avp-options

Description

This command excludes the user equipment information in the CCR message AVP. If exclude-user-equipment-info command is enabled, the PGW does not send any user equipment information. If exclude-user-equipment-info command is not enabled and the IMEI is present, the PGW will send the IMEI, unless the RAT type is WLAN or VIRTUAL and the MAC address is not zero, in which case the PGW sends the MAC address instead of the IMEI (regardless of the supported Vendor ID).

Default

no exclude-user-equipment-info

6.2.1.57.79 include-framed-ip-address-ccrt

Syntax

[no] include-framed-ip-address-ccrt

Context

config>mobile>profile>policy-options>gx-avp-options

Description

This command includes the Framed-IP-Address AVP in CCR-T messages over the Gx interface to the PCRF. The Gx AVP options profile can be associated with a specific message type (CCR-I/CCR-U/CCR-T) within the Gx profile. The Framed-IP-Address AVP is included by default with CCR-I and CCR-U messages. To send this AVP to the PCRF in a CCR-T message, the Gx AVP options profile must be associated with the CCR-T message.

The no form of this command reverts to the default (excludes the Framed-IP-Address AVP from CCR-T messages).

Default

no include-framed-ip-address-ccrt

6.2.1.57.80 include-framed-ipv6-prefix-ccrt

Syntax

```
[no] include-framed-ipv6-prefix-ccrt
```

Context

```
config>mobile>profile>policy-options>gx-avp-options
```

Description

This command includes the Framed-IPv6-Prefix AVP in CCR-T messages over the Gx interface to the PCRF. The Gx AVP options profile can be associated with a specific message type (CCR-I/CCR-U/CCR-T) within the Gx profile. The Framed-IPv6-Prefix AVP is included by default in CCR-I and CCR-U messages. To send this AVP to the PCRF in a CCR-T message, the Gx AVP options profile must be associated with the CCR-T message.

The no form of this command reverts to the default (excludes the Framed-IP-Address AVP from CCRT messages).

Default

```
no include-framed-ipv6-prefix-ccrt
```

6.2.1.57.81 include-ipdomain-upf-id

Syntax

```
[no] include-ipdomain-upf-id
```

Context

```
config>mobile>profile>policy-options>gx-avp-options
```

Description

This command includes the ipDomain proprietary AVP in the Gx interface.

The no form of this command excludes the ipDomain proprietary AVP from the Gx interface.

Default

```
no include-ipdomain-upf-id
```

6.2.1.57.82 include-ue-tcp-source-port

Syntax

```
[no] include-ue-tcp-source-port
```

Context

```
config>mobile>profile>policy-options>gx-avp-options
```

Description

This command includes the UE TCP source port AVP over the Gx interface to the PCRF.

The no form of this command excludes sending the UE TCP source port AVP over the Gx interface to the PCRF.

Default

```
no include-ue-tcp-source-port
```

6.2.1.57.83 policy-rule

Syntax

```
policy-rule policy-rule-name policy-rule-unit policy-rule-unit-name [charging-rule-unit  
charging-rule-unit-policy-name] [qci qci] [arp arp-value] precedence precedence [action-rule-  
unit aru-name] [trigger-rule-unit tru-name] [stat-rule-unit-list sru-list-name] [qos-  
rule-unit qru-name] [non-terminating]
```

```
no policy-rule policy-rule-name
```

Context

```
config>mobile>profile>policy-options
```

Description

This command configures the policy rule. By default, the system supports modifications to policies that are referenced under an APN or installed on a UE (via PCRF or APN configuration), without having to remove them from the APN or the UE session. Changes are applied to existing sessions based on the APN policy-on-the-fly configuration. When the application-id in the config>mobile>profile>policy-options>policy-rule-unit>flow-description>match context is configured as part of the policy rule, the precedence value is sent in the Create PDR of the PFCP Session Establishment Request.

**NOTE:**

Charging rule unit and SRU list of non-terminating rules are ignored.

Parameters

policy-rule-name

specifies the policy name, up to 64 characters

policy-rule-unit-name

specifies the policy rule unit name, up to 32 characters

charging-rule-unit-policy-name

specifies the charging policy name, up to 32 characters

qci

the value of the QCI

Values: 1 to 9, 65, 66, 69, 70, 128 to 254, *-wildcard

arp-value

specifies the value for the ARP

Values: 1 to 15, *-wildcard

precedence

specifies the precedence value

Values: 0 to 65535

aru-name

specifies the name of the ARU, up to 32 characters (configured under config>mobile>profile>policy-options prior to association with the rule)

tru-name

specifies the Trigger Rule Unit (TRU) name, up to 32 characters

sru-list-name

specifies the SRU name, up to 32 characters

qru-name

specifies the QRU name, up to 32 characters

non-terminating

specifies the policy rule as non-terminating

6.2.1.57.84 policy-rule-base

Syntax

[no] policy-rule-base *base-policy-name*

Context

config>mobile>profile>policy-options

Description

This command configures the base policy. By default, the system supports modifications to policies that are referenced under an APN or installed on a UE (via PCRF or APN configuration), without having to remove them from the APN or the UE session. Changes are applied to existing sessions based on the APN policy-on-the-fly configuration.

Parameters

base-policy-name

specifies the base policy name, up to 64 characters

6.2.1.57.85 aa-adv-functions

Syntax

aa-adv-functions

Context

config>mobile>profile>policy-options>policy-rule-base

Description

This command configures AA advanced functions for the base policy.

6.2.1.57.86 app-profile-name

Syntax

app-profile-name *profile-name*

no app-profile-name

Context

config>mobile>profile>policy-options>policy-rule-base>aa-adv-functions

Description

This command specifies the application assurance profile name.

Default

no app-profile-name

Parameters

profile-name

specifies the application assurance profile name, up to 32 characters

6.2.1.57.87 characteristic

Syntax

characteristic *characteristic-name* value *value-name*

no characteristic *characteristic-name*

Context

config>mobile>profile>policy-options>policy-rule-base>aa-adv-functions

Description

This command enables the application assurance advanced functions characteristics.

Parameters

characteristic-name

specifies the advanced functions characteristics name, up to 32 characters

value-name

specifies the advanced functions values name, up to 32 characters

6.2.1.57.88 apn-ambr-attr

Syntax

```
apn-ambr-attr
```

Context

```
config>mobile>profile>policy-options>policy-rule-base
```

Description

This command enables the APN AMBR attributes.

6.2.1.57.89 aggr-rate

Syntax

```
aggr-rate downlink-rate downlink-rate uplink-rate uplink-rate  
no aggr-rate
```

Context

```
config>mobile>profile>policy-options>policy-rule-base>apn-ambr-attr
```

Description

This command specifies the downlink and uplink APN-AMBR. When this rule base is instantiated for an IP-CAN session, this is the rate locally enforced at the PCEF (gateway). For local enforcement at the gateway, this value overrides the APN-AMBR authorized by the PCRF at command level over Gx.

The no form of this command removes the configured apn-ambr.

Default

```
no aggr-rate
```

Parameters

downlink-rate

specifies the downlink APN-AMBR, in kb/s

Values: 0 to 25000000

uplink-rate

specifies the uplink APN-AMBR, in kb/s

Values: 0 to 25000000

6.2.1.57.90 charging-profile-id

Syntax

```
charging-profile-id charging-profile-id  
no charging-profile-id
```

Context

```
config>mobile>profile>policy-options>policy-rule-base
```

Description

This command allows PCRF to overwrite the charging profile selected for the session by installing at CCA-Init a policy rulebase configured with a charging profile ID. If a policy rulebase is installed at CCA-U, this parameter is ignored.

The no form of this command reverts to the default.



NOTE:

The Charging-Characteristics value is not affected by this configuration.

Default

no charging-profile-id

Parameters

charging-profile-id

specifies the charging profile ID number

Values: 0 to 255

6.2.1.57.91 ded-bearer-on-app-start

Syntax

```
[no] ded-bearer-on-app-start
```

Context

config>mobile>profile>policy-options>policy-rule-base

Description

This command enables dedicated bearer creation for an ADC rule base. The rule base is associated with a QCI/ARP that is different from that of the default bearer. Upon detecting an application start for a configured AA (DPI) rule within the rule base, a dedicated bearer is created.

The no form of this command reverts to the default (disabled).

Default

no ded-bearer-on-app-start

6.2.1.57.92 disable-bearer-bind-verification

Syntax

[no] disable-bearer-bind-verification

Context

config>mobile>profile>policy-options>policy-rule-base

Description

This command disables bearer binding verification for uplink traffic on a bearer. By default, bearer binding verification is enabled as per standard behavior. If at least one of the policy rule base options installed for the bearer has this command enabled, the uplink bearer binding verification is disabled for that bearer. This command disables uplink bearer binding verification for the bearer on which the policy-rule-base is installed. The APN level command disable-bearer-bind-verification disables uplink bearer binding verification for the entire session.

The no form of this command reverts to the default.

Default

no disable-bearer-bind-verification (uplink bearer binding verification is enabled)

6.2.1.57.93 policy-rule

Syntax

[no] policy-rule *policy-rule-name*

Context

```
config>mobile>profile>policy-options>policy-rule-base
```

Description

This command specifies the policy rule for the base policy.

Parameters

policy-rule-name

specifies a policy rule, up to 64 characters

6.2.1.57.94 session-level-svc-steering

Syntax

```
[no] session-level-svc-steering
```

Context

```
config>mobile>profile>policy-options>policy-rule-base
```

Description

This command enables the session-level service steering. When the PCF or PCRF activates a policy rule base with this command enabled, the gateway steers all traffic for the subscriber session based on the default steering profile that is configured under the APN. When this policy rule base is deactivated by the PCR or PCRF, the gateway disables the steering that is based on the default steering profile configured under the APN.



NOTE:

When multiple policy rule bases are installed for a session, the session-level steering applies if at least one policy rule base has this command enabled.

Default

```
no session-level-svc-steering
```

6.2.1.57.95 policy-rule-unit

Syntax

[no] policy-rule-unit *policy-name*

Context

config>mobile>profile>policy-options

Description

This command configures the policy rule unit. By default, the system supports modifications to policies that are referenced under an APN or installed on a UE (via PCRF or APN configuration), without having to remove them from the APN or the UE session. Changes will be applied to existing sessions based on the APN policy-on-the-fly configuration. Also, references to profiles within a policy rule unit can change, but the contents of those profiles cannot change.

Parameters

policy-name

specifies a policy rule unit, up to 32 characters

6.2.1.57.96 aa-advanced-functions

Syntax

[no] aa-advanced-functions

Context

config>mobile>profile>policy-options>policy-rule-unit

Description

This command specifies that all flows matching to this policy rule unit (PRU) are diverted to AA, even though the PRU would include only Layer 3 and Layer 4 flow descriptions.

Default

no aa-advanced-functions

6.2.1.57.97 flow-description

Syntax

[no] flow-description *entry-id*

Context

config>mobile>profile>policy-options>policy-rule-unit

Description

This command configures an identifier to identify a specific flow for a PRU.

Parameters

entry-id

specifies the entry identifier

Values: 1 to 255

6.2.1.57.98 direction

Syntax

direction {up-link | down-link | both}

no direction

Context

config>mobile>profile>policy-options>policy-rule-unit>flow-description

Description

This command configures the applicable direction for the flow.

Parameters

up-link

specifies direction uplink

down-link

specifies direction downlink

both

specifies directions both uplink and downlink

Default: both

6.2.1.57.99 match

Syntax

match

Context

config>mobile>profile>policy-options>policy-rule-unit>flow-description

Description

This command enables the context to configure match options for the PRU flow description.

Parameters

entry-id

specifies the entry identifier

6.2.1.57.100 aa-charging-group

Syntax

aa-charging-group *aa-charging-group-name* [sru *sru-name*]

no aa-charging-group

Context

config>mobile>profile>policy-options>policy-rule-unit>flow-description>match

Description

This command configures AA charging group for this policy rule unit and associates it with an SRU.

Parameters

aa-charging-group-name

specifies the AA charging group name, up to 32 characters

sru-name

specifies the SRU, up to 32 characters

6.2.1.57.101 aa-inactive-charging-group

Syntax

```
[no] aa-inactive-charging-group
```

Context

```
config>mobile>profile>policy-options>policy-rule-unit>flow-description>match
```

Description

This command enables a default AA charging group that is used to match traffic when AA returns a charging group that is not referenced by any PRU configured in the session. For example, the IP-CAN session has PRUs to match “CG1” and “CG2”, but AA identifies traffic “CGx” that is not “CG1” or “CG2”. This traffic is then matched to the AA inactive charging group. Similar to any PRU, the AA inactive charging group may specify a drop action, charging to a specific RG, and so on.

The no form of this command reverts to the default (disabled).

Default

```
no aa-inactive-charging-group
```

6.2.1.57.102 application-id

Syntax

```
application-id application-id-name  
no application-id
```

Context

```
config>mobile>profile>policy-options>policy-rule-unit>flow-description>match
```

Description

This command configures the application ID for this policy rule. There is a one-to-one mapping between the Charging-Rule-Base-Name/Charging-Rule-Name and the application ID. When configured, the system gets the Charging-Rule-Base-Name/Charging-Rule-Name from the PCRF and sends it as an application ID name over the Sx/N4 interface in the PFCP Session Establishment Request > Create PDR > PDI > Application ID IE.

**NOTE:**

With the application ID configured, the Activate Predefined Rules IE and the Deactivate Predefined Rules IE are not used and are not sent over the Sx/N4 interface in the Create PDR IE of the PFCP Session Establishment Request message. This command is not applicable to non-CUPS deployments. The application ID must only be configured on the CP function.

The no form of this command removes the configuration.

Parameters***application-id-name***

specifies the application ID name, a string up to 32 characters

6.2.1.57.103 dns-fqdn-list**Syntax**

`dns-fqdn-list fqdn-list-name`

`no dns-fqdn-list`

Context

`config>mobile>profile>policy-options>policy-rule-unit>flow-description>match`

Description

This command configures the specified FQDN list as part of the flow description within a policy rule unit that includes the FQDNs to be matched against.

The no form of this command disables the FQDN list matching for the flow.

Default

`no dns-fqdn-list`

Parameters***fqdn-list-name***

specifies the name of the FQDN list, up to 32 characters

6.2.1.57.104 dns-snooping

Syntax

```
dns-snooping fqdn-list-name  
no dns-snooping
```

Context

```
config>mobile>profile>policy-options>policy-rule-unit>flow-description>match
```

Description

This command configures a list of FQDNs for DNS snooping. A FQDN list includes a set of regular expressions (FQDNs) used to match domain names in DNS responses. The resolved IP addresses are stored in any FQDN list containing a matching FQDN (regular expression). The FQDN list must be configured using the `config>mobile>profile>list>fqdn-list` command.



NOTE:

- The storage of the resolved IP addresses is performed per APN and per FQDN list. For example, if a resolved IP address in APN-1 matches to FQDN-List-Nokia, that resolved IP address is only available in APN-1 FQDN-List-Nokia internal table (not in APN-2 FQDN-List-Nokia internal table).
- DNS snooping functionality requires that DNS snooping is enabled at the APN level using the `config>mobile>pdn>apn>dns-snooping` command.

Parameters

fqdn-list-name

specifies the name of the FQDN list, up to 32 characters

6.2.1.57.105 fqdn-group

Syntax

```
fqdn-group fqdn-group-name  
no fqdn-group
```

Context

```
config>mobile>profile>policy-options>policy-rule-unit>flow-description>match
```

Description

This command configures the specified FQDN group as part of the flow description within a policy rule unit. The IP filters are derived with the resolved IP addresses of the specified FQDN group. The FQDN group and remote IP options are mutually exclusive with the same flow description. Traffic matching the IP addresses is diverted to the AA card, if the AA advanced functions option is enabled under that policy rule unit; see `aa-advanced-functions`.

The `no` form of this command disables the FQDN group matching for the flow.

Default

`no fqdn-group`

Parameters

fqdn-group-name

specifies the name of the FQDN group used to resolve the remote IP addresses, up to 32 characters

6.2.1.57.106 local-ip

Syntax

`local-ip ip-prefix/ip-prefix-length`

`no local-ip`

Context

`config>mobile>profile>policy-options>policy-rule-unit>flow-description>match`

Description

This command configures the source IP address/subnet as match criteria for traffic classification in policy rules. Source address is the IP address of the UE. The IP address can be either an IPv4 address/subnet or an IPv6 address/subnet. If the command is not configured, the source IP address is not used as match criteria for traffic classification.

The `no` form of this command reverts to the default (not configured).

Default

`no local-ip`

Parameters

ip-prefix
specifies the IPv4 address prefix

Values	ipv4-address	a.b.c.d (host bits must be 0)
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

ip-prefix-length

Values	ipv4-prefix-length	1 to 32
–	ipv6-prefix-length	1 to 128

6.2.1.57.107 local-port

Syntax

```
local-port {lt | gt | eq} lcl-port-number
local-port range start end
no local-port
```

Context

```
config>mobile>profile>policy-options>policy-rule-unit>flow-description>match
```

Description

This command configures local port match condition.

Parameters

{lt | gt | eq}
specifies the port operators

lcl-port-number
specifies the port number

Values	lt	2 to 65535
–	gt	0 to 65533
–	eq	0 to 65535

start
specifies the start port number. This number must be less than the end port number. For port range to be valid at least two port numbers need to match.
Values: 0 to 65534

end
specifies the end port number
Values: 1 to 65535

6.2.1.57.108 local-port-list

Syntax

local-port-list *list-name*
no local-port-list

Context

config>mobile>profile>policy-options>policy-rule-unit>flow-description>match

Description

This command configures the specified local port list within a policy rule unit that the flow will use for matching.

The no form of this command disables the local port list matching for the flow.

Default

no local-port-list

Parameters

list-name

specifies the name of the local port list profile, up to 32 characters

6.2.1.57.109 protocol

Syntax

```
protocol protocol-id  
no protocol
```

Context

config>mobile>profile>policy-options>policy-rule-unit>flow-description>match

Description

This command configures protocol for the policy.

Parameters

protocol-id

specifies the protocol identifier

Values: 1 to 255

Keywords: none | crtp | crudp | egp | eigrp | encap | ether-ip | gre | icmp | idrp | igmp | igp | ip | ipv6 | ipv6-frag | ipv6-icmp | ipv6-no-nxt | ipv6-opts | ipv6-route | isis | iso-ip | l2tp | ospf-igp | pim | pnni | ptp | rdp | rsvp | stp | tcp | udp | vrrp | sctp

6.2.1.57.110 remote-ip

Syntax

```
remote-ip ip-prefix/ip-prefix-length  
no remote-ip
```

Context

config>mobile>profile>policy-options>policy-rule-unit>flow-description>match

Description

This command configures the destination IP address for matching.

Parameters

ip-prefix

specifies the IP address

Values	ipv4-prefix	a.b.c.d (host bits must be 0)
–	ipv6-prefix	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

ip-prefix-length

specifies the IP prefix length

Values	ipv4-prefix-length	1 to 32
–	ipv6-prefix-length	1 to 128

6.2.1.57.111 remote-ip-address-group

Syntax

remote-ip-address-group {*ipv6-prefix/prefix-length*}

no remote-ip-address-group

Context

config>mobile>profile>policy-options>policy-rule-unit>flow-description>match

Description

This command enables the use of the asterisk (*) character as wildcard within the destination IPv6 address. Only full quartets can be wildcarded. At least one and up to eight consecutive wildcards are supported. Wildcards can be applied to any subnet. For example, 2608:7700:0:*:*:82d3:bc84/128, 2608:7700::*:*:*:*:1100:0/104.

The no form of this command unsets this configuration.

Parameters

ipv6-prefix/prefix-length
specifies the IPv6 address group

Values	IPv6 prefix	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x - [0..FFFF]H
–	Prefix length	1 to 128

6.2.1.57.112 remote-ip-list

Syntax

```
remote-ip-list list-name
no remote-ip-list
```

Context

```
config>mobile>profile>policy-options>policy-rule-unit>flow-description>match
```

Description

This command enables the use of an IP address list as the destination IP address in a policy rule flow description. The list name is configured in the config>mobile>profile>addr-list context.

The no form of this command unsets this parameter (no list configured as destination IP address).

Parameters

list-name
name of the list of IP addresses, up to 32 characters

6.2.1.57.113 remote-port

Syntax

```
remote-port {lt | gt | eq} rmt-port-number  
remote-port range start end  
no remote-port
```

Context

```
config>mobile>profile>policy-options>policy-rule-unit>flow-description>match
```

Description

This command configures the remote port match condition.

Parameters

- {lt | gt | eq}**
specifies the port operators
- rmt-port-number***
specifies the port number

Values	lt	2 to 65535
–	gt	0 to 65533
–	eq	0 to 65535

- start***
specifies the start port number. This number must be less than the end port number. For port range to be valid at least two port numbers need to match.
Values: 0 to 65535
- end***
specifies the end port number
Values: 1 to 65535

6.2.1.57.114 remote-port-list

Syntax

```
remote-port-list list-name  
no remote-port-list
```

Context

```
config>mobile>profile>policy-options>policy-rule-unit>flow-description>match
```

Description

This command configures the specified remote port list within a policy rule unit that the flow will use for matching.

The no form of this command disables the remote port list matching for the flow.

Default

```
no remote-port-list
```

Parameters

list-name

specifies the name of the remote port list profile, up to 32 characters

6.2.1.57.115 tos-tc

Syntax

```
tos-tc dscp-value  
no tos-tc
```

Context

```
config>mobile>profile>policy-options>policy-rule-unit>flow-description>match
```

Description

This command enhances the PCC rule match criteria to include the ToS/TC DSCP.

The no form of this command reverts to the default.

Default

no tos-tc

Parameters

dscp-value

specifies the DSCP code point value. The packet filter component value field for the ToS/TC is encoded as a sequence with a 1-octet ToS/TC field and a 1-octet ToS/TC mask field (3GPP TS 24.008).

Values: 0 to 63

6.2.1.57.116 flow-gate-status

Syntax

flow-gate-status *flow-gate-status*

flow-gate-status traffic-redirect *traffic-redirect-profile-id*

flow-gate-status http-redirect *url-string* [encryption {aes-cbc-128}] [passphrase *passphrase*] [hash | hash2]

flow-gate-status traffic-hash *traffic-hash-profile-name*

no flow-gate-status

Context

config>mobile>profile>policy-options>policy-rule-unit

Description

This command configures the flow gate status of an SDF.

Parameters

flow-gate-status

specifies the flow gate status

Values	allowUL	specifies the flow gate status as allow uplink
–	allowDL	specifies the flow gate status as allow downlink

- allow specifies the flow gate status as allow uplink and downlink
- drop specifies the flow gate status as drop
- auto-redirect specifies the flow gate status to auto-redirect for HTTP redirection

Default: allow

traffic-redirect-profile-id

specifies the traffic redirect profile applied to a PCC rule, up to 32 characters

url-string

specifies the web portal redirect server URL, up to 255 characters

Values	\$APN	Access Point Name
–	\$IMEI	International Mobile Station Equipment Identity
–	\$IMSI	International Mobile Subscriber Identity
–	\$IP	IP address
–	\$MSISDN	Mobile Station International Subscriber Directory Number
–	\$PLMN	Public Land Mobile Network
–	\$URL	The Request-URI in the HTTP GET Request received
–	\$ENCRYPTED_IMSI	Encrypted International Mobile Subscriber Identity
–	\$ENCRYPTED_IMEI	Encrypted International Mobile Station Equipment Identity

–	\$ENCRYPTED_MSISDN	Encrypted Mobile Station International Subscriber Directory Number
–	\$RID	Agent Remote-ID
–	\$MAC	User Equipment MAC address
–	\$CID	Agent Circuit-ID
–	\$SUB	User Equipment MAC address
–	\$SAP	Service Access Point Identifier

encryption {aes-cbc-128}

specifies AES CBC 128 encryption; if specified, a passphrase string must also be specified

passphrase

encryption passphrase string, up to 64 characters; mandatory if encryption {aes-cbc-128} is specified



NOTE:

The URL encryption configuration depends on the following HTTP traffic redirection scenarios:

- for Gx policy-rule based redirection, URL encryption is based on the http-redirect and encryption options in the config>mobile>profile>policy-options>policy-rule-unit>flow-gate-status context, in CUPS and non-CUPS scenarios
- for Gy charging response with FUI Action Restrict and a filter ID PRB, URL encryption is based on the http-redirect and encryption options in the config>mobile>profile>policy-options>action-rule-unit>gy-filter-id-action context, in CUPS and non-CUPS scenarios
- for Gy charging response FUI Action Redirect:
 - in CUPS scenarios, URL encryption is based on the url-encryption command (configured for the UP PDN) in the config>mobile>pdn context
 - in non-CUPS scenarios, URL encryption is based on the url-encryption command in the config>mobile>profile>dcca context

hash | hash2

specifies the hashing scheme used in the hashed key

traffic-hash-profile-name

a traffic-hash profile (under policy-options) associated with a specific SDF (under a PRU), up to 64 characters

6.2.1.57.117 mute-notification**Syntax**

[no] mute-notification

Context

config>mobile>profile>policy-options>policy-rule-unit

Description

This command enables or disables mute-notification for a TDF application, (aa-charging-group) for a predefined PCC rule. Mute-notification is enabled by default for a predefined layer-7 (AA) PCC rule. When disabled, Application Start/Stop events are sent to PCRF, for the AA (TDF) application identified by the aa-charging-group within the PCC rule.

The no form of this command reverts to the default.

Default

no mute-notification

6.2.1.57.118 pdr-id**Syntax**

pdr-id *pdr-id*

no pdr-id

Context

config>mobile>profile>policy-options>policy-rule-unit

Description

This command specifies a PDR identifier to be associated with a PRU. It is used for referencing a pre-defined rule for communication between the control and the user plane. Its function is similar to an SDF identifier.

The no form of this command removes the selected PDR identifier from the configuration.

Parameters

pdr-id

specifies the value of the PDR identifier

Values: 1 to 15000

6.2.1.57.119 qos

Syntax

qos

Context

config>mobile>profile>policy-options>policy-rule-unit

Description

This command configures the QoS options for the policy options.

6.2.1.57.120 down-link

Syntax

down-link gbr *gbr* mbr *mbr*

no down-link

Context

config>mobile>profile>policy-options>policy-rule-unit>qos

Description

This command configures the downlink GBR/MBR rates for the associated SDF in the PCC rule.

Parameters

gbr

specifies the guaranteed bit rate, in kbps

Values: 0 to 250000000

Default: 0

mbr

specifies the maximum bit rate, in kbps

Values: 0 to 250000000

Default: 0

6.2.1.57.121 up-link

Syntax

up-link gbr *gbr* mbr *mbr*

no up-link

Context

config>mobile>profile>policy-options>policy-rule-unit>qos

Description

This command configures the uplink GBR/MBR rates for the associated SDF in the PCC rule.

Parameters

gbr

specifies the guaranteed bit rate, in kbps

Values: 0 to 250000000

Default: 0

mbr

specifies the maximum bit rate, in kbps

Values: 0 to 250000000

Default: 0

6.2.1.57.122 service-class-indicator

Syntax

```
service-class-indicator service-class-id
```

Context

```
config>mobile>profile>policy-options>policy-rule-unit
```

Description

This command configures the service class indicator for the policy-rule-unit of a PCC rule.

Parameters

service-class-id

specifies an indicator number

Values: 1 to 15

6.2.1.57.123 shallow-inspection-only

Syntax

```
shallow-inspection-only
```

```
no shallow-inspection-only
```

Context

```
config>mobile>profile>policy-options>policy-rule-unit
```

Description

Currently, if a PRB contains aa-charging-group in any PRU, all traffic from the 5-tuple rules in the PRB is diverted to the AA card, irrespective of the aa-advanced-functions CLI knob. This knob, when enabled for a policy rule unit, disables aa-divert for traffic matching the flow description (IP 5-tuples) specified for this policy-rule-unit. This configuration is not supported for individual static policy rules when individually referenced DPI policy rules are used for the session.

The no form of this command reverts to the default (disabled).

6.2.1.57.124 svc-steering

Syntax

```
svc-steering [steering-profile profile-name]  
no svc-steering
```

Context

```
config>mobile>profile>policy-options>policy-rule-unit
```

Description

This command is used to enable service steering for the SDF. Optionally a steering profile can be associated with the SDF when service steering is enabled. An SDF-level steering profile overrides an APN-level steering profile. When no profile is specified at the SDF level, the profile is inherited from the APN level.

The no form of this command reverts to the default (disabled).

Default

```
no svc-steering
```

Parameters

profile-name

specifies the steering profile name to be associated with this SDF, up to 32 characters

6.2.1.57.125 qci-policy

Syntax

```
[no] qci-policy policy-name
```

Context

```
config>mobile>profile>policy-options
```

Description

This command configures the QCI policy.



NOTE:

When used in conjunction with the `ims-emps-priority` command, different DSCP marking values for eMPS sessions support only the `qci` and `dscp` options under the `qci-policy` command. Additionally, the DSCP preserve option must be disabled (`no dscp-preserve`) in the `qci-policy` profile used for with the `ims-emps-priority` command.

Parameters

policy-name

specifies a policy name, up to 32 characters

6.2.1.57.126 qci

Syntax

```
qci qci-value arp arp-value
```

Context

```
config>mobile>profile>policy-options>qci-policy
```

Description

This command configures the QCI values for the QCI policy. This command is also used for IMS eMPS sessions when a different DSCP marking is required for the eMPS sessions. The QCI values configured in this command, are used to set the DSCP values for this QCI value when the session is an eMPS session. For more information, see the `config>mobile>pdn>ims-emps-priority` command.

Parameters

qci-value

specifies a QCI value

Values: 1 to 255

arp-value

specifies the ARP priority level value for a specific QCI

Values: * | 1 to 15

Default: * (wildcard)

6.2.1.57.127 dscp

Syntax

```
dscp {dscp-name | in dscp-name [out dscp-name] | out dscp-name [in dscp-name]}
```

no dscp

Context

config>mobile>profile>policy-options>qci-policy>qci

Description

This command configures the QCI values for the DSCP. This command is also used for IMS eMPS sessions when a different DSCP marking is required for the eMPS sessions. The DSCP values configured in this command, are used when the session is an eMPS session. For more information, see the config>mobile>pdn>ims-emps-priority command.

Parameters

dscp-name

specifies a DSCP name

Values: be | ef | cp1 | cp2 | cp3 | cp4 | cp5 | cp6 | cp7 | cp9 | cs1 | cs2 | cs3 | cs4 | cs5 | nc1 | nc2 | af11 | af12 | af13 | af21 | af22 | af23 | af31 | af32 | af33 | af41 | af42 | af43 | cp11 | cp13 | cp15 | cp17 | cp19 | cp21 | cp23 | cp25 | cp27 | cp29 | cp31 | cp33 | cp35 | cp37 | cp39 | cp41 | cp42 | cp43 | cp44 | cp45 | cp47 | cp49 | cp50 | cp51 | cp52 | cp53 | cp54 | cp55 | cp57 | cp58 | cp59 | cp60 | cp61 | cp62 | cp63

Default values per qci (qci:dscpln:dscpOut):

- 1:nc2:nc2
- 2:ef:ef
- 3:af41:af42
- 4:nc1:nc1
- 5:af21:af22
- 6:af11:af12
- 7:af11:af12
- 8:cs1:cs1
- 9-255:be:be

6.2.1.57.128 dscp-preserve

Syntax

[no] dscp-preserve [preserve-user-plane]

Context

config>mobile>profile>policy-options>qci-policy>qci

Description

This command controls how the SGW and PGW marks the DSCP for GTP-U packets in the uplink and downlink direction as well as user-plane packets on SGi in the uplink direction.

For SGW:

- dscp-preserve: The SGW marks uplink and downlink GTP-U packets using the DSCP received on the corresponding incoming GTP-U packets.
- no dscp-preserve: The SGW uses the QCI to DSCP mapping table described elsewhere in this qci-policy to set the GTP-U packets.

The SGW is transparent to the user-plane DSCP values.

For PGW:

- dscp-preserve: The PGW marks uplink user-plane (SGi) packets using the DSCP received on the corresponding incoming GTP-U packets. The PGW is not transparent to user-plane DSCP values in the uplink direction. The PGW marks downlink GTP-U packets using the DSCP received on the corresponding incoming user-plane (SGi) packets. The PGW is transparent to the user-plane DSCP values in the downlink direction.
- dscp-preserve preserve-user-plane: The PGW is transparent to the user-plane DSCP in uplink and downlink directions.
- no dscp-preserve: The PGW marks uplink user-plane (SGi) packets using the QCI to DSCP mapping table described elsewhere in this qci-policy. The PGW is not transparent to user-plane DSCP values in the uplink direction. The PGW marks downlink GTP-U packets using the QCI to DSCP mapping table described elsewhere in this qci-policy. The PGW is transparent to the user-plane DSCP values in the downlink direction.

Default

dscp-preserve

6.2.1.57.129 fc

Syntax

fc *fc-name*

no fc

Context

config>mobile>profile>policy-options>qci-policy>qci

Description

This command configures the FC name.

Parameters

fc-name

specifies the FC name

Values: be | l2 | af | l1 | h2 | ef | h1 | nc

Default values per qci (qci:fc):

- 1:nc
- 2:ef
- 3:h2
- 4:h1
- 5:l1
- 6:af
- 7:af
- 8:l2
- 9-255:be

6.2.1.57.130 profile

Syntax

profile {in | out | apply-cir}

no profile

Context

config>mobile>profile>policy-options>qci-policy>qci

Description

This command configures the QCI profile.

Default

no profile

Parameters

in

specifies the in profile

out

specifies the out profile

apply-cir

specifies the cir profile

6.2.1.57.131 qos-rule-unit

Syntax

[no] qos-rule-unit *qru-name*

Context

config>mobile>profile>policy-options

Description

This command specifies a QRU.

The no form of this command removes the specified QRU.

Parameters

qru-name

specifies the name of the QRU, up to 32 characters

6.2.1.57.132 average-window

Syntax

average-window *average-window-value*

no average-window

Context

config>mobile>profile>policy-options>qos-rule-unit

Description

This command configures the average window (duration over which the guaranteed and maximum bitrate is calculated) for the corresponding service data flow. This command applies for 5G systems.

The no form of this command reverts to the default.

Default

no average-window

Parameters

average-window-value

specifies a value for the average window static mapping, in milliseconds

Values: 0 to 4095

Default: 2000

6.2.1.57.133 down-link

Syntax

down-link gbr *gbr-value* mbr *mbr-value*

no down-link

Context

config>mobile>profile>policy-options>qos-rule-unit

Description

This command configures the downlink GBR and the MBR for the associated SDF.

Default

no down-link

Parameters

gbr-value

specifies the GBR

Values: 0 to 300000

mbr-value

specifies the MBR

Values: 0 to 300000

6.2.1.57.134 max-data-burst-vol

Syntax

`max-data-burst-vol` *max-data-burst-vol*

`no` `max-data-burst-vol`

Context

`config>mobile>profile>policy-options>qos-rule-unit`

Description

This command specifies the maximum data burst volume (largest amount of data that is required to be transferred within a period of 5G-AN PDB) for the corresponding service data flow. This command applies for 5G systems.

The no form of this command reverts to the default.

Default

`no` `max-data-burst-vol` (disabled)

Parameters

max-data-burst-vol

specifies a value for the maximum data burst volume for the static mapping

Values: 0 to 4095

Default: 0

6.2.1.57.135 priority-level

Syntax

`priority-level` *priority-level*

`no` `priority-level`

Context

config>mobile>profile>policy-options>qos-rule-unit

Description

This command specifies the priority-level (priority in scheduling resources among QoS flows) for the corresponding service data flow. This command applies for 5G systems.

The no form of this command reverts to the default.

Default

no priority-level

Parameters

priority-level

specifies a value for the priority level static mapping

Values: 0 to 127

Default: 0

6.2.1.57.136 qer-id

Syntax

qer-id *qer-id-value*

no qer-id

Context

config>mobile>profile>policy-options>qos-rule-unit

Description

This command defines a QER identifier to be associated with this QRU.



NOTE:

Every QRU must have a QER identifier defined for it.

The no form of this command removes the specified QER identifier.

Parameters

qer-id-value

specifies the QER identifier

Values: 1 to 30000

6.2.1.57.137 qfi

Syntax

[no] qfi *qfi-value*

Context

config>mobile>profile>policy-options>qos-rule-unit

Description

This command configures the QFI for the corresponding service data flow. This is required only when multiple QFIs per rule base are supported. This command applies for 5G systems.

The no form of this command reverts to the default (disabled).

Default

no qfi

Parameters

qfi-value

specifies a value for the QFI

Values: 0 to 63

Default: 0

6.2.1.57.138 qnc

Syntax

[no] qnc

Context

config>mobile>profile>policy-options>qos-rule-unit

Description

This command enables notification requests from the NG-RAN when GBR can no longer (or again) be guaranteed for a QoS flow during the lifetime of the QoS flow, for the corresponding service data flow. This command is applicable for 5G systems.

The no form of this command reverts to the default (disabled).

Default

no qnc

6.2.1.57.139 rqi

Syntax

[no] rqi

Context

config>mobile>profile>policy-options>qos-rule-unit

Description

This command enables indication when the QoS information is reflective for the corresponding service data flow. This command applies for 5G systems.

The no form of this command reverts to the default (disabled).

Default

no rqi

6.2.1.57.140 sdf-policing

Syntax

sdf-policing direction *direction*

no sdf-policing

Context

config>mobile>profile>policy-options>qos-rule-unit

Description

This command specifies the direction at which the rule is policed. The rule can be policed in the downlink direction or bi-directionally.

Parameters

direction

specifies the direction in which the rule is policed

Values: dl | both

6.2.1.57.141 service-class-indicator

Syntax

```
service-class-indicator service-class-id
```

```
no service-class-indicator
```

Context

```
config>mobile>profile>policy-options>qos-rule-unit
```

Description

This command configures the service class indicator for the QRU.

The no form of this command removes the service class indicator from the QRU.

Parameters

service-class-id

specifies a service class indicator

Values: 1 to 15

6.2.1.57.142 tos-tc

Syntax

```
tos-tc dscp-value
```

```
no tos-tc
```

Context

config>mobile>profile>policy-options>qos-rule-unit

Description

This command enhances the PCC rule match criteria to include the ToS/TC DSCP.

The no form of this command reverts to the default.

Default

no tos-tc

Parameters

dscp-value

specifies the DSCP code point value

Values: 0 to 63

6.2.1.57.143 up-link

Syntax

up-link gbr *gbr-value* mbr *mbr-value*

no up-link

Context

config>mobile>profile>policy-options>qos-rule-unit

Description

This command configures the uplink GBR and the MBR for the associated SDF.

The no form of this command reverts to the default.

Default

no up-link

Parameters

gbr-value

specifies the GBR

Values: 0 to 300000

gbr-value

specifies the MBR

Values: 0 to 300000

6.2.1.57.144 **service-class-indicator**

Syntax

`service-class-indicator service-class-id`

Context

`config>mobile>profile>policy-options`

Description

This command configures the service class indicator.

Parameters

service-class-id

Specifies an indicator number.

Values: 1 to 15

6.2.1.57.145 **service-class-value**

Syntax

`service-class-value sci-value`

Context

`config>mobile>profile>policy-options>service-class-indicator`

Description

This command configures the service class value number.

Parameters

sci-value

specifies a service class value number

Values: 0 to 15

Default: 0

6.2.1.57.146 sru-list

Syntax

[no] sru-list *list-name*

Context

config>mobile>profile>policy-options

Description

This command configures a SRU list consisting of a set of SRUs. An SRU must be defined before it can be included in the SRU list. A maximum of four SRUs can be included in the SRU list. These SRUs are associated with the policy rule configured at the control plane, within its CRU. An SRU list is specified for the policy rule at the user plane and not for a policy rule at the control plane.

The no form of this command removes the specified SRU list from the configuration.

Parameters

list-name

specifies the name of the SRU list, up to 32 characters

6.2.1.57.147 stat-rule-unit

Syntax

[no] stat-rule-unit *sru-name*

Context

config>mobile>profile>policy-options>sru-list

Description

This command associates a SRU to an SRU list.

The no form of this command removes the specified SRU from the SRU list.

Parameters

sru-name

specifies the name of the SRU, up to 32 characters

6.2.1.57.148 stat-rule-unit

Syntax

```
[no] stat-rule-unit sru-name
```

Context

```
config>mobile>profile>policy-options
```

Description

This command creates an SRU. An SRU is linked to a charging rule unit, depending on the reporting level for charging purposes and with a monitoring key for Gx-usage monitoring.

The no form of this command removes the specified SRU from the configuration.

Parameters

sru-name

specifies the name of the SRU, up to 32 characters

6.2.1.57.149 urr-id

Syntax

```
urr-id urr-id-value urr-profile profile-name  
no urr-id
```

Context

```
config>mobile>profile>policy-options>stat-rule-unit
```

Description

This command creates a URR ID and associates it to a URR profile. When the *application-id* in the `config>mobile>profile>policy-options>policy-rule-unit>flow-description>match` context is configured as part of the policy rule, the URR ID value is sent in the Create PDR and the Create URR of the PFCP Session Establishment Request.

The no form of this command removes the specified URR ID and the association from the configuration.

Parameters

urr-id-value

specifies the URR ID value

Values: 1 to 2147483647

profile-name

specifies the URR profile name, up to 32 characters

6.2.1.57.150 steering-profile

Syntax

[no] steering-profile *steering-profile-name*

Context

config>mobile>profile>policy-options

Description

This command is used to define a steering profile. Contents of the profile include uplink and downlink next-hop information. The steering profile can be associated within a policy rule unit for SDF-level steering. It can also be associated at the APN level as a default steering profile for an APN. When an SDF-level profile is enabled, it overrides the APN-level specified profile. If no profile is specified at the SDF level, the profile is inherited from the APN level.

The no form of this command removes the specified profile.

Parameters

steering-profile-name

specifies the name of the steering profile, up to 32 characters

6.2.1.57.151 traffic-redirect

Syntax

traffic-redirect *traffic-redirect-profile-id*

no traffic-redirect

Context

config>mobile>profile>policy-options>steering-profile

Description

This command specifies the next-hop information within a steering profile. The traffic redirect profile needs to be defined under `config>mobile>profile>policy-options` prior to associating it within the steering profile. The traffic redirect profile includes uplink and/or downlink next-hop information.

The no form of this command removes the association of the traffic-redirect-profile with this steering profile.

Parameters

traffic-redirect-profile

the name of the traffic redirect profile, up to 32 characters

6.2.1.57.152 traffic-hash-profile

Syntax

[no] traffic-hash-profile *profile-name*

Context

config>mobile>profile>policy-options

Description

This command is used to define a traffic hash profile. The specified profile is applied to an SDF when instantiated within a Policy Rule.



NOTE:

A maximum of 24 profiles is supported.

Parameters

profile-name

specifies the profile name, up to 32 characters

6.2.1.57.153 traffic-distribution

Syntax

[no] traffic-distribution

Context

config>mobile>profile>policy-options>traffic-hash-profile

Description

This command enables the context to configure traffic distribution.

6.2.1.57.154 access-lte-percentage

Syntax

access-lte-percentage *lte-percent-value*

no access-lte-percentage

Context

config>mobile>profile>policy-options>traffic-hash-profile>traffic-distribution

Description

This command is used to specify the traffic distribution percentage for mobile (for example, LTE) access type. The sum of the values specified for mobile access type and fixed access type must be equal to 100. The sum of the values specified for mobile (for example, LTE) access type and fixed (for example, xDSL) access type must be equal to 100.

Parameters

lte-percent-value

specifies the traffic distribution percentage for mobile access

Values: integer; range: 0 - 100

6.2.1.57.155 access-xdsl-percentage

Syntax

access-xdsl-percentage *xdsl-percent-value*

no access-xdsl-percentage

Context

config>mobile>profile>policy-options>traffic-hash-profile>traffic-distribution

Description

This command is used to specify the traffic distribution percentage for fixed (for example, xDSL) access type. The sum of the values specified for mobile (for example, LTE) access type and fixed (for example, xDSL) access type must be equal to 100.

Parameters

xdsl-percent-value

specifies the traffic distribution percentage for fixed access

Values: integer; range: 0 - 100

6.2.1.57.156 traffic-redirect-profile

Syntax

[no] traffic-redirect-profile *traffic-redirection-profile-name*

Context

config>mobile>profile>policy-options

Description

This command configures the traffic redirection profile.

Parameters

traffic-redirection-profile-name

specifies a name for the redirection profile, up to 32 characters long

6.2.1.57.157 bfd-enable

Syntax

[no] bfd-enable

Context

config>mobile>profile>policy-options>traffic-redirect-profile

Description

This command enables the BFD session to track the status of the steering function instance to start or stop forwarding steering traffic toward the service function instances.

6.2.1.57.158 destination-nat

Syntax

```
destination-nat
```

Context

```
config>mobile>profile>policy-options>traffic-redirect-profile
```

Description

This command enables a context for destination NAT parameters. A traffic-redirect profile may be configured for destination NAT or for traffic-redirect, but not for both.

6.2.1.57.159 destination-ipv4

Syntax

```
destination-ipv4 ipv4-address
```

```
no destination-ipv4
```

Context

```
config>mobile>profile>policy-options>traffic-redirect-profile>destination-nat
```

Description

This command enables the destination-nat feature for the traffic-redirect-profile. The system rewrites the destination IPv4 address for all IPv4 user plane packets received from a UE. This feature does not apply to packets sent toward the UE. The IP address may not be set to:

- 255.255.255.255 broadcast
- 127.0.0.0/8 loopback
- 224.0.0.0/4 multicast range

Parameters

ipv4-address

specifies the new destination IPv4 address

Values: a.b.c.d

6.2.1.57.160 mg-group

Syntax

[no] mg-group *group-id*

Context

config>mobile>profile>policy-options>traffic-redirect-profile

Description

This command configures the traffic redirection profile for a specific MG group.

Parameters

group-id

specifies the MG group ID

Values: 1 to 15

6.2.1.57.161 router

Syntax

router [u|l *router-name*] [d|l *router-name*]

no router

Context

config>mobile>profile>policy-options>traffic-redirect-profile>mg-group

Description

This command configures the traffic redirection router instance per MG group. The router instance for uplink and downlink traffic are configured independently.

Parameters

router-name

specifies the virtual router name or service ID for uplink or downlink traffic

Values	router-name service-id		
–	router-name	Base	
–	service-id	1 to 2147483647	

Default: Base

6.2.1.57.162 traffic-redirect-target

Syntax

[no] traffic-redirect-target *traffic-redirect-target-id*

Context

config>mobile>profile>policy-options>traffic-redirect-profile
config>mobile>profile>policy-options>traffic-redirect-profile>mg-group

Description

This command configures the traffic redirection target. The MG group-specific context applies only to CMG deployed as LB-less mode.

Parameters

traffic-redirect-target-id
specifies a identifier for the redirection target
Values: 1 to 64

6.2.1.57.163 downlink-next-hop-ipv4

Syntax

downlink-next-hop-ipv4 *ipv4-address*
no downlink-next-hop-ipv4

Context

config>mobile>profile>policy-options>traffic-redirect-profile>traffic-redirect-target

config>mobile>profile>policy-options>traffic-redirect-profile>mg-group>traffic-redirect-target

Description

This command specifies the redirection target IPv4 address for the downlink direction. The MG group-specific context applies only to CMG deployed as LB-less mode.

The no form of this command reverts to the default.

Default

no downlink-next-hop-ipv4

Parameters

ip-address

specifies a next-hop IPv4 address

Values: a.b.c.d

6.2.1.57.164 downlink-next-hop-ipv6

Syntax

downlink-next-hop-ipv6 *ipv6-address*

no downlink-next-hop-ipv6

Context

config>mobile>profile>policy-options>traffic-redirect-profile>traffic-redirect-target

config>mobile>profile>policy-options>traffic-redirect-profile>mg-group>traffic-redirect-target

Description

This command specifies the redirection target IPv6 address for the downlink direction. The MG group-specific context applies only to CMG deployed as LB-less mode.

The no form of this command reverts to the default.

Default

no downlink-next-hop-ipv6

Parameters

ipv6-address

specifies a next-hop IPv6 address

Values: 64 characters maximum

6.2.1.57.165 uplink-next-hop-ipv4

Syntax

uplink-next-hop-ipv4 *ip-address*

no uplink-next-hop-ipv4

Context

config>mobile>profile>policy-options>traffic-redirect-profile>traffic-redirect-target

config>mobile>profile>policy-options>traffic-redirect-profile>mg-group>traffic-redirect-target

Description

This command specifies the redirection target IPv4 address for the uplink direction. The MG group-specific context applies only to CMG deployed as LB-less mode.

The no form of this command reverts to the default.

Default

no uplink-next-hop-ipv4

Parameters

ip-address

specifies a next-hop IPv4 address

Values: a.b.c.d

6.2.1.57.166 uplink-next-hop-ipv6

Syntax

uplink-next-hop-ipv6 *ipv6-address*

no uplink-next-hop-ipv6

Context

config>mobile>profile>policy-options>traffic-redirect-profile>traffic-redirect-target

config>mobile>profile>policy-options>traffic-redirect-profile>mg-group>traffic-redirect-target

Description

This command specifies the redirection target IPv6 address for the uplink direction. The MG group-specific context applies only to CMG deployed as LB-less mode.

The no form of this command reverts to the default.

Default

no uplink-next-hop-ipv6

Parameters

ipv6-address

specifies a next-hop IPv6 address

Values: 64 characters maximum

6.2.1.57.167 router

Syntax

router *router-instance*

no router

Context

config>mobile>profile>policy-options>traffic-redirect-profile

Description

This command configures the traffic redirection router instance.

Parameters

router-instance

specifies the virtual router name or service ID

Values	router-name service-id		
–	router-name	Base	
–	service-id	1 to 2147483647	

Default: Base

6.2.1.57.168 threshold-target

Syntax

```
threshold-target hold-time seconds start start-targets stop stop-targets  
no threshold-target
```

Context

```
config>mobile>profile>policy-options>traffic-redirect-profile
```

Description

This command configures the number of targets that must be available to start and stop steering toward service function instances.

Parameters

seconds
specifies the hold time, in seconds, for targets to be available before starting the traffic flow
Values: 0 to 4294967295

start-targets
specifies the number of targets that must be available to start steering
Values: 0 to 63

stop-targets
specifies the number of targets that must be available to stop steering
Values: 0 to 63

6.2.1.57.169 trigger-rule-unit

Syntax

```
[no] trigger-rule-unit tru-name
```

Context

```
config>mobile>profile>policy-options
```

Description

This command is used to define a TRU. Applicable access types can be specified within a trigger rule unit. A trigger rule unit can be optionally associated with a pre-defined PCC rule. The rule is then applied to a session that goes into Assume Positive (AP) mode while on the applicable RAT within the pre-defined policy. The no form of this command removes the specified trigger rule unit.

Parameters

tru-name

specifies the name of the trigger rule unit

6.2.1.57.170 access-type

Syntax

```
access-type {[e]utran} [wlan] [e]utran-nb-iot [eh]rpd [u]tran [g]eran [g]an [h]spa-e}}  
no access-type
```

Context

```
config>mobile>profile>policy-options>trigger-rule-unit
```

Description

This command is used to specify the access types within a trigger rule unit. All applicable access types can be specified within a trigger rule unit. A trigger rule unit can be optionally associated with a pre-defined PCC rule. The rule is then applied to a session that goes into Assume Positive mode while on the applicable RAT within the pre-defined policy.

The no form of this command reverts to the default.

Default

```
no access-type
```

Parameters

eutran

specifies a EUTRAN access type

wlan

specifies a WLAN access type

eutran-nb-iot

specifies a EUTRAN-NB-IOT access type

ehrpdp

specifies an eHRPD access type

utran

specifies a UTRAN access type

geran

specifies a GERAN access type

gan

specifies a GAN access type

hspa-e

specifies a HSPA-E access type

6.2.1.57.171 tai-lai-list

Syntax

```
tai-lai-list tai-lai-list-name
```

```
no tai-lai-list
```

Context

```
config>mobile>profile>policy-options>trigger-rule-unit
```

Description

This command matches the current location of the UE and installs the policy for the corresponding DNAI configured with the `action-rule-unit`. The TAI or LAI profile list must be configured with the `tai-lai-list` command.

The no form of this command removes the list from the configuration.

Parameters

tai-lai-list-name

specifies the name of the TAI or LAI profile list, up to 32 characters

6.2.1.57.172 **urr-profile**

Syntax

[no] **urr-profile** *urr-profile-name*

Context

config>mobile>profile>policy-options

Description

This command creates a URR profile. The profile is then associated with a specific URR identifier within an SRU.

Parameters

urr-profile-name

specifies the URR profile name, up to 32 characters

6.2.1.57.173 **discard-on-fpt**

Syntax

[no] **discard-on-fpt**

Context

config>mobile>profile>policy-options>urr-profile

Description

This command facilitates dropping packets upon first packet trigger detection atomically, for the associated SDF. This is equivalent to the **discard-on-fpt** option within the traffic handling options (config>gateway>profile>dcca>traffic-handling-options) when quota is unavailable.



NOTE:

- The URR profiles which have this feature enabled must only be associated with rating groups used for online charging.

- When the `service-id-level-credit-control` command is configured under the DCCA profile, the `discard-on-fpt` flag must be enabled under the URR profile associated with the service ID not the rating group.

Default

no discard-on-fpt

6.2.1.57.174 usage-monitoring-actions

Syntax

[no] usage-monitoring-actions

Context

config>mobile>profile>policy-options

Description

This command enables the context to configure usage monitoring actions. The `usage-monitoring-actions` define the percentage of change in traffic distribution that applies when fill-out has been triggered.

6.2.1.57.175 monitoring-action

Syntax

[no] monitoring-action *monitoring-actions-profile-name*

no monitoring-action

Context

config>mobile>profile>policy-options>usage-monitoring-actions

config>mobile>profile>policy-options>usage-monitoring-info>monitoring-key

Description

This command configures a monitoring action profile in the usage monitoring context, and specifies a monitoring action profile in the monitoring key context.

Parameters

monitoring-actions-profile-name

specifies the profile name, up to 32 characters

6.2.1.57.176 after-fill-delta-percent

Syntax

```
after-fill-delta-percent delta-percent-value  
no after-fill-delta-percent
```

Context

```
config>mobile>profile>policy-options>usage-monitoring-actions>monitoring-action
```

Description

The command specifies the after fill delta percent for this monitoring action profile. When specified, it overrides the after fill delta percent specified at the APN/gateway level. This is used in the context of the fixed access fill-out feature (see *fixed-access-fill-out*). When a low or high threshold is hit, the traffic distribution load is altered by the specified delta percent. For example, an initial LTE/fixed equal to 0/100 changes to 10/90.

The no form of this command reverts to the default (0).

Parameters

delta-percent-value

specifies the after fill delta percent

Values: 10 to 50 (integer in increments of 10)

Default: 0

6.2.1.57.177 usage-monitoring-info

Syntax

```
[no] usage-monitoring-info
```

Context

```
config>mobile>profile>policy-options>usage-monitoring-action
```

Description

This command enables the context to configure usage monitoring key information.

6.2.1.57.178 monitoring-key

Syntax

```
[no] monitoring-key monitoring-key-profile-name
```

Context

```
config>mobile>profile>policy-options>usage-monitoring-info
```

Description

This command specifies a monitoring key profile to support the fixed access fill out first feature. The monitoring key profile configuration components include the quota type (defaults to rate), low and high threshold percent, sustained-rate duration, and monitoring action. If specified, an optional traffic-hash profile is also applied at the APN level; see `monitoring-key`.

The no form of this command removes the specified monitoring key profile.

Parameters

monitoring-key-profile-name

specifies the profile name, up to 32 characters

6.2.1.57.179 fixed-access-fill-out

Syntax

```
fixed-access-fill-out low-threshold-percent low-threshold-value high-threshold-percent high-threshold-value
```

```
no fixed-access-fill-out
```

Context

```
config>mobile>profile>policy-options>usage-monitoring-info>monitoring-key
```

Description

This command specifies the low and high thresholds for the fixed access fill out feature. When the low threshold is hit the traffic-distribution percentage is altered by the delta percent specified by the *after-fill-delta-percent*. For example, if the low threshold is 70% of the APN-AMBR, upon breaching the threshold (sustained rate), an original distribution of 0/100 is altered to 10/90 for the fixed access bearer. If the specified high threshold is reached, the traffic distribution is altered immediately.

The no form of this command reverts to the default (disabled).

Parameters

low-threshold-value

specifies the low threshold percent for the fixed access fill out, in increments of five

Values: 50 to 90

Default: 0

high-threshold-value

specifies the high threshold percent for the fixed access fill out, in increments of five

Values: 50 to 95

Default: 0

6.2.1.57.180 fixed-access-sustained-rate

Syntax

`fixed-access-sustained-rate` `above-low-threshold` *above-low-threshold-value* `below-low-threshold` *below-low-threshold-value*

`no fixed-access-sustained-rate`

Context

`config>mobile>profile>policy-options>usage-monitoring-info>monitoring-key`

Description

This command specifies the sustained rate duration in minutes, before a change in policy is effective. For example, an 'above low threshold' of three minutes, implies that if the sustained rate is above the low threshold for three minutes, the traffic distribution percentage is altered (for example, 0/100 is altered to 10/90 for the fixed access bearer) by the specified *after-fill-delta-percent* (10). Similarly, if the sustained rate is below the low threshold for a period of time, for example five minutes, the traffic distribution is altered from 10/90 to 0/100.

Parameters

above-low-threshold-value

specifies the duration, in minutes for the sustained rate to be above low threshold

Values: 1 to 10

Default: 3

below-low-threshold-value

specifies the duration, in minutes for the sustained rate to be below low threshold

Values: 1 to 10

Default: 5

6.2.1.57.181 quota-type

Syntax

```
quota-type rate
no quota-type
```

Context

```
config>mobile>profile>policy-options>usage-monitoring-info>monitoring-key
```

Description

This command specifies a quota type for the monitoring key profile. Defaults to rate for the fixed access fill-out feature.

The no form of this command reverts to the default.

Parameters

rate

keyword that specifies the quota type

Values: rate

Default: rate (for the fixed access fill-out feature, see [fixed-access-fill-out](#))

6.2.1.57.182 traffic-hash

Syntax

```
[no] traffic-hash traffic-hash-profile-name
```

Context

```
config>mobile>profile>policy-options>usage-monitoring-info>monitoring-key
```

Description

This command configures a traffic-hash profile for the monitoring key. It is a read-only field that is defaulted to "defaultXdsIHashProfile" and no changes are permitted.

Parameters

traffic-has-profile-name

specifies the profile name, up to 32 characters

6.2.1.57.183 validate-aa-charging-grp

Syntax

```
validate-aa-charging-grp {enable | disable}
```

```
no validate-aa-charging-grp
```

Context

```
config>mobile>profile>policy-options
```

Description

This command cross-checks whether an AA charging group configured under the `policy-rule-unit` context exists under any AA partition. When an undefined AA charging group is referred under the `policy-rule-unit` context, the commit is blocked and the operator receives an alert.

The `no` form of the command reverts to default (`disable`).



NOTE:

- This command must be enabled on the UPF only.
- An AA charging group may not be configured for the partition that applies to this APN, even if configured under a different partition; a mis-configuration may still occur in this case.

Parameters

enable

specifies to enable cross-checking

disable

specifies to disable cross-checking

Default: `disable`

6.2.1.58 PPI mapping profile command descriptions

6.2.1.58.1 ppi-mapping-profile

Syntax

[no] ppi-mapping-profile *profile-name*

Context

config>mobile>profile

Description

This command creates a PPI mapping profile that is used to define the PPI when the SMF initiates paging in the AMF.

The no form of this command removes the specified PPI mapping profile from the configuration.

Parameters

profile-name

specifies a PPI mapping profile name, up to 32 characters

6.2.1.58.2 dscp-to-ppi

Syntax

dscp-to-ppi *dscp-value* ppi *ppi-value*

no dscp-to-ppi *dscp-value*

Context

config>mobile>profile>ppi-mapping-profile

Description

This command defines the mapping of a DSCP/TC value to a PPI value. If a DSCP/TC value is already defined in the profile, the command replaces the PPI mapping of the DSCP/TC value.

The no form of this command reverts to the default (a PPI value of 0 is used if no matching DSCP/TC value is found in the configuration).

Parameters

dscp-value

specifies the DSCP/TC value

Values: 0 to 63

Default: 0

ppi-value

specifies the PPI value

Values: 0 to 7

Default: 0

6.2.1.59 QFI mapping profile command descriptions

6.2.1.59.1 qfi-mapping-profile

Syntax

[no] qfi-mapping-profile *profile-name*

Context

config>mobile>profile

Description

This command defines an optional QoS flow binding profile. QoS parameters specified within the profile are used for QoS flow binding in 5G systems. When this optional profile is not configured, the QFI is generated dynamically (in range 1 to 64) using 5QI/ARP as default.

The no form of this command removes the specified QoS flow binding profile.

Parameters

profile-name

specifies a QFI profile name, up to 32 characters

6.2.1.59.2 5qi-as-qfi

Syntax

[no] 5qi-as-qfi

Context

config>mobile>profile>qfi-mapping-profile

Description

This command enables 5QI to be used as the QFI value. When enabled, this option supports only a single QoS flow for each 5QI specified and the allowed 5QI values range from 0 to 63.

The no form of this command reverts to the default (disabled).

Default

no 5qi-as-qfi

6.2.1.59.3 dynamic-mapping

Syntax

dynamic-mapping

Context

config>mobile>profile>qfi-mapping-profile

Description

This command enables the context to configure additional QoS parameters (QNC, Average-window, priority, and maximum data burst volume) used for dynamic mapping in QoS flow binding. By default, QFI is dynamically generated using the 5QI and ARP, and QFI values range from 0 to 255.

6.2.1.59.4 average-window

Syntax

[no] average-window

Context

config>mobile>profile>qfi-mapping-profile>dynamic-mapping

Description

This command enables the average window (duration over which the guaranteed and maximum bitrate is calculated) in addition to 5QI and ARP for QoS flow binding. By default, QFI is dynamically generated using 5QI and ARP.

The no form of this command reverts to the default (disabled).

Default

no average-window

6.2.1.59.5 max-data-burst-vol

Syntax

[no] max-data-burst-vol

Context

config>mobile>profile>qfi-mapping-profile>dynamic-mapping

Description

This command enables maximum data burst volume (largest amount of data that is required to be transferred within a period of 5G-AN PDB) for QoS flow binding, in addition to 5QI and ARP. By default, QFI is dynamically generated using 5QI and ARP.

The no form of this command reverts to the default (disabled).

Default

no max-data-burst-vol

6.2.1.59.6 priority-level

Syntax

[no] priority-level

Context

config>mobile>profile>qfi-mapping-profile>dynamic-mapping

Description

This command enables the priority level (priority in scheduling resources among QoS flows) for QoS flow binding, in addition to 5QI and ARP. By default, QFI is dynamically generated using 5QI and ARP.

The no form of this command reverts to the default (disabled).

Default

no priority-level

6.2.1.59.7 qnc

Syntax

[no] qnc

Context

config>mobile>profile>qfi-mapping-profile>dynamic-mapping

Description

This command enables QNC, in addition to 5QI and ARP for QoS flow binding. By default, QFI is dynamically generated using 5QI and ARP.

The no form of this command reverts to the default (disabled).

Default

no qnc

6.2.1.60 QoS profile command descriptions

6.2.1.60.1 qos-5g-profile

Syntax

[no] 5g-qos-profile *profile-name*

Context

config>mobile>profile

Description

This command configures a 5G QoS profile that can be used in the APN configuration to define the default QoS values.

The no form of this command removes the profile from the configuration.

Parameters

profile-name

specifies the name of the 5G QoS profile name, up to 32 characters

6.2.1.60.2 5qi

Syntax

5qi *value*

no 5qi

Context

config>mobile>profile>qos-5g-profile

Description

This command configures the 5QI of the QoS profile that can be used to define the default 5QI in the DNN.
The no form of this command reverts to the default.

Parameters

value

specifies the 5QI value

Values: 1 to 255

Default: 9

6.2.1.60.3 arp

Syntax

arp *arp-value*

no arp

Context

config>mobile>profile>qos-5g-profile

Description

This command defines the ARP priority level for the QoS profile that can be used to define the default ARP in the DNN.

The no form of this command reverts to the default.

Parameters

value

specifies the ARP value

Values: 1 to 15

Default: 15

6.2.1.60.4 dl-ambr

Syntax

dl-ambr *ambr-value*

no dl-ambr

Context

config>mobile>profile>qos-5g-profile

Description

This command defines the DL bitrate for the session AMBR for the QoS profile.

The no form of this command reverts to the default.

Parameters

value

specifies the DL bitrate for the session AMBR in kb/s

Values: 1 to 25000000

Default: 4096

6.2.1.60.5 ul-ambr

Syntax

ul-ambr *ambr-value*

no ul-ambr

Context

config>mobile>profile>qos-5g-profile

Description

This command defines the UL bitrate for the session AMBR for the QoS profile.

The no form of this command reverts to the default.

Parameters

value

specifies the UL bitrate for the session AMBR in kb/s

Values: 1 to 25000000

Default: 4096

6.2.1.60.6 qos-profile

Syntax

[no] qos-profile *profile-name*

Context

config>mobile>profile

Description

This command is used to specify a new QoS profile. A QoS profile consists of a default bearer ARP, QCI, downlink (DL) AMBR, and uplink (UL) AMBR. The QoS profile can be associated with a specific RAT (access type) under an APN context.

The no form of this command removes the specified QoS profile.

Parameters

profile-name

specifies the QoS profile name, up to 32 characters

6.2.1.60.7 arp

Syntax

arp *arp-value*

no arp

Context

config>mobile>profile>qos-profile

Description

This command is used to specify the default bearer ARP within the specified QoS profile.

The no form of this command removes the specified ARP value.

Parameters

arp-value

specifies the default bearer ARP value

Values: 1 to 15

6.2.1.60.8 dl-ambr

Syntax

dl-ambr *ambr-value*

no dl-ambr

Context

config>mobile>profile>qos-profile

Description

This command is used to specify the default bearer DL AMBR within the specified QoS profile.

The no form of this command removes the specified DL AMBR value.

Parameters

ambr-value

specifies the default bearer DL AMBR value in kilobits per second

Values: 1 to 25000000

6.2.1.60.9 qci

Syntax

qci *qci-value*


```
no qci
```

Context

```
config>mobile>profile>qos-profile
```

Description

This command is used to specify the default bearer QCI within the specified QoS profile.

The no form of this command removes the specified default bearer QCI value.

Parameters

qci-value

specifies the default bearer QCI value

Values: 1 to 9 | 65 | 66 | 69 | 70 | 128 to 254

6.2.1.60.10 ul-ambr

Syntax

```
ul-ambr ambr-value
```

```
no ul-ambr
```

Context

```
config>mobile>profile>qos-profile
```

Description

This command is used to specify the default bearer UL AMBR within the specified QoS profile.

The no form of this command removes the specified UL AMBR value.

Parameters

ambr-value

specifies the default bearer UL AMBR value in kilobits per second

Values: 1 to 25000000

6.2.1.61 RADIUS profile command descriptions

6.2.1.61.1 radius

Syntax

[no] radius *profile-name*

Context

config>mobile>profile

Description

This command creates a named RADIUS profile.

The no form of this command removes the profile from the configuration.

Parameters

profile-name

specifies the profile name, up to 32 characters

6.2.1.61.2 acct-retry-count

Syntax

acct-retry-count *value*

no acct-retry-count

Context

config>mobile>profile>radius

Description

This command sets a limit to the number of times the system attempts to retry sending a RADIUS accounting message to the server. When this limit is reached, the server is considered to be dead and another server is selected. This command requires a shutdown and a no shutdown action in the RADIUS group peer definition associated with the RADIUS profile before the change is effective for a RADIUS server peer.



NOTE:

- The system waits for a few more seconds (≤20) before finally marking the peer as dead to smooth out fluctuations caused by temporary effect such as network congestion or server overload.

- For a single message, the system only tries up to three peers, regardless of the number of peers in the RADIUS group. After the three peers have been exhausted (or less if the RADIUS group has less than three peers) the system will declare a final timeout for that message and discard it.

Parameters

value

the number of times the system attempts to send a RADIUS accounting message to the server

Values: 0 to 10

Default: 3

6.2.1.61.3 acct-retry-timeout

Syntax

acct-retry-timeout *seconds*

no acct-retry-timeout

Context

config>mobile>profile>radius

Description

This command sets the interval, in seconds, between retries for RADIUS accounting messages. This command requires a shutdown and a no shutdown action in the RADIUS group peer definition associated with the RADIUS profile before the change is effective for a RADIUS server peer.

The no form of this command reverts to the default.

Parameters

seconds

specifies the interval, in seconds, between retries

Values: 1 to 255

Default: 4

6.2.1.61.4 apn-acct-on-off-initial-interval

Syntax

```
apn-acct-on-off-initial-interval value
no apn-acct-on-off-initial-interval
```

Context

```
config>mobile>profile>radius
```

Description

This command configures the initial interval between sending RADIUS accounting ON and OFF messages to the accounting servers. It can be used to fine-tune the initial messaging sequence. The configuration applies to all servers within the RADIUS group regardless of peer priorities. Retries are only sent to other servers if the current server doesn't respond within the specified time interval. This command is effective only if the config>mobile>pdn>radius>accounting-on-off-level command is set to value apn. This command is effective only if wait-for-ack is configured in the context config>mobile>pdn>apn>radius-accounting-on-off.

The no form of this command reverts to the default.

Parameters

value

specifies the time interval, in seconds, between sending RADIUS accounting ON and OFF messages to the accounting servers

Values	0	sent immediately to accounting servers
–	1 to 60	sent to accounting servers in sequence with N second intervals

Default: 0

6.2.1.61.5 apn-acct-on-retry-interval

Syntax

```
apn-acct-on-retry-interval value
```

```
no apn-acct-on-retry-interval
```

Context

```
config>mobile>profile>radius
```

Description

This command configures the retry interval between RADIUS accounting ON messages per peer server. The timeout is counted from the first message sent toward this particular server. This command is effective only if the `config>mobile>pdn>radius>accounting-on-off-level` command is set to value `apn`. This command is effective only if the `config>mobile>pdn>apn>radius-accounting-on-off>wait-for-ack` is configured.

The `no` form of this command reverts to the default.

Parameters

value

specifies the time interval, in seconds, between RADIUS accounting ON messages per peer server

Values: 0 (no retries) | 1 to 120

Default: 0

6.2.1.61.6 auth-probe-interval

Syntax

```
auth-probe-interval value
```

```
no auth-probe-interval
```

Context

```
config>mobile>profile>radius
```

Description

This command sets the interval at which the system send authentication probe messages to an authentication server. An authentication probe is a “fake” Access-Request message for a certain username and password. If the server is responds to the Access-Request (even with an Access-Reject) the server will be considered operational. The authentication probe messages will not be sent to servers that are used for accounting only. This command requires a shutdown and a no shutdown action in the RADIUS group peer definition associated with the RADIUS profile before the change is effective for a RADIUS server peer.

Parameters

seconds

specifies the time interval, in seconds, at which the system sends authentication probe messages to an authentication server

Values: 20 to 600

Default: 0

6.2.1.61.7 auth-retry-count

Syntax

auth-retry-count *value*

no auth-retry-count

Context

config>mobile>profile>radius

Description

This command sets a limit to the number of times the system attempts to retry sending a RADIUS authentication message to the server. When this limit is reached, the server is considered to be dead and another server is selected. This command requires a shutdown and a no shutdown action in the RADIUS group peer definition associated with the RADIUS profile before the change is effective for a RADIUS server peer.



NOTE:

- The system waits for a few more seconds (≤ 20) before finally marking the peer as dead to smooth out fluctuations caused by temporary effect such as network congestion or server overload.
- For a single message, the system only tries up to three peers, regardless of the number of peers in the RADIUS group. After the three peers have been exhausted (or less if the RADIUS group has less than three peers) the system will declare a final timeout for that message and discard it.

Parameters

value

specifies the number of times the system attempts to send a RADIUS authentication message to the server

Values: 0 to 10

Default: 3

6.2.1.61.8 auth-retry-timeout

Syntax

```
auth-retry-timeout seconds  
no auth-retry-timeout
```

Context

```
config>mobile>profile>radius
```

Description

This command sets the interval, in seconds, between retries for RADIUS authentication. This command requires a shutdown and a no shutdown action in the RADIUS group peer definition associated with the RADIUS profile before the change is effective for a RADIUS server peer.

The no form of this command reverts to the default.

Parameters

seconds

specifies the interval, in seconds, between retries

Values: 1 to 255

Default: 4

6.2.1.61.9 deadtime

Syntax

```
deadtime deadtime [drop-acct-messages] [send-auth-messages]  
no deadtime
```

Context

```
config>mobile>profile>radius
```

Description

When the system fails to reach a particular server after a configurable number of retries, the server is marked dead and another server is selected. This command sets the time that a server continues to be considered dead before the system attempts to use it for normal authentication or accounting requests.

**NOTE:**

For authentication servers, the system will continuously attempt to verify the operational state of the dead server if the auth-probe-interval is set to be non-zero. If the authentication server responds to an auth-probe the system considers the server to be operational even if the dead timer has not yet expired.

Once the dead-timer expires the system may once again send the server authentication or accounting messages. The server is subject to the retry-timeout and retry-count mechanism to determine if the server is still in the failed state. If all servers are unreachable (operating state down), for authentication purposes, the system immediately declares an authentication failure. For accounting purposes, the system attempts to send messages to the highest priority server even if it is during the dead time period. It is possible to separately configure the system to also drop accounting messages destined for a dead server.

The no form of this command reverts to the default.

Parameters***deadtime***

specifies the time interval that a server is considered dead, in seconds

Values: 0 to 7200

Default: 300

drop-acct-messages

specifies to drop accounting messages, if no servers are currently reachable

send-auth-messages

specifies to attempt sending messages to the highest priority server, even if it is during the dead time period, in case all servers are unreachable (operating state down)

6.2.1.61.10 delay-tolerance

Syntax

delay-tolerance *dt-value*

no delay-tolerance

Context

config>mobile>profile>radius

Description

This command configures the maximum number of seconds that a message remains in the queue.

The no form of this command reverts to the default.

Default

no delay-tolerance

Parameters

dt-value

specifies the delay tolerance, in seconds

Values: 1 to 10

Default: 2

6.2.1.61.11 max-message-rate

Syntax

max-message-rate *max-rate*

no max-message-rate

Context

config>mobile>profile>radius

Description

This commands configures the maximum number of messages per second on a RADIUS interface toward a peer.



NOTE:

Applied per Lrdm task basis. The RADIUS retries (outstanding messages) are also counted in the applied throttling limit, that is the maximum message rate.

The no form of this command reverts to the default.

Default

no max-message-rate

Parameters

max-rate

specifies the maximum number of messages per second

Values: 160 to 1600

Default: none

6.2.1.61.12 max-peer-reselections

Syntax

`max-peer-reselections count[wrap]`

`no max-peer-reselections`

Context

`config>mobile>profile>radius`

Description

This command specifies the number of times in total a peer can be reselected after exhausting the retries toward the currently selected peer. The maximum number of re-selections is limited to the number of available peers in the RADIUS group. By default, for a single message, the system will only try up to three peers, regardless of the number of peers in the RADIUS group. After all the three peers have been exhausted (or less if the RADIUS group has less than three peers), the system will declare a final timeout for that message and will discard it. When the `transaction-based-load-balancing` command is enabled for the RADIUS group, the server selection can wrap around since the retries are going to different servers. If the parameter `wrap` is specified, the system tries up to the configured count, possibly wrapping around back to the original selected server, once all other available servers have been tried already. If this parameter is not specified, the maximum amount of re-selections is limited to the number of available peers in the RADIUS group.

The `no` form of this command reverts to the default.

Parameters

count

specifies the number of times a peer can be reselected

Values: 0 to 250

Default: 3

wrap

specifies if the system tries the configured count, wrapping around back to the original selected server, once all other available servers are tried

Default: disabled

6.2.1.61.13 max-queue-size

Syntax

```
max-queue-size queue-size  
no max-queue-size
```

Context

```
config>mobile>profile>radius
```

Description

This command configures the maximum queue size for storing the incoming messages.

**NOTE:**

Nokia recommends configuring the maximum queue size before configuring the maximum message rate to avoid re-allocation.

The no form of this command reverts to the default.

Default

```
no max-queue-size
```

Parameters

queue-size

specifies the maximum queue size for storing the incoming messages

Values: 160 to 800

Default: 160

6.2.1.61.14 message-type-drop-order

Syntax

```
message-type-drop-order message-order  
no message-type-drop-order
```

Context

```
config>mobile>profile>radius
```

Description

This command configures the order in which the incoming RADIUS messages are throttled (rejected).

The no form of this command reverts to the default.

Default

no message-type-drop-order

Parameters

message-order

specifies the order in which the RADIUS messages are throttled

Values: interim-auth-start | interim-start-auth | auth-interim-start

Default: interim-auth-start

6.2.1.61.15 nas-port-type

Syntax

nas-port-type *nas-port-type*

no nas-port-type

Context

config>mobile>profile>radius

Description

This command specifies the NAS port type, as defined in IETF RFC 2865.

The no form of this command reverts to the default.

Parameters

nas-port-type

specifies the NAS port type

Values: 0 to 65535

Default: 15 (Ethernet)

6.2.1.61.16 throttle-emergency

Syntax

[no] throttle-emergency

Context

config>mobile>profile>radius

Description

This command enables or disables the throttling of the emergency call signaling.

The no form of this command reverts to the default.

6.2.1.61.17 radius-group

Syntax

[no] radius-group *radius-group*

Context

config>mobile>profile

Description

This command creates a named radius-group and enters a radius-group context. A radius-group may only apply to a single pdn gateway ID. A radius-group defines a list of RADIUS servers and associated parameters. The radius-profile is applied to the entire group or to an individual server. The radius-group is used at the PDN level or at the APN level.

Parameters

radius-group

specifies the group name

6.2.1.61.18 acct-server-port

Syntax

acct-server-port *port*

no acct-server-port

Context

```
config>mobile>profile>radius-group
```

```
config>mobile>profile>radius-group>peer
```

Description

This command sets the destination UDP port for the RADIUS accounting server. When configured at the radius-group level it is the default for the entire group.

The no form of this command reverts to the default.

Parameters

port

specifies the destination UDP port

Values: 1 to 65535

Default: 1813

6.2.1.61.19 auth-server-port

Syntax

```
auth-server-port port
```

```
no auth-server-port
```

Context

```
config>mobile>profile>radius-group
```

```
config>mobile>profile>radius-group>peer
```

Description

This command sets the destination UDP port for the RADIUS authentication server. When configured at the radius-group level it is the default for the entire group.

The no form of this command reverts to the default.

Parameters

port

specifies the destination UDP port

Values: 1 to 65535

Default: 1812

6.2.1.61.20 failover-threshold

Syntax

```
failover-threshold count  
no failover-threshold
```

Context

```
config>mobile>profile>radius-group>peer
```

Description

This command configures the number of times an unresponsive peer is re-selected, before the RADIUS server is considered unresponsive.

The no form of this command reverts to the default value.

Parameters

count

specifies the number of re-selection attempts

Values: 0 to 255

Default: 0

6.2.1.61.21 failure-detection-time

Syntax

```
failure-detection-time value  
no failure-detection-time
```

Context

```
config>mobile>profile>radius-group>peer
```

Description

This command configures the failure detection timer defining the maximum duration in seconds the PGW/GGSN waits to receive any successful Auth/Acct response before the server is considered failed. A timer is maintained separately for each authentication and accounting server. This command is effective only with transaction-based load balancing.

The no form of this command reverts to the default.

**NOTE:**

The system will wait for a few more seconds (≤ 20) before finally marking the peer dead to smooth out fluctuations caused by temporary effect such as network congestion or server overload.

Parameters

value

specifies the failure detection timer, in seconds

Values: 1 to 60

Default: 10

6.2.1.61.22 interface

Syntax

```
interface [router router-instance] interface-name
```

```
no interface
```

Context

```
config>mobile>profile>radius-group
```

Description

This command names the interface that will be used to send and receive RADIUS messages. Therefore this command indirectly determines the source IP address. In case this interface belongs to a VPRN other than the master instance, this command must include the VPRN name. (Interface names have VPRN scope.)

The no form of this command removes the parameter values from the configuration.

Parameters

router-instance

specifies the name of a VPRN

Values	router-name service-id
--------	--------------------------

–	router-name	Base
–	service-id	1 to 2147483647

Default: Base

interface-name

specifies an interface name, up to 32 characters; must start with a letter

6.2.1.61.23 interim-update-interval

Syntax

```
interim-update-interval value  
no interim-update-interval
```

Context

config>mobile>profile>radius-group

Description

This command enables interim accounting and sets the interval for sending interim-update messages to the accounting server.

The no form of this command reverts to the default.

Parameters

value

specifies the time interval, in minutes, for sending interim-update messages to the accounting server

Values: 15 to 1440 (1 day)

Default: 0

6.2.1.61.24 limit-user-name-length

Syntax

```
[no] limit-user-name-length
```

Context

config>mobile>profile>radius-group

Description

This command configures the PGW/GGSN to limit the length of the User-Name attribute. The attribute length cannot exceed 64 bytes. When the length of User-Name attribute in a message exceeds 64 bytes, it is truncated to 64 bytes before the attribute is encoded to be sent over the RADIUS interface. When a message is received over the RADIUS interface, decoding is performed only on the first 64 bytes of the User-Name attribute.

The no form of this command reverts to the default.

Default

no limit-user-name-length

6.2.1.61.25 peer

Syntax

[no] peer *ipv4-address* | *ipv6-address*

Context

config>mobile>profile>radius-group

Description

This command identifies a RADIUS server and enters the server context to configure parameters specific to the identified server. This command may configure a single RADIUS server by its IPv4 or IPv6 address. The servers belonging to a radius-group must use the same IP version and the IP version must match that of the RADIUS source interface.



NOTE:

You must not use the same IP address for a RADIUS group that is used in the server-group or config>mobile>pdn>apn>accounting>duplicate-server-group under the same APN.

The no form of this command removes the IP address from the configuration.

Parameters

ipv4* | *ipv6

specifies the RADIUS server address, either an IPv4 address or an IPv6 address

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

6.2.1.61.26 priority

Syntax

priority *value*
no priority

Context

config>mobile>profile>radius-group>peer

Description

This command specifies the priority value. This is a number from 1 to 3 with 3 considered the highest priority. The system will attempt to use the highest priority server available at the time. The system will automatically use round-robin load-balancing among servers configured with the same priority.

The no form of this command reverts to the default.

Parameters

value
specifies the priority of the server
Values: 1 to 3
Default: 1

6.2.1.61.27 radius-profile

Syntax

```
radius-profile profile-name  
no radius-profile
```

Context

```
config>mobile>profile>radius-group  
config>mobile>profile>radius-group>peer
```

Description

This command applies a radius-profile to the radius-group or to an individual server. When configured at the radius-group level it is the default for the entire group.

The no form of this command removes the *name* parameter from the configuration.

Parameters

profile-name
specifies the group name

6.2.1.61.28 secret

Syntax

```
secret secret [hash | hash2]  
no secret
```

Context

```
config>mobile>profile>radius-group  
config>mobile>profile>radius-group>peer
```

Description

This is the shared secret key to be used with the server. When configured at the radius-group level it is the default for the entire group. If the hash/hash2 parameter is not used for the password, the key is assumed to be in a non-encrypted, clear text form. For security, all keys are anyway stored automatically in encrypted hash form in the configuration. The info/save uses the configuration defined in system>security>hash-control context.

Parameters

secret

specifies the shared secret key to be used with the server

Values	secret	up to 64 characters
–	hash-key	up to 33 characters
–	hash2-key	up to 96 characters

hash

specifies the key is entered in an encrypted form

hash2

specifies the key is entered in a more complex encrypted form

6.2.1.61.29 single-server-per-ue

Syntax

[no] single-server-per-ue

Context

config>mobile>profile>radius-group

Description

This command enables sending all RADIUS messages per UE to the same RADIUS server, and allows the requests per UE to be load-balanced among the servers in the RADIUS group.

The no form of the command reverts to the default.



NOTE:

This feature applies when the Transaction Based Load Balancing is not enabled under the RADIUS group.

Default

no single-server-per-ue

6.2.1.61.30 radius-avp-options

Syntax

```
radius-avp-options
```

Context

```
config>mobile>profile>radius-group
```

Description

This command enables the context to configure the RADIUS AVP options profile.

6.2.1.61.31 acct-interim-avp-options

Syntax

```
acct-interim-avp-options
```

Context

```
config>mobile>profile>radius-group>radius-avp-options
```

Description

This command enables the context to configure the RADIUS AVP options profile for interim accounting messages. If configured, this overrides the `config>mobile>profile>radius-group>radius-avp-options` context configuration within the same RADIUS group specifically for these messages. The usage is enabled/disabled using the `use-acct-interim-avp-options` command.

6.2.1.61.32 exclude-3gpp-charging-characteristics

Syntax

```
[no] exclude-3gpp-charging-characteristics
```

Context

```
config>mobile>profile>radius-group>radius-avp-options>acct-interim-avp-options
```

```
config>mobile>profile>radius-group>radius-avp-options
```

Description

This command includes or excludes the 3GPP charging characteristics IE in the RADIUS messages sent by the PGW/GGSN.

The no form of this command reverts to the default.

Default

no exclude-3gpp-charging-characteristics

6.2.1.61.33 exclude-3gpp-charging-Id

Syntax

[no] exclude-3gpp-charging-Id

Context

config>mobile>profile>radius-group>radius-avp-options>acct-interim-avp-options

Description

This command includes or excludes the 3GPP charging identifier IE in the RADIUS messages sent by the PGW/GGSN.

The no form of this command reverts to the default.

Default

no exclude-3gpp-charging-Id

6.2.1.61.34 exclude-3gpp-charging-Id

Syntax

exclude-3gpp-charging-Id [accounting | authentication | both]

no exclude-3gpp-charging-Id

Context

config>mobile>profile>radius-group>radius-avp-options

Description

This command includes or excludes the 3GPP charging identifier IE in the RADIUS messages sent by the PGW/GGSN.

The no form of this command reverts to the default.

Default

no exclude-3gpp-charging-ld

Parameters

accounting

if specified, the AVP is excluded only in Accounting-Request (START/STOP) messages

authentication

if specified, the AVP is excluded only in Access-Request (START/STOP) messages

both

if both keywords are specified, the AVP is excluded in both in Accounting-Request (START/STOP) messages and Access-Request (START/STOP) messages

Default: both

6.2.1.61.35 exclude-3gpp-ggsn-address

Syntax

[no] exclude-3gpp-ggsn-address

Context

config>mobile>profile>radius-group>radius-avp-options>acct-interim-avp-options

config>mobile>profile>radius-group>radius-avp-options

Description

This command includes or excludes the 3GPP GGSN IPv4 address IE in the RADIUS messages sent by the PGW/GGSN.

The no form of this command reverts to the default.

Default

no exclude-3gpp-ggsn-address

6.2.1.61.36 **exclude-3gpp-ggsn-ipv6-address**

Syntax

[no] `exclude-3gpp-ggsn-ipv6-address`

Context

`config>mobile>profile>radius-group>radius-avp-options>acct-interim-avp-options`

`config>mobile>profile>radius-group>radius-avp-options`

Description

This command includes or excludes the 3GPP GGSN IPv6 address IE in the RADIUS messages sent by the PGW/GGSN.

The no form of this command reverts to the default.

Default

`no exclude-3gpp-ggsn-ipv6-address`

6.2.1.61.37 **exclude-3gpp-ggsn-mcc-mnc**

Syntax

[no] `exclude-3gpp-ggsn-mcc-mnc`

Context

`config>mobile>profile>radius-group>radius-avp-options>acct-interim-avp-options`

`config>mobile>profile>radius-group>radius-avp-options`

Description

This command includes or excludes the 3GPP GGSN MCC and MNC IE in the RADIUS messages sent by the PGW/GGSN.

The no form of this command reverts to the default.

Default

`no exclude-3gpp-ggsn-mcc-mnc`

6.2.1.61.38 **exclude-3gpp-gprs-qos-negotiated-profile**

Syntax

[no] exclude-3gpp-gprs-qos-negotiated-profile

Context

config>mobile>profile>radius-group>radius-avp-options>acct-interim-avp-options

config>mobile>profile>radius-group>radius-avp-options

Description

This command includes or excludes the 3GPP GPRS negotiated Quality of Service profile IE in the RADIUS messages sent by the PGW/GGSN.

The no form of this command reverts to the default.

Default

no exclude-3gpp-gprs-qos-negotiated-profile

6.2.1.61.39 **exclude-3gpp-imeisv**

Syntax

[no] exclude-3gpp-imeisv

Context

config>mobile>profile>radius-group>radius-avp-options>acct-interim-avp-options

config>mobile>profile>radius-group>radius-avp-options

Description

This command includes or excludes the 3GPP IMEI IE in the RADIUS messages sent by the PGW/GGSN.

The no form of this command reverts to the default.

Default

no exclude-3gpp-imeisv

6.2.1.61.40 **exclude-3gpp-imsi**

Syntax

```
[no] exclude-3gpp-imsi
```

Context

```
config>mobile>profile>radius-group>radius-avp-options>acct-interim-avp-options
```

Description

This command includes or excludes the 3GPP IMSI IE in the RADIUS messages sent by the PGW/GGSN.

The no form of this command reverts to the default.

Default

```
no exclude-3gpp-imsi
```

6.2.1.61.41 **exclude-3gpp-imsi**

Syntax

```
exclude-3gpp-imsi [accounting | authentication | both]
```

```
no exclude-3gpp-imsi
```

Context

```
config>mobile>profile>radius-group>radius-avp-options
```

Description

This command includes or excludes the 3GPP IMSI IE in the RADIUS messages sent by the PGW/GGSN.

The no form of this command reverts to the default.

Default

```
no exclude-3gpp-imsi
```

Parameters

accounting

if specified, the AVP is excluded only in Accounting-Request (START/STOP) messages

authentication

if specified, the AVP is excluded only in Access-Request (START/STOP) messages

both

if both keywords are specified, the AVP is excluded in both in Accounting-Request (START/STOP) messages and Access-Request (START/STOP) messages

Default: both

6.2.1.61.42 **exclude-3gpp-imsi-mcc-mnc**

Syntax

```
[no] exclude-3gpp-imsi-mcc-mnc
```

Context

```
config>mobile>profile>radius-group>radius-avp-options>acct-interim-avp-options
```

```
config>mobile>profile>radius-group>radius-avp-options
```

Description

This command includes or excludes the 3GPP IMSI MCC and MNC IE in the RADIUS messages sent by the PGW/GGSN.

The no form of this command reverts to the default.

Default

```
no exclude-3gpp-imsi-mcc-mnc
```

6.2.1.61.43 **exclude-3gpp-ms-timezone**

Syntax

```
[no] exclude-3gpp-ms-timezone
```

Context

```
config>mobile>profile>radius-group>radius-avp-options>acct-interim-avp-options
```

```
config>mobile>profile>radius-group>radius-avp-options
```

Description

This command includes or excludes the 3GPP MS time zone IE in the RADIUS messages sent by the PGW/GGSN.

The no form of this command reverts to the default.

Default

no exclude-3gpp-ms-timezone

6.2.1.61.44 exclude-3gpp-nsapi

Syntax

[no] exclude-3gpp-nsapi

Context

config>mobile>profile>radius-group>radius-avp-options>acct-interim-avp-options

config>mobile>profile>radius-group>radius-avp-options

Description

This command includes or excludes the 3GPP NS API IE in the RADIUS messages sent by the PGW/GGSN.

The no form of this command reverts to the default.

Default

no exclude-3gpp-nsapi

6.2.1.61.45 exclude-3gpp-pdp-type

Syntax

[no] exclude-3gpp-pdp-type

Context

config>mobile>profile>radius-group>radius-avp-options>acct-interim-avp-options

config>mobile>profile>radius-group>radius-avp-options

Description

This command includes or excludes the 3GPP Packet Data Protocol type IE in the RADIUS messages sent by the PGW/GGSN.

The no form of this command reverts to the default.

Default

no exclude-3gpp-pdp-type

6.2.1.61.46 exclude-3gpp-rat-type

Syntax

[no] exclude-3gpp-rat-type

Context

config>mobile>profile>radius-group>radius-avp-options>acct-interim-avp-options

config>mobile>profile>radius-group>radius-avp-options

Description

This command includes or excludes the 3GPP radio access technology IE in the RADIUS messages sent by the PGW/GGSN.

The no form of this command reverts to the default.

Default

no exclude-3gpp-rat-type

6.2.1.61.47 exclude-3gpp-selection-mode

Syntax

[no] exclude-3gpp-selection-mode

Context

config>mobile>profile>radius-group>radius-avp-options>acct-interim-avp-options

config>mobile>profile>radius-group>radius-avp-options

Description

This command includes or excludes the 3GPP selection mode IE in the RADIUS messages sent by the PGW/GGSN.

The no form of this command reverts to the default.

Default

no exclude-3gpp-selection-mode

6.2.1.61.48 exclude-3gpp-session-stop-indicator

Syntax

[no] exclude-3gpp-session-stop-indicator

Context

config>mobile>profile>radius-group>radius-avp-options

Description

This command includes or excludes the 3GPP session stop IE in the RADIUS messages sent by the PGW/GGSN.

The no form of this command reverts to the default.

Default

no exclude-3gpp-session-stop-indicator

6.2.1.61.49 exclude-3gpp-sgsn-address

Syntax

[no] exclude-3gpp-sgsn-address

Context

config>mobile>profile>radius-group>radius-avp-options>acct-interim-avp-options

Description

This command includes or excludes the 3GPP SGSN IPv4 address IE in the RADIUS messages sent by the PGW/GGSN.

The no form of this command reverts to the default.

Default

no exclude-3gpp-sgsn-address

6.2.1.61.50 **exclude-3gpp-sgsn-address**

Syntax

```
exclude-3gpp-sgsn-address [accounting | authentication | both]  
no exclude-3gpp-sgsn-address
```

Context

```
config>mobile>profile>radius-group>radius-avp-options
```

Description

This command includes or excludes the 3GPP SGSN IPv4 address IE in the RADIUS messages sent by the PGW/GGSN.

The no form of this command reverts to the default.

Default

```
no exclude-3gpp-sgsn-address
```

Parameters

accounting

if specified, the AVP is excluded only in Accounting-Request (START/STOP) messages

authentication

if specified, the AVP is excluded only in Access-Request (START/STOP) messages

both

if both keywords are specified, the AVP is excluded in both in Accounting-Request (START/STOP) messages and Access-Request (START/STOP) messages

Default: both

6.2.1.61.51 **exclude-3gpp-sgsn-ipv6-address**

Syntax

```
[no] exclude-3gpp-sgsn-ipv6-address
```

Context

```
config>mobile>profile>radius-group>radius-avp-options>acct-interim-avp-options
```



```
config>mobile>profile>radius-group>radius-avp-options
```

Description

This command includes or excludes the 3GPP SGSN IPv6 address IE in the RADIUS messages sent by the PGW/GGSN.

The no form of this command reverts to the default.

Default

```
no exclude-3gpp-sgsn-ipv6-address
```

6.2.1.61.52 exclude-3gpp-sgsn-mcc-mnc

Syntax

```
[no] exclude-3gpp-sgsn-mcc-mnc
```

Context

```
config>mobile>profile>radius-group>radius-avp-options>acct-interim-avp-options
```

```
config>mobile>profile>radius-group>radius-avp-options
```

Description

This command includes or excludes the 3GPP SGSN MCC and MNC IE in the RADIUS messages sent by the PGW/GGSN.

The no form of this command reverts to the default.

Default

```
no exclude-3gpp-sgsn-mcc-mnc
```

6.2.1.61.53 exclude-3gpp-user-location-info

Syntax

```
[no] exclude-3gpp-user-location-info
```

Context

```
config>mobile>profile>radius-group>radius-avp-options>acct-interim-avp-options
```

```
config>mobile>profile>radius-group>radius-avp-options
```

Description

This command includes or excludes the 3GPP user location IE in the RADIUS messages sent by the PGW/GGSN.

The no form of this command reverts to the default.

Default

no exclude-3gpp-user-location-info

6.2.1.61.54 exclude-acct-authentic

Syntax

```
[no] exclude-acct-authentic
```

Context

```
config>mobile>profile>radius-group>radius-avp-options
```

```
config>mobile>profile>radius-group>radius-avp-options>acct-interim-avp-options
```

Description

This command includes or excludes the account authentication IE in the RADIUS messages sent by the PGW/GGSN.

The no form of this command reverts to the default.

Default

no exclude-acct-authentic

6.2.1.61.55 exclude-acct-delay-time

Syntax

```
[no] exclude-acct-delay-time
```

Context

```
config>mobile>profile>radius-group>radius-avp-options>acct-interim-avp-options
```

```
config>mobile>profile>radius-group>radius-avp-options
```

Description

This command includes or excludes the account delay timer IE in the RADIUS messages sent by the PGW/GGSN.

The no form of this command reverts to the default.

Default

no exclude-acct-delay-time

6.2.1.61.56 exclude-acct-input-gigawords

Syntax

[no] exclude-acct-input-gigawords

Context

config>mobile>profile>radius-group>radius-avp-options>acct-interim-avp-options

config>mobile>profile>radius-group>radius-avp-options

Description

This command includes or excludes the Acct-Input-Gigawords IE in the RADIUS messages sent by the PGW/GGSN.

The no form of this command reverts to the default.

Default

no exclude-acct-input-gigawords

6.2.1.61.57 exclude-acct-input-octets

Syntax

[no] exclude-acct-input-octets

Context

config>mobile>profile>radius-group>radius-avp-options>acct-interim-avp-options

config>mobile>profile>radius-group>radius-avp-options

Description

This command includes or excludes the Acct-Input-Octets IE in the RADIUS messages sent by the PGW/GGSN.

The no form of this command reverts to the default.

Default

no exclude-acct-input-octets

6.2.1.61.58 exclude-acct-input-packets

Syntax

[no] exclude-acct-input-packets

Context

config>mobile>profile>radius-group>radius-avp-options>acct-interim-avp-options

config>mobile>profile>radius-group>radius-avp-options

Description

This command includes or excludes the Acct-Input-Packets IE in the RADIUS messages sent by the PGW/GGSN.

The no form of this command reverts to the default.

Default

no exclude-acct-input-packets

6.2.1.61.59 exclude-acct-output-gigawords

Syntax

[no] exclude-acct-output-gigawords

Context

config>mobile>profile>radius-group>radius-avp-options>acct-interim-avp-options

config>mobile>profile>radius-group>radius-avp-options

Description

This command includes or excludes the Acct-Output-Gigawords IE in the RADIUS messages sent by the PGW/GGSN.

The no form of this command reverts to the default.

Default

no exclude-acct-output-gigawords

6.2.1.61.60 exclude-acct-output-octets

Syntax

```
[no] exclude-acct-output-octets
```

Context

```
config>mobile>profile>radius-group>radius-avp-options>acct-interim-avp-options
```

```
config>mobile>profile>radius-group>radius-avp-options
```

Description

This command includes or excludes the Acct-Output-Octets IE in the RADIUS messages sent by the PGW/GGSN.

The no form of this command reverts to the default.

Default

no exclude-acct-output-octets

6.2.1.61.61 exclude-acct-output-packets

Syntax

```
[no] exclude-acct-output-packets
```

Context

```
config>mobile>profile>radius-group>radius-avp-options>acct-interim-avp-options
```

```
config>mobile>profile>radius-group>radius-avp-options
```

Description

This command includes or excludes the Acct-Output-Packets IE in the RADIUS messages sent by the PGW/GGSN.

The no form of this command reverts to the default.

Default

no exclude-acct-output-packets

6.2.1.61.62 exclude-acct-session-id

Syntax

```
[no] exclude-acct-session-id
```

Context

```
config>mobile>profile>radius-group>radius-avp-options>acct-interim-avp-options
```

```
config>mobile>profile>radius-group>radius-avp-options
```

Description

This command includes or excludes the Acct-Session-Id in the Access-Request message.

Default

no exclude-acct-session-id

6.2.1.61.63 exclude-acct-session-time

Syntax

```
[no] exclude-acct-session-time
```

Context

```
config>mobile>profile>radius-group>radius-avp-options>acct-interim-avp-options
```

```
config>mobile>profile>radius-group>radius-avp-options
```

Description

This command includes or excludes the Acct-Session-Time IE in the RADIUS messages sent by the PGW/GGSN.

The no form of this command reverts to the default.

Default

no exclude-acct-session-time

6.2.1.61.64 exclude-called-station-id

Syntax

[no] exclude-called-station-id

Context

config>mobile>profile>radius-group>radius-avp-options>acct-interim-avp-options

config>mobile>profile>radius-group>radius-avp-options

Description

This command includes or excludes the Called-Station-Id IE in the RADIUS messages sent by the PGW/GGSN.

The no form of this command reverts to the default.

Default

no exclude-called-station-id

6.2.1.61.65 exclude-calling-station-id

Syntax

[no] exclude-calling-station-id

Context

config>mobile>profile>radius-group>radius-avp-options>acct-interim-avp-options

config>mobile>profile>radius-group>radius-avp-options

Description

This command includes or excludes the calling station identifier IE in the RADIUS messages sent by the PGW/GGSN.

The no form of this command reverts to the default.

Default

no exclude-calling-station-id

6.2.1.61.66 exclude-class

Syntax

[no] exclude-class

Context

config>mobile>profile>radius-group>radius-avp-options>acct-interim-avp-options

config>mobile>profile>radius-group>radius-avp-options

Description

This command includes or excludes the Class IE in the RADIUS messages sent by the PGW/GGSN.

The no form of this command reverts to the default.

Default

no exclude-class

6.2.1.61.67 exclude-framed-interface-id

Syntax

[no] exclude-framed-interface-id

Context

config>mobile>profile>radius-group>radius-avp-options>acct-interim-avp-options

config>mobile>profile>radius-group>radius-avp-options

Description

This command includes or excludes the Framed-Interface-Id IE in the RADIUS messages sent by the PGW/GGSN.

The no form of this command reverts to the default.

Default

no exclude-framed-interface-id

6.2.1.61.68 exclude-framed-ip-address

Syntax

[no] exclude-framed-ip-address

Context

config>mobile>profile>radius-group>radius-avp-options>acct-interim-avp-options

config>mobile>profile>radius-group>radius-avp-options

Description

This command includes or excludes the Framed-IP-Address IE in the RADIUS messages sent by the PGW/GGSN.

The no form of this command reverts to the default.

Default

no exclude-framed-ip-address

6.2.1.61.69 exclude-framed-ip-netmask

Syntax

[no] exclude-framed-ip-netmask

Context

config>mobile>profile>radius-group>radius-avp-options

Description

This command includes or excludes the Framed-IP-Netmask IE in the RADIUS messages sent by the PGW/GGSN.

The no form of this command reverts to the default.

Default

no exclude-framed-ip-netmask

6.2.1.61.70 **exclude-framed-ipv6-prefix**

Syntax

```
[no] exclude-framed-ipv6-prefix
```

Context

```
config>mobile>profile>radius-group>radius-avp-options>acct-interim-avp-options
```

```
config>mobile>profile>radius-group>radius-avp-options
```

Description

This command includes or excludes the Framed-IPv6-Prefix IE in the RADIUS messages sent by the PGW/GGSN.

The no form of this command reverts to the default.

Default

```
no exclude-framed-ipv6-prefix
```

6.2.1.61.71 **exclude-framed-protocol**

Syntax

```
[no] exclude-framed-protocol
```

Context

```
config>mobile>profile>radius-group>radius-avp-options>acct-interim-avp-options
```

```
config>mobile>profile>radius-group>radius-avp-options
```

Description

This command includes or excludes framed protocol IE in the RADIUS messages sent by the PGW/GGSN.

The no form of this command reverts to the default.

Default

```
no exclude-framed-protocol
```

6.2.1.61.72 exclude-nas-ip-address

Syntax

[no] exclude-nas-ip-address

Context

config>mobile>profile>radius-group>radius-avp-options>acct-interim-avp-options

Description

This command includes or excludes the NAS-IP-Address IE in the RADIUS messages sent by the PGW/GGSN.

The no form of this command reverts to the default.

Default

no exclude-nas-ip-address

6.2.1.61.73 exclude-nas-ipv6-address

Syntax

[no] exclude-nas-ipv6-address

Context

config>mobile>profile>radius-group>radius-avp-options>acct-interim-avp-options

Description

This command includes or excludes the NAS-IPv6-Address IE in the RADIUS messages sent by the PGW/GGSN.

The no form of this command reverts to the default.

Default

no exclude-nas-ipv6-address

6.2.1.61.74 exclude-nas-port-type

Syntax

[no] exclude-nas-port-type

Context

```
config>mobile>profile>radius-group>radius-avp-options>acct-interim-avp-options  
config>mobile>profile>radius-group>radius-avp-options
```

Description

This command includes or excludes the NAS port type IE in the RADIUS messages sent by the PGW/GGSN.
The no form of this command reverts to the default.

Default

no exclude-nas-port-type

6.2.1.61.75 exclude-service-type

Syntax

```
[no] exclude-service-type
```

Context

```
config>mobile>profile>radius-group>radius-avp-options>acct-interim-avp-options  
config>mobile>profile>radius-group>radius-avp-options
```

Description

This command includes or excludes service type IE in the RADIUS messages sent by the PGW/GGSN.
The no form of this command reverts to the default.

Default

no exclude-service-type

6.2.1.61.76 exclude-user-name

Syntax

```
[no] exclude-user-name
```

Context

```
config>mobile>profile>radius-group>radius-avp-options>acct-interim-avp-options
```

```
config>mobile>profile>radius-group>radius-avp-options
```

Description

This command includes or excludes the user name IE in the RADIUS message sent by the PGW/GGSN or SSG.

The no form of this command reverts to the default.

Default

no exclude-user-name

6.2.1.61.77 include-3gpp-ip-addr-pool-info

Syntax

```
[no] include-3gpp-ip-addr-pool-info
```

Context

```
config>mobile>profile>radius-group>radius-avp-options>acct-interim-avp-options
```

```
config>mobile>profile>radius-group>radius-avp-options
```

Description

This command includes or excludes the 3GPP IP address pool information. This attribute is applicable to the following messages sent from the SMF:

- Access-Request
- Accounting-Request (START/STOP/INTERIM)

The no form of this command reverts to the default.

Default

no include-3gpp-ip-addr-pool-info

6.2.1.61.78 include-3gpp-supported-features

Syntax

```
[no] include-3gpp-supported-features
```

Context

```
config>mobile>profile>radius-group>radius-avp-options
```

Description

This command includes or excludes the 3GPP supported features. This attribute is applicable to the following messages sent from the SMF.

- Access-Request
- Accounting-Request (START)

The no form of this command reverts to the default.

Default

```
no include-3gpp-supported-features
```

6.2.1.61.79 include-event-timestamp

Syntax

```
[no] include-event-timestamp
```

Context

```
config>mobile>profile>radius-group>radius-avp-options>acct-interim-avp-options  
config>mobile>profile>radius-group>radius-avp-options
```

Description

This command includes or excludes the event timestamp IE in the RADIUS messages sent by the TWAG, the 3GPP2 Home Agent or the PGW.

The no form of this command reverts to the default.

Default

```
no include-event-timestamp
```

6.2.1.61.80 include-nas-identifier

Syntax

```
[no] include-nas-identifier
```

Context

```
config>mobile>profile>radius-group>radius-avp-options>acct-interim-avp-options  
config>mobile>profile>radius-group>radius-avp-options
```

Description

This command includes or excludes the NAS identifier IE in the RADIUS messages sent by the PGW/GGSN.
The no form of this command reverts to the default.

Default

no include-nas-identifier

6.2.1.61.81 include-nas-port-id

Syntax

```
[no] include-nas-port-id
```

Context

```
config>mobile>profile>radius-group>radius-avp-options>acct-interim-avp-options  
config>mobile>profile>radius-group>radius-avp-options
```

Description

This command includes or excludes NAS port ID IE in the RADIUS messages sent by the excluded.
The no form of this command reverts to the default.

Default

no include-nas-port-id

6.2.1.61.82 disconnect-ack-avp-options

Syntax

```
disconnect-ack-avp-options
```

Context

```
config>mobile>profile>radius-group>radius-avp-options
```

Description

This command enables the context to configure the RADIUS AVP options profile for Disconnect-Ack messages. The usage is enabled/disabled using the `use-disconnect-ack-avp-options` command.

6.2.1.61.83 `exclude-acct-terminate-cause`

Syntax

[no] `exclude-acct-terminate-cause`

Context

`config>mobile>profile>radius-group>radius-avp-options>disconnect-ack-avp-options`

Description

This command includes or excludes the Acct-Terminate-Cause IE in the RADIUS Disconnect-Ack messages sent by the PGW/GGSN.

The no form of this command reverts to the default.

Default

`no exclude-acct-terminate-cause`

6.2.1.61.84 `exclude-event-timestamp`

Syntax

[no] `exclude-event-timestamp`

Context

`config>mobile>profile>radius-group>radius-avp-options>disconnect-ack-avp-options`

Description

This command includes or excludes the Event-Timestamp IE in the RADIUS Disconnect-Ack messages sent by the PGW/GGSN.

The no form of this command reverts to the default.

Default

`no exclude-event-timestamp`

6.2.1.61.85 exclude-proxy-state

Syntax

[no] exclude-proxy-state

Context

config>mobile>profile>radius-group>radius-avp-options>disconnect-ack-avp-options

Description

This command includes or excludes the Proxy-State IE in the RADIUS Disconnect-Ack messages sent by the PGW/GGSN. This IE can only be included if it was received in the previous message.

The no form of this command reverts to the default.

Default

no exclude-proxy-state

6.2.1.61.86 disconnect-nak-avp-options

Syntax

disconnect-ack-avp-options

Context

config>mobile>profile>radius-group>radius-avp-options

Description

This command enables the context to configure the RADIUS AVP options profile for Disconnect-Nak messages. The usage is enabled/disabled using the use-disconnect-nak-avp-options command.

6.2.1.61.87 exclude-error-cause

Syntax

[no] exclude-error-cause

Context

```
config>mobile>profile>radius-group>radius-avp-options>disconnect-nak-avp-options
```

Description

This command includes or excludes the Error-Cause IE in the RADIUS Disconnect-Nak messages sent by the PGW/GGSN. The Error-Cause attribute may be included in case the system generates a value that can be sent back to the disconnect server.

The no form of this command reverts to the default.

Default

```
no exclude-error-cause
```

6.2.1.61.88 exclude-proxy-state

Syntax

```
[no] exclude-proxy-state
```

Context

```
config>mobile>profile>radius-group>radius-avp-options>disconnect-nak-avp-options
```

Description

This command includes or excludes the Proxy-State IE in the RADIUS Disconnect-Nak messages sent by the PGW/GGSN. This IE can only be included if it was received in the previous message.

The no form of this command reverts to the default.

Default

```
no exclude-proxy-state
```

6.2.1.61.89 exclude-3gpp-allocate-ip-type

Syntax

```
[no] exclude-3gpp-allocate-ip-type
```

Context

```
config>mobile>profile>radius-group>radius-avp-options
```

Description

This command includes or excludes the IP type allocation value in the Access-Request message sent from the PGW. The IP type allocation value indicates whether message is sent for user authentication only and/or for allocation of IPv4 and/or IPv6 addresses.

The no form of this command reverts to the default.

Default

no exclude-3gpp-allocate-ip-type

6.2.1.61.90 include-acct-session-id-in-on-off-messages

Syntax

[no] include-acct-session-id-in-on-off-messages

Context

config>mobile>profile>radius-group>radius-avp-options

Description

This command includes or excludes Acct-Session-Id IE in the RADIUS accounting on/off messages sent by the PGW/GGSN. This setting only takes effect if the accounting-on-off-level command in context config>mobile>pdn>radius is set to apn; if not, it is silently ignored.

The no form of this command reverts to the default.

Default

no include-acct-session-id-in-on-off-messages

6.2.1.61.91 include-alc-sla-prof-str

Syntax

[no] include-alc-sla-prof-str

Context

config>mobile>profile>radius-group>radius-avp-options

Description

This command includes or excludes the ALC SLA profile string IE in the RADIUS Accounting messages sent by the TWAG in open SSID mode.

The no form of this command reverts to the default.

Default

no include-alc-sla-prof-str

6.2.1.61.92 include-alc-subsc-id-str

Syntax

```
[no] include-alc-subsc-id-str
```

Context

```
config>mobile>profile>radius-group>radius-avp-options
```

Description

This command includes or excludes the ALC subscriber ID string IE in the RADIUS Accounting messages sent by the TWAG in open SSID mode.

The no form of this command reverts to the default.

Default

no include-alc-subsc-id-str

6.2.1.61.93 include-alc-subsc-prof-str

Syntax

```
[no] include-alc-subsc-prof-str
```

Context

```
config>mobile>profile>radius-group>radius-avp-options
```

Description

This command includes or excludes the ALC subscriber profile string IE in the RADIUS Accounting messages sent by the TWAG in open SSID mode.

The no form of this command reverts to the default.

Default

no include-alc-subsc-prof-str

6.2.1.61.94 include-called-station-id-apn-in-on-off-messages

Syntax

[no] include-called-station-id-apn-in-on-off-messages

Context

config>mobile>profile>radius-group>radius-avp-options

Description

This command includes or excludes the Called-Station-ID (APN) IE in the RADIUS accounting on/off messages sent by the PGW/GGSN. This setting only takes effect if the accounting-on-off-level command in context config>mobile>pdn>radius is set to apn; if not, it is silently ignored.

The no form of this command reverts to the default.

Default

no include-called-station-id-apn-in-on-off-messages

6.2.1.61.95 use-acct-interim-avp-options

Syntax

[no] use-acct-interim-avp-options

Context

config>mobile>profile>radius-group>radius-avp-options

Description

This command specifies that the interim accounting message content must be constructed as configured in the config>mobile>profile>radius-group>radius-avp-options>acct-interim-avp-options context. This applies not only to the explicitly configured include/exclude commands, but also to the default values of those commands that will take effect.

The no form of this command reverts to the default.

Default

no use-acct-interim-avp-options

6.2.1.61.96 use-disconnect-ack-avp-options

Syntax

[no] use-disconnect-ack-avp-options

Context

config>mobile>profile>radius-group>radius-avp-options

Description

This command specifies that the Disconnect-Ack message content must be constructed as configured in the config>mobile>profile>radius-group>radius-avp-options>disconnect-ack-avp-options context. This applies not only to the explicitly configured include/exclude commands, but also to the default values of those commands that will take effect.

The no form of this command reverts to the default.

Default

no use-disconnect-ack-avp-options

6.2.1.61.97 use-disconnect-nak-avp-options

Syntax

[no] use-disconnect-nak-avp-options

Context

config>mobile>profile>radius-group>radius-avp-options

Description

This command specifies that the Disconnect-Nak message content must be constructed as configured in the config>mobile>profile>radius-group>radius-avp-options>disconnect-nck-avp-options context. This applies not only to the explicitly configured include/exclude commands, but also to the default values of those commands that will take effect.

The no form of this command reverts to the default.

Default

no use-disconnect-nak-avp-options

6.2.1.61.98 fh-session-no-response

Syntax

fh-session-no-response {terminate | continue}

no fh-session-no-response

Context

config>mobile>profile>radius-group

Description

This command configures the failure handling behavior for new sessions that fail to receive a response from the AAA RADIUS server. Before applying the configured behavior, the session attempts to get a response from other peers, if configured.

The no form of this command reverts to the default.

Parameters

terminate

specifies that the session is terminated after a failed response

continue

specifies that the session continues after a failed response

Default: terminate

6.2.1.61.99 fh-time-limit

Syntax

fh-time-limit *time-limit*

no fh-time-limit

Context

config>mobile>profile>radius-group

Description

This command configures the time limit for new PDN sessions to continue when failure handling mechanism has been applied upon no further communication with the AAA RADIUS server. The PDN session is terminated upon reaching the specified time limit.

The no form of this command reverts to the default (disabled) and the PDN session is either terminated based on the configured session timeout under the APN or by the RAN.

Parameters

time-limit

specifies the time, in minutes, that the session continues after no response

Values: 0 to 2880

Default: 0 (disabled)

6.2.1.61.100 fh-volume-limit

Syntax

`fh-volume-limit volume-limit`

`no fh-volume-limit`

Context

`config>mobile>profile>radius-group`

Description

This command configures the volume limit for new PDN sessions to continue when failure handling mechanism has been applied upon no further communication with the AAA RADIUS server. The PDN session is terminated upon reaching the specified volume limit.

The no form of this command reverts to the default (disabled) and the PDN session is either terminated based on the configured session timeout under the APN or by the RAN.

Parameters

volume-limit

specifies the data volume, in kbytes, that the session continues after no response

Values: 0 to 10 000 000

Default: 0 (disabled)

6.2.1.61.101 ignore-acct-response

Syntax

```
ignore-acct-response  
no ignore-acct-response
```

Context

```
config>mobile>profile>radius-group
```

Description

This command configures the PGW/GGSN to not wait for the RADIUS server to send an Accounting-Response. The PGW/GGSN will not support the normal retry and timeout mechanism associated with Accounting messages for this RADIUS group. This configuration is expected to be used in cases where the RADIUS server may not respond to accounting messages at all or may be too busy to respond to all messages. By ignoring (not waiting) for responses, the PGW/GGSN will send a single Accounting-Request for each event and will assume that the message has been received by the server. This command is expected to be useful for duplicate accounting. When set for the primary RADIUS group, the PGW/GGSN will no longer wait for a response before sending Accounting-Request messages to the duplicate servers. The PGW/GGSN will send a single Accounting-Request to each of the RADIUS groups configured for the APN (primary RADIUS group and all the duplicate RADIUS groups), and will assume that the message has been received by each server. When set for a duplicate/secondary RADIUS group (if it is not set for the primary group), it will impact this RADIUS group only.

The no form of this command reverts to the default.



NOTE:

- Enabling the ignore-acct-response command affects the PGW/GGSNs ability to support RADIUS server redundancy because the PGW/GGSN may not detect when the server is down. Therefore this command must not be used if multiple RADIUS servers are defined for the RADIUS group.
- When the ignore-acct-response command is enabled, then
 - The acct Resp is not visible in the call-insight.
 - The stats for accounting resp are not incremented.

Default

```
no ignore-acct-response
```

6.2.1.61.102 server-type

Syntax

server-type *server-type*
no server-type

Context

config>mobile>profile>radius-group

Description

This command configures the RADIUS group to be all authentication servers, accounting servers, or both.

Parameters

server-type
specifies the type of the RADIUS server group

Values	none	the servers cannot be used for authentication or accounting
–	auth	all servers can used for authentication
–	acct	all servers can be used for accounting
–	both	all servers can be used for both authentication and accounting

Default: none

6.2.1.61.103 supported-features

Syntax

supported-features

Context

config>mobile>profile>radius-group

Description

This command enables the context to configure the supported features.

6.2.1.61.104 e-sess-ambr

Syntax

[no] e-sess-ambr

Context

config>mobile>profile>radius-group>supported-features

Description

This command enables the support of the enhanced Session AMBR function.

The no form of this command reverts to the default.

Default

no e-sess-ambr

6.2.1.61.105 transaction-based-load-balancing

Syntax

[no] transaction-based-load-balancing

Context

config>mobile>profile>radius-group

Description

This command configures the PGW/GGSN load balancing behavior for RADIUS authentication and accounting messages. Instead of the default behavior (session-based load balancing), where in the case of a response timeout the messages are retried to the same server until the retry-count is exhausted, in the transaction-based load balancing the server is selected separately for each authentication and accounting request using round-robin distribution. There is no guarantee that all accounting requests related to the

same bearer are sent to the same RADIUS server. Authentication and accounting functions are handled separately from each other. Accounting ON/OFF messages (per APN) are not affected by this command. This setting also affects retries. The RADIUS profile containing the `retry-count` parameter must be linked to the RADIUS group level, in which case it is used for all peers in that group (differing values per server are not supported). The system maintains this counter on group level.

Default

no transaction-based-load-balancing

6.2.1.61.106 via-up-function

Syntax

```
via-up-function apn apn-name all
```

```
via-up-function apn apn-name [up-peer-ip-address ip-address | param-name]
```

```
no via-up-function
```

Context

```
config>mobile>profile>radius-group
```

Description

For CUPS deployments, this command specifies the server group RADIUS traffic to be tunnelled through PFCP-u and routed via a UP function to the RADIUS server. The specific UP peer function can be optionally configured with the `up-peer-ip-address` keyword. If the `up-peer-ip-address` keyword is not configured, the UP peer function is selected from the UP peer list (configured using the `config>mobile>profile>pfcp>up-peer-list` command) that this APN is configured with. If there is more than one UP peer functions configured with this APN, the first matched peer in the configured UP peer list with this APN is selected as the UP peer function to route all RADIUS traffic for this server group.



NOTE:

- To use the `via-up-function` command, there must be at least one peer configured under the RADIUS group.
- When the `up-peer-ip-address` is specified, this peer must also be configured using the `config>mobile>profile>pfcp>up-peer-list` command.
- The APN specified in this command must also be configured under a peer using the `config>mobile>profile>pfcp>up-peer-list` and the `config>mobile>pdn>apn` commands. If the APN is not configured, the default PFCP-u tunnel between the CP and UP functions is not established and the RADIUS traffic for this server group profile fails.

- If the server group profile is configured to use a reserved IP address for the RADIUS client VPRN/loopback address from one of the UP peers configured using the `config>mobile>profile>pfc>up-peer-list>peer>apn` and the `config>mobile>profile>pfc>up-peer-list>peer>network-realm>ip-pool>reserved-ip-list` commands, the `up-peer-ip-address` keyword must be specified with the same UP peer IP address or not be configured. In case the `up-peer-ip-address` keyword is not configured, the system finds the peer configured in the `up-peer-list` by matching both the APN and the reserved IP address configured using the `reserved-ip-list` command.

The no form of this command removes this configuration.

Default

no via-up-function

Parameters

apn-name

specifies the APN to associate with the PFCP-u tunnel on the UP function (it can be the same as the UE APN or a separate APN configured for this RADIUS server group when shared with other UE APNs), up to 80 characters

ip-address | param-name

specifies the IP address of the UP function

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x: [0..FFFF]H
–	–	d: [0..255]D

all

keyword to specify the RADIUS traffic from all UP peers configured with APN (`config>mobile>pdn>apn`) and override (`config>mobile>profile>pfc>up-peer-list>peer>override`)

6.2.1.62 S2b error mapping profile command descriptions

6.2.1.62.1 s2b-error-mapping-profile

Syntax

[no] s2b-error-mapping-profile *profile-name*

Context

config>mobile>profile

Description

This command creates an S2b error mapping profile. The maximum number of entries that can be configured under s2b-error-mapping-profile is 256.

Parameters

profile-name

specifies a profile name, up to 32 characters

6.2.1.62.2 s2b-cause-code

Syntax

[no] s2b-cause-code timer-value *timer-value* *timer-unit*

Context

config>mobile>profile>s2b-error-mapping-profile

Description

This command specifies an S2b cause code for this S2b error mapping profile.

Parameters

s2b-cause-code-value

specifies an S2b cause code

Values: 1 to 255

6.2.1.62.3 backoff-timer

Syntax

```
backoff-timer [timer-value timer-value] [timer-unit timer-unit]  
no backoff-timer
```

Context

```
config>mobile>profile>s2b-error-mapping-profile>s2b-cause-code  
config>mobile>profile>swm-profile>swm-error-mapping-profile>swm-error-code
```

Description

This command enables the context to configure the backoff timer status type of the notify payload. The backoff timer will be sent with any error code notify payload if both the unit and value parameters are configured. The unit and value fields can be configured independently since they have default values.

The no option removes the backoff timer from the existing mapping code.



NOTE:

3GPP allows including the BACKOFF_TIMER Notify payload only when the Private Notify Message Type sent to the UE (using the `swu-notify` command) is set to NETWORK_FAILURE (IKE error code 10500) or to NO_APN_SUBSCRIPTION (IKE error code 9002). Nokia ePDG is not restricted to the 3GPP specification and provides a flexible implementation that allows the backoff-timer transmission along with any Private Notify Message and Error Type. Nokia recommends to follow always the 3GPP guideline.

Parameters

timer-value

configures the value of the backoff timer

Values: 0 to 31

Default: 5



NOTE:

The zero value specifies to the UE that it may retry the authentication procedure to an ePDG from the same PLMN

timer-unit

defines the unit for the *timer-value*

Values	2-s	value is incremented in multiples of 2 seconds
–	30-s	value is incremented in multiples of 30 seconds
–	1-m	value is incremented in multiples of 1 minute
–	10-m	value is incremented in multiples of 10 minutes
–	1-h	value is incremented in multiples of 1 hour
–	10-h	value is incremented in multiples of 10 hours
–	320-h	value is incremented in multiples of 320 hours
–	Deact	value indicates that the timer is deactivated

Default: 1-m



NOTE:

The value deactivated specifies to the UE that it must not retry the authentication procedure to the same ePDG until the UE is switched off or the UICC containing the USIM is removed

6.2.1.62.4 gtpv2-result-code-trigger-ipsec-lockout

Syntax

[no] gtpv2-result-code-trigger-ipsec-lockout

Context

config>mobile>profile>s2b-error-mapping-profile>s2b-cause-code

Description

This command allows a specific GTPv2 error/result code to trigger the UE IPsec client lockout functionality. A valid IPsec lockout functionality must already be enabled under the global config>mobile>pdn>swu-ipsec-lockout command, or under the specific s2b-cause-code configuration in the

config>mobile>profile>s2b-error-mapping-profile>s2b-cause-code>swu-ipsec-lockout
context. The swu-ipsec-lockout function of s2b-cause-code always has priority over the global swu-ipsec-lockout configuration.

The no form of this command reverts to the default.

Default

no gtpv2-result-code-trigger-ipsec-lockout

6.2.1.62.5 swu-notify

Syntax

swu-notify *swu-notify-value* [*description description-string*]
no swu-notify

Context

config>mobile>profile>s2b-error-mapping-profile>s2b-cause-code

Description

This command configures the SWu notify error message type.

Parameters

swu-notify-value

specifies the SWu notify type

Values: 1 to 16383

description-string

configures a description for this S2b error code, up to 80 characters

6.2.1.62.6 swu-send-info

Syntax

swu-send-info [notify]
no swu-send-info

Context

config>mobile>profile>s2b-error-mapping-profile>s2b-cause-code

Description

This command specifies whether an information request will be sent.

Parameters

`notify`

if the `notify` keyword is configured with the `swu-send-info` command, it sends the `notify` value from the "`swu-notify <value>`" configured along with the INFORMATIONAL request with delete payload; if no `notify` keyword is configured with the `swu-send-info` command, only INFORMATIONAL request is sent with delete payload

6.2.1.63 S6b interface profile command descriptions

6.2.1.63.1 s6b-profile

Syntax

```
[no] s6b-profile s6b-profile-name
```

Context

```
config>mobile>profile
```

Description

This command specifies the S6b interface to configure AVP IE options.

The no form of this command removes the profile from the configuration.

Parameters

s6b-profile-name

specifies an S6B interface profile, up to 32 characters

6.2.1.63.2 drmp

Syntax

```
drmp
```

Context

```
config>mobile>profile>s6b-profile
```

Description

This command enables the context to configure DRMP.

6.2.1.63.3 ehrpd-initial-attach

Syntax

```
ehrpd-initial-attach priority-value  
no ehrpd-initial-attach
```

Context

```
config>mobile>profile>s6b-profile>drmp
```

Description

The command configures the DRMP AVP priority value in the case of an eHRPD initial attach call flow processing.

The no form of this command reverts to the default.

Parameters

priority-value

specifies the priority value

Values: 0 to 15

Default: 8

6.2.1.63.4 ehrpd-initiated-detach

Syntax

```
ehrpd-initiated-detach priority-value  
no ehrpd-initiated-detach
```

Context

```
config>mobile>profile>s6b-profile>drmp
```

Description

The command configures the DRMP AVP priority value in the case of an UE/HSGW-initiated detach call flow processing.

The no form of this command reverts to the default.

Parameters

priority-value

specifies the priority value

Values: 0 to 15

Default: 6

6.2.1.63.5 ehrpd-lte-handover

Syntax

ehrpd-lte-handover *priority-value*

no ehrpd-lte-handover

Context

config>mobile>profile>s6b-profile>drmp

Description

The command configures the DRMP AVP priority value in the case of an eHRPD-LTE handover call flow processing.

The no form of this command reverts to the default.

Parameters

priority-value

specifies the priority value

Values: 0 to 15

Default: 6

6.2.1.63.6 ehrpd-wifi-handover

Syntax

ehrpd-wifi-handover *priority-value*

```
no ehrpd-wifi-handover
```

Context

```
config>mobile>profile>s6b-profile>drmp
```

Description

The command configures the DRMP AVP priority value in the case of an eHRPD-WiFi handover call flow processing.

The no form of this command reverts to the default.

Parameters

priority-value

specifies the priority value

Values: 0 to 15

Default: 6

6.2.1.63.7 enable

Syntax

```
[no] enable
```

Context

```
config>mobile>profile>s6b-profile>drmp
```

Description

The command enables or disables the DRMP AVP for the S6b profile.

The no form of this command means that the DRMP AVP will not be sent.

Default

no enable

6.2.1.63.8 eutran-initial-attach

Syntax

```
eutran-initial-attach priority-value
```

```
no eutran-initial-attach
```

Context

```
config>mobile>profile>s6b-profile>drmp
```

Description

The command configures the DRMP AVP priority value in the case of an eUTRAN initial attach handover call flow processing.

The no form of this command reverts to the default.

Parameters

priority-value

specifies the priority value

Values: 0 to 15

Default: 8

6.2.1.63.9 eutran-initiated-detach

Syntax

```
eutran-initial-attach priority-value
```

```
no eutran-initial-attach
```

Context

```
config>mobile>profile>s6b-profile>drmp
```

Description

The command configures the DRMP AVP priority value in the case of an UE-initiated detach, an MME-initiated detach or an HSS-initiated detach call flow processing.

The no form of this command reverts to the default.

Parameters

priority-value

specifies the priority value

Values: 0 to 15

Default: 6

6.2.1.63.10 lte-wifi-handover

Syntax

```
lte-wifi-handover priority-value  
no lte-wifi-handover
```

Context

```
config>mobile>profile>s6b-profile>drmp
```

Description

The command configures the DRMP AVP priority value in the case of an LTE-WiFi call flow processing.
The no form of this command reverts to the default.

Parameters

priority-value

specifies the priority value

Values: 0 to 15

Default: 6

6.2.1.63.11 pgw-initiated-disconnect

Syntax

```
pgw-initiated-disconnect priority-value  
no pgw-initiated-disconnect
```

Context

```
config>mobile>profile>s6b-profile>drmp
```

Description

The command configures the DRMP AVP priority value in the case of a PGW disconnect case which includes AAA session termination or PCRF/OCS-initiated detach call flows.

The no form of this command reverts to the default.

Parameters

priority-value

specifies the priority value

Values: 0 to 15

Default: 6

6.2.1.63.12 reauth-initiated

Syntax

reauth-initiated *priority-value*

no reauth-initiated

Context

config>mobile>profile>s6b-profile>drmp

Description

The command configures the DRMP AVP priority value in the case of an AAA reauthentication-initiated call flow processing.

The no form of this command reverts to the default.

Parameters

priority-value

specifies the priority value

Values: 0 to 15

Default: 8

6.2.1.63.13 wifi-initial-attach

Syntax

wifi-initial-attach *priority-value*

no wifi-initial-attach

Context

config>mobile>profile>s6b-profile>drmp

Description

The command configures the DRMP AVP priority value in the case of a WiFi initial attach call flow processing.

The no form of this command reverts to the default.

Parameters

priority-value

specifies the priority value

Values: 0 to 15

Default: 8

6.2.1.63.14 wifi-initiated-detach

Syntax

```
wifi-initiated-detach priority-value
```

```
no wifi-initiated-detach
```

Context

```
config>mobile>profile>s6b-profile>drmp
```

Description

The command configures the DRMP AVP priority value in the case of an UE/WiFi-initiated detach call flow processing.

The no form of this command reverts to the default.

Parameters

priority-value

specifies the priority value

Values: 0 to 15

Default: 6

6.2.1.63.15 fh-session-no-response

Syntax

```
fh-session-no-response {terminate | continue-emps}
```

```
no fh-session-no-response
```

Context

```
config>mobile>profile>s6b-profile
```

Description

The command specifies the failure handling behavior for new sessions that fail to get a response from the 3GPP AAA server. Before applying the selected behavior, the session will attempt to get a response from a secondary AAA server, if configured.

The no form of this command reverts to the default.

Parameters

terminate

specifies that the session will be terminated after a failed response

continue-emps

specifies that only eMPS sessions will continue after a failed response

Default: terminate

6.2.1.63.16 fh-time-limit

Syntax

```
fh-time-limit time-limit
```

```
no fh-time-limit
```

Context

```
config>mobile>profile>s6b-profile
```

Description

This command specifies the time limit for new PDN sessions that are set up and continue without further communication with the 3GPP AAA server (S6b interface fall back to local configuration parameters) as a result of failure handling (for example, 3GPP AAA no response, and so on). The PDN session is terminated upon reaching the specified time limit.

The no form of this command reverts to the default value (disabled) and the PDN session is either terminated based on the configured session time-out under the APN or by the UTRAN.

Parameters

time-limit

specifies the time, in seconds, that the session will continue after a failed response

Values: 1800 to 4294967295

Default: disabled

6.2.1.63.17 message

Syntax

```
[no] message {aar -i | aar -u}
```

Context

```
config>mobile>profile>s6b-profile
```

Description

This command specifies the message type and enables the context to configure the failure handling options.

Parameters

aar -i

specifies the CCR initial message

aar -u

specifies the CCR update message

6.2.1.63.18 result-code

Syntax

```
result-code result-code-value {terminate | continue-emps}
```

```
no result-code result-code-value
```

Context

```
config>mobile>profile>s6b-profile>message
```

Description

This command specifies the failure handling behavior on a per error code basis.

The no form of this command removes the error-specific failure handling for the specified result code and reverts to the behavior specified in the generic configuration.

**NOTE:**

Specifying a result code overrides the generic configuration.

Default

no result-code

Parameters

result-code-value

specifies the result code value applicable at the command level; the result code values include 3GPP-specified values and some operator custom result codes

**NOTE:**

For all the other values, the generic configuration applies.

Values: 0000 to 9999

terminate

specifies that the PDN session is terminated

continue-emps

specifies that only the eMPS PDN session continues for AAR-I/AAR-U

Default: terminate

6.2.1.63.19 result-code-3xxx

Syntax

```
result-code-3xxx {terminate | continue-emps}
```

```
no result-code-3xxx
```

Context

```
config>mobile>profile>s6b-profile>message
```

Description

This command specifies the failure handling behavior for the 3xxx series of error codes.

The no form of this command reverts to the default.

**NOTE:**

Specifying a result code overrides the generic configuration.

Parameters

terminate

specifies that the PDN session is terminated

continue-emps

specifies that only the eMPS PDN session continues for AAR-I/AAR-U

Default: terminate

6.2.1.63.20 result-code-4xxx

Syntax

```
result-code-4xxx {terminate | continue-emps}
```

```
no result-code-4xxx
```

Context

```
config>mobile>profile>s6b-profile>message
```

Description

This command specifies the failure handling behavior for the 4xxx series of error codes.

The no form of this command reverts to the default.

**NOTE:**

Specifying a result code overrides the generic configuration.

Parameters

terminate

specifies that the PDN session is terminated

continue-emps

specifies that only the eMPS PDN session continues for AAR-I/AAR-U

Default: terminate

6.2.1.63.21 result-code-5xxx

Syntax

```
result-code-5xxx {terminate | continue-emps}  
no result-code-5xxx
```

Context

```
config>mobile>profile>s6b-profile>message
```

Description

This command specifies the failure handling behavior for the 5xxx series of error codes.
The no form of this command reverts to the default.



NOTE:

Specifying a result code overrides the generic configuration.

Parameters

terminate

specifies that the PDN session is terminated

continue-emps

specifies that only the eMPS PDN session continues for AAR-I/AAR-U

Default: terminate

6.2.1.63.22 s6b-avp-options

Syntax

```
s6b-avp-options
```

Context

```
config>mobile>profile>s6b-profile
```

Description

This command enables configuration of AVP IE options for the specified S6b interface profile.

6.2.1.63.23 **exclude-emergency-service**

Syntax

[no] exclude-emergency-service

Context

config>mobile>profile>s6b-profile>s6b-avp-options

Description

The command enables or disables sending the Emergency-Service AVP over S6b, in the case of an initial attachment for emergency PDN connection over GTP-based S2b.

Default

no exclude-emergency-service

6.2.1.63.24 **exclude-idle-timeout**

Syntax

[no] exclude-idle-timeout

Context

config>mobile>profile>s6b-profile>s6b-avp-options

Description

The command enables or disables sending of idle-timeout AVP to AAA provided idle timeout is configured for the APN.

Default

no exclude-idle-timeout

6.2.1.63.25 **exclude-MIP6-agent-info**

Syntax

[no] exclude-MIP6-Agent-Info

Context

```
config>mobile>profile>s6b-profile>s6b-avp-options
```

Description

This command includes or excludes the PGW FQDN/IP address.

Default

no exclude-MIP6-Agent-Info

6.2.1.63.26 exclude-MIP6-feature-vector

Syntax

```
[no] exclude-MIP6-feature-vector
```

Context

```
config>mobile>profile>s6b-profile>s6b-avp-options
```

Description

This command includes or excludes the Mip6-Feature Vector.

Default

no exclude-MIP6-feature-vector

6.2.1.63.27 exclude-rat-type

Syntax

```
[no] exclude-rat-type
```

Context

```
config>mobile>profile>s6b-profile>s6b-avp-options
```

Description

This command enables or disables the radio access technology that is serving the UE in the AAR message sent from the PGW.

Default

no exclude-rat-type

6.2.1.63.28 exclude-supported-feature

Syntax

[no] exclude-supported-feature

Context

config>mobile>profile>s6b-profile>s6b-avp-options

Description

This command enables or disables the list of features supported by the origin host for the lifetime of the Diameter session in the AAR message sent from the PGW.

Default

no exclude-supported-feature

6.2.1.63.29 exclude-visited-network-identifier

Syntax

[no] exclude-visited-network-identifier

Context

config>mobile>profile>s6b-profile>s6b-avp-options

Description

This command enables or disables the PLMN ID in the AAR message sent from the PGW.

Default

no exclude-visited-network-identifier

6.2.1.63.30 include-3gpp-allocate-ip-type

Syntax

[no] include-allocate-ip-type

Context

config>mobile>profile>s6b-profile>s6b-avp-options

Description

This command enables or disables the IP Type Allocation value in the AAR message sent from the PGW. The IP Type Allocation value indicates whether the Access-Request is sent for user authentication only and/or for allocation of IPv4 and/or IPv6 addresses.

Default

no include-allocate-ip-type

6.2.1.63.31 include-3gpp-charging-characteristics

Syntax

[no] include-3gpp-charging-characteristics

Context

config>mobile>profile>s6b-profile>s6b-avp-options

Description

This command includes or exclude 3GPP charging characteristics IE in the S6b messages sent by the PGW/GGSN. For PGW/GGSN this is the Charging-Characteristics received via GTP. For S2a this is the Charging-Characteristics appearing in the PBU.

Default

no include-3gpp-charging-characteristics

6.2.1.63.32 include-3gpp-charging-id

Syntax

[no] include-3gpp-charging-id

Context

config>mobile>profile>s6b-profile>s6b-avp-options

Description

This command includes or exclude 3GPP charging identifier IE in the S6b messages sent by the PGW/GGSN.

Default

no include-3gpp-charging-id

6.2.1.63.33 include-3gpp-ggsn-address

Syntax

[no] include-3gpp-ggsn-address

Context

config>mobile>profile>s6b-profile>s6b-avp-options

Description

This command includes or excludes the 3GPP GGSN IPv4 address IE in the messages sent by the PGW/GGSN. For GGSN, this is the IP address of the GTP control plane (Gn IPv4 address). For PGW, this is the IP address of the GTP control plane (S5 IPv4 address) or PMIP control plane (S2a IPv4 address). This AVP is applicable to AAR message sent from PGW/GGSN.

Default

no include-3gpp-ggsn-address

6.2.1.63.34 include-3gpp-ggsn-ipv6-address

Syntax

[no] include-3gpp-ggsn-ipv6-address

Context

```
config>mobile>profile>s6b-profile>s6b-avp-options
```

Description

This command includes or excludes the 3GPP GGSN IPv6 address IE in the messages sent by the PGW/GGSN. For GGSN, this is the IPv6 address of the GTP control plane (Gn IPv6 address.). For PGW, this is the IPv6 address of the GTP control plane (S5 IPv6 address) or PMIP control plane (S2a IPv6 address). This AVP is applicable to AAR Message sent from PGW/GGSN.

Default

```
no include-3gpp-ggsn-ipv6-address
```

6.2.1.63.35 include-3gpp-ggsn-mcc-mnc

Syntax

```
[no] include-3gpp-ggsn-mcc-mnc
```

Context

```
config>mobile>profile>s6b-profile>s6b-avp-options
```

Description

This command includes or excludes the 3GPP GGSN MCC and MNC IE in the S6b messages sent by the PGW/GGSN.

Default

```
no include-3gpp-ggsn-mcc-mnc
```

6.2.1.63.36 include-3gpp-gprs-negotiated-qos-profile

Syntax

```
[no] include-3gpp-gprs-negotiated-qos-profile
```

Context

```
config>mobile>profile>s6b-profile>s6b-avp-options
```

Description

This command includes or excludes the 3GPP GPRS negotiated QoS-profile IE in the S6b messages sent by the PGW/GGSN. For GGSN/PDP context, it includes the QoS-profile requested as specified in TS24.008 Release 7. For PGW/EPS bearer, the format differs for GBR and non-GBR bearers. For a GBR bearer, the QoS-profile contain the ARP, QCI, UL/DL MBR and UL/DL GBR. For a non-GBR bearer the QoS-profile contains the ARP, QCI and UL/DL APN-AMBR. This attribute is not included for S2a. This AVP is applicable to AAR message sent from the PGW/GGSN.

Default

no include-3gpp-gprs-negotiated-qos-profile

6.2.1.63.37 include-3gpp-imeisv

Syntax

[no] include-3gpp-imeisv

Context

config>mobile>profile>s6b-profile>s6b-avp-options

Description

This command includes or excludes 3GPP IMEI IE in the S6b messages sent by the PGW/GGSN.

Default

no include-3gpp-imeisv

6.2.1.63.38 include-3gpp-imsi

Syntax

[no] include-3gpp-imsi

Context

config>mobile>profile>s6b-profile>s6b-avp-options

Description

This command includes or excludes the 3GPP IMSI IE in the S6b messages sent by the PGW/GGSN.

Default

no include-3gpp-imsi

6.2.1.63.39 include-3gpp-ms-timezone

Syntax

[no] include-3gpp-ms-timezone

Context

config>mobile>profile>s6b-profile>s6b-avp-options

Description

This command includes or excludes the 3GPP MS time zone IE in the S6b messages sent by the PGW/GGSN.

Default

no include-3gpp-ms-timezone

6.2.1.63.40 include-3gpp-nsapi

Syntax

[no] include-3gpp-nsapi

Context

config>mobile>profile>s6b-profile>s6b-avp-options

Description

This command includes or excludes the 3GPP NS API IE in the S6b messages sent by the PGW/GGSN. For GGSN, this is the NSAPI received via GTP. For PGW, this is the EPS Bearer ID received via GTP.

Default

no include-3gpp-nsapi

6.2.1.63.41 include-3gpp-pdp-type

Syntax

[no] include-3gpp-pdp-type

Context

config>mobile>profile>s6b-profile>s6b-avp-options

Description

This command includes or excludes the 3GPP Packet Data Protocol type IE in the S6b messages sent by the PGW/GGSN.

Default

no include-3gpp-pdp-type

6.2.1.63.42 include-3gpp-rat-type

Syntax

[no] include-3gpp-rat-type

Context

config>mobile>profile>s6b-profile>s6b-avp-options

Description

This command includes or excludes the 3GPP radio access technology IE in the S6b messages sent by the PGW/GGSN.

Default

no include-3gpp-rat-type

6.2.1.63.43 include-3gpp-selection-mode

Syntax

[no] include-3gpp-selection-mode

Context

```
config>mobile>profile>s6b-profile>s6b-avp-options
```

Description

This command includes or excludes the 3GPP selection mode IE in the S6b messages sent by the PGW/GGSN. For GGSN it contains the Selection mode for this PDP Context received in the Create PDP Context Request message. For PGW, it contains the Selection mode for this EPS Bearer received in the Create Session Request message. This AVP is applicable to AAR Message sent from PGW/GGSN.

Default

```
no include-3gpp-selection-mode
```

6.2.1.63.44 include-3gpp-sgsn-address

Syntax

```
[no] include-3gpp-sgsn-address
```

Context

```
config>mobile>profile>s6b-profile>s6b-avp-options
```

Description

This command includes or excludes the 3GPP SGSN IPv4 address IE in the messages sent by the PGW/GGSN. For GGSN, this is the IPv4 address of the SGSN. For PGW, this is the IPv4 address of the SGW. For S2a this is the IPv4 address of an HSGW. This AVP is applicable to AAR message sent from the PGW/GGSN.

Default

```
no include-3gpp-sgsn-address
```

6.2.1.63.45 include-3gpp-sgsn-ipv6-address

Syntax

```
[no] include-3gpp-sgsn-ipv6-address
```

Context

```
config>mobile>profile>s6b-profile>s6b-avp-options
```


Description

This command includes or excludes the 3GPP SGSN IPv6 address IE in the messages sent by the PGW/GGSN. For GGSN, this is the IPv6 address of the SGSN. For PGW, this is the IPv6 address of the SGW. For S2a this is the IPv6 address of an HSGW. This AVP is applicable to AAR message sent from the PGW/GGSN.

Default

no include-3gpp-sgsn-ipv6-address

6.2.1.63.46 include-3gpp-sgsn-mcc-mnc

Syntax

[no] include-3gpp-sgsn-mcc-mnc

Context

config>mobile>profile>s6b-profile>s6b-avp-options

Description

This command includes or excludes the 3GPP SGSN MCC and MNC IE extracted from the RAI within the Create PDP Context Request or Update PDP Context Request message. For S2a the MCC and MNC is extracted from the Serving Network. This AVP is applicable to AAR message sent from the PGW/GGSN.

Default

no include-3gpp-sgsn-mcc-mnc

6.2.1.63.47 include-3gpp-user-location-info

Syntax

[no] include-3gpp-user-location-info

Context

config>mobile>profile>s6b-profile>s6b-avp-options

Description

This command includes or excludes the 3GPP user location IE in the S6b messages sent by the PGW/GGSN.

Default

no include-3gpp-user-location-info

6.2.1.63.48 include-auth-session-state**Syntax**

[no] include-auth-session-state

Context

config>mobile>profile>s6b-profile>s6b-avp-options

Description

This command includes or excludes the auth-session-state set to STATE_MAINTAINED. This AVP is sent from the PGW/GGSN.

Default

no include-auth-session-state

6.2.1.63.49 include-framed-ip-address**Syntax**

[no] include-framed-ip-address

Context

config>mobile>profile>s6b-profile>s6b-avp-options

Description

This command includes or excludes the IP address assigned by in the S6b messages sent by the PGW/GGSN.

Default

no include-framed-ip-address

6.2.1.63.50 include-framed-ip-netmask

Syntax

[no] include-framed-ip-netmask

Context

config>mobile>profile>s6b-profile>s6b-avp-options

Description

This command includes or excludes the IP address netmask assigned by in the S6b messages sent by the PGW/GGSN.

Default

no include-framed-ip-netmask

6.2.1.63.51 include-framed-ipv6-prefix

Syntax

[no] include-framed-ipv6-netmask

Context

config>mobile>profile>s6b-profile>s6b-avp-options

Description

This command includes or excludes the IPv6 address prefix assigned by in the S6b messages sent by the PGW/GGSN.

Default

no include-framed-ipv6-netmask

6.2.1.63.52 supported-features

Syntax

[no] supported-features

Context

```
config>mobile>profile>s6b-profile
```

Description

This command enables the context to configure supported features for the S6b profile. For more information about the S6b interface Supported-Features AVP, refer to the *S6b Interface Supported-Features AVP* section, in the *CMG Configuration Guide*.

6.2.1.63.53 clear-m-bit

Syntax

```
[no] clear-m-bit
```

Context

```
config>mobile>profile>s6b-profile>supported features
```

Description

This command specifies the M-Bit setting for the supported features AVP. When this command is enabled, the M-bit is cleared (0) in the supported features AVP.

The no form of this command reverts to the default.

Default

M bit is set

6.2.1.63.54 wlan-pcscf-restoration

Syntax

```
[no] wlan-pcscf-restoration
```

Context

```
config>mobile>profile>s6b-profile>supported features
```

Description

This command causes the system to send the "P-CSCF Restoration for WLAN" feature in the Supported-Features AVP.

The no form of this command reverts to the default.

Default

no wlan-pcscf-restoration

6.2.1.64 SBI error mapping profile command descriptions

6.2.1.64.1 sbi-error-mapping-profile

Syntax

[no] sbi-error-mapping-profile *profile-name*

Context

config>mobile>profile

Description

This command configures the SBI error mapping profile.

The no form of this command removes the specific profile from the configuration.

Parameters

profile-name

specifies an SBI error mapping profile name, up to 32 characters

6.2.1.64.2 http-to-5gsm

Syntax

http-to-5gsm *http-status-code* [application-error *app-error-code*] 5gsm-cause *5gsm-cause-code*

no http-to-5gsm *http-status-code* [application-error *app-error-code*]

Context

config>mobile>profile>sbi-error-mapping-profile

Description

This command configures the mapping between the HTTP status code and the 5GSM cause code.

The no form of this command removes the mapping.

Parameters

http-status-code

specifies the three-digit HTTP status code value

Values: 300 to 599

app-error-code

specifies the application error, up to 36 characters

Standard application errors: INVALID_API | INVALID_MSG_FORMAT |
INVALID_QUERY_PARAM | MANDATORY_QUERY_PARAM_INCORRECT |
OPTIONAL_QUERY_PARAM_INCORRECT | MANDATORY_QUERY_PARAM_MISSING |
MANDATORY_IE_INCORRECT | OPTIONAL_IE_INCORRECT | MANDATORY_IE_MISSING |
UNSPECIFIED_MSG_FAILURE | MODIFICATION_NOT_ALLOWED | SUBSCRIPTION_NOT_FOUND
| RESOURCE_URI_STRUCTURE_NOT_FOUND | INCORRECT_LENGTH | NF_CONGESTION_RISK
| INSUFFICIENT_RESOURCES | UNSPECIFIED_NF_FAILURE | SYSTEM_FAILURE |
USER_UNKNOWN | ERROR_INITIAL_PARAMETERS | ERROR_TRIGGER_EVENT |
ERROR_TRAFFIC_MAPPING_INFO_REJECTED | ERROR_CONFLICTING_REQUEST |
POLICY_CONTEXT_DENIED | any customer-specific application errors

5gsm-cause-code

specifies the 5GSM cause code

Values: 1 to 255

6.2.1.64.3 http-to-gtp

Syntax

```
http-to-gtp http-status-code [application-error app-error-code] gtp-error gtp-cause-code
```

```
no http-to-gtp http-status-code [application-error app-error-code]
```

Context

```
config>mobile>profile>sbi-error-mapping-profile
```

Description

This command configures the mapping between the HTTP status code and the GTP cause code.

The no form of this command removes the mapping.

Parameters

http-status-code

specifies the three-digit HTTP status code value

Values: 300 to 599

app-error-code

specifies the application error, up to 36 characters

Standard application errors: INVALID_API | INVALID_MSG_FORMAT |
INVALID_QUERY_PARAM | MANDATORY_QUERY_PARAM_INCORRECT |
OPTIONAL_QUERY_PARAM_INCORRECT | MANDATORY_QUERY_PARAM_MISSING |
MANDATORY_IE_INCORRECT | OPTIONAL_IE_INCORRECT | MANDATORY_IE_MISSING |
UNSPECIFIED_MSG_FAILURE | MODIFICATION_NOT_ALLOWED | SUBSCRIPTION_NOT_FOUND
| RESOURCE_URI_STRUCTURE_NOT_FOUND | INCORRECT_LENGTH | NF_CONGESTION_RISK
| INSUFFICIENT_RESOURCES | UNSPECIFIED_NF_FAILURE | SYSTEM_FAILURE |
USER_UNKNOWN | ERROR_INITIAL_PARAMETERS | ERROR_TRIGGER_EVENT |
ERROR_TRAFFIC_MAPPING_INFO_REJECTED | ERROR_CONFLICTING_REQUEST |
POLICY_CONTEXT_DENIED | any customer-specific application errors

gtp-cause-code

specifies the 5GSM cause code

Values: 1 to 15 | 64 to 253 | 255

6.2.1.64.4 http-to-http

Syntax

`http-to-http http-code-in [application-error-in app-error-code-in] http-error http-code-out application-error-out app-error-code-out`

`no http-to-http http-code-in [application-error-in app-error-code-in]`

Context

`config>mobile>profile>sbi-error-mapping-profile`

Description

This command configures the mapping between the HTTP cause codes and the application error. The incoming HTTP cause codes or the application error are received from the CHF, PCF or UDM. The outgoing HTTP cause code and application error are used in N11 when SMF rejects the request.

The no form of this command removes the mapping.

Parameters

http-code-in

specifies the three-digit HTTP status code value

Values: 300 to 999

app-error-code-in

specifies the application error, up to 36 characters

http-code-out

specifies the three-digit HTTP status code value

Values: 400 | 403 | 404 | 413 | 415 | 429 | 500 | 503 | 504

app-error-code-out

specifies the application error, up to 36 characters

6.2.1.65 SCTP profile command descriptions

6.2.1.65.1 sctp

Syntax

[no] sctp *profile-name*

Context

config>mobile>profile

Description

This command configures the name of the SCTP policy.

The no form of this command removes the policy name from the configuration.

Parameters

profile-name

specifies an SCTP policy name, up to 32 characters

6.2.1.65.2 adv-receiver-window-credit

Syntax

adv-receiver-window-credit *value*


```
no adv-receiver-window-credit
```

Context

```
config>mobile>profile>sctp
```

Description

This command configures the advertised receiver window credit value of the SCTP protocol.

The no form of this command reverts to the default.

Default

```
no adv-receiver-window-credit
```

Parameters

value

specifies the value of the advertised receiver window credit, in bytes

Values: 65536 to 1048576

Default: 300000

6.2.1.65.3 assoc-max-retrans

Syntax

```
assoc-max-retrans value
```

```
no assoc-max-retrans
```

Context

```
config>mobile>profile>sctp
```

Description

This command configures the maximum number of retransmissions for SCTP associations.

The no form of this command removes the maximum value from the configuration.

Default

```
no assoc-max-retrans
```

Parameters

value

specifies the maximum number of retransmissions, as an integer

Values: 0 to 255

Default: 10

6.2.1.65.4 cookie-life-timer

Syntax

```
cookie-life-timer seconds
```

```
no cookie-life-timer
```

Context

```
config>mobile>profile>sctp
```

Description

This command configures the cookie life timer.

The no form of this command removes the cookie life timer from the configuration.

Default

no cookie-life-timer

Parameters

seconds

specifies the timer for the cookie, in seconds

Values: 5 to 120

Default: 60

6.2.1.65.5 heartbeat-interval

Syntax

```
heartbeat-interval seconds
```

```
no heartbeat-interval
```

Context

config>mobile>profile>sctp

Description

This command configures the heartbeat interval.

The no form of this command removes the heartbeat interval from the configuration.

Default

no heartbeat-interval

Parameters

seconds

specifies the heartbeat interval, in seconds

Values: 1 to 300

Default: 30

6.2.1.65.6 initial-max-retrans

Syntax

initial-max-retrans *value*

no initial-max-retrans

Context

config>mobile>profile>sctp

Description

This command configures the maximum number of retransmissions of SCTP paths allowed.

The no form of this command removes the maximum number of retransmissions from the configuration.

Default

no initial-max-retrans

Parameters

value

specifies the maximum number of retransmissions, as an integer

Values: 0 to 255

Default: 8

6.2.1.65.7 mtu-size

Syntax

mtu-size *bytes*

no mtu-size

Context

config>mobile>profile>sctp

Description

This command configures the MTU size (in bytes) for SCTP streams.

The no form of this command removes the MTU size from the configuration.

Default

no mtu-size

Parameters

bytes

specifies the MTU size, in bytes

Values: 512 to 65535

Default: 1500

6.2.1.65.8 path-max-retrans

Syntax

path-max-retrans *value*

no path-max-retrans

Context

config>mobile>profile>sctp

Description

This command configures the maximum number of retransmissions of SCTP paths.

The no form of this command removes the maximum number of retransmissions of SCTP paths from the configuration.

Default

no path-max-retrans

Parameters

value

specifies the maximum number of retransmissions of SCTP paths, as an integer

Values: 0 to 255

Default: 5

6.2.1.65.9 rto-init-max-timer

Syntax

rto-init-max-timer *timer*

no rto-init-max-timer

Context

config>mobile>profile>sctp

Description

This command configures the RTO initial maximum timer. It represents the maximum RTO value is used when attempting an INIT. This timer is only used to bound the initial setup time, and does not affect the RTO maximum timer (configured using the rto-max-timer command), during data transmission.

The no form of this command resets the timer to the default value.

Default

no rto-init-max-timer

Parameters

timer

specifies the value of the RTO initial maximum timer, in milliseconds

Values: 500 to 100000

Default: 60000

6.2.1.65.10 rto-init-timer

Syntax

```
rto-init-timer timer
```

```
no rto-init-timer
```

Context

```
config>mobile>profile>sctp
```

Description

This command configures the RTO initial timer.

The no form of this command removes the RTO initial timer from the configuration.

Default

```
no rto-init-timer
```

Parameters

timer

specifies the value of the RTO initial timer, in milliseconds

Values: 100 to 100000

Default: 3000

6.2.1.65.11 rto-max-timer

Syntax

```
rto-max-timer timer
```

```
no rto-max-timer
```

Context

```
config>mobile>profile>sctp
```

Description

This command configures the RTO maximum timer.

The no form of this command removes the RTO maximum timer from the configuration.

Default

no rto-max-timer

Parameters

timer

specifies the value of the RTO maximum timer, in milliseconds

Values: 500 to 100000

Default: 60000

6.2.1.65.12 rto-min-timer

Syntax

```
rto-min-timer timer
```

```
no rto-min-timer
```

Context

```
config>mobile>profile>sctp
```

Description

This command configures the RTO minimum timer.

The no form of this command removes the RTO minimum timer from the configuration.

Default

no rto-min-timer

Parameters

timer

specifies the value of the RTO minimum timer, in milliseconds

Values: 50 to 50000

Default: 1000

6.2.1.65.13 sack-frequency

Syntax

sack-frequency *value*

no sack-frequency

Context

config>mobile>profile>sctp

Description

This command configures the frequency of SCTP SACKs allowed on the SCTP connection.

The no form of this command removes the frequency of SCTP SACKs allowed on the SCTP connection from the configuration.

Default

no sack-frequency

Parameters

value

specifies the frequency of SCTP SACKs allowed, as an integer

Values: 1 to 20

Default: 2

6.2.1.65.14 sack-period

Syntax

sack-period *timer*

no sack-period

Context

config>mobile>profile>sctp

Description

This command configures the period for SCTP SACK allowed on the SCTP connection. The no form of this command removes the period for SCTP SACK allowed on the SCTP connection from the configuration.

Default

no sack-period

Parameters

timer

specifies the period for SCTP SACK allowed, in 10-milliseconds

Values: 0 to 500

Default: 200

6.2.1.66 Sd profile command descriptions

6.2.1.66.1 sd

Syntax

[no] sd *profile-name*

Context

config>mobile>profile

Description

This command defines an Sd profile that can be applied on a PDN or APN basis; see the `sd-profile` command. It also supports enabling and disabling the optional features sent in the supported feature AVP. In the case of a Diameter race condition at the PCRF, the configured transaction timer and retry count values on the gateway is used.

Parameters

profile-name

specifies the Sd profile name, up to 32 characters

6.2.1.66.2 pending-transaction-retry-count

Syntax

```
pending-transaction-retry-count value  
no pending-transaction-retry-count
```

Context

```
config>mobile>profile>sd
```

Description

This command is used to specify the pending transaction retry count for the Diameter race condition handling feature. If the PCRF sends back a DIAMETER-PENDING-TRANSACTION response to a request message sent from the gateway, the gateway retries the request after the pending transaction timer expiration. The number of times the same request is re-attempted due to DIAMETER_PENDING_TRANSACTION is configured using this command. For the retries to apply for the Diameter race condition handling feature, command `pending-transaction` in the context `config>mobile>profile>sd>supported-features` must be enabled.

The `no` form of this command reverts to the default.

Default

```
no pending-transaction-retry-count
```

Parameters

value

specifies the number of retries

Values: 0 to 8

Default: 3

6.2.1.66.3 pending-transaction-timer

Syntax

```
pending-transaction-timer msec-value  
no pending-transaction-timer
```

Context

```
config>mobile>profile>sd
```

Description

This command is used to specify the pending transaction timer for the Diameter race condition handling feature. If the PCRF sends back a DIAMETER-PENDING-TRANSACTION response to a request message sent from the gateway, the gateway retries the request after the pending transaction timer expiration. A value of 0 disables the timer. For the timer to apply for the Diameter race condition handling feature, command `pending-transaction` in the context `config>mobile>profile>sd>supported-features` must be enabled.

The no form of this command reverts to the default.

Default

no pending-transaction-timer

Parameters

msec-value

timer value in milliseconds

Values: 0 to 180000

Default: 100

6.2.1.66.4 supported-features

Syntax

supported-features

Context

config>mobile>profile>sd

Description

This command enables the context to configure supported features for the Sd profile.

6.2.1.66.5 cno-uli

Syntax

[no] cno-uli

Context

config>mobile>profile>sd>supported-features

Description

This command facilitates the gateway in populating the CNO-ULI feature bit within the supported features AVP sent to the PCRF. The CNO-ULI feature indicates support for PRA information reporting.

The no form of this command reverts to the default (disabled).

Default

no cno-uli

6.2.1.66.6 pending-transaction

Syntax

[no] pending-transaction

Context

config>mobile>profile>sd>supported-features

Description

This command, when enabled, facilitates the gateway populating the PendingTransaction feature bit within the supported features AVP sent to the PCRF.

The no form of this command reverts to the default.

Default

no pending-transaction

6.2.1.67 SGSN profile command descriptions

6.2.1.67.1 sgsn-group

Syntax

[no] sgsn-group *SGSN-group-id*

Context

config>mobile>profile

Description

This command creates a new SGSN group for network-triggered service restoration.

Parameters

SGSN-group-id

specifies the group name ID

Values: 1 to 65535

6.2.1.67.2 peer

Syntax

[no] peer *ipv4* | *ipv6*

Context

config>mobile>profile>sgsn-group

Description

This command identifies the nodes which can be used for network-triggered service restoration. When a SGSN failure triggers the NTSR procedure, the SGSN group that contains the failed SGSN node, is selected.



NOTE:

When modifying an already referenced SGSN group, a peer IP that exists to another SGSN group under the same S4 referenced point, cannot be added.

Parameters

ipv4 | *ipv6*

specifies the peer IPv4 or IPv6 address

Values

ipv4-address

a.b.c.d

–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces
–	–	x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

6.2.1.68 Signaling rate limit profile command descriptions

6.2.1.68.1 sig-rate-limit-profile

Syntax

[no] sig-rate-limit-profile *profile-name*

Context

config>mobile>profile

Description

This command configures a signaling rate limiting profile.

The no form of this command removes the profile from the configuration.

Parameters

profile-name

specifies the signaling rate limiting profile name, up to 32 characters

6.2.1.68.2 delay-tolerance

Syntax

delay-tolerance *dt-value*

no delay-tolerance

Context

config>mobile>profile>sig-rate-limit-profile

Description

This command specifies the maximum number of seconds a message can be queued.

The no form of this command reverts to the default.

Parameters

dt-value

specifies the delay tolerance value in seconds

Values: 0 to 10

Default: 2

6.2.1.68.3 max-burst-size

Syntax

max-burst-size *size*

no max-burst-size

Context

config>mobile>profile>sig-rate-limit-profile

Description

This command specifies the maximum number of messages that can be sent out within a 50ms interval.

The default value specifies that the number of messages within a 50ms interval cannot exceed the configured value of the max-message-rate command divided by 20.

The no form of this command reverts to the default.

Parameters

size

specifies the burst size

Values: 1 to 25000

Default: 0 (that is, the configured value of the max-message-rate command divided by 20)

6.2.1.68.4 max-message-rate

Syntax

```
max-message-rate max-rate  
no max-message-rate
```

Context

```
config>mobile>profile>sig-rate-limit-profile
```

Description

This command specifies the maximum number of allowed transactions per second on a Diameter interface toward a peer. If the `single-diameter-host` command is enabled, the `max-message-rate` refers to the total rate, that is the maximum number of allowed transactions per second on a Diameter client for all the MG groups. If the `single-diameter-host` is not enabled, the `max-message-rate` is per MG group because each MG group acts as a single diameter client.

The no form of this command reverts to default.

Parameters

max-rate

specifies the maximum number of messages allowed per second

Values: 1 to 25000

Default: 0

6.2.1.68.5 queue-threshold

Syntax

```
queue-threshold low low-value high high-value  
no queue-threshold
```

Context

```
config>mobile>profile>sig-rate-limit-profile
```

Description

This command specifies the high and low thresholds for the queue. The high value indicates the threshold when the number of outstanding messages in the queue reaches that value. Beyond this threshold, it is an

overload condition. The low value indicates the threshold when the number of outstanding messages in the queue reaches that low value.

Parameters

low-value

specifies the low watermark threshold

Values: 10 to 100

Default: 30

high-value

specifies the high watermark threshold

Values: 10 to 100

Default: 80

6.2.1.68.6 throttle-emergency

Syntax

[no] throttle-emergency

Context

config>mobile>profile>sig-rate-limit-profile

Description

This command enables or disables throttling of emergency call signaling. The no form of this command returns throttling of emergency call signaling to the default (disabled).

Default

no throttle-emergency

6.2.1.69 SLAAC host profile command descriptions

6.2.1.69.1 slaac-host

Syntax

[no] slaac-host *profile name*

Context

config>mobile>profile

Description

This command configures an IPv6 SLAAC host profile.

The no form of this command removes the profile from the configuration.

Parameters

profile-name

specifies a profile name, up to 32 characters

6.2.1.69.2 router-advertisements

Syntax

router-advertisements

Context

config>mobile>profile>slaac-host

Description

This command enables the context to configure router advertisements.

6.2.1.69.3 force-unicast-ip

Syntax

[no] force-unicast-ip

Context

config>mobile>profile>slaac-host>router-advertisements

Description

This command configures the router advertisements on this interface to be sent either with destination IP multicast or with destination IP unicast. By default, the router advertisements are sent with destination multicast IP.

Default

no force-unicast-ip

6.2.1.69.4 include-rdnss

Syntax

[no] include-rdnss

Context

config>mobile>profile>slaac-host>router-advertisements

Description

This command specifies the inclusion of the RDNSS option as defined in RFC 6106 in IPv6 Router Advertisements for DNS name resolution of IPv6 SLAAC hosts.

Default

no include-rdnss

6.2.1.69.5 max-rtr-adv-interval

Syntax

max-rtr-adv-interval *seconds*

no max-rtr-adv-interval

Context

config>mobile>profile>slaac-host>router-advertisements

Description

This command specifies the maximum time allowed between sending unsolicited multicast router advertisements as defined in RFC 4861.

Parameters

seconds

specifies the time, in seconds

Values: 4 to 1800

Default: 600

6.2.1.69.6 min-rtr-adv-interval

Syntax

```
min-rtr-adv-interval seconds  
no min-rtr-adv-interval
```

Context

```
config>mobile>profile>slaac-host>router-advertisements
```

Description

This command specifies the minimum time allowed between sending unsolicited multicast router advertisements as defined in RFC 4861.

Parameters

seconds

specifies the time, in seconds

Values: 3 to 1350

Default: 200

6.2.1.69.7 mtu

Syntax

```
mtu bytes  
no mtu
```

Context

```
config>mobile>profile>slaac-host>router-advertisements
```

Description

This command specifies the value to be inserted in the link MTU options of the router advertisements messages.

The no form of this command means that the MTU option in the RA is not sent.

Parameters

bytes

specifies the advertised MTU value, in bytes

Values: 1280 to 9212

Default: 0

6.2.1.69.8 preferred-lifetime

Syntax

preferred-lifetime *seconds*

no preferred-lifetime

Context

config>mobile>profile>slaac-host>router-advertisements

Description

This command specifies the remaining time for this prefix to be preferred, that is the time until deprecation of the prefix.

Parameters

seconds

specifies the time for the prefix to remain preferred, in seconds

Values: 0 to 4294967295

Default: 3600

6.2.1.69.9 rdns-lifetime

Syntax

rdns-lifetime *seconds*

no rdns-lifetime

Context

config>mobile>profile>slaac-host>router-advertisements

Description

This command specifies the maximum time in seconds that the RDNSS address can be used for the sent name resolution.

Parameters

seconds

specifies the time for the prefix to remain preferred, in seconds

Values: 900 to 3600

Default: 3600

6.2.1.69.10 reachable-time

Syntax

reachable-time *milliseconds*

no reachable-time

Context

config>mobile>profile>slaac-host>router-advertisements

Description

This command configures the value to be inserted in the reachable time field in the router advertisement messages sent from this interface.

Parameters

milliseconds

specifies the reachable time, in milliseconds

Values: 0 to 4294967295

Default: 0

6.2.1.69.11 retransmit-time

Syntax

retransmit-time *milliseconds*

no retransmit-time

Context

config>mobile>profile>slaac-host>router-advertisements

Description

This command configures the value to be inserted in the retransmit timer field in the router advertisement messages sent from this interface.

Parameters

milliseconds

specifies the reachable time, in milliseconds

Values: 0 to 1800000

Default: 0

6.2.1.69.12 router-lifetime

Syntax

router-lifetime *seconds*

no router-lifetime

Context

config>mobile>profile>slaac-host>router-advertisements

Description

This command configures the value to be inserted in the router lifetime field in the router advertisement messages sent from this interface.

Parameters

seconds

specifies the reachable time, in seconds

Values: 2700 to 9000

Default: 4500

6.2.1.69.13 valid-lifetime

Syntax

```
valid-lifetime seconds  
no valid-lifetime
```

Context

```
config>mobile>profile>slaac-host>router-advertisements
```

Description

This command specifies the remaining time for this prefix to be valid for on-link determination.

Parameters

seconds

specifies the time for the prefix to remain valid, in seconds

Values: 0 to 4294967295

Default: 86400

6.2.1.70 STa profile command descriptions

6.2.1.70.1 sta-profile

Syntax

```
[no] sta-profile profile-name
```

Context

```
config>mobile>profile
```

Description

This command creates an STa profile.

The no form of this command removes the profile from the configuration.

Parameters

profile-name

specifies an STa profile name, up to 32 characters

6.2.1.70.2 diameter-optional-avps

Syntax

```
diameter-optional-avps
```

Context

```
config>mobile>profile>sta-profile
```

Description

This command enables the context to configure the optional Diameter AVPs.

6.2.1.70.3 exclude-der-flags-avp

Syntax

```
[no] exclude-der-flags-avp
```

Context

```
config>mobile>profile>sta-profile>diameter-optional-avps
```

Description

This command enables or disables sending the DER-FLAGS-AVP AVP to the AAA on STa.

Default

```
no exclude-der-flags-avp
```

6.2.1.71 Subscriber mapping profile command descriptions

6.2.1.71.1 subscriber-mapping

Syntax

```
subscriber-mapping profile-name
```

```
no subscriber-mapping
```

Context

config>mobile>profile

Description

This command enables configuration of a static table of subscribers and IP addresses used for Multi-Path TCP (MP-TCP) Hybrid Access Gateway (HAG) functionality. Configuration of the table takes precedence over the `transit-sub` protocol configuration in the `config>service>vprn>ip-local-pool` context. This table is used to create a context for MP-TCP subscribers when the RADIUS/PCRF is not available.

The `no` form of this command removes the table.

Parameters

profile-name

specifies the profile name, up to 32 characters

6.2.1.71.2 address

Syntax

[no] address *ip-address* subscriber-id *subscriber-string*

Context

config>mobile>profile>subscriber-mapping

Description

This command configures an entry for the subscriber ID mapping table. Each entry includes an IP address and the subscriber ID allocated to the IP address. It is used when statically configuring the IP addresses assigned to MP-TCP subscribers.

The `no` form of this command removes the entry from the table.

Parameters

ip-address

specifies the IP address of the subscriber

Values	ipv4-address:	a.b.c.d
--------	---------------	---------

–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

subscriber-string

specifies any string that identifies the subscriber (IMEI, MSISDN, IMSI, any other subscriber ID the operator assigns to a subscriber), up to 32 characters

6.2.1.72 SWm error mapping profile command descriptions**6.2.1.72.1 swm-error-mapping-profile****Syntax**

[no] swm-error-mapping-profile *profile-name*

Context

config>mobile>profile

Description

This command creates an SWm error mapping profile. The maximum number of entries that can be configured under swm-error-mapping-profile is 256.

Parameters***profile-name***

specifies a profile name, up to 32 characters

6.2.1.72.2 swm-error-code**Syntax**

[no] swm-error-code *swm-error-code-value* [experimental]

Context

```
config>mobile>profile>swm-error-mapping-profile
```

Description

This command specifies an SWm error code for an SWm error mapping profile.

Parameters

swm-error-code-value

specifies a SWm error code

Values: 1000 to 5999

experimental

If the *experimental* keyword is used, the error mapping is for the experimental error code, else by default it is for the standard error code.

6.2.1.72.3 diameter-result-code-trigger-lockout

Syntax

```
[no] diameter-result-code-trigger-lockout
```

Context

```
config>mobile>profile>swm-error-mapping-profile>swm-error-code
```

Description

This command configures a specific Diameter error/result code to trigger the UE IPsec client lockout functionality (it implies that the IPsec lockout functionality is already enabled under the `config>mobile>pdn>swu-ipsec-lockout` command).

The `no` form of this command reverts to the default.

Default

```
no diameter-result-code-trigger-lockout
```

6.2.1.72.4 swu-notify

Syntax

```
swu-notify swu-notify-value [description description-string]
```

no swu-notify

Context

config>mobile>profile>swm-error-mapping-profile>swm-error-code

Description

This command configures the SWu notify error message type.

Parameters

swu-notify-value

specifies the SWu notify type

Values: 1 to 16383

description-string

configures a description for this SWm error code, up to 80 characters

6.2.1.72.5 swu-send-info

Syntax

swu-send-info [notify]

no swu-send-info

Context

config>mobile>profile>swm-error-mapping-profile>swm-error-code

Description

This command specifies whether an information request will be sent.

Parameters

notify

If the *notify* keyword is configured with the *swu-send-info* command, it sends the *notify* value from the "swu-notify <value>" configured along with the INFORMATIONAL request with delete payload. If no *notify* keyword is configured with the *swu-send-info* command, only INFORMATIONAL request is sent with delete payload.

6.2.1.73 SWm profile command descriptions

6.2.1.73.1 swm-profile

Syntax

[no] swm-profile *profile-name*

Context

config>mobile>profile

Description

This command creates an SWm profile that is used to apply different parameters to the SWm reference point.

The no form of this command removes the profile.

Parameters

profile-name

specifies a profile name, up to 32 characters

6.2.1.73.2 swm-avp-options

Syntax

swm-avp-options

Context

config>mobile>profile>swm-profile

Description

This command enables the context to configure a set of SWm AVPs.

6.2.1.73.3 exclude-ue-local-ip-address

Syntax

[no] exclude-ue-local-ip-address

Context

```
config>mobile>profile>swm-profile>swm-avp-options
```

Description

This command excludes the address of the UE as seen by the ePDG. If there is a NAT between the UE and ePDG this is the NAT outside address.

The no form of this command causes the ePDG to include the address of the UE.

Default

```
no exclude-ue-local-ip-address
```

6.2.1.73.4 include-aar-flags

Syntax

```
[no] include-aar-flags
```

Context

```
config>mobile>profile>swm-profile>swm-avp-options
```

Description

This command includes the AAR flags AVP into AAR messages. When this command is configured, the AAR flags AVP contains the WLAN-Location-Info-Request AVP with bit=1.

The no form of this command excludes the AAR flags AVP in AAR messages.

Default

```
no include-aar-flags
```

6.2.1.73.5 include-terminal-information

Syntax

```
[no] include-terminal-information
```

Context

```
config>mobile>profile>swm-profile>swm-avp-options
```

Description

This command causes the ePDG to include the terminal information AVP with the IMEI if learned from the notify payload.

The no form causes the ePDG to exclude this AVP.

Default

no include-terminal-information

6.2.1.74 SWw profile command descriptions

6.2.1.74.1 sww-radius-profile

Syntax

```
[no] sww-radius-profile profile-name
```

Context

```
config>mobile>profile
```

Description

This command specifies the SWw RADIUS profile tied to this SWw reference point interface.

The no form of this command removes the profile name from the configuration.

Parameters

profile-name

specifies a profile name, up to 32 characters

6.2.1.74.2 access-accept-optional-attributes

Syntax

```
access-accept-optional-attributes
```

Context

```
config>mobile>profile>sww-radius-profile
```


Description

This command enables the context to configure the Access - Accept optional attributes.

6.2.1.74.3 include-3gpp-charging-characteristics

Syntax

```
[no] include-3gpp-charging-characteristics
```

Context

```
config>mobile>profile>swr-radius-profile>access-accept-optional-attributes
```

Description

This command includes the 3GPP charging characteristics attribute in the RADIUS Access – Accept message to the Wifi WLAN APs / WLCs.

Default

```
no include-3gpp-charging-characteristics
```

6.2.1.74.4 include-3gpp-ggsn-address

Syntax

```
[no] include-3gpp-ggsn-address
```

Context

```
config>mobile>profile>swr-radius-profile>access-accept-optional-attributes
```

Description

This command includes the 3GPP GGSN address attribute in the RADIUS Access – Accept message to the Wifi WLAN APs / WLCs.

Default

```
no include-3gpp-ggsn-address
```

6.2.1.74.5 include-3gpp-gprs-negotiated-qos-profile

Syntax

```
[no] include-3gpp-gprs-negotiated-qos-profile
```

Context

```
config>mobile>profile>swm-radius-profile>access-accept-optional-attributes
```

Description

This command includes the 3GPP GPRS negotiated QoS profile attribute in the RADIUS Access – Accept message to the Wifi WLAN APs / WLCs.

Default

```
no include-3gpp-gprs-negotiated-qos-profile
```

6.2.1.74.6 include-3gpp-imsi

Syntax

```
[no] include-3gpp-imsi
```

Context

```
config>mobile>profile>swm-radius-profile>access-accept-optional-attributes
```

Description

This command includes the 3GPP IMSI attribute in the RADIUS Access – Accept message to the Wifi WLAN APs / WLCs.

Default

```
no include-3gpp-imsi
```

6.2.1.74.7 include-3gpp-msisdn

Syntax

```
[no] include-3gpp-msisdn
```

Context

```
config>mobile>profile>swm-radius-profile>access-accept-optional-attributes
```

Description

This command includes the 3GPP MSISDN attribute in the RADIUS Access – Accept message to the Wifi WLAN APs / WLCs.

Default

```
no include-3gpp-msisdn
```

6.2.1.74.8 include-3gpp-pdp-type

Syntax

```
[no] include-3gpp-pdp-type
```

Context

```
config>mobile>profile>swm-radius-profile>access-accept-optional-attributes
```

Description

This command includes the 3GPP PDP type attribute in the RADIUS Access – Accept message to the Wifi WLAN APs / WLCs.

Default

```
no include-3gpp-pdp-type
```

6.2.1.74.9 include-alc-mgw-interface-type

Syntax

```
[no] include-alc-mgw-interface-type
```

Context

```
config>mobile>profile>swm-radius-profile>access-accept-optional-attributes
```

Description

This command includes the Alc-Mgw-Interface attribute in the RADIUS Access – Accept message to the Wifi WLAN APs / WLCs.

Default

no include-alc-mgw-interface-type

6.2.1.74.10 include-alc-msisdn

Syntax

[no] include-alc-msisdn

Context

config>mobile>profile>swm-radius-profile>access-accept-optional-attributes

Description

This command includes the Alc-MSISDN attribute in the RADIUS Access – Accept message to the Wifi WLAN APs / WLCs.

Default

no include-alc-msisdn

6.2.1.74.11 include-alc-subsc-id-str

Syntax

[no] include-alc-subsc-id-str

Context

config>mobile>profile>swm-radius-profile>access-accept-optional-attributes

Description

This command includes the Alc-Subsc-Id-Str attribute in the RADIUS Access – Accept message to the Wifi WLAN APs / WLCs.

Default

no include-alc-subsc-id-str

6.2.1.74.12 include-alc-wlan-apn-name

Syntax

[no] include-alc-wlan-apn-name

Context

config>mobile>profile>swm-radius-profile>access-accept-optional-attributes

Description

This command includes the Alc-WLAN-APN-name attribute in the RADIUS Access – Accept message to the Wifi WLAN APs / WLCs.

Default

no include-alc-wlan-apn-name

6.2.1.74.13 acct-triggered-mobility

Syntax

[no] acct-triggered-mobility

Context

config>mobile>profile>swm-radius-profile

Description

This command enables the TWAG to use the RADIUS accounting requests sent by WLAN APs to check if the UEs WLAN location has changed (due to mobility).

The no form of this command reverts to the default.

Default

no acct-triggered-mobility

6.2.1.74.14 disconnect-optional-attributes

Syntax

```
disconnect-optional-attributes
```

Context

```
config>mobile>profile>swm-radius-profile
```

Description

This command enables the context to configure the Disconnect optional attributes.

6.2.1.74.15 exclude-framed-ip-address

Syntax

```
[no] exclude-framed-ip-address
```

Context

```
config>mobile>profile>swm-radius-profile>disconnect-optional-attributes
```

Description

This command excludes the framed IP address attribute in the RADIUS disconnect to the Wifi WLAN APs / WLCs.

Default

```
no exclude-framed-ip-address
```

6.2.1.74.16 exclude-user-name

Syntax

```
[no] exclude-user-name
```

Context

```
config>mobile>profile>swm-radius-profile>disconnect-optional-attributes
```

Description

This command excludes the user name attribute in the RADIUS disconnect to the Wifi WLAN APs / WLCs.

Default

no exclude-user-name

6.2.1.74.17 disconnect-port

Syntax

disconnect-port *port*

no disconnect-port

Context

config>mobile>profile>swm-radius-profile

Description

This command configures the transmission of RADIUS Disconnect message toward the WLAN AP/WLC.

Default

no disconnect-port

Parameters

port

specifies the port

Values: 1 to 65535

6.2.1.74.18 fast-inter-ap-handover

Syntax

[no] fast-inter-ap-handover

Context

config>mobile>profile>swm-radius-profile

Description

This command enables the TWAG to use the RADIUS accounting requests sent by the WLAN APs, to check if the UE's WLAN location has changed (due to mobility).

The no form of this command reverts to the default (disabled).

Default

no fast-inter-ap-handover

6.2.1.74.19 retry-count

Syntax

retry-count *value*

no retry-count

Context

config>mobile>profile>swm-radius-profile

Description

This command sets a limit to the number of times the system tries to send a RADIUS Disconnect message.

Parameters

value

specifies the number of attempts

Values: 1 to 8

Default: 4

6.2.1.74.20 retry-timeout

Syntax

retry-timeout *value*

no retry-timeout

Context

config>mobile>profile>swm-radius-profile

Description

This command sets the interval in seconds between RADIUS Disconnect retries toward the WLAN AP/WLC.

Parameters

value

specifies the retry time, in seconds

Values: 3 to 20

Default: 4

6.2.1.74.21 sww-tunnel-profile

Syntax

```
[no] sww-tunnel-profile profile-name
```

Context

```
config>mobile>profile
```

Description

This command specifies the SWw tunnel profile tied to this SWw reference point interface.

The no form of this command removes the profile name from the configuration.

Parameters

profile-name

specifies a profile name, up to 32 characters

6.2.1.74.22 ap-mac-discovery

Syntax

```
[no] ap-mac-discovery
```

Context

```
config>mobile>profile>sww-tunnel-profile
```

Description

This command enables sending the ARP over GRE to verify the WLAN AP MAC@ which must be enabled when Soft-GRE and RADIUS Accounting-Interim location trigger updates are enabled to the AAA server when the TWAG detects a change in the UE's WLAN AP location (due to mobility).

The no form of this command reverts to the default.

Default

no ap-mac-discovery

6.2.1.74.23 default-access-type

Syntax

default-access-type [portal | eap]

no default-access-type

Context

config>mobile>profile>swt-tunnel-profile

Description

This command specifies the default fallback access type if none of the qtag / SSID entries match the SWt tunnel profile.

Parameters

portal

specifies the closed SSID as the fallback access type

eap

specifies the open SSID as the fallback access type

Default: portal

6.2.1.74.24 default-apn

Syntax

default-apn *apn-name* [portal | eap]

no default-apn [portal | eap]

Context

config>mobile>profile>swt-tunnel-profile

Description

This command specifies the default APN to use when no other match exist.

Parameters

apn-name

specifies the APN, up to 80 characters

portal

specifies that the portal authentication (open SSID) access is used

eap

specifies that the EAP authentication (closed SSID) access is used

Default: *portal*

6.2.1.74.25 dhcpv4-server

Syntax

dhcpv4-server *profile-name*

no dhcpv4-server

Context

config>mobile>profile>swt-tunnel-profile

Description

This command specifies the dhcp-server profile tied with to this SWt reference point name.

The no form of this command removes the profile name from the configuration.

Parameters

profile-name

specifies a profile name, up to 32 characters

6.2.1.74.26 qtag

Syntax

```
qtag qtag-value apn apn-name [portal | eap] [ssid ssid-name]  
no qtag qtag-value
```

Context

```
config>mobile>profile>swt-tunnel-profile
```

Description

This command specifies the authentication method to use for a defined VLAN-id. Additionally, APN and SSID names must be specified. For portal authentication (open SSID), the APN name needs to be configured to map the APN defined on the MG TWAG that holds the associated configuration. The SSID name may be configured in the case where IPv6 only UE's allow the SSID name string to be included in the accounting messages. For EAP authentication (closed SSID), the APN name can be used in cases where it is not received via AAA / STa.



NOTE:

The same VLAN-id cannot be configured for both open SSID (portal) and closed SSID (EAP) authentication.

Default

```
no qtag
```

Parameters

qtag-value

specifies the VLAN ID

Values: 0 to 4095

apn-name

specifies the APN which the VLAN-id will be mapped to, up to 80 characters

portal

keyword that specifies that portal authentication (open SSID) access is used

eap

keyword that specifies that EAP authentication (closed SSID) access is used

Default: portal

ssid-name

specifies the SSID name string (if configured), up to 32 characters

6.2.1.74.27 slaac-host

Syntax

```
[no] slaac-host profile name  
no slaac-host
```

Context

```
config>mobile>profile>swt-tunnel-profile
```

Description

This command specifies the IPv6 SLAAC host profile tied to this SWt reference point name.

The no form of this command removes the profile name from the configuration.

Parameters

profile-name

specifies a profile name, up to 32 characters

6.2.1.74.28 ssid

Syntax

```
ssid ssid-value apn apn-name [portal | eap]  
no ssid ssid-value
```

Context

```
config>mobile>profile>swt-tunnel-profile
```

Description

This command specifies (when no VLAN-id is used) the authentication method (EAP or portal) applied to the SSID-names. Additionally, the APN name may need to be specified. For portal authentication (open SSID), the APN name needs to be configured for the APNs defined on the MG TWAG that holds the associated configuration. For EAP authentication (closed SSID) the APN name can be used in cases where it is not received via AAA / STa.

**NOTE:**

The same VLAN-id cannot be configured for both open SSID (portal) and closed SSID (EAP) authentication.

Parameters***ssid-value***

specifies the SSID, up to 32 characters

apn-name

specifies the APN that the SSID will be mapped to, up to 80 characters

portal

keyword that specifies that portal authentication (open SSID) access is used

eap

keyword that specifies that EAP authentication (closed SSID) access is used

Default: portal

6.2.1.74.29 ue-cache-time**Syntax**

ue-cache-time *seconds*

no ue-cache-time

Context

config>mobile>profile>swt-tunnel-profile

Description

This command specifies the time in seconds that a UE session is cached on the TWAG in case the AAA EAP-Success answer does not include a session time. If session time is included in the AAA EAP-Success answer, the UE cache time is ignored.

Parameters***seconds***

specifies the time that a UE session is cached

Values: 0 to 28800

Default: 600

6.2.1.75 T6c profile command descriptions

6.2.1.75.1 t6c-profile

Syntax

[no] t6c-profile *profile-name*

Context

config>mobile>profile

Description

This command creates a T6c profile. It defines the optional parameters to be included in the responses. The no form of this command removes the specified profile name from the configuration.

Parameters

profile-name

specifies a profile name, up to 32 characters

6.2.1.75.2 include-apn

Syntax

[no] include-apn

Context

config>mobile>profile>t6c-profile

Description

This command includes the APN in the T6c response message. The no form of this command reverts to the default.

Default

no include-apn

6.2.1.75.3 include-ip

Syntax

[no] include-ip

Context

config>mobile>profile>t6c-profile

Description

This command includes the IP address of the UE in the T6c response message.

The no form of this command reverts to the default.

Default

no include-ip

6.2.1.75.4 include-rat-type

Syntax

[no] include-rat-type

Context

config>mobile>profile>t6c-profile

Description

This command includes the RAT type in the T6c response message.

The no form of this command reverts to the default.

Default

no include-rat-type

6.2.1.76 Tracking and location area identity profile commands descriptions

6.2.1.76.1 tai-lai-list

Syntax

```
[no] tai-lai-list tai-lai-list-name
```

Context

```
config>mobile>profile
```

Description

This command creates a new profile list of TAI and LAI. The list can contain TAI, LAI, TAI and LAI ranges. The no form of this command removes the specified list from the configuration.

Parameters

tai-lai-list-name

specifies the name of the TAI or LAI profile list, up to 32 characters

6.2.1.76.2 lai

Syntax

```
[no] lai mcc mcc-value mnc mnc-value lac lac-value
```

```
[no] lai mcc mcc-value mnc mnc-value lac-range-start lac-range-start value lac-range-end  
lac-range-end value
```

Context

```
config>mobile>profile>tai-lai-list
```

Description

This command adds a new LAI list to the specified TAI LAI list profile. The no form of this command removes the specified list from the configuration.

Parameters

mcc-value

specifies the MCC value

Values: 000 to 999

mnc-value

specifies the MNC value

Values: 00 to 999

lac-value

specifies the LAC value

Values: 4 digit hexadecimal number

lac-range-start value

specifies the start LAC number

Values: 4 digit hexadecimal number

lac-range-end value

specifies the end LAC number

Values: 4 digit hexadecimal number

6.2.1.76.3 tai

Syntax

```
[no] tai mcc mcc-value mnc mnc-value tac tac-value
```

```
[no] tai mcc mcc-value mnc mnc-value tac-range-start tac-range-start value tac-range-end  
tac-range-end value
```

Context

```
config>mobile>profile>tai-lai-list
```

Description

This command adds a new TAI or TAC range to the specified TAI LAI list profile.

The no form of this command removes the specified entry from the list.

Parameters

mcc-value

specifies the MCC value of the TAI

Values: 000 to 999

mnc-value

specifies the MNC value of the TAI

Values: 00 to 999

tac-value

specifies the TAC value of the TAI

Values: 6 digit hexadecimal number

tac-range-start value

specifies the starting value of the TAC range

Values: 6 digit hexadecimal number

tac-range-end value

specifies the end value of the TAC range

Values: 6 digit hexadecimal number

6.2.1.77 SDM-UDM profile command descriptions

6.2.1.77.1 udm-sdm-profile

Syntax

```
[no] udm-sdm-profile profile-name
```

Context

```
config>mobile>profile
```

Description

This command configures a profile for the Nudm_SubscriberDataManagement service.

The no form of this command removes the specified profile from the configuration.

Parameters

profile-name

specifies a profile name, up to 32 characters

6.2.1.77.2 fh-session-no-response

Syntax

```
fh-session-no-response {terminate | continue | retry-and-terminate | retry-and-continue} [block]
```

```
no fh-session-no-response
```

Context

```
config>mobile>profile>udm-sdm-profile
config>mobile>profile>udm-sdm-profile>create-subscription
config>mobile>profile>udm-sdm-profile>get-sm-subscription-data
config>mobile>profile>udm-uecm-profile
config>mobile>profile>udm-uecm-profile>register-smf
```

Description

This command defines the failure handling behavior if there is no response from the UDM. The optional `block` parameter blocks the UDM when there is no response.

The no form of this command reverts to the default behavior (continue).

Parameters

terminate

specifies that the PDU session is terminated

continue

specifies that the PDU session continues; that is, the rejection from the UDM is ignored

retry-and-terminate

specifies to reattempt the service operation with another UDM peer; if a maximum number of peer failovers, configured using the `max-peer-failover` command, is reached, the PDU session is terminated

retry-and-continue

specifies to reattempt the service operation with another UDM peer; if a maximum number of peer failovers, configured using the `max-peer-failover` command, is reached, the PDU session is not terminated

Default: `continue`

block

blocks the UDM

6.2.1.77.3 block-timer

Syntax

```
block-timer interval
no block-timer
```

Context

config> mobile > profile > udm-sdm-profile

config> mobile > profile > udm-uecm-profile

Description

This command specifies the period that the UDM remains blocked. When the UDM is blocked, the SMF does not send requests to the UDM.

Parameters

interval

specifies the block duration, in minutes

Values: 0 to 1440

Default: 0 (the UDM is immediately unblocked)

6.2.1.77.4 continue-timer

Syntax

continue-timer *interval* action *action*

no continue-timer

Context

config> mobile > profile > udm-sdm-profile

config> mobile > profile > udm-uecm-profile

Description

This command enables the continue timer, which starts when the failure action is set to continue. It defines the duration of the continue timer and the action taken when the timer expires.

Parameters

interval

specifies the continue timer duration, in minutes

Values: 0 to 1440

Default: 0 (continue timer is disabled)

action

specifies the continue action

Values:

release

The PDU session is released and the UE is requested to reactivate the session

retry

The failed service request is retried

6.2.1.77.5 create-subscription

Syntax

```
create-subscription
```

Context

```
config>mobile>profile>udm-sdm-profile
```

Description

This command enables the context to configure an SMF subscription to the UDM to get notifications about changes in session management subscription data.

6.2.1.77.6 http-status-code

Syntax

```
http-status-code status-code-value [application-error app-error-code] action  
{terminate | continue | retry-and-terminate | retry-and-continue} [block]  
no http-status-code status-code-value [application-error app-error-code]
```

Context

```
config>mobile>profile>udm-sdm-profile>create-subscription
```

```
config>mobile>profile>udm-uecm-profile>register-smf
```

Description

This command specifies the failure handling behavior. The defined action is applied, if the response from the UDM matches with the configured HTTP status code. An optional application error can be included and the response must have a matching application error. The optional `block` parameter blocks the UDM when the UDM rejects the request.

The `no` form of this command removes the error handling for the specified HTTP status code and reverts to the default.

Parameters

status-code-value

specifies the three-digit status code value

Values: 300 to 599

app-error-code

specifies the application error, up to 36 characters

action

specifies the action for failure handling

Values:

terminate

The PDU session is terminated

continue

The PDU session is not terminated, that is the rejection from the UDM is ignored

retry-and-terminate

The service operation is reattempted with another UDM peer. If a maximum number of peer failovers configured using the `max-peer-failover` command has been reached, the PDU session is terminated.

retry-and-continue

The service operation is reattempted with another UDM peer. If a maximum number of peer failovers configured using the `max-peer-failover` command has been reached, the PDU session is not terminated.

Default: terminate

block

blocks the UDM

6.2.1.77.7 retry-count

Syntax

retry-count *retry-count*

no retry-count

Context

config>mobile>profile>udm-sdm-profile>create-subscription

Description

This command configures the maximum number of retransmissions when the SMF creates a subscription to the UDM to get notifications about changes in session management subscription data.

The no form of this command reverts to the default value.

Parameters

retry-count

specifies the maximum number of retransmissions

Values: 0 to 8

Default: 3

6.2.1.77.8 transaction-timer

Syntax

transaction-timer *value*

no transaction-timer

Context

config>mobile>profile>udm-sdm-profile>create-subscription

Description

This command configures the time that the SMF waits for a response from the UDM when the SMF creates a subscription to the UDM to get notifications about changes in session management subscription data.

The no form of this command reverts to the default value.

Parameters

value

specifies the transaction timer, in milliseconds

Values: 1000 to 20000 (increments of 100)

Default: 5000

6.2.1.77.9 get-sm-subscription-data

Syntax

get-sm-subscription-data

Context

```
config>mobile>profile>udm-sdm-profile
```

Description

This command enables the context to configure the SMF to retrieve the session management subscription data from the UDM.

6.2.1.77.10 http-status-code

Syntax

```
http-status-code status-code-value [application-error app-error-code] [access-type {5g  
| eps | epdg}] action {terminate | continue | retry-and-terminate | retry-and-continue}  
[block]
```

```
no http-status-code status-code-value [application-error app-error-code] [access-type  
{5g | eps | epdg}]
```

Context

```
config>mobile>profile>udm-sdm-profile>get-sm-subscription-data
```

Description

This command specifies the failure handling behavior. The defined action is applied, if the response from the UDM matches with the configured HTTP status code. An optional application error can be included and the response must have a matching application error. If the rejection from the UDM is ignored, the default 5G QoS profile must be defined in the related APN/DNN and the PDU session continues in 5G access. If the default 5G QoS profile is not defined and the UDM did not provide a 5G QoS, the PDU session is terminated. The optional access type is used to define different failure handling. If the access type is not defined, the failure handling is used for those access types which have not access-specific failure handling.

The no form of this command removes the error handling for the specified HTTP status code and reverts to the default.

Parameters

status-code-value

specifies the three-digit status code value

Values: 300 to 599

app-error-code

specifies the application error, up to 36 characters

access-type

specifies the access type

Values: 5g | eps | epdg

action

specifies the action for failure handling

Values:

terminate

The PDU session is terminated

continue

The PDU session is not terminated, that is the rejection from the UDM is ignored

retry-and-terminate

The service operation is reattempted with another UDM peer. If a maximum number of peer failovers configured using the `max-peer-failover` command has been reached, the PDU session is terminated.

retry-and-continue

The service operation is reattempted with another UDM peer. If a maximum number of peer failovers configured using the `max-peer-failover` command has been reached, the PDU session is not terminated.

Default: terminate

block

blocks the UDM

6.2.1.77.11 exclude-plmn-id

Syntax

[no] `exclude-plmn-id`

Context

`config>mobile>profile>udm-sdm-profile>get-sm-subscription-data`

Description

This command controls whether the serving PLMN ID is included when the SMF retrieves the session management subscription data from the UDM.



NOTE:

Applies only in roaming.

Default

exclude-plmn-id

6.2.1.77.12 **retry-count**

Syntax

retry-count *retry-count*

no retry-count

Context

config>mobile>profile>udm-sdm-profile>get-sm-subscription-data

Description

This command configures the maximum number of retransmissions when the SMF retrieves the session management subscription data.

The no form of this command reverts to the default value.

Parameters

retry-count

specifies the maximum number of retransmissions

Values: 0 to 8

Default: 3

6.2.1.77.13 **send-sd-with-reserved-value**

Syntax

send-sd-with-reserved-value {enable | disable}

no send-sd-with-reserved-value

Context

config>mobile>profile>udm-sdm-profile>get-sm-subscription-data

Description

This command enables the CR 1151 of 3GPP TS 29.503. When enabled, the SMF uses the FFFFFFFF value in the SD when the subscription data is retrieved from the UDM for the S-NSSAI without an SD. The reserved SD value indicates to the UDM that it returns only subscription data for the S-NSSAI with matching SST without an SD. The CR is disabled by default, and the UDM returns the subscription data for all the subscribed S-NSSAIs which have the matching SST.

Parameters

enable

enables the use of reserved SD value when subscription data is retrieved from the UDM

disable

disables the use of reserved SD value when subscription data is retrieved from the UDM

Default: disable

6.2.1.77.14 transaction-timer

Syntax

transaction-timer *value*

no transaction-timer

Context

config>mobile>profile>udm-sdm-profile>get-sm-subscription-data

Description

This command configures the time that the SMF waits for a response from the UDM when the SMF retrieves the session management subscription data.

The no form of this command reverts to the default value.

Parameters

value

specifies the transaction timer, in milliseconds

Values: 1000 to 20000 (increments of 100)

Default: 5000

6.2.1.77.15 http-status-code

Syntax

```
http-status-code status-code-value [application-error app-error-code] action  
{terminate | continue | retry-and-terminate | retry-and-continue} [block]  
  
no http-status-code status-code-value [application-error app-error-code]
```

Context

```
config>mobile>profile>udm-sdm-profile  
config>mobile>profile>udm-uecm-profile
```

Description

This command specifies the failure handling behavior. The defined action is applied, if the response from the UDM matches with the configured HTTP status code. An optional application error can be included and the response must have a matching application error.

The no form of this command removes the error handling for the specified HTTP status code and reverts to the default.

Parameters

status-code-value

specifies the three-digit status code value

Values: 300 to 599

app-error-code

specifies the application error, up to 36 characters

action

specifies the action for failure handling

Values:

terminate

The PDU session is terminated

continue

The PDU session is not terminated, that is the rejection from the UDM is ignored

retry-and-terminate

The service operation is reattempted with another UDM peer. If a maximum number of peer failovers configured using the `max-peer-failover` command has been reached, the PDU session is terminated.

retry-and-continue

The service operation is reattempted with another UDM peer. If a maximum number of peer failovers configured using the `max-peer-failover` command has been reached, the PDU session is not terminated.

Default: terminate

block

blocks the UDM

6.2.1.77.16 max-peer-failover

Syntax

`max-peer-failover` *value*

`no max-peer-failover`

Context

`config>mobile>profile>udm-sdm-profile`

`config>mobile>profile>udm-uecm-profile`

Description

This command configures the number of times that a peer can be reselected after exhausting the retries toward the currently selected peer. If a non-zero value is configured, the SMF failovers to another UDM if the original UDM is not responding. The maximum number of failovers is limited to the number of UDM peers number of retransmissions when the SMF retrieves the session management subscription data.

The no form of this command reverts to the default value.

Parameters

value

specifies the maximum number of UDM peers

Values: 0 to 3

Default: 2

6.2.1.77.17 retry-count

Syntax

`retry-count` *retry-count*

`no retry-count`

Context

config>mobile>profile>udm-sdm-profile

config>mobile>profile>udm-uecm-profile

Description

This command configures the maximum number of retransmissions.

The no form of this command reverts to the default value.

Parameters

retry-count

specifies the maximum number of retransmissions

Values: 0 to 8

Default: 3

6.2.1.77.18 sbi-error-mapping-profile

Syntax

sbi-error-mapping-profile *profile-name*

no sbi-error-mapping-profile

Context

config>mobile>profile>udm-sdm-profile

Description

This command configures the SBI error mapping profile for rejections when the SMF uses the udm-sdm service.

The no form of this command removes the configuration.

Parameters

profile-name

specifies the SBI error mapping profile name, up to 32 characters

6.2.1.77.19 transaction-timer

Syntax

```
transaction-timer value  
no transaction-timer
```

Context

```
config>mobile>profile>udm-sdm-profile  
config>mobile>profile>udm-uecm-profile
```

Description

This command configures the time that the SMF waits for a response from the UDM.
The no form of this command reverts to the default value.

Parameters

value

specifies the transaction timer, in milliseconds
Values: 1000 to 20000 (increments of 100)
Default: 5000

6.2.1.78 UDM-UECM profile command descriptions

6.2.1.78.1 udm-uecm-profile

Syntax

```
[no] udm-uecm-profile profile-name
```

Context

```
config>mobile>profile
```

Description

This command configures a profile for the Nudm_UEContextManagement service.
The no form of this command removes the specified profile from the configuration.

Parameters

profile-name

specifies a profile name, up to 32 characters

6.2.1.78.2 register-smf

Syntax

```
register-smf
```

Context

```
config>mobile>profile>udm-uecm-profile
```

Description

This command enables the context to configure the SMF to register to the UDM.

6.2.1.78.3 retry-count

Syntax

```
retry-count retry-count
```

```
no retry-count
```

Context

```
config>mobile>profile>udm-uecm-profile>register-smf
```

Description

This command configures the maximum number of retransmissions when the SMF registers to the UDM.

The no form of this command reverts to the default value.

Parameters

retry-count

specifies the maximum number of retransmissions

Values: 0 to 8

Default: 3

6.2.1.78.4 transaction-timer

Syntax

```
transaction-timer value  
no transaction-timer
```

Context

```
config>mobile>profile>udm-uecm-profile>register-smf
```

Description

This command configures the time that the SMF waits for a response from the UDM when the SMF registers to the UDM.

The no form of this command reverts to the default value.

Parameters

value

specifies the transaction timer, in milliseconds

Values: 1000 to 20000 (increments of 100)

Default: 5000

6.2.1.79 sbi-error-mapping-profile

Syntax

```
sbi-error-mapping-profile profile-name  
no sbi-error-mapping-profile
```

Context

```
config>mobile>profile>udm-uecm-profile
```

Description

This command configures the SBI error mapping profile for rejections when SMF is using udm-uecm service.

The no form of this command removes the configuration.

Parameters

profile-name

specifies the SBI error mapping profile name, up to 32 characters

6.2.1.80 UDP tunnel profile command descriptions

6.2.1.80.1 udp-tunnel-group

Syntax

[no] udp-tunnel-group *tunnel-group-name*

Context

config>mobile>profile

Description

This command enables the context for a set of UDP tunnels used for non-IP PDN tunneling.

The no form of this command removes the specified profile.

Parameters

tunnel-group-name

specifies a tunnel group name, up to 32 characters

6.2.1.80.2 peer

Syntax

[no] peer *ipv4* | *ipv6*

Context

config>mobile>profile>udp-tunnel-group

Description

This command associates an application server IP address with the tunnel group. There may be more than one peer address. If more than one tunnel is listed, the system must use round-robin tunnel selection.

The no form of this command removes the peer from the configuration.

Parameters

ipv4 | *ipv6*
the remote tunnel endpoint IPv4 or IPv6 address

Values	ipv4-address	a.b.c.d (host bits must be 0)
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

6.2.1.80.3 destination-port

Syntax

destination-port *port*
no destination-port

Context

config>mobile>profile>udp-tunnel-group>peer

Description

This command sets the destination UDP port for the tunnel.
The no form of this command reverts to the default.



NOTE:

The default value (0) is a reserved port and cannot be used as the actual destination port.

Parameters

port
specifies the destination UDP port

Values: 1 to 65535

6.2.1.81 UMTS QoS policy profile commands descriptions

6.2.1.81.1 umts-qos-policy

Syntax

`umts-qos-policy umts-qos-policy`

`no umts-qos-policy`

Context

`config>mobile>profile`

Description

This command configures the name of the UMTS QoS policy.

The no form of this command removes the policy name from the configuration.

Parameters

umts-qos-policy

specifies a UMTS QoS policy name, up to 32 characters

6.2.1.81.2 background

Syntax

`background qci qci`

`no background`

Context

`config>mobile>profile>umts-qos-policy`

Description

This command sets the QCI value corresponding to the background traffic class.

Parameters

qci

specifies the QCI value to be assigned as an 8-bit integer

Values: 1 to 9

Default: 9



NOTE:

Unique QCI mappings are required per traffic class, which means that the same QCI value cannot be assigned to two different traffic classes.

6.2.1.81.3 conversational

Syntax

```
conversational ssd ssd qci qci [delay delay]
```

```
no conversational ssd ssd [delay delay]
```

Context

```
config>mobile>profile>umts-qos-policy
```

Description

This command sets the QCI value corresponding to the conversational traffic class. The source statistics descriptor value is also used to determine the QCI value.

Parameters

ssd

specifies the source statistics descriptor may be set to speech or unknown

Values: speech | unknown

qci

specifies the QCI value when the value for Source Statistics Descriptor (SSD) is speech and traffic class is conversational; specifies as an 8-bit integer

Values: 1 to 9



NOTE:

Unique QCI mappings are required per traffic class, which means that the same QCI value cannot be assigned to two different traffic classes.

delay

specifies the UMTS Transfer Delay is higher or lower than 150ms

Values: higher | lower

6.2.1.81.4 interactive

Syntax

```
interactive priority priority qci qci [signaling-indication]
```

```
no interactive priority priority [signaling-indication]
```

Context

```
config>mobile>profile>umts-qos-policy
```

Description

This command sets the qci value corresponding to the interactive traffic class. The UMTS Traffic Handling Priority must be specified as well. For completeness with table E.1 from TS 23.401 Annex E, we would need to distinguish between QCI 5 and QCI 6 using the UMTS Signaling-Indication.

Parameters

priority

specifies the UMTS Traffic Handling Priority

Values: 1 to 3

qci

specifies the QCI value to be assigned as an 8-bit integer

Values: 1 to 9

**NOTE:**

Unique QCI mappings are required per traffic class, which means that the same QCI value cannot be assigned to two different traffic classes.

signaling-indication

used if the UMTS signaling indication equals to yes. If it is absent, the signaling indication equals to no.

6.2.1.81.5 operator-qci

Syntax

```
operator-qci operator-qci traffic-class traffic-class [traffic-priority traffic-priority]
[signaling-indication signaling-indication] [ssd ssd] [delay delay]

no operator-qci operator-qci
```

Context

```
config>mobile>profile>umts-qos-policy
```

Description

This command configures a mapping of operator-specific QCI values to pre-release8 QoS parameters, including traffic class, traffic priority, signaling indication, SSD, and delay. The mapping is configured as part of the UMTS QoS policy, and can then be applied at the PDN or APN level. To allow multiple operator-specific QCI values to map to a specific traffic class, multiple entries are supported for each traffic class within the UMTS QoS policy profile; for example:

- QCI=128, TRAFFICCLASS=BACKGROUND, THP=NA, SSD=SPEECH, SI=TRUE, DELAY=4000
- QCI=129, TRAFFICCLASS=BACKGROUND, THP=NA, SSD=UNKNOWN, SI=FALSE, DELAY=2000
- QCI=150, TRAFFICCLASS=BACKGROUND, THP=NA, SSD=NA, SI=NA, DELAY=4000
- QCI=200, TRAFFICCLASS=INTERACTIVE, THP=HIGH, SSD=SPEECH, SI=NA, DELAY=4000

The no form of this command removes the mapping for the specified operator QCI.



NOTE:

- Different combinations of traffic class settings are supported, however *operator-qci* and *traffic-class* must not be set to NA.
- This functionality is supported for PCRF-specified QCI only.
- The operator-specific QoS profile is only used if the operator configures it; there is no default configuration.
- This functionality only considers the primary PDP QCI/ARP parameters and not the secondary PDP parameters; the gateway rejects secondary PDP Create Requests.
- Both the gateway and the PCRF must be configured to be aware of, and synchronized with, the operator-specific QCI values used.
- Show commands display the configuration and statistics for the operator QCI for the bearer that is applicable; for example, `show>mobile>profile>umts-qos-policy` and `show>mobile>pdn>bearer-context>bearer`.

Parameters

operator-qci

specifies the operator-specific QCI value for this profile

Values: 128 to 254

traffic-class

specifies the traffic class option for this profile

Values: conversational | streaming | interactive | background

traffic-priority

specifies the traffic priority option this profile

Values: high | normal | low | not-available

Default: not-available

signaling-indication

specifies the signaling indication optimize option for this profile

Values: true | false | not-available

Default: not-available

ssd

specifies the SSD option for this profile

Values: speech | unknown | not-available

Default: not-available

delay

specifies the delay for this profile, in milliseconds

Values: 0 to 4000

Default: 0

6.2.1.81.6 streaming

Syntax

```
streaming qci qci
```

```
no streaming
```

Context

```
config>mobile>profile>umts-qos-policy
```

Description

This command sets the qci value corresponding to the streaming traffic class.

Parameters

qci

specifies the QCI value to be assigned as an 8-bit integer

Values: 1 to 9

Default: 4



NOTE:

Unique QCI mappings are required per traffic class, which means that the same QCI value cannot be assigned to two different traffic classes.

6.2.1.82 UP security policy profile commands descriptions

6.2.1.82.1 up-security-policy

Syntax

[no] up-security-policy *profile-name*

Context

config>mobile>profile

Description

This command defines UP security policy profile.

The no form of this command removes the specified UP security policy profile from the configuration.

Parameters

profile-name

specifies the name of the UP security policy profile, up to 32 characters

6.2.1.82.2 override-udm-policy

Syntax

```
override-udm-policy  
no override-udm-policy
```

Context

```
config>mobile>profile>up-security-policy
```

Description

This command enables overriding of the UP security policy received from the UDM.
The no form of this command reverts to the default value.

6.2.1.82.3 up-confidentiality

Syntax

```
up-confidentiality {required | preferred | not-needed}  
no up-confidentiality
```

Context

```
config>mobile>profile>up-security-policy
```

Description

This command defines the UP confidentiality protection policy. The configuration is used to define the policy when it is not defined by the UDM, or when overriding the UDM policy has been enabled.

The no form of this command reverts to the default value.

Parameters

required

UP confidentiality protection is required

preferred

UP confidentiality protection is preferred

not-needed

UP confidentiality protection is not needed

Default: not-needed

6.2.1.82.4 up-integrity

Syntax

```
up-integrity {required | preferred | not-needed}  
no up-integrity
```

Context

```
config>mobile>profile>up-security-policy
```

Description

This command defines the UP integrity protection policy. The configuration is used to define the policy when it is not defined by the UDM, or when overriding the UDM policy has been enabled.

The no form of this command reverts to the default value.

Parameters

required

UP integrity protection is required

preferred

UP integrity protection is preferred

not-needed

UP integrity protection is not needed

Default: not-needed

6.2.1.83 V-SMF profile command descriptions

6.2.1.83.1 vsmf-profile

Syntax

```
[no] vsmf-profile profile-name
```

Context

```
config>mobile>profile
```

Description

This command creates a V-SMF profile that defines the V-SMF policies.

The no form of this command removes the V-SMF profile from the configuration.

Parameters

profile-name

specifies the V-SMF profile name, up to 32 characters

6.2.1.83.2 allow-always-on

Syntax

[no] allow-always-on

Context

config>mobile>profile>vsmf-profile

Description

This command enables always-on PDU sessions in the V-SMF.

The no form of this command reverts to the default value.

Default

no allow-always-on

6.2.1.83.3 up-confidentiality

Syntax

up-confidentiality {required | preferred | not-needed}

no up-confidentiality

Context

config>mobile>profile>vsmf-profile

Description

This command configures the UP confidentiality protection policy in the V-SMF.

The no form of this command reverts to the default value.

Default

no up-confidentiality

Parameters

required

UP confidentiality protection is required

preferred

UP confidentiality protection is preferred

not - needed

UP confidentiality protection is not needed

Default: not-needed

6.2.1.83.4 up-integrity

Syntax

up-integrity {required | preferred | not-needed}

no up-integrity

Context

config>mobile>profile>vsmf-profile

Description

This command defines the UP integrity protection policy in the V-SMF.

The no form of this command reverts to the default value.

Default

no up-integrity

Parameters

required

UP integrity protection is required

preferred

UP integrity protection is preferred

not -needed

UP integrity protection is not needed

Default: not-needed

6.2.1.83.5 vplmn-qos

Syntax

`vplmn-qos profile-name`

`no vplmn-qos`

Context

`config>mobile>profile>vsmf-profile`

Description

This command configures the V-PLMN QoS of the V-SMF. The profile used to define the V-PLMN QoS is configured in the `config>mobile>profile>qos-5g-profile` context.

The `no` form of this command removes the V-PLMN QoS and the V-SMF accepts any QoS.

Parameters

profile-name

specifies the 5G QoS profile name, up to 32 characters

6.2.1.84 WLAN emergency profile command descriptions

6.2.1.84.1 wlan-emergency

Syntax

`[no] wlan-emergency profile-name`

Context

`config>mobile>profile`

Description

This command creates multiple (up to 10) ECD profiles for the WLAN emergency functionality. ECD data includes APN name, QoS (QCI and ARP), and APN-AMBR PGW Identity. A WLAN emergency profile (ECD) has default values for the mandatory items but the operator can configure these items with different values.

The no form of this command deletes the entire ECD profile.

Default

no wlan-emergency

Parameters

profile-name

specifies the profile name, up to 32 characters

6.2.1.84.2 ecd-table

Syntax

[no] ecd-table

Context

config>mobile>profile>wlan-emergency

Description

This command creates multiple (up to 10) WLAN emergency profiles for the WLAN emergency functionality. ECD data includes the APN name, the QoS (QCI and ARP), the APN-AMBR PDN gateway identity, and the 3GPP HO emergency PDN gateway identity.

The no form of this command reverts to the default values of the entire ECD table. A WLAN emergency profile ECD table contains default values for the mandatory items; different values for these items can be configured.

Default

no ecd-table



NOTE:

When the ePDG is configured to support 5GS capable UEs and interaction with 5GS network, the ECD table entries “Emergency PDN GW identity (pgw-id)”, “3GPP HO Emergency PDN GW identity (3gpp-ho-

emergency-pgw-id)” and “Emergency Access Point Name (apn)” must point or be configured with the PGW-C/SMF gateways, and not only the PGW.

6.2.1.84.3 3gpp-ho-emergency-pgw-id

Syntax

```
3gpp-ho-emergency-pgw-id list-name  
no 3gpp-ho-emergency-pgw-id
```

Context

```
config>mobile>profile>wlan-emergency>ecd-table
```

Description

This command configures a 3GPP HO emergency PGW ID. It is recommended that the PGW selection candidate list contains only one IP address since the 3GPP has not yet determined whether the FQDN resolves to one or more IP addresses. If the FQDN is configured and the DNS provides more than one IP address, the ePDG uses the first IP address received by the DNS.

Default

```
no 3gpp-ho-emergency-pgw-id
```

Parameters

list-name

specifies a valid and existing PGW selection candidate list entry specified in the pgw-selection-candidate-list entry under the config>mobile>profile>list>pgw-selection-candidate-list context, up to 32 characters

6.2.1.84.4 apn

Syntax

```
apn apn-name  
no apn
```

Context

```
config>mobile>profile>wlan-emergency>ecd-table
```

Description

This command specifies the emergency APN name (APN-NI) to be used toward PGW. Only NI value is needed to be configured.

Parameters

apn-name

specifies the emergency APN name, up to 80 character; the wildcard "*" is not allowed

Default: sos

6.2.1.84.5 apn-ambr-dl

Syntax

```
apn-ambr-dl apn-ambr-dl
```

```
no apn-ambr-dl
```

Context

```
config>mobile>profile>wlan-emergency>ecd-table
```

Description

This command specifies the APN AMBR downlink rate in kb/s.

Parameters

apn-ambr-dl

specifies the APN AMBR downlink rate in kb/s

Values: 1 to 300000

Default: 4096

6.2.1.84.6 apn-ambr-ul

Syntax

```
apn-ambr-ul apn-ambr-ul
```

```
no apn-ambr-ul
```

Context

config>mobile>profile>wlan-emergency>ecd-table

Description

This command specifies the APN AMBR uplink rate in kb/s.

Parameters

apn-ambr-ul

specifies the APN AMBR uplink rate in kb/s

Values: 1 to 300000

Default: 4096

6.2.1.84.7 pgw-id

Syntax

pgw-id *list-name*

no pgw-id

Context

config>mobile>profile>wlan-emergency>ecd-table

Description

This command assigns a PGW selection candidate list. This PGW selection candidate list needs to be already available under the config>mobile>profile>list>pgw-selection-candidate-list context. The created PGW selection candidate list can have up to 20 entries of IPv4 or IPv6 or FQDN with their dedicated weight and priority. In case all the IPs or FQDNs of the pgw-selection-candidate-list have failed to reply, the ePDG could make use of the "Static PGW Selection Table".

Parameters

list-name

specifies a valid and existing pgw-selection-candidate-list entry under the config>mobile>profile>list>pgw-selection-candidate-list context, up to 32 characters

6.2.1.84.8 qos

Syntax

qos

Context

config>mobile>profile>wlan-emergency>ecd-table

Description

This command configures the QoS options for the wlan emergency profile.

6.2.1.84.9 arp

Syntax

arp *arp-value*

Context

config>mobile>profile>wlan-emergency>ecd-table>qos

Description

This command specifies the ARP value for the emergency call routing toward PGW.

Parameters

arp-value

specifies the ARP value for the emergency call routing toward the PGW

Values: 1 to 15

Default: 1

6.2.1.84.10 qci

Syntax

qci *qci*

Context

```
config>mobile>profile>wlan-emergency>ecd-table>qos
```

Description

This command specifies the QCI value for the emergency call routing toward the PGW.

Parameters

qci

specifies the QCI value for the emergency call routing toward the PGW

Values: 5 to 9

Default: 5

6.2.1.84.11 emergency-call-numbers-ikev2

Syntax

```
emergency-call-numbers-ikev2
```

Context

```
config>mobile>profile>wlan-emergency
```

Description

This command enables the context to locally configure emergency call numbers that can be provided to the UE, via IKEv2, depending on the UE local IP address. These numbers can be used by the UE to identify emergency call numbers dialed by the user. It offers the capability to import, from an external FTP server or CF card, database files that contain IP ranges with their associated country codes. This command also allows configuring emergency call digits with their associated category per country code. When the ePDG finds a UE local IP address that matches the configured IP range/country code, it provides the UE with the emergency call number digits defined under the country-code command.

6.2.1.84.12 country-code

Syntax

```
country-code country-code
```

```
no country-code {country-code | all}
```

Context

```
config>mobile>profile>wlan-emergency>emergency-call-numbers-ikev2
```

Description

This command configures the country codes and their associated emergency call digits with their category.

The no form of this command removes the specific country code entry (with its configured emergency digits and category) or removes all the configured country codes at once.

Default

no country-code

Parameters

country-code

Specifies a two letter country code (for example, US = USA, IT = Italy, CN = China). The two letter country code naming is compliant with the ISO 3166-1 alpha-2 specification.

6.2.1.84.13 digits

Syntax

```
digits digits [category category [category...(up to 5 max)]]
```

```
no digits digits
```

Context

```
config>mobile>profile>wlan-emergency>emergency-call-numbers-ikev2>country-code
```

Description

This command configures the local emergency numbers entry, for example, 112 and the associated category.

Default

no digits

Parameters

digits

Allowed values (up to 6 digits): 0-9, *, #, A, B, C

category

Specifies the type of category for the digits entry. More than one type of categories, with a maximum up to five, can be entered for the same digits entry.

Values: police | ambulance | fire-brigade | marine-guard | mountain-rescue

Default: police

6.2.1.84.14 ip-range-country-file-format

Syntax

```
ip-range-country-file-format dbip url-dbip url-string
```

```
ip-range-country-file-format geolite url-geolite-v4 url-string url-geolite-cc url-string [url-geolite-v6 url-string]
```

```
ip-range-country-file-format ip2loc url-ip2loc-v4 url-string [url-ip2loc-v6 url-string]
```

```
no ip-range-country-file-format
```

Context

```
config>mobile>profile>wlan-emergency>emergency-call-numbers-ikev2
```

Description

This command configures the type of source database for importing the IP ranges per country code file. The ePDG allows databases from three different sources, all of which are in a csv file format. These are:

- ip2location.com (<http://download.ip2location.com/lite/>)
- maxmind.com (<https://dev.maxmind.com/geoip/geoip2/geolite2/>)
- db-ip.com (<https://db-ip.com/db/download/country>)

All three different types of databases are free and distributed under the Creative Commons Attribution-ShareAlike 4.0 International License (legal code). The source file has to be downloaded to a local ftp server or copied to a CF card and then mounted onto the ePDG. The file is then imported to the ePDG, via the relevant ePDG CLI, and used accordingly. If there is already a valid files imported and used in the ePDG and the operator imports a new one, the old database is used until the new one is imported, processed and successfully activated. When a file is successfully imported and activated, the emergency call number functionality becomes active. The completion status of a file importation can be one of the following:

- EMPTY
- IN_PROGRESS
- LOADED
- LOADED_ERRORS

- **LOADING_FAILED**

To determine the reason for a loading failure or a loading failure with errors, use `thedebug>mobile>profile>emergency-db-errors csv-file-name` command.

The no form of this command removes any assigned file and deactivates the emergency call numbers function.

Parameters

dbip

specifies the entry of a csv file for url-dbip, which is downloaded from <https://db-ip.com/db/download/country>

url-dbip

specifies the IPv4/IPv6/country code file to be imported to the ePDG (mandatory parameter)

Values: a valid FTP or CF URL

geolite

specifies a keyword that allows to enter a csv file for url-geolite-v4, url-geolite-cc and url-geolite-v6, which is downloaded from <https://dev.maxmind.com/geoip/geoip2/geolite2/>

url-geolite-v4

specifies the IPv4 file to be imported to the ePDG (mandatory parameter)

Values: a valid FTP or CF URL

url-geolite-cc

specifies the country codes file to be imported to the ePDG (mandatory parameter)

Values: a valid FTP or CF URL

url-geolite-v6

specifies the IPv6 file to be imported to the ePDG (optional parameter)

Values: a valid FTP or CF URL

ip2loc

specifies a keyword that allows to enter a csv file for url-ip2loc-v4 and url-ip2loc-v6 which is downloaded from <http://download.ip2location.com/lite/>

url-ip2loc-v4

specifies the IPv4 file to be imported to the ePDG (mandatory parameter)

Values: a valid FTP or CF URL

url-ip2loc-v6

specifies the IPv6 file to be imported to the ePDG (mandatory parameter)

Values: a valid FTP or CF URL

6.2.1.84.15 **idr-emergency-apn-name-list**

Syntax

```
idr-emergency-apn-name-list apn-list  
no idr-emergency-apn-name-list
```

Context

```
config>mobile>profile>wlan-emergency
```

Description

This command configures a list that contains a number of APN entries that the ePDG considers as an emergency indication on top of the EMERGENCY IDr value specified by the 3GPP. The list to be applied, must be available under the config>mobile>profile>list>apn-list command.

The no form of this command removes any configured APN list.

Parameters

apn-list

specifies an APN list name, up to 32 characters

6.2.1.84.16 **unauthenticated-uiccless-support**

Syntax

```
[no] unauthenticated-uiccless-support
```

Context

```
config>mobile>profile>wlan-emergency
```

Description

This command enables the ePDG to support an emergency call attach service request for the UEs, or provide their subscriber ID (IMSI) but cannot be authenticated by the network during an initial or a handover attach service request or provide their identity based on the device IMEI (UICC-less).

The no form of this command reverts to the default.

Default

```
no unauthenticated-uiccless-support
```

6.2.1.85 WPS profile command descriptions

6.2.1.85.1 wps-profile

Syntax

[no] wps-profile *profile-name*

Context

config>mobile>profile

Description

This command configures a Wireless Priority Service (WPS) profile or a Multimedia Priority Service (MPS) profile.



NOTE:

The Multimedia Priority Service functionality is applicable only to ePDG. On-the-fly changes in the configuration are applied to existing and new sessions.

The no form of this command removes the profile from the configuration.

Parameters

profile-name

specifies the WPS or the MPS profile name, up to 32 characters

6.2.1.85.2 arp

Syntax

[no] arp *value*

Context

config>mobile>profile>wps-profile

Description

This command configures each ARP value to match as an WPS/MPS session.

The no form of this command removes the ARP value from the configuration.

Parameters

value

specifies the ARP value for WPS/MPS sessions

Values: 1 to 15

6.2.1.85.3 message-priority

Syntax

[no] message-priority

Context

config>mobile>profile>wps-profile>arp

Description

This command includes the MP-flag set and message-priority in the GTP-C, PFCP, or in the SBI header.



NOTE:

The PFCP must be configured with the message-priority value on the CP function either via the config>mobile-gateway>profile>wps-profile>arp>message-priority>all or the config>mobile-gateway>profile>wps-profile>arp>message-priority>pfcpc command for the UP function to identify any session as an eMPS/WPS session.

The no form of this command reverts to the default.

Default

no message-priority

6.2.1.85.4 all

Syntax

all *value*

no all

Context

```
config>mobile>profile>wps-profile>arp>message-priority
```

Description

This command sets the same message-priority value in all protocol headers (GTP-C, PFCP, and SBI message-priority) for the eMPS/WPS session subsequent messaging.

The no form of this command reverts to the default value.

Parameters

value

specifies the priority value in the message-priority header

Values: 0 to 15

Default: 0 (the highest priority value)

6.2.1.85.5 gtpc

Syntax

```
gtpc value
```

```
no gtpc
```

Context

```
config>mobile>profile>wps-profile>arp>message-priority
```

Description

This command configures the message-priority value for the ARP value and sets the same value in the message-priority in the GTPv2-C headers for the WPS session subsequent messaging over GTP-C.



NOTE:

This command and the all command, are applicable to the ePDG. When both the gtpc and the all commands are configured, the gtpc is applied to the ePDG.

The no form of this command removes this configuration.

Parameters

value

specifies the priority value in the message-priority GTPv2-C header

Values: 0 to 15 (0 is the highest priority value)

6.2.1.85.6 pfcpc

Syntax

pfcpc value

no pfcpc

Context

config>mobile>profile>wps-profile>arp>message-priority

Description

This command configures the message-priority value set in the PFCP protocol headers for the eMPS/WPS session subsequent messaging over PFCP. This command overrides the value set using the config>mobile>profile>wps-profile>arp>message-priority>all command for PFCP.



NOTE:

- PFCP must be configured with message-priority value either via the message-priority>all or via the message-priority>pfcpc command on the CP function for the UP function to be able to identify any session as an eMPS/WPS session.
- This command is not applicable to the ePDG.

The no form of this command removes this configuration.

Parameters

value

specifies the priority value set in the message-priority PFCP header

Values: 0 to 15 (0 is the highest priority value)

6.2.1.85.7 sbi

Syntax

sbi value

no sbi

Context

```
config>mobile>profile>wps-profile>arp>message-priority
```

Description

This command configures the message-priority value set in the SBI protocol headers for the eMPS/WPS session subsequent messaging over SBI. This command overrides the value set using the `config>mobile>profile>wps-profile>arp>message-priority>all` command for SBI.



NOTE:

This command is not applicable to the ePDG.

The no form of this command removes this configuration.

Parameters

value

specifies the priority value set in the the 3gpp-sbi-message-priority header

Values: 0 to 31 (0 is the highest priority value)

6.2.1.85.8 up-dscp-marking

Syntax

```
up-dscp-marking
```

Context

```
config>mobile>profile>wps-profile
```

Description

This command configures the DSCP transport marking value for the WPS/MPS sessions on the user plane. When configured, the WPS/MPS session packets on the transport are marked with the DSCP values specified for the UP DSCP marking on 4G and 5G UP uplink and UP downlink on the S1-U, S5-U, S2b-U, SWu-ESP, N3, and SGi/N6 reference points.



NOTE:

This configuration only applies to the WPS profile on the UP function.

6.2.1.85.9 down-link

Syntax

```
down-link qci-policy-name  
no down-link
```

Context

```
config>mobile>profile>wps-profile>up-dscp-marking
```

Description

This command specifies the QCI policy configured in the config>mobile>profile>policy-options>qci-policy context, used for the downlink direction (N3, S1-u, S5-u, SWu-ESP).

The no form of this command reverts to the default value.

Default

```
no down-link
```

Parameters

qci-policy-name

specifies the QCI policy name, up to 32 characters

6.2.1.85.10 up-link

Syntax

```
up-link qci-policy-name  
no up-link
```

Context

```
config>mobile>profile>wps-profile>up-dscp-marking
```

Description

This command specifies the QCI policy configured in the config>mobile>profile>policy-options>qci-policy context, used for the uplink direction (S2b-U, SGI/N6).

The no form of this command reverts to the default value.

Default

no up-link

Parameters

qci-policy-name

specifies the QCI policy name, up to 32 characters

6.2.2 Profile show command descriptions

6.2.2.1 profile

Syntax

profile

Context

show>mobile

Description

This command enables the context to display configured mobile gateway profiles.

6.2.2.2 apn-rate-control

Syntax

apn-rate-control

Context

show>mobile>profile

Description

This command displays the profile for APN rate control.

Output

The following output examples show APN rate control examples.

- *Example output APN rate control*

- *Example output APN rate control with UL policy enforcement*

Example output APN Rate Control

The following output example shows the profile for the APN rate control and *Show mobile profile APN rate control output fields* describes the output fields.

```
# show mobile-gateway profile apn-rate-control
=====
APN Rate Control profile
=====
Profile name           : Apn Rate Profile 1
ul-exception-reports   : Included    time-unit       : Hour
ul-rate-limit          : 10          dl-rate-limit    : 15
ul-rate-control-in-pgw : Disabled
additional-ul-rate-limit-for-exception-data : Disabled
-----
Number of profiles : 1
=====
```

Table 103: Show mobile profile APN rate control output fields

Label	Description
Profile name	The profile name of the APN rate control
UL exception reports	The uplink exception reports for the APN rate control in the CDRs
Time unit	Time unit limit for the APN rate control
UL rate limit	The uplink rate limit of packets sent
DL rate limit	The downlink rate limit of packets sent
UL rate control in PGW	The uplink rate limit control enabling in PGW
Additional UL rate limit for exception data	The additional uplink rate limit for exception data
Number of profiles	The number of profiles being viewed

Example output APN rate control with UL policy enforcement

The following example shows the profile for the APN rate control with UL packet policy enforcement where additional UL rate limit is applied for exception data.

```
# show mobile-gateway profile apn-rate-control
=====
APN Rate Control profile
=====
Profile name           : Apn Rate Profile 1
ul-exception-reports   : Included    time-unit           : Hour
ul-rate-limit          : 10          dl-rate-limit        : 15
ul-rate-control-in-pgw : Enabled
additional-ul-rate-limit-for-exception-data : 5
-----
```

6.2.2.3 call-insight

Syntax

call-insight

Context

show>mobile>profile

Description

This command displays call-insight profiles.

6.2.2.4 all-ues

Syntax

all-ues *profile-name*

Context

show>mobile>profile>call-insight

Description

This command displays call-trace profiles specified by name or all profiles when no name is specified.

Parameters

profile-name

displays information about the specified profile

Output

The following output example shows call trace information for all UEs, and *Show mobile profile call-insight all UEs output fields* describes the fields.

Example output call trace profile

```
# show mobile-gateway profile call-insight all-ues
=====
Call-insight All-UEs profile
=====
Profile name       : prof_all
Description        : none
Live output       : none
Reference points   : none
SBA services      : none
Events            : acctmgr, apm, bmc, dcm, dhcp, ga, gal, gtp, gtpv2, gxc, gxl, ike, l
                  : rdm, mip, msngr, pfcp, pmip, rfl, rol, s6b, sctp, sd, sessmgr, sig
                  : mgr, swm, sww, slaac, rpy, dpy, sta, n3iwf-ngap
Format            : simulated-pcap
Path mgmt ref-points : ga, gn, gp, gx, gxc, rf, ro, s1, s11, s12, s2a, s2b, s4, s5, s6b, s8,
                  : swm, sx-n4, n2
Path mgmt SBA service: none
Size limit        : 1000 MB
Time limit        : 86400 secs
-----
Number of profiles : 1
=====
```

Table 104: Show mobile profile call-insight all UEs output fields

Label	Description
Profile name	The name of the profile of UEs for call-insight
Description	The total number of profiles for call-insight
Live output	The live output name
Live output router	The live output router name
Reference points	The reference points used
Events	The names of entities which can generate certain call trace events to be logged

Table 104: Show mobile profile call-insight all UEs output fields (continued)

Label	Description
Format	The format in which the capture is done
Path mgmt ref-points	The path management reference points
Size limit	The limit on size permitted for the process
Time limit	The limit on time permitted for the process
Number of profiles	The number of profiles being viewed

6.2.2.5 ue

Syntax

ue profile-name

Context

show>mobile>profile>call-insight

Description

This command displays information for the call trace profile specified by the profile name or all profiles when no name is specified.

Parameters

profile-name

displays information about the specified profile

Output

The following output example shows call trace UE profile information, and *Show mobile profile call-insight UE output fields* describes the fields.

Example output call trace UE profile

```
# show mobile-gateway profile call-insight ue "default"
```

```

=====
Call-insight UE profile
=====
Profile name       : default
Description        : none
Live output        : none
Reference points   : dhcp, dns, ga, gn, gp, gx, gxc, pi, radius, rf, ro, s1, s11, s12, s2a
                   , s2b, s4, s5, s6b, s8, sd, swm, swu, swu-clear-text, sww, sta, sx-n4
User-traffic       : none
Events            : none
Format            : simulated-pcap
Size limit         : 10 MB
Time limit        : 86400 secs
user-traffic capture-type : packet-and-stats
user-traffic capture-location : inner-only
-----
Number of profiles : 1
=====

```

Table 105: Show mobile profile call-insight UE output fields

Label	Description
Profile name	The name of the current profile
Description	The profile, if necessary
Live output	Where to resend each captured message/event
Reference points	Which reference points are needed
User traffic	The user traffic
Events	The events that occur with the profile
Format	The format of the profile
Size limit	The space used in total
Time limit	The time used in total
User traffic capture type	The type of the captured user traffic

Table 105: Show mobile profile call-insight UE output fields (continued)

Label	Description
User traffic capture location	The location of the captured user traffic
Number of profiles	The number of profiles retrieved

6.2.2.6 charging

Syntax

charging

Context

show>mobile>profile

Description

This command enables the context to display charging profile information.

6.2.2.7 ccfh-profile

Syntax

ccfh-profile *profile-id*

Context

show>mobile>profile>charging

Description

This command displays CCFH profile information.

Parameters

profile-id

specifies the CCFH profile ID

Values: 0 to 255

Output

The following output examples show CCFH charging profile information, and *Show mobile profile charging CCFH profile output fields* describes the fields.

- *Example output CCFH charging profile*
- *Example output CCFH charging profile with FUI terminate session*

Example output CCFH charging profile

```
# show mobile-gateway profile charging ccfh-profile 1
=====
ccfh-profile
=====
ccfh-profile-id          : 1
Failure Handling
cc-failure-handling      : terminate
cc-session-failover      : enabled
fh-session-continue-timer *: 30      fh-volume-limit (kbytes) : 0
fh-continue-action       : allow      ccfh-fh-action          : terminate
session-restore-retry-time*: 0
store-and-replay         : disabled   buffer-playback-timer (min*: 5
gy-to-gtp-error-mapping-profile :
cc-failure-handling-enh   : disabled   service-restore-max-offset*: 0
result-code-3xxx : none                rc-3xxx-restore : disabled
result-code-4xxx : none                rc-4xxx-restore : disabled
result-code-5xxx : none                rc-5xxx-restore : disabled
result-code-4111 : terminate            rc-4111-restore : disabled
result-code-4222 : retry-and-terminate rc-4222-restore : disabled
SBI Error Mapping Profile : (Not Specified)
-----
Multiple Services Credit Control (MSCC) Result Codes
-----
Result Code      Service Delay Timer (mins)      FUI Terminate Session
-----
-----
Assume Positive
-----
assume-positive          : disabled   orig-session-id-ccr-i      : disabled
-----
nchf-cause-codes
-----
nchf-http-code
-----
message-type           : ccr-i
result-code-3111 : continue            rc-3111-restore : enabled
result-code-3222 : retry-and-continue  rc-3222-restore : enabled
-----
message-type           : ccr-u
result-code-3333 : retry-and-cont-allow rc-3333-restore : enabled
result-code-3444 : retry-and-cont-offl* rc-3444-restore : enabled
-----
```

```
nchf-mui-result-code
```

```
Number of profiles : 1
```

Example output CCFH charging profile with FUI terminate session

```
# show mobile-gateway profile charging ccfh-profile 1
=====
ccfh-profile
=====
ccfh-profile-id          : 1
Failure Handling
cc-failure-handling      : terminate
cc-session-failover      : enabled
fh-session-continue-timer *: 30      fh-volume-limit (kbytes) : 0
fh-continue-action       : allow      ccfh-fh-action          : terminate
session-restore-retry-time*: 0
gy-to-gtp-error-mapping-profile :
result-code-3xxx : none              result-code-4xxx : none
result-code-5xxx : none
=====
Multiple Services Credit Control (MSCC) Result Codes
=====
Result Code      Service Delay Timer (mins)      FUI Terminate Session
=====
4012             0                          enabled
=====
Assume Positive
=====
assume-positive      : disabled              orig-session-id-ccr-i : disabled
=====

Number of profiles : 1
=====
```

Table 106: Show mobile profile charging CCFH profile output fields

Label	Description
ccfh-profile-id	The CCFH profile being viewed
cc-failure-handling	The CCFH type
cc-session-failover	The CC session failover support

Table 106: Show mobile profile charging CCFH profile output fields (continued)

Label	Description
fh-session-continue-timer	The length of the failure handling session continue timer in minutes
fh-volume-limit	The FH maximum volume limit
gy-to-gtp-error-mapping-profile	The Gy to GTP error mapping profile
Result Code	The MSCC result code
Service Delay Timer	The length of the service delay in minutes
FUI Terminate Session	The FUI terminate session status: enabled or disabled
assume-positive	The assume positive function
orig-session-id-ccr-i	The original session ID used in the CCR-I
message-type	The CC message type, that is, CCR-I or CCR-U
result-code-xxxx	The diameter CC result code and the associated action
rc-xxxx-restore	The restore option for the specific result code
Number of profiles	The number of profiles being viewed

6.2.2.8 cdr-avp-options

Syntax

`cdr-avp-options [cdr-avp-options-profile-id]`

Context

`show>mobile>profile>charging`

Description

This command displays PDN PGW charging profile information.

Parameters

cdr-avp-options-profile-id

displays information about the specified profile

Output

The following output example shows CDR AVP options profile information, and *Show mobile profile CDR AVP options output fields* describes the fields.

CDR AVP options profile output

```
# show mobile-gateway profile charging cdr-avp-options 1
=====
cdr-avp-option profile
=====
Profile Id           : 1
Description          : N/A
nw-init-pdp-context  : exclude   served-imeisv       : include
imeisv-acr-interim   : exclude   imeisv-acr-stop     : exclude
imsi-unauthenticated-flag : exclude   node-id            : include
served-mn-nai        : exclude   pdn-connection-charging-id : exclude
sgw-change           : include    pgw-plmn-id         : include
apn-network-id       : exclude    virtual APN          : exclude
pdp-pdn-type         : include    served-pdp-pdn-addr    : include
served-pdp-pdn-addr-extens*: exclude   dynamic-addr-flag    : exclude
pdn-pdp-address-prefix-len*: exclude   dynamic-addr-flag-extensio*: exclude
list-traffic-data-volume : exclude   list-of-service-data : include
ms-time-zone         : include    diagnostics           : include
local-record-seq-no   : exclude    apn-selection-mode   : exclude
served-msisdn        : include    user-location-info    : include
pgw-address          : include    serving-node-plmn-id   : include
ps-furnish-charging-info : exclude   camel-information    : exclude
rat-type             : include    low-priority-indicator  : exclude
start-time           : include    stop-time              : include
charging-char-select-mode : include   ims-sig-context      : exclude
3gpp2-bsid           : include    sgw-address            : include
service-context-id    : N/A
epdg-change          : include
dualstack-sgw-address-ipv6 : exclude   dualstack-serving-node-ipv6*: exclude
dualstack-pgw-address-ipv6 : exclude   user-location-info-time : exclude
user-csg-info        : include
twan-user-location-info : exclude   dualstack-epdg-address-ipv6: exclude
tdv-charging-id       : exclude    sdc-charging-id      : exclude
sdc-qci              : include
dualstack-twag-address-ipv6: exclude   cp-ciot-eps-optimisation-i*: exclude
enhanced-diagnostics  : exclude
pra-info             : exclude    sdc-pra-status       : exclude
tdv-pra-status        : exclude
cn-operator-selection  : exclude
```

```

uwan-user-location-info      : exclude
uni-pdu-cp-only-flag        : exclude
apn-rate-control             : exclude
mo-exception-data-counter    : exclude
extended-bw-nr              : exclude
originating-bsid            : include
subscription-external-id     : exclude
ran-sec-rat-usage-reports    : exclude
sdc-result-code              : exclude
serving-plmn-rate-control    : exclude
pdp-pdn-type-extension       : exclude
sgi-ptp-tunnelling-method    : exclude
originating-user-loc-info    : include
subscription-imsi            : exclude
-----
Number of profiles : 1
=====
* indicates that the corresponding row element may have been truncated.

```

Table 107: Show mobile profile CDR AVP options output fields

Label	Description
Profile Id	The ID number of the profile being viewed
Description	The profile, if applicable
nw-init-pdp-context	The network-initiated PDP context IE
served-imeisv	The IMEISV of the UE IE
imeisv-acr-interim	The IMEISV in the ACR-Interim message
imeisv-acr-stop	The IMEISV in the ACR-Stop message
imsi-unauthenticated-flag	The IMSI unauthenticated flag IE in Gy messages
node-id	The ID of the node
served-mn-nai	The mobile node identifier IE
pdn-connection-charging-id	The PDN connection identifier IE
sgw-change	The first record from another serving node IE
pgw-plmn-id	The PLMN Identifier (MCC & MNC) of the PGW IE

Table 107: Show mobile profile CDR AVP options output fields (continued)

Label	Description
apn-network-id	The APN Network Identifier IE
virtual APN	The virtual APN
pdp-pdn-type	The PDN type (IPv4, IPv6 or IPv4v6) IE
served-pdp-pdn-addr	The IP address allocated to the PDN connection IE (IPv4 when PDN Type is IPv4 or IPv6 when PDN Type is IPv6 or IPv4v6)
served-pdp-pdn-addr-extension	The IP address allocated to the PDN connection IE (IPv4 address of the served IMSI, if available, when PDN type is IPv4v6)
dynamic-addr-flag	The PDN address that is dynamically allocated IE
dynamic-addr-flag-extensio*	The PDN address extension is dynamically allocated IE
pdn-pdp-address-prefix-len*	The prefix length of an IPv6-typed served PDP/PDN address IE
list-traffic-data-volume	The List of Traffic Data Volumes IE
list-of-service-data	The List of Service Data containers IE
ms-time-zone	The timezone the UE is located in IE
diagnostics	The Diagnostics IE
local-record-seq-no	The Local Record sequence Number IE
apn-selection-mode	The APN Selection Mode IE
served-msisdn	The MSISDN of the UE IE
user-location-info	The user location information IE

Table 107: Show mobile profile CDR AVP options output fields (continued)

Label	Description
pgw-address	The Control Plane IP address of the PGW used IE
serving-node-plmn-id	The SGW MCC and MNC used IE
ps-furnish-charging-info	The online charging session information IE
camel-information	The CAMEL information related to IP CAN bearer IE
rat-type	The RAT type used by the UE IE
low-priority-indicator	The IP-CAN session has a low priority IE
start-time	The start-time of the user IP-CAN session IE
stop-time	The stop-time of the user IP-CAN session IE
charging-char-select-mode	The Charging Characteristics selection mode IE
ims-sig-context	The IMS signaling context IE
3gpp2-bsid	The user location information for 3gpp2 access IE
sgw-address	The IP address of the service node used IE
service-context-id	The service context ID (sent to OCS) IE
epdg-change	The first record from another serving node IE
dualstack-sgw-address-ipv6	The SGW IPv6 address in case of IPv4v6 dual stack IE
dualstack-serving-node-ipv6*	The serving node IPv6 address in case of IPv4v6 dual stack IE
dualstack-pgw-address-ipv6	The PGW IPv6 address in case of IPv4v6 dual stack IE

Table 107: Show mobile profile CDR AVP options output fields (continued)

Label	Description
user-location-info-time	The user location information time IE
user-csg-info	The user CSG information IE
twan-user-location-info	The TWAN user location IE
dualstack-epdg-address-ipv6	The ePDG IPv6 address in case of IPv4v6 dual stack IE
tdv-charging-id	The 3GPP charging ID within the traffic data volumes container IE
sdv-charging-id	The 3GPP charging ID within a service data container IE
sdv-qci	The QoS class identifier within an service data container IE
dualstack-twap-address-ipv6	The TWAG control plane IPv6 address, in case of IPv4v6 dual stack, IE
cp-ciot-eps-optimisation-i*	The Cp Ciot EPS optimisation indicator IE
enhanced-diagnostics	The Enhanced-Diagnostics IE
pra-info	The PRA information toward an online charging server IE
sdv-pra-status	The PRA status within a Service Data Container toward an offline charging server IE
tdv-pra-status	The PRA status within a Traffic Data Volume container toward an offline charging server IE
uwan-user-location-info	The UWAN user location information IE
originating-bsid	Indicates whether the originating BSID is sent to the PFCS

Table 107: Show mobile profile CDR AVP options output fields (continued)

Label	Description
originating-user-loc-info	Indicates whether the originating user location information is sent to the PFCS
extended-bw-nr	Indicates whether extended information AVPs used for high data rates for option-3x as a part of the Qos-information NR are sent to the OCS
ran-sec-rat-usage-reports	Indicates whether secondary RAT usage data reports (5G NR) are sent to the offline charging server
Number of profiles	The total number of profiles being viewed

6.2.2.9 chf-profile

Syntax

```
chf-profile [profile-id]
```

Context

```
show>mobile>profile>charging
```

Description

This command displays profile information for the 5G charging function.

Parameters

profile-id

specifies the CHF profile for which to display information

Values: 0 to 255

Output

The following output examples show ePDG charging information, and [Show CHF charging profile output fields](#) describes the fields.

Example output CHF charging profile

```
# show mobile-gateway profile charging chf-profile 5
=====
CHF profile
=====
Profile Id           : 5
Description          : N/A
activity-threshold   : 0          ccfh-profile           : 0
request-quota-before-traff*: disabled  unit-count-inactivity-timer: 0 (secs)

UUC Options
no-quota-initial      : suppress
no-quota-mid-session  : suppress
with-quota            : none
uuc-reporting-enh     : disabled
implicit-vol-threshold : 0          reason-for-reporting      : threshold
supported-vendor      : none
ran-sec-rat-usage-reports : exclude

sbi-srvc-operation
transaction-timer     : 5000        retry-count              : 0

default-rg-list
-----
Number of profiles : 1
=====
* indicates that the corresponding row element may have been truncated.
```

Table 108: Show CHF charging profile output fields

Label	Description
CHF Profile	
Profile Id	The ID number of the profile being viewed
Description	The CHF profile description
activity-threshold	The configured activity threshold for the profile
ccfh-profile	The configured CCFH profile
request-quota-before-traffic	The status of the request quota before traffic option

Table 108: Show CHF charging profile output fields (continued)

Label	Description
unit-count-inactivity-timer	Unit count inactivity timer, in seconds
UUC Options	
no-quota-initial	Option for quota for which the CHF has not allocated any initial quota
no-quota-mid-session	Option for quota for which the CHF has allocated quota, but has run out of credit mid-stream
with-quota	Option for quota for which the CHF allocates quota
uuc-reporting-enh	UUC reporting enhancement status (enabled/disabled)
implicit-vol-threshold	An implicit-volume threshold applicable when the allocated quota does not include a quota threshold from the OCS
reason-for-reporting	The reporting reason for the implicit-volume threshold
supported-vendor	The number of the supported vendor
ran-sec-rat-usage-reports	Indicates whether secondary RAT usage data reports (5G NR) are sent from the SMF to the Nchf interface
sbi-srvc-operation	
transaction-timer	Configured transaction timer for SBI service operation for this profile, in milliseconds
retry-count	Configured retry count used to detect connection failures to the PCF
default-rg-list	Default rating group list
Number of profiles	Number of profiles reported

6.2.2.10 epdg-charging

Syntax

epdg-charging [*charging-profile-id*]

Context

show>mobile>profile>charging

Description

This command displays the ePDG charging profile information.

Parameters

charging-profile-id

displays information about the specified profile

Output

The following output examples show ePDG charging information, and *Show ePDG charging profile statistics output fields* describes the fields.

Example output ePDG charging profile statistics

```
# show mobile-gateway profile charging epdg-charging 0
=====
ePDG Charging profile
=====
Profile Id           : 0
Description          : N/A
Enable-offline       : enabled
exclude-zero-vol-cdr : disabled
exclude-zero-vol-complete-cdr : enabled
Charging-info-trigger
QoS change           : disabled   Tariff time change           : disabled
Tariff change list profile : (Not Specified)
UWAN location change  : disabled
Partial-record-closure-trigger
Max-change-condition  : 0           Time-limit/session (mins) : 0
Vol-limit/session (Kbytes) : 0
Min-Data-Threshold (Kbytes): 0
-----
Number of profiles : 1
=====
```

Table 109: Show ePDG charging profile statistics output fields

Label	Description
EPDG Charging profile	
Profile Id	The ID number of the profile being viewed
Description	The profile, if applicable
Enable offline	The profile can work offline
Exclude-zero-vol-cdr	The generation of CDRs that have zero volume
Exclude-zero-vol-complete-cdr	The generation of CDRs that have zero volume (with zero duration or non-zero)
Charging-info-trigger	The context to configure triggers
QoS change	The Quality of Service change
Tariff time change	The change in tariff time
Tariff time	The tariff time
uwan-loc-change	The uwan-loc-change event trigger
Partial-record-closure-trigger	The time limit (in seconds) applicable per session or bearer
Max-change-condition	The maximum number of change conditions that can occur
Time-limit/session (mins)	The amount of time permitted per session in minutes
Vol-limit/session (Kbytes)	The amount of volume permitted per session in Kbytes
Number of profiles	The number of profiles being viewed

Table 109: Show ePDG charging profile statistics output fields (continued)

Label	Description
CDR-AVP-Options Charging profile	
Profile Id	The ID number of the profile being viewed
Description	The profile, if applicable
CDR Profile	The CDR profile associated with the profile
nw-init-pdp-context	A network initiated pdp context
served-imeisv	The IMEISV of the UE, if available
imeisv-acr-interim	The IMEISV in the ACR-Interim message
imeisv-acr-stop	The IMEISV in the ACR-Stop message
imsi-unauthenticated-flag	The served IMSI is not authenticated (emergency bearer)
node-id	The Node ID in CDRs
served-mn-nai	The mobile node identifier in NAI format (based on IMSI)
pdn-connection-charging-id	The PDN connection charging ID
sgw-change	The first record after a change from another serving node
pgw-plmn-id	The PLMN Identifier (MCC & MNC) of the PGW used
apn-network-id	The Access Point Name Network Identifier
virtual APN	The virtual APN
pdp-pdn-type	The PDN type

Table 109: Show ePDG charging profile statistics output fields (continued)

Label	Description
served-pdp-pdn-addr	The IP address allocated to the PDN connection
served-pdp-pdn-addr-extens*	The IP address extension allocated to the PDN connection
dynamic-addr-flag	The dynamic address flag information elements
pdn-pdp-address-prefix-len*	The prefix length of an IPv6-typed served PDP/PDN address IE
dynamic-addr-flag-extensio*	The PDN address extension is dynamically allocated
list-traffic-data-volume	The List of Traffic Data Volumes
list-of-service-data	The List of Service Data containers
ms-time-zone	The timezone the UE is located in, if available (provided by MME)
diagnostics	The Diagnostics
local-record-seq-no	The Local Record sequence Number
apn-selection-mode	The APN Selection Mode
served-msisdn	The MSISDN of the UE, if available
user-location-info	The user location information IE
pgw-address	The Control Plane IP address of the PGW used IE
serving-node-plmn-id	The SGW MCC+MNC used
ps-furnish-charging-info	The online charging session information
camel-information	The CAMEL information related to the IP CAN bearer IE

Table 109: Show ePDG charging profile statistics output fields (continued)

Label	Description
rat-type	The RAT type used by the UE IE
low-priority-indicator	The IP-CAN session that has a low priority IE
start-time	The start-time of the user IP-CAN session IE
stop-time	The stop-time of the user IP-CAN session IE
charging-char-select-mode	The Charging Characteristics selection mode IE
ims-sig-context	The IMS signaling context IE
3gpp2-bsid	The user location information for 3gpp2 access IE
sgw-address	The IP address of the service node used IE
service-context-id	The service context ID (sent to OCS) IE
epdg-change	The first record from another serving node IE
dualstack-sgw-address-ipv6	The SGW IPv6 address in case of IPv4v6 dual stack IE
dualstack-serving-node-ipv6*	The serving node IPv6 address in case of IPv4v6 dual stack IE
dualstack-pgw-address-ipv6	The PGW IPv6 address in case of IPv4v6 dual stack IE
user-location-info-time	The user location information time IE
user-csg-info	The user CSG information IE
served-3gpp2-meid	—
twan-user-location-info	The TWAN user location information IE

Table 109: Show ePDG charging profile statistics output fields (continued)

Label	Description
dualstack-epdg-address-ipv6	The ePDG IPv6 address in case of IPv4v6 dual stack IE
Number of Profiles	The number of profiles being viewed

6.2.2.11 event-triggers-fbc

Syntax

```
event-triggers-fbc [profile-id]
```

Context

```
show>mobile>profile>charging
```

Description

This command displays the configuration information for the specified FBC event triggers profile for 5G systems.

Parameters

profile-id

displays the FBC event trigger information for the specified profile ID

Values: 0 to 255

Output

The following is example output and *Show mobile profile output fields (event triggers FBC)* describes the fields.

Show event triggers FBC output

```
# show mobile-gateway profile charging event-triggers-fbc 2
=====
Event Triggers FBC profile
=====
Profile Id           : 2
Description          : N/A
3gpp-ps-data-off-status-ch*: disabled  addition-of-upf      : disabled
plmn-change         : disabled  qos-change           : disabled
```

```

rat-type-change           : disabled  removal-of-upf           : disabled
session-ambr-change      : disabled  ue-in-pra-change        : disabled
ue-time-zone-change      : disabled  user-loc-change         : disabled
time-limit               : 60 (mins) time-limit-eutran       : 0 (mins)
time-limit-nr            : 0 (mins)  time-limit-per-rating-group: 30 (mins)
volume-limit             : 4096 (kby* volume-limit-eutran       : 0 (kbytes)
volume-limit-nr          : 0 (kbytes) volume-limit-per-rating-gr*: 2048 (kby*
max-change-condition      : 0        serving-node-change-limit : 15
default-qht              : 0 (mins)
tariff-change-list       :
trigger-override-option  : chf       serving-node-type        : r15
3gpp-ps-data-off-status-ch*: deferred addition-of-upf-category  : immediate
plmn-change-category     : immediate qos-change-category     : deferred
rat-type-change-category : immediate removal-of-upf-category  : immediate
session-ambr-change-catego*: immediate ue-in-pra-change-category : deferred
ue-time-zone-change-catego*: immediate user-loc-change-category : deferred
time-limit-per-rating-grou*: deferred volume-limit-per-rating-gr*: deferred
serving-node-change-catego*: deferred
handover-cancel          : disabled  handover-cancel-category : immediate
handover-start           : disabled  handover-start-category  : immediate
handover-complete        : disabled  handover-complete-category: immediate
insertion-of-ismf        : disabled  insertion-of-ismf-category: deferred
reallocation-of-ismf     : disabled  reallocation-of-ismf-categ*: deferred
removal-of-ismf          : disabled  removal-of-ismf-category : deferred
-----
Number of profiles : 1
=====
* indicates that the corresponding row element may have been truncated.

```

Table 110: Show mobile profile output fields (event triggers FBC)

Label	Description
Profile Id	The ID number of the profile being viewed
Description	The profile type (Event Trigger FBC Profile)
3gpp-ps-data-off-status-change	The configured status (enabled/disabled) for this trigger
3gpp-ps-data-off-status-change-category	The configured trigger category (immediate or deferred)
addition-of-upf	The configured status (enabled/disabled) for this trigger
addition-of-upf-category	The configured trigger category (immediate or deferred)

Table 110: Show mobile profile output fields (event triggers FBC) (continued)

Label	Description
default-qht	The configured status (disabled or the value if enabled) for this trigger
handover-cancel	The configured status (enabled/disabled) for this trigger
handover-cancel-category	The configured category for this trigger (immediate or deferred)
handover-complete	The configured status (enabled/disabled) for this trigger
handover-complete-category	The configured category for this trigger (immediate or deferred)
handover-start	The configured status (enabled/disabled) for this trigger
handover-start-category	The configured category for this trigger (immediate or deferred)
insertion-of-ismf	The configured status (enabled/disabled) for this trigger
insertion-of-ismf-category	The configured category for this trigger (immediate or deferred)
max-change-condition	The configured status (disabled or the value if enabled) for this trigger
plmn-change	The configured status (enabled/disabled) for this trigger
plmn-change-category	The configured trigger category (immediate or deferred)
qos-change	The configured status (enabled/disabled) for this trigger
qos-change-category	The configured trigger category (immediate or deferred)
rat-type-change	The configured status (enabled/disabled) for this trigger
rat-type-change-category	The configured trigger category (immediate or deferred)

Table 110: Show mobile profile output fields (event triggers FBC) (continued)

Label	Description
reallocation-of-ismf	The configured status (enabled/disabled) for this trigger
reallocation-of-ismf-category	The configured category for this trigger (immediate or deferred)
removal-of-ismf	The configured status (enabled/disabled) for this trigger
removal-of-ismf-category	The configured category for this trigger (immediate or deferred)
removal-of-upf	The configured status (enabled/disabled) for this trigger
removal-of-upf-category	The configured category for this trigger (immediate or deferred)
serving-node-change	The configured status (disabled/enabled) for this trigger
serving-node-change-category	The configured category for this trigger (immediate or deferred)
serving-node-change-limit	The configured status (disabled or the value if enabled) for this trigger
serving-node-type	The configured serving node type (r15 or r16)
session-ambr-change	The configured status (enabled/disabled) for this trigger
session-ambr-change-category	The configured category for this trigger (immediate or deferred)
tariff-change-list	The configured status (disabled or the name of the list if enabled) for this trigger
time-limit	The configured status (disabled or the value if enabled) for this trigger
time-limit-eutran	The configured status (disabled or the value if enabled) for this trigger

Table 110: Show mobile profile output fields (event triggers FBC) (continued)

Label	Description
time-limit-nr	The configured status (disabled or the value if enabled) for this trigger
time-limit-per-rating-group	The configured status (disabled or the value if enabled) for this trigger
time-limit-per-rating-group-category	The configured category for this trigger (immediate or deferred)
trigger-override-option	The configured status of the trigger override option (local, chf, or local-and-chf)
ue-in-pra-change	The configured status (enabled/disabled) for this trigger
ue-in-pra-change-category	The configured category for this trigger (immediate or deferred)
ue-time-zone-change	The configured status (enabled/disabled) for this trigger
ue-time-zone-change-category	The configured category for this trigger (immediate or deferred)
user-loc-change	The configured status (enabled/disabled) for this trigger
user-loc-change-category	The configured category for this trigger (immediate or deferred)
volume-limit	The configured status (disabled or the value if enabled) for this trigger
volume-limit-eutran	The configured status (disabled or the value if enabled) for this trigger
volume-limit-nr	The configured status (disabled or the value if enabled) for this trigger

Table 110: Show mobile profile output fields (event triggers FBC) (continued)

Label	Description
volume-limit-per-rating-group	The configured status (disabled or the value if enabled) for this trigger
volume-limit-per-rating-group-category	The configured category for this trigger (immediate or deferred)
Number of profiles	The number of profiles being viewed

6.2.2.12 gy-avp-options

Syntax

gy-avp-options [*gy-avp-options-profile-id*]

Context

show>mobile>profile>charging

Description

This command displays the configured contents of the specified gy-avp-options profile. When the optional profile-id is omitted, the list of configured profiles is displayed.

Parameters

gy-avp-options-profile-id

displays information about the specified profile

Output

The following is example output for Gy AVP options, and *Show mobile profile charging output fields (Gy AVP)* describes the fields.

Show Gy AVP options profile output

```
# show mobile-gateway profile charging gy-avp-options 1
=====
gy-avp-option profile
=====
```

```

Profile Id      : 1
Description     : N/A
user-name      : include   origin-state-id      : include
event-timestamp : include   subscription-id-imsi   : include
subscription-id-msisdn : include subscription-id-nai     : include
cc-correlation-id : include user-equipment-info    : include
3gpp-charging-id : include   pdn-connection-charging-id : include
node-id        : include   3gpp-pdp-type          : include
pdp-address     : include   pdp-address-extension    : include
pdp-address-prefix-length : include dynamic-addr-flag      : include
dynamic-address-flag-exten* : include qos-information       : exclude
3gpp-gprs-qos-negotiated-p* : exclude sgsn-address         : include
ggsn-address    : include   serving-node-type       : include
3gpp-imsi-mcc-mnc : include   imsi-unauthenticated-flag : include
3gpp-ggsn-mcc-mnc : include   3gpp-nsapi              : include
3gpp-session-stop-indicator : include 3gpp-selection-mode    : include
3gpp-charging-characterist* : include charging-characteristics-s* : include
3gpp-sgsn-mcc-mnc : include   3gpp-ms-timezone       : include
3gpp-user-location-info : include 3gpp2-bsid          : include
pdp-context-type : include   service-id             : include
3gpp-rat-type    : include   ims-information         : include
dualstack-sgsn-address-ipv6 : exclude dualstack-ggsn-address-ipv6 : exclude
cg-address       : exclude   user-location-info-time  : exclude
user-csg-info    : exclude   twan-user-location-info  : exclude
charging-rule-base-name : exclude enhanced-diagnostics : exclude
user-equipment-info-bcd-en* : exclude
pra-info        : include
mscc-qos-information : exclude
uwan-user-location-info : exclude
extended-bw-nr   : include
rg-in-raa        : exclude   user-name-raa           : exclude
user-name-msisdn-at-apn : include   uwan-user-location-info-r16 : exclude
twan-user-location-info-r16 : exclude
3gpp-user-location-info-5g : exclude
  
```

```

-----
Number of profiles : 1
=====
  
```

Table 111: Show mobile profile charging output fields (Gy AVP)

Label	Description
Profile Id	The ID number of the profile being viewed
Description	The profile, if applicable
user-name	The username

Table 111: Show mobile profile charging output fields (Gy AVP) (continued)

Label	Description
origin-state-id	The origin state ID
event-timestamp	The event timestamp
subscription-id-imsi	The subscription ID ISMI
subscription-id-msisdn	The subscription ID MSISDN
subscription-id-nai	The subscription ID NAI
cc-correlation-id	The CC Correlation ID
user-equipment-info	The user equipment information
3gpp-charging-id	The 3GPP charging ID
pdn-connection-charging-id	The PDN connection charging ID
node-id	The node ID
3gpp-pdp-type	The 3GPP PDP type
pdp-address	The PDP address
pdp-address-extension	The PDP address extension
pdp-address-prefix-length	The PDP address prefix length
dynamic-addr-flag	The dynamic address flag
dynamic-address-flag-exten*	The dynamic address flag extension
qos-information	The QoS information

Table 111: Show mobile profile charging output fields (Gy AVP) (continued)

Label	Description
3gpp-gprs-qos-negotiated-p*	The 3GPP GPRS negotiated Quality of Service profile Information Element in the Gy messages
sgsn-address	The SGSN address
ggsn-address	The GGSN address
serving-node-type	The serving node type
3gpp-imsi-mcc-mnc	The 3GPP IMSI MCC MNC
imsi-unauthenticated-flag	The IMSI unauthenticated flag
3gpp-ggsn-mcc-mnc	The 3GPP GGSN MCC MNC
3gpp-nsapi	The 3GPP NSAPI
3gpp-session-stop-indicator	The 3GPP session stop indicator
3gpp-selection-mode	The 3GPP selection mode
3gpp-charging-characterist*	The 3GPP charging characteristics
charging-characteristics-s*	The charging characteristics selection Information Element in the Gy messages
3gpp-sgsn-mcc-mnc	The 3GPP SGSN MCC MNC
3gpp-ms-timezone	The 3GPP MS timezone
3gpp-user-location-info	The 3GPP user location information
3gpp2-bsid	The 3GPP BSID

Table 111: Show mobile profile charging output fields (Gy AVP) (continued)

Label	Description
pdp-context-type	The PDP context type
service-id	The service ID
3gpp-rat-type	The 3GPP RAT type
ims-information	The IMS information
dualstack-sgsn-address-ipv6	The Dualstack SGSN address (IPv6)
dualstack-ggsn-address-ipv6	The Dualstack GGSN address (IPv6)
cg-address	The CG address
user-location-info-time	The user location information time
user-csg-info	The user CG information
twan-user-location-info	The TWAN user location information
charging-rule-base-name	The charging rule base name
enhanced-diagnostics	The enhanced diagnostics
user-equipment-info-bcd-en*	The user equipment information BCD encoding type
pra-info	The PRA information
uwan-user-location-info	The UWAN user location information
extended-bw-nr	Indicates whether extended information AVPs used for high data rates for option-3x as a part of the Qos-information NR are sent to the OCS

Table 111: Show mobile profile charging output fields (Gy AVP) (continued)

Label	Description
rg-in-raa	Indicates whether the Rating-Group AVP is included in the RAA reply
user-name-raa	Indicates whether the User-Name AVP is sent in the RAA message to the OCS
user-name-msisdn-at-apn	Indicates whether the User-Name AVP in MSISDN@APN format is sent to the OCS
3gpp-user-location-info-5g	Indicates whether the 5G attributes are included in the 3GPP user location IE
Number of profiles	The number of profiles being viewed

6.2.2.13 pgw-charging

Syntax

pgw-charging [*charging-profile-id*]

Context

show>mobile>profile>charging

Description

This command displays PGW charging profile information.

Parameters

charging-profile-id

displays information about the specified profile

Output

The following output example shows PGW charging profile information, and *PGW charging profile output fields* describes the fields.

Example output PGW charging profile

```
# show mobile-gateway profile charging pgw-charging 2
=====
PgW Charging profile
=====
Profile Id           : 2
Description          : N/A
Charging-info-trigger
QoS change           : disabled   User Loc change           : disabled
Tariff time change   : disabled   SGW change                : disabled
Vol-limit/rtg-grp (Kbytes) : 2048      Time-limit/rtg-grp (mins) : 30
Tariff change list profile : (Not Specified)
change-csg-cell      : disabled
Change-csg-subscribed-hybrid-cell : disabled
Change-csg-unsubscribed-hybrid-cell : disabled
plmn-rate-control-change : disabled   apn-rate-control-change : disabled
gy-final             : disabled   gy-validity-time        : disabled
gy-rating-condition-change : disabled   gy-forced-reauthorization : disabled
gy-qht               : disabled   gy-tariff-time-change   : disabled
gy-quota-exhausted   : disabled   gy-threshold-vqt        : disabled
ecgi-change          : disabled   tai-change              : disabled
cgi-sai-change       : disabled   rai-change              : disabled
change-ue-presence-pra : disabled
CC-User Loc Change   : disabled
uwan-loc-change      : disabled
fh-close-cdr-containers : disabled
Partial-record-closure-trigger
Max-change-condition : 0           Ms-time-zone-change      : disabled
PLMN Change          : disabled   Serving-node-change-limit : 15
RAT-change           : disabled   Time-limit/session (mins) : 60
Vol-limit/session (Kbytes) : 4096
Vol-limit-eutran/session (Kbytes) : 0
Vol-limit-utran/session (Kbytes) : 0
Vol-limit-eutran-nb-iot/session (Kbytes) : 1000
Time-limit-eutran/session (Mins) : 0
Time-limit-utran/session (Mins) : 0
Time-limit-eutran-nb-iot/session (Mins) : 60
Min-Data-Threshold (Kbytes) : 0
mo-exception-data-receipt : disabled

Service Data Container
zero-vol-sdc          : disabled   qos-info-first-sdc-of-cdr : disabled
exclude-zero-vol-cdr  : disabled
exclude-zero-vol-complete-cdr : disabled
Radius-accounting
RAT change            : disabled
QoS change            : disabled
Peer change           : disabled
Interim Volume Threshold : 0 (KB)
User Location Change   : disabled
eCGI Change           : disabled
TAI Change            : disabled
RAI Change            : disabled
CGI/SAI Change        : disabled
Off-line Charging     : enabled
```

```

On-line Charging      : enabled
DCCA Profile         : DCCA-Prof2
Primary OCS Diameter Peer Profile : Gy-Peer-Prof1
Secondary OCS Diameter Peer Profile : Gy-Peer-Prof2
Dynamic-PCC          : inherited
pcc-disable-per-rat   : none
pcc-disable-per-rat-timer : 30 secs
selection-type-id     : none      selection-type-mapping : none
include-bearer-charging : disabled
-----
Number of profiles : 1
=====

```

Table 112: PGW charging profile output fields

Label	Description
Profile Id	The ID number of the profile being viewed
Description	The profile, if applicable
Charging-info-trigger	The context to configure triggers
QoS change	The Quality of Service change
User Loc change	The change in user location
Tariff time change	The change in tariff time
SGW change	—
Vol-limit/rtg-grp (Kbytes)	The volume limit (in Kbytes) applicable per session or bearer
Time-limit/rtg-grp (mins)	The time limit (in minutes) applicable per session or bearer
Tariff change list profile	A tariff change list profile that can be applied to an offline charging information trigger
change-csg-cell	The gateway to report user CSG information change to the OFCS when the UE enters, leaves, or accesses via a CSG cell

Table 112: PGW charging profile output fields (continued)

Label	Description
Change-csg-subscribed-hybrid-cell	The gateway to report user CSG information change to the OFCS when the UE enters, leaves, or accesses via a hybrid cell in which the subscriber is a CSG member
Change-csg-unsubscribed-hybrid-cell	—
gy-final	—
gy-validity-time	—
gy-rating-condition-change	—
gy-forced-reauthorization	—
gy-qht	—
gy-tariff-time-change	The TTC online/offline tight integration trigger
gy-quota-exhausted	—
gy-threshold-vqt	—
ecgi-change	The eCGI change location event trigger
tai-change	The TAI change location event trigger
cgi-sai-change	The CGI-SAI change location event trigger
rai-change	The RAI change location event trigger
change-ue-presence-pra	The change in UE presence in a PRA event trigger for online charging

Table 112: PGW charging profile output fields (continued)

Label	Description
Offline Trigger CC User Lo*	—
uwan-loc-change	The uwan-loc-change event trigger
fh-close-cdr-containers	Enables the closing of the CDR containers by the PGW if the PDN session is under online charging control and a failure handling begins
Partial-record-closure-trigger	The time limit (in seconds) applicable per session or bearer
Max-change-condition	The maximum number of change conditions
Ms-time-zone-change	The MS time zone change partial record trigger
PLMN Change	The PLMN change partial record trigger
Serving-node-change-limit	The change limit partial record trigger
RAT-change	The RAT change partial record trigger
Time-limit/session (mins)	The amount of time permitted per session in minutes
Vol-limit/session (Kbytes)	The amount of volume permitted per session in Kbytes
Vol-limit-eutran/session (Kbytes)	The amount of volume permitted per session in Kbytes in eUTRAN
Vol-limit-utran/session (Kbytes)	The amount of volume Utran permitted per session in Kbytes in UTRAN
Vol-limit-eutran-nb-iot/session (Kbytes)	The amount of volume permitted per session in Kbytes in eUTRAN NB-IoT
Time-limit-eutran/session (Mins)	The amount of time permitted per session in minutes in eUTRAN

Table 112: PGW charging profile output fields (continued)

Label	Description
Time-limit-utran/session (Mins)	The amount of time permitted per session in minutes in UTRAN
Time-limit-eutran-nb-iot/session (Mins)	The amount of time permitted per session in minutes in eUTRAN NB IoT
Service Data Container	—
zero-vol-sdc	The sending of zero volume (SDC) in the PGW-CDRs for the OFCS
qos-info-first-sdc-of-cdr	The QoS-Information AVP which is always included in the first SDC for an RG/SVC-ID in a CDR/ACR
exclude-zero-vol-cdr	The generation of CDRs that have zero volume
exclude-zero-vol-complete-cdr	The generation of CDRs that have zero volume (with zero duration or non-zero)
Peer change	—
Interim Volume Threshold	—
Off-line Charging	The offline charging
On-line Charging	The online charging
DCCA Profile	The DCCA profile information
Primary OCS Diameter Peer Profile	The primary OCS Diameter peer profile
Secondary OCS Diameter Peer Profile	The secondary OCS Diameter peer profile
Dynamic-PCC	PCC interface (Gx) for a subscriber's IP-CAN session

Table 112: PGW charging profile output fields (continued)

Label	Description
pcc-disable-per-rat	The set of RAT types which are not allowed to be under the control of the PCRF
pcc-disable-per-rat-timer	The timer which is started when the UE moves into a RAT type which is not allowed under PCRF control
selection-type-id	The ID of the OCS selection criteria that is based on an IMSI or MSISDN range
user-range-name	The name for the user-range
diameter-peer-group-list	The Diameter peer group list profile
start-id	The start identifier for the user-range
end-id	The end identifier for the user-range
include-bearer-charging	Enables the reporting of IP CAN bearer charging in eG-CDRs and PGW-CDRs.
Number of profiles	The number of profiles being viewed

6.2.2.14 sgw-charging

Syntax

sgw-charging [*charging-profile-id*]

Context

show>mobile>profile>charging

Description

This command displays SGW charging profile information.

Parameters

charging-profile-id

displays information about the specified profile

Output

The following output example shows SGW charging profile information, and *SGW charging profile statistics output fields* describes the fields.

Example output SGW charging profile statistics

```
# show mobile-gateway profile charging sgw-charging 1
=====
Sgw Charging profile
=====
Profile Id           : 1
Description          : N/A
Primary Diameter Peer Profile: Peer-Profile2
Charging-info-trigger
QoS change           : disabled   User Loc change           : disabled
Tariff time change   : disabled
Tariff change list profile : (Not Specified)
change-csg-cell       : disabled
Change-csg-subscribed-hybrid-cell : disabled
Change-csg-unsubscribed-hybrid-cell : disabled
ecgi-change          : disabled   tai-change              : disabled
cgi-sai-change        : disabled   rai-change              : disabled
change-ue-presence-pra : disabled   change-in-user-plane    : disabled
plmn-rate-control-change : disabled
CC-User Loc Change    : disabled
Partial-record-closure-trigger
Max-change-condition   : 0         Ms-time-zone-change      : disabled
RAT-change             : disabled   Time-limit/session (mins) : 60
Vol-limit/session (Kbytes) : 4096
Min-Data-Threshold (Kbytes): 0
mo-exception-data-receipt : disabled
plmn-change            : disabled   serving-node-change-limit : 15
sgw-change-cause-record-clo: disabled
exclude-zero-vol-cdr    : disabled
enable-offline charging : enabled
exclude-zero-vol-complete-cdr : enabled
=====
Number of profiles : 1
=====
```


Table 113: SGW charging profile statistics output fields

Label	Description
Profile Id	The ID number of the profile being viewed
Description	The profile, if applicable
Charging-info-trigger	The context to configure triggers
QoS change	The Quality of Service change
User Local change	—
Tariff time change	The change in tariff time
Tariff change list profile	A tariff change list profile that can be applied to an offline charging information trigger
change-csg-cell	The gateway to report user CSG information change to the OFCS when the UE enters, leaves, or accesses via a CSG cell
Change-csg-subscribed-hybrid-cell	The gateway to report user CSG information change to the OFCS when the UE enters, leaves, or accesses via a hybrid cell in which the subscriber is a CSG member
Change-csg-unsubscribed-hybrid-cell	—
ecgi-change	The eCGI change location event trigger
tai-change	The TAI change location event trigger
cgi-sai-change	The CGI-SAI change location event trigger
rai-change	The RAI change location event trigger
change-ue-presence-pra	The change in UE presence in a PRA event trigger for online charging

Table 113: SGW charging profile statistics output fields (continued)

Label	Description
change-in-user-plane	A change in UP to UE trigger
Partial-record-closure-trigger	The time limit (in seconds) applicable per session or bearer
Max-change-condition	The maximum number of change conditions that can occur
Ms-time-zone-change	The MS time zone change partial record trigger
RAT-change	The RAT change partial record trigger
Time-limit/session (mins)	The amount of time permitted per session in minutes
Vol-limit/session (Kbytes)	The amount of volume permitted per session in Kbytes
sgw-change-cause-record-clo	The population of SGW Change as the reason for record closure from the old /source SGW in the case of SGW relocation, for Ga interface
exclude-zero-vol-cdr	The generation of CDRs that have zero volume
exclude-zero-vol-complete-cdr	The generation of CDRs that have zero volume (with zero duration or non-zero)
Number of profiles	The number of profiles being viewed

6.2.2.15 smf-charging

Syntax

smf-charging [*charging-profile-id*]

Context

show>mobile>profile>charging

Description

This command displays SMF charging profile information.

Parameters

charging-profile-id

displays information about the specified profile

Output

The following output example shows SMF charging profile information.

Example output SMF charging profile statistics

```
# show mobile-gateway profile charging smf-charging 1

=====
Smf Charging profile
=====
Profile Id           : 1
Description          : N/A
event-triggers-fbc   : 1          default-charging-method : online
event-triggers-qbc   : 0          qbc-charging             : disabled

chf-selection
nf-id-list           :
chf-profile           : 1

quota-management
final-unit-indication
redirect-append-url   : disabled
redirection-start-send-rr-*: enabled  redirection-start-send-rr-*: enabled7
redirect-filter-id    : N/A
traffic-handling-options
forced-re-authorization : allow      quota-exhausted-no-thresho*: allow
quota-exhausted-threshold-*: allow    quota-unavailable          : allow
rating-condition-change : allow      validity-time-expired      : allow
report-serving-node-suppli*: Disabled

radius-accounting
rat-change            : disabled    qos-change                 : disabled
ecgi-change           : disabled    tai-change                 : disabled
user-loc-change       : disabled    peer-change                : disabled
volume-limit (kbytes) : 0

-----
Number of profiles : 1
=====
* indicates that the corresponding row element may have been truncated.
=====
```

⁷ This is the configuration state of the redirection-start-send-rr-final-enh command.

6.2.2.16 smf-profile

Syntax

```
smf-profile [smf-profile-id]
```

Context

```
show>mobile>profile>charging
```

Description

This command displays SMF profile information for the 5G charging function.

Parameters

smf-profile-id

specifies the SMF profile for which to display information

Values: 0 to 255

6.2.2.17 tdf-charging

Syntax

```
tdf-charging [charging-profile-id]
```

Context

```
show>mobile>profile>charging
```

Description

This command displays TDF charging profile information.

Parameters

charging-profile-id

displays information about the specified profile

Output

The following output example shows TDF charging profile information and *TDF charging profile statistics output fields* describes the fields.

Example output TDF charging profile statistics

```
# show mobile-gateway profile charging tdf-charging 0
=====
TDF Charging profile
=====
Profile Id           : 0
Description          : N/A
Off-line charging    : enabled
Off-line charging type : fbc
charging-info-trigger
serving-node-change  : disabled
Tariff change list profile : (Not Specified)
tariff-time-change   : disabled   user-loc-change       : disabled
ecgi-change          : disabled   tai-change          : disabled
cgi-sai-change       : disabled   rai-change          : disabled
change-ue-presence-pra : disabled
CC-User Loc Change   : disabled
partial-record-closure-trigger
max-change-condition : 0          ms-time-zone-change   : disabled
plmn-change          : disabled   rat-change           : disabled
time-limit (mins)    : 60         volume-limit (Kbytes) : 0
Min-Data-Threshold (Kbytes): 0
On-line Charging     : disabled
On-line charging type : rg-level
vendor-specific-rg    : none       vendor-specific-svc-id : none
exclude-zero-vol-cdr  : disabled
exclude-zero-vol-complete-cdr : enabled
selection-type-id     : none       selection-type-mapping : none
-----
Number of profiles : 1
=====
```

Table 114: TDF charging profile statistics output fields

Label	Description
Profile Id	The ID number of the profile being viewed
Description	The profile, if applicable
Off-line charging	The offline charging
Off-line charging type	The type of offline charging method used
charging-info-trigger	The context to configure triggers

Table 114: TDF charging profile statistics output fields (continued)

Label	Description
serving-node-change	The serving node change event triggers locally for online charging
Tariff change list profile	A tariff change list profile that can be applied to an offline charging information trigger
tariff-time-change	The TTC
user-loc-change	The user location change trigger
ecgi-change	The eCGI change location event trigger
tai-change	The TAI change location event trigger
cgi-sai-change	The CGI-SAI change location event trigger
rai-change	The RAI change location event trigger
change-ue-presence-pra	The change in UE presence in a PRA event trigger for online charging
partial-record-closure-trigger	The time limit (in seconds) applicable per session or bearer
max-change-condition	The maximum number of change conditions
ms-time-zone-change	The MS time zone change partial record trigger
plmn-change	The PLMN change partial record trigger
rat-change	The RAT change partial record trigger
time-limit (mins)	The maximum time limit in minutes
volume-limit (Kbytes)	The data volume limit in Kbytes

Table 114: TDF charging profile statistics output fields (continued)

Label	Description
On-line Charging	The online charging
On-line charging type	The type of online charging used
vendor-specific-rg	—
vendor-specific-svc-id	—
exclude-zero-vol-cdr	The generation of CDRs that have zero volume
exclude-zero-vol-complete-cdr	The generation of CDRs that have zero volume (with zero duration or non-zero)
Number of profiles	The number of profiles being viewed

6.2.2.18 twag-charging

Syntax

twag-charging [*charging-profile-id*]

Context

show>mobile>profile>charging

Description

This command displays TWAG charging profile information.

Parameters

charging-profile-id

displays information about the specified profile

Output

The following output example shows TWAG charging profile information, and *TWAG charging profile statistics output fields* describes the fields.

Example output TWAG charging profile statistics

```
# show mobile-gateway profile charging twag-charging 1
=====
TWAG Charging profile
=====
Profile Id : 1
Description : TWAG-charging profile 1
Radius-accounting
Wlan-ap-location-change : enable/disable
-----
Number of profiles : 1
=====
```

Table 115: TWAG charging profile statistics output fields

Label	Description
Profile Id	The ID number of the profile being viewed
Description	The profile, if applicable
Radius-accounting	The context to configure RADIUS accounting
Wlan-ap-location-change	The triggering of RADIUS Accounting-Interim updates to the AAA server when the TWAG detects a change in the UE's WLAN AP location (due to mobility).

6.2.2.19 upf-acct-profile

Syntax

```
upf-acct-profile [profile-id]
```

Context

```
show>mobile>profile>charging
```


Description

This command displays UPF profile information.

Parameters

profile-id

displays information about the specified profile

Values: 0 to 255

Output

The following output example shows the profile information, and *UPF profile statistics output fields* describes the fields.

Example output UPF profile statistics

```
# show mobile-gateway profile charging upf-acct-profile 1
=====
UPF acct profile
=====
Profile Id : 1
faster-poll-remaining-quota-trigger : 24
-----
Number of profiles : 1
=====
```

Table 116: UPF profile statistics output fields

Label	Description
Profile Id	The ID of the profile being viewed
faster-poll-remaining-quota-trigger	The value of the faster poll remaining quota trigger, in Megabytes
Number of profiles	The number of profiles reported

6.2.2.20 dcca

Syntax

dcca [*profile-name*]

Context

```
show>mobile>profile
```

Description

This command displays DCCA profile information.

Parameters

profile-name

displays information about the specified existing profile

Output

The following output example shows DCCA Profile information, and *DCCA profile statistics output fields* describes the fields.

Example output DCCA profile statistics

```
# show mobile-gateway profile dcca "DCCA-Prof1"
=====
Dcca profile
=====
Profile name       : DCCA-Prof1
Description        : N/A
Application Tx Timer-msec : 6000 msec  Retry Count           : 0
CC Session Failover : enabled    CC Failover Handling    : terminate
fh-session-continue-timer : 30 mins  default-qht           : 0 mins
Command level FUJ   : disabled

Include AVP
3gpp-gprs-qos-neg-prof : disabled  called-station-id-virtual : real
QoS Information        : disabled  service-id               : enabled

Traffic Handling Options
forced-re-authorization : allow    quota-exhausted-no-thresho*: allow
quota-exhausted-threshold-*: allow
quota-unavailable       : allow
rating-condition-change  : allow    validity-time-expired     : allow

ps-fci-fh              : disabled  ccr-initial-on-first-pkt   : disabled
ps-fci-fh-container     : disabled
first-pkt-behavior      : enabled   pre-emptive-quota          : disabled
service-context-id      : N/A
supported-version       : 3gpp      gy-avp-options-profile    : 1
redirection-start-send-rr-*: disabled
activity-threshold      : 0

USU Options
with-quota              : none
no-quota-mid-session    : suppress
no-quota-initial        : suppress
```

```
ccrt-no-mscc           : enabled
usu-reporting-enh      : disabled  discard-overage       : disabled
implicit-vol-threshold : 0        reason-for-reporting  : threshold
ccr-mscc-limit        : 16
```

Local Event Triggers

```
rat-change             : disabled  user-loc-change       : disabled
qos-change             : disabled  serving-node-change   : disabled
qos-apn-ambr-change    : disabled  qos-max-bitrate-dl-change : disabled
qos-max-bitrate-ul-change : disabled change-ue-presence-pra : disabled
user-loc-change-cellid : disabled  user-loc-change-ecgi   : disabled
user-loc-change-lac    : disabled  user-loc-change-rac       : disabled
user-loc-change-tac    : disabled  trigger-override-option  : ocs
uwan-loc-change        : disabled
```

```
fh-volume-limit       : 0 (kilobytes)
ccfh-fh-action        : terminate
session-restore-retry-timer: 0 (minutes)
ccr-initial-on-first-online-rg : disabled
URL Encryption Type   : none
URL Passphrase        : (Not Specified)
supported-vendor      : none
```

```
ignore-invalid-thresholds : disabled
ignore-unsupported-triggers: disabled
redirect-filter-id       :
fui-validity-timer      : 0 (minutes)
ccfh-profile-id         : not-in-use
vol-threshold-reauth-enh : disabled
validity-timer-enh      : disabled
```

Failure Handling

```
result-code-3xxx : none          result-code-4xxx : none
result-code-5xxx : none
```

Multiple Services Credit Control (MSCC) Result Codes

```
Result Code      Service Delay Timer (mins)
```

```
append-url           : enabled  cca-init-no-mscc-answer-ig*: enabled
```

Default Rating Group List

```
Rating Group Id : 1
Rating Group Id : 2
Rating Group Id : 6
Rating Group Id : 7
Rating Group Id : 88
```

```
destination-realm      :
rating-group-in-ch-rule-re*: disabled
wait-for-first-packet-on-n*: disabled
serialize-ccr          : disabled
reauth-ccru-no-mscc    : disabled
```

```

suppress-multiple-ooc-notifications-per-rg : disabled
include-rpt-reason-on-forced-reaut       : disabled
wait-for-first-packet-on-fui-t-success   : disabled
diam-sess-end-serialize-ccr              : enabled

service-type          : all      ccfh-profile          : 1
-----
Number of profiles : 1
=====
•indicates that the corresponding row element may have been truncated.

```

Table 117: DCCA profile statistics output fields

Label	Description
Profile name	The ID number of the profile being viewed
Description	The profile, if applicable
Application Tx Timer-msec	The application transaction timer for CCR and CCA
msecs Retry Count	The limit to the number of times the system tries to send a RADIUS Disconnect message
CC Session Failover	The CC session failover support
CC Failover Handling	The CCFH support option
fh-session-continue-timer	The length of the failure handling session continue timer in minutes
default-qht	The default QHT
Command level FUI	The option to enable support for FUI AVP at the command level of a CCA message
Include AVP	The Include AVP context
3gpp-gprs-qos-neg-prof	The 3GPP-QoS-Negotiated-Profile AVP

Table 117: DCCA profile statistics output fields (continued)

Label	Description
called-station-id-virtual	—
QoS Information	The option to include QoS-Information AVP in the CCR
service-id	The option to include the service identifier information element in the DCCA messages sent by the PGW/GGSN
Traffic Handling Options	The Traffic Handling Options context
forced-re-authorization	The option to allow or discard the traffic for the service while waiting for a CCA response message from the OCS when a CCR message is sent to the OCS as a result of forced re-authorization by the OCS
quota-exhausted-no-thresho*	The option to allow or discard the traffic for the service while waiting for a CCA response message from the OCS when no threshold is active and the quota gets exhausted
quota-exhausted-threshold-*	The option to allow or discard the traffic for the service while waiting for a CCA response message from the OCS when there is an active threshold, but the quota gets exhausted while waiting for a CCA when the threshold is hit
quota-unavailable	The option to allow or discard the traffic for the service while waiting for a CCA response message from the OCS when no quota is available for a rating group
rating-condition-change	The online/offline tight integration trigger
validity-time-expired	The option to allow or discard the traffic for the service while waiting for a CCA response message from the OCS when the validity timer expires
ps-fci-fh	The custom failure handling applicable to a specific operator
ccr-initial-on-first-pkt	A CCR initial message on detection of a first packet

Table 117: DCCA profile statistics output fields (continued)

Label	Description
first-pkt-behavior	The first packet behavior
pre-emptive-quota	The support for pre-emptive quota handling
service-context-id	The service context ID (sent to OCS) IE
supported-version	The VF DCCA version
redirection-start-send-rr-*	The option to specify if an MSCC under redirection is reported when a DCCA session is terminated
gy-avp-options-profile	The profile charging identifier
max-number-of-pra	—
default-rating-group	The default rating group
activity-threshold	The activity-threshold for time-based metering
USU Options	The USU Options context
with-quota	The USU option for categories that the OCS has allocated quota
no-quota-mid-session	The option to include USU options for categories that the OCS has allocated quota to, but run out of credit mid-stream
no-quota-initial	The USU option for categories that the OCS has not allocated any quota to start with
ccrt-no-mscc	The option to enable reporting of MSCC instances in the CCR-U message, upon session release and not in the CCR-T message
usu-reporting-enh	Back to back reporting of USU with multiple CCR-U/CCA-U transactions, when the observed USU is greater than the GSU

Table 117: DCCA profile statistics output fields (continued)

Label	Description
discard-coverage	—
implicit-vol-threshold	An implicit-volume threshold applicable when the allocated quota does not include a quota threshold from the OCS
reason-for-reporting	The reporting reason for the implicit-volume threshold
ccr-mscc-limit	The maximum number of MSCC instances in a CCR message
Local Event Triggers	The Local Event Triggers Context
rat-change	The RAT change partial record trigger
user-loc-change	The user location change trigger
qos-change	The QoS change event trigger locally for online charging
serving-node-change	The serving node change event triggers locally for online charging
qos-apn-ambr-change	The qos-apn-ambr-change event trigger
qos-max-bitrate-dl-change	The qos-max-bitrate-downlink-change event trigger
qos-max-bitrate-ul-change	The qos-max-bitrate-uplink-change event trigger
change-ue-presence-pra	The change in UE presence in a PRA event trigger for online charging
user-loc-change-cellid	The change in location (cell identity) event trigger
user-loc-change-ecgi	The change in location (EUTRAN cell) global identifier event trigger
user-loc-change-lac	The change in location (location area code) event trigger

Table 117: DCCA profile statistics output fields (continued)

Label	Description
user-loc-change-rac	The change in location (routing area code) event trigger
user-loc-change-tac	The change in location (tracking area code) event trigger
trigger-override-option	The specified trigger override option
uwan-loc-change	The uwan-loc-change event trigger
fh-volume-limit	The FH maximum volume limit
ccfh-fh-action	The action to take, upon the expiration of the fh-session-continue-timer or when the limit set by fh-volume-limit is reached (whichever occurs first)
session-restore-retry-timer	The session restore retry timer
ccr-initial-on-first-online-rg	A CCR initial message on detection of a first online rating group
URL Encryption Type	The URL encryption type
URL Passphrase	The URL passphrase
supported-vendor	The number of the supported vendor
ignore-invalid-thresholds	The gateway ignores an invalid threshold sent by the OCS
ignore-unsupported-triggers	The gateway ignores unsupported event triggers to which the OCS subscribes
redirect-filter-id	A policy rule base that gets locally activated when a rating group is put into an FUI redirect state, provided that the OCS doesn't specify the filter ID.

Table 117: DCCA profile statistics output fields (continued)

Label	Description
fui-validity-timer	The time that the Final Unit Action (Terminate, Restrict-Access, and Redirect) is active
ccfh-profile-id	The ID of the CCFH profile
vol-threshold-reauth-enh	—
validity-timer-enh	The option to reduce the average amount of CCR-U messages
Failure Handling	The Failure Handling context
result-code-3xxx	The result code value 3xxx for the CCFH profile
result-code-4xxx	The result code value 4xxx for the CCFH profile
result-code-5xxx	The result code value 5xxx for the CCFH profile
Multiple Services Credit Control (MSCC) Result Codes	
Result Code Service Delay Timer (mins)	
append-url	Specifies if the UE URL is appended to the Redirect-Server URL in case of FUI redirection
cca-init-no-mscc-answer-ig*	Specifies the CMG behavior if no MSCC instance exists at the CCA-Init when a request for quota for that MSCC at the CCR-Init has been made
Default Rating Group List	The configured default rating-group list
service-type	A service type for Assume Positive behavior
ccfh-profile	A CCFH profile

Table 117: DCCA profile statistics output fields (continued)

Label	Description
serialize-ccr	The option to serialize CCR messages handling with OCS
suppress-multiple-ooc-notifications-per-rg	Specifies whether to send Out Of Credit notifications to the PCRF
reauth-ccru-no-mscc	Specifies whether to send an empty CCR-U message after a session-level RAR message
include-rpt-reason-on-forced-reaut	Includes the Reporting-Reason AVP with value Forced-Reauthorisation (7)
wait-for-first-packet-on-fui-t-success	Specifies the behavior for the traffic generated after the MSCC instance is terminated
diam-sess-end-serialize-ccr	Specifies whether the gateway waits for outstanding CCA-U messages (as a result of quota requests) before terminating the Diameter session.
Number of profiles	The number of profiles being viewed

Example output DCCA profile statistics discard on FPT traffic

```
# show mobile-gateway profile dcca "DCCA-Prof1"
=====
Dcca profile
=====
Profile name       : DCCA-Prof1
Description        : N/A
Application Tx Timer-msec : 6000 msecs  Retry Count           : 0
CC Session Failover : enabled    CC Failover Handling   : terminate
fh-session-continue-timer : 30 mins  default-qht           : 0 mins
Command level FUI   : disabled
Include AVP
3gpp-gprs-qos-neg-prof : disabled  called-station-id-virtual : real
QoS Information       : disabled  service-id               : enabled
Traffic Handling Options
forced-re-authorization : allow    quota-exhausted-no-thresho*: allow
quota-exhausted-threshold-*: allow
quota-unavailable       : discard-on-fpt
rating-condition-change  : allow    validity-time-expired    : allow
```

6.2.2.21 dhcp-server-group

Syntax

```
dhcp-server-group [profile-name]
```

Context

```
show>mobile>profile
```

Description

This command displays DHCP server group information.

Parameters

profile-name

displays information about the specified existing profile

Output

The following output example shows DHCP server group profile information.

Example output DHCP server group profile statistics

```
# show mobile-gateway profile dhcp-server-group "1"
=====
DHCP Server Group profile
=====
Profile name           : 1
Description            : 1
Interface              : N/A      Router                : Base
Dead Time              : 303      Retry Timeout         : 4
Retry Count            : 3        Renew Time Pct        : 44
Rebind Time Pct        : 69      Recommend Lease Time  : 3601
client-identifier      :          MSISDN-encoded:APN

Peer Address           : 1.2.3.4
Priority               : 4        Admin State           : down

via-up-function
APN                   : Nokia
up-peer-ip-address     : 0.0.0.0
-----
Number of profiles : 1
=====
```

6.2.2.22 dhcpv4-server

Syntax

```
dhcpv4-server [profile-name]
```

Context

```
show>mobile>profile
```

Description

This command displays DHCPv4 server information.

Parameters

profile-name

displays information about the specified existing profile

Output

The following output example shows DHCPv4 profile information, and *DHCPv4 server profile statistics output fields* describes the fields.

Example output DHCPv4 server profile statistics

```
# show mobile-gateway profile dhcpv4-server "dhcpProf_1"
=====
Dynamic Host Configuration Protocol version 4 (DHCPv4) server profile
=====
Profile name           : dhcpProf_1
Description            : (Not Specified)
Lease Time             : 2 hrs 0 mins 0 secs
Hold Time              : 1 mins 0 secs
Ignore rapid commit    : disabled
-----
Number of profiles : 1
=====
```

Table 118: DHCPv4 server profile statistics output fields

Label	Description
Profile name	The profile being viewed
Description	The profile, if necessary

Table 118: DHCPv4 server profile statistics output fields (continued)

Label	Description
Lease Time	The time in the lease
Hold Time	The time in the hold
Ignore rapid commit	The rapid commits should be ignored or not
Number of profiles	The number of profiles being viewed

6.2.2.23 diameter

Syntax

diameter [*profile-name*]

Context

show>mobile>profile

Description

This command displays Diameter profile information.

Parameters

profile-name

displays information about the specified existing profile

Output

The following output example shows Diameter profile information, and *Diameter profile statistics output fields* describes the fields.

Example output Diameter profile statistics

```
# show mobile-gateway profile diameter "diam-prof1"
=====
Diameter profile
=====
Profile name           : diam-prof1
```

```

Description          : N/A
Watchdog timer       : 30 secs   Connection timer      : 15 secs
Transaction timer    : 5 secs    DPR timeout           : 1800 secs
IP TTL               : 255       IP DSCP                : 56
Perm fail retry timeout : 30 mins DNS Refresh Interval  : 6 hrs
Fail num retries     : 3         Fail retry interval   : 45 decisec
-----
Number of profiles : 1
=====

```

Table 119: Diameter profile statistics output fields

Label	Description
Profile name	The profile being viewed
Watchdog timer	The maximum amount of time the node waits for a Diameter peer to respond to the watchdog packet
Connection timer	The maximum amount of time the node attempts to reconnect to a Diameter peer
Transaction timer	The maximum amount of time the node waits for a Diameter peer to respond before trying another peer
DPR timeout	The DPR timeout value
Retry count	The limit to the number of times the system tries to send a RADIUS Disconnect message
IP TTL	The IP TTL value
IP DSCP	The forwarding class for Diameter signaling packets sent
Perm fail retry timeout	The amount of time the node waits before retrying the Diameter connection setup
DNS Refresh Interval	The amount of time in hours that the node waits before sending a DNS query

Table 119: Diameter profile statistics output fields (continued)

Label	Description
Fail num retries	The number of attempts to re-establish the connection with a Diameter server
Fail retry interval	The time after which the mobile gateway re-attempts to establish the connection with a Diameter server
Number of profiles	The number of profiles being viewed

6.2.2.24 diameter-peer

Syntax

diameter-peer [*profile-name*]

Context

show>mobile>profile

Description

This command displays Diameter peer information.

Parameters

profile-name

displays information about the specified existing profile

Output

The following output examples show Diameter peer profile information, and *Show Diameter peer output fields* describes the output fields.

Example output Diameter peer profile statistics

```
# show mobile-gateway profile diameter-peer
=====
Diameter peer profile
=====
Profile name           : Peer-Profile1
```

```

Application type      : gx
Interface            : N/A
Diameter profile     : diam-prof1
Sig-Rate-Limit-Profile : Prof11
Description          : N/A
Diameter realm       : nokia.com
Custom Vendor Id     : N/A
Signaling            : TCP      Load balance      : disabled
SCTP multi-homing    : disabled
exclude-dest-host-after-switchover : included
Swo source port consistent : disabled
Validate Supported Host : disabled
disable-learnt-dest-realm : disabled   learnt-dest-realm-after-sw*: disabled
Peer address         : 10.10.43.2
Port                 : 3868
Enable-dra           : enabled
-----
Number of profiles : 1

```

```

# show mobile-gateway profile diameter-peer
=====
Diameter peer profile
=====
Profile name : aa
Application type : N/A
Interface : N/A
Diameter profile : default
Sig-Rate-Limit-Profile : N/A
Description : N/A
Diameter realm : N/A
Custom Vendor Id : N/A
Signaling : TCP Load balance : disabled
SCTP multi-homing : disabled
exclude-dest-host-after-switchover : included
Swo source port consistent : disabled
Validate Supported Host : disabled
disable-learnt-dest-realm : disabled
last-known-dest-realm-after-swo : disabled
Reconnect with new tcp port: disabled dest-host-table-threshold : 80
disable-cea-auth-app-id-check : disabled
disable-dwr-dwa-origin-host-length-check : disabled
Peer address : 2.2.2.2
Port : 2222
Enable-dra : disabled
Peer address : 1.1.1.1
Port : 1111
Enable-dra : disabled

```


Table 120: Show Diameter peer output fields

Label	Description
Profile name	The name of the profile
Application type	The type of application being used
Interface	The interface configured for this Diameter peer
Diameter profile	The Diameter profile being used
Sig-Rate-Limit-Profile	The rate limiting profile associated with this Diameter peer
Description	Description of the profile
Diameter realm	Location of the Diameter
Custom Vendor Id	The custom supported vendor-id associated with this Diameter peer
Signaling	The method of signaling being used; either TCP or SCTP based
Load balance	The option to enable load balancing
SCTP multi-homing	The status of SCTP multi-homing for diameter signaling
exclude-dest-host-after-switchover	The option to prevent the destination host from being populated after switchover
Swo source port consistent	The option to keep the same source port as previous on an intra-VNF or chassis-HA switchover
Validate supported host	The option to validate a received origin host from the server with the supported hosts configured within the Diameter peer profile
disable-learnt-dest-realm	The option to disable sending the learned realm of a Diameter peer within the destination-realm AVP in subsequent messages to a peer

Table 120: Show Diameter peer output fields (continued)

Label	Description
learnt-dest-realm-after-sw*	The option to enable or disable populating the learned destination realm after failover or switchover
disable-dwr-dwa-origin-host-length-check	The option to enable or disable the checking for invalid length of the origin-host and the origin-realm in DWR/DWA messages
Peer address	The IP address of the peer
Port	The port being used
Enable-dra	The option to enhance support for specific DRA-based network environments
Number of profiles	The number of Diameter peer profiles being viewed

6.2.2.25 diameter-error-mapping-profile

Syntax

diameter-error-mapping-profile *[profile-name]*

Context

show>mobile>profile

Description

This command is used to display the configured content of the specified Diameter error mapping profile.

Parameters

profile-name

specifies the name of the Diameter error mapping profile

Output

The following is a example output of Diameter error mapping profile statistics information, and [Show Diameter error mapping profile statistics output fields](#) describes the output fields.

Example output Diameter error mapping profile statistics

```
# show mobile-gateway profile diameter-error-mapping-profile "3412_diamap1"
=====
Diameter error map profile
=====
Profile                : 3412_diamap1
result-code            : 3xxx          gtp-error-code            : 73
result-code            : 4xxx          gtp-error-code            : 89
result-code            : 5xxx          gtp-error-code            : 93
result-code            : 3005          gtp-error-code            : 82
result-code            : 4002          gtp-error-code            : 84
result-code            : 5005          gtp-error-code            : 100
fh-tx-timeout          : 109
-----
Number of profiles : 1
=====
```

Table 121: Show Diameter error mapping profile statistics output fields

Label	Description
Profile	The profile being viewed
result-code	The result code value from the Diameter peer
gtp-error-code	The cause value sent back to the access node
fh-tx-timeout	The GTP error code that is sent back to the access node when no response is received from the peer node
Number of profiles	The number of profiles being viewed

6.2.2.26 diameter-peer-group-list

Syntax

diameter-peer-group-list [/list-name]

Context

show>mobile>profile

Description

This command displays Diameter peer group list information.

Parameters

list-name

displays information about the specified list

Output

The following output example shows Diameter peer group list information, and *Show Diameter peer group list output fields* describes the output fields.

Example output Diameter peer group list

```
# show mobile-gateway profile diameter-peer-group-list
=====
Diameter peer group list
=====
List name           : list1
Peer Group Id       : 1           Peer Group Priority       : 10
Primary Diameter Peer : Gy-Peer-Prof1
Secondary Diameter Peer : Gy-Peer-Prof2
Peer Group Id       : 2           Peer Group Priority       : 1
Primary Diameter Peer : Gy-Peer-Prof4
Secondary Diameter Peer : Gy-Peer-Prof5
-----
Number of profiles : 1
=====
```

Table 122: Show Diameter peer group list output fields

Label	Description
List name	The list being viewed
Peer Group Id	The peer group associated with this list
Peer Group Priority	The assigned priority level for a peer group
Primary Diameter Peer	The name of the primary Diameter peer

Table 122: Show Diameter peer group list output fields (continued)

Label	Description
Secondary Diameter Peer	The name of the secondary Diameter peer
Number of profiles	The number of profiles being viewed

6.2.2.27 dns-error-mapping-profile

Syntax

dns-error-mapping-profile *profile-name*

Context

show>mobile>profile

Description

This command displays the DNS error mapping profile information.

Parameters

profile-name

displays information about the existing specified DNS error mapping profile

Output

The following output example shows DNS error mapping profile information, and *Show DNS error mapping profile output fields* describes the output fields.

Example output DNS error mapping profile information

```
# show mobile-gateway profile dns-error-mapping-profile "dns-1"
=====
DNS error mapping profile
=====
Profile Name : dns-1
-----
SWu Action : send-info
Notify Error Type : 10500
Notify Error Description : -
Backoff-Timer Value : 1
Backoff-Timer Unit : 10-m
```

Number of DNS profiles : 1
=====

Table 123: Show DNS error mapping profile output fields

Label	Description
Profile Name	The name of the profile being viewed
SWu Action	The action taken related to the cause code
Notify Error Type	The notify error type
Notify Error Description	The notify error description, if applicable
Backoff-Timer Value	The value of the backoff timer
Backoff-Timer Unit	The unit for the <i>timer-value</i> parameter
Number of DNS profiles	The number of DNS profiles being viewed

6.2.2.28 dns-profile

Syntax

dns-profile [*profile-name*]

Context

show>mobile>profile

Description

This command displays DNS profile information.

Parameters

profile-name

displays information about the specified DNS profile

Output

The following output example shows DNS profile information, and *Show DNS profile output Fields* describes the output fields.

Example output DNS profile information

```
# show mobile-gateway profile dns-profile "prof1"
=====
DNS profile
=====
Profile                : prof1
Primary DNS            : 10.10.0.23
Secondary DNS          : 10.20.0.23
Tertiary DNS           : 10.30.0.23
IP time-to-live        : 5          IP DSCP                : 34
Msg Retransmit Timeout : 5          Msg Retransmit Retry Count : 3
-----
Number of DNS profiles : 1
=====
```

Table 124: Show DNS profile output Fields

Label	Description
Profile	The name of the profile being viewed
Primary DNS	The IP address of the primary DNS server
Secondary DNS	The IP address of the secondary DNS server
Tertiary DNS	The IP address of the tertiary DNS server
IP time-to-live	The IP TTL value for DNS lookup messages
IP DSCP	The IP DSCP/TOS value to be used in the IP header of a DNS request
Msg Retransmit Timeout	The time, in seconds, before a DNS request message is retransmitted
Msg Retransmit Retry Count	The number of retries before a DNS request message to the server fails

Table 124: Show DNS profile output Fields (continued)

Label	Description
Number of DNS profiles	The number of DNS profiles being viewed

6.2.2.29 dns-redirect-server

Syntax

```
dns-redirect-server [profile-name]
```

Context

```
show>mobile>profile
```

Description

This command displays DNS redirect server information.

Parameters

profile-name

displays information about the specified existing profile

6.2.2.30 emergency-profile

Syntax

```
emergency-profile [profile-name]
```

Context

```
show>mobile>profile
```

Description

This command displays information about the configured emergency profiles.

Parameters

profile-name

displays information about the specified emergency profile, up to 32 characters

Output

The following output example shows the configured emergency profile information, and *Show emergency profile output fields* describes the output fields.

Example output emergency profile information

```
# show mobile-gateway profile emergency-profile
=====
Emergency profile
=====
Profile Name           : emrprof1

Allow UDM SDM          : No           Block UDM UECM          : Yes
Default 5QI            : 5             Default AMBR UL Rate (kbps): 300000
Default AMBR DL Rate (kbps): 300000    Default ARP             : 1
-----
Profile Name           : emrprof2

Allow UDM SDM          : Yes          Block UDM UECM          : No
Default 5QI            : 5             Default AMBR UL Rate (kbps): 4096
Default AMBR DL Rate (kbps): 4096    Default ARP             : 1
-----
Number of emergency profiles : 2
=====
```

Table 125: Show emergency profile output fields

Label	Description
Profile Name	The name of the profile
Allow UDM SDM	If the UDM SDM service for emergency calls is enabled (Yes) or disabled (No)
Block UDM UECM	If the UDM UEContextManagement service for emergency calls is enabled (No) or disabled (Yes)
Default 5QI	The default 5QI value for the 5G emergency calls
Default AMBR UL Rate (kbps)	The default uplink bitrate for session AMBR
Default AMBR DL Rate (kbps)	The default downlink bitrate for session AMBR

Table 125: Show emergency profile output fields (continued)

Label	Description
Default ARP	The default ARP priority level for the 5G emergency calls
Number of emergency profiles	The number of emergency profiles

6.2.2.31 fa-ha-peer-list

Syntax

```
fa-ha-peer-list [profile-name]
```

Context

```
show>mobile>profile
```

Description

This command displays Fa peer list information.

Parameters

profile-name

displays information about the specified existing profile

6.2.2.32 fqdn-group

Syntax

```
fqdn-group [profile-name]
```

Context

```
show>mobile>profile
```

Description

This command display the configuration information for the specified FQDN group.

Parameters

profile-name

displays information about the specified existing profile

Output

The following output example shows FQDN group profile information, and *Show mobile profile output fields* describes the fields.

Example output FQDN profile statistics

```
# show mobile-gateway profile fqdn-group "fqdn-group1"
=====
FQDN Groups
=====
Group                : fqdn-group1
DNS Refresh Interval  : 00:10:00 (hr:min:sec)
IP Expiry Refresh Count : 3

FQDN 1                : host1.cm.com
FQDN 2                : host2.cm.com

Total FQDN Count      : 2
-----
FQDN Groups           : 1
=====
```

Table 126: Show mobile profile output fields

Label	Description
FQDN Groups	
Group	The FQDN group
DNS Refresh Interval	The amount of time between DNS Refreshes
IP Expiry Refresh Count	The times the IP expired and refreshed
FQDN 1	The location of FQDN 2
FQDN 2	The location of FQDN 2

Table 126: Show mobile profile output fields (continued)

Label	Description
Total FQDN Count	The number of FQDNs presented
FQDN Groups	The number of FQDN groups

6.2.2.33 gtp

Syntax

```
gtp [profile-name]
```

Context

```
show>mobile>profile
```

Description

This command enables the context to display PGW and SGW charging profile information.

Parameters

profile-name

displays information about the specified existing profile name

Output

The following output example shows PGW and SGW charging profile information, and [Show GTP profile output fields](#) describes the output fields.

Example output GTP profile statistics

```
# show mobile-gateway profile gtp "default"
=====
GTP profile
=====
Profile name           : default
Description            : N/A
Msg retransmit timeout-msec: 5000 ms   Msg retransmit retry count : 3
Keep alive interval    : 60 secs       Keep alive retry count    : 4
Keep alive T3 response  : 5 secs
IP TTL                 : 255           IP DSCP                   : 56
```

```

GTPv1 Configurable Cause Code IEs
APN Shutdown Cause Code      : 78      Username Missing Cause Code: 70
GW Suspend Cause Code        : 113     Sx MSG Timeout Cause Code  : 209

GTPv2 Configurable Cause Code IEs
APN Shutdown Cause Code      : 93      Username Missing Cause Code: 70
GW Suspend Cause Code        : 89      Sx MSG Timeout Cause Code  : 94

GTPv1 Configurable IEs
Recovery IE                  : Included  NSAPI IE                      : Excluded
PCO IE                       : Included  Charging GW Addr IE          : Included
Common Flags IE              : Excluded  APN Restriction IE           : Included
BCM IE                       : Included

GTPv2 Configurable IEs
PCO IE                       : Included  APCO IE                      : Included
Node Features IE             : Included  DDN IMSI IE                  : Included
DDN FTEID IE                 : Included  Access Point MAC address IE: Excluded
ARP IE                       : Included  Paging Priority               : Included
CSG Reporting Action          : Included  Change Reporting Action      : Included
Signaling Priority            : Included  RAN/NAS Cause IE             : Included
MO Exception Data Counter     : Excluded  Serving PLMN Rate Control    : Excluded
CN Operator Selection         : Included  ULI-Timestamp                : Included
TWAN Identifier              : Included  TWAN Id TimeStamp            : Included
Origination Timestamp         : Included  Maximum Wait Time            : Included
LCI IE                       : Included  OCI IE                      : Included
AAA Node Identifier           : Included
UP Selection Indication       : Included
non-3gpp-rat-type-pref       : wlan

GTPv2 Node Features
CIOT                          : Excluded

-----
Number of profiles : 1
=====

```

Table 127: Show GTP profile output fields

Label	Description
Profile name	The name of the profile being viewed
Description	Describes the profile, if applicable
Msg retransmit timeout	The time, in seconds, before a request message is retransmitted
Msg retransmit retry count	The number of retries until a request message to the server fails

Table 127: Show GTP profile output fields (continued)

Label	Description
Keep alive interval	The keep alive message interval, in seconds
Keep alive retry count	The number of retries until the path is declared to be down
Keep alive T3 response	The time, in seconds, until a message is retransmitted
IP TTL	The IP TTL value for GTP signaling messages
IP DSCP	The DSCP value in the IP header for GTP signaling messages
GTPv2 Configurable Cause Code IEs	The GTPv2 Configurable Cause Code IEs context
APN Shutdown Cause Code	The code for the Cause IE when the APN is in shutdown state
GW Suspend Cause Code	The code for the Cause IE when the gateway is in suspend state
Username Missing Cause Code	The code for the Cause IE when the PDN connection activation is rejected because a username or password is required for session activation
Sx MSG Timeout Cause Code	The code for the Cause IE in the following cases: • when the PDP Context activation is rejected due to Sx connection failure • the PDN Connection activation is rejected due to Sx connection failure
GTPv1 Configurable IEs	The GTPv1 Configurable IEs context
Recovery IE	The option to include or exclude the recovery IE from the PGW/GGSN create PDP context response, update PDP context request, and update PDP context response messages

Table 127: Show GTP profile output fields (continued)

Label	Description
NSAPI IE	The option to include or exclude the NSAPI IE in the PGW/GGSN create PDP context response messages
PCO IE	The option to include or exclude the PCO IE in the PGW/GGSN initiate PDP context request, create PDP context response, and update PDP context response messages
Charging GW Addr IE	The option to include or exclude the charging gateway address in the PGW/GGSN create PDP context response and update PDP context response messages
Common Flags IE	The option to include or exclude the common flags IE in the PGW/GGSN create PDP context response and update PDP context response messages
APN Restriction IE	The option to include or exclude the APN restriction IE in the PGW/GGSN create PDP context response, update PDP context request, and update PDP context response messages
BCM IE	The option to include or exclude the BCM IE in the PGW/GGSN messages
GTPv2 Configurable IEs	The GTPv2 Configurable IEs context
PCO IE	The option to include or exclude the PCO/ePCO IE in the GTPv2 messages
APCO IE	The option to include or exclude the APCO IE in the GTPv2 messages
Node Features IE	The option to include or exclude the node features IE in the PGW/GGSN messages
DDN IMSI IE	The option to include or exclude the DDN IMSI IE in the PGW/GGSN messages

Table 127: Show GTP profile output fields (continued)

Label	Description
DDN FTEID IE	The option to include or exclude the DDN FTEID IE in the PGW/GGSN
Access Point MAC Address IE	The option to cause the ePDG to send the WiFi Access Point MAC address if learned from the IDi format
ARP IE	The option to include or exclude the ARP IE in the PGW/GGSN messages
Paging Priority	The option to include or exclude of the paging and service information IE within a DDN message on an S11 interface
CSG Reporting Action	The option to include or exclude the CSG reporting action IE in GTPv2 messages
Change Reporting Action	The option to include or exclude the change reporting action IE in GTPv2 messages
Signaling Priority	The option to include or exclude the signaling priority indication IE on S5 and S8 interfaces
RAN/NAS Cause IE	The option to include or exclude the RAN/NAS cause IE in the GTPv2 messages
MO Exception Data Counter	The option to include or exclude the MO Exception Data Counter IE in GTPv2 messages
Serving PLMN Rate Control	The option to include or exclude the serving PLMN rate control IE in GTPv2 messages
CN Operator Selection	The option to include or exclude the sending of a CN Operator Selection Entity IE
ULI-Timestamp	The option to include or exclude the sending of a ULI Timestamp IE

Table 127: Show GTP profile output fields (continued)

Label	Description
TWAN Identifier	The TWAN identifier
TWAN Id Timestamp	The option to include or exclude the TWAN Timestamp Id
Origination Timestamp	The option to include or exclude the Origination Timestamp IE
Maximum Wait Time	The option to include or exclude the Maximum Wait Time IE
LCI	Load Control Indicator IE
OCI	Overload Control Indicator IE
AAA Node Identifier	The option to include or exclude the AAA node identifier IE
non-3gpp-rat-type-pref	The RAT type value that the ePDG sends to the PGW
GTPv2 Node Features	The GTPv2 Node Features context
CIOT	The option to include or exclude the CIOT flag in the Node Features IE
Number of Profiles	The number of profiles being viewed

6.2.2.34 gtp-prime-group

Syntax

```
gtp-prime-group [profile-name]
```

Context

```
show>mobile>profile
```

Description

This command displays GTP prime group profile information.

Parameters

profile-name

displays information about the specified existing profile

Output

The following example shows GTP prime group profile information, and *Show GTP prime group profile output fields* describes the output fields.

Example output GTP prime group profile information

```
# show mobile-gateway profile gtp-prime-group "gtp_server_group1"
=====
gtp-prime-group profile
=====
Profile name           : gtp_server_group1
Description            : N/A
Primary CF             : CF1
File closure-rec-num   : 50000      File closure-size       : 50
File closure-time      : 1 hours    File obsolete time      : 7 days
Max CDRs per PDU       : 0          Deadtime                : 0
Queue size             : 100000     Redirection             : disabled
CF1 limit              : 0          CF2 limit               : 0
Inactive time          : 10
Duplicate Prevention    : enabled    Dup Prevention Storage  : disabled
MTU                    : 1500       Fragment DRT            : NO
Round Robin load-balance : disabled  Cdr File Compression    : disabled
Gtp prime server address : 10.11.7.101
Port                   : 3386       Retries count           : 4
Timeout                : 20 secs    Echo-interval           : 60 secs
Max-requests           : 256        Priority                : 51
Path-protocol          : UDP        Node-alive              : disabled
Enable-gtpv2           : disabled
Gtp prime server address : 10.11.7.102
Port                   : 3386       Retries count           : 4
Timeout                : 20 secs    Echo-interval           : 60 secs
Max-requests           : 256        Priority                : 50
Path-protocol          : UDP        Node-alive              : disabled
Enable-gtpv2           : disabled
-----
Number of profiles : 1
=====
```

Table 128: Show GTP prime group profile output fields

Label	Description
Profile name	The name of the profile being viewed
Description	Describes the profile, if applicable
Primary CF	The primary compact flash
File closure-rec-num	The number of ACRs that can be stored in a file before the file is closed
File closure-size	The file closure size limit
File closure-time	The file closure lifetime before the file is closed
File obsolete time	The file duration before the file is deleted
Max CDRs per PDU	The maximum number of CDRs that can be placed in a single GTP Prime Server PDU
Deadtime	The time that a server is considered dead before it can be used again
Queue size	The maximum number of unsent GTP packets cached in the Mobile Gateway, waiting for at least one CGF to come up
Redirection	The option to enable the redirect function
CF1 limit	The limit, in megabytes, for an amount of memory on the compact flash
CF2 limit	The limit, in megabytes, for the amount of memory on the compact flash
Inactive time	The amount of time the peer must remain inactive to store all the cached GTP packets in the flash

Table 128: Show GTP prime group profile output fields (continued)

Label	Description
Duplicate prevention	The option to enable the duplicate prevention algorithm based on GTP prime signaling
Dup Prevention Storage	The option to specify whether duplicate CDRs are stored in permanent storage in case of failure
MTU	The size of the MTU for this GTP prime group profile
Fragment DRT	The option to enable fragmentation for this GTP prime group profile
Round Robin load-balance	The option to use round robin load-balance traffic among servers with equal priority
Cdr File Compression	The option to enable CDR file compression
GTP prime server address	The IP address of the GTP prime server
Port	The port being used
Retries count	The number of times the system tries to send a GTP Prime Server PDU to a CGF
Timeout	The interval between GTP Prime Server PDU retries
Echo-interval	The interval at which the system should send echo-requests for the GTP Prime Server PDUs used to send the CDRs
Max-requests	The maximum number of unacknowledged GTP PDUs before the system stops sending the CDRs
Priority	The priority of the GTP primary server
Path-protocol	The path protocol used for the GTP primary server

Table 128: Show GTP prime group profile output fields (continued)

Label	Description
Node-alive	The option to enable the GTP Node Alive mechanism
Enable-gtpv2	The option to enable GTPv2 to send initial messages
Number of profiles	The number of profiles being viewed

6.2.2.35 gx

Syntax

gx [*profile-name*]

Context

show>mobile>profile

Description

This command displays Gx profile information.

Parameters

profile-name

displays information about the specified existing profile

Output

The following example outputs show Gx profile information, and *Show Gx profile output fields* describes the output fields.

- *Example output Gx profile statistics*
- *Example output Gx profile statistics with disabled event triggers*
- *Example output Gx profile statistics — NetLoc*

Example output Gx profile statistics

```
# show mobile-gateway profile gx "CommonGxProfile"
=====
```

Gx profile

```

=====
Profile name           : CommonGxProfile
message               : ccr-i
gx-avp-options        : 0          retry-count           : 3
transaction-timer-msec : 5000      result-code-3xxx    : terminate
result-code-4xxx      : terminate  result-code-5xxx    : terminate
ap-action-profile-3xxx :
ap-action-profile-4xxx :
ap-action-profile-5xxx :
message               : ccr-u
gx-avp-options        : 0          retry-count           : 3
transaction-timer-msec : 5000      result-code-3xxx    : terminate
result-code-4xxx      : terminate  result-code-5xxx    : terminate
ap-action-profile-3xxx :
ap-action-profile-4xxx :
ap-action-profile-5xxx :
message               : ccr-t
gx-avp-options        : 0          retry-count           : 3
transaction-timer-msec : 5000      result-code-3xxx    : terminate
result-code-4xxx      : terminate  result-code-5xxx    : terminate
ap-action-profile-3xxx :
ap-action-profile-4xxx :
ap-action-profile-5xxx :
sponsored-connectivity : disabled  adc                : disabled
netloc                 : disabled  trusted-wlan        : disabled
ran-nas-cause          : disabled  mission-critical-qcis : disabled
conditional-policy-info : disabled
clear-m-bit             : disabled  fh-session-no-response : terminate
fh-volume-limit (kbytes) : 0       fh-time-limit (minutes) : 0
sgw-rest                : disabled  cno-uli             : disabled
pending-transaction     : disabled  pending-transaction-timer : 100 ms
pending-transaction-retry : 3
gx-to-gtp-error-mapping-profile :
pcscf-restoration      : disabled
netloc-untrusted-wlan   : disabled
app-based-charging      : disabled
rule-bound-to-def-bearer : enabled  cond-pol-info-default-qos : enabled
override-usage-report-value : disabled
qos-info-always-present : disabled
time-based-usg-monitoring : enabled  exclude-usage          : disabled
session-restore-retry-timer : 0 (mins)
-----
Monitoring-key          Threshold
-----
653321                  1000
-----
Event Triggers          :
tethering-start         : 1501      tethering-stop          : 1502
multiple-pra            : disabled
extended-bw-nr          : disabled
Forced Triggers
qos-change               : enabled
ambr-modification-failure : enabled
default-eps-bearer-qos-modification-failure : enabled
default-eps-bearer-qos-change : enabled

```

```

report-serving-node-supplied-cc          : disabled

ccr-i-on-csreq                          : enabled
serialize-ccr                          : disabled
include-last-known-uli                  : disabled

Validation Checks
raa-immediate                          : enabled
-----
Number of profiles : 1
=====

```

Table 129: Show Gx profile output fields

Label	Description
Gx profile	
Profile name	The name of the profile being viewed
message	The message type
gx-avp-options	The profile charging identifier
retry-count	The retry count for a message type
transaction-timer	The transaction timer, in seconds or milliseconds, for the Gx message profile
result-code-3xxx	The failure handling behavior for the 3xxx series of error codes
result-code-4xxx	The failure handling behavior for the 4xxx error codes
result-code-5xxx	The failure handling behavior for the 5xxx error codes
sponsored-connectivity	The option to enable the sponsored connectivity feature within the supported features AVP

Table 129: Show Gx profile output fields (continued)

Label	Description
adc	The option to enable the ADC feature within the supported features AVP
netloc	The option to enable inclusion of the netloc access support AVP over the Gx interface to the PCRF
trusted-wlan	The option to enable the trusted WLAN feature within the supported features AVP
ran-nas-cause	The option to enable the RAN-NAS cause feature within the supported features AVP
mission-critical-qcis	The option to enable the mission critical QCIs feature within the supported features AVP sent to the PCRF
conditional-policy-info	The option to enable the Conditional Policy Information feature within the supported features AVP sent to the PCRF
clear-m-bit	The option to enable clearing the M-Bit in the supported features AVP sent to the PCRF
fh-session-no-response	The failure handling behavior for new sessions that fail to get a response from the PCRF
fh-volume-limit (kbytes)	The volume limit, in kilobytes, for new sessions before the PDN session is terminated
fh-time-limit (minutes)	The time limit, in minutes, for new sessions before the PDN session is terminated
sgw-rest	The option to enable support for SGW restoration procedures
cno-uli	The option to enable the population of the Core Network Overload ULI feature bit within the supported features AVP sent to the PCRF

Table 129: Show Gx profile output fields (continued)

Label	Description
pending-transaction	The option to enable the population of the Pending Transaction feature bit within the supported features AVP sent to the PCRF
pending-transaction-timer	The pending transaction timer for the Diameter race condition handling feature
pending-transaction-retry	The pending transaction retry count for the Diameter race condition handling feature
gx-to-gtp-error-mapping-profile	The Gx to GTP error mapping profile name
pcscf-restoration	The option to enable the P-CSCF Restoration Enhancement feature bit setting within the supported features AVP sent to the PCRF
app-based-charging	The option to enable the Application Based Charging feature bit setting within the supported features AVP sent to the PCRF
time-based-usg-monitoring	The option to enable the time-based usage monitoring optional feature within the supported features AVP sent to the PCRF
exclude-usage	The option to exclude the corresponding service data flow for the volume and/or time measurement on an IP-CAN session within the supported features AVP sent to the PCRF
session-restore-retry-timer	Indicates the session restore retry timer in minutes
Monitoring-key	The option to enable specifying the activity threshold on a per monitoring-key basis
Threshold	The activity threshold that defines an average data rate under which the subscriber host is considered silent
serialize-ccr	The option to serialize CCR messages handling with PCRF

Table 129: Show Gx profile output fields (continued)

Label	Description
include-last-known-uli	The option to enable CMG to include the last known ULI in a CCR-U sent toward the PCRF
Event Triggers	
tethering-start	Indicates whether tethering start notification to the PCRF is enabled and the configured value
tethering-stop	Indicates whether tethering stop notification to the PCRF is enabled and the configured value
multiple-pra	Indicates whether the multiple PRA function is enabled
extended-bw-nr	Indicates whether support for high data rates for option-3x NR is enabled
Forced Triggers	
ambr-modifications-failure	Indicates whether reporting to the PCRF is enabled in case of an AMBR modification failure event
default-eps-bearer-qos-modification-failure	Indicates whether reporting to the PCRF is enabled in case of a default EPS bearer QoS modification failure
default-eps-bearer-qos-change	Indicates whether reporting to the PCRF is enabled in case of a default EPS bearer QoS change event
Validation checks	
raa-immediate	The option to enable upon receiving a RAR from the PCRF, the PGW to send a RAA immediately prior to validating the rules and/or rule bases sent in the RAR
Number of profiles	The number of profiles being viewed

Example output Gx profile statistics with disabled event triggers

```
# show mobile-gateway profile gx "forced-triggers-gx"
=====
Gx profile
=====
Profile name          : forced-triggers-gx
message              : ccr-i
gx-avp-options        : 0          retry-count          : 3
transaction-timer-msec : 5000      result-code-3xxx    : terminate
result-code-4xxx      : terminate  result-code-5xxx    : terminate
ap-action-profile-3xxx :
ap-action-profile-4xxx :
ap-action-profile-5xxx :
message              : ccr-u
gx-avp-options        : 0          retry-count          : 3
transaction-timer-msec : 5000      result-code-3xxx    : terminate
result-code-4xxx      : continue   result-code-5xxx    : terminate
ap-action-profile-3xxx :
ap-action-profile-4xxx :
ap-action-profile-5xxx :
message              : ccr-t
gx-avp-options        : 0          retry-count          : 3
transaction-timer-msec : 5000      result-code-3xxx    : terminate
result-code-4xxx      : terminate  result-code-5xxx    : terminate
ap-action-profile-3xxx :
ap-action-profile-4xxx :
ap-action-profile-5xxx :
sponsored-connectivity : disabled  adc                : disabled
netloc                 : disabled  trusted-wlan       : disabled
ran-nas-cause          : enabled   mission-critical-qcis : disabled
conditional-policy-info : enabled
clear-m-bit             : disabled  fh-session-no-response : terminate
fh-volume-limit (kbytes) : 0      fh-time-limit (minutes) : 0
sgw-rest                : enabled   cno-uli            : disabled
pending-transaction     : enabled   pending-transaction-timer : 100 ms
pending-transaction-retry : 1
gx-to-gtp-error-mapping-profile :
pcscf-restoration       : disabled
netloc-untrusted-wlan   : disabled
app-based-charging       : disabled
rule-bound-to-def-bearer : disabled  cond-pol-info-default-qos : disabled
override-usage-report-value : disabled
qos-info-always-present : disabled
time-based-usg-monitoring : disabled  exclude-usage        : disabled
Event Triggers          :
tethering-start         : 1501     tethering-stop        : 1502
multiple-pra             : disabled
extended-bw-nr           : disabled

Forced Triggers
qos-change                : disabled
ambr-modification-failure : disabled
default-eps-bearer-qos-modification-failure : disabled
default-eps-bearer-qos-change : disabled
```

Validation Checks

```
raa-immediate          : disabled
```

```
Number of profiles : 1
```

Example output Gx profile statistics — NetLoc

```
# show mobile-gateway profile gx "netloc"
```

Gx profile

```
=====
Profile name          : netloc
message              : ccr-i
gx-avp-options        : 1          retry-count          : 3
transaction-timer-msec : 5000      result-code-3xxx    : terminate
result-code-4xxx      : terminate result-code-5xxx      : terminate
ap-action-profile-3xxx :
ap-action-profile-4xxx :
ap-action-profile-5xxx :
message              : ccr-u
gx-avp-options        : 1          retry-count          : 3
transaction-timer-msec : 5000      result-code-3xxx    : terminate
result-code-4xxx      : terminate result-code-5xxx      : terminate
ap-action-profile-3xxx :
ap-action-profile-4xxx :
ap-action-profile-5xxx :
message              : ccr-t
gx-avp-options        : 1          retry-count          : 3
transaction-timer-msec : 5000      result-code-3xxx    : terminate
result-code-4xxx      : terminate result-code-5xxx      : terminate
ap-action-profile-3xxx :
ap-action-profile-4xxx :
ap-action-profile-5xxx :
sponsored-connectivity : disabled  adc              : disabled
netloc                 : enabled   trusted-wlan        : disabled
ran-nas-cause          : disabled  mission-critical-qcis : disabled
conditional-policy-info : disabled
clear-m-bit            : disabled   fh-session-no-response : terminate
fh-volume-limit (kbytes) : 0        fh-time-limit (minutes) : 0
sgw-rest               : disabled   cno-uli             : disabled
pending-transaction    : disabled   pending-transaction-timer : 100 ms
pending-transaction-retry : 3
gx-to-gtp-error-mapping-profile :
pcscf-restoration      : disabled
netloc-untrusted-wlan   : enabled
app-based-charging      : disabled
rule-bound-to-def-bearer : disabled cond-pol-info-default-qos : disabled
override-usage-report-value : disabled
qos-info-always-present : disabled
time-based-usg-monitoring : disabled exclude-usage        : disabled
Event Triggers         :
tethering-start         : 1501      tethering-stop        : 1502
multiple-pra            : disabled
extended-bw-nr          : disabled
Forced Triggers
```

```

qos-change                : enabled
ambr-modification-failure : enabled
default-eps-bearer-qos-modification-failure : enabled
default-eps-bearer-qos-change : enabled

```

Validation Checks

```

raa-immediate          : disabled

```

```

-----
Number of profiles : 1
=====

```

6.2.2.36 http2

Syntax

```
http2 [profile-name]
```

Context

```
show>mobile>profile
```

Description

This command displays HTTP/2 profile information.

Parameters

profile-name

displays information about the specified existing profile

Output

The following example output shows HTTP/2 profile information, and *Show HTTP/2 profile output fields* describes the output fields.

Example output HTTP/2 profile statistics

```

# show mobile-gateway profile http2 "http2prof"

=====
HTTP/2 profile
=====
Profile Name          : profhttp2

TLS Options:
Client TLS Profile    :
Server TLS Profile    :
Strict Validation     : dns-id uuid keyUsage extKeyUsage flags

HTTP2 Options:
Srv Max Streams Allowed : 256          Srv Stream Lifetime (secs) : 15

```

```

Ping period (secs)           : 60           Header Table Size (octets) : 4096
Init Window Size (octets)    : 524287       Max Hdr List Size (octets) : 2147483647
Max Frame Size (octets)      : 16384

IP Options:
IP DSCP                      : 256

Connection Timers:
Request Timeout (seconds)    : 20           Connection Timer (seconds) : 20
Fail Num Retries             : 20           Fail Retry Interval (secs) : 20
-----
Number of profiles : 1
=====

```

Table 130: Show HTTP/2 profile output fields

Label	Description
Profile name	The name of the profile being viewed
Client tls profile	The name of the client TLS profile
Server tls profile	The name of the server TLS profile
Strict Validation	–
HTTP2 Options	
Srv Max Streams Allowed	The maximum number of an HTTP/2 server streams allowed
Srv Stream Lifetime (secs)	The maximum number of lifetime of an HTTP/2 server streams in seconds
Ping period (secs)	The frequency of sending ping frames after a quiet period
Header Table Size (octets)	The header table size
Init Window Size (octets)	The initial window size
Max Hdr List Size (octets)	The maximum header list size

Table 130: Show HTTP/2 profile output fields (continued)

Label	Description
Max Frame Size (octets)	The maximum frame size
IP Options	
IP DSCP	The DSCP value or traffic class
Connection Timers	
request-timeout	The time that an HTTP/2 client waits for an HTTP/2 response from a server before timing out
srv-max-streams-allowed	The maximum number of concurrent streams sent to an HTTP/2 client in the SETTING message

6.2.2.37 ipsec-profile

Syntax

`ipsec-profile profile-name`

Context

`show>mobile>profile`

Description

This command displays IPsec profile information.

Parameters

profile-name

displays information about the specified existing profile name

Output

The following example output shows IPsec profile information, and *Show IPsec profile statistics output fields* describes the output fields.

Example output IPsec profile statistics

```
# show mobile-gateway profile ipsec-profile "ipsec-prof"
=====
IPsec profile
=====
Profile name          : ipsec-prof
Description           : none
ESP SA
-----
Lifetime              : 80000      NAT travers. dyn. updates: disabled
Rekey-margin          : 10         Replay-window           : 0
Proposal entry number 1
Authentication algorithm : sha1      Diffie-Hellman group      : 1
Encryption algorithm    : 3des       Extended sequence number : unsupported
Proposal entry number 2
Authentication algorithm : md5       Diffie-Hellman group      : none
Encryption algorithm    : 3des       Extended sequence number : unsupported
Proposal entry number 3
Authentication algorithm : md5       Diffie-Hellman group      : none
Encryption algorithm    : aes128ctr  Extended sequence number : unsupported
Proposal entry number 4
Authentication algorithm : sha256   Diffie-Hellman group      : none
Encryption algorithm    : aes192ctr  Extended sequence number : unsupported
IKE SA
-----
Cert entry            : 1
ePDG Private key      : epdgX.key
ePDG Certificate      : epdgX.cert
    Hash&Url           : https://epdg.nokia/ashes/
ddb7deaab1d65f07bde34ab5f937f2056863741c
Computed CA chain
    subCA-2            : https://epdg.nokia/ashes/
d1772dfdaa96e2ef1b8ff1513c251f6b6ef85ca2
    subCA-1            : https://epdg.nokia/ashes/
6715f9e559ebce2570f295d1df7d5ef5ea0e15c9
    rootCa             : https://epdg.nokia/ashes/
5397f8856afff5ed90ac814cc52f6c7e51c0d03d
Send chain            : 1
URL base address       : https://epdg.nokia/ashes
Admin state            : Up
Operation state        : Up
CERTREQ Preferred     : enabled
-----
Local identity         : vmg.ee.telcocloud.3gppnetwork.org
Lifetime              : 80000      Msg retransmit timeout   : 2
Rekey-margin          : 10         Msg retransmit retry cnt : 3
DPD interval          : 60         NAT traversal mode       : auto
Global half-open SA limit: 2000    Peer half-open SA limit  : 200
MOBIKE protocol support : enabled   Check return routability : disabled
NAT travers. dyn. updates: disabled
```



```

imsi-privacy-protection : disabled
PLMN BLACK/WHITE List   : enabled
Type                    : white
PLMN-list               : List-1
Error-code              : 11011
Backoff-Timer Value     : 10
Backoff-Timer Unit      : 10-h
IDi-realm-list          : realm-list-1
Proposal entry number 1
Authentication algorithm : md5           Diffie-Hellman group : 1
Encryption algorithm     : 3des
Proposal entry number 2
Authentication algorithm : sha1          Diffie-Hellman group : 2
Encryption algorithm     : 3des
Proposal entry number 3
Authentication algorithm : md5           Diffie-Hellman group : none
Encryption algorithm     : aes128ctr     Extended sequence number : unsupported
Proposal entry number 4
Authentication algorithm : sha256        Diffie-Hellman group : none
Encryption algorithm     : aes192ctr     Extended sequence number : unsupported
IKE-Redirect
Always-on                : enabled
Redirect-server-list      : redirect-list-1
established-sessions-redirect-rate : 2000
redirect-established-emergency : disabled
redirect-when-idle       : enabled
Overload-redirection     : enabled
Redirect-server-list      : redirect-list-1
redirect-established-emergency : enabled
Idi-realm-redirection    : enabled
Redirect-server-list      : idi-realm-redirect-list-1
Type-of-screening        : white
Source-ip-redirection    : enabled
white-list-redir-epdg    : 10.10.20.20
Redirect-server-list      : source-ip-list-1
Type-of-screening        : black
white-list-redir-epdg    : 10.10.20.20
-----
Number of IPsec profiles : 1
=====

```

Example output IPsec profile statistics ePDG with PLMN denylists/allowlists

```

=====
IPsec profile
=====
Profile name      : epdgProfile1
Description       : none

ESP SA
-----
Lifetime          : 84000          NAT travers. dyn. updates: disabled
Rekey-margin      : 10             Replay-window           : 0
Re-key            : enabled
Ignore-ue-rekey-requests : disabled
Proposal entry number 1

```

```

Integrity algorithm      : sha1      Diffie-Hellman group    : none
Encryption algorithm    : aes256cbc Extended sequence number : unsupported
Proposal entry number 2
Integrity algorithm      : sha1      Diffie-Hellman group    : none
Encryption algorithm    : aes128cbc Extended sequence number : unsupported
Proposal entry number 3
Integrity algorithm      : sha1      Diffie-Hellman group    : none
Encryption algorithm    : aes192ctr Extended sequence number : unsupported
Proposal entry number 4
Integrity algorithm      : sha1      Diffie-Hellman group    : none
Encryption algorithm    : aes256cbc Extended sequence number : unsupported

```

IKE SA

```

CERTREQ Preferred       : disabled
Cert entry              : 1
ePDG Private key        : second_server_key
ePDG Cert               : second_server
    Hash&Url            : -
Computed CA chain
    second4_root         : -
    second3_root         : -
    second2_root         : -
    second1_root         : -
    second_root          : -
Send chain              : 1
URL base address        : -
Admin state             : Up
Operational state       : Up
Local identity          : epdg.epc.mnc012.mcc310.pub.3gppnetwork.org
Lifetime                : 84000      Msg retransmit timeout   : 2
Rekey-margin            : 10         Msg retransmit retry cnt : 3
Re-key                  : enabled
Ignore-ue-rekey-requests : disabled
DPD interval           : 3600        NAT traversal mode      : auto
Global half-open SA limit: 10000     Peer half-open SA limit : 10000
MOBIKE protocol support : enabled   Check return routability : disabled
NAT travers. dyn. updates: disabled
imsi-privacy-protection : disabled
Fragmentation MTU Ipv4  : 0         Fragmentation MTU Ipv6  : 0
Device Id function       : enabled   Device Id type          : imeisv
Device Id reject if fail : false
Liveness Check Support   : disabled  Liveness Check Tmt (Sec): 1800

```

APN BLACK/WHITE List

```

Type                    : white
APN List                :
Error-code              : pdn-connect*
PLMN BLACK/WHITE List   : disabled
Type                    : white
PLMN-List               :
Error-code              : 11011
IDi-realm-list          :
Backoff-Timer           : disabled
Proposal entry number 1
Integrity algorithm      : sha1      Diffie-Hellman group    : 2
Encryption algorithm     : aes128cbc Pseudo-Random-Function : sha1

```

```

Proposal entry number 2
Integrity algorithm      : sha1          Diffie-Hellman group    : 2
Encryption algorithm    : aes128cbc     Pseudo-Random-Function : sha1
Proposal entry number 3
Integrity algorithm      : sha1          Diffie-Hellman group    : 2
Encryption algorithm    : aes192ctr     Pseudo-Random-Function : sha1
Proposal entry number 4
Integrity algorithm      : sha1          Diffie-Hellman group    : 2
Encryption algorithm    : aes256cbc     Pseudo-Random-Function : sha1

IKE-Redirect
-----
Always-on                : disabled
Redirect-server-list      :
Establish sess redirect rate: 500
Redirect Establish Emergency: disabled
Redirect When Idle        : disabled

Overload redirection      : disabled
Redirect-server-list      :
Redirect Establish Emergency: disabled

IDi Realm redirection     : disabled
Redirect-server-list      :
Type of screening         : white

Source IP redirection     : disabled
Redirect-server-list      :
Type of screening         : white
-----
Number of IPsec profiles : 1
=====
* indicates that the corresponding row element may have been truncated.

```

Table 131: Show IPsec profile statistics output fields

Label	Description
Profile name	The IPsec profile name
Description	Describes the profile, if applicable
ESP SA	The ESP SA context
Lifetime	The value, in seconds, of the ESP SA lifetime

Table 131: Show IPSec profile statistics output fields (continued)

Label	Description
NAT travers. dyn. updates	The option to enable the ePDG to dynamically update the source IP address and port for the ESP SA
Rekey-margin	The percentage of the lifetime to expire before the system attempts to re-key the SA
Replay-window	The size of the anti-replay window
Proposal entry number 1	The context for proposal entry number 1
Authentication algorithm	The algorithm to use for authentication
Diffie-Hellman group	The value of the Diffie-Hellman group
Encryption algorithm	The algorithm to use for encryption
Extended sequence number	The option to enable extended sequence numbers
Proposal entry number 2	The context for proposal entry number 2
Authentication algorithm	The algorithm to use for authentication
Diffie-Hellman group	The value of the Diffie-Hellman group
Encryption algorithm	The algorithm to use for encryption
Extended sequence number	The option to enable extended sequence numbers
Proposal entry number 3	The context for proposal entry number 3
Authentication algorithm	The algorithm to use for authentication
Diffie-Hellman group	The value of the Diffie-Hellman group

Table 131: Show IPSec profile statistics output fields (continued)

Label	Description
Encryption algorithm	The algorithm to use for encryption
Extended sequence number	The option to enable extended sequence numbers
Proposal entry number 4	The context for proposal entry number 4
Authentication algorithm	The algorithm to use for authentication
Diffie-Hellman group	The value of the Diffie-Hellman group
Encryption algorithm	The algorithm to use for encryption
Extended sequence number	The option to enable extended sequence numbers
IKE SA	The IKE SA context
Cert entry	The certificate entry ID
ePDG Private key	The ePDG Private key name
ePDG Cert	The ePDG certificate name
Hash&Url	The hash and URL string
Computed CA chain	The hash of each certificate of the ePDG certificate chain when the URL is used
Send chain	The number of chain certificates sent by the ePDG
URL base address	The URL string
Admin state	The administrator state of the certificate entry Id

Table 131: Show IPsec profile statistics output fields (continued)

Label	Description
Operational state	The operational state of the certificate entry Id
Local Identity	The local identity that the ePDG provides the UE
Lifetime	The value, in seconds, of the IKE SA lifetime
Msg retransmit timeout	The number of seconds the system waits before resending a message when no response is received
Rekey-margin	The percentage of the lifetime to expire before the system attempts to re-key the SA
Msg retransmit retry cnt	The number of times the system resends a control message when no response is received
DPD interval	The interval, in seconds, at which the ePDG sends DPD messages
NAT traversal mode	The IKE NAT traversal mode
Global half-open SA limit	The threshold for the global larval SA
Peer half-open SA limit	The threshold for the number of larval SA per peer
MOBIKE protocol support	The option to enable MOBIKE support on the ePDG
Check return routability	The option to enable check return routability MOBIKE
NAT travers. dyn. updates	The option to enable the ePDG to dynamically update the source IP address and port for the IKE SA
imsi-privacy-protection	The option to enable support for Apple's IMSI privacy protection in the SWu interface
PLMN BLACK/WHITE List	The option to enable the PLMN denylist/allowlist functionality

Table 131: Show IPSec profile statistics output fields (continued)

Label	Description
Type	The type of the list
PLMN-list	The name of the PLMN list
Error-code	The value of the error code that the ePDG sends when rejecting a UE attach request
Backoff-Timer Value	The value of the backoff timer
Backoff-Timer Unit	The unit for the <i>timer-value</i> parameter
IDI-realm-list	The name of the IDi realm list
Proposal entry number 1	The context for proposal entry number 1
Authentication algorithm	The algorithm to use for authentication
Diffie-Hellman group	The value of the Diffie-Hellman group
Encryption algorithm	The algorithm to use for encryption
Proposal entry number 2	The context for proposal entry number 2
Authentication algorithm	The algorithm to use for authentication
Diffie-Hellman group	The value of the Diffie-Hellman group
Encryption algorithm	The algorithm to use for encryption
Proposal entry number 3	The context for proposal entry number 3
Authentication algorithm	The algorithm to use for authentication

Table 131: Show IPSec profile statistics output fields (continued)

Label	Description
Diffie-Hellman group	The value of the Diffie-Hellman group
Encryption algorithm	The algorithm to use for encryption
Extended sequence number	The option to enable extended sequence numbers
Proposal entry number 4	The context for proposal entry number 4
Authentication algorithm	The algorithm to use for authentication
Diffie-Hellman group	The value of the Diffie-Hellman group
Encryption algorithm	The algorithm to use for encryption
Extended sequence number	The option to enable extended sequence numbers
IKE-Redirect	The IKE Redirect context
Always-on	The option to enable IKEv2 for all the sessions, new and already established
Redirect-server-list	The list name of the IKE redirect server list
established-sessions-redirect-rate	The redirection rate value of the established sessions in UEs per second
redirect-established-emergency	The option to enable the redirection of the established emergency calls
Idi-realm-redirection	The option to enable the ePDG to redirect a UE request based on the realm part of the provided IDi
Redirect-server-list	The list name of the IDi realm list

Table 131: Show IPSec profile statistics output fields (continued)

Label	Description
Type-of-screening	The type of IDi realm list screening
Source-ip-redirection	The option to enable redirection based on the source IP address of the incoming UE-originated IKE packets
Redirect-server-list	The list name of the source IP address list
Type-of-screening	The type of source IP address list screening
Number of IPSec profiles	The number of IPSec profiles being viewed
IPSec Profile Statistics ePDG with PLMN Denylists/Allowlists	
Profile name	The IPSec profile name
Description	Describes the profile, if applicable
ESP SA	The ESP SA context
Lifetime	The value, in seconds, of the ESP SA lifetime
NAT travers. dyn. updates	The option to enable the ePDG to dynamically update the source IP address and port for the ESP SA
Rekey-margin	The percentage of the lifetime to expire before the system attempts to re-key the SA
Replay-window	The size of the anti-replay window
Re-key	The option to enable the usage of re-key logic
Ignore-ue-rekey-requests	The option to enable the ePDG behavior when receiving UE initiated re-key requests for ESP SA

Table 131: Show IPSec profile statistics output fields (continued)

Label	Description
Proposal entry number 1	The context for proposal entry number 1
Integrity algorithm	The algorithm to use for authentication
Diffie-Hellman group	The value of the Diffie-Hellman group
Encryption algorithm	The algorithm to use for encryption
Extended sequence number	The option to enable extended sequence numbers
Proposal entry number 2	The context for proposal entry number 2
Integrity algorithm	The algorithm to use for authentication
Diffie-Hellman group	The value of the Diffie-Hellman group
Encryption algorithm	The algorithm to use for encryption
Extended sequence number	The option to enable extended sequence numbers
Proposal entry number 3	The context for proposal entry number 3
Integrity algorithm	The algorithm to use for authentication
Diffie-Hellman group	The value of the Diffie-Hellman group
Encryption algorithm	The algorithm to use for encryption
Extended sequence number	The option to enable extended sequence numbers
Proposal entry number 4	The context for proposal entry number 4
Integrity algorithm	The algorithm to use for authentication

Table 131: Show IPSec profile statistics output fields (continued)

Label	Description
Diffie-Hellman group	The value of the Diffie-Hellman group
Encryption algorithm	The algorithm to use for encryption
Extended sequence number	The option to enable extended sequence numbers
Proposal entry number 5	The context for proposal entry number 5
Integrity algorithm	The algorithm to use for authentication
Diffie-Hellman group	The value of the Diffie-Hellman group
Encryption algorithm	The algorithm to use for encryption
Extended sequence number	The option to enable extended sequence numbers
Proposal entry number 6	The context for proposal entry number 6
Integrity algorithm	The algorithm to use for authentication
Diffie-Hellman group	The value of the Diffie-Hellman group
Encryption algorithm	The algorithm to use for encryption
Extended sequence number	The option to enable extended sequence numbers
Proposal entry number 7	The context for proposal entry number 7
Integrity algorithm	The algorithm to use for authentication
Diffie-Hellman group	The value of the Diffie-Hellman group
Encryption algorithm	The algorithm to use for encryption

Table 131: Show IPSec profile statistics output fields (continued)

Label	Description
Extended sequence number	The option to enable extended sequence numbers
Proposal entry number 8	The context for proposal entry number 8
Integrity algorithm	The algorithm to use for authentication
Diffie-Hellman group	The value of the Diffie-Hellman group
Encryption algorithm	The algorithm to use for encryption
Extended sequence number	The option to enable extended sequence numbers
Proposal entry number 9	The context for proposal entry number 9
Integrity algorithm	The algorithm to use for authentication
Diffie-Hellman group	The value of the Diffie-Hellman group
Encryption algorithm	The algorithm to use for encryption
Extended sequence number	The option to enable extended sequence numbers
Proposal entry number 10	The context for proposal entry number 10
Integrity algorithm	The algorithm to use for authentication
Diffie-Hellman group	The value of the Diffie-Hellman group
Encryption algorithm	The algorithm to use for encryption
Extended sequence number	The option to enable extended sequence numbers
IKE SA	The IKE SA context

Table 131: Show IPSec profile statistics output fields (continued)

Label	Description
Local Identity	The local identity that the ePDG provides the UE
Lifetime	The value, in seconds, of the IKE SA lifetime
Msg retransmit timeout	The number of seconds the system waits before resending a message when no response is received
Rekey-margin	The percentage of the lifetime to expire before the system attempts to re-key the SA
Msg retransmit retry cnt	The number of times the system resends a control message when no response is received
Re-key	The option to enable the usage of re-key logic
Ignore-ue-rekey-requests	The option to enable the ePDG behavior when receiving UE initiated re-key requests for ESP SA
DPD interval	The interval, in seconds, at which the ePDG sends DPD messages
NAT traversal mode	The IKE NAT traversal mode
Global half-open SA limit	The threshold for the global larval SA
Peer half-open SA limit	The threshold for the number of larval SA per peer
MOBIKE protocol support	The option to enable MOBIKE support on the ePDG
Check return routability	The option to enable check return routability MOBIKE
NAT travers. dyn. updates	The option to enable the ePDG to dynamically update the source IP address and port for the IKE SA

Table 131: Show IPsec profile statistics output fields (continued)

Label	Description
imsi-privacy-protection	The option to enable support for Apple's IMSI privacy protection in the SWu interface
Fragmentation MTU IPv4	The IPv4 PDU size that governs the maximum size of an IKE fragment
Fragmentation MTU IPv6	The IPv6 PDU size that governs the maximum size of an IKE fragment
Device Id function	The option to enable the device identity functionality
Device Id type	The type of device identity asked from the ePDG toward the UE
Device Id reject if fail	The option to reject a call when device identity fails
Liveness Check Support	The value (enabled/disabled) of the option to provide Timeout Period to UE
Liveness Check Timeout	The timeout period in seconds
APN BLACK/WHITE List	The APN Denylist/Allowlist context
Type	The type of the list
APN List	The name of the APN list
Error-code	The value of the error code sent by the ePDG
PLMN BLACK/WHITE List	The option to enable the PLMN denylist/allowlist functionality
Type	The type of the list
PLMN-list	The name of the PLMN list

Table 131: Show IPSec profile statistics output fields (continued)

Label	Description
Error-code	The value of the error code that the ePDG sends when rejecting a UE attach request
Idi-realm-list	The name of the IDi realm list
Backoff-Timer	The option to enable the backoff timer
Proposal entry number 1	The context for proposal entry number 1
Integrity algorithm	The algorithm to use for authentication
Diffie-Hellman group	The value of the Diffie-Hellman group
Pseudo-Random-Function	The Pseudo-Random-Function algorithm
Proposal entry number 2	The context for proposal entry number 2
Integrity algorithm	The algorithm to use for authentication
Diffie-Hellman group	The value of the Diffie-Hellman group
Pseudo-Random-Function	The Pseudo-Random-Function algorithm
Proposal entry number 3	The context for proposal entry number 3
Integrity algorithm	The algorithm to use for authentication
Diffie-Hellman group	The value of the Diffie-Hellman group
Pseudo-Random-Function	The Pseudo-Random-Function algorithm
Proposal entry number 4	The context for proposal entry number 4

Table 131: Show IPSec profile statistics output fields (continued)

Label	Description
Integrity algorithm	The algorithm to use for authentication
Diffie-Hellman group	The value of the Diffie-Hellman group
Pseudo-Random-Function	The Pseudo-Random-Function algorithm
Proposal entry number 5	The context for proposal entry number 5
Integrity algorithm	The algorithm to use for authentication
Diffie-Hellman group	The value of the Diffie-Hellman group
Pseudo-Random-Function	The Pseudo-Random-Function algorithm
Proposal entry number 6	The context for proposal entry number 6
Integrity algorithm	The algorithm to use for authentication
Diffie-Hellman group	The value of the Diffie-Hellman group
Pseudo-Random-Function	The Pseudo-Random-Function algorithm
Proposal entry number 7	The context for proposal entry number 7
Integrity algorithm	The algorithm to use for authentication
Diffie-Hellman group	The value of the Diffie-Hellman group
Pseudo-Random-Function	The Pseudo-Random-Function algorithm
Proposal entry number 8	The context for proposal entry number 8
Integrity algorithm	The algorithm to use for authentication

Table 131: Show IPSec profile statistics output fields (continued)

Label	Description
Diffie-Hellman group	The value of the Diffie-Hellman group
Pseudo-Random-Function	The Pseudo-Random-Function algorithm
Proposal entry number 9	The context for proposal entry number 9
Integrity algorithm	The algorithm to use for authentication
Diffie-Hellman group	The value of the Diffie-Hellman group
Pseudo-Random-Function	The Pseudo-Random-Function algorithm
Proposal entry number 10	The context for proposal entry number 10
Integrity algorithm	The algorithm to use for authentication
Diffie-Hellman group	The value of the Diffie-Hellman group
Pseudo-Random-Function	The Pseudo-Random-Function algorithm
IKE-Redirect	The IKE Redirect context
Always-on	The option to enable IKEv2 for all the sessions, new and already established
Redirect-server-list	The list name of the IKE redirect server list
Establ sess redirect rate	The redirection rate value of the established sessions in UEs per second
Redirect Establ Emergency	The option to enable the redirection of the established emergency calls

Table 131: Show IPSec profile statistics output fields (continued)

Label	Description
Idi-realm-redirect	The option to enable the ePDG to redirect a UE request based on the realm part of the provided IDi
Redirect-server-list	The list name of the IDi realm list
Type-of-screening	The type of IDi realm list screening
Source-ip-redirect	The option to enable redirection based on the source IP address of the incoming UE-originated IKE packets
Redirect-server-list	The list name of the source IP address list
Type-of-screening	The type of source IP address list screening
Number of IPSec profiles	The number of IPSec profiles being viewed

6.2.2.38 list

Syntax

```
list
```

Context

```
show>mobile>profile
```

Description

This command displays mobile gateway list information.

6.2.2.39 aaa-host-realm

Syntax

```
aaa-host-realm [list-name]
```

Context

```
show>mobile>profile>list
```

Description

This command displays a list of configured AAA hosts and realms.

Parameters

list-name

displays information about the specified existing list

Output

The following output example shows the AAA host and realm information.

Example output AAA host and realm information

```
# show mobile-gateway profile list aaa-host-realm <set-1>
=====
AAA Hosts Realms
Profile name           : set-1
Entry Id   Host        Realm        diameter-peer-group-list
1          aaa-test.realm.com  realm.com  peer-group-aaa-1
2          aaa2-test2.realm.com realm.com  peer-group-aaa-2

# show mobile-gateway profile list aaa-host-realm
=====
AAA Hosts Realms
Profile name           : set-1
Entry Id   Host        Realm        diameter-peer-group-list
1          aaa-test.realm.com  realm.com  peer-group-aaa-1
2          aaa2-test2.realm.com realm.com  peer-group-aaa-2
Profile name           : set-2
Entry Id   Host        Realm        diameter-peer-group-list
1          aaa-test.dia.com    dia.com    peer-group-aaa-1
2          aaa2-test2.dia.com  dia.com    peer-group-aaa-2
```

6.2.2.40 apn-list

Syntax

```
apn-list [list-name]
```

Context

```
show>mobile>profile>list
```

Description

This command displays APN list information.

Parameters

list-name

displays information about the specified existing list

6.2.2.41 arp-list

Syntax

```
arp-list [arp-list-name]
```

Context

```
show>mobile>profile>list
```

Description

This command displays the ARP list information.

Parameters

arp-list-name

displays information about the specified existing list

Output

The following output examples show the ARP list information. *Show ARP list output fields with all the configured ARP Lists* describes the output fields with all the configured ARP lists, and *Show ARP list output fields with the excluded ARP names* describes the output fields with the excluded ARP names.

Example output APR list information with all the configured ARP lists

```
# show mobile-gateway profile list arp-list
=====
ARP List Profile
=====
ARP List Profile      : arpTestList
-----
ARP                   : 3
                      : 6
                      : 7
                      : 12
                      : 15
-----
ARP List Profile      : arpTestList2
```

```

-----
ARP                : 3
                  : 6
-----
ARP List Profile   : arpTestList3
-----
ARP                : 7
                  : 12
                  : 15
-----
Number of profiles : 3
=====

```

Table 132: Show ARP list output fields with all the configured ARP Lists

Label	Description
ARP List Profile	The profile of the ARP list
arpTestList	The list with all ARP lists
ARP	The number of ARPs
Number of profiles	The number of profiles

Example Output APR List information with the excluded ARP names

```

# show mobile-gateway profile list arp-list exclude-arp-list-name
=====
ARP List Profile
=====
ARP List Profile      : exclude-arp-list-name
-----
ARP                  : 6    message-priority : 6
                  : 7    message-priority : 7
-----
Number of profiles : 1
=====

```

Table 133: Show ARP list output fields with the excluded ARP names

Label	Description
ARP List Profile	The profile of the ARP list
exclude-arp-list-name	The list with the excluded ARP names
ARP	The number of ARPs
message-priority	Message priority only when configured for the ARP list
Number of profiles	The number of profiles

6.2.2.42 broadcast-plmn-list

Syntax

```
broadcast-plmn-list [list-name]
```

Context

```
show>mobile>profile>list
```

Description

This command displays the broadcast PLMN lists.

Parameters

list-name

displays information about the specified list, up to 32 characters

Output

The following output examples show the broadcast PLMN list information and [Show broadcast PLMN list information output fields](#) describes the output fields.

Example output broadcast PLMN list information

```
# show mobile-gateway profile list broadcast-plmn-list
```

```
=====
Broadcast PLMN List(s)
=====
Broadcast PLMN List : bpl1
-----
Broadcast PLMN Id      PLMN MCC      PLMN MNC      TAI Slice Support List
-----
1                      111          22            sliceList1
2                      222          22            sliceList2
-----

Broadcast PLMN List : bpl2
-----
Broadcast PLMN Id      PLMN MCC      PLMN MNC      TAI Slice Support List
-----
1                      111          11            sliceList1
2                      222          22            sliceList2
-----

Number of Broadcast PLMN lists : 2
=====
```

```
# show mobile-gateway profile list broadcast-plmn-list bplmn-list-1

=====
Broadcast PLMN List(s)
=====
Broadcast PLMN List: bplmn-list-1
-----
Broadcast PLMN Id      PLMN MCC      PLMN MNC      TAI Slice Support List
-----
1                      320          210           slice-list-1
2                      458          677           slice-list-2
-----

Number of Broadcast PLMN lists : 1
=====
```

Table 134: Show broadcast PLMN list information output fields

Label	Description
Broadcast PLMN List	The name of the broadcast PLMN list

Table 134: Show broadcast PLMN list information output fields (continued)

Label	Description
Broadcast PLMN Id	The ID of the broadcast PLMN item in the list
PLMN MCC	The MCC value of the PLMN
PLMN MNC	The MNC value of the PLMN
TAI Slice Support List	The name of the TAI slice support list

Example Output APR List information with the excluded ARP names

```
# show mobile-gateway profile list arp-list exclude-arp-list-name
=====
ARP List Profile
=====
ARP List Profile      : exclude-arp-list-name
-----
ARP                  : 6      message-priority : 6
                    : 7      message-priority : 7
-----
Number of profiles : 1
=====
```

Table 135: Show ARP list output fields with the excluded ARP names

Label	Description
ARP List Profile	The profile of the ARP list
exclude-arp-list-name	The list with the excluded ARP names
ARP	The number of ARPs
message-priority	Message priority only when configured for the ARP list
Number of profiles	The number of profiles

6.2.2.43 cell-id-list

Syntax

```
cell-id-list [cell-id-list-name]
```

Context

```
show>mobile>profile>list
```

Description

This command displays the cell-id-list(s).

Parameters

cell-id-list-name

the cell-id-list name, up to 32 characters

Output

The following output example shows the cell-id-list(s) and [Show cell-id list information output fields](#) describes the output fields.

Example output cell-id lists information

```
show mobile-gateway profile list cell-id-list
```

```
=====
Cell ID List(s)
=====
```

```
-----
List Name : cell1
-----
```

MCC	MNC	NCI	NCI Range Start	NCI Range End
111	11	0x123456789		

```
-----
Number of Cell ID entries : 1
-----
```

```
-----
List Name : cell2
-----
```

MCC	MNC	NCI	NCI Range Start	NCI Range End
-----	-----	-----	-----------------	---------------

```

111          11          0x123456789          0x123456789
-----
Number of Cell ID entries : 1
-----
Number of Cell ID lists   : 2
=====

```

Table 136: Show cell-id list information output fields

Label	Description
List Name	The name of the cell-id-list
MCC	The MCC value of the NCGI
MNC	The MNC value of the NCGI
NCI	The NR cell-id value of the NCGI
NCI Range Start	The start value of the NR cell-id range
NCI Range End	The end value of the NR cell-id range
Number of Cell ID entries	The number of the cell-id entries in the list
Number of Cell ID lists	The number of the cell-id lists displayed

6.2.2.44 fqdn-list

Syntax

```
fqdn-list [list-name]
```

Context

```
show>mobile>profile>list
```

Description

This command displays FQDN list information.

Parameters

list-name

displays information about the specified existing list

Output

The following output example shows FQDN list information, and [Show FQDN list output fields](#) describes the output fields.

Example output FQDN list information

```
# show mobile-gateway profile list fqdn-list
=====
FQDN-List
=====
List-Name       : fqdnList
FQDNs           : 1 yahoo.com
                  : 2 google.com
-----
List-Name       : fqdnList1
FQDNs           : 1 imdb.com
                  : 2 gatech.edu
-----
Number of FQDN-Lists : 2
=====
```

Table 137: Show FQDN list output fields

Label	Description
FQDN-List	The FQDN-List context
List-Name	The name of the FQDN list
FQDNs	The FQDN names in an FQDN list
Number of FQDN-Lists	The number of FQDN lists

6.2.2.45 nf-id-list

Syntax

```
nf-id-list list-name
```

Context

```
show>mobile>profile>list
```

Description

This command displays NF ID list information.

Parameters

list-name

specifies the NF ID list name, up to 32 characters

Output

The following output example shows NF ID list information, and [Show NF ID list output fields](#) describes the output fields.

Example output NF ID list information

```
# show mobile-gateway profile list nf-id-list
=====
Network Function List(s)
=====
List-Name       : list1

NF Id           : 1 (enabled)
  Uuid          : -
  Address List  : pList1
  Priority      : -
  API prefix    : -
  FQDN          :
  api-version   : 0.0.0
  NF set        : -
  Peer selection : enabled
-----
Number of lists: 1
=====
```

Table 138: Show NF ID list output fields

Label	Description
List-Name	The name of the NF ID list
NF Id	The NF ID
Uuid	The UUID associated with the NF ID
Address list	The name of the prioritized IP address list
Priority	The priority value
API prefix	The NF ID API prefix
FQDN	The FQDN name
api-version	The major API version
Peer selection	The status of the peer selection (enabled or disabled for selection from the list)
NF set	The name of the NF ID list with the backup NFs (AMF)
Number of lists	The number of lists displayed

6.2.2.46 oi-override-mapping

Syntax

```
oi-override-mapping [list-name]
```

Context

```
show>mobile>profile>list
```

Description

This command displays the contents of an Operator Identifier override mapping list.

Parameters

list-name

displays information about the specified list

Output

The following output example shows `oi-override-mapping` command list information, and [Show OI override mapping list output fields](#) describes the output fields.

Example output OI override mapping list information

```
# show mobile-gateway profile list oi-override-mapping
=====
OI Override list
=====
List name: list1
-----
Subscriber          Replacement          Serving Network
MCC                MNC                MCC                MNC                MCC                MNC
-----
111                111                111                111                111                111
222                222                222                222                222                222
-----
Number of lists : 2
```

Table 139: Show OI override mapping list output fields

Label	Description
OI Override list	The OI Override list context
List name	The name of the list
Subscriber MCC	The subscriber MCC value
Subscriber MNC	The subscriber MNC value
Replacement MCC	The replacement MCC value

Table 139: Show OI override mapping list output fields (continued)

Label	Description
Replacement MNC	The replacement MNC value
Serving Network MCC	The replacement MCC value for serving network GTPv2 IE
Serving Network MNC	The replacement MNC value for serving network GTPv2 IE
Number of lists	The number of lists

6.2.2.47 pgw-selection-candidate-list

Syntax

```
pgw-selection-candidate-list [list-name]
```

Context

```
show>mobile>profile>list
```

Description

This command displays PGW selection candidate list information.

Parameters

list-name

displays information about the specified existing list

6.2.2.48 plmn-list

Syntax

```
plmn-list [list-name]
```

Context

```
show>mobile>profile>list
```

Description

This command displays PLMN list information.

Parameters

list-name

displays information about the specified existing list

6.2.2.49 port-list

Syntax

```
port-list [list-name]
```

Context

```
show>mobile>profile>list
```

Description

This command displays port list information.

Parameters

list-name

displays information for all the port lists configured for the list profile, or for the particular port list, if specified

Output

The following example output shows port list information, and [Show port list output fields](#) describes the output fields.

Example output port list information

```
# show mobile-gateway profile list port-list
=====
port list profile
=====
port list profile: list-name-1
-----
port: 80, 8080
port range: 50 75
-----
Number of profiles : 1
=====
```


Table 140: Show port list output fields

Label	Description
port list profile	The port list profile context
port list profile	The name of the port list profile
port	The port number
port range	The port range
Number of profiles	The number of profiles

6.2.2.50 prioritized-ip-address-list

Syntax

prioritized-ip-address-list *list-name*

Context

show>mobile>profile>list

Description

This command displays prioritized IP address list information.

Parameters

list-name

specifies the prioritized IP address list name, up to 32 characters

Output

The following example output shows prioritized IP address list information, and *Prioritized IP address list output fields* describes the output fields.

Example output prioritized IP address list information

```
# show mobile-gateway profile list prioritized-ip-address-list
```

```

Prioritized IP Address List(s)
=====
-----
List Name : prio-list-1
-----
IP Address                                Port          Priority
-----
1.1.1.1                                  8080          1
2.2.2.2                                  8081          2

Number of IP Addresses      : 2
-----
-----
List Name : prio-list-2
-----
IP Address                                Port          Priority
-----
3.3.3.3                                  80            1

Number of IP Addresses      : 1
-----
Number of IP Address lists : 2
=====

```

Table 141: Prioritized IP address list output fields

Label	Description
List Name	The prioritized IP address list name
IP Address	The IP addresses in the list
Port	The port numbers in the list
Priority	The priorities in the list
Number of IP Addresses	The number of IP addresses in the list
Number of IP Address Lists	The number of prioritized IP address lists

6.2.2.51 qci-list

Syntax

```
qci-list [/list-name]
```

Context

```
show>mobile>profile>list
```

Description

This command displays QCI list information.

Parameters

list-name

displays information about the specified existing QCI list

Output

The following example output shows QCI list information.

Example output QCI list information

```
# show mobile-gateway profile list qci-list
=====
QCI list profile
=====
Qci List profile: qci-list-1
-----
QCI : 6
    :9
-----
Number of profiles : 1
=====
```

6.2.2.52 slice-instance-list

Syntax

```
slice-instance-list [/list-name]
```

Context

```
show>mobile>profile>list
```

Description

This command displays slice instance list information.

Parameters

list-name

specifies the slice instance list name, up to 32 characters

Output

The following output example shows slice instance list information, and [Show slice instance list output fields](#) describes the output fields.

Example output slice instance list information

```
# show mobile-gateway profile list slice-instance-list
=====
Network Slice Instance List(s)
=====
Slice Instance List      : nokia-smf-slices
-----
slice inst name          |      Suspended
-----
instance1                |      Yes
instance2                |      No
...                      |
-----

Number of Slice Instances : 3
-----
Nbr of Slice Instance lists: 1
=====
```

Table 142: Show slice instance list output fields

Label	Description
Slice Instance List	The name of the slice instance list
slice inst name	If the slice instance is in suspended state or not
Number of Slice Instances	The number of slice instances in the list
Number of Slice Instance lists	The number of slice instance lists displayed

6.2.2.53 slice-list

Syntax

```
slice-list [list-name]
```

Context

```
show>mobile>profile>list
```

Description

This command displays slice list information.

Parameters

list-name

specifies the slice list name, up to 32 characters

Output

The following output example shows slice list information, and [Show slice list output fields](#) describes the output fields.

Example output slice list information

```
# show mobile-gateway profile list slice-list
=====
Network Slice List(s)
=====
Slice List           : nokia-smf-slices
-----
Slice Name           : slice1
  Slice Service Type  : 1
  Slice Differentiator : 123456
  Slice Instances     : slice-instance-list1
  Suspended           : No

Slice Name           : slice2
  Slice Service Type  : 2
  Slice Differentiator : -
  Slice Instances     : -

Number of Slices      : 2
-----
Number of Slice lists : 1
=====
```

Table 143: Show slice list output fields

Label	Description
Slice List	The name of the slice list
Slice Name	The name of the slice
Slice Service Type	The slice service type
Slice Differentiator	The slice differentiator
Slice Instance	The slice instance
Suspended	If the slice instance is in suspended state or not
Number of Slices	The number of slices in the list
Number of Slice lists	The number of slice lists displayed

6.2.2.54 ue-usage-type-list

Syntax

```
ue-usage-type-list [list-name]
```

Context

```
show>mobile>profile>list
```

Description

This command displays the UE usage type list information.

Parameters

list-name

specifies information about a specified list, up to 32 characters

Output

The following output example shows UE usage type list information.

Example output usage type list information

```
# show mobile-gateway profile list ue-usage-type-list
=====
UE Usage Type List Profile
=====
UE Usage Type List Profile : usagetype1
-----
Usage-Type : 3
            : 6
            : 7
            : 12
            : 15
-----
UE Usage Type List Profile : usagetype2
-----
Usage-Type : 56
            : 78
            : 93
            : 220
-----
Number of profiles : 2
=====
```

```
# show mobile-gateway profile list ue-usage-type-list usagetype1
=====
UE Usage Type List Profile
=====
UE Usage Type List Profile : usagetype1
-----
Usage-Type : 3
            : 6
            : 7
            : 12
            : 15
-----
Number of profiles : 1
=====
```

6.2.2.55 mipv4

Syntax

mipv4 [*profile-name*]

Context

show>mobile>profile

Description

This command enables the context to display Pi profile information.

Parameters

profile-name

displays information about the specified existing profile

Output

The following output example shows MIPv4 profile information.

Example output MIPv4 profile statistics

```
# show mobile-gateway profile mipv4 "default"

=====
Mobile IPv4 (MIPv4) profile
=====
Profile Name           : default
Description            : (Not Specified)
IP DSCP                : 56
NEMO                   : disabled
MN-HA-AUTH             : always
MN-AAA-AUTH            : always
Registration Lifetime  : 7200 secs
Tunnel Mode            :

Revocation Settings:
Revocation              : disabled   I-Bit Negotiation      : disabled
Max Retransmission     : 3          Retransmission Timeout : 3 secs

DMU Settings:
DMU                     : disabled   DMU Tolerance          : 0 secs
DMU Hash Algorithm      : rfc2002-md5

-----
MN-HA Keys
-----
SPI Number  Hash Algorithm  Tolerance (sec)  Description
-----
-----
-----
Number of profiles : 1
=====
```


6.2.2.56 mme-group

Syntax

mme-group [*MME-group-id*]

Context

show>mobile>profile

Description

This command enables the context to display new MME group profile information.

Parameters

MME-group-id

specifies the group name identifier

Values: 1 to 65535

Output

The following output example shows MME group profile information, and *Show mobile profile MME group output fields* describes the fields.

Example output MME group ID profile

```
# show mobile-gateway profile mme-group
=====
MME Group profile
=====
Profile name           : 1
Peer Address           : 5.5.5.5
Peer Address           : 6.6.6.6
Peer Address           : ffff::1244:aa12
Peer Address           : ffff::1244:bb34
-----
Profile name           : 1043
Peer Address           : 1.1.1.1
Peer Address           : 2.2.2.2
Peer Address           : 3.3.3.3
Peer Address           : 4.4.4.4
-----
Number of profiles : 2
=====
```

Table 144: Show mobile profile MME group output fields

Label	Description
MME Group profile	
Profile name	The name of the profile being viewed
Peer Address	The IP address of a peer
Number of profiles	The number of profiles being viewed

6.2.2.57 n1-profile

Syntax

n1-profile *profile-name*

Context

show>mobile>profile

Description

This command enables the context to display N1 profile configuration.

Parameters

profile-name

specifies the name of the N1 profile, up to 32 characters

Output

The following output example shows N1 profile configuration, and *Show mobile N1 profile output fields* describes the fields.

Example output N1 profile

```
# show mobile-gateway profile n1-profile
=====
N1 profile
=====
Profile Name           : 1
```

```

n1-t3591 (deciseconds)      : 160      n1-t3592 (deciseconds)      : 160
n1-t3591 retry count        : 4         n1-t3592 retry count        : 4

```

Cause Codes

APN Shutdown

```

5GSM Cause Code           : 8
HTTP Status Code          : 403
Application Error          : DNN_NOT_SUPPORTED

```

DNN Subscription Removed

```

5GSM Cause Code           : 29

```

GW Suspend

```

5GSM Cause Code           : 8
HTTP Status Code          : 500
Application Error          : NF_FAILOVER

```

```

-----
Profile Name                : default

```

```

n1-t3591 (deciseconds)      : 160      n1-t3592 (deciseconds)      : 160
n1-t3591 retry count        : 4         n1-t3592 retry count        : 4

```

Cause Codes

APN Shutdown

```

5GSM Cause Code           : 8
HTTP Status Code          : 403
Application Error          : DNN_NOT_SUPPORTED

```

DNN Subscription Removed

```

5GSM Cause Code           : 29

```

GW Suspend

```

5GSM Cause Code           : 8
HTTP Status Code          : 500
Application Error          : NF_FAILOVER

```

```

-----
Number of N1 profiles : 2
=====

```

Table 145: Show mobile N1 profile output fields

Label	Description
N1 profile	
Profile name	The name of the profile being viewed

Table 145: Show mobile N1 profile output fields (continued)

Label	Description
n1-t3591	The T3591 timer, which defines the retransmission timeout for the PDU SESSION MODIFICATION COMMAND message
n1-t3592	The T3592 timer, which defines the retransmission timeout for the PDU SESSION RELEASE COMMAND message
n1-t3591 retry count	The number of retransmissions for a PDU SESSION MODIFICATION COMMAND message over the N11 interface
n1-t3592 retry count	The number of retransmissions for a PDU SESSION RELEASE COMMAND message over the N11 interface
5GSM Cause Code	The 5GSM Cause IE in the N1 PDU Session Establishment Reject message
HTTP Status Code	The HTTP status code in the N11 Create SM Context Response or N16/N16a Create Response messages
Application Error	The application error in the N11 Create SM Context Response or N16/N16a Create Response messages
Number of N1 profiles	The number of profiles being viewed

6.2.2.58 n2-peer

Syntax

n2-peer *profile-name*

Context

show>mobile>profile

Description

This command displays the N2 peer information.

Parameters

profile-name

specifies the name of the N2 peer profile, up to 32 characters

Output

The following output example shows the N2 peer profile configuration, and *Show N2 peer profile output fields* describes the fields.

Example output N2 peer profile

```
# show mobile-gateway profile n2-peer
```

```
=====
N2 Peer Profile
=====
```

```
Profile name   : n2-Peer-Profile1
SCTP Profile   : n2-prof1
PLMN List Name : plmn-list-1
Peer Address   : 10.10.43.2
```

```
-----
Number of profiles : 1
=====
```

Table 146: Show N2 peer profile output fields

Label	Description
N2 peer profile	
Profile name	The name of the profile
SCTP Profile	The name of the SCTP profile
PLMN List Name	The name of the PLMN list
Peer Address	The IP address of the peer

6.2.2.59 network-trace-profile

Syntax

```
network-trace-profile profile-name
```

Context

```
show>mobile>profile
```

Description

This command displays information of the network trace profile.

Parameters

profile-name

specifies the name of the network trace profile, up to 32 characters

Output

The following output example shows the network trace profile configuration, and *Show network trace profile output fields* describes the fields.

Example output network trace profile

```
# show mobile-gateway profile network-trace-profile
```

```
=====
Network Trace Profile
=====
```

```
Profile Name           : network_trace_1
```

```
Packet Broker Server Options
```

```
Packet Broker Server Address : 10.10.1.255
```

```
Packet Broker Server Port    : 80
```

```
Packet Broker Server Path    : server_path_1
```

```
HTTP/2 Options
```

```
HTTP/2 Profile Name       : http2prof1
```

```
-----
Number of profiles        : 1
=====
```

Table 147: Show network trace profile output fields

Label	Description
Network trace profile	
Packet Broker Server Address	The IP address of the Packet Broker server
Packet Broker Server Port	The port of the Packet Broker server
Packet Broker Server Path	The name of the server path
HTTP/2 Profile Name	The name of the HTTP2 profile

6.2.2.60 node-selection

Syntax

```
node-selection
```

Context

```
show>mobile>profile
```

Description

This command enables the context to display node selection information.

6.2.2.61 client-profile

Syntax

```
client-profile [profile-name]
```

Context

```
show>mobile>profile>node-selection
```

Description

This command enables the context to display node selection client profile information.

Parameters

profile-name

specifies the profile name, up to 32 characters

Output

The following output examples show node selection client profile information and *Show mobile profile client profile output fields* describes the fields.

Example output node selection client profile

```
# show mobile-gateway profile node-selection client-profile "client-profile-APN"
=====
Client profile
=====
Profile name          : client-profile-APN
Description           : -
Default action
  Target profile      : target-profile-APN
  Entry id            : 1
  Description         : -
  Action
    Target profile    : target-profile-APN
  Match criteria
    Serving node list : -
    TAI LAI list      : -
    Rat type          : none
  APN(s)
    "agatohelencups1.mnc008.mcc244.gprs"
    "ims.agamemnon.mg.mnc008.mcc244.gprs"
    "odytoagacups1.mnc008.mcc244.gprs"
    "odytoagacups2.mnc008.mcc244.gprs"
    "achtoagacups1.mnc008.mcc244.gprs"
  Slice               : (Not Specified)
  UE Usage Type List  : usagetype1
  NRI5GC              : false
  IWKI5GS             : false
  Cell ID list        : -
-----
Number of Client profiles : 1
=====
```

```
# show mobile-gateway profile node-selection client-profile "client-profile-TAI-ENB2"
=====
Client profile
=====
Profile name          : client-profile-TAI-ENB2
Description           : -
```



```

Default action
  Target profile      : target-profile-TAI-ENB2
  Entry id           : 1
  Description        : -
  Action
    Target profile    : target-profile-TAI-ENB2
  Match criteria
    Serving node list : -
    TAI LAI list      : tai-lai-1
    Rat type          : eutran
    APN(s)
      -
    Slice             : (Not Specified)
-----
Number of Client profiles : 1
=====
    
```

Table 148: Show mobile profile client profile output fields

Label	Description
Client profile	
Profile name	The name of the client profile
Description	The profile's description
Default action	
Target profile	The name of the target profile
Entry ID	The entry ID
Description	The entry's description
Action	
Target profile	The name of the target profile
Match criteria	
Service node list	The name of the service node list

Table 148: Show mobile profile client profile output fields (continued)

Label	Description
UE Usage Type List	The UE usage type list
NRI5GC	The 5GC Not Restricted Indication flag
IWKI5GS	The 5GS Interworking Indication flag
Cell ID list	The cell-id-list
TAI LAI list	The name of the TAI/LAI list
Rat type	The RAT type
APN(s)	The matching APNs
Slice	The network slice list names
Number of Client profiles	The number of client profiles retrieved

6.2.2.62 group-label

Syntax

```
group-label [profile-name]
```

Context

```
show>mobile>profile>node-selection
```

Description

This command enables the context to display node selection group label information.

Parameters

profile-name

specifies the profile name, up to 32 characters

Output

The following output example shows node selection group label profile information and *Show mobile profile group label output fields* describes the fields.

Example output node selection group label profile

```
# show mobile-gateway profile node-selection group-label "group-label-APN"
=====
node selection group label profile
=====
Profile name           : group-label-APN
Description            : (Not Specified)
-----
Number of policies : 1
=====
```

Table 149: Show mobile profile group label output fields

Label	Description
Profile name	The name of the group label profile
Description	The profile description
Number of policies	The number of group label profiles retrieved

6.2.2.63 target-profile

Syntax

```
target-profile [profile-name]
```

Context

```
show>mobile>profile>node-selection
```

Description

This command enables the context to display node selection target profile information.

Parameters

profile-name
specifies the profile name, up to 32 characters

Output

The following output example shows node selection target profile information and *Show mobile profile target profile output fields* describes the fields.

Example output node selection target profile

```
# show mobile-gateway profile node-selection target-profile "target-profile-APN"
=====
Target Profile
=====
Profile name           : target-profile-APN
Description            : -
Default Action
  Reject               : enabled
  Entry id             : 1
  Description          : -
Match Criteria
  Apn                  : -
  Slice                : -
  Group(s)             : group-label-APN
Action
  Reject               : disabled
-----
Number of Target Profiles : 1
=====
```

Table 150: Show mobile profile target profile output fields

Label	Description
Target profile	
Profile name	The name of the target profile
Description	The profile’s description
Default action	
Reject	Indicates if the reject action is enabled

Table 150: Show mobile profile target profile output fields (continued)

Label	Description
Entry ID	The entry ID
Description	The entry's description
Match Criteria	
APN	The matching APN
Slice	The matching network slice
Group(s)	The matching group of APNs
Action	
Reject	Indicates if the reject action is enabled
Number of Target profiles	The number of target profiles retrieved

6.2.2.64 npcf

Syntax

npcf [*profile-name*]

Context

show>mobile>profile

Description

This command displays information of one or all NPCF profiles.

Parameters

profile-name

specifies the name of the profile, up to 32 characters

Output

The following output example shows the output for one NPCF profile.

Example output NPCF profile

The following output example shows the scenario where one profile is selected, and *Show mobile profile NPCF profile output fields* describes the output fields.

```
# show mobile-gateway profile npcf "parentCntlNpcfProf"
=====
Npcf profile
=====
Profile Name          : parentCntlNpcfProf
Failure Handling:
fh-volume-limit (kbytes) : 0          fh-time-limit (minutes) : 0
fh-session-no-response  : terminate  ap-action-profile-no-resp :
npcf-error-mapping-profile :
supported-vendor        : custom-3
Supported Features:
adc                     : disabled    sponsored-connectivity    : disabled
netloc                  : disabled    umc                       : disabled
tsc                     : disabled    prov-af-signal-flow        : disabled
pcscf-restoration       : disabled    pra                       : disabled
policy-update-when-ue-susp* : disabled session-rule-error-handling: disabled
-----
Number of profiles: 1
=====
```

Table 151: Show mobile profile NPCF profile output fields

Label	Description
Failure Handling	
fh-volume-limit (kbytes)	The volume limit for new and ongoing sessions that are set up and continue without further PCF control (fall back to static policy) as a result of failure handling (for example, PCF no response)
fh-time-limit (minutes)	The time limit for new and ongoing sessions that get set up and continue without further PCF control (fall back to static policy) as a result of failure handling.(for example, PCF no response)
fh-session-no-response	The failure handling behavior for new and ongoing sessions that fail to get a response from the NPCF

Table 151: Show mobile profile NPCF profile output fields (continued)

Label	Description
ap-action-profile-no-resp	The AP action profile for sessions that fail to get a response from the NPCF
npcf-error-mapping-profile	The error mapping profile
supported-vendor	The number of the supported vendor
Supported Features	
adc	The status (enabled or disabled) of the ADC option within the Supported-Features AVP
sponsored-connectivity	The status (enabled or disabled) of the sponsored connectivity option within the Supported-Features AVP
netloc	The status (enabled or disabled) of the NetLoc option within the Supported-Features AVP
umc	The status (enabled or disabled) of the UMC option within the Supported-Features AVP
tsc	The status (enabled or disabled) of the TSC option within the Supported-Features AVP
prov-af-signal-flow	The status (enabled or disabled) of the provisional AF signal flow option within the Supported-Features AVP
pcscf-restoration	The status (enabled or disabled) of the P-CSCF Restoration Enhancement option within the Supported-Features AVP
pra	The status (enabled or disabled) of the PRA option within the Supported-Features AVP

Table 151: Show mobile profile NPCF profile output fields (continued)

Label	Description
policy-update-when-ue-suspends	The status (enabled or disabled) of the option for policy updates when the UE is suspended within the Supported-Features AVP
session-rule-error-handling	The status (enabled or disabled) of the session rule error handling option within the Supported-Features AVP

6.2.2.65 overload-control-profile

Syntax

```
overload-control-profile [profile-name]
```

Context

```
show>mobile>profile
```

Description

This command displays information of the overload control profile.

Parameters

profile-name

specifies the name of the profile, up to 32 characters

Output

The following 7750 SR MG output examples show the overload control profile used for two cases.

- *#unique_3312/unique_3312_Connect_42_p_jtj_k4p_g5b*
- *#unique_3312/unique_3312_Connect_42_p_jtj_k4p_g5b*

Example output overload control profile for all profiles

The following output example shows all profiles, and *Show mobile profile overload control profile output fields* describes the output fields.

```
# show mobile-gateway profile overload-control-profile
=====
Overload control profile
```



```
=====
Profile Name           : profile1
Session rejection rate : 0%
Excluded ARP prio level(s) : -
-----
Profile Name           : profile2
Session rejection rate : 18%
Excluded ARP prio level(s) : 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15
-----
Profile Name           : profile3
Session rejection rate : 0%
Excluded ARP prio level(s) : 14
-----
Number of overload control profiles: 3
=====
```

Example output overload control profile for one profile

The following output example shows the scenario where one profile is selected, and *Show mobile profile overload control profile output fields* describes the output fields.

```
# show mobile-gateway profile overload-control-profile "profile2"
=====
Overload control profile
=====
Profile Name           : profile2
Session rejection rate : 18%
Excluded ARP prio level(s) : 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15
-----
Number of overload control profiles: 1
=====
```

Table 152: Show mobile profile overload control profile output fields

Label	Description
profile name	The name of the profile
session rejection rate	The rejection rate as percentage for new sessions in the PGW
excluded ARP prio level(s)	The new ARP priority level to the overload control profile
number of overload control profiles	The number of the displayed profiles

6.2.2.66 pcmd

Syntax

```
pcmd [profile-name]
```

Context

```
show>mobile>profile
```

Description

This command displays PCMD profile information for all profiles, or for the profile name if specified.

Parameters

profile-name

specifies the name of a particular PCMD profile for which to display information

Output

The following output example shows PCMD profile information, and *Show mobile profile PCMD output fields* describes the fields.

Example output PCMD profile information

```
# show mobile-gateway profile pcmd profile-name
=====
PCMD profile
=====
Profile name      : test
Description       : PCMD profile description
Destination address : 10.10.7.10
Destination port   : 29780
Destination router : Base
Heartbeat         : 10/disabled
Session Report:
Verbosity         : standard
Destination address : 10.10.7.11
Destination port   : 29781
Destination router : Base
Data Report:
Generation        : event-based
Destination address : inherit
Destination port   : -
Destination router : -
-----
Number of profiles : 1
=====
```

Table 153: Show mobile profile PCMD output fields

Label	Description
PCMD profile	
Profile name	The profile that is being used
Description	The profile description
Destination address	The destination IP address of the PCMD stream
Destination port	The UDP port of the PCMD destination
Destination router	The router instance that the destination IP address belongs
Heartbeat	The status or the period of the heartbeat messages in seconds
Session Report	
Verbosity	The amount and type of session records being sent
Data Report	
Generation	The method by which data-reports are being generated
Number of profiles	The number of profiles being viewed

6.2.2.67 pcscf-group

Syntax

```
pcscf-group [group-name]
```

Context

```
show>mobile>profile
```

Description

This command displays P-CSCF server group information.

Parameters

group-name

displays information about the specified P-CSCF group

Output

The following output examples show P-CSCF group profile information.

Example output for P-CSCF group IP address configuration

```
# show mobile-gateway profile pcscf-group
=====
P-CSCF server group
=====
Group name           : pcscf-group1
FQDN                 : N/A
FQDN type            : N/A
Refresh interval     : 6 hours
FQDN resolution time : N/A
P-CSCF server list   : 10.10.10.10           Weight: 10
                    : 20.20.20.20           Weight: 20
-----
Number of groups : 1
=====
```

```
# show mobile-gateway profile pcscf-group
=====
P-CSCF server group
=====
Group name           : google.com
FQDN                 : google.com
FQDN Type            : ipv4
Refresh interval     : 6 hours
FQDN resolution time : Sep 24 21:25:28.455 2014
P-CSCF server list   : 173.194.40.194         Weight: 1
                    : 173.194.40.193         Weight: 1
                    : 173.194.40.196         Weight: 1
-----
Number of groups : 1
=====
```

6.2.2.68 peer-list

Syntax

peer - list [*profile-name*]

Context

show>mobile>profile

Description

This command enables the context to display mobile gateway peer list information.

Parameters

profile-name

displays information about the specified existing profile

6.2.2.69 pfcf

Syntax

pfcf

Context

show>mobile>profile

Description

This command enables the context to display PFCF profile information.

6.2.2.70 pfcf-association-peer-list

Syntax

pfcf-association-peer-list [*profile-name*]

Context

show>mobile>profile>pfcf

Description

This command displays PFCF association peer list profile information.

Parameters

profile-name

displays information about the specified existing profile

Output

The following output example shows PFCP association peer list profile information

Example output PFCP association peer list profile

```
# show mobile-gateway profile pfc pfc-association-peer-list
=====
PFCP association peer list profile
=====
Profile name           : pfcAPList1
Description            : Test List
Node ID                : 310.150.PGW.1.1.1
  BFD                  : Disabled
  Partial Failure      : Enabled
  UE IP Allocation     : Disabled
  Path-hold-timer      : 20
  Peer [Port]          : 33.34.1.1 [8001]
  Peer [Port]          : 3ffe::2122:101 [8005]
Node ID                : 510.180.PGW.1.1.2
  BFD                  : Disabled
  Partial Failure      : Disabled
  UE IP Allocation     : Disabled
  Path-hold-timer      : Disabled
  Peer                 : 33.34.1.2
  Peer                 : 3ffe::2122:102
-----
Number of PFCP-peers : 4
-----
Number of profiles   : 1
=====
```

6.2.2.71 pfc-association-peer-list

Syntax

`pfc-association-peer-list profile-name`

Context

`show>mobile>profile>pfc`

Description

This command displays the PFCP error mapping profile.

Parameters

profile-name

displays information about the specified PFCP error mapping profile

Output

The following output example shows PFCP error mapping profile information.

Example output PFCP profile

```
# show mobile-gateway profile pfcpc pfcpc-error-mapping-profile
=====
PFCP error mapping profile
=====
Profile Name           : pfcpcErrorMapProf

=====
PFCP to 5GSM cause code mapping
=====
PFCP      5GSM
-----
66         77
67         78
=====

=====
PFCP to GTP cause code mapping
=====
PFCP      GTP
-----
88         99
100        101
=====

=====
PFCP to HTTP cause code mapping
=====
PFCP      HTTP      APP ERROR OUT
-----
100        504      appError
101        500      appError1
=====

-----
Number of PFCP error mapping profiles : 1
=====
```

6.2.2.72 pfcpc-profile

Syntax

pfcpc-profile [*profile-name*]

Context

show>mobile>profile>pfcpc

Description

This command displays PFCP profile information.

Parameters

profile-name

displays information about the specified existing profile

Output

The following examples show PFCP profile information and *Show mobile profile PFCP output fields* describes the output fields.

Example output PFCP profile

```
# show mobile-gateway profile pfcf pfcf-profile
=====
PFCF profile
=====
Profile name           : pfcf_prof_1
Description            : N/A
Msg retransmit timeout : 5 secs      Msg retransmit retry count : 3
Heartbeat timeout      : 5 secs      Heartbeat retry count      : 4
Heartbeat T3 response  : 60 secs
IP TTL                 : 255          IP DSCP                     : 56
Association Retry Time  : 10 secs
PDI Optimization       : enabled     Audit IP DSCP              : 56
PFCF error mapping profile : (Not Specified)

PFCF Configurable IEs
LCI IE                 :Included/Excluded      OCI IE: Included/Excluded
PFCF-u Shared Tunnel
Re-attempt timer       : 600 secs
-----
Number of profiles : 1
=====
```

```
# show mobile-gateway profile pfcf pfcf-profile "prof_1"
=====
PFCF profile
=====
Profile name           : prof_1
Description            : N/A      Msg retransmit retry count : 3
Heartbeat interval     : 60 secs  Heartbeat retry count      : 4
Heartbeat T3 response  : 5 secs
IP TTL                 : 255      IP DSCP                     : 56
Association Retry Time  : 10 secs
PDI Optimization       : enabled  Audit IP DSCP              : 56
PFCF error mapping profile : (Not Specified)
Path Restoration Time   : disabled
SMF Set Support         : disabled
```



```

PCF-u Shared Tunnel
Re-attempt timer      : disabled
ext-urr-id            : disabled
LCI IE                : Included   OCI IE                : Included
-----
Number of profiles : 1
=====

```

Table 154: Show mobile profile PCF output fields

Label	Description
Profile name	The name of the PCF profile
Description	The description of the PCF profile
Msg retransmit timeout	The time, in seconds, that the PCF waits before resending a message if there is no response to an initial message
Msg retransmit retry count	The number of times the PCF attempts to send a message
Heartbeat timeout	The heartbeat timeout interval, in seconds
Heartbeat retry count	The heartbeat retry count
Heartbeat T3 response	The heartbeat response, in seconds
IP TTL	The IP TTL value used for PCF signaling messages
IP DSCP	The DSCP value in the IP Header used for PCF signaling messages
Association Retry Time	The association retry time
PDI Optimization	The PDI optimization status
SMF Set Support	The SMF set support on the UP function
Re-attempt Timer	The configured value of the re-attempt timer for failed PCF-u tunnel session establishment between the CP function and the

Table 154: Show mobile profile PFCP output fields (continued)

Label	Description
	UP function, when ICMPv6 RS/RA and NS/NA message handling is enabled for the CP function
ext-urr-id	The support for the extended URR ID range for pre-defined URR IDs
LCI IE	LCI information to be included or excluded
OCI IE	OCI information to be included or excluded
Number of profiles	The number of profiles retrieved

6.2.2.73 pfcpx-translate-profile

Syntax

```
pfcpx-translate-profile [profile-name]
```

Context

```
show>mobile>profile>pfcpx
```

Description

This command displays PFCP translation profile information.

Parameters

profile-name

displays information about the specified existing profile

Output

The following example shows PFCP translation profile information and *Show mobile profile PFCP translation output fields* describes the output fields.

Example output PFCP translation profile

```
# show mobile-gateway profile pfcpx pfcpx-translate-profile
```

```
=====
```

```
PFPCP translate profile
=====
Profile name           : pfcsp-translation-profile-1
Description            : test
Field                  : hdr-enrich-imsi
Field                  : hdr-enrich-billing-type
-----
Number of profiles : 1
=====
```

Table 155: Show mobile profile PFPCP translation output fields

Label	Description
Profile name	The name of the PFPCP translation profile
Description	The description of the PFPCP translation profile
Field	The values that are forwarded to the UP
Number of profiles	The number of profiles retrieved

6.2.2.74 up-peer-list

Syntax

```
up-peer-list [profile-name]
```

Context

```
show>mobile>profile>pfcsp
```

Description

This command displays user plane peer list profile information.

Parameters

profile-name
displays information about the specified existing profile

Output

The following example shows user plane peer list profile information and *Show mobile profile PFCP UP peer list output fields* describes the output fields.

Example output user plane peer list profile

```
# show mobile-gateway profile pfc up-peer-list "listTest"
=====
UP peer list profile
=====
-----
Profile name          : listTest
Description           : N/A
-----
Peer                  : 1.2.3.4
UPF Selection         : True
upf-id                : -
-----
APN                   : test
tai-lai-list          : TaiLaiListTest
Block                 : true
EPS interworking      : Enabled
Slice                 : (Not Specified)
-----
Number of profiles : 1
=====
```

```
# show mobile-gateway profile pfc up-peer-list "upf_peers"
=====
UP peer list profile
=====
-----
Profile name : upf_peers
Description  : N/A
-----
Peer : 33.34.1.1
UPF Selection : True
upf-id : 69010000-eb45-418e-9694-40000eb4518e
Disable EPS interworking : Enabled
Slice : (Not Specified)
APN : apn.interntSgiRedirect.1.1.com
Network Realm : realmForinterntSgi
pool-Name : pool1Grp1
APN : apn.interntSgiEntpRedirect.1.1.com
PCO option
DNS Addr IPv4 : 20.20.20.20
DNS Sec Addr IPv6 : 3ffe:567::
Network Realm : realmForinterntSgi
pool-Name : pool2Grp1
APN : apn.imsSgiRedirect.1.1.com
PCO option
DNS Addr IPv4 : 20.20.20.20
DNS Sec Addr IPv6 : 3ffe:567::
Network Realm : realmForimsSgi
pool-Name : pool1Grp1
```

```

pool-Name : pool2Grp1
pool-Name : pool3Grp1
Network Realm : realmForinterntSgi
ip-pool-distribution : enabled
pool-Name : pool1Grp1
IPv4/IPv6 Prefix pool : 6.0.0.0/9
3ffe:60::/48
Remote : No supp-dotzero-broadcast-add : No
Hold-timer : 3 Hold-for-pdn : No
Round-Robin : No
Priority : none
pool-Name : pool2Grp1
IPv4/IPv6 Prefix pool : 6.128.0.0/9
3ffe:188::/48
Remote : No supp-dotzero-broadcast-add : No
Hold-timer : 3 Hold-for-pdn : No
Round-Robin : No
Priority : none
Network Realm : realmForimsSgi
ip-pool-distribution : enabled
pool-Name : pool1Grp1
IPv4/IPv6 Prefix pool : 6.0.0.0/23
3ffe:60::/55
Remote : No supp-dotzero-broadcast-add : No
Hold-timer : 0 Hold-for-pdn : No
Round-Robin : No
Priority : none
pool-Name : pool2Grp1
IPv4/IPv6 Prefix pool : 6.1.0.0/23
3ffe:60:1000::/55
Remote : No supp-dotzero-broadcast-add : No
Hold-timer : 0 Hold-for-pdn : No
Round-Robin : No
Priority : low
pool-Name : pool3Grp1
IPv4/IPv6 Prefix pool : 6.2.0.0/23
3ffe:60:2000::/55
Remote : No supp-dotzero-broadcast-add : No
Hold-timer : 0 Hold-for-pdn : No
Round-Robin : No
Priority : high
-----
Number of profiles : 1
=====

```

Table 156: Show mobile profile PFCP UP peer list output fields

Label	Description
UP peer list profile	

Table 156: Show mobile profile PFCP UP peer list output fields (continued)

Label	Description
Profile name	The name of the UP peer list
Peer	IPv4 or IPv6 address of the UP peer list peer
UPF Selection	The peer selection status is allowed (True) or not allowed (False)
upf-id	The UPF ID used by the SMF in charging records to report charging information over the Nchf SBI, to allocate quota and report usage per UPF Configured locally per peer, but is overridden by the UPF ID received in the NRF discovery, if NRF is used
Block	The blocking state of the UP peer according to the block configuration. "True" is displayed when the peer is blocked at the APN level or when all the APN prefixes for this peer are blocked
EPS interworking	The EPS interworking is enabled or disabled
Slice	The slice specifies the network slice name associated with the user plane peer
Suspend-peer	The peer suspend status is enabled or disabled
Group-label profile	The group-label profile name associated with the group for the UP instance; the group association creates a logical grouping of target nodes with selection criteria based on different parameters (APN, TAI, LAI, serving node, IP address, and so on)
APN	The APN name associated with the UP peer
PCO option	The UE PCO options
DNS Addr IPv4	The IPv4 address of the primary DNS server used by a UE to resolve domain names to IP addresses

Table 156: Show mobile profile PFCP UP peer list output fields (continued)

Label	Description
DNS Sec Addr IPv4	The IPv4 address of the secondary DNS server used by a UE to resolve domain names to IP addresses
DNS Addr IPv6	The IPv6 address of the primary DNS server used by a UE to resolve domain names to IP addresses
DNS Sec Addr IPv6	The IPv6 address of the secondary DNS server used by a UE to resolve domain names to IP addresses
Network Realm	The network realm of the APN
ip-pool-distribution	Indicates if the CP function distributes the allocation of the UE IP addresses across the configured IP address pools and the prefixes configured under these IP address pools for the network realm
pool-Name	The IP pool name
IPv4/IPv6 Prefix pool	The IPv4/IPv6 prefix pool
Remote	Indicates a remote IP pool allocation
supp-dotzero-broadcast-add	The IP address pool to suppress allocation of 0 and x.x.x.255 IP addresses from the local IP address pool to mobile customers
Hold-timer	The value configured for the IP address pool hold-timer, after which the PDN session is released and the gateway assigns the IP address to another PDN session
Hold-for-pdn	Indicates if the hold for PDN option is enabled, in which case the gateway reassigns the held IP address to the PDN connection that released it, if the PDN connection reattaches before the timer expires
Round-Robin	Indicates if the round robin selection mechanism to allocate the IP address is enabled

Table 156: Show mobile profile PFCP UP peer list output fields (continued)

Label	Description
Priority	Indicates the priority value of an IP address pool

6.2.2.75 pmipv6

Syntax

```
pmipv6 [profile-name]
```

Context

```
show>mobile>profile
```

Description

This command enables the context to display IPv6 information.

Parameters

profile-name

displays information about the specified existing profile

6.2.2.76 policy-options

Syntax

```
policy-options
```

Context

```
show>mobile>profile
```

Description

This command enables the context to display mobile gateway policy information.

Output

The following example shows policy options profile information, and *Show mobile profile policy rule unit output fields* describes the fields.

Example output policy options policy rule units

```
# show mobile-gateway profile policy-options policy-rule-unit
=====
Policy rule unit
=====
Policy rule unit name      : test
Uplink GBR                 : 0 kbps
Uplink MBR                 : 0 kbps
Downlink GBR               : 0 kbps
Downlink MBR               : 0 kbps
In use                     : NO          Reference count      : 0
Flow gate status           : allow
AA advanced fns enabled    : NO
Shallow Inspection Only    : NO
Mute Notification          : YES
Steering enabled           : NO
URL Encryption Type        : none
URL Passphrase             : (Not Specified)
PDR id                     :
Flow Id                    : 1
Direction                  : both       Protocol           : any
Local port                 : any
Remote port                : any
Local address              : 10.0.0.0/24
Remote address             : any
AA Charging-Group          :
FQDN-Group                 :
Type of Srvc/Traffic Class : disabled
DNS FQDN List              : (Not Specified)
-----
Number of policy rule units : 1
=====
```

Table 157: Show mobile profile policy rule unit output fields

Label	Description
Policy rule unit	
Policy rule unit name	The policy rule unit name
Local address	The IP address of the UE

Table 157: Show mobile profile policy rule unit output fields (continued)

Label	Description
Remote address	The destination IP address
FQDN-Group	The name of the FQDN group
Number of policy rule units	The number of configured policy rule units

6.2.2.77 action-rule-unit

Syntax

action-rule-unit [*action-rule-unit-name*]

Context

show>mobile>profile>policy-options

Description

This command displays ARU information.

Parameters

action-rule-unit-name

displays information about the specified existing policy

Output

The following example shows information for the ARU profile.

Example output policy options action rule unit profile

```
# show mobile-gateway profile policy-options action-rule-unit "Prof1"
=====
Action Rule Unit profile
=====
Profile name       : Prof1
Custom PCO        : none
Disable UBR signaling: no
Gy filter-id action : none
DNS-Redirect      : All Access
```

```

In use           : no           Reference count      : 0
App profile      : (Not Specified)
modify-qos-default-b*: disabled

data-driven-dedicate*: disabled
sdf-level-policing : disabled
sdf-lvl-plc direction: downlink
sdf-lvl-plc group   : sdf

src-dest-ports-in-ue*: exclude

Total Number of Action Rule Unit profiles: 1
=====

```

Example output policy options action rule unit profile with FQDN redirect

```

# show mobile-gateway profile policy-options action-rule-unit "aru-fqdn-redirect"
=====
Action Rule Unit profile
=====
Profile name       : aru-fqdn-redirect
Custom PCO        : none
Disable UBR signaling: no
Gy filter-id action : none
DNS-Redirect       : All Access (or HRPD Access, as shown below)
In use            : yes           Reference count      : 1
aa application-profi*:

modify-qos-default-b*: enabled
Total Number of Action Rule Unit profiles: 1
=====

```

6.2.2.78 ap-action-profile

Syntax

```
ap-action-profile [profile-name]
```

Context

```
show>mobile>profile>policy-options
```

Description

This command is used to display the contents of the specified Assume Positive action profile (configured with the `config>mobile>profile>policy-options>ap-action-profile` command).

Parameters

profile-name

name of the assume action profile

Output

The following example output shows information for the Assume Positive action profile.

Example output policy options AP action profile

```
# show mobile-gateway profile policy-options ap-action-profile "actionProf1"
=====
Assume Positive Action profile
=====
ap-action-profile nam: actionProf1
policy-action      : pre-defined-policy
-----
Total Number of Assume Positive Action profiles: 1
=====
```

6.2.2.79 charging-rule-unit

Syntax

charging-rule-unit [*policy-name*]

Context

show>mobile>profile>policy-options

Description

This command displays charging rule unit information.

Parameters

policy-name

displays information about the specified existing policy

Output

The following example shows information for the charging rule unit profile.

Example output policy options charging rule profile

```
# show mobile-gateway profile policy-options charging-rule-unit "CRU_SelRadAcct"
=====
Charging rule unit
```

```

=====
Charging rule unit name      : "CRU_SelRadAcct"
::
::
Selective RADIUS Accounting : Enabled/Disabled
-----
Number of charging rule units : 1
=====

```

6.2.2.80 gx-avp-options

Syntax

gx-avp-options *profile-id*

Context

show>mobile>profile>policy-options

Description

This command displays the configured contents of the specified gx-avp-options profile. When the optional profile ID is omitted, the list of configured profiles is displayed.

Parameters

profile-id

displays information about the specified AVP profile (optional)

Output

The following example shows configuration information for the Gx AVP options of the policy option profile and *Policy options profile Gx AVP options output fields* describes the output fields.

Example output policy options profile Gx AVP options

```

# show mobile-gateway profile policy-options gx-avp-options 1
=====
gx-avp-option profile
=====
Profile Id           : 1
Description          : N/A
subscription-id-imsi : include   subscription-id-msisdn : include
subscription-id-nai  : include   user-equipment-info    : include
ue-local-ip-address  : include   ue-udp-port            : include
pdn-connection-id    : include
3GPP-charging-characterist*: include
an-trusted           : include   twan-identifier            : include
ue-tcp-source-port   : exclude
framed-ip-address-ccrt : include  framed-ipv6-prefix-ccrt : include

```

```
-----
Number of profiles : 1
=====
```

```
* indicates that the corresponding row element may have been truncated.
```

Table 158: Policy options profile Gx AVP options output fields

Label	Description
Profile Id	Gx AVP options profile ID
Description	Optional user configured profile description
subscription-id-imsi	Included or excluded in Gx CCR messages AVP
subscription-id-msisdn	Included or excluded in Gx CCR messages AVP
subscription-id-nai	Included or excluded in Gx CCR messages AVP
user-equipment-info	Included or excluded in Gx CCR messages AVP
ue-local-ip-address	Included or excluded in Gx CCR messages AVP
ue-udp-port	Included or excluded in Gx CCR messages AVP
pdn-connection-id	Included or excluded in Gx CCR messages AVP
3GPP-charging-characteristics	Included or excluded in Gx CCR messages AVP
an-trusted	Included or excluded in Gx CCR messages sent to the PCRF, which indicates whether the AN is trusted or un-trusted for non-3GPP access networks
twan-identifier	Included or excluded in Gx CCR messages sent to the PCRF
ue-tcp-source-port	Included or excluded in the Gx CCR messages AVP sent to the PCRF
framed-ip-address-ccrt	Included or excluded in the Gx CCR-T message AVP sent to the PCRF

Table 158: Policy options profile Gx AVP options output fields (continued)

Label	Description
framed-ipv6-prefix-ccrt	Included or excluded in the Gx CCR-T message AVP sent to the PCRF

6.2.2.81 policy-rule

Syntax

policy-rule [*policy-name*]

Context

show>mobile>profile>policy-options

Description

This command displays policy information.

Parameters

policy-name

displays information about the specified existing policy

Output

The following example shows policy rule information for the policy option profile.

Example output policy options policy rule profile

```
# show mobile-gateway profile policy-options policy-rule "pr1"
=====
Policy rule
=====
Policy rule name       : pr1
QCI                   : 9          ARP                   : 11
Precedence            : 100
Policy rule unit name  : pru1
Charging rule unit name : cru1
Action rule unit name:  : (Not Specified)
Trigger rule unit name : (Not Specified)
Stat rule unit list name : (Not Specified)
QOS rule unit name     : otfQru1
In Use                 : YES
Session Reference Count : 1
```

```

on older Versions      : 0
Configuration Ref Count : 1
-----
Number of policies : 1
=====

```

6.2.2.82 policy-rule-base

Syntax

```
policy-rule-base [policy-name]
```

Context

```
show>mobile>profile>policy-options
```

Description

This command displays base policy information.

Parameters

policy-name

displays information about the specified existing policy

Output

The following examples show policy rule base information for the policy option profile.

Example output policy options policy rule base

```

# show>mobile>profile>policy-options# policy-rule-base
=====
Policy rule bases
=====
Policy rule base name      : prb1
In use                     : YES
Session references        : 1
  on older Versions       : 0
Configuration references   : 0
Bearer binding verification: enabled
CC Group                   : none
Ded Bearer On App-start    : enabled
Policy rules               :
  policyRule1
Number of policy rules     : 1
-----
Number of policy rule bases : 1

```


Example output policy options policy rule base

```
# show>mobile>profile>policy-options# policy-rule-base "pol1"
=====
Policy rule bases
=====
Policy rule base name      : pol1
In use                     : NO
Session references        : 0
    on older Versions      : 0
Configuration references   : 0
Bearer binding verification: enabled
CC Group                   : none
Ded Bearer On App-start    : disabled
Charging Profile Id        : 0
-----
Number of policy rule bases : 1
```

6.2.2.83 policy-rule-unit

Syntax

policy-rule [*policy-name*]

Context

show>mobile>profile>policy-options

Description

This command displays policy rule unit information.

Parameters

policy-name

displays information about the specified existing policy

Output

The following output example shows policy rule unit information for the policy option profile.

Example output policy options policy rule unit

```
# show>mobile>profile>policy-options# policy-rule-unit
=====
Policy rule unit
=====
Policy rule unit name      : fqdn-redirect
Uplink GBR                 : 0 kbps
Uplink MBR                 : 0 kbps
```

```

Downlink GBR          : 0 kbps
Downlink MBR          : 0 kbps
In use                : YES      Reference count      : 1
Flow gate status      : allow
AA advanced fns enabled : NO
Shallow Inspection Only : NO
Mute Notification     : YES
Steering enabled      : NO
URL Encryption Type   : none
URL Passphrase        : (Not Specified)
Flow Id              : 1
Direction            : both      Protocol            : any
Local port           : list-name-local
Remote port          : list-name-remote
Local address        : any
Remote address       : any
AA Charging-Group    :
FQDN-Group           :
Type of Srvc/Traffic Class : disabled
DNS FQDN List        : fqdnList1
-----
Number of policy rule units : 1
=====

```

6.2.2.84 qci-policy

Syntax

```
qci-policy [policy-name]
```

Context

```
show>mobile>profile>policy-options
```

Description

This command displays QCI policy information.

Parameters

policy-name

displays information about the specified existing policy

Output

The following output example shows QCI policy information for the policy option profile.

Example output policy options QCI policy profile

```
# show>mobile>profile>policy-options# qci-policy
```

```

=====
QCI policy
=====
Policy rule name      : default
QCI                  : 1          ARP                  : 1
In-profile DSCP       : nc2        Out-profile DSCP      : nc2
FC                   : nc
Profile              : None
QCI                  : 1          ARP                  : 2
In-profile DSCP       : nc2        Out-profile DSCP      : nc2
FC                   : nc
Profile              : None
QCI                  : 1          ARP                  : 3
In-profile DSCP       : nc2        Out-profile DSCP      : nc2
FC                   : nc
Profile              : None
QCI                  : 1          ARP                  : 4
In-profile DSCP       : nc2        Out-profile DSCP      : nc2
FC                   : nc
Profile              : None
QCI                  : 1          ARP                  : 5
In-profile DSCP       : nc2        Out-profile DSCP      : nc2
FC                   : nc
Profile              : None
QCI                  : 1          ARP                  : 6
In-profile DSCP       : nc2        Out-profile DSCP      : nc2
....
QCI                  : 9          ARP                  : 13
In-profile DSCP       : be         Out-profile DSCP      : be
FC                   : be
Profile              : None
QCI                  : 9          ARP                  : 14
In-profile DSCP       : be         Out-profile DSCP      : be
FC                   : be
Profile              : None
QCI                  : 9          ARP                  : 15
In-profile DSCP       : be         Out-profile DSCP      : be
FC                   : be
Profile              : None
-----
Number of policies : 1
=====

```

```

# show mobile-gateway profile policy-options qci-policy "qciOption_1"
=====

```

```

QCI policy
=====
Policy rule name      : qciOption_1
Description           : N/A
QCI                  : 1          ARP                  : *
In-profile DSCP       : nc2        Out-profile DSCP      : nc2
DSCP preserve         : enabled    FC                  : nc
Profile              : None
QCI                  : 1          ARP                  : 15
In-profile DSCP       : af42       Out-profile DSCP      : af42
DSCP preserve         : disabled   FC                  : nc

```

```

Profile          : None
QCI              : 2          ARP          : *
In-profile DSCP  : ef          Out-profile DSCP : ef
DSCP preserve    : enabled    FC          : ef
Profile          : None
QCI              : 2          ARP          : 15
In-profile DSCP  : af42        Out-profile DSCP : af42
DSCP preserve    : disabled   FC          : ef
Profile          : None
QCI              : 3          ARP          : *
In-profile DSCP  : ef          Out-profile DSCP : ef
DSCP preserve    : enabled    FC          : h2
Profile          : None
QCI              : 4          ARP          : *
In-profile DSCP  : af32        Out-profile DSCP : af32
DSCP preserve    : enabled    FC          : h1
Profile          : None
QCI              : 5          ARP          : *
In-profile DSCP  : af41        Out-profile DSCP : af41
DSCP preserve    : enabled    FC          : l1
Profile          : None
QCI              : 6          ARP          : *
In-profile DSCP  : af21        Out-profile DSCP : af21
DSCP preserve    : enabled    FC          : af
Profile          : None
QCI              : 7          ARP          : *
In-profile DSCP  : af22        Out-profile DSCP : af22
DSCP preserve    : enabled    FC          : af
Profile          : None
QCI              : 8          ARP          : *
In-profile DSCP  : af11        Out-profile DSCP : af11
DSCP preserve    : enabled    FC          : l2
Profile          : None
QCI              : 9          ARP          : *
In-profile DSCP  : be          Out-profile DSCP : be
DSCP preserve    : enabled    FC          : be
Profile          : None
QCI              : 10         ARP          : *
In-profile DSCP  : be          Out-profile DSCP : be
DSCP preserve    : enabled    FC          : be
Profile          : None
....
QCI              : 252        ARP          : *
In-profile DSCP  : be          Out-profile DSCP : be
DSCP preserve    : enabled    FC          : be
Profile          : None
QCI              : 253        ARP          : *
In-profile DSCP  : be          Out-profile DSCP : be
DSCP preserve    : enabled    FC          : be
Profile          : None

```

```

-----
Number of policies : 1
=====

```

6.2.2.85 qos-rule-unit

Syntax

```
qos-rule-unit [profile-name]
```

Context

```
show>mobile>profile>policy-options
```

Description

This command displays the configured contents of a QRU.

Parameters

profile-name

displays information about the specified QRU

Output

The following output example shows information for the specified QRU.

Example output policy options QRU

```
# show mobile-gateway profile policy-options qos-rule-unit "otfQru1"
=====
QoS Rule Unit profile
=====
QoS rule unit name   : otfQru1
Qer-id              : 1                Sdf-policing directi*: -
Downlink GBR        : 100000           Downlink MBR          : 100000
Uplink GBR          : 100000           Uplink MBR            : 100000
Service class id    : 0                tos-tc                 : 0
rqi                 : disabled          qfi                   : 0
qnc                 : disabled          average-window         : 2000 ms
priority-level      : 0                max-data-burst-vol     : 0 bytes

Total Number of QoS Rule Unit profiles: 1
=====
```

6.2.2.86 stat-rule-unit

Syntax

```
stat-rule-unit [profile-name]
```

Context

show>mobile>profile>policy-options

Description

This command displays the configured contents of a SRU.

Parameters

profile-name

displays information about the specified SRU

Output

The following output example shows information for the specified SRU.

Example output policy options SRU

```
# show mobile-gateway profile policy-options stat-rule-unit "otfSru1"
=====
Stat Rule Unit profile
=====
Stat Rule unit name   : otfSru1
URR-id                : 1                URR-profile          : urrProf1
Total Number of Stat Rule Unit profiles: 1
=====
```

6.2.2.87 steering-profile

Syntax

steering-profile [*steering-profile-name*]

Context

show>mobile>profile>policy-options

Description

This command displays steering profile information.

Parameters

steering-profile-name

displays information about the specified steering profile

Values: 1 to 32

Output

The following output example shows steering profile information.

Example output steering profile

```
# show mobile-gateway profile policy-options steering-profile
=====
Steering profile
=====
steering-profile           : ssg-profile1
nsh-enabled                : disabled
traffic-redirect           : traffic-profile1
-----
Number of profiles: 1
=====
```

6.2.2.88 traffic-redirect-profile

Syntax

traffic-redirect-profile [*traffic-redirection-profile-id*]

Context

show>mobile>profile>policy-options

Description

This command displays traffic redirection information.

Parameters

traffic-redirection-profile-id

displays information about the specified traffic redirection profile ID

Values: 1 to 32

Output

The following output example shows traffic redirection information.

Example output traffic redirection profile

```
#show mobile-gateway profile policy-options traffic-redirect-profile
=====
Traffic redirection profile
=====
MG Group id                : 1
Uplink Router               : Base
=====
```

```

Downlink Router      : Base

Redirect target id    : 1
Uplink nexthop ipv4 addr : 7.7.7.7      Dnlink nexthop ipv4 addr : 7.7.7.7
Number of sessions    : 1000
Status                : UP/DOWN          Number of UP/DOWN events : 10
Last UP event         : 12:00:01 05/19/2020
Last DOWN event       : 12:00:01 05/19/2020

Uplink nexthop ipv6 addr : ff::7        Dnlink nexthop ipv6 addr : ff::7
Number of sessions      : 1000
Status                  : UP/DOWN          Number of UP/DOWN events : 10
Last UP event : 12:00:01 05/19/2020
Last DOWN event: 12:00:01 05/19/2020
=====

```

6.2.2.89 trigger-rule-unit

Syntax

```
trigger-rule-unit [tru-name]
```

Context

```
show>mobile>profile>policy-options
```

Description

This command is used to display the contents of the specified trigger rule unit.

Parameters

tru-name

specifies the name of the trigger rule unit

Output

The following output example shows trigger rule unit information.

Example output trigger rule unit

```

# show mobile-gateway profile policy-options trigger-rule-unit "tru1"
=====
Trigger Rule Unit profile
=====
TRU name          : tru1
access-type       : ehrpd geran
-----
Total Number of Action Rule Unit profiles: 1

```


6.2.2.90 ppi-mapping-profile

Syntax

```
ppi-mapping-profile [profile-name]
```

Context

```
show>mobile>profile
```

Description

This command displays the contents of the specified PPI mapping profile.

Parameters

profile-name

specifies the name of the PPI mapping profile

Output

The following output example shows PPI mapping profile information.

Example output PPI mapping profile

```
# show mobile-gateway profile ppi-mapping-profile
=====
Paging Policy Indicator (PPI) mapping profile
=====

Profile Name          : ppiProf
-----
DSCP      PPI
-----
63         5
62         1
61         1
-----

Profile Name          : ppiProf2
-----
DSCP      PPI
-----
20         1
-----

Number of PPI mapping profiles : 2
=====
```

6.2.2.91 qfi-mapping-profile

Syntax

```
qfi-mapping-profile [profile-name]
```

Context

```
show>mobile>profile
```

Description

This command displays the contents of the specified QFI mapping profile.

Parameters

profile-name

specifies the name of the QFI mapping profile

Output

The following output example shows QFI mapping profile information.

Example output QFI mapping profile

```
# show mobile-gateway profile qfi-mapping-profile
=====
Qfi Mapping Profile
=====
Profile-Name      : aa
5qi-as-qfi       : disabled
Dynamic-mapping
qnc              : enabled      average-window      : enabled
priority-level   : enabled      max-data-burst-vol : enabled
-----
Profile-Name      : bb
5qi-as-qfi       : disabled
Dynamic-mapping
qnc              : disabled     average-window      : disabled
priority-level   : disabled     max-data-burst-vol : disabled
-----
Number of Qfi-Mapping-Profile : 2
=====
```

6.2.2.92 qos-profile

Syntax

qos-profile [*profile-name*]

Context

show>mobile>profile

Description

This command is used to display the contents of the specified QoS profile.

Parameters

profile-name

specifies the name of the QoS profile

Output

The following output example shows QoS profile information.

Example output QoS profile

```
# show mobile-gateway profile qos-profile "qosProf1"
=====
Qos profile
=====
Profile name           : qosProf1
qci                    : 9          arp                      : 11
ul-ambr                : 500000 kb* dl-ambr                  : 600000 kb*
-----
Number of profiles : 1
=====
```

6.2.2.93 radius

Syntax

radius [*profile-name*]

Context

show>mobile>profile

Description

This command displays RADIUS profile information.

Parameters

profile-name

displays information about the specified existing RADIUS profile

Output

The following output example shows RADIUS profile information.

Example output RADIUS profile

```
# show mobile-gateway profile radius
=====
Radius profile
=====
Profile name           : default
Description            : N/A
Auth Prob Interval     : 0 secs      Server Dead Time       : 300 secs
Auth Retry Timeout     : 4 secs      Auth Retry Count        : 3
Acct Retry Timeout     : 4 secs      Acct Retry Count        : 3
Apn Acct On/Off Init Intrvl: 0        Apn Acct On Retry Interval : 0
Max Peer Reselections Count: 3
NAS Port Type          : 15
-----
Number of Radius profiles : 1
=====
```

6.2.2.94 radius-group

Syntax

radius-group [*radius-group-name*]

Context

show>mobile>profile

Description

This command displays RADIUS group information.

Parameters

radius-group-name

displays information about the specified existing RADIUS group

Output

The following output example shows RADIUS group information.

Example output RADIUS group profile

```
# show mobile-gateway profile radius-group "Radius Group"
=====
Radius Group profile
=====
Group name           : Radius Group
Description          : acctAuthBasic
Interface            : pgwicr-ra* Router                : vprn20005
Auth Server Port     : 1812      Acct Server Port       : 1813
Interim Update Interval : disabled  Server Type       : both
Radius Profile       : Profile1 Ignore Acct Resp        : disabled
Transaction Based LB  : disabled
Use Acct-interim-avp-optio*: enabled  Use Disconnect-ack-avp-opt*: enabled
Use Disconnect-nak-avp-opt*: enabled
Limit User Name Length : enabled

via-up-function
APN                  : Nokia
up-peer-ip-address   : 0.0.0.0

radius-avp-options
-----
service-type         : include   framed-protocol       : include
calling-station-id   : include   nas-port-type         : include
acct-delay-time      : exclude   acct-authentic        : include
3gpp-imsi            : include   3gpp-charging-Id        : include
3gpp-pdp-type        : include   3gpp-gprs-negotiated-qos-p*: include
3gpp-sgsn-address    : include   3gpp-ggsn-address        : include
3gpp-imsi-mcc-mnc    : exclude   3gpp-ggsn-mcc-mnc        : include
3gpp-nsapi           : include   3gpp-session-stop-indicator: include
3gpp-selection-mode  : include   3gpp-charging-characterist*: include
3gpp-sgsn-ipv6-address : include 3gpp-ggsn-ipv6-address : include
3gpp-sgsn-mcc-mnc    : include   3gpp-imeisv             : include
3gpp-rat-type        : include   3gpp-user-location-info    : include
3gpp-ms-timezone     : include   served-mdn                : exclude
alc-subsc-id-str     : exclude   alc-subsc-prof-str        : exclude
alc-sla-prof-str     : exclude   nas-port-id               : exclude
acct-session-id      : include   event-timestamp           : exclude
user-name            : include
nas-identifier        : exclude   class                     : exclude
framed-ip-address    : include   framed-ip-netmask         : include
framed-ipv6-prefix   : include   framed-interface-id        : exclude
acct-session-time    : include
acct-input-gigawords  : include   acct-output-gigawords      : include
acct-input-octets     : include   acct-output-octets         : include
acct-input-packets    : exclude   acct-output-packets        : exclude
called-station-id-apn-in-o*: include
acct-session-id-in-on-off-*: include
called-station-id    : include   3gpp-allocate-ip-type      : include
```

acct-interim-avp-options

```

-----
service-type           : exclude   framed-protocol       : exclude
calling-station-id    : include    nas-port-type         : include
acct-delay-time       : exclude    acct-authentic        : exclude
3gpp-imsi             : include    3gpp-charging-Id       : include
3gpp-pdp-type         : include    3gpp-gprs-negotiated-qos-p*: include
3gpp-sgsn-address     : include    3gpp-ggsn-address       : include
3gpp-imsi-mcc-mnc     : exclude    3gpp-ggsn-mcc-mnc       : exclude
3gpp-nsapi            : include
3gpp-selection-mode   : include    3gpp-charging-characterist*: include
3gpp-sgsn-ipv6-address : exclude    3gpp-ggsn-ipv6-address   : exclude
3gpp-sgsn-mcc-mnc     : exclude    3gpp-imeisv            : exclude
3gpp-rat-type         : exclude    3gpp-user-location-info   : exclude
3gpp-ms-timezone      : exclude    nas-port-id             : exclude
user-name             : include    event-timestamp        : exclude
nas-identifier        : exclude    class                   : exclude
nas-ip-address        : include    nas-ipv6-address         : exclude
framed-ip-address     : include
framed-ipv6-prefix    : include    framed-interface-id      : exclude
acct-session-time     : include
acct-input-gigawords  : include    acct-output-gigawords     : include
acct-input-octets     : include    acct-output-octets        : include
acct-input-packets    : exclude    acct-output-packets       : exclude
called-station-id     : include

```

disconnect-ack-avp-options

```

-----
proxy-state           : include    acct-terminate-cause   : include
event-timestamp      : include

```

disconnect-nak-avp-options

```

-----
proxy-state           : include    error-cause              : include

Peer Address          : 10.23.12.2
Priority               : 3          Radius Profile          : Profile1
Auth Server Port      : 1812       Acct Server Port      : 1813
Admin State           : up         Failure Detection Time  : 10 secs

Peer Address          : 10.23.14.2
Priority               : 2          Radius Profile          : Profile1
Auth Server Port      : 1812       Acct Server Port      : 1813
Admin State           : up         Failure Detection Time  : 10 secs

Peer Address          : 10.23.11.2
Priority               : 1          Radius Profile          : Profile1
Auth Server Port      : 1812       Acct Server Port      : 1813
Admin State           : up         Failure Detection Time  : 10 secs

```

```

-----
Number of Radius Groups: 1
=====

```

* indicates that the corresponding row element may have been truncated.

6.2.2.95 s2b-error-mapping-profile

Syntax

```
s2b-error-mapping-profile [profile-name]
```

Context

```
show>mobile>profile
```

Description

This command displays S2b error mapping profile information.

Parameters

profile-name

displays information about the specified existing S2b error mapping profile

Output

The following output example shows S2b error mapping profile information, and *Show mobile profile error mapping output fields* describes the fields.

Example output S2b error mapping profile

```
# show mobile-gateway profile s2b-error-mapping-profile "p1"
=====
S2b error mapping profile
=====
Profile Name           : p1
-----
S2b Cause Code         : 10
Cause Code Description : -
SWu Action              : send-info
Notify Error Type       : 36
Notify Error Description : -
Backoff-Timer Value     : 1
Backoff-Timer Unit      : 2-s
GTPV2 Rslt Code Trig Lockout: disabled
-----
S2b Cause Code         : 20
Cause Code Description : -
SWu Action              : send-info-notify
Notify Error Type       : 36
Notify Error Description : -
Backoff-Timer Value     : -
Backoff-Timer Unit      : -
GTPV2 Rslt Code Trig Lockout: disabled
-----
S2b Cause Code         : 100
Cause Code Description : -
```

```

SWu Action          : -
Notify Error Type   : 4444
Notify Error Description : s2b cfg test
Backoff-Timer Value : -
Backoff-Timer Unit   : -
GTPV2 Rslt Code Trig Lockout: disabled
-----
S2b Cause Code      : 111
Cause Code Description : -
SWu Action          : -
Notify Error Type   : 10000
Notify Error Description : This is a long string for test
Backoff-Timer Value : 3
Backoff-Timer Unit   : 10-m
GTPV2 Rslt Code Trig Lockout: disabled
-----
Number of profiles : 1
=====

```

Table 159: Show mobile profile error mapping output fields

Label	Description
Profile Name	The profile being viewed
S2b Cause Code	The cause code
SWu Action	The action taken related to the cause code
Notify Error Type	The notify error type
Notify Error Description	The notify error description, if applicable
Backoff Timer Value	The value of the backoff timer
Backoff Timer Unit	The units of the value of the backoff timer
Number of profiles	The number of profiles being viewed

6.2.2.96 s6b-profile

Syntax

s6b-profile [*profile-name*]

Context

show>mobile>profile

Description

This command displays S6b profile information.

Parameters

profile-name

displays information about the specified existing S6b profile

Output

The following output examples show s6b profile information.

- [Example output S6b profile](#)
- [Example output S6b profile bypass for eMPS advance priority calls](#)

Example output S6b profile

```
# show mobile-gateway profile s6b-profile
=====
S6b profile
=====
Profile name           : default
Description            : N/A
AVP Options
Idle Timeout           : Included   RAT Type           : Included
Supp Features          : Included   Mip6 Agent Info      : Included
Mip6 Feature Vector    : Included   Auth Session State   : Excluded
IMSI                   : Excluded   PDP Type             : Excluded
Charging ID            : Excluded   GPRS Neg Qos Prof    : Excluded
SGSN Address           : Excluded   GGSN Address         : Excluded
IMSI MCC MNC           : Excluded   GGSN MCC MNC         : Excluded
NSAPI                  : Excluded   Selection Mode        : Excluded
Charging Characteristics : Excluded GGSN IPv6 Address    : Excluded
SGSN IPv6 Address      : Excluded   SGSN MCC MNC         : Excluded
IMEISV                 : Excluded   3GPP RAT Type        : Excluded
User Location Info     : Excluded   MS Time Zone         : Excluded
Visited Network Identifier : Included Framed IP Address     : Excluded
Framed IP Netmask      : Excluded   Framed IP IPv6 Prefix : Excluded
3GPP Allocate IP Type  : Excluded
Supported Features
```

```
wlan-pcscf-restoration      : Disabled  clear-m-bit              : Disabled
```

```
DRMP Options : disabled
```

```
eutran-initial-attach      : 8          eutran-initiated-detach    : 6
ehrpd-initial-attach       : 8          ehrpd-initiated-detach    : 6
wifi-initial-attach        : 8          wifi-initiated-detach     : 6
reauth-initiated           : 8          lte-wifi-handover         : 6
ehrpd-lte-handover         : 6          ehrpd-wifi-handover       : 6
pgw-initiated-disconnect   : 6
```

```
-----
Number of S6b profiles : 1
=====
```

Example output S6b profile bypass for eMPS advance priority calls

```
# show mobile-gateway profile s6b-profile
```

```
=====
S6b profile
=====
```

```
Profile name      : s65
Description       : N/A
AVP Options
Idle Timeout      : Included  RAT Type           : Included
Supp Features     : Included  Mip6 Agent Info    : Included
Mip6 Feature Vector : Included Auth Session State  : Excluded
IMSI              : Excluded  PDP Type           : Excluded
Charging ID       : Excluded  GPRS Neg Qos Prof  : Excluded
SGSN Address      : Excluded  GGSN Address       : Excluded
IMSI MCC MNC      : Excluded  GGSN MCC MNC       : Excluded
NSAPI             : Excluded  Selection Mode      : Excluded
Charging Characteristics : Excluded GGSN IPv6 Address  : Excluded
SGSN IPv6 Address  : Excluded  SGSN MCC MNC       : Excluded
IMEISV           : Excluded  3GPP RAT Type       : Excluded
User Location Info : Excluded  MS Time Zone        : Excluded
Visited Network Identifier : Included Framed IP Address  : Excluded
Framed IP Netmask  : Excluded  Framed IP IPv6 Prefix : Excluded
3GPP Allocate IP Type : Excluded
Supported Features
wlan-pcscf-restoration : Enabled  clear-m-bit        : Enabled
fh-session-no-response : continue-* fh-time-limit    : 0
```

```
message          : aar-i
result-code-3xxx : terminate
result-code-4xxx : terminate
result-code-5xxx : terminate
```

```
DRMP Options : disabled
```

```
eutran-initial-attach      : 8          eutran-initiated-detach    : 6
ehrpd-initial-attach       : 8          ehrpd-initiated-detach    : 6
wifi-initial-attach        : 8          wifi-initiated-detach     : 6
reauth-initiated           : 8          lte-wifi-handover         : 6
ehrpd-lte-handover         : 6          ehrpd-wifi-handover       : 6
pgw-initiated-disconnect   : 6
```

```
-----
Number of S6b profiles : 1
=====
```

```
* indicates that the corresponding row element may have been truncated.
```

6.2.2.97 sctp-profile

Syntax

```
sctp-profile [profile-name]
```

Context

```
show>mobile>profile
```

Description

This command displays SCTP profile information.

Parameters

profile-name

displays information about the specified existing SCTP profile

Output

The following output example shows SCTP profile information and *Show mobile profile STCP profile output fields* describes the output fields.

Example output SCTP profile

```
# show>mobile>profile# sctp-profile
=====
SCTP Profile
=====
Profile name           : SctpProfile1
RT0 Min Timer          : 400           RT0 Max Timer           : 1000
RT0 Init Timer         : 400           Cookie Life Timer        : 60
Heart Beat Interval    : 30           SACK Period              : 200
SACK Frequency         : 2             MTU Size                 : 1500
Associated Max ReTx    : 10           Init Max ReTx            : 8
In Max Streams         : 1             path-max-retrans         : 5
Out Max Streams        : 1             RT0 Init Max Timer       : 2000
adv-receiver-window-credit : 3000000
-----
Number of profiles : 1
=====
```

Table 160: Show mobile profile STCP profile output fields

Label	Description
Profile Name	The profile being viewed
RTO Min Timer	The minimum RTO used during data transmission
RTO Max Timer	The maximum RTO used during data transmission
RTO Init Timer	The RTO for the initial setup
Cookie Life Timer	The cookie life timer
Heart Beat Interval	The heart Beat Interval
SACK Period	The period of SCTP SACKs on the SCTP connection
SACK Frequency	The frequency of SCTP SACKs on the SCTP connection
MTU Size	The MTU size for SCTP streams
Associated Max ReTx	The maximum number of retransmissions for SCTP associations
Path Max ReTx	The maximum number of STCP path retransmissions
Init Max ReTx	The maximum number of initial SCTP retransmissions
In Max Streams	The maximum number of inbound SCTP streams
Out Max Streams	The maximum number of outbound SCTP streams
RTO Init Max Timer	The maximum RTO used when attempting an initial setup
adv-receiver-window-credit	The value of the advertised receiver window credit

6.2.2.98 sd

Syntax

sd [*profile-name*]

Context

show>mobile>profile

Description

This command displays Sd information.

Parameters

profile-name

displays information about the specified existing Sd profile

Output

The output example shows Sd profile information.

Example output Sd profile

```
# show mobile-gateway profile sd "1"
=====
Sd profile
=====
Profile name           : 1
Description            : N/A
cno-uli                : enabled
pending-transaction    : disabled
pending-transaction-timer : 100 ms    pending-transaction-retry : 3
-----
Number of Sd profiles : 1
=====
```

6.2.2.99 sgsn-group

Syntax

sgsn [*SGSN-group-id*]

Context

show>mobile>profile

Description

This command displays SGSN group information.

Parameters

SGSN-group-id

displays the ID of the specified SGSN group

Values: 1 to 65535

Output

The output example shows SGSN profile information.

Example output SGSN group

```
# show mobile-gateway profile sgsn-group
=====
SGSN Group profile
=====
Profile name           : 65
-----
Profile name           : 89
Peer Address           : 1.1.2.2
Peer Address           : 1.1.2.3
Peer Address           : 1.1.2.4
Peer Address           : ffff::1244:aa12
Peer Address           : ffff::1244:bb34
Peer Address           : 1.1.2.7
-----
Profile name           : 65535
Peer Address           : 1.1.1.0
Peer Address           : 1.1.1.1
Peer Address           : 1.1.1.2
-----
Number of profiles : 3
=====
```

6.2.2.100 sig-rate-limit-profile

Syntax

```
sig-rate-limit-profile [profile-name]
```

Context

```
show>mobile>profile
```

Description

This command displays signaling message rate limiting profile information.

Parameters

profile-name

displays information about the specified existing signaling message rate limiting profile

Output

The following output example shows signaling message rate limiting profile information.

Example output signaling message rate limiting profile

```
# show>mobile>profile# sig-rate-limit-profile
=====
Signaling rate limit profile
=====
Profile name           : Prof10
Maximum burst size     : 0           Queue low threshold      : 30
Maximum message rate   : 25000       Queue high threshold     : 80
Delay tolerance        : 10 secs
Throttle-Emergency     : disabled
-----
Number of profiles : 1
```

6.2.2.101 slaac-host

Syntax

```
slaac-host [profile-name]
```

Context

```
show>mobile>profile
```

Description

This command displays SLAAC host profile information.

Parameters

profile-name

displays information about the specified existing SLAAC host profile

Output

The following output example shows SLAAC host profile information.

Example output SLAAC host profile

```
# show>mobile>profile# slaac-host
=====
Slaacv6 host profile
=====
Profile name           : default
Description            : N/A
Force-Mcast type      : ip
Preferred-lifetime     : 3600 (seconds)
Valid-lifetime         : 86400 (seconds)
Reachable-time         : 0 (milli-seconds)
Retransmit-time        : 0 (milli-seconds)
Router-lifetime        : 4500 (seconds)
Mtu                    : 0 (bytes)
Min-rtr-adv-interval   : 200 (seconds)
Max-rtr-adv-interval   : 600 (seconds)
Include-rdnss          : disabled
Rdnss-lifetime         : 3600 (seconds)
-----
Number of Slaac host profiles : 1
=====
```

6.2.2.102 swm-error-mapping-profile

Syntax

swm-error-mapping-profile [*profile-name*]

Context

show>mobile>profile

Description

This command displays SWm error mapping profile information.

Parameters

profile-name

displays information about the specified existing SWm error mapping profile

Output

The following output example shows Swm error mapping profile information, and *Show mobile profile SWm error mapping profile output fields* describes the fields.

Example output SWm error mapping profile

```
# show mobile-gateway profile swm-error-mapping-profile
=====
SWm error mapping profile
=====
Profile Name                : test
-----
Rx Error Code               : 5002
Error Code Description      : test
SWu Action                  : -
Notify Error Type          : 24
Notify Error Description    : N/A
Backoff Timer Value        : 4
Backoff Timer Unit         : 10-m
diameter-result-code-trigger-lockout: disabled
-----
Rx Experimental Error Code  : 3003
Error Code Description      : N/A
SWu Action                  : send-info-notify
Notify Error Type          : 24
Notify Error Description    : N/A
Backoff Timer Value        : -
Backoff Timer Unit         : -
diameter-result-code-trigger-lockout: disabled
-----
Number of profiles : 1
=====
```

Table 161: Show mobile profile SWm error mapping profile output fields

Label	Description
Profile Name	The profile being viewed
Rx Error Code	The error code number
Rx Experimental Error Code	The experimental error code number

Table 161: Show mobile profile SWm error mapping profile output fields (continued)

Label	Description
Error Code Description	The error code, if applicable
SWu Action	The action taken related to the cause code
Notify Error Type	The notify error type
Notify Error Description	The notify error description, if applicable
Backoff Timer Value	The value of the backoff timer
Backoff Timer Unit	The units of the value of the backoff timer
diameter-result-code-trigger-lockout	The Diameter error/result code used to trigger the UE IPsec client lockout functionality
Number of profiles	The number of profiles being viewed

6.2.2.103 sww-radius-profile

Syntax

```
sww-radius-profile [profile-name]
```

Context

```
show>mobile>profile
```

Description

This command displays SWw RADIUS profile information.

Parameters

profile-name

displays information about the specified existing SWw RADIUS profile

Output

The following output example shows SWw RADIUS profile information.

Example output SWw RADIUS profile

```
# show mobile-gateway profile sww-radius-profile "default"
=====
SWw radius profile
=====
Profile name           : default
Description            : N/A
Disconnect port       : 3799
retry-count           : 4          retry-timeout           : 4
acct-triggered-mobility : disabled
fast-inter-ap-handover : disabled
disconnect-optional-attributes
user-name             : include   framed-ip-address       : include
access-accept-optional-attributes
alc-wlan-apn-name     : exclude   alc-subsc-id-str         : exclude
alc-mgw-interface-type : exclude   alc-msisdn              : exclude
3gpp-pdp-type         : exclude   3gpp-ggsn-address        : exclude
3gpp-gprs-negotiated-qos-p*: exclude   3gpp-charging-characterist*: exclude
3gpp-imsi             : exclude   3gpp-msisdn             : exclude
-----
Number of SWw radius profiles : 1
=====
* indicates that the corresponding row element may have been truncated.
```

6.2.2.104 sww-tunnel-profile

Syntax

```
sww-tunnel-profile [profile-name]
```

Context

```
show>mobile>profile
```

Description

This command displays SWw tunnel profile information.

Parameters

profile-name

displays information about the specified existing SWw tunnel profile

Output

The following output example shows SWw tunnel profile information.

Example output SWw tunnel profile

```
# show mobile-gateway profile sww-tunnel-profile "swwTunnel_1"
=====
Sww tunnel profile
=====
Profile name           : swwTunnel_1
Description            : (Not Specified)
Default APN name (portal) : ipd.nokia.1.1.com
Default APN name (eap)   : (Not Specified)
DHCPv4 server profile name : dhcpProf_1
Slaac host profile name  : default
UE cache time          : 600 secs
wlan-ap-mac-discovery    : enabled
shcv-on-handovers       : disabled
Default Access type      : portal

qtag value             : 101
APN name                : ipd.nokia.2.1.com
Access Type             : portal
SSID name               : (Not Specified)
ssid name               : nokia.wifi2.hotspot.com
APN name                 : ipd.nokia.3.1.com
Access Type             : portal
-----
Number of Sww tunnel profiles : 1
=====
```

6.2.2.105 tai-lai-list

Syntax

```
tai-lai-list [tai-lai-list-name]
```

Context

```
show>mobile>profile
```

Description

This command displays TAI LAI list profile information.

Parameters

tai-lai-list-name

displays information about the specified existing TAI LAI list profile, up to 32 characters

Output

The following output example shows TAI LAI list profile information, and *Show mobile profile TAI LAI list output fields* describes the output fields.

Example output TAI LAI list profile

```
# show mobile-gateway profile tai-lai-list nokia-smf-tai-list
=====
TAI LAI List
=====
-----
List Name : Nokia-smf-tai-list
-----
MCC      MNC      TAC/LAI      TAC-RANGE      TAC-RANGE
          START      END
-----
202      010      0x02ae      0x03ae
202      010      0x01ae      0x01ae

Number of TAI-LAI entries : 2
-----
Number of TAI-LAI lists   : 1
=====
```

Table 162: Show mobile profile TAI LAI list output fields

Label	Description
List Name	The TAI LAI list name
MCC	The MCC value
MNC	The MNC value
TAC/LAI	The TAC/LAI value
TAC-RANGE START	The TAC range start value
TAC-RANGE END	The TAC range end value
Number of TAI-LAI entries	The number of TAI LAI entries for the list
Number of TAI-LAI lists	The number of TAI LAI lists displayed

6.2.2.106 udm-sdm-profile

Syntax

```
udm-sdm-profile [profile-name]
```

Context

```
show>mobile>profile
```

Description

This command displays Nudm_SubscriberDataManagement service profile information.

Parameters

profile-name

displays information about the specified existing profile, up to 32 characters

Output

The following output example shows Nudm_SubscriberDataManagement service information.

Example output udm-sdm profile

```
# show mobile-gateway profile udm-sdm-profile
```

```
=====
UDM SDM Profile(s)
=====
Profile name           : udmSdmProf1
max-peer-failover      : 2
retry-count            : 3
transaction-timer(ms)  : 5000

get-sm-subscription-data
  retry-count           : 3
  transaction-timer(ms) : 5000

create-subscription
  retry-count           : 3
  transaction-timer(ms) : 5000

SBI Error Mapping Profile : (Not Specified)
-----
Number of profiles: 1
=====
```

6.2.2.107 udm-uecm-profile

Syntax

```
udm-uecm-profile [profile-name]
```

Context

```
show>mobile>profile
```

Description

This command displays the Nudm_UEContextManagement service profile information.

Parameters

profile-name

displays information about the specified existing profile, up to 32 characters

Output

The following output example shows Nudm_UEContextManagement service information.

Example output udm-uecm profile

```
*show mobile-gateway profile udm-uecm-profile
```

```
=====
UDM UECM Profile(s)
=====
Profile name           : udmUecmProf1
max-peer-failover      : 2
retry-count            : 3
transaction-timer(ms)  : 5000

register-smf
  retry-count           : 3
  transaction-timer(ms) : 5000

SBI Error Mapping Profile : (Not Specified)
-----
Number of profiles: 1
=====
```

6.2.2.108 udp-tunnel-group

Syntax

```
udp-tunnel-group [tunnel-group-name]
```

Context

show>mobile>profile

Description

This command displays UDP tunnel group profile information.

Parameters

tunnel-group-name

displays information about the specified existing tunnel group profile

Output

The following output example shows UDP tunnel group profile information, and *Show mobile profile UDP tunnel group output fields* describes the fields.

Example output UDP tunnel group profile

```
# show mobile-gateway profile udp-tunnel-group
=====
UDP Tunnel Group profile
=====
Profile name           : Tunnel1
Description            : IPV4 Tunnel Group

Peer Address           : 15.1.2.1
Destination Port       : 4567

Peer Address           : 15.1.3.1
Destination Port       : 9876

Peer Address           : 15.1.4.1
Destination Port       : 5544

Peer Address           : 15.1.5.1
Destination Port       : 6677
-----
Number of profiles : 2
=====
```

Table 163: Show mobile profile UDP tunnel group output fields

Label	Description
Profile name	The profile being viewed

Table 163: Show mobile profile UDP tunnel group output fields (continued)

Label	Description
Description	The profile, if applicable
Peer Address	The IP address for each peer
Destination Port	The destination port
Number of profiles	The number of profiles being viewed

6.2.2.109 umts-qos-policy

Syntax

```
umts-qos-policy [profile-name]
```

Context

```
show>mobile>profile
```

Description

This command displays UMTS QoS policy information.

Output

The following output example shows UMTS QoS policy profile information and *Show mobile profile UMTS QoS policy output fields* describes the output fields.

Example output UMTS QoS policy profile

```
# show mobile-gateway profile umts-qos-policy
=====
umts-qos-policy profile (Traffic-class to QCI mapping)
=====
Profile name           : profile1
Conversational speech  : 1
Conversational unknown (transfer delay < 150ms) : 3
Conversational unknown (transfer delay >= 150ms) : 2
Streaming              : 4           Background              : 9
Interactive P1         : 6           Interactive P1 signaling: 5
Interactive P2         : 7           Interactive P3         : 8
```

```

Operator Specific QCI : 129
Traffic class       : conversation*   Traffic priority      : not-available
Signalling indication : not-available  SSD                   : not-available
Delay               : 3

```

```

Operator Specific QCI : 130
Traffic class       : streaming        Traffic priority      : not-available
Signalling indication : not-available  SSD                   : not-available
Delay               : 0

```

```

-----
Number of profiles : 1
=====

```

Table 164: Show mobile profile UMTS QoS policy output fields

Label	Description
Profile name	The profile name The default profile with the default UMTS to QCI mappings per the standards is displayed first Any custom profile added via <code>config>mobile>profile>umts-qos-policy</code> is displayed after the default profile
Conversational	The QCI value (1 to 9) that is mapped to the conversational traffic class, with the SSD (speech or unknown) and an optional transfer delay (higher or lower than 150 m/s)
Streaming	The QCI value (1 to 9) that is mapped to the streaming traffic class
Background	The QCI value (1 to 9) that is mapped to the background traffic class
Interactive	The QCI value (1 to 9) that is mapped to the Interactive traffic class, with the priority (P1 to P3) and the signaling indication option
Operator Specific QCI	The specific operator QCI value
Traffic class	The traffic class option configured for the specific operator QCI

Table 164: Show mobile profile UMTS QoS policy output fields (continued)

Label	Description
Traffic priority	The configured traffic priority option configured for the specific operator QCI
Signalling indication	The configured signaling indication option configured for the specific operator QCI
SSD	The configured SSD option configured for the specific operator QCI
Delay	The configured delay value configured for the specific operator QCI
Number of profiles	Total number of profiles

6.2.2.110 up-security-policy

Syntax

up-security-policy [*profile-name*]

Context

show>mobile>profile

Description

This command displays the UP security policy information.

Output

The following output example shows the UP security policy profile information.

Example output UP security policy profile

```
# show mobile-gateway profile up-security-policy
=====
UP-security-policy profile
=====
Profile name : pf1
UP confidentiality : preferred
UP integrity : required
```

```
UP override-udm-policy : enabled
-----
Profile name : pf2
UP confidentiality : required
UP integrity : preferred
UP override-udm-policy : enabled
-----
Profile name : pf3
UP confidentiality : not-needed
UP integrity : not-needed
UP override-udm-policy : enabled
-----
Profile name : default
UP confidentiality : not-needed
UP integrity : not-needed
UP override-udm-policy : disabled
-----
Number of profiles : 4
=====
```

6.2.2.111 wlan-emergency

Syntax

```
wlan-emergency [profile-name]
wlan-emergency profile-name country-code country-code
```

Context

```
show>mobile>profile
```

Description

This command displays WLAN emergency profile configuration data such as the Emergency APN name, QoS, APN AMBR, and PGW Identity.

Parameters

profile-name

displays information about the specified existing WLAN emergency profile, up to 32 characters

country-code

displays the configured emergency digits for a specific country code, up to 2 characters

Output

The following output examples show WLAN emergency profile information.

Example output WLAN emergency profile

```
# show mobile-gateway profile wlan-emergency "ecd"
=====
WLAN Emergency profile
=====

Profile name           : ecd
APN                    : ehrrpd
QCI                    : 6          ARP                      : 2
APN-AMBR DL            : 100000 kb* APN-AMBR UL              : 100000 kb*
PGW-ID                 : myList    3GPP-HO-emergency-PGW-ID  :
Unauth-Uicless Support : Enabled   Idr-emergency APN-name list:
Ip Range Country File(s)
File format : geolite
File(s) used :
ftp://*:~@135.243.193.162/./ws/gash/testsuites/mgEpdg/GeoLite2-Country-Blocks-
IPv4.csv -
MD5: eedbd11d12ed8d695d81b8a29440fcf4
ftp://*:~@135.243.193.162/./ws/gash/testsuites/mgEpdg/GeoLite2-Country-
Locations-en.csv -
MD5: 1db878ada248ee6cea1aea191e336d43
ftp://*:~@135.243.193.162/./ws/gash/testsuites/mgEpdg/GeoLite2-Country-Blocks-
IPv6.csv -
MD5: 1607262ac8925e84fe26801a959b214e
DB State Loaded - Entries in IPv4-DB 282878 IPv6-DB 83006
Country Code
Countries :
-----
Number of profiles : 1
=====
```

Example output WLAN emergency country code profile

```
# show mobile-gateway profile wlan-emergency "ecd" country-code "AU"
=====
Emergency Call Digits
=====
-----
Country Code: AU
-----
Digits          Category
-----
112             police ambulance fireBrigade marineGuard mountai*
-----
* indicates that the corresponding row element may have been truncated.
Number of Entries : 1
=====
```

6.2.2.112 wps-profile

Syntax

wps-profile [*profile-name*]

Context

show>mobile>profile

Description

This command displays WPS profile configuration data.

Parameters

profile-name

displays information about the specified WPS profile, up to 32 characters

7 IP tunnel command reference

This section describes the IP tunnel-related commands.

7.1 IP tunnel command hierarchies

- *IP tunnels configuration command hierarchies*
- *Services configuration command hierarchies*
- *IP tunnels show command hierarchy*
- *IP tunnels debug command hierarchy*

7.1.1 IP tunnels configuration command hierarchies

- *Hardware command hierarchy*
- *ISA command hierarchy*
- *IPSec command hierarchies*

7.1.1.1 Hardware command hierarchy

```
configure
- card slot-number
- mda mda-slot
  - mda-type isa-tunnel
  - no mda-type
```

7.1.1.2 ISA command hierarchy

```
configure
- isa
- [no] ipsmg-group ipsmg-group-id
  - description description-string
  - no description
- [no] mda mda-id [create]
- tunnel-group tunnel-group-id [create]
- no tunnel-group tunnel-group-id
  - active-mda-number [1..16]
  - no active-mda-number
```

```
– backup mda-id
– no backup
– description description-string
– no description
– mda mda-id
– [no] mda
– multi-active
– primary mda-id
– no primary
– reassembly [wait-msecs]
– no reassembly
– [no] shutdown
```

7.1.1.3 IPsec command hierarchies

- *Certificate profile command hierarchy*
- *Internet Key Exchange (IKE) policy command hierarchy*
- *Internet Key Exchange (IKE) transform command hierarchy*
- *IPsec transform command hierarchy*
- *RADIUS accounting policy command hierarchy*
- *IPsec static security association command hierarchy*
- *Trust anchor profile/TS list command hierarchy*
- *Tunnel template command hierarchy*

7.1.1.3.1 Certificate profile command hierarchy

```
configure
– ipsec
– cert-profile profile-name [create]
– no cert-profile profile-name
– entry entry-id [create]
– no entry entry-id
– cert cert-filename
– no cert
– key key-filename
– no key
– [no] send-chain
– [no] send-chain
– [no] ca-profile name
– [no] shutdown
```


7.1.1.3.2 Internet Key Exchange (IKE) policy command hierarchy

```

configure
- ipsec
- ike-policy ike-policy-id [create]
- no ike-policy ike-policy-id
- auth-algorithm auth-algorithm
- no auth-algorithm
- auth-method {psk | plain-psk-xauth | cert-auth | psk-radius | cert-radius | eap
| auto-eap-radius}
- no auth-method
- auto-eap-method {psk | cert | psk-or-cert}
- auto-eap-own-method {psk | cert}
- description description-string
- no description
- dh-group {1 | 2 | 5 | 14 | 15}
- no dh-group
- dpd [interval interval] [max-retries max-retries] [reply-only]
- no dpd
- encryption-algorithm {des | 3des | aes128 | aes192 | aes256}
- no encryption-algorithm
- ike-mode {main | aggressive}
- no ike-mode
- ike-version [1..2]
- ipsec-lifetime ipsec-lifetime
- no ipsec-lifetime
- isakmp-lifetime isakmp-lifetime
- no isakmp-lifetime
- [no] match-peer-id-to-cert
- nat-traversal [force] [keep-alive-interval keep-alive-interval] [force-keep-
alive]
- no nat-traversal
- own-auth-method {psk | cert | eap-only}
- no own-auth-method
- pfs [dh-group {1 | 2 | 5 | 14 | 15}]
- no pfs
- relay-unsolicited-cfg-attribute
- [no] internal-ip4-dns
- [no] internal-ip4-netmask
- [no] internal-ip6-dns

```

7.1.1.3.3 Internet Key Exchange (IKE) transform command hierarchy

```
configure
- ipsec
- ike-transform ike-transform-id [create]
- no ike-transform ike-transform-id
- dh-group {1 | 2 | 5 | 14 | 15 | 19 | 20 | 21}
- ike-auth-algorithm {md5 | sha1 | sha256 | sha384 | sha512 | aes-xcbc | auth-
encryption}
- ike-encryption-algorithm {des | 3des | aes128 | aes192 | aes256 | aes128-gcm8 |
aes128-gcm16 | aes256-gcm8 | aes256-gcm16}
- ike-prf-algorithm {md5 | sha1 | sha256 | sha384 | sha512 | aes-xcbc | same-as-
auth}
- isakmp-lifetime seconds
```

7.1.1.3.4 IPSec transform command hierarchy

```
configure
- ipsec
- ipsec-transform transform-id [create]
- no ipsec-transform transform-id
- esp-auth-algorithm {null | md5 | sha1 | sha256 | sha384 | sha512 | aes-xcbc}
- no esp-auth-algorithm
- esp-encryption-algorithm {null | des | 3des | aes128 | aes192 | aes256}
- no esp-encryption-algorithm
```

7.1.1.3.5 RADIUS accounting policy command hierarchy

```
configure
- ipsec
- radius-accounting-policy name [create]
- no radius-accounting-policy name
- [no] include-radius-attribute
- [no] called-station-id
- [no] calling-station-id
- [no] framed-ipv6-prefix
- [no] nas-identifier
- [no] nas-ip-addr
- [no] nas-port-id
- radius-server-policy radius-server-policy-name
- no radius-server-policy
- radius-authentication-policy name [create]
```

- no *radius-authentication-policy* *name*
- [no] *include-radius-attribute*
 - [no] *called-station-id*
 - [no] *calling-station-id*
 - [no] *client-cert-subject-key-id*
 - [no] *nas-identifier*
 - [no] *nas-ip-addr*
 - [no] *nas-port-id*
- *password* *password* [hash|hash2]
- no *password*
- *radius-server-policy* *radius-server-policy-name*
- no *radius-server-policy*

7.1.1.3.6 IPsec static security association command hierarchy

configure

- *ipsec*
 - [no] *static-sa* *sa-name*
 - *authentication* *auth-algorithm* *ascii-key* *ascii-string*
 - *authentication* *auth-algorithm* *hex-key* *hex-string* [hash|hash2]
 - no *authentication*
 - *description* *description-string*
 - no *description*
 - *direction* *ipsec-direction*
 - no *direction*
 - *protocol* *ipsec-protocol*
 - no *protocol*
 - *spi* *spi*
 - no *spi*

7.1.1.3.7 Trust anchor profile/TS list command hierarchy

configure

- *ipsec*
 - *trust-anchor-profile* *name* [create]
 - no *trust-anchor-profile*
 - *trust-anchor* *ca-profile-name*
 - no *trust-anchor*
 - *ts-list* *list-name* [create]
 - no *ts-list* *list-name*
 - *local*
 - *entry* *entry-id* [create]

- no *entry entry-id*
- *address* prefix *ip-prefix/ip-prefix-len*
- *address* from *begin-ip-address* to *end-ip-address*
- no *address*

7.1.1.3.8 Tunnel template command hierarchy

configure

- *ipsec*
 - *tunnel-template ipsec-template-identifier* [create]
 - no *tunnel-template ipsec-template-identifier*
 - [no] *clear-df-bit*
 - *description description-string*
 - no *description*
 - *encapsulated-ip-mtu octets*
 - no *encapsulated-ip-mtu*
 - *icmp6-generation*
 - *packet-too-big* number [10..1000] seconds [1..60]
 - *packet-too-big*
 - no *packet-too-big*
 - *ip-mtu octets*
 - no *ip-mtu*
 - *replay-window* {32 | 64 | 128 | 256 | 512}
 - no *replay-window*
 - [no] *sp-reverse-route*
 - *transform transform-id* [*transform-id...*(up to 4 max)]
 - no *transform*

7.1.2 Services configuration command hierarchies

- *IES service interface command hierarchy*
- *VPRN service command hierarchy*
- *CMPv2 command hierarchies*

7.1.2.1 IES service interface command hierarchy

configure

- *service*
 - *ies service-id* [customer *customer-id*] [vpn *vpn-id*]
 - [no] *interface ip-int-name* [tunnel]
 - *dynamic-tunnel-redundant-next-hop ip-address*

- no *dynamic-tunnel-redundant-next-hop*
- *static-tunnel-redundant-next-hop* *ip-address*
- no *static-tunnel-redundant-next-hop*
- [no] *sap sap-id* [create]
 - *ip-tunnel ip-tunnel-name* [create]
 - *backup-remote-ip ip-address*
 - no *backup-remote-ip*
 - [no] *clear-df-bit*
 - *delivery-service* {*service-id* | *svc-name*}
 - no *delivery-service*
 - *description* *description-string*
 - no *description*
 - *dscp dscp-name*
 - no *dscp*
 - [no] *dest-ip ip-address*
 - *gre-header*
 - *gre-header* *send-key send-key* *receive-key receive-key*
 - no *gre-header*
 - *ip-mtu octets*
 - no *ip-mtu*
 - *reassembly* [*wait-msecs*]
 - no *reassembly*
 - *remote-ip ip-address*
 - no *remote-ip*
 - [no] *shutdown*
 - *source ip-address*
 - no *source*
- [no] *ipsec-gw*
 - *cert*
 - *cert filename*
 - no *cert*
 - *cert-profile profile*
 - no *cert-profile*
 - *key filename*
 - no *key*
 - *status-verify*
 - *default-result* {*revoked* | *good*}
 - no *default-result*
 - *primary* {*ocsp* | *crl*}
 - no *primary*
 - *secondary* {*ocsp* | *crl*}
 - no *secondary*
 - *trust-anchor ca-profile-name*
 - no *trust-anchor*
 - *trust-anchor-profile profile-name*

```

    – no trust-anchor-profile
    – trust-anchor ca-profile-name
    – no trust-anchor
    – trust-anchor-profile profile-name
    – no trust-anchor-profile
    – default-secure-service service-id ipsec-interface ip-int-name
    – no default-secure-service
    – default-tunnel-template ipsec template identifier
    – no default-tunnel-template
    – ike-policy ike-policy-id
    – no ike-policy
    – local-gateway-address ip-address
    – no local-gateway-address
    – local-id {ipv4 | fqdn} [value value]
    – no local-id
    – pre-shared-key key
    – no pre-shared-key
    – radius-accounting-policy policy-name
    – no radius-accounting-policy
    – radius-accounting-policy name
    – no radius-accounting-policy
    – radius-authentication-policy policy-name
    – no radius-authentication-policy
    – [no] shutdown
    – ts-negotiation ts-list list-name
    – no ts-negotiation

```

7.1.2.2 VPRN service command hierarchy

- *VPRN Service IPSec Command Hierarchy*
- *VPRN Service Interface Command Hierarchy*
- *SAP IPSec Gateway Command Hierarchy*
- *SAP IPSec Tunnel Command Hierarchy*

7.1.2.2.1 VPRN service IPSec command hierarchy

```

configure
– service
  – vprn service-id [customer customer-id]
  – no vprn service-id
  – ipsec
    – security-policy security-policy-id [create]

```

- no *security-policy* *security-policy-id*
- *entry* *entry-id* [create] [user-name *user-name*]
- no *entry* *entry-id*
 - *local-ip* {*ip-prefix/prefix-length* | *ip-prefix netmask* | any}
 - *local-v6-ip* *ipv6-prefix/prefix-length*
 - *local-v6-ip* any
 - no *local-v6-ip*
 - *remote-ip* {*ip-prefix/prefix-length* | *ip-prefix netmask* | any}
 - *remote-v6-ip* any
 - *remote-v6-ip* *ipv6-prefix/prefix-length*
 - no *remote-v6-ip*

7.1.2.2.2 VPRN service interface command hierarchy

configure

- service
 - *vprn service-id* [customer *customer-id*]
 - no *vprn service-id*
 - [no] *interface ip-int-name* [create] [tunnel]
 - *dynamic-tunnel-redundant-next-hop ip-address*
 - no *dynamic-tunnel-redundant-next-hop*
 - *ipv6*
 - *address ipv6-address/prefix-length* [eui-64] [preferred] [track-srrp *srrp-instance*]
 - no *address ipv6-address/prefix-length*
 - *link-local-address ipv6-address* [preferred]
 - *static-tunnel-redundant-next-hop ip-address*
 - no *static-tunnel-redundant-next-hop*
 - [no] *sap sap-id* [create]
 - *ip-tunnel ip-tunnel-name* [create]
 - *backup-remote-ip ip-address*
 - no *backup-remote-ip*
 - [no] *clear-df-bit*
 - *delivery-service* {*service-id* | *svc-name*}
 - no *delivery-service*
 - *description description-string*
 - no *description*
 - *dscp dscp-name*
 - no *dscp*
 - [no] *dest-ip ip-address*
 - [no] *gre-header*
 - *ip-mtu octets*
 - no *ip-mtu*
 - *reassembly* [wait-msecs]

- no *reassemble*
- *remote-ip* *ip-address*
- no *remote-ip*
- [no] *shutdown*
- *source* *ip-address*
- no *source*

7.1.2.2.3 SAP IPSec gateway command hierarchy

- [no] *ipsec-gw*
 - *cert*
 - *cert filename*
 - no *cert*
 - *cert-profile* *profile*
 - no *cert-profile*
 - *key filename*
 - no *key*
 - *status-verify*
 - *default-result* {revoked | good}
 - no *default-result*
 - *primary* {ocsp | crl}
 - no *primary*
 - *secondary* {ocsp | crl}
 - no *secondary*
 - *trust-anchor* *ca-profile-name*
 - no *trust-anchor*
 - *trust-anchor-profile* *profile-name*
 - no *trust-anchor-profile*
- *default-secure-service* *service-id* *ipsec-interface* *ip-int-name*
- no *default-secure-service*
- *default-tunnel-template* *ipsec* *template* *identifier*
- no *default-tunnel-template*
- *ike-policy* *ike-policy-id*
- no *ike-policy*
- *local-gateway-address* *ip-address*
- no *local-gateway-address*
- *local-id* {ipv4 | fqdn} [value *value*]
- no *local-id*
- *pre-shared-key* *key*
- no *pre-shared-key*
- *radius-accounting-policy* *policy-name*
- no *radius-accounting-policy*
- *radius-authentication-policy* *policy-name*

- no *radius-authentication-policy*
- [no] *shutdown*
- *ts-negotiation* *ts-list list-name*
- no *ts-negotiation*

7.1.2.2.4 SAP IPSec tunnel command hierarchy

- *ipsec-tunnel ipsec-tunnel-name* [create]
- no *ipsec-tunnel ipsec-tunnel-name*
 - [no] *bfd-designate*
 - *bfd-enable* *service service-id* *interface interface-name* *dst-ip ip-address*
 - [no] *clear-df-bit*
 - *description* *description-string*
 - no *description*
 - [no] *dest-ip ip-address*
 - [no] *dynamic-keying*
 - [no] *auto-establish*
 - *cert*
 - *cert filename*
 - no *cert*
 - *cert-profile profile*
 - no *cert-profile*
 - *key filename*
 - no *key*
 - *status-verify*
 - *default-result* {revoked | good}
 - no *default-result*
 - *primary* {ocsp | crl}
 - no *primary*
 - *secondary* {ocsp | crl}
 - no *secondary*
 - *trust-anchor ca-profile-name*
 - no *trust-anchor*
 - *trust-anchor-profile profile-name*
 - no *trust-anchor-profile*
 - *ike-policy ike-policy-id*
 - no *ike-policy*
 - *local-id* {ipv4 | fqdn} [value [255 chars max]]
 - no *local-id*
 - *pre-shared-key key*
 - no *pre-shared-key*
 - *transform transform-id* [*transform-id...*(up to 4 max)]
 - no *transform*

```

    - encapsulated-ip-mtu octets
    - no encapsulated-ip-mtu
    - icmp6-generation
      - packet-too-big
      - packet-too-big number [10..1000] seconds [1..60]
      - no packet-too-big
    - ip-mtu octets
    - no ip-mtu
    - local-gateway-address ip-address peer ip-address delivery-
service service-id
    - no local-gateway-address
    - [no] manual-keying
      - replay-window replay-window-size
      - no replay-window
      - security-association security-entry-id authentication-
key authentication-key encryption-key encryption-key spi spi transform transform-id
    direction {inbound | outbound}
      - no security-association security-entry-id direction {inbound |
outbound}
      - security-policy security-policy-id
      - no security-policy

```

7.1.2.3 CMPv2 command hierarchies

```

configure
- system
- security
  - pki
    - certificate-display-format {ascii | utf8}
    - ca-profile name [create]
    - no ca-profile name
    - cmpv2
      - [no] accept-unprotected-errormsg
      - [no] accept-unprotected-pkiconf
      - [no] always-set-sender-for-ir
      - http-response-timeout timeout
      - no http-response-timeout
      - http-version [1.0 | 1.1]
      - key-list
        - key password [hash | hash2] reference reference-number
        - no key reference reference-number
      - response-signing-cert filename
      - no response-signing-cert
      - [no] same-recipnonce-for-pollreq

```

- *url* *url-string* [*service-id* *service-id*]
- *no url*
- *revocation-check* {*crl* | *crl-optional*}

admin

- *certificate*
 - *cmpv2*
 - *cert-request* *ca* *ca-profile-name* *current-key* *key-filename* *current-cert* *cert-filename* [*hash-alg* *hash-algorithm*] *newkey* *key-filename* *subject-dn* *subject-dn* *save-as* *save-path-of-result-cert*
 - *clear-request* *ca* *ca-profile-name*
 - *initial-registration* *ca* *ca-profile-name* *key-to-certify* *key-filename* *protection-alg* {*password* *password* *reference* *ref-number* | *signature* [*cert* *cert-file-name* [*send-chain* [*with-ca* *ca-profile-name*]]] [*protection-key* *key-file-name*] [*hash-alg* {*md5* | *sha1* | *sha224* | *sha256* | *sha384* | *sha512*}]}
 - *key-update* *ca* *ca-profile-name* *newkey* *key-filename* *oldkey* *key-filename* *oldcert* *cert-filename* [*hash-alg* *hash-algorithm*] *save-as* *save-path-of-result-cert*
 - *poll* *ca* *ca-profile-name*
 - *show-request* [*ca* *ca-profile-name*]

7.1.3 IP tunnels show command hierarchy

show

- *ipsec*
 - *cert-profile* *name* *association*
 - *cert-profile* [*name*]
 - *cert-profile* *name* *entry* [1..8]
 - *certificate* *filename* *association*
 - *gateway* *name* *name*
 - *gateway* [*service* *service-id*]
 - *gateway* *tunnel* [*ip-address:port*]
 - *gateway* *name* *name* *tunnel* *ip-address:port*
 - *gateway* *name* *name* *tunnel*
 - *gateway* *tunnel* *count*
 - *ike-policy* *ike-policy-id*
 - *ike-policy*
 - *radius-accounting-policy* [*name*]
 - *radius-authentication-policy* [*name*]
 - *security-policy* *service* *service-id* [*security-policy-id* *security-policy-id*]
 - *security-policy*
 - *static-sa*
 - *static-sa* *name* *sa-name*
 - *static-sa* *spi* *spi*

- *transform* [*transform-id*]
- *trust-anchor-profile* *trust-anchor-profile* association
- *trust-anchor-profile* [*trust-anchor-profile*]
- *ts-list* [*list-name*]
- *ts-list* *list-name* association
- *ts-list* *list-name* entry [1 to 32]
- *tunnel* *ipsec-tunnel-name*
- *tunnel*
- *tunnel-template* [*ipsec-template-identifier*]

7.1.4 IP tunnels debug command hierarchy

```
debug
  – ipsec
    – [no] gateway name name tunnel ip-address[:port] [nat-ip nat-ip[:port]] [detail]
    – tunnel ipsec-tunnel-name [detail]
    – no tunnel ipsec-tunnel-name
    – [no] certificate filename
  – cmpv2
    – [no] ca-profile profile-name
  – ocsp
```

7.2 IP tunnel command descriptions

- *IP tunnels configuration command descriptions*
- *IP tunnel show command descriptions*
- *IP tunnel debug command descriptions*

7.2.1 IP tunnels configuration command descriptions

- *Generic command descriptions*
- *Hardware command descriptions*
- *ISA command descriptions*
- *IPSec command descriptions*
 - *Certificate profile command descriptions*
 - *Internet Key Exchange (IKE) command descriptions*
 - *IPSec transform command descriptions*

- *RADIUS policy command descriptions*
- *IPSec static security association command descriptions*
- *Trust anchor profile/TS list command descriptions*
- *Tunnel template command descriptions*
- *IES and VPRN service command descriptions*
 - *VPRN IPSec commands*
 - *SAP IPSec gateway commands*
 - *SAP IPSec tunnel commands*
- *CMPv2 commands*

7.2.1.1 Generic command descriptions

7.2.1.1.1 description

Syntax

`description description-string`

Context

`config>ipsec-static-sa`

`config>isa`

`config>isa>ipsec-group`

Description

This command creates a text description which is stored in the configuration file to help identify the content of the entity.

The no form of this command removes the string from the configuration.

Parameters

description-string

the description character string. Allowed values are any string up to 80 characters long composed of printable, 7-bit ASCII characters. If the string contains special characters (#, \$, spaces, and so on), the entire string must be enclosed within double quotes.

7.2.1.1.2 shutdown

Syntax

[no] shutdown

Context

config>isa

config>isa>aa-group

config>isa>tunnel-grp

config>ipsec>cert-profile

Description

This command administratively disables the entity. When disabled, an entity does not change, reset, or remove any configuration settings or statistics. Many entities must be explicitly enabled using the no shutdown command.

The shutdown command administratively disables an entity. The operational state of the entity is disabled as well as the operational state of any entities contained within. Many objects must be shut down before they can be deleted.

7.2.1.2 Hardware command descriptions

7.2.1.2.1 mda-type

Syntax

mda-type *isa-tunnel*

no mda-type

Context

config>card>mda

Description

This command provisions or de-provisions an MDA to or from the device configuration for the slot.

Parameters

isa-tunnel

specifies the ISA tunnel

7.2.1.3 ISA command descriptions

7.2.1.3.1 isa

Syntax

isa

Context

config

Description

This command enables the context to configure Integrated Services Adapter (ISA) parameters.

7.2.1.3.2 ipsmg-group

Syntax

[no] ipsmg-group *ipsmg-group-id* [create]

Context

config>isa

Description

This command creates an IPsec MG (IPSMG) group that is used to terminate IPsec tunnels for the ePDG. The IPSMG group must contain all ISAs that are to be assigned to the IPSMG function as part of the ePDG. Currently only the second ISA in an ISM-MG can be used for IPSMG.

The no form of this command deletes the specified group from the configuration.

Default

no ipsmg-group

Parameters

ipsmg-group-id

an integer value that uniquely identifies the tunnel-group

Values: 1 to 8

create

mandatory keyword used when creating tunnel group in the ISA context. The create keyword requirement can be enabled/disabled in the `environment>create` context.

7.2.1.3.3 mda

Syntax

[no] mda *mda-id* [create]

Context

config>isa>ipsmg-group

Description

This command identifies an ISA that is to belong to the IPSMG group. The IPSMG group must contain all ISAs that are to be assigned to the IPSMG function as part of the ePDG. Currently, only the second ISA in an ISM-MG can be used for IPSMG.

The no form of this command deletes the specified group from the configuration.

Parameters

mda-id

Specifies the slot/mda identifying a provisioned IPSMG ISA.

Values: slot/mda

create

mandatory keyword used when creating tunnel group in the ISA context. The create keyword requirement can be enabled/disabled in the `environment>create` context.

7.2.1.3.4 tunnel-group

Syntax

tunnel-group *tunnel-group-id* [create]

no tunnel-group *tunnel-group-id*

Context

config>isa

Description

This command allows a tunnel group to be created or edited. A tunnel group is a set of one or more MS-ISAs that support the origination and termination of IPSec and IP/GRE tunnels. All of the MS-ISAs in a tunnel group must have isa-tunnel as their configured mda-type.

The no form of this command deletes the specified tunnel group from the configuration.

Parameters

tunnel-group-id

an integer value that uniquely identifies the tunnel-group

Values: 1 to 16

create

mandatory keyword used when creating tunnel group in the ISA context. The create keyword requirement can be enabled/disabled in the environment>create context.

7.2.1.3.5 active-mda-number

Syntax

active-mda-number *number*

no active-mda-number

Context

config>isa>tunnel-grp

Description

This command specifies the number of active MS-ISA within all configured MS-ISA in the tunnel-group with multi-active enabled. IPSec traffic will be load balanced across all active MS-ISAs. If the number of configured MS-ISA is greater than the active-mda-number then the delta number of MS-ISA will be backup.

Default

no active-mda-number

Parameters

number

specifies the number of active MDAs

Values: 1 to 16

7.2.1.3.6 backup

Syntax

backup *mda-id*

no backup

Context

config>isa>tunnel-grp

Description

This command assigns an ISA IPSec module configured in the specified slot to this IPSec group. The backup module provides the IPSec group with warm redundancy when the primary module in the group is configured. An IPSec group must always have a primary configured.

Primary and backup modules have equal operational status and when both modules are coming up, the one that becomes operational first becomes the active module. An IPSec module can serve as a backup for multiple IPSec groups but the backup can become active for only one ISA IPSec group at a time.

All configuration information is pushed down to the backup MDA from the CPM once the CPM gets notice that the primary module has gone down. This allows multiple IPSec groups to use the same backup module. Any statistics not yet spooled will be lost. Auto-switching from the backup to primary, once the primary becomes available again, is supported.

The operator is notified through SNMP events when:

- When the ISA IPSec service goes down (all modules in the group are down) or comes back up (a module in the group becomes active).
- When ISA IPSec redundancy fails (one of the modules in the group is down) or recovers (the failed module comes back up).
- When an ISA IPSec activity switch took place.

The no form of this command removes the specified module from the IPSec group.

Default

no backup

Parameters

mda-id

specifies the card/slot identifying a provisioned module to be used as a backup module

Values	mda-id	slot/mda
–	–	slot: 1 to 10 depending on chassis model
–	–	mda: 1 to 2

7.2.1.3.7 mda

Syntax

```
mda mda-id
no mda
```

Context

```
config>isa>tunnel-grp
```

Description

This command specifies the MDA id of the MS-ISA as the member of tunnel-group with multi-active enabled. Up to 16 MDA could be configured under the same tunnel-group.

Default

```
no mda
```

Parameters

```
mda-id
    specifies the id of MS-ISA
    Values: iom-slot-id | mda-slot-id
```

7.2.1.3.8 multi-active

Syntax

```
[no] multi-active
```

Context

config>isa>tunnel-grp

Description

This command enables configuring multiple active MS-ISA in the tunnel-group. IPSec traffic will be load balanced to configured active MS-ISAs.



NOTE:

- A shutdown of group and removal of all existing configured tunnels of the tunnel-group are needed before provisioning the multi-active command.
- If the tunnel-group is admin-up with multi-active configured then the configuration of "primary" and "backup" are not allowed.
- The active-mda-number must be $\leq$ total number of ISA configured.
 - If active-mda-number is less than the total number of ISA configured then the delta number of ISA will become backup ISA.

Default

no multi-active

7.2.1.3.9 primary

Syntax

primary *mda-id*

no primary

Context

config>isa>tunnel-grp

Description

This command assigns an ISA IPSec module configured in the specified slot to this IPSec group. The backup ISA IPSec provides the IPSec group with warm redundancy when the primary ISA IPSec in the group is configured. Primary and backup ISA IPSec have equal operational status and when both MDAs are coming up, the one that becomes operational first becomes the active ISA IPSec.

All configuration information is pushed down to the backup MDA from the CPM once the CPM gets notice that the primary module has gone down. This allows multiple IPSec groups to use the same backup module.

Any statistics not yet spooled will be lost. Auto-switching from the backup to primary, once the primary becomes available again, is supported.

The operator is notified through SNMP events when:

- When the ISA IPSec service goes down (all modules in the group are down) or comes back up (a module in the group becomes active).
- When ISA IPSec redundancy fails (one of the modules in the group is down) or recovers (the failed module comes back up).
- When an ISA IPSec activity switch took place.

The no form of this command removes the specified primary ID from the group's configuration.

Default

no primary

Parameters

mda-id

specifies the card/slot identifying a provisioned IPSec ISAA

7.2.1.3.10 reassembly

Syntax

reassembly [*wait-msecs*]

no reassembly

Context

config>isa>tunnel-group

config>service>ies>interface>sap>ip-tunnel

config>service>vpn>interface>sap>ip-tunnel

Description

This command configures IP packet reassembly for IPSec and GRE tunnels supported by an MS-ISA. The reassembly command at the tunnel-group level configures IP packet reassembly for all IPSec and GRE tunnels associated with the tunnel-group. The reassembly command at the GRE tunnel level configures IP packet reassembly for that one specific GRE tunnel, overriding the tunnel-group configuration.

The no form of this command disables IP packet reassembly.

Default

no reassembly (tunnel-group level)

reassembly (ip-tunnel level)

Parameters

wait-msecs

specifies the maximum number of milliseconds that the ISA tunnel application will wait to receive all fragments of a particular IPSec or GRE packet. If one or more fragments are still missing when this limit is reached the partially reassembled datagram is discarded and an ICMP time exceeded message is sent to the source host (if allowed by the ICMP configuration of the sending interface). Internally, the configured value is rounded up to the nearest multiple of 100 ms.

Values: 100 to 5000

Default: 2000 (tunnel-group level)

7.2.1.4 IPSec command descriptions

7.2.1.4.1 Certificate profile command descriptions

7.2.1.4.1.1 ipsec

Syntax

ipsec

Context

config

Description

This command enables the context to configure IPSec parameters. IPSec is a structure of open standards to ensure private, secure communications over IP networks by using cryptographic security services.

7.2.1.4.1.2 cert-profile

Syntax

cert-profile *profile-name* [create]

```
no cert-profile profile-name
```

Context

```
config>ipsec
```

Description

This command creates a new certificate profile or enters the configuration context of an existing certificate profile.

The no form of this command removes the profile name from the certificate profile configuration.

Parameters

profile-name

specifies the name of the certificate profile, up to 32 characters

7.2.1.4.1.3 entry

Syntax

```
entry entry-id[create]
```

```
no entry entry-id
```

Context

```
config>ipsec>cert-profile
```

Description

This command configures the certificate profile entry information

The no form of this command removes the entry ID from the certificate profile configuration.

Parameters

entry-id

specifies the entry ID

Values: 1 to 8

7.2.1.4.1.4 cert

Syntax

```
cert cert-filename  
no cert
```

Context

```
config>ipsec>cert-profile>entry
```

Description

This command specifies the filename of an imported certificate for the certificate profile entry. The no form of this command removes the certificate filename from the entry configuration.

Default

no cert

7.2.1.4.1.5 key

Syntax

```
key key-filename  
no key
```

Context

```
config>ipsec>cert-profile>entry
```

Description

This command specifies the filename of an imported key for the certificate profile entry. The no form of this command removes the key filename from the entry configuration.

Default

no key

Parameters

key-filename

specifies the filename of an imported key

7.2.1.4.1.6 send-chain

Syntax

[no] send-chain

Context

config>ipsec>cert-profile>entry

Description

This command enters the configuration context of send-chain in the certificate profile entry.

The configuration of this command is optional, by default system will only send the certificate specified by `cert` command in the selected entry to the peer. This command allows system to send additional CA certificates to the peer. These additional CA certificates must be in the certificate chain of the certificate specified by the `cert` command in the same entry.

7.2.1.4.1.7 ca-profile

Syntax

[no] ca-profile *name*

Context

config>ipsec>cert-profile>entry>send-chain

Description

This command specifies a CA certificate in the specified CA profile to be sent to the peer.

Multiple configurations (up to seven) of this command are allowed in the same entry.

Default

no ca-profile

Parameters

name

specifies the profile name, up to 32 characters

7.2.1.4.2 Internet Key Exchange (IKE) command descriptions

7.2.1.4.2.1 ike-policy

Syntax

```
ike-policy ike-policy-id [create]
```

```
no ike-policy ike-policy-id
```

Context

```
config>ipsec
```

Description

This command enables the context to configured an IKE policy.

Parameters

ike-policy-id

specifies a policy ID value to identify the IKE policy

Values: 1 to 2048

7.2.1.4.2.2 auth-algorithm

Syntax

```
auth-algorithm auth-algorithm
```

```
no auth-algorithm
```

Context

```
config>ipsec>ike-policy
```

Description

The command specifies which hashing algorithm to use for the IKE authentication function.

The no form of this command removes the parameter from the configuration.

Parameters

auth-algorithm

specifies which hashing algorithm to use

Values	md5	specifies the hmac-md5 algorithm for authentication
–	sha1	specifies the hmac-sha1 algorithm for authentication
–	sha256	specifies the sha256 algorithm for authentication
–	sha384	specifies the sha384 algorithm for authentication
–	sha512	specifies the sha512 algorithm for authentication
–	aes-xcbc	specifies the aes-xcbc algorithm for authentication

7.2.1.4.2.3 auth-method

Syntax

```
auth-method {psk | plain-psk-xauth | cert-auth | psk-radius | cert-radius | eap | auto-  
eap-radius}
```

```
no auth-method
```

Context

```
config>ipsec>ike-policy
```

Description

This command specifies the authentication method used with this IKE policy.

The no form of this command removes the parameter from the configuration.

Default

```
no auth-method
```

Parameters

psk

Both client and gateway authenticate each other by a hash derived from a pre-shared secret. Both client and gateway must have the PSK. This work with both IKEv1 and IKEv2

plain-psk-xauth

Both client and gateway authenticate each other by pre-shared key and RADIUS. This work with IKEv1 only.

psk-radius

Use the pre-shared-key and RADIUS to authenticate. IKEv2 remote-access tunnel only.

cert-radius

Use the certificate, public/private key and RADIUS to authenticate. IKEv2 remote-access tunnel only.

eap

Use the EAP to authenticate peer. IKEv2 remote-access tunnel only

auto-eap-radius

Use EAP or potentially other method to authenticate peer. IKEv2 remote-access tunnel only. See also the auto-eap-method and the auto-eap-own-method commands.

7.2.1.4.2.4 auto-eap-method

Syntax

```
auto-eap-method {psk | cert | psk-or-cert}
```

Context

```
config>ipsec>ike-policy
```

Description

This command enables the following behavior for IKEv2 remote-access tunnel when auth-method is configured as auto-eap-radius:

- If there is no AUTH payload in IKE_AUTH request, the system uses EAP to authenticate client and uses own-auth-method to generate AUTH payload.
- If there is AUTH payload in IKE_AUTH request:
 - if auto-eap-method is psk, the system proceeds as auth-method:psk-radius
 - if auto-eap-method is cert, the system proceeds as auth-method:cert-radius
 - if auto-eap-method is psk-or-cert:
 - if the "Auth Method" field of AUTH payload is PSK, the system proceeds as auth-method:psk-radius

- if the "Auth Method" field of AUTH payload is RSA or DSS, the system proceeds as auth-method:cert-radius
- The system uses auto-eap-own-method to generate AUTH payload.

This command only applies when auth-method is configured as auto-eap-radius.

Parameters

psk

Uses the pre-shared-key as the authentication method.

cert

Uses the certificate as the authentication method.

psk-or-cert

Uses either the pre-shared-key or certificate based on the "Auth Method" field of the received AUTH payload.

Default: cert

7.2.1.4.2.5 auto-eap-own-method

Syntax

```
auto-eap-own-method {psk | cert}
```

Context

```
config>ipsec>ike-policy
```

Description

This command enables following behavior for IKEv2 remote-access tunnel when auth-method is configured as auto-eap-radius:

- If there is no AUTH payload in IKE_AUTH request, the system uses EAP to authenticate client and uses own-auth-method to generate AUTH payload.
- If there is AUTH payload in IKE_AUTH request:
 - if auto-eap-method is psk, the system proceeds as auth-method:psk-radius.
 - if auto-eap-method is cert, the system proceeds as auth-method:cert-radius.
 - if auto-eap-method is psk-or-cert:
 - if the "Auth Method" field of AUTH payload is PSK, the system proceeds as auth-method:psk-radius.

- if the "Auth Method" field of AUTH payload is RSA or DSS, the system proceeds as auth-method:cert-radius.

- The system will use auto-eap-own-method to generate AUTH payload.

This command only applies when auth-method is configured as auto-eap-radius.

Parameters

psk

uses a pre-shared-key to generate AUTH payload

cert

uses a public/private key to generate AUTH payload

Default: cert

7.2.1.4.2.6 dh-group

Syntax

dh-group {1 | 2 | 5 | 14 | 15}

no dh-group

Context

config>ipsec>ike-policy

Description

This command specifies which Diffie-Hellman group to calculate session keys. The following groups are supported with IKE-v1:

- Group 1: 768 bits
- Group 2: 1024 bits
- Group 5: 1536 bits
- Group 14: 2048 bits
- Group 15: 3072 bits

More bits provide a higher level of security, but require more processing.

The no form of this command removes the Diffie-Hellman group specification.

Default

5

7.2.1.4.2.7 dpd

Syntax

```
dpd [interval interval] [max-retries max-retries] [reply-only]  
no dpd
```

Context

```
config>ipsec>ike-policy
```

Description

This command controls the dead peer detection mechanism.

The no form of this command removes the parameters from the configuration.

Parameters

interval

specifies the interval used to test connectivity to the tunnel peer, in seconds. If the peer initiates the connectivity check before the interval timer, it will be reset.

Values: 10 to 300

Default: 30

max-retries

specifies the maximum number of retries before the tunnel is removed

Values: 2 to 5

Default: 3

reply-only

specifies to only reply to DPD keepalives. Issuing the command without the reply-only keyword disables the behavior

7.2.1.4.2.8 encryption-algorithm

Syntax

```
encryption-algorithm {des | 3des | aes128 | aes192 | aes256}  
no encryption-algorithm
```

Context

```
config>ipsec>ike-policy
```

Description

This command specifies the encryption algorithm to use for the IKE session.

The no form of this command removes the encryption algorithm from the configuration.

Parameters

des

This parameter configures the 56-bit des algorithm for encryption. This is an older algorithm, with relatively weak security. While better than nothing, it should only be used where a strong algorithm is not available on both ends at an acceptable performance level.

3des

This parameter configures the 3-des algorithm for encryption. This is a modified application of the des algorithm which uses multiple des operations for more security.

aes128

This parameter configures the aes algorithm with a block size of 128 bits. This is the mandatory implementation size for aes.

aes192

This parameter configures the aes algorithm with a block size of 192 bits. This is a stronger version of aes.

aes256

This parameter configures the aes algorithm with a block size of 256 bits. This is the strongest available version of aes.

Default: aes128

7.2.1.4.2.9 ike-mode

Syntax

```
ike-mode {main | aggressive}
```

```
no ike-mode
```

Context

```
config>ipsec>ike-policy
```


Description

This command specifies one of either two modes of operation. IKE version 1 can support main mode and aggressive mode. The difference lies in the number of messages used to establish the session.

The no form of this command removes the mode of operation from the configuration.

Parameters

main

specifies identity protection for the hosts initiating the IPSec session. This mode takes slightly longer to complete.

aggressive

aggressive mode provides no identity protection but is faster

Default: main

7.2.1.4.2.10 ike-version

Syntax

```
ike-version[1..2]
```

```
no ike-version
```

Context

```
config>ipsec>ike-policy
```

Description

This command sets the IKE version (1 or 2) that the IKE policy will use.

Parameters

1 | 2

the version of IKE protocol

Default: 1

7.2.1.4.2.11 ipsec-lifetime

Syntax

```
ipsec-lifetime ipsec-lifetime
```

```
no ipsec-lifetime
```

Context

config>ipsec>ike-policy

Description

This parameter specifies the lifetime of a phase two SA.

The no form of this command reverts the *ipsec-lifetime* value to the default.

Parameters

ipsec-lifetime

specifies the lifetime of the phase two IKE key in seconds

Values: 1200 to 172800

Default: 3600

7.2.1.4.2.12 isakmp-lifetime

Syntax

isakmp-lifetime *isakmp-lifetime*

no isakmp-lifetime

Context

config>ipsec>ike-policy

Description

This command specifies the lifetime of a phase one SA. ISAKMP stands for Internet Security Association and Key Management Protocol.

The no form of this command reverts the *isakmp-lifetime* value to the default.

Parameters

isakmp-lifetime

specifies the lifetime of the phase one IKE key in seconds

Values: 1200 to 172800

Default: 86400

7.2.1.4.2.13 match-peer-id-to-cert

Syntax

```
[no] match-peer-id-to-cert
```

Context

```
config>ipsec>ike-policy
```

Description

This command enables checking the IKE peer's ID matches the peer's certificate when performing certificate authentication.

7.2.1.4.2.14 nat-traversal

Syntax

```
nat-traversal [force] [keep-alive-interval keep-alive-interval] [force-keep-alive]  
no nat-traversal
```

Context

```
config>ipsec>ike-policy
```

Description

This command specifies whether NAT-T (Network Address Translation Traversal) is enabled, disabled or in forced mode.

The no form of this command reverts the parameters to the default.

Default

```
no nat-traversal
```

Parameters

force

forces to enable NAT-T

keep-alive-interval

specifies the keep-alive interval, in seconds

Values: 10 to 3600

force-keep-alive

When specified, the keep alive does not expire.

7.2.1.4.2.15 own-auth-method**Syntax**

```
own-auth-method {psk | cert | eap-only}
no own-auth-method
```

Context

```
config>ipsec>ike-policy
```

Description

This command configures the authentication method used with this IKE policy on its own side.

7.2.1.4.2.16 pfs**Syntax**

```
pfs [dh-group {1 | 2 | 5 | 14 | 15}]
no pfs
```

Context

```
config>ipsec>ike-policy
```

Description

This command enables perfect forward secrecy on the IPSec tunnel using this policy. PFS provides for a new Diffie-hellman key exchange each time the SA key is renegotiated. After that SA expires, the key is forgotten and another key is generated (if the SA remains up). This means that an attacker who cracks part of the exchange can only read the part that used the key before the key changed. There is no advantage in cracking the other parts if they attacker has already cracked one.

The no form of this command disables PFS. If this it turned off during an active SA, when the SA expires and it is time to re-key the session, the original Diffie-hellman primes will be used to generate the new keys.

Parameters

dh-group {1 | 2 | 5 | 14 | 15}

specifies which Diffie-Hellman group to use for calculating session keys. More bits provide a higher level of security, but require more processing. The following groups are supported with IKE-v1:

Group 1: 768 bits

Group 2: 1024 bits

Group 5: 1536 bits

Group 14: 2048 bits

Group 15: 3072 bits

Default: 15

7.2.1.4.2.17 relay-unsolicited-cfg-attribute

Syntax

```
relay-unsolicited-cfg-attribute
```

Context

```
config>ipsec>ike-policy
```

Description

This command enters relay unsolicited configuration attributes context. With this configuration, the configured attributes returned from source (such as a RADIUS server) will be returned to IKEv2 remote-access tunnel client regardless if the client has requested it in the CFG_REQUEST payload.

7.2.1.4.2.18 internal-ip4-dns

Syntax

```
[no] internal-ip4-dns
```

Context

```
config>ipsec>ike-policy>relay-unsol-attr
```

Description

This command returns the IPv4 DNS server address from the source (such as a RADIUS server) to the IKEv2 remote-access tunnel client regardless if the client has requested it in the CFG_REQUEST payload.

7.2.1.4.2.19 internal-ip4-netmask

Syntax

```
[no] internal-ip4-netmask
```

Context

```
config>ipsec>ike-policy>relay-unsol-attr
```

Description

This command returns the IPv4 netmask from the source (such as a RADIUS server) to the IKEv2 remote-access tunnel client regardless if the client has requested it in the CFG_REQUEST payload.

7.2.1.4.2.20 internal-ip6-dns

Syntax

```
[no] internal-ip6-dns
```

Context

```
config>ipsec>ike-policy>relay-unsol-attr
```

Description

This command returns the IPv6 DNS server address from the source (for example, RADIUS server) to the IKEv2 remote-access tunnel client regardless if the client has requested it in the CFG_REQUEST payload.

7.2.1.4.3 Internet Key Exchange (IKE) transform command descriptions

7.2.1.4.3.1 ike-transform

Syntax

```
ike-transform ike-transform-id [create]
```

```
no ike-transform ike-transform-id
```

Context

```
config>ipsec
```

Description

This command creates a new or enters an existing IKE transform instance. The IKE transform includes the following configuration for IKE SA:

- DH Group
- IKE authentication algorithm
- IKE encryption algorithm
- IKE SA lifetime

The *ike-transform-id* is referenced in the *ike-policy* configuration.

The no form of this command removes the ID from the configuration.

Parameters

ike-transform-id

specifies a number used to uniquely identify an IKE transform instance

Values: 1 to 4096

create

keyword used to create the *ike-transform* instance. The *create* keyword requirement can be enabled or disabled in the *environment>create* context.

7.2.1.4.3.2 dh-group

Syntax

```
dh-group {1 | 2 | 5 | 14 | 15 | 19 | 20 | 21}
```

Context

```
config>ipsec>ike-transform
```

Description

This command specifies which Diffie-Hellman group to be used in this IKE transform instance.

Parameters

dh-group {1 | 2 | 5 | 14 | 15 | 19 | 20 | 21}

specifies which Diffie-Hellman group to use for calculating session keys. More bits provide a higher level of security, but require more processing. Three groups are supported with IKE-v1:

- Group 1: 768 bits

- Group 2: 1024 bits
- Group 5: 1536 bits
- Group 14: 2048 bits
- Group 15: 3072 bits
- Group 19: P-256 ECC Curve, 256 bits
- Group 20: P-384 ECC Curve, 384 bits
- Group 21: P-512 ECC Curve, 512 bits

Default: 2 (1024-bit — More Modular Exponential (MODP))

7.2.1.4.3.3 ike-auth-algorithm

Syntax

```
ike-auth-algorithm {md5 | sha1 | sha256 | sha384 | sha512 | aes-xcbc | auth-encryption}
```

Context

```
config>ipsec>ike-transform
```

Description

This command specifies the IKE authentication algorithm for the IKE transform.

Parameters

md5	specifies the hmac-md5 algorithm for authentication
sha1	specifies the hmac-sha1 algorithm for authentication
sha256	specifies the hmac-sha256 algorithm for authentication
sha384	specifies the hmac-sha384 algorithm for authentication
sha512	specifies the hmac-sha512 algorithm for authentication
aes-xcbc	specifies the aes-xcbc (RFC 3566, <i>The AESXCBC-MAC-96 Algorithm and Its Use With IPsec</i>) algorithm for authentication

auth-encryption specifies the auth-encryption algorithm for authentication

Default: sha1

7.2.1.4.3.4 ike-encryption-algorithm

Syntax

```
encryption-algorithm {des | 3des | aes128 | aes192 | aes256 | aes128-gcm8 | aes128-gcm16  
| aes256-gcm8 | aes256-gcm16}
```

Context

```
config>ipsec>ike-transform
```

Description

This command specifies the IKE encryption algorithm to be used in the IKE transform instance.

Parameters

des

configures the 56-bit des algorithm for encryption. This is an older algorithm, with relatively weak security. While better than nothing, it should only be used where a strong algorithm is not available on both ends at an acceptable performance level.

3des

configures the 3-des algorithm for encryption. This is a modified application of the des algorithm which uses multiple des operations to make information more secure.

aes128

configures the aes algorithm with a block size of 128 bits. This is a mandatory implementation size for aes. This is a very strong algorithm choice.

aes192

configures the aes algorithm with a block size of 192 bits. This is a stronger version of aes.

aes256

configures the aes algorithm with a block size of 256 bits. This is the strongest available version of aes.

aes128-gcm8

configures ESP to use aes-gcm with a 128-bit key size and an 8-byte ICV for encryption and authentication

aes128-gcm16

configures ESP to use aes-gcm with a 128-bit key size and an 16-byte ICV for encryption and authentication

aes256-gcm8

configures ESP to use aes-gcm with a 256-bit key size and a 8-byte ICV for encryption and authentication

aes256-gcm16

configures ESP to use aes-gcm with a 256-bit key size and a 16-byte ICV for encryption and authentication

Default: aes128

7.2.1.4.3.5 ike-prf-algorithm

Syntax

```
ike-prf-algorithm {md5 | sha1 | sha256 | sha384 | sha512 | aes-xcbc | auth-encryption}
```

Context

```
config>ipsec>ike-transform
```

Description

This command specifies the Pseudo-Random-Function (PRF) algorithm for the IKE security association.

**NOTE:**

If an authenticated encryption algorithm like AES-GCM is used for IKE encryption algorithm, same-as-auth cannot be used for ike-prf-algorithm.

Parameters

md5

configures IKE to use the hmac-md5 algorithm for PRF

sha1

configures IKE to use the hmac-sha1 algorithm for PRF

sha256

configures IKE to use the hmac-sha256 algorithm for PRF

sha384

configures IKE to use the hmac-sha384 algorithm for PRF

sha512

configures IKE to use the hmac-sha512 algorithm for PRF

aes-xcbc

configures IKE to use the aes128-xcbc algorithm for PRF

same-as-auth

configures the same algorithm as IKE authentication algorithm

Default: same-as-auth

7.2.1.4.3.6 isakmp-lifetime

Syntax

`isakmp-lifetime seconds`

Context

`config>ipsec>ike-transform`

Description

This command specifies the lifetime of the IKE SA.

Parameters***seconds***

specifies the Phase 1 life time for this IKE transform

Values: 1200 to 31536000

Default: 86400

7.2.1.4.4 IPsec transform command descriptions

7.2.1.4.4.1 ipsec-transform

Syntax

`ipsec-transform transform-id [create]`

`no ipsec-transform transform-id`

Context

`config>ipsec`

Description

This command enables the context to create an ipsec-transform policy. IPSec transforms policies can be shared. A change to the ipsec-transform is allowed at any time. The change does not impact tunnels that have been established until they are renegotiated. If the change is required immediately the tunnel must be cleared (reset) for force renegotiation.

IPSec transform policy assignments to a tunnel require the tunnel to be shutdown.

The no form of this command removes the ID from the configuration.

Parameters

transform-id

specifies a policy ID value to identify the IPSec transform policy

Values: 1 to 2048

create

This keyword is mandatory when creating an ipsec-transform policy. The create keyword requirement can be enabled/disabled in the environment>create context.

7.2.1.4.4.2 esp-auth-algorithm

Syntax

```
esp-auth-algorithm {null | md5 | sha1 | sha256 | sha384 | sha512| aes-xcbc}
```

```
no esp-auth-algorithm
```

Context

```
config>ipsec>transform
```

Description

The command specifies which hashing algorithm should be used for the authentication function ESP. Both ends of a manually configured tunnel must share the same configuration parameters for the IPSec tunnel to enter the operational state.

The no form of this command disables the authentication.

Parameters

null

This is a very fast algorithm specified in RFC 2410, which provides no authentication.

md5

This parameter configures ESP to use the hmac-md5 algorithm for authentication.

sha1

This parameter configures ESP to use the hmac-sha1 algorithm for authentication.

sha256

This parameter configures ESP to use the sha256 algorithm for authentication.

sha384

This parameter configures ESP to use the sha384 algorithm for authentication.

sha512

This parameter configures ESP to use the sha512 algorithm for authentication.

aes-xcbc

specifies the aes-xcbc algorithm for authentication

7.2.1.4.4.3 esp-encryption-algorithm

Syntax

```
esp-encryption-algorithm {null | des | 3des | aes128 | aes192 | aes256}  
no esp-encryption-algorithm
```

Context

```
config>ipsec>transform
```

Description

This command specifies the encryption algorithm to use for the IPSec session. Encryption only applies to esp configurations. If encryption is not defined, ESP will not be used.

For IPSec tunnels to come up, both ends need to be configured with the same encryption algorithm.

Parameters**null**

This parameter configures the high-speed null algorithm, which does nothing. This is the same as not having encryption turned on.

des

This parameter configures the 56-bit des algorithm for encryption. This is an older algorithm, with relatively weak security. Although slightly better than no encryption, it should only be used where a strong algorithm is not available on both ends at an acceptable performance level.

3des

This parameter configures the 3-des algorithm for encryption. This is a modified application of the des algorithm which uses multiple des operations to make things more secure.

aes128

This parameter configures the aes algorithm with a block size of 128 bits. This is the mandatory implementation size for aes. As of today, this is a very strong algorithm choice.

aes192

This parameter configures the aes algorithm with a block size of 192 bits. This is a stronger version of aes.

aes256

This parameter configures the aes algorithm with a block size of 256 bits. This is the strongest available version of aes.

Default: aes128

7.2.1.4.5 RADIUS accounting policy command descriptions

7.2.1.4.5.1 radius-accounting-policy

Syntax

```
radius-accounting-policy name [create]
```

```
no radius-accounting-policy name
```

Context

```
config>ipsec
```

Description

This command specifies an existing RADIUS accounting policy to use to collect accounting statistics on this subscriber profile by RADIUS. This command is used independently of the `collect - stats` command.

Parameters

name

specifies an existing RADIUS based accounting policy

create

keyword used to create the RADIUS accounting policy. The `create` keyword requirement can be enabled/disabled in the `environment>create` context.

7.2.1.4.5.2 include-radius-attribute

Syntax

```
[no] include-radius-attribute
```

Context

```
config>ipsec>rad-acct-plcy>include
```

```
config>ipsec>rad-auth-plcy>include
```

Description

This command enables the context to specify the RADIUS parameters that the system should include into RADIUS authentication-request messages.

7.2.1.4.5.3 called-station-id

Syntax

```
[no] called-station-id
```

Context

```
config>ipsec>rad-acct-plcy>include
```

```
config>ipsec>rad-auth-plcy>include
```

Description

This command includes called station ID attributes.

The no form of this command excludes called station id attributes.

7.2.1.4.5.4 calling-station-id

Syntax

```
[no] calling-station-id
```

Context

```
config>ipsec>rad-acct-plcy>include
```

```
config>ipsec>rad-auth-plcy>include
```

Description

This command enables the inclusion of the calling-station-id attribute in the RADIUS authentication requests and RADIUS accounting messages.

Default

no calling-station-id

7.2.1.4.5.5 framed-ip-addr

Syntax

[no] framed-ipv6-prefix

Context

config>ipsec>rad-acct-plcy>include

Description

This command enables the inclusion of the framed-ip-addr attribute.

7.2.1.4.5.6 framed-ipv6-prefix

Syntax

[no] framed-ipv6-prefix

Context

config>ipsec>rad-acct-plcy>include

Description

This command enables the inclusion of the framed IPv6 prefix attribute.

7.2.1.4.5.7 nas-identifier

Syntax

[no] nas-identifier

Context

```
config>ipsec>rad-acct-plcy>include
```

```
config>ipsec>rad-auth-plcy>include
```

Description

This command enables the generation of the nas-identifier RADIUS attribute.

7.2.1.4.5.8 nas-ip-addr

Syntax

```
[no] nas-ip-addr
```

Context

```
config>ipsec>rad-acct-plcy>include
```

```
config>ipsec>rad-auth-plcy>include
```

Description

This command enables the generation of the nas-ip-address attribute.

7.2.1.4.5.9 nas-port-id

Syntax

```
[no] nas-port-id
```

Context

```
config>ipsec>rad-acct-plcy>include
```

```
config>ipsec>rad-auth-plcy>include
```

Description

This command enables the generation of the nas-port-id RADIUS attribute. Optionally, the value of this attribute (the SAP-id) can be prefixed by a fixed string and suffixed by the circuit-id or the remote-id of the client connection. If a suffix is configured, but no corresponding data is available, the suffix used will be 0/0/0/0/0/0.

7.2.1.4.5.10 radius-server-policy

Syntax

```
radius-server-policy radius-server-policy-name  
no radius-server-policy
```

Context

```
config>ipsec>rad-acct-plcy>include  
config>ipsec>rad-auth-plcy>include
```

Description

This command references an existing radius-server-policy (available under the config>aaa context) for use in subscriber management authentication and accounting.

When configured in an authentication-policy, the following CLI commands are ignored in the policy to avoid conflicts:

- all commands in the radius-authentication-server context
- accept-authorization-change
- coa-script-policy
- accept-script-policy
- request-script-policy

When configured in a radius-accounting-policy, the following CLI commands are ignored in the policy to avoid conflicts:

- all commands in the radius-accounting-server context
- acct-request-script-policy

The no form of this command removes the radius-server-policy reference from the configuration

Default

```
no radius-server-policy
```

Parameters

radius-server-policy-name

specifies the RADIUS server policy

7.2.1.4.5.11 radius-authentication-policy

Syntax

```
radius-authentication-policy name [create]  
no radius-authentication-policy name
```

Context

```
config>ipsec
```

Description

This command specifies the RADIUS authentication policy associated with this IPSec gateway.

Parameters

name

specifies an existing RADIUS based authentication policy

create

keyword used to create the RADIUS authentication policy. The create keyword requirement can be enabled/disabled in the environment>create context.

7.2.1.4.5.12 client-cert-subject-key-id

Syntax

```
[no] client-cert-subject-key-id
```

Context

```
config>ipsec>rad-auth-plcy>include
```

Description

This command enables the inclusion of the Subject Key Identifier of the peer's certificate in the RADIUS Access-Request packet as VSA: Alc-Subject-Key-Identifier. For more information, refer to the *7750 SR and VSR RADIUS Attributes Reference Guide*.

Default

```
no client-cert-subject-key-id
```

7.2.1.4.5.13 password

Syntax

```
password password [hash | hash2]  
no password
```

Context

```
config>ipsec>rad-auth-plcy>include
```

Description

This command specifies the password that is used in the RADIUS access requests. It shall be specified as a string, up to 32 characters.

The no form of this command resets the password to its default of NOKIA and will be stored using hash/hash2 encryption.

Parameters

password

specifies a password string, up to 32 characters

hash

specifies the key is entered in an encrypted form. If the hash parameter is not used, the key is assumed to be in a non-encrypted, clear text form. For security, all keys are stored in encrypted form in the configuration file with the hash parameter specified.

hash2

specifies the key is entered in a more complex encrypted form. If the hash2 parameter is not used, the less encrypted hash form is assumed.

Default: NOKIA

7.2.1.4.6 IPSec static security association command descriptions

7.2.1.4.6.1 static-sa

Syntax

```
static-sa  
static-sa name sa-name  
static-sa spi spi
```

Context

show>ipsec

Description

This command displays IPSec static-SA information.

Parameters

sa-name

specifies the SA name

Values: up to 32 characters

spi

specifies the spi

Values: 256 to 16383

7.2.1.4.6.2 static-sa

Syntax

[no] static-sa *sa-name*

Context

config>ipsec

Description

This command configures an IPSec static SA.

7.2.1.4.6.3 authentication

Syntax

authentication *auth-algorithm* *ascii-key* *ascii-string*

authentication *auth-algorithm* *hex-key* *hex-string* [hash | hash2]

no authentication

Context

config>ipsec>static-sa

Description

This command configures the authentication algorithm to use for an IPsec manual SA.

The no form of this command reverts to the default.

Parameters

auth-algorithm

specifies the authentication algorithm

Values: md5 | sha1

Default: sha1

ascii-string

specifies an ASCII key

hex-string

specifies a HEX key

7.2.1.4.6.4 direction

Syntax

direction *ipsec-direction*

no direction

Context

config>ipsec>static-sa

Description

This command configures the direction for an IPsec manual SA.

The no form of this command reverts to the default.

Parameters

ipsec-direction

identifies the direction to which this static SA entry can be applied

Values: inbound | outbound | bidirectional

Default: bidirectional

7.2.1.4.6.5 protocol

Syntax

```
protocol ipsec-protocol  
no protocol
```

Context

```
config>ipsec>static-sa
```

Description

This command configures the security protocol to use for an IPSec manual SA.

The no form of this command reverts to the default.

Parameters

ipsec-protocol

identifies the IPSec protocol used with this static SA

Values		
ah		specifies the Authentication Header protocol
–	esp	specifies the Encapsulation Security Payload protocol

Default: esp

7.2.1.4.6.6 spi

Syntax

```
spi spi  
no spi
```

Context

```
config>ipsec>static-sa
```

Description

This command configures the SPI key value for an IPsec manual SA.

This command specifies the SPI (Security Parameter Index) used to lookup the instruction to verify and decrypt the incoming IPsec packets when the value of the `direction` command is `inbound`.

The SPI value specifies the SPI used in the encoding of the outgoing packets when the value of the `direction` command is `outbound`. The remote node can use this SPI to lookup the instruction to verify and decrypt the packet.

If no `spi` is selected, this static SA cannot be used.

The no form of this command reverts to the default value.

Default

no spi

Parameters

spi

specifies the security parameter index for this SA

Values: 256 to 16383

7.2.1.4.7 Trust anchor profile/TS list command descriptions

7.2.1.4.7.1 trust-anchor

Syntax

trust-anchor *profile-name*

Context

config>ipsec

Description

This command specifies a CA profile as a trust-anchor CA. Multiple trust-anchors (up to 8) could be specified in a single trust-anchor-profile.

Parameters

profile-name

the name of the CA profile

7.2.1.4.7.2 ts-list

Syntax

```
ts-list list-name [create]  
no ts-list list-name
```

Context

```
config>ipsec
```

Description

This command creates a new TS list.

The no form of this command removes the list name from the configuration.

Parameters

list-name

specifies the name of the ts-list

create

keyword used to create the TS list. The create keyword requirement can be enabled/disabled in the environment>create context.

7.2.1.4.7.3 local

Syntax

```
local
```

Context

```
config>ipsec>ts-list
```

Description

This command enables the context to configure local TS list parameters. The traffic selector of the local system, such as TSr when the system acts as a IKEv2 responder.

7.2.1.4.7.4 entry

Syntax

```
entry entry-id [create]  
no entry entry-id
```

Context

```
config>ipsec>ts-list>local
```

Description

This command specifies a TS list entry.

The no form of this command removes the entry from the local configuration.

Parameters

entry-id

specifies the entry id

Values: 1 to 32

create

keyword used to create the TS list entry. The create keyword requirement can be enabled/disabled in the environment>create context.

7.2.1.4.7.5 address

Syntax

```
address prefix ip-prefix/ip-prefix-len  
address from begin-ip-address to end-ip-address  
no address
```

Context

```
config>ipsec>ts-list>local>entry
```

Description

This command specifies the address range in the IKEv2 traffic selector.

Parameters

ip-prefix/ip-prefix-len

specifies the IP subnet and prefix

begin-ip-address

specifies the beginning address of the range for this entry

end-ip-address

specifies the ending address of the range for this entry

7.2.1.4.8 Tunnel template command descriptions

7.2.1.4.8.1 tunnel-template

Syntax

```
tunnel-template ipsec-template-identifier [create]
```

```
no tunnel-template ipsec-template-identifier
```

Context

```
config>ipsec
```

Description

This command creates a tunnel template. Up to 2,000 templates are allowed.

Parameters

ipsec-template-identifier

specifies the template identifier

Values: 1 to 2048

create

mandatory keyword used when creating a tunnel-template in the IPSec context. The create keyword requirement can be enabled/disabled in the environment>create context.

7.2.1.4.8.2 clear-df-bit

Syntax

```
[no] clear-df-bit
```

Context

```
config>ipsec>tnl-temp
```

Description

This command enables clearing of the Do-not-Fragment bit.

7.2.1.4.8.3 encapsulated-ip-mtu

Syntax

```
encapsulated-ip-mtu octets
```

```
no encapsulated-ip-mtu
```

Context

```
config>service>vprn>if>sap>ipsec-tun
```

```
config>ipsec>tnl-temp
```

```
config>service>vprn>if>sap>ip-tunnel
```

```
config>service>ies>if>sap>ip-tunnel
```

Description

This command specifies the max size of encapsulated tunnel packet for the ipsec-tunnel/ip-tunnel or the dynamic tunnels terminated on the ipsec-gw. If the encapsulated v4/v6 tunnel packet exceeds the encapsulated-ip-mtu, the system fragments the packet against the encapsulated-ip-mtu.

Parameters

octets

specifies the max size in octets

Values: 512 to 9000

7.2.1.4.8.4 icmp6-generation

Syntax

```
icmp6-generation
```

Context

```
config>service>vprn>if>sap>ipsec-tun
config>ipsec>tnl-temp
config>service>vprn>if>sap>ip-tunnel
config>service>ies>if>sap>ip-tunnel
```

Description

This command enters ICMPv6 packet generation configuration context.

7.2.1.4.8.5 packet-too-big

Syntax

```
packet-too-big number [10..1000] seconds [1..60]
packet-too-big
no packet-too-big
```

Context

```
config>service>vprn>if>sap>ipsec-tun
config>ipsec>tnl-temp
config>service>vprn>if>sap>ip-tunnel
config>service>ies>if>sap>ip-tunnel
```

Description

This command enables system to send ICMPv6 PTB (Packet Too Big) message on private side and optionally specifies the rate.

With this command configured, system sends PTB back if the received v6 packet on the private side is bigger than 1280 bytes and also exceeds the private MTU of the tunnel.



NOTE:

The `ip-mtu` command (under `ipsec-tunnel` or `tunnel-template`) specifies the private MTU for the `ipsec-tunnel` or dynamic tunnel.

Parameters

number

specifies the number of PTB messages

seconds

specifies the number of seconds

7.2.1.4.8.6 ip-mtu

Syntax

`ip-mtu octets`

`no ip-mtu`

Context

`config>ipsec>tnl-temp`

Description

This command configures the template IP MTU.

Parameters

octets

specifies the maximum size in octets

Values: 512 to 9000

7.2.1.4.8.7 replay-window

Syntax

`replay-window {32 | 64 | 128 | 256 | 512}`

`no replay-window`

Context

`config>ipsec>tnl-temp`

Description

This command sets the anti-replay window.

The no form of this command removes the parameter from the configuration.

Default

no replay-window

Parameters

32 | 64 | 128 | 256 | 512

specifies the size of the anti-replay window

7.2.1.4.8.8 sp-reverse-route

Syntax

[no] sp-reverse-route

Context

config>ipsec>tnl-temp

Description

This command specifies whether the node using this template accepts framed-routes sent by the RADIUS server and installs them for the lifetime of the tunnel as managed routes.

The no form of this command disables sp-reverse-route.

Default

no sp-reverse-route

7.2.1.4.8.9 transform

Syntax

transform *transform-id* [*transform-id*...(up to 4 max)]

no transform

Context

config>ipsec>tnl-temp

config>service>ies>if>sap>ipsec-gateway

config>service>vprn>if>sap>ipsec-gateway

Description

This command configures IPSec transform.

Parameters

transform-id

specifies the IPSec transformation ID

7.2.1.5 IES and VPRN service command descriptions

7.2.1.5.1 interface

Syntax

```
interface ip-int-name [create]
interface ip-int-name [create] tunnel
no interface ip-int-name
```

Context

```
config>service>ies
config>service>vprn
```

Description

This command creates a logical IP routing interface for an IES or VPRN. Once created, attributes like an IP address and SAP can be associated with the IP interface.

The `interface` command, under the context of services, is used to create and maintain IP routing interfaces within VPRN service IDs. The `interface` command can be executed in the context of an VPRN service ID. The IP interface created is associated with the service core network routing instance and default routing table. The typical use for IP interfaces created in this manner is for subscriber Internet access.

Interface names are case sensitive and must be unique within the group of defined IP interfaces defined for `config router interface` and `config service vprn interface`. Interface names must not be in the dotted decimal notation of an IP address. For example, the name "1.1.1.1" is not allowed, but "int-1.1.1.1" is allowed. Show commands for router interfaces use either interface names or the IP addresses. Use unique IP address values and IP address names to maintain clarity. It could be unclear to the user if the same IP address and IP address name values are used. Although not recommended, duplicate interface names can exist in different router instances.

The available IP address space for local subnets and routes is controlled with the `config router service-prefix` command. The `service-prefix` command administers the allowed subnets that can

be defined on service IP interfaces. It also controls the prefixes that may be learned or statically defined with the service IP interface as the egress interface. This allows segmenting the IP address space into `config router` and `config service` domains.

When a new name is entered, a new logical router interface is created. When an existing interface name is entered, the user enters the router interface context for editing and configuration.

By default, there are no default IP interface names defined within the system. All VPRN IP interfaces must be explicitly defined. Interfaces are created in an enabled state.

The `no` form of this command removes IP the interface and all the associated configuration. The interface must be administratively shutdown before issuing the `no interface` command.

For VPRN services, the IP interface must be shutdown before the SAP on that interface may be removed. VPRN services do not have the shutdown command in the SAP CLI context. VPRN service SAPs rely on the interface status to enable and disable them.

Parameters

ip-int-name

specifies the name of the IP interface. Interface names can be from 1 to 32 alphanumeric characters. If the string contains special characters (`#`, `$`, spaces, and so on), the entire string must be enclosed within double quotes.

Values: 1 to 32 characters

tunnel

specifies that the interface is configured as tunnel interface, which could be used to terminate IPSec or GRE tunnels in the private service

create

keyword used to create the IPSec interface instance. The `create` keyword requirement can be enabled/disabled in the `environment>create` context.

7.2.1.5.2 dynamic-tunnel-redundant-next-hop

Syntax

`dynamic-tunnel-redundant-next-hop ip-address`

`no dynamic-tunnel-redundant-next-hop`

Context

`config>service>ies>if`

`config>service>vprn>if`

Description

This command configures the dynamic ISA tunnel redundant next-hop address.

Default

no dynamic-tunnel-redundant-next-hop

Parameters

ip-address
specifies the IP address of the next hop

7.2.1.5.3 address

Syntax

address *ipv6-address/prefix-length* [eui-64] [preferred] [track-srrp *srrp-instance*]
no address *ipv6-address/prefix-length*

Context

config>service>vprn>if>ipv6

Description

This command add an IPv6 address to the tunnel interface. The prefix length must be 96 or higher.

Parameters

ipv6-address/prefix-length
specifies the IPv6 address on the interface

Values	ipv6-address prefix-length	
–	ipv6-address	ipv6-address x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H

–	–	d - [0..255]D
–	prefix-length	1 to 128

eui-64

When the `eui-64` keyword is specified, a complete IPv6 address from the supplied prefix and 64-bit interface identifier is formed. The 64-bit interface identifier is derived from MAC address on Ethernet interfaces. For interfaces without a MAC address, for example ATM interfaces, the Base MAC address of the chassis is used.

preferred

specifies that the IPv6 address is the preferred IPv6 address for this interface. Preferred address is an address assigned to an interface whose use by upper layer protocols is unrestricted. Preferred addresses maybe used as the source (or destination) address of packets sent from (or to) the interface. Preferred address doesn't go through the DAD process.

srrp-instance

specifies an SRRP instance to track

7.2.1.5.4 link-local-address

Syntax

```
link-local-address ipv6-address [preferred]
```

Context

```
config>service>vprn>if>ipv6
```

Description

This command specifies the link-local-address for the tunnel interface. Only one link-local-address is allowed per interface

Parameters***ipv6-address***

specifies the IPv6 address on the interface

Values	ipv6-address	x::x::x::x::x::x (eight 16-bit pieces)
–	–	x::x::x::x::d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

preferred

specifies that the IPv6 address is the preferred IPv6 address for this interface. Preferred address is an address assigned to an interface whose use by upper layer protocols is unrestricted. Preferred addresses maybe used as the source (or destination) address of packets sent from (or to) the interface. Preferred address doesn't go through the DAD process.

7.2.1.5.5 static-tunnel-redundant-next-hop**Syntax**

```
static-tunnel-redundant-next-hop ip-address
no static-tunnel-redundant-next-hop
```

Context

```
config>service>ies>if
config>service>vprn>if
```

Description

This command specifies redundant next-hop address on public or private IPsec interface (with public or private tunnel-sap) for static IPsec tunnel. The specified next-hop address will be used by standby node to shunt IPsec traffic to master in case of it receives them.

The next-hop address will be resolved in routing table of corresponding service.

Default

```
no static-tunnel-redundant-next-hop
```

Parameters

ip-address

specifies the IP address of the next hop

7.2.1.5.6 sap

Syntax

```
sap sap-id [create]
```

```
no sap sap-id
```

Context

```
config>service>ies>if
```

```
config>service>vpn>if
```

Description

This command creates a SAP within a service. A SAP is a combination of port and encapsulation parameters which identifies the service access point on the interface and within the router. Each SAP must be unique.

All SAPs must be explicitly created. If no SAPs are created within a service or on an IP interface, a SAP will not exist on that object.

Enter an existing SAP without the `create` keyword to edit SAP parameters. The SAP is owned by the service in which it was created.

A SAP can only be associated with a single service. A SAP can only be defined on a port that has been configured as an access port using the `config interface port-type port-id mode access` command. Channelized TDM ports are always access ports.

If a port is shutdown, all SAPs on that port become operationally down. When a service is shutdown, SAPs for the service are not displayed as operationally down although all traffic traversing the service will be discarded. The operational state of a SAP is relative to the operational state of the port on which the SAP is defined.

The `no` form of this command deletes the SAP with the specified port. When a SAP is deleted, all configuration parameters for the SAP will also be deleted.

Default

```
no sap
```

`sap tunnel-id.private | public:tag`— This parameter associates a tunnel group SAP with this interface.

This context provides a SAP to the tunnel. The operator may associate an ingress and egress QoS policies as well as filters and virtual scheduling contexts. Internally this creates an Ethernet SAP that is used to send and receive encrypted traffic to and from the MDA. Multiple tunnels can be associated with this SAP. The "tag" is set to the dot1q value. The operator may see it as an identifier. The range is from 1 to 4094.

Parameters

sap-id

specifies the physical port identifier portion of the SAP definition

port-id

specifies the physical port ID in the *slot/mda/port* format

If the card in the slot has Media Dependent Adapters (MDAs) installed, the *port-id* must be in the *slot_number/MDA_number/port_number* format. For example 61/2/3 specifies port 3 on MDA 2 in slot 61.

The *port-id* must reference a valid port type. When the *port-id* parameter represents SONET/SDH and TDM channels the port ID must include the channel ID. A period "." separates the physical port from the *channel-id*. The port must be configured as an access port.

If the SONET/SDH port is configured as clear-channel then only the port is specified.

create

keyword used to create a SAP instance

7.2.1.5.7 ip-tunnel

Syntax

```
ip-tunnel ip-tunnel-name [create]
```

```
no ip-tunnel ip-tunnel-name
```

Context

```
config>service>ies>sap
```

```
config>service>vprn>sap
```

Description

This command is used to configure an IP-GRE or IP-IP tunnel and associate it with a private tunnel SAP within an IES or VPRN service.

The no form of this command deletes the specified IP/GRE or IP-IP tunnel from the configuration. The tunnel must be administratively shutdown before issuing the no `ip-tunnel` command.

Parameters

ip-tunnel-name

specifies the name of the IP tunnel. Tunnel names can be from 1 to 32 alphanumeric characters. If the string contains special characters (for example, #, \$, spaces), the entire string must be enclosed within double quotes.

create

keyword used to create the tunnel. The create keyword requirement can be enabled/disabled in the `environment>create` context.

7.2.1.5.8 backup-remote-ip

Syntax

`backup-remote-ip ip-address`

`no backup-remote-ip`

Context

`config>service>interface>ies>sap>ip-tunnel`

`config>service>interface>vprn>sap>ip-tunnel`

Description

This command sets the backup destination IPv4 address of GRE encapsulated packets associated with a particular GRE tunnel. If the primary destination address is not reachable in the delivery service (there is no route) or not defined then this is the destination IPv4 address of GRE encapsulated packets sent by the delivery service.

The no form of this command deletes the backup-destination address from the GRE tunnel configuration.

Parameters

ip-address

specifies the destination IPv4 address of the GRE tunnel

Values: 1.0.0.0 to 223.255.255.255

7.2.1.5.9 delivery-service

Syntax

`delivery-service {service-id / svc-name}`

```
no delivery-service
```

Context

```
config>service>interface>ies>sap>ip-tunnel
```

```
config>service>interface>vprn>sap>ip-tunnel
```

Description

This command sets the delivery service for GRE encapsulated packets associated with a particular GRE tunnel. This is the IES or VPRN service where the GRE encapsulated packets are injected and terminated. The delivery service may be the same service that owns the private tunnel SAP associated with the GRE tunnel. The GRE tunnel does not come up until a valid delivery service is configured.

The no form of this command deletes the delivery-service from the GRE tunnel configuration.

Parameters

service-id

identifies the service used to originate and terminate the GRE encapsulated packets belonging to the GRE tunnel

Values: 1 to 2147483648

svc-name

identifies the service used to originate and terminate the GRE encapsulated packets belonging to the GRE tunnel

Values: 1 to 64 characters

7.2.1.5.10 dscp

Syntax

```
dscp dscp-name
```

```
no dscp
```

Context

```
config>service>interface>ies>sap>ip-tunnel
```

```
config>service>interface>vprn>sap>ip-tunnel
```

Description

This command sets the DSCP code-point in the outer IP header of GRE encapsulated packets associated with a particular GRE tunnel. The default, set using the no form of this command, is to copy the DSCP

value from the inner IP header (after remarking by the private tunnel SAP egress QoS policy) to the outer IP header.

Default

no dscp

Parameters

dscp-name

specifies the DSCP code-point to be used

Values: be | cp1 | cp2 | cp3 | cp4 | cp5 | cp6 | cp7 | cs1 | cp9 | af11 | cp11 | af12 | cp13 | af13 | cp15 | cs2 | cp17 | af21 | cp19 | af22 | cp21 | af23 | cp23 | cs3 | cp25 | af31 | cp27 | af32 | cp29 | af33 | cp31 | cs4 | cp33 | af41 | cp35 | af42 | cp37 | af43 | cp39 | cs5 | cp41 | cp42 | cp43 | cp44 | cp45 | ef | cp47 | nc1 | cp49 | cp50 | cp51 | cp52 | cp53 | cp54 | cp55 | nc2 | cp57 | cp58 | cp59 | cp60 | cp61 | cp62 | cp63

7.2.1.5.11 gre-header

Syntax

gre-header send-key *send-key* receive-key *receive-key*

Context

config>service>ies>sap>ip-tunnel

config>service>vprn>sap>ip-tunnel

Description

This command configures the type of the IP tunnel. If the gre-header command is configured, the tunnel is a GRE tunnel with a GRE header inserted between the outer and inner IP headers. If the no form of this command is configured then the tunnel is a simple IP-IP tunnel.

Default

no gre-header

Parameters

send-key

specifies a 32-bit unsigned integer

Values: 0 to 4294967295

receive-key

specifies a 32-bit unsigned integer

Values: 0 to 4294967295

7.2.1.5.12 ip-mtu

Syntax

```
ip-mtu octets
```

```
no ip-mtu
```

Context

```
config>service>ies>if>sap>ip-tunnel
```

```
config>service>vprn>if>sap>ip-tunnel
```

```
config>service>vprn>if>sap>ipsec-tunnel
```

Description

This command configures the IP maximum transmit unit (packet) for this interface.

This connects a Layer 2 to a Layer 3 service, this parameter can be adjusted under the IES interface.

The MTU that is advertised from the IES size is:

$\text{MINIMUM}((\text{SdpOperPathMtu} - \text{EtherHeaderSize}), (\text{Configured ip-mtu}))$

By default (for Ethernet network interface) if no ip-mtu is configured it is $(1568 - 14) = 1554$.

The `ip-mtu` command instructs the MS-ISA to perform IP packet fragmentation, prior to IPSec encryption and encapsulation, based on the configured MTU value. In particular:

- If the length of a payload IP packet (including its header) exceeds the configured MTU value and the DF flag is clear (due to the presence of the `clear-df-bit` command or because the original DF value was 0) then the MS-ISA fragments the payload packet as efficiently as possible (that is, it creates the minimum number of fragments each less than or equal to the configured MTU size); in each created fragment the DF bit shall be 0.

If the length of a payload IP packet (including its header) exceeds the configured MTU value and the DF flag is set (because the original DF value was 1 and the tunnel has no `clear-df-bit` in its configuration) then the MS-ISA discards the payload packet without sending an ICMP type 3/code 4 message back to the packet's source address.

The `no ip-mtu` command, corresponding to the default behavior, disables fragmentation of IP packets by the MS-ISA; all IP packets, regardless of size or DF bit setting, are allowed into the tunnel.



NOTE:

The effective MTU for packets entering a tunnel is the minimum of the private tunnel SAP interface IP MTU value (used by the IOM) and the tunnel IP MTU value (configured using the above command and used by the MS-ISA). So if it desired to fragment IP packets larger than X bytes with DF set, rather than discarding them, the tunnel IP MTU should be set to X and the private tunnel SAP interface IP MTU should be set to a value larger than X.

Default

no ip-mtu

7.2.1.5.13 reassembly

Syntax

reassembly [*wait-msecs*]

no reassembly

Context

config>service>ies>if>sap>ip-tunnel

config>service>vprn>if>sap>ip-tunnel

Description

This command configures the reassembly wait time.

7.2.1.5.14 remote-ip

Syntax

remote-ip *ip-address*

no remote-ip

Context

config>service>interface>ies>sap>ip-tunnel

config>service>interface>vprn>sap>ip-tunnel

Description

This command sets the primary destination IPv4 address of GRE encapsulated packets associated with a particular GRE tunnel. If this address is reachable in the delivery service (there is a route) then this is the destination IPv4 address of GRE encapsulated packets sent by the delivery service.

The no form of this command deletes the destination address from the GRE tunnel configuration.

Parameters

ip-address

specifies the destination IPv4 address of the GRE tunnel

Values: 1.0.0.0 to 223.255.255.255

7.2.1.5.15 source

Syntax

source *ip-address*

no source

Context

config>service>interface>ies>sap>ip-tunnel

config>service>interface>vprn>sap>ip-tunnel

Description

This command sets the source IPv4 address of GRE encapsulated packets associated with a particular GRE tunnel. It must be an address in the subnet of the associated public tunnel SAP interface. The GRE tunnel does not come up until a valid source address is configured.

The no form of this command deletes the source address from the GRE tunnel configuration. The tunnel must be administratively shutdown before issuing the no source command.

Parameters

ip-address

specifies the source IPv4 address of the GRE tunnel

Values: 1.0.0.0 to 223.255.255.255

7.2.1.5.16 VPRN IPSec Commands

7.2.1.5.16.1 ipsec

Syntax

`ipsec`

Context

`config>service>vprn`

Description

This command enables the context to configure IPSec policies.

7.2.1.5.16.2 security-policy

Syntax

`security-policy security-policy-id [create]`

`no security-policy security-policy-id`

Context

`config>service>vprn>ipsec`

Description

This command configures a security policy to use for an IPSec tunnel.

Parameters

security-policy-id

specifies a value to be assigned to a security policy

Values: 1 to 8192

create

keyword used to create the security policy instance. The create keyword requirement can be enabled/disabled in the `environment>create` context.

7.2.1.5.16.3 entry

Syntax

`entry entry-id [create] [user-name user-name]`

no entry *entry-id*

Context

config>service>vpn>ipsec>sec-plcy

Description

This command configures an IPSec security policy entry.

Parameters

entry-id

Specifies the IPSec security policy entry.

Values: 1 to 16

create

keyword used to create the security policy entry instance. The create keyword requirement can be enabled/disabled in the environment>create context.

user-name

specifies the owner of the entry, up to 32 characters

7.2.1.5.16.4 local-ip

Syntax

local-ip {*ip-prefix/prefix-length* | *ip-prefix netmask* | any}

Context

config>service>vpn>ipsec>sec-plcy>entry

Description

This command configures the local (from the VPN) IP prefix/mask for the policy parameter entry.

Only one entry is necessary to describe a potential flow. The local-ip and remote-ip commands can be defined only once. The system evaluates the local IP as the source IP when traffic is examined in the direction of VPN to the tunnel, and as the destination IP when traffic flows from the tunnel to the VPN. The remote IP is evaluated as the source IP when traffic flows from the tunnel to the VPN and when traffic flows from the VPN to the tunnel.

Parameters

ip-prefix/prefix-length

the destination address of the aggregate route in dotted decimal notation

Values	ip-prefix	a.b.c.d
		host bits must be 0
–	prefix-length	1 to 32

netmask

the subnet mask in dotted decimal notation

any

keyword to specify that it can be any address

7.2.1.5.16.5 local-v6-ip

Syntax

```
local-v6-ip ipv6-prefix/prefix-length
local-v6-ip any
no local-v6-ip
```

Context

```
config>service>vprn>ipsec>sec-plcy>entry
```

Description

This command specifies the local v6 prefix for the security-policy entry.

Parameters

ipv6-prefix/prefix-length

specifies the local v6 prefix and length

Values	ipv6-prefix prefix-length	
–	ipv6-prefix	x::x::x::x::x::x (eight 16-bit pieces)

–	–	x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D
–	–	host bits must be 0
–	–	:: not allowed
–	prefix-length	1 to 128

any

keyword to specify that it can be any address

7.2.1.5.16.6 remote-ip**Syntax**

```
remote-ip {ip-prefix/prefix-length | ip-prefix netmask | any}
```

Context

```
config>service>vpn>ipsec>sec-plcy>entry
```

Description

This command configures the remote (from the tunnel) IP prefix/mask for the policy parameter entry.

Only one entry is necessary to describe a potential flow. The `local-ip` and `remote-ip` commands can be defined only once. The system evaluates the local IP as the source IP when traffic is examined in the direction of VPN to the tunnel and as the destination IP when traffic flows from the tunnel to the VPN. The remote IP is evaluated as the source IP when traffic flows from the tunnel to the VPN and when traffic flows from the VPN to the tunnel.

Parameters***ip-prefix/prefix-length***

the destination address of the aggregate route in dotted decimal notation

Values	ip-prefix	a.b.c.d (host bits must be 0)
–	–	0.0.0.0 not allowed
–	prefix-length	1 to 32

netmask

the subnet mask in dotted decimal notation

Values	netmask	a.b.c.d (network bits all 1 and host bits all 0)
–	–	0.0.0.0 not allowed

any

keyword to specify that it can be any address

7.2.1.5.16.7 remote-v6-ip

Syntax

```
remote-v6-ip any
remote-v6-ip ipv6-prefix/prefix-length
no remote-v6-ip
```

Context

```
config>service>vprn>ipsec>sec-plcy>entry
```

Description

This command specifies the remote v6 prefix for the security-policy entry.

Parameters

ipv6-prefix/prefix-length

specifies the local v6 prefix and length

Values	ipv6-prefix	x::x::x::x::x::x (eight 16-bit pieces)
–	–	x::x::x::x::x::x.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D
–	–	host bits must be 0
–	–	:: not allowed
–	prefix-length	1 to 128

any
keyword to specify that it can be any address

7.2.1.5.17 SAP IPsec gateway commands

7.2.1.5.17.1 ipsec-gw

Syntax

[no] ipsec-gw

Context

config>service>ies>if>sap
config>service>vprn>if>sap

Description

This command configures an IPsec gateway.

7.2.1.5.17.2 cert

Syntax

cert t

Context

config>service>ies>if>sap>ipsec-gw

config>service>vprn>if>sap>ipsec-gw

config>service>vprn>if>sap>ipsec-tunnel>dynamic-keying

Description

This command configures certificate parameters used by this SAP IPSec gateway. The certificate is loaded from the configured URL when the IPSec gateway is no shutdown. When system is loading the certificate, it checks if it is a valid X.509v3 certificate. The following requirements apply:

- The key file must be already configured.
- The configured cert file must be a DER formatted X.509v3 certificate file .
- All non-optional fields defined in section 4.1 of RFC5280 must exist in the cert-file and conform to the RFC5280 defined format.

7.2.1.5.17.3 cert

Syntax

cert *filename*

Context

config>service>ies>if>sap>ipsec-gw>cert

config>service>vprn>if>sap>ipsec-gw>cert

config>service>vprn>if>sap>ipsec-tun>dyn>cert

Description

This command configures cert parameters used by this SAP IPSec gateway.

Parameters

filename

specifies the name of the certificate that is used to identify itself if required, up to 95 characters maximum

7.2.1.5.17.4 cert-profile

Syntax

cert-profile *profile-name*

no cert-profile

Context

config>service>ies>if>sap>ipsec-gw>cert

config>service>vprn>if>sap>ipsec-gw>cert

config>service>vprn>if>sap>ipsec-tun>dyn>cert

Description

This command specifies the certificate profile for the IPsec tunnel or IPsec gateway. This command overrides the cert and key configuration under the ipsec-tunnel or ipsec-gw contexts.

Parameters

profile-name

specifies the name of certificate profile

7.2.1.5.17.5 key

Syntax

[no] key *local-file-url*

Context

config>service>vprn>if>sap>ipsec-tun>dynamic-keying>cert

config>service>vprn>if>sap>ipsec-gw>cert

config>service>ies>if>sap>ipsec-gw>cert

Description

This command specifies the key pair file that 7750 uses for X.509 certificate authentication. The system loads the key file when the ipsec-tunnel/gw is “no shutdown”

When the system is loading the key file, it checks if it is a valid 7750 formatted key file.

Key file URL can only be changed or removed when the tunnel or gateway is shutdown.

The same key can be used for multiple ipsec-tunnels or ipsec-gws. However, for each key file there is only one memory instance. If a key file has been updated, “no shutdown” in any of the tunnels that use the key file will cause update of the memory instance, which will not impact the current up and running tunnels that use the key file. The new authentication will use the updated memory instance.

**NOTE:**

From 12.0.R1 release onward, use the `cert-profile` command instead.

Parameters

local-file-url

URL for input file; this URL is the local CF card URL

7.2.1.5.17.6 status-verify

Syntax

`status-verify`

Context

`config>service>ies>if>sap>ipsec-gw>cert`

`config>service>vprn>if>sap>ipsec-gw>cert`

`config>service>vprn>if>sap>ipsec-tun>dyn>cert`

Description

This command enables the context to configure certificate revocation status verification parameters.

7.2.1.5.17.7 default-result

Syntax

`default-result {revoked | good}`

`no default-result`

Context

```
config>service>ies>if>sap>ipsec-gw>cert>cert>status-verify  
config>service>vprn>if>sap>ipsec-gw>cert>cert>status-verify  
config>service>vprn>if>sap>ipsec-tun>dyn>cert>status-verify
```

Description

This command specifies the default result when both the primary and secondary method failed to provide an answer.

Parameters

good

specifies that the certificate is considered as acceptable

revoked

specifies that the certificate is considered as revoked

Default: revoked

7.2.1.5.17.8 primary

Syntax

```
primary {ocsp | crl}  
no primary
```

Context

```
config>service>ies>if>sap>ipsec-gw>cert>cert-status-verify  
config>service>vprn>if>sap>ipsec-gw>cert>cert-status-verify  
config>service>vprn>if>sap>ipsec-tun>dyn>cert>cert-status-verify
```

Description

This command specifies the primary method that used to verify revocation status of the peer's certificate; could be either CRL or OCSP

OCSP or CRL use the corresponding configuration in the CA profile of the issuer of the certificate in question.

Parameters

ocsp

specifies to use the OCSP protocol. The OCSP server is configured in the corresponding ca-profile command

crl

specifies to use the local CRL file. The CRL file is configured in the corresponding ca-profile command

Default: crl

7.2.1.5.17.9 secondary

Syntax

secondary {ocsp | crl}

no secondary

Context

config>service>ies>if>sap>ipsec-gw>cert>cert-status-verify

config>service>vprn>if>sap>ipsec-gw>cert>cert-status-verify

config>service>vprn>if>sap>ipsec-tun>dyn>cert>cert-status-verify

Description

This command specifies the secondary method that is used to verify revocation status of the peer's certificate; could be either CRL or OCSP.

OCSP or CRL use the corresponding configuration in the CA profile of the issuer of the certificate in question.

The secondary method is only used when the primary method fails to provide an answer:

- OCSP: unreachable / any answer other than "good" or "revoked: / ocsp is NOT configured in the ca-profile/ OCSP response is not signed / Invalid nextUpdate
- CRL: CRL expired

Default

no secondary

Parameters**ocsp**

specifies to use the OCSP protocol, the OCSP server is configured in the corresponding ca-profile command

crl

specifies to use the local CRL file, the CRL file is configured in the corresponding `ca-profile` command

7.2.1.5.17.10 `trust-anchor`

Syntax

```
trust-anchor ca-profile-name  
no trust-anchor
```

Context

```
config>service>ies>if>sap>ipsec-gateway>cert  
config>service>vprn>if>sap>ipsec-gw>cert  
config>service>vprn>if>sap>ipsec-tun>dyn>cert
```

Description

This command configures trust anchor with a CA profile used by this SAP IPsec gateway.



NOTE:

From 12.0.R1 release onward, use the `cert-profile` command instead.

Parameters

ca-profile-name

specifies the CA profile name to use in the trust anchor, up to 95 characters

7.2.1.5.17.11 `trust-anchor-profile`

Syntax

```
trust-anchor-profile profile-name  
no trust-anchor-profile
```

Context

```
config>service>ies>if>sap>ipsec-gw>cert  
config>service>vprn>if>sap>ipsec-gw>cert  
config>service>vprn>if>sap>ipsec-tun>dyn>cert
```


Description

This command specifies the trust-anchor-profile for the ipsec-tunnel or ipsec-gw. This command overrides the “trust-anchor” configuration under the ipsec-tunnel or ipsec-gw.

Parameters

profile-name

specifies the name of the trust anchor profile, up to 32 characters

7.2.1.5.17.12 default-secure-service

Syntax

```
default-secure-service service-id interface ip-int-name  
default-secure-service name service-name interface ip-int-name  
no default-secure-service
```

Context

```
config>service>ies>if>sap>ipsec-gateway  
config>service>vprn>if>sap>ipsec-gateway
```

Description

This command specifies a service ID or service name of the default security service used by this SAP IPSec gateway.

Parameters

service-id

specifies a default secure service

Values	service-id svc-name	
–	service-id	1 to 2147483648
–	svc-name	specifies an existing service name, up to 64 characters

ip-int-name

specifies the name of the IP interface that must start with a letter, up to 32 characters

7.2.1.5.17.13 default-tunnel-template

Syntax

```
default-tunnel-template ipsec-template-identifier
```

```
no default-tunnel-template
```

Context

```
config>service>ies>if>sap>ipsec-gateway
```

```
config>service>vpn>if>sap>ipsec-gateway
```

Description

This command configures a default tunnel policy template for the gateway.

Parameters***ipsec-template-identifier***

specifies the tunnel policy template to use

7.2.1.5.17.14 ike-policy

Syntax

```
ike-policy ike-policy-id
```

```
no ike-policy
```

Context

```
config>service>ies>if>sap>ipsec-gateway
```

```
config>service>vpn>if>sap>ipsec-gateway
```

```
config>service>vpn>if>sap>ipsec-tunnel>dynamic-keying
```

Description

This command configures IKE policy for the gateway.

Parameters

ike-policy-id
specifies the IKE policy ID
Values: 1 to 2048

7.2.1.5.17.15 local-gateway-address

Syntax

local-gateway-address *ip-address* peer *ip-address* delivery-service *service-id*
no local-gateway-address

Context

config>service>ies>if>sap>ipsec-gw
config>service>vprn>if>sap>ipsec-gw
config>service>vprn>if>sap>ipsec-tunnel

Description

This command specifies the local gateway address used for the tunnel and the address of the remote security gateway at the other end of the tunnel remote peer IP address to use.

Default

The base routing context is used if the delivery-router option is not specified.

Parameters

ip-address
specifies IP address of the local end of the tunnel

service-id
specifies the ID of the IES or VPRN (front-door) delivery service of this tunnel. Use this service-id to find the VPRN used for delivery.

Values	service-id svc-name	
–	service-id	1 to 2147483648

–	svc-name	specifies an existing service name, up to 64 characters
---	----------	---

7.2.1.5.17.16 local-id

Syntax

```
local-id type {ipv4 | fqdn} [value value]  
no local-id
```

Context

```
config>service>ies>if>sap>ipsec-gateway  
config>service>vprn>if>sap>ipsec-gateway  
service>vprn>if>sap>ipsec-tunnel
```

Description

This command specifies the local ID for 7750 SR routers used for IDi or IDr for IKEv2 tunnels.
The no form of this command removes the parameters from the configuration.

Default

Depends on local-auth-method like following:

- Psk: local tunnel IP address
- Cert-auth: subject of the local certificate

Parameters

type

specifies the type of local ID payload, it could be ipv4 address/FQDN domain name, distinguish name of subject in X.509 certificate

ipv4

specifies to use ipv4 as the local ID type, the default value is the local tunnel end-point address

fqdn

specifies to use FQDN as the local ID type

value

specifies to use FQDN as the local ID type. The value must be configured. Maximum 255 characters.

7.2.1.5.17.17 pre-shared-key

Syntax

```
pre-shared-key key  
no pre-shared-key
```

Context

```
config>service>ies>if>sap>ipsec-gateway  
config>service>vprn>if>sap>ipsec-gateway  
config>service>vprn>if>sap>ipsec-tunnel>dynamic-keying
```

Description

This command specifies the shared secret between the two peers forming the tunnel.

Parameters

key
specifies a pre-shared-key for dynamic-keying

7.2.1.5.17.18 radius-accounting-policy

Syntax

```
radius-accounting-policy policy-name  
no radius-accounting-policy
```

Context

```
config>service>ies>if>sap>ipsec-gw  
config>service>vprn>if>sap>ipsec-gw
```

Description

This command specifies the radius-accounting-policy to be used for the IKEv2 remote-access tunnels terminated on the ipsec-gw. The radius-accounting-policy is defined under config>ipsec context.

Default

no radius-accounting-policy

Parameters

policy-name

specifies the name of an existing radius-accounting-policy

7.2.1.5.17.19 radius-authentication-policy

Syntax

radius-authentication-policy *policy-name*

no radius-authentication-policy

Context

config>service>ies>if>sap>ipsec-gw

config>service>vprn>if>sap>ipsec-gw

Description

This command specifies the radius-authentication-policy to be used for the IKEv2 remote-access tunnels terminated on the ipsec-gw. The radius-authentication-policy is defined under the config>ipsec context.

Default

no radius-authentication-policy

Parameters

policy-name

specifies the name of an existing radius-authentication-policy

7.2.1.5.17.20 ts-negotiation

Syntax

ts-negotiation ts-list *list-name*

no ts-negotiation

Context

```
config>service>ies>if>sap>ipsec-gw
```

```
config>service>vprn>if>sap>ipsec-gw
```

Description

This command enables the IKEv2 traffic selector negotiation with the specified ts-list.

Parameters

list-name

specifies the TS list name, up to 32 characters

7.2.1.5.18 SAP IPsec tunnel commands

7.2.1.5.18.1 ipsec-tunnel

Syntax

```
ipsec-tunnel ipsec-tunnel-name [create]
```

```
no ipsec-tunnel ipsec-tunnel-name
```

Context

```
config>service>vprn>if>sap
```

Description

This command specifies an IPsec tunnel name. An IPsec client sets up the encrypted tunnel across public network. The 7750 SR IPsec MDA acts as a concentrator gathering, and terminating these IPsec tunnels into an IES or VPRN service. This mechanism allows as service provider to offer a global VPRN service even if node of the VPRN are on an uncontrolled or insecure portion of the network.

Parameters

ipsec-tunnel-name

specifies an IPsec tunnel name, up to 32 characters

create

keyword used to create the IPsec tunnel instance. The create keyword requirement can be enabled/disabled in the `environment>create` context.

7.2.1.5.18.2 bfd-designate

Syntax

[no] bfd-designate

Context

config>service>vprn>if>sap>ipsec-tunnel

Description

This command specifies whether this IPsec tunnel is the BFD designated tunnel.

7.2.1.5.18.3 bfd-enable

Syntax

[no] bfd-enable service *service-id* interface *interface-name* dst-ip *ip-address*

Context

config>service>vprn>if>sap>ipsec-tunnel

Description

This command assigns a BFD session and provides heart-beat mechanism for given IPsec tunnel. There can be only one BFD session assigned to any given IPsec tunnel, but there can be multiple IPsec tunnels using same BFD session. BFD control the state of the associated tunnel; if BFD session goes down, the system will also bring down the associated non-designated IPsec tunnel.

Default

no bfd-enable

Parameters

service-id

specifies where the service-id that the BFD session resides

interface-name

specifies the name of the interface used by the BFD session

ip-address

specifies the destination address to be used for the BFD session

7.2.1.5.18.4 clear-df-bit

Syntax

```
[no] clear-df-bit
```

Context

```
config>service>vprn>interface>sap>ipsec-tunnel
```

```
config>service>vprn>interface>sap>ip-tunnel
```

```
config>service>ies>interface>sap>ip-tunnel
```

Description

This command instructs the MS-ISA to reset the DF bit to 0 in all payload IP packets associated with the GRE or IPsec tunnel, before any potential fragmentation resulting from the `ip-mtu` command. (This requires a modification of the header checksum.) The `no clear-df-bit` command, corresponding to the default behavior, leaves the DF bit unchanged.

The `no` form of this command disables the DF bit reset.

Default

```
no clear-df-bit
```

7.2.1.5.18.5 dest-ip

Syntax

```
dest-ip ip-address
```

Context

```
config>service>ies>sap>ip-tunnel
```

```
config>service>vprn>sap>ip-tunnel
```

```
config>service>vprn>sap>ipsec-tunnel
```

Description

This command configures the private address of the remote tunnel endpoint. The configuration of this address is mandatory in the configuration of every IP-IP or IP-GRE tunnel.



NOTE:

Unnumbered interfaces are not supported.

Parameters

ip-address

specifies the private IP address of the remote IP tunnel endpoint. If this remote IP address is not within the subnet of the IP interface associated with the tunnel, the tunnel will not come up.

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF] H
–	–	d - [0..255] D

7.2.1.5.18.6 dynamic-keying

Syntax

[no] dynamic-keying

Context

config>service>vpn>if>sap>ipsec-tunnel

Description

This command enables dynamic keying for the IPSec tunnel.

Default

no dynamic-keying

7.2.1.5.18.7 auto-establish

Syntax

```
[no] auto-establish
```

Context

```
config>service>vprn>if>sap>ipsec-tun>dynamic-keyig
```

Description

The system automatically establishes phase 1 SA as soon as the tunnel is provisioned and enabled (no shutdown). This option must only be configured on one side of the tunnel.

Any associated static routes remain up as long as the tunnel is up, even though it may actually be operationally down according to the CLI.

Default

no auto-establish

7.2.1.5.18.8 transform

Syntax

```
transform transform-id [transform-id...(up to 4 max)]
```

```
no transform
```

Context

```
config>service>vprn>if>ipsec-tunnel>dynamic-keying
```

Description

This command associates the IPSec transform sets allowed for this tunnel. A maximum of four transforms can be specified. The transforms are listed in decreasing order of preference (the first one specified is the most preferred).

Parameters

transform-id

specifies the value used for transforms for dynamic keying

Values: 1 to 2048

7.2.1.5.18.9 manual-keying

Syntax

```
[no] manual-keying
```

Context

```
config>service>vprn>if>sap>ipsec-tunnel
```

Description

This command configures Security Association (SA) for manual keying. When enabled, the command specifies whether this SA entry is created manually by the user or dynamically by the IPSec sub-system.

Default

```
no manual-keying
```

7.2.1.5.18.10 replay-window

Syntax

```
replay-window {32 | 64 | 128 | 256 | 512}  
no replay-window
```

Context

```
config>service>vprn>if>ipsec-tunnel>manual-keying
```

Description

This command specifies the size of the anti-replay window. The anti-replay window protocol secures IP against an entity that can inject messages in a message stream from a source to a destination computer on the Internet.

Default

```
no replay-window
```

Parameters

32 | 64 | 128 | 256 | 512

specifies the size of the SA anti-replay window

7.2.1.5.18.11 security-association

Syntax

```
security-association security-entry-id authentication-key authentication-key  
encryption-key encryption-key spi spi transform transform-id direction {inbound |  
outbound}
```

```
no security-association security-entry-id direction {inbound | outbound}
```

Context

```
config>service>vpn>if>ipsec-tunnel>manual-keying
```

Description

This command configures the information required for manual keying SA creation.

Parameters

security-entry-id

specifies the ID of an SA entry

Values: 1 to 16

encryption-key

specifies the key used for the encryption algorithm

Values: none or 0x0..0xFFFFFFFF...(max 64 hex nibbles)

authentication-key

specifies the authentication key

Values: none or 0x0..0xFFFFFFFF...(max 40 hex nibbles)

spi

specifies the SPI (Security Parameter Index) used to look up the instruction to verify and decrypt the incoming IPSec packets when the direction is inbound. When the direction is outbound, the SPI is used in the encoding of the outgoing packets. The remote node can use this SPI to lookup the instruction to verify and decrypt the packet.

Values: 256 to 16383

transform-id

specifies the transform entry used by this SA entry. This object should be specified for all the entries created which are manual SAs. If the value is dynamic, this value is irrelevant and will be zero.

Values: 1 to 2048

inbound | outbound

specifies the direction of an IPSec tunnel

7.2.1.5.18.12 security-policy

Syntax

```
security-policy security-policy-id  
no security-policy
```

Context

```
config>service>vpn>ipsec-if>tunnel
```

Description

This command configures an IPSec security policy. The policy may then be associated with tunnels defined in the same context.

Parameters

security-policy-id

specifies the IPSec security policy entry that the tunnel uses

Values: 1 to 8192

7.2.1.6 CMPv2 commands

7.2.1.6.1 pki

Syntax

```
pki
```

Context

```
config>system>security
```

Description

This command enables the context to configure PKI related parameters.

7.2.1.6.2 ca-profile

Syntax

```
ca-profile name [create]
no ca-profile name
```

Context

```
config>system>security>pki
```

Description

This command creates a new `ca-profile` or enter the configuration context of an existing `ca-profile`. Up to 128 `ca-profiles` could be created in the system. A shutdown to the `ca-profile` does not affect the current up and running of the `ipsec-tunnel` or the `ipsec-gw` that associated with the `ca-profile`. But authentication afterwards will fail with a shutdown `ca-profile`.

Executing a `no shutdown` command in this context will cause system to reload the configured `cert-file` and `crl-file`.

The `ca-profile` command can be applied under the `ipsec-tunnel` or `ipsec-gw` configuration.

The `no` form of this command removes the `name` parameter from the configuration. A `ca-profile` can not be removed until all the associations (`ipsec-tunnel/gw`) have been removed.

Parameters

name

specifies the name of the `ca-profile`, a string up to 32 characters

create

keyword used to create a new `ca-profile` command. The `create` keyword requirement can be enabled/disabled in the `environment>create` context.

7.2.1.6.3 certificate

Syntax

```
certificate
```

Context

```
admin
```

Description

This command enables the context to configure X.509 certificate related operational parameters.

7.2.1.6.4 certificate-display-format

Syntax

```
certificate-display-format {ascii|utf8}
```

Context

```
config>system>security>pki
```

Description

This command specifies the certificate subject display format.

Parameters

ascii

Use ascii encoding.

utf8

Use utf8 encoding.

Default: ascii

7.2.1.6.5 cmpv2

Syntax

```
cmpv2
```

Context

```
admin>certificate
```

```
config>system>security>pki>ca-profile
```

Description

This command enables the context to configure CMPv2 parameters. Changes are not allowed when the CA profile is enabled (no shutdown).

7.2.1.6.6 accept-unprotected-errormsg

Syntax

[no] accept-unprotected-errormsg

Context

config>system>security>pki>ca-profile>cmpv2

Description

This command enables the system to accept both protected and unprotected CMPv2 error message. Without this command, the system only accepts protected error messages.

The no form of this command causes the system to only accept protected PKI confirmation message.

Default

no accept-unprotected-errormsg

7.2.1.6.7 accept-unprotected-pkiconf

Syntax

[no] accept-unprotected-pkiconf

Context

config>system>security>pki>ca-profile>cmpv2

Description

This command enables the system to accept both protected and unprotected CMPv2 PKI confirmation messages. Without this command, the system only accepts protected PKI confirmation message.

The no form of this command causes the system to only accept protected PKI confirmation message.

7.2.1.6.8 always-set-sender-for-ir

Syntax

[no] always-set-sender-for-ir

Context

```
config>system>security>pki>ca-profile>cmpv2
```

Description

This command specifies to always set the sender field in CMPv2 header of all Initial Registration (IR) messages with the subject name. By default, the sender field is only set if an optional certificate is specified in the CMPv2 request.

Default

no always-set-sender-for-ir

7.2.1.6.9 key-list

Syntax

```
cmp-key-list
```

Context

```
config>system>security>pki>ca-profile>cmpv2
```

Description

This command enables the context to configure pre-shared key list parameters.

7.2.1.6.10 key

Syntax

```
key password [hash | hash2] reference reference-number  
no key reference reference-number
```

Context

```
config>system>security>pki>ca-profile>cmpv2>key-list
```

Description

This command specifies a pre-shared key used for CMPv2 initial registration. Multiples of key commands are allowed to be configured under this context.

The password and reference-number is distributed by the CA via out-of-band means.

The configured password is stored in configuration file in an encrypted form by using SR OS hash2 algorithm.

The no form of this command removes the parameters from the configuration.

Parameters

password

specifies a printable ASCII string, up to 64 characters

hash

Specifies that the given password is already hashed using hashing algorithm version 1. A semantic check is performed on the given password field to verify if it is a valid hash 1 key to store in the database.

hash2

Specifies that the given password is already hashed using hashing algorithm version 2. A semantic check is performed on the given password field to verify if it is a valid hash 2 key to store in the database.

reference-number

specifies a printable ASCII string, up to 64 characters

7.2.1.6.11 url

Syntax

```
url url-string [service-id service-id]  
url url-string [service-name service-name]  
no url
```

Context

```
config>system>security>pki>ca-profile>cmpv2
```

Description

This command specifies HTTP URL of the CMPv2 server. The URL must be unique across all configured ca-profiles.

The URL will be resolved by the DNS server configured (if configured) in the corresponding router context.

If the *service-id* is 0 or omitted, the system tries to resolve the FQDN via DNS server configured in `bof.cfg`. After resolution, the system connects to the address in management routing instance first, and then to the base routing instance.

If the service is VPRN, the system only allows HTTP ports 80 and 8080.

Default

no url

Parameters

url-string
specifies the HTTP URL of the CMPv2 server, up to 180 characters

service-id
specifies the service instance that used to reach CMPv2 server

Values	service-id service-name		
–	service-id	1 to 2147483647	
–	service-name	specifies an existing service name, up to 64 characters	
–	–	If both service-id and service-name are not configured, the server is reachable in the mgmt router (checked first) or the base router	

7.2.1.6.12 revocation-check

Syntax

revocation-check {crl | crl-optional}

Context

config>system>security>pki>ca-profile

Description

This command specifies the revocation method system used to check the revocation status of certificate issued by the CA, the default value is `crl`, which will use CRL. But if it is `crl-optional`, it means when the user disables the ca-profile, the system tries to load the configured CRL (specified by the `crl-file` command). But if the system fails to load it for following reasons, the system still brings ca-profile oper-up, but leave the CRL as non-exist.

- CRL file does not exist

- CRL is not properly encoded - maybe due to interrupted file transfer
- CRL does not match cert
- Wrong CRL version
- CRL expired

If the system needs to use the CRL of a specific ca-profile to check the revocation status of an end-entity cert, and the CRL is non-existent due to the above reasons, the system treats it as being unable to get an answer from CRL and fall back to the next status-verify method or default-result.

If the system needs to check the revocation of a CA cert in cert chain, and if the CRL is non-existent due to the above reasons, the system skips checking the revocation status of the CA cert. For example, if CA1 is issued by CA2, if CA2 revocation-check is `crl-optional` parameter and the CA2 CRL is non-existent, the system will not check CA1 cert revocation status and consider it as “good”.

The users must shutdown the ca-profile to change the revocation-check configuration.

Default

revocation-check `crl`

Parameters

`crl`

specifies to use the configured CRL

`crl-optional`

specifies that the CRL is optional

7.2.1.6.13 http-response-timeout

Syntax

`http-response-timeout` *timeout*

`no http-response-timeout`

Context

`config>system>security>pki>ca-profile>cmpv2`

Description

This command specifies the timeout value for HTTP response that is used by CMPv2.

The no form of this command reverts to the default.

Parameters

timeout

specifies the HTTP response timeout in seconds

Values: 1 to 3600

Default: 30

7.2.1.6.14 http-version

Syntax

```
http-version [1.0 | 1.1]
```

Context

```
config>system>security>pki>ca-profile>cmpv2
```

Description

This command configures the the HTTP version for CMPv2 messages.

Default

1.1

7.2.1.6.15 response-signing-cert

Syntax

```
response-signing-cert filename
```

```
no response-signing-cert
```

Context

```
config>system>security>pki>ca-profile>cmpv2
```

Description

This command specifies a imported certificate that is used to verify the CMP response message if they are protected by signature. If this command is not configured, the CA certificate is used.

Parameters

filename

specifies the filename of the imported certificate

7.2.1.6.16 same-recipnonce-for-pollreq

Syntax

[no] same-recipnonce-for-pollreq

Context

config>system>security>pki>ca-profile>cmpv2

Description

This command enables the system to use same recipNonce as the last CMPv2 response for poll request.

Default

no same-recipnonce-for-pollreq

7.2.1.6.17 cert-request

Syntax

cert-request ca *ca-profile-name* current-key *key-filename* current-cert *cert-filename*
[hash-alg *hash-algorithm*] newkey *key-filename* subject-dn *subject-dn* save-as *save-path-of-result-cert*

Context

admin>certificate>cmpv2

Description

This command requests an additional certificate after the system has obtained the initial certificate from the CA.

The request is authenticated by a signature signed by the current-key, along with the current-cert. The hash algorithm used for signature is depends on the key type:

- DSA key: SHA1
- RSA key: MD5/SHA1/SHA224 | SHA256 | SHA384 | SHA512, by default is SHA1

In some cases, the CA may not return a certificate immediately, due to reasons such as request processing need manual intervention. In such cases, the `admin certificate cmpv2 poll` command can be used to poll the status of the request.

Parameters

ca-profile-name

specifies a ca-profile name which includes CMP server information, up to 32 characters maximum

key-filename

specifies corresponding certificate issued by the CA, up to 95 characters maximum

cert-filename

specifies the file name of an imported certificate that is attached to the certificate request, up to 95 characters maximum

key-filename

specifies the file name of the imported key, up to 95 characters maximum

hash-algorithm

specifies the hash algorithm for RSA key

Values: md5 | sha1 | sha224 | sha256 | sha384 | sha512

dn

specifies the subject of the requesting certificate, up to 256 characters maximum

Values: attr1=val1,attr2=val2 where: attrN = {C | ST | O | OU | CN}

save-path-of-result-cert

specifies the save full path name of saving the result certificate, up to 200 characters maximum

7.2.1.6.18 clear-request

Syntax

```
clear-request ca ca-profile-name
```

Context

```
admin>certificate>cmpv2
```

Description

This command clears current pending CMPv2 requests toward the specified CA. If there are no pending requests, it will clear the saved result of prior request.

Parameters

ca-profile-name

specifies a ca-profile name, up to 32 characters

7.2.1.6.19 initial-registration

Syntax

```
initial-registration ca ca-profile-name key-to-certify key-filename protection-alg
{password password reference ref-number | signature [cert cert-file-name [send-chain
[with-ca ca-profile-name]]] [protection-key key-file-name] [hash-alg {md5 | sha1 | sha224 |
sha256 | sha384 | sha512}]} subject-dn dn save-as save-path-of-result-cert
```

Context

admin>certificate>cmpv2

Description

This command request initial certificate from CA by using CMPv2 initial registration procedure.

The ca parameter specifies a CA-profile which includes CMP server information.

The key-to-certify keyword is an imported key file to be certified by the CA.

The protection-key is an imported key file used to for message protection if protection-alg is signature.

The request is authenticated either of following methods:

- A password and a reference number that pre-distributed by CA via out-of-band means. The specified password and reference number are not necessarily in the cmp-keylist configured in the corresponding CA-Profile
- A signature signed by the protection-key or key-to-certify, optionally along with the corresponding certificate. If the protection-key is not specified, system will use the key-to-certify for message protection. The hash algorithm used for signature is depends on key type: DSA key: SHA1 RSA key: MD5/SHA1/SHA224 | SHA256 | SHA384 | SHA512, by default is SHA1

Optionally, the system could also send a certificate or a chain of certificates in extraCerts field. Certificate is specified by the “cert” parameter, it must include the public key of the key used for message protection.

Sending a chain is enabled by specify the send-chain parameter.

The subject-dn parameter specifies the subject of the requesting certificate.

The save-as parameter specifies full path name of saving the result certificate.

In some cases, CA may not return certificate immediately, due to reason like request processing need manual intervention. In such cases, the admin certificate cmpv2 poll command could be used to poll

the status of the request. If `key-list` is not configured in the corresponding `ca-profile`, the system uses the existing password to authenticate the CMPv2 packets from server if it is in password protection.

If `key-list` is configured in the corresponding `ca-profile` and server doesn't send `SenderKID`, the system uses lexicographical first key in the `key-list` to authenticate the CMPv2 packets from server in case it is in password protection.

Parameters

ca ca-profile-name

specifies a `ca-profile` name which includes CMP server information up to 32 characters max

key-to-certify key-filename

specifies the file name of the key to certify up to 95 characters max

password

specifies an ASCII string, up to 64 characters

ref-number

specifies the reference number for this CA initial authentication key up to 64 characters max

cert-file-name

specifies the certificate file up to 95 characters max

send-chain with-ca ca-profile-name

specifies to send the chain

protection-key key-file-name

specifies the protection key associated with the action on the CA profile

hash-algorithm

specifies the hash algorithm for RSA key

Values: `md5` | `sha1` | `sha224` | `sha256` | `sha384` | `sha512`

dn

specifies the subject of the requesting certificate up to 256 chars max

Values: `attr1=val1,attr2=val2` where: `attrN = {C | ST | O | OU | CN}`

save-path-of-result-cert

specifies the save full path name of saving the result certificate up to 200 characters max

7.2.1.6.20 key-update

Syntax

```
key-update ca ca-profile-name newkey key-filename oldkey key-filename oldcert cert-
filename [hash-alg hash-algorithm] save-as save-path-of-result-cert
```

Context

admin>certificate>cmpv2

Description

This command requests a new certificate from the CA to update an existing certificate due to reasons such as key refresh or replacing compromised key.

In some cases, the CA may not return certificate immediately, due to reasons such as request processing need manual intervention. In such cases, the admin certificate cmpv2 poll command can be used to poll the status of the request.

Parameters

ca-profile-name

specifies a ca-profile name which includes CMP server information up to 32 characters max

newkey key-filename

specifies the key file of the requesting certificate up to 95 characters max

oldkey key-filename

specifies the key to be replaced up to 95 characters max

cert-filename

specifies the file name of an imported certificate to be replaced up to 95 characters max

hash-algorithm

specifies the hash algorithm for RSA key

Values: md5 | sha1 | sha224 | sha256 | sha384 | sha512

save-path-of-result-cert

specifies the save full path name of saving the result certificate, up to 200 characters

7.2.1.6.21 poll

Syntax

poll ca *ca-profile-name*

Context

admin>certificate>cmpv2

Description

This command polls the status of the pending CMPv2 request toward the specified CA.

If the response is ready, this command resumes the CMPv2 protocol exchange with server as the original command would do. The requests could be also still be pending as a result, this command could be used again to poll the status.

SR OS allows only one pending CMP request per CA, which means no new request is allowed when a pending request is present.

Parameters

ca-profile-name

specifies a ca-profile name, up to 32 characters

7.2.1.6.22 show-request

Syntax

```
show-request [ca ca-profile-name]
```

Context

```
admin>certificate>cmpv2
```

Description

This command displays current the CMPv2 pending request toward the specified CA. If there is no pending request, the last pending request is displayed including the status (success/fail/rejected) and the receive time of last CMPv2 message from server.

The following information is included in the output:

Request type, original input parameter (password is not displayed), checkAfter and reason in of last PollRepContent, time of original command input.

Parameters

ca-profile-name

specifies a ca-profile name, up to 32 characters. If not specified, the system displays the pending requests of all the CA profiles.

7.2.2 IP tunnel show command descriptions

7.2.2.1 cert-profile

Syntax

```
cert-profile name association
cert-profile [name]
cert-profile name entry [1..8]
```

Context

```
show>ipsec
```

Description

This command displays IPsec certificate profile information.

Parameters

name

specifies an existing cert-profile name

association

displays information for which this IPsec certificate profile is associated

entry [1..8]

displays information for the specified entry

Output

The following output examples show IPsec certificate profile information and *Show IPsec certificate profile output fields* and *Show IPsec certificate profile association output fields* describe the fields.

- *Output example IPsec certificate profile statistics*
- *Output example IPsec certificate profile entry statistics*
- *Output example IPsec certificate profile association statistics*

Output example IPsec certificate profile statistics

The following output example shows IPsec certificate profile statistics and *Show IPsec certificate profile output fields* describes the fields.

```
# show ipsec cert-profile cert "cert-1.der"
=====
Certificate Profile Entry
=====
Id Cert                               Key                               Status Flags
-----
1  cert-1.der                         key-1.der
```

Table 165: Show IPsec certificate profile output fields

Label	Description
Certificate Profile Entry	
Id	The certificate profile entry identifier
Cert/Cert File	The certificate file name
Key/Key File	The key file name
Status Flags	The status flags of the certificate profile
IPsec Certificate Profile	The IPsec certificate profile name
Entry	The IPsec certificate entry number
Comp Chain	The compute chain status
Compute Chain CA Profiles	The compute chain CA profiles associated with this certificate profile

Output example IPsec certificate profile entry statistics

The following output example shows IPsec certificate profile entry statistics and *Show IPsec certificate profile output fields* describes the fields.

```
# show ipsec cert-profile "cert-1.der" entry 1
=====
IPsec Certificate Profile: cert-1.der Entry: 1 Detail
=====
Cert File       : cert-1.der
Key File        : key-1.der
Status Flags    : (Not Specified)
Comp Chain      : complete

Compute Chain CA Profiles
-----
CA10
CA9
CA8
```

```

CA7
CA6
=====
# exit

```

Output example IPsec certificate profile association statistics

The following output example shows IPsec certificate profile association statistics and *Show IPsec certificate profile association output fields* describes the fields.

In the following example, the “cert-1.der” is the certificate-profile name.

```

# show ipsec cert-profile association "cert-1.der"
=====
IPsec tunnels using certificate profile
=====
SvcId      Type   SAP                Tunnel
-----
3          vprn   tunnel-1.private:3  tun-1-s-cert-v2
8          vprn   tunnel-1.private:7  tun-1-s-cert-MTA-v2
42         vprn   tunnel-1.private:10 tun-1-s-cert-i_op-ss-v2
48         vprn   tunnel-1.private:11 tun-1-s-cert-MTA-i_op-ss-v2
=====
Number of tunnel entries: 4
=====
IPsec gateways using certificate profile
=====
SvcId      Type   SAP                Gateway
-----
1057       vprn   tunnel-1.public:18  d-cert-MTA-g1-1-v2
1092       vprn   tunnel-1.public:21  d-cert-i_op-ss-g1-1-v2
=====
Number of gateway entries: 2
=====

```

Table 166: Show IPsec certificate profile association output fields

Label	Description
IPsec tunnels using certificate profile	
SvcId	The service ID of the IPsec tunnel that is using the certificate profile
Type	The service type
SAP	The SAP associated with the IPsec tunnel

Table 166: Show IPsec certificate profile association output fields (continued)

Label	Description
Tunnel	The IPsec tunnel name
Number of tunnel entries	The total number of IPsec tunnel entries
IPsec gateways using certificate profile	
SvcId	The service ID of the IPsec gateway that is using the certificate profile
Type	The service type
SAP	The SAP associated with the IPsec gateway
Gateway	The gateway name
Number of gateway entries	The total number of IPsec gateway entries

7.2.2.2 certificate

Syntax

`certificate filename association`

Context

`show>ipsec`

Description

This command displays certificate-related information.

Parameters

filename

specifies the certificate file name

association

displays information for which this IPsec certificate is associated

Output

The following output examples show certificate information and *Show IPsec certificate output fields* and *Show certificate output fields* describe the fields.

- *Output example IPsec certificate association statistics*
- *Output example certificate statistics*

Output example IPsec certificate association statistics

The following output example shows IPsec certificate association statistics and *Show IPsec certificate output fields* describes the fields.

```
# show ipsec certificate cert-1.der association
=====
Associated Tunnels
=====
Tunnel                SvcId      Sap                Admin
-----
tun-1-s-cert-v2       3          tunnel-1.private:3  Up
tun-1-s-cert-MTA-v2   8          tunnel-1.private:7  Up
tun-1-s-cert-i_op-ss-v2 42         tunnel-1.private:10 Up
tun-1-s-cert-MTA-i_op-ss-v2 48        tunnel-1.private:11 Up
-----
IPsec Tunnels: 4
=====
```

Table 167: Show IPsec certificate output fields

Label	Description
Associated Tunnels	
Tunnel	The tunnel name associated with the certificate
SvcId	The service ID associated with the tunnel
Sap	The SAP associated with the tunnel
Admin	The administrative state of the tunnel

Table 167: Show IPsec certificate output fields (continued)

Label	Description
IPsec Tunnels	The total number of IPsec tunnels associated with the certificate

Output example certificate statistics

The following output example shows certificate statistics and *Show certificate output fields* describes the fields.

In the following example, “cert-1.der” is the actual file in use.

```
# show certificate ca-profile
-----
Max Cert Chain Depth: 7 (default)
-----
Certificate Display Format: 1 ASCII
=====
CA Profile
=====
CA Profile      Admin Oper  Cert File      CRL File
                State State
-----
CA0             up    up    CA1-00cert.der    CA1-00crl.der
CA1             up    up    CA1-01cert.der    CA1-01crl.der
CA2             up    up    CA1-02cert.der    CA1-02crl.der
CA3             up    up    CA1-03cert.der    CA1-03crl.der
CA4             up    up    CA1-04cert.der    CA1-04crl.der
CA5             up    up    rsa_sha512_1024_0cert.d*  rsa_sha512_1024_0crl.der
CA6             up    up    rsa_sha512_1024_1cert.d*  rsa_sha512_1024_1crl.der
CA7             up    up    rsa_sha512_1024_2cert.d*  rsa_sha512_1024_2crl.der
CA8             up    up    rsa_sha512_1024_3cert.d*  rsa_sha512_1024_3crl.der
CA9             up    up    rsa_sha512_1024_4cert.d*  rsa_sha512_1024_4crl.der
CA10            up    up    rsa_sha512_1024_5cert.d*  rsa_sha512_1024_5crl.der
CA11            up    up    rsa_sha384_1024_0cert.d*  rsa_sha384_1024_0crl.der
CA12            up    up    rsa_sha384_1024_1cert.d*  rsa_sha384_1024_1crl.der
CA13            up    up    rsa_sha384_1024_2cert.d*  rsa_sha384_1024_2crl.der
CA14            up    up    rsa_sha384_1024_3cert.d*  rsa_sha384_1024_3crl.der
CA15            up    up    rsa_sha384_1024_4cert.d*  rsa_sha384_1024_4crl.der
CA16            up    up    rsa_sha384_1024_5cert.d*  rsa_sha384_1024_5crl.der
CMPv2           up    up    rsaCMPv2cert.der    rsaCMPv2CRL.der
-----
Entries found: 18
=====
* indicates that the corresponding row element may have been truncated.
```

Table 168: Show certificate output fields

Label	Description
Max Cert Chain Depth	The maximum certificate chain depth
Certificate Display Format	The certificate display format
CA Profile	
CA Profile	The CA profile name
Admin State	The administrative state of the CA profile
Oper State	The operating state of the CA profile
Cert file	The CA profile certificate file name
CRL file	The CA profile CRL file name
Entries found	The number of CA profile entries found

7.2.2.3 gateway

Syntax

```
gateway name name  
gateway [service service-id]  
gateway tunnel [ip-address:port]  
gateway name name tunnel ip-address:port  
gateway name name tunnel  
gateway tunnel count
```

Context

```
show>ipsec
```

Description

This command displays IPSec gateway information.

Parameters

name

specifies an IPSec gateway name

service-id

specifies the service ID of the default security service used by the IPSec gateway

Values	service-id service-name	
–	service-id	1 to 214748364
–	service-name	specifies an existing service name, up to 64 characters

ip-address:port

specifies to display the IP address and UDP port of the SAP IPSec gateway to the tunnel

Values: port: 0 to 65535

count

specifies to display the number of IPSec gateway tunnels with the `ike-policy>auth-method` command set to `psk`

Output

The following output examples show gateway statistics and *Show IPSec gateway name statistics output fields*, *Show IPSec gateway service statistics output fields*, *Show IPSec gateway tunnel detail statistics output fields*, *Show IPSec gateway name and tunnel statistics output fields*, and *Show IPSec gateway tunnel count statistics output fields* describe the fields.

- *Output example gateway name statistics*
- *Output example IPSec gateway service statistics*
- *Output example gateway tunnel detail statistics*
- *Output example gateway name and tunnel detail statistics*
- *Output example gateway name and tunnels statistics*
- *Output example gateway tunnel count statistics*

Output example gateway name statistics

The following output example shows gateway name statistics and *Show IPSec gateway name statistics output fields* describes the fields.

```
# show ipsec gateway name "d-psk-g1-1-v2"
=====
IPSec Gateway (SAP)
=====
-----
IPSec Gateway ( d-psk-g1-1-v2 )
-----
Sap                : tunnel-1.public:1      Service           : 1052
Local GW           : 141.1.1.2
Admin State        : Up                    Oper State        : Up
Def Secure Svc     : 52
Def Secure Svc If  : priv1-t1-d-psk-v2
Ike Policy Id      : 1
Ike Version        : 2                    Ike Policy Auth   : psk
Pre Shared Key     : dynamic-psk-v2
X509 Cert          : (Not Specified)
Key                : (Not Specified)
Local Id Type      : none
Cert Profile       : N/A
Trust Anchor       : None
Radius Acct Plcy   : None
Radius Auth Plcy   : None
TS-List            : 1052
Certificate Status Verify
-----
Primary            : crl                    Secondary         : none
Default Result     : revoked
-----
Template Id: 1
-----
Transform Id1      : 1                    Transform Id2     : None
Transform Id3      : None                  Transform Id4     : None
Reverse Route      : useSecurityPolicy     Replay Window     : None
IP MTU             : max                   Encap IP MTU      : max
Pkt Too Big        : true                  Clear DF BIT      : false
Pkt Too Big Number : 100                   Pkt Too Big Intvl : 10 secs
=====
```

Table 169: Show IPSec gateway name statistics output fields

Label	Description
IPSec Gateway <i>name</i>)	The IPSec gateway for which information is being displayed
Sap	The SAP name

Table 169: Show IPSec gateway name statistics output fields (continued)

Label	Description
Local GW	The local gateway address
Admin State	The administrative state of the gateway: up or down
Def Secure Svc	The default secure service ID
Def Secure Svc If	The default secure service interface name
Ike Policy Id	The IKE policy ID
Ike Version	The IKE version
Pre Shared Key	The pre-shared key
X509 Cert	The certificate used for verification
Key	The private key associated with the certificate
Local Id Type	The local ID type
Cert Profile	The certificate profile
Trust Anchor	The trust anchor
Radius Acct Plcy	The RADIUS accounting policy
Radius Auth Plcy	The RADIUS authentication policy
TS-List	The traffic selector list
Certificate Status Verify	
Primary	The primary certificate status verify method: OCSP or CRL

Table 169: Show IPsec gateway name statistics output fields (continued)

Label	Description
Secondary	The secondary certificate status verify method: OCSP or CRL
Default Result	The default certificate status verify result: good or revoked
Template Id	The tunnel template ID
Transform Id1...Transform Id4	The IKE transformation IDs
Reverse Route	The reverse route
IP MTU	The Maximum Transmission Unit (MTU) size of IP packets
Pkt TooBig	Specifies whether packets are too big: true or false
Pkt Too Big Number	The number of packets that were too big
Pkt Too Big Intvl	The interval of the packets that were too big
Replay Window	The anti-replay window for the tunnel template
Encap IP MTU	The encapsulated IP MTU
Clear DF BIT	Whether clearing of the Do-not-Fragment bit is enabled: true or false.

Output example IPsec gateway service statistics

The following output example shows gateway service statistics and *Show IPsec gateway service statistics output fields* describes the fields.

```
# show ipsec gateway service 1052
=====
IPSec Gateway 1052
=====
Name                LclGwAddr      Adm  Opr  Ike  Auth
SAP                 Service
-----
```

```
d-psk-g1-1-v2          141.1.1.2      Up   Up   1   psk
tunnel-1.public:1      1052
```

```
-----
Number of gateways: 1
=====
```

Table 170: Show IPsec gateway service statistics output fields

Label	Description
IPSec Gateway (<i>name</i>)	The IPsec gateway for which information is being displayed
Name	The gateway name
SAP	The SAP name
LclGwAddr	The local gateway address
Service	The service name
Adm	The administrative state: up or down
Opr	The operational state: up or down
Ike	The IKE version
Auth	The authorization code
Number of Gateways	The total number of gateways displayed

Output example gateway tunnel detail statistics

The following output example shows gateway tunnel statistics and *Show IPsec gateway tunnel detail statistics output fields* describes the fields.

```
# show ipsec gateway tunnel 41.1.1.2:500
=====
IPsec Remote Users Tunnel Detail
=====
IP Addr: 41.1.1.2:500, port: 500
-----
Service Id      : 1052                Sap Id          : tunnel-1.public:1
```



```

Address       : 41.1.1.2:500
Private If    : priv1-t1-d-psk-v2
Private Service : 52
Replay Window : None
Host MDA      : 1/1
Match TrustAnchor: None
Last Oper Changed: 11/27/2017 22:12:28

```

Dynamic Keying Parameters

```

Transform Id1 : 1
Transform Id2 : None
Transform Id3 : None
Transform Id4 : None
IPsec GW Name : d-psk-g1-1-v2
Local GW Address : 141.1.1.2
Ike Policy Id : 1
Ike Pol Auth : psk
Pre Shared Key : dynamic-psk-v2
Cert Profile : N/A
Trust Anchor : N/A
Local Id Type : none
Radius Acct Plcy : None
Radius Auth Plcy : None
TS-List : 1052
Certificate Status Verify

```

```

Primary : crl
Secondary : none
Default Result : revoked

```

ISAKMP-SA

```

State : Up
Established : 11/27/2017 22:12:29
Expires : 11/28/2017 22:12:29
Lifetime : 86400

```

ISAKMP Statistics

```

Tx Packets : 82
Tx Errors : 0
Tx DPD : 0
Tx DPD ACK : 79
DPD Timeouts : 0
Rx Packets : 82
Rx Errors : 0
Rx DPD : 79
Rx DPD ACK : 0
Rx DPD Errors : 0

```

IPsec-SA : 1, Inbound (index 2)

```

SPI : 399494
Auth Algorithm : Sha384
Installed : 11/27/2017 22:12:28
Encr Algorithm : 3Des
Lifetime : 3600

```

Local Traffic Selectors:

172.16.0.0-172.16.0.255

Remote Traffic Selectors:

192.168.0.0-192.168.0.255

Aggregate Statistics

```

Bytes Processed : 10567336
Crypto Errors : 0
SA Errors : 0
Packets Processed: 38012
Replay Errors : 0
Policy Errors : 0

```

IPsec-SA : 1, Outbound (index 1)

```

SPI                : 96442
Auth Algorithm     : Sha384           Encr Algorithm    : 3Des
Installed         : 11/27/2017 22:12:28 Lifetime         : 3600
Local Traffic Selectors:
172.16.0.0-172.16.0.255
Remote Traffic Selectors:
192.168.0.0-192.168.0.255
Aggregate Statistics
-----
Bytes Processed   : 10557050          Packets Processed: 37975
Crypto Errors     : 0                 Replay Errors     : 0
SA Errors         : 0                 Policy Errors     : 0

```

```

=====
Fragmentation Statistics
=====

```

```

Encapsulation Overhead      : 69
Pre-Encapsulation
  Fragmentation Count       : 0
  Last Fragmented Packet Size : 0
Post-Encapsulation
  Fragmentation Count       : 0
  Last Fragmented Packet Size : 0

```

```

-----
IPsec-SA : 2, Inbound (index 2)
-----

```

```

SPI                : 495076
Auth Algorithm     : Sha384           Encr Algorithm    : 3Des
Installed         : 11/27/2017 22:12:29 Lifetime         : 3600
Local Traffic Selectors:
172.16.1.0-172.16.1.255
Remote Traffic Selectors:
192.168.1.0-192.168.1.255
Aggregate Statistics
-----
Bytes Processed   : 10551212          Packets Processed: 37954
Crypto Errors     : 0                 Replay Errors     : 0
SA Errors         : 0                 Policy Errors     : 0

```

```

-----
IPsec-SA : 2, Outbound (index 1)
-----

```

```

SPI                : 355180
Auth Algorithm     : Sha384           Encr Algorithm    : 3Des
Installed         : 11/27/2017 22:12:29 Lifetime         : 3600
Local Traffic Selectors:
172.16.1.0-172.16.1.255
Remote Traffic Selectors:
192.168.1.0-192.168.1.255
Aggregate Statistics
-----
Bytes Processed   : 10531752          Packets Processed: 37884
Crypto Errors     : 0                 Replay Errors     : 0
SA Errors         : 0                 Policy Errors     : 0

```

```

=====
Fragmentation Statistics
=====

```

```

Encapsulation Overhead      : 69

```

```

Pre-Encapsulation
  Fragmentation Count      : 0
  Last Fragmented Packet Size : 0
Post-Encapsulation
  Fragmentation Count      : 0
  Last Fragmented Packet Size : 0
=====

```

Table 171: Show IPSec gateway tunnel detail statistics output fields

Label	Description
IPSec Remote Users Tunnel Detail	
IP Addr	The IP address of the SAP IPSec gateway to the tunnel
Port	The UDP port of the SAP IPSec gateway to the tunnel
Service Id	The service ID of the SAP IPSec gateway to the tunnel
Sap Id	The SAP ID of the SAP IPSec gateway to the tunnel
Address	The IP address of the SAP IPSec gateway to the tunnel
Private If	The private interface name
Private Service	The private service number
Template Id	The template ID
Replay Window	The replay window
Bi Direction SA	The bi-directional SA
Host MDA	The host MDA
Match TrustAnchor	The match trust anchor
Last Oper Changed	The date and time of the last operational change to the tunnel

Table 171: Show IPsec gateway tunnel detail statistics output fields (continued)

Label	Description
Dynamic Keying Parameters	
Transform Id1... Transform Id4	The IKE transformation IDs
IPsec GW Name	The IPsec gateway name
Local GW Address	The local gateway IP address
Ike Policy Id	The IKE policy ID
Ike Pol Auth	The IKE policy authorization
Pre Shared Key	The pre-shared key
Cert Profile	The certificate profile
Trust Anchor	The trust anchor
Local Id Type	The local ID type
Radius Acct Plcy	The RADIUS account policy
Radius Auth Plcy	The RADIUS authorization policy
TS-List	The traffic selector list
Certificate Status Verify	
Primary	The primary certificate status verify method: OCSP or CRL
Secondary	The secondary certificate status verify method: OCSP or CRL
Default Results	The default result of the certificate status verify: good or revoked

Table 171: Show IPsec gateway tunnel detail statistics output fields (continued)

Label	Description
ISAKMP-SA	
State	The state of the ISAKMP-SA
Established	The date and time that the ISAKMP-SA was established
Lifetime	The ISAKMP-SA lifetime
Expires	The date and time that the ISAKMP-SA will expire
ISAKMP Statistics	
Tx Packets	The number of transmitted ISAKMP packets
Tx Errors	The number of transmission ISAKMP errors
Tx DPD	The number of transmitted ISAKMP DPDs
Tx DPD ACK	The number of transmitted ISAKMP DPD acknowledgments
DPD Timeouts	The number of ISAKMP DPD timeouts
Rx Packets	The number of received ISAKMP packets
Rx Errors	The number of received ISAKMP errors
Rx DPD	The number of received ISAKMP DPDs
Rx DPD ACK	The number of received ISAKMP DPD acknowledgments
Rx DPD Errors	The number of received ISAKMP DPD errors
IPsec-SA	

Table 171: Show IPSec gateway tunnel detail statistics output fields (continued)

Label	Description
SPI	The Security Parameter Index
Auth Algorithm	The authorization algorithm
Encr Algorithm	The encryption algorithm
Installed	The installed time of the IPsec SA
Lifetime	The lifetime of the IPsec SA
Local Traffic Selectors	The local traffic selectors
Remote Traffic Selectors	The remote traffic selectors
Aggregate Statistics	
Bytes Processed	The number of bytes processed
Packets Processed	The number of packets processed
Crypto Errors	The number of cryptographic errors
Replay Errors	The number of replay errors
SA Errors	The number of SA errors
Policy Errors	The number of policy errors
Fragmentation Statistics	
Encapsulation Overhead	The encapsulation overhead
Pre-Encapsulation	

Table 171: Show IPsec gateway tunnel detail statistics output fields (continued)

Label	Description
Fragmentation Count	The pre-encapsulation fragmentation count
Last Fragmented Packet Size	The pre-encapsulation last fragmentation packet size
Post-Encapsulation	
Fragmentation Count	The post-encapsulation fragmentation count
Last Fragmented Packet Size	The post-encapsulation last fragmentation packet size

Output example gateway name and tunnel detail statistics

The following output example shows gateway name and detail tunnel statistics and *Show IPsec gateway tunnel detail statistics output fields* describes the fields.

```
# show ipsec gateway name d-psk-g1-1-v2 tunnel 41.1.1.2:500
=====
IPsec Remote Users Tunnel Detail
=====
-----
IP Addr: 41.1.1.2, port: 500
-----
Service Id       : 1052                Sap Id           : tunnel-1.public:1
Address          : 41.1.1.2
Private If       : priv1-t1-d-psk-v2
Private Service  : 52                  Template Id      : 1
Replay Window    : None                Bi Direction SA  : true
Host MDA         : 1/1
Match TrustAnchor: None
Last Oper Changed: 11/27/2017 22:12:28
-----
Dynamic Keying Parameters
-----
Transform Id1    : 1                   Transform Id2     : None
Transform Id3    : None                Transform Id4     : None
IPsec GW Name    : d-psk-g1-1-v2
Local GW Address : 141.1.1.2
Ike Policy Id    : 1                   Ike Pol Auth      : psk
Pre Shared Key   : dynamic-psk-v2
Cert Profile     : N/A
Trust Anchor     : N/A
Local Id Type    : none
Radius Acct Plcy : None
Radius Auth Plcy : None
```

```

TS-List      : 1052
Certificate Status Verify
-----
Primary       : crl                Secondary       : none
Default Result : revoked
-----
ISAKMP-SA
-----
State         : Up
Established    : 11/27/2017 22:12:29  Lifetime     : 86400
Expires       : 11/28/2017 22:12:29
ISAKMP Statistics
-----
Tx Packets    : 86                  Rx Packets    : 86
Tx Errors     : 0                   Rx Errors     : 0
Tx DPD        : 0                   Rx DPD        : 83
Tx DPD ACK    : 83                  Rx DPD ACK    : 0
DPD Timeouts  : 0                   Rx DPD Errors : 0
-----
IPsec-SA : 1, Inbound (index 2)
-----
SPI           : 399494
Auth Algorithm : Sha384              Encr Algorithm : 3Des
Installed     : 11/27/2017 22:12:28 Lifetime       : 3600
Local Traffic Selectors:
172.16.0.0-172.16.0.255
Remote Traffic Selectors:
192.168.0.0-192.168.0.255
Aggregate Statistics
-----
Bytes Processed : 10567336           Packets Processed: 38012
Crypto Errors   : 0                  Replay Errors    : 0
SA Errors       : 0                  Policy Errors    : 0
-----
IPsec-SA : 1, Outbound (index 1)
-----
SPI           : 96442
Auth Algorithm : Sha384              Encr Algorithm : 3Des
Installed     : 11/27/2017 22:12:28 Lifetime       : 3600
Local Traffic Selectors:
172.16.0.0-172.16.0.255
Remote Traffic Selectors:
192.168.0.0-192.168.0.255
Aggregate Statistics
-----
Bytes Processed : 10557050           Packets Processed: 37975
Crypto Errors   : 0                  Replay Errors    : 0
SA Errors       : 0                  Policy Errors    : 0
=====
Fragmentation Statistics
=====
Encapsulation Overhead      : 69
Pre-Encapsulation
  Fragmentation Count       : 0
  Last Fragmented Packet Size : 0
Post-Encapsulation
  Fragmentation Count       : 0

```



```

Last Fragmented Packet Size      : 0
=====
-----
IPsec-SA : 2, Inbound (index 2)
-----
SPI              : 495076
Auth Algorithm    : Sha384          Encr Algorithm    : 3Des
Installed        : 11/27/2017 22:12:29 Lifetime        : 3600
Local Traffic Selectors:
172.16.1.0-172.16.1.255
Remote Traffic Selectors:
192.168.1.0-192.168.1.255
Aggregate Statistics
-----
Bytes Processed   : 10551212          Packets Processed: 37954
Crypto Errors     : 0                  Replay Errors     : 0
SA Errors         : 0                  Policy Errors     : 0
-----
IPsec-SA : 2, Outbound (index 1)
-----
SPI              : 355180
Auth Algorithm    : Sha384          Encr Algorithm    : 3Des
Installed        : 11/27/2017 22:12:29 Lifetime        : 3600
Local Traffic Selectors:
172.16.1.0-172.16.1.255
Remote Traffic Selectors:
192.168.1.0-192.168.1.255
Aggregate Statistics
-----
Bytes Processed   : 10531752          Packets Processed: 37884
Crypto Errors     : 0                  Replay Errors     : 0
SA Errors         : 0                  Policy Errors     : 0
=====
Fragmentation Statistics
=====
Encapsulation Overhead      : 69
Pre-Encapsulation
  Fragmentation Count       : 0
  Last Fragmented Packet Size : 0
Post-Encapsulation
  Fragmentation Count       : 0
  Last Fragmented Packet Size : 0
=====
=====

```

Output example gateway name and tunnel statistics

The following output example shows gateway name and tunnel statistics and *Show IPsec gateway name and tunnel statistics output fields* describes the fields.

```

# show ipsec gateway name d-psk-g1-1-v2 tunnel
=====
IPsec Remote User Tunnels
=====
Remote Endpoint Address      GW Name      TnlType
GW Lcl Address              SvcId

```

Private Address	Secure SvcId	BiDirSA
41.1.1.2:500	d-psk-g1-1-v2	
141.1.1.2	1052	psk
	52	true

IPsec Gateway Tunnels: 1		
=====		

Table 172: Show IPsec gateway name and tunnel statistics output fields

Label	Description
IPSec Remote Users Tunnel	
Remote Endpoint Address	The remote endpoint IP address
GW Name	The gateway name
GW Lcl Address	The local gateway IP address
Svc Id	The gateway service ID
Private Address	The private gateway IP address
Secure SvcId	The gateway secure service ID
IPSec Gateway Tunnels	The total number of tunnels on the IPSec gateway

Output example gateway tunnel count statistics

The following output example shows gateway tunnel count statistics and *Show IPsec gateway tunnel count statistics output fields* describes the fields.

```
# show ipsec gateway tunnel count
=====
IPsec Tunnel Count
=====
Total IPsec Tunnels           : 0
-----
IPsec Gateway Psk Tunnels     : 1
IPsec Gateway plainPskXAuth Tunnels : 0
IPsec Gateway Cert Tunnels    : 0
IPsec Gateway Psk-Radius Tunnels : 0
```

```

IPsec Gateway Cert-Radius Tunnels      : 0
IPsec Gateway EAP Tunnels              : 0
IPsec Gateway Auto-EAP Radius Tunnels  : 0
Total IPsec Gateway Tunnels            : 1
=====

```

Table 173: Show IPsec gateway tunnel count statistics output fields

Label	Description
IPsec Tunnel Count	
Total IPsec Tunnels	The total number of IPsec tunnels on the gateway
IPsec Gateway Psk Tunnels	The number of PSK tunnels on the gateway
IPsec Gateway plainPskXAuth Tunnels	The number of plain PSK authorization tunnels on the gateway
IPsec Gateway Cert Tunnels	The number of certificate tunnels on the gateway
IPsec Gateway Psk-Radius Tunnels	The number of PSK RADIUS tunnels on the gateway
IPsec Gateway Cert-Radius Tunnels	The number of certificate RADIUS tunnels on the gateway
IPsec Gateway EAP Tunnels	The number of EAP tunnels on the gateway
IPsec Gateway Auto-EAP Radius Tunnels	The number of auto-EAP RADIUS tunnels on the gateway
Total IPsec Gateway Tunnels	The total number of tunnels on the IPsec gateway

7.2.2.4 ike-policy

Syntax

```
ike-policy ike-policy-id  
ike-policy
```

Context

```
show>ipsec
```

Description

This command displays IKE policy information.

Parameters

ike-policy-id

specifies the ID of an IKE policy entry

Values: 1 to 2048

Output

The following output example shows IKE policy information.

```
# show ipsec ike-policy 10  
=====
```

IPsec IKE policy Configuration Detail			
=====			
Policy Id	: 10	IKE Mode	: main
DH Group	: Group2	Auth Method	: psk
PFS	: False	PFS DH Group	: Group2
Auth Algorithm	: Sha1	Encr Algorithm	: Aes128
ISAKMP Lifetime	: 86400	IPsec Lifetime	: 3600
NAT Traversal	: Disabled		
NAT-T Keep Alive	: 0	Behind NAT Only	: True
DPD	: Disabled		
DPD Interval	: 30	DPD Max Retries	: 3
Description	: (Not Specified)		

```
=====
```

7.2.2.5 radius-accounting-policy

Syntax

```
radius-accounting-policy [name]
```

Context

show>ipsec

Description

This command displays IPSec RADIUS accounting policy information.

Parameters

name

specifies an existing RADIUS accounting policy

7.2.2.6 radius-authentication-policy

Syntax

radius-authentication-policy [*name*]

Context

show>ipsec

Description

This command displays IPSec RADIUS authentication policy information.

Parameters

name

specifies an existing RADIUS authentication policy

7.2.2.7 security-policy

Syntax

security-policy service *service-id*[security-policy-id *security-policy-id*]

security-policy

Context

show>ipsec

Description

This command displays IPSec security policy information.

Parameters

service-id

specifies the service-id of the tunnel delivery service

Values	service-id service-name	
–	service-id	1 to 214748364
–	service-name	specifies an existing service name, up to 64 characters

security-policy-id

specifies the IPSec security policy entry that this tunnel will use

Values: 1 to 8192

Output

The following output example shows IPSec security policy information and *Show IPSec security policy output fields* describes the output fields.

Output example

```
# show ipsec security-policy service 100
=====
Security Policy Param Entries
=====
Service Id      Security Plcy  Local Ip
                Plcy ParamsId Remote Ip
-----
100             1             198.51.100.255/32
                1             198.51.100.255/32
                2001:db8:abcd:12::/128
                2001:db8:abcd:12::/128
100             1             ****
                2             ****
-----
No. of IPsec Security Policy Param Entries: 2
=====
```

Table 174: Show IPSec security policy output fields

Field	Description
Service Id	The IPSec security policy service ID
Security Plcy	The security policy
Local Ip	The local IP address
Plcy ParamsId	The policy parameters ID
Remote Ip	The Remote IP address

7.2.2.8 tunnel-template

Syntax

```
tunnel-template [ipsec-template-identifier]
```

Context

```
show>ipsec
```

Description

This command displays tunnel template information.

Parameters

ipsec-template-identifier

displays an existing IPSec tunnel template ID

Values: 1 to 2048

Output

The following output example shows tunnel template information.

Output example

```
# config>ipsec# show ipsec tunnel-template 1
```

```
=====
```

IPSec Tunnel Template

```
=====
Id      Trnsfrm1 Trnsfrm2 Trnsfrm3 Trnsfrm4 ReverseRoute ReplayWnd
-----
1       1      none   none    none    useSecurityPolicy 128
-----
```

```
Number of templates: 1
=====
```

7.2.2.9 transform

Syntax

```
transform [transform-id]
```

Context

```
show>ipsec
```

Description

This command displays IPSec transforms.

Parameters

transform-id

specifies an IPSec transform entry

Values: 1 to 2048

Output

The following output example shows IPSec transform information.

Output example

```
# config>ipsec# show ipsec transform 1
=====
IPsec Transforms
=====
TransformId   EspAuthAlgorithm   EspEncryptionAlgorithm
-----
1             Sha1               Aes128
-----
No. of IPsec Transforms: 1
=====
```


7.2.2.10 trust-anchor-profile

Syntax

```
trust-anchor-profile [trust-anchor-profile] association
trust-anchor-profile [trust-anchor-profile]
```

Context

```
show>ipsec
```

Description

This command displays trust anchor profile information.

Parameters

trust-anchor-profile

Specifies the trust anchor profile name up, to 32 characters

association

displays information for which this trust anchor profile is associated

Output

The following output example shows trust anchor profile information.

Output example

```
# show ipsec trust-anchor-profile
=====
Trust Anchor Profile Information
=====
Name                               CA Profiles Down
-----
CA0wCMPv2                          0
CA1wCMPv2                          0
CA2wCMPv2                          0
CA3wCMPv2                          0
CA4wCMPv2                          0
CA5wCMPv2                          0
CA6wCMPv2                          0
CA7wCMPv2                          0
CA8wCMPv2                          0
CA9wCMPv2                          0
CA10wCMPv2                         0
=====

# show ipsec trust-anchor-profile
=====
```

```
Trust Anchor CA-profile List
```

=====	
CA Profile	Admin/Oper State

CA6	up/up
CMPv2	up/up
=====	

7.2.2.11 ts-list

Syntax

```
ts-list [list-name]  
ts-list list-name association  
ts-list list-name entry [1..32]
```

Context

```
show>ipsec
```

Description

This command displays IPsec traffic-selector list information.

Parameters

list-name

specifies the traffic-selector list name

association

displays information for which this traffic-selector list is associated

entry [*1..32*]

displays information for the specified entry

Output

The following output example shows IPsec traffic-selector list information.

Output example

```
show ipsec ts-list "ts1"
```

=====	
TS-List Local Entry	
=====	
Entry Id	IP Address Range or Prefix/Prefix-Len

1	192.168.1.0/24

```
2          192.168.2.0/24
=====
show ipsec ts-list "ts1" association

=====
IPsec gateway using traffic-selection-list
=====
SvcId      Type      SAP
-----
300         ies      tunnel-1.public:100
=====
Number of entries: 1
=====
```

7.2.2.12 tunnel

Syntax

tunnel *ipsec-tunnel-name*

tunnel

Context

show>ipsec

Description

This command displays tunnel information.

Parameters

ipsec-tunnel-name

specifies the name of the tunnel, up to 32 characters

Output

The following output examples show tunnel statistics and *Show IPsec tunnel statistics output fields*, *Show IPsec tunnel detail statistics output fields*, and *Show IPsec gateway tunnel count statistics output fields* describe the fields.

- *Output example tunnel statistics*
- *Output example tunnel detail statistics*
- *Output example tunnel count statistics*

Output example tunnels statistics

The following output example shows tunnel statistics and *Show IPsec tunnel statistics output fields* describes the fields.

```
# show ipsec tunnel
=====
IPsec Tunnels
=====
TunnelName      LocalAddress      SvcId      Admn  Keying
 SapId          RemoteAddress     DlvrySvcId Oper   Sec
                                   Plcy
-----
LI              ****             100        Down  None
  tunnel-1.private:1      ****             100        Down  None
nonLi           2001:db8:abcd:12* 100        Down  None
  tunnel-1.private:1      2001:db8:abcd:*  100        Down  None
-----
IPsec Tunnels: 2
=====
```

Table 175: Show IPsec tunnel statistics output fields

Label	Description
IPsec Tunnels	
TunnelName	The IPsec tunnel name
Sap Id	The tunnel SAP ID
LocalAddress	The local IP address of the IPsec tunnel
RemoteAddress	The remote IP address of the IPsec tunnel
SvcId	The service ID of the IPsec tunnel
DlvrySvcId	The delivery service ID of the IPsec tunnel
Admn	The administrative state of the IPsec tunnel: Up or Down
Oper	The operational state of the IPsec tunnel
Keying	The type of encryption keying: manual or dynamic

Table 175: Show IPsec tunnel statistics output fields (continued)

Label	Description
Sec Plcy	The security policy ID used for the IPsec tunnel
IPsec Tunnels	The total number of IPsec tunnels

Output example tunnel detail statistics

The following output example shows detail tunnel statistics and *Show IPsec tunnel detail statistics output fields* describes the fields.

```
# show ipsec tunnel "LI"
=====
IPsec Tunnel Configuration Detail
=====
Service Id       : 100                Sap Id           : tunnel-1.private:1
Tunnel Name      : LI
Description      : None
Local Address    : ****
Remote Address   : ****
Delivery Service : 100
Security Policy  : None
Admin State      : Down              Oper State       : Down
Last Oper Change : 06/25/2019 01:17:14
Keying Type      : None              Replay Window    : None
IP MTU           : max               Clear DF Bit     : false
Encap IP MTU     : max
Pkt Too Big      : true
Pkt Too Big Count: 100              Pkt Too Big Intv : 10 secs
Oper Flags       : unresolvedLocalIp tunnelAdminDown sapDown
                  unresolvedPublicSvc
Host MDA         : 0/0
TCP MSS Adjust
  Public         : Disabled
  Private        : Disabled
Max Num of Keys
  Phase 1        : 0
  Phase 2        : 0

-----
Target Address Table
-----
Destination IP           IP Resolved Status
-----
No Entries Found
-----

BFD Interface
-----
```

```
BFD Designate      : no
=====
```

Table 176: Show IPsec tunnel detail statistics output fields

Label	Description
IPsec Tunnels Configuration Detail	
Service Id	The service ID of the IPsec tunnel
Sap Id	The SAP ID of the IPsec tunnel
Tunnel Name	The IPsec tunnel name
Description	The IPsec tunnel description
Local Address	The local IP address of the IPsec tunnel
Remote Address	The remote IP address of the IPsec tunnel
Delivery Service	The delivery service number of the IPsec tunnel
Security Policy	The security policy of the IPsec tunnel
Admin State	The administrative state of the IPsec tunnel
Oper State	The operational state of the IPsec tunnel
Last Oper Change	The date and time of the last operational change to the IPsec tunnel
Keying Type	The type of encryption keying: manual or dynamic
Replay Window	The replay window
Match TrustAnchor	The match trust anchor

Table 176: Show IPsec tunnel detail statistics output fields (continued)

Label	Description
TrustAnchor	The trust anchor
Cert File	The certificate file
Key File	The key file
Local Id Type	The local ID type
Clear DF Bit	Whether clearing of the Do-not-Fragment bit is enabled: true or false
IP MTU	The Maximum Transmission Unit (MTU) size of IP packets
Oper Flags	The operational flags for the IPsec tunnel
BFD Interface	
BFD Designate	Specifies whether this IPsec tunnel is the BFD designated tunnel: yes or no
Dynamic Keying Parameters	
Transform Id1...Id4	The IKE transformation IDs
Ike Policy Id	The IKE policy ID
Auto Establish	Specifies whether the dynamic keying is auto-established
Pre Shared Key	The pre-shared key for the IPsec tunnel
Certificate Status Verify	
Primary	The primary certificate status verify method: OCSP or CRL

Table 176: Show IPsec tunnel detail statistics output fields (continued)

Label	Description
Secondary	The secondary certificate status verify method: OCSP or CRL
Default Result	The default result of the certificate status verify: good or revoked
ISAKMP-SA	
State	The state of the ISAKMP-SA
Established	The date and time that the ISAKMP-SA was established
Lifetime	The ISAKMP-SA lifetime
Expires	The date and time that the ISAKMP-SA will expire
ISAKMP Statistics	
Tx Packets	The number of transmitted ISAKMP packets
Tx Errors	The number of transmission ISAKMP errors
Tx DPD	The number of transmitted ISAKMP DPDs
Tx DPD ACK	The number of transmitted ISAKMP DPD acknowledgments
DPD Timeouts	The number of ISAKMP DPD timeouts
Rx Packets	The number of received ISAKMP packets
Rx Errors	The number of received ISAKMP errors
Rx DPD	The number of received ISAKMP DPDs
Rx DPD ACK	The number of received ISAKMP DPD acknowledgments

Table 176: Show IPsec tunnel detail statistics output fields (continued)

Label	Description
Rx DPD Errors	The number of received ISAKMP DPD errors
IPsec-SA	
Type	The IPsec SA type
SPI	The security parameter index
Auth Algorithm	The authorization algorithm
Encr Algorithm	The encryption algorithm
Installed	The date and time that the IPsec-SA was installed
Lifetime	The IPsec SA lifetime
Aggregate Statistics	
Bytes Processed	The number of bytes processed
Packets Processed	The number of packets processed
Crypto Errors	The number of cryptographic errors
Replay Errors	The number of replay errors
SA Errors	The number of SA errors
Policy Errors	The number of policy errors

Output example tunnel count statistics

The following output example shows tunnel count statistics and *Show IPsec gateway tunnel count statistics output fields* describes the fields.

```
# show ipsec tunnel count
=====
IPsec Tunnel Count
=====
Total IPsec Tunnels                : 1
-----
IPsec Gateway Psk Tunnels          : 0
IPsec Gateway plainPskXAuth Tunnels : 0
IPsec Gateway Cert Tunnels         : 0
IPsec Gateway Psk-Radius Tunnels   : 0
IPsec Gateway Cert-Radius Tunnels  : 0
IPsec Gateway EAP Tunnels          : 0
IPsec Gateway Auto-EAP Radius Tunnels : 0
Total IPsec Gateway Tunnels        : 0
=====
```

Table 177: Show IPsec gateway tunnel count statistics output fields

Label	Description
IPsec Tunnel Count	
Total IPsec Tunnels	The total number of IPsec tunnels on the gateway
IPsec Gateway Psk Tunnels	The number of PSK tunnels on the gateway
IPsec Gateway plainPskXAuth Tunnels	The number of plain PSK authorization tunnels on the gateway
IPsec Gateway Cert Tunnels	The number of certificate tunnels on the gateway
IPsec Gateway Psk-Radius Tunnels	The number of PSK RADIUS tunnels on the gateway
IPsec Gateway Cert-Radius Tunnels	The number of certificate RADIUS tunnels on the gateway
IPsec Gateway EAP Tunnels	The number of EAP tunnels on the gateway

Table 177: Show IPsec gateway tunnel count statistics output fields (continued)

Label	Description
IPsec Gateway Auto-EAP Radius Tunnels	The number of auto-EAP RADIUS tunnels on the gateway
Total IPsec Gateway Tunnels	The total number of tunnels on the IPsec gateway

7.2.3 IP tunnel debug command descriptions

7.2.3.1 gateway

Syntax

```
gateway name name tunnel ip-address [:port] [nat-ip nat-ip[:port]] [detail] [no-dpd-debug]  
no gateway name name tunnel ip-address [:port]
```

Context

```
debug>ipsec
```

Description

This command enables debugging for specified IPsec tunnel terminated on specified ipsec-gw. Only one IPsec tunnel is allowed to enable debugging at a time.

Parameters

name

specifies the name of ipsec-gw

ip-address

the tunnel IP address of remote peer

port

the remote UDP port of IKE

nat-ip:port

specifies inside IP address and optionally port for NATed tunnel

detail

displays detailed debug information

no-dpd-debug

stops logging IKEv1 and IKEv2 DPD events for less noise during debug

7.2.3.2 tunnel

Syntax

```
tunnel ipsec-tunnel-name [detail] [no-dpd-debug]
```

```
no tunnel ipsec-tunnel-name
```

Context

```
debug>ipsec
```

Description

This command enables debugging for specified IPSec tunnel. Only one IPSec tunnel is allowed to enable debugging at a time.

Parameters***ipsec-tunnel-name***

specifies the name of ipsec-tunnel

detail

displays detailed debug information

no-dpd-debug

stops logging IKEv1 and IKEv2 DPD events for less noise during debug

7.2.3.3 certificate

Syntax

```
certificate filename
```

Context

```
debug>ipsec
```

Description

This command enables debug for certificate chain computation in cert-profile.

Parameters

filename

displays the filename of imported certificate

7.2.3.4 cmpv2

Syntax

cmpv2

Context

debug

Description

This command enables the context to perform CMPv2 operations.

7.2.3.5 ca-profile

Syntax

[no] ca-profile *profile-name*

Context

debug>cmpv2

Description

This command debugs output of the specified CA profile.

- Protection method of each message is logged.
- All HTTP messages are logged. Format allows offline analysis using Wireshark.
- In the event of failed transactions, saved certificates are not deleted from file system for further debug and analysis.
- The system allows CMPv2 debugging for multiple ca-profile at the same time.

Parameters

profile-name

specifies the CA profile name

7.2.3.6 ocs

Syntax

[no] ocs *ca-profile-name*

Context

debug

Description

This command enable debug output of OCS protocol for the specified CA

Default

no ocs

Parameters

ca-profile-name

specifies the name of an existing ca-profile

8 NAT command reference

This section describes the NAT-related commands.

8.1 NAT command hierarchies

- *NAT configuration command hierarchies*
- *NAT show command hierarchies*
- *NAT clear command hierarchy*
- *NAT tools command hierarchies*
- *NAT filter command hierarchy*

8.1.1 NAT configuration command hierarchies

- *ISA command hierarchy*
- *NAT service command hierarchy*
- *NAT syslog command hierarchy*
- *IPFlow information export protocol command hierarchy*
- *NAT VPRN command hierarchy*
- *NAT router command hierarchy*
- *NAT DNAT command hierarchy*
- *NAT admin command hierarchy*

8.1.1.1 ISA command hierarchy

```
configure
- isa
- nat-group nat-group-id [create]
- no nat-group
- acct-policy-id acct-policy-id
- no acct-policy-id
- active-mda-limit number
- no active-mda-limit
- description description-string
- no description
```

- [no] *dynamic-mda dynamic-mda-id*
- [no] *local-nat*
- [no] *mda mda-id*
- *radius-accounting-policy nat-accounting-policy*
- no *radius-accounting-policy*
- *session-limits*
 - *reserved num-sessions*
 - no *reserved*
 - *watermarks high percentage low percentage*
 - no *watermarks*
- [no] *shutdown*

8.1.1.2 NAT service command hierarchy

configure

- service
 - nat
 - *deterministic-script*
 - *location remote-url*
 - no *location*
 - *nat-policy nat-policy-name [create]*
 - no *nat-policy nat-policy-name*
 - *block-limit [1..40]*
 - no *block-limit*
 - *description description-string*
 - no *description*
 - *filtering filtering-mode*
 - no *filtering*
 - *ipfix-export-policy [32 chars max]*
 - no *ipfix-export-policy*
 - *pool nat-pool-name service-name service-name*
 - *pool nat-pool-name router router-instance*
 - no *pool*
 - *port-limits*
 - *forwarding limit*
 - no *forwarding*
 - *reserved num-ports*
 - no *reserved*
 - *watermarks high percentage-high low percentage-low*
 - no *watermarks*
 - [no] *priority-sessions*
 - [no] *fc fc-name*
 - *session-limits*
 - *max num-sessions*

- no *max*
- *reserved num-sessions*
- no *reserved*
- *watermarks high percentage-high low percentage-low*
- no *watermarks*
- *tcp-mss-adjust segment-size*
- no *tcp-mss-adjust*
- [no] *timeouts*
 - *icmp-query* [min minutes] [sec seconds]
 - no *icmp-query*
 - *sip* min minutes] [sec seconds]
 - no *sip*
 - *subscriber-retention* [hrs hours] [min minutes]
 - no *subscriber-retention*
 - *tcp-established* [hrs hours] [min minutes] [sec seconds]
 - no *tcp-established*
 - *tcp-syn* [hrs hours] [min minutes] [sec seconds]
 - no *tcp-syn*
 - *tcp-time-wait* [min minutes] [sec seconds]
 - no *tcp-time-wait*
 - *tcp-transitory* [hrs hours] [min minutes] [sec seconds]
 - no *tcp-transitory*
 - *udp* [hrs hours] [min minutes] [sec seconds]
 - no *udp*
 - *udp-dns* [hrs hours] [min minutes] [sec seconds]
 - no *udp-dns*
 - *udp-initial* [min minutes] [sec seconds]
 - no *udp-initial*
- [no] *udp-inbound-refresh*

8.1.1.3 NAT syslog command hierarchy

configure

- service
 - nat
 - *syslog*
 - *syslog-export-policy name* [create]
 - no *syslog-export-policy name*
 - *collector router router-name ip ip-address* [create]
 - *collector service-name service-name ip ip-address* [create]
 - no *collector router router-name ip ip-address*
 - no *collector service-name service-name ip ip-address*
 - *destination-port* [destination-port]
 - *ipv4-source-address ip-address*

- no *ipv4-source-address*
- [no] *shutdown*
- *description* *description-string*
- no *description*
- *facility* *syslog-facility*
- [no] *include*
 - [no] *destination-ip*
 - [no] *foreign-ip*
 - [no] *foreign-port*
 - [no] *nat-policy-name*
 - [no] *sub-id*
- *log-prefix* *prefix-text*
- *max-tx-delay* *deciseconds*
- no *max-tx-delay*
- *mtu* *mtu-size*
- *rate-limit* *packets-per-second*
- no *rate-limit*
- *severity-level* *syslog-level*

8.1.1.4 IPFlow information export protocol command hierarchy

configure

- service
 - *ipfix*
 - *ipfix-export-policy* *policy-name* [create]
 - no *ipfix-export-policy* *policy-name*
 - *collector* *router* *router-instance* *ip* *ip-address* [create]
 - no *collector* *router* *router-instance* *ip* *ip-address*
 - *mtu* [512..9212]
 - [no] *shutdown*
 - *source-address* *ip-address*
 - no *source-address*
 - *template-refresh-timeout* [hrs *hours*] [min *minutes*] [sec *seconds*]
 - no *template-refresh-timeout*
 - *description* *description-string*
 - no *description*

8.1.1.5 NAT VPRN command hierarchy

configure

- service
 - *vprn* *service-id* *customer* *cust-id* create

```

- nat
- inside
  - [no] destination-prefix ip-prefix/length
  - deterministic
    - classic-lsn-max-subscriber-limit max
    - no classic-lsn-max-subscriber-limit
    - prefix ip-prefix/length subscriber-type nat-sub-type nat-policy nat-policy-name [create]
    - prefix ip-prefix/length subscriber-type nat-sub-type
    - no prefix ip-prefix/length subscriber-type nat-sub-type
    - map start inside-ip-address end inside-ip-address to outside-ip-address
    - no map start inside-ip-address end inside-ip-address
    - [no] shutdown
  - nat-policy nat-policy-name
  - no nat-policy
  - |[no] nat64
    - destination-prefix ipv6-prefix/prefix-length [nat-policy policy-name]
    - no destination-prefix ipv6-prefix/prefix-length
    - [no] drop-zero-ipv4-checksum
    - [no] ignore-tos
    - [no] insert-ipv6-fragment-header
    - ip-fragmentation {disabled | fragment-ipv6 | fragment-ipv6-unless-ipv4-
df-set}
    - no ip-fragmentation
    - ipv6-mtu [1280..9212]
    - no ipv6-mtu
    - prefix ipv6-prefix/prefix-length
    - no prefix
    - set-tos [0..255]
    - no set-tos
    - [no] shutdown
    - subscriber-prefix-length prefix-length
    - no subscriber-prefix-length
  - redundancy
    - peer ip-address
    - no peer
    - peer6 ipv6-address
    - no peer6
    - steering-route ip-prefix/length
    - no steering-route
- outside
  - mtu [512..9000]
  - no mtu
  - pool nat-pool-name [nat-group nat-group-id type pool-type create]
  - no pool nat-pool-name

```

- *address-range start-ip-address end-ip-address [create]*
- *no address-range start-ip-address end-ip-address*
 - *description description-string*
 - *no description*
 - *[no] drain*
- *description description-string*
- *no description*
- *deterministic*
 - *port-reservation num-ports*
 - *no port-reservation*
- *mode {auto | napt | one-to-one}*
- *no mode*
- *[no] port-forwarding-dyn-block-reservation*
- *port-forwarding-range range-end*
- *no port-forwarding-range*
- *port-reservation blocks num-blocks*
- *port-reservation ports num-ports*
- *no port-reservation*
- *redundancy*
 - *export ip-prefix/length*
 - *no export*
 - *follow router router-instance pool name*
 - *no follow*
 - *monitor ip-prefix/length*
 - *no monitor*
- *[no] shutdown*
- *subscriber-limit [1..65535]*
- *no subscriber-limit*
- *watermarks high percentage-high low percentage-low*
- *no watermarks*
- *upstream-ip-filter filter-id*
- *no upstream-ip-filter*

8.1.1.6 NAT router command hierarchy

```

configure
– router
  – nat
    – inside
      – [no] destination-prefix ip-prefix/length
      – deterministic
        – classic-lsn-max-subscriber-limit max
        – no classic-lsn-max-subscriber-limit

```

```

    - prefix ip-prefix/length subscriber-type nat-sub-type nat-policy nat-policy-
name [create]
    - prefix p-prefix/length subscriber-type nat-sub-type
    - no prefix ip-prefix/length subscriber-type nat-sub-type
- nat-policy nat-policy-name
- no nat-policy
- |[no] nat64
    - destination-prefix ipv6-prefix/prefix-length [nat-policy policy-name]
    - no destination-prefix ipv6-prefix/prefix-length
    - [no] drop-zero-ipv4-checksum
    - [no] ignore-tos
    - [no] insert-ipv6-fragment-header
    - ip-fragmentation {disabled | fragment-ipv6 | fragment-ipv6-unless-ipv4-df-
set}
    - no ip-fragmentation
    - ipv6-mtu [1280..9212]
    - no ipv6-mtu
    - prefix ipv6-prefix/prefix-length
    - no prefix
    - set-tos [0..255]
    - no set-tos
    - [no] shutdown
    - subscriber-prefix-length prefix-length
    - no subscriber-prefix-length
- redundancy
    - peer ip-address
    - no peer
    - steering-route ip-prefix/length
    - no steering-route
- subscriber-identification
    - attribute [vendor vendor-id] attribute-type attribute-type
    - no attribute
    - description description-string
    - no description
    - [no] drop-unidentified-traffic
    - radius-proxy-server router router-instance name server-name
    - no radius-proxy-server
    - [no] shutdown
- outside
    - mtu [512..9000]
    - no mtu
    - pool nat-pool-name [nat-group nat-group-id type pool-type create]
    - no pool nat-pool-name
    - address-range start-ip-address end-ip-address [create]
    - no address-range start-ip-address end-ip-address

```

- *description* *description-string*
- *no description*
- *[no] drain*
- *description* *description-string*
- *no description*
- *mode* {*auto* | *napt* | *one-to-one*}
- *no mode*
- *[no] port-forwarding-dyn-block-reservation*
- *port-forwarding-range* *range-end*
- *no port-forwarding-range*
- *port-reservation* *blocks num-blocks*
- *port-reservation* *ports num-ports*
- *no port-reservation*
- *redundancy*
 - *export ip-prefix/length*
 - *no export*
 - *follow* *router router-instance pool name*
 - *no follow*
 - *monitor ip-prefix/length*
 - *no monitor*
- *[no] shutdown*
- *subscriber-limit* [*1..65535*]
- *no subscriber-limit*
- *watermarks* *high percentage-high low percentage-low*
- *no watermarks*

8.1.1.7 NAT DNAT command hierarchy

```
configure
– service
– nat
– nat-classifier classifier-name [create]
– no nat-classifier classifier-name
– default-action {dnat | forward} [ip-address ip-address]
– default-dnat-ip-address ip-address
– no default-dnat-ip-address
– description description-string
– no description
– entry entry-id [create]
– no entry entry-id
– action {dnat | forward} [ip-address ip-address]
– no action
– description description-string
– no description
```

- *match* protocol *ip-protocol*
- no *match*
 - *dst-port-range* start *port-number* end *port-number*
 - no *dst-port-range*
- *nat-policy* *nat-policy-name* [create]
- no *nat-policy* *nat-policy-name*
 - [no] *dnat*
 - *dnat-only* router *router-instance* nat-group *nat-group-id*
 - no *dnat-only*
 - *nat-classifier* *classifier-name*
 - no *nat-classifier*
- *nat-prefix-list* name [create] [application *application-choice*]
- no *nat-prefix-list* name
 - *prefix* *ip-prefix/length* [nat-policy *nat-policy-name*]
 - no *prefix* *ip-prefix/length*configure
- router
 - nat
 - *inside*
 - *classic-lsn-max-subscriber-limit* max
 - no *classic-lsn-max-subscriber-limit*
 - *dnat-only*
 - *source-prefix-list* *prefix-list-name*
 - no *source-prefix-list*

8.1.1.8 NAT admin command hierarchy

- ```
admin
 – nat
 – save-deterministic-script
```

## 8.1.2 NAT show command hierarchies

- ```
show
  – isa
    – nat-group
    – nat-group nat-group-id [associations]
    – nat-group nat-group-id member [1..255] [statistics]
    – nat-group [nat-group-id] members
  – service
    – nat
      – nat-policy nat-policy-name associations
      – nat-policy nat-policy-name statistics
```

- *nat-policy nat-policy-name*
- *nat-policy*
- *port-forwarding-entries*

show

- router
 - *nat*
 - *lsn-blocks* [inside-router *router-instance*] [inside-ip *ip-address*] [outside-ip-prefix *ip-prefix/length*] [outside-port [1..65535]] [pool *pool-name*]
 - *pool pool-name*
 - *pool*
 - *summary*

8.1.3 NAT clear command hierarchy

clear

- nat
 - isa
 - *nat-group nat-group-id member* [1..255] l2-aware-subscribers
 - *nat-group nat-group-id member* [1..255] statistics

8.1.4 NAT tools command hierarchies

tools

- dump
 - *nat*
 - *histogram* router *router-instance* pool *pool-name* bucket-size [1..65536] num-buckets [2..50]
 - *isa*
 - *resources* mda *mda-id*
 - *sessions* [nat-group *nat-group-id*] [mda *mda-id*] [protocol {icmp | tcp | udp}] [inside-ip *ip-address*] [inside-router *router-instance*] [inside-port *port-number*] [outside-ip *ip-address*] [outside-port *port-number*] [foreign-ip *ip-address*] [foreign-port *port-number*]
 - perform
 - *nat*
 - *port-forwarding-action*
 - *lsn* create router *router-instance* [b4 *ipv6-address*] [aftr *ipv6-address*] ip *ip-address* protocol {tcp | udp} [port *port*] lifetime *lifetime* [outside-ip *ipv4-address*] [outside-port *port*]
 - *lsn* delete router *router-instance* [b4 *ipv6-address*] ip *ip-address* protocol {tcp | udp} port *port*


```

    - lsn modify router router-instance [b4 ipv6-address] ip ip-address protocol
    {tcp | udp} port port lifetime lifetime

```

8.1.5 NAT filter command hierarchy



NOTE:

For filter command descriptions, refer to the *7450 ESS, 7750 SR, 7950 XRS, and VSR Router Configuration Guide*.

```

configure
- filter
- ip-filter filter-id create
- no ip-filter filter-id
- ipv6-filter filter-id create
- no ipv6-filter filter-id
- entry entry-id create
- no entry entry-id
- match [protocol protocol-id]
- no match
- dst-ip {ip-address/mask | ip-address netmask | ip-prefix-list prefix-list-
name}
- no dst-ip
- dst-port {lt | gt | eq} dst-port-number
- dst-port port-list port-list-name
- dst-port range dst-port-number dst-port-number
- no dst-port
- port {lt | gt | eq} port-number
- port port-list port-list-name
- port range port-number port-number
- no port
- src-ip {ip-address/mask | ip-address netmask | ip-prefix-list prefix-list-
name}
- no src-ip
- src-port {lt | gt | eq} src-port-number
- src-port port-list port-list-name
- src-port range src-port-number src-port-number
- no src-port

```

8.2 NAT command descriptions

- *NAT configuration command descriptions*
- *NAT show command descriptions*
- *NAT clear command descriptions*
- *NAT tools command descriptions*

8.2.1 NAT configuration command descriptions

- *Generic NAT command descriptions*
- *ISA NAT command descriptions*
- *Service NAT command descriptions*
- *Syslog NAT command descriptions*
- *IPFlow information export protocol NAT command descriptions*
- *VPRN and router NAT command descriptions*
- *DNAT NAT command descriptions*
- *Admin NAT command descriptions*

8.2.1.1 Generic NAT command descriptions

8.2.1.1.1 description

Syntax

`description description-string`

`no description`

Context

`config>service>nat>syslog>syslog-export-policy`

`config>service>vprn>nat>outside>pool>address-range`

`config>service>vprn>nat>outside>pool`

`config>router>nat>outside>pool>address-range`

`config>router>nat>outside>pool`

`config>router>nat>inside>subscriber-id`

```
config>service>ipfix>export-policy
```

Description

This command creates a text description which is stored in the configuration file to help identify the content of the entity.

The no form of this command removes the string from the configuration.

Parameters

description-string

the description character string

Allowed values are any string composed of printable, 7-bit ASCII characters. If the string contains special characters (#, \$, spaces, and so on), the entire string must be enclosed within double quotes.

8.2.1.1.2 shutdown

Syntax

```
[no] shutdown
```

Context

```
config>service>ipfix>ipfix-export-policy>collector
```

```
config>service>nat>syslog>syslog-export-policy>collector context
```

```
config>service>vprn>nat>outside>pool
```

```
config>router>nat>outside>pool
```

```
config>router>nat>inside>nat64
```

```
config>router>nat>inside>redundancy>subscriber-identification
```

```
config>service>vprn>nat>inside>nat64
```

```
config>router>nat>inside>subscriber-id
```

Description

This command administratively disables the entity. When disabled, an entity does not change, reset, or remove any configuration settings or statistics. Many entities must be explicitly enabled using the no shutdown command.

The shutdown command administratively disables an entity. The operational state of the entity is disabled as well as the operational state of any entities contained within. Many objects must be shut down before they may be deleted.

8.2.1.2 ISA NAT command descriptions

8.2.1.2.1 nat-group

Syntax

```
nat-group nat-group-id [create]
no nat-group nat-group-id
```

Context

```
config>isa
```

Description

This command configures an ISA NAT group.

Parameters

nat-group-id

an integer value that uniquely identifies the ISA NAT group

Values: 1 to 15

create

mandatory keyword used when creating a NAT group in the ISA context

The create keyword requirement can be enabled/disabled in the environment>create context.

8.2.1.2.2 acct-policy-id

Syntax

```
acct-policy-id acct-policy-id
no acct-policy-id
```

Context

```
config>isa>nat-group
```

Description

This command applies the accounting policy to each MDA in an ISA group to generate NAT binding records in CSV format.

Configure the accounting policy in the `config>log>accounting-policy>record` context with the record type `nat-binding-port-block` to enable the NAT port-block level binding assignment.

Default

no acct-policy-id

Parameters

acct-policy-id

specifies the accounting policy for generating NAT binding records in CSV format

Values: 1 to 99

8.2.1.2.3 active-mda-limit

Syntax

`active-mda-limit` *number*

`no active-mda-limit`

Context

`config>isa>nat-group`

Description

This command configures the number of MDAs in the NAT ISA group that are intended for active use.

Parameters

number

specifies the active MDA limit

8.2.1.2.4 dynamic-mda

Syntax

`[no] dynamic-mda` *dynamic-mda-id*

Context

config>isa>nat-group

Description

This command configures a dynamic MDA that is associated with a NAT group. A dynamic MDA is configured without any card or a VM associated with it. The dynamic MDA is mapped to a physical MDA based on a VM related to the MG group. For the N:K redundancy method, this allows the physical MDA to move to a VM that becomes active after failure.



NOTE:

- Dynamic MDA and MDA are mutually exclusive.
- A mobile-gateway system group that is used in an N:K redundant system requires a dynamic MDA in its NAT groups and a dynamic SAP on its VPRNs. For more information, see the config>mobile>system>group, config>isa>nat-group, and the config>service>vprn>if>dynamic-sap commands.

Parameters

dynamic-mda-id

specifies the MDA ID that is associated with the MG group

Values	mg-group-<group-id>/<mda>		
–	group-id	1 to 15	
–	mda	1 to 4	

8.2.1.2.5 local-nat

Syntax

[no] local-nat

Context

config>isa>nat-group

Description

This command enables the NAT function to be located on the MG-VM for PGW/GGSN deployment. This command is only supported in CMG.

Default

no local-nat

8.2.1.2.6 mda

Syntax

[no] mda *mda-id*

Context

config>isa>nat-group

Description

This command configures an ISA NAT group MDA.

Parameters

mda-id
specifies the MDA ID in the *slot/mda* format

Values	slot	1 to 20
–	mda	1 to 4

8.2.1.2.7 radius-accounting-policy

Syntax

radius-accounting-policy *nat-accounting-policy*
no radius-accounting-policy

Context

```
config>isa>nat-group
```

Description

This command specifies the RADIUS accounting policy to use for each MDA in this ISA group.

The no form of this command removes the policy ID from the configuration.

Parameters

nat-accounting-policy

reference to the nat-accounting-policy which defines:

- the source IP addresses that will be assigned to BB-ISA cards
- the parameters related to RADIUS server itself
- the list of RADIUS attributes that will be included in accounting messages

8.2.1.2.8 session-limits

Syntax

```
session-limits
```

Context

```
config>isa>nat-group
```

```
config>service>nat>nat-policy
```

Description

This command configures the ISA NAT group or service NAT policy session limits.

8.2.1.2.9 reserved

Syntax

```
reserved num-sessions
```

```
no reserved
```

Context

```
config>isa>nat-group>session-limits
```



```
config>service>nat>nat-policy>session-limits
```

Description

This command configures the number of sessions per block that will be reserved for prioritized sessions.

Parameters

num-sessions

specifies the number of sessions reserved for prioritized sessions

Values	0 to 6291456 (ISA NAT group)
–	1 to 65534 (service NAT policy)

8.2.1.2.10 watermarks

Syntax

```
watermarks high percentage low percentage  
no watermarks
```

Context

```
config>isa>nat-group>session-limits
```

```
config>service>nat>nat-policy>session-limits
```

Description

This command configures the ISA NAT group or service NAT policy watermarks.

Parameters

high percentage

specifies the high watermark of the number of sessions for each MDA

Values: 1 to 100

low percentage

specifies the low watermark of the number of sessions for each MDA

Values: 0 to 99

8.2.1.3 Service NAT command descriptions

8.2.1.3.1 deterministic-script

Syntax

```
deterministic-script
```

Context

```
config>service>nat
```

Description

This command configures the script generated for deterministic NAT.

8.2.1.3.2 location

Syntax

```
location remote-url  
no location
```

Context

```
config>service>nat>deterministic-script
```

Description

This command configures the remote location where the Python script will be exported. The Python script is then used offline to perform a reverse query. If this command is configured, the Python script generation is triggered by any modification of the deterministic NAT configuration. The new script reflects the change in mappings caused by configuration change. However, the script must be manually exported to the outside location with the `admin na save-deterministic-nat` command. The script cannot be stored locally on the system.

The script allows two forms of queries:

Forward

input is NAT inside parameters, output is NAT outside parameters.

Backward

input is NAT outside parameters, output is NAT inside parameters.

Forward query:

```
user@external-server:/home/ftp/pub/det-nat-script$ ./det-nat.py -f -s 10 -  
a 20.0.5.10
```

Output:

```
subscriber has public ip address 85.0.0.1 from service 0 and is using ports [1324 -  
1353]
```

Reverse query:

```
user@external-server:/home/ftp/pub/det-nat-script$ ./det-nat.py -b -s 0 -a 85.0.0.1 -  
p 3020
```

Output:

```
subscriber has private ip address 20.0.5.66 from service 10
```

Parameters***remote-url***

a remote location where the script is stored:

[{ftp://|tftp://}login:pswd@remote-locn] [file-path], up to 180 characters

8.2.1.3.3 nat-policy**Syntax**

nat-policy *nat-policy-name* [create]

no nat-policy *nat-policy-name*

Context

config>service>nat

Description

This command configures a NAT policy.

Parameters***nat-policy-name***

specifies the NAT policy name

Values: up to 32 characters

8.2.1.3.4 block-limit

Syntax

```
block-limit [1..40]
```

```
no block-limit
```

Context

```
config>service>nat>nat-policy
```

Description

This command configures the maximum number of port blocks per subscriber.

Parameters

[1..40]

specifies the block limit

Values: 1 to 40

8.2.1.3.5 filtering

Syntax

```
filtering filtering-mode
```

```
no filtering
```

Context

```
config>service>nat>nat-policy
```

Description

This command configures the filtering of the NAT policy.

Parameters

filtering-mode

specifies the way that inbound traffic is filtered

Values: address-and-port-dependent | endpoint-independent

8.2.1.3.6 ipfix-export-policy

Syntax

```
ipfix-export-policy [32 chars max]
```

```
no ipfix-export-policy
```

Context

```
config>service>nat>nat-policy
```

Description

This command configures the IP flow information export protocol.

The no form of this command removes the IP flow information export protocol.

8.2.1.3.7 pool

Syntax

```
pool nat-pool-name service-name service-name
```

```
pool nat-pool-name router router-instance
```

```
no pool
```

Context

```
config>service>nat>nat-policy
```

Description

This command configures the NAT pool of this policy.

Parameters

nat-pool-name

specifies the name of the NAT pool, up to 32 characters

router-instance

specifies the router instance the pool belongs to, either by router name or service ID

Values: router-name: “Base” | “management”

Values	router-name vprn-svc-id		
–	service-id	1 to 2147483648	
–	svc-name	specifies a string, up to 64 characters	

Default: Base

service-name
specifies the name of the service, up to 64 characters

8.2.1.3.8 port-limits

Syntax
port-limits

Context
config>service>nat>nat-policy

Description
This command configures the port limits of this policy.

8.2.1.3.9 forwarding

Syntax
forwarding *limit*
no forwarding

Context
config>service>nat>nat-policy>port-limits

Description

This command configures the maximum number of port forwarding entries.

Parameters

limit

specifies the maximum number of port forwarding entries per subscriber

Default: 0

8.2.1.3.10 reserved

Syntax

reserved *num-ports*

no reserved

Context

config>service>nat>nat-policy>port-limits

Description

This command configures the number of ports per block that will be reserved for prioritized sessions.

Parameters

num-ports

specifies the number of ports to reserve for prioritized sessions

Values: 1 to 65534

8.2.1.3.11 watermarks

Syntax

watermarks high *percentage-high* low *percentage-low*

no watermarks

Context

config>service>nat>nat-policy>port-limits

Description

This command configures the port usage watermarks for the NAT policy.

Parameters

percentage-high

specifies the high percentage

Values: 1 to 100

percentage-low

specifies the low percentage

Values: 0 to 99

8.2.1.3.12 priority-sessions

Syntax

[no] priority-sessions

Context

config>service>nat>nat-policy

Description

This command configures the prioritized sessions of this NAT policy.

8.2.1.3.13 fc

Syntax

[no] fc *fc-name*

Context

config>service>nat>nat-policy>priority-sessions

Description

This command configures the forwarding classes that have their sessions prioritized.

Parameters

fc-name

specifies the forwarding class

Values: be | l2 | af | l1 | h2 | ef | h1 | nc

8.2.1.3.14 max

Syntax

max *num-sessions*

no max

Context

config>service>nat>nat-policy>session-limits

Description

This command configures the session limit of this policy. The session limit is the maximum number of sessions allowed for a subscriber associated with this policy

Parameters

num-sessions

specifies the session limit

Values: 1 to 65535

8.2.1.3.15 tcp-mss-adjust

Syntax

tcp-mss-adjust *segment-size*

no tcp-mss-adjust

Context

config>service>nat>nat-policy

Description

This command configures the value to adjust the TCP MSS option. This triggers the interception of transit TCP packets with the SYN bit set and consequently rewrites the MSS field of the intercepted packet. The

MSS field will only be rewritten if the configured TCP MSS adjust value is smaller than the value inside the intercepted packet. If the given TCP option field is not present in the TCP SYN packet, it will be inserted with the preconfigured value.

The no form of this command returns the segment size to the default.

Parameters

segment-size

specifies the value to put into the TCP Maximum Segment Size (MSS) option if not already present, or if the present value is higher

Values: 0 | 160 to 10240

Default: 0

8.2.1.3.16 timeouts

Syntax

[no] timeouts

Context

config>service>nat>nat-policy

Description

This command configures session idle timeouts for this policy.

8.2.1.3.17 icmp-query

Syntax

icmp-query [min *minutes*] [sec *seconds*]

no icmp-query

Context

config>service>nat>nat-policy>timeouts

Description

This command configures the timeout applied to an ICMP query session.

Parameters

minutes

specifies the timeout, in minutes, applied to an ICMP query session

Values: 1 to 4

Default: 1

seconds

specifies the timeout, in seconds, applied to an ICMP query session

Values: 1 to 59

8.2.1.3.18 sip

Syntax

sip [min *minutes*] [sec *seconds*]

no sip

Context

config>service>nat>nat-policy>timeouts

Description

This command configures the SIP inactive media timeout.

Parameters

minutes

specifies the SIP inactive media timeout, in minutes

Values: 1 to 4

Default: 1

seconds

specifies the SIP inactive media timeout, in seconds

Values: 1 to 59

8.2.1.3.19 subscriber-retention

Syntax

subscriber-retention [hrs *hours*] [min *minutes*]

```
no subscriber-retention
```

Context

```
config>service>nat>nat-policy>timeouts
```

Description

This command specifies the subscriber retention timeout, the time a NAT subscriber and its associated IP address is kept after all hosts and associated port blocks have expired.

If a NAT subscriber host appears before the retention timeout has elapsed, it will be given the same outside IP address.

Parameters

hours

configures the hours a subscriber's IP address is kept after all hosts and port blocks have expired

Values: 1 to 24

minutes

configures the minutes a subscriber's IP address is kept after all hosts and port blocks have expired

Values: 1 to 59

8.2.1.3.20 tcp-established

Syntax

```
tcp-established [hrs hours] [min minutes] [sec seconds]
```

```
no tcp-established
```

Context

```
config>service>nat>nat-policy>timeouts
```

Description

This command configures the idle timeout applied to a TCP session in the established state.

Parameters

hours

specifies the timeout hours field

Values: 1 to 24

minutes

specifies the timeout minutes field

Values: 1 to 59

seconds

specifies the timeout seconds field

Values: 1 to 59

8.2.1.3.21 tcp-syn

Syntax

tcp-syn [hrs *hours*] [min *minutes*] [sec *seconds*]

no tcp-syn

Context

config>service>nat>nat-policy>timeouts

Description

This command configures the timeout applied to a TCP session in the SYN state.

Parameters

hours

specifies the timeout hours field

Values: 1 to 24

minutes

specifies the timeout minutes field

Values: 1 to 59

seconds

specifies the timeout seconds field

Values: 1 to 59

8.2.1.3.22 tcp-time-wait

Syntax

```
tcp-time-wait [min minutes] [sec seconds]  
no tcp-time-wait
```

Context

```
config>service>nat>nat-policy>timeouts
```

Description

This command configures the timeout applied to a TCP session in a time wait state.

Parameters

minutes

specifies the timeout minutes field
Values: 1 to 4

seconds

specifies the timeout seconds field
Values: 1 to 59

8.2.1.3.23 tcp-transitory

Syntax

```
tcp-transitory [hrs hours] [min minutes] [sec seconds]  
no tcp-transitory
```

Context

```
config>service>nat>nat-policy>timeouts
```

Description

This command configures the idle timeout applied to a TCP session in a transitory state.

Parameters

hours

specifies the timeout hours field

Values: 1 to 24

minutes

specifies the timeout minutes field

Values: 1 to 59

seconds

specifies the timeout seconds field

Values: 1 to 59

8.2.1.3.24 **udp**

Syntax

udp [hrs *hours*] [min *minutes*] [sec *seconds*]

no udp

Context

config>service>nat>nat-policy>timeouts

Description

This command configures the UDP mapping timeout.

Parameters

hours

specifies the timeout hours field

Values: 1 to 24

minutes

specifies the timeout minutes field

Values: 1 to 59

seconds

specifies the timeout seconds field

Values: 1 to 59

8.2.1.3.25 udp-dns

Syntax

```
udp-dns [hrs hours] [min minutes] [sec seconds]  
no udp-dns
```

Context

```
config>service>nat>nat-policy>timeouts
```

Description

This command configures the timeout applied to a UDP session with destination port 53.

Parameters

hours

specifies the timeout hours field
Values: 1 to 24

minutes

specifies the timeout minutes field
Values: 1 to 59

seconds

specifies the timeout seconds field
Values: 1 to 59

8.2.1.3.26 udp-initial

Syntax

```
udp-initial [min minutes] [sec seconds]  
no udp-initial
```

Context

```
config>service>nat>nat-policy>timeouts
```

Description

This command configures the UDP mapping timeout applied to new sessions.

Parameters

minutes

specifies the timeout minutes field

Values: 1 to 4

seconds

specifies the timeout seconds field

Values: 1 to 59

8.2.1.3.27 udp-inbound-refresh

Syntax

[no] udp-inbound-refresh

Context

config>service>nat>nat-policy

Description

This command specifies the NAT inbound refresh behavior.

Default

no udp-inbound-refresh

8.2.1.4 Syslog NAT command descriptions

8.2.1.4.1 syslog

Syntax

syslog

Context

config>service>nat

Description

This command enables the context to configure syslog reporting of NAT flow parameters.

8.2.1.4.2 syslog-export-policy

Syntax

```
syslog-export-policy name [create]  
no syslog-export-policy name
```

Context

```
config>service>nat>syslog
```

Description

This command creates a syslog export policy with a set of transport parameters that are used to transmit NAT flow records in syslog format to an external collector node. This policy name is then referenced from the NAT policy applied to an inside routing context.

The no form of this command removes the policy name from the configuration.

Parameters

name

specifies the syslog export policy name, up to 32 characters

create

keyword used to create the syslog export policy

8.2.1.4.3 collector

Syntax

```
collector router router-name ip ip-address [create]  
collector service-name service-name ip ip-address [create]  
no collector router router-name ip ip-address  
no collector service-name service-name ip ip-address
```

Context

```
config>service>nat>syslog>syslog-export-policy
```

Description

This command defines an external collector node that collects syslog records. The syslog records are streamed to the collector node using UDP transport. Traffic is originated from a random ephemeral UDP port to the destination port 514. Up to two collector nodes can be defined for redundancy purposes.

Stateless UDP streams are used as transport due to the significant volume of transactions. However, they do contain 32-bit sequence numbers so packet loss can be identified. The sequence numbers are generated per BB-ISA per collector, and within each stream they are monotonically increased by 1. Overlapping sequence numbers between BB-ISAs can be differentiated by the MDA ID field carried in the syslog message.

Multiple syslog records are sent in a single UDP packet. UDP packet transmission is triggered when the packet size containing syslog records exceeds the configured MTU value or the configurable timer, whichever occurs first.

The no form of this command removes the parameters from the configuration.

Parameters

router-name

specifies the router instances from which the collector node is reachable

ip-address

specifies the IPv4 address of the external collector node to which the syslog records are sent

service-name

specifies the service name from which the collector node is reachable

create

keyword used to create the collector instance

8.2.1.4.4 destination-port

Syntax

```
destination-port [destination-port]
```

Context

```
config>service>nat>syslog>syslog-export-policy>collector
```

Description

This command configures the destination port (collector port) to which UDP stream containing the syslog flow records are sent.

Default

destination-port 514

Parameters

destination-port

specifies the destination port to which UDP streams are sent

Values: 1 to 65535

8.2.1.4.5 ipv4-source-address

Syntax

ipv4-source-address *ip-address*

no ipv4-source-address

Context

config>service>nat>syslog>syslog-export-policy>collector

Description

This command configures the IPv4 source address from which the UDP streams containing syslog flow records are sourced.

The no form of this command removes the IPv4 address from the configuration.

Parameters

ip-address

specifies the source IPv4 address from which UDP streams are sent

Values: a.b.c.d

8.2.1.4.6 facility

Syntax

facility *syslog-facility*

Context

config>service>nat>syslog>syslog-export-policy

Description

This command configures a syslog facility. For more information, refer to the *7450 ESS, 7750 SR, 7950 XRS, and VSR System Management Guide*. The `config>log>syslog>level` hierarchy also applies to this context.

Default

facility local0

Parameters

syslog-facility

specifies a syslog facility name which represents a specific numeric facility code. The code must be entered in accordance with the syslog RFC. However, the software does not validate if the facility code configured is appropriate for the event type being sent to the syslog target host.

Values: kernel | user | mail | systemd | auth | syslogd | printer | netnews | uucp | cron | authpriv | ftp | ntp | logaudit | logalert | cron2 | local0 | local1 | local2 | local3 | local4 | local5 | local6 | local7

8.2.1.4.7 include

Syntax

[no] include

Context

config>service>nat>syslog>syslog-export-policy

Description

This command enables the context to specify the optional fields to include in the NAT syslog messages.

8.2.1.4.8 destination-ip

Syntax

[no] destination-ip

Context

config>service>nat>syslog>syslog-export-policy>include

Description

This command includes the destination IP address in the flow log. The `destination-ip` is significant in Destination Based NAT (DNAT) where the foreign IP address is translated. A foreign IP address is the original IP address toward the destination node and in DNAT it is replaced by the `destination-ip`. More clearly, on the inside (private side), the IP address of the destination node is referred to as foreign IP (original destination IP), and once this address is translated by DNAT, it is referred to as destination IP (translated destination IP) on the outside (public side).

The `no` form of this command disables the feature.

8.2.1.4.9 `foreign-ip`

Syntax

```
[no] foreign-ip
```

Context

```
config>service>nat>syslog>syslog-export-policy>include
```

Description

This command includes the foreign IP address in the flow log. A foreign IP address is the original IP address toward the destination node and in DNAT it is replaced by the destination IP.

If DNAT is not used, the foreign IP address (the IP address of the destination node) is the same on both sides of NAT.

The `no` form of this command disables the feature.

8.2.1.4.10 `foreign-port`

Syntax

```
[no] foreign-port
```

Context

```
config>service>nat>syslog>syslog-export-policy>include
```

Description

This command includes the foreign port address in the flow log. A foreign port is the port toward the destination node and this port is not translated by any form of NAT.

A foreign port (the port toward the destination node) is the same on both sides of NAT.

The no form of this command disables the feature.

8.2.1.4.11 nat-policy-name

Syntax

[no] nat-policy-name

Context

config>service>nat>syslog>syslog-export-policy>include

Description

This command includes the NAT policy name in the flow log.

The no form of this command disables the feature.

8.2.1.4.12 sub-id

Syntax

[no] sub-id

Context

config>service>nat>syslog>syslog-export-policy>include

Description

This command includes the sub-id string in the flow log. The sub-id is applicable only in subscriber-aware NAT. If subscriber-aware NAT is not enabled, the sub-id string is set to '-'.

Mobile subscriber information (IMSI, IMEI, MSDISN, and APN) is embedded in sub-id string.

The no form of this command disables the feature.

8.2.1.4.13 log-prefix

Syntax

log-prefix *prefix-text*

Context

config>service>nat>syslog>syslog-export-policy

Description

This command configures the syslog log prefix. For more information, refer to the *7450 ESS, 7750 SR, 7950 XRS, and VSR System Management Guide*. The config>log>syslog>level hierarchy also applies to this context.

Default

log-prefix "TMNX"

Parameters

prefix-text

specifies an alphanumeric string, up to 32 characters (spaces and colons cannot be used in the string)

8.2.1.4.14 max-tx-delay

Syntax

max-tx-delay *deciseconds*

no max-tx-delay

Context

config>service>nat>syslog>syslog-export-policy

Description

This command enables aggregation of flow log messages within a syslog frame. It introduces a delay during which logs are collected in each BB-ISA so they can be sent in a single syslog message to conserve system resources and network bandwidth.

When aggregation is enabled, generation of a syslog frame carrying multiple flow logs is triggered by one of the two events (whichever occurs first):

- expiry of the max-tx-delay timer
- exceeding MTU size

The no form of this command reverts to the default.

Default

max-tx-delay 3

Parameters

deciseconds

specifies the maximum time a syslog message is delayed in the system's output buffer

Values: 1 to 100

8.2.1.4.15 mtu

Syntax

mtu *mtu-size*

Context

config>service>nat>syslog>syslog-export-policy

Description

The command defines the MTU, the maximum size of the IP frame that can be transmitted. The size of the frame includes the syslog message and the IP header. This is an IP-MTU value.

When aggregation is enabled (the max-tx-delay command), generation of a syslog frame carrying multiple flow logs is triggered by one of the two events (whichever occurs first):

- expiry of the max-tx-delay timer
- exceeding MTU size

Default

mtu 1500

Parameters

mtu-size

specifies the MTU size in bytes

Values: 512 to 9000

8.2.1.4.16 rate-limit

Syntax

```
rate-limit packets-per-second  
no rate-limit
```

Context

```
config>service>nat>syslog>syslog-export-policy
```

Description

This command configures the maximum rate limit at which syslog messages are sent. Once the rate limit is exceeded, NAT flow logs will be buffered. Overload condition is characterized by exhaustion of this buffer space. This condition can occur due to imposed rate limit or the software speed limit. Once the buffer space is exhausted, new flow creation will be denied, and the teardown of the existing flows will be delayed until the buffer space becomes available.

The no form of this command removes the maximum rate limit from the configuration.

Parameters

packets-per-second

specifies the packet rate limit in seconds

Values: 10 to 2147483647

8.2.1.4.17 severity-level

Syntax

```
severity-level syslog-level
```

Context

```
config>service>nat>syslog>syslog-export-policy
```

Description

This command configures the severity level.

For more information, refer to the *7450 ESS, 7750 SR, 7950 XRS, and VSR System Management Guide*. The `config>log>syslog>level` hierarchy also applies to this context.

Default

severity-level info

Parameters

syslog-level

specifies the severity level

Values: emergency, alert, critical, error, warning, notice, info, debug

8.2.1.5 IPFlow information export protocol NAT command descriptions

8.2.1.5.1 ipfix

Syntax

```
ipfix
```

Context

```
config>service
```

Description

This command enables the context to configure IPFIX parameters.

8.2.1.5.2 ipfix-export-policy

Syntax

```
ipfix-export-policy policy-name [create]
```

```
no ipfix-export-policy policy-name
```

Context

```
config>service>ipfix
```

Description

This command creates an IPFIX export policy with a set of transport parameters that will be used to transmit IPFIX records generated by an application within 7750 SR MG and CMG node to an external collector node. This policy name can be referenced from each application within 7750 SR MG and CMG that requires flow logging.

Parameters

- policy-name*

specifies the name of the policy that can be referenced within an application in 7750 SR MG and CMG node that requires flow logging
- create**

mandatory keyword used when creating an IPFIX export policy in the IPFIX context. The create keyword requirement can be enabled/disabled in the environment>create context.

8.2.1.5.3 collector

Syntax

```
collector router router-instance ip ip-address [create]
no collector router router-instance ip ip-address
```

Context

```
config>service>ipfix>ipfix-export-policy
```

Description

This command defines an external collector node that will collect IPFIX records sent by 7750 SR MG and CMG node. The IPFIX records will be streamed to the collector node using UDP transport. Traffic is originated from a random ephemeral UDP port to the destination port 4739. Up to two collector nodes can be defined for redundancy purposes.

UDP streams are stateless due to the significant volume of transactions. However they do contain 32bit sequence numbers such that packet loss can be identified.

Multiple IPFIX records are sent in a single UDP packet. UDP packet transmission is triggered when the packet size containing IPFIX records exceeds the configured MTU value or the internal timer which is set to 250ms, whichever occurs first.

Parameters

- router-instance*

router instance from which the collector node is reachable

Values	router-name vprn-svc-id	
—	router-name	"Base"

– service-id 1 to 2147483647

ip-address

IPv4 address of the external collector node to which IPFIX records will be sent

Values: a.b.c.d.

create

mandatory keyword used when creating a collector in the IPFIX export policy context. The create keyword requirement can be enabled/disabled in the `environment>create` context.

8.2.1.5.4 mtu

Syntax

mtu *mtu-size*

no mtu

Context

config>service>ipfix>export-policy>collector

Description

This command sets the MTU size of the UDP packet containing IPFIX records destined for the collector node. Multiple records will be stuffed into a single IP packet until stuffing an additional data record would exceed MTU or the internal timer of 250ms expires.

Parameters

mtu-size

specifies the MTU range

Values 512 to 9000

Default: 1500

8.2.1.5.5 source-address

Syntax

source-address *ip-address*

```
no source-address
```

Context

```
config>service>ipfix>export-policy>collector
```

Description

This command configures the source address from which UDP streams containing IPFIX flow records will be sourced.

Parameters

ip-address

source IPv4 address from which UDP streams are sent

Values: a.b.c.d.

8.2.1.5.6 template-refresh-timeout

Syntax

```
template-refresh-timeout [hrs hours] [min minutes] [sec seconds]
```

```
no template-refresh-timeout
```

Context

```
config>service>ipfix>export-policy>collector
```

Description

This command configures the time interval in which Template Set messages are sent to the collector node. Template sets is an IPFIX message that defines fields for subsequent IPFIX messages but contains no data of its own. In other words, IPFIX data is NOT passed as set of TLVs, but instead data is encoded with a scheme defined through the Template Set message.

Parameters

hours

specifies the time interval, in hours, after which IPFIX templates are resent to this collector

Values: 1 to 24

minutes

specifies the time interval, in minutes, after which IPFIX templates are resent to this collector

Values: 1 to 59

Default: 10

seconds

specifies the time interval, in seconds, after which IPFIX templates are resent to this collector

Values: 1 to 59

8.2.1.6 VPRN and router NAT command descriptions

8.2.1.6.1 nat

Syntax

[no] nat

Context

config>service>vprn

config>router

Description

This command configures, creates or deletes a NAT instance.

8.2.1.6.2 inside

Syntax

inside

Context

config>service>vprn>nat

config>router>nat

Description

This command enters the inside context to configure the inside NAT instance.

8.2.1.6.3 outside

Syntax

```
outside
```

Context

```
config>service>vpn>nat
```

```
config>router>nat
```

Description

This command enters the outside context to configure the outside NAT instance.

8.2.1.6.4 mtu

Syntax

```
mtu [512..9000]
```

```
no mtu
```

Context

```
config>service>vpn>nat>outside
```

Description

This command configures the MTU for downstream traffic flowing through this router (as outside NAT router). The system fragments IP datagrams exceeding the MTU.

The no form of this command reverts to the default.

Parameters

512..9000

specifies the MTU for downstream traffic

Default: 0

8.2.1.6.5 nat-policy

Syntax

```
nat-policy nat-policy-name  
no nat-policy
```

Context

```
config>services>vprn>nat>inside  
config>router>nat>inside
```

Description

This command configures the NAT policy that will be used for large-scale NAT in this service.

The no form of this command removes the policy name from the configuration.

Parameters

nat-policy-name

specifies the NAT policy name, up to 32 characters

8.2.1.6.6 destination-prefix

Syntax

```
[no] destination-prefix ip-prefix/length
```

Context

```
config>service>vprn>nat>inside  
config>router>nat>inside
```

Description

This command configures a destination prefix. An (internal) static route will be created for this prefix. All traffic that hits this route will be subject to NAT. The system will not allow a destination-prefix to be configured if the configured nat-policy refers to an IP pool that resides in the same service (as this would result in a routing loop).

Parameters

ip-prefix

specifies the IP prefix; host bits must be zero (0)

Values: a.b.c.d

length

specifies the prefix length

Values: 0 to 32

8.2.1.6.7 deterministic

Syntax

```
deterministic
```

Context

```
config>service>vprn>nat>inside
```

```
config>router>nat>inside
```

Description

This command enables the context to configure deterministic NAT.

8.2.1.6.8 classic-lsn-max-subscriber-limit

Syntax

```
classic-lsn-max-subscriber-limit max
```

```
no classic-lsn-max-subscriber-limit
```

Context

```
config>service>vprn>nat>inside>deterministic
```

```
config>router>nat>inside>deterministic
```

Description

This command affects ingress hashing of the subscribers for deterministic NAT. It will also affect hashing of the subscribers for non-deterministic NAT if the both types of NAT are configured simultaneously. The hashing will ensure that traffic load is distributed over multiple MS-ISAs in the system. For deterministic LSN44, $(32 - n)$ bits of the source IP address will be considered for hashing, where $2^n = \text{classic-lsn-max-subscriber-limit}$.

The scope of this command is the inside routing instance. This command must match the largest subscriber limit of all pools that are referenced by nat-policies configured within the corresponding inside routing instance.

This parameter must be configured before any prefix is configured and can be modified only if there are no prefixes configured under the deterministic NAT CLI hierarchy.

If non-deterministic NAT is not used simultaneously with deterministic NAT within a routing context, the hashing for non-deterministic NAT is performed based on the subscriber.

Parameters

max

The power of 2 (2^n) number that must match the largest subscriber limit number in a deterministic pool referenced from this inside routing instance. The range for this command is the same as the subscriber-limit command under the pool hierarchy.

8.2.1.6.9 prefix

Syntax

```
prefix ip-prefix/length subscriber-type nat-sub-type nat-policy nat-policy-name [create]
prefix p-prefix/length subscriber-type nat-sub-type
no prefix ip-prefix/length subscriber-type nat-sub-type
```

Context

```
config>service>vprn>nat>inside>deterministic
config>router>nat>inside>deterministic
```

Description

This command is applicable only to deterministic NAT (LSN44). It configures prefixes on the inside and their association with outside deterministic pools via the nat-policy. Subscribers within the prefix will be deterministically mapped to outside IP addresses and corresponding port-ranges in the associated pool.

Multiple prefixes within an inside routing instance can be defined and they can reference different NAT policies (and therefore outside pools and routing instances). Moreover, prefixes from multiple routing instances can share the same deterministic pool.

Non-deterministic NAT can be used simultaneously with deterministic NAT within the same inside routing instance. However, they cannot share the same pool.

Prefixes can be added/removed under the condition that the associated deterministic pool is in a no shutdown mode.

Removing a prefix or modifying the map statement under it requires that the prefix be in a shutdown mode.

The subscribers under the prefix are mapped deterministically into the outside IPv4 addresses and port ranges. The subscribers in LSN44 are the IPv4 addresses under the configured prefix.

Default

no prefix

Parameters

ip-prefix/length

specifies a prefix on the inside encompassing subscribers that will be deterministically mapped to an outside IP address and port block in the corresponding pool

Values	ipv4-prefix/ipv4-prefix-length ipv6-prefix/ipv6-prefix-length	
–	ipv4-prefix	a.b.c.d (host bits must be 0)
–	ipv4-prefix-length	[0..32]
–	ipv6-prefix	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D
–	ipv6-prefix-length	[0..128]

nat-sub-type

specifies the classic-lsn-sub | dslite-lsn-sub

nat-policy-name

specifies a NAT policy that points to an outside pool and outside routing instance; up to 32 characters

8.2.1.6.10 map

Syntax

```
map start lsn-sub-address end lsn-sub-address to outside-ip-address  
no map start lsn-sub-address end lsn-sub-address
```

Context

```
config>service>vprn>nat>inside>deterministic
```

```
config>router>nat>inside>deterministic>prefix
```

Description

This command is applicable to prefixes in deterministic NAT (LSN44). Its purpose is to split the number of subscribers within the configured prefix over available sequence of outside IP addresses.

There are several rules guiding the usage of the map statement:

- If the number of subscribers per configured prefix is greater than the subscriber-limit per outside IP parameter (2^n), the lowest n bits of the map start *inside-addr-start* must be set to 0.



NOTE:

Subscriber in LSN44 is equals to an inside IPv4 address. Otherwise it will be an IPv6 prefix with length in the range [32..64] as set by the subscriber-prefix-length command.

- If the number of subscribers per configured prefix is equal or less than the subscriber-limit per outside IP parameter (2^n), only one map command for this prefix is allowed. In this case there is no restriction on the lower n bits of the map start *inside-ip-address*. The range of the inside IP addresses in such map statement represents the prefix itself.
- *outside-ip-address* in the map statements must be unique amongst all map statements referencing the same pool. In other words, two map statements cannot reference the same *outside-ip-address* in a pool.

To modify map statements, the corresponding prefix must be in a shutdown mode.

Map statements can be configured automatically by the system, as soon as the prefix is enabled (no shutdown state) or they can be configured manually by the operator while the prefix is disabled.

The following is an example of the map statement for the LSN44 case:

- The subscriber-limit in the pool is 128
- The pool has an address range 128.251.0.1 - 128.251.0.10
- The prefix is 10.0.0.0/24

- The map statement is configured as follows:

```
map start 10.0.0.0 end 10.0.0.255 to 128.251.0.1
```

Since each outside IP address can accommodate only 128 hosts, the subscribers (IPv4 addresses in LSN44) from the 10.0.0.0/24 prefix will be split and mapped into two outside IP addresses as follows:

```
10.0.0.0 - 10.0.0.127 (10.0.0.0/25) - 128.251.0.1
```

```
10.0.0.128 - 10.0.0.255 (10.0.0.128/25) - 128.251.0.2
```

The first IP address range will be mapped to the ‘to’ address in the map statement => 128.251.0.1. The second IP address range will be mapped into the next consecutive IP address in the pool assuming that this IP address is free. In this case this consecutive address (128.251.0,2) would not be shown in the map statement.

Default

By default, the system will automatically divide the prefix and create the map statements when the prefix command is enabled (no shutdown). However, this automatic map provisioning can be overruled by manual configuration.

Parameters

start *lsn-sub-address*

start IPv4/v6 address or IPv6 prefix on the inside

Values	ipv4-address ipv6-prefix/ipv6-prefix-length	
–	ipv4-address	d.d.d.d
–	ipv6-prefix	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D
–	ipv6-prefix-length	[0..128]

end lsn-sub-address

End IPv4/v6 address or IPv6 prefix on the inside. The number of subscribers (range of inside IPv4 addresses in LSN44) in the map statement does not have to be a power of 2. Rather it has to be a multiple of a power of two $m * 2^n$, where m is the number of consecutive outside IP addresses to which the subscribers are mapped and the 2^n is the subscriber-limit per outside IP.

Values	ipv4-address ipv6-prefix/ipv6-prefix-length	
–	ipv4-address	d.d.d.d
–	ipv6-prefix	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D
–	ipv6-prefix-length	[0..128]

outside-ip-address

The first outside IPv4 address in the pool to which the subscribers are mapped. In case that the number of subscribers in the map statement is larger than the subscriber-limit for the outside-ip address, the consecutive outside IP addresses will be used for additional mappings. Those additional (consecutive) outside IP addresses are not shown in the map statement (only the first address is shown in the map statement).

Values: a.b.c.d

8.2.1.6.11 ip-fragmentation

Syntax

```
ip-fragmentation {disabled | fragment-ipv6 | fragment-ipv6-unless-ipv4-df-set}
no ip-fragmentation
```

Context

```
config>router>nat>inside>nat64
```

```
config>service>vpn>nat>inside>nat64
```

Description

This command configures downstream IPv6 fragmentation behavior in NAT64. IPv6 fragmentation is performed in the ISA. IPv4 fragmentation is not affected by this command. If desired, downstream IPv4 packet can be fragmented in the carrier IOM before the packet reaches ISA (and the NAT function). The IPv4 fragmentation in the downstream direction can be set by the `config>router /vpn>nat>outside>mtu` command.

NAT64 IPv6 Fragmentation in Downstream Direction (IPv4to IPv6)

NAT64 transport is not based on tunneling. Instead, IP headers are translated between IPv4 and IPv6. Consequently, NAT64 fragmentation operates based on the `ipv6-mtu`.

In case that the length of the translated IPv6 packet exceeds the size of the configured `ipv6-mtu` value while fragmentation is allowed, the resulting IPv6 packet will be fragmented. The maximum size of the of the fragmented IPv6 packet will be the configured `ipv6-mtu` value.

In case that fragmentation is not allowed while the translated IPv6 packet size is larger than configured `ipv6-mtu` size, the IPv4 packet (that is supposed to be translated into IPv6) will be dropped and an ICMPv4 Datagram Too Big message will be generated toward the source. The advertised mtu size in that ICMP message will be set to the `ipv6-mtu` value minus 28bytes. The 28bytes comes from the size of the IPv6 overhead of the translated packet (20bytes difference between the IP header sizes # 40bytes in IPv6 vs 20bytes in IPv4; 8 bytes for extended IPv6 fragmentation header).

Parameters

disabled

IPv6 Fragmentation is disabled. In case that the packet size is larger than what is set by the `mtu` value (`tunnel-mtu` or `ipv6-mtu`), the IPv4 packet will be dropped and ICMPv4 Datagram Too Big messages will be sent back to the source.

fragment -ipv6

IPv6 fragmentation will be performed in all cases, regardless of the DF bit setting in the tunneled/translated IPv4 packet.

fragment -ipv6 -unless -ipv4 -df -set

IPv6 Fragmentation will be performed only in cases when DF bit in tunneled/translated IPv4 packet is cleared.

Default: disabled

8.2.1.6.12 nat64

Syntax

[no] nat64

Context

config>service>vprn>inside

Description

This command enables the context to configure NAT64.

The no form of this command disables NAT64.

8.2.1.6.13 destination-prefix

Syntax

destination-prefix *ipv6-prefix/prefix-length* [nat-policy *policy-name*]

no destination-prefix *ipv6-prefix/prefix-length*

Context

config>service>vprn>inside>nat64

config>router>nat>inside>nat64

Description

This command configures a destination prefix. An (internal) static route is created for this prefix. All traffic that hits this route is subject to NAT. The system will not allow a destination-prefix to be configured, if the configured NAT policy refers to an IP pool that resides in the same service (as this would result in a routing loop).

Parameters

ipv6-prefix/prefix-length

specifies the NAT64 destination prefix

Values	ipv6-prefix	x::x::x::x::x::x (eight 16-bit pieces)
--------	-------------	--

–	–	x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D
–	prefix-length	[0..128]

policy-name

specifies the NAT policy name associated with the destination prefix, up to 32 characters

8.2.1.6.14 drop-zero-ipv4-checksum**Syntax**

[no] drop-zero-ipv4-checksum

Context

config>service>vprn>inside>nat64

Description

This command specifies if UDP datagrams with zero IPv4 checksum are dropped.

If this command is disabled, the system calculates the IPv6 checksum for each such datagram.

8.2.1.6.15 ignore-tos**Syntax**

[no] ignore-tos

Context

config>router>nat>inside>nat64

config>service>vprn>inside>nat64

config>service>vprn>nat>inside>nat64

Description

This command specifies if the IPv4 Type Of Service (TOS) is ignored and the IPv6 traffic class bits set to zero.

When disabled, the system copies the IPv4 TOS into the IPv6 traffic class.

The no form of this command recognizes the IPv4 Type Of Service (TOS).

Default

no ignore-tos

8.2.1.6.16 insert-ipv6-fragment-header

Syntax

[no] insert-ipv6-fragment-header

Context

config>service>vprn>inside>nat64

Description

This command specifies if the system always inserts an IPv6 fragment header, to indicate that the sender allows fragmentation.

The no form of this command does not allow the system to insert an IPv6 fragment header.

8.2.1.6.17 nat64

Syntax

[no] nat64

Context

config>service>vprn>nat>inside

config>router>nat>inside

Description

This command enables the context to configure NAT64 parameters.

The no form of this command disables NAT64.

8.2.1.6.18 drop-zero-ipv4-checksum

Syntax

```
[no] drop-zero-ipv4-checksum
```

Context

```
config>service>vprn>nat>inside>nat64
```

```
config>router>nat>inside>nat64
```

Description

This command enables the NAT64 node to drop received UDP datagrams with zero IPv4 checksum. By default, checksum is re-calculated for non-fragmented datagrams.

The no form of this command disables the command.

Default

```
no drop-zero-ipv4-checksum
```

8.2.1.6.19 insert-ipv6-fragment-header

Syntax

```
[no] insert-ipv6-fragment-header
```

Context

```
config>service>vprn>nat>inside>nat64
```

```
config>router>nat>inside>nat64
```

Description

This command specifies whether the NAT64 node will insert IPv6 fragment header to IPv6 packets for which the DF bit is not set in the corresponding IPv4 packet, and is not already a fragment.

The no form of this command disables the insertion.

8.2.1.6.20 ipv6-mtu

Syntax

```
ipv6-mtu [1280..9212]
no ipv6-mtu
```

Context

```
config>service>vprn>nat>inside>nat64
config>router>nat>inside>nat64
```

Description

This command sets the size of the IPv6 downstream packet in NAT64. This packet is translated from IPv4. The no form of this command reverts to the default.

Parameters

[1280..9212]

specifies the IPv6 MTU
Values: 1280 to 9212
Default: 11520

8.2.1.6.21 prefix

Syntax

```
prefix ipv6-prefix/prefix-length
no prefix
```

Context

```
config>service>vprn>nat>inside>nat64
config>router>nat>inside>nat64
```

Description

This command configures the IPv6 prefix used to derive the IPv6 address from the IPv4 address, and is same as the prefix used by DNS64 to generate AAAA record returned for IPv4 endpoint resolution. NAT64 node announces this prefix in routing to attract traffic from IPv6 hosts. If the prefix is not configured, a well-known prefix, 64:FF9B::/96, is used.

The no form of this command removes the prefix from the NAT64 configuration.

Parameters

ipv6-prefix/prefix-length
specifies the NAT64 destination prefix

Values	ipv6-prefix	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D
–	prefix-length	[0..128]

8.2.1.6.22 set-tos

Syntax

set - tos [0..255]
no set - tos

Context

config>service>vprn>nat>inside>nat64
config>router>nat>inside>nat64

Description

This command specifies the value of the IPv4 Type Of Service (TOS) field. When enabled, the NAT64 node ignores IPv6 traffic-class and sets IPv4 TOS to supplied tos-value in the translated IPv4 packet.

The no form of this command reverts to the default.

Parameters

[0..255]

sets the IPv4 TOS to a fixed value the IPv6 Traffic Class and set the IPv4 TOS to a fixed value and ignores the IPv6 traffic class

Default: 0

8.2.1.6.23 subscriber-prefix-length

Syntax

`subscriber-prefix-length prefix-length`

`no subscriber-prefix-length`

Context

`config>service>vprn>nat>inside>nat64`

`config>router>nat>inside>nat64`

Description

This command specifies the IPv6 address prefix length to be used for the NAT64 subscribers in this virtual router instance.

The no form of this command

Parameters

prefix-length

specifies the subscriber identification for Large Scale NAT

Values: 32 to 64 | 128

Default: 128

8.2.1.6.24 redundancy

Syntax

`redundancy`

Context

`config>router>nat>inside`

`config>service>vprn>nat>inside`

Description

This command enables the context to configure redundancy parameters.

8.2.1.6.25 **peer**

Syntax

peer ip-address

no peer

Context

config>router>nat>inside>redundancy

config>service>vprn>nat>inside>redundancy

Description

This command configures the IP address of the NAT redundancy peer in the realm of this virtual router instance.

While the import prefix of the outside NAT router instance associated with this virtual router instance is present, this system redirects the traffic received for the NAT function in this virtual router instance to the NAT peer.

Parameters

ip-address

specifies the IP address of the NAT redundancy peer

Values: a.b.c.d

8.2.1.6.26 **peer6**

Syntax

peer6 ipv6-address

no peer6

Context

config>router>nat>inside>redundancy

config>service>vprn>nat>inside>redundancy

Description

This command configures the IPv6 address of the NAT redundancy peer in the realm of this virtual router instance.

While the import prefix of the outside NAT router instance associated with this virtual router instance is present, this system redirects the traffic received for the NAT function in this virtual router instance to the NAT peer.

Parameters

ipv6-address

specifies the IPv6 address of the NAT redundancy peer

Values	x:x:x:x:x:x:x (eight 16-bit pieces)
–	x:x:x:x:x:d.d.d.d
–	x - [0..FFFF]H
–	d - [0..255]D

8.2.1.6.27 steering-route

Syntax

steering-route *ip-prefix/length*

no steering-route

Context

config>router>nat>inside>redundancy

config>service>vprn>nat>inside>redundancy

Description

This command configures the IP address of the steering route.

The steering route is used in the realm of this virtual router instance as an indirect next-hop for all the traffic that must be routed to the Large Scale NAT function.

The no form of this command removes the ip-prefix/length from the configuration.

Parameters

ip-prefix/length

specifies the IP address and length of the steering route

Values	ip-prefix	a.b.c.d
–	ip-prefix-length	0 to 32

8.2.1.6.28 subscriber-identification

Syntax

```
subscriber-identification
```

Context

```
config>router>nat>inside
```

Description

This command enables the context to configure subscriber identification for Large Scale NAT.

8.2.1.6.29 attribute

Syntax

```
attribute [vendor vendor-id] attribute-type attribute-type
```

```
no attribute
```

Context

```
config>router>nat>inside>subscriber-id
```

```
config>service>vprn>nat>inside>subscriber-identification
```

Description

This command defines the attribute that will in addition to framed-ip-address (inside IP address) and service-id be used for correlating BNG subscriber with the NAT subscriber.

Only a single attribute at the time can be configured. The attribute will be extracted from the BNG accounting start and/or interim-update messages via RADIUS accounting proxy server. This attribute can be then optionally passed to the Large Scale NAT44 accounting server. User-name attribute (if included) in Large Scale NAT44 accounting messages is automatically set to the subscriber-id string.

The attribute parameter can be changed at any given time and the change will be reflected automatically when the next interim-update message from the BNG host is received by RADIUS accounting proxy.

In case that the BNG accounting message in RADIUS accounting proxy does not contain this attribute, subscriber aware Large Scale NAT44 functionality for this particular subscriber will be disabled.

Default

attribute vendor "nokia" attribute-type "nok-sub-string"

Parameters

vendor-id

specifies the RADIUS vendor ID

Values: standard, nokia, 3gpp

Default: nokia

attribute-type

specifies the RADIUS attribute to be used as subscriber identifier

Values	nok-sub-string (nokia)	Subscriber-id string (Alc-Subsc-ID-Str) is cached in Large Scale NAT44 application and used to correlate Large Scale NAT44 subscriber to BNG subscriber.
–	user-name (stnd)	User-Name standard RADIUS attribute is cached in Large Scale NAT44 application and is used to correlate Large Scale NAT44 subscriber to BNG subscriber.
–	class (stnd)	Class standard RADIUS attribute is cached in Large Scale NAT44 application and is used to correlate Large Scale NAT44 subscriber to BNG subscriber. Class attribute is initially set and send by RADIUS server. As such it must be echoed by BNG in all accounting messages.

–	station-id (stnd)	Calling-Station-Id RADIUS attribute is cached in Large Scale NAT44 application and is used to correlate Large Scale NAT44 subscriber to BNG subscriber.
–	imsi (3gpp)	International Mobile Subscriber Identification is used in WiFi Offload applications as a SIM card identifier.
–	imei (3gpp)	International Mobile Equipment Identification is used in WiFi Offload applications as a physical phone device identifier.

8.2.1.6.30 drop-unidentified-traffic

Syntax

[no] drop-unidentified-traffic

Context

config>router>nat>inside>subscriber-id

Description

When this command denies address translation to subscribers that have not been identified via accounting messages sent by BNG and received by RADIUS accounting proxy. This command has effect only in Subscriber Aware Application.

Default

no drop-unidentified-traffic

8.2.1.6.31 radius-proxy-server

Syntax

radius-proxy-server router *router-instance* name *server-name*

no radius-proxy-server

Context

config>router>nat>inside>subscriber-id

config>service>vprn>nat>inside>subscriber-identification

Description

This command configures RADIUS proxy server parameters. This is a reference to a RADIUS accounting proxy server in Subscriber Aware Large Scale NAT44 application. RADIUS accounting proxy server will cache attributes related to a BNG subscriber as they are received in standard accounting messages (RFC 2866). RADIUS accounting proxy server can be configured in any routing instance within 7750 SR MG and CMG.

Parameters

router-instance

specifies the routing instance in which the RADIUS accounting proxy is configured

server-name

specifies the name reference to the RADIUS accounting proxy server that is instantiated in 7750 SR MG and CMG

8.2.1.6.32 mtu

Syntax

mtu *[512..9000]*

no mtu

Context

config>router>nat>outside

Description

This command configures the MTU for downstream traffic flowing through this router (as outside NAT router). The system fragments IP datagrams exceeding the MTU.

Parameters

[512..9000]

specifies the MTU for downstream traffic

8.2.1.6.33 pool

Syntax

```
pool nat-pool-name [nat-group nat-group-id type pool-type create]  
no pool nat-pool-name
```

Context

```
config>service>vprn>nat>outside  
config>router>nat>outside
```

Description

This command configures a NAT pool.

Parameters

nat-pool-name

specifies the NAT pool name

Values: 32 chars max

nat-group-id

specifies the NAT group ID

Values: 1 to 4

create

This parameter must be specified to create the instance.

pool-type

specifies the pool type, large-scale

8.2.1.6.34 address-range

Syntax

```
address-range start-ip-address end-ip-address [create]  
no address-range start-ip-address end-ip-address
```

Context

```
config>service>vprn>nat>outside>pool  
config>router>nat>outside>pool
```

Description

This command configures a NAT address range.

Parameters

start-ip-address

specifies the beginning IP address in a.b.c.d form

end-ip-address

specifies the ending IP address in a.b.c.d. form

create

This parameter must be specified to create the instance.

8.2.1.6.35 drain

Syntax

[no] drain

Context

config>service>vprn>nat>outside>pool>address-range

config>router>nat>outside>pool>address-range

Description

This command starts or stops draining this NAT address range. When an address-range is being drained, it will not be used to serve new hosts. Existing hosts, however, will still be able to use the address that was assigned to them even if it is being drained. An address-range can only be deleted if the parent pool is shut down or if the range itself is effectively drained (no hosts are using the addresses anymore).

8.2.1.6.36 mode

Syntax

mode {auto | napt | one-to-one}

no mode

Context

config>router>nat>outside>pool

Description

This command specifies the mode of operation of this NAT address pool.

The no form of this command reverts to the default.

Parameters

auto | napt | one-to-one

specifies the mode of operation of this NAT pool

Default: auto

8.2.1.6.37 port-forwarding-range

Syntax

port-forwarding-range *range-end*

no port-forwarding-range

Context

config>router>nat>outside>pool

config>service>vprn>nat>outside>pool

Description

This command specifies the end of the port range available for port forwarding. The start of the range is always equal to one.

The number of ports that can be configured is half of the available block => $64512 : 2 = 32256$

In combination with port-forwarding-range the formulas are:

"max port-reservation blocks" = $65535 - \text{"port-forwarding-range"}$

"max port-reservation ports" = $(65535 - \text{"port-forwarding-range"}) / 2$

with:

the default min value for "port-forwarding-range" = 1023

Also, the same applies for max port-forwarding-range if the port-reservation is already configured:

"max port-forwarding-range" = $65535 - \text{"port-reservation blocks"}$

"max port-forwarding-range" = $65535 - (\text{"port-reservation ports"} * 2)$

The no form of this command reverts to the default.

Parameters

range-end

specifies the end of the port range available for port forwarding

Values: 1023 to 65535

8.2.1.6.38 deterministic

Syntax

deterministic

Context

config>service>vprn>nat>outside>pool

Description

This command configures deterministic NAT for this pool.

8.2.1.6.39 port-reservation

Syntax

port-reservation *num-ports*

no port-reservation

Context

config>service>vprn>nat>outside>pool>deterministic

Description

This command is applicable only to deterministic NAT. It configures the number of deterministic ports per subscriber (for example a subscriber is an inside IP address in LSN44 or IPv6 address). Once this command is enabled, the pool will transition into deterministic mode of operation. This means that the subscribers can use dynamic port-blocks in the pool only as a mean to expand the range of originally assigned deterministic ports. A pool with such property is referred to as deterministic pool. However, deterministic NAT and non-deterministic NAT cannot use the same pool simultaneously.

All subscribers in deterministic pool are pre-mapped during the configuration phase to outside IP addresses and deterministic port-blocks. Because of this, the deterministic pool cannot be oversubscribed with subscribers (first-come, first-served).

Once the deterministic pool becomes operational (no shutdown) a log is created. The same applies if the pool is disabled (shutdown). As a result of this “one time” logging, there will be no additional logging when a subscriber starts using ports from the pre-assigned deterministic port block. This drastically reduces the logging overhead. However, when a deterministic port block is expanded by a dynamic port block, a log will be created on any allocation/de-allocation of the dynamic port block. The logs are also created for static port forwards (including PCP).

The number of subscribers per outside IP address (subscriber-limit) multiplied by the number of deterministic ports per subscriber (port-reservation) will determine the port range of an outside IP address that will be dedicated to deterministic mappings. The number of subscribers per outside IP address in deterministic NAT must be power of 2 (2^n). Once the deterministic ports are allocated, the dynamic ports are carved out of the remaining port space of the same outside IP address according to the existing port-reservation command under the same hierarchy.

Parameters

num-ports

specifies the number of ports in a deterministic port block that is allocated and dedicated to a single subscribers during the configuration phase

Values: 1 to 65535

8.2.1.6.40 port-reservation

Syntax

```
port-reservation blocks num-blocks
```

```
port-reservation ports num-ports
```

```
no port-reservation
```

Context

```
config>service>vprn>nat>outside>pool
```

```
config>router>nat>outside>pool
```

Description

This command configures the size of the port-block that will be assigned to a host that is served by this pool. The number of ports configured here will be available to UDP, TCP, and ICMP (as identifiers).

Parameters

num-blocks

specifies the number of port-blocks per IP address. Setting num-blocks to one (1) for large scale NAT will enable 1:1 NAT for IP addresses in this pool.

Values: 1 to 65535

num-ports

Specifies the number of ports per block.

Values: 1 to 32256

8.2.1.6.41 mode

Syntax

mode {auto | napt | one-to-one}

no mode

Context

config>service>vprn>nat>outside>pool

Description

This command configures the mode of operation of this NAT pool.

Parameters

napt

specifies NAPT (Network Address Port Translation)

auto

The system selects the actual mode based upon other configuration parameters; the actual mode can be NAPT or 1:1 NAT (also known as 'Basic NAT').

one-to-one

indicates 1:1 NAT (also known as 'Basic NAT')

8.2.1.6.42 port-forwarding-dyn-block-reservation

Syntax

[no] port-forwarding-dyn-block-reservation

Context

config>service>vprn>nat>outside>pool

config>service>router>nat>outside>pool

Description

This command will enable the reservation of the dynamic port blocks when the first port forward for the subscriber is created. The dynamic port block allocation is logged only if the block is being utilized (mapping are created). In other words, dynamic port block reservation due to the port forward creation but without any dynamic mapping, will not be logged.

The reserved port block will be released only when the last mapping in the block expires AND there is not port forward associated with the subscriber. The de-allocation log (syslog or RADIUS) will be generated when the dynamic port block is completely released.

Dynamic port block reservation can be enabled only if the configured maximum number of subscriber per outside IP address is less or equal then the maximum number of configured port blocks per outside IP address.

Default

port-forwarding-dyn-block-reservation

8.2.1.6.43 redundancy

Syntax

redundancy

Context

config>router>nat>outside>pool

Description

This command enables the context to configure NAT pool redundancy parameters.

8.2.1.6.44 export

Syntax

export *ip-prefix/length*

no export

Context

config>router>nat>outside>pool>redundancy

Description

This command configures the route to export to the peer. While the export prefix is configured and the value of the object `tmnxNatPILsnRedActive` is equal to true, the system exports this prefix in the realm of the virtual router instance associated with this pool; to the NAT redundancy peer, the presence of this prefix is an indication that the Large Scale NAT function in this virtual router instance is active; hence, the export prefix of this system is the monitor prefix of the peer.

The export prefix must be different from the monitor prefix.

Parameters

ip-prefix/length

specifies the IP address and length of the prefix to be exported

Values	ip-prefix	a.b.c.d
–	ip-prefix-length	0 to 32

8.2.1.6.45 follow

Syntax

`follow router router-instance pool name`

`no follow`

Context

`config>service>vprn>nat>outside>pool>redundancy`

`config>router>nat>outside>pool>redundancy`

Description

This command implicitly enables Pool Fate-Sharing Group (PFSG) which is required in case of multiple NAT policies per inside routing context. A NAT pool configured with this command will not advertise or monitor any route to change its (activity) state but instead it will directly follow the state of the lead pool in the PFSG. Once the lead pool changes its (activity) state, all the remaining pools following the lead pool will change their state accordingly.

Default

no follow

Parameters

router-instance

specifies the routing instance where the lead pool resides

Values	router-name vprn-svc-id	
–	router-name	"Base"
–	service-id	[1..2147483647]

name

specifies the pool whose activity state is being shared, up to 32 characters

8.2.1.6.46 monitor

Syntax

monitor *ip-prefix/length*

no monitor

Context

config>router>nat>outside>pool>redundancy

Description

This command configures the IP address of the prefix to be monitored.

While the monitor prefix is configured, the system monitors the presence of this prefix in the routing table of the virtual router instance associated with this pool; the presence of this prefix is an indication that the NAT redundancy peer is active; the monitor prefix of this system is the export prefix of the peer.

The monitor prefix must be different from the export prefix.

Parameters

ip-prefix/length

specifies the peer route to monitor

Values	ip-prefix	a.b.c.d
–	ip-prefix-length	0 to 32

8.2.1.6.47 subscriber-limit

Syntax

```
subscriber-limit [1..65535]  
no subscriber-limit
```

Context

```
config>service>vprn>nat>outside  
config>nat>outside>pool
```

Description

This command configures the maximum number of subscribers per outside IP address. In case multiple port blocks per subscriber are used, the block size is typically small; all blocks assigned to a given subscriber belong to the same IP address; the subscriber limit guarantees that any subscriber can get a minimum number of ports.

Parameters

limit

specify the maximum number of subscribers per IP address

Values: 1 to 65535

Default: 65535

8.2.1.6.48 watermarks

Syntax

```
watermarks high percentage-high low percentage-low  
no watermarks
```

Context

config>service>vpn>nat>outside>pool

config>router>nat>outside>pool

Description

This command configures the watermarks for this NAT pool.

Parameters

percentage-high

specifies the high percentage

Values: 1 to 100

percentage-low

specifies the low percentage

Values: 0 to 99

8.2.1.6.49 upstream-ip-filter

Syntax

upstream-ip-filter *filter-id*

no upstream-ip-filter

Context

config>service>vpn>nat>outside

Description

This command configures the ip-filter for upstream traffic. The filter is applied to upstream traffic after the NAT function and before it enters outside virtual router instance; it is useful for traffic that bypasses ingress filters applied in the inside virtual router instance, such as DSLite traffic.

Parameters

filter-id

specifies the identifier of an IP filter

8.2.1.7 DNAT NAT command descriptions

8.2.1.7.1 dnat

Syntax

[no] dnat

Context

config>service>nat>nat-policy

Description

This command defines context for destination NAT (DNAT) specific configuration under the nat-policy.

8.2.1.7.2 dnat-only

Syntax

dnat-only router *router-instance* nat-group *nat-group-id*

no dnat-only

Context

config>service>nat>nat-policy>dnat

Description

This command configures outside routing context and nat-group in which DNAT translation should take place. This command is mutually exclusive with the pool command in nat-policy. When DNAT-only is enabled, no source and port NAT (SNAPT) is performed. In other words, only the destination IP address (going from inside to outside) is translated while the source IP address and port are not translated.

Parameters

router-instance

specifies the routing context on the outside (public side)

nat-group-id

specifies the NAT group IP

Values: 1 to 4

8.2.1.7.3 dnat-only

Syntax

```
dnat-only
```

Context

```
config>router>nat>inside
```

```
config>service>vprn>nat>inside
```

Description

This command enables the context on the NAT inside context where dnat-only parameters are configured.

8.2.1.7.4 source-prefix-list

Syntax

```
source-prefix-list prefix-list-name
```

```
no source-prefix-list
```

Context

```
config>router>nat>inside>dnat-only
```

```
config>service>vprn>nat>inside>dnat-only
```

Description

This command references the `nat-prefix-list` that contains source IP addresses on the inside (private side).

The source IP addresses on the inside must be known in advance in a DNAT only instance. This is required so the corresponding routes can be installed in the routing table and thus the downstream traffic is properly routed toward the MS-ISAs where the original translation was performed (and state is kept).

In the DNAT-only case, it is mandatory that the inside (private side) and the outside (public side) are in separated VPRNs.

Parameters

prefix-list-name

specifies the name of the NAT prefix list that contains the source IP addresses (original IP addresses), up to 32 characters.

8.2.1.7.5 route-limit

Syntax

```
route-limit [1..131072]
```

Context

```
config>router>nat>outside>dnat-only
```

```
config>service>vprn>nat>outside>dnat-only
```

Description

The source IP addresses on the inside must be known in advance in a dnat-only instance. This is required so that the corresponding routes can be installed in the routing table and thus the downstream traffic is properly routed toward the MS-ISAs where the original translation was performed (and state is kept).

In the dnat-only case, it is mandatory that the inside (private side) and the outside (public side) are in separated VPRNs.

This command limits the number of source routes (inside routes) that are installed on the outside in DNAT only case. In case that the number of actual routes is larger than the number of configured routes, the excess of the routes will not be installed in the routing table and a log will be raised.

Parameters

[1..131072]

specifies the maximum number of source routes installed on the outside the DNAT only scenario

Default: 32768

8.2.1.7.6 classic-lsn-max-subscriber-limit

Syntax

```
classic-lsn-max-subscriber-limit max
```

```
no classic-lsn-max-subscriber-limit
```

Context

```
config>router>nat>inside
```

```
config>service>vprn>nat>inside
```

Description

This command sets the granularity of traffic distribution in the upstream direction across the MS-ISA within the scope of an inside routing context. Traffic distribution mechanism is based on the source IPv4 addresses/prefixes. More granular distribution is based on the IPv4 address, while distribution based on the IPv4 prefix (determined by prefix length) will be less granular. The granularity will further decrease with shorter prefix length.

For example, a prefix length of 32 will distribute individual /32 IPv4 addresses over multiple MS-ISAs in an ISA group. This will ensure better traffic load balancing at the expense of forwarding table utilization on the outside (public side) where each /32 is installed in the forwarding table. On the contrary, shorter prefixes will ensure better utilization of the forwarding table on the outside, at the expense of coarser spread of IP addresses over multiple MS-ISAs.

This command affects all flavors of LSN44 within the inside routing contexts, although its primary use is intended for deterministic NAT and dnat-only.

The length of the prefix that is used for distribution purposes is $(32-n)$, where $2^n = \text{classic-lsn-max-subscriber-limit}$. For example, if traffic distribution is based on the IPv4 address (prefix length = 32), n must be 0. From here, it follows that *classic-lsn-max-subscriber-limit* must be set to 1:

Prefix length = 32 $\rightarrow 32-n = 32 \rightarrow n=0 \rightarrow 2^0 = 1 = \text{classic-lsn-max-subscriber-limit}$
classic-lsn-max-subscriber-limit = 1

The implicit method given by this command uses power of 2 calculations to provide prefix length for traffic distribution purposes. This roundabout approach to determine the prefix-length has roots in deterministic NAT where this command was originally introduced.

Even though deterministic NAT and dnat-only have very little in common, the method (and CLI syntax) for calculating the prefix length using the *classic-lsn-max-subscriber-limit* parameter for traffic distribution purposes is shared between the two. In dnat-only, this parameter is important from an operational perspective since it affects traffic load balancing over MS-ISA and the size of the routing table.

This command must be configured before any prefix is configured and can be modified only if there are no prefixes configured under the deterministic NAT.

Parameters

max

The power of 2 (2^n) value which in deterministic NAT must match the largest subscriber-limit value in any deterministic pool referenced from this inside routing instance.

In dnat-only, this value can be set to any value from the allowed range.

In both cases, this value will determine the prefix-length (17-32) that will directly influence load distribution between the MS-ISAs and the size of the routing table.

Values: 1 | 2 | 4 | 8 to 32768

8.2.1.7.7 nat-classifier

Syntax

```
nat-classifier nat-classifier-name  
no nat-classifier
```

Context

```
config>service>nat>nat-policy>dnat
```

Description

This command when configured within the nat-policy, references a nat-classifier and consequently activates DNAT functionality. Unless this command is provisioned, the destination IP address translation will not take place. The `nat-classifier` command identifies the traffic (in a filter-like fashion) that is subjected to DNAT.

The no form of this command removes the *nat-classifier-name* from the configuration.

Parameters

nat-classifier-name

specifies the name of the NAT classifier, up to 32 characters

8.2.1.7.8 nat-classifier

Syntax

```
nat-classifier nat-classifier-name [create]  
no nat-classifier
```

Context

```
config>service>nat
```

Description

This command creates a nat-classifier. Traffic can be identified in nat-classifier based on the protocol type, source ports and IP addresses. Once the traffic is identified, an action associated with identified traffic, such as destination NAT (DNAT), can be taken.

The no form of this command removes the *nat-classifier-name* from the configuration.

Parameters

nat-classifier-name

specifies the name of the referenced NAT classifier, up to 32 characters

create

specifies the keyword used to create the NAT classifier

8.2.1.7.9 nat-prefix-list

Syntax

```
nat-prefix-list name
```

```
no nat-prefix-list
```

Context

```
config>subscrib-mgmt>sub-profile
```

Description

This command specifies the nat-prefix-list referenced within the subscriber-profile is used to associate L2-aware subscriber traffic with additional nat-policies based on the destination IPv4 address of the traffic.

The no form of this command removes the prefix list name from the configuration.

Parameters

name

specifies the NAT prefix list name

Allowed values are any string up to 32 characters long composed of printable, 7-bit ASCII characters. If the string contains special characters (#, \$, spaces, and so on), the entire string must be enclosed within double quotes.

8.2.1.7.10 nat-prefix-list

Syntax

```
nat-prefix-list name [create] [application application-choice]
```

```
no nat-prefix-list name
```

Context

```
config>service>nat
```

Description

This command is used to create configuration context for:

- IP prefixes that are used select multiple nat-policies per subscriber in L2-aware NAT.
- Inside IP prefixes in DNAT-only scenario. The inside IP prefixes are then setup as downstream routes used to steer the return (downstream) traffic to the proper MS-ISA.

The no form of this command removes the prefix list name from the configuration.

Parameters

name

Specifies the nat prefix list name. Allowed values are any string up to 32 characters long composed of printable, 7-bit ASCII characters. If the string contains special characters (#, \$, spaces, and so on), the entire string must be enclosed within double quotes.

application-choice

Specifies how this NAT prefix list is to be applied.

Values: : Specifies that the nat-prefix-list can be applied only within the sub-profile for L2-aware subscribers. It will contain mapping between the destination prefix and a nat-policy.

Specifies that the nat-prefix-list can be applied only to dnat-only-subscribers. It will contain the source-prefix that needs to be install in outside routing context so that the return traffic from the outside can be directed to proper MS-ISA.

8.2.1.7.11 prefix

Syntax

```
prefix ip-prefix/length [nat-policy nat-policy-name]
```

```
no prefix ip-prefix/length
```

Context

```
config>service>nat>nat-prefix-list
```

Description

This command creates a prefix entry in the nat-prefix-list.

This prefix can be used to identify traffic with specific destination IP that needs to be associated with corresponding nat-policy (and implicitly the NAT pool) for L2-aware subscribers. In this fashion, a single L2-aware subscriber can direct traffic to multiple NAT pools, depending on the traffic destination.

Another use for a prefix is in DNAT-only application (DNAT without SNAPT). In this case the prefix identifies the inside source IP range that will be explicitly configured to ensure proper downstream routing in DNAT-only case.

The `nat-prefix-list` command cannot reference the default `nat-policy` command (the one that is referenced in the subscriber-profile).

The no form of this command reverts to the default.

Parameters

ip-prefix/length

Specifies the IP prefix for nat prefix list entry in dotted decimal notation.

Values: a.b.c.d (host bits must be 0) 0 to 32

nat-policy-name

specifies the NAT policy name

Allowed values are any string up to 32 characters long composed of printable, 7-bit ASCII characters. If the string contains special characters (#, \$, spaces, and so on), the entire string must be enclosed within double quotes.

8.2.1.7.12 default-dnat-ip-address

Syntax

`default-dnat-ip-address ip-address`

`no default-dnat-ip-address`

Context

`config>service>nat>nat-classifier`

Description

This command configures default destination ip address for action DNAT in cases where the destination IP address is not explicitly stated as part of the action or default-action statement.

The no form of this command reverts to the default.

Parameters

ip-address

specifies the IPv4 address in dotted decimal notation

Values: a.b.c.d

8.2.1.7.13 entry

Syntax

```
entry entry-id [create]  
no entry entry-id
```

Context

```
config>service>nat>nat-classifier
```

Description

This command creates or edits a nat-classifier entry. Multiple entries can be created using unique entry-id numbers within the nat-classifier. Entries must be sequenced from most to least explicit. An entry may not have any match criteria defined, in which case all UDP traffic will be matched. In case that the action is not explicitly configured, a default-action will be applied.

The no form of this command removes the specified entry from the filter. Entries removed from the nat-classifier are immediately removed from all entities to which the nat-classifier is applied.

Parameters

entry-id

Specifies an entry ID that uniquely identifies a match criteria and the corresponding action. It is recommended that multiple entries be given entry-ids in staggered increments. This allows users to insert a new entry in an existing policy without requiring renumbering of all the existing entries.

Values: 1 to 1000

8.2.1.7.14 match

Syntax

```
match protocol ip-protocol  
no match
```

Context

```
config>service>nat>nat-classifier>entry
```

Description

This command configures an IP protocol to be used as a nat-classifier match criterion. When the match criteria have been satisfied the action associated with the match criteria is executed. The no form of this command removes the match criteria for the entry-id.

Parameters

ip-protocol

Specifies the text value representing the IP protocol to be used as a match criterion.

Values: udp | tcp

Default: udp

8.2.1.7.15 dst-port-range

Syntax

dst-port-range start *port-number* end *port-number*

no dst-port-range

Context

config>service>nat>nat-classifier>entry

Description

This command configures a destination TCP or UDP port number or port range.

An entry containing Layer 4 match criteria will not match non-initial (2nd, 3rd, and so on) fragments of a fragmented packet since only the first fragment contains the Layer 4 information.

The no form of this command removes the destination port match criterion.

Parameters

start port-number

Specifies the start of the port range expressed as a decimal integer.

Values: 0 to 65535

end port-number

Specifies the end of the port range expressed as a decimal integer.

Values: 0 to 65535

Default: start 0 end 65535

8.2.1.7.16 action

Syntax

```
action {dnat | forward} [ip-address ip-address]  
no action
```

Context

```
config>service>nat>nat-classifier>entry
```

Description

This command specifies the action to take for packets that match this nat-classifier entry.

The no form of this command removes the specified action statement.

Default

no action

This means that this entry is ignored (skipped). Consequently, the action from another matching entry will be applied. If there are no other matching entries found, the default-action will be applied.

Parameters

dnat

specifies the DNAT function to be performed; the destination IP address of the packet traversing the router in the direction from inside to outside is replaced by the configured IP address, and the destination port is not translated; in the opposite direction (from outside to inside), the source address in the returning packet is restored to the original value

forward

specifies the forward action to ensure that the packet is transparently passed through the nat-classifier

ip-address

specifies the destination IP address that replaces the original IP address in the packet traveling from inside to outside

Values: a.b.c.d

8.2.1.7.17 default-action

Syntax

```
default-action {dnat | forward} [ip-address ip-address]
```

Context

config>service>nat>nat-classifier

Description

This command specifies the default action to take for packets in this nat-classifier. The default-action will apply to packet that do not match any configured criteria within nat-classifier.

The no form of this command equals action forward.

Parameters

dnat

Performs the DNAT function. The destination IP address of the packet traversing the router in the direction from inside to outside is replaced by the configured IP address. Destination port is not translated. In the opposite direction (from outside to inside), the source address in the returning packet is restored to the original value.

forward

The forward action will ensure that the packet is transparently passed through the nat-classifier.

ip-address

The destination IP address that will replace the original IP address in the packet traveling from inside to outside.

Values: a.b.c.d

Default: forward

8.2.1.718 default-dnat-ip-address

Syntax

default-dnat-ip-address *ip-address*

no default-dnat-ip-address

Context

config>service>nat>nat-classifier

Description

This command configures the IP address to substitute for the destination IP address of the packets

Default

no default-dnat-ip-address

Parameters

ip-address

specifies the default DNAP IP address

8.2.1.8 Admin NAT command descriptions

8.2.1.8.1 save-deterministic-script

Syntax

save-deterministic-script

Context

admin>nat

Description

This command saves the script that calculates deterministic NAT map entries.

Once the location for the Python deterministic NAT script is configured, the script is generated/updated every time deterministic NAT configuration is modified. However, the script must be manually exported to the remote location. This command triggers the export of the script to a remote location.

8.2.2 NAT show command descriptions

8.2.2.1 nat-group

Syntax

nat-group

nat-group *nat-group-id* [associations]

nat-group *nat-group-id* member [1..255] [statistics]

nat-group [*nat-group-id*] members

Context

show>isa

Description

This command displays ISA NAT group information.

Parameters

nat-group-id

specifies the NAT group ID

Values: 1 to 15

associations

keyword to display router instances and/or subscriber profiles associated with the ISA NAT group

member

keyword to display ISA NAT group information for a specific member

Values: 1 to 255

statistics

displays the NAT group statistics

members

displays the ISA NAT group information for all members

Output

The following output example shows ISA NAT group information.

Example output

```
# show isa nat-group
=====
ISA NAT Group Summary
=====
Mda Group 1 Group 2 Group 3
-----
3/1 active - -
3/2 - active busy
4/1 - busy active
4/2 - standby standby
=====
*A:SR12_PPPoE>config>isa>nat-group# show isa nat-group 1
=====
ISA NAT Group 1
=====
Admin state           : inService
Operational state     : inService
Active MDA limit      : 2
```

 NAT specific information for ISA group 1

Reserved sessions : 0
 High Watermark (%) : (Not Specified)
 Low Watermark (%) : (Not Specified)
 Accounting policy : my-acct-plcy
 Last Mgmt Change : 01/28/2012 14:47:59

=====

ISA Group 1 members

=====

Group	Member	State	Mda	Addresses	Blocks	Se-%	Hi	Se-Prio
1	1	active	3/1	3	3	< 1	N	0
1	2	active	3/2	4	4	< 1	N	0

No. of members: 2
 =====

A:SR12_PPPOE#

config>isa>nat-group# show isa nat-group

=====

ISA NAT Group Summary

=====

Mda	Group 1	Group 2	Group 3	Group 4
2/1	-	provisioned	-	-
3/1	active	-	up	-
3/1	active	-	up	-
3/2	active	-	up	-
3/2	active	-	up	-

=====

show isa nat-group 1

=====

ISA NAT Group 1

=====

Description : (Not Specified)
 Admin state : inService
 Operational state : inService
 Degraded : false
 Redundancy : active-standby
 Active MDA limit : 1
 Failed MDA limit : 0
 Local NAT : true

NAT specific information for ISA group 1

Reserved sessions : 0
 High Watermark (%) : (Not Specified)
 Low Watermark (%) : (Not Specified)
 Accounting policy : (Not Specified)
 UPnP mapping limit : 524288
 Suppress LsnSubBlksFree : false
 LSN support : enabled

Last Mgmt Change : 09/07/2018 12:36:16

ISA Group 1 members

Group	Member	State	Mda	Addresses	Blocks	Se-%	Hi	Se-Prio
1	1	active	3/2	65534	131068	< 1	N	0

No. of members: 1

show isa nat-group 3 member 1 statistics

ISA NAT Group 3 Member 1

no resource	: 0
pkt rx on wrong port	: 0
unsupported protocol	: 0
no host or host group	: 0
no ip or port	: 0
no matching flow	: 3
max flow exceeded	: 0
TCP no flow for RST	: 0
TCP no flow for FIN	: 0
TCP no flow	: 0
addr. dep. filtering	: 0
ICMP type unsupported	: 0
ICMP local unsupported	: 0
ICMP checksum error	: 0
ICMP embedded checksum error	: 0
ICMP unsupported L4	: 0
ICMP too short	: 0
ICMP length error	: 0
Pkt not IPv4 or IPv6	: 0
Pkt rcv error	: 0
Pkt error	: 0
IPv4 header checksum violation	: 0
IPv4 header malformed	: 0
IPv4 malformed packet	: 0
IPv4 ttl zero	: 0
IPv4 opt /IPv6 ext headers	: 0
IPv4 undefined error	: 0
IPv6 fragments unsupported	: 0
TCP/UDP malformed	: 0
TCP/UDP checksum failure	: 0
TCP/UDP length error	: 0
Pkt send error	: 0
no buf to copy pkt	: 0
no policy	: 0
locked by mgmt core	: 0
port range log failed	: 0
MTU exceeded	: 0
DS Lite unrecognized next hdr	: 0
DS Lite unknown AFTR	: 0
too many fragments for IP packet	: 0


```

too many fragmented packets      : 0
too many fragment holes         : 0
too many frags buffered         : 0
fragment list expired           : 0
fragment rate too high          : 0
flow log failed                 : 0
no multiple host or subscr. IPs allowed : 0
to local                        : 1
to local ignored                : 0
NAT64 disabled                  : 0
NAT64 invalid src addr         : 0
NAT64 frag has zero checksum    : 0
NAT64 v4 has zero checksum      : 0
NAT64 ICMP frag unsupported     : 0
CPM out of memory              : 0
new flow                       : 1
TCP closed                     : 1
TCP expired                    : 0
UDP expired                    : 0
ICMP expired                   : 0
ICMP local                     : 0
found flow                     : 34
ARPs ignored                   : 4
Fragments RX L2A               : 0
Fragments RX LSN               : 0
Fragments RX DSL               : 0
Fragments RX OUT               : 0
Fragments TX L2A               : 0
Fragments TX LSN               : 0
Fragments TX DSL               : 0
Fragments TX NAT64             : 0
Fragments TX OUT               : 0
flow create logged              : 0
flow delete logged              : 0
flow log pkt tx                 : 0
=====
A:SR12_PPP0E#

# show isa nat-group 1 member 1 statistics
=====
ISA NAT Group 1 Member 1
=====
no resource                      : 0
[eNatFlowNoResource]             "no resource",\
    ->the default, all errors without more specific reason

[eNatFlowWrongPort]              "pkt rx on wrong port",\
    -> packet came in on wrong port on ISA

[eNatFlowWrongProt]              "unsupported protocol",\
    -> protocol is not UDMP/TCP/ICMP

[eNatFlowNoHostGrp]              "no host or host group",\
    -> can not create new host group because out of resources, or
current host group is not usable at the moment (because in a transient
state)

```

```

[eNatFlowNoIpOrPort]          "no ip or port",\
    -> no Ip or port range available

[eNatFlowNoMatchingFlow]      "no matching flow",\
    -> no matching flow found

[eNatFlowMaxExceeded]         "max flow exceeded",\
    -> max flows for subscriber exceeded

[eNatFlowTcpUnexpectedRst]     "TCP no flow for RST",\
[eNatFlowTcpUnexpectedFin]     "TCP no flow for FIN",\
[eNatFlowTcpUnexpected]       "TCP no flow",\
    -> TCP state machine problem

[eNatFlowAddressDependentFiltering] "addr. dep. filtering",\
    -> pkt dropped because of addr. dependent filtering

[eNatFlowUnsupportedICMP]     "ICMP type unsupported",\
    -> unsupported icmp type

[eNatFlowUnsupportedLocalICMP] "ICMP local unsupported",\
    -> packet to ip address on ISA is not an echo request

[eNatFlowIcmpChecksumError]    "ICMP checksum error",\
    -> ICMP checksum error

[eNatFlowIcmpEmbeddedPktChecksumError] "ICMP embedded checksum
error",\
    -> checksum error on embedded IP header

[eNatFlowIcmpEmbeddedPktUnsupportedL4] "ICMP unsupported L4",\
    -> embedded IP packet is not UDP/TCP

[eNatFlowIcmpTooShort]        "ICMP too short",\
    -> packet too short to include the ICMP header

[eNatFlowIcmpLengthError]     "ICMP length error",\
    -> packet too short to include the embedded header

[eNatFlowPacketErrorNotIp]     "Pkt not IPv4 or IPv6",\
[eNatFlowPacketErrorRecv]     "Pkt rcv error",\
[eNatFlowPacketError]         "Pkt error",\
[eNatFlowPacketErrorIpv4HdrChk] "IPv4 header checksum
violation",\
[eNatFlowPacketErrorIpv4HdrMal] "IPv4 header malformed",\
[eNatFlowPacketErrorIpv4PktMal] "IPv4 malformed packet",\
[eNatFlowPacketErrorIpv4TtlZero] "IPv4 ttl zero",\
[eNatFlowPacketErrorIpv4Optv6Ext] "IPv4 opt /IPv6 ext headers",\
[eNatFlowPacketErrorIpv4Bad]   "IPv4 undefined error", \
[eNatFlowPacketErrorIpv6Frag]  "IPv6 fragments unsupported",\
[eNatFlowPacketErrorTcpUdpMal] "TCP/UDP malformed",\
[eNatFlowPacketErrorTcpUdpChk] "TCP/UDP checksum failure",\
[eNatFlowPacketErrorTcpUdpLen] "TCP/UDP length error",\
    -> malformed incoming packet

[eNatFlowPacketSendError]     "Pkt send error",\
    -> failed to tx the packet

```

```

[eNatFlowPacketNoCpyBuf]          "no buf to copy pkt",\
-> failed to copy the packet to another buffer needed for
correct processing

[eNatFlowLockedByMgmtCore]        "locked by mgmt core",\
-> resources temp. locked by the mgmt core

[eNatFlowPRLogFailed]            "port range log failed",\
-> port range log failed

[eNatFlowDslUnknownAFTR]         "DS Lite unknown AFTR",\
-> AFTR address is unrecognised

[eNatFlowTooManyFrgsForIpPkt]     "too many fragments for IP
packet",\
[eNatFlowTooManyFragmentedPkts]  "too many fragmented
packets",\
[eNatFlowTooManyFragHoles]       "too many fragment holes",\
[eNatFlowFragListExpire]         "fragment list expired",\
[eNatFlowTooManyFragBufs]        "too many frags buffered",\
[eNatFlowFragRateTooHigh]        "fragment rate too high",\
-> various fragment problems

[eNatFlowNoPolicy]               "no policy",\
->vrf not mapped to a policy

[eNatFlowLogFailed]              "flow log failed",\
-> flow logging can not follow the setup rate

[eNatFlowMultiHostOrSubscrIp]    "no multiple host or
subscr. IPs allowed",\
->multiple hosts or subscribers on the inside in use without
port translation

[eNatFlowToLocalError]           "to local ignored",\
-> radius authentication failure (?)

[eNatFlow64Disabled]             "NAT64 disabled",\
-> nat64 was disabled

[eNatFlow64InvalidSource]        "NAT64 invalid src addr",\
-> source address matches pref64

[eNatFlow64FragZeroChecksum]     "NAT64 frag has zero
checksum",\
-> v4 UDP frag has zero checksum

[eNatFlow64ZeroChecksum]         "NAT64 v4 has zero checksum",\
-> v4 UDP has zero checksum, and policy configured to drop

[eNatFlow64FragIcmp]             "NAT64 ICMP frag unsupported"\
->v4 fragmented ICMP

```

8.2.2.2 nat-policy

Syntax

```
nat-policy nat-policy-name associations
nat-policy nat-policy-name
nat-policy nat-policy-name statistics
nat-policy
```

Context

```
show>service>nat
```

Description

This command displays NAT policy information.

Parameters

nat-policy-name

specifies the NAT Policy name

associations

specifies the keyword that displays the router instances and/or subscriber profiles associated with the NAT policy, up to 32 characters

statistics

specifies the keyword that displays statistics of the specified NAT policy

Output

The following output example shows NAT policy information.

Example output

```
# show service nat nat-policy
=====
NAT policies
=====
Policy Description
-----
outPolicy
outPolicy2
outPolicy3
-----
No. of NAT policies: 3
=====

*A:SR12_PPP0E>show>router>nat#  show service nat nat-policy "priv-nat-policy"
```

```
=====
NAT Policy priv-nat-policy
=====
```

```
Pool : privpool
Router : Base
Filtering : endpointIndependent
Block limit : 4
Reserved ports : 0
Port usage High Watermark (%) : (Not Specified)
Port usage Low Watermark (%) : (Not Specified)
Port forwarding limit : 64
Session limit : 65535
Reserved sessions : 0
Session usage High Watermark (%) : (Not Specified)
Session usage Low Watermark (%) : (Not Specified)
Prioritized forwarding classes : (Not Specified)
Timeout TCP established (s) : 7440
Timeout TCP transitory (s) : 240
Timeout TCP SYN (s) : 15
Timeout TCP TIME-WAIT (s) : 0
Timeout UDP mapping (s) : 300
Timeout UDP initial (s) : 15
Timeout UDP DNS (s) : 15
Timeout ICMP Query (s) : 60
Timeout SIP Inactive Media (s) : 120
Subscriber retention (s) : 0
UDP inbound refresh : false
TCP MSS Adjust : (Not Specified)
Destination-NAT IP : (Not Specified)
IPFIX export policy : (Not Specified)
Last Mgmt Change : 01/28/2012 14:47:59
=====
```

```
*A:SR12_PPP0E>show>router>nat#
```

```
# show service nat nat-policy "outPolicy2" associations
```

```
=====
NAT Policy outPolicy2 Subscriber Profile Associations
=====
```

```
sub_prof_B_3
-----
```

```
No. of subscriber profiles: 1
=====
```

```
# show service nat nat-policy "outPolicy2" statistics
```

```
=====
NAT Policy outPolicy2 Statistics
=====
```

```
mda 3/1
-----
```

```
hostsActive : 1
hostsPeak : 1
sessionsTcpCreated : 0
sessionsTcpDestroyed : 0
sessionsUdpCreated : 0
sessionsUdpDestroyed : 0
sessionsIcmpQueryCreated : 0
sessionsIcmpQueryDestroyed : 0
```

8.2.2.3 port-forwarding-entries

Syntax

```
port-forwarding-entries
```

Context

```
show>router>nat
```

Description

This command displays port forwarding entries.

Output

The following output example shows port forwarding entry information.

Example output

```
# show service nat port-forwarding-entries
=====
NAT port forwarding entries
=====
Subscriber
iRtr      iAddress                prot iPort type
oRtr      oAddress                persist-id oPort expiry
=====
100       1.2.3.4                 tcp   666   classic-lsn-sub
Base      13.0.0.6                N/A   666   N/A

100       1.2.3.4                 udp   666   classic-lsn-sub
Base      13.0.0.6                N/A   666   N/A

-----
No. of entries: 2
=====
```

8.2.2.4 lsn-blocks

Syntax

```
lsn-blocks [inside-router router-instance] [inside-ip ip-address] [outside-ip-prefix ip-
prefix/length] [outside-port [1..65535]] [pool pool-name]
```

Context

show>router>nat

Description

This command displays large scale NAT blocks.

Parameters

router-instance

specifies the router instance name and service ID

Values	router-name service-id	
–	router-name	Base, management
–	service-id	1 to 2147483647
–	svc-name	specifies a string up to 64 characters

ip-address

specifies the IP address

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D
–	ip-prefix/length	

- ip-prefix a.b.c.d (host bits must be 0)
- length [1..32]

pool-name

specifies the pool name, up to 32 characters

Output

The following output example shows large scale NAT block information.

Example output

```
# show>router>nat# show router Base nat lsn-blocks
=====
Large-Scale NAT blocks for Base
=====
13.0.0.5 [1024..1527]
Pool                : privpool
Policy              : priv-nat-policy
Started             : 2012/01/28 19:10:17
Inside router       : vprn100
Inside IP address   : 2001:470:1F00:FFFF::189
-----
Number of blocks: 1
=====
```

8.2.2.5 pool**Syntax**

`pool pool-name`

`pool`

Context

`show>router>nat`

Description

This command displays NAT pool information.

Parameters

pool-name

specifies the pool name, up to 32 characters

Output

The following output example shows NAT pool information.

Example output

```
# show router nat pool
=====
NAT pools
=====
Pool NAT-group Type Admin-state
-----
MyPool 1 l2Aware inService
MyPool2 1 l2Aware inService
-----
No. of pools: 2
=====

# show>router>nat# show router "Base" nat pool "privpool"
=====
NAT Pool privpool
=====
ISA NAT Group           : 3
Pool type                : largeScale
Admin state              : inService
Mode                    : auto (napt)
Port forwarding range    : 1 - 1023
Port reservation        : 128 blocks
Block usage High Watermark (%) : (Not Specified)
Block usage Low Watermark (%)  : (Not Specified)
Subscriber limit per IP address : 65535
Active                   : true
Last Mgmt Change        : 01/28/2012 14:47:59
=====
NAT address ranges of pool privpool
=====
Range                                Drain Num-blk
-----
13.0.0.5 - 13.0.0.6                  1
-----
No. of ranges: 1
=====
NAT members of pool privpool ISA NAT group 3
=====
Member                                Block-Usage-% Hi
-----
1                                     < 1          N
2                                     < 1          N
-----
No. of members: 2
```

8.2.2.6 summary

Syntax

```
summary
```

Context

```
show>router>nat
```

Description

This command displays the NAT information summary.

Output

The following output example shows NAT information.

Example output

```
# show>router>nat# show router Base nat summary
=====
NAT pools
=====
Pool                NAT-group  Type           Admin-state
-----
privpool            3          largeScale    inService
pubpool             1          largeScale    inService
-----
No. of pools: 2
=====
```

8.2.3 NAT clear command descriptions

8.2.3.1 nat-group

Syntax

```
nat-group nat-group-id member [1..255] l2-aware-subscribers
nat-group nat-group-id member [1..255] statistics
```

Context

clear>nat>isa

Description

This command clears ISA NAT group commands related statistics or removes all the subscribers that are associated with a specific NAT group member.

Parameters

nat-group-id

specifies the NAT group ID to clear

Values: 1 to 15

statistics

specifies to clear the statistics for the NAT group ID

l2-aware-subscribers

specifies to clear the l2-aware subscribers for the NAT group ID (l2-aware is not supported)

8.2.4 NAT tools command descriptions

8.2.4.1 nat

Syntax

nat

Context

tools>dump

tools>perform

Description

This command enables the dump or perform tools for NAT.

8.2.4.2 isa

Syntax

isa

Context

```
tools>dump>nat
```

Description

This command enables the dump tools for NAT ISA.

8.2.4.3 resources

Syntax

```
resources mda mda-id
```

Context

```
tools>dump>nat>isa
```

Description

This command enables dump ISA resources for an MDA.

Output

The following output example shows MDA ISA resource information.

Example output

```
# tools dump nat isa resources mda 3/1
```

```
Resource Usage for Slot #3 Mda #1:
```

	Total	Allocated	Free
Flows	6291456	0	6291456
Policies	256	2	254
Port-ranges	1310720	128	1310592
Ports	12884901888	0	12884901888
IP-addresses	65536	1	65535
Large-scale hosts	524288	0	524288
Delayed ICMP's	200	0	200
LI entries	8191	0	8191
Upstream fragment lists	16384	0	16384
Downstream fragment lists	16384	0	16384
Upstream fragment holes	131072	0	131072
Downstream fragment holes	131072	0	131072
Upstream fragment bufs	13824	0	13824
Downstream fragment bufs	13824	0	13824
flow log dest. set 0	2	0	2
flow log packets set 0	50	0	50
flow log dest. set 1	2	0	2

flow log packets set 1	50	0	50
flow log dest. set 2	1	0	1
flow log packets set 2	50	0	50

8.2.4.4 sessions

Syntax

```
sessions [nat-group nat-group-id] [mda mda-id] [protocol {icmp|tcp|udp}] [inside-ip ip-address] [inside-router router-instance] [inside-port port-number] [outside-ip ip-address] [outside-port port-number] [foreign-ip ip-address] [foreign-port port-number]
```

Context

```
tools>dump>nat
```

Description

This command dumps ISA sessions.

Output

The following output example shows ISA session information.

Example output

```
# tools dump nat sessions
=====
Matched 2 sessions on Slot #3 MDA #1
=====
Owner          : LSN-Host@1.2.3.4
Router         : 100
FlowType       : UDP PortFwd
Inside IP Addr : 1.2.3.4      Inside Port      : 666
Outside IP Addr : 13.0.0.6   Outside Port     : 666
Foreign IP Addr : *          Foreign Port      : *
Dest IP Addr   : *          Dest Port           : *
-----
Owner          : LSN-Host@1.2.3.4
Router         : 100
FlowType       : TCP PortFwd
Inside IP Addr : 1.2.3.4      Inside Port      : 666
Outside IP Addr : 13.0.0.6   Outside Port     : 666
Foreign IP Addr : *          Foreign Port      : *
Dest IP Addr   : *          Dest Port           : *
-----
=====

=====
Matched 1 session on Slot #3 MDA #2
=====
```

```
Owner      : LSN-Host@2001:470:1F00:FFFF::189
Router     : 100
FlowType   : TCP                               Timeout (sec)   : 6769
Inside IP Addr : 138.203.16.218             Inside Port    : 41555
Outside IP Addr : 13.0.0.5                 Outside Port   : 1529
Foreign IP Addr : 15.0.0.1                 Foreign Port   : 22
Dest IP Addr  : 15.0.0.1                   Dest Port      : 22
-----
=====
```

8.2.4.5 histogram

Syntax

```
histogram router router-instance pool pool-name bucket-size [1..65536] num-buckets [2..50]
```

Context

```
tools>dump>nat
```

Description

This command displays a NAT pool port usage histogram.

Parameters

router-instance

specifies the router instance

Values	router-name vprn-svc-id		
–	router-name	Base	
–	vprn-svc-id	[1..2147483647]	

pool-name

specifies the identification of the NAT pool

bucket-size [*1..65536*]

specifies the unit of the X-axis of the histogram; a value of ten, for example, will return in a histogram with results for [0-9], [10-19], [20-29],... ports

num-buckets [2..50]

specifies the size of the histogram; a value of five, for example, will return five results: [0-9], [10-19], [20-29], [30-39], [40-infinite]

8.2.4.6 port-forwarding-action

Syntax

```
port-forwarding-action
```

Context

```
tools>perform>nat
```

Description

This command displays NAT port forwarding actions.

8.2.4.7 lsn

Syntax

```
lsn create router router-instance [b4 ipv6-address] [afr ipv6-address] ip ip-address  
protocol {tcp | udp} [port port] lifetime lifetime [outside-ip ipv4-address] [outside-port  
port]
```

```
lsn delete router router-instance [b4 ipv6-address] ip ip-address protocol {tcp | udp} port  
port
```

```
lsn modify router router-instance [b4 ipv6-address] ip ip-address protocol {tcp | udp} port  
port lifetime lifetime
```

Context

```
tools>perform>nat>port-forwarding-action
```

Description

This command enables large-scale NAT port forwarding actions.

Output

The following output example shows large-scale NAT port forwarding information.

Example output

```
# tools perform nat port-forwarding-action lsn create router 100
ip 1.2.3.4 protocol tcp lifetime infinite outside-port 666
# tools perform nat port-forwarding-action lsn create router 100
ip 1.2.3.4 protocol udp lifetime infinite outside-port 666
  configure system persistence nat-port-forwarding location cf3:
# tools dump persistence nat-port-forwarding
-----
Persistence Information
-----
Client                : nat-fwds
File Info :
  Filename             : cf3:\nat_fwds.002
  File State           : CLOSED (Not enough space on disk)
Subsystem Info :
  Nbr Of Registrations : 524288
  Registrations In Use : 2
  Subsystem State      : NOK

show+service+nat
|  |  |  +---lsn-subscribers
|  |  |  +---nat-policy
|  |  |  +---port-forwarding-entries
|  |  |  |  +---classic-lsn-sub
|  |  |  |  +---dslite-lsn-sub
|  |  |  |  +---nat64-lsn-sub
```


9 Lawful Interception command reference

This section describes the Lawful interception-related commands.

9.1 Lawful Interception command hierarchies

- *Lawful Interception configuration command hierarchies*
- *Lawful Interception show command hierarchies*

9.1.1 Lawful Interception configuration command hierarchies

The following LI commands are described in this guide and the *7450 ESS, 7750 SR, 7950 XRS, and VSR OAM and Diagnostics Guide*.

```
configure
- li
- li-filter
- li-ip-filter li-filter-name [create]
- no li-ip-filter li-filter-name
- description description-string
- no description
- entry li-entry-id [create]
- no entry li-entry-id
- description description-string
- no description
- match [protocol protocol-id]
- no match
- dst-ip {ipv4-address/mask | ip-address ipv4-address-mask}
- no dst-ip
- dst-port {lt | gt | eq} dst-port-number
- dst-port range start end
- no dst-port
- src-ip ipv4-address/mask | ip-address ipv4-address-mask
- no src-ip
- src-port {lt | gt | eq} src-port-number
- src-port range start end
- no src-port
- li-ipv6-filter li-filter-name [create]
- no li-ipv6-filter li-filter-name
- description description-string
- no description
```

```

- entry li-entry-id
- no entry
  - description description-string
  - no description
  - match [next-header next-header]
  - no match
    - dst-ip ipv6-address/prefix-length
    - no dst-ip
    - dst-port {lt | gt | eq} dst-port-number
    - dst-port range start end
    - no dst-port
    - src-ip ipv6-address/prefix-length
    - no src-ip
    - src-port {lt | gt | eq} src-port-number
    - src-port range start end
    - no src-port
- li-mac-filter li-filter-name [create]
- no li-mac-filter li-filter-name
  - description description-string
  - no description
  - entry li-entry-id
  - no entry
    - description description-string
    - no description
    - match [frame-type {802dot3 | 802dot2-llc | 802dot2-snap | ethernet_II}]
    - no match
      - dst-mac ieee-address [ieee-address-mask]
      - no dst-mac
      - src-mac ieee-address [ieee-address-mask]
      - no src-mac
- li-filter-associations
  - [no] li-ip-filter li-filter-name
  - [no] ip-filter ip-filter-id
  - [no] li-ipv6-filter li-filter-name
  - [no] ipv6-filter ipv6-filter-id
  - [no] li-mac-filter li-filter-name
  - [no] mac-filter mac-filter-id
- li-filter-block-reservation
  - li-reserved-block block-name [create]
  - no li-reserved-block block-name
    - description description-string
    - no description
    - [no] ip-filter ip-filter-id
    - [no] ipv6-filter ipv6-filter-id
    - [no] mac-filter mac-filter-id

```

```

    - [no] start-entry entry-id count count
  - li-filter-lock-state {locked | unlocked-for-li-users | unlocked-for-all-users}
  - no li-filter-lock-state
  - [no] li-source mirror-service-id [name mirror-service-name]
    - ip-filter ip-filter-id entry entry-id [entry-id...(up to 8 max)] [intercept-id intercept-id [intercept-id...up to 8 max]] [session-id session-id [session-id...up to 8 max]]
    - no ip-filter ip-filter-id [entry entry-id [entry-id...(up to 8 max)]]
    - ipv6-filter ipv6-filter-id entry entry-id [entry-id...(up to 8 max)] [intercept-id intercept-id [intercept-id...up to 8 max]] [session-id session-id [session-id...(up to 8 max)]]
    - no ipv6-filter ipv6-filter-id [entry entry-id [entry-id...(up to 8 max)]]
    - li-ip-filter li-filter-name [entry entry-id...] [intercept-id id] [session-id id]
    - no li-ip-filter li-filter-name [entry entry-id...]
    - li-ipv6-filter li-filter-name [entry entry-id...] [intercept-id intercept-id...] [session-id session-id...]
    - no li-ipv6-filter li-filter-name [entry entry-id...]
    - li-mac-filter li-filter-name entry entry-id [entry-id...] [intercept-id id] [session-id id]
    - no li-mac-filter li-filter-name entry entry-id [entry-id...]
    - mac-filter mac-filter-id entry entry-id [entry-id...] [intercept-id intercept-id] [session-id id]
    - no mac-filter li-filter-name entry entry-id [entry-id...]
  - nat
    - [no] classic-lsn-sub router router-instance ip ip-address
      - intercept-id id
      - no intercept-id
      - session-id session-id
      - no session-id
    - [no] dslite-lsn-sub router router-instance b4 ipv6-prefix
      - intercept-id id
      - no intercept-id
      - session-id session-id
      - no session-id
    - ethernet-header [da ieee-address] [sa ieee-address] [etype ethertype]
    - no ethernet-header
    - [no] l2-aware-sub sub-ident-string
      - intercept-id id
      - no intercept-id
      - session-id session-id
      - no session-id
    - [no] dslite-lsn-sub router router-instance ip ipv6-prefix
      - intercept-id id
      - no intercept-id

```

```

    - session-id id
    - no session-id
  - sap sap-id {[ingress] [egress]} [intercept-id id] [session-id session-id]
  - no sap sap-id {[ingress] [egress]}
  - [no] shutdown
  - subscriber sub-ident-string [sap sap-id [ip ip-address] [mac ieee-address] |
```

sla-profile *sla-profile-name*] [*fc* {[*be*] [*l2*] [*af*] [*l1*] [*h2*] [*ef*] [*h1*] [*nc*]}}] {[*ingress*] [*egress*]} [*intercept-id* *intercept-id*] [*session-id* *session-id*]

```

    - no subscriber sub-ident-string
  - log
    - [no] log-id log-id
    - description description-string
    - no description
    - filter filter-id
    - no filter
    - from li
    - no from
    - [no] shutdown
    - time-format {local | utc}
    - to memory [size]
    - to session
    - to snmp [size]

```

The following mobile gateway LI commands are described in this guide.

```

configure
- li
  - mobile-gateway
    - 3gpp-5g-release release-num
    - no 3gpp-5g-release
    - 3gpp-release release
    - no 3gpp-release
    - [no] allow-duplicate-msisdn-or-imei
    - client-tls-profile tls-profile-name
    - no client-tls-profile
    - custom-correlation-id-format {enable | disable}
    - no custom-correlation-id-format
    - df-peer df-peer-id [df2-addr ip-address df2-port port] [df3-addr ip-address
df3-port port] [df2-tls-profile profile-name] [df3-tls-profile profile-name]
    - no df-peer df-peer-id
    - df3-tcp-connections {true | false}
    - no df3-tcp-connections
    - [no] fan-out
    - [no] include-rai
    - iri

```

```

    - [no] gw-function-iri-mode
    - [no] iri-per-mb-request
    - neid neid
    - no neid
    - [no] te-id
    - [no] x2-iri-ice-type
- li-t3
  - alpn {enable | disable}
  - no alpn
  - li-t3-local-interface ip-address [router router-instance] [local-port port-
num]
  - no li-t3-local-interface
  - li-t3-mgmt-local-interface ip-address [router router-instance] [local-
port port-num]
  - no li-t3-mgmt-local-interface
  - li-t3-up-peer df-peer-id upf-addr upf-peer-address [t3-port t3-port-num] upf-
id uuid
  - no li-t3-up-peer df-peer-id
  - sni {enable | disable}
  - no sni
- li-x1
  - admf-peer df-peer-id admf-addr admf-peer-address [x1-port x1-port-num]
  - no admf-peer df-peer-id
  - alpn {enable | disable}
  - no alpn
  - li-x1-keep-alive-time-p2 x1-time-p2-value
  - no li-x1-keep-alive-time-p2
  - li-x1-local-interface ip-address [router router-instance] [local-port port-
num]
  - no li-x1-local-interface
  - sni {enable | disable}
  - no sni
- li-x1-mgmt
  - alpn {enable | disable}
  - no alpn
  - li-x1-mgmt-admf-peer df-peer-id admf-addr admf-peer-address [x1-port x1-port-
num]
  - no li-x1-mgmt-admf-peer df-peer-id
  - li-x1-mgmt-local-interface ip-address [router router-instance] [local-
port port-num]
  - no li-x1-mgmt-local-interface
  - sni {enable | disable}
  - no sni
- li-x2
  - li-x2-keep-alive-time-p1 x2-time-p1-value

```

```

    - no li-x2-keep-alive-time-p1
    - li-x2-keep-alive-time-p2 x2-time-p2-value
    - no li-x2-keep-alive-time-p2
  - li-x3
    - li-x3-keep-alive-time-p1 x3-time-p1-value
    - no li-x3-keep-alive-time-p1
    - li-x3-keep-alive-time-p2 x3-time-p2-value
    - no li-x3-keep-alive-time-p2
    - li-x3-sequence {enable | disable}
    - no li-x3-sequence
    - li-x3-tls {enable | disable}
    - no li-x3-tls
  - local-interface ip-address [router router-instance] [override-x2-interface x2-
ip-address [x2-router x2-router-instance]]
    - no local-interface
    - nf-id-value {uuid | ip-addr | ip-addr-hex-string}
    - no nf-id-value
    - num-conn-retries num-of-retries
    - no num-conn-retries
    - operator-id id-string
    - no operator-id
    - pcrf-shared-key key-value
    - no pcrf-shared-key
    - [no] per-bearer-seq-num
    - pfcpl-shared-key key-value
    - no pfcpl-shared-key
    - server-tls-profile tls-profile-name
    - no server-tls-profile
    - target target-type id target-value intercept intercept peer df-peer-id
[liid liid-string] [secondary-peer sec-df-peer-id] [ha-peer-id ha-df-peer-id]
[correlation-id correlation-id] [bearer-id bearer-id]
    - no target target-type id target-value [liid liid-string]
    - tcp-keepalive
      - idle-time idle
      - no idle-time
      - interval interval
      - no interval
      - retries count
      - no retries
      - [no] shutdown
    - [no] tls
    - x2-iri-cache-size cache-size
    - no x2-iri-cache-size
    - x2-iri-qos dscp {dscp-value | dscp-name}
    - no x2-iri-qos

```

- `x2-keep-alive-time` `keep-alive-time`
- `no x2-keep-alive-time`
- `x3-cc-cache-size` `cache-size`
- `no x3-cc-cache-size`
- `x3-cc-qos` `dscp` {`dscp-value` | `dscp-name`}
- `no x3-cc-qos`
- `x3-keep-alive-time` `keep-alive-time`
- `no x3-keep-alive-time`
- `x3-transport` {`tcp` | `udp`} [`ulic-header` {`v0` | `v1`}]
- `save`
- `tech-support` {`standard` | `limited-li-content`}
- `no tech-support`

The following commands that may be used for LI configuration are described in this guide and in the *7450 ESS, 7750 SR, 7950 XRS, and VSR OAM and Diagnostics Guide*

- bof
- [no] `li-local-save`
 - [no] `li-memory-encrypt`
 - [no] `li-separate`
 - [no] `save` [`cflash-id`]

The following commands are described in this guide and in the *7450 ESS, 7750 SR, 7950 XRS, and VSR System Management Guide*.

- configure
- system
 - security
 - [no] `profile` `user-profile-name`
 - [no] `li`
 - user
 - [no] `access` [`ftp`] [`snmp`] [`console`] [`li`]

- bof
- [no] `li-local-save`
 - [no] `li-separate`

9.1.2 Lawful Interception show command hierarchies

The following commands are described in this guide and the *7450 ESS, 7750 SR, 7950 XRS, and VSR OAM and Diagnostics Guide*.

- show
- `li`

```

– filter li-ip filter-name {counter | associations}
– filter li-ip filter-name entry entry-id [counters]
– filter li-ipv6 filter-name {counter | associations}
– filter li-ipv6 filter-name entry entry-id [counters]
– li-source [service-id]
– log
  – log-id [log-id] [severity severity-level] [application application]
[sequence from-seq [to-seq]] [count count] [router router-instance [expression]]
[subject subject [regexp]] [ascending | descending]
  – status
– service
  – active-subscribers summary
  – active-subscribers [subscriber sub-ident-string [sap sap-id sla-profile sla-
profile-name]] [detail | mirror]
  – active-subscribers hierarchy [subscriber sub-ident-string]
  – service-using mirror

```

The following commands are described in this guide:

```

show
– li
– mobile-gateway df-peer [df-peer-id]
– mobile-gateway df-peer ip-addr ip-address | ipv6-address
– mobile-gateway summary
– mobile-gateway target [target-type] [id target-value] [gx] [liid liid-string]
– mobile-gateway target target-type id target-value detail
– mobile-gateway target target-type id target-value stats

```

9.2 Lawful Interception command descriptions

- [Lawful Interception configuration command descriptions](#)
- [Lawful Interception show command descriptions](#)

9.2.1 Lawful Interception configuration command descriptions

9.2.1.1 Generic Lawful Interception Commands

9.2.1.1.1 description

Syntax

```
description description-string  
no description
```

Context

```
config>li>li-filter>li-ip-filter  
config>li>li-filter>li-ip-filter>entry  
config>li>li-filter>li-ipv6-filter  
config>li>li-filter>li-ipv6-filter>entry  
config>li>li-filter>li-mac-filter  
config>li>li-filter>li-mac-filter>entry  
config>li>li-filter-block-reservation>li-reserved-block  
config>li>log>log-id  
config>mirror>mirror-dest
```

Description

This command creates a text description stored in the configuration file for a configuration context to help the administrator identify the content of the file.

The no form of this command removes the description string from the configuration.

Parameters

description-string

The description character string. Allowed values are any string up to 80 characters long composed of printable, 7-bit ASCII characters. If the string contains special characters (#, \$, spaces, and so on), the entire string must be enclosed within double quotes.

9.2.1.1.2 shutdown

Syntax

```
[no] shutdown
```

Context

config>li>li-source

config>li>log>log-id

config>mirror>mirror-dest

config>mirror>mirror-dest>spoke-sdp>egress

debug>mirror-source

Description

The shutdown command administratively disables an entity. When disabled, an entity does not change, reset, or remove any configuration settings or statistics. Many entities must be explicitly enabled using the `no shutdown` command.

The operational state of the entity is disabled as well as the operational state of any entities contained within. Many objects must be shut down before they may be deleted.

Unlike other commands and parameters where the default state is not indicated in the configuration file, shutdown and `no shutdown` are always indicated in system generated configuration files.

The `no` form of this command puts an entity into the administratively enabled state.

Default

See Special Cases below.

Mirror Destination

When a mirror destination service ID is shutdown, mirrored packets associated with the service ID are not accepted from the mirror source or remote source router. The associated mirror source is put into an operationally down mode. Mirrored packets are not transmitted out of the SAP or SDP. Each mirrored packet is silently discarded. If the mirror destination is a SAP, the SAP discard counters are incremented.

The shutdown command places the mirror destination service or mirror source into an administratively down state. The `mirror-dest` service ID command must be shut down in to delete the service ID, SAP or SDP association from the system.

The default state for a mirror destination service ID is shutdown. A `no shutdown` command is required to enable the service.

Mirror Source

Mirror sources do not need to be shutdown to remove them from the system.

When a mirror source is shutdown, mirroring is terminated for all sources defined locally for the `mirror-dest` service ID. If the `remote-source` command has been executed on the `mirror-dest` associated with the shutdown mirror-source mirroring continues for remote sources.

The default state for a mirror source for a given `mirror-dest` service ID is `no shutdown`. A shutdown command is required to disable mirroring from that mirror-source.

9.2.1.2 Lawful Interception command descriptions

The following commands are also described in the *7450 ESS, 7750 SR, 7950 XRS, and VSR OAM and Diagnostics Guide*.

9.2.1.2.1 li

Syntax

```
li
```

Context

```
config
```

Description

This command configures the context to configure the LI parameters.

9.2.1.2.2 li-filter

Syntax

```
li-filter
```

Context

```
config>li
```

Description

This command enables the context to create lawful intercept filter lists and entries.

9.2.1.2.3 li-ip-filter

Syntax

```
li-ip-filter li-filter-name [create]
```

```
no li-ip-filter li-filter-name
```

Context

```
config>li>li-filter
```

Description

This command creates a Lawful Interception (LI) IPv4 filter list, or enters the CLI context for a LI IPv4 filter list. LI IPv4 filters are used as a manner to create confidential IPv4 filter-based li-source entries. The LI IPv4 filter entries are inserted/merged into normal IPv4 filters as configured via the `li-filter-associations` and `li-filter-block-reservation` commands, but the LI IPv4 filter entries are not visible to users without LI permissions.

Parameters

li-filter-name

specifies the name of the IPv4 address filter; filter names cannot start with an underscore character (for example, “_my-filter”) and cannot use the name “default”; up to 32 characters

9.2.1.2.4 entry

Syntax

```
entry li-entry-id [create]
```

```
no entry li-entry-id
```

Context

```
config>li>li-filter>li-ip-filter
```

```
config>li>li-filter>li-ipv6-filter
```

```
config>li>li-filter>li-mac-filter
```

Description

This command creates or edits a Lawful Interception filter entry. Multiple entries can be created using unique entry-id numbers within the filter.

An entry in a LI filter always has an implicit action of “forward”.

The no form of this command removes the specified entry from the filter. Entries removed from the filter are immediately removed from all services or network ports where the associated filter is applied.

LI filter entries can be used as li-source entries.

The entry numbers for li filters serve purely as keys for managing the entries (deleting entries, and so on). The order of LI filter entries is not guaranteed to match the entry numbers and s/w may reorder entries.

Operators must use LI entries in a manner such that relative order of the LI entries amongst themselves is not important.

Parameters

li-entry-id

identifies the Lawful Interception filter entry

Values: 1 to 65536

9.2.1.2.5 match

Syntax

```
match [protocol protocol-id]
```

```
no match
```

Context

```
config>li>li-filter>li-ip-filter>entry
```

Description

This command enables context to enter match criteria for LI IPv4 filter and optionally allows specifying protocol value to match on.

If more than one match criterion are configured then all criteria must be satisfied for a match to occur (logical “AND”). Multiple criteria must be configured within a single match context for a given entry.

The no form of this command removes the match criteria for the entry.

Parameters

protocol

The **protocol** keyword configures an IP protocol to be used as an IP filter match criterion.

The protocol type such as TCP or UDP is identified by its respective protocol number.

protocol-id

configures the decimal value representing the IP protocol to be used as an IP filter match criterion. Well known protocol numbers include ICMP(1), TCP(6), UDP(17)

Values: 0 to 255 (values can be expressed in decimal, hexadecimal, or binary - DHB)

Keywords: none | icmp | igmp | ip | tcp | egp | igp | udp | rdp | ipv6 | ipv6-route | ipv6-frag | idrp | rsvp | gre | ipv6-icmp | ipv6-no-nxt | ipv6-opts | iso-ip | eigrp | ospf-igp | ether-ip | encap | pnni | pim | vrrp | l2tp | stp | ptp | isis | crtp | crudp | sctp | * - udp/tcp wildcard

Match configuration protocols lists the match configuration protocols.

Table 178: Match configuration protocols

Protocol	Protocol ID	Description
icmp	1	Internet Control Message
igmp	2	Internet Group Management
ip	4	IP in IP (encapsulation)
tcp	6	Transmission Control
egp	8	Exterior Gateway Protocol
igp	9	Any private interior gateway (used by Cisco for IGRP)
udp	17	User Datagram
rdp	27	Reliable Data Protocol
ipv6	41	IPv6
ipv6-route	43	Routing Header for IPv6
ipv6-frag	44	Fragment Header for IPv6
idrp	45	Inter-Domain Routing Protocol
rsvp	46	Reservation Protocol
gre	47	General Routing Encapsulation
ipv6-icmp	58	ICMP for IPv6
ipv6-no-nxt	59	No Next Header for IPv6

Table 178: Match configuration protocols (continued)

Protocol	Protocol ID	Description
ipv6-opts	60	Destination Options for IPv6
iso-ip	80	ISO Internet Protocol
eigrp	88	EIGRP
ospf-igp	89	OSPFIGP
ether-ip	97	Ethernet-within-IP Encapsulation
encap	98	Encapsulation Header
pnni	102	PNNI over IP
pim	103	Protocol Independent Multicast
vrrp	112	Virtual Router Redundancy Protocol
l2tp	115	Layer Two Tunneling Protocol
stp	118	Spanning Tree Protocol
ptp	123	Performance Transparency Protocol
isis	124	ISIS over IPv4
crtip	126	Combat Radio Transport Protocol
crudp	127	Combat Radio User Datagram

9.2.1.2.6 dst-ip

Syntax

```
dst-ip {ip-address/mask | ip-address ipv4-address-mask}
```

Context

```
config>li>li-filter>li-ip-filter>entry>match
```

Description

This command configures destination IP address LI filter match criterion.

The no form of this command removes configured destination IP address; match criterion is ignored.

Parameters

ip-address

any address specified in dotted quad notation

Values: a.b.c.d

mask

eight 16-bit hexadecimal pieces representing bit match criteria

Values: 1 to 32

ipv4-address-mask

any mask expressed in dotted quad notation

Values: 0.0.0.0 to 255.255.255.255

9.2.1.2.7 dst-port

Syntax

```
dst-port {lt | gt | eq} dst-port-number
```

```
dst-port range start end
```

```
no dst-port
```

Context

```
config>li>li-filter>li-ip-filter>entry>match
```

```
config>li>li-filter>li-ipv6-filter>entry>match
```


Description

This command configures a destination TCP or UDP port number or port range for an IP LI filter match criterion. An entry containing Layer 4 match criteria will not match non-initial (2nd, 3rd, and so on) fragments of a fragmented packet since only the first fragment contains the Layer 4 information.

The no form of this command removes the destination port match criterion.

Parameters

lt | gt | eq

specifies the operator to use relative to *dst-port-number* for specifying the port number match criteria

lt

specifies all port numbers less than *dst-port-number* match

gt

specifies all port numbers greater than *dst-port-number* match

eq

specifies that *dst-port-number* must be an exact match

dst-port-number

the destination port number to be used as a match criteria expressed as a decimal integer

Values: 0 to 65535

range *start end*

specifies an inclusive range of port numbers to be used as a match criteria. The destination port numbers *start* and *end* are expressed as decimal integers.

Values: 0 to 65535

9.2.1.2.8 src-ip

Syntax

```
src-ip {ip-address/ mask | ip-address ipv4-address-mask}
```

```
no src-ip
```

Context

```
config>li>li-filter>li-ip-filter>entry>match
```

```
config>li>li-filter>li-ipv6-filter>entry>match
```

Description

This command configures source IP address LI filter match criterion.

The no form of this command removes any configured source IP. The match criterion is ignored.

Default

no src-ip

Parameters

ip-address

any address specified as dotted quad

Values: a.b.c.d

mask

eight 16-bit hexadecimal pieces representing bit match criteria

Values: 1 to 32

ipv4-address-mask

any mask expressed in dotted quad notation

Values: 0.0.0.0 to 255.255.255.255

9.2.1.2.9 src-port

Syntax

src-port {lt | gt | eq} *src-port-number*

src-port range *start end*

no src-port

Context

config>li>li-filter>li-ip-filter>entry>match

config>li>li-filter>li-ipv6-filter>entry>match

Description

This command configures a source TCP or UDP port number or port range for an IP LI filter match criterion. An entry containing Layer 4 match criteria will not match non-initial (2nd, 3rd, and so on) fragments of a fragmented packet since only the first fragment contains the Layer 4 information.

The no form of this command removes the source port match criterion.

Default

no src-port

Parameters

lt | gt | eq

specifies the operator to use relative to *src-port-number* for specifying the port number match criteria

lt specifies all port numbers less than *src-port-number* match.

gt specifies all port numbers greater than *src-port-number* match.

eq specifies that *src-port-number* must be an exact match.

src-port-number

the source port number to be used as a match criteria expressed as a decimal integer

Values: 0 to 65535

start end

specifies an inclusive range of port numbers to be used as a match criteria. The source port numbers *start-port* and *end-port* are expressed as decimal integers.

Values: 0 to 65535

9.2.1.2.10 li-ipv6-filter

Syntax

```
li-ipv6-filter filter-name [create]
```

```
no li-ipv6-filter filter-name
```

Context

```
config>li>li-filter
```

Description

This command creates a Lawful Interception (LI) IPv6 filter list, or enables the CLI context for a LI IPv6 filter list. LI IPv6 filters are used as a manner to create confidential IPv6 filter-based li-source entries. The LI IPv6 filter entries are inserted/merged into normal IPv6 filters as configured via the li-filter-associations and li-filter-block-reservation commands, but the LI IPv6 filter entries are not visible to users without LI permissions.

Parameters

filter-name

specifies the name of the IPv6 address filter; filter names cannot start with an underscore character (for example, “_my-filter”) and cannot use the name “default”

9.2.1.2.11 match

Syntax

```
match [next-header next-header]  
no match
```

Context

```
config>li>li-filter>li-ipv6-filter>entry
```

Description

This command enables context to enter match criteria for LI IPv6 filter and optionally allows specifying IPv6 next-header value to match on.

If more than one match criterion are configured then all criteria must be satisfied for a match to occur (logical “AND”). Multiple criteria must be configured within a single match context for a given entry.

The no form of this command removes the match criteria for the entry.

Parameters

next-header

Specifies the IPv6 next header to match. This parameter is analogous to the protocol parameter used in IP-Filter match criteria.

Values: 1 to 42 | 45 to 49 | 52 to 59 | 61 to 255 - protocol numbers accepted in decimal, hexadecimal, or binary - DHB

Keywords: none | icmp | igmp | ip | tcp | egp | igp | udp | rdp | ipv6 | idrp | rsvp | gre | ipv6-icmp | ipv6-no-nxt | iso-ip | eigrp | ospf-igp | ether-ip | encap | pnni | pim | vrrp | l2tp | stp | ptp | isis | crtp | crudp | sctp | * - udp/tcp wildcard

9.2.1.2.12 li-mac-filter

Syntax

```
li-mac-filter li-filter-name [create]  
no li-mac-filter li-filter-name
```

Context

```
config>li>li-filter
```

Description

This command creates a Lawful Interception (LI) MAC filter list, or enables the CLI context for a LI MAC filter list. LI MAC filters are used as a manner to create confidential MAC filter-based li-source entries. The LI MAC filter entries are inserted/merged into normal MAC filters as configured via the li-filter-associations and li-filter-block-reservation commands, but the LI MAC filter entries are not visible to users without LI permissions.

Parameters

li-filter-name

specifies the name of the MAC filter; filter names cannot start with an underscore character (for example, “_my-filter”) and cannot use the name “default”; up to 32 characters

9.2.1.2.13 match

Syntax

```
match [frame-type {802dot3 | 802dot2-llc | 802dot2-snap | ethernet_II}]
no match
```

Context

```
config>li>li-filter>li-mac-filter>entry
```

Description

This command enables the context for entering/editing match criteria for the filter entry and specifies an Ethernet frame type for the entry.

If more than one match criteria (within one match statement) are configured then all criteria must be satisfied (AND function) for a match to occur. A match context may consist of multiple match criteria, but multiple match statements cannot be entered per entry.

The no form of this command removes the match criteria for the entry.

Parameters

frame-type

filters can continue to be edited by all users even when an li-source references an entry in that filter

Values:

802dot3

specifies the frame type is Ethernet IEEE 802.3

802dot2-llc

specifies the frame type is Ethernet IEEE 802.2 LLC

802dot2-snap

specifies the frame type is Ethernet IEEE 802.2 SNAP

ethernet_II

specifies the frame type is Ethernet Type II

Default: 802dot3

9.2.1.2.14 dst-mac

Syntax

`dst-mac ieee-address [ieee-address-mask]`

`no dst-mac`

Context

`config>li>li-filter>li-mac-filter>entry>match`

Description

Configures a destination MAC address or range to be used as a MAC filter match criterion.

The no form of this command removes the destination mac address as the match criterion.

Default

`no dst-mac`

Parameters

ieee-address

Enter the 48-bit IEEE mac address to be used as a match criterion.

Values: HH:HH:HH:HH:HH:HH or HH-HH-HH-HH-HH-HH where H is a hexadecimal digit

ieee-address-mask

This 48-bit mask can be configured using the values from *Destination MAC address mask formats*:

Table 179: Destination MAC address mask formats

Format style	Format syntax	Example
Decimal	DDDDDDDDDDDDDDDD	281474959933440
Hexadecimal	0xHHHHHHHHHHHHH	0x0FFFFFF000000
Binary	0bBBBBBBB...B	0b11110000...B

To configure so that all packets with a destination MAC OUI value of 00-03-FA are subject to a match condition then the entry should be specified as: 003FA000000 0xFFFFFFFF000000

Default: 0xFFFFFFFFFFFF (exact match)

Values: 0x0000000000000000 to 0xFFFFFFFFFFFFFFF

9.2.1.2.15 src-mac

Syntax

```
src-mac ieee-address [ieee-address-mask]
```

```
no src-mac
```

Context

```
config>li>li-filter>li-mac-filter>entry>match
```

Description

Configures a source MAC address or range to be used as a MAC filter match criterion.

The no form of this command removes the source mac as the match criteria.

Default

```
no src-mac
```

Parameters

ieee-address

Enter the 48-bit IEEE mac address to be used as a match criterion.

Values: HH:HH:HH:HH:HH:HH or HH-HH-HH-HH-HH-HH where H is a hexadecimal digit

ieee-address-mask

This 48-bit mask can be configured using the values from *Source MAC address mask formats*:

Table 180: Source MAC address mask formats

Format style	Format syntax	Example
Decimal	DDDDDDDDDDDDDDDD	281474959933440
Hexadecimal	0xHHHHHHHHHHHHH	0x0FFFFFF000000
Binary	0bBBBBBBB...B	0b11110000...B

To configure so that all packets with a source MAC OUI value of 00-03-FA are subject to a match condition then the entry should be specified as: 003FA000000 0xFFFFFFFF000000

Default: 0xFFFFFFFFFFFF (exact match)

Values: 0x0000000000000000 to 0xFFFFFFFFFFFFFF

9.2.1.2.16 li-filter-associations

Syntax

li-filter-associations

Context

config>li

Description

This command enables context to define which LI filter entries get inserted into which normal filters.

9.2.1.2.17 li-ip-filter

Syntax

[no] li-ip-filter *li-filter-name*

Context

```
config>li>li-filter-assoc
```

Description

This command configures the LI IP filter that will have its entries inserted into a list of normal IP filters.

Parameters

li-filter-name

specifies an existing LI IP filter, up to 32 characters

9.2.1.2.18 ip-filter

Syntax

```
[no] ip-filter filter-id
```

Context

```
config>li>li-filter-assoc>li-ip-filter
```

Description

This command specifies the IP-filters into which the entries from the specified li-ip-filter are to be inserted. The LI IP filter and IP filter must already exist before the association is made. If the normal IP filter is deleted then the association is also removed (and not re-created if the IP filter comes into existence in the future).

Parameters

filter-id

an existing IP filter policy

Values: 1 to 65536 | name: up to 64 characters

9.2.1.2.19 li-ipv6-filter

Syntax

```
[no] li-ipv6-filter li-filter-name
```

Context

```
config>li>li-fltr-assoc
```

Description

Specifies the LI IPv6 filter that will have its entries inserted into a list of normal IPv6 filters.

Parameters

li-filter-name

an existing LI IPv6 filter, up to 32 characters

9.2.1.2.20 ipv6-filter

Syntax

```
[no] ipv6-filter ipv6-filter-id
```

Context

```
config>li>li-fltr-assoc>li-ipv6-filter
```

Description

Specifies the IP filters into which the entries from the specified li-ipv6-filter are to be inserted. The LI IPv6 filter and IPv6 filter must already exist before the association is made. If the normal IPv6 filter is deleted then the association is also removed (and not re-created if the IPv6 filter comes into existence in the future).

Parameters

ipv6-filter-id

An existing IPv6 filter policy

Values: 1 to 65536 | name: up to 64 characters

9.2.1.2.21 li-mac-filter

Syntax

```
[no] li-mac-filter li-filter-name
```

Context

```
config>li>li-filter-assoc
```

Description

This command specifies the LI MAC filter that has its entries inserted into a list of normal mac filters.

Parameters

li-filter-name

specifies the name of the LI MAC filter. Filter names cannot start with an underscore character (For example, “_my-filter”) and cannot use the name “default”, up to 32 characters.

9.2.1.2.22 mac-filter

Syntax

```
[no] mac-filter filter-id
```

Context

```
config>li>li-filter-assoc>li-mac-fltr
```

Description

Specifies the MAC filters into which the entries from the specified LI MAC filter are to be inserted. The LI MAC filter and MAC filter must already exist before the association is made. If the normal MAC filter is deleted then the association is also removed (and not re-created if the MAC filter comes into existence in the future).

Parameters

filter-id

The filter identification identifies the MAC filters, up to 64 characters

Values: 1 to 65536

9.2.1.2.23 li-filter-block-reservation

Syntax

```
li-filter-block-reservation
```

Context

```
config>li
```

Description

This command enters the li-filter-block-reservation branch to create lawful intercept filter reservations.

9.2.1.2.24 li-reserved-block

Syntax

```
li-reserved-block block-name [create]
```

```
no li-reserved-block block-name
```

Context

```
config>li>li-filter-block-reservation
```

Description

This command creates or edits an LI reserved block. The reserved block allows an operator to define where entries from an LI filter will be inserted into a normal filter. The block reserves a configurable number of entries in the normal filter that can only be used for entries inserted from associated LI filters. The LI filter entries that get inserted into the reserved block in each normal filter are not visible to non-LI operators. The block also defines normal filters to which the reservation will be applied.

Parameters

block-name

specifies the name of the LI reserved block; block names cannot start with an underscore character (For example, “_my-block”) and cannot use the name “default”; up to 32 characters

9.2.1.2.25 ip-filter

Syntax

```
[no] ip-filter ip-filter-id
```

Context

```
config>li>li-filter-block-reservation>li-reserved-block
```

Description

This command configures to which normal IPv4 address filters the entry reservation is applied.

Parameters

ip-filter-id

The filter identification identifies the normal IPv4 address filters

Values: 1 to 65536 for IDs; up to 64 characters for names

9.2.1.2.26 ipv6-filter

Syntax

```
[no] ipv6-filter ipv6-filter-id
```

Context

```
config>li>li-filter-block-reservation>li-reserved-block
```

Description

This command configures to which normal IPv6 address filters the entry reservation is applied.

Parameters

ipv6-filter-id

The filter identification identifies the normal IPv6 address filters, up to 64 characters

Values: 1 to 65536 for IDs; up to 64 characters for names

9.2.1.2.27 mac-filter

Syntax

```
[no] mac-filter mac-filter-id
```

Context

```
config>li>li-filter-block-reservation>li-reserved-block
```

Description

This command configures to which normal MAC filters the entry reservation is applied.

Parameters

mac-filter-id

The filter identification identifies the normal MAC filters, up to 64 characters

Values: 1 to 65536 for IDs; up to 64 characters for names

9.2.1.2.28 start-entry

Syntax

start-entry *entry-id* count *count*

no start-entry

Context

config>li>li-filter-block-reservation>li-reserved-block

Description

This command defines a block of reserved filter entries that are used to insert LI filter entries into a normal filter.

Default

no start-entry

Parameters

entry-id

The entry identification identifies the start of a block of reserved filter entries.

Values: 1 to 2097151

count

This parameter identifies the number of entries in the block.

Values: 1 to 65535

9.2.1.2.29 li-filter-lock-state

Syntax

li-filter-lock-state {locked | unlocked-for-li-users | unlocked-for-all-users}

no li-filter-lock-state

Context

config>li

Description

This command configures the lock state of the filters used by LI. With the configurable filter lock for LI feature an LI user can control the behavior of filters when they are used for LI.

Prior to 12.0.R1 release, when a filter entry was used as an LI mirror source criteria, all subsequent attempts to modify the filter were then blocked to avoid having the LI session impacted by a non-LI user.

The no form of this command reverts to the default.

Parameters

locked

When an li-source criteria is configured that references any entry of filter Y, the filter Y can no longer be changed (until there are no longer any li-sources references to entries of filter Y).

unlocked-for-li-users

Filters can continue to be edited by LI users only even when an li-source references an entry in that filter.

unlocked-for-all-users

Filters can continue to be edited by all users even when an li-source references an entry in that filter.

Default: locked

9.2.1.2.30 li-source

Syntax

[no] li-source *mirror-service-id* [name *mirror-service-name*]

Context

config>li

Description

This command configures a lawful intercept (LI) mirror source.

Parameters

mirror-service-id | *svc-name*

The service identification identifies the service in the service domain. This ID is unique to this service and cannot be used by any other service, regardless of service type. The same service ID must be configured on every router that this particular service is defined on.

Values: 1 to 2147483647 | up to 64 characters

mirror-service-name

Values: up to 64 characters

9.2.1.2.31 ip-filter

Syntax

```
ip-filter ip-filter-id entry entry-id [entry-id... (up to 8 max)] [intercept-id intercept-id [intercept-id... (up to 8 max)]] [session-id session-id [session-id... (up to 8 max)]]  
no ip-filter ip-filter-id [entry entry-id [entry-id... (up to 8 max)]]
```

Context

config>li>li-source

Description

This command enables lawful interception (LI) of packets that match specific entries in an existing IP filter.

The `ip-filter` command directs packets which match the defined list of entry IDs to be intercepted to the destination referenced by the `mirror-dest-service-id` of the `mirror-source` command.

The IP filter must already exist in order for the command to execute. Filters are configured in the `config>filter` context. If the IP filter does not exist, an error will occur. If the filter exists but has not been associated with a SAP or IP interface, an error is not generated but mirroring will not be enabled (there are no packets to mirror). Once the IP filter is defined to a SAP, IP interface or subscriber, mirroring is enabled.

If the IP filter is defined as ingress, only ingress packets are intercepted. Ingress packets are sent to the destination prior to any ingress packet modifications.

If the IP filter is defined as egress, only egress packets are intercepted. Egress packets are sent to the destination after all egress packet modifications.

An *entry-id* within an IP filter can only be intercepted to a single destination. If the same *entry-id* is defined multiple times, an error occurs and only the first definition is in effect.

By default, no packets matching any IP filters are intercepted. Interception of IP filter entries must be explicitly defined.

When the `no` command is executed with the `entry` keyword and one or more *entry-id*s, interception of that list of *entry-id*s is terminated within the *ip-filter-id*. If an *entry-id* is listed that does not exist, an error will occur and the command will not execute. If an *entry-id* is listed that is not currently being intercepted, no error will occur for that *entry-id* and the command will execute normally.

Parameters

ip-filter-id

The IP filter ID whose entries are to be intercepted. If the *ip-filter-id* does not exist, an error will occur and the command will not execute. Intercepting packets will commence when the *ip-filter-id* is defined on a SAP or IP interface.

Values: 1 to 65535 for IDs; up to 64 characters for names

entry-id

The IP filter entries to use as match criteria for lawful intercept (LI). The *entry* keyword begins a list of *entry-id*'s for interception. Multiple *entry-id* entries can be specified with a single command. Each *entry-id* must be separated by a space. Up to <N><n> 8 entry IDs may be specified in a single command.

If an *entry-id* does not exist within the IP filter, an error occurs and the command will not execute.

If the filter's *entry-id* is renumbered within the IP filter definition, the old *entry-id* is removed but the new *entry-id* must be manually added to the configuration to include the new (renumbered) entry's criteria.

Values: 1 to 2097151

intercept-id

This command configures the *intercept-id* that is inserted into the packet header for all mirrored packets of the associated li-source entry. This *intercept-id* can be used (for example by a downstream LI Gateway) to identify the particular LI session to which the packet belongs. For all types of LI source entries (filter, nat, sap, subscriber), when the mirror service is configured with *ip-udp-shim* routable encap command, an *intercept-id* field (as part of the routable encap) is always present in the mirrored packets. If there is no *intercept-id* configured for an LI source entry, the default value is inserted. When the mirror service is configured with *ip-gre* routable encap, no *intercept-id* is inserted and none should be specified against the LI source entries.

Values	1 to 4294967295 (32b)	For NAT li-source entries that are using a mirror service that is not configured with routable encap.
–	1 to 1073741824 (30b)	For all types of li-source entries that are using a mirror service with routable ip-udp-shim encap and no direction-bit.
–	1 to 536870912 (29b)	For all types of li-source entries that are using a mirror service with routable ip-udp-shim encap and with the direction-bit enabled.

session-id

This command configures the *session-id* that is inserted into the packet header for all mirrored packets of the associated LI source entry. This *session-id* can be used (for example by a downstream LI Gateway) to identify the particular LI session to which the packet belongs. The *session-id* is only valid and used for mirror services that are configured with *ip-udp-shim* routable encap (config>mirror>mirror-dest>encap#ip-udp-shim). For all types of *li-source* entries (filter, nat, sap, subscriber), when the mirror service is configured with *ip-udp-shim* routable encap, a *session-id* field (as part of the routable encap) is always present in the mirrored packets. If there is no *session-id* configured for an LI source entry, the default value is inserted. When a mirror service is configured with *ip-gre* routable encap, no *session-id* is inserted and none should be specified against the LI source entries.

Values: 1 to 4,294,967,295 (32b)

9.2.1.2.32 ipv6-filter

Syntax

```
ipv6-filter ipv6-filter-id entry entry-id [entry-id... (up to 8 max)] [intercept-id
intercept-id [intercept-id... (up to 8 max)]] [session-id session-id [session-id... (up to 8
max)]]

no ipv6-filter ipv6-filter-id [entry entry-id [entry-id... (up to 8 max)]]
```

Context

```
config>li>li-source
```

Description

This command enables lawful interception (LI) of packets that match specific entries in an existing IPv6 filter.

The *ipv6-filter* command directs packets which match the defined list of entry IDs to be intercepted to the destination referenced by the *mirror-dest-service-id* of the *mirror-source* command.

The IPv6 filter must already exist in order for the command to execute. Filters are configured in the *config>filter* context. If the IPv6 filter does not exist, an error will occur. If the filter exists but has not been associated with a SAP or IPv6 interface, an error is not generated but mirroring will not be enabled (there are no packets to mirror). Once the IPv6 filter is defined to a SAP, IPv6 interface or subscriber, mirroring is enabled.

If the IPv6 filter is defined as ingress, only ingress packets are intercepted. Ingress packets are sent to the destination prior to any ingress packet modifications.

If the IPv6 filter is defined as egress, only egress packets are intercepted. Egress packets are sent to the destination after all egress packet modifications.

An *entry-id* within an IPv6 filter can only be intercepted to a single destination. If the same *entry-id* is defined multiple times, an error occurs and only the first definition is in effect.

By default, no packets matching any IPv6 filters are intercepted. Interception of IPv6 filter entries must be explicitly defined.

When the `no` command is executed with the `entry` keyword and one or more *entry-id*s, interception of that list of *entry-id*s is terminated within the *ipv6-filter-id*. If an *entry-id* is listed that does not exist, an error will occur and the command will not execute. If an *entry-id* is listed that is not currently being intercepted, no error will occur for that *entry-id* and the command will execute normally.

Parameters

ipv6-filter-id

The IPv6 filter ID whose entries are to be intercepted. If the *ipv6-filter-id* does not exist, an error will occur and the command will not execute. Intercepting packets will commence when the *ipv6-filter-id* is defined on a SAP or IPv6 interface.

Values: 1 to 65535 for IDs; up to 64 characters for names

entry-id

The IPv6 filter entries to use as match criteria for lawful intercept (LI). The `entry` keyword begins a list of *entry-id*s for interception. Multiple *entry-id* entries can be specified with a single command. Each *entry-id* must be separated by a space. Up to <N><n> 8 entry IDs may be specified in a single command.

If an *entry-id* does not exist within the IPv6 filter, an error occurs and the command will not execute.

If the filter's *entry-id* is renumbered within the IPv6 filter definition, the old *entry-id* is removed but the new *entry-id* must be manually added to the configuration to include the new (renumbered) entry's criteria.

Values: 1 to 2097151

intercept-id

This command configures the *intercept-id* that is inserted into the packet header for all mirrored packets of the associated li-source entry. This *intercept-id* can be used (for example by a downstream LI Gateway) to identify the particular LI session to which the packet belongs. For all types of LI source entries (filter, nat, sap, subscriber), when the mirror service is configured with `ip-udp-shim` routable encaps, an *intercept-id* field (as part of the routable encaps) is always present in the mirrored packets. If there is no *intercept-id* configured for an LI source entry, the default value is inserted. When the mirror service is configured with `ip-gre` routable encaps, no *intercept-id* is inserted and none should be specified against the LI source entries.

Values	1 to 4294967295 (32b)	For NAT li-source entries that are using a mirror service that is not configured with routable encap.
–	1 to 1073741824 (30b)	For all types of li-source entries that are using a mirror service with routable ip-udp-shim encap and no direction-bit.
–	1 to 536870912 (29b)	For all types of li-source entries that are using a mirror service with routable ip-udp-shim encap and with the direction-bit enabled.

session-id

This command configures the *session-id* that is inserted into the packet header for all mirrored packets of the associated li-source entry. This *session-id* can be used (for example by a downstream LI Gateway) to identify the particular LI session to which the packet belongs. The *session-id* is only valid and used for mirror services that are configured with ip-udp-shim routable encap (config>mirror>mirror-dest>encap#ip-udp-shim). For all types of li-source entries (filter, nat, sap, subscriber), when the mirror service is configured with ip-udp-shim routable encap, a *session-id* field (as part of the routable encap) is always present in the mirrored packets. If there is no *session-id* configured for an li-source entry, the default value is inserted.. When a mirror service is configured with ip-gre routable encap, no *session-id* is inserted and none should be specified against the li-source entries.

session-id

The session-id value to insert into the header of the mirrored packets.

Values: 1 to 4,294,967,295 (32b)

9.2.1.2.33 li-ip-filter

Syntax

```
li-ip-filter li-filter-name entry li-entry-id [li-entry-id...(up to 8 max)] [intercept-id
intercept-id [intercept-id...(up to 8 max)]] [session-id session-id [session-id...(up to 8 max)]]
no li-ip-filter filter-name [entry li-entry-id [li-entry-id...(up to 8 max)]]
```

Context

```
config>li>li-source
```

Description

This command enables lawful interception (LI) of packets that match specific entries in an existing LI IP filter that has been associated with a normal IP filter. The specification of an li-ip-filter entry as an li-source means that packets matching the li-ip-filter entry will be intercepted on all interfaces/saps/and so on. where the associated normal ip-filters are applied.

Parameters

li-filter-name

the name of the li-ip-filter, up to 32 characters

li-entry-id

the entry id in the li-ip-filter that is to be used as an li-source criteria

Values: 1 to 65535

intercept-id

This command configures the intercept-id that is inserted into the packet header for all mirrored packets of the associated li-source entry. This *intercept-id* can be used (for example by a downstream LI Gateway) to identify the particular LI session to which the packet belongs. For all types of LI source entries (filter, nat, sap, subscriber), when the mirror service is configured with ip-udp-shim routable encaps, an *intercept-id* field (as part of the routable encaps) is always present in the mirrored packets. If there is no *intercept-id* configured for an LI source entry, the default value is inserted. When the mirror service is configured with ip-gre routable encaps, no *intercept-id* is inserted and none should be specified against the LI source entries.

Values: 1 to 1073741823

session-id

The session-id that is inserted into the packet header for all mirrored packets of the associated li-source entry. This session-id can be used (for example by a downstream LI Gateway) to identify the particular LI session to which the packet belongs. The session-id is only valid and used for mirror services that are configured with ip-udp-shim routable encaps (config>mirror>mirror-dest>encap#ip-udp-shim). For all types of li-source entries (filter, nat, sap, subscriber), when the mirror service is configured with ip-udp-shim routable encaps, a session-id field (as part of the routable encaps) is always present in the mirrored packets. If there is no session-id configured for an li-source entry, the default value is inserted. When a mirror service is configured with ip-gre routable encaps, no session-id is inserted and none should be specified against the li-source entries.

Values: 1 to 4294967295

9.2.1.2.34 li-ipv6-filter

Syntax

```
li-ipv6-filter li-filter-name entry li-entry-id [li-entry-id...(up to 8 max)] [intercept-id  
intercept-id [intercept-id...(up to 8 max)]] [session-id session-id [session-id...(up to 8 max)]]  
no li-ipv6-filter filter-name [entry li-entry-id [li-entry-id...(up to 8 max)]]
```

Context

config>li>li-source

Description

This command enables lawful interception (LI) of packets that match specific entries in an existing LI IPv6 filter that has been associated with a normal IPv6 filter. The specification of an li-ipv6-filter entry as an li-source means that packets matching the li-ipv6-filter entry will be intercepted on all interfaces/saps/and so on where the associated normal ip-filters are applied.

Parameters

li-filter-name

the name of the li-ipv6-filter, up to 32 characters

li-entry-id

the entry id in the li-ipv6-filter that is to be used as an li-source criteria

Values: 1 to 65535

intercept-id

The intercept-id that is inserted into the packet header for all mirrored packets of the associated li-source entry. This intercept-id can be used (for example by a downstream LI Gateway) to identify the particular LI session to which the packet belongs. For all types of LI source entries (filter, nat, sap, subscriber), when the mirror service is configured with ip-udp-shim routable encap, an intercept-id field (as part of the routable encap) is always present in the mirrored packets. If there is no intercept-id configured for an LI source entry, the default value is inserted. When the mirror service is configured with ip-gre routable encap, no intercept-id is inserted and none should be specified against the LI source entries.

Values: 1 to 1073741823

session-id

The session ID that is inserted into the packet header for all mirrored packets of the associated li-source entry. This session ID can be used (for example by a downstream LI Gateway) to identify the particular LI session to which the packet belongs. The session ID is only valid and used for mirror services that are configured with ip-udp-shim routable encap (config>mirror>mirror-dest>encap#ip-udp-shim). For all types of LI source entries (filter, nat, sap, subscriber), when the mirror service is configured with ip-udp-shim routable encap, a

session-id field (as part of the routable encaps) is always present in the mirrored packets. If there is no session-id configured for an LI source entry, the default value is inserted. When a mirror service is configured with ip-gre routable encaps, no session-id is inserted and none should be specified against the LI source entries.

Values: 1 to 4294967295

9.2.1.2.35 li-mac-filter

Syntax

```
li-mac-filter li-filter-name entry li-entry-id [li-entry-id...(up to 8 max)] [intercept-id  
intercept-id [intercept-id...(up to 8 max)]] [session-id session-id [session-id...(up to 8 max)]]  
no li-mac-filter filter-name [entry li-entry-id [li-entry-id...(up to 8 max)]]
```

Context

config>li>li-source

Description

This command enables lawful interception (LI) of packets that match specific entries in an existing LI MAC filter that has been associated with a normal MAC filter. The specification of an LI MAC filter entry as an LI source means that packets matching the LI MAC filter entry will be intercepted on all interfaces/saps/and so on where the associated normal MAC filters are applied.

Parameters

li-filter-name

the name of the LI MAC filter, up to 32 characters

li-entry-id

the entry id in the LI MAC filter that is to be used as an LI source criteria

Values: 1 to 65535

intercept-id

This command configures the intercept-id that is inserted into the packet header for all mirrored packets of the associated li-source entry. This *intercept-id* can be used (for example by a downstream LI Gateway) to identify the particular LI session to which the packet belongs. For all types of LI source entries (filter, nat, sap, subscriber), when the mirror service is configured with ip-udp-shim routable encaps, an *intercept-id* field (as part of the routable encaps) is always present in the mirrored packets. If there is no *intercept-id* configured for an LI source entry, the default value is inserted. When the mirror service is configured with ip-gre routable encaps, no *intercept-id* is inserted and none should be specified against the LI source entries.

Values: 1 to 1073741823

session-id

This command configures the *session-id* that is inserted into the packet header for all mirrored packets of the associated *li-source* entry. This *session-id* can be used (for example by a downstream LI gateway) to identify the particular LI session to which the packet belongs. The *session-id* is only valid and used for mirror services that are configured with *ip-udp-shim* routable encap (`config>mirror>mirror-dest>encap#ip-udp-shim`). For all types of LI source entries (filter, nat, sap, subscriber), when the mirror service is configured with *ip-udp-shim* routable encap, a *session-id* field (as part of the routable encap) is always present in the mirrored packets. If there is no *session-id* configured for an LI source entry, the default value is inserted. When a mirror service is configured with *ip-gre* routable encap, no *session-id* is inserted and none should be specified against the LI source entries.

Values: 1 to 4294967295

9.2.1.2.36 mac-filter

Syntax

```
mac-filter mac-filter-id entry [entry-id...] [intercept-id intercept-id...] [session-id session-id...]
```

```
no mac-filter mac-filter-id
```

Context

```
config>li>li-source
```

Description

This command enables lawful interception (LI) of packets that match specific entries in an existing MAC filter. Multiple entries can be created using unique *entry-id* numbers within the filter. The router implementation exits the filter on the first match found and executes the actions in accordance with the accompanying action command. For this reason, entries must be sequenced correctly from most to least explicit.

An entry may not have any match criteria defined (in which case, everything matches) but must have at least the *action* keyword for it to be considered complete. Entries without the *action* keyword will be considered incomplete and, therefore, will be rendered inactive.

An *entry-id* within an MAC filter can only be intercepted to a single destination. If the same *entry-id* is defined multiple times, an error occurs and only the first definition is in effect.

The *no* form of this command removes the specified entry from the IP or MAC filter. Entries removed from the IP or MAC filter are immediately removed from all services or network ports where that filter is applied.

Parameters

mac-filter-id

specifies the MAC filter ID. If the *mac-filter-id* does not exist, an error will occur and the command will not execute

entry-id

specifies the MAC filter entries to use as match criteria

intercept-id

configures the *intercept-id* that is inserted into the packet header for all mirrored packets of the associated li-source entry. This *intercept-id* can be used (for example by a downstream LI Gateway) to identify the particular LI session to which the packet belongs. For all types of LI source entries (filter, nat, sap, subscriber), when the mirror service is configured with *ip-udp-shim* routable encap, an *intercept-id* field (as part of the routable encap) is always present in the mirrored packets. If there is no *intercept-id* configured for an LI source entry, the default value is inserted. When the mirror service is configured with *ip-gre* routable encap, no *intercept-id* is inserted and none should be specified against the LI source entries.

Values	1 to 4294967295 (32b)	For NAT li-source entries that are using a mirror service that is not configured with routable encap.
–	1 to 1073741824 (30b)	For all types of li-source entries that are using a mirror service with routable ip-udp-shim encap and no direction-bit.
–	1 to 536870912 (29b)	For all types of li-source entries that are using a mirror service with routable ip-udp-shim encap and with the direction-bit enabled.

session-id

configures the *session-id* that is inserted into the packet header for all mirrored packets of the associated LI source entry. This *session-id* can be used (for example by a downstream LI Gateway) to identify the particular LI session to which the packet belongs. The *session-id* is only valid and used for mirror services that are configured with *ip-udp-shim* routable encap (config>mirror>mirror-dest>encap#ip-udp-shim). For all types of LI source entries (filter, nat, sap, subscriber), when the mirror service is configured with *ip-udp-shim* routable encap, a *session-id* field (as part of the routable encap) is always present in the mirrored packets. If there is no *session-id* configured for an LI source entry, the default value is inserted. When a mirror service is configured with *ip-gre* routable encap, no *session-id* is inserted and none should be specified against the LI source entries.

Values: 1 to 4,294,967,295 (32b)

9.2.1.2.37 nat

Syntax

nat

Context

config>li>li-source

Description

This command enables the context to configure LI NAT parameters.

9.2.1.2.38 classic-lsn-sub

Syntax

[no] classic-lsn-sub router *router-instance* ip *ip-address*

Context

config>li>li-source>nat

Description

This command configures a classic LSN subscriber sources.

The no form of this command removes the parameter from the configuration.

Parameters

router-instance

specifies the router instance the pool belongs to, either by router name or service ID

Values: router-name: “Base” | “management”

ip-address

specifies the IP address in the a.b.c.d format

9.2.1.2.39 intercept-id

Syntax

intercept-id *id*

```
no intercept-id
```

Context

```
config>li>li-source>nat>classic-lsn-sub
```

```
config>li>li-source>nat>dslite-lsn-sub
```

```
config>li>li-source>nat>ethernet-header
```

```
config>li>li-source>nat>l2-aware-sub
```

```
config>li>li-source>nat>nat64-lsn-sub
```

Description

This command configures the intercept-id that is inserted into the packet header for all mirrored packets of the associated li-source entry. This intercept-id can be used (for example by a downstream LI Gateway) to identify the particular LI session to which the packet belongs.

For nat mirroring (a nat li-source entry type), when the mirror service is not configured with any routable encap (for example, no ip-udp-shim or ip-gre configured under config>mirror>mirror-dest>encap), the presence of a configured intercept-id against an li-source (nat) entry will cause the insertion of the intercept-id after a configurable mac-da, mac-sa and etype (configured under li-source>nat>ethernet-header), at the front of each packet mirrored for that particular li-source entry. If there is no intercept-id configured (for a nat entry using a mirror service without routable encap), a configurable mac-da and mac-sa are added to the front of the packets (but no intercept-id). In both cases a non-configurable etype is also added immediately before the mirrored customer packet. The routable encapsulation configured in the mirror-dest takes precedence over the ethernet-header configuration in the li-source nat entries. If routable encapsulation is configured, the ethernet-header config is ignored and no mac header is added to the packet (the encap is determined by the mirror-dest in this case).

For all types of li-source entries (filter, nat, sap, subscriber), when the mirror service is configured with ip-udp-shim routable encap, an intercept-id field (as part of the routable encap) is always present in the mirrored packets. If there is no intercept-id configured for an li-source entry, the default value is inserted. When the mirror service is configured with ip-gre routable encap, no intercept-id is inserted and none should be specified against the li-source entries.

The no form of this command removes the value from the configuration.

Default

no intercept-id (an ID of 0, or no ID)

Parameters

id

the intercept ID value to insert into the header of the mirrored packets

Values	1 to 4294967295 (32b)	For NAT li-source entries that are using a mirror service that is not configured with routable encap
–	1 to 1073741824 (30b)	For all types of li-source entries that are using a mirror service with routable ip-udp-shim encap and no direction-bit.
–	1 to 536870912 (29b)	For all types of li-source entries that are using a mirror service with routable ip-udp-shim encap and with the direction-bit enabled.

9.2.1.2.40 session-id

Syntax

```
session-id session-id
no session-id
```

Context

```
config>li>li-source>nat>classic-lsn-sub
config>li>li-source>nat>dslite-lsn-sub
config>li>li-source>nat>ethernet-header
config>li>li-source>nat>l2-aware-sub
config>li>li-source>nat>nat64-lsn-sub
```

Description

This command configures the session-id that is inserted into the packet header for all mirrored packets of the associated LI source entry. This session ID can be used (for example by a downstream LI Gateway) to identify the particular LI session to which the packet belongs.

The session-id is only valid and used for mirror services that are configured with ip-udp-shim routable encap (config>mirror>mirror-dest>encap# ip-gre-shim).

For all types of LI source entries (filter, nat, sap, subscriber), when the mirror service is configured with ip-udp-shim routable encap, a session-id field (as part of the routable encap) is always present in the mirrored packets. If there is no session ID configured for an LI source entry, the default value is inserted. When a mirror service is configured with ip-gre routable encap, no session-id is inserted and none should be specified against the LI source entries.

The no form of this command removes the session-id from the configuration which results in the default value being used.

Default

no session-id (an id of 0, or no id)

Parameters

id

the session ID value to insert into the header of the mirrored packets

Values: 1 to 4,294,967,295 (32b)

9.2.1.2.41 dslite-lsn-sub

Syntax

```
[no] dslite-lsn-sub router router-instance b4 ipv6-prefix
```

Context

```
config>li>li-source>nat
```

Description

This command configures the Dual Stack Lite LSN subscriber source.

The no form of this command removes the value from the configuration.

Parameters

router-instance

specifies the router instance the pool belongs to, either by router name or service ID

Values: router-name: “Base” | “management”

Default: Base

ipv6-prefix

specifies the IPv6 address

Values	ipv6-prefix	prefix/length
–	prefix	x:x:x:x:x:x (eight 16-bit pieces)

–	–	x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D
–	length	[0..128]

9.2.1.2.42 ethernet-header

Syntax

```
ethernet-header [da ieee-address] [sa ieee-address] [etype ethertype]  
no ethernet-header
```

Context

```
config>li>li-source>nat
```

Description

This command configures the Ethernet header for the NAT sources.

The no form of this command removes the values from the configuration.

Parameters

da *ieee-address*

specifies the DA IEEE address, up to 30 characters

sa *ieee-address*

specifies the SA IEEE address, up to 30 characters

ethertype

specifies the Ethernet type

Values: 1536 to 65535

9.2.1.2.43 l2-aware-sub

Syntax

[no] l2-aware-sub *sub-ident-string*

Context

config>li>li-source>nat

Description

This command configures a Layer-2-Aware subscriber source.

The no form of this command removes the values from the configuration.

Parameters

sub-ident-string

specifies a source name, up to 32 characters

9.2.1.2.44 sap

Syntax

sap *sap-id* {[ingress] [egress]} [intercept-id *intercept-id...*] [session-id *session-id...*]

no sap *sap-id*

Context

config>li>li-source

Description

This command creates a service access point (SAP) within an LI configuration. The specified SAP must define a FastE, GigE, or XGigE, or XGigE access port with a dot1q, null, or q-in-q encapsulation type.

The *intercept-id* parameter configures the intercept IDs that is inserted into the packet header for all mirrored packets of the associated LI source entry.

The *session-id* parameter inserts the specified IDs into the packet header for all mirrored packets of the associated LI source entry.

When the no form of this command is used on a SAP, the SAP with the specified port and encapsulation parameters is deleted.

Parameters

sap-id

specifies the physical port identifier portion of the SAP definition; refer to the *7450 ESS*, *7750 SR*, *7950 XRS*, and *VSR OAM and Diagnostics Guide* for the command syntax

egress

specifies that packets egressing the SAP should be mirrored; egress packets are mirrored to the mirror destination after egress packet modification

ingress

specifies that packets ingressing the SAP should be mirrored; ingress packets are mirrored to the mirror destination prior to ingress packet modification

intercept-id

configures the *intercept-id* that is inserted into the packet header for all mirrored packets of the associated li-source entry; this *intercept-id* can be used (for example by a downstream LI Gateway) to identify the particular LI session to which the packet belongs.

For all types of LI source entries (filter, nat, sap, subscriber), when the mirror service is configured with *ip-udp-shim* routable encap, an *intercept-id* field (as part of the routable encap) is always present in the mirrored packets. If there is no *intercept-id* configured for an LI source entry, the default value is inserted. When the mirror service is configured with *ip-gre* routable encap, no *intercept-id* is inserted and none should be specified against the LI source entries.

Values	1 to 4294967295 (32b)	For NAT li-source entries that are using a mirror service that is not configured with routable encap
–	1 to 1073741824 (30b)	For all types of li-source entries that are using a mirror service with routable ip-udp-shim encap and no direction-bit.
–	1 to 536870912 (29b)	For all types of li-source entries that are using a mirror service with routable ip-udp-shim encap and with the direction-bit enabled.

session-id

configures the *session-id* that is inserted into the packet header for all mirrored packets of the associated LI source entry; this *session-id* can be used (for example by a downstream LI Gateway) to identify the particular LI session to which the packet belongs.

The *session-id* is only valid and used for mirror services that are configured with *ip-udp-shim* routable encap (config>mirror>mirror-dest>encap#ip-udp-shim).

For all types of LI source entries (filter, nat, sap, subscriber), when the mirror service is configured with `ip-udp-shim` routable encap, a *session-id* field (as part of the routable encap) is always present in the mirrored packets. If there is no *session-id* configured for an LI source entry, the default value is inserted. When a mirror service is configured with `ip-gre` routable encap, no *session-id* is inserted and none should be specified against the LI source entries.

Values: 1 to 4, 294, 967, 295 (32b)

9.2.1.2.45 subscriber

Syntax

```
subscriber sub-ident-string [sap sap-id [ip ip-address] [mac ieee-address] [sla-profile sla-profile-name] [fc {[be] [l2] [af] [l1] [h2] [ef] [h1] [nc]]} {[ingress] [egress]} [intercept-id intercept-id...]] [session-id session-id...]
```

```
no subscriber sub-ident-string
```

Context

```
config>li>li-source
```

Description

This command adds hosts of a subscriber to mirroring service.

Parameters

sub-ident-string

specifies the name of the subscriber identification policy

sap-id

specifies the physical port identifier portion of the SAP definition. Refer to the *7450 ESS*, *7750 SR*, *7950 XRS*, and *VSR OAM and Diagnostics Guide* for the command syntax.

ip-address

the service IP address (system IP address) of the remote device sending LI traffic. If 0.0.0.0 is specified, any remote router is allowed to send to this service.

Values: 1.0.0.1 to 223.255.255.254

mac mac-address

specifies this optional parameter when defining a static host. The MAC address must be specified for `anti-spoof`, `ip-mac` and `arp-populate`. Multiple static hosts may be configured with the same MAC address given that each definition is distinguished by a unique IP address.

sla-profile-name

specifies the SLA profile name, up to 32 characters

fc

the name of the forwarding class with which to associate LI traffic. The forwarding class name must already be defined within the system. If the *fc-name* does not exist, an error will be returned and the *fc* command will have no effect. If the *fc-name* does exist, the forwarding class associated with *fc-name* will override the default forwarding class.

Values: be, l2, af, l1, h2, ef, h1, nc

ingress

specifies information for the ingress policy

egress

specifies information for the egress policy

intercept-id

configures the *intercept-id* that is inserted into the packet header for all mirrored packets of the associated li-source entry. This *intercept-id* can be used (for example by a downstream LI Gateway) to identify the particular LI session to which the packet belongs.

For all types of LI source entries (filter, nat, sap, subscriber), when the mirror service is configured with *ip-udp-shim* routable encap, an *intercept-id* field (as part of the routable encap) is always present in the mirrored packets. If there is no *intercept-id* configured for an LI source entry, the default value is inserted. When the mirror service is configured with *ip-gre* routable encap, no *intercept-id* is inserted and none should be specified against the LI source entries.

Values	1 to 4294967295 (32b)	For NAT li-source entries that are using a mirror service that is not configured with routable encap.
–	1 to 1073741824 (30b)	For all types of li-source entries that are using a mirror service with routable ip-udp-shim encap and no direction-bit.
–	1 to 536870912 (29b)	For all types of li-source entries that are using a mirror service with routable ip-udp-shim encap and with the direction-bit enabled.

session-id

This command configures the *session-id* that is inserted into the packet header for all mirrored packets of the associated LI source entry. This *session-id* can be used (for example by a downstream LI Gateway) to identify the particular LI session to which the packet

belongs. The *session-id* is only valid and used for mirror services that are configured with `ip-udp-shim` routable encaps (`config>mirror>mirror-dest>encap#ip-udp-shim`).

For all types of LI source entries (filter, nat, sap, subscriber), when the mirror service is configured with `ip-udp-shim` routable encaps, a *session-id* field (as part of the routable encaps) is present in the mirrored packets. If there is no *session-id* configured for an LI source entry, the default value is inserted. When a mirror service is configured with `ip-gre` routable encaps, no *session-id* is inserted and none should be specified against the LI source entries.

Values: 1 to 4,294,967,295 (32b)

9.2.1.2.46 log

Syntax

log

Context

config>li

Description

This command enables the context to configure an event log for Lawful Interception.

9.2.1.2.47 log-id

Syntax

[no] log-id *log-id*

Context

config>li>log

Description

This command configures an LI event log destination. The log ID is used to direct events, alarms/traps, and debug information to respective destinations.

Parameters

log-id

the log ID number, expressed as a decimal integer

Values: 1 to 100

9.2.1.2.48 filter

Syntax

```
filter filter-id  
no filter
```

Context

```
config>li>log>log-id
```

Description

This command adds an event filter policy with the log destination.

The `filter` command is optional. If no event filter is configured, all events, alarms and traps generated by the source stream will be forwarded to the destination.

An event filter policy defines (limits) the events that are forwarded to the destination configured in the `log-id`. The event filter policy can also be used to select the alarms and traps to be forwarded to a destination `snmp-trap-group`.

The application of filters for debug messages is limited to application and subject only.

Accounting records cannot be filtered using the `filter` command.

Only one `filter-id` can be configured per log destination.

The `no` form of this command removes the specified event filter from the *log-id*.

Default

no filter

Parameters

filter-id

specifies the event filter policy ID that is used to associate the filter with the *log-id* configuration; the event filter policy ID must already be defined in `config>log>filter filter-id` command

Values: 1 to 1500

9.2.1.2.49 from

Syntax

```
from li
```

no from

Context

config>li>log>log-id

Description

This command configures a bit mask that specifies the log event source streams to be forwarded to the destination specified in the log destination (memory, session, SNMP). Events from more than one source can be forwarded to the log destination.

Parameters

li

specifies the LI event stream that contains all events configured for LI activities

If the requester does not have access to the LI context, the event stream fails.

9.2.1.2.50 time-format

Syntax

time-format {local | utc}

Context

config>li>log>log-id

Description

This command specifies whether the time should be displayed in local or Coordinated Universal Time (UTC) format.

Parameters

local

specifies that timestamps are written in the system's local time

utc

Specifies that timestamps are written using the UTC value. This was formerly called Greenwich Mean Time (GMT) and Zulu time.

Default: utc

9.2.1.2.51 to

Syntax

```
to memory [size]  
to session  
to snmp [size]
```

Context

```
config>li>log>log-id
```

Description

This command enables the context to configure the destination type for the event log.

The source of the data stream must be specified in the `from` command prior to configuring the destination with the `to` command.

The `to` command cannot be modified or re-entered. If the destination or maximum size of an SNMP or memory log needs to be modified, the log ID must be removed and then re-created.

Parameters

size

indicates the number of events that can be stored into memory

Values: 50 to 1024

Default: 100

9.2.1.2.52 3gpp-5g-release

Syntax

```
3gpp-5g-release release-num  
no 3gpp-5g-release
```

Context

```
config>li>mobile
```

Description

This command enables the 3GPP release used for the 5G LI interfaces.

The `no` form of this command reverts to the default (base release).

Parameters

release-num
specifies the 3GPP release for the 5G LI interfaces

Values	rel-base	specifies the base release (release 15)
–	rel-16	specifies the release 16
–	rel-17	specifies the release 17

Default: rel-base

9.2.1.2.53 3gpp-release

Syntax

3gpp-release *release*
no 3gpp-release

Context

config>li>mobile

Description

This command enables the specified 3GPP release.
The no form of this command reverts to the default (base release).

Parameters

release
specifies the 3GPP release

Values	rel-base	specifies the base release, per existing functionality
–	rel-13	specifies the release 13

Default: rel-base

9.2.1.2.54 allow-duplicate-msisdn-or-imei

Syntax

```
[no] allow-duplicate-msisdn-or-imei
```

Context

```
config>li>mobile
```

Description

This command enables LI support for duplicate MSISDN or IMEI.

The no form of this command disables LI support for duplicate MSOISDN or IMEI.

9.2.1.2.55 client-tls-profile

Syntax

```
client-tls-profile tls-profile-name
```

```
no client-tls-profile
```

Context

```
config>li>mobile
```

Description

This command configures the default client TLS profile.

The no form of this command removes the client TLS profile from the configuration.

Parameters

tls-profile-name

specifies the client TLS profile name, up to 32 characters

This is the name of the profile created under the

config>system>security>tls>client-tls-profile context.

9.2.1.2.56 custom-correlation-id-format

Syntax

```
custom-correlation-id-format {enable | disable}  
no custom-correlation-id-format
```

Context

```
config>li>mobile
```

Description

This command facilitates sending the eight octets correlation ID with the first four octets set to 0 and the last four octets including the charging ID information (without including the GW IP address) for the 5G LI. The correlation ID is included in IRI as well as in CC messages so that the LIG can correlate IRI and CC messages.

The no form of this command reverts to default (disable).

Parameters

enable

specifies to facilitate sending the correlation ID with only the charging identifier

disable

specifies not to facilitate sending the correlation ID with only the charging identifier

Default: disable

9.2.1.2.57 df-peer

Syntax

```
df-peer df-peer-id {[df2-addr ip-address df2-port port] [df3-addr ip-address df3-port port]}  
[df2-tls-profile profile-name] [df3-tls-profile profile-name]  
no df-peer df-peer-id
```

Context

```
config>li>mobile
```

Description

This command configures the DF peer of an LI gateway as DF2 (X2) and/or DF3 (X3). It supports the following configuration options:

- DF2 IP address and port only
- DF3 IP address and port only
- both DF2 and DF3 IP addresses and ports

The no form of this command removes the DF peer information from the configuration.

Parameters

df-peer-id
specifies the DF peer ID
Values: 1 to 254 for CMG, 1 to 15 for 7750 SR MG

df2-addr ip-address
specifies the DF2 peer IP address where the X2 is sent

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit group)
–	–	x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

df2-port port
specifies the DF2 port number where the X2 is sent
Values: 1 to 65535
Default: 0

df3-addr ip-address
specifies the DF3 IP address where the X3 is sent

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit group)

–	–	x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

df3-port *port*

specifies the DF3 port number, which is the TCP/UDP port where the X3 is sent

Values: 1 to 65535

Default: 0

**NOTE:**

If the DF3 information is configured, the `df3-tcp-connections` command must be set to true (default), to ensure that the DF3 peer connections are made using the configured IP address and port.

profile-name

specifies the DF2 or DF3 TLS profile name, up to 32 characters

**NOTE:**

- The DF2 and DF3 peers can use the same or different TLS client profiles.
- To use a TLS profile, the TLS protocol must be enabled (using the `tls` command).
- The TLS client profiles must be in the 'no shutdown' state.

9.2.1.2.58 df3-tcp-connections**Syntax**

```
df3-tcp-connections {true | false}
```

```
no df3-tcp-connections
```

Context

```
config>li>mobile
```

Description

This command configures the status of TCP connections (true/false) to DF3 peers from the CP function/SMF. When set to false, there can be no TCP connections to DF3 peers. For legacy X1/CLI, the DF3 configuration on the CP function is used to validate and propagate the targets and associated DF peers on to the UP function.

The no form of this command reverts to default.



NOTE:

- This command is applicable only on the CP function/SMF.
- For CUPS and 5G-LI, the UP function supports X3.

Parameters

true

specifies that TCP connections are set up to DF3 peers from the SMF/CP function

false

specifies that TCP connections are not set up to DF3 peers from the SMF/CP function

Default: true

9.2.1.2.59 fan-out

Syntax

[no] fan-out

Context

config>li>mobile

Description

This command enables IRI and IRI+CC fan-out functionality for lawful interception on the mobile gateway. Each target activation must be uniquely identified by the Lawful Interception Identifier (LIID) specified using the config>li>mobile>target command. A maximum of 10 target activations are supported for a UE. There could be, for example, 5 activations using IMSI, 3 activations using MSISDN, and 2 activations using IMEI. A maximum of 7 activations are supported using the same target type (for example, IMSI). Each activation must be deactivated separately using the same LIID used at the time of activation.

The no form of this command reverts to the default.

**NOTE:**

- When this command is enabled, the LIID (specified using the `config>li>mobile>target` command) is mandatory for target activation and de-activation.
- Enabling this functionality in an existing deployment requires re-provisioning the targets (equivalent to a target database reset).
- Disabling fan-out after it has been enabled requires re-provisioning all the targets.

Default

no fan-out

9.2.1.2.60 include-rai**Syntax**

[no] include-rai

Context

config>li>mobile

Description

This command supports the inclusion of the Routing Area Identifier (RAI) from the message level along with the ULI received from the SGSN, within the Location of the Target Information Element that is sent in the IRI messages to the LIG.

The no form of this command reverts to the default.

Default

no include-rai

9.2.1.2.61 iri**Syntax**

iri

Context

config>li>mobile

Description

This command enables the context to configure LI IRI parameters.

9.2.1.2.62 gw-function-iri-mode

Syntax

```
[no] gw-function-iri-mode
```

Context

```
config>li>mobile>iri
```

Description

This command controls the IRI mode of generating events for a combined SGW/PGW session. For a combined session, by default, only PGW IRI events are generated. When this command is enabled, separate events are generated for the combined session from both the SGW and the PGW function.

The no form of this command reverts to the default (only PGW events are generated for the combined SGW/PGW session).

Default

```
no gw-function-iri-mode
```

9.2.1.2.63 iri-per-mb-request

Syntax

```
[no] iri-per-mb-request
```

Context

```
config>li>mobile>iri
```

Description

This command enables the additional MBReq event on the SGW.

The no form of this command reverts to the default.

Default

```
no iri-per-mb-request
```

9.2.1.2.64 neid

Syntax

`neid neid`

`no neid`

Context

`config>li>mobile>iri`

Description

This command controls the population of the NEID (local endpoint IP address of the X2 or X3 interface) information element carried, for example, in the E-UTRAN Attach IRI event from the SGW/PGW.

The no form of this command reverts to the default.

Parameters

default

specifies the S5 interface address

li-local-interface

specifies the local endpoint IP address

Default: default

9.2.1.2.65 te-id

Syntax

`[no] te-id`

Context

`config>li>mobile>iri`

Description

This command controls the population of the TEID (S5/S8-U PGW F-TEID: TEID+IP address) information element in a bearer activation IRI event from the SGW/PGW.

The no form of this command reverts to the default.

Default

no te-id

9.2.1.2.66 x2-iri-ice-type

Syntax

[no] x2-iri-ice-type

Context

config>li>mobile>iri

Description

This command is used to enable inclusion of the ICE-Type in GTP signaling-related IRI messages sent by the mobile gateway. Applicable gateway types include SGW, PGW, and combined SGW/PGW.

The no form of this command reverts to the default (that is, ICE-Type is not included in GTP signaling related IRI messages).

Default

no x2-iri-ice-type

9.2.1.2.67 li-t3

Syntax

li-t3

Context

config>li>mobile

Description

This command enables the context to configure the LI_T3 interface.

9.2.1.2.68 alpn

Syntax

```
alpn {enable | disable}
no alpn
```

Context

```
config>li>mobile>li-t3
config>li>mobile>li-x1
config>li>mobile>li-x1-mgmt
```

Description

This command enables or disables the ALPN extension handling for TLS over the LI_T3, the LI_X1, and the LI_X1 Management interface respectively.

The no form of this command reverts to the default (enable).

Parameters

enable

enables the ALPN extension handling for TLS over the interface

disable

disables the ALPN extension handling for TLS over the interface

Default: enable

9.2.1.2.69 li-t3-local-interface

Syntax

```
li-t3-local-interface ip-address [router router-instance] [local-port port-num]
no li-t3-local-interface
```

Context

```
config>li>mobile>li-t3
```

Description

This command configures the IP address for the LI_T3 local endpoint and optionally a port, if other than port 443 (default) is to be used

The no form of this command removes the IP address for the LI_T3 local endpoint from the configuration.

Parameters

ip-address
specifies the IP address for the LI_T3 local endpoint

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

router-instance
specifies the router instance
Values: Router instance
Default: Base

port-num
specifies the port number for the LI_T3 local endpoint
Values: 1 to 65535
Default: 443

9.2.1.2.70 li-t3-mgmt-local-interface

Syntax

li-t3-mgmt-local-interface *ip-address* [router *router-instance*] [local-port *port-num*]
no li-t3-mgmt-local-interface

Context

config>li>mobile>li-t3

Description

This command configures the IP address for the LI_T3 management local endpoint and optionally a port, if other than port 443 (default) is to be used

The no form of this command removes the IP address for the LI_T3 management local endpoint from the configuration.

Parameters

ip-address
specifies the IP address for the LI_T3 management local endpoint

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:x.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

router-instance
specifies the router instance

Values: Router instance

Default: Base

port-num
specifies the port number for the LI_T3 management local endpoint

Values: 1 to 65535

Default: 443

9.2.1.2.71 li-t3-up-peer

Syntax

```
li-t3-up-peer df-peer-id upf-addr upf-peer-address [t3-port t3-port-num] upf-id uuid
no li-t3-up-peer df-peer-id
```

Context

config>li>mobile>li-t3

Description

This command configures the UPF peer IP address/port, and associates a TLS profile to be used.



NOTE:

When the 5G LI functionality is enabled, the configuration of the upf -id is mandatory.

The no form of this command removes the UPF peer information from the configuration.

Parameters

df-peer-id

specifies the UPF peer ID
Values: 1 to 254 for CMG, 1 to 15 for 7750 SR MG

upf-peer-address

specifies the UPF peer IP address

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

t3-port-num

specifies the UPF port number for the X1 interface
Values: 1 to 65535
Default: 443

uuid

specifies the UUID of the UPF

Values: xxxxxxxx-xxxx-xxxx-xxxx-xxxxxxxxxxxx (x - [0..F]H)

9.2.1.2.72 sni

Syntax

```
sni {enable | disable}
no sni
```

Context

```
config>li>mobile>li-t3
config>li>mobile>li-x1
config>li>mobile>li-x1-mgmt
```

Description

This command enables or disables the SNI extension handling for TLS over the LI_T3, the LI_X1, and the LI_X1 Management interface respectively.

The no form of this command reverts to the default (enable).

Parameters

enable

enables the SNI extension handling for TLS over the interface

disable

disables the SNI extension handling for TLS over the interface

Default: enable

9.2.1.2.73 li-x1

Syntax

```
li-x1
```

Context

```
config>li>mobile
```

Description

This command enables the context to configure the LI_X1 interface.

9.2.1.2.74 admf-peer

Syntax

```
admf-peer df-peer-id admf-addr admf-peer-address [x1-port x1-port-num]
no admf-peer df-peer-id
```

Context

```
config>li>mobile>li-x1
```

Description

This command configures the ADMF peer IP address, and optionally a port if other than 443 (default) is to be used.

The no form of this command removes the ADMF peer information from the configuration.



NOTE:

- For DF2 and DF3 the following apply:
- The default client profile specified under the config>li>mobile context is used.
 - If you need to use a different TLS client profile for a specific DF-peer, reuse the existing DF-peer configuration to associate the TLS profiles.

Parameters

df-peer-id
specifies the ADMF peer ID
Values: 1 to 254 for CMG, 1 to 15 for 7750 SR MG

admf-peer-address
specifies the ADMF peer IP address

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)

–	–	x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

x1-port-num

specifies the ADMF port number

Values: 1 to 65535

Default: 443

9.2.1.2.75 li-x1-keep-alive-time-p2

Syntax

li-x1-keep-alive-time-p2 *x1-time-p2-value*

no li-x1-keep-alive-time-p2

Context

config>li>mobile>li-x1

Description

This command enables the keep-alive time for the X1 interface. MG responds with an OK for each keep-alive that it receives from the ADMF. When CMG does not receive a keep-alive message for *x1-time-p2-value* seconds, it deactivates all the tasks (all XIDs) that are provided on the network function by the ADMF. This timer is reset when an X1 request is received from the ADMF (including a keep-alive request).

The no form of this command reverts to the default (disables the keep-alive mechanism).

Parameters***x1-time-p2-value***

specifies the keep-alive time, in seconds

Values: 0 | 3600 to 86400

Default: 0

9.2.1.2.76 li-x1-local-interface

Syntax

```
li-x1-local-interface ip-address [router router-instance] [local-port port-num]
no li-x1-local-interface
```

Context

```
config>li>mobile>li-x1
```

Description

This command configures the IP address for the LI_X1 local endpoint.

The no form of this command removes the IP address for the LI_X1 local endpoint from the configuration.

Parameters

ip-address
specifies the IP address for the LI_X1 local endpoint

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

router-instance
specifies the router instance

Values: Router instance

Default: Base

port-num
specifies the port number

Values: 1 to 65535

Default: 443

9.2.1.2.77 li-x1-mgmt

Syntax

```
li-x1-mgmt
```

Context

```
config>li>mobile
```

Description

This command enables the context to configure the LI_X1 Management interface.

9.2.1.2.78 li-x1-mgmt-admf-peer

Syntax

```
li-x1-mgmt-admf-peer df-peer-id admf-addr admf-peer-address [x1-port x1-port-num]  
no li-x1-mgmt-admf-peer df-peer-id
```

Context

```
config>li>mobile>li-x1-mgmt
```

Description

This command configures the ADMF peer for the LI_X1 interface.

The no form of this command removes the ADMF peer information from the configuration.

Parameters

df-peer-id

specifies the ADMF peer ID

Values: 1 to 254 for CMG, 1 to 15 for 7750 SR MG

*adm**f*-*peer-address*

specifies the ADMF peer IP address

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

x1-port-num

specifies the ADMF port number for the X1 interface

Values: 1 to 65535

Default: 443

9.2.1.2.79 li-x1-mgmt-local-interface

Syntax

```
li-x1-mgmt-local-interface ip-address [router router-instance] [local-port port-num]  
no li-x1-mgmt-local-interface
```

Context

```
config>li>mobile>li-x1-mgmt
```

Description

This command configures the IP address for the LI_X1 Management local endpoint.

The no form of this command removes the IP address for the LI_X1 Management local endpoint from the configuration.

Parameters

ip-address

specifies the IP address for the LI_X1 Management local endpoint

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

router-instance
specifies the router instance
Values: Router instance
Default: Base

local-port
specifies the port number
Values: 1 to 65535
Default: 443

9.2.1.2.80 li-x2

Syntax

li-x2

Context

config>li>mobile

Description

This command enables the context to configure the LI_X2 interface.

9.2.1.2.81 li-x2-keep-alive-time-p1

Syntax

```
li-x2-keep-alive-time-p1 x2-time-p1-value  
no li-x2-keep-alive-time-p1
```

Context

```
config>li>mobile>li-x2
```

Description

This command specifies the keep-alive time for the LI_X2 interface on the SMF. The POI sends a keep-alive PDU at least every *x2-time-p1-value* seconds (the default is 60 seconds). If the POI does not receive a keep-alive Acknowledgment PDU within *x2-time-p2-value* seconds, it disconnects the connection and attempts to reconnect to the MDF while reporting an error through the X1 interface as defined in ETSI TS 103 221-1. The default *x2-time-p2-value* is 180 seconds and it can be modified using the `config>li>mobile>li-x2>li-x2-keep-alive-time-p2` command.

The no form of this command reverts to the default. If the timer is set to 0 value, the keep-alive mechanism is disabled.

Parameters

x2-time-p1-value

specifies the keep-alive time, in seconds

Values: 0 to 300

Default: 60

9.2.1.2.82 li-x2-keep-alive-time-p2

Syntax

```
li-x2-keep-alive-time-p2 x2-time-p2-value  
no li-x2-keep-alive-time-p2
```

Context

```
config>li>mobile>li-x2
```

Description

This command specifies the keep-alive time for the LI_X2 interface on the SMF. If the POI does not receive a keep-alive Acknowledgment PDU within *x2-time-p2-value* seconds (the default is 180 seconds), it disconnects the connection and attempts to reconnect to the MDF while reporting an error through the X1 interface as defined in ETSI TS 103 221-1.

The no form of this command reverts to the default. If the timer is set to 0 value, the keep-alive mechanism is disabled.

Parameters

x2-time-p2-value

specifies the keep-alive time, in seconds

Values: 0 to 1800

Default: 180

9.2.1.2.83 li-x3

Syntax

```
li-x3
```

Context

```
config>li>mobile
```

Description

This command enables the context to configure the LI_X3 interface.

9.2.1.2.84 li-x3-keep-alive-time-p1

Syntax

```
li-x3-keep-alive-time-p1 x3-time-p1-value
```

```
no li-x3-keep-alive-time-p1
```

Context

```
config>li>mobile>li-x3
```

Description

This command specifies the keep-alive time for the LI_X3 interface on the UPF. The POI sends a keep-alive PDU at least every *x3-time-p1-value* seconds (the default is 60 seconds). If the POI does not receive a keep-alive Acknowledgment PDU within *x2-time-p2-value* seconds, it disconnects the connection and attempts to reconnect to the MDF while reporting an error through the X1 interface as defined in ETSI TS 103 221-1. The default *x2-time-p2-value* is 180 seconds and it can be modified using the `config>li>mobile>li-x3>li-x3-keep-alive-time-p2` command.

The no form of this command reverts to the default. If the timer is set to 0 value, the keep-alive mechanism is disabled.

Parameters

x3-time-p1-value

specifies the keep-alive time, in seconds

Values: 0 to 300

Default: 60

9.2.1.2.85 li-x3-keep-alive-time-p2

Syntax

```
li-x3-keep-alive-time-p2 x3-time-p2-value
```

```
no li-x3-keep-alive-time-p2
```

Context

```
config>li>mobile>li-x3
```

Description

This command specifies the keep-alive time for the LI_X3 interface on the UPF. If the POI does not receive a keep-alive Acknowledgment PDU within *x3-time-p2-value* seconds (the default is 180 seconds), it disconnects the connection and attempts to reconnect to the MDF while reporting an error through the X1 interface as defined in ETSI TS 103 221-1.

The no form of this command reverts to the default. If the timer is set to 0 value, the keep-alive mechanism is disabled.

Parameters

x3-time-p2-value

specifies the keep-alive time, in seconds

Values: 0 to 1800

Default: 180

9.2.1.2.86 li-x3-sequence

Syntax

```
li-x3-sequence {enable | disable}  
no li-x3-sequence
```

Context

```
config>li>mobile>li-x3
```

Description

This command enables or disables the UPF support of the X3 sequence numbers. By default, this feature is enabled and the X3 Sequence Number IE is filled with the sequence number of the X3 PDU. The sequence number starts from zero and increments by one for each X3 PDU with the same XID, DID, NFID, IPID, and Correlation ID context. A separate sequence number is maintained for the X2 and X3 PDUs within the same context.



NOTE:

The sequence number is encoded as a four-octet unsigned integer in the network byte order. When the maximum sequence number is reached, the UPF restarts the sequence number from zero.

The no form of this command reverts to the default (enable).

Parameters

enable

enables the support of the X3 sequence numbers

disable

disables the support of the X3 sequence numbers (the X3 Sequence Number IE is not included)

Default: enable

9.2.1.2.87 li-x3-tls

Syntax

```
li-x3-tls {enable | disable}
```

```
no li-x3-tls
```

Context

```
config>li>mobile>li-x3
```

Description

This command controls the usage of TLS for the LI_X3 interface used for the xCC messages. TLS must be used for the 5G LI interfaces according to the standards, but it can be disabled with this command.

The no form of this command reverts to the default (enable).

Parameters

enable

enables the usage of TLS for the LI_X3 interface used for the xCC messages

disable

disables the usage of TLS for the LI_X3 interface used for the xCC messages

Default: enable

9.2.1.2.88 local-interface

Syntax

```
local-interface ip-address [router router-instance] [override-x2-interface x2-ip-address  
[x2-router x2-router-instance]]
```

```
no local-interface
```

Context

```
config>li>mobile
```

Description

This command configures the source IP address used by the xGW/GGSN for a Lawful Interception (LI) interface. When the `override-x2-interface` option is specified, X2 and X3 interfaces can each use a different VPRN context. When the `override-x2-interface` option is not specified, X2 and X3 interfaces use the same IP address and VPRN context.

The no form of this command reverts to the default.

Default

```
no local-interface
```


Parameters

ip-address

specifies the source IP address

Values		
	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:x.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

router-instance

specifies the router instance, up to 32 characters

x2-ip-address

specifies the source IP address

Values		
	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:x.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

x2-router-instance

specifies the router instance, up to 32 characters

9.2.1.2.89 nf-id-value

Syntax

```
nf-id-value {uuid | ip-addr | ip-addr-hex-string}
no nf-id-value
```

Context

```
config>li>mobile
```

Description

This command specifies the network function identifier choice of uuid or ip-addr to be used for IRI and CC messages over the LI_X1, LI_X2 and LI_X3 interfaces. When the ip-addr is selected, the configured local/source IP address for the LI_X1, LI_X2 and LI_X3 interfaces is used as the network function identifier in IRI and CC messages respectively.

The no form of this command reverts to the default (uuid).

Default

```
no nf-id-value
```

Parameters

uuid

specifies the network function identifier

ip-addr

specifies the local/source IP address configured for the LI_X1, LI_X2 and the LI_X3 interfaces for the IRI and CC messages respectively

ip-addr-hex-string

specifies the local/source IP address configured for the LI_X1, LI_X2 and the LI_X3 interfaces for the IRI and CC messages respectively encoded as a hex string (individual digits and the character '.' are converted to the hex form)

Default: uuid

9.2.1.2.90 num-conn-retries

Syntax

```
num-conn-retries num-of-retries
no num-conn-retries
```

Context

config>li>mobile

Description

This command is used to specify the number of (TCP) connection retries to the LIG server, before it is deemed to be down.

The no form of this command reverts to the default.

Parameters

num-of-retries

specifies the number of retries

Values: 1 to 10

Default: 3

9.2.1.2.91 operator-id

Syntax

operator-id *id-string*

no operator-id

Context

config>li>mobile

Description

This command is used to configure the operator identifier for an operator's deployment. The configured value is used to populate the operator-identifier field of the Network-Identifier IE.

The no form of this command removes the configuration.

Parameters

id-string

specifies the operator identifier, up to 5 alphanumeric characters

Default: oprId

9.2.1.2.92 pcrf-shared-key

Syntax

```
pcrf-shared-key key-value  
no pcrf-shared-key
```

Context

```
config>li>mobile
```

Description

This command configures the shared key used between the policy and charging rules function (PCRF) and the PGW. The shared key decrypts the LI-Indicator AVP at the PGW.

The no form of this command returns the configuration to the default setting.

Default

```
no pcrf-shared-key
```

Parameters

key-value

specifies the PCRF shared key value, up to 128 alphanumeric characters

9.2.1.2.93 per-bearer-seq-num

Syntax

```
[no] per-bearer-seq-num
```

Context

```
config>li>mobile
```

Description

This command facilitates generation of a sequence number for X3 CC packets on a per bearer (correlation number) basis. When this command is enabled, each new bearer with a unique correlation number is assigned a new separate sequence number that starts with 0 and is incremented for each packet flowing through that bearer.

For an already existing bearer, the sequence number increments by 1 for new packets forwarded on the bearer.

When this command is disabled, the sequence number is generated on a per PDN session basis and is shared across bearers belonging to the PDN session.

The no form of this command reverts to the default (that is, the sequence number is generated on a session basis).

Default

no per-bearer-seq-num

9.2.1.2.94 pfcpl-shared-key

Syntax

pfcpl-shared-key *key-value*

no pfcpl-shared-key

Context

config>li>mobile

Description

This command configures the shared key used between the CP and the UP over the Sx/N4 interface for the LI container IE sent from the SGW-C/PGW-C/SMF to the SGW-U/PGW-U/UPF. It is used to encrypt the LI container IE at the CP and decrypt the LI container IE at the SGW-U/PGW-U/UPF.

The no form of this command reverts to the default.

Parameters

key-value

specifies the Sx/N4 interface shared key value, up to 128 characters

9.2.1.2.95 server-tls-profile

Syntax

server-tls-profile *tls-profile-name*

no server-tls-profile

Context

config>li>mobile

Description

This command configures the default server TLS profile.

The no form of this command removes the default server TLS profile from the configuration.

Parameters

tls-profile-name

specifies the server TLS profile name, up to 32 characters.

This is the name of the profile created under the
config>system>security>tls>server-tls-profile context.

9.2.1.2.96 target

Syntax

```
target target-type id target-value intercept intercept peer df-peer-id [liid liid-string]  
[secondary-peer sec-df-peer-id] [ha-peer-id ha-df-peer-id] [correlation-id correlation-id]  
[bearer-id bearer-id]
```

```
no target target-type id target-value [liid liid-string]
```

Context

```
config>li>mobile
```

Description

This command configures a target for interception and assigns the DF Peer that receives the IRI and CC for this target.

LIIDs can be specified for a target and are used in conjunction with the fan-out command to enable separate IRI or IRI+CC streams to the specified DF peer when there are multiple activations of the target.

If a target is activated without using a unique LIID, all IRI and CC messages for this target are sent to the newly specified DF peer, subsequent to target modifications.

A total of up to 10 target activations are allowed for a UE. There could be, for example, 5 activations using IMSI, 3 activations using MSISDN, and 2 activations using IMEI. A maximum of 7 activations are supported using the same target type (for example, IMSI).

Modifying the target command's parameters does not require a gateway shutdown.

The no form of this command deactivates a target that is being intercepted. If more than one activation exists for a UE, each activation must be deactivated separately using the same LIID used at the time of activation.

Parameters

target-type
specifies the type of surveillance target identifier to be provisioned. A maximum number of 80K targets (irrespective of target-type) is supported.
Values: imei | imsi | msisdn | mac| nai | s8hr-apn| s8hr-media

target-value
uniquely identifies a target for the interception

Values	for imsi, imei, msisdn	up to 16 digits
–	for nai	username@realm, up to 253 characters
–	for mac	an address in the format of xx:xx:xx:xx:xx:xx or xx-xx-xx-xx-xx-xx
–	for s8hr-apn	up to 80 characters
–	for 8hr-media	up to 16 digits

liid-string
uniquely identifies the Lawful Interception Identifier (LIID), up to 25 characters

intercept
specifies the interception type for the target; the intercept type is allowed to change from IRI to IRI+CC and from IRI+CC to IRI

Values	iri	Interception Related Information
–	iricc	Interception Related Information (IRI) and Content of Communication (CC)
–	cc	used for HA interception only; this option is supported when the Target Type = NAI

Default: iri

df-peer-id

specifies the DF peer associated with the target

Values: 1 to 254 for CMG, 1 to 15 for 7750 SR MG

Default: 1

sec-df-peer-id

specifies an optional secondary DF peer associated with the target that is used when the primary DF peer is Down

Values: 1 to 254 for CMG, 1 to 15 for 7750 SR MG

Default: 0

**NOTE:**

The secondary DF peer must be different from the primary peer and the configuration must match the primary DF peer; see the `config>mobile>li>df-peer` command.

ha-df-peer-id

specifies the Home Agent DF peer associated with the target

Values: 1 to 254 for CMG, 1 to 15 for 7750 SR MG

correlation-id

specifies the CC correlation ID

Values: 8 or 20 octets correlation identifier without separator

bearer-id

specifies the bearer ID

Values: 1 to 15

9.2.1.2.97 tcp-keepalive

Syntax

tcp-keepalive

Context

config>li>mobile

Description

This command enables the context to configure the TCP keep-alive parameters for the X2 and X3 interfaces.

9.2.1.2.98 idle-time

Syntax

`idle-time idle`

`no idle-time`

Context

`config>li>mobile>tcp-keepalive`

Description

This command configures the time before sending the first TCP keep-alive message.

The no form of this command reverts to the default value.

Parameters

idle

specifies the time before sending the first TCP keep-alive message, in seconds

Values: 1 to 100000

Default: 600

9.2.1.2.99 interval

Syntax

`interval interval`

`no interval`

Context

`config>li>mobile>tcp-keepalive`

Description

This command configures the time between two TCP keep-alive messages.

The no form of this command reverts to the default value.

Parameters

interval

specifies the time between two TCP keep-alive messages, in seconds

Values: 1 to 100000

Default: 15

9.2.1.2.100 retries

Syntax

`retries count`

`no retries`

Context

`config>li>mobile>tcp-keepalive`

Description

This command configures the number of missed keep-alive messages before the TCP connection is declared down.

The no form of this command reverts to the default value.

Parameters

count

specifies the number of missed keep-alive messages before the TCP connection is declared down

Values: 3 to 100

Default: 4

9.2.1.2.101 shutdown

Syntax

`[no] shutdown`

Context

`config>li>mobile>tcp-keepalive`

Description

This command enables or disables the TCP keep-alive messages.

The no form of this command reverts to the default (disable).

Default

shutdown

9.2.1.2.102 `tls`

Syntax

[no] `tls`

Context

config>li>mobile

Description

This command enables the TLS protocol in the LI X2 and X3 interfaces.

To enable the TLS protocol, make sure that:

- valid client TLS profiles in all DF peers is configured
- the client TLS profiles are in no shutdown state
- the X3 interface has TCP as the transport option

The no form of this command reverts to the default (TLS disabled).

Default

no `tls`

9.2.1.2.103 `x2-iri-cache-size`

Syntax

`x2-iri-cache-size` *cache-size*

no `x2-iri-cache-size`

Context

config>li>mobile

Description

This command is used to specify the size of the IRI buffer cache. (IRI messages buffered when connectivity to the LIG is down). Setting the cache size to a value of 0 disables buffering.

The no form of this command reverts to the default.

Parameters

cache-size

specifies the size of the IRI buffer cache, in Mbytes

Values: 0 to 1

Default: 1

9.2.1.2.104 x2-iri-qos

Syntax

```
x2-iri-qos dscp {dscp-value | dscp-name}
```

```
no x2-iri-qos
```

Context

```
config>li>mobile
```

Description

This command specifies the DSCP to be set for IRI (Interception Related Information) messages sent to a LIG (Lawful Interception Gateway).

The no form of this command reverts to the default.

Parameters

dscp-value

specifies the value of the DSCP to be set for IRI

Values: 0 to 63

dscp-name

specifies the name of the DSCP to be set for IRI

Values: none | be | ef | cp1 | cp2 | cp3 | cp4 | cp5 | cp6 | cp7 | cp9 | cs1 | cs2 | cs3 | cs4 | cs5 | nc1 | nc2 | af11 | af12 | af13 | af21 | af22 | af23 | af31 | af32 | af33 | af41 | af42 | af43 | cp11 | cp13 | cp15 | cp17 | cp19 | cp21 | cp23 | cp25 | cp27 | cp29 | cp31 | cp33 | cp35 | cp37 | cp39 | cp41 | cp42 | cp43 | cp44 | cp45 | cp47 | cp49 | cp50 | cp51 | cp52 | cp53 | cp54 | cp55 | cp57 | cp58 | cp59 | cp60 | cp61 | cp62 | cp63

Default: af41

9.2.1.2.105 x2-keep-alive-time

Syntax

x2-keep-alive-time *keep-alive-time*
no x2-keep-alive-time

Context

config>li>mobile

Description

This command is used to specify the keep-alive time for x2 interface. An LI message including a keep-alive parameter is sent to the LIG when no LI message has been sent for the configured amount of time (for example, five minutes), indicating to the LIG that the LI connection is still up.

The no form of this command reverts to the default. A value of 0 disables the keep-alive mechanism.

Parameters

keep-alive-time

specifies the keep alive time, in minutes

Values: 0 to 5

Default: 0

9.2.1.2.106 x3-cc-cache-size

Syntax

x3-cc-cache-size *cache-size*
no x3-cc-cache-size

Context

config>li>mobile

Description

This command specifies the size of the CC buffer cache (CC packets buffered when connectivity to LIG is down). Setting the cache size to a value of 0 disables buffering.

The no form of this command reverts to the default.

**NOTE:**

From release 23.11.R1 onwards, this command applies only to 4G. For prior releases, the command is also applicable to 5G.

Parameters

cache-size

specifies the size of the CC buffer cache, in Mbytes

Values: 0 to 10

Default: 1

9.2.1.2.107 x3-cc-qos

Syntax

```
x3-cc-qos dscp {dscp-value | dscp-name}
```

```
no x3-cc-qos dscp
```

Context

```
config>li>mobile
```

Description

This command specifies the DSCP to be set for CC (content of Communication) traffic sent to a LIG (Lawful Interception Gateway). No form of this command reverts to the default. For ULICv0, the value is "be" and cannot be changed.

Applies to Transport Protocol and ULIC-Header versions:

- TCP with ULICv1
- UDP with ULICv1
- UDP with ULICv0

Parameters

dscp-value

specifies the DSCP value

Values: 0 to 63

dscp-name

specifies the DSCP name

Values: none | be | ef | cp1 | cp2 | cp3 | cp4 | cp5 | cp6 | cp7 | cp9 | cs1 | cs2 | cs3 | cs4 | cs5 | nc1 | nc2 | af11 | af12 | af13 | af21 | af22 | af23 | af31 | af32 | af33 | af41 | af42 | af43 | cp11 | cp13 | cp15 | cp17 | cp19 | cp21 | cp23 | cp25 | cp27 | cp29 | cp31 | cp33 | cp35 | cp37 | cp39 | cp41 | cp42 | cp43 | cp44 | cp45 | cp47 | cp49 | cp50 | cp51 | cp52 | cp53 | cp54 | cp55 | cp57 | cp58 | cp59 | cp60 | cp61 | cp62 | cp63

Default: be (if ulic-header is v0) | af41 (if ulic-header is v1)

9.2.1.2.108 x3-keep-alive-time

Syntax

x3-keep-alive-time *keep-alive-time*

no x3-keep-alive-time

Context

config>li>mobile

Description

This command specifies the keep alive time for x3 interface. A connection status message including a keep alive parameter is sent when no CC message has been sent for the configured amount of time (for example, 5 minutes), indicating to the LIG that the LI connection is still up.

The no form of this command reverts to the default. A value of 0 disables the keep-alive mechanism.

Parameters

keep-alive-time

specifies the keep alive time, in minutes

Values: 0 to 5

Default: 0

9.2.1.2.109 x3-transport

Syntax

x3-transport {tcp | udp} [ulic-header {v0 | v1}]

Context

config>li>mobile

Description

This command specifies the transport option for an X3 interface, along with the ULIC Header version to be used. The same transport option is supported to all the Delivery Function (DF) peers in a service provider network. Changing the option requires a gateway shutdown/no shutdown.

The following are the valid combinations of Transport protocol and supported ULIC Header versions:

- TCP with ULIC Header v1
- UDP with ULIC Header v1
- UDP with ULIC Header v0

Parameters

tcp

specifies TCP as the transport option for an X3 interface; also applies to Home Agent

udp

specifies UDP as the transport option for an X3 interface

ulic-header v0 | v1

specifies the header option

Default: v0 if x3-transport is udp | v1 if x3-transport is tcp

for Home Agent, the cdma2000® Lawful Interception Correlation (CLIC) header, per J-STD-025-B, is used

9.2.1.2.110 save

Syntax

save

Context

config>li

Description

This command is required to save LI configuration parameters.

9.2.1.2.111 tech-support

Syntax

tech-support {standard | limited-li-content}

no tech-support

Context

config>li

Description

This command includes or excludes the LI information in the generated tech support file.

The no form of this command reverts to the default value.

Parameters

standard

specifies to include obfuscated LI information in the generated tech support file

limited-li-content

specifies to include obfuscated LI information if the tech support file is generated by users with access to LI configuration; the obfuscated LI information is excluded if the file is generated by non-LI users

Default: standard

9.2.1.2.112 Other LI configuration commands

The following commands are also described in the *7450 ESS, 7750 SR, 7950 XRS, and VSR Basic System Configuration Guide* or the *7450 ESS, 7750 SR, 7950 XRS, and VSR System Management Guide*.

9.2.1.2.112.1 profile

Syntax

[no] profile *user-profile-name*

Context

config>system>security

Description

This command enables a context to create user profiles for CLI command tree permissions. Profiles are used to either deny or permit user console access to a hierarchical branch or to specific commands. After the profiles are created, the user command assigns users to one or more profiles. You can define up to 16 user profiles but a maximum of 8 profiles can be assigned to a user. The *user-profile-name* can consist of up to 32 alphanumeric characters.

The no form of this command deletes a user profile.

Default

user-profile default

Parameters

user-profile-name

The user profile name entered as a character string. The string is case sensitive and limited to 32 ASCII 7-bit printable characters with no spaces.

9.2.1.2.112.2 li

Syntax

li

Context

config>system>security>profile

Description

This command enables the Lawful Interception (LI) profile identifier.

Default

no li

9.2.1.2.112.3 access

Syntax

[no] access [ftp] [snmp] [console] [li]

Context

config>system>security>user

Description

This command grants a user permission for FTP, SNMP, console or lawful intercept (LI) access. If a user requires access to more than one application, the multiple applications can be specified in a single command. Multiple commands are treated additively.

The no form of this command removes access for a specific application and denies permission for all management access methods. To deny a single access method, enter the no form of this command followed by the method to be denied, for example, `no access ftp` denies FTP access.

Default

no access

Parameters

ftp

specifies FTP permission

snmp

specifies SNMP permission; this keyword is only configurable in the `config>system>security>user` context

console

specifies console access (serial port or Telnet) permission

li

allows user to access CLI commands in the lawful intercept (LI) context

9.2.1.2.112.4 li-local-save

Syntax

[no] `li-local-save`

Context

bof

Description

This command specifies whether or not lawful intercept (LI) configuration is allowed to be save to a local file. Modifying this command will not take affect until the system is rebooted.

Default

li-local-save

9.2.1.2.112.5 li-memory-encrypt

Syntax

[no] li-memory-encrypt

Context

bof

Description

This command configures the LI in sensitivity mode. To enable it, execute the bof save command and reboot the system.

Default

no li-memory-encrypt

9.2.1.2.112.6 li-separate

Syntax

[no] li-separate

Context

bof

Description

This command specifies whether or not a non-LI user has access to lawful intercept (LI) information. When this command is enabled, a user who does not have LI access will not be allowed to access CLI or SNMP objects in the li context. Modifying this command will not take affect until the system is rebooted.

When the no li-separate command is set (the default mode), those who are allowed access to the config>system>security>profile context and user command nodes are allowed to modify the configuration of the LI parameters. In this mode, a user that has a profile allowing access to the config>li and/or show>li command contexts can enter and use the commands under those nodes.

When the li-separate command is configured, only users that have the LI access capabilities set in the config>system>security>user>access li context are allowed to access the config>li and/or show>li command contexts. A user who does not have LI access is not allowed to enter the config>li and show>li contexts even though they have a profile that allows access to these nodes. When in the li-separate mode, only users with config>system>security>user>access li set in their user

account have the ability modify the setting LI parameters in either their own or others profiles and user configurations.

Default

no li-separate

9.2.1.2.112.7 save

Syntax

save [*cflash-id*]

Context

bof

Description

This command saves the LI sensitivity mode configured using the `li-memory-encrypt` command. To apply the changes, reboot the system.

Parameters

cflash-id

specifies the flash slot ID

Values: cf1: | cf1-A: | cf1-B: | cf2: | cf2-A: | cf2-B: | cf3: | cf3-A: | cf3-B:

9.2.2 Lawful Interception show command descriptions

9.2.2.1 Lawful Interception show command descriptions

9.2.2.1.1 debug

Syntax

debug [*application*]

Context

show

Description

This command displays set debug points.

Parameters

application

displays which debug points have been set

Values: Some examples of applications include service, ip, ospf, ospf3, bgp, mtrace, isis, mpls, rsvp, ldp, mirror, vrrp, system, filter, lag and oam.

Output

The following output example shows debug point information.

Example output debug information

```
# show debug
debug
  mirror-source 100
    subscriber "user1" ingress
    subscriber "user2" fc be h2 h1 nc egress
    subscriber "user3" ingress egress
    subscriber "user4" sap 1/1/2:1 fc af ef nc ingress
    subscriber "user5" sap 1/1/2:1 egress
    subscriber "user6" sap 1/1/2:1 fc be l2 af h2 ef nc ingress egress
    subscriber "user7" sap 1/1/2:1 ip 1.1.0.7 fc l1 h2 ingress
    subscriber "user8" sap 1/1/2:1 ip 1.1.0.8 fc af l1 h2 ef nc egress
    subscriber "user9" sap 1/1/2:1 ip 1.1.0.9 ingress egress
    subscriber "user10" sap 1/1/2:1 mac 00:00:01:00:00:01 fc be l2 l1 h1 nc ingress
    subscriber "user11" sap 1/1/2:1 mac 00:00:01:00:00:02 fc be l1 h2 ef h1 egress
    subscriber "user12" sap 1/1/2:1 mac 00:00:01:00:00:03 fc be ef ingress egress
    subscriber "user13" sap 1/1/2:1 ip 1.1.0.13 mac 00:00:01:00:00:01 fc be ef h1 ingress
    subscriber "user14" sap 1/1/2:1 ip 1.1.0.14 mac 00:00:01:00:00:02 egress
    subscriber "user15" sap 1/1/2:1 ip 1.1.0.15 mac 00:00:01:00:00:03 fc af l1 ef nc ingress egress
    subscriber "user16" sla-profile "sla1" ingress
    subscriber "user17" sla-profile "sla2" egress
    subscriber "user18" sla-profile "sla3" fc be af h2 ingress egress
  no shutdown
exit
exit

# show debug
debug
  mirror-source 101
    port 1/1/1 ingress
    no shutdown
  exit
  mirror-source 102
    port 1/1/3 egress
    no shutdown
  exit
```

```
exit
```

9.2.2.1.2 active-subscribers

Syntax

```
active-subscribers summary  
  
active-subscribers [subscriber sub-ident-string [sap sap-id sla-profile sla-profile-name]]  
[detail | mirror]  
  
active-subscribers hierarchy [subscriber sub-ident-string]
```

Context

```
show>service
```

Description

This command displays active subscriber information.

Parameters

sub-ident-string

specifies an existing subscriber identification string

sap-id

specifies the physical port identifier portion of the SAP definition

sla-profile-name

displays an existing SLA profile name

hierarchy

keyword to display the subscriber hierarchy

summary

keyword to display a subscriber summary

detail

keyword to display detailed information

mirror

keyword to include ingress and egress mirror information

Output

The following output example shows active subscriber information.

Example output active subscribers

```
# show service active-subscribers mirror
=====
Active Subscribers
=====
Subscriber user1 (sub1)
-----
SLA Profile Instance sap:lag-8:1 - sla:sla1
-----
IP Address      MAC Address      Origin
-----
1.1.0.1         00:00:01:00:00:01 Static
                  Ingress mirror:    100   l2 af l1 nc
-----
SLA Profile Instance sap:lag-8:11 - sla:sla1
-----
IP Address      MAC Address      Origin
-----
11.1.0.1        00:00:01:00:00:01 Static
                  Ingress mirror:    100   l2 af l1 nc
-----
Subscriber user10 (sub1)
-----
SLA Profile Instance sap:lag-8:1 - sla:sla1
-----
IP Address      MAC Address      Origin
-----
1.1.0.10        00:00:01:00:00:01 Static
                  Ingress mirror:    100   af ef h1 nc
-----
Subscriber user11 (sub1)
-----
SLA Profile Instance sap:lag-8:1 - sla:sla1
-----
IP Address      MAC Address      Origin
-----
1.1.0.11        00:00:01:00:00:02 Static
                  Egress mirror:     100   l2 ef h1
-----
Subscriber user12 (sub1)
-----
SLA Profile Instance sap:lag-8:1 - sla:sla1
-----
IP Address      MAC Address      Origin
-----
1.1.0.12        00:00:01:00:00:03 Static
                  Ingress mirror:    100   be l2 af l1 h2 ef h1 nc
                  Egress mirror:    100   be l2 af l1 h2 ef h1 nc
-----
Subscriber user13 (sub1)
-----
SLA Profile Instance sap:lag-8:1 - sla:sla1
-----
IP Address      MAC Address      Origin
-----
```



```

1.1.0.13      00:00:01:00:00:01 Static
              Ingress mirror:      100   l1 ef h1
-----
Subscriber user14 (sub1)
-----
SLA Profile Instance sap:lag-8:1 - sla:sla1
-----
IP Address      MAC Address      Origin
-----
1.1.0.14      00:00:01:00:00:02 Static
              Egress mirror:      100   l2 h2 ef h1
-----
Subscriber user15 (sub1)
-----
SLA Profile Instance sap:lag-8:1 - sla:sla1
-----
IP Address      MAC Address      Origin
-----
1.1.0.15      00:00:01:00:00:03 Static
              Ingress mirror:      100   l1 nc
              Egress mirror:      100   l1 nc
-----
Subscriber user16 (sub1)
-----
SLA Profile Instance sap:lag-8:1 - sla:sla1
-----
IP Address      MAC Address      Origin
-----
1.1.0.16      00:00:01:00:00:01 Static
              Ingress mirror:      100   be l2 af nc
-----
SLA Profile Instance sap:lag-8:11 - sla:sla1
-----
IP Address      MAC Address      Origin
-----
11.1.0.16     00:00:01:00:00:01 Static
              Ingress mirror:      100   be l2 af nc
-----
Subscriber user17 (sub1)
-----
SLA Profile Instance sap:lag-8:1 - sla:sla2
-----
IP Address      MAC Address      Origin
-----
1.1.0.17      00:00:01:00:00:01 Static
              Egress mirror:      100   af l1 h1
-----
SLA Profile Instance sap:lag-8:11 - sla:sla2
-----
IP Address      MAC Address      Origin
-----
11.1.0.17     00:00:01:00:00:01 Static
              Egress mirror:      100   af l1 h1
-----
Subscriber user18 (sub1)
-----
SLA Profile Instance sap:lag-8:1 - sla:sla3

```

```

-----
IP Address      MAC Address      Origin
-----
1.1.0.18        00:00:01:00:00:01 Static
                  Ingress mirror:    100   h2
                  Egress  mirror:    100   h2
-----

SLA Profile Instance sap:lag-8:11 - sla:sla3
-----
IP Address      MAC Address      Origin
-----
11.1.0.18        00:00:01:00:00:01 Static
                  Ingress mirror:    100   h2
                  Egress  mirror:    100   h2
-----

Subscriber user2 (sub1)
-----
SLA Profile Instance sap:lag-8:1 - sla:sla1
-----
IP Address      MAC Address      Origin
-----
1.1.0.2          00:00:01:00:00:01 Static
                  Egress  mirror:    100   be l2 af l1 h2 ef h1 nc
-----

SLA Profile Instance sap:lag-8:11 - sla:sla1
-----
IP Address      MAC Address      Origin
-----
11.1.0.2          00:00:01:00:00:01 Static
                  Egress  mirror:    100   be l2 af l1 h2 ef h1 nc
-----

Subscriber user3 (sub1)
-----
SLA Profile Instance sap:lag-8:1 - sla:sla1
-----
IP Address      MAC Address      Origin
-----
1.1.0.3          00:00:01:00:00:01 Static
                  Ingress mirror:    100   be l2 af l1 h2 ef h1 nc
                  Egress  mirror:    100   be l2 af l1 h2 ef h1 nc
-----

SLA Profile Instance sap:lag-8:11 - sla:sla1
-----
IP Address      MAC Address      Origin
-----
11.1.0.3          00:00:01:00:00:01 Static
                  Ingress mirror:    100   be l2 af l1 h2 ef h1 nc
                  Egress  mirror:    100   be l2 af l1 h2 ef h1 nc
-----

Subscriber user4 (sub1)
-----
SLA Profile Instance sap:lag-8:1 - sla:sla1
-----
IP Address      MAC Address      Origin
-----
1.1.0.4          00:00:01:00:00:01 Static
                  Ingress mirror:    100   be l2 af l1 h2 ef h1 nc

```

```

-----
Subscriber user5 (sub1)
-----
SLA Profile Instance sap:lag-8:1 - sla:sla1
-----
IP Address      MAC Address      Origin
-----
1.1.0.5         00:00:01:00:00:01 Static
                  Egress mirror:    100   be l2 af l1 h2 ef h1 nc
-----
Subscriber user6 (sub1)
-----
SLA Profile Instance sap:lag-8:1 - sla:sla1
-----
IP Address      MAC Address      Origin
-----
1.1.0.6         00:00:01:00:00:01 Static
                  Ingress mirror:    100   be af l1 h2
                  Egress mirror:    100   be af l1 h2
-----
Subscriber user7 (sub1)
-----
SLA Profile Instance sap:lag-8:1 - sla:sla1
-----
IP Address      MAC Address      Origin
-----
1.1.0.7         00:00:01:00:00:01 Static
                  Ingress mirror:    100   be l2 af l1 h2 ef h1 nc
-----
Subscriber user8 (sub1)
-----
SLA Profile Instance sap:lag-8:1 - sla:sla1
-----
IP Address      MAC Address      Origin
-----
1.1.0.8         00:00:01:00:00:01 Static
                  Egress mirror:    100   be af l1 h1 nc
-----
Subscriber user9 (sub1)
-----
SLA Profile Instance sap:lag-8:1 - sla:sla1
-----
IP Address      MAC Address      Origin
-----
1.1.0.9         00:00:01:00:00:01 Static
                  Ingress mirror:    100   be l2 af l1 h2 ef h1 nc
                  Egress mirror:    100   be l2 af l1 h2 ef h1 nc
=====

```

9.2.2.1.3 service-using

Syntax

service-using [mirror]

Context

show>service

Description

This command displays mirror services.

If no optional parameters are specified, all services defined on the system are displayed.

Parameters

mirror

displays mirror services

Output

Service using output fields describes service-using mirror output fields:

Table 181: Service using output fields

Label	Description
Service Id	The service identifier
Type	The service type configured for the service ID
Adm	The desired state of the service
Opr	The operating state of the service
CustomerID	The ID of the customer who owns this service
Last Mgmt Change	The date and time of the most recent management-initiated change to this service

The following output example shows mirror service information.

Example output service using mirror

```
# show service service-using mirror
```

```
=====
Services [mirror]
```

```
=====
```

ServiceId	Type	Adm	Opr	CustomerId	Last Mgmt Change

218	Mirror	Up	Down	1	04/08/2007 13:49:57
318	Mirror	Down	Down	1	04/08/2007 13:49:57
319	Mirror	Up	Down	1	04/08/2007 13:49:57
320	Mirror	Up	Down	1	04/08/2007 13:49:57
1000	Mirror	Down	Down	1	04/08/2007 13:49:57
1216	Mirror	Up	Down	1	04/08/2007 13:49:57
1412412	Mirror	Down	Down	1	04/08/2007 13:49:57

Matching Services : 7					
=====					

9.2.2.1.4 li

Syntax

```
li
```

Context

```
show
```

Description

Displays LI information.

9.2.2.1.5 li-source

Syntax

```
li-source [service-id]
```

Context

```
show>li
```

Description

Displays LI mirror configuration and operation information.

Parameters

service-id

specifies the service ID

Values: 1 to 2147483647

Output

The following output example shows LI mirror configuration and operation information.

Example output LI source

```
# show li li-source 2
=====
Mirror Service
=====
Service Id       : 2                Type           : Ether
Admin State      : Up               Oper State      : Up
Forwarding Class : be              Remote Sources: No
Slice           : 0
Destination SDP  : 1000 (100.1.1.2) Egress Label   : 4000
Signaling        : None

-----
Local Sources
-----
Admin State      : Up

- IP Filter      1          Entry 1
=====
```

9.2.2.1.6 filter

Syntax

```
li-ip filter-name {counter | associations}
li-ip filter-name entry entry-id [counters]
li-ipv6 filter-name {counter | associations}
li-ipv6 filter-name entry entry-id [counters]
```

Context

```
show>li
```

Description

Displays LI mirror IPv4 or IPv6 address filter configuration and operation information.

Parameters

filter-name

specifies the LI filter name

entry-id

specifies the LI filter entry ID number

counter

specifies LI filter counter information

associations

specifies LI filter association information

9.2.2.1.7 log

Syntax

log

Context

show>li

Description

Displays LI event log information.

9.2.2.1.8 mobile-gateway

Syntax

mobile-gateway df-peer [*df-peer-id*]

mobile-gateway df-peer ip-addr *ip-address* | *ipV6-address*

mobile-gateway summary

mobile-gateway target [*target-type*] [id *target-value*] [gx] [*liid liid-string*]

mobile-gateway target *target-type* id *target-value* detail

mobile-gateway target *target-type* id *target-value* stats

Context

show>li

Description

Displays LI configuration and operation information.

When retrieving target information for an individual UE, the following are displayed:

- all known target identities, regardless of the *target-type* specified; for example, if IMSI is specified as the *target-type*, the displayed information includes all other known target identities for the subscriber (such as MSISDN and IMEI)
- latest location information of the UE known to the gateway and the ULI time
- UE State (idle, connected), Attach Time, and Failure Reason (Session/Context not found)

This command supports retrieving this information regardless of whether the specified identifier is a target for interception.

Parameters

df-peer-id
specifies the ID of the DF peer
Values: 1 to 254 for CMG, 1 to 15 for 7750 SR MG

ip-address | ipv6-address
specifies the IP address for 5G peers

Values	ip-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x: [0..FFFF]H
–	–	d: [0..255]D

target-type
specifies the specific type of surveillance target identifier
Values: imsi | imei | msisdn | mac | nai | s8hr-apn | s8hr-media

target-value
specifies a unique target for the interception

Values	for imsi, imei, msisdn	up to 16 digits
--------	------------------------	-----------------

- for nai username@realm, up to 253 chars max
- for mac an address in the format of xx:xx:xx:xx:xx:xx or xx-xx-xx-xx-xx-xx
- for s8hr-apn up to 80 characters
- for 8hr-media up to 16 digits

gx

specifies to display Gx information

summary

specifies to display a summary of the LI configuration and operation information

liid-string

Specifies the LIID for which target activation information is displayed; the LI identifier can be up to 25 characters.

If this parameter is not specified and there is more than one target activation for a target type (for example, target type IMSI), information pertaining to all activations for the target type will be displayed, including the LIIDs of each activation. If a target type is not specified, information pertaining to all activations will be displayed.

detail

specifies to display additional information of the UE, such as other identities and location information

stats

specifies to display the IRI counts per relevant IRI message type

Output

The following output examples show LI configuration and operation information.

- *Example output LI DF peer*
- *Example output LI DF peer for 5G*
- *Example output LI summary*
- *Example output LI target*
- *Example output LI target IMSI*
- *Example output LI target IMEI*
- *Example output LI target MSISDN*

- *Example output LI target Gx*
- *Example output LI target IMSI Gx*
- *Example output LI target IMSI ID Detail*
- *Example output LI target IMSI ID stats*
- *Example output LI target S8HR APN*
- *Example output LI target S8HR APN ID stats*
- *Example output LI target S8HR media stats*

Example output LI DF peer

```
# show li mobile-gateway df-peer 1
=====
LI DF peer information
=====
DF peer id           : 1           Peer type: Pre 5G
DF2 peer address      : 10.13.8.10
DF2 peer port         : 8315        DF2 pkts tx           : 0
DF2 Client TLS Profile : (Not Specified)
DF3 peer address      : 10.13.8.10
DF3 peer port         : 8317        DF3 pkts tx           : 2
DF3 Client TLS Profile : (Not Specified)
Total X1 targets per peer : 0           Total Gx targets per peer : 0
Number of xCC octets    : 352

Group ID              : 1
DF2 Oper state        : Peer connected
DF2 pkts tx           : 0
DF3 Oper state        : Peer connected
DF3 pkts tx           : 2
-----
Number of peers : 1
=====
```

Example output LI DF peer for 5G

The following example output applies to 5G peers when the X2 and X3 interfaces are defined.

```
# show li mobile-gateway df-peer ip-addr 10.23.20.10
=====
LI DF peer information
=====
Peer id              : 1           Peer type              : 5G
Destination Identifier : 4f675db7-5c4f-2757-9361-17c45a809791
Intercept type       : iri
Peer address         : 10.23.20.10
Peer port            : 8316
Client TLS Profile    : client
```

```

Total X1 targets per peer : 0          Number of xCC octets      : 0

Group ID                  : 1
DF2 Oper state           : Peer connected
DF2 pkts tx              : 0
DF3 Oper state           : Down
DF3 pkts tx              : 0
-----
Peer id                   : 2          Peer type                 : 5G
Destination Identifier    : 64f785a1-23da-2874-8f6a-f5574dff73e2
Intercept type            : CC
Peer address              : 10.23.20.10
Peer port                 : 8316
Client TLS Profile        : client
Total X1 targets per peer : 0          Number of xCC octets      : 0

Group ID                  : 1
DF2 Oper state           : Down
DF2 pkts tx              : 0
DF3 Oper state           : Peer connected
DF3 pkts tx              : 0
-----
Number of peers : 2
=====

```

Example output LI summary

The following example output applies to LI SMF when the X2 and X3 interfaces are in use.



NOTE:

- The Total peers field is incremented for peers that are client connections.
 - For the SMF, this includes 4G, 5G, 5G ADMF and LI_T3 UP peers.
 - For the UPF, this includes 4G, 5G and 5G ADMF peers.
- In 5G LI the following apply for the counters related to LI targets, part of this example output:
 - For the SMF, the LI targets are measured in two counters. Depending on the delivery type the counters are:
 - Total X1 targets and Total IRI-CC targets when X2 and X3 apply
 - Total X1 targets and Total IRI targets when only X2 applies
 - Total X1 targets and Total CC targets when only X3 applies

- For the UPF, the LI targets are measured in one counter, that is, Total X1 targets; this is actually all the CC targets. The Total IRI targets and Total CC targets counters are not used in the UPF.

```
# show li mobile-gateway summary
=====
LI summary
=====
Total X1 targets      : 32768          Total peers          : 16
Total Gx targets      : 0
Total IRI targets     : 0              Total IRI-CC targets : 32768
Total CC targets      : 0
X3 transport type     : TCP            ULIC header          : v1
Router context X3     : vprn20005      Local interface X3    : 10.23.8.1
Router context X2     : vprn20005      Local interface X2    : 10.23.8.1
TLS                   : enabled
Operator ID           : 31001
IRI-DSCP               : af41          CC-DSCP              : af41
Num of conn retries   : 3              X2 Keep alive time   : disabled
X2 IRI Cache size     : 1 MB           X3 Keep alive time   : disabled
X3 CC Cache size      : 1 MB
PCRF Shared Key       : mySecret       3GPP Release         : rel-base
Gw Function Iri Mode  : enabled        IRI per MB Request   : enabled
Fan-out               : disabled       TEID                 : disabled
PFCP LI Shared Key    : mySecret
=====
```

Example output LI target

```
# show li mobile-gateway target
=====
LI Target Information
=====
Target id (imsi)      : 123456789099001
Intercept type        : iricc          Df peer              : 1
Li ID                 : liid1
Sec Df peer           : 0
-----
Target id (imsi)      : 123456789099012
Intercept type        : iri            Df peer              : 1
Li ID                 : not used
Sec Df peer           : 0
-----
Target id (imsi)      : 123456789099013
Intercept type        : cc             Df peer              : 1
Li ID                 : N/A
Sec Df peer           : N/A
Gx-enabled :          : yes
-----
Number of targets : 3
=====
```

Example output LI target IMSI

```
# show li mobile-gateway target imsi id 123456789011111
=====
LI Target Information
=====
Target id (imsi)   : 123456789011111
Intercept type    : iricc                Df peer          : 1
Li ID             : liid1
Sec Df Peer       : 0                    HA Df peer        : 0
-----
Number of targets : 1
=====
```

Example output LI target IMEI

```
# show li mobile-gateway target imei id 123456789022222
=====
LI Target Information
=====
Target id (imei)   : 123456789022222
Intercept type     : iri                 Df peer          : 1
Li ID              : liid2
Sec Df Peer        : 0                    HA Df peer        : 0
-----
Number of targets : 1
=====
```

Example output LI target MSISDN

```
# show li mobile-gateway target msisdn id 123456789033333
=====
LI Target Information
=====
Target id (msisdn) : 123456789033333
Intercept type     : iricc                Df peer          : 1
Li ID              : liid3
Sec Df Peer        : 0                    HA Df peer        : 0
-----
Number of targets : 1
=====
```

Example output LI target Gx

```
# show li mobile-gateway target gx
=====
LI Target Information
=====
Target id (imsi)    : 123456789099001
Intercept type      : cc                  Gx-enabled       : <yes>
Gx-LI-Correlation-ID : 11:22:33:44:55:66:7a:01
                   : 11:22:33:44:55:66:7a:02
-----
Number of targets : 1
=====
```

Example output LI target IMSI Gx

```
# show li mobile-gateway target gx imsi id 123456789099001
=====
LI Target Information
=====
Target id (imsi)      : 123456789099001
Intercept type       : cc                Gx-enabled          : <yes>
-----
Session ID           : 1
-----
Gx-LI-Correlation-ID : 11:22:33:44:55:66:7a:01
Gx Target ID         : 01:02:0a:01      DF peer             : 1
Charging Rule: dynRule2
-----
Gx-LI-Correlation-ID : 11:22:33:44:55:66:7a:02
Gx Target ID         : 01:02:0a:02      DF peer             : 1
Charging Rule: dynRule3
-----
Number of targets : 1
=====
```

LI target Gx and Gx IMSI ID output fields describes the LI target Gx and Gx IMSI ID output fields.

Table 182: LI target Gx and Gx IMSI ID output fields

Label	Description
Target Id	The IMSI target identifier
Intercept type	The intercept type configured for the target ID
Gx-LI-Correlation-ID	The correlation identifier associated with the LI
Gx Target ID	The Gx target identifier
Charging Rule	The charging rule associated with the Gx-LI-Correlation-ID
Number of targets	The total number of targets
Session ID	The session identifier

Example output LI target IMSI ID detail

```
# show li mobile-gateway target imsi id 123456789011111 detail
```

```

=====
LI Target Information
=====
Target id (imsi)   : 123456789011111
Intercept type    : iricc                Df peer          : 1
Li ID             : liid1
Sec Df Peer       : 0                    HA Df peer        : 0
-----
Number of targets : 1
-----
IMSI              : 123456789011111
IMEI              : 123456789022222
MSISDN           : 123456789033333
UE State          : connected             Failure Reason    : noError
Attach Time       : 2018/04/13 21:39:40
ULI Type          : taiAndEcgi
ULI Time          : 2018/04/13 21:40:43
ULI               : 310-012-eeee-1234568
=====

```

LI target IMSI ID detail output fields describes the LI target IMSI ID detail output fields.

Table 183: LI target IMSI ID detail output fields

Label	Description
Target Id	The IMSI target identifier
Intercept type	The intercept type configured for the target ID
Df peer	The Delivery Function peer
Li ID	The LI identifier
DF2 Oper state/DF3 Oper state	The DF2 or DF3 operational state; either: Peer connected, Retrying peer, Peer retries exceeded
Sec Df Peer	The secondary Delivery Function peer
HA Df peer	The HA Delivery Function peer
Number of targets	The total number of targets
IMSI	The IMSI identifier

Table 183: LI target IMSI ID detail output fields (continued)

Label	Description
IMEI	The IMEI identifier
MSISDN	The MSISDN identifier
UE State	The UE state: Connected, Idle, or None
Failure Reason	The failure reason, for example, Session/Context not found noError is displayed if there is no failure.
Attach Time	The attach time
ULI Type	The User Location Information type
ULI Time	The User Location Information time; refers to the reception of the ULI timestamp information element.
ULI	The User Location Information

Example output LI target IMSI ID statistics

```
# show li mobile-gateway target imsi id 123456789012345 stats
```

```
=====
```

LI Target Information

```
=====
```

```
Target id (imsi)   : 123456789012345
Intercept type    : iricc                Df peer           : 1
Li ID             : liid1
Sec Df Peer       : 0                    HA Df peer         : 0
-----
```

```
Target id (imsi)   : 123456789012345
Intercept type    : iricc                Df peer           : 1
Li ID             : liid2
Sec Df Peer       : 0                    HA Df peer         : 0
-----
```

```
Target id (imsi)   : 123456789012345
Intercept type    : iricc                Df peer           : 1
Li ID             : liid3
Sec Df Peer       : 0                    HA Df peer         : 0
-----
```



```

Bearer Activation (successful)           : 1
Bearer Activation (unsuccessful)         : 1
Bearer Deactivation                      : 1
Bearer Modification (successful)         : 1
Bearer Modification (unsuccessful)       : 1
UE Requested bearer Resource Modification (successful) : 1
UE Requested bearer Resource Modification (unsuccessful) : 1
Start of Interception with active bearer : 1
PMIP Attach/Tunnel Activation (successful) : 1
PMIP Attach/Tunnel Activation (unsuccessful) : 1
PMIP Attach/Tunnel De-activation        : 1
Start of Interception with Active PMIP Tunnel : 1
PMIP Resource Allocation Deactivation    : 1
PDP context Activation(successful)       : 1
PDP context Activation(unsuccessful)     : 1
PDP Context De-activation                : 1
PDP Context Modification                : 1
Start of Interception with PDP context active : 1
I-WLAN Tunnel Establishment (successful) : 1
I-WLAN Tunnel Establishment (unsuccessful) : 1
I-WLAN Tunnel Disconnect                 : 1
Start of Intercept with I-WLAN Communication Active : 1

```

```

Number of targets : 3
=====

```

Example output LI target S8HR APN

```

# show li mobile-gateway target s8hr-apn id apn.roaming.com
=====
LI Target Information
=====
Target id (s8hr-apn) : apn.roaming.com
Intercept type       : iricc                      LMISF peer : 1
Sec LMISF Peer       : 0
-----
Number of targets : 1
=====

```

Example output LI target S8HR APN ID statistics

```

# show li mobile-gateway target s8hr-apn id apn.roaming.com stats
=====
LI Target Information
=====
Target id (s8hr-apn) : apn.roaming.com
Intercept type       : iricc                      LMISF peer : 1
Sec LMISF Peer       : 0
-----
Bearer Activation (successful)           :
Bearer Activation (unsuccessful)         :
Bearer Deactivation                      :
Bearer Modification (successful)         :
Bearer Modification (unsuccessful)       :
-----
Number of targets : 1

```

Example output LI target S8HR media stats

```
# show li mobile-gateway target s8hr-media stats
=====
LI Target Information
=====
Target id (s8hr-media) : <correlation-id/gtp-tunnel id>
Intercept type         : cc                      LMISF peer : 1
Sec LMISF Peer         : 0
-----
Target id (s8hr-media) : <correlation-id/gtp-tunnel id>
Intercept type         : cc                      LMISF peer : 1
Sec LMISF Peer         : 0
-----
Target id (s8hr-media) : <correlation-id/gtp-tunnel id>
Intercept type         : cc                      LMISF peer : 1
Sec LMISF Peer         : 0
-----
Number of targets : 3
=====
```

9.2.2.1.9 log-id

Syntax

```
log-id [log-id] [severity severity-level] [application application] [sequence from-seq
[to-seq]] [count count] [router router-instance [expression]] [subject subject [regex]]
[ascending | descending]
```

Context

```
show>li>log
```

Description

Displays information for specified log.

Parameters

log-id

specifies the log ID

Values: 1 to 100

severity-level

specifies the severity level

Values: cleared, indeterminate, critical, major, minor, warning

application

specifies the application name

Values: bgp, cflowd, chassis, debug, igmp, lldp, mirror, ospf, pim, port, snmp, system, user, vrtr

from-seq [to-seq]

specifies the sequence value

Values: 1 to 4294967295

count

specifies the count

Values: 1 to 4294967295

subject

specifies a subject string to match

regex

specifies to use a regular expression match for the subject string

ascending | descending

specifies the sort direction

expression

keyword to use an expression match for the router instance

router-instance

specifies the router instance

Output

The following output example shows LI log information.

Example output LI log

```
# show li log log-id 10
=====
Event Log 10
=====
Description : (Not Specified)
Memory Log contents [size=500  next event=9  (not wrapped)]

8 2019/08/01 19:37:35.396 UTC MINOR: LI #2101 Base liadmin
"User liadmin from CONSOLE logged in"

7 2019/08/01 19:37:29.895 UTC WARNING: LI #3004 Base 'allow-duplicate-msisdn'
"'allow-duplicate-msisdn' is set to false on Gateway 1"

6 2019/08/01 19:37:25.887 UTC WARNING: LI #3003 Base 'allow-duplicate-imei'
"'allow-duplicate-imei' is set to false on Gateway 1"

5 2019/08/01 19:37:21.542 UTC WARNING: LI #3004 Base 'allow-duplicate-msisdn'
"'allow-duplicate-msisdn' is set to true on Gateway 1"
```

```
4 2019/08/01 19:37:18.149 UTC WARNING: LI #3003 Base 'allow-duplicate-imei'
"'allow-duplicate-imei' is set to true on Gateway 1"

3 2019/08/01 19:37:10.957 UTC MINOR: LI #2102 Base liadmin
"User liadmin from CONSOLE logged out"

2 2019/08/01 19:37:02.988 UTC MINOR: LI #2115 Base liadmin
"User from CONSOLE: Dut-C>config>li>log>log-id$ to memory 500"

1 2019/08/01 19:37:02.988 UTC WARNING: LI #2206 Base LOGGER
"Log Id 10 configuration modified"
```

9.2.2.1.10 status

Syntax

status

Context

show>li

Description

Displays LI status information.

Output

The following output example shows LI status information.

Example output LI status

```
# show li status
=====
Lawful Intercept Status Information
=====
LI Booted Config Status      : fail
LI Local Save Allowed       : yes
Separate LI administration  : no
Last LI Config Save Time    : N/A
Last Config Save Result     : none
Changes Since Last Save     : yes
Last LI Config Modified Time : 2008/01/11 10:24:30
=====
```

10 Application Assurance command reference

This section describes the Application Assurance-related commands.

10.1 Application Assurance command hierarchies

- *Application Assurance configuration command hierarchies*
 - *Hardware command hierarchy*
 - *Admin command hierarchy*
 - *ISA command hierarchy*
 - *Application Assurance command hierarchy*
 - *Application Assurance interface command hierarchy*
 - *Application Assurance group command hierarchy*
 - *Application Assurance group cflowd command hierarchy*
 - *Application Assurance group policer command hierarchy*
 - *Application Assurance group policy command hierarchy*
 - *Application Assurance group session filter command hierarchy*
 - *Application Assurance group statistics command hierarchy*
 - *Application Assurance group URL filter and URL list command hierarchy*
- *Application Assurance show command hierarchy*
- *Application Assurance tools command hierarchy*
- *Application Assurance clear command hierarchy*
- *Application Assurance debug command hierarchy*

10.1.1 Application Assurance configuration command hierarchies

10.1.1.1 Hardware command hierarchy

configure

- *card slot-number*
- *mda mda-slot*
- *isa2-aa (mda-type) mda-type*

10.1.1.2 Admin command hierarchy

admin

- *application-assurance*
- *group aa-group-id*
 - *url-list url-list-name upgrade*
- *upgrade*

10.1.1.3 ISA command hierarchy

configure

- *isa*
 - *application-assurance-group aa-group-id [create] [aa-sub-scale sub-scale]*
 - *no application-assurance-group aa-group-id*
 - *[no] description description-string*
 - *dynamic-rating-expiry timeout-in-secs*
 - *no dynamic-rating-expiry*
 - *[no] fail-to-open*
 - *[no] flow-attribute flow-attribute name*
 - *http-enrich-max-pkt size*
 - *isa-capacity-cost-high-threshold threshold*
 - *no isa-capacity-cost-high-threshold*
 - *isa-capacity-cost-low-threshold threshold*
 - *no isa-capacity-cost-low-threshold*
 - *[no] isa-overload-cut-through*
 - *minimum-isa-generation min-isa-generation*
 - *[no] partitions*
 - *[no] primary mda-id*
 - *qos*
 - *egress*
 - *from-subscriber*
 - *pool [pool-name]*
 - *no pool*
 - *resv-cbs percent-or-default*
 - *no resv-cbs*
 - *slope-policy slope-policy name*
 - *no slope-policy*
 - *port-scheduler-policy port-scheduler-policy-name*
 - *no port-scheduler-policy*
 - *queue-policy network-queue-policy-name*
 - *no queue-policy*
 - *wa-shared-high-wmark percent*
 - *no wa-shared-high-wmark*

- *wa-shared-low-wmark percent*
- *no wa-shared-low-wmark*
- *to-subscriber*
 - *pool [pool-name]*
 - *no pool*
 - *resv-cbs percent-or-default*
 - *no resv-cbs*
 - *slope-policy slope-policy-name*
 - *no slope-policy*
 - *port-scheduler-policy port-scheduler-policy-name*
 - *no port-scheduler-policy*
 - *wa-shared-high-wmark percent*
 - *no wa-shared-high-wmark*
 - *wa-shared-low-wmark percent*
 - *no wa-shared-low-wmark*
 - *queue-policy network-queue-policy-name*
 - *no queue-policy*
- *shared-resources*
 - *bandwidth-shaper-buffer size*
 - *multi-device-detection size*
 - *tcp-adv-func size*
 - *web-service-url-filter size*
- *tcp-adv-func size*
- *[no] shutdown*
- *statistics*
- *performance*
- *accounting-policy acct-policy-id*
- *no accounting-policy*
- *[no] collect-stats*
- *transit-prefix-ipv4-entries entries*
- *no transit-prefix-ipv4-entries*
- *transit-prefix-ipv4-remote-entries entries*
- *no transit-prefix-ipv4-entries*
- *transit-prefix-ipv6-entries entries*
- *no transit-prefix-ipv6-entries*
- *transit-prefix-ipv6-remote-entries entries*
- *no transit-prefix-ipv6-remote-entries*
- *[no] vm-traffic-distribution-by-ip*

10.1.1.4 Application Assurance command hierarchy

configure

- *application-assurance*
 - *bit-rate-high-wmark high-watermark*

- no *bit-rate-high-wmark*
- *bit-rate-low-wmark* low-watermark
- no *bit-rate-low-wmark*
- *datapath-cpu-high-wmark* high-watermark
- *datapath-cpu-low-wmark* low-watermark
- *flow-setup-high-wmark* high-watermark
- no *flow-setup-high-wmark*
- *flow-setup-low-wmark* low-watermark
- no *flow-setup-low-wmark*
- *flow-table-high-wmark* high-watermark
- no *flow-table-high-wmark*
- *flow-table-low-wmark* low-watermark
- no *flow-table-low-wmark*
- *group* aa-group-id [:partition-id [create]]
- no *group* aa-group-id:partition-id
- *packet-rate-high-wmark* high-watermark
- no *packet-rate-high-wmark*
- *packet-rate-low-wmark* low-watermark
- no *packet-rate-low-wmark*
- *protocol* protocol-name
 - [no] *shutdown*

10.1.1.5 Application Assurance interface command hierarchy

configure

- service
 - *ies/vprn* service-id
 - *aa-interface* aa-if-name [create]
 - no *aa-interface* aa-if-name
 - *address* {ip-address/mask | ip-address netmask}
 - no *address* [ip-address/mask | ip-address netmask]
 - *description* description-string
 - no *description*
 - *ip-mtu* octets
 - no *ip-mtu*
 - *sap* sap-id [create]
 - no *sap* sap-id
 - *description* long-description-string
 - no *description*
 - *egress*
 - *filter* ip ip-filter-id
 - no *filter* [ip ip-filter-id]
 - *qos* policy-id
 - no *qos* [policy-id]

- *ingress*
- *qos policy-id*
- *no qos [policy-id]*
- *[no] shutdown*
- *[no] shutdown*

10.1.1.6 Application Assurance group command hierarchy

configure

- *application-assurance*
 - *group aa-group-id [:partition-id [create]]*
 - *no group aa-group-id:partition-id*
 - *aa-sub-congestion-detection*
 - *rtt-threshold threshold*
 - *no rtt-threshold*
 - *rtt-threshold-rat rat-type rat-type rtt-threshold rtt-threshold*
 - *no rtt-threshold-rat rat-type rat-type*
 - *rtt-threshold-tolerance tolerance*
 - *no rtt-threshold-tolerance*
 - *[no] shutdown*
 - *access-network-location*
 - *source source-type level level*
 - *rtt-threshold threshold*
 - *no rtt-threshold*
 - *rtt-threshold-tolerance tolerance*
 - *no rtt-threshold-tolerance*
 - *no source source-type*
 - *alternate-cdn-host-matching ip-prefix-list-name*
 - *no alternate-cdn-host-matching*
 - *certificate-profile cert-prof-name create*
 - *no certificate-profile cert-prof-name*
 - *certificate certificate-file*
 - *no certificate*
 - *key-file key-filename*
 - *no key-file*
 - *description description-string*
 - *no description*
 - *[no] shutdown*
 - *cflowd*
 - *[no] description description-string*
 - *[no] disable-deferred-billing*
 - *dns-enrich dns-enrich-name [create]*
 - *no dns-enrich dns-enrich-name*
 - *[no] description description-string*

- *gateway gateway-id*
- *no gateway*
- *option option* [create]
- *no option*
- *packet-size-limit packet-size*
- *no packet-size-limit*
- *[no] shutdown*
- *dns-ip-cache dns-ip-cache-name* [create]
- *no dns-ip-cache dns-ip-cache-name*
 - *description description-string*
 - *no description*
- *dns-match*
 - *description description-string*
 - *no description*
 - *domain domain-name* expression expression
 - *no domain*
 - *server-address server-address* [name server-name]
 - *no server-address server-address*
- *ip-cache*
 - *high-watermark percent*
 - *low-watermark percent*
 - *size cache-size*
 - *[no] static-address static-ip-address*
- *[no] shutdown*
- *event-log event-log-name* [create]
- *no event-log event-log-name*
 - *buffer-type buffer-type*
 - *max-entries max-entries*
 - *[no] shutdown*
 - *syslog*
 - *address ip-address*
 - *no address*
 - *description description-string*
 - *no description*
 - *facility syslog-facility*
 - *port port*
 - *severity syslog-severity*
 - *vlan-id service-port-vlan-id*
 - *no vlan-id*
- *[no] http-anti-fraud-cr-eol*
- *[no] http-anti-fraud-policy*
- *[no] http-enrich enrich_name* [create]
 - *description description-string*
 - *no description*
- *[no] field field-name*

- *aes-initialization-vector* *hex-string*
- *no aes-initialization-vector*
- [no] *anti-spoof*
- [no] *calling-line-id*
- *encode* type type cert-profile *cert-profile-name*
- *encode* type type key *key*
- *encode* type type key *hash-key* *hash*
- *encode* type type key *hash2-key* *hash2*
- *no encode*
- *md5-salt* *string*
- *no md5-salt*
- *name* *header-name*
- *static-string* *string*
- *no static-string*
- *rat-enrichment*
 - *rat-type* *rat-type* *rat-value*
 - *no rat-type* *rat-type*
- [no] *shutdown*
- *tls-extension*
 - *extension-id* *extension-id*
 - *no extension-id*
 - [no] *subtype* *tls* *extension* *subtype*
 - *extension-id* *extension-id*
 - *no extension-id*
 - *static-string* *string*
 - *no static-string*
 - *encode* type type key *key|hash-key|hash2-key|custom-key* [*hash|hash2|*

custom]

- *no encode*
- *aes-initialization-vector* *hex-string*
- *no aes-initialization-vector*
- [no] *http-error-redirect* *redirect_name* [create]
- [no] *description* *description-string*
- [no] *error-code* *error-code* [custom-msg-size *custom-msg-size*]
- [no] *http-host* *http-host*
- [no] *participant-id* *participant-id*
- [no] *shutdown*
- *template* *template-id*
- [no] *template*
- [no] *http-match-all-requests*
- *http-notification* *http-notification-name* [create]
- [no] *http-notification* *http-notification-name*
 - *description* *description-string*
 - *no description*
 - *interval* {one-time | *minimum-interval*}

```
– script-url script-url-name [create]
– no script-url
– template template-id
– no template
– http-proxy http-proxy-name mode mode create
– http-proxy http-proxy-name create
– [no] http-proxy http-proxy-name
– description description-string
– no description
– dns-server ip-address
– no dns-server
– https
– intermediate-certificate-1 cert-profile
– no intermediate-certificate-1
– intermediate-certificate-2 cert-profile
– no intermediate-certificate-2
– root-certificate cert-profile
– no root-certificate
– tls-profile cert-profile
– no tls-profile
– url-mapping
– entry entry-id [create]
– no entry entry-id
– action
– http-enrich name
– no http-enrich
– target-server fqdn
– no target-server
– uri-transform transform-type
– no uri-transform
– match
– expression expr-index type type operator expression-str
– [no] shutdown
– vlan-id vlan-id
– no vlan-id
– [no] http-redirect http-redirect_name [create]
– [no] description description-string
– [no] redirect-https
– redirect-url redirect-url
– no redirect-url
– [no] tcp-client-reset
– template template-id
– no template
– [no] shutdown
– [no] http-x-online-host
```

- *ip-identification-assist*
 - *passive-dns*
 - *monitor*
 - *trusted-server-address* *address* [create]
 - no *trusted-server-address* *address*
 - [no] *shutdown*
- [no] *ip-identification-contribute*
- *ip-prefix-list* *ip-prefix-list-name* [create]
- no *ip-prefix-list*
 - [no] *description* *description-string*
 - prefix *ip-prefix/ip-prefix-length* [name *prefix-name*]
 - no prefix *ip-prefix/ip-prefix-length*
- *mp-tcp-proxy* *mp-tcp-proxy-name* [create]
- no *mp-tcp-proxy* *mp-tcp-proxy-name*
 - [no] *bonding-assist*
 - *congestion-control* *congestion-algorithm*
 - no *congestion-control*
 - *default-scheduler-policy* *scheduler-policy*
 - no *default-scheduler-policy*
 - *description* *description-string*
 - no *description*
 - [no] *force-dss-checksums*
 - *ip-address-profile* *profile-name*
 - no *ip-address-profile*
 - *path*
 - *fixed* *admin-cost*
 - no *fixed*
 - *mobile* *admin-cost*
 - no *mobile*
 - [no] *subflow-failure-indication*
- *multi-device-detection*
 - *max-devices* *count*
 - no *max-devices*
- *policer* *policer-name* *type* *granularity* [create]
- *policer* *policer-name*
- no *policer* *policer-name*
- *policy*
- *port-list* *port-list-name* [create]
- no *port-list* *port-list-name*
 - [no] *description* *description-string*
 - [no] *port* *port-number*
 - [no] *port* range *start-port-num* *end-port-num*
- *session-filter* *session-filter-name* [create]
- no *session-filter* *session-filter-name*
- [no] *shallow-inspection*

- *statistics*
- *tcp-optimizer tcp-optimizer-name [create]*
- *no tcp-optimizer tcp-optimizer-name*
 - *dack-timeout dack-timeout*
 - *no dack-timeout*
 - *description description-string*
 - *no description*
 - *[no] high-cpu-backoff*
 - *init-cwnd-size init-cwnd-size*
 - *init-ss-threshold init-ss-threshold*
 - *network-rtt-threshold network-rtt-threshold*
 - *no network-rtt-threshold*
 - *tcp-stack tcp-stack-type*
- *tcp-validate tcp-validate-name [create]*
- *no tcp-validate tcp-validate-name*
 - *description description-string*
 - *no description*
 - *event-log event-log-name [all]*
 - *no event-log*
 - *[no] strict*
- *tethering-detection*
 - *[no] shutdown*
 - *single-device*
 - *[no] expected-ttl ttl-value*
 - *[no] invert-match*
 - *tll-monitoring*
 - *tcp-protocols protocol-set*
 - *udp-protocols protocol-set*
- *url-filter url-filter-name*
- *url-list url-list-name [create]*
- *no url-list url-list-name*
- *wap1x*
 - *[no] shutdown*

10.1.1.7 Application Assurance group cflowd command hierarchy

configure

- application-assurance
 - *group aa-group-id [:partition-id [create]]*
 - *no group aa-group-id:partition-id*
 - *cflowd*
 - *collector ip-address [:port] [create]*
 - *no collector ip-address [:port]*
 - *[no] description description-string*

```

    - [no] shutdown
- no comprehensive
    - app-group app-group-name {flow-rate | flow-rate2}
    - no app-group app-group-name
    - application application-name {flow-rate | flow-rate2}
    - no application application-name
    - flow-rate sample-rate
    - no flow-rate
    - flow-rate2 sample-rate
    - no flow-rate2
    - [no] shutdown
- template
    - dynamic-fields
        - [no] field field-name
        - [no] shutdown
    - field-selection field-selection
    - no field-selection
- direct-export
    - collector collector-id [create]
    - no collector collector-id
        - [no] address ip-address [:port]
        - [no] shutdown
        - [no] description description-string
        - no description
    - vlan-id service-port-vlan-id
    - no vlan-id
- export-override mode
- no export-override
- interim-report-mode mode
- no interim-report-mode
- obfuscation
    - aes-128-encryption plaintext key type key-value {hash | hash2 | custom}
    - no aes-128-encryption
    - aes-256-encryption plaintext key type key-value {hash | hash2 | custom}
    - no aes-256-encryption
- pgw-edr
    - flow-rate sample-rate
    - no flow-rate
- [no] shutdown
- system-name name
- no system-name
- tcp-performance
- no tcp-performance
    - app-group app-group-name {flow-rate | flow-rate2}
    - no app-group app-group-name

```

- *application application-name {flow-rate | flow-rate2}*
- *no application application-name*
- *flow-rate sample-rate*
- *no flow-rate*
- *flow-rate2 sample-rate*
- *no flow-rate2*
- *[no] shutdown*
- *template*
 - *dynamic-fields*
 - *[no] field field-name*
 - *[no] shutdown*
 - *field-selection field-selection*
 - *no field-selection*
- *template-retransmit seconds*
- *no template-retransmit*
- *volume*
 - *rate sample-rate*
 - *no rate*
 - *[no] shutdown*
 - *template*
 - *dynamic-fields*
 - *[no] field field-name*
 - *[no] shutdown*
 - *field-selection field-selection*
 - *no field-selection*

10.1.1.8 Application Assurance group policer command hierarchy

configure

- *application-assurance*
 - *group aa-group-id [:partition-id [create]]*
 - *no group aa-group-id:partition-id*
 - *policer policer-name type type granularity granularity [create]*
 - *policer policer-name*
 - *no policer policer-name*
 - *action policer-action*
 - *adaptation-rule pir {max | min | closest} [cir {max | min | closest}]*
 - *no adaptation-rule*
 - *cbs committed-burst-size*
 - *no cbs*
 - *congestion-override*
 - *cbs congested-cbs*
 - *no cbs*
 - *cir congested-cir*


```

    - no cir
    - mbs congested-mbs
    - no mbs
    - pir congested-pir
    - no pir
  - congestion-override-stage2
    - cbs congested-cbs
    - no cbs
    - cir congested-cir
    - no cir
    - mbs congested-mbs
    - no mbs
    - pir congested-pir
    - no pir
  - [no] description description-string
  - flow-count flow-count
  - no flow-count
  - mbs maximum-burst-size
  - no mbs
  - rate pir-rate [cir cir-rate]
  - no rate
  - rate-percentage rate-percentage
  - no rate-percentage
  - time-range daily start start-time end end-time [on day [day...(upto 7 max)]]
  - time-range weekly start start-time end end-time
  - no time-range
  - tod-override tod-override-id [create]
  - no tod-override tod-override-id
    - cbs committed-burst-size
    - no cbs
    - [no] description description-string
    - flow-count flow-count
    - no flow-count
    - mbs maximum-burst-size
    - no mbs
    - rate pir-rate [cir cir-rate]
    - no rate
    - [no] shutdown
    - time-range daily start start-time end end-time [on day [day...(upto 7
max)]]
    - time-range weekly start start-time end end-time
    - no time-range

```

10.1.1.9 Application Assurance group policy command hierarchy

```

configure
- application-assurance
- group aa-group-id [:partition-id [create]]
- no group aa-group-id:partition-id
- policy
- abort
- app-filter
- entry entry-id [create]
- no entry entry-id
- application application-name
- description description-string
- no description
- dynamic-rating-action action
- no dynamic-rating-action
- expression expr-index expr-type {eq | neq} expr-string
- no expression expr-index
- flow-setup-direction {subscriber-to-network | network-to-subscriber |
both}
- http-match-all-requests
- no http-match-all-requests
- http-port {eq | neq} port-num
- http-port {eq | neq} port-list port-list-name
- no http-port
- [no] ip-identification-assist
- ip-protocol-num {eq | neq} protocol-id
- no ip-protocol-num
- protocol {eq | neq} protocol-name
- no protocol
- server-address {eq | neq} ip-address
- server-address {eq | neq} ip-address ip-address-match netmask ip-address-
mask
- no server-address
- server-port {eq | neq | gt | lt} server-port-number
- server-port {eq} server-port-number [first-packet-trusted | first-packet-
validate]
- server-port {eq | neq} port-list port-list-name
- server-port {eq} port-list port-list-name {first-packet-trusted | first-
packet-validate}
- no server-port
- [no] shutdown
- app-group application-group-name [create]
- no app-group application-group-name
- charging-group charging-group-name

```

- no *charging-group* *application-group-name*
- *app-profile* *app-profile-name* [create]
- no *app-profile* *app-profile-name*
 - *capacity-cost* *cost*
 - no *capacity-cost*
 - *characteristic* *characteristic-name* *value* *value-name*
 - no *characteristic* *characteristic-name*
 - [no] *description* *description-string*
 - [no] *divert*
- *app-qos-policy*
 - *entry* *entry-id* [create]
 - no *entry* *entry-id*
 - *action*
 - [no] *abandon-tcp-optimization*
 - *bandwidth-policer* *policer-name*
 - no *bandwidth-policer*
 - *dns-enrich* *dns-enrich-name*
 - no *dns-enrich*
 - *dns-ip-cache* *dns-ip-cache-name*
 - [no] *drop*
 - *error-drop* [event-log *event-log-name*]
 - no *error-drop*
 - *overload-drop* *policer-name*
 - no *overload-drop*
 - *overload-drop* *policer-name*
 - no *overload-drop*
 - *fragment-drop* {all | out-of-order} [event-log *event-log-name*]
 - no *fragment-drop*
 - *http-enrich* *http-enrich-name*
 - no *http-enrich*
 - *http-error-redirect* *http-error-name*
 - no *http-error-redirect*
 - *http-notification* *http-notification*
 - no *http-notification*
 - *http-redirect* *http-redirect-name* *flow-type* *flow-type*
 - no *http-redirect*
 - *mirror-source* *mirror-service-id* [all-inclusive]
 - no *mirror-source*
 - *mp-tcp-proxy* *mp-tcp-proxy-name*
 - no *mp-tcp-proxy*
 - *mp-tcp-scheduler-policy-override* *scheduler-policy*
 - no *mp-tcp-scheduler-policy-override*
 - [no] *offload*
 - *overload-drop* [event-log *event-log-name*]
 - no *overload-drop*

- *remark*
- *dscp* in-profile *dscp-name* out-profile *dscp-name*
- no *dscp*
- *fc* *fc-name*
- no *fc*
- *priority* *priority-level*
- no *priority*
- *session-filter* *session-filter-name*
- no *session-filter*
- *tcp-mss-adjust* *segment-size*
- no *tcp-mss-adjust*
- *tcp-validate* *tcp-validate-name*
- no *tcp-validate*
- *tls-enrich* *tls-enrich-name*
- no *tls-enrich*
- *url-filter* *url-filter-name* [*characteristic* *characteristic-name*]
- no *url-filter*
- [no] *description* *description-string*
- *match*
 - *aa-sub* *sap* {eq | neq} *sap-id*
 - *aa-sub* *esm* {eq | neq} *sub-ident-string*
 - *aa-sub* *spoke-sdp* {eq | neq} *sdp-id:vc-id*
 - *aa-sub* *transit* {eq | neq} *transit-aasub-name*
 - no *aa-sub*
 - *aa-sub-tethering-state* {detected | not-detected}
 - no *aa-sub-tethering-state*
 - *app-group* {eq | neq} *application-group-name*
 - no *app-group*
 - *application* {eq | neq} *application-group-name*
 - no *application*
 - *characteristic* *characteristic-name* {eq | neq} *value-name*
 - no *characteristic*
 - *charging-group* {eq | neq} *charging-group-name*
 - no *charging-group*
 - *dscp* {eq | neq} *dscp-name*
 - no *dscp*
- *dst-ip* {eq | neq} {*ip-address* [/mask] | *ip-prefix-list* *prefix-list-name*}
- no *dst-ip*
- *dst-port* {eq | neq} *port-num*
- *dst-port* {eq | neq} *port-list* *port-list-name*
- *dst-port* {eq | neq} *range* *start-port-num* *end-port-num*
- no *dst-port*
- [no] *flow-attribute* *flow-attribute name*
 - *confidence* eq *equal-value*
 - *confidence* gte *greater-than-or-equal-value*

```

    - confidence lt less-than-value
    - ip-protocol-num {eq | neq} protocol-id
    - no ip-protocol-num
    - multi-device-detection {eq | neq} device-type device-type
    - no multi-device-detection
    - presence-reporting-area {eq | neq} PRA-value
    - no presence-reporting-area
    - rat-type {eq | neq} {utran | geran | wlan | gan | hspa-evol | eutran
| virtual | eutran-nb | ehrpd | hrpd | cdma-1x | umb | wifi | nr | lte-m | wireline |
wireline-cable | wireline-bbf | nr-u | eutra-u | trusted-n3ga | trusted-wlan}
    - no rat-type
    - roaming-status {eq | neq}{home | roaming | visiting}
    - no roaming-status
    - src-ip {eq | neq} {ip-address [/mask] | ip-prefix-list prefix-list-name}
    - no src-ip
    - src-port {eq | neq} port-num
    - src-port {eq | neq} port-list port-list-name
    - src-port {eq | neq} range start-port-num end-port-num
    - no src-port
    - traffic-direction {subscriber-to-network | network-to-subscriber |
both}
    - wireless-device-os {eq | neq} {other | android | ios | windows}
[{other | android | ios | windows} ... (up to 4 max)]
    - no wireless-device-os
    - wireless-device-type {eq | neq} {other | tablet | phone} [{other |
tablet | phone}... (up to 3 max)]
    - no wireless-device-type
    - [no] shutdown
- app-service-options
- characteristic characteristic-name [create]
- no characteristic characteristic-name
- [no] default-value
- [no] value
- application application-name [create]
- no application application-name
- app-group app-group-name
- [no] charging-group description-string
- [no] description description-string
- begin
- charging-filter
- entry entry-id [create]
- no entry entry-id
- charging-group charging-group-name
- no charging-group
- description dedcription-string

```

```

    - no description
    - match
      - app-group {eq | neq} application-group-name
      - no app-group
      - application {eq | neq} application-name
      - no application
      - [no] flow-attribute flow-attribute-name
        - confidence confidence {eq equal-value |
gte greater-than-or-equal-value | lt less-than-value}
        - [no] tethered-flow
      - charging-group charging-group-name [create]
      - no charging-group charging-group-name
      - [no] description description-string
      - notify-start-stop
      - notify-start-stop notify-type
      - no notify-start-stop
    - commit
    - custom-protocol custom-protocol-id ip-protocol-num protocol-id [create]
    - custom-protocol custom-protocol-id
    - no custom-protocol custom-protocol-id
      - [no] description description-string
      - expression expr-index eq expr-string offset payload-octet-offset
direction direction
    - no expression expr-index
    - [no] shutdown
    - default-charging-group charging-group-name
    - no default-charging-group
    - default-tethered-charging-group charging-group-name
    - no default-tethered-charging-group
    - [no] description description-string

```

10.1.1.10 Application Assurance group session filter command hierarchy

```

configure
- application-assurance
- group aa-group-id [:partition-id] [create]]
- no group aa-group-id:partition-id
- session-filter session-filter-name [create]
- no session-filter session-filter-name
- default-action {permit | deny}
- [no] description description-string
- [no] entry entry-id [create]
- action {permit | deny | tcp-strict-order} [event-log event-log-name]

```

```

    - action redirect [dst-ip ip-address] [dst-port tcp/udp-port] [event-
log event-log-name]
    - action tcp-optimizer tcp-optimizer-name
    - [no] description description-string
    - match
      - dst-ip ip-address[/mask]
      - dst-ip ip-prefix-list ip-prefix-list-name
      - no dst-ip
      - dst-port {eq | gt | lt} port-num
      - dst-port port-list port-list-name
      - dst-port range start-port-num end-port-num
      - no dst-port
      - ip-protocol-num {eq | neq} protocol-id
      - no ip-protocol-num
      - src-ip ip-address[/mask]
      - src-ip ip-prefix-list ip-prefix-list-name
      - no src-ip
      - src-port {eq | gt | lt} port-num
      - src-port port-list port-list-name
      - src-port range start-port-num end-port-num
      - no src-port

```

10.1.1.11 Application Assurance group statistics command hierarchy

```

configure
- application-assurance
- group aa-group-id[:partition-id [create]]
- no group aa-group-id:partition-id
  - statistics
    - aa-admit-deny
      - accounting-policy acct-policy-id
      - no accounting-policy
      - [no] collect-stats
      - [no] policer-stats
      - [no] policer-stats-resources
      - [no] session-filter-stats
      - [no] tcp-validate-stats
    - aa-partition
      - accounting-policy acct-policy-id
      - no accounting-policy
      - [no] collect-stats
      - [no] traffic-type
  - aa-sub
    - [no] exclude-tcp-retrans

```

- *app-group*
 - *accounting-policy acct-policy-id*
 - *no accounting-policy*
 - *[no] collect-stats*
- *application*
 - *accounting-policy acct-policy-id*
 - *no accounting-policy*
 - *[no] collect-stats*
- *protocol*
 - *accounting-policy acct-policy-id*
 - *no accounting-policy*
 - *[no] collect-stats*
- *threshold-crossing-alert*
 - *error-drop direction direction [create]*
 - *no error-drop direction direction*
 - *high-wmark high-watermark low-wmark low-watermark*
 - *fragment-drop-all direction direction [create]*
 - *no fragment-drop-all direction direction*
 - *high-wmark high-watermark low-wmark low-watermark*
 - *fragment-drop-out-of-order direction direction [create]*
 - *no fragment-drop-out-of-order direction direction*
 - *high-wmark high-watermark low-wmark low-watermark*
 - *overload-drop direction direction [create]*
 - *no overload-drop direction direction*
 - *high-wmark high-watermark low-wmark low-watermark*
 - *policer policer-name direction direction [create]*
 - *no policer policer-name direction direction*
 - *high-wmark high-watermark low-wmark low-watermark*
- *session-filter session-filter-name*
 - *default-action direction direction [create]*
 - *no default-action direction direction*
 - *high-wmark high-watermark low-wmark low-watermark*
 - *entry entry-id direction direction [create]*
 - *no entry entry-id direction direction*
 - *high-wmark high-watermark low-wmark low-watermark*
- *tcp-validate tcp-validate-name direction direction [create]*
- *[no] tcp-validate tcp-validate-name direction direction*
 - *high-wmark high-watermark low-wmark low-watermark*

10.1.1.12 Application Assurance group URL filter and URL list command hierarchy

configure

- application-assurance
 - *group aa-group-id[:partition-id [create]]*

- no *group aa-group-id:partition-id*
- *url-filter url-filter-name*
 - [no] *apply-function-specific-behavior*
 - *default-action* allow
 - *default-action* block-all
 - *default-action* block-http-redirect *http-redirect-name*
 - no *default-action*
 - *description* *description-string*
 - no *description*
 - *http-redirect* *http-redirect-name*
 - no *http-redirect*
 - *http-request-filtering* {all | first}
- *icap*
 - *custom-x-header* *custom-x-header-name*
 - no *custom-x-header*
 - *default-action* allow
 - *default-action* block-all
 - *default-action* block-http-redirect *http-redirect-name*
 - no *default-action*
 - *http-redirect* *http-redirect-name*
 - no *http-redirect*
 - *server ip-address* [:port] [create]
 - no *server ip-address*[:port]
 - *description* *description-string*
 - no *description*
 - [no] *shutdown*
 - *vlan-id* *service-port-vlan-id*
 - no *vlan-id*
- *local-filtering*
 - *allow-list* *allow-list-name*
 - no *allow-list*
 - [no] *deny-list* *deny-list-name*
 - *default-action* allow
 - *default-action* block-all
 - *default-action* block-http-redirect *http-redirect-name*
 - no *default-action*
 - *http-redirect* *http-redirect-name*
 - no *http-redirect*
- *web-service*
 - *classification-overrides*
 - *entry* *entry-id* *expression* *hostname* *category* *category*
 - no *entry* *entry-id*
 - *classifier* *classifier* *category-set-id* *category-set*
 - no *classifier*
 - *default-action* allow

- *default-action* block-all
- *default-action* block-http-redirect *http-redirect-name*
- no *default-action*
- *default-profile* *profile-name*
- no *default-profile*
- *dns-server* *ip-address*
- no *dns-server*
- *fqdn* *fully-qualified-domain-name*
- no *fqdn*
- *http-redirect* *http-redirect-name*
- no *http-redirect*
- *profile* *profile-name* [create]
- no *profile* *profile-name*
 - *category* *category* block
 - no *category*
 - *description* *description*
 - no *description*
- *server* *ip-address* [:port] [create]
- no *server* *ip-address* [:port]
 - [no] *description* *description-string*
- *vlan-id* *vlan-id*
- no *vlan-id*
- [no] *shutdown*
- *url-list* *url-list-name* [create]
- no *url-list* *url-list-name*
 - *description* *description-string*
 - no *description*
 - *decrypt-key* *key* | *hash-key* | *hash2-key* | *custom-key* [hash | hash2 | custom]
 - no *decrypt-key*
 - [no] *expression-match*
 - *file* *file-url*
 - no *file*
 - [no] *shutdown*
 - *size* *url-list-size*

10.1.2 Application Assurance show command hierarchy

show

- isa
 - *application-assurance-group* [*aa-group-id* [load-balance [unassigned]]]
 - *ip-reas-stats* [*port-name*] [ipv4 | ipv6]
- application-assurance
 - *group* *aa-group-id* [:*partition-id*]
 - *aa-interface* *isa* *mda-id*

```

- aa-sub mobile [imsi imsi-msisdn | msisdn imsi-msisdn | imei imei] apn apn-name
[snapshot]
- aa-sub-list [isa mda-id]
- aa-sub-list summary
- app-group [app-group-name] count [detail]
- app-group count [detail]
- app-group count top granularity [max-count max-count]
- application [application-name] count [detail]
- application count top granularity [max-count max-count]
- certificate-profile
- cflowd
- collector [detail]
- direct-export
  - collector collector-id
  - isa-collector
- status
- dns-enrich dns-enrich-name detail
- dns-ip-cache [isa mda-id]
- dns-ip-cache cache-name
- event-log event-log-name syslog
- http-enrich enrich-name
  - detail [partition]
  - field field-name
  - fields
- http-error-redirect redirect-name
- http-notification http-notification-name [summary]
- http-proxy http-proxy-name isa mda-id
- http-redirect redirect-name [detail]
- ip-identification-assist [isa mda-id]
- mp-tcp-proxy proxy-name [isa mda-id]
- partition summary
- policer
- policer policer-name [detail]
- policer summary
- policy
  - admin
  - app-filter [entry-id]
  - app-group [app-group-name]
  - app-profile [app-prof-name]
  - app-profile app-prof-name associations
  - app-qos-policy [entry-id]
  - app-qos-policy non-zero
  - app-service-option [characteristic-name]
  - application [app-name]
  - charging-filter id

```

- *charging-group* [*charging-group-name*]
- *custom-protocol*
- *summary*
- *protocol* [*protocol-name*] count [*detail*]
- *protocol* count top *granularity* [*max-count* *max-count*]
- *session-filter* [*session-filter-name*]
- *status* [*isa mda-id*] cflowd
- *status* [*isa mda-id*]
- *status* [*isa mda-id*] detail
- *status* [*isa mda-id*] overload
- *status* *isa mda-id* cpu [*sample-period* *seconds*]
- *status* *isa mda-id* qos count
- *status* *isa mda-id* qos pools
- *tcp-optimizer* *tcp-optimizer-name* [*isa mda-id*]
- *tcp-validate* *tcp-validate-name* [*isa mda-id*]
- *traffic-type* [*detail* | *ip-family* | *ip-protocol* | *tls-family*]
- *url-filter* *url-filter-name*
- *url-filter* *url-filter-name* *isa card/mda* [*detail*]
- *url-filter* *url-filter-name* *web-service* [*profile* *profile-name*]
- *url-list* *url-list-name*
- *http-enrich*
 - *field* *field-name*
 - *fields*
- *http-error-redirect*
 - *error-codes*
 - *template*
- *protocol* [*protocol-name*] [*detail*]
- *threshold-crossing-alert* [*detail*]
- *version*
- *web-service*
 - *category-set-id* *category-set*

10.1.3 Application Assurance tools command hierarchy

```
tools
  – dump
  – application-assurance
    – resources
    – group aa-group-id [:partition-id]
      – aa-anl-list [congested-only] [sort-type {top-by-subs | top-by-rate}] [isa
mda-id]
        – aa-anl-list summary
        – aa-sub-list [filter-by-type sub-type] [isa mda-id]
        – aa-sub-list [filter-by-type sub-type] [isa mda-id] policers-exceeded
```

```

– aa-sub-list summary
– aa-sub-search search-type granularity [direction direction] [max-count max-
count]
– aa-sub-search tethering summary
– admit-deny-stats
– cflowd
– template template
– dns-ip-cache cache-name isa mda-id [url file-url] [new-ip-count]
– dynamic-rating
– statistics isa mda-id
– event-log event-log-name isa mda-id
– event-log event-log-name [url file-url] isa mda-id
– flow-record-search isa mda-id [protocol protocol-name] [application app-name]
[app-group app-group-name] [flow-status flow-status] [start-flowid start-flowid] [max-
count max-count] [search-type search-type] [url file-url] [flow-modified flow-modified]
– flow-record-search aa-sub {esm sub-ident-string | sap sap-id | spoke-sdp sdp-
id:vc-id | transit transit-aasub-name} [protocol protocol-name] [application app-name]
[app-group app-group name] [flow-status flow-status] [start-flowid start-flowid]
[classified classified] [server-ip ip-address] [server-port port-num] [client-ip ip-
address] [bytes-tx kbytes] [flow-duration minutes] [max-count max-count] [search-
type search-type] [url file-url] [flow-modified flow-modified]
– http-host-recorder detail [isa mda-id] url file-url
– http-host-recorder status [isa mda-id]
– http-host-recorder top {bytes | flows} [max-count {1..25}] [isa mda-id]
– http-proxy session isa mda-id flow-id flow-id
– ip-identification-assist
– applications [isa mda-id]
– cache-contents [isa mda-id] url file-url
– passive-dns
– monitor [isa mda-id]
– mp-tcp-proxy session aa-sub esm esm-sub-id-str flow-id flow-id
– mp-tcp-proxy session aa-sub mobile imei imei apn apn flow-id flow-id
– mp-tcp-proxy session aa-sub mobile imsi imsi-msisdn apn apn flow-idflow-id
– mp-tcp-proxy session aa-sub mobile msisdn imsi-msisdn apn apn flow-idflow-id
– mp-tcp-proxy session aa-sub mobile apn apn flow-id flow-id subscriber-
id subscriber-id
– mp-tcp-proxy session aa-sub sap sap-id flow-id flow-id
– mp-tcp-proxy session aa-sub spoke-sdp sdp-id:vc-id flow-id flow-id
– mp-tcp-proxy session aa-sub transit trnst-sub-nm flow-id flow-id
– mp-tcp-proxy session isa mda-id flow-id flow-id
– port-recorder detail [flow-count flows] [byte-count kbytes] [isa mda-id]
url file-url
– port-recorder status [isa mda-id]
– port-recorder top granularity [max-count max-count] [isa mda-id]
– policer policer-name day day time time-of-day

```

- *resources*
- *tethering* detail
- *traffic-capture* detail url *file-url*
- *traffic-capture* status

10.1.4 Application Assurance clear command hierarchy

clear

- application-assurance
 - *group* *aa-group-id* cflowd
 - *group* *aa-group-id* statistics
 - *group* *aa-group-id* status

10.1.5 Application Assurance debug command hierarchy

debug

- application-assurance
 - *group* *aa-group-id* [:*partition-id*]
 - [no] *http-host-recorder*
 - [no] *filter*
 - *default-filter-action* *default-action*
 - *expression* *expr-index* *expr-type* *eq* *expr-string* {*record* | *no-record*}
 - no *expression* *expr-index*
 - *record* {*all-hosts* | *http-host-app-filter-candidates*}
 - *rate* *sample-rate*
 - no *rate*
 - [no] *shutdown*
 - [no] *port-recorder*
 - [no] *application* *application-name*
 - *rate* *sample-rate*
 - no *rate*
 - [no] *shutdown*
 - [no] *traffic-capture*
 - [no] *match*
 - *application* {*eq* | *neq*} *application-name*
 - no *application*
 - *client-ip* {*eq* | *neq*} *ip-address*
 - no *client-ip*
 - *client-port* {*eq* | *neq*} *port-num*
 - no *client-port*
 - *dst-ip* {*eq* | *neq*} *ip-address*
 - no *dst-ip*

```
– dst-port {eq | neq} port-num
– no dst-port
– ip-addr1 {eq | neq} ip-address
– no ip-addr1
– ip-addr2 {eq | neq} ip-address
– no ip-addr2
– ip-protocol-num {eq | neq} protocol-id
– no ip-protocol-num
– port1 {eq | neq} port-num
– no port1
– port2 {eq | neq} port-num
– no port2
– server-ip {eq | neq} ip-address
– no server-ip
– server-port {eq | neq} port-num
– no server-port
– src-ip {eq | neq} ip-address
– no src-ip
– src-port {eq | neq} port-num
– no src-port
– record
– cut-through-packets cut-through-packets
– limit {all-packet-matches | first-session-match}
– start {immediate | on-new-session}
– [no] shutdown
```

10.2 Application Assurance command descriptions

This section provides the following AA command descriptions:

- *Application Assurance configuration command descriptions*
- *Application Assurance show command descriptions*
- *Application Assurance tools command descriptions*
- *Application Assurance clear command descriptions*
- *Application Assurance debug command descriptions*

10.2.1 Application Assurance configuration command descriptions

- *Generic Application Assurance command descriptions*
- *Hardware Application Assurance command descriptions*

- [Admin Application Assurance command descriptions](#)
- [ISA Application Assurance command descriptions](#)
- [Application Assurance command descriptions](#)
- [Application Assurance interface command descriptions](#)
- [Application Assurance group command descriptions](#)
- [Application Assurance group cflowd command descriptions](#)
- [Application Assurance group policer command descriptions](#)
- [Application Assurance group policy command descriptions](#)
- [Application Assurance group session filter command descriptions](#)
- [Application Assurance group statistics command descriptions](#)
- [Application Assurance group URL filter and URL list command descriptions](#)

10.2.1.1 Generic Application Assurance command descriptions

10.2.1.1.1 description

Syntax

[no] description *description-string*

Context

```
config>isa>aa-grp
config>app-assure>group
config>app-assure>group>dns-enrich
config>app-assure>group>dns-ip-cache
config>app-assure>group>certificate-profile
config>app-assure>group>cflowd>collector
config>app-assure>group>cflowd>direct-export>collector
config>app-assure>group>event-log>syslog
config>app-assure>group>http-enrich
config>app-assure>group>http-error-redirect
config>app-assure>group>http-notification
config>app-assure>group>http-redirect
```



```
config>app-assure>group>ip-prefix-list
config>app-assure>group>mp-tcp-proxy
config>app-assure>group>policer
config>app-assure>group>policer>tod-override
config>app-assure>group>policy>aqp>entry
config>app-assure>group>policy>app-filter>entry
config>app-assure>group>policy>app-group
config>app-assure>group>policy>app-profile
config>app-assure>group>policy>application
config>app-assure>group>policy>app-profile
config>app-assure>group>policy>charging-filter>entry
config>app-assure>group>policy>custom-protocol
config>app-assure>group>port-list
config>app-assure>group>tcp-validate
config>app-assure>group>url-filter
config>app-assure>group>url-filter>icap>server
config>app-assure>group>url-filter>web-service>profile
config>app-assure>group>url-filter>web-service>server
config>app-assure>group>url-list
config>service>ies>aa-interface
config>service>vpn>aa-interface
```

Description

This command creates a text description that is stored in the configuration file to help identify the content of the entity.

The no form of this command removes the string from the configuration.

Parameters

string

the description character string. Allowed values are any string composed of printable, 7-bit ASCII characters. If the string contains special characters (#, \$, spaces, and so on), the entire string must be enclosed within double quotes.

10.2.1.1.2 description

Syntax

```
description long-description-string  
no description
```

Context

```
config>service>ies>aa-interface>sap  
config>service>vprn>aa-interface>sap
```

Description

This command creates a text description which is stored in the configuration file to help identify the content of the entity.

The no form of this command removes the string from the configuration.

Parameters

string

specifies the description character string, up to 160 characters. Allowed values are any string composed of printable, 7-bit ASCII characters

If the string contains special characters (#, \$, spaces, and so on), the entire string must be enclosed within double quotes.

10.2.1.1.3 shutdown

Syntax

```
[no] shutdown
```

Context

```
config>app-assure>group  
config>app-assure>group>aa-sub-congest  
config>app-assure>group>certificate-profile  
config>app-assure>group>cflowd  
config>app-assure>group>cflowd>collector
```

```
config>app-assure>group>cflowd>comprehensive
config>app-assure>group>cflowd>comprehensive>template>dynamic-fields
config>app-assure>group>cflowd>direct-export>collector>address
config>app-assure>group>cflowd>tcp-performance
config>app-assure>group>dns-enrich
config>app-assure>group>dns-ip-cache
config>app-assure>group>event-log
config>app-assure>group>http-enrich
config>app-assure>group>http-error-redirect
config>app-assure>group>http-redirect
config>app-assure>group>ip-identification-assist
config>app-assure>group>policer>tod-override
config>app-assure>group>policy>aqp>entry
config>app-assure>group>policy>app-filter>entry
config>app-assure>group>policy>app-qos-policy>entry
config>app-assure>group>policy>custom-protocol
config>app-assure>group>tethering-detection
config>app-assure>group>url-list
config>app-assure>group>wap1x
config>app-assure>protocol
config>service>ies>aa-interface
config>service>ies>aa-interface>sap
config>service>vprn>aa-interface
config>service>vprn>aa-interface>sap
```

Description

This command administratively disables the entity. When disabled, an entity does not change, reset, or remove any configuration settings or statistics. Many entities must be explicitly enabled using the no shutdown command.

The shutdown command administratively disables an entity. The operational state of the entity is disabled as well as the operational state of any entities contained within. Many objects must be shut down before they can be deleted.

10.2.1.2 Hardware Application Assurance command descriptions

10.2.1.2.1 isa2-aa (mda-type)

Syntax

mda-type *isa2-aa*

no mda-type

Context

config>card>mda

Description

This command provisions an adapter into an MDA position on an ISM-MG slot. The AA ISA is provisioned into the system in the same manner as all other MDA type. Once an AA ISA is provisioned, the AA ISA can be defined as a member of an application assurance group.

The no form of this command removes the module from the configuration. The module must be administratively shut down before it can be deleted from the configuration.

For more information about command usage and syntax for the AA ISA and other MDA and ISA card types, refer to the *7450 ESS, 7750 SR, 7950 XRS, and VSR Interface Configuration Guide*.

Default

no mda-type

Parameters

isa2-aa

specifies the Application Assurance Integrated Services Adapter for the slot position

10.2.1.3 Admin Application Assurance command descriptions

10.2.1.3.1 application-assurance

Syntax

application-assurance

Context

admin

Description

This command enables the context to perform Application Assurance (AA) configuration operations.

10.2.1.3.2 group

Syntax

group *aa-group-id*

Context

admin>app-assure

Description

This command specifies the ISA AA group for which you want to upgrade the URL list.

Parameters

aa-group-id

represents a group of ISA MDAs

Values: 1 to 255

10.2.1.3.3 url-list

Syntax

url-list *url-list-name* upgrade

Context

admin>app-assure>group

Description

Use this command to upgrade the URL list loaded in the system using the file referenced in the URL list policy. To upgrade a list, the operator downloads a new file to the system compact flash of the 7750 SR MG, (this overwrites the previous URL list file) and then executes the `url-list` command to load the list on the ISA cards.

Parameters

url-list-name

specifies the Application Assurance URL list to upgrade

upgrade

keyword to upgrade the list

10.2.1.3.4 upgrade

Syntax

upgrade

Context

admin>app-assure

Description

Use this command to load a new `isa-aa.tim` file as part of a router-independent signature upgrade. An AA ISA reboot is required.

10.2.1.4 ISA Application Assurance command descriptions

10.2.1.4.1 application-assurance-group

Syntax

application-assurance-group *aa-group-id* [create] [aa-sub-scale *sub-scale*]

no application-assurance-group *aa-group-id*

Context

config>isa

Description

This command enables the context to create an application assurance group with the specified system-unique index and enables the context to configure that group's parameters.

The no form of this command deletes the specified application assurance group from the system. The group must be shutdown first.

Parameters

aa-group-id

specifies an integer to identify the AA group

Values: 1 to 255

create

mandatory keyword used when creating an application assurance group in the ISA context

The create keyword requirement can be enabled/disabled in the environment>create context

sub-scale

specifies the set of scaling limits that are supported with regards to the maximum number of AA subscribers per ISA and the corresponding policies that can be specified

Values: mobile-gateway | mobile-gateway-1m | mobile-gateway-2m

mobile-gateway

7750 SR MG hardware systems ISM-MG-C (ISA2): scaling limits with up to 2M mobile AA subscribers and 12M flows; CMG: scaling limits with up to 1M mobile AA subscribers and 8M flows

mobile-gateway-1m

CMG: scaling limits with up to 1M mobile AA subscribers and 8M flows

mobile-gateway-2m

CMG: scaling limits with up to 2M mobile AA subscribers and 16M flows

Default: mobile-gateway

10.2.1.4.2 dynamic-rating-expiry

Syntax

dynamic-rating-expiry *timeout-in-secs*

no dynamic-rating-expiry

Context

config>isa>aa-grp

Description

This command configures (in seconds) the timeout value of the dynamic rating. When the timeout expires, AA triggers an application-stop.

Parameters

timeout-in-secs

specifies the value, in seconds, of the timeout

Values: 5 to 120

Default: 120

10.2.1.4.3 fail-to-open

Syntax

[no] fail-to-open

Context

config>isa>aa-grp

Description

This command configures the mode of operation during an operational failure of this application assurance group when no application assurance engines are available to service traffic. When enabled, all traffic that was to be inspected will be dropped. When disabled, all traffic that was to be inspected will be forwarded without any inspection as if the group was not configured at all.

Default

no fail-to-open

10.2.1.4.4 flow-attribute

Syntax

[no] flow-attribute *flow-attribute name*

Context

config>isa>aa-grp

Description

This command enables the specified flow attribute.

The no form of this command disables the attribute.

Parameters

flow-attribute name

specifies the name of the flow attribute, up to 256 characters

Values:

video

Specifies streaming or real-time video media traffic transferred between a sender and receiver. It does not differentiate adaptive and non-adaptive video streaming. It is assigned to flows based on packet payload inspection (many protocols) or behavioral mechanisms, and may be used together with the RTC attribute to identify video call traffic.

abr_service

Assigned to adaptive bit rate traffic exchanges where the traffic rate or behavior can be automatically adjusted based on changes in network conditions. It is assigned by application filter configuration or by behavioral mechanisms.

audio

Assigned to streaming or real-time audio media traffic transferred between a sender and receiver. It is assigned to flows based on packet payload inspection or behavioral mechanisms, and may be used together with the RTC attribute to identify voice call traffic.

encrypted

Assigned to traffic exchanges where the initial payload or the entirety of the exchange is encrypted. It is assigned to sessions based on packet payload inspection or by behavioral mechanisms.

download

Assigned to traffic that has a high likelihood of exchanging data predominantly in the network to subscriber direction over the lifetime of a session. It may be assigned to sessions based on behavioral mechanisms.

upload

Assigned to traffic that has a high likelihood of exchanging data predominantly in the subscriber to network direction over the lifetime of the session. The upload and download attributes are mutually exclusive. It is assigned to sessions based on behavioral mechanisms.

real_time_communication

Assigned to traffic that provides a low latency or real-time exchange of information between two or more communicating endpoints. It is assigned by packet payload inspection of an RTP protocol being used or by behavioral mechanisms.

esni

Assigned to traffic that uses an encrypted server name indication (eSNI), as part of the TLS layer negotiation.

10.2.1.4.5 http-enrich-max-pkt

Syntax

http-enrich-max-pkt *size*

Context

config>isa>aa-grp

Description

This command configures the maximum HTTP enriched packet size.

Parameters

size

specifies the maximum HTTP enriched packet size in octets

Values: 576 to 9212

Default: 1500

10.2.1.4.6 isa-capacity-cost-high-threshold

Syntax

isa-capacity-cost-high-threshold *threshold*

no isa-capacity-cost-high-threshold

Context

config>isa>aa-grp

Description

This command configures the ISA-AA capacity cost high threshold.

The no form of this command reverts the threshold to the default value.

Default

isa-capacity-cost-high-threshold 4294967295

Parameters

threshold

specifies the capacity cost high threshold for the ISA-AA group

Values: 0 to 4294967295

10.2.1.4.7 isa-capacity-cost-low-threshold

Syntax

isa-capacity-cost-low-threshold *threshold*

no isa-capacity-cost-low-threshold

Context

config>isa>aa-grp

Description

This command configures the ISA-AA capacity cost low threshold.

The no form of this command reverts the threshold to the default value.

Default

isa-capacity-cost-low-threshold 0

Parameters

threshold

specifies the capacity cost low threshold for the ISA-AA group

Values: 0 to 4294967295

10.2.1.4.8 isa-overload-cut-through

Syntax

[no] isa-overload-cut-through

Context

config>isa>aa-grp

Description

This command configures the ISA group to enable cut-through of traffic if an overload event occurs, triggered when the IOM weighted average queues depth exceeds the `wa-shared-high-wmark`. In this ISA state, packets are cut-through from application analysis but retain subscriber context with default subscriber policy applied.

The no form of this command disables cut-through processing on overload.

Default

no isa-overload-cut-through

10.2.1.4.9 minimum-isa-generation

Syntax

minimum-isa-generation *min-isa-generation*

Context

config>isa>aa-grp

Description

This command configures the scale parameters for the ISA group. When *min-isa-generation* is configured as 1, the group and per-ISA limits are the MS-ISA scale. When *min-isa-generation* is configured as 2, the per-ISA resource limits shown in the `show isa application-assurance-group 1 load-balance` output will increase to show ISA2 limits.

If there is a mix of ISA1 and ISA2, the *min-isa-generation* parameter must be left as 1.

Parameters

min-isa-generation

specifies the minimum ISA generation allowed in this group

Values	1	ISA (ISA1)
–	2	ISA2

Default: 1

10.2.1.4.10 partitions

Syntax

[no] partitions

Context

config>isa>aa-grp

Description

This command enables partitions within an ISA-AA group. When enabled, partitions can be created

The no form of this command disables partitions within an ISA-AA group.

Default

no partitions

10.2.1.4.11 primary

Syntax

[no] primary *mda-id*

Context

config>isa>aa-grp

Description

This command assigns an AA ISA module configured in the specified slot to this application assurance group. Primary and backup ISAs have equal operational status and when both ISAs are coming up, the one that becomes operational first becomes the active ISA.

On an activity switch from the primary ISA, all configurations are already on the backup ISA but flow state information must be re-learned. Any statistics not yet spooled will be lost. Auto-switching from the backup to primary, once the primary becomes available again, is not supported.

Operator is notified through SNMP events when:

- the AA service goes down (all ISAs in the group are down) or comes back up (an ISA in the group becomes active)
- the AA redundancy fails (one of the ISAs in the group is down) or recovers (the failed MDA comes back up)

- an AA activity switch occurred.

The no form of this command removes the specified ISA from the application assurance group.

Default

no primary

Parameters

mda-id

specifies the slot and MDA identifying a provisioned AA-ISA

Values	<i>mda-id:/slot/mda</i>
–	slot 1 to up to 10 depending on chassis model
–	mda 1 to 2

10.2.1.4.12 qos

Syntax

qos

Context

config>isa>aa-grp

Description

This command enables the context for Quality of Service configuration for this application assurance group.

10.2.1.4.13 egress

Syntax

egress

Context

```
config>isa>aa-grp>qos
```

Description

This command enables the context for IOM port-level Quality of Service configuration for this application assurance group in the egress direction (traffic entering an application assurance engine).

10.2.1.4.14 from-subscriber

Syntax

```
from-subscriber
```

Context

```
config>isa>aa-grp>qos>egress
```

Description

This command enables the context for Quality of Service configuration for this application assurance group from-subscriber logical port, traffic entering the system from AA subscribers and entering an application assurance engine.

10.2.1.4.15 pool

Syntax

```
pool [pool-name]
```

```
no pool
```

Context

```
config>isa>aa-grp>qos>egress>from-subscriber
```

```
config>isa>aa-grp>qos>egress>to-subscriber
```

```
config>isa>aa-grp>qos>ingress
```

Description

This command enables the context to configure an IOM pool as applicable to the specific application assurance group traffic. The user can configure resv-cbs (as percentage) values and slope-policy similarly to other IOM pool commands.

Parameters

pool-name

the name of the pool

10.2.1.4.16 resv-cbs

Syntax

`resv-cbs percent-or-default`

`no resv-cbs`

Context

`config>isa>aa-grp>qos>egress>from-subscriber>pool`

`config>isa>aa-grp>qos>egress>to-subscriber>pool`

`config>isa>aa-grp>qos>ingress>pool`

Description

This command defines the percentage or specifies the sum of the pool buffers that are used as a guideline for CBS calculations for access and network ingress and egress queues. Two actions are accomplished by this command.

- A reference point is established to compare the currently assigned (provisioned) total CBS with the amount the buffer pool considers to be reserved. Based on the percentage of the pool reserved that has been provisioned, the over provisioning factor can be calculated.
- The size of the shared portion of the buffer pool is indirectly established. The shared size is important to the calculation of the instantaneous-shared-buffer-utilization and the average-shared-buffer-utilization variables used in Random Early Detection (RED) per packet slope plotting.



NOTE:

This command does not actually set aside buffers within the buffer pool for CBS reservation. The CBS value per queue only determines the point at which enqueueing packets are subject to a RED slope. Oversubscription of CBS could result in a queue operating within its CBS size and still not able to enqueue a packet due to unavailable buffers. The `resv-cbs` parameter can be changed at any time.

If the total pool size is 10 MB and the `resv-cbs` set to 5, the 'reserved size' is 500 KB.

The no form of this command reverts to the default.

Parameters

percent-or-default

specifies the pool buffer size percentage

Values: 0 to 100, default

Default: default (30%)

10.2.1.4.17 slope-policy

Syntax

slope-policy *name*

no slope-policy

Context

config>isa>aa-grp>qos>egress>from-subscriber>pool

config>isa>aa-grp>qos>egress>to-subscriber>pool

config>isa>aa-grp>qos>ingress>pool

Description

This command specifies an existing slope policy which defines high and low priority RED slope parameters and the time average factor. The slope policy is defined in the config>qos>slope-policy context.

10.2.1.4.18 queue-policy

Syntax

queue-policy *network-queue-policy-name*

no queue-policy

Context

config>isa>aa-grp>qos>egress>from-subscriber

config>isa>aa-grp>qos>egress>to-subscriber

config>isa>aa-grp>qos>ingress

Description

This command assigns an IOM network queue policy as applicable to specific application assurance group traffic.

Parameters

network-queue-policy-name

the name of the network queue policy defined in the system

10.2.1.4.19 wa-shared-high-wmark

Syntax

wa-shared-high-wmark *percent*

no wa-shared-high-wmark

Context

config>isa>aa-grp>qos>egress>from-sub

config>isa>aa-grp>qos>egress>to-sub

Description

This command configures the high watermark for the weighted average utilization of the shared buffer space in the from-subscriber buffer pool for each ISA. When a buffer pool is not in the overload state and the wa-shared buffer utilization for an ISA crosses above the high watermark value in the ISA from-subscriber buffer pool enters an overload state and an overload notification is raised.

The no form of this command reverts to the default.

Default

wa-shared-high-wmark max

Parameters

percent

specifies the weighted average shared buffer utilization high watermark

Values: 1 to 100 | max percent

Default: max (disabled)

10.2.1.4.20 wa-shared-low-wmark

Syntax

```
wa-shared-low-wmark percent  
no wa-shared-low-wmark
```

Context

```
config>isa>aa-grp>qos>egress>from-sub  
config>isa>aa-grp>qos>egress>to-sub
```

Description

This command configures the low watermark for the weighted average utilization of the shared buffer space in the from-subscriber buffer pool. When a buffer pool is in an overloaded state and the wa-shared buffer utilization for an ISA drops below low watermark value ISA from-subscriber buffer pool leaves the overload state and a is sent to indicate the overload state has cleared.

The no form of this command reverts to the default.

Default

wa-shared-low-wmark 0

Parameters

percent

specifies the weighted average shared buffer utilization low watermark

Values: 0 to 99

10.2.1.4.21 port-scheduler-policy

Syntax

```
port-scheduler-policy port-scheduler-policy-name  
no port-scheduler-policy
```

Context

```
config>isa>aa-grp>qos>egress>from-subscriber  
config>isa>aa-grp>qos>egress>to-subscriber
```

Description

This command assigns an existing port scheduler policy as applicable to the specific application assurance group traffic.

Parameters

port-scheduler-policy-name

specifies the name of an existing port scheduler policy

10.2.1.4.22 to-subscriber

Syntax

to-subscriber

Context

config>isa>aa-grp>qos>egress

Description

This command enables the context for Quality of Service configuration for this application assurance group to-subscriber logical port, traffic destined to AA subscribers and entering an application assurance engine.

10.2.1.4.23 shared-resources

Syntax

shared-resources

Context

config>isa>aa-grp

Description

This command enables the context to configure the shared resources pool.

10.2.1.4.24 bandwidth-shaper-buffer

Syntax

bandwidth-shaper-buffer *size*

Context

config>isa>aa-grp>shared-resources

Description

This command configures the amount of shared memory to be used by the bandwidth shaper. It is not applicable to 7750 MG hardware deployments.

Parameters

size

specifies the amount of shared memory as a percentage

Values: 0 or 100

Default: 0

10.2.1.4.25 multi-device-detection

Syntax

multi-device-detection *size*

Context

config>isa>aa-grp>shared-resources

Description

This command configures the allocation of shared resource pool for buffering for multi device detection.

Parameters

size

specifies the amount of shared memory as a percentage

Values: 0 or 100

Default: 0

10.2.1.4.26 tcp-adv-func

Syntax

tcp-adv-func *size*

Context

```
config>isa>aa-grp>shared-resources
```

Description

This command configures the allocation of shared resource pool for TCP advanced functions.



NOTE:

By default, and before configuring the `tcp-adv-func` command, no shared resources are allocated to any function.

Parameters

size

specifies the allocation of the shared resource pool

Values: 0 to 100

Default: 100

10.2.1.4.27 web-service-url-filter

Syntax

```
web-service-url-filter size
```

Context

```
config>isa>aa-grp>shared-resources
```

Description

This command configures the amount of shared memory to be used by the web-service URL filter cache.

Parameters

size

specifies the amount of shared memory as a percentage

Values: 0 to 100

Default: 0

10.2.1.4.28 transit-prefix-ipv4-entries

Syntax

```
transit-prefix-ipv4-entries entries  
no transit-prefix-ipv4-entries
```

Context

```
config>isa>aa-grp
```

Description

This command defines the number of transit-prefix IPv4 entries for an ISA.

The no form of this command removes the assignment of entries space from the configuration. All entries must be removed to delete the configuration.

Default

```
no transit-prefix-ipv4-entries
```

Parameters

entries

specifies an integer that determines the number of transit-prefix-ipv4 entries

Values: 0 to 16383

10.2.1.4.29 transit-prefix-ipv4-remote-entries

Syntax

```
transit-prefix-ipv4-remote-entries entries  
no transit-prefix-ipv4-remote-entries
```

Context

```
config>isa>aa-grp
```

Description

This command configures the ISA-AA-group transit prefix IPv4 remote entry limit. This entry space is allocated on the IOM within a common area with the second MDA/ISA position of the IOM and also used for

IPv4filter entries for system SDPs. The per-ISA size allocated for transit-prefix-ipv4 entries should be set to allow sufficient space on the IOM for SDP IPv4 filters.

The no form of this command removes the assignment of entries space from the configuration. All entries must be removed to delete the configuration.

Default

no transit-prefix-ipv4-remote-entries

Parameters

entries

Specifies the ISA-AA-Group transit prefix IPv4 remote entry limit.

Values: 0 to 2047

10.2.1.4.30 transit-prefix-ipv6-entries

Syntax

transit-prefix-ipv6-entries *entries*

no transit-prefix-ipv6-entries

Context

config>isa>aa-grp

Description

This command configures the ISA-AA-group transit prefix IPv6 entry limit for each ISA in the group. This entry space is allocated on the IOM within a common area with the second MDA / ISA position of the IOM and also used for ipv6-filter entries for system SDPs. The per-ISA size allocated for transit-prefix-ipv6 entries should be set to allow sufficient space on the IOM for SDP ipv6-filters.

The no form of this command removes the assignment of entries space from the configuration. All entries must be removed to delete the configuration.

Default

no transit-prefix-ipv6-entries

Parameters

entries

specifies the ISA-AA-Group transit prefix IPv6 entry limit

Values: 0 to 8191

10.2.1.4.31 transit-prefix-ipv6-remote-entries

Syntax

```
transit-prefix-ipv6-remote-entries entries  
no transit-prefix-ipv6-remote-entries
```

Context

```
config>isa>aa-grp
```

Description

This command configures the ISA-AA-group transit prefix IPv6 remote entry limit. This entry space is allocated on the IOM within a common area with the second MDA/ISA position of the IOM and also used for IPv6filter entries for system SDPs. The per-ISA size allocated for transit-prefix-ipv6 entries should be set to allow sufficient space on the IOM for SDP IPv6 filters.

The no form of this command removes the assignment of entries space from the configuration. All entries must be removed to delete the configuration.

Default

```
no transit-prefix-ipv6-remote-entries
```

Parameters

entries

specifies the ISA-AA-Group transit prefix IPv6 remote entry limit

Values: 0 to 1023

10.2.1.4.32 vm-traffic-distribution-by-ip

Syntax

```
[no] vm-traffic-distribution-by-ip
```

Context

```
config>isa>aa-grp
```

Description

On CMG systems, this command enables the inclusion of packets by IP addresses in the data plane VM traffic distribution to CPU cores for AA processing.

The no form of this command uses traffic distribution by AA subscriber.

Default

no vm-traffic-distribution-by-ip

10.2.1.4.33 ingress

Syntax

ingress

Context

config>card>mda>network>ingress

Description

This command enables the context for MDA-level IOM Quality of Service configuration.

10.2.1.5 Application Assurance command descriptions

10.2.1.5.1 application-assurance

Syntax

application-assurance

Context

config

Description

This command enables the context to perform Application Assurance (AA) configuration operations.

10.2.1.5.2 bit-rate-high-wmark

Syntax

```
bit-rate-high-wmark high-watermark  
no bit-rate-high-wmark
```

Context

```
config>app-assure
```

Description

The command configures the high-watermark value for bit rate alarms. The value must be larger than or equal to the bit-rate-low-wmark parameter.

The no form of this command reverts to the default.

Default

```
no bit-rate-high-wmark
```

Parameters

high-watermark

specifies the high watermark for bit rate alarms, in Mbits/sec

Values: 1 to 10000

10.2.1.5.3 bit-rate-low-wmark

Syntax

```
bit-rate-low-wmark low-watermark  
no bit-rate-low-wmark
```

Context

```
config>app-assure
```

Description

The command configures the low watermark value for bit rate alarms. The value must be lower than or equal to the bit-rate-high-wmark command.

Parameters

low-watermark

specifies the low watermark for bit rate alarms, in mbits/sec

Values: 0 to 9999

Default: 0

10.2.1.5.4 datapath-cpu-high-wmark

Syntax

`datapath-cpu-high-wmark high-watermark`

Context

`config>app-assure`

Description

This command configures the system-wide high watermark threshold as a percentage of the per-ISA datapath core CPU utilization, where an alarm is raised by the agent. CPU usage is measured as an average across all datapath cores.

Parameters

high-watermark

specifies the high watermark for datapath CPU usage alarms

Values: 1 to 100

Default: 95

10.2.1.5.5 datapath-cpu-low-wmark

Syntax

`datapath-cpu-low-wmark low-watermark`

Context

`config>app-assure`

Description

This command configures the system-wide low watermark threshold as a percentage of the per-ISA datapath core CPU utilization, where an alarm is raised by the agent. CPU usage is measured as an average across all datapath cores.

Parameters

low-watermark

specifies the low watermark for datapath CPU usage alarms

Values: 1 to 100

Default: 90

10.2.1.5.6 flow-setup-high-wmark

Syntax

flow-setup-high-wmark *high-watermark*

no flow-table-high-wmark

Context

config>app-assure

Description

The command configures the high-watermark value for flow setup rate alarms. The value must be larger than or equal to the flow-setup-low-wmark parameter.

Default

no flow-setup-high-wmark

Parameters

high-watermark

specifies the high watermark for flow setup rate alarms, in flows/sec

Values: 1 to 200000

10.2.1.5.7 flow-setup-low-wmark

Syntax

```
flow-setup-low-wmark low-watermark  
no flow-table-low-wmark
```

Context

```
config>app-assure
```

Description

The command configures the low-watermark value for flow setup rate alarms. The value must be lower than or equal to the flow-setup-high-wmark parameter.

Parameters

low-watermark

specifies the low watermark for flow setup rate alarms, in flows/sec

Values: 0 to 199999

Default: 0

10.2.1.5.8 flow-table-high-wmark

Syntax

```
flow-table-high-wmark high-watermark  
no flow-table-high-wmark
```

Context

```
config>app-assure
```

Description

The command configures the utilization of the flow records on the AA ISA group when a full alarm will be raised by the agent. The value must be larger than or equal to the flow-table-low-wmark low- watermark parameter.

Parameters

high-watermark

specifies the high watermark for flow table full alarms

Values: 0 to 100

10.2.1.5.9 flow-table-low-wmark

Syntax

flow-table-low-wmark *low-watermark*

no flow-table-low-wmark

Context

config>app-assure

Description

The command configures the utilization of the flow records on the AA ISA group when the full alarm will be cleared by the agent. The value must be lower than or equal to the flow-table-high-wmark high-watermark parameter.

Parameters

low-watermark

specifies the low watermark for flow table full alarms

Values: 0 to 100

10.2.1.5.10 packet-rate-high-wmark

Syntax

packet-rate-high-wmark *high-watermark*

no packet-rate-high-wmark

Context

config>app-assure

Description

The command configures the high-watermark value for packet rate alarms. The value must be larger than or equal to the packet-rate-low-wmark parameter.

Default

no packet-rate-high-wmark

Parameters

high-watermark

specifies the high watermark for packet rate alarms, in packets/sec

Values: 1 to 14880952

10.2.1.5.11 packet-rate-low-wmark

Syntax

packet-rate-low-wmark *low-watermark*

no packet-rate-low-wmark

Context

config>app-assure

Description

The command configures the low-watermark value for packet rate alarms. The value must be lower than or equal to the packet-rate-high-wmark parameter.

Parameters

low-watermark

specifies the low watermark for packet rate alarms, in packets/sec

Values: 0 to 14880951

Default: 0

10.2.1.5.12 protocol

Syntax

protocol *protocol-name*

Context

config>app-assure

Description

This command configures the shutdown of protocols system-wide.

Parameters

protocol-name

a string of up to 32 characters identifying a predefined protocol

10.2.1.6 Application Assurance interface command descriptions

10.2.1.6.1 aa-interface

Syntax

```
aa-interface aa-if-name [create]
```

```
no aa-interface aa-if-name
```

Context

```
config>service>ies
```

```
config>service>vpn
```

Description

This commands creates a new AA interface within an IES or VPRN service. It is used by the aa-isa to send/receive IPv4 traffic. In the context of ICAP url-filtering this interface is used by the ISA to establish ICAP TCP connections to the ICAP servers.

This interface supports /31 subnet only, and uses by default, 1q encapsulation.

The system automatically configures the ISA IP address based on the address configured by the operator under the aa-interface object (which represents the ISA sap facing interface on the ISA).

Parameters

aa-if-name

specifies the name of the AA Interface

create

keyword that specifies to create the interface

10.2.1.6.2 address

Syntax

```
address {ip-address/mask | ip-address netmask}  
no address [ip-address/mask | ip-address netmask]
```

Context

```
config>service>ies>aa-interface  
config>service>vprn>aa-interface
```

Description

This command assigns an IP address to the interface.

Default

no address

Parameters

ip-address/mask

specifies an IP address/IP subnet format to the interface

ip-address netmask

specifies a string of 0s and 1s that mask or screen out the network part of an IP address, so that only the host computer part of the address remains

create

keyword that specifies to create the interface

10.2.1.6.3 ip-mtu

Syntax

```
ip-mtu octets  
no ip-mtu
```

Context

```
config>service>ies>aa-interface  
config>service>vprn>aa-interface
```

Description

This command configures the AA interface IP MTU.

Default

no ip-mtu

Parameters

octets

specifies the MTU value

Values: 512 to 9000

10.2.1.6.4 sap

Syntax

sap *sap-id* [create]

no sap *sap-id*

Context

config>service>ies>aa-interface

config>service>vprn>aa-interface

Description

This command configures the AA interface SAP.

Parameters

sap-id

specifies the physical port identifier portion of the SAP definition

create

creates the SAP instance

10.2.1.6.5 egress

Syntax

egress

Context

```
config>service>ies>aa-interface>sap
```

```
config>service>vprn>aa-interface>sap
```

Description

This command enables the context to configure egress parameters.

10.2.1.6.6 ingress

Syntax

```
ingress
```

Context

```
config>service>ies>aa-interface>sap
```

```
config>service>vprn>aa-interface>sap
```

Description

This command enables the context to configure ingress parameters.

10.2.1.6.7 filter

Syntax

```
filter ip ip-filter-id
```

```
no filter [ip ip-filter-id]
```

Context

```
config>service>ies>aa-if>sap>egress
```

```
config>service>vprn>aa-if>sap>egress
```

Description

This command applies an IP filter to the SAP.

Default

```
no filter
```

Parameters

ip-filter-id

specifies an existing IP filter ID value or name, up to 64 characters

Values: 1 to 65535

10.2.1.6.8 qos

Syntax

qos *policy-id*

no qos [*policy-id*]

Context

config>service>ies>aa-if>sap>egress

config>service>ies>aa-if>sap>ingress

config>service>vprn>aa-if>sap>egress

config>service>vprn>aa-if>sap>ingress

Description

This command applies an QoS policy to the SAP.

Default

qos 1

Parameters

policy-id

specifies an existing QoS policy ID

10.2.1.7 Application Assurance group command descriptions

10.2.1.7.1 group

Syntax

group *aa-group-id*[:*partition-id*][create]]

```
no group aa-group-id:partition-id
```

Context

```
config>app-assure
```

Description

This command configures and enables the context to configure an application assurance group and partition parameters.

Parameters

aa-group-id

represents a group of ISA MDAs

Values: 1 to 255

partition-id

specifies a partition within a group

Values: 1 to 65535

create

keyword used to create the partition in the group

10.2.1.7.2 aa-sub-congestion-detection

Syntax

```
aa-sub-congestion-detection
```

Context

```
config>app-assure>group
```

Description

This command enables the context to configure congestion detection parameters related to Non-location Based Dynamic Experience Management (NLB-DEM).



NOTE:

NLB-DEM cannot be enabled if Access-Network Location (ANL) DEM mode is enabled; the two are mutually exclusive.

10.2.1.7.3 rtt-threshold

Syntax

```
rtt-threshold threshold  
no rtt-threshold
```

Context

```
config>app-assure>group>aa-sub-cong  
config>app-assure>group>anl>source
```

Description

This command configures the maximum acceptable Round-Trip-Time (RTT) for TCP connections under no congestion. This is used by the by the DEM gateway (DEM-GW) congestion detection algorithm (if applicable) to determine subscriber congestion. Any measured RTT above the configured threshold is considered as an indication of possible congestion.



NOTE:

The specified threshold level is not used when the *source level* is set to transport-network-link; set in the in the config>app-assure>group>anl>source command.

Parameters

threshold

specifies the threshold value, in milliseconds, used by the DEM gateway (DEM-GW) algorithm to determine subscriber or ANL congestion

Values: 0 to 500 (a value of 0 indicates no congestion detection via RTT is to be used)

Default: 173

10.2.1.7.4 rtt-threshold-rat

Syntax

```
rtt-threshold-rat rat-type rat-type rtt-threshold rtt-threshold  
no rtt-threshold-rat rat-type rat-type
```

Context

```
config>app-assure>group>aa-sub-cong
```

```
config>app-assure>group>anl>source
```

Description

This command configures the roundtrip delay threshold for each RAT Type to be used for congestion detection algorithm (if applicable). The value of `rtt - threshold` keyword is used by the DEM-GW algorithm to determine ANL congestion or subscriber congestion in the case of NLB-DEM. Any RTT value higher than this threshold is considered an indication of possible congestion.

Parameters

rat-type

specifies the 3GPP RAT type

Values: `utran` | `geran` | `wlan` | `gan` | `hspa-evol` | `eutran` | `virtual` | `eutran-nb` | `ehrpd` | `hrpd` | `cdma-1x` | `umb` | `wifi` | `nr`

rtt-threshold

specifies the maximum acceptable RTT under no congestion, in milliseconds

Values: 0 to 500

10.2.1.7.5 rtt-threshold-tolerance

Syntax

```
rtt-threshold-tolerance tolerance
```

```
no rtt-threshold-tolerance
```

Context

```
config>app-assure>group>aa-sub-cong
```

```
config>app-assure>group>anl>source
```

Description

This command configures the roundtrip delay threshold tolerance to be used by the congestion detection algorithm (if applicable) to determine subscriber or ANL congestion. The threshold represents the ratio of Round-Trip-Times (RTTs) above the configured RTT threshold over the total RTT measurements. That is:

$$\text{rtt-threshold-tolerance} = (\text{RTTs} > \text{rtt-threshold}) / (\text{total number of RTTs})$$
 measured across a one minute period

If the `rtt-threshold-tolerance` ratio is exceeded, all of the ANL subscribers (or just one subscriber in the case of NLB-DEM) will be declared congested.

**NOTE:**

The specified threshold level is not used when the source level is set to `transport-network-link`; set in the `config>app-assure>group>anl>source` command.

Parameters

tolerance

specifies the tolerance threshold value, in milliseconds, used by the DEM gateway (DEM-GW) algorithm to determine subscriber or ANL congestion

Values: 0 to 100% (a value of 100 effectively turns off congestion detection using RTT)

Default: 50%

10.2.1.7.6 access-network-location

Syntax

```
access-network-location
```

Context

```
config>app-assure>group
```

Description

This command provides the context to configure parameters related to Dynamic Experience Management (DEM), also known as Access Network Location (ANL).

The parameters that can be configured include the location source type (for example, mobile 3G), congestion point, and congestion detection threshold and tolerance levels; such as the round-trip delay threshold.

10.2.1.7.7 source

Syntax

```
source source-type level level
```

```
no source source-type
```

Context

```
config>app-assure>group>anl
```

Description

This command configures the location source for Dynamic Experience Management (DEM).

Default

no source

Parameters

source-type

specifies the location or access technology

Values: `uli-3gpp`

Provides DEM for 3GPP Fixed Wireless Access network. DEM uses the User Location Information (ULI) passed at bearer setup as the fixed location of the subscriber.

level

specifies which congestion point within the specified source type to monitor for congestion

Values: `uli-3gpp`

Radio or wireless cell. The cell level is the only option supported when the *source-type* is `uli-3gpp`.

10.2.1.7.8 alternate-cdn-host-matching

Syntax

`alternate-cdn-host-matching ip-prefix-list-name`

`no alternate-cdn-host-matching`

Context

`config>app-assure>group`

Description

This command retrieves and uses for classification, the host name from the URI path if the host in the URL is an IP address found in the IP list.

The `no` form of this command removes the matching.

Default

`no alternate-cdn-host-matching`

Parameters

ip-prefix-list-name

specifies an IP address prefix list that is used for alternate Content Delivery Network (CDN) host matching, up to 32 characters

10.2.1.7.9 certificate-profile

Syntax

```
certificate-profile cert-prof-name create  
no certificate-profile cert-prof-name
```

Context

```
config>app-assure>group
```

Description

This command creates a certificate profile to be used for certificate-based encryption in HTTP header enrichment.

The no form of this command removes the certificate from the profile.

Parameters

cert-prof-name

specifies the file name of the certificate, up to 32 characters

create

specifies to create the certificate profile

10.2.1.7.10 certificate

Syntax

```
certificate certificate-file  
no certificate
```

Context

```
config>app-assure>group>certificate-profile
```

Description

This command configures a certificate file for the certificate profile.

The no form of this command removes the certificate file from the profile.

Default

no certificate

Parameters

certificate-file

specifies the certificate file name, up to 95 characters

10.2.1.7.11 key-file

Syntax

key-file *key-filename*

no key-file

Context

config>app-assure>group>certificate-profile

Description

This command configures the private key for the certificate profile.

The no form of this command removes the private key from the profile.

Parameters

key-filename

specifies the private key file name, up to 95 characters

10.2.1.7.12 disable-deferred-billing

Syntax

[no] disable-deferred-billing

Context

config>app-assure>group

Description

This command disables deferred billing for the AA partition (for a partitioned group) and AA group (for a non-partitioned group). AA deferred billing allows charging of all the packets in a flow according to a detected flow application or charging group; including flow packets received before a positive identification. When deferred billing is disabled, flow packets received before flow application identification can be charged differently from those packets received after flow identification.

The no form of this command enables deferred billing.

Default

no disable-deferred-billing

10.2.1.7.13 dns-enrich

Syntax

dns-enrich *dns-enrich-name* [create]

no dns-enrich *dns-enrich-name*

Context

config>app-assure>group

Description

This command enables the context to configure a DNS enrichment template and assigns a name to it.

Parameters

dns-enrich-name

specifies the name of the DNS enrichment policy name, up to 32 characters

create

mandatory keyword used when creating a new DNS enrichment template

10.2.1.7.14 gateway

Syntax

gateway *gateway-id*

no gateway

Context

config>app-assure>group>dns-enrich

Description

This command configures a unique id for an AA node used along with a per sub policy-ID passed from the PCF to create a device-id that is inserted into the DNS request flows.

Parameters

gateway-id

a 32-bit value that represents a unique ID for the AA node

Values: 0 to 4294967295

10.2.1.7.15 option

Syntax

option *option* [create]

no option *option*

Context

config>app-assure>group>dns-enrich

Description

This command configures the DNS enrichment type.

Parameters

option

the DNS option for the enriched device-id

Value: device-id

10.2.1.7.16 packet-size-limit

Syntax

```
packet-size-limit packet-size  
no packet-size-limit
```

Context

```
config>app-assure>group>dns-enrich
```

Description

This command configures the maximum DNS packet size. When the packet size exceeds the configured maximum value because of the enrichment, AA does not enrich the DNS packet.

Parameters

packet-size

specifies the maximum DNS packet size, in bytes

Values: 256 to 9212

Default: 1500

10.2.1.7.17 dns-ip-cache

Syntax

```
dns-ip-cache dns-ip-cache-name [create]  
no dns-ip-cache dns-ip-cache-name
```

Context

```
config>app-assure>group
```

Description

This command configures a DNS IP cache used to snoop DNS requests generated by subscribers to populate a cache of IP addresses matching a specified list of domain names. In the context of URL content charging strengthening, it is also mandatory to specify a list of trusted DNS servers to populate the DNS IP cache.

Parameters

dns-ip-cache-name

specifies the name of the Application Assurance DNS IP cache, up to 32 characters

create

mandatory keyword used when creating a new DNS IP cache; the create keyword requirement can be enabled or disabled in the `environment>create` context

10.2.1.7.18 dns-match

Syntax

```
dns-match
```

Context

```
config>app-assure>group>dns-ip-cache
```

Description

This command enables the context to configure match criteria for the DNS IP cache.

10.2.1.7.19 domain

Syntax

```
domain domain-name expression expression
```

```
no domain
```

Context

```
config>app-assure>group>dns-ip-cache>dns-match
```

Description

This command configures a domain expression to populate the DNS IP cache match criteria.

Parameters

domain-name

specifies the domain name up to 32 characters

expression

specifies a domain name expression string, up to 64 characters long used to define a pattern match. This domain expression uses the same syntax as the expressions used in app-filters.

10.2.1.7.20 **server-address**

Syntax

```
server-address server-address [name server-name]  
no server-address server-address
```

Context

```
config>app-assure>group>dns-ip-cache>dns-match
```

Description

This command configures a DNS server-address. DNS responses from this DNS server are used to populate the DNS IP cache. Up to 64 server addresses can be configured.

Parameters

server-address

specifies the IPv4 or IPv6 address of the DNS server

Values	ipv4 - a.b.c.d [mask]
–	ipv6 - x:x:x:x:x:x:x
–	x - [0..FFFF]H
–	d - [0..255]D
–	ipv6-prefix-length: [1 to 128]

server-name

specifies an optional server name for the given server address, up to 32 characters

10.2.1.7.21 ip-cache

Syntax

ip-cache

Context

config>app-assure>group>dns-ip-cache

Description

This command enables the context to configure the parameters for the DNS IP cache including its maximum size and high and low watermarks.

10.2.1.7.22 high-watermark

Syntax

high-watermark *percent*

Context

config>app-assure>group>dns-ip-cache>ip-cache

Description

This command configures the high watermark value for the DNS IP cache. When the number of IP addresses stored in the cache crosses above this threshold, the system will generate a trap.

Parameters

percent

specifies the high watermark value in percent

Values: 0 to 100

Default: 90

10.2.1.7.23 low-watermark

Syntax

low-watermark *percent*

Context

```
config>app-assure>group>dns-ip-cache>ip-cache
```

Description

This command configures the low watermark value for the DNS IP cache. If the cache has previously crossed the low watermark value, the system will clear the trap in case the number of IP addresses stored in the cache crosses below the low watermark value.

Parameters

percent

specifies the low watermark value in percent

Values: 0 to 100

Default: 90

10.2.1.7.24 size

Syntax

```
size cache-size
```

Context

```
config>app-assure>group>dns-ip-cache>ip-cache
```

Description

This command configures the maximum number of IP addresses that can be stored in the cache. A value of 0 indicates no IP addresses are stored. In addition, decreasing the size of the cache will cause the cache to be cleared.

Parameters

cache-size

specifies the maximum number of IP addresses that can be stored in the cache

Values: 10 to 64000

Default: 10

10.2.1.7.25 static-address

Syntax

[no] static-address *static-ip-address*

Context

config>app-assure>group>dns-ip-cache>ip-cache

Description

This parameter is used to configure an IP address to be used as a trusted IP address. For packets whose destination IP address matches that of the trusted IP address, a cache hit is assumed.

The no form of this command reverts to the default (disabled or undefined).

Parameters

static-ip-address

specifies the IPv4 or IPv6 address to be used as the trusted IP address

Values	ipv4 - a.b.c.d [mask]
–	ipv6 - x:x:x:x:x:x:x
–	x - [0..FFFF]H
–	d - [0..255]D
–	ipv6-prefix-length: [1 to 128]

10.2.1.7.26 event-log

Syntax

event-log *event-log-name* [create]

no event-log *event-log-name*

Context

config>app-assure>group

Description

This command configures an event log for the AA group.

The no form of this command removes the event log.

Parameters

event-log-name

specifies the name of the event log, up to 32 characters

create

mandatory keyword used when creating a new event log

10.2.1.7.27 buffer-type

Syntax

buffer-type *buffer-type*

Context

config>app-assure>group>event-log

Description

This command specifies the type of buffer to be used in the event log.

Parameters

buffer-type

specifies the type of buffer to be used in the event log

Values:

linear

specifies a linear buffer which, when full, will stop recording events until it is cleared

circular

specifies a circular buffer where older entries will be overwritten by newer entries

syslog

specifies that events are stored offline on a syslog host

Default: linear

10.2.1.7.28 max-entries

Syntax

`max-entries` *max-entries*

Context

`config>app-assure>group>event-log`

Description

This command specifies the maximum number of entries in the event log buffer.

Parameters

max-entries

specifies the permitted entries in the event log

Values: 1 to 100000

Default: 500

10.2.1.7.29 syslog

Syntax

`syslog`

Context

`config>app-assure>group>event-log`

Description

This command enables the context for configuring the target syslog server.

Default

shutdown

10.2.1.7.30 address

Syntax

```
address ip-address  
no address
```

Context

```
config>app-assure>group>event-log>syslog
```

Description

This command configures the target syslog host IP address.

Default

```
no address
```

Parameters

ip-address

specifies the IP address of the target syslog host, either IPv4 or IPv6

Values	ipv4 - a.b.c.d [mask]
–	ipv6 - x:x:x:x:x:x:x
–	x - [0..FFFF]H
–	d - [0..255]D
–	ipv6-prefix-length: [1 to 128]

10.2.1.7.31 facility

Syntax

```
facility syslog-facility
```

Context

```
config>app-assure>group>event-log>syslog
```

Description

This command configures the syslog facility. The syslog facility is an information field associated with a syslog message. It is defined by the syslog protocol and provides an indication of which part of the system originated the message.

Parameters

syslog-facility

specifies the syslog facility keyword

Values: kernel | user | mail | systemd | auth | syslogd | printer | netnews | uucp | cron | authpriv | ftp | ntp | logaudit | logalert | cron2 | local0 | local1 | local2 | local3 | local4 | local5 | local6 | local7

Default: local7

10.2.1.7.32 port

Syntax

```
port port
```

Context

```
config>app-assure>group>event-log>syslog
```

Description

This command specifies the UDP port used by AA to inject the syslog events in-band.

Parameters

port

specifies the UDP port number

Values: 0 to 65535

Default: 514

10.2.1.7.33 severity

Syntax

`severity syslog-severity`

Context

`config>app-assure>group>event-log>syslog`

Description

This command configures the syslog message severity level threshold.

Parameters

syslog-severity

specifies the severity level for the syslog message

Values: emergency | alert | critical | error | warning | notice | info | debug

Default: info

10.2.1.7.34 vlan-id

Syntax

`vlan-id service-port-vlan-id`

`no vlan-id`

Context

`config>app-assure>group>event-log>syslog`

Description

This command configures the service port VLAN ID to be used by AA to inject the syslog events in-band. This VLAN ID must also be configured for the AA interface.

Default

`no vlan-id`

Parameters

port

specifies the service port VLAN identifier

Values: 1 to 4094

10.2.1.7.35 http-anti-fraud-cr-eol

Syntax

```
[no] http-anti-fraud-cr-eol
```

Context

```
config>app-assure>group
```

Description

For a network with a non-conformant HTTP/WAP proxy, this command enables accepting a carriage return (the character "/r") as an end-of-line marker in HTTP request headers for anti-fraud policies for the group partition.



NOTE:

Enabling this command will affect AA performance and must only be enabled if there are non-conformant HTTP/WAP proxies in the network.

The no form of this command reverts to the default (disabled).

Default

```
no http-anti-fraud-cr-eol
```

10.2.1.7.36 http-anti-fraud-policy

Syntax

```
[no] http-anti-fraud-policy
```

Context

```
config>app-assure>group
```

Description

For a network with a non-conformant HTTP/WAP proxy, this command triggers AA to look for situations where the proxy might follow an improperly formatted HTTP request containing multiple sets of HTTP

strings. When a non-conformant situation is identified, AA prevents billing fraud by avoiding the use of all HTTP strings in the HTTP request.

**NOTE:**

Enabling this command will affect AA performance and must only be enabled if there are non-conformant HTTP/WAP proxies in the network.

The no form of this command reverts to the default (disabled).

Default

no http-anti-fraud-policy

10.2.1.7.37 http-enrich

Syntax

```
http-enrich enrich_name [create]  
no http-enrich enrich_name
```

Context

```
config>app-assure>group
```

Description

This command configures an HTTP enrichment policy.

The no form of this command removes the HTTP enrichment policy from the configuration.

Parameters***enrich-name***

specifies the name of the HTTP enrichment policy, up to 32 characters

create

mandatory keyword used when creating an application profile; the create keyword requirement can be enabled/disabled in the environment>create context

10.2.1.7.38 field

Syntax

[no] field *field-name*

Context

config>app-assure>group>http-enrich

Description

This command configures the fields that will be inserted into the HTTP header. The command is repeated for each field to be inserted with a maximum of four fields. The same field cannot be inserted twice into the header under different names.

The no form of this command removes the specified parameter so that it is not inserted into the HTTP header.

Parameters

field-name

specifies the parameters to insert into the header for the field name

Values:

imsi

UE IMSI

imsi-2

subscriber IMSI

imsi-5g

subscriber IMSI in 5G (prefixed with "imsi-", if the user is in 5G)

msisdn

UE MSISDN

msisdn-ts

MSISDN with timestamp

msisdn-without-cc

UE MSISDN without a country code

msisdn-5g

UE MSISDN in 5G (prefixed with "msisdn-", if the user is in 5G)

subscriber-ip

UE IP address

imei

subscriber IMEI

imei-5g

subscriber IMEI in 5G (prefixed with "imei-", if the user is in 5G)

rat-type

Radio Access Technology

charging-id

subscriber charging ID

charging-id-2

subscriber charging ID

pgw-ggsn-address

PGW/GGSN/UPF IP address serving the UE

sgw-sgsn-address

SGW/SGSN IP address serving the UE

sgw-sgsn-address-2

SGW/SGSN IP address serving the UE

msisdn-apn

MSISDN-APN followed by APN

apn

APN used by the UE

apn-ni

APN-NI (Network Identifier) used by the UE

timestamp

timestamp (inserted in Unix time format; for example: 1531204313)

user-location

UE location (ULI)

static-string

configured static string

static-string-2

configured static string

billing-type

UE charging type

roaming-status

subscriber's roaming status

plmn-id

PLMN ID of the SGSN/MME

customer-id

Nokia customer ID

imei-hyphenated

subscriber IMEI with format AABBBBBB-CCCCC-EE

imei-hyphenated-2

subscriber IMEI with format AABBBBBB-CCCCC-EE

user-location-raw

ULI, in raw fmt <uli-type1>[+<uli-type2>]=<ULI hex>

user-location-raw-2

ULI, in raw fmt <uli-type1>[+<uli-type2>]=<ULI hex>

static-acr

static ACR

dynamic-acr

dynamic ACR

user-location-3gpp

specifies the subscriber's location using the format defined in 3GPP29.061, section 16.4.7.2

msisdn-not-enriched

specifies that the software searches for the MSISDN field, but does not perform any enrichment; this value should be used together with anti-spoof, resulting in anti-spoofing the HTTP header, but not inserting the (correct) MSISDN as an HTTP header

sm-app-sid-md5-b64

operator and session ID (MD5 encoding of Session ID)

sm-app-sid-sha256-b64

operator and session ID (sha256 encoding of session id)

smms-tv

vf tv user ID

sess-id-md5-b64

user session ID (MD5 encoding)

sess-id-sha256-b64

user session ID (sha256 encoding)

msisdn-2 - msisdn-8

MSISDN

rat-type-2

Radio Access Technology

rat-type-gtpc

Radio Access Technology (value as defined in 3GPP29.274)

static-acr-2

static ACR

dynamic-acr-2

dynamic ACR

subscriber-ip-2

UE IP address

static-string-3 – static-string-12

configured static string

apn-2 – apn-3

APN used by the UE

pgw-ggsn-address-2, pgw-ggsn-address -3

PGW IP address

customer-id-2

Nokia customer ID

gw-instance-id

ID of the VM/pod/card that send the enriched request

pcc-rulebase

name in the HTTP header for the PCC rulebase name

10.2.1.739 aes-initialization-vector

Syntax

```
aes-initialization-vector hex-string
```

```
no aes-initialization-vector
```

Context

```
config>app-assure>group>http-enrich>field
```

```
config>app-assure>group> http-enrich>tls-extension>subtype
```

Description

This command configures the initialization vector (in HEX) used for the AES CBC encryption.

The no form of this command removes the initialization vector.

Default

no aes-initialization-vector

Parameters

hex-string

specifies the initialization vector in HEX

Values: 0x01234567890abcdef...(32 HEX nibbles)

10.2.1.7.40 anti-spoof

Syntax

[no] anti-spoof

Context

config>app-assure>group>http-enrich>field

Description

This command configures the HTTP header enrichment anti-spoofing functionality.

The no form of this command disables anti-spoofing functionality.

Default

no anti-spoof

10.2.1.7.41 calling-line-id

Syntax

[no] calling-line-id

Context

config>app-assure>group>http-enrich>field

Description

This command configures the HTTP header for “x-up-calling-line-id” anti-spoofing.

**NOTE:**

For the anti-spoofing mechanism to work, the header name has to be configured to “x-up-calling-line-id” using the `config>app-assure>group>http-enrich>field>name` command.

Default

no calling-line-id

10.2.1.7.42 encode**Syntax**

encode type *type* cert-profile *cert-profile-name*

encode type *type* key *key*

encode type *type* key *hash-key* hash

encode type *type* key *hash2-key* hash2

no encode

Context

config>app-assure>group>http-enrich>field

Description

This command defines the encoding to be done for the configured parameter.

The no form of this command removes the encoding.

Default

no encode

Parameters***cert-profile-name***

specifies the name of the certificate profile, up to 32 characters

type

specifies to encode the parameters with certificate values, hash the parameters with MD5, or encrypt the parameters with RC4 using the configured key

Values: md5 | rc4 | certificate | cert-base64 | rc4md5-base64 (fields encrypted with RC4 using MD5 hash key and encoded in base64) | aes128 | aes256 | aes128cbc | aes256cbc |

rc4-base64 | aes128-base64 | aes256-base64 | aes128cbc-base64 | aes256cbc-base64 |
aes256gcm-sha384-base64

key

specifies the key string, 64 characters maximum

hash-key

specifies the first hashed key

hash-key2

specifies the second hashed key

hash

specifies the key is entered in an encrypted form

If the hash or hash2 parameter is not used, the key is assumed to be in an unencrypted, clear text form. For security, all keys are stored in encrypted form in the configuration file with the hash or hash2 parameter specified.

hash2

specifies the key is entered in a more complex encrypted form that involves more variables than the key value alone, meaning that the hash2 encrypted variable cannot be copied and pasted

If the hash or hash2 parameter is not used, the key is assumed to be in an unencrypted, clear text form. For security, all keys are stored in encrypted form in the configuration file with the hash or hash2 parameter specified.

10.2.1.7.43 md5-salt

Syntax

md5-salt *string*

no md5-salt

Context

config>app-assure>group>http-enrich>field

Description

This command appends the configured string prior to performing MD5 hashing of the field.

Default

no md5-salt

Parameters

string

specifies the string which is appended to the parameter prior to being hashed, up to 16 characters

10.2.1.7.44 name

Syntax

name *header-name*

Context

config>app-assure>group>http-enrich>field

Description

This command configures the header name of the field to be inserted.

Parameters

header-name

specifies an operator-defined string up to 32 characters, which is inserted before the actual field name (for example, “x-MSISDN” = MSISDN)

10.2.1.7.45 static-string

Syntax

static-string *string*

no static-string

Context

config>app-assure>group>http-enrich>field

Description

This command configures the value of an operator-defined “static” string that is inserted after “static-string” header-field name as defined above.

10.2.1.7.46 rat-enrichment

Syntax

```
rat-enrichment
```

Context

```
config>app-assure>group>http-enrich
```

Description

This command enables the context to configure RAT enrichment values.

10.2.1.7.47 rat-type

Syntax

```
rat-type rat-type rat-value
```

```
no rat-type rat-type
```

Context

```
config>app-assure>group>http-enrich>rat-enrichment
```

Description

This command specifies the customized RAT values to be used for header enrichment.

The no form of this command uses the default value of the specified RAT type.

Parameters

rat-type

specifies the RAT type

Values: utran | geran | wlan | gan | hspa-evol | eutran | virtual | eutran-nb | ehrpd | hrpd |
cdma-1x | umb | wifi

rat-value

specifies the RAT value, up to 31 characters

10.2.1.7.48 **tls-extension**

Syntax

```
tls-extension
```

Context

```
config>app-assure>group>http-enrich
```

Description

This command enables the context to configure a TLS extension for HTTPS enrichment.

10.2.1.7.49 **extension-id**

Syntax

```
extension-id extension-id
```

```
no extension-id
```

Context

```
config>app-assure>group>http-enrich>tls-extension
```

Description

This command configures the extension ID that is used for customized extension.

The no form of this command removes the extension ID from the TLS extension.

Parameters

extension-id

specifies the TLS extension ID

Values: 0 to 65535

Default: 0

10.2.1.7.50 **subtype**

Syntax

```
[no] subtype tls extension subtype
```

Context

```
config>app-assure>group>http-enrich>tls-extension
```

Description

This command configures the TLS extension name that is used for HTTPS enrichment.

The no form of this command removes the subtype from the TLS extension.

To configure a TLS enrichment action for flows matching an AQP entry, use the `config>app-assure>group>policy>aqp>entry>action>tls-enrich` command.

Parameters

tls extension subtype

specifies the TLS extension subtype, up to 32 characters

Values:

msisdn

specifies the insertion of the MSISDN in the TLS extension

msisdn-5g

specifies the insertion of the subscriber's MSISDN in 5G (prefixed with the string "msisdn-", if the user is in 5G)

msisdn-no-insert

specifies that the software searches for the MSISDN field, but does not perform any enrichment; if the MSISDN sub-extension is found (sub-extension ID = 1), the AA replaces the sub-extension ID with the 255 value

subscriber-ip

specifies the insertion of the UE IP address in the TLS extension

imsi

specifies the insertion of the subscriber's IMSI

imsi-5g

specifies the insertion of the subscriber's IMSI in 5G (prefixed with the string "imsi-", if the user is in 5G)

imei

specifies the insertion of the subscriber's IMEI

imei-5g

specifies the insertion of the subscriber's IMEI in 5G (prefixed with the string "imei-" if the user is in 5G)

user-location

specifies the insertion of the subscriber's location using the format defined in 3GPP29.274, section 8.21

user - location - 3GPP

specifies the insertion of the subscriber's location using the format defined in 3GPP29.061, section 16.4.7.2

sgw - sgsn - address

specifies the insertion of the SGW (or SGSN) IP address serving the UE

pgw - ggsn - address

specifies the insertion of the PGW (GGSN or UPF) IP address serving the UE

timestamp

specifies the insertion of the timestamp (UNIX time) at the time of encryption

rat - type

specifies the insertion of the RAT

apn - ni

specifies the insertion of the APN-NI (Network Identifier)

sgw - sgsn - address - non - 5g

specifies the SGW (or SGSN) IP address serving the UE when the user is not in 5G, otherwise the sub-extension is not present

plmn - id

specifies the PLMN ID of the SGSN/MME

static - string

configured static string, up to 63 characters

10.2.1.7.51 encode

Syntax

`encode type type key | hash-key | hash2-key | custom-key [hash | hash2 | custom]`

`no encode`

Context

`config>app-assure>group>http-enrich>tls-extension>subtype`

Description

This command defines the encoding performed for the extension.

The no form of this command removes the encoding.

Default

`no encode`

Parameters

type

specifies whether the parameters are hashed with MD5, encrypted with RC4 or AES using the configured key

Values: md5 | rc4 | aes256cbc

key | hash-key | hash2-key | custom-key

specifies the key used for the encoding

hash | hash2 | custom

specifies the hashing scheme used in the hashed key

10.2.1.7.52 extension-id

Syntax

extension-id *tls-extension-id*

no extension-id

Context

config>app-assure>group>http-enrich>tls-extension>subtype

Description

This command configures the extension ID that is used for customized extension for the subtype.

The no form of this command removes the extension ID from the TLS extension.

Parameters

extension-id

specifies the TLS extension ID

Values: 1 to 65535

Default: 0

10.2.1.7.53 static-string

Syntax

static-string *string*

no static-string

Context

```
config>app-assure>group>http-enrich>tls-extension>subtype
```

Description

This command configures an HTTP header enrichment TLS extension for the subtype static string.

The no form of this command removes the static string.

Parameters

string

configures a string, up to 16 characters

10.2.1.7.54 http-error-redirect

Syntax

```
http-error-redirect redirect-name [create]
```

```
no http-error-redirect redirect-name
```

Context

```
config>app-assure>group
```

Description

This command configures an HTTP error redirect policy. The policy contains important information relevant to the redirect server.

The no form of this command removes the redirect name from the group configuration.

Parameters

redirect-name

a string of up to 32 characters that identifies the HTTP error redirect policy

create

mandatory keyword used when creating the HTTP error redirect policy

10.2.1.755 error-code

Syntax

```
error-code error-code [custom-msg-size custom-msg-size]  
no error-code error-code
```

Context

```
config>app-assure>group>http-error-redirect
```

Description

This command refers to the HTTP status codes to which a redirect action is applied. Only messages with sizes less than that configured here (*custom-msg-size*) are eligible for redirect action.

The no form of this command removes the parameters from the configuration.

Parameters

error-code

specifies the error code for a HTTP error redirect

Values: 0 to 4294967295, of which 400, 401, 402, 403, 404, 405, 406, 407, 408, 409, 410, 411, 412, 413, 414, 415, 416, 417, 421, 422, 423, 424, 425, 426, 427, 428, 429, 430, 431, 451, 500, 501, 502, 503, 504, 505, 506, 507, 508, 509, 510, 511, 730, 731, and 735 are supported for redirect

custom-msg-size

specifies the maximum message size above which redirect will not be done

Values: 0 to 4294967295

10.2.1.756 http-host

Syntax

```
http-host http-host  
no http-host
```

Context

```
config>app-assure>group>http-error-redirect
```

Description

This is a string that refers to the HTTP host name of the landing server. It is used in the HTTP GET operation from the client (which is being redirected) to the redirect search landing server. It must contain a valid IP address or HTTP host name / URI for the HTTP GET from the client to the landing server to work.

The no form of this command removes the HTTP host string from the configuration.

Parameters

http-host

specifies a string of up to 255 characters, that refers to the HTTP host name of the landing server

10.2.1.7.57 participant-id

Syntax

participant-id *participant-id*

no participant-id

Context

config>app-assure>group>http-error-redirect

Description

This command specifies a 32-character string assigned to the operator by Barefruit. It is used by Barefruit landing servers (applies to template # 1 only).

Default

no participant-id

Parameters

participant-id

32-character string supplied by the Barefruit

10.2.1.7.58 template

Syntax

template *template-id*

no template

Context

config>app-assure>group>http-error-redirect

Description

The redirect template refers to the template of parameters passed from the AA-ISA to the redirect server via JavaScript in the redirect packet. The template is specific to the redirect server being used in the network. Currently, two partners are used and tested with AA-ISA redirect solution, Barefruit and Xerocole. The no form of this command reverts to the default.

Default

no template

Parameters

template-id
specifies an HTTP error redirect template

Values	1	Barefruit specific template
–	2	xerocole.specific template

10.2.1.7.59 http-match-all-requests

Syntax

[no] http-match-all-requests

Context

config>app-assure>group

Description

This command enables constant monitoring performed for HTTP request strings for application changes in regular HTTP flows. The no form of this command disables the monitoring.

10.2.1.7.60 http-notification

Syntax

```
http-notification http-notification-name [create]  
no http-notification http-notification-name
```

Context

```
config>app-assure>group
```

Description

This command configures an HTTP notification object for a subscriber in browser notification.
The no form of this command removes the HTTP notification policy from the configuration.

Parameters

http-notification-name

specifies the name of the HTTP notification policy

create

mandatory keyword used when creating the policy

10.2.1.7.61 interval

Syntax

```
interval {one-time | minimum-interval}
```

Context

```
config>app-assure>group>http-notif
```

Description

This command configures the minimum interval between notification messages. It can be set to one-time or a value in minutes from 1 to 1440.

The no form of this command removes the interval from the HTTP notification policy.

Parameters

one-time

keyword used to configure a one-time notification message

minimum-interval

represents the minimum interval value in minutes between two HTTP notifications

Values: 1 to 1440

10.2.1.7.62 template

Syntax

template *template-id*
no template

Context

config>app-assure>group>http-notif

Description

This command configures the template which defines the format and parameters included in the HTTP notification message.

The no form of this command removes the template from the configuration.

Parameters

template-id

specifies the template ID for this HTTP notification template

Values	1	appends both the subscriber ID and the HTTP URL parameter VSA as parameters of the redirect URL
–	2	appends only the HTTP URL parameter VSA as a parameter of the redirect-URL

10.2.1.7.63 http-proxy

Syntax

http-proxy *http-proxy-name* mode *mode* create

```
http-proxy http-proxy-name create  
no http-proxy http-proxy-name
```

Context

```
config>app-assure>group
```

Description

This command configures an explicit HTTP or HTTPS proxy.

Parameters

http-proxy-name

specifies the name of the HTTP proxy, up to 32 characters

mode

specifies either an HTTP or HTTPS proxy

Values: HTTP | HTTPS

create

mandatory keyword used when creating an HTTP or HTTPS proxy

10.2.1.7.64 description

Syntax

```
description description-string  
no description
```

Context

```
config>app-assure>group>http-proxy
```

Description

This command provides a description for the configured HTTP proxy.

Parameters

description-string

specifies the description of the HTTP proxy, up to 80 characters

10.2.1.7.65 dns-server

Syntax

```
dns-server ip-address  
no dns-server
```

Context

```
config>app-assure>group>http-proxy
```

Description

This command configures a DNS server for the HTTP proxy. The DNS server resolves the target hostname.

Parameters

ip-address

specifies the IP address of the DNS server

Values: a.b.c.d (IPv4-address)

10.2.1.7.66 https

Syntax

```
https
```

Context

```
config>app-assure>group>http-proxy
```

Description

This command enables the context for the HTTPS proxy.

10.2.1.7.67 intermediate-certificate-1

Syntax

```
intermediate-certificate-1 cert-profile  
no intermediate-certificate-1
```


Context

```
config>app-assure>group>http-proxy>https
```

Description

This command configures the intermediate certificate profile that the HTTPS proxy uses to validate the network-side connections.

The no form of this command removes the intermediate certificate profile.

Parameters

cert-profile

specifies the intermediate certificate profile, up to 32 characters

10.2.1.7.68 intermediate-certificate-2

Syntax

```
intermediate-certificate-2 cert-profile  
no intermediate-certificate-2
```

Context

```
config>app-assure>group>http-proxy>https
```

Description

This command configures the intermediate certificate profile that the HTTPS proxy uses to validate the network-side connections.

The no form of this command removes the intermediate certificate profile.

Parameters

cert-profile

specifies the intermediate certificate profile, up to 32 characters

10.2.1.7.69 root-certificate

Syntax

```
root-certificate cert-profile  
no root-certificate
```

Context

```
config>app-assure>group>http-proxy>https
```

Description

This command configures the root certificate profile that the HTTPS proxy uses to validate the network-side connections.

The no form of this command removes the root certificate profile.

Parameters

cert-profile

specifies the root certificate profile, up to 32 characters

10.2.1.7.70 *tls-profile*

Syntax

```
tls-profile cert-profile
```

```
no tls-profile
```

Context

```
config>app-assure>group>http-proxy>https
```

Description

This command configures the TLS profile to be used for client-side connections; AA acts as a web server.

The no form of this command removes the TLS profile .

Parameters

cert-profile

specifies the TLS profile, up to 32 characters

10.2.1.7.71 *url-mapping*

Syntax

```
url-mapping
```

Context

```
config>app-assure>group>http-proxy>https
```

Description

This command enables the context to configure the match conditions and the actions for the HTTPS proxy.

10.2.1.7.72 entry

Syntax

```
entry entry-id [create]
```

```
no entry entry-id
```

Context

```
config>app-assure>group>http-proxy>https>url-mapping
```

Description

This command configures a url-mapping entry. The configured entries are an ordered list where the lowest numerical entry has the highest priority.

The no form of this command deletes the specific url-mapping entry.

Parameters

entry-id

an integer that identifies a url-mapping entry

Values: 1 to 65535

create

keyword to create the url-mapping entry

10.2.1.7.73 action

Syntax

```
action
```

Context

```
config>app-assure>group>http-proxy>https>url-mapping>entry
```

Description

This command enables the context to configure a url-mapping action for the flows that match the criteria of the url-mapping entry.

10.2.1.7.74 http-enrich

Syntax

```
http-enrich name  
no http-enrich
```

Context

```
config>app-assure>group>http-proxy>https>url-mapping>entry>action
```

Description

This command configures an HTTP enrichment action for the flows that match this entry.

The no form of this command removes the HTTP enrichment action from the actions on the flows that match this url-mapping entry.

Default

```
no http-enrich
```

Parameters

name

specifies the name of the existing HTTP enrichment action for this AA profile, up to 32 characters

10.2.1.7.75 target-server

Syntax

```
target-server fqdn  
no target-server
```

Context

```
config>app-assure>group>http-proxy>https>url-mapping>entry>action
```

Description

This command configures the hostname of the destination server (to which AA sends requests).

The no form of this command removes the FQDN action from the actions on the flows that match this url-mapping entry.

Default

no target-server

Parameters

fqdn

specifies the hostname of the destination server (not including a scheme), up to 255 characters

10.2.1.7.76 uri-transform

Syntax

uri-transform *transform-type*

no uri-transform

Context

config>app-assure>group>http-proxy>https>url-mapping>entry>action

Description

This command configures the uri-transform action to perform on the URI toward the destination server.

The no form of this command removes the uri-transform action from the actions on the flows that match this url-mapping entry.

Default

no uri-transform

Parameters

transform-type

specifies the transform-type of the destination server

Values	none	performs no transformation; the original URI is used for the request to the destination server
–	remove-all	removes the complete URI; the request to the destination server has no URI
–	remove-query	removes the query string — all characters starting from the '?' character (including the '?' character)

10.2.1.777 match

Syntax

match

Context

config>app-assure>group>http-proxy>https>url-mapping>entry

Description

This command enables the context to configure the url-mapping condition(s) for the flows that match this url-mapping entry.

10.2.1.778 expression

Syntax

expression *expr-index* type *type* operator *expression-str*

no expression *expr-index*

Context

config>app-assure>group>http-proxy>https>url-mapping>entry>match

Description

This command configures the match condition for the url-mapping entry.

The no form of this command removes the match condition from the url-mapping entry.

Default

no expression

Parameters

expr-index

specifies an index value that represents the expression substrings
Values: 1 to 3

type

specifies the expression type (including the expression substring)

Values	http-host	matches the string against the HTTP host field
–	http-uri	matches the string against the HTTP URI field
–	http-query	specifies the string against the HTTP URI query field

operator

specifies the operator of the expression
Values: eq | neq

expression-str

specifies the expression string, up to 255 characters

10.2.1.7.79 shutdown

Syntax

[no] shutdown

Context

config>app-assure>group>http-proxy>https>url-mapping>entry

Description

This command enables or disables this url-mapping entry. The command must be enabled to modify the url-mapping entry. Also, the command must be disabled to enable the url-mapping entry.

10.2.1.7.80 **vlan-id**

Syntax

`vlan-id vlan-id`

`no vlan-id`

Context

`config>app-assure>group>http-proxy`

Description

This command configures the VLAN ID on which the AA ISA emits the traffic mapping to a preconfigured AA interface.

Parameters

vlan-id

specifies the VLAN ID to connect to the web service

Values: 1 to 4094

10.2.1.7.81 **script-url**

Syntax

`script-url script-url-name [create]`

`no script-url`

Context

`config>app-assure>group>http-notif`

Description

This command configures the URL of the script used by the HTTP notification policy.

The no form of this command removes the script URL from the HTTP notification policy.

Parameters

script-url-name

specifies the string, up to 255 characters long, representing the URL of the script used in the HTTP notification policy

create

mandatory keyword used when creating the script URL

10.2.1.7.82 template

Syntax

template *value*

no template

Context

config>app-assure>group>http-notif

Description

This command configures the template which defines the format and parameters included in the HTTP notification message.

The no form of this command removes the template from the configuration.

Parameters

value

specifies the template values

Values	1	automatically appends the aa-sub-name as a parameter of the URL along with the “Alc-AA-Sub-Http-url-Param” VSA, if present
–	2	only appends the “Alc-AA-Sub-Http-Url-Param” VSA as a parameter of the URL, if present

10.2.1.7.83 http-redirect

Syntax

http-redirect *redirect-name* [create]

```
no http-redirect redirect-name
```

Context

```
config>app-assure>group
```

Description

This command creates or modifies an HTTP redirect policy.

The no form of this command removes the HTTP redirect policy from the configuration.

Parameters

redirect-name

specifies the name of the HTTP redirect policy, up to 32 characters

create

mandatory keyword used when creating the HTTP redirect policy. The create keyword requirement can be enabled/disabled in the environment>create context.

10.2.1.7.84 redirect-https

Syntax

```
[no] redirect-https
```

Context

```
config>app-assure>group>http-redirect
```

Description

This command configures the HTTP redirect policy to redirect HTTPS sessions to the configured redirect URL.

The no form of this command removes the redirect HTTPS.

10.2.1.7.85 redirect-url

Syntax

```
redirect-url redirect-url
```

```
no redirect-url
```

Context

config>app-assure>group>http-redirect

Description

This command configures the HTTP redirect URL which is the URL (page) that the user is redirected to when an HTTP redirect takes effect.

The operator can select the URL arguments to include in the redirect URL by using either a specific template ID or by configuring the redirect URL using one of the supported macro substitution keywords.

The no form of this command removes the redirect-url field from the configuration.

Parameters

redirect-url

specifies the URL of the landing page
macro substitutions:

Values	\$CATID	the category ID
–	\$CATNAME	the category name of the URL
–	\$IP	a string that represents the IP address of the subscriber host
–	\$RTRID	a string that represents the router ID
–	\$SUB	a string that represents the subscriber ID
–	\$URL	the Request-URI in the HTTP GET Request received
–	\$URLPRM	the HTTP URL parameter associated with the subscriber

10.2.1.7.86 tcp-client-reset

Syntax

[no] tcp-client-reset

Context

config>app-assure>group>http-redirect

Description

This command enables a HTTP-redirect policy to initiate a TCP reset toward the client if the AA policy results in a redirect with packet drop but the HTTP redirect cannot be delivered. Scenarios for this include HTTPs (TLS) sessions, blocking of non-HTTP TCP traffic, and blocking of existing flows after a policy re-evaluate of an existing subscriber.

The no form of this command disables the command.

10.2.1.7.87 template

Syntax

template *template-id*

no template

Context

config>app-assure>group>http-redirect

Description

This command configures the template that defines which parameters are appended to the HTTP host redirect field in the redirect message.

The HTTP redirect template provides HTTP 302 redirect containing only the URL specified in the redirect policy, with no other parameters.

The no form of this command removes the template from the configuration.

Parameters

template-id

specifies the HTTP Policy Redirect template

Values	1	Javascript based redirect embedded in HTTP 200 OK response with a predefined number of arguments automatically appended to the redirect URL
–	2	HTTP 302 Redirect with a predefined number of arguments automatically appended to the redirect URL
–	3	HTTP 302 Redirect with no parameters appended to the URL (empty)
–	4	empty Redirect format using Javascript
–	5	redirect supporting macro substitution using HTTP 302
–	6	redirect supporting macro substitution using Javascript

10.2.1.7.88 http-x-online-host

Syntax

[no] http-x-online-host

Context

config>app-assure>group

Description

This command specifies whether the X-Online-Host header field is used as a replacement for the HTTP Host header field.

The no form of this command disables the use of X-Online-Host header field used as a replacement.

10.2.1.7.89 ip-identification-assist

Syntax

```
ip-identification-assist
```

Context

```
config>app-assure>group
```

Description

This command enables the context to configure the parameters for the IP addresses that are used for traffic identification. By default, the system uses the IP addresses identification assist. The IP identification assist is an optional mechanism that is used by default to look through an internally generated and stored application IP database if the app-filters fail to classify the traffic into one of the configured applications from the application database (other than unknown TCP/UDP/TLS). To disable the IP identification assist, the config>app-assure>group>ip-identification-assist>shutdown command must be used.

10.2.1.7.90 passive-dns

Syntax

```
passive-dns
```

Context

```
config>app-assure>group>ip-identification-assist
```

Description

This command enables the context to configure the passive DNS that is related to the IP identification assist.

10.2.1.7.91 monitor

Syntax

```
monitor
```

Context

```
config>app-assure>group>ip-identification-assist>passive-dns
```

Description

This command enables the system to collect IP addresses by passively monitoring the DNS traffic to build its internal application IP database. This command is enabled by default.

10.2.1.792 trusted-server-address

Syntax

```
trusted-server-address address [create]  
no trusted-server-address address
```

Context

```
config>app-assure>group>ip-identification-assist>passive-dns
```

Description

This command configures a DNS server address. The responses from this DNS server are used by the IP identification assist mechanism. Up to 64 server addresses can be configured.

Parameters

address

specifies the IPv4 or IPv6 address of the DNS server, up to 64 characters

create

keyword used when creating a new IP address

10.2.1.793 ip-identification-contribute

Syntax

```
[no] ip-identification-contribute
```

Context

```
config>app-assure>group
```

Description

This command enables the system to collect information from traffic in this partition to contribute to the group level of the IP identification assist function. The command is enabled by default.

10.2.1.7.94 ip-prefix-list

Syntax

```
ip-prefix-list ip-prefix-list-name [create]
no ip-prefix-list
```

Context

```
config>app-assure>group
```

Description

This command creates or enables the context to modify an IP prefix list for the application assurance group. The list displays IP prefix list information.

The no form of this command removes the list from the configuration.

Parameters

ip-prefix-list-name

specifies the name of the IP prefix list, up to 32 characters

create

mandatory keyword used when creating a new IP prefix list

The create keyword requirement can be enabled/disabled in the environment>create context.

10.2.1.7.95 prefix

Syntax

```
prefix ip-prefix/ip-prefix-length [name prefix-name]
no prefix ip-prefix/ip-prefix-length
```

Context

```
config>app-assure>group>ip-prefix-list
```

Description

This command specifies an IP Prefix within the IP prefix list.

The no form of this command removes the IP prefix from the list.

Parameters

ip-prefix/ip-prefix-length

specifies the IP prefix and IP prefix length for the entry

Values	<i>ip-prefix/ip-pref*</i> <i>ipv4-prefix/ipv4-prefix-length</i> <i>ipv6-prefix/ipv6-prefix-length</i>	
–	<i>ipv4-prefix</i>	a.b.c.d (host bits must be 0)
–	<i>ipv4-prefix-length</i>	[0..32]
–	–	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D
–	<i>ipv6-prefix-length</i>	[0..128]

name prefix-name

specifies the name of an IP prefix, up to 32 characters

10.2.1.7.96 mp-tcp-proxy

Syntax

```
mp-tcp-proxy mp-tcp-proxy-name [create]
```

```
no mp-tcp-proxy mp-tcp-proxy-name
```

Context

```
config>app-assure>group
```

Description

This command configures the MP-TCP proxy policy.

The no form of this command removes the specified MP-TCP proxy policy.

When an MP-TCP proxy policy is removed or deleted, the existing flows using this policy will be abandoned, and the MP-TCP proxy will be stopped. If, however, the MP-TCP proxy action is removed from an AQP entry (in the context `config>app-assure>group>aqp>entry>action`), the existing flows will not be affected.

Parameters

mp-tcp-proxy-name

specifies the name of the MP-TCP proxy policy, up to 32 characters

create

specifies to create the MP-TCP proxy policy; this keyword is mandatory when creating an MP-TCP proxy policy

10.2.1.7.97 bonding-assist

Syntax

[no] bonding-assist

Context

`config>app-assure>group>mp-tcp-proxy`

Description

This command enables application assurance to extract an auto-learned secondary path IP address for an MP-TCP subscriber from the MPTCP Add Addr option (defined in RFC 6824). This is used with MG subscriber auto-create in both Radius-less HAG subscriber auto-create model, and in the DSL SPR Radius subscriber create model.

The no form of this command disables auto-learning of the secondary path IP address.

Default

no bonding-assist

10.2.1.7.98 congestion-control

Syntax

`congestion-control` *congestion-algorithm*

`no congestion-control`

Context

config>app-assure>group>mp-tcp-proxy

Description

This command configures the congestion control algorithm for MP TCP.

The no form of this command reverts to the default.

Parameters

congestion-algorithm

specifies the congestion control algorithm for MP TCP

Values:

olia

specifies the opportunistic linked increases algorithm (OLIA), which is the default congestion algorithm for MP-TCP traffic and is an adaptation of TCP New Reno to support multiple paths

olia-plus

specifies the OLIA-plus congestion control algorithm for the MP-TCP proxy, which exhibits reduced transmission burstiness versus the default OLIA algorithm

bbr

specifies the Bottleneck Bandwidth and Round-trip propagation time (BBR) congestion control algorithm for the MP-TCP proxy

10.2.1.7.99 default-scheduler-policy

Syntax

default-scheduler-policy *scheduler-policy*

no default-scheduler-policy

Context

config>app-assure>group>mp-tcp-proxy

Description

This command configures the MP-TCP proxy default scheduling policy for downstream path selection.

The no form of this command removes the specified configuration.

Parameters

scheduler-policy

specifies the scheduler policy

Values:

min-rtt-prefer

If TCP subflow congestion windows (CWINs) have space available on both paths or subflows, the path or subflow with a lower RTT value will be preferred for traffic transmission. Otherwise, the path or subflow that has CWIN space available (assuming there is room in the TCP receiver window) will be used for transmission. In the case of bulk data transmission, if the application is able to fill the congestion windows of all subflows, the packets will be scheduled as soon as space is available in each subflow's congestion window ("ack-clock"). The RTT calculation is based on standard TCP algorithms

min-cost-prefer

If TCP subflow congestion windows (CWINs) have space available on both paths or subflows, the path or subflow with the lower administrative cost will be preferred. Otherwise, the path or subflow that has CWIN space available (assuming there is room in the TCP receiver window) will be used for transmission. In case of bulk data transmission, if the application is able to fill the congestion windows of all subflows then packets will be scheduled as soon as space is available in each subflow's congestion window ("ack-clock").

min-cost-handover

The available path or subflow with the lower administrative cost is selected for transmission; transmission is done when a lower cost TCP subflow congestion window (CWIN) has space available (assuming there is room in the TCP receiver window). This provides path handover for protection, based on cost and path availability.

min-cost-forbid

Transmission on the path with the higher administrative cost is prevented by preventing subflow establishment on that path. The HAG transmits data to the low cost path only and prevents the CPE from transmitting on the maximum cost path.

min-cost-blest

if the round-trip time performance of the higher cost path is considerably worse than the lower cost path, a blocking estimation calculation is performed to check if the transmission attempts on the higher cost path should be paused to avoid a head of line blocking effect that would reduce performance of the lower cost path

Default: min-cost-prefer

10.2.1.7.100 force-dss-checksums

Syntax

[no] force-dss-checksums

Context

config>app-assure>group>mp-tcp-proxy

Description

This command specifies the force DSS checksums of the MP-TCP proxy.

The no form of this command disables force DSS checksums.

Default

no force-dss-checksums

10.2.1.7.101 ip-address-profile

Syntax

ip-address-profile *profile-name*

no ip-address-profile

Context

config>app-assure>group>mp-tcp-proxy

Description

This command configures the IP address profile to be associated with the MP-TCP proxy.

The no form of this command removes the IP address profile association with the MP-TCP proxy.

Parameters

profile-name

specifies the name of the IP address profile, up to 32 characters

10.2.1.7.102 path

Syntax

path

Context

config>app-assure>group>mp-tcp-proxy

Description

This command enables the context to configure the path associated with the MP-TCP proxy.

The no form of this command disables the path association with the MP-TCP proxy.

10.2.1.7.103 fixed

Syntax

fixed *admin-cost*

no fixed

Context

config>app-assure>group>mp-tcp-proxy>path

Description

This command configures the fixed path attributes associated with the MP-TCP proxy.

The no form of this command removes the fixed path attributes association with the MP-TCP proxy.

Parameters

admin-cost

specifies the administrative cost

Values: 1 to 10

Default: 5

10.2.1.7.104 mobile

Syntax

```
mobile admin-cost  
no mobile
```

Context

```
config>app-assure>group>mp-tcp-proxy>path
```

Description

This command configures the mobile path attributes associated with the MP-TCP proxy.

The no form of this command removes the mobile path attributes association with the MP-TCP proxy.

Parameters

admin-cost

specifies the administrative cost

Values: 1 to 10

Default: 5

10.2.1.7.105 subflow-failure-indication

Syntax

```
[no] subflow-failure-indication
```

Context

```
config>app-assure>group>mp-tcp-proxy
```

Description

This command enables additional resiliency when a subflow of a MPTCP session has failed. Upon failure and removal of a subflow, the HAG sends a MPTCP ADD_ADDR on the working subflow to signal that the HCPE should attempt to re-establish the failed subflow.

The no form of this command disables the auto-learning of the secondary path IP address.

Default

```
no subflow-failure-indication
```

10.2.1.7.106 multi-device-detection

Syntax

```
multi-device-detection
```

Context

```
config>app-assure>group
```

Description

This command enables the context for multi device detection.

10.2.1.7.107 max-devices

Syntax

```
max-devices count
```

```
no max-devices
```

Context

```
config>app-assure>group>multi-device-detection
```

Description

This command configures the maximum number of devices that are tracked when the user has activated tethering. The value includes the host device (the device that has activated tethering).

Parameters

count

specifies the number of devices tracked in device tethering detection

Values: 1 to 16

10.2.1.7.108 port-list

Syntax

```
port-list port-list-name [create]
```

```
no port-list port-list-name
```


Context

config>app-assure>group

Description

This command defines an AA group or partition named port-list, which contains a list of port numbers or port ranges. The port list is then referenced in AA policy app-filters, allowing increased flexibility in the use of server ports for application definition.

The no form of this command removes the list.

Default

no port-list

Parameters

port-list-name

specifies the name of the port list

10.2.1.7.109 port

Syntax

[no] port *port-number*

[no] port range *start-port-num end-port-num*

Context

config>app-assure>group>port-list

Description

This command specifies the server TCP or UDP port number to use in the port list definition.

The no form of this command restores the default by removing the port number from the port list.

Default

no port

Parameters

port-number

specifies the port number

Values: 0 to 65535

start-port-number

specifies the start port number

Values: 0 to 65535

end-port-number

specifies the start port number

Values: 0 to 65535

10.2.1.7.110 shallow-inspection

Syntax

[no] shallow-inspection

Context

config>app-assure>group

Description

This command enables a shallow-flow inspection. When this command is enabled, the AA Layer 7 classification will be disabled to increase throughput performance for deployments that can operate using only AA Layer 3 and Layer 4 shallow-flow inspections. This configuration may be used with TCP optimization, DEM, Layer 3 and Layer 4 application-filter classification, MP-TCP HAG, TCP MSS adjust, and session-filter (without ALG).

The no form of this command reverts to the default.

Default

no shallow-inspection

10.2.1.7.111 tcp-optimizer

Syntax

tcp-optimizer *tcp-optimizer-name* [create]

no tcp-optimizer *tcp-optimizer-name*

Context

config>app-assure>group

Description

This command configures the TCP optimizer policy.

The no form of this command removes the specified TCP optimizer policy.

When a TCP optimizer policy is removed/deleted, the existing flows using this policy will be abandoned, and optimization is stopped. If however, the TCP optimizer action is removed from a session-filter entry (in the context `config>app-assure>group>sess-fltr>entry>action`), the existing flows are not affected.

Parameters

tcp-optimizer-name

specifies the name of the TCP optimizer policy, up to 32 characters

create

this keyword is mandatory when creating a TCP optimizer policy

10.2.1.7.112 dack-timeout

Syntax

`dack-timeout dack-timeout`

`no dack-timeout`

Context

`config>app-assure>group>tcp-optimizer`

Description

This command configures a delayed acknowledgment timeout for the TCP optimizer.

By entering this command a default of 200 msec timeout is enabled for delayed acknowledgment. This value is not configurable.

The no form of this command disables the delayed acknowledgment timeout.

Default

`no dack-timeout`

Parameters

dack-timeout

specifies the delayed acknowledgment timeout in msec

Default: 200

10.2.1.7.113 high-cpu-backoff

Syntax

```
[no] high-cpu-backoff
```

Context

```
config>app-assure>group>tcp-optimizer
```

Description

This command enables AA to stop optimizing sessions when the CPU capacity is limited.

AA implements three non-configurable CPU utilization cut-off thresholds, that is, level 1 (80%), level 2 (90%), and level 3 (95%). These levels are the following:

Early random bypass stage	When the CPU utilization is between cut-off thresholds level 1 and level 2, AA performs an early random bypass of new sessions proportional to the position of the CPU utilization within the range between level 1 and level 2.
Stop optimization of new flows stage	When the CPU utilization is at or above level 2, AA stops optimizing new sessions.
Abandon stage	When the CPU utilization is at or above level 3, AA abandons the existing optimized sessions.

**NOTE:**

The CPU utilization is measured once per second per core/worker (the CPU utilization may differ between cores).

Default

```
no high-cpu-backoff
```

10.2.1.7.114 init-cwnd-size

Syntax

```
init-cwnd-size init-cwnd-size
```

Context

config>app-assure>group>tcp-optimizer

Description

This command configures the initial TCP congestion window (cwin) used during TCP slow start period.

Parameters

init-cwnd-size

specifies the initial TCP congestion window size value in mss (max segment size)

Values: 1 to 256

Default: 10

10.2.1.7.115 *init-ss-threshold*

Syntax

init-ss-threshold *init-ss-threshold*

Context

config>app-assure>group>tcp-optimizer

Description

This command configures the initial Slow Start (SS) threshold for a given TCP optimizer policy. It is recommended to set the threshold close to the access network bandwidth delay product.

Parameters

init-ss-threshold

specifies the initial SS threshold value in kilobytes

Values: auto | 0 to 1000000

The auto value instructs the system to use an intelligent mechanism to set the optimum value for the initial SS threshold to dynamically track the bandwidth delay product of the access network.

Default: 1000000

10.2.1.7.116 network-rtt-threshold

Syntax

```
network-rtt-threshold network-rtt-threshold  
no network-rtt-threshold
```

Context

```
config>app-assure>group>tcp-optimizer
```

Description

This command configures the threshold of the round trip delay of the network side (between AA and the content provider) above which TCP optimization is performed. This enables to operator to disable optimization for content that is served from a location close to the optimizer.

Default

```
no network-rtt-threshold
```

Parameters

network-rtt-threshold

specifies the network side round trip delay threshold, in milliseconds

Values: 1 to 100

10.2.1.7.117 tcp-stack

Syntax

```
tcp-stack tcp-stack-type
```

Context

```
config>app-assure>group>tcp-optimizer
```

Description

This command configures the TCP stack used toward the subscriber.



NOTE:

- The TCP stack used toward the core network is new-reno, and it is not configurable.

- TCP BBR, TCP Illinois, and TCP Westwood implement a sender-side modification of the TCP congestion window algorithm that improves upon the performance of TCP Reno in wireless networks with lossy links.

Parameters

tcp-stack-type

specifies the TCP stack used toward the subscriber

Values: tcp-bbr | tcp-bbr-v2 | tcp-illinois | tcp-new-reno | tcp-westwood

Default: tcp-westwood

10.2.1.7.118 tcp-validate

Syntax

```
tcp-validate tcp-validate-name [create]
```

```
no tcp-validate tcp-validate-name
```

Context

```
config>app-assure>group
```

Description

This command configures a TCP validation policy.

The no form of this command removes the specified TCP validation policy.

Parameters

tcp-validate-name

specifies the name of the TCP validation policy, up to 32 characters

create

This keyword is mandatory when creating a TCP validation policy.

10.2.1.7.119 event-log

Syntax

```
event-log event-log-name [all]
```

```
no event-log
```

Context

config>app-assure>group>tcp-validate

Description

This command enables logging of traffic dropped by TCP validation.

The no form of this command disables logging of traffic dropped by TCP validation.

Default

no event-log

Parameters

event-log-name

specifies the name of the event log, up to 32 characters

all

logs all dropped traffic. Using the *all* option allows the operator to capture all discards made by the TCP validation policy, including those related to:

- packets that were received after an RST and discarded
- packets received before TCP session establishment (before SYN) and discarded

Without the *all* option, discards related to these cases are not captured in any event log.

10.2.1.7.120 strict

Syntax

[no] strict

Context

config>app-assure>group>tcp-validate

Description

This command specifies whether enforcement of TCP sequence and acknowledgment numbers is applied. If a packet does not meet the expected sequence or acknowledgment number, it is dropped.

This command must only be enabled if the expected bit error rate or packet loss is low. For example, if acknowledgments are lost before being detected by AA, the server timeouts are triggered and retransmissions occur. If *strict* is enabled, these retransmissions would resemble a reply attack and would be dropped by AA.

The no form of this command removes TCP sequence and acknowledgment number enforcement.

Default

no strict

10.2.1.7.121 tethering-detection

Syntax

```
tethering-detection
```

Context

```
config>app-assure>group
```

Description

This command enables the context to configure tethering detection for the group. The shutdown and no shutdown commands are used in this context to enable or disable tethering detection.

Default

no tethering detection

10.2.1.7.122 single-device

Syntax

```
single-device
```

Context

```
config>app-assure>group>tethering-detection
```

Description

This command enables the context to configure expected TTL values for flow-level tethering detection. A maximum of 16 expected TTL values can be configured per group partition.

The TTL values are used for flow-level tethering detection. Flow-level tethering detection checks whether flows have a TTL value that matches one of the configured TTL values. If there is a match, the flows are considered to be non-tethered flows. All other flows are considered to be tethered flows. Flow-level tethering detection is performed for HTTP and TLS traffic.

10.2.1.7.123 expected-ttl

Syntax

[no] expected-ttl *ttl-value*

Context

config>app-assure>group>tethering-detection>single-device

Description

This command configures the expected TTL values to be used for flow-based tethering detection. A maximum of 16 expected TTLs can be configured per group partition.

The no form of this command disables the TTL value for flow-based tethering detection.

Default

no expected-ttl

Parameters

ttl-value

specifies a TTL value for a non-tethered flow

Values: 1 to 255

10.2.1.7.124 invert-match

Syntax

[no] invert-match

Context

config>app-assure>group>tethering-detection>single-device

Description

This command defines if the values configured in the expected-ttl table , are the ones expected to appear either on the host or on the connected device(s) .

Default

no invert-match

10.2.1.7.125 ttl-monitoring

Syntax

```
ttl-monitoring
```

Context

```
config>app-assure>group>tethering-detection
```

Description

This command enables the context to configure the TCP and UDP protocols.

10.2.1.7.126 tcp-protocols

Syntax

```
tcp-protocols protocol-set
```

Context

```
config>app-assure>group>tethering-detection>ttl-monitoring
```

Description

This command configures whether AA will analyze all TCP traffic or only the traffic from the standard applications that generate consistent TTL values.

Parameters

protocol-set

specifies which traffic to analyze

Values:

standard

specifies that AA must analyze only the traffic from the supported applications

all

specifies that AA must analyze all TCP traffic

10.2.1.7.127 udp-protocols

Syntax

`udp-protocols protocol-set`

Context

`config>app-assure>group>tethering-detection>tll-monitoring`

Description

This command configures whether AA will analyze all UDP traffic or only the traffic from the standard applications that generate consistent TTL values.

Parameters

protocol-set

specifies which traffic to analyze

Values:

standard

specifies that AA must analyze only the traffic from the supported applications

all

specifies that AA must analyze all UDP traffic

10.2.1.7.128 wap1x

Syntax

`wap1x`

Context

`config>app-assure>group`

Description

This command administratively enables/disables AA support for Wireless Application Protocol (WAP) 1.X. When enabled, the system trusts UDP server port numbers to be WAP/WTP/WSP. When disabled, the system does not detect WAP1.x traffic as it uses server ports numbers to detect such traffic.

10.2.1.8 Application Assurance group cflowd command descriptions

10.2.1.8.1 cflowd

Syntax

`cflowd`

Context

`config>app-assure>group`

Description

This command enables the context to configure Cflowd parameters for the application assurance group.

10.2.1.8.2 collector

Syntax

`collector ip-address [:port] [create]`

`no collector ip-address [:port]`

Context

`config>app-assure>group>cflowd`

Description

This command defines a flow data collector for Cflowd data. The IP address of the flow collector must be specified. The UDP port number is an optional parameter. If it is not set, the default of 2055 is used.

Parameters

ip-address

the IP address of the flow data collector in dotted decimal notation

port

the UDP port of flow data collector

Default: 2055

Values: 1 to 65535

create

mandatory keyword used when creating a new flow data collector

10.2.1.8.3 comprehensive

Syntax

```
comprehensive
```

Context

```
config>app-assure>group>cflowd
```

Description

This command configures Cflowd comprehensive exports.

10.2.1.8.4 app-group

Syntax

```
app-group app-group-name [flow-rate | flow-rate2]
```

```
no app-group app-group-name
```

Context

```
config>app-assure>group>cflowd>comprehensive
```

Description

This command configures application groups for the comprehensive records Cflowd export.

The no form of this command reverts to the default.

Default

```
no app-group
```

Parameters

app-group-name

specifies the application group name

flow-rate

specifies the primary Cflowd comprehensive flow rate

flow-rate2

specifies the secondary Cflowd comprehensive flow rate

10.2.1.8.5 application

Syntax

`application application-name [flow-rate | flow-rate2]`

`no application application-name`

Context

`config>app-assure>group>cflowd>comprehensive`

Description

This command configures the application for the comprehensive records Cflowd export.

The no form of this command reverts to the default.

Default

no application

Parameters

application-name

specifies the application name

flow-rate

specifies the primary Cflowd comprehensive flow rate

flow-rate2

specifies the secondary Cflowd comprehensive flow rate

10.2.1.8.6 flow-rate

Syntax

`flow-rate sample-rate`

`no flow-rate`

Context

`config>app-assure>group>cflowd>comprehensive`

`config>app-assure>group>cflowd>tcp-performance`

Description

This command configures the per-flow sampling rate for the comprehensive or TCP Cflowd export of AA performance statistics.

The no form of this command reverts to the default.

Default

no flow-rate

Parameters

sample-rate

specifies the rate at which to sample flows that are eligible for comprehensive or TCP Cflowd performance measurement

Values: 1 to 1000

10.2.1.8.7 flow-rate2

Syntax

flow-rate2 *sample-rate*

no flow-rate2

Context

config>app-assure>group>cflowd>comprehensive

config>app-assure>group>cflowd>tcp-performance

Description

This command configures the per-flow second sampling rate for the comprehensive or TCP Cflowd export of AA performance statistics.

The no form of this command reverts to the default.

Default

no flow-rate2

Parameters

sample-rate

specifies the rate at which to sample flows that are eligible for comprehensive or TCP Cflowd performance measurement

Values: 1 to 1000

10.2.1.8.8 template

Syntax

```
template
```

Context

```
config>app-assure>group>cflowd>comprehensive
```

```
config>app-assure>group>cflowd>tcp-performance
```

```
config>app-assure>group>cflowd>volume
```

Description

This command enables the context to configure the template for Cflowd comprehensive, TCP performance, or volume fields.

10.2.1.8.9 dynamic-fields

Syntax

```
dynamic-fields
```

Context

```
config>app-assure>group>cflowd>comprehensive>template
```

```
config>app-assure>group>cflowd>tcp-performance>template
```

```
config>app-assure>group>cflowd>volume>template
```

Description

This command enables the context to configure which fields are included in the exported Cflowd template.

The no form of this command removes all configured dynamic fields from the template.



NOTE:

This command is only supported if the `dynamic` option is configured in the `field-selection` command.

10.2.1.8.10 `field`

Syntax

[no] `field field-name`

Context

`config>app-assure>group>cflowd>comprehensive>template>dynamic-fields`

`config>app-assure>group>cflowd>tcp-performance>template>dynamic-fields`

`config>app-assure>group>cflowd>pgw-edr>template>dynamic-fields`

`config>app-assure>group>cflowd>volume>template>dynamic-fields`

Description

This command configures the fields to be included in the exported Cflowd templates.

The `no` form of this command removes the specified fields from the template.



NOTE:

Certain fields (for example, source, destination, IP addresses, and ports) may not be allowed in certain templates.

Parameters

field-name

specifies the name of the field to be included in the exported Cflowd template, up to 256 characters

Values:

- Common to all templates are the following values: `flowStartSeconds` | `flowDurationMilliseconds` | `postIppPrecedence` | `aaProt` | `aaApp` | `aaAppGrp` | `hostName` | `hostNameExtended` | `deviceId` | `deviceMfgId` | `deviceOsId` | `ipFamily` | `deviceOsVer1` | `deviceOsVer2` | `deviceOsVer3` | `anIType` | `anITopology` | `anICongestionState` | `apn` | `chargId` | `mnc` | `imsi` | `msisdN` | `sgw-sgsnAddr` | `pgw-ggsnAddr` | `bsid` | `ratType` | `regionId` | `cellId` | `timeZone` | `aaChargingGrp` | `flowAttr_video` | `flowAttr_abr_service` | `flowAttr_audio` | `flowAttr_encrypted` | `flowAttr_download` | `flowAttr_upload` | `flowAttr_realtime_communication` | `aaSubTetheringState` | `imei` | `ipTTL` | `imeiRoamingStatus` | `mcc` | `uli` | `chargingChar` | `plmnid` | `customerId`

- When AA is deployed in mobile gateways (MG-SROS), the following values apply: apn | ApnExtended | chargeld | mnc | imsi | msisdn | sgsn | ggsn | bsid | ratType | regionId | cellId | timeZone | msisdn-sha-1 | imsi-sha-1 | imei-sha-1 | msisdn-aes-128 | imsi-aes-128 | imei-aes-128 | msisdn-aes-256 | imsi-aes-256 | imei-aes-256 | msisdn-sha-256 | imsi-sha-256 | imei-sha-256 | wireless-device-os-id | wireless-device-type-id
- When AA is deployed in mobile gateways (MG-SROS), selectable only for comprehensive /volume cflowd templates the following values apply: chargingRatingGrp | policyRuleBaseName | policyRuleName | policyRuleUnitName | policyChargingRuleUnitName | policyActionRuleUnitName | qci | chargingServiceId
- When AA is deployed in FWA SR, the following value applies: ApnExtended
- For the comprehensive template only, the following values apply: http-url-len | http-uri | http-referer | http-ua | http-ua-len | http-request-method | http-content-type | http-content-len | Session-average-throughput | Session-Peak-throughput (the use of per flow policer must be configured in *monitor* mode) | TCPSessionEstDelayUsec | TCPSession-average-delay | TCPSession-min-delay | TCPSession-max-delay | MSS (TCP-optimization MUST be enabled) | dnsanswername | dnstid | dnsstate | dnstype | flowactivetime (the use of per flow policer must be configured in *monitor* mode) | flowactivebytes (the use of per flow policer must be configured in *monitor* mode) | chargevolumebytes | tlsname | cellidext | regionidext | amfmmeaddr | sliceid | aasystemname | sessionstartmilliseconds | sessionstopmilliseconds | flowendreason | ednsSubPolicyId
- For volume and comprehensive templates, the following values apply: tcpSessionEstDelay | tcpRetransmittedBytes | tcpRetransmittedPackets

10.2.1.8.11 field-selection

Syntax

field-selection *field-selection*

no field-selection

Context

config>app-assure>group>cflowd>comprehensive>template

config>app-assure>group>cflowd>tcp-performance>template

config>app-assure>group>cflowd>volume>template

Description

This command configures how fields included in the exported Cflowd template are selected.

The no form of this command reverts to the legacy field selection type.

Default

field-selection legacy

Parameters

field-selection

specifies how fields are selected

Values:

legacy

specifies that the fields within the Cflowd template are set and fixed as per MG SR OS release 10 (or earlier) and as per SR OS release 17 (or earlier)

dynamic

specifies that the operator can select which fields are included in the Cflowd template

10.2.1.8.12 direct-export

Syntax

direct-export

Context

config>app-assure>group>cflowd

Description

This command enables the context to perform configurations related to the export of AA Cflowd records directly in-band from AA instead of going through the CPM.

10.2.1.8.13 collector

Syntax

collector *collector-id* [create]

no collector *collector-id*

Context

config>app-assure>group>cflowd>direct-export

Description

This command configures the Cflowd direct export collector.

Parameters

collector-id a

specifies the ID of the Cflowd direct export collector

Values: 1 to 65535



NOTE:

In addition, a Cflowd collector assignment to an MG-group /AA-ISA is done only in the following conditions: If an MG group/AA-ISA is assigned a collector, shutting down the associated PDN or the MG group will unassign the MG group from the cflowd collector.

- The system will make use of newly configured collectors when the Cflowd admin state shuts down and then re-enabled (no-shutdown state). The system will re-assign the collectors to the MG-Groups/AA-ISAs or when any additional Cflowd collector ID is created. The new collector IDs are used when a new MG-Group is added later.
- When a collector ID is removed, the MG-Groups (AA-ISAs) that are assigned to this collector are removed and assigned to another available collector. The affected ISAs reset their collector statistics as they change to the new collector.
- the admin state is in no - shutdown state for:
 - the MG group
 - the PDN to which the MG group belongs
 - the AA group Cflowd
- a collector is available under the AA group with at least one address in a no - shutdown admin state

10.2.1.8.14 address

Syntax

[no] address *ip-address* [:*port*]

Context

config>app-assure>group>cflowd>direct-export>collector

Description

This command configures the Cflowd direct export collector remote address. Two addresses can be configured for each collector for redundancy. AA sends the same records to both at the same time.

Parameters

ip-address

specifies the IP address of the Cflowd direct export collector, in the format a.b.c.d

port

specifies the port of the Cflowd direct export collector

Values: 1 to 65535

Default: 4739

10.2.1.8.15 *vlan-id*

Syntax

vlan-id *service-port-vlan-id*

no *vlan-id*

Context

config>app-assure>group>cflowd>dir-exp

Description

This command configures the VLAN ID on which the ISA-AA is expected to be emitting traffic.

Default

no *vlan-id*

Parameters

service-port-vlan-id

specifies the VLAN ID of the service port

Values: 1 to 4094

10.2.1.8.16 export-override

Syntax

```
export-override mode  
no export-override
```

Context

```
config>app-assure>group>cflowd
```

Description

This command enables the context to encrypt the interface name field that is exported to the Cflowd.

The interface name has the following format: IMSI (/MSISDN) xxxxxxx. After encryption, the field has the following format: IMSI (/MSISDN) AES-encrypted value of xxxxxx. All AA Cflowd record types are affected by this command. To change any of the export override, the Cflowd must be first shutdown.

The no form of this command reverts to the default.

Default

no export-override

Parameters

mode

specifies the system type emulated by statistics export

Values: ifname-obfuscate (encrypts the cflowd interface-name field name using AES128)



NOTE:

The encryption must be configured by using the config>app-assure>group>cflowd>obfuscation aes-128-encryption command.

10.2.1.8.17 interim-report-mode

Syntax

```
interim-report-mode mode  
no interim-report-mode
```

Context

config>app-assure>group>cflowd

Description

This commands allows the operator to configure the system to generate interim Cflows records triggered by the UE changing PRA id, ULI or RAT-Type.

The no form of this command reverts to the default.



NOTE:

This command is applicable only to the volume and comprehensive Cflowd templates.

Default

no interim-report-mode

Parameters

mode

specifies the trigger type for interim Cflowd generation

Values:

pra-id

generates interim Cflowd records when the associated presence reporting area is changed

user-location

generates interim Cflowd records triggered by change of the associated ULI

rat-type

generates interim Cflowd records when the associated RAT-Type is changed

10.2.1.8.18 obfuscation

Syntax

obfuscation

Context

config>app-assure>group>cflowd

Description

This command enables the context to configure the various obfuscation/encryption keys used to encrypt cflowd fields.

10.2.1.8.19 aes-128-encryption

Syntax

```
aes-128-encryption plaintext key type key-value {hash | hash2 | custom}  
no aes-128-encryption
```

Context

```
config>app-assure>group>cflowd>obfuscation
```

Description

This command allows the operator to enter the AES-128 encryption key used to encrypt exported cflowd fields.

Parameters

plaintext key type

specifies the encryption key value in hexadecimal format or ASCII format

Values: hex | ascii

key-value

specifies the key value, 32 hexadecimal characters or 16 ASCII characters

hash | hash2 | custom

Specifies how the entered key is stored in the system. For security, all keys are stored encrypted in the configuration file using hash, hash2 or custom encryption.

hash

specifies that the key is stored in the system encrypted using standard hash mechanism

hash2

specifies that the key is stored in a more complex encrypted form that involves more variables than the key value alone, meaning that the hash2 encrypted variable cannot be copied and pasted

custom

specifies the custom encryption to the management interface

Default: hash2

10.2.1.8.20 aes-256-encryption

Syntax

```
aes-256-encryption plaintext key type key-value {hash | hash2 | custom}  
no aes-256-encryption
```

Context

```
config>app-assure>group>cflowd>obfuscation
```

Description

This command allows the operator to enter the AES-256 encryption key used to encrypt exported cflowd fields.

Parameters

plaintext key type

specifies the encryption key value in hexadecimal format or ASCII format

Values: hex | ascii

key-value

specifies the key value, 64 hexadecimal characters or 32 ASCII characters

hash | hash2 | custom

Specifies how the entered key is stored in the system. For security, all keys are stored in encrypted form in the configuration file using hash, hash2 or custom encryption.

hash

specifies that the key is stored in the system encrypted using standard hash mechanism

hash2

specifies that the key is stored in a more complex encrypted form that involves more variables than the key value alone, meaning that the hash2 encrypted variable cannot be copied and pasted

custom

specifies the custom encryption to the management interface

Default: hash2

10.2.1.8.21 pgw-edr

Syntax

pgw-edr

Context

config>app-assure>group>cflowd

Description

This command enables or disables Cflowd PGW-EDR export for an AA group partition.

10.2.1.8.22 flow-rate

Syntax

flow-rate *sample-rate*

no flow-rate

Context

config>app-assure>group>cflowd>pgw-edr

Description

This command configure the Cflowd PGW-EDR flow sampling rate.

The no form of this command reverts to the default value.

Default

no flow-rate

Parameters

sample-rate

specifies the export rate of sample flows that are eligible for a PGW-EDR Cflowd session

Values: 1 to 10000

10.2.1.8.23 **template-retransmit**

Syntax

```
template-retransmit seconds  
no template-retransmit
```

Context

```
config>app-assure>group>cflowd
```

Description

This command configures the period of time, in seconds, for the template to be retransmitted.

Parameters

seconds

specifies the time period for the template to be retransmitted

Values: 10 to 600

Default: 600

10.2.1.8.24 **system-name**

Syntax

```
system-name name  
no system-name
```

Context

```
config>app-assure>group>cflowd
```

Description

This command configures the system name of the Cflowd AA group. It is exported to the Cflowd template as aaSystemName.

The no form of this command removes the Cflowd system name.

Default

```
no system-name
```

Parameters

name

specifies the name of the system, up to 32 characters

10.2.1.8.25 tcp-performance

Syntax

tcp-performance

Context

config>app-assure>group>cflowd

Description

This command enables the context to configure Cflowd TCP performance export parameters.

10.2.1.8.26 app-group

Syntax

app-group *app-group-name* [flow-rate | flow-rate2]

no app-group *app-group-name*

Context

config>app-assure>group>cflowd>tcp-performance

Description

This command configures application groups for the TCP performance records Cflowd export.

The no form of this command reverts to the default.

Parameters

app-group-name

specifies the application group name

flow-rate

specifies the primary Cflowd TCP performance flow rate

flow-rate2

specifies the secondary Cflowd TCP performance flow rate

Default: no flow-rate

10.2.1.8.27 application

Syntax

application *application-name* [flow-rate | flow-rate2]

no application *application-name*

Context

config>app-assure>group>cflowd>tcp-performance

Description

This command configures application for the TCP performance records Cflowd export.

The no form of this command reverts to the default.

Parameters

application-name

specifies the application name

flow-rate

specifies the primary Cflowd TCP performance flow rate

flow-rate2

specifies the secondary Cflowd TCP performance flow rate

Default: no flow-rate2

10.2.1.8.28 volume

Syntax

volume

Context

config>app-assure>group>cflowd

Description

This command configures the Cflowd volume export.

10.2.1.8.29 rate

Syntax

rate *sample-rate*

no rate

Context

config>app-assure>group>cflowd>volume

Description

This command configures the sampling rate of packets for the Cflowd export of application assurance volume statistics.

The no form of this command reverts to the default value.

Parameters

sample-rate

specifies the rate at which to sample flows that are eligible for TCP performance measurement

Values: 1 to 10000

10.2.1.9 Application Assurance group policer command descriptions

10.2.1.9.1 policer

Syntax

policer *policer-name* type *type* granularity *granularity* [create]

policer *policer-name*

no policer *policer-name*

Context

config>app-assure>group

Description

This command creates application assurance policer profile of a specified type. Policers can be bandwidth policer, a bandwidth shaper or flow limiting and can have a system scope (limits traffic entering AA ISA for all or a subset of AA subscribers), subscriber scope or granularity (limits apply to each AA subscriber traffic). The bandwidth-shaper can only be applied at the subscriber scope or granularity.

The policer type and granularity can only be configured during creation. They cannot be modified. The policer profile must be removed from all AQPs to be removed. Changes to policer profile parameters take effect immediately for policers instantiated as result of AQP actions using this profile.

The no form of this command deletes the specified policer from the configuration.

Parameters

type

specifies the policer type

Values:

single-bucket-bandwidth

creates a profile for a single bucket (PIR) bandwidth limiting policer

dual-bucket-bandwidth

creates a profile for a dual bucket (PIR, CIR) bandwidth limiting policer

flow-rate-limit

creates a profile for a policer limiting rate of flow set-ups

flow-count-limit

creates a profile for a policer limiting total flow count

bandwidth-shaper

creates a profile for a bandwidth shaper that can only be used at the subscriber level

granularity

specifies the granularity type

Values:

system

Creates a system policer profile for a policer that limits the traffic in the scope of all or a subset of AA subscribers on a given AA ISA. No bandwidth-shaper is allowed at the system granularity.

subscriber

Creates a policer profile for a policer for each AA subscriber that limits the traffic in the scope of that subscriber

access-network-location

Creates a policer profile for a policer instance for each Access Network Location (ANL) that limits traffic bandwidth in the scope of that ANL. For ANL, only single-bucket bandwidth policers can be configured.

flow

Creates a policer profile for a policer for each IP flow that limits the traffic in the scope of that individual flow. Only single-bucket bandwidth policers with permit-deny or Monitor can be configured.

create

keyword used to create the policer name and parameters

policer-name

a string that identifies the policer, up to 32 characters

10.2.1.9.2 action

Syntax

action policer-action

Context

config>app-assure>group>policer

Description

This command configures the action to be performed by single-bucket bandwidth policers for non-conformant traffic.

Dual bucket bandwidth policers cannot have their action configured and always mark traffic below CIR in profile, between CIR and PIR out of profile, and drop traffic above PIR. Flow policers always discard non-conformant traffic.

When multiple application assurance policers are configured against a single flow (including policers at both subscriber and system), the final action done to the flow/packet will be a logical OR of all policer actions. For example, if only of the policers requires the packet to be discarded, the packet will be dropped regardless of the action of the other policers.

Parameters

policer-action

Specifies the policer action

Values:

priority-mark	Non-conformant traffic will be marked out of profile and the conformant traffic will be marked in profile. The new marking will overwrite any previous IOM QoS marking done to a packet.
permit-deny	Non-conformant traffic will be dropped. Default: permit-deny
monitor	Record the duration and volume of non-conformant traffic (applies only to single-rate flow level policers) . Non-conformant traffic is not dropped.

10.2.1.9.3 adaptation-rule

Syntax

```
adaptation-rule pir {max | min | closest} [cir {max | min | closest}]  
no adaptation-rule
```

Context

```
config>app-assure>group>policer
```

Description

This command defines the method used by the system to derive the operational CIR and PIR settings when the queue is provisioned in hardware. For the CIR and PIR parameters individually, the system attempts to find the best operational rate depending on the defined option. To change the CIR adaptation rule only, the current PIR rule must be part of the command executed.

The no form of this command removes any explicitly defined constraints used to derive the operational CIR and PIR created by the application of the policy. When a specific adaptation-rule is removed, the default constraints for rate and CIR apply.

Parameters

max

The operational PIR or CIR for the queue will be equal to or less than the administrative rate specified using the rate command.

min

The operational PIR or CIR for the queue will be equal to or greater than the administrative rate specified using the rate command.

closest

The operational PIR or CIR for the queue will be the rate closest to the rate specified using the rate command.

Default: closest

10.2.1.9.4 congestion-override

Syntax

```
congestion-override
```

Context

```
config>app-assure>group>policer
```

Description

This command enables the context to configure per subscriber congestion bandwidth policer override rates.

10.2.1.9.5 congestion-override-stage2

Syntax

```
congestion-override-stage2
```

Context

```
config>app-assure>group>policer
```

Description

This command enables the context to configure per subscriber stage 2 congestion bandwidth policer override rates.

10.2.1.9.6 cbs

Syntax

`cbs committed-burst-size`

`cbs congested-cbs`

`no cbs`

Context

`config>app-assure>group>policer`

`config>app-assure>group>policer>congestion-override`

`config>app-assure>group>policer>congestion-override-stage2`

`config>app-assure>group>policer>tod-override`

Description

This command enables a mechanism to configure the committed burst size congestion override. It is recommended that the CBS is configured larger than twice the maximum MTU for the traffic handled by the policer to allow for some burstiness of the traffic. The CBS is configurable for dual-bucket bandwidth policers only.

The no form of this command resets the CBS value to its default.

Parameters

committed-burst-size / congested-cbs

specifies an integer value defining size, in kilobytes, for the CBS of the policer

Values: 0 to 131071

Default: 0

10.2.1.9.7 cir

Syntax

`cir congested-cir`

`no cir`

Context

`config>app-assure>group>policer>congestion-override`

```
config>app-assure>group>policer>congestion-override-stage2
```

Description

This command provides a mechanism to configure the CIR for the congestion override policer. It is recommended that the CIR is configured larger than twice the maximum MTU for the traffic handled by the policer to allow for some burstiness of the traffic. The CIR is configurable for dual-bucket bandwidth policers only.

The no form of this command resets the CIR value to its default.

Parameters

congested-cir

specifies an integer value defining size, in kilobytes, for the CIR of the policer

Values: 0 to 100000000

Default: 0

10.2.1.9.8 mbs

Syntax

```
mbs congested-mbs
```

```
mbs maximum-burst-size
```

```
no mbs
```

Context

```
config>app-assure>group>policer
```

```
config>app-assure>group>policer>congestion-override
```

```
config>app-assure>group>policer>congestion-override-stage2
```

```
config>app-assure>group>policer>tod-override
```

Description

This command provides a mechanism to configure the maximum burst size or the MBS for the policer. It is recommended that the MBS is configured larger than twice the MTU for the traffic handled by the policer to allow for some burstiness of the traffic. The MBS is configurable for single-bucket, dual-bucket bandwidth, bandwidth-shaper, and flow setup rate policers only. For bandwidth-shaper, the MBS indicates the maximum size of the shaping queue.

The no form of this command resets the MBS value to its default.

Parameters

maximum-burst-size | *congested-mbs*

specifies an integer value defining either size, in kbytes, for the MBS of the bandwidth policer, shaper, or flow count for the MBS of the flow setup rate policers

Values: 0 to 131071

Default: 0

10.2.1.9.9 pir

Syntax

`pir congested-pir`

`no pir`

Context

`config>app-assure>group>policer>congestion-override`

`config>app-assure>group>policer>congestion-override-stage2`

Description

This command provides a mechanism to configure the PIR for the congestion override policer. It is recommended that the PIR is configured larger than twice the maximum MTU for the traffic handled by the policer to allow for some burstiness of the traffic.

The no form of this command resets the PIR value to its default.

Parameters

congested-pir

specifies an integer value defining size, in kbytes, for the PIR of the policer

Values: 0 to 100000000

Default: 0

10.2.1.9.10 flow-count

Syntax

`flow-count flow-count`

`no flow-count`

Context

```
config>app-assure>group>policer
```

```
config>app-assure>group>policer>tod-override
```

Description

This command configures the flow count for the flow-count-limit policer. It is recommended to configure flow count subscriber-level policer for AA subscribers to ensure fair usage of flow resources between AA subscribers.

Parameters

flow-count

specifies the flow count for the flow-count-limit policer

10.2.1.9.11 rate

Syntax

```
rate pir-rate [cir cir-rate]
```

```
no rate
```

Context

```
config>app-assure>group>policer
```

```
config>app-assure>group>policer>tod-override
```

Description

This command configures the administrative PIR and CIR for bandwidth policers and flow setup rate limits for flow policers. The actual rate sustained by the flow can be limited by other policers that may be applied to that flow traffic. This command does not apply to flow-count-limit policers. PIR for bandwidth-shaper specifies the shaping rate.

The *cir* option is applicable only to dual-bucket bandwidth policers. It is recommended to configure flow setup rate subscriber-level policer for AA subscribers to ensure fair usage of flow resources between AA subscribers. CIR does not apply to bandwidth-shapers.

The *no* form of this command reverts to the default.

Parameters

pir-rate

an integer specifying either the PIR rate in Kbps for bandwidth policers

Values: 1 to 100000000, max or flows

cir-rate

an integer specifying the CIR rate in Kbps

Values: 0 to 100000000, max

Default: 0

10.2.1.9.12 rate-percentage

Syntax

rate-percentage *rate-percentage*

no rate-percentage

Context

config>app-assure>group>policer

Description

This command provides a mechanism to indirectly configure the rate used by Access-Network-Location (ANL) policers. Given that the ANL total bandwidth is dynamically measured and estimated by AA, this command allows the operator to configure the ratio of that measured bandwidth to be used by the ANL policer as the policer rate.

The rate is configurable for single-bucket ANL policers only.

The no form of this command resets the rate percentage value to its default.

Default

no rate-percentage

Parameters

rate-percentage

an integer value that specifies a percentage that is applied against the ANL estimated maximum bandwidth to produce the actual rate that is used by the policer when ANL congestion occurs

Values: 0 to 200, where 0 means drop all traffic

10.2.1.9.13 tod-override

Syntax

```
tod-override tod-override-id [create]  
no tod-override tod-override-id
```

Context

```
config>app-assure>group>policer
```

Description

This command specifies the time-of-day policer override.

Parameters

tod-override-id

The time-of-day override identifier.

Values: 1 to 255

create

keyword used when creating a new time-of-day override. The create keyword requirement can be enabled/disabled in the environment>create context.

10.2.1.9.14 time-range

Syntax

```
time-range daily start start-time end end-time [on day [day...(up to 7 max)]]  
time-range weekly start start-time end end-time  
no time-range
```

Context

```
config>app-assure>group>policer  
config>app-assure>group>policer>tod-override
```

Description

This command specifies the time-of-day policer override.

Parameters

start daily start-time

the daily time to start override

Values: <hh>:<mm> where <hh>: 0 to 23 and <mm>: 0 | 15 | 30 | 45

start weekly start-time

the weekly time to start override

Values: *day*, *hh:mm* where *hh*: 0 to 24 and *mm*: 0 | 15 | 30 | 45

end daily end-time

the daily time to end override

Values: *hh:mm* where *hh*: 0 to 23 and *mm*: 0 | 15 | 30 | 45

end weekly end-time

the weekly time to end override

Values: *day*, *hh:mm* where *hh*: 0 to 24 and *mm*: 0 | 15 | 30 | 45

day

the day of the week

Values: sunday | monday | tuesday | wednesday | thursday | friday | saturday

daily

schedule the override as a daily occurrence

weekly

schedule the override as a weekly occurrence

10.2.1.10 Application Assurance group policy command descriptions

10.2.1.10.1 policy

Syntax

policy

Context

config>app-assure>group>policy

Description

This command enables the context to configure parameters for application assurance policy. To edit any policy content, the `begin` command must be executed first to enter editing mode. The editing mode is exited when the `abort` or `commit` commands are issued.

10.2.1.10.2 abort

Syntax

`abort`

Context

`config>app-assure>group>policy`

Description

This command ends the current editing session and aborts any changes entered during this policy editing session.

10.2.1.10.3 begin

Syntax

`begin`

Context

`config>app-assure>group>policy`

Description

This command begins a policy editing session.

The editing session continues until one of the following conditions takes place:

- `abort` or `commit` is issued
- Control complex resets

The editing session is not interrupted by:

- High-availability (HA) activity switch
- CLI session termination (for example, as result of closing a Telnet session)

10.2.1.10.4 charging-filter

Syntax

`charging-filter`

Context

```
config>app-assure>group>policy
```

Description

This command enables the context to configure a charging filter for application assurance.

10.2.1.10.5 entry

Syntax

```
entry entry-id[create]
```

```
no entry entry-id
```

Context

```
config>app-assure>group>policy>charging-filter
```

Description

This command creates a charging-filter entry. Charging filter entries are an ordered list; the lowest numerical entry that matches the flow, defines the charging-filter for this flow.

The no form of this command reverts to the default.

Default

```
no entry
```

10.2.1.10.6 charging-group

Syntax

```
charging-group charging-group-name
```

```
no charging-group
```

Context

```
config>app-assure>group>policy>charging-filter>entry
```

Description

This command associates a charging group to the flows that match the charging filter entry.

The no form of this command reverts to the default.

Default

no charging-group

Parameters

charging-group-name

specifies a string that uniquely identifies the charging group in the system, up to 32 characters

10.2.1.10.7 match

Syntax

match

Context

config>app-assure>group>policy>charging-filter>entry

Description

This command enables the context to configure the match criteria for the charging filter.

10.2.1.10.8 app-group

Syntax

app-group {eq | neq} *application-group-name*

no app-group

Context

config>app-assure>group>policy>charging-filter>entry>match

Description

This command adds app-group to the match criteria used by this charging filter entry.

The no form of this command reverts to the default.

Default

no app-group

Parameters

eq

specifies that the value configured and the value in the flow are equal

neq

specifies that the value configured differs from the value in the flow

application-group-name

specifies the name of the existing application group entry, up to 32 characters

10.2.10.9 application

Syntax

application {eq | neq} *application-name*

no application

Context

config>app-assure>group>policy>charging-filter>entry>match

Description

This command adds an application to the match criteria used by this charging filter entry.

The no form of this command reverts to the default.

Default

no application

Parameters

eq

specifies that the value configured and the value in the flow are equal

neq

specifies that the value configured differs from the value in the flow

application-name

specifies the name of the existing application, up to 32 characters

10.2.1.10.10 flow-attribute

Syntax

[no] flow-attribute *flow-attribute-name*

Context

config>app-assure>group>policy>charging-filter>entry>match

Description

This command adds a flow attribute to the match criteria used by this charging filter entry.

The no form of this command reverts to the default.

Default

no flow-attribute

Parameters

flow-attribute-name

specifies the name of the flow-attribute, up to 32 characters

10.2.1.10.11 tethered-flow

Syntax

[no] tethered-flow

Context

config>app-assure>group>policy>charging-filter>entry>match

Description

This command adds the tethering status to the match criteria used by this charging filter entry.

The no form of this command reverts to the default.

Default

no tethered-flow

10.2.1.10.12 commit

Syntax

```
commit
```

Context

```
config>app-assure>group>policy
```

Description

This command commits changes made during the current editing session. None of the policy changes done will take effect until the `commit` command is issued. If the changes can be successfully committed with no errors detected during the commit during cross-reference verification against exiting application assurance configuration, the editing session will also be closed.

The newly committed policy takes affect immediately for all new flows, existing flows will transition onto the new policy shortly after the commit.

10.2.1.10.13 admin

Syntax

```
admin
```

Context

```
config>app-assure>group>policy
```

Description

This command displays the application-assurance policy uncommitted changes.

10.2.1.10.14 app-filter

Syntax

```
app-filter
```

Context

```
config>app-assure>group>policy
```


Description

This command enables the context to configure an application filter for application assurance.

10.2.1.10.15 aa-sub-ip

Syntax

```
aa-sub-ip ip-address[/mask]
no aa-sub-ip
```

Context

```
config>app-assure>group>policy>app-filter>entry>match
```

Description

This command configures the IP address of the transit prefix subscriber.

The no form of this command restores the default (removes the address from criteria defined by this entry).

Default

```
no aa-sub-ip
```

Parameters

ip-address

specifies a valid unicast IPv4 or IPv6 address

Values	ipv4 - a.b.c.d [mask]
–	ipv6 - x:x:x:x:x:x:x
–	x - [0..FFFF]H
–	d - [0..255]D
–	ipv6-prefix-length: [1 to 128]

mask

the mask associated with the IP address as a mask length, for example, /16 for a sixteen-bit mask

Values	1 to 32 (IPv4)	–
–	1 to 128 (IPv6)	–

10.2.10.16 network-ip

Syntax

```
network-ip ip-address[/mask]
no network-ip
```

Context

```
config>app-assure>group>policy>app-filter>entry>match
```

Description

This command configures the IP address of the transit prefix network.

The no form of this command restores the default (removes the address from criteria defined by this entry).

Default

```
no network-ip
```

Parameters

ip-address

specifies a valid unicast IPv4 or IPv6 address

Values	ipv4 - a.b.c.d [mask]
–	ipv6 - x:x:x:x:x:x:x

- x - [0..FFFF]H
- d - [0..255]D
- ipv6-prefix-length: [1 to 128]

mask

the mask associated with the IP address as a mask length, for example, /16 for a sixteen-bit mask

Values	1 to 32 (IPv4)	-
-	1 to 128 (IPv6)	-

10.2.10.17 app-group

Syntax

```
app-group application-group-name [create]
no app-group application-group-name
```

Context

```
config>app-assure>group>policy
```

Description

This command creates an application group for an application assurance policy.

The no form of this command deletes the application group from the configuration. All associations must be removed to delete a group.

Default

```
no app-group
```

Parameters

application-group-name

a string of up to 32 characters uniquely identifying this application group in the system

create

mandatory keyword used when creating an application group. The create keyword requirement can be enabled/disabled in the `environment>create` context.

10.2.1.10.18 app-profile

Syntax

```
app-profile app-profile-name [create]
```

```
no app-profile app-profile-name
```

Context

```
config>app-assure>group>policy
```

Description

This command creates an application profile and enables the context to configure the profile parameters.

The no form of this command removes the application profile from the configuration.

Parameters

app-profile-name

specifies the name of the application profile, up to 32 characters

create

mandatory keyword used when creating an application profile. The create keyword requirement can be enabled/disabled in the `environment>create` context.

10.2.1.10.19 app-qos-policy

Syntax

```
app-qos-policy
```

Context

```
config>app-assure>group>policy
```

Description

This command enables the context to configure an application QoS policy.

10.2.1.10.20 app-service-options

Syntax

```
app-service-options
```

Context

```
config>app-assure>group>policy
```

Description

This command enables the context to configure application service option parameters.

10.2.1.10.21 charging-group

Syntax

```
charging-group charging-group-name  
no charging-group charging-group-name
```

Context

```
config>app-assure>group>policy>app-group
```

Description

This command associates a charging group with this application group in an application assurance policy.

The no form of this command deletes the charging group from the configuration. All associations must be removed to delete a group.

Default

```
no charging-group
```

Parameters

charging-group-name

a string of up to 32 characters uniquely identifying this application group in the system

10.2.1.10.22 characteristic

Syntax

```
characteristic characteristic-name [create]  
no characteristic characteristic-name
```

Context

```
config>app-assure>group>policy>app-service-options
```

Description

This command configures an application service option characteristic.

The no form of this command removes the characteristic from the application service.

Parameters

characteristic-name

specifies the name of an existing application service option characteristic

create

keyword used to create the characteristic

10.2.1.10.23 value

Syntax

```
[no] value value-name
```

Context

```
config>app-assure>group>policy>app-service-options>characteristic
```

Description

This command configures an application service option characteristic value.

The no form of this command removes the characteristic value from the application service.

Parameters

value-name

specifies the value name of the characteristic

10.2.1.10.24 default-value

Syntax

```
[no] default-value value-name
```

Context

```
config>app-assure>group>policy>app-service-options>characteristic
```

Description

This command configures an application service option characteristic default value.

The no form of this command removes the characteristic default value from the application service.

Parameters

value

specifies the default value name of the characteristic

10.2.1.10.25 application

Syntax

```
application application-name [create]
```

```
no application application-name
```

Context

```
config>app-assure>group>policy
```

Description

This command creates an application of an application assurance policy.

The no form of this command deletes the application. To delete an application, all associations to the application must be removed.

Parameters

application-name

specifies a string of up to 32 characters uniquely identifying this application in the system

create

mandatory keyword used when creating an application. The `create` keyword requirement can be enabled/disabled in the `environment>create` context.

10.2.1.10.26 app-group

Syntax

`app-group` *application-group-name*

Context

`config>app-assure>group>policy>application`

Description

This command associates an application with an application group of an application assurance policy.

Parameters

application-name

a string of up to 32 characters uniquely identifying an existing application in the system

10.2.1.10.27 charging-group

Syntax

`charging-group` *charging-group-name*

`no charging-group` *charging-group-name*

Context

`config>app-assure>group>policy>application`

Description

This command associates a charging group with this application in an Application Assurance policy.

The `no` form of this command deletes the charging group from the configuration. All associations must be removed to delete a group.

Default

`no charging-group`

Parameters

charging-group-name

a string of up to 32 characters uniquely identifying this charging group in the system

10.2.10.28 charging-group

Syntax

charging-group *charging-group-name* [create]

no charging-group

Context

config>app-assure>group>policy

Description

This command creates a charging group, or enables the context to modify a charging group for the Application Assurance group policy.

The no form of this command deletes the charging group from the configuration. All associations must be removed to delete a charging group.

Parameters

charging-group-name

a string of up to 32 characters uniquely identifying a new or existing charging group in the system

create

a mandatory keyword used when creating a new charging group

10.2.10.29 notify-start-stop

Syntax

notify-start-stop

notify-start-stop *notify-type*

no notify-start-stop

Context

config>app-assure>group>policy>charging-group

Description

This command enables start and stop notifications for the charging group for the AA group policy.

The no form of this command disables notifications.



NOTE:

The granularity level can be overridden and can dynamically change for a session by configuring an action rule unit and by activating the corresponding PCC rule.

Default

no notify-start-stop

Parameters

notify-type

specifies the type of the notifications

Values	flow-based	enables start and stop notifications for each flow matching the charging-group, that is, start-stop notifications are sent for each flow of this charging group. When the flow-based option is not selected, the AA-ISA sends start notification when the first flow of the charging group is detected and sends stop notifications when no flows exist for the charging group, that is, notifications are at the charging-group level.
–	dynamic-rating	enables the start and stop notifications. An application-start is sent when AA detects traffic matching to the application filter with the dynamic-rating-action parameter set to start (using the dynamic-rating-action command). An application-stop is sent when AA detects traffic matching to the application filter with the dynamic-rating-action parameter set to stop (using the dynamic-rating-action command), or when the dynamic rating timeout (configured in the dynamic-rating-expiry command) has expired, and AA has not detected traffic that matches to an application filter with the dynamic-rating-action parameter

set to keep-alive (using the dynamic-rating-action command).

10.2.1.10.30 default-charging-group

Syntax

```
default-charging-group charging-group-name  
no charging-group
```

Context

```
config>app-assure>group>policy
```

Description

This command associates a default charging group with all application groups of an application assurance policy.

Parameters

charging-group-name

a string of up to 32 characters uniquely identifying an existing charging group in the system to be the default charging group

10.2.1.10.31 default-tethered-charging-group

Syntax

```
default-tethered-charging-group charging-group-name  
no default-tethered-charging-group
```

Context

```
config>app-assure>group>policy
```

Description

This command configures the default charging group to be assigned to all flows which are classified as tethered.

For flows that have been identified as tethered, the AA will replace the charging group configured at the application level with the charging group configured for tethered traffic (that is, configured with default-tethered-charging-group).

The no form of this command removes the configured default charging group for tethered traffic.

Default

no default-tethered-charging-group

Parameters

charging-group-name

specifies a string of up to 32 characters uniquely identifying an existing charging group in the system to be used as the default tethered charging group

10.2.10.32 entry

Syntax

entry *entry-id* [create]

no entry *entry-id*

Context

config>app-assure>group>policy>app-filter

Description

This command creates an application filter entry.

App filter entries are an ordered list, the lowest numerical entry that matches the flow defines the application for that flow.

An application filter entry or entries configures match attributes of an application.

The no form of this command deletes the specified application filter entry.

Parameters

entry-id

an integer that identifies an app-filter entry

Values: 1 to 65535

create

keyword used to create the entry

10.2.1.10.33 application

Syntax

`application application-name`

Context

`config>app-assure>group>policy>app-filter>entry`

Description

This command assigns this application filter entry to an existing application. Assigning the entry to unknown application restores the default configuration.

Default

unknown application

Parameters

application-name

specifies an existing application name

10.2.1.10.34 dynamic-rating-action

Syntax

`dynamic-rating-action action`

`no dynamic-rating-action`

Context

`config>app-assure>group>policy>app-filter>entry`

Description

This command configures the dynamic rating action of the application filter entry.

Default

no dynamic-rating-action

Parameters

action

specifies the dynamic rating action

Values	start	specifies that the application filter entry is used for triggering to activate the dynamic rating
–	keep-alive	specifies that the application filter entry is used for the keep-alive dynamic rating
–	end	specifies that the application filter entry is used for triggering to deactivate the dynamic rating

10.2.1.10.35 expression

Syntax

expression *expr-index* *expr-type* {eq | neq} *expr-string*

no expression *expr-index*

Context

config>app-assure>group>policy>app-filter>entry

Description

This command configures string values to use in the application definition.

Parameters

expr-index

specifies an index value which represents expression substrings

Values: 1 to 4

expr-type

represents a type (and thereby the expression substring)

Values:

http-host

matches the string against the HTTP Host field

http-uri

matches the string against the HTTP URI field

http-referer

matches the string against the HTTP Referer field

http-user-agent

matches the string against the HTTP User Agent field

sip-ua

matches the string against the SIP UA field

sip-uri

matches the string against the SIP URI field

sip-mt

matches the string against the SIP MT field

citrix-app

matches the string against the Citrix app field

h323-product-id

matches the string against the h323-product-id field

tls-cert-subj-org-name

matches the string against the TLS certificate subject organization name field

tls-cert-subj-common-name

matches the string against the TLS certificate subject common name field

rtsp-host

matches the string against the RTSP host field

rtsp-uri

matches the string against the RTSP URI field

rtsp-ua

matches the string against the RTSP UA field

dns-domain-name

matches the string against the DNS NAME field

eq

specifies the equal to comparison operator to match the specified HTTP string

neq

specifies the not equal to comparison operator to match the specified HTTP string

expr-string

specifies an expression string, up to 64 characters, used to define a pattern match; denotes a printable ASCII substring used as input to an application assurance filter match criteria object

The following syntax is permitted within the substring to define the pattern match criteria:

`^<substring>*` — matches when `<substring>` is at the beginning of the object.

`*<substring>*` — matches when `<substring>` is at any place within the object.

`*<substring>$` — matches when `<substring>` is at the end of the object.

`^<substring>$` — matches when `<substring>` is the entire object.

`<substring1>*<substring2>` - matches wildcard as an infix.

`*` — matches zero to many of any character. A single wildcard as infix in the expression is allowed.

`\.` — matches any single character

`\d` — matches any single decimal digit [0-9]

`\I` — forces case sensitivity (by default, the expression match are case insensitive), the `\I` can be specified anywhere between the leading `[^*]` and trailing `[$*]`

`\*` — matches the asterisk character

Rules for `<substring>` characters:

`<substring>` must contain printable ASCII characters.

`<substring>` must not contain the “double quote” character or the space character on its own.

`<substring>` match is case sensitive.

`<substring>` must not include any regular expression meta-characters.

The “\” (slash) character is used as an ESCAPE sequence. The following ESCAPE sequences are permitted within the `<substring>`:

Character to match `<substring>` input

Hexadecimal Octet `YY \xYY`



NOTE:

- A `<substring>` that uses the ‘\’ (backslash) ESCAPE character which is not followed by a “\” or “\x” and a 2-digit hex octet is not valid.
- When matching a TCP flow against HTTP-string based applications, the HTTP header fields are collected from the first HTTP request (for example a GET or a POST) for a given TCP flow. The collected strings are then evaluated against each HTTP flow created within the given TCP flow to determine whether a given HTTP flow matches the application. By not specifying a protocol, the HTTP expressions are matched against all protocols in the HTTP family. By specifying a specific HTTP protocol (for example, `http_video`) the expression match can be constrained to a subset of the HTTP protocols.
- To uniquely identify a SIP-based application a protocol match is not required in the app-filter entry with the SIP expression. The SIP expression match is performed against any protocol in the SIP family (such as `sip` and `rtp_sip`). By

specifying a specific SIP protocol (like `rtp_sip`) the expression match can be constrained to a subset of the SIP protocols.

10.2.1.10.36 flow-setup-direction

Syntax

```
flow-setup-direction {subscriber-to-network | network-to-subscriber | both}
```

Context

```
config>app-assure>group>policy>app-filter>entry
```

Description

This command configures the direction of flow setup to which the application filter entry is to be applied.

Parameters

subscriber-to-network

specifies that the app-filter entry will be applied to flows initiated by a local subscriber

network-to-subscriber

specifies that the app-filter entry will be applied to flows initiated from a remote destination toward a local subscriber

both

specifies that the app filter entry will be applied for subscriber-to-network and network-to-subscriber traffic

Default: both

10.2.1.10.37 http-match-all-requests

Syntax

```
[no] http-match-all-requests
```

Context

```
config>app-assure>group>policy>app-filter>entry
```

Description

This command enables constant monitoring performed for HTTP request strings for application changes in regular HTTP flows.

The no form of this command disables the monitoring.

10.2.1.10.38 http-port

Syntax

```
http-port {eq | neq} port-num
```

```
http-port {eq | neq} port-list port-list-name
```

```
no http-port
```

Context

```
config>app-assure>group>policy>app-filter>entry
```

Description

This command specifies an HTTP server TCP or UDP port number or port list to use in the application definition.

The no form of this command restores the default by removing the HTTP port or port list from the application criteria defined by this app-filter entry.

Default

```
no http-port
```

Parameters

eq

specifies that the value configured and the value in the flow are equal

neq

specifies that the value configured differs from the value in the flow

port-list *port-list-name*

specifies a named port list containing a set or range of ports

port-num

specifies a valid server port number

Values: 0 to 65535

10.2.1.10.38.1 ip-identification-assist

Syntax

```
[no] ip-identification-assist
```

Context

```
config>app-assure>group>policy>app-filter>entry
```

Description

This command enables triggering a network-IP address look-up that can override the assigned application if the look-up finds the network IP in the longest-prefix-match look-up in the IP identification table. If an IP match is found, the application assigned from the app-filter is overridden with the application from the IP look-up. The app-group and charging-group can also be affected. If an IP match is not found, the application assigned from the app-filter remains (including the app-group and charging-group).

10.2.1.10.39 ip-protocol-num

Syntax

```
ip-protocol-num {eq | neq} protocol-id  
no ip-protocol-num
```

Context

```
config>app-assure>group>policy>app-filter>entry
```

Description

This command configures the IP protocol to use in the application definition.

The no form of this command restores the default (removes IP protocol number from application criteria defined by this app-filter entry).

Parameters

eq

specifies that the value configured and the value in the flow must be equal

neq

specifies that the value configured differs from the value in the flow

protocol-id

specifies the decimal value representing the IP protocol to be used as an IP filter match criterion. Well known protocol numbers include ICMP (1), TCP (6), UDP (17).

The no form the command removes the protocol from the match criteria.

Values: 0 to 255 (values can be expressed in decimal, hexadecimal, or binary - DHB)

keywords: none, crtp, crudp, egp, eigrp, encap, ether-ip, gre, icmp, idrp, igmp, igp, ip, ipv6, ipv6-frag, ipv6-icmp, ipv6-no-nxt, ipv6-opts, ipv6-route, isis, iso-ip, l2tp, ospf-igp, pim, pnni, ptp, rdp, rsvp, stp, tcp, udp, vrrp * — udp/tcp wildcard

10.2.1.10.40 server-address

Syntax

```
server-address {eq | neq} ip-address
```

```
server-address {eq | neq} ip-address ip-address-match netmask ip-address-mask
```

```
no server-address
```

Context

```
config>app-assure>group>policy>app-filter>entry
```

Description

This command configures the server address to use in application definition. The server IP address may be the source or destination, network or subscriber IP address, and may include the use of netmasks.

The no form of this command restores the default (removes the server address from application criteria defined by this entry).

Default

```
no server-address
```

Parameters

eq

specifies a comparison operator that the value configured and the value in the flow are equal

neq

specifies a comparison operator that the value configured differs from the value in the flow

ip-address-match

specifies a valid unicast IPv4 or IPv6 address

ip-address-mask

specifies an IPv4 or IPv6 address mask

Values	ipv4 - a.b.c.d [mask]
–	ipv6 - x:x:x:x:x:x:x
–	x - [0..FFFF]H
–	d - [0..255]D
–	ipv6-prefix-length: [1 to 128]

10.2.1.10.41 server-port

Syntax

```

server-port {eq | neq | gt | lt} server-port-number
server-port {eq | neq} range start-port-num end-port-num
server-port {eq} {port-num | range start-port-num end-port-num} {first-packet-trusted |
first-packet-validate}
server-port {eq | neq} port-list port-list-name name
server-port {eq} port-list port-list-name {first-packet-trusted | first-packet-
validate}
no server-port

```

Context

```
config>app-assure>group>policy>app-filter>entry
```

Description

This command specifies the server TCP or UDP port number to use in the application definition.

The no form of this command restores the default (removes server port number from application criteria defined by this app-filter entry).

The following are the server port options:

No option specified

TCP/UDP port applications with full signature verification:

- AA ensures that other applications that can be identified do not run over a well-known port.

- Application-aware policy applied once signature-based identification completes (likely requiring several packets).

first-packet-validate

TCP/UDP trusted port applications with signature verification:

- Application identified using well known TCP/UDP port based filters and re-identified once signature identification completes.
- AA policy applied from the first packet of a flow while continuing signature-based application identification. Policy re-evaluated once the signature identification completes, allowing to detect improper/unexpected applications on a well-known port.

first-packet-trusted –

TCP/UDP trusted port applications — no signature verification:

- Application identified using well known TCP/UDP port based filters only.
- Application Aware policy applied from the first packet of a flow.
- No signature processing assumes operator/customer trusts that no other applications can run on the well-known TCP/UDP port (statistics collected against trusted_tcp or trusted_udp protocol).

Default

no server-port (the server port is not used in the application definition)

Parameters

eq

specifies that the value configured and the value in the flow are equal

first-packet-trusted

specifies TCP/UDP trusted port applications; no signature verification

first-packet-validate

specifies TCP/UDP trusted port applications with signature verification

gt

specifies all port numbers greater than server-port-number match

lt

specifies all port numbers less than server-port-number match

neq

specifies that the value configured differs from the value in the flow

port-list

specifies a named port list containing a set or range of ports

server-port-num

specifies a valid server port number

Values: 0 to 65535

start-port-num

specifies the starting port number

Values: 0 to 65535

end-port-num

specifies the ending port number

Values: 0 to 65535

10.2.10.42 protocol

Syntax

```
protocol {eq | neq} protocol-name
```

```
no protocol
```

Context

```
config>app-assure>group>policy>app-filter>entry
```

Description

This command configures the protocol signature in the application definition.

The no form of this command restores the default (removes the protocol from the match application defined by this app-filter entry).

Default

```
no protocol
```

Parameters

eq

specifies that the value configured and the value in the flow are equal

neq

specifies that the value configured differs from the value in the flow

protocol-name

a string of up to 32 characters identifying a predefined protocol

Values: Use the `show>application-assurance>protocol` command to display a full list of protocols available based on the `isa-aa.tim` image used with the applicable release software.

10.2.1.10.43 **characteristic**

Syntax

`characteristic characteristic-name value value-name`

`no characteristic characteristic-name`

Context

`config>app-assure>group>policy>app-profile`

Description

This command assigns one of the existing values of an existing application service option characteristic to the application profile.

The `no` form of this command removes the characteristic from the application profile.

Parameters

characteristic-name

specifies the name of an existing ASO characteristic

value-name

specifies the name for the application profile characteristic, up to 32 characters

10.2.1.10.44 **capacity-cost**

Syntax

`capacity-cost cost`

`no capacity-cost`

Context

`config>app-assure>group>policy>app-profile`

Description

This command assigns a capacity cost value to the application profile.

The no form of this command removes the characteristic from the application profile.

Parameters

cost

specifies the value of the capacity cost

Values: 1 to 65535

10.2.1.10.45 divert

Syntax

characteristic *characteristic-name* value *value-name*

no characteristic *characteristic-name*

Context

config>app-assure>group>policy>app-profile

Description

This command configures the application profile flag to enable or disable subscriber traffic to be sent to an ISA-AA.

The no form of this command removes the characteristic from the application profile.

10.2.1.10.46 entry

Syntax

[no] entry *entry-id* [create]

Context

config>app-assure>group>policy>aqp

Description

This command creates an application QoS policy entry. A flow that matches multiple Application QoS policies (AQP) entries will have multiple AQP entries actions applied. When a conflict occurs for two or more actions, the action from the AQP entry with the lowest numerical value takes precedence.

The no form of this command deletes the specified application QoS policy entry.

Parameters

entry-id

an integer identifying the AQP entry

Values: 1 to 65535

create

mandatory keyword creates the entry

The create keyword requirement can be enabled/disabled in the environment>create context.

10.2.1.10.47 action

Syntax

action

Context

config>app-assure>group>policy>aqp>entry

Description

This command enables the context to configure AQP actions to be performed on flows that match the AQP entry's match criteria.

10.2.1.10.48 abandon-tcp-optimization

Syntax

[no] abandon-tcp-optimization

Context

config>app-assure>group>policy>aqp>entry>action

Description

This command causes TCPO to stop for flows matching this AQP entry. The flows are counted as TCPO abandoned by policy flows.

The no form of this command removes abandon TCPO from actions on flows matching this AQP entry.

Default

no abandon-tcp-optimization

10.2.1.10.49 bandwidth-policer

Syntax

bandwidth-policer *policer-name*

no bandwidth-policer

Context

config>app-assure>group>policy>aqp>entry>action

Description

This command assigns an existing bandwidth policer as an action on flows matching this AQP entry. The match criteria for the AQP entry must specify a uni-directional traffic direction before a policer action can be configured. If a policer is used in one direction in an AQP match entry the same policer name cannot be used by another AQP entry which uses a different traffic direction match criteria.

When multiple policers apply to a single flow, the final action on a packet is the worse case of all policer outcome (for example, if one of the policers marks packet out of profile, the final marking will reflect that).

The no form of this command removes bandwidth policer from actions on flows matching this AQP entry.

Default

no bandwidth-policer

Parameters

policer-name

the name of the existing flow setup rate policer for this application assurance profile

The *policer-name* is configured in the config>app-assure>group>policer context.

10.2.1.10.50 dns-enrich

Syntax

dns-enrich *dns-enrich-name*

no dns-enrich

Context

```
config>app-assure>group>policy>aqp>entry>action
```

Description

This command assigns an existing DNS enrichment template as an action on flows matching this AQP entry. This action is applicable only to the matching DNS flows.

The DNS enrichment template is configured using the `dns-enrichment` command in the `config>app-assure>group>dns-enrichment` context.

Parameters

dns-enrich-name

specifies the name of the DNS enrichment template, up to 256 characters

10.2.10.51 dns-ip-cache

Syntax

```
dns-ip-cache dns-ip-cache-name
```

Context

```
config>app-assure>group>policy>aqp>entry>action
```

Description

This command specifies the DNS IP cache for this application QoS policy entry. The DNS IP cache is configured in the Application Assurance group context.

Parameters

dns-ip-cache-name

specifies the name of the Application Assurance DNS IP cache, up to 32 characters

10.2.10.52 drop

Syntax

```
[no] drop
```

Context

```
config>app-assure>group>policy>aqp>entry>action
```

Description

This command configures the drop action on flows matching this AQP entry. When enabled, all flow traffic matching this AQP entry will be dropped. When drop action is part of a set of multiple actions to be applied to a single flow as result of one or more AQP entry match, drop action will be performed first and no other action will be invoked on that flow.

The no form of this command disables the drop action on flows matching this AQP entry.

Default

```
no drop
```

10.2.10.53 error-drop

Syntax

```
[no] error-drop [event - log event-log-name]
```

```
no error-drop
```

Context

```
config>app-assure>group>policy>aqp>entry>action
```

Description

This command specifies the drop action for error flow events (bad IP checksums, TCP and, or UDP port 0, and so on).

The no form of this command disables the drop action on the error flow events.

Default

```
no error-drop
```

Parameters

event-log-name

The dropped error events will be logged to the specified event log.

10.2.10.54 flow-count-limit

Syntax

```
flow-count-limit policer-name  
no flow-count-limit
```

Context

```
config>app-assure>group>policy>aqp>entry>action
```

Description

This command assigns an existing flow count limit policer as an action on flows matching this AQP entry.

The match criteria for the AQP entry must specify a uni-directional traffic direction before a policer action can be configured. If a policer is used in one direction in an AQP match entry the same policer name cannot be used by another AQP entry which uses a different traffic direction match criteria.

When multiple policers apply to a single flow, the final action on a packet is the worse case of all policer outcome (for example, if one of the policers marks packet out of profile, the final marking will reflect that).

The no form of this command removes this flow policer from actions on flows matching this AQP entry.

Default

```
no flow-count-limit
```

Parameters

policer-name

the name of the existing flow setup rate policer for this application assurance profile

The *policer-name* is configured in the config>app-assure>group>policer context.

10.2.10.55 flow-rate-limit

Syntax

```
flow-rate-limit policer-name  
no flow-rate-limit
```

Context

```
config>app-assure>group>policy>aqp>entry>action
```

Description

This command assigns an existing flow setup rate limit policer as an action on flows matching this AQP entry.

The match criteria for the AQP entry must specify a uni-directional traffic direction before a policer action can be configured. If a policer is used in one direction in an AQP match entry the same policer name cannot be used by another AQP entry which uses a different traffic direction match criteria.

When multiple policers apply to a single flow, the final action on a packet is the worse case of all policer outcome (for example, if one of the policers marks packet out of profile, the final marking will reflect that).

The no form of this command removes this flow policer from actions on flows matching this AQP entry.

Default

no flow-rate-limit

Parameters

policer-name

the name of the existing flow setup rate policer for this application assurance profile

The *policer-name* is configured in the config>app-assure>group>policer context.

10.2.10.56 fragment-drop

Syntax

fragment-drop {all | out-of-order} [event-log *event-log-name*]

no fragment-drop

Context

config>app-assure>group>policy>aqp>entry>action

Description

This command specifies the action to apply to fragments.

The no form of this command removes this fragment drop from actions on this application QoS policy entry.

Default

no fragment-drop

Parameters

all

specifies that all the fragments are dropped

out-of-order

specifies that all out of order fragments in the first packet are dropped

For example, if fragments are received in this order: 4, 5, 1, 3, 2, fragments 4 and 5 are dropped.

event-log-name

specifies that the dropping of fragments are logged to the specified event log

10.2.10.57 http-enrich

Syntax

```
http-enrich http-enrich-name
```

```
no http-enrich
```

Context

```
config>app-assure>group>policy>aqp>entry>action
```

Description

This command configures an HTTP enrichment action for flows matching this entry.

The no form of this command removes the HTTP enrichment action from actions on flows matching this AQP entry.

Default

```
no http-enrich
```

Parameters

http-enrich-name

the name of the existing HTTP enrichment action for this application assurance profile

10.2.10.58 http-error-redirect

Syntax

```
http-error-redirect redirect-name
```



```
no http-redirect
```

Context

```
config>app-assure>group>policy>aqp>entry>action
```

Description

This command configure a HTTP redirect action for flows of a specific type matching this entry.

The no form of this command removes the HTTP redirect action from actions on flows matching this AQP entry.

Default

```
no http-enrich
```

Parameters

redirect-name

the name of the existing HTTP error redirect action for this application assurance profile

10.2.10.59 http-notification

Syntax

```
http-notification http-notification
```

```
no http-notification
```

Context

```
config>app-assure>group>policy>aqp>entry>action
```

Description

This command configures an HTTP notification action for flows matching this entry.

Parameters

http-notification

specifies the Application-Assurance HTTP Notification that will be applied as defined in the tmnxBsxHttpNotifTable

If no string is configured, no HTTP notification occurs.

10.2.10.60 http-redirect

Syntax

```
http-redirect http-redirect-name flow-type flow-type  
no http-redirect
```

Context

```
config>app-assure>group>policy>aqp>entry>action
```

Description

This command assigns an existing HTTP redirect policy as an action on flows matching this AQP entry.

The redirect takes effect only if the matching flows are HTTP/HTTPS (depending on the policy configuration) and the condition specified by the *flow-type*, either *dropped-flows* or *admitted-flows*, is met. The condition specified by *dropped-flows* means that the flow is dropped due to AQP actions such as "flow rate/count policers" or "drop" actions. HTTP policy redirect on *admitted-flows* allows the operator to redirect HTTP/HTTPS traffic to a web portal while allowing non-HTTP matching the same AQP rule to be forwarded.



NOTE:

No HTTP redirect takes place if HTTP redirect action and a "drop/flow-police" action are part of the default AQP policy, because in this case, any flow drop actions take place before identification of the application/application-group.

The no form of this command removes HTTP redirect from actions on flows matching this AQP entry.

Default

```
no http-redirect
```

Parameters

http-redirect-name

specifies the name of the existing HTTP redirect policy for this application assurance profile. The HTTP redirect name is configured in the `config>app-assure>group>http-redirect` command.

flow-type

specifies the flow type

Values:

dropped-flows

redirects those HTTP flows that are dropped due to other AQP actions

admitted-flows

redirects HTTP flows matching the AQP criteria

10.2.1.10.61 mirror-source

Syntax

`mirror-source mirror-service-id [all-inclusive]`

`no mirror-source`

Context

`config>app-assure>group>policy>aqp>entry>action`

Description

This command configures a mirror source action for flows matching this entry.

The no form of this command removes the mirror source action from actions on flows matching this AQP entry.

Default

`no mirror-source`

Parameters

mirror-service-id

the name of the existing mirror source action for this application assurance profile

all-inclusive

enables the mirror source action to be comprehensive

10.2.1.10.62 mp-tcp-proxy

Syntax

`mp-tcp-proxy mp-tcp-proxy-name`

`no mp-tcp-proxy`

Context

`config>app-assure>group>policy>aqp>entry>action`

Description

This command assigns an existing Multi-Path TCP (MP-TCP) proxy policy as an action on flows matching this AQP entry.

The MP-TCP proxy can only be called from AQP entries that:

- have no matching conditions relating to information extracted from AA flow classification; for example, no application, application group, or charging group
- allow the following match conditions:
 - none
 - aa-sub
 - characteristic
 - src-ip
 - dst-ip
 - dst-port
 - ip-protocol-num
 - traffic-direction (both only)
 - traffic-direction cannot be unidirectional (from or to sub). It can either be set to both or left unspecified.

The no form of this command removes the MP-TCP proxy action from flows matching this AQP entry.

Default

no mp-tcp-proxy

Parameters

mp-tcp-proxy-name

specifies the name of the MP-TCP proxy policy for this application assurance profile, up to 32 characters

The MP-TCP proxy policy is configured using the `config>app-assure>group>mp-tcp-proxy` command

10.2.10.63 mp-tcp-scheduler-policy-override

Syntax

`mp-tcp-scheduler-policy-override scheduler-policy`

`no mp-tcp-scheduler-policy-override`

Context

config>app-assure>group>policy>aqp>entry>action

Description

This command configures an override of the MP-TCP proxy scheduling policy for downstream path selection. The no form of this command removes the specified configuration.

Default

no mp-tcp-proxy

Parameters

scheduler-policy

specifies the scheduler policy

Values:

min-rtt-prefer

If TCP subflow congestion windows (CWINs) have space available on both paths or subflows, the path or subflow with a lower RTT value will be preferred for transmission. Otherwise, the path or subflow that has CWIN space available (assuming there is room in the TCP receiver window) will be used for transmission. In the case of bulk data transmission, if the application is able to fill the congestion windows of all subflows, the packets will be scheduled as soon as space is available in each subflow's congestion window ("ack-clock"). The RTT calculation is based on standard TCP algorithms.

min-cost-prefer

If TCP subflow congestion windows (CWINs) have space available on both paths or subflows, the path/subflow with the lower administrative cost will be preferred for transmission. Otherwise, the path or subflow that has CWIN space available will be used for transmission (assuming there is room in the TCP receiver window). In the case of bulk data transmission, if the application is able to fill the congestion windows of all subflows, the packets will be scheduled as soon as space is available in each subflow's congestion window ("ack-clock").

min-cost-handover

The available path or subflow with the lower administrative cost is used for transmission. Transmission occurs when the lower-cost subflow congestion window (CWIN) has space available (assuming there is room in the TCP receiver window). This provides path handover for protection, based on cost and path availability.

max-cost-forbid

Transmission on the path with the higher administrative cost is prevented by preventing subflow establishment on that path. The HAG transmits only to the low cost path and prevents CPE from sending on the maximum cost path.

Default: min-cost-prefer

10.2.10.64 offload

Syntax

[no] offload

Context

config>app-assure>group>policy>aqp>entry>action

Description

This command configures the offload action on flows matching this AQP entry. When enabled, all flows matching this AQP entry are offloaded to the mobile gateway for flow-level processing by fast path (FP) or virtual fast path (vFP) only. Offloading prevents the AA from processing traffic after classification of the flow. Processing of flows by fast path is done to achieve higher throughput performance.

The no form of this command disables the offload action on flows matching this AQP entry.

Default

no offload

10.2.10.65 overload-drop

Syntax

[no] error-drop [event - log *event-log-name*]

no error-drop

Context

config>app-assure>group>policy>aqp>entry>action

Description

This command specifies the drop action for cases where flow records are not created (overload).

The no form of this command disables the overload drop action for the application QoS policy entry.

Default

no overload-drop

Parameters

event-log-name

the dropped flow record events are logged to the specified event log

10.2.1.10.66 remark

Syntax

remark

Context

config>app-assure>group>policy>aqp>entry>action

Description

This command configures remark action on flows matching this AQP entry.

10.2.1.10.67 dscp

Syntax

dscp in-profile *dscp-name* out-profile *dscp-name*

no dscp

Context

config>app-assure>group>policy>aqp>entry>action>remark

Description

This command enables the context to configure DSCP remark action or actions on flows matching this AQP entry. When enabled, all packets for all flows matching this AQP entry will be remarked to the configured DSCP name.

DSCP remark can only be applied when the entry remarks forwarding class or forwarding class and priority. In-profile and out-of profile of a given packet for DSCP remark is assessed after all AQP policing and priority remarking actions took place.

The no form of this command stops DSCP remarking action on flows matching this AQP entry.

Parameters

in-profile dscp-name

specifies the DSCP name to use to remark in-profile flows that match this policy

out-profile *dscp-name*

specifies the DSCP name to use to remark out-of-profile flows that match this policy

Values: be | cp1 | cp2 | cp3 | cp4 | cp5 | cp6 | cp7 | cs1 | cp9 | af11 | cp11 | af12 | cp13 | af13 | cp15 | cs2 | cp17 | af21 | cp19 | af22 | cp21 | af23 | cp23 | cs3 | cp25 | af31 | cp27 | af32 | cp29 | af33 | cp31 | cs4 | cp33 | af41 | cp35 | af42 | cp37 | af43 | cp39 | cs5 | cp41 | cp42 | cp43 | cp44 | cp45 | ef | cp47 | nc1 | cp49 | cp50 | cp51 | cp52 | cp53 | cp54 | cp55 | nc2 | cp57 | cp58 | cp59 | cp60 | cp61 | cp62 | cp63

10.2.10.68 fc

Syntax

`fc fc-name`

`no fc`

Context

`config>app-assure>group>policy>aqp>entry>action>remark`

Description

This command configures remark FC action on flows matching this AQP entry. When enabled, all packets for all flows matching this AQP entry will be remarked to the configured forwarding class.

The no form of this command stops FC remarking action on packets belonging to flows matching this AQP entry

Parameters

fc-name

Configure the FC remark action for flows matching this entry.

Values: be | l2 | af | l1 | h2 | ef | h1 | nc

10.2.10.69 priority

Syntax

`priority priority-level`

`no priority`

Context

config>app-assure>group>policy>aqp>entry>action>remark

Description

This command configures remark discard priority action on flows matching this AQP entry. When enabled, all packets for all flows matching this AQP entry will be remarked to the configured discard priority.

Default

no priority

Parameters

priority-level

specifies the priority to apply to a packet

Values: high, low

10.2.1.10.70 session-filter

Syntax

session-filter *session-filter-name*

no session-filter

Context

config>app-assure>group>policy>aqp>entry>action

Description

This command configures a session filter for flows matching this entry.

AQPs with session filter action can have the following as matching conditions:

- traffic direction
- ASOs
- and/or subscriber name

The no form of this command removes the session filter from actions on flows matching this AQP entry.

Default

no session-filter

Parameters

session-filter-name

the name of the existing session filter for this application assurance profile

10.2.10.71 tcp-mss-adjust

Syntax

tcp-mss-adjust *segment-size*

no tcp-mss-adjust

Context

config>app-assure>group>policy>aqp>entry>action

Description

This command configures the value to adjust the TCP Maximum Segment Size (MSS) option. This enables the TCP MSS field to be set for the AA group policy to prevent packet fragmentation. This triggers the interception of transit TCP packets with the SYN bit set and consequently rewrites the MSS field of the intercepted packet. The MSS field will only be rewritten if the configured TCP MSS adjust value is smaller than the value inside the IPv4 intercepted packet, or if the configured TCP MSS adjust value minus 20 is smaller than the value inside an IPv6 intercepted packet. If the given TCP option field is not present in the TCP SYN packet, it will be inserted with the preconfigured value for IPv4 packets or with the preconfigured value minus 20 for IPv6 packets.

The no form of this command reverts the segment size to the default.

Parameters

segment-size

specifies the value to put into the TCP Maximum Segment Size (MSS) option if not already present, or if the present value is higher for IPv4 TCP flows. For IPv6, the value is automatically adjusted down to account for the 20 Bytes of extra IP header, used by IPv6. Specifically, for IPv6 packets, the applied MSS value is equal to segment-size meaning 20 Bytes.

Values: 0 | 160 to 10240

Default: 0

10.2.10.72 tcp-validate

Syntax

```
tcp-validate tcp-validate-name  
no tcp-validate
```

Context

```
config>app-assure>group>policy>aqp>entry>action
```

Description

This command assigns an existing TCP validation policy as an action on flows matching this AQP entry.

`tcp-validate` can only be called from AQP entries that:

- have no matching conditions that relate to information extracted from the incoming IP packets; for example, no application or IP address.
- allow the following match conditions:
 - none
 - aa-sub
 - characteristic
 - traffic-direction (both only) traffic-direction cannot be unidirectional (from or to sub). It can either be set to both or left unspecified.

The `no` form of this command removes the `tcp-validate` action from flows matching this AQP entry.

Default

```
no tcp-validate
```

Parameters

tcp-validate-name

specifies the name of the TCP validation policy for this application assurance profile

The TCP validation policy is configured using the `config>app-assure>group>tcp-validate tcp-validate-name` command.

10.2.1.10.73 **tls-enrich**

Syntax

```
tls-enrich tls-enrich-name  
no tls-enrich
```

Context

```
config>app-assure>group>policy>aqp>entry>action
```

Description

This command configures a TLS enrichment action for flows matching this AQP entry.

The no form of this command removes the TLS enrichment action from actions on flows matching this AQP entry.

The TLS extension to be used for HTTPS enrichment is defined using the `config>aa>group>http-enrich>tls-extension` and `config>aa>group>http-enrich>tls-extension>subtype` commands.

Default

```
no tls-enrich
```

Parameters

tls-enrich-name

specifies the name of the HTTP enrichment action for this application assurance profile, up to 32 characters

10.2.1.10.74 **url-filter**

Syntax

```
url-filter url-filter-name characteristic characteristic-name  
no url-filter
```

Context

```
config>app-assure>group>policy>aqp>entry>action
```

Description

This command configures a URL filter action for flows matching this entry.

Parameters

url-filter-name

specifies the name of the URL filter, up to 32 characters

characteristic-name

optional characteristic used to select the profile to be applied for the web-service

10.2.1.10.75 match

Syntax

match

Context

config>app-assure>group>policy>aqp>entry

Description

This command enables the context to configure flow match rules for this AQP entry. A flow matches this AQP entry only if it matches all the match rules defined (logical and of all rules). If no match rule is specified, the entry will match all flows.

10.2.1.10.76 aa-sub

Syntax

aa-sub esm {eq | neq} *sub-ident-string*

aa-sub sap {eq | neq} *sap-id*

aa-sub spoke-sdp {eq | neq} *sdp-id:vc-id*

aa-sub transit {eq | neq} *transit-aasub-name*

no aa-sub

Context

config>app-assure>group>policy>aqp>entry>match

Description

This command specifies a Service Access Point (SAP) or an ESM subscriber as matching criteria.

The no form of this command removes the SAP or ESM matching criteria.

Parameters

eq

specifies that the value configured and the value in the flow are equal

neq

specifies that the value configured differs from the value in the flow

sub-ident-string

specifies the name of an existing application assurance subscriber

sap-id

specifies the SAP ID

sap sap-id

specifies the physical port identifier portion of the SAP definition

sdp-id:vc-id

specifies the spoke SDP ID and VC ID

Values: 1 to 17407 : 1 to 4294967295

transit-aa-sub-name

specifies the name of a transit AA subscriber

10.2.10.77 aa-sub-tethering-state

Syntax

aa-sub-tethering-state {detected | not-detected}

no aa-sub-tethering-state

Context

config>app-assure>group>policy>aqp>entry>match

Description

This command specifies the Tethering state of the subscriber where the AQP match entry will be applied.

The tethering state match condition is meaningful when configured in non-default subscriber policy AQP. Default subscriber policy consists of those AQPs that include match criteria based on subscriber's

configuration. On the other hand, tethering state match condition is applicable in those AQPs that include matching criteria that are derived from actual subscriber's traffic.

Default

no aa-sub-tethering-state

Parameters

detected

specifies subscriber is in tethering state

not-detected

specifies subscriber is not in tethering state

10.2.10.78 app-group

Syntax

app-group {eq | neq} *application-group-name*

no app-group

Context

config>app-assure>group>policy>aqp>entry>match

Description

This command adds app-group to match criteria used by this AQP entry.

The no form of this command removes the app-group from match criteria for this AQP entry.

Default

no app-group

Parameters

eq

specifies that the value configured and the value in the flow are equal

neq

specifies that the value configured differs from the value in the flow

application-group-name

the name of the existing application group entry

The application-group-name is configured in the `config>app-assure>group>policy>aqp>entry>match` context.

10.2.1.10.79 application

Syntax

```
application {eq | neq} application-name  
no application
```

Context

```
config>app-assure>group>policy>aqp>entry>match
```

Description

This command adds an application to match criteria used by this AQP entry.

The no form of this command removes the application from match criteria for this AQP entry.

Default

```
no application
```

Parameters

eq

specifies that the value configured and the value in the flow are equal

neq

specifies that the value configured differs from the value in the flow

application-name

specifies the name of name existing application name

The application-group-name is configured in the `config>app-assure>group>policy>aqp>entry>match` context.

10.2.1.10.80 characteristic

Syntax

```
characteristic characteristic-name {eq | neq} value-name  
no characteristic
```


Context

```
config>app-assure>group>policy>aqp>entry>match
```

Description

This command adds an existing characteristic and its value to the match criteria used by this AQP entry.

The no form of this command removes the characteristic from match criteria for this AQP entry.

Default

no characteristic

Parameters

eq

specifies that the value configured and the value in the ASO characteristic are equal

neq

specifies that the value configured differs from the value in the ASO characteristic

characteristic-name

specifies the name of the existing ASO characteristic, up to 32 characters

value-name

specifies the name of an existing value for the characteristic, up to 32 characters

10.2.10.81 charging-group

Syntax

```
charging-group {eq | neq} charging-group-name
```

```
no charging-group
```

Context

```
config>app-assure>group>policy>aqp>entry>match
```

Description

This command configures a charging group to match against to resolve to actions.

The no form of this command removes the charging group from match criteria for this AQP entry.

Default

no charging-group

Parameters

eq

specifies that the value configured and the value in the flow are equal

neq

specifies that the value configured differs from the value in the flow

charging-group-name

the charging group name to be used in match, up to 32 characters

10.2.10.82 dscp

Syntax

```
dscp {eq | neq} dscp-name
```

```
no dscp
```

Context

```
config>app-assure>group>policy>aqp>entry>match
```

Description

This command adds a DSCP name to the match criteria used by this AQP entry.

The no form of this command removes dscp from match criteria for this AQP entry.

Default

```
no dscp
```

Parameters

eq

specifies that the value configured and the value in the flow are equal

neq

specifies that the value configured differs from the value in the flow

dscp-name

the DSCP name to be used in match

Values: be | cp1 | cp2 | cp3 | cp4 | cp5 | cp6 | cp7 | cs1 | cp9 | af11 | cp11 | af12 | cp13 | af13 | cp15 | cs2 | cp17 | af21 | cp19 | af22 | cp21 | af23 | cp23 | cs3 | cp25 | af31 | cp27 | af32 | cp29 | af33 | cp31 | cs4 | cp33 | af41 | cp35 | af42 | cp37 | af43 | cp39 | cs5 | cp41 | cp42 | cp43 | cp44 | cp45 | ef | cp47 | nc1 | cp49 | cp50 | cp51 | cp52 | cp53 | cp54 | cp55 | nc2 | cp57 | cp58 | cp59 | cp60 | cp61 | cp62 | cp63

10.2.1.10.83 dst-ip

Syntax

```
dst-ip {eq | neq} {<ip-address [/mask]> | ip-prefix-list <prefix-list-name>}  
no dst-ip
```

Context

```
config>app-assure>group>policy>aqp>entry>match
```

Description

This command specifies a destination IP address to use as match criteria.

Parameters

eq

specifies a that a successful match occurs when the flow matches the specified address or prefix

neq

specifies that a successful match occurs when the flow does not match the specified address or prefix

ip-address

specifies a valid unicast address

mask

specifies the mask associated with the IP address as a mask length, for example, /16 for a sixteen-bit mask

Values: 1 to 32

ip-prefix-list-name

specifies an IP address prefix list for the destination IP address match criteria, up to 32 characters

10.2.1.10.84 dst-port

Syntax

```
dst-port {eq | neq} port-num  
dst-port {eq | neq} port-list port-list-name  
dst-port {eq | neq} range start-port-num end-port-num
```

no dst-port

Context

config>app-assure>group>policy>aqp>entry>match

Description

This command specifies a destination TCP/UDP port, destination port list, or destination range to use as match criteria.

The no form of this command removes the parameters from the configuration.

Parameters

eq

specifies that a successful match occurs when the flow matches the specified port

neq

specifies that a successful match occurs when the flow does not match the specified port

port-num

specifies the destination port number

Values: 0 to 65535

port-list-name

specifies a named port-list, up to 32 characters, containing a set of ports or ranges of ports

start-port-num

specifies the start destination port number

Values: 0 to 65535

end-port-num

specifies the end destination port number

Values: 0 to 65535

10.2.1.10.85 flow-attribute

Syntax

[no] flow-attribute *flow-attribute name*

Context

config>app-assure>group>policy>aqp>entry>match

Description

This command configures a flow attribute to use as match criteria.

The no form of this command reverts to the default value.

Default

no flow-attribute

Parameters

flow-attribute name

specifies the name of the flow attribute, up to 256 characters

10.2.1.10.86 confidence

Syntax

confidence eq *equal-value* | gte *greater-than-or-equal-value* | lt *less-than-value*

Context

config>app-assure>group>policy>aqp>entry>match>flow-attribute

config>app-assure>group>policy>charging-filter>entry>match>flow-attribute

Description

This command configure the confidence level of the flow attribute to use as match criteria.

Parameters

equal-value

specifies that a successful match occurs when the flow attribute confidence level is equal to the specified value

Values: 0 to 100

greater-than-or-equal-value

specifies that a successful match occurs when the flow attribute confidence level is greater than or equal to the specified value

Values: 0 to 100

less-than-value

specifies that a successful match occurs when the flow attribute confidence level is less than the specified value

Values: 1 to 100

10.2.1.10.87 ip-protocol-num

Syntax

```
ip-protocol-num {eq | neq} protocol-id  
no ip-protocol-num
```

Context

```
config>app-assure>group>policy>aqp>entry>match
```

Description

This command configures the IP protocol number to match against to resolve to actions.
The no form of this command removes protocol from match criteria for this AQP entry.

Default

```
no ip-protocol-num
```

Parameters

- eq

specifies that the value configured and the value in the flow are equal
- neq

specifies that the value configured differs from the value in the flow
- protocol-id*

the IP protocol identifier to be used in match

Values	1 to 255 (decimal, hexadecimal, or binary representation).
–	Supported IANA IP protocol names: crtp crudp egp eigrp encap ether-ip gre icmp idrp igmp igp ip ipv6 ipv6-frag ipv6-icmp ipv6-no-nxt ipv6-opts ipv6-route isis iso-ip l2tp ospf-igp pim pnni ptp rdp rsvp sctp stp tcp udp vrrp

10.2.10.88 multi-device-detection

Syntax

```
multi-device-detection {eq | neq} device-type device-type  
no multi-device-detection
```

Context

```
config>app-assure>group>policy>aqp>entry>match
```

Description

This command specifies a source TCP/UDP address as match criteria.

Parameters

eq

specifies that the configured value and the value in the flow are equal

neq

specifies that the configured value differs from the value in the flow

device-type

specifies the device type used

Values:

host

the host device

tracked

a device within the max-devices range

excess

a device exceeding the configured maximum devices

10.2.10.89 presence-reporting-area

Syntax

```
presence-reporting-area {eq | neq} PRA-value  
no presence-reporting-area
```

Context

```
config>app-assure>group>policy>aqp>entry>match
```

Description

This command specifies the subscriber PRA to match the criteria used by this AQP entry.

The no form of this command removes the PRA from the match criteria for this AQP entry.



NOTE:

This command must not be used in case of overlapping PRAs.

Parameters

eq

specifies that the value configured and the value in the flow are equal

neq

specifies that the value configured differs from the value in the flow

PRA-value

specifies the value of the PRA for the AQP entry

Values: 1 to 16777215

10.2.10.90 rat-type

Syntax

```
rat-type {eq | neq} {utran | geran | wlan | gan | hspa-evol | eutran | virtual | eutran-nb |  
ehrpd | hrrpd | cdma-1x | umb | wifi | nr | lte-m | wireline | wireline-cable | wireline-bbf  
| nr-u | eutra-u | trusted-n3ga | trusted-wlan}  
no rat-type
```

Context

```
config>app-assure>group>policy>aqp>entry>match
```

Description

This command specifies the RAT type to match the criteria used by this AQP entry.

The no form of this command removes the RAT type from the match criteria for this AQP entry.

Parameters

eq

specifies that the value configured and the value in the flow are equal

neq

specifies that the value configured differs from the value in the flow

keyword

specifies the RAT type

Values: utran | geran | wlan | gan | hspa-evol | eutran | virtual | eutran-nb | ehrpd | hrpd |
cdma-1x | umb | wifi | nr | lte-m | wireline | wireline-cable | wireline-bbf | nr-u | eutra-u |
trusted-n3ga | trusted-wlan

10.2.1.10.91 roaming-status

Syntax

```
roaming-status {eq | neq} {home | roaming | visiting}
```

```
roaming-status
```

Context

```
config>app-assure>group>policy>aqp>entry>match
```

Description

This command specifies the subscriber roaming status to match the criteria used by this AQP entry.

The no form of this command removes the roaming status from the match criteria for this AQP entry.

Parameters**eq**

specifies the comparison operator where the value configured and the value in the flow are equal

neq

specifies the comparison operator where the value configured differs from the value in the flow

Values home | roaming | visiting

10.2.1.10.92 src-ip

Syntax

```
src-ip {eq | neq} {ip-address [/mask] | ip-prefix-list prefix-list-name}  
no src-ip
```

Context

```
config>app-assure>group>policy>aqp>entry>match
```

Description

This command specifies a source TCP/UDP address to use as match criteria.

Parameters

eq

specifies that a successful match occurs when the flow matches the specified address or prefix

neq

specifies that a successful match occurs when the flow does not match the specified address or prefix

ip-address

specifies a valid IPv4 or IPv6 unicast address

mask

the mask associated with the IP address as a mask length, for example, /16 for a sixteen-bit mask

Values: 1 to 32

ip-prefix-list-name

specifies an IP address prefix list for the source IP address match criteria, up to 32 characters

10.2.1.10.93 src-port

Syntax

```
src-port {eq | neq} port-num  
src-port {eq | neq} port-list port-list-name  
src-port {eq | neq} range start-port-num end-port-num
```

```
no src-port
```

Context

```
config>app-assure>group>policy>aqp>entry>match
```

Description

This command specifies a source IP port, source port list, or source range to use as match criteria.

The no form of this command removes the parameters from the configuration.

Parameters

eq

specifies that a successful match occurs when the flow matches the specified port

neq

specifies that a successful match occurs when the flow does not match the specified port

port-num

specifies the source port number

Values: 0 to 65535

start-port-num

specifies the start source port number

Values: 0 to 65535

end-port-num

specifies the end source port number

Values: 0 to 65535

port-list-name

specifies a named port-list, up to 32 characters, containing a set of ports or ranges of ports

10.2.10.94 traffic-direction

Syntax

```
traffic-direction {subscriber-to-network | network-to-subscriber | both}
```

Context

```
config>app-assure>group>policy>aqp>entry>match
```

Description

This command specifies the direction of traffic where the AQP match entry will be applied.

To use a policer action with the AQP entry the match criteria must specify a traffic-direction of either subscriber-to-network or network-to-subscriber.

Parameters

subscriber-to-network

traffic from a local subscriber will match this AQP entry

network-to-subscriber

traffic to a local subscriber will match this AQP entry

both

combines subscriber-to-network and network-to-subscriber

Default: both

10.2.1.10.95 wireless-device-os

Syntax

```
wireless-device-os {eq | neq} {other | android | ios | windows} [{other | android | ios | windows}...(up to 4 max)]  
no wireless-device-os
```

Context

```
config>app-assure>group>policy>aqp>entry>match
```

Description

This command specifies the subscribers device OS as a match criteria to be used by this AQP entry.

The no form of the command removes the device OS from the AQP entry match criteria.

Parameters

eq

specifies that the value configured and the value in the flow are equal

neq

specifies that the value configured differs from the value in the flow

other

specifies a value for the subscriber's device OS other than Android, IOS, or Windows

android

specifies that the subscriber's device OS is Android

ios

specifies that the subscriber's device OS is IOS

windows

specifies that the subscriber's device is Windows

**NOTE:**

The operator can list up to four device type values on the same CLI match condition line with an IMPLICIT "or" operation among them.

10.2.1.10.96 wireless-device-type

Syntax

```
wireless-device-type {eq | neq} {other | tablet | phone} [{other | tablet | phone}...(up to 3 max)]
```

```
no wireless-device-type
```

Context

```
config>app-assure>group>policy>aqp>entry>match
```

Description

This command specifies the subscribers device type as a match criteria to be used by this AQP entry.

The no form of the command removes the device type from the AQP entry match criteria.

Parameters

eq

specifies that the value configured and the value in the flow are equal

neq

specifies that the value configured differs from the value in the flow

other

specifies a value for the subscriber's device type other than tablet or phone

tablet

specifies that the subscriber's device type is tablet

phone

specifies that the subscriber's device type is phone

**NOTE:**

The operator can list up to three device type values on the same CLI match condition line with an IMPLICIT "or" operation among them.

10.2.10.97 custom-protocol

Syntax

`custom-protocol custom-protocol-id ip-protocol-num protocol-id [create]`

`custom-protocol custom-protocol-id`

`no custom-protocol custom-protocol-id`

Context

`config>app-assure>group>policy`

Description

This command creates and enters configuration context for custom protocols. Custom protocols allow the creation of TCP and UDP-based custom protocols (based on the *ip-protocol-num* option) that employ pattern-match at offset in protocol signature definition.

Operator-configurable custom-protocols are evaluated ahead of any Nokia provided protocol signature in order of `custom-protocol-id` (the lower ID is matched first in case of flow matching multiple custom-protocols) within the context the protocol is defined.

Custom protocols must be created before they can be used in application definition but do not have to be enabled. To reference a custom protocol in application definition, or any other CLI configuration one must use protocol name that is a concatenation of "custom_" and <custom-protocol-id>, (for example custom_01, custom_02,..., custom_10, and so on). This concatenation is also used when reporting custom protocol statistics.

Parameters

custom-protocol-id

specifies the index into the protocol list that defines a custom protocol for application assurance

Values: 1 to 10

protocol-id

specifies the IP protocol to match against for the custom protocol

Values: 0 to 255, protocol numbers accepted in DHB

Keywords: none | crtp | crudp | egp | eigrp | encap | ether-ip | gre | icmp | idrp | igmp | igp | ip | ipv6 | ipv6-frag | ipv6-icmp | ipv6-no-nxt | ipv6-opts | ipv6-route | isis | iso-ip | l2tp | ospf-igp | pim | pnni | ptp | rdp | rsvp | stp | tcp | udp | vrrp | * - udp/tcp wildcard

create

mandatory keyword used when creating custom protocol. The create keyword requirement can be enabled/disabled in the environment>create context.

10.2.10.98 expression

Syntax

```
expression expr-index eq expr-string offset payload-octet-offset direction direction
no expression expr-index
```

Context

```
config>app-assure>group>policy>custom-protocol
```

Description

This command configures an expression string value for pattern-based custom protocols match. A flow matches a custom protocol if the specified string is found at an offset of a TCP/UDP of the first payload packet.

Options:

client-to-server	a pattern will be matched against a flow from a TCP client
server-to-client	a pattern will be matched against a flow from a TCP server
any	a pattern will be matched against a TCP/UDP flow in any direction (toward or from AA subscriber)

The no form of this command deletes a specified string expression from the definition.

Parameters

expr-index

specifies the expression substring index

Values: 1

expr-string

denotes a printable ASCII string, up to 16 characters, used to define a custom protocol match. Rules for expr-string characters:

- Must contain printable ASCII characters.
- Must not contain the “double quote” character or the space character on its own.
- Match is case sensitive.
- Must not include any regular expression meta-characters.

The “\” (slash) character is used as an ESCAPE sequence. The following ESCAPE sequences are permitted within the expr-string:

Character to match expr-string input

Hexadecimal Octet YY \xYY

**NOTE:**

An expr-string that uses the '\' (backslash) ESCAPE character which is not followed by a “\” or “\x” and a 2-digit hex octet is not valid.

payload-octet-offset

specifies the offset (in octets) into the protocol payload, where the expr-string match criteria will start.

Values: 0 to 127

direction

specifies the protocol direction to match against to resolve to a custom protocol

Values: client-to-server | server-to-client | any

10.2.1.11 Application Assurance group session filter command descriptions

10.2.1.11.1 session-filter

Syntax

```
session-filter session-filter-name [create]
```

```
no session-filter
```

Context

```
config>app-assure>group[:partition]
```


Description

This command configures a session filter for flows matching this entry.

The no form of this command removes the session filter from actions on flows matching this group partition entry.

Default

no session-filter

Parameters

session-filter-name

the name of the existing session filter for this application assurance profile

create

keyword used to create the filter

10.2.1.11.2 default-action

Syntax

```
default-action {permit | deny}
```

Context

```
config>app-assure>group[:partition]>session-filter
```

Description

This command configures a default action.

The no form of this command removes the default action from actions on flows matching this group partition entry.

Default

no default-action

Parameters

permit

permits the flow

deny

blocks the flow

10.2.111.3 entry

Syntax

```
[no] entry entry-id [create]
```

Context

```
config>app-assure>group[:partition]>session-filter
```

Description

This command creates a policy entry. A flow that matches multiple policies entries will have multiple entries actions applied. When a conflict occurs for two or more actions, the action from the entry with the lowest numerical value takes precedence.

The no form of this command deletes the specified policy entry.

Parameters

entry-id

an integer identifying the entry

Values: 1 to 65535

create

mandatory keyword creates the entry

The create keyword requirement can be enabled/disabled in the environment>create context.

10.2.111.4 action

Syntax

```
action {permit | deny | tcp-strict-order} [event - log event-log-name]
```

Context

```
config>app-assure>group[:partition]>session-filter>entry
```

Description

This command enables the context to configure actions to be performed on flows that match the entry's match criteria.

Parameters

permit

permits the flow

deny

blocks the flow

tcp-strict-order

drops out-of-order segments that match the session-filter entry

event - log *event-log-name*

specifies the event log name, up to 32 characters

10.2.11.5 action

Syntax

```
action redirect [dst-ip ip-address] [dst-port tcp/udp-port] [event - log event-log-name]
```

Context

```
config>app-assure>group>sess-fltr>entry
```

Description

This command configures a session filter entry action to perform L3/L4 traffic redirection. For the matching flows, AA replaces the destination address in the uplink direction (IP and/or port) with the values configured in this command. In the reverse direction (toward the UE), AA restores the original IP address/port numbers (as the source IP address/port fields).

If the redirect action is removed from the session filter, the existing redirected flows are not affected, and redirection will continue for those flows. However, no redirect will be applied to newly established flows.



NOTE:

In CMG deployments, for L3/L4 traffic redirection to work, load balancing among CPU cores cannot use an IP address. To enable L3/L4 redirection, the operator must disable `vm-traffic-distribution-by-ip` under the `config>isa>app-assure-group aa-sub-scale` context.

Parameters

dst-ip *ip-address*

specifies a valid unicast server IPv4 or IPv6 address to which to forward traffic

dst-port *tcp/udp-port*

specifies a valid TCP/UDP server destination port number

event - log *event-log-name*

specifies the event log name, up to 32 characters

10.2.111.6 action

Syntax

```
action tcp-optimizer tcp-optimizer-name
```

Context

```
config>app-assure>group>sess-fltr>entry
```

Description

This command configures a session filter entry action to TCP-Optimize the matching flows.

The TCP optimizer policy referenced within this session filter entry is configured under the AA group.

If the TCP-optimizer action is removed from a session-filter entry, the existing flows are not affected.

However, no new TCP flows are optimized.

Parameters

tcp-optimizer-name

specifies the name of the TCP optimizer policy, up to 32 characters

10.2.111.7 match

Syntax

```
match
```

Context

```
config>app-assure>group[:partition]>session-filter>entry
```

Description

This command enables the context to configure flow match rules for this entry. A flow matches this entry only if it matches all the match rules defined. If no match rule is specified, the entry will match all flows.

10.2.111.8 dst-ip

Syntax

```
dst-ip ip-address[/mask]  
dst-ip ip-prefix-list ip-prefix-list-name  
no dst-ip
```

Context

```
config>app-assure>group[:partition]>session-filter>entry>match
```

Description

This command specifies a destination IP address to use as match criteria.

Parameters

ip-address

specifies a valid unicast address

mask

the mask associated with the IP address as a mask length; for example, /16 for a sixteen-bit mask

Values: 1 to 32

ip-prefix-list-name

specifies an IP address prefix list for the destination IP address match criteria

10.2.111.9 dst-port

Syntax

```
dst-port {eq |gt | lt} port-num  
dst-port port-list port-list-name  
dst-port range start-port-num end-port-num  
no dst-port
```

Context

```
config>app-assure>group[:partition]>session-filter>entry>match
```

Description

This command specifies a destination TCP/UDP port, destination port list, or destination range to use as match criteria.

The no form of this command removes the parameters from the configuration.

Parameters

eq

specifies that a successful match occurs when the flow matches the specified port

gt

specifies all port numbers greater than the port-number match

lt

specifies all port numbers less than the port-number match

port-num

specifies the destination port number

Values: 0 to 65535

start-port-num

specifies the start destination port number

Values: 0 to 65535

end-port-num

specifies the end destination port number

Values: 0 to 65535

port-list-name

specifies a named port-list, up to 32 characters, containing a set of ports or ranges of ports

10.2.1.11.10 ip-protocol-num

Syntax

`ip-protocol-num {eq | neq} protocol-id`

`no ip-protocol-num`

Context

`config>app-assure>group[:partition]>session-filter>entry>match`

Description

This command configures the IP protocol number to match against to resolve to actions.

The no form of this command removes protocol from match criteria for this AQP entry.

Default

no ip-protocol-num

Parameters

eq

specifies that the value configured and the value in the flow are equal

neq

specifies that the value configured differs from the value in the flow

protocol-id

the IP protocol identifier to be used in match

Values: 1 to 255 (decimal, hexadecimal, or binary representation)

Supported IANA IP protocol names: crtp | crudp | egp | eigrp | encap | ether-ip | gre | icmp | idrp | igmp | igp | ip | ipv6 | ipv6-frag | ipv6-icmp | ipv6-no-nxt | ipv6-opts | ipv6-route | isis | iso-ip | l2tp | ospf-igp | pim | pnni | ptp | rdp | rsvp | sctp | stp | tcp | udp | vrrp

10.2.1.11.11 src-ip

Syntax

src-ip *ip-address*[/*mask*]

src-ip ip-prefix-list *ip-prefix-list-name*

no src-ip

Context

config>app-assure>group[:partition]>session-filter>entry>match

Description

This command specifies a source TCP/UDP address to use as match criteria.

Parameters

ip-address

specifies a valid IPv4 or IPv6 unicast address

mask

the mask associated with the IP address as a mask length, for example /16 for a sixteen-bit mask

Values: 1 to 32

ip-prefix-list-name

specifies an IP address prefix list for the source IP address match criteria

10.2.1.11.12 **src-port**

Syntax

```
src-port {eq | gt | lt} port-num  
src-port port-list port-list-name  
src-port range start-port-num end-port-num  
no src-port
```

Context

```
config>app-assure>group[:partition]>session-filter>entry>match
```

Description

This command specifies a source IP port, source port list, or source range to use as match criteria. The no form of this command removes the parameters from the configuration.

Parameters

eq

specifies that a successful match occurs when the flow matches the specified port

gt

specifies all port numbers greater than the port-number match

lt

specifies all port numbers less than the port-number match

port-num

specifies the source port number

Values: 0 to 65535

start-port-num

specifies the start source port number

Values: 0 to 65535

end-port-num

specifies the end source port number

Values: 0 to 65535

port-list-name

specifies a named port-list, up to 32 characters, containing a set of ports or ranges of ports

10.2.1.12 Application Assurance group statistics command descriptions

10.2.1.12.1 statistics

Syntax

```
statistics
```

Context

```
config>app-assure>group
```

Description

This command enables the context to configure accounting and billing statistics for this AA ISA group.

10.2.1.12.2 performance

Syntax

```
performance
```

Context

```
config>isa>aa-grp>statistics
```

Description

This command configures the ISA group to enable the aa-performance statistic record. This record contains information on the traffic load and resource consumption for each ISA in the group, to allow tracking of ISA load for long term capacity planning and short term anomalies. The user can configure the accounting policy to be used, and enables the record using the `[no] collect-stats` command.

10.2.1.12.3 aa-admit-deny

Syntax

```
aa-admit-deny
```

Context

```
config>app-assure>group>statistics
```

Description

This command enables the context to configure admit-deny statistics generation.

10.2.1.12.4 accounting-policy

Syntax

```
accounting-policy acct-policy-id  
no accounting-policy
```

Context

```
config>app-assure>group>statistics>aa-admit-deny  
config>app-assure>group>statistics>aa-partition  
config>app-assure>group>statistics>app  
config>app-assure>group>statistics>app-grp  
config>app-assure>group>statistics>protocol  
config>isa>aa-grp>statistics
```

Description

This command specifies the existing accounting policy to use for AA. Accounting policies are configured in the config>log>accounting-policy context.

Parameters

acct-policy-id

specifies the existing accounting policy to use for applications

Values: 1 to 99

10.2.1.12.5 collect-stats

Syntax

```
[no] collect-stats
```

Context

```
config>app-assure>group>statistics>aa-admit-deny
config>app-assure>group>statistics>aa-partition
config>app-assure>group>statistics>app-grp
config>app-assure>group>statistics>application
config>app-assure>group>statistics>protocol
config>isa>aa-grp>statistics
```

Description

This command enables statistics collection within the applicable context.

Default

no collect-stats

10.2.1.12.6 policer-stats

Syntax

```
[no] policer-stats
```

Context

```
config>app-assure>group>statistics>aa-admit-deny
```

Description

This command configures whether to include or exclude system and subscriber-level flow count and flow setup rate policer admit-deny statistics in accounting records.

Default

no policer-stats

10.2.1.12.7 policer-stats-resources

Syntax

```
[no] policer-stats-resources
```

Context

```
config>app-assure>group>statistics>aa-admit-deny
```

Description

This command allows the operator to allocate or deallocate AA partition resources for policer admit-deny statistics.

Default

no policer-stats-resources

10.2.1.12.8 session-filter-stats

Syntax

```
[no] session-filter-stats
```

Context

```
config>app-assure>group>statistics>aa-admit-deny
```

Description

This command configures whether to include or exclude session filter admit-deny statistics in accounting records.

Default

no session-filter-stats

10.2.1.12.9 tcp-validate-stats

Syntax

```
[no] tcp-validate-stats
```

Context

```
config>app-assure>group>statistics>aa-admit-deny
```

Description

This command configures whether to include or exclude TCP validation admit-deny statistics in accounting records.

Default

no tcp-validate-stats

10.2.1.12.10 aa-partition

Syntax

aa-partition

Context

config>app-assure>group>statistics

Description

Enables the context to configure accounting and statistics collection parameters per system for application groups of application assurance for the AA partition.

The no form of this command removes AA partition statistics collection.

10.2.1.12.11 aa-sub

Syntax

app-sub

Context

config>app-assure>group>statistics

Description

This command enables the context to configure per-subscriber statistics generation.

10.2.1.12.12 **exclude-tcp-retrans**

Syntax

[no] exclude-tcp-retrans

Context

config>app-assure>group>statistics>aa-sub

Description

This command enables excluding billing of TCP retransmissions.

The no form of this command disables excluding billing of TCP retransmissions.

Default

no exclude-tcp-retrans

10.2.1.12.13 **app-group**

Syntax

[no] app-group *app-group-name*

Context

config>app-assure>group>statistics

Description

This command enables the context to configure accounting and statistics collection parameters per system for application groups of application assurance for a given AA ISA group/partition.

The no form of this command removes the application group name.

Parameters

app-group-name

specifies an existing application group name, up to 32 characters

10.2.12.14 application

Syntax

[no] application *application-name*

Context

config>app-assure>group>statistics

Description

This command enables the context to configure accounting and statistics collection parameters per system for application groups of application assurance for a given AA ISA group/partition.

The no form of this command removes the application name.

Parameters

application-name

specifies an existing application name, up to 32 characters

10.2.12.15 protocol

Syntax

protocol

Context

config>app-assure>group>statistics

Description

This command enables the context to configure accounting and statistics collection parameters per-system for protocols of application assurance for a given AA ISA group/partition.

10.2.12.16 traffic-type

Syntax

[no] traffic-type

Context

```
config>app-assure>group>statistics>aa-partition
```

Description

This command enables or disables traffic type statistics collection within the AA partition. For more information about traffic type statistics, refer to the section *AA Partition Traffic Type Statistics* in the *CMG Configuration Guide*.

Default

no traffic-type

10.2.12.17 threshold-crossing-alert

Syntax

```
threshold-crossing-alert
```

Context

```
config>app-assure>group>statistics
```

Description

This command enables the context to configure the generation of threshold crossing alerts (TCAs).

10.2.12.18 error-drop

Syntax

```
error-drop direction direction [create]
```

```
no error-drop direction direction
```

Context

```
config>app-assure>group>statistics>tca
```

Description

This command configures a TCA for the counter that is capturing error drops. An error drop TCA can be created for traffic generated from the subscriber side of AA (from-sub) or for traffic generated from the

network toward the AA subscriber (to-sub). The create keyword is mandatory when creating an error-drop TCA.

Parameters

direction

specifies the traffic direction

Values: from-sub | to-sub

10.2.1.12.19 high-wmark

Syntax

high-wmark *high-watermark* low-wmark *low-watermark*

Context

config>app-assure>group>statistics>tca>error-drop

config>app-assure>group>statistics>tca>fragment-drop-all

config>app-assure>group>statistics>tca>fragment-drop-out-of-order

config>app-assure>group>statistics>tca>overload-drop

config>app-assure>group>statistics>tca>policer

config>app-assure>group>statistics>tca>session-filter>default-action

config>app-assure>group>statistics>tca>session-filter>entry

config>app-assure>group>statistics>tca>tcp-validate

Description

This command configures the high-water mark and low-water mark thresholds for the specified TCA.

Parameters

high-watermark

specifies the TCA high-water mark

Values: 1 to 4294967295

Default: 4294967295

low-watermark

specifies the TCA low-water mark

Values: 0 to 4294967294

Default: 0

10.2.12.20 fragment-drop-all

Syntax

```
fragment-drop-all direction direction [create]
no fragment-drop-all direction direction
```

Context

```
config>app-assure>group>statistics>tca
```

Description

This command configures a TCA for the counter that is capturing all dropped fragments due to the fragment-drop-all AQP command. A fragment-drop-all TCA can be created for traffic generated from the subscriber side of AA (from-sub) or for traffic generated from the network toward the AA subscriber (to-sub). The create keyword is mandatory when creating a fragment-drop-all TCA.

Parameters

direction

specifies the traffic direction

Values: from-sub | to-sub

10.2.12.21 fragment-drop-out-of-order

Syntax

```
fragment-drop-out-of-order direction direction [create]
no fragment-drop-out-of-order direction direction
```

Context

```
config>app-assure>group>statistics>tca
```

Description

This command configures a TCA for the counter that is capturing out-of-order fragments due to the fragment-drop-out-of-order AQP command. A fragment-drop-out-of-order TCA can be created for traffic generated from the subscriber side of AA (from-sub) or for traffic generated from the network toward the

AA subscriber (to-sub). The create keyword is mandatory when creating a fragment-drop-out-of-order TCA.

Parameters

direction

specifies the traffic direction

Values: from-sub | to-sub

10.2.112.22 overload-drop

Syntax

```
overload-drop direction direction [create]
```

```
no overload-drop direction direction
```

Context

```
config>app-assure>group>statistics>tca
```

Description

This command configures a TCA for the counter that is capturing overload fragments due to the overload-drop AQP command. An overload-drop TCA can be created for traffic generated from the subscriber side of AA (from-sub) or for traffic generated from the network toward the AA subscriber (to-sub). The create keyword is mandatory when creating an overload-drop TCA.

Parameters

direction

specifies the traffic direction

Values: from-sub | to-sub

10.2.112.23 policer

Syntax

```
policer policer-name direction direction [create]
```

```
no policer policer-name direction direction
```

Context

```
config>app-assure>group>statistics>tca
```

Description

This command configures a TCA for the counter that is capturing drops or admit events due to the specified flow policer. A policer TCA can be created for traffic generated from the subscriber side of AA (from-sub) or for traffic generated from the network toward the AA subscriber (to-sub). The create keyword is mandatory when creating a policer TCA.

Parameters

policer-name

specifies the name of the flow policer, up to 32 characters

direction

specifies the traffic direction

Values: from-sub | to-sub

10.2.112.24 session-filter

Syntax

```
session-filter session-filter-name
```

Context

```
config>app-assure>group>statistics>tca
```

Description

This command configures TCA generation for a session filter.

Parameters

session-filter-name

specifies the name of the session filter, up to 32 characters

10.2.112.25 default-action

Syntax

```
default-action direction direction [create]
```

```
no default-action direction direction
```

Context

```
config>app-assure>group>statistics>tca>session-filter
```

Description

This command configures a TCA for the counter that is capturing hits for the specified session filter default action. A default action TCA can be created for traffic generated from the subscriber side of AA (from-sub) or for traffic generated from the network toward the AA subscriber (to-sub). The create keyword is mandatory when creating a default action TCA.

Parameters

direction

specifies the traffic direction

Values: from-sub | to-sub

10.2.1.12.26 entry

Syntax

```
entry entry-id direction direction [create]
```

```
no entry entry-id direction direction
```

Context

```
config>app-assure>group>statistics>tca>session-filter
```

Description

This command configures a TCA for the counter that is capturing hits for the specified session filter entry. A session filter entry TCA can be created for traffic generated from the subscriber side of AA (from-sub) or for traffic generated from the network toward the AA subscriber (to-sub). The create keyword is mandatory when creating a TCA.

Parameters

entry-id

specifies the session filter entry identifier

Values: 1 to 65535

direction

specifies the traffic direction

Values: from-sub | to-sub

10.2.1.12.27 tcp-validate

Syntax

```
tcp-validate tcp-validate-name direction direction [create]
```

```
no tcp-validate tcp-validate-name direction direction
```

Context

```
config>app-assure>group>statistics>tca
```

Description

This command configures TCA for the counter, and enables the capture of drop or admit events due to the specified TCP validation function.

Parameters

tcp-validate-name

specifies the name of the TCP validation policy, 32 characters maximum

direction

specifies the traffic direction in relation to the AA subscriber

Values: from-sub | to-sub

create

this keyword is mandatory when creating a TCA instance

10.2.1.13 Application Assurance group URL filter and URL list command descriptions

10.2.1.13.1 url-filter

Syntax

```
url-filter url-filter-name [create]
```

```
no url-filter url-filter-name
```

Context

config>app-assure>group

Description

This command specifies the name of the URL filter for the group.

Parameters

url-filter-name

the name of the filter

Values: 1 to 255 characters

create

keyword used when creating a new URL filter

The create keyword requirement can be enabled/disabled in the environment>create context.

10.2.13.2 apply-function-specific-behavior

Syntax

[no] apply-function-specific-behavior

Context

config>app-assure>group>url-filter

Description

This command applies if the default-action, default-http-redirect and http-redirect configured commands on the url-filter function level (ICAP / local filtering / web service) are enabled. Otherwise, the configuration on url-filter level applies to all the url-filter functions configured.

Default

no apply-function-specific-behavior

10.2.1.13.3 default-action

Syntax

```
default-action allow
default-action block-all
default-action block-http-redirect http-redirect-name
no default-action
```

Context

```
config>app-assure>group>url-filter
config>app-assure>group>url-filter>icap
config>app-assure>group>url-filter>web-service
config>app-assure>group>url-filter>local-filtering>deny-list
```

Description

This command configures a default action for the URL filter. The default action takes effect when the URL filter cannot be used. This may occur in the following cases:

- for the local URL list when the URL list is shutdown or the file is not loaded due to an error
- In the case of ICAP or web filtering when all TCP connections are busy or down

The no form of this command removes the default action for the URL filter.

Default

no default-action

Parameters

allow

specifies to allow traffic

block-all

specifies to block traffic

block-http-redirect *http-redirect-name*

specifies to block traffic and redirect the user to an information page that can be different from the page the user is redirected to when the site that the user tried to access is found in the URL filter; this page is configured using the `config>aa>group>url-filter>http-redirect` command

10.2.1.13.4 http-redirect

Syntax

```
http-redirect http-redirect-name  
no http-redirect
```

Context

```
config>app-assure>group>url-filter  
config>app-assure>group>url-filter>icap  
config>app-assure>group>url-filter>web-service  
config>app-assure>group>url-filter>local-filtering>deny-list
```

Description

This command creates or modifies an HTTP redirect policy.

The no form of this command removes the HTTP redirect policy from the configuration.

Parameters

http-redirect-name

specifies the name of the HTTP redirect policy, up to 32 characters

10.2.1.13.5 http-request-filtering

Syntax

```
http-request-filtering {all | first}
```

Context

```
config>app-assure>group>url-filter
```

Description

This command enables HTTP filtering for all HTTP request within a flow or for only the first HTTP request in a flow.

Parameters

all

specifies all HTTP requests within a flow

first

specifies the first HTTP request within a flow

10.2.13.6 icap

Syntax

icap

Context

config>app-assure>group>url-filter

Description

This command configures the URL filter policy for ICAP filtering. Within the ICAP context the operator needs to specify the ICAP server IP addresses and the AA interface VLAN ID.

10.2.13.7 custom-x-header

Syntax

custom-x-header *custom-x-header-name*

no custom-x-header

Context

config>app-assure>group>url-filter>icap

Description

This command configures the URL filter ICAP policy to include a new x-header field; the content of the x-header is populated based on AQP URL filter action which can optionally specify the ASO characteristic value to include in the x-header.

Parameters***custom-x-header-name***

the name of the x-header added to the ICAP request

10.2.13.8 server

Syntax

```
server ip-address [:port] [create]  
no server ip-address [:port]
```

Context

```
config>app-assure>group>url-filter>icap  
config>app-assure>group>url-filter>web-service
```

Description

This command configures the IP address and optionally the server port of the ICAP server.

Parameters

ip-address[:port]

specifies the ICAP server IP address and port, up to 25 characters

create

keyword used to create the URL classification database server

10.2.13.9 vlan-id

Syntax

```
vlan-id service-port-vlan-id  
no vlan-id
```

Context

```
config>app-assure>group>url-filter>icap
```

Description

This command configures the VLAN ID on which the ISA-AA is expected to be emitting traffic mapping to a preconfigured AA interface.

10.2.13.10 local-filtering

Syntax

```
local-filtering
```

Context

```
config>app-assure>group>url-filter
```

Description

This command configures a URL filter policy for local filtering to filter traffic based on a list of URLs located on a file stored in the router compact flash.

10.2.13.11 allow-list

Syntax

```
allow-list allow-list-name
```

```
no allow-list
```

Context

```
config>app-assure>group>url-filter>local-filtering
```

Description

This command adds a URL allow-list to the local filtering URL filter policy.

The no form of this command removes the URL list object.

Parameters

allow-list-name

specifies the URL list

10.2.13.12 deny-list

Syntax

```
[no] deny-list deny-list-name
```

Context

```
config>app-assure>group>url-filter>local-filtering
```

Description

This command adds a URL deny-list to the local filtering URL filter policy.

The no form of this command removes the URL list object.

Parameters

deny-list-name

specifies the URL list

10.2.13.13 web-service

Syntax

```
web-service
```

Context

```
config>app-assure>group>url-filter
```

Description

This command enables the context to configure the URL filter policy using web-service filtering. The operator must configure the web service, host name, DNS server to use, the AA interface VLAN ID, and provision the category profiles.

10.2.13.14 classification-overrides

Syntax

```
classification-overrides
```

Context

```
config>app-assure>group>url-filter>web-service
```

Description

This command enables the context to configure hostname classification overrides.

10.2.1.13.15 entry

Syntax

```
entry entry-id expression hostname category category  
no entry entry-id
```

Context

```
config>app-assure>group>url-filter>web-service>classification-overrides
```

Description

This command configures a classification override. The category of a hostname is set manually.

Parameters

entry-id

specifies the classification of the override entry

Values: 1 to 65535

hostname

specifies the hostname of the configured override category, up to 255 characters

category

specifies the override category, up to 256 characters

10.2.1.13.16 classifier

Syntax

```
classifier classifier category-set-id category-set  
no classifier
```

Context

```
config>app-assure>group>url-filter>web-service
```

Description

This command selects the web service to use from the supported web services and the category set to perform URL categorization.

The no form of this command removes the selected web service.

Default

no classifier

Parameters

classifier

specifies the web service to use

Values: web-service-1 | web-service-2

category-set

specifies the category set ID to use for URL categorization; a category set ID defines the list of categories that the web service uses to perform URL categorization

Values: 1 to 3

10.2.13.17 default-profile

Syntax

default-profile *profile-name*

no default-profile

Context

config>app-assure>group>url-filter>web-service

Description

This command configures the default category profile to use when no category profile is explicitly selected for the subscriber.

Default

no default-profile

Parameters

profile-name

specifies a configured category profile to use as the default profile name, up to 256 characters

10.2.13.18 dns-server

Syntax

```
dns-server ip-address  
no dns-server
```

Context

```
config>app-assure>group>url-filter>web-service
```

Description

This command configures the DNS server that is used to resolve the web-service host name.

Default

```
no dns-server
```

Parameters

ip-address
specifies the IP address of the DNS server to use

Values	ipv4 - a.b.c.d [mask]
–	ipv6 - x:x:x:x:x:x:x
–	x - [0..FFFF]H
–	d - [0..255]D
–	ipv6-prefix-length: [1 to 128]

10.2.13.19 fqdn

Syntax

```
fqdn fully-qualified-domain-name
```


no fqdn

Context

config>app-assure>group>url-filter>web-service

Description

This command configures the host name of the web service.

Default

no fqdn

Parameters

fully-qualified-domain-name

specifies the host name of the web service, up to 256 characters

10.2.13.20 profile

Syntax

profile *profile-name* [create]

no profile *profile-name*

Context

config>app-assure>group>url-filter>web-service

Description

This command configures the category profiles of the web service.

Parameters

profile-name

specifies the name of the category profile, up to 256 characters

create

keyword that specifies to create a category profile

10.2.1.13.21 category

Syntax

```
category category block
no category category
```

Context

```
config>app-assure>group>url-filter>web-service>profile
```

Description

This command configures the category that will be blocked in the category profile.

Parameters

category

specifies the URL category name for the configured web service, up to 256 characters

10.2.1.13.22 server

Syntax

```
server ip-address [:port] [create]
no server ip-address [:port]
```

Context

```
config>app-assure>group>url-filter>web-service
```

Description

This command configures the IP address and optionally the port of the URL classification database server.

Default

```
no server ip-address [:port]
```

Parameters

ip-address [:*port*]

specifies the server IP address and port, up to 25 characters

create

keyword used to create the URL classification database server

10.2.13.23 **vlan-id**

Syntax

`vlan-id vlan-id`

`no vlan-id`

Context

`config>app-assure>group>url-filter>web-service`

Description

This command configures the VLAN ID on which the AA ISA emits the traffic mapping to a preconfigured AA interface.

Default

`no vlan-id`

Parameters

vlan-id

specifies the VLAN ID to connect to the web service

Values: 1 to 4094

10.2.13.24 **url-list**

Syntax

`url-list url-list-name [create]`

`no url-list url-list-name`

Context

`config>app-assure>group`

Description

This command configures a URL list object. The URL list points to a file containing a list of URLs located on the system compact flash. The URL list is then referenced in a URL filter object to filter and redirect subscribers when a URL from this file is accessed.

The no form of this command removes the URL list object.

Parameters

url-list-name

specifies the name of the URL list

10.2.13.25 decrypt-key

Syntax

decrypt-key *key* | *hash-key* | *hash2-key* | *custom-key* [*hash* | *hash2* | *custom*]

no decrypt-key

Context

config>app-assure>group>url-list

Description

In case the file is encrypted this command is used to configure the decryption key used to read the file.

The no form of this command removes the decrypt-key.

Parameters

key* | *hash-key* | *hash2-key

specifies the Application-Assurance url-list decryption key

hash* | *hash2* | *custom

specifies the hashing scheme used in the hashed key

10.2.13.26 expression-match

Syntax

[no] expression-match

Context

config>app-assure>group>url-list

Description

This command configures a URL list that contains hostnames with wildcards.

10.2.1.13.27 file

Syntax

file *file-url*

no file

Context

config>app-assure>group>url-list

Description

This command specifies the file location that contains the list of URLs to filter. This file must be located locally on the router compact flash. The no form of this command removes the file reference.

Parameters

file-url

specifies the URL of the file which contains the URL list to filter

Values	:[<i>cflash-id</i> /] <i>file-path</i> - [200 chars max]
–	cflash-id - cf1: cf1-A: cf1-B: cf2: cf2-A: cf2- B: cf3: cf3-A: cf3-B:

10.2.1.13.28 size

Syntax

size *url-list-size*

Context

config>app-assure>group>url-list

Description

This command specifies the size of the URL list that can be filtered. The size can be set to either standard or extended. Configuring the specified URL list as extended provides support for filtering on a larger number of URLs.

Parameters

url-list-size

specifies the size of the AA URL list for URL filtering

Values: standard | extended

Default: standard

10.2.2 Application Assurance show command descriptions

10.2.2.1 application-assurance-group

Syntax

application-assurance-group [*aa-group-id*] [load-balance [unassigned]]

Context

show>isa

Description

This command displays ISA group information.

Parameters

aa-group-id

specifies the AA ISA group ID

load-balance

specifies the load balancing information

unassigned

specifies load balancing unassigned aa-sub information

Output

The following output examples show AA ISA group information, and the *Show AA ISA group statistics output fields*, *Show ISA AA group load-balancing output fields*, and *Show ISA AA group load-balancing unassigned output fields* describe the output fields.

- [Example output AA group statistics](#)
- [Example output AA group load-balancing](#)
- [Example output AA group load-balancing unassigned](#)
- [Example output AA group shared resources](#)

Example output AA group statistics

```
# show isa application-assurance-group 1
=====
ISA Application-assurance-groups
=====
ISA-AA Group Index      : 1
Description             : (Not Specified)
Subscriber Scale        : mobile-gateway
WLAN GW Group Index     : N/A
Primary ISA-AA          : 3/2 up/active
Backup ISA-AA           : none
Minimum ISA Generation  : 1
Last Active change      : 01/17/2019 21:22:35
Admin State             : Up
Oper State              : Up
Diverted FCs            : be
Fail to mode            : fail-to-wire
Partitions              : enabled
QoS
  Egress from subscriber
    Pool                 : default
    Reserved Cbs         : default
    Slope Policy          : default
    Queue Policy          : default
    Scheduler Policy      :
  Egress to subscriber
    Pool                 : default
    Reserved Cbs         : default
    Slope Policy          : default
    Queue Policy          : default
    Scheduler Policy      :
Capacity Cost
  High Threshold         : 4294967295
  Low Threshold          : 0
Overload Cut Through     : disabled
Transit Prefix
  Max IPv4 entries       : 0
  Max IPv6 entries       : 0
  Max IPv6 remote entries : 0
HTTP Enrichment
  Max Packet Size        : 1500 octets
Shared Resources
```

```

TCP Advanced Functions      : 0%
Web Service Cache           : 0%
Bandwidth Shaper Buffer     : 100%
Multi Device Detect         : 100%
VM Traffic Distribution by IP : disabled
=====

```

Table 184: Show AA ISA group statistics output fields

Label	Description
ISA Application-assurance-groups	
ISA-AA Group Index	The Integrated Service Adapter's Application Assurance group index
Description	The Application Assurance Group type
Primary ISA-AA	The primary ISA-AA
Backup ISA-AA	The ISA-AA being used as a backup
Last Active Change	Date when last active change was made
Admin State	Desired state of the service
Oper State	Operating state of the service
Diverted FCs	—
Fail to Mode	—
Pool	—
Reserved Cbs	—
Slope Policy	—
Queue Policy	—

Table 184: Show AA ISA group statistics output fields (continued)

Label	Description
Scheduler Policy	—
Egress State to/from Subscriber	—
VM Traffic Distribution by IP	Indicates for the CMG whether data plane VM traffic distribution to CPU cores for AA processing is based on IP addresses

Example output AA group load-balancing

```
# show isa application-assurance-group 84 load-balance
=====
ISA Application-assurance-group 84
=====
load-balance status : Complete
isa-capacity-cost-threshold : low 0
: high 4294967295
-----
capacity-cost aa-sub aa-sub stats
count count
-----
3/1 1024 1024 1024
Mda Limit NA 65535 2097120
=====
aa-sub type count for group 84
=====
all esm sap spoke-sdp
-----
3/1 1024 20 1000 4
Unassigned 14 2 8 4
=====
```

Table 185: Show ISA AA group load-balancing output fields

Label	Description
load-balance status	Load balancing information

Table 185: Show ISA AA group load-balancing output fields (continued)

Label	Description
isa-capacity-cost-threshold	—
capacity-cost	—
aa-sub	—
stats	—
count	—
type	—
all	—
esm	—
sap	—
spoke-sdp	—

Example output AA group load-balancing unassigned

```
# show isa application-assurance-group 84 load-balance unassigned
=====
ISA Application-assurance-group 84 unassigned
=====
type SvcId aa-sub App-Profile
-----
esm 2 Sub1 Cost30
esm 50 Sub2 Cost31
sap 29 2/1/10:527 Cost29
sap 30 2/1/10:528 Cost29
sap 31 2/1/10:529 Cost29
sap 31 2/1/10:530 Cost29
sap 31 2/1/10:531 Cost29
sap 32 2/1/10:546 Cost29
sap 32 2/1/10:547 Cost29
sap 33 2/1/10:548 Cost29
spoke 201 199:10 Cost27
```

```

spoke 202 199:17 Cost10
spoke 202 199:18 Cost10
spoke 202 199:19 Cost10
=====

```

Table 186: Show ISA AA group load-balancing unassigned output fields

Label	Description
Svc Id	—
aa-sub	—
App-profile	—

Example output AA group shared resources

```

# show isa application-assurance-group 1 shared-resources
=====
-----
MDA 3/2
-----
Tcp Advanced Functions (tcp-optimizer, mp-tcp-proxy, http-proxy)
  Configured                               : 100%
  TCP Buffer Memory                         : 37 MB
  TCP Buffer Utilization                    : 0%
  Max Number of Possible
    Concurrent Sessions TCP Optimizer      : 20000
    MP-TCP Proxy                           : 10000
    HTTP Proxy                             : 1000
  Session Resource Utilization              : 0%
Web Service Cache
  Configured                               : 0%
  Cache Utilization                        : 0%
Bandwidth Shaping Policer
  Configured                               : 100%
Multi Device Detect
  Configured                               : 100%
=====

```

10.2.2.2 ip-reas-stats

Syntax

ip-reas-stats [*port-name*] [ipv4 | ipv6]

Context

show>isa

Description

This command displays the UE fragment reassembly packet statistics.

Parameters

port-name

specifies the port name

Values	port-id	[:context-val]
–	context-val	[1 to 31]

port-id

specifies the configured port ID

Values	slot/mda/port	–
–	lag-id	[:lag-<id>] (the configured lag-id)
–	lag	–
–	id	[1 to 200]

ipv4 | ipv6

specifies the IPv4 or IPv6 address

Output

The following output example shows the IP reassembly statistics.

Example output IP reassembly statistics

```
# show isa ip-reas-stats
=====
IP Reassembly statistics
=====
-----
Interface          : 1-3/2/1:1 (Reassembly)
Selective Reassembly : disabled
-----
IPv4 Reassembly statistics
-----
Total Pkts Rx      : 0          Total Bytes Rx      : 0
Total Pkts Tx      : 0          Total Bytes Tx      : 0
Fragmented Pkts Rcvd : 0        Fragmented Bytes Rcvd : 0
Packets Reassembled : 0          Bytes Reassembled   : 0
Fragments Timed Out : 0          Fragments Discarded  : 0
-----
IPv6 Reassembly statistics
-----
Total Pkts Rx      : 0          Total Bytes Rx      : 0
Total Pkts Tx      : 0          Total Bytes Tx      : 0
Fragmented Pkts Rcvd : 0        Fragmented Bytes Rcvd : 0
Packets Reassembled : 0          Bytes Reassembled   : 0
Fragments Timed Out : 0          Fragments Discarded  : 0
=====
UE Fragmented Packets Handling statistics
=====
-----
IPv4 Reassembly statistics
-----
Fragmented Pkts Rcvd : 0          Fragments Discarded : 0
Fragments Timed Out  : 0          Packet Reass. Failures : 0
Packets Reassembled  : 0
-----
IPv6 Reassembly statistics
-----
Fragmented Pkts Rcvd : 0          Fragments Discarded : 0
Fragments Timed Out  : 0          Packet Reass. Failures : 0
Packets Reassembled  : 0
-----
Number of entries : 1
=====
```

10.2.2.3 group

Syntax

group *aa-group-id* [:*partition-id*]

Context

show>app-assure

Description

This command enables the context to display application-assurance group information.

Parameters

- aa-group-id**
specifies an AA ISA group ID
Values: 1 to 255
- partition-id**
specifies a partition within a group
Values: 1 to 65535

10.2.2.4 aa-interface

Syntax

aa-interface isa mda-id

Context

show>app-assure>group

Description

This command displays AA interface information.

Parameters

- mda-id**
specifies the MDA ID

Values	<mda-id>	<slot>/<mda>
–	slot	[1 to 10] (depending on platform)

– mda [1 to 2]

10.2.2.5 aa-sub

Syntax

```
aa-sub mobile [imsi imsi-msisdn | msisdn imsi-msisdn | imei imei] apn apn-name [snapshot]
```

Context

```
show>app-assure>group
```

Description

This command displays per-subscriber statistics.

Parameters

imsi-msisdn

specifies the IMSI-MSIDN, 15 digits maximum

imei

specifies the IMEI, 16 digits maximum

apn-name

specifies the APN name, up to 80 characters

snapshot

specifies displaying snapshot statistics

Output

The following output example shows per-subscriber statistics.

Example output AA group application subscriber

```
# show application-assurance group 1 aa-sub mobile imsi 123457789012362 apn
"ipd.nokia.1.1.com" summary
=====
Application-Assurance Subscriber Summary (realtime)
=====
Mobile Gateway Id      : 1
Subscriber name        : IMSI 310012567489000
5G Subscriber          : No
IMSI                   : 310012567489000
MSISDN                 : 12345098761
```

```

IMEI           : 987654321012399
Session-Id     : 0x03440001
Customer Id    : 0
APN            : ipd.nokia.com
Virtual APN    : N/A
ISA assigned   : 3/2
App-Profile    : APP_PROFILE_1
RAT Type       : EUTRAN
Bearer Id      : 5
HTTP URL Parameters : (Not Specified)
Mobile Charging Id : 3506438144
Mobile Billing Type : 0x0100
PGW/GGSN address : 10.20.1.3
SGW/SGSN address(es) : 10.20.1.1
User Location   : N/A
PLMN ID        : 310012
PRA ID         : 2105376
Roaming Status  : HOME
Wireless Device Type : tablet
Wireless Device OS : ios
Access Ntwk Location : None
ANL Type       : N/A
Sub Congestion State : Not Congested
Tethering Status : not-detected (disabled)
Devices Detected : N/A
Tracked Devices : tracked: 10 (max: 10)  excess: yes (hist: yes)

```

Traffic	Octets	Packets	Flows
From subscriber:			
Admitted	0	0	0
Denied	0	0	0
Active flows			0
To subscriber:			
Admitted	0	0	0
Denied	0	0	0
Active flows			0
Flow counts:			
Terminated			0
Short duration			0
Med duration			0
Long duration			0

Total flow duration : 0 seconds

Application Service Options (ASO)

Characteristic	Value	Derived from
----------------	-------	--------------

None

Notify-Start-Stop

Charging-Group	Notify-Type	Enabled	Active
CG_1	flow	no	n/a
CG_2	group	yes	no
CG_3	dyn-rating	yes	yes

10.2.2.6 aa-sub-list

Syntax

```
aa-sub-list [isa mda-id]
```

```
aa-sub-list summary
```

Context

```
show>app-assure>group
```

Description

This command displays AA subscriber lists.

Parameters

isa mda-id

displays the slot and MDA ID

Values	<i>isa</i>	1 to 10 (depending on chassis model)
–	<i>mda-id</i>	1, 2

summary

displays summary information

Output

The following output example shows per-subscriber statistics, and *Show AA group application subscriber list output fields* describes the fields.

Example output AA group application subscriber list

```
# show application-assurance group 1 aa-sub-list
* Use CTRL-C to terminate command if it takes too long to return the prompt.
* Consider using 'tools>dump>application-assurance>group>aa-sub-list' instead.
```

```
=====
Application-Assurance Subscriber List for Group 1
```

```

=====
Type      Id                APN                ISA      App-Profile
              assigned
-----
imsi      123456789012001    ipd.nokia.com      3/2      APP_PROFILE_1
imsi      123456789012002    ipd.nokia.com      3/2      APP_PROFILE_1
imsi      123456789012003    ipd.nokia.com      3/2      APP_PROFILE_1
imsi      123456789012004    ipd.nokia.com      3/2      APP_PROFILE_1
imsi      123456789012005    ipd.nokia.com      3/2      APP_PROFILE_1
imsi      123456789012006    ipd.nokia.com      3/2      APP_PROFILE_1
imsi      123456789012007    ipd.nokia.com      3/2      APP_PROFILE_1
imsi      123456789012008    ipd.nokia.com      3/2      APP_PROFILE_1
imsi      123456789012009    ipd.nokia.com      3/2      APP_PROFILE_1
imsi      123456789012010    ipd.nokia.com      3/2      APP_PROFILE_1
msisdn    11122233401         ipd.nokia.com      3/2      APP_PROFILE_1
msisdn    11122233402         ipd.nokia.com      3/2      APP_PROFILE_1
msisdn    11122233403         ipd.nokia.com      3/2      APP_PROFILE_1
msisdn    11122233404         ipd.nokia.com      3/2      APP_PROFILE_1
msisdn    11122233405         ipd.nokia.com      3/2      APP_PROFILE_1
msisdn    11122233406         ipd.nokia.com      3/2      APP_PROFILE_1
msisdn    11122233407         ipd.nokia.com      3/2      APP_PROFILE_1
msisdn    11122233408         ipd.nokia.com      3/2      APP_PROFILE_1
msisdn    11122233409         ipd.nokia.com      3/2      APP_PROFILE_1
msisdn    11122233410         ipd.nokia.com      3/2      APP_PROFILE_1
imei      123459789012201     ipd.nokia.com      3/2      APP_PROFILE_1
imei      123459789012211     ipd.nokia.com      3/2      APP_PROFILE_1
imei      123459789012221     ipd.nokia.com      3/2      APP_PROFILE_1
imei      123459789012231     ipd.nokia.com      3/2      APP_PROFILE_1
imei      123459789012241     ipd.nokia.com      3/2      APP_PROFILE_1
imei      123459789012251     ipd.nokia.com      3/2      APP_PROFILE_1
imei      123459789012261     ipd.nokia.com      3/2      APP_PROFILE_1
imei      123459789012271     ipd.nokia.com      3/2      APP_PROFILE_1
imei      123459789012281     ipd.nokia.com      3/2      APP_PROFILE_1
imei      123459789012291     ipd.nokia.com      3/2      APP_PROFILE_1
-----
Total number of aa-subs found          : 30
=====

```

Table 187: Show AA group application subscriber list output fields

Label	Description
Application-Assurance Subscriber List for Group 1	
Type	The type of AA subscriber
Id	The ID number for the AA subscriber

Table 187: Show AA group application subscriber list output fields (continued)

Label	Description
APN	The access point node for the AA subscriber
ISA assigned	Shows which ISA is assigned to the AA subscriber
App-Profile	The AA subscriber's application profile
Total number of aa-subs found	The number of AA subscribers being viewed

10.2.2.7 app-group

Syntax

```
app-group [app-group-name] count [detail]
```

Context

```
show>app-assure>group
```

Description

This command displays per-application-group statistics. System-wide statistics displayed account for all flows completed and the last internal snapshot of the active flows.

Parameters

app-group-name

Displays information about the specified application group name.

count

displays the counters for the application group

detail

displays detailed information

Output

The following output examples show AA group information, and *Show AA group application group output fields* describes the fields.

Example output AA group application group

```
# show>app-assure>group# app-group count
```

```
=====
```

App-group Statistics

```
=====
```

Application Group	Disc Octets	Packets	Flows
File Transfer	0% 0	0	0
Games	0% 3865532	4952	144
Infrastructure	0% 174524	1217	1177
Instant Messaging	0% 2979117	9930	97
Local Content	0% 10581539	10942	74
Mail	0% 57940	346	24
MultiMedia	0% 76911464	79417	198
NNTP	0% 0	0	0
Peer to Peer	0% 10903442	13901	485
Premium Partner	0% 0	0	0
Remote Connectivity	0% 0	0	0
Server	0% 1097	8	2
Suspect	72% 1012	11	11
Tunneling	0% 19872617	33989	204
Unknown	0% 5243395	27510	2648
Web	0% 82135303	91828	2152

```
-----
```

```
# show>app-assure>group# app-group "MultiMedia" count detail
```

```
=====
```

App-group "MultiMedia" Statistics

```
=====
```

```
Application Group:
```

Type	Octets	Packets	Flows

MultiMedia:			
Admitted from subscriber:	193605	1797	23
Denied from subscriber:	0	0	0
Active flows from subscriber:			0
Admitted to subscriber:	4835822	3366	23
Denied to subscriber:	0	0	0
Active flows to subscriber:			0
Total flow duration:	433 seconds		
Terminated flows:			46
Short Duration flows:			36
Medium Duration flows:			10
Long Duration flows:			0
Active subscribers:	0		

```
=====
```

Table 188: Show AA group application group output fields

Label	Description
App-group Statistics	
Application Group	The application group
Disc	Specifies which percentage of the disc is being used
Octets	The total number of octets received
Packets	The total number of packets received
Flows	The total number of flows received
App-group "MultiMedia" Statistics	
Type	Specifies which category holds which octets, packets, and flows
Octets	The total number of octets received in a type
Packets	The total number of packets received in a type
Flows	The total number of flows received in a type
Admitted from subscriber	The number of octets, packets, and flows admitted from the subscriber
Denied from subscriber	The number of octets, packets, and flows denied from the subscriber
Active flows from subscriber	The number of active flows coming from the subscriber
Admitted to subscriber	The number of octets, packets, and flows admitted to the subscriber
Denied to subscriber	The number of octets, packets, and flows denied to the subscriber

Table 188: Show AA group application group output fields (continued)

Label	Description
Active flows to subscriber	The number of active flows going to the subscriber
Total flow duration	The amount of time in the flow duration
Terminated flows	The total number of terminated flows
Short Duration flows	The number of short duration flows
Medium Duration flows	The number of medium duration flows
Long Duration flows	The number of long duration flows
Active subscribers	The total number of active subscribers

10.2.2.8 application

Syntax

```
application [application-name] count [detail]
```

Context

```
show>app-assure>group
```

```
show>app-assure>group>aa-sub-study
```

Description

This command displays per-application statistics. The system-wide statistics displayed account for all flows completed and the last internal snapshot of the active flows.

Subscriber statistics are available for special-study subscribers and account for all completed and active flows at the moment of this statistics request.

Parameters

application-name

displays information about the specified application name

count

displays counter information

detail

displays detailed information

Output

The following output examples show per-application statistic information, and *Show AA group application output fields* describes the fields.

Example output AA group application

```
# show>app-assure>group# application count
=====
Application Statistics
=====
Application                Disc Octets                Packets                Flows
-----
...
DHT                        0% 0                      0                      0
DNS_53                    0% 96781                  627                    627
DNS_Local                 0% 0                      0                      0
DNS_Server                0% 276                    3                      3
DNS_Suspect               100% 736                  8                      8
FTP                       0% 0                      0                      0
...
=====

# show>app-assure>group# application "POP3" count detail
=====
Application "POP3" Statistics
=====
Application:
Type                Octets                Packets                Flows
-----
POP3:
Admitted from subscriber: 14095                149                10
Denied from subscriber:    0                    0                    0
Active flows from subscriber:                    0
Admitted to subscriber:   30707                128                10
Denied to subscriber:     0                    0                    0
Active flows to subscriber:                    0
Total flow duration:      7 seconds
Terminated flows:                    20
Active subscribers:       0

A:NOK>show>app-assure>group# application "HTTP_Video" count detail
=====
```

Application "HTTP_Video" Statistics

```

=====
Application:
Type           Octets           Packets           Flows
-----
HTTP_Video:
Admitted from subscriber: 369528           5404           36
Denied from subscriber:    0                0            0
Active flows from subscriber:                1
Admitted to subscriber:  15387734        10629           36
Denied to subscriber:    0                0            0
Active flows to subscriber:                1
Total flow duration:      463 seconds
Terminated flows:                72
Short Duration flows:          66
Medium Duration flows:         6
Long Duration flows:           0
Active subscribers:         1
=====

```

Table 189: Show AA group application output fields

Label	Description
Application Statistics	
Application	Specifies which application is being used
Disc	Specifies which percentage of the disc is being used
Octets	The total number of octets received
Packets	The total number of packets received
Flows	The total number of flows received
Application "POP3" or "HTTP_Video" Statistics	
Type	Specifies which category holds which octets, packets, and flows
Octets	The total number of octets received in a type
Packets	The total number of packets received in a type

Table 189: Show AA group application output fields (continued)

Label	Description
Flows	The total number of flows received in a type
Admitted from subscriber	The number of octets, packets, and flows admitted from the subscriber
Denied from subscriber	The number of octets, packets, and flows denied from the subscriber
Active flows from subscriber	The number of active flows coming from the subscriber
Admitted to subscriber	The number of octets, packets, and flows admitted to the subscriber
Denied to subscriber	The number of octets, packets, and flows denied to the subscriber
Active flows to subscriber	The number of active flows going to the subscriber
Total flow duration	The amount of time in the flow duration
Terminated flows	The total number of terminated flows
Active subscribers	The total number of active subscribers

10.2.2.9 certificate-profile

Syntax

```
certificate-profile
```

Context

```
show>app-assure>group
```

Description

This command displays certificate profile group statistics.

Output

The following output example shows certificate profile information, and *Show AA group certificate profile output fields* describes the fields.

Example output AA group certificate profile

```
# show application-assurance group 1 cert-profile Nokia_CERT_1
=====
Application Assurance Group cert-profile "Nokia_CERT_1"
=====
Description   : (Not Specified)
File Name     : nokia_certificate.crt
Admin State   : Up
Oper State    : Down                      Reason: Invalid Certificate File
Enrich Ref    : 3
=====
```

Table 190: Show AA group certificate profile output fields

Label	Description
Description	Describes the certificate profile
File Name	The certificate profile file name
Admin State	The administrative state of the certificate profile: Up or Down
Oper State	The operating state of the certificate profile: Up or Down
Reason	Possible reasons: Admin Down Invalid Certificate File Invalid / No Public Key Invalid Public Key Modulus Invalid Public Key Exponent Unsupported Cert File Algorithm
Enrich Ref	The enrichment reference number

10.2.2.10 cflowd

Syntax

```
cflowd status
```

Context

```
show>app-assure>group
```

Description

This command enables the context to display Cflowd output.

10.2.2.11 collector

Syntax

```
collector [detail]
```

Context

```
show>app-assure>group>cflowd
```

Description

This command displays Cflowd Collector statistics.

Output

The following output examples show Cflowd output information, and *Show AA group Cflowd collector output fields* describes the fields.

Example output AA group Cflowd collector

```
# show application-assurance group 1 cflowd collector
=====
Application Assurance Cflowd Collectors for group 1
=====
Host Address      Port  Version  Admin   Oper    Recs Sent
-----
192.168.7.7       2055   10       up      up      0
192.168.7.8       2055   10       up      up      0
-----
Collectors : 2
-----

# show application-assurance group 1 cflowd collector detail
```

```
=====
Application Assurance Cflowd Collectors for group 1
=====
```

```
Address          : 192.168.7.7
Port             : 2055
Description      : AA Collector 1
Version          : 10
Admin State      : up
Oper State       : up
Records Sent     : 0
Last Changed     : 07/27/2009 13:36:50
```

```
Address          : 192.168.7.8
Port             : 2055
Description      : AA Collector 2
Version          : 10
Admin State      : up
Oper State       : up
Records Sent     : 0
Last Changed     : 07/27/2009 13:37:10
=====
```

Table 191: Show AA group Cflowd collector output fields

Label	Description
Application Assurance Cflowd Collectors for group 1	
Host Address	The IP address of the host
Port	The port of the collector
Version	The version number of the collector
Admin	The desired state of operation for the collector
Oper	The active state of operation for the collector
Recs Sent	The number of records sent
Collectors	The number of collectors being viewed
Description	Describes the collector, if applicable

Table 191: Show AA group Cflowd collector output fields (continued)

Label	Description
Last Changed	Indicates when the collector was most recently changed

10.2.2.12 direct-export

Syntax

```
direct-export
```

Context

```
show>app-assure>group>cflowd
```

Description

This command enables the context to display Cflowd direct-export output.

10.2.2.13 collector

Syntax

```
collector collector-id
```

Context

```
show>app-assure>group>cflowd>dir-exp
```

Description

This command enables the context to display Cflowd direct-export collector output.

Output

The following output is an example of collector Cflowd direct-export information, and *Show AA group Cflowd collector output fields* describes the fields.

Example output AA group Cflowd collector

```
# show application-assurance group 1 cflowd direct-export collector 1
=====
```

```

Application Assurance Group 1 Cflowd Direct-Export Collector 1
=====
Collector Status : Inactive
VLAN Id         : N/A
Cflowd Version  : 10
Admin State     : down
Description     : (Not Specified
ISA using       : 3/2, 4/2, 5/2, 6/2, esa-3/4, esa-4/1
-----
Host Address      Port      Admin   Oper      Records Sent
-----
192.168.2.1       4739    down    down      0
192.168.2.2       4739    down    down      0
-----
No. of Cflowd Direct Export Collector Addresses: 2
=====
Application Assurance Group 1 Cflowd Collectors
Collector Status      : Active
=====
Address      : 192.168.2.1
Port         : 4739
Description  :
Version      : 10
Admin State  : down
Oper State   : down
Records Sent : 5400
Last Changed : 08/12/2015 19:12:28
Address      : 192.168.2.2
Port         : 4739
Description  :
Version      : 10
Admin State  : down
Oper State   : down
Records Sent : 5400
Last Changed : 08/12/2015 19:12:35
=====
status [isa <slot/mda>] cflowd

```

Table 192: Show AA group Cflowd collector output fields

Label	Description
Application Assurance Group 1 Cflowd Direct-Export Collector 1	
Collector Status	Indicates if the collector is active
VLAN Id	The ID of the VLAN associated with the collector
Cflowd Version	The number of the Cflowd version

Table 192: Show AA group Cflowd collector output fields (continued)

Label	Description
Admin State	The desired state of operation for the collector
Description	Describes the collector, if applicable
Host Address	The IP address of the host
Port	The port of the collector
Version	The version number of the collector
Oper	The active state of operation for the collector
Recs Sent	The number of records sent
No. of Cflowd Direct Export Collector Addresses	The number of Cflowd Direct Export Collector Addresses being viewed
Last Changed	Indicates when the collector was most recently changed

10.2.2.14 isa-collector

Syntax

```
isa-collector
```

Context

```
show>app-assure>group>cflowd>dir-exp
```

Description

This command displays the Cflowd collector assignments to ISAs.

10.2.2.15 status

Syntax

status

Context

show>app-assure>group>cflowd

Description

This command display status information.

Output

The following output examples show status information, and *Show AA group Cflowd status output fields* and *Show AA group partition Cflowd status output fields* describe the fields.

- *Example output AA group Cflowd Status*
- *Example output AA group partition Cflowd status*

Example output AA group Cflowd status

```
# show application-assurance group 1 status [isa 1/2] cflowd
=====
Application-Assurance Group Cflowd Status
=====
Cflowd Admin Status   : Enabled
Cflowd Oper Status    : Enabled
-----
Volume :
-----
Sample Rate           : <Disabled> or <1 in 500 packets>
Active Flows           : 23102
Records Reported       : 12345
Records Dropped        : 10
Records Per Second     : 45
Packets Sent           : 1763
Packets Sent Per Sec   : 7
-----
TCP Performance :
-----
Sample Rate           : <Disabled> or <1 in 1000 flows>
Active Flows           : 32103
Flows Not Allocated    : 33
Records Reported       : 12345678
Records Dropped        : 100
Records Per Second     : 456
Packets Sent           : 2057613
Packets Sent Per Sec   : 76
=====
```


Table 193: Show AA group Cflowd status output fields

Label	Description
Cflowd Admin Status	The desired state of the Cflowd
Cflowd Oper Status	The current state of the Cflowd
Volume	
Sample Rate	The sample rate
Active Flows	The number of active flows
Records Reported	The number of records that have been reported
Records Dropped	The number of records that have been dropped
Records per second	The number of records made per second
Packets Sent	The number of packets sent
Packets Sent Per Sec	The number of packets sent per second
TCP Performance	
Sample Rate	The sample rate
Active Flows	The number of active flows
Records Reported	The number of records that have been reported
Records Dropped	The number of records that have been dropped
Records per second	The number of records made per second
Packets Sent	The number of packets sent

Table 193: Show AA group Cflowd status output fields (continued)

Label	Description
Packets Sent Per Sec	The number of packets sent per second

Example output AA group partition Cflowd status

```
# show application-assurance group <aa-group-id:[partition]> cflowd status
=====
Application-Assurance Group:Partition Cflowd Status
=====
-----
Volume :
-----
Admin State      : Up
Records Reported : 12345
Records Dropped  : 10
-----
TCP Performance :
-----
Admin State      : Up
Flows Not Allocated : 33
Records Reported : 12345678
Records Dropped  : 100
-----
PGW-EDR
-----
ADMIN STATE      : OUTOFSERVICE
RECORDS REPORTED : 0
RECORDS DROPPED  : 0
=====
```

Table 194: Show AA group partition Cflowd status output fields

Label	Description
Volume	
Admin State	The desired state of the Cflowd Volume
Records Reported	The number of records that have been reported
Records Dropped	The number of records that have been dropped

Table 194: Show AA group partition Cflowd status output fields (continued)

Label	Description
TCP Performance	
Admin State	The desired state of the Cflowd TCP performance
Flows Not Allocated	The number of flows that are not allocated
Records Reported	The number of records that have been reported
Records Dropped	The number of records that have been dropped
PGW-EDR	
ADMIN STATE	The desired state of the Cflowd PGW-EDR
RECORDS REPORTED	The number of records that have been reported
RECORDS DROPPED	The number of records that have been dropped

10.2.2.16 dns-enrich

Syntax

```
dns-enrich dns-enrich-name detail
```

Context

```
show>app-assure>group
```

Description

This command displays DNS enrichment information.

Parameters

dns-enrich-name

specifies the DNS enrichment object name, up to 32 characters

detail

keyword to display detailed information

Output

The following output example shows DNS enrichment information, and table [Show DNS Enrichment output fields](#) describes the DNS enrich show command output fields.

Example output DNS IP cache

```
# show application-assurance group 1:0 dns-enrich "dns_enrich_01" detail
=====
Application Assurance Group 1:0 DNS Enrichment "dns_enrich_01"
=====
Description : (Not Specified)
Admin Status : Down
AQP Referenced : No
=====
Application Assurance Group 1:0 DNS Enrichment "dns_enrich_01" Statistics
=====
UDP
-----
Enriched Requests : 0

Missing Subscriber ID : 0
Template Not Enabled : 0
Traffic Characteristics : 0
Exceeded Max Packet Size : 0

-----
TCP
-----
Enriched Requests : 0

Missing Subscriber ID : 0
Template Not Enabled : 0
Out Of Resources : 0
Traffic Characteristics : 0
Exceeded Max Packet Size : 0

-----
Total : 0
=====
```

Table 195: Show DNS Enrichment output fields

Label	Description
Application Assurance Group	Displays the name of the application assurance group
Description	Displays the description of the group (if specified)

Table 195: Show DNS Enrichment output fields (continued)

Label	Description
Admin Status	Displays the administrative status of the group
AQP Referenced	Displays whether an AQP is referenced
Enriched Requests (UDP)	Displays the number of the enriched DNS UDP requests processed by the group
Missing Subscriber ID (UDP)	Displays the number of the DNS UDP requests in which the subscriber ID is missing
Template Not Enabled (UDP)	Displays the number of the DNS UDP requests in which the DNS enrichment policy is not enabled
Traffic Characteristics (UDP)	Displays the number of the DNS UDP requests dropped due to traffic characteristics errors such as fragmented, malformed, or tunneled (IP in IP) requests
Exceeded Max Packet Size (UDP)	Displays the number of DNS UDP requests dropped due to packet size exceeding the limit
Enriched Requests (TCP)	Displays the number of enriched DNS TCP requests processed by the group
Missing Subscriber ID (TCP)	Displays the number of DNS TCP requests in which the subscriber ID is missing
Template Not Enabled (TCP)	Displays the number of DNS TCP requests in which the DNS enrichment policy is not enabled
Out Of Resources (TCP)	Displays the number of DNS UDP requests dropped due to insufficient resources
Traffic Characteristics (TCP)	Displays the number of DNS UDP requests dropped due to traffic characteristics errors such as fragmented, malformed, or tunneled (IP in IP) requests in addition to other errors such as exceeded TCP window size or exceeded MTU size
Exceeded Max Packet Size (TCP)	Displays the number of DNS TCP requests dropped due to packet size exceeding the limit

Table 195: Show DNS Enrichment output fields (continued)

Label	Description
Total	Displays the total number of the enriched DNS requests (both UDP and TCP) processed by the group

10.2.2.17 dns-ip-cache

Syntax

```
dns-ip-cache [isa mda-id]
```

Context

```
show>app-assure>group
```

Description

This command displays information about the DNS IP cache for a particular ISA AA card.

Parameters

isa mda-id

specifies the ISA AA card for which DNS IP cache will be displayed

Output

The following output examples show DNS IP cache information, and [Show DNS IP cache output fields](#) describes the fields.

Example output DNS IP cache

```
# show application-assurance group 1 dns-ip-cache "testCache"
=====
Application Assurance Group 1 dns-ip-cache "testCache"
=====
Admin Status           : Down
AQP Ref                : No
Domain expressions     : 0 (out of 2000)
Server addresses       : 0 (out of 64)
Static addresses       : 0 (out of 8)
High watermark         : 90%
Low watermark          : 80%
Cache size             : 10
-----
```

```

ISA                               Usage          (%)          Alarm          Hit Count
                               State
-----
1/1                               0           0.00          clear           0
=====

# show application-assurance group 1 dns-ip-cache "Default DNS IP Cache" isa 5/1
=====
Application Assurance Group 1 dns-ip-cache "Default DNS IP Cache" ISA 5/1
=====
Admin Status                     : Up
AQP Ref                         : Yes
Domain expressions               : 1 (out of 2000)
Server addresses                 : 0 (out of 64)
Static addresses                 : 0 (out of 8)
High watermark                   : 75%
Low watermark                    : 50%
Cache size                       : 100
-----
ISA 5/1 DNS Stats
-----
DNS
  Total responses                : 93
  Domain name matched            : 93
  Domain & server matched        : 93
Cache
  Total entries added            : 179
  Total entries removed          : 113
Usage
  Usage                          : 66 (66.00%) threshold alarm clear
  Cache full count               : 0
  Static addresses matched       : 0
  Cached addresses matched       : 189
  Address miss count             : 55
=====

```

Show DNS IP cache output fields describes the DNS IP cache show command output fields.

Table 196: Show DNS IP cache output fields

Label	Description
Admin Status	[Up down] - Administrative status of the dns-ip-cache
Domain expressions	Number of DNS domain expressions configured
Server addresses	Number of server addresses configured

Table 196: Show DNS IP cache output fields (continued)

Label	Description
Static addresses	Number of static addresses configured
High-Watermark	Value in percent for the configured high-watermark
Low-Watermark	Value in percent for the configured low-watermark
Cache-size	Value for the configured maximum cache size
Usage	Value in percent of the total for the number of entries in the cache
Hit-Count	The number of times an IP address lookup in this cache was successful
DNS Total Responses	Total number of DNS responses analyzed
Domain name matched	Number of times a domain name defined in the dns-match criteria matched a DNS response
Domain and server matched	Number of times both the domain name and server-address defined in the dns-match criteria matched a DNS response
Cache Total entries added	Total number of IP entries added in the cache
Cache Total entries removed	Total number of IP entries removed in the cache after the entry expired
Full Count	Total number of IP entries in the cache
Hit Count	The number of times an IP address lookup in this cache was successful. The IP address lookup is performed in app-filters and is successful if the server-address dns-ip-cache criteria is met.
Miss Count	The number of times an IP address lookup in this cache was unsuccessful. The IP address lookup is performed in app-filters and

Table 196: Show DNS IP cache output fields (continued)

Label	Description
	is unsuccessful if the server-address dns-ip-cache criteria is not met.

10.2.2.18 event-log

Syntax

```
event - log event-log-name syslog
```

Context

```
show>app-assure>group
```

Description

This command displays information about event logs stored on a syslog server.

Parameters

event-log-name

displays information about the specified event log

Output

The following output shows an example of event information, and *Show AA group event log output fields* describes the fields.

Example output AA group event log

```
# show application-assurance group 1:1 event-log "e11" syslog
=====
Event-log admin status : up
Description             : (Not Specified)
Address                 : 3.3.3.3
Port                   : 514
VLAN ID                : 10
Facility                : local7
Severity               : info
-----
ISA                     Events Sent      Events Failed
-----
2/1                     0                0
```

Table 197: Show AA group event log output fields

Label	Description
Event Log	
Event-log admin status	[Up down] - Administrative status of the event-log
Description	Describes the log, if applicable
Address	The IP address of the event-log
Port	The port of the event-log
VLAN ID	The VLAN ID of the event-log
Facility	The facility of the event-log
Severity	Indicates the purpose of the event-log
ISA	Indicates the ISA
Events Sent	The number of events sent
Events Failed	The number of events that failed

10.2.2.19 http-enrich

Syntax

http-enrich *enrich-name*

Context

show>app-assure>group

Description

This command displays HTTP enrichment statistics.

Parameters

enrich-name

displays information about the specified enrichment name

Output

The following output examples show HTTP enrichment information.

Example output AA group HTTP enrichment statistics

```
# show application-assurance group 1 http-enrich "test"
=====
Application Assurance Group 1 HTTP Enrichment test
=====
Description : (Not Specified)
Admin Status: Down
-----
Grp:Part   Enriched           Out Of Resources    Missing Sub Data
          Tmpl Not Enabled       Traffic Char
-----
1:1        0                   0                   0
          0
1:2        0                   0                   0
          0
1:3        0                   0                   0
          0
-----
Total      0                   0                   0
          0
-----
=====
```

```
# show application-assurance group 1 http-enrich "Template" detail
=====
Application Assurance Group 1:1 HTTP Enrichment "Template" Statistics
=====
Enriched
-----
Total                      : 0
Not Enriched
-----
Out Of Resources           : 0
Missing Subscriber Data    : 0
Templates Not Enabled      : 0
Traffic Characteristics    : 0
Exceeded Max Packet Size   : 0
Session-dropped            : 0
Total                      : 0
Anti-Spoof
```

```
-----
Header Mods           : 0
No Header Mods (Too Short): 0
=====
```

10.2.2.20 detail

Syntax

detail [*partition*]

Context

show>app-assure>group>http-enrich

Description

This command displays detailed HTTP enrichment information.

10.2.2.21 field

Syntax

field *field-name*

Context

show>app-assure>http-enrich

show>app-assure>group>http-enrich

Description

This command displays HTTP enrichment field information.

Parameters

field-name

specifies the field name for which to display information, up to 32 characters

Output

The following output examples show HTTP enrichment field information.

Example output AA group HTTP enrichment field

```
# show application-assurance group 1 http-enrich "Template1" field
"customer-id"
=====
Application Assurance Group 1 HTTP Enrichment "Template1" Field "customer-id"
=====
Header          : IDNUM
Encoding        : none

Dump of all the fields:
>config>app-assure>group>http-enrich$ show application-assurance group 1
http-enrich "Template1" field
<field-name>
"imsi"          "msisdn"          "subscriber-ip"    "imei"
"rat-type"      "static-string"
"charging-id"   "pgw-ggsn-address" "sgw-sgsn-address" "apn"
"user-location" "billing-type"
"plmn-id"       "roaming-status"   "customer-id"      "timestamp"
"apn-ni"        "msisdn-without-cc"
"charging-id-2" "imsi-2"           "sgw-sgsn-address-2" "static-string-2"
"imei-hyphenated" "imei-hyphenated-2"
"user-location-raw" "user-location-raw-2" "pcc-rulebase"
```

10.2.2.22 fields

Syntax

fields

Context

show>app-assure>http-enrich

Description

This command displays HTTP enrichment fields.

10.2.2.23 http-notification

Syntax

http-notification *http-notification-name* [summary]

Context

show>app-assure>group

Description

This command displays information about the configured http-notification policy with associated raw statistics:

- Template: Template Id in use
- Script URL: URL address of the script used in the notification message
- Admin Status: Up / Down
- AQP Referenced: Yes/No
- Notified: Total number of notifications sent
- Notification criteria selection not matched: Number of HTTP request not matching the selection criteria for in browser notification

Parameters

http-notification-name

displays the name of the http-notification policy

summary

keyword to display summarized information

Output

The following output example shows HTTP notification information, and *Show AA group HTTP notification output fields* describes the fields.

Example output AA group HTTP notification

```
# show application-assurance group 1 http-notification "in-browser-notification"
=====
Application Assurance Group 1 HTTP Notification "in-browser-notification"
=====
Description   : IBN Demo NOK Message
Template      : 1 - Javascript-url with subId and optional Http-Url-Param
Script URL    : http://1.1.1.1/In-Browser-Notification/script.js
Admin Status  : Up
AQP Ref       : Yes
-----

```

	Notified	Notification Selection Criteria Not Matched
1:1	3	0
1:2	2	0
1:3	0	0
1:4	0	0
1:5	0	0
Total	5	0

```
-----
```

Table 198: Show AA group HTTP notification output fields

Label	Description
Application Assurance Group 1 HTTP Notification "in-browser-notification"	
Description	Describes the HTTP notification
Template	The type of template being used
Script URL	Provides the URL for the script
Admin Status	The preferred status of the HTTP notification
AQP Ref	Indicates if there is an AQP reference
Notified	Indicates whether the user is notified
Notification Selection Criteria Not Matched	Indicates how many criteria do not match the notification selection

10.2.2.24 http-proxy

Syntax

```
http-proxy http-proxy-name isa mda-id
```

Context

```
show>app-assure>group
```

Description

This command displays information about the HTTP proxy.

Parameters

http-proxy-name

displays the name of the configured proxy

mda-id

specifies the slot and MDA in the format slot/mda

Output

The following output examples show information about the HTTP proxy and the URL mapping.

Example output AA group HTTP-proxy

```
# show application-assurance group 1 http-proxy "h1"
=====
Application Assurance Group 1 HTTP-Proxy h1
=====
Description                               : (Not Specified)
DNS Server Address                         : 10.1.0.2
VLAN ID                                   : 100
-----
HTTP Proxy Statistics
-----
Sessions Initiated                        : 1
Sessions Established                      : 1
Sessions Closed                          : 1
Sessions Aborted                         : 0
Time-Out                                 : 0
Unsupported Version                      : 0
Other                                    : 0
-----
HTTP Host Requests
-----
Sent                                     : 1
Resent                                  : 0
Resolved                                : 1
Failed                                  : 0
-----
TCP Connection Statistics                 Subscriber      Network
-----
Active Connections                       0               0
Closed Connections                       1               1
-----
DNS Client Statistics
-----
Requests Sent                           : 1
Requests Sent Cache Refresh              : 0
Requests Failed                         : 0
Responses Received                       : 1
Responses Failed                        : 0
Average Round Trip                       : 2379 microseconds
```

DNS Cache Statistics

```

Entries           : 0
Lookups           : 1
Matches           : 0
Misses            : 1

```

Traffic Statistics Sub-to-Net Net-to-Sub

```

Packets Received      40091      72554
  Octets Received      2160259    108817958
Packets Transmitted    37588      73342
  Octets Transmitted    1955717    109997101
Packets Retransmitted    0         786
  Octets Retransmitted    0      1179000

```

TLS Counts Subscriber Network

```

Handshakes Initiated      1         0
Handshakes Completed      0         0
Handshakes Aborted        1         0
  Invalid Message          0         0
  Authentication Error      0         0
  Unsupported Encryption    0         0
  Other                    1         0
Decryption Errors          0         0
=====
-----

```

Example output AA group http-proxy HTTPS url-mapping

```
# show application-assurance group 1 http-proxy H1 https url-mapping
```

```
=====
Application Assurance Group 1 HTTP-Proxy HP1 URL Mapping Entry 1
=====
```

```
Entry ID                      Hit Counts
-----
```

```
1                              1
=====
```

10.2.2.25 count**Syntax**

```
count [detail]
```

Context

```
show>app-assure>group>aa-sub
```

Description

This command displays per-subscriber app-group application and protocol statistics.

Parameters

detail

displays detailed information

Output

The following output examples show per subscriber application group application and protocol information and *Show AA group AA-sub count detail output fields* describes the fields.

Example output AA group AA-sub count detail

```
# show>app-assure>group>aa-sub# count
=====
Application-Assurance Subscriber TestSubscriberName
Application Group, Application and Protocol Statistics
=====
Application Group          Disc Octets          Packets          Flows
-----
Database                   0% 0                0                0
File Transfer              0% 27243            169              22
Games                     0% 0                0                0
Infrastructure             0% 71494            555              515
Instant Messaging         0% 4947792          25587            411
Local Content             0% 923              8                2
Mail                     0% 53729            318              22
Mail Server               0% 0                0                0
MultiMedia                0% 31670667         33087            142
NNTP                     0% 0                0                0
Peer to Peer              .45% 11096224       16339            2431
Premium Partner           0% 0                0                0
Remote Connectivity       0% 15321            171              2
Server                   0% 0                0                0
Suspect                   72% 1012            11               11
Tunneling                 0% 19659289         33535            164
Unknown                   0% 1945164          6317             287
Web                       0% 29538078         34873            1022
Web Server                0% 0                0                0
=====
Application                Disc Octets          Packets          Flows
-----
HTTP_Local                0% 923              8                2
=====
Protocol                  Disc Octets          Packets          Flows
-----
dns                       1.8% 40010          277              277
```

```
>show>app-assure>group>aa-sub# count detail
```

```
Application-Assurance Subscriber TestSubscriberName
Application Group, Application and Protocol Statistics
```

```
=====
Subscriber                               Application Group:
Type                               Octets           Packets           Flows
-----
TestSubscriberName                    Instant Messaging:
Admitted from subscriber: 2558576           12720             229
Denied from subscriber:    0                 0                 0
Active flows from subscriber:                0
Admitted to subscriber:   2389216           12867             182
Denied to subscriber:     0                 0                 0
Active flows to subscriber:                0
Total flow duration:      2912 seconds
Terminated flows:                411
Short Duration flows:            387
Medium Duration flows:           22
Long Duration flows:             2
...
TestSubscriberName                    Web:
Admitted from subscriber: 2343429           22806             511
Denied from subscriber:    0                 0                 0
Active flows from subscriber:                1
Admitted to subscriber:   56359191          40528             511
Denied to subscriber:     0                 0                 0
Active flows to subscriber:                1
Total flow duration:      4783 seconds
Terminated flows:                1020
Short Duration flows:            989
Medium Duration flows:           31
Long Duration flows:             0
=====
```

```
=====
Subscriber                               Application:
Type                               Octets           Packets           Flows
-----
TestSubscriberName                    HTTP_Local:
Admitted from subscriber: 0                 0                 0
Denied from subscriber:    0                 0                 0
Active flows from subscriber:                0
Admitted to subscriber:    0                 0                 0
Denied to subscriber:     0                 0                 0
Active flows to subscriber:                0
Total flow duration:      0 seconds
Terminated flows:                0
Short Duration flows:            0
Medium Duration flows:           0
Long Duration flows:             0
=====
```

```
=====
Subscriber                               Protocol:
Type                               Octets           Packets           Flows
-----
```

```

TestSubscriberName      dns:
Admitted from subscriber: 0          0          0
Denied from subscriber:  0          0          0
Active flows from subscriber:          0
Admitted to subscriber:  0          0          0
Denied to subscriber:    0          0          0
Active flows to subscriber:          0
Total flow duration:      0 seconds
Terminated flows:          0
Short Duration flows:      0
Medium Duration flows:     0
Long Duration flows:       0
=====

```

Table 199: Show AA group AA-sub count detail output fields

Label	Description
Application-Assurance Subscriber TestSubscriberName Application Group, Application and Protocol Statistics	
Subscriber	Specifies which subscriber is being viewed
Application Group	Specifies the application group
Type	Specifies the AA type
Octets	The total number of octets received
Packets	The total number of packets received
Flows	The total number of flows received
Application	The application
Protocol	The protocol
Admitted from subscriber	The number of octets, packets, and flows admitted from the subscriber
Denied from subscriber	The number of octets, packets, and flows denied from the subscriber

Table 199: Show AA group AA-sub count detail output fields (continued)

Label	Description
Active flows from subscriber	The number of active flows coming from the subscriber
Admitted to subscriber	The number of octets, packets, and flows admitted to the subscriber
Denied to subscriber	The number of octets, packets, and flows denied to the subscriber
Active flows to subscriber	The number of active flows going to the subscriber
Total flow duration	Identifies the amount of time in the flow duration
Terminated flows	The total number of terminated flows
Short Duration flows	The number of short duration flows
Medium Duration flows	The number of medium duration flows
Long Duration flows	The number of long duration flows

10.2.2.26 admin

Syntax

admin

Context

show>app-assure>group>policy

Description

This command displays the application-assurance policy uncommitted changes.

Output

The following output example shows uncommitted changes for AA policy.

Example output AA group administrator

```
# show>app-assure>group>policy# admin
begin
app-filter
    entry 10 create
        shutdown
    exit
exit
app-qos-policy
    entry 10 create
        shutdown
    exit
exit
commit
```

10.2.2.27 app-filter

Syntax

app-filter [*entry-id*]

Context

show>app-assure>group>policy

Description

This command displays application-assurance policy filter information.

Parameters

entry-id

specifies an existing application filter entry

Values: 1 to 65535

Output

The following output example shows about information for AA policy filter.

Example output AA policy filter

```
# show application-assurance group 1:0 policy app-filter 35505
app-filter
    entry 35505 create (487 flows, 136330 kB)
        ip-identification-assist (388 flows, 134410 kB)
        description "HTTP over SSL/TLS"
        protocol eq "tls"
        server-port eq 443
```

```
        application "HTTPS"  
        no shutdown  
    exit  
exit
```

10.2.2.28 app-group

Syntax

```
app-group [app-group-name]
```

Context

```
show>app-assure>group>policy
```

Description

This command displays application-assurance policy application group information.

10.2.2.29 app-profile

Syntax

```
app-profile [app-prof-name]  
app-profile app-prof-name associations
```

Context

```
show>app-assure>group>policy
```

Description

This command displays application-assurance policy application profile information.

Parameters

app-prof-name

specifies an existing application profile name

associations

displays subscriber management associations

10.2.2.30 app-qos-policy

Syntax

```
app-qos-policy [entry-id]
```

Context

```
show>app-assure>group>policy
```

Description

This command displays application-assurance policy application QoS policy information.

Parameters

entry-id

specifies an existing application QoS policy entry id

Values: 1 to 65535

10.2.2.31 app-service-option

Syntax

```
app-service-option [characteristic-name]
```

Context

```
show>app-assure>group>policy
```

Description

This command displays application-assurance policy application service option information.

10.2.2.32 application

Syntax

```
application [app-name]
```

Context

```
show>app-assure>group>policy
```


Description

This command displays application-assurance policy application information.

10.2.2.33 charging-filter

Syntax

charging-filter *id*

Context

show>app-assure>group>policy

Description

This command displays information about the charging filter.

Parameters

id

specifies the ID for the existing charging filter entry

Output

The following output example shows the charging filter information, and *Show AA group policy charging filter output fields* describes the fields.

Example output AA group policy charging filter

```
# show application-assurance group 1:0 policy charging-filter 1

=====
Charging Filter Entry 1
=====
Description      : "Charging-filter entry for WhatsApp Video"
Admin State      : in-service
Charging Group    : "cgVideo100"
Hits             : 0 flows

Match :
  Application      : eq "Whats App"
  Flow Attribute   : flow-attribute video confidence gte 100
=====
```

Table 200: Show AA group policy charging filter output fields

Label	Description
Charging Filter Entry	The charging filter
Description	The description of the charging filter
Admin State	Indicates the administrative state of the filter
Charging Group	Indicates the charging group
Hits	The number of hits
Application	Indicates the matched Application
Flow Attribute	Indicates the flow attribute used by the match criteria

10.2.2.34 charging-group

Syntax

charging-group *charging-group-name*

Context

show>app-assure>group>policy

Description

This command displays a charging group with this application group in an application assurance policy.

Parameters

charging-group-name

a string of up to 32 characters uniquely identifying this application group in the system

Output

The following output examples show charging group information, and *Show AA group policy charging group output fields* describes the fields.

Example output AA group policy charging group

The following output shows a charging group that has been configured as a default-tethered charging group.

```
# show>app-assure>group>policy# charging-group "bob"
=====
Charging-group Instance "bob" (default-tethered)
=====
Description      : (Not Specified)
Export Id        : 12
References
-----
AA Sub Stat : none
Application :
App Group   :
AQP Entry   :

Reference Counts
-----
No. of application references : 0
No. of app group references   : 0
No. of AQP entry references   : 0
=====
```

The following output shows a charging group that has been configured both as a default charging group and as a default-tethered charging group.

```
# show>app-assure>group>policy# charging-group "bob"
=====
Charging-group Instance "bob" (default, default-tethered)
=====
Description      : (Not Specified)
Export Id        : 12
References
-----
AA Sub Stat : none
Application :
App Group   :
AQP Entry   :

Reference Counts
-----
No. of application references : 0
No. of app group references   : 0
No. of AQP entry references   : 0
=====
```

The following output shows a charging group that is referenced by a charging filter:

```
show application-assurance group 1:0 policy charging-group "cgAudio"

=====
Charging-group Instance "cgAudio"
=====
Description      : (Not Specified)
Export Id       : none
Notify-start-stop : none

References
-----
AA Sub Stat      : none
Application      :
App Group       :
AQP Entry       :
Charging Filter Entry :
                  4

Reference Counts
-----
No. of application references      : 0
No. of app group references       : 0
No. of AQP entry references       : 0
No. of charging-filter entry references : 1
```

Table 201: Show AA group policy charging group output fields

Label	Description
Charging-group Instance	The charging group instance name and whether it is the default or the default-tethered charging group, or both
Description	The charging filter description
Export Id	The export ID
References	References
AA Sub Stat	The AA subscriber statistic

Table 201: Show AA group policy charging group output fields (continued)

Label	Description
Application	The application
App Group	The application group
AQP Entry	The AQP entry
Reference Counts	Reference counts
No. of application references	The number of application references
No. of app group references	The number of application group references
No. of AQP entry references	The number of AQP entry references

10.2.2.35 policers

Syntax

policers

Context

show>app-assure>group>policy>aa-sub

Description

This command displays policer configuration information.

Output

The following output example shows policer configuration information, and *Show AA group AA-sub policers output fields* describes the fields.

Example output AA group AA-sub policers

```
# show>app-assure>group>aa-sub# policers
=====
```

Application-Assurance Subscriber Policer Summary

```
=====
AA-Subscriber      : Alex (esm)
-----
```

```
Type: single-bucket-bandwidth Direction: subscriber-to-network
-----
```

```
AQP  Policer                      Resources Exceeded?
-----
```

```
61   SuspectUp_policer           N
-----
```

```
Type: single-bucket-bandwidth Direction: network-to-subscriber
-----
```

```
AQP  Policer                      Resources Exceeded?
-----
```

```
62   SuspectDown_policer        N
-----
```

```
Policer usage counts:
```

```
single-bucket-bandwidth
```

```
  subscriber-to-network  1      out of   32
```

```
  network-to-subscriber  1      out of   32
```

```
dual-bucket-bandwidth
```

```
  subscriber-to-network  0      out of    1
```

```
  network-to-subscriber  0      out of    1
```

```
flow-count-limit        0      out of    8
```

```
flow-rate-limit         0      out of    8
=====
```

Table 202: Show AA group AA-sub policers output fields

Label	Description
Application-Assurance Subscriber Policer Summary	
AA Subscriber	Indicates which AA Subscriber is being viewed
Type	Specifies the type of policer
Direction	The direction of the flow of the policer
AQP	Specifies the AQP
Policer	Specifies which policer is being viewed
Resources Exceeded?	Indicates if the resources have been exceeded or not

Table 202: Show AA group AA-sub policers output fields (continued)

Label	Description
Policer usage counts	Indicates how much a policer has used in specified categories

10.2.2.36 custom-protocol

Syntax

```
custom-protocol
```

Context

```
show>app-assure>group>policy
```

Description

This command displays application-assurance policy custom protocol information.

10.2.2.37 summary

Syntax

```
summary
```

Context

```
show>app-assure>group>policy
```

Description

This command displays application-assurance policy summary information.

10.2.2.38 protocol

Syntax

```
protocol [protocol-name] count [detail]
```

Context

```
show>app-assure>group>aa-sub
```

```
show>app-assure>group
```

Description

This command displays per-protocol statistics. The system-wide statistics displayed account for all flows completed and the last internal snapshot of the active flows.

Subscriber statistics are available for special study subscribers and account for all completed and active flows at the moment of this statistics request.

Parameters

protocol-name

displays information about the specified protocol name

count

displays protocol counters

detail

displays detailed information

Output

The following output examples show per-protocol statistics information.

- [Example output AA group protocol count](#)
- [Example output group protocol count details](#)

Example output AA group protocol count

```
# show>app-assure>group# protocol count
=====
Protocol Statistics
=====
Protocol                               Disc Octets           Packets           Flows
-----
aim_oscar                             0% 0                 0                 0
aim_oscar_file_xfer                   0% 0                 0                 0
aim_oscar_video_voice                 0% 0                 0                 0
aim_toc                               0% 0                 0                 0
bittorrent                            0% 0                 0                 0
```

Example output group protocol count details

```
# show>app-assure>group# protocol "http_audio" count detail
=====
Protocol "http_audio" Statistics
=====
```



```

Protocol:
Type           Octets           Packets           Flows
-----
http_audio:
Admitted from subscriber: 14958           201           2
Denied from subscriber:    0             0           0
Active flows from subscriber:                0
Admitted to subscriber:   587590          396           2
Denied to subscriber:     0             0           0
Active flows to subscriber:                0
Total flow duration:      21 seconds
Terminated flows:                4
Short Duration flows:          4
Medium Duration flows:         0
Long Duration flows:           0
Active subscribers:         1
=====

```

10.2.2.39 summary

Syntax

```
summary
```

Context

```
show>app-assure>group>aa-sub
```

Description

This command displays a summary of statistics for a specific aa-sub.

Output

The following output examples show statistics summary information.

Example output AA group summary statistics

```

# show application-assurance group 1 aa-sub mobile imsi 123456789012001 apn "ipd.nokia.com" summary
=====
Application-Assurance Subscriber Summary (realtime)
=====
Mobile Gateway Id      : 1
Subscriber name        : IMSI 310012567489000
5G Subscriber          : No
IMSI                   : 310012567489000
MSISDN                 : 12345098761
IMEI                   : 987654321012399
Session-Id             : 0x03440001
Customer Id            : 0
APN                    : ipd.nokia.com

```

```

Virtual APN      : N/A
ISA assigned     : 3/2
App-Profile      : APP_PROFILE_1
RAT Type        : EUTRAN
Bearer Id       : 5
HTTP URL Parameters : (Not Specified)
Mobile Charging Id : 3506438144
Mobile Billing Type : 0x0100
PGW/GGSN address : 10.20.1.3
SGW/SGSN address(es) : 10.20.1.1
User Location    : N/A
PLMN ID         : 310012
Roaming Status   : HOME
Access Ntwk Location : None
ANL Type        : N/A
Sub Congestion State : Not Congested
Tethering Status : not-detected (disabled)
Devices Detected : N/A
Tracked Devices  : tracked: 10 (max: 10)  excess: yes (hist: yes)

```

Traffic	Octets	Packets	Flows
From subscriber:			
Admitted	0	0	0
Denied	0	0	0
Active flows			0
To subscriber:			
Admitted	0	0	0
Denied	0	0	0
Active flows			0
Flow counts:			
Terminated			0
Short duration			0
Med duration			0
Long duration			0

Total flow duration : 0 seconds

Application Service Options (ASO)

Characteristic	Value	Derived from
----------------	-------	--------------

None

Notify-Start-Stop

Charging-Group	Notify-Type	Enabled	Active
----------------	-------------	---------	--------

None

Notify-Start-Stop

Charging-Group	Notify-Type	Enabled	Active
----------------	-------------	---------	--------

CG_1	flow	no	n/a
CG_2	group	yes	no
CG_3	dyn-rating	yes	yes

10.2.2.40 ip-identification-assist

Syntax

```
ip-identification-assist [isa mda-id]
```

Context

```
show>app-assure>group
```

Description

This command displays the statistics and status of IP identification assist feature.

Parameters

mda-id

specifies the slot and MDA in the format *slot/mda*

Values	<i>slot</i>	1 to 20
–	<i>mda</i>	1 or 4

Output

The following example output shows information about the IP identification assist.

Example output IP identification assist

```
# show application-assurance group 1 ip-identification-assist isa 1/1
```

```
=====
Application Assurance Group 1 IP Identification Assist
=====
```

```
ip-identification-assist      : Up
passive-dns-monitor           : Enabled
```

```
-----
IP Entry Statistics
-----
```

```

Maximum IP entries           : 100000
Free IP entries              : 99866
Active IP entries            : 134
    application               : 126
    parent application        : 6
    conflict (apps/parent)    : 0
    conflict (other)          : 2

```

Active IP entry match status

```

IP entry matched             : 85
IP entry not matched         : 49

```

Cumulative IP lookup results

```

total misses                  : 1383
total matches                  : 594
    match - application        : 528
    match - parent application : 56
    match - app/parent conflict : 0
    match - other conflict      : 10

```

10.2.2.41 mp-tcp-proxy

Syntax

```
mp-tcp-proxy proxy-name [isa mda-id]
```

Context

```
show>app-assure>group
```

Description

This command displays Multi-Path TCP (MP-TCP) proxy information.

Parameters

proxy-name

specifies the MP-TCP proxy name

mda-id

specifies the slot and MDA in the format *slot/mda*

Values	<i>slot</i>	1 to 10
–	<i>mda</i>	1 or 2

Output

The following is an output example of the `mp-tcp-proxy show` command.

Example output AA group multi-path TCP proxy

```
# show application-assurance group 1 mp-tcp-proxy "MPTCP1"
=====
Application-Assurance Group 1 mp-tcp-proxy MPTCP1
=====
Description      : (Not Specified)
Def-sched-plcy   : min-cost-prefer
Fixed path cost  : 5
Mobile path cost : 5
DSS Checksum     : Disabled
Subflow Fail Ind : Disabled
IP Profile       : IPProfile1
IPv4 Address     : 11.12.13.14
IPv6 Address     : (Not Specified)
-----
Session Error Counts
-----
No Buffers      : 0
No Sessions     : 0
-----
Session Counts   single-path   multi-path   total
-----
Active          0             0            0
Closed          17            291          308
Aborted         106            0            106
  Network       0             0            0
  Other         106            0            106
-----
Component Counts fixed path   mobile path
-----
Initial Accepted 414          0
Initial Denied   0            0
Join Accepted    0            291
Join Denied      0            0
  Duplicate      0            0
  Bad MSS        0            0
  Other          0            0
Closed Flows     308          291
Aborted Flows    106          0
-----
```

Traffic Counts	fixed path	mobile path	network

sub-to-net octets	1086001	1258423	1847291
sub-to-net packets	16362	8066	19630
net-to-sub octets	16411989	4221042	19394639
net-to-sub packets	16974	8582	18256
retransmit octets	285082	146505	31455
retransmit packets	206	101	450
dropped error octs	0	0	0
dropped error pkts	0	0	0
=====			

10.2.2.42 partition

Syntax

```
partition summary
```

Context

```
show>app-assure>group
```

Description

This command displays partition information.

Parameters

summary

displays partition summary information

10.2.2.43 policer

Syntax

```
policer [policer-name]
```

```
policer summary
```

Context

```
show>app-assure>group
```

Description

This command displays application-assurance policer information.

Parameters

policer-name

displays information about the specified policer

summary

displays summarized information about policers on this node

Output

The following output examples show AA policer information and *Show policer output fields* describes the show output fields.

Example output AA group policer

```
# show>app-assure>group# policer
policer "100k_policer" type single-bucket-bandwidth granularity subscriber create
    rate 100
    mbs 10
    exit
policer "200FlowsDown_Policer" type flow-count-limit granularity subscriber create
    flow-count 200
    exit
policer "ServerDown_policer" type single-bucket-bandwidth granularity subscriber
create
    action priority-mark
    rate 500
    mbs 50
    exit
policer "ServerUp_policer" type single-bucket-
bandwidth granularity subscriber create
    action priority-mark
    rate 500
    mbs 50
    exit
...
policer "SuspectUp_policer" type single-bucket-bandwidth granularity subscriber
create
    rate 100
    mbs 10
    exit
policer "Video_channel_policer" type single-bucket-bandwidth granularity subscriber
create
    action priority-mark
    rate 2000
    mbs 200
    exit

Number of policers          : 26

>show>app-assure>group# policer "P2PAggrBwDown"
policer "P2PAggrBwDown" type single-bucket-bandwidth granularity system create
    rate 1000000
    mbs 100000
```

```
exit
```

```
# show>app-assure>group# policer summary
System-level single-bucket-bandwidth policers : 3 out of 64
System-level flow-count-limit policers       : 1 out of 64
System-level flow-rate-limit policers        : 1 out of 64
Subscriber-level single-bucket-bandwidth to-subscriber policers
  Base non-subscriber-specific entries       : 1 out of 32
  Worst case                                : 3 out of 32
  occurs with app-profile "Super"
Subscriber-level single-bucket-bandwidth from-subscriber policers
  Base non-subscriber-specific entries       : 1 out of 32
  Worst case                                : 2 out of 32
  occurs with app-profile "Lite"
Subscriber-level flow-count-limit policers
  Base non-subscriber-specific entries       : 0 out of 8
  Worst case                                : 2 out of 8
  occurs with app-profile "Lite"
Subscriber-level flow-rate-limit policers
  Base non-subscriber-specific entries       : 0 out of 8
  Worst case                                : 0 out of 8
```

Example output for a specific policer instance

```
# show>app-assure>group# policer "test"
=====
Policer Instance "test"
=====
Description      : (Not Specified)
Type             : single-bucket-bandwidth
Granularity      : access-network-location
Action           : bandwidth-shaper
Adaptation Rule  : pir closest
Rate-Percentage  : 10
=====
```

Table 203: Show policer output fields

Label	Description
Policer Instance	The name of the group for which the policer information is being shown
Description	The description entered by the user
Type	The policer type, either:

Table 203: Show policer output fields (continued)

Label	Description
	single-bucket-bandwidth a single bucket (PIR) bandwidth limiting policer dual-bucket-bandwidth a dual bucket (PIR, CIR) bandwidth limiting policer flow-rate-limit a limiting rate of flow set-ups policer flow-count-limit a limiting total flow count policer system a policer that limits the traffic in the scope of all or a subset of AA subscribers on a given AA ISA subscriber a policer for each AA subscriber that limits the traffic in the scope of that subscriber access-network-location a policer instance for each Access Network Location (ANL) that limits traffic bandwidth in the scope of that ANL. For ANL, only single-bucket bandwidth policers can be configured.
Granularity	The granularity type of the policer, either:
Action	The bandwidth-shaper policer type
Adaptation Rule	Defines the method used by the system to derive the operational CIR and PIR settings
Rate Percentage	The rate used by Access-Network-Location (ANL) policers

10.2.2.44 policy

Syntax

```
policy
```

Context

```
show>app-assure>group
```

Description

This command enables the context to display application-assurance policy configuration information.

10.2.2.45 http-redirect

Syntax

```
http-redirect redirect-name
```

Context

```
show>app-assure
```

```
show>app-assure>group
```

Description

This command displays information about the configured HTTP redirect policies along with some associated raw statistics. These output statistics are:

Redirects Sent: total number of redirects handled.

Redirect not sent out of resource: Redirects that were suppose to take effect but didn't due to running out of memory allocated for HTTP redirects.

Error: This parameter captures of redirect cases that were suppose to take effect but didn't due to any of the following reasons:

- fragmented response packets
- circular redirects
- disabled or incomplete redirect configuration

Parameters

redirect-name

the HTTP redirect policy

Output

The following output example shows information about configured HTTP redirect policies, and *Show AA group HTTP redirect output fields* describes the fields.

Example output AA group HTTP redirect

```
# show application-assurance group 1 http-redirect "ispredirect"
=====
Application-Assurance Group 1 http-redirect ispredirect
=====
description      : The ISP common http redirect
template         : 1
                  : Default redirect format
http-host        : www.isp.com/redirectpage
admin status     : Down
-----
Grp:Part  Redirects      Redirects Not Sent
          Sent           Out of
                           Resource
-----
1:1       0              0
1:9000    0              0
-----
Total     0              0
-----
=====
```

Table 204: Show AA group HTTP redirect output fields

Label	Description
description	Describes the http-redirect
template	Indicates which template is being used for the http-redirect
http-host	The HTTP host
admin status	Indicates the desired status of the http-redirect
Grp:Part	Indicates the group to part ratio
Redirects Sent	Indicates the number of redirects successfully sent

Table 204: Show AA group HTTP redirect output fields (continued)

Label	Description
Redirects Not Sent - Out of Resource	Indicates the number of redirects that were not sent due to a lack of resources
Redirects Not Sent - Error	Indicates the number of redirects that were not sent due to an error

10.2.2.46 status

Syntax

```
status [isa mda-id] cflowd
status [isa mda-id]
status [isa mda-id] detail
status [isa mda-id] overload
status {isa mda-id} qos count
status {isa mda-id} qos pools
```

Context

```
show>app-assure>group
```

Description

This command displays system statistics.

Parameters

isa

displays information about the specified AA ISA

cflowd

displays cflowd status information

detail

displays detailed status information

overload

displays overload information

qos count

displays information about queue statistics. The *isa mda-id* must be specified.

qos pools

displays information about pool utilization. The *isa mda-id* must be specified.

Output

The following output examples show AA ISA information.

Example output AA group status statistics

```
# show>app-assure>group# status
=====
Application-assurance Status
=====
Last time change affecting status: 01/30/2009 20:14:37
Active Subs                : 1
-----

```

	Packets	Octets
Diverted traffic	: 58783	46140537
Diverted discards	: 4	0
Entered ISA-AAs	: 58784	46140614
Discarded in ISA-AAs	: 60	4620
Exited ISA-AAs	: 58724	46135994
Returned discards	: 0	0
Returned traffic	: 58724	46135994

```
=====

# show application-assurance group 1 status detail
=====
Application-Assurance Status
=====
Last time change affecting status : 11/09/2018 20:28:30
Number of Active ISAs            : 1
Flows                            : 156506
Flow Resources In Use            : 0
AA Subs Created                  : 514
AA Subs Deleted                  : 0
AA Subs Modified                 : 0
Seen IP Requests Sent            : 0
Seen IP Requests Dropped         : 0
-----

```

	Current	Average	Peak
Active Flows	: 0	372	11656
Flow Setup Rate (per second)	: 0	43	1368
Traffic Rate (Mbps)	: 0	8	254
Packet Rate (per second)	: 0	1394	44164
AA-Subs Downloaded	: 514	96	514
Active Subs	: 0	11	256

```
-----
from-sub packets      from-sub octets
```

	to-sub packets	to-sub octets

Diverted traffic	: 2124423	471664562
	2936935	3129010444
Diverted discards	: 0	0
	0	0
Congestion	: 0	0
	0	0
Errors	: 0	N/A
	0	N/A
Entered ISA-AAs	: 2124423	471664562
	2936935	3129010444
Buffered in ISA-AAs	: 0	0
	0	0
Discarded in ISA-AAs	: 0	0
	0	0
Policy	: 0	0
	0	0
Congestion	: 0	0
	0	0
Errors	: 0	0
	0	0
TCP Optimization	: 0	0
	0	0
MP-TCP Proxy	: 0	0
	0	0
Modified in ISA-AAs	: 0	0
	0	0
Packet size increased	: 0	0
	0	0
Packet size decreased	: 0	0
	0	0
Errors (policy bypass)	: 0	0
	0	0
Errors (cut-through)	: 0	0
	0	0
Generated in ISA-AAs	: 0	0
	0	0
TCP Optimization	: 0	0
	0	0
MP-TCP Proxy	: 0	0
	0	0
HTTP Redirect	: 0	0
	0	0
Exited ISA-AAs	: 2124423	471664562
	2936935	3129010444
Returned discards	: 0	0
	0	0
Congestion	: 0	0
	0	0
Errors	: 0	N/A
	0	N/A
Returned traffic	: 2124423	471664562
	2936935	3129010444
=====		

```
# show>app-assure>group# status isa 3/2 qos count
```

```
=====
Application-assurance Queue Statistics for ISA-AA Group: 1, isa 3/2
=====
```

```
-----
Egress From-Subscriber
```

Queue		Packets	Octets
1			
In Profile	forwarded :	0	0
In Profile	dropped :	0	0
Out Profile	forwarded :	28940	3767233
Out Profile	dropped :	0	0
2			
In Profile	forwarded :	0	0
In Profile	dropped :	0	0
Out Profile	forwarded :	0	0
Out Profile	dropped :	0	0

```
Egress To-Subscriber
```

Queue		Packets	Octets
1			
In Profile	forwarded :	0	0
In Profile	dropped :	0	0
Out Profile	forwarded :	44499	53066848
Out Profile	dropped :	0	0
2			
In Profile	forwarded :	0	0
In Profile	dropped :	0	0
Out Profile	forwarded :	0	0
Out Profile	dropped :	0	0

```
Ingress From-Subscriber
```

Queue		Packets	Octets
1			
In Profile	forwarded :	25548	3361023
In Profile	dropped :	0	0
Out Profile	forwarded :	1	60
Out Profile	dropped :	0	0
2			
In Profile	forwarded :	2921	365606
In Profile	dropped :	0	0
Out Profile	forwarded :	0	0
Out Profile	dropped :	0	0
9			
In Profile	forwarded :	0	0
In Profile	dropped :	0	0
Out Profile	forwarded :	0	0
Out Profile	dropped :	0	0
10			
In Profile	forwarded :	0	0
In Profile	dropped :	0	0
Out Profile	forwarded :	0	0
Out Profile	dropped :	0	0

```
Ingress To-Subscriber
```

Queue		Packets	Octets
1			

```

In Profile forwarded : 39541          46899769
In Profile dropped   : 0              0
Out Profile forwarded : 1             92
Out Profile dropped   : 0              0
Queue 2              Packets          Octets
In Profile forwarded : 5050           6291204
In Profile dropped   : 0              0
Out Profile forwarded : 0              0
Out Profile dropped   : 0              0
Queue 9              Packets          Octets
In Profile forwarded : 0              0
In Profile dropped   : 0              0
Out Profile forwarded : 0              0
Out Profile dropped   : 0              0
Queue 10             Packets          Octets
In Profile forwarded : 0              0
In Profile dropped   : 0              0
Out Profile forwarded : 0              0
Out Profile dropped   : 0              0

```

```

=====
# show>app-assure>group# status isa 3/2 qos pools
=====

```

Pool Information

```

=====
MDA           : 3/2
Application    : Net-Ing          Pool Name       : default
Resv CBS       : 50%

```

```

-----
Utilization      State      Start-Avg  Max-Avg  Max-Prob
-----
High-Slope       Up          70%       90%      80%
Low-Slope        Up          50%       75%      80%

```

```

Time Avg Factor  : 7
Pool Total       : 40960 KB
Pool Shared      : 20480 KB      Pool Resv       : 20480 KB

```

```

High Slope Start Avg : 12288 KB      High slope Max Avg : 16384 KB
Low Slope Start Avg  : 10240 KB      Low slope Max Avg  : 14336 KB

```

```

Pool Total In Use  : 0 KB
Pool Shared In Use : 0 KB      Pool Resv In Use  : 0 KB
WA Shared In Use   : 0 KB

```

```

Hi-Slope Drop Prob : 0      Lo-Slope Drop Prob : 0

```

```

-----
FC-Maps          Dest      MBS      Depth  A.CIR  A.PIR
Q-Grp            Q-Id      CBS
-----
be af l1 h2 ef h1 nc  5/*      20480    0      8000000 20000000
                      1        1280    8000000 Max
be af l1 h2 ef h1 nc  4/*      20480    0      8000000 20000000
                      1        1280    8000000 Max
be af l1 h2 ef h1 nc  3/1      20480    0      8000000 20000000

```



```

1          1280          8000000 Max
be af l1 h2 ef h1 nc 2/1    20480          0    8000000 20000000
1          1280          8000000 Max
be af l1 h2 ef h1 nc 1/1    20480          0    8000000 20000000
1          1280          8000000 Max
be af l1 h2 ef h1 nc 5/*    20480          0    8000000 20000000
1          1280          8000000 Max
be af l1 h2 ef h1 nc 4/*    20480          0    8000000 20000000
1          1280          8000000 Max
...

```

Pool Information

```

Port          : 3/2/fm-sub
Application    : Net-Egr          Pool Name      : default
Resv CBS      : 50%

```

Queue-Groups

Utilization	State	Start-Avg	Max-Avg	Max-Prob
High-Slope	Up	70%	90%	80%
Low-Slope	Up	50%	75%	80%

```

Time Avg Factor : 7
Pool Total      : 12288 KB
Pool Shared     : 6144 KB          Pool Resv      : 6144 KB

High Slope Start Avg : 4096 KB          High slope Max Avg : 5120 KB
Low Slope Start Avg  : 3072 KB          Low slope Max Avg  : 4096 KB

Pool Total In Use   : 0 KB
Pool Shared In Use  : 0 KB          Pool Resv In Use   : 0 KB
WA Shared In Use    : 0 KB

```

```

Hi-Slope Drop Prob : 0          Lo-Slope Drop Prob : 0

```

FC-Maps	ID	MBS	Depth	A.CIR	A.PIR
Q-Grp	Q-Id	CBS		O.CIR	O.PIR
be af l1 h2 ef h1 nc	3/2/fm-*	8192	0	4000000	10000000
	1	5120		4000000	5000000
l2	3/2/fm-*	6144	0	6000000	10000000
	2	3584		5000000	5000000

Pool Information

```

Port          : 3/2/to-sub
Application    : Net-Egr          Pool Name      : default
Resv CBS      : 50%

```

Queue-Groups

Utilization	State	Start-Avg	Max-Avg	Max-Prob
High-Slope	Up	70%	90%	80%
Low-Slope	Up	50%	75%	80%

```

Time Avg Factor      : 7
Pool Total           : 24576 KB
Pool Shared          : 12288 KB      Pool Resv           : 12288 KB

High Slope Start Avg : 8192 KB      High slope Max Avg : 10240 KB
Low Slope Start Avg  : 6144 KB      Low slope Max Avg  : 8192 KB

Pool Total In Use    : 0 KB
Pool Shared In Use   : 0 KB      Pool Resv In Use    : 0 KB
WA Shared In Use     : 0 KB

Hi-Slope Drop Prob   : 0           Lo-Slope Drop Prob : 0

```

```

-----
FC-Maps              ID      MBS      Depth  A.CIR  A.PIR
Q-Grp                Q-Id   CBS                      O.CIR  O.PIR
-----
be af l1 h2 ef h1 nc      3/2/to-* 16384      0      4000000 10000000
                          1      10240      4000000  Max
l2                        3/2/to-* 12288      0      6000000 10000000
                          2      7168      6000000  Max
=====

```

```
# show application-assurance group 1 status isa 4/1 overload
```

```
=====
Application Assurance Group 1 overload
=====
```

```

-----
Trap Configuration          High Watermark          Low Watermark
-----
Flow usage                  95%                  90%
Flow setup rate (fps)      60000                55000
Bit rate (Mbps)            9000                 8500
Packet rate (pps)          12345678901234567890 12345678901234567890
Datapath CPU usage         95%                  90%
-----

```

```
ISA 4/1                      Total up-time (s) : 12345678901234567890
```

```
TCA Type
```

```

      Current Level          Average Level          Peak Level
      State                Count                Duration (s)
-----

```

```
Flow resources
```

```

      75.1%                  50.0%                  82.8%
      cleared                12345678901234567890 12345678901234567890

```

```
Flow setup rate
```

```

12345678901234567890 12345678901234567890 12345678901234567890
      cleared                0                0

```

```
Bit rate
```

```

      11000                  5000                  12000
      raised                1                123456789

```

```
Packet rate
```

```

      1500000                750000                1600000
      cleared                2                12345

```

```
Datapath CPU usage
```

```

      15.1%                  10.5%                  32.1%

```

```

      cleared                                0                                0
Management CPU usage
      65.2%                                50.0%                                77.5%
      N/A                                  N/A                                  N/A
-----
Overload Trap Configuration      High Watermark      Low Watermark
-----
From-sub WA buffer depth        80%              75%
To-sub WA buffer depth          80%              75%
-----
Isa-overload-cut-through : enabled
Overload cut-through TCA
Current from-sub WA : 30%
Current to-sub WA   : 40%
State               : cleared
Count               : 3
Duration (s)        : 123
=====

```

10.2.2.47 tcp-optimizer

Syntax

```
tcp-optimizer tcp-optimizer-name [isa mda-id]
```

Context

```
show>app-assure>group
```

Description

This command displays the TCP optimizer information.

Parameters

tcp-optimizer-name

specifies the name of the TCP validation policy, up to 32 characters

mda-id

specifies the slot number of an adapter card, in the format *slot/mda*

Output

The following output is an example of TCP optimizer policy information.

- [Example output for AA group TCP optimizer policy](#)
- [Example output for AA group TCP optimizer ISA MDA](#)

Example output for AA group TCP optimizer policy

```
# show>app-assure>group# tcp-optimizer "T01"
=====
Application Assurance Group 1 tcp-optimizer "OP1"
=====
Description          : (Not Specified) TCP Stack : tcp-bbr
Initial CWND Size    : 8 MSS
Initial SS Thresh    : 1000000 KB
DACK Timeout         : None Session Filter Ref   : No
Network RTT Threshold : None
Cumulative Session Statistics
Active Sessions      : 0
Optimized Sessions   : 0
Unoptimized Sessions : 0
Abandoned Sessions   : 0
-----
Session Statistics           Abandoned           Unoptimized
-----
Unsupported Packet          0                  0
Packet Error                0                  0
Inactivity                  0                  0
No Buffers                  N/A                0
Existing Session            N/A                0
Low Network RTT             N/A                0
Policy                      0                  N/A
Other                       0                  0
-----
Traffic Statistics           Sub-to-net           Net-to-sub
-----
Octets Received             0                  0
Packets Received            0                  0
Octets Transmitted          0                  0
Retransmitted              0                  0
Packets Transmitted         0                  0
Retransmitted              0                  0
=====
```

Example output for AA group TCP optimizer ISA MDA

```
# show>app-assure>group# tcp-optimizer "T01"
=====
Application Assurance Group 1 tcp-optimizer "T01" ISA 1/1
=====
Description          : (Not Specified)
TCP Stack            : tcp-westwood
Initial CWND Size    : 8 MSS
Initial SS Thresh    : 1000000 KB
DACK Timeout         : None
Session Filter Ref   : No
Network RTT Threshold : None
Buffer Utilization   : 0 of 6786 buffers

Cumulative Session Statistics
Active Sessions      : 0
```

Optimized Sessions	: 0	
Unoptimized Sessions	: 0	
Abandoned Sessions	: 0	

Session Statistics	Abandoned	Unoptimized

Unsupported Packet	0	0
Packet Error	0	0
Inactivity	0	0
No Buffers	N/A	0
Existing Session	N/A	0
Low Network RTT	N/A	0
Resource Exhaustion	0	0
Other	0	0

Traffic Statistics	Sub-to-net	Net-to-sub

Octets Received	0	0
Packets Received	0	0
Octets Transmitted	0	0
Retransmitted	0	0
Packets Transmitted	0	0
Retransmitted	0	0
=====		

10.2.2.48 tcp-validate

Syntax

```
tcp-validate tcp-validate-name [isa mda-id]
```

Context

```
show>app-assure>group
```

Description

This command displays TCP validation policy information.

When the *mda-id* parameter is included, only TCP validation policy information for the specified adapter card is displayed.

Parameters

tcp-validate-name

specifies the name of the TCP validation policy, 32 characters maximum

mda-id

specifies the slot number of an adapter card, in the format *slot/mda*

Output

The following output is an example of TCP validation policy information.

Example output AA group TCP validation

```
# show application-assurance group 1 tcp-validate "test" isa 1/1
=====
Application Assurance Group 1 tcp-validate "test"
=====
Description      : (Not Specified)
Event log        : (Not Specified)
Strict Validation: Yes
AQP referenced   : No
-----
Decision Statistics          sub-to-net          net-to-sub
-----
Total
-----
Allowed
  Octets                0                0
  Packets               0                0
Dropped
  Octets                0                0
  Packets               0                0
Dropped Reason
-----
Bad Flags
  Octets                0                0
  Packets               0                0
Bad Options
  Octets                0                0
  Packets               0                0
Bad Sequence Number
  Octets                0                0
  Packets               0                0
Bad Acknowledgment Number
  Octets                0                0
  Packets               0                0
No Establishment
  Octets                0                0
  Packets               0                0
SYN After Conn Establishment
  Octets                0                0
  Packets               0                0
Asymmetric Traffic
  Octets                0                0
  Packets               0                0
Traffic After Reset
  Octets                0                0
  Packets               0                0
Fragmented
  Octets                0                0
  Packets               0                0
```

10.2.2.49 traffic-type

Syntax

```
traffic-type detail
traffic-type ip-family
traffic-type ip-protocol
traffic-type tls-family
```

Context

```
show>app-assure>group
```

Description

This command displays per traffic type statistics.

Parameters

detail

displays detailed statistics

ip-family

displays IP family statistics

ip-protocol

displays IP protocol statistics

tls-family

displays TLS family statistics

Output

The following output example shows traffic type TLS family information and *Show AA traffic type TLS family output fields* describes the output fields.

Example output AA traffic type TLS family

```
# show application-assurance group 1:2 traffic-type tls-family
=====
Application-Assurance Traffic Type Statistics Per tls-family
=====
tls Variant  Disc      Octets    Packets    Flows
-----
quic         0%         0         0         0
quic esni    0%         0         0         0
quic existing 0%         0         0         0
quic unknown 0%         0         0         0
tls 1.0      0%         0         0         0
```

tls 1.1	0%	0	0	0
tls 1.2	0%	1475583	2205	150
tls 1.3	0%	0	0	0
tls 1.3 esni	0%	432189	459	34
tls existing	0%	1234	12	2
tls unknown	0%	0	0	0

Table 205: Show AA traffic type TLS family output fields

Label	Description
tls Variant	The TLS variant
Disc	The Disc statistics
Octets	The octets statistics
Packets	The packets statistics
Flows	The flow statistics

10.2.2.50 url-filter

Syntax

`url-filter url-filter-name`

`url-filter url-filter-name isa card/mda [detail]`

`url-filter url-filter-name web-service [profile profile-name]`

Context

`show>app-assure>group`

Description

This command displays information about the configured url-filter policy along with some associated raw statistics. These output statistics are:

- Vlan Id: VLAN IS used by the AA interfaces
- Admin Status: Up / Down

- Oper Status: Up / Down
- Oper Flags: adminDown, no-aa-if, aa-if-down, icap-server-down
- Default Action: default policy action taken by the URL filter
- ICAP HTTP Redirect: HTTP redirect Policy
- AQP Referenced: Yes/No
- Web service filtering information
- HTTP Request: Number of subscriber HTTP requests
- HTTP Errors: Impossible to send an ICAP request, this can be caused by either no TCP connection available, associated flow with a drop action due to another AQP policy, system resource exhausted
- ICAP Request: Number of ICAP request sent
- ICAP Errors: ICAP request timeout, unexpected ICAP response, internal TCP errors.
- Custom-x-header: Name of the custom-x-header, if configured. If it is not configured, the value is "Not Specified".

In addition to these counters, the system will count the type of action taken by the URL filter policy (allow, block, redirect, and default) as well as the type of responses received from the ICAP server (allow, block, redirect, and late).

Parameters

url-filter-name

specifies the name of the URL filter policy

card/mda

specifies the card and MDA reference of the ISA card

detail

specifies detailed statistics related to the ISA card

profile-name

specifies the name of a configured profile for web service, up to 256 characters

Output

The following output examples show URL filter information.

- *Example output AA URL Filter*
- *Example output AA web-service URL filter*

Example output AA URL filter

```
show application-assurance group 1 url-filter "web-serv1" isa 7/2
=====
Application Assurance Group 1 URL Filter " web-serv1" ISA 7/2
```

```

=====
Description          : Url filter WebService
Admin Status         : Up
Oper Status          : Down
Oper Flags           : web-service-down
DNS Oper Status      : Down
DNS Oper Flags       : unreachable
HTTP Request Filtering : all
AQP Referenced       : Yes
-----

```

URL Stats Summary

```

-----
Total Requests   : 0                Default Action : 0
Requests Allowed: 0                Reqs Block/Redir: 0
-----

```

Local Filter

```

-----
allow-list        : allowlist1
  Admin Status    : Up
  Oper Status     : Up
  Oper Flags      : <none>
  Number of URLs  : 1
  URL-List Lookups : 0
    Match         : 0
    Miss          : 0

deny-list         : denylist1
  Admin Status    : Up
  Oper Status     : Up
  Oper Flags      : <none>
  Number of URLs  : 1
  Default Action   : block-all
  HTTP Redirect    : (Not Specified)
  URL-List Lookups : 0
    Match         : 0
    Miss          : 0
  Default Action   : 0

deny-list         : denylist2
  Admin Status    : Up
  Oper Status     : Up
  Oper Flags      : <none>
  Number of URLs  : 1
  Default Action   : block-all
  HTTP Redirect    : (Not Specified)
  URL-List Lookups : 0
    Match         : 0
    Miss          : 0
  Default Action   : 0
-----

```

Web Service Filter

```

-----
Classifier        : web-service-1
Category-set-id   : 1
default-profile    : prof1
VLAN Id           : 1
FQDN              : nokia-api.webtitancloud.com
-----

```

```

DNS Address      : 12.0.0.100
Default Action   : allow
HTTP Redirect    : ws
AA Interface     : to_isa_7/2
Service         : VPRN 1
SAP Id          : 7/2/aa-svc:1
Client IP       : 10.1.1.2/31

Web Service Filter Stats
Class Ovrld Lkup : 0                Class Ovrld match: 0
Cache Lookup     : 0                Cache size       : 0
Cache match      : 0                Cache max size   : 20971520
Cache miss       : 0

URL Request      : 0                Default Action   : 0
URL Response     : 0                Late Response    : 0
  Allow          : 0                Packet Discard   : 0
  Block          : 0                Octet Discard    : 0
  Redirect       : 0

-----
ISA 7/2 Web Service Connection Stats
-----
Established Connections : 0 of 0 connections
Request Rate           : 0 per second
Round Trip Time        : 0 microseconds
=====

```

Example output AA web-service URL filter

The following output examples display detailed information about a web service and a web-service profile, and *Show web-service URL filter output fields* describes the output fields.

```

# show application-assurance group 1 url-filter "testFilter" web-service
-----
Web services for url-filter "testFilter"
-----
Category-set-id    : 1
-----
=====
Profile Categories
=====
Profile Name          Blocked Categories  Allowed Categories
-----
testProfile           1                54
=====

# show application-assurance group 1 url-filter "testFilter" web-
service profile "testProfile"
-----
Web service profile "testProfile"
-----
Category-set-id     : 1
Description          :
-----
=====
Blocked Categories

```

Category Id	Category Name
25	Personal Webpages
=====	
Allowed Categories	
=====	
Category Id	Category Name
1	Compromised
2	Criminal Skills/Hacking
3	Hate Speech
4	Illegal Drugs
5	Phishing/Fraud
6	Spyware and Malicious Sites
7	Nudity
8	Mature
9	Pornography/Sex
10	Violence
11	Weapons
12	Anonymizer
13	Computers and Technology
14	Download Sites
15	Translator
16	Alcohol
17	Health
18	Pharmacy
19	Tobacco
20	Gambling
21	Games
22	Cars/Transportation
23	Dating & Relationships
24	Home/Leisure
26	Restaurants
27	Sports and Recreation
28	Travel
29	Government
30	Military
31	Non-profits
32	Politics and Law
33	Religion
34	Education
35	Art
36	Entertainment and Videos
37	Humor
38	Music
39	News
40	Finance
41	Internet Watch Foundation List
42	Shopping
43	Chat/IM
44	Community Sites
45	Social Networking
46	Web-based Email
47	Portal Sites
48	Search Engines

```

49      Online Ads
50      Business/Services
51      Job Search
52      Real Estate
53      Spam
54      Miscellaneous
55      Uncategorized
=====

```

Table 206: Show web-service URL filter output fields

Label	Description
Category-set-id	The ID of the category set used for URL categorization
Description	The description of the web-service profile
Profile Name	The name of the web-service profile
Blocked Categories	The blocked categories for the web-service profile
Allowed Categories	The allowed categories for the web-service profile
Category Id	The ID of the blocked/allowed category
Category Name	The name of the blocked/allowed category

10.2.2.51 url-list

Syntax

`url-list url-list-name`

Context

`show>app-assure>group`

Description

This command displays information about the configured URL list.

Parameters

url-list-name

specifies the name of the URL list

Output

The following output examples show URL list information, and *Show AA URL list output fields* describes the output fields.

Example output AA URL list

```
# show application-assurance group 1 url-list "allow-list1"
=====
Application Assurance Group 1 url-list "allow-list1"
=====
Description           : Default Url-list object
Size                  : standard
Host Expressions      : enabled
Admin Status          : Up
Oper Status           : Up
Oper Flags            : <none>
File deployed to ISAs : Yes
-----
Upgrade Statistics
-----
Last Success          : 06/19/2020 02:57:06
  Deployed
    File Name          : cf1:/expr_15000.vm.only.expr
    URL Entries        : 15000 (100.00% full)
    URL Characters     : 249390 ( 5.08% full)
    URL Host Expr Entries: 15000 (100.00% full)
    Blank/Comment Lines : 60519
Last Attempt          : 06/19/2020 02:57:06
  Result              : Success
    File Name          : cf1:/expr_15000.vm.only.expr
=====
```

Table 207: Show AA URL list output fields

Label	Description
Description	The description of the URL list
Size	[standard extended] — Specifies the size parameter for the URL list
Host Expressions	If the host is enabled or disabled

Table 207: Show AA URL list output fields (continued)

Label	Description
Admin Status	[Up down] — Administrative status of the URL list
Oper Status	[Up down] — Operational status of the URL list
Oper Flags	[admin-down file-does-not-exist invalid-file-format too-many-urls switch-over-error]
File Deployed to ISA	[Yes No] — This flag describes if the file located in the compact flash is the one deployed in the ISA. In the event the file is overwritten and before the admin upgrade command is used this flag will display "No".
Upgrade Statistics	
Last Success	The last time the list was successfully upgraded
File Name	The file name for the last successful upgrade
URL Entries	Number of URLs loaded at the last success and percentage to full
URL Characters	Number of characters loaded at the last success and percentage to full
URL Host Expr Entries	The URL host expression entries
Blank/CommentLines	Number of blank or commented out lines
Last Attempt	The last time the operator tried to upgrade the list
Result	[Success Failure] — Result of the last upgrade
File Name	File name for the last upgrade attempt

10.2.2.52 http-enrich

Syntax

```
http-enrich
```

Context

```
show>app-assure>http-enrich
```

Description

This command displays http-enrichment status information.

Parameters

enrich-name

specifies an http-enrichment policy, up to 32 characters for flows matching this entry

Output

The following output example shows HTTP enrichment status information.

Example output AA HTTP enrichment status

```
# show application-assurance group 1 http-enrich "test"
=====
Application Assurance Group 1 HTTP Enrichment test
=====
Description : (Not Specified)
Admin Status: Down
-----
Grp:Part  Enriched          Out Of Resources      Missing Sub Data
          Tmpl Not Enabled      Traffic Char
-----
1:1       0                  0                    0
          0                  0
1:2       0                  0                    0
          0                  0
1:3       0                  0                    0
          0                  0
-----
Total    0                  0                    0
          0                  0
-----
=====
```


10.2.2.53 http-error-redirect

Syntax

```
http-error-redirect
```

Context

```
show>app-assure
```

```
show>app-assure>group
```

Description

This command enables the context to display HTTP error redirect static definitions.

Output

The following output example shows information about HTTP error redirect static definition.

Example output AA HTTP error redirect

```
# show application-assurance group 1 http-error-redirect <redirect-name>
=====
Application-Assurance Group 1 http-error-redirect <redirect-name>
=====
description      : <description-string>
template         : <template-id>
                  : text description of template
participant-id   : <string>
http-host        : <http-host-name>
error code       : <http-error-code>    custom-msg-size : <msg size>
admin status     : Up
-----
Grp:Part  Error   Redirects   Redirects Not Sent
          Code    Sent        > Custom    Out ofFile   Error
                   size  Resource type
-----
1:1       404     1250         52          10          10
1:56789   404     2000         952          81          01
-----
Total           3250    1004         91           1      1
-----
=====
```

10.2.2.54 error-codes

Syntax

```
error-codes
```

Context

```
show>app-assure>http-redirect
```

Description

This command displays HTTP error redirect error codes.

Output

The following output example shows HTTP error redirect error code information, and *Show AA HTTP redirect error codes output fields* describes the fields.

Example output AA HTTP error redirect error codes

```
# show application-assurance http-error-redirect error-codes
=====
Application-Assurance http-error-redirect error-codes
=====
```

Code	Description	Default custom-msg-size
400	Bad Request	1024
401	Unauthorized	1024
402	Payment Required	1024
403	Forbidden	1024
404	Not Found	1024
405	Method Not Allowed	1024
406	Not Acceptable	1024
407	Proxy Authentication Required	1024
408	Request Timeout	1024
409	Conflict	1024
410	Gone	1024
411	Length Required	1024
412	Precondition Failed	1024
413	Payload Too Large	1024
414	URI Too Long	1024
415	Unsupported Media Type	1024
416	Range Not Satisfiable	1024
417	Expectation Failed	1024
421	Misdirected Request	1024
422	Unprocessable Entity	1024
423	Locked	1024
424	Failed Dependency	1024
425	Too Early	1024
426	Upgrade Required	1024
427	Unassigned	1024
428	Precondition Required	1024
429	Too Many Requests	1024
430	Unassigned	1024
431	Request Header Fields Too Large	1024
451	Unavailable For Legal Reasons	1024
500	Internal Server Error	1024
501	Not Implemented	1024
502	Bad Gateway	1024
503	Service Unavailable	1024

504	Gateway Timeout	1024
505	HTTP Version Not Supported	1024
506	Variant Also Negotiates	1024
507	Insufficient Storage	1024
508	Loop Detected	1024
509	Unassigned	1024
510	Not Extended	1024
511	Network Authentication Required	1024
730	Custom Error Code	1024
731	Custom Error Code	1024
735	Custom Error Code	1024

=====

Table 208: Show AA HTTP redirect error codes output fields

Label	Description
Description	Describes the error
Default	Indicates what the default is for the error
custom-msg-size (Custom Message Size)	The size of the message

10.2.2.55 template

Syntax

template

Context

show>app-assure>http-redirect

Description

This command displays HTTP error redirect template information.

Output

The following output example shows HTTP error redirect template information.

Example output AA HTTP redirect

```
# show application-assurance http-error-redirect template
=====
Application-Assurance http-error-redirect templates
=====
ID      Description
-----
1       Template suited for Barefruit landing server. Includes participant-id.
2       Template suited for Xerocole landing server.
=====
```

10.2.2.56 protocol

Syntax

```
protocol [protocol-name]
protocol [protocol-name] detail
```

Context

```
show>app-assure
```

Description

This command displays application-assurance policy protocols loaded from the isa-aa.tim file.

Parameters

protocol-name

displays all protocols from the isa-aa.tim file

detail

displays detailed information about the specified protocol name

Output

The following output example shows application assurance protocol information based on the isa-aa.time file.



NOTE:

The file is updated frequently with new and updated AA protocols and the output varies accordingly.

Example output AA policy protocols

```
# show application-assurance protocol
```

```
=====
Application Assurance Protocols
=====
```

```
Protocol : Description
```

```
-----
advanced_direct_connect : Advanced Direct Connect peer to peer
                           protocol
aim : America Online instant messaging and media
      traffic.
amazon_video : Amazon Video RTMPE traffic.
ares : Ares P2P File Sharing Protocol
bbm : BBM STUN and RTP flows.
betamax_voip : Betamax VoIP Protocol traffic.
bgp : IETF RFC 4271: Border Gateway Protocol
bittorrent : BitTorrent peer to peer protocol.
cccam : CCcam satellite card sharing protocol
citrix_ica : Citrix ICA protocol.
citrix_ima : Citrix IMA protocol.
cnnlive : CNN Live Streaming Video
cups : Common Unix Printing Service.
cut_through : Traffic that cannot be categorized. Only
              default subscriber policy is applied.
cut_through_by_default_policy : Traffic that has been cut-through due to a
                               subscriber default policy.
cvs : Concurrent Versions System.
daap : iTunes Digital Audio Access Protocol media
       sharing protocol.
dcerpc : DCERPC Remote Procedure Call.
denied_by_default_policy : Traffic that was denied by a default
                           subscriber flow policer.
dhcp : Dynamic Host Configuration Protocol
       traffic.
dht : Peer to Peer Distributed Hash Table
      exchange.
direct_connect : Direct Connect peer to peer protocol
dns : IETF RFC 1035: Domain Name System.
domino : IBM Domino-Notes.
dtls : IETF RFC 6347: Datagram Transport Layer
       Security protocol.
empty_tcp : TCP flows that close without ever having
            exchanged any data.
emule : eMule/eDonkey ENC Server Client.
existing : Traffic that was in progress or with no
          start of flow.
fix : FIX (Financial Information eXchange)
      protocol.
flow_export : Flow information export protocols IPFIX/
              NetFlow and sFlow
fring : Fring Mobile traffic.
ftp_control : IETF RFC 959: File Transfer Protocol
              control traffic.
ftp_data : IETF RFC 959: File Transfer Protocol data
           traffic.
funshion : Funshion Streaming Video
gamecenter : Apple Game Center
gnutella : Gnutella/Gnutella2 peer to peer protocol.
google_talk_file_xfer : Google Talk Instant Messaging file
```

```
transfer.
google_talk_im : Google Talk Instant Messaging.
google_talk_voicemail : Google Talk Instant Messaging voice mail.
gtp : GTP (GPRS Tunneling Protocol).
h225 : ITU H.225 Multimedia Call Signalling
Protocol
h245 : ITU H.245 Control Protocol for MultiMedia
Communication
headcall : Headcall Protocol traffic.
hotline : Hotline Communications: A client-server
protocol for file sharing and chatting.
http : IETF RFC 2616: Hypertext transfer protocol.
http_audio : HTTP transported Audio content.
http_shockwaveflash : HTTP transported Shockwave Flash content.
http_video : HTTP transported Video content.
http_webfeed : RSS or ATOM Web Feed
hulu : HULU media traffic.
iax2 : InterAsterisk Exchange Protocol.
ibmdb2 : IBM DB2 Database Server.
icq : ICQ protocol traffic.
ident : IETF RFC 1413 Identification Protocol.
iiop : CORBA IIOP Network Protocol.
imap4 : IETF RFC 3501: Internet Message Access
Protocol V.4.
ipp : Internet Printing Protocol.
ipsec_nat_t : IETF RFC 3948: UDP Encapsulated IPsec ESP.
irc : RFC 1459 Internet Relay Chat
isakmp : IETF RFC 2408 4306: Internet Security
Association and Key Management Protocol.
iscsi : iSCSI Protocol.
jolt : Oracle JOLT (Java OnLine Transactions)
Protocol.
justintv : Justin.tv media traffic.
kerberos : Kerberos Version 5 Network Authentication
ldap : IETF RFC 4510: Lightweight Directory
Access Protocol.
line : LINE secure text messaging.
llmnr : LLNMR Protocol.
lync : LYNC TLS Control
mail_ru : mail.ru messaging protocol.
manolito : Manolito P2P File Sharing Protocol
megaco : Media Gateway Control Protocol.
mgcp : Media Gateway Control Protocol.
mms : Multimedia Messaging Service over HTTP.
mms_retrieve : Multimedia Messaging Service Retrieve over
HTTP.
mms_send : Multimedia Messaging Service Send over
HTTP.
ms_communicator : Microsoft Communicator Client.
msexchange : MS Exchange MAPI Interface.
msn_msgr : MSN Messenger client/server protocol.
msn_msgr_file_xfer : MSN Messenger initiated P2P file transfer.
msn_msgr_video : MSN Messenger Video Chat.
mssql_smb : MS SQL Server Named Pipe traffic.
mssql_tcp : MS SQL Server over TCP.
mssql_udp : MS SQL Server Monitoring Service.
mysql : MySQL Network Protocol.
```

```
net2phone : Net2Phone protocol.
net2phone_voip : Net2Phone VOIP
netbios : IETF RFC 1001: Network Basic Input Output
          System.
newcamd : Newcamd satellite card sharing protocol
nimbuzz : Nimbuzz Protocol.
nntp : IETF RFC 3977: Network News Transfer
       Protocol.
non_tcp_udp : Non TCP or UDP traffic.
ntp : IETF RFC1305 RFC2030: Network Time
      Protocol.
octoshape : Octoshape Streaming Video
onlive : OnLive Cloud Streaming Services
oovoo : ooVoo Protocol.
openft : openft peer to peer protocol.
openvpn : OpenVPN: open source virtual private
          network protocol.
opera_mini : Opera Mini mobile web browser.
oracle_net : Oracle TNS (Transparent Network Subtrate)
            Protocol.
pcanywhere : Symantec PcAnywhere.
pcoip : PCoIP remote desktop protocol.
pop3 : IETF RFC 1939: Post Office Protocol V.3.
postgresql : PostgreSQL Network Protocol.
pplive : PPLive Peer to Peer Video Streaming
         Protocol
ppstream : PPStream Chinese P2P streaming video.
pptp : Point-to-Point Tunneling Protocol.
q931 : ITU Q.931 Call Signalling Protocol
qq : QQ Instant Messaging Protocol
qvod : QVOD: Streaming media on demand.
rdp : Remote Desktop Protocol.
rdt : Realnetworks Data Transport protocol.
rfb : Remote Framebuffer protocol.
rlogin : IETF RFC 1282 rlogin Virtual Terminal
         Protocol.
rsh : Unix Remote Shell Command.
rsync : rsync File Transfer Protocol.
rtmfp : IETF RFC 7016 Adobe Secure Real-Time Media
        Flow Protocol.
rtmp : RTMP: Adobe Real Time Messaging Protocol.
rtmpe : RTMPE: Encrypted Adobe Real Time Messaging
        Protocol.
rtmpt : RTMPT: HTTP Tunneled Adobe Real Time
        Messaging Protocol.
rtp : IETF RFC 3550: Real-time Transport
      Protocol.
rtp_h323 : H323 RTP Voice.
rtp_lynx : LYNC STUN RTP
rtp_msn_msgr : MSN Messenger RTP Voice.
rtp_rtsp : RTSP RTP Data
rtp_sip : SIP RTP Data
rtp_skinny : Skinny RTP Data
rtp_yahoo_im : Yahoo Instant Messenger RTP Voice.
rtsp : IETF RFC 2326: Real Time Streaming
        Protocol.
sap : SAP Protocol.
```

shoutcast : SHOUTcast audio streaming protocol.
siebel : Siebel Suite.
sip : IETF RFC 3261: Session Initiation Protocol.
skinny : Skinny Call Control Protocol.
skype : Skype
skype_av : Skype Voice and Video
slingbox : SlingBox: TV video streaming and remote control
smb : Server Message Block protocol over TCP.
smb_netbios : Server Message Block protocol over NetBIOS.
smtp : IETF RFC 2821: Simple Mail Transfer Protocol.
snapchat : Snapchat secure traffic to Google.
snmp : Simple Network Management Protocol traffic.
socks : SOCKS UDP
soulseek : SoulSeek P2P File Sharing Protocol
spotify : Spotify Protocol.
ssh : IETF RFC 4251: Secure shell protocol.
starcraft2 : Starcraft II Protocol
steam : Steam Gaming Protocol.
steam_gaming : Steam Online Gaming Protocol.
stun : IETF RFC 5389: Session Traversal Utilities for NAT.
sunrpc : SUNRPC Remote Procedure Call.
svn : Subversion Version Control System.
sybase_db : SYBASE Database Network Protocol.
syslog : IETF RFC 3164: syslog protocol.
t125 : ITU T.125 Multipoint communication service protocol
tango : Tango voice and video.
teamspeak : TeamSpeak Protocol traffic.
telnet : IETF RFC 854: Telnet Network Virtual Terminal protocol.
teredo : Teredo: IPv6 packets in IPv4 UDP datagrams tunneling protocol.
tftp : IETF RFC 1350: Trivial File Transfer Protocol.
tivo : TiVo Service
tls : IETF RFC 4346: Transport Layer Security protocol.
tn3270 : IETF RFC1576 RFC2355: TN3270 terminal emulation via telnet.
tor : Tor internet anonymity protocol.
trusted_tcp : Traffic identified using a trusted TCP port number.
trusted_udp : Traffic identified using a trusted UDP port number.
tuxedo : Oracle TUXEDO Protocol.
ultrasurf : Ultrasurf TLS traffic.
ultravox : Ultravox streaming media protocol.
unknown_tcp : Unknown or unidentified TCP traffic.
unknown_udp : Unknown or unidentified UDP traffic.
ustream : Ustream media traffic.
utp : uTP: Micro Transport Protocol.
ventrilo : Ventrilo Protocol traffic.
viber : Viber Mobile traffic.
vine : Vine secure traffic to Amazon.


```

vmware : VMware Traffic.
vudu : VUDU on-demand video distribution
webex : Cisco Webex web conferencing
weixin : Weixin Instant Messaging Protocol
whatsapp : WhatsApp Protocol.
winmx : WinMX P2P File Sharing Protocol
wow : World of Warcraft Protocol
wsp_http : WSP transported HTTP traffic.
xboxlive : Xbox Live: Microsoft online game and media
           delivery service.
xmpp : IETF RFC 3920: Extensible Messaging and
       Presence Protocol.
xmpp_facebook : Facebook XMPP traffic.
xunlei : Xunlei Client.
xwindows : X Window System: A graphical user
           interface for networked computers
yahoo_file_xfer : Yahoo Instant Messaging Protocol File
                 Transfer.
yahoo_im : Yahoo Instant Messaging Protocol.
yahoo_video : Yahoo Instant Messaging Protocol Webcam
              Video.
youtube : YouTube RTMP/RTMPE traffic.

```

```

=====
Number of protocols      : 192

```

10.2.2.57 session-filter

Syntax

```
session-filter
```

```
session-filter session-filter-name
```

Context

```
show>app-assure>group
```

Description

This command displays session filter information.

Parameters

session-filter-name

specifies a session filter name, up to 32 characters

Output

The following output example shows session filter information, and *Show AA group session filter output fields* describes the fields.

Example output AA group session filter with L3/L4 redirect action

```
# show application-assurance group 1 session-filter SF1
=====
AA Session Filter Instance "SF1"
=====
Description      : Optimizer's Session Filter
Default Action   : permit
    Event Log    : (Not Specified)
AQP Entries      : 1

-----
Filter Match Criteria
-----
Entry           : 1
Description      : (Not Specified)
IP Protocol      : *
Dest IP         : 2011::1:8a78:fc54/128
Dest Port       : eq 8080
Action          : redirect 2011::19:d106:da33:c0a8:b2 80
Event Log       : event_log
Hits            : 1 flows
-----
No. of entries   : 1
```

Table 209: Show AA group session filter output fields

Label	Description
Default Action	The default action
Description	Describes what the Session Filter does
Event Log	The event log
AQP Entries	The number of AQP entries
Filter Match Criteria	
Entry	Indicates which entry matches the criteria
Description	Describes the entry, if applicable
Dest IP	The destination IP address

Table 209: Show AA group session filter output fields (continued)

Label	Description
Dest Port	The destination port
IP Protocol	The IP protocol
Protocol	The protocol
Action	Indicates whether to permit or deny the entry
Event Log	The event log
Hits	The number of hits

10.2.2.58 threshold-crossing-alert

Syntax

```
threshold-crossing-alert [detail]
```

Context

```
show>app-assure
```

Description

This command displays information about threshold crossing alerts.

Output

The following output example shows application assurance threshold crossing alert information.

Example output AA threshold crossing alerts

```
# show application-assurance threshold-crossing-alert
=====
Grp:Part  Type          Name                               Entry    Dir
-----
1         error-drop      N/A                               N/A      to-sub
1         error-drop      N/A                               N/A      from-sub
```

1	frag-drop-ooo	N/A	N/A	to-sub
1	overload-drop	N/A	N/A	from-sub
1	overload-drop	N/A	N/A	to-sub
1	gtp-filter	gtpFilter	def-action	from-sub
1	gtp-filter	gtpfilter	max-payload	from-sub
1	gtp-filter	gtpfilter	def-action	from-sub
1	gtp-filter	red	2	to-sub
1	gtp-filter	red	def-action	from-sub
1	sctp-filter	blue	ppid-range	to-sub
1	sctp-filter	red	1	from-sub
1	sctp-filter	red	2	to-sub
1	sctp-filter	red	ppid-range	from-sub
1	sctp-filter	red	def-action	to-sub
1	sctp-filter	red	def-action	from-sub
1	session-filter	blue	def-action	from-sub
1	session-filter	red	1	from-sub
1	session-filter	red	2	to-sub
1	session-filter	red	def-action	from-sub
1	tcp-validate	green	N/A	to-sub
2:50	error-drop	N/A	N/A	to-sub
2:32656	error-drop	N/A	N/A	from-sub
2:32656	policer	12345679801234567890123456789012	N/A	from-sub
2:32656	policer	red	N/A	from-sub

No. of TCAs: 24

show application-assurance threshold-crossing-alert detail

=====

Application Assurance Threshold Crossing Alerts

=====

error-drop from-sub

Group:Part	: 1	Trigger on	: denied-traffic
High watermark	: 4294967295	Low watermark	: 0
Last raised	: N/A	Last cleared	: N/A
State	: cleared		

error-drop from-sub

Group:Part	: 2:1	Trigger on	: denied-traffic
High watermark	: 4294967295	Low watermark	: 0
Last raised	: N/A	Last cleared	: N/A
State	: cleared		

fragment-drop-out-of-order from-sub

Group:Part	: 2:1	Trigger on	: denied-traffic
High watermark	: 4294967295	Low watermark	: 0
Last raised	: N/A	Last cleared	: N/A
State	: cleared		

fragment-drop-all to-sub

Group:Part	: 2:1	Trigger on	: denied-traffic
High watermark	: 4294967295	Low watermark	: 0

```
Last raised      : N/A                Last cleared   : N/A
State           : cleared
No. of TCAs     : 4
=====
```

10.2.2.59 version

Syntax

```
version
```

Context

```
show>app-assure
```

Description

This command displays the versions of the isa-aa.tim used by the CPM and the AA ISAs.

Output

The following output example shows information about the version of the isa-aa.tim file used by the CPM and the AA ISAs.

Example output AA versions

```
# show>app-assure# version
=====
Versions of isa-aa.tim in use
=====
CPM           : TiMOS-M-7.0.R4
1/1           : TiMOS-I-7.0.R1
2/1           : TiMOS-I-7.0.R1
3/2           : TiMOS-I-7.0.R1
=====
```

10.2.2.60 web-service

Syntax

```
web-service
```

Context

```
show>app-assure
```

Description

This command displays web-service information.

10.2.2.61 category-set-id

Syntax

`category-set-id category-set`

Context

`show>app-assure>web-service`

Description

This command displays information for the web-service category set.

Parameters

category-set

specifies the category set ID

Values: 0 to 65535

Output

The following output examples show information about SAPs using the application profile.

Example output AA web-service category set

```
# show application-assurance web-service category-set-id 1
=====
Compatible Classifier-ids
=====
Name                Description
-----
web-service-1       WebTitan Web Filtering
=====
```

10.2.2.62 aa-sub-using

Syntax

`aa-sub-using`

`aa-sub-using app-profile app-profile-name`

Context

show>service

Description

This command displays application subscriber information.

Parameters

app-profile-name

specifies the application profile name

10.2.2.63 app-profile

Syntax

app-profile *app-profile-name*

Context

show>service>sap-using

Description

This command displays information about SAPs using the specified application profile.

Parameters

app-profile-name

specifies an existing application profile name created in the config>app-assure>group>policy context

Output

The following output examples show information about SAPs using the application profile.

Example output AA application profile

```
# show service sap-using app-profile test
=====
Service Access Point Using Application Profile 'test'
=====
PortId                SvcId      Ing.   Ing.   Egr.   Egr.   Adm  Opr
                   QoS    Fltr   QoS    Fltr
-----
1/1/18:0              89         1    none    1    none   Up   Down
-----
```

```
Number of SAPs : 1
-----
=====
```

10.2.3 Application Assurance tools command descriptions

10.2.3.1 resources

Syntax

```
resources
```

Context

```
tools>dump>application-assurance
```

Description

This command displays a view of all the AA system limits, along with current usage, for the configured scale mode and ISA-AA versions.

10.2.3.2 group

Syntax

```
group aa-group-id[:partition-id]
```

Context

```
tools>dump>application-assurance
```

Description

This command dumps application-assurance information within a group/partition.

Parameters

aa-group-id[:partition-id]

specifies the group/partition

Values	<i>aa-group-id:partition:aa-group-id[:partition-id]</i>	
–	<i>aa-group-id</i>	1 to 255
–	<i>partition-id</i>	1 to 65535

10.2.3.3 aa-anl-list

Syntax

```
aa-anl-list [congested only] [sort-type {top-by-subs | top-by-rate}] [isa mda-id]  
aa-anl-list summary
```

Context

```
tools>dump>application-assurance>group
```

Description

This command displays a list of active Active Network Locations (ANLs) detected by AA along with the associated information; for example, congestion, measured rate, and number of subscribers.

Parameters

congested only

displays only congested ANLs

top-by-subs

displays the top ANLs sorted by those ANLs that have the most number of subscribers

top-by-rate

displays the top ANLs sorted by measured ANL bandwidth

mda-id

displays ANLs with the specified ISA MDAs in the format *slot/mda*

Values	<i>slot</i>	1 to 10
–	<i>mda</i>	1 to 2

summary

displays a summary of detected ANLs

Output

The following output shows `aa-anl-list` with no parameters specified. All ANLs detected by AA are displayed.

Example output AA active network locations list

```
# tools dump application-assurance group 250:101 aa-anl-list
=====
Application-Assurance Access Network Location List: Group 250:101, ISA 1/2
=====
Type      Location          Congestion State      Rate (kbps)  Subscribers
-----
3G-TNL    0001-0000-0003-0000  Not Congested         1000         2
3G-TNL    0001-0000-0004-0000  CEA Reported Delay Buildup  5000         5
3G-TNL    0001-0000-0005-0000  CEA Reported Frame Loss  3000         10
Number of access network locations : 3
```

The following output shows only congested ANLs.

```
# tools dump application-assurance group 250:101 aa-anl-list congested-only
=====
Application-Assurance Access Network Location List: Group 250:101, ISA 1/2
=====
Type      Location          Congestion State      Rate (kbps)  Subscribers
-----
3G-TNL    0001-0000-0004-0000  CEA Reported Delay Buildup  5000         5
3G-TNL    0001-0000-0005-0000  CEA Reported Frame Loss  3000         10
Number of congested access network locations : 2
-----
```

The following output shows the top congested-only ANLs sorted by measured ANL bandwidth.

```
# tools dump application-assurance group 250:101 aa-anl-list congested-
only sort-type top-by-rate
=====
Application-Assurance Access Network Location List: Group 250:101, ISA 3/2
=====
Type      Location          Congestion State      Rate (kbps)  Subscribers
-----
3G-TNL    0001-0000-0003-0000  CEA Reported Delay Buildup  4185         2
3G-TNL    0001-0000-0004-0000  CEA Reported Delay Buildup  2185         6
...
Displaying top 50 congested access network locations by rate
Number of congested access network locations : 96
-----
```

The following output shows the top ANLs sorted by those ANLs that have the most number of subscribers.

```
# tools dump application-assurance group 1:1 aa-anl-list top-by-sub
=====
```

```
Application-Assurance Access Network Location List: Group 1:1, ISA 3/2
=====
Type      Location      Congestion State      Rate (kbps)  Subscribers
-----
3G-CELL   3001-0000-0000-1234  RTT Limits Exceeded   8123         10
3G-CELL   3001-0000-0000-4567  RTT Limits Exceeded   2185         26
...
Displaying top 50 congested access network locations by subscribers
Number of congested access network locations : 15
-----
```

10.2.3.4 aa-sub-list

Syntax

```
aa-sub-list [filter-by-type sub-type] [isa mda-id]
aa-sub-list [filter-by-type sub-type] [isa mda-id] policers-exceeded
aa-sub-list summary
```

Context

```
tools>dump>app-assure>group
```

Description

This command displays the AA subscriber list for a specific ISA.

Parameters

```
sub-type
    filters by the specified sub-type
    Values: dsm, esm, esm-mac, sap, spoke-sdp, transit

mda-id
    specifies the slot and MDA in the format slot/mda
```

Values	slot	1 to 10
–	mda	1 or 2

policers-exceeded

displays subscribers that have exceeded policer resources

summary

displays summary information

Output

The following output example shows summary statistics for AA subscriber lists.

Example output AA subscriber list for specific ISA

```
# tools dump application-assurance group 74:40346 aa-sub-list summary
=====
Application-Assurance Subscriber Summary for Group 74:40346, ISA 1/2
=====
      all      esm      sap      spoke-  transit dsm      esm-mac
      sd
-----
Total      100      0      0      0      0      0      100
Overrides  100      0      0      0      0      0      100
Pol Exceeded 0      0      0      0      0      0      0
-----
Total number of aa-subs found                      : 100
Total number of aa-subs with overrides found          : 100
Total number of aa-subs with policer resources exceeded found : 0
=====
```

10.2.3.5 aa-sub-search**Syntax**

aa-sub-search *search-type granularity* [direction *direction*] [max-count *max-count*]

aa-sub-search tethering summary

Context

tools>dump>app-assure>group

Description

This command displays application-assurance aa-sub information.

Parameters***search-type***

specifies the type of search

Values: top

granularity

specifies the granularity of the search

Values: bytes | packets | flows

direction

specifies the network/subscriber direction

Values: from-sub | to-sub | both

max-count

specifies the maximum flows to display

Values: 1 to 100

tethering

specifies tethering statistics

summary

specifies summary statistics for tethering

Output

The following example outputs show AA subscriber search statistics:

- [Example output AA subscriber search](#)
- [Example output AA subscriber tethering summary](#)

Example output AA subscriber search

```
# tools dump application-assurance group 74 aa-sub-search top bytes
=====
Application-Assurance aa-sub search for Group 74: Top 10 by bytes (both)
=====
```

Entry	isa	SubType	SubName	From-sub-count	To-sub-count	Group:Partition	Interval(UTC)
1	3/2	esm-mac	"diameter_esm-000100000001"	40	42	74:40346	"01/13/2017 14:39:18"
2	3/2	esm-mac	"diameter_esm-000100000002"	40	42	74:40346	"01/13/2017 14:39:18"
3	3/2	esm-mac	"diameter_esm-000100000003"	40	42	74:40346	"01/13/2017 14:39:18"
4	3/2	esm-mac	"diameter_esm-000100000004"	40	42	74:40346	"01/13/2017 14:39:18"
5	3/2	esm-mac	"diameter_esm-000100000005"	40	42	74:40346	"01/13/2017 14:39:18"
6	3/2	esm-mac	"diameter_esm-000100000006"	40	42	74:40346	"01/13/2017 14:39:18"
7	3/2	esm-mac	"diameter_esm-000100000007"	40	42	74:40346	"01/13/2017 14:39:18"
8	3/2	esm-mac	"diameter_esm-000100000008"	40	42	74:40346	"01/13/2017 14:39:18"
9	3/2	esm-mac	"diameter_esm-000100000009"	40	42	74:40346	"01/13/2017 14:39:18"
10	3/2	esm-mac	"diameter_esm-00010000000a"	40	42	74:40346	"01/13/2017 14:39:18"

```
Total number of aa-subs found: 10
```

Example output AA subscriber tethering summary

```
# tools dump application-assurance group 1 aa-sub-search tethering summary
=====
Application-Assurance aa-sub tethering search for Group 1
=====
Partition      Detection      Tethering      Not Tethering      Total
-----
1:49           disabled        0              4321              4321
1:50           enabled       123456         567890            691346
-----
```

10.2.3.6 admit-deny-stats

Syntax

```
admit-deny-stats
```

Context

```
tools>dump>application-assurance>group
```

Description

This command displays AA admit-deny statistics.

Output

The following output shows AA admit-deny statistics information.

Example output AA admin-deny statistics

```
# tools dump application-assurance group 1 admit-deny-stats
=====
Application-Assurance Group 1 Admit-Deny Statistics
=====
-----
Packet Validation Statistics
  Admitted Sub-To-Net  Denied Sub-To-Net  Admitted Net-To-Sub  Denied Net-To-Sub
  (Packets)           (Packets)          (Packets)            (Packets)
-----
Error
  0                   0                   0                   0
Fragments: Out-Of-Order
  0                   0                   0                   0
Fragments: All
  0                   0                   0                   0
Overload
  N/A                 0                   N/A                 0
-----
TCP Validation Statistics
```

	Admitted Sub-To-Net (Packets)	Denied Sub-To-Net (Packets)	Admitted Net-To-Sub (Packets)	Denied Net-To-Sub (Packets)
2	0	0	0	0
net	14	1	15	1

10.2.3.7 cflowd

Syntax

cflowd

Context

tools>dump>application-assurance>group

Description

This command enables the context to display Cflowd template information.

10.2.3.8 template

Syntax

template *template*

Context

tools>dump>application-assurance>group>cflowd

Description

This command displays AA Cflowd template information.

Parameters

template

specifies the template to display
Values: comprehensive | tcp-performance | volume

Output

The following output shows Cflowd template information.

Example output AA Cflowd template

```
# show>app-assure>group>cflowd# /tools dump application-
assurance group 46 cflowd template volume
=====
Application-Assurance group 46 cflowd volume template
=====
-----
Optional Fields
-----
Ent Id      Name                                     Size Description
-----
*   2      aaApp                               32  AA Application
-----
Mandatory Fields
-----
Ent Id      Name                                     Size Description
-----
   1      octetDeltaCount                     8   Octet delta count
   2      packetDeltaCount                     8   Packet delta count
   4      protocolIdentifier                   1   Protocol Identifier
   7      sourceTransportPort                  2   Source transport port
   8      sourceIPv4Address                     4   Source IPv4 address
  11      destinationTransportPort              2   Destination transport port
  12      destinationIPv4Address                 4   Destination IPv4 address
  27      sourceIPv6Address                     16  Source IPv6 address
  28      destinationIPv6Address                 16  Destination IPv6 address
  61      flowDirection                         1   Flow direction
  82      interfaceName                         32  Interface name
 138      observationPointId                     4   Observation point Id
*   12      aaSubType                           1   AA Subscriber Type
-----
Legend:
Ent value '*' specifies an enterprise field, 'R' specifies reverse PEN field,
otherwise it is a standard IANA field.
Note: Either IPv4 or IPv6 source and destination IP address
fields are in the cflowd record based on traffic type.
=====
```


10.2.3.9 dns-ip-cache

Syntax

```
dns-ip-cache cache-name isa mda-id [url file-url] [new-ip-count]
```

Context

```
tools>dump>application-assurance>group
```

Description

This command displays the list of IP addresses stored in a dns-ip-cache.

Parameters

cache-name

specifies the cache name, up to 32 characters

mda-id

specifies the dns-ip-cache for a particular ISA AA card.

file-url

Specifies the URL of the file to which the search output will be directed. The file may be local or remote.

Values	local-url remote-url	
–	local-url [<cfld-id>/][<file-path>]	
–	remote-url[{{ftp:// tftp://}<login>:<pswd>@<remote-locn>/}[<file-path>]	
–	remote-locn [<hostname> <ipv4-address> <ipv6-address>]	
–	ipv4-address	a.b.c.d
–	ipv6-address	x::x::x::x::x::x[-interface]
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

- interface up to 32 characters, for link local addresses
- cflash-id flash slot ID

new-ip-count

specifies the number of new IP Addresses added to the cache.

Output

The following output example shows the DNS IP cache information, and *Show DNS IP cache debug output fields* describes the DNS IP cache debug output fields.

Example output AA DNS IP cache

```
# tools dump application-assurance group 1 dns-ip-
cache "Default DNS IP Cache"
isa 3/2
=====
Application-Assurance dns-ip-cache "Default DNS IP Cache"
Current Time:          "01/21/2015 16:44:00" (UTC)
  group:                1
  isa:                  3/2
  admin state:          no shutdown
  max-entries:          150
=====
ip-address              creationTime(UTC)    lastUpdated(sec)    numDNSResponses
lastMatchTime(UTC)    numTimesMatched
2600:5:3d40:3::3fa8:3d59      "01/21/2015 16:42:49" 71      5
"01/21/2015 16:42:49" 0
2600:5:3d40:3::3fa8:3d0b      "01/21/2015 16:42:49" 71      1
"01/21/2015 16:42:49" 0
2600:5:3d40:3::3fa8:3d2b      "01/21/
2015 16:42:49" 71      1
"01/21/2015 16:42:49" 0
157.238.74.203              "01/21/
2015 16:42:36" 84      67
"01/21/2015 16:42:36" 0
207.152.124.91              "01/21/
2015 16:42:36" 84      15
"01/21/2015 16:42:36" 0
207.152.124.97              "01/21/
2015 16:42:36" 84      15
"01/21/2015 16:42:36" 0
77.67.86.136                "01/21/
2015 16:42:36" 84      5
"01/21/2015 16:42:36" 0
157.238.74.176              "01/21/
2015 16:42:36" 84      67
"01/21/2015 16:42:36" 0
77.67.86.99                  "01/21/
```

```
2015 16:42:36" 84          11
"01/21/2015 16:42:36" 0
```

```
# tools dump application-assurance group 1 dns-ip-
cache "Default DNS IP Cache" isa 5/2 new-ip-count
=====
Application-Assurance dns-ip-cache "Default DNS IP Cache"
Current Time:          "08/06/2020 15:16:50" (UTC)
  group:                1
  isa:                  5/2
  admin state:          no shutdown
  num-domains:          2
=====
domain      hostname      new-ip-count
"Tango"     "*.tango.me$"           37
"Google"    "*.google.*"            6
=====
```

Table 210: Show DNS IP cache debug output fields

Label	Description
ip-address	IP Address stored in the dns-ip-cache, the address is added in the cache if the DNS response meets the dns-ip-cache dns-match criteria (domain-name and dns server-address)
creationTime	The time at which the entry was created. The entry is created by a DNS response meeting the dns-ip-cache dns-match criteria (domain-name and dns server-address)
lastUpdated(UTC)	The time at which the entry was last updated, either from a new IP flow (fully classified) using the same IP address or a new DNS response meeting the dns-ip-cache dns-match criteria
numDNSResponses	Number of DNS responses including this IP address meeting the dns-ip-cache dns-match criteria
lastMatchTime(UTC)	Last time the IP address matched an app-filter with a server-address dns-ip-cache criteria
numTimesMatched	Number of times the IP address matched an app-filter with a server-address dns-ip-cache

Table 210: Show DNS IP cache debug output fields (continued)

Label	Description
Full Count	Total number of IP entries in the cache
Hit Count	The number of times an IP address lookup in this cache was successful. The IP address lookup is performed in app-filters and is successful if the server-address dns-ip-cache criteria is met.
Miss Count	The number of times an IP address lookup in this cache was unsuccessful. The IP address lookup is performed in app-filters and is unsuccessful if the server-address dns-ip-cache criteria is not met.
domain	The domain as configured in dns-ip-cache
hostname	The hostname as configured in dns-ip-cache
new-ip-count	The number of new IP Addresses added in the cache for the domain

10.2.3.10 dynamic-rating

Syntax

dynamic-rating

Context

tools>dump>application-assurance>group

Description

This command enables the context to dump the statistics of the dynamic rating.

10.2.3.11 statistics

Syntax

```
statistics
statistics isa mda-id
```

Context

```
tools>dump>application-assurance>group>dynamic-rating
```

Description

This command displays the statistics of the dynamic rating.

Parameters

mda-id

specifies the slot and MDA in the format *slot/mda*

Values	<mda-id>	<slot>/<mda>
–	slot	1 to 20
–	mda	1 to 4

Output

The following output shows the statistics of the dynamic rating and the *Show output fields of the dynamic rating statistics* describes the output fields.



NOTE:

The invalid numbers come from:

- Start/keep/end received for deleted sub
- Start/keep/end received for inactive (according to the active sub bitmap) sub
- Start for sub already on timer wheel, keep or end for sub not on timer wheel

Example output of the dynamic rating statistics

```
# tools dump application-assurance group 1 dynamic-rating
=====
Application-Assurance Group 1 Dynamic Zero Rating stats
=====
Start Counts
    Number of Start indications recvd          : 200
    Number of invalid Start indications         : 15

Keepalive Counts
    Number of Keepalive indications recvd       : 555
    Number of invalid Keepalive indications     : 20
    Number of valid Keepalive indications       : 535
    Number of Keepalive timer expiry            : 90

End Counts
    Number of End indications recvd             : 110
    Number of invalid End indications           : 10

Messaging Counts to MG:
    Number of Start msgs sent to MG             : 184
    Number of Start messaging errors             : 1
    Number of End msgs sent to MG               : 97
    Number of End messaging errors              : 3
```

Table 211: Show output fields of the dynamic rating statistics

Label	Description
Start Counts	Specifies the number of the received Start indications and the number of the invalid Start indications
Keepalive Counts	Specifies the number of the following: - received keepalive indications - invalid keepalive indications - valid keepalive indications - keepalive timer expiry
End Counts	Specifies the number of the received End indications and the number of the invalid End indications
Messaging Counts to MG	Specifies the number of the following: - Start messages sent to the mobile gateway - Start messaging errors

Table 211: Show output fields of the dynamic rating statistics (continued)

Label	Description
	• End messages sent to mobile gateway • End messaging errors

10.2.3.12 event-log

Syntax

```
event - log event-log-name url file-url
event - log event-log-name isa mda-id [url file-url]
```

Context

```
tools>dump>app-assure>group
```

Description

This command displays Application Assurance event log information.

Parameters

- event-log-name**
specifies the name of the event log, up to 32 characters
- mda-id**
specifies the slot and MDA in the format *slot/mda*

Values	<i>slot</i>	1 to 20
–	<i>mda</i>	1 to 4

- file-url**
specifies the URL for the file to direct the event-log output to

Values	local-url	[cflash-id] [file- path] up to 200 characters, including cflash-id directory length, up to 99 characters each
–	remote-url	[[ftp:// tftp://]login:pswd@remote-locn] [file-path] 255 characters maximum directory length, up to 99 characters each
–	remote-locn	[hostname ipv4-address ipv6-address]
–	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x[-interface] x:x:x:x:x:x:d.d.d.d[-interface] x - [0..FFFF]H d - [0..255]D
–	interface	up to 32 characters, for link local addresses
–	cflash-id	cf1: cf1-A: cf1-B: cf2: cf2-A: cf2-B: cf3: cf3-A: cf3-B:

Output

The following shows the event log information for an aborted L3/L4 redirect action.

Example output AA event log

```
# tools dump application-assurance group 1 event-log "event_log" isa 3/2
=====
Application-Assurance event-log "event_log"
Current Time:      "04/12/2019 23:38:41" (UTC)
  group[:partition]: 1
  isa:                3/2
  admin state:        no shutdown
  buffer-type:        linear
  max-entries:        50000
=====
Event-source

Action SubType      SubName    Direction Src-ip  Dst-ip
Ip-protocol Src-port Dst-port Timestamp "session-
filter SF1 entry 1 reason: same dst ip and port"

abort      mobile      "IMSI 123456789012001"      from-
```



```

sub 5.127.255.0 216.58.216.198
tcp 36008 443 "04/12/2019 23:38:17"
Total Records: 1
=====
total_from_sub: 1

```

10.2.3.13 flow-record-search

Syntax

```

flow-record-search aa-sub {esm sub-ident-string | sap sap-id | spoke-sdp sdp-id:vc-id
| transit transit-aasub-name} [protocol protocol-name] [application app-name] [app-
group app-group-name] [flow-status flow-status] [start-flowid start-flowid] [classified
classified] [server-ip ip-address] [server-port port-num] [client-ip ip-address] [bytes-tx
kbytes] [flow-duration minutes] [max-count max-count] [search-type search-type] [url file-
url] [flow-modified flow-modified]

```

```

flow-record-search isa mda-id [protocol protocol-name] [application app-name] [app-
group app-group-name] [flow-status flow-status] [start-flowid start-flowid] [classified
classified] [server-ip ip-address] [server-port port-num] [client-ip ip-address] [bytes-tx
kbytes] [flow-duration minutes] [max-count max-count] [search-type search-type] [url file-
url] [flow-modified flow-modified]

```

Context

```
tools>dump>app-assure>group
```

Description

This command displays Application Assurance flow records matching the specified criteria for a specific AA subscriber.

Parameters

app-name

displays flows for the specified application name

app-group-name

displays flows for the specified application group

kbytes

displays flows with the specified minimum kilobytes

Values: 0 to 4294967295

classified

specifies the starting flow ID

Values: yes, no

server -ip ip-address

display flows with the specified client IP address

Values	ipv4- a.b.c.d address
–	ipv6- x:x:x:x:x:x:x (eight 16-bit pieces) address

sub-ident-string

displays flows for the specified subscriber

minutes

display flows with the specified minimum duration in minutes

Values: 0 to 4294967295

flow-status

displays only flows that are active or closed

Values: active, closed

max-count

specifies the maximum count of flows to display

Values: 1 to 4294967295

protocol-name

displays flows for the specified protocol

sap-id

displays flows for the specified SAP

search-type

specifies the level of detail displayed for flows that match the search criteria

Values	default	displays some per flow information
–	count	displays the number of matching flows
–	detail	displays all per flow information available

client-ip *ip-address*
displays flows for the specified client IP address

server-ip *ip-address*
displays flows with the specified server IP address

Values	ipv4- a.b.c.d address
–	ipv6- x:x:x:x:x:x:x (eight 16-bit pieces) address

port-num
displays flows with the specified server port number
Values: 0 to 65535

sdp-id:vc-id
displays flows for the specified spoke SDP

start-flowid
specifies the starting flow ID
Values: 0 to 4294967295

transit-aasub-name
displays flows for the specified transit subscriber

file-url
specifies the URL for the local or remote file to which the HTTP host recorder output will be directed

Values	local-url remote-url
–	local-url[<cflash-id>/][<file-path>]
–	up to 200 characters, including cflash-id directory length up to 99 characters each

Values	remote-url[{{ftp:// tftp://}<login>:<pswd>@<remote-locn>/}][<file-path>]	
–	255 chars max	
	directory length up to 99 characters each	
–	remote-locn [<hostname> <ipv4-address> <ipv6-address>]	
–	ipv4-address a.b.c.d	
–	ipv6-address x:x:x:x:x:x:x[-interface]	
–	x:x:x:x:x:d.d.d.d[-interface]	
–	–	x - [0..FFFF]H
–	–	d - [0..255]D
–	interface	up to 32 characters, for link local addresses
–	cflash-id	flash slot ID

flow-modified

displays information about proxied flows

Output

The following are examples of the tools dump application assurance group flow record search.

- [Example output AA event log for aborted L3/L4 redirect action](#)
- [Example output AA flow record search for redirected flow](#)

Example output AA event log for aborted L3/L4 redirect action

```
# tools dump application-assurance group 1:1 flow-record-search isa 8/2 search-
type detail
=====
Application-Assurance flow record search
Search Start Time:      "02/26/2019 16:37:53" (UTC)
Search Criteria:
  group[:partition]:    1:1
  isa:                  8/2
  protocol name:        none specified
```

```

application name:      none specified
app-group name:        none specified
flow-status:           none specified
start-flowId:          none specified
classified:             none specified
server-ip:             none specified
server-port:           none specified
client-ip:             none specified
bytes-tx:              none specified
flow-duration:         none specified
max-count:             none specified
flow-modified:         none specified
search-type:           detail
=====
FlowId SubType SubName Initiator Direction Src-ip Dst-ip
Ip-protocol Ip-family Src-port Dst-port Flow-key
FC DSCP Classified Protocol Application AppGroup ChrgGroup
Packets-tx Bytes-tx Packets-discarded Bytes-
discarded Time-of-first-packet(UTC) Time-of-last-packet(UTC) Flow-
status Cflowd-Volume-packets-sampled Cflowd-Volume-bytes-sampled Cflowd-
Performance Flow-modified Device-Mfg Device-Id Device-0s OS-Version Flow-
attribute(conf)3 mobile "SESSION-
ID 0x00448001" yes from-
sub 1.0.0.2 69.165.166.23
tcp IPv4 59202 1723 "re-
dir: 0x00000005" be be yes "existing" "Existing T
CP" "Existing/
Empty TCP" "ChargedData" 1 52
0 0 "02/26/
2019 16:12:11" "02/26/2019 16:12:12" closed n/
a n/
a "no" redir "NotHTTP"
"NotHTTP" "NotHTTP" ""
7 mobile "SESSION-ID 0x00448001" yes from-
sub 1.0.0.2 69.165.166.23
gre IPv4 0 0 "" be be yes "
non_tcp_udp" "GRE" "Tunnel and Rem
ote Access" "ChargedData" 1293 171405
0 0 "02/26/
2019 16:12:11" "02/26/2019 16:12:55" closed n/
a n/
a "no" unchnng "NotHTTP"
"NotHTTP" "NotHTTP" ""
9 mobile "SESSION-ID 0x00448001" yes from-
sub 1.0.0.2 69.165.166.23
tcp IPv4 59202 1723 "re-
dir: 0x0000000b" nc be yes "pptp" "PPTP"
"Tunnel and Remote Access" "ChargedData"
7 696 0 0
"02/26/2019 16:12:12" "02/26/2019 16:12:55" closed n/a n/
a "no" redir "NotHTTP" "NotHTTP" "NotHTTP" ""

```

Example output AA flow record search for redirected flow

```

# tools dump application-assurance group 1:1 flow-record-search isa 3/2 flow-
modified redirected search-type detail

```

=====

Application-Assurance flow record search

Search Start Time: "04/15/2019 18:25:48" (UTC)

Search Criteria:

```

group[:partition]: 1:1
isa: 3/2
protocol name: none specified
application name: none specified
app-group name: none specified
flow-status: none specified
start-flowId: none specified
classified: none specified
server-ip: none specified
server-port: none specified
client-ip: none specified
bytes-tx: none specified
flow-duration: none specified
max-count: none specified
flow-modified: "redirected"
search-type: detail

```

```

=====
FlowId      SubType  SubName                               Initiator Direction Src-
ip          ip        ip        Dst-ip        Ip-
protocol Ip-family Src-port Dst-port Flow-
key          FC DSCP Classified Protocol          Application
          AppGroup                      ChrgGroup
          Packets-tx      Bytes-tx      Packets-discarded  Bytes-
discarded    Time-of-first-packet(UTC) Time-of-last-packet(UTC) Flow-
status Cflowd-Volume-packets-sampled Cflowd-Volume-bytes-sampled Cflowd-
Performance Flow-modified Device-Mfg      Device-
Id          Device-OS      OS-Version  Flow-attribute(conf)
3          mobile      "IMSI 310012567489944"      yes      from-
sub 1.0.0.2      216.58.192.196
tcp        IPv4        54249      80      "re-
dir: 0x00000005" be be yes      "http"      "Unknown"
          "Unknown"      "uTcpCG"
          6          318          0          0
          "04/15/2019 18:22:55" "04/15/2019 18:22:55" closed n/
a          n/
a          "no"      redir      ""
          ""      ""
7          mobile      "IMSI 310012567489944"      yes      from-
sub 1.0.0.2      216.58.192.196
tcp        IPv4        54314      80      "re-
dir: 0x00000009" be be yes      "http"      "Unknown"
          "Unknown"      "uTcpCG"
          6          322          0          0
          "04/15/2019 18:22:55" "04/15/2019 18:22:55" closed n/
a          n/
a          "no"      redir      ""
          ""      ""
=====

```

10.2.3.14 http-host-recorder detail

Syntax

http-host-recorder detail [isa *mda-id*] url *file-url*

Context

tools>dump>app-assure>group

Description

This command saves the HTTP host values recorded by the tool into a file. The http-host-recorder is configured using debug commands.

Parameters

mda-id

specifies the slot and MDA in the format *slot/mda*

Values	<i>slot</i>	1 to 10
–	<i>mda</i>	1 to 2

file-url

specifies the URL for the file to direct the event-log output to

Values	local-url	[cflash-id] [file- path] up to 200 characters, including cflash-id
–	directory length, up to 99 characters each	
–	remote-url	[[ftp:// tftp://]login:pswd@remote-locn] [file-path]
–	255 characters maximum	
–	directory length, up to 99 characters each	

- remote-locn [hostname | ipv4-address | ipv6-address]
- ipv4-address a.b.c.d
- ipv6-address x:x:x:x:x:x[-interface]
- x:x:x:x:x:d.d.d.d[-interface]
- x - [0..FFFF]H
- d - [0..255]D
- interface up to 32 characters, for link local address
- cflash-id flash slot ID

10.2.3.15 http-host-recorder status

Syntax

```
http-host-recorder status [isa mda-id]
```

Context

```
tools>dump>app-assure>group
```

Description

This command displays the current status of the http-host-recorder with current-time, start-time, stop-time, sample-rates, filters, buffer as well as number of bytes and flows recorded for the specified AA ISA. The http-host-recorder is configured using debug commands.

Parameters

mda-id

specifies the slot and MDA in the format *slot/mda*

Values	<i>slot</i>	1 to 10
–	<i>mda</i>	1 to 2

10.2.3.16 http-host-recorder top

Syntax

```
http-host-recorder top granularity [max-count {1..25}] [isa mda-id]
```

Context

```
tools>dump>app-assure>group
```

Description

This command displays application-assurance http-host-recorder information.

Parameters

granularity

specifies if the output for top is sorted by bytes or flows

max-count

specifies the maximum count of flows to display

Values: 1 to 25

mda-id

specifies the slot and MDA in the format *slot/mda*

Values	<i>slot</i>	1 to 10
–	<i>mda</i>	1 to 2

10.2.3.17 http-proxy

Syntax

http-proxy session isa *mda-id* flow-id *flow-id*

Context

tools>dump>app-assure>group

Description

This command displays connection information for the proxy sessions of the specified ISA and flow ID.

Parameters

mda-id
specifies the slot and MDA in the format slot/mda

Values	<i>slot</i>	1 to 10
–	<i>mda</i>	1 to 2

flow-id
specifies the flow ID
Values: 0 to 4294967295

Output

The following outputs show examples of an HTTP-Proxy session and HTTP-Proxy status .

Example output AA HTTP-proxy session

```
/tools dump application-assurance group 1:1 http-proxy session isa 8/2 flow-id 3
DEBUG: |19:16:04 +00:00.038|
=====
Application-Assurance Group 1:1 http-proxy session
=====
group[:partition]      : 1:1
isa                   : 8/2
aa-sub                : "SESSION-ID 0x00044001" (mobile)
-----
Subscriber (flowIds 2, 3)
-----
subscriber-ip         : 1.0.0.2
```

```

network-ip      : 13.0.0.1
ip-protocol     : tcp
subscriber-port : 32824
network-port    : 5080
proxy session-id : 0x00010002
minimum rtt     : 1 ms
average rtt     : 20 ms
tcp state       : ESTABLISHED
flow modification : proxy-sub
congestion window : 21940
receiver window  : 128
bandwidth estimate : 227576 bytes/sec
bytes in-flight  : 0
bytes ready to read : 0
Traffic Counts   Packets           Octets
-----
sub-to-net       252               13802
net-to-sub       1096             625225
retransmitted    0                0
-----
Network (flowIds 4, 5)
-----
subscriber-ip    : 1.0.0.2
network-ip       : 12.0.0.2
ip-protocol      : tcp
subscriber-port  : 32824
network-port     : 444
proxy session-id : 0x00010002
minimum rtt     : 2 ms
average rtt     : 2 ms
tcp state       : ESTABLISHED
flow modification : proxy-net
congestion window : 23360
receiver window  : 16640
bandwidth estimate : 7629 bytes/sec
bytes in-flight  : 0
bytes ready to read : 112256
Traffic Counts   Packets           Octets
-----
sub-to-net       372               19843
net-to-sub       496             708909
retransmitted    0                0
=====
-----

```

Example output AA HTTP-proxy status

```

/tools dump application-assurance group 1 http-proxy status HP1
=====
Application Assurance Group 1 HTTP-Proxy HP1
=====
proxy mode           : https
session-filter references : 1
dns-server           : 3.3.3.100

```

```

vlan-id                : 100
num enabled url-mappings : 1

aa-interfaces :
  to_AA7/2
  Admin State      : Up
  Service          : VPRN 1
  Oper State       : Up
  SAP              : 7/2/aa-svc:100
  Address          : 5.5.5.3/31
  ISA Address      : 5.5.5.2/31

tls-profile
  Certificate Profile : tlsProfile (Oper State: Up)
  Certificate         : end_cert.der
  Key File            : end_key.der

intermediate-certificate-1
  Certificate Profile : interCert1Profile (Oper State: Up)
  Certificate         : inter1_cert.der

intermediate-certificate-2
  Certificate Profile : interCert2Profile (Oper State: Up)

  Certificate         : inter2_cert.der

root-certificate
  Certificate Profile : rootProfile (Oper State: Up)

  Certificate         : root_cacert_nointmd.der

=====

```

10.2.3.18 ip-identification-assist

Syntax

```
ip-identification-assist
```

Context

```
tools>dump>app-assure>group
```

Description

This command enables the context to dump the statistics detected from the IP identification assist.

10.2.3.19 applications

Syntax

```
applications [isa mda-id]
```

Context

```
tools>dump>app-assure>group>ip-identification-assist
```

Description

This command dumps the statistics that show how much traffic is detected by the IP identification assist for top applications.

Parameters

mda-id

specifies the slot and MDA in the format *slot/mda*

Values	<i>slot</i>	1 to 20
–	<i>mda</i>	1 or 4

Output

The following example output shows information about the traffic detected by the IP identification assist.

Example output applications statistics

```
# tools dump application-assurance group 1:0 ip-identification-assist applications isa 1/1
=====
Application-Assurance group 1:0 ip-identification-assist application statistics
=====

isa 1/1
-----
Application Name          Detection          flows          packets
          octets
-----
-----
```

"Apple"		app-filter	0 (0.0%)	0 (0.0%)
	0 (0.0%)	ip-assist	32 (100.0%)	43768 (100.0%)
	39252130 (100.0%)			
"Apple App Store and iTunes"		app-filter	0 (0.0%)	0 (0.0%)
	0 (0.0%)	ip-assist	0 (0.0%)	0 (0.0%)
	0 (0.0%)			
"Apple Music"		app-filter	0 (0.0%)	0 (0.0%)
	0 (0.0%)	ip-assist	0 (0.0%)	0 (0.0%)
	0 (0.0%)			
"Apple iCloud"		app-filter	0 (0.0%)	0 (0.0%)
	0 (0.0%)	ip-assist	0 (0.0%)	0 (0.0%)
	0 (0.0%)			
"Facebook"		app-filter	0 (0.0%)	0 (0.0%)
	0 (0.0%)	ip-assist	84 (100.0%)	90580 (100.0%)
	96430386 (100.0%)			
"Google"		app-filter	0 (0.0%)	0 (0.0%)
	0 (0.0%)	ip-assist	382 (100.0%)	25980 (100.0%)
	21150315 (100.0%)			
"Google Play Store"		app-filter	0 (0.0%)	0 (0.0%)
	0 (0.0%)	ip-assist	0 (0.0%)	0 (0.0%)
	0 (0.0%)			
"Instagram"		app-filter	0 (0.0%)	0 (0.0%)
	0 (0.0%)	ip-assist	52 (100.0%)	18714 (100.0%)
	18655139 (100.0%)			
"Netflix"		app-filter	0 (0.0%)	0 (0.0%)
	0 (0.0%)	ip-assist	78 (100.0%)	59059 (100.0%)
	52961501 (100.0%)			
"Snapchat"		app-filter	0 (0.0%)	0 (0.0%)
	0 (0.0%)	ip-assist	0 (0.0%)	0 (0.0%)
	0 (0.0%)			
"Spotify"		app-filter	0 (0.0%)	0 (0.0%)
	0 (0.0%)	ip-assist	85 (100.0%)	35703 (100.0%)
	32556052 (100.0%)			
"Twitter"		app-filter	0 (0.0%)	0 (0.0%)
	0 (0.0%)			

	16997835 (100.0%)	ip-assist	38 (100.0%)	20555 (100.0%)
"Whats App"	0 (0.0%)	app-filter	0 (0.0%)	0 (0.0%)
	0 (0.0%)	ip-assist	0 (0.0%)	0 (0.0%)
"YouTube"	0 (0.0%)	app-filter	0 (0.0%)	0 (0.0%)
	31582683 (100.0%)	ip-assist	6 (100.0%)	27562 (100.0%)

TOTAL (14 applications)	0 (0.0%)	app-filter	0 (0.0%)	0 (0.0%)
		ip-assist	757 (100.0%)	321921 (100.0%)
	309586041 (100.0%)			
=====				

10.2.3.20 cache-contents

Syntax

```
cache-contents [isa mda-id] url file-url
```

Context

```
tools>dump>app-assure>group>ip-identification-assist
```

Description

This command dumps the content of the cache used by the IP identification assist feature.

Parameters

mda-id
specifies the slot and MDA in the format *slot/mda*

Values	<i>slot</i>	1 to 20
–	<i>mda</i>	1 or 4

file-url

specifies the URL for the local or remote file to which the cache content output is directed

Values	local-url	[<i>cflash-id</i>] <i>file-path</i> up to 200 characters, including <i>cflash-id</i> directory up to 99 characters each
–	remote-url	{ftp tftp}://[<i>login:pswd@</i>] <i>remote-locn</i> /[<i>file-path</i>] up to 255 characters, <i>file-path</i> up to 199 characters
–	remote-locn	{ <i>hostname</i> <i>ipv4-address</i> "[" <i>ipv6-address</i> "]" }[: <i>port</i>]
–	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x[- <i>interface</i>] x:x:x:x:x:x:d.d.d.d[- <i>interface</i>] x - [0..FFFF]H d - [0..255]D interface: up to 32 characters, for link local addresses
–	port	[0 to 65535]
–	cflash-id	cf1: cf1-A: cf1-B: cf2: cf2-A: cf2-B: cf3: cf3-A: cf3-B:

Output

The following example output shows information about the content of the cache used by the IP identification assist .

Example output cache contents statistics

```
# tools dump application-assurance group 1 ip-identification-assist cache-contents isa 1/1 url ftp://
bsx:siglab@12.12.12.1/../../../../tftpboot/SIM15/sample-cache

=====
Application-Assurance Group 1 ip-identification-assist cache contents
=====

Writing to file ftp://bsx:siglab@12.12.12.1/../../../../tftpboot/SIM15/sample-cache_1_1.ip-ident-assist-cache ...
File closed
```



```
-rw-r--r-- 1 bsx  bsx  16553 Aug 24 09:22 sample-cache_1_1.ip-ident-assist-cache
```

10.2.3.21 passive-dns

Syntax

```
passive-dns
```

Context

```
tools>dump>app-assure>group>ip-identification-assist
```

Description

This command enables the context to dump the statistics of the DNS traffic for top applications.

10.2.3.22 monitor

Syntax

```
monitor [isa mda-id]
```

Context

```
tools>dump>app-assure>group>ip-identification-assist>passive-dns
```

Description

This command dumps the statistics that show the matches found from passive snooping of the DNS traffic for top applications.

Parameters

mda-id

specifies the slot and MDA in the format *slot/mda*

Values	<i>slot</i>	1 to 20
–	<i>mda</i>	1 or 4

Output

The following example output shows information about the matches found from passive snooping of the DNS traffic.

Example output passive-dns-monitor statistics

```
# tools dump application-assurance group 1 ip-identification-assist passive-dns monitor isa 1/1
```

```
=====
Application-Assurance passive-dns-monitor for ip-identification-assist
=====
```

```
group                1
ip-identification-assist  no shutdown
passive-dns-monitor    enabled
-----
```

```
isa 1/1
-----
```

Application Name	DNS domains	DNS domain matches	IPs in cache	IPs added
IPs updated	IP lookups found	Parent Application Name		
"Apple"	6	18	21	21
23	31 "Apple"			
"Apple App Store and iTunes"	4	0	0	0
0	0 "Apple"			
"Apple Music"	8	0	0	0
0	0 "Apple"			
"Apple iCloud"	12	0	0	0
0	0 "Apple"			
"Facebook"	12	21	9	9
12	70 "Facebook"			
"Google"	53	147	55	56
132	252 "Google"			
"Google Play Store"	3	6	1	0
6	0 "Google"			
"Instagram"	6	14	4	4
10	42 "Facebook"			
"Netflix"	6	10	15	15
2	78 "Netflix"			
"Snapchat"	16	0	0	0
0	0 "Snapchat"			
"Spotify"	8	25	9	9
19	70 "Spotify"			
"Twitter"	6	10	14	14
7	38 "Twitter"			
"Whats App"	2	0	0	0
0	0 "Facebook"			
"YouTube"	8	8	7	7
21	3 "Google"			

TOTAL (14 applications)	150	259	135	135
232	584			
DNS responses monitored		365		

10.2.3.23 mp-tcp-proxy

Syntax

```
mp-tcp-proxy session aa-sub esm esm-sub-id-str flow-id flow-id
mp-tcp-proxy session aa-sub mobile imei imei apn apn flow-id flow-id
mp-tcp-proxy session aa-sub mobile imsi imsi-msisdn apn apn flow-id flow-id
mp-tcp-proxy session aa-sub mobile msisdn imsi-msisdn apn apn flow-id flow-id
mp-tcp-proxy session aa-sub mobile apn apn flow-id flow-id subscriber-id subscriber-id
mp-tcp-proxy session aa-sub sap sap-id flow-id flow-id
mp-tcp-proxy session aa-sub spoke-sdp sdp-id:vc-id flow-id flow-id
mp-tcp-proxy session aa-sub transit trnst-sub-nm flow-id flow-id
mp-tcp-proxy session isa mda-id flow-id flow-id
```

Context

```
tools>dump>app-assure>group
```

Description

This command displays Application Assurance (AA) MP-TCP flow records matching the specified criteria for a specific AA subscriber.

To use this command to examine MP-TCP proxy flows, first find the MP proxied flow IDs of the network side using the following tools dump command:

```
tools>dump>app-assure>group >flow-record-search isa card/mda search-type match mp-net
```

Parameters

apn

specifies the APN, up to 80 characters

esm-sub-id-str

specifies the ESM subscriber ID, up to 32 characters

flow-id

specifies the flow ID

Values: 0 to 4294967295

imei
displays flows for the specified subscriber, up to 16 characters

imsi-msidsn
displays flows for the specified subscriber, up to 15 digits

mda-id
specifies the MP-TCP proxy for a particular ISA AA card in the format *slot/mda*

Values	<i>slot</i>	1 to 10
–	<i>mda</i>	1 to 2

imsi-msisdn
displays flows for the specified subscriber, up to 15 digits

sap-id
specifies the SAP ID, up to 32 characters

sdp-id:vc-id
specifies the SDP and VC IDs

Values	<i>sdp-id</i>	1 to 32767
–	<i>vc-id</i>	1 to 4294967295

subscriber-id
specifies the subscriber identifier, up to 32 characters

trnst-sub-nm
specifies the transit subscriber name, up to 32 characters

10.2.3.24 port-recorder detail

Syntax

port-recorder detail [flow-count *flows*] [byte-count *kbytes*] [isa *mda-id*] url *file-url*

Context

tools>dump>app-assure>group

Description

This command saves the port recorded by the tool into a file. The port-recorder is configured using debug commands.

Parameters

flows

Match ports with flow count greater than the specified value.

Values: slot — 1 to 4294967295

kilobytes

Match ports with bytes count greater than the specified value.

Values: slot — 1 to 4294967295

mda-id

specifies the slot and MDA in the format *slot/mda*

Values	<i>slot</i>	1 to 10
–	<i>mda</i>	1 to 2

file-url

specifies the URL for the local or remote file to which the port recorder output will be directed

Values	local-url	[cflash-id] [file- path] up to 200 characters, including cflash-id
–	directory length, up to 99 characters each	
–	remote-url	[[ftp:// tftp://]login:pswd@remote-locn] [file-path]
–	255 characters maximum	

- directory length, up to 99 characters each
- remote-locn [hostname | ipv4-address | ipv6-address]
- ipv4-address a.b.c.d
- ipv6-address x:x:x:x:x:x:x[-interface]
- x:x:x:x:x:x.d.d.d[-interface]
- x - [0..FFFF]H
- d - [0..255]D –
- interface up to 32 characters, for link local address
- cflash-id flash slot ID

10.2.3.25 port-recorder status

Syntax

```
port-recorder status [isa mda-id]
```

Context

```
tools>dump>app-assure>group
```

Description

This command displays the current status of the port-recorder with current-time, start-time, stop-time, sample-rates as well as number of bytes and flows for UDP and TCP traffic on the specified AA ISA card. The port-recorder is configured using debug commands.

Parameters

mda-id

specifies the slot and MDA in the format *slot/mda*

Values	<i>slot</i>	1 to 10
–	<i>mda</i>	1 to 2

10.2.3.26 port-recorder top

Syntax

```
port-recorder top granularity [max-count max-count] [isa mda-id]
```

Context

```
tools>dump>app-assure>group
```

Description

This command displays by bytes or flows the top ports recorded by the tool on a particular AA ISA.

Parameters

- granularity***
specifies if the output is sorted by bytes or flows
- count-value***
specifies the maximum number of values to display.
Values: 1 to 25
- mda-id***
specifies the slot and MDA in the format *slot/mda*

Values	<i>slot</i>	1 to 10
–	<i>mda</i>	1 to 2

10.2.3.27 policer

Syntax

`policer policer-name day daytime time-of-day`

Context

`tools>dump>app-assure>group`

Description

This command displays rates for the policer for a specific day and time.

Parameters

- policer-name***
specifies an existing policer name, up to 256 characters
- day***
specifies a day to display policer rates
Values: sunday, monday, tuesday, wednesday, thursday, friday, saturday
- time-of-day***
specifies a time of day (in hours and minutes) to display policer rates

Values	<i>hh</i>	0 to 24
–	<i>mm</i>	0 15 30 45

10.2.3.28 resources

Syntax

`resources`

Context

`tools>dump>application-assurance>group`

Description

This command displays a view of all the ISA group or group partition limits, along with current usage, for the configured scale mode and ISA-AA versions.

10.2.3.29 tethering

Syntax

```
tethering detail
```

Context

```
tools>dump>application-assurance>group
```

Description

This command displays AA current subscribers' tethering statistics.

Parameters

detail

specifies detailed tethering statistics

Output

The following output example shows tethering detail information.

Example output AA tethering statistics

```
# tools dump application-assurance group 1:1 tethering detail
=====
Application-Assurance tethering detail - Group 1:1
=====
      TTL           Expected      Flow Count
-----
      62             no           215
      63             yes        1501255
     126             no           7612
     127             yes        1078002
=====
```

10.2.3.30 traffic-capture

Syntax

```
traffic-capture detail url file-url
```

traffic-capture status

Context

tools>dump>app-assure>group

Description

This command displays application-assurance traffic-capture information.

Parameters

file-url

specifies the URL for the local or remote file to which the traffic capture output will be directed

Values	local-url	[cflash-id] [file- path] up to 200 characters, including cflash-id
–	directory length, up to 99 characters each	–
–	remote-url	[[ftp:// tftp://]login:pswd@remote-locn] [file-path]
–	255 characters maximum	
–	directory length, up to 99 characters each	
–	remote-locn	[hostname ipv4-address ipv6-address]
–	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:[-interface]
–	x:x:x:x:x:d.d.d.d[-interface]	
–	x - [0..FFFF]H	
–	d - [0..255]D	–

- | | | |
|---|-----------|---|
| – | interface | up to 32 characters, for link local address |
| – | cflash-id | flash slot ID |

10.2.4 Application Assurance clear command descriptions

10.2.4.1 group

Syntax

```
group aa-group-id cflowd
group aa-group-id statistics
group aa-group-id status
```

Context

```
clear>app-assure
```

Description

This command clears application assurance group statistics or status.

Parameters

aa-group-id

clears data for the specified AA ISA group

cflowd

clears application assurance Cflowd statistics

statistics

clears application assurance system and subscriber statistics

status

clears application assurance status statistics

10.2.5 Application Assurance debug command descriptions

10.2.5.1 group

Syntax

```
group aa-group-id [: partition-id]
```

Context

```
debug>app-assure
```

Description

This command configures application-assurance within a group/partition debugging.

Parameters

***aa-group-id* [:*partition-id*]**

Specifies the existing application assurance group and partition id.

Values	<i>aa-group-id</i> . <i>parti</i> *: <i>aa-group-id</i> [: <i>partition-id</i>]	
–	<i>aa-group-id</i>	1 to 255
–	<i>partition-id</i>	1 to 65535

10.2.5.2 http-host-recorder

Syntax

```
[no] http-host-recorder
```

Context

```
debug>app-assur>group
```

Description

This command enables the http-host-recorder feature on a particular group:partition.

The no form of this command disables the http-host-recorder feature.

10.2.5.3 filter

Syntax

```
filter
```

Context

```
debug>app-assur>group>http-host-recorder
```

Description

This command configures recorder filter settings. This command specifies the filtering parameter for the http-host-recorder feature.

10.2.5.4 default-filter-action

Syntax

```
default-filter-action default-action
```

Context

```
debug>app-assur>group>http-host-recorder>filter
```

Description

This command configures the recorder filter default action to either record or no-record. This parameter applies to http-host values not matching any expressions defined in the filter context.

default-action

specifies the default action

Values: record | no-record

10.2.5.5 expression

Syntax

```
expression expr-index expr-type eq expr-string{record | no-record}
```

```
no expression expr-index
```

Context

```
debug>app-assur>group>http-host-recorder>filter
```

Description

This command configures the recorder filter expressions.

Parameters

expr-index

Specifies the expression index value.

Values: 1 to 4

expr-type

Specifies the expression type.

Values: http-host

expr-string

Specifies the HTTP host filter expression string.

Values: format *<expression>\$ (33 chars max)

10.2.5.6 record

Syntax

```
record {all-hosts | http-host-app-filter-candidates}
```

Context

```
debug>app-assur>group>http-host-recorder>filter
```

Description

This command configures which http-host are selected for the http-host-recorder. It is either any http-host values going through the AA ISA or the http-host corresponding to flows not matching a string based app-filter.

For the feature to work it is required to configure at least one app-filter to catch the HTTP protocol signature.

Parameters

all-hosts | http-host-app-filter-candidates

Specifies which hosts the recorder will record

Default: http-host-app-filter-candidates

10.2.5.7 rate

Syntax

rate *sample-rate*

no rate

Context

debug>app-assure>group>http-host-recorder

debug>app-assure>group>port-recorder

Description

This command configures the sampling rate for the recorded http host, a sampling rate of 10 will sample one out of 10 http-host.

Parameters

sample-rate

Specifies the sample rate.

Values: 1 to 10000

Default: 100

Output

The following configuration records http-host entries ending with “.com” as a result of the expression filter configuration. It will not record any other HTTP host values since the default-filter-action set to no-record. The http-host entries analyzed by the recorder in the first place are http-host-app-filter-candidates.

Example Output

```
# show debug
debug
  application-assurance
    group 1:1
      http-host-recorder
        filter
          default-filter-action no-record
          expression 1 http-host eq "*.com$" record
          record http-host-app-filter-candidates
        exit
      rate 100
      no shutdown
    exit
```

```
exit
exit
exit
```

10.2.5.8 shutdown

Syntax

[no] shutdown

Context

debug>app-assure>group>http-host-recorder

debug>app-assure>group>port-recorder

Description

This commands stops or starts the http-host-recorder. To reset the recorded values execute shutdown followed by no shutdown.

10.2.5.9 port-recorder

Syntax

[no] port - recorder

Context

debug>app-assure>group

Description

This commands allows to stop or start the http-host-recorder. To reset the recorded values execute shutdown followed by no shutdown.

10.2.5.10 application

Syntax

[no] application *application-name*

Context

debug>app-assure>group>port-recorder

Description

This commands specifies the applications used as input by the port-recorder. Applications responsible for unknown or unidentified traffic are meant to be used by this tool.

Output

The following sample configuration records TCP and UDP port numbers for the application “Unidentified TCP”.

Example Output

```
# show debug
debug
  application-assurance
    group 1:1
      port-recorder
        application "Unidentified TCP"
        rate 100
        no shutdown
      exit
    exit
  exit
exit
```

10.2.5.11 traffic-capture

Syntax

[no] traffic-capture

Context

debug>app-assure>group

Description

This command configures debugging for traffic capture.

10.2.5.12 match

Syntax

```
[no] match
```

Context

```
debug>app-assure>group>traffic-capture
```

Description

This command configures debugging for traffic match criteria.

10.2.5.13 application

Syntax

```
application {eq | neq} application-name  
no application
```

Context

```
debug>app-assure>group>traffic-capture>match
```

Description

This command configures debugging on an application.

10.2.5.14 client-ip

Syntax

```
client-ip {eq | neq} ip-address  
no client-ip
```

Context

```
debug>app-assure>group>traffic-capture>match
```

Description

This command configures debugging of a client IP.

10.2.5.15 client-port

Syntax

```
client-port {eq | neq} port-num  
no client-port
```

Context

```
debug>app-assure>group>traffic-capture>match
```

Description

This command configures debugging of a client port.

10.2.5.16 dst-ip

Syntax

```
dst-ip {eq | neq} ip-address  
no dst-ip
```

Context

```
debug>app-assure>group>traffic-capture>match
```

Description

This command configures debugging on a destination IP address.

10.2.5.17 dst-port

Syntax

```
dst-port {eq | neq} port-num  
no dst-port
```

Context

```
debug>app-assure>group>traffic-capture>match
```

Description

This command configures debugging on a destination port.

10.2.5.18 ip-addr1

Syntax

```
ip-addr1 {eq | neq} ip-address  
no ip-addr1
```

Context

```
debug>app-assure>group>traffic-capture>match
```

Description

This command configures debugging on IP address 1.

10.2.5.19 ip-addr2

Syntax

```
ip-addr2 {eq | neq} ip-address  
no ip-addr2
```

Context

```
debug>app-assure>group>traffic-capture>match
```

Description

This command configures debugging on IP address 2.

10.2.5.20 ip-protocol-num

Syntax

```
ip-protocol-num {eq | neq} protocol-id  
no ip-protocol-num
```

Context

```
debug>app-assure>group>traffic-capture>match
```

Description

This command configures debugging on an IP protocol number.

10.2.5.21 port1

Syntax

```
port1 {eq | neq} port-num  
no port1
```

Context

```
debug>app-assure>group>traffic-capture>match
```

Description

This command configures debugging on port 1.

10.2.5.22 port2

Syntax

```
port2 {eq | neq} port-num  
no port2
```

Context

```
debug>app-assure>group>traffic-capture>match
```

Description

This command configures debugging on port 2.

10.2.5.23 server-ip

Syntax

```
server-ip {eq | neq} ip-address
```

```
no server-ip
```

Context

```
debug>app-assure>group>traffic-capture>match
```

Description

This command configures debugging on a server IP address.

10.2.5.24 server-port

Syntax

```
server-port {eq | neq} port-num
```

```
no server-port
```

Context

```
debug>app-assure>group>traffic-capture>match
```

Description

This command configures debugging on a server port.

10.2.5.25 src-ip

Syntax

```
src-ip {eq | neq} ip-address
```

```
no src-ip
```

Context

```
debug>app-assure>group>traffic-capture>match
```

Description

This command configures debugging on a source IP address.

10.2.5.26 src-port

Syntax

```
src-port {eq | neq} port-num  
no src-port
```

Context

```
debug>app-assure>group>traffic-capture>match
```

Description

This command configures debugging on a source port.

10.2.5.27 record

Syntax

```
record
```

Context

```
debug>app-assure>group>traffic-capture
```

Description

This command configures traffic recording options.

10.2.5.28 cut-through-packets

Syntax

```
cut-through-packets cut-through-packets
```

Context

```
debug>app-assure>group>traffic-capture>record
```

Description

This command records cut-through packet conditions.

Parameters

cut-through-packets

Indicates whether to capture cut-through only packets or cut-through and other packets, or to exclude them all together.

Values: exclude | include | only

Default: include

10.2.5.29 limit

Syntax

```
limit {all-packet-matches | first-session-match}
```

Context

```
debug>app-assure>group>traffic-capture>record
```

Description

This command records limit conditions.

Parameters

all-packet-matches

Records all the packets matching the condition.

first-session-match

Records only the first session matching the condition.

10.2.5.30 start

Syntax

```
start {immediate | on-new-session}
```

Context

```
debug>app-assure>group>traffic-capture>record
```

Description

This command records limit conditions.

Parameters

immediate

Start recording immediately for new or existing flows or sessions.

on-new-session

Only start recording record for new flows or sessions.

10.2.5.31 shutdown

Syntax

[no] shutdown

Context

debug>app-assur>group>traffic-capture

Description

This command administratively disables traffic capture.

11 Tools, OAM, and ping command reference

This section describes the tools-, OAM-, and ping-related commands.

11.1 Tools, OAM, and ping command hierarchies

- *Tools command hierarchy*
- *OAM command hierarchy*
- *Ping command hierarchy*

11.1.1 Tools command hierarchy



NOTICE:

The commands under the tools perform>mobile>pdn>force-clear context must be used when necessary and as instructed by Nokia Technical Support.

tools

- *dump*
- *dhcp*
 - *test* dhcp-server-group *dhcp-group* [detail]
- *micronets* [local-pool-name] {VPRN *name* | {network-realm *name* peer *ipv4/ipv6}}*
- *mobile-gateway* *gw-id*
 - *dedicated-bearer-monitoring* [{imei | imsi | msisdn} *string*] [apn *apn*] [status]
 - *diameter-peer-realm* {s6b | gx | gy} peer *ipv4* | *ipv6* [port *port*] group *group*
 - *dns-client-cache* {apn | pgw-id | p-cscf | candidate-list} *name* [access-type *access-type*]
 - *dns-client-cache-candidates* {pgw-selection | p-cscf | upf-selection} [access-type *access-type*]
 - *dns-client-cache-remove* type *name*
 - *nf-profile* [bsf]
 - *pdn-session-db* [imsi *imsi*] [attribute {unacked | del-in-prog | ho-in-prog}]
 - *pfcf-audit* remote-peer *ip-address* | *ipv6-address* [router *router-name*]
 - *pfcf-stale-session-audit-status*
 - *policy-on-the-fly-update-stats* [apn *apn*]
- *radius*
 - *test*
 - *accounting* radius-group *radius-group* [detail]
 - *authentication* radius-group *radius-group* [detail]
- *perform*

```

- dhcp
  - test dhcp-server-group dhcp-group [dhcp-server peer] imsi imsi apn apn
- mobile-gateway
  - pdn gw-id
    - apn apn-name
      - dedicated-bearer-monitoring {enable | disable | inherit}
      - no dedicated-bearer-monitoring
    - canary-test
      - user-range user-range-name target targets
    - dedicated-bearer-monitoring
      - [no] imsi imsi
      - [no] shutdown
    - force-clear
      - bearer-context {imsi | imei | msisdn} string [local-only] [apn apn] [slave]
      - diameter-peer-destination-host ref-point-type {s6b | gx | gy} group group-
number peer ipv4 | ipv6 [port port-number] destination-host destination-host
      realm realm-name
        - path {router router-name [local-peer ipv4 | ipv6] remote-peer {ipv4 | ipv6}
protocol-type {gtp | pmip | pfcf}
        - force-gtpu-echo-req-action {drop | forward}
        - nf-register [bsf]
        - pfcf-stale-session-audit-cancel
        - pfcf-stale-session-audit-init [cleanup-on-no-resp]
    - profile
      - diameter-peer diameter-peer-profile-name
      - refresh-dns
      - fqdn-group name router vprn-id
      - clear-list
      - refresh-list
      - pcscf-group pcscf-group
      - refresh-list
- radius
  - test
    - accounting {on|off} radius-group radius-group [radius-server peer] [apn apn-
name]
    - accounting start radius-group radius-group [radius-server peer] [user-
name user-name] apn apn-name
    - authentication radius-group radius-group [radius-server peer] [user-
name user-name] [password password] [apn apn-name]
  - router
    - ip-local-pool
      - reserve-ipv4 ipv4-address
      - no reserve-ipv4
      - reserve-ipv6 ipv6-address
      - no reserve-ipv6

```

- *security*
- *authentication-server-check*

11.1.2 OAM command hierarchy

This section describes OAM commands that are unique to the 7750 SR MG and CMG and thus are not described in the 7750 SR documentation. Additional supported OAM commands are described in the *7450 ESS, 7750 SR, 7950 XRS, and VSR OAM and Diagnostics Guide*.

```
oam
  – dns target-addr dns-name name-server ip-address {[source ip-address] |
    [interface interface-name]][router-instance router-instance] [count send-count]
    [timeout timeout] [interval interval] [record-type {ipv4-a-record | ipv6-aaaa-record |
    naptr}] [service-label service-label-string] [replacement replacement-string] [detail]
  – gtp-ping {s11 | s5c | s1u | s2b | s5u | s8c | s8u | gnc | gpc | s2bc | s2bu | s4c
    | s4u | s12 | s2ac | s2au | n3 | n9} [router router-instance] [source ip-address]
    destination ip-address port port-number gateway gw-id [retry-count count] [time-
    out seconds]
  – radius-ping apn [tos type-of-service] source ip-address [size bytes]
    [count requests] router router-or-service destination ip-address [gateway gw-id]
  – ue-ping {imsi | msisdn | ue-ip} value apn apn source-ip ip-address [size bytes]
    [count requests] [timeout timeout] gateway gw-id
```

11.1.3 Ping command hierarchy

This section describes ping commands that are unique to the 7750 SR MG and CMG and thus are not described in the 7750 SR documentation. Additional supported Ping commands are described in the *7450 ESS, 7750 SR, 7950 XRS, and VSR OAM and Diagnostics Guide*.

```
– ping [ip-address | dns-name] {next-hop ip-address} [interface interface-name]
  bypass-routing [timeout-type timeout-type] [ip-address] subscriber-id [sub-
  ident-string] [count requests] [detail | rapid] [do-not-fragment] [fc fc-name]
  [interval centisecs | secs] [pattern pattern] [{router router-or-service} | {router-
  instance router-instance} | {service-name service-name}] [size bytes] [source ip-
  address] [timeout timeout] [tos type-of-service] [ttl time-to-live]
```

11.2 Tools, OAM and ping command descriptions

- *Tools command descriptions*
- *OAM command descriptions*

- *Ping command descriptions*

11.2.1 Tools command descriptions

11.2.1.1 tools

Syntax

tools

Context

root

Description

This command enables the context to use tools for debugging purposes.

11.2.1.2 dump

Syntax

dump

Context

tools

Description

This command enables the context to display information for debugging purposes.

11.2.1.3 dhcp

Syntax

dhcp

Context

tools>dump

Description

This command enables the context to execute DHCP test commands.

11.2.1.4 test

Syntax

```
test dhcp-server-group dhcp-group [detail]
```

Context

```
tools>dump>dhcp
```

Description

This command dumps the test results of the DHCP servers.

Parameters

dhcp-group

specifies the DHCP server group name, up to 32 characters

detail

specifies to display the event history

Output

The following output example shows a DHCP server group test.

Example output DHCP server group test

```
# tools dump dhcp test dhcp-server-group "dhcpv4-grp1" detail
=====
Dump DHCP Tests      :
=====
Server Type          : DHCP
Server Preference    : Priority 1
Admin State          : Up
Server OP State       : Up
Peer Address         : 11.23.10.2
Peer Port            : 67
DHCP Server Group    : dhcpv4-grp1
Event History         : Up           02/03/2024 02:11:17
                     : Down        02/03/2024 02:11:16
                     : Up           02/03/2024 02:10:45
-----
Number of Servers    : 1
=====
```

11.2.1.5 micronets

Syntax

```
micronets [local-pool-name] {VPRN name | {network-realm name peer ipv4/ipv6}}
```

Context

```
tools>dump
```

Description

This command enables micronet usage monitoring by querying the allocated and available micronets. You can monitor the micronet usage per local IP address pool (optional), routing instance, or network realm and peer address.

Parameters

local-pool-name

specifies the local IP address pool name, up to 256 characters

name

specifies the VPRN name, up to 256 characters

network-realm *name*

specifies the network realm name, up to 256 characters

ipv4 / ipv6

specifies the IP address of the peer

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

Output

The following output example shows micronet usage monitoring.

Example output micronet usage monitoring

```
# tools dump micronets ims61 peer 33.34.1.1 network-realm realm-for-ims-61

VRID = 4 Pool = ims61
  Subnet = 2600:1002:1000::
    pfx len = 44 asgn len = 64 block=0 Default GW = ::
    Non-zero per MG group micronet count:
      MG Group 1: micronet count=1973
      MG Group 2: micronet count=2123
  Subnet = 2600:1002:1010::
    pfx len = 44 asgn len = 64 block=0 Default GW = ::
    Non-zero per MG group micronet count:
      MG Group 1: micronet count=2081
      MG Group 2: micronet count=2015
  Subnet = 2600:1002:1040::
    pfx len = 44 asgn len = 64 block=0 Default GW = ::
    Non-zero per MG group micronet count:
      MG Group 1: micronet count=2048
      MG Group 2: micronet count=2048
  Mscp Grps that own v6 Micronets in this pool:
    mscpGrpId = 1 count = 6102
    mscpGrpId = 2 count = 6186
  Number of available micronets = 0
  Number of allocated micronets = 12288
```

11.2.1.6 mobile-gateway

Syntax

mobile-gateway *gw-id*

Context

tools>dump

Description

This command enables the context to execute mobile gateway performance commands.

Parameters

gw-id

specifies a mobile gateway identifier

Values: 1 to 8

11.2.1.7 dedicated-bearer-monitoring

Syntax

```
dedicated-bearer-monitoring [{imei | imsi | msisdn} string] [apn apn] [status]
```

Context

```
tools>dump>mobile
```

Description

This command dumps the detailed information of a dedicated bearer for a UE when the IMEI, IMSI, or MSISDN are specified in the command. If the UE identity is not used, the command lists the bearer contexts for which there is currently information on the detailed dedicated bearer.

Parameters

string

specifies the IMEI (up to 16 characters), IMSI (up to 15 characters), or MSISDN (up to 15 characters)

apn

specifies an APN name, up to 80 characters

status

displays the dedicated bearer monitoring status at the APN and PDN level



NOTE:

The status parameter cannot be combined with IMSI or APN.

Output

The following output examples show the dedicated bearer monitoring, the dedicated bearer monitoring with IMSI, IMEI, MSISDN and APN, and the dedicated bearer monitoring status. [Dedicated dearer monitoring output fields](#) describes the output fields.

- [Example output dedicated bearer monitoring](#)
- [Example output dedicated bearer monitoring with IMEI, IMSI, MSISDN, and APN](#)
- [Example output dedicated bearer monitoring status](#)

Example output dedicated bearer monitoring

```
# tools dump mobile-gateway 1 dedicated-bearer-monitoring
=====
```

Dedicated Bearer Context Monitoring

```
=====
IMSI/IMEI(#)      ebi  Status group  timestamp    Persistent
/MSISDN(^)/MAC(`)
/SubId
-----
311480017833417   1    Activated      1      05/30 05:59  Y
311480017833918   3    Activated      1      05/30 05:59  Y
311480017835667   5    Deactivated     1      05/30 05:59  Y
311480017835168   7    Activation Failed 2      05/30 05:59  Y
311480017835417   2    Activation Failed 2      05/30 05:59
311480017835668   4    Deactivated     1      05/30 05:59
311480017835917   6    Activation Failed 1      05/30 05:59
Number of bearer contexts : 7
=====
```

Example output dedicated bearer monitoring with IMEI, IMSI, MSISDN, and APN

```
# tools dump mobile-gateway <gw-id> dedicated-
bearer monitoring {imsi|imei|msisdn} <string> [apn <apn>]
=====
Dedicated Bearer Context Monitoring
=====
IMSI              : 123456789012953      Persistent : Y/N
APN               : ipd.nokia.com
PDN Type          : IPv6                LBI        : 5
Serving MSCP group : 1
Bearer Id         : 4
Status            :
Timestamp         :
Setup duration    :
Active Duration   :
P-CSCF IP Address :
Deactivation Reason :
Deactivation CC-GTP : GTP sent / received CC
Deactivation CC-Dia*: Diameter sent received CC
Setup Fail Gx CC   :
Detailed Failure Gx :
Setup Fail GTP CC  :
Detailed Failure GTP:
-----
Number of bearer contexts : 1
```

Example output dedicated bearer monitoring status

```
# tools dump mobile-gateway 1 dedicated-bearer-monitoring status
=====
Dedicated Bearer Context Monitoring Status
=====
PDN status:  enabled/disabled
Enabled APN list:
    apn1.nokia.com
    apn2.nokia.co
Number of APNs that status enabled: 2
=====
```

Table 212: Dedicated bearer monitoring output fields

Label	Description
Serving MSCP Group	The MSCP group or MG group that is or was served by the bearer
Bearer Id	The EBI of the bearer
LBI	The EBI of the default bearer that this dedicated bearer is linked to
Status	The feature status at the APN and PDN level (Activated/Deactivated/Activation Failed)
Timestamp	Timestamp of the activation/deactivation/activation failure
Setup Duration	The duration of the activation procedure
Active Duration	The time between activation and deactivation, or up to present time if the dedicated bearer still exists
P-CSCF IP Address	The last used P-CSCF IP address
Deactivation Reason	The reason of deactivation (PCRF-init/MME-init/UE-init/internal GW)
Deactivation CC-GTP	The CC value that is sent or received in GTPv1 or GTPv2 in the release signaling
Deactivation CC-Dia	The Diameter CC value sent or received in the deactivation procedure
Setup Failure Gx CC	The GX Diameter failure CC value in case of activation failure; for example, in PGW rejection of RAR
Detailed Failure Gx	The detailed failure information on Gx, in case of activation failure; for example, the reason of resource allocation, the reason of failure in installing policy rules, and so on

Table 212: Dedicated dearer monitoring output fields (continued)

Label	Description
Setup Failure GTP CC	The GTP failure CC value used in case of activation failure; for example, in CBR rejection from the MME
Detailed Failure GTP	The detailed failure information on GTP in case of activation failure; for example, no response from the MME

11.2.1.8 diameter-peer-realm

Syntax

diameter-peer {s6b | gx | gy} peer *ipv4* | *ipv6* [port *port*] group *group*

Context

tools>dump>mobile

Description

This command displays Diameter peer realm information for the specified s6b, Gx, or Gy peer.

Parameters

s6b | gx | gy

keyword that specifies the interface type for the Diameter peer

ipv4 | ipv6

specifies the IP address of the Diameter peer

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d

-	-	x - [0..FFFF]H
-	-	d - [0..255]D

port

specifies the port number

Values: 0 to 65535

Default: 3868

group

specifies the group number

Values: 1 to 15

11.2.1.9 dns-client-cache

Syntax

```
dns-client-cache {apn | pgw-id | p-cscf | candidate-list} name [access-type access-type]
```

Context

```
tools>dump>mobile
```

Description

This command displays the DNS resolution cache for the specified APN, PGW ID, P-CSCF FQDN or a candidate list.

Parameters

apn

specifies that the cache will be displayed for the APN name

Values: 0 to 255

pgw-id

specifies that the cache will be displayed for the PGW name

Values: 0 to 255

p-cscf

specifies that the cache will be displayed for the P-CSCF name

Values: 0 to 255

candidate-list

specifies that the cache will be displayed for the candidate list name

access-type

specifies the access type

Values: s2a | s2b | s2b+nc-smf | sxa | sxa+nc-nr | sxb | sxb+nc-nr

Output

The following output examples show DNS resolution cache information.

Example output APN cache

```
# tools dump mobile-gateway 1 dns-client-cache apn ehrpd.apn.epc.mnc012.mcc310.3gppnetwork.org

GwId: 1
Type: apn
Application Unique String: ehrpd.apn.epc.mnc012.mcc310.3gppnetwork.org
Service parameters: x-3gpp-pgw:x-s2b-gtp

Resolve time:      02/13/2015 16:50:53
Expire time:       02/13/2015 22:50:53

NAPTR records:
String
  TTL      Order  Preference  Flag      Service      Regexp
Replacement
-----
ehrpdp.apn.epc.mnc012.mcc310.3gppnetwork.org
   0       10     10         s        x-3gpp-pgw:x-s2b-gtp
ehrpdp.apn.epc.mnc012.mcc310.srv1.3gppnetwork.org

SRV records:
Service
  TTL      Priority  Weight    Port
Target
-----
ehrpdp.apn.epc.mnc012.mcc310.srv1.3gppnetwork.org
   0        20      100       2123
ehrpdp.apn.epc.mnc012.mcc310.srv1.a1.3gppnetwork.org

A/AAAA records:
Hostname
  TTL      Type      Address
-----
ehrpdp.apn.epc.mnc012.mcc310.srv1.a1.3gppnetwork.org
   0        A        10.20.1.3
```

Example output PGW identity cache

```
# tools dump mobile-gateway 1 dns-client-cache pgw-id ehrpd.apn.epc.mnc012.mcc310.srv1.a1.3gppnetwork.org

GwId: 1
```

```
Type: pgw-id
Application Unique String: ehrrpd.apn.epc.mnc012.mcc310.srv1.a1.3gppnetwork.org

Resolve time:      02/13/2015 16:55:25
Expire  time:      02/13/2015 22:55:25

A/AAAA records:
Hostname
  TTL      Type      Address
-----
ehrrpd.apn.epc.mnc012.mcc310.srv1.a1.3gppnetwork.org
    0        A        10.20.1.3
```

Example output P-CSCF FQDN cache

```
# tools dump mobile-gateway 1 dns-client-cache p-cscf nok.3.ipv4.3gpp.pcscf
```

```
GwId: 1
Type: p-cscf
Application Unique String: nok.3.ipv4.3gpp.pcscf
```

```
Resolve time:      03/17/2015 10:50:52
Expire  time:      03/17/2015 16:50:52
```

```
A/AAAA records:
Hostname
  TTL      Type      Address
-----
nok.3.ipv4.3gpp.pcscf
    0        A        51.3.1.1
nok.3.ipv4.3gpp.pcscf
    0        A        51.3.1.2
nok.3.ipv4.3gpp.pcscf
    0        A        51.3.1.3
nok.3.ipv4.3gpp.pcscf
    0        A        51.3.1.4
nok.3.ipv4.3gpp.pcscf
    0        A        51.3.1.5
nok.3.ipv4.3gpp.pcscf
    0        A        51.3.1.6
nok.3.ipv4.3gpp.pcscf
    0        A        51.3.1.7
nok.3.ipv4.3gpp.pcscf
    0        A        51.3.1.8
nok.3.ipv4.3gpp.pcscf
    0        A        51.3.1.9
nok.3.ipv4.3gpp.pcscf
    0        A        51.3.1.10
nok.3.ipv4.3gpp.pcscf
    0        A        51.3.1.11
nok.3.ipv4.3gpp.pcscf
    0        A        51.3.1.12
nok.3.ipv4.3gpp.pcscf
    0        A        51.3.1.13
nok.3.ipv4.3gpp.pcscf
    0        A        51.3.1.14
```

```
nok.3.ipv4.3gpp.pcscf
0      A      51.3.1.15
nok.3.ipv4.3gpp.pcscf
0      A      51.3.1.16
```

11.2.1.10 dns-client-cache-candidates

Syntax

```
dns-client-cache-candidates {pgw-selection | p-cscf | upf-selection} [access-type
access-type]
```

Context

```
tools>dump>mobile
```

Description

This command provides a list of PGWs or SMFs, or P-CSCFs or UPFs that the system has compiled from DNS records.

Parameters

pgw-selection

specifies that the list of APNs and the associated PGW or SMF candidates will be displayed

Values: 0 to 255

p-cscf

specifies that the list of P-CSCFs and the associated IP addresses will be displayed

Values: 0 to 255

upf-selection

specifies that the list of UPFs and the associated IP addresses will be displayed

access-type

specifies the access type

Values: s2a | s2b | s2b+nc-smf | sxa | sxa+nc-nr | sxb | sxb+nc-nr

Output

The following output example shows DNS client cache candidate information.

Example output DNS client cache candidates

```
# tools dump mobile-gateway 1 dns-client-cache-candidates pgw-selection

APN ehrpd.apn.epc.mnc012.mcc310.3gppnetwork.org candidates:
```



```
10.20.1.3          apn ehrpd.apn.epc.mnc012.mcc310.3gppnetwork.org
```

```
PGW ehrpd.apn.epc.mnc012.mcc310.srv1.a1.3gppnetwork.org candidates:  
10.20.1.3
```

11.2.1.11 dns-client-cache-remove

Syntax

```
dns-client-cache-remove type name
```

Context

```
tools>dump>mobile
```

Description

This command allows the manual removal of a specific APN or PGW-ID FQDN entry from the gateway cached DNS results. When a FQDN is removed, its associated second level FQDNs and IP addresses are also removed.

Parameters

type

specifies if an APN or PGW_ID FQDN entry is removed

Values: apn | pgw-id

name

specifies the APN or PGW_ID FQDN entry name, up to 256 characters

11.2.1.12 nf-profile

Syntax

```
nf-profile [bsf]
```

Context

```
tools>dump>mobile
```

Description

This command displays the NF profile of the SMF, BSF, or UPF that is received by the NRF after a successful registration operation. If CMG is not registered to the NRF, the command displays the registration status and the NFInstanceId that are used for registration.

Parameters

bsf

specifies to enable the context to configure BSF attributes

Output

The following output example shows NF profile information.

Example output NF profile

```
# tools dump mobile-gateway 1 nf-profile
Registered NfProfile:
NfInstanceID : 01300000-c1ba-4e3d-9813-20000c1bae3d
NfType: SMF
Nf Status: Registered
PLMN List Elements 2
Home Plmn: mcc 310, mnc 12
Home Plmn: mcc 311, mnc 480
fqdn :
capacity :
priority :
sNssai List (1):
  sst : 12, sd : 0x10203
IPv4 addresses (1):
  ip: 16.70.16.16
IPv6 addresses (1):
  ip: 3ffe::1046:1010
HeartBeat Timer: 60
Printing SMFInfo:
Num of slices (1)
  sst : 12, sd : 0x10203
  dnn[0]: apn.0.1.com
  dnn[1]: apn.0.2.com
  dnn[2]: apn.0.3.com
  dnn[3]: apn.0.4.com
  dnn[4]: apn.1.1.com
  dnn[5]: apn.1.2.com
  dnn[6]: apn.1.3.com
  dnn[7]: apn.1.4.com
  dnn[8]: apn.nonip.1.1.com
  dnn[9]: apn.nonip.1.2.com
  dnn[10]: apndynamicrulebase
  dnn[11]: apnstaticpccnorules
  dnn[12]: apnstaticpccrules
  dnn[13]: ipd.nokia.com
  dnn[14]: ipd.nokia.0.1.com
  dnn[15]: ipd.nokia.0.2.com
  dnn[16]: ipd.nokia.0.3.com
  dnn[17]: ipd.nokia.0.4.com
  dnn[18]: ipd.nokia.1.1.com
  dnn[19]: ipd.nokia.1.1.com.mnc012.mcc310.gprs
  dnn[20]: ipd.nokia.1.1.com.mnc012.mcc345.gprs
  dnn[21]: ipd.nokia.1.2.com
  dnn[22]: ipd.nokia.1.2.com.mnc012.mcc310.gprs
```

```
dnn[23]: ipd.nokia.1.2.com.mnc012.mcc345.gprs
dnn[24]: ipd.nokia.1.3.com
dnn[25]: ipd.nokia.1.4.com
dnn[26]: ipd.nokia.1.5.com
dnn[27]: ipd.nokia.1.6.com
dnn[28]: ipd.nokia.1.7.com
dnn[29]: ipd.nokia.1.8.com
dnn[30]: ipd.nokia.sccg.scale.0.1.com
dnn[31]: ipd.nokia.sccg.scale.0.10.com
dnn[32]: ipd.nokia.sccg.scale.0.2.com
dnn[33]: ipd.nokia.sccg.scale.0.3.com
dnn[34]: ipd.nokia.sccg.scale.0.4.com
dnn[35]: ipd.nokia.sccg.scale.0.5.com
dnn[36]: ipd.nokia.sccg.scale.0.6.com
dnn[37]: ipd.nokia.sccg.scale.0.7.com
dnn[38]: ipd.nokia.sccg.scale.0.8.com
dnn[39]: ipd.nokia.sccg.scale.0.9.com
Tai list (0):
Tai Range list (0):
pgwFqdn: pgw1.nokia.com
NfServices (1)
NfService[0]:
  ServiceInstanceId: nsmf-pdusession
  ServiceName: nsmf-pdusession
  Scheme: HTTP
  ServiceStatus : Registered
  ApiVersionInUri: v1
  ApiFullVersion: 1.R15.2.0
  Ip endpoints (2):
    ip: 16.70.16.16, port 80
    ipv6: 3ffe::1046:1010, port 80
  allowedNfTypes (1):
    nfType: AMF
  priority: 10
  Allowed PLMN IDs List Elements 0
  Allowed Nssais List (0):
```

11.2.1.13 pdn-session-db

Syntax

```
pdn-session-db [imsi imsi] [attribute {unacked | del-in-prog | ho-in-prog}]
```

Context

```
tools>dump>mobile
```

Description

This command displays PDN session database information for the gateway or specified IMSI.

Parameters

imsi

specifies the IMSI, up to a maximum of 15 digits

attribute

optional keyword that can be used to specify a specific attribute for the PDN session information

unacked

keyword that specifies the UE type

del-in-prog

keyword that specifies the deletes in process

ho-in-prog

keyword that specifies the UEs that are currently in the process of doing a handover between 3GPP and non-3GPP

11.2.1.14 pfcpc-audit

Syntax

```
pfcpc-audit remote-peer ip-address | ipv6-address [router router-name]
```

Context

```
tools>dump>mobile
```

Description

This command displays the progress of the ongoing bulk audit for the paths of a specified PFCPC peer.



NOTE:

This command is only applicable when the PFCPC path-hold-timer is configured. The total number of sessions and the number of audited sessions are only displayed when the command is executed from the CP function.



NOTICE:

This command is available but not supported.

Parameters

ip-address* | *ipv6-address

IPv4 or IPv6 address of a remote PFCP peer

Values	ip-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces) x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

router-name

router-name | *service-id*, specifies the routing context

Values	router-name	Base
–	service-id	1 to 2147483647

Default: Base

Output

The following output example shows the progress of an ongoing PFCP audit.

Example output PFCP audit

```
tools dump mobile-gateway 1 pfcg-audit remote-peer 33.34.1.1 router "vprn20091"
=====
PFCP Audit
=====
UPF: VR: 8 Peer Address: 33.34.1.1 Local Address: 33.33.1.1
Session Audit: In Progress
Total number of sessions: 337  Number of audited sessions: 300
-----
No. of Entries: 1
```

11.2.1.15 pfcf-stale-session-audit-status

Syntax

```
pfcf-stale-session-audit-status
```

Context

```
tools>dump>mobile
```

Description

This command displays the progress of a PFCF stale session audit and clean-up initiated by the `pfcf-stale-session-audit-init` command. It displays the current status of in-progress or not in-progress and the number of audit sessions completed.

Output

The following output example shows PFCF stale session audit and clean-up information.

Example output PFCF stale session audit status

```
# tools dump mobile-gateway pfcf-stale-session-audit-status
=====
PFCF Stale Session Audit
=====
Stale session Audit: not in progress
Total number of session:0 Number of audited session:0
=====
```

11.2.1.16 policy-on-the-fly-update-stats

Syntax

```
policy-on-the-fly-update-stats [apn apn]
```

Context

```
tools>dump>mobile
```

Description

This command provides policy update statistics for all APNs or for the specified APN.

Parameters

apn *apn*

specifies an APN name

Output

The following output example shows policy update statistics information.

Example output policy on the fly update statistics

```
# tools dump mobile-gateway 1 policy-on-the-fly-update-stats
Policy Update Start time      049 m 09/26/21 02:52:15.612
Policy Update End time       049 m 09/26/21 02:52:16.883
Commit version:              10
Number of UE Processed:      1
Number of Pdn Processed:     1
Pdn New Session: count of sessions that did not get updated under apn which configur
ed policy-on-the-fly = new-sessions
Pdn Update Session: count of sessions that get updated under apn which configured po
licy-on-the-fly = existing-sessions
Pdn No Update Req: count of sessions that did not get updated under apn which confi
gured policy-on-the-fly = existing-sessions
Pdn Pended: count of sessions that pended the update because of sessions is busy wh
en walk through
Pdn Ignored: count of sessions that created and then deleted during policy committ
ing under apn which configured policy-on-the-fly = new-sessions
Number of Pdn Update Failed: 0
Pdn Rollback: count of sessions that can not apply the new policy (for example, th
e session is ipv4 while the policy is ipv6) and then Rollback to the old policy unde
r apn which did not configure policy-on-the-fly existing-sessions = reactivate-if-
fail
Pdn Reattach: count of sessions that can not apply the new policy (for example, th
e session is ipv4 while the policy is ipv6) and then reactivate under apn which c
onfigured policy-on-the-fly existing-sessions = reactivate-if-fail
Number of PlcyMod Delay ABS On: count of sessions that get update delayed because o
f ABS on
Processed: MG scan sessions (both in new-session and existing-
session mode) to determine which sessions should apply the new committed policy
```

11.2.1.17 radius

Syntax

radius

Context

tools>dump

Description

This command enables the context to execute RADIUS test commands.

11.2.1.18 test

Syntax

```
test
```

Context

```
tools>dump>radius
```

Description

This command enables the context to dump the RADIUS test information.

11.2.1.19 accounting

Syntax

```
accounting radius-group radius-group [detail]
```

Context

```
tools>dump>radius>test
```

Description

This command dumps the test results of the RADIUS accounting servers.

Parameters

radius-group

specifies the RADIUS accounting server group, up to 32 characters

detail

specifies to display the event history

Output

The following output example shows a RADIUS accounting server group test.

Example output RADIUS accounting server group test

```
# tools dump radius test accounting radius-group "upfRadiusGroup1" detail
=====
Dump Radius Tests : Accounting
=====
```



```

Server Type      : Both
Server Preference : lowest 1
Admin State      : Up
Server OP State  : Up
Peer Address     : 10.23.11.3
Peer Port        : 1813
Radius Group     : upfRadiusGroup1
Event History    : Up          06/16/2023 01:06:24
                  : Unknown    06/16/2023 00:40:59
                  : Down       06/16/2023 00:30:59

```

```

-----
Server Type      : Both
Server Preference : highest 3
Admin State      : Up
Server OP State  : Up
Peer Address     : 10.23.11.5
Peer Port        : 1813
Radius Group     : upfRadiusGroup1
Event History    : Up          06/16/2023 01:06:24
                  : Unknown    06/16/2023 00:40:59
                  : Down       06/16/2023 00:30:59

```

```

-----
Server Type      : Both
Server Preference : medium 2
Admin State      : Up
Server OP State  : Up
Peer Address     : 10.23.11.4
Peer Port        : 1813
Radius Group     : upfRadiusGroup1
Event History    : Up          06/16/2023 01:06:24
                  : Unknown    06/16/2023 00:40:59
                  : Down       06/16/2023 00:30:59

```

```

-----
Number of Servers : 3
=====

```

11.2.1.20 authentication

Syntax

```
authentication radius-group radius-group [detail]
```

Context

```
tools>dump>radius>test
```

Description

This command dumps the test results of the RADIUS authentication servers.

Parameters

radius-group

specifies the RADIUS authentication server group, up to 32 characters

detail

specifies to display the event history

Output

The following output example shows RADIUS authentication server group test.

Example output RADIUS authentication server group test

```
# tools dump radius test authentication radius-group "upfRadiusGroup1"
=====
Dump Radius Tests : Authentication
=====
Server Type      : Both
Server Preference : lowest 1
Admin State      : Up
Server OP State  : Up
Peer Address     : 10.23.11.3
Peer Port        : 1812
Radius Group     : upfRadiusGroup1
Event History    : Up          06/09/2023 20:42:37
-----
Server Type      : Both
Server Preference : highest 3
Admin State      : Up
Server OP State  : Up
Peer Address     : 10.23.11.5
Peer Port        : 1812
Radius Group     : upfRadiusGroup1
Event History    : Up          06/09/2023 20:42:37
-----
Server Type      : Both
Server Preference : medium 2
Admin State      : Up
Server OP State  : Up
Peer Address     : 10.23.11.4
Peer Port        : 1812
Radius Group     : upfRadiusGroup1
Event History    : Up          06/09/2023 20:42:37
-----
Number of Servers : 3
=====
```

11.2.1.21 perform

Syntax

perform

Context

tools

Description

This command enables the context to use performance measuring tools.

11.2.1.22 dhcp

Syntax

dhcp

Context

tools>perform

Description

This command enables the context to execute the DHCP performance commands.

11.2.1.23 test

Syntax

test dhcp-server-group *dhcp-group* [dhcp-server *peer*] imsi *imsi* apn *apn*

Context

tools>perform>dhcp

Description

This command tests DHCP servers.



NOTE:

To run the `tools perform dhcp test` command, the `client-identifier-msisdn-encoded` command must be disabled.

Parameters

- dhcp-group**
specifies the DHCP server group name, up to 32 characters
- peer**
specifies the IP address of a DHCP server

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces) x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

- imsi**
specifies the IMSI supplied to the DHCP server, up to 15 characters
- apn-name**
specifies the APN name supplied to the DHCP server, up to 80 characters

Output

The following output examples show the DHCP server performance.

- *Example output DHCP server performance (IPv4 - successful)*
- *Example output DHCP server performance (IPv4 - failed)*
- *Example output DHCP server performance (IPv6 - successful)*
- *Example output DHCP server performance (IPv6 - failed)*

Example output DHCP server performance (IPv4 - successful)

DHCP Discover Signaling to dhcp server 11.23.10.2 port 67 (MSCP 1:1)
DHCP response received: dhcp-ack. Round-trip time for response was 9.1 ms.
Release requested.

Example output DHCP server performance (IPv4 - failed)

```
DHCP Discover Signaling to dhcp server 11.23.10.2 port 67 (MSCP 2:1)
Negative or no response. Error 113(9).
```

Example output DHCP server performance (IPv6 - successful)

```
DHCP Solicit Signaling to dhcp server 3ffe::b17:a02 port 547 (MSCP 2:1)
DHCP response received: reply. Round-trip time for response was 9.7 ms.
Release requested.
```

Example output DHCP server performance (IPv6 - failed)

```
DHCP Solicit Signaling to dhcp server 3ffe::b17:a02 port 547 (MSCP 2:2)
Negative or no response. Error 113(22).
```

11.2.1.24 mobile-gateway

Syntax

```
mobile-gateway
```

Context

```
tools>perform
```

Description

This command enables the context to execute mobile gateway performance commands.

11.2.1.25 pdn

Syntax

```
pdn gw-id
```

Context

```
tools>perform>mobile
```

Description

This command enables the context to execute tools commands on a given mobile gateway instance.

Parameters

gw-id

specifies the gateway ID

Values: 1 to 8

11.2.1.26 apn

Syntax

```
apn apn-name
```

Context

```
tools>perform>mobile>pdn
```

Description

This command enables the context to perform actions specific to the APN.

Parameters

apn-name

specifies the APN name

11.2.1.27 dedicated-bearer-monitoring

Syntax

```
dedicated-bearer-monitoring {enable | disable | inherit}
```

```
no dedicated-bearer-monitoring
```

Context

```
tools>perform>mobile>pdn>apn
```

Description

This command enables or disables the dedicated bearer troubleshooting functionality for dumping detailed dedicated bearer information for the last 10 000 processed dedicated bearers at the APN level.

The no form of the command reverts to the default value (*inherit*)

Default

inherit

Parameters

enable

enables the functionality for dumping detailed dedicated bearer information

disable

disables the functionality for dumping detailed dedicated bearer information

inherit

inherits the setting for the APN from PDN level in the following context:

```
tools>perform>mobile>pdn>dedicated-bearer-monitoring>shutdown
```

11.2.1.28 canary-test

Syntax

canary-test

Context

tools>perform>mobile>pdn

Description

This command enables the context to perform canary testing.

11.2.1.29 user-range

Syntax

user-range *user-range-name* target *targets*

Context

tools>perform>mobile>pdn>canary-test

Description

This command configures a number of subscribers for canary testing in a specified user range.

Parameters

user-range-name

specifies the user range name for the canary testing, up to 32 characters

targets

specifies the maximum number of configured subscribers for the canary testing, up to 32 characters

Values: 0 to 4294967295

11.2.1.30 dedicated-bearer-monitoring

Syntax

```
dedicated-bearer-monitoring
```

Context

```
tools>perform>mobile>pdn
```

Description

This command enables the context to perform actions for the dedicated bearer monitoring functionality on the gateway level.

11.2.1.31 imsi

Syntax

```
[no] imsi imsi
```

Context

```
tools>perform>mobile>pdn>dedicated-bearer-monitoring
```

Description

This command specifies the IMSIs for which the extra bearer data is not overwritten but kept persistently. An IMSI can be configured before the UE is attached. When the UE is detached and set to be persistent, the data related to the UE is persistently kept. If the UE is re-attached, new data is persistently added to the memory .

The no form of this command removes the specified IMSI from this functionality.

Parameters

imsi

specifies the IMSI, up to 15 characters

11.2.1.32 shutdown

Syntax

[no] shutdown

Context

tools>perform>mobile>pdn>dedicated-bearer-monitoring

Description

This command enables or disables the dedicated bearer troubleshooting functionality for dumping the detailed dedicated bearer information for the last 10 000 processed dedicated bearers. The APN level command overrides the PDN level command.

11.2.1.33 force-clear

Syntax

force-clear

Context

tools>perform>mobile>pdn

Description

This command enables the context to force execution of clear commands during troubleshooting.



NOTICE:

Commands in the tools>perform>force-clear context must only be used when necessary and as instructed by Nokia Technical Support.

11.2.1.34 bearer-context

Syntax

```
bearer-context {imsi | imei | msisdn} string [local-only] [apn apn] [slave]
```

Context

```
tools>perform>mobile>pdn>force-clear
```

Description

This command forcefully clears the bearer context for a specific UE. All sessions of the UE at the PDN level are cleared when the APN option is not used. A rate limiting is applied to the command, by only accepting 16 commands in parallel and only allowing one command per second to proceed internally for execution.

**NOTICE:**

This command performs an exhaustive search of the UE identity on all the cards/VMs of the mobile gateway and should only be used if necessary and as instructed by Nokia Technical Support.

Parameters

string

specifies the value of the address string for the UE key (IMEI, IMSI, or MSISDN) used to locate the bearer context, up to 15 digits for IMSI or MSISDN or 16 digits for IMEI

apn

specifies the name of the APN; when used, only the UE sessions on this APN are cleared.

**NOTE:**

Virtual APNs are not supported in this command, use only real APNs.

local-only

specifies an internal cleanup

slave

specifies that the clearing is needed on the slave in an ICR deployment

11.2.1.35 diameter-peer-destination-host

Syntax

```
diameter-peer-destination-host ref-point-type {s6b | gx | gy} group group-number  
peer {ipv4 | ipv6} [port port-number] destination-host destination-host realm realm-name
```

Context

tools>perform>mobile>pdn>force-clear

Description

This command forces removal of entries from the Destination-Host table for Gx, Gy, and S6b interfaces when the entries are no longer required, instead of waiting for them to automatically age out.



NOTE:

- Aging out of entries in the Destination Host table occurs automatically if the entries have been inactive (no sessions) for more than five minutes (this value is not configurable).
- For locally configured peers, after establishing the connection, the respective related entries are not aging out and removed from the table, they are cleared only when the peer is shutdown.



NOTICE:

This command must only be used when necessary and as instructed by Nokia Technical Support.

Parameters

ref-point-type

specifies the reference point type for which entries will be cleared

Values: s6b | gx | gy

group-number

specifies the configured group number for the reference point interface

Values: 1 to 15

ipv4 | ipv6

specifies the IPv4 or IPv6 IP address of the destination host server

Values	ipv4-address	a.b.c.d
--------	--------------	---------

–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces) x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

port-number

specifies the port number used by the destination host server

Values: 1 to 65535

Default: 3868

destination-host

specifies the name of the destination host where entries will be deleted, up to 255 characters

realm-name

specifies the name of the realm to which the destination host belongs, up to 255 characters

11.2.1.36 path**Syntax**

```
path router router-name [local-peer ipv4 | ipv6] [remote-peer ipv4 | ipv6] protocol-type
{gtp | pmip | pfcf}
```

Context

```
tools>perform>mobile>pdn>force-clear
```

Description

This command can be used to force removal of path record entries in the mobile gateway path-management databases, instead of waiting for them to automatically age out. The purpose is to release a path that is stuck as a result of inconsistencies between the mobile gateway path-management databases. As part of the path record deletion, sessions associated with the path are released, and no new requests (for example, Echo Req, CS Req, MB Req, and so on) are accepted from that path until it is deleted in all databases.

**NOTE:**

When you remove the path record entries using this command, explicit session deletion messages for the sessions on this peer/reference point are not sent to inform the remote-end of the peer/reference point about the session deletions. This allows the peer to restore the sessions if required, as per TS23.007. Path record entries for GTP and PMIP automatically age out in all databases if the path is inactive (no sessions) for more than 24 hours (this value is not configurable). During the deletion process, new requests are not accepted until the entries are deleted in all databases. Any requests that may be dropped during the brief period required to delete the entries are processed in a second retry. The path record aging out functionality does not apply for PFCP. This tools command and the path record entry aging out functionality only affect peers for the control plane function; user plane peers and S2a-PGW/S2b-PGW peers (TWAG/ePDG) are not impacted.



NOTICE:

This command must only be used when necessary and as instructed by Nokia Technical Support.

Parameters

router-name

specifies the router name or service ID

Values	router-name	base
–	service-id	1 to 2147483647

Default: base

ipv4 | ipv6

specifies the IPv4 or IPv6 IP address of the local and remote peers



NOTE:

The local peer is optional for PMIP.

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces) x:x:x:x:x:d.d.d.d

–	–	x - [0..FFFF]H
–	–	d - [0..255]D

protocol-type

specifies the protocol type of the path record entry to be deleted

Values: gtp | pmip | pfcf

11.2.1.37 force-gtpu-echo-req-action

Syntax

```
force-gtpu-echo-req-action {drop | forward}
```

Context

```
tools>perform>mobile>pdn
```

Description

This command forcefully sets the drop status of the GTP-U Echo Requests to true, or clears the drop status from the GTP-U Echo Requests.

**NOTICE:**

This command must only be used when necessary and as instructed by Nokia Technical Support.

Parameters**drop**

forcefully sets the drop status of the GTP-U Echo Requests to true

forward

forcefully clears the drop status from the GTP-U Echo Requests

11.2.1.38 nf-register

Syntax

```
nf-register [bsf]
```

Context

tools>perform>mobile>pdn

Description

This command triggers an NF registration message toward the NRF, as SMF, UPF, or BSF. If the `bsf` keyword is not specified, registration is performed for SMF or UPF.

Parameters

bsf

specifies to trigger registration of BSF

11.2.1.39 pfcf-stale-session-audit-cancel

Syntax

pfcf-stale-session-audit-cancel

Context

tools>perform>mobile>pdn

Description

This command cancels an ongoing manual audit and cleanup of stale sessions on the CP function or the UP function that was initiated using the `pfcf-stale-session-audit-init` command.

11.2.1.40 pfcf-stale-session-audit-init

Syntax

pfcf-stale-session-audit-init [cleanup-on-no-resp]

Context

tools>perform>mobile>pdn

Description

This command performs a manual audit and cleanup of any stale sessions on the CP function or the UP function.

**NOTICE:**

- Run this command only during a maintenance window.
- Do not run this command when a `clear` command is in progress or vice versa; attempting to do so causes an alarm notification to be generated.

To monitor the cleanup in process, use the `tools dump mobile gateway pfc-p-stale-session-audit-status` command.

To cancel an ongoing audit, use the `pfc-p-stale-session-audit-cancel` command.

**NOTE:**

The specific audit related PFCP Session Modification Request/Response and the PFCP Session Report Request/Response messages that are issued due to the `pfc-p-stale-session-audit-init` command only affect the "PFCP session audit related messages" counters of the `ref-point-stats sx-n4` command in the `show>mobile>pdn` context.

Parameters

`cleanup-on-no-resp`

optional keyword that triggers cleanup of no responsive sessions on the UPF side

11.2.1.41 `profile`

Syntax

`[no] profile profile-id`

Context

`tools>perform>mobile`

Description

This command enables the context to execute profile performance management commands.

Parameters

profile-id

specifies a profile number identifier

Values: 0 to 255

11.2.1.42 diameter-peer

Syntax

diameter-peer *profile-name*

Context

tools>perform>mobile>profile

Description

This command enables performance tools for the Diameter peer profile.

Parameters

profile-name

specifies a profile name

11.2.1.43 refresh-dns

Syntax

refresh-dns

Context

tools>perform>mobile>profile>diameter-peer

Description

This command executes a manual refresh of the specified Diameter peer.

11.2.1.44 fqdn-group

Syntax

fqdn-group *profile-name* [router *vprn-id*]

Context

tools>perform>mobile>profile

Description

This command enables the context to use performance tools for the specified FQDN group.

Parameters

profile-name

specifies the name of the FQDN group

Values: Alphanumeric string of 1 to 32 characters

router

keyword to specify a VPRN instance for the router

vprn-id

specifies the VPRN ID

Values: 1 to 2147483690 | svc-name: 64 characters maximum

11.2.1.45 clear-list

Syntax

```
clear-list
```

Context

```
tools>perform>mobile>profile>fqdn-group
```

Description

This command clears the list of resolved IP addresses for the FQDN group on the specified VPRN.

11.2.1.46 refresh-list

Syntax

```
refresh-list
```

Context

```
tools>perform>mobile>profile>fqdn-group
```

Description

This command is used to perform a manual DNS refresh, for the specified FQDN group. When specified, the list of URLs is refreshed by sending a DNS query. The DNS-refresh-interval, if enabled, applies for the manual refresh.

11.2.1.47 pcscf-group

Syntax

```
pcscf-group pcscf-group
```

Context

```
tools>perform>mobile>profile
```

Description

This command enables the context to access tools for the specified P-CSCF server group.

Parameters

pcscf-group

name of the P-CSCF server group

refresh-list

optional keyword to manually refresh the DNS for the pcscf-group

11.2.1.48 refresh-list

Syntax

```
refresh-list
```

Context

```
tools>perform>mobile>profile>pcscf-group
```

Description

This command is used to perform a manual DNS refresh, for the specified P-CSCF group.

11.2.1.49 radius

Syntax

```
radius
```

Context

```
tools>perform
```

Description

This command enables the context to execute the RADIUS performance commands.

11.2.1.50 test

Syntax

```
test
```

Context

```
tools>perform>radius
```

Description

This command enables the context to test the RADIUS performance.

11.2.1.51 accounting

Syntax

```
accounting {on | off} radius-group radius-group [radius-server peer] [apn apn-name]
```

```
accounting start radius-group radius-group [radius-server peer] [user-name user-name]  
apn apn-name
```

Context

```
tools>perform>radius>test
```

Description

This command tests the network connectivity and application availability of either one specific RADIUS accounting server or of all the configured RADIUS accounting servers. The outputs show the accounting on

or off messages (command `accounting {on | off}`) and the accounting request start and accounting request stop pair (command `accounting start`).

Parameters

on | off

specifies the RADIUS accounting ON or accounting OFF messages

radius-group

specifies the RADIUS accounting server group, up to 32 characters

peer

specifies the IP address of a RADIUS accounting server

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces) x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

apn-name

specifies the APN name supplied to the RADIUS accounting server, up to 80 characters

user-name

specifies the username supplied to the RADIUS accounting server, up to 32 characters

Output

The following output examples show the RADIUS accounting servers performance.

- *Example output RADIUS accounting server ON performance*
- *Example output RADIUS accounting server OFF performance*
- *Example output RADIUS accounting server START performance*

Example output RADIUS accounting server ON performance

```
RADIUS Acct-on to accounting server 10.23.11.3 port 1813
Accounting response received: Success. Round-trip time for response was 3.4 ms.
RADIUS Acct-on to accounting server 10.23.11.5 port 1813
Accounting response received: Success. Round-trip time for response was 3.4 ms.
```

```
RADIUS Acct-on to accounting server 10.23.11.4 port 1813
Accounting response received: Success. Round-trip time for response was 3.5 ms.
```

Example output RADIUS accounting server OFF performance

```
RADIUS Acct-off to accounting server 10.23.11.3 port 1813
Accounting response received: Success. Round-trip time for response was 3.0 ms.
RADIUS Acct-off to accounting server 10.23.11.5 port 1813
Accounting response received: Success. Round-trip time for response was 3.0 ms.
RADIUS Acct-off to accounting server 10.23.11.4 port 1813
Accounting response received: Success. Round-trip time for response was 3.0 ms.
```

Example output RADIUS accounting server START performance

```
RADIUS Acct-start to accounting server 10.23.11.5 port 1813 (MSCP 1:4)
Accounting response received: Success. Round-trip time for response was 9.8 ms.
RADIUS Acct-stop to accounting server 10.23.11.5 port 1813 (MSCP 1:4)
Accounting response received: Success. Round-trip time for response was 9.9 ms.
```

11.2.1.52 authentication

Syntax

```
authentication radius-group radius-group [radius-server peer] [user-name user-name]
[password password] [apn apn-name]
```

Context

```
tools>dump>radius>test
```

Description

This command tests the network connectivity and application availability of either one specific RADIUS authentication server or of all the configured RADIUS authentication servers. The output generates the authentication messages (Access-Request) for specific user-name.

Parameters

radius-group

specifies the RADIUS authentication server group, up to 32 characters

peer

specifies the IP address of a RADIUS authentication server

Values	ipv4-address	a.b.c.d
--------	--------------	---------

–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces) x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

user-name

specifies the username sent to the RADIUS authentication server, up to 32 characters

password

specifies the password associated with the username that is supplied to the RADIUS authentication server, up to 32 characters

apn-name

specifies the APN name sent to the RADIUS authentication server, up to 80 characters

Output

The following output example shows the RADIUS authentication server performance.

Example output RADIUS authentication server performance

```
Authentication to authentication server 10.23.11.5 port 1812 (MSCP 1:3)
Authentication response received: Access-Accept. Round-trip time for response was 9.9 ms
```

11.2.1.53 router**Syntax**

```
router
```

Context

```
tools>perform
```

Description

This command enables the context to configure router-specific tools.

11.2.1.54 ip-local-pool

Syntax

```
ip-local-pool
```

Context

```
tools>perform>router
```

Description

This command enables the context to configure IP address pool-specific tools.

11.2.1.55 reserve-ipv4

Syntax

```
reserve-ipv4 ipv4-address
```

```
no reserve-ipv4
```

Context

```
tools>perform>router>ip-local-pool
```

Description

This command reserves an IPv4 address from the local IP address pool to be used as the IP address loopback for ping commands. Only one IPv4 address can be reserved as the loopback IP address.

The loopback interface must be separately configured and assigned with the reserved IP address. The IP address must belong to some of the configured prefixes of the IP local pool. When an IP address is reserved, it cannot be allocated to a UE and the other way around. When an IP address has been already assigned already to a UE, it cannot be reserved.

The no form of this command releases the reserved IP address back to the IP address pool.

Default

```
no reserve-ipv4
```

Parameters

ipv4-address

specifies a valid IPv4 address that belongs to this IP address pool

Values: a.b.c.d

11.2.1.56 **reserve-ipv6**

Syntax

```
reserve-ipv6 ipv6-address  
no reserve-ipv6
```

Context

```
tools>perform>router>ip-local-pool
```

Description

This command reserves an IPv6 address from the local IP address pool to be used as the IP address loopback for ping commands. Only one IPv6 address can be reserved as the loopback IP address.

The loopback interface must be separately configured and be assigned with the reserved IP address. The IP address must belong to some of the configured prefixes of the IP local pool. When an IP address is reserved, it cannot be allocated to a UE and the other way around. When an IP address has been already assigned to a UE, it cannot be reserved.



NOTE:

Only a 64-bit prefix can be reserved.

The no form of this command releases the reserved IP address back to the IP address pool.

Default

```
no reserve-ipv6
```

Parameters

ipv6-address

specifies a valid IPv6/64 prefix belonging to this IP pool

11.2.1.57 **security**

Syntax

```
security
```

Context

tools>perform

Description

This command enables the context to perform security checks.

11.2.1.58 authentication-server-check

Syntax

authentication-server-check server-address *ip-address* [port *port*] {{user-name *user-name* password *password*} | attr-from-file *file-url*} secret *key* [source-address *ip-address*] [timeout *seconds*] [router *router-instance* | service-name *service-name*] [debug]

Context

tools>perform>security

Description

This command performs an authentication check on the RADIUS server.

Parameters

server-address *ip-address*
specifies the IP address of the RADIUS server

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces) x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

port
specifies the port of the RADIUS server

Values: 1 to 65535

user-name
specifies the user name, up to 253 characters

password
specifies the user password, up to 64 characters

file-url
specifies to attach additional VSAs from a file

Values	local-url remote-url	
–	local-url	[cflash-id/] [file-path] 200 characters maximum, including cflash-id directory length, up to 99 characters each
–	remote-url	[{ftp:// tftp://} login:pswd@remote-locn/] [file-path]
–	remote-locn	[hostname ipv4-address ipv6-address]
–	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces) x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D
–	interface	up to 32 characters, for link local addresses
–	cflash-id	cf1: cf1-A: cf1-B: cf2: cf2-A: cf2-B: cf3: cf3-A: cf3-B:

key
specifies the secret key shared between the RADIUS client and server, up to 64 characters

source-address ip-address

specifies the source IP address used for the Access-Request message

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

seconds

specifies the time to wait for the RADIUS server response

Values: 1 to 90

router-instance

specifies the router instance that the Access-Request message uses

Values	router-name vprn-svc-id	
–	router-name	Base management
–	vprn-svc-id	1 to 2147483647

Default: Base

service-name

specifies the service name that the Access-Request message uses, up to 64 characters

debug

specifies to display the full debug message, including all VSAs and the full hexadecimal string of the RADIUS reply message

11.2.2 OAM command descriptions

11.2.2.1 oam

Syntax

oam

Context

root

Description

This command enables the context to use OAM commands.

11.2.2.2 dns

Syntax

```
dns target-addr dns-name name-server ip-address [[source ip-address] | [interface  
interface-name]] [router-instance router-instance] [count send-count] [timeout timeout]  
[interval interval] [record-type {ipv4-a-record | ipv6-aaaa-record | naptr}] [service-  
label service-label-string] [replacement replacement-string] [detail]
```

Context

oam

Description

This command performs a DNS name resolution. When you select the `ipv4-a-record` option, DNS names are queried for A-records only. Also, the NAPTR record type is supported.

Parameters

dns-name

specifies the DNS name to be resolved to an IP address, up to 255 characters

ip-address

specifies the IP address of the primary DNS server

Values	ipv4-address	a.b.c.d
--------	--------------	---------

–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces) x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

source *ip-address*

specifies the IP address to be used as the source IP address for performing an OAM operation

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces) x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

interface-name

specifies an interface name that is used to perform the DNS query. If no interface name is specified, the system uses the default interface of the defined router. The interface name must begin with a letter, and be up to 32 characters in length.

router-instance

specifies the router name or service name or service ID

Values	router-name vprn-svc-name vprn-svc-id	
–	router-name	Base _Base_and_management management
–	vprn-svc-name	up to 64 characters

– vprn-svc-id 1 to 2147483647

Default: _Base_and_management

send-count

specifies the number of messages to send, expressed as a decimal integer. The send-count parameter is used to override the default number of message requests sent. Each message request must either time out or receive a reply before the next message request is sent. The message interval value must expire before the next message request is sent.

Values: 1 to 100

Default: 1

timeout

specifies the timeout parameter in seconds, expressed as a decimal integer. This value is used to override the default timeout value, and it is the amount of time that the router waits for a message reply after sending the message request. When the message timeout expires, the requesting router assumes that the message response is not received. Any response received after the request times out, is silently discarded.

Values: 1 to 120

Default: 5

interval

specifies the interval parameter in seconds, expressed as a decimal integer. This parameter is used to override the default request message send interval and defines the minimum amount of time that must expire before the next message request is sent.

If the interval is set to 1 second, and the timeout value is set to 10 seconds, the maximum time between message requests is 10 seconds and the minimum is 1 second. This depends on the receipt of a message reply that corresponds to the outstanding message request.

Values: 1 to 10

Default: 1

record-type

specifies the record type

Values:

ipv4-a-record

a specific record that maps a host name to an IPv4 address

ipv6-aaaa-record a record specific to the Internet class that stores a single IPv6 address

naptr record type of the DNS query

service-label-string

specifies the filtering of the printout of the returned DNS answers. These answers are based on the `naptr` option that matches to the entered service label. The parameter must be up to ten characters in length. The service label option is applicable only when the `naptr` record type is selected.

replacement-string

specifies the filtering of the printout of the DNS returned answers where a part or the whole of the `naptr` replacement string matches the entered string. The parameter must be up to 255 characters in length.

detail

specifies more details. The parameter displays either all the resolved IP addresses for a specified FQDN when `record-type` is set to `ipv6-aaaa-record`, or a detailed output of the returned DNS results when `record-type` is set to `naptr`.

11.2.2.3 gtp-ping

Syntax

```
gtp-ping {s11 | s5c | s1u | s2b | s5u | s8c | s8u | gnc | gpc | s2bc | s2bu | s4c | s4u | s12 | s2ac | s2au | n3 | n9} [router router-instance] [source ip-address] destination ip-address port port-number gateway gw-id [retry-count count] [time-out seconds]
```

Context

oam

Description

This command tests the connectivity with GTP peers. It generates and transmits GTP echo requests according to the specified reference point toward the specified destination address. The GTP echo requests can be transmitted even if no bearers exist between the two peers.



NOTE:

The destination port numbers for GTP-C (2123) and GTP-U (2152) are fixed.

Parameters

s11 | s5c | s1u | s2b | s5u | s8c | s8u | gnc | gpc | s2bc | s2bu | s4c | s4u | s12 | s2ac | s2au | n3 | n9

specifies the reference point

gw-id

specifies the gateway ID

ip-address

specifies the IP address to be used as the source IP

Values		
	ipv4-address	a.b.c.d
–	ipv6-address	x::x::x::x::x::x (eight 16-bit pieces) x::x::x::x::d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

count

specifies the number of packets transmitted

Values: 1 to 10

Default: 3

seconds

specifies the timeout, in seconds, before sending the next packet or exiting

Values: 1 to 10

Default: 1

router-instance

specifies the router instance VPRN ID

Values	router-name	Base
–	service-id	1 to 2147483647

port-number
specifies the port number for the destination IP address
Values: 1 to 65535

destination ip-address
specifies the IP address of the remote peer

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces) x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

Output
The following output example shows GTP ping information.
Example output successful GTP ping

```
# oam gtp-ping s11 gateway 1 destination 7.50.4.13 port 2123 router 1
-----
Ping Successful.
-----
```

11.2.2.4 radius-ping

Syntax
radius-ping *apn* [*tos type-of-service*] source *ip-address* [*size bytes*] [*count requests*] router
router-or-service destination *ip-address* [*gateway gw-id*]

Context
oam

Description
This command pings a RADIUS peer IP address.

Parameters

apn

specifies the tunnel APN configured on the respective RADIUS group, up to 80 characters

type-of-service

specifies the type of service, bits in the IP header

Values: 0 to 255

Default: 184

ip-address

specifies the IP address of the source interface from where the ICMP packet is initiated

Values: a.b.c.d

bytes

specifies the request packet size including the ICMP header and payload

Values: 0 to 16384

Default: 56

requests

specifies the number of ping requests to send to the remote RADIUS peer

Values: 1 to 100000

Default: 5

router-or-service

specifies the router or service

Values	router-name vprn-svc-id	
–	router-name	Base management cpm-vr-name vpls-management
–	vprn-svc-id	1 to 2147483647
–	cpm-vr-name	up to 32 characters

Default: router-name - Base

destination ip-address

specifies the IP address to use as the source IP

Values: a.b.c.d

gw-id

specifies the gateway ID

Values: 1 to 8

Output

The following output example shows the GTP ping information.

Example output RADIUS ping

```
# oam radius gateway 1
ping ims.mnc480.mcc311.gprs destination 203.0.0.115 source 10.10.50.2 router Vprn200
05
PING 203.0.0.115 56 data bytes
64 bytes from 203.0.0.115: icmp_seq=1 ttl=64 time=0.114ms.
64 bytes from 203.0.0.115: icmp_seq=2 ttl=64 time=0.123ms.
64 bytes from 203.0.0.115: icmp_seq=3 ttl=64 time=0.111ms.
64 bytes from 203.0.0.115: icmp_seq=4 ttl=64 time=0.108ms.
64 bytes from 203.0.0.115: icmp_seq=5 ttl=64 time=0.227ms.

---- 203.0.0.115 PING Statistics ----
5 packets transmitted, 5 packets received, 0.00% packet loss
round-trip min = 0.108ms, avg = 0.136ms, max = 0.227ms, stddev = 0.045ms
```

11.2.2.5 ue-ping

Syntax

ue-ping {imsi | msisdn | ue-ip} value apn apn source-ip ip-address [size bytes] [count requests] [timeout timeout] gateway gw-id

Context

oam

Description

This command initiates a ping from the PGW or GGSN toward a UE and presents statistics from the replies received. The UE is specified by the IMSI, MSISDN, or IP address. The UE PDN session that is used to convey the ping packet (default bearer) is the one that matches the IP address type given as the source IP address for the given APN. If the UE has more than one session of the same IP type in the same APN, the UE's IP address must be used as the key in the command.

**NOTE:**

PMIP is not supported for UE ping.

Parameters

value

specifies the IMSI, MSISDN, or IP address of the target UE to ping

Values	IMSI MSISDN	up to 15 digits
–	ipv4-address	a.b.c.d
–	ipv6-address	x::x::x::x::x::x (eight 16-bit pieces) x::x::x::x::d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

apn

specifies the APN to which the UE is connected, up to 80 characters. The APN specifies the PDN connection and the default bearer that will be used for conveying the ping packets.

ip-address

specifies the source IPv4 or IPv6 address of the ICMP Echo Request packet

Values	ipv4-address	a.b.c.d
–	ipv6-address	x::x::x::x::x::x (eight 16-bit pieces) x::x::x::x::d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

bytes

specifies the number of data bytes to be sent. The default is 56, which translates into 64 ICMP data bytes when combined with the 8 bytes of the ICMP header data. The maximum value is 8184, which translates to 8192 (8 kb) with the ICMP header.

Values: 0 to 8184

Default: 56

requests

specifies the number of ping requests to send to the UE. Each request is sent right after the reply for the previous request is received or after the previous request times out.

Values: 1 to 100

Default: 5

timeout

specifies the time, in seconds, to wait for each ICMP Echo Reply before the timeout is hit. The maximum timeout for a single ping command is 30 seconds.

Values: 1 to 10

Default: 1

gw-id

specifies the gateway ID

Values: 1 to 8

Output

Example output UE ping information

```
# oam ue-ping imsi XXXXX apn ipd.nokia.com source-ip A.B.C.D gateway 1
PING UE.IP.xx.xx4 (IMSI: XXXXX, APN: ipd.nokia.com) from A.B.C.D : 56 data bytes of
data
64 bytes from UE.IP.xx.xx4 via fei0: icmp_seq=0 ttl=64 time=0.000 ms.
64 bytes from UE.IP.xx.xx4 via fei0: icmp_seq=1 ttl=64 time=0.000 ms.
64 bytes from UE.IP.xx.xx4 via fei0: icmp_seq=2 ttl=64 time=0.000 ms.
64 bytes from UE.IP.xx.xx4 via fei0: icmp_seq=3 ttl=64 time=0.000 ms.
64 bytes from UE.IP.xx.xx4 via fei0: icmp_seq=4 ttl=64 time=0.000 ms.

---- UE.IP.xx.xx4 PING Statistics ----
5 packets transmitted, 5 packets received, 0.00% packet loss
round-trip min/avg/max/stddev = 0.000/0.000/0.000/0.000 ms
```

11.2.3 Ping command descriptions

11.2.3.1 ping

Syntax

ping [*ip-address* | *dns-name*] {next-hop *ip-address*} [interface *interface-name*] bypass-routing] [timeout-type *timeout-type*] [*ip-address*] subscriber-id [*sub-ident-string*] [count

requests] [*detail* | *rapid*] [*do-not-fragment*] [*fc fc-name*] [*interval centisecs* | *secs*]
[*pattern pattern*] [{*router router-or-service*} | {*router-instance router-instance*} | {*service-*
name service-name}] [*size bytes*] [*source ip-address*] [*timeout timeout*] [*tos type-of-service*]
[*tll time-to-live*]

Context

<global>

Description

This command is the TCP/IP utility that is used to verify IP address reachability.

Parameters

ip-address | *dns-name*

Specifies the remote host to ping. The IP address or the DNS name (if DNS name resolution is configured) can be specified.

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces) x:x:x:x:x:d.d.d.d
–	–	x - [0..FFFF]H
–	–	d - [0..255]D

ip-address

Specifies to disregard the routing table and sends this packet to the specified next hop address. This address must be on an adjacent router that is attached to a subnet that is common between this and the next-hop router.

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces) x:x:x:x:x:d.d.d.d

-	-	x - [0..FFFF]H
-	-	d - [0..255]D

Default: Per the routing table

interface-name

specifies the interface name, up to 32 characters

bypass-routing

Specifies to send the ping request to a host on a directly attached network bypassing the routing table. The host must be on a directly attached network or an error is returned.

timeout-type

specifies the timeout type

Values: per-probe | per-test-run

per - probe

The provided timeout value is treated as a timeout per probe (ICMP Echo Request). This is the default timeout type for on-demand ping executions.

per - test - run

The provided timeout value is treated as a timeout for the test (command execution), regardless of how many ICMP Echo packets have been sent or received. With per - test - run timeout type, the ping does not stop after count packet are sent, it waits either for timeout to expire, or until count probes are answered or for some error notification from the network.

Default: per-probe

sub-ident-string

specifies an existing subscriber ID

requests

specifies the number of ping requests to send to the remote host, expressed as a decimal integer

Values: 1 to 10000

Default: 5

detail | rapid

Specifies to send ping requests rapidly. The results are reported in a single message, not in individual messages for each ping request. By default, five ping requests are sent before the results are reported. To change the number of requests, include the count option.

The detail parameter includes in the output the interface on which the ping reply was received.

Example output:

```
# ping 192.168.25.114 detail
PING 192.168.25.114: 56 data bytes
64 bytes from 192.168.25.114 via fei0: icmp_seq=0 ttl=64 time=0.000 ms.
64 bytes from 192.168.25.114 via fei0: icmp_seq=1 ttl=64 time=0.000 ms.
64 bytes from 192.168.25.114 via fei0: icmp_seq=2 ttl=64 time=0.000 ms.
64 bytes from 192.168.25.114 via fei0: icmp_seq=3 ttl=64 time=0.000 ms.
64 bytes from 192.168.25.xx1144 via fei0: icmp_seq=4 ttl=64 time=0.000 ms.

---- 192.168.xx.xx4 PING Statistics ----
5 packets transmitted, 5 packets received, 0.00% packet loss
round-trip min/avg/max/stddev = 0.000/0.000/0.000/0.000 ms
```

do-not-fragment

Specifies that the request frame must not be fragmented. This option is particularly useful in combination with the size parameter for maximum MTU determination (does not apply to ICMPv6).

fc-name

specifies the forwarding class

Values: be | l2 | af | l1 | h2 | ef | h1 | nc

Default: nc

centisecs | secs

specifies the interval between consecutive ping requests, expressed as a decimal integer

Values: 1 to 10000 seconds; if is selected, 1 to 10000 centiseconds

Default: 1 second; if rapid is selected, 1 centisecond

pattern

specifies a 16-bit pattern string to include in the ping packet, expressed as a decimal integer

Values: 0 to 65535

Default: system-generated sequential pattern

router-or-service

specifies the router name or service ID

Values <router-name> | <vprn-svc-id>

–	router-name	Base, management, vpls-management
–	Default: Base	–
–	vprn-svc-id	1 to 2147483647
–	Router-instance is the preferred parameter for specifying the router or service	

router-instance

specifies the router name, CPM router instance, or service ID

Values	router-name	“Base” “management”
–	cpm-vr-name	up to 32 characters
–	service-id	1 to 2147483647

Default: Base

service-name

specifies the service name, up to 64 characters

bytes

specifies the size, in bytes, of the ping request packets

Values: 0 to 16384

Default: 56 bytes (actually 64 bytes because 8 bytes of ICMP header data are added to the packet)

source ip-address

specifies the source IP address to use in the ping requests, in dotted decimal notation

Values	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)

x:x:x:x:x:d.d.d.d

– – x - [0..FFFF]H

– – d - [0..255]D

Default: The IP address of the egress IP interface.

timeout

specifies the timeout in seconds

Values 1 to 10, when timeout -type is set to per -probe

– 1 to 60, when timeout -type is set to per -test -run

Default: 5

type-of-service

specifies the TOS bits in the IP address header of the ping packets, expressed as a decimal integer

Values: 0 to 255

Default: 0

time-to-live

specifies the IP address TTL value to include in the ping request, expressed as a decimal integer

Values: 0 to 128

Default: 64

12 Debug command reference

This section describes the debug-related commands.

12.1 Debug command hierarchies

- *Call trace debug command hierarchy*
- *Profile debug command hierarchy*



NOTICE:

Debug commands must be used with caution. Nokia recommends disabling debugging after any debug operation has been finished. The `no debug` command disables all the enabled debug commands.

12.1.1 Call trace debug command hierarchy

```
debug
- mobile-gateway
- call-insight
  - all-ues profile profile-name
  - no all-ues
  - ue {imei | imsi | msisdn} value [profile profile-name]
  - no ue {imei | imsi | msisdn} value
  - no ue all
  - no ue mac-address mac-address
  - no ue {imei | imsi | msisdn} mask pattern
  - no ue mask-name mask-name
  - no ue-mask all
  - ue mac-address mac-address [profile profile-name]
  - ue-mask {imei | imsi | msisdn} pattern [name mask-name] [profile profile-name]
[max-ues num]
  - no ue-mask {imei | imsi | msisdn} pattern
  - no ue-mask name mask-name
  - ue {imei | imsi | msisdn} value reset-user-traffic-stats
  - ue mac-address mac-address reset-user-traffic-stats
```

12.1.2 Profile debug command hierarchy

```
debug
  - mobile-gateway
    - profile
      - emergency-db-errors profile-name [csv-file-name csv-file-name]
```

12.2 Debug command descriptions



NOTICE:

Debug commands must be used with caution. Nokia recommends disabling debugging after any debug operation has been finished. The `no debug` command disables all the enabled debug commands.

12.2.1 Debug commands

- *Call trace debug command descriptions*
- *Profile debug command descriptions*

12.2.2 Call trace debug command descriptions

12.2.2.1 call-insight

Syntax

```
call-insight
```

Context

```
debug>mobile
```

Description

This command enables the context for debugging call trace.

12.2.2.2 all-ues

Syntax

```
all-ues profile profile-name  
no all-ues
```

Context

```
debug>mobile>call-insight
```

Description

This command enables the call-insight debugging for all UEs .



NOTE:

Nokia does not recommend to use this command in live deployments with more than 50 UEs because it may cause performance degradation.

Parameters

profile-name

specifies an existing profile name. Allowed values are any string up to 32 characters long composed of printable, 7-bit ASCII characters. If the string contains special characters (#, \$, spaces, and so on), the entire string must be enclosed within double quotes.

12.2.2.3 ue

Syntax

```
ue {imei | imsi | msisdn} value [profile profile-name]  
no ue {imei | imsi | msisdn} value  
no ue all  
no ue mac-address mac-address  
no ue {imei | imsi | msisdn} mask pattern  
no ue mask-name mask-name  
ue mac-address mac-address [profile profile-name]  
ue {imei | imsi | msisdn} value reset-user-traffic-stats  
ue mac-address mac-address reset-user-traffic-stats
```

Context

debug>mobile>call-insight

Description

This command enables adding and removing of specified call trace jobs.

For the `ue mac-address` command, the command enables the packet capture based on the UE MAC-address. This is a configuration option for the TWAG.

Parameters

all

removes all call trace jobs

imei

adds or removes an imei call trace job, up to 16 digits

imsi

adds or removes an imsi call trace job. up to 15 digits

msisdn

adds or removes an msisdn call trace job, up to 15 digits

mask-name

removes call trace jobs matching a specified mask name. The mask name is up to 16 characters. The default mask name is "default".

mac-address

specifies the UE MAC-address that the packet capture is based on

Values: xx:xx:xx:xx:xx:xx or xx-xx-xx-xx-xx-xx (where x is an hexadecimal digit)

pattern

removes call trace jobs matching a unique mask type and pattern. The pattern is up to 16 characters long composed of digits 0-9, '*' and '.'.

profile-name

specifies an existing profile name. Allowed values are any string up to 32 characters long composed of printable, 7-bit ASCII characters. If the string contains special characters (#, \$, spaces, and so on), the entire string must be enclosed within double quotes.

value

removes call trace jobs matching a specified IMEI, IMSI, or MSISDN value

reset-user-traffic-stats

resets user plane statistics, closes running PCAP file and opens a new PCAP file for a specific UE

**NOTE:**

The parameters of the following commands are not supported with metadata for call trace:

- `no ue {imei | imsi | msisdn} mask pattern`
- `no ue mac-address mac-address`
- `no ue mask-name mask-name`
- `ue mac-address mac-address [profile profile-name]`

12.2.2.4 ue-mask

Syntax

```
ue-mask {imei | imsi | msisdn} pattern [name mask-name] [profile profile-name] [max-ues  
num]  
  
no ue-mask {imei | imsi | msisdn} pattern  
  
no ue-mask name mask-name  
  
no ue-mask all
```

Context

```
debug>mobile>call-insight
```

Description

This command adds or removes a mask template for call-trace jobs to be generated automatically. If the name is not specified, a default name is used. If the profile is not specified, a default profile is used. If the max-ues is not specified, a default value of 50 is used.

Parameters

all

removes all mask templates

imei

adds or removes an IMEI mask template

imsi

adds or removes an IMSI mask template

msisdn

adds or removes an MSISDN mask template

mask-name

specifies the mask name, up to 16 characters. The default mask name is “default”.

num

specifies the maximum number of UEs. The default is 50.

Values: 1 to 50

pattern

The pattern is a maximum of 16 characters composed of digits 0-9, '*' and '.'

profile-name

specifies an existing profile name. Allowed values are any string up to 32 characters long composed of printable, 7-bit ASCII characters. If the string contains special characters (#, \$, spaces, and so on), the entire string must be enclosed within double quotes.

12.2.3 Profile debug command descriptions

12.2.3.1 profile

Syntax

```
profile
```

Context

```
debug>mobile
```

Description

This command enables the context for debugging mobile gateway profiles.

12.2.3.2 emergency-db-errors

Syntax

```
emergency-db-errors profile-name [csv-file-name csv-file-name]
```

Context

```
debug>mobile>profile
```

Description

This command enables debugging of emergency DB errors. It allows viewing log files that contain errors generated by the CSV file importation via the `ip-range-country-file-format` command.

Parameters

profile-name

specifies the name of the emergency profile available under the `show>mobile>profile>wlan-emergency` command, up to 32 characters

csv-file-name

specifies the name of the CSV file, up to 256 characters

Output

The following example output shows emergency debug profile information.

Example output debug emergency errors

```
# debug mobile-gateway profile emergency-db-errors "ecd" csv-file-name IP2LOCATION-
LITE-DB1-ER.CSV
=====
Profile: ecd
=====
-----
File : ftp://*:~@135.243.193.165/./ws/gash/testsuites/mgEpdg/IP2LOCATION-LITE-DB1-
ER.CSV
MD5   : 13f8bc9a22e760282b2745c916d6f3bd
-----
THU MAR 29 11:33:48 2018 UTC - IP2LOC-
IP4 line 39, "Syntax error" cause 0 error dump: 0x00000000 0x0000055b
THU MAR 29 11:33:48 2018 UTC - IP2LOC-
IP4 line 58, "Syntax error" cause 0 error dump: 0x00000000 0x0000055b
THU MAR 29 11:33:48 2018 UTC - IP2LOC-
IP4 line 59, "Syntax error" cause 0 error dump: 0x00000000 0x0000055b
THU MAR 29 11:33:48 2018 UTC - IP2LOC-
IP4 line 60, "Syntax error" cause 0 error dump: 0x00000000 0x0000055b
THU MAR 29 11:33:48 2018 UTC - IP2LOC-
IP4 line 132, "Syntax error" cause 0 error dump: 0x00000000 0x0000055b
THU MAR 29 11:33:48 2018 UTC - IP2LOC-
IP4 line 163, "Syntax error" cause 0 error dump: 0x00000000 0x0000055b
THU MAR 29 11:33:48 2018 UTC - IP2LOC-
IP4 line 164, "Duplicate record (line)" cause 0 error dump: 0x00004b52 0x00000247
THU MAR 29 11:33:48 2018 UTC - IP2LOC-
IP4 line 164, "Duplicate record (line)" cause 0 error dump: 0x00004b52 0x00000247
THU MAR 29 11:33:48 2018 UTC - IP2LOC-
IP4 line 164, "Duplicate record (line)" cause 0 error dump: 0x00004b52 0x00000247
-----
Number of Files with errors : 1
=====
```

13 VPRN and router command reference

This section describes the VPRN- and router-related commands.

13.1 VPRN and router command hierarchies

This section describes VPRN and router commands that are specifically supported for the 7750 SR MG and the Cloud Mobile Gateway. The 7750 SR MG and Cloud Mobile Gateway also support most of the 7750 SR VPRN and router commands. For descriptions of all 7750 SR VPRN and router commands, refer to the 7750 SR documentation.

- *VPRN command hierarchy*
- *Router command hierarchy*

13.1.1 VPRN command hierarchy

```
configure
- service
- vprn service-id [customer customer-id]
- no vprn service-id
- [no] bgp
- rapid-update [ipv4] [ipv6]
- no rapid-update
- [no] enable-l2-access
- interface ip-int-name [create] [tunnel]
- no interface ip-int-name
- [no] dynamic-sap dynamic-sap-id
- egress
- filter ip ip-filter-id
- filter ipv6 ipv6-filter-id
- no filter [ip ip-filter-id] [ipv6 ipv6-filter-id]
- qos policy-id
- no qos [policy-id]
- ingress
- filter ip ip-filter-id
- filter ipv6 ipv6-filter-id
- no filter [ip ip-filter-id] [ipv6 ipv6-filter-id]
- qos policy-id
- no qos [policy-id]
- steering [ul-from-gw] [dl-from-sfc]
- [no] ip-local-pool name
```

```

    - [no] exclusive
    - group-name group-name
    - no group-name
    - hold-timer-sec hold-timer-in-sec [hold-for-pdn]
    - no hold-timer-sec
    - ipv4-prefix ipv4-address/mask [block [force]] [prefix-fragment-size prefix-fragment-size-value]
    - no ipv4-prefix ipv4-address/mask
    - ipv6-prefix ipv6-address/prefix-length [block [force]] [prefix-fragment-size prefix-fragment-size-value]
    - no ipv6-prefix ipv6-address/prefix-length
    - [no] network-determined
    - next-hop-gw-ipv4 ip-address
    - no next-hop-gw-ipv4
    - next-hop-gw-ipv6 ip-address
    - no next-hop-gw-ipv6
    - pool-usage-threshold high-threshold high-threshold [low-threshold low-threshold]
    - no pool-usage-threshold
    - [no] remote
    - reserved-ip-list ip-address-list-name
    - no reserved-ip-list
    - reserved-ipv4 ipv4-address
    - no reserved-ipv4
    - reserved-ipv6 ipv6-address
    - no reserved-ipv6
    - [no] round-robin
    - [no] suppress-dotzero-broadcast-address
    - [no] suppress-dotzero-broadcast-address
    - transit-sub {fixed | mobile} [protocol {gx | radius | autocreate | mptcp}]
    - no transit-sub
    - [no] use-prefixes-immediately
    - [no] ip-pool-distribution
    - [no] l2tp
    - [no] local-address ip-address
    - group tunnel-group-name [create] [protocol protocol]
    - no group tunnel-group-name
    - [no] local-address ip-address
    - tunnel tunnel-name [create]
    - no tunnel tunnel-name
    - [no] local-address ip-address
    - peer ip-address
    - no peer
    - mg-group group-id [local-ecmp-egress]
    - parent-router router-instance

```

- no *parent-router*
- *twamp-light*
 - *reflector* [udp-port *udp-port-number*] [create]
 - no *reflector*
 - *prefix* *ip-prefix/prefix-length* [create]
 - no *prefix* *ip-prefix/prefix-length*
 - *description* *description-string*
 - no *description*

show

- service
 - *dynamic-sap-using*
 - *mg-group-using*

13.1.2 Router command hierarchy

configure

- *router*
 - [no] *bgp*
 - *rapid-update* [ipv4] [ipv6]
 - no *rapid-update*
 - [no] *interface* *ip-int-name* [unnumbered-mpls-tp]
 - *mc-redirect-mode* [epdg] [home-agent]
 - no *mc-redirect-mode*
 - *steering* [ul-from-gw] [dl-from-sfc]
 - [no] *ip-local-pool* *name*
 - *group-name* *group-name*
 - no *group-name*
 - *hold-timer-sec* *hold-timer-in-sec* [hold-for-pdn]
 - no *hold-timer-sec*
 - *ipv4-prefix* *ipv4-address/mask* [block [force]] [prefix-fragment-size *prefix-fragment-size-value*]
 - no *ipv4-prefix* *ipv4-address/mask*
 - *ipv6-prefix* *ipv6-address/prefix-length* [block [force]] [prefix-fragment-size *prefix-fragment-size-value*]
 - no *ipv6-prefix* *ipv6-address/prefix-length*
 - [no] *network-determined*
 - *next-hop-gw-ipv4* *ip-address*
 - no *next-hop-gw-ipv4*
 - *next-hop-gw-ipv6* *ip-address*
 - no *next-hop-gw-ipv6*
 - *pool-usage-threshold* *high-threshold* *high-threshold* [low-threshold *low-threshold*]

- no *pool-usage-threshold*
- [no] *remote*
- *reserved-ip-list ip-address-list-name*
- no *reserved-ip-list*
- *reserved-ipv4 ipv4-address*
- no *reserved-ipv4*
- *reserved-ipv6 ipv6-address*
- no *reserved-ipv6*
- [no] *round-robin*
- [no] *suppress-dotzero-broadcast-address*
- [no] *suppress-network-broadcast-address*
- *transit-sub* {fixed | mobile} protocol {gx | radius | auto-create}
- no *transit-sub*
- [no] *use-prefixes-immediately*
- [no] *ip-pool-distribution*
- [no] *l2tp*
 - [no] *local-address ip-address*
 - *group tunnel-group-name* [create] [protocol protocol]
 - no *group tunnel-group-name*
 - [no] *local-address ip-address*
 - *tunnel tunnel-name* [create]
 - no *tunnel tunnel-name*
 - [no] *local-address ip-address*
 - *peer ip-address*
 - no *peer*
- [no] *policy-options*
 - [no] *policy-statement name*
 - [no] *entry entry-id*
 - [no] *from*
 - *active-mg vm-id*
 - no *active-mg*
 - *state state*
 - no *state*
- [no] *selective-reassembly*
- *twamp-light*
 - *reflector* [udp-port udp-port-number] [create]
 - no *reflector*
 - *prefix ip-prefix/prefix-length* [create]
 - no *prefix ip-prefix/prefix-length*
 - *description description-string*
 - no *description*

show

- router

– *ip-local-pool-stats name*

13.2 VPRN and router command descriptions

This section describes VPRN and router commands that are specifically supported for the 7750 SR MG and the Cloud Mobile Gateway. The 7750 SR MG and Cloud Mobile Gateway also support most of the 7750 SR VPRN and router commands. For descriptions of all 7750 SR VPRN and router commands, refer to the 7750 SR documentation.

- [VPRN and router configuration command descriptions](#)
- [VPRN and router show command descriptions](#)

13.2.1 VPRN and router configuration command descriptions

13.2.1.1 router

Syntax

router *router-name*

no router

Context

config

Description

This command enables the context to configure router parameters including interfaces, route policies, and protocols.

Default

no router

Parameters

router-name

specifies the router name

Values: base | management

Default: base

13.2.1.2 vprn

Syntax

```
vprn service-id [customer customer-id] [inter-as-mvpn] [create]  
no vprn service-id
```

Context

```
config>service
```

Description

This command creates or edits a VPRN service instance.

If the *service-id* does not exist, a context for the service is created. If the *service-id* exists, the context for editing the service is entered.

VPRN services allow the creation of customer-facing IP interfaces in the same routing instance used for service network core routing connectivity. VPRN services require that the IP addressing scheme used by the subscriber must be unique between it and other addressing schemes used by the provider and potentially the entire Internet.

IP interfaces defined within the context of an VPRN service ID must have a SAP created as the access point to the subscriber network.

When a service is created, the *customer* keyword and *customer-id* must be specified and associates the service with a customer. The *customer-id* must already exist having been created using the *customer* command in the service context. When a service is created with a customer association, it is not possible to edit the customer association. The service must be deleted and re-created with a new customer association.

When a service is created, the use of the *customer customer-id* is optional to navigate into the service configuration context. If attempting to edit a service with the incorrect *customer-id* results in an error.

Multiple VPRN services are created to separate customer-owned IP interfaces. More than one VPRN service can be created for a single customer ID. More than one IP interface can be created within a single VPRN service ID. All IP interfaces created within an VPRN service ID belongs to the same customer.

The *no* form of this command deletes the VPRN service instance with the specified *service-id*. The service cannot be deleted until all the IP interfaces and all routing protocol configurations defined within the service ID have been shutdown and deleted.

Parameters

service-id

the unique service identification number or string identifying the service in the service domain. This ID must be unique to this service and may not be used for any other service of

any type. The *service-id* must be the same number used for every 7750 SR and 7710 SR on which this service is defined.

Values	<i>service-id</i>	1 to 2147483648
–	<i>svc-name</i>	up to 64 characters

customer-id

specifies an existing customer identification number to be associated with the service. This parameter is required on service creation and optional for service editing or deleting.

Values: 1 to 2147483647

inter-as-mvpn

This is a creation-time parameter, indicating that the VPRN service being configured is used for inter-AS MVPN only.

13.2.1.3 bgp

Syntax

[no] bgp

Context

config>service>vprn

Description

This command enables the BGP protocol with the VPRN service.

The no form of this command disables the BGP protocol from the specified VPRN service.

Default

no bgp

13.2.1.4 rapid-update

Syntax

rapid-update [ipv4] [ipv6]

```
no rapid-update
```

Context

```
config>service>vprn>bgp
```

Description

This command configures the BGP rapid update for the specified address families.

When rapid update is enabled for a set of address families, and a route belonging to a family in that set is received by the router and chosen for propagation to certain BGP peers, the remaining time on the MRAI timer of these peers is ignored and the route is transmitted immediately, along with all other pending routes for these peers (including routes of IP address families not specified in the `rapid-update` command).

The `rapid-update` command overrides the peer-level time and applies the minimum setting (0 seconds) to routes belonging to specified address families; routes of other address families continue to be advertised according to the session-level MRAI setting.

The `no` form of this command disables rapid update for all IP address families.

Default

```
no rapid-update
```

Parameters

ipv4

specifies BGP rapid update for the IPv4 address family

ipv6

specifies BGP rapid update for the IPv6 address family

13.2.1.5 enable-l2-access

Syntax

```
[no] enable-l2-access
```

Context

```
config>service>vprn
```

Description

This command enables or disables the L2 access for CMG TWAG.

Default

no enable-l2-access

13.2.1.6 interface

Syntax

```
interface ip-int-name [create]
```

Context

```
config>service>vprn
```

Description

This command creates a logical IP address routing interface for a VPRN. Once created, attributes like an IP address and SAP can be associated with the IP address interface.

The interface command, under the context of services, is used to create and maintain IP address routing interfaces within VPRN service IDs. The interface command can be executed in the context of an VPRN service ID. The IP address interface created is associated with the service core network routing instance and default routing table. The typical use for IP address interfaces created in this manner is for subscriber Internet access.

Parameters

ip-int-name

specifies the name of the IP address interface. Interface names must be unique within the group of defined IP address interfaces for config>router>interface and config>service>vprn>interface commands. An interface name cannot be in the form of an IP address. Interface names can be from 1 to 32 alphanumeric characters. If the string contains special characters (#, \$, spaces, and so on), the entire string must be enclosed within double quotes.

13.2.1.7 mc-redirect-mode

Syntax

```
mc-redirect-mode [epdg] [home-agent]
```

```
no mc-redirect-mode
```

Context

config>router>if

Description

This command specifies the multi-chassis redirect mode.

Parameters

epdg

specifies ePDG multi-chassis redirect mode

home-agent

specifies Home Agent multi-chassis redirect mode

13.2.1.8 dynamic-sap

Syntax

[no] dynamic-sap *dynamic-sap-id*

Context

config>service>vprn>if

Description

This command configures the dynamic SAP that is associated with an interface. A dynamic SAP is configured without any card or VM associated with it. Also, it is mapped to a physical SAP that is based on a VM related to an MG group. For the case of N:K redundancy method, the dynamic SAP allows the physical SAP to move to a VM that becomes active after failure.



NOTE:

- Dynamic SAP and SAP are mutually exclusive.
- A mobile-gateway system group that is used in an N:K redundant system requires a dynamic SAP on its VPRNs and a dynamic MDA in its NAT groups. For more information, see config>mobile>system>group, config>isa>nat-group, and config>isa>nat-group>if>dynamic-mds.

Parameters

dynamic-sap-id

specifies the SAP ID that is associated with the MG group

Values	<i>mg-group-group-id/mda/port-id:encap-val</i>		
–	<i>group-id</i>		1 to 15
–	<i>mda-slot-num</i>		1 to 4
–	<i>port-id</i>		1 to 20
–	<i>encap-val</i>		0 to 4094

13.2.1.9 egress

Syntax

egress

Context

config>service>vpn>if>dynamic-sap

Description

This command enables the context to configure the egress QoS policy for the dynamic SAP.

13.2.1.10 filter

Syntax

filter ip *ip-filter-id*

filter ipv6 *ipv6-filter-id*

no filter [ip *ip-filter-id*] [ipv6 *ipv6-filter-id*]

Context

config>service>vpn>if>dynamic-sap>egress

Description

This command configures an egress IP address or an IPv6 filter for the dynamic SAP. The filter ID or name must first be created in the `config>filter>ip-filter` or in the `config>filter>ipv6-filter` context.

The `no` form of this command removes the specified filter ID or IDs.

Parameters

ip-filter-id

specifies the IP filter ID or name, created in the `config>filter>ip-filter` context

Values	<i>IP filter ID</i>	1 to 65535
–	<i>name</i>	up to 64 characters

ipv6-filter-id

specifies the IPv6 filter ID or name, created in the `config>filter>ipv6-filter` context

Values	<i>IPv6 filter ID</i>	1 to 65535
–	<i>name</i>	up to 64 characters

13.2.1.11 qos

Syntax

```
qos policy-id  
no qos [policy-id]
```

Context

```
config>service>vprn>if>dynamic-sap>egress
```

Description

This command configures the egress QoS policy for the dynamic SAP. The QoS policy must first be created in the `config>qos>sap-egress` context.

Parameters

policy-id

specifies the QoS policy ID or name, created in the `config>qos>sap-egress` context

Values	<i>policy ID</i>	1 to 65535
–	<i>name</i>	up to 65 characters

13.2.1.12 ingress

Syntax

`ingress`

Context

`config>service>vprn>if>dynamic-sap`

Description

This command enables the context to configure the ingress QoS policy for the dynamic SAP.

13.2.1.13 filter

Syntax

`filter ip ip-filter-id`

`filter ipv6 ipv6-filter-id`

`no filter [ip ip-filter-id] [ipv6 ipv6-filter-id]`

Context

`config>service>vprn>if>dynamic-sap>ingress`

Description

This command configures an ingress IP address or IPv6 filter for the dynamic SAP. The filter ID or name must first be created in the `config>filter>ip-filter` or in the `config>filter>ipv6-filter` context.

The `no` form of this command removes the specified filter ID or IDs.

Parameters

ip-filter-id

specifies the IP filter ID or name, created in the `config>filter>ip-filter` context

Values	<i>IP filter ID</i>	1 to 65535
–	<i>name</i>	up to 64 characters

ipv6-filter-id

specifies the IPv6 filter ID or name, created in the `config>filter>ipv6-filter` context

Values	<i>IPv6 filter ID</i>	1 to 65535
–	<i>name</i>	up to 64 characters

13.2.1.14 qos

Syntax

```
qos policy-id  
no qos [policy-id]
```

Context

```
config>service>vprn>if>dynamic-sap>ingress
```


Description

This command configures the ingress QoS policy for the dynamic SAP. The QoS policy must first be created in the `config>qos>sap-ingress` context.

Parameters

policy-id

specifies the QoS policy ID or name, created in the `config>qos>sap-ingress` context

Values	<i>policy ID</i>	1 to 65535
–	<i>name</i>	up to 64 characters

13.2.1.15 steering

Syntax

```
steering [ul-from-gw] [dl-from-sfc]
```

Context

```
config>service>vprn>if
```

```
config>router>if
```

Description

This command enables the steering function based on the direction of the user traffic flow and connectivity to peering nodes.

Parameters

ul-from-gw

configures the interface between a standalone SSG and PGW/GGSN. The *ul-from-gw* keyword enables the source IP-based packet processing for uplink user traffic received from the PGW/GGSN. User traffic received from the PGW/GGSN is processed on the anchor ISM-MG card. Source IP-based packet processing allows locating the anchor ISM-MG card on IOM.

dl-from-sfc

configures the interface between a standalone SSG node (or PGW/GGSN combined with SSG) and the service function chain (SFC). The *dl-from-sfc* keyword enables packet processing

for downlink user traffic received from SFC. User traffic received from the SFC is processed on the anchor ISM-MG card. To avoid sending user traffic from the ISM-MG back to the SFC, a flag is required to indicate that the SFC is done and user traffic must be forwarded based on the destination IP address.

13.2.1.16 ip-local-pool

Syntax

```
[no] ip - local - pool pool-name
```

Context

```
config>service>vprn
```

```
config>router
```

Description

This command configures an IP pool address that can be referenced from an APN context to provide IP addresses for the UEs.

A maximum of eight IP pools can be configured for each APN.

The no form of this command removes the pool name from the configuration.

Parameters

pool-name

specifies a pool name, up to 32 characters

13.2.1.17 exclusive

Syntax

```
[no] exclusive
```

Context

```
config>service>vprn>ip-local-pool
```

```
config>router>ip-local-pool
```

Description

This command configures the `ip - pool` to be APN or tai-lai exclusive.

Default

no exclusive

13.2.1.18 group-name

Syntax

[no] group-name *group-name*

Context

config>router>pool

config>service>vprn>pool

Description

This command configures the group name required for selecting IP address pools used for load balancing. The IP address pools which are configured with the same group name belong to the same IP address group. The no form of this command removes the group name from the configuration.

Parameters

group-name

specifies a group name, up to 32 characters

13.2.1.19 hold-timer-sec

Syntax

hold-timer-sec *hold-timer-in-sec* [hold-for-pdn]

no hold-timer-sec

Context

config>service>vprn>ip-local-pool

config>router>ip-local-pool

Description

This command configures the IP address pool hold timer. This is the time, after the UE or the session is released, that the gateway waits before assigning the IP address to another UE or the session. When

you specify the `hold-for-pdn` keyword, the gateway reassigns the reserved IP address to the UE or the session connection that released it if the UE or the session connection reattaches before the timer expires. The no form of this command removes the configured hold timer value from the configuration.

Parameters

hold-timer-in-sec

specifies the hold timer, in seconds

Values: 0 to 864000

Default: 180

hold-for-pdn

keyword to reallocate a held IP address to the UE/session that released it



NOTICE:

Configuring a high value for the hold timer can lead to IP address exhaustion. If you configure a high value for the hold timer, make sure you have several IP addresses available or enable the `hold-for-pdn` keyword.

13.2.1.20 **ipv4-prefix**

Syntax

```
ipv4-prefix ipv4-address/mask [block [force]] [prefix-fragment-size prefix-fragment-size-value]
```

```
no ipv4-prefix ipv4-address/mask
```

Context

```
config>service>vprn>ip-local-pool
```

```
config>router>ip-local-pool
```

Description

This command adds an IPv4 prefix address to the IP address pool. Multiple prefixes can be added to one IP address pool. The prefix fragment size is defined for parent IP pools in LB-less mode deployments (CUPS and non-CUPS). The IP pools are divided into fragments of this length/size. The system can internally lease fragments to individual MG groups at runtime to optimize IP address usage. A mobile host route is installed in each MG group specific child VPRN for each different fragment assigned to each MG group. Also, this is visible/advertised to site routers for attracting DL traffic.

Parameters

ipv4-address

specifies an IPv4 address

Values: a.b.c.d (host bits must be 0)

mask

specifies the mask value

Values: 9 to 30

block

blocks any new IP address allocation from this prefix

force

clears all sessions that run in the blocked IP pool

prefix-fragment-size-value

specifies the value of the prefix fragment size

Values: 0 | 14 to 24

Default: 0 (disabled)

13.2.1.21 ipv6-prefix

Syntax

```
ipv6-prefix ipv6-address/prefix-length [block [force]] [prefix-fragment-size prefix-fragment-size-value]
```

```
no ipv6-prefix ipv6-address/prefix-length
```

Context

```
config>service>vprn>ip-local-pool
```

```
config>router>ip-local-pool
```

Description

This command adds an IPv6 prefix to the IP address pool. Multiple prefixes can be added to one IP pool. The prefix fragment size is defined for parent IP pools in LB-less mode deployments (CUPS and non-CUPS). The IP pool(s) are divided into fragments of this length/size. The system can internally lease fragments to individual MG groups at runtime to optimize IP address usage. A mobile host route is installed in each MG group specific child VPRN for each different fragment assigned to each MG group. Also, this is visible/advertised to site routers for attracting DL traffic.

Parameters

ipv6-address/prefix-length
specifies an IPv6 address

Values	<i>ipv6-address</i>	x::x::x::x::x::x (eight 16-bit pieces)
–	–	x::x::x::x::d.d.d.d
–	–	x: [0..FFFF]H
–	–	d: [0..255]D
–	<i>prefix-length</i>	41 to 56

block
blocks any new IP address allocation from this prefix

force
clears all sessions that run in the blocked IP pool

prefix-fragment-size-value
specifies the value of the prefix fragment size
Values: 0 | 46 to 56
Default: 0 (disabled)

13.2.1.22 next-hop-gw-ipv4

Syntax

next-hop-gw-ipv4 *ip-address*
no next-hop-gw-ipv4

Context

config>service>vpn>ip-local-pool
config>router>ip-local-pool

Description

This command configures an IPv4 address of a PGW/GGSN to forward downlink traffic on a standalone SSG. Downlink user traffic received on a standalone SSG requires a next-hop gateway address to forward traffic based on the destination IP address of the UE. The PGW/GGSN address can be configured per IP address pool. This command is only applicable when the UE IP address is allocated from a locally configured IP address pool on the PGW/GGSN.

Default

no next-hop-gw-ipv4

Parameters

ip-address

specifies the IPv4 address for the next-hop-gateway IPv4

Values: a.b.c.d

13.2.1.23 next-hop-gw-ipv6

Syntax

next-hop-gw-ipv6 *ip-address*

no next-hop-gw-ipv6

Context

config>service>vprn>ip-local-pool

config>router>ip-local-pool

Description

This command configures the IPv6 address of a PGW/GGSN to forward downlink traffic on a standalone SSG. Downlink user traffic received on a standalone SSG requires a next-hop gateway address to forward traffic based on destination IP address of the UE. The PGW/GGSN address can be configured per IP address pool. This command is only applicable when the UE IP address is allocated from a locally configured IP address pool on the PGW/GGSN.

Default

no next-hop-gw-ipv6

Parameters

ip-address

specifies the IPv6 address for the next-hop-gateway IPv6

Values: up to 64 characters

13.2.1.24 remote

Syntax

[no] remote

Context

config>service>vprn>ip-local-pool

config>router>ip-local-pool

Description

This command configures an IP pool for remote allocation.

The no form of this command removes the value from the configuration.

Default

no remote

13.2.1.25 reserved-ip-list

Syntax

reserved-ip-list *ip-address-list-name*

no reserved-ip-list

Context

config>service>vprn>ip-local-pool

config>router>ip-local-pool

Description

This command configures a list of IP addresses as reserved addresses in the local IP address pool; it is not used for IP address allocation. Up to three IPv4 addresses and/or IPv6 addresses can be reserved from

the configured IP address pool. The reserved IP addresses can then be used for RADIUS or DHCP loopback under the same `apn>vprn-id/base`. This command can also reserve IP addresses to be used with the `ping` command.

The `config>profile>ip-profile` list must be configured first. The list cannot be empty.

The IP addresses configured in the `ip-address-list-name` parameter must belong to the configured prefixes in the local IP address pool and must not be in use.

The `no` form of this command removes the IP address list and releases the reserved IP addresses back to the IP address pool.

**NOTE:**

- When the reserved IP addresses are configured and used for RADIUS or DHCP loopback/client IP addresses, or used with the `ping` command, the `no` form of this command must not be used.
- This command is not allowed if any IP address in the `reserved-ip-list` is already allocated.

Parameters

ip-profile-list-name

specifies the list name of the reserved IP addresses under the `config>profile>ip-profile` command

Values: up to 32 characters

13.2.1.26 reserved-ipv4

Syntax

```
reserved-ipv4 ipv4-address
```

```
no reserved-ipv4
```

Context

```
config>service>vprn>ip-local-pool
```

```
config>router>ip-local-pool
```

Description

This command configures the IPv4 address, as the reserved address of this IP address pool. Only one IPv4 address can be configured as a reserved address in the IP address pool.

The loopback interface must be configured separately with the reserved IP address. The IP address must belong to some of the configured prefixes of the IP address local pool. When the IP address is reserved, it

cannot be allocated to a UE and the other way around. When an IP address has been already assigned to a UE, it cannot be configured as a reserved IP address.

The no form of this command removes the IP address from the configuration and releases the reserved IP address back to the IP address pool. The `tools>perform> router>ip-local-pool>no reserve-ip` command is not needed to release the reserved IP address.

Default

no reserved-ipv4

Parameters

ipv4-address

specifies a valid IPv4 address that belongs to this IP address pool

13.2.1.27 reserved-ipv6

Syntax

`reserved-ipv6 ipv6-address`

`no reserved-ipv6`

Context

`config>service>vprn>ip-local-pool`

`config>router>ip-local-pool`

Description

This command configures the IPv6 address, which is previously reserved by `tools>perform>router>ip-local-pool>reserve-ip`, as the reserved address of this IP address pool. Only one IPv6 address can be configured as a reserved address in the IP address pool.

The loopback interface must be configured separately with the reserved IP address. The IP address must belong to some of the configured prefixes of the IP address local pool. When the IP address is reserved, it cannot be allocated to a UE and the other way around. When an IP address has been already assigned to a UE, it cannot be configured as a reserved IP address.



NOTE:

Only a 64-bit prefix can be reserved.

The no form of this command removes the IP address from the configuration and releases the reserved IP address back to the IP address pool. The `tools>perform>router>ip-local-pool>no reserve-ip` command is not needed to release the reserved IP address.

Default

no reserved-ipv6

Parameters

ipv6-address

specifies a valid IPv6/64 prefix belonging to this IP pool

13.2.1.28 round-robin

Syntax

[no] round-robin

Context

config>service>vprn>ip-local-pool

config>router>ip-local-pool

Description

This command enables the round robin selection mechanism to allocate the IP address. The round robin algorithm is performed across all IP pools and prefixes configured on the APN/VPRN so that an IP address is not reused unless all other IPs under that APN/VPRN are already used.

The no form of this command reverts to the default.

Default

no round-robin

13.2.1.29 suppress-dotzero-broadcast-address

Syntax

[no] suppress-dotzero-broadcast-address

Context

```
config>service>vprn>ip-local-pool
```

```
config>router>ip-local-pool
```

Description

This command configures an IP address pool to suppress allocation of all 0 and 255 IP addresses from the local IP pool to mobile customers.

The no form of this command removes the configuration.

Default

```
no suppress-dotzero-broadcast-address
```

13.2.1.30 suppress-network-broadcast-address

Syntax

```
[no] suppress-network-broadcast-address
```

Context

```
config>service>vprn>ip-local-pool
```

```
config>router>ip-local-pool
```

Description

This command configures an IP address pool to suppress allocation of network (1st IP address of the prefix) and broadcast (last IP address of the prefix) addresses from the local IP address pool to mobile subscribers for IPv4 prefixes which have 128 or fewer IP addresses.

The no form of this command reverts to the default.

Default

```
no suppress-network-broadcast-address
```

13.2.1.31 transit-sub

Syntax

```
transit-sub {fixed | mobile} [protocol {gx | radius | auto-create | mptcp}]
```

```
no transit-sub
```

Context

```
config>service>vprn>ip-local-pool
```

Description

This command is used in SSG to indicate if the local pool configuration is used by fixed or mobile subscribers and to indicate if the subscriber context is created via RADIUS or PCRF.

The no form of this command reverts to the default (not configured).

Parameters

fixed

specifies a fixed transit subscriber type

mobile

specifies a mobile transit subscriber type

gx

specifies a Gx protocol used for creating the subscriber context

radius

specifies a RADIUS protocol used for creating the subscriber context

auto-create

specifies the automatic creation of the subscriber context

mptcp

specifies use of MP-TCP ADD_ADDR for bonding with the primary subscriber context

13.2.1.32 ip-pool-distribution

Syntax

```
[no] ip-pool-distribution
```

Context

```
config>router
```

```
config>service>vprn
```

Description

This command enables IP address load balancing. All the configured IP addresses (pools/prefixes) are allocated according to the round robin algorithm.

The no form of this command reverts to the default.

Default

no ip-pool-distribution

13.2.1.33 l2tp

Syntax

[no] l2tp

Context

config>router

config>service>vprn

Description

This command enables the context to configure L2TP parameters. L2TP extends the PPP model by allowing Layer 2 and PPP endpoints to reside on different devices interconnected by a packet-switched network.

13.2.1.34 local-address

Syntax

local-address *ip-address*

no local-address

Context

config>router>l2tp

config>router>l2tp>group

config>router>l2tp>group>tunnel

config>service>vprn>l2tp>

config>service>vprn>l2tp>group

config>service>vprn>l2tp>group>tunnel

Description

This command configures the local address.

The no form of this command removes the local IP address from the configuration.

Default

no local-address

Parameters

ip-address

specifies the IP address used during L2TP authentication

13.2.1.35 group

Syntax

group *tunnel-group-name* [create] [protocol *protocol*]

no group *tunnel-group-name*

Context

config>router>l2tp

config>service>vprn>l2tp

Description

This command configures an L2TP tunnel group.

Parameters

tunnel-group-name

specifies the of the L2TP tunnel group

Values: up to 63 characters

protocol

specifies the L2TP protocol for use

Values: v2 | v3 | v3draft

Default: v2

13.2.1.36 tunnel

Syntax

```
tunnel tunnel-name [create]
```

```
no tunnel tunnel-name
```

Context

```
config>router>l2tp>group
```

```
config>service>vprn>l2tp>group
```

Description

This command configures an L2TP tunnel. A tunnel exists between a LAC-LNS pair and consists of a Control Connection and zero or more L2TP sessions. The tunnel carries encapsulated PPP datagrams and control messages between the LAC and the L2TP Network Server (LNS).

The no form of this command removes the tunnel name from the configuration.

Parameters

tunnel-name

specifies a valid string to identify an L2TP

Values: up to 32 characters

13.2.1.37 peer

Syntax

```
peer ip-address
```

```
no peer
```

Context

```
config>router>l2tp>group>tunnel
```

```
config>service>vprn>l2tp>group>tunnel
```

Description

This command configures the peer address.

The no form of this command removes the IP address from the tunnel configuration.

Default

no peer

Parameters

ip-address

specifies the IP address of the peer

Values		
	ipv4-address	a.b.c.d
–	ipv6-address	x:x:x:x:x:x:x (eight 16-bit pieces)
–	–	x:x:x:x:x:d.d.d
–	–	x: [0..FFFF]H
–	–	d: [0..255]D

13.2.1.38 mg-group

Syntax

mg-group *group-id* [local-ecmp-address]

no mg-group

Context

config>service>vpn

Description

This command configures an MG group to use with the VPRN routes.

The no form of this command removes the MG group associated with the VPRN.



NOTE:

This command applies only to CMG.

Default

no mg-group

Parameters

group-id

specifies the MG group to associate with the VPRN

Values: 1 to 15

local-ecmp-egress

enables the option to program the local ECMP routes in FIB

13.2.1.39 parent-router

Syntax

parent-router *router-instance*

no parent-router

Context

config>service>vprn

Description

This command configures the router instance of the VPRN parent for this VPRN child.

The no form of this command reverts to the default (disabled).

Default

no parent-router

Parameters

router-instance

specifies the router instance specifies the router instance either by the router name (Base), or by the service ID

Values	router-name vprn-svc-id
--------	---------------------------

–	router-name	Base
–	vprn-svc-id	1 to 2147483647

13.2.1.40 twamp-light

Syntax

twamp-light

Context

config>service>vprn

config>router

Description

This command enables the context to configure the TWAMP Light parameters.

13.2.1.41 reflector

Syntax

reflector [udp-port *udp-port-number*] [create]

no reflector

Context

config>service>vprn>twamp-light

config>router>twamp-light

Description

This command configures a TWAMP Light session reflector parameters and enables TWAMP Light functionality with the no shutdown command. The udp-port keyword and value must be specified with the create keyword. An error message is generated if the specific UDP port is unavailable.

Parameters

udp-port-number

specifies the UDP port number. A strictly enforced restricted range has been introduced. The TWAMP Light session reflector must be brought in line with this new restriction prior upgrading or rebooting from any previous release if there is an active TWAMP Light session reflector configured. Failure to do so prevents an ISSU operation from proceeding and fails to activate any reflector outside of the enforced range. This parameter is required and specifies the destination udp-port that the session reflector uses to listen for TWAMP Light packets. The session controller launching the TWAMP Light packets must be configured with the same destination UDP port as part of the TWAMP Light test. The IES service uses the destination UDP port that is configured under the router context. Only one UDP port can be configured per unique context.

Values: 862 | 63078 to 63087 (CMG), 862 | 64364 to 64373 (7750 SR MG)

create

mandatory keyword when creating a TWAMP Light reflector

13.2.1.42 prefix

Syntax

prefix *ip-prefix/prefix-length* [create]

no prefix *ip-prefix/prefix-length*

Context

config>service>vprn>twamp-light>reflector

config>router>twamp-light>reflector

Description

This command creates or configures a prefix for the TWAMP Light reflector.

The no form of this command removes the prefix from the configuration.

Parameters

ip-prefix/prefix-length

specifies the IP prefix

Values	ipv4-prefix	a.b.c.d (host bits must be 0)
--------	-------------	-------------------------------

–	ipv4-prefix-length	0 to 32
–	ipv6-prefix	x::x::x::x::x::x (eight 16-bit pieces) x::x::x::x::d.d.d.d x - [0..FFFF]H d - [0..255]D
–	ipv6-prefix-length	0 to 128

create

mandatory keyword when creating a TWAMP Light reflector prefix

13.2.1.42.1 description

Syntax

description *description-string*
no description

Context

config>service>vprn>twamp-light>reflector>prefix
config>router>twamp-light>reflector>prefix

Description

This command configures a description for the prefix of the TWAMP Light reflector.

Parameters

description-string

specifies a description string for the reflector prefix
Values: up to 80 characters

13.2.1.43 bgp

Syntax

[no] bgp

Context

config>router

Description

This command enables the context to configure of the base router BGP parameters in the system.

13.2.1.44 rapid-update

Syntax

rapid-update [ipv4] [ipv6]

no rapid-update

Context

config>router>bgp

Description

This command configures the BGP rapid update for the specified IP address families.

When rapid update is enabled for a set of address families, and a route belonging to a family in that set is received by the router and chosen for propagation to certain BGP peers, the remaining time on the MRAI timer of these peers is ignored and the route is transmitted immediately, along with all other pending routes for these peers (including routes of IP address families not specified in the rapid-update command).

The rapid-update command overrides the peer-level time and applies the minimum setting (0 seconds) to routes belonging to specified address families; routes of other address families continue to be advertised according to the session-level MRAI setting.

The no form of this command disables rapid update for all IP address families.

Default

no rapid-update

Parameters

ipv4

specifies BGP rapid update for the IPv4 address family

ipv6

specifies BGP rapid update for the IPv6 address family

13.2.1.45 interface

Syntax

```
interface ip-int-name [unnumbered-mpls-tp]
```

```
no interface ip-int-name
```

Context

```
config>router
```

Description

This command creates a logical IP address routing or unnumbered MPLS-TP interface. Once created, attributes like IP address, port, or system can be associated with the IP address interface.

Interface names are case-sensitive and must be unique within the group of IP address interfaces defined in the `config>router>interface` and `config>service>ies>interface` contexts. Interface names must not be in the dotted decimal notation of an IP address. For example, the name "1.1.1.1" is not allowed, but "int-1.1.1.1" is allowed. Show commands for the router interfaces use either the interface names or the IP addresses. Ambiguity can exist if an IP address is used as an IP address as well as an interface name.

Parameters

ip-int-name

specifies the name of the IP address interface. Interface names must be unique within the group of defined IP address interfaces for `config>router>interface` and `config>service>ies>interface` commands. An interface name cannot be in the form of an IP address. If the string contains special characters (#, \$, spaces, and so on), the entire string must be enclosed within double quotes.

unnumbered-mpls-tp

specifies that an interface is of type Unnumbered MPLS-TP. An unnumbered MPLS-TP interface is only intended for MPLS-TP LSPs. IP routing protocols are blocked on this type of interface. If an interface is configured as an Unnumbered MPLS-TP, it can only be associated with an Ethernet port or VLAN, using the port command. Either a unicast, multicast or

broadcast remote MAC address may be configured using the `static-arp` command. Only static ARP is supported.

13.2.1.46 policy-options

Syntax

[no] `policy-options`

Context

`config>router`

Description

This command enables the context to configure route policies. Route policies are applied to the routing protocol.

The `no` form of this command deletes the route policy configuration.

13.2.1.47 policy-statement

Syntax

[no] `policy-statement` *name*

Context

`config>router>policy-options`

Description

This command creates the context to configure a route policy statement.

Route policy statements control the flow of routing information to and from a specific protocol, set of protocols, or to a specific BGP neighbor.

The `policy-statement` is a logical grouping of match and action criteria. A single `policy-statement` can affect routing in one or more protocols and/or one or more protocols peers/neighbors. A single `policy-statement` can also affect both the import and export of routing information.

The `no` form of this command deletes the policy statement.

Default

`no policy-statement`

Parameters

name

specifies the route policy statement name

Values: up to 32 characters

13.2.1.48 **entry**

Syntax

[no] entry *entry-id*

Context

config>router>policy-options>policy-statement

Description

This command creates the context to edit route policy entries within the route policy statement.

Multiple entries can be created using unique entries. The router exits the filter when the first match is found and executes the action specified. For this reason, entries must be sequenced correctly from most to least explicit.

An entry does not require matching criteria defined (in which case, everything matches) but must have at least define an action to be considered complete. Entries without an action are considered incomplete and will be rendered inactive.

The no form of this command removes the specified entry from the route policy statement.

Parameters

entry-id

The entry ID expressed as a decimal integer. An *entry-id* uniquely identifies match criteria and the corresponding action. It is recommended that multiple entries be given *entry-ids* in staggered increments. This allows users to insert a new entry in an existing policy without requiring renumbering of all the existing entries.

Values: 1 to 4294967295

13.2.1.49 **from**

Syntax

[no] from

Context

config>router>policy-options>policy-statement>entry

Description

This command creates the context to configure policy match criteria based on a route's source or the protocol from which the route is received.

If no condition is specified, all route sources are considered to match.

The no form of this command deletes the source match criteria for the route policy statement entry.

13.2.1.50 active-mg

Syntax

active-mg *vm-id*

no active-mg

Context

config>router>policy-options>policy-statement>entry>from

Description

This command configures the route policy entry to include active VMs. An MG group-specific VM is used to advertise and update the next-hop of an active VM during inter-VM failover.

The no form of this command removes the active MG-VM associated with the route policy entry.



NOTE:

This command applies only to CMG.

Default

no active-mg

Parameters

vm-id

specifies the MG-VM ID to include for the route policy

Values: 1 to 20

13.2.1.51 state

Syntax

state *state*

no state

Context

config>router>policy-options>policy-statement>entry>from

Description

This command configures a criterion that identifies in resilient gateways which routes are associated with an active context and which routes are associated with a standby context.

The no form of this command deletes the criterion.

Parameters

state

specifies the state attribute

Values	srrp-master	Track routes with the state attribute carrying srrp-master state
–	srrp-non-master	Track routes with the state attribute carrying srrp-non-master state
–	ipsec-master-with-peer	Track routes with the state attribute carrying ipsec-master-with-peer state
–	ipsec-non-master	Track routes with the state attribute carrying ipsec-non-master state
–	ipsec-master-without-peer	Track routes with the state attribute carrying ipsec-master-without-peer state
–	mobile-master	Track routes with the state attribute carrying mobile-master state

–	mobile-slave	Track routes with the state attribute carrying mobile-slave state
–	mobile-pre-slave	Track routes with the state attribute carrying mobile-pre-slave state

13.2.1.52 selective-reassembly

Syntax

[no] selective-reassembly

Context

config>router

Description

This command configures the system to reassemble fragmented tunneled packets before forwarding them to the appropriate ISM-MG group for processing. The following is the supported behavior:

- When selective reassembly is enabled, fragmented packets will be sent to the reassembly ISAs, (if IP address reassembly is configured for the ISAs). Non-fragmented packets will not be sent to the reassembly ISAs. Selective reassembly is not supported for the Home Agent. Both fragmented and non-fragmented packets destined to the Home Agent will be sent to the reassembly ISAs, if configured to do so.
- When selective reassembly is disabled, both fragmented and non-fragmented packets will be sent to the reassembly ISAs (if IP address reassembly is configured for the ISAs).

Default

no selective-reassembly

13.2.2 VPRN and router show command descriptions

13.2.2.1 dynamic-sap-using

Syntax

dynamic-sap-using

Context

show>service

Description

This command displays for each MG group, the dynamic SAPs that are associated with the VPRN interface, and the ingress and egress policies and filters that have been applied.

A dynamic SAP is configured without any associated card or VM. Also, it is mapped to a physical SAP that is based on a VM related to an MG group. For the case of N:K redundancy method, the dynamic SAP allows the physical SAP to move to a VM that becomes active after failure.



NOTE:

This command applies only to CMG.

Output

The following output example shows the allocation of dynamic SAPs at VPRN interfaces for specific MG groups and *Show service dynamic-sap-using output fields* describes the output fields.

Example output MG groups with dynamic SAPs

```
# show service dynamic-sap-using
=====
Service Access Points
=====
```

Dynamic Sap id	SvcId	Sap id	Ing. QoS	Ing. Fltr	Egr. QoS	Egr. Fltr	Adm	Opr
mg-group-1/3/1:101	101	[4/3/1:101]	1	none	1	none	Up	Up
mg-group-1/3/2:102	101	[4/3/2:102]	1	none	1	none	Up	Up
mg-group-2/3/1:101	102	[5/3/1:101]	1	none	1	none	Up	Up
mg-group-2/3/2:102	102	[5/3/2:102]	1	none	1	none	Up	Up
mg-group-1/3/1:151	151	[4/3/1:151]	1	none	1	none	Up	Up
mg-group-1/3/2:152	151	[4/3/2:152]	1	none	1	none	Up	Up
mg-group-2/3/1:151	152	[5/3/1:151]	1	none	1	none	Up	Up
mg-group-2/3/2:152	152	[5/3/2:152]	1	none	1	none	Up	Up
mg-group-1/3/1:2001	2001	[4/3/ 1:2001]	1	none	1	none	Up	Up
mg-group-1/3/2:2002	2001	[4/3/ 2:2002]	1	none	1	none	Up	Up
mg-group-2/3/1:2001	2002	[5/3/ 1:2001]	1	none	1	none	Up	Up

```

mg-group-2/3/2:2002 2002      [5/3/      1      none      1      none      Up      Up
                             2:2002]
mg-group-1/3/1:2051 2051      [4/3/      1      none      1      none      Up      Up
                             1:2051]
mg-group-1/3/2:2052 2051      [4/3/      1      none      1      none      Up      Up
                             2:2052]
mg-group-2/3/1:2051 2052      [5/3/      1      none      1      none      Up      Up
                             1:2051]
mg-group-2/3/2:2052 2052      [5/3/      1      none      1      none      Up      Up
                             2:2052]
-----
Number of Dynamic SAPs : 16
-----

```

Table 213: Show service dynamic-sap-using output fields

Label	Description
Dynamic SAP ID	Indicates the dynamic SAP ID associated with the MG system group and VPRN interface The format is: <i>mg-group-group-id/mda/port-id: encap-val</i> <i>group-id</i> : 1 to 15 <i>mda</i> : 1 to 4
SvcId	Indicates the VPRN instance that is associated with a specific MG system group
SAP ID	Indicates the physical SAP ID
Ing-QoS	Indicates the ingress QoS policy ID
Ing-Fltr	Indicates the ingress QoS filter
Egr-QoS	Indicates the egress QoS policy ID
Egr-Fltr	Indicates the egress QoS filter
Adm	Indicates the administratively applied operational state of the SAP (up or down)
Opr	Indicates the operational state of the SAP (up or down)

13.2.2.2 mg-group-using

Syntax

```
mg-group-using
```

Context

```
show>service
```

Description

This command displays for each MG group the VPRNs which are configured to redirect the GTP-U traffic and the SGI VPRNs which allocate all the IP addresses from an IP pool to the respective MG host VPRN.



NOTE:

This command applies only to CMG.

Output

The following output example shows the allocation of VPRN routes at the SGI interface to specific MG groups and *Show service mg-group-using output fields* describes the output fields.

Example output MG groups and VPRNs

```
# show service mg-group-using
=====
MG Group Using
=====
MG-Group      VPRN                      SGI/No of APNs
-----
1             vprn101                  non-SGI
1             vprn151                  non-SGI
1             vprn2001                 SGI/2
1             vprn2051                 SGI/1
-----
2             vprn102                  non-SGI
2             vprn152                  non-SGI
2             vprn2002                 SGI/2
2             vprn2052                 SGI/1
-----
3             vprn103                  non-SGI
3             vprn153                  non-SGI
3             vprn2003                 SGI/2
3             vprn2053                 SGI/1
-----
MG-groupEntries found: 3
=====
```

Table 214: Show service mg-group-using output fields

Label	Description
MG-Group	Indicates the MG group that is associated with a specific VPRN instance at the SGi interface.
VPRN	Indicates the VPRN instance which is associated with a specific MG Group.
SGI/No of APNs	Indicates the number of APNs at the SGi interface which are associated with a specific MG group.

13.2.2.3 ip-local-pool-stats

Syntax

`ip-local-pool-stats name`

Context

`show>router`

Description

This command displays the local IP pool utilization statistics.

Parameters

name

specifies the local IP pool name

Output

The following output examples show local IP pool utilization statistics information.

- *Example output local IP pool utilization statistics for VPRN parent*
- *Example output local IP pool utilization statistics for VPRN child*

Example output local IP pool utilization statistics for VPRN parent

```
=====
IP Local Pool Stats (Router: Base)
```



```

=====
Pool Name       : cups1
Apn-Exclusive  : disabled
Priority        : none
Parent-Router  : -
Suppress-netw* : disabled
Remote-Pool     : enabled
Network-Determ : disabled
=====
  
```

Prefix	Block	Force	Used	Free	Held	Util%	Fragm Size
10.0.0.0/10	No	No	0	4194304	0	0.00	16
3ffe:60:60::/46	No	No	0	252144	0	0.00	47
Total			Used	Free	Held	Util%	
Ipv4			0	270332	0	0.00	
Ipv6			0	262144	0	0.00	

Prefix	No. of Free Fragments	List of Free Fragments
10.0.0.0/22	4	10.0.0.0 10.0.1.0 10.0.2.0 10.0.3.0
3ffe:60:60::/46	2	3ffe:60:60:: 3ffe:60:62:

CLI Reserved Addresses

```

-----
IPv4 Address  : -
IPv6 Address  : -
  
```

```

Reserv IP List: -
-----
  
```

No IP Addresses configured

```

=====
Number of IP pools : 1
=====
  
```

* indicates that the corresponding row element may have been truncated.

Example output local IP pool utilization statistics for VPRN child

```

# show router 121 ip-local-pool-stats "cups1"
=====
  
```

IP Local Pool Stats (Router: vprn121)

```

=====
Pool Name       : cups1
Apn-Exclusive   : disabled           Remote-Pool    : enabled
Network-Deter* : disabled
Parent-Router   : Base
=====

```

```

-----
Prefix
Block Force Used   Free   Held   Util% Fragm Size
-----
21.21.51.0/24
      No    No    0       256    0       0.00   -
-----
Total
      Used   Free   Held   Util%
-----
Ipv4      0     256    0       0.00
Ipv6      0      0      0       0.00
-----

```

CLI Reserved Addresses

```

-----
IPv4 Address : -
IPv6 Address : -
Reserv IP List: -
-----

```

```

No IP Addresses configured
-----

```

```

-----
Gateway  APN
-----
1        agamemnon1
1        agamemnon5.mnc071.mcc588.gprs
1        achtoagacups1.mnc008.mcc244.gprs
1        odytoagacups1.mnc008.mcc244.gprs
1        odytoagacups2.mnc008.mcc244.gprs
-----

```

```

=====
Number of IP pools : 1
=====

```

```

* indicates that the corresponding row element may have been truncated.

```